Oracle® Database

Database Reference



19c
E96196-45
September 2026

Oracle Database Database Reference, 19c

E96196-45

Copyright © 2002, 2026, Oracle and/or its affiliates.

Primary Authors: Bert Rich, Mary Beth Roeser

Contributing Authors: David McDermid, Sarika Surampudi

Contributors: Nipun Agarwal, Rick Anderson, Mark Bauer, Neerja Bhatt, Tudor Bosman, George Candea, Wilson Chan, Sumanta Chatterjee, Eugene Chong, Connie Dialeris Green, Harvey Eneman, Bruce Ernst, Ira Greenberg, Steve Harris, Thuvan Hoang, Lilian Hobbs, Namit Jain, Hakkan Jakobsson, Bob Jenkins, Mark Johnson, Jonathan Klein, Sushil Kumar, Tirthankar Lahiri, Bill Lee, Yunrui Li, Juan Loaiza, Rich Long, Diana Lorentz, Catherine Luu, Neil MacNaughton, Ari Mozes, Gopal Mulagund, Subramanian Muralidhar, Ravi Murthy, Karuna Muthiah, Arvind Nithrakashyap, Ronald Obermarck, Kant Patel, Christopher Racicot, Mark Ramacher, Shankar Raman, Suvendu Ray, Ann Rhee, Kathy Rich, Vivian Schupmann, Debbie Steiner, Malai Stalin, Seema Sundara, Juan Tellez, Alex Tsukerman, Kothanda Umamageswaran, Randy Urbano, Steve Vivian, Eric Voss, Tak Wang, Steve Wertheimer, Andy Witkowski, Daniel Wong, Graham Wood, Brian Wright, Aravind Yalamanchi, Qin Yu, Mohamed Ziauddin

This software and related documentation are provided under a license agreement containing restrictions on use and disclosure and are protected by intellectual property laws. Except as expressly permitted in your license agreement or allowed by law, you may not use, copy, reproduce, translate, broadcast, modify, license, transmit, distribute, exhibit, perform, publish, or display any part, in any form, or by any means. Reverse engineering, disassembly, or decompilation of this software, unless required by law for interoperability, is prohibited.

The information contained herein is subject to change without notice and is not warranted to be error-free. If you find any errors, please report them to us in writing.

If this is software, software documentation, data (as defined in the Federal Acquisition Regulation), or related documentation that is delivered to the U.S. Government or anyone licensing it on behalf of the U.S. Government, then the following notice is applicable:

U.S. GOVERNMENT END USERS: Oracle programs (including any operating system, integrated software, any programs embedded, installed, or activated on delivered hardware, and modifications of such programs) and Oracle computer documentation or other Oracle data delivered to or accessed by U.S. Government end users are "commercial computer software," "commercial computer software documentation," or "limited rights data" pursuant to the applicable Federal Acquisition Regulation and agency-specific supplemental regulations. As such, the use, reproduction, duplication, release, display, disclosure, modification, preparation of derivative works, and/or adaptation of i) Oracle programs (including any operating system, integrated software, any programs embedded, installed, or activated on delivered hardware, and modifications of such programs), ii) Oracle computer documentation and/or iii) other Oracle data, is subject to the rights and limitations specified in the license contained in the applicable contract. The terms governing the U.S. Government's use of Oracle cloud services are defined by the applicable contract for such services. No other rights are granted to the U.S. Government.

This software or hardware is developed for general use in a variety of information management applications. It is not developed or intended for use in any inherently dangerous applications, including applications that may create a risk of personal injury. If you use this software or hardware in dangerous applications, then you shall be responsible to take all appropriate fail-safe, backup, redundancy, and other measures to ensure its safe use. Oracle Corporation and its affiliates disclaim any liability for any damages caused by use of this software or hardware in dangerous applications.

Oracle®, Java, MySQL, and NetSuite are registered trademarks of Oracle and/or its affiliates. Other names may be trademarks of their respective owners.

Intel and Intel Inside are trademarks or registered trademarks of Intel Corporation. All SPARC trademarks are used under license and are trademarks or registered trademarks of SPARC International, Inc. AMD, Epyc, and the AMD logo are trademarks or registered trademarks of Advanced Micro Devices. UNIX is a registered trademark of The Open Group.

This software or hardware and documentation may provide access to or information about content, products, and services from third parties. Oracle Corporation and its affiliates are not responsible for and expressly disclaim all warranties of any kind with respect to third-party content, products, and services unless otherwise set forth in an applicable agreement between you and Oracle. Oracle Corporation and its affiliates will not be responsible for any loss, costs, or damages incurred due to your access to or use of third-party content, products, or services, except as set forth in an applicable agreement between you and Oracle.

For information about Oracle's commitment to accessibility, visit the Oracle Accessibility Program website at http://www.oracle.com/pls/topic/lookup?ctx=acc&id=docacc.

Oracle customer access to and use of Oracle support services will be pursuant to the terms and conditions specified in their Oracle order for the applicable services.

Contents

Preface

Audience i
Documentation Accessibility i
Diversity and Inclusion i
Related Documents ii
Conventions ii

Changes in This Release for Oracle Database Reference

Changes in Oracle Database 19c, Release Update 19.29 i
Changes in Oracle Database 19c, Release Update 19.28 i
Changes in Oracle Database 19c, Release Update 19.26 ii
Changes in Oracle Database 19c, Release Update 19.23 ii
Changes in Oracle Database 19c, Release Update 19.22 ii
Changes in Oracle Database 19c, Release Update 19.21 ii
Changes in Oracle Database 19c, Release Update 19.16 ii
Changes in Oracle Database 19c, Release Update 19.13 ii
Changes in Oracle Database 19c, Release Update 19.11 iii
Changes in Oracle Database 19c, Release Update 19.10 iii
Changes in Oracle Database 19c, Release Update 19.6 iv
Changes in Oracle Database 19c, Release Update 19.4 iv
Changes in Oracle Database 19c iv

Part I Initialization Parameters

1 Initialization Parameters

1.1 Uses of Initialization Parameters 1
1.1.1 Types of Initialization Parameters 2
1.1.1.1 Derived Parameters 2
1.1.1.2 Operating System-Dependent Parameters 2
1.1.1.3 Variable Parameters 2
1.2 Basic Initialization Parameters 3

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.3 Parameter Files 3
1.3.1 Server Parameter Files 3
1.3.2 Initialization Parameter Files 4
1.3.2.1 About the Character Set of Parameter Values 4
1.3.2.2 Specifying Values in an Initialization Parameter File 5
1.4 Changing Parameter Values in a Parameter File 7
1.4.1 Parameters by Functional Category 7
1.4.2 Modifiable Parameters 16
1.4.3 Displaying Current Parameter Values 23
1.4.4 Parameters You Should Not Specify in the Parameter File 24
1.4.5 When Parameters Are Set Incorrectly 24
1.5 Reading the Parameter Descriptions 24
1.6 Initialization Parameter Descriptions 25
1.7 ACTIVE_INSTANCE_COUNT 26
1.8 ADG_ACCOUNT_INFO_TRACKING 27
1.9 ADG_REDIRECT_DML 27
1.10 ALLOW_GLOBAL_DBLINKS 28
1.11 ALLOW_GROUP_ACCESS_TO_SGA 29
1.12 ALLOW_ROWID_COLUMN_TYPE 29
1.13 APPROX_FOR_AGGREGATION 30
1.14 APPROX_FOR_COUNT_DISTINCT 31
1.15 APPROX_FOR_PERCENTILE 31
1.16 AQ_TM_PROCESSES 32
1.17 ARCHIVE_LAG_TARGET 33
1.18 ASM_DISKGROUPS 34
1.19 ASM_DISKSTRING 35
1.20 ASM_IO_PROCESSES 36
1.21 ASM_POWER_LIMIT 36
1.22 ASM_PREFERRED_READ_FAILURE_GROUPS 37
1.23 AUDIT_FILE_DEST 38
1.24 AUDIT_SYS_OPERATIONS 39
1.25 AUDIT_SYSLOG_LEVEL 40
1.26 AUDIT_TRAIL 41
1.27 AUTOTASK_MAX_ACTIVE_PDBS 43
1.28 AWR_PDB_AUTOFLUSH_ENABLED 43
1.29 AWR_PDB_MAX_PARALLEL_SLAVES 45
1.30 AWR_SNAPSHOT_TIME_OFFSET 45
1.31 BACKGROUND_CORE_DUMP 46
1.32 BACKGROUND_DUMP_DEST 47
1.33 BACKUP_TAPE_IO_SLAVES 47
1.34 BITMAP_MERGE_AREA_SIZE 48
1.35 BLANK_TRIMMING 49

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.36 BLOCKCHAIN_TABLE_MAX_NO_DROP 50
1.37 CIRCUITS 51
1.38 CLIENT_RESULT_CACHE_LAG 51
1.39 CLIENT_RESULT_CACHE_SIZE 52
1.40 CLIENT_STATISTICS_LEVEL 52
1.41 CLONEDB 54
1.42 CLONEDB_DIR 54
1.43 CLUSTER_DATABASE 55
1.44 CLUSTER_DATABASE_INSTANCES 55
1.45 CLUSTER_INTERCONNECTS 56
1.46 COMMIT_LOGGING 57
1.47 COMMIT_POINT_STRENGTH 57
1.48 COMMIT_WAIT 58
1.49 COMMIT_WRITE 59
1.50 COMMON_USER_PREFIX 59
1.51 COMPATIBLE 61
1.52 CONNECTION_BROKERS 62
1.53 CONTAINER_DATA 63
1.54 CONTAINERS_PARALLEL_DEGREE 64
1.55 CONTROL_FILE_RECORD_KEEP_TIME 65
1.56 CONTROL_FILES 66
1.57 CONTROL_MANAGEMENT_PACK_ACCESS 66
1.58 CORE_DUMP_DEST 67
1.59 CPU_COUNT 68
1.60 CPU_MIN_COUNT 69
1.61 CREATE_BITMAP_AREA_SIZE 70
1.62 CREATE_STORED_OUTLINES 71
1.63 CURSOR_BIND_CAPTURE_DESTINATION 71
1.64 CURSOR_INVALIDATION 72
1.65 CURSOR_SHARING 73
1.66 CURSOR_SPACE_FOR_TIME 73
1.67 DATA_GUARD_MAX_IO_TIME 75
1.68 DATA_GUARD_MAX_LONGIO_TIME 75
1.69 DATA_GUARD_SYNC_LATENCY 76
1.70 DATA_TRANSFER_CACHE_SIZE 77
1.71 DB_nK_CACHE_SIZE 77
1.72 DB_BIG_TABLE_CACHE_PERCENT_TARGET 78
1.73 DB_BLOCK_BUFFERS 79
1.74 DB_BLOCK_CHECKING 80
1.75 DB_BLOCK_CHECKSUM 82
1.76 DB_BLOCK_SIZE 83
1.77 DB_CACHE_ADVICE 83

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.78 DB_CACHE_SIZE 84
1.79 DB_CREATE_FILE_DEST 85
1.80 DB_CREATE_ONLINE_LOG_DEST_n 86
1.81 DB_DOMAIN 86
1.82 DB_FILE_MULTIBLOCK_READ_COUNT 87
1.83 DB_FILE_NAME_CONVERT 88
1.84 DB_FILES 90
1.85 DB_FLASH_CACHE_FILE 90
1.86 DB_FLASH_CACHE_SIZE 91
1.87 DB_FLASHBACK_RETENTION_TARGET 92
1.88 DB_INDEX_COMPRESSION_INHERITANCE 92
1.89 DB_KEEP_CACHE_SIZE 94
1.90 DB_LOST_WRITE_PROTECT 94
1.91 DB_NAME 95
1.92 DB_PERFORMANCE_PROFILE 96
1.93 DB_RECOVERY_FILE_DEST 97
1.94 DB_RECOVERY_FILE_DEST_SIZE 98
1.95 DB_RECYCLE_CACHE_SIZE 98
1.96 DB_SECUREFILE 99
1.97 DB_ULTRA_SAFE 100
1.98 DB_UNIQUE_NAME 101
1.99 DB_UNRECOVERABLE_SCN_TRACKING 102
1.100 DB_WRITER_PROCESSES 102
1.101 DBFIPS_140 103
1.102 DBWR_IO_SLAVES 104
1.103 DDL_LOCK_TIMEOUT 105
1.104 DEFAULT_SHARING 105
1.105 DEFERRED_SEGMENT_CREATION 106
1.106 DG_BROKER_CONFIG_FILEn 106
1.107 DG_BROKER_START 107
1.108 DIAGNOSTIC_DEST 107
1.109 DISK_ASYNCH_IO 108
1.110 DISPATCHERS 109
1.111 DISTRIBUTED_LOCK_TIMEOUT 111
1.112 DML_LOCKS 112
1.113 DNFS_BATCH_SIZE 113
1.114 DRCP_DEDICATED_OPT 113
1.115 DST_UPGRADE_INSERT_CONV 114
1.116 ENABLE_AUTOMATIC_MAINTENANCE_PDB 115
1.117 ENABLE_DDL_LOGGING 116
1.118 ENABLE_DNFS_DISPATCHER 117
1.119 ENABLE_GOLDENGATE_REPLICATION 118

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.120 ENABLE_IMC_WITH_MIRA 119
1.121 ENABLE_PLUGGABLE_DATABASE 119
1.122 ENABLED_PDBS_ON_STANDBY 120
1.123 ENCRYPT_NEW_TABLESPACES 121
1.124 EVENT 122
1.125 EXTERNAL_KEYSTORE_CREDENTIAL_LOCATION 122
1.126 FAL_CLIENT 123
1.127 FAL_SERVER 124
1.128 FAST_START_MTTR_TARGET 124
1.129 FAST_START_PARALLEL_ROLLBACK 125
1.130 FILE_MAPPING 126
1.131 FILEIO_NETWORK_ADAPTERS 126
1.132 FILESYSTEMIO_OPTIONS 127
1.133 FIXED_DATE 127
1.134 FORWARD_LISTENER 128
1.135 GCS_SERVER_PROCESSES 129
1.136 GLOBAL_NAMES 130
1.137 GLOBAL_TXN_PROCESSES 130
1.138 HASH_AREA_SIZE 131
1.139 HEARTBEAT_BATCH_SIZE 131
1.140 HEAT_MAP 132
1.141 HI_SHARED_MEMORY_ADDRESS 132
1.142 HS_AUTOREGISTER 133
1.143 IDENTITY_PROVIDER_CONFIG 134
1.144 IDENTITY_PROVIDER_TYPE 135
1.145 IFILE 136
1.146 IGNORE_SESSION_SET_PARAM_ERRORS 137
1.147 INMEMORY_ADG_ENABLED 138
1.148 INMEMORY_AUTOMATIC_LEVEL 139
1.149 INMEMORY_CLAUSE_DEFAULT 140
1.150 INMEMORY_EXPRESSIONS_USAGE 144
1.151 INMEMORY_FORCE 145
1.152 INMEMORY_MAX_POPULATE_SERVERS 146
1.153 INMEMORY_OPTIMIZED_ARITHMETIC 148
1.154 INMEMORY_QUERY 148
1.155 INMEMORY_SIZE 149
1.156 INMEMORY_TRICKLE_REPOPULATE_SERVERS_PERCENT 150
1.157 INMEMORY_VIRTUAL_COLUMNS 151
1.158 INSTANCE_ABORT_DELAY_TIME 152
1.159 INSTANCE_GROUPS 153
1.160 INSTANCE_MODE 153
1.161 INSTANCE_NAME 154

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.162 INSTANCE_NUMBER 155
1.163 INSTANCE_TYPE 155
1.164 JAVA_JIT_ENABLED 156
1.165 JAVA_MAX_SESSIONSPACE_SIZE 157
1.166 JAVA_POOL_SIZE 157
1.167 JAVA_SOFT_SESSIONSPACE_LIMIT 158
1.168 JOB_QUEUE_PROCESSES 158
1.169 LARGE_POOL_SIZE 160
1.170 LDAP_DIRECTORY_ACCESS 161
1.171 LDAP_DIRECTORY_SYSAUTH 163
1.172 LICENSE_MAX_SESSIONS 163
1.173 LICENSE_MAX_USERS 164
1.174 LICENSE_SESSIONS_WARNING 165
1.175 LISTENER_NETWORKS 166
1.176 LOB_SIGNATURE_ENABLE 167
1.177 LOCAL_LISTENER 167
1.178 LOCK_NAME_SPACE 168
1.179 LOCK_SGA 168
1.180 LOG_ARCHIVE_CONFIG 169
1.181 LOG_ARCHIVE_DEST 170
1.182 LOG_ARCHIVE_DEST_n 171
1.183 LOG_ARCHIVE_DEST_STATE_n 175
1.184 LOG_ARCHIVE_DUPLEX_DEST 176
1.185 LOG_ARCHIVE_FORMAT 177
1.186 LOG_ARCHIVE_MAX_PROCESSES 179
1.187 LOG_ARCHIVE_MIN_SUCCEED_DEST 179
1.188 LOG_ARCHIVE_TRACE 180
1.189 LOG_BUFFER 183
1.190 LOG_CHECKPOINT_INTERVAL 183
1.191 LOG_CHECKPOINT_TIMEOUT 184
1.192 LOG_CHECKPOINTS_TO_ALERT 185
1.193 LOG_FILE_NAME_CONVERT 185
1.194 LONG_MODULE_ACTION 187
1.195 MAIN_WORKLOAD_TYPE 187
1.196 MAX_AUTH_SERVERS 188
1.197 MAX_DATAPUMP_JOBS_PER_PDB 188
1.198 MAX_DATAPUMP_PARALLEL_PER_JOB 189
1.199 MAX_DISPATCHERS 190
1.200 MAX_DUMP_FILE_SIZE 190
1.201 MAX_IDLE_BLOCKER_TIME 191
1.202 MAX_IDLE_TIME 192
1.203 MAX_IOPS 193

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.204 MAX_MBPS 194
1.205 MAX_PDBS 195
1.206 MAX_SHARED_SERVERS 195
1.207 MAX_STRING_SIZE 196
1.208 MEMOPTIMIZE_POOL_SIZE 201
1.209 MEMORY_MAX_TARGET 202
1.210 MEMORY_TARGET 203
1.211 MFA_DUO_API_HOST 204
1.212 MFA_OMA_IAM_DOMAIN_URL 205
1.213 MFA_SENDER_EMAIL_DISPLAYNAME 206
1.214 MFA_SENDER_EMAIL_ID 206
1.215 MFA_SMTP_HOST 207
1.216 MFA_SMTP_PORT 208
1.217 MIN_AUTH_SERVERS 209
1.218 MULTISHARD_QUERY_DATA_CONSISTENCY 209
1.219 NLS_CALENDAR 210
1.220 NLS_COMP 211
1.221 NLS_CURRENCY 212
1.222 NLS_DATE_FORMAT 213
1.223 NLS_DATE_LANGUAGE 214
1.224 NLS_DUAL_CURRENCY 215
1.225 NLS_ISO_CURRENCY 215
1.226 NLS_LANGUAGE 216
1.227 NLS_LENGTH_SEMANTICS 217
1.228 NLS_NCHAR_CONV_EXCP 218
1.229 NLS_NUMERIC_CHARACTERS 219
1.230 NLS_SORT 219
1.231 NLS_TERRITORY 221
1.232 NLS_TIMESTAMP_FORMAT 221
1.233 NLS_TIMESTAMP_TZ_FORMAT 222
1.234 NONCDB_COMPATIBLE 223
1.235 OBJECT_CACHE_MAX_SIZE_PERCENT 224
1.236 OBJECT_CACHE_OPTIMAL_SIZE 225
1.237 OFS_THREADS 225
1.238 OGG_ONLINE_DDL 226
1.239 OLAP_PAGE_POOL_SIZE 226
1.240 ONE_STEP_PLUGIN_FOR_PDB_WITH_TDE 227
1.241 OPEN_CURSORS 228
1.242 OPEN_LINKS 228
1.243 OPEN_LINKS_PER_INSTANCE 229
1.244 OPTIMIZER_ADAPTIVE_PLANS 230
1.245 OPTIMIZER_ADAPTIVE_REPORTING_ONLY 230

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.246 OPTIMIZER_ADAPTIVE_STATISTICS 231
1.247 OPTIMIZER_CAPTURE_SQL_PLAN_BASELINES 232
1.248 OPTIMIZER_CAPTURE_SQL_QUARANTINE 232
1.249 OPTIMIZER_DYNAMIC_SAMPLING 233
1.250 OPTIMIZER_FEATURES_ENABLE 234
1.251 OPTIMIZER_IGNORE_HINTS 236
1.252 OPTIMIZER_IGNORE_PARALLEL_HINTS 236
1.253 OPTIMIZER_INDEX_CACHING 237
1.254 OPTIMIZER_INDEX_COST_ADJ 237
1.255 OPTIMIZER_INMEMORY_AWARE 238
1.256 OPTIMIZER_MODE 238
1.257 OPTIMIZER_REAL_TIME_STATISTICS 239
1.258 OPTIMIZER_SECURE_VIEW_MERGING 240
1.259 OPTIMIZER_SESSION_TYPE 241
1.260 OPTIMIZER_USE_INVISIBLE_INDEXES 241
1.261 OPTIMIZER_USE_PENDING_STATISTICS 242
1.262 OPTIMIZER_USE_SQL_PLAN_BASELINES 242
1.263 OPTIMIZER_USE_SQL_QUARANTINE 243
1.264 OS_AUTHENT_PREFIX 244
1.265 OS_ROLES 245
1.266 OUTBOUND_DBLINK_PROTOCOLS 245
1.267 PARALLEL_ADAPTIVE_MULTI_USER 246
1.268 PARALLEL_DEGREE_LIMIT 247
1.269 PARALLEL_DEGREE_POLICY 248
1.270 PARALLEL_EXECUTION_MESSAGE_SIZE 249
1.271 PARALLEL_FORCE_LOCAL 250
1.272 PARALLEL_INSTANCE_GROUP 250
1.273 PARALLEL_MAX_SERVERS 251
1.274 PARALLEL_MIN_DEGREE 252
1.275 PARALLEL_MIN_PERCENT 253
1.276 PARALLEL_MIN_SERVERS 254
1.277 PARALLEL_MIN_TIME_THRESHOLD 255
1.278 PARALLEL_SERVERS_TARGET 255
1.279 PARALLEL_THREADS_PER_CPU 257
1.280 PDB_FILE_NAME_CONVERT 258
1.281 PDB_LOCKDOWN 259
1.282 PDB_OS_CREDENTIAL 261
1.283 PERMIT_92_WRAP_FORMAT 262
1.284 PGA_AGGREGATE_LIMIT 262
1.285 PGA_AGGREGATE_TARGET 264
1.286 PLSCOPE_SETTINGS 266
1.287 PLSQL_CCFLAGS 267

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.288 PLSQL_CODE_TYPE 268
1.289 PLSQL_DEBUG 269
1.290 PLSQL_OPTIMIZE_LEVEL 270
1.291 PLSQL_V2_COMPATIBILITY 271
1.292 PLSQL_WARNINGS 271
1.293 PRE_PAGE_SGA 273
1.294 PRIVATE_TEMP_TABLE_PREFIX 274
1.295 PROCESSES 275
1.296 PROCESSOR_GROUP_NAME 275
1.297 QUERY_REWRITE_ENABLED 276
1.298 QUERY_REWRITE_INTEGRITY 277
1.299 RDBMS_SERVER_DN 278
1.300 READ_ONLY_OPEN_DELAYED 279
1.301 RECOVERY_PARALLELISM 280
1.302 RECYCLEBIN 280
1.303 REDO_TRANSPORT_USER 281
1.304 REMOTE_DEPENDENCIES_MODE 281
1.305 REMOTE_LISTENER 282
1.306 REMOTE_LOGIN_PASSWORDFILE 282
1.307 REMOTE_OS_AUTHENT 284
1.308 REMOTE_OS_ROLES 284
1.309 REMOTE_RECOVERY_FILE_DEST 285
1.310 REPLICATION_DEPENDENCY_TRACKING 285
1.311 RESOURCE_LIMIT 286
1.312 RESOURCE_MANAGE_GOLDENGATE 286
1.313 RESOURCE_MANAGER_CPU_ALLOCATION 287
1.314 RESOURCE_MANAGER_PLAN 288
1.315 RESULT_CACHE_MAX_RESULT 289
1.316 RESULT_CACHE_MAX_SIZE 290
1.317 RESULT_CACHE_MODE 290
1.318 RESULT_CACHE_REMOTE_EXPIRATION 291
1.319 RESUMABLE_TIMEOUT 292
1.320 ROLLBACK_SEGMENTS 292
1.321 SEC_CASE_SENSITIVE_LOGON 293
1.322 SEC_MAX_FAILED_LOGIN_ATTEMPTS 294
1.323 SEC_PROTOCOL_ERROR_FURTHER_ACTION 295
1.324 SEC_PROTOCOL_ERROR_TRACE_ACTION 295
1.325 SEC_RETURN_SERVER_RELEASE_BANNER 296
1.326 SERIAL_REUSE 297
1.327 SERVICE_NAMES 298
1.328 SESSION_CACHED_CURSORS 299
1.329 SESSION_EXIT_ON_PACKAGE_STATE_ERROR 300

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.330 SESSION_MAX_OPEN_FILES 300
1.331 SESSIONS 301
1.332 SGA_MAX_SIZE 302
1.333 SGA_MIN_SIZE 303
1.334 SGA_TARGET 304
1.335 SHADOW_CORE_DUMP 306
1.336 SHARED_MEMORY_ADDRESS 307
1.337 SHARED_POOL_RESERVED_SIZE 307
1.338 SHARED_POOL_SIZE 308
1.339 SHARED_SERVER_SESSIONS 309
1.340 SHARED_SERVERS 310
1.341 SHRD_DUPL_TABLE_REFRESH_RATE 311
1.342 SKIP_UNUSABLE_INDEXES 311
1.343 SMTP_OUT_SERVER 312
1.344 SORT_AREA_RETAINED_SIZE 313
1.345 SORT_AREA_SIZE 314
1.346 SPATIAL_VECTOR_ACCELERATION 315
1.347 SPFILE 315
1.348 SQL_TRACE 316
1.349 SQL92_SECURITY 317
1.350 SQLTUNE_CATEGORY 318
1.351 STANDBY_DB_PRESERVE_STATES 318
1.352 STANDBY_FILE_MANAGEMENT 319
1.353 STANDBY_PDB_SOURCE_FILE_DBLINK 320
1.354 STANDBY_PDB_SOURCE_FILE_DIRECTORY 321
1.355 STAR_TRANSFORMATION_ENABLED 321
1.356 STATISTICS_LEVEL 322
1.357 STREAMS_POOL_SIZE 323
1.358 TABLESPACE_ENCRYPTION 324
1.359 TAPE_ASYNCH_IO 326
1.360 TDE_CONFIGURATION 327
1.361 TEMP_UNDO_ENABLED 329
1.362 THREAD 330
1.363 THREADED_EXECUTION 331
1.364 TIMED_OS_STATISTICS 332
1.365 TIMED_STATISTICS 333
1.366 TRACE_ENABLED 333
1.367 TRACEFILE_IDENTIFIER 334
1.368 TRANSACTIONS 335
1.369 TRANSACTIONS_PER_ROLLBACK_SEGMENT 335
1.370 UNDO_MANAGEMENT 336
1.371 UNDO_RETENTION 336

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.372 UNDO_TABLESPACE 338
1.373 UNIFIED_AUDIT_COMMON_SYSTEMLOG 339
1.374 UNIFIED_AUDIT_SGA_QUEUE_SIZE 340
1.375 UNIFIED_AUDIT_SYSTEMLOG 341
1.376 UNIFORM_LOG_TIMESTAMP_FORMAT 342
1.377 USE_DEDICATED_BROKER 342
1.378 USE_LARGE_PAGES 343
1.379 USER_DUMP_DEST 345
1.380 WALLET_ROOT 346
1.381 WORKAREA_SIZE_POLICY 349
1.382 XML_DB_EVENTS 350

Part II Static Data Dictionary Views

2 Static Data Dictionary Views: ALL_ALL_TABLES to ALL_OUTLINES

2.1 About Static Data Dictionary Views 1
2.2 AWR_PDB_* Views 2
2.3 AWR_ROOT_* Views 2
2.4 CDB_* Views 3
2.5 DBA_HIST_* Views 3
2.6 Oracle Database Real Application Security Views 4
2.7 Oracle Database Vault Views 4
2.8 Oracle Label Security Views 4
2.9 Oracle Workspace Manager Views 5
2.10 Recovery Catalog Views 6
2.11 Static Data Dictionary View Descriptions 7
2.12 ALL_ALL_TABLES 7
2.13 ALL_ANALYTIC_VIEW_ATTR_CLASS 13
2.14 ALL_ANALYTIC_VIEW_ATTR_CLS 14
2.15 ALL_ANALYTIC_VIEW_ATTR_CLS_AE 14
2.16 ALL_ANALYTIC_VIEW_BAS_MEAS 15
2.17 ALL_ANALYTIC_VIEW_BAS_MEAS_AE 15
2.18 ALL_ANALYTIC_VIEW_BASE_MEAS 16
2.19 ALL_ANALYTIC_VIEW_CALC_MEAS 17
2.20 ALL_ANALYTIC_VIEW_CLASS 18
2.21 ALL_ANALYTIC_VIEW_CLASS_AE 19
2.22 ALL_ANALYTIC_VIEW_CLC_MEAS 20
2.23 ALL_ANALYTIC_VIEW_CLC_MEAS_AE 20
2.24 ALL_ANALYTIC_VIEW_COLUMNS 21
2.25 ALL_ANALYTIC_VIEW_COLUMNS_AE 22

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.26 ALL_ANALYTIC_VIEW_DIM_CLASS 24
2.27 ALL_ANALYTIC_VIEW_DIM_CLS 25
2.28 ALL_ANALYTIC_VIEW_DIM_CLS_AE 25
2.29 ALL_ANALYTIC_VIEW_DIMENSIONS 26
2.30 ALL_ANALYTIC_VIEW_DIMS 27
2.31 ALL_ANALYTIC_VIEW_DIMS_AE 27
2.32 ALL_ANALYTIC_VIEW_HIER_CLASS 29
2.33 ALL_ANALYTIC_VIEW_HIER_CLS 29
2.34 ALL_ANALYTIC_VIEW_HIER_CLS_AE 30
2.35 ALL_ANALYTIC_VIEW_HIERS 31
2.36 ALL_ANALYTIC_VIEW_HIERS_AE 32
2.37 ALL_ANALYTIC_VIEW_KEYS 33
2.38 ALL_ANALYTIC_VIEW_KEYS_AE 33
2.39 ALL_ANALYTIC_VIEW_LEVEL_CLASS 34
2.40 ALL_ANALYTIC_VIEW_LEVELS 35
2.41 ALL_ANALYTIC_VIEW_LEVELS_AE 36
2.42 ALL_ANALYTIC_VIEW_LVL_CLS 37
2.43 ALL_ANALYTIC_VIEW_LVL_CLS_AE 37
2.44 ALL_ANALYTIC_VIEW_LVLGRPS 38
2.45 ALL_ANALYTIC_VIEW_LVLGRPS_AE 39
2.46 ALL_ANALYTIC_VIEW_MEAS_CLASS 40
2.47 ALL_ANALYTIC_VIEW_MEAS_CLS 41
2.48 ALL_ANALYTIC_VIEW_MEAS_CLS_AE 41
2.49 ALL_ANALYTIC_VIEWS 42
2.50 ALL_ANALYTIC_VIEWS_AE 43
2.51 ALL_ANNOTATION_VALUES 44
2.52 ALL_ANNOTATIONS 45
2.53 ALL_ANNOTATIONS_USAGE 45
2.54 ALL_APPLY 46
2.55 ALL_APPLY_CHANGE_HANDLERS 48
2.56 ALL_APPLY_CONFLICT_COLUMNS 49
2.57 ALL_APPLY_DML_CONF_HANDLERS 50
2.58 ALL_APPLY_DML_HANDLERS 50
2.59 ALL_APPLY_ENQUEUE 51
2.60 ALL_APPLY_ERROR 52
2.61 ALL_APPLY_ERROR_MESSAGES 53
2.62 ALL_APPLY_EXECUTE 55
2.63 ALL_APPLY_HANDLE_COLLISIONS 55
2.64 ALL_APPLY_INSTANTIATED_GLOBAL 56
2.65 ALL_APPLY_INSTANTIATED_OBJECTS 57
2.66 ALL_APPLY_INSTANTIATED_SCHEMAS 57
2.67 ALL_APPLY_KEY_COLUMNS 58

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.68 ALL_APPLY_PARAMETERS 58
2.69 ALL_APPLY_PROGRESS 59
2.70 ALL_APPLY_REPERROR_HANDLERS 60
2.71 ALL_APPLY_TABLE_COLUMNS 61
2.72 ALL_ARGUMENTS 62
2.73 ALL_ASSEMBLIES 64
2.74 ALL_ASSOCIATIONS 65
2.75 ALL_ATTRIBUTE_DIM_ATTR_CLASS 66
2.76 ALL_ATTRIBUTE_DIM_ATTR_CLS 66
2.77 ALL_ATTRIBUTE_DIM_ATTR_CLS_AE 67
2.78 ALL_ATTRIBUTE_DIM_ATTRS 68
2.79 ALL_ATTRIBUTE_DIM_ATTRS_AE 68
2.80 ALL_ATTRIBUTE_DIM_CLASS 69
2.81 ALL_ATTRIBUTE_DIM_CLASS_AE 70
2.82 ALL_ATTRIBUTE_DIM_JN_PTHS 71
2.83 ALL_ATTRIBUTE_DIM_JN_PTHS_AE 71
2.84 ALL_ATTRIBUTE_DIM_JOIN_PATHS 72
2.85 ALL_ATTRIBUTE_DIM_KEYS 73
2.86 ALL_ATTRIBUTE_DIM_KEYS_AE 74
2.87 ALL_ATTRIBUTE_DIM_LEVEL_ATTRS 75
2.88 ALL_ATTRIBUTE_DIM_LEVELS 76
2.89 ALL_ATTRIBUTE_DIM_LEVELS_AE 77
2.90 ALL_ATTRIBUTE_DIM_LVL_ATRS 78
2.91 ALL_ATTRIBUTE_DIM_LVL_ATRS_AE 78
2.92 ALL_ATTRIBUTE_DIM_LVL_CLASS 79
2.93 ALL_ATTRIBUTE_DIM_LVL_CLS 80
2.94 ALL_ATTRIBUTE_DIM_LVL_CLS_AE 80
2.95 ALL_ATTRIBUTE_DIM_ORD_ATRS 81
2.96 ALL_ATTRIBUTE_DIM_ORD_ATRS_AE 81
2.97 ALL_ATTRIBUTE_DIM_ORDER_ATTRS 82
2.98 ALL_ATTRIBUTE_DIM_TABLES 83
2.99 ALL_ATTRIBUTE_DIM_TABLES_AE 84
2.100 ALL_ATTRIBUTE_DIMENSIONS 85
2.101 ALL_ATTRIBUTE_DIMENSIONS_AE 86
2.102 ALL_ATTRIBUTE_TRANSFORMATIONS 87
2.103 ALL_AUDIT_POLICIES 88
2.104 ALL_AUDIT_POLICY_COLUMNS 89
2.105 ALL_AW_PS 90
2.106 ALL_AWS 90
2.107 ALL_BASE_TABLE_MVIEWS 91
2.108 ALL_BLOCKCHAIN_TABLES 92
2.109 ALL_CAPTURE 93

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.110 ALL_CAPTURE_EXTRA_ATTRIBUTES 95
2.111 ALL_CAPTURE_PARAMETERS 96
2.112 ALL_CAPTURE_PREPARED_DATABASE 96
2.113 ALL_CAPTURE_PREPARED_SCHEMAS 97
2.114 ALL_CAPTURE_PREPARED_TABLES 98
2.115 ALL_CATALOG 99
2.116 ALL_CERTIFICATES 100
2.117 ALL_CLUSTER_HASH_EXPRESSIONS 100
2.118 ALL_CLUSTERING_DIMENSIONS 101
2.119 ALL_CLUSTERING_JOINS 102
2.120 ALL_CLUSTERING_KEYS 103
2.121 ALL_CLUSTERING_TABLES 104
2.122 ALL_CLUSTERS 105
2.123 ALL_CODE_ROLE_PRIVS 107
2.124 ALL_COL_COMMENTS 107
2.125 ALL_COL_PENDING_STATS 108
2.126 ALL_COL_PRIVS 109
2.127 ALL_COL_PRIVS_MADE 110
2.128 ALL_COL_PRIVS_RECD 110
2.129 ALL_COLL_TYPES 111
2.130 ALL_CONS_COLUMNS 112
2.131 ALL_CONS_OBJ_COLUMNS 113
2.132 ALL_CONSTRAINTS 114
2.133 ALL_CONTEXT 116
2.134 ALL_CREDENTIALS 116
2.135 ALL_CUBE_ATTR_VISIBILITY 117
2.136 ALL_CUBE_ATTRIBUTES 118
2.137 ALL_CUBE_BUILD_PROCESSES 120
2.138 ALL_CUBE_CALCULATED_MEMBERS 120
2.139 ALL_CUBE_DIM_LEVELS 121
2.140 ALL_CUBE_DIM_MODELS 122
2.141 ALL_CUBE_DIM_VIEW_COLUMNS 122
2.142 ALL_CUBE_DIM_VIEWS 123
2.143 ALL_CUBE_DIMENSIONALITY 124
2.144 ALL_CUBE_DIMENSIONS 124
2.145 ALL_CUBE_HIER_LEVELS 125
2.146 ALL_CUBE_HIER_VIEW_COLUMNS 126
2.147 ALL_CUBE_HIER_VIEWS 127
2.148 ALL_CUBE_HIERARCHIES 128
2.149 ALL_CUBE_MEASURES 129
2.150 ALL_CUBE_NAMED_BUILD_SPECS 130
2.151 ALL_CUBE_SUB_PARTITION_LEVELS 130

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.152 ALL_CUBE_VIEW_COLUMNS 131
2.153 ALL_CUBE_VIEWS 132
2.154 ALL_CUBES 132
2.155 ALL_DB_LINKS 133
2.156 ALL_DEF_AUDIT_OPTS 134
2.157 ALL_DEPENDENCIES 135
2.158 ALL_DEQUEUE_QUEUES 136
2.159 ALL_DIM_ATTRIBUTES 137
2.160 ALL_DIM_CHILD_OF 137
2.161 ALL_DIM_HIERARCHIES 138
2.162 ALL_DIM_JOIN_KEY 138
2.163 ALL_DIM_LEVEL_KEY 139
2.164 ALL_DIM_LEVELS 139
2.165 ALL_DIMENSIONS 140
2.166 ALL_DIRECTORIES 141
2.167 ALL_EDITION_COMMENTS 141
2.168 ALL_EDITIONING_VIEW_COLS 142
2.169 ALL_EDITIONING_VIEW_COLS_AE 142
2.170 ALL_EDITIONING_VIEWS 143
2.171 ALL_EDITIONING_VIEWS_AE 143
2.172 ALL_EDITIONS 144
2.173 ALL_ENCRYPTED_COLUMNS 144
2.174 ALL_ERROR_TRANSLATIONS 145
2.175 ALL_ERRORS 146
2.176 ALL_ERRORS_AE 147
2.177 ALL_EVALUATION_CONTEXT_TABLES 148
2.178 ALL_EVALUATION_CONTEXT_VARS 148
2.179 ALL_EVALUATION_CONTEXTS 149
2.180 ALL_EXPRESSION_STATISTICS 150
2.181 ALL_EXTERNAL_LOCATIONS 150
2.182 ALL_EXTERNAL_TABLES 151
2.183 ALL_FILE_GROUP_EXPORT_INFO 152
2.184 ALL_FILE_GROUP_FILES 153
2.185 ALL_FILE_GROUP_TABLES 153
2.186 ALL_FILE_GROUP_TABLESPACES 154
2.187 ALL_FILE_GROUP_VERSIONS 155
2.188 ALL_FILE_GROUPS 155
2.189 ALL_GG_AUTO_CDR_COLUMN_GROUPS 156
2.190 ALL_GG_AUTO_CDR_COLUMNS 157
2.191 ALL_GG_AUTO_CDR_TABLES 157
2.192 ALL_GG_INBOUND_PROGRESS 158
2.193 ALL_GOLDENGATE_INBOUND 159

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.194 ALL_GOLDENGATE_PRIVILEGES 160
2.195 ALL_GOLDENGATE_RULES 160
2.196 ALL_HEAT_MAP_SEG_HISTOGRAM 162
2.197 ALL_HEAT_MAP_SEGMENT 162
2.198 ALL_HIER_CLASS 163
2.199 ALL_HIER_CLASS_AE 164
2.200 ALL_HIER_COLUMNS 165
2.201 ALL_HIER_COLUMNS_AE 166
2.202 ALL_HIER_HIER_ATTR_CLASS 167
2.203 ALL_HIER_HIER_ATTR_CLASS_AE 168
2.204 ALL_HIER_HIER_ATTRIBUTES 169
2.205 ALL_HIER_HIER_ATTRIBUTES_AE 170
2.206 ALL_HIER_JOIN_PATHS 171
2.207 ALL_HIER_JOIN_PATHS_AE 172
2.208 ALL_HIER_LEVEL_ID_ATTRS 172
2.209 ALL_HIER_LEVEL_ID_ATTRS_AE 173
2.210 ALL_HIER_LEVELS 174
2.211 ALL_HIER_LEVELS_AE 175
2.212 ALL_HIERARCHIES 176
2.213 ALL_HIERARCHIES_AE 176
2.214 ALL_HISTOGRAMS 177
2.215 ALL_HIVE_COLUMNS 177
2.216 ALL_HIVE_DATABASES 178
2.217 ALL_HIVE_PART_KEY_COLUMNS 179
2.218 ALL_HIVE_TAB_PARTITIONS 179
2.219 ALL_HIVE_TABLES 180
2.220 ALL_IDENTIFIERS 181
2.221 ALL_IMMUTABLE_TABLES 183
2.222 ALL_IND_COLUMNS 184
2.223 ALL_IND_EXPRESSIONS 185
2.224 ALL_IND_PARTITIONS 185
2.225 ALL_IND_PENDING_STATS 188
2.226 ALL_IND_STATISTICS 189
2.227 ALL_IND_SUBPARTITIONS 191
2.228 ALL_INDEXES 193
2.229 ALL_INDEXTYPE_ARRAYTYPES 198
2.230 ALL_INDEXTYPE_COMMENTS 198
2.231 ALL_INDEXTYPE_OPERATORS 199
2.232 ALL_INDEXTYPES 199
2.233 ALL_INTERNAL_TRIGGERS 200
2.234 ALL_JAVA_ARGUMENTS 201
2.235 ALL_JAVA_CLASSES 202

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.236 ALL_JAVA_COMPILER_OPTIONS 203
2.237 ALL_JAVA_DERIVATIONS 203
2.238 ALL_JAVA_FIELDS 204
2.239 ALL_JAVA_IMPLEMENTS 205
2.240 ALL_JAVA_INNERS 205
2.241 ALL_JAVA_LAYOUTS 206
2.242 ALL_JAVA_METHODS 207
2.243 ALL_JAVA_NCOMPS 209
2.244 ALL_JAVA_RESOLVERS 209
2.245 ALL_JAVA_THROWS 210
2.246 ALL_JOBS 210
2.247 ALL_JOIN_IND_COLUMNS 210
2.248 ALL_JSON_COLUMNS 211
2.249 ALL_JSON_DATAGUIDE_FIELDS 212
2.250 ALL_JSON_DATAGUIDES 213
2.251 ALL_LIBRARIES 213
2.252 ALL_LOB_PARTITIONS 214
2.253 ALL_LOB_SUBPARTITIONS 217
2.254 ALL_LOB_TEMPLATES 219
2.255 ALL_LOBS 220
2.256 ALL_LOG_GROUP_COLUMNS 222
2.257 ALL_LOG_GROUPS 223
2.258 ALL_MEASURE_FOLDER_CONTENTS 223
2.259 ALL_MEASURE_FOLDER_SUBFOLDERS 224
2.260 ALL_MEASURE_FOLDERS 225
2.261 ALL_METADATA_PROPERTIES 225
2.262 ALL_METHOD_PARAMS 226
2.263 ALL_METHOD_RESULTS 227
2.264 ALL_MINING_ALGORITHMS 227
2.265 ALL_MINING_MODEL_ATTRIBUTES 228
2.266 ALL_MINING_MODEL_PARTITIONS 230
2.267 ALL_MINING_MODEL_SETTINGS 230
2.268 ALL_MINING_MODEL_VIEWS 231
2.269 ALL_MINING_MODEL_XFORMS 231
2.270 ALL_MINING_MODELS 232
2.271 ALL_MVIEW_AGGREGATES 233
2.272 ALL_MVIEW_ANALYSIS 234
2.273 ALL_MVIEW_COMMENTS 235
2.274 ALL_MVIEW_DETAIL_PARTITION 236
2.275 ALL_MVIEW_DETAIL_RELATIONS 237
2.276 ALL_MVIEW_DETAIL_SUBPARTITION 238
2.277 ALL_MVIEW_JOINS 238

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.278 ALL_MVIEW_KEYS 239
2.279 ALL_MVIEW_LOGS 240
2.280 ALL_MVIEW_REFRESH_TIMES 241
2.281 ALL_MVIEWS 242
2.282 ALL_NESTED_TABLE_COLS 246
2.283 ALL_NESTED_TABLES 248
2.284 ALL_OBJ_COLATTRS 249
2.285 ALL_OBJECT_TABLES 250
2.286 ALL_OBJECTS 255
2.287 ALL_OBJECTS_AE 257
2.288 ALL_OPANCILLARY 259
2.289 ALL_OPARGUMENTS 259
2.290 ALL_OPBINDINGS 260
2.291 ALL_OPERATOR_COMMENTS 261
2.292 ALL_OPERATORS 261
2.293 ALL_OUTLINE_HINTS 262
2.294 ALL_OUTLINES 262

3 Static Data Dictionary Views: ALL_PART_COL_STATISTICS to DATABASE_PROPERTIES

3.1 ALL_PART_COL_STATISTICS 1
3.2 ALL_PART_HISTOGRAMS 2
3.3 ALL_PART_INDEXES 3
3.4 ALL_PART_KEY_COLUMNS 6
3.5 ALL_PART_LOBS 6
3.6 ALL_PART_TABLES 9
3.7 ALL_PARTIAL_DROP_TABS 14
3.8 ALL_PENDING_CONV_TABLES 14
3.9 ALL_PLSQL_COLL_TYPES 15
3.10 ALL_PLSQL_OBJECT_SETTINGS 16
3.11 ALL_PLSQL_TYPE_ATTRS 17
3.12 ALL_PLSQL_TYPES 17
3.13 ALL_POLICIES 18
3.14 ALL_POLICY_ATTRIBUTES 19
3.15 ALL_POLICY_CONTEXTS 20
3.16 ALL_POLICY_GROUPS 21
3.17 ALL_PROCEDURES 21
3.18 ALL_PROPAGATION 23
3.19 ALL_QUEUE_SCHEDULES 24
3.20 ALL_QUEUE_SUBSCRIBERS 26
3.21 ALL_QUEUE_TABLES 27

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.22 ALL_QUEUES 28
3.23 ALL_REFRESH 29
3.24 ALL_REFRESH_CHILDREN 30
3.25 ALL_REFRESH_DEPENDENCIES 31
3.26 ALL_REFS 31
3.27 ALL_REGISTERED_MVIEWS 32
3.28 ALL_REGISTRY_BANNERS 33
3.29 ALL_REPL_DBNAME_MAPPING 33
3.30 ALL_REPLICATION_PROCESS_EVENTS 33
3.31 ALL_REWRITE_EQUIVALENCES 34
3.32 ALL_RULE_SET_RULES 35
3.33 ALL_RULE_SETS 36
3.34 ALL_RULES 36
3.35 ALL_SCHEDULER_CHAIN_RULES 37
3.36 ALL_SCHEDULER_CHAIN_STEPS 37
3.37 ALL_SCHEDULER_CHAINS 39
3.38 ALL_SCHEDULER_CREDENTIALS 39
3.39 ALL_SCHEDULER_DB_DESTS 40
3.40 ALL_SCHEDULER_DESTS 41
3.41 ALL_SCHEDULER_EXTERNAL_DESTS 42
3.42 ALL_SCHEDULER_FILE_WATCHERS 42
3.43 ALL_SCHEDULER_GLOBAL_ATTRIBUTE 43
3.44 ALL_SCHEDULER_GROUP_MEMBERS 44
3.45 ALL_SCHEDULER_GROUPS 44
3.46 ALL_SCHEDULER_INCOMPAT_MEMBER 45
3.47 ALL_SCHEDULER_INCOMPATS 45
3.48 ALL_SCHEDULER_JOB_ARGS 46
3.49 ALL_SCHEDULER_JOB_CLASSES 47
3.50 ALL_SCHEDULER_JOB_DESTS 47
3.51 ALL_SCHEDULER_JOB_LOG 49
3.52 ALL_SCHEDULER_JOB_RUN_DETAILS 50
3.53 ALL_SCHEDULER_JOBS 51
3.54 ALL_SCHEDULER_NOTIFICATIONS 55
3.55 ALL_SCHEDULER_PROGRAM_ARGS 56
3.56 ALL_SCHEDULER_PROGRAMS 57
3.57 ALL_SCHEDULER_REMOTE_DATABASES 58
3.58 ALL_SCHEDULER_REMOTE_JOBSTATE 59
3.59 ALL_SCHEDULER_RESOURCES 60
3.60 ALL_SCHEDULER_RSC_CONSTRAINTS 60
3.61 ALL_SCHEDULER_RUNNING_CHAINS 61
3.62 ALL_SCHEDULER_RUNNING_JOBS 62
3.63 ALL_SCHEDULER_SCHEDULES 63

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.64 ALL_SCHEDULER_WINDOW_DETAILS 64
3.65 ALL_SCHEDULER_WINDOW_GROUPS 65
3.66 ALL_SCHEDULER_WINDOW_LOG 66
3.67 ALL_SCHEDULER_WINDOWS 66
3.68 ALL_SCHEDULER_WINGROUP_MEMBERS 68
3.69 ALL_SEC_RELEVANT_COLS 68
3.70 ALL_SECONDARY_OBJECTS 69
3.71 ALL_SEQUENCES 69
3.72 ALL_SERVICES 70
3.73 ALL_SOURCE 74
3.74 ALL_SOURCE_AE 75
3.75 ALL_SQL_TRANSLATION_PROFILES 75
3.76 ALL_SQL_TRANSLATIONS 76
3.77 ALL_SQLJ_TYPE_ATTRS 77
3.78 ALL_SQLJ_TYPE_METHODS 78
3.79 ALL_SQLJ_TYPES 79
3.80 ALL_SQLSET 80
3.81 ALL_SQLSET_BINDS 81
3.82 ALL_SQLSET_PLANS 82
3.83 ALL_SQLSET_REFERENCES 86
3.84 ALL_SQLSET_STATEMENTS 86
3.85 ALL_STAT_EXTENSIONS 88
3.86 ALL_STATEMENTS 89
3.87 ALL_STORED_SETTINGS 90
3.88 ALL_STREAMS_GLOBAL_RULES 91
3.89 ALL_STREAMS_MESSAGE_CONSUMERS 92
3.90 ALL_STREAMS_NEWLY_SUPPORTED 93
3.91 ALL_STREAMS_SCHEMA_RULES 94
3.92 ALL_STREAMS_TABLE_RULES 95
3.93 ALL_STREAMS_TRANSFORM_FUNCTION 97
3.94 ALL_SUBPART_COL_STATISTICS 97
3.95 ALL_SUBPART_HISTOGRAMS 98
3.96 ALL_SUBPART_KEY_COLUMNS 99
3.97 ALL_SUBPARTITION_TEMPLATES 100
3.98 ALL_SUMDELTA 101
3.99 ALL_SYNC_CAPTURE 101
3.100 ALL_SYNC_CAPTURE_PREPARED_TABS 102
3.101 ALL_SYNC_CAPTURE_TABLES 102
3.102 ALL_SYNONYMS 103
3.103 ALL_TAB_COL_STATISTICS 104
3.104 ALL_TAB_COLS 106
3.105 ALL_TAB_COLUMNS 109

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.106 ALL_TAB_COMMENTS 111
3.107 ALL_TAB_HISTGRM_PENDING_STATS 112
3.108 ALL_TAB_HISTOGRAMS 113
3.109 ALL_TAB_IDENTITY_COLS 114
3.110 ALL_TAB_MODIFICATIONS 115
3.111 ALL_TAB_PARTITIONS 115
3.112 ALL_TAB_PENDING_STATS 121
3.113 ALL_TAB_PRIVS 122
3.114 ALL_TAB_PRIVS_MADE 123
3.115 ALL_TAB_PRIVS_RECD 123
3.116 ALL_TAB_STAT_PREFS 124
3.117 ALL_TAB_STATISTICS 125
3.118 ALL_TAB_STATS_HISTORY 126
3.119 ALL_TAB_SUBPARTITIONS 127
3.120 ALL_TABLES 131
3.121 ALL_TRANSFORMATIONS 137
3.122 ALL_TRIGGER_COLS 138
3.123 ALL_TRIGGER_ORDERING 139
3.124 ALL_TRIGGERS 139
3.125 ALL_TRIGGERS_AE 141
3.126 ALL_TSTZ_TAB_COLS 143
3.127 ALL_TSTZ_TABLES 144
3.128 ALL_TYPE_ATTRS 145
3.129 ALL_TYPE_METHODS 146
3.130 ALL_TYPE_VERSIONS 146
3.131 ALL_TYPES 147
3.132 ALL_UNIFIED_AUDIT_ACTIONS 148
3.133 ALL_UNUSED_COL_TABS 149
3.134 ALL_UPDATABLE_COLUMNS 149
3.135 ALL_USERS 150
3.136 ALL_USTATS 151
3.137 ALL_VARRAYS 152
3.138 ALL_VIEWS 153
3.139 ALL_VIEWS_AE 154
3.140 ALL_WARNING_SETTINGS 156
3.141 ALL_XML_INDEXES 157
3.142 ALL_XML_NESTED_TABLES 158
3.143 ALL_XML_OUT_OF_LINE_TABLES 158
3.144 ALL_XML_SCHEMA_ATTRIBUTES 159
3.145 ALL_XML_SCHEMA_COMPLEX_TYPES 160
3.146 ALL_XML_SCHEMA_ELEMENTS 161
3.147 ALL_XML_SCHEMA_NAMESPACES 162

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.148 ALL_XML_SCHEMA_SIMPLE_TYPES 163
3.149 ALL_XML_SCHEMA_SUBSTGRP_HEAD 164
3.150 ALL_XML_SCHEMA_SUBSTGRP_MBRS 164
3.151 ALL_XML_SCHEMAS 165
3.152 ALL_XML_TAB_COLS 166
3.153 ALL_XML_TABLES 167
3.154 ALL_XML_VIEW_COLS 168
3.155 ALL_XML_VIEWS 169
3.156 ALL_XSTREAM_ADMINISTRATOR 169
3.157 ALL_XSTREAM_INBOUND 170
3.158 ALL_XSTREAM_INBOUND_PROGRESS 171
3.159 ALL_XSTREAM_OUT_SUPPORT_MODE 172
3.160 ALL_XSTREAM_OUTBOUND 173
3.161 ALL_XSTREAM_OUTBOUND_PROGRESS 174
3.162 ALL_XSTREAM_RULES 174
3.163 ALL_XSTREAM_TRANSFORMATIONS 176
3.164 ALL_XTERNAL_LOC_PARTITIONS 177
3.165 ALL_XTERNAL_LOC_SUBPARTITIONS 178
3.166 ALL_XTERNAL_PART_TABLES 178
3.167 ALL_XTERNAL_TAB_PARTITIONS 179
3.168 ALL_XTERNAL_TAB_SUBPARTITIONS 180
3.169 ALL_ZONEMAP_MEASURES 180
3.170 ALL_ZONEMAPS 181
3.171 AUDIT_ACTIONS 183
3.172 AUDIT_UNIFIED_CONTEXTS 183
3.173 AUDIT_UNIFIED_ENABLED_POLICIES 184
3.174 AUDIT_UNIFIED_POLICIES 185
3.175 AUDIT_UNIFIED_POLICY_COMMENTS 186
3.176 AUDITABLE_SYSTEM_ACTIONS 187
3.177 CAT 188
3.178 CATALOG 188
3.179 CHAINED_ROWS 189
3.180 CLIENT_RESULT_CACHE_STATS$ 189
3.181 CLU 190
3.182 COL 190
3.183 COLS 190
3.184 DATABASE_EXPORT_OBJECTS 190
3.185 DATABASE_PROPERTIES 191

Copyright © 2002, 2026, Oracle and/or its affiliates.

4 Static Data Dictionary Views: DBA_2PC_NEIGHBORS to DBA_HIST_JAVA_POOL_ADVICE

4.1 DBA_2PC_NEIGHBORS 1
4.2 DBA_2PC_PENDING 1
4.3 DBA_ACCHK_EVENTS 2
4.4 DBA_ACCHK_EVENTS_SUMMARY 4
4.5 DBA_ACCHK_STATISTICS 5
4.6 DBA_ACCHK_STATISTICS_SUMMARY 7
4.7 DBA_ACL_NAME_MAP 9
4.8 DBA_ADDM_FDG_BREAKDOWN 9
4.9 DBA_ADDM_FINDINGS 10
4.10 DBA_ADDM_INSTANCES 11
4.11 DBA_ADDM_SYSTEM_DIRECTIVES 12
4.12 DBA_ADDM_TASK_DIRECTIVES 13
4.13 DBA_ADDM_TASKS 13
4.14 DBA_ADVISOR_ACTIONS 16
4.15 DBA_ADVISOR_COMMANDS 18
4.16 DBA_ADVISOR_DEF_PARAMETERS 18
4.17 DBA_ADVISOR_DEFINITIONS 19
4.18 DBA_ADVISOR_DIR_DEFINITIONS 20
4.19 DBA_ADVISOR_DIR_INSTANCES 21
4.20 DBA_ADVISOR_DIR_TASK_INST 21
4.21 DBA_ADVISOR_EXEC_PARAMETERS 22
4.22 DBA_ADVISOR_EXECUTION_TYPES 22
4.23 DBA_ADVISOR_EXECUTIONS 23
4.24 DBA_ADVISOR_FDG_BREAKDOWN 24
4.25 DBA_ADVISOR_FINDING_NAMES 25
4.26 DBA_ADVISOR_FINDINGS 25
4.27 DBA_ADVISOR_JOURNAL 26
4.28 DBA_ADVISOR_LOG 26
4.29 DBA_ADVISOR_OBJECT_TYPES 27
4.30 DBA_ADVISOR_OBJECTS 28
4.31 DBA_ADVISOR_PARAMETERS 29
4.32 DBA_ADVISOR_RATIONALE 31
4.33 DBA_ADVISOR_RECOMMENDATIONS 32
4.34 DBA_ADVISOR_SQLA_REC_SUM 34
4.35 DBA_ADVISOR_SQLA_TABLES 34
4.36 DBA_ADVISOR_SQLA_WK_MAP 35
4.37 DBA_ADVISOR_SQLA_WK_STMTS 35
4.38 DBA_ADVISOR_SQLPLANS 37
4.39 DBA_ADVISOR_SQLSTATS 40

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.40 DBA_ADVISOR_SQLW_JOURNAL 42
4.41 DBA_ADVISOR_SQLW_PARAMETERS 42
4.42 DBA_ADVISOR_SQLW_STMTS 43
4.43 DBA_ADVISOR_SQLW_SUM 45
4.44 DBA_ADVISOR_SQLW_TABLES 45
4.45 DBA_ADVISOR_SQLW_TEMPLATES 46
4.46 DBA_ADVISOR_TASKS 46
4.47 DBA_ADVISOR_TEMPLATES 48
4.48 DBA_ADVISOR_USAGE 49
4.49 DBA_ALERT_HISTORY 49
4.50 DBA_ALERT_HISTORY_DETAIL 50
4.51 DBA_ALL_TABLES 51
4.52 DBA_ANALYTIC_VIEW_ATTR_CLASS 52
4.53 DBA_ANALYTIC_VIEW_ATTR_CLS 52
4.54 DBA_ANALYTIC_VIEW_ATTR_CLS_AE 52
4.55 DBA_ANALYTIC_VIEW_BAS_MEAS 53
4.56 DBA_ANALYTIC_VIEW_BAS_MEAS_AE 53
4.57 DBA_ANALYTIC_VIEW_BASE_MEAS 53
4.58 DBA_ANALYTIC_VIEW_CALC_MEAS 54
4.59 DBA_ANALYTIC_VIEW_CLASS 54
4.60 DBA_ANALYTIC_VIEW_CLASS_AE 54
4.61 DBA_ANALYTIC_VIEW_CLC_MEAS 54
4.62 DBA_ANALYTIC_VIEW_CLC_MEAS_AE 55
4.63 DBA_ANALYTIC_VIEW_COLUMNS 55
4.64 DBA_ANALYTIC_VIEW_COLUMNS_AE 55
4.65 DBA_ANALYTIC_VIEW_DIM_CLASS 56
4.66 DBA_ANALYTIC_VIEW_DIM_CLS 56
4.67 DBA_ANALYTIC_VIEW_DIM_CLS_AE 56
4.68 DBA_ANALYTIC_VIEW_DIMENSIONS 57
4.69 DBA_ANALYTIC_VIEW_DIMS 57
4.70 DBA_ANALYTIC_VIEW_DIMS_AE 57
4.71 DBA_ANALYTIC_VIEW_HIER_CLASS 58
4.72 DBA_ANALYTIC_VIEW_HIER_CLS 58
4.73 DBA_ANALYTIC_VIEW_HIER_CLS_AE 58
4.74 DBA_ANALYTIC_VIEW_HIERS 59
4.75 DBA_ANALYTIC_VIEW_HIERS_AE 59
4.76 DBA_ANALYTIC_VIEW_KEYS 59
4.77 DBA_ANALYTIC_VIEW_KEYS_AE 59
4.78 DBA_ANALYTIC_VIEW_LEVEL_CLASS 60
4.79 DBA_ANALYTIC_VIEW_LEVELS 60
4.80 DBA_ANALYTIC_VIEW_LEVELS_AE 60
4.81 DBA_ANALYTIC_VIEW_LVL_CLS 61

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.82 DBA_ANALYTIC_VIEW_LVL_CLS_AE 61
4.83 DBA_ANALYTIC_VIEW_LVLGRPS 61
4.84 DBA_ANALYTIC_VIEW_LVLGRPS_AE 62
4.85 DBA_ANALYTIC_VIEW_MEAS_CLASS 62
4.86 DBA_ANALYTIC_VIEW_MEAS_CLS 62
4.87 DBA_ANALYTIC_VIEW_MEAS_CLS_AE 63
4.88 DBA_ANALYTIC_VIEWS 63
4.89 DBA_ANALYTIC_VIEWS_AE 63
4.90 DBA_ANNOTATION_VALUES 64
4.91 DBA_ANNOTATIONS 64
4.92 DBA_ANNOTATIONS_USAGE 64
4.93 DBA_APP_ERRORS 65
4.94 DBA_APP_PATCHES 65
4.95 DBA_APP_PDB_STATUS 65
4.96 DBA_APP_STATEMENTS 66
4.97 DBA_APP_VERSIONS 66
4.98 DBA_APPLICATION_ROLES 67
4.99 DBA_APPLICATIONS 67
4.100 DBA_APPLY 68
4.101 DBA_APPLY_CHANGE_HANDLERS 68
4.102 DBA_APPLY_CONFLICT_COLUMNS 68
4.103 DBA_APPLY_DML_CONF_HANDLERS 68
4.104 DBA_APPLY_DML_HANDLERS 69
4.105 DBA_APPLY_ENQUEUE 69
4.106 DBA_APPLY_ERROR 69
4.107 DBA_APPLY_ERROR_MESSAGES 69
4.108 DBA_APPLY_EXECUTE 70
4.109 DBA_APPLY_HANDLE_COLLISIONS 70
4.110 DBA_APPLY_INSTANTIATED_GLOBAL 70
4.111 DBA_APPLY_INSTANTIATED_OBJECTS 70
4.112 DBA_APPLY_INSTANTIATED_SCHEMAS 71
4.113 DBA_APPLY_KEY_COLUMNS 71
4.114 DBA_APPLY_OBJECT_DEPENDENCIES 71
4.115 DBA_APPLY_PARAMETERS 71
4.116 DBA_APPLY_PROGRESS 72
4.117 DBA_APPLY_REPERROR_HANDLERS 72
4.118 DBA_APPLY_SPILL_TXN 72
4.119 DBA_APPLY_TABLE_COLUMNS 73
4.120 DBA_APPLY_VALUE_DEPENDENCIES 73
4.121 DBA_AQ_AGENT_PRIVS 73
4.122 DBA_AQ_AGENTS 74
4.123 DBA_ARGUMENTS 74

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.124 DBA_ASSEMBLIES 74
4.125 DBA_ASSOCIATIONS 74
4.126 DBA_ATTRIBUTE_DIM_ATTR_CLASS 75
4.127 DBA_ATTRIBUTE_DIM_ATTR_CLS 75
4.128 DBA_ATTRIBUTE_DIM_ATTR_CLS_AE 75
4.129 DBA_ATTRIBUTE_DIM_ATTRS 76
4.130 DBA_ATTRIBUTE_DIM_ATTRS_AE 76
4.131 DBA_ATTRIBUTE_DIM_CLASS 76
4.132 DBA_ATTRIBUTE_DIM_CLASS_AE 76
4.133 DBA_ATTRIBUTE_DIM_JN_PTHS 77
4.134 DBA_ATTRIBUTE_DIM_JN_PTHS_AE 77
4.135 DBA_ATTRIBUTE_DIM_JOIN_PATHS 77
4.136 DBA_ATTRIBUTE_DIM_KEYS 78
4.137 DBA_ATTRIBUTE_DIM_KEYS_AE 78
4.138 DBA_ATTRIBUTE_DIM_LEVEL_ATTRS 78
4.139 DBA_ATTRIBUTE_DIM_LEVELS 78
4.140 DBA_ATTRIBUTE_DIM_LEVELS_AE 79
4.141 DBA_ATTRIBUTE_DIM_LVL_ATRS 79
4.142 DBA_ATTRIBUTE_DIM_LVL_ATRS_AE 79
4.143 DBA_ATTRIBUTE_DIM_LVL_CLASS 80
4.144 DBA_ATTRIBUTE_DIM_LVL_CLS 80
4.145 DBA_ATTRIBUTE_DIM_LVL_CLS_AE 80
4.146 DBA_ATTRIBUTE_DIM_ORD_ATRS 81
4.147 DBA_ATTRIBUTE_DIM_ORD_ATRS_AE 81
4.148 DBA_ATTRIBUTE_DIM_ORDER_ATTRS 81
4.149 DBA_ATTRIBUTE_DIM_TABLES 82
4.150 DBA_ATTRIBUTE_DIM_TABLES_AE 82
4.151 DBA_ATTRIBUTE_DIMENSIONS 82
4.152 DBA_ATTRIBUTE_DIMENSIONS_AE 82
4.153 DBA_ATTRIBUTE_TRANSFORMATIONS 83
4.154 DBA_AUDIT_EXISTS 83
4.155 DBA_AUDIT_MGMT_CLEAN_EVENTS 86
4.156 DBA_AUDIT_MGMT_CLEANUP_JOBS 87
4.157 DBA_AUDIT_MGMT_CONFIG_PARAMS 88
4.158 DBA_AUDIT_MGMT_LAST_ARCH_TS 89
4.159 DBA_AUDIT_OBJECT 90
4.160 DBA_AUDIT_POLICIES 92
4.161 DBA_AUDIT_POLICY_COLUMNS 92
4.162 DBA_AUDIT_SESSION 92
4.163 DBA_AUDIT_STATEMENT 94
4.164 DBA_AUDIT_TRAIL 97
4.165 DBA_AUTO_INDEX_CONFIG 101

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.166 DBA_AUTO_SEGADV_CTL 101
4.167 DBA_AUTO_SEGADV_SUMMARY 102
4.168 DBA_AUTO_STAT_EXECUTIONS 102
4.169 DBA_AUTOTASK_CLIENT 103
4.170 DBA_AUTOTASK_CLIENT_HISTORY 104
4.171 DBA_AUTOTASK_CLIENT_JOB 105
4.172 DBA_AUTOTASK_JOB_HISTORY 106
4.173 DBA_AUTOTASK_OPERATION 106
4.174 DBA_AUTOTASK_SCHEDULE 107
4.175 DBA_AUTOTASK_STATUS 107
4.176 DBA_AUTOTASK_TASK 107
4.177 DBA_AUTOTASK_WINDOW_CLIENTS 110
4.178 DBA_AUTOTASK_WINDOW_HISTORY 111
4.179 DBA_AW_PS 111
4.180 DBA_AWS 111
4.181 DBA_BASE_TABLE_MVIEWS 112
4.182 DBA_BLOCKCHAIN_TABLES 112
4.183 DBA_BLOCKERS 112
4.184 DBA_CAPTURE 113
4.185 DBA_CAPTURE_EXTRA_ATTRIBUTES 113
4.186 DBA_CAPTURE_PARAMETERS 113
4.187 DBA_CAPTURE_PREPARED_DATABASE 113
4.188 DBA_CAPTURE_PREPARED_SCHEMAS 114
4.189 DBA_CAPTURE_PREPARED_TABLES 114
4.190 DBA_CATALOG 114
4.191 DBA_CDB_RSRC_PLAN_DIRECTIVES 114
4.192 DBA_CDB_RSRC_PLANS 115
4.193 DBA_CERTIFICATES 116
4.194 DBA_CHANGE_NOTIFICATION_REGS 116
4.195 DBA_CHECKED_ROLES 117
4.196 DBA_CHECKED_ROLES_PATH 117
4.197 DBA_CLU_COLUMNS 118
4.198 DBA_CLUSTER_HASH_EXPRESSIONS 118
4.199 DBA_CLUSTERING_DIMENSIONS 119
4.200 DBA_CLUSTERING_JOINS 119
4.201 DBA_CLUSTERING_KEYS 119
4.202 DBA_CLUSTERING_TABLES 120
4.203 DBA_CLUSTERS 120
4.204 DBA_CODE_ROLE_PRIVS 120
4.205 DBA_COL_COMMENTS 120
4.206 DBA_COL_PENDING_STATS 121
4.207 DBA_COL_PRIVS 121

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.208 DBA_COLL_TYPES 122
4.209 DBA_COMMON_AUDIT_TRAIL 122
4.210 DBA_COMPARISON 126
4.211 DBA_COMPARISON_COLUMNS 127
4.212 DBA_COMPARISON_ROW_DIF 127
4.213 DBA_COMPARISON_SCAN 128
4.214 DBA_COMPARISON_SCAN_VALUES 129
4.215 DBA_CONNECT_ROLE_GRANTEES 129
4.216 DBA_CONNECTION_TESTS 129
4.217 DBA_CONS_COLUMNS 131
4.218 DBA_CONS_OBJ_COLUMNS 131
4.219 DBA_CONSTRAINTS 131
4.220 DBA_CONTAINER_DATA 131
4.221 DBA_CONTEXT 132
4.222 DBA_CPOOL_INFO 133
4.223 DBA_CPU_USAGE_STATISTICS 134
4.224 DBA_CQ_NOTIFICATION_QUERIES 135
4.225 DBA_CREDENTIALS 135
4.226 DBA_CUBE_ATTR_VISIBILITY 135
4.227 DBA_CUBE_ATTRIBUTES 136
4.228 DBA_CUBE_BUILD_PROCESSES 136
4.229 DBA_CUBE_CALCULATED_MEMBERS 136
4.230 DBA_CUBE_DIM_LEVELS 136
4.231 DBA_CUBE_DIM_MODELS 137
4.232 DBA_CUBE_DIM_VIEW_COLUMNS 137
4.233 DBA_CUBE_DIM_VIEWS 137
4.234 DBA_CUBE_DIMENSIONALITY 137
4.235 DBA_CUBE_DIMENSIONS 138
4.236 DBA_CUBE_HIER_LEVELS 138
4.237 DBA_CUBE_HIER_VIEW_COLUMNS 138
4.238 DBA_CUBE_HIER_VIEWS 138
4.239 DBA_CUBE_HIERARCHIES 139
4.240 DBA_CUBE_MEASURES 139
4.241 DBA_CUBE_NAMED_BUILD_SPECS 139
4.242 DBA_CUBE_SUB_PARTITION_LEVELS 139
4.243 DBA_CUBE_VIEW_COLUMNS 140
4.244 DBA_CUBE_VIEWS 140
4.245 DBA_CUBES 140
4.246 DBA_DATA_FILES 140
4.247 DBA_DATAPUMP_JOBS 142
4.248 DBA_DATAPUMP_SESSIONS 142
4.249 DBA_DB_LINK_SOURCES 143

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.250 DBA_DB_LINKS 144
4.251 DBA_DBFS_HS 145
4.252 DBA_DBFS_HS_COMMANDS 145
4.253 DBA_DBFS_HS_FIXED_PROPERTIES 146
4.254 DBA_DBFS_HS_PROPERTIES 146
4.255 DBA_DDL_LOCKS 147
4.256 DBA_DEPENDENCIES 147
4.257 DBA_DIGEST_VERIFIERS 148
4.258 DBA_DIM_ATTRIBUTES 148
4.259 DBA_DIM_CHILD_OF 148
4.260 DBA_DIM_HIERARCHIES 148
4.261 DBA_DIM_JOIN_KEY 149
4.262 DBA_DIM_LEVEL_KEY 149
4.263 DBA_DIM_LEVELS 149
4.264 DBA_DIMENSIONS 149
4.265 DBA_DIRECTORIES 150
4.266 DBA_DISCOVERY_SOURCE 150
4.267 DBA_DML_LOCKS 150
4.268 DBA_DMT_FREE_SPACE 151
4.269 DBA_DMT_USED_EXTENTS 151
4.270 DBA_EDITION_COMMENTS 152
4.271 DBA_EDITIONED_TYPES 152
4.272 DBA_EDITIONING_VIEW_COLS 152
4.273 DBA_EDITIONING_VIEW_COLS_AE 153
4.274 DBA_EDITIONING_VIEWS 153
4.275 DBA_EDITIONING_VIEWS_AE 153
4.276 DBA_EDITIONS 153
4.277 DBA_ENABLED_AGGREGATIONS 154
4.278 DBA_ENABLED_TRACES 154
4.279 DBA_ENCRYPTED_COLUMNS 155
4.280 DBA_EPG_DAD_AUTHORIZATION 155
4.281 DBA_ERROR_TRANSLATIONS 155
4.282 DBA_ERRORS 156
4.283 DBA_ERRORS_AE 156
4.284 DBA_EVALUATION_CONTEXT_TABLES 156
4.285 DBA_EVALUATION_CONTEXT_VARS 156
4.286 DBA_EVALUATION_CONTEXTS 157
4.287 DBA_EXP_FILES 157
4.288 DBA_EXP_OBJECTS 157
4.289 DBA_EXP_VERSION 157
4.290 DBA_EXPRESSION_STATISTICS 158
4.291 DBA_EXTENTS 158

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.292 DBA_EXTERNAL_LOCATIONS 159
4.293 DBA_EXTERNAL_SCN_ACTIVITY 159
4.294 DBA_EXTERNAL_TABLES 161
4.295 DBA_FEATURE_USAGE_STATISTICS 161
4.296 DBA_FGA_AUDIT_TRAIL 162
4.297 DBA_FILE_GROUP_EXPORT_INFO 164
4.298 DBA_FILE_GROUP_FILES 164
4.299 DBA_FILE_GROUP_TABLES 164
4.300 DBA_FILE_GROUP_TABLESPACES 164
4.301 DBA_FILE_GROUP_VERSIONS 165
4.302 DBA_FILE_GROUPS 165
4.303 DBA_FLASHBACK_ARCHIVE 165
4.304 DBA_FLASHBACK_ARCHIVE_TABLES 166
4.305 DBA_FLASHBACK_ARCHIVE_TS 166
4.306 DBA_FLASHBACK_TXN_REPORT 166
4.307 DBA_FLASHBACK_TXN_STATE 167
4.308 DBA_FREE_SPACE 168
4.309 DBA_FREE_SPACE_COALESCED 168
4.310 DBA_GG_AUTO_CDR_COLUMN_GROUPS 169
4.311 DBA_GG_AUTO_CDR_COLUMNS 169
4.312 DBA_GG_AUTO_CDR_TABLES 169
4.313 DBA_GG_INBOUND_PROGRESS 170
4.314 DBA_GG_PROC_OBJECT_EXCLUSION 170
4.315 DBA_GG_PROCEDURE_ANNOTATION 170
4.316 DBA_GG_SUPPORTED_PACKAGES 171
4.317 DBA_GG_SUPPORTED_PROCEDURES 171
4.318 DBA_GLOBAL_CONTEXT 171
4.319 DBA_GOLDENGATE_INBOUND 172
4.320 DBA_GOLDENGATE_NOT_UNIQUE 172
4.321 DBA_GOLDENGATE_PRIVILEGES 173
4.322 DBA_GOLDENGATE_RULES 173
4.323 DBA_GOLDENGATE_SUPPORT_MODE 173
4.324 DBA_HANG_MANAGER_PARAMETERS 174
4.325 DBA_HEAT_MAP_SEG_HISTOGRAM 175
4.326 DBA_HEAT_MAP_SEGMENT 175
4.327 DBA_HEATMAP_TOP_OBJECTS 175
4.328 DBA_HEATMAP_TOP_TABLESPACES 176
4.329 DBA_HIER_CLASS 176
4.330 DBA_HIER_CLASS_AE 177
4.331 DBA_HIER_COLUMNS 177
4.332 DBA_HIER_COLUMNS_AE 177
4.333 DBA_HIER_HIER_ATTR_CLASS 178

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.334 DBA_HIER_HIER_ATTR_CLASS_AE 178
4.335 DBA_HIER_HIER_ATTRIBUTES 178
4.336 DBA_HIER_HIER_ATTRIBUTES_AE 179
4.337 DBA_HIER_JOIN_PATHS 179
4.338 DBA_HIER_JOIN_PATHS_AE 179
4.339 DBA_HIER_LEVEL_ID_ATTRS 180
4.340 DBA_HIER_LEVEL_ID_ATTRS_AE 180
4.341 DBA_HIER_LEVELS 180
4.342 DBA_HIER_LEVELS_AE 181
4.343 DBA_HIERARCHIES 181
4.344 DBA_HIERARCHIES_AE 181
4.345 DBA_HIGH_WATER_MARK_STATISTICS 181
4.346 DBA_HIST_ACTIVE_SESS_HISTORY 182
4.347 DBA_HIST_APPLY_SUMMARY 188
4.348 DBA_HIST_ASH_SNAPSHOT 189
4.349 DBA_HIST_BASELINE 190
4.350 DBA_HIST_BASELINE_DETAILS 191
4.351 DBA_HIST_BASELINE_METADATA 192
4.352 DBA_HIST_BASELINE_TEMPLATE 193
4.353 DBA_HIST_BG_EVENT_SUMMARY 194
4.354 DBA_HIST_BUFFER_POOL_STAT 195
4.355 DBA_HIST_BUFFERED_QUEUES 196
4.356 DBA_HIST_BUFFERED_SUBSCRIBERS 197
4.357 DBA_HIST_CAPTURE 199
4.358 DBA_HIST_CHANNEL_WAITS 200
4.359 DBA_HIST_CLUSTER_INTERCON 201
4.360 DBA_HIST_COLORED_SQL 201
4.361 DBA_HIST_COMP_IOSTAT 202
4.362 DBA_HIST_CON_SYS_TIME_MODEL 203
4.363 DBA_HIST_CON_SYSMETRIC_HIST 203
4.364 DBA_HIST_CON_SYSMETRIC_SUMM 204
4.365 DBA_HIST_CON_SYSSTAT 205
4.366 DBA_HIST_CON_SYSTEM_EVENT 206
4.367 DBA_HIST_CR_BLOCK_SERVER 207
4.368 DBA_HIST_CURRENT_BLOCK_SERVER 208
4.369 DBA_HIST_DATABASE_INSTANCE 209
4.370 DBA_HIST_DATAFILE 210
4.371 DBA_HIST_DB_CACHE_ADVICE 211
4.372 DBA_HIST_DISPATCHER 212
4.373 DBA_HIST_DLM_MISC 213
4.374 DBA_HIST_DYN_REMASTER_STATS 214
4.375 DBA_HIST_ENQUEUE_STAT 215

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.376 DBA_HIST_EVENT_HISTOGRAM 216
4.377 DBA_HIST_EVENT_NAME 217
4.378 DBA_HIST_FILEMETRIC_HISTORY 217
4.379 DBA_HIST_FILESTATXS 218
4.380 DBA_HIST_IC_CLIENT_STATS 219
4.381 DBA_HIST_IC_DEVICE_STATS 220
4.382 DBA_HIST_IM_SEG_STAT 221
4.383 DBA_HIST_IM_SEG_STAT_OBJ 222
4.384 DBA_HIST_INST_CACHE_TRANSFER 222
4.385 DBA_HIST_INSTANCE_RECOVERY 224
4.386 DBA_HIST_INTERCONNECT_PINGS 226
4.387 DBA_HIST_IOSTAT_DETAIL 227
4.388 DBA_HIST_IOSTAT_FILETYPE 228
4.389 DBA_HIST_IOSTAT_FILETYPE_NAME 229
4.390 DBA_HIST_IOSTAT_FUNCTION 230
4.391 DBA_HIST_IOSTAT_FUNCTION_NAME 231
4.392 DBA_HIST_JAVA_POOL_ADVICE 231

5 Static Data Dictionary Views: DBA_HIST_LATCH to DBA_STORED_SETTINGS

5.1 DBA_HIST_LATCH 1
5.2 DBA_HIST_LATCH_CHILDREN 2
5.3 DBA_HIST_LATCH_MISSES_SUMMARY 3
5.4 DBA_HIST_LATCH_NAME 4
5.5 DBA_HIST_LATCH_PARENT 5
5.6 DBA_HIST_LIBRARYCACHE 6
5.7 DBA_HIST_LOG 7
5.8 DBA_HIST_MEM_DYNAMIC_COMP 8
5.9 DBA_HIST_MEMORY_RESIZE_OPS 9
5.10 DBA_HIST_MEMORY_TARGET_ADVICE 10
5.11 DBA_HIST_METRIC_NAME 11
5.12 DBA_HIST_MTTR_TARGET_ADVICE 12
5.13 DBA_HIST_MUTEX_SLEEP 13
5.14 DBA_HIST_OPTIMIZER_ENV 13
5.15 DBA_HIST_OSSTAT 14
5.16 DBA_HIST_OSSTAT_NAME 15
5.17 DBA_HIST_PARAMETER 15
5.18 DBA_HIST_PARAMETER_NAME 16
5.19 DBA_HIST_PDB_IN_SNAP 17
5.20 DBA_HIST_PDB_INSTANCE 17
5.21 DBA_HIST_PERSISTENT_QMN_CACHE 18

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.22 DBA_HIST_PERSISTENT_QUEUES 19
5.23 DBA_HIST_PERSISTENT_SUBS 21
5.24 DBA_HIST_PGA_TARGET_ADVICE 22
5.25 DBA_HIST_PGASTAT 23
5.26 DBA_HIST_PLAN_OPERATION_NAME 24
5.27 DBA_HIST_PLAN_OPTION_NAME 25
5.28 DBA_HIST_PROCESS_MEM_SUMMARY 25
5.29 DBA_HIST_PROCESS_WAITTIME 26
5.30 DBA_HIST_RECOVERY_PROGRESS 27
5.31 DBA_HIST_REPLICATION_TBL_STATS 28
5.32 DBA_HIST_REPLICATION_TXN_STATS 29
5.33 DBA_HIST_REPORTS 29
5.34 DBA_HIST_REPORTS_CONTROL 31
5.35 DBA_HIST_REPORTS_DETAILS 31
5.36 DBA_HIST_REPORTS_TIMEBANDS 32
5.37 DBA_HIST_RESOURCE_LIMIT 33
5.38 DBA_HIST_ROWCACHE_SUMMARY 34
5.39 DBA_HIST_RSRC_CONSUMER_GROUP 35
5.40 DBA_HIST_RSRC_METRIC 37
5.41 DBA_HIST_RSRC_PDB_METRIC 39
5.42 DBA_HIST_RSRC_PLAN 41
5.43 DBA_HIST_RULE_SET 42
5.44 DBA_HIST_SEG_STAT 43
5.45 DBA_HIST_SEG_STAT_OBJ 45
5.46 DBA_HIST_SERVICE_NAME 46
5.47 DBA_HIST_SERVICE_STAT 47
5.48 DBA_HIST_SERVICE_WAIT_CLASS 47
5.49 DBA_HIST_SESS_SGA_STATS 48
5.50 DBA_HIST_SESS_TIME_STATS 49
5.51 DBA_HIST_SESSMETRIC_HISTORY 50
5.52 DBA_HIST_SGA 50
5.53 DBA_HIST_SGA_TARGET_ADVICE 51
5.54 DBA_HIST_SGASTAT 52
5.55 DBA_HIST_SHARED_POOL_ADVICE 53
5.56 DBA_HIST_SHARED_SERVER_SUMMARY 54
5.57 DBA_HIST_SNAP_ERROR 55
5.58 DBA_HIST_SNAPSHOT 56
5.59 DBA_HIST_SNAPSHOT_DETAILS 57
5.60 DBA_HIST_SQL_BIND_METADATA 58
5.61 DBA_HIST_SQL_PLAN 58
5.62 DBA_HIST_SQL_SUMMARY 61
5.63 DBA_HIST_SQL_WORKAREA_HSTGRM 61

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.64 DBA_HIST_SQLBIND 62
5.65 DBA_HIST_SQLCOMMAND_NAME 64
5.66 DBA_HIST_SQLSTAT 64
5.67 DBA_HIST_SQLTEXT 68
5.68 DBA_HIST_STAT_NAME 69
5.69 DBA_HIST_STREAMS_APPLY_SUM 70
5.70 DBA_HIST_STREAMS_CAPTURE 71
5.71 DBA_HIST_STREAMS_POOL_ADVICE 72
5.72 DBA_HIST_SYS_TIME_MODEL 73
5.73 DBA_HIST_SYSMETRIC_HISTORY 73
5.74 DBA_HIST_SYSMETRIC_SUMMARY 74
5.75 DBA_HIST_SYSSTAT 75
5.76 DBA_HIST_SYSTEM_EVENT 76
5.77 DBA_HIST_TABLESPACE 77
5.78 DBA_HIST_TABLESPACE_STAT 77
5.79 DBA_HIST_TBSPC_SPACE_USAGE 78
5.80 DBA_HIST_TEMPFILE 79
5.81 DBA_HIST_TEMPSTATXS 79
5.82 DBA_HIST_THREAD 81
5.83 DBA_HIST_TOPLEVELCALL_NAME 81
5.84 DBA_HIST_UNDOSTAT 82
5.85 DBA_HIST_WAITCLASSMET_HISTORY 83
5.86 DBA_HIST_WAITSTAT 84
5.87 DBA_HIST_WR_CONTROL 85
5.88 DBA_HIST_WR_SETTINGS 85
5.89 DBA_HISTOGRAMS 86
5.90 DBA_HIVE_COLUMNS 86
5.91 DBA_HIVE_DATABASES 86
5.92 DBA_HIVE_PART_KEY_COLUMNS 87
5.93 DBA_HIVE_TAB_PARTITIONS 87
5.94 DBA_HIVE_TABLES 87
5.95 DBA_HOST_ACES 87
5.96 DBA_HOST_ACLS 88
5.97 DBA_IDENTIFIERS 88
5.98 DBA_ILMDATAMOVEMENTPOLICIES 89
5.99 DBA_ILMEVALUATIONDETAILS 90
5.100 DBA_ILMOBJECTS 91
5.101 DBA_ILMPARAMETERS 92
5.102 DBA_ILMPOLICIES 93
5.103 DBA_ILMRESULTS 94
5.104 DBA_ILMTASKS 95
5.105 DBA_IM_EXPRESSIONS 95

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.106 DBA_IMMUTABLE_TABLES 96
5.107 DBA_IND_COLUMNS 96
5.108 DBA_IND_EXPRESSIONS 97
5.109 DBA_IND_PARTITIONS 97
5.110 DBA_IND_PENDING_STATS 97
5.111 DBA_IND_STATISTICS 97
5.112 DBA_IND_SUBPARTITIONS 98
5.113 DBA_INDEX_USAGE 98
5.114 DBA_INDEXES 99
5.115 DBA_INDEXTYPE_ARRAYTYPES 99
5.116 DBA_INDEXTYPE_COMMENTS 99
5.117 DBA_INDEXTYPE_OPERATORS 99
5.118 DBA_INDEXTYPES 100
5.119 DBA_INMEMORY_AIMTASKDETAILS 100
5.120 DBA_INMEMORY_AIMTASKS 100
5.121 DBA_INTERNAL_TRIGGERS 101
5.122 DBA_INVALID_OBJECTS 101
5.123 DBA_JAVA_ARGUMENTS 103
5.124 DBA_JAVA_CLASSES 103
5.125 DBA_JAVA_COMPILER_OPTIONS 103
5.126 DBA_JAVA_DERIVATIONS 103
5.127 DBA_JAVA_FIELDS 104
5.128 DBA_JAVA_IMPLEMENTS 104
5.129 DBA_JAVA_INNERS 104
5.130 DBA_JAVA_LAYOUTS 104
5.131 DBA_JAVA_METHODS 105
5.132 DBA_JAVA_NCOMPS 105
5.133 DBA_JAVA_POLICY 105
5.134 DBA_JAVA_RESOLVERS 106
5.135 DBA_JAVA_THROWS 106
5.136 DBA_JOBS 106
5.137 DBA_JOBS_RUNNING 107
5.138 DBA_JOIN_IND_COLUMNS 108
5.139 DBA_JOINGROUPS 108
5.140 DBA_JSON_COLUMNS 109
5.141 DBA_JSON_DATAGUIDE_FIELDS 110
5.142 DBA_JSON_DATAGUIDES 110
5.143 DBA_KGLLOCK 110
5.144 DBA_LIBRARIES 110
5.145 DBA_LMT_FREE_SPACE 111
5.146 DBA_LMT_USED_EXTENTS 111
5.147 DBA_LOB_PARTITIONS 111

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.148 DBA_LOB_SUBPARTITIONS 112
5.149 DBA_LOB_TEMPLATES 112
5.150 DBA_LOBS 112
5.151 DBA_LOCK 112
5.152 DBA_LOCK_INTERNAL 113
5.153 DBA_LOCKDOWN_PROFILES 113
5.154 DBA_LOCKS 115
5.155 DBA_LOG_GROUP_COLUMNS 115
5.156 DBA_LOG_GROUPS 115
5.157 DBA_LOGMNR_DICTIONARY_BUILDLOG 115
5.158 DBA_LOGMNR_LOG 116
5.159 DBA_LOGMNR_PURGED_LOG 117
5.160 DBA_LOGMNR_SESSION 117
5.161 DBA_LOGSTDBY_EVENTS 118
5.162 DBA_LOGSTDBY_HISTORY 119
5.163 DBA_LOGSTDBY_LOG 121
5.164 DBA_LOGSTDBY_NOT_UNIQUE 122
5.165 DBA_LOGSTDBY_PARAMETERS 122
5.166 DBA_LOGSTDBY_PLSQL_MAP 124
5.167 DBA_LOGSTDBY_PLSQL_SUPPORT 124
5.168 DBA_LOGSTDBY_PROGRESS 125
5.169 DBA_LOGSTDBY_SKIP 125
5.170 DBA_LOGSTDBY_SKIP_TRANSACTION 126
5.171 DBA_LOGSTDBY_UNSUPPORTED 126
5.172 DBA_LOGSTDBY_UNSUPPORTED_TABLE 127
5.173 DBA_MEASURE_FOLDER_CONTENTS 127
5.174 DBA_MEASURE_FOLDER_SUBFOLDERS 127
5.175 DBA_MEASURE_FOLDERS 128
5.176 DBA_METADATA_PROPERTIES 128
5.177 DBA_METHOD_PARAMS 128
5.178 DBA_METHOD_RESULTS 128
5.179 DBA_MINING_MODEL_ATTRIBUTES 129
5.180 DBA_MINING_MODEL_PARTITIONS 129
5.181 DBA_MINING_MODEL_SETTINGS 129
5.182 DBA_MINING_MODEL_TABLES 129
5.183 DBA_MINING_MODEL_VIEWS 130
5.184 DBA_MINING_MODEL_XFORMS 130
5.185 DBA_MINING_MODELS 130
5.186 DBA_MVIEW_AGGREGATES 130
5.187 DBA_MVIEW_ANALYSIS 131
5.188 DBA_MVIEW_COMMENTS 131
5.189 DBA_MVIEW_DETAIL_PARTITION 131

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.190 DBA_MVIEW_DETAIL_RELATIONS 132
5.191 DBA_MVIEW_DETAIL_SUBPARTITION 132
5.192 DBA_MVIEW_JOINS 132
5.193 DBA_MVIEW_KEYS 132
5.194 DBA_MVIEW_LOG_FILTER_COLS 133
5.195 DBA_MVIEW_LOGS 133
5.196 DBA_MVIEW_REFRESH_TIMES 133
5.197 DBA_MVIEWS 133
5.198 DBA_MVREF_CHANGE_STATS 134
5.199 DBA_MVREF_RUN_STATS 135
5.200 DBA_MVREF_STATS 136
5.201 DBA_MVREF_STATS_PARAMS 137
5.202 DBA_MVREF_STATS_SYS_DEFAULTS 137
5.203 DBA_MVREF_STMT_STATS 138
5.204 DBA_NESTED_TABLE_COLS 139
5.205 DBA_NESTED_TABLES 139
5.206 DBA_NETWORK_ACL_PRIVILEGES 139
5.207 DBA_NETWORK_ACLS 140
5.208 DBA_OBJ_AUDIT_OPTS 141
5.209 DBA_OBJ_COLATTRS 142
5.210 DBA_OBJECT_SIZE 142
5.211 DBA_OBJECT_TABLES 143
5.212 DBA_OBJECT_USAGE 143
5.213 DBA_OBJECTS 144
5.214 DBA_OBJECTS_AE 144
5.215 DBA_OPANCILLARY 144
5.216 DBA_OPARGUMENTS 145
5.217 DBA_OPBINDINGS 145
5.218 DBA_OPERATOR_COMMENTS 145
5.219 DBA_OPERATORS 145
5.220 DBA_OPTSTAT_OPERATION_TASKS 145
5.221 DBA_OPTSTAT_OPERATIONS 147
5.222 DBA_ORPHAN_KEY_TABLE 148
5.223 DBA_OUTLINE_HINTS 148
5.224 DBA_OUTLINES 149
5.225 DBA_OUTSTANDING_ALERTS 150
5.226 DBA_PARALLEL_EXECUTE_CHUNKS 151
5.227 DBA_PARALLEL_EXECUTE_TASKS 152
5.228 DBA_PART_COL_STATISTICS 153
5.229 DBA_PART_HISTOGRAMS 153
5.230 DBA_PART_INDEXES 153
5.231 DBA_PART_KEY_COLUMNS 154

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.232 DBA_PART_LOBS 154
5.233 DBA_PART_TABLES 154
5.234 DBA_PARTIAL_DROP_TABS 154
5.235 DBA_PDB_HISTORY 154
5.236 DBA_PDB_SAVED_STATES 155
5.237 DBA_PDB_SNAPSHOTFILE 156
5.238 DBA_PDB_SNAPSHOTS 156
5.239 DBA_PDBS 157
5.240 DBA_PENDING_CONV_TABLES 159
5.241 DBA_PENDING_TRANSACTIONS 159
5.242 DBA_PLSQL_COLL_TYPES 160
5.243 DBA_PLSQL_OBJECT_SETTINGS 160
5.244 DBA_PLSQL_TYPE_ATTRS 160
5.245 DBA_PLSQL_TYPES 160
5.246 DBA_POLICIES 161
5.247 DBA_POLICY_ATTRIBUTES 161
5.248 DBA_POLICY_CONTEXTS 161
5.249 DBA_POLICY_GROUPS 161
5.250 DBA_PRIV_AUDIT_OPTS 162
5.251 DBA_PRIV_CAPTURES 162
5.252 DBA_PRIVATE_TEMP_TABLES 163
5.253 DBA_PROCEDURES 164
5.254 DBA_PROFILES 164
5.255 DBA_PROPAGATION 164
5.256 DBA_PROXIES 165
5.257 DBA_QUEUE_SCHEDULES 165
5.258 DBA_QUEUE_SUBSCRIBERS 166
5.259 DBA_QUEUE_TABLES 166
5.260 DBA_QUEUES 166
5.261 DBA_RAT_CAPTURE_SCHEMA_INFO 166
5.262 DBA_RCHILD 167
5.263 DBA_RECOVERABLE_SCRIPT 167
5.264 DBA_RECOVERABLE_SCRIPT_BLOCKS 168
5.265 DBA_RECOVERABLE_SCRIPT_ERRORS 168
5.266 DBA_RECOVERABLE_SCRIPT_HIST 169
5.267 DBA_RECOVERABLE_SCRIPT_PARAMS 169
5.268 DBA_RECYCLEBIN 169
5.269 DBA_REDEFINITION_ERRORS 171
5.270 DBA_REDEFINITION_OBJECTS 171
5.271 DBA_REDEFINITION_STATUS 172
5.272 DBA_REFRESH 173
5.273 DBA_REFRESH_CHILDREN 173

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.274 DBA_REFS 173
5.275 DBA_REGISTERED_ARCHIVED_LOG 173
5.276 DBA_REGISTERED_MVIEWS 174
5.277 DBA_REGISTRY 174
5.278 DBA_REGISTRY_BACKPORTS 175
5.279 DBA_REGISTRY_HIERARCHY 176
5.280 DBA_REGISTRY_HISTORY 176
5.281 DBA_REGISTRY_LOG 177
5.282 DBA_REGISTRY_SCHEMAS 177
5.283 DBA_REGISTRY_SQLPATCH 178
5.284 DBA_REPAIR_TABLE 179
5.285 DBA_REPL_DBNAME_MAPPING 180
5.286 DBA_REPLICATION_PROCESS_EVENTS 180
5.287 DBA_RESOURCE_INCARNATIONS 180
5.288 DBA_RESUMABLE 181
5.289 DBA_REWRITE_EQUIVALENCES 182
5.290 DBA_RGROUP 182
5.291 DBA_ROLE_PRIVS 183
5.292 DBA_ROLES 183
5.293 DBA_ROLLBACK_SEGS 184
5.294 DBA_ROLLING_DATABASES 185
5.295 DBA_ROLLING_EVENTS 186
5.296 DBA_ROLLING_PARAMETERS 186
5.297 DBA_ROLLING_PLAN 187
5.298 DBA_ROLLING_STATISTICS 188
5.299 DBA_ROLLING_STATUS 188
5.300 DBA_ROLLING_UNSUPPORTED 189
5.301 DBA_RSRC_CATEGORIES 190
5.302 DBA_RSRC_CONSUMER_GROUP_PRIVS 190
5.303 DBA_RSRC_CONSUMER_GROUPS 191
5.304 DBA_RSRC_GROUP_MAPPINGS 191
5.305 DBA_RSRC_IO_CALIBRATE 191
5.306 DBA_RSRC_MANAGER_SYSTEM_PRIVS 192
5.307 DBA_RSRC_MAPPING_PRIORITY 192
5.308 DBA_RSRC_PLAN_DIRECTIVES 193
5.309 DBA_RSRC_PLANS 196
5.310 DBA_RULE_SET_RULES 197
5.311 DBA_RULE_SETS 197
5.312 DBA_RULES 197
5.313 DBA_SCHEDULER_CHAIN_RULES 198
5.314 DBA_SCHEDULER_CHAIN_STEPS 198
5.315 DBA_SCHEDULER_CHAINS 198

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.316 DBA_SCHEDULER_CREDENTIALS 198
5.317 DBA_SCHEDULER_DB_DESTS 199
5.318 DBA_SCHEDULER_DESTS 199
5.319 DBA_SCHEDULER_EXTERNAL_DESTS 199
5.320 DBA_SCHEDULER_FILE_WATCHERS 199
5.321 DBA_SCHEDULER_GLOBAL_ATTRIBUTE 200
5.322 DBA_SCHEDULER_GROUP_MEMBERS 200
5.323 DBA_SCHEDULER_GROUPS 200
5.324 DBA_SCHEDULER_INCOMPAT_MEMBER 200
5.325 DBA_SCHEDULER_INCOMPATS 201
5.326 DBA_SCHEDULER_JOB_ARGS 201
5.327 DBA_SCHEDULER_JOB_CLASSES 201
5.328 DBA_SCHEDULER_JOB_DESTS 201
5.329 DBA_SCHEDULER_JOB_LOG 202
5.330 DBA_SCHEDULER_JOB_ROLES 202
5.331 DBA_SCHEDULER_JOB_RUN_DETAILS 203
5.332 DBA_SCHEDULER_JOBS 203
5.333 DBA_SCHEDULER_NOTIFICATIONS 204
5.334 DBA_SCHEDULER_PROGRAM_ARGS 204
5.335 DBA_SCHEDULER_PROGRAMS 204
5.336 DBA_SCHEDULER_REMOTE_DATABASES 204
5.337 DBA_SCHEDULER_REMOTE_JOBSTATE 205
5.338 DBA_SCHEDULER_RESOURCES 205
5.339 DBA_SCHEDULER_RSC_CONSTRAINTS 205
5.340 DBA_SCHEDULER_RUNNING_CHAINS 205
5.341 DBA_SCHEDULER_RUNNING_JOBS 206
5.342 DBA_SCHEDULER_SCHEDULES 206
5.343 DBA_SCHEDULER_WINDOW_DETAILS 206
5.344 DBA_SCHEDULER_WINDOW_GROUPS 206
5.345 DBA_SCHEDULER_WINDOW_LOG 207
5.346 DBA_SCHEDULER_WINDOWS 207
5.347 DBA_SCHEDULER_WINGROUP_MEMBERS 207
5.348 DBA_SEC_RELEVANT_COLS 207
5.349 DBA_SECONDARY_OBJECTS 208
5.350 DBA_SEGMENTS 208
5.351 DBA_SEGMENTS_OLD 210
5.352 DBA_SENSITIVE_COLUMN_TYPES 211
5.353 DBA_SENSITIVE_DATA 212
5.354 DBA_SEQUENCES 212
5.355 DBA_SERVER_REGISTRY 212
5.356 DBA_SERVICES 213
5.357 DBA_SODA_COLLECTIONS 213

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.358 DBA_SOURCE 214
5.359 DBA_SOURCE_AE 215
5.360 DBA_SQL_MANAGEMENT_CONFIG 215
5.361 DBA_SQL_QUARANTINE 215
5.362 DBA_SQL_PATCHES 216
5.363 DBA_SQL_PLAN_BASELINES 217
5.364 DBA_SQL_PLAN_DIR_OBJECTS 219
5.365 DBA_SQL_PLAN_DIRECTIVES 219
5.366 DBA_SQL_PROFILES 220
5.367 DBA_SQL_TRANSLATION_PROFILES 221
5.368 DBA_SQL_TRANSLATIONS 222
5.369 DBA_SQLJ_TYPE_ATTRS 222
5.370 DBA_SQLJ_TYPE_METHODS 222
5.371 DBA_SQLJ_TYPES 222
5.372 DBA_SQLSET 223
5.373 DBA_SQLSET_BINDS 223
5.374 DBA_SQLSET_PLANS 223
5.375 DBA_SQLSET_REFERENCES 223
5.376 DBA_SQLSET_STATEMENTS 223
5.377 DBA_SQLTUNE_BINDS 225
5.378 DBA_SQLTUNE_PLANS 226
5.379 DBA_SQLTUNE_RATIONALE_PLAN 228
5.380 DBA_SQLTUNE_STATISTICS 229
5.381 DBA_SR_GRP_STATUS 229
5.382 DBA_SR_GRP_STATUS_ALL 231
5.383 DBA_SR_OBJ 232
5.384 DBA_SR_OBJ_ALL 233
5.385 DBA_SR_OBJ_STATUS 234
5.386 DBA_SR_OBJ_STATUS_ALL 235
5.387 DBA_SR_PARTN_OPS 236
5.388 DBA_SR_STLOG_EXCEPTIONS 236
5.389 DBA_SR_STLOG_STATS 237
5.390 DBA_SSCR_CAPTURE 238
5.391 DBA_SSCR_RESTORE 238
5.392 DBA_STAT_EXTENSIONS 239
5.393 DBA_STATEMENTS 239
5.394 DBA_STMT_AUDIT_OPTS 240
5.395 DBA_STORED_SETTINGS 240

Copyright © 2002, 2026, Oracle and/or its affiliates.

6 Static Data Dictionary Views: DBA_STREAMS_ADD_COLUMN to USER_ZONEMAPS

6.1 DBA_STREAMS_ADD_COLUMN 1
6.2 DBA_STREAMS_DELETE_COLUMN 1
6.3 DBA_STREAMS_GLOBAL_RULES 2
6.4 DBA_STREAMS_KEEP_COLUMNS 2
6.5 DBA_STREAMS_MESSAGE_CONSUMERS 3
6.6 DBA_STREAMS_NEWLY_SUPPORTED 3
6.7 DBA_STREAMS_RENAME_COLUMN 3
6.8 DBA_STREAMS_RENAME_SCHEMA 3
6.9 DBA_STREAMS_RENAME_TABLE 4
6.10 DBA_STREAMS_SCHEMA_RULES 4
6.11 DBA_STREAMS_TABLE_RULES 5
6.12 DBA_STREAMS_TP_COMPONENT 5
6.13 DBA_STREAMS_TP_COMPONENT_LINK 5
6.14 DBA_STREAMS_TP_COMPONENT_STAT 6
6.15 DBA_STREAMS_TP_DATABASE 7
6.16 DBA_STREAMS_TP_PATH_BOTTLENECK 7
6.17 DBA_STREAMS_TP_PATH_STAT 8
6.18 DBA_STREAMS_TRANSFORM_FUNCTION 8
6.19 DBA_SUBPART_COL_STATISTICS 8
6.20 DBA_SUBPART_HISTOGRAMS 9
6.21 DBA_SUBPART_KEY_COLUMNS 9
6.22 DBA_SUBPARTITION_TEMPLATES 9
6.23 DBA_SUBSCR_REGISTRATIONS 9
6.24 DBA_SUPPLEMENTAL_LOGGING 11
6.25 DBA_SYNC_CAPTURE 12
6.26 DBA_SYNC_CAPTURE_PREPARED_TABS 12
6.27 DBA_SYNC_CAPTURE_TABLES 12
6.28 DBA_SYNONYMS 12
6.29 DBA_SYS_PRIVS 12
6.30 DBA_TAB_COL_STATISTICS 13
6.31 DBA_TAB_COLS 13
6.32 DBA_TAB_COLUMNS 16
6.33 DBA_TAB_COMMENTS 18
6.34 DBA_TAB_HISTGRM_PENDING_STATS 19
6.35 DBA_TAB_HISTOGRAMS 19
6.36 DBA_TAB_IDENTITY_COLS 19
6.37 DBA_TAB_MODIFICATIONS 19
6.38 DBA_TAB_PARTITIONS 20
6.39 DBA_TAB_PENDING_STATS 20

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.40 DBA_TAB_PRIVS 20
6.41 DBA_TAB_STATISTICS 21
6.42 DBA_TAB_STAT_PREFS 21
6.43 DBA_TAB_STATS_HISTORY 21
6.44 DBA_TAB_SUBPARTITIONS 22
6.45 DBA_TABLES 22
6.46 DBA_TABLESPACE_GROUPS 22
6.47 DBA_TABLESPACE_THRESHOLDS 22
6.48 DBA_TABLESPACE_USAGE_METRICS 23
6.49 DBA_TABLESPACES 24
6.50 DBA_TEMP_FILES 28
6.51 DBA_TEMP_FREE_SPACE 29
6.52 DBA_THRESHOLDS 29
6.53 DBA_TRANSFORMATIONS 30
6.54 DBA_TRIGGER_COLS 31
6.55 DBA_TRIGGER_ORDERING 31
6.56 DBA_TRIGGERS 31
6.57 DBA_TRIGGERS_AE 32
6.58 DBA_TS_QUOTAS 32
6.59 DBA_TSDP_IMPORT_ERRORS 32
6.60 DBA_TSDP_POLICY_CONDITION 33
6.61 DBA_TSDP_POLICY_FEATURE 33
6.62 DBA_TSDP_POLICY_PARAMETER 34
6.63 DBA_TSDP_POLICY_PROTECTION 34
6.64 DBA_TSDP_POLICY_TYPE 35
6.65 DBA_TSM_DESTINATION 35
6.66 DBA_TSM_SOURCE 36
6.67 DBA_TSTZ_TAB_COLS 37
6.68 DBA_TSTZ_TABLES 37
6.69 DBA_TUNE_MVIEW 37
6.70 DBA_TYPE_ATTRS 38
6.71 DBA_TYPE_METHODS 38
6.72 DBA_TYPE_VERSIONS 38
6.73 DBA_TYPES 38
6.74 DBA_UMF_LINK 39
6.75 DBA_UMF_REGISTRATION 39
6.76 DBA_UMF_SERVICE 40
6.77 DBA_UMF_TOPOLOGY 40
6.78 DBA_UNDO_EXTENTS 41
6.79 DBA_UNUSED_COL_TABS 42
6.80 DBA_UNUSED_GRANTS 42
6.81 DBA_UNUSED_OBJPRIVS 43

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.82 DBA_UNUSED_OBJPRIVS_PATH 44
6.83 DBA_UNUSED_PRIVS 44
6.84 DBA_UNUSED_SYSPRIVS 45
6.85 DBA_UNUSED_SYSPRIVS_PATH 46
6.86 DBA_UNUSED_USERPRIVS 47
6.87 DBA_UNUSED_USERPRIVS_PATH 48
6.88 DBA_UPDATABLE_COLUMNS 48
6.89 DBA_USED_OBJPRIVS 49
6.90 DBA_USED_OBJPRIVS_PATH 49
6.91 DBA_USED_PRIVS 50
6.92 DBA_USED_PUBPRIVS 51
6.93 DBA_USED_SYSPRIVS 52
6.94 DBA_USED_SYSPRIVS_PATH 53
6.95 DBA_USED_USERPRIVS 54
6.96 DBA_USED_USERPRIVS_PATH 55
6.97 DBA_USERS 55
6.98 DBA_USERS_WITH_DEFPWD 61
6.99 DBA_USTATS 62
6.100 DBA_VARRAYS 62
6.101 DBA_VIEWS 62
6.102 DBA_VIEWS_AE 62
6.103 DBA_WAITERS 62
6.104 DBA_WALLET_ACES 63
6.105 DBA_WALLET_ACLS 63
6.106 DBA_WARNING_SETTINGS 64
6.107 DBA_WI_CAPTURE_FILES 64
6.108 DBA_WI_JOBS 64
6.109 DBA_WI_OBJECTS 65
6.110 DBA_WI_PATTERN_ITEMS 65
6.111 DBA_WI_PATTERNS 65
6.112 DBA_WI_STATEMENTS 66
6.113 DBA_WI_TEMPLATE_EXECUTIONS 66
6.114 DBA_WI_TEMPLATES 67
6.115 DBA_WORKLOAD_ACTIVE_USER_MAP 67
6.116 DBA_WORKLOAD_CAPTURE_SQLTEXT 68
6.117 DBA_WORKLOAD_CAPTURES 68
6.118 DBA_WORKLOAD_CONNECTION_MAP 71
6.119 DBA_WORKLOAD_DIV_SUMMARY 71
6.120 DBA_WORKLOAD_FILTERS 73
6.121 DBA_WORKLOAD_GROUP_ASSIGNMENTS 73
6.122 DBA_WORKLOAD_LONG_SQLTEXT 74
6.123 DBA_WORKLOAD_REPLAY_CLIENTS 74

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.124 DBA_WORKLOAD_REPLAY_DIVERGENCE 75
6.125 DBA_WORKLOAD_REPLAY_SCHEDULES 76
6.126 DBA_WORKLOAD_REPLAYS 77
6.127 DBA_WORKLOAD_SCHEDULE_CAPTURES 81
6.128 DBA_WORKLOAD_SCHEDULE_ORDERING 82
6.129 DBA_WORKLOAD_SQL_MAP 83
6.130 DBA_WORKLOAD_TRACKED_COMMITS 83
6.131 DBA_WORKLOAD_USER_MAP 84
6.132 DBA_XML_INDEXES 85
6.133 DBA_XML_NESTED_TABLES 85
6.134 DBA_XML_OUT_OF_LINE_TABLES 85
6.135 DBA_XML_SCHEMA_ATTRIBUTES 85
6.136 DBA_XML_SCHEMA_COMPLEX_TYPES 86
6.137 DBA_XML_SCHEMA_ELEMENTS 86
6.138 DBA_XML_SCHEMA_NAMESPACES 86
6.139 DBA_XML_SCHEMA_SIMPLE_TYPES 86
6.140 DBA_XML_SCHEMA_SUBSTGRP_HEAD 87
6.141 DBA_XML_SCHEMA_SUBSTGRP_MBRS 87
6.142 DBA_XML_SCHEMAS 87
6.143 DBA_XML_TAB_COLS 87
6.144 DBA_XML_TABLES 88
6.145 DBA_XML_VIEW_COLS 88
6.146 DBA_XML_VIEWS 88
6.147 DBA_XS_AUDIT_POLICY_OPTIONS 88
6.148 DBA_XS_AUDIT_TRAIL 89
6.149 DBA_XS_ENABLED_AUDIT_POLICIES 90
6.150 DBA_XS_ENB_AUDIT_POLICIES 91
6.151 DBA_XSTREAM_ADMINISTRATOR 91
6.152 DBA_XSTREAM_INBOUND 91
6.153 DBA_XSTREAM_INBOUND_PROGRESS 92
6.154 DBA_XSTREAM_OUT_SUPPORT_MODE 92
6.155 DBA_XSTREAM_OUTBOUND 92
6.156 DBA_XSTREAM_OUTBOUND_PROGRESS 92
6.157 DBA_XSTREAM_RULES 93
6.158 DBA_XSTREAM_SPLIT_MERGE 93
6.159 DBA_XSTREAM_SPLIT_MERGE_HIST 95
6.160 DBA_XSTREAM_STMT_HANDLERS 96
6.161 DBA_XSTREAM_STMTS 96
6.162 DBA_XSTREAM_TRANSFORMATIONS 97
6.163 DBA_XTERNAL_LOC_PARTITIONS 97
6.164 DBA_XTERNAL_LOC_SUBPARTITIONS 97
6.165 DBA_XTERNAL_PART_TABLES 98

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.166 DBA_XTERNAL_TAB_PARTITIONS 98
6.167 DBA_XTERNAL_TAB_SUBPARTITIONS 98
6.168 DBA_ZONEMAP_MEASURES 98
6.169 DBA_ZONEMAPS 99
6.170 DBFS_CONTENT 99
6.171 DBFS_CONTENT_PROPERTIES 101
6.172 DBMS_ALERT_INFO 101
6.173 DBMS_LOCK_ALLOCATED 101
6.174 DBMS_METADATA_PARSE_ITEMS 102
6.175 DBMS_METADATA_TRANSFORM_PARAMS 102
6.176 DBMS_METADATA_TRANSFORMS 103
6.177 DEPTREE 103
6.178 DICT 104
6.179 DICT_COLUMNS 104
6.180 DICTIONARY 104
6.181 DICTIONARY_CREDENTIALS_ENCRYPT 104
6.182 DM_USER_MODELS 105
6.183 DOCUMENT_LINKS 106
6.184 ERROR_SIZE 107
6.185 EXCEPTIONS 107
6.186 FLASHBACK_TRANSACTION_QUERY 107
6.187 GLOBAL_CONTEXT 108
6.188 GLOBAL_NAME 109
6.189 HS_ALL_CAPS 109
6.190 HS_ALL_DD 109
6.191 HS_ALL_INITS 109
6.192 HS_BASE_CAPS 110
6.193 HS_BASE_DD 110
6.194 HS_CLASS_CAPS 110
6.195 HS_CLASS_DD 111
6.196 HS_CLASS_INIT 111
6.197 HS_FDS_CLASS 111
6.198 HS_FDS_INST 112
6.199 HS_INST_CAPS 112
6.200 HS_INST_DD 112
6.201 HS_INST_INIT 113
6.202 IDEPTREE 113
6.203 IND 113
6.204 INDEX_HISTOGRAM 114
6.205 INDEX_STATS 114
6.206 LOGSTDBY_UNSUPPORTED_TABLES 116
6.207 MAP_OBJECT 116

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.208 NLS_DATABASE_PARAMETERS 116
6.209 NLS_INSTANCE_PARAMETERS 117
6.210 NLS_SESSION_PARAMETERS 117
6.211 OBJ 117
6.212 PATH_VIEW 117
6.213 PDB_ALERTS 118
6.214 PDB_PLUG_IN_VIOLATIONS 118
6.215 PLAN_TABLE 119
6.216 PLUGGABLE_SET_CHECK 122
6.217 PRODUCT_COMPONENT_VERSION 123
6.218 PROXY_USERS 123
6.219 PSTUBTBL 124
6.220 PUBLIC_DEPENDENCY 124
6.221 PUBLICSYN 124
6.222 QUEUE_PRIVILEGES 125
6.223 RECYCLEBIN 125
6.224 REDACTION_COLUMNS 125
6.225 REDACTION_EXPRESSIONS 126
6.226 REDACTION_POLICIES 126
6.227 REDACTION_VALUES_FOR_TYPE_FULL 127
6.228 REPORT_COMPONENTS 128
6.229 REPORT_FILES 128
6.230 REPORT_FORMATS 129
6.231 RESOURCE_COST 129
6.232 RESOURCE_MAP 129
6.233 RESOURCE_VIEW 130
6.234 ROLE_ROLE_PRIVS 130
6.235 ROLE_SYS_PRIVS 131
6.236 ROLE_TAB_PRIVS 131
6.237 SCHEDULER_BATCH_ERRORS 132
6.238 SCHEMA_EXPORT_OBJECTS 132
6.239 SEQ 133
6.240 SESSION_CONTEXT 133
6.241 SESSION_PRIVS 133
6.242 SESSION_ROLES 133
6.243 SOURCE_SIZE 133
6.244 STMT_AUDIT_OPTION_MAP 134
6.245 SYN 134
6.246 SYNONYMS 134
6.247 SYS_OBJECTS 135
6.248 SYSCATALOG 135
6.249 SYSFILES 135

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.250 SYSSEGOBJ 135
6.251 SYSTEM_PRIVILEGE_MAP 135
6.252 TAB 136
6.253 TABLE_EXPORT_OBJECTS 136
6.254 TABLE_PRIVILEGE_MAP 136
6.255 TABQUOTAS 136
6.256 TABS 137
6.257 TRUSTED_SERVERS 137
6.258 TS_PITR_CHECK 137
6.259 TS_PITR_OBJECTS_TO_BE_DROPPED 138
6.260 UNI_PLUGGABLE_SET_CHECK 139
6.261 UNIFIED_AUDIT_TRAIL 139
6.262 USABLE_EDITIONS 151
6.263 USER_ADDM_FDG_BREAKDOWN 151
6.264 USER_ADDM_FINDINGS 151
6.265 USER_ADDM_INSTANCES 152
6.266 USER_ADDM_TASK_DIRECTIVES 152
6.267 USER_ADDM_TASKS 152
6.268 USER_ADVISOR_ACTIONS 152
6.269 USER_ADVISOR_DIR_TASK_INST 153
6.270 USER_ADVISOR_EXEC_PARAMETERS 153
6.271 USER_ADVISOR_EXECUTIONS 153
6.272 USER_ADVISOR_FDG_BREAKDOWN 153
6.273 USER_ADVISOR_FINDINGS 154
6.274 USER_ADVISOR_JOURNAL 154
6.275 USER_ADVISOR_LOG 154
6.276 USER_ADVISOR_OBJECTS 154
6.277 USER_ADVISOR_PARAMETERS 155
6.278 USER_ADVISOR_RATIONALE 155
6.279 USER_ADVISOR_RECOMMENDATIONS 155
6.280 USER_ADVISOR_SQLA_REC_SUM 155
6.281 USER_ADVISOR_SQLA_TABLES 156
6.282 USER_ADVISOR_SQLA_WK_MAP 156
6.283 USER_ADVISOR_SQLA_WK_STMTS 156
6.284 USER_ADVISOR_SQLPLANS 156
6.285 USER_ADVISOR_SQLSTATS 157
6.286 USER_ADVISOR_SQLW_JOURNAL 157
6.287 USER_ADVISOR_SQLW_PARAMETERS 157
6.288 USER_ADVISOR_SQLW_STMTS 157
6.289 USER_ADVISOR_SQLW_SUM 158
6.290 USER_ADVISOR_SQLW_TABLES 158
6.291 USER_ADVISOR_SQLW_TEMPLATES 158

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.292 USER_ADVISOR_TASKS 158
6.293 USER_ADVISOR_TEMPLATES 159
6.294 USER_ALL_TABLES 159
6.295 USER_ANALYTIC_VIEW_ATTR_CLASS 159
6.296 USER_ANALYTIC_VIEW_ATTR_CLS 159
6.297 USER_ANALYTIC_VIEW_ATTR_CLS_AE 160
6.298 USER_ANALYTIC_VIEW_BAS_MEAS 160
6.299 USER_ANALYTIC_VIEW_BAS_MEAS_AE 160
6.300 USER_ANALYTIC_VIEW_BASE_MEAS 161
6.301 USER_ANALYTIC_VIEW_CALC_MEAS 161
6.302 USER_ANALYTIC_VIEW_CLASS 161
6.303 USER_ANALYTIC_VIEW_CLASS_AE 161
6.304 USER_ANALYTIC_VIEW_CLC_MEAS 162
6.305 USER_ANALYTIC_VIEW_CLC_MEAS_AE 162
6.306 USER_ANALYTIC_VIEW_COLUMNS 162
6.307 USER_ANALYTIC_VIEW_COLUMNS_AE 163
6.308 USER_ANALYTIC_VIEW_DIM_CLASS 163
6.309 USER_ANALYTIC_VIEW_DIM_CLS 163
6.310 USER_ANALYTIC_VIEW_DIM_CLS_AE 164
6.311 USER_ANALYTIC_VIEW_DIMENSIONS 164
6.312 USER_ANALYTIC_VIEW_DIMS 164
6.313 USER_ANALYTIC_VIEW_DIMS_AE 165
6.314 USER_ANALYTIC_VIEW_HIER_CLASS 165
6.315 USER_ANALYTIC_VIEW_HIER_CLS 165
6.316 USER_ANALYTIC_VIEW_HIER_CLS_AE 166
6.317 USER_ANALYTIC_VIEW_HIERS 166
6.318 USER_ANALYTIC_VIEW_HIERS_AE 166
6.319 USER_ANALYTIC_VIEW_KEYS 167
6.320 USER_ANALYTIC_VIEW_KEYS_AE 167
6.321 USER_ANALYTIC_VIEW_LEVEL_CLASS 167
6.322 USER_ANALYTIC_VIEW_LEVELS 168
6.323 USER_ANALYTIC_VIEW_LEVELS_AE 168
6.324 USER_ANALYTIC_VIEW_LVL_CLS 168
6.325 USER_ANALYTIC_VIEW_LVL_CLS_AE 169
6.326 USER_ANALYTIC_VIEW_LVLGRPS 169
6.327 USER_ANALYTIC_VIEW_LVLGRPS_AE 169
6.328 USER_ANALYTIC_VIEW_MEAS_CLASS 170
6.329 USER_ANALYTIC_VIEW_MEAS_CLS 170
6.330 USER_ANALYTIC_VIEW_MEAS_CLS_AE 170
6.331 USER_ANALYTIC_VIEWS 171
6.332 USER_ANALYTIC_VIEWS_AE 171
6.333 USER_ANNOTATION_VALUES 171

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.334 USER_ANNOTATIONS 172
6.335 USER_ANNOTATIONS_USAGE 172
6.336 USER_APPLY_ERROR 172
6.337 USER_AQ_AGENT_PRIVS 173
6.338 USER_ARGUMENTS 173
6.339 USER_ASSEMBLIES 173
6.340 USER_ASSOCIATIONS 173
6.341 USER_ATTRIBUTE_DIM_ATTR_CLASS 174
6.342 USER_ATTRIBUTE_DIM_ATTR_CLS 174
6.343 USER_ATTRIBUTE_DIM_ATTR_CLS_AE 174
6.344 USER_ATTRIBUTE_DIM_ATTRS 175
6.345 USER_ATTRIBUTE_DIM_ATTRS_AE 175
6.346 USER_ATTRIBUTE_DIM_CLASS 175
6.347 USER_ATTRIBUTE_DIM_CLASS_AE 176
6.348 USER_ATTRIBUTE_DIM_JN_PTHS 176
6.349 USER_ATTRIBUTE_DIM_JN_PTHS_AE 176
6.350 USER_ATTRIBUTE_DIM_JOIN_PATHS 177
6.351 USER_ATTRIBUTE_DIM_KEYS 177
6.352 USER_ATTRIBUTE_DIM_KEYS_AE 177
6.353 USER_ATTRIBUTE_DIM_LEVEL_ATTRS 178
6.354 USER_ATTRIBUTE_DIM_LEVELS 178
6.355 USER_ATTRIBUTE_DIM_LEVELS_AE 178
6.356 USER_ATTRIBUTE_DIM_LVL_ATRS 179
6.357 USER_ATTRIBUTE_DIM_LVL_ATRS_AE 179
6.358 USER_ATTRIBUTE_DIM_LVL_CLASS 179
6.359 USER_ATTRIBUTE_DIM_LVL_CLS 180
6.360 USER_ATTRIBUTE_DIM_LVL_CLS_AE 180
6.361 USER_ATTRIBUTE_DIM_ORD_ATRS 180
6.362 USER_ATTRIBUTE_DIM_ORD_ATRS_AE 181
6.363 USER_ATTRIBUTE_DIM_ORDER_ATTRS 181
6.364 USER_ATTRIBUTE_DIM_TABLES 181
6.365 USER_ATTRIBUTE_DIM_TABLES_AE 182
6.366 USER_ATTRIBUTE_DIMENSIONS 182
6.367 USER_ATTRIBUTE_DIMENSIONS_AE 182
6.368 USER_ATTRIBUTE_TRANSFORMATIONS 183
6.369 USER_AUDIT_OBJECT 183
6.370 USER_AUDIT_POLICIES 183
6.371 USER_AUDIT_POLICY_COLUMNS 184
6.372 USER_AUDIT_SESSION 184
6.373 USER_AUDIT_STATEMENT 184
6.374 USER_AUDIT_TRAIL 185
6.375 USER_AW_PS 185

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.376 USER_AWS 185
6.377 USER_BASE_TABLE_MVIEWS 186
6.378 USER_BLOCKCHAIN_TABLES 186
6.379 USER_CATALOG 186
6.380 USER_CERTIFICATES 186
6.381 USER_CHANGE_NOTIFICATION_REGS 187
6.382 USER_CLU_COLUMNS 187
6.383 USER_CLUSTER_HASH_EXPRESSIONS 187
6.384 USER_CLUSTERING_DIMENSIONS 188
6.385 USER_CLUSTERING_JOINS 188
6.386 USER_CLUSTERING_KEYS 188
6.387 USER_CLUSTERING_TABLES 189
6.388 USER_CLUSTERS 189
6.389 USER_CODE_ROLE_PRIVS 189
6.390 USER_COL_COMMENTS 189
6.391 USER_COL_PENDING_STATS 190
6.392 USER_COL_PRIVS 190
6.393 USER_COL_PRIVS_MADE 190
6.394 USER_COL_PRIVS_RECD 190
6.395 USER_COLL_TYPES 191
6.396 USER_COMPARISON 191
6.397 USER_COMPARISON_COLUMNS 191
6.398 USER_COMPARISON_ROW_DIF 191
6.399 USER_COMPARISON_SCAN 192
6.400 USER_COMPARISON_SCAN_VALUES 192
6.401 USER_CONS_COLUMNS 192
6.402 USER_CONS_OBJ_COLUMNS 192
6.403 USER_CONSTRAINTS 193
6.404 USER_CQ_NOTIFICATION_QUERIES 193
6.405 USER_CREDENTIALS 193
6.406 USER_CUBE_ATTR_VISIBILITY 193
6.407 USER_CUBE_ATTRIBUTES 194
6.408 USER_CUBE_BUILD_PROCESSES 194
6.409 USER_CUBE_CALCULATED_MEMBERS 194
6.410 USER_CUBE_DIM_LEVELS 194
6.411 USER_CUBE_DIM_MODELS 195
6.412 USER_CUBE_DIM_VIEW_COLUMNS 195
6.413 USER_CUBE_DIM_VIEWS 195
6.414 USER_CUBE_DIMENSIONALITY 195
6.415 USER_CUBE_DIMENSIONS 196
6.416 USER_CUBE_HIER_LEVELS 196
6.417 USER_CUBE_HIER_VIEW_COLUMNS 196

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.418 USER_CUBE_HIER_VIEWS 196
6.419 USER_CUBE_HIERARCHIES 197
6.420 USER_CUBE_MEASURES 197
6.421 USER_CUBE_NAMED_BUILD_SPECS 197
6.422 USER_CUBE_SUB_PARTITION_LEVELS 197
6.423 USER_CUBE_VIEW_COLUMNS 198
6.424 USER_CUBE_VIEWS 198
6.425 USER_CUBES 198
6.426 USER_DATAPUMP_JOBS 198
6.427 USER_DB_LINKS 199
6.428 USER_DBFS_HS 199
6.429 USER_DBFS_HS_COMMANDS 199
6.430 USER_DBFS_HS_FILES 199
6.431 USER_DBFS_HS_FIXED_PROPERTIES 200
6.432 USER_DBFS_HS_PROPERTIES 200
6.433 USER_DEPENDENCIES 200
6.434 USER_DIM_ATTRIBUTES 200
6.435 USER_DIM_CHILD_OF 201
6.436 USER_DIM_HIERARCHIES 201
6.437 USER_DIM_JOIN_KEY 201
6.438 USER_DIM_LEVEL_KEY 201
6.439 USER_DIM_LEVELS 202
6.440 USER_DIMENSIONS 202
6.441 USER_EDITIONED_TYPES 202
6.442 USER_EDITIONING_VIEW_COLS 202
6.443 USER_EDITIONING_VIEW_COLS_AE 202
6.444 USER_EDITIONING_VIEWS 203
6.445 USER_EDITIONING_VIEWS_AE 203
6.446 USER_ENCRYPTED_COLUMNS 203
6.447 USER_EPG_DAD_AUTHORIZATION 204
6.448 USER_ERROR_TRANSLATIONS 204
6.449 USER_ERRORS 204
6.450 USER_ERRORS_AE 204
6.451 USER_EVALUATION_CONTEXT_TABLES 205
6.452 USER_EVALUATION_CONTEXT_VARS 205
6.453 USER_EVALUATION_CONTEXTS 205
6.454 USER_EXPRESSION_STATISTICS 205
6.455 USER_EXTENTS 206
6.456 USER_EXTERNAL_LOCATIONS 206
6.457 USER_EXTERNAL_TABLES 206
6.458 USER_FILE_GROUP_EXPORT_INFO 206
6.459 USER_FILE_GROUP_FILES 207

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.460 USER_FILE_GROUP_TABLES 207
6.461 USER_FILE_GROUP_TABLESPACES 207
6.462 USER_FILE_GROUP_VERSIONS 207
6.463 USER_FILE_GROUPS 208
6.464 USER_FLASHBACK_ARCHIVE 208
6.465 USER_FLASHBACK_ARCHIVE_TABLES 208
6.466 USER_FLASHBACK_TXN_REPORT 209
6.467 USER_FLASHBACK_TXN_STATE 209
6.468 USER_FREE_SPACE 209
6.469 USER_GOLDENGATE_PRIVILEGES 209
6.470 USER_HEAT_MAP_SEG_HISTOGRAM 210
6.471 USER_HEAT_MAP_SEGMENT 210
6.472 USER_HIER_CLASS 210
6.473 USER_HIER_CLASS_AE 210
6.474 USER_HIER_COLUMNS 211
6.475 USER_HIER_COLUMNS_AE 211
6.476 USER_HIER_HIER_ATTR_CLASS 211
6.477 USER_HIER_HIER_ATTR_CLASS_AE 212
6.478 USER_HIER_HIER_ATTRIBUTES 212
6.479 USER_HIER_HIER_ATTRIBUTES_AE 212
6.480 USER_HIER_JOIN_PATHS 213
6.481 USER_HIER_JOIN_PATHS_AE 213
6.482 USER_HIER_LEVEL_ID_ATTRS 213
6.483 USER_HIER_LEVEL_ID_ATTRS_AE 214
6.484 USER_HIER_LEVELS 214
6.485 USER_HIER_LEVELS_AE 214
6.486 USER_HIERARCHIES 215
6.487 USER_HIERARCHIES_AE 215
6.488 USER_HISTOGRAMS 215
6.489 USER_HIVE_COLUMNS 215
6.490 USER_HIVE_DATABASES 216
6.491 USER_HIVE_PART_KEY_COLUMNS 216
6.492 USER_HIVE_TAB_PARTITIONS 216
6.493 USER_HIVE_TABLES 216
6.494 USER_HOST_ACES 217
6.495 USER_IDENTIFIERS 217
6.496 USER_ILMDATAMOVEMENTPOLICIES 217
6.497 USER_ILMEVALUATIONDETAILS 218
6.498 USER_ILMOBJECTS 218
6.499 USER_ILMPOLICIES 218
6.500 USER_ILMRESULTS 219
6.501 USER_ILMTASKS 219

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.502 USER_IM_EXPRESSIONS 220
6.503 USER_IMMUTABLE_TABLES 220
6.504 USER_IND_COLUMNS 220
6.505 USER_IND_EXPRESSIONS 221
6.506 USER_IND_PARTITIONS 221
6.507 USER_IND_PENDING_STATS 221
6.508 USER_IND_STATISTICS 221
6.509 USER_IND_SUBPARTITIONS 222
6.510 USER_INDEXES 222
6.511 USER_INDEXTYPE_ARRAYTYPES 222
6.512 USER_INDEXTYPE_COMMENTS 222
6.513 USER_INDEXTYPE_OPERATORS 223
6.514 USER_INDEXTYPES 223
6.515 USER_INTERNAL_TRIGGERS 223
6.516 USER_JAVA_ARGUMENTS 223
6.517 USER_JAVA_CLASSES 224
6.518 USER_JAVA_COMPILER_OPTIONS 224
6.519 USER_JAVA_DERIVATIONS 224
6.520 USER_JAVA_FIELDS 224
6.521 USER_JAVA_IMPLEMENTS 225
6.522 USER_JAVA_INNERS 225
6.523 USER_JAVA_LAYOUTS 225
6.524 USER_JAVA_METHODS 225
6.525 USER_JAVA_NCOMPS 226
6.526 USER_JAVA_POLICY 226
6.527 USER_JAVA_RESOLVERS 226
6.528 USER_JAVA_THROWS 226
6.529 USER_JOBS 227
6.530 USER_JOIN_IND_COLUMNS 227
6.531 USER_JOINGROUPS 227
6.532 USER_JSON_COLUMNS 227
6.533 USER_JSON_DATAGUIDE_FIELDS 228
6.534 USER_JSON_DATAGUIDES 228
6.535 USER_LIBRARIES 228
6.536 USER_LOB_PARTITIONS 229
6.537 USER_LOB_SUBPARTITIONS 229
6.538 USER_LOB_TEMPLATES 229
6.539 USER_LOBS 229
6.540 USER_LOG_GROUP_COLUMNS 230
6.541 USER_LOG_GROUPS 230
6.542 USER_MEASURE_FOLDER_CONTENTS 230
6.543 USER_MEASURE_FOLDER_SUBFOLDERS 230

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.544 USER_MEASURE_FOLDERS 231
6.545 USER_METADATA_PROPERTIES 231
6.546 USER_METHOD_PARAMS 231
6.547 USER_METHOD_RESULTS 231
6.548 USER_MINING_MODEL_ATTRIBUTES 232
6.549 USER_MINING_MODEL_PARTITIONS 232
6.550 USER_MINING_MODEL_SETTINGS 232
6.551 USER_MINING_MODEL_VIEWS 232
6.552 USER_MINING_MODEL_XFORMS 233
6.553 USER_MINING_MODELS 233
6.554 USER_MVIEW_AGGREGATES 233
6.555 USER_MVIEW_ANALYSIS 233
6.556 USER_MVIEW_COMMENTS 234
6.557 USER_MVIEW_DETAIL_PARTITION 234
6.558 USER_MVIEW_DETAIL_RELATIONS 234
6.559 USER_MVIEW_DETAIL_SUBPARTITION 235
6.560 USER_MVIEW_JOINS 235
6.561 USER_MVIEW_KEYS 235
6.562 USER_MVIEW_LOGS 235
6.563 USER_MVIEW_REFRESH_TIMES 236
6.564 USER_MVIEWS 236
6.565 USER_MVREF_CHANGE_STATS 236
6.566 USER_MVREF_RUN_STATS 236
6.567 USER_MVREF_STATS 237
6.568 USER_MVREF_STATS_PARAMS 237
6.569 USER_MVREF_STATS_SYS_DEFAULTS 237
6.570 USER_MVREF_STMT_STATS 237
6.571 USER_NESTED_TABLE_COLS 238
6.572 USER_NESTED_TABLES 238
6.573 USER_NETWORK_ACL_PRIVILEGES 238
6.574 USER_OBJ_AUDIT_OPTS 239
6.575 USER_OBJ_COLATTRS 239
6.576 USER_OBJECT_SIZE 239
6.577 USER_OBJECT_TABLES 240
6.578 USER_OBJECT_USAGE 240
6.579 USER_OBJECTS 240
6.580 USER_OBJECTS_AE 240
6.581 USER_OPANCILLARY 241
6.582 USER_OPARGUMENTS 241
6.583 USER_OPBINDINGS 241
6.584 USER_OPERATOR_COMMENTS 241
6.585 USER_OPERATORS 242

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.586 USER_OUTLINE_HINTS 242
6.587 USER_OUTLINES 242
6.588 USER_PARALLEL_EXECUTE_CHUNKS 242
6.589 USER_PARALLEL_EXECUTE_TASKS 243
6.590 USER_PART_COL_STATISTICS 243
6.591 USER_PART_HISTOGRAMS 243
6.592 USER_PART_INDEXES 243
6.593 USER_PART_KEY_COLUMNS 244
6.594 USER_PART_LOBS 244
6.595 USER_PART_TABLES 244
6.596 USER_PARTIAL_DROP_TABS 244
6.597 USER_PASSWORD_LIMITS 245
6.598 USER_PENDING_CONV_TABLES 245
6.599 USER_PLSQL_COLL_TYPES 245
6.600 USER_PLSQL_OBJECT_SETTINGS 245
6.601 USER_PLSQL_TYPE_ATTRS 246
6.602 USER_PLSQL_TYPES 246
6.603 USER_POLICIES 246
6.604 USER_POLICY_ATTRIBUTES 246
6.605 USER_POLICY_CONTEXTS 247
6.606 USER_POLICY_GROUPS 247
6.607 USER_PRIVATE_TEMP_TABLES 247
6.608 USER_PRIVILEGE_MAP 247
6.609 USER_PROCEDURES 248
6.610 USER_PROXIES 248
6.611 USER_QUEUE_SCHEDULES 248
6.612 USER_QUEUE_SUBSCRIBERS 248
6.613 USER_QUEUE_TABLES 249
6.614 USER_QUEUES 249
6.615 USER_RECYCLEBIN 249
6.616 USER_REFRESH 249
6.617 USER_REFRESH_CHILDREN 250
6.618 USER_REFS 250
6.619 USER_REGISTERED_MVIEWS 250
6.620 USER_REGISTRY 250
6.621 USER_RESOURCE_LIMITS 250
6.622 USER_RESUMABLE 251
6.623 USER_REWRITE_EQUIVALENCES 251
6.624 USER_ROLE_PRIVS 251
6.625 USER_RSRC_CONSUMER_GROUP_PRIVS 252
6.626 USER_RSRC_MANAGER_SYSTEM_PRIVS 252
6.627 USER_RULE_SET_RULES 252

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.628 USER_RULE_SETS 253
6.629 USER_RULES 253
6.630 USER_SCHEDULER_CHAIN_RULES 253
6.631 USER_SCHEDULER_CHAIN_STEPS 253
6.632 USER_SCHEDULER_CHAINS 254
6.633 USER_SCHEDULER_CREDENTIALS 254
6.634 USER_SCHEDULER_DB_DESTS 254
6.635 USER_SCHEDULER_DESTS 255
6.636 USER_SCHEDULER_FILE_WATCHERS 255
6.637 USER_SCHEDULER_GROUP_MEMBERS 255
6.638 USER_SCHEDULER_GROUPS 255
6.639 USER_SCHEDULER_INCOMPAT_MEMBER 256
6.640 USER_SCHEDULER_INCOMPATS 256
6.641 USER_SCHEDULER_JOB_ARGS 256
6.642 USER_SCHEDULER_JOB_DESTS 256
6.643 USER_SCHEDULER_JOB_LOG 257
6.644 USER_SCHEDULER_JOB_RUN_DETAILS 257
6.645 USER_SCHEDULER_JOBS 257
6.646 USER_SCHEDULER_NOTIFICATIONS 257
6.647 USER_SCHEDULER_PROGRAM_ARGS 258
6.648 USER_SCHEDULER_PROGRAMS 258
6.649 USER_SCHEDULER_REMOTE_JOBSTATE 258
6.650 USER_SCHEDULER_RESOURCES 258
6.651 USER_SCHEDULER_RSC_CONSTRAINTS 259
6.652 USER_SCHEDULER_RUNNING_CHAINS 259
6.653 USER_SCHEDULER_RUNNING_JOBS 259
6.654 USER_SCHEDULER_SCHEDULES 259
6.655 USER_SEC_RELEVANT_COLS 260
6.656 USER_SECONDARY_OBJECTS 260
6.657 USER_SEGMENTS 260
6.658 USER_SEQUENCES 260
6.659 USER_SODA_COLLECTIONS 261
6.660 USER_SOURCE 261
6.661 USER_SOURCE_AE 261
6.662 USER_SQL_TRANSLATION_PROFILES 262
6.663 USER_SQL_TRANSLATIONS 262
6.664 USER_SQLJ_TYPE_ATTRS 262
6.665 USER_SQLJ_TYPE_METHODS 262
6.666 USER_SQLJ_TYPES 263
6.667 USER_SQLSET 263
6.668 USER_SQLSET_BINDS 263
6.669 USER_SQLSET_PLANS 263

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.670 USER_SQLSET_REFERENCES 264
6.671 USER_SQLSET_STATEMENTS 264
6.672 USER_SQLTUNE_BINDS 264
6.673 USER_SQLTUNE_PLANS 264
6.674 USER_SQLTUNE_RATIONALE_PLAN 265
6.675 USER_SQLTUNE_STATISTICS 265
6.676 USER_SR_GRP_STATUS 265
6.677 USER_SR_GRP_STATUS_ALL 265
6.678 USER_SR_OBJ 266
6.679 USER_SR_OBJ_ALL 266
6.680 USER_SR_OBJ_STATUS 266
6.681 USER_SR_OBJ_STATUS_ALL 266
6.682 USER_SR_PARTN_OPS 267
6.683 USER_SR_STLOG_EXCEPTIONS 267
6.684 USER_SR_STLOG_STATS 267
6.685 USER_STAT_EXTENSIONS 267
6.686 USER_STATEMENTS 268
6.687 USER_STORED_SETTINGS 268
6.688 USER_SUBPART_COL_STATISTICS 268
6.689 USER_SUBPART_HISTOGRAMS 268
6.690 USER_SUBPART_KEY_COLUMNS 269
6.691 USER_SUBPARTITION_TEMPLATES 269
6.692 USER_SUBSCR_REGISTRATIONS 269
6.693 USER_SYNONYMS 269
6.694 USER_SYS_PRIVS 269
6.695 USER_TAB_COL_STATISTICS 270
6.696 USER_TAB_COLS 270
6.697 USER_TAB_COLUMNS 271
6.698 USER_TAB_COMMENTS 271
6.699 USER_TAB_HISTGRM_PENDING_STATS 271
6.700 USER_TAB_HISTOGRAMS 272
6.701 USER_TAB_IDENTITY_COLS 272
6.702 USER_TAB_MODIFICATIONS 272
6.703 USER_TAB_PARTITIONS 272
6.704 USER_TAB_PENDING_STATS 273
6.705 USER_TAB_PRIVS 273
6.706 USER_TAB_PRIVS_MADE 273
6.707 USER_TAB_PRIVS_RECD 273
6.708 USER_TAB_STAT_PREFS 274
6.709 USER_TAB_STATISTICS 274
6.710 USER_TAB_STATS_HISTORY 274
6.711 USER_TAB_SUBPARTITIONS 274

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.712 USER_TABLES 275
6.713 USER_TABLESPACES 275
6.714 USER_TRANSFORMATIONS 275
6.715 USER_TRIGGER_COLS 275
6.716 USER_TRIGGER_ORDERING 276
6.717 USER_TRIGGERS 276
6.718 USER_TRIGGERS_AE 276
6.719 USER_TS_QUOTAS 276
6.720 USER_TSTZ_TAB_COLS 277
6.721 USER_TSTZ_TABLES 277
6.722 USER_TUNE_MVIEW 277
6.723 USER_TYPE_ATTRS 277
6.724 USER_TYPE_METHODS 278
6.725 USER_TYPE_VERSIONS 278
6.726 USER_TYPES 278
6.727 USER_UNUSED_COL_TABS 278
6.728 USER_UPDATABLE_COLUMNS 279
6.729 USER_USERS 279
6.730 USER_USTATS 283
6.731 USER_VARRAYS 284
6.732 USER_VIEWS 284
6.733 USER_VIEWS_AE 284
6.734 USER_WALLET_ACES 284
6.735 USER_WARNING_SETTINGS 285
6.736 USER_XML_INDEXES 285
6.737 USER_XML_NESTED_TABLES 285
6.738 USER_XML_OUT_OF_LINE_TABLES 285
6.739 USER_XML_SCHEMA_ATTRIBUTES 286
6.740 USER_XML_SCHEMA_COMPLEX_TYPES 286
6.741 USER_XML_SCHEMA_ELEMENTS 286
6.742 USER_XML_SCHEMA_NAMESPACES 286
6.743 USER_XML_SCHEMA_SIMPLE_TYPES 287
6.744 USER_XML_SCHEMA_SUBSTGRP_HEAD 287
6.745 USER_XML_SCHEMA_SUBSTGRP_MBRS 287
6.746 USER_XML_SCHEMAS 287
6.747 USER_XML_TAB_COLS 288
6.748 USER_XML_TABLES 288
6.749 USER_XML_VIEW_COLS 288
6.750 USER_XML_VIEWS 288
6.751 USER_XTERNAL_LOC_PARTITIONS 289
6.752 USER_XTERNAL_LOC_SUBPARTITIONS 289
6.753 USER_XTERNAL_PART_TABLES 289

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.754 USER_XTERNAL_TAB_PARTITIONS 289
6.755 USER_XTERNAL_TAB_SUBPARTITIONS 290
6.756 USER_ZONEMAP_MEASURES 290
6.757 USER_ZONEMAPS 290

Part III Dynamic Performance Views

7 Dynamic Performance (V$) Views: V$ACCESS to V$HVMASTER_INFO

7.1 About Dynamic Performance Views 1
7.1.1 V$ Views 2
7.1.2 GV$ Views 2
7.2 Oracle Exadata Database Machine Views 3
7.3 Dynamic Performance View Descriptions 3
7.4 V$ACCESS 3
7.5 V$ACTIVE_INSTANCES 4
7.6 V$ACTIVE_SERVICES 4
7.7 V$ACTIVE_SESS_POOL_MTH 6
7.8 V$ACTIVE_SESSION_HISTORY 7
7.9 V$ADVISOR_PROGRESS 12
7.10 V$ALERT_TYPES 13
7.11 V$ALL_ACTIVE_SESSION_HISTORY 14
7.12 V$ALL_SQL_MONITOR 19
7.13 V$ALL_SQL_PLAN_MONITOR 24
7.14 V$AQ 27
7.15 V$AQ_BACKGROUND_COORDINATOR 28
7.16 V$AQ_BMAP_NONDUR_SUBSCRIBERS 28
7.17 V$AQ_CROSS_INSTANCE_JOBS 29
7.18 V$AQ_IPC_ACTIVE_MSGS 31
7.19 V$AQ_IPC_MSG_STATS 31
7.20 V$AQ_IPC_PENDING_MSGS 32
7.21 V$AQ_JOB_COORDINATOR 33
7.22 V$AQ_MESSAGE_CACHE 33
7.23 V$AQ_MESSAGE_CACHE_ADVICE 34
7.24 V$AQ_MESSAGE_CACHE_STAT 35
7.25 V$AQ_NONDUR_REGISTRATIONS 37
7.26 V$AQ_NONDUR_SUBSCRIBER 38
7.27 V$AQ_NONDUR_SUBSCRIBER_LWM 39
7.28 V$AQ_NOTIFICATION_CLIENTS 39
7.29 V$AQ_PARTITION_STATS 40
7.30 V$AQ_REMOTE_DEQUEUE_AFFINITY 42

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.31 V$AQ_SERVER_POOL 42
7.32 V$AQ_SHARDED_SUBSCRIBER_STAT 43
7.33 V$AQ_SUBSCRIBER_LOAD 44
7.34 V$ARCHIVE 45
7.35 V$ARCHIVE_DEST 46
7.36 V$ARCHIVE_DEST_STATUS 49
7.37 V$ARCHIVE_GAP 52
7.38 V$ARCHIVE_PROCESSES 52
7.39 V$ARCHIVED_LOG 53
7.40 V$ASM_ACFS_ENCRYPTION_INFO 55
7.41 V$ASM_ACFS_SEC_ADMIN 56
7.42 V$ASM_ACFS_SEC_CMDRULE 57
7.43 V$ASM_ACFS_SEC_REALM 58
7.44 V$ASM_ACFS_SEC_REALM_FILTER 59
7.45 V$ASM_ACFS_SEC_REALM_GROUP 60
7.46 V$ASM_ACFS_SEC_REALM_USER 61
7.47 V$ASM_ACFS_SEC_RULE 62
7.48 V$ASM_ACFS_SEC_RULESET 63
7.49 V$ASM_ACFS_SEC_RULESET_RULE 64
7.50 V$ASM_ACFS_SECURITY_INFO 64
7.51 V$ASM_ACFSAUTORESIZE 65
7.52 V$ASM_ACFSREPL 66
7.53 V$ASM_ACFSREPLTAG 68
7.54 V$ASM_ACFSSNAPSHOTS 69
7.55 V$ASM_ACFSTAG 70
7.56 V$ASM_ACFSVOLUMES 71
7.57 V$ASM_ALIAS 72
7.58 V$ASM_ATTRIBUTE 73
7.59 V$ASM_AUDIT_CLEANUP_JOBS 74
7.60 V$ASM_AUDIT_CONFIG_PARAMS 75
7.61 V$ASM_AUDIT_LAST_ARCH_TS 75
7.62 V$ASM_CLIENT 76
7.63 V$ASM_DBCLONE_INFO 77
7.64 V$ASM_DISK 78
7.65 V$ASM_DISK_IOSTAT 83
7.66 V$ASM_DISK_STAT 84
7.67 V$ASM_DISKGROUP 84
7.68 V$ASM_DISKGROUP_STAT 86
7.69 V$ASM_ESTIMATE 87
7.70 V$ASM_FILE 87
7.71 V$ASM_FILEGROUP 89
7.72 V$ASM_FILEGROUP_FILE 91

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.73 V$ASM_FILEGROUP_PROPERTY 91
7.74 V$ASM_FILESYSTEM 93
7.75 V$ASM_OPERATION 94
7.76 V$ASM_QUOTAGROUP 96
7.77 V$ASM_TEMPLATE 97
7.78 V$ASM_USER 98
7.79 V$ASM_USERGROUP 98
7.80 V$ASM_USERGROUP_MEMBER 99
7.81 V$ASM_VOLUME 100
7.82 V$ASM_VOLUME_STAT 101
7.83 V$AW_AGGREGATE_OP 101
7.84 V$AW_ALLOCATE_OP 102
7.85 V$AW_CALC 102
7.86 V$AW_LONGOPS 103
7.87 V$AW_OLAP 104
7.88 V$AW_SESSION_INFO 105
7.89 V$BACKUP 106
7.90 V$BACKUP_ARCHIVELOG_DETAILS 106
7.91 V$BACKUP_ARCHIVELOG_SUMMARY 107
7.92 V$BACKUP_ASYNC_IO 108
7.93 V$BACKUP_CONTROLFILE_DETAILS 109
7.94 V$BACKUP_CONTROLFILE_SUMMARY 110
7.95 V$BACKUP_COPY_DETAILS 111
7.96 V$BACKUP_COPY_SUMMARY 112
7.97 V$BACKUP_CORRUPTION 112
7.98 V$BACKUP_DATAFILE 113
7.99 V$BACKUP_DATAFILE_DETAILS 115
7.100 V$BACKUP_DATAFILE_SUMMARY 116
7.101 V$BACKUP_DEVICE 117
7.102 V$BACKUP_FILES 117
7.103 V$BACKUP_NONLOGGED 120
7.104 V$BACKUP_PIECE 121
7.105 V$BACKUP_PIECE_DETAILS 122
7.106 V$BACKUP_REDOLOG 124
7.107 V$BACKUP_SET 125
7.108 V$BACKUP_SET_DETAILS 126
7.109 V$BACKUP_SET_SUMMARY 128
7.110 V$BACKUP_SPFILE 129
7.111 V$BACKUP_SPFILE_DETAILS 129
7.112 V$BACKUP_SPFILE_SUMMARY 130
7.113 V$BACKUP_SYNC_IO 130
7.114 V$BGPROCESS 131

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.115 V$BH 132
7.116 V$BLOCK_CHANGE_TRACKING 133
7.117 V$BLOCKING_QUIESCE 134
7.118 V$BT_SCAN_CACHE 134
7.119 V$BT_SCAN_OBJ_TEMPS 135
7.120 V$BUFFER_POOL 136
7.121 V$BUFFER_POOL_STATISTICS 137
7.122 V$BUFFERED_PUBLISHERS 138
7.123 V$BUFFERED_QUEUES 139
7.124 V$BUFFERED_SUBSCRIBERS 140
7.125 V$CACHE 141
7.126 V$CACHE_LOCK 143
7.127 V$CACHE_TRANSFER 144
7.128 V$CHUNK_METRIC 146
7.129 V$CIRCUIT 146
7.130 V$CLASS_CACHE_TRANSFER 147
7.131 V$CLEANUP_PROCESS 148
7.132 V$CLIENT_SECRETS 149
7.133 V$CLIENT_STATS 150
7.134 V$CLONEDFILE 151
7.135 V$CLUSTER_INTERCONNECTS 151
7.136 V$CODE_CLAUSE 152
7.137 V$CON_EVENT_HISTOGRAM_MICRO 153
7.138 V$CON_SYS_TIME_MODEL 153
7.139 V$CON_SYSMETRIC 154
7.140 V$CON_SYSMETRIC_HISTORY 155
7.141 V$CON_SYSMETRIC_SUMMARY 155
7.142 V$CON_SYSSTAT 156
7.143 V$CON_SYSTEM_EVENT 157
7.144 V$CON_SYSTEM_WAIT_CLASS 158
7.145 V$CONFIGURED_INTERCONNECTS 158
7.146 V$CONTAINERS 159
7.147 V$CONTEXT 161
7.148 V$CONTROLFILE 161
7.149 V$CONTROLFILE_RECORD_SECTION 162
7.150 V$COPY_CORRUPTION 163
7.151 V$COPY_NONLOGGED 164
7.152 V$CORRUPT_XID_LIST 165
7.153 V$CPOOL_CC_INFO 165
7.154 V$CPOOL_CC_STATS 165
7.155 V$CPOOL_CONN_INFO 166
7.156 V$CPOOL_STATS 167

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.157 V$CR_BLOCK_SERVER 168
7.158 V$CURRENT_BLOCK_SERVER 169
7.159 V$DATABASE 170
7.160 V$DATABASE_BLOCK_CORRUPTION 178
7.161 V$DATABASE_INCARNATION 179
7.162 V$DATABASE_KEY_INFO 180
7.163 V$DATAFILE 181
7.164 V$DATAFILE_COPY 183
7.165 V$DATAFILE_HEADER 185
7.166 V$DATAGUARD_CONFIG 186
7.167 V$DATAGUARD_PROCESS 187
7.168 V$DATAGUARD_STATS 190
7.169 V$DATAGUARD_STATUS 191
7.170 V$DB_CACHE_ADVICE 192
7.171 V$DB_OBJECT_CACHE 193
7.172 V$DB_PIPES 195
7.173 V$DB_TRANSPORTABLE_PLATFORM 195
7.174 V$DBFILE 196
7.175 V$DBLINK 196
7.176 V$DEAD_CLEANUP 197
7.177 V$DELETED_OBJECT 198
7.178 V$DG_BROKER_CONFIG 199
7.179 V$DIAG_ALERT_EXT 200
7.180 V$DIAG_APP_TRACE_FILE 202
7.181 V$DIAG_INCIDENT 203
7.182 V$DIAG_INFO 204
7.183 V$DIAG_OPT_TRACE_RECORDS 205
7.184 V$DIAG_PROBLEM 206
7.185 V$DIAG_SESS_OPT_TRACE_RECORDS 207
7.186 V$DIAG_SESS_SQL_TRACE_RECORDS 208
7.187 V$DIAG_SQL_TRACE_RECORDS 209
7.188 V$DIAG_TRACE_FILE 210
7.189 V$DIAG_TRACE_FILE_CONTENTS 210
7.190 V$DISPATCHER 211
7.191 V$DISPATCHER_CONFIG 212
7.192 V$DISPATCHER_RATE 213
7.193 V$DNFS_CHANNELS 217
7.194 V$DNFS_FILES 218
7.195 V$DNFS_SERVERS 218
7.196 V$DNFS_STATS 219
7.197 V$DYNAMIC_REMASTER_STATS 220
7.198 V$EDITIONABLE_TYPES 221

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.199 V$EMON 221
7.200 V$EMX_USAGE_STATS 223
7.201 V$ENABLEDPRIVS 223
7.202 V$ENCRYPTED_TABLESPACES 224
7.203 V$ENCRYPTION_KEYS 225
7.204 V$ENCRYPTION_WALLET 227
7.205 V$ENQUEUE_LOCK 229
7.206 V$ENQUEUE_STAT 230
7.207 V$ENQUEUE_STATISTICS 230
7.208 V$EVENT_HISTOGRAM 231
7.209 V$EVENT_HISTOGRAM_MICRO 232
7.210 V$EVENT_NAME 232
7.211 V$EVENTMETRIC 233
7.212 V$EXADIRECT_ACL 233
7.213 V$EXECUTION 234
7.214 V$EXP_STATS 234
7.215 V$FALSE_PING 235
7.216 V$FAST_START_SERVERS 236
7.217 V$FAST_START_TRANSACTIONS 236
7.218 V$FILE_CACHE_TRANSFER 237
7.219 V$FILE_HISTOGRAM 238
7.220 V$FILEMETRIC 239
7.221 V$FILEMETRIC_HISTORY 239
7.222 V$FILESPACE_USAGE 240
7.223 V$FILESTAT 240
7.224 V$FIXED_TABLE 241
7.225 V$FIXED_VIEW_DEFINITION 242
7.226 V$FLASHBACK_DATABASE_LOG 242
7.227 V$FLASHBACK_DATABASE_LOGFILE 243
7.228 V$FLASHBACK_DATABASE_STAT 243
7.229 V$FLASHBACK_TXN_GRAPH 244
7.230 V$FLASHBACK_TXN_MODS 245
7.231 V$FLASHFILESTAT 246
7.232 V$FOREIGN_ARCHIVED_LOG 247
7.233 V$FS_FAILOVER_OBSERVERS 249
7.234 V$FS_FAILOVER_STATS 250
7.235 V$FS_OBSERVER_HISTOGRAM 250
7.236 V$GC_ELEMENT 251
7.237 V$GC_ELEMENTS_WITH_COLLISIONS 252
7.238 V$GCR_ACTIONS 253
7.239 V$GCR_LOG 253
7.240 V$GCR_METRICS 254

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.241 V$GCR_STATUS 255
7.242 V$GCSHVMASTER_INFO 256
7.243 V$GCSPFMASTER_INFO 256
7.244 V$GES_BLOCKING_ENQUEUE 257
7.245 V$GES_CONVERT_LOCAL 258
7.246 V$GES_CONVERT_REMOTE 259
7.247 V$GES_DEADLOCK_SESSIONS 260
7.248 V$GES_DEADLOCKS 260
7.249 V$GES_ENQUEUE 261
7.250 V$GES_LATCH 262
7.251 V$GES_RESOURCE 262
7.252 V$GES_STATISTICS 263
7.253 V$GG_APPLY_COORDINATOR 263
7.254 V$GG_APPLY_READER 265
7.255 V$GG_APPLY_RECEIVER 267
7.256 V$GG_APPLY_SERVER 268
7.257 V$GLOBAL_BLOCKED_LOCKS 271
7.258 V$GLOBAL_TRANSACTION 271
7.259 V$GOLDENGATE_CAPABILITIES 272
7.260 V$GOLDENGATE_CAPTURE 273
7.261 V$GOLDENGATE_MESSAGE_TRACKING 276
7.262 V$GOLDENGATE_PROCEDURE_STATS 277
7.263 V$GOLDENGATE_TABLE_STATS 278
7.264 V$GOLDENGATE_TRANSACTION 279
7.265 V$HANG_INFO 281
7.266 V$HANG_SESSION_INFO 282
7.267 V$HANG_STATISTICS 283
7.268 V$HEAT_MAP_SEGMENT 284
7.269 V$HM_CHECK 285
7.270 V$HM_CHECK_PARAM 285
7.271 V$HM_FINDING 286
7.272 V$HM_INFO 287
7.273 V$HM_RECOMMENDATION 287
7.274 V$HM_RUN 288
7.275 V$HS_AGENT 289
7.276 V$HS_PARAMETER 290
7.277 V$HS_SESSION 290
7.278 V$HVMASTER_INFO 291

Copyright © 2002, 2026, Oracle and/or its affiliates.

8 Dynamic Performance (V$) Views: V$IM_COLUMN_LEVEL to V$RULE_SET_AGGREGATE_STATS

8.1 V$IM_COLUMN_LEVEL 1
8.2 V$IM_SEGMENTS 2
8.3 V$IM_USER_SEGMENTS 4
8.4 V$INDEX_USAGE_INFO 5
8.5 V$INDEXED_FIXED_COLUMN 6
8.6 V$INMEMORY_AREA 7
8.7 V$INMEMORY_FASTSTART_AREA 8
8.8 V$INSTANCE 8
8.9 V$INSTANCE_CACHE_TRANSFER 11
8.10 V$INSTANCE_PING 12
8.11 V$INSTANCE_RECOVERY 13
8.12 V$IO_CALIBRATION_STATUS 14
8.13 V$IO_OUTLIER 15
8.14 V$IOFUNCMETRIC 16
8.15 V$IOFUNCMETRIC_HISTORY 16
8.16 V$IOS_CLIENT 17
8.17 V$IOSTAT_CONSUMER_GROUP 17
8.18 V$IOSTAT_FILE 18
8.19 V$IOSTAT_FUNCTION 19
8.20 V$IOSTAT_FUNCTION_DETAIL 20
8.21 V$IOSTAT_NETWORK 21
8.22 V$IP_ACL 21
8.23 V$JAVA_LIBRARY_CACHE_MEMORY 22
8.24 V$JAVA_POOL_ADVICE 22
8.25 V$KERNEL_IO_OUTLIER 23
8.26 V$KEY_VECTOR 24
8.27 V$LATCH 26
8.28 V$LATCH_CHILDREN 27
8.29 V$LATCH_MISSES 28
8.30 V$LATCH_PARENT 29
8.31 V$LATCHHOLDER 29
8.32 V$LATCHNAME 30
8.33 V$LGWRIO_OUTLIER 30
8.34 V$LIBCACHE_LOCKS 31
8.35 V$LIBRARY_CACHE_MEMORY 32
8.36 V$LIBRARYCACHE 33
8.37 V$LICENSE 33
8.38 V$LOADISTAT 34
8.39 V$LOADPSTAT 35

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.40 V$LOCK 35
8.41 V$LOCK_ACTIVITY 37
8.42 V$LOCK_TYPE 38
8.43 V$LOCKDOWN_RULES 38
8.44 V$LOCKED_OBJECT 39
8.45 V$LOG 40
8.46 V$LOG_HISTORY 41
8.47 V$LOGFILE 41
8.48 V$LOGHIST 42
8.49 V$LOGMNR_CONTENTS 42
8.50 V$LOGMNR_DICTIONARY 48
8.51 V$LOGMNR_DICTIONARY_LOAD 49
8.52 V$LOGMNR_LATCH 49
8.53 V$LOGMNR_LOGS 50
8.54 V$LOGMNR_PARAMETERS 51
8.55 V$LOGMNR_PROCESS 52
8.56 V$LOGMNR_SESSION 53
8.57 V$LOGMNR_STATS 54
8.58 V$LOGSTDBY 56
8.59 V$LOGSTDBY_PROCESS 56
8.60 V$LOGSTDBY_PROGRESS 57
8.61 V$LOGSTDBY_STATE 58
8.62 V$LOGSTDBY_STATS 59
8.63 V$LOGSTDBY_TRANSACTION 61
8.64 V$MANAGED_STANDBY 62
8.65 V$MAP_COMP_LIST 64
8.66 V$MAP_ELEMENT 65
8.67 V$MAP_EXT_ELEMENT 65
8.68 V$MAP_FILE 66
8.69 V$MAP_FILE_EXTENT 67
8.70 V$MAP_FILE_IO_STACK 67
8.71 V$MAP_LIBRARY 68
8.72 V$MAP_SUBELEMENT 69
8.73 V$MAPPED_SQL 69
8.74 V$MEMOPTIMIZE_WRITE_AREA 70
8.75 V$MEMORY_CURRENT_RESIZE_OPS 71
8.76 V$MEMORY_DYNAMIC_COMPONENTS 72
8.77 V$MEMORY_RESIZE_OPS 73
8.78 V$MEMORY_TARGET_ADVICE 73
8.79 V$METRIC 75
8.80 V$METRICGROUP 76
8.81 V$METRIC_HISTORY 76

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.82 V$METRICNAME 77
8.83 V$MTTR_TARGET_ADVICE 77
8.84 V$MUTEX_SLEEP 78
8.85 V$MUTEX_SLEEP_HISTORY 79
8.86 V$MVREFRESH 80
8.87 V$MYSTAT 80
8.88 V$NFS_CLIENTS 81
8.89 V$NFS_LOCKS 81
8.90 V$NFS_OPEN_FILES 82
8.91 V$NLS_PARAMETERS 82
8.92 V$NLS_VALID_VALUES 83
8.93 V$NONLOGGED_BLOCK 83
8.94 V$OBJECT_DEPENDENCY 84
8.95 V$OBJECT_PRIVILEGE 85
8.96 V$OBJECT_USAGE 85
8.97 V$OBSOLETE_BACKUP_FILES 86
8.98 V$OBSOLETE_PARAMETER 88
8.99 V$OFFLINE_RANGE 88
8.100 V$OFS_STATS 89
8.101 V$OFSMOUNT 91
8.102 V$ONLINE_REDEF 91
8.103 V$OPEN_CURSOR 92
8.104 V$OPTIMIZER_PROCESSING_RATE 93
8.105 V$OPTION 94
8.106 V$OSSTAT 95
8.107 V$PARALLEL_DEGREE_LIMIT_MTH 96
8.108 V$PARAMETER 97
8.109 V$PARAMETER_VALID_VALUES 99
8.110 V$PARAMETER2 100
8.111 V$PASSWORDFILE_INFO 102
8.112 V$PATCHES 102
8.113 V$PDB_INCARNATION 103
8.114 V$PDBS 103
8.115 V$PERSISTENT_PUBLISHERS 105
8.116 V$PERSISTENT_QMN_CACHE 106
8.117 V$PERSISTENT_QUEUES 107
8.118 V$PERSISTENT_SUBSCRIBERS 109
8.119 V$PGA_TARGET_ADVICE 110
8.120 V$PGA_TARGET_ADVICE_HISTOGRAM 112
8.121 V$PGASTAT 113
8.122 V$PLSQL_DEBUGGABLE_SESSIONS 115
8.123 V$PQ_SESSTAT 116

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.124 V$PQ_SLAVE 117
8.125 V$PQ_SYSSTAT 117
8.126 V$PQ_TQSTAT 118
8.127 V$PROCESS 119
8.128 V$PROCESS_MEMORY 120
8.129 V$PROCESS_MEMORY_DETAIL 121
8.130 V$PROCESS_POOL 122
8.131 V$PROPAGATION_RECEIVER 122
8.132 V$PROPAGATION_SENDER 123
8.133 V$PROXY_ARCHIVEDLOG 125
8.134 V$PROXY_ARCHIVELOG_DETAILS 127
8.135 V$PROXY_ARCHIVELOG_SUMMARY 128
8.136 V$PROXY_COPY_DETAILS 128
8.137 V$PROXY_COPY_SUMMARY 129
8.138 V$PROXY_DATAFILE 130
8.139 V$PROXY_PDB_TARGETS 132
8.140 V$PWFILE_USERS 133
8.141 V$PX_INSTANCE_GROUP 134
8.142 V$PX_PROCESS 134
8.143 V$PX_PROCESS_SYSSTAT 135
8.144 V$PX_SESSION 137
8.145 V$PX_SESSTAT 137
8.146 V$QMON_COORDINATOR_STATS 138
8.147 V$QMON_SERVER_STATS 139
8.148 V$QMON_TASK_STATS 140
8.149 V$QMON_TASKS 140
8.150 V$QUARANTINE 141
8.151 V$QUARANTINE_SUMMARY 141
8.152 V$QUEUE 142
8.153 V$QUEUEING_MTH 142
8.154 V$RECOVER_FILE 143
8.155 V$RECOVERY_AREA_USAGE 143
8.156 V$RECOVERY_FILE_DEST 144
8.157 V$RECOVERY_FILE_STATUS 144
8.158 V$RECOVERY_LOG 145
8.159 V$RECOVERY_PROGRESS 146
8.160 V$RECOVERY_SLAVE 147
8.161 V$RECOVERY_STATUS 149
8.162 V$REDO_DEST_RESP_HISTOGRAM 150
8.163 V$REQDIST 150
8.164 V$RESERVED_WORDS 151
8.165 V$RESOURCE 151

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.166 V$RESOURCE_LIMIT 152
8.167 V$RESTORE_POINT 153
8.168 V$RESULT_CACHE_DEPENDENCY 154
8.169 V$RESULT_CACHE_MEMORY 154
8.170 V$RESULT_CACHE_OBJECTS 155
8.171 V$RESULT_CACHE_STATISTICS 156
8.172 V$RMAN_BACKUP_JOB_DETAILS 157
8.173 V$RMAN_BACKUP_SUBJOB_DETAILS 158
8.174 V$RMAN_BACKUP_TYPE 160
8.175 V$RMAN_COMPRESSION_ALGORITHM 160
8.176 V$RMAN_CONFIGURATION 161
8.177 V$RMAN_ENCRYPTION_ALGORITHMS 161
8.178 V$RMAN_OUTPUT 162
8.179 V$RMAN_STATUS 163
8.180 V$RO_USER_ACCOUNT 164
8.181 V$ROLLNAME 165
8.182 V$ROLLSTAT 165
8.183 V$ROWCACHE 166
8.184 V$ROWCACHE_PARENT 167
8.185 V$ROWCACHE_SUBORDINATE 168
8.186 V$RSRC_CONS_GROUP_HISTORY 168
8.187 V$RSRC_CONSUMER_GROUP 171
8.188 V$RSRC_CONSUMER_GROUP_CPU_MTH 174
8.189 V$RSRC_PDB 175
8.190 V$RSRC_PDB_HISTORY 177
8.191 V$RSRC_PLAN 178
8.192 V$RSRC_PLAN_CPU_MTH 180
8.193 V$RSRC_PLAN_HISTORY 180
8.194 V$RSRC_SESSION_INFO 182
8.195 V$RSRCMGRMETRIC 186
8.196 V$RSRCMGRMETRIC_HISTORY 188
8.197 V$RSRCPDBMETRIC 188
8.198 V$RSRCPDBMETRIC_HISTORY 190
8.199 V$RULE 190
8.200 V$RULE_SET 191
8.201 V$RULE_SET_AGGREGATE_STATS 192

9 Dynamic Performance (V$) Views: V$SCHEDULER_RUNNING_JOBS to V$ZONEMAP_USAGE_STATS

9.1 V$SCHEDULER_RUNNING_JOBS 1
9.2 V$SECUREFILE_TIMER 1

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.3 V$SEGMENT_STATISTICS 2
9.4 V$SEGSTAT 3
9.5 V$SEGSTAT_NAME 3
9.6 V$SERV_MOD_ACT_STATS 4
9.7 V$SERVICE_EVENT 4
9.8 V$SERVICE_REGION_METRIC 5
9.9 V$SERVICE_STATS 6
9.10 V$SERVICE_WAIT_CLASS 6
9.11 V$SERVICEMETRIC 7
9.12 V$SERVICEMETRIC_HISTORY 8
9.13 V$SERVICES 8
9.14 V$SES_OPTIMIZER_ENV 10
9.15 V$SESS_IO 11
9.16 V$SESS_TIME_MODEL 12
9.17 V$SESSION 14
9.18 V$SESSION_BLOCKERS 20
9.19 V$SESSION_CONNECT_INFO 21
9.20 V$SESSION_CURSOR_CACHE 22
9.21 V$SESSION_EVENT 23
9.22 V$SESSION_FIX_CONTROL 24
9.23 V$SESSION_LONGOPS 24
9.24 V$SESSION_OBJECT_CACHE 26
9.25 V$SESSION_WAIT 27
9.26 V$SESSION_WAIT_CLASS 28
9.27 V$SESSION_WAIT_HISTORY 29
9.28 V$SESSIONS_COUNT 29
9.29 V$SESSMETRIC 30
9.30 V$SESSTAT 30
9.31 V$SGA 31
9.32 V$SGA_CURRENT_RESIZE_OPS 31
9.33 V$SGA_DYNAMIC_COMPONENTS 32
9.34 V$SGA_DYNAMIC_FREE_MEMORY 33
9.35 V$SGA_RESIZE_OPS 33
9.36 V$SGA_TARGET_ADVICE 34
9.37 V$SGAINFO 35
9.38 V$SGASTAT 35
9.39 V$SHARED_POOL_ADVICE 36
9.40 V$SHARED_POOL_RESERVED 36
9.41 V$SHARED_SERVER 37
9.42 V$SHARED_SERVER_MONITOR 38
9.43 V$SHARED_SERVER_STAT 39
9.44 V$SORT_SEGMENT 40

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.45 V$SPPARAMETER 41
9.46 V$SQL 41
9.47 V$SQL_BIND_CAPTURE 47
9.48 V$SQL_BIND_DATA 48
9.49 V$SQL_BIND_METADATA 49
9.50 V$SQL_CS_HISTOGRAM 50
9.51 V$SQL_CS_SELECTIVITY 51
9.52 V$SQL_CS_STATISTICS 51
9.53 V$SQL_CURSOR 52
9.54 V$SQL_JOIN_FILTER 53
9.55 V$SQL_MONITOR 53
9.56 V$SQL_MONITOR_SESSTAT 58
9.57 V$SQL_MONITOR_STATNAME 59
9.58 V$SQL_OPTIMIZER_ENV 60
9.59 V$SQL_PLAN 60
9.60 V$SQL_PLAN_MONITOR 62
9.61 V$SQL_PLAN_STATISTICS 66
9.62 V$SQL_PLAN_STATISTICS_ALL 68
9.63 V$SQL_REDIRECTION 72
9.64 V$SQL_SHARD 72
9.65 V$SQL_SHARED_CURSOR 73
9.66 V$SQL_SHARED_MEMORY 76
9.67 V$SQL_TESTCASES 77
9.68 V$SQL_WORKAREA 78
9.69 V$SQL_WORKAREA_ACTIVE 79
9.70 V$SQL_WORKAREA_HISTOGRAM 81
9.71 V$SQLAREA 82
9.72 V$SQLAREA_PLAN_HASH 86
9.73 V$SQLCOMMAND 89
9.74 V$SQLFN_ARG_METADATA 90
9.75 V$SQLFN_METADATA 90
9.76 V$SQLSTATS 91
9.77 V$SQLSTATS_PLAN_HASH 95
9.78 V$SQLTEXT 95
9.79 V$SQLTEXT_WITH_NEWLINES 96
9.80 V$STANDBY_EVENT_HISTOGRAM 96
9.81 V$STANDBY_LOG 97
9.82 V$STATISTICS_LEVEL 98
9.83 V$STATNAME 99
9.84 V$STATS_ADVISOR_RULES 100
9.85 V$STREAMS_APPLY_COORDINATOR 100
9.86 V$STREAMS_APPLY_READER 102

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.87 V$STREAMS_APPLY_SERVER 105
9.88 V$STREAMS_POOL_ADVICE 108
9.89 V$STREAMS_POOL_STATISTICS 108
9.90 V$SUBCACHE 109
9.91 V$SUBSCR_REGISTRATION_STATS 110
9.92 V$SYS_OPTIMIZER_ENV 111
9.93 V$SYS_TIME_MODEL 112
9.94 V$SYSAUX_OCCUPANTS 112
9.95 V$SYSMETRIC 113
9.96 V$SYSMETRIC_HISTORY 113
9.97 V$SYSMETRIC_SUMMARY 114
9.98 V$SYSSTAT 115
9.99 V$SYSTEM_CURSOR_CACHE 116
9.100 V$SYSTEM_EVENT 116
9.101 V$SYSTEM_FIX_CONTROL 117
9.102 V$SYSTEM_PARAMETER 118
9.103 V$SYSTEM_PARAMETER2 120
9.104 V$SYSTEM_WAIT_CLASS 121
9.105 V$TABLESPACE 122
9.106 V$TEMP_CACHE_TRANSFER 122
9.107 V$TEMP_EXTENT_MAP 123
9.108 V$TEMP_EXTENT_POOL 124
9.109 V$TEMP_SPACE_HEADER 124
9.110 V$TEMPFILE 125
9.111 V$TEMPORARY_LOBS 125
9.112 V$TEMPSEG_USAGE 126
9.113 V$TEMPSTAT 127
9.114 V$TEMPUNDOSTAT 128
9.115 V$THREAD 129
9.116 V$THRESHOLD_TYPES 129
9.117 V$TIMER 130
9.118 V$TIMEZONE_FILE 130
9.119 V$TIMEZONE_NAMES 131
9.120 V$TOPLEVELCALL 131
9.121 V$TRANSACTION 131
9.122 V$TRANSACTION_ENQUEUE 133
9.123 V$TRANSPORTABLE_PLATFORM 134
9.124 V$TSDP_SUPPORTED_FEATURE 134
9.125 V$TYPE_SIZE 135
9.126 V$UNDOSTAT 135
9.127 V$UNUSABLE_BACKUPFILE_DETAILS 137
9.128 V$VERSION 138

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.129 V$VPD_POLICY 138
9.130 V$WAIT_CHAINS 139
9.131 V$WAITCLASSMETRIC 141
9.132 V$WAITCLASSMETRIC_HISTORY 141
9.133 V$WAITSTAT 142
9.134 V$WALLET 142
9.135 V$WORKLOAD_REPLAY_THREAD 143
9.136 V$XML_AUDIT_TRAIL 145
9.137 V$XSTREAM_APPLY_COORDINATOR 147
9.138 V$XSTREAM_APPLY_READER 149
9.139 V$XSTREAM_APPLY_RECEIVER 151
9.140 V$XSTREAM_APPLY_SERVER 152
9.141 V$XSTREAM_CAPTURE 155
9.142 V$XSTREAM_MESSAGE_TRACKING 158
9.143 V$XSTREAM_OUTBOUND_SERVER 159
9.144 V$XSTREAM_TABLE_STATS 162
9.145 V$XSTREAM_TRANSACTION 163
9.146 V$ZONEMAP_USAGE_STATS 165

Part IV Appendixes

A Database Limits

A.1 Datatype Limits A-1
A.2 Physical Database Limits A-2
A.3 Logical Database Limits A-4
A.4 Process and Runtime Limits A-6

B SQL Scripts

B.1 Creating the Data Dictionary B-1
B.2 Creating Additional Data Dictionary Structures B-2
B.3 The "NO" Scripts B-5
B.4 Upgrade Scripts B-6
B.5 Java Scripts B-6

C Oracle Wait Events

C.1 Classes of Wait Events C-1
C.2 Descriptions of Common Wait Event Parameters C-2
C.3 Descriptions of Wait Events C-5
C.3.1 alter system set dispatcher C-5

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.2 ARCH Remote Write C-5
C.3.3 ASYNC Remote Write C-5
C.3.4 BFILE check if exists C-5
C.3.5 BFILE check if open C-6
C.3.6 BFILE closure C-6
C.3.7 BFILE get length C-6
C.3.8 BFILE get name object C-6
C.3.9 BFILE get path object C-7
C.3.10 BFILE internal seek C-7
C.3.11 BFILE open C-7
C.3.12 BFILE read C-7
C.3.13 broadcast mesg queue transition C-7
C.3.14 broadcast mesg recovery queue transition C-8
C.3.15 buffer busy waits C-8
C.3.16 buffer deadlock C-9
C.3.17 buffer latch C-9
C.3.18 buffer read retry C-10
C.3.19 checkpoint completed C-10
C.3.20 cleanup of aborted process C-10
C.3.21 control file parallel write C-10
C.3.22 control file sequential read C-11
C.3.23 control file single write C-11
C.3.24 cursor: mutex S C-11
C.3.25 cursor: mutex X C-12
C.3.26 cursor: pin S C-12
C.3.27 cursor: pin S wait on X C-12
C.3.28 cursor: pin X C-12
C.3.29 db file async I/O submit C-13
C.3.30 db file parallel read C-13
C.3.31 db file parallel write C-13
C.3.32 db file scattered read C-14
C.3.33 db file sequential read C-14
C.3.34 db file single write C-15
C.3.35 DFS db file lock C-15
C.3.36 DFS lock handle C-16
C.3.37 direct path read C-16
C.3.38 direct path sync C-16
C.3.39 direct path write C-17
C.3.40 Disk file operations I/O C-17
C.3.41 dispatcher shutdown C-17
C.3.42 dispatcher timer C-18
C.3.43 dupl. cluster key C-18

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.44 enq: OW - initialization C-18
C.3.45 enq: OW - termination C-19
C.3.46 enq: SV - contention C-19
C.3.47 enq: TX - index contention C-19
C.3.48 enq: TX - row lock contention C-19
C.3.49 enqueue C-20
C.3.50 flashback buf free by RVWR C-20
C.3.51 flashback log file sync C-20
C.3.52 free buffer waits C-21
C.3.53 free global transaction table entry C-21
C.3.54 free process state object C-21
C.3.55 gc buffer busy acquire C-22
C.3.56 gc buffer busy release C-22
C.3.57 gc cr block 2-way C-22
C.3.58 gc cr block 3-way C-22
C.3.59 gc cr block busy C-23
C.3.60 gc cr block congested C-23
C.3.61 gc cr block lost C-24
C.3.62 gc cr grant read-mostly invalidation C-24
C.3.63 gc cr multi block grant C-24
C.3.64 gc cr multi block mixed C-24
C.3.65 gc cr request C-25
C.3.66 gc current block 2-way C-25
C.3.67 gc current block 3-way C-25
C.3.68 gc current block busy C-26
C.3.69 gc current block congested C-26
C.3.70 gc current block lost C-26
C.3.71 gc current request C-27
C.3.72 gc index operation C-27
C.3.73 gc recovery quiesce C-27
C.3.74 GCS lock open S C-28
C.3.75 GCS lock open X C-28
C.3.76 global cache busy C-28
C.3.77 inactive session C-28
C.3.78 inactive transaction branch C-29
C.3.79 index block split C-29
C.3.80 instance state change C-29
C.3.81 io done C-30
C.3.82 ksxr wait for mount shared C-30
C.3.83 ktm: instance recovery C-30
C.3.84 latch activity C-30
C.3.85 latch free C-31

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.86 latch: redo copy C-31
C.3.87 latch: row cache objects C-31
C.3.88 library cache load lock C-31
C.3.89 library cache lock C-32
C.3.90 library cache pin C-32
C.3.91 library cache shutdown C-33
C.3.92 library cache: mutex X C-33
C.3.93 LMON global data update C-33
C.3.94 Log archive I/O C-33
C.3.95 log buffer space C-33
C.3.96 log file parallel write C-34
C.3.97 log file sequential read C-34
C.3.98 log file single write C-34
C.3.99 log file switch (archiving needed) C-35
C.3.100 log file switch (checkpoint incomplete) C-35
C.3.101 log file switch (clearing log file) C-35
C.3.102 log file switch (private strand flush incomplete) C-35
C.3.103 log file switch completion C-35
C.3.104 log file sync C-36
C.3.105 log switch/archive C-36
C.3.106 optimizer stats update retry C-36
C.3.107 parallel recovery change buffer free C-36
C.3.108 parallel recovery control message reply C-37
C.3.109 parallel recovery coord send blocked C-37
C.3.110 parallel recovery coord wait for reply C-37
C.3.111 parallel recovery coordinator waits for slave cleanup C-37
C.3.112 parallel recovery read buffer free C-37
C.3.113 parallel recovery slave next change C-37
C.3.114 pending global transaction(s) C-38
C.3.115 pipe get C-38
C.3.116 pipe put C-38
C.3.117 PL/SQL lock timer C-38
C.3.118 pmon timer C-39
C.3.119 prior spawner clean up C-39
C.3.120 process startup C-39
C.3.121 PX Deque wait C-40
C.3.122 PX qref latch C-40
C.3.123 PX server shutdown C-40
C.3.124 PX signal server C-40
C.3.125 rdbms ipc message C-41
C.3.126 rdbms ipc message block C-41
C.3.127 rdbms ipc reply C-41

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.128 read by other session C-41
C.3.129 recovery active instance mapping setup C-42
C.3.130 recovery apply pending C-42
C.3.131 recovery cancel C-42
C.3.132 recovery checkpoint C-42
C.3.133 recovery file header update for checkpoint C-42
C.3.134 recovery file header update for fuzziness C-42
C.3.135 recovery marker apply C-42
C.3.136 recovery merge pending C-42
C.3.137 recovery metadata latch C-42
C.3.138 recovery move influx buffers C-43
C.3.139 recovery read C-43
C.3.140 recovery receive buffer free C-43
C.3.141 recovery remote file verification C-43
C.3.142 recovery send buffer free C-43
C.3.143 recovery shutdown C-43
C.3.144 Redo Transport Attach C-43
C.3.145 Redo Transport Close C-44
C.3.146 Redo Transport Detach C-44
C.3.147 Redo Transport Open C-44
C.3.148 Redo Transport Ping C-44
C.3.149 Redo Transport Slave Shutdown C-45
C.3.150 Redo Transport Slave Startup C-45
C.3.151 Redo Writer Remote Sync Complete C-45
C.3.152 Redo Writer Remote Sync Notify C-45
C.3.153 Remote SYNC Ping C-45
C.3.154 resmgr:become active C-45
C.3.155 resmgr:cpu quantum C-45
C.3.156 resmgr: I/O rate limit C-46
C.3.157 resmgr:pq queued C-46
C.3.158 rolling migration: cluster quiesce C-47
C.3.159 row cache lock C-47
C.3.160 RVWR wait for flashback copy C-47
C.3.161 sbtbufinfo C-47
C.3.162 sbtgetbuf C-48
C.3.163 sbtmapbuf C-48
C.3.164 sbtrelbuf C-48
C.3.165 scginq AST call C-48
C.3.166 SGA: allocation forcing component growth C-48
C.3.167 SGA: MMAN sleep for component shrink C-48
C.3.168 SGA: sga_target resize C-49
C.3.169 Shared IO Pool Memory C-49

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.170 shared server idle wait C-49
C.3.171 single-task message C-49
C.3.172 smon timer C-49
C.3.173 SQL*Net break/reset to client C-50
C.3.174 SQL*Net break/reset to dblink C-50
C.3.175 SQL*Net message from client C-50
C.3.176 SQL*Net message from dblink C-51
C.3.177 SQL*Net message to client C-51
C.3.178 SQL*Net message to dblink C-51
C.3.179 SQL*Net more data from client C-52
C.3.180 SQL*Net more data from dblink C-52
C.3.181 SQL*Net more data to client C-52
C.3.182 SQL*Net more data to dblink C-53
C.3.183 Streams AQ: waiting for messages in the queue C-53
C.3.184 switch logfile command C-53
C.3.185 SYNC Remote Write C-53
C.3.186 TCP Socket (KGAS) C-53
C.3.187 timer in sksawat C-54
C.3.188 transaction C-54
C.3.189 unbound tx C-54
C.3.190 undo_retention publish retry C-54
C.3.191 undo segment extension C-55
C.3.192 undo segment recovery C-55
C.3.193 undo segment tx slot C-55
C.3.194 utl_file I/O C-55
C.3.195 virtual circuit wait C-56
C.3.196 WCR: replay client notify C-56
C.3.197 WCR: replay clock C-56
C.3.198 WCR: replay lock order C-56
C.3.199 WCR: replay paused C-57
C.3.200 WCR: Sync context busy C-57
C.3.201 write complete waits C-57
C.3.202 writes stopped by instance recovery or database suspension C-57

D Oracle Enqueue Names

E Statistics Descriptions

E.1 Displaying Statistics E-1
E.2 Statistics Descriptions E-1

Copyright © 2002, 2026, Oracle and/or its affiliates.

F Background Processes

Index

Copyright © 2002, 2026, Oracle and/or its affiliates.

Preface

This manual provides reference information about database initialization parameters, static data dictionary views, dynamic performance views, database limits, and SQL scripts that are part of the Oracle Database.

This preface contains these topics:

- Audience
- Documentation Accessibility
- Diversity and Inclusion
- Related Documents
- Conventions

Audience

Oracle Database Reference is intended for database administrators, system administrators, and database application developers.

To use this document, you need to be familiar with the following:

- Oracle database management system (DBMS) concepts
- Your operating system environment

Documentation Accessibility

For information about Oracle's commitment to accessibility, visit the Oracle Accessibility Program website at http://www.oracle.com/pls/topic/lookup?ctx=acc&id=docacc.

Access to Oracle Support

Oracle customers that have purchased support have access to electronic support through My Oracle Support. For information, visit http://www.oracle.com/pls/topic/lookup?ctx=acc&id=info or visit http://www.oracle.com/pls/topic/lookup?ctx=acc&id=trs if you are hearing impaired.

Diversity and Inclusion

Oracle is fully committed to diversity and inclusion. Oracle respects and values having a diverse workforce that increases thought leadership and innovation. As part of our initiative to build a more inclusive culture that positively impacts our employees, customers, and partners, we are working to remove insensitive terms from our products and documentation. We are also mindful of the necessity to maintain compatibility with our customers' existing technologies and the need to ensure continuity of service as Oracle's offerings and industry standards evolve. Because of these technical constraints, our effort to remove insensitive terms is ongoing and will take time and external cooperation.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Related Documents

For more information, see these Oracle resources:

- *Oracle Database Concepts* for a comprehensive introduction to the concepts and terminology used in this manual
- *Oracle Database Administrator's Guide* for information about administering Oracle Database
- *Oracle Database Upgrade Guide* for the procedures for upgrading a previous release of Oracle to the new release
- *Oracle Database SQL Language Reference* for information on Oracle's SQL commands and functions
- *Oracle Database Development Guide* for information about developing database applications within Oracle Database

The complete documentation set for Oracle Database is available on the Oracle Help Center: http://docs.oracle.com/database/

Conventions

The following text conventions are used in this document:

Convention	Meaning
boldface	Boldface type indicates graphical user interface elements associated with an action, or terms defined in text or the glossary.
italic	Italic type indicates book titles, emphasis, or placeholder variables for which you supply particular values.
`monospace`	Monospace type indicates commands within a paragraph, URLs, code in examples, text that appears on the screen, or text that you enter.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Changes in This Release for Oracle Database Reference

This preface lists changes in *Oracle Database Reference*:

- Changes in Oracle Database 19c, Release Update 19.29
- Changes in Oracle Database 19c, Release Update 19.28
- Changes in Oracle Database 19c, Release Update 19.26
- Changes in Oracle Database 19c, Release Update 19.23
- Changes in Oracle Database 19c, Release Update 19.22
- Changes in Oracle Database 19c, Release Update 19.16
- Changes in Oracle Database 19c, Release Update 19.13
- Changes in Oracle Database 19c, Release Update 19.11
- Changes in Oracle Database 19c, Release Update 19.10
- Changes in Oracle Database 19c, Release Update 19.6
- Changes in Oracle Database 19c, Release Update 19.4
- Changes in Oracle Database 19c

Changes in Oracle Database 19c, Release Update 19.29

The following initialization parameter is new in Oracle Database 19c, Release Update 19.29:

- HEARTBEAT_BATCH_SIZE

Changes in Oracle Database 19c, Release Update 19.28

The following initialization parameters are new in Oracle Database 19c, Release Update 19.28:

- MFA_DUO_API_HOST
- MFA_OMA_IAM_DOMAIN_URL
- MFA_SENDER_EMAIL_DISPLAYNAME
- MFA_SENDER_EMAIL_ID
- MFA_SMTP_HOST
- MFA_SMTP_PORT

The following static data dictionary views are new in Oracle Database 19c, Release Update 19.28:

- ALL_ANNOTATION_VALUES, DBA_ANNOTATION_VALUES, and USER_ANNOTATION_VALUES

Copyright © 2002, 2026, Oracle and/or its affiliates.

- ALL_ANNOTATIONS, DBA_ANNOTATIONS, and USER_ANNOTATIONS
- ALL_ANNOTATIONS_USAGE, DBA_ANNOTATIONS_USAGE, and USER_ANNOTATIONS_USAGE
- DBA_HIST_SNAPSHOT_DETAILS

Changes in Oracle Database 19c, Release Update 19.26

The following initialization parameter is changed in Oracle Database 19c, Release Update 19.26:

- A new AUTO setting is available for the DB_LOST_WRITE_PROTECT parameter and it is the default setting.

Changes in Oracle Database 19c, Release Update 19.23

The following initialization parameter is new in Oracle Database 19c, Release Update 19.23:

- SESSION_EXIT_ON_PACKAGE_STATE_ERROR

Changes in Oracle Database 19c, Release Update 19.22

The following initialization parameter is new in Oracle Database 19c, Release Update 19.22:

- OGG_ONLINE_DDL

The following initialization parameter is changed in Oracle Database 19c, Release Update 19.22:

- The PRE_PAGE_SGA initialization parameter is deprecated. The value of PRE_PAGE_SGA is always set to TRUE on Exadata systems. Any attempt to set this parameter to FALSE on an Exadata system will not take effect.

Changes in Oracle Database 19c, Release Update 19.21

The following initialization parameter is new in Oracle Database 19c, Release Update 19.21:

- MAIN_WORKLOAD_TYPE

Changes in Oracle Database 19c, Release Update 19.16

The following initialization parameters are new in Oracle Database 19c, Release Update 19.16:

- IDENTITY_PROVIDER_CONFIG
- IDENTITY_PROVIDER_TYPE
- TABLESPACE_ENCRYPTION

Changes in Oracle Database 19c, Release Update 19.13

New AE Analytic Views

Oracle Database 19c, Release Update 19.13, introduces AE ("all edition") analytic views. Each new AE analytic view corresponds to an existing non-AE analytic view. AE views have the same

Copyright © 2002, 2026, Oracle and/or its affiliates.

columns as their non-AE counterparts, plus a column that displays the name of the application edition where the editioned object is defined.

The following table lists the new AE analytic views, divided into three categories: analytic view views, attribute dimension views, and hierarchy views. For brevity, only the ALL_ views are listed. The corresponding DBA_ and USER_ views are also available and are documented in this guide.

Analytic View Views	Attribute Dimension Views	Hierarchy Views
ALL_ANALYTIC_VIEW_ATTR_CLS_AE ALL_ANALYTIC_VIEW_BAS_MEAS_AE ALL_ANALYTIC_VIEW_CLASS_AE ALL_ANALYTIC_VIEW_CLC_MEAS_AE ALL_ANALYTIC_VIEW_COLUMNS_AE ALL_ANALYTIC_VIEW_DIM_CLS_AE ALL_ANALYTIC_VIEW_DIMS_AE ALL_ANALYTIC_VIEW_HIER_CLS_AE ALL_ANALYTIC_VIEW_HIERS_AE ALL_ANALYTIC_VIEW_KEYS_AE ALL_ANALYTIC_VIEW_LEVELS_AE ALL_ANALYTIC_VIEW_LVL_CLS_AE ALL_ANALYTIC_VIEW_LVLGRPS_AE ALL_ANALYTIC_VIEW_MEAS_CLS_AE ALL_ANALYTIC_VIEWS_AE	ALL_ATTRIBUTE_DIM_ATTR_CLS_AE ALL_ATTRIBUTE_DIM_ATTRS_AE ALL_ATTRIBUTE_DIM_CLASS_AE ALL_ATTRIBUTE_DIM_JN_PTHS_AE ALL_ATTRIBUTE_DIM_KEYS_AE ALL_ATTRIBUTE_DIM_LEVELS_AE ALL_ATTRIBUTE_DIM_LVL_ATRS_AE ALL_ATTRIBUTE_DIM_LVL_CLS_AE ALL_ATTRIBUTE_DIM_ORD_ATRS_AE ALL_ATTRIBUTE_DIM_TABLES_AE ALL_ATTRIBUTE_DIMENSIONS_AE	ALL_HIER_CLASS_AE ALL_HIER_COLUMNS_AE ALL_HIER_HIER_ATTR_CLASS_AE ALL_HIER_HIER_ATTRIBUTES_AE ALL_HIER_JOIN_PATHS_AE ALL_HIER_LEVEL_ID_ATTRS_AE ALL_HIER_LEVELS_AE ALL_HIERARCHIES_AE

Changes in Oracle Database 19c, Release Update 19.11

The following initialization parameter is new in Oracle Database 19c, Release Update 19.11:

- DRCP_DEDICATED_OPT

The following static data dictionary views are new in Oracle Database 19c, Release Update 19.11:

- DBA_ACCHK_EVENTS
- DBA_ACCHK_EVENTS_SUMMARY
- DBA_ACCHK_STATISTICS
- DBA_ACCHK_STATISTICS_SUMMARY
- ALL_IMMUTABLE_TABLES, DBA_IMMUTABLE_TABLES, and USER_IMMUTABLE_TABLES

Changes in Oracle Database 19c, Release Update 19.10

The following initialization parameters are new in Oracle Database 19c, Release Update 19.10:

- BLOCKCHAIN_TABLE_MAX_NO_DROP
- CONTAINER_DATA
- MAX_AUTH_SERVERS
- MIN_AUTH_SERVERS
- OPTIMIZER_REAL_TIME_STATISTICS

The following static data dictionary views are new in Oracle Database 19c, Release Update 19.10:

Copyright © 2002, 2026, Oracle and/or its affiliates.

- ALL_BLOCKCHAIN_TABLES, DBA_BLOCKCHAIN_TABLES, and USER_BLOCKCHAIN_TABLES
- ALL_CERTIFICATES, DBA_CERTIFICATES, and USER_CERTIFICATES
- DBA_LOGMNR_DICTIONARY_BUILDLOG

Changes in Oracle Database 19c, Release Update 19.6

The following initialization parameter is new in Oracle Database 19c, Release Update 19.6:

- OPTIMIZER_SESSION_TYPE

Changes in Oracle Database 19c, Release Update 19.4

The following initialization parameter is new in Oracle Database 19c, Release Update 19.4:

- IGNORE_SESSION_SET_PARAM_ERRORS

Changes in Oracle Database 19c

This section lists changes in *Oracle Database Reference* for Oracle Database 19c.

See *Oracle Database New Features Guide* for a complete list of new features for this release.

New Features

This section lists new features in this manual for Oracle Database 19c.

The following changes to initialization parameters are new in Oracle Database 19c:

- On Linux platforms only, Oracle recommends against explicitly setting values for `MEMORY_TARGET` and `MEMORY_MAX_TARGET` in the initialization parameter file at the time of instance startup. Doing so may result in suboptimal small page allocation in the SGA. Furthermore, if the `USE_LARGE_PAGES` parameter is set to `ONLY` and you attempt to set values for `MEMORY_TARGET` and `MEMORY_MAX_TARGET`, the instance will fail to start.

  See MEMORY_TARGET, MEMORY_MAX_TARGET, and USE_LARGE_PAGES.

The following initialization parameters are new in Oracle Database 19c:

- ADG_REDIRECT_DML
- ALLOW_ROWID_COLUMN_TYPE
- CLIENT_STATISTICS_LEVEL
- CPU_MIN_COUNT
- DATA_GUARD_MAX_IO_TIME
- DATA_GUARD_MAX_LONGIO_TIME
- ENABLE_IMC_WITH_MIRA
- LOB_SIGNATURE_ENABLE
- MAX_DATAPUMP_PARALLEL_PER_JOB
- MAX_IDLE_BLOCKER_TIME
- OPTIMIZER_CAPTURE_SQL_QUARANTINE

Copyright © 2002, 2026, Oracle and/or its affiliates.

- OPTIMIZER_USE_SQL_QUARANTINE
- UNIFIED_AUDIT_COMMON_SYSTEMLOG

> **Note**
>
> See Initialization Parameters for more information about these initialization parameters.

The following static data dictionary views are new in Oracle Database 19c:

- ALL_TRIGGERS_AE, DBA_TRIGGERS_AE, and USER_TRIGGERS_AE
- DBA_AUTO_INDEX_CONFIG
- DBA_AUTO_STAT_EXECUTIONS
- DBA_PDB_SNAPSHOTFILE
- DBA_RAT_CAPTURE_SCHEMA_INFO
- DBA_REGISTRY_BACKPORTS
- DBA_SQL_QUARANTINE

> **Note**
>
> Static Data Dictionary Views for more information about these static data dictionary views

The following dynamic performance views are new in Oracle Database 19c:

- GV$AQ_PARTITION_STATS and V$AQ_PARTITION_STATS
- GV$ASM_ACFSAUTORESIZE and V$ASM_ACFSAUTORESIZE
- GV$MEMOPTIMIZE_WRITE_AREA and V$MEMOPTIMIZE_WRITE_AREA
- GV$SHARED_SERVER_STAT and V$SHARED_SERVER_STAT
- GV$SQL_TESTCASES and V$SQL_TESTCASES

> **Note**
>
> GV$ Views and Dynamic Performance Views for more information about these dynamic performance views

Deprecated Features

The following features are deprecated in Oracle Database 19c, and may be desupported in a future release:

- The CLUSTER_DATABASE_INSTANCES initialization parameter
- The PRE_PAGE_SGA initialization parameter (starting with Oracle Database 19c, Release Update 19.22)

Copyright © 2002, 2026, Oracle and/or its affiliates.

- The SERVICE_NAMES initialization parameter

Desupported Features

The following features are desupported in Oracle Database 19c:

- The `MAX_CONNECTIONS` attribute of the `LOG_ARCHIVE_DEST_`*n* initialization parameter, for Oracle Data Guard redo transport
- The `EXAFUSION_ENABLED` initialization parameter
- The `O7_DICTIONARY_ACCESSIBILITY` initialization parameter
- Oracle Streams is desupported. As a result, the following views are desupported:

  ```
  ALL_STREAMS_COLUMNS
  ALL_STREAMS_MESSAGE_RULES
  ALL_STREAMS_RULES
  ALL_STREAMS_UNSUPPORTED
  DBA_STREAMS_ADMINISTRATOR
  DBA_STREAMS_COLUMNS
  DBA_STREAMS_MESSAGE_RULES
  DBA_STREAMS_RULES
  DBA_STREAMS_SPLIT_MERGE
  DBA_STREAMS_SPLIT_MERGE_HIST
  DBA_STREAMS_STMTS
  DBA_STREAMS_STMT_HANDLERS
  DBA_STREAMS_TRANSFORMATIONS
  DBA_STREAMS_UNSUPPORTED
  V$STREAMS_CAPTURE
  V$STREAMS_MESSAGE_TRACKING
  V$STREAMS_TRANSACTION
  ```

- Extended Datatype Support (EDS) is desupported. As a result, the following views are desupported:

  ```
  DBA_LOGSTDBY_EDS_SUPPORTED
  DBA_LOGSTDBY_EDS_TABLES
  ```

See Also

Oracle Database Upgrade Guide

Copyright © 2002, 2026, Oracle and/or its affiliates.

Part I
Initialization Parameters

Database initialization parameters can be specified in a parameter file to start or configure an instance.

This part contains the following chapter:

- Initialization Parameters

Copyright © 2002, 2026, Oracle and/or its affiliates.

1

Initialization Parameters

This chapter contains detailed descriptions (in alphabetical order) of the database initialization parameters.

This chapter contains the following topics:

- Uses of Initialization Parameters
- Basic Initialization Parameters
- Parameter Files
- Changing Parameter Values in a Parameter File
- Reading the Parameter Descriptions
- Initialization Parameter Descriptions

1.1 Uses of Initialization Parameters

Initialization parameters are used to perform various functions.

For example, initialization parameters are used to:

- Set limits for the entire database
- Set user or process limits
- Set limits on database resources
- Affect performance (these are called **variable parameters**)

  Variable parameters are of particular interest to database administrators, because these parameters are used primarily to improve database performance.

Database administrators can use initialization parameters to:

- Optimize performance by adjusting memory structures, such as the number of database buffers in memory
- Set database-wide defaults, such as the amount of space initially allocated for a context area when it is created
- Set database limits, such as the maximum number of database users
- Specify names of files or directories required by the database

Many initialization parameters can be fine-tuned to improve database performance. Other parameters should never be altered or should be altered only under the supervision of Oracle Support Services.

All initialization parameters are optional. Oracle has a default value for each parameter. This value may be operating system-dependent, depending on the parameter.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.1.1 Types of Initialization Parameters

The Oracle database server has derived parameters, operating system-dependent parameters, and variable parameters.

- Derived Parameters
- Operating System-Dependent Parameters
- Variable Parameters (these can be dynamic parameters or any of the preceding ones)

1.1.1.1 Derived Parameters

Some initialization parameters are **derived**, meaning that their values are calculated from the values of other parameters. Normally, you should not alter values for derived parameters, but if you do, then the value you specify will override the calculated value.

For example, the default value of the `SESSIONS` parameter is derived from the value of the `PROCESSES` parameter. If the value of `PROCESSES` changes, then the default value of `SESSIONS` changes as well, unless you override it with a specified value.

1.1.1.2 Operating System-Dependent Parameters

The valid values or value ranges of some initialization parameters depend upon the host operating system. For example, the parameter `DB_BLOCK_BUFFERS` indicates the number of data buffers in main memory, and its maximum value depends on the operating system. The size of those buffers, set by `DB_BLOCK_SIZE`, has an operating system-dependent default value.

> **See Also**
>
> Your operating system-specific Oracle documentation for more information on operating system-dependent Oracle parameters and operating system parameters

1.1.1.3 Variable Parameters

The variable initialization parameters offer the most potential for improving system performance. Some variable parameters set capacity limits but do not affect performance. For example, when the value of `OPEN_CURSORS` is 10, a user process attempting to open its eleventh cursor receives an error. Other variable parameters affect performance but do not impose absolute limits. For example, reducing the value of `DB_BLOCK_BUFFERS` does not prevent work even though it may slow down performance.

Increasing the values of variable parameters may improve your system's performance, but increasing most parameters also increases the system global area (SGA) size. A larger SGA can improve database performance up to a point. In virtual memory operating systems, an SGA that is too large can degrade performance if it is swapped in and out of memory. Operating system parameters that control virtual memory working areas should be set with the SGA size in mind. The operating system configuration can also limit the maximum size of the SGA.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.2 Basic Initialization Parameters

Most databases should only need to have the database basic initialization parameters set to run properly and efficiently.

Oracle advises you to become familiar with the basic parameters and only use other parameters when directed to by feature documentation or in special circumstances. The following is a list of the database basic initialization parameters:

CLUSTER_DATABASE
COMPATIBLE
CONTROL_FILES
DB_BLOCK_SIZE
DB_CREATE_FILE_DEST
DB_CREATE_ONLINE_LOG_DEST_*n*
DB_DOMAIN
DB_NAME
DB_RECOVERY_FILE_DEST
DB_RECOVERY_FILE_DEST_SIZE
DB_UNIQUE_NAME
INSTANCE_NUMBER
LDAP_DIRECTORY_SYSAUTH
LOG_ARCHIVE_DEST_*n*
LOG_ARCHIVE_DEST_STATE_*n*
NLS_DATE_LANGUAGE
NLS_TERRITORY
OPEN_CURSORS
PGA_AGGREGATE_TARGET
PROCESSES
REMOTE_LISTENER
REMOTE_LOGIN_PASSWORDFILE
SESSIONS
SGA_TARGET
SHARED_SERVERS
STAR_TRANSFORMATION_ENABLED
UNDO_TABLESPACE

1.3 Parameter Files

A **parameter file** is a file that contains a list of initialization parameters and a value for each parameter. You specify initialization parameters in a parameter file that reflect your particular installation.

Oracle supports the following two types of parameter files:

- Server Parameter Files
- Initialization Parameter Files

1.3.1 Server Parameter Files

A **server parameter file** is a binary file that acts as a repository for initialization parameters.

Copyright © 2002, 2026, Oracle and/or its affiliates.

The server parameter file can reside on the computer where the Oracle database server executes. Initialization parameters stored in a server parameter file are persistent, in that any changes made to the parameters while an instance is running can persist across instance shutdown and startup.

> **See Also**
>
> *Oracle Database Administrator's Guide* for an introduction to and detailed information about managing and using server parameter files and initialization parameter files

1.3.2 Initialization Parameter Files

An **initialization parameter file** is a text file that contains a list of initialization parameters.

> **Note**
>
> See "About the Character Set of Parameter Values" for more information about the character set to use when specifying values in an initialization parameter file.

The following are sample entries in an initialization parameter file:

```
PROCESSES = 100
OPEN_LINKS = 12
GLOBAL_NAMES = true
```

The name of the initialization parameter file varies depending on the operating system. For example, it can be in mixed case or lowercase, or it can have a logical name or a variation of the name `init.ora`. Also supplied is an `initdw.ora` file, which contains suggested parameter settings for data warehouses and data marts. The database administrator can choose a different file name for the initialization parameter file.

Refer to your operating system-specific Oracle documentation for the default locations and filenames for initialization parameter files on your operating system. The initialization parameter file is read by the client-side tool used to start the server (such as SQL*Plus).

Sample initialization parameter files are provided on the Oracle distribution medium for each operating system. A sample file is sufficient for initial use, but you will probably want to modify the file to tune the database for best performance. Any changes will take effect after you completely shut down and restart the instance.

1.3.2.1 About the Character Set of Parameter Values

Only characters from the default character set of the database platform are supported in parameter values. For all platforms except IBM z/OS and Fujitsu BS2000 the default platform character set is US7ASCII (7-bit ASCII).

The database behavior is undefined when parameter values with unsupported characters are specified. When using the `ALTER SYSTEM` statement to set a parameter value in a server parameter file, make sure only supported characters are used. The database does not always report an error when a value with unsupported characters is specified.

The initialization parameter file is a client-side file. When used, it is located on the host on which you start SQL*Plus to start up a database instance. The initialization parameter file must

Copyright © 2002, 2026, Oracle and/or its affiliates.

be written in the client character set as specified by the NLS_LANG client setting. However, parameter values should not contain characters not supported by the default character set of the database.

1.3.2.2 Specifying Values in an Initialization Parameter File

This section describes several aspects of setting parameter values in an initialization parameter file.

1.3.2.2.1 Rules Governing Initialization Parameter Files

This section describes the rules that govern the specification of parameters in initialization parameter files.

- An initialization parameter file should contain only parameters and comments. A pound sign (#) starts a comment line. The rest of the line is ignored.
- Comments must use the same character set as parameter values.
- You can specify parameters in any order.
- Case (upper or lower) in filenames is significant only if case is significant on the host operating system.
- To enter several parameters on one line, use spaces between parameter names and values, as in the following example:

  ```
  PROCESSES = 100 CPU_COUNT = 1 OPEN_CURSORS = 10
  ```

- Some parameters, such as ROLLBACK_SEGMENTS, accept multiple values. Any of the following represent valid syntax.
  - Enter multiple values enclosed in parentheses and separated by commas. For example:

    ```
    ROLLBACK_SEGMENTS = (SEG1, SEG2, SEG3, SEG4, SEG5)
    ```

  - Enter multiple values without parentheses and commas. For example:

    ```
    ROLLBACK_SEGMENTS = SEG1 SEG2 SEG3 SEG4 SEG5
    ```

  - Enter multiple values, one per line. For example:

    ```
    ROLLBACK_SEGMENTS = SEG1
    ROLLBACK_SEGMENTS = SEG2
    ROLLBACK_SEGMENTS = SEG3
    ROLLBACK_SEGMENTS = SEG4
    ROLLBACK_SEGMENTS = SEG5
    ```

  If you enter values for one parameter on multiple lines, then the entries must be on consecutive lines. If they are not, then the first entry will not be processed properly. For example, in the following entry the setting for SEG3 and SEG4 will override the setting for SEG1 and SEG2:

  ```
  ROLLBACK_SEGMENTS = SEG1 SEG2
  OPEN_CURSORS = 10
  ROLLBACK_SEGMENTS = SEG3 SEG4
  ```

- A backslash (\), also known as an escape character, indicates continuation of the parameter specification. If a backslash continues a line, then the continued line must have no leading spaces. For example:

  ```
  ROLLBACK_SEGMENTS = (SEG1, SEG2, \
  SEG3, SEG4, SEG5)
  ```


Copyright © 2002, 2026, Oracle and/or its affiliates.

- You can use the `IFILE` initialization parameter to embed the contents of another initialization parameter file into the current initialization parameter file.
- Enclose in quotation marks any parameter values that contain spaces or tabs. You can use either single or double quotation marks unless otherwise indicated. For example:

```
NLS_TERRITORY = 'CZECH REPUBLIC'
```

> **Note**
>
> Listing parameters in alphabetical order in the initialization parameter file can help you to find them and can help ensure that each parameter is specified only once.

- Enclose in quotation marks any parameter value that contains a special character.

> **See Also**
>
> - Your operating system-specific Oracle documentation for more information on initialization parameter files
> - "IFILE"

1.3.2.2.2 Using Special Characters in Parameter Values

If a parameter value contains a special character, then the special character must be preceded by a backslash or the entire parameter value must be enclosed in quotation marks.

For example, you can specify special characters using either of the following:

```
DB_DOMAIN = 'JAPAN.ACME#.COM'

DB_DOMAIN = JAPAN.ACME\#.COM
```

Table 1-1 lists the special characters you can use in initialization parameter files.

Table 1-1 Special Characters in Initialization Parameter Files

Character	Name	Description
#	Number sign	Comment
(	Left parenthesis	Start list of values
)	Right parenthesis	End list of values
"	Double quotation mark	Start or end of quoted string
'	Single quotation mark	Start or end of quoted string
=	Equal sign	Separator of keyword and values
,	Comma	Separator of elements
-	Minus sign	Precedes UNIX-style keywords
\	Backslash	Escape character

If a special character must be treated literally in an initialization parameter file, then it must either be preceded by the backslash character, or the entire string containing the special character must be enclosed in quotation marks.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.3.2.2.3 Using the Escape Character

In an initialization parameter file, a backslash (\) can also signify a line continuation. If the backslash is followed by an alphanumeric character, then the backslash is treated as a normal character in the input.

If it is not followed by an alphanumeric character, then the backslash is treated either as a backslash or as a continuation character.

> **See Also**
>
> "Rules Governing Initialization Parameter Files"

1.3.2.2.4 Using Quotation Marks

Quotation marks can be nested in two ways in an initialization parameter file.

The first method is to double the quotation marks in the nested string. For example:

```
NLS_DATE_FORMAT = '''Today is'' MM/DD/YYYY'
```

The second method is to alternate single and double quotation marks. For example:

```
NLS_DATE_FORMAT = '"Today is" MM/DD/YYYY'
```

1.4 Changing Parameter Values in a Parameter File

You can change the value of a parameter in a parameter file in several ways.

- By editing an initialization parameter file

  In most cases, the new value takes effect the next time you start an instance of the database.
- By issuing an `ALTER SYSTEM SET ... SCOPE=SPFILE` statement to update a server parameter file
- By issuing an `ALTER SYSTEM RESET` statement to clear an initialization parameter value.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information about using the `ALTER SYSTEM RESET` statement to clear initialization parameter values

1.4.1 Parameters by Functional Category

This section lists initialization parameters by their functional category.

- **ANSI Compliance**

  BLANK_TRIMMING
- **Backup and Restore**

  BACKUP_TAPE_IO_SLAVES

Copyright © 2002, 2026, Oracle and/or its affiliates.

DATA_TRANSFER_CACHE_SIZE
CLONEDB
CLONEDB_DIR
RECYCLEBIN
TAPE_ASYNCH_IO

- **BFILEs**

  SESSION_MAX_OPEN_FILES

- **Buffer Cache and I/O**

  CLIENT_RESULT_CACHE_LAG
  CLIENT_RESULT_CACHE_SIZE
  DB_*n*K_CACHE_SIZE
  DB_BIG_TABLE_CACHE_PERCENT_TARGET
  DB_BLOCK_BUFFERS
  DB_BLOCK_SIZE
  DB_CACHE_ADVICE
  DB_CACHE_SIZE
  DB_FILE_MULTIBLOCK_READ_COUNT
  DB_KEEP_CACHE_SIZE
  DB_RECYCLE_CACHE_SIZE
  DB_WRITER_PROCESSES
  DBWR_IO_SLAVES
  DISK_ASYNCH_IO
  DNFS_BATCH_SIZE
  FILESYSTEMIO_OPTIONS
  READ_ONLY_OPEN_DELAYED
  RESULT_CACHE_MAX_RESULT
  RESULT_CACHE_MAX_SIZE
  RESULT_CACHE_MODE

- **Cursors and Library Cache**

  CURSOR_BIND_CAPTURE_DESTINATION
  CURSOR_INVALIDATION
  CURSOR_SHARING
  CURSOR_SPACE_FOR_TIME
  OPEN_CURSORS
  SESSION_CACHED_CURSORS

- **Database/Instance Identification**

  DB_DOMAIN
  DB_NAME
  INSTANCE_NAME

- **Diagnostics and Statistics**

  BACKGROUND_CORE_DUMP
  BACKGROUND_DUMP_DEST
  CLIENT_STATISTICS_LEVEL
  CORE_DUMP_DEST
  DB_BLOCK_CHECKING
  DB_BLOCK_CHECKSUM
  DIAGNOSTIC_DEST

Copyright © 2002, 2026, Oracle and/or its affiliates.

EVENT
HEAT_MAP
MAX_DUMP_FILE_SIZE
SHADOW_CORE_DUMP
STATISTICS_LEVEL
TIMED_OS_STATISTICS
TIMED_STATISTICS
TRACE_ENABLED
TRACEFILE_IDENTIFIER
USER_DUMP_DEST

- **Distributed, Replication**

  COMMIT_POINT_STRENGTH
  DISTRIBUTED_LOCK_TIMEOUT
  ENABLE_GOLDENGATE_REPLICATION
  GLOBAL_NAMES
  HS_AUTOREGISTER
  OGG_ONLINE_DDL
  OPEN_LINKS
  OPEN_LINKS_PER_INSTANCE
  REPLICATION_DEPENDENCY_TRACKING

- **File Locations, Names, and Sizes**

  AUDIT_FILE_DEST
  BACKGROUND_CORE_DUMP
  BACKGROUND_DUMP_DEST
  CONTROL_FILES
  CORE_DUMP_DEST
  DB_CREATE_FILE_DEST
  DB_CREATE_ONLINE_LOG_DEST_*n*
  DB_FILES
  DB_RECOVERY_FILE_DEST
  DB_RECOVERY_FILE_DEST_SIZE
  FILE_MAPPING
  IFILE
  LOG_ARCHIVE_DEST_*n*
  SPFILE

- **Globalization**

  NLS_CALENDAR
  NLS_COMP
  NLS_CURRENCY
  NLS_DATE_FORMAT
  NLS_DATE_LANGUAGE
  NLS_DUAL_CURRENCY
  NLS_ISO_CURRENCY
  NLS_LANGUAGE
  NLS_LENGTH_SEMANTICS
  NLS_NCHAR_CONV_EXCP
  NLS_NUMERIC_CHARACTERS
  NLS_SORT

Copyright © 2002, 2026, Oracle and/or its affiliates.

NLS_TERRITORY
NLS_TIMESTAMP_FORMAT
NLS_TIMESTAMP_TZ_FORMAT

- **Java**

  JAVA_JIT_ENABLED
  JAVA_MAX_SESSIONSPACE_SIZE
  JAVA_POOL_SIZE
  JAVA_SOFT_SESSIONSPACE_LIMIT

- **Job Queues**

  JOB_QUEUE_PROCESSES

- **License Limits**

  LICENSE_MAX_SESSIONS
  LICENSE_MAX_USERS
  LICENSE_SESSIONS_WARNING

- **Memory**

  INMEMORY_ADG_ENABLED
  INMEMORY_AUTOMATIC_LEVEL
  INMEMORY_CLAUSE_DEFAULT
  INMEMORY_EXPRESSIONS_USAGE
  INMEMORY_FORCE
  INMEMORY_MAX_POPULATE_SERVERS
  INMEMORY_OPTIMIZED_ARITHMETIC
  INMEMORY_QUERY
  INMEMORY_SIZE
  INMEMORY_TRICKLE_REPOPULATE_SERVERS_PERCENT
  INMEMORY_VIRTUAL_COLUMNS
  MEMOPTIMIZE_POOL_SIZE
  MEMORY_MAX_TARGET
  MEMORY_TARGET

- **Miscellaneous**

  ALLOW_ROWID_COLUMN_TYPE
  AQ_TM_PROCESSES
  ASM_IO_PROCESSES
  ASM_PREFERRED_READ_FAILURE_GROUPS
  AWR_SNAPSHOT_TIME_OFFSET
  COMPATIBLE
  DB_INDEX_COMPRESSION_INHERITANCE
  ENABLE_DNFS_DISPATCHER
  FIXED_DATE
  IGNORE_SESSION_SET_PARAM_ERRORS
  INSTANCE_ABORT_DELAY_TIME
  LDAP_DIRECTORY_SYSAUTH
  MAX_DATAPUMP_JOBS_PER_PDB
  MAX_DATAPUMP_PARALLEL_PER_JOB
  MAX_STRING_SIZE
  MULTISHARD_QUERY_DATA_CONSISTENCY
  PRIVATE_TEMP_TABLE_PREFIX

Copyright © 2002, 2026, Oracle and/or its affiliates.

SHRD_DUPL_TABLE_REFRESH_RATE
SPATIAL_VECTOR_ACCELERATION
UNIFORM_LOG_TIMESTAMP_FORMAT
XML_DB_EVENTS

- **Multitenant Architecture**

  AUTOTASK_MAX_ACTIVE_PDBS
  AWR_PDB_AUTOFLUSH_ENABLED
  COMMON_USER_PREFIX
  CONTAINER_DATA
  CONTAINERS_PARALLEL_DEGREE
  CPU_MIN_COUNT
  DB_PERFORMANCE_PROFILE
  DEFAULT_SHARING
  ENABLE_AUTOMATIC_MAINTENANCE_PDB
  ENABLED_PDBS_ON_STANDBY
  EXTERNAL_KEYSTORE_CREDENTIAL_LOCATION
  MAX_IOPS
  MAX_MBPS
  MAX_PDBS
  NONCDB_COMPATIBLE
  PDB_FILE_NAME_CONVERT
  PDB_LOCKDOWN
  PDB_OS_CREDENTIAL
  SGA_MIN_SIZE

- **Networking**

  CONNECTION_BROKERS
  DRCP_DEDICATED_OPT
  FORWARD_LISTENER
  LOCAL_LISTENER
  MAX_AUTH_SERVERS
  MIN_AUTH_SERVERS
  OFS_THREADS
  REMOTE_LISTENER
  SERVICE_NAMES

- **Objects and LOBs**

  OBJECT_CACHE_MAX_SIZE_PERCENT
  OBJECT_CACHE_OPTIMAL_SIZE

- **OLAP**

  OLAP_PAGE_POOL_SIZE

- **Optimizer**

  APPROX_FOR_AGGREGATION
  APPROX_FOR_COUNT_DISTINCT
  APPROX_FOR_PERCENTILE
  OPTIMIZER_ADAPTIVE_PLANS
  OPTIMIZER_ADAPTIVE_REPORTING_ONLY
  OPTIMIZER_ADAPTIVE_STATISTICS
  OPTIMIZER_CAPTURE_SQL_PLAN_BASELINES

Copyright © 2002, 2026, Oracle and/or its affiliates.

OPTIMIZER_CAPTURE_SQL_QUARANTINE
OPTIMIZER_DYNAMIC_SAMPLING
OPTIMIZER_FEATURES_ENABLE
OPTIMIZER_IGNORE_HINTS
OPTIMIZER_IGNORE_PARALLEL_HINTS
OPTIMIZER_INDEX_CACHING
OPTIMIZER_INDEX_COST_ADJ
OPTIMIZER_INMEMORY_AWARE
OPTIMIZER_MODE
OPTIMIZER_REAL_TIME_STATISTICS
OPTIMIZER_SECURE_VIEW_MERGING
OPTIMIZER_SESSION_TYPE
OPTIMIZER_USE_PENDING_STATISTICS
OPTIMIZER_USE_SQL_PLAN_BASELINES
OPTIMIZER_USE_SQL_QUARANTINE
QUERY_REWRITE_ENABLED
QUERY_REWRITE_INTEGRITY
STAR_TRANSFORMATION_ENABLED

- **Oracle RAC**

  ACTIVE_INSTANCE_COUNT
  CLUSTER_DATABASE
  CLUSTER_DATABASE_INSTANCES
  CLUSTER_INTERCONNECTS
  INSTANCE_NUMBER
  PARALLEL_INSTANCE_GROUP
  THREAD

- **Parallel Execution**

  PARALLEL_ADAPTIVE_MULTI_USER
  PARALLEL_EXECUTION_MESSAGE_SIZE
  PARALLEL_MAX_SERVERS
  PARALLEL_MIN_DEGREE
  PARALLEL_MIN_PERCENT
  PARALLEL_MIN_SERVERS
  PARALLEL_THREADS_PER_CPU

- **PL/SQL**

  LONG_MODULE_ACTION
  PLSQL_V2_COMPATIBILITY
  REMOTE_DEPENDENCIES_MODE
  SESSION_EXIT_ON_PACKAGE_STATE_ERROR

- **PL/SQL Compiler**

  NLS_LENGTH_SEMANTICS
  PERMIT_92_WRAP_FORMAT
  PLSCOPE_SETTINGS
  PLSQL_CCFLAGS
  PLSQL_CODE_TYPE
  PLSQL_DEBUG
  PLSQL_OPTIMIZE_LEVEL

Copyright © 2002, 2026, Oracle and/or its affiliates.

PLSQL_WARNINGS

- **Redo Logs, Archiving, and Recovery**

  ADG_ACCOUNT_INFO_TRACKING
  ADG_REDIRECT_DML
  CONTROL_FILE_RECORD_KEEP_TIME
  DATA_GUARD_MAX_IO_TIME
  DATA_GUARD_MAX_LONGIO_TIME
  DATA_GUARD_SYNC_LATENCY
  DB_CREATE_ONLINE_LOG_DEST_*n*
  DB_RECOVERY_FILE_DEST
  DB_RECOVERY_FILE_DEST_SIZE
  DB_UNRECOVERABLE_SCN_TRACKING
  FAST_START_MTTR_TARGET
  LOG_ARCHIVE_CONFIG
  LOG_ARCHIVE_DEST_*n*
  LOG_ARCHIVE_DEST_STATE_*n*
  LOG_ARCHIVE_DUPLEX_DEST
  LOG_ARCHIVE_FORMAT
  LOG_ARCHIVE_MAX_PROCESSES
  LOG_ARCHIVE_MIN_SUCCEED_DEST
  LOG_ARCHIVE_TRACE
  LOG_BUFFER
  LOG_CHECKPOINT_INTERVAL
  LOG_CHECKPOINT_TIMEOUT
  LOG_CHECKPOINTS_TO_ALERT
  RECOVERY_PARALLELISM
  REDO_TRANSPORT_USER
  REMOTE_RECOVERY_FILE_DEST

- **Resource Manager**

  MAX_IDLE_BLOCKER_TIME
  MAX_IDLE_TIME
  PROCESSOR_GROUP_NAME
  RESOURCE_LIMIT
  RESOURCE_MANAGER_CPU_ALLOCATION
  RESOURCE_MANAGER_PLAN
  SGA_MIN_SIZE
  STANDBY_DB_PRESERVE_STATES

- **Security and Auditing**

  ALLOW_GLOBAL_DBLINKS
  ALLOW_GROUP_ACCESS_TO_SGA
  AUDIT_FILE_DEST
  AUDIT_SYS_OPERATIONS
  AUDIT_SYSLOG_LEVEL
  AUDIT_TRAIL
  BLOCKCHAIN_TABLE_MAX_NO_DROP
  COMMIT_LOGGING
  COMMIT_WAIT
  DBFIPS_140

Copyright © 2002, 2026, Oracle and/or its affiliates.

ENCRYPT_NEW_TABLESPACES
IDENTITY_PROVIDER_CONFIG
IDENTITY_PROVIDER_TYPE
LOB_SIGNATURE_ENABLE
OS_AUTHENT_PREFIX
OS_ROLES
OUTBOUND_DBLINK_PROTOCOLS
RDBMS_SERVER_DN
REMOTE_LOGIN_PASSWORDFILE
REMOTE_OS_AUTHENT
REMOTE_OS_ROLES
SEC_CASE_SENSITIVE_LOGON
SEC_MAX_FAILED_LOGIN_ATTEMPTS
SEC_PROTOCOL_ERROR_FURTHER_ACTION
SEC_PROTOCOL_ERROR_TRACE_ACTION
SEC_RETURN_SERVER_RELEASE_BANNER
SQL92_SECURITY
TABLESPACE_ENCRYPTION
TDE_CONFIGURATION
UNIFIED_AUDIT_COMMON_SYSTEMLOG
UNIFIED_AUDIT_SGA_QUEUE_SIZE
UNIFIED_AUDIT_SYSTEMLOG
WALLET_ROOT

- **Sessions and Processes**

  CPU_COUNT
  CPU_MIN_COUNT
  PROCESSES
  SESSIONS
  THREADED_EXECUTION
  USE_DEDICATED_BROKER

- **SGA Memory**

  DB_*n*K_CACHE_SIZE
  DB_CACHE_SIZE
  HI_SHARED_MEMORY_ADDRESS
  JAVA_POOL_SIZE
  LARGE_POOL_SIZE
  LOCK_SGA
  OLAP_PAGE_POOL_SIZE
  PRE_PAGE_SGA
  SGA_MAX_SIZE
  SGA_MIN_SIZE
  SGA_TARGET
  SHARED_MEMORY_ADDRESS
  SHARED_POOL_RESERVED_SIZE
  SHARED_POOL_SIZE
  STREAMS_POOL_SIZE
  USE_LARGE_PAGES

- **Shared Server Architecture**

Copyright © 2002, 2026, Oracle and/or its affiliates.

CIRCUITS
DISPATCHERS
MAX_DISPATCHERS
MAX_SHARED_SERVERS
SHARED_SERVER_SESSIONS
SHARED_SERVERS

- **Standby Database**

  ARCHIVE_LAG_TARGET
  DB_FILE_NAME_CONVERT
  DB_UNIQUE_NAME
  DG_BROKER_CONFIG_FILE*n*
  DG_BROKER_START
  ENABLE_IMC_WITH_MIRA
  FAL_CLIENT
  FAL_SERVER
  LOG_FILE_NAME_CONVERT
  STANDBY_FILE_MANAGEMENT
  STANDBY_PDB_SOURCE_FILE_DBLINK
  STANDBY_PDB_SOURCE_FILE_DIRECTORY

- **Temporary Sort Space**

  BITMAP_MERGE_AREA_SIZE
  CREATE_BITMAP_AREA_SIZE
  HASH_AREA_SIZE
  PGA_AGGREGATE_LIMIT
  PGA_AGGREGATE_TARGET
  SORT_AREA_RETAINED_SIZE
  SORT_AREA_SIZE
  WORKAREA_SIZE_POLICY

- **Transactions**

  COMMIT_LOGGING
  COMMIT_WAIT
  DB_LOST_WRITE_PROTECT
  DDL_LOCK_TIMEOUT
  DML_LOCKS
  FAST_START_PARALLEL_ROLLBACK
  GLOBAL_TXN_PROCESSES
  TRANSACTIONS

- **Undo Management**

  RESUMABLE_TIMEOUT
  ROLLBACK_SEGMENTS
  TEMP_UNDO_ENABLED
  TRANSACTIONS_PER_ROLLBACK_SEGMENT
  UNDO_MANAGEMENT
  UNDO_RETENTION
  UNDO_TABLESPACE

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.4.2 Modifiable Parameters

You can modify some initialization parameters using the `ALTER SESSION` or `ALTER SYSTEM` statements while an instance is running.

Use the following syntax to modify initialization parameters:

```
ALTER SESSION SET parameter_name = value
ALTER SYSTEM SET parameter_name = value [DEFERRED]
```

Whenever a parameter is modified using the `ALTER SYSTEM` statement, the Oracle Database records the statement that modifies the parameter in the alert log.

The `ALTER SESSION` statement changes the value of the specified parameter for the duration of the session that invokes the statement. The value of the parameter does not change for other sessions in the instance. The value of the following initialization parameters can be changed with `ALTER SESSION`:

- ALLOW_ROWID_COLUMN_TYPE
- APPROX_FOR_AGGREGATION
- APPROX_FOR_COUNT_DISTINCT
- APPROX_FOR_PERCENTILE
- ASM_DISKSTRING
- ASM_POWER_LIMIT
- COMMIT_LOGGING
- COMMIT_WAIT
- COMMIT_WRITE
- CONTAINER_DATA
- CONTAINERS_PARALLEL_DEGREE
- CREATE_STORED_OUTLINES
- CURSOR_BIND_CAPTURE_DESTINATION
- CURSOR_INVALIDATION
- CURSOR_SHARING
- DB_CREATE_FILE_DEST
- DB_CREATE_ONLINE_LOG_DEST_*n*
- DB_FILE_MULTIBLOCK_READ_COUNT
- DB_FILE_NAME_CONVERT
- DB_INDEX_COMPRESSION_INHERITANCE
- DB_SECUREFILE
- DB_UNRECOVERABLE_SCN_TRACKING
- DDL_LOCK_TIMEOUT
- DEFAULT_SHARING
- DEFERRED_SEGMENT_CREATION
- DST_UPGRADE_INSERT_CONV
- ENABLE_DDL_LOGGING
- GLOBAL_NAMES
- HASH_AREA_SIZE
- HEAT_MAP
- IGNORE_SESSION_SET_PARAM_ERRORS
- INMEMORY_CLAUSE_DEFAULT
- INMEMORY_QUERY
- JAVA_JIT_ENABLED

Copyright © 2002, 2026, Oracle and/or its affiliates.

LOG_ARCHIVE_DEST_*n*
LOG_ARCHIVE_DEST_STATE_*n*
LOG_ARCHIVE_MIN_SUCCEED_DEST
MAX_DUMP_FILE_SIZE
MULTISHARD_QUERY_DATA_CONSISTENCY
NLS_CALENDAR
NLS_COMP
NLS_CURRENCY
NLS_DATE_FORMAT
NLS_DATE_LANGUAGE
NLS_DUAL_CURRENCY
NLS_ISO_CURRENCY
NLS_LANGUAGE
NLS_LENGTH_SEMANTICS
NLS_NCHAR_CONV_EXCP
NLS_NUMERIC_CHARACTERS
NLS_SORT
NLS_TERRITORY
NLS_TIMESTAMP_FORMAT
NLS_TIMESTAMP_TZ_FORMAT
OBJECT_CACHE_MAX_SIZE_PERCENT
OBJECT_CACHE_OPTIMAL_SIZE
OLAP_PAGE_POOL_SIZE
OPTIMIZER_ADAPTIVE_PLANS
OPTIMIZER_ADAPTIVE_REPORTING_ONLY
OPTIMIZER_ADAPTIVE_STATISTICS
OPTIMIZER_CAPTURE_SQL_PLAN_BASELINES
OPTIMIZER_CAPTURE_SQL_QUARANTINE
OPTIMIZER_DYNAMIC_SAMPLING
OPTIMIZER_FEATURES_ENABLE
OPTIMIZER_IGNORE_HINTS
OPTIMIZER_IGNORE_PARALLEL_HINTS
OPTIMIZER_INDEX_CACHING
OPTIMIZER_INDEX_COST_ADJ
OPTIMIZER_INMEMORY_AWARE
OPTIMIZER_MODE
OPTIMIZER_REAL_TIME_STATISTICS
OPTIMIZER_SESSION_TYPE
OPTIMIZER_USE_INVISIBLE_INDEXES
OPTIMIZER_USE_PENDING_STATISTICS
OPTIMIZER_USE_SQL_PLAN_BASELINES
OPTIMIZER_USE_SQL_QUARANTINE
PARALLEL_DEGREE_LIMIT
PARALLEL_DEGREE_POLICY
PARALLEL_FORCE_LOCAL
PARALLEL_INSTANCE_GROUP
PARALLEL_MIN_DEGREE
PARALLEL_MIN_PERCENT
PARALLEL_MIN_TIME_THRESHOLD
PDB_FILE_NAME_CONVERT

Copyright © 2002, 2026, Oracle and/or its affiliates.

- PDB_LOCKDOWN
- PLSCOPE_SETTINGS
- PLSQL_CCFLAGS
- PLSQL_CODE_TYPE
- PLSQL_DEBUG
- PLSQL_OPTIMIZE_LEVEL
- PLSQL_V2_COMPATIBILITY
- PLSQL_WARNINGS
- QUERY_REWRITE_ENABLED
- QUERY_REWRITE_INTEGRITY
- RECYCLEBIN
- REMOTE_DEPENDENCIES_MODE
- RESULT_CACHE_MODE
- RESULT_CACHE_REMOTE_EXPIRATION
- RESUMABLE_TIMEOUT
- SESSION_CACHED_CURSORS
- SESSION_EXIT_ON_PACKAGE_STATE_ERROR
- SKIP_UNUSABLE_INDEXES
- SMTP_OUT_SERVER
- SORT_AREA_RETAINED_SIZE
- SORT_AREA_SIZE
- SPATIAL_VECTOR_ACCELERATION
- SQL_TRACE
- SQLTUNE_CATEGORY
- STAR_TRANSFORMATION_ENABLED
- STATISTICS_LEVEL
- TEMP_UNDO_ENABLED
- TIMED_OS_STATISTICS
- TIMED_STATISTICS
- TRACEFILE_IDENTIFIER
- WORKAREA_SIZE_POLICY
- XML_DB_EVENTS

The `ALTER SYSTEM` statement without the `DEFERRED` keyword modifies the global value of the parameter for all sessions in the instance, for the duration of the instance (until the database is shut down). The value of the following initialization parameters can be changed with `ALTER SYSTEM`:

- ADG_REDIRECT_DML
- ALLOW_GLOBAL_DBLINKS
- ALLOW_ROWID_COLUMN_TYPE
- APPROX_FOR_AGGREGATION
- APPROX_FOR_COUNT_DISTINCT
- APPROX_FOR_PERCENTILE
- AQ_TM_PROCESSES
- ARCHIVE_LAG_TARGET
- ASM_DISKGROUPS
- ASM_DISKSTRING
- ASM_IO_PROCESSES
- ASM_POWER_LIMIT
- ASM_PREFERRED_READ_FAILURE_GROUPS

Copyright © 2002, 2026, Oracle and/or its affiliates.

AUTOTASK_MAX_ACTIVE_PDBS
AWR_PDB_AUTOFLUSH_ENABLED
AWR_SNAPSHOT_TIME_OFFSET
BACKGROUND_CORE_DUMP
BACKGROUND_DUMP_DEST
BLOCKCHAIN_TABLE_MAX_NO_DROP
CIRCUITS
COMMIT_LOGGING
COMMIT_WAIT
COMMIT_WRITE
CONNECTION_BROKERS
CONTAINERS_PARALLEL_DEGREE
CONTROL_FILE_RECORD_KEEP_TIME
CONTROL_MANAGEMENT_PACK_ACCESS
CORE_DUMP_DEST
CPU_COUNT
CPU_MIN_COUNT
CREATE_STORED_OUTLINES
CURSOR_BIND_CAPTURE_DESTINATION
CURSOR_INVALIDATION
CURSOR_SHARING
DATA_GUARD_MAX_IO_TIME
DATA_GUARD_MAX_LONGIO_TIME
DATA_GUARD_SYNC_LATENCY
DATA_TRANSFER_CACHE_SIZE
DB_*n*K_CACHE_SIZE
DB_BIG_TABLE_CACHE_PERCENT_TARGET
DB_BLOCK_CHECKING
DB_BLOCK_CHECKSUM
DB_CACHE_ADVICE
DB_CACHE_SIZE
DB_CREATE_FILE_DEST
DB_CREATE_ONLINE_LOG_DEST_*n*
DB_FILE_MULTIBLOCK_READ_COUNT
DB_FLASH_CACHE_FILE
DB_FLASH_CACHE_SIZE
DB_FLASHBACK_RETENTION_TARGET
DB_INDEX_COMPRESSION_INHERITANCE
DB_KEEP_CACHE_SIZE
DB_LOST_WRITE_PROTECT
DB_RECOVERY_FILE_DEST
DB_RECOVERY_FILE_DEST_SIZE
DB_RECYCLE_CACHE_SIZE
DB_SECUREFILE
DB_UNRECOVERABLE_SCN_TRACKING
DDL_LOCK_TIMEOUT
DEFAULT_SHARING
DEFERRED_SEGMENT_CREATION
DG_BROKER_CONFIG_FILE*n*
DG_BROKER_START

Copyright © 2002, 2026, Oracle and/or its affiliates.

DIAGNOSTIC_DEST
DISPATCHERS
DRCP_DEDICATED_OPT
DST_UPGRADE_INSERT_CONV
ENABLE_AUTOMATIC_MAINTENANCE_PDB
ENABLE_DDL_LOGGING
ENABLE_GOLDENGATE_REPLICATION
ENABLE_IMC_WITH_MIRA
ENABLED_PDBS_ON_STANDBY
ENCRYPT_NEW_TABLESPACES
FAL_CLIENT
FAL_SERVER
FAST_START_MTTR_TARGET
FAST_START_PARALLEL_ROLLBACK
FILE_MAPPING
FIXED_DATE
FORWARD_LISTENER
GLOBAL_NAMES
GLOBAL_TXN_PROCESSES
HEAT_MAP
HS_AUTOREGISTER
IDENTITY_PROVIDER_CONFIG
IDENTITY_PROVIDER_TYPE
IGNORE_SESSION_SET_PARAM_ERRORS
INMEMORY_ADG_ENABLED
INMEMORY_AUTOMATIC_LEVEL
INMEMORY_CLAUSE_DEFAULT
INMEMORY_EXPRESSIONS_USAGE
INMEMORY_FORCE
INMEMORY_MAX_POPULATE_SERVERS
INMEMORY_OPTIMIZED_ARITHMETIC
INMEMORY_QUERY
INMEMORY_SIZE
INMEMORY_TRICKLE_REPOPULATE_SERVERS_PERCENT
INMEMORY_VIRTUAL_COLUMNS
JAVA_JIT_ENABLED
JAVA_POOL_SIZE
JOB_QUEUE_PROCESSES
LARGE_POOL_SIZE
LDAP_DIRECTORY_ACCESS
LICENSE_MAX_SESSIONS
LICENSE_MAX_USERS
LICENSE_SESSIONS_WARNING
LISTENER_NETWORKS
LOB_SIGNATURE_ENABLE
LOCAL_LISTENER
LOG_ARCHIVE_CONFIG
LOG_ARCHIVE_DEST
LOG_ARCHIVE_DEST_*n*
LOG_ARCHIVE_DEST_STATE_*n*

Copyright © 2002, 2026, Oracle and/or its affiliates.

LOG_ARCHIVE_DUPLEX_DEST
LOG_ARCHIVE_MAX_PROCESSES
LOG_ARCHIVE_MIN_SUCCEED_DEST
LOG_ARCHIVE_TRACE
LOG_CHECKPOINT_INTERVAL
LOG_CHECKPOINT_TIMEOUT
LOG_CHECKPOINTS_TO_ALERT
LONG_MODULE_ACTION
MAIN_WORKLOAD_TYPE
MAX_AUTH_SERVERS
MAX_DATAPUMP_JOBS_PER_PDB
MAX_DATAPUMP_PARALLEL_PER_JOB
MAX_DISPATCHERS
MAX_DUMP_FILE_SIZE
MAX_IDLE_BLOCKER_TIME
MAX_IDLE_TIME
MAX_IOPS
MAX_MBPS
MAX_PDBS
MAX_SHARED_SERVERS
MEMOPTIMIZE_POOL_SIZE
MEMORY_TARGET
MIN_AUTH_SERVERS
MULTISHARD_QUERY_DATA_CONSISTENCY
NLS_LENGTH_SEMANTICS
NLS_NCHAR_CONV_EXCP
OFS_THREADS
OGG_ONLINE_DDL
OPEN_CURSORS
OPTIMIZER_ADAPTIVE_PLANS
OPTIMIZER_ADAPTIVE_REPORTING_ONLY
OPTIMIZER_ADAPTIVE_STATISTICS
OPTIMIZER_CAPTURE_SQL_PLAN_BASELINES
OPTIMIZER_CAPTURE_SQL_QUARANTINE
OPTIMIZER_DYNAMIC_SAMPLING
OPTIMIZER_FEATURES_ENABLE
OPTIMIZER_IGNORE_HINTS
OPTIMIZER_IGNORE_PARALLEL_HINTS
OPTIMIZER_INDEX_CACHING
OPTIMIZER_INDEX_COST_ADJ
OPTIMIZER_INMEMORY_AWARE
OPTIMIZER_MODE
OPTIMIZER_REAL_TIME_STATISTICS
OPTIMIZER_SECURE_VIEW_MERGING
OPTIMIZER_USE_INVISIBLE_INDEXES
OPTIMIZER_USE_PENDING_STATISTICS
OPTIMIZER_USE_SQL_PLAN_BASELINES
OPTIMIZER_USE_SQL_QUARANTINE
OUTBOUND_DBLINK_PROTOCOLS
PARALLEL_ADAPTIVE_MULTI_USER

Copyright © 2002, 2026, Oracle and/or its affiliates.

PARALLEL_DEGREE_LIMIT
PARALLEL_DEGREE_POLICY
PARALLEL_FORCE_LOCAL
PARALLEL_INSTANCE_GROUP
PARALLEL_MAX_SERVERS
PARALLEL_MIN_DEGREE
PARALLEL_MIN_SERVERS
PARALLEL_MIN_TIME_THRESHOLD
PARALLEL_SERVERS_TARGET
PARALLEL_THREADS_PER_CPU
PDB_FILE_NAME_CONVERT
PDB_LOCKDOWN
PGA_AGGREGATE_LIMIT
PGA_AGGREGATE_TARGET
PLSCOPE_SETTINGS
PLSQL_CCFLAGS
PLSQL_CODE_TYPE
PLSQL_DEBUG
PLSQL_OPTIMIZE_LEVEL
PLSQL_V2_COMPATIBILITY
PLSQL_WARNINGS
QUERY_REWRITE_ENABLED
QUERY_REWRITE_INTEGRITY
RECOVERY_PARALLELISM
REDO_TRANSPORT_USER
REMOTE_DEPENDENCIES_MODE
REMOTE_LISTENER
REMOTE_RECOVERY_FILE_DEST
RESOURCE_LIMIT
RESOURCE_MANAGER_CPU_ALLOCATION
RESOURCE_MANAGER_PLAN
RESULT_CACHE_MAX_RESULT
RESULT_CACHE_MAX_SIZE
RESULT_CACHE_MODE
RESULT_CACHE_REMOTE_EXPIRATION
RESUMABLE_TIMEOUT
SEC_CASE_SENSITIVE_LOGON
SEC_PROTOCOL_ERROR_FURTHER_ACTION
SEC_PROTOCOL_ERROR_TRACE_ACTION
SERVICE_NAMES
SESSION_EXIT_ON_PACKAGE_STATE_ERROR
SGA_MIN_SIZE
SGA_TARGET
SHADOW_CORE_DUMP
SHARED_POOL_SIZE
SHARED_SERVER_SESSIONS
SHARED_SERVERS
SHRD_DUPL_TABLE_REFRESH_RATE
SKIP_UNUSABLE_INDEXES
SMTP_OUT_SERVER

Copyright © 2002, 2026, Oracle and/or its affiliates.

SPATIAL_VECTOR_ACCELERATION
SPFILE
SQL_TRACE
SQLTUNE_CATEGORY
STANDBY_FILE_MANAGEMENT
STANDBY_PDB_SOURCE_FILE_DBLINK
STANDBY_PDB_SOURCE_FILE_DIRECTORY
STAR_TRANSFORMATION_ENABLED
STATISTICS_LEVEL
STREAMS_POOL_SIZE
TDE_CONFIGURATION
TEMP_UNDO_ENABLED
THREAD
TIMED_OS_STATISTICS
TIMED_STATISTICS
TRACE_ENABLED
UNDO_RETENTION
UNDO_TABLESPACE
UNIFORM_LOG_TIMESTAMP_FORMAT
USE_DEDICATED_BROKER
USER_DUMP_DEST
WORKAREA_SIZE_POLICY
XML_DB_EVENTS

The `ALTER SYSTEM ... DEFERRED` statement does not modify the global value of the parameter for existing sessions, but the value will be modified for future sessions that connect to the database. The value of the following initialization parameters can be changed with `ALTER SYSTEM ... DEFERRED`:

AUDIT_FILE_DEST
BACKUP_TAPE_IO_SLAVES
CLIENT_STATISTICS_LEVEL
OBJECT_CACHE_MAX_SIZE_PERCENT
OBJECT_CACHE_OPTIMAL_SIZE
OLAP_PAGE_POOL_SIZE
PRIVATE_TEMP_TABLE_PREFIX
RECYCLEBIN
SESSION_CACHED_CURSORS
SORT_AREA_RETAINED_SIZE
SORT_AREA_SIZE

1.4.3 Displaying Current Parameter Values

You can use the SQL*Plus `SHOW PARAMETERS` statement to see the current settings for initialization parameters.

For example:

```
SQL> SHOW PARAMETERS
```

This statement displays all parameters in alphabetical order, along with their current values.

Enter the following text string to display all parameters having `BLOCK` in their names:

Copyright © 2002, 2026, Oracle and/or its affiliates.

```
SQL> SHOW PARAMETERS BLOCK
```

You can use the SPOOL command to write the output to a file.

1.4.4 Parameters You Should Not Specify in the Parameter File

You should not specify these types of parameters in your parameter files:

- Parameters that you never alter unless instructed to do so by Oracle to resolve a problem
- Derived parameters, which normally do not need altering because their values are calculated automatically by the Oracle database server

1.4.5 When Parameters Are Set Incorrectly

When a parameter is set incorrectly, the effect can be different for different parameters, or based on how low or high the parameter is set.

Some parameters have a minimum setting below which an Oracle instance will not start. For other parameters, setting the value too low or too high may cause Oracle to perform badly, but it will still run. Also, Oracle may convert some values outside the acceptable range to usable levels.

If a parameter value is too low or too high, or you have reached the maximum for some resource, then Oracle returns an error. Frequently, you can wait a short while and retry the operation when the system is not as busy. If a message occurs repeatedly, then you should shut down the instance, adjust the relevant parameter, and restart the instance.

1.5 Reading the Parameter Descriptions

This section describes the properties that are documented in the initialization parameter descriptions.

PARAMETER_NAME

Property	Description
Parameter type	Specifies the type of the parameter: • A Boolean parameter accepts either `true` or `false` as its value. • A string parameter accepts any sequence of characters as its value, subject to the syntax for the parameter. • An integer parameter accepts a 4-byte value that can range from 0 to 2^{32} - 1. • A parameter file parameter accepts an initialization parameter file specification as its value. • A **big integer** parameter accepts an 8-byte value that can range from 0 to 2^{64} - 1. You specify a value for a big integer as an integer together with an optional modifier such as K, M, or G, which respectively denotes kilobytes, megabytes, or gigabytes. For example, 1000, 100 KB, 50 MB and 2 GB are valid specifications for big integers.
Syntax	For string and big integer parameters, specifies the valid syntax for specifying the parameter.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	Specifies the value that the parameter assumes if a value is not explicitly specified. For most initialization parameters, the value that is documented for the **Default value** property is obtained by querying the `DEFAULT_VALUE` column in the `V$PARAMETER` view for the parameter.
Modifiable	Specifies whether the parameter can be changed for the current session (by an `ALTER SESSION` statement) or for all sessions in the current instance (by an `ALTER SYSTEM` statement): • `ALTER SESSION` overrides the instance-wide setting of the parameter for the current session only. You can restore the instance-wide setting for that session only by issuing another `ALTER SESSION` statement. • `ALTER SYSTEM` can be used to change the value in the server parameter file (SPFILE) of any initialization parameter. Such a change takes effect only in subsequent instances. The parameter descriptions indicate only those parameters that can be modified for the current instance.
Modifiable in a PDB	Specifies whether the parameter can be modified in a PDB (Yes) or not (No) For most initialization parameters, the value that is documented for the **Modifiable in a PDB** property is obtained by querying the `ISPDB_MODIFIABLE` column in the `V$PARAMETER` view for the parameter.
Range of values	Specifies the valid range of values that this parameter can assume, shown as a minimum and maximum value. Not applicable to all parameters.
Basic	Specifies whether the parameter is a basic initialization parameter or not
Oracle RAC	Specifies how the values for this parameter must be specified for multiple instances in an Oracle Real Application Clusters environment. Not applicable to all parameters.

For each parameter, paragraphs following these details further describe the parameter and the effects of different settings.

See Also

"V$PARAMETER"

1.6 Initialization Parameter Descriptions

The remainder of this chapter describes the initialization parameters in alphabetical order.

Initialization parameter values apply to the entire database, not to an individual user, unless otherwise specified.

Note

Parameters that have become obsolete are not documented.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database Upgrade Guide* for information about obsolete parameters
- Your system release bulletins or other operating system-specific Oracle documentation

1.7 ACTIVE_INSTANCE_COUNT

`ACTIVE_INSTANCE_COUNT` enables you to designate one instance in a two-instance cluster as the primary instance and the other instance as the secondary instance. This parameter has no functionality in a cluster with more than two instances.

Property	Description
Parameter type	Integer
Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	No
Range of values	1 or >= the number of instances in the cluster. (Values other than 1 have no effect on the active or standby status of any instances.)
Basic	No
Oracle RAC	You must set this parameter for every instance, and multiple instances must have the same value.

Note

The `ACTIVE_INSTANCE_COUNT` parameter is deprecated. It is retained for backward compatibility only.

When you set this parameter to 1, the first instance you start up becomes the primary instance and accepts client connections. The second instance starts up as a secondary instance and can accept client connections only if the first instance fails. In such an event, the secondary instance becomes the primary instance.

When the failed instance can once again be started up, it starts up as the secondary instance, and will not accept client connections unless the current primary instance fails.

Note

This parameter functions only in a cluster with only two instances.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.8 ADG_ACCOUNT_INFO_TRACKING

ADG_ACCOUNT_INFO_TRACKING controls login attempts of users on Oracle Active Data Guard standby databases. It extends the control of user account security information.

Property	Description
Parameter type	String
Syntax	ADG_ACCOUNT_INFO_TRACKING = { LOCAL | GLOBAL }
Default value	LOCAL
Modifiable	No
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The same value must be used on all instances.

Setting ADG_ACCOUNT_INFO_TRACKING to LOCAL (the default value) continues to enforce the default behavior, by maintaining a local copy of users account information in the standby's in-memory view. Login failures are only tracked locally on a per database basis, and login is denied when the failure maximum is reached.

Setting the parameter to GLOBAL triggers a more secure behavior, by maintaining a single global copy of users account information across all Data Guard databases (primary and standby). Login failures across all databases in the Data Guard configuration count towards the maximum count and logins anywhere will be denied when the count is reached. This setting improves security against login attacks across a production database and all Active Data Guard standby databases.

1.9 ADG_REDIRECT_DML

Use ADG_REDIRECT_DML to enable or disable automatic redirection of DML operations from a standby to the primary in Oracle Active Data Guard environments.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	ALTER SYSTEM Can be modified at the session level with a special ALTER SESSION command. See details below.
Modifiable in a PDB	No
Range of values	true | false
Basic	No
Oracle RAC	Different instances can use different values.

Automatic redirection of DML operations to the primary can be configured at the system level or the session level. The session level setting overrides the system level setting.

- To enable automatic redirection of DML operations to the primary at the system level, set ADG_REDIRECT_DML to true.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- To disable automatic redirection of DML operations to the primary at the system level, set `ADG_REDIRECT_DML` to `false`.
- To enable automatic redirection of DML operations from a standby to the primary for the current session, issue the following SQL statement:

```
ALTER SESSION ENABLE ADG_REDIRECT_DML;
```

- To disable automatic redirection of DML operations from a standby to the primary for the current session, issue the following SQL statement:

```
ALTER SESSION DISABLE ADG_REDIRECT_DML;
```

See Also

Oracle Data Guard Concepts and Administration for more information about automatic redirection of DML operations

Note

This parameter is available starting with Oracle Database 19c.

1.10 ALLOW_GLOBAL_DBLINKS

`ALLOW_GLOBAL_DBLINKS` specifies whether LDAP lookup for database links is allowed for the database.

Property	Description
Parameter type	Boolean
Syntax	`ALLOW_GLOBAL_DBLINKS = { FALSE | TRUE }`
Default value	`FALSE`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Basic	No
Oracle RAC	The same value must be used on all instances.

The following values can be set:

- `FALSE`: LDAP lookup for a database link's definition is not allowed for the database.
- `TRUE`: LDAP lookup for a database link's definition is allowed for the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.11 ALLOW_GROUP_ACCESS_TO_SGA

ALLOW_GROUP_ACCESS_TO_SGA controls group access to shared memory on UNIX platforms.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	No
Modifiable in a PDB	No
Range of values	true | false
Basic	No
Oracle RAC	The same value must be used on all instances.

The default value is false, which means that database shared memory is created with owner access only. In Oracle Database releases prior to Oracle Database 12c Release 2 (12.2.0.1), database shared memory was created with owner and group access.

When this parameter is set to true, database shared memory is created with owner and group access. This behavior grants permissions to DBAs to manage shared memory outside the database, but also allows DBAs to read and write to shared memory, which may not be desirable for certain installations.

1.12 ALLOW_ROWID_COLUMN_TYPE

ALLOW_ROWID_COLUMN_TYPE determines whether table columns of data type ROWID are allowed to be created.

> **Note**
>
> This parameter is valid only for Oracle Autonomous Database on dedicated Exadata infrastructure. On other types of Oracle databases, the value of this parameter is ignored and ROWID columns are always allowed to be created.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	true | false
Basic	No
Oracle RAC	Multiple instances should use the same value

Values:

- false

Copyright © 2002, 2026, Oracle and/or its affiliates.

Columns of data type ROWID are not allowed.

You cannot specify ROWID columns when creating tables, nor can you add ROWID columns to existing tables. If the value of this parameter is changed from true to false at a time when ROWID columns exist in the database, then those columns are allowed to remain. However, Oracle recommends that you leave this parameter set to its default value of false and avoid introducing ROWID columns into Oracle Autonomous Database on dedicated Exadata infrastructure.

- true

  Columns of data type ROWID are allowed.

  If you choose this setting, be aware that ROWID columns are incompatible with rolling upgrade operations and other internal operations that physically move table rows. At a minimum, database activities involving ROWID columns should be suspended during upgrades. Applications using ROWID columns should introduce correctness validation to mitigate against logical errors in the application if a row relocates.

> **Note**
>
> This parameter is available starting with Oracle Database 19c.

1.13 APPROX_FOR_AGGREGATION

APPROX_FOR_AGGREGATION replaces exact query processing for aggregation queries with approximate query processing.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	true | false
Basic	No
Oracle RAC	Different instances can use different values.

Data analysis applications heavily use aggregate function and analytic function queries. Aggregation functions and analytic functions require sorting of large volumes of data, and exact query answering requires lots of memory, and can be time consuming. With approximate query processing, the results of aggregate function and analytic function queries are returned much faster than with exact query processing. Approximate query processing is useful for situations where a tolerable amount of error is acceptable in order to obtain faster query results.

The values that can be set are:

- false:

  Approximate query processing is not used for aggregation queries and analytic queries.

- true: Approximate query processing is used for aggregation queries and analytic queries.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "APPROX_FOR_COUNT_DISTINCT"
- "APPROX_FOR_PERCENTILE"

1.14 APPROX_FOR_COUNT_DISTINCT

`APPROX_FOR_COUNT_DISTINCT` automatically replaces `COUNT (DISTINCT expr)` queries with `APPROX_COUNT_DISTINCT` queries.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SESSION, ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No
Oracle RAC	Different instances can use different values.

Query results for `APPROX_COUNT_DISTINCT` queries are returned faster than the equivalent `COUNT (DISTINCT expr)` queries. `APPROX_COUNT_DISTINCT` queries are useful for situations where a tolerable amount of error is acceptable in order to obtain faster query results than with a `COUNT (DISTINCT expr)` query.

See Also

- "APPROX_FOR_AGGREGATION"
- "APPROX_FOR_PERCENTILE"

1.15 APPROX_FOR_PERCENTILE

`APPROX_FOR_PERCENTILE` converts exact percentile functions to their approximate percentile function counterparts.

Property	Description
Parameter type	String
Syntax	`APPROX_FOR_PERCENTILE = { NONE | PERCENTILE_CONT | PERCENTILE_CONT DETERMINISTIC | PERCENTILE_DISC | PERCENTILE_DISC DETERMINISTIC | ALL | ALL DETERMINISTIC }`
Default value	`none`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Oracle RAC	Different instances can use different values.

Approximate percentile function queries are faster than their exact percentile function query counterparts, so they can be useful in situations where a tolerable amount of error is acceptable in order to obtain faster query results.

The values that can be set are:

- `NONE`:

  No queries are converted. This is the default value.

- `PERCENTILE_CONT`:

  Converts `PERCENTILE_CONT` queries to `APPROX_PERCENTILE` queries.

- `PERCENTILE_CONT DETERMINISTIC`:

  Converts `PERCENTILE_CONT` queries to `APPROX_PERCENTILE DETERMINSTIC` queries.

- `PERCENTILE_DISC`:

  Converts `PERCENTILE_DISC` queries to `APPROX_PERCENTILE` queries.

- `PERCENTILE DISC DETERMINISTIC`:

  Converts `PERCENTILE_DISC` queries to `APPROX_PERCENTILE DETERMINISTIC` queries.

- `ALL`:

  Converts both `PERCENTILE_CONT` queries and `PERCENTILE_DISC` queries to `APPROX_PERCENTILE` queries.

- `ALL DETERMINISTIC`:

  Converts both `PERCENTILE_CONT` and `PERCENTILE_DISC` queries to `APPROX_PERCENTILE DETERMINISTIC` queries.

See Also

- "APPROX_FOR_AGGREGATION"
- "APPROX_FOR_COUNT_DISTINCT"
- *Oracle Database SQL Language Reference* for information about `APPROX_MEDIAN` aggregate functions
- *Oracle Database SQL Language Reference* for information about `APPROX_PERCENTILE` aggregate functions

1.16 AQ_TM_PROCESSES

`AQ_TM_PROCESSES` controls time monitoring on queue messages and controls processing of messages with delay and expiration properties specified.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	Integer
Default value	1
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to 40
Basic	No

You do not need to specify a value for this parameter because Oracle Database automatically determines the number of processes and autotunes them, as necessary. Therefore, Oracle highly recommends that you leave the AQ_TM_PROCESSES parameter unspecified and let the system autotune.

The default value for AQ_TM_PROCESSES is used if the client does not explicitly set a value for the parameter in the init.ora file or using the ALTER SYSTEM statement.

> **Note**
>
> If you want to disable the Queue Monitor Coordinator, then you must set AQ_TM_PROCESSES to 0 in your parameter file. Oracle strongly recommends that you do NOT set AQ_TM_PROCESSES to 0.

> **See Also**
>
> *Oracle Database Advanced Queuing User's Guide* for more information about this parameter

1.17 ARCHIVE_LAG_TARGET

ARCHIVE_LAG_TARGET forces a log switch after the specified amount of time elapses.

Property	Description
Parameter type	Integer
Default value	0 (disabled)
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	0 or any integer in [60, 7200]
Basic	No
Oracle RAC	Multiple instances should use the same value

A 0 value disables the time-based thread advance feature; otherwise, the value represents the number of seconds. Values larger than 7200 seconds are not of much use in maintaining a reasonable lag in the standby database. The typical, or recommended value is 1800 (30 minutes). Extremely low values can result in frequent log switches, which could degrade

Copyright © 2002, 2026, Oracle and/or its affiliates.

performance; such values can also make the archiver process too busy to archive the continuously generated logs.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information about setting this parameter

1.18 ASM_DISKGROUPS

`ASM_DISKGROUPS` specifies a list of disk group names that an Oracle ASM instance mounts at startup when the `ALTER DISKGROUP ALL MOUNT` statement is issued.

Property	Description
Parameter type	String
Syntax	`ASM_DISKGROUPS = diskgroup [, diskgroup ] ...`
Default value	There is no default value.
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	Comma-separated list of strings, up to 30 characters
Basic	No
Oracle RAC	Multiple instances can have different values.

> **Note**
>
> This parameter may only be specified in an Oracle Automatic Storage Management (Oracle ASM) instance.

The Oracle ASM instance startup process executes `ALTER DISKGROUP ALL MOUNT` unless the `NOMOUNT` startup option is specified.

The `ASM_DISKGROUPS` parameter is dynamic. If you are using a server parameter file (SPFILE), then you do not have to manually alter the value of `ASM_DISKGROUPS` in most situations. Oracle ASM automatically adds a disk group to the parameter when the disk group is successfully created or mounted. Oracle ASM also automatically removes a disk group from the parameter when the disk group is dropped. However, the SPFILE is not updated on a manual dismount.

Issuing the `ALTER DISKGROUP...ALL MOUNT` or `ALTER DISKGROUP...ALL DISMOUNT` command does not affect the value of this parameter.

Supporting Up to 511 Disk Groups for an Oracle ASM Instance

In Oracle Database 12*c* Release 1 or later, Oracle ASM configurations support up to 511 disk groups. Oracle ASM configurations with Oracle Database releases before 12*c* Release 1 can only support up to 63 disk groups.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Automatic Storage Management Administrator's Guide* for more information about and examples of using this parameter

1.19 ASM_DISKSTRING

`ASM_DISKSTRING` specifies an operating system-dependent value used by Oracle ASM to limit the set of disks considered for discovery.

Property	Description
Parameter type	String
Syntax	`ASM_DISKSTRING = discovery_string [, discovery_string ] ...`
Default value	The null string; Oracle Automatic Storage Management discovery finds all disks in an operating system-specific location to which the Oracle Automatic Storage Management instance has read/write access.
Modifiable	`ALTER SESSION, ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Multiple instances can have different values. Different nodes might see the same disks under different names; however, each instance must be able to use its `ASM_DISKSTRING` to discover the same physical media as the other nodes in the cluster.

> **Note**
>
> This parameter may only be specified in an Oracle Automatic Storage Management (Oracle ASM) instance.

When a new disk is added to a disk group, each Oracle ASM instance that has the disk group mounted must be able to discover the new disk using the value of `ASM_DISKSTRING`.

In most cases, the default value will be sufficient. Using a more restrictive value may reduce the time required for Oracle ASM to perform discovery, and thus improve disk group mount time or the time for adding a disk to a disk group. A "?" at the beginning of the string gets expanded to the Oracle home directory. Depending on the operating system, wildcard characters can be used. It may be necessary to dynamically change `ASM_DISKSTRING` before adding a disk so that the new disk will be discovered.

An attempt to dynamically modify `ASM_DISKSTRING` will be rejected and the old value retained if the new value cannot be used to discover a disk that is in a disk group that is already mounted.

> **See Also**
>
> *Oracle Automatic Storage Management Administrator's Guide* for more information and examples of using this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.20 ASM_IO_PROCESSES

ASM_IO_PROCESSES specifies the number of I/O worker processes to be started in an Oracle IOServer instance.

Property	Description
Parameter type	Integer
Default value	20
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	1 – 32
Basic	No
Oracle RAC	Multiple instances can have different values.

This parameter is applicable only in an Oracle IOServer instance, which runs out of an Oracle Grid Infrastructure home.

The default value should work in most cases. However, under heavy I/O load, there may be some delays associated with posting processes out of I/O waits. In this case, a slightly higher value than the default may be appropriate.

1.21 ASM_POWER_LIMIT

ASM_POWER_LIMIT specifies the maximum power on an Oracle ASM instance for disk rebalancing.

Property	Description
Parameter type	Integer
Default value	1
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	No
Range of values	0 to 11[1]
Basic	No
Oracle RAC	Multiple instances can have different values.

[1] Beginning with Oracle Database 11*g* Release 2 (11.2.0.2), if the COMPATIBLE.ASM disk group attribute is set to 11.2.0.2 or higher, then the range of values is 0 to 1024.

Note

- This parameter may only be specified in an Oracle Automatic Storage Management (Oracle ASM) instance.
- In an Oracle Exadata or Oracle Data Appliance environment, you cannot set this parameter to 0.

Copyright © 2002, 2026, Oracle and/or its affiliates.

The higher the limit, the faster rebalancing will complete. Lower values will take longer, but consume fewer processing and I/O resources.

If the POWER clause of a rebalance operation is not specified, then the default power will be the value of ASM_POWER_LIMIT.

See Also

Oracle Automatic Storage Management Administrator's Guide for more information about using this parameter

1.22 ASM_PREFERRED_READ_FAILURE_GROUPS

ASM_PREFERRED_READ_FAILURE_GROUPS specifies the failure groups that contain preferred read disks. Preferred disks are instance specific.

Property	Description
Parameter type	String
Syntax	ASM_PREFERRED_READ_FAILURE_GROUPS = '<diskgroup_name>.<failure_group_name>, ...'
Default value	NULL
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No
Oracle RAC	The value is different on different nodes

Note

The ASM_PREFERRED_READ_FAILURE_GROUPS initialization parameter is deprecated in Oracle Database 12*c* Release 2 (12.2.0.1) and may be desupported in a future release. It is replaced by the PREFERRED_READ.ENABLED disk group attribute.

See *Oracle Automatic Storage Management Administrator's Guide* for more information about the PREFERRED_READ.ENABLED disk group attribute.

To reset the parameter value to null, issue the following statement, which updates the server parameter file (SP file), then reboot Oracle ASM so that the change takes effect:

```
alter system reset asm_preferred_read_failure_groups;
```

See Also

Oracle Automatic Storage Management Administrator's Guide for more information about this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.23 AUDIT_FILE_DEST

`AUDIT_FILE_DEST` specifies the operating system directory into which the audit trail is written when the `AUDIT_TRAIL` initialization parameter is set to **os**, **xml**, or **xml,extended**.

> **Note**
>
> In an Oracle database that has migrated to unified auditing, the setting of this parameter has no effect.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Property	Description
Parameter type	String
Syntax	`AUDIT_FILE_DEST = '`*`directory`*`'`
Default value	The first default value is: *`ORACLE_BASE`*`/admin/`*`ORACLE_SID`*`/adump` The second default value, which is used if the first default value does not exist or is unusable, is: *`ORACLE_HOME`*`/rdbms/audit` Both of these default values are for UNIX systems. Other platforms may have different defaults. In a multitenant container database (CDB), both of these default values will be appended with the GUID of the pluggable database (PDB) to store audit records that belong to the PDB. For example, if the PDB's GUID is `03E1F908EE04252CE053B280E80AAAA3`, the first default directory will be: *`ORACLE_BASE`*`/admin/`*`ORACLE_SID`*`/adump/` `03E1F908EE04252CE053B280E80AAAA3` You can use the `V$CONTAINERS` view to query a PDB's GUID.
Modifiable	`ALTER SYSTEM ... DEFERRED`
Modifiable in a PDB	No
Basic	No

If you do not explicitly set a value for this parameter in the initialization file, then at least one of the two default operating system directories mentioned in the preceding table must exist and be writeable. Otherwise, the instance will fail to start.

If you explicitly set a value for this parameter in the initialization file, then the directory you specify must exist and be writeable. Otherwise, the instance will fail to start.

The audit records will be written in XML format if the `AUDIT_TRAIL` initialization parameter is set to **xml** or **xml, extended**. It is also the location to which mandatory auditing information is written and, if so specified by the `AUDIT_SYS_OPERATIONS` initialization parameter, audit records for user `SYS`.

In a multitenant container database (CDB), the scope of the settings for this initialization parameter is the CDB. Although the audit trail is provided per pluggable database (PDB) in a CDB, this initialization parameter cannot be configured for individual PDBs.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Multitenant Administrator's Guide* for conceptual information about CDBs and PDBs
- *Oracle Multitenant Administrator's Guide* for information about managing CDBs and PDBs
- "V$CONTAINERS"
- "V$PDBS"

1.24 AUDIT_SYS_OPERATIONS

`AUDIT_SYS_OPERATIONS` enables or disables the auditing of directly issued user SQL statements with `SYS` authorization. These include SQL statements directly issued by users when connected with the `SYSASM`, `SYSBACKUP`, `SYSDBA`, `SYSDG`, `SYSKM`, or `SYSOPER` privileges, as well as SQL statements that have been executed with `SYS` authorization using the PL/SQL package `DBMS_SYS_SQL`.

Note

In an Oracle database that has migrated to unified auditing, the setting of this parameter has no effect.

- See *Oracle Database Security Guide* for more information about unified auditing.
- See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Property	Description
Parameter type	Boolean
Default value	`TRUE`
Modifiable	No
Modifiable in a PDB	No
Range of values	`TRUE | FALSE`
Basic	No

The audit records are written to the operating system's audit trail. The audit records will be written in XML format if the `AUDIT_TRAIL` initialization parameter is set to `xml` or `xml, extended`

On UNIX platforms, if the `AUDIT_SYSLOG_LEVEL` parameter has also been set, then it overrides the `AUDIT_TRAIL` parameter and SYS audit records are written to the system audit log using the `SYSLOG` utility.

In a CDB, the scope of the settings for this initialization parameter is the CDB. Although the audit trail is provided per PDB in a CDB, this initialization parameter cannot be configured for individual PDBs.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.25 AUDIT_SYSLOG_LEVEL

AUDIT_SYSLOG_LEVEL allows SYS and standard OS audit records to be written to the system audit log using the SYSLOG utility.

> **Note**
>
> In an Oracle database that has migrated to unified auditing, the setting of this parameter has no effect.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Property	Description
Parameter type	String
Syntax	AUDIT_SYSLOG_LEVEL = '*facility_clause.priority_clause*'
Syntax	**facility_clause::=** { USER | LOCAL[0 | 1 | 2 | 3 | 4 | 5 | 6 | 7] | SYSLOG | DAEMON | KERN | MAIL | AUTH | LPR | NEWS | UUCP | CRON }
Syntax	**priority_clause::=** { NOTICE | INFO | DEBUG | WARNING | ERR | CRIT | ALERT | EMERG }
Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	No
Basic	No
Examples	AUDIT_SYSLOG_LEVEL = 'KERN.EMERG'; AUDIT_SYSLOG_LEVEL = 'LOCAL1.WARNING';

If you use this parameter, it is best to assign a file corresponding to every combination of facility and priority (especially KERN.EMERG) in syslog.conf . Sometimes these are assigned to print to the console in the default syslog.conf file. This can become annoying and will be useless as audit logs. Also, if you use this parameter, it is best to set the maximum length of syslog messages in the system to 512 bytes.

> **Note**
>
> Audit records written to the system audit log could get truncated to 512 bytes, and different parts of the same audit record may not be joined to get the original complete audit record.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Security Guide* for information about configuring syslog auditing

If `AUDIT_SYSLOG_LEVEL` is set and `SYS` auditing is enabled (`AUDIT_SYS_OPERATIONS = TRUE`), then `SYS` audit records are written to the system audit log. If `AUDIT_SYSLOG_LEVEL` is set and standard audit records are being sent to the operating system (`AUDIT_TRAIL = os`), then standard audit records are written to the system audit log.

In a CDB, the scope of the settings for this initialization parameter is the CDB. Although the audit trail is provided per PDB in a CDB, this initialization parameter cannot be configured for individual PDBs.

1.26 AUDIT_TRAIL

`AUDIT_TRAIL` enables or disables database auditing.

> **Note**
>
> In an Oracle database that has migrated to unified auditing, the setting of this parameter has no effect.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Property	Description
Parameter type	String
Syntax	`AUDIT_TRAIL = { none | os | db [, extended] | xml [, extended] }`
Default value	`none`
Modifiable	No
Modifiable in a PDB	No
Basic	No

Values

- `none`

  Disables standard auditing. This value is the default if the `AUDIT_TRAIL` parameter was not set in the initialization parameter file or if you created the database using a method other than Database Configuration Assistant. If you created the database using Database Configuration Assistant, then the default is `db`.

- `os`

  Directs all audit records to an operating system file. Oracle recommends that you use the `os` setting, particularly if you are using an ultra-secure database configuration.

- `db`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Directs audit records to the database audit trail (the SYS.AUD$ table), except for records that are always written to the operating system audit trail. Use this setting for a general database for manageability.

If the database was started in read-only mode with AUDIT_TRAIL set to db, then Oracle Database internally sets AUDIT_TRAIL to os. Check the alert log for details.

- db, extended

  Performs all actions of AUDIT_TRAIL=db, and also populates the SQL bind and SQL text CLOB-type columns of the SYS.AUD$ table, when available. These two columns are populated only when this parameter is specified. When standard auditing is used with DB, EXTENDED, then virtual private database (VPD) predicates and policy names are also populated in the SYS.AUD$ table.

  If the database was started in read-only mode with AUDIT_TRAIL set to db, extended, then Oracle Database internally sets AUDIT_TRAIL to os. Check the alert log for details.

- xml

  Writes to the operating system audit record file in XML format. Records all elements of the AuditRecord node except Sql_Text and Sql_Bind to the operating system XML audit file.

- xml, extended

  Performs all actions of AUDIT_TRAIL=xml, and includes SQL text and SQL bind information in the audit trail.

You can use the SQL AUDIT statement to set auditing options regardless of the setting of this parameter.

In a CDB, the scope of the settings for this initialization parameter is the CDB. Although the audit trail is provided per PDB in a CDB, this initialization parameter cannot be configured for individual PDBs.

Examples

The following statement sets the db, extended value for the AUDIT_TRAIL parameter. The new value takes effect after the database is restarted.

```
SQL> alter system set AUDIT_TRAIL=db, extended scope=spfile;

System altered.

SQL>
```

The following statement sets the xml, extended value for the AUDIT_TRAIL parameter. The new value takes effect after the database is restarted.

```
SQL> alter system set AUDIT_TRAIL=xml, extended scope=spfile;

System altered.

SQL>
```

The following statement sets the db value for the AUDIT_TRAIL parameter. The new value takes effect after the database is restarted.

```
SQL> alter system set AUDIT_TRAIL=db scope=spfile;

System altered.

SQL>
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database Security Guide* for information about configuring unified audit policies
- *Oracle Database Upgrade Guide* to learn more about traditional non-unified auditing

1.27 AUTOTASK_MAX_ACTIVE_PDBS

`AUTOTASK_MAX_ACTIVE_PDBS` enables you to specify the maximum number of PDBs that can schedule automated maintenance tasks at the same time (during a maintenance window).

Property	Description
Parameter type	Integer
Default value	2
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`0` to the number of PDBs in the CDB
Basic	No
Oracle RAC	The same value should be used on all instances.

This parameter only affects PDBs. The CDB$ROOT container (CDB root) for a CDB can always schedule and run maintenance tasks during a maintenance window.

The default value is 2. Therefore, by default, two PDBs and the CDB root can run tasks at the same time during a maintenance window.

This parameter can be set only in the CDB root, not in a PDB.

See Also

- "ENABLE_AUTOMATIC_MAINTENANCE_PDB" for information on disabling or enabling the running of automated maintenance tasks for specific PDBs or for all the PDBs in CDB
- *Oracle Database Administrator's Guide* for more information about managing automated database maintenance tasks

1.28 AWR_PDB_AUTOFLUSH_ENABLED

`AWR_PDB_AUTOFLUSH_ENABLED` enables you to specify whether to enable or disable automatic Automatic Workload Repository (AWR) snapshots for all the PDBs in a CDB or for individual PDBs in a CDB.

Property	Description
Parameter type	Boolean

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	`false`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No
Oracle RAC	The same value must be used on all instances.

Note

The value of this parameter in CDB$ROOT (the root of a CDB) has no effect in the root. Automatic AWR snapshots are always enabled in the root, regardless of the setting of this parameter.

The default value of `AWR_PDB_AUTOFLUSH_ENABLED` is `false`. Thus, by default, automatic AWR snapshots are disabled for all the PDBs in a CDB.

When you change the value of `AWR_PDB_AUTOFLUSH_ENABLED` in the CDB root, the new value takes effect in all the PDBs in the CDB.

Therefore, if you change the value of `AWR_PDB_AUTOFLUSH_ENABLED` in the CDB root to `true`, the value of `AWR_PDB_AUTOFLUSH_ENABLED` is also changed to `true` in all of the PDBs, so that automatic AWR snapshots are enabled for all the PDBs.

You can also change the value of `AWR_PDB_AUTOFLUSH_ENABLED` in any of the individual PDBs in a CDB, and the value that is set for each individual PDB will be honored. This enables you to enable or disable automatic AWR snapshots for individual PDBs.

When a new PDB is created, or a PDB from a previous database release is upgraded to the current database release, automatic AWR snapshots are enabled or disabled for the PDB based on the current value of `AWR_PDB_AUTOFLUSH_ENABLED` in the root.

See Also

- *Oracle Database Performance Tuning Guide* for more information about Automatic Workload Repository (AWR)
- *Oracle Database Performance Tuning Guide* for more information about AWR snapshots

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.29 AWR_PDB_MAX_PARALLEL_SLAVES

AWR_PDB_MAX_PARALLEL_SLAVES enables a DBA to allocate the correct amount of resources to enable quick and timely Automatic Workload Repository (AWR) flushes for multitenant container databases (CDBs).

Property	Description
Parameter type	Integer
Default value	10
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	1 – 30
Basic	No
Oracle RAC	Different instances can use different values.

Use AWR_PDB_MAX_PARALLEL_SLAVES to control the amount of resources dedicated to AWR snapshot flushing in pluggable databases (PDBs). With this parameter, a DBA can modify the maximum number of MMON worker processes (M*nnn* background processes) that can concurrently be used to handle AWR flush operations for the entire CDB.

For CDBs with a large number of PDBs enabled to create automatic AWR snapshots, a DBA can increase this parameter to enable timely AWR flushing. For CDBs with a small number of PDBs enabled, the value of this parameter can be decreased to reduce the concurrency and the chance of performance spikes.

Note that this parameter is set in the root of a CDB and determines the maximum degree of parallelism used to create AWR automatic snapshots for PDBs.

See Also

- "AWR_PDB_AUTOFLUSH_ENABLED" for information about enabling the automatic creation of AWR PDB snapshots
- *Oracle Database Concepts* for an introduction to AWR
- "Background Processes" for more information about the MMON and M*nnn* background processes

1.30 AWR_SNAPSHOT_TIME_OFFSET

AWR_SNAPSHOT_TIME_OFFSET specifies an offset for the Automatic Workload Repository (AWR) snapshot start time.

Property	Description
Parameter type	Integer
Default value	There is no offset by default.
Modifiable	ALTER SYSTEM

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable in a PDB	No
Range of values	0 - 3599, or the special value 1000000
Basic	No
Oracle RAC	Multiple instances should use the same value

AWR snapshots normally start at the top of the hour (12:00, 1:00, 2:00, and so on). This parameter allows DBAs to specify an offset for the AWR snapshot start time.

This is a useful parameter to avoid CPU spikes from multiple instances all starting their AWR snapshots at the same time. If you have a large system with many instances on it (like many Exadata installations), and you are experiencing such CPU spikes, this parameter can be very useful.

The parameter is specified in seconds. Normally, you set it to a value less than 3600. If you set the special value 1000000 (1,000,000), you get an automatic mode, in which the offset is based on the database name.

The automatic mode is an effective way of getting a reasonable distribution of offset times when you have a very large number of instances running on the same node.

1.31 BACKGROUND_CORE_DUMP

`BACKGROUND_CORE_DUMP` specifies whether Oracle includes the SGA in the core file for Oracle background processes.

Property	Description
Parameter type	String
Syntax	`BACKGROUND_CORE_DUMP = { partial | full }`
Default value	`partial`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Basic	No

Values

- `partial`

  Oracle does not include the SGA in the core dump.
- `full`

  Oracle includes the SGA in the core dump.

> **See Also**
>
> "SHADOW_CORE_DUMP"

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.32 BACKGROUND_DUMP_DEST

`BACKGROUND_DUMP_DEST` specifies the pathname (directory or disc) where debugging trace files for the background processes (LGWR, DBW*n*, and so on) are written during Oracle operations.

Property	Description
Parameter type	String
Syntax	`BACKGROUND_DUMP_DEST = { pathname | directory }`
Default value	Operating system-dependent
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	Any valid local path, directory, or disk
Basic	No

Note

The `BACKGROUND_DUMP_DEST` parameter is deprecated in Oracle Database 12*c* Release 1 (12.1.0.1).

An **alert log** in the directory specified by `BACKGROUND_DUMP_DEST` logs significant database events and messages. Anything that affects the database instance or global database is recorded here. The alert log is a normal text file. Its file name is operating system-dependent. For platforms that support multiple instances, it takes the form `alert_`*`sid`*`.log`, where *sid* is the system identifier. This file grows slowly, but without limit, so you might want to delete it periodically. You can delete the file even when the database is running.

Note

This parameter is ignored by the diagnosability infrastructure introduced in Oracle Database 11g Release 1 (11.1), which places trace and core files in a location controlled by the `DIAGNOSTIC_DEST` initialization parameter.

See Also

- *Oracle Database Administrator's Guide* for more information on the DIAGNOSTIC_DEST initialization parameter
- "USER_DUMP_DEST" for information on setting a destination for server process trace files

1.33 BACKUP_TAPE_IO_SLAVES

`BACKUP_TAPE_IO_SLAVES` specifies whether I/O server processes are used by Recovery Manager to back up, copy, or restore data to tape.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SYSTEM ... DEFERRED`
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

When the value is set to `true`, Oracle uses an I/O server process to write to or read from a tape device. When the value is set to `false` (the default), Oracle does not use I/O server process for backups. Instead, the shadow process engaged in the backup accesses the tape device.

Note

You cannot perform duplexed backups unless you enable this parameter. Otherwise, Oracle returns an error. When this parameter is enabled, Recovery Manager will configure as many server processes as needed for the number of backup copies requested.

See Also

- *Oracle Database Backup and Recovery User's Guide* for more information on duplexed backups
- "DBWR_IO_SLAVES"

1.34 BITMAP_MERGE_AREA_SIZE

`BITMAP_MERGE_AREA_SIZE` specifies the amount of memory Oracle uses to merge bitmaps retrieved from a range scan of the index.

Property	Description
Parameter type	Integer
Default value	`1048576` (1 MB)
Modifiable	No
Modifiable in a PDB	Yes
Range of values	Operating system-dependent
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> Oracle does not recommend using the `BITMAP_MERGE_AREA_SIZE` parameter unless the instance is configured with the shared server option. Oracle recommends that you enable automatic sizing of SQL working areas by setting `PGA_AGGREGATE_TARGET` instead. `BITMAP_MERGE_AREA_SIZE` is retained for backward compatibility.

`BITMAP_MERGE_AREA_SIZE` is relevant only for systems containing bitmap indexes. A larger value usually improves performance, because the bitmap segments must be sorted before being merged into a single bitmap.

> **See Also**
>
> *Oracle Database SQL Tuning Guide* for more information on using bitmap indexes for performance

1.35 BLANK_TRIMMING

`BLANK_TRIMMING` specifies the data assignment semantics of character datatypes.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	No
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No

Values

- `TRUE`

  Allows the data assignment of a source character string or variable to a destination character column or variable even though the source length is longer than the destination length. In this case, however, the additional length over the destination length must be all blanks, else an exception condition is raised. This value complies with the semantics of SQL-92 Transitional Level and above.

- `FALSE`

  Disallows the data assignment if the source length is longer than the destination length and reverts to SQL92 Entry Level semantics.

> **See Also**
>
> *Oracle Database Globalization Support Guide* for more information on how using this parameter can help prevent data truncation issues during character set migration

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.36 BLOCKCHAIN_TABLE_MAX_NO_DROP

`BLOCKCHAIN_TABLE_MAX_NO_DROP` lets you control the maximum amount of idle time that can be specified when creating a blockchain table or an immutable table. The idle time represents the length of time the table must be inactive before it can be dropped. When explicitly set by a user, the only value allowed for this parameter is `0`.

Property	Description
Parameter type	Integer
Default value	None
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	None or `0`
Basic	No
Oracle RAC	The same value must be used on all instances.

When creating a blockchain table or an immutable table with the SQL statement `CREATE TABLE`, you can specify the clause `NO DROP UNTIL` *`number`* `DAYS IDLE`. This clause specifies the number of days the table must be inactive before it can be dropped, that is, the number of days that must pass after the most recent row insertion before the table can be dropped.

- If `BLOCKCHAIN_TABLE_MAX_NO_DROP` is set to `0`, then when a user creates a blockchain table or an immutable table, and specifies the `NO DROP UNTIL` *`number`* `DAYS IDLE` clause, the value of *`number`* must be `0`:

  ```
  CREATE BLOCKCHAIN TABLE ... NO DROP UNTIL 0 DAYS IDLE ...
  CREATE IMMUTABLE TABLE ... NO DROP UNTIL 0 DAYS IDLE ...
  ```

  This allows the table to be dropped at any time, regardless of how long the table has been inactive.
- If `BLOCKCHAIN_TABLE_MAX_NO_DROP` is not set, then when a user creates a blockchain table or an immutable table, and specifies the `NO DROP UNTIL` *`number`* `DAYS IDLE` clause, any number value can be specified for *`number`*.

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.10.

> **See Also**
>
> *Oracle Database SQL Language Reference* for information about the `CREATE TABLE` statement

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.37 CIRCUITS

`CIRCUITS` specifies the total number of virtual circuits that are available for inbound and outbound network sessions.

Property	Description
Parameter type	Integer
Default value	`4294967295`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Basic	No

It is one of several parameters that contribute to the total SGA requirements of an instance.

You should not specify a value for this parameter unless you want to limit the number of virtual circuits.

See Also

- *Oracle Database Concepts* for more information on memory structures
- *Oracle Database Concepts* for more information on processes

1.38 CLIENT_RESULT_CACHE_LAG

`CLIENT_RESULT_CACHE_LAG` specifies the maximum time (in milliseconds) since the last round trip to the server, before which the OCI client query execute makes a round trip to get any database changes related to the queries cached on the client.

Property	Description
Parameter type	Big integer
Syntax	`CLIENT_RESULT_CACHE_LAG = integer`
Default value	`3000`
Modifiable	No
Modifiable in a PDB	Yes
Range of values	`0` to operating system-dependent
Basic	No

See Also

Oracle Call Interface Programmer's Guide for more information about the client query cache feature

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.39 CLIENT_RESULT_CACHE_SIZE

CLIENT_RESULT_CACHE_SIZE specifies the maximum size of the client per-process result set cache (in bytes).

Property	Description
Parameter type	Big integer
Syntax	CLIENT_RESULT_CACHE_SIZE = *integer* [K | M | G]
Default value	0
Modifiable	No
Modifiable in a PDB	Yes
Range of values	0 to operating system-dependent
Basic	No

All OCI client processes inherit this maximum size. Setting a nonzero value enables the client query cache feature. This can be overridden by the client configuration parameter OCI_RESULT_CACHE_MAX_SIZE.

See Also

Oracle Call Interface Programmer's Guide for more information about the client query cache feature

1.40 CLIENT_STATISTICS_LEVEL

CLIENT_STATISTICS_LEVEL controls whether database clients report network statistics to the database.

This parameter applies only to clients that use Oracle Call Interface (OCI) to connect to the database.

Property	Description
Parameter type	String
Syntax	CLIENT_STATISTICS_LEVEL = { TYPICAL | OFF }
Default value	TYPICAL
Modifiable	ALTER SYSTEM ... DEFERRED
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The same value must be used on all instances.

Database clients sometimes have performance issues when interacting with the database over a network (LAN or WAN). You can use this parameter to enable clients to collect network statistics and periodically send them to the database, where the statistics are displayed in AWR and V$ views. These statistics include TCP and SQL*Net statistics. They complement existing

Copyright © 2002, 2026, Oracle and/or its affiliates.

database statistics and can be used to analyze, troubleshoot, and tune potential network performance or configuration issues.

Possible values:

- `TYPICAL`

  This setting enables clients to collect network statistics and periodically report them to the database.

- `OFF`

  This setting disables this feature. Clients will not report network statistics to the database.

When you modify this parameter, the new setting applies only to subsequent new connections to database clients.

Note

To view a complete listing of the statistics that are collected and reported when this feature is enabled, issue the following SQL statement:

```
SELECT name FROM V$STATNAME
  WHERE name LIKE 'Client%' ORDER BY name;

NAME
----------------------------------------------------
Client Advertised Receive Window
Client Advertised Send Window
Client Data Segments In
Client Data Segments Out
Client Lost Packets
Client Path Maximum Transmission Unit(MTU)
Client Send Congestion Window
Client Time (usec) Busy Sending Data
Client Time (usec) Busy Sending Data under Congestion
Client Time (usec) Last Ack Received
Client Time (usec) Last Ack Sent
Client Time (usec) Last Data Received
Client Time (usec) Last Data Sent
Client Time (usec) Limited by Receive Window
Client Time (usec) Limited by Send Buffer
Client Time (usec) Round Trip Time
Client Time (usec) Round Trip Time Variance
Client Total Bytes Acked
Client Total Bytes Received
Client Total Number of Retransmitted Packets
```

Note

This parameter is available starting with Oracle Database 19c.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.41 CLONEDB

`CLONEDB` should be set on Direct NFS Client CloneDB databases. When this parameter is set, the CloneDB database uses the database backup as the backing store for the datafiles.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	No
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No
Oracle RAC	The same value should be set for all instances.

See Also

- See *Oracle Database Administrator's Guide* for more information about cloning databases on network attached storage (NAS).
- "CLONEDB_DIR"

1.42 CLONEDB_DIR

`CLONEDB_DIR` sets the directory path where CloneDB bitmap files should be created and accessed.

Property	Description
Parameter type	String
Syntax	`CLONEDB_DIR = string`
Default value	`$ORACLE_HOME/dbs`
Modifiable	No
Modifiable in a PDB	No
Basic	No
Oracle RAC	In an Oracle RAC environment, this parameter should be set to a shared location that is accessible from all the instances.

By default the CloneDB bitmap file is created under the `$ORACLE_HOME/dbs` directory. This directory may not be in a shared location in an Oracle RAC environment, and therefore this parameter is provided to identify a shared location where CloneDB specific files can be created.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"CLONEDB"

1.43 CLUSTER_DATABASE

CLUSTER_DATABASE is an Oracle RAC parameter that specifies whether Oracle RAC is enabled.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	No
Modifiable in a PDB	No
Range of values	true | false
Basic	Yes
Oracle RAC	For all instances, the value must be set to true.

See Also

Oracle Real Application Clusters Administration and Deployment Guide for an introduction to Oracle RAC

1.44 CLUSTER_DATABASE_INSTANCES

CLUSTER_DATABASE_INSTANCES is an Oracle RAC parameter that specifies the number of instances that are configured as part of the cluster database.

Property	Description
Parameter type	Integer
Default value	If CLUSTER_DATABASE is set to false, then 1 If CLUSTER_DATABASE is set to true, the number of configured Oracle RAC instances
Modifiable	No
Modifiable in a PDB	No
Range of values	Any nonzero value
Basic	No
Oracle RAC	Multiple instances should have the same value.

You must set this parameter for every instance. Normally you should set this parameter to the number of instances in your Oracle RAC environment. A proper setting for this parameter can improve memory use.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The `CLUSTER_DATABASE_INSTANCES` parameter is deprecated in Oracle Database 19c, and may be desupported in a future release. Starting with Oracle Database 19c, the number of configurable Oracle RAC instances is derived from Oracle Clusterware.

> **See Also**
>
> - *Oracle Database SQL Tuning Guide* for more information about parallel execution
> - *Oracle Real Application Clusters Administration and Deployment Guide* for information on Oracle Real Application Clusters

1.45 CLUSTER_INTERCONNECTS

`CLUSTER_INTERCONNECTS` can be used in Oracle Real Application Clusters environments to indicate cluster interconnects available for use for the database traffic.

Property	Description
Parameter type	String
Syntax	`CLUSTER_INTERCONNECTS = ifn [: ifn ] ...`
Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	No
Range of values	One or more IP addresses, separated by colons
Basic	No

Use this parameter to override the default interconnect configured for the database traffic, which is stored in the cluster registry. This procedure also may be useful with Data Warehouse systems that have reduced availability requirements and high interconnect bandwidth demands.

`CLUSTER_INTERCONNECTS` specifically overrides the following:

- Network classifications stored by `oifcfg` in the OCR.
- The default interconnect chosen by Oracle.

If you want to load-balance the interconnect, then Oracle recommends that you use link-bonding at the operating system level, even if you have two databases on the same server, so that multiple interconnects use the same address. Note that multiple private addresses provide load balancing, but do not provide failover unless bonded. If you specify multiple addresses in init.ora using `CLUSTER_INTERCONNECTS`, instead of bonding multiple addresses at the operating system level, then typically availability is reduced, because each network interface card failure will take down that instance.

Refer to your vendor documentation for information about bonding interfaces. Some vendor bonding architectures may require the use of this parameter.

Copyright © 2002, 2026, Oracle and/or its affiliates.

If you have multiple database instances on Oracle Real Application Clusters nodes and want to use a specific interface for each instance, then you can set the `CLUSTER_INTERCONNECTS` initialization parameter to the IP address for each database instance. For example:

```
hr1.init.ora.cluster_interconnects="192.0.2.111"
oltp3.init.ora.cluster_interconnects="192.0.2.112"
```

If the Oracle RAC interconnect is configured to run on a different interface than the Oracle Clusterware interconnect, then this configuration can cause reduced availability, as failovers or instance evictions can be delayed if the Oracle RAC interconnect fails while the Oracle Clusterware NIC remains up.

See Also

Oracle Real Application Clusters Administration and Deployment Guide for additional information about using `CLUSTER_INTERCONNECTS`

1.46 COMMIT_LOGGING

`COMMIT_LOGGING` is an advanced parameter used to control how redo is batched by Log Writer.

Property	Description
Parameter type	String
Syntax	`COMMIT_LOGGING = { IMMEDIATE | BATCH }`
Default value	There is no default value.
Modifiable	Yes (at both session-level and system-level)
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Each instance may have its own setting

If `COMMIT_LOGGING` is altered after setting `COMMIT_WAIT` to `FORCE_WAIT`, then the `FORCE_WAIT` option is no longer valid.

1.47 COMMIT_POINT_STRENGTH

`COMMIT_POINT_STRENGTH` specifies a value that determines the **commit point site** in a distributed transaction.

Property	Description
Parameter type	Integer
Default value	1
Modifiable	No
Modifiable in a PDB	Yes
Range of values	`0` to 255
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This parameter is relevant only in distributed database systems.

The node in the transaction with the highest value for `COMMIT_POINT_STRENGTH` will be the commit point site.

The commit point site of a database should reflect the amount of critical shared data in the database. For example, a database on a mainframe computer typically shares more data among users than one on a personal computer. Therefore, `COMMIT_POINT_STRENGTH` should be set to a higher value for the mainframe computer.

The commit point site stores information about the status of transactions. Other computers in a distributed transaction require this information during Oracle's two-phase commit, so it is desirable to have machines that are always available as commit point sites. Therefore, set `COMMIT_POINT_STRENGTH` to a higher value on your more available machines.

> **See Also**
>
> - *Oracle Database Concepts* and *Oracle Database Administrator's Guide* for information on two-phase commit
> - Your operating system-specific Oracle documentation for the default value of this parameter

1.48 COMMIT_WAIT

`COMMIT_WAIT` is an advanced parameter used to control when the redo for a commit is flushed to the redo logs.

Property	Description
Parameter type	String
Syntax	`COMMIT_WAIT = { NOWAIT | WAIT | FORCE_WAIT }`
Default value	There is no default value.
Modifiable	Yes (at both session-level and system-level)
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Each instance may have its own setting

Be aware that the `NOWAIT` option can cause a failure that occurs after the database receives the commit message, but before the redo log records are written. This can falsely indicate to a transaction that its changes are persistent. Also, it can violate the durability of ACID (Atomicity, Consistency, Isolation, Durability) transactions if the database shuts down unexpectedly.

If the parameter is set to `FORCE_WAIT`, the default behavior (immediate flushing of the redo log buffer with wait) is used. If this is a system setting, the session level and transaction level options will be ignored. If this is a session level setting, the transaction level options will be ignored. If `COMMIT_WAIT` is altered after it has been set to `FORCE_WAIT`, then the `FORCE_WAIT` option is no longer valid.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.49 COMMIT_WRITE

COMMIT_WRITE is an advanced parameter used to control how redo for transaction commits is written to the redo logs.

Property	Description
Parameter type	String
Syntax	COMMIT_WRITE = '{IMMEDIATE | BATCH},{WAIT |NOWAIT}'
Default value	If this parameter is not explicitly specified, then database commit behavior defaults to writing commit records to disk before control is returned to the client. If only IMMEDIATE or BATCH is specified, but not WAIT or NOWAIT, then WAIT mode is assumed. If only WAIT or NOWAIT is specified, but not IMMEDIATE or BATCH, then IMMEDIATE mode is assumed. Be aware that the NOWAIT option can cause a failure that occurs after the database receives the commit message, but before the redo log records are written. This can falsely indicate to a transaction that its changes are persistent. Also, it can violate the durability of ACID (Atomicity, Consistency, Isolation, Durability) transactions if the database shuts down unexpectedly.
Modifiable	Yes (at both session-level and system-level). Values supplied for COMMIT_WRITE in an ALTER SYSTEM or ALTER SESSION statement must be separated by a comma.
Modifiable in a PDB	Yes
Range of values	Single-quoted, comma-separated list of either IMMEDIATE or BATCH, and either WAIT or NOWAIT.
Basic	No
Oracle RAC	Each instance may have its own setting

The IMMEDIATE and BATCH options control how redo is batched by Log Writer. The WAIT and NOWAIT options control when the redo for a commit is flushed to the redo logs.

> **Note**
>
> The COMMIT_WRITE parameter is deprecated. It is retained for backward compatibility only. It is replaced by the COMMIT_LOGGING and COMMIT_WAIT parameters.

1.50 COMMON_USER_PREFIX

COMMON_USER_PREFIX specifies a prefix that the names of common users, roles, and profiles in a multitenant container database (CDB) must start with.

Property	Description
Parameter type	String
Syntax	COMMON_USER_PREFIX = prefix

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	In a CDB root, C## is the default value. In an application root, the empty string is the default value.
Modifiable	No
Modifiable in a PDB	No However, you can modify this parameter in an application root.
Basic	No
Oracle RAC	Multiple instances must have the same value

Names of local users, roles, and profiles must not start with the same prefix. A case-insensitive comparison of the prefixes for user, role, and profile names to the value of this parameter is done.

When a value is set for COMMON_USER_PREFIX, Oracle will require that the names of common users, roles, and profiles start with the string assigned to this parameter.

The names of users, roles, and profiles created using this prefix must be valid names.

Regardless of the value of the COMMON_USER_PREFIX parameter, the names of local users, roles, and profiles cannot start with C##. Similarly, the names of application common users and local users in an application PDB cannot start with C##.

> ⚠ **Caution**
>
> You can change the value of the COMMON_USER_PREFIX parameter, but do so only with great care.

If COMMON_USER_PREFIX is set to an empty string, Oracle will *not* enforce any restrictions on the names of common or local users, roles, and profiles.

Setting this parameter to an empty string will result in no restrictions being placed on names of common and local users, roles, and profiles, which could lead to conflicts between the names of local and common users, roles, and profiles when a PDB is plugged into a different CDB, or when opening a PDB that was closed when a common user was created.

> ⓘ **Note**
>
> COMMON_USER_PREFIX can be set in the context of an application container.
>
> If you set COMMON_USER_PREFIX in the application root to a non-null value, then the application common user should start with that prefix, and a local user created in an application PDB cannot start with that prefix.
>
> By default, COMMON_USER_PREFIX is the empty string in an application root.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Note

When you query the V$PARAMETER or V$SYSTEM_PARAMETER view, a value of NONE is returned for the DEFAULT_VALUE column for the COMMON_USER_PREFIX parameter.

However, the database enforces a default value of C## for a CDB root and a default value of the empty string for an application root.

See Also

- *Oracle Database SQL Language Reference* for information about valid user names
- *Oracle Database SQL Language Reference* for information about valid role names
- *Oracle Database SQL Language Reference* for information about valid profile names
- *Oracle Multitenant Administrator's Guide* for more information about the multitenant architecture

1.51 COMPATIBLE

COMPATIBLE enables you to use a new release of Oracle while ensuring the ability to downgrade the database to an earlier release.

Property	Description
Parameter type	String
Syntax	COMPATIBLE = *release_number*
Default value	19.0.0
Modifiable	No
Modifiable in a PDB	No
Range of values	11.2.0 to default release Values must be specified as at least three decimal numbers with each pair separated by a dot, such as 19.0.0 or 19.0.0.0.0
Basic	Yes
Oracle RAC	Multiple instances must have the same value.

The COMPATIBLE parameter specifies the Oracle version number that the database disk format must be compatible with. The database can be downgraded to the version specified in the COMPATIBLE parameter or any later version.

Setting COMPATIBLE ensures that new features do not write data formats or structures to disk that are not compatible with the earlier release, preventing a future downgrade. Features that require a higher value of COMPATIBLE to work correctly may be restricted or disabled to ensure downgrades are possible.

Typically, users keep COMPATIBLE unchanged when upgrading their Oracle software. After upgrade, users will run the new release of the Oracle software for a few weeks to ensure that

Copyright © 2002, 2026, Oracle and/or its affiliates.

the new release is working correctly. Afterwards, users can choose to update COMPATIBLE to the latest version to take advantage of the new features.

Notes

- The value of the COMPATIBLE parameter can be increased to a higher version, but it can never be decreased to a lower version.
- The value of the COMPATIBLE parameter should not be changed for a Release Update (RU) or Release Update Revision (RUR). For example, assume you are running Oracle Database 19c and the value of COMPATIBLE is 19.0.0. You then apply Oracle Database Release Update 19.6.0.0.0. Do not set the value of COMPATIBLE to 19.6.0; leave it set to 19.0.0.
- When using a standby database, this parameter must have the same value on both the primary and standby databases.
- When you set the value of the COMPATIBLE parameter to 12.2.0, the maximum size of SQL identifiers is increased from 30 bytes to 128 bytes.

See Also

Oracle Database Upgrade Guide for more information on setting this parameter

1.52 CONNECTION_BROKERS

CONNECTION_BROKERS is used to specify connection broker types, the number of connection brokers of each type, and the maximum number of connections per broker.

Property	Description
Parameter type	String
Syntax	CONNECTION_BROKERS = broker_description[,...]
Syntax	**broker_description::=** ((type_clause)(brokers_clause)[(broker_options)])
Syntax	**type_clause::=** TYPE={ DEDICATED | EMON | POOLED } Indicates the broker type. This is a mandatory parameter.
Syntax	**brokers_clause::=** BROKERS=*integer* Indicates the number of brokers of the specified type. This is a mandatory parameter. The specified integer value must be greater than or equal to 0. If set to 0, brokers will not accept new connections, but existing connections will continue as normal.
Syntax	**broker_options::=** CONNECTIONS=*integer* The maximum number of connections per broker. This is an optional parameter. If the specified integer value is greater than the maximum allowed for the platform, the maximum allowed for the platform is used.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	The default value for the initialization parameter will contain the DEDICATED and EMON broker types. Certain types of brokers are only started when certain features are enabled, so their configuration will only apply if the broker is needed.
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No
Oracle RAC	The same value should be set on all instances

The connection brokers are used in these cases:

- DEDICATED brokers are used by the multiprocess and multithreaded Oracle features. See *Oracle Database Concepts* for more information about these features.

  DEDICATED brokers are also used when the USE_DEDICATED_BROKER initialization parameter is set to true. See "USE_DEDICATED_BROKER" for more information about the USE_DEDICATED_BROKER initialization parameter.
- EMON brokers are used when there is a client registered for notification over 12*c* AQ queues. It is not on by default. See *Oracle Database Advanced Queuing User's Guide* for more information about AQ queues.
- POOLED brokers are used whenever a Database Resident Connection Pooling (DRCP) pool is active. See *Oracle Database Concepts* for more information about DRCP pooling.

  The POOLED broker does not appear in the parameter by default. In this case, the DRCP fields are used. If you specify the POOLED broker in the parameter, then that specification will override the DRCP fields, and further attempts to set the DRCP fields will then throw an error. Oracle recommends that the parameter be used instead of the DRCP fields.

To make it easier to modify the list, individual entries are indexed using the TYPE so it is not necessary to re-specify the entire list. For example, to adjust the number of DEDICATED brokers:

```
alter system set connection_brokers = '((TYPE=DEDICATED)(BROKERS=2))'
```

1.53 CONTAINER_DATA

Use CONTAINER_DATA to control whether queries on extended data-linked objects from within a PDB return rows from both the root and the PDB, or rows from only the PDB.

Property	Description
Parameter type	String
Syntax	CONTAINER_DATA = { ALL | CURRENT | CURRENT_DICTIONARY }
Default value	ALL
Modifiable	ALTER SESSION
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Different instances can use different values.

This parameter controls the behavior of queries on the following types of objects:

Copyright © 2002, 2026, Oracle and/or its affiliates.

- Extended data-linked application common objects

  This type of object stores shared data in an application root, and also allows application PDBs to store local data that is unique to each application PDB.

  When this type of object is queried from an application root, only the shared data in the application root is returned.

  When this type of object is queried from an application PDB, the value of the `CONTAINER_DATA` initialization parameter controls the query result as follows:

  - `ALL`: The query returns both the shared data in the application root and the local data pertaining to the application PDB
  - `CURRENT`: The query returns only the local data pertaining to the application PDB
  - `CURRENT_DICTIONARY`: The query returns both the shared data in the application root and the local data pertaining to the application PDB
- Extended data-linked Oracle-supplied data dictionary objects

  This type of object stores data pertaining to the CDB root, as well as data pertaining to individual PDBs.

  When this type of object is queried from the CDB root, only data pertaining to the CDB root is returned.

  When this type of object is queried from a PDB, the value of the `CONTAINER_DATA` initialization parameter controls the query result as follows:

  - `ALL`: The query returns data pertaining to both the CDB root and the PDB
  - `CURRENT`: The query returns only data pertaining to the PDB
  - `CURRENT_DICTIONARY`: The query returns only data pertaining to the PDB

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.10.

1.54 CONTAINERS_PARALLEL_DEGREE

`CONTAINERS_PARALLEL_DEGREE` can be used to control the degree of parallelism of a query involving containers().

Property	Description
Parameter type	Integer
Default value	65535
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	Cannot be set to a value lower than 2 or higher than 65535.
Basic	No
Oracle RAC	Different values can be set on different instances.

The value of `CONTAINERS_PARALLEL_DEGREE`, if set, will override the default DOP for a containers() query.

Copyright © 2002, 2026, Oracle and/or its affiliates.

By default, a containers() query uses a degree of parallelism equal to (1 + number of open PDBs) in the case of CDB root and (1 + number of open application PDBs) in the case of application root.

If the value of CONTAINERS_PARALLEL_DEGREE is lower than 65535, then this value is used as the degree of parallelism of a query involving containers(). Otherwise (when the value is 65535), the default degree of parallelism is (1 + number of open PDBs) or (1 + number of open application PDBs) as described above.

See Also

- *Oracle Multitenant Administrator's Guide* for more information about CDB roots
- *Oracle Multitenant Administrator's Guide* for more information about application roots

1.55 CONTROL_FILE_RECORD_KEEP_TIME

CONTROL_FILE_RECORD_KEEP_TIME specifies the minimum number of days before a reusable record in the control file can be reused.

Property	Description
Parameter type	Integer
Default value	7 (days)
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	0 to 365 (days)
Basic	No

In the event a new record must be added to a reusable section and the oldest record has not aged enough, the record section expands. If this parameter is set to 0, then reusable sections never expand, and records are reused as needed.

If the number of reusable records in the control file exceeds the circular reuse record limit UB4MAXVAL, then reusable records will be overwritten even if CONTROL_FILE_RECORD_KEEP_TIME has not elapsed. UB4MAXVAL is defined in the oratypes.h header file, which is found in the public directory. Its value may vary according to the operating system you are using.

Note

This parameter applies only to records in the control file that are circularly reusable (such as archive log records and various backup records). It does not apply to records such as data file, tablespace, and redo thread records, which are never reused unless the corresponding object is dropped from the tablespace.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> - "CONTROL_FILES"
> - *Oracle Database Backup and Recovery User's Guide* for more information about this parameter

1.56 CONTROL_FILES

`CONTROL_FILES` specifies one or more names of control files, separated by commas.

Property	Description
Parameter type	String
Syntax	`CONTROL_FILES = filename [, filename ] ...` **Note:** The control file name can be an OMF (Oracle Managed Files) name. This occurs when the control file is re-created using the `CREATE CONTROLFILE REUSE` statement.
Default value	Operating system-dependent
Modifiable	No
Modifiable in a PDB	No
Range of values	1 to 8 filenames
Basic	Yes
Oracle RAC	Multiple instances must have the same value.

Every database has a **control file**, which contains entries that describe the structure of the database (such as its name, the timestamp of its creation, and the names and locations of its data files and redo files).

Oracle recommends that you multiplex multiple control files on different devices or mirror the file at the operating system level.

> **See Also**
>
> *Oracle Database Administrator's Guide* for information about specifying control files at database creation

1.57 CONTROL_MANAGEMENT_PACK_ACCESS

`CONTROL_MANAGEMENT_PACK_ACCESS` specifies which of the Server Manageability Packs should be active.

Property	Description
Parameter type	String
Syntax	`CONTROL_MANAGEMENT_PACK_ACCESS = { NONE | DIAGNOSTIC | DIAGNOSTIC+TUNING }`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	Enterprise Edition: DIAGNOSTIC+TUNING All other editions: NONE
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No

The following packs are available:

- The DIAGNOSTIC pack includes AWR, ADDM, and so on.
- The TUNING pack includes SQL Tuning Advisor, SQLAccess Advisor, and so on.

  A license for DIAGNOSTIC is required for enabling the TUNING pack.

Values

- NONE

  Both packs are not available.
- DIAGNOSTIC

  Only the DIAGNOSTIC pack is available.
- DIAGNOSTIC+TUNING

  Both packs are available. This is the default.

> **See Also**
>
> *Oracle Database Licensing Information User Manual*

1.58 CORE_DUMP_DEST

CORE_DUMP_DEST specifies the directory where Oracle dumps core files.

Property	Description
Parameter type	String
Syntax	CORE_DUMP_DEST = *directory*
Default value	*ORACLE_HOME*/DBS
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No

CORE_DUMP_DEST is primarily a UNIX parameter and may not be supported on your platform.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This parameter is ignored by the new diagnosability infrastructure introduced in Oracle Database 11g Release 1 (11.1), which places trace and core files in a location controlled by the DIAGNOSTIC_DEST initialization parameter.

> **See Also**
>
> "DIAGNOSTIC_DEST"

1.59 CPU_COUNT

CPU_COUNT specifies the number of CPUs available for Oracle Database to use.

Property	Description
Parameter type	Integer
Default value	0
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to unlimited
Basic	No
Oracle RAC	The same value should be used on all instances.

On CPUs with multiple CPU threads, it specifies the total number of available CPU threads. Various components of Oracle Database are configured based on the number of CPUs, such as the Optimizer, Parallel Query, and Resource Manager.

If CPU_COUNT is set to 0 (its default setting), then Oracle Database continuously monitors the number of CPUs reported by the operating system and uses the current count. If CPU_COUNT is set to a value other than 0, then Oracle Database will use this count rather than the actual number of CPUs, thus disabling dynamic CPU reconfiguration.

When Resource Manager is managing CPU (RESOURCE_MANAGER_PLAN is set), then the database's CPU utilization is limited to CPU_COUNT CPU threads. This feature is called Instance Caging. If Resource Manager is enabled at the CDB level, then the PDB's CPU utilization is limited to the PDB's CPU_COUNT.

> **Note**
>
> Setting CPU_COUNT to a value greater than the current number of CPUs results in an error. However, if CPU_COUNT is set to a value greater than the current number of CPUs in the initialization parameter file, then CPU_COUNT is capped to the current number of CPUs.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> When a value is not explicitly set for CPU_COUNT, the maximum default value for CPU_COUNT is 2 for an Oracle ASM Proxy instance, 4 for an Oracle ASM instance, and 8 for an Oracle IOServer instance.

> **See Also**
>
> - "CPU_MIN_COUNT"
> - *Oracle Database VLDB and Partitioning Guide* for information about how CPU_COUNT is used to determine the default degree of parallelism for a single instance or Oracle RAC configuration when the PARALLEL clause is specified but no degree of parallelism is listed
> - *Oracle Database Administrator's Guide* for an example of how CPU resources are allocated if you enable instance caging and set a maximum utilization limit in a resource plan

1.60 CPU_MIN_COUNT

CPU_MIN_COUNT specifies the minimum number of CPUs required by a pluggable database (PDB) at any given time.

Property	Description
Parameter type	String
Syntax	CPU_MIN_COUNT = '*value*'
Default value	The value of CPU_COUNT
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of Values	Decimal values from 0.1 through 0.95, where the decimal value is a multiple of 0.05 Integer values from 1 up to and including the value of CPU_COUNT
Basic	No
Oracle RAC	The same value should be used on all instances.

This parameter specifies the minimum number of CPUs required by a PDB at any given time. For multi-threaded CPUs, this number corresponds to CPU threads, not CPU cores.

You can set this parameter at the CDB level, and for each individual PDB. This enables you to control each PDBs minimum share of CPU utilization within a CDB. If the sum of the CPU_MIN_COUNT values across all open PDBs in a CDB is equal to the value of CPU_MIN_COUNT for the CDB, then the CDB instance is considered full. If the sum exceeds the value of CPU_MIN_COUNT for the CDB, then the CDB instance is over-provisioned. Oracle does not prevent you from over-provisioning a CDB.

Resource Manager is enabled at the CDB level by setting the RESOURCE_MANAGER_PLAN at the root level to the name of a CDB resource plan. If the CDB resource plan has no configured CPU directives, that is, the SHARES and UTILIZATION_LIMIT directives are unset, then

Copyright © 2002, 2026, Oracle and/or its affiliates.

Resource Manager uses the CPU_COUNT and CPU_MIN_COUNT settings for the PDB to manage CPU utilization.

> **Note**
>
> Setting this parameter to a very low value can result in insufficient CPU allocation and poor performance.

> **Note**
>
> This parameter is available starting with Oracle Database 19c.

> **See Also**
>
> "CPU_COUNT"

1.61 CREATE_BITMAP_AREA_SIZE

CREATE_BITMAP_AREA_SIZE specifies the amount of memory (in bytes) allocated for bitmap creation.

Property	Description
Parameter type	Integer
Default value	0
Modifiable	No
Modifiable in a PDB	Yes
Range of values	Operating system-dependent
Basic	No

This parameter is relevant only for systems containing bitmap indexes.

> **Note**
>
> Oracle does not recommend using the CREATE_BITMAP_AREA_SIZE parameter unless the instance is configured with the shared server option. Oracle recommends that you enable automatic sizing of SQL working areas by setting PGA_AGGREGATE_TARGET instead. CREATE_BITMAP_AREA_SIZE is retained for backward compatibility.

A larger value may speed up index creation.

Cardinality is the number of unique values in a column in relation to the number of rows in the table. If cardinality is very small, you can set a small value for this parameter. For example, if cardinality is only 2, then the value can be in kilobytes rather than megabytes. As a general rule, the higher the cardinality, the more memory is needed for optimal performance.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database SQL Tuning Guide* for more information on using bitmap indexes

1.62 CREATE_STORED_OUTLINES

`CREATE_STORED_OUTLINES` determines whether Oracle automatically creates and stores an outline for each query submitted during the session.

Property	Description
Parameter type	String
Syntax	`CREATE_STORED_OUTLINES = { true | false | category_name } [NOOVERRIDE]`
Default value	There is no default value.
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

Values

- `true`

  Enables automatic outline creation for subsequent queries in the same session. These outlines receive a unique system-generated name and are stored in the `DEFAULT` category. If a particular query already has an outline defined for it in the `DEFAULT` category, then that outline will remain and a new outline will not be created.

- `false`

  Disables automatic outline creation during the session. This is the default.

- *`category_name`*

  Enables the same behavior as `true` except that any outline created during the session is stored in the *`category_name`* category.

- `NOOVERRIDE`

  `NOOVERRIDE` specifies that this system setting will not override the setting for any session in which this parameter was explicitly set. If you do not specify `NOOVERRIDE`, then this setting takes effect in all sessions.

> **See Also**
>
> *Oracle Database SQL Tuning Guide* for more information on setting this parameter

1.63 CURSOR_BIND_CAPTURE_DESTINATION

`CURSOR_BIND_CAPTURE_DESTINATION` determines the location at which bind variables that are captured from SQL cursors are available.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	String
Syntax	`CURSOR_BIND_CAPTURE_DESTINATION = { off | memory | memory+disk }`
Default value	`memory+disk`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

Values

- `off`

  Bind variables are not captured from SQL cursors.

- `memory`

  Bind variables are captured from SQL cursors, and are available only in memory (`V$` views).

- `memory+disk`

  Bind variables are captured from SQL cursors, and are available in memory (`V$` views) and disk (Automatic Workload Repository tables, SQL Tuning Set tables, and so on).

  When you specify this value with a SQL `ALTER SESSION` or `ALTER SYSTEM` statement, enclose the value in single quotes or double quotes. For example:

```
ALTER SESSION SET CURSOR_BIND_CAPTURE_DESTINATION = 'memory+disk';
ALTER SYSTEM SET CURSOR_BIND_CAPTURE_DESTINATION = "memory+disk";
```

1.64 CURSOR_INVALIDATION

`CURSOR_INVALIDATION` controls whether deferred cursor invalidation or immediate cursor invalidation is used for DDL statements by default.

Property	Description
Parameter type	String
Syntax	`CURSOR_INVALIDATION = { DEFERRED | IMMEDIATE }`
Default value	`IMMEDIATE`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Different instances can have different values.

Deferred invalidation reduces the number of cursor invalidations and spreads the recompilation workload over time. Note that when the recompilation workload is spread over time, a cursor may run with a sub-optimal plan until it is recompiled, and may incur small execution-time overhead.

Prior to Oracle Database 12*c* Release 2 (12.2.0.1), immediate cursor invalidation was used.

Copyright © 2002, 2026, Oracle and/or its affiliates.

This parameter provides system or session level default for the DEFERRED or IMMEDIATE option for the INVALIDATION clause in DDL statements.

When this parameter is set to DEFERRED, an application can take advantage of reduced cursor invalidation without making any other application changes.

When this parameter is set to IMMEDIATE, the application will experience the same cursor invalidation behavior as in Oracle Database 12*c* Release 1 (12.1).

1.65 CURSOR_SHARING

CURSOR_SHARING determines what kind of SQL statements can share the same cursors.

Property	Description
Parameter type	String
Syntax	CURSOR_SHARING = { EXACT | FORCE }
Default value	EXACT
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No

Values

- FORCE

  Allows the creation of a new cursor if sharing an existing cursor, or if the cursor plan is not optimal.

- EXACT

  Only allows statements with identical text to share the same cursor.

> **See Also**
>
> *Oracle Database SQL Tuning Guide* before setting the CURSOR_SHARING parameter to learn about the performance implications

1.66 CURSOR_SPACE_FOR_TIME

CURSOR_SPACE_FOR_TIME lets you use more space for cursors to save time.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	No
Modifiable in a PDB	No
Range of values	true | false

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Basic	No

> **Note**
>
> The `CURSOR_SPACE_FOR_TIME` parameter is deprecated. It is retained for backward compatibility only.

This parameter affects both the shared SQL area and the client's private SQL area.

Most users will not need to set this parameter because of the significantly enhanced concurrency modifications introduced in Oracle Database 10*g* Release 2 (10.2.0.2) and later.

Values

- `true`

  Shared SQL areas are kept pinned in the shared pool. As a result, shared SQL areas are not aged out of the pool as long as an open cursor references them. Because each active cursor's SQL area is present in memory, execution is faster. However, the shared SQL areas never leave memory while they are in use. Therefore, you should set this parameter to `true` only when the shared pool is large enough to hold all open cursors simultaneously.

  In addition, a setting of `true` retains the private SQL area allocated for each cursor between executions instead of discarding it after cursor execution, saving cursor allocation and initialization time.

- `false`

  Shared SQL areas can be deallocated from the library cache to make room for new SQL statements.

> **Note**
>
> If this parameter is set to `true`, then the `SERIAL_REUSE` parameter is disabled.

> **See Also**
>
> - "SERIAL_REUSE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.67 DATA_GUARD_MAX_IO_TIME

DATA_GUARD_MAX_IO_TIME sets the maximum number of seconds that can elapse before a process is considered hung while performing a regular I/O operation in an Oracle Data Guard environment. Regular I/O operations include read, write, and status operations.

See Also

"DATA_GUARD_MAX_LONGIO_TIME," which enables you to set the timeout for long I/O operations, such as open and close operations, in an Oracle Data Guard environment

Property	Description
Parameter type	Integer
Default value	240
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	10 to 7200
Basic	No
Oracle RAC	Different instances can use different values.

Note

This parameter is available starting with Oracle Database 19c.

1.68 DATA_GUARD_MAX_LONGIO_TIME

DATA_GUARD_MAX_LONGIO_TIME sets the maximum number of seconds that can elapse before a process is considered hung while performing a long I/O operation in an Oracle Data Guard environment. Long I/O operations include open and close operations.

Property	Description
Parameter type	Integer
Default value	240
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	10 to 7200
Basic	No
Oracle RAC	Different instances can use different values.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This parameter is available starting with Oracle Database 19c.

> **See Also**
>
> "DATA_GUARD_MAX_IO_TIME"

1.69 DATA_GUARD_SYNC_LATENCY

DATA_GUARD_SYNC_LATENCY controls how many seconds the Log Writer (LGWR) process waits beyond the response of the first in a series of Oracle Data Guard SYNC redo transport mode connections.

Property	Description
Parameter type	Integer
Default value	0
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	0 to the number of seconds specified by the NET_TIMEOUT attribute for the LOG_ARCHIVE_DEST_n parameter
Basic	No
Oracle RAC	The same value must be used on all instances.

The default value is 0, which means that the LGWR will wait up to the number of seconds specified by the NET_TIMEOUT attribute of the LOG_ARCHIVE_DEST_n parameter for each SYNC standby destination.

For example, in an Oracle Data Guard configuration that has two standbys which receive redo in SYNC mode, you would define LOG_ARCHIVE_DEST_2 and LOG_ARCHIVE_DEST_3 to use SYNC transport with DATA_GUARD_SYNC_LATENCY set to a value of 2 seconds. When redo needs to be sent to the standbys, it is sent to both the LOG_ARCHIVE_DEST_2 and LOG_ARCHIVE_DEST_3 connections in parallel and the LGWR will wait for acknowledgement. If LOG_ARCHIVE_DEST_2 responds with its message acknowledging that the redo has been received first, the LGWR will not wait for LOG_ARCHIVE_DEST_3 to respond for more than 2 extra seconds (DATA_GUARD_SYNC_LATENCY). If LOG_ARCHIVE_DEST_3 does not respond within those 2 seconds the LGWR disconnects from LOG_ARCHIVE_DEST_3, putting it into the error state. Redo continues to be sent to LOG_ARCHIVE_DEST_2 as usual. After the duration of the REOPEN attribute for LOG_ARCHIVE_DEST_3 has elapsed, the LGWR reconnects to LOG_ARCHIVE_DEST_3 and redo transfer continues. In this manner, the maximum impact of having the second SYNC standby would be restricted to 2 seconds more than having only one SYNC standby, while allowing the Oracle Data Guard configuration to maintain the desired protection level.

LGWR will never wait longer than the value of the NET_TIMEOUT attribute of the LOG_ARCHIVE_DEST_n parameter, regardless of the value of DATA_GUARD_SYNC_LATENCY.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "LOG_ARCHIVE_DEST_*n*"

1.70 DATA_TRANSFER_CACHE_SIZE

`DATA_TRANSFER_CACHE_SIZE` sets the size of the data transfer cache (in bytes) used to receive data blocks (typically from a primary database in an Oracle Data Guard environment) for consumption by an instance during execution of an `RMAN RECOVER ... NONLOGGED BLOCK` command.

Property	Description
Parameter type	Big integer
Syntax	`DATA_TRANSFER_CACHE_SIZE = integer [K | M | G]`
Default value	If `SGA_TARGET` is set, then if `DATA_TRANSFER_CACHE_SIZE` is not specified, the default size of the data transfer cache is 0 (internally determined by Oracle Database). If `SGA_TARGET` is set and `DATA_TRANSFER_CACHE_SIZE` is specified, then the user-specified value indicates the minimum value for the data transfer cache. If `SGA_TARGET` is not set, the data transfer cache will not be available.
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	0 – 512M, rounded up to the next granule size
Basic	No
Oracle RAC	Multiple instances can have different values.

This initialization parameter needs to be set only for databases that do not use Automatic Memory Management and that use the `RMAN RECOVER ... NONLOGGED BLOCK` command.

1.71 DB_*n*K_CACHE_SIZE

`DB_nK_CACHE_SIZE` (where *n* = 2, 4, 8, 16, 32) specifies the size of the cache for the *n*K buffers.

Property	Description
Parameter type	Big integer
Syntax	`DB_[2 | 4 | 8 | 16 | 32]K_CACHE_SIZE = integer [K | M | G]`
Default value	`0` (additional block size caches are not configured by default)
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	Minimum: `0` (values greater than zero are automatically modified to be either the granule size * number of processor groups, or 4 MB * number of CPUs, whichever is greater) Maximum: operating system-dependent
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

You can set this parameter only when DB_BLOCK_SIZE has a value other than *n*K. For example, if DB_BLOCK_SIZE=4096, then you cannot specify the parameter DB_4K_CACHE_SIZE (because the size for the 4 KB block cache is already specified by DB_CACHE_SIZE).

Do not set this parameter to zero if there are any online tablespaces with an *n*K block size.

Operating system-specific block size restrictions apply. For example, you cannot set DB_32K_CACHE_SIZE if the operating system's maximum block size is less than 32 KB. Also, you cannot set DB_2K_CACHE_SIZE if the minimum block size is greater than 2 KB.

See Also

Your operating system-specific Oracle documentation for more information on block size restrictions

1.72 DB_BIG_TABLE_CACHE_PERCENT_TARGET

DB_BIG_TABLE_CACHE_PERCENT_TARGET specifies the cache section target size for automatic big table caching, as a percentage of the buffer cache.

Property	Description
Parameter type	String
Syntax	DB_BIG_TABLE_CACHE_PERCENT_TARGET = *string*
Default value	0
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No
Oracle RAC	Multiple instances can have different values, but it is recommended to keep the big table cache section size uniform.

Automatic big table caching enables parallel queries and serial queries to use the buffer cache, which enhances the in-memory query capabilities of Oracle Database. Automatic big table caching is designed primarily to enhance performance for data warehouse workloads, but it also improves performance in mixed workloads.

Starting in Oracle Database 12*c* Release 1 (12.1.0.2), table scans can use a different algorithm in the following scenarios:

- Parallel queries:

  In single-instance and Oracle Real Application Clusters (Oracle RAC) databases, parallel queries can use the automatic big table cache when the DB_BIG_TABLE_CACHE_PERCENT_TARGET initialization parameter is set to a nonzero value, and the PARALLEL_DEGREE_POLICY initialization parameter is set to AUTO or ADAPTIVE.

- Serial queries:

  In a single-instance configuration only, serial queries can use the automatic big table cache when the DB_BIG_TABLE_CACHE_PERCENT_TARGET initialization parameter is set to a nonzero value.

Copyright © 2002, 2026, Oracle and/or its affiliates.

When a nonzero value is specified for the DB_BIG_TABLE_CACHE_PERCENT_TARGET parameter, the value indicates the percentage of the buffer cache to reserve for the big table cache. The largest value that can be specified with the DB_BIG_TABLE_CACHE_PERCENT_TARGET parameter is 90, which reserves 10% of the buffer cache for usage besides table scans.

The default value of this parameter is 0. Therefore, automatic big table caching is not enabled by default. When automatic big table caching is not enabled, queries might run using the least recently used (LRU) mechanism for cached reads, or might decide to use direct reads for the table scan.

If a large table is about the size of the combined size of the big table cache of all instances, the table will be partitioned and cached or mostly cached on all instances. With in-memory parallel query, this could eliminate most disk reads for queries on the table, or the database could intelligently read from disk only for the portion of the table that does not fit in the big table cache. If the big table cache cannot cache all the tables to be scanned, only the most frequently accessed tables will be cached, and the rest will be read via direct read automatically.

Use these guidelines when setting the parameter:

- If you do not enable automatic degree of parallelism (DOP) in your Oracle RAC environment, do not set this parameter because the big table cache is not used in that situation.
- When setting this parameter, consider the workload mix: how much of the workload is for OLTP; insert, update, and random access; and how much of the workload involves table scans. Because data warehouse workloads often perform large table scans, you may consider giving the big table cache section a higher percentage of buffer cache space for data warehouses.
- This parameter can be dynamically changed if the workload changes. The change could take some time to reach the target (depending on the current workload) because buffer cache memory might be actively used at that time.

Note

Automatic big table caching uses temperature and object-based algorithms to track medium and big tables. Oracle will cache very small tables, but they will not be tracked by automatic big table caching.

See Also

- "PARALLEL_DEGREE_POLICY"
- See "V$BT_SCAN_CACHE" and "V$BT_SCAN_OBJ_TEMPS" for more information about the big table cache
- *Oracle Database VLDB and Partitioning Guide* for more information about this parameter and about automatic big table caching

1.73 DB_BLOCK_BUFFERS

DB_BLOCK_BUFFERS specifies the number of database buffers in the buffer cache.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	No
Modifiable in a PDB	No
Range of values	`50` to an operating system-specific maximum
Basic	No
Oracle RAC	Multiple instances can have different values, and you can change the values as needed.

> **Note**
>
> `DB_BLOCK_BUFFERS` cannot be combined with the dynamic `DB_CACHE_SIZE` parameter; combining these parameters in the same parameter file will produce an error.

`DB_BLOCK_BUFFERS` is one of several parameters that contribute to the total memory requirements of the SGA of an instance.

This parameter, together with `DB_BLOCK_SIZE`, determines the total size of the buffer cache. Effective use of the buffer cache can greatly reduce the I/O load on the database. `DB_BLOCK_SIZE` can be specified only when the database is first created, so you use `DB_BLOCK_BUFFERS` to control the size of the buffer cache.

> **Note**
>
> The `DB_BLOCK_BUFFERS` initialization parameter is deprecated. It is maintained for backward compatibility.
>
> Oracle recommends that you use `DB_CACHE_SIZE` instead.

> **See Also**
>
> "DB_CACHE_SIZE"

1.74 DB_BLOCK_CHECKING

`DB_BLOCK_CHECKING` specifies whether Oracle Database performs block checking for database blocks.

Property	Description
Parameter type	String
Syntax	`DB_BLOCK_CHECKING = { FALSE | OFF | LOW | MEDIUM | TRUE | FULL }`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	FALSE
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes, with the following restriction: If block checking is enabled for a CDB, then you cannot subsequently disable block checking in any of its PDBs. That is, if the value of DB_BLOCK_CHECKING in a CDB is LOW, MEDIUM, TRUE, or FULL, and you then attempt to set the value of DB_BLOCK_CHECKING in one of its PDBs to FALSE or OFF, an error will occur.
Basic	No

Values

- OFF or FALSE

  No block checking is performed for blocks in user tablespaces. However, semantic block checking for SYSTEM tablespace blocks is always turned on.

- LOW

  Basic block header checks are performed after block contents change in memory (for example, after UPDATE, INSERT or DELETE statements, or after inter-instance block transfers in Oracle RAC).

- MEDIUM

  All LOW checks and full semantic checks are performed for all objects except indexes (whose contents can be reconstructed by a drop+rebuild on encountering a corruption).

- FULL or TRUE

  All LOW and MEDIUM checks and full semantic checks are performed for all objects.

Oracle checks a block by going through the data in the block, making sure it is logically self-consistent. Block checking can often prevent memory and data corruption. Block checking typically causes 1% to 10% overhead in most applications, depending on workload and the parameter value. Specific DML overhead may be higher. The more updates or inserts in a workload, the more expensive it is to turn on block checking. You should set DB_BLOCK_CHECKING to FULL if the performance overhead is acceptable.

For backward compatibility, the use of FALSE (implying OFF) and TRUE (implying FULL) is preserved.

> ⚠ **Caution**
>
> Before enabling block checking with this parameter, Oracle recommends that you detect and repair any logical corruptions in the database. Otherwise, a block that contains logical corruption will be marked as "soft corrupt" after block checking is enabled and the block is modified by a DML statement. This will result in ORA-1578 errors and the block will be unreadable. For more information about detecting and repairing logical corruptions, see *Oracle Database Backup and Recovery User's Guide*.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information about this parameter

1.75 DB_BLOCK_CHECKSUM

`DB_BLOCK_CHECKSUM` determines whether `DBWn` and the direct loader will calculate a checksum (a number calculated from all the bytes stored in the block) and store it in the cache header of every data block when writing it to disk.

Property	Description
Parameter type	String
Syntax	`DB_BLOCK_CHECKSUM = { OFF | FALSE | TYPICAL | TRUE | FULL }`
Default value	`TYPICAL`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Basic	No

Checksums are verified when a block is read - only if this parameter is `TYPICAL` or `FULL` and the last write of the block stored a checksum. In `FULL` mode, Oracle also verifies the checksum before a change application from update/delete statements and recomputes it after the change is applied. In addition, Oracle gives every log block a checksum before writing it to the current log.

Most of the log block checksum is done by the generating foreground processes, while the LGWR or the LGWR worker processes (LG*nn* processes) perform the rest of the work, for better CPU and cache efficiency.

If this parameter is set to `OFF`, `DBWn` calculates checksums only for the `SYSTEM` tablespace, but not for user tablespaces. In addition, no log checksum is performed when this parameter is set to `OFF`.

Checksums allow Oracle to detect corruption caused by underlying disks, storage systems, or I/O systems. If set to `FULL`, `DB_BLOCK_CHECKSUM` also catches in-memory corruptions and stops them from making it to the disk. Turning on this feature in `TYPICAL` mode causes only an additional 1% to 2% overhead. In the `FULL` mode it causes 4% to 5% overhead. Oracle recommends that you set `DB_BLOCK_CHECKSUM` to `TYPICAL`.

For backward compatibility the use of `TRUE` (implying `TYPICAL`) and `FALSE` (implying `OFF`) values is preserved.

> **See Also**
>
> *Oracle Database Backup and Recovery Reference* for more information about this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.76 DB_BLOCK_SIZE

DB_BLOCK_SIZE specifies (in bytes) the size of Oracle database blocks.

Property	Description
Parameter type	Integer
Default value	8192
Modifiable	No
Modifiable in a PDB	No
Range of values	2048 to 32768, but your operating system may have a narrower range
Basic	Yes
Oracle RAC	You must set this parameter for every instance, and multiple instances must have the same value.

> **Note**
>
> Set this parameter at the time of database creation. Do not alter it afterward.

Typical values for DB_BLOCK_SIZE are 4096 and 8192. The value of this parameter must be a multiple of the physical block size at the device level.

The value for DB_BLOCK_SIZE in effect at the time you create the database determines the size of the blocks. The value must remain set to its initial value.

For Oracle Real Application Clusters (Oracle RAC), this parameter affects the maximum value of the FREELISTS storage parameter for tables and indexes. Oracle uses one database block for each freelist group. Decision support system (DSS) and data warehouse database environments tend to benefit from larger block size values.

> **Note**
>
> 64-bit operating systems support a maximum DB_BLOCK_SIZE value of 32768

> **See Also**
>
> - *Oracle Database Administrator's Guide* for information on setting this parameter
> - *Oracle Database SQL Language Reference* for information on freelist groups

1.77 DB_CACHE_ADVICE

DB_CACHE_ADVICE enables or disables statistics gathering used for predicting behavior with different cache sizes through the V$DB_CACHE_ADVICE performance view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	String
Syntax	DB_CACHE_ADVICE = { ON | READY | OFF }
Default value	If STATISTICS_LEVEL is set to TYPICAL or ALL, then ON If STATISTICS_LEVEL is set to BASIC, then OFF
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No

Values

- OFF

  Advisory is turned off and the memory for the advisory is not allocated.

- READY

  Advisory is turned off but the memory for the advisory remains allocated. Allocating the memory before the advisory is actually turned on avoids the risk of an error when you switch the parameter to ON.

  If the parameter is switched to this state from ON, the contents of the view are preserved and the memory for the advisory is retained.

  If the parameter is switched to this state from OFF, you may get an error.

- ON

  Advisory is turned on. CPU and memory overheads are incurred. Attempting to set the parameter to this state when it is already in the OFF state may result in an error. Otherwise, the view (V$DB_CACHE_ADVICE) is reset and statistics are gathered to the newly refreshed view.

  If the parameter is in the READY state, you can set it to ON without any errors because the memory is already allocated. The view is reset and statistics are displayed in the newly refreshed view.

See Also

"V$DB_CACHE_ADVICE"

1.78 DB_CACHE_SIZE

DB_CACHE_SIZE specifies the size of the DEFAULT buffer pool for buffers with the primary block size (the block size defined by the DB_BLOCK_SIZE initialization parameter).

Note

Oracle recommends against setting this parameter in a PDB. The only shared memory sizing parameter that should be set in a PDB is SGA_TARGET, which specifies the maximum SGA that the PDB can use at any time.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	Big integer
Syntax	`DB_CACHE_SIZE = integer [K | M | G]`
Default value	If `SGA_TARGET` is set: If the parameter is not specified, then the default is `0` (internally determined by the Oracle Database). If the parameter is specified, then the user-specified value indicates a minimum value for the memory pool. If `SGA_TARGET` is not set, then the default is either 48 MB or 4 MB * number of CPUs, whichever is greater
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

The value must be at least `4M * number of cpus` (smaller values are automatically rounded up to this value). A user-specified value larger than this is rounded up to the nearest granule size. A value of zero is invalid because it is needed for the `DEFAULT` memory pool of the primary block size, which is the block size for the `SYSTEM` tablespace.

See Also

- "SGA_TARGET"
- *Oracle Database Performance Tuning Guide* and *Oracle Database Administrator's Guide* for more information on setting this parameter
- *Oracle Multitenant Administrator's Guide* for more information about the initialization parameters that control the memory usage of PDBs

1.79 DB_CREATE_FILE_DEST

`DB_CREATE_FILE_DEST` specifies the default location for Oracle-managed datafiles.

Property	Description
Parameter type	String
Syntax	`DB_CREATE_FILE_DEST = directory | disk group`
Default value	There is no default value.
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	Yes

This location is also used as the default location for Oracle-managed control files and online redo logs if none of the `DB_CREATE_ONLINE_LOG_DEST_n` initialization parameters are specified.

If a file system directory is specified as the default location, then the directory must already exist; Oracle does not create it. The directory must have appropriate permissions that allow Oracle to create files in it. Oracle generates unique names for the files, and a file thus created is an Oracle Managed File.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information on setting this parameter and on Oracle Managed Files

1.80 DB_CREATE_ONLINE_LOG_DEST_*n*

`DB_CREATE_ONLINE_LOG_DEST_`*n* (where *n* = 1, 2, 3, ... 5) specifies the default location for Oracle-managed control files and online redo logs.

Property	Description
Parameter type	String
Syntax	`DB_CREATE_ONLINE_LOG_DEST_[1 | 2 | 3 | 4 | 5] = directory | disk group`
Default value	There is no default value.
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	Yes

If more than one `DB_CREATE_ONLINE_LOG_DEST_`*n* parameter is specified, then the control file or online redo log is multiplexed across the locations of the other `DB_CREATE_ONLINE_LOG_DEST_`*n* parameters. One member of each online redo log is created in each location, and one control file is created in each location.

Specifying at least two parameters provides greater fault tolerance for the control files and online redo logs if one of the locations should fail.

If a file system directory is specified as the default location, then the directory must already exist; Oracle does not create it. The directory must have appropriate permissions that allow Oracle to create files in it. Oracle generates unique names for the files, and a file thus created is an Oracle Managed File.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information on setting this parameter and on Oracle Managed Files

1.81 DB_DOMAIN

In a distributed database system, `DB_DOMAIN` specifies the logical location of the database within the network structure.

Property	Description
Parameter type	String
Syntax	`DB_DOMAIN = domain_name`
Default value	There is no default value.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable	No
Modifiable in a PDB	Yes
Range of values	Any legal string of name components, separated by periods and up to 128 characters long (including the periods).
Basic	Yes
Oracle RAC	You must set this parameter for every instance, and multiple instances must have the same value.

You should set this parameter if this database is or ever will be part of a distributed system. The value consists of the extension components of a global database name, consisting of valid identifiers (any alphanumeric ASCII characters), separated by periods.

> **Note**
>
> Oracle recommends that you specify `DB_DOMAIN` as a unique string for all databases in a domain.

This parameter allows one department to create a database without worrying that it might have the same name as a database created by another department. If one sales department's `DB_DOMAIN` is `JAPAN.EXAMPLE.COM`, then their `SALES` database (`SALES.JAPAN.EXAMPLE.COM`) is uniquely distinguished from another database with `DB_NAME = SALES` but with `DB_DOMAIN = US.EXAMPLE.COM`.

If you omit the domains from the name of a database link, Oracle expands the name by qualifying the database with the domain of your local database as it currently exists in the data dictionary, and then stores the link name in the data dictionary. `DB_DOMAIN` should start with an alphabetic character and exclude these characters:

```
"!@%^&*()+=\\|`~[{]};:'\",<>/?"
```

> **See Also**
>
> - *Oracle Database Administrator's Guide* for more information on setting this parameter
> - The data dictionary view "GLOBAL_NAME"

1.82 DB_FILE_MULTIBLOCK_READ_COUNT

`DB_FILE_MULTIBLOCK_READ_COUNT` specifies the maximum number of blocks read in one I/O operation during a sequential scan.

Property	Description
Parameter type	Integer

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	The default value corresponds to the maximum I/O size that can be efficiently performed and is platform-dependent
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	Operating system-dependent
Basic	No

`DB_FILE_MULTIBLOCK_READ_COUNT` is one of the parameters you can use to minimize I/O during table scans. The total number of I/Os needed to perform a full table scan depends on such factors as the size of the table, the multiblock read count, and whether parallel execution is being utilized for the operation.

The default value of this parameter is a value that corresponds to the maximum I/O size that can be performed efficiently. This value is platform-dependent and is 1MB for most platforms.

Because the parameter is expressed in blocks, it will be set to a value that is equal to the maximum I/O size that can be performed efficiently divided by the standard block size. Note that if the number of sessions is extremely large the multiblock read count value is decreased to avoid the buffer cache getting flooded with too many table scan buffers.

Even though the default value may be a large value, the optimizer will not favor large I/Os if you do not set this parameter. It will favor large I/Os only if you explicitly set this parameter to a large value.

Online transaction processing (OLTP) and batch environments typically have values in the range of 4 to 16 for this parameter. DSS and data warehouse environments tend to benefit most from maximizing the value of this parameter. The optimizer is more likely to choose a full table scan over an index if the value of this parameter is high.

The maximum value is the operating system's maximum I/O size expressed as Oracle blocks ((max I/O size)/`DB_BLOCK_SIZE`). If you set this parameter to a value greater than the maximum, then Oracle uses the maximum.

See Also

Oracle Database Performance Tuning Guide for information about how setting this parameter incorrectly can cause excessive I/O waits for some execution plans

1.83 DB_FILE_NAME_CONVERT

`DB_FILE_NAME_CONVERT` is useful for creating a duplicate database for recovery purposes. It converts the filename of a new datafile on the primary database to a filename on the standby database.

Property	Description
Parameter type	String

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description

Syntax	`DB_FILE_NAME_CONVERT = 'string1' , 'string2' , 'string3' , 'string4' , ...`
	Where:
	• *string1* is the pattern of the primary database filename
	• *string2* is the pattern of the standby database filename
	• *string3* is the pattern of the primary database filename
	• *string4* is the pattern of the standby database filename
	You can enclose each string in single or double quotation marks.
	You can specify as many pairs of primary and standby replacement strings as required. However, starting with Oracle Database 12c Release 2 (12.2), Oracle recommends that you limit the number of pairs to 99.
	Example:
	`DB_FILE_NAME_CONVERT = '/dbs/t1/','/dbs/t1/s','dbs/t2/','dbs/t2/s'`

Default value	There is no default value.
Modifiable	`ALTER SESSION`
Modifiable in a PDB	No
Basic	No

If you add a datafile to the primary database, you must add a corresponding file to the standby database. When the standby database is updated, this parameter converts the datafile name on the primary database to the datafile name on the standby database. The file on the standby database must exist and be writable, or the recovery process will halt with an error.

Set the value of this parameter to one or more pairs of strings. The first string in a pair is the pattern found in the datafile names on the primary database. The second string in a pair is the pattern found in the datafile names on the standby database.

If you specify an odd number of strings (the last string has no corresponding replacement string), an error is returned during startup. If the filename being converted matches more than one pattern in the pattern/replace string list, the first matched pattern takes effect. There is no limit on the number of pairs that you can specify in this parameter (other than the hard limit of the maximum length of multivalue parameters). However, starting with Oracle Database 12c Release 2 (12.2), Oracle recommends that you limit the number of pairs to 99.

You can also use `DB_FILE_NAME_CONVERT` to rename the datafiles in the clone control file when setting up a clone database during tablespace point-in-time recovery.

ⓘ See Also

- *Oracle Database Backup and Recovery User's Guide* for more information about database backup and recovery
- *Oracle Data Guard Concepts and Administration* for more information about Oracle Data Guard

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.84 DB_FILES

DB_FILES specifies the maximum number of database files that can be opened for this database.

Property	Description
Parameter type	Integer
Default value	200
Modifiable	No
Modifiable in a PDB	Yes
Range of values	Minimum: the largest among the absolute file numbers of the datafiles in the database Maximum: operating system-dependent
Basic	No
Oracle RAC	Multiple instances must have the same value.

The maximum valid value is the maximum number of files, subject to operating system constraint, that will ever be specified for the database, including files to be added by ADD DATAFILE statements.

If you increase the value of DB_FILES, then you must shut down and restart all instances accessing the database before the new value can take effect. If you have a primary and standby database, then they should have the same value for this parameter.

See Also

- *Oracle Real Application Clusters Administration and Deployment Guide* for information on setting this parameter in an Oracle RAC environment
- Your operating system-specific Oracle documentation for the default value of this parameter

1.85 DB_FLASH_CACHE_FILE

DB_FLASH_CACHE_FILE specifies file name(s) for the flash memory or disk group representing a collection of flash memory, for use with Database Smart Flash Cache.

Property	Description
Parameter type	String
Syntax	DB_FLASH_CACHE_FILE = *filename [,filename]... | disk group*
Default value	There is no default value.
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

You can specify up to 16 file names for flash memory devices. For example, if there are three flash raw devices:

```
db_flash_cache_file = /dev/raw/sda, /dev/raw/sdb, /dev/raw/sdc
```

Specifying this parameter without also specifying the DB_FLASH_CACHE_SIZE initialization parameter is not allowed.

See Also

"DB_FLASH_CACHE_SIZE"

1.86 DB_FLASH_CACHE_SIZE

DB_FLASH_CACHE_SIZE specifies the size of the Database Smart Flash Cache (flash cache). This parameter may only be specified at instance startup.

Property	Description
Parameter type	Big integer
Syntax	DB_FLASH_CACHE_SIZE = *integer* [K | M | G | T] [, *integer* [K | M | G | T]]...
Default value	0
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	0 to (DB_BLOCK_SIZE * 256 MB) • If DB_BLOCK_SIZE = 2 KB, then 0 to 512 GB • If DB_BLOCK_SIZE = 4 KB, then 0 to 1 TB • If DB_BLOCK_SIZE = 8 KB, then 0 to 2 TB • If DB_BLOCK_SIZE = 16 KB, then 0 to 4 TB • If DB_BLOCK_SIZE = 32 KB, then 0 to 8 TB
Basic	No

You can specify up to 16 file sizes, for each of the flash memory devices specified with DB_FLASH_CACHE_FILE. For example, if there are three flash raw devices, you can specify the sizes of each device as follows:

```
db_flash_cache_file = /dev/raw/sda, /dev/raw/sdb, /dev/raw/sdc
db_flash_cache_size = 32G, 32G, 64G
```

If your flash cache consists of one flash cache device, you can dynamically change this parameter to 0 for that flash cache device (disabling the flash cache) after the database is started. You can then reenable the flash cache by setting this parameter for the device back to the original value when the database was started. Dynamic resizing of DB_FLASH_CACHE_SIZE or reenabling flash cache to a different size is not supported.

If your flash cache includes multiple flash cache devices, you can dynamically change the parameter to 0 for a particular flash cache device (turning it off) after the database is started. You can then reenable that flash cache device by setting this parameter for the device back to the original value it had when the database was started (turning it back on).

Copyright © 2002, 2026, Oracle and/or its affiliates.

For example, to turn off the `/dev/raw/sdb` flash cache device:

```
db_flash_cache_file = /dev/raw/sda, /dev/raw/sdb, /dev/raw/sdc
db_flash_cache_size = 32G, 0, 64G
```

And, to turn the `/dev/raw/sdb` flash cache device back on again:

```
db_flash_cache_file = /dev/raw/sda, /dev/raw/sdb, /dev/raw/sdc
db_flash_cache_size = 32G, 32G, 64G
```

> **See Also**
>
> "DB_FLASH_CACHE_FILE"

1.87 DB_FLASHBACK_RETENTION_TARGET

`DB_FLASHBACK_RETENTION_TARGET` specifies the upper limit (in minutes) on how far back in time the database may be flashed back.

Property	Description
Parameter type	Integer
Default value	`1440` (minutes)
Modifiable	`ALTER SYSTEM ... SID='*'`
Modifiable in a PDB	No
Range of values	`0` to 2^{31} - 1
Basic	No

How far back one can flashback a database depends on how much flashback data Oracle has kept in the fast recovery area.

> **See Also**
>
> *Oracle Data Guard Broker* for enabling Flashback Database on Oracle Data Guard primary and target standby databases

1.88 DB_INDEX_COMPRESSION_INHERITANCE

`DB_INDEX_COMPRESSION_INHERITANCE` dictates how index creation inherits compression attributes.

Property	Description
Parameter type	String
Syntax	`DB_INDEX_COMPRESSION_INHERITANCE = { TABLESPACE | TABLE | ALL | NONE }`
Default value	`NONE`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	You must set this parameter for every instance, and multiple instances must have the same value.

Values:

- TABLESPACE

  Index inheritance is based on tablespace attributes.

- TABLE

  Index inheritance is based on table attributes.

- ALL

  Index inheritance is based on table settings. However, if the table does not have default compression attributes, then index inheritance is based on tablespace attributes.

- NONE

  There is no index inheritance from the table or tablespace.

If the CREATE INDEX statement specifies compression attributes, then the value of DB_INDEX_COMPRESSION_INHERITANCE is ignored, and there is no inheritance from the table or tablespace. When there is table inheritance or tablespace inheritance (and the tablespace does not have an explicit index compression clause), then the following matrix is used:

Table/Tablespace Level Compression	Inherited Compression for Index
OLTP	ADVANCED LOW
QUERY LOW, QUERY HIGH	ADVANCED HIGH
ARCHIVE LOW, ARCHIVE HIGH	ADVANCED HIGH

Note

The value of DB_INDEX_COMPRESSION_INHERITANCE is ignored when creating an index owned by SYS. Such indexes do not inherit index compression attributes from tables or tablespaces. If you want to compress an index owned by SYS, you must explicitly specify the index compression attributes in the CREATE INDEX or ALTER INDEX statement.

See Also

Oracle Database Administrator's Guide for more information about index compression

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.89 DB_KEEP_CACHE_SIZE

DB_KEEP_CACHE_SIZE specifies the size of the KEEP buffer pool.

Property	Description
Parameter type	Big integer
Syntax	DB_KEEP_CACHE_SIZE = *integer* [K | M | G]
Default value	0 (DB_KEEP_CACHE_SIZE is not configured by default)
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	Minimum: 0 (values greater than zero are automatically modified to be either the granule size * number of processor groups, or 4 MB * number of CPUs, whichever is greater) Maximum: operating system-dependent
Basic	No

The size of the buffers in the KEEP buffer pool is the primary block size (the block size defined by the DB_BLOCK_SIZE initialization parameter).

See Also

- "DB_RECYCLE_CACHE_SIZE"
- *Oracle Database Performance Tuning Guide* for information on setting these parameters and on using multiple buffer pools

1.90 DB_LOST_WRITE_PROTECT

DB_LOST_WRITE_PROTECT enables or disables lost write detection.

Property	Description
Parameter type	String
Syntax	DB_LOST_WRITE_PROTECT = { AUTO | TYPICAL | FULL | NONE }
Default value	Starting with Oracle Database 19c, Release Update 19.26: AUTO Oracle Database 19c, Release Update 19.25 or earlier: NONE
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No
Oracle RAC	In Oracle RAC instances, the parameter value is systemwide.

A data block lost write occurs when an I/O subsystem acknowledges the completion of the block write, while in fact the write did not occur in the persistent storage.

- AUTO

Copyright © 2002, 2026, Oracle and/or its affiliates.

This setting is available starting with Oracle Database 19c, Release Update 19.26. When this parameter is set to AUTO on a primary database, the instance automatically decides whether it logs buffer cache reads in the redo log or not, depending on the status of the standby databases. Specifically, the primary database only logs buffer cache reads if physical standby databases with real time redo apply exist.

When this parameter is set to AUTO on a standby database, the instance will automatically decide whether it incurs additional performance overhead to perform lost write detection or not, depending on whether apply is keeping up. If apply lag is beyond the reasonable threshold, the standby database will skip lost write protection temporarily until redo apply catches up with primary again, to ensure the lowest Data Guard role transition timings.

The reasonable threshold is determined as follows:

- If Fast Start Failover (FSFO) is configured through Data Guard Broker, based on apply lag threshold, then the threshold is the lesser of 60 seconds and two-thirds of the FSFO apply lag threshold.
- Otherwise, the threshold is 60 seconds.

- TYPICAL

  When this parameter is set to TYPICAL on a primary database, the instance logs buffer cache reads for read/write tablespaces in the redo log, which is necessary for detection of lost writes.

  When this parameter is set to TYPICAL on a standby database, or on a primary database during media recovery, the instance performs lost write detection, regardless of higher redo apply lags that may impact Data Guard role transition timings.

- FULL

  When this parameter is set to FULL on a primary database, the instance logs reads for read-only tablespaces and read/write tablespaces.

  When this parameter is set to FULL on a standby database, or on a primary database during media recovery, the instance performs lost write detection.

- NONE

  When this parameter is set to NONE on either a primary database or a standby database, no lost write detection functionality is enabled.

1.91 DB_NAME

DB_NAME specifies a database identifier of up to 8 characters.

Property	Description
Parameter type	String
Syntax	DB_NAME = *database_name*
Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	No
Basic	Yes
Oracle RAC	You must set this parameter for every instance. Multiple instances must have the same value.

Copyright © 2002, 2026, Oracle and/or its affiliates.

This parameter must be specified and must correspond to the name specified in the `CREATE DATABASE` statement.

If you have multiple databases, the value of this parameter should match the Oracle instance identifier of each one to avoid confusion with other databases running on the system. The value of `DB_NAME` must be the same in both the standby and production initialization parameter files.

The database name specified in either the `STARTUP` command or the `ALTER DATABASE ... MOUNT` statement for each instance of the cluster database must correspond to the `DB_NAME` initialization parameter setting.

The following characters are valid in a database name: alphanumeric characters, underscore (_), number sign (#), and dollar sign ($). No other characters are valid. The database name must start with an alphabetic character. Oracle removes double quotation marks before processing the database name. Therefore you cannot use double quotation marks to embed other characters in the name. The database name is case insensitive.

See Also

Oracle Database Administrator's Guide and *Oracle Real Application Clusters Administration and Deployment Guide* for more information on setting this parameter

1.92 DB_PERFORMANCE_PROFILE

`DB_PERFORMANCE_PROFILE` is used by Resource Manager and Exadata I/O Resource Management (IORM) to provide easier resource management for CDBs and PDBs.

Property	Description
Parameter type	String
Syntax	`DB_PERFORMANCE_PROFILE = string`
Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The same value should be set on all instances.

On non-Exadata systems, this parameter can be set at the PDB level for use by Resource Manager. On Exadata systems, this parameter can be set at the PDB level for use by Resource Manager, and at the CDB level for use by IORM.

- On non-Exadata systems:
  - You can set this parameter at the PDB level to specify a **performance profile** for Resource Manager. In this case, a CDB resource plan specifies different resource controls for a performance profile using the `DBMS_RESOURCE_MANAGER.CREATE_CDB_PROFILE_DIRECTIVE` procedure, and a PDB with a matching performance profile will use the specified controls.
  - There is no practical use for setting this parameter at the CDB level on a non-Exadata system.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- On Exadata systems:
  - Similar to non-Exadata systems, you can set this parameter at the PDB level to specify a **performance profile** for Resource Manager.
  - You can also set this parameter at the CDB level to specify an **IORM profile**. This profile is pushed down to the Exadata storage cells, where it is used by IORM to manage I/O allocation.
  - Note that setting this parameter at the CDB level identifies the entire database to IORM and has no effect on how Exadata Smart Flash Cache is divided among PDBs. To control flash cache allocation between PDBs, you must use Resource Manager, that is, you must either create a CDB resource plan and enable that plan, or, if using profile directives, specify a performance profile for each PDB. This instructs Resource Manager to enforce per-PDB flash cache minimums and limits, while IORM continues to manage I/O resources at the CDB level.

The value of this parameter can be up to 30 characters and is not case sensitive.

- *Oracle Multitenant Administrator's Guide* to learn more about how Resource Manager uses this parameter
- *Oracle Exadata System Software User's Guide* to learn more about how IORM uses this parameter

1.93 DB_RECOVERY_FILE_DEST

`DB_RECOVERY_FILE_DEST` specifies the default location for the fast recovery area.

Property	Description
Parameter type	String
Syntax	`DB_RECOVERY_FILE_DEST = directory | disk group`
Default value	There is no default value.
Modifiable	`ALTER SYSTEM ... SID='*'`
Modifiable in a PDB	No
Basic	Yes
Oracle RAC	You must set this parameter for every instance, and multiple instances must have the same value.

The fast recovery area contains multiplexed copies of current control files and online redo logs, as well as archived redo logs, flashback logs, and RMAN backups.

Specifying this parameter without also specifying the `DB_RECOVERY_FILE_DEST_SIZE` initialization parameter is not allowed.

See Also

- *Oracle Database Backup and Recovery User's Guide* for information on setting up and configuring the fast recovery area
- "DB_RECOVERY_FILE_DEST_SIZE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.94 DB_RECOVERY_FILE_DEST_SIZE

DB_RECOVERY_FILE_DEST_SIZE specifies (in bytes) the hard limit on the total space to be used by target database recovery files created in the fast recovery area.

Property	Description
Parameter type	Big integer
Syntax	DB_RECOVERY_FILE_DEST_SIZE = *integer* [K | M | G]
Default value	0
Modifiable	ALTER SYSTEM ... SID='*'
Modifiable in a PDB	No
Basic	Yes
Oracle RAC	You must set this parameter for every instance, and multiple instances must have the same value.

Note that neither block 0 nor the OS block header of each Oracle file is included in this size. Allow an extra 10% for this data when computing the actual disk usage required for the fast recovery area.

See Also

- *Oracle Database Backup and Recovery User's Guide* for information on setting up and configuring the fast recovery area
- "DB_RECOVERY_FILE_DEST"

1.95 DB_RECYCLE_CACHE_SIZE

DB_RECYCLE_CACHE_SIZE specifies the size of the RECYCLE buffer pool.

Property	Description
Parameter type	Big integer
Syntax	DB_RECYCLE_CACHE_SIZE = *integer* [K | M | G]
Default value	0 (DB_RECYCLE_CACHE_SIZE is not configured by default)
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	Minimum: 0 (values greater than zero are automatically modified to be either the granule size * number of processor groups, or 4 MB * number of CPUs, whichever is greater) Maximum: operating system-dependent
Basic	No

The size of the buffers in the RECYCLE pool is the primary block size (the block size defined by the DB_BLOCK_SIZE initialization parameter).

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DB_KEEP_CACHE_SIZE"
- *Oracle Database Performance Tuning Guide* for information on setting these parameters and on using multiple buffer pools

1.96 DB_SECUREFILE

`DB_SECUREFILE` specifies whether to treat LOB files as SecureFiles.

Property	Description
Parameter type	String
Syntax	`DB_SECUREFILE = { NEVER | PERMITTED | PREFERRED | ALWAYS | IGNORE }`
Default value	`PERMITTED if the COMPATIBLE initialization parameter is set to 11.2.0.1, 11.2.0.2, or 11.2.0.3, or PREFERRED if the COMPATIBLE initialization parameter is set to 12.0.0.0 or higher`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

Values

- `NEVER`

  Any LOBs that are specified as SecureFiles are created as BasicFiles LOBs. All SecureFiles-specific storage options and features (for example, compress, encrypt, deduplicate) will cause an exception. The BasicFiles LOB defaults will be used for storage options not specified.

- `PERMITTED`

  LOBs are allowed to be created as SecureFiles.

- `PREFERRED`

  All LOBs are created as SecureFiles unless `BASICFILE` is explicitly specified in the LOB storage clause or the tablespace is a Manual Segment Space Management tablespace. When `PREFERRED` is set, cases where `BASICFILE` would otherwise be inherited from the partition or column level LOB storage are ignored; the LOBs will be created as SecureFiles instead.

- `ALWAYS`

  Attempts to create all LOBs as SecureFiles LOBs but creates any LOBs not in an Automatic Segment Space Managed (ASSM) tablespace as BasicFiles LOBs, unless `SECUREFILE` is explicitly specified. Any BasicFiles LOB storage options that are specified will be ignored and the SecureFiles LOB defaults will be used for all storage options not specified.

- `IGNORE`

  The `SECUREFILE` keyword and all SecureFiles options are ignored.

Copyright © 2002, 2026, Oracle and/or its affiliates.

If the COMPATIBLE initialization parameter is not set to 11.1.0 or higher, then LOBs are not treated as SecureFiles.

If there is a LOB column with two partitions (one that has a tablespace for which ASSM is enabled and one that has a tablespace for which ASSM is not enabled), then LOBs in the partition with the ASSM-enabled tablespace will be treated as SecureFiles and LOBs in the other partition will be treated as BasicFiles.

If an application is not certified with SecureFiles, the value of the DB_SECUREFILE initialization parameter can be set to PERMITTED to get the default behavior of Oracle Database 11*g*.

See Also

Oracle Database SecureFiles and Large Objects Developer's Guide for an example of setting this parameter using the ALTER SYSTEM statement

1.97 DB_ULTRA_SAFE

DB_ULTRA_SAFE sets the default values for other parameters that control protection levels.

Property	Description
Parameter type	String
Syntax	DB_ULTRA_SAFE = { OFF | DATA_ONLY | DATA_AND_INDEX }
Default value	OFF
Modifiable	No
Modifiable in a PDB	No
Basic	No

Values

- OFF

  When any of DB_BLOCK_CHECKING, DB_BLOCK_CHECKSUM, or DB_LOST_WRITE_PROTECT are explicitly set, no changes are made.
- DATA_ONLY
  - DB_BLOCK_CHECKING will be set to MEDIUM.
  - DB_LOST_WRITE_PROTECT will be set to TYPICAL.
  - DB_BLOCK_CHECKSUM will be set to FULL.
- DATA_AND_INDEX
  - DB_BLOCK_CHECKING will be set to FULL.
  - DB_LOST_WRITE_PROTECT will be set to TYPICAL.
  - DB_BLOCK_CHECKSUM will be set to FULL.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DB_BLOCK_CHECKING"
- "DB_BLOCK_CHECKSUM"
- "DB_LOST_WRITE_PROTECT"

1.98 DB_UNIQUE_NAME

DB_UNIQUE_NAME specifies a globally unique name for the database.

Property	Description
Parameter type	String
Syntax	DB_UNIQUE_NAME = *database_unique_name*
Default value	Database instances: the value of DB_NAME Oracle Automatic Storage Management instances: +ASM
Modifiable	No
Modifiable in a PDB	No
Basic	Yes
Oracle RAC	Multiple instances must have the same value.

Databases with the same DB_NAME within the same DB_DOMAIN (for example, copies of a database created for reporting or a physical standby) must have a unique DB_UNIQUE_NAME. Every database's DB_UNIQUE_NAME must be unique within the enterprise.

The value of DB_UNIQUE_NAME can be up to 30 characters and is case insensitive. The following characters are valid in a database name: alphanumeric characters, underscore (_), number sign (#), and dollar sign ($).

Note

DB_UNIQUE_NAME is used by several components within an Oracle instance to default file names or file paths, such as the default for the DG_BROKER_CONFIG_FILE*n* initialization parameter. In these cases, on UNIX platforms, the dollar sign ($) character will be removed from the path or file name, because the character is used by Oracle to define the start of an environment variable substitution within a path or file name.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> As part of their operations, some database tools or utilities create a string that uniquely identifies a database. The string may include the DB_UNIQUE_NAME for a database, and other identifying information for the database, such as the database SID. Oracle Database restricts some identifiers to 30 characters, so using a short DB_UNIQUE_NAME can help prevent ORA-00972 "identifier is too long" messages from database tools and utilities that create a string that includes the DB_UNIQUE_NAME.

> **See Also**
>
> *Oracle Data Guard Concepts and Administration* and *Oracle Database Administrator's Guide* for more information on setting this parameter

1.99 DB_UNRECOVERABLE_SCN_TRACKING

DB_UNRECOVERABLE_SCN_TRACKING enables or disables the tracking of unrecoverable (NOLOGGING) direct-path INSERT and direct-path load operations.

Property	Description
Parameter type	Boolean
Default value	true
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	true | false
Basic	No
Oracle RAC	Multiple instances must have the same value

When the value is set to true, updates are made to the controlfile that maintains the V$DATAFILE.UNRECOVERABLE_CHANGE# and V$DATAFILE.UNRECOVERABLE_TIME columns. When the value is set to false, updates are not made to the controlfile. Setting this parameter to false may improve performance of direct-path NOLOGGING operations.

1.100 DB_WRITER_PROCESSES

DB_WRITER_PROCESSES specifies the initial number of Database Writer Processes for an instance. This parameter is useful for systems that modify data heavily.

Property	Description
Parameter type	Integer

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	1 or CPU_COUNT / 8, whichever is greater. If the number of processor groups is less than 100 but greater than the number of Database Writer Processes, then the number of Database Writer Processes is adjusted to be a multiple of the number of processor groups. If the number of Database Writer Processes is greater than or equal to the number of processor groups, then there is no adjustment. For Oracle Solaris on SPARC systems, the default value of this parameter is calculated using an internal algorithm.
Modifiable	No
Modifiable in a PDB	No
Range of values	1 to 100
Basic	No

There can be 1 to 100 Database Writer Processes. The names of the first 36 Database Writer Processes are DBW0-DBW9 and DBWa-DBWz. The names of the 37th through 100th Database Writer Processes are BW36-BW99.

See Also

- Background Processes for additional information about the Database Writer Process
- *Oracle Database Concepts* for information on when the Database Writer Process writes dirty buffers to disk

1.101 DBFIPS_140

DBFIPS_140 enables Transparent Data Encryption (TDE) and DBMS_CRYPTO PL/SQL package program units to run in a mode compliant to the Federal Information Processing Standard (subsequently known as "FIPS mode").

Property	Description
Parameter type	Boolean
Default value	FALSE
Modifiable	No
Modifiable in a PDB	No
Range of values	TRUE | FALSE
Basic	No
Oracle RAC	All instances must use the same value.

Values:

- TRUE

  Set this parameter to TRUE to use TDE and DBMS_CRYPTO in FIPS mode. This means that only FIPS-compliant algorithms may be used.

Copyright © 2002, 2026, Oracle and/or its affiliates.

This method of configuring FIPS mode is considered a legacy configuration, but it is still supported. Oracle recommends that you instead use the consolidated `FIPS_140` parameter in the `fips.ora` file. See *Oracle Database Security Guide* for more information.

This parameter can be set to `TRUE` during a rolling RAC upgrade. However, verify that TDE is using a FIPS-compliant algorithm before making the change. If you are not using a FIPS-compliant algorithm and you set this parameter to `TRUE`, you will not be able to read the data.

- `FALSE`

  When this parameter is set to `FALSE`, all algorithms (FIPS-compliant or not) may be used. This is the default.

See Also

Oracle Database Security Guide for a table that describes the effect of setting the value of `DBFIPS_140` to `TRUE` or `FALSE` on different platforms

1.102 DBWR_IO_SLAVES

`DBWR_IO_SLAVES` specifies the number of I/O server processes used by the `DBW0` process.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	No
Modifiable in a PDB	No
Range of values	0 to operating system-dependent
Basic	No

`DBWR_IO_SLAVES` is relevant only on systems with only one database writer process (`DBW0`). The `DBW0` process and its server processes always write to disk. By default, the value is `0` and I/O server processes are not used.

If you set `DBWR_IO_SLAVES` to a nonzero value, the number of I/O server processes used by the `ARCH` and `LGWR` processes is set to `4`. However, the number of I/O server processes used by Recovery Manager is set to `4` only if asynchronous I/O is disabled (either your platform does not support asynchronous I/O or `disk_asynch_io` is set to `false`).

Typically, I/O server processes are used to simulate asynchronous I/O on platforms that do not support asynchronous I/O or that implement it inefficiently. However, you can use I/O server processes even when asynchronous I/O is being used. In that case the I/O server processes will use asynchronous I/O.

I/O server processes are also useful in database environments with very large I/O throughput, even if asynchronous I/O is enabled.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "BACKUP_TAPE_IO_SLAVES"
- *Oracle Database Performance Tuning Guide* for more information about this parameter

1.103 DDL_LOCK_TIMEOUT

`DDL_LOCK_TIMEOUT` specifies a time limit for how long DDL statements will wait in a DML lock queue.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	0 to 1,000,000 (in seconds)
Basic	No

A value of zero indicates a status of `NOWAIT`. The maximum value of 1,000,000 seconds will result in the DDL statement waiting forever to acquire a DML lock.

If a lock is not acquired before the timeout period expires, then an error is returned.

See Also

- *Oracle Database Administrator's Guide* for more information about the `DDL_LOCK_TIMEOUT` parameter
- *Oracle Database Development Guide* for information about nonblocking and blocking DDL

1.104 DEFAULT_SHARING

`DEFAULT_SHARING` sets the value of the sharing clause in statements creating objects in an application root.

Property	Description
Parameter type	String
Syntax	`DEFAULT_SHARING = { NONE | METADATA | DATA | EXTENDED DATA }`
Default value	For types of objects that support sharing, `METADATA` is the default. For types of objects that do not support sharing, `NONE` is the default.
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Basic	No
Oracle RAC	All instances must have the same value

Specifying `SHARING=` in the create DDL overrides the value of the `DEFAULT_SHARING` parameter.

Example

Issuing the following `ALTER SYSTEM` statement in an application root sets the default value of the sharing clause to `NONE` in the server parameter file for the application root:

```
ALTER SYSTEM SET DEFAULT_SHARING = NONE SCOPE = SPFILE;
```

1.105 DEFERRED_SEGMENT_CREATION

`DEFERRED_SEGMENT_CREATION` specifies the semantics of deferred segment creation.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true	false`
Basic	No

If set to `true`, then segments for tables and their dependent objects (LOBs, indexes) will not be created until the first row is inserted into the table.

Before creating a set of tables, if it is known that a significant number of them will not be populated, then consider setting this parameter to `true`. This saves disk space and minimizes install time.

1.106 DG_BROKER_CONFIG_FILE*n*

`DG_BROKER_CONFIG_FILE`*n* (where *n* = 1, 2) specifies the names for the Data Guard broker configuration files.

Property	Description
Parameter type	String
Syntax	`DG_BROKER_CONFIG_FILE[1	2] = filename`
Default value	Operating system-dependent
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	One filename
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Every database that is part of a Data Guard broker configuration has two broker configuration files, which contain entries that describe the state and properties of the configuration (such as the sites and databases that are part of the configuration, the roles and properties of each of the databases, and the state of each of the elements of the configuration). Two files are provided to always maintain the last known good state of the configuration.

If DG_BROKER_CONFIG_FILE*n* is not explicitly defined, then it is set to an operating system-specific default value at instance startup. This parameter can be altered only when the Data Guard broker is not running. See "DG_BROKER_START" for information on how to stop and start the broker.

See Also

Oracle Data Guard Broker for more information about setting this parameter

1.107 DG_BROKER_START

DG_BROKER_START enables Oracle to determine whether the Oracle Data Guard broker should be started.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	ALTER SYSTEM...SID='*'
Modifiable in a PDB	No
Range of values	true | false
Basic	No

The Oracle Data Guard broker consists of background processes, foreground processes, and the system global area (SGA). The background and foreground processes exist as long as the instance exists, whenever this parameter is set to true.

If the database is never going to be configured in an Oracle Data Guard broker configuration, then you can leave this parameter unspecified and accept the default value of false. If the database is part of an Oracle Data Guard broker configuration, then you can simplify automatic startup of the broker by setting this parameter to true in the initialization parameter file.

See Also

Oracle Data Guard Broker for examples of using this parameter to start the Oracle Data Guard broker

1.108 DIAGNOSTIC_DEST

As of Oracle Database 11g Release 1 (11.1), the diagnostics for each database instance are located in a dedicated directory that is specified by the DIAGNOSTIC_DEST initialization parameter.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	String
Syntax	`DIAGNOSTIC_DEST = { pathname | directory }`
Default value	Derived from the value of the `$ORACLE_BASE` environment variable. If `$ORACLE_BASE` is not set, then derived from `ORACLE_BASE` as set by the Oracle Universal Installer. If `ORACLE_BASE` is not set, then `$ORACLE_HOME/rdbms/log` is used.
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Basic	No
Oracle RAC	This parameter can be set on each instance. Oracle recommends that each instance in a cluster specify a `DIAGNOSTIC_DEST` directory location that is located on shared disk and that the same value for `DIAGNOSTIC_DEST` be specified for each instance.

The structure of the directory specified by `DIAGNOSTIC_DEST` is as follows:

`<diagnostic_dest>/diag/rdbms/<dbname>/<instname>`

This location is known as the Automatic Diagnostic Repository (ADR) Home. For example, if the database name is `proddb` and the instance name is `proddb1`, the ADR home directory would be `<diagnostic_dest>/diag/rdbms/proddb/proddb1`.

The following files are located under the ADR home directory:

- Trace files - located in subdirectory `<adr-home>/trace`
- Alert logs - located in subdirectory `<adr-home>/alert`. In addition, the `alert.log` file is now in XML format, which conforms to the Oracle ARB logging standard.
- Core files - located in the subdirectory `<adr-home>/cdump`
- Incident files - the occurrence of each serious error (for example, ORA-600, ORA-1578, ORA-7445) causes an incident to be created. Each incident is assigned an ID and dumping for each incident (error stack, call stack, block dumps, and so on) is stored in its own file, separated from process trace files. Incident dump files are located in `<adr-home>/incident/<incdir#>`. You can find the incident dump file location inside the process trace file.

See Also

Oracle Automatic Storage Management Administrator's Guide for an example of the diagnostic directory for an Oracle ASM instance

1.109 DISK_ASYNCH_IO

`DISK_ASYNCH_IO` controls whether I/O to datafiles, control files, and logfiles is asynchronous (that is, whether parallel server processes can overlap I/O requests with CPU processing during table scans).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	No
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

If your platform supports asynchronous I/O to disk, Oracle recommends that you leave this parameter set to its default value. However, if the asynchronous I/O implementation is not stable, you can set this parameter to `false` to disable asynchronous I/O. If your platform does not support asynchronous I/O to disk, this parameter has no effect.

If you set `DISK_ASYNCH_IO` to `false`, then you can increase `DB_WRITER_PROCESSES` or use `DBWR_IO_SLAVES` to simulate asynchronous I/O.

See Also

- "DB_WRITER_PROCESSES"
- "DBWR_IO_SLAVES"
- *Oracle Database Performance Tuning Guide* for more information

1.110 DISPATCHERS

`DISPATCHERS` configures dispatcher processes in the shared server architecture.

Property	Description
Parameter type	String
Syntax	`DISPATCHERS = '`*`dispatch_clause`*`'`

Syntax	**`dispatch_clause::=`**
	`(PROTOCOL =` *`protocol`*`) |`
	`(ADDRESS =` *`address`*`) |`
	`(DESCRIPTION =` *`description`* `)`
	`[options_clause]`

Syntax	**`options_clause::=`**
	`(DISPATCHERS =` *`integer`* `|`
	`SESSIONS =` *`integer`* `|`
	`CONNECTIONS =` *`integer`* `|`
	`MULTIPLEX = {1 | ON | YES | TRUE |`
	`0 | OFF | NO | FALSE | BOTH | IN | OUT} |`
	`LISTENER =` *`tnsname`* `|`
	`SERVICE =` *`service`* `|`
	`INDEX =` *`integer`*`)`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	If SHARED_SERVERS is greater than 0, then DISPATCHERS defaults to '(PROTOCOL=tcp)', causing 1 TCP/IP dispatcher to be created.
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No

The parsing software supports a name-value syntax to enable the specification of attributes in a position-independent, case-insensitive manner. For example:

```
DISPATCHERS = '(PROTOCOL=TCP)(DISPATCHERS=3)'
```

Attributes may be specified using the full attribute name or any substring beginning with the first 3 characters. For example, SESSIONS can be specified as SES, SESS, SESSI, and so on.

Specify only one of the following attributes: PROTOCOL, ADDRESS, or DESCRIPTION. If you specify either ADDRESS or DESCRIPTION, then you can specify additional network attributes. Doing so supports multi-homed hosts.

dispatch_clause

- PROTOCOL

  The network protocol for which the dispatcher generates a listening endpoint.

- ADDRESS

  The network protocol address of the endpoint on which the dispatchers listen.

- DESCRIPTION

  The network description of the endpoint on which the dispatchers listen, including the protocol address.

options_clause

- DISPATCHERS

  The initial number of dispatchers to start. The default is 1.

- SESSIONS

  The maximum number of network sessions to allow for each dispatcher. The default is operating system-specific. Most operating systems have a default of 16 KB.

- CONNECTIONS

  The maximum number of network connections to allow for each dispatcher. The default is operating system-specific.

- MULTIPLEX

  Enables the Oracle Connection Manager session multiplexing feature.

  - The values 1, ON, YES, TRUE, and BOTH indicate that Network Session Multiplex is enabled for both incoming and outgoing network connections.
  - The value IN indicates that Network Session Multiplex is enabled for incoming network connections.
  - The value OUT indicates that Network Session Multiplexing is enabled for outgoing network connections.

Copyright © 2002, 2026, Oracle and/or its affiliates.

  - The values `0`, `NO`, `OFF`, and `FALSE` indicate that Network Session Multiplexing is disabled for both incoming and outgoing network connections. This is the default.

- `LISTENER`

  Specifies the network name of an address or address list of the Oracle Net listeners with which the dispatchers will register.

  The `LISTENER` attribute facilitates administration of multi-homed hosts. This attribute specifies the appropriate listeners with which the dispatchers will register. The `LISTENER` attribute takes precedence over the `LOCAL_LISTENER` and `REMOTE_LISTENER` parameters. See "LOCAL_LISTENER" and "REMOTE_LISTENER".

- `SERVICE`

  Specifies one or more names by which clients can connect to the dispatchers. The `SERVICE` attribute takes precedence over any settings configured by using the `SRVCTL` command-line utility, the `GDSCTL` command-line utility, or the `DBMS_SERVICE` PL/SQL package.

- `INDEX`

  Use this attribute in an `ALTER SYSTEM SET DISPATCHERS` statement to indicate which dispatcher configuration you want to modify. (If you specify `INDEX` in the initialization parameter file, the Oracle Database ignores it.) In an `ALTER SYSTEM` statement, `INDEX` specifies the order in which the parameter's values were initialized. The value ranges from `0` (for the first dispatcher configuration) to one less than the total number of dispatcher configurations you define.

  For example, if you specify 3 dispatcher configurations in the initialization parameter file, you would modify the third dispatcher configuration by specifying `INDEX=2` in the `ALTER SYSTEM` statement. You could also add another dispatcher configuration in the `ALTER SYSTEM` statement by specifying `INDEX=3`.

  If `INDEX` is not specified in the `ALTER SYSTEM` statement, then the `PROTOCOL`, `ADDRESS`, or `DESCRIPTION` attributes must be specified, and if a dispatcher configuration matching this `PROTOCOL`, `ADDRESS`, or `DESCRIPTION` exists, then that configuration will be modified. Otherwise, a new configuration will be added.

See Also

- "SHARED_SERVERS"
- *Oracle Database Net Services Administrator's Guide* and *Oracle Database Administrator's Guide* for more information on setting this parameter

1.111 DISTRIBUTED_LOCK_TIMEOUT

`DISTRIBUTED_LOCK_TIMEOUT` specifies the amount of time (in seconds) for distributed transactions to wait for locked resources.

Property	Description
Parameter type	Integer
Default value	`60`
Modifiable	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable in a PDB	No
Range of values	1 to 2^{31} - 1
Basic	No

> **See Also**
>
> *Oracle Database Concepts* for more information on data concurrency

1.112 DML_LOCKS

`DML_LOCKS` specifies the maximum number of DML locks—one for each table modified in a transaction.

Property	Description
Parameter type	Integer
Default value	Derived: 4 * `TRANSACTIONS`
Modifiable	No
Modifiable in a PDB	No
Range of values	20 to unlimited; a setting of 0 disables enqueues
Basic	No
Oracle RAC	You must set this parameter for every instance, and all instances must have positive values or all must be 0.

A **DML lock** is a lock obtained on a table that is undergoing a DML operation (insert, update, delete). The `DML_LOCKS` value should equal the grand total of locks on tables currently referenced by all users. For example, if three users are modifying data in one table, then three entries would be required. If three users are modifying data in two tables, then six entries would be required.

The default value assumes an average of four tables referenced for each transaction. For some systems, this value may not be enough.

Enqueues are shared memory structures that serialize access to database resources. If you set the value of `DML_LOCKS` to `0`, enqueues are disabled and performance is slightly increased. However, you should be aware of the following restrictions when you set you `DML_LOCKS` to `0`:

- You cannot use `DROP TABLE`, `CREATE INDEX` statements
- You cannot use explicit lock statements such as `LOCK TABLE IN EXCLUSIVE MODE`
- Enterprise Manager cannot run on any instances for which `DML_LOCKS` is set to `0`

Oracle holds more locks during parallel DML than during serial execution. Therefore, if your database supports a lot of parallel DML, you may need to increase the value of this parameter.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database Concepts* for a discussion of lock and enqueue resources needed for parallel DML
- *Oracle Database Concepts* for more information on data concurrency

1.113 DNFS_BATCH_SIZE

`DNFS_BATCH_SIZE` controls the number of asynchronous I/O's that can be queued by an Oracle process when Direct NFS Client is enabled.

Property	Description
Parameter type	Integer
Default value	`4096`
Modifiable	No
Modifiable in a PDB	No
Range of values	0 - 4096
Basic	No

In environments where the NFS server cannot handle a large number of outstanding asynchronous I/O requests, use this parameter to limit the number of I/O's issued by an Oracle foreground process. The recommended setting for this parameter is to start at 128 and increase or decrease it based on NFS server performance.

See Also

Oracle Database Performance Tuning Guide for additional information about the `DNFS_BATCH_SIZE` initialization parameter

1.114 DRCP_DEDICATED_OPT

Use `DRCP_DEDICATED_OPT` to enable or disable the use of dedicated optimization with Database Resident Connection Pooling (DRCP).

Dedicated optimization makes DRCP behave like a dedicated server when the number of connections to the DRCP broker is less than the DRCP maximum size.

Property	Description
Parameter type	String
Syntax	`DRCP_DEDICATED_OPT = { YES | NO }`
Default value	`YES`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Oracle RAC	Different instances can use different values.

When the value of this parameter is set to `YES`, the use of dedicated optimization with DRCP is enabled. When it is set to `NO`, the use of dedicated optimization with DRCP is disabled.

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.11.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information about setting this parameter

1.115 DST_UPGRADE_INSERT_CONV

`DST_UPGRADE_INSERT_CONV` specifies whether internal operators will be allocated on top of `TIMESTAMP WITH TIME ZONE` (TSTZ) columns of tables which have not been upgraded during the upgrade window of daylight saving time patching for `TIMESTAMP WITH TIME ZONE` data.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true	false`
Basic	No

Values

- `true`

  Internal operators will be allocated on top of TSTZ columns of tables which have not been upgraded. This is the default.

- `false`

  Internal operators will not be allocated on top of TSTZ columns of tables which have not been upgraded.

When DST_UPGRADE_INSERT_CONV is set to true during the upgrade window of the daylight saving time patching process:

- `SELECT` queries on tables with TSTZ data which have not been upgraded will use internal operators on top of TSTZ columns to present TSTZ data as if they were recorded using the new time zone translation rules.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- DML on tables with TSTZ data which have not been upgraded will use internal operators on top of TSTZ columns to ensure that the TSTZ data is recorded using the old time zone translation rules in order to be consistent with the existing TSTZ data in the same tables.

> **Note**
>
> Oracle strongly recommends that this parameter is set to `true` throughout the upgrade window of the daylight saving time patching process. This parameter reduces the performance impact since indexes on TSTZ columns will be disabled whenever internal operators are allocated. If the parameter is set to `false`, then indexes will be used and this may affect performance of queries against TSTZ data during the DST upgrade window. Turning off this parameter during the upgrade window may corrupt data on disk when DMLs occur for tables with TSTZ data which have not yet been upgraded.

> **See Also**
>
> *Oracle Database Globalization Support Guide* for more information about this parameter

1.116 ENABLE_AUTOMATIC_MAINTENANCE_PDB

`ENABLE_AUTOMATIC_MAINTENANCE_PDB` can be used to enable or disable the running of automated maintenance tasks for all the PDBs in a CDB or for individual PDBs in a CDB.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No
Oracle RAC	The same value should be used for all instances.

> **Note**
>
> The value of this parameter in CDB$ROOT (the root of a CDB) has no effect in the root. Automated maintenance tasks are always run in the root, regardless of the setting of this parameter.

By default, the value of `ENABLE_AUTOMATIC_MAINTENANCE_PDB` is `true` in CDB$ROOT (the root container in the CDB) and in the individual PDBs in a CDB. This means that by default, automated maintenance tasks are run for the CDB root and all the PDBs in the CDB.

Copyright © 2002, 2026, Oracle and/or its affiliates.

When you change the value of ENABLE_AUTOMATIC_MAINTENANCE_PDB in the CDB root, the new value takes effect in the root and in all the PDBs in the CDB.

Therefore, if you change the value of ENABLE_AUTOMATIC_MAINTENANCE_PDB in the CDB root to false, the value of ENABLE_AUTOMATIC_MAINTENANCE_PDB is also changed to false in all of the PDBs in the CDB.

You can also change the value of ENABLE_AUTOMATIC_MAINTENANCE_PDB in any of the individual PDBs in a CDB, and the value that is set for each individual PDB will be honored. This enables you to enable or disable automated maintenance tasks for individual PDBs.

See Also

- "AUTOTASK_MAX_ACTIVE_PDBS" for information about specifying the maximum number of PDBs that can schedule automated maintenance tasks at the same time
- *Oracle Database Administrator's Guide* for more information about managing automated database maintenance tasks

1.117 ENABLE_DDL_LOGGING

ENABLE_DDL_LOGGING enables or disables the writing of a subset of data definition language (DDL) statements to a DDL log.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	true | false
Basic	No

The DDL log is a file that has the same format and basic behavior as the alert log, but it only contains the DDL statements issued by the database. The DDL log is created only for the RDBMS component and only if the ENABLE_DDL_LOGGING initialization parameter is set to true. When this parameter is set to false, DDL statements are not included in the DDL log, but they are included in the alert log.

The DDL log contains one log record for each DDL statement issued by the database. The DDL log is included in IPS incident packages.

There are two DDL logs that contain the same information. One is an XML file, and the other is a text file. The DDL log is stored in the log/ddl subdirectory of the ADR home.

When ENABLE_DDL_LOGGING is set to true, the following DDL statements are written to the log:

- ALTER/CREATE/DROP/TRUNCATE CLUSTER
- ALTER/CREATE/DROP FUNCTION
- GRANT

Copyright © 2002, 2026, Oracle and/or its affiliates.

- ALTER/CREATE/DROP INDEX
- ALTER/CREATE/DROP OUTLINE
- ALTER/CREATE/DROP PACKAGE
- ALTER/CREATE/DROP PACKAGE BODY
- ALTER/CREATE/DROP PROCEDURE
- ALTER/CREATE/DROP PROFILE
- REVOKE
- ALTER/CREATE/DROP SEQUENCE
- CREATE/DROP SYNONYM
- ALTER/CREATE/DROP/RENAME/TRUNCATE TABLE
- ALTER/CREATE/DROP TRIGGER
- ALTER/CREATE/DROP TYPE
- ALTER/CREATE/DROP TYPE BODY
- DROP USER
- ALTER/CREATE/DROP VIEW

Note

The DDL statement written to the log may be truncated. You can use DDL triggers to view the entire DDL statement. See *Oracle Database PL/SQL Language Reference* for more information about DDL triggers.

See Also

Oracle Database Licensing Information User Manual for licensing information for the `ENABLE_DDL_LOGGING` initialization parameter.

1.118 ENABLE_DNFS_DISPATCHER

`ENABLE_DNFS_DISPATCHER` enables dispatcher support for the Oracle Direct NFS client.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	No
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No
Oracle RAC	All instances should use the same value.

Copyright © 2002, 2026, Oracle and/or its affiliates.

When this parameter is set to `true` in the initialization parameter file, dispatcher processes will be spawned for the Oracle Direct NFS client.

The number of dispatcher processes spawned is calculated as one-eighth of the value of the `CPU_COUNT` parameter.

See Also

"CPU_COUNT"

1.119 ENABLE_GOLDENGATE_REPLICATION

`ENABLE_GOLDENGATE_REPLICATION` controls services provided by the RDBMS for Oracle GoldenGate (both capture and apply services).

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No
Oracle RAC	All instances must have the same setting

Set this to `true` to enable RDBMS services used by Oracle GoldenGate.

This parameter primarily controls supplemental logging required to support logical replication of new data types and operations. The redo log file is designed to be applied physically to a database, therefore the default contents of the redo log file often do not contain sufficient information to allow logged changes to be converted into SQL statements. Supplemental logging adds extra information into the redo log files so that replication can convert logged changes into SQL statements without having to access the database for each change. Previously these extra changes were controlled by the supplemental logging DDL. Now the `ENABLE_GOLDENGATE_REPLICATION` parameter must also be set to enable the required supplemental logging for any new data types or operations.

All enhancements to supplemental logging required to support logical replication are also controlled by this parameter.

The RDBMS services controlled by this parameter also include (but are not limited to):

- Transparent Data Encryption (including Tablespace Encryption) utilities used by GoldenGate Extract
- Service to read redo logs used by GoldenGate Extract
- Service to suppress triggers used by GoldenGate Replicat
- Service to handle transient duplicate handling used by GoldenGate Replicat
- Service to bypass referential integrity checking used by GoldenGate Replicat
- Services required to run Oracle GoldenGate in Integrated Extract and Integrated Replicat

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> When this database parameter is set to `true`, you must have a valid Oracle GoldenGate license or an authorized managed cloud client to use the enabled features. GoldenGate licenses or authorized managed cloud clients are required for all 3rd party or end-user tools that are used with this feature. Refer to *Oracle GoldenGate Licensing Information* and ensure that you have obtained the appropriate licensing before setting this parameter to `true`.

1.120 ENABLE_IMC_WITH_MIRA

`ENABLE_IMC_WITH_MIRA` enables or disables the In-Memory Column Store and Oracle Data Guard Multi-Instance Redo Apply, at the same time, on an Active Data Guard standby database.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No
Oracle RAC	Different instances can use different values.

The value of this parameter is relevant only on the instance where MRP0 is spawned at the start of Redo Apply on a standby database.

> **Note**
>
> This parameter is available starting with Oracle Database 19c.

1.121 ENABLE_PLUGGABLE_DATABASE

`ENABLE_PLUGGABLE_DATABASE` is a bootstrap initialization parameter to create a CDB.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	No
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

This parameter enables a database in NOMOUNT startup mode to know that it is a CDB.

This parameter must be set in init.ora before creating a CDB.

> **Note**
>
> See *Oracle Multitenant Administrator's Guide* for more information about CDBs and PDBs.

1.122 ENABLED_PDBS_ON_STANDBY

ENABLED_PDBS_ON_STANDBY specifies which pluggable databases (PDBs) to replicate on an Oracle Data Guard standby database.

Property	Description
Parameter type	String
Syntax	ENABLED_PDBS_ON_STANDBY = *PDB-list*
Default value	* if no value is specified for this parameter in the init.ora file
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No
Oracle RAC	All instances should use the same value.

> **Note**
>
> This parameter is meaningful only on standby databases. Its settings are ignored on a primary database.

PDB-list accepts a list of PDB names represented by glob patterns such as “PDB?” or “PDB*a”, “PDB2”. The glob pattern rules are like those of UNIX shells. The asterisk (*) and question mark (?) wildcard characters are supported. The ? wildcard character matches exactly one unknown character, and the * wildcard character matches any number of unknown characters. Also, the minus sign (-) character can be used as the first character in a PDB name to indicate that the PDB should be excluded on the standby database.

The following characters are valid in a PDB name: alphanumeric characters, underscore (_), number sign (#), and dollar sign ($). No other characters are valid. Oracle removes double quotation before processing the PDB name. Therefore you cannot use double quotation marks to embed other characters in the name. The PDB name is case insensitive. These are the same naming conventions as for a database name (as described in the DB_NAME initialization parameter description).

These rules apply for the ENABLED_PDBS_ON_STANDBY parameter:

- Any PDB names that match glob patterns starting with minus sign (-) will not be part of the standby database.
- If a PDB name matches several patterns in the *PDB-list* at the same time, the rightmost matched pattern has the highest precedence.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- If this parameter is not specified in the `init.ora` file, it is assumed that "*" is specified by default.
- If one or more patterns are specified in this parameter, it is assumed that "-*" is implicitly specified in the leftmost position. For example, `ENABLED_PDBS_ON_STANDBY="PDB1", "PDB2"` is equivalent to `ENABLED_PDBS_ON_STANDBY="-*", "PDB1", "PDB2"`. It means that a PDB name that does not match any patterns is not allowed to be part of the standby.

Examples

These examples assumes that nine new PDBs named PDB1A, PDB1B, PDB1C, PDB2A, PDB2B, PDB2C, PDB3A, PDB3B, and PDB3C are being added to the primary database.

1. If STANDBY1 sets `ENABLED_PDBS_ON_STANDBY="*"` on STANDBY1, then all nine PDBs will be created on STANDBY1.
2. If STANDBY1 sets `ENABLED_PDBS_ON_STANDBY="PDB1*"` on STANDBY1, then PDB1A, PDB1B, and PDB1C will be created on STANDBY1.
3. If STANDBY1 sets `ENABLED_PDBS_ON_STANDBY="PDB*A"` on STANDBY1, then PDB1A, PDB2A, and PDB3A will be created on STANDBY1.
4. If STANDBY1 sets `ENABLED_PDBS_ON_STANDBY="PDB1*", "-PDB*A"` on STANDBY1, then PDB1B and PDB1C will be created on STANDBY1.
5. If STANDBY1 sets `ENABLED_PDBS_ON_STANDBY="*", "-PDB*A", "-PDB*B"` on STANDBY1, then PDB1C, PDB2C and PDB3C will be created on STANDBY1. All other PDBs match "*", and the rightmost pattern has the higher precedence, so they are excluded on the standby.
6. If STANDBY1 sets `ENABLED_PDBS_ON_STANDBY="*", "-PDB*A", "PDB2A"` on STANDBY1, then PDB1A and PDB3A are excluded, but all other PDBs including PDB2A will be created.

1.123 ENCRYPT_NEW_TABLESPACES

`ENCRYPT_NEW_TABLESPACES` specifies whether to encrypt newly created user tablespaces.

Property	Description
Parameter type	String
Syntax	`ENCRYPT_NEW_TABLESPACES = { CLOUD_ONLY | ALWAYS | DDL }`
Default value	`CLOUD_ONLY`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The same value should be specified for all instances.

The values that can be specified for the `ENCRYPT_NEW_TABLESPACES` parameter have the following meanings:

- `CLOUD_ONLY`:

  When a user tablespace is created in the Oracle Cloud, it is transparently encrypted with Advanced Encryption Standard 128 (AES 128) if the `ENCRYPTION … ENCRYPT` clause for the SQL `CREATE TABLESPACE` statement is not specified. When a user tablespace is created in an on-premises database, the `ENCRYPTION` clause of the `CREATE TABLESPACE` statement determines if the tablespace is encrypted.

Copyright © 2002, 2026, Oracle and/or its affiliates.

CLOUD_ONLY is the default value.

- ALWAYS:

  Whether the user tablespace is created in the Oracle Cloud or in an on-premises database, the tablespace will be transparently encrypted with AES128 if the ENCRYPTION … ENCRYPT clause is not specified in the CREATE TABLESPACE statement.

- DDL:

  Whether the user tablespace is created in the Oracle Cloud or in an on-premises database, the CREATE TABLESPACE statement will follow the specified DDL. If no ENCRYPTION clause is specified, then the tablespace will not be encrypted. If the ENCRYPTION USING … ENCRYPT clause is specified, then the specified algorithm will be used to encrypt the tablespace. If ENCRYPTION ENCRYPT is specified, but no algorithm is specified, then the tablespace will be encrypted with the default AES128 algorithm.

See Also

- *Oracle Database Advanced Security Guide* for more information about tablespace encryption
- *Oracle Database SQL Language Reference* for more information about the CREATE TABLESPACE statement

1.124 EVENT

EVENT is a parameter used only to debug the system.

Property	Description
Parameter type	String
Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	No
Basic	No

Do not alter the value of this parameter except under the supervision of Oracle Support Services staff.

1.125 EXTERNAL_KEYSTORE_CREDENTIAL_LOCATION

EXTERNAL_KEYSTORE_CREDENTIAL_LOCATION specifies the location of the secure external keystore.

Property	Description
Parameter type	String
Syntax	EXTERNAL_KEYSTORE_CREDENTIAL_LOCATION = file-location
Default value	No default value
Modifiable	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable in a PDB	No
Basic	No
Oracle RAC	This parameter can be set on each instance. Oracle recommends that if the instances have a shared location, then use a directory on the shared location for this parameter and set the same value on all Oracle RAC instances. If the database does not have a shared location, then each instance of the database will have its own directory and the value should be set per-instance.

TDE keystore credentials can be stored in a secure external keystore to automate Transparent Data Encryption (TDE) operations.

To automate the setting of the Master Key in a newly-provisioned PDB, set `EXTERNAL_KEYSTORE_CREDENTIAL_LOCATION` so that the CDB knows the location of the secure external keystore.

When a file specification is set with this parameter, the secure external keystore is looked for in that location. If the secure external keystore is not found in that location, then this parameter is ignored.

See Also

Oracle Database Advanced Security Guide for more information about using TDE

1.126 FAL_CLIENT

`FAL_CLIENT` specifies the FAL (fetch archive log) client name that is used by the FAL service, configured through the `FAL_SERVER` initialization parameter, to refer to the FAL client.

Property	Description
Parameter type	String
Syntax	`FAL_CLIENT = string`
Default value	There is no default value.
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Basic	No

The value is an Oracle Net service name, which is assumed to be configured properly on the FAL server system to point to the FAL client (standby database).

See Also

Oracle Data Guard Concepts and Administration for more information about FAL server

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.127 FAL_SERVER

FAL_SERVER specifies the FAL (fetch archive log) server for a standby database.

Property	Description
Parameter type	String
Syntax	FAL_SERVER = *string*
Default value	There is no default value.
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No

The value is an Oracle Net service name, which is assumed to be configured properly on the standby database system to point to the desired FAL server.

See Also

Oracle Data Guard Concepts and Administration for more information about FAL server

1.128 FAST_START_MTTR_TARGET

FAST_START_MTTR_TARGET enables you to specify the number of seconds the database takes to perform crash recovery of a single instance.

Property	Description
Parameter type	Integer
Default value	0
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	0 to 3600 seconds
Basic	No
Oracle RAC	Multiple instances can have different values, and you can change the values at run time.

When specified, FAST_START_MTTR_TARGET is overridden by LOG_CHECKPOINT_INTERVAL.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database Performance Tuning Guide* for an example of setting the lower bound for this parameter
- *Oracle Database Performance Tuning Guide* for an example of setting the upper bound for this parameter
- *Oracle Database Performance Tuning Guide* for an example of selecting a preliminary value for this parameter

1.129 FAST_START_PARALLEL_ROLLBACK

`FAST_START_PARALLEL_ROLLBACK` specifies the degree of parallelism used when recovering terminated transactions.

Property	Description
Parameter type	String
Syntax	`FAST_START_PARALLEL_ROLLBACK = { HIGH | LOW | FALSE }`
Default value	`LOW`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes This parameter is modifiable in a PDB in Oracle Database 19c, starting with Release Update 19.17.
Basic	No

Terminated transactions are transactions that are active before a system failure. If a system fails when there are uncommitted parallel DML or DDL transactions, then you can speed up transaction recovery during startup by using this parameter.

Values

- `FALSE`

  Parallel rollback is disabled

- `LOW`

  Limits the maximum degree of parallelism to 2 * `CPU_COUNT`

- `HIGH`

  Limits the maximum degree of parallelism to 4 * `CPU_COUNT`

If you change the value of this parameter, then transaction recovery will be stopped and restarted with the new implied degree of parallelism.

See Also

Oracle Database VLDB and Partitioning Guide for more information about this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.130 FILE_MAPPING

`FILE_MAPPING` specifies whether file mapping is enabled.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

> **Note**
>
> The `FILE_MAPPING` initialization parameter is deprecated. It is still supported for backward compatibility.

The FMON background process is started to manage the mapping information when file mapping is enabled.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information about the file mapping feature

1.131 FILEIO_NETWORK_ADAPTERS

`FILEIO_NETWORK_ADAPTERS` specifies a list of network adapters that can be used to access the disk storage.

Property	Description
Parameter type	String
Syntax	`FILEIO_NETWORK_ADAPTERS = adapter_name [, adapter_name ] ...`
Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	No
Range of values	One or more network adapter names, separated by commas
Basic	No

On platforms where the database files reside in network attached storage, this parameter provides the storage access library the list of network adapters that can be used to access the storage.

Copyright © 2002, 2026, Oracle and/or its affiliates.

The network adapter name is a fully qualified address name of the network card that can be accessed through the host name database or using the Network Information Service. The components of the adapter name are separated by periods. For example, the following is a fully qualified adapter name:

```
ib1.oracle.com
```

1.132 FILESYSTEMIO_OPTIONS

`FILESYSTEMIO_OPTIONS` specifies I/O operations for file system files. This parameter is platform-specific and has a default value that is suitable for a particular platform.

Property	Description
Parameter type	String
Syntax	`FILESYSTEMIO_OPTIONS = { ASYNCH | DIRECTIO | SETALL | NONE }`
Default value	Varies by database version and operating system.
Modifiable	No
Modifiable in a PDB	No
Basic	No

Values:

- `ASYNCH` - Enable asynchronous I/O on file system files, which has no timing requirement for transmission.
- `DIRECTIO` - Enable direct I/O on file system files, which bypasses the buffer cache.
- `SETALL` - Enable both asynchronous and direct I/O on file system files.
- `NONE` - Disable both asynchronous and direct I/O on file system files.

See Also

- Your platform-specific documentation for more details
- *Oracle Database Performance Tuning Guide* for information about setting this parameter before running I/O calibration

1.133 FIXED_DATE

`FIXED_DATE` enables you to set a constant date that `SYSDATE` will always return instead of the current date.

Property	Description
Parameter type	String
Syntax	`FIXED_DATE = [ YYYY-MM-DD-HH24:MI:SS` (or the default Oracle date format) `| NONE ]`
Default value	There is no default value.
Modifiable	`ALTER SYSTEM`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable in a PDB	Yes
Basic	No

To undo a fixed date setting, specify `FIXED_DATE=NONE`. This parameter is useful primarily for testing. The value can be in the format shown above or in the default Oracle date format, without a time.

1.134 FORWARD_LISTENER

`FORWARD_LISTENER` specifies the name of a listener to which a connection must be forwarded by an existing set of remote listeners.

Property	Description
Parameter type	String
Syntax	`FORWARD_LISTENER = listener-name`
Default value	NULL
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	A different value can be set on different instances.

The parameter is useful when it is difficult to change an existing client connect string, for example, after an offsite database has been moved into the Oracle Cloud. With the `FORWARD_LISTENER` parameter, clients can continue to connect to their offsite listener, and the offsite listener forwards the connection to the Oracle Cloud listener.

Once a forward listener has been configured through the `FORWARD_LISTENER` parameter, the `LOCAL_LISTENER` parameter can be cleared by setting its value to `-oracle-none-` so that all the connections coming to an existing set of remote listeners configured through the `REMOTE_LISTENER` parameter are forwarded only to listeners configured through `FORWARD_LISTENER`.

The `FORWARD_LISTENER` parameter can also be specified in the `LISTENER_NETWORKS` parameter.

Examples

The following setup can be used to forward all the database connections coming to an existing customer SCAN listener to the Oracle Cloud SCAN listener.

```
REMOTE_LISTENER=customer-scan
FORWARD_LISTENER=oracle-cloud-scan
LOCAL_LISTENER=oracle-none-
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "LOCAL_LISTENER"
- "REMOTE_LISTENER"
- "LISTENER_NETWORKS"
- *Oracle Clusterware Administration and Deployment Guide* for information about SCAN listeners

1.135 GCS_SERVER_PROCESSES

`GCS_SERVER_PROCESSES` specifies the number of background GCS server processes (LMS*n* and LM*nn*) to serve the inter-instance traffic among Oracle RAC instances.

If there is high Cache Fusion congestion on the system, the number of GCS server processes could be increased to reduce Cluster wait time. Additionally, the number of GCS server processes could be increased to reduce the application brownout time associated with Cluster reconfiguration and Dynamic Re-mastering (DRM).

Property	Description
Parameter type	Integer
Default value	Oracle calculates the default value as follows (in order of precedence): 1. If `CLUSTER_DATABASE` is set to `false`, then `0` 2. If Oracle ASM, then 1 3. If 1 - 3 CPUS, then 1 4. If 4 - 15 CPUs, then 2 5. If 128 or more CPUs and SGA is 100 GB or more, then (CPUs / 6). If the result includes a fraction, then the fraction is disregarded. 6. Otherwise, the value is 2 + (CPUs / 32). If the result includes a fraction, then the fraction is disregarded. For example, if you have 20 CPUs, then 2 + (20 / 32) would equal 2 GCS processes. 7. On NUMA-enabled systems with 32 or more CPUs, the value is rounded up to a multiple of the number of NUMA processor groups, with a limit of (CPUs / 4) rounded down to a multiple of the number of NUMA processor groups.
Modifiable	No
Modifiable in a PDB	No
Range of values	`0` if Oracle RAC is disabled (`CLUSTER_DATABASE` is set to `false`) `1` to `100` if Oracle RAC is enabled (`CLUSTER_DATABASE` is set to `true`)
Basic	No
Oracle RAC	Multiple instances can have different values.

GCS server processes are only seen in an Oracle RAC environment.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.136 GLOBAL_NAMES

`GLOBAL_NAMES` specifies whether a database link is required to have the same name as the database to which it connects.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No

If the value of `GLOBAL_NAMES` is `false`, then no check is performed. If you use or plan to use distributed processing, then Oracle recommends that you set this parameter to `true` to ensure the use of consistent naming conventions for databases and links in a networked environment.

See Also

Oracle Database Administrator's Guide for more information on setting this parameter

1.137 GLOBAL_TXN_PROCESSES

`GLOBAL_TXN_PROCESSES` specifies the initial number of GTX*n* background processes (GTX0, ... GTX9 and GTXa, ... GTXj) per instance to support global (XA) transactions in an Oracle RAC environment.

Property	Description
Parameter type	Integer
Default value	`1`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`0` to `20`
Basic	No
Oracle RAC	Multiple instances can have different values.

If you want to disable the GTX*n* background processes, then you must set `GLOBAL_TXN_PROCESSES` to `0` in your parameter file. Setting this parameter to `0` will disable the XA support on an Oracle RAC database. Error `ORA-55712` will be returned if you try to run XA transactions on an Oracle RAC database with this parameter set to `0`. You can change the setting to a nonzero value at run time to turn on the support for XA.

`GLOBAL_TXN_PROCESSES` is useful for systems that process global (XA) transactions heavily. You do not need to specify a value for this parameter since Oracle Database automatically

Copyright © 2002, 2026, Oracle and/or its affiliates.

determines the number of processes and autotunes them, as necessary. GTX*n* background processes are only seen in an Oracle RAC environment.

> **See Also**
>
> *Oracle Database Development Guide* for more information about this parameter

1.138 HASH_AREA_SIZE

`HASH_AREA_SIZE` specifies the maximum amount of memory, in bytes, to be used for hash joins.

Property	Description
Parameter type	Integer
Default value	Derived: 2 * `SORT_AREA_SIZE`
Modifiable	`ALTER SESSION`
Modifiable in a PDB	No
Range of values	0 to operating system-dependent
Basic	No

> **Note**
>
> Oracle does not recommend using the `HASH_AREA_SIZE` parameter unless the instance is configured with the shared server option. Oracle recommends that you enable automatic sizing of SQL working areas by setting `PGA_AGGREGATE_TARGET` instead. `HASH_AREA_SIZE` is retained for backward compatibility.

`HASH_AREA_SIZE` is relevant to parallel execution operations and to the query portion of DML or DDL statements.

> **See Also**
>
> - *Oracle Database Concepts* for information on hash joins in general

1.139 HEARTBEAT_BATCH_SIZE

Use `HEARTBEAT_BATCH_SIZE` to configure the heartbeat batch size, which is the number of heartbeats sent during one heartbeat period.

Property	Description
Parameter type	Integer
Default value	5
Modifiable	`ALTER SYSTEM`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable in a PDB	Yes
Range of values	2 to 100
Basic	No
Oracle RAC	Multiple instances must have the same value.

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.29.

1.140 HEAT_MAP

Use HEAT_MAP to enable or disable both the Heat Map and Automatic Data Optimization (ADO) features.

Property	Description
Parameter type	String
Syntax	HEAT_MAP = { ON | OFF }
Default value	OFF
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	If specified, use the same value on all instances

Setting the HEAT_MAP initialization parameter to ON causes the database to track read and write access of all segments, as well as modification of database blocks, due to DMLs and DDLs. These activities are not tracked for objects in the SYSTEM and SYSAUX tablespaces.

> **See Also**
>
> - *Oracle Database VLDB and Partitioning Guide* for more information about enabling and disabling the Heat Map feature
> - *Oracle Database VLDB and Partitioning Guide* for more information about Heat Map tracking

1.141 HI_SHARED_MEMORY_ADDRESS

HI_SHARED_MEMORY_ADDRESS specifies the starting address at run time of the system global area (SGA).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	No
Modifiable in a PDB	No
Basic	No

This parameter is ignored on platforms that specify the SGA's starting address at linktime.

On 64-bit platforms, use `HI_SHARED_MEMORY_ADDRESS` to specify the high-order 32 bits of a 64-bit address. Use `SHARED_MEMORY_ADDRESS` to specify the low-order 32 bits of the address (see "SHARED_MEMORY_ADDRESS"). If both parameters are 0 or unspecified, the SGA address defaults to a platform-specific location.

1.142 HS_AUTOREGISTER

`HS_AUTOREGISTER` enables or disables automatic self-registration of Heterogeneous Services (HS) agents.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

When enabled, information is uploaded into the server's data dictionary to describe a previously unknown agent class or a new agent version.

Oracle recommends that you set this parameter to `true`. Oracle incurs less overhead when establishing subsequent connections through the same agent if self-registered information is available in the server's data dictionary.

See Also

Oracle Database Heterogeneous Connectivity User's Guide for more information about using HS agents

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.143 IDENTITY_PROVIDER_CONFIG

Use `IDENTITY_PROVIDER_CONFIG` to configure the database to use Microsoft Azure Active Directory (Azure AD) as an external identity provider.

Property	Description
Parameter type	String
Syntax	`IDENTITY_PROVIDER_CONFIG = '{ "application_id_uri" : "string", "tenant_id" : "string", "app_id" : "string" }'` Where: • The `application_id_uri` value is obtained from the registered application, to be mapped in the JWT "aud" (audience) claim. It must be domain qualified and start with "https://" to specify the OCI database to access. • The `tenant_id` value is obtained from the tenant configuration and is also available on the database application registration overview page. • The `app_id` value is obtained from the registered resource application and is also available on the database application registration overview page.
Default value	None
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The same value must be used on all instances.

This parameter is effective only when the database is enabled to use Azure AD as an external identity provider, that is, when the `IDENTITY_PROVIDER_TYPE` initialization parameter is set to `AZURE_AD`. You must use this parameter to specify Azure AD configuration settings.

Example:

```
ALTER SYSTEM SET IDENTITY_PROVIDER_CONFIG =
  '{ "application_id_uri" : "https://example.com/pdb1",
     "tenant_id" : "111a1111-a11a-111a-111111111111a",
     "app_id" : "11aa1a1-aaaa-1111-1111-1111aa111111" }'
   SCOPE=BOTH;
```

> **Note**
>
> Setting the value of this parameter is only one step in a series of steps for configuring the database to use Azure AD. Refer to *Oracle Database Security Guide* for the complete steps before you set this parameter.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.16.

> **Note**
>
> This parameter is not available to update directly for Oracle Autonomous Database Serverless.

> **See Also**
>
> "IDENTITY_PROVIDER_TYPE"

1.144 IDENTITY_PROVIDER_TYPE

`IDENTITY_PROVIDER_TYPE` specifies an external identity provider for the database.

Property	Description
Parameter type	String
Syntax	`IDENTITY_PROVIDER_TYPE = { AZURE_AD | OCI_IAM | NONE }`
Default value	`NONE`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The same value must be used on all instances.

Values:

- `AZURE_AD` - Enables the database to use Microsoft Azure Active Directory (Azure AD) as an external identity provider.

  When you specify this value, you must also specify configuration settings for Azure AD by setting the `IDENTITY_PROVIDER_CONFIG` initialization parameter.
- `OCI_IAM` - Enables the database to use Oracle Cloud Infrastructure Identity and Access Management (OCI IAM) as an external identity provider.

  When you specify this value, there is no need to specify configuration settings with the `IDENTITY_PROVIDER_CONFIG` initialization parameter.
- `NONE` - The database will not use Azure AD or OCI IAM as an external identity provider.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> Setting the value of this parameter is only one step in a series of steps for configuring an external identity provider for the database. Refer to *Oracle Database Security Guide* for the complete set of steps for configuring Azure AD or OCI IAM before you set this parameter value.

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.16.

> **Note**
>
> This parameter is not available to update directly for Oracle Autonomous Database Serverless.

> **See Also**
>
> "IDENTITY_PROVIDER_CONFIG"

1.145 IFILE

Use `IFILE` to embed another parameter file within the current parameter file.

Property	Description
Parameter type	Parameter file
Syntax	`IFILE = parameter_file_name`
Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	No
Range of values	Valid parameter filenames
Basic	No
Oracle RAC	Multiple instances can have different values.

For example:

```
IFILE = COMMON.ORA
```

You can have up to three levels of nesting. In this example, the file `COMMON.ORA` could contain a second `IFILE` parameter for the file `COMMON2.ORA`, which could contain a third `IFILE` parameter for the file `GCPARMS.ORA`. You can also include multiple parameter files in one parameter file by listing `IFILE` several times with different values:

Copyright © 2002, 2026, Oracle and/or its affiliates.

```
IFILE = DBPARMS.ORA
IFILE = GCPARMS.ORA
IFILE = LOGPARMS.ORA
```

> **Note**
>
> You must list multiple entries on contiguous lines of the parameter file.

1.146 IGNORE_SESSION_SET_PARAM_ERRORS

`IGNORE_SESSION_SET_PARAM_ERRORS` controls whether the database ignores errors when clients attempt to modify session parameter values.

Property	Description
Parameter type	String
Syntax	`IGNORE_SESSION_SET_PARAM_ERRORS = parameter_name [, parameter_name ] ...`
Default value	None
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Multiple instances should use the same value.

The `IGNORE_SESSION_SET_PARAM_ERRORS` parameter allows you to change the outcome of the following scenario:

1. An Oracle database uses PDB lockdown profiles to prevent certain initialization parameters from being modified with the `ALTER SESSION` statement.
2. A database client attempts to modify one of those parameters with the `ALTER SESSION` statement. The client has no knowledge that the parameter is locked down.
3. The database returns an error to the client.

To prevent the database from returning an error in the preceding scenario, set the value of `IGNORE_SESSION_SET_PARAM_ERRORS` to a comma-separated list of locked down initialization parameters. Thereafter, if a client attempts to modify one of those parameters with the `ALTER SESSION` statement, the database will ignore the modification attempt and indicate to the client that the operation was successful.

You can set the `IGNORE_SESSION_SET_PARAM_ERRORS` parameter while connected to a specific PDB, or you can apply it to all PDBs by setting it in the CDB root.

Example

The `CURSOR_INVALIDATION` and `CURSOR_SHARING` parameters are locked down in one of the PDBs in your database. If a client application attempts to modify either of these parameters, you want the database to ignore the request and allow the application to continue running

Copyright © 2002, 2026, Oracle and/or its affiliates.

without incurring an error. To achieve this, you can connect to the PDB and set the IGNORE_SESSION_SET_PARAM_ERRORS parameter. For example:

```
ALTER SYSTEM SET IGNORE_SESSION_SET_PARAM_ERRORS = CURSOR_INVALIDATION,
CURSOR_SHARING;
```

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.4.

1.147 INMEMORY_ADG_ENABLED

INMEMORY_ADG_ENABLED indicates whether in-memory for Active Data Guard is enabled in addition to the in-memory cache size.

Property	Description
Parameter type	Boolean
Default value	true
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	true | false
Basic	No
Oracle RAC	The same value should be used on all instances. If the value for this parameter is changed on one instance, the parameter's value should be changed to the same value on the other instances.

For Active Data Guard, media recovery needs to retrieve in-memory objects when applying redo and to invalidate the related objects after the query advance. This parameter controls whether media recovery does the retrieving and invalidating.

This parameter should be set on standby databases. The value of this parameter is meaningless on a primary database.

If the standby is an Oracle RAC instance, then all the different standby instances must have this parameter set to the same value.

This parameter is a dynamic system-modifiable parameter whose value can be changed only when Active Data Guard recovery is not running.

If you start multi-instance redo apply (MIRA) Active Data Guard recovery, all instances involved in MIRA must have the parameter set to the same value, otherwise MIRA will signal an error.

Example

If the standby is an Oracle RAC instance, this parameter should be set to the same value on all the Oracle RAC instances:

```
SQL> alter system set inmemory_adg_enabled=true sid='*';
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "INMEMORY_SIZE"
- *Oracle Database In-Memory Guide* for more information about deploying an In-Memory Column Store (IM column store) in an Active Data Guard environment

1.148 INMEMORY_AUTOMATIC_LEVEL

`INMEMORY_AUTOMATIC_LEVEL` is used to enable the Automatic In-Memory feature, which automates the management of the In-Memory Column Store (IM column store) to help ensure that the working data set is in the IM column store at all times.

Property	Description
Parameter type	String
Syntax	`INMEMORY_AUTOMATIC_LEVEL = { LOW | MEDIUM | OFF }`
Default value	`OFF`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	All instances should use the same value.

Typically, among all the IM enabled segments, only a subset is actively queried at any time. This subset is known as the working data set. The working data set is expected to change over time for many applications.

`INMEMORY_AUTOMATIC_LEVEL` attempts to keep the working data set in the IM column store at all times by moving segments in and out of the IM column store based on access patterns.

Oracle recommends that you provision enough memory for the working data set to fit in the IM column store.

The following values can be set:

- `LOW`: When this value is set, the database evicts cold segments from the IM column store when it is under memory pressure.
- `MEDIUM`: When this value is set, the database evicts cold segments from the IM column store when it is under memory pressure. This level includes an additional optimization that ensures that any hot segment that was not populated because of memory pressure is populated first.
- `OFF`: When this value is set, Automatic In-Memory is disabled. This value returns the IM column store to the behavior that existed prior to Oracle Database 18c. If you do not expect a stable working data set, set the parameter to `OFF`. This is the default value.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> - *Oracle Database Licensing Information User Manual* to learn which database offerings support the Automatic In-Memory feature
> - *Oracle Database In-Memory Guide* for more information about configuring the Automatic In-Memory feature

1.149 INMEMORY_CLAUSE_DEFAULT

`INMEMORY_CLAUSE_DEFAULT` enables you to specify a default In-Memory Column Store (IM column store) clause for new tables and materialized views.

> **Note**
>
> The setting of his parameter has no effect on In-Memory external tables or hybrid partitioned tables.

Property	Description
Parameter type	String
Syntax	`INMEMORY_CLAUSE_DEFAULT = '[INMEMORY] [NO INMEMORY] [other-clauses]'`
Syntax	**other-clauses::=** `[compression-clause] [priority-clause] [rac-clause]`
Syntax	**compression-clause::=** `NO MEMCOMPRESS | MEMCOMPRESS FOR { DML | QUERY [ LOW | HIGH ] | CAPACITY [LOW | HIGH] }`[1]
Syntax	**priority-clause::=** `PRIORITY { LOW | MEDIUM | HIGH | CRITICAL | NONE }`[1]
Syntax	**rac-clause::=** `[distribute-clause] [duplicate-clause]`[1]
Syntax	**distribute-clause::=** `DISTRIBUTE [ AUTO | BY ROWID RANGE ]`[1]
Syntax	**duplicate-clause::=** `NO DUPLICATE | DUPLICATE [ ALL ]`[1]
Default value	An empty string
Modifiable	`ALTER SESSION, ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	All instances should use the same value

[1] See Table 1-2 for more information about this clause.

If the `INMEMORY_CLAUSE_DEFAULT` parameter is unset or set to an empty string (the default), only tables and materialized views explicitly specified as `INMEMORY` will be populated into the IM

Copyright © 2002, 2026, Oracle and/or its affiliates.

column store. Setting the value of the INMEMORY_CLAUSE_DEFAULT parameter to NO INMEMORY has the same effect as setting it to the default value.

If the INMEMORY_CLAUSE_DEFAULT parameter is set, then any *newly created* table or materialized view specified as INMEMORY will inherit unspecified attributes from this parameter. This can force certain in-memory options by default that are not explicitly specified in the syntax. For example, if the INMEMORY_CLAUSE_DEFAULT parameter is set to MEMCOMPRESS FOR CAPACITY LOW and a table is created as INMEMORY PRIORITY HIGH, then the table is treated as if it was declared as INMEMORY MEMCOMPRESS FOR CAPACITY LOW PRIORITY HIGH.

If INMEMORY is specified as part of this parameter, then all newly created tables and materialized views will be populated into the IM column store, except tables and materialized views explicitly specified as NO INMEMORY. For example, if this parameter is set to INMEMORY MEMCOMPRESS FOR CAPACITY HIGH, then all new tables will be created as if this clause were present in the SQL CREATE TABLE statement. If there is a default INMEMORY value for the tablespace for a given segment, then it will override the value for this parameter.

Table 1-2 Meaning of INMEMORY_CLAUSE_DEFAULT Parameter Values

Syntax	Description
INMEMORY	Specifies that all newly-created tables and materialized views populate the IM column store unless they are specified as NO INMEMORY in the SQL CREATE TABLE or CREATE MATERIALIZED VIEW statement
NO INMEMORY	Specifies that only tables and materialized views explicitly specified as INMEMORY in the SQL CREATE TABLE or CREATE MATERIALIZED VIEW statements populate the IM column store
compression-clause	Specifies that in-memory compression should be used for the instance. Use the MEMCOMPRESS FOR values to specify the in-memory compression level.
NO MEMCOMPRESS	When NO MEMCOMPRESS is specified, no in-memory compression is done in the IM column store.
MEMCOMPRESS FOR	MEMCOMPRESS FOR is used to indicate the in-memory compression level for the IM column store.
DML	When DML is specified, the IM column store is optimized for DML operations, and some lightweight in-memory compression may be done.
QUERY	When QUERY is specified, the in-memory compression level is for high performance. If QUERY is specified without LOW or HIGH, it defaults to QUERY LOW.
QUERY LOW	When QUERY LOW is specified, the in-memory compression level provides the highest performance.
QUERY HIGH	When QUERY HIGH is specified, the in-memory compression level provides a balance between compression and performance, weighted toward performance.
CAPACITY	When CAPACITY is specified without LOW or HIGH, it defaults to CAPACITY LOW.
CAPACITY LOW	When CAPACITY LOW is specified, the in-memory compression level is a balance between compression and performance, weighted toward capacity.
CAPACITY HIGH	When CAPACITY HIGH is specified, the in-memory compression level is for highest capacity.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 1-2 (Cont.) Meaning of INMEMORY_CLAUSE_DEFAULT Parameter Values

Syntax	Description
`priority-clause`	Specifies the priority to use when populating tables in the IM column store. Use the `PRIORITY` values to specify the priority. By default, the population of a table in the IM column store can be delayed until the database determines it is useful. On database instance startup, tables are populated in priority order.
`PRIORITY NONE`	When `PRIORITY NONE` is specified, the population of a table in the IM column store can be delayed until the database determines it is useful. This is the default value when no priority is specified.
`PRIORITY LOW`	When `PRIORITY LOW` is specified for a table or tables, the population of those tables in the IM column store is done before tables that have no priority specified.
`PRIORITY MEDIUM`	When `PRIORITY MEDIUM` is specified for a table or tables, the population of those tables in the IM column store is done before tables that have no priority and `PRIORITY LOW` specified.
`PRIORITY HIGH`	When `PRIORITY HIGH` is specified for a table or tables, the population of those tables in the IM column store is done before tables that have no priority, `PRIORITY LOW`, and `PRIORITY MEDIUM` specified.
`PRIORITY CRITICAL`	When `PRIORITY CRITICAL` is specified for a table or tables, the population of those tables in the IM column store is done before tables that have no priority, `PRIORITY LOW`, `PRIORITY MEDIUM`, and `PRIORITY HIGH` specified.
`rac-clause`	Specifies how tables in the IM column store will be managed among Oracle Real Application Clusters (Oracle RAC) instances. Use the `distribute-clause` and `duplicate-clause` to specify how tables in the IM store will be managed in Oracle RAC instances. For a non-Oracle RAC database, these settings have no effect, because the whole table or partition has to be on the single instance.
`distribute-clause`	Specifies how a table is distributed among Oracle RAC instances.
`DISTRIBUTE AUTO`	Specifies that the database will automatically decide how to distribute tables in the IM column store across the Oracle RAC instances based on the type of partitioning and the value of the `duplicate-clause`. `DISTRIBUTE AUTO` is the default, and it is also used when `DISTRIBUTE` is specified by itself.
`DISTRIBUTE BY ROWID RANGE`	Specifies that the tables in the IM column store will be distributed by rowid range to different Oracle RAC instances.
`duplicate-clause`	Specifies how many copies of each In-Memory Compression Unit (IMCU) of the tables in the IM column store will be spread across all the Oracle RAC instances. **Note:** The `duplicate-clause` is only applicable if you are using Oracle Real Application Clusters (Oracle RAC) on an engineered system. Otherwise, the `duplicate-clause` is ignored and there is only one copy of each IMCU in memory.
`NO DUPLICATE`	Data is not duplicated across Oracle RAC instances. This is the default.
`DUPLICATE`	Data is duplicated on another Oracle RAC instance, resulting in data existing on a total of two Oracle RAC instances.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 1-2 (Cont.) Meaning of INMEMORY_CLAUSE_DEFAULT Parameter Values

Syntax	Description
`DUPLICATE ALL`	Data is duplicated across all Oracle RAC instances. If you specify `DUPLICATE ALL`, then the database uses the `DISTRIBUTE AUTO` setting, regardless of whether or how you specify the `distribute-clause`.

Examples

The following statement causes no tables to populate the IM column store:

```
alter system set INMEMORY_CLAUSE_DEFAULT='NO INMEMORY'
scope=both;
```

The following statement causes new tables and materialized views (except those specified as `NO INMEMORY`) to populate the IM column store at the high capacity compression level:

```
alter system set
INMEMORY_CLAUSE_DEFAULT='INMEMORY MEMCOMPRESS FOR CAPACITY HIGH'
scope=both;
```

The following statement causes new tables (even those specified as `NO INMEMORY`) to populate the IM column store at the highest performance compression level at `LOW` priority:

```
alter system set
INMEMORY_CLAUSE_DEFAULT='INMEMORY MEMCOMPRESS FOR QUERY LOW PRIORITY LOW'
scope=both;
```

The following statement causes new tables (even those specified as `NO INMEMORY`) to populate the IM column store without any in-memory compression:

```
alter system set
INMEMORY_CLAUSE_DEFAULT='INMEMORY NO MEMCOMPRESS'
scope=both;
```

The following statement causes tables in the IM column store to be duplicated on every Oracle RAC instance, unless on a non-engineered system. For a non-engineered system, the `duplicate-clause` (`DUPLICATE ALL`) will be ignored and tables in the column store will be automatically distributed across the Oracle RAC instance, with only one copy of each IMCU in the tables in the IM column store:

```
alter system set
INMEMORY_CLAUSE_DEFAULT='INMEMORY MEMCOMPRESS FOR QUERY
DISTRIBUTE AUTO DUPLICATE ALL'
scope=both;
```

The following statement sets the value of the `INMEMORY_CLAUSE_DEFAULT` parameter back to its default value, the empty string:

```
alter system set
INMEMORY_CLAUSE_DEFAULT=''
scope=both;
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "INMEMORY_FORCE"
- "INMEMORY_QUERY"
- "INMEMORY_SIZE"
- "V$IM_SEGMENTS"
- "V$IM_USER_SEGMENTS"
- "QUERY_REWRITE_INTEGRITY"
- *Oracle Database In-Memory Guide* for an introduction to the IM column store
- *Oracle Database In-Memory Guide* for more information about the IM column store
- *Oracle Database In-Memory Guide* for more information on IM column store compression methods
- *Oracle Database In-Memory Guide* for more information on priority levels for populating a database object in the IM column store
- *Oracle Database SQL Language Reference* for more information on the CREATE TABLE statement
- *Oracle Database SQL Language Reference* for more information on the CREATE MATERIALIZED VIEW statement

1.150 INMEMORY_EXPRESSIONS_USAGE

`INMEMORY_EXPRESSIONS_USAGE` controls which In-Memory Expressions (IM expressions) are populated into the In-Memory Column Store (IM column store) and are available for queries.

Property	Description
Parameter type	String
Syntax	`INMEMORY_EXPRESSIONS_USAGE = { STATIC_ONLY | DYNAMIC_ONLY | ENABLE | DISABLE }`
Default value	`ENABLE`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The same value must be used on all instances.

The four values for this parameter are:

- `STATIC_ONLY`: Tables enabled for in-memory and containing certain data types such as Oracle numbers or JSON will have these columns populated in the IM column store using a more efficient representation. Note that this setting will increase the in-memory footprint for some tables. A static configuration enables the IM column store to cache OSON (binary JSON) columns, which are marked with an IS_JSON check constraint. Internally, an OSON column is a hidden virtual column named `SYS_IME_OSON`. In contrast, a dynamic configuration automatically creates and populates frequently used expressions.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- `DYNAMIC_ONLY`: IM expressions will be automatically created and populated into the IM column store, if used in conjunction with PL/SQL procedure `DBMS_INMEMORY.IME_CAPTURE_EXPRESSIONS`. Note that setting this value will increase the in-memory footprint for some tables.
- `ENABLE`: Both static and dynamic IM expressions will be populated into the IM column store and available to be used by queries. Note that setting this value will increase the in-memory footprint for some tables. This is the default value.
- `DISABLE`: No IM expressions of any kind will be populated into the IM column store.

Changing the mode of a parameter does not have an immediate effect on existing in-memory data. For example, if the `inmemory_expressions_usage` clause is changed from `DYNAMIC_ONLY` to `DISABLE`, the stored IM expressions are not immediately removed. The next repopulation will not bring back the expressions in memory, thereby effectively removing the expressions.

See Also

- "INMEMORY_VIRTUAL_COLUMNS"
- *Oracle Database In-Memory Guide* for more information about IM expressions
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_INMEMORY` PL/SQL package

1.151 INMEMORY_FORCE

`INMEMORY_FORCE` allows you to specify whether tables and materialized views that are specified as `INMEMORY` are populated into the In-Memory Column Store (IM column store) or not.

Property	Description
Parameter type	String
Syntax	`INMEMORY_FORCE = { DEFAULT | OFF | BASE_LEVEL | CELLMEMORY_LEVEL }`
Default value	`DEFAULT`
Modifiable	You can use `ALTER SYSTEM` to dynamically modify this parameter setting as follows: • From `DEFAULT` to `OFF` • From `OFF` to `DEFAULT` • From `BASE_LEVEL` to `DEFAULT` • From `CELLMEMORY_LEVEL` to `DEFAULT` • From `CELLMEMORY_LEVEL` to `OFF` All other modifications of this parameter require you to update the initialization file and restart the instance.
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	All instances should use the same value.

The following values can be set:

- `DEFAULT`

Copyright © 2002, 2026, Oracle and/or its affiliates.

The IM column store is populated with objects that are specified as `INMEMORY`.

- `OFF`

  Even if the IM column store is configured on this instance, no objects are populated in memory.

- `BASE_LEVEL`

  Enables Database In-Memory Base Level. The Base Level lets you to experiment with Oracle Database In-Memory features without purchasing the Oracle Database In-Memory option. This setting is available starting with Oracle Database 19c, Release Update 19.8.

  When the Base Level is enabled, the value of the `INMEMORY_SIZE` initialization parameter is limited to 16 GB for a CDB. In an Oracle RAC environment, the value of `INMEMORY_SIZE` is limited to 16 GB for each instance. The compression level for all objects and columns is automatically and transparently set to `QUERY LOW`. Automatic In-Memory is disabled, and In-Memory Column Store feature tracking is tracked for "In-Memory Base Level" rather than "In-Memory Column Store." The CellMemory feature is disabled for Oracle Exadata.

- `CELLMEMORY_LEVEL`

  CellMemory is an Exadata feature that allows you to use Exadata Smart Flash Cache to populate data in in-memory columnar format. In older releases of Oracle Database, in order to use the CellMemory feature, you were required to enable the IM column store (by setting `INMEMORY_SIZE` to a value greater than `0`), even if you had no intention of using the IM column store. This incurred the overhead of enabling the IM column store without any benefit. Starting with Oracle Database 19c, Release Update 19.8, you can use the CellMemory feature without enabling the IM column store by setting `INMEMORY_FORCE=CELLMEMORY_LEVEL` and `INMEMORY_SIZE=0`. With these settings, the IM column store is not enabled and queries can use CellMemory to scan objects.

  Note that if the value of `INMEMORY_SIZE` is greater than `0`, then setting `INMEMORY_FORCE=CELLMEMORY_LEVEL` is equivalent to setting `INMEMORY_FORCE=DEFAULT`. The IM column store is enabled, even if you only use CellMemory.

See Also

- "INMEMORY_CLAUSE_DEFAULT"
- "INMEMORY_MAX_POPULATE_SERVERS"
- "INMEMORY_QUERY"
- "INMEMORY_SIZE"
- *Oracle Database In-Memory Guide* for an introduction to the IM column store
- *Oracle Database In-Memory Guide* for more information about the IM column store
- *Oracle Database Licensing Information User Manual* to learn which database offerings support the Database In-Memory Base Level feature

1.152 INMEMORY_MAX_POPULATE_SERVERS

`INMEMORY_MAX_POPULATE_SERVERS` specifies the maximum number of background populate servers to use for In-Memory Column Store (IM column store) population, so that these servers do not overload the rest of the system.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	Integer
Default value	Half of the value of CPU_COUNT or the PGA_AGGREGATE_TARGET value divided by 512M, whichever is less. See the "CPU_COUNT" description for information about how CPU_COUNT is calculated.
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	0 to a value based on the number of cores in the system.
Basic	No
Oracle RAC	All instances should use the same value.

This parameter has meaning only if the INMEMORY_SIZE parameter is also set to a positive value.

The value to use for this parameter depends on the number of cores in the system. A certain percentage of CPU should be allocated for in-memory background population, and this parameter should be set accordingly. You can also set this parameter to 0 to temporarily disable populate tasks on the system from executing.

Note

Be careful not to set the value of this parameter too high. If it is set close to the number of cores or higher, no CPU could be left for the rest of the system to run.

Note

The IM column store is not populated if this parameter is set to 0.

See Also

- "INMEMORY_CLAUSE_DEFAULT"
- "INMEMORY_FORCE"
- "INMEMORY_QUERY"
- "INMEMORY_SIZE"
- *Oracle Database In-Memory Guide* for an introduction to the IM column store
- *Oracle Database In-Memory Guide* for more information about the IM column store

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.153 INMEMORY_OPTIMIZED_ARITHMETIC

INMEMORY_OPTIMIZED_ARITHMETIC encodes the NUMBER data type in in-memory tables compressed with QUERY LOW as a fixed-width native integer scaled by a common exponent.

Property	Description
Parameter type	String
Syntax	INMEMORY_OPTIMIZED_ARITHMETIC = { ENABLE | DISABLE }
Default value	DISABLE
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Different values can be specified on different instances.

The Oracle Database NUMBER format can incur a significant performance overhead when executing queries because arithmetic operations cannot be performed natively in hardware.

When INMEMORY_OPTIMIZED_ARITHMETIC is set to ENABLE, for tables compressed with QUERY LOW, NUMBER columns are encoded as a fixed-width native integer scaled by a common exponent. This In-Memory optimized number format enables fast calculations using SIMD hardware. By using SIMD vector processing, arithmetic operations, simple aggregations, and group-by aggregations can benefit significantly.

Not all row sources in the query processing engine have support for the In-Memory optimized number format. Therefore, the IM column store must store both the traditional Oracle Database NUMBER data type and the In-Memory optimized number type. This means that the acceleration in analytic query performance comes at a cost of increased space overhead.

When INMEMORY_OPTIMIZED_ARITHMETIC is set to DISABLE (the default), the database does not use the optimized encoding.

If INMEMORY_OPTIMIZED_ARITHMETIC is set to ENABLE and then to DISABLE, the optimized number format for existing IMCUs is not dropped immediately. Instead, as the IM column store repopulates IMCUs, the new IMCUs do not use the optimized encoding.

See Also

Oracle Database In-Memory Guide for details about the improved performance and increased space overhead when INMEMORY_OPTIMIZED_ARITHMETIC is enabled

1.154 INMEMORY_QUERY

INMEMORY_QUERY is used to enable or disable in-memory queries for the entire database at the session or system level.

Property	Description
Parameter type	String

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Syntax	`INMEMORY_QUERY = { ENABLE | DISABLE }`
Default value	`ENABLE`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	All instances should use the same value.

This parameter is helpful when you want to test workloads with and without the use of the In-Memory Column Store (IM column store).

This parameter enables in-memory queries for the entire database by default when the `INMEMORY_SIZE` parameter is specified.

Set this parameter to `DISABLE` if you want to disable in-memory queries.

See Also

- "INMEMORY_CLAUSE_DEFAULT"
- "INMEMORY_FORCE"
- "INMEMORY_MAX_POPULATE_SERVERS"
- "INMEMORY_SIZE"
- *Oracle Database In-Memory Guide* for an introduction to the IM column store
- *Oracle Database In-Memory Guide* for more information about the IM column store

1.155 INMEMORY_SIZE

`INMEMORY_SIZE` sets the size of the In-Memory Area, which contains the IM Column Store (IM column store) on a database instance.

Property	Description
Parameter type	Big integer
Syntax	`INMEMORY_SIZE = integer [K | M | G]`
Default value	`0`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	0 to the amount of memory left in the SGA after other allocations
Basic	No
Oracle RAC	All instances should use the same value.

The default value is 0, which means that the IM column store is not used.

The database must be restarted after setting this parameter to enable the IM column store.

Copyright © 2002, 2026, Oracle and/or its affiliates.

The minimum size to which this parameter can be set is 100 MB.

Typically this parameter should be set to at least the size needed to accommodate all the tables that will use the IM column store. It can be set higher to allow for growth of those tables or to accommodate other tables that will use the IM column store in the future.

This parameter can also be set per PDB to limit the maximum size of the IM column store for that PDB. Note that the sum of the PDB values do not have to equal the CDB value, and the sum of the PDB values may even be greater than the CDB value.

Unless this parameter is specifically set on a PDB, each PDB inherits the CDB value, which means they can use all of the available IM column store.

The value specified for this parameter counts toward `SGA_TARGET`. For example, if you set `SGA_TARGET` to 10 GB and you set `INMEMORY_SIZE` to 2 GB, then 20% of the `SGA_TARGET` setting is allocated to the In-Memory Area.

Unlike other SGA components such as the buffer cache and shared pool, the In-Memory Area size is not controlled by automatic memory management. The database does not automatically shrink the In-Memory Area when the buffer cache or shared pool requires more memory, or increase the In-Memory Area when it runs out of space. You can only increase the size of the In-Memory Area by manually adjusting the `INMEMORY_SIZE` initialization parameter.

In-Memory Dynamic Scans require the Resource Manager. Therefore, the Resource Manager is automatically enabled when you change the value of `INMEMORY_SIZE` from `0` to a non-zero value. No specific resource plan is required.

Database In-Memory Base Level allows a maximum of 16 GB for the size of the In-Memory area. Therefore, if Database In-Memory Base Level is enabled (the value of the `INMEMORY_FORCE` initialization parameter is set to `BASE_LEVEL`), then the value of `INMEMORY_SIZE` cannot exceed 16 GB for a CDB. In an Oracle RAC environment, the value of `INMEMORY_SIZE` is limited to 16 GB for each instance.

See Also

- "INMEMORY_CLAUSE_DEFAULT"
- "INMEMORY_FORCE"
- "INMEMORY_MAX_POPULATE_SERVERS"
- "INMEMORY_QUERY"
- *Oracle Database In-Memory Guide* for an introduction to the IM column store
- *Oracle Database In-Memory Guide* for more information about the IM column store
- *Oracle Database In-Memory Guide* for an example of using the `INMEMORY_SIZE` parameter

1.156 INMEMORY_TRICKLE_REPOPULATE_SERVERS_PERCENT

`INMEMORY_TRICKLE_REPOPULATE_SERVERS_PERCENT` limits the maximum number of background populate servers used for In-Memory Column Store (IM column store) repopulation, as trickle repopulation is designed to use only a small percentage of the populate servers.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	Integer
Default value	1
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	0 to 50
Basic	No
Oracle RAC	All instances should use the same value.

The value for this parameter is a percentage of the INMEMORY_MAX_POPULATE_SERVERS initialization parameter value.

For example, if this parameter is set to 5 and INMEMORY_MAX_POPULATE_SERVERS is set to 10, then on average half of a core is used for trickle repopulation.

The default value of 1 is good in most cases. In some cases, if you want to disable trickle repopulate, this parameter can be set to 0. If you want to keep the system more aggressively up to date (at the expense of more background CPU), you can set the parameter to higher values such as 5 or 10.

A value of greater than 50 is not allowed, so that at least half of the populate servers are available for other (re)populate tasks. On some systems, a value of less than 50 can be problematic, depending on other workload.

This parameter has meaning only if the INMEMORY_SIZE parameter is also set to a positive value.

1.157 INMEMORY_VIRTUAL_COLUMNS

INMEMORY_VIRTUAL_COLUMNS controls which user-defined virtual columns are stored as In-Memory virtual columns (IM columns).

Property	Description
Parameter type	String
Syntax	INMEMORY_VIRTUAL_COLUMNS = { ENABLE | MANUAL | DISABLE }
Default value	MANUAL
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The same value must be used on all instances.

IM virtual columns improve query performance by avoiding the necessity of repeated calculations. Also, the database can scan and filter IM virtual columns using techniques such as SIMD vector processing.

These values can be set for the parameter:

- ENABLE: For a table or partition that has been enabled for in-memory storage, all virtual columns will be stored in-memory at the default table or partition memcompress level unless:

Copyright © 2002, 2026, Oracle and/or its affiliates.

  - They have been explicitly excluded using the no inmemory syntax.
  - They have been altered to have a different memcompress level than the base table or partition, in which case they will be stored at the specified memcompress level.
- `MANUAL`: This is the default value for the parameter. For a table or partition that has been enabled for in-memory storage, no virtual columns will be stored in-memory unless:
  - They have been explicitly marked for inmemory, in which case they will be stored in-memory at the table or partition memcompress level.
  - They have been marked for inmemory with a different memcompress level than the base table or partition, in which case they will be stored at the specified memcompress level.
- `DISABLE`: For a table or partition that has been enabled for in-memory storage, no virtual columns will ever be stored in-memory. Any changes to the inmemory_column_clause for a virtual column including changes in memcompress level will be recorded, but not acted upon with regards to population of virtual columns.

See Also

- "INMEMORY_EXPRESSIONS_USAGE"
- *Oracle Database In-Memory Guide* for more information about IM virtual columns

1.158 INSTANCE_ABORT_DELAY_TIME

`INSTANCE_ABORT_DELAY_TIME` specifies how much time to delay an internal initiated instance termination (in seconds), such as when an irrecoverable process dies or an irrecoverable instance error occurs.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`0` and higher
Basic	No
Oracle RAC	Different instances should use the same value.

This parameter does not apply to a `SHUTDOWN ABORT` operation or any instance termination necessary to implement a user command.

Setting this parameter to a value greater than 0 enables a DBA to take some actions before a termination occurs due to an irrecoverable error. Note that since the instance is in an irrecoverable state, the DBA should not be too ambitious with the actions taken because some processes and/ or resources may be corrupted or unavailable, making complex actions impossible. Oracle does not guarantee what it is possible when an instance is in this state. A message is written to the alert log when the delayed termination is initiated. The value will not apply in the case of PMON death.

Copyright © 2002, 2026, Oracle and/or its affiliates.

The larger the specified value, the longer the instance stays up, and the potential increases for other problems to occur. If you set this parameter, Oracle recommends setting it to a value between 0 and 60.

1.159 INSTANCE_GROUPS

Used with the PARALLEL_INSTANCE_GROUP parameter, INSTANCE_GROUPS lets you restrict parallel query operations to a limited number of instances.

Property	Description
Parameter type	String
Syntax	INSTANCE_GROUPS = *group_name* [, *group_name*] ...
Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	No
Range of values	One or more instance group names, separated by commas
Basic	No
Oracle RAC	Multiple instances can have different values.

Note

The INSTANCE_GROUPS parameter is deprecated. It is retained for backward compatibility only.

INSTANCE_GROUPS is an Oracle RAC parameter that you can specify only in parallel mode.

This parameter specifies one or more instance groups and assigns the current instance to those groups. If one of the specified groups is also specified in the PARALLEL_INSTANCE_GROUP parameter, then Oracle allocates query processes for a parallel operation from this instance.

See Also

- *Oracle Real Application Clusters Administration and Deployment Guide* for more information on parallel query execution in a Real Application Clusters environment
- "PARALLEL_INSTANCE_GROUP"

1.160 INSTANCE_MODE

INSTANCE_MODE indicates whether the instance is read-write, read-only, or read-mostly.

Property	Description
Parameter type	String
Syntax	INSTANCE_MODE = { READ-WRITE | READ-ONLY | READ-MOSTLY }
Default value	READ-WRITE

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable	No
Basic	No
Oracle RAC	See the Oracle RAC restrictions in the initialization parameter description below.

A `READ-WRITE` instance is a regular Oracle instance.

A `READ-ONLY` instance is an Oracle instance that can only be opened in read-only mode.

A `READ-MOSTLY` instance is an Oracle instance that performs very few database writes.

These restrictions apply when setting the parameter on different Oracle Real Application Clusters (Oracle RAC) instances:

1. There must be at least one running instance with `INSTANCE_MODE` set to `READ-WRITE` at any given time. If the only instance with `INSTANCE_MODE` set to `READ-WRITE` goes down, all other instances will be brought down by Oracle automatically.
2. An instance with `INSTANCE_MODE` set to `READ-ONLY` cannot be the first instance to open a database or any pluggable database (PDB) among all instances in an Oracle RAC cluster.
3. Oracle does not support co-existence of an instance with `INSTANCE_MODE` set to `READ-ONLY` and an instance with `INSTANCE_MODE` set to `READ-MOSTLY` in the same Oracle RAC cluster.

1.161 INSTANCE_NAME

In an Oracle Real Application Clusters environment, multiple instances can be associated with a single database service. Clients can override Oracle's connection load balancing by specifying a particular instance by which to connect to the database. `INSTANCE_NAME` specifies the unique name of this instance.

Property	Description
Parameter type	String
Syntax	`INSTANCE_NAME = instance_id`
Default value	The instance's SID **Note:** The SID identifies the instance's shared memory on a host, but may not uniquely distinguish this instance from other instances.
Modifiable	No
Modifiable in a PDB	No
Range of values	Alphanumeric ASCII characters and the underscore (_) character, up to a maximum of 255 characters
Basic	No

In a single-instance database system, the instance name is usually the same as the database name.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Real Application Clusters Administration and Deployment Guide* and *Oracle Database Net Services Administrator's Guide* for more information

1.162 INSTANCE_NUMBER

`INSTANCE_NUMBER` specifies a unique number that maps the instance to one free list group for each database object created with storage parameter `FREELIST GROUPS`.

Property	Description
Parameter type	Integer
Default value	`0` (zero)
Modifiable	No
Modifiable in a PDB	No
Range of values	1 to maximum number of instances specified when the database was created
Basic	Yes
Oracle RAC	You must set this parameter for every instance, and all instances must have different values.

`INSTANCE_NUMBER` is an Oracle RAC parameter that can be specified in parallel mode or exclusive mode.

The `INSTANCE` parameter of the `ALTER TABLE ... ALLOCATE EXTENT` statement assigns an extent to a particular free list group. If you set `INSTANCE_NUMBER` to the value specified for the `INSTANCE` parameter, the instance uses that extent for inserts and for updates that expand rows.

The practical maximum value of this parameter is the maximum number of instances specified in the `CREATE DATABASE` statement. The absolute maximum is operating system-dependent.

> **See Also**
>
> *Oracle Real Application Clusters Administration and Deployment Guide* for more information

1.163 INSTANCE_TYPE

`INSTANCE_TYPE` specifies whether the instance is a database instance, an Oracle Automatic Storage Management (Oracle ASM) instance, or an Oracle ASM Proxy instance.

Property	Description
Parameter type	String
Syntax	`INSTANCE_TYPE = { RDBMS | ASM | ASMPROXY }`
Default value	`RDBMS`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable	No
Modifiable in a PDB	No
Basic	No
Oracle RAC	Multiple instances must have the same value.

Values

- `RDBMS`

  The instance is a database instance.
- `ASM`

  The instance is an Oracle ASM instance.
- `ASMPROXY`

  The instance is an Oracle ASM proxy instance.

> **See Also**
>
> *Oracle Automatic Storage Management Administrator's Guide* for information about managing Oracle Flex ASM

1.164 JAVA_JIT_ENABLED

`JAVA_JIT_ENABLED` enables or disables the Just-in-Time (JIT) compiler for the Oracle Java Virtual Machine (OracleJVM) environment.

Property	Description
Parameter type	Boolean
Default value	Operating system-dependent
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true	false`
Basic	No

For platforms that support the JIT compiler, the default value of this parameter is `true`; otherwise the default value is `false`. Attempting to set this parameter to `true` on unsupported platforms will result in an error.

> **See Also**
>
> *Oracle Database Java Developer's Guide* for more information about setting this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.165 JAVA_MAX_SESSIONSPACE_SIZE

JAVA_MAX_SESSIONSPACE_SIZE specifies (in bytes) the maximum amount of session space made available to a Java program executing in the server.

Property	Description
Parameter type	Integer
Default value	0
Modifiable	No
Modifiable in a PDB	No
Range of values	0 to 2 GB - 1
Basic	No

Java session space is the memory that holds Java state from one database call to another. When a user's session-duration Java state attempts to exceed the amount specified by JAVA_MAX_SESSIONSPACE_SIZE, the Java virtual machine terminates the session with an out-of-memory failure.

See Also

- *Oracle Database Java Developer's Guide* for more information about this parameter
- "JAVA_SOFT_SESSIONSPACE_LIMIT"

1.166 JAVA_POOL_SIZE

JAVA_POOL_SIZE specifies (in bytes) the size of the Java pool, from which the Java memory manager allocates most Java state during run-time execution. This memory includes the shared in-memory representation of Java method and class definitions, as well as the Java objects that are migrated to the Java session space at end-of-call.

Property	Description
Parameter type	Big integer
Syntax	JAVA_POOL_SIZE = *integer* [K | M | G]
Default value	If SGA_TARGET is set: If the parameter is not specified, then the default is 0 (internally determined by the Oracle Database). If the parameter is specified, then the user-specified value indicates a minimum value for the memory pool. If SGA_TARGET is not set: 24 MB, rounded up to the nearest granule size
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	Minimum: 0 (values greater than zero are rounded up to the nearest granule size) Maximum: operating system-dependent
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Java Developer's Guide* for more information about this parameter

1.167 JAVA_SOFT_SESSIONSPACE_LIMIT

`JAVA_SOFT_SESSIONSPACE_LIMIT` specifies (in bytes) a **soft limit** on Java memory usage in a session, as a means to warn you if a user's session-duration Java state is using too much memory.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	No
Modifiable in a PDB	No
Range of values	0 to 2 GB - 1
Basic	No

Java session space is the memory that holds Java state from one database call to another. When a user's session-duration Java state exceeds the size specified by `JAVA_SOFT_SESSIONSPACE_LIMIT`, Oracle generates a warning that goes into the trace files.

> **See Also**
>
> - *Oracle Database Java Developer's Guide* for more information on this parameter
> - "JAVA_MAX_SESSIONSPACE_SIZE"

1.168 JOB_QUEUE_PROCESSES

`JOB_QUEUE_PROCESSES` specifies the maximum number of job workers per instance that can be created for the execution of `DBMS_JOB` jobs and Oracle Scheduler (`DBMS_SCHEDULER`) jobs.

Property	Description
Parameter type	Integer

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	Derived. The lesser value of: • CPU_COUNT * 20 • SESSIONS / 4 For a CDB, if the result of the above derivation is less than twice the number of open containers in the CDB, then the value of this parameter is adjusted to equal twice the number of open containers in the CDB. Containers include CDB$ROOT, PDB$SEED, PDBs, application roots, application seeds, and application PDBs. You can obtain the number of open containers in a CDB with the following query: `SELECT COUNT(*) FROM V$CONTAINERS WHERE open_mode != 'MOUNTED';`
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to 4000
Basic	No
Oracle RAC	Multiple instances can have different values.

DBMS_JOB and Oracle Scheduler share the same job coordinator and job workers, and they are both controlled by the JOB_QUEUE_PROCESSES parameter. The actual number of job workers created for Oracle Scheduler jobs is auto-tuned by the Scheduler depending on several factors, including available resources, Resource Manager settings, and currently running jobs. However, the combined total number of job workers running DBMS_JOB jobs and Oracle Scheduler jobs in a non-CDB, CDB, or PDB can never exceed the value of JOB_QUEUE_PROCESSES for that non-CDB, CDB, or PDB.

The default value for JOB_QUEUE_PROCESSES provides a compromise between quality of service for applications and reasonable use of system resources. However, it is possible that the default value does not suit every environment. In such cases, you can use the following guidelines to fine tune this parameter:

- In a non-CDB:
  Set JOB_QUEUE_PROCESSES to the maximum number of job workers that can be used simultaneously in the entire database instance. If JOB_QUEUE_PROCESSES is 0, then DBMS_JOB jobs and Oracle Scheduler jobs will not run in the database instance.
- In a CDB root:
  Set JOB_QUEUE_PROCESSES to the maximum number of job workers that can be used simultaneously in the entire CDB. Oracle recommends that you set the value of this parameter to at least twice the number of open containers in the CDB, otherwise, there might be severe starvation between PDBs trying to run multiple jobs. If JOB_QUEUE_PROCESSES is set to 0 in a CDB root, then DBMS_JOB and Oracle Scheduler jobs cannot run in the CDB root or in any PDB, regardless of the JOB_QUEUE_PROCESSES setting at the PDB level.
- In a PDB:
  Set JOB_QUEUE_PROCESSES to the maximum number of job workers that can be used simultaneously in the PDB. The actual number depends on the resources assigned by Resource Manager and the demand in other containers. When multiple PDBs request jobs, Oracle Scheduler attempts to give all PDBs a fair share of the processes. Oracle recommends that you set the value of this parameter to at least 2 in a PDB. However, if

Copyright © 2002, 2026, Oracle and/or its affiliates.

you do not want to run DBMS_JOB and Oracle Scheduler jobs in a PDB, then set JOB_QUEUE_PROCESSES to 0 in the PDB.

Materialized views and AutoTask use Oracle Scheduler for automatic refreshes. Setting JOB_QUEUE_PROCESS to 0 will disable these features and any other features that use Oracle Scheduler or DBMS_JOB.

Note

DBMS_JOB is deprecated in Oracle Database 12*c* Release 2 (12.2.0.1) and may be removed in a future release. Oracle recommends that you use DBMS_SCHEDULER instead.

See Also

- *Oracle Database PL/SQL Packages and Types Reference* for more information on the DBMS_SCHEDULER package
- *Oracle Database Data Warehousing Guide* for more information on managing materialized views
- *Oracle Database Advanced Queuing User's Guide* for more information about job queue processes
- *Oracle Database Administrator's Guide* for more information about the maximum number of scheduler job processes

1.169 LARGE_POOL_SIZE

LARGE_POOL_SIZE specifies (in bytes) the size of the large pool allocation heap.

Property	Description
Parameter type	Big integer
Syntax	LARGE_POOL_SIZE = *integer* [K | M | G]
Default value	If SGA_TARGET is set, but a value is not specified for LARGE_POOL_SIZE, then the default is 0 (internally determined by the Oracle database). If LARGE_POOL_SIZE is specified, then the user-specified value indicates a minimum value for the memory pool. If SGA_TARGET is not set, then the default is 0.
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	Minimum: the granule size Maximum: operating system-dependent
Basic	No

The large pool allocation heap is used in shared server systems for session memory, by parallel execution for message buffers, and by backup processes for disk I/O buffers. Parallel execution allocates buffers out of the large pool only when SGA_TARGET is set.

Copyright © 2002, 2026, Oracle and/or its affiliates.

You can specify the value of this parameter using a number, optionally followed by K or M to specify kilobytes or megabytes, respectively. If you do not specify K or M, then the number is taken as bytes.

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information on setting this parameter

1.170 LDAP_DIRECTORY_ACCESS

`LDAP_DIRECTORY_ACCESS` specifies whether Oracle refers to Oracle Internet Directory for user authentication information.

Property	Description
Parameter type	String
Syntax	`LDAP_DIRECTORY_ACCESS = { NONE | PASSWORD | SSL }`
Default value	`NONE`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

If directory access is turned on, then this parameter also specifies how users are authenticated.

Values

- `NONE`

  Oracle does not refer to Oracle Internet Directory for Enterprise User Security information.
- `PASSWORD`

  Oracle tries to connect to the enterprise directory service using the database password stored in the database wallet. If that fails, then the Oracle Internet Directory connection fails and the database will not be able to retrieve enterprise roles and schema mappings upon enterprise user login.
- `SSL`

  Oracle tries to connect to Oracle Internet Directory using SSL.

> **See Also**
>
> *Oracle Database Enterprise User Security Administrator's Guide* for more information on Enterprise User Security

Using LDAP_DIRECTORY_ACCESS with PDBs

PDBs can use password and SSL authentication with Oracle Internet Directory when the default database wallet location is used.

Copyright © 2002, 2026, Oracle and/or its affiliates.

The LDAP_DIRECTORY_ACCESS initialization parameter is PDB-specific and can be set as follows:

- Prior to Oracle Database 19c, Release Update 19.10, if you set this parameter in a CDB root container, then all PDBs in the CDB will use that setting of the parameter. If you set this parameter in a PDB, then the parameter setting affects only that PDB. You can use ALTER SYSTEM to set this parameter in a PDB.
- Starting with Oracle Database 19c, Release Update 19.10:
  - When you use the ALTER SYSTEM command to set the value of LDAP_DIRECTORY_ACCESS while connected to the CDB root:
    * If you specify the CONTAINER=ALL clause, then the setting applies to the CDB root and all PDBs.
    * If you omit the CONTAINER=ALL clause, or specify the CONTAINER=CURRENT clause, then the setting applies only to the CDB root.
  - When you use the ALTER SYSTEM command to set the value of LDAP_DIRECTORY_ACCESS while connected to a PDB, the setting applies only to that PDB.
  - When you set the value of LDAP_DIRECTORY_ACCESS in an initialization parameter file, the setting applies only to the CDB root; it does not apply to the PDBs.

For a CDB, if the wallet location is not specified in sqlnet.ora, then the default database wallet path is:

ORACLE_BASE/admin/*db-unique-name/pdb-GUID*/wallet (if ORACLE_BASE is set)

or:

ORACLE_HOME/admin/*db-unique-name/pdb-GUID*/wallet (if ORACLE_BASE is not set)

The exception is for the root database, which has a default wallet path of:

ORACLE_BASE/admin/*db-unique-name*/wallet (if ORACLE_BASE is set)

or:

ORACLE_HOME/admin/*db-unique-name*/wallet (if ORACLE_BASE is not set)

All PDBs in a CDB have the same database unique name. By placing wallets in the default location, each PDB can have its own identity.

Note that because there is only one sqlnet.ora file for a CDB, the wallet location in sqlnet.ora is only used by the CDB root container. Because each PDB must have its own wallet, a PDB wallet will be specified by the *pdb-GUID* under the wallet location in sqlnet.ora.

For the root container of a CDB, the wallet location is:

WALLET_LOCATION_specified_in_sqlnet.ora

For each PDB of the CDB, the wallet location is:

WALLET_LOCATION_specified_in_sqlnet.ora/pdb-GUID/

> **Note**
>
> Oracle databases are registered with Oracle Internet Directory using Database Configuration Assistant (DBCA). For registration with Oracle Internet Directory to work, all the PDBs for a CDB must be registered using DBCA.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Using LDAP_DIRECTORY_ACCESS with Non-CDBs

For non-CDBs, if the wallet location is not specified in `sqlnet.ora`, then the default database wallet path is:

`ORACLE_BASE/admin/`*db-unique-name*`/wallet` (if `ORACLE_BASE` is set)

or:

`ORACLE_HOME/admin/`*db-unique-name*`/wallet` (if `ORACLE_BASE` is not set)

> **See Also**
>
> *Oracle Database Enterprise User Security Administrator's Guide* for an example of setting the value of this parameter to `SSL` in the server parameter file using `ALTER SYSTEM`

1.171 LDAP_DIRECTORY_SYSAUTH

`LDAP_DIRECTORY_SYSAUTH` allows or disallows directory-based authorization for users granted administrative privileges, such as `SYSDBA`, `SYSOPER`, `SYSBACKUP`, `SYSDG`, and `SYSKM`.

Property	Description
Parameter type	String
Syntax	`LDAP_DIRECTORY_SYSAUTH = { yes | no }`
Default value	`no`
Modifiable	No
Modifiable in a PDB	Yes
Basic	Yes

When `LDAP_DIRECTORY_SYSAUTH` is set to `yes`, directory users are allowed to connect to the database as `SYSDBA`, `SYSOPER`, `SYSBACKUP`, `SYSDG`, or `SYSKM`, if they have mapped database global users that are granted corresponding administrative privileges such as `SYSDBA`, `SYSOPER`, `SYSBACKUP`, `SYSDG`, and `SYSKM`.

When `LDAP_DIRECTORY_SYSAUTH` is set to no, directory users are not allowed to connect to the database as `SYSDBA`, `SYSOPER`, `SYSBACKUP`, `SYSDG`, or `SYSKM`, even if they have mapped database global users that are granted corresponding administrative privileges such as `SYSDBA`, `SYSOPER`, `SYSBACKUP`, `SYSDG`, and `SYSKM`.

When modifying this parameter in a PDB, use the `ALTER SYSTEM` command with `SCOPE=SPFILE`.

1.172 LICENSE_MAX_SESSIONS

`LICENSE_MAX_SESSIONS` specifies the maximum number of concurrent user sessions allowed.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	0 to number of session licenses
Basic	No
Oracle RAC	Multiple instances can have different values, but the total for all instances mounting a database should be less than or equal to the total number of sessions licensed for that database.

> **Note**
>
> Oracle no longer offers licensing by the number of concurrent sessions. Therefore the `LICENSE_MAX_SESSIONS` and `LICENSE_SESSIONS_WARNING` initialization parameters have been deprecated.

When the limit specified by `LICENSE_MAX_SESSIONS` is reached, only users with the `RESTRICTED SESSION` privilege can connect to the database. Users who are not able to connect receive a warning message indicating that the system has reached maximum capacity.

A zero value indicates that concurrent usage (session) licensing is not enforced. If you set this parameter to a nonzero number, you might also want to set `LICENSE_SESSIONS_WARNING` (see "LICENSE_SESSIONS_WARNING").

Do not enable both concurrent usage licensing and user licensing. Set either `LICENSE_MAX_SESSIONS` or `LICENSE_MAX_USERS` to zero.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information about this parameter

1.173 LICENSE_MAX_USERS

`LICENSE_MAX_USERS` specifies the maximum number of users you can create in the database.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	0 to number of user licenses
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Oracle RAC	Oracle recommends that multiple instances have the same value.

When you reach the limit specified by `LICENSE_MAX_USERS`, you cannot create more users. You can, however, increase the limit.

> **Note**
>
> Oracle no longer offers licensing by the number of concurrent sessions. Therefore the `LICENSE_MAX_SESSIONS` and `LICENSE_SESSIONS_WARNING` initialization parameters have been deprecated.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information about this parameter

1.174 LICENSE_SESSIONS_WARNING

`LICENSE_SESSIONS_WARNING` specifies a warning limit on the number of concurrent user sessions.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	0 to value of `LICENSE_MAX_SESSIONS` parameter
Basic	No
Oracle RAC	Multiple instances can have different values.

> **Note**
>
> Oracle no longer offers licensing by the number of concurrent sessions. Therefore the `LICENSE_MAX_SESSIONS` and `LICENSE_SESSIONS_WARNING` initialization parameters have been deprecated.

When this limit specified by `LICENSE_SESSIONS_WARNING` is reached, additional users can connect, but Oracle writes a message in the alert log for each new connection. Users with `RESTRICTED SESSION` privilege who connect after the limit is reached receive a warning message stating that the system is nearing its maximum capacity.

Copyright © 2002, 2026, Oracle and/or its affiliates.

If this parameter is set to zero, no warning is given as you approach the concurrent usage (session) limit. If you set this parameter to a nonzero number, you should also set `LICENSE_MAX_SESSIONS` (see "LICENSE_MAX_SESSIONS").

See Also

- *Oracle Database Administrator's Guide* for more information on setting this parameter

1.175 LISTENER_NETWORKS

`LISTENER_NETWORKS` specifies one or more sets of local, forward, and remote listeners for cross-registration. All listeners within the same *network_name* will cross-register.

Property	Description
Parameter type	String

Syntax	`LISTENER_NETWORKS =`
	`'((NAME=network_name)`
	`(LOCAL_LISTENER=["]listener_address[,...]["])`
	`(FORWARD_LISTENER=["]listener_address[,...]["])`
	`[(REMOTE_LISTENER=["]listener_address[,...]["])])'`
	`[,...]`

Default value	There is no default value.
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

If a *network_name* is specified multiple times, then the resulting listener set is the union of each specification. This can be used to specify sets that require more than 255 characters, which is the per element limit.

The *listener_address* specifies a string that is an address, address list, or an alias that resolves to an address or address list of Oracle Net listeners. If an alias, the address or address list is specified in the TNSNAMES.ORA file or another address repository as configured for your system.

If a comma is used to specify a *listener_address* list, then the set of addresses must be surrounded by double quotes.

See Also

- "LOCAL_LISTENER"
- "FORWARD_LISTENER"
- "REMOTE_LISTENER"
- *Oracle Database Net Services Administrator's Guide* for more information on setting this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.176 LOB_SIGNATURE_ENABLE

LOB_SIGNATURE_ENABLE is used to enable or disable the LOB locator signature feature.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	true | false
Basic	No
Oracle RAC	Multiple instances must have the same value.

You can secure your LOBs by enabling the LOB locator signature feature. A LOB locator is a pointer to the location of a large object (LOB) value. If the LOB_SIGNATURE_ENABLED initialization parameter is set to true, then when you create a LOB, Oracle Database automatically assigns a signature to the LOB locator. When Oracle Database receives a request from a client, it uses the signature to determine if any tampering with the LOB locator has occurred.

When this parameter is set to true, you also have the option of further securing your LOBs by encrypting your LOB locator signature keys.

Note

This parameter is available starting with Oracle Database 19c, Release Update 19.1.

See Also

Oracle Database Security Guide for more information on LOB locator signatures and encrypting LOB locator signature keys

1.177 LOCAL_LISTENER

LOCAL_LISTENER specifies a network name that resolves to an address or address list of Oracle Net local listeners (that is, listeners that run on the same system as this instance). The address or address list is specified in the TNSNAMES.ORA file or other address repository as configured for your system.

Property	Description
Parameter type	String
Syntax	LOCAL_LISTENER = *network_name*
Default value	(ADDRESS = (PROTOCOL=TCP)(HOST=*hostname*)(PORT=1521)) where *hostname* is the network name of the local host.
Modifiable	ALTER SYSTEM

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable in a PDB	Yes
Basic	No

See Also

- *Oracle Database Concepts* for more information about listener processes and dispatcher processes
- *Oracle Database Net Services Administrator's Guide* and your operating system-specific Oracle documentation for more information about specifying network addresses for the protocols on your system

1.178 LOCK_NAME_SPACE

`LOCK_NAME_SPACE` specifies the namespace that the distributed lock manager (DLM) uses to generate lock names.

Property	Description
Parameter type	String
Syntax	`LOCK_NAME_SPACE = namespace`
Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	No
Range of values	Up to 8 alphanumeric characters. No special characters allowed.
Basic	No

Note

The `LOCK_NAME_SPACE` parameter is deprecated. It is retained for backward compatibility only.

Consider setting this parameter if a standby or clone database has the same database name on the same cluster as the primary database.

If the standby database resides on the same file system as the primary database, then set `LOCK_NAME_SPACE` in the standby parameter file to a distinct value such as the following:

```
LOCK_NAME_SPACE = standby
```

1.179 LOCK_SGA

`LOCK_SGA` locks the entire SGA into physical memory.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	No
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

It is usually advisable to lock the SGA into real (physical) memory, especially if the use of virtual memory would include storing some of the SGA using disk space. This parameter is ignored on platforms that do not support it.

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about using this parameter

1.180 LOG_ARCHIVE_CONFIG

`LOG_ARCHIVE_CONFIG` enables or disables the sending of redo logs to remote destinations and the receipt of remote redo logs, and specifies the unique database names (`DB_UNIQUE_NAME`) for each database in the Data Guard configuration.

Property	Description
Parameter type	String

Syntax	`LOG_ARCHIVE_CONFIG =`
	`{`
	`[ SEND | NOSEND ]`
	`[ RECEIVE | NORECEIVE ]`
	`[ DG_CONFIG=(`*`remote_db_unique_name1`*` [, ...`
	`remote_db_unique_name30``) | NODG_CONFIG ]`
	`}`

Default value	`NULL`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Basic	No

Values

- `SEND`

  Enables the sending of redo logs to remote destinations
- `NOSEND`

  Disables the sending of redo logs to remote destinations

Copyright © 2002, 2026, Oracle and/or its affiliates.

- RECEIVE

  Enables the receipt of remotely archived redo logs

- NORECEIVE

  Disables the receipt of remotely archived redo logs

- DG_CONFIG

  Specifies a list of up to 30 unique database names (defined with the DB_UNIQUE_NAME initialization parameter) for all of the databases in the Data Guard configuration.

- NODG_CONFIG

  Eliminates the list of service provider names previously specified with the DG_CONFIG option.

When this parameter has not been set, its default value is NULL. However, the NULL value is treated as if the parameter has been set to 'SEND, RECEIVE, NODG_CONFIG'.

1.181 LOG_ARCHIVE_DEST

Use LOG_ARCHIVE_DEST to specify the destination to which redo log files will be archived.

Property	Description
Parameter type	String
Syntax	LOG_ARCHIVE_DEST = *filespec*
Default value	Null
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	Any valid path or device name, except raw partitions
Basic	No
Oracle RAC	Multiple instances can have different values.

> **Note**
>
> For Enterprise Edition users, this parameter has been deprecated in favor of the LOG_ARCHIVE_DEST_*n* parameters. If Oracle Enterprise Edition is not installed or it is installed, but you have not specified any LOG_ARCHIVE_DEST_*n* parameters, this parameter is valid.

LOG_ARCHIVE_DEST is applicable only if you are running the database in ARCHIVELOG mode or are recovering a database from archived redo logs. LOG_ARCHIVE_DEST is incompatible with the LOG_ARCHIVE_DEST_*n* parameters, and must be defined as the null string ("") or (' ') when any LOG_ARCHIVE_DEST_*n* parameter has a value other than a null string. Use a text string to specify the default location and root of the disk file or tape device when archiving redo log files. (Archiving to tape is not supported on all operating systems.) The value cannot be a raw partition.

If LOG_ARCHIVE_DEST is not explicitly defined and all the LOG_ARCHIVE_DEST_*n* parameters have null string values, LOG_ARCHIVE_DEST is set to an operating system-specific default value on instance startup.

Copyright © 2002, 2026, Oracle and/or its affiliates.

To override the destination that this parameter specifies, either specify a different destination for manual archiving or use the SQL*Plus statement `ARCHIVE LOG START filespec` for automatic archiving, where *filespec* is the new archive destination. To permanently change the destination, use the statement `ALTER SYSTEM SET LOG_ARCHIVE_DEST = filespec`, where *filespec* is the new archive destination.

Neither `LOG_ARCHIVE_DEST` nor `LOG_ARCHIVE_FORMAT` have to be complete file or directory specifiers themselves; they only need to form a valid file path after the variables are substituted into `LOG_ARCHIVE_FORMAT` and the two parameters are concatenated together.

See Also

- *Oracle Database Backup and Recovery User's Guide*
- "LOG_ARCHIVE_DUPLEX_DEST", "LOG_ARCHIVE_MIN_SUCCEED_DEST", and "V$ARCHIVE_DEST" for more information on setting this parameter
- Your Oracle operating system-specific documentation for the default value and for an example of how to specify the destination path or file name using `LOG_ARCHIVE_DEST`

1.182 LOG_ARCHIVE_DEST_*n*

The `LOG_ARCHIVE_DEST_n` initialization parameter defines up to 31 (where *n* = 1, 2, 3, ... 31) destinations, each of which *must* specify either the `LOCATION` or the `SERVICE` attribute to specify where to archive the redo data.

Property	Description
Parameter type	String

Syntax	`LOG_ARCHIVE_DEST_[1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31] =`
	`{ null_string |`
	`{ LOCATION=path_name | SERVICE=service_name }`
	`[ MANDATORY ]`
	`[ REOPEN[=seconds] ]`
	`[ DELAY[=minutes] ]`
	`[ ENCRYPTION=ENABLED|DISABLED ]`
	`[ GROUP=group ]`
	`[ NOREGISTER ]`
	`[ PRIORITY=priority ]`
	`[ TEMPLATE=template ]`
	`[ ALTERNATE=destination ]`
	`[ MAX_FAILURE=count ]`
	`[ SYNC | ASYNC ]`
	`[ AFFIRM | NOAFFIRM ]`
	`[ NET_TIMEOUT=seconds ]`
	`[ VALID_FOR=(redo_log_type,database_role) ]`
	`[ DB_UNIQUE_NAME ]`
	`[ COMPRESSION={ENABLE|DISABLE|ZLIB|LZO} ]`
	`}`

Default value	There is no default value.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	No
Basic	Yes

All other attributes except the LOCATION or SERVICE attributes are optional. Note that whether you are specifying the LOCATION attribute or the SERVICE attribute, it must be the first attribute supplied in the list of attributes.

If you choose not to enter any attributes, then you can specify a NULL string by entering the following:

```
LOG_ARCHIVE_DEST_n=' ';
```

You set the attributes for the LOG_ARCHIVE_DEST_*n* initialization parameter to control different aspects of how redo transport services transfer redo data from a production or primary database destination to another (standby) database destination. You can query the V$ARCHIVE_DEST view to see the current attribute settings for each destination (n).

> **Note**
>
> Some of the attributes for this parameter are deprecated, but are being retained for backward compatibility. See "Deprecated Attributes for LOG_ARCHIVE_DEST_n".

For every LOG_ARCHIVE_DEST_n initialization parameter that you define, you must specify a corresponding LOG_ARCHIVE_DEST_STATE_n parameter. The LOG_ARCHIVE_DEST_STATE_n (where n is an integer from 1 to 31) initialization parameter specifies whether the corresponding destination is currently enabled or disabled.

> **See Also**
>
> "LOG_ARCHIVE_DEST_STATE_*n*"

Destinations LOG_ARCHIVE_DEST_11 through LOG_ARCHIVE_DEST_31 do not support the SYNC, ARCH, LOCATION, MANDATORY, or ALTERNATE attributes, and cannot be specified as the target of the ALTERNATE attribute.

LOG_ARCHIVE_DEST_11 through LOG_ARCHIVE_DEST_31 can only be used when the COMPATIBLE initialization parameter is set to 11.2.0 or higher.

Values

> **See Also**
>
> *Oracle Data Guard Concepts and Administration* for detailed descriptions of all the values listed in this section

Copyright © 2002, 2026, Oracle and/or its affiliates.

- AFFIRM and NOAFFIRM

  Control whether a redo transport destination acknowledges received redo data before or after writing it to the standby redo log. The default is NOAFFIRM.

- ALTERNATE=LOG_ARCHIVE_DEST_*n*

  Specifies an alternate archiving destination to be used when the original destination fails. There is no default value; if an alternate destination is not specified, then archiving does not automatically change to another destination if the original destination fails.

- ASYNC

  The redo data generated by a transaction need not have been received at a destination which has this attribute before that transaction can commit. This is the default behavior if neither SYNC nor ASYNC is specified.

- COMPRESSION

  Indicates whether network compression is enabled or disabled, or disabled, or whether the ZLIB or LZO algorithm is used. The possible values include:

  – DISABLE: Compression is disabled.
  – ENABLE: Compression is enabled. The ZLIB compression algorithm is used.
  – ZLIB: ZLIB compression is used.
  – LZO: LZO compression is used.

  The Advanced Compression option is required in order to use the COMPRESSION attribute.

- DB_UNIQUE_NAME=*name*

  Specifies a unique name for the database at this destination. You must specify a name; there is no default value.

- DELAY[=*minutes*]

  Specifies a minimum time lag between when redo data is archived on a standby site retrieving redo from a primary and when the archived redo log file is applied to the standby database or any standbys cascading from it. If you specify the DELAY attribute without a time interval, the default is 30 minutes.

- ENCRYPTION=DISABLE|ENABLE

  Controls whether encryption of the redo stream sent to Zero Data Loss Recovery Appliance (Recovery Appliance) is enabled or disabled. The default value for the attribute is DISABLE.

  This attribute cannot be used with the COMPRESSION, LOCATION, or SYNC attributes.

  To use the ENCRYPTION attribute, you must set the COMPATIBLE initialization parameter to 11.2.0.4 or higher on the protected database.

  > **See Also**
  >
  > *Zero Data Loss Recovery Appliance Administrator's Guide* for introductory information about Recovery Appliance

- GROUP=*group*

  The GROUP attribute is used to specify membership in a specific collection of log archive destinations. Groups are numbered 1 through 8. The default group (GROUP=0) is special in

Copyright © 2002, 2026, Oracle and/or its affiliates.

that it cannot be assigned. The default group is populated with all destinations that are not explicitly assigned to a group. All groups other than the default group must consist of a set of remote destinations. No group (other than the default group) can contain local destinations.

- `LOCATION=`*local_disk_directory* or `USE_DB_RECOVERY_FILE_DEST`

  Specifies either a local file system destination or the directory, file system, or Automatic Storage Management disk group that will serve as the fast recovery area. You must specify this attribute for at least one destination. You can specify either a local disk directory or fast recovery area with the `LOCATION` attribute. You *must* include either the `LOCATION` or the `SERVICE` attribute for each destination to specify where to archive the redo data.

- `MANDATORY`

  Specifies that the transmission of redo data to the destination must succeed before the local online redo log file can be made available for reuse. If the `MANDATORY` attribute is not specified, then the destination is optional.

- `MAX_FAILURE`

  Controls the consecutive number of times redo transport services attempt to reestablish communication and transmit redo data to a failed destination before the primary database gives up on the destination. See the `MAX_FAILURE` description in *Oracle Data Guard Concepts and Administration* for usage notes about how this attribute is handled differently in Oracle Database 12*c* Release 2 (12.2.0.1) and in Oracle Database 12*c* Release 1 (12.1.0.2).

- `NET_TIMEOUT=`*seconds*

  Specifies the number of seconds the log writer process on the primary system waits for status from the SYNC (NSS*n*) process before terminating the network connection. The default is 30 seconds.

- `NOREGISTER`

  Indicates that the location of the archived redo log file is not to be recorded at the corresponding destination.

- `PRIORITY=`*priority*

  The `PRIORITY` attribute is used to specify preference within a group of log archive destinations. Priorities are numbered 1 through 8. A lower value represents a higher priority. The lowest priority (`PRIORITY=8`) is special in the sense that if that priority is active then all destinations at that priority will be made active. If any higher priority destination returns to service then that destination will be made active and all low priority destinations will be made inactive because they will be receiving redo from one of the other redo destinations, either through cascading or a Far Sync.

- `REOPEN`[=*seconds*]

  Specifies the minimum number of seconds before the archiver processes (ARC*n*) or the log writer process (LGWR) should try again to access a previously failed destination. The default is 300 seconds.

- `SERVICE=`*net_service_name*

  Specifies a valid Oracle Net service name (`SERVICE=`*net_service_name*) that identifies the remote Oracle database instance to which redo data will be sent. Each destination *must* specify either the `LOCATION` or the `SERVICE` attribute. There is no default net service name.

- `SYNC`

  The redo data generated by a transaction must have been received by every enabled destination which has this attribute before that transaction can commit.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- TEMPLATE=*filename_template*_%t_%s_%r

  Specifies a path name and a file name template for archived redo log files created at a redo transport destination that contain redo data from the database where this attribute is specified. This attribute overrides the value of the LOG_ARCHIVE_FORMAT initialization parameter at a redo transport destination. This attribute does not have a default value.

- VALID_FOR=(redo_log_type, database_role)

  Identifies when redo transport services can transmit redo data to destinations based on the following factors:

  – redo_log_type—whether online redo log files, standby redo log files, or both are *currently* being archived on the database at this destination

  – database_role—whether the database is *currently* running in the primary or the standby role

Deprecated Attributes for LOG_ARCHIVE_DEST_n

The following attributes are deprecated for the LOG_ARCHIVE_DEST_*n* parameter. They are retained for backward compatibility only.

Table 1-3 Deprecated Attributes on the LOG_ARCHIVE_DEST_n Initialization Parameter

Deprecated Attribute	Alternative
ARCH	Specify SYNC or ASYNC. ASYNC is the default if neither attribute is specified.
LGWR	Specify SYNC or ASYNC. ASYNC is the default if neither attribute is specified.
OPTIONAL	Destinations are optional by default.
VERIFY	None. This attribute is only used with the deprecated ARCH attribute.

In addition, note the following changes to the ASYNC and SYNC attributes:

- The BLOCKS keyword on the ASYNC attribute is no longer needed.

  It is no longer necessary to set this keyword because Data Guard dynamically adjusts the block count up or down to an appropriate number of blocks, as necessary.

- The PARALLEL and NOPARALLEL keywords on the SYNC attribute are no longer needed.

1.183 LOG_ARCHIVE_DEST_STATE_*n*

The LOG_ARCHIVE_DEST_STATE_*n* parameters (where *n* = 1, 2, 3, ... 31) specify the availability state of the corresponding destination.

Property	Description
Parameter type	String
Syntax	LOG_ARCHIVE_DEST_STATE_[1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 31] = { enable | defer | alternate | reset }

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	`enable`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	No
Basic	Yes

The parameter suffix (1 through 31) specifies one of the corresponding `LOG_ARCHIVE_DEST_`*n* destination parameters.

Values

- `enable`

  Specifies that a valid log archive destination can be used for a subsequent archiving operation (automatic or manual). This is the default.

- `defer`

  Specifies that valid destination information and attributes are preserved, but the destination is excluded from archiving operations until reenabled.

- `alternate`

  Specifies that a log archive destination is not enabled but will become enabled if communications to another destination fail.

- `reset`

  Functions the same as `defer`, but clears any error messages for the destination if it had previously failed.

The `LOG_ARCHIVE_DEST_STATE_`*n* parameters have no effect on the `ENABLE` state for the `LOG_ARCHIVE_DEST` or `LOG_ARCHIVE_DUPLEX_DEST` parameters.

The `V$ARCHIVE_DEST` dynamic performance view shows values in use for the current session. The `DEST_ID` column of that view corresponds to the archive destination suffix *n*.

See Also

- *Oracle Data Guard Concepts and Administration* for more information about this parameter
- "V$ARCHIVE_DEST"

1.184 LOG_ARCHIVE_DUPLEX_DEST

`LOG_ARCHIVE_DUPLEX_DEST` is similar to the initialization parameter `LOG_ARCHIVE_DEST`. This parameter specifies a second archive destination: the **duplex** archive destination. This duplex archive destination can be either a must-succeed or a best-effort archive destination, depending on how many archive destinations must succeed (as specified in the `LOG_ARCHIVE_MIN_SUCCEED_DEST` parameter).

Property	Description
Parameter type	String

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Syntax	`LOG_ARCHIVE_DUPLEX_DEST = filespec`
Default value	There is no default value.
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	Either a null string or any valid path or device name, except raw partitions
Basic	No

Note

If you are using Oracle Enterprise Edition, this parameter is deprecated in favor of the `LOG_ARCHIVE_DEST_n` parameters. If Oracle Enterprise Edition is not installed or it is installed but you have not specified any `LOG_ARCHIVE_DEST_n` parameters, this parameter is valid.

The default setting of a null string ("") or (' ') indicates that a duplex archive destination does not exist.

See Also

- "LOG_ARCHIVE_DEST_*n*"
- "LOG_ARCHIVE_MIN_SUCCEED_DEST"
- "V$ARCHIVE_DEST"
- *Oracle Database Administrator's Guide* for an example of using this parameter to specify an optional secondary archive destination

1.185 LOG_ARCHIVE_FORMAT

Use `LOG_ARCHIVE_FORMAT` to specify the default filename format when archiving redo log files.

Property	Description
Parameter type	String
Syntax	`LOG_ARCHIVE_FORMAT = filename`
Default value	Operating system-dependent
Modifiable	No
Modifiable in a PDB	No
Range of values	Any string that resolves to a valid filename
Basic	No
Oracle RAC	Multiple instances can have different values, but identical values are recommended.

Copyright © 2002, 2026, Oracle and/or its affiliates.

LOG_ARCHIVE_FORMAT is applicable only if you are using the redo log in ARCHIVELOG mode. Use a text string and variables to specify the default filename format when archiving redo log files. The string generated from this format is appended to the string specified in the LOG_ARCHIVE_DEST parameter.

The following variables can be used in the format:

%s log sequence number

%S log sequence number, zero filled

%t thread number

%T thread number, zero filled

%a activation ID

%d database ID

%r resetlogs ID that ensures unique names are constructed for the archived log files across multiple incarnations of the database

Using uppercase letters for the variables (for example, %S) causes the value to be fixed length and padded to the left with zeros. An example of specifying the archive redo log filename format follows:

```
LOG_ARCHIVE_FORMAT = 'log%t_%s_%r.arc'
```

Archive log file names must contain each of the elements %s (sequence), %t (thread), and %r (resetlogs ID) to ensure that all archive log file names are unique. If the LOG_ARCHIVE_FORMAT initialization parameter is set in the parameter file, then make sure the parameter value contains the %s, %t, and %r elements. Otherwise, the following error is displayed at the time of instance startup:

```
ORA-19905: log_archive_format must contain %s, %t and %r
```

Neither LOG_ARCHIVE_DEST nor LOG_ARCHIVE_FORMAT have to be complete file or directory specifiers themselves; they only need to form a valid file path after the variables are substituted into LOG_ARCHIVE_FORMAT and the two parameters are concatenated together.

LOG_ARCHIVE_FORMAT is ignored in these cases:

- For archived log files that go to the fast recovery area
- When LOG_ARCHIVE_DEST[_n] points to the root of an Oracle ASM disk group (for example, +DATA). The directory of a disk group (for example, +DATA/logs) must be specified for the parameter to be honored.

In these cases where LOG_ARCHIVE_FORMAT is ignored, an Oracle ASM file name is used. See *Oracle Automatic Storage Management Administrator's Guide* for more information on Oracle ASM file names.

ⓘ See Also

- *Oracle Database Backup and Recovery User's Guide*, *Oracle Data Guard Concepts and Administration*, and *Oracle Real Application Clusters Administration and Deployment Guide* for more information about this parameter
- Your operating system- specific Oracle documentation for the default value and range of values for LOG_ARCHIVE_FORMAT

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.186 LOG_ARCHIVE_MAX_PROCESSES

`LOG_ARCHIVE_MAX_PROCESSES` specifies the maximum number of ARC*n* processes that can be created.

Property	Description
Parameter type	Integer
Default value	4
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	1 to 30
Basic	No

See Also

- "Background Processes" for more information about ARC*n* processes
- *Oracle Database Administrator's Guide* for more information about `LOG_ARCHIVE_MAX_PROCESSES`

1.187 LOG_ARCHIVE_MIN_SUCCEED_DEST

`LOG_ARCHIVE_MIN_SUCCEED_DEST` defines the minimum number of destinations that must succeed in order for the online logfile to be available for reuse.

Property	Description
Parameter type	Integer
Default value	1
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	1 to 10 if you are using `LOG_ARCHIVE_DEST_`*n* 1 or 2 if you are using `LOG_ARCHIVE_DEST` and `LOG_ARCHIVE_DUPLEX_DEST`
Basic	No

If you are using the `LOG_ARCHIVE_DEST_`*n* parameters and automatic archiving is enabled, then the value of this parameter cannot exceed the number of enabled, valid destinations specified as `MANDATORY` plus the number of enabled, valid destinations that are configured with the `OPTIONAL` and `LOCATION` attributes.

If you are using `LOG_ARCHIVE_DEST` and `LOG_ARCHIVE_DUPLEX_DEST` and automatic archiving is enabled, a value of `1` specifies that the destination specified in `LOG_ARCHIVE_DEST` must succeed. A value of `2` specifies that the destinations specified in both parameters must succeed.

If the value of this parameter is less than the number of enabled, valid `MANDATORY` destinations, this parameter is ignored in favor of the `MANDATORY` destination count. If the value is more than

Copyright © 2002, 2026, Oracle and/or its affiliates.

the number of enabled, valid MANDATORY destinations, then some of the enabled, valid destinations configured with the OPTIONAL and LOCATION attributes are treated as MANDATORY.

You can switch dynamically from using the older parameters to the LOG_ARCHIVE_DEST_*n* parameter using ALTER SYSTEM, as follows:

1. Set LOG_ARCHIVE_MIN_SUCCEED_DEST to 1.
2. Set the value of LOG_ARCHIVE_DEST and LOG_ARCHIVE_DUPLEX_DEST to the null string.
3. Set the desired number of destinations for the LOG_ARCHIVE_DEST_*n* parameters.
4. Reset LOG_ARCHIVE_MIN_SUCCEED_DEST to the desired value.

See Also

- *Oracle Database Administrator's Guide* for more information on setting this parameter
- "LOG_ARCHIVE_DEST_*n*", "LOG_ARCHIVE_DUPLEX_DEST", and "V$ARCHIVE_DEST" for information on related parameters

1.188 LOG_ARCHIVE_TRACE

LOG_ARCHIVE_TRACE enables and controls the generation of comprehensive trace information for log archiving and redo transport activity.

Property	Description
Parameter type	Integer
Default value	0
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	0, 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096, 8192, 16384, 32768
Basic	No
Oracle RAC	Multiple instances can have different values.

The additional tracing that is output when setting LOG_ARCHIVE_TRACE to a non-zero value can appear in trace files for an archive process, RFS process, LGWR process, SYNC process, ASYNC process, foreground process, MRP process, recovery process, log apply process, startup process, shutdown process, and other processes that use redo transport services.

Table 1-4 describes the valid LOG_ARCHIVE_TRACE values.

Table 1-4 LOG_ARCHIVE_TRACE Values

Value	Description	More Information
0	Disables archivelog tracing (this is the default)	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 1-4 (Cont.) LOG_ARCHIVE_TRACE Values

Value	Description	More Information
1	High-level tracing	High-level tracing provides a small amount of tracing from most redo transport processes. When a value is specified that is not high-level (not 1), the high-level tracing for that value will be output to the trace file even if the high-level tracing has not been specifically requested. For example, if you set LOG_ARCHIVE_TRACE to 128, you will get all the trace messages pertaining to the FAL engine, *including* the high-level tracing from the FAL engine.
2	Tracks Data Guard interfaces	This tracing level is for the various Data Guard configuration parameters, the various Data Guard related SQL commands, the various Data Guard table accesses, and some Data Guard internal interfaces.
4	Tracks common redo transport services	This tracing level is for a number of common services provided by redo transport to various internal users (such as the RFS process, the LGWR process, and the ASYNC process). These include locking primitives, use of common VOS (Virtual Operating System) interfaces, task scheduling, process monitoring, control file access, and other miscellaneous common services.
8	Tracks database protection mode	
16	Tracks disk and network I/O requests	
32	Tracks redo transport destinations	
64	Tracks redo transport archive engine	
128	Tracks redo transport FAL (fetch archive log) engine	
256	Tracks Physical, Logical, and Availability Machine clients of RFS process	
512	Tracks redo transport of LGWR, SYNC, and ASYNC processes	
1024	Tracks redo transport RFS process	
2048	Tracks redo transport gap resolution	
4096	Tracks real-time apply	
8192	Tracks recovery process	
16384	Tracks redo transport buffer management	
32768	Tracks LogMiner dictionary	

You can combine tracing levels by adding together the values of the desired tracing levels. For example, a setting of 3 will generate level 1 and level 2 trace output. You can set different values for the primary and standby database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

When this parameter is set to the default value of 0, Oracle will still generate appropriate alert and trace entries in response to error conditions. If you change the value of this parameter dynamically in an `ALTER SYSTEM` statement, the change takes effect immediately.

Many of the trace messages from redo transport processes include a routine name at the beginning of the trace message. The first part of the routine name specifies the module for that routine. Table 1-5 shows the tracing value used for each redo transport module. Note that the tracing value for transport modules can change from release to release.

Table 1-5 Tracing Values Used for Redo Transport Modules

Module Name	Purpose	Tracing Value
krsa	Locking primitives	4
krsb	Buffer management	16384
krsc	Crash recovery	64
krsd	Destination	32
krse	Archive engine	64
krsf	FAL engine	128
krsg	Gap resolution	2048
krsh	Helper	4
krsi	Input/output	16
krsj	Protection mode	8
krsk	Control file access	4
krsl	Log writer	512
krsm	MRP process	8192
krsn	Configuration	2
krso	Process monitor	4
krsp	Physical standby and Availability Machine RFS client	256
krsq	SQL commands	2
krsr	RFS process	1024
krst	Tables	2
krsu	Network I/O	16
krsv	VOS front end	4
krsw	ASYNC process	512
krsx	Internal interfaces	2

See Also

Oracle Database Administrator's Guide

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.189 LOG_BUFFER

`LOG_BUFFER` specifies the amount of memory (in bytes) that Oracle uses when buffering redo entries to a redo log file.

Property	Description
Parameter type	Big integer
Default value	2 MB to 32 MB, depending on the SGA size and CPU count
Modifiable	No
Modifiable in a PDB	No
Range of values	2 MB to operating system-dependent
Basic	No

Redo log entries contain a record of the changes that have been made to the database block buffers. The LGWR process writes redo log entries from the log buffer to a redo log file.

The log buffer size depends on the number of redo strands in the system. One redo strand is allocated for every 16 CPUs and has a default size of 2 MB. Oracle allocates a minimum of 2 redo strands per instance. When the log buffer size is not specified, any remaining memory in the redo granules is given to the log buffer.

See Also

- Your operating system-specific Oracle documentation for the default value and range of values
- *Oracle Database Performance Tuning Guide* for information on resizing the redo log buffer using this parameter

1.190 LOG_CHECKPOINT_INTERVAL

`LOG_CHECKPOINT_INTERVAL` specifies the frequency of checkpoints in terms of the number of redo log file blocks that can exist between an incremental checkpoint and the last block written to the redo log. This number refers to physical operating system blocks, not database blocks.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`0` to 2^{31} - 1
Basic	No
Oracle RAC	Multiple instances can have different values.

Regardless of this value, a checkpoint always occurs when switching from one online redo log file to another. Therefore, if the value exceeds the actual redo log file size, checkpoints occur

Copyright © 2002, 2026, Oracle and/or its affiliates.

only when switching logs. Checkpoint frequency is one of the factors that influence the time required for the database to recover from an unexpected failure.

Note

- Specifying a value of `0` (zero) for `LOG_CHECKPOINT_INTERVAL` has the same effect as setting the parameter to infinity and causes the parameter to be ignored. Only nonzero values of this parameter are considered meaningful.
- Recovery I/O can also be limited by setting the `LOG_CHECKPOINT_TIMEOUT` parameter or by the size specified for the smallest redo log. For information on which mechanism is controlling checkpointing behavior, query the `V$INSTANCE_RECOVERY` view.

See Also

- "LOG_CHECKPOINT_TIMEOUT"
- "V$INSTANCE_RECOVERY"
- *Oracle Database Performance Tuning Guide* for information on disabling or removing this parameter when the `FAST_START_MTTR_TARGET` parameter is set

1.191 LOG_CHECKPOINT_TIMEOUT

`LOG_CHECKPOINT_TIMEOUT` specifies (in seconds) the amount of time that has passed since the incremental checkpoint at the position where the last write to the redo log (sometimes called the **tail of the log**) occurred. This parameter also signifies that no buffer will remain dirty (in the cache) for more than *integer* seconds.

Property	Description
Parameter type	Integer
Default value	`1800`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`0` to 2^{31} - 1
Basic	No
Oracle RAC	Multiple instances can have different values.

Specifying a value of `0` for the timeout disables time-based checkpoints. Hence, setting the value to `0` is not recommended unless `FAST_START_MTTR_TARGET` is set.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Note

- A checkpoint scheduled to occur because of this parameter is delayed until the completion of the previous checkpoint if the previous checkpoint has not yet completed.
- Recovery I/O can also be limited by setting the `LOG_CHECKPOINT_INTERVAL` parameter or by the size specified for the smallest redo log. For information on which mechanism is controlling checkpointing behavior, query the `V$INSTANCE_RECOVERY` view.

See Also

- "LOG_CHECKPOINT_TIMEOUT"
- "V$INSTANCE_RECOVERY"
- *Oracle Database Performance Tuning Guide* for information on disabling or removing this parameter when the `FAST_START_MTTR_TARGET` parameter is set

1.192 LOG_CHECKPOINTS_TO_ALERT

`LOG_CHECKPOINTS_TO_ALERT` lets you log your checkpoints to the alert log. Doing so is useful for determining whether checkpoints are occurring at the desired frequency.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

1.193 LOG_FILE_NAME_CONVERT

`LOG_FILE_NAME_CONVERT` converts the filename of a new log file on the primary database to the filename of a log file on the standby database.

Property	Description
Parameter type	String

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description

Syntax	`LOG_FILE_NAME_CONVERT = 'string1' , 'string2' , 'string3' , 'string4' , ...`
	Where:
	• *string1* is the pattern of the primary database filename
	• *string2* is the pattern of the standby database filename
	• *string3* is the pattern of the primary database filename
	• *string4* is the pattern of the standby database filename
	You can enclose each string in single or double quotation marks.
	You can specify as many pairs of primary and standby replacement strings as required.
	Example:
	`LOG_FILE_NAME_CONVERT = '/dbs/t1/','/dbs/t1/s','dbs/t2/','dbs/t2/s'`

Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	No
Basic	No

If you add a log file to the primary database, you must add a corresponding file to the standby database.

Set the value of this parameter to one or more pairs of strings. The first string is the pattern found in the log file names on the primary database. The second string is the pattern found in the log file names on the standby database.

When the standby database is updated, this parameter converts the log file name on the primary database to the log file name on the standby database. The file must exist on the standby database and must be writable or the recovery process will halt with an error.

If you specify an odd number of strings (the last string has no corresponding replacement string), an error is signalled during startup. If the filename being converted matches more than one pattern in the pattern/replace string list, the first matched pattern takes effect. There is no limit on the number of pairs that you can specify in this parameter (other than the hard limit of the maximum length of multivalue parameters).

You should also use `LOG_FILE_NAME_CONVERT` to rename the logfiles in the clone control file when setting up the clone database during tablespace point-in-time recovery.

Note

The `LOG_FILE_NAME_CONVERT` parameter applies only to online logs (not to archived logs).

See Also

Oracle Data Guard Concepts and Administration

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.194 LONG_MODULE_ACTION

LONG_MODULE_ACTION enables the use of longer lengths for modules and actions.

Property	Description
Parameter type	Boolean
Default value	true
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	true | false
Basic	No
Oracle RAC	The same value must be used on all instances.

Module length was 48 bytes and action length was 32 bytes in Oracle Database releases prior to Oracle Database 12*c* Release 2 (12.2.0.1).

Starting with Oracle Database 12*c* Release 2 (12.2.0.1):

- If LONG_MODULE_ACTION is set to TRUE (the default value), then the length of modules and actions will be 64 bytes each.
- If LONG_MODULE_ACTION is set to FALSE, then the length of modules will be 48 bytes, and the length of actions will be 32 bytes.

1.195 MAIN_WORKLOAD_TYPE

MAIN_WORKLOAD_TYPE specifies the main workload type for a database or PDB.

Property	Description
Parameter type	String
Syntax	MAIN_WORKLOAD_TYPE = { OLTP | ANALYTICS }
Default value	OLTP
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Different instances can use different values.

This parameter is supported only for Oracle Exadata. See *Oracle Exadata System Software User's Guide* for more information about setting this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.196 MAX_AUTH_SERVERS

MAX_AUTH_SERVERS specifies the maximum number of authentication servers in the authentication pool. This pool authenticates user connections when client applications connect to Database Resident Connection Pooling (DRCP).

Property	Description
Parameter type	Integer
Default value	25
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	Minimum: MIN_AUTH_SERVERS or 1, whichever is the greater value Maximum: operating system-dependent
Basic	No
Oracle RAC	Different instances can use different values.

The value of this parameter must be greater than or equal to the value of the MIN_AUTH_SERVERS initialization parameter, which specifies the minimum number of authentication servers in the authentication pool. The exception is when MIN_AUTH_SERVERS = 0. In this case, the value of MAX_AUTH_SERVERS must be greater than or equal to 1.

Note

This parameter is available starting with Oracle Database 19c, Release Update 19.10.

See Also

- *Oracle Database Administrator's Guide* for more information about setting this parameter
- "MIN_AUTH_SERVERS"

1.197 MAX_DATAPUMP_JOBS_PER_PDB

MAX_DATAPUMP_JOBS_PER_PDB determines the maximum number of concurrent Oracle Data Pump jobs per PDB.

Property	Description
Parameter type	String
Syntax	MAX_DATAPUMP_JOBS_PER_PDB = { AUTO | integer }
Default value	100
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Range of values	`0` to `250` or `AUTO`
Basic	No
Oracle RAC	The same value must be used on all instances.

The default value will not work for all databases. Database administrators will have to determine if the default value works well for their database.

When this parameter has a value of `AUTO`, Oracle Data Pump will derive its actual value to be 50% of the `SESSIONS` initialization parameter.

A value that is too large could cause Oracle Data Pump to consume too many system resources, while a value that is too small could prevent users from performing their Oracle Data Pump tasks.

The main resource Oracle Data Pump uses is shared pool in the System Global Area (SGA) for the database. Parallel jobs increase the number of sessions and, depending on the job, the number of PX servers used.

See Also

- *Oracle Database Utilities* for more information about using Oracle Data Pump with CDBs
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_DATAPUMP` PL/SQL package

1.198 MAX_DATAPUMP_PARALLEL_PER_JOB

`MAX_DATAPUMP_PARALLEL_PER_JOB` specifies the maximum number of parallel processes allowed per Oracle Data Pump job.

Property	Description
Parameter type	String
Syntax	`MAX_DATAPUMP_PARALLEL_PER_JOB = { integer | AUTO }`
Default value	`50`
Range of values	`1` to `1024`, or `AUTO`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Different instances can use different values.

When this parameter has a value of `AUTO`, Oracle Data Pump derives its value to be 50 percent of the value of the `SESSIONS` initialization parameter.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.1.

1.199 MAX_DISPATCHERS

`MAX_DISPATCHERS` specifies the maximum number of dispatcher processes allowed to be running simultaneously.

Property	Description
Parameter type	Integer
Default value	There is no default value.
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	If `MAX_DISPATCHERS` is specified, then it should be greater than or equal to the number of dispatchers specified by the `DISPATCHERS` parameter and less than the number of processes specified by the `PROCESSES` parameter.
Basic	No

This parameter can be overridden by the `DISPATCHERS` parameter and is maintained for backward compatibility with older releases.

> **See Also**
>
> - *Oracle Database Administrator's Guide* for more information about this parameter
> - Your operating system-specific Oracle documentation for the default value and range of values

1.200 MAX_DUMP_FILE_SIZE

`MAX_DUMP_FILE_SIZE` specifies the maximum size of trace files (excluding the alert log).

Property	Description
Parameter type	String
Syntax	`MAX_DUMP_FILE_SIZE = { integer [K | M | G] | UNLIMITED }`
Default value	`UNLIMITED`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	0 to unlimited, or `UNLIMITED`
Basic	No

You can change this limit if you are concerned that trace files may use too much space.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- A numeric value for MAX_DUMP_FILE_SIZE specifies the maximum size in operating system blocks.
- A numeric value followed by a K or M or G suffix specifies the file size in kilobytes, megabytes, or gigabytes.
- The special value string UNLIMITED means that there is no upper limit on trace file size. Thus, dump files can be as large as the operating system permits.

When the trace file is limited in size, it may be automatically split into multiple files, called segments, if needed. This process is call trace file segmentation. The segments will have the same file name as the active trace file, but with an extra segment number appended.

The trace file can be split into a maximum of 5 segments, and the size of each segment will typically be 1/5th of the trace file limit.

When the combined size of all the trace file segments exceeds the specified limit, the oldest segment is deleted, and a new, empty segment is created. Thus, the trace file always contains the most recent trace information. The first segment is never deleted, because it may contain relevant information about the initial state of the process.

Note that trace file segmentation will not occur when this parameter is set to a value less than 25 MB.

See Also

Oracle Database Administrator's Guide and *Oracle Database SQL Tuning Guide* for more information on setting this parameter

1.201 MAX_IDLE_BLOCKER_TIME

MAX_IDLE_BLOCKER_TIME specifies the maximum number of minutes that a blocking session can be idle. After that point, the session is automatically terminated.

Property	Description
Parameter type	Integer
Default value	0
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to the maximum integer. The value of 0 indicates that there is no limit.
Basic	No
Oracle RAC	Different instances can use different values.

A session is considered to be a blocking session when it is holding resources required by other sessions. For example:

- The session is holding a lock required by another session.
- The session is a parallel operation and its consumer group, PDB, or database has either reached its maximum parallel server limit or has queued parallel operations.
- The session's PDB or database instance is about to reach its SESSIONS or PROCESSES limit.

Copyright © 2002, 2026, Oracle and/or its affiliates.

This parameter differs from the MAX_IDLE_TIME parameter in that MAX_IDLE_TIME applies to all sessions (blocking and non-blocking), whereas MAX_IDLE_BLOCKING_TIME applies only to blocking sessions. Therefore, in order for MAX_IDLE_BLOCKING_TIME to be effective, its limit must be less than the MAX_IDLE_TIME limit.

Note

This parameter does not have an effect on parallel query processes, nor on SYS user sessions.

Note

This parameter is available starting with Oracle Database 19c.

See Also

"MAX_IDLE_TIME"

1.202 MAX_IDLE_TIME

MAX_IDLE_TIME specifies the maximum number of minutes that a session can be idle. After that point, the session is automatically terminated.

Property	Description
Parameter type	Integer
Default value	0
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to the maximum integer. The value of 0 indicates that there is no limit.
Basic	No
Oracle RAC	Different instances can use different values.

See Also

"MAX_IDLE_BLOCKER_TIME" to learn how to set a lower idle time limit for sessions that are holding resources required by other sessions

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.203 MAX_IOPS

`MAX_IOPS` enables you to set the maximum number of I/Os that can be issued per second on a per pluggable database (PDB) basis. This parameter is used to throttle PDB I/Os.

Property	Description
Parameter type	Integer
Default value	0
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`0` to the maximum `Integer` value. A very low value (for example, under 100 I/Os per second) is not recommended.
Basic	No
Oracle RAC	Different values can be set on different instances.

DBWR I/Os, control file I/Os, password file I/Os and other critical I/Os are exempted from the rate limit set by this parameter, but their I/Os are accounted for while throttling. Because of these exemptions, the PDB's actual I/O rate may sometimes exceed the limit.

This feature is enabled for multitenant container database (CDB) only. The feature is not supported on Oracle Exadata.

This parameter can be set from inside a PDB. If the parameter is set in CDB$ROOT, all the PDBs in that CDB will inherit the parameter value from CDB$ROOT. This parameter cannot be set in a non-CDB environment.

The default value of `0` means that no limits are set on the maximum number of I/Os that can be issued per second in a PDB.

If Oracle processes need to wait because of this IO rate limit, the wait event is resmgr: I/O rate limit.

See Also

- "MAX_MBPS"
- "resmgr: I/O rate limit"

Examples

This example shows how to use SQL statements to set a maximum of 3000 I/Os per second on a PDB named CDB1_PDB1:

```
alter session set container = cdb1_pdb1;
alter system set max_iops = 3000;
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

1.204 MAX_MBPS

MAX_MBPS enables you to set the maximum number of megabytes (MB) of I/Os issued per second on a per pluggable database (PDB) basis. This parameter is used to throttle PDB I/Os.

Property	Description
Parameter type	Integer
Default value	0
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to the maximum Integer value. A very low value (for example, under 25 MB per second) is not recommended.
Basic	No
Oracle RAC	Different values can be set on different instances.

DBWR I/Os, control file I/Os, password file I/Os and other critical I/Os are exempted from the throughput limit set by this parameter, but their I/Os are accounted for while throttling. Because of these exemptions, the PDB's actual I/O rate may sometimes exceed the limit.

This feature is enabled for multitenant container database (CDB) only. The feature is not supported on Oracle Exadata.

This parameter can be set from inside a PDB. If the parameter is set in CDB$ROOT, all the PDBs in that CDB will inherit the parameter value from CDB$ROOT. This parameter cannot be set in a non-CDB environment.

The default value of 0 means that no limits are set on the maximum number of megabytes (MB) of I/Os that can be issued per second in a PDB.

If Oracle processes need to wait because of this IO rate limit, the wait event is resmgr: I/O rate limit.

See Also

- "MAX_IOPS"
- "resmgr: I/O rate limit"

Examples

This example shows how to use SQL statements to set a maximum of 200 megabytes per second on a PDB named CDB1_PDB1:

```
alter session set container = cdb1_pdb1;
alter system set max_mbps = 200;
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

1.205 MAX_PDBS

MAX_PDBS allows you to limit the number of pluggable databases (PDBs) that can be created in a CDB or an application container.

Property	Description
Parameter type	Integer
Default value	Depends on the Oracle Database edition Possible values: 5, 254, or 4098
Modifiable	ALTER SYSTEM ... SID='*'
Modifiable in a PDB	No. However, this parameter can be modified in the application root of an application container.
Range of values	Minimum: 0 Maximum: The default value for the Oracle Database edition (5, 254, or 4098)
Basic	No
Oracle RAC	The same value must be used on all instances.

This parameter can only be set while connected to the CDB root or an application root.

This parameter applies to PDBs, application containers, and application PDBs. An application PDB is a PDB that resides in an application container.

This parameter does not apply to the PDB seed (PDB$SEED), application seeds, or application root clones.

> **Note**
>
> The maximum number of PDBs that you are allowed to create in a CDB depends on the licensing policy for your Oracle Database edition. To learn more, refer to the Oracle Multitenant information in the "Consolidation" table in *Oracle Database Licensing Information User Manual*.

1.206 MAX_SHARED_SERVERS

MAX_SHARED_SERVERS specifies the maximum number of shared server processes allowed to be running simultaneously. Setting this parameter enables you to reserve process slots for other processes, such as dedicated servers.

Property	Description
Parameter type	Integer
Default value	There is no default value.
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	If MAX_SHARED_SERVERS is specified, then it should be greater than or equal to SHARED_SERVERS and less than PROCESSES.
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

When you want to reduce the range of shared servers, you can reduce MAX_SHARED_SERVERS before reducing SHARED_SERVERS. If MAX_SHARED_SERVERS is lower than SHARED_SERVERS, then the number of shared servers will not vary but will remain at the constant level specified by SHARED_SERVERS. If MAX_SHARED_SERVERS is not specified, then a shared server process may be spawned as long as the number of free process slots is greater than 1 / 8 the maximum number of processes, or 2 if PROCESSES is less than 24.

See Also

- "SHARED_SERVERS"
- "PROCESSES"
- *Oracle Database Administrator's Guide* for more information on setting this parameter
- *Oracle Database Concepts* for information on processes
- Your operating system-specific Oracle documentation for the default value and range of values

1.207 MAX_STRING_SIZE

MAX_STRING_SIZE controls the maximum size of VARCHAR2, NVARCHAR2, and RAW data types in SQL.

Property	Description
Parameter type	String
Syntax	MAX_STRING_SIZE = { STANDARD | EXTENDED }
Default value	STANDARD
Modifiable	ALTER SYSTEM ... SID='*'[1]
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Multiple instances must use the same value.

1 Use ALTER SYSTEM only when the database is in UPGRADE mode, and run the utl32k.sql script afterward, as explained in this section.

STANDARD means that the length limits for Oracle Database releases prior to Oracle Database 12*c* apply (for example, 4000 bytes for VARCHAR2 and NVARCHAR2, and 2000 bytes for RAW).

EXTENDED means that the 32767 byte limit introduced in Oracle Database 12*c* applies.

The COMPATIBLE initialization parameter must be set to 12.0.0.0 or higher to set MAX_STRING_SIZE = EXTENDED.

You can change the value of MAX_STRING_SIZE from STANDARD to EXTENDED. However, you cannot change the value of MAX_STRING_SIZE from EXTENDED to STANDARD.

By setting MAX_STRING_SIZE = EXTENDED, users are taking an explicit action that could introduce application incompatibility in their database. Applications that do not want to use the expanded data types can be rewritten for compatibility with either setting; for example, these applications

Copyright © 2002, 2026, Oracle and/or its affiliates.

could use explicit CASTs to fix the length of VARCHAR2 expressions during CREATE TABLE AS SELECT.

Altering MAX_STRING_SIZE will update database objects and possibly invalidate them, as follows:

- Tables with virtual columns will be updated with new data type metadata for virtual columns of VARCHAR2(4000), 4000-byte NVARCHAR2, or RAW(2000) type.
  - Functional indexes will become unusable if a change to their associated virtual columns causes the index key to exceed index key length limits. Attempts to rebuild such indexes will fail with ORA-01450: maximum key length exceeded.
- Views will be invalidated if they contain VARCHAR2(4000), 4000-byte NVARCHAR2, or RAW(2000) typed expression columns.
- Materialized views will be updated with new metadata VARCHAR2(4000), 4000-byte NVARCHAR2, and RAW(2000) typed expression columns

Increasing the Maximum Size of VARCHAR2, NVARCHAR2, and RAW Columns in a Non-CDB

To increase the maximum size of VARCHAR2, NVARCHAR2, and RAW columns in a non-CDB:

1. Shut down the database.
2. Restart the database in UPGRADE mode.
3. Change the setting of MAX_STRING_SIZE to EXTENDED.
4. Run the rdbms/admin/utl32k.sql script. You must be connected AS SYSDBA to run the script.
5. Restart the database in NORMAL mode.

> **Note**
>
> The utl32k.sql script increases the maximum size of the VARCHAR2, NVARCHAR2, and RAW columns for the views where this is required. The script does not increase the maximum size of the VARCHAR2, NVARCHAR2, and RAW columns in some views because of the way the SQL for those views is written.

6. Run the rdbms/admin/utlrp.sql script to recompile invalid objects. You must be connected AS SYSDBA to run the script.

Increasing the Maximum Size of VARCHAR2, NVARCHAR2, and RAW Columns in a CDB

To increase the maximum size of VARCHAR2, NVARCHAR2, and RAW columns in a CDB and in all the PDBs in the CDB:

1. Connect to the CDB AS SYSDBA.
2. In the root, change the setting of MAX_STRING_SIZE to EXTENDED:

```
ALTER SESSION SET CONTAINER=CDB$ROOT;
ALTER SYSTEM SET max_string_size=extended SCOPE=SPFILE;
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The root continues to use STANDARD semantics even after MAX_STRING_SIZE is set to EXTENDED. The reason for setting MAX_STRING_SIZE to EXTENDED in the root is so all the PDBs in the CDB can inherit the EXTENDED setting from the root.

3. Shut down the CDB.
4. Restart the CDB in UPGRADE mode.

```
startup upgrade;
```

5. Use the catcon.pl script to run the rdbms/admin/utl32k.sql script in the root and in all the PDBs in the CDB to increase the maximum size of the VARCHAR2, NVARCHAR2, and RAW columns. The --force_pdb_mode 'UPGRADE' option is used to ensure that all PDBs, including application root clones, are opened in migrate mode. Enter the SYS password when prompted:

```
$ cd $ORACLE_HOME/rdbms/admin
$ mkdir /scratch/mydir/utl32k_cdb_pdbs_output
$ $ORACLE_HOME/perl/bin/perl $ORACLE_HOME/rdbms/admin/catcon.pl -u SYS --
force_pdb_mode 'UPGRADE' -d $ORACLE_HOME/rdbms/admin -l '/scratch/mydir/
utl32k_cdb_pdbs_output' -b
utl32k_cdb_pdbs_output utl32k.sql
catcon: ALL catcon-related output will be written to [/scratch/mydir/
utl32k_cdb_pdbs_output/utl32k_cdb_pdbs_output_catcon_23172.lst]
catcon: See [/scratch/mydir/utl32k_cdb_pdbs_output/
utl32k_cdb_pdbs_output*.log] files for output generated by scripts
catcon: See [/scratch/mydir/utl32k_cdb_pdbs_output/
utl32k_cdb_pdbs_output_*.lst] files for spool files, if any
Enter Password:
catcon.pl: completed successfully
$
```

> **Note**
>
> The utl32k.sql script increases the maximum size of the VARCHAR2, NVARCHAR2, and RAW columns for the views where this is required. The script does not increase the maximum size of the VARCHAR2, NVARCHAR2, and RAW columns in some views because of the way the SQL for those views is written.

6. Connect to the CDB AS SYSDBA and shut down the database.
7. Restart the CDB in NORMAL mode.

```
startup;
```

8. Use the catcon.pl script to run the rdbms/admin/utlrp.sql script to recompile invalid objects in the root and in all the PDBs in the CDB. The --force_pdb_mode 'READ WRITE'

Copyright © 2002, 2026, Oracle and/or its affiliates.

option is used to ensure that all the PDBs (including application root clones) are opened in read write mode. Enter the SYS password when prompted:

```
$ cd $ORACLE_HOME/rdbms/admin
$ mkdir /scratch/mydir/utlrp_cdb_pdbs_output
$ $ORACLE_HOME/perl/bin/perl $ORACLE_HOME/rdbms/admin/catcon.pl -u SYS --
force_pdb_mode 'READ WRITE' -d $ORACLE_HOME/rdbms/admin -l '/scratch/mydir/
utlrp_cdb_pdbs_output' -b utlrp_cdb_pdbs_output utlrp.sql
catcon: ALL catcon-related output will be written to [/scratch/mydir/
utlrp_cdb_pdbs_output/utlrp_cdb_pdbs_output_catcon_24271.lst]
catcon: See [/scratch/mydir/utlrp_cdb_pdbs_output/
utlrp_cdb_pdbs_output*.log] files for output generated by scripts
catcon: See [/scratch/mydir/utlrp_cdb_pdbs_output/
utlrp_cdb_pdbs_output_*.lst] files for spool files, if any
Enter Password:
catcon.pl: completed successfully
$
```

See Also

Oracle Multitenant Administrator's Guide for information about using the `catcon.pl` script to run Oracle-supplied scripts in a CDB and PDBs.

Increasing the Maximum Size of VARCHAR2, NVARCHAR2, and RAW Columns in a PDB

To increase the maximum size of `VARCHAR2`, `NVARCHAR2`, and `RAW` columns in a PDB:

1. Shut down the PDB.
2. Reopen the PDB in migrate mode.

Note

The following SQL statement can be used to reopen a PDB in migrate mode when the current container is the PDB:

`ALTER PLUGGABLE DATABASE` *pdb-name* `OPEN UPGRADE;`

3. Change the setting of `MAX_STRING_SIZE` in the PDB to `EXTENDED`.
4. Run the `rdbms/admin/utl32k.sql` script in the PDB. You must be connected `AS SYSDBA` to run the `utl32k.sql` script.
5. Reopen the PDB in `NORMAL` mode.

Note

The `utl32k.sql` script increases the maximum size of the `VARCHAR2`, `NVARCHAR2`, and `RAW` columns for the views where this is required. The script does not increase the maximum size of the `VARCHAR2`, `NVARCHAR2`, and `RAW` columns in some views because of the way the SQL for those views is written.

Copyright © 2002, 2026, Oracle and/or its affiliates.

6. Run the `rdbms/admin/utlrp.sql` script in the PDB to recompile invalid objects. You must be connected `AS SYSDBA` to run the script.

> **See Also**
>
> *Oracle Multitenant Administrator's Guide* for more information about modifying the open mode of PDBs.

Increasing the Maximum Size of VARCHAR2, NVARCHAR2, and RAW Columns in an Oracle RAC Database

To increase the maximum size of `VARCHAR2`, `NVARCHAR2`, and `RAW` columns in an Oracle RAC database:

1. Shut down all of the Oracle RAC database instances, except one.
2. Restart the Oracle RAC database instance in `UPGRADE` mode.
3. Change the setting of `MAX_STRING_SIZE` to `EXTENDED`.
4. Run the `rdbms/admin/utl32k.sql` script in the Oracle RAC database instance. You must be connected `AS SYSDBA` to run the script.
5. Restart all Oracle RAC database instances in `NORMAL` mode.

> **Note**
>
> The `utl32k.sql` script increases the maximum size of the `VARCHAR2`, `NVARCHAR2`, and `RAW` columns for the views where this is required. The script does not increase the maximum size of the `VARCHAR2`, `NVARCHAR2`, and `RAW` columns in some views because of the way the SQL for those views is written.

6. Run the `rdbms/admin/utlrp.sql` script to recompile invalid objects. You must be connected `AS SYSDBA` to run the script.

Increasing the Maximum Size of VARCHAR2, NVARCHAR2, and RAW Columns in an Oracle Data Guard Logical Standby Database

To increase the maximum size of `VARCHAR2`, `NVARCHAR2`, and `RAW` columns in an Oracle Data Guard logical standby database:

1. Shut down the Oracle Data Guard primary database and logical standby database.
2. Restart the primary database and logical standby database in `UPGRADE` mode.
3. Change the setting of `MAX_STRING_SIZE` to `EXTENDED` on the primary database and logical standby database.
4. Run the `rdbms/admin/utl32k.sql` script on both the primary database and the logical standby database. You must be connected `AS SYSDBA` to run the script.
5. Restart the primary database and logical standby database in `NORMAL` mode.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The `utl32k.sql` script increases the maximum size of the `VARCHAR2`, `NVARCHAR2`, and `RAW` columns for the views where this is required. The script does not increase the maximum size of the `VARCHAR2`, `NVARCHAR2`, and `RAW` columns in some views because of the way the SQL for those views is written.

6. Run the `rdbms/admin/utlrp.sql` script on the primary database and logical standby database to recompile invalid objects. You must be connected `AS SYSDBA` to run the script.
7. Restart SQL Apply.

> **See Also**
>
> *Oracle Database Globalization Support Guide* for more information about the `MAX_STRING_SIZE` parameter

1.208 MEMOPTIMIZE_POOL_SIZE

`MEMOPTIMIZE_POOL_SIZE` sets the size of the memoptimize pool, a memory area in the system global area (SGA) used by the Memoptimized Rowstore.

Property	Description
Parameter type	Big integer
Syntax	`MEMOPTIMIZE_POOL_SIZE = integer [K | M | G]`
Default value	0
Modifiable	No
Modifiable in a PDB	No
Range of values	0 to no maximum
Basic	No
Oracle RAC	Different values can be used on different instances.

The Memoptimized Rowstore improves the data query performance of applications, such as Internet of Things (IoT), that frequently query tables based on primary key values.

The Memoptimized Rowstore provides the capability of fast lookup of data for the tables that are mainly queried based on primary key columns.

This parameter specifies an integer value to indicate the amount of SGA to use for allocating the following structures for the memoptimize pool:

- The size of the buffer cache region: This is the total number of blocks for all `MEMOPTIMIZE FOR READ` tables.
- The size of the hash index segmented data structure pointing to the special blocks of `MEMOPTIMIZE FOR READ` tables.

Calculate the buffer cache requirement for the table being considered for `MEMOPTIMIZE FOR READ`, and include an additional 25% memory requirement for the hash index segmented data structure.

Copyright © 2002, 2026, Oracle and/or its affiliates.

These structures are allocated from SGA at instance startup.

The value specified for this parameter counts toward SGA_TARGET. For example, if you set SGA_TARGET to 10 GB and you set MEMOPTIMIZE_POOL_SIZE to 2 GB, then 20% of the SGA_TARGET setting is allocated to the memoptimize pool.

Unlike other SGA components such as the buffer cache and shared pool, the memoptimize pool size is not controlled by automatic memory management. The database does not automatically shrink the memoptimize pool when the buffer cache or shared pool requires more memory, or increase the memoptimize pool when it runs out of space. You can only increase the size of the memoptimize pool by manually adjusting the MEMOPTIMIZE_POOL_SIZE initialization parameter.

See Also

Oracle Database Performance Tuning Guide for more information about the Memoptimized Rowstore and the memoptimize pool.

1.209 MEMORY_MAX_TARGET

MEMORY_MAX_TARGET specifies the maximum value to which a DBA can set the MEMORY_TARGET initialization parameter.

Property	Description
Parameter type	Big integer
Syntax	MEMORY_MAX_TARGET = *integer* [K | M | G]
Default value	0
Modifiable	No
Modifiable in a PDB	No
Range of values	0 to the physical memory size available to the Oracle Database
Basic	No

Note

On Linux platforms only, Oracle recommends against explicitly setting values for MEMORY_TARGET and MEMORY_MAX_TARGET in the initialization parameter file at the time of instance startup. Doing so may result in suboptimal small page allocation in the SGA. Furthermore, if the USE_LARGE_PAGES parameter is set to ONLY and you attempt to set values for MEMORY_TARGET and MEMORY_MAX_TARGET, the instance will fail to start.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database Administrator's Guide* for more information about managing memory
- "MEMORY_TARGET"
- "USE_LARGE_PAGES"

1.210 MEMORY_TARGET

MEMORY_TARGET specifies the Oracle system-wide usable memory.

The database tunes memory to the MEMORY_TARGET value, reducing or enlarging the SGA and PGA as needed.

Property	Description
Parameter type	Big integer
Syntax	MEMORY_TARGET = *integer* [K | M | G]
Default value	0 (SGA autotuning is disabled for DEFERRED mode autotuning requests, but allowed for IMMEDIATE mode autotuning requests)
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	152 MB to MEMORY_MAX_TARGET
Basic	No

MEMORY_TARGET should be set higher than or equal to the sum of the current sizes of the SGA and PGA.

In a text-based initialization parameter file, if you omit MEMORY_MAX_TARGET and include a value for MEMORY_TARGET, then the database automatically sets MEMORY_MAX_TARGET to the value of MEMORY_TARGET. If you omit the line for MEMORY_TARGET and include a value for MEMORY_MAX_TARGET, the MEMORY_TARGET parameter defaults to zero. After startup, you can then dynamically change MEMORY_TARGET to a nonzero value, provided that it does not exceed the value of MEMORY_MAX_TARGET.

Total memory usage can grow beyond the value of MEMORY_TARGET. For example, memory is allocated to PL/SQL tables and varrays regardless of the value of MEMORY_TARGET as long as memory is available at the operating system level.

In the **Default value** field, IMMEDIATE mode autotuning requests are necessary to avoid ORA-04031 errors. The DEFERRED and IMMEDIATE modes are reflected in the OPER_MODE column of the V$MEMORY_RESIZE_OPS view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> On Linux platforms only, Oracle recommends against explicitly setting values for `MEMORY_TARGET` and `MEMORY_MAX_TARGET` in the initialization parameter file at the time of instance startup. Doing so may result in suboptimal small page allocation in the SGA. Furthermore, if the `USE_LARGE_PAGES` parameter is set to `ONLY` and you attempt to set values for `MEMORY_TARGET` and `MEMORY_MAX_TARGET`, the instance will fail to start.

> **Note**
>
> The default value of `SGA_MAX_SIZE` depends on the values of `MEMORY_TARGET` and `MEMORY_MAX_TARGET`.

> **See Also**
>
> - "MEMORY_MAX_TARGET"
> - "USE_LARGE_PAGES"
> - *Oracle Database Administrator's Guide* for more information about managing memory
> - *Oracle Multitenant Administrator's Guide* for information about the initialization parameters that control the memory usage of PDBs

1.211 MFA_DUO_API_HOST

`MFA_DUO_API_HOST` specifies the host name of the Cisco Duo security API to be used for multi-factor authentication using Cisco Duo push notifications.

Property	Description
Parameter type	String
Syntax	`MFA_DUO_API_HOST = string`
Default value	None
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Multiple instances can have different values.

This parameter is configured at the PDB level. Setting this parameter allows users in a PDB to be registered for multi-factor authentication using Cisco Duo push notifications.

Set this parameter to a valid Duo API host. For example:

```
ALTER SYSTEM SET MFA_DUO_API_HOST = "api-855fd6a0.duosecurity.com";
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.28.

> **See Also**
>
> *Oracle Database Security Guide* for more information about this parameter

1.212 MFA_OMA_IAM_DOMAIN_URL

`MFA_OMA_IAM_DOMAIN_URL` specifies the OCI IAM domain URL for multi-factor authentication using Oracle Mobile Authenticator (OMA) push notifications.

Property	Description
Parameter type	String
Syntax	`MFA_OMA_IAM_DOMAIN_URL = string`
Default value	None
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Multiple instances can have different values.

This parameter is configured at the PDB level. Setting this parameter allows users in a PDB to be registered for multi-factor authentication using OMA push notifications.

Set this parameter to a valid OCI IAM domain URL. For example:

```
ALTER SYSTEM SET MFA_OMA_IAM_DOMAIN_URL =
   "https://idcs-cbcb9aaca84945e78466ab5b2e576360.identity.oraclecloud.com";
```

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.28.

> **See Also**
>
> *Oracle Database Security Guide* for more information about this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.213 MFA_SENDER_EMAIL_DISPLAYNAME

`MFA_SENDER_EMAIL_DISPLAYNAME` specifies the `From:` display name for emails sent to users when configuring multi-factor authentication using Oracle Mobile Authenticator (OMA) or Cisco Duo push notifications.

Property	Description
Parameter type	String
Syntax	`MFA_SENDER_EMAIL_DISPLAYNAME = string`
Default value	None
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Multiple instances can have different values.

This parameter is configured at the PDB level. When adding OMA or Duo as an additional authentication method for a user in a PDB, the user is sent an email with instructions on how to register a device. This parameter specifies the display name used in those emails.

Set this parameter to the desired email display name. For example:

```
ALTER SYSTEM SET MFA_SENDER_EMAIL_DISPLAYNAME = "Database MFA Registration";
```

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.28.

> **See Also**
>
> *Oracle Database Security Guide* for more information about this parameter

1.214 MFA_SENDER_EMAIL_ID

`MFA_SENDER_EMAIL_ID` specifies the `From:` email ID for emails sent to users when configuring multi-factor authentication using Oracle Mobile Authenticator (OMA) or Cisco Duo push notifications.

Property	Description
Parameter type	String
Syntax	`MFA_SENDER_EMAIL_ID = string`
Default value	None
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Basic	No
Oracle RAC	Multiple instances can have different values.

This parameter is configured at the PDB level. When adding OMA or Duo as an additional authentication method for a user in a PDB, the user is sent an email with instructions on how to register a device. This parameter specifies the email ID that is used in those emails.

Set this parameter to the desired email ID. For example:

```
ALTER SYSTEM SET MFA_SENDER_EMAIL_ID =
   "database_mfa_registration@example.com";
```

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.28.

> **See Also**
>
> *Oracle Database Security Guide* for more information about this parameter

1.215 MFA_SMTP_HOST

`MFA_SMTP_HOST` specifies the SMTP host for emails sent to users when configuring multi-factor authentication using Oracle Mobile Authenticator (OMA) or Cisco Duo push notifications.

Property	Description
Parameter type	String
Syntax	`MFA_SMTP_HOST = string`
Default value	None
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Multiple instances can have different values.

This parameter is configured at the PDB level. When adding OMA or Duo as an additional authentication method for a user in a PDB, the user is sent an email with instructions on how to register a device. This parameter specifies the SMTP host name for those emails.

Set this parameter to a valid SMTP host name. For example:

```
ALTER SYSTEM SET MFA_SMTP_HOST =
   "smtp.email.us-sanjose-1.oci.oraclecloud.com";
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.28.

> **See Also**
>
> *Oracle Database Security Guide* for more information about this parameter

1.216 MFA_SMTP_PORT

`MFA_SMTP_PORT` specifies the SMTP port for emails sent to users when configuring multi-factor authentication using Oracle Mobile Authenticator (OMA) or Cicco Duo push notifications.

Property	Description
Parameter type	Integer
Default value	587
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	25 to 587
Basic	No
Oracle RAC	Multiple instances can have different values.

This parameter is configured at the PDB level. When adding OMA or Duo as an additional authentication method for a user in a PDB, the user is sent an email with instructions on how to register a device. This parameter specifies the SMTP port for those emails.

Set this parameter to to a valid SMTP port. For example:

```
ALTER SYSTEM SET MFA_SMTP_PORT = 587;
```

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.28.

> **See Also**
>
> *Oracle Database Security Guide* for more information about this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.217 MIN_AUTH_SERVERS

MIN_AUTH_SERVERS specifies the minimum number of authentication servers in the authentication pool. This pool authenticates user connections when client applications connect to Database Resident Connection Pooling (DRCP).

Property	Description
Parameter type	Integer
Default value	1
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	Minimum: 0 Maximum: MAX_AUTH_SERVERS or PROCESSES, whichever is the lesser value
Basic	No
Oracle RAC	Different instances can use different values.

The value of this parameter must be less than or equal to the values of both the MAX_AUTH_SERVERS initialization parameter and the PROCESSES initialization parameter. MAX_AUTH_SERVERS specifies the maximum number of authentication servers in the authentication pool. PROCESSES specifies the maximum number of operating system user processes that can simultaneously connect to Oracle.

Note

This parameter is available starting with Oracle Database 19c, Release Update 19.10.

See Also

- *Oracle Database Administrator's Guide* for more information about setting this parameter
- "MAX_AUTH_SERVERS"
- "PROCESSES"

1.218 MULTISHARD_QUERY_DATA_CONSISTENCY

MULTISHARD_QUERY_DATA_CONSISTENCY enables you to specify a data consistency setting for multi-shard queries.

Property	Description
Parameter type	String
Syntax	MULTISHARD_QUERY_DATA_CONSISTENCY = { STRONG | SHARD_LOCAL | DELAYED_STANDBY_ALLOWED }
Default value	STRONG

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The value of this parameter can be different on different Oracle RAC instances.

You can use `MULTISHARD_QUERY_DATA_CONSISTENCY` to avoid the cost of SCN synchronization when executing multi-shard queries across shards, which can be globally distributed.

`MULTISHARD_QUERY_DATA_CONSISTENCY` can be set at the system level or the session level.

The values that can be set for `MULTISHARD_QUERY_DATA_CONSISTENCY` are:

- `STRONG`: With this setting, SCN synchronization is performed across all shards, and data is consistent across all shards. This setting provides global consistent read capability. This is the default value.
- `SHARD_LOCAL`: With this setting, SCN synchronization is not performed across all shards. Data is consistent within each shard. This setting provides the most current data.
- `DELAYED_STANDBY_ALLOWED`: With this setting, SCN synchronization is not performed across all shards. Data is consistent within each shard. This setting allows data to be fetched from Data Guard standby databases when possible (for example, depending on load balancing), and may return stale data from standby databases.

See Also

Using Oracle Sharding for more information about specifying consistency levels in a multi-shard query

1.219 NLS_CALENDAR

`NLS_CALENDAR` specifies which calendar system Oracle uses.

Property	Description
Parameter type	String
Syntax	`NLS_CALENDAR = "`*`calendar_system`*`"`
Default value	None, implies `GREGORIAN`
Modifiable	`ALTER SESSION`
Modifiable in a PDB	Yes
Range of values	Any valid calendar format name
Basic	No

`NLS_CALENDAR` can have one of the following values:

- `Arabic Hijrah`
- `English Hijrah`

Copyright © 2002, 2026, Oracle and/or its affiliates.

- Ethiopian
- GREGORIAN
- Japanese Imperial
- Persian
- ROC Official (Republic of China)
- Thai Buddha

For example, suppose NLS_CALENDAR is set to English Hijrah. The date format is DD Month YYYY. If the date is March 7, 2025, then the SYSDATE is displayed as follows:

```
SELECT SYSDATE FROM DUAL;
SYSDATE
-------------------------
07 Ramadan            1446
```

> **Note**
>
> The value of the initialization parameter NLS_CALENDER is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. If the initialization parameter is not specified, the initial session value becomes GREGORIAN. This initial value is overridden by a client-side value if the client is OCI-based and the NLS_LANG client setting (environment variable) is defined.

> **See Also**
>
> *Oracle Database Globalization Support Guide* for a listing of available calendar systems

1.220 NLS_COMP

NLS_COMP specifies the collation behavior of the database session.

Property	Description
Parameter type	String
Syntax	NLS_COMP = { BINARY | LINGUISTIC | ANSI }
Default value	BINARY
Modifiable	ALTER SESSION
Modifiable in a PDB	Yes
Basic	No

Values

- BINARY

  Normally, comparisons in the WHERE clause and in PL/SQL blocks is binary unless you specify the NLSSORT function.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- `LINGUISTIC`

  Comparisons for all SQL operations in the `WHERE` clause and in PL/SQL blocks should use the linguistic sort specified in the `NLS_SORT` parameter. To improve the performance, you can also define a linguistic index on the column for which you want linguistic comparisons.

- `ANSI`

  A setting of `ANSI` is for backward compatibility; in general, you should set `NLS_COMP` to `LINGUISTIC`

> **Note**
>
> Unless you explicitly set the value for `NLS_COMP` in your initialization parameter file, a default value of NULL is shown in the following views: `V$PARAMETER`, `V$SYSTEM_PARAMETER`, `V$PARAMETER2`, `V$SYSTEM_PARAMETER2`, and `NLS_INSTANCE_PARAMETERS`. However, the actual default value, and behavior, is `BINARY`. Note that you cannot change the default to NULL, because NULL is not among the valid values.

Examples

See *Oracle Database Globalization Support Guide* for examples of using this parameter.

> **Note**
>
> The value of this initialization parameter `NLS_COMP` is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. This initial value is overridden by a client-side value if the client uses the Oracle JDBC driver or if the client is OCI-based and the `NLS_LANG` client setting (environment variable) is defined. The initialization parameter value is, therefore, usually ignored.

1.221 NLS_CURRENCY

`NLS_CURRENCY` specifies the string to use as the local currency symbol for the L number format element. The default value of this parameter is determined by `NLS_TERRITORY`.

Property	Description
Parameter type	String
Syntax	`NLS_CURRENCY = currency_symbol`
Default value	Derived from `NLS_TERRITORY`
Modifiable	`ALTER SESSION`
Modifiable in a PDB	Yes
Range of values	Any valid character string, with a maximum of 10 bytes (not including null)
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The value of this initialization parameter `NLS_CURRENCY` is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. This initial value is overridden by a client-side value if the client uses the Oracle JDBC driver or if the client is OCI-based and the `NLS_LANG` client setting (environment variable) is defined. The initialization parameter value is, therefore, usually ignored.

> **See Also**
>
> - *Oracle Database Globalization Support Guide* for more information about this parameter
> - *Oracle Database SQL Language Reference* for information on number format elements

1.222 NLS_DATE_FORMAT

`NLS_DATE_FORMAT` specifies the default datetime format model to use with the `TO_CHAR` and `TO_DATE` functions. The default value of this parameter is determined by `NLS_TERRITORY`.

Property	Description
Parameter type	String
Syntax	`NLS_DATE_FORMAT = "format"`
Default value	Derived from `NLS_TERRITORY`
Modifiable	`ALTER SESSION`
Modifiable in a PDB	Yes
Range of values	Any valid datetime format model
Basic	No

The value of this parameter can be any valid datetime format model, and the value must be surrounded by double quotation marks. For example:

```
NLS_DATE_FORMAT = "MM/DD/YYYY"
```

> **Note**
>
> The value of this initialization parameter `NLS_DATE_FORMAT` is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. This initial value is overridden by a client-side value if the client uses the Oracle JDBC driver or if the client is OCI-based and the `NLS_LANG` client setting (environment variable) is defined. The initialization parameter value is, therefore, usually ignored.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database SQL Language Reference* for more information on datetime format models
- *Oracle Database Globalization Support Guide* for more information about this parameter

1.223 NLS_DATE_LANGUAGE

`NLS_DATE_LANGUAGE` specifies the language to use for the spelling of day and month names and date abbreviations (a.m., p.m., AD, BC) returned by the `TO_DATE` and `TO_CHAR` functions.

Property	Description
Parameter type	String
Syntax	`NLS_DATE_LANGUAGE = language`
Default value	Derived from `NLS_LANGUAGE`
Modifiable	`ALTER SESSION`
Modifiable in a PDB	Yes
Range of values	Any valid `NLS_LANGUAGE` value
Basic	No

Note

The value of this initialization parameter `NLS_DATE_LANGUAGE` is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. This initial value is overridden by a client-side value if the client uses the Oracle JDBC driver or if the client is OCI-based and the `NLS_LANG` client setting (environment variable) is defined. The initialization parameter value is, therefore, usually ignored.

See Also

- *Oracle Database Globalization Support Guide* for more information about this parameter
- *Oracle Database SQL Language Reference* for information on the `TO_DATE` function.
- *Oracle Database SQL Language Reference* for information on the `TO_CHAR` function.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.224 NLS_DUAL_CURRENCY

NLS_DUAL_CURRENCY specifies the dual currency symbol (such as "Euro") for the territory. The default is the dual currency symbol defined in the territory of your current language environment.

Property	Description
Parameter type	String
Syntax	NLS_DUAL_CURRENCY = *currency_symbol*
Default value	Derived from NLS_TERRITORY
Modifiable	ALTER SESSION
Modifiable in a PDB	Yes
Range of values	Any valid format name up to 10 characters
Basic	No

Note

The value of this initialization parameter NLS_DUAL_CURRENCY is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. This initial value is overridden by a client-side value if the client uses the Oracle JDBC driver or if the client is OCI-based and the NLS_LANG client setting (environment variable) is defined. The initialization parameter value is, therefore, usually ignored.

See Also

Oracle Database Globalization Support Guide for more information about this parameter

1.225 NLS_ISO_CURRENCY

NLS_ISO_CURRENCY determines the string to use as the international currency symbol corresponding to the C number format element in a call to the TO_CHAR function.

Property	Description
Parameter type	String
Syntax	NLS_ISO_CURRENCY = *territory*
Default value	Derived from NLS_TERRITORY
Modifiable	ALTER SESSION
Modifiable in a PDB	Yes
Range of values	Any valid NLS_TERRITORY value
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Local currency symbols can be ambiguous. For example, a dollar sign ($) can refer to U.S. dollars or Australian dollars. ISO Specification 4217 defines unique "international" currency symbols for the currencies of specific territories or countries. The value of the NLS_ISO_CURRENCY parameter is the Oracle name of the territory whose ISO currency symbol is returned in place of the C number format element. For example, if NLS_ISO_CURRENCY is set to AMERICA, the string 'USD' is returned by TO_CHAR where the C element is specified in the format.

Note

The value of this initialization parameter NLS_ISO_CURRENCY is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. This initial value is overridden by a client-side value if the client uses the Oracle JDBC driver or if the client is OCI-based and the NLS_LANG client setting (environment variable) is defined. The initialization parameter value is, therefore, usually ignored.

See Also

- *Oracle Database Globalization Support Guide* for more information about this parameter
- *Oracle Database SQL Language Reference* for information on number format elements

1.226 NLS_LANGUAGE

NLS_LANGUAGE specifies the default language of the database.

Property	Description
Parameter type	String
Syntax	NLS_LANGUAGE = *language*
Default value	Operating system-dependent, derived from the NLS_LANG environment variable
Modifiable	ALTER SESSION
Modifiable in a PDB	Yes
Range of values	Any valid language name
Basic	Yes

This language specified by NLS_LANGUAGE is used for messages, day and month names, symbols for AD, BC, a.m., and p.m., and the default sorting mechanism. This parameter also determines the default values of the parameters NLS_DATE_LANGUAGE and NLS_SORT.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The value of this initialization parameter NLS_LANGUAGE is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. This initial value is overridden by a client-side value if the client uses the Oracle JDBC driver or if the client is OCI-based and the NLS_LANG client setting (environment variable) is defined. The initialization parameter value is, therefore, usually ignored.

Examples

See these examples of using the NLS_LANGUAGE parameter:

- For an example of setting NLS_LANGUAGE to Italian, see *Oracle Database Globalization Support Guide*.
- For an example of overriding default values for NLS_LANGUAGE and NLS_TERRITORY during a session, see *Oracle Database Globalization Support Guide*.

> **See Also**
>
> - *Oracle Database Globalization Support Guide* for more information about this parameter.
> - *Oracle Database Globalization Support Guide* for a complete list of languages that can be specified using this parameter
> - *Oracle Database Globalization Support Guide* for information on overriding the default values for this parameter
> - Your operating system-specific Oracle documentation and the release notes for your country

1.227 NLS_LENGTH_SEMANTICS

NLS_LENGTH_SEMANTICS is used to specify length semantics.

Property	Description
Parameter type	String
Syntax	NLS_LENGTH_SEMANTICS = *string* Example: NLS_LENGTH_SEMANTICS = 'CHAR'
Default value	BYTE
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	BYTE | CHAR
Basic	No

The session-level value of NLS_LENGTH_SEMANTICS specifies the default length semantics to use for VARCHAR2 and CHAR table columns, user-defined object attributes, and PL/SQL variables in

Copyright © 2002, 2026, Oracle and/or its affiliates.

database objects created in the session. This default may be overridden by the explicit length semantics qualifiers BYTE and CHAR in column, attribute, and variable definitions.

The instance-level value of NLS_LENGTH_SEMANTICS provides a default for the session-level value if NLS_LENGTH_SEMANTICS is not set explicitly by the database client through the NLS_LENGTH_SEMANTICS client environment variable (does not apply to JDBC Thin clients), or the ALTER SESSION SET NLS_LENGTH_SEMANTICS statement.

NCHAR, NVARCHAR2, CLOB, and NCLOB columns are always character-based.

Sessions logged in as SYS do not use the NLS_LENGTH_SEMANTICS parameter. They use BYTE length semantics for all created objects unless overridden by the explicit BYTE and CHAR qualifiers in object definitions (SQL DDL statements).

Note

Oracle strongly recommends that you do NOT set the NLS_LENGTH_SEMANTICS parameter to CHAR in the instance or server parameter file. This may cause many existing installation scripts to unexpectedly create columns with character length semantics, resulting in run-time errors, including buffer overflows.

See Also

Oracle Database Globalization Support Guide for more information about this parameter

1.228 NLS_NCHAR_CONV_EXCP

NLS_NCHAR_CONV_EXCP determines whether an error is reported when there is data loss during an implicit or explicit character type conversion between NCHAR/NVARCHAR2 and CHAR/VARCHAR2.

Property	Description
Parameter type	String
Syntax	NLS_NCHAR_CONV_EXCP = { TRUE | FALSE }
Default value	FALSE
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No

The default value results in no error being reported.

See Also

Oracle Database Globalization Support Guide for more information about this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.229 NLS_NUMERIC_CHARACTERS

NLS_NUMERIC_CHARACTERS specifies the characters to use as the group separator and decimal character.

Property	Description
Parameter type	String
Syntax	NLS_NUMERIC_CHARACTERS = "*decimal_character group_separator*"
Default value	Derived from NLS_TERRITORY
Modifiable	ALTER SESSION
Modifiable in a PDB	Yes
Basic	No

NLS_NUMERIC_CHARACTERS overrides those characters defined implicitly by NLS_TERRITORY. The group separator separates integer groups (that is, thousands, millions, billions, and so on). The decimal separates the integer portion of a number from the decimal portion.

You can specify any character as the decimal or group separator. The two characters specified must be single-byte and must be different from each other. The characters cannot be any numeric character or any of the following characters: plus (+), minus sign (-), less than sign (<), greater than sign (>). Either character can be a space.

For example, if you want to specify a comma as the decimal character and a space as the group separator, you would set this parameter as follows:

```
NLS_NUMERIC_CHARACTERS = ", "
```

Note

The value of this initialization parameter NLS_NUMERIC_CHARACTERS is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. This initial value is overridden by a client-side value if the client uses the Oracle JDBC driver or if the client is OCI-based and the NLS_LANG client setting (environment variable) is defined. The initialization parameter value is, therefore, usually ignored.

See Also

Oracle Database Globalization Support Guide for more information about this parameter

1.230 NLS_SORT

NLS_SORT specifies the collating sequence for character value comparison in various SQL operators and clauses.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	String
Syntax	`NLS_SORT = { BINARY | linguistic_definition }`
Default value	Derived from `NLS_LANGUAGE`
Modifiable	`ALTER SESSION`
Modifiable in a PDB	Yes
Range of values	`BINARY` or any valid linguistic definition name
Basic	No

For example, `NLS_SORT` specifies the collating sequence for character value comparison in these SQL operators and clauses: `ORDER BY`, `GROUP BY`, comparison conditions (=, <>, <=, >=), `IN`, `BETWEEN`, `LIKE`, `MIN/MAX`, `GREATEST/LEAST`, and `INSTR`.

- If the value is `BINARY`, then comparison is based directly on byte values in the binary encoding of the character values being compared. The ordering depends on the character set of the compared values, which is either the database character set (for `VARCHAR2`, `CHAR`, `LONG`, and `CLOB`) or the national character set (for `NVARCHAR2`, `NCHAR`, and `NCLOB`).
- If the value is a named linguistic sort, then comparison is defined by this sort. A linguistic sort uses various rules to achieve ordering expected by speakers of one or more natural languages. This is usually the same ordering that is used in dictionaries and telephone directories in those languages.

The exact operators and query clauses that obey the `NLS_SORT` parameter depend on the value of the `NLS_COMP` parameter. If an operator or clause does not obey the `NLS_SORT` value, as determined by `NLS_COMP`, the collation used is `BINARY`.

The `BINARY` comparison is faster and uses less resources than any linguistic comparison but for text in a natural language, it does not provide ordering expected by users.

The value of `NLS_SORT` affects execution plans of queries. Because a standard index cannot be used as a source of values sorted in a linguistic order, an explicit sort operation must usually be performed instead of an index range scan. A functional index on the `NLSSORT` function may be defined to provide values sorted in a linguistic order and reintroduce the index range scan to the execution plan.

Note

The value of the initialization parameter `NLS_SORT` is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. This initial value is overriden by a client-side value if the client uses the Oracle JDBC driver or if the client is OCI-based and the `NLS_LANG` client setting (environment variable) is defined. The initialization parameter value is, therefore, usually ignored.

See Also

- *Oracle Database Globalization Support Guide* for more information about this parameter and a current listing of values you can specify

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.231 NLS_TERRITORY

`NLS_TERRITORY` specifies the name of the territory whose conventions are to be followed for day and week numbering.

Property	Description
Parameter type	String
Syntax	`NLS_TERRITORY = territory`
Default value	Operating system-dependent
Modifiable	`ALTER SESSION`
Modifiable in a PDB	Yes
Range of values	Any valid territory name
Basic	Yes

This parameter also establishes the default date format, the default decimal character and group separator, and the default ISO and local currency symbols.

For information on these settings, see "NLS_DATE_FORMAT", "NLS_NUMERIC_CHARACTERS", "NLS_CURRENCY", and "NLS_ISO_CURRENCY".

> **Note**
>
> The value of this initialization parameter `NLS_TERRITORY` is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. This initial value is overridden by a client-side value if the client uses the Oracle JDBC driver or if the client is OCI-based and the `NLS_LANG` client setting (environment variable) is defined. The initialization parameter value is, therefore, usually ignored.

Examples

For an example of overriding the default value for the `NLS_TERRITORY` parameter, see *Oracle Database Globalization Support Guide*.

> **See Also**
>
> - *Oracle Database Globalization Support Guide* for a complete list of territories
> - Your operating system-specific Oracle documentation for the territory-dependent default values for these parameters

1.232 NLS_TIMESTAMP_FORMAT

`NLS_TIMESTAMP_FORMAT` defines the datetime format model to use with the `TO_CHAR` and `TO_TIMESTAMP` functions.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	String
Syntax	NLS_TIMESTAMP_FORMAT = "*format*"
Default value	Derived from NLS_TERRITORY
Modifiable	ALTER SESSION
Modifiable in a PDB	Yes
Range of values	Any valid datetime format model
Basic	No

The value must be surrounded by quotation marks as follows:

```
NLS_TIMESTAMP_FORMAT = 'YYYY-MM-DD HH:MI:SS.FF'
```

You can specify the value of NLS_TIMESTAMP_FORMAT by setting it in the initialization parameter file. You can specify its value for a client as a client environment variable.

You can also alter the value of NLS_TIMESTAMP_FORMAT by changing its value in the initialization parameter and then restarting the instance. To alter the value during a session use the ALTER SESSION SET statement.

Note

The value of this initialization parameter NLS_TIMESTAMP_FORMAT is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. This initial value is overridden by a client-side value if the client uses the Oracle JDBC driver or if the client is OCI-based and the NLS_LANG client setting (environment variable) is defined. The initialization parameter value is, therefore, usually ignored.

See Also

- *Oracle Database SQL Language Reference* for more information on datetime format models
- *Oracle Database Globalization Support Guide* for more information about this parameter

1.233 NLS_TIMESTAMP_TZ_FORMAT

NLS_TIMESTAMP_TZ_FORMAT defines the default datetime format model to use with the TO_CHAR and TO_TIMESTAMP_TZ functions.

Property	Description
Parameter type	String
Syntax	NLS_TIMESTAMP_TZ_FORMAT = "*format*"
Default value	Derived from NLS_TERRITORY

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable	`ALTER SESSION`
Modifiable in a PDB	Yes
Range of values	Any valid datetime format model
Basic	No

The value must be surrounded by quotation marks as follows:

```
NLS_TIMESTAMP_TZ_FORMAT = 'YYYY-MM-DD HH:MI:SS.FF TZH:TZM'
```

You can specify the value of `NLS_TIMESTAMP_TZ_FORMAT` by setting it in the initialization parameter file. You can specify its value for a client as a client environment variable.

You can also alter the value of `NLS_TIMESTAMP_TZ_FORMAT` by changing its value in the initialization parameter and then restarting the instance. To alter the value during a session use the `ALTER SESSION SET` statement.

Note

The value of this initialization parameter `NLS_TIMESTAMP_TZ_FORMAT` is used to initialize the session value of this parameter, which is the actual value referenced by the SQL query processing. This initial value is overridden by a client-side value if the client uses the Oracle JDBC driver or if the client is OCI-based and the `NLS_LANG` client setting (environment variable) is defined. The initialization parameter value is, therefore, usually ignored.

See Also

- *Oracle Database SQL Language Reference* for more information on datetime format models
- *Oracle Database Globalization Support Guide* for more information about this parameter

1.234 NONCDB_COMPATIBLE

`NONCDB_COMPATIBLE` enables you to get behavior similar to a non-CDB when issuing SQL commands inside a PDB in a CDB.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	No
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Set this parameter if you are using a single PDB in your CDB configuration and you have legacy code that causes ORA-65040 when you run it in the PDB.

Values

- TRUE

  Indicates the behavior for SQL statements will be like a non-CDB although the statements are issued in a PDB in a CDB.

- FALSE

  Indicates the behavior will for SQL statements will be like a CDB. This is the default value of the parameter.

There are some statements (such as ALTER DB BACKUP CONTROLFILE) which can be issued in a non-CDB, but in a CDB they must be issued in the root (because they affect the whole CDB) and would result in an error if issued in a PDB.

Some ALTER DATABASE or ALTER SYSTEM statements are not permitted inside a PDB, and they will fail if NONCDB_COMPATIBLE=FALSE is set in init.ora. However, these statements will succeed if NONCDB_COMPATIBLE=TRUE is set.

See Also

- *Oracle Multitenant Administrator's Guide* for an introduction to PDBs and CDBs
- *Oracle Multitenant Administrator's Guide* for information about the initialization parameters that control the memory usage of PDBs
- *Oracle Multitenant Administrator's Guide* for information about the ALTER SYSTEM statements that can be run in a PDB.

1.235 OBJECT_CACHE_MAX_SIZE_PERCENT

The **object cache** is a memory block on the client that allows applications to store entire objects and to navigate among them without round trips to the server. OBJECT_CACHE_MAX_SIZE_PERCENT specifies the percentage of the optimal cache size that the session object cache can grow past the optimal size.

Property	Description
Parameter type	Integer
Default value	10
Modifiable	ALTER SESSION, ALTER SYSTEM ... DEFERRED
Modifiable in a PDB	Yes
Range of values	0 to operating system-dependent maximum
Basic	No

The maximum size is equal to the optimal size plus the product of this percentage and the optimal size. When the cache size exceeds this maximum size, the system will attempt to shrink the cache to the optimal size.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> - "OBJECT_CACHE_OPTIMAL_SIZE" for a description of the object cache
> - *Pro*C/C++ Programmer's Guide* and *Oracle Call Interface Programmer's Guide* for information on precompiler use of the object cache

1.236 OBJECT_CACHE_OPTIMAL_SIZE

`OBJECT_CACHE_OPTIMAL_SIZE` specifies (in bytes) the size to which the session object cache is reduced when the size of the cache exceeds the maximum size.

Property	Description
Parameter type	Integer
Default value	`10240000` (10M)
Modifiable	`ALTER SESSION`, `ALTER SYSTEM ... DEFERRED`
Modifiable in a PDB	Yes
Range of values	10 KB to operating system-dependent maximum
Basic	No

The **object cache** is a memory block on the client that allows applications to store entire objects and to navigate among them without round trips to the server. On the server, it is used to cache frequently used objects requested as a result of client requests to help in performance.

> **See Also**
>
> *Pro*C/C++ Programmer's Guide* and *Oracle Call Interface Programmer's Guide* for information on precompiler use of the object cache

1.237 OFS_THREADS

`OFS_THREADS` sets the maximum number of Oracle file system (OFS) threads that can be started to service Oracle file system requests.

Property	Description
Parameter type	Integer
Default value	4
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	2 to 128
Basic	No
Oracle RAC	The same value should be specified on all instances.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This initialization parameter is supported only on the Linux operating system.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information about the Oracle Database NFS server feature

1.238 OGG_ONLINE_DDL

Use `OGG_ONLINE_DDL` to enable or disable GoldenGate replication of online DDLs.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SYSTEM ... SID='*'`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No
Oracle RAC	The same value must be used on all instances.

To enable GoldenGate replication of online DDLs, set this parameter to `true`; to disable GoldenGate replication of online DDLs, set this parameter to `false`.

Prior to the introduction of the `OGG_ONLINE_DDL` parameter, online DDLs were not allowed while using GoldenGate replication.

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.22.

1.239 OLAP_PAGE_POOL_SIZE

`OLAP_PAGE_POOL_SIZE` specifies (in bytes) the size of the OLAP page pool.

Property	Description
Parameter type	Big integer
Syntax	`OLAP_PAGE_POOL_SIZE = integer [K | M | G]`
Default value	`0`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM ... DEFERRED`
Modifiable in a PDB	Yes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Range of values	0, 2097152 (2 MB) to 2^{31} - 1
Basic	No

⚠ Caution

Oracle strongly recommends that you leave this parameter set to its default value of `0`. This instructs the database to autonomously adjust the OLAP page pool size to serve the requirements of the analytic workspaces. Setting this parameter to a value other than `0` can lead to errors in OLAP processing or wasted memory.

ⓘ See Also

Oracle OLAP User's Guide for more information about the OLAP option for Oracle Database

1.240 ONE_STEP_PLUGIN_FOR_PDB_WITH_TDE

If a pluggable database (PDB) has Transparent Data Encryption-encrypted (TDE-encrypted) tables or tablespaces, you can enable `ONE_STEP_PLUGIN_FOR_PDB_WITH_TDE` on the target CDB to simplify the move of TDE keys in a single step PDB move operation.

`ONE_STEP_PLUGIN_FOR_PDB_WITH_TDE` eliminates the need of having to manually provide a keystore password when you import the TDE keys into the PDB after it has moved to the target CDB.

Property	Description
Parameter type	Boolean
Default value	`FALSE`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No
Oracle RAC	A different value can be set for this parameter on different Oracle RAC instances.

The default for `ONE_STEP_PLUGIN_FOR_PDB_WITH_TDE` is `FALSE`.

When `ONE_STEP_PLUGIN_FOR_PDB_WITH_TDE` is set to `TRUE` on the target CDB, the plug in of the PDB does not require a keystore password.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.241 OPEN_CURSORS

OPEN_CURSORS specifies the maximum number of open cursors (handles to private SQL areas) a session can have at once. You can use this parameter to prevent a session from opening an excessive number of cursors.

Property	Description
Parameter type	Integer
Default value	50
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to 65535
Basic	Yes

It is important to set the value of OPEN_CURSORS high enough to prevent your application from running out of open cursors. The number will vary from one application to another. Assuming that a session does not open the number of cursors specified by OPEN_CURSORS, there is no added overhead to setting this value higher than actually needed.

See Also

- *Oracle Database Performance Tuning Guide* for more information on setting this parameter
- Your operating system-specific Oracle documentation for the range of values

1.242 OPEN_LINKS

OPEN_LINKS specifies the maximum number of concurrent open connections to remote databases in one session. These connections include database links, as well as external procedures and cartridges, each of which uses a separate process.

Property	Description
Parameter type	Integer
Default value	4
Modifiable	No
Modifiable in a PDB	Yes
Range of values	0 to 32768
Basic	No

Oracle counts one open link for the following:

- For each user that references a public or private database link
- For each external procedure or cartridge connection when it is executed for the first time

Both types of connections close when the session ends. You can also close a database link connection explicitly by issuing an ALTER SESSION CLOSE DATABASE LINK statement.

Copyright © 2002, 2026, Oracle and/or its affiliates.

You should set this parameter to allow for the external procedure and cartridge connections expected during the session plus the number of databases referred to in typical distributed transactions (that is, a single SQL statement that references multiple databases), so that all the databases can be open to execute the statement. For example, if queries alternately access databases A, B, and C, and `OPEN_LINKS` is set to `2`, time will be lost waiting while one connection is broken and another made. Increase the value if many different databases are accessed over time.

This parameter refers only to connections used for distributed transactions. Direct connections to a remote database specified as an application connects are not counted.

If you set `OPEN_LINKS` to `0`, then no distributed transactions are allowed.

In a multitenant container database (CDB), the `OPEN_LINKS` parameter can be set at both the root and at the PDB level using either an initialization parameter file (PFILE) or server parameter file (SPFILE). You need to restart the instance or reopen the PDB for a new `OPEN_LINKS` value to become effective.

See Also

"OPEN_LINKS_PER_INSTANCE" for information on setting open connections globally for a database instance

1.243 OPEN_LINKS_PER_INSTANCE

`OPEN_LINKS_PER_INSTANCE` specifies the maximum number of migratable open connections globally for each database instance.

Property	Description
Parameter type	Integer
Default value	4
Modifiable	No
Modifiable in a PDB	No
Range of values	0 to 2^{31} - 1
Basic	No
Oracle RAC	Multiple instances can have different values.

XA transactions use migratable open connections so that the connections are cached after a transaction is committed. Another transaction can use the connection, provided the user who created the connection is the same as the user who owns the transaction.

`OPEN_LINKS_PER_INSTANCE` is different from `OPEN_LINKS`, which indicates the number of connections from a session. The `OPEN_LINKS` parameter is not applicable to XA applications.

See Also

- "OPEN_LINKS"
- *Oracle Database Development Guide* for more information about using this parameter in Oracle XA applications

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.244 OPTIMIZER_ADAPTIVE_PLANS

OPTIMIZER_ADAPTIVE_PLANS controls adaptive plans. Adaptive plans are execution plans built with alternative choices that are decided at run time based on statistics collected as the query executes.

Property	Description
Parameter type	Boolean
Default value	true
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	true | false
Basic	No
Oracle RAC	The same value must be set on all instances

Setting this parameter to false disables the following adaptive features:

- Nested loop join/hash join selection
- Star transformation bitmap pruning
- Adaptive parallel distribution method

> **See Also**
>
> *Oracle Database SQL Tuning Guide* for information about adaptive plans

1.245 OPTIMIZER_ADAPTIVE_REPORTING_ONLY

OPTIMIZER_ADAPTIVE_REPORTING_ONLY controls reporting-only mode for adaptive optimizations.

Property	Description
Parameter type	Boolean
Default value	FALSE
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	TRUE | FALSE
Basic	No

When OPTIMIZER_ADAPTIVE_REPORTING_ONLY is set to FALSE, reporting-only mode is off, and the adaptive optimizations are enabled as usual.

When OPTIMIZER_ADAPTIVE_REPORTING_ONLY is set to TRUE, adaptive optimizations run in reporting-only mode. With this setting, the information required for an adaptive optimization is gathered, but no action is taken to change the plan. For instance, an adaptive plan will always

Copyright © 2002, 2026, Oracle and/or its affiliates.

choose the default (optimizer-chosen) plan, but information is collected on what plan to adapt to in non-reporting mode. This information can be viewed in the adaptive plan report.

This parameter affects only adaptive optimizations that are enabled.

See Also

- *Oracle Database SQL Tuning Guide* for more information about adaptive plans and automatic reoptimization
- *Oracle Database SQL Tuning Guide* for information on controlling adaptive optimization

1.246 OPTIMIZER_ADAPTIVE_STATISTICS

`OPTIMIZER_ADAPTIVE_STATISTICS` controls adaptive statistics. Some query shapes are too complex to rely on base table statistics alone, so the optimizer augments these statistics with adaptive statistics.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true	false`
Basic	No
Oracle RAC	The same value must be set on all instances

Setting this parameter to `false` disables the following adaptive features:

- SQL plan directives
- Statistics feedback for joins
- Adaptive dynamic sampling for parallel execution

Note

Setting `OPTIMIZER_ADAPTIVE_STATISTICS` to `false` preserves the statistics feedback functionality that was introduced in Oracle Database 11*g*.

`OPTIMIZER_ADAPTIVE_STATISTICS` does not control the creation of SQL plan directives. SQL plan directives will be created even if this parameter is `false`, but they will not be used to refine SQL execution plans with dynamic sampling.

See Also

Oracle Database SQL Tuning Guide for information about adaptive plans

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.247 OPTIMIZER_CAPTURE_SQL_PLAN_BASELINES

`OPTIMIZER_CAPTURE_SQL_PLAN_BASELINES` enables or disables the automatic recognition of repeatable SQL statements, as well as the generation of SQL plan baselines for such statements.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No

See Also

Oracle Database SQL Tuning Guide for more information about the optimizer

1.248 OPTIMIZER_CAPTURE_SQL_QUARANTINE

`OPTIMIZER_CAPTURE_SQL_QUARANTINE` enables or disables the automatic creation of SQL Quarantine configurations.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No
Oracle RAC	Different instances can use different values.

Values:

- `true`

  Enables the automatic creation of SQL Quarantine configurations. If the Resource Manager terminates a SQL statement because the statement has exceeded resource limits, then the database automatically creates a SQL Quarantine configuration for the execution plan used by the terminated SQL statement. Note that the *plan* for a terminated SQL statement is quarantined, not the statement itself.

- `false`

  Disables the automatic creation of SQL Quarantine configurations.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This parameter is available on Oracle Autonomous Database starting with Oracle Database 19c. It is available on other types of Oracle databases starting with Oracle Database 19c, Release Update 19.27.

> **See Also**
>
> - "DBA_SQL_QUARANTINE"
> - "OPTIMIZER_USE_SQL_QUARANTINE"
> - *Oracle Database SQL Tuning Guide* for more information about SQL Quarantine

1.249 OPTIMIZER_DYNAMIC_SAMPLING

`OPTIMIZER_DYNAMIC_SAMPLING` controls both when the database gathers dynamic statistics, and the size of the sample that the optimizer uses to gather the statistics.

Property	Description
Parameter type	Integer
Default value	If `OPTIMIZER_FEATURES_ENABLE` is set to `10.0.0` or higher, then 2 If `OPTIMIZER_FEATURES_ENABLE` is set to `9.2.0`, then 1 If `OPTIMIZER_FEATURES_ENABLE` is set to `9.0.1` or lower, then `0`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`0` to 11
Basic	No

> **Note**
>
> Dynamic statistics were called dynamic sampling in releases earlier than Oracle Database 12*c* Release 1 (12.1).

If the value of `OPTIMIZER_DYNAMIC_SAMPLING` is set to `11`, the `OPTIMIZER_FEATURES_ENABLE` setting has no effect on the `OPTIMIZER_DYNAMIC_SAMPLING` setting.

> **See Also**
>
> *Oracle Database SQL Tuning Guide* for detailed information about the values (0 – 11) that can be set for the `OPTIMIZER_DYNAMIC_SAMPLING` parameter.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.250 OPTIMIZER_FEATURES_ENABLE

OPTIMIZER_FEATURES_ENABLE acts as an umbrella parameter for enabling a series of optimizer features based on an Oracle release number.

Property	Description
Parameter type	String
Syntax	OPTIMIZER_FEATURES_ENABLE = { 8.0.0 | 8.0.3 | 8.0.4 | 8.0.5 | 8.0.6 | 8.0.7 | 8.1.0 | 8.1.3 | 8.1.4 | 8.1.5 | 8.1.6 | 8.1.7 | 9.0.0 | 9.0.1 | 9.2.0 | 9.2.0.8 | 10.1.0 | 10.1.0.3 | 10.1.0.4 | 10.1.0.5 | 10.2.0.1 | 10.2.0.2 | 10.2.0.3 | 10.2.0.4 | 10.2.0.5 | 11.1.0.6 | 11.1.0.7 | 11.2.0.1 | 11.2.0.2 | 11.2.0.3 | 11.2.0.4 | 12.1.0.1 | 12.1.0.2 | 12.2.0.1 | 18.1.0 | 19.1.0 }
Default value	19.1.0
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No

For example, if you upgrade your database from release 11.1 to release 12.1, but you want to keep the release 11.1 optimizer behavior, you can do so by setting this parameter to 11.1.0.6. At a later time, you can try the enhancements introduced in releases up to and including release 12.1 by setting the parameter to 12.1.0.2.

> **Note**
>
> When setting this parameter to a value representing Oracle Database 18c or later, you must specify three numeric values separated by periods, such as 18.1.0 or 19.1.0.

Table 1-6 describes some of the optimizer features that are enabled when you set the OPTIMIZER_FEATURES_ENABLE parameter to an 11.1 or 11.2 release.

Table 1-7 describes some of the optimizer features that are enabled when you set the OPTIMIZER_FEATURES_ENABLE parameter to a 12.1 or later release.

> **See Also**
>
> *Oracle Database SQL Tuning Guide* for more information about the optimizer and for information about the features listed in the following tables

Table 1-6 Optimizer Features for Oracle Database 11g Releases

Features	11.1.0.6	11.1.0.7	11.2.0.1	11.2.0.2	11.2.0.3	11.2.0.4
Adaptive cursor sharing	X	X	X	X	X	X
Join predicate pushdown	X	X	X	X	X	X
Use extended statistics to estimate selectivity	X	X	X	X	X	X

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 1-6 (Cont.) Optimizer Features for Oracle Database 11g Releases

Features	11.1.0.6	11.1.0.7	11.2.0.1	11.2.0.2	11.2.0.3	11.2.0.4
Use native implementation for full outer joins	X	X	X	X	X	X
Partition pruning using join filtering	X	X	X	X	X	X
Group by placement optimization	X	X	X	X	X	X
Null aware antijoins	X	X	X	X	X	X
Join predicate pushdown	X	X	X	X	X	X
Join Factorization			X	X	X	X
Cardinality Feedback			X	X	X	X
Subquery Unnesting			X	X	X	X
Subquery Coalescing			X	X	X	X
Table Expansion			X	X	X	X
Filtering Join Elimination			X	X	X	X
Dynamic statistics enhancements						X

Table 1-7 Optimizer Features for Oracle Database 12c and Later Releases

Features	12.1.0.1	12.1.0.2	12.2.0.1	18c	19c
All optimizer features listed in Table 1-6	X	X	X	X	X
Adaptive Query Optimization	X	X	X	X	X
Online statistics gathering for bulk loads	X	X	X	X	X
Session level statistics for Global Temporary Tables	X	X	X	X	X
Multi-table left outer joins	X	X	X	X	X
Lateral views	X	X	X	X	X
Batch table access by rowid	X	X	X	X	X
Null accepting semi joins	X	X	X	X	X
Scalar subquery unnesting	X	X	X	X	X
Conversion of joins that produce unnecessary duplicates to semi-joins	X	X	X	X	X
Parallel Union and Parallel Union All operations	X	X	X	X	X
Enhance Auto DOP	X	X	X	X	X
Approximate count distinct		X	X	X	X
Support for Oracle Database In-Memory		X	X	X	X
Group-by and aggregation elimination		X	X	X	X
Query rewrite for approximate query processing			X	X	X
Statistics advisor			X	X	X
Support for sharded databases			X	X	X
Expression tracking			X	X	X
Space-saving algorithm for partition synopses			X	X	X
Oracle In-Memory Database statistics			X	X	X
Support for sharding			X	X	X

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 1-7 (Cont.) Optimizer Features for Oracle Database 12c and Later Releases

Features	12.1.0.1	12.1.0.2	12.2.0.1	18c	19c
Cost-based OR expansion			X	X	X
Sub-query elimination			X	X	X
Multi-column key join elimination			X	X	X
SQL Quarantine					X
Gathering and use of real-time statistics					X
Use of automatic indexes					X

1.251 OPTIMIZER_IGNORE_HINTS

`OPTIMIZER_IGNORE_HINTS` enables embedded hints to be ignored.

Property	Description
Parameter type	Boolean
Default value	`FALSE`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No
Oracle RAC	Different values can be set on different instances.

When this parameter is set to `TRUE`, the optimizer ignores embedded hints.

The default value is `FALSE`. When this parameter's value is `FALSE`, the optimizer does not ignore embedded hints.

1.252 OPTIMIZER_IGNORE_PARALLEL_HINTS

`OPTIMIZER_IGNORE_PARALLEL_HINTS` enables embedded parallel hints to be ignored.

Property	Description
Parameter type	Boolean
Default value	`FALSE`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No
Oracle RAC	Different values can be set on different instances.

When this parameter is set to `TRUE`, the optimizer ignores embedded parallel hints.

The default value is `FALSE`. When this parameter's value is `FALSE`, the optimizer does not ignore parallel embedded hints.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.253 OPTIMIZER_INDEX_CACHING

`OPTIMIZER_INDEX_CACHING` lets you adjust the behavior of cost-based optimization to favor nested loops joins and IN-list iterators.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	0 to 100
Basic	No

The cost of executing an index using an IN-list iterator or of executing a nested loops join when an index is used to access the inner table depends on the caching of that index in the buffer cache. The amount of caching depends on factors that the optimizer cannot predict, such as the load on the system and the block access patterns of different users.

You can modify the optimizer's assumptions about index caching for nested loops joins and IN-list iterators by setting this parameter to a value between 0 and 100 to indicate the percentage of the index blocks the optimizer should assume are in the cache. Setting this parameter to a higher value makes nested loops joins and IN-list iterators look less expensive to the optimizer. As a result, it will be more likely to pick nested loops joins over hash or sort-merge joins and to pick indexes using IN-list iterators over other indexes or full table scans. The default for this parameter is 0, which results in default optimizer behavior.

See Also

Oracle Database SQL Language Reference for additional information about this initialization parameter

1.254 OPTIMIZER_INDEX_COST_ADJ

`OPTIMIZER_INDEX_COST_ADJ` lets you tune optimizer behavior for access path selection to be more or less index friendly—that is, to make the optimizer more or less prone to selecting an index access path over a full table scan.

Property	Description
Parameter type	Integer
Default value	`100`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	1 to 10000
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

The default for this parameter is 100 percent, at which the optimizer evaluates index access paths at the regular cost. Any other value makes the optimizer evaluate the access path at that percentage of the regular cost. For example, a setting of 50 makes the index access path look half as expensive as normal.

Note

The adjustment does not apply to user-defined cost functions for domain indexes.

See Also

Oracle Database SQL Language Reference for additional information about this initialization parameter

1.255 OPTIMIZER_INMEMORY_AWARE

`OPTIMIZER_INMEMORY_AWARE` enables or disables all of the optimizer cost model enhancements for in-memory.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true	false`
Basic	No
Oracle RAC	All instances should use the same value

Setting the parameter to `false` causes the optimizer to ignore the in-memory property of tables during the optimization of SQL statements. This behavior can also be achieved by setting the `OPTIMIZER_FEATURES_ENABLE` initialization parameter to values lower than 12.1.0.2.

See Also

- "OPTIMIZER_FEATURES_ENABLE"
- *Oracle Database Concepts* and *Oracle Database SQL Tuning Guide* for more information about the optimizer

1.256 OPTIMIZER_MODE

`OPTIMIZER_MODE` establishes the default behavior for choosing an optimization approach for the instance.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	String
Syntax	`OPTIMIZER_MODE = { FIRST_ROWS_[1 | 10 | 100 | 1000] | FIRST_ROWS | ALL_ROWS }`
Default value	`ALL_ROWS`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

Values

- `FIRST_ROWS_`*n*

  The optimizer uses a cost-based approach and optimizes with a goal of best response time to return the first *n* rows (where *n* = 1, 10, 100, 1000).

- `FIRST_ROWS`

  The optimizer uses a mix of costs and heuristics to find a best plan for fast delivery of the first few rows.

  `FIRST_ROWS` is available for backward compatibility and plan stability; use `FIRST_ROWS_`*n* instead.

- `ALL_ROWS`

  The optimizer uses a cost-based approach for all SQL statements in the session and optimizes with a goal of best throughput (minimum resource use to complete the entire statement).

See Also

- *Oracle Database SQL Tuning Guide* for more information on setting this parameter
- *Oracle Database Concepts* and *Oracle Database SQL Tuning Guide* for more information about the optimizer

1.257 OPTIMIZER_REAL_TIME_STATISTICS

`OPTIMIZER_REAL_TIME_STATISTICS` controls whether the database automatically gathers real-time statistics during conventional DML operations.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Oracle RAC	Different values can be set on different instances.

When this parameter is set to `true`, the database automatically gathers real-time statistics during conventional DML operations. The default setting is `false`, which means real-time statistics are disabled.

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.10.

1.258 OPTIMIZER_SECURE_VIEW_MERGING

`OPTIMIZER_SECURE_VIEW_MERGING` enables the optimizer to use view merging to improve query performance without performing the checks that would otherwise be performed to ensure that view merging does not violate any security intentions of the view creator.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No
Oracle RAC	Multiple instances can have different values

Values

- `false`

  Oracle Database does not perform security checks that may prevent view merging and predicate move-around.

- `true`

  Oracle Database performs checks to ensure that view merging and predicate move-around do not violate any security intentions of the view creator.

To enable the optimizer to use view merging for any query issued by a user, you must grant the `MERGE ANY VIEW` privilege to the user. Grant the `MERGE VIEW` privilege to a user on specific views to enable the optimizer to use view merging for queries on these views. These privileges are required only under specific conditions, such as when a view is not merged because the security checks fail.

> **See Also**
>
> *Oracle Database SQL Tuning Guide* for more information about view merging

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.259 OPTIMIZER_SESSION_TYPE

`OPTIMIZER_SESSION_TYPE` controls how the database performs automatic indexing for SQL statements in the session.

Property	Description
Parameter type	String
Syntax	`OPTIMIZER_SESSION_TYPE = { NORMAL | CRITICAL | ADHOC }`
Default value	`NORMAL`
Modifiable	`ALTER SESSION`
Modifiable in a PDB	No
Basic	No

Values:

- `NORMAL`

  Automatic indexing is performed normally for SQL statements in the session. The automatic indexing process identifies auto index candidates, creates auto indexes, and verifies them against SQL statements.

- `CRITICAL`

  Automatic indexing is performed for SQL statements in the session, with a higher priority given to long-running statements. The automatic indexing process identifies auto index candidates and creates auto indexes. However, auto indexes are verified against SQL statements in descending order of elapsed execution time for each statement.

- `ADHOC`

  Automatic indexing is suspended for SQL statements in the session. The automatic indexing process does not identify auto index candidates, create auto indexes, or verify auto indexes against SQL statements. This setting is useful when running ad hoc queries or testing new functionality.

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.6.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information about how automatic indexing works

1.260 OPTIMIZER_USE_INVISIBLE_INDEXES

`OPTIMIZER_USE_INVISIBLE_INDEXES` enables or disables the use of invisible indexes.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true	false`
Basic	No

Values

- `true`

  Invisible indexes are treated as visible (normal) indexes.
- `false`

  Invisible indexes will not be considered by the optimizer but will still be maintained by DML operations.

1.261 OPTIMIZER_USE_PENDING_STATISTICS

`OPTIMIZER_USE_PENDING_STATISTICS` specifies whether the optimizer uses pending statistics when compiling SQL statements.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true	false`
Basic	No

See Also

Oracle Database SQL Tuning Guide for more information on setting this parameter

1.262 OPTIMIZER_USE_SQL_PLAN_BASELINES

`OPTIMIZER_USE_SQL_PLAN_BASELINES` enables or disables the use of SQL plan baselines stored in SQL Management Base.

Property	Description
Parameter type	Boolean
Default value	`true`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No

When enabled, the optimizer looks for a SQL plan baseline for the SQL statement being compiled. If one is found in SQL Management Base, then the optimizer will cost each of the baseline plans and pick one with the lowest cost.

See Also

- *Oracle Database SQL Tuning Guide* for information on enabling automatic initial plan capture
- *Oracle Database SQL Tuning Guide* for information about configuring the capture and use of SQL plan baselines

1.263 OPTIMIZER_USE_SQL_QUARANTINE

`OPTIMIZER_USE_SQL_QUARANTINE` determines whether the optimizer considers SQL Quarantine configurations when choosing an execution plan for a SQL statement.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No
Oracle RAC	Different instances can use different values.

Values:

- `true`

  The optimizer considers SQL Quarantine configurations when choosing an execution plan for a SQL statement. The statement will be terminated if it would otherwise use a quarantined execution plan.

- `false`

  The optimizer ignores SQL Quarantine configurations when choosing an execution plan for a SQL statement. It may use any execution plan, regardless of its quarantine status.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This parameter is available on Oracle Autonomous Database starting with Oracle Database 19c. It is available on other types of Oracle databases starting with Oracle Database 19c, Release Update 19.27.

> **See Also**
>
> - "DBA_SQL_QUARANTINE"
> - "OPTIMIZER_CAPTURE_SQL_QUARANTINE"
> - *Oracle Database SQL Tuning Guide* for more information about SQL Quarantine

1.264 OS_AUTHENT_PREFIX

`OS_AUTHENT_PREFIX` specifies a prefix that Oracle Database uses to authenticate users attempting to connect to the server.

Property	Description
Parameter type	String
Syntax	`OS_AUTHENT_PREFIX = authentication_prefix`
Default value	`OPS$`
Modifiable	No
Modifiable in a PDB	No
Basic	No

Oracle Database concatenates the value of this parameter to the beginning of the user's operating system account name. When a connection request is attempted, Oracle Database compares the prefixed username with Oracle user names in the database.

The default value of this parameter is `OPS$` for backward compatibility with previous versions. However, you might prefer to set the prefix value to "" (a null string), thereby eliminating the addition of any prefix to operating system account names.

> **Note**
>
> The text of the `OS_AUTHENT_PREFIX` parameter is case sensitive on some operating systems.

> **See Also**
>
> - *Oracle Database Security Guide* for more information on setting this parameter
> - Your operating system-specific Oracle documentation for the default value

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.265 OS_ROLES

OS_ROLES determines whether Oracle or the operating system identifies and manages the roles of each username.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	No
Modifiable in a PDB	No
Range of values	true | false
Basic	No

Values

- TRUE

  The operating system completely manages the role grants for all database usernames. When a user attempts to create a session, the username's security domain is initialized using the roles identified by the operating system.

  Revocation by Oracle of roles granted by the operating system is ignored, as are any roles previously granted by Oracle.

- FALSE

  Oracle identifies and manages the roles.

See Also

- *Oracle Database Administrator's Guide* and *Oracle Database Enterprise User Security Administrator's Guide* for more information on roles and on setting this parameter
- "REMOTE_OS_ROLES"

1.266 OUTBOUND_DBLINK_PROTOCOLS

OUTBOUND_DBLINK_PROTOCOLS specifies the network protocols allowed for communicating for outbound database links in the database.

Property	Description
Parameter type	String
Syntax	OUTBOUND_DBLINK_PROTOCOLS = { ALL | NONE | [TCP | [,] | TCPS | [,] | IPC] }
Default value	ALL
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Basic	No
Oracle RAC	The same value must be used on all instances.

Specify a value of `ALL` to allow all network protocols, and a value of `NONE` to disallow all network communication protocols for database link communication. Specify a single value or a list of comma separated network communication protocols to allow only certain network protocols for outbound database link communication.

1.267 PARALLEL_ADAPTIVE_MULTI_USER

`PARALLEL_ADAPTIVE_MULTI_USER`, when set to `true`, enables an adaptive algorithm designed to improve performance in multiuser environments that use parallel execution.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

The algorithm automatically reduces the requested degree of parallelism based on the system load at query startup time. The effective degree of parallelism is based on the default degree of parallelism, or the degree from the table or hints, divided by a reduction factor.

The algorithm assumes that the system has been tuned for optimal performance in a single-user environment.

Tables and hints use the default degree of parallelism.

> **Note**
>
> The `PARALLEL_ADAPTIVE_MULTI_USER` initialization parameter is deprecated in Oracle Database 12*c* Release 2 (12.2.0.1) and may be removed in a future release. Oracle recommends that you use the parallel statement queuing feature instead.

> **See Also**
>
> - *Oracle Database SQL Language Reference* for more information about optimizer hints
> - *Oracle Database VLDB and Partitioning Guide* for more information about parallel statement queuing

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.268 PARALLEL_DEGREE_LIMIT

`PARALLEL_DEGREE_LIMIT` limits the degree of parallelism used by the optimizer to ensure that parallel server processes do not flood the system.

Property	Description
Parameter type	String
Syntax	`PARALLEL_DEGREE_LIMIT = { CPU | AUTO | IO | integer }`
Default value	`CPU`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

With automatic degree of parallelism, Oracle automatically decides whether a statement should execute in parallel and what degree of parallelism the statement should use. The optimizer automatically determines the degree of parallelism for a statement based on the resource requirements of the statement. However, `PARALLEL_DEGREE_LIMIT` enforces the limit for the degree of parallelism used by the optimizer.

Values

- `CPU`

  The maximum degree of parallelism is limited by the number of CPUs in the system. The formula used to calculate the limit is `PARALLEL_THREADS_PER_CPU` * `CPU_COUNT` * the number of instances available (by default, all the opened instances on the cluster but can be constrained using `PARALLEL_INSTANCE_GROUP` or service specification). This is the default.

- `AUTO`

  This value is equivalent to the `CPU` value.

- `IO`

  The maximum degree of parallelism the optimizer can use is limited by the I/O capacity of the system. The value is calculated by dividing the total system throughput by the maximum I/O bandwidth per process. You must run the `DBMS_RESOURCE_MANAGER.CALIBRATE_IO` procedure on the system to use the `IO` setting. This procedure will calculate the total system throughput and the maximum I/O bandwidth per process.

- *`integer`*

  A numeric value for this parameter specifies the maximum degree of parallelism the optimizer can choose for a SQL statement when automatic degree of parallelism is active. Automatic degree of parallelism is only enabled if `PARALLEL_DEGREE_POLICY` is set to `ADAPTIVE`, `AUTO`, or `LIMITED`.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database VLDB and Partitioning Guide* for information about automatic degree of parallelism
- *Oracle Database PL/SQL Packages and Types Reference* for information on the `DBMS_RESOURCE_MANAGER.CALIBRATE_IO` procedure

1.269 PARALLEL_DEGREE_POLICY

`PARALLEL_DEGREE_POLICY` specifies whether automatic degree of parallelism, statement queuing, and in-memory parallel execution will be enabled.

Property	Description
Parameter type	String
Syntax	`PARALLEL_DEGREE_POLICY = { MANUAL | LIMITED | AUTO | ADAPTIVE }`
Default value	`MANUAL`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

Values

Note

Automatic degree of parallelism will be enabled regardless of the value of `PARALLEL_DEGREE_POLICY` if a `PARALLEL` hint is used at the SQL statement level.

- `MANUAL`

  Disables automatic degree of parallelism, statement queuing, and in-memory parallel execution. This reverts the behavior of parallel execution to what it was prior to Oracle Database 11*g* Release 2 (11.2). This is the default.

- `LIMITED`

  Enables automatic degree of parallelism for some statements but statement queuing and in-memory Parallel Execution are disabled. Automatic degree of parallelism is only applied to those statements that access tables or indexes decorated explicitly with the `DEFAULT` degree of parallelism using the `PARALLEL` clause. Statements that do not access any tables or indexes decorated with the `DEFAULT` degree of parallelism will retain the `MANUAL` behavior.

- `AUTO`

  Enables automatic degree of parallelism, statement queuing, and in-memory parallel execution.

- `ADAPTIVE`

Copyright © 2002, 2026, Oracle and/or its affiliates.

This value enables automatic degree of parallelism, statement queuing and in-memory parallel execution, similar to the AUTO value. In addition, performance feedback is enabled. Performance feedback helps to improve the degree of parallelism automatically chosen for repeated SQL statements. After the initial execution of a statement, the degree of parallelism chosen by the optimizer is compared to the degree of parallelism computed based on the actual execution performance. If they vary significantly, then the statement is marked for re-parse and the initial execution performance statistics (for example, CPU-time) are provided as feedback for subsequent executions. The optimizer uses the initial execution performance statistics to better determine a degree of parallelism for subsequent executions.

See Also

- *Oracle Database SQL Language Reference* for information about PARALLEL hints
- *Oracle Database VLDB and Partitioning Guide* for information about automatic degree of parallelism

1.270 PARALLEL_EXECUTION_MESSAGE_SIZE

PARALLEL_EXECUTION_MESSAGE_SIZE specifies the size of messages used for parallel execution (formerly referred to as parallel query, PDML, Parallel Recovery, replication).

Property	Description
Parameter type	Integer
Default value	Operating system-dependent
Modifiable	No
Modifiable in a PDB	No
Range of values	Minimum: 2148 Maximum: 65536, but some operating systems may have a smaller value
Basic	No
Oracle RAC	Multiple instances must have the same value.

On most platforms, the default value is as follows:

- 16384 bytes if COMPATIBLE is set to 11.2.0 or higher
- 2148 bytes if COMPATIBLE is less than 11.2.0

The default value is adequate for most applications. Larger values require a larger shared pool. Larger values result in better performance at the cost of higher memory use. For this reason, replication gets no benefit from increasing the size.

See Also

Oracle Database VLDB and Partitioning Guide to learn how this parameter affects memory consumption for parallel operations, including parallel execution

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.271 PARALLEL_FORCE_LOCAL

PARALLEL_FORCE_LOCAL controls parallel execution in an Oracle RAC environment.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	true | false
Basic	No

By default, the parallel server processes selected to execute a SQL statement can operate on any or all Oracle RAC nodes in the cluster. By setting PARALLEL_FORCE_LOCAL to true, the parallel server processes are restricted so that they can only operate on the same Oracle RAC node where the query coordinator resides (the node on which the SQL statement was executed).

Note that, regardless of the value of this parameter, parallel server worker processes are spawned across Oracle RAC nodes for queries accessing GV$ views and CDB_* views.

See Also

Oracle Database VLDB and Partitioning Guide for more information about this parameter

1.272 PARALLEL_INSTANCE_GROUP

Used in conjunction with services or with the INSTANCE_GROUPS parameter, PARALLEL_INSTANCE_GROUP lets you restrict parallel query operations to a limited number of instances.

Property	Description
Parameter type	String
Syntax	PARALLEL_INSTANCE_GROUP = *service_name* | *group_name*
Default value	There is no default value; parallel execution is enabled across all currently active instances.
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	Any service name or any group name specified in the INSTANCE_GROUPS parameter of any active instance
Basic	No
Oracle RAC	Multiple instances can have different values.

Copyright © 2002, 2026, Oracle and/or its affiliates.

PARALLEL_INSTANCE_GROUP is an Oracle RAC parameter that you can specify in parallel mode only. Note that the INSTANCE_GROUPS parameter has been deprecated.

This parameter identifies the parallel instance group Oracle will use for spawning parallel execution processes. If used in conjunction with services, then parallel operations will spawn parallel execution processes only on instances defined in the service. If used in conjunction with INSTANCE_GROUPS, then parallel operations will spawn parallel execution processes only on instances that specify a matching group in their INSTANCE_GROUPS parameter.

If the value assigned to PARALLEL_INSTANCE_GROUP is the name of a service or group that does not exist, then the operation runs serially. No parallelism is used.

1.273 PARALLEL_MAX_SERVERS

PARALLEL_MAX_SERVERS specifies the maximum number of parallel execution processes and parallel recovery processes for an instance. As demand increases, Oracle Database increases the number of processes from the number created at instance startup up to this value.

Property	Description
Parameter type	Integer
Default value	PARALLEL_THREADS_PER_CPU * CPU_COUNT * *concurrent_parallel_users* * 5
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to 32767
Basic	No
Oracle RAC	Multiple instances can have different values.

> **Note**
>
> This parameter applies to parallel execution in exclusive mode as well as in a Real Application Clusters environment.

The number of concurrent parallel users running at default degree of parallelism on an instance depends on the memory initialization parameter settings for the instance. For example, if the MEMORY_TARGET or SGA_TARGET initialization parameter is set, then the number of concurrent_parallel_users = 4. If neither MEMORY_TARGET or SGA_TARGET is set, then PGA_AGGREGATE_TARGET is examined. If a value is set for PGA_AGGREGATE_TARGET, then concurrent_parallel_users = 2. If a value is not set for PGA_AGGREGATE_TARGET, then concurrent_parallel_users = 1.

The database system always reserves a certain number of reserved processes. Therefore, the following two values are taken into account when setting the PARALLEL_MAX_SERVERS initialization parameter:

- The default value of PARALLEL_MAX_SERVERS determined using the calculation in the table above
- The value of the PROCESSES initialization parameter minus the number of reserved processes

Copyright © 2002, 2026, Oracle and/or its affiliates.

The lower of the two values is used as the default value of PARALLEL_MAX_SERVERS, and if you attempt to explicitly set PARALLEL_MAX_SERVERS to a value that is higher than either of the values, then the setting is adjusted to the lower of the two values.

When the PARALLEL_MAX_SERVERS parameter is set to a value lower than the value in the table above, the lower default value enables the database to start service processes and allows user processes to connect to the database.

If the FAST_START_PARALLEL_ROLLBACK initialization parameter is set to LOW or HIGH, then its value should be taken into account when setting PARALLEL_MAX_SERVERS. If FAST_START_PARALLEL_ROLLBACK = LOW, then PARALLEL_MAX_SERVERS should not be set to a value lower than CPU_COUNT * 2. If FAST_START_PARALLEL_ROLLBACK = HIGH, then PARALLEL_MAX_SERVERS should not be set to a value lower than CPU_COUNT * 4.

The default value for PARALLEL_MAX_SERVERS for a PDB is determined using the calculation in the table above with the PDB's CPU_COUNT value.

Note

If you set this parameter too low, then some queries may not have a parallel execution process available to them during query processing. If you set it too high, then memory resource shortages may occur during peak periods, which can degrade performance.

See Also

Oracle Database SQL Tuning Guide for more information about parallel execution

1.274 PARALLEL_MIN_DEGREE

PARALLEL_MIN_DEGREE controls the minimum degree of parallelism computed by automatic degree of parallelism.

Property	Description
Parameter type	String
Syntax	PARALLEL_MIN_DEGREE = [n | CPU]
Default value	1
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	A different value can be set on different instances.

The value of PARALLEL_MIN_DEGREE is either a number that corresponds to the lower bound on the degree of parallelism computed by automatic degree of parallelism, or the string value CPU, which is computed by the function CPU_COUNT * PARALLEL_THREADS_PER_CPU.

The default value of PARALLEL_MIN_DEGREE is 1.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Note

`PARALLEL_MIN_DEGREE` has no impact in either of these cases:

- When the value of `PARALLEL_MIN_DEGREE` is greater than the value of `CPU_COUNT`
- When the object is Oracle-owned, such as a dictionary table or view created on a dictionary table

See Also

- "PARALLEL_DEGREE_LIMIT"
- *Oracle Database VLDB and Partitioning Guide* for information about how the optimizer automatically determines the degree of parallelism for a statement

1.275 PARALLEL_MIN_PERCENT

`PARALLEL_MIN_PERCENT` lets you specify the minimum percentage of the requested number of parallel execution processes required for parallel execution.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	`ALTER SESSION`
Modifiable in a PDB	No
Range of values	0 to 100
Basic	No
Oracle RAC	Multiple instances can have different values.

This parameter controls the behavior for parallel operations when parallel statement queuing is not enabled (when `PARALLEL_DEGREE_POLICY` is set to `manual` or `limited`). It ensures that an operation always gets a minimum percentage of parallel execution servers or errors out. Setting this parameter ensures that parallel operations will not execute unless adequate resources are available. The default value of `0` means that no minimum percentage of processes has been set.

Consider the following settings:

```
PARALLEL_MIN_PERCENT = 50
PARALLEL_MIN_SERVERS = 5
PARALLEL_MAX_SERVERS = 10
```

If 8 of the 10 parallel execution processes are busy, only 2 processes are available. If you then request a query with a degree of parallelism of 8, the minimum 50% will not be met.

You can use this parameter with `PARALLEL_ADAPTIVE_MULTI_USER`. In a multi-user environment, an individual user or application can set `PARALLEL_MIN_PERCENT` to a minimum value until sufficient resources are available on the system and an acceptable degree of parallelism is returned.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database SQL Tuning Guide* for more information about parallel execution
- "PARALLEL_DEGREE_POLICY", "PARALLEL_MAX_SERVERS", "PARALLEL_MIN_SERVERS", and "PARALLEL_ADAPTIVE_MULTI_USER"

1.276 PARALLEL_MIN_SERVERS

PARALLEL_MIN_SERVERS is the number of parallel execution processes Oracle creates when the instance is started. These processes will be kept active to service parallel statements.

Property	Description
Parameter type	Integer
Default value	CPU_COUNT * PARALLEL_THREADS_PER_CPU * 2
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	Default value to the value of PARALLEL_MAX_SERVERS
Basic	No
Oracle RAC	Multiple instances can have different values.

Note

This parameter applies to parallel execution in exclusive mode as well as in an Oracle Real Application Clusters environment.

Note

When the PROCESSES initialization parameter is set to a value that is lower than the documented default value for the PARALLEL_MIN_SERVERS parameter in the table above, the database sets the default value of PARALLEL_MIN_SERVERS to a value that is lower than the documented default value. The lower default value enables the database to start service processes and allows user processes to connect to the database.

See Also

Oracle Database Administrator's Guide for more information about parallel execution servers

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.277 PARALLEL_MIN_TIME_THRESHOLD

PARALLEL_MIN_TIME_THRESHOLD specifies the minimum execution time a statement should have before the statement is considered for automatic degree of parallelism.

Property	Description
Parameter type	String
Syntax	PARALLEL_MIN_TIME_THRESHOLD = { AUTO | *integer* }
Default value	AUTO
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No

By default, this parameter is set to 10 seconds. Automatic degree of parallelism is only enabled if PARALLEL_DEGREE_POLICY is set to ADAPTIVE, AUTO, or LIMITED.

If all tables referenced by a SQL statement use In-Memory Column Store (IM column store), then PARALLEL_MIN_TIME_THRESHOLD defaults to 1.

See Also

Oracle Database VLDB and Partitioning Guide for information about automatic degree of parallelism

1.278 PARALLEL_SERVERS_TARGET

PARALLEL_SERVERS_TARGET specifies the number of parallel server processes allowed to run parallel statements before statement queuing will be used.

Property	Description
Parameter type	Integer
Default value	For a CDB: Equal to the PARALLEL_MAX_SERVERS value for the CDB. For a PDB or non-CDB: PARALLEL_THREADS_PER_CPU * CPU_COUNT * *concurrent_parallel_users* * 2
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to PARALLEL_MAX_SERVERS
Basic	No

When the parameter PARALLEL_DEGREE_POLICY is set to ADAPTIVE or AUTO, Oracle will queue SQL statements that require parallel execution, if the necessary parallel server processes are not available. In a single instance database, statement queuing will begin once the number of parallel server processes active on the system is equal to or greater than

Copyright © 2002, 2026, Oracle and/or its affiliates.

PARALLEL_SERVERS_TARGET. In an Oracle RAC database, a statement running on one instance can allocate parallel server processes on another instance when necessary. For example, if a statement running on one instance requires parallel execution, but the number of parallel server processes active on that instance is equal to or greater than the value of PARALLEL_SERVERS_TARGET for that instance, then the statement can allocate parallel server processes on a different instance. Therefore, statement queuing in an Oracle RAC database will begin only when every instance has reached its PARALLEL_SERVER_TARGET threshold.

> **Note**
>
> Consumer groups that have been marked with the PARALLEL_STMT_CRITICAL directive set to BYPASS_QUEUE are allowed to bypass the parallel statement queue, and therefore may drive the total number of active parallel server processes beyond PARALLEL_SERVERS_TARGET. Parallel statements issued with PARALLEL_DEGREE_POLICY not set to ADAPTIVE and AUTO can also drive the total number of active parallel server processes beyond PARALLEL_SERVERS_TARGET.

By default, PARALLEL_SERVERS_TARGET is set lower than the maximum number of parallel server processes allowed on the system (PARALLEL_MAX_SERVERS) to ensure each parallel statement will get all of the parallel server resources required and to prevent overloading the system with parallel server processes.

The number of concurrent parallel users running at default degree of parallelism on an instance depends on the memory initialization parameter settings for the instance. For example, if the MEMORY_TARGET or SGA_TARGET initialization parameter is set, then the number of concurrent_parallel_users = 4. If neither MEMORY_TARGET or SGA_TARGET is set, then PGA_AGGREGATE_TARGET is examined. If a value is set for PGA_AGGREGATE_TARGET, then concurrent_parallel_users = 2. If a value is not set for PGA_AGGREGATE_TARGET, then concurrent_parallel_users = 1.

Note that all serial (non-parallel) statements will execute immediately even if statement queuing has been activated.

By default, all PDB queries are subjected to queuing at the PDB level first by the PDB's PARALLEL_SERVERS_TARGET value, and then at the CDB level by the CDB's PARALLEL_SERVERS_TARGET value. This default behavior prevents any SQL statement inside a PDB from getting downgraded if parallel servers are exhausted in the CDB.

Parallel statement queuing is enabled by default at the CDB level because the CDB has a default value for PARALLEL_SERVERS_TARGET. You can disable parallel statement queuing at the CDB level by using ALTER SYSTEM to set PARALLEL_SERVERS_TARGET to 0 for the CDB.

The default value for PARALLEL_SERVERS_TARGET for a PDB is determined using the calculation in the table above with the PDB's CPU_COUNT value.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Note

A PDB can set a lower limit for parallel execution servers than the limit specified in the CDB resource plan. When the PARALLEL_SERVERS_TARGET initialization parameter is set in a PDB, and parallel execution server limit is specified for a PDB in the CDB resource plan, then the lower of the two limits is enforced. For example, assume that the PARALLEL_SERVERS_TARGET initialization parameter is set to 100 in the CDB root. Also assume that hrpdb has its PARALLEL_SERVERS_TARGET initialization parameter set to 50 and the CDB plan has a directive for hrpdb with parallel_server_limit set to 70%. In this case, the limit for parallel execution servers for hrpdb is 50, because 50 is lower than the CDB resource plan limit of 70 for hrpdb.

See Also

- "PARALLEL_DEGREE_POLICY"
- "PARALLEL_MAX_SERVERS"
- "MEMORY_TARGET"
- "SGA_TARGET"
- "PGA_AGGREGATE_TARGET"
- *Oracle Database VLDB and Partitioning Guide* for more information about this parameter

1.279 PARALLEL_THREADS_PER_CPU

PARALLEL_THREADS_PER_CPU describes the number of parallel execution processes or threads that a CPU can handle during parallel execution.

Property	Description
Parameter type	Integer
Default value	1
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	Any nonzero number
Basic	No

Note

This parameter applies to parallel execution in exclusive mode as well as in an Oracle Real Application Clusters environment.

This parameter is used in determining the default values of other parallel execution related parameters, for example, PARALLEL_MAX_SERVERS. It is also used in determining the default

Copyright © 2002, 2026, Oracle and/or its affiliates.

degree of parallelism for SQL statements, and determining the upper bound for the degree of parallelism in automatic degree of parallelism. The default value is adequate in most cases.

See Also

- *Oracle Database SQL Tuning Guide* for more information about parallel execution
- *Oracle Database VLDB and Partitioning Guide* for information about how the optimizer automatically determines the degree of parallelism for a statement

1.280 PDB_FILE_NAME_CONVERT

`PDB_FILE_NAME_CONVERT` maps names of existing files to new file names when processing a `CREATE PLUGGABLE DATABASE` statement, as well as when processing the `ENABLE PLUGGABLE DATABASE` clause of the `CREATE DATABASE` statement, if the *`file_name_convert_clause`* is not specified and Oracle Managed Files is not enabled.

Property	Description
Parameter type	String

Syntax	`PDB_FILE_NAME_CONVERT = '`*`string1`*`' , '`*`string2`*`' , '`*`string3`*`' , '`*`string4`*`' , ...`
	Where:
	• *`string1`* is the pattern of the existing filename
	• *`string2`* is the pattern of the new filename
	• *`string3`* is the pattern of the existing filename
	• *`string4`* is the pattern of the new filename
	You can enclose each string in single or double quotation marks.
	You can specify as many pairs of existing and new replacement strings as required.
	Example:
	`PDB_FILE_NAME_CONVERT = '/dbs/t1/','/dbs/t1/s','dbs/t2/','dbs/t2/s'`

Default value	There is no default value
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	You must set this parameter for every instance, and multiple instances must have the same value.

File name patterns specified in this initialization parameter cannot match files or directories managed by Oracle Managed Files.

See Also

Oracle Multitenant Administrator's Guide for more information about this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.281 PDB_LOCKDOWN

PDB_LOCKDOWN determines the PDB lockdown profile that applies to a PDB.

Property	Description
Parameter type	String
Syntax	PDB_LOCKDOWN = pdb-lockdown-profile-name
Default value	NULL
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	All instances should have the same value

A PDB lockdown profile is a mechanism to restrict operations (such as setting values of certain parameters and using certain options) that can be performed by users connected to a given PDB. You can also restrict execution of any packages that allow network access, for example, UTL_SMTP.

You create lockdown profiles using the SQL CREATE LOCKDOWN PROFILE statement. Then you can set a profile using the SQL ALTER SESSION or ALTER SYSTEM statement. See the Examples section.

This parameter can be set using the ALTER SYSTEM statement with scope set to MEMORY, SPFILE, or BOTH.

The lockdown profile for PDBs can be specified by a common user with common ALTER SYSTEM or common SYSDBA privilege.

If a PDB lockdown profile is dropped, any PDB to which the dropped profile was assigned (by means of storing the dropped profile name in the PDB_LOCKDOWN parameter) will continue to have its PDB_LOCKDOWN parameter set to the dropped lockdown profile name. However, the PDB will not have any restrictions imposed by the dropped lockdown profile.

Lockdown profiles can now be created in an application root and are referred to as application lockdown profiles.

A CDB common user with common SYSDBA or common ALTER SYSTEM privilege can only set PDB_LOCKDOWN to a CDB lockdown profile. Similarly, an application common user with application common SYSDBA or application common ALTER SYSTEM privilege can only set PDB_LOCKDOWN to an application lockdown profile.

An application common user cannot overwrite PDB_LOCKDOWN if PDB_LOCKDOWN is already set to a CDB lockdown profile in an application root or application PDB.

If the PDB_LOCKDOWN parameter in a PDB is set to the name of a lockdown profile different from that in its ancestor (for a CDB, the CDB root or, for application PDBs, the application root), the following will govern the interaction between restrictions imposed by these profiles:

- If the PDB_LOCKDOWN parameter in a PDB (including an application PDB) is set to a CDB lockdown profile, lockdown profiles specified by the PDB_LOCKDOWN parameter in CDB root (and for application PDBs, the application root) are ignored.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- If the PDB_LOCKDOWN parameter in an application PDB is set to an application lockdown profile while the PDB_LOCKDOWN parameter in the application root or CDB root is set to a CDB lockdown profile, in addition to the rules stipulated in the application lockdown profile, the DISABLE rules from the CDB lockdown profile set in its nearest ancestor (that is, an application root or CDB root) are inherited.
- If there are conflicts between rules comprising the CDB lockdown profile and the application lockdown profile, the rules in the CDB lockdown profile will take precedence (for example, the OPTION_VALUE clause of a CDB lockdown profile will take precedence over the OPTION_VALUE clause of an application lockdown profile).

Examples

This example shows how the SYS user can connect to the database AS SYSDBA and use the CREATE LOCKDOWN PROFILE statement in the root of a CDB to define a new lockdown profile. After defining the new lockdown profile, the SYS user can assign the new lockdown profile to a PDB using the PDB_LOCKDOWN parameter:

```
SQL> ALTER SESSION SET CONTAINER=CDB$ROOT;

Session altered.

SQL> CREATE LOCKDOWN PROFILE MYPROFILE;

Lockdown Profile created.

SQL> ALTER SESSION SET CONTAINER=CDB1_PDB1;

Session altered.

SQL> ALTER SYSTEM SET PDB_LOCKDOWN=MYPROFILE;

System altered.

SQL> SHOW PARAMETER PDB_LOCKDOWN

NAME           TYPE        VALUE
-------------- ----------- ---------
pdb_lockdown   string      MYPROFILE
SQL>
```

See Also

- "DBA_LOCKDOWN_PROFILES"
- *Oracle Multitenant Administrator's Guide* for an introduction to PDB lockdown profiles
- *Oracle Database SQL Language Reference* for more information about the CREATE LOCKDOWN PROFILE statement

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.282 PDB_OS_CREDENTIAL

PDB_OS_CREDENTIAL determines the identity of the operating system user (OS user) employed when interacting with the operating system from a PDB.

Property	Description
Parameter type	String
Syntax	PDB_OS_CREDENTIAL = *credential*
Default value	None
Modifiable	No
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The same value should be used for all instances

> **Note**
>
> The Oracle OS user will continue to be used when interacting with the operating system from the root.

The Oracle OS user is usually a highly privileged user, and using the same user for operating system interactions for every PDB is not recommended. Also, using the same OS user for operating system interactions from different PDBs may compromise data belonging to a given PDB.

In contrast, using an OS user described by a credential whose name is specified as a value of the PDB_OS_CREDENTIAL parameter helps ensure that operating system interactions are performed as a less powerful user and provides the ability to protect data belonging to one PDB from being accessed by users connected to another PDB. A credential is an object that is created using the CREATE_CREDENTIAL procedure for the DBMS_CREDENTIAL package.

The operating system interactions that are done as the OS user name specified in the credential include:

- External jobs that do not already have an operating system credential specified
- External table pre-processors
- PL/SQL library executions

This parameter can be specified for all the PDBs in a CDB but the CDB-wide value can be overridden for a specific PDB and can be modified *only* by a common administrative user with the EXECUTE privilege for the DBMS_CREDENTIAL PL/SQL package and the ALTER SYSTEM system privilege.

If a value is not set for this parameter for a given PDB, the Oracle OS User will continue to be used when interacting with the operating system from that PDB.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database Security Guide* for an example of setting an OS user for a PDB using this parameter
- *Oracle Multitenant Administrator's Guide* for conceptual information about CDBs and PDBs
- *Oracle Multitenant Administrator's Guide* for information about managing CDBs and PDBs
- *Oracle Database PL/SQL Packages and Types Reference* for information about creating a credential using the `DBMS_CREDENTIAL.CREATE_CREDENTIAL` procedure

1.283 PERMIT_92_WRAP_FORMAT

`PERMIT_92_WRAP_FORMAT` allows Oracle Database release 9.2 wrapped versions of PL/SQL source text to be used in Oracle Database releases 10.2 and later when this parameter is set to `true`.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	No
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No
Oracle RAC	Multiple instances should use the same value

Oracle recommends that wrapped files be created using the PL/SQL Wrapper utility from Oracle Database release 10 or later.

See Also

Oracle Database PL/SQL Language Reference for more information about PL/SQL source text wrapping

1.284 PGA_AGGREGATE_LIMIT

`PGA_AGGREGATE_LIMIT` specifies a limit on the aggregate PGA memory consumed by the instance.

Property	Description
Parameter type	Big integer
Syntax	`PGA_AGGREGATE_LIMIT = integer [K | M | G]`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	If MEMORY_TARGET is set, then PGA_AGGREGATE_LIMIT defaults to the MEMORY_MAX_TARGET value. If MEMORY_TARGET is not set, then PGA_AGGREGATE_LIMIT defaults to 200% of PGA_AGGREGATE_TARGET. If MEMORY_TARGET is not set, and PGA_AGGREGATE_TARGET is explicitly set to 0, then the value of PGA_AGGREGATE_LIMIT is set to 90% of the physical memory size minus the total SGA size. In all cases, the default PGA_AGGREGATE_LIMIT is at least 2GB and at least 3MB times the PROCESSES parameter (and at least 5MB times the PROCESSES parameter for an Oracle RAC instance). For a PDB, the default value is the same as the CDB's default value.
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	Do not attempt to set PGA_AGGREGATE_LIMIT below its default value, even in a parameter file (pfile), or instance startup will fail. However, PGA_AGGREGATE_LIMIT can be set to 0 either in a parameter file or dynamically after startup. If a value of 0 is specified, it means there is no limit to the aggregate PGA memory consumed by the instance.
Basic	No

Actions Taken When PGA_AGGREGATE_LIMIT is Exceeded

Parallel queries will be treated as a unit. First, the sessions that are using the most untunable memory will have their calls terminated. Then, if the total PGA memory usage is still over the limit, the sessions that are using the most untunable memory will be terminated.

SYS processes and background processes other than job queue processes will not be subjected to any of the actions described in this section. Instead, if they are using the most untunable memory, they will periodically write a brief summary of their PGA usage to a trace file.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Note

This parameter is optional for pluggable databases (PDBs). When this parameter is set for a PDB, it specifies the maximum PGA size for the PDB.

To be able to use Resource Manager in a CDB to control the amount of memory each PDB can use:

- The `NONCDB_COMPATIBLE` initialization parameter must be set to `FALSE` at the CDB level (in the root of the CDB).
- The `MEMORY_TARGET` initialization parameter must not be set at the CDB level.
- You must set the `PGA_AGGREGATE_LIMIT` initialization parameter in a PDB to a value that meets these requirements:
  - Less than or equal to the `PGA_AGGREGATE_LIMIT` value set at the CDB level
  - Greater than or equal to twice the value of `PGA_AGGREGATE_TARGET` set in the PDB

  When you set `PGA_AGGREGATE_LIMIT` in a PDB to a value that does not meet these requirements, you receive an error. If these requirements are violated after the PDB's parameter is set (for example, if the `PGA_AGGREGATE_LIMIT` value is changed at the CDB level), Oracle will adjust the PDB's value to meet these requirements.

See Also

- "MEMORY_TARGET"
- "PGA_AGGREGATE_TARGET"
- *Oracle Multitenant Administrator's Guide* for more information about the initialization parameters that control the memory usage of PDBs
- *Oracle Database Performance Tuning Guide* for more information about this parameter

1.285 PGA_AGGREGATE_TARGET

`PGA_AGGREGATE_TARGET` specifies the target aggregate PGA memory available to all server processes attached to the instance.

Property	Description
Parameter type	Big integer
Syntax	`PGA_AGGREGATE_TARGET = integer [K | M | G]`
Default value	`10` MB or 20% of the size of the SGA, whichever is greater For a PDB, the default value is the same as the CDB's default value.
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Range of values	Minimum: 10 MB Maximum: 4096 GB - 1
Basic	Yes

To set a hard limit for aggregate PGA memory, use the `PGA_AGGREGATE_LIMIT` parameter.

Setting `PGA_AGGREGATE_TARGET` to a nonzero value has the effect of automatically setting the `WORKAREA_SIZE_POLICY` parameter to `AUTO`. With this setting, SQL working areas used by memory-intensive SQL operators (such as sort, group-by, hash-join, bitmap merge, and bitmap create) will be automatically sized. A nonzero value for this parameter is the default since, unless you specify otherwise, Oracle sets it to 20% of the SGA or 10 MB, whichever is greater.

Setting `PGA_AGGREGATE_TARGET` to 0 automatically sets the `WORKAREA_SIZE_POLICY` parameter to `MANUAL`. With this setting, SQL working areas are sized using the `*_AREA_SIZE` parameters.

Oracle attempts to keep the amount of private memory below the target specified by this parameter by adapting the size of the working areas to private memory. When increasing the value of this parameter, you indirectly increase the memory allotted to working areas. Consequently, more memory-intensive operations are able to run fully in memory and fewer will work their way over to disk.

If Automatic Memory Management is enabled (`MEMORY_TARGET` is set to a positive value) and `PGA_AGGREGATE_TARGET` is also set to a positive value, the `PGA_AGGREGATE_TARGET` value acts as the minimum value for the size of the instance PGA.

Note

This parameter is optional for pluggable databases (PDBs). When this parameter is set for a PDB, it specifies the target aggregate PGA size for the PDB.

To be able to use Resource Manager in a CDB to control the amount of memory each PDB can use:

- The `NONCDB_COMPATIBLE` initialization parameter must be set to `FALSE` at the CDB level (in the root of the CDB).
- The `MEMORY_TARGET` initialization parameter must not be set at the CDB level.
- You must set the `PGA_AGGREGATE_TARGET` initialization parameter in a PDB to a value that meets these requirements:
  - Less than or equal to the `PGA_AGGREGATE_TARGET` value set at the CDB level
  - Less than or equal to 50% of the `PGA_AGGREGATE_LIMIT` initialization parameter value set at the CDB level
  - Less than or equal to 50% of the `PGA_AGGREGATE_LIMIT` value set in the PDB

  When you set `PGA_AGGREGATE_TARGET` in a PDB to a value that does not meet these requirements, you receive an error. If these requirements are violated after the PDB's parameter is set (for example, if the `PGA_AGGREGATE_TARGET` value is changed at the CDB level), Oracle will adjust the PDB's value to meet these requirements.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "WORKAREA_SIZE_POLICY"
- *Oracle Database Performance Tuning Guide* for more information about configuring memory
- "PGA_AGGREGATE_LIMIT"

1.286 PLSCOPE_SETTINGS

PLSCOPE_SETTINGS controls the compile time collection, cross-reference, and storage of PL/SQL source code identifier data.

Property	Description
Parameter type	String
Syntax	PLSCOPE_SETTINGS = 'value_clause [, value_clause]' **value_clause::=** { IDENTIFIERS | STATEMENTS } : { ALL | NONE | PUBLIC (for IDENTIFIERS only) | SQL (for IDENTIFIERS only) | PLSQL (for IDENTIFIERS only) }
Default value	'IDENTIFIERS:NONE'
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No

Values

- IDENTIFIERS:ALL

  Enables the collection of all source code identifier data.
- IDENTIFIERS:NONE

  Disables the collection of all source code identifier data.
- IDENTIFIERS: PUBLIC

  Enables the collection of all PUBLIC user identifier data (except for DEFINITION)
- IDENTIFIERS:SQL

  Enables the collection of all SQL identifier data.
- IDENTIFIERS:PLSQL

  Enables the collection of all PLSQL identifier data.
- STATEMENTS:ALL

  Enables the collection of all SQL statements used in PL/SQL.
- STATEMENTS:NONE

  Disables the collection of all statements.

PLSCOPE_SETTINGS can be set on a session, system, or per-library unit (ALTER COMPILE) basis. The current setting of PLSCOPE_SETTINGS for any library unit can be attained by querying the

Copyright © 2002, 2026, Oracle and/or its affiliates.

*_PLSQL_OBJECT_SETTINGS views. Any identifier data collected by setting this parameter can be accessed using the *_IDENTIFIERS views.

When a STATEMENTS setting is not specified, and IDENTIFIERS is specified but set to a value other than NONE, STATEMENTS defaults to a setting of ALL, which is equal to:

```
IDENTIFIERS: [ALL|PLSQL|PLSQL|PUBLIC]
```

Examples

The following parameter setting causes PL/SQL and SQL identifiers and statements to be collected:

```
PLSCOPE_SETTINGS = 'IDENTIFIERS:ALL, STATEMENTS:ALL'
```

The following parameter setting causes only SQL identifiers and statements to be collected:

```
PLSCOPE_SETTINGS = 'IDENTIFIERS:SQL, STATEMENTS:ALL'
```

The following parameter setting causes only PL/SQL identifiers to be collected:

```
PLSCOPE_SETTINGS = 'IDENTIFIERS:PLSQL, STATEMENTS:NONE'
```

The following parameter setting causes no identifier data to be collected, and drops any existing identifier data:

```
PLSCOPE_SETTINGS = 'IDENTIFIERS: NONE'
```

See Also

- "ALL_PLSQL_OBJECT_SETTINGS," "DBA_PLSQL_OBJECT_SETTINGS," and "USER_PLSQL_OBJECT_SETTINGS"
- "ALL_IDENTIFIERS,""DBA_IDENTIFIERS," and "USER_IDENTIFIERS"
- *Oracle Database PL/SQL Language Reference* for more information about this parameter

1.287 PLSQL_CCFLAGS

PLSQL_CCFLAGS provides a mechanism that allows PL/SQL programmers to control conditional compilation of each PL/SQL library unit independently.

Property	Description
Parameter type	String
Syntax	PLSQL_CCFLAGS = '<*v1*>:<*c1*>,<*v2*>:<*c2*>,...,<*vn*>:<*cn*>'
Default value	Empty string
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Range of values	Any string literal that satisfies the internal syntax
Basic	No
Examples	`ALTER SESSION SET PLSQL_CCFLAGS = 'DeBug:TruE';` `ALTER SESSION SET PLSQL_CCFLAGS = 'debug:TRUE';`

Values

- <*vi*> has the form of an unquoted PL/SQL identifier. It is unrestricted and can be a reserved word or a keyword. The text is insensitive to case. Each one is known as a flag or flag name. Each <*vi*> can occur more than once in the string, each occurrence can have a different flag value, and the flag values can be of different kinds.
- <*ci*> is one of the following: a PL/SQL boolean literal, a `PLS_INTEGER` literal, or the literal NULL. The text is insensitive to case. Each one is known as a flag value and corresponds to a flag name.

You can define any allowable value for `PLSQL_CCFLAGS`. However, Oracle recommends that this parameter be used for controlling the conditional compilation of debugging or tracing code. It is recommended that the following identifiers not be used as flag name values:

- Names of Oracle parameters (for example, `NLS_LENGTH_SEMANTICS`)
- Identifiers with any of the following prefixes: `PLS_`, `PLSQL_`, `PLSCC_`, `ORA_`, `ORACLE_`, `DBMS_`, `SYS_`

> **See Also**
>
> *Oracle Database PL/SQL Language Reference* for more information about this parameter

1.288 PLSQL_CODE_TYPE

`PLSQL_CODE_TYPE` specifies the compilation mode for PL/SQL library units.

Property	Description
Parameter type	String
Syntax	`PLSQL_CODE_TYPE = { INTERPRETED | NATIVE }`
Default value	`INTERPRETED`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

Values

- `INTERPRETED`

  PL/SQL library units will be compiled to PL/SQL bytecode format. Such modules are executed by the PL/SQL interpreter engine.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- `NATIVE`

  PL/SQL library units (with the possible exception of top-level anonymous PL/SQL blocks) will be compiled to native (machine) code. Such modules will be executed natively without incurring any interpreter overhead.

When the value of this parameter is changed, it has no effect on PL/SQL library units that have already been compiled. The value of this parameter is stored persistently with each library unit.

If a PL/SQL library unit is compiled native, all subsequent automatic recompilations of that library unit will use native compilation.

> **See Also**
>
> *Oracle Database PL/SQL Language Reference* for more information about this parameter

1.289 PLSQL_DEBUG

`PLSQL_DEBUG` specifies whether or not PL/SQL library units will be compiled for debugging.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No

> **Note**
>
> The `PLSQL_DEBUG` parameter is deprecated. It is retained for backward compatibility only.

Values

- `true`

  PL/SQL library units will be compiled for debugging

- `false`

  PL/SQL library units will be compiled for normal execution

When `PLSQL_DEBUG` is set to `true`, PL/SQL library units are always compiled `INTERPRETED` in order to be debuggable.

When the value of this parameter is changed, it has no effect on PL/SQL library units that have already been compiled. The value of this parameter is stored persistently with each library unit.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.290 PLSQL_OPTIMIZE_LEVEL

PLSQL_OPTIMIZE_LEVEL specifies the optimization level that will be used to compile PL/SQL library units. The higher the setting of this parameter, the more effort the compiler makes to optimize PL/SQL library units.

Property	Description
Parameter type	Integer
Default value	2
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to 3
Basic	No

Values

- 0

  Maintains the evaluation order and hence the pattern of side effects, exceptions, and package initializations of Oracle9*i* and earlier releases. Also removes the new semantic identity of BINARY_INTEGER and PLS_INTEGER and restores the earlier rules for the evaluation of integer expressions. Although code will run somewhat faster than it did in Oracle9*i*, use of level 0 will forfeit most of the performance gains of PL/SQL in Oracle Database 10*g*.

- 1

  Applies a wide range of optimizations to PL/SQL programs including the elimination of unnecessary computations and exceptions, but generally does not move source code out of its original source order.

- 2

  Applies a wide range of modern optimization techniques beyond those of level 1 including changes which may move source code relatively far from its original location.

- 3

  Applies a wide range of optimization techniques beyond those of level 2, automatically including techniques not specifically requested.

Generally, setting this parameter to 2 pays off in better execution performance. If, however, the compiler runs slowly on a particular source module or if optimization does not make sense for some reason (for example, during rapid turnaround development), then setting this parameter to 1 will result in almost as good a compilation with less use of compile-time resources.

The value of this parameter is stored persistently with the library unit.

See Also

- *Oracle Database PL/SQL Language Reference* for more information about this parameter
- *Oracle Database Development Guide* for an example of using this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.291 PLSQL_V2_COMPATIBILITY

`PLSQL_V2_COMPATIBILITY` is used to specify whether nonstandard behavior that PL/SQL Version 2 allows will be allowed in PL/SQL Version 8.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true	false`
Basic	No

> **Note**
>
> The `PLSQL_V2_COMPATIBILITY` parameter is deprecated. It is retained for backward compatibility only.

PL/SQL Version 2 allows some nonstandard behavior that Version 8 disallows. If you want to retain that behavior for backward compatibility, set `PLSQL_V2_COMPATIBILITY` to `true`. If you set it to `false`, then PL/SQL Version 8 behavior is enforced and Version 2 behavior is not allowed.

> **See Also**
>
> *Oracle Database PL/SQL Language Reference* for a description of the differences between PL/SQL Version 2 and Version 8, and for more information on setting this parameter

1.292 PLSQL_WARNINGS

`PLSQL_WARNINGS` enables or disables the reporting of warning messages by the PL/SQL compiler, and specifies which warning messages to show as errors.

Property	Description
Parameter type	String
Syntax	`PLSQL_WARNINGS = '`*`value_clause`*`' [, '`*`value_clause`*`' ] ...`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Syntax	**value_clause::=** { ENABLE | DISABLE | ERROR }: { ALL | SEVERE | INFORMATIONAL | PERFORMANCE | { *integer* | (*integer* [, *integer*] ...) } }
Default value	'DISABLE:ALL'
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No
Examples	PLSQL_WARNINGS = 'ENABLE:SEVERE', 'DISABLE:INFORMATIONAL'; PLSQL_WARNINGS = 'DISABLE:ALL'; PLSQL_WARNINGS = 'DISABLE:5000', 'ENABLE:5001', 'ERROR:5002'; PLSQL_WARNINGS = 'ENABLE:(5000,5001,5002)', 'DISABLE:(6000,6001)';

value_clause

Multiple value clauses may be specified, enclosed in quotes and separated by commas. Each value clause is composed of a qualifier, a colon (:), and a modifier.

Qualifier values:

- ENABLE

  Enable a specific warning or a set of warnings
- DISABLE

  Disable a specific warning or a set of warnings
- ERROR

  Treat a specific warning or a set of warnings as errors

Modifier values:

- ALL

  Apply the qualifier to all warning messages
- SEVERE

  Apply the qualifier to only those warning messages in the SEVERE category
- INFORMATIONAL

  Apply the qualifier to only those warning messages in the INFORMATIONAL category
- PERFORMANCE

  Apply the qualifier to only those warning messages in the PERFORMANCE category

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database PL/SQL Language Reference* for more information about this parameter

1.293 PRE_PAGE_SGA

`PRE_PAGE_SGA` determines whether Oracle accesses the entire SGA at instance startup.

Operating system page table entries are then prebuilt for each page of the SGA.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	No
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

When this parameter is set to `true`, the database server touches the entire SGA after the instance startup. These accesses are done by a background process and are initiated after the instance startup. The accesses cause the OS to allocate memory to the SGA and populate the page tables, making the SGA memory resident and increasing database performance.

> **Note**
>
> This setting does not prevent your operating system from paging or swapping the SGA after it is initially read into memory.

> **Note**
>
> The Oracle Database design for SGA packaging has evolved over time, so that process start-up effects on the SGA are initiated after instance startup, and there is little to no benefit in changing the value for `PRE_PAGE_SGA`. The only use case for this parameter is on Oracle Exadata systems, which should have `PRE_PAGE_SGA` set to `TRUE`. Therefore, starting with Oracle Database 19c, Release Update 19.22, the value of this parameter is always set to `TRUE` on Exadata systems. Any attempt to set this parameter to `FALSE` on an Exadata system will not take effect.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The PRE_PAGE_SGA parameter was deprecated in Oracle Database 19c, Release Update 19.22. However, this deprecation was rescinded in Oracle Database 19c, Release Update 19.30, and PRE_PAGE_SGA remains a supported initialization parameter. Removing its deprecated flag keeps automated duplicate and standby creation workflows from failing when the parameter is inherited from a source SPFILE.
>
> Restoring full support for PRE_PAGE_SGA lets you keep legacy configurations in place without disrupting DBCA duplicate, Data Guard standby, or RMAN clone operations, eliminating ORA-32006 interruptions for fleets that cannot immediately drop the parameter.

1.294 PRIVATE_TEMP_TABLE_PREFIX

PRIVATE_TEMP_TABLE_PREFIX specifies the prefix that the database uses for private temporary tables.

Property	Description
Parameter type	String
Syntax	PRIVATE_TEMP_TABLE_PREFIX = *string*
Default value	ORA$PTT_
Modifiable	ALTER SYSTEM ... DEFERRED
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The same value must be specified on each instance.

The default value used for the PRIVATE_TEMP_TABLE_PREFIX parameter is ORA$PTT_.

If you choose to specify a different prefix value, it must begin with the string ORA$ and must be unique across the database. Use the following SQL query to determine if the prefix value is unique (in the query, replace prefix-value with your actual prefix value):

```
select count(*) from obj$ where name like '<prefix-value>%';
```

If the above query returns 0, then the prefix you specified is an appropriate value to set.

> **Note**
>
> After setting the prefix, regular table, view, and object names cannot use the same prefix.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_PRIVATE_TEMP_TABLES"
- "USER_PRIVATE_TEMP_TABLES"
- *Oracle Database Administrator's Guide* for an introduction to private temporary tables

1.295 PROCESSES

`PROCESSES` specifies the maximum number of operating system user processes that can simultaneously connect to Oracle.

Property	Description
Parameter type	Integer
Default value	The value is derived, and it typically depends on the number of cores reported in the alert log.
Modifiable	No
Modifiable in a PDB	No
Range of values	40 to operating system dependent
Basic	Yes
Oracle RAC	Multiple instances can have different values.

The value for this parameter should allow for all background processes such as locks, job queue processes, and parallel execution processes.

The default values of the `SESSIONS` and `TRANSACTIONS` parameters are derived from this parameter. Therefore, if you change the value of `PROCESSES`, you should evaluate whether to adjust the values of those derived parameters.

See Also

- Your operating system-specific Oracle documentation for the range of values
- *Oracle Database Concepts* for an introduction to Oracle database instance processes

1.296 PROCESSOR_GROUP_NAME

`PROCESSOR_GROUP_NAME` specifies the name of the processor group that this instance should run in.

Property	Description
Parameter type	String
Syntax	`PROCESSOR_GROUP_NAME = string`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	There is no default value
Modifiable	No
Modifiable in a PDB	No
Basic	No
Oracle RAC	For Oracle RAC databases, it is recommended that the operating system processor groups for each database instance are all named the same, so that the same PROCESSOR_GROUP_NAME value can be used. However, you can choose to use different names. In this case, you would set the PROCESSOR_GROUP_NAME explicitly for each database instance.

This initialization parameter is supported on the Linux and Solaris operating systems.

PROCESSOR_GROUP_NAME instructs the database instance to run itself within the specified operating system processor group. All Oracle processes will be bound to the CPUs in this group and will only run on these CPUs.

Notes

- Oracle recommends that this parameter be set only for databases on which the USE_DEDICATED_BROKER initialization parameter is also set to TRUE.
- For NUMA systems, all SGA and PGA memory allocated by the database instance will be allocated from NUMA nodes within the group.
- Starting with Oracle Database Appliance (ODA) release 19.9, you can use PROCESSOR_GROUP_NAME to specify the set of CPU cores on which the database on ODA will run. For more information, refer to *Oracle Database Appliance X8-2 Deployment and User's Guide*.

See Also

- "USE_DEDICATED_BROKER"
- *Oracle Database Administrator's Reference for Linux and UNIX-Based Operating Systems* for more information about using the PROCESSOR_GROUP_NAME initialization parameter

1.297 QUERY_REWRITE_ENABLED

Use QUERY_REWRITE_ENABLED to enable or disable query rewriting globally for the database.

Property	Description
Parameter type	String
Syntax	QUERY_REWRITE_ENABLED = { false | true | force }
Default value	If OPTIMIZER_FEATURES_ENABLE is set to 10.0.0 or higher, then true If OPTIMIZER_FEATURES_ENABLE is set to 9.2.0 or lower, then false

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Multiple instances can have different values.

Values

- `false`

  Oracle does not use rewrite.

- `true`

  Oracle costs the query with rewrite and without rewrite and chooses the method with the lower cost.

- `force`

  Oracle always uses rewrite and does not evaluate the cost before doing so. Use `force` when you know that the query will always benefit from rewrite and when reduction in compile time is important.

To take advantage of query rewrite for a particular materialized view, you must enable query rewrite for that materialized view, and you must enable cost-based optimization.

See Also

- *Oracle Database Data Warehousing Guide* for information on query rewrite of materialized views
- *Oracle Database SQL Tuning Guide* and "OPTIMIZER_MODE" for information on cost-based optimization

1.298 QUERY_REWRITE_INTEGRITY

`QUERY_REWRITE_INTEGRITY` determines the degree to which Oracle must enforce query rewriting. At the safest level, Oracle does not use query rewrite transformations that rely on unenforced relationships.

Property	Description
Parameter type	String
Syntax	`QUERY_REWRITE_INTEGRITY = { enforced | trusted | stale_tolerated }`
Default value	`enforced`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Multiple instances can have different values.

Copyright © 2002, 2026, Oracle and/or its affiliates.

QUERY_REWRITE_INTEGRITY is relevant for materialized views as well as for foreign key constraints in NOVALIDATE state.

Values

- enforced

  Oracle enforces and guarantees consistency and integrity.

- trusted

  Oracle allows rewrites using relationships that have been declared, but that are not enforced by Oracle.

- stale_tolerated

  Oracle allows rewrites using unenforced relationships. Materialized views are eligible for rewrite even if they are known to be inconsistent with the underlying detail data. You must set the QUERY_REWRITE_INTEGRITY initialization parameter to stale_tolerated before querying an external table in the In-Memory Column Store (IM column store).

If a foreign key constraint is in NOVALIDATE state, join elimination is not done when QUERY_REWRITE_INTEGRITY=enforced. This means that queries with joins over a foreign key constraint that is in RELY NOVALIDATE state can potentially take longer to parse and execute as the optimizer does not trust the RELY.

See Also

- "INMEMORY_CLAUSE_DEFAULT"
- *Oracle Database Data Warehousing Guide* for more information about query rewrite for materialized views
- *Oracle Database Data Warehousing Guide* for more information about the QUERY_REWRITE_INTEGRITY parameter

1.299 RDBMS_SERVER_DN

RDBMS_SERVER_DN specifies the Distinguished Name (DN) of the Oracle server. It is used for retrieving Enterprise Roles from an enterprise directory service.

Property	Description
Parameter type	X.500 Distinguished Name
Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	No
Range of values	All X.500 Distinguished Name format values
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The `RDBMS_SERVER_DN` parameter is deprecated in Oracle Database 12*c* Release 1 (12.1.0.2). It is replaced by the `LDAP_DIRECTORY_ACCESS` initialization parameter.

If you do not want to use a directory for enterprise user and privilege management, but prefer to use SSL authentication alone, do not set this parameter.

> **See Also**
>
> - *Oracle Database Enterprise User Security Administrator's Guide* for more information on enterprise roles and the enterprise directory service
> - "LDAP_DIRECTORY_ACCESS"

1.300 READ_ONLY_OPEN_DELAYED

`READ_ONLY_OPEN_DELAYED` determines when datafiles in read-only tablespaces are accessed.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	No
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

Values

- `true`

  The datafiles are accessed for the first time only when an attempt is made to read data stored within them.

- `false`

  The datafiles are accessed at database open time.

You can use this parameter to speed up some operations (primarily opening the database) for very large databases when substantial portions of the database are stored in read-only tablespaces. Consider setting this parameter to `true` for such databases, especially if portions of the read-only data are stored on slow-access devices or hierarchical storage.

> **See Also**
>
> *Oracle Database Administrator's Guide* for information on the consequences of delaying access of datafiles in read-only tablespaces

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.301 RECOVERY_PARALLELISM

RECOVERY_PARALLELISM specifies the number of processes to participate in instance or crash recovery.

Property	Description
Parameter type	Integer
Default value	System-determined parallel recovery
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	System-determined, but cannot exceed PARALLEL_MAX_SERVERS initialization parameter setting
Basic	No

To force serial crash and instance recovery, set the RECOVERY_PARALLELISM parameter to 0 or 1. 0 or 1 disable parallel instance and crash recovery on a system that has multiple CPUs. An alert log displays the degree of parallelism that was chosen when the database instance/ recovery starts.

See Also

- *Oracle Real Application Clusters Administration and Deployment Guide* for information on setting this parameter in an Oracle RAC environment

1.302 RECYCLEBIN

RECYCLEBIN is used to control whether the Flashback Drop capability is turned on or off.

Property	Description
Parameter type	String
Syntax	RECYCLEBIN = { on | off }
Default value	on
Modifiable	ALTER SESSION, ALTER SYSTEM ... DEFERRED
Modifiable in a PDB	Yes
Basic	No

If the parameter is set to off, then dropped tables do not go into the recycle bin. If this parameter is set to on, then dropped tables go into the recycle bin and can be recovered.

See Also

Oracle Database Administrator's Guide for more information about using Flashback Drop and managing the recycle bin

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.303 REDO_TRANSPORT_USER

REDO_TRANSPORT_USER specifies the name of the user whose password verifier is used when a remote login password file is used for redo transport authentication. This user must have the SYSOPER privilege and must have the same password in the database that initiates the redo transport session and in the database that is the target of the redo transport session.

Property	Description
Parameter type	String
Syntax	REDO_TRANSPORT_USER = *user_name*
Default value	There is no default value.
Range of values	Any character string that matches the name of a user who has been granted the SYSOPER privilege
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No
Oracle RAC	Every instance should use the same value

The value of this parameter is case sensitive and must exactly match the value of the USERNAME column of a row in the V$PWFILE_USERS view. The value of the SYSOPER column of the row must also be TRUE.

If this parameter is not specified, then the password verifier of the SYS user will be used when a remote login password file is used for redo transport authentication.

See Also

Oracle Data Guard Concepts and Administration for more information about this parameter

1.304 REMOTE_DEPENDENCIES_MODE

REMOTE_DEPENDENCIES_MODE specifies how Oracle should handle dependencies upon remote PL/SQL stored procedures.

Property	Description
Parameter type	String
Syntax	REMOTE_DEPENDENCIES_MODE = { TIMESTAMP | SIGNATURE }
Default value	TIMESTAMP
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Values

- `TIMESTAMP`

  The client running the procedure compares the timestamp recorded on the server-side procedure with the current timestamp of the local procedure and executes the procedure only if the timestamps match.

- `SIGNATURE`

  Oracle allows the procedure to execute as long as the signatures are considered safe. This setting allows client PL/SQL applications to be run without recompilation.

> **See Also**
>
> *Oracle Database Development Guide* for suggestions for managing dependencies

1.305 REMOTE_LISTENER

`REMOTE_LISTENER` specifies a network name that resolves to an address or address list of Oracle Net remote listeners (that is, listeners that are not running on the same system as this instance). The address or address list is specified in the `TNSNAMES.ORA` file or other address repository as configured for your system.

Property	Description
Parameter type	String
Syntax	`REMOTE_LISTENER = network_name`
Default value	There is no default value.
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	Yes

> **See Also**
>
> - *Oracle Database Concepts* for more information about listener processes and dispatcher processes
> - *Oracle Database Net Services Administrator's Guide* and your operating system-specific Oracle documentation for more information about specifying network addresses for the protocols on your system
> - *Oracle Clusterware Administration and Deployment Guide* for information about SCAN addresses

1.306 REMOTE_LOGIN_PASSWORDFILE

`REMOTE_LOGIN_PASSWORDFILE` specifies whether Oracle checks for a password file.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	String
Syntax	`REMOTE_LOGIN_PASSWORDFILE = { SHARED | EXCLUSIVE | NONE }`
Default value	`EXCLUSIVE`
Modifiable	No
Modifiable in a PDB	No
Basic	Yes
Oracle RAC	Multiple instances must have the same value.

Values

- `SHARED`

  One or more databases can use the password file. The password file can contain `SYS` and non-`SYS` users.

  When `REMOTE_LOGIN_PASSWORDFILE` is set to `SHARED`:

  – The database disables modification of the password file by an ASM or RDBMS instance.

  – If you attempt to change the password of a user that is in the password file, the operation fails with "ORA-01999: password file cannot be updated in SHARED mode.

  – If you attempt to change the password of an administrative user (except SYS) that is *not* in the password file, the password change operation will succeed, but the new password will not go into effect. This is because the password is updated only in the data dictionary and not in the password file.

  – Grants and revokes of administrative privileges (`SYSDBA`, `SYSOPER`, `SYSASM`, `SYSBACKUP`, `SYSDG`, `SYSKM`) to or from individual users are not allowed. For example, `GRANT SYSDBA TO SCOTT` fails with "ORA-01999: password file cannot be updated in SHARED mode."

  – If the password file does not exist, then the behavior is the same as setting `REMOTE_LOGIN_PASSWORDFILE` to `NONE`.

- `EXCLUSIVE`

  The password file is intended for use by ASM instances in the same cluster or by one or more instances of the same Oracle database.

  The password file can contain `SYS` and non-`SYS` users.

  If the password file does not exist, then the behavior is the same as setting `REMOTE_LOGIN_PASSWORDFILE` to `NONE`.

- `NONE`

  ASM and RDBMS instances ignore the password file. Therefore, privileged users must be authenticated by the operating system.

> **Note**
>
> If you change `REMOTE_LOGIN_PASSWORDFILE` to `EXCLUSIVE` or `SHARED` from `NONE`, then ensure that the password file is synchronized with the dictionary passwords.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.307 REMOTE_OS_AUTHENT

`REMOTE_OS_AUTHENT` specifies whether remote clients will be authenticated with the value of the `OS_AUTHENT_PREFIX` parameter.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	No
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

> **Note**
>
> The `REMOTE_OS_AUTHENT` parameter is deprecated. It is retained for backward compatibility only.

> **See Also**
>
> "OS_AUTHENT_PREFIX"

1.308 REMOTE_OS_ROLES

`REMOTE_OS_ROLES` specifies whether operating system roles are allowed for remote clients.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	No
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

The default value, `false`, causes Oracle to identify and manage roles for remote clients.

> **See Also**
>
> - *Oracle Database Security Guide* for more information on setting this parameter
> - "OS_ROLES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.309 REMOTE_RECOVERY_FILE_DEST

`REMOTE_RECOVERY_FILE_DEST` specifies a directory from which to read archive log files during a pluggable database (PDB) refresh operation if the source is not available.

Property	Description
Parameter type	String
Syntax	`REMOTE_RECOVERY_FILE_DEST = string`
Default value	None
Modifiable	`ALTER SYSTEM ... SID='*'`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The directory location where archive redo logs are accessible can be different on different instances, so this parameter can be set to different values on different instances.

For a PDB refresh copy to be in sync with its source, redo is accessed from the source PDB over a database link. Sometimes the source PDB or the CDB to which the source PDB belongs is not accessible when the refresh copy needs to be updated. In those cases, if this parameter is set, an attempt will be made to read archive log files from the directory specified by this parameter.

1.310 REPLICATION_DEPENDENCY_TRACKING

`REPLICATION_DEPENDENCY_TRACKING` enables or disables dependency tracking for read/write operations to the database.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	No
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No

Dependency tracking is essential for propagating changes in a replicated environment in parallel.

Values

- `TRUE`

  Enables dependency tracking.

- `FALSE`

  Allows read/write operations to the database to run faster, but does not produce dependency information for Oracle to perform parallel propagation.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> Do not specify this value unless you are sure that your application will not perform any read/write operations to the replicated tables.

1.311 RESOURCE_LIMIT

`RESOURCE_LIMIT` determines whether resource limits are enforced in database profiles.

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No

Values

- `TRUE`

  Enables the enforcement of resource limits
- `FALSE`

  Disables the enforcement of resource limits

> **See Also**
>
> *Oracle Database Administrator's Guide* and *Oracle Database SQL Language Reference* for more information on setting resource limits for profiles

1.312 RESOURCE_MANAGE_GOLDENGATE

`RESOURCE_MANAGE_GOLDENGATE` determines whether Oracle GoldenGate apply processes in the database are resource managed.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No
Oracle RAC	All instances should use the same value.

Copyright © 2002, 2026, Oracle and/or its affiliates.

To enable Resource Manager, set the RESOURCE_MANAGER_PLAN parameter.

By default, Oracle GoldenGate apply processes in the database are not resource managed. Given that replication to a PDB requires a separate Oracle GoldenGate apply process, it is possible that the apply processes for one PDB could end up consuming most of the CPU on the machine, even if there is a CPU resource management plan in place to limit CPU usage per PDB.

You can set the following values for the RESOURCE_MANAGE_GOLDENGATE parameter:

- TRUE: With this setting, Oracle GoldenGate apply processes in the database are resource managed based on the resources allocated to the GoldenGate apply user.
- FALSE: With this setting, Oracle GoldenGate apply processes are not resource managed.

1.313 RESOURCE_MANAGER_CPU_ALLOCATION

RESOURCE_MANAGER_CPU_ALLOCATION specifies the number of CPUs that the Resource Manager should use. The Resource Manager controls how a system's CPUs are utilized by its database's sessions.

Note

The RESOURCE_MANAGER_CPU_ALLOCATION parameter is deprecated and is retained only for backward compatibility. Furthermore, starting with Oracle Database 19c, Release Update 19.10, the value of this parameter is ignored and its default value is set to 0. Use the CPU_COUNT parameter instead.

Property	Description
Parameter type	Integer
Default value	The number of logical CPUs reported by the operating system.
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	0 to operating system-specific
Basic	No

The Resource Manager schedules database sessions on the CPUs according to a resource plan that has been configured and enabled by the DBA. Normally, the Resource Manager schedules enough database sessions to keep all CPUs utilized. However, in some scenarios, a DBA may only want to schedule enough database sessions to keep a subset of the CPUs utilized.

See Also

"CPU_COUNT"

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.314 RESOURCE_MANAGER_PLAN

RESOURCE_MANAGER_PLAN specifies the resource plan to use for a database (CDB or non-CDB).

Property	Description
Parameter type	String
Syntax	RESOURCE_MANAGER_PLAN = *resource_plan_name*
Default value	There is no default value.
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	Any valid character string
Basic	No

In a CDB

In the root for a CDB, RESOURCE_MANAGER_PLAN specifies the CDB resource plan. A CDB resource plan allocates resources among PDBs.

A CDB resource plan is created using DBMS_RESOURCE_MANAGER.CREATE_CDB_PLAN and CREATE_CDB_PLAN_DIRECTIVE.

> **See Also**
>
> - *Oracle Multitenant Administrator's Guide* for more information about using DBMS_RESOURCE_MANAGER.CREATE_CDB_PLAN and DBMS_RESOURCE_MANAGER.CREATE_CDB_PLAN_DIRECTIVE
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the CREATE_CDB_PLAN procedure and CREATE_CDB_PLAN_DIRECTIVE procedure for the DBMS_RESOURCE_MANAGER package

A session must be root to change the value of RESOURCE_MANAGER_PLAN for a CDB using the ALTER SYSTEM statement. For example, to enable and disable a CDB resource plan:

```
SQL> ALTER SYSTEM SET RESOURCE_MANAGER_PLAN = CDB_resource_plan_name;
SQL> ALTER SYSTEM SET RESOURCE_MANAGER_PLAN = '';
```

In a PDB, RESOURCE_MANAGER_PLAN specifies the PDB resource plan to use for the PDB.

A session must be in the PDB to enable or disable a PDB resource plan for that PDB. For example, to enable and disable a PDB resource plan:

```
SQL> ALTER SYSTEM SET RESOURCE_MANAGER_PLAN = PDB_resource_plan_name;
SQL> ALTER SYSTEM SET RESOURCE_MANAGER_PLAN = '';
```

In a PDB, the PDB resource plan has some restrictions compared to the resource plan of a non-CDB. The following restrictions apply to PDB resource plans:

- A PDB resource plan cannot have subplans.
- A PDB resource plan cannot have a multiple-level scheduling policy.

Copyright © 2002, 2026, Oracle and/or its affiliates.

To enforce certain PDB resource plan policies, policies regarding resource allocation among PDBs should exist in the CDB resource plan. Without a CDB resource plan, certain PDB resource plan policies will not be enforced. If a PDB resource plan contains CPU or parallel statement queuing directives and a CDB resource plan is not specified, then Resource Manager will automatically enable the `DEFAULT_CDB_PLAN` plan. To prevent this behavior, set the `RESOURCE_MANAGER_PLAN` parameter at the root level to `ORA$INTERNAL_CDB_PLAN`.

> **Note**
>
> See *Oracle Multitenant Administrator's Guide* for information on CPU, I/O bandwidth, and parallel execution servers requirements in CDB resource plans, and for a description of the results in the PDB resource plans when those requirements are not met.

In a Non-CDB

`RESOURCE_MANAGER_PLAN` specifies the top-level resource plan to use for an instance in a non-CDB. The resource manager will load this top-level resource plan along with all its descendants (subplans, directives, and consumer groups). If you do not specify this parameter, the resource manager is off by default.

You can change the setting of this parameter using the `ALTER SYSTEM` statement to turn on the resource manager (if it was previously off) or to turn off the resource manager or change the current resource plan (if it was previously on). If you specify a resource plan that does not exist in the data dictionary, Oracle returns an error message.

> **See Also**
>
> - *Oracle Database Administrator's Guide* for information on resource plans
> - *Oracle Multitenant Administrator's Guide* for information on using Oracle Resource Manager with a CDB and PDBs
> - *Oracle Database PL/SQL Packages and Types Reference* for information on the `DBMS_RESOURCE_MANAGER` package
> - *Oracle Database PL/SQL Packages and Types Reference* for information on the `DBMS_RESOURCE_MANAGER_PRIVS` package
> - "DBA_RSRC_PLANS", "DBA_RSRC_PLAN_DIRECTIVES", and the various `V$RSRC_*` dynamic performance views in Dynamic Performance Views for information on existing resource plans

1.315 RESULT_CACHE_MAX_RESULT

`RESULT_CACHE_MAX_RESULT` specifies the percentage of `RESULT_CACHE_MAX_SIZE` that any single result can use.

Property	Description
Parameter type	Integer
Default value	5 percent

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to 100
Basic	No
Oracle RAC	Multiple instances can have different values.

1.316 RESULT_CACHE_MAX_SIZE

RESULT_CACHE_MAX_SIZE specifies the maximum amount of SGA memory (in bytes) that can be used by the Result Cache.

Property	Description
Parameter type	Big integer
Syntax	RESULT_CACHE_MAX_SIZE = *integer* [K | M | G]
Default value	Derived from the values of SHARED_POOL_SIZE, SGA_TARGET, and MEMORY_TARGET
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to operating system-dependent
Basic	No
Oracle RAC	You must either set this parameter to 0 on all instances to disable the result cache, or use a nonzero value on all instances. Disabling the result cache on some instances may lead to incorrect results.

Values of this parameter greater than 0 are rounded up to the next multiple of 32 KB. If the value of this parameter is 0, then the feature is disabled.

See Also

Oracle Database Performance Tuning Guide for information about tuning the result cache

1.317 RESULT_CACHE_MODE

RESULT_CACHE_MODE specifies when a ResultCache operator is spliced into a query's execution plan.

Property	Description
Parameter type	String
Syntax	RESULT_CACHE_MODE = { MANUAL | FORCE }
Default value	MANUAL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Multiple instances can have different values.

Values

- `MANUAL`

  The `ResultCache` operator is added only when the query is annotated (that is, hints).

- `FORCE`

  The `ResultCache` operator is added to the root of all `SELECT` statements (when it is valid to do so).

  For the `FORCE` setting, if the statement contains a `NO_RESULT_CACHE` hint, then the hint takes precedence over the parameter setting.

> **Note**
>
> `FORCE` mode is not generally recommended because queries that call non-deterministic PL/SQL functions are cached, potentially causing material changes to the results. Enabling the result cache in such a broad-based manner can also impact performance in mixed workload environments, where the same tables are continuously updated and retrieved.

> **See Also**
>
> - *Oracle Database SQL Tuning Guide* for more information on how the result cache handles PL/SQL functions before changing the value of this initialization parameter
> - *Oracle Database SQL Language Reference* for more information about the NO_RESULT_CACHE hint

1.318 RESULT_CACHE_REMOTE_EXPIRATION

`RESULT_CACHE_REMOTE_EXPIRATION` specifies the number of minutes that a result using a remote object is allowed to remain valid.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`0` to operating system-dependent

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Basic	No

Setting this parameter to `0` implies that results using remote objects should not be cached. Setting this parameter to a nonzero value can produce stale answers (for example, if the remote table used by a result is modified at the remote database).

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for information about tuning the result cache

1.319 RESUMABLE_TIMEOUT

`RESUMABLE_TIMEOUT` enables or disables resumable statements and specifies resumable timeout at the system level.

Property	Description
Parameter type	Integer
Default value	`0` (seconds)
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`0` to 2^{31} - 1 (in seconds)
Basic	No
Oracle RAC	Multiple instances can have different values.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information about enabling resumable space allocation, what conditions are correctable, and what statements can be made resumable

1.320 ROLLBACK_SEGMENTS

`ROLLBACK_SEGMENTS` allocates one or more rollback segments by name to this instance.

Property	Description
Parameter type	String
Syntax	`ROLLBACK_SEGMENTS =` `(segment_name [, segment_name] ... )`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	If you do not specify this parameter, the instance uses public rollback segments by default, unless the UNDO_MANAGEMENT initialization parameter is set to AUTO. In that case, the ROLLBACK_SEGMENTS parameter is ignored and automatic undo management is used.
Modifiable	No
Modifiable in a PDB	Yes
Range of values	Any rollback segment names listed in DBA_ROLLBACK_SEGS except SYSTEM
Basic	No
Oracle RAC	Multiple instances must have different values.

If you set this parameter, the instance acquires all of the rollback segments named in this parameter, even if the number of rollback segments exceeds the minimum number required by the instance (calculated as TRANSACTIONS / TRANSACTIONS_PER_ROLLBACK_SEGMENT).

You cannot change the value of this parameter dynamically, but you can change its value and then restart the instance. Although this parameter usually specifies private rollback segments, it can also specify public rollback segments if they are not already in use.

To find the name, segment ID number, and status of each rollback segment in the database, query the data dictionary view DBA_ROLLBACK_SEGS.

When UNDO_MANAGEMENT is set to AUTO, ROLLBACK_SEGMENTS is ignored.

See Also

- *Oracle Real Application Clusters Administration and Deployment Guide* for information on setting this parameter in an Oracle RAC environment
- "DBA_ROLLBACK_SEGS"

1.321 SEC_CASE_SENSITIVE_LOGON

SEC_CASE_SENSITIVE_LOGON enables or disables password case sensitivity in the database.

Property	Description
Parameter type	Boolean
Default value	true
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	true | false
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The `SEC_CASE_SENSITIVE_LOGON` parameter is deprecated. It is retained for backward compatibility only. For more information about the deprecation of this parameter, see *Oracle Database Security Guide*.

Values

- `true`

  Database logon passwords are case sensitive.
- `false`

  Database logon passwords are not case sensitive.

> **See Also**
>
> *Oracle Database Security Guide* for more details about the SEC_CASE_SENSITIVE_LOGON parameter

1.322 SEC_MAX_FAILED_LOGIN_ATTEMPTS

`SEC_MAX_FAILED_LOGIN_ATTEMPTS` specifies the number of authentication attempts that can be made by a client on a connection to the server process.

Property	Description
Parameter type	Integer
Default value	3
Modifiable	No
Modifiable in a PDB	No
Range of values	An integer greater than or equal to 1.
Basic	No

These login attempts can be for multiple user accounts in the same connection. After the specified number of failure attempts, the connection will be automatically dropped by the server process, and the server process is terminated.

> **See Also**
>
> *Oracle Database Security Guide* for more information about this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.323 SEC_PROTOCOL_ERROR_FURTHER_ACTION

`SEC_PROTOCOL_ERROR_FURTHER_ACTION` specifies the further execution of a server process when receiving bad packets from a possibly malicious client.

Property	Description
Parameter type	String
Syntax	`SEC_PROTOCOL_ERROR_FURTHER_ACTION = { CONTINUE	(DELAY,`*`integer`*`)	(DROP,`*`integer`*`) }`
Default value	`(DROP,3)`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Basic	No

Values

- `CONTINUE`

  The server process continues execution. The database server may be subject to a Denial of Service (DoS) if bad packets continue to be sent by a malicious client.

- `(DELAY,`*`integer`*`)`

  The client experiences a delay of *integer* seconds before the server process accepts the next request from the same client connection. Malicious clients are prevented from excessive consumption of server resources while legitimate clients experience a degradation in performance but can continue to function.

- `(DROP,`*`integer`*`)`

  The server forcefully terminates the client connection after *integer* cumulative bad packets. The server protects itself at the expense of the client (for example, a client transaction may be lost). The client may reconnect and attempt the same operation.

> **See Also**
>
> *Oracle Database Security Guide* for examples of using this parameter

1.324 SEC_PROTOCOL_ERROR_TRACE_ACTION

`SEC_PROTOCOL_ERROR_TRACE_ACTION` specifies the kind of logging the database server does when bad packets are received from a possibly malicious client, apart from the client receiving the error.

Property	Description
Parameter type	String
Syntax	`SEC_PROTOCOL_ERROR_TRACE_ACTION = { NONE	TRACE	LOG	ALERT }`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Default value	TRACE
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No

Values

- NONE

  The database server does not record it in any of the trace files.

- TRACE

  A short entry is made in the alert log file and a detailed incident file is generated. The server trace file also will have an entry about the protocol error and incident file.

- LOG

  The server trace file will have an entry about the protocol violation.

- ALERT

  A short entry is made in the alert log file and in the server trace file about the protocol violation.

1.325 SEC_RETURN_SERVER_RELEASE_BANNER

SEC_RETURN_SERVER_RELEASE_BANNER specifies whether the server returns complete database software information to unauthenticated clients.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	No
Modifiable in a PDB	No
Range of values	true | false
Basic	No

Values

- true

  Returns complete database version information to the client.

- false

  Returns a generic version string to the client.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Call Interface Programmer's Guide* and *Oracle Database Security Guide* for more information on controlling the display of the database version banner.

1.326 SERIAL_REUSE

`SERIAL_REUSE` specifies which types of cursors make use of the serial-reusable memory feature. This feature allocates private cursor memory in the SGA so that it can be reused (serially, not concurrently) by sessions executing the same cursor.

Property	Description
Parameter type	String
Syntax	`SERIAL_REUSE = { disable | all | select | dml | plsql | force }`
Default value	`disable`
Modifiable	No
Modifiable in a PDB	No
Basic	No

> **Note**
>
> The `SERIAL_REUSE` parameter is deprecated. It is retained for backward compatibility only.

Values

- `disable`

  Disables the option for all SQL statement types. This value overrides any other values included in the list.

- `all`

  Enables the option for both DML and `SELECT` statements. Equivalent to setting `select`, `dml`, and `plsql`.

- `select`

  Enables the option for `SELECT` statements.

- `dml`

  Enables the option for DML statements.

- `plsql`

  Currently has no effect (although PL/SQL packages do support the serial-reuse memory option using PL/SQL pragmas).

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> If CURSOR_SPACE_FOR_TIME is set to true, then the value of SERIAL_REUSE is ignored and treated as if it were set to disable.

> **See Also**
>
> "CURSOR_SPACE_FOR_TIME"

1.327 SERVICE_NAMES

SERVICE_NAMES specifies one or more names by which clients can connect to the instance.

Property	Description
Parameter type	String
Syntax	SERVICE_NAMES = *db_service_name* [, *db_service_name* [...]]
Default value	DB_UNIQUE_NAME.DB_DOMAIN if defined
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	Any ASCII string or comma-separated list of string names
Basic	No
Oracle RAC	Do not set the SERVICE_NAMES parameter for Oracle RAC environments. Instead, define services using Oracle Enterprise Manager and manage those services using Server Control (SRVCTL) utility.

The instance registers its service names with the listener. When a client requests a service, the listener determines which instances offer the requested service and routes the client to the appropriate instance.

You can specify multiple service names to distinguish among different uses of the same database. For example:

```
SERVICE_NAMES = sales.example.com, widgetsales.example.com
```

You can also use service names to identify a single service that is available from two different databases through the use of replication.

If you do not qualify the names in this parameter with a domain, Oracle qualifies them with the value of the DB_DOMAIN parameter. If DB_DOMAIN is not specified, then no domain will be applied to the non-qualified SERVICE_NAMES values.

When you specify additional service names with this parameter, the default service name is not overridden. The default service name plus the additional service names specified with this parameter are the service names that clients can use to connect to the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Note

The SERVICE_NAMES initialization parameter is deprecated in Oracle Database 19c and may be desupported in a future release.

Use of the SERVICE_NAMES parameter is no longer actively supported. It must not be used for high availability (HA) deployments and it is not supported for HA operations. This restriction includes FAN, load balancing, FAILOVER_TYPE, FAILOVER_RESTORE, SESSION_STATE_CONSISTENCY, and any other uses.

To manage your services, Oracle recommends that you instead use the SRVCTL command-line utility, the GDSCTL command-line utility, or the DBMS_SERVICE PL/SQL package.

See Also

- *Oracle Database Net Services Administrator's Guide* for more information on this parameter and its settings
- *Oracle Real Application Clusters Administration and Deployment Guide* for information about services administration in an Oracle RAC environment
- "DB_DOMAIN"

1.328 SESSION_CACHED_CURSORS

SESSION_CACHED_CURSORS specifies the number of session cursors to cache.

Property	Description
Parameter type	Integer
Default value	50
Modifiable	ALTER SESSION, ALTER SYSTEM ... DEFERRED
Modifiable in a PDB	Yes
Range of values	0 to operating system-dependent
Basic	No
Oracle RAC	Multiple instances can have different values.

Repeated parse calls of the same SQL (including recursive SQL) or PL/SQL statement cause the session cursor for that statement to be moved into the session cursor cache. Oracle uses a least recently used algorithm to remove entries in the session cursor cache to make room for new entries when needed.

See Also

Oracle Database Performance Tuning Guide for information about enabling the session cursor cache

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.329 SESSION_EXIT_ON_PACKAGE_STATE_ERROR

Use SESSION_EXIT_ON_PACKAGE_STATE_ERROR to force a hard session exit when a session's state has been invalidated.

Exiting sessions after state invalidation avoids errors that can occur when applications mishandle an invalid state.

Property	Description
Parameter type	Boolean
Default value	FALSE
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	TRUE | FALSE
Basic	No
Oracle RAC	Different instances can use different values.

When a stateful PL/SQL package undergoes modification, the sessions that have an active instantiation of the package receive an ORA-4068 error when they attempt to run the package. When SESSION_EXIT_ON_PACKAGE_STATE_ERROR is set to TRUE, the session immediately exits instead of raising ORA-04068. This can be advantageous because many applications are better equipped to handle a session being discarded, and failing to catch errors related to session state, such as ORA-04068, can lead to silent data corruption.

> **Note**
>
> This parameter is available starting with Oracle Database 19c, Release Update 19.23.

1.330 SESSION_MAX_OPEN_FILES

SESSION_MAX_OPEN_FILES specifies the maximum number of BFILEs that can be opened in any session. Once this number is reached, subsequent attempts to open more files in the session by using DBMS_LOB.FILEOPEN() or OCILobFileOpen() will fail. The maximum value for this parameter depends on the equivalent parameter defined for the underlying operating system.

Property	Description
Parameter type	Integer
Default value	10
Modifiable	No
Modifiable in a PDB	No
Range of values	1 to either 50 or the value of MAX_OPEN_FILES defined at the operating system level, whichever is less
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database SecureFiles and Large Objects Developer's Guide* for information on LOBs in general
- *Oracle Database SecureFiles and Large Objects Developer's Guide* for information on BFILEs
- *Oracle Database PL/SQL Packages and Types Reference* for information on the `DBMS_LOB.FILEOPEN()` procedure
- *Oracle Call Interface Programmer's Guide* for information on the `OCILobFileOpen()` procedure

1.331 SESSIONS

`SESSIONS` specifies the maximum number of sessions that can be created in the system.

Property	Description
Parameter type	Integer
Default value	Derived: (1.5 * `PROCESSES`) + 22
Modifiable	`ALTER SYSTEM` can be used in a PDB only to change the value of the `SESSIONS` parameter for that PDB. `ALTER SYSTEM` *cannot* be used to change the value of the `SESSIONS` parameter in a non-CDB or in `CDB$ROOT`.
Modifiable in a PDB	Yes
Range of values	If you explicitly set a value for this parameter, then the range of values that you can specify is 1 to (2^{16} - 1), which is 1 to 65535. If you do not explicitly set a value for this parameter, then it is set to the default value. If `COMPATIBLE` >= `19.0.0`, then the default value will not exceed a maximum of (2^{18} - 1), which is 262143. If `COMPATIBLE` < `19.0.0`, then the default value will not exceed a maximum of (2^{16} - 1), which is 65535.
Basic	Yes

Because every login requires a session, this parameter effectively determines the maximum number of concurrent users in the system. You should always set this parameter explicitly to a value equivalent to your estimate of the maximum number of concurrent users, plus the number of background processes, plus approximately 10% for recursive sessions.

Oracle uses the default value of this parameter as its minimum. Values between 1 and the default do not trigger errors, but Oracle ignores them and uses the default instead.

The default values of the `ENQUEUE_RESOURCES` and `TRANSACTIONS` parameters are derived from `SESSIONS`. Therefore, if you increase the value of `SESSIONS`, you should consider whether to adjust the values of `ENQUEUE_RESOURCES` and `TRANSACTIONS` as well. (Note that `ENQUEUE_RESOURCES` is obsolete as of Oracle Database 10g release 2 (10.2).)

In a shared server environment, the value of `PROCESSES` can be quite small. Therefore, Oracle recommends that you adjust the value of `SESSIONS` to approximately 1.1 * total number of *connections*.

Copyright © 2002, 2026, Oracle and/or its affiliates.

For a CDB, the root container's SESSIONS parameter specifies the total number of sessions for the database.

The SESSIONS parameter for a PDB specifies the total number of sessions that can be used by that PDB. Its value defaults to the root container's SESSIONS value. If the PDB tries to use more sessions than configured by its SESSIONS parameter, an ORA-00018 error message is generated. For PDBs, the SESSIONS parameter does not count recursive sessions and hence does not require the 10% adjustment.

The SESSIONS parameter for a PDB can only be modified by the PDB. It cannot be set higher than the CDB's SESSIONS value.

See Also

- *Oracle Database Concepts* for more information on memory structures
- *Oracle Database Concepts* for more information on processes
- *Oracle Multitenant Administrator's Guide* for more information about CDBs and PDBs

1.332 SGA_MAX_SIZE

SGA_MAX_SIZE specifies the maximum size of the SGA for the lifetime of the instance.

Property	Description
Parameter type	Big integer
Syntax	SGA_MAX_SIZE = *integer* [K | M | G]
Default value	Initial size of SGA at startup, dependent on the sizes of different pools in the SGA, such as buffer cache, shared pool, large pool, and so on.
Modifiable	No
Modifiable in a PDB	No
Range of values	0 to operating system-dependent

On 64-bit platforms and non-Windows 32-bit platforms, when either MEMORY_TARGET or MEMORY_MAX_TARGET is specified, the default value of SGA_MAX_SIZE is set to the larger of the two parameters. This causes more address space to be reserved for expansion of the SGA.

On Windows 32-bit platforms, the default value of SGA_MAX_SIZE is the largest of the following values:

- 60% of MEMORY_TARGET, if specified
- 60% of MEMORY_MAX_TARGET, if specified
- 25% of the total available virtual address space

See Also

Oracle Database Performance Tuning Guide for more information about this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.333 SGA_MIN_SIZE

SGA_MIN_SIZE indicates a possible minimum value for SGA usage of a pluggable database (PDB).

Property	Description
Parameter type	Big integer
Syntax	SGA_MIN_SIZE = *integer* [K | M | G]
Default value	0
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	0 to 50% of SGA_TARGET
Basic	No
Oracle RAC	The same value must be used for all instances.

Setting this parameter at the CDB level has no effect.

Note

To be able to use Resource Manager in a CDB to control the amount of memory each PDB can use:

- The NONCDB_COMPATIBLE initialization parameter must be set to FALSE at the CDB level (in the root of the CDB).
- The MEMORY_TARGET initialization parameter must not be set at the CDB level.
- A value for SGA_TARGET must be set at the CDB level.

  If SGA_TARGET is not set at the CDB level, then setting SGA_MIN_SIZE in a PDB has no effect. You will not receive an error message, and the PDB's SGA_MIN_SIZE value will not be enforced.
- You must set the SGA_MIN_SIZE value to a value that meets these requirements:
  - In a PDB, to a value that is less than or equal to 50% of the value of SGA_TARGET in the PDB
  - In a PDB, to a value that is less than or equal to 50% of the value of SGA_TARGET at the CDB level
  - In a PDB, to a value that is less than the value of SGA_MAX_SIZE at the CDB level
  - Across all the open PDBs in a CDB, the sum of SGA_MIN_SIZE values must be less than or equal to 50% of the managed SGA total at the CDB level. Managed SGA is DB_CACHE_SIZE plus SHARED_POOL_SIZE.

  When you set SGA_MIN_SIZE in a PDB to a value that does not meet these requirements, you receive an error.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Multitenant Administrator's Guide* for more information about the initialization parameters that control the memory usage of PDBs

1.334 SGA_TARGET

SGA_TARGET specifies the total size of all SGA components.

Property	Description
Parameter type	Big integer
Syntax	SGA_TARGET = *integer* [K | M | G]
Default value	0 (SGA autotuning is disabled for DEFERRED mode autotuning requests, but allowed for IMMEDIATE mode autotuning requests)
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	64 MB to SGA_MAX_SIZE Note that if MEMORY_MAX_TARGET > 0, then the value of SGA_TARGET also cannot exceed the value of MEMORY_MAX_TARGET.
Basic	Yes

If SGA_TARGET is specified, then the following memory pools are automatically sized:

- Buffer cache (DB_CACHE_SIZE)
- Shared pool (SHARED_POOL_SIZE)
- Large pool (LARGE_POOL_SIZE)
- Java pool (JAVA_POOL_SIZE)
- Streams pool (STREAMS_POOL_SIZE)
- Data transfer cache (DATA_TRANSFER_CACHE_SIZE)

If these automatically tuned memory pools are set to nonzero values, then those values are used as minimum levels by Automatic Shared Memory Management. You would set minimum values if an application component needs a minimum amount of memory to function properly.

The following pools are manually sized components and are not affected by Automatic Shared Memory Management:

- Log buffer
- Other buffer caches, such as KEEP, RECYCLE, and other block sizes
- Fixed SGA and other internal allocations

The memory allocated to these pools is deducted from the total available for SGA_TARGET when Automatic Shared Memory Management computes the values of the automatically tuned memory pools.

In the **Default value** field, IMMEDIATE mode autotuning requests are necessary to avoid ORA-04031 errors. The DEFERRED and IMMEDIATE modes are reflected in the OPER_MODE column of the V$MEMORY_RESIZE_OPS view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

If Automatic Memory Management is enabled (MEMORY_TARGET is set to a positive value) and SGA_TARGET is also set to a positive value, the SGA_TARGET value acts as the minimum value for the size of the SGA.

Note

This parameter is optional for pluggable databases (PDBs). When this parameter is set for a PDB, it specifies the maximum SGA that the PDB can use at any time. When this parameter is not set at the PDB level, the PDB has no limit for the amount of SGA it can use, other than the CDB's SGA size.

To be able to use Resource Manager in a CDB to control the amount of memory each PDB can use:

- The NONCDB_COMPATIBLE initialization parameter must be set to FALSE at the CDB level (in the root of the CDB).
- The MEMORY_TARGET initialization parameter must not be set at the CDB level.
- You must set the SGA_TARGET initialization parameter at the CDB level.
- You must set SGA_TARGET in a PDB.

  If you set a PDB's SGA_TARGET value and SGA_TARGET is not set at the CDB level, you will not receive an error message and the PDB's SGA_TARGET value will not be enforced.
- When setting SGA_TARGET in a PDB, the following requirements must be met:
  - The value of SGA_TARGET in the PDB must be less than or equal to the SGA_TARGET value at the CDB level.
  - The value of SGA_TARGET in the PDB must be less than or equal to the SGA_MAX_SIZE value at the CDB level.
  - The value of SGA_TARGET in the PDB must be twice the PDB's DB_CACHE_SIZE value, if the PDB's DB_CACHE_SIZE value is set.
  - The value of SGA_TARGET in the PDB must be twice the PDB's SHARED_POOL_SIZE value, if the PDB's SHARED_POOL_SIZE value is set.
  - The value of SGA_TARGET in the PDB must be twice the PDB's SGA_MIN_SIZE value, if the PDB's SGA_MIN_SIZE value is set.

  When you set SGA_TARGET in a PDB to a value that does not meet these requirements, you receive an error.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DB_CACHE_SIZE"
- "SHARED_POOL_SIZE"
- "LARGE_POOL_SIZE"
- "JAVA_POOL_SIZE"
- "STREAMS_POOL_SIZE"
- "V$MEMORY_RESIZE_OPS"
- "MEMORY_TARGET"
- "PGA_AGGREGATE_TARGET"
- *Oracle Multitenant Administrator's Guide* for more information about the initialization parameters that control the memory usage of PDBs
- *Oracle Database Administrator's Guide* for information on automatic memory management
- *Oracle Database Administrator's Guide* for information on managing the SGA manually

1.335 SHADOW_CORE_DUMP

`SHADOW_CORE_DUMP` specifies whether Oracle includes the SGA in the core file for foreground (client) processes.

Property	Description
Parameter type	String
Syntax	`SHADOW_CORE_DUMP = { partial | full | none }`
Default value	`partial`. On Windows, the default value is `none`.
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

Values

- `partial`

  Oracle does not include the SGA in the core dump.
- `full`

  Oracle includes the SGA in the core dump.
- `none`

  No core files will be generated for foreground processes.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "BACKGROUND_CORE_DUMP"

1.336 SHARED_MEMORY_ADDRESS

SHARED_MEMORY_ADDRESS and HI_SHARED_MEMORY_ADDRESS specify the starting address at run time of the system global area (SGA). This parameter is ignored on the many platforms that specify the SGA's starting address at linktime.

Property	Description
Parameter type	Integer
Default value	0
Modifiable	No
Modifiable in a PDB	No
Basic	No

Use this parameter to specify the entire address on 32-bit platforms and to specify the low-order 32 bits of a 64-bit address on 64-bit platforms. Use HI_SHARED_MEMORY_ADDRESS to specify the high-order 32 bits of a 64-bit address on 64-bit platforms. If both parameters are 0 or unspecified, the SGA address defaults to a platform-specific location.

> **See Also**
>
> "HI_SHARED_MEMORY_ADDRESS"

1.337 SHARED_POOL_RESERVED_SIZE

SHARED_POOL_RESERVED_SIZE specifies (in bytes) the shared pool space that is reserved for large contiguous requests for shared pool memory.

Property	Description
Parameter type	Big integer
Syntax	SHARED_POOL_RESERVED_SIZE = *integer* [K | M | G]
Default value	5% of the value of SHARED_POOL_SIZE
Modifiable	No
Modifiable in a PDB	No
Range of values	Minimum: 5000 Maximum: half of the value of SHARED_POOL_SIZE
Basic	No

You can use this parameter to avoid performance degradation in the shared pool in situations where pool fragmentation forces Oracle to search for and free chunks of unused pool to satisfy the current request.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "SHARED_POOL_SIZE"
- *Oracle Database Performance Tuning Guide* for information on sizing the shared pool

1.338 SHARED_POOL_SIZE

`SHARED_POOL_SIZE` specifies (in bytes) the size of the shared pool.

Note

Oracle recommends against setting this parameter in a PDB. The only shared memory sizing parameter that should be set in a PDB is `SGA_TARGET`, which specifies the maximum SGA that the PDB can use at any time.

Property	Description
Parameter type	Big integer
Syntax	`SHARED_POOL_SIZE = integer [K | M | G]`
Default value	If `SGA_TARGET` is set: If the parameter is not specified, then the default is `0` (internally determined by the Oracle Database). If the parameter is specified, then the user-specified value indicates a minimum value for the memory pool. If `SGA_TARGET` is not set (32-bit platforms): 64 MB, rounded up to the nearest granule size. If `SGA_TARGET` is not set (64-bit platforms): 128 MB, rounded up to the nearest granule size. For considerations when dealing with database instances using Oracle ASM, see "SHARED_POOL_SIZE and Automatic Storage Management".
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	Minimum: the granule size Maximum: operating system-dependent
Basic	No

The shared pool contains shared cursors, stored procedures, control structures, and other structures. If `SGA_TARGET` is not set, then Oracle also allocates parallel execution message buffers from the shared pool. Larger values improve performance in multiuser systems. Smaller values use less memory.

You can monitor utilization of the shared pool by querying the view `V$SGASTAT`.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "SGA_TARGET"
- "V$SGASTAT"
- *Oracle Multitenant Administrator's Guide* for more information about the initialization parameters that control the memory usage of PDBs
- *Oracle Database Performance Tuning Guide* for more information on setting this parameter

SHARED_POOL_SIZE and Automatic Storage Management

On a database instance using Oracle Automatic Storage Management (Oracle ASM), additional memory is required to store extent maps. As a general guideline, you can aggregate the values from the following queries to obtain current database storage size that is either already on Oracle ASM or will be stored in Oracle ASM. Then determine the redundancy type that is used (or will be used), and calculate the value for `SHARED_POOL_SIZE`, using the aggregated value as input.

```
SELECT SUM(BYTES)/(1024*1024*1024) FROM V$DATAFILE;
SELECT SUM(BYTES)/(1024*1024*1024) FROM V$LOGFILE a, V$LOG b
WHERE a.group#=b.group#;
SELECT SUM(BYTES)/(1024*1024*1024) FROM V$TEMPFILE WHERE
status='ONLINE';
```

Additionally, keep the following guidelines in mind:

- For disk groups using external redundancy:

  (Every 100G of space needs 1M of extra shared pool) + 2M
- For disk groups using normal redundancy:

  (Every 50G of space needs 1M of extra shared pool) + 4M
- For disk groups using high redundancy:

  (Every 33G of space needs 1M of extra shared pool) + 6M

1.339 SHARED_SERVER_SESSIONS

`SHARED_SERVER_SESSIONS` specifies the total number of shared server sessions to allow.

Property	Description
Parameter type	Integer
Default value	There is no default value.
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	If `SHARED_SERVER_SESSIONS` is specified, then it should be less than `SESSIONS`. If `SHARED_SERVER_SESSIONS` is not specified, then a shared server session may be created as long as there is a free session slot.
Basic	No

Setting this parameter enables you to reserve user sessions for dedicated servers.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Concepts* for more information on sessions

1.340 SHARED_SERVERS

SHARED_SERVERS specifies the number of server processes that you want to create when an instance is started. If system load decreases, then this minimum number of servers is maintained. Therefore, you should take care not to set SHARED_SERVERS too high at system startup.

Property	Description
Parameter type	Integer
Default value	0, meaning that shared server is not on. If you are using shared server architecture or if the DISPATCHERS parameter is set such that the total number of dispatchers is more than 0, then the default value is 1.
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	The value of this parameter should be less than MAX_SHARED_SERVERS. If it is greater than or equal to MAX_SHARED_SERVERS, then the number of servers will not be self-tuned but will remain constant, as specified by SHARED_SERVERS.
Basic	Yes

Starting with Oracle Database 12*c* Release 1 (12.1.0.2), the SHARED_SERVERS parameter can be set on PDBs. However, unlike most other parameters that can be set on a PDB, this parameter can only be used within a PDB to enable or disable use of shared servers for that PDB. Therefore, in a PDB, the DBA can either set SHARED_SERVERS to 0 to disable use of shared servers for that PDB or use ALTER SYSTEM RESET SHARED_SERVERS to re-enable shared servers for the PDB. The configuration of shared servers for the CDB can only be done in the root.

> **See Also**
>
> - "DISPATCHERS"
> - "MAX_SHARED_SERVERS"
> - *Oracle Database Administrator's Guide* for more information on setting this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.341 SHRD_DUPL_TABLE_REFRESH_RATE

SHRD_DUPL_TABLE_REFRESH_RATE displays the duplicated table refresh rate (in seconds).

Property	Description
Parameter type	Integer
Default value	60
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	20 to 2^{31} - 1
Basic	No
Oracle RAC	All instances should use the same instance.

You can change the value of this parameter to change the duplicated table refresh rate.

A duplicated table is a table that has the same contents on all shards in a sharded database. Duplicated tables are implemented using materialized views that are refreshed at a certain rate.

See Also

- *Using Oracle Sharding* for an introduction to sharding
- *Using Oracle Sharding* for more information about duplicated tables

1.342 SKIP_UNUSABLE_INDEXES

SKIP_UNUSABLE_INDEXES enables or disables the use and reporting of tables with unusable indexes or index partitions.

Property	Description
Parameter type	Boolean
Default value	true
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	true | false
Basic	No

If a SQL statement uses a hint that forces the usage of an unusable index, then this hint takes precedence over initialization parameter settings, including SKIP_UNUSABLE_INDEXES. If the optimizer chooses an unusable index, then an ORA-01502 error will result. (See *Oracle Database Administrator's Guide* for more information about using hints.)

Values

- true

Copyright © 2002, 2026, Oracle and/or its affiliates.

Disables error reporting of indexes and index partitions marked UNUSABLE. This setting allows all operations (inserts, deletes, updates, and selects) on tables with unusable indexes or index partitions.

> **Note**
>
> If an index is used to enforce a UNIQUE constraint on a table, then allowing insert and update operations on the table might violate the constraint. Therefore, this setting does not disable error reporting for unusable indexes that are unique.

- `false`

  Enables error reporting of indexes marked UNUSABLE. This setting does not allow inserts, deletes, and updates on tables with unusable indexes or index partitions.

> **See Also**
>
> *Oracle Database SQL Language Reference* for more information about hints

1.343 SMTP_OUT_SERVER

`SMTP_OUT_SERVER` specifies the SMTP host and port to which `UTL_MAIL` delivers out-bound E-mail.

Property	Description
Parameter type	String
Syntax	`SMTP_OUT_SERVER = server_clause [, server_clause ] ...`
Syntax	**`server_clause::=`** *`host_name`*`[:port]`
Default value	There is no default value.
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

Multiple servers may be specified, separated by commas.

If the first server in the list is unavailable, then `UTL_MAIL` tries the second server, and so on.

If `SMTP_OUT_SERVER` is not specified, then the SMTP server name defaults to the value of `DB_DOMAIN`, the port number defaults to `25`, and the SMTP domain defaults to the suffix of `DB_DOMAIN`.

> **See Also**
>
> *Oracle Database PL/SQL Packages and Types Reference* for information on the `UTL_MAIL` package

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.344 SORT_AREA_RETAINED_SIZE

SORT_AREA_RETAINED_SIZE specifies (in bytes) the maximum amount of the user global area (UGA) memory retained after a sort run completes. The retained size controls the size of the read buffer, which Oracle uses to maintain a portion of the sort in memory. This memory is released back to the UGA, not to the operating system, after the last row is fetched from the sort space.

Property	Description
Parameter type	Integer
Default value	Derived from SORT_AREA_SIZE
Modifiable	ALTER SESSION, ALTER SYSTEM ... DEFERRED
Modifiable in a PDB	Yes
Range of values	From the value equivalent of two database blocks to the value of SORT_AREA_SIZE
Basic	No

Note

Oracle does not recommend using the SORT_AREA_RETAINED_SIZE parameter unless the instance is configured with the shared server option. Oracle recommends that you enable automatic sizing of SQL working areas by setting PGA_AGGREGATE_TARGET instead. SORT_AREA_RETAINED_SIZE is retained for backward compatibility.

Oracle may allocate multiple sort spaces of this size for each query. Usually, only one or two sorts occur simultaneously, even for complex queries. In some cases, however, additional concurrent sorts are required, and each sort keeps its own memory area. If the shared server is used, allocation is to the SGA until the value in SORT_AREA_RETAINED_SIZE is reached. The difference between SORT_AREA_RETAINED_SIZE and SORT_AREA_SIZE is allocated to the PGA.

Note

The default value as reflected in the V$PARAMETER dynamic performance view is 0. However, if you do not explicitly set this parameter, Oracle actually uses the value of the SORT_AREA_SIZE parameter.

See Also

- "SORT_AREA_SIZE"
- *Oracle Database Performance Tuning Guide* for information on setting the values of this parameter and the SORT_AREA_SIZE parameter to tune sort operations using shared servers

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.345 SORT_AREA_SIZE

SORT_AREA_SIZE specifies (in bytes) the maximum amount of memory Oracle will use for a sort.

Property	Description
Parameter type	Integer
Default value	65536
Modifiable	ALTER SESSION, ALTER SYSTEM ... DEFERRED
Modifiable in a PDB	Yes
Range of values	Minimum: the value equivalent of six database blocks Maximum: operating system-dependent
Basic	No

> **Note**
>
> Oracle does not recommend using the SORT_AREA_SIZE parameter unless the instance is configured with the shared server option. Oracle recommends that you enable automatic sizing of SQL working areas by setting PGA_AGGREGATE_TARGET instead. SORT_AREA_SIZE is retained for backward compatibility.

After the sort is complete, but before the rows are returned, Oracle releases all of the memory allocated for the sort, except the amount specified by the SORT_AREA_RETAINED_SIZE parameter. After the last row is returned, Oracle releases the remainder of the memory.

Increasing SORT_AREA_SIZE size improves the efficiency of large sorts.

Each sort in a query can consume memory up to the amount specified by SORT_AREA_SIZE, and there can be multiple sorts in a query. Also, if a query is executed in parallel, each PX server can consume memory up to the amount specified by SORT_AREA_SIZE for each sort it does.

SORT_AREA_SIZE is also used for inserts and updates to bitmap indexes. Setting this value appropriately results in a bitmap segment being updated only once for each DML operation, even if more than one row in that segment changes.

Larger values of SORT_AREA_SIZE permit more sorts to be performed in memory. If more space is required to complete the sort than will fit into the memory provided, then temporary segments on disk are used to hold the intermediate sort runs.

The default is adequate for most OLTP operations. You might want to adjust this parameter for decision support systems, batch jobs, or large CREATE INDEX operations.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database Concepts* for information on sort areas
- Your operating system-specific Oracle documentation for the default value on your system
- "SORT_AREA_RETAINED_SIZE"

1.346 SPATIAL_VECTOR_ACCELERATION

`SPATIAL_VECTOR_ACCELERATION` enables or disables the spatial vector acceleration.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true | false`
Basic	No
Oracle RAC	Multiple instances should use the same value

Setting this parameter to `true` improves spatial vector query performance.

See Also

Oracle Spatial and Graph Developer's Guide for more information about this parameter

1.347 SPFILE

The value of this parameter is the name of the current server parameter file (`SPFILE`) in use.

Property	Description
Parameter type	String
Syntax	`SPFILE = spfile_name`
Default value	*`ORACLE_HOME`*`/dbs/spfile.ora`
Modifiable	No
Modifiable in a PDB	No
Range of values	Any valid `SPFILE`
Basic	No
Oracle RAC	Multiple instances must have the same value.

Copyright © 2002, 2026, Oracle and/or its affiliates.

This parameter can be defined in a client side PFILE to indicate the name of the server parameter file to use.

When the default server parameter file is used by the server, the value of SPFILE is internally set by the server.

The SPFILE resides in the *ORACLE_HOME*/dbs directory; however, users can place it anywhere on their system as long as it is specified in an initialization parameter file.

See Also

Oracle Database Administrator's Guide for more information about creating the server parameter file

1.348 SQL_TRACE

SQL_TRACE enables or disables the SQL trace facility.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	true | false
Basic	No

Setting this parameter to true provides information on tuning that you can use to improve performance.

Note

Using this initialization parameter to enable the SQL trace facility for the entire instance can have a severe performance impact. Enable the facility for specific sessions using the ALTER SESSION statement. If you must enable the facility on an entire production environment, then you can minimize performance impact by:

- Maintaining at least 25% idle CPU capacity
- Maintaining adequate disk space for the USER_DUMP_DEST location
- Striping disk space over sufficient disks

Note

The SQL_TRACE parameter is deprecated. Oracle recommends that you use the DBMS_MONITOR and DBMS_SESSION packages instead. SQL_TRACE is retained for backward compatibility only.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "USER_DUMP_DEST"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_MONITOR package
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SESSION package
- *Oracle Database Performance Tuning Guide* for more information about performance diagnostic tools

1.349 SQL92_SECURITY

SQL92_SECURITY specifies whether users must have been granted the SELECT privilege on a table in order to specify certain clauses in a DELETE, INSERT, or UPDATE statement.

Property	Description
Parameter type	Boolean
Default value	TRUE
Modifiable	No
Modifiable in a PDB	Yes
Range of values	TRUE | FALSE
Basic	No

The SQL standard specifies that security administrators should be able to require that users have the SELECT privilege on a table when executing a DELETE, INSERT, or UPDATE statement that references table column values in a RETURNING, SET, or WHERE clause.

Values

- TRUE

  A user must have the SELECT privilege on the target table in order to reference its columns in:

  - The WHERE clause of a DELETE or UPDATE statement
  - The right hand side of an assignment in the SET clause of an UPDATE statement
  - The RETURNING clause of a DELETE, INSERT, or UPDATE statement
- FALSE
  - A user with the DELETE privilege on the target table of a DELETE statement may reference any column of that target table in the WHERE clause of the DELETE statement.
  - A user with the UPDATE privilege on the target table of an UPDATE statement may reference any column of that target table in the WHERE clause or on the right hand side of any assignment in the SET clause of the UPDATE statement.
  - A user with the READ or SELECT privilege on the target table of a DELETE, INSERT, or UPDATE statement may reference any column of that target table in the RETURNING clause.

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.350 SQLTUNE_CATEGORY

SQLTUNE_CATEGORY specifies the category name for use by sessions to qualify the lookup of SQL profiles during SQL compilation.

Property	Description
Parameter type	String
Syntax	SQLTUNE_CATEGORY = *category_name*
Default value	DEFAULT
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No

See Also

Oracle Database PL/SQL Packages and Types Reference for information on the DBMS_SQLTUNE package

1.351 STANDBY_DB_PRESERVE_STATES

STANDBY_DB_PRESERVE_STATES controls whether user sessions, buffers, and other internal states of the instance are retained when a readable physical standby database is converted to a primary database. This parameter is meaningful on a physical standby database that is open in real-time query mode.

Property	Description
Parameter type	String
Syntax	STANDBY_DB_PRESERVE_STATES = { NONE | SESSION | BUFFER | ALL }
Default value	NONE
Modifiable	No
Modifiable in a PDB	No
Basic	No
Oracle RAC	The same value must be used on all instances.

Values

- NONE

  Nothing is retained. All sessions are disconnected and all buffers are flushed. This is the default value.

- SESSION

  The role transition waits 10 seconds for active sessions to finish the current statement or top-level PL/SQL call. The sessions that are still active after 10 seconds are terminated. The sessions that complete their work in this lap of time are retained, along with inactive

Copyright © 2002, 2026, Oracle and/or its affiliates.

sessions. If the database is reopened as the primary within 300 seconds, the retained sessions resume their operations as if nothing had happened. If the database (or individual PDB) is not opened in the primary role within 300 seconds, the sessions will be terminated.

- `BUFFER`

  All current buffers are retained. Media recovery buffers are converted to current buffers and retained, if possible. When the database is reopened as the primary, the retained buffers are available for use, which may enable queries to run faster during the first few minutes of operation. If the database (or an individual PDB) is not opened in the primary role, the buffers will be flushed.

- `ALL`

  This value is equivalent to setting both the `SESSION` and `BUFFER` values.

> **Note**
>
> Sessions that have long running queries or are using database links will not be retained regardless of the setting of this parameter.

> **See Also**
>
> - *Oracle Data Guard Concepts and Administration* for more information about real-time query mode
> - *Oracle Data Guard Concepts and Administration* for more information about preserving user sessions and buffers when a standby database is converted to a primary database

1.352 STANDBY_FILE_MANAGEMENT

`STANDBY_FILE_MANAGEMENT` enables or disables automatic standby file management.

Property	Description
Parameter type	String
Syntax	`STANDBY_FILE_MANAGEMENT = { MANUAL | AUTO }`
Default value	`MANUAL`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Basic	No

When automatic standby file management is enabled, operating system file additions and deletions on the primary database are replicated on the standby database. `STANDBY_FILE_MANAGEMENT` is only applicable to physical standby databases.

Values

- `MANUAL`

  Disables automatic standby file management

Copyright © 2002, 2026, Oracle and/or its affiliates.

- AUTO

  Enables automatic standby file management

Setting STANDBY_FILE_MANAGEMENT to AUTO causes Oracle to automatically create files on the standby database and, in some cases, overwrite existing files. Care must be taken when setting STANDBY_FILE_MANAGEMENT and DB_FILE_NAME_CONVERT so that existing standby files will not be accidentally overwritten.

If the standby database is on the same system as the primary database, then ensure that the primary and standby systems do not point to the same files.

See Also

Oracle Data Guard Concepts and Administration for more information about setting this parameter

1.353 STANDBY_PDB_SOURCE_FILE_DBLINK

STANDBY_PDB_SOURCE_FILE_DBLINK specifies the name of a database link that will be used to try to copy the datafiles from a source PDB to which the database link points.

Property	Description
Parameter type	String
Syntax	STANDBY_PDB_SOURCE_FILE_DBLINK = database-link
Default value	NULL
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No
Oracle RAC	Different values can be set on different instances.

In Oracle Database releases prior to Oracle Database 18c, if a PDB was created by cloning a PDB from the same CDB (a local clone), on a standby the datafiles were automatically copied from the source PDB. However, when the PDB was created as a remote clone, the user was responsible for copying datafiles to the Oracle Managed Files (OMF) location on the standby.

To address that deficiency, the STANDBY_PDB_SOURCE_FILE_DBLINK parameter specifies the name of a database link that will be used to try to copy the datafiles from a source PDB to which the database link points. The file copy is done only if the database link points to the source PDB and the source PDB is open in read only mode. Otherwise, the user is still responsible for copying datafiles to the OMF location on the standby.

This parameter can also be used to try to copy files in cases where the source PDB's files are not present on the standby. For example, the source PDB could have been created with standbys=NONE. In this case, the STANDBY_PDB_SOURCE_FILE_DBLINK parameter can enable the standby to copy files from the primary if there is a database link set up to the primary.

See Also

"STANDBY_PDB_SOURCE_FILE_DIRECTORY"

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.354 STANDBY_PDB_SOURCE_FILE_DIRECTORY

STANDBY_PDB_SOURCE_FILE_DIRECTORY specifies a directory location on the standby where source datafiles for instantiating the standby PDB may be found.

Property	Description
Parameter type	String
Syntax	STANDBY_PDB_SOURCE_FILE_DIRECTORY = datafiles-directory
Default value	NULL
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Basic	No
Oracle RAC	Different values can be set on different instances.

In Oracle Database releases prior to Oracle Database 18c, if a PDB was plugged into a CDB, on a standby, the datafiles were expected to be in PDB's Oracle Managed Files (OMF) directory location. If they were not found there, the user had to copy the datafiles to the OMF location and then restart redo apply on the standby.

To address that deficiency, the STANDBY_PDB_SOURCE_FILE_DIRECTORY specifies a directory location on the standby where source datafiles for instantiating the PDB may be found. If the datafiles are not found there, an attempt will be made to locate them in the OMF location on the standby.

This parameter can also be used to try to copy files in cases where the source PDB's files are not present on the standby. For example, the source PDB could have been created with standbys=NONE. In this case, the STANDBY_PDB_SOURCE_FILE_DIRECTORY parameter can enable the standby to copy files from a location if they are made available.

See Also

"STANDBY_PDB_SOURCE_FILE_DBLINK"

1.355 STAR_TRANSFORMATION_ENABLED

STAR_TRANSFORMATION_ENABLED determines whether a cost-based query transformation will be applied to star queries.

Property	Description
Parameter type	String
Syntax	STAR_TRANSFORMATION_ENABLED = { FALSE | TRUE | TEMP_DISABLE }
Default value	FALSE
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	Yes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Values

- FALSE

  The transformation will not be applied.

- TRUE

  The optimizer will consider performing a cost-based query transformation on the star query.

- TEMP_DISABLE

  The optimizer will consider performing a cost-based query transformation on the star query but will not use temporary tables in the star transformation.

See Also

- *Oracle Database SQL Tuning Guide* for information on enabling star queries

1.356 STATISTICS_LEVEL

STATISTICS_LEVEL specifies the level of collection for database and operating system statistics. The Oracle Database collects these statistics for a variety of purposes, including making self-management decisions.

Property	Description
Parameter type	String
Syntax	STATISTICS_LEVEL = { ALL | TYPICAL | BASIC }
Default value	TYPICAL
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No

The default setting of TYPICAL ensures collection of all major statistics required for database self-management functionality and provides best overall performance. The default value should be adequate for most environments.

When the STATISTICS_LEVEL parameter is set to ALL, additional statistics are added to the set of statistics collected with the TYPICAL setting. The additional statistics are timed operating system statistics and plan execution statistics.

Setting the STATISTICS_LEVEL parameter to BASIC disables the collection of many of the important statistics required by Oracle Database features and functionality, including:

- Automatic Workload Repository (AWR) Snapshots
- Automatic Database Diagnostic Monitor (ADDM)
- All server-generated alerts
- Automatic SGA Memory Management
- Automatic optimizer statistics collection

Copyright © 2002, 2026, Oracle and/or its affiliates.

- Object level statistics
- End to End Application Tracing (V$CLIENT_STATS)
- Database time distribution statistics (V$SESS_TIME_MODEL and V$SYS_TIME_MODEL)
- Service level statistics
- Buffer cache advisory
- MTTR advisory
- Shared pool sizing advisory
- Segment level statistics
- PGA Target advisory
- Timed statistics
- Monitoring of statistics

> **Note**
>
> Oracle strongly recommends that you do not disable these important features and functionality.

When the STATISTICS_LEVEL parameter is modified by ALTER SYSTEM, all advisories or statistics are dynamically turned on or off, depending on the new value of STATISTICS_LEVEL. When modified by ALTER SESSION, the following advisories or statistics are turned on or off in the local session only. Their systemwide state is not changed:

- Timed statistics
- Timed operating system statistics
- Plan execution statistics

The V$STATISTICS_LEVEL view displays information about the status of the statistics or advisories controlled by the STATISTICS_LEVEL parameter. See "V$STATISTICS_LEVEL".

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this parameter

1.357 STREAMS_POOL_SIZE

The STREAMS_POOL_SIZE value helps determine the size of the Streams pool.

Property	Description
Parameter type	Big integer
Syntax	STREAMS_POOL_SIZE = *integer* [K | M | G]
Default value	0

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No
Range of values	Minimum: 0 (values greater than zero are rounded up to the nearest granule size) Maximum: operating system-dependent
Basic	No

Oracle's Automatic Shared Memory Management feature manages the size of the Streams pool when the SGA_TARGET initialization parameter is set to a nonzero value. If the STREAMS_POOL_SIZE initialization parameter also is set to a nonzero value, then Automatic Shared Memory Management uses this value as a minimum for the Streams pool.

If SGA_TARGET is set to a nonzero value and STREAMS_POOL_SIZE is not specified or is set to a null value, Automatic Shared Memory Management uses 0 (zero) bytes as a minimum for the Streams pool.

If the STREAMS_POOL_SIZE initialization parameter is set to a nonzero value, and the SGA_TARGET parameter is set to 0 (zero), then the Streams pool size is the value specified by the STREAMS_POOL_SIZE parameter, in bytes.

If both the STREAMS_POOL_SIZE and the SGA_TARGET initialization parameters are set to 0 (zero), then, by default, on the first request for Streams pool memory in a database, an amount of memory equal to 10% of the shared pool is transferred from the buffer cache to the Streams pool. Products and features that use the Streams pool include Oracle GoldenGate, XStream, Oracle Advanced Queuing, and Oracle Data Pump.

The Streams pool is a shared resource, and the amount of memory a process can use from the Streams pool is determined by the application. The capture or apply parameter MAX_SGA_SIZE can be controlled for Oracle GoldenGate or XStream. For Oracle Advanced Queuing, use the procedures in the dbms_aqadm package to control the amount of Streams Pool needed.

See Also

- *Oracle Database XStream Guide* for information on configuring the Streams pool for an XStream Out configuration
- *Oracle Database XStream Guide* for information on configuring the Streams pool for an XStream In configuration
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the dbms_aqadm package

1.358 TABLESPACE_ENCRYPTION

TABLESPACE_ENCRYPTION specifies the tablespace encryption policy for a database.

Property	Description
Parameter type	String

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Syntax	TABLESPACE_ENCRYPTION = { AUTO_ENABLE | MANUAL_ENABLE | DECRYPT_ONLY }
Default value	Cloud databases: AUTO_ENABLE On-premises databases: MANUAL_ENABLE
Modifiable	No
Modifiable in a PDB	No
Basic	No
Oracle RAC	The same value should be specified for all instances.

This parameter is introduced in Oracle Database 19c, Release Update 19.16, as an alternative to the ENCRYPT_NEW_TABLESPACES parameter. Similar to ENCRYPT_NEW_TABLESPACES, this parameter allows you to specify whether to encrypt newly created user tablespaces. However, the TABLESPACE_ENCRYPTION parameter offers an additional benefit in Data Guard environments. In previous releases, tablespaces in Data Guard environments were required to use the same encryption policy on the primary and standby databases. For example, if all tablespaces were encrypted on the primary database, then all tablespaces were required to be encrypted on the standby database. The TABLESPACE_ENCRYPTION parameter eliminates this requirement and allows you to use different tablespace encryption policies on primary and standby databases. See *Oracle Database Advanced Security Guide* for more information.

If the behavior specified by the ENCRYPT_NEW_TABLESPACES setting conflicts with the behavior specified by the TABLESPACE_ENCRYPTION setting, then the TABLESPACE_ENCRYPTION behavior takes precedence.

Values:

- AUTO_ENABLE

  Use this setting if you would like to encrypt all tablespaces in the database.

  All newly created tablespaces will be encrypted. In the CREATE TABLESPACE statement, if you specify the DECRYPT clause, it will be ignored and the tablespace will be encrypted. If you specify the ENCRYPTION USING *algorithm* ENCRYPT clause, the specified algorithm will be used to encrypt the tablespace. If you specify the ENCRYPTION ENCRYPT clause and do not specify an algorithm, the tablespace will be encrypted with Advanced Encryption Standard 128 (AES128).

  If an existing tablespace is unencrypted, the database writes a warning to the alert log. See *Oracle Database Advanced Security Guide* for information about encrypting existing tablespaces.

  If you attempt to decrypt an existing encrypted tablespace, an error will occur.

  This is the default setting for Cloud databases. Moreover, because all tablespaces must be encrypted in the Cloud, setting this parameter to MANUAL_ENABLE or DECRYPT_ONLY for a Cloud database will be ignored and the Cloud database will behave as if the setting is AUTO_ENABLE.

- MANUAL_ENABLE

  Use this setting if you would like to manually control which tablespaces are encrypted.

  To encrypt a tablespace, specify the ENCRYPTION ... ENCRYPT clause in the CREATE TABLESPACE statement. If you specify the ENCRYPTION USING *algorithm* ENCRYPT clause, the specified algorithm will be used to encrypt the tablespace. If you specify the ENCRYPTION

Copyright © 2002, 2026, Oracle and/or its affiliates.

ENCRYPT clause and do not specify an algorithm, the tablespace will be encrypted with AES128.

This is the default setting for on-premises databases. This setting is ignored for Cloud databases.

- DECRYPT_ONLY

  Use this setting if you do not want any encrypted tablespaces in the database.

  All newly created tablespaces must be unencrypted. If you specify the ENCRYPT clause in the CREATE TABLESPACE statement, an error will occur.

  If an existing tablespace is encrypted, the database writes a warning to the alert log. See *Oracle Database Advanced Security Guide* for information about decrypting existing tablespaces.

  If you attempt to encrypt an existing unencrypted tablespace, an error will occur.

  This setting is valid only for on-premises databases. This setting is ignored for Cloud databases.

Note

This parameter is available starting with Oracle Database 19c, Release Update 19.16.

See Also

"ENCRYPT_NEW_TABLESPACES"

1.359 TAPE_ASYNCH_IO

TAPE_ASYNCH_IO controls whether I/O to sequential devices (for example, backup or restore of Oracle data to or from tape) is asynchronous—that is, whether parallel server processes can overlap I/O requests with CPU processing during table scans.

Property	Description
Parameter type	Boolean
Default value	true
Modifiable	No
Modifiable in a PDB	No
Range of values	true | false
Basic	No

If your platform supports asynchronous I/O to sequential devices, Oracle recommends that you leave this parameter set to its default. However, if the asynchronous I/O implementation is not stable, you can set TAPE_ASYNCH_IO to false to disable asynchronous I/O. If your platform does not support asynchronous I/O to sequential devices, this parameter has no effect.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database VLDB and Partitioning Guide* for more information about this parameter

1.360 TDE_CONFIGURATION

Use `TDE_CONFIGURATION` to set the type of keystore that is used for Transparent Data Encryption (TDE) by the root container; united PDBs inherit the value from the root container, isolated PDBs can be set individually.

Before Oracle Database 18c, each PDB stored its separate encryption keys in the CDB's keystore (united mode). Starting with Oracle Database 18c Cloud environments, a PDB can optionally store its encryption keys in a separate keystore (isolated mode), thus allowing protection by a separate keystore password. Starting with Oracle Database 19c, Release Update 19.14, isolated mode is available for all Oracle database deployments, including on-premises databases, legacy deployments, and engineered systems.

The `WALLET_ROOT` initialization parameter must be set in order for `TDE_CONFIGURATION` to take effect.

Property	Description
Parameter type	String
Syntax	`TDE_CONFIGURATION = "{ KEYSTORE_CONFIGURATION = value [; CONTAINER = pdb-name] }"`

Syntax	**value ::=**
	`{`
	`FILE |`
	`OKV |`
	`HSM |`
	`FILE|OKV |`
	`FILE|HSM |`
	`OKV|FILE |`
	`HSM|FILE`
	`}`
	Notes:
	• The `KEYSTORE_CONFIGURATION` value is case-insensitive. For example, you can specify `FILE` or `file`.
	• `FILE|OKV`, `FILE|HSM`, `OKV|FILE`, and `HSM|FILE` are values. The vertical bars they contain are not separators in the syntax shown above.

Default value	None
Modifiable	`ALTER SYSTEM`[1]
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The same value must be specified on all instances using the `ALTER SYSTEM SET TDE_CONFIGURATION="KEYSTORE_CONFIGURATION=value" SCOPE=BOTH SID='*';` statement.

Copyright © 2002, 2026, Oracle and/or its affiliates.

[1] In some cases, when this parameter is set using `ALTER SYSTEM SCOPE=SPFILE`, the `SHOW PARAMETER TDE_CONFIGURATION` statement does not show the correct value. However, the value set for `TDE_CONFIGURATION` can be derived from information shown in the `V$ENCRYPTION_WALLET` view.

The following attributes can be specified:

- `KEYSTORE_CONFIGURATION` attribute. This attribute is required. The value specified with this attribute configures the keystore type for the specified PDB. The following values can be specified for this attribute:
  - `FILE`: This value configures a wallet keystore.
  - `OKV`: This value configures the database to use Oracle Key Vault (OKV) for TDE key management.

    This value is also used to disable an auto-login OKV configuration and cause any existing `cwallet.sso` files, containing the credentials to the OKV server as the `OKV_PASSWORD` client secret, to be ignored.
  - `HSM`: This value configures a hardware security module (HSM).

    Oracle does not support the use of HSMs for TDE key management. For more information, see My Oracle Support note KB593570 "Oracle TDE Support With 3rd Party HSM Vendors" at the following URL: https://support.oracle.com/support/?documentId=KB593570
  - `FILE|OKV`: This value configures a reverse migration from OKV to a wallet keystore.
  - `FILE|HSM`: This value configures a reverse migration from a HSM to a wallet keystore.
  - `OKV|FILE`: This value configures a migration from a wallet to OKV.

    This value is also used in an auto-login OKV configuration, because in this configuration a `cwallet.sso` file, containing the `OKV_PASSWORD` client secret, must be used by the Oracle server to obtain the credentials to log in to the OKV server.
  - `HSM|FILE`: This value configures a migration from a wallet to a HSM.

    Oracle does not support the use of HSMs for TDE key management. For more information, see My Oracle Support note KB593570 "Oracle TDE Support With 3rd Party HSM Vendors" at the following URL: https://support.oracle.com/support/?documentId=KB593570

  Some of the `KEYSTORE_CONFIGURATION` attribute values consist of a single word, for example, the `FILE`, `OKV`, and `HSM` values. The other `KEYSTORE_CONFIGURATION` attribute values consist of two words separated by the "|" character that is a required part of the value's syntax, for example, the `FILE|OKV`, `FILE|HSM`, `OKV|FILE`, and `HSM|FILE` values.

  In Oracle Database releases prior to Oracle Database 18.1, keystore types were configured in `sqlnet.ora` using the `METHOD` attribute of the `SQLNET.ENCRYPTION_WALLET_LOCATION` parameter.
- `CONTAINER` attribute: This optional attribute can be used only when setting the parameter in the `CDB$ROOT` of a CDB. The `CONTAINER` attribute can be specified only when the `CDB$ROOT` is in `MOUNTED` state. With this attribute, you must specify the name of the PDB for which you are setting the parameter. When you specify the `CONTAINER` attribute, you must use a semicolon ";" as the separation character between the `KEYSTORE_CONFIGURATION` and `CONTAINER` attributes.

Examples

The following statement configures a wallet keystore for the open PDB from which the statement is issued:

Copyright © 2002, 2026, Oracle and/or its affiliates.

```
ALTER SYSTEM SET TDE_CONFIGURATION="KEYSTORE_CONFIGURATION=FILE" SCOPE=BOTH SID='*';
```

The following statement configures an OKV keystore for the PDB in `MOUNTED` state from which the statement is issued:

```
ALTER SYSTEM SET TDE_CONFIGURATION="KEYSTORE_CONFIGURATION=OKV" SCOPE=SPFILE SID='*';
```

The following statement configures an isolated standby PDB for an auto-open Oracle Key Vault setup; standby PDBs are `MOUNTED`, so the command must be executed in the standby root CDB:

```
ALTER SYSTEM SET TDE_CONFIGURATION="KEYSTORE_CONFIGURATION=OKV|FILE;
CONTAINER=FINANCIALS" SCOPE=both SID='*';
```

Note

In an Oracle Data Guard configuration, during keystore migration to OKV or OCI Key Management Service (KMS), or reverse migration to FILE, after the configuration change to `OKV|FILE` or `FILE|OKV` respectively, the wallet must be closed and reopened on the standby.

See Also

- WALLET_ROOT
- V$ENCRYPTION_WALLET
- *Oracle Database Advanced Security Guide* for information about managing keystores and encryption keys in united mode
- *Oracle Database Advanced Security Guide* for information about managing keystores and encryption keys in isolated mode

1.361 TEMP_UNDO_ENABLED

`TEMP_UNDO_ENABLED` determines whether transactions within a particular session can have a temporary undo log.

Property	Description
Parameter type	Boolean
Default value	`false`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	`true	false`
Basic	No
Oracle RAC	Each session of each instance can have its own value or not set any value at all

The default choice for database transactions has been to have a single undo log per transaction. This parameter, at the session level / system level scope, lets a transaction split its

Copyright © 2002, 2026, Oracle and/or its affiliates.

undo log into temporary undo log (for changes on temporary objects) and permanent undo log (for changes on persistent objects).

By splitting the undo stream of a transaction into two streams (temporary and permanent), a database can provide separate storage and retention model for these. This results in overall reduction in the size of undo log and redo log in the database

If database applications make use of temporary objects (using global temporary tables or temporary table transformations), it is advisable to set this parameter's value to `true`.

When `TEMP_UNDO_ENABLED` is set to `true` and the COMPATIBLE initialization parameter is set to `12.0.0`, this feature is enabled. The temporary undo feature is enabled for the session in which it is set. Setting it across the system will affect all existing and upcoming sessions. If the value is set in the `init.ora` file, all upcoming sessions will inherit this value unless overwritten by an explicit ALTER SESSION or ALTER SYSTEM statement. All undo for operations on temporary objects is deemed temporary.

If `TEMP_UNDO_ENABLED` is not set to `true`, existing applications that make use of temporary objects run as is without any change.

Once the value of the parameter is set, it cannot be changed for the lifetime of the session. If the session has temporary objects using temporary undo, the parameter cannot be disabled for the session. Similarly, if the session already has temporary objects using regular undo, setting this parameter will have no effect.

This parameter is only applicable for the primary database. For a standby database, this parameter is ignored because temporary undo is enabled by default on the standby database.

See Also

Oracle Database Administrator's Guide for information on managing temporary undo

1.362 THREAD

`THREAD` has been superseded by the `INSTANCE_NAME` and `INSTANCE_NUMBER` parameters, and will be made obsolete in a future release of the Oracle Database.

Property	Description
Parameter type	Integer
Default value	`0`
Modifiable	No
Modifiable in a PDB	No
Range of values	0 to the maximum number of enabled threads
Basic	No
Oracle RAC	If specified, multiple instances must have different values.

See Also

"INSTANCE_NAME" and "INSTANCE_NUMBER"

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.363 THREADED_EXECUTION

THREADED_EXECUTION specifies whether to enable the multithreaded Oracle model.

Property	Description
Parameter type	Boolean
Default value	false
Modifiable	No
Modifiable in a PDB	No
Range of values	true | false
Basic	No
Oracle RAC	If specified, all instances must use the same value

Starting in Oracle Database 12*c*, the multithreaded Oracle model enables Oracle processes on UNIX and Linux to run as operating system threads in separate address spaces.

By default, some background processes on UNIX and Linux always use threaded execution; the remaining Oracle processes run as operating system processes. Thus, an "Oracle process" is not always equivalent to an "operating system process."

Note

When this initialization parameter is set to TRUE, which enables the multithreaded Oracle model, operating system authentication is not supported. Attempts to connect to the database using operating system authentication (for example, CONNECT / AS SYSDBA or CONNECT /) when this initialization parameter is set to TRUE receive an ORA-01031 "insufficient privileges" error.

The solution to this error is to always use the password when connecting to the database.

Also, when this initialization parameter is set to TRUE, the DEDICATED_THROUGH_BROKER_listener-name=ON parameter should be added to the listener.ora file, where *listener-name* is the name of the Oracle Net listener and the LOCAL_LISTENER initialization parameter should be set to a TNS name entry corresponding to your instance service. This enables the server to spawn threads when connections to the database are requested through the listener.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database Concepts* for more information about multithreaded Oracle
- Table F-1 for more information about background processes that run as threads instead of as operating system processes when multithreaded Oracle is enabled
- *Oracle Database Net Services Reference* for more information about the `DEDICATED_THROUGH_BROKER_`*listener-name* parameter in the listener.ora file
- *Oracle Database Net Services Administrator's Guide* for an overview of Oracle Net listener

1.364 TIMED_OS_STATISTICS

`TIMED_OS_STATISTICS` specifies (in seconds) the interval at which Oracle collects operating system statistics when a request is made from the client to the server or when a request completes.

Property	Description
Parameter type	Integer
Default value	If `STATISTICS_LEVEL` is set to `ALL`, then `60` If `STATISTICS_LEVEL` is set to `BASIC` or `TYPICAL`, then `0`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Range of values	Unlimited
Basic	No

On dedicated servers, Oracle collects operating system statistics at user logon and after each subsequent client invocation through the OCI into the Oracle server as a remote procedure call message.

On shared servers, Oracle collects statistics when client calls to Oracle are processed.

A value of zero specifies that operating system statistics are not gathered. To collect statistics, set a value meaningful for your application and site needs.

Note

Gathering operating system statistics is very expensive. Oracle recommends that you set this parameter in an `ALTER SYSTEM` statement rather than in the initialization parameter file, and that you reset the value to zero as soon as the needed statistics have been gathered.

See Also

Oracle Database Performance Tuning Guide for more information about this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

1.365 TIMED_STATISTICS

TIMED_STATISTICS specifies whether statistics related to time are collected.

Property	Description
Parameter type	Boolean
Default value	If STATISTICS_LEVEL is set to TYPICAL or ALL, then true If STATISTICS_LEVEL is set to BASIC, then false
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	true | false
Basic	No

Values

- true

  The statistics are collected and stored in trace files or displayed in the V$SESSTATS and V$SYSSTATS dynamic performance views.

- false

  The value of all time-related statistics is set to zero. This setting lets Oracle avoid the overhead of requesting the time from the operating system.

Starting with release 11.1.0.7.0, the value of the TIMED_STATISTICS parameter cannot be set to false if the value of STATISTICS_LEVEL is set to TYPICAL or ALL.

On some systems with very fast timer access, Oracle might enable timing even if this parameter is set to false. On these systems, setting the parameter to true can sometimes produce more accurate statistics for long-running operations.

See Also

- *Oracle Database SQL Tuning Guide* for more information on setting this parameter
- Statistics Descriptions indicates which statistics depend on the setting of this parameter.

1.366 TRACE_ENABLED

TRACE_ENABLED controls tracing of the execution history, or code path, of Oracle. Oracle Support Services uses this information for debugging.

Property	Description
Parameter type	Boolean
Default value	true
Modifiable	ALTER SYSTEM
Modifiable in a PDB	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Range of values	`true | false`
Basic	No
Oracle RAC	The default value is TRUE. Oracle recommends that multiple instances have the same value.

When `TRACE_ENABLED` is set to `true`, Oracle records information in specific files when errors occur.

Oracle records this information for all instances, even if only one instance terminates. This allows Oracle to retain diagnostics for an entire cluster.

Although the overhead incurred from this processing is not excessive, you can improve performance by setting `TRACE_ENABLED` to `false`. You might do this, for example, to meet high-end benchmark requirements. However, if you leave this parameter set to `false`, you may lose valuable diagnostic information. Therefore, always set `TRACE_ENABLED` to `true` to trace system problems and to reduce diagnostic efforts when unexplained instance failures occur.

1.367 TRACEFILE_IDENTIFIER

`TRACEFILE_IDENTIFIER` specifies a custom identifier that becomes part of the Oracle Trace file name. Such a custom identifier is used to identify a trace file simply from its name and without having to open it or view its contents.

Property	Description
Parameter type	String
Syntax	`TRACEFILE_IDENTIFIER = "traceid"`
Default value	There is no default value.
Modifiable	`ALTER SESSION`
Modifiable in a PDB	No
Range of values	Any characters that can occur as part of a file name on the customer platform
Basic	No

Each time this parameter is dynamically modified, the next trace dump will be written to a trace file which has the new parameter value embedded in its name. Trace file continuity information is automatically added to both the old and new trace files to indicate that these trace files belong to the same process.

This parameter can only be used to change the name of the foreground process' trace file; the background processes continue to have their trace files named in the regular format. For foreground processes, the `TRACEID` column of the `V$PROCESS` view contains the current value of the `TRACEFILE_IDENTIFIER` parameter. When this parameter value is set, the trace file name has the following format:

*sid*_ora_*pid*_*traceid*.trc

In this example, *sid* is the Oracle instance ID, *pid* is the process ID, and *traceid* is the value of the `TRACEFILE_IDENTIFIER` parameter.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database SQL Tuning Guide* for more information about this parameter
- This parameter is not supported on all operating systems. See your operating system-specific Oracle documentation for more information.

1.368 TRANSACTIONS

`TRANSACTIONS` specifies how many rollback segments to online when `UNDO_MANAGEMENT = MANUAL`.

Property	Description
Parameter type	Integer
Default value	Derived: (1.1 * `SESSIONS`)
Modifiable	No
Modifiable in a PDB	No
Range of values	4 to 2^{31} - 1
Basic	No
Oracle RAC	Multiple instances can have different values.

The maximum number of concurrent transactions is now restricted by undo tablespace size (`UNDO_MANAGEMENT = AUTO`) or the number of online rollback segments (`UNDO_MANAGEMENT = MANUAL`).

1.369 TRANSACTIONS_PER_ROLLBACK_SEGMENT

`TRANSACTIONS_PER_ROLLBACK_SEGMENT` specifies the number of concurrent transactions you expect each rollback segment to have to handle.

Property	Description
Parameter type	Integer
Default value	5
Modifiable	No
Modifiable in a PDB	No
Range of values	1 to operating system-dependent
Basic	No
Oracle RAC	Multiple instances can have different values.

The minimum number of rollback segments acquired at startup is `TRANSACTIONS` divided by the value for this parameter. For example, if `TRANSACTIONS` is 101 and this parameter is 10, then the minimum number of rollback segments acquired would be the ratio 101/10, rounded up to 11.

You can acquire more rollback segments by naming them in the parameter `ROLLBACK_SEGMENTS`.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

Your operating system-specific Oracle documentation for the range of values for this parameter.

1.370 UNDO_MANAGEMENT

`UNDO_MANAGEMENT` specifies which undo space management mode the system should use.

Property	Description
Parameter type	String
Syntax	`UNDO_MANAGEMENT = { MANUAL | AUTO }`
Default value	`AUTO`
Modifiable	No
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	Multiple instances must have the same value.

When `UNDO_MANAGEMENT` is set to `AUTO`, the instance starts in automatic undo management mode. In manual undo management mode, undo space is allocated externally as rollback segments.

Note

In a CDB, the `UNDO_MANAGEMENT` initialization parameter must be set to `AUTO`, and an undo tablespace is required to be created to manage the undo data.

See Also

- *Oracle Database Administrator's Guide* for more information about this parameter
- *Oracle Database Administrator's Guide* for more information about managing undo data

1.371 UNDO_RETENTION

`UNDO_RETENTION` specifies (in seconds) the low threshold value of undo retention.

Property	Description
Parameter type	Integer
Default value	`900`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	Yes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Range of values	0 to 2^{31} - 1
Basic	No
Oracle RAC	Oracle recommends that multiple instances have the same value.

Starting with Oracle Database 19c, Release Update 19.7, for both AUTOEXTEND undo tablespaces and fixed-size undo tablespaces, the system retains undo for at least the time specified in this parameter, and automatically tunes the undo retention period to satisfy the undo requirements of the queries.

In earlier versions of Oracle Database 19c, the behavior is different for the two types of undo tablespaces. For AUTOEXTEND undo tablespaces, the system retains undo for at least the time specified in this parameter, and automatically tunes the undo retention period to satisfy the undo requirements of the queries. For fixed-size undo tablespaces, the system automatically tunes for the maximum possible undo retention period, based on undo tablespace size and usage history, and ignores UNDO_RETENTION unless retention guarantee is enabled for the tablespace.

Automatic tuning of undo retention is not supported for LOBs. The RETENTION value for LOB columns is set to the value of the UNDO_RETENTION parameter.

The UNDO_RETENTION parameter is honored only if the current undo tablespace has enough space. If an active transaction requires undo space and the undo tablespace does not have available space, then the system starts reusing unexpired undo space. This action can potentially cause some queries to fail with a "snapshot too old" message.

The amount of time for which undo is retained for the Oracle Database for the current undo tablespace can be obtained by querying the TUNED_UNDORETENTION column of the V$UNDOSTAT dynamic performance view.

Recommendations for Modifying UNDO_RETENTION

Oracle generally recommends that you leave UNDO_RETENTION set to its default value. Modifying this parameter is recommended only in the following situations:

- You may want to increase the value of UNDO_RETENTION when using flashback features, such as Oracle Flashback Query, which require undo to be retained for longer than the longest running query in the system.
- In Oracle Active Data Guard environments, you may want to increase the value of UNDO_RETENTION on the primary instance in order to accommodate undo retention requirements on the standby instances. This allows the primary instance to retain undo for a longer period of time to serve queries on the standby instances. For more information, see *Oracle Database Administrator's Guide* and *Oracle Data Guard Concepts and Administration*.

Modifying UNDO_RETENTION in a CDB

You can set the value of UNDO_RETENTION in the CDB root (CDB$ROOT) and in individual PDBs. The CDB root and PDBs can have the same value or different values. Note that you can modify this parameter in a PDB only if the PDB is in local undo mode; you cannot modify this parameter if the PDB is in shared undo mode.

Starting with Oracle Database 19c, Release Update 19.9, the value of UNDO_RETENTION is not inherited in a CDB. Therefore, if you change the value for UNDO_RETENTION in the CDB root, the PDBs do not inherit the new value. If you want to change the value for UNDO_RETENTION at the

Copyright © 2002, 2026, Oracle and/or its affiliates.

PDB level, you must explicitly do so. For example, if your CDB contains two PDBs (PDB1 and PDB2), you can set UNDO_RETENTION to 2000 in the CDB root and both PDBs as follows:

```
ALTER SESSION SET CONTAINER = CDB$ROOT;
ALTER SYSTEM SET UNDO_RETENTION=2000 SCOPE=BOTH;

ALTER SESSION SET CONTAINER = PDB1;
ALTER SYSTEM SET UNDO_RETENTION=2000 SCOPE=BOTH;

ALTER SESSION SET CONTAINER = PDB2;
ALTER SYSTEM SET UNDO_RETENTION=2000 SCOPE=BOTH;
```

See Also

- *Oracle Database SQL Language Reference* for information about creating undo tablespaces
- *Oracle Database Administrator's Guide* for information about managing undo data

1.372 UNDO_TABLESPACE

UNDO_TABLESPACE specifies the undo tablespace to be used when an instance starts. If this parameter is specified when the instance is in manual undo management mode, then an error will occur and startup will fail.

Property	Description
Parameter type	String
Syntax	UNDO_TABLESPACE = *undoname*
Default value	The first available undo tablespace in the database.
Modifiable	ALTER SYSTEM
Modifiable in a PDB	Yes
Range of values	Legal name of an existing undo tablespace
Basic	Yes
Oracle RAC	Each instance must have a unique value for this parameter, when it is set.

If the UNDO_TABLESPACE parameter is omitted, the first available undo tablespace in the database is chosen. If no undo tablespace is available, the instance will start without an undo tablespace. In such cases, user transactions will be executed using the SYSTEM rollback segment. You should avoid running in this mode under normal circumstances.

You can replace an undo tablespace with another undo tablespace while the instance is running.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> When you update this parameter on the primary database in an Oracle Data Guard configuration, you must also update it on all the physical standby databases in the configuration. This ensures that the standby databases can find the undo tablespace when they become the primary database.

> **See Also**
>
> *Oracle Database SQL Language Reference* for information about creating undo tablespaces

1.373 UNIFIED_AUDIT_COMMON_SYSTEMLOG

`UNIFIED_AUDIT_COMMON_SYSTEMLOG` specifies whether key fields of unified audit records generated due to common audit policies will be written to the SYSLOG utility.

> **Note**
>
> This initialization parameter is supported only on UNIX platforms.

Property	Description
Parameter type	String

Syntax	`UNIFIED_AUDIT_COMMON_SYSTEMLOG = '`*`facility_clause.priority_clause`*`'`
	`facility_clause``::=`
	`{ USER | LOCAL[ 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 ] }`
	`priority_clause``::=`
	`{ NOTICE | INFO | DEBUG | WARNING | ERR | CRIT | ALERT | EMERG }`

Default value	None
Modifiable	No
Modifiable in a PDB	No
Basic	No
Oracle RAC	The same value must be used on all instances.

When this parameter is set, key fields of unified audit records generated due to common audit policies are written to SYSLOG. These fields uniquely identify the detailed unified audit records in the `UNIFIED_AUDIT_TRAIL` view. Only a subset of unified audit record fields are written to ensure that the audit record entries do not exceed the maximum allowed size for a SYSLOG entry (typically 1024 bytes).

Do not set this parameter if you do not want key fields of unified audit records generated due to common audit policies written to SYSLOG.

Copyright © 2002, 2026, Oracle and/or its affiliates.

This parameter differs from the UNIFIED_AUDIT_SYSTEMLOG parameter in that it is set at the CDB level and enables all unified audit records from common unified audit policies to be consolidated into a single destination, whereas UNIFIED_AUDIT_SYSTEMLOG is set at the PDB level and enables the logging of unified audit records on a per-PDB basis.

Note

This parameter is available starting with Oracle Database 19c.

See Also

- "UNIFIED_AUDIT_SYSTEMLOG"
- "UNIFIED_AUDIT_TRAIL"
- *Oracle Database Security Guide* for a table that maps the names given to the unified audit records fields that are written to SYSLOG to the corresponding column names in the UNIFIED_AUDIT_TRAIL view

1.374 UNIFIED_AUDIT_SGA_QUEUE_SIZE

UNIFIED_AUDIT_SGA_QUEUE_SIZE specifies the size of SGA queue for unified auditing.

Property	Description
Parameter type	Integer
Default value	1 MB
Modifiable	No
Modifiable in a PDB	No
Range of values	1 MB to 30 MB
Basic	No
Oracle RAC	Multiple instances can have different values

UNIFIED_AUDIT_SGA_QUEUE_SIZE can be useful when queued-write mode is used for unified auditing. In queued-write mode, audit records are first written to the SGA queue. When the SGA queue reaches a particular threshold, the audit records are flushed to the AUDSYS schema table. The SGA queue size should be tuned according to the audit data generation. If large numbers of audit records are generated very frequently, you can increase the size of SGA queue, so that frequent flushes of queue data can be prevented. Similarly, if fewer audit records are generated, a smaller size can be used for the SGA queue.

Note

The UNIFIED_AUDIT_SGA_QUEUE_SIZE initialization parameter is deprecated in Oracle Database 12*c* Release 2 (12.2.0.1), and may be desupported in a future release.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Security Guide* for more information about writing unified audit trail records to the AUDSYS schema.

1.375 UNIFIED_AUDIT_SYSTEMLOG

UNIFIED_AUDIT_SYSTEMLOG specifies whether key fields of unified audit records will be written to the SYSLOG utility (on UNIX platforms) or to the Windows Event Viewer (on Windows). In a CDB, this parameter is a per-PDB static initialization parameter.

Property	Description
Parameter type	String for UNIX platforms, Boolean for Windows
Syntax	**On UNIX:** UNIFIED_AUDIT_SYSTEMLOG = '*facility_clause.priority_clause*' *facility_clause*::= { USER | LOCAL[0 | 1 | 2 | 3 | 4 | 5 | 6 | 7] } *priority_clause*::= { NOTICE | INFO | DEBUG | WARNING | ERR | CRIT | ALERT | EMERG } **On Windows:** UNIFIED_AUDIT_SYSTEMLOG = { FALSE | TRUE }
Default value	No default on UNIX platforms FALSE on Windows
Modifiable	No
Modifiable in a PDB	Yes
Basic	No
Oracle RAC	The same value must be used on all instances.

When this parameter is set on UNIX, key fields of unified audit records are written to SYSLOG. When this parameter is set on Windows, key fields of unified audit records are written to the Windows Event Viewer.

Do not set this parameter (or set it to FALSE on Windows) if you do not want key fields of unified audit records written to SYSLOG or the Windows Event Viewer.

When UNIFIED_AUDIT_SYSTEMLOG is enabled, the key fields of the unified audit records that are written to SYSLOG or Windows Event Viewer uniquely identify the detailed unified audit records in the UNIFIED_AUDIT_TRAIL view. Only a subset of the unified audit record fields are written to ensure that the audit record entries do not exceed the maximum allowed size for a SYSLOG entry (typically 1024 bytes).

This parameter differs from the UNIFIED_AUDIT_COMMON_SYSTEMLOG parameter in that it is set at the PDB level and enables the logging of unified audit records on a per-PDB basis, whereas UNIFIED_AUDIT_COMMON_SYSTEMLOG is set at the CDB level and enables all unified audit records from common unified audit policies to be consolidated into a single destination.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "UNIFIED_AUDIT_COMMON_SYSTEMLOG"
- "UNIFIED_AUDIT_TRAIL"
- *Oracle Database Security Guide* for a table that maps the names given to the unified audit records fields that are written to SYSLOG and the Windows Event Viewer to the corresponding column names in the `UNIFIED_AUDIT_TRAIL` view

1.376 UNIFORM_LOG_TIMESTAMP_FORMAT

`UNIFORM_LOG_TIMESTAMP_FORMAT` specifies that a uniform timestamp format be used in Oracle Database trace (.trc) files and log files (such as the alert log).

Property	Description
Parameter type	Boolean
Default value	`true`
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No
Oracle RAC	Multiple instances should use the same value.

When the value of `UNIFORM_LOG_TIMESTAMP_FORMAT` is `TRUE`, the format used for timestamps in trace files is standardized on universal time with millisecond precision. For example:

```
2012-09-26 00:16:47.154
```

When the value of `UNIFORM_LOG_TIMESTAMP_FORMAT` is `FALSE`, trace files include a mix of timestamps using different precisions, with some timestamps showing local time and other timestamps showing universal time.

1.377 USE_DEDICATED_BROKER

`USE_DEDICATED_BROKER` determines how dedicated servers are spawned.

Property	Description
Parameter type	Boolean
Default value	`False` unless multithreaded Oracle is enabled[1]
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	`true | false`
Basic	No
Oracle RAC	If specified, all instances must use the same value

Copyright © 2002, 2026, Oracle and/or its affiliates.

[1] Multithreaded Oracle is enabled by setting the THREADED_EXECUTION initialization parameter to true.

When this parameter is set to false, the listener spawns a dedicated server directly.

When this parameter is set to true, the listener hands the connection to a dedicated connection broker that spawns the dedicated server. Unlike the listener, the dedicated connection broker is a database process, and hence it can implement policies leveraging database information before the spawn.

To enable the dedicated connection broker using the USE_DEDICATED_BROKER initialization parameter, you must:

1. Set the USE_DEDICATED_BROKER initialization parameter to true.
2. Set the DEDICATED_THROUGH_BROKER_*listener-name* parameter to on in the listener.ora file.

Performing these two steps configures the Connection Broker Process (N*nnn*).

> **Note**
>
> When multithreaded Oracle is enabled, the dedicated connection broker process is also enabled and used.

Configuration of the brokers is controlled by the CONNECTION_BROKERS initialization parameter.

> **See Also**
>
> - "CONNECTION_BROKERS" for more information about configuring brokers
> - Table F-1 for more information about the Connection Broker Process (N*nnn*)
> - "THREADED_EXECUTION" for more information about enabling multithreaded Oracle
> - *Oracle Database Net Services Reference* for more information about the DEDICATED_THROUGH_BROKER_*listener-name* parameter in the listener.ora file

1.378 USE_LARGE_PAGES

USE_LARGE_PAGES is used to manage the database's use of large pages for SGA memory.

> **Note**
>
> This parameter is applicable only on the Linux operating system. However, setting this parameter to FALSE can cause performance degradation on any platform.
>
> The exception is when USE_LARGE_PAGES is set to FALSE automatically in an Oracle ASM instance because MEMORY_TARGET is enabled (set to a non-zero value). In this case, the FALSE setting does not cause performance degradation.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Property	Description
Parameter type	String
Syntax	`USE_LARGE_PAGES = { TRUE | FALSE | AUTO | ONLY | AUTO_ONLY }`
Default value	`ONLY` for Exadata Database Service in the cloud, Base Database Service in the cloud, and databases created using Exadata automation tools such as Oracle Exadata Deployment Assistant (OEDA) and OEDA Command Line Utility (OEDACLI) `AUTO_ONLY` for all other Exadata systems `TRUE` for all other systems
Modifiable	No
Modifiable in a PDB	No
Basic	No
Oracle RAC	Multiple instances can use different values

This parameter does not affect process-private memory allocations.

Values

These values can be specified for `USE_LARGE_PAGES`:

- `TRUE`

  Specifies that the instance can use large pages if large pages are configured on the system.

  In Oracle Database 11*g* Release 2 (11.2.0.2), if there are not enough large pages configured on the system, then regular sized pages will be used to allocate SGA memory. This can cause the free large pages to go unused, and the operating system can allocate a huge amount of memory to create page tables to map SGA into physical pages for the Oracle processes. This may lead to `ORA-04030` errors and severe performance degradation on an instance.

  In Oracle Database 11*g* Release 2 (11.2.0.3) and later releases, Oracle allocates as much of the SGA as it can in large pages, and if it runs out, it will allocate the rest of the SGA using regular sized pages. This can cause the instance to create additional shared memory segments for the SGA, but the total SGA size will be unchanged. In this supported mixed page mode allocation, the database will exhaust the available large pages before switching to regular sized pages.

- `FALSE`

  Specifies that the instance will not use large pages. This setting is not recommended because it can cause severe performance degradation for the instance. Although the `USE_LARGE_PAGES` initialization parameter applies only to Linux systems, setting this parameter to `FALSE` can cause performance degradation on any platform.

- `AUTO`

  Specifies that, during startup, the instance will calculate and request the number of large pages it requires. If the operating system cannot fulfill this request, then the instance will start with a combination of large and regular pages.

- `ONLY`

  Specifies that the instance will fail to start if large pages cannot be used for the entire SGA memory. Oracle recommends this setting for consistent performance. This setting is the

Copyright © 2002, 2026, Oracle and/or its affiliates.

default setting for Exadata Database Service in the cloud and Base Database Service in the cloud.

When `USE_LARGE_PAGES` is set to `ONLY`, you must not explicitly set values for `MEMORY_TARGET` and `MEMORY_MAX_TARGET` in the initialization parameter file. Doing so will prevent the instance from starting.

- `AUTO_ONLY`

  Specifies that, during startup, the instance will calculate and request the number of large pages it requires. If the operating system can fulfill this request, then the instance will start successfully. If the operating system cannot fulfill this request, then the instance will fail to start.

  This setting is available starting with Oracle Database 19c.

See Also

- "MEMORY_TARGET"
- "MEMORY_MAX_TARGET"

1.379 USER_DUMP_DEST

`USER_DUMP_DEST` specifies the pathname for a directory where the server will write debugging trace files on behalf of a user process.

Property	Description
Parameter type	String
Syntax	`USER_DUMP_DEST = { pathname | directory }`
Default value	Operating system-dependent
Modifiable	`ALTER SYSTEM`
Modifiable in a PDB	No
Range of values	Any valid local path, directory, or disk
Basic	No

Note

The `USER_DUMP_DEST` initialization parameter is deprecated.

For example, this directory might be set as follows:

- On MS-DOS: `C:\ORACLE\UTRC`
- On UNIX: `/oracle/utrc`
- On VMS: `DISK$UR3:[ORACLE.UTRC]`

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This parameter is ignored by the diagnosability infrastructure introduced in Oracle Database 11g Release 1 (11.1), which places trace and core files in a location controlled by the DIAGNOSTIC_DEST initialization parameter.

> **See Also**
>
> - *Oracle Database SQL Tuning Guide* for more information about the use of trace files
> - Your operating system-specific Oracle documentation for the range of values

1.380 WALLET_ROOT

WALLET_ROOT specifies the path to the root of a directory tree containing a subdirectory for each pluggable database (PDB).

Property	Description
Parameter type	String
Syntax	WALLET_ROOT = wallet-root-directory-path
Default value	There is no default value.
Modifiable	No
Modifiable in a PDB	No
Basic	No
Oracle RAC	Multiple instances must have the same value.

The name of the various wallet files is always the same, regardless of the component they are associated with. The wallets for each component are stored under each PDB GUID directory within the WALLET_ROOT directory structure in a directory whose name is based on the component name. For example, for the TDE component, the subdirectory name is tde.

If the WALLET_ROOT parameter is not set, the SQLNET.ENCRYPTION_WALLET_LOCATION parameter is used (as in Oracle Database releases prior to Oracle Database 18c), but no isolated keystore can be used unless the WALLET_ROOT parameter is set. The TDE_CONFIGURATION initialization parameter cannot be used to configure any PDB to run in isolated mode unless the WALLET_ROOT parameter is also set.

> **Note**
>
> The SQLNET.ENCRYPTION_WALLET_LOCATION parameter is deprecated in Oracle Database 18c.

Copyright © 2002, 2026, Oracle and/or its affiliates.

For example, the contents of the directory at the location specified by the WALLET_ROOT initialization parameter could look as follows, where *wallet-root* is the directory specified by the WALLET_ROOT parameter:

```
wallet-root/eus/ewallet.p12
wallet-root/tde/ewallet.p12
wallet-root/tde/ewallet_2016120918333644.p12
wallet-root/tde_seps/cwallet.sso
wallet-root/tls/ewallet.p12
wallet-root/xdb_wallet/ewallet.p12
wallet-root/3FD1C95B48205D0FE053C5A0E40AEF8C/tde/ewallet.p12
wallet-root/3FD1C95B48205D0FE053C5A0E40AEF8C/tde/ewallet_2016110918331622.p12
wallet-root/3FD1C95B48205D0FE053C5A0E40AEF8C/tde/ewallet_2016110918332363.p12
wallet-root/3FD1C95B48205D0FE053C5A0E40AEF8C/tde_seps/cwallet.sso
wallet-root/3FD1C95B48205D0FE053C5A0E40AEF8C/tls/cwallet.sso
wallet-root/3FD1C95B48205D0FE053C5A0E40AEF8C/tls/ewallet.p12
```

When the WALLET_ROOT parameter is set, you can omit the path from some ADMINISTER KEY MANAGEMENT commands.

The WALLET_ROOT value can include references to environment variables. The following example shows how to use WALLET_ROOT when multiple TDE-enabled databases are installed into the same ORACLE_HOME:

```
WALLET_ROOT=/etc/ORACLE/KEYSTORES/$ORACLE_SID
```

The ORACLE_SID environment variable (or the DB_UNIQUE_NAME environment variable for Oracle RAC) makes sure that each database that is installed into the same ORACLE_HOME has its own set of wallets and TDE keys. This sets the root of the wallet directory hierarchy to the directory specified by wallet-root-directory-path.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The normalized length of the `wallet-root-directory-name` that is specified with the `WALLET_ROOT` parameter cannot exceed 255 characters, otherwise one of the following sets of error messages is displayed:
>
> ```
> ORA-46693: The WALLET_ROOT location is missing or invalid.
> ORA-32021: parameter value longer than 255 characters
> ORA-01078: failure in processing system parameters
>
>
> ORA-46693: The WALLET_ROOT location is missing or invalid.
> ORA-07204: sltln: name translation failed due to lack of output buffer
> space.
> ORA-01078: failure in processing system parameters
> ```
>
> The normalized length includes the length of expanded environment variables specified with the `WALLET_ROOT` parameter. The values of the environment variables of the user who starts the instance are used in the normalization of the `WALLET_ROOT` parameter.
>
> The `SHOW PARAMETER WALLET_ROOT` command always displays the normalized value (with all the environment variables expanded).

For non-ASM file systems, the PDB GUID-extended paths for the TDE component are created automatically under the directory specified by the `WALLET_ROOT` parameter when any Transparent Data Encryption (TDE) wallet is created for a PDB.

Enabling Automatic Creation of Directories Under WALLET_ROOT

By using the specific configuration of `WALLET_ROOT` described in each of the following sub-sections, Oracle Database can be configured to automatically create the necessary *pdb-guid* and *component name* directories under the `WALLET_ROOT` directory path. Other settings of `WALLET_ROOT` are allowed, but would not result in the automatic creation of the necessary sub-directories by the ASM OMF layer.

Required setting to enable auto-directory creation for a database not using Oracle ASM

For a database not using Oracle ASM filesystems, the `WALLET_ROOT` parameter needs to be set as follows:

```
WALLET_ROOT=wallet-root-directory-path
```

This sets the root of the wallet directory hierarchy to the directory specified by `wallet-root-directory-path`. For example:

```
/etc/ORACLE/KEYSTORES/FINANCE
```

When this is done, Oracle Database automatically creates the directory for the TDE wallet of a CDB$ROOT at the following location:

```
/etc/ORACLE/KEYSTORES/FINANCE/tde
```

The directories, that Oracle Database automatically creates for holding the TDE wallets of isolated PDBs, will include the *pdb-guid*. For example:

Copyright © 2002, 2026, Oracle and/or its affiliates.

```
/etc/ORACLE/KEYSTORES/FINANCE/3FD1C95B48205D0FE053C5A0E40AEF8C/tde
```

Required setting to enable auto-directory creation for a database using Oracle ASM with Oracle Managed Files

For a database using ASM with OMF, the `WALLET_ROOT` parameter needs to begin with a plus sign followed by a disk group name and the value of the `DB_UNIQUE_NAME` initialization parameter. In the example below, `DATA` is the name of a disk group and `FINRAC` is the value of the `DB_UNIQUE_NAME` initialization parameter:

```
WALLET_ROOT=+DATA/FINRAC
```

When this is done, Oracle Database automatically creates the necessary directory within the ASM filesystem at the following location when the `ADMINISTER KEY MANAGEMENT CREATE KEYSTORE` command is run:

```
+DATA/FINRAC/tde
```

For isolated PDBs, the directories that Oracle Database automatically creates for holding the TDE wallets of PDBs will include the *pdb-guid*. For example:

```
+DATA/FINRAC/3FD1C95B48205D0FE053C5A0E40AEF8C/tde
```

Required setting to enable auto-directory creation for RAC-enabled databases

For a RAC-enabled database, only shared TDE wallets are supported (as opposed to individual TDE-wallets per RAC instance). `WALLET_ROOT` can either point to an ASM disk group, or a directory in ACFS. If the `WALLET_ROOT` parameter points to `+diskgroup/dbname`, then the `/tde` sub-directory is automatically created when issuing an `ADMINISTER KEY MANAGEMENT CREATE KEYSTORE` command. For example, when`WALLET_ROOT` is set to `+DATA/FINANCE`, the directory `+DATA/FINANCE/tde` is automatically created. This guarantees that when multiple databases are installed, their TDE-wallets are kept separate.

> **See Also**
>
> "TDE_CONFIGURATION"

1.381 WORKAREA_SIZE_POLICY

`WORKAREA_SIZE_POLICY` specifies the policy for sizing work areas. This parameter controls the mode in which working areas are tuned.

Property	Description
Parameter type	String
Syntax	`WORKAREA_SIZE_POLICY = { AUTO | MANUAL }`
Default value	`AUTO`
Modifiable	`ALTER SESSION`, `ALTER SYSTEM`
Modifiable in a PDB	Yes
Basic	No

Copyright © 2002, 2026, Oracle and/or its affiliates.

Values

You can specify the following values for WORKAREA_SIZE_POLICY:

- AUTO

  When AUTO is specified, work areas used by memory-intensive operators are sized automatically, based on the PGA memory used by the system, the target PGA memory set in PGA_AGGREGATE_TARGET, and the requirement of each individual operator.

- MANUAL

  When MANUAL is specified, the sizing of work areas is manual and based on the values of the *_AREA_SIZE parameter corresponding to the operation (for example, a sort uses SORT_AREA_SIZE). Specifying MANUAL may result in sub-optimal performance and poor PGA memory utilization.

See Also

Oracle Database Performance Tuning Guide for additional information on setting this parameter

1.382 XML_DB_EVENTS

XML_DB_EVENTS enables or disables XML DB events.

Property	Description
Parameter type	String
Syntax	XML_DB_EVENTS = { enable | disable }
Default value	enable
Modifiable	ALTER SESSION, ALTER SYSTEM
Modifiable in a PDB	Yes
Basic	No

Changing this parameter through an ALTER SESSION statement affects only the current session. Only users with the XDBADMIN role are allowed to change this parameter in a session.

Changing this parameter through an ALTER SYSTEM statement will make a systemwide change of this parameter; however, the change is only registered by new sessions started after the change. Currently running sessions need to be restarted to pick up the new change. Users must have the ALTER SYSTEM privilege to make a systemwide change.

See Also

Oracle XML DB Developer's Guide for more information about this parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

Part II
Static Data Dictionary Views

This part describes data dictionary tables and views. These tables and views are called **static**, because they change only when a change is made to the data dictionary (for example, when a new table is created or when a user is granted new privileges).
This part contains the following chapters:

- Static Data Dictionary Views: ALL_ALL_TABLES to ALL_OUTLINES
- Static Data Dictionary Views: ALL_PART_COL_STATISTICS to DATABASE_PROPERTIES
- Static Data Dictionary Views: DBA_2PC_NEIGHBORS to DBA_HIST_JAVA_POOL_ADVICE
- Static Data Dictionary Views: DBA_HIST_LATCH to DBA_STORED_SETTINGS
- Static Data Dictionary Views: DBA_STREAMS_ADD_COLUMN to USER_ZONEMAPS

> **Note**
>
> Oracle also maintains views that monitor ongoing database activity. These **dynamic performance views** are described in Dynamic Performance Views .

Copyright © 2002, 2026, Oracle and/or its affiliates.

2

Static Data Dictionary Views: ALL_ALL_TABLES to ALL_OUTLINES

This chapter describes the first set (in alphabetical order) of static data dictionary views.

The remaining static data dictionary views appear in alphabetical order in Static Data Dictionary Views: ALL_PART_COL_STATISTICS to DATABASE_PROPERTIES through Static Data Dictionary Views: DBA_STREAMS_ADD_COLUMN to USER_ZONEMAPS.

This chapter contains the following topics:

- About Static Data Dictionary Views
- AWR_PDB_* Views
- AWR_ROOT_* Views
- CDB_* Views
- DBA_HIST_* Views
- Oracle Database Real Application Security Views
- Oracle Database Vault Views
- Oracle Label Security Views
- Oracle Workspace Manager Views
- Recovery Catalog Views
- Static Data Dictionary View Descriptions

2.1 About Static Data Dictionary Views

Data dictionary tables are not directly accessible, but you can access information in them through data dictionary views. To list the data dictionary views available to you, query the view `DICTIONARY`.
Many data dictionary tables have three corresponding views:

- An `ALL_` view displays all the information accessible to the current user, including information from the current user's schema as well as information from objects in other schemas, if the current user has access to those objects by way of grants of privileges or roles.
- A `DBA_` view displays all relevant information in the entire database. `DBA_` views are intended only for administrators. They can be queried only by users with the `SYSDBA` system privilege or `SELECT ANY DICTIONARY` privilege, or `SELECT_CATALOG_ROLE` role, or by users with direct privileges granted to them. The `SELECT ANY DICTIONARY` privilege is assigned to the `DBA` role when the system is initially installed.
- A `USER_` view displays all the information from the schema of the current user. No special privileges are required to query these views.

The columns of the `ALL_`, `DBA_`, and `USER_` views corresponding to a single data dictionary table are usually nearly identical. Therefore, these views are described in full only once in this

Copyright © 2002, 2026, Oracle and/or its affiliates.

chapter, at their first occurrence alphabetically, and are listed without full descriptions at their other occurrences.

> **See Also**
>
> "Static Data Dictionary View Descriptions" introduces the alphabetical listing of view descriptions that are documented in this manual

2.2 AWR_PDB_* Views

The AWR_PDB views show the local Automatic Workload Repository (AWR) data present on a CDB root or a PDB from where the AWR_PDB views are accessed.

The AWR_PDB views on a CDB root show the AWR data stored on the CDB root.

The AWR_PDB views on a PDB show the AWR data stored on that PDB.

> **See Also**
>
> - "AWR_ROOT_* Views"
> - *Oracle Database Performance Tuning Guide* for an introduction to AWR
> - *Oracle Database Performance Tuning Guide* for more information about AWR_PDB views and related views for accessing AWR in a multitenant environment

2.3 AWR_ROOT_* Views

The AWR_ROOT views show the AWR data stored only on a CDB root. In general, the AWR_ROOT views are equivalent to the DBA_HIST views.

When the AWR_ROOT views are accessed from a CDB root, they show the AWR data specific to the CDB root.

When the AWR_ROOT views are accessed from a PDB, they show the AWR data specific to that PDB.

> **See Also**
>
> - "AWR_PDB_* Views"
> - *Oracle Database Performance Tuning Guide* for an introduction to AWR
> - *Oracle Database Performance Tuning Guide* for more information about AWR_ROOT views and related views for accessing AWR in a multitenant environment

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.4 CDB_* Views

For every DBA_* view, a CDB_* view is defined. In the root of a multitenant container database (CDB), CDB_* views can be used to obtain information about tables, tablespaces, users, privileges, parameters, and so on contained in the root and in pluggable databases (PDBs). The CDB_* views can be queried only by users with the `SYSDBA` system privilege or `SELECT ANY DICTIONARY` privilege, or `SELECT_CATALOG_ROLE` role, or by users with direct privileges granted to them.

CDB_* views are container data objects. When a user connected to the root queries a CDB_* view, the query results will depend on the `CONTAINER_DATA` attribute for users for the view. The `CONTAINER_DATA` clause of the SQL `ALTER USER` statement is used to set and modify users' `CONTAINER_DATA` attribute.

The CDB_* views are owned by `SYS`, regardless of who owns the underlying DBA_* view.

By default, a user connected to the root will only see data pertaining to the root.

See Also

- *Oracle Database Security Guide* for more information about container data objects
- *Oracle Database SQL Language Reference* for more information about the `CONTAINER_DATA` clause for the SQL `ALTER USER` statement

CDB_* views include these hidden columns:

- `CON$NAME`: This column includes the name of the container whose data a given CDB_* row represents
- `CDB$NAME`: This column displays the name of the CDB whose data a given CDB_* row represents

In a PDB, the CDB_* views only show objects visible through a corresponding DBA_* view.

In addition to all the columns found in a given DBA_* view, the corresponding CDB_* view also contains the `CON_ID` column, which identifies a container whose data a given CDB_* row represents. In a non-CDB, the value of a `CON_ID` column will be `0`.

CDB views can return data from different containers in a CDB when queried from the root container. These objects will implicitly convert data to the character set of the root container (AL32UTF8) and then return the result to the user. Some character sets may have character expansion (more bytes needed to represent a character) when converted to AL32UTF8, so there may be data truncation if the view column width is not able to accommodate data from a given PDB.

Data is returned by these views from all open PDBs at the time the query is issued, except for PDBs that are open in RESTRICTED mode. In an Oracle RAC environment, data returned by these views may vary according to the instance to which a session is connected.

2.5 DBA_HIST_* Views

The DBA_HIST views show the local Automatic Workload Repository (AWR) data present on a CDB root or a PDB.

Copyright © 2002, 2026, Oracle and/or its affiliates.

If a DBA_HIST view is queried from a CDB root, the view shows all the AWR data stored on the CDB root.

If a DBA_HIST view is queried from a PDB, the view shows the subset of the CDB root AWR data that is specific to that PDB. Also, the view shows the PDB level snapshots at that PDB, if they exist.

See Also

- *Oracle Database Performance Tuning Guide* for an introduction to AWR
- *Oracle Database Performance Tuning Guide* for an introduction to Oracle Database views for accessing AWR data stored on the CDB root and individual PDBs in a multitenant environment

2.6 Oracle Database Real Application Security Views

This manual describes these Oracle Database Real Application Security auditing views:

- DBA_XS_AUDIT_POLICY_OPTIONS
- DBA_XS_AUDIT_TRAIL
- DBA_XS_ENB_AUDIT_POLICIES

Descriptions of the other Oracle Database Real Application Security views are provided in *Oracle Database Real Application Security Administrator's and Developer's Guide*.

Note

Oracle Database Real Application Security views include `_XS_` in the view name.

2.7 Oracle Database Vault Views

Descriptions of Oracle Database Vault views are not provided in this manual.

See *Oracle Database Vault Administrator's Guide* for descriptions of Oracle Database Vault views.

Note

Oracle Database Vault views include `_DV_` in the view name.

2.8 Oracle Label Security Views

Descriptions of Oracle Label Security views are not provided in this manual.

See *Oracle Label Security Administrator's Guide* for descriptions of Oracle Label Security views.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> Oracle Label Security views include _SA_ in the view name.

2.9 Oracle Workspace Manager Views

A number of data dictionary views are relevant only if you are using Oracle Workspace Manager:

- ALL_MP_GRAPH_WORKSPACES and USER_MP_GRAPH_WORKSPACES
- ALL_MP_PARENT_WORKSPACES and USER_MP_PARENT_WORKSPACES
- ALL_REMOVED_WORKSPACES and USER_REMOVED_WORKSPACES
- ALL_VERSION_HVIEW
- ALL_WM_CONS_COLUMNS and USER_WM_CONS_COLUMNS
- ALL_WM_CONSTRAINTS and USER_WM_CONSTRAINTS
- ALL_WM_IND_COLUMNS and USER_WM_IND_COLUMNS
- ALL_WM_IND_EXPRESSIONS and USER_WM_IND_EXPRESSIONS
- ALL_WM_LOCKED_TABLES and USER_WM_LOCKED_TABLES
- ALL_WM_MODIFIED_TABLES and USER_WM_MODIFIED_TABLES
- ALL_WM_RIC_INFO and USER_WM_RIC_INFO
- ALL_WM_TAB_TRIGGERS and USER_WM_TAB_TRIGGERS
- ALL_WM_VERSIONED_TABLES and USER_WM_VERSIONED_TABLES
- ALL_WM_VT_ERRORS, DBA_WM_VT_ERRORS, and USER_WM_VT_ERRORS
- ALL_WORKSPACE_PRIVS and USER_WORKSPACE_PRIVS
- ALL_WORKSPACE_SAVEPOINTS and USER_WORKSPACE_SAVEPOINTS
- ALL_WORKSPACES, DBA_WORKSPACES, and USER_WORKSPACES
- DBA_WM_SYS_PRIVS
- DBA_WORKSPACE_SESSIONS
- ROLE_WM_PRIVS
- USER_WM_PRIVS
- WM_COMPRESS_BATCH_SIZES
- WM_COMPRESSIBLE_TABLES
- WM_EVENTS_INFO
- WM_INSTALLATION

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Workspace Manager Developer's Guide* for information about these views

2.10 Recovery Catalog Views

The following data dictionary views are only available after you create an optional recovery catalog (which contains schemas containing information about backups) for use with Recovery Manager:

- RC_ARCHIVED_LOG
- RC_BACKUP_ARCHIVELOG_DETAILS
- RC_BACKUP_ARCHIVELOG_SUMMARY
- RC_BACKUP_CONTROLFILE
- RC_BACKUP_CONTROLFILE_DETAILS
- RC_BACKUP_CONTROLFILE_SUMMARY
- RC_BACKUP_COPY_DETAILS
- RC_BACKUP_COPY_SUMMARY
- RC_BACKUP_CORRUPTION
- RC_BACKUP_DATAFILE
- RC_BACKUP_DATAFILE_DETAILS
- RC_BACKUP_DATAFILE_SUMMARY
- RC_BACKUP_FILES
- RC_BACKUP_PIECE
- RC_BACKUP_PIECE_DETAILS
- RC_BACKUP_REDOLOG
- RC_BACKUP_SET
- RC_BACKUP_SET_DETAILS
- RC_BACKUP_SET_SUMMARY
- RC_BACKUP_SPFILE
- RC_BACKUP_SPFILE_DETAILS
- RC_BACKUP_SPFILE_SUMMARY
- RC_CHECKPOINT
- RC_CONTROLFILE_COPY
- RC_COPY_CORRUPTION
- RC_DATABASE
- RC_DATABASE_BLOCK_CORRUPTION
- RC_DATABASE_INCARNATION
- RC_DATAFILE
- RC_DATAFILE_COPY
- RC_LOG_HISTORY
- RC_OFFLINE_RANGE
- RC_PROXY_ARCHIVEDLOG
- RC_PROXY_ARCHIVELOG_DETAILS
- RC_PROXY_ARCHIVELOG_SUMMARY
- RC_PROXY_CONTROLFILE
- RC_PROXY_COPY_DETAILS

Copyright © 2002, 2026, Oracle and/or its affiliates.

RC_PROXY_COPY_SUMMARY
RC_PROXY_DATAFILE
RC_REDO_LOG
RC_REDO_THREAD
RC_RESTORE_POINT
RC_RESYNC
RC_RMAN_BACKUP_JOB_DETAILS
RC_RMAN_BACKUP_SUBJOB_DETAILS
RC_RMAN_BACKUP_TYPE
RC_RMAN_CONFIGURATION
RC_RMAN_OUTPUT
RC_RMAN_STATUS
RC_SITE
RC_STORED_SCRIPT
RC_STORED_SCRIPT_LINE
RC_TABLESPACE
RC_TEMPFILE
RC_UNUSABLE_BACKUPFILE_DETAILS

See Also

Oracle Database Backup and Recovery Reference for information about these views

2.11 Static Data Dictionary View Descriptions

The remainder of this chapter describes the static data dictionary views in alphabetical order.

2.12 ALL_ALL_TABLES

ALL_ALL_TABLES describes the object tables and relational tables accessible to the current user.

Related Views

- DBA_ALL_TABLES describes all object tables and relational tables in the database.
- USER_ALL_TABLES describes the object tables and relational tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the table; NULL for partitioned, temporary, and index-organized tables
CLUSTER_NAME	VARCHAR2(128)		Name of the cluster, if any, to which the table belongs

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
IOT_NAME	VARCHAR2(128)		Name of the index-organized table, if any, to which the overflow or mapping table entry belongs. If the IOT_TYPE column is not NULL, then this column contains the base table name.
STATUS	VARCHAR2(8)		If a previous DROP TABLE operation failed, indicates whether the table is unusable (UNUSABLE) or valid (VALID)
PCT_FREE	NUMBER		Minimum percentage of free space in a block; NULL for partitioned tables
PCT_USED	NUMBER		Minimum percentage of used space in a block; NULL for partitioned tables
INI_TRANS	NUMBER		Initial number of transactions; NULL for partitioned tables
MAX_TRANS	NUMBER		Maximum number of transactions; NULL for partitioned tables
INITIAL_EXTENT	NUMBER		Size of the initial extent (in bytes); NULL for partitioned tables
NEXT_EXTENT	NUMBER		Size of secondary extents (in bytes); NULL for partitioned tables
MIN_EXTENTS	NUMBER		Minimum number of extents allowed in the segment; NULL for partitioned tables
MAX_EXTENTS	NUMBER		Maximum number of extents allowed in the segment; NULL for partitioned tables
PCT_INCREASE	NUMBER		Percentage increase in extent size; NULL for partitioned tables
FREELISTS	NUMBER		Number of process freelists allocated to the segment; NULL for partitioned tables
FREELIST_GROUPS	NUMBER		Number of freelist groups allocated to the segment
LOGGING	VARCHAR2(3)		Indicates whether or not changes to the table are logged: • YES • NO
BACKED_UP	VARCHAR2(1)		Indicates whether the table has been backed up since the last modification (Y) or not (N)
NUM_ROWS	NUMBER		Number of rows in the table
BLOCKS	NUMBER		Number of used blocks in the table
EMPTY_BLOCKS	NUMBER		Number of empty (never used) blocks in the table
AVG_SPACE	NUMBER		Average available free space in the table
CHAIN_CNT	NUMBER		Number of rows in the table that are chained from one data block to another or that have migrated to a new block, requiring a link to preserve the old rowid. This column is updated only after you analyze the table.
AVG_ROW_LEN	NUMBER		Average row length, including row overhead
AVG_SPACE_FREELIST_BLOCKS	NUMBER		Average freespace of all blocks on a freelist
NUM_FREELIST_BLOCKS	NUMBER		Number of blocks on the freelist

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEGREE	VARCHAR2(10)		Number of threads per instance for scanning the table, or DEFAULT
INSTANCES	VARCHAR2(10)		Number of instances across which the table is to be scanned, or DEFAULT
CACHE	VARCHAR2(5)		Indicates whether the table is to be cached in the buffer cache (Y) or not (N)
TABLE_LOCK	VARCHAR2(8)		Indicates whether table locking is enabled (ENABLED) or disabled (DISABLED)
SAMPLE_SIZE	NUMBER		Sample size used in analyzing the table
LAST_ANALYZED	DATE		Date on which the table was most recently analyzed
PARTITIONED	VARCHAR2(3)		Indicates whether the table is partitioned (YES) or not (NO)
IOT_TYPE	VARCHAR2(12)		If the table is an index-organized table, then IOT_TYPE is IOT, IOT_OVERFLOW, or IOT_MAPPING. If the table is not an index-organized table, then IOT_TYPE is NULL.
OBJECT_ID_TYPE	VARCHAR2(16)		Indicates whether the object ID (OID) is USER-DEFINED or SYSTEM GENERATED
TABLE_TYPE_OWNER	VARCHAR2(128)		If an object table, owner of the type from which the table is created
TABLE_TYPE	VARCHAR2(128)		If an object table, type of the table
TEMPORARY	VARCHAR2(1)		Indicates whether the table is temporary (Y) or not (N)
SECONDARY	VARCHAR2(1)		Indicates whether the table is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge to contain the contents of a domain index (Y) or not (N)
NESTED	VARCHAR2(3)		Indicates whether the table is a nested table (YES) or not (NO)
BUFFER_POOL	VARCHAR2(7)		Buffer pool to be used for table blocks: • DEFAULT • KEEP • RECYCLE • NULL
FLASH_CACHE	VARCHAR2(7)		Database Smart Flash Cache hint to be used for table blocks: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)		Cell flash cache hint to be used for table blocks: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
ROW_MOVEMENT	VARCHAR2(8)		If a partitioned table, indicates whether row movement is enabled (ENABLED) or disabled (DISABLED)
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics are gathered or incrementally maintained, otherwise it will be NO

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
DURATION	VARCHAR2(15)		Indicates the duration of a temporary table: SYS$SESSION - Rows are preserved for the duration of the session SYS$TRANSACTION - Rows are deleted after COMMIT Null - Permanent table
SKIP_CORRUPT	VARCHAR2(8)		Indicates whether Oracle Database ignores blocks marked corrupt during table and index scans (ENABLED) or raises an error (DISABLED). To enable this feature, run the DBMS_REPAIR.skip_corrupt_blocks procedure.
MONITORING	VARCHAR2(3)		Indicates whether the table has the MONITORING attribute set (YES) or not (NO)
CLUSTER_OWNER	VARCHAR2(128)		Owner of the cluster, if any, to which the table belongs
DEPENDENCIES	VARCHAR2(8)		Indicates whether row-level dependency tracking is enabled (ENABLED) or disabled (DISABLED)
COMPRESSION	VARCHAR2(8)		Indicates whether table compression is enabled (ENABLED) or not (DISABLED); NULL for partitioned tables
COMPRESS_FOR	VARCHAR2(30)		Default compression for what kind of operations: • BASIC • ADVANCED • QUERY LOW • QUERY HIGH • ARCHIVE LOW • ARCHIVE HIGH • NULL The QUERY LOW, QUERY HIGH, ARCHIVE LOW, and ARCHIVE HIGH values are associated with Hybrid Columnar Compression, a feature of the Enterprise Edition of Oracle Database that is dependent on the underlying storage system. See *Oracle Database Concepts* for more information.
DROPPED	VARCHAR2(3)		Indicates whether the table has been dropped and is in the recycle bin (YES) or not (NO); NULL for partitioned tables This view does not return the names of tables that have been dropped.
SEGMENT_CREATED	VARCHAR2(3)		Indicates whether the table segment has been created (YES) or not (NO)
INMEMORY	VARCHAR2(8)		Indicates whether the In-Memory Column Store (IM column store) is enabled (ENABLED) or disabled (DISABLED) for this table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INMEMORY_PRIORITY	VARCHAR2(8)		Indicates the priority at which this table is populated into the In-Memory Column Store (IM column store). Possible values: • LOW • MEDIUM • HIGH • CRITICAL • NONE • NULL This column has a value based on where the segments lie for a table. For example, if the table is partitioned and is enabled for the IM column store, the value is NULL for ALL_TABLES but non-NULL for ALL_TAB_PARTITIONS.
INMEMORY_DISTRIBUTE	VARCHAR2(15)		Indicates how the table will be distributed in the IM column stores in an Oracle Real Application Clusters (Oracle RAC) environment: • AUTO • BY ROWID RANGE • BY PARTITION • BY SUBPARTITION
INMEMORY_COMPRESSION	VARCHAR2(17)		Compression level for the in-memory store: • NO MEMCOMPRESS • FOR DML • FOR QUERY [LOW | HIGH] • FOR CAPACITY [LOW | HIGH] • NULL This column has a value based on where the segments lie for a table. For example, if the table is partitioned and is enabled for the IM column store, the value is NULL for ALL_TABLES but non-NULL for ALL_TAB_PARTITIONS.
INMEMORY_DUPLICATE	VARCHAR2(13)		Indicates the duplicate setting for the In-Memory Column Store (IM column store) in an Oracle RAC environment: • NO DUPLICATE • DUPLICATE • DUPLICATE ALL
EXTERNAL	VARCHAR2(3)		Indicates whether the table is an external table (YES) or not (NO)
HYBRID[1]	VARCHAR2(3)		Indicates whether the table is a hybrid partitioned table (YES) or not (NO). A hybrid partitioned table can contain a mixture of partitions stored in segments and partitions stored externally.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CELLMEMORY[2]	VARCHAR2(24)		The value for columnar compression in the storage cell flash cache. Possible values: • ENABLED: Oracle Exadata Storage will decide automatically whether to cache in columnar form • DISABLED: Oracle Exadata Storage is prevented from caching in columnar form • NO CACHECOMPRESS: Oracle Exadata Storage will cache in HCC format (no recompression) • FOR QUERY: Oracle Exadata Storage will recompress and cache in INMEMORY query high format • FOR CAPACITY: Oracle Exadata Storage will recompress and cache in INMEMORY capacity low format
INMEMORY_SERVICE	VARCHAR2(12)		Indicates how the IM column store is populated on various instances. The possible values are: • DEFAULT: Data is populated on all instances specified with the PARALLEL_INSTANCE_GROUP initialization parameter. If that parameter is not set, then the data is populated on all instances. This is the default. • NONE: Data is not populated on any instance. • ALL: Data is populated on all instances, regardless of the value of the PARALLEL_INSTANCE_GROUP initialization parameter. • USER_DEFINED: Data is populated only on the instances on which the user-specified service is active. The service name corresponding to this is stored in the INMEMORY_SERVICE_NAME column.
INMEMORY_SERVICE_NAME	VARCHAR2(1000)		Indicates the service name for the service on which the IM column store should be populated. This column has a value only when the corresponding INMEMORY_SERVICE is USER_DEFINED. In all other cases, this column is null.
MEMOPTIMIZE_READ	VARCHAR2(8)		Indicates whether the table is enabled for Fast Key Based Access (ENABLED) or not (DISABLED)
MEMOPTIMIZE_WRITE	VARCHAR2(8)		For internal use only
HAS_SENSITIVE_COLUMN	VARCHAR2(3)		Indicates whether the table has one or more sensitive columns (YES) or not (NO)
LOGICAL_REPLICATION[1]	VARCHAR2(8)		Indicates whether the table is enabled for logical replication (ENABLED) or not (DISABLED). This setting is ignored if database-wide column data supplemental logging is enabled.

1 This column is available starting with Oracle Database 19c.

2 This column is intended for use with Oracle Exadata.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_ALL_TABLES"
- "USER_ALL_TABLES"
- "PARALLEL_INSTANCE_GROUP"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_REPAIR.SKIP_CORRUPT_BLOCKS` procedure
- *Oracle Database In-Memory Guide* for an introduction to the IM column store
- *Oracle Database In-Memory Guide* for more information about the IM column store

2.13 ALL_ANALYTIC_VIEW_ATTR_CLASS

`ALL_ANALYTIC_VIEW_ATTR_CLASS` describes the attribute classifications of the analytic views accessible to the current user.

Related Views

- `DBA_ANALYTIC_VIEW_ATTR_CLASS` describes the attribute classifications of all analytic views in the database.
- `USER_ANALYTIC_VIEW_ATTR_CLASS` describes the attribute classifications of the analytic views owned by the current user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the analytic view
`ANALYTIC_VIEW_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of analytic view
`DIMENSION_ALIAS`	`VARCHAR2(128)`		Alias of the attribute dimension in the analytic view
`HIER_ALIAS`	`VARCHAR2(128)`		Alias of the hierarchy specified by an attribute dimension in the analytic view
`ATTRIBUTE_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the attribute within the analytic view
`CLASSIFICATION`	`VARCHAR2(128)`		Name of analytic view attribute classification
`VALUE`	`CLOB`		Value of attribute classification
`LANGUAGE`	`VARCHAR2(64)`		Language of attribute classification
`ORDER_NUM`	`NUMBER`	`NOT NULL`	Order number of the attribute classification
`ORIGIN_CON_ID`	`NUMBER`		The ID of the container where the data originates. Possible values include: • `0`: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_ANALYTIC_VIEW_ATTR_CLASS"
- "USER_ANALYTIC_VIEW_ATTR_CLASS"

2.14 ALL_ANALYTIC_VIEW_ATTR_CLS

ALL_ANALYTIC_VIEW_ATTR_CLS is identical to ALL_ANALYTIC_VIEW_ATTR_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_ATTR_CLASS"

2.15 ALL_ANALYTIC_VIEW_ATTR_CLS_AE

ALL_ANALYTIC_VIEW_ATTR_CLS_AE describes the attribute classifications of the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_ATTR_CLS_AE describes the attribute classifications of all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEW_ATTR_CLS_AE describes the attribute classifications of the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of analytic view
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the analytic view
HIER_ALIAS	VARCHAR2(128)		Alias of the hierarchy specified by an attribute dimension in the analytic view
ATTRIBUTE_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute within the analytic view
CLASSIFICATION	VARCHAR2(128)		Name of analytic view attribute classification
VALUE	CLOB		Value of attribute classification
LANGUAGE	VARCHAR2(64)		Language of attribute classification
ORDER_NUM	NUMBER	NOT NULL	Order number of the attribute classification

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ANALYTIC_VIEW_ATTR_CLS_AE"
- "USER_ANALYTIC_VIEW_ATTR_CLS_AE"

2.16 ALL_ANALYTIC_VIEW_BAS_MEAS

ALL_ANALYTIC_VIEW_BAS_MEAS is identical to ALL_ANALYTIC_VIEW_BASE_MEAS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_BASE_MEAS"

2.17 ALL_ANALYTIC_VIEW_BAS_MEAS_AE

ALL_ANALYTIC_VIEW_BAS_MEAS_AE describes the base measures in the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_BAS_MEAS_AE describes the base measures in all analytic views (across all editions) in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_ANALYTIC_VIEW_BAS_MEAS_AE describes the base measures in the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
MEASURE_NAME	VARCHAR2(128)		Name of the analytic view base measure
TABLE_ALIAS	VARCHAR2(128)		Alias of the table or view in the USING clause to which the column belongs
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Column name in the table or view on which this measure is defined
AGGR_FUNCTION	VARCHAR2(128)		Aggregation operator specified for this measure or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order number of the base measure in the list of measures in the analytic view
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ANALYTIC_VIEW_BAS_MEAS_AE"
- "USER_ANALYTIC_VIEW_BAS_MEAS_AE"

2.18 ALL_ANALYTIC_VIEW_BASE_MEAS

ALL_ANALYTIC_VIEW_BASE_MEAS describes the base measures in the analytic views accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_BASE_MEAS describes the base measures in all analytic views in the database.
- USER_ANALYTIC_VIEW_BASE_MEAS describes the base measures in the analytic views owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
MEASURE_NAME	VARCHAR2(128)		Name of the analytic view base measure
TABLE_ALIAS	VARCHAR2(128)		Alias of the table or view in the USING clause to which the column belongs
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Column name in the table or view on which this measure is defined
AGGR_FUNCTION	VARCHAR2(128)		Aggregation operator specified for this measure or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order number of the base measure in the list of measures in the analytic view
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ANALYTIC_VIEW_BASE_MEAS"
- "USER_ANALYTIC_VIEW_BASE_MEAS"

2.19 ALL_ANALYTIC_VIEW_CALC_MEAS

ALL_ANALYTIC_VIEW_CALC_MEAS describes the calculated measures in the analytic views accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_CALC_MEAS describes the calculated measures in all analytic views in the database.
- USER_ANALYTIC_VIEW_CALC_MEAS describes the calculated measures in the analytic views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
MEASURE_NAME	VARCHAR2(128)		Name of the analytic view calculated measure
MEAS_EXPRESSION	CLOB		Text of the expression for the measure
ORDER_NUM	NUMBER	NOT NULL	Order number of the calculated measure in the list of the measures in the analytic view

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ANALYTIC_VIEW_CALC_MEAS"
- "USER_ANALYTIC_VIEW_CALC_MEAS"

2.20 ALL_ANALYTIC_VIEW_CLASS

ALL_ANALYTIC_VIEW_CLASS describes the classifications of the analytic views accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_CLASS describes the classifications of all analytic views in the database.
- USER_ANALYTIC_VIEW_CLASS describes the classifications of the analytic views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
CLASSIFICATION	VARCHAR2(128)		Classification associated with the analytic view
VALUE	CLOB		Value of the classification or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the analytic view
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_ANALYTIC_VIEW_CLASS"
- "USER_ANALYTIC_VIEW_CLASS"

2.21 ALL_ANALYTIC_VIEW_CLASS_AE

ALL_ANALYTIC_VIEW_CLASS_AE describes the classifications of the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_CLASS_AE describes the classifications of all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEW_CLASS_AE describes the classifications of the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
CLASSIFICATION	VARCHAR2(128)		Classification associated with the analytic view
VALUE	CLOB		Value of the classification or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the analytic view
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ANALYTIC_VIEW_CLASS_AE"
- "USER_ANALYTIC_VIEW_CLASS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.22 ALL_ANALYTIC_VIEW_CLC_MEAS

ALL_ANALYTIC_VIEW_CLC_MEAS is identical to ALL_ANALYTIC_VIEW_CALC_MEAS.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_CALC_MEAS"

2.23 ALL_ANALYTIC_VIEW_CLC_MEAS_AE

ALL_ANALYTIC_VIEW_CLC_MEAS_AE describes the calculated measures in the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_CLC_MEAS_AE describes the calculated measures in all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEW_CLC_MEAS_AE describes the calculated measures in the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
MEASURE_NAME	VARCHAR2(128)		Name of the analytic view calculated measure
MEAS_EXPRESSION	CLOB		Text of the expression for the measure
ORDER_NUM	NUMBER	NOT NULL	Order number of the calculated measure in the list of the measures in the analytic view
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> - "DBA_ANALYTIC_VIEW_CLC_MEAS_AE"
> - "USER_ANALYTIC_VIEW_CLC_MEAS_AE"

2.24 ALL_ANALYTIC_VIEW_COLUMNS

ALL_ANALYTIC_VIEW_COLUMNS describes the columns of the analytic views accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_COLUMNS describes the columns of all analytic views in the database.
- USER_ANALYTIC_VIEW_COLUMNS describes the columns of the analytic views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
DIMENSION_NAME	VARCHAR2(128)		Alias of the analytic view dimension in the analytic view; for a measure the value is MEASURES
HIER_NAME	VARCHAR2(128)		Alias of the analytic view hierarchy within DIMENSION_NAME in the analytic view; for a measure the value is MEASURES
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column
ROLE	VARCHAR2(4)	NOT NULL	The role the attribute plays in the analytic view: • KEY • AKEY • HIER • PROP • MEAS
DATA_TYPE	VARCHAR2(106)	NOT NULL	Datatype of the column
DATA_LENGTH	NUMBER	NOT NULL	Length of the column (in bytes)
DATA_PRECISION	NUMBER		Decimal precision for the NUMBER datatype; binary precision for the FLOAT datatype, NULL for all other datatypes
DATA_SCALE	NUMBER		Number of digits to the right of the decimal point in a number
NULLABLE	CHAR(1)	NOT NULL	Indicates whether a column allows NULL values; the value is N if there is a NOT NULL constraint on the column or if the column is part of a PRIMARY KEY

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CHARACTER_SET_NAME	VARCHAR2(44)		Name of the character set: • CHAR_CS • NCHAR_CS
CHAR_COL_DECL_LENGTH	NUMBER		Declaration length of the character type column
CHAR_USED	VARCHAR2(1)		Indicates that the column uses BYTE length semantics (B) or CHAR length semantics (C), or whether the datatype is not any of the following (NULL): • CHAR • VARCHAR2 • NCHAR • NVARCHAR2
ORDER_NUM	NUMBER	NOT NULL	Order of the column, with the hierarchy columns first followed by measure columns. The columns for a hierarchy are grouped together, listed in their order in the HIERARCHIES clause of the analytic view definition. Within a hierarchy, attributes are listed first in order of their definition in the ATTRIBUTES clause of the attribute dimension definition followed by hierarchical attributes in the DIMENSION BY clause of the analytic view.
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ANALYTIC_VIEW_COLUMNS"
- "USER_ANALYTIC_VIEW_COLUMNS"

2.25 ALL_ANALYTIC_VIEW_COLUMNS_AE

ALL_ANALYTIC_VIEW_COLUMNS_AE describes the columns of the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_COLUMNS_AE describes the columns of all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEW_COLUMNS_AE describes the columns of the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DIMENSION_NAME	VARCHAR2(128)		Alias of the analytic view dimension in the analytic view; for a measure the value is MEASURES
HIER_NAME	VARCHAR2(128)		Alias of the analytic view hierarchy within DIMENSION_NAME in the analytic view; for a measure the value is MEASURES
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column
ROLE	VARCHAR2(4)	NOT NULL	The role the attribute plays in the analytic view: • KEY • AKEY • HIER • PROP • MEAS
DATA_TYPE	VARCHAR2(106)	NOT NULL	Datatype of the column
DATA_LENGTH	NUMBER	NOT NULL	Length of the column (in bytes)
DATA_PRECISION	NUMBER		Decimal precision for the NUMBER datatype; binary precision for the FLOAT datatype, NULL for all other datatypes
DATA_SCALE	NUMBER		Number of digits to the right of the decimal point in a number
NULLABLE	CHAR(1)	NOT NULL	Indicates whether a column allows NULL values; the value is N if there is a NOT NULL constraint on the column or if the column is part of a PRIMARY KEY
CHARACTER_SET_NAME	VARCHAR2(44)		Name of the character set: • CHAR_CS • NCHAR_CS
CHAR_COL_DECL_LENGTH	NUMBER		Declaration length of the character type column
CHAR_USED	VARCHAR2(1)		Indicates that the column uses BYTE length semantics (B) or CHAR length semantics (C), or whether the datatype is not any of the following (NULL): • CHAR • VARCHAR2 • NCHAR • NVARCHAR2
ORDER_NUM	NUMBER	NOT NULL	Order of the column, with the hierarchy columns first followed by measure columns. The columns for a hierarchy are grouped together, listed in their order in the HIERARCHIES clause of the analytic view definition. Within a hierarchy, attributes are listed first in order of their definition in the ATTRIBUTES clause of the attribute dimension definition followed by hierarchical attributes in the DIMENSION BY clause of the analytic view.
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> - "DBA_ANALYTIC_VIEW_COLUMNS_AE"
> - "USER_ANALYTIC_VIEW_COLUMNS_AE"

2.26 ALL_ANALYTIC_VIEW_DIM_CLASS

`ALL_ANALYTIC_VIEW_DIM_CLASS` describes the classifications of the attribute dimensions in the analytic views accessible to the current user.

Related Views

- `DBA_ANALYTIC_VIEW_DIM_CLASS` describes the classifications of the attribute dimensions in all analytic views in the database.
- `USER_ANALYTIC_VIEW_DIM_CLASS` describes the classifications of the attribute dimensions in the analytic views owned by the current user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the analytic view
`ANALYTIC_VIEW_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the analytic view
`DIMENSION_ALIAS`	`VARCHAR2(128)`		Alias of the attribute dimension in the analytic view
`CLASSIFICATION`	`VARCHAR2(128)`		Classification associated with the attribute dimension
`VALUE`	`CLOB`		Value of the classification or NULL if not specified
`LANGUAGE`	`VARCHAR2(64)`		`NLS_LANGUAGE` value associated with the classification or NULL if not specified
`ORDER_NUM`	`NUMBER`	`NOT NULL`	Order of the classification in the list of classifications associated with the attribute dimension
`ORIGIN_CON_ID`	`NUMBER`		The ID of the container where the data originates. Possible values include: • `0`: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

> **See Also**
>
> - "DBA_ANALYTIC_VIEW_DIM_CLASS"
> - "USER_ANALYTIC_VIEW_DIM_CLASS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.27 ALL_ANALYTIC_VIEW_DIM_CLS

ALL_ANALYTIC_VIEW_DIM_CLS is identical to ALL_ANALYTIC_VIEW_DIM_CLASS.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_DIM_CLASS"

2.28 ALL_ANALYTIC_VIEW_DIM_CLS_AE

ALL_ANALYTIC_VIEW_DIM_CLS_AE describes the classifications of the attribute dimensions in the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_DIM_CLS_AE describes the classifications of the attribute dimensions in all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEW_DIM_CLS_AE describes the classifications of the attribute dimensions in the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the analytic view
CLASSIFICATION	VARCHAR2(128)		Classification associated with the attribute dimension
VALUE	CLOB		Value of the classification or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the attribute dimension
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> - "DBA_ANALYTIC_VIEW_DIM_CLS_AE"
> - "USER_ANALYTIC_VIEW_DIM_CLS_AE"

2.29 ALL_ANALYTIC_VIEW_DIMENSIONS

`ALL_ANALYTIC_VIEW_DIMENSIONS` describes the attribute dimensions in the analytic views accessible to the current user.

Related Views

- `DBA_ANALYTIC_VIEW_DIMENSIONS` describes the attribute dimensions in all analytic views in the database.
- `USER_ANALYTIC_VIEW_DIMENSIONS` describes the attribute dimensions in the analytic views owned by the current user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the analytic view
`ANALYTIC_VIEW_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the analytic view
`DIMENSION_OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the schema containing the attribute dimension
`DIMENSION_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the attribute dimension
`DIMENSION_ALIAS`	`VARCHAR2(128)`		Alias of the attribute dimension in the analytic view
`DIMENSION_TYPE`	`VARCHAR2(8)`		Type of the attribute dimension: • `TIME` • `STANDARD`
`ALL_MEMBER_NAME`	`CLOB`		An expression for the name of the ALL member for the attribute dimension
`ALL_MEMBER_CAPTION`	`CLOB`		An expression for the caption for the ALL member of the attribute dimension, or NULL if not specified
`ALL_MEMBER_DESCRIPTION`	`CLOB`		An expression for the description for the ALL member of the attribute dimension, or NULL if not specified
`REFERENCES_DISTINCT`	`VARCHAR2(1)`		Indicates whether the reference between the fact table key and the attribute dimension attribute specifies the `DISTINCT` keyword. Possible values are: • `Y`: The reference specifies the `DISTINCT` keyword. • `N`: The reference does not specify the `DISTINCT` keyword.
`ORDER_NUM`	`NUMBER`	`NOT NULL`	Order number of the attribute dimension in the analytic view

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ANALYTIC_VIEW_DIMENSIONS"
- "USER_ANALYTIC_VIEW_DIMENSIONS"

2.30 ALL_ANALYTIC_VIEW_DIMS

ALL_ANALYTIC_VIEW_DIMS is identical to ALL_ANALYTIC_VIEW_DIMENSIONS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_DIMENSIONS"

2.31 ALL_ANALYTIC_VIEW_DIMS_AE

ALL_ANALYTIC_VIEW_DIMS_AE describes the attribute dimensions in the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_DIMS_AE describes the attribute dimensions in all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEW_DIMS_AE describes the attribute dimensions in the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DIMENSION_OWNER	VARCHAR2(128)	NOT NULL	Owner of the schema containing the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the analytic view
DIMENSION_TYPE	VARCHAR2(8)		Type of the attribute dimension: • TIME • STANDARD
ALL_MEMBER_NAME	CLOB		An expression for the name of the ALL member for the attribute dimension
ALL_MEMBER_CAPTION	CLOB		An expression for the caption for the ALL member of the attribute dimension, or NULL if not specified
ALL_MEMBER_DESCRIPTION	CLOB		An expression for the description for the ALL member of the attribute dimension, or NULL if not specified
REFERENCES_DISTINCT	VARCHAR2(1)		Indicates whether the reference between the fact table key and the attribute dimension attribute specifies the DISTINCT keyword. Possible values are: • Y: The reference specifies the DISTINCT keyword. • N: The reference does not specify the DISTINCT keyword.
ORDER_NUM	NUMBER	NOT NULL	Order number of the attribute dimension in the analytic view
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ANALYTIC_VIEW_DIMS_AE"
- "USER_ANALYTIC_VIEW_DIMS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.32 ALL_ANALYTIC_VIEW_HIER_CLASS

ALL_ANALYTIC_VIEW_HIER_CLASS describes the classifications of the hierarchies in the analytic views accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_HIER_CLASS describes the classifications of the hierarchies in all analytic views in the database.
- USER_ANALYTIC_VIEW_HIER_CLASS describes the classifications of the hierarchies in the analytic views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the analytic view
HIER_ALIAS	VARCHAR2(128)		Alias of the hierarchy in the attribute dimension in the analytic view
CLASSIFICATION	VARCHAR2(128)		Classification associated with the hierarchy
VALUE	CLOB		Value of the classification, or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the hierarchy
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ANALYTIC_VIEW_HIER_CLASS"
- "USER_ANALYTIC_VIEW_HIER_CLASS"

2.33 ALL_ANALYTIC_VIEW_HIER_CLS

ALL_ANALYTIC_VIEW_HIER_CLS is identical to ALL_ANALYTIC_VIEW_HIER_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_HIER_CLASS"

2.34 ALL_ANALYTIC_VIEW_HIER_CLS_AE

ALL_ANALYTIC_VIEW_HIER_CLS_AE describes the classifications of the hierarchies in the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_HIER_CLS_AE describes the classifications of the hierarchies in all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEW_HIER_CLS_AE describes the classifications of the hierarchies in the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the analytic view
HIER_ALIAS	VARCHAR2(128)		Alias of the hierarchy in the attribute dimension in the analytic view
CLASSIFICATION	VARCHAR2(128)		Classification associated with the hierarchy
VALUE	CLOB		Value of the classification, or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the hierarchy
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_ANALYTIC_VIEW_HIER_CLS_AE"
- "USER_ANALYTIC_VIEW_HIER_CLS_AE"

2.35 ALL_ANALYTIC_VIEW_HIERS

ALL_ANALYTIC_VIEW_HIERS describes the hierarchies in the analytic views accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_HIERS describes the hierarchies in all analytic views in the database.
- USER_ANALYTIC_VIEW_HIERS describes the hierarchies in the analytic views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view hierarchy
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the analytic view
HIER_OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
HIER_ALIAS	VARCHAR2(128)		Alias specified for the hierarchy
IS_DEFAULT	VARCHAR2(1)		Indicates whether this is the default hierarchy for the analytic view dimension in the analytic view (Y) or not (N)
ORDER_NUM	NUMBER	NOT NULL	Order of the hierarchy in the list of hierarchies in the analytic view
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ANALYTIC_VIEW_HIERS"
- "USER_ANALYTIC_VIEW_HIERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.36 ALL_ANALYTIC_VIEW_HIERS_AE

ALL_ANALYTIC_VIEW_HIERS_AE describes the hierarchies in the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_HIERS_AE describes the hierarchies in all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEW_HIERS_AE describes the hierarchies in the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view hierarchy
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the analytic view
HIER_OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
HIER_ALIAS	VARCHAR2(128)		Alias specified for the hierarchy
IS_DEFAULT	VARCHAR2(1)		Indicates whether this is the default hierarchy for the analytic view dimension in the analytic view (Y) or not (N)
ORDER_NUM	NUMBER	NOT NULL	Order of the hierarchy in the list of hierarchies in the analytic view
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ANALYTIC_VIEW_HIERS_AE"
- "USER_ANALYTIC_VIEW_HIERS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.37 ALL_ANALYTIC_VIEW_KEYS

ALL_ANALYTIC_VIEW_KEYS describes the key columns of the attribute dimensions in the analytic views accessible to the current user.

The keys reference attributes of the attribute dimensions of the analytic view.

Related Views

- DBA_ANALYTIC_VIEW_KEYS describes the key columns of the attribute dimensions in all analytic views in the database.
- USER_ANALYTIC_VIEW_KEYS describes the key columns of the attribute dimensions in the analytic views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the analytic view
AV_KEY_TABLE_ALIAS	VARCHAR2(128)		Table alias of the key column
AV_KEY_COLUMN	VARCHAR2(128)	NOT NULL	Name of the column for the key
REF_DIMENSION_ATTR	VARCHAR2(128)		Name of the referenced attribute dimension attribute
ORDER_NUM	NUMBER	NOT NULL	Order number of the key in the list of keys in the analytic view
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

> **See Also**
>
> - "DBA_ANALYTIC_VIEW_KEYS"
> - "USER_ANALYTIC_VIEW_KEYS"

2.38 ALL_ANALYTIC_VIEW_KEYS_AE

ALL_ANALYTIC_VIEW_KEYS_AE describes the key columns of the attribute dimensions in the analytic views (across all editions) accessible to the current user.

The keys reference attributes of the attribute dimensions of the analytic view.

Related Views

- DBA_ANALYTIC_VIEW_KEYS_AE describes the key columns of the attribute dimensions in all analytic views (across all editions) in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_ANALYTIC_VIEW_KEYS_AE describes the key columns of the attribute dimensions in the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the analytic view
AV_KEY_TABLE_ALIAS	VARCHAR2(128)		Table alias of the key column
AV_KEY_COLUMN	VARCHAR2(128)	NOT NULL	Name of the column for the key
REF_DIMENSION_ATTR	VARCHAR2(128)		Name of the referenced attribute dimension attribute
ORDER_NUM	NUMBER	NOT NULL	Order number of the key in the list of keys in the analytic view
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ANALYTIC_VIEW_KEYS_AE"
- "USER_ANALYTIC_VIEW_KEYS_AE"

2.39 ALL_ANALYTIC_VIEW_LEVEL_CLASS

ALL_ANALYTIC_VIEW_LEVEL_CLASS describes the level classifications of the analytic views accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_LEVEL_CLASS describes the level classifications of all analytic views in the database.
- USER_ANALYTIC_VIEW_LEVEL_CLASS describes the level classifications of the analytic views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of analytic view
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the analytic view dimension in the analytic view
HIER_ALIAS	VARCHAR2(128)		Alias of the hierarchy in the analytic view
LEVEL_NAME	VARCHAR2(128)	NOT NULL	Name of the level in the analytic view
CLASSIFICATION	VARCHAR2(128)		Classification associated with the level
VALUE	CLOB		Value of the classification, or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the level
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ANALYTIC_VIEW_LEVEL_CLASS"
- "USER_ANALYTIC_VIEW_LEVEL_CLASS"

2.40 ALL_ANALYTIC_VIEW_LEVELS

ALL_ANALYTIC_VIEW_LEVELS describes the levels in the hierarchies in the analytic views accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_LEVELS describes the levels in the hierarchies in all analytic views in the database.
- USER_ANALYTIC_VIEW_LEVELS describes the levels in the hierarchies in the analytic views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the analytic view
HIER_ALIAS	VARCHAR2(128)		Alias of the hierarchy in the attribute dimension in the analytic view
LEVEL_NAME	VARCHAR2(128)	NOT NULL	Name of the level within the attribute dimension in the analytic view

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORDER_NUM	NUMBER	NOT NULL	Order number of the level in the list of levels in the analytic view hierarchy
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ANALYTIC_VIEW_LEVELS"
- "USER_ANALYTIC_VIEW_LEVELS"

2.41 ALL_ANALYTIC_VIEW_LEVELS_AE

ALL_ANALYTIC_VIEW_LEVELS_AE describes the levels in the hierarchies in the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_LEVELS_AE describes the levels in the hierarchies in all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEW_LEVELS_AE describes the levels in the hierarchies in the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the analytic view
HIER_ALIAS	VARCHAR2(128)		Alias of the hierarchy in the attribute dimension in the analytic view
LEVEL_NAME	VARCHAR2(128)	NOT NULL	Name of the level within the attribute dimension in the analytic view
ORDER_NUM	NUMBER	NOT NULL	Order number of the level in the list of levels in the analytic view hierarchy
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ANALYTIC_VIEW_LEVELS_AE"
- "USER_ANALYTIC_VIEW_LEVELS_AE"

2.42 ALL_ANALYTIC_VIEW_LVL_CLS

ALL_ANALYTIC_VIEW_LVL_CLS is identical to ALL_ANALYTIC_VIEW_LEVEL_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_LEVEL_CLASS"

2.43 ALL_ANALYTIC_VIEW_LVL_CLS_AE

ALL_ANALYTIC_VIEW_LVL_CLS_AE describes the level classifications of the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_LVL_CLS_AE describes the level classifications of all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEW_LVL_CLS_AE describes the level classifications of the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of analytic view

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the analytic view dimension in the analytic view
HIER_ALIAS	VARCHAR2(128)		Alias of the hierarchy in the analytic view
LEVEL_NAME	VARCHAR2(128)	NOT NULL	Name of the level in the analytic view
CLASSIFICATION	VARCHAR2(128)		Classification associated with the level
VALUE	CLOB		Value of the classification, or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the level
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ANALYTIC_VIEW_LVL_CLS_AE"
- "USER_ANALYTIC_VIEW_LVL_CLS_AE"

2.44 ALL_ANALYTIC_VIEW_LVLGRPS

ALL_ANALYTIC_VIEW_LVLGRPS describes the analytic view measure and level groups of the analytic views accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_LVLGRPS describes the analytic view measure and level groups of all analytic views in the database.
- USER_ANALYTIC_VIEW_LVLGRPS describes the analytic view measure and level groups of the analytic views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)		Name of the analytic view

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CACHE_TYPE	VARCHAR2(128)		Type of the materialized view; one of the following: • DYNAMIC • MATERIALIZED (the default value)
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the group
HIER_ALIAS	VARCHAR2(128)		Alias of the hierarchy associated with the attribute dimension in the group
LEVEL_NAME	VARCHAR2(128)		Name of the level in the hierarchy in the group
MEASURE_NAME	VARCHAR2(128)		Names of the measures in the group
AV_LVLGRP_ORDER	NUMBER		Order of the groups in the analytic view
LEVEL_MEAS_ORDER	NUMBER		Order of the levels and measures in the group
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ANALYTIC_VIEW_LVLGRPS"
- "USER_ANALYTIC_VIEW_LVLGRPS"

2.45 ALL_ANALYTIC_VIEW_LVLGRPS_AE

ALL_ANALYTIC_VIEW_LVLGRPS_AE describes the analytic view measure and level groups of the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_LVLGRPS_AE describes the analytic view measure and level groups of all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEW_LVLGRPS_AE describes the analytic view measure and level groups of the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)		Name of the analytic view
CACHE_TYPE	VARCHAR2(128)		Type of the materialized view; one of the following: • DYNAMIC • MATERIALIZED (the default value)
DIMENSION_ALIAS	VARCHAR2(128)		Alias of the attribute dimension in the group
HIER_ALIAS	VARCHAR2(128)		Alias of the hierarchy associated with the attribute dimension in the group

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LEVEL_NAME	VARCHAR2(128)		Name of the level in the hierarchy in the group
MEASURE_NAME	VARCHAR2(128)		Names of the measures in the group
AV_LVLGRP_ORDER	NUMBER		Order of the groups in the analytic view
LEVEL_MEAS_ORDER	NUMBER		Order of the levels and measures in the group
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ANALYTIC_VIEW_LVLGRPS_AE"
- "USER_ANALYTIC_VIEW_LVLGRPS_AE"

2.46 ALL_ANALYTIC_VIEW_MEAS_CLASS

ALL_ANALYTIC_VIEW_MEAS_CLASS describes the classifications of the measures of the analytic views accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_MEAS_CLASS describes the classifications of the measures of all analytic views in the database.
- USER_ANALYTIC_VIEW_MEAS_CLASS describes the classifications of the measures of the analytic views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
MEASURE_NAME	VARCHAR2(128)		Name of the measure associated with the classification
CLASSIFICATION	VARCHAR2(128)		Classification associated with the measure of the analytic view
VALUE	CLOB		Value of the classification, or NULL if not specified

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the measure
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ANALYTIC_VIEW_MEAS_CLASS"
- "USER_ANALYTIC_VIEW_MEAS_CLASS"

2.47 ALL_ANALYTIC_VIEW_MEAS_CLS

ALL_ANALYTIC_VIEW_MEAS_CLS is identical to ALL_ANALYTIC_VIEW_MEAS_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_MEAS_CLASS"

2.48 ALL_ANALYTIC_VIEW_MEAS_CLS_AE

ALL_ANALYTIC_VIEW_MEAS_CLS_AE describes the classifications of the measures of the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEW_MEAS_CLS_AE describes the classifications of the measures of all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEW_MEAS_CLS_AE describes the classifications of the measures of the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
MEASURE_NAME	VARCHAR2(128)		Name of the measure associated with the classification
CLASSIFICATION	VARCHAR2(128)		Classification associated with the measure of the analytic view
VALUE	CLOB		Value of the classification, or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the measure
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ANALYTIC_VIEW_MEAS_CLS_AE"
- "USER_ANALYTIC_VIEW_MEAS_CLS_AE"

2.49 ALL_ANALYTIC_VIEWS

ALL_ANALYTIC_VIEWS describes the analytic views accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEWS describes all analytic views in the database.
- USER_ANALYTIC_VIEWS describes the analytic views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the fact table or view on which the analytic view is defined
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the fact table or view on which the analytic view is defined
TABLE_ALIAS	VARCHAR2(128)		Alias of the fact table or view on which the analytic view is defined; the default is TABLE_NAME
DEFAULT_AGGR	VARCHAR2(128)		Default aggregation of the analytic view
DEFAULT_MEASURE	VARCHAR2(128)		Name of the default measure of the analytic view
COMPILE_STATE	VARCHAR2(7)		Compile status of the analytic view: • VALID • INVALID
DYN_ALL_CACHE	VARCHAR2(1)		The value of this column is always N
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ANALYTIC_VIEWS"
- "USER_ANALYTIC_VIEWS"

2.50 ALL_ANALYTIC_VIEWS_AE

ALL_ANALYTIC_VIEWS_AE describes the analytic views (across all editions) accessible to the current user.

Related Views

- DBA_ANALYTIC_VIEWS_AE describes all analytic views (across all editions) in the database.
- USER_ANALYTIC_VIEWS_AE describes the analytic views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic view
ANALYTIC_VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the analytic view
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the fact table or view on which the analytic view is defined
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the fact table or view on which the analytic view is defined
TABLE_ALIAS	VARCHAR2(128)		Alias of the fact table or view on which the analytic view is defined; the default is TABLE_NAME

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEFAULT_AGGR	VARCHAR2(128)		Default aggregation of the analytic view
DEFAULT_MEASURE	VARCHAR2(128)		Name of the default measure of the analytic view
COMPILE_STATE	VARCHAR2(7)		Compile status of the analytic view: • VALID • INVALID
DYN_ALL_CACHE	VARCHAR2(1)		The value of this column is always N
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the analytic view is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ANALYTIC_VIEWS_AE"
- "USER_ANALYTIC_VIEWS_AE"

2.51 ALL_ANNOTATION_VALUES

ALL_ANNOTATION_VALUES displays the values for schema annotations accessible to the current user.

Related Views

- DBA_ANNOTATION_VALUES displays the values for all schema annotations in the database.
- USER_ANNOTATION_VALUES displays the values for schema annotations owned by the current user. This view does not display the ANNOTATION_OWNER column.

Column	Datatype	NULL	Description
ANNOTATION_OWNER	VARCHAR2(128)	NOT NULL	Annotation owner
ANNOTATION_NAME	VARCHAR2(1024)	NOT NULL	Annotation name
ANNOTATION_VALUE	VARCHAR2(4000)		Annotation value

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.28.

> **See Also**
>
> - "DBA_ANNOTATION_VALUES"
> - "USER_ANNOTATION_VALUES"

2.52 ALL_ANNOTATIONS

`ALL_ANNOTATIONS` displays schema annotations accessible to the current user.

Related Views

- `DBA_ANNOTATIONS` displays all schema annotations in the database.
- `USER_ANNOTATIONS` displays schema annotations owned by the current user. This view does not display the `ANNOTATION_OWNER` column.

Column	Datatype	NULL	Description
`ANNOTATION_OWNER`	`VARCHAR2(128)`	`NOT NULL`	Annotation owner
`ANNOTATION_NAME`	`VARCHAR2(1024)`	`NOT NULL`	Annotation name

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.28.

> **See Also**
>
> - "DBA_ANNOTATIONS"
> - "USER_ANNOTATIONS"

2.53 ALL_ANNOTATIONS_USAGE

`ALL_ANNOTATIONS_USAGE` provides usage information about schema annotations accessible to the current user.

Related Views

- `DBA_ANNOTATIONS_USAGE` provides usage information about all schema annotations in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_ANNOTATIONS_USAGE provides usage information about schema annotations owned by the current user. This view does not display the ANNOTATION_OWNER column.

Column	Datatype	NULL	Description
OBJECT_NAME	VARCHAR2(128)		Object name
OBJECT_TYPE	VARCHAR2(9)		Object type
COLUMN_NAME	VARCHAR2(128)		If the annotation is specified for a table column, view column, or domain column, the name of the column; otherwise NULL
DOMAIN_NAME	VARCHAR2(128)		If the annotation is carried over from a domain or a domain column, the name of the domain; otherwise NULL Note that domains carry over their annotations to objects that reference them.
DOMAIN_OWNER	VARCHAR2(128)		If the annotation is carried over from a domain or domain column, the owner of the domain; otherwise NULL Note that domains carry over their annotations to objects that reference them.
ANNOTATION_OWNER	VARCHAR2(128)		Annotation owner
ANNOTATION_NAME	VARCHAR2(1024)		Annotation name
ANNOTATION_VALUE	VARCHAR2(4000)		Optional annotation value A NULL value is possible when the annotation does not require a value, that is, when the annotation is defined solely by its name.

Note

This view is available starting with Oracle Database 19c, Release Update 19.28.

See Also

- "DBA_ANNOTATIONS_USAGE"
- "USER_ANNOTATIONS_USAGE"

2.54 ALL_APPLY

ALL_APPLY displays information about the apply processes that dequeue messages from queues accessible to the current user.

Related View

DBA_APPLY displays information about all apply processes in the database.

Column	Datatype	NULL	Description
APPLY_NAME	VARCHAR2(128)	NOT NULL	Name of the apply process

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
QUEUE_NAME	VARCHAR2(128)	NOT NULL	Name of the queue from which the apply process dequeues
QUEUE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the queue from which the apply process dequeues
APPLY_CAPTURED	VARCHAR2(3)		Indicates whether the apply process applies captured messages (YES) or user-enqueued messages (NO)
RULE_SET_NAME	VARCHAR2(128)		Name of the positive rule set used by the apply process for filtering
RULE_SET_OWNER	VARCHAR2(128)		Owner of the positive rule set used by the apply process for filtering
APPLY_USER	VARCHAR2(128)		User who is applying messages
APPLY_DATABASE_LINK	VARCHAR2(128)		Database link to which changes are applied. If NULL, then changes are applied to the local database.
APPLY_TAG	RAW(2000)		Tag associated with redo log records that are generated when changes are made by the apply process
DDL_HANDLER	VARCHAR2(98)		Name of the user-specified data definition language (DDL) handler, which handles DDL logical change records (LCRs)
PRECOMMIT_HANDLER	VARCHAR2(98)		Name of the user-specified pre-commit handler
MESSAGE_HANDLER	VARCHAR2(98)		Name of the user-specified procedure that handles dequeued messages other than logical change records (LCRs)
STATUS	VARCHAR2(8)		Status of the apply process: • DISABLED • ENABLED • ABORTED
MAX_APPLIED_MESSAGE_NUMBER	NUMBER		System change number (SCN) corresponding to the apply process high watermark for the last time the apply process was stopped using the DBMS_APPLY_ADM.STOP_APPLY procedure with the force parameter set to false. The apply process high watermark is the SCN beyond which no messages have been applied.
NEGATIVE_RULE_SET_NAME	VARCHAR2(128)		Name of the negative rule set used by the apply process for filtering
NEGATIVE_RULE_SET_OWNER	VARCHAR2(128)		Owner of the negative rule set used by the apply process for filtering
STATUS_CHANGE_TIME	DATE		Time that the STATUS of the apply process was changed
ERROR_NUMBER	NUMBER		Error number if the apply process was terminated
ERROR_MESSAGE	VARCHAR2(4000)		Error message if the apply process was terminated
MESSAGE_DELIVERY_MODE	VARCHAR2(10)		Reserved for internal use

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PURPOSE	VARCHAR2(19)		Purpose of the apply process: • GoldenGate Apply - An Oracle GoldenGate Inbound server configured by Oracle GoldenGate integrated replicat • XStream Out - An XStream outbound server in an XStream Out configuration • XStream In - An XStream inbound server in an XStream In configuration • AUDIT VAULT - An apply process in an audit vault configuration • CHANGE DATA CAPTURE - An apply process in a change data capture configuration
LCRID_VERSION	NUMBER		LCR ID format currently being used

See Also

- "DBA_APPLY"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLY_ADM.STOP_APPLY procedure
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_XSTREAM_ADM.ENABLE_GG_XSTREAM_FOR_STREAMS procedure

2.55 ALL_APPLY_CHANGE_HANDLERS

ALL_APPLY_CHANGE_HANDLERS displays information about the change handlers on the tables accessible to the current user.

Related View

DBA_APPLY_CHANGE_HANDLERS displays information about the change handlers on all tables in the database.

Column	Datatype	NULL	Description
CHANGE_TABLE_OWNER	VARCHAR2(128)		Owner of the change table
CHANGE_TABLE_NAME	VARCHAR2(128)		Name of the change table
SOURCE_TABLE_OWNER	VARCHAR2(128)		Owner of the source table
SOURCE_TABLE_NAME	VARCHAR2(128)		Name of the source table
HANDLER_NAME	VARCHAR2(128)		Name of the statement-based change handler
CAPTURE_VALUES	VARCHAR2(3)		Indicates whether to capture the old (OLD), new (NEW), or both (*) values
APPLY_NAME	VARCHAR2(128)		Name of the apply process

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OPERATION_NAME	VARCHAR2(10)		Name of the DML operation to which the DML handler is set: • DEFAULT • INSERT • UPDATE • DELETE • LOB_UPDATE
CREATION_TIME	TIMESTAMP(6)		Change handler creation time
MODIFICATION_TIME	TIMESTAMP(6)		Change handler modification time

See Also

"DBA_APPLY_CHANGE_HANDLERS"

2.56 ALL_APPLY_CONFLICT_COLUMNS

ALL_APPLY_CONFLICT_COLUMNS displays information about the conflict handlers on the tables accessible to the current user.

Related View

DBA_APPLY_CONFLICT_COLUMNS displays information about the conflict handlers on all tables in the database.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)		Owner of the object on which the update conflict handler is defined
OBJECT_NAME	VARCHAR2(128)		Name of the object on which the update conflict handler is defined
METHOD_NAME	VARCHAR2(92)		Name of the update conflict handler used to resolve conflicts
RESOLUTION_COLUMN	VARCHAR2(4000)		Name of the column used to resolve conflicts
COLUMN_NAME	VARCHAR2(128)		Name of a column in the column list for the update conflict handler
APPLY_DATABASE_LINK	VARCHAR2(128)		Database link to which changes are applied. If null, then changes are applied to the local database.

See Also

"DBA_APPLY_CONFLICT_COLUMNS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.57 ALL_APPLY_DML_CONF_HANDLERS

ALL_APPLY_DML_CONF_HANDLERS provides details about DML conflict handlers on objects visible to the current user.

Related View

DBA_APPLY_DML_CONF_HANDLERS provides details about DML conflict handlers.

Column	Datatype	NULL	Description
APPLY_NAME	VARCHAR2(128)		Name of the apply process
OBJECT_OWNER	VARCHAR2(128)		Owner of the target object
OBJECT_NAME	VARCHAR2(128)		Name of the target object
SOURCE_OBJECT_OWNER	VARCHAR2(128)		Source database owner of the object
SOURCE_OBJECT_NAME	VARCHAR2(128)		Source database name of the object
COMMAND_TYPE	VARCHAR2(6)		Type of the DML operation: INSERT, UPDATE, or DELETE
CONFLICT_TYPE	VARCHAR2(11)		Type of conflict: • ROW EXISTS • ROW MISSING
METHOD_NAME	VARCHAR2(9)		Method used for resolving the error, depending on the conflict type: • OVERWRITE • RECORD • IGNORE • MAXIMUM • MINIMUM • DELTA
CONFLICT_HANDLER_NAME	VARCHAR2(128)		Name of the conflict handler
RESOLUTION_COLUMN	VARCHAR2(128)		Name of the column used to resolve the conflict for MAXIMUM, MINIMUM, and DELTA
SET_BY	VARCHAR2(10)		Entity that set up the handler: • USER • GOLDENGATE

See Also

"DBA_APPLY_DML_CONF_HANDLERS"

2.58 ALL_APPLY_DML_HANDLERS

ALL_APPLY_DML_HANDLERS displays information about the DML handlers on the tables accessible to the current user.

Related View

DBA_APPLY_DML_HANDLERS displays information about the DML handlers on all tables in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the object on which the DML handler is specified
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object on which the DML handler is specified
OPERATION_NAME	VARCHAR2(13)		Name of the DML operation for which the DML handler is used: • DEFAULT • INSERT • UPDATE • DELETE • LOB_UPDATE • ASSEMBLE_LOBS
USER_PROCEDURE	VARCHAR2(98)		Name of the user-specified DML handler, which handles row logical change records that contain the DML operation in the OPERATION_NAME column on the object
ERROR_HANDLER	VARCHAR2(1)		Indicates whether the DML handler handles only the relevant row logical change records that result in apply errors (Y) or all relevant row logical change records (N)
APPLY_DATABASE_LINK	VARCHAR2(128)		Database link to which changes are applied. If null, then changes are applied to the local database.
APPLY_NAME	VARCHAR2(128)		Name of the apply process for the given object
ASSEMBLE_LOBS	VARCHAR2(1)		Indicates whether LOB assembly is used for LOB columns in logical change records (LCRs) processed by the handler (Y) or not (N) LOB assembly combines multiple LCRs for a LOB column resulting from a single row change into one row LCR before passing the LCR to the handler.
SET_BY	VARCHAR2(10)		Entity that set up the handler. Possible values include: • GOLDENGATE • USER

See Also

- "DBA_APPLY_DML_HANDLERS"
- *Oracle Database XStream Guide* for more information about DML handlers in an Oracle XStream environment

2.59 ALL_APPLY_ENQUEUE

ALL_APPLY_ENQUEUE displays information about the apply enqueue actions for the rules where the destination queue exists and is accessible to the current user.

Related View

DBA_APPLY_ENQUEUE displays information about the apply enqueue actions for all rules in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
RULE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule
RULE_NAME	VARCHAR2(128)	NOT NULL	Name of the rule
DESTINATION_QUEUE_NAME	VARCHAR2(4000)		Name of the queue where events satisfying the rule will be enqueued

See Also

"DBA_APPLY_ENQUEUE"

2.60 ALL_APPLY_ERROR

ALL_APPLY_ERROR displays information about the error transactions generated by the apply processes that dequeue messages from queues accessible to the current user.

Related Views

- DBA_APPLY_ERROR displays information about the error transactions generated by all apply processes in the database.
- USER_APPLY_ERROR displays information about the error transactions generated by apply processes visible to the current user. This view does not display the SOURCE_ROOT_NAME column.

Column	Datatype	NULL	Description
APPLY_NAME	VARCHAR2(128)		Name of the apply process at the local database which processed the transaction
QUEUE_NAME	VARCHAR2(128)		Name of the queue at the local database from which the transaction was dequeued
QUEUE_OWNER	VARCHAR2(128)		Owner of the queue at the local database from which the transaction was dequeued
LOCAL_TRANSACTION_ID	VARCHAR2(22)		Local transaction ID for the error transaction
SOURCE_DATABASE	VARCHAR2(128)		Database where the transaction originated
SOURCE_TRANSACTION_ID	VARCHAR2(128)		Original transaction ID at the source database
SOURCE_COMMIT_SCN	NUMBER		Original commit system change number (SCN) for the transaction at the source database
MESSAGE_NUMBER	NUMBER		Identifier for the message in the transaction that raised an error
ERROR_NUMBER	NUMBER		Error number of the error raised by the transaction
ERROR_MESSAGE	VARCHAR2(4000)		Error message of the error raised by the transaction
RECIPIENT_ID	NUMBER		User ID of the original user that applied the transaction
RECIPIENT_NAME	VARCHAR2(128)		Name of the original user that applied the transaction
MESSAGE_COUNT	NUMBER		Total number of messages inside the error transaction
ERROR_CREATION_TIME	DATE		Time that the error was created
SOURCE_COMMIT_POSITION	RAW(64)		Original commit position for the transaction

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ERROR_TYPE	VARCHAR2(11)		NULL if the apply process can access all of the LCRs in the error transaction. EAGER ERROR if the apply process cannot access all of the LCRs in the error transaction. This error type typically means that the apply process was applying LCRs in a large transaction. When the ERROR_TYPE is EAGER ERROR, manage the error transaction using the instructions in *Oracle Database XStream Guide*. RECORD LCR indicates that a single LCR has been recorded as requested by user-specified error handling configuration RECORD TXN NO LCRS indicates that the identified transaction encountered an error and only the transaction ID is recorded as requested by user-specified error handling configuration RECORD TXN WITH LCRS indicates that the identified transaction encountered and error. The entire transaction is recorded as requested by user-specified error handling configuration. UNHANDLED ERRORS NO LCR indicates that the identified transaction encountered an error and there was no error handling specified for this handler. No LCRs are recorded for this transaction. DISCARDED is used to mark recorded and discarded LCRs.
SOURCE_ROOT_NAME	VARCHAR2(128)		The global name of the source root database
ERROR_POSITION	RAW(64)		LCR position at which the error occurred

See Also

- "DBA_APPLY_ERROR"
- "USER_APPLY_ERROR"
- *Oracle Database XStream Guide* for information on how to display detailed information about apply errors

2.61 ALL_APPLY_ERROR_MESSAGES

ALL_APPLY_ERROR_MESSAGES displays information about the individual messages in an error transaction generated by the apply processes that dequeue messages from queues accessible to the current user.

For XStream inbound servers, each message in an error transaction is an LCR.

Note

- Messages that were spilled from memory to hard disk do not appear in this view.
- This view does not contain information related to XStream outbound servers.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Related View

DBA_APPLY_ERROR_MESSAGES displays information about the individual messages in all of the error transactions generated by all apply processes in the database.

Column	Datatype	NULL	Description
MESSAGE_ID	RAW(16)		Unique identifier of the message stored in the error queue
LOCAL_TRANSACTION_ID	VARCHAR2(22)		Local transaction ID for the error transaction
TRANSACTION_MESSAGE_NUMBER	NUMBER		Message number of the message that raised the error. The message number is a sequence number for the messages in the transaction, starting with 1.
ERROR_NUMBER	NUMBER		Error number of the error raised by the transaction. The error number is populated only for the LCR that raised the error. This field is NULL for the other LCRs in the transaction.
ERROR_MESSAGE	VARCHAR2(4000)		Error message of the error raised by the transaction. The error message is populated only for the LCR that raised the error. This field is NULL for the other LCRs in the transaction.
SOURCE_OBJECT_OWNER	VARCHAR2(128)		Owner of the object at the source database
SOURCE_OBJECT_NAME	VARCHAR2(128)		Name of the object at the source database
OBJECT_OWNER	VARCHAR2(128)		Owner of the target table
OBJECT_NAME	VARCHAR2(128)		Object name of the target table
PRIMARY_KEY	VARCHAR2(4000)		Primary key of the table row that caused the source transaction to fail at the target
POSITION	RAW(64)		The LCR position
OPERATION	VARCHAR2(100)		The DML or DDL operation represented in the LCR
CONFLICT_TYPE	VARCHAR2(18)		Conflict type. Possible values: • INSERT ROW EXISTS: DML operation is INSERT and a row already exists with the specified key value. • UPDATE ROW EXISTS: DML operation is UPDATE. A row with the specified key exists but has conflicting values for some columns. • UPDATE ROW MISSING: DML operation is UPDATE and no row with the specified key value exists. • DELETE ROW EXISTS: DML operation is DELETE. A row with the specified key exists but has conflicting values for some columns • DELETE ROW MISSING: DML operation is DELETE and no row with the specified key value exists.
APPLIED_STATE	VARCHAR2(7)		Conflict applied state. Possible values: • WON: Incoming Logical Change Record was applied in its entirety. • PARTIAL: Incoming Logical Change Record was applied for one or more conflict groups. • LOST: Incoming Logical Change Record was not applied.
SEQ#	NUMBER		Trail file number (Oracle GoldenGate)
RBA	NUMBER		Position with Trail file (Oracle GoldenGate)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CONFLICT_INFO	VARCHAR2(4000)		Identifies the conflict group information
SOURCE_PACKAGE_NAME	VARCHAR2(128)		Package name of the source for procedural replication
PACKAGE_NAME	VARCHAR2(128)		Package name of the destination for procedural replication
MESSAGE	CLOB		The content of the LCR. Content includes column name and value for old and/or new values in DML LCRs. For DDL LCRs, the content is the text of the DDL SQL.

See Also

- "DBA_APPLY_ERROR_MESSAGES"
- *Oracle Database XStream Guide* for information on managing eager errors encountered by an inbound server

2.62 ALL_APPLY_EXECUTE

ALL_APPLY_EXECUTE displays information about the apply execute actions for the rules visible to the current user.

Related View

DBA_APPLY_EXECUTE displays information about the apply execute actions for all rules in the database.

Column	Datatype	NULL	Description
RULE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule
RULE_NAME	VARCHAR2(128)	NOT NULL	Name of the rule
EXECUTE_EVENT	VARCHAR2(2)		Indicates whether the event satisfying the rule is executed

See Also

"DBA_APPLY_EXECUTE"

2.63 ALL_APPLY_HANDLE_COLLISIONS

ALL_APPLY_HANDLE_COLLISIONS provides details about apply handlers for collisions on objects visible to the user at the table level.

Related View

DBA_APPLY_HANDLE_COLLISIONS provides details about apply handlers for collisions at the table level.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
APPLY_NAME	VARCHAR2(128)	NOT NULL	Name of the apply process
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the target object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the target object
SOURCE_OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Source database owner of the object
SOURCE_OBJECT_NAME	VARCHAR2(128)	NOT NULL	Source database name of the object
ENABLED	VARCHAR2(1)	NOT NULL	State of the collision handlers: Y for enabled, N for disabled
SET_BY	VARCHAR2(10)		Entity that set up the handler: • USER • GOLDENGATE

See Also

"DBA_APPLY_HANDLE_COLLISIONS"

2.64 ALL_APPLY_INSTANTIATED_GLOBAL

ALL_APPLY_INSTANTIATED_GLOBAL displays information for the current user about databases for which an instantiation SCN has been set.

Related View

DBA_APPLY_INSTANTIATED_GLOBAL displays information about databases for which an instantiation SCN has been set.

Column	Datatype	NULL	Description
SOURCE_DATABASE	VARCHAR2(128)	NOT NULL	Name of the database that was instantiated
INSTANTIATION_SCN	NUMBER		Instantiation SCN for the database. Only changes committed after this SCN are applied by an apply process.
APPLY_DATABASE_LINK	VARCHAR2(128)		Database link to which changes are applied. If null, then changes are applied to the local database.
SOURCE_ROOT_NAME	VARCHAR2(128)		The global name of the source root database

See Also

"DBA_APPLY_INSTANTIATED_GLOBAL"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.65 ALL_APPLY_INSTANTIATED_OBJECTS

ALL_APPLY_INSTANTIATED_OBJECTS displays information about objects accessible to the current user for which an instantiation SCN has been set.

Related View

DBA_APPLY_INSTANTIATED_OBJECTS displays information about objects for which an instantiation SCN has been set.

Column	Datatype	NULL	Description
SOURCE_DATABASE	VARCHAR2(128)	NOT NULL	Name of the database where the object originated
SOURCE_OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the object at the source database
SOURCE_OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object at the source database
SOURCE_OBJECT_TYPE	VARCHAR2(11)		Type of the object at the source database
INSTANTIATION_SCN	NUMBER		Instantiation SCN for the object. Only changes committed after this SCN are applied by an apply process.
IGNORE_SCN	NUMBER		SCN below which the instantiation SCN cannot be set. This value corresponds to the SCN value at the source database at the time when the object was prepared for instantiation.
APPLY_DATABASE_LINK	VARCHAR2(128)		Database link to which changes are applied. If null, then changes are applied to the local database.
SOURCE_ROOT_NAME	VARCHAR2(128)		The global name of the source root database

See Also

"DBA_APPLY_INSTANTIATED_OBJECTS"

2.66 ALL_APPLY_INSTANTIATED_SCHEMAS

ALL_APPLY_INSTANTIATED_SCHEMAS displays information about schemas accessible to the current user for which an instantiation SCN has been set.

Related View

DBA_APPLY_INSTANTIATED_SCHEMAS displays information about schemas for which an instantiation SCN has been set.

Column	Datatype	NULL	Description
SOURCE_DATABASE	VARCHAR2(128)	NOT NULL	Name of the database where the schema originated
SOURCE_SCHEMA	VARCHAR2(128)		Name of the schema at the source database
INSTANTIATION_SCN	NUMBER		Instantiation SCN for the schema. Only changes committed after this SCN are applied by an apply process.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
APPLY_DATABASE_LINK	VARCHAR2(128)		Database link to which changes are applied. If null, then changes are applied to the local database.
SOURCE_ROOT_NAME	VARCHAR2(128)		The global name of the source root database

See Also

"DBA_APPLY_INSTANTIATED_SCHEMAS"

2.67 ALL_APPLY_KEY_COLUMNS

ALL_APPLY_KEY_COLUMNS displays information about the substitute key columns for the tables accessible to the current user. Substitute key columns are set using the DBMS_APPLY_ADM.SET_KEY_COLUMNS procedure.

Related View

DBA_APPLY_KEY_COLUMNS displays information about the substitute key columns for all tables in the database.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the object on which substitute key columns are set
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object on which substitute key columns are set
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Column name of a column specified as a substitute key column
APPLY_DATABASE_LINK	VARCHAR2(128)		Database link to which changes are applied. If null, then changes are applied to the local database.

See Also

- "DBA_APPLY_KEY_COLUMNS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLY_ADM.SET_KEY_COLUMNS procedure

2.68 ALL_APPLY_PARAMETERS

ALL_APPLY_PARAMETERS displays information about the parameters for the apply processes that dequeue events from queues accessible to the current user.

Related View

DBA_APPLY_PARAMETERS displays information about the parameters for all apply processes in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
APPLY_NAME	VARCHAR2(128)	NOT NULL	Name of the apply process
PARAMETER	VARCHAR2(128)	NOT NULL	Name of the parameter
VALUE	VARCHAR2(4000)		Parameter value
SET_BY_USER	VARCHAR2(3)		Indicates whether the parameter value was set by the user (YES) or was not set by the user (NO). If NO for a parameter, then the parameter is set to its default value. If YES for a parameter, then the parameter may or may not be set to its default value.

See Also

"DBA_APPLY_PARAMETERS"

2.69 ALL_APPLY_PROGRESS

ALL_APPLY_PROGRESS displays information about the progress made by the apply processes that dequeue events from queues accessible to the current user. This view only contains information about captured events. It does not contain information about user-enqueued events.

Related View

DBA_APPLY_PROGRESS displays information about the progress made by all apply processes in the database.

Column	Datatype	NULL	Description
APPLY_NAME	VARCHAR2(128)	NOT NULL	Name of the apply process
SOURCE_DATABASE	VARCHAR2(128)	NOT NULL	Global name of the source database of the changes that are applied by the apply process
APPLIED_MESSAGE_NUMBER	NUMBER	NOT NULL	Message number up to which all transactions have definitely been applied. This value is the low watermark for the apply process. That is, messages with a commit message number less than or equal to this message number have definitely been applied, but some messages with a higher commit message number may also have been applied.
OLDEST_MESSAGE_NUMBER	NUMBER	NOT NULL	Earliest message number of the transactions currently being dequeued and applied
APPLY_TIME	DATE		Time at which the message with the message number displayed in the APPLIED_MESSAGE_NUMBER column was applied
APPLIED_MESSAGE_CREATE_TIME	DATE		Time at which the message with the message number displayed in the APPLIED_MESSAGE_NUMBER column was created at its source database
OLDEST_TRANSACTION_ID	VARCHAR2(128)		Oldest transaction ID of interest. (useful for detecting long-running or large transactions)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SPILL_MESSAGE_NUMBER	NUMBER		Spill low watermark. Any message with a lower SCN has either been applied or spilled to disk (it will be dequeued from the Streams queue and capture will not need to resend any logical change records (LCRs) with a lower SCN). Spilled messages may not have been applied yet.
SOURCE_ROOT_NAME	VARCHAR2(128)		The global name of the source root database

See Also

"DBA_APPLY_PROGRESS"

2.70 ALL_APPLY_REPERROR_HANDLERS

ALL_APPLY_REPERROR_HANDLERS provides details about apply reperror handlers on objects visible to the user.

Related View

DBA_APPLY_REPERROR_HANDLERS provides details about apply reperror handlers.

Column	Datatype	NULL	Description
APPLY_NAME	VARCHAR2(128)	NOT NULL	Name of the apply process
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object
SOURCE_OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Source database owner of the source object
SOURCE_OBJECT_NAME	VARCHAR2(128)	NOT NULL	Source database name of the object
ERROR_NUMBER	NUMBER	NOT NULL	Error number for the handler
METHOD	VARCHAR2(18)		Error handling method: • ABEND • RECORD • IGNORE • RETRY • RETRY_TRANSACTION • RECORD_TRANSACTION
MAX_RETRIES	NUMBER		Maximum number of times to retry for the method RETRY and RETRY_TRANSACTION
DELAY_CSECS	NUMBER		Number of centiseconds to wait between retries for RETRY and RETRY_TRANSACTION
SET_BY	VARCHAR2(10)		Entity that set up the handler: • USER • GOLDENGATE

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "DBA_APPLY_REPERROR_HANDLERS"

2.71 ALL_APPLY_TABLE_COLUMNS

ALL_APPLY_TABLE_COLUMNS displays, for the tables accessible to the current user, information about the nonkey table columns for which apply process conflict detection has been stopped for update and delete operations.

Conflict detection for nonkey columns can be stopped using the DBMS_APPLY_ADM.COMPARE_OLD_VALUES procedure.

Related View

DBA_APPLY_TABLE_COLUMNS displays, for all tables in the database, information about the nonkey table columns for which apply process conflict detection has been stopped for update and delete operations.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)		Owner of the table
OBJECT_NAME	VARCHAR2(128)		Name of the table
COLUMN_NAME	VARCHAR2(4000)		Name of the column
COMPARE_OLD_ON_DELETE	VARCHAR2(3)		Indicates whether to Compare the old value of the column on deletes (YES) or not (NO)
COMPARE_OLD_ON_UPDATE	VARCHAR2(3)		Indicates whether to Compare the old value of the column on updates (YES) or not (NO)
APPLY_DATABASE_LINK	VARCHAR2(128)		For remote tables, name of the database link pointing to the remote database

> **See Also**
>
> - "DBA_APPLY_TABLE_COLUMNS"
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLY_ADM.COMPARE_OLD_VALUES procedure

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.72 ALL_ARGUMENTS

ALL_ARGUMENTS lists the arguments of the functions and procedures that are accessible to the current user.

> **Note**
>
> The following changes have been made to this view:
>
> - Starting with Oracle Database 12c release 1 (12.1.0.2), this view omits procedures with no arguments. Prior to Oracle Database 12c release 1 (12.1.0.2), a procedure with no arguments was presented as a single row in this view.
> - Starting with Oracle Database 18c, this view displays only one row for an argument that is a composite type. Prior to Oracle Database 18c, this view displayed multiple rows for composite types.
>
> To obtain information about composite type arguments, use the value of the TYPE_NAME column in this view to query the ALL_PLSQL_TYPES, ALL_PLSQL_TYPE_ATTRS, and ALL_PLSQL_COLL_TYPES views, which fully describe composite types.
>
> See *Oracle Database Upgrade Guide* for more information about these changes.

Related Views

- DBA_ARGUMENTS lists the arguments of the functions and procedures that are available in the database.
- USER_ARGUMENTS lists the arguments of the functions and procedures that are owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
OBJECT_NAME	VARCHAR2(128)		Name of the procedure or function
PACKAGE_NAME	VARCHAR2(128)		Name of the package
OBJECT_ID	NUMBER	NOT NULL	Object number of the object
OVERLOAD	VARCHAR2(40)		Indicates the *n*th overloading ordered by its appearance in the source; otherwise, it is NULL.
SUBPROGRAM_ID	NUMBER		Unique subprogram identifier
ARGUMENT_NAME	VARCHAR2(128)		Name of the argument A null argument name is used to denote a function return.
POSITION	NUMBER	NOT NULL	This column holds the position of this item in the argument list, or 0 for a function return value.
SEQUENCE	NUMBER	NOT NULL	Defines the sequential order of the argument. Argument sequence starts from 1. Return type comes first, and each argument will follow.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DATA_LEVEL	NUMBER	NOT NULL	Nesting depth of the argument for composite types **Note:** Starting with Oracle Database 18c, the value of this column is always 0, because this view displays only one row for each argument. This view no longer displays multiple rows for composite type arguments.
DATA_TYPE	VARCHAR2(30)		Datatype of the argument
DEFAULTED	VARCHAR2(1)		Specifies whether or not the argument is defaulted
DEFAULT_VALUE	LONG		Reserved for future use
DEFAULT_LENGTH	NUMBER		Reserved for future use
IN_OUT	VARCHAR2(9)		Direction of the argument: • IN • OUT • IN/OUT
DATA_LENGTH	NUMBER		Length of the column (in bytes)
DATA_PRECISION	NUMBER		Length in decimal digits (NUMBER) or binary digits (FLOAT)
DATA_SCALE	NUMBER		Digits to the right of the decimal point in a number
RADIX	NUMBER		Argument radix for a number
CHARACTER_SET_NAME	VARCHAR2(44)		Character set name for the argument
TYPE_OWNER	VARCHAR2(128)		Owner of the type of the argument
TYPE_NAME	VARCHAR2(128)		Name of the type of the argument. If the type is a package local type (that is, it is declared in a package specification), then this column displays the name of the package.
TYPE_SUBNAME	VARCHAR2(128)		Relevant only for package local types. Displays the name of the type declared in the package identified in the TYPE_NAME column.
TYPE_LINK	VARCHAR2(128)		Relevant only for package local types when the package identified in the TYPE_NAME column is a remote package. This column displays the database link used to refer to the remote package.
TYPE_OBJECT_TYPE	VARCHAR2(7)		Displays the type of the type described by the TYPE_OWNER, TYPE_NAME, and TYPE_SUBNAME columns. Possible values are: • TABLE • VIEW • PACKAGE • TYPE
PLS_TYPE	VARCHAR2(128)		For numeric arguments, the name of the PL/SQL type of the argument. Null otherwise.
CHAR_LENGTH	NUMBER		Character limit for string datatypes
CHAR_USED	VARCHAR2(1)		Indicates whether the byte limit (B) or char limit (C) is official for the string

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORIGIN_CON_ID	VARCHAR2(256)		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)

Note

To list the procedure names in a package, use the ALL_PROCEDURES view.

See Also

- "DBA_ARGUMENTS"
- "USER_ARGUMENTS"
- "ALL_PLSQL_TYPES"
- "ALL_PLSQL_TYPE_ATTRS"
- "ALL_PLSQL_COLL_TYPES"
- "ALL_PROCEDURES" for information about the functions and procedures that are accessible to the current user

2.73 ALL_ASSEMBLIES

ALL_ASSEMBLIES provides information about assemblies accessible to the current user.

Related Views

- DBA_ASSEMBLIES provides information about all assemblies in the database.
- USER_ASSEMBLIES provides information about all assemblies owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the assembly
ASSEMBLY_NAME	VARCHAR2(128)	NOT NULL	Name of the assembly
FILE_SPEC	VARCHAR2(4000)		Operating system file specification of the assembly
SECURITY_LEVEL	VARCHAR2(10)		The maximum security level of the assembly
IDENTITY	VARCHAR2(4000)		The identity of the assembly
STATUS	VARCHAR2(7)		Status of the assembly

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_ASSEMBLIES"
- "USER_ASSEMBLIES"

2.74 ALL_ASSOCIATIONS

ALL_ASSOCIATIONS describes user-defined statistics associated with objects accessible to the current user.

Related Views

- DBA_ASSOCIATIONS describes all user-defined statistics in the database.
- USER_ASSOCIATIONS describes user-defined statistics associated with objects owned by the current user.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the object for which the association is being defined
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object for which the association is being defined
COLUMN_NAME	VARCHAR2(128)		Column name in the object for which the association is being defined
OBJECT_TYPE	VARCHAR2(9)		Kind of object with which statistics are being associated: column, type, package or function, indextype, or domain index.
STATSTYPE_SCHEMA	VARCHAR2(128)		Owner of the statistics type
STATSTYPE_NAME	VARCHAR2(128)		Name of statistics type that contains the cost, selectivity or statistics functions
DEF_SELECTIVITY	NUMBER		Default selectivity of the object, if any
DEF_CPU_COST	NUMBER		Default CPU cost of the object, if any
DEF_IO_COST	NUMBER		Default I/O cost of the object, if any
DEF_NET_COST	NUMBER		Default networking cost of the object, if any
INTERFACE_VERSION	NUMBER		Identifies the version number of the ODCIStats interface. Value is 1 for statistics type implementing Oracle8*i* 8.1; 0 for types implementing Oracle9*i* 9.0.0.
MAINTENANCE_TYPE	VARCHAR2(14)		Specifies whether the object is system-managed or user-managed

See Also

- "DBA_ASSOCIATIONS"
- "USER_ASSOCIATIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.75 ALL_ATTRIBUTE_DIM_ATTR_CLASS

ALL_ATTRIBUTE_DIM_ATTR_CLASS describes the attribute classifications of the attribute dimensions accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_ATTR_CLASS describes the attribute classifications of all attribute dimensions in the database.
- USER_ATTRIBUTE_DIM_ATTR_CLASS describes the attribute classifications of the attribute dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
ATTRIBUTE_NAME	VARCHAR2(128)		Name of the attribute
CLASSIFICATION	VARCHAR2(128)		Classification associated with the attribute
VALUE	CLOB		Value of the classification, or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the attribute
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ATTRIBUTE_DIM_ATTR_CLASS"
- "USER_ATTRIBUTE_DIM_ATTR_CLASS"

2.76 ALL_ATTRIBUTE_DIM_ATTR_CLS

ALL_ATTRIBUTE_DIM_ATTR_CLS is identical to ALL_ATTRIBUTE_DIM_ATTR_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "ALL_ATTRIBUTE_DIM_ATTR_CLASS"

2.77 ALL_ATTRIBUTE_DIM_ATTR_CLS_AE

ALL_ATTRIBUTE_DIM_ATTR_CLS_AE describes the attribute classifications of the attribute dimensions (across all editions) accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_ATTR_CLS_AE describes the attribute classifications of all attribute dimensions (across all editions) in the database.
- USER_ATTRIBUTE_DIM_ATTR_CLS_AE describes the attribute classifications of the attribute dimensions (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
ATTRIBUTE_NAME	VARCHAR2(128)		Name of the attribute
CLASSIFICATION	VARCHAR2(128)		Classification associated with the attribute
VALUE	CLOB		Value of the classification, or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the attribute
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the attribute dimension is defined

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> - "DBA_ATTRIBUTE_DIM_ATTR_CLS_AE"
> - "USER_ATTRIBUTE_DIM_ATTR_CLS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.78 ALL_ATTRIBUTE_DIM_ATTRS

ALL_ATTRIBUTE_DIM_ATTRS describes the attributes of the attribute dimensions accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_ATTRS describes the attributes of all attribute dimensions in the database.
- USER_ATTRIBUTE_DIM_ATTRS describes the attributes of the attribute dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
ATTRIBUTE_NAME	VARCHAR2(128)		Name of the attribute dimension attribute
TABLE_ALIAS	VARCHAR2(128)		Alias of the table or view in the USING clause to which the column belongs
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column in the table or view on which the attribute is defined
ORDER_NUM	NUMBER	NOT NULL	Order of the attribute in the list of attribute dimension attributes
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ATTRIBUTE_DIM_ATTRS"
- "USER_ATTRIBUTE_DIM_ATTRS"

2.79 ALL_ATTRIBUTE_DIM_ATTRS_AE

ALL_ATTRIBUTE_DIM_ATTRS_AE describes the attributes of the attribute dimensions (across all editions) accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_ATTRS_AE describes the attributes of all attribute dimensions (across all editions) in the database.
- USER_ATTRIBUTE_DIM_ATTRS_AE describes the attributes of the attribute dimensions (across all editions) owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
ATTRIBUTE_NAME	VARCHAR2(128)		Name of the attribute dimension attribute
TABLE_ALIAS	VARCHAR2(128)		Alias of the table or view in the USING clause to which the column belongs
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column in the table or view on which the attribute is defined
ORDER_NUM	NUMBER	NOT NULL	Order of the attribute in the list of attribute dimension attributes
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the attribute dimension is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ATTRIBUTE_DIM_ATTRS_AE"
- "USER_ATTRIBUTE_DIM_ATTRS_AE"

2.80 ALL_ATTRIBUTE_DIM_CLASS

ALL_ATTRIBUTE_DIM_CLASS describes the classifications of the attribute dimensions accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_CLASS describes the classifications of all attribute dimensions in the database.
- USER_ATTRIBUTE_DIM_CLASS describes the classifications of the attribute dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
CLASSIFICATION	VARCHAR2(128)		Classification associated with the attribute dimension

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
VALUE	CLOB		Value of the classification, or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the attribute dimension
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ATTRIBUTE_DIM_CLASS"
- " USER_ATTRIBUTE_DIM_CLASS"

2.81 ALL_ATTRIBUTE_DIM_CLASS_AE

ALL_ATTRIBUTE_DIM_CLASS_AE describes the classifications of the attribute dimensions (across all editions) accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_CLASS_AE describes the classifications of all attribute dimensions (across all editions) in the database.
- USER_ATTRIBUTE_DIM_CLASS_AE describes the classifications of the attribute dimensions (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
CLASSIFICATION	VARCHAR2(128)		Classification associated with the attribute dimension
VALUE	CLOB		Value of the classification, or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the attribute dimension

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the attribute dimension is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ATTRIBUTE_DIM_CLASS_AE"
- "USER_ATTRIBUTE_DIM_CLASS_AE"

2.82 ALL_ATTRIBUTE_DIM_JN_PTHS

ALL_ATTRIBUTE_DIM_JN_PTHS is identical to ALL_ATTRIBUTE_DIM_JOIN_PATHS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_JOIN_PATHS"

2.83 ALL_ATTRIBUTE_DIM_JN_PTHS_AE

ALL_ATTRIBUTE_DIM_JN_PTHS_AE describes the join paths for the attribute dimensions (across all editions) accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_JN_PTHS_AE describes the join paths for all attribute dimensions (across all editions) in the database.
- USER_ATTRIBUTE_DIM_JN_PTHS_AE describes the join paths for the attribute dimensions (across all editions) owned by the current user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
JOIN_PATH_NAME	VARCHAR2(128)	NOT NULL	Name of the join path
ON_CONDITION	VARCHAR2(4000)		Join condition specified in the ON clause
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the attribute dimension
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the attribute dimension is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ATTRIBUTE_DIM_JN_PTHS_AE"
- "USER_ATTRIBUTE_DIM_JN_PTHS_AE"

2.84 ALL_ATTRIBUTE_DIM_JOIN_PATHS

ALL_ATTRIBUTE_DIM_JOIN_PATHS describes the join paths for the attribute dimensions accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_JOIN_PATHS describes the join paths for all attribute dimensions in the database.
- USER_ATTRIBUTE_DIM_JOIN_PATHS describes the join paths for the attribute dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
JOIN_PATH_NAME	VARCHAR2(128)	NOT NULL	Name of the join path
ON_CONDITION	VARCHAR2(4000)		Join condition specified in the ON clause

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the attribute dimension
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ATTRIBUTE_DIM_JOIN_PATHS"
- "USER_ATTRIBUTE_DIM_JOIN_PATHS"

2.85 ALL_ATTRIBUTE_DIM_KEYS

ALL_ATTRIBUTE_DIM_KEYS describes the keys of the attribute dimensions accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_KEYS describes the keys of all attribute dimensions in the database.
- USER_ATTRIBUTE_DIM_KEYS describes the keys of the attribute dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
LEVEL_NAME	VARCHAR2(128)		Name of the level of the key
IS_ALTERNATE	VARCHAR2(1)		Indicates whether the attribute dimension key is an alternate key (Y) or not (N)
ATTRIBUTE_NAME	VARCHAR2(128)		Name of the key attribute
ATTR_ORDER_NUM	NUMBER	NOT NULL	Order of the attribute in the list of attributes comprising the key
KEY_ORDER_NUM	NUMBER	NOT NULL	Order of the key in the list of keys (if alternate keys are specified)
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> - "DBA_ATTRIBUTE_DIM_KEYS"
> - "USER_ATTRIBUTE_DIM_KEYS"

2.86 ALL_ATTRIBUTE_DIM_KEYS_AE

`ALL_ATTRIBUTE_DIM_KEYS_AE` describes the keys of the attribute dimensions (across all editions) accessible to the current user.

Related Views

- `DBA_ATTRIBUTE_DIM_KEYS_AE` describes the keys of all attribute dimensions (across all editions) in the database.
- `USER_ATTRIBUTE_DIM_KEYS_AE` describes the keys of the attribute dimensions (across all editions) owned by the current user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the attribute dimension
`DIMENSION_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the attribute dimension
`LEVEL_NAME`	`VARCHAR2(128)`		Name of the level of the key
`IS_ALTERNATE`	`VARCHAR2(1)`		Indicates whether the attribute dimension key is an alternate key (`Y`) or not (`N`)
`ATTRIBUTE_NAME`	`VARCHAR2(128)`		Name of the key attribute
`ATTR_ORDER_NUM`	`NUMBER`	`NOT NULL`	Order of the attribute in the list of attributes comprising the key
`KEY_ORDER_NUM`	`NUMBER`	`NOT NULL`	Order of the key in the list of keys (if alternate keys are specified)
`ORIGIN_CON_ID`	`NUMBER`		The ID of the container where the data originates. Possible values include: • `0`: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
`EDITION_NAME`	`VARCHAR2(128)`		Name of the application edition where the attribute dimension is defined

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_ATTRIBUTE_DIM_KEYS_AE"
- "USER_ATTRIBUTE_DIM_KEYS_AE"

2.87 ALL_ATTRIBUTE_DIM_LEVEL_ATTRS

ALL_ATTRIBUTE_DIM_LEVEL_ATTRS describes the attributes of the levels of the attribute dimensions accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_LEVEL_ATTRS describes the attributes of the levels of all attribute dimensions in the database.
- USER_ATTRIBUTE_DIM_LEVEL_ATTRS describes the attributes of the levels of the attribute dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)		Name of the attribute dimension
LEVEL_NAME	VARCHAR2(128)		Name of the attribute dimension level
ATTRIBUTE_NAME	VARCHAR2(128)		Name of the attribute determined by the level
ROLE	VARCHAR2(4)		Role of the attribute determined by the level
IS_MINIMAL_DTM	VARCHAR2(1)		Indicates whether the attribute is minimally determined (Y) or not (N)
ORDER_NUM	NUMBER		Order of the attribute in the list of attributes determined by the level
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ATTRIBUTE_DIM_LEVEL_ATTRS"
- "USER_ATTRIBUTE_DIM_LEVEL_ATTRS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.88 ALL_ATTRIBUTE_DIM_LEVELS

ALL_ATTRIBUTE_DIM_LEVELS describes the levels of the attribute dimensions accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_LEVELS describes the levels of all attribute dimensions in the database.
- USER_ATTRIBUTE_DIM_LEVELS describes the levels of the attribute dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
LEVEL_NAME	VARCHAR2(128)		Name of the attribute dimension level
SKIP_WHEN_NULL	VARCHAR2(1)		Indicates whether to skip the level when the key is NULL; the value can be Y or N.
LEVEL_TYPE	VARCHAR2(10)		Type of attribute dimension level
MEMBER_NAME_EXPR	CLOB	NOT NULL	Expression representing the level member name
MEMBER_CAPTION_EXPR	CLOB		Expression representing the level member caption, or NULL if not specified
MEMBER_DESCRIPTION_EXPR	CLOB		Expression representing the level member description, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the level in the list of attribute dimension levels
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ATTRIBUTE_DIM_LEVELS"
- "USER_ATTRIBUTE_DIM_LEVELS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.89 ALL_ATTRIBUTE_DIM_LEVELS_AE

ALL_ATTRIBUTE_DIM_LEVELS_AE describes the levels of the attribute dimensions (across all editions) accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_LEVELS_AE describes the levels of all attribute dimensions (across all editions) in the database.
- USER_ATTRIBUTE_DIM_LEVELS_AE describes the levels of the attribute dimensions (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
LEVEL_NAME	VARCHAR2(128)		Name of the attribute dimension level
SKIP_WHEN_NULL	VARCHAR2(1)		Indicates whether to skip the level when the key is NULL; the value can be Y or N.
LEVEL_TYPE	VARCHAR2(10)		Type of attribute dimension level
MEMBER_NAME_EXPR	CLOB	NOT NULL	Expression representing the level member name
MEMBER_CAPTION_EXPR	CLOB		Expression representing the level member caption, or NULL if not specified
MEMBER_DESCRIPTION_EXPR	CLOB		Expression representing the level member description, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the level in the list of attribute dimension levels
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the attribute dimension is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ATTRIBUTE_DIM_LEVELS_AE"
- "USER_ATTRIBUTE_DIM_LEVELS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.90 ALL_ATTRIBUTE_DIM_LVL_ATRS

ALL_ATTRIBUTE_DIM_LVL_ATRS is identical to ALL_ATTRIBUTE_DIM_LEVEL_ATTRS.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ATTRIBUTE_DIM_LEVEL_ATTRS"

2.91 ALL_ATTRIBUTE_DIM_LVL_ATRS_AE

ALL_ATTRIBUTE_DIM_LVL_ATRS_AE describes the attributes of the levels of the attribute dimensions (across all editions) accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_LVL_ATRS_AE describes the attributes of the levels of all attribute dimensions (across all editions) in the database.
- USER_ATTRIBUTE_DIM_LVL_ATRS_AE describes the attributes of the levels of the attribute dimensions (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)		Name of the attribute dimension
LEVEL_NAME	VARCHAR2(128)		Name of the attribute dimension level
ATTRIBUTE_NAME	VARCHAR2(128)		Name of the attribute determined by the level
ROLE	VARCHAR2(4)		Role of the attribute determined by the level
IS_MINIMAL_DTM	VARCHAR2(1)		Indicates whether the attribute is minimally determined (Y) or not (N)
ORDER_NUM	NUMBER		Order of the attribute in the list of attributes determined by the level
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the attribute dimension is defined

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> - "DBA_ATTRIBUTE_DIM_LVL_ATRS_AE"
> - "USER_ATTRIBUTE_DIM_LVL_ATRS_AE"

2.92 ALL_ATTRIBUTE_DIM_LVL_CLASS

`ALL_ATTRIBUTE_DIM_LVL_CLASS` describes the level classifications of the attribute dimensions accessible to the current user.

Related Views

- `DBA_ATTRIBUTE_DIM_LVL_CLASS` describes the level classifications of all attribute dimensions in the database.
- `USER_ATTRIBUTE_DIM_LVL_CLASS` describes the level classifications of the attribute dimensions owned by the current user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the attribute dimension
`DIMENSION_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the attribute dimension
`LEVEL_NAME`	`VARCHAR2(128)`		Name of the level
`CLASSIFICATION`	`VARCHAR2(128)`		Classification associated with the level
`VALUE`	`CLOB`		Value of the classification, or NULL if not specified
`LANGUAGE`	`VARCHAR2(64)`		`NLS_LANGUAGE` value associated with the classification, or NULL if not specified
`ORDER_NUM`	`NUMBER`	`NOT NULL`	Order of the classification in the list of classifications associated with the level
`ORIGIN_CON_ID`	`NUMBER`		The ID of the container where the data originates. Possible values include: • `0`: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

> **See Also**
>
> - "DBA_ATTRIBUTE_DIM_LVL_CLASS"
> - "USER_ATTRIBUTE_DIM_LVL_CLASS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.93 ALL_ATTRIBUTE_DIM_LVL_CLS

ALL_ATTRIBUTE_DIM_LVL_CLS is identical to ALL_ATTRIBUTE_DIM_LVL_CLASS.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ATTRIBUTE_DIM_LVL_CLASS"

2.94 ALL_ATTRIBUTE_DIM_LVL_CLS_AE

ALL_ATTRIBUTE_DIM_LVL_CLS_AE describes the level classifications of the attribute dimensions (across all editions) accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_LVL_CLS_AE describes the level classifications of all attribute dimensions (across all editions) in the database.
- USER_ATTRIBUTE_DIM_LVL_CLS_AE describes the level classifications of the attribute dimensions (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
LEVEL_NAME	VARCHAR2(128)		Name of the level
CLASSIFICATION	VARCHAR2(128)		Classification associated with the level
VALUE	CLOB		Value of the classification, or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification, or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the level
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the attribute dimension is defined

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> - "DBA_ATTRIBUTE_DIM_LVL_CLS_AE"
> - "USER_ATTRIBUTE_DIM_LVL_CLS_AE"

2.95 ALL_ATTRIBUTE_DIM_ORD_ATRS

`ALL_ATTRIBUTE_DIM_ORD_ATRS` is identical to `ALL_ATTRIBUTE_DIM_ORDER_ATTRS`.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ATTRIBUTE_DIM_ORDER_ATTRS"

2.96 ALL_ATTRIBUTE_DIM_ORD_ATRS_AE

`ALL_ATTRIBUTE_DIM_ORD_ATRS_AE` describes the order attributes of the attribute dimensions (across all editions) accessible to the current user.

Related Views

- `DBA_ATTRIBUTE_DIM_ORD_ATRS_AE` describes the order attributes of all attribute dimensions (across all editions) in the database.
- `USER_ATTRIBUTE_DIM_ORD_ATRS_AE` describes the order attributes of the attribute dimensions (across all editions) owned by the current user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the attribute dimension
`DIMENSION_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the attribute dimension
`LEVEL_NAME`	`VARCHAR2(128)`		Name of the level to order or the name of the level that has the `ORDER BY` clause
`AGG_FUNC`	`VARCHAR2(3)`		Aggregation function of the `ORDER BY` clause: • `MIN` • `MAX`
`ATTRIBUTE_NAME`	`VARCHAR2(128)`		Name of the order attribute

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORDER_NUM	NUMBER	NOT NULL	Order number of the attribute in the list of order attributes
CRITERIA	VARCHAR2(4)		Criteria of the ordering, either ascending or descending: • ASC • DESC
NULLS_POSITION	VARCHAR2(5)		Position of ORDER BY values in the orderings: • FIRST • LAST
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the attribute dimension is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ATTRIBUTE_DIM_ORD_ATRS_AE"
- "USER_ATTRIBUTE_DIM_ORD_ATRS_AE"

2.97 ALL_ATTRIBUTE_DIM_ORDER_ATTRS

ALL_ATTRIBUTE_DIM_ORDER_ATTRS describes the order attributes of the attribute dimensions accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_ORDER_ATTRS describes the order attributes of all attribute dimensions in the database.
- USER_ATTRIBUTE_DIM_ORDER_ATTRS describes the order attributes of the attribute dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LEVEL_NAME	VARCHAR2(128)		Name of the level to order or the name of the level that has the ORDER BY clause
AGG_FUNC	VARCHAR2(3)		Aggregation function of the ORDER BY clause: • MIN • MAX
ATTRIBUTE_NAME	VARCHAR2(128)		Name of the order attribute
ORDER_NUM	NUMBER	NOT NULL	Order number of the attribute in the list of order attributes
CRITERIA	VARCHAR2(4)		Criteria of the ordering, either ascending or descending: • ASC • DESC
NULLS_POSITION	VARCHAR2(5)		Position of ORDER BY values in the orderings: • FIRST • LAST
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ATTRIBUTE_DIM_ORDER_ATTRS"
- "USER_ATTRIBUTE_DIM_ORDER_ATTRS"

2.98 ALL_ATTRIBUTE_DIM_TABLES

ALL_ATTRIBUTE_DIM_TABLES describes the tables used by the attribute dimensions accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_TABLES describes the tables used by all attribute dimensions in the database.
- USER_ATTRIBUTE_DIM_TABLES describes the tables used by the attribute dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of table used by the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the table or view used by the attribute dimension

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table or view used by the attribute dimension
TABLE_ALIAS	VARCHAR2(128)		Alias specified for the table or view; if not specified, the name of the table or view
ORDER_NUM	NUMBER	NOT NULL	Order of the table in the list of tables in the USING clause
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ATTRIBUTE_DIM_TABLES"
- "USER_ATTRIBUTE_DIM_TABLES"

2.99 ALL_ATTRIBUTE_DIM_TABLES_AE

ALL_ATTRIBUTE_DIM_TABLES_AE describes the tables used by the attribute dimensions (across all editions) accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIM_TABLES_AE describes the tables used by all attribute dimensions (across all editions) in the database.
- USER_ATTRIBUTE_DIM_TABLES_AE describes the tables used by the attribute dimensions (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of table used by the attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the table or view used by the attribute dimension
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table or view used by the attribute dimension
TABLE_ALIAS	VARCHAR2(128)		Alias specified for the table or view; if not specified, the name of the table or view
ORDER_NUM	NUMBER	NOT NULL	Order of the table in the list of tables in the USING clause

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the attribute dimension is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ATTRIBUTE_DIM_TABLES_AE"
- "USER_ATTRIBUTE_DIM_TABLES_AE"

2.100 ALL_ATTRIBUTE_DIMENSIONS

ALL_ATTRIBUTE_DIMENSIONS describes the attribute dimensions accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIMENSIONS describes all attribute dimensions in the database.
- USER_ATTRIBUTE_DIMENSIONS describes the attribute dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
DIMENSION_TYPE	VARCHAR2(8)		Type of the attribute dimension: • TIME • STANDARD
ALL_MEMBER_NAME	CLOB)	NOT NULL	An expression for the name of the ALL member for the attribute dimension
ALL_MEMBER_CAPTION	CLOB		An expression for the caption for the ALL member of the attribute dimension, or NULL if not specified
ALL_MEMBER_DESCRIPTION	CLOB		An expression for the description for the ALL member of the attribute dimension
COMPILE_STATE	VARCHAR2(7)		Compile status of the attribute dimension: • VALID • INVALID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_ATTRIBUTE_DIMENSIONS"
- "USER_ATTRIBUTE_DIMENSIONS"

2.101 ALL_ATTRIBUTE_DIMENSIONS_AE

ALL_ATTRIBUTE_DIMENSIONS_AE describes the attribute dimensions (across all editions) accessible to the current user.

Related Views

- DBA_ATTRIBUTE_DIMENSIONS_AE describes all attribute dimensions (across all editions) in the database.
- USER_ATTRIBUTE_DIMENSIONS_AE describes the attribute dimensions (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of attribute dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension
DIMENSION_TYPE	VARCHAR2(8)		Type of the attribute dimension: • TIME • STANDARD
ALL_MEMBER_NAME	CLOB)	NOT NULL	An expression for the name of the ALL member for the attribute dimension
ALL_MEMBER_CAPTION	CLOB		An expression for the caption for the ALL member of the attribute dimension, or NULL if not specified
ALL_MEMBER_DESCRIPTION	CLOB		An expression for the description for the ALL member of the attribute dimension
COMPILE_STATE	VARCHAR2(7)		Compile status of the attribute dimension: • VALID • INVALID
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the attribute dimension is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_ATTRIBUTE_DIMENSIONS_AE"
- "USER_ATTRIBUTE_DIMENSIONS_AE"

2.102 ALL_ATTRIBUTE_TRANSFORMATIONS

ALL_ATTRIBUTE_TRANSFORMATIONS describes the transformation functions for the transformations accessible to the current user.

Related Views

- DBA_ATTRIBUTE_TRANSFORMATIONS describes the transformation functions for all transformations in the database.
- USER_ATTRIBUTE_TRANSFORMATIONS describes the transformation functions for the transformations owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
TRANSFORMATION_ID	NUMBER	NOT NULL	Unique identifier for the transformation
OWNER	VARCHAR2(128)	NOT NULL	Owning user of the transformation
NAME	VARCHAR2(128)	NOT NULL	Transformation name
FROM_TYPE	VARCHAR2(257)		Source type name
TO_TYPE	VARCHAR2(385)		Target type name
ATTRIBUTE	NUMBER	NOT NULL	Target type attribute number
ATTRIBUTE_TRANSFORMATION	VARCHAR2(4000)		Transformation function for the attribute

See Also

- "DBA_ATTRIBUTE_TRANSFORMATIONS"
- "USER_ATTRIBUTE_TRANSFORMATIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.103 ALL_AUDIT_POLICIES

ALL_AUDIT_POLICIES describes the fine-grained auditing policies on the tables and views accessible to the current user.

Related Views

- DBA_AUDIT_POLICIES describes all fine-grained auditing policies in the database.
- USER_AUDIT_POLICIES describes the fine-grained auditing policies on the tables and views owned by the current user. This view does not display the OBJECT_SCHEMA column.

Column	Datatype	NULL	Description
OBJECT_SCHEMA	VARCHAR2(128)		Name of the schema that includes the table or view
OBJECT_NAME	VARCHAR2(128)		Name of the table or view
POLICY_OWNER	VARCHAR2(128)		Owner of the policy
POLICY_NAME	VARCHAR2(128)		Name of the policy
POLICY_TEXT	VARCHAR2(4000)		Audit condition
POLICY_COLUMN	VARCHAR2(128)		This column is deprecated. Query the ALL_AUDIT_POLICY_COLUMNS view for fine-grained auditing policy column information.
PF_SCHEMA	VARCHAR2(128)		Owner of the audit handler function
PF_PACKAGE	VARCHAR2(128)		Name of the package containing the audit handler function
PF_FUNCTION	VARCHAR2(128)		Name of the audit handler function
ENABLED	VARCHAR2(3)		Indicates whether the policy is enabled (YES) or disabled (NO)
SEL	VARCHAR2(3)		Indicates whether the policy is applied to queries on the object (YES) or not (NO)
INS	VARCHAR2(3)		Indicates whether the policy is applied to INSERT statements on the object (YES) or not (NO)
UPD	VARCHAR2(3)		Indicates whether the policy is applied to UPDATE statements on the object (YES) or not (NO)
DEL	VARCHAR2(3)		Indicates whether the policy is applied to DELETE statements on the object (YES) or not (NO)
AUDIT_TRAIL	VARCHAR2(12)		Indicates the audit trail to which the audit records generated by this audit policy will be written: • DB - Audit records are written to DBA_FGA_AUDIT_TRAIL (fine-grained audit trail) • DB+EXTENDED - Audit records are written to DBA_FGA_AUDIT_TRAIL (fine-grained audit trail) and the SQL_TEXT and SQL_BIND columns are populated for this policy • XML - Audit records are written to V$XML_AUDIT_TRAIL (XML audit files) • XML+EXTENDED - Audit records are written to V$XML_AUDIT_TRAIL (XML audit files) and the SQL_TEXT and SQL_BIND columns are populated for this policy

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
POLICY_COLUMN_OPTIONS	VARCHAR2(11)		Indicates whether all columns in the AUDIT_COLUMN parameter (ALL_COLUMNS) or any of the columns in the AUDIT_COLUMN parameter (ANY_COLUMNS) are considered for triggering fine-grained auditing
COMMON	VARCHAR2(3)		Indicates whether the policy applies across multiple containers (YES) or not (NO)
INHERITED	VARCHAR2(3)		Indicates whether the policy was inherited from another container (YES) or not (NO)

See Also

- "DBA_AUDIT_POLICIES"
- "USER_AUDIT_POLICIES"

2.104 ALL_AUDIT_POLICY_COLUMNS

ALL_AUDIT_POLICY_COLUMNS describes the fine-grained auditing policy columns on the tables and views accessible to the current user.

Related Views

- DBA_AUDIT_POLICY_COLUMNS describes all fine-grained auditing policy columns in the database.
- USER_AUDIT_POLICY_COLUMNS describes the fine-grained auditing policy columns on the tables and views owned by the current user.

Column	Datatype	NULL	Description
OBJECT_SCHEMA	VARCHAR2(128)		Owner of the table or view
OBJECT_NAME	VARCHAR2(128)		Name of the table or view
POLICY_NAME	VARCHAR2(128)		Name of the policy
POLICY_COLUMN	VARCHAR2(128)		Relevant column of the policy

See Also

- "DBA_AUDIT_POLICY_COLUMNS"
- "USER_AUDIT_POLICY_COLUMNS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.105 ALL_AW_PS

ALL_AW_PS describes the page spaces in the analytic workspaces accessible to the current user.

Related Views

- DBA_AW_PS describes the page spaces in all analytic workspaces in the database.
- USER_AW_PS describes the page spaces in the analytic workspaces owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic workspace
AW_NUMBER	NUMBER	NOT NULL	Number of the analytic workspace
AW_NAME	VARCHAR2(128)		Name of the analytic workspace
PSNUMBER	NUMBER(10)		Number of the page space
GENERATIONS	NUMBER		Number of active generations in the page space
MAXPAGES	NUMBER		Maximum pages allocated in the page space

See Also

- "DBA_AW_PS"
- "USER_AW_PS"
- *Oracle OLAP User's Guide* for more information about the OLAP option for Oracle Database

2.106 ALL_AWS

ALL_AWS describes the analytic workspaces accessible to the current user.

Related Views

- DBA_AWS describes all analytic workspaces in the database.
- USER_AWS describes the analytic workspaces owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the analytic workspace
AW_NUMBER	NUMBER	NOT NULL	Number of the analytic workspace
AW_NAME	VARCHAR2(128)		Name of the analytic workspace
AW_VERSION	VARCHAR2(4)		Format version of the analytic workspace: • 9.1 • 10.1 • 10.2 • 11.1

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PAGESPACES	NUMBER		Number of pagespaces in the analytic workspace
GENERATIONS	NUMBER		Number of active generations in the analytic workspace
FROZEN	VARCHAR2(6)		Freeze state of the analytic workspace: • Frozen • NoThaw

See Also

- "DBA_AWS"
- "USER_AWS"
- *Oracle OLAP User's Guide* for more information about the OLAP option for Oracle Database

2.107 ALL_BASE_TABLE_MVIEWS

ALL_BASE_TABLE_MVIEWS describes the materialized views using materialized view logs accessible to the current user. A materialized view log can be created for a master, base table, or master materialized view. Query this view at the master site or the master materialized view site to show one row for each materialized view using a materialized view log.

Related Views

- DBA_BASE_TABLE_MVIEWS describes all materialized views using materialized view logs in the database.
- USER_BASE_TABLE_MVIEWS describes the materialized views using materialized view logs owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Schema in which the master table or the master materialized view was created
MASTER	VARCHAR2(128)	NOT NULL	Name of the master table or the master materialized view
MVIEW_LAST_REFRESH_TIME	DATE	NOT NULL	Date when the materialized view based on the master was last refreshed
MVIEW_ID	NUMBER(38)		Unique identifier of the materialized view that is based on the master

See Also

- "DBA_BASE_TABLE_MVIEWS"
- "USER_BASE_TABLE_MVIEWS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.108 ALL_BLOCKCHAIN_TABLES

ALL_BLOCKCHAIN_TABLES describes the blockchain tables accessible to the current user.

Related Views

- DBA_BLOCKCHAIN_TABLES describes all blockchain tables in the database.
- USER_BLOCKCHAIN_TABLES describes the blockchain tables owned by the current user. This view does not display the SCHEMA_NAME column.

Column	Datatype	NULL	Description
SCHEMA_NAME	VARCHAR2(128)	NOT NULL	The schema containing the blockchain table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the blockchain table
ROW_RETENTION	NUMBER		Row retention period for the blockchain table, that is, the minimum number of days a row must be retained and cannot be deleted after it is inserted into the table If the value of this column is NULL, then rows can never be deleted from the table.
ROW_RETENTION_LOCKED	VARCHAR2(3)		Indicates whether the row retention period for the blockchain table is locked. Possible values: • YES: The row retention period is locked. You cannot change the row retention period • NO: The row retention period is not locked. You can change the row retention period to a value higher than the current value with the SQL statement ALTER TABLE … NO DELETE UNTIL *n* DAYS AFTER INSERT
TABLE_INACTIVITY_RETENTION	NUMBER		Number of days for which the blockchain table must be inactive before it can be dropped, that is, the number of days that must pass after the most recent row insertion before the table can be dropped A table with no rows can be dropped at any time, regardless of this column value.
HASH_ALGORITHM	VARCHAR2(8)		Algorithm used for computing the hash value for each table row

Note

This view is available starting with Oracle Database 19c, Release Update 19.10.

See Also

- "DBA_BLOCKCHAIN_TABLES"
- "USER_BLOCKCHAIN_TABLES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.109 ALL_CAPTURE

ALL_CAPTURE displays information about the capture processes that enqueue the captured changes into queues accessible to the current user.

Related View

DBA_CAPTURE displays information about all capture processes in the database.

Column	Datatype	NULL	Description
CAPTURE_NAME	VARCHAR2(128)		Name of the capture process
QUEUE_NAME	VARCHAR2(128)		Name of the queue used for staging captured changes
QUEUE_OWNER	VARCHAR2(128)		Owner of the queue used for staging captured changes
RULE_SET_NAME	VARCHAR2(128)		Name of the positive rule set used by the capture process for filtering
RULE_SET_OWNER	VARCHAR2(128)		Owner of the positive rule set
CAPTURE_USER	VARCHAR2(128)		Current user who is enqueuing captured messages
START_SCN	NUMBER		System change number (SCN) from which the capture process will start to capture changes. START_SCN is only modified as the result of an ALTER_CAPTURE statement or if the FIRST_SCN moves beyond the existing START_SCN.
STATUS	VARCHAR2(8)		Status of the capture process: • DISABLED • ENABLED • ABORTED
CAPTURED_SCN	NUMBER		System change number (SCN) of the last redo log record scanned
APPLIED_SCN	NUMBER		System change number (SCN) of the most recent message dequeued by the relevant apply processes. All changes below this SCN have been dequeued by all apply processes that apply changes captured by this capture process.
USE_DATABASE_LINK	VARCHAR2(3)		Indicates whether the source database name is used as the database link to connect to the source database from the downstream database (YES) or not (NO). If the capture process was created at the source database, then this column will be NULL.
FIRST_SCN	NUMBER		System change number (SCN) from which the capture process can be restarted. FIRST_SCN indicates the lowest SCN to which the capture can be repositioned.
SOURCE_DATABASE	VARCHAR2(128)		Global name of the source database
SOURCE_DBID	NUMBER		Database ID of the source database
SOURCE_RESETLOGS_SCN	NUMBER		Resetlogs system change number (SCN) of the source database
SOURCE_RESETLOGS_TIME	NUMBER		Resetlogs time of the source database
LOGMINER_ID	NUMBER		Session ID of the Oracle LogMiner session associated with the capture process

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
NEGATIVE_RULE_SET_NAME	VARCHAR2(128)		Name of the negative rule set used by the capture process for filtering
NEGATIVE_RULE_SET_OWNER	VARCHAR2(128)		Owner of the negative rule set used by the capture process for filtering
MAX_CHECKPOINT_SCN	NUMBER		System change number (SCN) at which the last checkpoint was taken by the capture process
REQUIRED_CHECKPOINT_SCN	NUMBER		Lowest checkpoint SCN for which the capture process requires redo information. **Note:** This SCN value does not necessarily correspond with a checkpoint SCN value.
LOGFILE_ASSIGNMENT	VARCHAR2(8)		Logfile assignment type for the capture process: • IMPLICIT • EXPLICIT
STATUS_CHANGE_TIME	DATE		Time that the STATUS of the capture process was changed
ERROR_NUMBER	NUMBER		Error number if the capture process was terminated
ERROR_MESSAGE	VARCHAR2(4000)		Error message if the capture process was terminated
VERSION	VARCHAR2(64)		Version number of the capture process
CAPTURE_TYPE	VARCHAR2(10)		Type of the capture process: • DOWNSTREAM • LOCAL
LAST_ENQUEUED_SCN	NUMBER		Last enqueued system change number (SCN)
CHECKPOINT_RETENTION_TIME	NUMBER		Checkpoint retention time **Note:** When the checkpoint retention time for a capture process is set to INFINITE, then the value displayed in this column is 4294967295.
START_TIME	TIMESTAMP(6)		Time from which the capture process will start to capture changes. START_TIME is related to START_SCN and can only be modified by an ALTER_CAPTURE statement. You can modify either START_SCN or START_TIME, but not both at the same time.
PURPOSE	VARCHAR2(19)		Purpose of the capture process: • GoldenGate Capture - A capture process configured using Oracle GoldenGate Extract in integrated capture mode • XStream Out - A capture process in an XStream Out configuration • AUDIT VAULT - A capture process in an audit vault configuration
SOURCE_ROOT_NAME	VARCHAR2(128)		The global name of the source root database
CLIENT_NAME	VARCHAR2(4000)		Client name of the capture process. This is the outbound name for XStream Out, and the extract name for GoldenGate.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CLIENT_STATUS	VARCHAR2(8)		Status of the client process: • DISABLED - For XStream Out if the outbound server is not running; for GoldenGate if the capture process is not running • DETACHED - For XStream Out if the outbound server is running, but the XStream client application is not attached to it; For GoldenGate if the capture process is running, but the extract process is not attached to it • ATTACHED - For XStream out if the outbound server is running and the XStream client application is attached to it; For GoldenGate if the capture process is running and the extract process is attached to it • ABORTED - For XStream out if the outbound server became disabled because it encountered an error; for GoldenGate if the capture process became disabled because it encountered an error
OLDEST_SCN	NUMBER		Oldest SCN of the transactions currently being processed
FILTERED_SCN	NUMBER		SCN of the low watermark transaction processed

See Also

- "DBA_CAPTURE"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_XSTREAM_ADM.ENABLE_GG_XSTREAM_FOR_STREAMS procedure

2.110 ALL_CAPTURE_EXTRA_ATTRIBUTES

ALL_CAPTURE_EXTRA_ATTRIBUTES displays information about the extra attributes for the capture processes accessible to the current user.

Related View

DBA_CAPTURE_EXTRA_ATTRIBUTES displays information about the extra attributes for all capture processes in the database.

Column	Datatype	NULL	Description
CAPTURE_NAME	VARCHAR2(128)	NOT NULL	Name of the capture process
ATTRIBUTE_NAME	VARCHAR2(128)	NOT NULL	Name of the extra attribute
INCLUDE	VARCHAR2(3)		Indicates whether the extra attribute is included (YES) or not (NO)
ROW_ATTRIBUTE	VARCHAR2(3)		Indicates whether the extra attribute is a row LCR attribute (YES) or not (NO)
DDL_ATTRIBUTE	VARCHAR2(3)		Indicates whether the extra attribute is a DDL LCR attribute (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "DBA_CAPTURE_EXTRA_ATTRIBUTES"

2.111 ALL_CAPTURE_PARAMETERS

ALL_CAPTURE_PARAMETERS displays information about the parameters for the capture processes that enqueue the captured changes into queues accessible to the current user.

Related View

DBA_CAPTURE_PARAMETERS displays information about the parameters for all capture processes in the database.

Column	Datatype	NULL	Description
CAPTURE_NAME	VARCHAR2(128)	NOT NULL	Name of the capture process
PARAMETER	VARCHAR2(128)	NOT NULL	Name of the parameter
VALUE	VARCHAR2(4000)		Parameter value
SET_BY_USER	VARCHAR2(3)		Indicates whether the parameter value was set by the user (YES) or was not set by the user (NO). If NO, then the parameter is set to its default value. If YES, then the parameter may or may not be set to its default value.
SOURCE_DATABASE	VARCHAR2(128)		Global name of the container for which the capture parameter is defined

> **See Also**
>
> "DBA_CAPTURE_PARAMETERS"

2.112 ALL_CAPTURE_PREPARED_DATABASE

ALL_CAPTURE_PREPARED_DATABASE displays information about when the local database was prepared for instantiation. If the local database was not prepared for instantiation, then this view contains no rows.

Related View

DBA_CAPTURE_PREPARED_DATABASE displays information about when the local database was prepared for instantiation.

Column	Datatype	NULL	Description
TIMESTAMP	DATE		Date and time at which the local database was ready to be instantiated

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SUPPLEMENTAL_LOG_DATA_PK	VARCHAR2(8)		Status of database-level PRIMARY KEY COLUMNS supplemental logging: • IMPLICIT • EXPLICIT • NO
SUPPLEMENTAL_LOG_DATA_UI	VARCHAR2(8)		Status of database-level UNIQUE INDEX COLUMNS supplemental logging: • IMPLICIT • EXPLICIT • NO
SUPPLEMENTAL_LOG_DATA_FK	VARCHAR2(8)		Status of database-level FOREIGN KEY COLUMNS supplemental logging: • IMPLICIT • EXPLICIT • NO
SUPPLEMENTAL_LOG_DATA_ALL	VARCHAR2(8)		Status of database-level ALL COLUMNS supplemental logging: • IMPLICIT • EXPLICIT • NO

See Also

"DBA_CAPTURE_PREPARED_DATABASE"

2.113 ALL_CAPTURE_PREPARED_SCHEMAS

ALL_CAPTURE_PREPARED_SCHEMAS displays information about the schemas prepared for instantiation that are accessible to the current user at the local database.

Related View

DBA_CAPTURE_PREPARED_SCHEMAS displays information about all schemas prepared for instantiation at the local database.

Column	Datatype	NULL	Description
SCHEMA_NAME	VARCHAR2(128)	NOT NULL	Name of the schema that is ready to be instantiated
TIMESTAMP	DATE		Date and time at which the schema was ready to be instantiated
SUPPLEMENTAL_LOG_DATA_PK	VARCHAR2(8)		Status of schema-level PRIMARY KEY COLUMNS supplemental logging: • IMPLICIT • EXPLICIT • NO

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SUPPLEMENTAL_LOG_DATA_UI	VARCHAR2(8)		Status of schema-level UNIQUE INDEX COLUMNS supplemental logging: • IMPLICIT • EXPLICIT • NO
SUPPLEMENTAL_LOG_DATA_FK	VARCHAR2(8)		Status of schema-level FOREIGN KEY COLUMNS supplemental logging: • IMPLICIT • EXPLICIT • NO
SUPPLEMENTAL_LOG_DATA_ALL	VARCHAR2(8)		Status of schema-level ALL COLUMNS supplemental logging: • IMPLICIT • EXPLICIT • NO

See Also

"DBA_CAPTURE_PREPARED_SCHEMAS"

2.114 ALL_CAPTURE_PREPARED_TABLES

ALL_CAPTURE_PREPARED_TABLES displays information about the tables prepared for instantiation that are accessible to the current user at the local database.

Related View

DBA_CAPTURE_PREPARED_TABLES displays information about all tables prepared for instantiation at the local database.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the table that is ready to be instantiated
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table that is ready to be instantiated
SCN	NUMBER	NOT NULL	Smallest system change number (SCN) for which the table can be instantiated
TIMESTAMP	DATE		Date and time at which the table was ready to be instantiated
SUPPLEMENTAL_LOG_DATA_PK	VARCHAR2(8)		Status of table-level PRIMARY KEY COLUMNS supplemental logging: • IMPLICIT • EXPLICIT • NO
SUPPLEMENTAL_LOG_DATA_UI	VARCHAR2(8)		Status of table-level UNIQUE INDEX COLUMNS supplemental logging: • IMPLICIT • EXPLICIT • NO

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SUPPLEMENTAL_LOG_DATA_FK	VARCHAR2(8)		Status of table-level FOREIGN KEY COLUMNS supplemental logging: • IMPLICIT • EXPLICIT • NO
SUPPLEMENTAL_LOG_DATA_ALL	VARCHAR2(8)		Status of table-level ALL COLUMNS supplemental logging: • IMPLICIT • EXPLICIT • NO

See Also

"DBA_CAPTURE_PREPARED_TABLES"

2.115 ALL_CATALOG

ALL_CATALOG displays the tables, clusters, views, synonyms, and sequences accessible to the current user.

Related Views

- DBA_CATALOG displays all tables, clusters, views, synonyms, and sequences in the entire database.
- USER_CATALOG displays the tables, clusters, views, synonyms, and sequences in the current user's schema. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the TABLE, CLUSTER, VIEW, SYNONYM, SEQUENCE, or UNDEFINED
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the TABLE, CLUSTER, VIEW, SYNONYM, SEQUENCE, or UNDEFINED
TABLE_TYPE	VARCHAR2(11)		Type of the TABLE, CLUSTER, VIEW, SYNONYM, SEQUENCE, or UNDEFINED

See Also

- "DBA_CATALOG"
- "USER_CATALOG"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.116 ALL_CERTIFICATES

ALL_CERTIFICATES displays the certificates accessible to the current user which are used for signature verification for blockchain tables.

Related Views

- DBA_CERTIFICATES displays all certificates in the database which are used for signature verification for blockchain tables.
- USER_CERTIFICATES displays the certificates added by the current user which are used for signature verification for blockchain tables. This view does not display the USER_NAME column.

Column	Datatype	NULL	Description
CERTIFICATE_ID	RAW(16)	NOT NULL	ID for the certificate
USER_NAME	VARCHAR2(128)	NOT NULL	User who added the certificate
DISTINGUISHED_NAME	VARCHAR2(2000)		Uniquely identifies the entity that owns the certificate
CERTIFICATE	BLOB		Contents of the certificate

Note

This view is available starting with Oracle Database 19c, Release Update 19.10.

See Also

- "DBA_CERTIFICATES"
- "USER_CERTIFICATES"

2.117 ALL_CLUSTER_HASH_EXPRESSIONS

ALL_CLUSTER_HASH_EXPRESSIONS displays hash functions for all hash clusters accessible to the current user.

Related Views

- DBA_CLUSTER_HASH_EXPRESSIONS displays hash functions for all hash clusters in the database.
- USER_CLUSTER_HASH_EXPRESSIONS displays hash functions for all hash clusters owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the cluster
CLUSTER_NAME	VARCHAR2(128)	NOT NULL	Name of the cluster
HASH_EXPRESSION	LONG		Text of the hash function of the hash cluster

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_CLUSTER_HASH_EXPRESSIONS"
- "USER_CLUSTER_HASH_EXPRESSIONS"

2.118 ALL_CLUSTERING_DIMENSIONS

`ALL_CLUSTERING_DIMENSIONS` describes dimension tables associated with tables with an attribute clustering clause that the user owns or has system privileges for.

Related Views

- `DBA_CLUSTERING_DIMENSIONS` describes dimension tables associated with all tables with an attribute clustering clause in the database.
- `USER_CLUSTERING_DIMENSIONS` describes dimension tables associated with tables with an attribute clustering clause owned by the user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the attribute clustering table
`TABLE_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the attribute clustering table
`DIMENSION_OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the dimension table
`DIMENSION_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the dimension table

See Also

- "DBA_CLUSTERING_DIMENSIONS"
- "USER_CLUSTERING_DIMENSIONS"
- The `ALTER TABLE` section in *Oracle Database SQL Language Reference* for information about using the `CLUSTERING` clause to create an attribute clustering table
- The `CREATE TABLE` section in *Oracle Database SQL Language Reference* for information about using the `CLUSTERING` clause to create an attribute clustering table
- *Oracle Database Data Warehousing Guide* for information about dimension tables
- *Oracle Database Data Warehousing Guide* for information about attribute clustering with zone maps

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.119 ALL_CLUSTERING_JOINS

`ALL_CLUSTERING_JOINS` describes joins to the dimension tables associated with tables with an attribute clustering clause the user owns or has system privileges for.

Related Views

- `DBA_CLUSTERING_JOINS` describes joins to the dimension tables associated with all tables with an attribute clustering clause in the database.
- `USER_CLUSTERING_JOINS` describes joins to the dimension tables associated with tables with an attribute clustering clause owned by the user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the attribute clustering table
`TABLE_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the attribute clustering table
`TAB1_OWNER`	`VARCHAR2(128)`	`NOT NULL`	Table 1 owner of the join
`TAB1_NAME`	`VARCHAR2(128)`	`NOT NULL`	Table 1 name of the join
`TAB1_COLUMN`	`VARCHAR2(128)`	`NOT NULL`	Table 1 column name of the join
`TAB2_OWNER`	`VARCHAR2(128)`	`NOT NULL`	Table 2 owner of the join
`TAB2_NAME`	`VARCHAR2(128)`	`NOT NULL`	Table 2 name of the join
`TAB2_COLUMN`	`VARCHAR2(128)`	`NOT NULL`	Table 2 column name of the join

See Also

- "DBA_CLUSTERING_JOINS"
- "USER_CLUSTERING_JOINS"
- The `ALTER TABLE` section in *Oracle Database SQL Language Reference* for information about using the `CLUSTERING` clause to create an attribute clustering table
- The `CREATE TABLE` section in *Oracle Database SQL Language Reference* for information about using the `CLUSTERING` clause to create an attribute clustering table
- *Oracle Database Data Warehousing Guide* for information about dimension tables
- *Oracle Database Data Warehousing Guide* for information about attribute clustering with zone maps

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.120 ALL_CLUSTERING_KEYS

ALL_CLUSTERING_KEYS describes clustering keys for tables with an attribute clustering clause accessible to the user.

Related Views

- DBA_CLUSTERING_KEYS describes clustering keys for all tables with an attribute clustering clause.
- USER_CLUSTERING_KEYS describes clustering keys for tables with an attribute clustering clause owned by the user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table on which the clustering clause is defined
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table on which the clustering clause is defined
DETAIL_OWNER	VARCHAR2(128)	NOT NULL	Owner of the detailed table contributing to the clustering keys
DETAIL_NAME	VARCHAR2(128)	NOT NULL	Name of the detailed table contributing to the clustering keys
DETAIL_COLUMN	VARCHAR2(128)	NOT NULL	Name of the detail column
POSITION	NUMBER	NOT NULL	Position of the column in the clustering clause
GROUPID	NUMBER	NOT NULL	Group ID of the column in the clustering clause

See Also

- "DBA_CLUSTERING_KEYS"
- "USER_CLUSTERING_KEYS"
- The ALTER TABLE section in *Oracle Database SQL Language Reference* for information about using the CLUSTERING clause to create an attribute clustering table
- The CREATE TABLE section in *Oracle Database SQL Language Reference* for information about using the CLUSTERING clause to create an attribute clustering table
- *Oracle Database Data Warehousing Guide* for information about dimension tables
- *Oracle Database Data Warehousing Guide* for information about attribute clustering with zone maps

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.121 ALL_CLUSTERING_TABLES

ALL_CLUSTERING_TABLES describes tables with an attribute clustering clause that are accessible to the user.

Related Views

- DBA_CLUSTERING_TABLES describes all the tables with an attribute clustering clause.
- USER_CLUSTERING_TABLES describes the tables with an attribute clustering clause owned by the user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
CLUSTERING_TYPE	VARCHAR2(11)		Clustering type: • INTERLEAVED • LINEAR
ON_LOAD	VARCHAR2(3)		Indicates whether Oracle will cluster data on load (YES) or not (NO)
ON_DATAMOVEMENT	VARCHAR2(3)		Indicates whether Oracle will cluster data on data movement, for example, partition move (YES), or not (NO)
VALID	VARCHAR2(3)		Indicates if clustering is valid (YES) or not (NO). For clustering with dimension tables, it is required that the joins of the fact table to the dimensions is via primary key or unique key on the dimension table. Therefore, dimension join keys must have a valid primary key or unique key constraint. If the primary key or unique key constraint is not valid, then clustering will not occur. If there are no joins in the CLUSTERING clause, then the value of this column is YES.
WITH_ZONEMAP	VARCHAR2(3)		Indicates if a zonemap was created with clustering (YES) or not (NO).
LAST_LOAD_CLST	TIMESTAMP(6)		Last time the clustering occurred on load
LAST_DATAMOVE_CLST	TIMESTAMP(6)		Last time the clustering occurred on data movement, for example, partition move

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_CLUSTERING_TABLES"
- "USER_CLUSTERING_TABLES"
- The `ALTER TABLE` section in *Oracle Database SQL Language Reference* for information about using the `CLUSTERING` clause to create an attribute clustering table
- The `CREATE TABLE` section in *Oracle Database SQL Language Reference* for information about using the `CLUSTERING` clause to create an attribute clustering table
- *Oracle Database Data Warehousing Guide* for information about dimension tables
- *Oracle Database Data Warehousing Guide* for information about attribute clustering with zone maps

2.122 ALL_CLUSTERS

`ALL_CLUSTERS` describes all clusters accessible to the current user.

Related Views

- `DBA_CLUSTERS` describes all clusters in the database.
- `USER_CLUSTERS` describes all clusters owned by the current user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the cluster
`CLUSTER_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the cluster
`TABLESPACE_NAME`	`VARCHAR2(30)`	`NOT NULL`	Name of the tablespace containing the cluster
`PCT_FREE`	`NUMBER`		Minimum percentage of free space in a block
`PCT_USED`	`NUMBER`		Minimum percentage of used space in a block
`KEY_SIZE`	`NUMBER`		Estimated size of cluster key plus associated rows
`INI_TRANS`	`NUMBER`	`NOT NULL`	Initial number of transactions
`MAX_TRANS`	`NUMBER`	`NOT NULL`	Maximum number of transactions
`INITIAL_EXTENT`	`NUMBER`		Size of the initial extent in bytes
`NEXT_EXTENT`	`NUMBER`		Size of secondary extents in bytes
`MIN_EXTENTS`	`NUMBER`	`NOT NULL`	Minimum number of extents allowed in the segment
`MAX_EXTENTS`	`NUMBER`	`NOT NULL`	Maximum number of extents allowed in the segment
`PCT_INCREASE`	`NUMBER`		Percentage increase in extent size
`FREELISTS`	`NUMBER`		Number of process freelists allocated to this segment
`FREELIST_GROUPS`	`NUMBER`		Number of freelist groups allocated to this segment
`AVG_BLOCKS_PER_KEY`	`NUMBER`		Number of blocks in the table divided by number of cluster keys

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CLUSTER_TYPE	VARCHAR2(5)		Type of the cluster: • INDEX - B*-Tree index • HASH - Hash
FUNCTION	VARCHAR2(15)		If the cluster is a hash cluster, the hash function
HASHKEYS	NUMBER		If the cluster is a hash cluster, the number of hash keys (hash buckets)
DEGREE	VARCHAR2(10)		Number of threads per instance for scanning the cluster, or DEFAULT
INSTANCES	VARCHAR2(10)		Number of instances across which the cluster is to be scanned , or DEFAULT
CACHE	VARCHAR2(5)		Indicates whether the cluster is to be cached in the buffer cache (Y) or not (N)
BUFFER_POOL	VARCHAR2(7)		Buffer pool to be used for cluster blocks: • DEFAULT • KEEP • RECYCLE • NULL
FLASH_CACHE	VARCHAR2(7)		Database Smart Flash Cache hint to be used for cluster blocks: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)		Cell flash cache hint to be used for cluster blocks: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
SINGLE_TABLE	VARCHAR2(5)		Indicates whether this is a single-table cluster (Y) or not (N)
DEPENDENCIES	VARCHAR2(8)		Indicates whether row-level dependency tracking is enabled (ENABLED) or disabled (DISABLED)

See Also

- "DBA_CLUSTERS"
- "USER_CLUSTERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.123 ALL_CODE_ROLE_PRIVS

ALL_CODE_ROLE_PRIVS describes all the roles that are associated with program units owned or accessible by the current user.

Related Views

- DBA_CODE_ROLE_PRIVS describes all the roles that are associated with program units in the database.
- USER_CODE_ROLE_PRIVS describes all the roles that are associated with program units owned by current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Username of the owner of the object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object
OBJECT_TYPE	VARCHAR2(9)		Type of the object
ROLE	VARCHAR2(128)	NOT NULL	The role associated with the object

See Also

- "DBA_CODE_ROLE_PRIVS"
- "USER_CODE_ROLE_PRIVS"
- *Oracle Database Security Guide* for more information about granting and revoking roles to and from program units

2.124 ALL_COL_COMMENTS

ALL_COL_COMMENTS displays comments on the columns of the tables and views accessible to the current user.

Related Views

- DBA_COL_COMMENTS displays comments on the columns of all tables and views in the database.
- USER_COL_COMMENTS displays comments on the columns of the tables and views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the object
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column
COMMENTS	VARCHAR2(4000)		Comment on the column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs • *n*: This value is used for rows containing data that originate in the container with the ID *n* (*n*=1 if the data originates in root)

See Also

- "DBA_COL_COMMENTS"
- "USER_COL_COMMENTS"

2.125 ALL_COL_PENDING_STATS

ALL_COL_PENDING_STATS describes the pending statistics of the columns accessible to the current user.

Related Views

- DBA_COL_PENDING_STATS describes the pending statistics of all columns in the database.
- USER_COL_PENDING_STATS describes the pending statistics of the columns owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table
PARTITION_NAME	VARCHAR2(128)		Name of the partition
SUBPARTITION_NAME	VARCHAR2(128)		Name of the subpartition
COLUMN_NAME	VARCHAR2(128)		Name of the column
NUM_DISTINCT	NUMBER		Number of distinct values in the column
LOW_VALUE	RAW(32)		Low value in the column
HIGH_VALUE	RAW(32)		High value in the column
DENSITY	NUMBER		If a histogram is available on COLUMN_NAME, then this column displays the selectivity of a value that spans fewer than 2 endpoints in the histogram. It does not represent the selectivity of values that span 2 or more endpoints. If a histogram is not available on COLUMN_NAME, then the value of this column is 1/NUM_DISTINCT.
NUM_NULLS	NUMBER		Number of NULLs in the column
AVG_COL_LEN	NUMBER		Average length of the column (in bytes)
SAMPLE_SIZE	NUMBER		Sample size used in analyzing the column
LAST_ANALYZED	DATE		Most recent date on which the column was analyzed

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_COL_PENDING_STATS"
- "USER_COL_PENDING_STATS"

2.126 ALL_COL_PRIVS

ALL_COL_PRIVS describes the following types of grants:

- Column object grants for which the current user is the object owner, grantor, or grantee
- Column object grants for which an enabled role or PUBLIC is the grantee

Related Views

- DBA_COL_PRIVS describes all column object grants in the database.
- USER_COL_PRIVS describes the column object grants for which the current user is the object owner, grantor, or grantee.

Column	Datatype	NULL	Description
GRANTOR	VARCHAR2(128)		Name of the user who performed the grant
GRANTEE	VARCHAR2(128)		Name of the user or role to whom access was granted
TABLE_SCHEMA	VARCHAR2(128)		Schema of the object
TABLE_NAME	VARCHAR2(128)		Name of the object
COLUMN_NAME	VARCHAR2(128)		Name of the column
PRIVILEGE	VARCHAR2(40)		Privilege on the column: • INSERT • REFERENCES • UPDATE
GRANTABLE	VARCHAR2(3)		Indicates whether the privilege was granted with the GRANT OPTION (YES) or not (NO)
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the privilege was granted commonly (CONTAINER=ALL was used) • NO if the privilege was granted locally (CONTAINER=ALL was not used)
INHERITED	VARCHAR2(3)		Indicates whether the privilege grant was inherited from another container (YES) or not (NO)

See Also

- "DBA_COL_PRIVS"
- "USER_COL_PRIVS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.127 ALL_COL_PRIVS_MADE

ALL_COL_PRIVS_MADE describes the column object grants for which the current user is the object owner or grantor.

Related View

USER_COL_PRIVS_MADE describes the column object grants for which the current user is the object owner. This view does not display the OWNER column.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)		Name of the user or role to whom access was granted
OWNER	VARCHAR2(128)		Owner of the object
TABLE_NAME	VARCHAR2(128)		Name of the object
COLUMN_NAME	VARCHAR2(128)		Name of the column
GRANTOR	VARCHAR2(128)		Name of the user who performed the grant
PRIVILEGE	VARCHAR2(40)		Privilege on the column: • INSERT • REFERENCES • UPDATE
GRANTABLE	VARCHAR2(3)		Indicates whether the privilege was granted with the GRANT OPTION (YES) or not (NO)
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the privilege was granted commonly (CONTAINER=ALL was used) • NO if the privilege was granted locally (CONTAINER=ALL was not used)
INHERITED	VARCHAR2(3)		Indicates whether the privilege grant was inherited from another container (YES) or not (NO)

See Also

"USER_COL_PRIVS_MADE"

2.128 ALL_COL_PRIVS_RECD

ALL_COL_PRIVS_RECD describes the following types of grants:

- Column object grants for which the current user is the grantee
- Column object grants for which an enabled role or PUBLIC is the grantee

Related View

USER_COL_PRIVS_RECD describes the column object grants for which the current user is the grantee. This view does not display the GRANTEE column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)		Name of the user or role to whom access was granted
OWNER	VARCHAR2(128)		Owner of the object
TABLE_NAME	VARCHAR2(128)		Name of the object
COLUMN_NAME	VARCHAR2(128)		Name of the column
GRANTOR	VARCHAR2(128)		Name of the user who performed the grant
PRIVILEGE	VARCHAR2(40)		Privilege on the column: • INSERT • REFERENCES • UPDATE
GRANTABLE	VARCHAR2(3)		Indicates whether the privilege was granted with the GRANT OPTION (YES) or not (NO)
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the privilege was granted commonly (CONTAINER=ALL was used) • NO if the privilege was granted locally (CONTAINER=ALL was not used)
INHERITED	VARCHAR2(3)		Indicates whether the privilege grant was inherited from another container (YES) or not (NO)

See Also

"USER_COL_PRIVS_RECD"

2.129 ALL_COLL_TYPES

ALL_COLL_TYPES describes all named collection types (varrays and nested tables) accessible to the current user.

Related Views

- DBA_COLL_TYPES describes all named collection types in the database.
- USER_COLL_TYPES describes all named collection types owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the collection
TYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the collection
COLL_TYPE	VARCHAR2(128)	NOT NULL	Description of the collection, such as VARYING ARRAY, [nested] TABLE
UPPER_BOUND	NUMBER		For varrays only, maximum size
ELEM_TYPE_MOD	VARCHAR2(7)		Type modifier of the element
ELEM_TYPE_OWNER	VARCHAR2(128)		Owner of the type upon which the collection is based. This value is useful primarily for a user-defined type.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ELEM_TYPE_NAME	VARCHAR2(128)		Name of the data type or user-defined type upon which the collection is based
LENGTH	NUMBER		Length of CHAR elements or maximum length of VARCHAR or VARCHAR2 elements
PRECISION	NUMBER		Decimal precision of NUMBER or DECIMAL elements; binary precision of FLOAT elements
SCALE	NUMBER		Scale of NUMBER or DECIMAL elements
CHARACTER_SET_NAME	VARCHAR2(44)		Name of the character set (CHAR_CS | NCHAR_CS)
ELEM_STORAGE	VARCHAR2(7)		Obsolete column
NULLS_STORED	VARCHAR2(3)		Obsolete column
CHAR_USED	VARCHAR2(1)		Indicates whether the attribute uses BYTE length semantics (B) or CHAR length semantics (C). For NCHAR and NVARCHAR2 attribute types, this value is always C.

See Also

- "DBA_COLL_TYPES"
- "USER_COLL_TYPES"

2.130 ALL_CONS_COLUMNS

ALL_CONS_COLUMNS describes columns that are accessible to the current user and that are specified in constraints.

Related Views

- DBA_CONS_COLUMNS describes all columns in the database that are specified in constraints.
- USER_CONS_COLUMNS describes columns that are owned by the current user and that are specified in constraints.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the constraint definition
CONSTRAINT_NAME	VARCHAR2(128)	NOT NULL	Name of the constraint definition
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table with the constraint definition
COLUMN_NAME	VARCHAR2(4000)		Name of the column or attribute of the object type column specified in the constraint definition **Note:** If you create a constraint on a user-defined REF column, the system creates the constraint on the attributes that make up the REF column. Therefore, the column names displayed in this view are the attribute names, with the REF column name as a prefix, in the following form: "REF_name"."attribute"

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
POSITION	NUMBER		Original position of the column or attribute in the definition of the object

See Also

- "DBA_CONS_COLUMNS"
- "USER_CONS_COLUMNS"

2.131 ALL_CONS_OBJ_COLUMNS

ALL_CONS_OBJ_COLUMNS displays information about the types that object columns (or attributes) or collection elements have been constrained to, in the tables accessible to the current user.

Related Views

- DBA_CONS_OBJ_COLUMNS displays information about the types that object columns (or attributes) or collection elements have been constrained to, in all tables in the database.
- USER_CONS_OBJ_COLUMNS displays information about the types that object columns (or attributes) or collection elements have been constrained to, in the tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table containing the object column or attribute
COLUMN_NAME	VARCHAR2(4000)		Fully qualified name of the object column or attribute
CONS_TYPE_OWNER	VARCHAR2(128)		Owner of the type that the column (or element) is constrained to
CONS_TYPE_NAME	VARCHAR2(128)		Name of the type that the column (or element) is constrained to
CONS_TYPE_ONLY	VARCHAR2(15)		Indicates whether the column (or element) is constrained to ONLY type (Y) or not (N)

See Also

- "DBA_CONS_OBJ_COLUMNS"
- "USER_CONS_OBJ_COLUMNS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.132 ALL_CONSTRAINTS

ALL_CONSTRAINTS describes constraint definitions on tables accessible to the current user.

Related Views

- DBA_CONSTRAINTS describes all constraint definitions in the database.
- USER_CONSTRAINTS describes constraint definitions on tables in the current user's schema.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the constraint definition
CONSTRAINT_NAME	VARCHAR2(128)		Name of the constraint definition
CONSTRAINT_TYPE	VARCHAR2(1)		Type of the constraint definition: • C - Check constraint on a table • P - Primary key • U - Unique key • R - Referential integrity • V - With check option, on a view • O - With read only, on a view • H - Hash expression • F - Constraint that involves a REF column • S - Supplemental logging
TABLE_NAME	VARCHAR2(128)		Name associated with the table (or view) with the constraint definition
SEARCH_CONDITION	LONG		Text of search condition for a check constraint. This column returns the correct value only when the row originates from the current container.
SEARCH_CONDITION_VC	VARCHAR2(4000)		Text of search condition for a check constraint. This column may truncate the search condition.
R_OWNER	VARCHAR2(128)		Owner of the table referred to in a referential constraint
R_CONSTRAINT_NAME	VARCHAR2(128)		Name of the unique constraint definition for the referenced table
DELETE_RULE	VARCHAR2(9)		Delete rule for a referential constraint: • CASCADE • SET NULL • NO ACTION
STATUS	VARCHAR2(8)		Enforcement status of the constraint: • ENABLED • DISABLED
DEFERRABLE	VARCHAR2(14)		Indicates whether the constraint is deferrable (DEFERRABLE) or not (NOT DEFERRABLE)
DEFERRED	VARCHAR2(9)		Indicates whether the constraint was initially deferred (DEFERRED) or not (IMMEDIATE)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
VALIDATED	VARCHAR2(13)		When STATUS = ENABLED, possible values are: • VALIDATED - All data obeys the constraint (that is, the existing data in the table was validated when the constraint was enabled, as well as any subsequent data entered into the table) • NOT VALIDATED - All data may not obey the constraint (that is, the existing data in the table was not validated when the constraint was enabled, but subsequent data entered into the table was validated) When STATUS = DISABLED, possible values are: • VALIDATED - All data obeys the constraint, but the unique index on the constraint has been dropped. This setting is useful in data warehousing environments, but has some restrictions. Refer to *Oracle Database Data Warehousing Guide* for more information on this setting. • NOT VALIDATED - All data may not obey the constraint
GENERATED	VARCHAR2(14)		Indicates whether the name of the constraint is user-generated (USER NAME) or system-generated (GENERATED NAME)
BAD	VARCHAR2(3)		Indicates whether this constraint specifies a century in an ambiguous manner (BAD) or not (NULL). To avoid errors resulting from this ambiguity, rewrite the constraint using the TO_DATE function with a four-digit year. **See Also:** the TO_DATE function in *Oracle Database SQL Language Reference* and *Oracle Database Development Guide*
RELY	VARCHAR2(4)		When VALIDATED = NOT VALIDATED, this column indicates whether the constraint is to be taken into account for query rewrite (RELY) or not (NULL). When VALIDATED = VALIDATED, this column is not meaningful. **See Also:** *constraints* in *Oracle Database SQL Language Reference*
LAST_CHANGE	DATE		When the constraint was last enabled or disabled
INDEX_OWNER	VARCHAR2(128)		Name of the user owning the index
INDEX_NAME	VARCHAR2(128)		Name of the index (only shown for unique and primary-key constraints)
INVALID	VARCHAR2(7)		Indicates whether the constraint is invalid (INVALID) or not (NULL)
VIEW_RELATED	VARCHAR2(14)		Indicates whether the constraint depends on a view (DEPEND ON VIEW) or not (NULL)
ORIGIN_CON_ID	VARCHAR2(256)		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_CONSTRAINTS"
- "USER_CONSTRAINTS"

2.133 ALL_CONTEXT

ALL_CONTEXT describes all context namespaces in the current session for which attributes and values have been specified using the DBMS_SESSION.SET_CONTEXT procedure. This view does not display the TYPE and ORIGIN_CON_ID columns.

Related View

DBA_CONTEXT describes all context namespaces defined in the database, regardless whether any attributes have been specified for them using the DBMS_SESSION.SET_CONTEXT procedure.

Column	Datatype	NULL	Description
NAMESPACE	VARCHAR2(128)	NOT NULL	Name of the context namespace
SCHEMA	VARCHAR2(128)	NOT NULL	Schema name of the designated package that can set attributes using this namespace
PACKAGE	VARCHAR2(128)	NOT NULL	Package name of the designated package that can set attributes using this namespace

See Also

- "DBA_CONTEXT"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SESSION.SET_CONTEXT procedure

2.134 ALL_CREDENTIALS

ALL_CREDENTIALS lists all credentials visible to the user.

Related Views

- DBA_CREDENTIALS lists all credentials in the database.
- USER_CREDENTIALS lists credentials owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the credential
CREDENTIAL_NAME	VARCHAR2(128)	NOT NULL	Name of the credential
USERNAME	VARCHAR2(128)		Name of the user that will be used to log in to the remote database or the remote or local operating system

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
WINDOWS_DOMAIN	VARCHAR2(30)		For a Windows target, the Windows domain to use when logging in
COMMENTS	VARCHAR2(4000)		Comments on the credential
ENABLED	VARCHAR2(5)		Indicates whether this credential is enabled (TRUE) or not (FALSE)

Note

DBMS_CREDENTIAL lists credentials that can be used to run external procedures, or by DBMS_SCHEDULER for remote or external jobs, or for storing or retrieving files from the operating system.

If a credential is disabled, then any of the actions above that attempts to use the credential will fail.

See Also

- "DBA_CREDENTIALS"
- "USER_CREDENTIALS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_CREDENTIAL package
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SCHEDULER package

2.135 ALL_CUBE_ATTR_VISIBILITY

ALL_CUBE_ATTR_VISIBILITY describes the OLAP attributes visible for the dimensions, hierarchies, and levels accessible to the current user.

Related Views

- DBA_CUBE_ATTR_VISIBILITY describes all OLAP attributes visible for the dimensions, hierarchies, and levels in the database.
- USER_CUBE_ATTR_VISIBILITY describes the OLAP attributes visible for the dimensions, hierarchies, and levels owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the cube dimension
DIMENSION_NAME	VARCHAR2(128)		Name of a cube dimension (such as TIME)
ATTRIBUTE_NAME	VARCHAR2(128)		Name of an attribute of the dimension (such as LONG_DESCRIPTION or END_DATE)
HIERARCHY_NAME	VARCHAR2(128)		Name of a hierarchy for the dimension (such as CALENDAR)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LEVEL_NAME	VARCHAR2(128)		Name of the dimension level (such as MONTH)
FROM_TYPE	VARCHAR2(10)		Identifies the dimension type that the current row derives the attribute visibility from. Possible values: • DIMENSION - Derives the attribute visibility from itself. • HIERARCHY - Derives the attribute visibility from the VisibleAttributes explicitly set on the associated DIMENSION or itself. • DIM_LEVEL - Derives the attribute visibility from the VisibleAttributes explicitly set on the associated DIMENSION or itself. • HIER_LEVEL - Derives the attribute visibility from the VisibleAttributes explicitly set on the associated DIMENSION, HIERARCHY, DIM_LEVEL, or itself.
TO_TYPE	VARCHAR2(10)		Identifies the dimension type for the current row. Possible values: • DIMENSION - When the TO_TYPE is DIMENSION, then only the DIMENSION_NAME is populated. • HIERARCHY - When the TO_TYPE is HIERARCHY, then only the DIMENSION_NAME and HIERARCHY_NAME are populated. • DIM_LEVEL - When the TO_TYPE is DIM_LEVEL, then only the DIMENSION_NAME and LEVEL_NAME are populated. • HIER_LEVEL - When the TO_TYPE is HIER_LEVEL, then only the HIERARCHY_NAME and LEVEL_NAME are populated.

See Also

- "DBA_CUBE_ATTR_VISIBILITY"
- "USER_CUBE_ATTR_VISIBILITY"

2.136 ALL_CUBE_ATTRIBUTES

ALL_CUBE_ATTRIBUTES describes the attributes for the OLAP cube dimensions accessible to the current user.

Related Views

- DBA_CUBE_ATTRIBUTES describes the attributes for all OLAP cube dimensions in the database.
- USER_CUBE_ATTRIBUTES describes the attributes for the OLAP cube dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the cube dimension

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of a cube dimension (such as TIME)
ATTRIBUTE_NAME	VARCHAR2(128)	NOT NULL	Name of an attribute of the dimension (such as LONG_DESCRIPTION or END_DATE)
ATTRIBUTE_ID	NUMBER	NOT NULL	ID of the attribute of the dimension
TARGET_DIMENSION_NAME	VARCHAR2(128)		Name of the target dimension of the attribute
ATTRIBUTE_ROLE	VARCHAR2(17)		Special role this attribute plays; NULL if none: • SHORT_DESCRIPTION • LONG_DESCRIPTION • DESCRIPTION • TIME_SPAN • END_DATE
DESCRIPTION	NVARCHAR2(300)		Description of the attribute in the session language
ATTRIBUTE_GROUP_NAME	VARCHAR2(200)		Name of the attribute group
DATA_TYPE	VARCHAR2(106)		Data type of the attribute, (such as VARCHAR2 or FLOAT)
DATA_LENGTH	NUMBER	NOT NULL	Length of a text data type
DATA_PRECISION	NUMBER		Precision of a numeric data type
DATA_SCALE	NUMBER		Scale of a numeric data type
CREATE_INDEX	VARCHAR2(3)		Create index flag of the OLAP attribute. Possible values: • YES: The attribute is represented in the AW as a relation. Setting CreateIndex="True" in the metadata guarantees that it will be represented in the AW as a relation. • NO: The attribute is not represented in the AW as a relation. Setting CreateIndex="False" in the metadata does not guarantee that it will be represented in the AW a a variable; the system will make that determination.
IS_MULTI_LINGUAL	VARCHAR2(3)		Shows the setting for the IsMultiLingual flag of the OLAP Atttribute. Possible values: • YES: The attribute is set as multilingual. Setting IsMultiLingual to True in the metadata means that the attribute can have a value per language instead of a single value. • NO: The attribute is not set as multilingual. Setting IsMultiLingual to False in the metadata means that the attribute has only one value, independent of language.

See Also

- "DBA_CUBE_ATTRIBUTES"
- "USER_CUBE_ATTRIBUTES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.137 ALL_CUBE_BUILD_PROCESSES

ALL_CUBE_BUILD_PROCESSES describes the OLAP build processes and maintenance scripts accessible to the current user.

Related Views

- DBA_CUBE_BUILD_PROCESSES describes all OLAP build processes and maintenance scripts in the database.
- USER_CUBE_BUILD_PROCESSES describes the OLAP build processes and maintenance scripts owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the build process
BUILD_PROCESS_NAME	VARCHAR2(128)	NOT NULL	Name of the build process
BUILD_PROCESS_ID	NUMBER	NOT NULL	ID of the build process
BUILD_PROCESS	CLOB		Syntax of the build process
DESCRIPTION	NVARCHAR2(300)		Description of the build process in the session language

See Also

- "DBA_CUBE_BUILD_PROCESSES"
- "USER_CUBE_BUILD_PROCESSES"

2.138 ALL_CUBE_CALCULATED_MEMBERS

ALL_CUBE_CALCULATED_MEMBERS describes the calculated members for the OLAP cube dimensions accessible to the current user.

Related Views

- DBA_CUBE_CALCULATED_MEMBERS describes the calculated members for all OLAP cube dimensions in the database.
- USER_CUBE_CALCULATED_MEMBERS describes the calculated members for the OLAP cube dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the cube dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of a cube dimension
MEMBER_NAME	VARCHAR2(128)	NOT NULL	Name of a calculated member in the cube dimension
IS_CUSTOM_AGGREGATE	VARCHAR2(3)		Indicates whether the calculated member is a custom aggregate (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STORAGE_TYPE	VARCHAR2(10)		Storage type of the calculated member: • DYNAMIC - Value of the member is calculated for a query • PRECOMPUTE - Value of the member is calculated and stored during data maintenance
EXPRESSION	CLOB		Expression used to generate the value of the calculated member

See Also

- "DBA_CUBE_CALCULATED_MEMBERS"
- "USER_CUBE_CALCULATED_MEMBERS"

2.139 ALL_CUBE_DIM_LEVELS

ALL_CUBE_DIM_LEVELS describes the OLAP dimension levels accessible to the current user.

Related Views

- DBA_CUBE_DIM_LEVELS describes all OLAP dimension levels in the database.
- USER_CUBE_DIM_LEVELS describes the OLAP dimension levels owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of a dimension, such as CUSTOMER
LEVEL_NAME	VARCHAR2(128)	NOT NULL	Name of a level in the dimension, such as WAREHOUSE
LEVEL_ID	NUMBER	NOT NULL	ID of the dimension level
DESCRIPTION	NVARCHAR2(300)		Description of the dimension level in the session language

See Also

- "DBA_CUBE_DIM_LEVELS"
- "USER_CUBE_DIM_LEVELS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.140 ALL_CUBE_DIM_MODELS

ALL_CUBE_DIM_MODELS describes the models for the OLAP dimensions accessible to the current user.

Related Views

- DBA_CUBE_DIM_MODELS describes the models for all OLAP dimensions in the database.
- USER_CUBE_DIM_MODELS describes the models for the OLAP dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the cube dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of a cube dimension
MODEL_NAME	VARCHAR2(128)	NOT NULL	Name of a model for the cube dimension
MODEL_ID	NUMBER	NOT NULL	ID of the model
DESCRIPTION	NVARCHAR2(300)		Description of the model in the session language

See Also

- "DBA_CUBE_DIM_MODELS"
- "USER_CUBE_DIM_MODELS"

2.141 ALL_CUBE_DIM_VIEW_COLUMNS

ALL_CUBE_DIM_VIEW_COLUMNS describes the columns of the relational views of the OLAP cube dimensions accessible to the current user.

Related Views

- DBA_CUBE_DIM_VIEW_COLUMNS describes the columns of the relational views of all OLAP cube dimensions in the database.
- USER_CUBE_DIM_VIEW_COLUMNS describes the columns of the relational views of the OLAP cube dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the cube dimension
DIMENSION_NAME	VARCHAR2(128)		Name of a cube dimension, such as PRODUCT
VIEW_NAME	VARCHAR2(128)		Name of a view of the dimension, such as PRODUCT_VIEW
COLUMN_NAME	VARCHAR2(128)		Name of a column in the view, such as LONG_DESCRIPTION or WAREHOUSE_ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COLUMN_TYPE	VARCHAR2(11)		Type of the column: • KEY - A key of the dimension view (that is, the dimension value itself) • LEVEL_NAME - Name of the level (if any) corresponding to a row in the view • DIM_ORDER - A column by which the results may be ordered (if present) • MEMBER_TYPE • ATTRIBUTE - An attribute owned by the dimension
OBJECT_NAME	VARCHAR2(128)		Name of the level or attribute represented in the column, such as LONG_DESCRIPTION or WAREHOUSE_ID

See Also

- "DBA_CUBE_DIM_VIEW_COLUMNS"
- "USER_CUBE_DIM_VIEW_COLUMNS"

2.142 ALL_CUBE_DIM_VIEWS

ALL_CUBE_DIM_VIEWS describes the relational views of the OLAP dimensions accessible to the current user.

Related Views

- DBA_CUBE_DIM_VIEWS describes the relational views of all OLAP dimensions in the database.
- USER_CUBE_DIM_VIEWS describes the relational views of the OLAP dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the cube dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of a cube dimension, such as PRODUCT
VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of a view of the cube dimension, such as PRODUCT_VIEW

See Also

- "DBA_CUBE_DIM_VIEWS"
- "USER_CUBE_DIM_VIEWS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.143 ALL_CUBE_DIMENSIONALITY

ALL_CUBE_DIMENSIONALITY describes the dimension order for the OLAP cubes accessible to the current user.

Related Views

- DBA_CUBE_DIMENSIONALITY describes the dimension order for all OLAP cubes in the database.
- USER_CUBE_DIMENSIONALITY describes the dimension order for the OLAP cubes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the cube
CUBE_NAME	VARCHAR2(128)	NOT NULL	Name of a cube, such as UNITS_CUBE
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of a dimension of the cube, such as PRODUCT
DIMENSIONALITY_NAME	VARCHAR2(200)		The name of a dimensionality of the cube. For example, a cube dimensioned by the PRODUCT dimension can have a product dimension named PRODUCT_DIM.
DIMENSIONALITY_ID	NUMBER	NOT NULL	ID of the cube dimensionality
ORDER_NUM	NUMBER	NOT NULL	Order number of the dimension in the cube
IS_SPARSE	NUMBER		Indicates whether the dimension is sparse in the cube (1) or not sparse (0)
ET_ATTR_PREFIX	VARCHAR2(200)		Specifies the prefix that will be added to the column names in the Materialized Views to ensure uniqueness. If the user does not specify an ET_ATTR_PREFIX for any dimensions in a cube, then they default in the pattern D1_, D2_, and so on.

See Also

- "DBA_CUBE_DIMENSIONALITY"
- "USER_CUBE_DIMENSIONALITY"

2.144 ALL_CUBE_DIMENSIONS

ALL_CUBE_DIMENSIONS describes the OLAP cube dimensions accessible to the current user.

Related Views

- DBA_CUBE_DIMENSIONS describes all OLAP cube dimensions in the database.
- USER_CUBE_DIMENSIONS describes the OLAP cube dimensions owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the cube dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of a cube dimension, such as TIME
DIMENSION_ID	NUMBER	NOT NULL	ID of the cube dimension
DIMENSION_TYPE	VARCHAR2(17)		Type of the OLAP cube dimension: • STANDARD • TIME
AW_NAME	VARCHAR2(128)		Name of the analytic workspace that contains the cube dimension, such as GLOBAL
DEFAULT_HIERARCHY_NAME	VARCHAR2(128)		Name of the default hierarchy for the cube dimension, such as FISCAL
DESCRIPTION	NVARCHAR2(300)		Description of the cube dimension in the session language
HIERARCHY_CONSISTENCY_RULE	VARCHAR2(200)		Hierarchy consistency rule of the OLAP cube dimension. Possible values: • CONSISTENT • STAR_CONSISTENT • SOLVE_CONSISTENT
ADD_UNIQUE_KEY_PREFIX	VARCHAR2(3)		Add_Unique_Key_Prefix flag of the OLAP cube dimension. Possible values: • YES: This is the value if AddUniqueKeyPrefix="True" was set in the metadata. This tells the system to add the level name prefix to the dimension members. This should be done when a dimension member can have the same value across different levels, for example, New York (state) and New York (city). • NO: This is the value if AddUniqueKeyPrefix="True" was not set in the metadata.
CUSTOM_ORDER	CLOB		The textual representation of the sort orderby clause used to load dimension members into the AW

See Also

- "DBA_CUBE_DIMENSIONS"
- "USER_CUBE_DIMENSIONS"

2.145 ALL_CUBE_HIER_LEVELS

ALL_CUBE_HIER_LEVELS describes the hierarchy levels for the OLAP cube dimensions accessible to the current user.

Related Views

- DBA_CUBE_HIER_LEVELS describes the hierarchy levels for all OLAP cube dimensions in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_CUBE_HIER_LEVELS describes the hierarchy levels for the OLAP cube dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the cube dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of a cube dimension, such as TIME
HIERARCHY_NAME	VARCHAR2(128)	NOT NULL	Name of a hierarchy for the dimension, such as CALENDAR
LEVEL_NAME	VARCHAR2(128)	NOT NULL	Name of the dimension level, such as MONTH
HIERARCHY_LEVEL_ID	NUMBER	NOT NULL	ID of the hierarchy level
ORDER_NUM	NUMBER	NOT NULL	Order number of the level within the hierarchy; 0 is the top level
DESCRIPTION	NVARCHAR2(300)		Description of the level in the session language

See Also

- "DBA_CUBE_HIER_LEVELS"
- "USER_CUBE_HIER_LEVELS"

2.146 ALL_CUBE_HIER_VIEW_COLUMNS

ALL_CUBE_HIER_VIEW_COLUMNS describes the columns of the relational hierarchy views of the OLAP cube dimensions accessible to the current user.

Related Views

- DBA_CUBE_HIER_VIEW_COLUMNS describes the columns of the relational hierarchy views of all OLAP cube dimensions in the database.
- USER_CUBE_HIER_VIEW_COLUMNS describes the columns of the relational hierarchy views of the OLAP cube dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the cube dimension
DIMENSION_NAME	VARCHAR2(128)		Name of a cube dimension, such as TIME
HIERARCHY_NAME	VARCHAR2(128)		Name of a hierarchy for the cube dimension, such as CALENDAR
VIEW_NAME	VARCHAR2(128)		Name of a view of the hierarchy, such as TIME_CALENDAR_VIEW
COLUMN_NAME	VARCHAR2(128)		Name of a column in the view, such as CALENDAR_QUARTER or PARENT

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COLUMN_TYPE	VARCHAR2(11)		Type of the column: • KEY - A key of the hierarchy view (that is, the hierarchy value itself) • PARENT - Dimension value of the parent of the current row in the view (or NULL if no parent) • LEVEL_NAME - Name of the level (if any) corresponding to a row in the view • DEPTH - Depth in the hierarchy tree of the current row in the view • HIER_ORDER - A column by which the results may be ordered (if present) • MEMBER_TYPE • ATTRIBUTE - An attribute owned by the hierarchy • LEVEL - One of the level columns comprising the hierarchy
OBJECT_NAME	VARCHAR2(128)		Name of a level or attribute for the dimension

See Also

- "DBA_CUBE_HIER_VIEW_COLUMNS"
- "USER_CUBE_HIER_VIEW_COLUMNS"

2.147 ALL_CUBE_HIER_VIEWS

ALL_CUBE_HIER_VIEWS describes the hierarchies for the OLAP cube dimensions accessible to the current user.

Related Views

- DBA_CUBE_HIER_VIEWS describes the hierarchies for all OLAP cube dimensions in the database.
- USER_CUBE_HIER_VIEWS describes the hierarchies for the OLAP cube dimensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the cube dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of a cube dimension, such as TIME
HIERARCHY_NAME	VARCHAR2(128)	NOT NULL	Name of a hierarchy for the cube dimension, such as CALENDAR
VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of a view of the hierarchy, such as TIME_CALENDAR_VIEW

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_CUBE_HIER_VIEWS"
- "USER_CUBE_HIER_VIEWS"

2.148 ALL_CUBE_HIERARCHIES

ALL_CUBE_HIERARCHIES describes the OLAP dimension hierarchies accessible to the current user.

Related Views

- DBA_CUBE_HIERARCHIES describes all OLAP dimension hierarchies in the database.
- USER_CUBE_HIERARCHIES describes the OLAP dimension hierarchies owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of a dimension, such as TIME
HIERARCHY_NAME	VARCHAR2(128)	NOT NULL	Name of a hierarchy for the dimension, such as CALENDAR
HIERARCHY_ID	NUMBER	NOT NULL	ID of the hierarchy
HIERARCHY_TYPE	VARCHAR2(5)		Type of the hierarchy: • LEVEL • VALUE
DESCRIPTION	NVARCHAR2(300)		Description of the hierarchy in the session language
IS_RAGGED	NUMBER		Indicates whether ragged hierarchies are permitted in subsequent builds. User dimensions that are enabled for materialized views and Time dimensions are set to 0. Builds then check the data for ragged hierarchies and fail if one is detected. When User dimensions are set to 1, the builds do not check for ragged hierarchies.
IS_SKIP_LEVEL	NUMBER		Indicates whether skip-level hierarchies are permitted in subsequent builds. User dimensions that are enabled for materialized views and Time dimensions are set to 0. Builds then check the data for skip-level hierarchies and fail if one is detected. When User dimensions are set to 1, the builds do not check for skip-level hierarchies.
REFRESH_MVIEW_NAME	VARCHAR2(200)		Name of the Refresh Materialized View associated with the hierarchy
CUSTOM_ORDER	CLOB		The textual representation of the sort orderby clause used to load dimension members of the hierarchy into the AW

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_CUBE_HIERARCHIES"
- "USER_CUBE_HIERARCHIES"

2.149 ALL_CUBE_MEASURES

ALL_CUBE_MEASURES describes the measures for the OLAP cubes accessible to the current user.

Related Views

- DBA_CUBE_MEASURES describes the measures for all OLAP cubes in the database.
- USER_CUBE_MEASURES describes the measures for the OLAP cubes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the cube
CUBE_NAME	VARCHAR2(128)	NOT NULL	Name of a cube, such as UNITS_CUBE
MEASURE_NAME	VARCHAR2(128)	NOT NULL	Name of a measure in the cube, such as SALES
MEASURE_ID	NUMBER	NOT NULL	ID of a measure
OVERRIDE_SOLVE_SPEC	CLOB		Syntax text for the measure's consistent solve specification that overrides that of its cube
MEASURE_TYPE	VARCHAR2(7)		Type of the OLAP measure: • BASE - Base measures store the data • DERIVED - Derived measures calculate the data from base measures; also called calculated measures
EXPRESSION	CLOB		Expression that provides the values of the measure
DESCRIPTION	NVARCHAR2(300)		Description of the measure in the session language
DATA_TYPE	VARCHAR2(106)		Data type of the measure, such as NUMBER
DATA_LENGTH	NUMBER	NOT NULL	Length of a character data type
DATA_PRECISION	NUMBER		Precision of a numeric data type
DATA_SCALE	NUMBER		Scale of a numeric data type
LOOP_VAR_OVERRIDE	VARCHAR2(200)		The value that overrides the $LOOP_VAR property of the OLAP derived measure
LOOP_DENSE_OVERRIDE	VARCHAR2(200)		The value that overrides the $LOOP_DENSE property of the OLAP derived measure
LOOP_TYPE	VARCHAR2(200)		The $LOOP_TYPE property of the OLAP derived measure. Possible values: • INNER • OUTER • DENSE

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_CUBE_MEASURES"
- "USER_CUBE_MEASURES"

2.150 ALL_CUBE_NAMED_BUILD_SPECS

ALL_CUBE_NAMED_BUILD_SPECS describes the OLAP cube named build specifications in the database that are accessible by the user.

Related Views

- DBA_CUBE_NAMED_BUILD_SPECS describes the OLAP cube named build specifications in the database.
- USER_CUBE_NAMED_BUILD_SPECS describes the OLAP cube named build specifications in the database that are owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the OLAP named build specification
CUBE_NAME	VARCHAR2(128)	NOT NULL	Name of the OLAP cube
NAMED_BUILD_SPEC	CLOB		Name of the OLAP cube named build specification

See Also

- "DBA_CUBE_NAMED_BUILD_SPECS"
- "USER_CUBE_NAMED_BUILD_SPECS"

2.151 ALL_CUBE_SUB_PARTITION_LEVELS

ALL_CUBE_SUB_PARTITION_LEVELS describes the OLAP secondary partition levels in the database that are accessible by the user.

Related Views

- DBA_CUBE_SUB_PARTITION_LEVELS describes the OLAP secondary partition levels in the database.
- USER_CUBE_SUB_PARTITION_LEVELS describes the OLAP secondary partition levels in the database that are owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the OLAP secondary partition level
CUBE_NAME	VARCHAR2(128)		Name of the OLAP cube
SUB_PARTITION_LEVEL_NAME	VARCHAR2(200)		Name of the secondary partition level of the OLAP cube

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PRECOMPUTE_PERCENT	NUMBER		Precompute percent of the secondary partition level of the OLAP cube
PARTITION_DIMENSION_NAME	VARCHAR2(128)		Name of the cube dimension for which there is a secondary partition level on the OLAP cube
PARTITION_HIERARCHY_NAME	VARCHAR2(128)		Name of the hierarchy for which there is a secondary partition level on the OLAP cube
PARTITION_LEVEL_NAME	VARCHAR2(128)		Name of the hierarchy level for which there is a secondary partition level on the OLAP cube
SUB_PARTITION_LEVEL_ORDER	NUMBER		Order number of the secondary partition level on the OLAP cube

See Also

- "DBA_CUBE_SUB_PARTITION_LEVELS"
- "USER_CUBE_SUB_PARTITION_LEVELS"

2.152 ALL_CUBE_VIEW_COLUMNS

ALL_CUBE_VIEW_COLUMNS describes the columns of the relational views of the OLAP cubes accessible to the current user.

Related Views

- DBA_CUBE_VIEW_COLUMNS describes the columns of relational views of all OLAP cubes in the database.
- USER_CUBE_VIEW_COLUMNS describes the columns of relational views of OLAP cubes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the cube
CUBE_NAME	VARCHAR2(128)		Name of a cube, such as UNITS_CUBE
VIEW_NAME	VARCHAR2(128)		Name of a view of the cube, such as PRODUCT_VIEW
COLUMN_NAME	VARCHAR2(128)		Name of a column in the view, such as DIM_KEY or LEVEL_NAME
COLUMN_TYPE	VARCHAR2(7)		Type of the column: • MEASURE • KEY
OBJECT_NAME	VARCHAR2(128)		Name of the measure or dimension represented in the column

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_CUBE_VIEW_COLUMNS"
- "USER_CUBE_VIEW_COLUMNS"

2.153 ALL_CUBE_VIEWS

ALL_CUBE_VIEWS describes the relational views of the OLAP cubes accessible to the current user.

Related Views

- DBA_CUBE_VIEWS describes the relational views of all OLAP cubes in the database.
- USER_CUBE_VIEWS describes the relational views of the OLAP cubes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the cube
CUBE_NAME	VARCHAR2(128)	NOT NULL	Name of a cube, such as UNITS_CUBE
VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of a view of the cube, such as UNITS_CUBE_VIEW

See Also

- "DBA_CUBE_VIEWS"
- "USER_CUBE_VIEWS"

2.154 ALL_CUBES

ALL_CUBES describes the OLAP cubes accessible to the current user.

Related Views

- DBA_CUBES describes all OLAP cubes in the database.
- USER_CUBES describes the OLAP cubes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the cube
CUBE_NAME	VARCHAR2(128)	NOT NULL	Name of a cube, such as UNITS_CUBE
CUBE_ID	NUMBER	NOT NULL	ID of a cube
AW_NAME	VARCHAR2(128)		Name of the analytic workspace that contains the cube, such as GLOBAL
CONSISTENT_SOLVE_SPEC	CLOB		Default aggregation rules for the cube
DESCRIPTION	NVARCHAR2(300)		Description of the cube in the session language

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SPARSE_TYPE	VARCHAR2(200)		Text value indicating the type of sparsity for the OLAP cube
PRECOMPUTE_CONDITION	CLOB		Condition syntax representing the precompute condition of the OLAP cube
PRECOMPUTE_PERCENT	NUMBER		Percentage of aggregate data values that are calculated and stored during data maintenance. If the cube is partitioned, then this percentage is for the bottom partitions.
PRECOMPUTE_PERCENT_TOP	NUMBER		Percentage of aggregate data values in the top partition that are calculated and stored during data maintenance
PARTITION_DIMENSION_NAME	VARCHAR2(128)		Name of the dimension used to partition the cube, such as TIME
PARTITION_HIERARCHY_NAME	VARCHAR2(128)		Name of the dimension hierarchy used to partition the cube, such as CALENDAR
PARTITION_LEVEL_NAME	VARCHAR2(128)		Name of the level used to partition the cube, such as QUARTER
REFRESH_MVIEW_NAME	VARCHAR2(200)		Name of the refresh materialized view for the OLAP cube
REWRITE_MVIEW_NAME	VARCHAR2(200)		Name of the rewrite materialized view for the OLAP cube
DEFAULT_BUILD_SPEC	CLOB		The default build specification for the OLAP cube
MEASURE_STORAGE	VARCHAR2(200)		The measure storage for the OLAP cube. Possible values: • INDEPENDENT • SHARED
SQL_CUBE_STORAGE_TYPE	CLOB		The SQL cube storage type for the OLAP cube. This value represents a SQL data type.
CUBE_STORAGE_TYPE	VARCHAR2(200)		The cube storage type for the OLAP cube. This value represents a DML data type.

See Also

- "DBA_CUBES"
- "USER_CUBES"

2.155 ALL_DB_LINKS

ALL_DB_LINKS describes the database links accessible to the current user.

Related Views

- DBA_DB_LINKS describes all database links in the database.
- USER_DB_LINKS describes the database links owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the database link
DB_LINK	VARCHAR2(128)	NOT NULL	Name of the database link
USERNAME	VARCHAR2(128)		Name of the user when logging in
HOST	VARCHAR2(2000)		Oracle Net connect string
CREATED	DATE	NOT NULL	Creation time of the database link
HIDDEN	VARCHAR2(3)		For internal use only
SHARD_INTERNAL	VARCHAR2(3)		Indicates whether the database link is used to support operations across sharded databases. Possible values: • YES: The database link is used and managed for to support sharded databases • NO: The database link is not used and managed to support sharded databases Users should not alter or delete database links that are used and managed to support sharded databases.
VALID	VARCHAR2(3)		Indicates whether the database link is valid and usable. Possible values: • YES: The database link is valid and usable. • NO: The database link is invalid and unusable.
INTRA_CDB[1]	VARCHAR2(3)		For internal use only

1 This column is available starting with Oracle Database 19c.

See Also

- "DBA_DB_LINKS"
- "USER_DB_LINKS"
- "DBA_DB_LINK_SOURCES"
- "DBA_EXTERNAL_SCN_ACTIVITY"

2.156 ALL_DEF_AUDIT_OPTS

ALL_DEF_AUDIT_OPTS contains default object-auditing options that will be applied when objects are created.

The output for each column takes one of the following forms:

- -/-: no default auditing
- S/-: auditing whenever successful
- -/S: auditing whenever not successful

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from `UNIFIED_AUDIT_TRAIL`.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
ALT	VARCHAR2(3)		Auditing ALTER WHENEVER SUCCESSFUL / UNSUCCESSFUL
AUD	VARCHAR2(3)		Auditing AUDIT WHENEVER SUCCESSFUL / UNSUCCESSFUL
COM	VARCHAR2(3)		Auditing COMMENT WHENEVER SUCCESSFUL / UNSUCCESSFUL
DEL	VARCHAR2(3)		Auditing DELETE WHENEVER SUCCESSFUL / UNSUCCESSFUL
GRA	VARCHAR2(3)		Auditing GRANT WHENEVER SUCCESSFUL / UNSUCCESSFUL
IND	VARCHAR2(3)		Auditing INDEX WHENEVER SUCCESSFUL / UNSUCCESSFUL
INS	VARCHAR2(3)		Auditing INSERT WHENEVER SUCCESSFUL / UNSUCCESSFUL
LOC	VARCHAR2(3)		Auditing LOCK WHENEVER SUCCESSFUL / UNSUCCESSFUL
REN	VARCHAR2(3)		Auditing RENAME WHENEVER SUCCESSFUL / UNSUCCESSFUL
SEL	VARCHAR2(3)		Auditing SELECT WHENEVER SUCCESSFUL / UNSUCCESSFUL
UPD	VARCHAR2(3)		Auditing UPDATE WHENEVER SUCCESSFUL / UNSUCCESSFUL
EXE	VARCHAR2(3)		Auditing EXECUTE WHENEVER SUCCESSFUL / UNSUCCESSFUL
FBK	VARCHAR2(3)		Auditing FLASHBACK WHENEVER SUCCESSFUL / UNSUCCESSFUL
REA	VARCHAR2(3)		Auditing READ WHENEVER SUCCESSFUL / UNSUCCESSFUL

2.157 ALL_DEPENDENCIES

`ALL_DEPENDENCIES` describes dependencies between procedures, packages, functions, package bodies, and triggers accessible to the current user, including dependencies on views created without any database links. This view does not display the `SCHEMAID` column.

Related Views

- `DBA_DEPENDENCIES` describes all dependencies between objects in the database. This view does not display the `SCHEMAID` column.
- `USER_DEPENDENCIES` describes dependencies between objects in the current user's schema. This view does not display the `OWNER` column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
NAME	VARCHAR2(128)	NOT NULL	Name of the object
TYPE	VARCHAR2(18)		Type of the object
REFERENCED_OWNER	VARCHAR2(128)		Owner of the referenced object (remote owner if remote object)
REFERENCED_NAME	VARCHAR2(128)		Name of the referenced object
REFERENCED_TYPE	VARCHAR2(18)		Type of the referenced object
REFERENCED_LINK_NAME	VARCHAR2(128)		Name of the link to the parent object (if remote)
DEPENDENCY_TYPE	VARCHAR2(4)		Indicates whether the dependency is a REF dependency (REF) or not (HARD)

See Also

- "DBA_DEPENDENCIES"
- "USER_DEPENDENCIES"

2.158 ALL_DEQUEUE_QUEUES

ALL_DEQUEUE_QUEUES describes all queues on which the current user has dequeue privileges. If the user has any Advanced Queuing system privileges, such as DEQUEUE ANY QUEUE or MANAGE ANY QUEUE, then this view describes all queues in the database.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the queue
NAME	VARCHAR2(128)	NOT NULL	Name of the queue
QUEUE_TABLE	VARCHAR2(128)	NOT NULL	Name of the table in which the queue data resides
QID	NUMBER	NOT NULL	Object number of the queue
QUEUE_TYPE	VARCHAR2(20)		Type of the queue: • EXCEPTION_QUEUE • NORMAL_QUEUE
MAX_RETRIES	NUMBER		Maximum number of retries allowed when dequeuing from the queue
RETRY_DELAY	NUMBER		Time interval between retries
ENQUEUE_ENABLED	VARCHAR2(7)		Indicates whether the queue is enabled for enqueue (YES) or not (NO)
DEQUEUE_ENABLED	VARCHAR2(7)		Indicates whether the queue is enabled for dequeue (YES) or not (NO)
RETENTION	VARCHAR2(40)		Time interval that processed messages are retained in the queue, or FOREVER
USER_COMMENT	VARCHAR2(50)		User-specified comment
NETWORK_NAME	VARCHAR2(512)		Network name of the queue service
SHARDED	VARCHAR2(5)		TRUE if the queue is sharded, FALSE otherwise

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.159 ALL_DIM_ATTRIBUTES

ALL_DIM_ATTRIBUTES describes the relationship between a dimension level and a functionally dependent column. The level columns and the dependent column must be in the same table.

Related Views

- DBA_DIM_ATTRIBUTES describes all such dimension relationships in the database.
- USER_DIM_ATTRIBUTES describes all such dimension attributes in the current user's schema.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the dimension
ATTRIBUTE_NAME	VARCHAR2(128)		Name of the attribute
LEVEL_NAME	VARCHAR2(128)		Name of the hierarchy level
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Dependent column name
INFERRED	CHAR(1)		Indicates whether the attribute is inferred from a JOIN KEY specification (Y) or not (N)

See Also

- "DBA_DIM_ATTRIBUTES"
- "USER_DIM_ATTRIBUTES"

2.160 ALL_DIM_CHILD_OF

ALL_DIM_CHILD_OF describes hierarchical relationships of 1 to *n* between the pairs of levels in the dimensions accessible to the current user.

Related Views

- DBA_DIM_CHILD_OF describes all such hierarchical relationships in the database.
- USER_DIM_CHILD_OF describes all such hierarchical attributes in the current user's schema.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the dimension
HIERARCHY_NAME	VARCHAR2(128)		Hierarchy name
POSITION	NUMBER	NOT NULL	Hierarchical position within this hierarchy, position 1 being the most detailed
CHILD_LEVEL_NAME	VARCHAR2(128)		Child side of 1:n relationship
JOIN_KEY_ID	VARCHAR2(40)		If non-null, then the child joins to the parent
PARENT_LEVEL_NAME	VARCHAR2(128)		Parent side of 1:n relationship in relation to the CHILD_LEVEL_NAME

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_DIM_CHILD_OF"
- "USER_DIM_CHILD_OF"

2.161 ALL_DIM_HIERARCHIES

ALL_DIM_HIERARCHIES describes all dimension hierarchies accessible to the current user.

Related Views

- DBA_DIM_HIERARCHIES describes all such hierarchies in the database.
- USER_DIM_HIERARCHIES describes all such hierarchies owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the dimension
HIERARCHY_NAME	VARCHAR2(128)		Hierarchy name

See Also

- "DBA_DIM_HIERARCHIES"
- "USER_DIM_HIERARCHIES"

2.162 ALL_DIM_JOIN_KEY

ALL_DIM_JOIN_KEY describes the joins between two dimension tables that are accessible to the current user. The join is always specified between a parent dimension level column and a child column.

Related Views

- DBA_DIM_JOIN_KEY describes all such joins in the database.
- USER_DIM_JOIN_KEY describes all such joins owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the dimension
DIM_KEY_ID	NUMBER	NOT NULL	Join key ID (unique within a dimension)
LEVEL_NAME	VARCHAR2(128)		Name of the hierarchy level
KEY_POSITION	NUMBER	NOT NULL	Ordinal position of the key column within the level
HIERARCHY_NAME	VARCHAR2(128)		Name of the hierarchy
CHILD_JOIN_OWNER	VARCHAR2(128)	NOT NULL	Owner of the join column table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CHILD_JOIN_TABLE	VARCHAR2(128)	NOT NULL	Name of the join column table
CHILD_JOIN_COLUMN	VARCHAR2(128)	NOT NULL	Name of the join column
CHILD_LEVEL_NAME	VARCHAR2(128)		Name of the child hierarchy level of the join key

See Also

- "DBA_DIM_JOIN_KEY"
- "USER_DIM_JOIN_KEY"

2.163 ALL_DIM_LEVEL_KEY

ALL_DIM_LEVEL_KEY describes a column of a dimension level accessible to the current user. The position of a column within a level is specified by KEY_POSITION.

Related Views

- DBA_DIM_LEVEL_KEY describes all columns of dimension levels in the database.
- USER_DIM_LEVEL_KEY describes all columns of dimension levels owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the dimension
LEVEL_NAME	VARCHAR2(128)		Name of the hierarchy level
KEY_POSITION	NUMBER	NOT NULL	Ordinal position of the key column within the level
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the key column

See Also

- "DBA_DIM_LEVEL_KEY"
- "USER_DIM_LEVEL_KEY"

2.164 ALL_DIM_LEVELS

ALL_DIM_LEVELS describes the dimension levels accessible to the current user. All columns of a dimension level must come from the same relation.

Related Views

- DBA_DIM_LEVELS describes all dimension levels in the database.
- USER_DIM_LEVELS describes the levels of all dimensions owned by the current user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the dimension
LEVEL_NAME	VARCHAR2(128)		Unique within a dimension
NUM_COLUMNS	NUMBER		Number of columns in the level definition
DETAILOBJ_OWNER	VARCHAR2(128)	NOT NULL	Owner of the detail object that the keys of this level come from
DETAILOBJ_NAME	VARCHAR2(128)	NOT NULL	Name of the table that the keys of this level come from
SKIP_WHEN_NULL	VARCHAR2(1)		Indicates whether the level is declared with the SKIP WHEN NULL clause (Y) or not (N)

See Also

- "DBA_DIM_LEVELS"
- "USER_DIM_LEVELS"

2.165 ALL_DIMENSIONS

ALL_DIMENSIONS describes the dimension objects accessible to the current user.

Related Views

- DBA_DIMENSIONS describes all dimensions in the database.
- USER_DIMENSIONS describes the dimensions in the current user's schema.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the dimension
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the dimension
INVALID	VARCHAR2(1)		Indicates whether the dimension is invalid (Y) or valid (N)
COMPILE_STATE	VARCHAR2(13)		Compile status of the dimension: • INVALID • NEEDS_COMPILE • ERROR
REVISION	NUMBER		Dimension revision level

See Also

- "DBA_DIMENSIONS"
- "USER_DIMENSIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.166 ALL_DIRECTORIES

ALL_DIRECTORIES describes all directories accessible to the current user.

Related View

DBA_DIRECTORIES describes all directories in the database.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the directory (always SYS)
DIRECTORY_NAME	VARCHAR2(128)	NOT NULL	Name of the directory
DIRECTORY_PATH	VARCHAR2(4000)		Operating system pathname for the directory
ORIGIN_CON_ID	VARCHAR2(256)		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)

See Also

"DBA_DIRECTORIES"

2.167 ALL_EDITION_COMMENTS

ALL_EDITION_COMMENTS describes the comments on the editions accessible to the current user.

Related View

DBA_EDITION_COMMENTS describes the comments on all editions in the database.

Column	Datatype	NULL	Description
EDITION_NAME	VARCHAR2(128)	NOT NULL	Name of the edition
COMMENTS	VARCHAR2(4000)		Edition comments

See Also

"DBA_EDITION_COMMENTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.168 ALL_EDITIONING_VIEW_COLS

ALL_EDITIONING_VIEW_COLS describes the relationship between the columns of the editioning views accessible to the current user and the table columns to which they map.

Related Views

- DBA_EDITIONING_VIEW_COLS describes the relationship between the columns of all editioning views in the database and the table columns to which they map.
- USER_EDITIONING_VIEW_COLS describes the relationship between the columns of the editioning views owned by the current user and the table columns to which they map. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of an editioning view
VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of an editioning view
VIEW_COLUMN_ID	NUMBER	NOT NULL	Column number within the editioning view
VIEW_COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column in the editioning view
TABLE_COLUMN_ID	NUMBER	NOT NULL	Column number of a table column to which this editioning view column maps
TABLE_COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of a table column to which this editioning view column maps

See Also

- "DBA_EDITIONING_VIEW_COLS"
- "USER_EDITIONING_VIEW_COLS"

2.169 ALL_EDITIONING_VIEW_COLS_AE

ALL_EDITIONING_VIEW_COLS_AE describes the relationship between the columns of the editioning views (across all editions) accessible to the current user and the table columns to which they map.

Related Views

- DBA_EDITIONING_VIEW_COLS_AE describes the relationship between the columns of all editioning views (across all editions) in the database and the table columns to which they map.
- USER_EDITIONING_VIEW_COLS_AE describes the relationship between the columns of the editioning views (across all editions) owned by the current user and the table columns to which they map. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of an editioning view
VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of an editioning view

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
VIEW_COLUMN_ID	NUMBER	NOT NULL	Column number within the editioning view
VIEW_COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column in the editioning view
TABLE_COLUMN_ID	NUMBER	NOT NULL	Column number of a table column to which this editioning view column maps
TABLE_COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of a table column to which this editioning view column maps
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the editioning view is defined

See Also

- "DBA_EDITIONING_VIEW_COLS_AE"
- "USER_EDITIONING_VIEW_COLS_AE"

2.170 ALL_EDITIONING_VIEWS

ALL_EDITIONING_VIEWS describes the editioning views accessible to the current user.

Related Views

- DBA_EDITIONING_VIEWS describes all editioning views in the database.
- USER_EDITIONING_VIEWS describes the editioning views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of an editioning view
VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of an editioning view
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of an editioning view's base table

See Also

- "DBA_EDITIONING_VIEWS"
- "USER_EDITIONING_VIEWS"

2.171 ALL_EDITIONING_VIEWS_AE

ALL_EDITIONING_VIEWS_AE describes the editioning views (across all editions) accessible to the current user.

Related Views

- DBA_EDITIONING_VIEWS_AE describes all editioning views (across all editions) in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_EDITIONING_VIEWS_AE describes the editioning views (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of an editioning view
VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of an editioning view
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of an editioning view's base table
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the editioning view is defined

See Also

- "DBA_EDITIONING_VIEWS_AE"
- "USER_EDITIONING_VIEWS_AE"

2.172 ALL_EDITIONS

ALL_EDITIONS describes the editions accessible to the current user.

Related View

DBA_EDITIONS describes all editions in the database.

Column	Datatype	NULL	Description
EDITION_NAME	VARCHAR2(128)	NOT NULL	Name of the edition
PARENT_EDITION_NAME	VARCHAR2(128)		Name of the parent edition for this edition
USABLE	VARCHAR2(3)		Indicates whether the edition is usable (YES) or unusable (NO)

See Also

- "DBA_EDITIONS"
- *Oracle Database Development Guide* for more information about editions

2.173 ALL_ENCRYPTED_COLUMNS

ALL_ENCRYPTED_COLUMNS displays encryption algorithm information for the encrypted columns in the tables accessible to the current user.

Related Views

- DBA_ENCRYPTED_COLUMNS displays encryption algorithm information for all encrypted columns in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_ENCRYPTED_COLUMNS displays encryption algorithm information for the encrypted columns in the tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column
ENCRYPTION_ALG	VARCHAR2(29)		Encryption algorithm used to protect secrecy of data in this column: • 3 Key Triple DES 168 bits key • AES 128 bits key • AES 192 bits key • AES 256 bits key
SALT	VARCHAR2(3)		Indicates whether the column is encrypted with SALT (YES) or not (NO)
INTEGRITY_ALG	VARCHAR2(12)		Integrity algorithm used for the column: • SHA-1 • NOMAC

See Also

- "DBA_ENCRYPTED_COLUMNS"
- "USER_ENCRYPTED_COLUMNS"

2.174 ALL_ERROR_TRANSLATIONS

ALL_ERROR_TRANSLATIONS describes all error translations accessible to the user.

Related Views

- DBA_ERROR_TRANSLATIONS describes all error translations in the database.
- USER_ERROR_TRANSLATIONS describes all error translations owned by the user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the SQL translation profile
PROFILE_NAME	VARCHAR2(128)	NOT NULL	Name of the SQL translation profile
ERROR_CODE	NUMBER	NOT NULL	The Oracle error code
TRANSLATED_CODE	NUMBER		The translated error code
TRANSLATED_SQLSTATE	VARCHAR2(5)		The translated SQLSTATE
ENABLED	VARCHAR2(5)		TRUE if the translation is enabled, FALSE otherwise.
REGISTRATION_TIME	TIMESTAMP(6)		Time the translation was registered
COMMENTS	VARCHAR2(4000)		Comment on the translation

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_ERROR_TRANSLATIONS"
- "USER_ERROR_TRANSLATIONS"

2.175 ALL_ERRORS

ALL_ERRORS describes the current errors on the stored objects accessible to the current user.

Related Views

- DBA_ERRORS describes the current errors on all stored objects in the database.
- USER_ERRORS describes the current errors on the stored objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
NAME	VARCHAR2(128)	NOT NULL	Name of the object
TYPE	VARCHAR2(12)		Type of the object: • ANALYTIC VIEW • ASSEMBLY • ATTRIBUTE DIMENSION • DIMENSION • FUNCTION • HIERARCHY • JAVA CLASS • JAVA SOURCE • LIBRARY • PACKAGE • PACKAGE BODY • PROCEDURE • QUEUE • TRIGGER • TYPE • TYPE BODY • VIEW
SEQUENCE	NUMBER	NOT NULL	Sequence number (for ordering purposes)
LINE	NUMBER	NOT NULL	Line number at which the error occurred
POSITION	NUMBER	NOT NULL	Position in the line at which the error occurred
TEXT	VARCHAR2(4000)	NOT NULL	Text of the error
ATTRIBUTE	VARCHAR2(9)		Indicates whether the error is an error (ERROR) or a warning (WARNING)
MESSAGE_NUMBER	NUMBER		Numeric error number (without any prefix)

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_ERRORS"

  "USER_ERRORS"

2.176 ALL_ERRORS_AE

ALL_ERRORS_AE describes the current errors on the stored objects (across all editions) accessible to the current user.

Related Views

- DBA_ERRORS_AE describes the current errors on all stored objects (across all editions) in the database.
- USER_ERRORS_AE describes the current errors on the stored objects (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
NAME	VARCHAR2(128)	NOT NULL	Name of the object
TYPE	VARCHAR2(12)		Type of the object: • ANALYTIC VIEW • ASSEMBLY • ATTRIBUTE DIMENSION • DIMENSION • FUNCTION • HIERARCHY • JAVA CLASS • JAVA SOURCE • LIBRARY • PACKAGE • PACKAGE BODY • PROCEDURE • QUEUE • TRIGGER • TYPE • TYPE BODY • VIEW
SEQUENCE	NUMBER	NOT NULL	Sequence number (for ordering purposes)
LINE	NUMBER	NOT NULL	Line number at which this error occurred
POSITION	NUMBER	NOT NULL	Position in the line at which this error occurred
TEXT	VARCHAR2(4000)	NOT NULL	Text of the error
ATTRIBUTE	VARCHAR2(9)		Indicates whether the error is an error (ERROR) or a warning (WARNING)
MESSAGE_NUMBER	NUMBER		Numeric error number (without any prefix)
EDITION_NAME	VARCHAR2(128)		Name of the edition in which the object is actual

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_ERRORS_AE"
- "USER_ERRORS_AE"

2.177 ALL_EVALUATION_CONTEXT_TABLES

ALL_EVALUATION_CONTEXT_TABLES describes the tables in the rule evaluation contexts accessible to the current user.

Related Views

- DBA_EVALUATION_CONTEXT_TABLES describes the tables in all rule evaluation contexts in the database.
- USER_EVALUATION_CONTEXT_TABLES describes the tables in the rule evaluation contexts owned by the current user. This view does not display the EVALUATION_CONTEXT_OWNER column.

Column	Datatype	NULL	Description
EVALUATION_CONTEXT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the evaluation context
EVALUATION_CONTEXT_NAME	VARCHAR2(128)	NOT NULL	Name of the evaluation context
TABLE_ALIAS	VARCHAR2(128)		Alias for a table in the evaluation context
TABLE_NAME	VARCHAR2(4000)		Name of the table referred to by the table alias

See Also

- "DBA_EVALUATION_CONTEXT_TABLES"
- "USER_EVALUATION_CONTEXT_TABLES"

2.178 ALL_EVALUATION_CONTEXT_VARS

ALL_EVALUATION_CONTEXT_VARS describes the variables in the rule evaluation contexts accessible to the current user.

Related Views

- DBA_EVALUATION_CONTEXT_VARS describes the variables in all rule evaluation contexts in the database.
- USER_EVALUATION_CONTEXT_VARS describes the variables in the rule evaluation contexts owned by the current user. This view does not display the EVALUATION_CONTEXT_OWNER column.

Column	Datatype	NULL	Description
EVALUATION_CONTEXT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the evaluation context
EVALUATION_CONTEXT_NAME	VARCHAR2(128)	NOT NULL	Name of the evaluation context

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
VARIABLE_NAME	VARCHAR2(128)		Name of a variable in the evaluation context
VARIABLE_TYPE	VARCHAR2(4000)		Datatype of the variable
VARIABLE_VALUE_FUNCTION	VARCHAR2(4000)		Function used to retrieve the value of the variable; NULL for variables that are not implicit
VARIABLE_METHOD_FUNCTION	VARCHAR2(228)		Function used to retrieve the result of method invocation on the variable. Such a function can speed up evaluation, if there are many simple rules that invoke the method on the variable.

See Also

- "DBA_EVALUATION_CONTEXT_VARS"
- "USER_EVALUATION_CONTEXT_VARS"

2.179 ALL_EVALUATION_CONTEXTS

ALL_EVALUATION_CONTEXTS describes the rule evaluation contexts accessible to the current user.

Related Views

- DBA_EVALUATION_CONTEXTS describes all rule evaluation contexts in the database.
- USER_EVALUATION_CONTEXTS describes the rule evaluation contexts owned by the current user. This view does not display the EVALUATION_CONTEXT_OWNER column.

Column	Datatype	NULL	Description
EVALUATION_CONTEXT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the evaluation context
EVALUATION_CONTEXT_NAME	VARCHAR2(128)	NOT NULL	Name of the evaluation context
EVALUATION_FUNCTION	VARCHAR2(4000)		Evaluation function associated with the evaluation context, if any
EVALUATION_CONTEXT_COMMENT	VARCHAR2(4000)		Comment specified with the evaluation context, if any

See Also

- "DBA_EVALUATION_CONTEXTS"
- "USER_EVALUATION_CONTEXTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.180 ALL_EXPRESSION_STATISTICS

ALL_EXPRESSION_STATISTICS provides expression usage tracking statistics for tables that are accessible to the current user.

Related Views

- DBA_EXPRESSION_STATISTICS provides expression usage tracking statistics for all the tables in the database.
- USER_EXPRESSION_STATISTICS provides expression usage tracking statistics for tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table contained in the expression
EXPRESSION_ID	NUMBER		Expression ID of the current expression
SNAPSHOT	VARCHAR2(10)		Type of snapshot for the expression: • LATEST: Latest snapshot • CUMULATIVE: Cumulative snapshot • WINDOW: Window snapshot
EVALUATION_COUNT	NUMBER		Number of times the expression has been evaluated
FIXED_COST	NUMBER	NOT NULL	Optimizer fixed cost of evaluating the expression
DYNAMIC_COST	NUMBER		Optimizer dynamic cost of evaluating the expression
EXPRESSION_TEXT	VARCHAR2(4000)	NOT NULL	Text of the expression
CREATED	DATE	NOT NULL	Time this expression is first evaluated
LAST_MODIFIED	DATE		Time this expression is last evaluated

See Also

- "DBA_EXPRESSION_STATISTICS"
- "USER_EXPRESSION_STATISTICS"
- "V$EXP_STATS"

2.181 ALL_EXTERNAL_LOCATIONS

ALL_EXTERNAL_LOCATIONS describes the locations (data sources) of the external tables accessible to the current user.

Related Views

- DBA_EXTERNAL_LOCATIONS describes the locations (data sources) of all external tables in the database.
- USER_EXTERNAL_LOCATIONS describes the locations (data sources) of the external tables owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the external table location
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the corresponding external table
LOCATION	VARCHAR2(4000)		External table location clause
DIRECTORY_OWNER	CHAR(3)		Owner of the directory containing the external table location
DIRECTORY_NAME	VARCHAR2(128)		Name of the directory containing the external table location

See Also

- "DBA_EXTERNAL_LOCATIONS"
- "USER_EXTERNAL_LOCATIONS"

2.182 ALL_EXTERNAL_TABLES

ALL_EXTERNAL_TABLES describes the external tables accessible to the current user.

Related Views

- DBA_EXTERNAL_TABLES describes all external tables in the database.
- USER_EXTERNAL_TABLES describes the external tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the external table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the external table
TYPE_OWNER	CHAR(3)		Owner of the implementation type for the external table access driver
TYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the implementation type for the external table access driver
DEFAULT_DIRECTORY_OWNER	CHAR(3)		Owner of the default directory for the external table
DEFAULT_DIRECTORY_NAME	VARCHAR2(128)	NOT NULL	Name of the default directory for the external table
REJECT_LIMIT	VARCHAR2(40)		Reject limit for the external table, or UNLIMITED
ACCESS_TYPE	VARCHAR2(7)		Type of access parameters for the external table: • BLOB • CLOB
ACCESS_PARAMETERS	CLOB		Access parameters for the external table
PROPERTY	VARCHAR2(10)		Property of the external table: • REFERENCED - Referenced columns • ALL - All columns
INMEMORY	VARCHAR2(8)		Indicates whether the In-Memory Column Store (IM column store) is enabled (ENABLED) or disabled (DISABLED) for this table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INMEMORY_COMPRESSION	VARCHAR2(17)		Indicates the compression level for the IM column store: • NO MEMCOMPRESS • FOR DML • FOR CAPACITY [HIGH | LOW] • FOR QUERY [HIGH | LOW] • NULL This column has a value based on where the segments lie for a table. For example, if the table is partitioned and is enabled for the IM column store, the value is NULL for ALL_EXTERNAL_TABLES but non-NULL for ALL_XTERNAL_TAB_PARTITIONS.

See Also

- "DBA_EXTERNAL_TABLES"
- "USER_EXTERNAL_TABLES"

2.183 ALL_FILE_GROUP_EXPORT_INFO

ALL_FILE_GROUP_EXPORT_INFO shows export-related information for each version accessible to the current user. There will only be information in this view for versions that have a valid Data Pump dump file.

Related Views

- DBA_FILE_GROUP_EXPORT_INFO shows export-related information for each version in the database that has a valid Data Pump dump file.
- USER_FILE_GROUP_EXPORT_INFO shows export-related information for all file groups owned by the current user. This view does not display the FILE_GROUP_OWNER column.

Column	Datatype	NULL	Description
FILE_GROUP_OWNER	VARCHAR2(128)	NOT NULL	Owner of the file group
FILE_GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the file group
VERSION_NAME	VARCHAR2(128)	NOT NULL	User-specified name for the version
VERSION	NUMBER	NOT NULL	Internal version number
EXPORT_VERSION	VARCHAR2(128)	NOT NULL	Version of exported objects
PLATFORM_NAME	VARCHAR2(101)	NOT NULL	Platform on which the export was performed
EXPORT_TIME	DATE	NOT NULL	Time at which the export job was performed
EXPORT_SCN	NUMBER		SCN of the export job
SOURCE_GLOBAL_NAME	VARCHAR2(128)		Global name of the exporting database

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_FILE_GROUP_EXPORT_INFO"
- "USER_FILE_GROUP_EXPORT_INFO"

2.184 ALL_FILE_GROUP_FILES

ALL_FILE_GROUP_FILES shows the file set for each versioned file group accessible to the current user.

Related Views

- DBA_FILE_GROUP_FILES shows the file set for each versioned group in the database.
- USER_FILE_GROUP_FILES shows the file set for each versioned group owned by the current user. This view does not display the FILE_GROUP_OWNER column.

Column	Datatype	NULL	Description
FILE_GROUP_OWNER	VARCHAR2(128)	NOT NULL	Owner of the file group
FILE_GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the file group
VERSION_NAME	VARCHAR2(128)	NOT NULL	Name of the version to which the file belongs
VERSION	NUMBER	NOT NULL	Internal version number of the file group version to which the file belongs
FILE_NAME	VARCHAR2(512)	NOT NULL	Name of the file
FILE_DIRECTORY	VARCHAR2(128)	NOT NULL	Directory object for the directory where the file is stored
FILE_TYPE	VARCHAR2(32)		User-specified file type
FILE_SIZE	NUMBER		Size of the file
FILE_BLOCK_SIZE	NUMBER		Block size for the file
COMMENTS	VARCHAR2(4000)		Comments about the file group

See Also

- "DBA_FILE_GROUP_FILES"
- "USER_FILE_GROUP_FILES"

2.185 ALL_FILE_GROUP_TABLES

ALL_FILE_GROUP_TABLES shows information about the tables accessible to the current user that can be imported using the file set.

Related Views

- DBA_FILE_GROUP_TABLES shows information about all the tables in the database that can be imported using the file set.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_FILE_GROUP_TABLES shows information about tables owned by the current user that can be imported using the file set. This view does not display the FILE_GROUP_OWNER column.

Column	Datatype	NULL	Description
FILE_GROUP_OWNER	VARCHAR2(128)	NOT NULL	Owner of the file group
FILE_GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the file group
VERSION_NAME	VARCHAR2(128)	NOT NULL	Version of the file group that contains the table
VERSION	NUMBER	NOT NULL	Internal version number
OWNER	VARCHAR2(128)	NOT NULL	Schema to which the table belongs
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace to which the table belongs
SCN	NUMBER		SCN at which the table was exported (available only for Streams-prepared tables)

See Also

- "DBA_FILE_GROUP_TABLES"
- "USER_FILE_GROUP_TABLES"

2.186 ALL_FILE_GROUP_TABLESPACES

ALL_FILE_GROUP_TABLESPACES shows information about the transportable tablespaces present (partially or completely) in the file groups accessible to the current user (when the file groups contain dump files).

Related Views

- DBA_FILE_GROUP_TABLESPACES shows information about the transportable tablespaces present (partially or completely) in all file groups in the database (when the file groups contain dump files).
- USER_FILE_GROUP_TABLESPACES shows information about the transportable tablespaces present (partially or completely) in the file groups owned by the current user (when the file groups contain dump files). This view does not display the FILE_GROUP_OWNER column.

Column	Datatype	NULL	Description
FILE_GROUP_OWNER	VARCHAR2(128)	NOT NULL	Owner of the file group
FILE_GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the file group
VERSION_NAME	VARCHAR2(128)	NOT NULL	Version of the file group that contains the tablespace
VERSION	NUMBER	NOT NULL	Internal version number
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of the tablespace

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_FILE_GROUP_TABLESPACES"
- "USER_FILE_GROUP_TABLESPACES"

2.187 ALL_FILE_GROUP_VERSIONS

ALL_FILE_GROUP_VERSIONS shows top-level version information for the file groups accessible to the current user.

Related Views

- DBA_FILE_GROUP_VERSIONS shows top-level version information for all file groups in the database.
- USER_FILE_GROUP_VERSIONS shows top-level version information for all file groups owned by the current user. This view does not display the FILE_GROUP_OWNER column.

Column	Datatype	NULL	Description
FILE_GROUP_OWNER	VARCHAR2(128)	NOT NULL	Owner of the file group
FILE_GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the file group
VERSION_NAME	VARCHAR2(128)	NOT NULL	User-specified name for the version
VERSION	NUMBER	NOT NULL	Internal version number
CREATOR	VARCHAR2(128)	NOT NULL	User who created the version
CREATED	TIMESTAMP(6) WITH TIME ZONE	NOT NULL	Time at which the version was created
COMMENTS	VARCHAR2(4000)		Comments about the file group
DEFAULT_DIRECTORY	VARCHAR2(128)		Default directory object for this version, if specified

See Also

- "DBA_FILE_GROUP_VERSIONS"
- "USER_FILE_GROUP_VERSIONS"

2.188 ALL_FILE_GROUPS

ALL_FILE_GROUPS shows top-level metadata about the file groups accessible to the current user.

Related Views

- DBA_FILE_GROUPS shows top-level metadata about all file groups in the database.
- USER_FILE_GROUPS shows top-level metadata about file groups owned by the current user. This view does not display the FILE_GROUP_OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FILE_GROUP_OWNER	VARCHAR2(128)	NOT NULL	Owner of the file group
FILE_GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the file group
KEEP_FILES	VARCHAR2(1)	NOT NULL	A value of Y or N to indicate whether or not files should be deleted when a version is purged
MIN_VERSIONS	NUMBER	NOT NULL	Autopurge should not drop a version if this condition will become violated
MAX_VERSIONS	NUMBER	NOT NULL	Autopurge will drop the oldest version when this condition is violated
RETENTION_DAYS	NUMBER	NOT NULL	Autopurge will drop versions older than this if doing so does not violate MIN_VERSIONS
CREATED	TIMESTAMP(6) WITH TIME ZONE	NOT NULL	Time at which the file group was created
COMMENTS	VARCHAR2(4000)		Comments about the file group
DEFAULT_DIRECTORY	VARCHAR2(128)		Name of the default directory object

See Also

- "DBA_FILE_GROUPS"
- "USER_FILE_GROUPS"

2.189 ALL_GG_AUTO_CDR_COLUMN_GROUPS

ALL_GG_AUTO_CDR_COLUMN_GROUPS provides details about Oracle GoldenGate automatic conflict detection and resolution (CDR) column groups owned by the current user.

Related View

DBA_GG_AUTO_CDR_COLUMN_GROUPS provides details about all of the Oracle GoldenGate automatic CDR column groups in the database.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Table name
COLUMN_GROUP_NAME	VARCHAR2(128)	NOT NULL	Column group name
RESOLUTION_COLUMN	VARCHAR2(128)	NOT NULL	Timestamp resolution column

See Also

"DBA_GG_AUTO_CDR_COLUMN_GROUPS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.190 ALL_GG_AUTO_CDR_COLUMNS

ALL_GG_AUTO_CDR_COLUMNS provides details about Oracle GoldenGate automatic conflict detection and resolution (CDR) columns owned by the current user.

Related View

DBA_GG_AUTO_CDR_COLUMNS provides details about all of the Oracle GoldenGate automatic CDR columns in the database.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Table name
COLUMN_GROUP_NAME	VARCHAR2(128)		Column group name
COLUMN_NAME	VARCHAR2(128)		Column name
RESOLUTION_COLUMN	VARCHAR2(128)		Timestamp resolution column

See Also

"DBA_GG_AUTO_CDR_COLUMNS"

2.191 ALL_GG_AUTO_CDR_TABLES

ALL_GG_AUTO_CDR_TABLES provides details about tables configured for Oracle GoldenGate automatic conflict detection and resolution (CDR) that are owned by the current user.

Related View

DBA_GG_AUTO_CDR_TABLES provides details about all the tables configured for Oracle GoldenGate automatic conflict detection and resolution (CDR).

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Table name
RESOLUTION_GRANULARITY	VARCHAR2(6)		Resolution granularity: • ROW • COLUMN
FETCHCOLS	VARCHAR2(3)		Extract fetchcols configuration: • Yes: Extract will fetch non-scalar data • No: Extract will not fetch non-scalar data
RECORD_CONFLICTS	VARCHAR2(3)		Monitoring of conflicts: • Yes: Conflict info is recorded • No: Conflict info is not recorded
USE_CUSTOM_HANDLERS	VARCHAR2(4)		Use of customized or automatic conflict handlers: • All: If using custom handlers • None: If using automatic handlers

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TOMBSTONE_TABLE	VARCHAR2(128)		Tombstone table name (if table has delete tombstoning enabled)
ROW_RESOLUTION_COLUMN	VARCHAR2(128)	NOT NULL	Name of row-level timestamp column
EXISTING_DATA_TIMESTAMP	TIMESTAMP(6)		Timestamp to give existing rows when a new timestamp column is added

See Also

"DBA_GG_AUTO_CDR_TABLES"

2.192 ALL_GG_INBOUND_PROGRESS

ALL_GG_INBOUND_PROGRESS displays information about the progress made by the GoldenGate inbound servers accessible to the current user.

Related View

DBA_GG_INBOUND_PROGRESS displays information about the progress made by all GoldenGate inbound servers in the database.

Column	Datatype	NULL	Description
SERVER_NAME	VARCHAR2(128)	NOT NULL	Name of the inbound server
PROCESSED_LOW_POSITION	VARCHAR2(4000)		Position of the processed low transaction
APPLIED_LOW_POSITION	VARCHAR2(4000)		All messages with commit position less than this value have been applied. This column should be used to view the progress of the GoldenGate apply. This column will hold an Oracle SCN numeric value in text format for an Oracle source database. For a non-Oracle source database, this column will hold the apply low position in GoldenGate CSN text format for that specific source database.
APPLIED_HIGH_POSITION	VARCHAR2(4000)		Highest commit position of a transaction that has been applied
SPILL_POSITION	VARCHAR2(4000)		Position of the spill low watermark of the transactions currently being applied
OLDEST_POSITION	VARCHAR2(4000)		Earliest position of the transactions currently being applied
APPLIED_LOW_SCN	NUMBER	NOT NULL	All SCN below or equal to this number have been successfully applied. This column is not applicable for GoldenGate replication since the source database may be non-Oracle.
APPLIED_TIME	DATE		Time at which the APPLIED_MESSAGE_NUMBER message was applied
APPLIED_MESSAGE_CREATE_TIME	DATE		Time at which the APPLIED_MESSAGE_NUMBER message was created
SOURCE_DATABASE	VARCHAR2(128)		Database where the transaction originated

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SOURCE_ROOT_NAME	VARCHAR2(128)		The global name of the source root database where all transactions originated
LOGBSN	VARCHAR2(4000)		Log BSN value from the GoldenGate trail file

See Also

"DBA_GG_INBOUND_PROGRESS"

2.193 ALL_GOLDENGATE_INBOUND

ALL_GOLDENGATE_INBOUND displays information about the GoldenGate inbound servers accessible to the current user.

Related View

DBA_GOLDENGATE_INBOUND displays information about all GoldenGate inbound servers in the database.

Column	Datatype	NULL	Description
REPLICAT_NAME	VARCHAR2(4000)		The name of the replicat group created from GGSCI using GoldenGate
SERVER_NAME	VARCHAR2(128)	NOT NULL	Name of the inbound server
APPLY_USER	VARCHAR2(128)		Name of the user who can connect to the inbound server and apply messages
USER_COMMENT	VARCHAR2(4000)		User comment
CREATE_DATE	TIMESTAMP(6)		Date when inbound server was created
STATUS	VARCHAR2(8)		Status of the inbound server: • DISABLED - The inbound server is not running. • DETACHED - The inbound server is running, but the GoldenGate client application is not attached to it. • ATTACHED - The inbound server is running, and the GoldenGate client application is attached to it. • ABORTED - The inbound server became disabled because it encountered an error.

See Also

"DBA_GOLDENGATE_INBOUND"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.194 ALL_GOLDENGATE_PRIVILEGES

ALL_GOLDENGATE_PRIVILEGES displays details about Oracle GoldenGate privileges for the user. Oracle GoldenGate privileges are granted using the DBMS_GOLDENGATE_AUTH package.

Related Views

- DBA_GOLDENGATE_PRIVILEGES displays details about Oracle GoldenGate privileges for all users who have been granted Oracle GoldenGate privileges.
- USER_GOLDENGATE_PRIVILEGES displays details about Oracle GoldenGate privileges. This view does not display the USERNAME column.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user that is granted the privilege
PRIVILEGE_TYPE	VARCHAR2(7)		Type of privilege granted: • APPLY • CAPTURE • *: Both APPLY and CAPTURE
GRANT_SELECT_PRIVILEGES	VARCHAR2(3)		Shows whether the set of privileges granted to the administrator make the administrator a full privilege administrator or a minimum privilege administrator: • YES: The administrator has the SELECT_CATALOG_ROLE role and other privileges, is considered a full privilege administrator, and can manage any Oracle GoldenGate configuration. • NO: The administrator is considered a minimum privilege administrator, and can only manage Oracle GoldenGate configurations where the apply_user or capture_user (based on the PRIVILEGE_TYPE column) matches the username.
CREATE_TIME	TIMESTAMP(6)		Time at which the privilege was granted

See Also

- "DBA_GOLDENGATE_PRIVILEGES"
- "USER_GOLDENGATE_PRIVILEGES"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_GOLDENGATE_AUTH package

2.195 ALL_GOLDENGATE_RULES

ALL_GOLDENGATE_RULES displays information about the GoldenGate rules accessible to the current user.

Related View

DBA_GOLDENGATE_RULES displays information about all GoldenGate server rules in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COMPONENT_NAME	VARCHAR2(128)		Name of the GoldenGate process
COMPONENT_TYPE	VARCHAR2(12)		Type of the GoldenGate process: • CAPTURE • APPLY
COMPONENT_RULE_TYPE	VARCHAR2(9)		For global, schema or table rules, the GoldenGate type of the rule: • TABLE • SCHEMA • GLOBAL
RULE_SET_OWNER	VARCHAR2(128)		Owner of the rule set
RULE_SET_NAME	VARCHAR2(128)		Name of the rule set
RULE_SET_TYPE	CHAR(8)		Type of the rule set: • POSITIVE • NEGATIVE
RULE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule
RULE_NAME	VARCHAR2(128)	NOT NULL	Name of the rule
RULE_TYPE	VARCHAR2(9)		For global, schema or table rules, the type of the rule: • DML • DDL
RULE_CONDITION	VARCHAR2(1)		This column is obsolete
SCHEMA_NAME	VARCHAR2(128)		For table and schema rules, the schema name
OBJECT_NAME	VARCHAR2(128)		For table rules, the table name
INCLUDE_TAGGED_LCR	VARCHAR2(3)		For global, schema or table rules, indicates whether to include tagged LCRs (YES) or not (NO)
SUBSETTING_OPERATION	VARCHAR2(6)		For subset rules, the type of operation: • INSERT • UPDATE • DELETE
DML_CONDITION	VARCHAR2(4000)		For subset rules, the row subsetting condition
SOURCE_DATABASE	VARCHAR2(128)		For global, schema or table rules, the name of the database where the LCRs originated
ORIGINAL_RULE_CONDITION	VARCHAR2(4000)		For rules created by GoldenGate administrative APIs, the original rule condition when the rule was created
SAME_RULE_CONDITION	VARCHAR2(3)		For rules created by GoldenGate administrative APIs, indicates whether the current rule condition is the same as the original rule condition (YES) or not (NO)
SOURCE_ROOT_NAME	VARCHAR2(128)		The global name of the source root database where the transactions originated
SOURCE_CONTAINER_NAME	VARCHAR2(128)		The container name of the database where the transactions originated

See Also

"DBA_GOLDENGATE_RULES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.196 ALL_HEAT_MAP_SEG_HISTOGRAM

ALL_HEAT_MAP_SEG_HISTOGRAM displays segment access information for all segments visible to the user.

Related Views

- DBA_HEAT_MAP_SEG_HISTOGRAM displays segment access information for all segments.
- USER_HEAT_MAP_SEG_HISTOGRAM displays segment access information for segments owned by the user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Table owner
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object
SUBOBJECT_NAME	VARCHAR2(128)		Name of the sub-object
TRACK_TIME	DATE		System time when the segment access was tracked
SEGMENT_WRITE	VARCHAR2(3)		Indicates whether the segment has write access (YES or NO)
FULL_SCAN	VARCHAR2(3)		Indicates whether the segment has full table scan (YES or NO)
LOOKUP_SCAN	VARCHAR2(3)		Indicates whether the segment has lookup scan (YES or NO)

See Also

- "DBA_HEAT_MAP_SEG_HISTOGRAM"
- "USER_HEAT_MAP_SEG_HISTOGRAM"

2.197 ALL_HEAT_MAP_SEGMENT

ALL_HEAT_MAP_SEGMENT displays the latest segment access time for all segments visible to the user. The timestamps in the view are coarse with a granularity of a day reflecting the flush times of the heat map.

Related Views

- DBA_HEAT_MAP_SEGMENT displays the latest segment access time for all segments.
- USER_HEAT_MAP_SEGMENT displays the latest segment access time for all segments owned by the user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Table owner
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object
SUBOBJECT_NAME	VARCHAR2(128)		Name of the sub-object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SEGMENT_WRITE_TIME	DATE		Latest timestamp on which the segment has write access
SEGMENT_READ_TIME	DATE		Latest timestamp on which the segment has read access
FULL_SCAN	DATE		Latest timestamp on which the segment has full table scan
LOOKUP_SCAN	DATE		Latest timestamp on which the segment has index scan

See Also

- "DBA_HEAT_MAP_SEGMENT"
- "USER_HEAT_MAP_SEGMENT"

2.198 ALL_HIER_CLASS

ALL_HIER_CLASS describes the classifications of the hierarchies accessible to the current user.

Related Views

- DBA_HIER_CLASS describes the classifications of all hierarchies in the database.
- USER_HIER_CLASS describes the classifications of the hierarchies owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
CLASSIFICATION	VARCHAR2(128)		Classification associated with the hierarchy
VALUE	CLOB		Value of the classification or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the hierarchy
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_HIER_CLASS"
- "USER_HIER_CLASS"

2.199 ALL_HIER_CLASS_AE

ALL_HIER_CLASS_AE describes the classifications of the hierarchies (across all editions) accessible to the current user.

Related Views

- DBA_HIER_CLASS_AE describes the classifications of all hierarchies (across all editions) in the database.
- USER_HIER_CLASS_AE describes the classifications of the hierarchies (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
CLASSIFICATION	VARCHAR2(128)		Classification associated with the hierarchy
VALUE	CLOB		Value of the classification or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the hierarchy
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the hierarchy is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_HIER_CLASS_AE"
- "USER_HIER_CLASS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.200 ALL_HIER_COLUMNS

ALL_HIER_COLUMNS describes the columns of the hierarchies accessible to the current user.

Related Views

- DBA_HIER_COLUMNS describes the columns of all hierarchies in the database.
- USER_HIER_COLUMNS describes the columns of the hierarchies owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column
ROLE	VARCHAR2(4)		The role the attribute plays in the hierarchy: • KEY • AKEY • HIER • PROP
DATA_TYPE	VARCHAR2(106)		Datatype of the column
DATA_LENGTH	NUMBER	NOT NULL	Length of the column (in bytes)
DATA_PRECISION	NUMBER		Decimal precision for the NUMBER datatype; binary precision for the FLOAT datatype, NULL for all other datatypes
DATA_SCALE	NUMBER		Number of digits to the right of the decimal point in a number
NULLABLE	CHAR(1)		Indicates whether a column allows NULL values; the value is N if there is a NOT NULL constraint on the column or if the column is part of a PRIMARY KEY
CHARACTER_SET_NAME	VARCHAR2(44)		Name of the character set: • CHAR_CS • NCHAR_CS
CHAR_COL_DECL_LENGTH	NUMBER		Declaration length of the character type column
CHAR_USED	VARCHAR2(1)		Indicates that the column uses BYTE length semantics (B) or CHAR length semantics (C), or whether the datatype is not any of the following (NULL): • CHAR • VARCHAR2 • NCHAR • NVARCHAR2
ORDER_NUM	NUMBER	NOT NULL	Order of the column, with attributes first in the order specified in the definition of the hierarchy followed by hierarchical attributes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_HIER_COLUMNS"
- "USER_HIER_COLUMNS"

2.201 ALL_HIER_COLUMNS_AE

ALL_HIER_COLUMNS_AE describes the columns of the hierarchies (across all editions) accessible to the current user.

Related Views

- DBA_HIER_COLUMNS_AE describes the columns of all hierarchies (across all editions) in the database.
- USER_HIER_COLUMNS_AE describes the columns of the hierarchies (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column
ROLE	VARCHAR2(4)		The role the attribute plays in the hierarchy: • KEY • AKEY • HIER • PROP
DATA_TYPE	VARCHAR2(106)		Datatype of the column
DATA_LENGTH	NUMBER	NOT NULL	Length of the column (in bytes)
DATA_PRECISION	NUMBER		Decimal precision for the NUMBER datatype; binary precision for the FLOAT datatype, NULL for all other datatypes
DATA_SCALE	NUMBER		Number of digits to the right of the decimal point in a number
NULLABLE	CHAR(1)		Indicates whether a column allows NULL values; the value is N if there is a NOT NULL constraint on the column or if the column is part of a PRIMARY KEY

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CHARACTER_SET_NAME	VARCHAR2(44)		Name of the character set: • CHAR_CS • NCHAR_CS
CHAR_COL_DECL_LENGTH	NUMBER		Declaration length of the character type column
CHAR_USED	VARCHAR2(1)		Indicates that the column uses BYTE length semantics (B) or CHAR length semantics (C), or whether the datatype is not any of the following (NULL): • CHAR • VARCHAR2 • NCHAR • NVARCHAR2
ORDER_NUM	NUMBER	NOT NULL	Order of the column, with attributes first in the order specified in the definition of the hierarchy followed by hierarchical attributes
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the hierarchy is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_HIER_COLUMNS_AE"
- "USER_HIER_COLUMNS_AE"

2.202 ALL_HIER_HIER_ATTR_CLASS

ALL_HIER_HIER_ATTR_CLASS describes the classifications of the hierarchical attributes of the hierarchies accessible to the current user.

Related Views

- DBA_HIER_HIER_ATTR_CLASS describes the classifications of the hierarchical attributes of all hierarchies in the database.
- USER_HIER_HIER_ATTR_CLASS describes the classifications of the hierarchical attributes of the hierarchies owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
HIER_ATTR_NAME	VARCHAR2(128)		Name of the hierarchical attribute
CLASSIFICATION	VARCHAR2(128)		Classification associated with the hierarchical attribute
VALUE	CLOB		Value associated with the classification or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification or NULL if not specified
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the hierarchical attribute
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_HIER_HIER_ATTR_CLASS"
- "USER_HIER_HIER_ATTR_CLASS"

2.203 ALL_HIER_HIER_ATTR_CLASS_AE

ALL_HIER_HIER_ATTR_CLASS_AE describes the classifications of the hierarchical attributes of the hierarchies (across all editions) accessible to the current user.

Related Views

- DBA_HIER_HIER_ATTR_CLASS_AE describes the classifications of the hierarchical attributes of all hierarchies (across all editions) in the database.
- USER_HIER_HIER_ATTR_CLASS_AE describes the classifications of the hierarchical attributes of the hierarchies (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
HIER_ATTR_NAME	VARCHAR2(128)		Name of the hierarchical attribute
CLASSIFICATION	VARCHAR2(128)		Classification associated with the hierarchical attribute
VALUE	CLOB		Value associated with the classification or NULL if not specified
LANGUAGE	VARCHAR2(64)		NLS_LANGUAGE value associated with the classification or NULL if not specified

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of classifications associated with the hierarchical attribute
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the hierarchy is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_HIER_HIER_ATTR_CLASS_AE"
- "USER_HIER_HIER_ATTR_CLASS_AE"

2.204 ALL_HIER_HIER_ATTRIBUTES

ALL_HIER_HIER_ATTRIBUTES describes the hierarchical attributes of the hierarchies accessible to the current user.

Related Views

- DBA_HIER_HIER_ATTRIBUTES describes the hierarchical attributes of all hierarchies in the database.
- USER_HIER_HIER_ATTRIBUTES describes the hierarchical attributes of the hierarchies owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
HIER_ATTR_NAME	VARCHAR2(128)		Name of the hierarchical attribute
EXPRESSION	CLOB		The expression defining the hierarchical attribute value
ORDER_NUM	NUMBER	NOT NULL	Order of the hierarchical attribute in the list of hierarchical attributes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_HIER_HIER_ATTRIBUTES"
- "USER_HIER_HIER_ATTRIBUTES"

2.205 ALL_HIER_HIER_ATTRIBUTES_AE

ALL_HIER_HIER_ATTRIBUTES_AE describes the hierarchical attributes of the hierarchies (across all editions) accessible to the current user.

Related Views

- DBA_HIER_HIER_ATTRIBUTES_AE describes the hierarchical attributes of all hierarchies (across all editions) in the database.
- USER_HIER_HIER_ATTRIBUTES_AE describes the hierarchical attributes of the hierarchies (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
HIER_ATTR_NAME	VARCHAR2(128)		Name of the hierarchical attribute
EXPRESSION	CLOB		The expression defining the hierarchical attribute value
ORDER_NUM	NUMBER	NOT NULL	Order of the hierarchical attribute in the list of hierarchical attributes
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the hierarchy is defined

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> - "DBA_HIER_HIER_ATTRIBUTES_AE"
> - "USER_HIER_HIER_ATTRIBUTES_AE"

2.206 ALL_HIER_JOIN_PATHS

`ALL_HIER_JOIN_PATHS` describes the join paths for the hierarchies accessible to the current user.

Related Views

- `DBA_HIER_JOIN_PATHS` describes the join paths for all hierarchies in the database.
- `USER_HIER_JOIN_PATHS` describes the join paths for the hierarchies owned by the current user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the hierarchy
`HIER_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the hierarchy
`JOIN_PATH_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the join path
`ORDER_NUM`	`NUMBER`	`NOT NULL`	Order of the classification in the list of join paths associated with the hierarchy
`ORIGIN_CON_ID`	`NUMBER`		The ID of the container where the data originates. Possible values include: • `0`: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

> **See Also**
>
> - "DBA_HIER_JOIN_PATHS"
> - "USER_HIER_JOIN_PATHS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.207 ALL_HIER_JOIN_PATHS_AE

ALL_HIER_JOIN_PATHS_AE describes the join paths for the hierarchies (across all editions) accessible to the current user.

Related Views

- DBA_HIER_JOIN_PATHS_AE describes the join paths for all hierarchies (across all editions) in the database.
- USER_HIER_JOIN_PATHS_AE describes the join paths for the hierarchies (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
JOIN_PATH_NAME	VARCHAR2(128)	NOT NULL	Name of the join path
ORDER_NUM	NUMBER	NOT NULL	Order of the classification in the list of join paths associated with the hierarchy
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the hierarchy is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_HIER_JOIN_PATHS_AE"
- "USER_HIER_JOIN_PATHS_AE"

2.208 ALL_HIER_LEVEL_ID_ATTRS

ALL_HIER_LEVEL_ID_ATTRS describes the attributes that uniquely identify members of the levels of the hierarchies accessible to the current user.

Related Views

- DBA_HIER_LEVEL_ID_ATTRS describes the attributes that uniquely identify members of the levels of all hierarchies in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_HIER_LEVEL_ID_ATTRS describes the attributes that uniquely identify members of the levels of the hierarchies owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
LEVEL_NAME	VARCHAR2(128)	NOT NULL	Name of hierarchy level
ATTRIBUTE_NAME	VARCHAR2(128)	NOT NULL	Name of the unique identifier attribute
ORDER_NUM	NUMBER	NOT NULL	Order of the level in the list of unique identifier attributes for the level
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_HIER_LEVEL_ID_ATTRS"
- "USER_HIER_LEVEL_ID_ATTRS"

2.209 ALL_HIER_LEVEL_ID_ATTRS_AE

ALL_HIER_LEVEL_ID_ATTRS_AE describes the attributes that uniquely identify members of the levels of the hierarchies (across all editions) accessible to the current user.

Related Views

- DBA_HIER_LEVEL_ID_ATTRS_AE describes the attributes that uniquely identify members of the levels of all hierarchies (across all editions) in the database.
- USER_HIER_LEVEL_ID_ATTRS_AE describes the attributes that uniquely identify members of the levels of the hierarchies (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
LEVEL_NAME	VARCHAR2(128)	NOT NULL	Name of hierarchy level
ATTRIBUTE_NAME	VARCHAR2(128)	NOT NULL	Name of the unique identifier attribute
ORDER_NUM	NUMBER	NOT NULL	Order of the level in the list of unique identifier attributes for the level

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the hierarchy is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_HIER_LEVEL_ID_ATTRS_AE"
- "USER_HIER_LEVEL_ID_ATTRS_AE"

2.210 ALL_HIER_LEVELS

ALL_HIER_LEVELS describes the levels of the hierarchies accessible to the current user.

Related Views

- DBA_HIER_LEVELS describes the levels of all hierarchies in the database.
- USER_HIER_LEVELS describes the levels of the hierarchies owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
LEVEL_NAME	VARCHAR2(128)	NOT NULL	Name of hierarchy level
ORDER_NUM	NUMBER	NOT NULL	Order of the level in the list of hierarchy levels
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> - "DBA_HIER_LEVELS"
> - "USER_HIER_LEVELS"

2.211 ALL_HIER_LEVELS_AE

`ALL_HIER_LEVELS_AE` describes the levels of the hierarchies (across all editions) accessible to the current user.

Related Views

- `DBA_HIER_LEVELS_AE` describes the levels of all hierarchies (across all editions) in the database.
- `USER_HIER_LEVELS_AE` describes the levels of the hierarchies (across all editions) owned by the current user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the hierarchy
`HIER_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the hierarchy
`LEVEL_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of hierarchy level
`ORDER_NUM`	`NUMBER`	`NOT NULL`	Order of the level in the list of hierarchy levels
`ORIGIN_CON_ID`	`NUMBER`		The ID of the container where the data originates. Possible values include: • `0`: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
`EDITION_NAME`	`VARCHAR2(128)`		Name of the application edition where the hierarchy is defined

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> - "DBA_HIER_LEVELS_AE"
> - "USER_HIER_LEVELS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.212 ALL_HIERARCHIES

ALL_HIERARCHIES describes the hierarchies accessible to the current user.

Related Views

- DBA_HIERARCHIES describes all hierarchies in the database.
- USER_HIERARCHIES describes the hierarchies owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy
DIMENSION_OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension used by the hierarchy
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension used by the hierarchy
PARENT_ATTR	VARCHAR2		The value of this column is always NULL
COMPILE_STATE	VARCHAR2(7)		Compile status of the hierarchy: • VALID • INVALID
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).

See Also

- "DBA_HIERARCHIES"
- "USER_HIERARCHIES"

2.213 ALL_HIERARCHIES_AE

ALL_HIERARCHIES_AE describes the hierarchies (across all editions) accessible to the current user.

Related Views

- DBA_HIERARCHIES_AE describes all hierarchies (across all editions) in the database.
- USER_HIERARCHIES_AE describes the hierarchies (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the hierarchy
HIER_NAME	VARCHAR2(128)	NOT NULL	Name of the hierarchy

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DIMENSION_OWNER	VARCHAR2(128)	NOT NULL	Owner of the attribute dimension used by the hierarchy
DIMENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute dimension used by the hierarchy
PARENT_ATTR	VARCHAR2		The value of this column is always NULL
COMPILE_STATE	VARCHAR2(7)		Compile status of the hierarchy: • VALID • INVALID
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root).
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the hierarchy is defined

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

- "DBA_HIERARCHIES_AE"
- "USER_HIERARCHIES_AE"

2.214 ALL_HISTOGRAMS

ALL_HISTOGRAMS is a synonym for ALL_TAB_HISTOGRAMS.

See Also

"ALL_TAB_HISTOGRAMS"

2.215 ALL_HIVE_COLUMNS

ALL_HIVE_COLUMNS describes all Hive columns accessible to the current user in a Hive metastore.

Related Views

- DBA_HIVE_COLUMNS describes all Hive columns in a Hive metastore.
- USER_HIVE_COLUMNS describes all Hive columns owned by the current user in a Hive metastore.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CLUSTER_ID	VARCHAR2(4000)		Identifier for the Hadoop cluster
DATABASE_NAME	VARCHAR2(4000)		Hive database where the owning Hive table resides
TABLE_NAME	VARCHAR2(4000)		Hive table name that the column belongs to
COLUMN_NAME	VARCHAR2(4000)		Hive column name
HIVE_COLUMN_TYPE	VARCHAR2(4000)		Data type of the Hive column
ORACLE_COLUMN_TYPE	VARCHAR2(4000)		Equivalent Oracle data type of the Hive column
LOCATION	VARCHAR2(4000)		Physical location of the Hive table
OWNER	VARCHAR2(4000)		Owner of the Hive table
CREATION_TIME	DATE		Time that the Hive column was created
HIVE_URI	VARCHAR2(4000)		The connection string (URI and port number) for the metastore database

See Also

- "DBA_HIVE_COLUMNS"
- "USER_HIVE_COLUMNS"

2.216 ALL_HIVE_DATABASES

ALL_HIVE_DATABASES describes all the Hive schemas accessible to the current user in a Hadoop cluster.

Related Views

- DBA_HIVE_DATABASES describes all the Hive schemas in a Hadoop cluster.
- USER_HIVE_DATABASES describes all the Hive schemas owned by the current user in a Hadoop cluster.

Column	Datatype	NULL	Description
CLUSTER_ID	VARCHAR2(4000)		Hadoop cluster name
DATABASE_NAME	VARCHAR2(4000)		Name of the Hive database
DESCRIPTION	VARCHAR2(4000)		Description of the Hive database
DB_LOCATION	VARCHAR2(4000)		Physical location of the Hive database
HIVE_URI	VARCHAR2(4000)		The connection string (URI and port number) for the metastore database

See Also

- "DBA_HIVE_DATABASES"
- "USER_HIVE_DATABASES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.217 ALL_HIVE_PART_KEY_COLUMNS

ALL_HIVE_PART_KEY_COLUMNS provides information about all Hive table partition columns accessible to the current user in the database.

Related Views

- DBA_HIVE_PART_KEY_COLUMNS provides information about all Hive table partition columns in the database.
- USER_HIVE_PART_KEY_COLUMNS provides information about all Hive table partition columns owned by the current user in the database.

Column	Datatype	NULL	Description
CLUSTER_ID	VARCHAR2(4000)		Hadoop cluster name
DATABASE_NAME	VARCHAR2(4000)		Hive database where the Hive table resides
TABLE_NAME	VARCHAR2(4000)		Hive table name
OWNER	VARCHAR2(4000)		Owner of the Hive table
COLUMN_NAME	VARCHAR2(4000)		Partition column name
COLUMN_TYPE	VARCHAR2(4000)		Partition column type
COLUMN_POSITION	NUMBER		Partition column position in the Hive partition specification
ORACLE_COLUMN_TYPE	VARCHAR2(4000)		Equivalent Oracle data type of the Hive column

See Also

- "DBA_HIVE_PART_KEY_COLUMNS"
- "USER_HIVE_PART_KEY_COLUMNS"

2.218 ALL_HIVE_TAB_PARTITIONS

ALL_HIVE_TAB_PARTITIONS provides information about all Hive table partitions accessible to the current user in the database.

Related Views

- DBA_HIVE_TAB_PARTITIONS provides information about all Hive table partitions in the database.
- USER_HIVE_TAB_PARTITIONS provides information about all Hive table partitions owned by the current user in the database.

Column	Datatype	NULL	Description
CLUSTER_ID	VARCHAR2(4000)		Hadoop cluster name
DATABASE_NAME	VARCHAR2(4000)		Hive database where the Hive table resides
TABLE_NAME	VARCHAR2(4000)		Hive table name
LOCATION	VARCHAR2(4000)		Physical location of the Hive partition

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(4000)		Owner of the Hive table
PARTITION_SPECS	VARCHAR2(4000)		The current Hive partition specification
PART_SIZE	NUMBER		Partition size in bytes
CREATION_TIME	DATE		Time that the partition was created

See Also

- "DBA_HIVE_TAB_PARTITIONS"
- "USER_HIVE_TAB_PARTITIONS"

2.219 ALL_HIVE_TABLES

ALL_HIVE_TABLES provides information about all the Hive tables accessible to the current user in the Hive metastore.

Related Views

- DBA_HIVE_TABLES provides information about all Hive tables in the Hive metastore.
- USER_HIVE_TABLES provides information about all Hive tables owned by the current user in the Hive metastore.

Column	Datatype	NULL	Description
CLUSTER_ID	VARCHAR2(4000)		Hadoop cluster name
DATABASE_NAME	VARCHAR2(4000)		Hive database where the Hive table resides
TABLE_NAME	VARCHAR2(4000)		Hive table name
LOCATION	VARCHAR2(4000)		Physical location of the Hive table
NO_OF_COLS	NUMBER		Number of columns in the Hive table
CREATION_TIME	DATE		Creation time of the Hive table
LAST_ACCESSED_TIME	DATE		Time that the Hive table was last accessed
OWNER	VARCHAR2(4000)		Owner of the Hive table
TABLE_TYPE	VARCHAR2(4000)		Type of the Hive table
PARTITIONED	VARCHAR2(4000)		Is this Hive table partitioned?
NO_OF_PART_KEYS	NUMBER		Number of partition keys in the Hive table
INPUT_FORMAT	VARCHAR2(4000)		Hive table input format
OUTPUT_FORMAT	VARCHAR2(4000)		Hive table output format
SERIALIZATION	VARCHAR2(4000)		Hive table serialization
COMPRESSED	NUMBER		Is this Hive table compressed?
HIVE_URI	VARCHAR2(4000)		The connection string (URI and port number) for the metastore database

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_HIVE_TABLES"
- "USER_HIVE_TABLES"

2.220 ALL_IDENTIFIERS

ALL_IDENTIFIERS displays information about the identifiers in the stored objects accessible to the current user.

Related Views

- DBA_IDENTIFIERS displays information about the identifiers in all stored objects in the database.
- USER_IDENTIFIERS displays information about the identifiers in the stored objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the identifier
NAME	VARCHAR2(128)		Name of the identifier
SIGNATURE	VARCHAR2(32)		Signature of the identifier
TYPE	VARCHAR2(18)		Type of the identifier. For SQL identifiers, the types include: • TABLE • VIEW • SEQUENCE • ALIAS • COLUMN • MATERIALIZED VIEW • OPERATOR For PL/SQL identifiers, the types include: • FUNCTION • PROCEDURE • PACKAGE
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object where the identifier action occurred
OBJECT_TYPE	VARCHAR2(13)		Type of the object where the identifier action occurred
USAGE	VARCHAR2(11)		Type of the identifier usage: • DECLARATION • DEFINITION • CALL • REFERENCE • ASSIGNMENT
USAGE_ID	NUMBER		Unique key for the identifier usage within the object
LINE	NUMBER		Line number of the identifier action
COL	NUMBER		Column number of the identifier action
USAGE_CONTEXT_ID	NUMBER		Context USAGE_ID of the identifier usage

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CHARACTER_SET	VARCHAR2(10)		Contains the value of the `character set` clause when it is used in a variable identifier declaration. These are the possible values when the character set is derived from another variable identifier: • CHAR_CS • NCHAR_CS • IDENTIFIER
ATTRIBUTE	VARCHAR2(7)		Column contains the attribute value when `%attribute` is used in a variable declaration. Possible values: • ROWTYPE • TYPE • CHARSET
CHAR_USED	VARCHAR2(4)		Contains the type of the length constraint when it is used in a string length constraint declaration. Possible values: • CHAR • BYTE
LENGTH	NUMBER		Contains the numeric length constraint value for a string length constraint declaration
PRECISION	NUMBER		Contains the numeric precision when it is used in a variable declaration
PRECISION2	NUMBER		Contains the numeric second precision value (for instance, interval types) used in a variable declaration
SCALE	NUMBER		Contains the numeric scale value used in a variable declaration.
LOWER_RANGE	NUMBER		Contains the numeric lower range value used by a variable declaration with a range constraint
UPPER_RANGE	NUMBER		Contains the numeric upper range value used by a variable declaration with a range constraint
NULL_CONSTRAINT	VARCHAR2(8)		This column is set when a `NULL` constraint is used by a variable declaration. Possible values: • NULL • NOT NULL
SQL_BUILTIN	VARCHAR2(3)		Is set to `YES` when an identifier is a SQL builtin used in a SQL statement issued from PL/SQL. Otherwise, this column is set to `NO`.
IMPLICIT[1]	VARCHAR2(3)		Indicates whether the identifier is an implicit identifier that does not appear in the source (`YES`) or not (`NO`)
DECLARED_OWNER[1]	VARCHAR2(128)	NOT NULL	Owner of the object in which this identifier was declared
DECLARED_OBJECT_NAME[1]	VARCHAR2(128)	NOT NULL	Name of the object in which this identifier was declared
DECLARED_OBJECT_TYPE[1]	VARCHAR2(12)		Type of the object in which this identifier was declared
ORIGIN_CON_ID	VARCHAR2(256)		The ID of the container where the data originates. Possible values include: • `0`: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)

Copyright © 2002, 2026, Oracle and/or its affiliates.

[1] This column is available starting with Oracle Database 19c.

See Also

- "DBA_IDENTIFIERS"
- "USER_IDENTIFIERS"

2.221 ALL_IMMUTABLE_TABLES

ALL_IMMUTABLE_TABLES describes the immutable tables accessible to the current user.

Related Views

- DBA_IMMUTABLE_TABLES describes all immutable tables in the database.
- USER_IMMUTABLE_TABLES describes the immutable tables owned by the current user. This view does not display the SCHEMA_NAME column.

Column	Datatype	NULL	Description
SCHEMA_NAME	VARCHAR2(128)	NOT NULL	The schema containing the immutable table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the immutable table
ROW_RETENTION	NUMBER		Row retention period for the immutable table, that is, the minimum number of days a row must be retained and cannot be deleted after it is inserted into the table If the value of this column is NULL, then rows can never be deleted from the table.
ROW_RETENTION_LOCKED	VARCHAR2(3)		Indicates whether the row retention period for the immutable table is locked. Possible values: • YES: The row retention period is locked. You cannot change the row retention period • NO: The row retention period is not locked. You can change the row retention period to a value higher than the current value with the SQL statement ALTER TABLE … NO DELETE UNTIL *n* DAYS AFTER INSERT
TABLE_INACTIVITY_RETENTION	NUMBER		Number of days for which the immutable table must be inactive before it can be dropped, that is, the number of days that must pass after the most recent row insertion before the table can be dropped A table with no rows can be dropped at any time, regardless of this column value.

Note

This view is available starting with Oracle Database 19c, Release Update 19.11.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_IMMUTABLE_TABLES"
- "USER_IMMUTABLE_TABLES"

2.222 ALL_IND_COLUMNS

ALL_IND_COLUMNS describes the columns of indexes on all tables accessible to the current user.

Related Views

- DBA_IND_COLUMNS describes the columns of indexes on all tables in the database.
- USER_IND_COLUMNS describes the columns of indexes owned by the current user and columns of indexes on tables owned by the current user. This view does not display the INDEX_OWNER or TABLE_OWNER columns.

Note

For join indexes, the TABLE_NAME and TABLE_OWNER columns in this view may not match the TABLE_NAME and TABLE_OWNER columns you find in the *_INDEXES (and other similar) data dictionary views.

Column	Datatype	NULL	Description
INDEX_OWNER	VARCHAR2(128)	NOT NULL	Owner of the index
INDEX_NAME	VARCHAR2(128)	NOT NULL	Name of the index
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the table or cluster
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table or cluster
COLUMN_NAME	VARCHAR2(4000)		Column name or attribute of the object type column **Note:** If you create an index on a user-defined REF column, the system creates the index on the attributes that make up the REF column. Therefore, the column names displayed in this view are the attribute names, with the REF column name as a prefix, in the following form: "REF_name"."attribute"
COLUMN_POSITION	NUMBER	NOT NULL	Position of the column or attribute within the index
COLUMN_LENGTH	NUMBER	NOT NULL	Indexed length of the column
CHAR_LENGTH	NUMBER		Maximum codepoint length of the column
DESCEND	VARCHAR2(4)		Indicates whether the column is sorted in descending order (DESC) or ascending order (ASC)
COLLATED_COLUMN_ID	NUMBER		Internal sequence number of the column for which this column provides linguistic ordering

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_IND_COLUMNS"
- "USER_IND_COLUMNS"

2.223 ALL_IND_EXPRESSIONS

ALL_IND_EXPRESSIONS describes the expressions of function-based indexes on tables accessible to the current user.

Related Views

- DBA_IND_EXPRESSIONS describes the expressions of all function-based indexes in the database.
- USER_IND_EXPRESSIONS describes the expressions of function-based indexes on tables owned by the current user. This view does not display the INDEX_OWNER or TABLE_OWNER columns.

Column	Datatype	NULL	Description
INDEX_OWNER	VARCHAR2(128)	NOT NULL	Owner of the index
INDEX_NAME	VARCHAR2(128)	NOT NULL	Name of the index
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the table or cluster
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table or cluster
COLUMN_EXPRESSION	LONG		Function-based index expression defining the column
COLUMN_POSITION	NUMBER	NOT NULL	Position of the column or attribute within the index

See Also

- "DBA_IND_EXPRESSIONS"
- "USER_IND_EXPRESSIONS"

2.224 ALL_IND_PARTITIONS

ALL_IND_PARTITIONS displays, for each index partition accessible to the current user, the partition-level partitioning information, the storage parameters for the partition, and various partition statistics generated by the DBMS_STATS package.

Related Views

- DBA_IND_PARTITIONS describes all index partitions in the database.
- USER_IND_PARTITIONS describes the index partitions owned by the current user. This view does not display the INDEX_OWNER column.

Column	Datatype	NULL	Description
INDEX_OWNER	VARCHAR2(128)		Owner of the index

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INDEX_NAME	VARCHAR2(128)		Name of the index
COMPOSITE	VARCHAR2(3)		Indicates whether the partition belongs to a local index on a composite-partitioned table (YES) or not (NO)
PARTITION_NAME	VARCHAR2(128)		Name of the partition
SUBPARTITION_COUNT	NUMBER		If a local index on a composite-partitioned table, the number of subpartitions in the partition
HIGH_VALUE	LONG		Partition bound value expression
HIGH_VALUE_LENGTH	NUMBER		Length of the partition bound value expression
PARTITION_POSITION	NUMBER		Position of the partition within the index
STATUS	VARCHAR2(8)		Indicates whether the index partition is usable (USABLE) or not (UNUSABLE)
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the partition
PCT_FREE	NUMBER		Minimum percentage of free space in a block
INI_TRANS	NUMBER		Initial number of transactions
MAX_TRANS	NUMBER		Maximum number of transactions
INITIAL_EXTENT	NUMBER		Size of the initial extent in bytes
NEXT_EXTENT	NUMBER		Size of secondary extents in bytes
MIN_EXTENT	NUMBER		Minimum number of extents allowed in the segment
MAX_EXTENT	NUMBER		Maximum number of extents allowed in the segment
MAX_SIZE	NUMBER		Maximum number of blocks allowed in the segment
PCT_INCREASE	NUMBER		Percentage increase in extent size
FREELISTS	NUMBER		Number of process freelists allocated in this segment
FREELIST_GROUPS	NUMBER		Number of process freelist groups allocated in this segment
LOGGING	VARCHAR2(7)		Indicates whether or not changes to the index are logged: • NONE - Not specified **See Also:** the *_IND_SUBPARTITIONS view • YES • NO
COMPRESSION	VARCHAR2(13)		Type of compression being used for the partition: • ENABLED - Prefix compression • ADVANCED HIGH - Advanced high compression • ADVANCED LOW - Advanced low compression • DISABLED - No compression is present
BLEVEL	NUMBER		B*-Tree level (depth of the index from its root block to its leaf blocks). A depth of 0 indicates that the root block and leaf block are the same.
LEAF_BLOCKS	NUMBER		Number of leaf blocks in the index partition
DISTINCT_KEYS	NUMBER		Number of distinct keys in the index partition
AVG_LEAF_BLOCKS_PER_KEY	NUMBER		Average number of leaf blocks in which each distinct value in the index appears, rounded to the nearest integer. For indexes that enforce UNIQUE and PRIMARY KEY constraints, this value is always 1.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
AVG_DATA_BLOCKS_PER_KEY	NUMBER		Average number of data blocks in the table that are pointed to by a distinct value in the index rounded to the nearest integer. This statistic is the average number of data blocks that contain rows that contain a given value for the indexed columns.
CLUSTERING_FACTOR	NUMBER		Indicates the amount of order of the rows in the table based on the values of the index. • If the value is near the number of blocks, then the table is very well ordered. In this case, the index entries in a single leaf block tend to point to rows in the same data blocks. • If the value is near the number of rows, then the table is very randomly ordered. In this case, it is unlikely that index entries in the same leaf block point to rows in the same data blocks.
NUM_ROWS	NUMBER		Number of rows returned
SAMPLE_SIZE	NUMBER		Sample size used in analyzing this partition
LAST_ANALYZED	DATE		Date on which this partition was most recently analyzed
BUFFER_POOL	VARCHAR2(7)		Actual buffer pool for the partition: • DEFAULT • KEEP • RECYCLE • NULL
FLASH_CACHE	VARCHAR2(7)		Database Smart Flash Cache hint to be used for partition blocks: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)		Cell flash cache hint to be used for partition blocks: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
PCT_DIRECT_ACCESS	NUMBER		If a secondary index on index-organized table, the percentage of rows with VALID guess
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics have been gathered or NO if statistics have been aggregated from subpartitions or have not been gathered
DOMIDX_OPSTATUS	VARCHAR2(6)		Status of the operation on a domain index: • NULL - Index is not a domain index • VALID - Operation performed without errors • FAILED - Operation failed with an error
PARAMETERS	VARCHAR2(1000)		For a domain index, the parameter string

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INTERVAL	VARCHAR2(3)		Indicates whether the partition is in the interval section of an interval partitioned table (YES) or whether the partition is in the range section (NO)
SEGMENT_CREATED	VARCHAR2(3)		Indicates whether the index partition segment has been created (YES) or not (NO); N/A indicates that this index is subpartitioned and no segment exists at the partition level
ORPHANED_ENTRIES	VARCHAR2(3)		Indicates whether a global index partition contains stale entries because of deferred index maintenance during DROP/TRUNCATE PARTITION, or MODIFY PARTITION INDEXING OFF operations. Possible values: • YES - the index partition contains orphaned entries • NO - the index partition does not contain orphaned entries

See Also

- "DBA_IND_PARTITIONS"
- "USER_IND_PARTITIONS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

2.225 ALL_IND_PENDING_STATS

ALL_IND_PENDING_STATS describes the pending statistics for tables, partitions, and subpartitions accessible to the current user collected using the DBMS_STATS package.

Related Views

- DBA_IND_PENDING_STATS describes pending statistics for all tables, partitions, and subpartitions in the database.
- USER_IND_PENDING_STATS describes pending statistics for tables, partitions, and subpartitions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Name of the index owner
INDEX_NAME	VARCHAR2(128)		Index name
TABLE_OWNER	VARCHAR2(128)		Table owner name
TABLE_NAME	VARCHAR2(128)		Name of the table
PARTITION_NAME	VARCHAR2(128)		Name of the partition
SUBPARTITION_NAME	VARCHAR2(128)		Name of the subpartition
BLEVEL	NUMBER		Number of levels in the index
LEAF_BLOCKS	NUMBER		Number of leaf blocks in the index
DISTINCT_KEYS	NUMBER		Number of distinct keys in the index

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
AVG_LEAF_BLOCKS_PER_KEY	NUMBER		Average number of leaf blocks per key
AVG_DATA_BLOCKS_PER_KEY	NUMBER		Average number of data blocks per key
CLUSTERING_FACTOR	NUMBER		Clustering factor
NUM_ROWS	NUMBER		Number of rows in the index
SAMPLE_SIZE	NUMBER		Sample size
LAST_ANALYZED	DATE		Time of the last analysis

See Also

- "DBA_IND_PENDING_STATS"
- "USER_IND_PENDING_STATS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

2.226 ALL_IND_STATISTICS

ALL_IND_STATISTICS displays optimizer statistics for the indexes on the tables accessible to the current user collected using the DBMS_STATS package.

Related Views

- DBA_IND_STATISTICS displays optimizer statistics for all indexes in the database.
- USER_IND_STATISTICS displays optimizer statistics for the indexes on the tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the index
INDEX_NAME	VARCHAR2(128)		Name of the index
TABLE_OWNER	VARCHAR2(128)		Owner of the indexed object
TABLE_NAME	VARCHAR2(128)		Name of the indexed object
PARTITION_NAME	VARCHAR2(128)		Name of the partition
PARTITION_POSITION	NUMBER		Position of the partition within the index
SUBPARTITION_NAME	VARCHAR2(128)		Name of the subpartition
SUBPARTITION_POSITION	NUMBER		Position of the subpartition within the partition
OBJECT_TYPE	VARCHAR2(12)		Type of the object: • INDEX • PARTITION • SUBPARTITION
BLEVEL	NUMBER		B-Tree level
LEAF_BLOCKS	NUMBER		Number of leaf blocks in the index
DISTINCT_KEYS	NUMBER		Number of distinct keys in the index
AVG_LEAF_BLOCKS_PER_KEY	NUMBER		Average number of leaf blocks per key

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
AVG_DATA_BLOCKS_PER_KEY	NUMBER		Average number of data blocks per key
CLUSTERING_FACTOR	NUMBER		Indicates the amount of order of the rows in the table based on the values of the index. • If the value is near the number of blocks, then the table is very well ordered. In this case, the index entries in a single leaf block tend to point to rows in the same data blocks. • If the value is near the number of rows, then the table is very randomly ordered. In this case, it is unlikely that index entries in the same leaf block point to rows in the same data blocks.
NUM_ROWS	NUMBER		Number of rows in the index
AVG_CACHED_BLOCKS	NUMBER		Average number of blocks in the buffer cache
AVG_CACHE_HIT_RATIO	NUMBER		Average cache hit ratio for the object
SAMPLE_SIZE	NUMBER		Sample size used in analyzing the index
LAST_ANALYZED	DATE		Date of the most recent time the index was analyzed
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics are gathered or incrementally maintained, otherwise it will be NO
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
STATTYPE_LOCKED	VARCHAR2(5)		Type of statistics lock
STALE_STATS	VARCHAR2(3)		Whether statistics for the object are stale or not
SCOPE	VARCHAR2(7)		The value is SHARED for statistics gathered on any table other than global temporary tables. For a global tempoary table, the possible values are: • SESSION - Indicates that the statistics are session-specific • SHARED - Indicates that the statistics are shared across all sessions See *Oracle Database PL/SQL Packages and Types Reference* for information about using the GLOBAL_TEMP_TABLE_STATS preference of the DBMS_STATS package to control whether to gather session or shared statistics for global temporary tables.

See Also

- "DBA_IND_STATISTICS"
- "USER_IND_STATISTICS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.227 ALL_IND_SUBPARTITIONS

ALL_IND_SUBPARTITIONS displays, for each index subpartition accessible to the current user, the subpartition-level partitioning information, the storage parameters for the subpartition, and various subpartition statistics generated by the DBMS_STATS package.

Related Views

- DBA_IND_SUBPARTITIONS describes all index subpartitions in the database.
- USER_IND_SUBPARTITIONS describes the index subpartitions owned by the current user. This view does not display the INDEX_OWNER column.

Column	Datatype	NULL	Description
INDEX_OWNER	VARCHAR2(128)	NOT NULL	Owner of the index
INDEX_NAME	VARCHAR2(128)	NOT NULL	Name of the index
PARTITION_NAME	VARCHAR2(128)		Name of the partition
SUBPARTITION_NAME	VARCHAR2(128)		Name of the subpartition
HIGH_VALUE	LONG		Subpartition bound value expression
HIGH_VALUE_LENGTH	NUMBER	NOT NULL	Length of the subpartition bound value expression
PARTITION_POSITION	NUMBER		Position of the partition within the index
SUBPARTITION_POSITION	NUMBER		Position of a subpartition within a partition
STATUS	VARCHAR2(8)		Indicates whether the index partition is usable (USABLE) or not (UNUSABLE)
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of the tablespace containing the partition
PCT_FREE	NUMBER	NOT NULL	Minimum percentage of free space in a block
INI_TRANS	NUMBER	NOT NULL	Initial number of transactions
MAX_TRANS	NUMBER	NOT NULL	Maximum number of transactions
INITIAL_EXTENT	NUMBER		Size of the initial extent in bytes
NEXT_EXTENT	NUMBER		Size of secondary extents in bytes
MIN_EXTENT	NUMBER		Minimum number of extents allowed in the segment
MAX_EXTENT	NUMBER		Maximum number of extents allowed in the segment
MAX_SIZE	NUMBER		Maximum number of blocks allowed in the segment
PCT_INCREASE	NUMBER	NOT NULL	Percentage increase in extent size
FREELISTS	NUMBER		Number of process freelists allocated in this segment
FREELIST_GROUPS	NUMBER		Number of process freelist groups allocated in this segment
LOGGING	VARCHAR2(3)		Indicates whether or not changes to the index are logged: • YES • NO
COMPRESSION	VARCHAR2(13)		Type of compression being used for the subpartition: • ENABLED - Prefix compression • ADVANCED HIGH - Advanced high compression • ADVANCED LOW - Advanced low compression • DISABLED - No compression is present

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BLEVEL	NUMBER		B-Tree level (depth of the index from its root block to its leaf blocks). A depth of 0 indicates that the root block and leaf block are the same.
LEAF_BLOCKS	NUMBER		Number of leaf blocks in the index
DISTINCT_KEYS	NUMBER		Number of distinct keys in the index partition
AVG_LEAF_BLOCKS_PER_KEY	NUMBER		Average number of leaf blocks in which each distinct value in the index appears, rounded to the nearest integer. For indexes that enforce UNIQUE and PRIMARY KEY constraints, this value is always 1.
AVG_DATA_BLOCKS_PER_KEY	NUMBER		Average number of data blocks in the table that are pointed to by a distinct value in the index rounded to the nearest integer. This statistic is the average number of data blocks that contain rows that contain a given value for the indexed columns.
CLUSTERING_FACTOR	NUMBER		Indicates the amount of order of the rows in the table based on the values of the index. • If the value is near the number of blocks, then the table is very well ordered. In this case, the index entries in a single leaf block tend to point to rows in the same data blocks. • If the value is near the number of rows, then the table is very randomly ordered. In this case, it is unlikely that index entries in the same leaf block point to rows in the same data blocks.
NUM_ROWS	NUMBER		Number of rows in this index subpartition
SAMPLE_SIZE	NUMBER		Sample size used in analyzing this subpartition
LAST_ANALYZED	DATE		Date on which this partition was most recently analyzed
BUFFER_POOL	VARCHAR2(7)		Buffer pool for the subpartition: • DEFAULT • KEEP • RECYCLE • NULL
FLASH_CACHE	VARCHAR2(7)		Database Smart Flash Cache hint to be used for subpartition blocks: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)		Cell flash cache hint to be used for subpartition blocks: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics have been gathered or NO if statistics have not been gathered

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INTERVAL	VARCHAR2(3)		Indicates whether the partition is in the interval section of an interval partitioned table (YES) or whether the partition is in the range section (NO)
SEGMENT_CREATED	VARCHAR2(3)		Indicates whether the index subpartition segment has been created (YES) or not (NO); N/A indicates that this index is not subpartitioned
DOMIDX_OPSTATUS	VARCHAR2(6)		Status of the operation on the domain index: • NULL - Index is not a domain index • VALID - Operation performed without errors • FAILED - Operation failed with an error
PARAMETERS	VARCHAR2(1000)		For a domain index, the parameter string

See Also

- "DBA_IND_SUBPARTITIONS"
- "USER_IND_SUBPARTITIONS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

2.228 ALL_INDEXES

ALL_INDEXES describes the indexes on the tables accessible to the current user.

To gather statistics for this view and the related views DBA_INDEXES and USER_INDEXES, use the DBMS_STATS package.

Related Views

- DBA_INDEXES describes all indexes in the database.
- USER_INDEXES describes the indexes owned by the current user. This view does not display the OWNER column.

Note

Column names followed by an asterisk are populated only if you collect statistics on the index using the DBMS_STATS package.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the index
INDEX_NAME	VARCHAR2(128)	NOT NULL	Name of the index

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INDEX_TYPE	VARCHAR2(27)		Type of the index: • BITMAP • CLUSTER • DOMAIN • FUNCTION-BASED BITMAP • FUNCTION-BASED DOMAIN • FUNCTION-BASED NORMAL • FUNCTION-BASED NORMAL/REV • IOT - TOP • LOB • NORMAL • NORMAL/REV
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the indexed object
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the indexed object
TABLE_TYPE	VARCHAR2(11)		Type of the indexed object In the DBA_INDEXES and USER_INDEXES views, the possible values are: • CLUSTER • INDEX • NEXT OBJECT • SEQUENCE • SYNONYM • TABLE • VIEW In the ALL_INDEXES view, the value of this column is always TABLE and the data type is CHAR(5).
UNIQUENESS	VARCHAR2(9)		Indicates whether the index is unique (UNIQUE) or nonunique (NONUNIQUE)
COMPRESSION	VARCHAR2(13)		Type of compression being used for the index: • ENABLED - Prefix compression • ADVANCED HIGH - Advanced high compression • ADVANCED LOW - Advanced low compression • DISABLED - No compression is present
PREFIX_LENGTH	NUMBER		Number of columns in the prefix of the compression key
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the index
INI_TRANS	NUMBER		Initial number of transactions
MAX_TRANS	NUMBER		Maximum number of transactions
INITIAL_EXTENT	NUMBER		Size of the initial extent
NEXT_EXTENT	NUMBER		Size of secondary extents
MIN_EXTENTS	NUMBER		Minimum number of extents allowed in the segment
MAX_EXTENTS	NUMBER		Maximum number of extents allowed in the segment
PCT_INCREASE	NUMBER		Percentage increase in extent size
PCT_THRESHOLD	NUMBER		Threshold percentage of block space allowed per index entry

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INCLUDE_COLUMN	NUMBER		Column ID of the last column to be included in index-organized table primary key (non-overflow) index. This column maps to the COLUMN_ID column of the *_TAB_COLUMNS view.
FREELISTS	NUMBER		Number of process freelists allocated to this segment
FREELIST_GROUPS	NUMBER		Number of freelist groups allocated to this segment
PCT_FREE	NUMBER		Minimum percentage of free space in a block
LOGGING	VARCHAR2(3)		Indicates whether or not changes to the index are logged: • YES • NO
BLEVEL*	NUMBER		B*-Tree level (depth of the index from its root block to its leaf blocks). A depth of 0 indicates that the root block and leaf block are the same.
LEAF_BLOCKS*	NUMBER		Number of leaf blocks in the index
DISTINCT_KEYS*	NUMBER		Number of distinct indexed values. For indexes that enforce UNIQUE and PRIMARY KEY constraints, this value is the same as the number of rows in the table (*_TABLES.NUM_ROWS)
AVG_LEAF_BLOCKS_PER_KEY*	NUMBER		Average number of leaf blocks in which each distinct value in the index appears, rounded to the nearest integer. For indexes that enforce UNIQUE and PRIMARY KEY constraints, this value is always 1.
AVG_DATA_BLOCKS_PER_KEY*	NUMBER		Average number of data blocks in the table that are pointed to by a distinct value in the index rounded to the nearest integer. This statistic is the average number of data blocks that contain rows that contain a given value for the indexed columns.
CLUSTERING_FACTOR*	NUMBER		Indicates the amount of order of the rows in the table based on the values of the index. • If the value is near the number of blocks, then the table is very well ordered. In this case, the index entries in a single leaf block tend to point to rows in the same data blocks. • If the value is near the number of rows, then the table is very randomly ordered. In this case, it is unlikely that index entries in the same leaf block point to rows in the same data blocks. For bitmap indexes, this column is not applicable.
STATUS	VARCHAR2(8)		Indicates whether a nonpartitioned index is VALID or UNUSABLE
NUM_ROWS	NUMBER		Number of rows in the index. For bitmap indexes, this column is the number of distinct keys, so its value is the same as the DISTINCT_KEYS column.
SAMPLE_SIZE	NUMBER		Size of the sample used to analyze the index
LAST_ANALYZED	DATE		Date on which this index was most recently analyzed
DEGREE	VARCHAR2(40)		Number of threads per instance for scanning the index, or DEFAULT

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INSTANCES	VARCHAR2(40)		Number of instances across which the indexes to be scanned, or DEFAULT
PARTITIONED	VARCHAR2(3)		Indicates whether the index is partitioned (YES) or not (NO)
TEMPORARY	VARCHAR2(1)		Indicates whether the index is on a temporary table (Y) or not (N)
GENERATED	VARCHAR2(1)		Indicates whether the name of the index is system-generated (Y) or not (N)
SECONDARY	VARCHAR2(1)		Indicates whether the index is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge (Y) or not (N)
BUFFER_POOL	VARCHAR2(7)		Buffer pool to be used for index blocks: • DEFAULT • KEEP • RECYCLE • NULL
FLASH_CACHE	VARCHAR2(7)		Database Smart Flash Cache hint to be used for index blocks: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)		Cell flash cache hint to be used for index blocks: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
DURATION	VARCHAR2(15)		Indicates the duration of a temporary table: • SYS$SESSION - Rows are preserved for the duration of the session • SYS$TRANSACTION - Rows are deleted after COMMIT NULL - Permanent table
PCT_DIRECT_ACCESS	NUMBER		For a secondary index on an index-organized table, the percentage of rows with VALID guess
ITYP_OWNER	VARCHAR2(128)		For a domain index, the owner of the indextype
ITYP_NAME	VARCHAR2(128)		For a domain index, the name of the indextype
PARAMETERS	VARCHAR2(1000)		For a domain index, the parameter string
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics are gathered or incrementally maintained, otherwise it will be NO
DOMIDX_STATUS	VARCHAR2(12)		Status of a domain index: • NULL - Index is not a domain index • VALID - Index is a valid domain index • IDXTYP_INVLD - Indextype of the domain index is invalid

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DOMIDX_OPSTATUS	VARCHAR2(6)		Status of the operation on a domain index: • NULL - Index is not a domain index • VALID - Operation performed without errors • FAILED - Operation failed with an error
FUNCIDX_STATUS	VARCHAR2(8)		Status of a function-based index: • NULL - Index is not a function-based index • ENABLED - Function-based index is enabled • DISABLED - Function-based index is disabled
JOIN_INDEX	VARCHAR2(3)		Indicates whether the index is a join index (YES) or not (NO)
IOT_REDUNDANT_PKEY_ELIM	VARCHAR2(3)		Indicates whether redundant primary key columns are eliminated from secondary indexes on index-organized tables (YES) or not (NO)
DROPPED	VARCHAR2(3)		Indicates whether the index has been dropped and is in the recycle bin (YES) or not (NO); NULL for partitioned tables This view does not return the names of indexes that have been dropped.
VISIBILITY	VARCHAR2(9)		Indicates whether the index is VISIBLE or INVISIBLE to the optimizer
DOMIDX_MANAGEMENT	VARCHAR2(14)		If this is a domain index, indicates whether the domain index is system-managed (SYSTEM_MANAGED) or user-managed (USER_MANAGED)
SEGMENT_CREATED	VARCHAR2(3)		Indicates whether the index segment has been created (YES) or not (NO)
ORPHANED_ENTRIES	VARCHAR2(3)		Indicates whether a global index contains stale entries because of deferred index maintenance during DROP/ TRUNCATE PARTITION, or MODIFY PARTITION INDEXING OFF operations. Possible values: • YES - The index contains orphaned entries • NO - The index does not contain orphaned entries
INDEXING	VARCHAR2(7)		Indicates whether a global index is decoupled from the underlying table. Possible values: • PARTIAL - The index is partial, that is, it will follow the table's indexing property. • FULL - The index will include all partitions of the table.
AUTO[1]	VARCHAR2(3)		Indicates whether the index is an auto index (YES) or not (NO)
CONSTRAINT_INDEX[1]	VARCHAR2(3)		Indicates whether the index was created as part of a constraint (YES) or not (NO)

[1] This column is available starting with Oracle Database 19c.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_INDEXES"
- "USER_INDEXES"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

2.229 ALL_INDEXTYPE_ARRAYTYPES

ALL_INDEXTYPE_ARRAYTYPES displays information about the array types specified by the indextypes accessible to the current user.

Related Views

- DBA_INDEXTYPE_ARRAYTYPES displays information about the array types specified by all indextypes in the database.
- USER_INDEXTYPE_ARRAYTYPES displays information about the array types specified by the indextypes owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the indextype
INDEXTYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the indextype
BASE_TYPE_SCHEMA	VARCHAR2(128)		Name of the base type schema
BASE_TYPE_NAME	VARCHAR2(128)		Name of the base type name
BASE_TYPE	VARCHAR2(30)		Datatype of the base type
ARRAY_TYPE_SCHEMA	VARCHAR2(128)	NOT NULL	Name of the array type schema
ARRAY_TYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the array type name

See Also

- "DBA_INDEXTYPE_ARRAYTYPES"
- "USER_INDEXTYPE_ARRAYTYPES"

2.230 ALL_INDEXTYPE_COMMENTS

ALL_INDEXTYPE_COMMENTS displays comments for the user-defined indextypes accessible to the current user.

Related Views

- DBA_INDEXTYPE_COMMENTS displays comments for all user-defined indextypes in the database.
- USER_INDEXTYPE_COMMENTS displays comments for the user-defined indextypes owned by the current user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the user-defined indextype
INDEXTYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the user-defined indextype
COMMENTS	VARCHAR2(4000)		Comment for the user-defined indextype

See Also

- "DBA_INDEXTYPE_COMMENTS"
- "USER_INDEXTYPE_COMMENTS"

2.231 ALL_INDEXTYPE_OPERATORS

ALL_INDEXTYPE_OPERATORS lists all operators supported by indextypes accessible to the current user.

Related Views

- DBA_INDEXTYPE_OPERATORS lists all operators supported by indextypes in the database.
- USER_INDEXTYPE_OPERATORS lists all operators supported by indextypes owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the indextype
INDEXTYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the indextype
OPERATOR_SCHEMA	VARCHAR2(128)	NOT NULL	Name of the operator schema
OPERATOR_NAME	VARCHAR2(128)	NOT NULL	Name of the operator for which the indextype is defined
BINDING#	NUMBER	NOT NULL	Binding number associated with the operator

See Also

- "DBA_INDEXTYPE_OPERATORS"
- "USER_INDEXTYPE_OPERATORS"

2.232 ALL_INDEXTYPES

ALL_INDEXTYPES displays information about the indextypes accessible to the current user.

Related Views

- DBA_INDEXTYPES displays information about all indextypes in the database.
- USER_INDEXTYPES displays information about the indextypes owned by the current user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the indextype
INDEXTYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the indextype
IMPLEMENTATION_SCHEMA	VARCHAR2(128)	NOT NULL	Name of the schema for the indextype implementation (that is, containing the indextype operators)
IMPLEMENTATION_NAME	VARCHAR2(128)	NOT NULL	Name of the indextype implementation type
INTERFACE_VERSION	NUMBER		Version of the indextype interface
IMPLEMENTATION_VERSION	NUMBER	NOT NULL	Version of the indextype implementation
NUMBER_OF_OPERATORS	NUMBER		Number of operators associated with the indextype
PARTITIONING	VARCHAR2(10)		Kinds of local partitioning supported by the indextype: • NONE - Indextype does not support local domain indexes • RANGE - Indextype can support range partitioned local user managed domain indexes • LOCAL - Indextype can support local system managed domain indexes (range, list, hash, or interval)
ARRAY_DML	VARCHAR2(3)		Indicates whether the indextype supports array DML (YES) or not (NO)
MAINTENANCE_TYPE	VARCHAR2(14)		Indicates whether the indextype is system-managed (SYSTEM_MANAGED) or user-managed (USER_MANAGED)

See Also

- "DBA_INDEXTYPES"
- "USER_INDEXTYPES"

2.233 ALL_INTERNAL_TRIGGERS

ALL_INTERNAL_TRIGGERS describes internal triggers on tables accessible to the current user. Internal triggers are internal pieces of code executed when a particular flag is set for a table. This view does not display the OWNER_NAME column.

Related Views

- DBA_INTERNAL_TRIGGERS describes internal triggers on all tables in the database.
- USER_INTERNAL_TRIGGERS describes all internal triggers on tables owned by the current user. This view does not display the OWNER_NAME column.

Column	Datatype	NULL	Description
TABLE_NAME	VARCHAR2(128)		Name of the table on which the trigger is defined
INTERNAL_TRIGGER_TYPE	VARCHAR2(19)		Indicates the type of internal trigger on the table

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_INTERNAL_TRIGGERS"
- "USER_INTERNAL_TRIGGERS"

2.234 ALL_JAVA_ARGUMENTS

ALL_JAVA_ARGUMENTS displays argument information about the stored Java classes accessible to the current user.

Related Views

- DBA_JAVA_ARGUMENTS displays argument information about all stored Java classes in the database.
- USER_JAVA_ARGUMENTS displays argument information about the stored Java classes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Java class
NAME	VARCHAR2(4000)		Name of the Java class
METHOD_INDEX	NUMBER		Index of the hosting method of the argument
METHOD_NAME	VARCHAR2(4000)		Name of the hosting method of the argument
ARGUMENT_POSITION	NUMBER		Position of the argument, starting from 0
ARRAY_DEPTH	NUMBER		Array depth of the type of the argument
BASE_TYPE	VARCHAR2(7)		Base type of the type of the argument: • int • long • float • double • boolean • byte • char • short • class
ARGUMENT_CLASS	VARCHAR2(4000)		Actual class name of the argument if the base type is class

See Also

- "DBA_JAVA_ARGUMENTS"
- "USER_JAVA_ARGUMENTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.235 ALL_JAVA_CLASSES

ALL_JAVA_CLASSES displays class level information about the stored Java classes accessible to the current user.

Related Views

- DBA_JAVA_CLASSES displays class level information about all stored Java classes in the database.
- USER_JAVA_CLASSES displays class level information about the stored Java classes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Java class
NAME	VARCHAR2(4000)		Name of the Java class
MAJOR	NUMBER		Major version number of the Java class, as defined in the JVM specification
MINOR	NUMBER		Minor version number of the Java class, as defined in the JVM specification
KIND	VARCHAR2(50)		Indicates whether the stored object is a Java class (CLASS) or a Java interface (INTERFACE)
ACCESSIBILITY	VARCHAR2(9)		Accessibility of the Java class
IS_INNER	VARCHAR2(3)		Indicates whether this Java class is an inner class (YES) or not (NO)
IS_ABSTRACT	VARCHAR2(3)		Indicates whether this Java class is an abstract class (YES) or not (NO)
IS_FINAL	VARCHAR2(3)		Indicates whether this Java class is a final class (YES) or not (NO)
IS_STATIC	VARCHAR2(3)		Indicates whether this is a static inner class (YES) or not (NO)
IS_STRICTFP	VARCHAR2(3)		Indicates whether the class is declared strict floating point for portability (YES) or not (NO)
IS_SYNTHETIC	VARCHAR2(3)		Indicates whether this is an inner class generated by the compiler (YES) or not (NO)
IS_DEBUG	VARCHAR2(3)		Indicates whether this Java class contains debug information (YES) or not (NO)
SOURCE	VARCHAR2(4000)		Source designation of the Java class
SUPER	VARCHAR2(4000)		Super class of this Java class
OUTER	VARCHAR2(4000)		Outer class of this Java class if this Java class is an inner class

See Also

- "DBA_JAVA_CLASSES"
- "USER_JAVA_CLASSES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.236 ALL_JAVA_COMPILER_OPTIONS

ALL_JAVA_COMPILER_OPTIONS displays information about the native compiler options accessible to the current user.

Related Views

- DBA_JAVA_COMPILER_OPTIONS displays information about all native compiler options in the database.
- USER_JAVA_COMPILER_OPTIONS displays information about the native compiler options owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the native compiler option
OPTION_NAME	VARCHAR2(64)		Name of the native compiler option
VALUE	VARCHAR2(4000)		Value of the native compiler option

See Also

- "DBA_JAVA_COMPILER_OPTIONS"
- "USER_JAVA_COMPILER_OPTIONS"

2.237 ALL_JAVA_DERIVATIONS

ALL_JAVA_DERIVATIONS displays mapping information about Java source objects and their derived Java class objects and Java resource objects for the Java classes accessible to the current user.

Related Views

- DBA_JAVA_DERIVATIONS displays mapping information about Java source objects and their derived Java class objects and Java resource objects for all Java classes in the database.
- USER_JAVA_DERIVATIONS displays mapping information about Java source objects and their derived Java class objects and Java resource objects for the Java classes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Java source object
SOURCE_NAME	VARCHAR2(4000)		Name of the Java source object
CLASS_INDEX	NUMBER		Index of the derived Java class object
CLASS_NAME	VARCHAR2(4000)		Name of the derived Java class object

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_JAVA_DERIVATIONS"
- "USER_JAVA_DERIVATIONS"

2.238 ALL_JAVA_FIELDS

ALL_JAVA_FIELDS displays field information about the stored Java classes accessible to the current user.

Related Views

- DBA_JAVA_FIELDS displays field information about all stored Java classes in the database.
- USER_JAVA_FIELDS displays field information about the stored Java classes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Java class
NAME	VARCHAR2(4000)		Name of the Java class
FIELD_INDEX	NUMBER		Index of the field
FIELD_NAME	VARCHAR2(4000)		Name of the field identified by the FIELD_INDEX column
ACCESSIBILITY	VARCHAR2(9)		Accessibility of the field: • PUBLIC • PRIVATE • PROTECTED
IS_STATIC	VARCHAR2(3)		Indicates whether the field is a static field (YES) or not (NO)
IS_FINAL	VARCHAR2(3)		Indicates whether the field is a final field (YES) or not (NO)
IS_VOLATILE	VARCHAR2(3)		Indicates whether the field is volatile (YES) or not (NO)
IS_TRANSIENT	VARCHAR2(3)		Indicates whether the field is transient (YES) or not (NO)
IS_SYNTHETIC	VARCHAR2(3)		Indicates whether the field is generated by the compiler (YES) or not (NO)
IS_ENUM	VARCHAR2(3)		Indicates whether the field is a member of an enum (YES) or not (NO)
ARRAY_DEPTH	NUMBER		Array depth of the type of the field
BASE_TYPE	VARCHAR2(7)		Base type of the type of the field: • int • long • float • double • boolean • byte • char • short • class

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FIELD_CLASS	VARCHAR2(4000)		Actual class name of the base object if the base type is class

See Also

- "DBA_JAVA_FIELDS"
- "USER_JAVA_FIELDS"

2.239 ALL_JAVA_IMPLEMENTS

ALL_JAVA_IMPLEMENTS describes interfaces implemented by the stored Java classes accessible to the current user.

Related Views

- DBA_JAVA_IMPLEMENTS describes interfaces implemented by all stored Java classes in the database.
- USER_JAVA_IMPLEMENTS describes interfaces implemented by the stored Java classes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Java class
NAME	VARCHAR2(4000)		Name of the Java class
INTERFACE_INDEX	NUMBER		Index of the interfaces implemented by the Java class
INTERFACE_NAME	VARCHAR2(4000)		Name of the interface identified by the INTERFACE_INDEX column

See Also

- "DBA_JAVA_IMPLEMENTS"
- "USER_JAVA_IMPLEMENTS"

2.240 ALL_JAVA_INNERS

ALL_JAVA_INNERS displays information about inner classes referred to by the stored Java classes accessible to the current user.

Related Views

- DBA_JAVA_INNERS displays information about inner classes referred to by all stored Java classes in the database.
- USER_JAVA_INNERS displays information about inner classes referred to by the stored Java classes owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Java class
NAME	VARCHAR2(4000)		Name of the Java class
INNER_INDEX	NUMBER		Index of the referred inner class
SIMPLE_NAME	VARCHAR2(4000)		Simple name of the referred inner class
FULL_NAME	VARCHAR2(4000)		Full name of the referred inner class
ACCESSIBILITY	VARCHAR2(9)		Accessibility of the referred inner class: • PUBLIC • PRIVATE • PROTECTED
IS_STATIC	VARCHAR2(3)		Indicates whether the referred inner class is declared static in the source file (YES) or not (NO)
IS_FINAL	VARCHAR2(3)		Indicates whether the referred inner class is declared final in the source file (YES) or not (NO)
IS_ABSTRACT	VARCHAR2(3)		Indicates whether the referred inner class is declared abstract in the source file (YES) or not (NO)
IS_INTERFACE	VARCHAR2(3)		Indicates whether the referred inner class is declared interface in the source file (YES) or not (NO)
IS_STRICTFP	VARCHAR2(3)		Indicates whether the inner class is declared strictfp (YES) or not (NO)
IS_SYNTHETIC	VARCHAR2(3)		Indicates whether the inner class is generated by the compiler (YES) or not (NO)
IS_ENUM	VARCHAR2(3)		Indicates whether the inner class is an enum (YES) or not (NO)
IS_ANNOTATION	VARCHAR2(3)		Indicates whether the inner class is an annotation (YES) or not (NO)
KIND	VARCHAR2(50)		Type (class, interface, enum or annotation) of the inner class
ALL_QUALIFIERS	VARCHAR2(101)		Concatenation of accessibility, type and other attributes of the inner class

See Also

- "DBA_JAVA_INNERS"
- "USER_JAVA_INNERS"

2.241 ALL_JAVA_LAYOUTS

ALL_JAVA_LAYOUTS displays class layout information about the stored Java classes accessible to the current user.

Related Views

- DBA_JAVA_LAYOUTS displays class layout information about all stored Java classes in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_JAVA_LAYOUTS displays class layout information about the stored Java classes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Java class
NAME	VARCHAR2(4000)		Name of the Java class
INTERFACES	NUMBER		Number of interfaces that this Java class implements
INNER_CLASSES	NUMBER		Number of inner classes that this Java class contains
FIELDS	NUMBER		Number of locally declared fields that this Java class contains
STATIC_FIELDS	NUMBER		Number of locally declared static fields that this Java class contains
METHODS	NUMBER		Number of locally declared methods that this Java class contains
STATIC_METHODS	NUMBER		Number of locally declared static methods that this Java class contains
NATIVE_METHODS	NUMBER		Number of locally declared native methods that this Java class contains

See Also

- "DBA_JAVA_LAYOUTS"
- "USER_JAVA_LAYOUTS"

2.242 ALL_JAVA_METHODS

ALL_JAVA_METHODS displays method information about the stored Java classes accessible to the current user.

Related Views

- DBA_JAVA_METHODS displays method information about all stored Java classes in the database.
- USER_JAVA_METHODS displays method information about the stored Java classes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Java class
NAME	VARCHAR2(4000)		Name of the Java class
METHOD_INDEX	NUMBER		Index of the method
METHOD_NAME	VARCHAR2(4000)		Name of the method identified by the METHOD_INDEX column
ACCESSIBILITY	VARCHAR2(9)		Accessibility of the method: • PUBLIC • PRIVATE • PROTECTED

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
IS_STATIC	VARCHAR2(3)		Indicates whether the method is a static method (YES) or not (NO)
IS_FINAL	VARCHAR2(3)		Indicates whether the method is a final method (YES) or not (NO)
IS_SYNCHRONIZED	VARCHAR2(3)		Indicates whether the method is a synchronized method (YES) or not (NO)
HAS_VARARGS	VARCHAR2(3)		Indicates whether the method has a variable number of arguments (YES) or not (NO)
IS_NATIVE	VARCHAR2(3)		Indicates whether the method is a native method (YES) or not (NO)
IS_ABSTRACT	VARCHAR2(3)		Indicates whether the method is an abstract method (YES) or not (NO)
IS_STRICTFP	VARCHAR2(3)		Indicates whether the method is a strict method (YES) or not (NO)
IS_SYNTHETIC	VARCHAR2(3)		Indicates whether the method is generated by the compiler (YES) or not (NO)
ARGUMENTS	NUMBER		Number of arguments of the method
THROWS	NUMBER		Number of exceptions thrown by the method
ARRAY_DEPTH	NUMBER		Array depth of the return type of the method
BASE_TYPE	VARCHAR2(7)		Base type of the return type of the method: • int • long • float • double • boolean • byte • char • short • class • void
RETURN_CLASS	VARCHAR2(4000)		Actual class name of the return value if the base type is class
IS_COMPILED	VARCHAR2(3)		Indicates whether the Java method has been natively compiled by the JIT compiler (YES) or not (NO)

See Also

- "DBA_JAVA_METHODS"
- "USER_JAVA_METHODS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.243 ALL_JAVA_NCOMPS

ALL_JAVA_NCOMPS displays ncomp-related information about the Java classes accessible to the current user.

Related Views

- DBA_JAVA_NCOMPS displays ncomp-related information about all Java classes in the database.
- USER_JAVA_NCOMPS displays ncomp-related information about the Java classes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Java class object
NAME	VARCHAR2(4000)		Name of the Java class object
SOURCE	VARCHAR2(4000)		ncomp source shown in this row
INITIALIZER	VARCHAR2(4000)		ncomp initializer shown in this row
LIBRARYFILE	VARCHAR2(4000)		ncomp library file shown in this row
LIBRARY	VARCHAR2(4000)		ncomp library shown in this row

See Also

- "DBA_JAVA_NCOMPS"
- "USER_JAVA_NCOMPS"

2.244 ALL_JAVA_RESOLVERS

ALL_JAVA_RESOLVERS displays information about resolvers of the Java classes accessible to the current user.

Related Views

- DBA_JAVA_RESOLVERS displays information about resolvers of all Java classes in the database.
- USER_JAVA_RESOLVERS displays information about resolvers of the Java classes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Java class object
NAME	VARCHAR2(4000)		Name of the Java class object
TERM_INDEX	NUMBER		Index of the resolver term in this row
PATTERN	VARCHAR2(4000)		Resolver pattern of the resolver term identified by the TERM_INDEX column
SCHEMA	VARCHAR2(64)		Resolver schema of the resolver term identified by the TERM_INDEX column

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> - "DBA_JAVA_RESOLVERS"
> - "USER_JAVA_RESOLVERS"

2.245 ALL_JAVA_THROWS

ALL_JAVA_THROWS displays information about exceptions thrown from methods of the Java classes accessible to the current user.

Related Views

- DBA_JAVA_THROWS displays information about exceptions thrown from methods of all Java classes in the database.
- USER_JAVA_THROWS displays information about exceptions thrown from methods of the Java classes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Java class
NAME	VARCHAR2(4000)		Name of the Java class
METHOD_INDEX	NUMBER		Index of the throwing method of the exception
METHOD_NAME	VARCHAR2(4000)		Name of the throwing method of the exception
EXCEPTION_INDEX	NUMBER		Index of the exception
EXCEPTION_CLASS	VARCHAR2(4000)		Class of the exception

> **See Also**
>
> - "DBA_JAVA_THROWS"
> - "USER_JAVA_THROWS"

2.246 ALL_JOBS

ALL_JOBS is a synonym for USER_JOBS.

> **See Also**
>
> "USER_JOBS"

2.247 ALL_JOIN_IND_COLUMNS

ALL_JOIN_IND_COLUMNS describes the join conditions of bitmap join indexes accessible to the current user. Bitmap join indexes are indexes built on a child table with an index key containing

Copyright © 2002, 2026, Oracle and/or its affiliates.

columns from associated parent tables, where all of the tables are connected through join conditions. There is one row for each join condition.

Related Views

- `DBA_JOIN_IND_COLUMNS` describes all join conditions in the database.
- `USER_JOIN_IND_COLUMNS` describes the join conditions owned by the current user. This view does not display the `INDEX_OWNER` column.

Column	Datatype	NULL	Description
INDEX_OWNER	VARCHAR2(128)	NOT NULL	Owner of the bitmap join index
INDEX_NAME	VARCHAR2(128)	NOT NULL	Name of the bitmap join index
INNER_TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the fact table
INNER_TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the fact table
INNER_TABLE_COLUMN	VARCHAR2(128)	NOT NULL	Name of the fact table join column
OUTER_TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the dimension table
OUTER_TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the dimension table
OUTER_TABLE_COLUMN	VARCHAR2(128)	NOT NULL	Name of the dimension table join column

See Also

- "DBA_JOIN_IND_COLUMNS"
- "USER_JOIN_IND_COLUMNS"

2.248 ALL_JSON_COLUMNS

`ALL_JSON_COLUMNS` provides information on the JavaScript Object Notation (JSON) columns accessible to the current user. Each column that has an IS JSON check constraint in an AND condition and is accessible to the user appears in this view. This view enables a user to find all the JSON columns that are accessible to him or her.
For example, if a check constraint combines the IS JSON condition with another condition using logical condition OR, then the column is not listed in this view. In this case, it is not certain that the data in the column is JSON data. For example, the following constraint does not ensure that the data in column jcol is JSON data:

```
jcol is json OR length(jcol) < 1000
```

Related Views

- `DBA_JSON_COLUMNS` provides information on all JSON columns.
- `USER_JSON_COLUMNS` provides information on the JSON columns for which the user is the owner. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table with the JSON column
TABLE_NAME	VARCHAR2(128)		Name of the table with the JSON column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OBJECT_TYPE	VARCHAR2(5)		Object type: • TABLE • VIEW
COLUMN_NAME	VARCHAR2(128)		Name of the JSON column
FORMAT	VARCHAR2(9)		Format of the JSON data
DATA_TYPE	VARCHAR2(13)		Data type of the JSON column

See Also

- "DBA_JSON_COLUMNS"
- "USER_JSON_COLUMNS"
- *Oracle Database JSON Developer's Guide* for more information about using JSON with Oracle Database

2.249 ALL_JSON_DATAGUIDE_FIELDS

ALL_JSON_DATAGUIDE_FIELDS extracts the path and type information from the data guides accessible to the current user, which are the data guides returned to the user by the ALL_JSON_DATAGUIDE view.

Related Views

- DBA_JSON_DATAGUIDE_FIELDS extracts the path and type information from all the data guides in the database, which are the data guides returned by the DBA_JSON_DATAGUIDE view.
- USER_JSON_DATAGUIDE_FIELDS extracts the path and type information from all the data guides in the current user's schema, which are the data guides returned to the user by the USER_JSON_DATAGUIDE view. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table containing the JSON column
TABLE_NAME	VARCHAR2(128)		Name of the table containing the JSON column
COLUMN_NAME	VARCHAR2(128)		Name of the JSON column that has data guide enabled
PATH	VARCHAR2(4000)		Path to the JSON field in the data guide
TYPE	VARCHAR2(40)		Type of the JSON field in the data guide
LENGTH	NUMBER		Maximum length of the JSON field value, in bytes

See Also

- "DBA_JSON_DATAGUIDE_FIELDS"
- "USER_JSON_DATAGUIDE_FIELDS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.250 ALL_JSON_DATAGUIDES

ALL_JSON_DATAGUIDES provides information on the JavaScript Object Notation (JSON) columns accessible to the current user that have data guide enabled.

Related Views

- DBA_JSON_DATAGUIDES provides information on all JavaScript Object Notation (JSON) columns in the database that have data guide enabled.
- USER_JSON_DATAGUIDES provides information on the JavaScript Object Notation (JSON) columns owned by the current user that have data guide enabled. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table containing the JSON column
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table containing the JSON column
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the JSON column that has data guide enabled
DATAGUIDE	CLOB		The data guide of the JSON column in flat format

See Also

- "DBA_JSON_DATAGUIDES"
- "USER_JSON_DATAGUIDES"

2.251 ALL_LIBRARIES

ALL_LIBRARIES describes the libraries accessible to the current user.

Related Views

- DBA_LIBRARIES describes all libraries in the database.
- USER_LIBRARIES describes the libraries owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the library
LIBRARY_NAME	VARCHAR2(128)	NOT NULL	Library name
FILE_SPEC	VARCHAR2(2000)		Operating system file specification associated with the library
DYNAMIC	VARCHAR2(1)		Indicates whether the library is dynamically loadable (Y) or not (N)
STATUS	VARCHAR2(7)		Status of the library: • N/A • VALID • INVALID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
AGENT	VARCHAR2(128)		Agent of the library
LEAF_FILENAME	VARCHAR2(2000)		Leaf filename of the library
ORIGIN_CON_ID	VARCHAR2(256)		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)

See Also

- "DBA_LIBRARIES"
- "USER_LIBRARIES"

2.252 ALL_LOB_PARTITIONS

ALL_LOB_PARTITIONS displays the LOB partitions in the tables accessible to the current user.

Related Views

- DBA_LOB_PARTITIONS displays all LOB partitions in the database.
- USER_LOB_PARTITIONS displays the LOB partitions owned by the current user. This view does not display the TABLE_OWNER column.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table
COLUMN_NAME	VARCHAR2(4000)		Name of the LOB column
LOB_NAME	VARCHAR2(128)		Name of the partitioned LOB item
PARTITION_NAME	VARCHAR2(128)		Name of the table partition
LOB_PARTITION_NAME	VARCHAR2(128)		Name of the LOB data partition
LOB_INDPART_NAME	VARCHAR2(128)		Name of the corresponding LOB index partition
PARTITION_POSITION	NUMBER		Position of the LOB data partition within the LOB item
COMPOSITE	VARCHAR2(3)		Indicates whether the partition is composite (YES) or not (NO)
CHUNK	NUMBER		Value of the CHUNK attribute of the LOB data partition
PCTVERSION	NUMBER		Value of the PCTVERSION attribute of the LOB data partition

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CACHE	VARCHAR2(10)		Indicates whether and how the LOB data is to be cached in the buffer cache: • YES - LOB data is placed in the buffer cache • NO - LOB data either is not brought into the buffer cache or is brought into the buffer cache and placed at the least recently used end of the LRU list • CACHEREADS - LOB data is brought into the buffer cache only during read operations but not during write operations
IN_ROW	VARCHAR2(3)		Indicates whether the STORAGE IN ROW attribute is enabled for the LOB data partition (YES) or not (NO)
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the LOB data partition
INITIAL_EXTENT	VARCHAR2(40)		Size (in bytes) of the initial extent of the LOB data partition, or DEFAULT
NEXT_EXTENT	VARCHAR2(40)		Size (in bytes) of secondary extents of the LOB data partition, or DEFAULT
MIN_EXTENTS	VARCHAR2(40)		Minimum number of extents allowed in the segment of the LOB data partition, or DEFAULT
MAX_EXTENTS	VARCHAR2(40)		Maximum number of extents allowed in the segment of the LOB data partition, or DEFAULT
MAX_SIZE	VARCHAR2(40)		Maximum number of blocks allowed in the segment of the LOB data partition, or DEFAULT
RETENTION	VARCHAR2(7)		Retention option. Possible values for a SecureFiles segment: • NONE • AUTO • MIN • MAX • DEFAULT • INVALID Possible values for a BasicFiles segment: • YES • NO
MINRETENTION	VARCHAR2(40)		Minimum retention duration for a SecureFiles segment, or DEFAULT
PCT_INCREASE	VARCHAR2(40)		Percentage increase in extent size for the LOB data partition, or DEFAULT
FREELISTS	VARCHAR2(40)		Number of process freelists allocated in the segment of the LOB data partition, or DEFAULT
FREELIST_GROUPS	VARCHAR2(40)		Number of freelist groups allocated in the segment of the LOB data partition, or DEFAULT
LOGGING	VARCHAR2(7)		Indicates whether or not changes to the LOB are logged: • NONE - Not specified **See Also:** the *_LOB_SUBPARTITIONS view • YES • NO

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BUFFER_POOL	VARCHAR2(7)		Buffer pool for the LOB partition blocks: • DEFAULT • KEEP • RECYCLE • NULL
FLASH_CACHE	VARCHAR2(7)		Database Smart Flash Cache hint to be used for partition blocks: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)		Cell flash cache hint to be used for partition blocks: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
ENCRYPT	VARCHAR2(4)		Indicates whether or not the LOB is encrypted. Possible values for SecureFiles: • YES • NO Possible value for BasicFiles: • NONE - Not applicable
COMPRESSION	VARCHAR2(6)		Level of compression used for this LOB. Possible values for SecureFiles: • LOW • MEDIUM • HIGH • NO - Compression is off Possible value for BasicFiles: • NONE - Not applicable
DEDUPLICATION	VARCHAR2(15)		Kind of deduplication used for this LOB. Possible values for SecureFiles: • LOB - Deduplicate • NO - Keep duplicates Possible value for BasicFiles: • NONE - Not applicable
SECUREFILE	VARCHAR2(3)		Indicates whether the LOB is SecureFiles (YES) or not (NO)
SEGMENT_CREATED	VARCHAR2(3)		Indicates whether the LOB partition segment has been created (YES) or not (NO); N/A indicates that this LOB is subpartitioned and no segment exists at the partition level

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_LOB_PARTITIONS"
- "USER_LOB_PARTITIONS"

2.253 ALL_LOB_SUBPARTITIONS

ALL_LOB_SUBPARTITIONS displays partition-level attributes of the LOB data subpartitions accessible to the current user.

Related Views

- DBA_LOB_SUBPARTITIONS displays partition-level attributes of all LOB data subpartitions in the database.
- USER_LOB_SUBPARTITIONS displays the LOB subpartitions owned by the current user. This view does not display the TABLE_OWNER column.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
COLUMN_NAME	VARCHAR2(4000)		Name of the LOB column
LOB_NAME	VARCHAR2(128)	NOT NULL	Name of the partitioned LOB item
LOB_PARTITION_NAME	VARCHAR2(128)		Name of the LOB data partition to which this LOB data subpartition belongs
SUBPARTITION_NAME	VARCHAR2(128)		Name of the table subpartition to which this LOB subpartition corresponds
LOB_SUBPARTITION_NAME	VARCHAR2(128)		Name of the LOB subpartition
LOB_INDSUBPART_NAME	VARCHAR2(128)		Name of the corresponding LOB index subpartition
SUBPARTITION_POSITION	NUMBER		Position of the LOB data partition within the LOB item
CHUNK	NUMBER		Value of the CHUNK attribute of the LOB data partition
PCTVERSION	NUMBER	NOT NULL	Value of the PCTVERSION attribute of the LOB data partition
CACHE	VARCHAR2(10)		Indicates whether and how the LOB data is to be cached in the buffer cache: • YES - LOB data is placed in the buffer cache • NO - LOB data either is not brought into the buffer cache or is brought into the buffer cache and placed at the least recently used end of the LRU list • CACHEREADS - LOB data is brought into the buffer cache only during read operations but not during write operations
IN_ROW	VARCHAR2(3)		Indicates whether the STORAGE IN ROW attribute of the LOB data partition is enabled (YES) or not (NO)
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of the tablespace containing the LOB data partition
INITIAL_EXTENT	VARCHAR2(40)		Size in bytes of the initial extent for the LOB data partition

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
NEXT_EXTENT	VARCHAR2(40)		Size in bytes of secondary extents for the LOB data partition
MIN_EXTENTS	VARCHAR2(40)	NOT NULL	Minimum number of extents allowed in the segment of the LOB data partition
MAX_EXTENTS	VARCHAR2(40)	NOT NULL	Maximum number of extents allowed in the segment of the LOB data partition
MAX_SIZE	VARCHAR2(40)		Maximum number of blocks allowed in the segment of the LOB data partition
RETENTION	VARCHAR2(7)		Retention option. Possible values for a SecureFiles segment: • NONE • AUTO • MIN • MAX • DEFAULT • INVALID Possible values for a BasicFiles segment: • YES • NO
MINRETENTION	VARCHAR2(40)		Minimum retention duration for a SecureFiles segment
PCT_INCREASE	VARCHAR2(40)		Percentage increase in extent size for the LOB data partition
FREELISTS	VARCHAR2(40)		Number of process freelists allocated in the segment of the LOB data partition
FREELIST_GROUPS	VARCHAR2(40)		Number of freelist groups allocated in the segment of the LOB data partition
LOGGING	VARCHAR2(3)		Indicates whether or not changes to the LOB are logged: • YES • NO
BUFFER_POOL	VARCHAR2(7)		Buffer pool to be used for the LOB data partition blocks: • DEFAULT • KEEP • RECYCLE • NULL
FLASH_CACHE	VARCHAR2(7)		Database Smart Flash Cache hint to be used for subpartition blocks: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)		Cell flash cache hint to be used for subpartition blocks: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ENCRYPT	VARCHAR2(4)		Indicates whether or not the LOB is encrypted. Possible values for SecureFiles: • YES • NO Possible value for BasicFiles: • NONE - Not applicable
COMPRESSION	VARCHAR2(6)		Level of compression used for this LOB. Possible values for SecureFiles: • LOW • MEDIUM • HIGH • NO - Compression is off Possible value for BasicFiles: • NONE - Not applicable
DEDUPLICATION	VARCHAR2(15)		Kind of deduplication used for this LOB. Possible values for SecureFiles: • LOB - Deduplicate • NO - Keep duplicates Possible value for BasicFiles: • NONE - Not applicable
SECUREFILE	VARCHAR2(3)		Indicates whether the LOB is SecureFiles (YES) or not (NO)
SEGMENT_CREATED	VARCHAR2(3)		Indicates whether the LOB subpartition segment has been created (YES) or not (NO); N/A indicates that this LOB is not subpartitioned

See Also

- "DBA_LOB_SUBPARTITIONS"
- "USER_LOB_SUBPARTITIONS"

2.254 ALL_LOB_TEMPLATES

ALL_LOB_TEMPLATES describes the LOB subpartition templates accessible to the current user.

Related Views

- DBA_LOB_TEMPLATES describes all LOB subpartition templates in the database.
- USER_LOB_TEMPLATES describes the LOB subpartition templates owned by the current user. This view does not display the USER_NAME column.

Column	Datatype	NULL	Description
USER_NAME	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
LOB_COL_NAME	VARCHAR2(4000)		Name of the LOB column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SUBPARTITION_NAME	VARCHAR2(132)	NOT NULL	Name of the subpartition
LOB_SEGMENT_NAME	VARCHAR2(132)	NOT NULL	Name of the LOB segment
TABLESPACE_NAME	VARCHAR2(30)		Tablespace name of the subpartition

See Also

- "DBA_LOB_TEMPLATES"
- "USER_LOB_TEMPLATES"

2.255 ALL_LOBS

ALL_LOBS displays the large objects (LOBs) contained in tables accessible to the current user. LOBs include binary large objects (BLOBs) and character large objects (CLOBs). Binary files (BFILEs) are stored outside the database, so they are not displayed by this view or the related views.

Related Views

- DBA_LOBS describes all LOBs in the database.
- USER_LOBS describes the LOBs owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the object containing the LOB
TABLE_NAME	VARCHAR2(128)		Name of the object containing the LOB
COLUMN_NAME	VARCHAR2(4000)		Name of the LOB column or attribute
SEGMENT_NAME	VARCHAR2(128)		Name of the LOB segment
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the LOB segment
INDEX_NAME	VARCHAR2(128)		Name of the LOB index
CHUNK	NUMBER		Size (in bytes) of the LOB chunk as a unit of allocation or manipulation
PCTVERSION	NUMBER		Maximum percentage of the LOB space used for versioning
RETENTION	NUMBER		Maximum time duration for versioning of the LOB space
FREEPOOLS	NUMBER		Number of freepools for this LOB segment

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CACHE	VARCHAR2(10)		Indicates whether and how the LOB data is to be cached in the buffer cache: • YES - LOB data is placed in the buffer cache • NO - LOB data either is not brought into the buffer cache or is brought into the buffer cache and placed at the least recently used end of the LRU list • CACHEREADS - LOB data is brought into the buffer cache only during read operations but not during write operations
LOGGING	VARCHAR2(7)		Indicates whether or not changes to the LOB are logged: • NONE - Not specified **See Also:** the *_LOB_SUBPARTITIONS view • YES • NO
ENCRYPT	VARCHAR2(4)		Indicates whether or not the LOB is encrypted. Possible values for SecureFiles: • YES • NO Possible value for BasicFiles: • NONE - Not applicable
COMPRESSION	VARCHAR2(6)		Level of compression used for this LOB. Possible values for SecureFiles: • LOW • MEDIUM • HIGH • NO - Compression is off Possible value for BasicFiles: • NONE - Not applicable
DEDUPLICATION	VARCHAR2(15)		Kind of deduplication used for this LOB. Possible values for SecureFiles: • LOB - Deduplicate • NO - Keep duplicates Possible value for BasicFiles: • NONE - Not applicable
IN_ROW	VARCHAR2(3)		Indicates whether some LOBs are stored inline with the base row (YES) or not (NO). For partitioned objects, refer to the *_LOB_PARTITIONS and *_PART_LOBS views.
FORMAT	VARCHAR2(15)		Indicates whether the LOB storage format depends on the endianness of the platform: • NOT APPLICABLE • ENDIAN SPECIFIC • ENDIAN NEUTRAL
PARTITIONED	VARCHAR2(3)		Indicates whether the LOB column is in a partitioned table (YES) or not (NO)
SECUREFILE	VARCHAR2(3)		Indicates whether the LOB is SecureFiles (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SEGMENT_CREATED	VARCHAR2(3)		Indicates whether the LOB segment has been created (YES) or not (NO)
RETENTION_TYPE	VARCHAR2(7)		Type of retention used for this LOB. Possible values for SecureFiles: • NONE • AUTO • MIN • MAX • DEFAULT • INVALID Possible values for BasicFiles: • YES • NO
RETENTION_VALUE	NUMBER		Minimum retention time (in seconds). This column is only meaningful for SecureFiles with RETENTION_TYPE set to MIN.

See Also

- "DBA_LOBS"
- "USER_LOBS"

2.256 ALL_LOG_GROUP_COLUMNS

ALL_LOG_GROUP_COLUMNS describes columns that are accessible to the current user and that are specified in log groups.

Related Views

- DBA_LOG_GROUP_COLUMNS describes all columns in the database that are specified in log groups.
- USER_LOG_GROUP_COLUMNS describes columns that are owned by the current user and that are specified in log groups.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the log group definition
LOG_GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the log group definition
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table in which the log group is defined
COLUMN_NAME	VARCHAR2(4000)		Name of the column or attribute of the object type column specified in the log group definition
POSITION	NUMBER		Original position of the column or attribute in the definition of the object
LOGGING_PROPERTY	VARCHAR2(6)		Indicates whether the column or attribute would be supplementally logged (LOG) or not (NO LOG)

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_LOG_GROUP_COLUMNS"
- "USER_LOG_GROUP_COLUMNS"

2.257 ALL_LOG_GROUPS

ALL_LOG_GROUPS describes the log group definitions on the tables accessible to the current user.

Related Views

- DBA_LOG_GROUPS describes the log group definitions on all tables in the database.
- USER_LOG_GROUPS describes the log group definitions on the tables owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the log group definition
LOG_GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the log group definition
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table on which the log group is defined
LOG_GROUP_TYPE	VARCHAR2(28)		Type of the log group: • ALL COLUMN LOGGING • FOREIGN KEY LOGGING • PRIMARY KEY LOGGING • UNIQUE KEY LOGGING • USER LOG GROUP
ALWAYS	VARCHAR2(11)		Indicates whether the log group is an unconditional log group (ALWAYS) or a conditional log group (CONDITIONAL)
GENERATED	VARCHAR2(14)		Indicates whether the name of the supplemental log group was system generated (GENERATED NAME) or not (USER NAME)

See Also

- "DBA_LOG_GROUPS"
- "USER_LOG_GROUPS"

2.258 ALL_MEASURE_FOLDER_CONTENTS

ALL_MEASURE_FOLDER_CONTENTS describes the contents of the OLAP measure folders accessible to the current user.

Related Views

- DBA_MEASURE_FOLDER_CONTENTS describes the contents of all OLAP measure folders in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_MEASURE_FOLDER_CONTENTS describes the contents of the OLAP measure folders owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the measure folder
MEASURE_FOLDER_NAME	VARCHAR2(128)	NOT NULL	Name of a measure folder
CUBE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the cube
CUBE_NAME	VARCHAR2(128)	NOT NULL	Name of a cube included in the measure folder
MEASURE_NAME	VARCHAR2(128)	NOT NULL	Name of a measure in the cube
ORDER_NUM	NUMBER	NOT NULL	Order number of the measure in the folder

See Also

- "DBA_MEASURE_FOLDER_CONTENTS"
- "USER_MEASURE_FOLDER_CONTENTS"

2.259 ALL_MEASURE_FOLDER_SUBFOLDERS

ALL_MEASURE_FOLDER_SUBFOLDERS describes the OLAP measure folders contained within the OLAP measure folders accessible to the user.

Related Views

- DBA_MEASURE_FOLDER_SUBFOLDERS describes the OLAP measure folders contained within the database OLAP measure folders.
- USER_MEASURE_FOLDER_SUBFOLDERS describes the OLAP measure folders contained within the OLAP measure folders owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the OLAP measure folder that contains a subfolder
MEASURE_FOLDER_NAME	VARCHAR2(128)	NOT NULL	Name of the OLAP measure folder that contains a subfolder
MEASURE_SUBFOLDER_OWNER	VARCHAR2(128)	NOT NULL	Owner of the OLAP measure folder subfolder
MEASURE_SUBFOLDER_NAME	VARCHAR2(128)	NOT NULL	Name of the owning OLAP measure folder subfolder

See Also

- "DBA_MEASURE_FOLDER_SUBFOLDERS"
- "USER_MEASURE_FOLDER_SUBFOLDERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.260 ALL_MEASURE_FOLDERS

ALL_MEASURE_FOLDERS describes the OLAP measure folders accessible to the current user.

Related Views

- DBA_MEASURE_FOLDERS describes all OLAP measure folders in the database.
- USER_MEASURE_FOLDERS describes the OLAP measure folders owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the measure folder
MEASURE_FOLDER_NAME	VARCHAR2(128)	NOT NULL	Name of a measure folder
MEASURE_FOLDER_ID	NUMBER	NOT NULL	ID of a measure folder
DESCRIPTION	NVARCHAR2(300)		Description of the measure folder in the session language

See Also

- "DBA_MEASURE_FOLDERS"
- "USER_MEASURE_FOLDERS"

2.261 ALL_METADATA_PROPERTIES

ALL_METADATA_PROPERTIES describes OLAP metadata properties in the database that are accessible to the current user.

Related Views

- DBA_METADATA_PROPERTIES describes OLAP metadata properties in the database.
- USER_METADATA_PROPERTIES describes OLAP metadata properties in the current user's schema. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the OLAP metadata property
OWNING_OBJECT_ID	NUMBER		Dictionary ID of the OLAP metadata property owner
OWNING_TYPE	VARCHAR2(23)		Owning type of the OLAP metadata property
PROPERTY_ID	NUMBER		Dictionary Id of the OLAP metadata property
PROPERTY_KEY	VARCHAR2(128)		Key of the OLAP metadata property
PROPERTY_VALUE	CLOB		Value of the OLAP metadata property
PROPERTY_ORDER	NUMBER		Order number of the OLAP metadata property

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_METADATA_PROPERTIES"
- "USER_METADATA_PROPERTIES"

2.262 ALL_METHOD_PARAMS

ALL_METHOD_PARAMS describes the method parameters of the object types accessible to the current user.

Related Views

- DBA_METHOD_PARAMS describes the method parameters of all object types in the database.
- USER_METHOD_PARAMS describes the method parameters of the object types owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the type
TYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the type
METHOD_NAME	VARCHAR2(128)	NOT NULL	Name of the method
METHOD_NO	NUMBER	NOT NULL	For an overloaded method, a number distinguishing this method from others of the same. Do not confuse this number with the object ID.
PARAM_NAME	VARCHAR2(128)	NOT NULL	Name of the parameter
PARAM_NO	NUMBER	NOT NULL	Parameter number (position)
PARAM_MODE	VARCHAR2(6)		Mode of the parameter (IN, OUT, IN OUT)
PARAM_TYPE_MOD	VARCHAR2(7)		Whether this parameter is a REF to another object
PARAM_TYPE_OWNER	VARCHAR2(128)		Owner of the type of the parameter
PARAM_TYPE_NAME	VARCHAR2(128)		Name of the type of the parameter
CHARACTER_SET_NAME	VARCHAR2(44)		Whether the character set or the method is fixed-length character set (CHAR_CS) or fixed-length national character set (NCHAR_CS), or a particular character set specified by the user

See Also

- "DBA_METHOD_PARAMS"
- "USER_METHOD_PARAMS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.263 ALL_METHOD_RESULTS

ALL_METHOD_RESULTS describes the method results of the object types accessible to the current user.

Related Views

- DBA_METHOD_RESULTS describes the method results of all object types in the database.
- USER_METHOD_RESULTS describes the method results of the object types owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the type
TYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the type
METHOD_NAME	VARCHAR2(128)	NOT NULL	Name of the method
METHOD_NO	NUMBER	NOT NULL	For an overloaded method, a number distinguishing this method from others of the same. Do not confuse this number with the object ID.
RESULT_TYPE_MOD	VARCHAR2(7)		Whether this result is a REF to another object
RESULT_TYPE_OWNER	VARCHAR2(128)		Owner of the type of the result
RESULT_TYPE_NAME	VARCHAR2(128)		Name of the type of the result
CHARACTER_SET_NAME	VARCHAR2(44)		Whether the character set or the method is fixed-length character set (CHAR_CS) or fixed-length national character set (NCHAR_CS), or a particular character set specified by the user

See Also

- "DBA_METHOD_RESULTS"
- "USER_METHOD_RESULTS"

2.264 ALL_MINING_ALGORITHMS

ALL_MINING_ALGORITHMS describes the settings for a current user.

Column	Datatype	NULL	Description
ALGORITHM_NAME	VARCHAR2(128)		Algorithm used by the model.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MINING_FUNCTION	VARCHAR2(30)		Function of the mining model. The mining function is specified when the model is built: • CLASSIFICATION • REGRESSION • CLUSTERING • FEATURE_EXTRACTION • ASSOCIATION_RULES • ATTRIBUTE_IMPORTANCE • ANOMALY DETECTION
ALGORITHM_TYPE	VARCHAR2(20)		Algorithm type of the model
ALGORITHM_METADATA	CLOB		Metadata of the algorithm
DESCRIPTION	VARCHAR2(4000)		Description of the algorithm

2.265 ALL_MINING_MODEL_ATTRIBUTES

ALL_MINING_MODEL_ATTRIBUTES describes the attributes of the mining models accessible to the current user. Only the attributes in the model signature are included in this view. The attributes in the model signature correspond to the columns in the training data that were used to build the model.
Mining models are schema objects created by Oracle Data Mining.

Related Views

- DBA_MINING_MODEL_ATTRIBUTES describes the attributes of all mining models in the database.
- USER_MINING_MODEL_ATTRIBUTES describes the attributes of the mining models owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the mining model
MODEL_NAME	VARCHAR2(128)	NOT NULL	Name of the mining model
ATTRIBUTE_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute
ATTRIBUTE_TYPE	VARCHAR2(11)	–	Logical type of the attribute. The type is identified during the model build or apply process: • NUMERICAL: Numeric data • CATEGORICAL: Character data • TEXT: Unstructured text data • PARTITION: The input signature column is used for the partitioning key • MIXED: The input signature column takes on more than one attribute type. This is due to user-defined embedded transformations that allow an input column to be transformed into multiple independent mining attributes, including mining attributes of different types.
DATA_TYPE	VARCHAR2(106)	–	Data type of the attribute

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DATA_LENGTH	NUMBER	–	Length of the data type
DATA_PRECISION	NUMBER	–	Precision of a fixed point number. Precision, which is the total number of significant decimal digits, is represented as *p* in the data type NUMBER(p,s).
DATA_SCALE	NUMBER	–	Scale of a fixed point number. Scale, which is the number of digits from the decimal to the least significant digit, is represented as *s* in the data type NUMBER(p,s).
USAGE_TYPE	VARCHAR2(8)	–	Indicates whether the attribute was used to construct the model (ACTIVE) or not (INACTIVE). Some attributes may be eliminated by transformations or algorithmic processing. The *_MINING_MODEL_ATTRIBUTES view only lists the attributes used by the model, therefore the value of this column is always ACTIVE.
TARGET	VARCHAR2(3)	–	Indicates whether the attribute is the target of a predictive model (YES) or not (NO). The target describes the result that is produced when the model is applied.
ATTRIBUTE_SPEC	VARCHAR2(4000)	–	One or more keywords that identify special treatment for the attribute during model build. Values are: • FORCE_IN: (GLM only) When feature selection is enabled, forces the inclusion of the attribute in the model build. Feature selection is disabled by default. If the model is not using GLM with feature selection enabled, this value is ignored. • NOPREP: When ADP is on, prevents automatic transformation of the attribute. If ADP is OFF, this value is ignored. • TEXT: Causes the attribute to be treated as unstructured text data. The TEXT value supports three subsettings: POLICY_NAME, MAX_FEATURES, TOKEN_TYPE, and MIN_DOCUMENTS. Subsettings are specified as name:value pairs within parentheses. For example: (POLICY_NAME:mypolicy)(MAX_FEATURES:2000)(TOKEN_TYPE:THEME). See *Oracle Data Mining User's Guide* for details. • NULL: The ATTRIBUTE_SPEC for this attribute is NULL. ATTRIBUTE_SPEC is a parameter to the PL/SQL procedure DBMS_DATA_MINING_TRANSFORM.SET_TRANSFORM. See *Oracle Database PL/SQL Packages and Types Reference* for details.

See Also

Oracle Data Mining User's Guide for more information about the attributes of machine learning models

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.266 ALL_MINING_MODEL_PARTITIONS

ALL_MINING_MODEL_PARTITIONS describes all the model partitions accessible to the user.

Related Views

- DBA_MINING_MODEL_PARTITIONS describes all the model partitions accessible to the system.
- USER_MINING_MODEL_PARTITIONS describes the user's own model partitions. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Name of the model owner
MODEL_NAME	VARCHAR2(128)	NOT NULL	Name of the model
PARTITION_NAME	VARCHAR2(128)	–	Name of the model partition
POSITION	NUMBER	–	Column position number for partitioning column. Column position represents the position of the column in a multi-column partitioning key, or 1 for a unary column partitioning key.
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column used for partitioning
COLUMN_VALUE	VARCHAR2(4000)	–	Value of the column for this partition

2.267 ALL_MINING_MODEL_SETTINGS

ALL_MINING_MODEL_SETTINGS describes the settings of the mining models accessible to the current user.

Mining models are schema objects created by Oracle Data Mining.

Related Views

- DBA_MINING_MODEL_SETTINGS describes the settings of all mining models in the database.
- USER_MINING_MODEL_SETTINGS describes the settings of the mining models owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the mining model
MODEL_NAME	VARCHAR2(128)	NOT NULL	Name of the mining model
SETTING_NAME	VARCHAR2(30)	NOT NULL	Name of the setting
SETTING_VALUE	VARCHAR2(4000)	–	Value of the setting
SETTING_TYPE	VARCHAR2(7)	–	Indicates whether the default value (DEFAULT) or a user-specified value (INPUT) is used by the model

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database PL/SQL Packages and Types Reference* for descriptions of model settings

2.268 ALL_MINING_MODEL_VIEWS

ALL_MINING_MODEL_VIEWS provides a description of all the model views accessible to the user.

Related Views

- DBA_MINING_MODEL_VIEWS provides a description of all the model views in the database.
- USER_MINING_MODEL_VIEWS provides a description of the user's own model views. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the model view
MODEL_NAME	VARCHAR2(128)	NOT NULL	Name of the model to which model views belongs
VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the model view
VIEW_TYPE	VARCHAR2(128)	–	Type of the model view

> **See Also**
>
> "ALL_MINING_MODEL_VIEWS" in *Oracle Data Mining User's Guide*

2.269 ALL_MINING_MODEL_XFORMS

ALL_MINING_MODEL_XFORMS describes the user-specified transformations embedded in all models accessible to the user.

Related Views

- DBA_MINING_MODEL_XFORMS describes the user-specified transformations embedded in all models accessible in the system.
- USER_MINING_MODEL_XFORMS describes the user-specified transformations embedded with the user's own models. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Name of the model owner
MODEL_NAME	VARCHAR2(128)	NOT NULL	Name of the model
ATTRIBUTE_NAME	VARCHAR2(128)		Name of the attribute used in the transformation
ATTRIBUTE_SUBNAME	VARCHAR2(4000)		Subname of the attribute used in the transformation
ATTRIBUTE_SPEC	VARCHAR2(4000)		Attribute specification provided to model training
EXPRESSION	CLOB		Transformation expression provided to model training

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
REVERSE	VARCHAR2(3)		Indicates whether the specified transformation is a reverse transformation (YES) or a forward expression (NO)

2.270 ALL_MINING_MODELS

ALL_MINING_MODELS describes the mining models accessible to the current user.

Mining models are schema objects created by Oracle Data Mining.

Related Views

- DBA_MINING_MODELS describes all mining models in the database.
- USER_MINING_MODELS describes the mining models owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the mining model
MODEL_NAME	VARCHAR2(128)	NOT NULL	Name of the mining model
MINING_FUNCTION	VARCHAR2(30)	NOT NULL	Function of the mining model. The function identifies the class of problems that can be solved by this model. The mining function is specified when the model is built: • CLASSIFICATION • REGRESSION • CLUSTERING • FEATURE_EXTRACTION • ASSOCIATION_RULES • ATTRIBUTE_IMPORTANCE
ALGORITHM	VARCHAR2(30)	NOT NULL	Algorithm used by the model. Each mining function has a default algorithm. The default can be overridden with a model setting (see *_MINING_MODEL_SETTINGS): • CUR_DECOMPOSITION • NAIVE_BAYES • DECISION_TREE • EXPLICIT_SEMANTIC_ANALYS • EXPONENTIAL_ SMOOTHING • SUPPORT_VECTOR_MACHINES • KMEANS • O_CLUSTER • NONNEGATIVE_MATRIX_FACTOR • NEURAL_NETWORK • GENERALIZED_LINEAR_MODEL • APRIORI_ASSOCIATION_RULES • MINIMUM_DESCRIPTION_LENGTH • EXPECTATION_MAXIMIZATION • RANDOM_FOREST • SINGULAR_VALUE_DECOMP • R_EXTENSIBLE

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ALGORITHM_TYPE	VARCHAR2(10)	NOT NULL	Algorithm type of the model
CREATION_DATE	DATE	NOT NULL	Date that the model was created
BUILD_DURATION	NUMBER	NOT NULL	Time (in seconds) of the model build process
MODEL_SIZE	NUMBER	NOT NULL	Size of the model (in megabytes)
PARTITIONED	VARCHAR2(3)	NOT NULL	Indicates whether the model is partitioned or not. Possible values: • YES: The model is partitioned. • NO: The model is not partitioned
COMMENTS	VARCHAR2(4000)	NOT NULL	Comment applied to the model with a SQL COMMENT statement

See Also

- *Oracle Data Mining User's Guide* for information about mining model schema objects
- *Oracle Data Mining Concepts* for an introduction to Data Mining

2.271 ALL_MVIEW_AGGREGATES

ALL_MVIEW_AGGREGATES describes the grouping functions (aggregate operations) that appear in the SELECT list of materialized aggregate views accessible to the current user.

Related Views

- DBA_MVIEW_AGGREGATES describes all such grouping functions defined for all materialized views in the database.
- USER_MVIEW_AGGREGATES describes all such grouping functions defined for all materialized views owned by the current user.

Note

All three views exclude materialized views that reference remote tables or that include references to a nonstatic value such as SYSDATE or USER. These views also exclude materialized views that were created as "snapshots" before Oracle8*i* and that were never altered to enable query rewrite.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
MVIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
POSITION_IN_SELECT	NUMBER	NOT NULL	Ordinal position of this aggregation within the SELECT list. For the position of nonaggregate elements of the select list, see "ALL_MVIEW_KEYS".
CONTAINER_COLUMN	VARCHAR2(128)	NOT NULL	Name of this column in the container table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
AGG_FUNCTION	VARCHAR2(8)		Aggregation function
DISTINCTFLAG	VARCHAR2(1)		Indicates whether this aggregation is distinct (Y) or not (N)
MEASURE	LONG		SQL text of the measure, excluding the aggregation function. Equal to * for COUNT(*).

See Also

- "DBA_MVIEW_AGGREGATES"
- "USER_MVIEW_AGGREGATES"

2.272 ALL_MVIEW_ANALYSIS

ALL_MVIEW_ANALYSIS describes the materialized views accessible to the current user. It provides additional information for analysis by applications. Minimal information is displayed for materialized views that do not support query rewrite (such as materialized views with remote master tables or nondeterministic functions).

Related Views

- DBA_MVIEW_ANALYSIS describes all such materialized views in the database.
- USER_MVIEW_ANALYSIS describes all such materialized views owned by the current user.

Note

All of the information in these views is also displayed in ALL_MVIEWS and its related views. Oracle recommends that you refer to ALL_MVIEWS for this information instead of these views.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
MVIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
MVIEW_TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the container table (see next column)
CONTAINER_NAME	VARCHAR2(128)		Name of the internal container in which the materialized view data is held. Normally this is the same as MVIEW_NAME. For materialized views created before Oracle8*i*, Oracle Database attaches the 6-byte prefix SNAP$_. If MVIEW_NAME has more than 19 bytes, then Oracle Database truncates the name to 19 bytes and adds a 4-byte sequence number as a suffix to produce a nonambiguous CONTAINER_NAME.
LAST_REFRESH_SCN	NUMBER		System change number (SCN) of the last refresh operation
LAST_REFRESH_DATE	DATE		SYSDATE of the last refresh

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
REFRESH_METHOD	VARCHAR2(8)		Default refresh method: • FORCE • FAST • COMPLETE • NEVER
SUMMARY	VARCHAR2(1)		Indicates whether this materialized view includes a GROUP BY clause or aggregation (Y) or not (N)
FULLREFRESHTIM	NUMBER		Approximate refresh time, in seconds, for full refresh
INCREFRESHTIM	NUMBER		Approximate refresh time, in seconds, for fast refresh
CONTAINS_VIEWS	VARCHAR2(1)		Indicates whether this materialized view contains a view in its definition (Y) or not (N)
UNUSABLE	VARCHAR2(1)		Indicates whether this materialized view is UNUSABLE (inconsistent data) (Y) or not (N). A materialized view can be UNUSABLE if a system failure occurs during a full refresh.
RESTRICTED_SYNTAX	VARCHAR2(1)		Indicates whether this materialized view had a restriction in its defining query that limits the use of query rewrite (Y) or not (N). More complete information is provided by the REWRITE_CAPABILITY column of the *_MVIEWS view.
INC_REFRESHABLE	VARCHAR2(1)		Indicates whether this materialized view can be fast refreshed (Y) or not (N)
KNOWN_STALE	VARCHAR2(1)		Indicates whether the data contained in the materialized view is known to be inconsistent with the master table data because that has been updated since the last successful refresh (Y) or not (N)
INVALID	VARCHAR2(1)		Indicates whether this materialized view is in an invalid state (inconsistent metadata) (Y) or not (N)
REWRITE_ENABLED	VARCHAR2(1)		Indicates whether this materialized view is currently enabled for query rewrite (Y) or not (N)
QUERY_LEN	NUMBER		Length (in bytes) of the query field
QUERY	LONG		SELECT expression of the materialized view definition
REVISION	NUMBER	NOT NULL	Reserved for internal use

See Also

- "DBA_MVIEW_ANALYSIS"
- "USER_MVIEW_ANALYSIS"

2.273 ALL_MVIEW_COMMENTS

ALL_MVIEW_COMMENTS displays comments on the materialized views accessible to the current user.

Related Views

- DBA_MVIEW_COMMENTS displays comments on the materialized views in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_MVIEW_COMMENTS displays comments on the materialized views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
MVIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
COMMENTS	VARCHAR2(4000)		Comment on the materialized view

See Also

- "DBA_MVIEW_COMMENTS"
- "USER_MVIEW_COMMENTS"

2.274 ALL_MVIEW_DETAIL_PARTITION

ALL_MVIEW_DETAIL_PARTITION displays freshness information, with respect to partition change tracking (PCT) detail partitions, for the materialized views accessible to the current user.

Related Views

- DBA_MVIEW_DETAIL_PARTITION displays freshness information, with respect to PCT detail partitions, for all materialized views in the database.
- USER_MVIEW_DETAIL_PARTITION displays freshness information, with respect to PCT detail partitions, for the materialized views owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
MVIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
DETAILOBJ_OWNER	VARCHAR2(128)	NOT NULL	Owner of the detail object
DETAILOBJ_NAME	VARCHAR2(128)	NOT NULL	Name of the detail object
DETAIL_PARTITION_NAME	VARCHAR2(128)		Name of the detail object partition
DETAIL_PARTITION_POSITION	NUMBER		Position of the detail object partition
FRESHNESS	VARCHAR2(7)		Freshness state of the detail object logical partition. Possible values: • FRESH • STALE • UNKNOWN
LAST_REFRESH_TIME[1]	DATE		Date and time at which the materialized view was last refreshed

1 This column is available starting with Oracle Database 19c.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_MVIEW_DETAIL_PARTITION"
- "USER_MVIEW_DETAIL_PARTITION"

2.275 ALL_MVIEW_DETAIL_RELATIONS

ALL_MVIEW_DETAIL_RELATIONS describes the named detail relations that are either specified in the FROM list of the subquery that defines a materialized view accessible to the current user, or that are indirectly referenced through views in that FROM list. Inline views in the materialized view definition are not represented in this view or the related views.

Related Views

- DBA_MVIEW_DETAIL_RELATIONS describes all such detail relations defined for all materialized views in the database.
- USER_MVIEW_DETAIL_RELATIONS describes such detail relations defined for all materialized views owned by the current user.

Note

All three views exclude materialized views that reference remote tables or that includes references to a nonstatic value such as SYSDATE or USER. These views also exclude materialized views that were created as *snapshots* before Oracle8*i* and that were never altered to enable query rewrite.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
MVIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
DETAILOBJ_OWNER	VARCHAR2(128)	NOT NULL	Detail object owner
DETAILOBJ_NAME	VARCHAR2(128)	NOT NULL	Detail object name (that is, the name of a table or view)
DETAILOBJ_TYPE	VARCHAR2(9)		Detail object type: • TABLE • VIEW • SNAPSHOT • CONTAINER • UNDEFINED
DETAILOBJ_ALIAS	VARCHAR2(128)		Implicit or explicit alias for detail relation
DETAILOBJ_PCT	VARCHAR2(1)		Indicates whether the detail object PCT is supported (Y) or not (N)
NUM_FRESH_PCT_PARTITIONS	NUMBER		Number of fresh PCT partitions
NUM_STALE_PCT_PARTITIONS	NUMBER		Number of stale PCT partitions

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_MVIEW_DETAIL_RELATIONS"
- "USER_MVIEW_DETAIL_RELATIONS"

2.276 ALL_MVIEW_DETAIL_SUBPARTITION

ALL_MVIEW_DETAIL_SUBPARTITION displays freshness information, with respect to partition change tracking (PCT) detail subpartitions, for the materialized views accessible to the current user.

Related Views

- DBA_MVIEW_DETAIL_SUBPARTITION displays freshness information, with respect to PCT detail subpartitions, for all materialized views in the database.
- USER_MVIEW_DETAIL_SUBPARTITION displays freshness information, with respect to PCT detail subpartitions, for the materialized views owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
MVIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
DETAILOBJ_OWNER	VARCHAR2(128)	NOT NULL	Owner of the detail object
DETAILOBJ_NAME	VARCHAR2(128)	NOT NULL	Name of the detail object
DETAIL_PARTITION_NAME	VARCHAR2(128)		Name of the detail object partition
DETAIL_SUBPARTITION_NAME	VARCHAR2(128)		Name of the detail object subpartition
DETAIL_SUBPARTITION_POSITION	NUMBER		Position of the detail object subpartition
FRESHNESS	CHAR(5)		Freshness state of the detail object subpartition (FRESH or STALE)

See Also

- "DBA_MVIEW_DETAIL_SUBPARTITION"
- "USER_MVIEW_DETAIL_SUBPARTITION"

2.277 ALL_MVIEW_JOINS

ALL_MVIEW_JOINS describes joins between two columns in the WHERE clause of the subquery that defines a materialized view accessible to the current user.

Related Views

- DBA_MVIEW_JOINS describes all such joins for all materialized views in the database.
- USER_MVIEW_JOINS describes such joins for all materialized views owned by the current user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> All three views exclude materialized views that reference remote tables or that includes references to a nonstatic value such as SYSDATE or USER. These views also exclude materialized views that were created as "snapshots" before Oracle8*i* and that were never altered to enable query rewrite.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
MVIEW_NAME	VARCHAR2(128)	NOT NULL	Materialized view name
DETAILOBJ1_OWNER	VARCHAR2(128)	NOT NULL	Owner of the first object in the join[1]
DETAILOBJ1_RELATION	VARCHAR2(128)	NOT NULL	Name of the first object in the join[1]
DETAILOBJ1_COLUMN	VARCHAR2(128)	NOT NULL	Join column of the first object in the join[1]
OPERATOR	CHAR(1)		Join operator[1]
OPERATOR_TYPE	VARCHAR2(1)		Indicates whether the join is an inner join (I) or the DETAILOBJ1 table is the left side of an outer join (L)[1]
DETAILOBJ2_OWNER	VARCHAR2(128)	NOT NULL	Owner of the second object in the join[1]
DETAILOBJ2_RELATION	VARCHAR2(128)	NOT NULL	Name of the second object in the join[1]
DETAILOBJ2_COLUMN	VARCHAR2(128)	NOT NULL	Join column of the second object in the join[1]

1 These rows relate only to materialized join views and materialized aggregate views. They describe the two detail objects of a materialized view join.

> **See Also**
>
> - "DBA_MVIEW_JOINS"
> - "USER_MVIEW_JOINS"

2.278 ALL_MVIEW_KEYS

ALL_MVIEW_KEYS describes the columns or expressions in the SELECT list upon which materialized views accessible to the current user are based.

Related Views

- DBA_MVIEW_KEYS describes such columns and expressions for all materialized views in the database.
- USER_MVIEW_KEYS describes such columns and expressions for all materialized views owned by the current user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> All three views exclude materialized views that reference remote tables or that includes references to a nonstatic value such as SYSDATE or USER. These views also exclude materialized views that were created as *snapshots* before Oracle8*i* and that were never altered to enable query rewrite.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
MVIEW_NAME	VARCHAR2(128)	NOT NULL	Materialized view name
POSITION_IN_SELECT	NUMBER	NOT NULL	Ordinal position of this key within the SELECT list
CONTAINER_COLUMN	VARCHAR2(128)	NOT NULL	Name of the column in the container table
DETAILOBJ_OWNER	VARCHAR2(128)	NOT NULL	Detail object owner
DETAILOBJ_NAME	VARCHAR2(128)	NOT NULL	Detail object name (for example, the name of a table or view)
DETAILOBJ_ALIAS	VARCHAR2(128)		Implicit or explicit alias for detail relation
DETAILOBJ_TYPE	VARCHAR2(5)		Detail object type: • TABLE • VIEW
DETAILOBJ_COLUMN	VARCHAR2(128)	NOT NULL	Name of the detail relation column

> **See Also**
>
> - "DBA_MVIEW_KEYS"
> - "USER_MVIEW_KEYS"

2.279 ALL_MVIEW_LOGS

ALL_MVIEW_LOGS describes all materialized view logs accessible to the current user.

Related Views

- DBA_MVIEW_LOGS describes all materialized view logs in the database.
- USER_MVIEW_LOGS describes all materialized view logs owned by the current user.

Column	Datatype	NULL	Description
LOG_OWNER	VARCHAR2(128)		Owner of the materialized view log
MASTER	VARCHAR2(128)		Name of the master table or master materialized view whose changes are logged
LOG_TABLE	VARCHAR2(128)		Name of the table where the changes to the master table or master materialized view are logged
LOG_TRIGGER	VARCHAR2(128)		Obsolete with Oracle8*i* and later. Set to NULL. Formerly, this parameter was an after-row trigger on the master which inserted rows into the log.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ROWIDS	VARCHAR2(3)		Indicates whether rowid information is recorded (YES) or not (NO)
PRIMARY_KEY	VARCHAR2(3)		Indicates whether primary key information is recorded (YES) or not (NO)
OBJECT_ID	VARCHAR2(3)		Indicates whether object identifier information in an object table is recorded (YES) or not (NO)
FILTER_COLUMNS	VARCHAR2(3)		Indicates whether filter column information is recorded (YES) or not (NO)
SEQUENCE	VARCHAR2(3)		Indicates whether the sequence value, which provides additional ordering information, is recorded (YES) or not (NO)
INCLUDE_NEW_VALUES	VARCHAR2(3)		Indicates whether both old and new values are recorded (YES) or old values are recorded but new values are not recorded (NO)
PURGE_ASYNCHRONOUS	VARCHAR2(3)		Indicates whether the materialized view log is purged asynchronously (YES) or not (NO)
PURGE_DEFERRED	VARCHAR2(3)		Indicates whether the materialized view log is purged in a deferred manner (YES) or not (NO)
PURGE_START	DATE		For deferred purge, the purge start date
PURGE_INTERVAL	VARCHAR2(200)		For deferred purge, the purge interval
LAST_PURGE_DATE	DATE		Date of the last purge
LAST_PURGE_STATUS	NUMBER		Status of the last purge (error code or 0 for success)
NUM_ROWS_PURGED	NUMBER		Number of rows purged in the last purge
COMMIT_SCN_BASED	VARCHAR2(3)		Indicates whether the materialized view log is commit SCN-based (YES) or not (NO)
STAGING_LOG	VARCHAR2(3)		Indicates whether the materialized view log is a staging log for synchronous refresh (YES) or not (NO)

See Also

- "DBA_MVIEW_LOGS"
- "USER_MVIEW_LOGS"

2.280 ALL_MVIEW_REFRESH_TIMES

ALL_MVIEW_REFRESH_TIMES describes refresh times of the materialized views accessible to the current user.

Related Views

- DBA_MVIEW_REFRESH_TIMES describes refresh times of all materialized views in the database.
- USER_MVIEW_REFRESH_TIMES describes refresh times of the materialized views owned by the current user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
MASTER_OWNER	VARCHAR2(128)		Owner of the master table
MASTER	VARCHAR2(128)		Name of the master table
LAST_REFRESH	DATE		SYSDATE from the master site at the time of the last refresh

See Also

- "DBA_MVIEW_REFRESH_TIMES"
- "USER_MVIEW_REFRESH_TIMES"

2.281 ALL_MVIEWS

ALL_MVIEWS describes all materialized views accessible to the current user.

Related Views

- DBA_MVIEWS describes all materialized views in the database.
- USER_MVIEWS describes all materialized views owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Schema in which the materialized view was created
MVIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
CONTAINER_NAME	VARCHAR2(128)	NOT NULL	Name of the container in which the materialized view's data is held. Normally this is the same as MVIEW_NAME. For materialized views created before Oracle8*i*, Oracle Database attaches the 6-byte prefix SNAP$_. If MVIEW_NAME has more than 19 bytes, then Oracle Database truncates the name to 19 bytes and may add a 4-byte sequence number as a suffix to produce a nonambiguous CONTAINER_NAME.
QUERY	LONG		Query that defines the materialized view
QUERY_LEN	NUMBER(38)		Length (in bytes) of the defining query
UPDATABLE	VARCHAR2(1)		Indicates whether the materialized view is updatable (Y) or not (N)
UPDATE_LOG	VARCHAR2(128)		For updatable materialized views, the filename of the update log
MASTER_ROLLBACK_SEG	VARCHAR2(128)		Rollback segment for the master site or the master materialized view site
MASTER_LINK	VARCHAR2(128)		Database link for the master site or the master materialized view site
REWRITE_ENABLED	VARCHAR2(1)		Indicates whether rewrite is enabled (Y) or not (N)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
REWRITE_CAPABILITY	VARCHAR2(9)		Indicates whether the materialized view is eligible for rewrite, and if so, what rules must be followed: • NONE - Materialized view cannot be used for rewrite, because rewrite is disallowed or prevented • TEXTMATCH - Defining query of the materialized view contained restrictions on the use of query rewrite • GENERAL - Defining query of the materialized view contained no restrictions on the use of query rewrite, so Oracle Database can apply any rewrite rule that is supported
REFRESH_MODE	VARCHAR2(9)		Refresh mode of the materialized view: • COMMIT - Oracle Database refreshes this materialized view when a transaction on one of the materialized view's masters commits • DEMAND - Oracle Database refreshes this materialized view whenever an appropriate refresh procedure is called • STATEMENT - Oracle Database refreshes this materialized view whenever a DML operation is performed on any of the materialized view's base tables • NEVER - Oracle Database never refreshes this materialized view
REFRESH_METHOD	VARCHAR2(8)		Default method used to refresh the materialized view (can be overridden through the API): • COMPLETE - Materialized view is completely refreshed from the masters • FORCE - Oracle Database performs a fast refresh if possible, otherwise a complete refresh • FAST - Oracle Database performs an incremental refresh applying changes that correspond to changes in the masters since the last refresh • NEVER - User specified that Oracle Database should not refresh this materialized view
BUILD_MODE	VARCHAR2(9)		Indicates how the materialized view was populated during creation: • IMMEDIATE - Populated from the masters during creation • DEFERRED - Not populated during creation. Must be explicitly populated later by the user. • PREBUILT - Populated with an existing table during creation. The relationship of the contents of this prebuilt table to the materialized view's masters is unknown to Oracle Database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FAST_REFRESHABLE	VARCHAR2(18)		Indicates whether the materialized view is eligible for incremental (fast) refresh. Oracle Database calculates this value statically, based on the materialized view definition query: • NO - Materialized view is not fast refreshable, and hence is complex • DML - Fast refresh is supported only for DML operations • DIRLOAD_DML - Fast refresh is supported for both direct loads and DML operations • DIRLOAD_LIMITEDDML - Fast refresh is supported for direct loads and a subset of DML operations
LAST_REFRESH_TYPE	VARCHAR2(8)		Method used for the most recent refresh: • COMPLETE - Most recent refresh was complete • FAST - Most recent refresh was fast (incremental) • NA - Materialized view has not yet been refreshed (for example, if it was created DEFERRED)
LAST_REFRESH_DATE	DATE		Date on which the materialized view was most recently refreshed (blank if not yet populated)
LAST_REFRESH_END_TIME	DATE		End time of the most recent refresh on the materialized view (blank if not yet populated)
STALENESS	VARCHAR2(19)		Relationship between the contents of the materialized view and the contents of the materialized view's masters: • FRESH - Materialized view is a read-consistent view of the current state of its masters • IMPORT - Materialized view is imported from another database (the value of the UNKNOWN_IMPORT column is Y). Therefore, it is unknown whether the materialized view is in a read-consistent view of its masters from any point in time. The STALENESS of this view will turn to FRESH on a complete refresh. • NEEDS_COMPILE - Some object upon which the materialized view depends has changed. An ALTER MATERIALIZED VIEW...COMPILE statement is required to validate this materialized view and compute the staleness of the contents. • STALE - Materialized view is out of date because one or more of its masters has changed. If the materialized view was FRESH before it became STALE, then it is a read-consistent view of a former state of its masters. • UNDEFINED - Materialized view has remote masters. The concept of staleness is not defined for such materialized views. • UNKNOWN - Oracle Database does not know whether the materialized view is in a read-consistent view of its masters from any point in time (this is the case for materialized views created on prebuilt tables) • UNUSABLE - Materialized view is not a read-consistent view of its masters from any point in time

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
AFTER_FAST_REFRESH	VARCHAR2(19)		Specifies the staleness value that will occur if a fast refresh is applied to this materialized view. Its values are the same as for the STALENESS column, plus the value NA, which is used when fast refresh is not applicable to this materialized view.
UNKNOWN_PREBUILT	VARCHAR2(1)		Indicates whether the materialized view is prebuilt (Y) or not (N)
UNKNOWN_PLSQL_FUNC	VARCHAR2(1)		Indicates whether the materialized view contains PL/SQL functions (Y) or not (N)
UNKNOWN_EXTERNAL_TABLE	VARCHAR2(1)		Indicates whether the materialized view contains external tables (Y) or not (N)
UNKNOWN_CONSIDER_FRESH	VARCHAR2(1)		Indicates whether the materialized view is considered fresh (Y) or not (N)
UNKNOWN_IMPORT	VARCHAR2(1)		Indicates whether the materialized view is imported (Y) or not (N)
UNKNOWN_TRUSTED_FD	VARCHAR2(1)		Indicates whether the materialized view uses trusted constraints for refresh (Y) or not (N)
COMPILE_STATE	VARCHAR2(19)		Validity of the materialized view with respect to the objects upon which it depends: • VALID - Materialized view has been validated without error, and no object upon which it depends has changed since the last validation • NEEDS_COMPILE - Some object upon which the materialized view depends has changed. An ALTER MATERIALIZED VIEW...COMPILE statement is required to validate this materialized view. • ERROR - Materialized view has been validated with one or more errors
USE_NO_INDEX	VARCHAR2(1)		Indicates whether the materialized view was created using the USING NO INDEX clause (Y) or the materialized view was created with the default index (N). The USING NO INDEX clause suppresses the creation of the default index.
STALE_SINCE	DATE		Time from when the materialized view became stale
NUM_PCT_TABLES	NUMBER		Number of PCT detail tables
NUM_FRESH_PCT_REGIONS	NUMBER		Number of fresh PCT partition regions
NUM_STALE_PCT_REGIONS	NUMBER		Number of stale PCT partition regions
SEGMENT_CREATED	VARCHAR2(3)		Indicates whether the materialized view was created using the SEGMENT CREATION DEFERRED clause. The value is YES if the segment for the materialized view is created and NO if it is not.
EVALUATION_EDITION	VARCHAR2(128)		Name of the edition in which editioned objects referenced in an expression column are resolved
UNUSABLE_BEFORE	VARCHAR2(128)		Name of the oldest edition in which the stored results of the materialized view's subquery may be used for query rewrite. In editions before the specified edition, the stored results of the materialized view's data are considered unusable. This value is NULL if no such edition is specified.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
UNUSABLE_BEGINNING	VARCHAR2(128)		Name of the oldest edition in which the stored results of the materialized view's subquery may *not* be used for query rewrite. The data is unusable for query rewrite in the specified edition and in any descendants of this edition. This value is NULL if no such edition is specified.
DEFAULT_COLLATION	VARCHAR2(100)		Default collation for the materialized view
ON_QUERY_COMPUTATION	VARCHAR2(1)		Indicates whether the materialized view is a real-time materialized view (Y) or not (N)

See Also

- "DBA_MVIEWS"
- "USER_MVIEWS"
- *Oracle Database Data Warehousing Guide* for more information on materialized views to support data warehousing

2.282 ALL_NESTED_TABLE_COLS

ALL_NESTED_TABLE_COLS describes the columns of the nested tables accessible to the current user. To gather statistics for this view, use the DBMS_STATS package.

Related Views

- DBA_NESTED_TABLE_COLS describes the columns of all nested tables in the database.
- USER_NESTED_TABLE_COLS describes the columns of the nested tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the nested table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the nested table
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Column name
DATA_TYPE	VARCHAR2(128)		Data type of the column
DATA_TYPE_MOD	VARCHAR2(3)		Data type modifier of the column
DATA_TYPE_OWNER	VARCHAR2(128)		Owner of the data type of the column
DATA_LENGTH	NUMBER	NOT NULL	Length of the column (in bytes)
DATA_PRECISION	NUMBER		Decimal precision for NUMBER data type; binary precision for FLOAT data type; NULL for all other data types
DATA_SCALE	NUMBER		Digits to the right of the decimal point in a number
NULLABLE	VARCHAR2(1)		Indicates whether a column allows NULLs. The value is N if there is a NOT NULL constraint on the column or if the column is part of a PRIMARY KEY.
COLUMN_ID	NUMBER		Sequence number of the column as created

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEFAULT_LENGTH	NUMBER		Length of the default value for the column
DATA_DEFAULT	LONG		Default value for the column
NUM_DISTINCT	NUMBER		Number of distinct values in the column. This column remains for backward compatibility with Oracle7. This information is now in the {TAB|PART}_COL_STATISTICS views.
LOW_VALUE	RAW(1000)		Low value in the column. This column remains for backward compatibility with Oracle7. This information is now in the {TAB|PART}_COL_STATISTICS views.
HIGH_VALUE	RAW(1000)		High value in the column. This column remains for backward compatibility with Oracle7. This information is now in the {TAB|PART}_COL_STATISTICS views.
DENSITY	NUMBER		If a histogram is available on COLUMN_NAME, then this column displays the selectivity of a value that spans fewer than 2 endpoints in the histogram. It does not represent the selectivity of values that span 2 or more endpoints. If a histogram is not available on COLUMN_NAME, then the value of this column is 1/NUM_DISTINCT. This column remains for backward compatibility with Oracle7. This information is now in the {TAB|PART}_COL_STATISTICS views.
NUM_NULLS	NUMBER		Number of NULLs in the column
NUM_BUCKETS	NUMBER		Number of buckets in the histogram for the column **Note:** The number of buckets in a histogram is specified in the SIZE parameter of the SQL ANALYZE statement. However, Oracle Database does not create a histogram with more buckets than the number of rows in the sample. Also, if the sample contains any values that are very repetitious, Oracle Database creates the specified number of buckets, but the value indicated by this column may be smaller because of an internal compression algorithm.
LAST_ANALYZED	DATE		Date on which this column was most recently analyzed
SAMPLE_SIZE	NUMBER		Sample size used in analyzing this column
CHARACTER_SET_NAME	VARCHAR2(44)		Name of the character set: • CHAR_CS • NCHAR_CS
CHAR_COL_DECL_LENGTH	NUMBER		Declaration length of the character type column
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics are gathered or incrementally maintained, otherwise it will be NO
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
AVG_COL_LEN	NUMBER		Average length of the column (in bytes)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CHAR_LENGTH	NUMBER		Displays the length of the column in characters. This value only applies to the following data types: • CHAR • VARCHAR2 • NCHAR • NVARCHAR2
CHAR_USED	VARCHAR2(1)		Indicates that the column uses BYTE length semantics (B) or CHAR length semantics (C), or whether the data type is not any of the following (NULL): • CHAR • VARCHAR2 • NCHAR • NVARCHAR2
V80_FMT_IMAGE	VARCHAR2(3)		Indicates whether the column data is in release 8.0 image format (YES) or not (NO)
DATA_UPGRADED	VARCHAR2(3)		Indicates whether the column data has been upgraded to the latest type version format (YES) or not (NO)
HIDDEN_COLUMN	VARCHAR2(3)		Indicates whether the column is a hidden column (YES) or not (NO)
VIRTUAL_COLUMN	VARCHAR2(3)		Indicates whether the column is a virtual column (YES) or not (NO)
SEGMENT_COLUMN_ID	NUMBER		Sequence number of the column in the segment
INTERNAL_COLUMN_ID	NUMBER	NOT NULL	Internal sequence number of the column
HISTOGRAM	VARCHAR2(15)		Indicates existence/type of histogram: • NONE • FREQUENCY • HEIGHT BALANCED
QUALIFIED_COL_NAME	VARCHAR2(4000)		Qualified column name

See Also

- "DBA_NESTED_TABLE_COLS"
- "USER_NESTED_TABLE_COLS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

2.283 ALL_NESTED_TABLES

ALL_NESTED_TABLES describes the nested tables in tables accessible to the current user.

Related Views

- DBA_NESTED_TABLES describes all nested tables in the database.
- USER_NESTED_TABLES describes nested tables owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the nested table
TABLE_NAME	VARCHAR2(128)		Name of the nested table
TABLE_TYPE_OWNER	VARCHAR2(128)		Owner of the type of which the nested table was created
TABLE_TYPE_NAME	VARCHAR2(128)		Name of the type of the nested table
PARENT_TABLE_NAME	VARCHAR2(128)		Name of the parent table containing the nested table
PARENT_TABLE_COLUMN	VARCHAR2(4000)		Column name of the parent table that corresponds to the nested table
STORAGE_SPEC	VARCHAR2(30)		Indicates whether storage for the nested table is USER_SPECIFIED or DEFAULT
RETURN_TYPE	VARCHAR2(20)		Return type of the varray column (LOCATOR) or (VALUE)
ELEMENT_SUBSTITUTABLE	VARCHAR2(25)		Indicates whether the nested table element is substitutable (Y) or not (N)

See Also

- "DBA_NESTED_TABLES"
- "USER_NESTED_TABLES"

2.284 ALL_OBJ_COLATTRS

ALL_OBJ_COLATTRS describes object columns and attributes contained in the tables accessible to the current user.

Related Views

- DBA_OBJ_COLATTRS describes object columns and attributes contained in all tables in the database.
- USER_OBJ_COLATTRS describes object columns and attributes contained in the tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table containing the object column or attribute
COLUMN_NAME	VARCHAR2(4000)		Fully qualified name of the object column or attribute
SUBSTITUTABLE	VARCHAR2(15)		Indicates whether the column is substitutable (Y) or not (N)

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_OBJ_COLATTRS"
- "USER_OBJ_COLATTRS"

2.285 ALL_OBJECT_TABLES

ALL_OBJECT_TABLES describes the object tables accessible to the current user.

Related Views

- DBA_OBJECT_TABLES describes all object tables in the database.
- USER_OBJECT_TABLES describes the object tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the table; NULL for partitioned, temporary, and index-organized tables
CLUSTER_NAME	VARCHAR2(128)		Name of the cluster, if any, to which the table belongs
IOT_NAME	VARCHAR2(128)		Name of the index-organized table, if any, to which the overflow or mapping table entry belongs. If the IOT_TYPE column is not NULL, then this column contains the base table name.
STATUS	VARCHAR2(8)		If a previous DROP TABLE operation failed, indicates whether the table is unusable (UNUSABLE) or valid (VALID)
PCT_FREE	NUMBER		Minimum percentage of free space in a block; NULL for partitioned tables
PCT_USED	NUMBER		Minimum percentage of used space in a block; NULL for partitioned tables
INI_TRANS	NUMBER		Initial number of transactions; NULL for partitioned tables
MAX_TRANS	NUMBER		Maximum number of transactions; NULL for partitioned tables
INITIAL_EXTENT	NUMBER		Size of the initial extent (in bytes); NULL for partitioned tables
NEXT_EXTENT	NUMBER		Size of secondary extents (in bytes); NULL for partitioned tables
MIN_EXTENTS	NUMBER		Minimum number of extents allowed in the segment; NULL for partitioned tables
MAX_EXTENTS	NUMBER		Maximum number of extents allowed in the segment; NULL for partitioned tables
PCT_INCREASE	NUMBER		Percentage increase in extent size; NULL for partitioned tables
FREELISTS	NUMBER		Number of process freelists allocated to the segment; NULL for partitioned tables

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FREELIST_GROUPS	NUMBER		Number of freelist groups allocated to the segment; NULL for partitioned tables
LOGGING	VARCHAR2(3)		Indicates whether or not changes to the table are logged: • YES • NO
BACKED_UP	VARCHAR2(1)		Indicates whether the table has been backed up since the last modification (Y) or not (N)
NUM_ROWS	NUMBER		Number of rows in the table
BLOCKS	NUMBER		Number of used blocks in the table
EMPTY_BLOCKS	NUMBER		Number of empty (never used) blocks in the table
AVG_SPACE	NUMBER		Average available free space in the table
CHAIN_CNT	NUMBER		Number of chained rows in the table
AVG_ROW_LEN	NUMBER		Average row length, including row overhead
AVG_SPACE_FREELIST_BLOCKS	NUMBER		Average free space of all blocks on a freelist
NUM_FREELIST_BLOCKS	NUMBER		Number of blocks on the freelist
DEGREE	VARCHAR2(10)		Number of parallel execution processes per instance for scanning the table, or DEFAULT
INSTANCES	VARCHAR2(10)		Number of instances across which the table is to be scanned, or DEFAULT
CACHE	VARCHAR2(5)		Indicates whether the table is to be cached in the buffer cache (Y) or not (N)
TABLE_LOCK	VARCHAR2(8)		Indicates whether table locking is enabled (ENABLED) or disabled (DISABLED)
SAMPLE_SIZE	NUMBER		Sample size used in analyzing this table
LAST_ANALYZED	DATE		Date on which this table was most recently analyzed
PARTITIONED	VARCHAR2(3)		Indicates whether the table is partitioned (YES) or not (NO)
IOT_TYPE	VARCHAR2(12)		If the table is an index-organized table, then IOT_TYPE is IOT, IOT_OVERFLOW, or IOT_MAPPING. If the table is not an index-organized table, then IOT_TYPE is NULL.
OBJECT_ID_TYPE	VARCHAR2(16)		Indicates whether the object ID (OID) is USER-DEFINED or SYSTEM GENERATED
TABLE_TYPE_OWNER	VARCHAR2(128)		Owner of the type of the table
TABLE_TYPE	VARCHAR2(128)		Type of the table
TEMPORARY	VARCHAR2(1)		Indicates whether this is a temporary table (Y) or not (N)
SECONDARY	VARCHAR2(1)		Indicates whether the object table is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge (Y) or not (N)
NESTED	VARCHAR2(3)		Indicates whether the table is a nested table (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BUFFER_POOL	VARCHAR2(7)		Buffer pool to be used for table blocks: • DEFAULT • KEEP • RECYCLE • NULL
FLASH_CACHE	VARCHAR2(7)		Database Smart Flash Cache hint to be used for table blocks: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)		Cell flash cache hint to be used for table blocks: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
ROW_MOVEMENT	VARCHAR2(8)		Indicates whether partitioned row movement is enabled (ENABLED) or disabled (DISABLED)
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics are gathered or incrementally maintained, otherwise it will be NO
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
DURATION	VARCHAR2(15)		Indicates the duration of a temporary table: • SYS$SESSION - Rows are preserved for the duration of the session • SYS$TRANSACTION - Rows are deleted after COMMIT Null - Permanent table
SKIP_CORRUPT	VARCHAR2(8)		Indicates whether Oracle Database ignores blocks marked corrupt during table and index scans (ENABLED) or raises an error (DISABLED). To enable this feature, run the DBMS_REPAIR.SKIP_CORRUPT_BLOCKS procedure.
MONITORING	VARCHAR2(3)		Indicates whether the table has the MONITORING attribute set (YES) or not (NO)
CLUSTER_OWNER	VARCHAR2(128)		Owner of the cluster, if any, to which the table belongs
DEPENDENCIES	VARCHAR2(8)		Indicates whether row-level dependency tracking is enabled (ENABLED) or disabled (DISABLED)
COMPRESSION	VARCHAR2(8)		Indicates whether table compression is enabled (ENABLED) or not (DISABLED); NULL for partitioned tables

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COMPRESS_FOR	VARCHAR2(30)		Default compression for what kind of operations: • BASIC • ADVANCED • QUERY LOW • QUERY HIGH • ARCHIVE LOW • ARCHIVE HIGH • NULL The QUERY LOW, QUERY HIGH, ARCHIVE LOW, and ARCHIVE HIGH values are associated with Hybrid Columnar Compression, a feature of the Enterprise Edition of Oracle Database. See *Oracle Database Concepts* for more information.
DROPPED	VARCHAR2(3)		Indicates whether the table has been dropped and is in the recycle bin (YES) or not (NO); NULL for partitioned tables This view does not return the names of tables that have been dropped.
SEGMENT_CREATED	VARCHAR2(3)		Indicates whether the table segment has been created (YES) or not (NO)
INMEMORY	VARCHAR2(8)		Indicates whether the In-Memory Column Store (IM column store) is enabled (ENABLED) or disabled (DISABLED) for this segment
INMEMORY_PRIORITY	VARCHAR2(8)		Indicates the priority for In-Memory Column Store (IM column store) population. Possible values: • LOW • MEDIUM • HIGH • CRITICAL • NONE • NULL This column has a value based on where the segments lie for a table. For example, if the table is partitioned and is enabled for the IM column store, the value is NULL for ALL_TABLES but non-NULL for ALL_TAB_PARTITIONS.
INMEMORY_DISTRIBUTE	VARCHAR2(15)		Indicates how the IM column store is distributed in an Oracle Real Application Clusters (Oracle RAC) environment: • AUTO • BY ROWID RANGE • BY PARTITION • BY SUBPARTITION

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INMEMORY_COMPRESSION	VARCHAR2(17)		Indicates the compression level for the IM column store: • NO MEMCOMPRESS • FOR DML • FOR QUERY [LOW | HIGH] • FOR CAPACITY [LOW | HIGH] • NULL This column has a value based on where the segments lie for a table. For example, if the table is partitioned and is enabled for the IM column store, the value is NULL for ALL_TABLES but non-NULL for ALL_TAB_PARTITIONS.
INMEMORY_DUPLICATE	VARCHAR2(13)		Indicates the duplicate setting for the IM column store in an Oracle RAC environment: • NO DUPLICATE • DUPLICATE • DUPLICATE ALL
EXTERNAL	VARCHAR2(3)		Indicates whether the table is an external table (YES) or not (NO).
CELLMEMORY[1]	VARCHAR2(24)		The value for columnar compression in the storage cell flash cache. Possible values: • ENABLED: Oracle Exadata Storage will decide automatically whether to cache in columnar form • DISABLED: Oracle Exadata Storage is prevented from caching in columnar form • NO CACHECOMPRESS: Oracle Exadata Storage will cache in HCC format (no recompression) • FOR QUERY: Oracle Exadata Storage will recompress and cache in INMEMORY query high format • FOR CAPACITY: Oracle Exadata Storage will recompress and cache in INMEMORY capacity low format
HYBRID[2]	VARCHAR2(3)		Indicates whether the table is a hybrid partitioned table (YES) or not (NO). A hybrid partitioned table can contain a mixture of partitions stored in segments and partitions stored externally.
INMEMORY_SERVICE	VARCHAR2(12)		Indicates how the IM column store is populated on various instances. The possible values are: • DEFAULT: Data is populated on all instances specified with the PARALLEL_INSTANCE_GROUP initialization parameter. If that parameter is not set, then the data is populated on all instances. This is the default. • NONE: Data is not populated on any instance. • ALL: Data is populated on all instances, regardless of the value of the PARALLEL_INSTANCE_GROUP initialization parameter. • USER_DEFINED: Data is populated only on the instances on which the user-specified service is active. The service name corresponding to this is stored in the INMEMORY_SERVICE_NAME column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INMEMORY_SERVICE_NAME	VARCHAR2(1000)		Indicates the service name for the service on which the IM column store should be populated. This column has a value only when the corresponding INMEMORY_SERVICE is USER_DEFINED. In all other cases, this column is null.
MEMOPTIMIZE_READ	VARCHAR2(8)		Indicates whether the table is enabled for Fast Key Based Access (ENABLED) or not (DISABLED)
MEMOPTIMIZE_WRITE	VARCHAR2(8)		For internal use only
HAS_SENSITIVE_COLUMN	VARCHAR2(3)		Indicates whether the table has one or more sensitive columns (YES) or not (NO)
LOGICAL_REPLICATION[2]	VARCHAR2(8)		Indicates whether the table is enabled for logical replication (ENABLED) or not (DISABLED). This setting is ignored if database-wide column data supplemental logging is enabled.

1 This column is intended for use with Oracle Exadata.

2 This column is available starting with Oracle Database 19c.

See Also

- "DBA_OBJECT_TABLES"
- "USER_OBJECT_TABLES"
- "PARALLEL_INSTANCE_GROUP"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_REPAIR.SKIP_CORRUPT_BLOCKS procedure

2.286 ALL_OBJECTS

ALL_OBJECTS describes all objects accessible to the current user.

Related Views

- DBA_OBJECTS describes all objects in the database.
- USER_OBJECTS describes all objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object
SUBOBJECT_NAME	VARCHAR2(128)		Name of the subobject (for example, partition)
OBJECT_ID	NUMBER	NOT NULL	Dictionary object number of the object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DATA_OBJECT_ID	NUMBER		Dictionary object number of the segment that contains the object. **Note:** OBJECT_ID and DATA_OBJECT_ID display data dictionary metadata. Do not confuse these numbers with the unique 16-byte object identifier (*object ID*) that Oracle Database assigns to row objects in object tables in the system.
OBJECT_TYPE	VARCHAR2(23)		Type of the object (such as TABLE, INDEX)
CREATED	DATE	NOT NULL	Timestamp for the creation of the object
LAST_DDL_TIME	DATE	NOT NULL	Timestamp for the last modification of the object and dependent objects resulting from a DDL statement (including grants and revokes)
TIMESTAMP	VARCHAR2(19)		Timestamp for the specification of the object (character data)
STATUS	VARCHAR2(7)		Status of the object: • VALID • INVALID • N/A
TEMPORARY	VARCHAR2(1)		Indicates whether the object is temporary (the current session can see only data that it placed in this object itself) (Y) or not (N)
GENERATED	VARCHAR2(1)		Indicates whether the name of this object was system-generated (Y) or not (N)
SECONDARY	VARCHAR2(1)		Indicates whether this is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge (Y) or not (N)
NAMESPACE	NUMBER	NOT NULL	Namespace for the object
EDITION_NAME	VARCHAR2(128)		Name of the edition in which the object is actual
SHARING	VARCHAR2(18)		Values: • DATA LINK - If the object is data-linked or a data link to an object in the root • METADATA LINK - If the object is metadata-linked or a metadata link to an object in the root • EXTENDED DATA LINK - If the object is extended-data-linked or an extended data link to an object in the root • NONE - If none of the above applies
EDITIONABLE	VARCHAR2(1)		Values: • Y - For objects marked EDITIONABLE • N - For objects marked NONEDITIONABLE • NULL - For objects whose type is not editionable in the database
ORACLE_MAINTAINED	VARCHAR2(1)		Denotes whether the object was created, and is maintained, by Oracle-supplied scripts (such as catalog.sql or catproc.sql). An object for which this column has the value Y must not be changed in any way except by running an Oracle-supplied script.
APPLICATION	VARCHAR2(1)		Indicates whether the object is an Application common object (Y) or not (N)
DEFAULT_COLLATION	VARCHAR2(100)		Default collation for the object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DUPLICATED	VARCHAR2(1)		Indicates whether this object is duplicated on this shard (Y) or not (N)
SHARDED	VARCHAR2(1)		Indicates whether this object is sharded (Y) or not (N)
CREATED_APPID	NUMBER		ID of the Application that created the object
CREATED_VSNID	NUMBER		ID of the Application Version that created the object
MODIFIED_APPID	NUMBER		ID of the Application that last modified the object
MODIFIED_VSNID	NUMBER		ID of the Application Version that last modified the object

See Also

- "DBA_OBJECTS"
- "USER_OBJECTS"

2.287 ALL_OBJECTS_AE

ALL_OBJECTS_AE describes the objects (across all editions) accessible to the current user. Dropped objects appear in this view with OBJECT_TYPE = NON-EXISTENT.

Related Views

- DBA_OBJECTS_AE describes all objects (across all editions) in the database.
- USER_OBJECTS_AE describes the objects (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object
SUBOBJECT_NAME	VARCHAR2(128)		Name of the subobject (for example, partition)
OBJECT_ID	NUMBER	NOT NULL	Dictionary object number of the object
DATA_OBJECT_ID	NUMBER		Dictionary object number of the segment which contains the object
OBJECT_TYPE	VARCHAR2(23)		Type of the object
CREATED	DATE	NOT NULL	Timestamp for the creation of the object
LAST_DDL_TIME	DATE	NOT NULL	Timestamp for the last modification of the object and dependent objects resulting from a DDL statement (including grants and revokes)
TIMESTAMP	VARCHAR2(19)		Timestamp for the specification of the object (character data)
STATUS	VARCHAR2(7)		Status of the object: • VALID • INVALID • N/A

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TEMPORARY	VARCHAR2(1)		Indicates whether the object is temporary (the current session can see only data that it placed in this object itself) (Y) or not (N)
GENERATED	VARCHAR2(1)		Indicates whether the name of this object was system-generated (Y) or not (N)
SECONDARY	VARCHAR2(1)		Indicates whether this is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge (Y) or not (N)
NAMESPACE	NUMBER	NOT NULL	Namespace for the object
EDITION_NAME	VARCHAR2(128)		Name of the edition in which the object is actual
SHARING	VARCHAR2(18)		Values: • DATA LINK - If the object is data-linked or a data link to an object in the root • METADATA LINK - If the object is metadata-linked or a metadata link to an object in the root • EXTENDED DATA LINK - If the object is extended-data-linked or an extended data link to an object in the root • NONE - If none of the above applies
EDITIONABLE	VARCHAR2(1)		Values: • Y - For objects marked EDITIONABLE • N - For objects marked NONEDITIONABLE • NULL - For objects whose type is not editionable in the database
ORACLE_MAINTAINED	VARCHAR2(1)		Denotes whether the object was created, and is maintained, by Oracle-supplied scripts (such as catalog.sql or catproc.sql). An object for which this column has the value Y must not be changed in any way except by running an Oracle-supplied script.
APPLICATION	VARCHAR2(1)		Indicates whether the object is an Application common object (Y) or not (N)
DEFAULT_COLLATION	VARCHAR2(100)		Default collation for the object
DUPLICATED	VARCHAR2(1)		Indicates whether this object is duplicated on this shard (Y) or not (N)
SHARDED	VARCHAR2(1)		Indicates whether this object is sharded (Y) or not (N)
CREATED_APPID	NUMBER		ID of the Application that created the object
CREATED_VSNID	NUMBER		ID of the Application Version that created the object
MODIFIED_APPID	NUMBER		ID of the Application that last modified the object
MODIFIED_VSNID	NUMBER		ID of the Application Version that last modified the object

See Also

- "DBA_OBJECTS_AE"
- "USER_OBJECTS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

2.288 ALL_OPANCILLARY

ALL_OPANCILLARY describes operators whose bindings are ancillary to other (primary) operators.

Related Views

- DBA_OPANCILLARY describes such information about all operators in the database.
- USER_OPANCILLARY describes such information about operators owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the ancillary operator
OPERATOR_NAME	VARCHAR2(128)	NOT NULL	Name of the ancillary operator
BINDING#	NUMBER	NOT NULL	Binding number of the ancillary operator
PRIMOP_OWNER	VARCHAR2(128)	NOT NULL	Owner of the primary operator
PRIMOP_NAME	VARCHAR2(128)	NOT NULL	Name of the primary operator
PRIMOP_BIND#	NUMBER	NOT NULL	Binding number of the primary operator

See Also

- "DBA_OPANCILLARY"
- "USER_OPANCILLARY"

2.289 ALL_OPARGUMENTS

ALL_OPARGUMENTS describes arguments for each operator binding accessible to the current user.

Related Views

- DBA_OPARGUMENTS describes arguments of all operator bindings in the database.
- USER_OPARGUMENTS describes arguments of all operator bindings owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the operator argument
OPERATOR_NAME	VARCHAR2(128)	NOT NULL	Name of the operator argument
BINDING#	NUMBER	NOT NULL	Binding number of the operator argument
POSITION	NUMBER	NOT NULL	Position of the operator argument (1, 2, 3, ...)
ARGUMENT_TYPE	VARCHAR2(61)		Datatype of the operator argument

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_OPARGUMENTS"
- "USER_OPARGUMENTS"

2.290 ALL_OPBINDINGS

ALL_OPBINDINGS describes the binding functions and methods on the operators accessible to the current user.

Related Views

- DBA_OPBINDINGS describes the binding functions and methods on all operators in the database.
- USER_OPBINDINGS describes the binding functions and methods on the operators owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the operator
OPERATOR_NAME	VARCHAR2(128)	NOT NULL	Name of the operator
BINDING#	NUMBER	NOT NULL	Binding number of the operator
FUNCTION_NAME	VARCHAR2(92)		Name of the binding function or method as specified by the user
RETURN_SCHEMA	VARCHAR2(128)		Name of the schema of the return type if the return type of the binding is an object type
RETURN_TYPE	VARCHAR2(128)		Name of the return type
IMPLEMENTATION_TYPE_SCHEMA	VARCHAR2(128)		If the operator was created WITH INDEX CONTEXT or SCAN CONTEXT, then this column displays the schema of the implementation type used by the functional implementation of the operator as a scan context (null if the operator was created without this syntax). **See Also:** the CREATE OPERATOR statement in *Oracle Database SQL Language Reference*
IMPLEMENTATION_TYPE	VARCHAR2(128)		If the operator was created WITH INDEX CONTEXT or SCAN CONTEXT, then this column displays the name of the implementation type used by the functional implementation of the operator as a scan context (null if the operator was created without this syntax). **See Also:** the CREATE OPERATOR statement in *Oracle Database SQL Language Reference*
PROPERTY	VARCHAR2(43)		Property of the operator binding: • WITH INDEX CONTEXT • COMPUTE ANCILLARY DATA • ANCILLARY TO • WITH COLUMN CONTEXT • WITH INDEX, COLUMN CONTEXT • COMPUTE ANCILLARY DATA, WITH COLUMN CONTEXT

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_OPBINDINGS"
- "USER_OPBINDINGS"

2.291 ALL_OPERATOR_COMMENTS

ALL_OPERATOR_COMMENTS displays comments for the user-defined operators accessible to the current user.

Related Views

- DBA_OPERATOR_COMMENTS displays comments for all user-defined operators in the database.
- USER_OPERATOR_COMMENTS displays comments for the user-defined operators owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the user-defined operator
OPERATOR_NAME	VARCHAR2(128)	NOT NULL	Name of the user-defined operator
COMMENTS	VARCHAR2(4000)		Comment for the user-defined operator

See Also

- "DBA_OPERATOR_COMMENTS"
- "USER_OPERATOR_COMMENTS"

2.292 ALL_OPERATORS

ALL_OPERATORS describes the operators accessible to the current user.

Related Views

- DBA_OPERATORS describes all operators in the database.
- USER_OPERATORS describes the operators owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the operator
OPERATOR_NAME	VARCHAR2(128)	NOT NULL	Name of the operator
NUMBER_OF_BINDS	NUMBER	NOT NULL	Number of bindings associated with the operator

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_OPERATORS"
- "USER_OPERATORS"

2.293 ALL_OUTLINE_HINTS

`ALL_OUTLINE_HINTS` is a synonym for `USER_OUTLINE_HINTS`.

See Also

"USER_OUTLINE_HINTS"

2.294 ALL_OUTLINES

`ALL_OUTLINES` is a synonym for `USER_OUTLINES`.

See Also

"USER_OUTLINES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3

Static Data Dictionary Views: ALL_PART_COL_STATISTICS to DATABASE_PROPERTIES

This chapter contains the static data dictionary views ALL_PART_COL_STATISTICS through DATABASE_PROPERTIES.

3.1 ALL_PART_COL_STATISTICS

ALL_PART_COL_STATISTICS displays column statistics and histogram information for the table partitions accessible to the current user.

Related Views

- DBA_PART_COL_STATISTICS displays column statistics and histogram information for all table partitions in the database.
- USER_PART_COL_STATISTICS displays column statistics and histogram information for the table partitions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the partitioned table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
PARTITION_NAME	VARCHAR2(128)		Name of the table partition
COLUMN_NAME	VARCHAR2(4000)		Name of the column
NUM_DISTINCT	NUMBER		Number of distinct values in the column
LOW_VALUE	RAW(2000)		Low value in the column
HIGH_VALUE	RAW(2000)		High value in the column
DENSITY	NUMBER		If a histogram is available on COLUMN_NAME, then this column displays the selectivity of a value that spans fewer than 2 endpoints in the histogram. It does not represent the selectivity of values that span 2 or more endpoints. If a histogram is not available on COLUMN_NAME, then the value of this column is 1/NUM_DISTINCT.
NUM_NULLS	NUMBER		Number of NULLs in the column
NUM_BUCKETS	NUMBER		Number of buckets in histogram for the column
SAMPLE_SIZE	NUMBER		Sample size used in analyzing the column
LAST_ANALYZED	DATE		Date on which the column was most recently analyzed
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics have been gathered or NO if statistics have been aggregated from subpartitions or have not been gathered

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
NOTES	VARCHAR2(81)		Additional properties of the statistics, if any. Possible values: • ADAPTIVE_SAMPLING: The column has synopses, and the synopses are in the adaptive sampling format introduced in Oracle Database 11*g* Release 1 (11.1). • HIST_FOR_INCREM_STATS: A histogram used to support incremental statistics has been created and it is not used for optimization. • HISTOGRAM_ONLY: Table statistics were gathered using the GATHER AUTO option and histograms were gathered without re-gathering other table and column statistics. • HYPERLOGLOG: The column has synopses, and the synopses are in the HyperLogLog format introduced in Oracle Database 12*c* Release 2 (12.2.0.1). • INCREMENTAL: The column has synopses. • STATS_ON_LOAD: Online statistics were gathered during direct path load. This column can be used to determine whether synopses in the adaptive sampling format have been phased out entirely and purged properly.
AVG_COL_LEN	NUMBER		Average length of the column (in bytes)
HISTOGRAM	VARCHAR2(15)		Indicates existence/type of histogram: • NONE • FREQUENCY • HEIGHT BALANCED • HYBRID • TOP-FREQUENCY

See Also

- "DBA_PART_COL_STATISTICS"
- "USER_PART_COL_STATISTICS"

3.2 ALL_PART_HISTOGRAMS

ALL_PART_HISTOGRAMS displays the histogram data (endpoints per histogram) for the histograms on the table partitions accessible to the current user.

Related Views

- DBA_PART_HISTOGRAMS displays the histogram data for the histograms on all table partitions in the database.
- USER_PART_HISTOGRAMS displays the histogram data for the histograms on the table partitions owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> These views are populated only if you collect statistics on the index using the DBMS_STATS package.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table
PARTITION_NAME	VARCHAR2(128)		Name of the table partition
COLUMN_NAME	VARCHAR2(4000)		Name of the column
BUCKET_NUMBER	NUMBER		Bucket number of the histogram
ENDPOINT_VALUE	NUMBER		Normalized endpoint values for the bucket
ENDPOINT_ACTUAL_VALUE	VARCHAR2(4000)		Actual (not normalized) string value of the endpoint for the bucket
ENDPOINT_ACTUAL_VALUE_RAW	RAW(1000)		Endpoint actual value in raw format
ENDPOINT_REPEAT_COUNT	NUMBER		Frequency of the endpoint (applies only to hybrid histograms, and is set to 0 for other histogram types)

> **See Also**
>
> - "DBA_PART_HISTOGRAMS"
> - "USER_PART_HISTOGRAMS"
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

3.3 ALL_PART_INDEXES

ALL_PART_INDEXES displays the object-level partitioning information for the partitioned indexes accessible to the current user.

Related Views

- DBA_PART_INDEXES displays the object-level partitioning information for all partitioned indexes in the database.
- USER_PART_INDEXES displays the object-level partitioning information for the partitioned indexes owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the partitioned index
INDEX_NAME	VARCHAR2(128)	NOT NULL	Name of the partitioned index
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the partitioned table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PARTITIONING_TYPE	VARCHAR2(9)		Type of the partitioning method: • NONE - Not specified **See Also:** the *_INDEXES view • RANGE • HASH • SYSTEM • LIST • REFERENCE
SUBPARTITIONING_TYPE	VARCHAR2(9)		Type of the composite partitioning method: • NONE - Not specified **See Also:** the *_INDEXES view • RANGE • HASH • SYSTEM • LIST • REFERENCE
PARTITION_COUNT	NUMBER	NOT NULL	Number of partitions in the index
DEF_SUBPARTITION_COUNT	NUMBER		For a composite-partitioned index, the default number of subpartitions, if specified
PARTITIONING_KEY_COUNT	NUMBER	NOT NULL	Number of columns in the partitioning key
SUBPARTITIONING_KEY_COUNT	NUMBER		For a composite-partitioned index, the number of columns in the subpartitioning key
LOCALITY	VARCHAR2(6)		Indicates whether the partitioned index is local (LOCAL) or global (GLOBAL)
ALIGNMENT	VARCHAR2(12)		Indicates whether the partitioned index is prefixed (PREFIXED) or non-prefixed (NON_PREFIXED)
DEF_TABLESPACE_NAME	VARCHAR2(30)		For a local index, the default tablespace to be used when adding or splitting a table partition
DEF_PCT_FREE	NUMBER	NOT NULL	For a local index, the default PCTFREE value to be used when adding a table partition
DEF_INI_TRANS	NUMBER	NOT NULL	For a local index, the default INITRANS value to be used when adding a table partition
DEF_MAX_TRANS	NUMBER	NOT NULL	For a local index, the default MAXTRANS value to be used when adding a table partition
DEF_INITIAL_EXTENT	VARCHAR2(40)		For a local index, the default INITIAL value (in Oracle blocks) to be used when adding a table partition, or DEFAULT if no INITIAL value was specified
DEF_NEXT_EXTENT	VARCHAR2(40)		For a local index, the default NEXT value (in Oracle blocks) to be used when adding a table partition, or DEFAULT if no NEXT value was specified
DEF_MIN_EXTENTS	VARCHAR2(40)		For a local index, the default MINEXTENTS value to be used when adding a table partition, or DEFAULT if no MINEXTENTS value was specified
DEF_MAX_EXTENTS	VARCHAR2(40)		For a local index, the default MAXEXTENTS value to be used when adding a table partition, or DEFAULT if no MAXEXTENTS value was specified

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEF_MAX_SIZE	VARCHAR2(40)		For a local index, the default MAXSIZE value to be used when adding a table partition, or DEFAULT if no MAXSIZE value was specified
DEF_PCT_INCREASE	VARCHAR2(40)		For a local index, the default PCTINCREASE value to be used when adding a table partition, or DEFAULT if no PCTINCREASE value was specified
DEF_FREELISTS	NUMBER	NOT NULL	For a local index, the default FREELISTS value to be used when adding a table partition
DEF_FREELIST_GROUPS	NUMBER	NOT NULL	For a local index, the default FREELIST GROUPS value to be used when adding a table partition
DEF_LOGGING	VARCHAR2(7)		For a local index, the default LOGGING attribute to be used when adding a table partition: • NONE - Not specified **See Also:** the *_INDEXES view • YES • NO
DEF_BUFFER_POOL	VARCHAR2(7)		For a local index, the default buffer pool to be used when adding a table partition: • DEFAULT • KEEP • RECYCLE • NULL
DEF_FLASH_CACHE	VARCHAR2(7)		For a local index, the default Database Smart Flash Cache hint to be used when adding a table partition: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
DEF_CELL_FLASH_CACHE	VARCHAR2(7)		For a local index, the default cell flash cache hint to be used when adding a table partition: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
DEF_PARAMETERS	VARCHAR2(1000)		Default parameter string for domain indexes
INTERVAL	VARCHAR2(1000)		String of the interval value
AUTOLIST	VARCHAR2(3)		Indicates whether a local index is partitioned by auto list partitioning (YES) or not (NO)
INTERVAL_SUBPARTITION	VARCHAR2(1000)		String of the subpartition interval value
AUTOLIST_SUBPARTITION	VARCHAR2(3)		Indicates whether a local index is subpartitioned by auto list partitioning (YES) or not (NO)

See Also

- "DBA_PART_INDEXES"
- "USER_PART_INDEXES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.4 ALL_PART_KEY_COLUMNS

ALL_PART_KEY_COLUMNS describes the partitioning key columns for the partitioned objects accessible to the current user.

Related Views

- DBA_PART_KEY_COLUMNS describes the partitioning key columns for all partitioned objects in the database.
- USER_PART_KEY_COLUMNS describes the partitioning key columns for the partitioned objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the partitioned table or index
NAME	VARCHAR2(128)		Name of the partitioned table or index
OBJECT_TYPE	CHAR(5)		Object type: • TABLE • INDEX
COLUMN_NAME	VARCHAR2(4000)		Name of the column
COLUMN_POSITION	NUMBER		Position of the column within the partitioning key
COLLATED_COLUMN_ID	NUMBER		Internal sequence number of the column for which this column provides linguistic ordering

See Also

- "DBA_PART_KEY_COLUMNS"
- "USER_PART_KEY_COLUMNS"

3.5 ALL_PART_LOBS

ALL_PART_LOBS displays table-level information about the partitioned LOBs accessible to the current user, including default attributes for LOB data partitions.

Related Views

- DBA_PART_LOBS displays table-level information about all partitioned LOBs in the database.
- USER_PART_LOBS displays table-level information about the partitioned LOBs owned by the current user. This view does not display the TABLE_OWNER column.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the partitioned table containing the LOBs
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the partitioned table containing the LOBs
COLUMN_NAME	VARCHAR2(4000)		Name of the LOB column
LOB_NAME	VARCHAR2(128)	NOT NULL	Name of the partitioned LOB
LOB_INDEX_NAME	VARCHAR2(128)	NOT NULL	Name of the partitioned LOB index

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEF_CHUNK	NUMBER	NOT NULL	Default value of CHUNK for a LOB data partition to be used when adding a partition
DEF_PCTVERSION	NUMBER	NOT NULL	Default value of PCTVERSION for a LOB data partition to be used when adding a partition
DEF_CACHE	VARCHAR2(10)		Indicates whether and how the LOB data is cached by default in the buffer cache: • YES - LOB data is placed in the buffer cache • NO - LOB data either is not brought into the buffer cache or is brought into the buffer cache and placed at the least recently used end of the LRU list • CACHEREADS - LOB data is brought into the buffer cache only during read operations but not during write operations
DEF_IN_ROW	VARCHAR2(3)		Indicates whether LOB data < 4000 bytes is stored by default inline (in the row) (YES) or not (NO); that is, whether or not ENABLE STORAGE IN ROW was specified when the LOB column was created or last altered
DEF_TABLESPACE_NAME	VARCHAR2(30)		Default tablespace for a LOB data partition to be used when adding a partition
DEF_INITIAL_EXTENT	VARCHAR2(40)		Default value of INITIAL for a LOB data partition to be used when adding a partition, or DEFAULT if no INITIAL value was specified
DEF_NEXT_EXTENT	VARCHAR2(40)		Default value of NEXT for a LOB data partition to be used when adding a partition, or DEFAULT if no NEXT value was specified
DEF_MIN_EXTENTS	VARCHAR2(40)		Default value of MINEXTENTS for a LOB data partition to be used when adding a partition, or DEFAULT if no MINEXTENTS value was specified
DEF_MAX_EXTENTS	VARCHAR2(40)		Default value of MAXEXTENTS for a LOB data partition to be used when adding a partition, or DEFAULT if no MAXEXTENTS value was specified
DEF_MAX_SIZE	VARCHAR2(40)		Default value of MAXSIZE for a LOB data partition to be used when adding a partition, or DEFAULT if no MAXSIZE value was specified
DEF_RETENTION	VARCHAR2(7)		Default value of RETENTION for a LOB data partition to be used when adding a partition. Possible values for SecureFiles: • NONE • AUTO • MIN • MAX • DEFAULT • INVALID Possible values for BasicFiles: • YES • NO
DEF_MINRET	VARCHAR2(40)		Default value of RETENTION MIN for a LOB data partition to be used when adding a partition, or DEFAULT if no RETENTION MIN value was specified

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEF_PCT_INCREASE	VARCHAR2(40)		Default value of PCTINCREASE for a LOB data partition to be used when adding a partition, or DEFAULT if no PCTINCREASE value was specified
DEF_FREELISTS	VARCHAR2(40)		Default value of FREELISTS for a LOB data partition to be used when adding a partition, or DEFAULT if no FREELISTS value was specified
DEF_FREELIST_GROUPS	VARCHAR2(40)		Default value of FREELIST GROUPS for a LOB data partition to be used when adding a partition, or DEFAULT if no FREELIST GROUPS value was specified
DEF_LOGGING	VARCHAR2(7)		Default LOGGING attribute for a LOB data partition to be used when adding a partition: • NONE - Not specified **See Also:** the *_LOBS and *_LOB_PARTITIONS views • YES • NO
DEF_BUFFER_POOL	VARCHAR2(7)		Default buffer pool for a LOB data partition to be used when adding a partition: • DEFAULT • KEEP • RECYCLE • NULL
DEF_FLASH_CACHE	VARCHAR2(7)		Default Database Smart Flash Cache hint to be used when adding a partition: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
DEF_CELL_FLASH_CACHE	VARCHAR2(7)		Default cell flash cache hint to be used when adding a partition: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
DEF_ENCRYPT	VARCHAR2(4)		Default value of ENCRYPT for a LOB data partition to be used when adding a partition. Possible values for SecureFiles: • YES • NO Possible value for BasicFiles: • NONE - Not applicable

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEF_COMPRESS	VARCHAR2(6)		Default value of COMPRESS for a LOB data partition to be used when adding a partition. Possible values for SecureFiles: • LOW • MEDIUM • HIGH • NO - Compression is off Possible value for BasicFiles: • NONE - Not applicable
DEF_DEDUPLICATE	VARCHAR2(15)		Default value of DEDUPLICATE for a LOB data partition to be used when adding a partition. Possible values for SecureFiles: • LOB - Deduplicate • NO - Keep duplicates Possible values for BasicFiles: • NONE - Not applicable
DEF_SECUREFILE	VARCHAR2(3)		Indicates whether the LOB is SecureFiles (YES) or not (NO)

See Also

- "DBA_PART_LOBS"
- "USER_PART_LOBS"

3.6 ALL_PART_TABLES

ALL_PART_TABLES displays the object-level partitioning information for the partitioned tables accessible to the current user.

Related Views

- DBA_PART_TABLES displays the object-level partitioning information for all partitioned tables in the database.
- USER_PART_TABLES displays the object-level partitioning information for the partitioned tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the partitioned table
TABLE_NAME	VARCHAR2(128)		Name of the partitioned table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PARTITIONING_TYPE	VARCHAR2(9)		Type of the partitioning method: • UNKNOWN - Not specified **See Also:** the *_TABLES view • RANGE • HASH • SYSTEM • LIST • REFERENCE
SUBPARTITIONING_TYPE	VARCHAR2(9)		Type of the composite partitioning method: • NONE - Not subpartitioned **See Also:** the *_TABLES view • RANGE • HASH • SYSTEM • LIST • REFERENCE
PARTITION_COUNT	NUMBER		Number of partitions in the table. For interval partitioned tables, the value of this column is always 1048575.
DEF_SUBPARTITION_COUNT	NUMBER		For a composite-partitioned table, the default number of subpartitions, if specified
PARTITIONING_KEY_COUNT	NUMBER		Number of columns in the partitioning key
SUBPARTITIONING_KEY_COUNT	NUMBER		For a composite-partitioned table, the number of columns in the subpartitioning key
STATUS	VARCHAR2(8)		If a previous DROP TABLE operation failed, indicates whether the table is unusable (UNUSABLE) or valid (VALID)
DEF_TABLESPACE_NAME	VARCHAR2(30)		Default tablespace to be used when adding a partition
DEF_PCT_FREE	NUMBER		Default value of PCTFREE to be used when adding a partition
DEF_PCT_USED	NUMBER		Default value of PCTUSED to be used when adding a partition
DEF_INI_TRANS	NUMBER		Default value of INITRANS to be used when adding a partition
DEF_MAX_TRANS	NUMBER		Default value of MAXTRANS to be used when adding a partition
DEF_INITIAL_EXTENT	VARCHAR2(40)		Default value of INITIAL (in Oracle blocks) to be used when adding a partition, or DEFAULT if no INITIAL value was specified
DEF_NEXT_EXTENT	VARCHAR2(40)		Default value of NEXT (in Oracle blocks) to be used when adding a partition, or DEFAULT if no NEXT value was specified
DEF_MIN_EXTENTS	VARCHAR2(40)		Default value of MINEXTENTS to be used when adding a partition, or DEFAULT if no MINEXTENTS value was specified
DEF_MAX_EXTENTS	VARCHAR2(40)		Default value of MAXEXTENTS to be used when adding a partition, or DEFAULT if no MAXEXTENTS value was specified

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEF_MAX_SIZE	VARCHAR2(40)		Default value of MAXSIZE to be used when adding a partition, or DEFAULT if no MAXSIZE value was specified
DEF_PCT_INCREASE	VARCHAR2(40)		Default value of PCTINCREASE to be used when adding a partition, or DEFAULT if no PCTINCREASE value was specified
DEF_FREELISTS	NUMBER		Default value of FREELISTS to be used when adding a partition
DEF_FREELIST_GROUPS	NUMBER		Default value of FREELIST GROUPS to be used when adding a partition
DEF_LOGGING	VARCHAR2(7)		Default LOGGING attribute to be used when adding a partition: • NONE - Not specified **See Also:** the *_TABLES view • YES • NO
DEF_COMPRESSION	VARCHAR2(8)		Default compression to be used when adding a partition: • NONE - Not specified **See Also:** the *_TABLES view • ENABLED • DISABLED
DEF_COMPRESS_FOR	VARCHAR2(30)		Default compression for what kind of operations to be used when adding a partition: • BASIC • ADVANCED • QUERY LOW[1] • QUERY HIGH[1] • ARCHIVE LOW[1] • ARCHIVE HIGH[1] • UNKNOWN • NULL
DEF_BUFFER_POOL	VARCHAR2(7)		Default buffer pool to be used when adding a partition: • DEFAULT • KEEP • RECYCLE • NULL
DEF_FLASH_CACHE	VARCHAR2(7)		Default Database Smart Flash Cache hint to be used when adding a partition: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
DEF_CELL_FLASH_CACHE	VARCHAR2(7)		Default cell flash cache hint to be used when adding a partition: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
REF_PTN_CONSTRAINT_NAME	VARCHAR2(128)		Name of the partitioning referential constraint for reference-partitioned tables
INTERVAL	VARCHAR2(1000)		String of the interval value
AUTOLIST	VARCHAR2(3)		Indicates whether a table is partitioned by auto list partitioning (YES) or not (NO)
INTERVAL_SUBPARTITION	VARCHAR2(1000)		String of the subpartition interval value
AUTOLIST_SUBPARTITION	VARCHAR2(3)		Indicates whether auto list partitioning is being used (YES) or not (NO) for this subpartition
IS_NESTED	VARCHAR2(3)		Indicates whether the partitioned table is a nested table (YES) or not (NO) **See Also:** the *_NESTED_TABLES view for the parent table name/column
DEF_SEGMENT_CREATION	VARCHAR2(4)		Specifies whether the default for segment creation was specified on the table level: • NO - deferred was specified • YES - immediate was specified • NONE - a default for segment creation was not specified
DEF_INDEXING	VARCHAR2(3)		Indicates the indexing property specified for the table. Possible values: • ON - INDEXING on was specified explicitly, or no indexing property was specified • OFF - INDEXING off was specified.
DEF_INMEMORY	VARCHAR2(8)		Indicates whether the In-Memory Column Store (IM column store) is by default enabled (ENABLED), disabled (DISABLED), or not specified (NONE) for partitions in this table
DEF_INMEMORY_PRIORITY	VARCHAR2(8)		Indicates the default priority for In-Memory Column Store (IM column store) population. Possible values: • LOW • MEDIUM • HIGH • CRITICAL • NONE • NULL
DEF_INMEMORY_DISTRIBUTE	VARCHAR2(15)		Indicates how the IM column store is distributed by default for partitions of the table in an Oracle Real Application Clusters (Oracle RAC) environment: • AUTO • BY ROWID RANGE • BY PARTITION • BY SUBPARTITION
DEF_INMEMORY_COMPRESSION	VARCHAR2(17)		Default compression level for the IM column store: • NO MEMCOMPRESS • FOR DML • FOR QUERY [LOW | HIGH] • FOR CAPACITY [LOW | HIGH] • NULL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEF_INMEMORY_DUPLICATE	VARCHAR2(13)		Indicates the default duplicate setting for the IM column store in an Oracle RAC environment: • NO DUPLICATE • DUPLICATE • DUPLICATE ALL
DEF_READ_ONLY	VARCHAR2(3)		Indicates the default setting for new partitions: • YES: The default setting for new partitions is read-only. • NO: The default setting for new partitions is read/write.
DEF_CELLMEMORY[2]	VARCHAR2(24)		Shows the default value for the CELLMEMORY attribute that new partitions in the parent table will inherit unless the behavior is overridden explicitly
DEF_INMEMORY_SERVICE	VARCHAR2(12)		Indicates how the IM column store is populated on various instances by default for partitions of the table. The possible values are: • DEFAULT: Data is populated on all instances specified with the PARALLEL_INSTANCE_GROUP initialization parameter. If that parameter is not set, then the data is populated on all instances. This is the default. • NONE: Data is not populated on any instance. • ALL: Data is populated on all instances, regardless of the value of the PARALLEL_INSTANCE_GROUP initialization parameter. • USER_DEFINED: Data is populated only on the instances on which the user-specified service is active. The service name corresponding to this is stored in the DEF_INMEMORY_SERVICE_NAME column.
DEF_INMEMORY_SERVICE_NAME	VARCHAR2(1000)		Specifies the service name for the service on which the IM column store should be populated by default for partitions of the table. This column has a value only when the corresponding DEF_INMEMORY_SERVICE is USER_DEFINED. In all other cases, this column is null.

1 Hybrid Columnar Compression is a feature of the Enterprise Edition of Oracle Database that is dependent on the underlying storage system. See *Oracle Database Concepts* for more information.
2 This column is intended for use with Oracle Exadata

See Also

- "DBA_PART_TABLES"
- "USER_PART_TABLES"
- "PARALLEL_INSTANCE_GROUP"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.7 ALL_PARTIAL_DROP_TABS

ALL_PARTIAL_DROP_TABS describes tables accessible to the current user that have partially completed DROP COLUMN operations. Such operations might have been interrupted by the user or by a system crash.

Related Views

- DBA_PARTIAL_DROP_TABS describes all tables in the database that have partially completed DROP COLUMN operations.
- USER_PARTIAL_DROP_TABS describes tables in the schema of the current user that have partially completed DROP COLUMN operations. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table

See Also

- "DBA_PARTIAL_DROP_TABS"
- "USER_PARTIAL_DROP_TABS"

3.8 ALL_PENDING_CONV_TABLES

ALL_PENDING_CONV_TABLES describes the pending conversion tables (tables which are not upgraded to the latest type version) accessible to the current user.

Related Views

- DBA_PENDING_CONV_TABLES describes all pending conversion tables in the database.
- USER_PENDING_CONV_TABLES describes the pending conversion tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table

See Also

- "DBA_PENDING_CONV_TABLES"
- "USER_PENDING_CONV_TABLES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.9 ALL_PLSQL_COLL_TYPES

ALL_PLSQL_COLL_TYPES describes named PL/SQL collection types accessible to the user.

Related Views

- DBA_PLSQL_COLL_TYPES describes all named PL/SQL collection types in the database. This view does not display the CHAR_USED column.
- USER_PLSQL_COLL_TYPES describes the user's own named PL/SQL collection types. This view does not display the OWNER or CHAR_USED columns.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the type
TYPE_NAME	VARCHAR2(128)		Name of the type
PACKAGE_NAME	VARCHAR2(128)		Name of the package containing the collection
COLL_TYPE	VARCHAR2(128)		Collection type
UPPER_BOUND	NUMBER		The upper bound of a varray or length constraint of an index by VARCHAR2 table
ELEM_TYPE_OWNER	VARCHAR2(128)		Owner of the type of the element
ELEM_TYPE_NAME	VARCHAR2(136)		Name of the type of the element
ELEM_TYPE_PACKAGE	VARCHAR2(128)		Name of the package containing the element
LENGTH	NUMBER		Length of the CHAR element or maximum length of the VARCHAR or VARCHAR2 element
PRECISION	NUMBER		Decimal precision of the NUMBER or DECIMAL element or binary precision of the FLOAT element
SCALE	NUMBER		Scale of the NUMBER or DECIMAL element
CHARACTER_SET_NAME	VARCHAR2(44)		Character set name of the element
ELEM_STORAGE	VARCHAR2(7)		Storage optimization specification for VARRAY of numeric elements
NULLS_STORED	VARCHAR2(3)		Indicates whether null information is stored with each VARRAY element (YES) or not (NO)
CHAR_USED	VARCHAR2(1)		C if the width was specified in characters, B if in bytes
INDEX_BY	VARCHAR2(14)		Index by BINARY_INTEGER or VARCHAR2
ELEM_TYPE_MOD	VARCHAR2(7)		Type modifier of the element

See Also

- "DBA_PLSQL_COLL_TYPES"
- "USER_PLSQL_COLL_TYPES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.10 ALL_PLSQL_OBJECT_SETTINGS

ALL_PLSQL_OBJECT_SETTINGS displays information about the compiler settings for the stored objects accessible to the current user.

Related Views

- DBA_PLSQL_OBJECT_SETTINGS displays information about the compiler settings for all stored objects in the database.
- USER_PLSQL_OBJECT_SETTINGS displays information about the compiler settings for the stored objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
NAME	VARCHAR2(128)	NOT NULL	Name of the object
TYPE	VARCHAR2(12)		Type of the object: • PROCEDURE • FUNCTION • PACKAGE • PACKAGE BODY • TRIGGER • TYPE • TYPE BODY
PLSQL_OPTIMIZE_LEVEL	NUMBER		Optimization level that was used to compile the object
PLSQL_CODE_TYPE	VARCHAR2(4000)		Compilation mode for the object
PLSQL_DEBUG	VARCHAR2(4000)		Indicates whether the object was compiled with debug information or not
PLSQL_WARNINGS	VARCHAR2(4000)		Compiler warning settings that were used to compile the object
NLS_LENGTH_SEMANTICS	VARCHAR2(4000)		NLS length semantics that were used to compile the object
PLSQL_CCFLAGS	VARCHAR2(4000)		Conditional compilation flag settings that were used to compile the object
PLSCOPE_SETTINGS	VARCHAR2(4000)		Settings for using PL/Scope
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)

See Also

- "DBA_PLSQL_OBJECT_SETTINGS"
- "USER_PLSQL_OBJECT_SETTINGS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.11 ALL_PLSQL_TYPE_ATTRS

ALL_PLSQL_TYPE_ATTRS describes the attributes of PL/SQL types accessible to the user.

Related Views

- DBA_PLSQL_TYPE_ATTRS describes the attributes of all PL/SQL types in the database.
- USER_PLSQL_TYPE_ATTRS describes the attributes of the user's own PL/SQL types. This view does not display the OWNER or CHAR_USED columns.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the type
TYPE_NAME	VARCHAR2(136)		Name of the type
PACKAGE_NAME	VARCHAR2(128)		Name of the package containing the type
ATTR_NAME	VARCHAR2(128)		Name of the attribute
ATTR_TYPE_MOD	VARCHAR2(7)		Type modifier of the attribute
ATTR_TYPE_OWNER	VARCHAR2(128)		Owner of the type of the attribute
ATTR_TYPE_NAME	VARCHAR2(136)		Name of the type of the attribute
ATTR_TYPE_PACKAGE	VARCHAR2(128)		Name of the package containing the attribute type
LENGTH	NUMBER		Length of the CHAR attribute or maximum length of the VARCHAR or VARCHAR2 attribute
PRECISION	NUMBER		Decimal precision of the NUMBER or DECIMAL attribute or binary precision of the FLOAT attribute
SCALE	NUMBER		Scale of the NUMBER or DECIMAL attribute
CHARACTER_SET_NAME	VARCHAR2(44)		Character set name of the attribute
ATTR_NO	NUMBER		Syntactical order number or position of the attribute as specified in the type specification or CREATE TYPE statement (not to be used as ID number)
CHAR_USED	VARCHAR2(1)		C if the width was specified in characters, B if in bytes

See Also

- "DBA_PLSQL_TYPE_ATTRS"
- "USER_PLSQL_TYPE_ATTRS"

3.12 ALL_PLSQL_TYPES

ALL_PLSQL_TYPES describes the PL/SQL types accessible to the user.

Related Views

- DBA_PLSQL_TYPES describes all the PL/SQL types in the database.
- USER_PLSQL_TYPES describes the user's own PL/SQL types. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the type
TYPE_NAME	VARCHAR2(136)		Name of the type
PACKAGE_NAME	VARCHAR2(128)	NOT NULL	Name of the package containing the type
TYPE_OID	RAW(16)	NOT NULL	Object identifier (OID) of the type
TYPECODE	VARCHAR2(58)		Typecode of the type
ATTRIBUTES	NUMBER		Number of attributes in the type
CONTAINS_PLSQL	VARCHAR2(3)		Indicates whether the type contains PL/SQL-specific data types (YES) or not (NO)

See Also

- "DBA_PLSQL_TYPES"
- "USER_PLSQL_TYPES"

3.13 ALL_POLICIES

ALL_POLICIES describes all Oracle Virtual Private Database (VPD) security policies for objects accessible to the current user. A security policy is a list of security requirements and rules that regulate row level access to those database objects.

Related Views

- DBA_POLICIES describes all Oracle Virtual Private Database (VPD) security policies in the database.
- USER_POLICIES describes all Oracle Virtual Private Database (VPD) security policies associated with objects owned by the current user. This view does not display the OBJECT_OWNER column.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)		Owner of the synonym, table, or view
OBJECT_NAME	VARCHAR2(128)		Name of the synonym, table, or view
POLICY_GROUP	VARCHAR2(128)		Name of the policy group
POLICY_NAME	VARCHAR2(128)		Name of the policy
PF_OWNER	VARCHAR2(128)		Owner of the policy function
PACKAGE	VARCHAR2(128)		Name of the package containing the policy function
FUNCTION	VARCHAR2(128)		Name of the policy function
SEL	VARCHAR2(3)		Indicates whether the policy is applied to queries on the object (YES) or not (NO)
INS	VARCHAR2(3)		Indicates whether the policy is applied to INSERT statements on the object (YES) or not (NO)
UPD	VARCHAR2(3)		Indicates whether the policy is applied to UPDATE statements on the object (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEL	VARCHAR2(3)		Indicates whether the policy is applied to DELETE statements on the object (YES) or not (NO)
IDX	VARCHAR2(3)		Indicates whether the policy is enforced for index maintenance on the object (YES) or not (NO)
CHK_OPTION	VARCHAR2(3)		Indicates whether the check option is enforced for the policy (YES) or not (NO)
ENABLE	VARCHAR2(3)		Indicates whether the policy is enabled (YES) or disabled (NO)
STATIC_POLICY	VARCHAR2(3)		Indicates whether the policy is static (YES) or not (NO). This column is obsolete because information about static policies is shown in the POLICY_TYPE column.
POLICY_TYPE	VARCHAR2(24)		Policy type: • STATIC • SHARED_STATIC • CONTEXT_SENSITIVE • SHARED_CONTEXT_SENSITIVE • DYNAMIC
LONG_PREDICATE	VARCHAR2(3)		Indicates whether the policy function can return a maximum of 32 KB of predicate (YES) or not (NO). If NO, the default maximum predicate size is 4000 bytes.
COMMON	VARCHAR2(3)		Indicates whether the policy is applied and enforced in all application PDBs (YES) or only in the local PDB (NO)
INHERITED	VARCHAR2(3)		Indicates whether the policy is inherited from the root (YES) or not (NO)

See Also

- "DBA_POLICIES"
- "USER_POLICIES"
- *Oracle Database Concepts* for an overview of security policies and fine-grained access control
- *Oracle Database Security Guide* for more information about security policies
- The DBMS_RLS package in *Oracle Database PL/SQL Packages and Types Reference* for information on administering security policies

3.14 ALL_POLICY_ATTRIBUTES

ALL_POLICY_ATTRIBUTES lists the attribute associations {Namespaces, Attributes} of context-sensitive and shared context-sensitive Oracle Virtual Private Database (VPD) policies for objects accessible to the current user.

Related Views

- DBA_POLICY_ATTRIBUTES lists the attribute associations {Namespaces, Attributes} of all context-sensitive and shared context-sensitive Oracle Virtual Private Database (VPD) policies in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_POLICY_ATTRIBUTES lists the attribute associations {Namespaces, Attributes} of all context-sensitive and shared-context sensitive Oracle Virtual Private Database (VPD) policies for synonyms, tables, or views owned by the user.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)		Owner of the synonym, table, or view
OBJECT_NAME	VARCHAR2(128)		Name of the synonym, table, or view
POLICY_GROUP	VARCHAR2(128)		Name of the policy group
POLICY_NAME	VARCHAR2(128)		Name of the policy
NAMESPACE	VARCHAR2(128)		Name of the local application context
ATTRIBUTE	VARCHAR2(128)		Name of the attribute
COMMON	VARCHAR2(3)		Indicates whether the policy attribute is applied and enforced in all application PDBs (YES) or only in the local PDB (NO)
INHERITED	VARCHAR2(3)		Indicates whether the policy attribute is inherited from the root (YES) or not (NO)

See Also

- "DBA_POLICY_ATTRIBUTES"
- "USER_POLICY_ATTRIBUTES"

3.15 ALL_POLICY_CONTEXTS

ALL_POLICY_CONTEXTS describes the driving contexts defined for the synonyms, tables, and views accessible to the current user.

Related Views

- DBA_POLICY_CONTEXTS describes all driving contexts in the database.
- USER_POLICY_CONTEXTS describes the driving contexts defined for the synonyms, tables, and views owned by the current user. This view does not display the OBJECT_OWNER column.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the synonym, table, or view
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the synonym, table, or view
NAMESPACE	VARCHAR2(128)	NOT NULL	Namespace of the driving context
ATTRIBUTE	VARCHAR2(128)	NOT NULL	Attribute of the driving context
COMMON	VARCHAR2(3)		Indicates whether the policy context is applied and enforced in all application PDBs (YES) or only in the local PDB (NO)
INHERITED	VARCHAR2(3)		Indicates whether the policy context is inherited from the root (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> - "DBA_POLICY_CONTEXTS"
> - "USER_POLICY_CONTEXTS"

3.16 ALL_POLICY_GROUPS

ALL_POLICY_GROUPS describes the policy groups defined for the synonyms, tables, and views accessible to the current user.

Related Views

- DBA_POLICY_GROUPS describes all policy groups in the database.
- USER_POLICY_GROUPS describes the policy groups defined for the synonyms, tables, and views owned by the current user. This view does not display the OBJECT_OWNER column.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)		Owner of the synonym, table, or view
OBJECT_NAME	VARCHAR2(128)		Name of the synonym, table, or view
POLICY_GROUP	VARCHAR2(128)		Name of the policy group
COMMON	VARCHAR2(3)		Indicates whether the policy group is applied and enforced in all application PDBs (YES) or only in the local PDB (NO)
INHERITED	VARCHAR2(3)		Indicates whether the policy group is inherited from the root (YES) or not (NO)

> **See Also**
>
> - "DBA_POLICY_GROUPS"
> - "USER_POLICY_GROUPS"

3.17 ALL_PROCEDURES

ALL_PROCEDURES lists all functions and procedures that are accessible to the current user, along with associated properties. For example, ALL_PROCEDURES indicates whether or not a function is pipelined, parallel enabled or an aggregate function. If a function is pipelined or an aggregate function, the associated implementation type (if any) is also identified.

Related Views

- DBA_PROCEDURES lists all functions and procedures available in the database, along with associated properties.
- USER_PROCEDURES lists all functions and procedures owned by the current user, along with associated properties. It does not contain the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the procedure
OBJECT_NAME	VARCHAR2(128)		Name of the object: top-level function, procedure, or package name
PROCEDURE_NAME	VARCHAR2(128)		Name of the procedure
OBJECT_ID	NUMBER		Object number of the object
SUBPROGRAM_ID	NUMBER		Unique subprogram identifier
OVERLOAD	VARCHAR2(40)		Overload unique identifier
OBJECT_TYPE	VARCHAR2(13)		The typename of the object
AGGREGATE	VARCHAR2(3)		Indicates whether the procedure is an aggregate function (YES) or not (NO)
PIPELINED	VARCHAR2(3)		Indicates whether the procedure is a pipelined table function (YES) or not (NO)
IMPLTYPEOWNER	VARCHAR2(128)		Owner of the implementation type, if any
IMPLTYPENAME	VARCHAR2(128)		Name of the implementation type, if any
PARALLEL	VARCHAR2(3)		Indicates whether the procedure or function is parallel-enabled (YES) or not (NO)
INTERFACE	VARCHAR2(3)		YES, if the procedure/function is a table function implemented using the ODCI interface; otherwise NO
DETERMINISTIC	VARCHAR2(3)		YES, if the procedure/function is declared to be deterministic; otherwise NO
AUTHID	VARCHAR2(12)		Indicates whether the procedure/function is declared to execute as DEFINER or CURRENT_USER (invoker)
RESULT_CACHE	VARCHAR2(3)		Indicates whether the function is result–cached (YES) or not (NO)
ORIGIN_CON_ID	VARCHAR2(256)		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)
POLYMORPHIC	VARCHAR2(5)		The type of polymorphic table function: • ROW • TABLE • LEAF • NULL

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> SQL macros for table expressions are supported starting with Oracle Database 19c, Release Update 19.7. In future releases, the `ALL_`, `DBA_`, and `USER_PROCEDURES` data dictionary views contain a column called `SQL_MACRO`, which enables you to identify the SQL macros in your database. However, this column is not available in Oracle Database 19c. Instead, refer to My Oracle Support note KB125926 "How To Identify the SQL Macros in Oracle Data Dictionary 19.7 Onwards" at the following URL to learn how to identify SQL macros in Oracle Database 19c:
>
> https://support.oracle.com/support/?documentId=KB125926

> **See Also**
>
> - "DBA_PROCEDURES"
> - "USER_PROCEDURES"
> - "ALL_ARGUMENTS" for information about the arguments of the functions and procedures that are accessible to the current user

3.18 ALL_PROPAGATION

`ALL_PROPAGATION` displays information about the propagations that have a source queue accessible to the current user.

Related View

`DBA_PROPAGATION` displays information about all propagations in the database.

Column	Datatype	NULL	Description
`PROPAGATION_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the propagation
`SOURCE_QUEUE_OWNER`	`VARCHAR2(128)`		Owner of the source queue of the propagation
`SOURCE_QUEUE_NAME`	`VARCHAR2(128)`		Name of the source queue of the propagation
`DESTINATION_QUEUE_OWNER`	`VARCHAR2(128)`		Owner of the destination queue of the propagation
`DESTINATION_QUEUE_NAME`	`VARCHAR2(128)`		Name of the destination queue of the propagation
`DESTINATION_DBLINK`	`VARCHAR2(128)`		Database link to propagate events from the source queue to the destination queue
`RULE_SET_OWNER`	`VARCHAR2(128)`		Owner of the propagation positive rule set
`RULE_SET_NAME`	`VARCHAR2(128)`		Name of the propagation positive rule set
`NEGATIVE_RULE_SET_OWNER`	`VARCHAR2(128)`		Owner of the propagation negative rule set
`NEGATIVE_RULE_SET_NAME`	`VARCHAR2(128)`		Name of the propagation negative rule set
`QUEUE_TO_QUEUE`	`VARCHAR2(5)`		Indicates whether the propagation is a queue-to-queue propagation (`TRUE`) or not (`FALSE`). A queue-to-queue propagation always has its own exclusive propagation job to propagate messages from the source queue to the destination queue.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(8)		Status of the propagation: • DISABLED • ENABLED • ABORTED
ERROR_MESSAGE	VARCHAR2(4000)		Error message last encountered by propagation
ERROR_DATE	DATE		Time that propagation last encountered an error
ORIGINAL_PROPAGATION_NAME	VARCHAR2(128)		Original propagation from which the propagation is cloned
ORIGINAL_SOURCE_QUEUE_OWNER	VARCHAR2(128)		Source queue owner of the original propagation
ORIGINAL_SOURCE_QUEUE_NAME	VARCHAR2(128)		Source queue name of the original propagation
ACKED_SCN	NUMBER		Acknowledged SCN of the subscribers of captured messages in the destination queue for the propagation
AUTO_MERGE_THRESHOLD	NUMBER		Merge threshold value for merging the propagation back to the original source queue

See Also

"DBA_PROPAGATION"

3.19 ALL_QUEUE_SCHEDULES

ALL_QUEUE_SCHEDULES describes the propagation schedules whose source queues are accessible to the current user.

Related Views

- DBA_QUEUE_SCHEDULES describes all propagation schedules in the database.
- USER_QUEUE_SCHEDULES describes the propagation schedules whose source queues are owned by the current user. This view does not display the SCHEMA column.

Column	Datatype	NULL	Description
SCHEMA	VARCHAR2(128)		Source queue owner
QNAME	VARCHAR2(128)		Source queue name
DESTINATION	VARCHAR2(395)		Destination name, currently limited to be a DBLINK name
START_DATE	TIMESTAMP(6) WITH TIME ZONE		Date at which to start propagation
START_TIME	VARCHAR2(8)		Time of day at which to start propagation (in HH:MI:SS format)
PROPAGATION_WINDOW	NUMBER		Duration for the propagation window (in seconds)
NEXT_TIME	VARCHAR2(4000)		Function to compute the start of the next propagation window

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LATENCY	NUMBER		Maximum wait time to propagate a message during the propagation window
SCHEDULE_DISABLED	VARCHAR2(1)		Indicates whether the schedule is disabled (Y) or enabled (N). If disabled, then the schedule will not be executed.
PROCESS_NAME	VARCHAR2(4)		Name of the process executing the schedule; NULL if not currently executing
SESSION_ID	VARCHAR2(82)		Session ID and session serial number of the job executing this schedule (SID, SERIAL#); NULL if not currently executing
INSTANCE	NUMBER		Cluster database instance number executing the schedule
LAST_RUN_DATE	TIMESTAMP(6) WITH TIME ZONE		Date of the last successful execution
LAST_RUN_TIME	VARCHAR2(8)		Time of day of the last successful execution (in HH:MI:SS format)
CURRENT_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Date at which the current window of this schedule was started
CURRENT_START_TIME	VARCHAR2(8)		Time of day at which the current window of this schedule was started (in HH:MI:SS format)
NEXT_RUN_DATE	TIMESTAMP(6) WITH TIME ZONE		Date at which the next window of this schedule will be started
NEXT_RUN_TIME	VARCHAR2(8)		Time of day at which the next window of this schedule will be started (in HH:MI:SS format)
TOTAL_TIME	NUMBER		Total time spent by the system in executing this schedule (in seconds)
TOTAL_NUMBER	NUMBER		Total number of messages propagated in this schedule
TOTAL_BYTES	NUMBER		Total number of bytes propagated in this schedule
MAX_NUMBER	NUMBER		Maximum number of messages propagated in a propagation window
MAX_BYTES	NUMBER		Maximum number of bytes propagated in a propagation window
AVG_NUMBER	NUMBER		Average number of messages propagated in a propagation window
AVG_SIZE	NUMBER		Average size of a propagated message (in bytes)
AVG_TIME	NUMBER		Average time to propagate a message (in seconds)
FAILURES	NUMBER		Number of consecutive times schedule execution has failed, if any. After 16 consecutive failures, a propagation job becomes disabled automatically.
LAST_ERROR_DATE	DATE		Date of the last unsuccessful execution
LAST_ERROR_TIME	VARCHAR2(8)		Time of day of the last unsuccessful execution (in HH:MI:SS format)
LAST_ERROR_MSG	VARCHAR2(4000)		Error number and error message text of the last unsuccessful execution
MESSAGE_DELIVERY_MODE	VARCHAR2(10)		Message delivery mode: • PERSISTENT • BUFFERED

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ELAPSED_DEQUEUE_TIME	NUMBER		Elapsed dequeue time (in hundredths of a second)
ELAPSED_PICKLE_TIME	NUMBER		Elapsed pickle time (time taken to linearize a logical change record (LCR) into a stream of bytes that can be sent over the network) (in hundredths of a second)
JOB_NAME	VARCHAR2(128)		Name of the Scheduler job

See Also

- "DBA_QUEUE_SCHEDULES"
- "USER_QUEUE_SCHEDULES"

3.20 ALL_QUEUE_SUBSCRIBERS

ALL_QUEUE_SUBSCRIBERS displays the list of subscribers that the current user has privilege to dequeue from.

Related Views

- DBA_QUEUE_SUBSCRIBERS displays the list of subscribers on all queues in the database.
- USER_QUEUE_SUBSCRIBERS displays the list of subscribers on queues that are under the current user's schema. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the queue
QUEUE_NAME	VARCHAR2(128)	NOT NULL	Name of the queue
QUEUE_TABLE	VARCHAR2(128)	NOT NULL	Name of the queue table on which the queue is defined
CONSUMER_NAME	VARCHAR2(512)		Name of the subscriber
ADDRESS	VARCHAR2(1024)		Address of the subscriber
PROTOCOL	NUMBER		Protocol of the subscriber
TRANSFORMATION	VARCHAR2(61)		Transformation for the subscriber
RULE	CLOB		Rule condition for the subscriber
DELIVERY_MODE	VARCHAR2(22)		Message delivery mode for the subscriber: • PERSISTENT • BUFFERED • PERSISTENT_OR_BUFFERED
IF_NONDURABLE_SUBSCRIBER	VARCHAR2(3)		Indicates whether the subscriber is a non-durable subscriber (YES) or not (NO)
QUEUE_TO_QUEUE	VARCHAR2(5)		Indicates whether the subscriber is a queue-to-queue subscriber (TRUE) or not (FALSE)
SUBSCRIBER_ID	NUMBER		ID of the subscriber
POS_BITMAP	NUMBER		Position of the subscriber in the bitmap

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_QUEUE_SUBSCRIBERS"
- "USER_QUEUE_SUBSCRIBERS"

3.21 ALL_QUEUE_TABLES

ALL_QUEUE_TABLES describes the queues in the queue tables accessible to the current user.

Related Views

- DBA_QUEUE_TABLES describes the queues in all queue tables in the database.
- USER_QUEUE_TABLES describes the queues in the queue tables created in the current user's schema. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the queue table
QUEUE_TABLE	VARCHAR2(128)		Name of the queue table
TYPE	VARCHAR2(9)		Type of user data: • RAW - Raw type • OBJECT - User-defined object type • VARIANT - Variant type (for internal use only)
OBJECT_TYPE	VARCHAR2(257)		Object type of the payload when TYPE is OBJECT
SORT_ORDER	VARCHAR2(22)		User-specified sort order
RECIPIENTS	VARCHAR2(8)		SINGLE or MULTIPLE recipients
MESSAGE_GROUPING	VARCHAR2(13)		NONE or TRANSACTIONAL
REPLICATION_MODE	VARCHAR2(22)		Indicates whether the queue tables are enabled for replication through Oracle GoldenGate. If the queue tables are replicated, these values appear in the column: • REPLICATED_SOURCE: This value is displayed for a source queue table. • REPLICATED_DESTINATION: This value is displayed for a destination queue table. If replication is not enabled on the queue tables, then this column is empty.
COMPATIBLE	VARCHAR2(6)		Lowest release level which the queue table is compatible with (for example, 8.0.3)
PRIMARY_INSTANCE	NUMBER		Indicates the instance number of the instance which is the primary owner of the queue table. A value of 0 indicates that there is no primary owner.
SECONDARY_INSTANCE	NUMBER		Indicates the instance number of the instance which is the secondary owner of the queue table. This instance becomes the owner of the queue table if the primary owner goes down. A value of 0 indicates that there is no secondary owner.
OWNER_INSTANCE	NUMBER		Instance number of the instance which currently owns the queue table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
USER_COMMENT	VARCHAR2(50)		Comment supplied by the user
SECURE	VARCHAR2(3)		Indicates whether the queue table is secure (YES) or not (NO)

See Also

- "DBA_QUEUE_TABLES"
- "USER_QUEUE_TABLES"
- *Oracle Database Advanced Queuing User's Guide* for more information Advanced Queuing

3.22 ALL_QUEUES

ALL_QUEUES describes all queues on which the current user has enqueue or dequeue privileges. If the user has any Advanced Queuing system privileges, like MANAGE ANY QUEUE, ENQUEUE ANY QUEUE or DEQUEUE ANY QUEUE, then this view describes all queues in the database.

Related Views

- DBA_QUEUES describes all queues in the database.
- USER_QUEUES describes the operational characteristics of every queue owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the queue
NAME	VARCHAR2(128)	NOT NULL	Name of the queue
QUEUE_TABLE	VARCHAR2(128)	NOT NULL	Name of the table the queue data resides in
QID	NUMBER	NOT NULL	Object number of the queue
QUEUE_TYPE	VARCHAR2(20)		Type of the queue: • EXCEPTION_QUEUE • NON_PERSISTENT_QUEUE • NORMAL_QUEUE
MAX_RETRIES	NUMBER		Maximum number of retries allowed when dequeuing from the queue
RETRY_DELAY	NUMBER		Time interval between retries
ENQUEUE_ENABLED	VARCHAR2(7)		Indicates whether the queue is enabled for enqueue (YES) or not (NO)
DEQUEUE_ENABLED	VARCHAR2(7)		Indicates whether the queue is enabled for dequeue (YES) or not (NO)
RETENTION	VARCHAR2(40)		Time interval (in seconds) processed messages are retained in the queue, or FOREVER
USER_COMMENT	VARCHAR2(50)		User specified comment
NETWORK_NAME	VARCHAR2(512)		Network name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SHARDED	VARCHAR2(5)		TRUE if queue is sharded, FALSE otherwise

See Also

- "DBA_QUEUES"
- "USER_QUEUES"
- *Oracle Database Advanced Queuing User's Guide* for more information Advanced Queuing

3.23 ALL_REFRESH

ALL_REFRESH describes all the refresh groups accessible to the current user.

Related Views

- DBA_REFRESH describes all refresh groups in the database.
- USER_REFRESH describes all refresh groups owned by the current user.

Column	Datatype	NULL	Description
ROWNER	VARCHAR2(128)	NOT NULL	Owner of the refresh group
RNAME	VARCHAR2(128)	NOT NULL	Name of the refresh group
REFGROUP	NUMBER		Internal identifier of the refresh group
IMPLICIT_DESTROY	VARCHAR2(1)		Indicates whether the refresh group is destroyed when its last item is subtracted (Y) or not (N)
PUSH_DEFERRED_RPC	VARCHAR2(1)		Indicates whether changes are pushed from the snapshot to the master before refresh (Y) or not (N)
REFRESH_AFTER _ERRORS	VARCHAR2(1)		Indicates whether to proceed with refresh despite errors when pushing deferred RPCs (Y) or not (N)
ROLLBACK_SEG	VARCHAR2(128)		Name of the rollback segment to use while refreshing
JOB	NUMBER		Identifier of the job used to refresh the group automatically
NEXT_DATE	DATE		Date that this job will next be refreshed automatically, if not broken
INTERVAL	VARCHAR2(200)		A date function used to compute the next NEXT_DATE
BROKEN	VARCHAR2(1)		Indicates whether the job is broken and will never be run (Y) or not (N)
PURGE_OPTION	NUMBER(38)		Method for purging the transaction queue after each push (1 indicates quick purge option; 2 indicates precise purge option)
PARALLELISM	NUMBER(38)		Level of parallelism for transaction propagation
HEAP_SIZE	NUMBER(38)		Size of the heap
JOB_NAME	VARCHAR2(128)		The name of the job used to automatically refresh the group

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_REFRESH"
- "USER_REFRESH"

3.24 ALL_REFRESH_CHILDREN

ALL_REFRESH_CHILDREN describes all the objects in refresh groups that are accessible to the current user.

Related Views

- DBA_REFRESH_CHILDREN describes the objects in all refresh groups in the database.
- USER_REFRESH_CHILDREN describes the objects in all refresh groups owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object in the refresh group
NAME	VARCHAR2(128)	NOT NULL	Name of the object in the refresh group
TYPE	VARCHAR2(128)		Type of the object in the refresh group
ROWNER	VARCHAR2(128)	NOT NULL	Owner of the refresh group
RNAME	VARCHAR2(128)	NOT NULL	Name of the refresh group
REFGROUP	NUMBER		Internal identifier of the refresh group
IMPLICIT_DESTROY	VARCHAR2(1)		Indicates whether the refresh group is destroyed when its last item is subtracted (Y) or not (N)
PUSH_DEFERRED_RPC	VARCHAR2(1)		Indicates whether changes are pushed from the snapshot to the master before refresh (Y) or not (N)
REFRESH_AFTER _ERRORS	VARCHAR2(1)		Indicates whether to proceed with refresh despite errors when pushing deferred RPCs (Y) or not (N)
ROLLBACK_SEG	VARCHAR2(128)		Name of the rollback segment to use while refreshing
JOB	NUMBER		Identifier of the job used to refresh the group automatically
NEXT_DATE	DATE		Date that this job will next be refreshed automatically, if not broken
INTERVAL	VARCHAR2(200)		A date function used to compute the next NEXT_DATE
BROKEN	VARCHAR2(1)		Indicates whether the job is broken and will never be run (Y) or not (N)
PURGE_OPTION	NUMBER(38)		Method for purging the transaction queue after each push. 1 indicates quick purge option; 2 indicates precise purge option
PARALLELISM	NUMBER(38)		Level of parallelism for transaction propagation
HEAP_SIZE	NUMBER(38)		Size of the heap
JOB_NAME	VARCHAR2(128)		The name of the job used to automatically refresh the group

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_REFRESH_CHILDREN"
- "USER_REFRESH_CHILDREN"

3.25 ALL_REFRESH_DEPENDENCIES

ALL_REFRESH_DEPENDENCIES displays the names of the dependent detail or container tables of all the materialized views in the current schema.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Table name, unique within this schema
PARENT_OBJECT_TYPE	CHAR(17)		MATERIALIZED VIEW
OLDEST_REFRESH_SCN	NUMBER		Minimum SCN of any summary or materialized view that has TABLE_NAME as a detail table
OLDEST_REFRESH_DATE	DATE		SYSDATE when last refreshed

3.26 ALL_REFS

ALL_REFS describes the REF columns and REF attributes in object type columns accessible to the current user.

Related Views

- DBA_REFS describes all REF columns and REF attributes in the database.
- USER_REFS describes the REF columns and REF attributes in object type columns owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
COLUMN_NAME	VARCHAR2(4000)		Name of the REF column or attribute. If it is not a top-level attribute, the value of COLUMN_NAME should be a path name starting with the column name.
WITH_ROWID	VARCHAR2(3)		Indicates whether the REF value is stored with ROWID (YES) or not (NO)
IS_SCOPED	VARCHAR2(3)		Indicates whether the REF column is scoped (YES) or not (NO)
SCOPE_TABLE_OWNER	VARCHAR2(128)		Owner of the scope table, if it exists and is accessible by the user
SCOPE_TABLE_NAME	VARCHAR2(128)		Name of the scope table, if it exists and is accessible by the user
OBJECT_ID_TYPE	VARCHAR2(33)		Indicates whether the object ID (OID) is USER-DEFINED or SYSTEM GENERATED

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_REFS"
- "USER_REFS"

3.27 ALL_REGISTERED_MVIEWS

ALL_REGISTERED_MVIEWS describes all registered materialized views (registered at a master site or a master materialized view site) accessible to the current user.

A materialized view created with the BUILD DEFERRED option of the CREATE MATERIALIZED VIEW statement is only registered with ALL_REGISTERED_MVIEWS if that materialized view has been completely refreshed at least once.

Related Views

- DBA_REGISTERED_MVIEWS describes all registered materialized views in the database.
- USER_REGISTERED_MVIEWS describes all registered materialized views owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
MVIEW_SITE	VARCHAR2(128)	NOT NULL	Global name of the materialized view site
CAN_USE_LOG	VARCHAR2(3)		Indicates whether the materialized view can use a materialized view log (YES) or the materialized view is too complex to use a log (NO)
UPDATABLE	VARCHAR2(3)		Indicates whether the materialized view is updatable (YES) or not and the materialized view is read only (NO)
REFRESH_METHOD	VARCHAR2(11)		Indicates whether the materialized view uses primary key (PRIMARY KEY), rowids (ROWID), or object identifiers (OBJECT ID) for fast refresh
MVIEW_ID	NUMBER(38)		Identifier for the materialized view used by the masters for fast refresh
VERSION	VARCHAR2(26)		Oracle version of the materialized view
QUERY_TXT	LONG		Query that defines the materialized view

See Also

- "DBA_REGISTERED_MVIEWS"
- "USER_REGISTERED_MVIEWS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.28 ALL_REGISTRY_BANNERS

ALL_REGISTRY_BANNERS displays the valid components loaded into the database.

Column	Datatype	NULL	Description
BANNER	VARCHAR2(80)		Component display banner
BANNER_FULL	VARCHAR2(80)		Component display banner with full version

3.29 ALL_REPL_DBNAME_MAPPING

ALL_REPL_DBNAME_MAPPING provides details about the database name mapping in replication for the current user.

Related View

DBA_REPL_DBNAME_MAPPING provides details about the database name mapping in replication.

Column	Datatype	NULL	Description
SOURCE_ROOT_NAME	VARCHAR2(128)		The fully qualified global name of the root in a multitenant container database (CDB) where the changes originated
SOURCE_DATABASE_NAME	VARCHAR2(128)		The fully qualified global name of the pluggable database (PDB) where the changes originated
SOURCE_CONTAINER_NAME	VARCHAR2(128)		The container name of the database where the changes originated

See Also

"DBA_REPL_DBNAME_MAPPING"

3.30 ALL_REPLICATION_PROCESS_EVENTS

ALL_REPLICATION_PROCESS_EVENTS provides information about the replication processes events accessible to the current user.

Related View

DBA_REPLICATION_PROCESS_EVENTS provides information about the replication processes events in the database.

Column	Datatype	NULL	Description
STREAMS_TYPE	VARCHAR2(10)		Streams type: • XStream • GoldenGate

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PROCESS_TYPE	VARCHAR2(27)		Process type: • Capture • Capture server • Apply Coordinator • Apply Server • Apply Network Receiver • Apply Reader • Apply Hash server
STREAMS_NAME	VARCHAR2(128)	NOT NULL	Streams name
EVENT_NAME	VARCHAR2(128)		Event name: • START • STOP • ABORT • CREATE • DROP • PARAMETER CHANGE • HANDLER CREATE • HANDLER REMOVE • ALTER
DESCRIPTION	VARCHAR2(2000)		Event description
EVENT_TIME	TIMESTAMP(6)		Time when the event occurred
ERROR_NUMBER	NUMBER		Error number (valid when event is Error)
ERROR_MESSAGE	VARCHAR2(2000)		Error Message (valid when event is an error)

See Also

"DBA_REPLICATION_PROCESS_EVENTS"

3.31 ALL_REWRITE_EQUIVALENCES

ALL_REWRITE_EQUIVALENCES describes the rewrite equivalences accessible to the current user.

Related Views

- DBA_REWRITE_EQUIVALENCES describes all rewrite equivalences in the database.
- USER_REWRITE_EQUIVALENCES describes the rewrite equivalences owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the rewrite equivalence
NAME	VARCHAR2(128)	NOT NULL	Name of the rewrite equivalence
SOURCE_STMT	CLOB		Source statement of the rewrite equivalence
DESTINATION_STMT	CLOB		Destination of the rewrite equivalence

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
REWRITE_MODE	VARCHAR2(10)		Rewrite mode of the rewrite equivalence: • DISABLED • TEXT_MATCH • GENERAL • RECURSIVE

See Also

- "DBA_REWRITE_EQUIVALENCES"
- "USER_REWRITE_EQUIVALENCES"

3.32 ALL_RULE_SET_RULES

ALL_RULE_SET_RULES describes the rules in the rule sets accessible to the current user.

Related Views

- DBA_RULE_SET_RULES describes the rules in all rule sets in the database.
- USER_RULE_SET_RULES describes the rules in the rule sets owned by the current user. This view does not display the RULE_SET_OWNER column.

Column	Datatype	NULL	Description
RULE_SET_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule set
RULE_SET_NAME	VARCHAR2(128)	NOT NULL	Name of the rule set
RULE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule
RULE_NAME	VARCHAR2(128)	NOT NULL	Name of the rule
RULE_SET_RULE_ENABLED	VARCHAR2(8)		Indicates whether the rule is enabled in the rule set (ENABLED) or not (DISABLED)
RULE_SET_RULE_EVAL_CTX_OWNER	VARCHAR2(128)		Owner of the evaluation context specified when the rule was added to the rule set, if any
RULE_SET_RULE_EVAL_CTX_NAME	VARCHAR2(128)		Name of the evaluation context specified when the rule was added to the rule set, if any
RULE_SET_RULE_COMMENT	VARCHAR2(4000)		Comment specified when the rule was added to the rule set, if any

See Also

- "DBA_RULE_SET_RULES"
- "USER_RULE_SET_RULES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.33 ALL_RULE_SETS

ALL_RULE_SETS describes the rule sets accessible to the current user.

Related Views

- DBA_RULE_SETS describes all rule sets in the database.
- USER_RULE_SETS describes the rule sets owned by the current user. This view does not display the RULE_SET_OWNER column.

Column	Datatype	NULL	Description
RULE_SET_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule set
RULE_SET_NAME	VARCHAR2(128)	NOT NULL	Name of the rule set
RULE_SET_EVAL_CONTEXT_OWNER	VARCHAR2(128)		Owner of the evaluation context associated with the rule set, if any
RULE_SET_EVAL_CONTEXT_NAME	VARCHAR2(128)		Name of the evaluation context associated with the rule set, if any
RULE_SET_COMMENT	VARCHAR2(4000)		Comment specified with the rule set, if any

See Also

- "DBA_RULE_SETS"
- "USER_RULE_SETS"

3.34 ALL_RULES

ALL_RULES describes the rules accessible to the current user.

Related Views

- DBA_RULES describes all rules in the database.
- USER_RULES describes the rules owned by the current user. This view does not display the RULE_OWNER column.

Column	Datatype	NULL	Description
RULE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule
RULE_NAME	VARCHAR2(128)	NOT NULL	Name of the rule
RULE_CONDITION	CLOB		Expressions and operators that constitute the rule condition
RULE_EVALUATION_CONTEXT_OWNER	VARCHAR2(128)		Owner of the evaluation context associated with the rule, if any
RULE_EVALUATION_CONTEXT_NAME	VARCHAR2(128)		Name of the evaluation context associated with the rule, if any
RULE_ACTION_CONTEXT	RE$NV_LIST		Action context associated with the rule, if any
RULE_COMMENT	VARCHAR2(4000)		Comment specified with the rule, if any

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_RULES"
- "USER_RULES"

3.35 ALL_SCHEDULER_CHAIN_RULES

ALL_SCHEDULER_CHAIN_RULES displays information about the rules for the chains accessible to the current user (that is, those chains that the user has ALTER or EXECUTE privileges for).

Related Views

- DBA_SCHEDULER_CHAIN_RULES displays information about the rules for all chains in the database.
- USER_SCHEDULER_CHAIN_RULES displays information about the rules for the chains owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Scheduler chain that the rule is in
CHAIN_NAME	VARCHAR2(128)	NOT NULL	Name of the Scheduler chain that the rule is in
RULE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule
RULE_NAME	VARCHAR2(128)		Name of the rule
CONDITION	VARCHAR2(4000)		Boolean condition triggering the rule
ACTION	VARCHAR2(4000)		Action to be performed when the rule is triggered
COMMENTS	VARCHAR2(4000)		User-specified comments about the rule

See Also

- "DBA_SCHEDULER_CHAIN_RULES"
- "USER_SCHEDULER_CHAIN_RULES"

3.36 ALL_SCHEDULER_CHAIN_STEPS

ALL_SCHEDULER_CHAIN_STEPS displays information about the defined steps of the chains accessible to the current user (that is, those chains that the user has ALTER or EXECUTE privileges for).

Related Views

- DBA_SCHEDULER_CHAIN_STEPS displays information about the defined steps of all chains in the database.
- USER_SCHEDULER_CHAIN_STEPS displays information about the defined steps of the chains owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Scheduler chain the step is in
CHAIN_NAME	VARCHAR2(128)	NOT NULL	Name of the Scheduler chain the step is in
STEP_NAME	VARCHAR2(128)	NOT NULL	Name of the chain step
PROGRAM_OWNER	VARCHAR2(392)		Owner of the program that runs during the step
PROGRAM_NAME	VARCHAR2(392)		Name of the program that runs during the step
EVENT_SCHEDULE_OWNER	VARCHAR2(392)		Owner of the event schedule that this step waits for
EVENT_SCHEDULE_NAME	VARCHAR2(392)		Name of the event schedule that this step waits for
EVENT_QUEUE_OWNER	VARCHAR2(128)		Owner of the source queue into which the event will be raised
EVENT_QUEUE_NAME	VARCHAR2(128)		Name of the source queue into which the event will be raised
EVENT_QUEUE_AGENT	VARCHAR2(523)		Name of the AQ agent used by the user on the event source queue (for a secure queue)
EVENT_CONDITION	VARCHAR2(4000)		Boolean expression used as the subscription rule for an event on the source queue
CREDENTIAL_OWNER	VARCHAR2(128)		Owner of the credential to be used for an external step job
CREDENTIAL_NAME	VARCHAR2(128)		Name of the credential to be used for an external step job
DESTINATION	VARCHAR2(261)		Destination host on which a remote step job will run
SKIP	VARCHAR2(5)		Indicates whether the step should be skipped (TRUE) or not (FALSE)
PAUSE	VARCHAR2(5)		Indicates whether the step should be paused after running (TRUE) or not (FALSE)
PAUSE_BEFORE	VARCHAR2(5)		Indicates whether the step should be paused before running (TRUE) or not (FALSE)
RESTART_ON_RECOVERY	VARCHAR2(5)		Indicates whether the step should be restarted on database recovery (TRUE) or not (FALSE)
RESTART_ON_FAILURE	VARCHAR2(5)		Indicates whether the step should be restarted on application failure (TRUE) or not (FALSE)
STEP_TYPE	VARCHAR2(21)		Type of the step: • EVENT_SCHEDULE • INLINE_EVENT • SUBCHAIN • PROGRAM
TIMEOUT	INTERVAL DAY(3) TO SECOND(0)		Timeout for waiting on an event schedule

See Also

- "DBA_SCHEDULER_CHAIN_STEPS"
- "USER_SCHEDULER_CHAIN_STEPS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.37 ALL_SCHEDULER_CHAINS

ALL_SCHEDULER_CHAINS displays information about the chains accessible to the current user (that is, those chains that the user has ALTER or EXECUTE privileges for).

Related Views

- DBA_SCHEDULER_CHAINS displays information about all chains in the database.
- USER_SCHEDULER_CHAINS displays information about the chains owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Scheduler chain
CHAIN_NAME	VARCHAR2(128)	NOT NULL	Name of the Scheduler chain
RULE_SET_OWNER	VARCHAR2(128)		Owner of the rule set describing the dependencies
RULE_SET_NAME	VARCHAR2(128)		Name of the rule set describing the dependencies
NUMBER_OF_RULES	NUMBER		Number of rules in the chain
NUMBER_OF_STEPS	NUMBER		Number of defined steps in the chain
ENABLED	VARCHAR2(5)		Indicates whether the chain is enabled (TRUE) or disabled (FALSE)
EVALUATION_INTERVAL	INTERVAL DAY(3) TO SECOND(0)		Periodic interval at which to reevaluate rules for the chain
USER_RULE_SET	VARCHAR2(5)		Indicates whether the chain uses a user-specified rule set (TRUE) or not (FALSE)
COMMENTS	VARCHAR2(4000)		Comments on the chain

See Also

- "DBA_SCHEDULER_CHAINS"
- "USER_SCHEDULER_CHAINS"

3.38 ALL_SCHEDULER_CREDENTIALS

ALL_SCHEDULER_CREDENTIALS displays information about the credentials accessible to the current user (that is, those credentials that the user has ALTER or EXECUTE privileges for).

Note

This view is deprecated in favor of the ALL_CREDENTIALS view. Oracle recommends that you use ALL_CREDENTIALS instead. ALL_SCHEDULER_CREDENTIALS is retained for backward compatibility only.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Related Views

- DBA_SCHEDULER_CREDENTIALS displays information about all credentials in the database.
- USER_SCHEDULER_CREDENTIALS displays information about the credentials owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Scheduler credential
CREDENTIAL_NAME	VARCHAR2(128)	NOT NULL	Name of the Scheduler credential
USERNAME	VARCHAR2(128)		Name of the user that will be used to log in to the remote database or operating system
DATABASE_ROLE	VARCHAR2(9)		For a database target, the database role to use when logging in: • SYSDBA • SYSOPER
WINDOWS_DOMAIN	VARCHAR2(30)		For a Windows target, the Windows domain to use when logging in
COMMENTS	VARCHAR2(4000)		Comments on the credential

See Also

- "ALL_CREDENTIALS"
- "DBA_SCHEDULER_CREDENTIALS"
- "USER_SCHEDULER_CREDENTIALS"

3.39 ALL_SCHEDULER_DB_DESTS

ALL_SCHEDULER_DB_DESTS displays information about the destination objects accessible to the current user pointing to remote databases.

Related Views

- DBA_SCHEDULER_DB_DESTS displays information about all destination objects in the database pointing to remote databases.
- USER_SCHEDULER_DB_DESTS displays information about the destination objects owned by the current user pointing to remote databases. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of this destination object
DESTINATION_NAME	VARCHAR2(128)	NOT NULL	Name of this destination object
CONNECT_INFO	VARCHAR2(4000)		Connect string to connect to the remote database
AGENT	VARCHAR2(128)		Name of the agent through which the connection to the remote database is being made
ENABLED	VARCHAR2(5)		Indicates whether this destination object is enabled (TRUE) or disabled (FALSE)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
REFS_ENABLED	VARCHAR2(5)		Indicates whether all referenced objects are enabled (TRUE) or disabled (FALSE)
COMMENTS	VARCHAR2(4000)		Optional comment

See Also

- "DBA_SCHEDULER_DB_DESTS"
- "USER_SCHEDULER_DB_DESTS"

3.40 ALL_SCHEDULER_DESTS

ALL_SCHEDULER_DESTS displays information about the destination objects for jobs accessible to the current user.

Related Views

- DBA_SCHEDULER_DESTS displays information about all destination objects for jobs in the database.
- USER_SCHEDULER_DESTS displays information about the destination objects for jobs owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of this destination object
DESTINATION_NAME	VARCHAR2(128)	NOT NULL	Name of this destination object
DESTINATION_TYPE	VARCHAR2(8)		Type of this destination object: • EXTERNAL • DATABASE
ENABLED	VARCHAR2(5)		Indicates whether this destination object is enabled (TRUE) or disabled (FALSE)
COMMENTS	VARCHAR2(4000)		Optional comment

See Also

- "DBA_SCHEDULER_DESTS"
- "USER_SCHEDULER_DESTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.41 ALL_SCHEDULER_EXTERNAL_DESTS

ALL_SCHEDULER_EXTERNAL_DESTS displays information about the destination objects accessible to the current user pointing to remote agents.

Related View

DBA_SCHEDULER_EXTERNAL_DESTS displays information about all destination objects in the database pointing to remote agents.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of this destination object
DESTINATION_NAME	VARCHAR2(128)	NOT NULL	Name of this destination object
HOSTNAME	VARCHAR2(256)		Name or IP address of the host on which the agent is located
PORT	NUMBER		Port that the agent is listening on
IP_ADDRESS	VARCHAR2(64)		IP address of the host on which the agent is located
ENABLED	VARCHAR2(5)		Indicates whether this destination object is enabled (TRUE) or disabled (FALSE)
COMMENTS	VARCHAR2(4000)		Optional comment

See Also

"DBA_SCHEDULER_EXTERNAL_DESTS"

3.42 ALL_SCHEDULER_FILE_WATCHERS

ALL_SCHEDULER_FILE_WATCHERS displays information about the Scheduler file watch requests accessible to the current user.

Related Views

- DBA_SCHEDULER_FILE_WATCHERS displays information about all Scheduler file watch requests in the database.
- USER_SCHEDULER_FILE_WATCHERS displays information about the Scheduler file watch requests owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the file watch request
FILE_WATCHER_NAME	VARCHAR2(128)	NOT NULL	Name of the file watch request
ENABLED	VARCHAR2(5)		Indicates whether this file watch request is enabled (TRUE) or disabled (FALSE)
DESTINATION_OWNER	VARCHAR2(261)		Owner of the named destination object
DESTINATION	VARCHAR2(261)		Name of the destination object
DIRECTORY_PATH	VARCHAR2(4000)	NOT NULL	Name of the directory path where the file will arrive

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FILE_NAME	VARCHAR2(512)	NOT NULL	Name or pattern specifying the files that need to be monitored
CREDENTIAL_OWNER	VARCHAR2(128)		Owner of the credential that should be used to authorize the file watch
CREDENTIAL_NAME	VARCHAR2(128)		Name of the credential that should be used to authorize the file watch
MIN_FILE_SIZE	NUMBER	NOT NULL	Minimum size of the file being monitored
STEADY_STATE_DURATION	INTERVAL DAY(3) TO SECOND(0)		Time to wait before concluding that the file has stopped growing
LAST_MODIFIED_TIME	TIMESTAMP(6) WITH TIME ZONE		Time at which this file watcher was last modified
COMMENTS	VARCHAR2(4000)		Comments on the file watch request

See Also

- "DBA_SCHEDULER_FILE_WATCHERS"
- "USER_SCHEDULER_FILE_WATCHERS"

3.43 ALL_SCHEDULER_GLOBAL_ATTRIBUTE

ALL_SCHEDULER_GLOBAL_ATTRIBUTE displays the values of all scheduler attributes (for example, DEFAULT_TIMEZONE and CURRENT_OPEN_WINDOW).

Related View

DBA_SCHEDULER_GLOBAL_ATTRIBUTE displays the values of all scheduler attributes in the database.

Column	Datatype	NULL	Description
ATTRIBUTE_NAME	VARCHAR2(128)	NOT NULL	Name of the Scheduler attribute
VALUE	VARCHAR2(256)		Value of the Scheduler attribute

See Also

"DBA_SCHEDULER_GLOBAL_ATTRIBUTE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.44 ALL_SCHEDULER_GROUP_MEMBERS

ALL_SCHEDULER_GROUP_MEMBERS displays information about the members of the Scheduler object groups accessible to the current user.

Related Views

- DBA_SCHEDULER_GROUP_MEMBERS displays information about the members of all Scheduler object groups in the database.
- USER_SCHEDULER_GROUP_MEMBERS displays information about the members of the Scheduler object groups owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the group
GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the group
MEMBER_NAME	VARCHAR2(523)		Name of the member of this group

See Also

- "DBA_SCHEDULER_GROUP_MEMBERS"
- "USER_SCHEDULER_GROUP_MEMBERS"

3.45 ALL_SCHEDULER_GROUPS

ALL_SCHEDULER_GROUPS displays information about the Scheduler object groups accessible to the current user.

Related Views

- DBA_SCHEDULER_GROUPS displays information about all Scheduler object groups in the database.
- USER_SCHEDULER_GROUPS displays information about the Scheduler object groups owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the group
GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the group
GROUP_TYPE	VARCHAR2(13)		Type of object contained in the group: • WINDOW • JOB • DB_DEST • EXTERNAL_DEST
ENABLED	VARCHAR2(5)		Indicates whether the group is enabled (TRUE) or disabled (FALSE)
NUMBER_OF_MEMBERS	NUMBER		Number of members in this group

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COMMENTS	VARCHAR2(4000)		An optional comment about this group

See Also

- "DBA_SCHEDULER_GROUPS"
- "USER_SCHEDULER_GROUPS"

3.46 ALL_SCHEDULER_INCOMPAT_MEMBER

ALL_SCHEDULER_INCOMPAT_MEMBER displays all Scheduler incompatibility resource objects members accessible to the current user.

Related Views

- DBA_SCHEDULER_INCOMPAT_MEMBER displays all Scheduler incompatibility resource objects members in the database.
- USER_SCHEDULER_INCOMPAT_MEMBER displays all Scheduler incompatibility resource objects members owned by the current user.

Column	Datatype	NULL	Description
INCOMPATIBILITY_OWNER	VARCHAR2(128)	NOT NULL	Owner of the incompatibility resource object containing this member
INCOMPATIBILITY_NAME	VARCHAR2(128)	NOT NULL	Name of the incompatibility resource object containing this member
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the incompatibility resource member
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the incompatibility resource member

See Also

- "DBA_SCHEDULER_INCOMPAT_MEMBER"
- "USER_SCHEDULER_INCOMPAT_MEMBER"

3.47 ALL_SCHEDULER_INCOMPATS

ALL_SCHEDULER_INCOMPATS displays all Scheduler incompatibility resource objects accessible to the current user.

Related Views

- DBA_SCHEDULER_INCOMPATS displays all Scheduler incompatibility resource objects in the database.
- USER_SCHEDULER_INCOMPATS displays all Scheduler incompatibility resource objects owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the incompatibility resource object
INCOMPATIBILITY_NAME	VARCHAR2(128)	NOT NULL	Name of the incompatibility resource object
CONSTRAINT_LEVEL	VARCHAR2(13)		JOB_LEVEL or PROGRAM_LEVEL The default value JOB_LEVEL indicates that only a single job that is based on the program (or programs) mentioned in the object_name argument of the DBMS_SCHEDULER.CREATE_INCOMPATIBILITY procedure can run at one time. The value PROGRAM_LEVEL indicates that the programs are incompatible, but the jobs based on the same program are not incompatible.
ENABLED	VARCHAR2(5)		Indicates whether the incompatibility is enabled (TRUE) or not (FALSE)
JOBS_RUNNING_COUNT	NUMBER		Current number of running jobs using the incompatibility resource object
COMMENTS	VARCHAR2(256)		Comments for the resource incompatibility object

See Also

- "DBA_SCHEDULER_INCOMPATS"
- "USER_SCHEDULER_INCOMPATS"

3.48 ALL_SCHEDULER_JOB_ARGS

ALL_SCHEDULER_JOB_ARGS displays information about the arguments of the Scheduler jobs accessible to the current user.

Related Views

- DBA_SCHEDULER_JOB_ARGS displays information about the arguments of all Scheduler jobs in the database.
- USER_SCHEDULER_JOB_ARGS displays information about the arguments of the Scheduler jobs owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the job to which the argument belongs
JOB_NAME	VARCHAR2(128)		Name of the job to which the argument belongs
ARGUMENT_NAME	VARCHAR2(128)		Optional name of the argument
ARGUMENT_POSITION	NUMBER		Position of the argument in the argument list
ARGUMENT_TYPE	VARCHAR2(257)		Data type of the argument
VALUE	VARCHAR2(4000)		Value of the argument (in string format) if the argument is a string
ANYDATA_VALUE	ANYDATA		Value of the argument (in AnyData format)
OUT_ARGUMENT	VARCHAR2(5)		Reserved for future use

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> - "DBA_SCHEDULER_JOB_ARGS"
> - "USER_SCHEDULER_JOB_ARGS"

3.49 ALL_SCHEDULER_JOB_CLASSES

ALL_SCHEDULER_JOB_CLASSES displays information about the Scheduler job classes accessible to the current user.

Related View

DBA_SCHEDULER_JOB_CLASSES displays information about all Scheduler job classes in the database.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Scheduler job class
JOB_CLASS_NAME	VARCHAR2(128)	NOT NULL	Name of the Scheduler job class
RESOURCE_CONSUMER_GROUP	VARCHAR2(128)		Resource consumer group associated with the class
SERVICE	VARCHAR2(64)		Name of the service the class is associated with
LOGGING_LEVEL	VARCHAR2(11)		Amount of logging that will be done pertaining to the class: • OFF • RUNS • FAILED RUNS • FULL
LOG_HISTORY	NUMBER		History (in days) to maintain in the job log for the class
COMMENTS	VARCHAR2(4000)		Comments on the class

> **See Also**
>
> "DBA_SCHEDULER_JOB_CLASSES"

3.50 ALL_SCHEDULER_JOB_DESTS

ALL_SCHEDULER_JOB_DESTS displays information about the state of the jobs accessible to the current user at each of their destinations.

Related Views

- DBA_SCHEDULER_JOB_DESTS displays information about the state of all jobs in the database at each of their destinations.
- USER_SCHEDULER_JOB_DESTS displays information about the state of the jobs owned by the current user at each of their destinations. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the Scheduler job
JOB_NAME	VARCHAR2(128)		Name of the Scheduler job
JOB_SUBNAME	VARCHAR2(128)		Subname of the Scheduler job
CREDENTIAL_OWNER	VARCHAR2(128)		Owner of the credential used for the remote destination
CREDENTIAL_NAME	VARCHAR2(128)		Name of the credential used for the remote destination
DESTINATION_OWNER	VARCHAR2(261)		Owner of the destination object that points to the destination
DESTINATION	VARCHAR2(261)		Name of the destination object or the name of the destination itself
JOB_DEST_ID	NUMBER		Numerical ID assigned to the job at this destination
ENABLED	VARCHAR2(5)		Indicates whether the parent job is enabled (TRUE) or disabled (FALSE)
REFS_ENABLED	VARCHAR2(5)		Indicates whether this destination and its agent are enabled (TRUE) or disabled (FALSE)
STATE	VARCHAR2(15)		State of this job at this destination: • DISABLED • RUNNING • CHAIN_STALLED • SCHEDULED • RETRY SCHEDULED • READY TO RUN • COMPLETED • BROKEN • FAILED • SUCCEEDED • REMOTE • STOPPED
NEXT_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Next start time of this job at this destination
RUN_COUNT	NUMBER		Number of times this job has run at this destination
RETRY_COUNT	NUMBER		Number of times this job has been retried at this destination
FAILURE_COUNT	NUMBER		Number of times this job has failed at this destination
LAST_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Last time this job started at this destination
LAST_END_DATE	TIMESTAMP(6) WITH TIME ZONE		Last time this job ended at this destination

See Also

- "DBA_SCHEDULER_JOB_DESTS"
- "USER_SCHEDULER_JOB_DESTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.51 ALL_SCHEDULER_JOB_LOG

ALL_SCHEDULER_JOB_LOG displays log information for the Scheduler jobs accessible to the current user.

Related Views

- DBA_SCHEDULER_JOB_LOG displays log information for all Scheduler jobs in the database.
- USER_SCHEDULER_JOB_LOG displays log information for the Scheduler jobs owned by the current user.

Column	Datatype	NULL	Description
LOG_ID	NUMBER	NOT NULL	Unique identifier that identifies a row
LOG_DATE	TIMESTAMP(6) WITH TIME ZONE		Date of the log entry
OWNER	VARCHAR2(128)		Owner of the Scheduler job
JOB_NAME	VARCHAR2(261)		Name of the Scheduler job
JOB_SUBNAME	VARCHAR2(261)		Subname of the Scheduler job (for a chain step job)
JOB_CLASS	VARCHAR2(128)		Class that the job belonged to at the time of entry
OPERATION	VARCHAR2(30)		Operation corresponding to the log entry
STATUS	VARCHAR2(30)		Status of the operation, if applicable. Possible values for this column are dependent on the value in the OPERATION column. In most cases, STATUS will be NULL. Only for job run operations will it have a value. STATUS will be NULL when OPERATION is one of the following: • CREATE - Job was created • UPDATE - One or more job attributes have been modified • ENABLE - Job has been enabled • DISABLE - Job has been disabled • COMPLETED - For repeating jobs only, job has reached its end date or maximum number of runs • BROKEN - Job has reached its maximum number of failures STATUS can be SUCCEEDED (job run completed successfully), FAILED (job run failed), or STOPPED (job run was stopped) when OPERATION is one of the following: • RUN - Regular job run • RETRY_RUN - Job is being retried because the previous run resulted in an error and RESTARTABLE is set to TRUE • RECOVERY_RUN - Job is being rerun because the database went down, or the job worker process crashed and RESTARTABLE is set to TRUE
USER_NAME	VARCHAR2(128)		Name of the user who performed the operation, if applicable
CLIENT_ID	VARCHAR2(64)		Client identifier of the user who performed the operation, if applicable

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
GLOBAL_UID	VARCHAR2(32)		Global user identifier of the user who performed the operation, if applicable
CREDENTIAL_OWNER	VARCHAR2(261)		Owner of the credential used for this remote job run
CREDENTIAL_NAME	VARCHAR2(261)		Name of the credential used for this remote job run
DESTINATION_OWNER	VARCHAR2(261)		Owner of the destination object used in this remote job run; NULL if no object used
DESTINATION	VARCHAR2(261)		Destination for a remote job operation
ADDITIONAL_INFO	CLOB		Additional information on the entry, if applicable

See Also

- "DBA_SCHEDULER_JOB_LOG"
- "USER_SCHEDULER_JOB_LOG"

3.52 ALL_SCHEDULER_JOB_RUN_DETAILS

ALL_SCHEDULER_JOB_RUN_DETAILS displays log run details for the Scheduler jobs accessible to the current user.

Related Views

- DBA_SCHEDULER_JOB_RUN_DETAILS displays log run details for all Scheduler jobs in the database.
- USER_SCHEDULER_JOB_RUN_DETAILS displays log run details for the Scheduler jobs owned by the current user.

Column	Datatype	NULL	Description
LOG_ID	NUMBER	NOT NULL	Unique identifier of the log entry (foreign key of the *_SCHEDULER_JOB_LOG views)
LOG_DATE	TIMESTAMP(6) WITH TIME ZONE		Date of the log entry
OWNER	VARCHAR2(128)		Owner of the Scheduler job
JOB_NAME	VARCHAR2(261)		Name of the Scheduler job
JOB_SUBNAME	VARCHAR2(261)		Subname of the Scheduler job (for a chain step job)
STATUS	VARCHAR2(30)		Status of the job run
ERROR#	NUMBER		Error number in the case of an error
REQ_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Requested start date of the job run
ACTUAL_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Actual date on which the job was run
RUN_DURATION	INTERVAL DAY(3) TO SECOND(0)		Duration of the job run
INSTANCE_ID	NUMBER		Identifier of the instance on which the job was run

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SESSION_ID	VARCHAR2(128)		Session identifier of the job run
SLAVE_PID	VARCHAR2(30)		Process identifier of the worker on which the job was run
CPU_USED	INTERVAL DAY(3) TO SECOND(2)		Amount of CPU used for the job run
CREDENTIAL_OWNER	VARCHAR2(261)		Owner of the credential used for this remote job run
CREDENTIAL_NAME	VARCHAR2(261)		Name of the credential used for this remote job run
DESTINATION_OWNER	VARCHAR2(261)		Owner of the destination object used in this remote job run; NULL if no object used
DESTINATION	VARCHAR2(261)		Destination for a remote job operation
ADDITIONAL_INFO	VARCHAR2(4000)		Additional information on the job run, if applicable
ERRORS	VARCHAR2(4000)		Error messages generated by this job run
OUTPUT	VARCHAR2(4000)		Output messages generated by this job run
BINARY_ERRORS	BLOB		Error messages generated by this job run in a binary format
BINARY_OUTPUT	BLOB		Binary output messages generated by this job run

See Also

- "DBA_SCHEDULER_JOB_RUN_DETAILS"
- "USER_SCHEDULER_JOB_RUN_DETAILS"

3.53 ALL_SCHEDULER_JOBS

ALL_SCHEDULER_JOBS displays information about the Scheduler jobs accessible to the current user.

Related Views

- DBA_SCHEDULER_JOBS displays information about all Scheduler jobs in the database.
- USER_SCHEDULER_JOBS displays information about the Scheduler jobs owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the Scheduler job
JOB_NAME	VARCHAR2(128)		Name of the Scheduler job
JOB_SUBNAME	VARCHAR2(128)		Subname of the Scheduler job (for a job running a chain step)
JOB_STYLE	VARCHAR2(17)		Job style: • REGULAR • LIGHTWEIGHT • IN_MEMORY_RUNTIME • IN_MEMORY_FULL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
JOB_CREATOR	VARCHAR2(128)		Original creator of the job
CLIENT_ID	VARCHAR2(65)		Client identifier of the user creating the job
GLOBAL_UID	VARCHAR2(33)		Global user identifier of the user creating the job
PROGRAM_OWNER	VARCHAR2(4000)		Owner of the program associated with the job
PROGRAM_NAME	VARCHAR2(4000)		Name of the program associated with the job
JOB_TYPE	VARCHAR2(16)		Inline job action type: • PLSQL_BLOCK • STORED_PROCEDURE • EXECUTABLE • CHAIN • SQL_SCRIPT • BACKUP_SCRIPT • EXTERNAL_SCRIPT
JOB_ACTION	VARCHAR2(4000)		Inline job action
NUMBER_OF_ARGUMENTS	NUMBER		Inline number of job arguments
SCHEDULE_OWNER	VARCHAR2(4000)		Owner of the schedule that the job uses (can be a window or a window group)
SCHEDULE_NAME	VARCHAR2(4000)		Name of the schedule that the job uses (can be a window or a window group)
SCHEDULE_TYPE	VARCHAR2(12)		Type of the schedule that the job uses: • IMMEDIATE - Start date and repeat interval are NULL • ONCE - Repeat interval is NULL • PLSQL - PL/SQL expression used as schedule • CALENDAR - Oracle calendaring expression used as schedule • EVENT - Event schedule • NAMED - Named schedule • WINDOW - Window used as schedule • WINDOW_GROUP - Window group used as schedule
START_DATE	TIMESTAMP(6) WITH TIME ZONE		Original scheduled start date of the job (for an inline schedule)
REPEAT_INTERVAL	VARCHAR2(4000)		Inline schedule PL/SQL expression or calendar string
EVENT_QUEUE_OWNER	VARCHAR2(128)		Owner of the source queue into which the event will be raised
EVENT_QUEUE_NAME	VARCHAR2(128)		Name of the source queue into which the event will be raised
EVENT_QUEUE_AGENT	VARCHAR2(523)		Name of the AQ agent used by the user on the event source queue (if it is a secure queue)
EVENT_CONDITION	VARCHAR2(4000)		Boolean expression used as the subscription rule for the event on the source queue
EVENT_RULE	VARCHAR2(261)		Name of the rule used by the coordinator to trigger the event-based job
FILE_WATCHER_OWNER	VARCHAR2(261)		Owner of the file watcher on which this job is based
FILE_WATCHER_NAME	VARCHAR2(261)		Name of the file watcher on which this job is based
END_DATE	TIMESTAMP(6) WITH TIME ZONE		Date after which the job will no longer run (for an inline schedule)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
JOB_CLASS	VARCHAR2(128)		Name of the job class associated with the job
ENABLED	VARCHAR2(5)		Indicates whether the job is enabled (TRUE) or disabled (FALSE)
AUTO_DROP	VARCHAR2(5)		Indicates whether the job will be dropped when it has completed (TRUE) or not (FALSE)
RESTART_ON_RECOVERY	VARCHAR2(5)		Indicates whether the step should be restarted on database recovery (TRUE) or not (FALSE)
RESTART_ON_FAILURE	VARCHAR2(5)		Indicates whether the step should be restarted on application failure (TRUE) or not (FALSE)
STATE	VARCHAR2(20)		Current state of the job: • BLOCKED • BROKEN • CHAIN_STALLED • COMPLETED • DISABLED • FAILED • READY TO RUN • REMOTE • RESOURCE_UNAVAILABLE • RETRY SCHEDULED • RUNNING • SCHEDULED • SOME FAILED • STOPPED • SUCCEEDED
JOB_PRIORITY	NUMBER		Priority of the job relative to other jobs in the same class
RUN_COUNT	NUMBER		Number of times the job has run
UPTIME_RUN_COUNT	NUMBER		Number of runs since the database last restarted. For in-memory jobs, this column is populated, but the RUN_COUNT column is not populated. For all other jobs, this column is NULL.
MAX_RUNS	NUMBER		Maximum number of times the job is scheduled to run
FAILURE_COUNT	NUMBER		Number of times the job has failed to run
UPTIME_FAILURE_COUNT	NUMBER		Number of failures since the database last restarted. For in-memory jobs, this column is populated, but the FAILURE_COUNT column is not populated. For all other jobs, this column is NULL.
MAX_FAILURES	NUMBER		Number of times the job will be allowed to fail before being marked broken
RETRY_COUNT	NUMBER		Number of times the job has retried, if it is retrying
LAST_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Last date on which the job started running
LAST_RUN_DURATION	INTERVAL DAY(9) TO SECOND(6)		Amount of time the job took to complete during the last run
NEXT_RUN_DATE	TIMESTAMP(6) WITH TIME ZONE		Next date on which the job is scheduled to run

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SCHEDULE_LIMIT	INTERVAL DAY(3) TO SECOND(0)		Time after which a job which has not run yet will be rescheduled
MAX_RUN_DURATION	INTERVAL DAY(3) TO SECOND(0)		Maximum amount of time for which the job will be allowed to run
LOGGING_LEVEL	VARCHAR2(11)		Amount of logging that will be done pertaining to the job: • OFF • RUNS • FAILED RUNS • FULL
STORE_OUTPUT	VARCHAR2(5)		Indicates whether all job output messages for the job are stored in the OUTPUT column of the *_JOB_RUN_DETAILS views for job runs that are logged. Possible values: • TRUE: All job output messages for the job are stored in the OUTPUT column of the *_JOB_RUN_DETAILS views for job runs that are logged. This is the default for new jobs. A new job is a job created using Oracle Database 12*c* software. • FALSE: Job output messages for the job are not stored in the OUTPUT column of the *_JOB_RUN_DETAILS views. This is the default for existing jobs. An existing job is a job created using pre-Oracle Database 12*c* software.
STOP_ON_WINDOW_CLOSE	VARCHAR2(5)		Indicates whether the job will stop if a window associated with the job closes (TRUE) or not (FALSE)
INSTANCE_STICKINESS	VARCHAR2(5)		Indicates whether the job is sticky (TRUE) or not (FALSE)
RAISE_EVENTS	VARCHAR2(4000)		List of job events to raise for the job: • JOB_STARTED • JOB_SUCCEEDED • JOB_FAILED • JOB_BROKEN • JOB_COMPLETED • JOB_STOPPED • JOB_SCH_LIM_REACHED • JOB_DISABLED • JOB_CHAIN_STALLED • JOB_OVER_MAX_DUR
SYSTEM	VARCHAR2(5)		Indicates whether the job is a system job (TRUE) or not (FALSE)
JOB_WEIGHT	NUMBER		Weight of the job
NLS_ENV	VARCHAR2(4000)		NLS environment of the job
SOURCE	VARCHAR2(128)		Source global database identifier
NUMBER_OF_DESTINATIONS	NUMBER		Number of destinations associated with this job
DESTINATION_OWNER	VARCHAR2(261)		Owner of the destination object (if used), else NULL
DESTINATION	VARCHAR2(261)		Destination that this job will run on
CREDENTIAL_OWNER	VARCHAR2(128)		Owner of the credential to be used for an external job
CREDENTIAL_NAME	VARCHAR2(128)		Name of the credential to be used for an external job

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INSTANCE_ID	NUMBER		Instance on which the user requests the job to run
DEFERRED_DROP	VARCHAR2(5)		Indicates whether the job will be dropped when completed due to user request (TRUE) or not (FALSE)
ALLOW_RUNS_IN_RESTRICTED_MODE	VARCHAR2(5)		Indicates whether the job is allowed to run in restricted session mode (TRUE) or not (FALSE)
COMMENTS	VARCHAR2(4000)		Comments on the job
FLAGS	NUMBER		This column is for internal use
RESTARTABLE	VARCHAR2(5)		Indicates whether the job can be restarted (TRUE) or not (FALSE)
HAS_CONSTRAINTS	VARCHAR2(5)		Indicates whether the job (not including the program of the job) is part of a resource constraint or incompatibility (TRUE) or not (FALSE)
CONNECT_CREDENTIAL_OWNER	VARCHAR2(128)		Owner of connect credential
CONNECT_CREDENTIAL_NAME	VARCHAR2(128)		Name of connect credential
FAIL_ON_SCRIPT_ERROR	VARCHAR2(5)		Indicates whether this job fails on script error (TRUE) or not (FALSE)

See Also

- "DBA_SCHEDULER_JOBS"
- "USER_SCHEDULER_JOBS"

3.54 ALL_SCHEDULER_NOTIFICATIONS

ALL_SCHEDULER_NOTIFICATIONS displays information about the E-mail notifications for the jobs accessible to the current user.

Related Views

- DBA_SCHEDULER_NOTIFICATIONS displays information about the E-mail notifications for all jobs in the database.
- USER_SCHEDULER_NOTIFICATIONS displays information about the E-mail notifications for the jobs owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
NOTIFICATION_OWNER	VARCHAR2(128)	NOT NULL	Owner of this notification
OWNER	VARCHAR2(128)	NOT NULL	Owner of the job this notification is for
JOB_NAME	VARCHAR2(128)	NOT NULL	Name of the job this notification is for
JOB_SUBNAME	VARCHAR2(128)		Subname of the job this notification is for
RECIPIENT	VARCHAR2(4000)	NOT NULL	E-mail address to send this E-mail notification to
SENDER	VARCHAR2(4000)		E-mail address to send this E-mail notification from
SUBJECT	VARCHAR2(4000)		Subject of the notification E-mail
BODY	VARCHAR2(4000)		Body of the notification E-mail

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FILTER_CONDITION	VARCHAR2(4000)		Filter specifying which job events to send notifications for
EVENT	VARCHAR2(19)		Job event to send notifications for: • JOB_STARTED • JOB_SUCCEEDED • JOB_FAILED • JOB_BROKEN • JOB_COMPLETED • JOB_STOPPED • JOB_SCH_LIM_REACHED • JOB_DISABLED • JOB_CHAIN_STALLED • JOB_OVER_MAX_DUR
EVENT_FLAG	NUMBER	NOT NULL	Event number of the job event to send notifications for

See Also

- "DBA_SCHEDULER_NOTIFICATIONS"
- "USER_SCHEDULER_NOTIFICATIONS"

3.55 ALL_SCHEDULER_PROGRAM_ARGS

ALL_SCHEDULER_PROGRAM_ARGS displays information about the arguments of the Scheduler programs accessible to the current user.

Related Views

- DBA_SCHEDULER_PROGRAM_ARGS displays information about the arguments of all Scheduler programs in the database.
- USER_SCHEDULER_PROGRAM_ARGS displays information about the arguments of the Scheduler programs owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the program to which the argument belongs
PROGRAM_NAME	VARCHAR2(128)	NOT NULL	Name of the program to which the argument belongs
ARGUMENT_NAME	VARCHAR2(128)		Optional name of the argument
ARGUMENT_POSITION	NUMBER	NOT NULL	Position of the argument in the argument list
ARGUMENT_TYPE	VARCHAR2(257)		Data type of the argument

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
METADATA_ATTRIBUTE	VARCHAR2(19)		Metadata attribute: • JOB_NAME • JOB_OWNER • JOB_START • WINDOW_START • WINDOW_END • JOB_SUBNAME • EVENT_MESSAGE • JOB_SCHEDULED_START
DEFAULT_VALUE	VARCHAR2(4000)		Default value taken by the argument (in string format) if the argument is a string
DEFAULT_ANYDATA_VALUE	ANYDATA		Default value taken by the argument (in AnyData format)
OUT_ARGUMENT	VARCHAR2(5)		Reserved for future use

See Also

- "DBA_SCHEDULER_PROGRAM_ARGS"
- "USER_SCHEDULER_PROGRAM_ARGS"

3.56 ALL_SCHEDULER_PROGRAMS

ALL_SCHEDULER_PROGRAMS displays information about the Scheduler programs accessible to the current user.

Related Views

- DBA_SCHEDULER_PROGRAMS displays information about all Scheduler programs in the database.
- USER_SCHEDULER_PROGRAMS displays information about the Scheduler programs owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Scheduler program
PROGRAM_NAME	VARCHAR2(128)	NOT NULL	Name of the Scheduler program
PROGRAM_TYPE	VARCHAR2(16)		Type of the program action: • PLSQL_BLOCK • STORED_PROCEDURE • EXECUTABLE
PROGRAM_ACTION	VARCHAR2(4000)		String specifying the program action
NUMBER_OF_ARGUMENTS	NUMBER		Number of arguments accepted by the program
ENABLED	VARCHAR2(5)		Indicates whether the program is enabled (TRUE) or disabled (FALSE)
DETACHED	VARCHAR2(5)		This column is for internal use

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SCHEDULE_LIMIT	INTERVAL DAY(3) TO SECOND(0)		Maximum delay in running the program after the scheduled start
PRIORITY	NUMBER		Priority of the program
WEIGHT	NUMBER		Weight of the program
MAX_RUNS	NUMBER		Maximum number of runs of any job based on this program
MAX_FAILURES	NUMBER		Maximum number of failures of any job based on this program
MAX_RUN_DURATION	INTERVAL DAY(3) TO SECOND(0)		Maximum amount of time this program can run
HAS_CONSTRAINTS	VARCHAR2(5)		Indicates whether the job (not including the program of the job) is part of a resource constraint or incompatibility (TRUE) or not (FALSE)
NLS_ENV	VARCHAR2(4000)		NLS environment in which the program was created
COMMENTS	VARCHAR2(4000)		Comments on the program

See Also

- "DBA_SCHEDULER_PROGRAMS"
- "USER_SCHEDULER_PROGRAMS"

3.57 ALL_SCHEDULER_REMOTE_DATABASES

ALL_SCHEDULER_REMOTE_DATABASES displays information about the remote databases accessible to the current user that have been registered as sources and destinations for remote database jobs.

Related View

DBA_SCHEDULER_REMOTE_DATABASES displays information about all remote databases that have been registered as sources and destinations for remote database jobs.

Column	Datatype	NULL	Description
DATABASE_NAME	VARCHAR2(512)	NOT NULL	Global name of the remote database
REGISTERED_AS	VARCHAR2(11)		Indicates whether the database is registered as a source (SOURCE) or as a destination (DESTINATION)
DATABASE_LINK	VARCHAR2(512)	NOT NULL	Name of a valid database link to the remote database

See Also

"DBA_SCHEDULER_REMOTE_DATABASES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.58 ALL_SCHEDULER_REMOTE_JOBSTATE

ALL_SCHEDULER_REMOTE_JOBSTATE displays information about the state of the jobs accessible to the current user at remote databases.

Related Views

- DBA_SCHEDULER_REMOTE_JOBSTATE displays information about the state of all jobs at remote databases.
- USER_SCHEDULER_REMOTE_JOBSTATE displays information about the state of the jobs owned by the current user at remote databases. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Scheduler job
JOB_NAME	VARCHAR2(128)	NOT NULL	Name of the Scheduler job
DESTINATION	VARCHAR2(512)	NOT NULL	Name of the job destination
STATE	VARCHAR2(15)		State of the job at the destination: • DISABLED • RETRY SCHEDULED • SCHEDULED • RUNNING • COMPLETED • BROKEN • FAILED • SUCCEEDED • STOPPED
NEXT_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Next start date of the job at the destination
RUN_COUNT	NUMBER		Run count of the job at the destination
FAILURE_COUNT	NUMBER		Failure count of the job at the destination
RETRY_COUNT	NUMBER		Retry count of the job at the destination
LAST_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Last start date of the job at the destination
LAST_END_DATE	TIMESTAMP(6) WITH TIME ZONE		Last end date of the job at the destination

See Also

- "DBA_SCHEDULER_REMOTE_JOBSTATE"
- "USER_SCHEDULER_REMOTE_JOBSTATE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.59 ALL_SCHEDULER_RESOURCES

ALL_SCHEDULER_RESOURCES displays all scheduler resource objects in the database that are accessible to the current user.

Related Views

- DBA_SCHEDULER_RESOURCES displays all scheduler resource objects in the database.
- USER_SCHEDULER_RESOURCES displays all scheduler resource objects in the database from the schema of the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the resource object
RESOURCE_NAME	VARCHAR2(128)	NOT NULL	Name of the resource object
STATUS	VARCHAR2(19)		Resource status for resource object.
RESOURCE_UNITS	NUMBER		Maximum number of available units for the resource object
UNITS_USED	NUMBER		Current number of resource units in use for the resource object
JOBS_RUNNING_COUNT	NUMBER		Current number of running jobs using the resource object
COMMENTS	VARCHAR2(256)		Comments for the resource object

See Also

- "DBA_SCHEDULER_RESOURCES"
- "USER_SCHEDULER_RESOURCES"

3.60 ALL_SCHEDULER_RSC_CONSTRAINTS

ALL_SCHEDULER_RSC_CONSTRAINTS lists all Oracle Scheduler resource constraint members accessible to the current user.

Related Views

- DBA_SCHEDULER_RSC_CONSTRAINTS lists all Oracle Scheduler resource constraint members in the database.
- USER_SCHEDULER_RSC_CONSTRAINTS lists all Oracle Scheduler resource constraint members owned by the current user.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the resource object the member is in
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the resource object the member is in
RESOURCE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the resource constraint resource member
RESOURCE_NAME	VARCHAR2(128)	NOT NULL	Name of the resource constraint resource member

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
UNITS_USED	NUMBER		Number of units used of the resource by this constraint resource member

See Also

- "DBA_SCHEDULER_RSC_CONSTRAINTS"
- "USER_SCHEDULER_RSC_CONSTRAINTS"

3.61 ALL_SCHEDULER_RUNNING_CHAINS

ALL_SCHEDULER_RUNNING_CHAINS displays information about the chain steps of the running chains accessible to the current user (that is, those chains that the user has ALTER privileges for). In the case of nested chains, this view also enables you to traverse the hierarchy of the chain with a SQL statement that contains a CONNECT BY clause linking up the JOB_SUBNAME and STEP_JOB_SUBNAME columns.

Related Views

- DBA_SCHEDULER_RUNNING_CHAINS displays information about the chain steps of all running chains in the database.
- USER_SCHEDULER_RUNNING_CHAINS displays information about the chain steps of the running chains owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the job which is running the chain
JOB_NAME	VARCHAR2(128)	NOT NULL	Name of the job which is running the chain
JOB_SUBNAME	VARCHAR2(128)		Subname of the job which is running the chain (for a nested chain), else NULL
CHAIN_OWNER	VARCHAR2(128)	NOT NULL	Owner of the chain being run
CHAIN_NAME	VARCHAR2(128)	NOT NULL	Name of the chain being run
STEP_NAME	VARCHAR2(128)	NOT NULL	Name of the step of the running chain
STATE	VARCHAR2(15)		State of the running chain step: • NOT_STARTED • RUNNING • SUCCEEDED • STOPPED • FAILED • SCHEDULED • RETRY SCHEDULED • PAUSED • STALLED
ERROR_CODE	NUMBER		Error code with which the step completed (if it has completed)
COMPLETED	VARCHAR2(5)		Indicates whether the running chain step has completed (TRUE) or not (FALSE)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
START_DATE	TIMESTAMP(6) WITH TIME ZONE		Date when the running chain step started (if it has started)
END_DATE	TIMESTAMP(6) WITH TIME ZONE		Date when the running chain step stopped (if it has stopped)
DURATION	INTERVAL DAY(9) TO SECOND(6)		Amount of time it took the chain step to complete (if it has completed)
SKIP	VARCHAR2(5)		Indicates whether the chain step should be skipped (TRUE) or not (FALSE)
PAUSE	VARCHAR2(5)		Indicates whether the chain step should be paused after running (TRUE) or not (FALSE)
PAUSE_BEFORE	VARCHAR2(5)		Indicates whether the chain step should be paused before running (TRUE) or not (FALSE)
RESTART_ON_RECOVERY	VARCHAR2(5)		Indicates whether the chain step will be restarted on database recovery (TRUE) or not (FALSE)
RESTART_ON_FAILURE	VARCHAR2(5)		Indicates whether the chain step will be restarted on application failure (TRUE) or not (FALSE)
STEP_JOB_SUBNAME	VARCHAR2(128)		Subname of the job running the step
STEP_JOB_LOG_ID	NUMBER		Log ID of the job running the step

See Also

- "DBA_SCHEDULER_RUNNING_CHAINS"
- "USER_SCHEDULER_RUNNING_CHAINS"

3.62 ALL_SCHEDULER_RUNNING_JOBS

ALL_SCHEDULER_RUNNING_JOBS displays information about the running Scheduler jobs accessible to the current user.

Related Views

- DBA_SCHEDULER_RUNNING_JOBS displays information about all running Scheduler jobs in the database.
- USER_SCHEDULER_RUNNING_JOBS displays information about the running Scheduler jobs owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the running Scheduler job
JOB_NAME	VARCHAR2(128)		Name of the running Scheduler job
JOB_SUBNAME	VARCHAR2(128)		Subname of the running Scheduler job (for a job running a chain step)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
JOB_STYLE	VARCHAR2(17)		Job style: • REGULAR • LIGHTWEIGHT • IN_MEMORY_RUNTIME • IN_MEMORY_FULL
DETACHED	VARCHAR2(5)		Indicates whether the detached attribute is set for the job (TRUE) or not (FALSE). If the detached attribute is set, then the job will remain running even after the job action has completed.
SESSION_ID	NUMBER		Identifier of the session running the Scheduler job
SLAVE_PROCESS_ID	NUMBER		Process number of the worker process running the Scheduler job
SLAVE_OS_PROCESS_ID	VARCHAR2(12)		Process number of the operating system worker process running the scheduler job
RUNNING_INSTANCE	NUMBER		Database instance number of the worker process running the Scheduler job
RESOURCE_CONSUMER_GROUP	VARCHAR2(32)		Resource consumer group of the session in which the Scheduler job is running
ELAPSED_TIME	INTERVAL DAY(3) TO SECOND(2)		Elapsed time since the Scheduler job was started
CPU_USED	INTERVAL DAY(3) TO SECOND(2)		CPU time consumed by the running Scheduler job, if available
DESTINATION_OWNER	VARCHAR2(261)		Owner of the destination object (if used), else NULL
DESTINATION	VARCHAR2(261)		Destination that this job is running on
CREDENTIAL_OWNER	VARCHAR2(128)		Owner of the login credential used for this running job, if any
CREDENTIAL_NAME	VARCHAR2(128)		Name of the login credential used for this running job, if any
LOG_ID	NUMBER		Log ID used for this running job. This column maps to the LOG_ID column of the *_SCHEDULER_JOB_LOG and *_SCHEDULER_JOB_RUN_DETAILS views.

See Also

- "DBA_SCHEDULER_RUNNING_JOBS"
- "USER_SCHEDULER_RUNNING_JOBS"

3.63 ALL_SCHEDULER_SCHEDULES

ALL_SCHEDULER_SCHEDULES displays information about the Scheduler schedules accessible to the current user.

Related Views

- DBA_SCHEDULER_SCHEDULES displays information about all Scheduler schedules in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_SCHEDULER_SCHEDULES displays information about the Scheduler schedules owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the schedule
SCHEDULE_NAME	VARCHAR2(128)	NOT NULL	Name of the schedule
SCHEDULE_TYPE	VARCHAR2(12)		Type of the schedule: • ONCE - Repeat interval is NULL • CALENDAR - Oracle calendaring expression used as schedule • EVENT - Event schedule
START_DATE	TIMESTAMP(6) WITH TIME ZONE		Start date for the repeat interval
REPEAT_INTERVAL	VARCHAR2(4000)		Calendar syntax expression for the schedule
EVENT_QUEUE_OWNER	VARCHAR2(128)		Owner of the source queue into which the event will be raised
EVENT_QUEUE_NAME	VARCHAR2(128)		Name of the source queue into which the event will be raised
EVENT_QUEUE_AGENT	VARCHAR2(523)		Name of the AQ agent used by the user on the event source queue (if it is a secure queue)
EVENT_CONDITION	VARCHAR2(4000)		Boolean expression used as the subscription rule for the event on the source queue
FILE_WATCHER_OWNER	VARCHAR2(261)		Owner of the file watcher on which this schedule is based
FILE_WATCHER_NAME	VARCHAR2(261)		Name of the file watcher on which this schedule is based
END_DATE	TIMESTAMP(6) WITH TIME ZONE		Cutoff date after which the schedule will not specify any dates
COMMENTS	VARCHAR2(4000)		Comments on the schedule

See Also

- "DBA_SCHEDULER_SCHEDULES"
- "USER_SCHEDULER_SCHEDULES"

3.64 ALL_SCHEDULER_WINDOW_DETAILS

ALL_SCHEDULER_WINDOW_DETAILS displays log details for the Scheduler windows accessible to the current user.

Related View

DBA_SCHEDULER_WINDOW_DETAILS displays log details for all Scheduler windows in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LOG_ID	NUMBER		Unique identifier of the log entry (foreign key of the *_SCHEDULER_WINDOW_LOG views)
LOG_DATE	TIMESTAMP(6) WITH TIME ZONE		Date of the log entry
OWNER	VARCHAR2(128)		Owner of the Scheduler window
WINDOW_NAME	VARCHAR2(261)		Name of the Scheduler window
REQ_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Requested start date for the Scheduler window
ACTUAL_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Actual start date of the Scheduler window
WINDOW_DURATION	INTERVAL DAY(3) TO SECOND(0)		Requested duration of the Scheduler window
ACTUAL_DURATION	INTERVAL DAY(3) TO SECOND(0)		Actual duration for which the Scheduler window lasted
INSTANCE_ID	NUMBER		Identifier of the instance on which the window was run
ADDITIONAL_INFO	VARCHAR2(4000)		Additional information on the entry, if applicable

See Also

"DBA_SCHEDULER_WINDOW_DETAILS"

3.65 ALL_SCHEDULER_WINDOW_GROUPS

ALL_SCHEDULER_WINDOW_GROUPS displays information about the Scheduler window groups accessible to the current user.

Related View

DBA_SCHEDULER_WINDOW_GROUPS displays information about all Scheduler window groups in the database.

Column	Datatype	NULL	Description
WINDOW_GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the window group
ENABLED	VARCHAR2(5)		Indicates whether the window group is enabled (TRUE) or disabled (FALSE)
NUMBER_OF_WINDOWS	NUMBER		Number of members in the window group
NEXT_START_DATE	VARCHAR2(64)		If a window group is disabled, then this column will be NULL. Otherwise, it will be set to the earliest NEXT_START_DATE from the enabled windows in the group.
COMMENTS	VARCHAR2(4000)		Optional comment about the window group

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "DBA_SCHEDULER_WINDOW_GROUPS"

3.66 ALL_SCHEDULER_WINDOW_LOG

ALL_SCHEDULER_WINDOW_LOG displays log information for the Scheduler windows accessible to the current user.

Related View

DBA_SCHEDULER_WINDOW_LOG displays log information for all Scheduler windows in the database.

Column	Datatype	NULL	Description
LOG_ID	NUMBER	NOT NULL	Unique identifier of the log entry
LOG_DATE	TIMESTAMP(6) WITH TIME ZONE		Date of the log entry
OWNER	VARCHAR2(128)		Owner of the Scheduler window
WINDOW_NAME	VARCHAR2(261)		Name of the Scheduler window
OPERATION	VARCHAR2(30)		Operation corresponding to the log entry
STATUS	VARCHAR2(30)		Status of the operation, if applicable
USER_NAME	VARCHAR2(128)		Name of the user who performed the operation, if applicable
CLIENT_ID	VARCHAR2(64)		Client identifier of the user who performed the operation, if applicable
GLOBAL_UID	VARCHAR2(32)		Global user identifier of the user who performed the operation, if applicable
ADDITIONAL_INFO	CLOB		Additional information on the entry, if applicable

> **See Also**
>
> "DBA_SCHEDULER_WINDOW_LOG"

3.67 ALL_SCHEDULER_WINDOWS

ALL_SCHEDULER_WINDOWS displays information about the Scheduler windows accessible to the current user.

Related View

DBA_SCHEDULER_WINDOWS displays information about all Scheduler windows in the database.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Scheduler window

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
WINDOW_NAME	VARCHAR2(128)	NOT NULL	Name of the Scheduler window
RESOURCE_PLAN	VARCHAR2(128)		Resource plan associated with the window
SCHEDULE_OWNER	VARCHAR2(4000)		Owner of the schedule of the window
SCHEDULE_NAME	VARCHAR2(4000)		Name of the schedule of the window
SCHEDULE_TYPE	VARCHAR2(8)		Type of the schedule of the window: • ONCE - Repeat interval is NULL • NAMED - Named schedule • CALENDAR - Oracle calendaring expression used as schedule
START_DATE	TIMESTAMP(6) WITH TIME ZONE		Start date of the window (for an inline schedule)
REPEAT_INTERVAL	VARCHAR2(4000)		Calendar string for the window (for an inline schedule)
END_DATE	TIMESTAMP(6) WITH TIME ZONE		Date after which the window will no longer open (for an inline schedule)
DURATION	INTERVAL DAY(3) TO SECOND(0)		Duration of the window
WINDOW_PRIORITY	VARCHAR2(4)		Priority of the job relative to other windows: • HIGH • LOW
NEXT_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Next date on which the window is scheduled to start
LAST_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Last date on which the window opened
ENABLED	VARCHAR2(5)		Indicates whether the window is enabled (TRUE) or disabled (FALSE)
ACTIVE	VARCHAR2(5)		Indicates whether the window is open (TRUE) or not (FALSE)
MANUAL_OPEN_TIME	TIMESTAMP(6) WITH TIME ZONE		Open time of the window if it was manually opened, else NULL
MANUAL_DURATION	INTERVAL DAY(3) TO SECOND(0)		Duration of the window if it was manually opened, else NULL
COMMENTS	VARCHAR2(4000)		Comments on the window

See Also

"DBA_SCHEDULER_WINDOWS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.68 ALL_SCHEDULER_WINGROUP_MEMBERS

ALL_SCHEDULER_WINGROUP_MEMBERS displays the members of the Scheduler window groups accessible to the current user.

Related View

DBA_SCHEDULER_WINGROUP_MEMBERS displays the members of all Scheduler window groups in the database.

Column	Datatype	NULL	Description
WINDOW_GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the window group
WINDOW_NAME	VARCHAR2(128)	NOT NULL	Name of the window member of the window group

See Also

"DBA_SCHEDULER_WINGROUP_MEMBERS"

3.69 ALL_SEC_RELEVANT_COLS

ALL_SEC_RELEVANT_COLS describes the security relevant columns of the security policies for the tables and views accessible to the current user.

Related Views

- DBA_SEC_RELEVANT_COLS describes the security relevant columns of all security policies in the database.
- USER_SEC_RELEVANT_COLS describes the security relevant columns of the security policies for the tables and views owned by the current user. This view does not display the OBJECT_OWNER column.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)		Owner of the table or view
OBJECT_NAME	VARCHAR2(128)		Name of the table or view
POLICY_GROUP	VARCHAR2(128)		Name of the policy group
POLICY_NAME	VARCHAR2(128)		Name of the policy
SEC_REL_COLUMN	VARCHAR2(128)		Name of the security relevant column
COLUMN_OPTION	VARCHAR2(8)		Option of the security relevant column: • NONE • ALL_ROWS
COMMON	VARCHAR2(3)		Indicates whether the policy security relevant column is applied and enforced in all application PDBs (YES) or only in the local PDB (NO)
INHERITED	VARCHAR2(3)		Indicates whether the policy security relevant column is inherited from the root (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> - "DBA_SEC_RELEVANT_COLS"
> - "USER_SEC_RELEVANT_COLS"

3.70 ALL_SECONDARY_OBJECTS

ALL_SECONDARY_OBJECTS provides information about secondary objects associated with domain indexes accessible to the user.

This view is only relevant for domain indexes. And currently, the secondary objects can only be tables.

Related Views

- DBA_SECONDARY_OBJECTS provides information about all secondary objects that are associated with domain indexes in the database.
- USER_SECONDARY_OBJECTS provides information about secondary objects associated with domain indexes owned by the current user.

Column	Datatype	NULL	Description
INDEX_OWNER	VARCHAR2(128)	NOT NULL	Owner of the domain index
INDEX_NAME	VARCHAR2(128)	NOT NULL	Name of the domain index
SECONDARY_OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the secondary object created by the domain index
SECONDARY_OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the secondary object created by the domain index
SECONDARY_OBJDATA_TYPE	VARCHAR2(20)		Type of the secondary object created by the domain index

> **See Also**
>
> - "DBA_SECONDARY_OBJECTS"
> - "USER_SECONDARY_OBJECTS"

3.71 ALL_SEQUENCES

ALL_SEQUENCES describes all sequences accessible to the current user.

Related Views

- DBA_SEQUENCES describes all sequences in the database.
- USER_SEQUENCES describes all sequences owned by the current user. This view does not display the SEQUENCE_OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SEQUENCE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the sequence
SEQUENCE_NAME	VARCHAR2(128)	NOT NULL	Sequence name
MIN_VALUE	NUMBER		Minimum value of the sequence
MAX_VALUE	NUMBER		Maximum value of the sequence
INCREMENT_BY	NUMBER	NOT NULL	Value by which sequence is incremented
CYCLE_FLAG	VARCHAR2(1)		Indicates whether the sequence wraps around on reaching the limit (Y) or not (N)
ORDER_FLAG	VARCHAR2(1)		Indicates whether sequence numbers are generated in order (Y) or not (N)
CACHE_SIZE	NUMBER	NOT NULL	Number of sequence numbers to cache
LAST_NUMBER	NUMBER	NOT NULL	Last sequence number written to disk. If a sequence uses caching, the number written to disk is the last number placed in the sequence cache. This number is likely to be greater than the last sequence number that was used. For session sequences, the value in this column should be ignored.
SCALE_FLAG	VARCHAR2(1)		Indicates whether this is a scalable sequence (Y) or not (N)
EXTEND_FLAG	VARCHAR2(1)		Indicates whether this scalable sequence's generated values extend beyond MAX_VALUE or MIN_VALUE (Y) or not (N)
SHARDED_FLAG [1]	VARCHAR2(1)		Indicates whether this is a sharded sequence (Y) or not (N)
SESSION_FLAG	VARCHAR2(1)		Indicates whether sequence values are session private (Y) or not (N)
KEEP_VALUE	VARCHAR2(1)		Indicates whether sequence values are kept during replay after a failure (Y) or not (N)

1 This column is available starting with Oracle Database 19c.

See Also

- "DBA_SEQUENCES"
- "USER_SEQUENCES"

3.72 ALL_SERVICES

ALL_SERVICES displays all services in the database. The view excludes rows marked for deletion.

Related View

DBA_SERVICES displays all services in the database. The view excludes rows marked for deletion.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SERVICE_ID	NUMBER		Unique ID for the service
NAME	VARCHAR2(64)		Name describing the workload
NAME_HASH	NUMBER		Hash of the short name for the service
NETWORK_NAME	VARCHAR2(512)		Network name used to connect to the service
CREATION_DATE	DATE		Date the service was created
CREATION_DATE_HASH	NUMBER		Hash of the creation date
FAILOVER_METHOD	VARCHAR2(64)		TAF only for compatibility - BASIC or NONE
FAILOVER_TYPE	VARCHAR2(64)		AUTO for Transparent Application Continuity, TRANSACTION for Application Continuity, SESSION or SELECT for TAF.
FAILOVER_RETRIES	NUMBER(10)		For Application Continuity and TAF, when reconnecting after a failure, number of attempts to re-connect per incident
FAILOVER_DELAY	NUMBER(10)		For Application Continuity and TAF, when reconnecting after a failure, delay between each connection retry (in seconds)
MIN_CARDINALITY	NUMBER		Reserved for internal use
MAX_CARDINALITY	NUMBER		Reserved for internal use
GOAL	VARCHAR2(12)		Runtime Load Balancing Goal being used to create run-time load balancing and connection load balancing advice: • NONE • SERVICE_TIME - Connections are balanced by response time • THROUGHPUT - Connections are balanced by throughput
DTP	VARCHAR2(1)		DTP (distributed transaction processing) enforces all sessions for a service at one instance. This is a requirement for XA before 11*g*, and is a requirement if resuming and suspending the same XA branch.
ENABLED	VARCHAR2(3)		Reserved for internal use
AQ_HA_NOTIFICATIONS	VARCHAR2(3)		To enable FAN for OCI connections, set AQ HA Notifications to True. For Oracle Database 12*c*, FAN uses ONS (Oracle Notification Service)
CLB_GOAL	VARCHAR2(5)		Connection load balancing goal used with statistics that are sent to the listeners to determine how new connections are distributed: • LONG • SHORT
EDITION	VARCHAR2(128)		A non-NULL value specifies the initial session edition for subsequent database connections that use the service and do not specify an edition. A NULL value has no effect.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COMMIT_OUTCOME	VARCHAR2(3)		For Transaction Guard, indicates whether the database service associated with the user session has the COMMIT_OUTCOME service attribute enabled (YES) or not (NO). This attribute applies on a per session basis and is set at connect time. When COMMIT_OUTCOME = YES: • Transaction Guard manages the commit status for all supported transaction types. The outcome of a COMMIT transaction is known. If there is an outage, the application can use DBMS_APP_CONT.GET_LTXID_OUTCOME to return a reliable status for the last in-flight work. • A logical transaction ID (LTXID) is set for each user session at login and at each successful commit. **See Also:** For information about preserving the commit outcome, see *Oracle Database Development Guide*. For information about logical transaction IDs, see *Oracle Database Development Guide*.
RETENTION_TIMEOUT	NUMBER		For Transaction Guard, when COMMIT_OUTCOME = YES, this value indicates the amount of time (in seconds) that the commit outcome is retained in the database.
REPLAY_INITIATION_TIMEOUT	NUMBER		For Application Continuity, indicates a time period (in seconds) after which the request will not be replayed. The time period starts at the first request submission. The default value is 300 seconds, which is 5 minutes.
SESSION_STATE_CONSISTENCY	VARCHAR2(128)		Describes how non-transactional is changed during a request. This parameter is considered only if failover_type is set to TRANSACTION for Application Continuity. Examples of session state are NLS settings, optimizer preferences, event settings, PL/SQL global variables, temporary tables, advanced queues, LOBs and result cache. If non-transactional values change after the request starts, the default value of DYNAMIC should be set. Almost all applications should use DYNAMIC mode. If you are unsure, use DYNAMIC mode.
GLOBAL_SERVICE	VARCHAR2(3)		Indicates whether the service is global. A global service is managed by Global Data Services (GDS) and can be provided by multiple databases that contain replicated data. Possible values: • YES: Indicates the service is global • NO: Indicates the service is not global This attribute is set when using GDS. It cannot be set by a user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PDB	VARCHAR2(128)		Name of a PDB associated with a given service. Will contain NULL for a non-CDB or if the service is not associated with a PDB (that is, connecting to a CDB using this service will cause a user to connect to the root.) When managing services for a PDB, use SRVCTL for Oracle RAC and Oracle RAC One Node databases, or connect to that PDB if it is a single instance (not RAC). The PDB attribute shows which PDB has the service. It cannot be set or modified explicitly.
SQL_TRANSLATION_PROFILE	VARCHAR2(261)		A non-NULL value specifies the initial SQL translation profile for subsequent database connections that use the service and do not specify a SQL translation profile. A NULL value has no effect.
MAX_LAG_TIME	VARCHAR2(128)		The maximum replication lag (in seconds) that is acceptable for a data replica to be used for providing the database service. Can only be specified for global services using the Global Data Services (GDS) interfaces. It is not supported to change this value at local databases.
GSM_FLAGS	NUMBER		Flags specific to Global Data Services (GDS). Can only be specified for global services using the GDS interfaces. It is not supported to change these values at local databases.
PQ_SVC	VARCHAR2(64)		Name of the associated parallel query rim service
STOP_OPTION	VARCHAR2(13)		Stop option for sessions of this service for planned maintenance
FAILOVER_RESTORE	VARCHAR2(6)		Indicates whether sessions restore their commonly used session state (such as NLS and schema settings) when they are failed over. Possible values: • AUTO - Sessions restore their commonly used session state when they are failed over with Transparent Application Continuity • LEVEL1 - Sessions restore their commonly used session state when they are failed over with Application Continuity • NONE - Sessions do not restore their commonly used session state when they are failed over
DRAIN_TIMEOUT	NUMBER		Number of seconds to wait for sessions to be drained
TABLE_FAMILY_ID	NUMBER		Sharded table family ID associated with the service
PLACEMENT_POLICY	NUMBER		Placement policy for the service. Possible values: • 0: PDB-NONE • 1: PDB-SINGLETON • 2: PDB-UNIFORM **Note:** Values other than 0 are applicable only in the ATP-Dedicated Cloud in an Oracle RAC environment.
RESET_STATE	VARCHAR2(6)		Reset state for the service. Possible values: • LEVEL1 • NONE
VCSPARE1	VARCHAR2(1024)		Reserved for internal use
NSPARE1	NUMBER		Reserved for internal use

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_SERVICES"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_APP_CONT.GET_LTXID_OUTCOME` procedure

3.73 ALL_SOURCE

`ALL_SOURCE` describes the text source of the stored objects accessible to the current user.

Related Views

- `DBA_SOURCE` describes the text source of all stored objects in the database.
- `USER_SOURCE` describes the text source of the stored objects owned by the current user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the object
`NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the object
`TYPE`	`VARCHAR2(12)`		Type of object: • `ASSEMBLEY` • `FUNCTION` • `LIBRARY` • `PACKAGE` • `PACKAGE BODY` • `PROCEDURE` • `TRIGGER` • `TYPE` • `TYPE BODY`
`LINE`	`NUMBER`	`NOT NULL`	Line number of this line of source
`TEXT`	`VARCHAR2(4000)`		Text source of the stored object
`ORIGIN_CON_ID`	`VARCHAR2(256)`		The ID of the container where the data originates. Possible values include: • `0`: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)

See Also

- "DBA_SOURCE"
- "USER_SOURCE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.74 ALL_SOURCE_AE

ALL_SOURCE_AE describes the text source of the stored objects (across all editions) accessible to the current user.

Related Views

- DBA_SOURCE_AE describes the text source of all stored objects (across all editions) in the database.
- USER_SOURCE_AE describes the text source of the stored objects (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the object
NAME	VARCHAR2(128)		Name of the object
TYPE	VARCHAR2(12)		Type of the object: • ASSEMBLEY • FUNCTION • LIBRARY • PACKAGE • PACKAGE BODY • PROCEDURE • TRIGGER • TYPE • TYPE BODY
LINE	NUMBER		Line number of this line of source
TEXT	VARCHAR2(4000)		Source text
EDITION_NAME	VARCHAR2(128)		Name of the edition
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)

See Also

- "DBA_SOURCE_AE"
- "USER_SOURCE_AE"

3.75 ALL_SQL_TRANSLATION_PROFILES

ALL_SQL_TRANSLATION_PROFILES describes all SQL translation profiles accessible to the user.

Related Views

- DBA_SQL_TRANSLATION_PROFILES describes all SQL translation profiles in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_SQL_TRANSLATION_PROFILES describes all SQL translation profiles owned by the user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the SQL translation profile
PROFILE_NAME	VARCHAR2(128)	NOT NULL	Name of the SQL translation profile
TRANSLATOR	VARCHAR2(261)		The translator package
FOREIGN_SQL_SYNTAX	VARCHAR2(5)		Indicates whether the SQL syntax is foreign. Possible values: • TRUE • FALSE
TRANSLATE_NEW_SQL	VARCHAR2(5)		Indicates whether to translate new SQL statements and errors using the translator. Possible values: • TRUE • FALSE
RAISE_TRANSLATION_ERROR	VARCHAR2(5)		Indicates whether to raise translation error. Possible values: • TRUE • FALSE
LOG_TRANSLATION_ERROR	VARCHAR2(5)		Indicates whether to log translation error. Possible values: • TRUE • FALSE
TRACE_TRANSLATION	VARCHAR2(5)		Indicates whether to trace translation. Possible values: • TRUE • FALSE
LOG_ERRORS	VARCHAR2(5)		Indicates whether there are log errors (TRUE) or not (FALSE)

See Also

- "DBA_SQL_TRANSLATION_PROFILES"
- "USER_SQL_TRANSLATION_PROFILES"

3.76 ALL_SQL_TRANSLATIONS

ALL_SQL_TRANSLATIONS describes all SQL translations accessible to the user.

Related Views

- DBA_SQL_TRANSLATIONS describes all SQL translations in the database.
- USER_SQL_TRANSLATIONS describes all SQL translations owned by the user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the SQL translation profile
PROFILE_NAME	VARCHAR2(128)	NOT NULL	Name of the SQL translation profile
SQL_TEXT	CLOB	NOT NULL	The SQL statement
TRANSLATED_TEXT	CLOB		The translated SQL statement
SQL_ID	VARCHAR2(13)	NOT NULL	SQL identifier of the SQL statement
HASH_VALUE	NUMBER	NOT NULL	Hash value of the SQL statement
ENABLED	VARCHAR2(5)		Displays whether the translation is enabled. Possible values: • TRUE • FALSE
REGISTRATION_TIME	TIMESTAMP(6)		Time the translation was registered
CLIENT_INFO	VARCHAR2(64)		Client information when the SQL was parsed and the translation was registered
MODULE	VARCHAR2(64)		Module when the SQL was parsed and the translation was registered
ACTION	VARCHAR2(64)		Action when the SQL was parsed and the translation was registered
PARSING_USER_ID	NUMBER		Current user ID when the SQL was parsed and the translation was registered
PARSING_SCHEMA_ID	NUMBER		Current schema ID when the SQL was parsed and the translation was registered
COMMENTS	VARCHAR2(4000)		Comment on the translation
ERROR_CODE	NUMBER		Last error code when the SQL was run
ERROR_SOURCE	VARCHAR2(9)		Source of the last error
TRANSLATION_METHOD	VARCHAR2(10)		Method used to translate the SQL during the last error
DICTIONARY_SQL_ID	VARCHAR2(13)		SQL identifier of the SQL text in the translation dictionary used to translate the SQL during the last error

See Also

- "DBA_SQL_TRANSLATIONS"
- "USER_SQL_TRANSLATIONS"

3.77 ALL_SQLJ_TYPE_ATTRS

ALL_SQLJ_TYPE_ATTRS describes the attributes of the SQLJ object types accessible to the current user.

Related Views

- DBA_SQLJ_TYPE_ATTRS describes the attributes of all SQLJ object types in the database.
- USER_SQLJ_TYPE_ATTRS describes the attributes of the object types owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the type
TYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the type
ATTR_NAME	VARCHAR2(128)	NOT NULL	Name of the attribute
EXTERNAL_ATTR_NAME	VARCHAR2(4000)		External name of the attribute
ATTR_TYPE_MOD	VARCHAR2(7)		Type modifier of the attribute: • REF • POINTER
ATTR_TYPE_OWNER	VARCHAR2(128)		Owner of the type of the attribute
ATTR_TYPE_NAME	VARCHAR2(128)		Name of the type of the attribute
LENGTH	NUMBER		Length of the CHAR attribute, or maximum length of the VARCHAR or VARCHAR2 attribute.
PRECISION	NUMBER		Decimal precision of the NUMBER or DECIMAL attribute, or binary precision of the FLOAT attribute.
SCALE	NUMBER		Scale of the NUMBER or DECIMAL attribute
CHARACTER_SET _NAME	VARCHAR2(44)		Character set name of the attribute (CHAR_CS or NCHAR_CS)
ATTR_NO	NUMBER	NOT NULL	Syntactical order number or position of the attribute as specified in the type specification or CREATE TYPE statement (not to be used as an ID number)
INHERITED	VARCHAR2(3)		Indicates whether the attribute is inherited from a supertype (YES) or not (NO)

See Also

- "DBA_SQLJ_TYPE_ATTRS"
- "USER_SQLJ_TYPE_ATTRS"

3.78 ALL_SQLJ_TYPE_METHODS

ALL_SQLJ_TYPE_METHODS describes the methods of the SQLJ object types accessible to the current user.

Related Views

- DBA_SQLJ_TYPE_METHODS describes the methods of all SQLJ object types in the database.
- USER_SQLJ_TYPE_METHODS describes the methods of the SQLJ object types owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the type
TYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the type
METHOD_NAME	VARCHAR2(128)	NOT NULL	Name of the method
EXTERNAL_VAR_NAME	VARCHAR2(4000)		Name of the external variable

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
METHOD_NO	NUMBER	NOT NULL	Method number that distinguishes overloaded methods (not to be used as an ID number)
METHOD_TYPE	VARCHAR2(6)		Type of the method: • MAP • ORDER • PUBLIC
PARAMETERS	NUMBER	NOT NULL	Number of parameters to the method
RESULTS	NUMBER	NOT NULL	Number of results returned by the method
FINAL	VARCHAR2(3)		Indicates whether the method is final (YES) or not (NO)
INSTANTIABLE	VARCHAR2(3)		Indicates whether the method is instantiable (YES) or not (NO)
OVERRIDING	VARCHAR2(3)		Indicates whether the method is overriding a supertype method (YES) or not (NO)
INHERITED	VARCHAR2(3)		Indicates whether the method is inherited from a supertype (YES) or not (NO)

See Also

- "DBA_SQLJ_TYPE_METHODS"
- "USER_SQLJ_TYPE_METHODS"

3.79 ALL_SQLJ_TYPES

ALL_SQLJ_TYPES describes the SQLJ object types accessible to the current user.

Related Views

- DBA_SQLJ_TYPES describes all SQLJ object types in the database.
- USER_SQLJ_TYPES describes the SQLJ object types owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the type
TYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the type
TYPE_OID	RAW(16)	NOT NULL	Object identifier (OID) of the type
EXTERNAL_NAME	VARCHAR2(4000)		External class name of the type
USING	VARCHAR2(21)		Representation of the type: • SQLData • CustomDatum • Serializable • Serializable Internal • ORAData
TYPECODE	VARCHAR2(128)		Typecode of the type
ATTRIBUTES	NUMBER		Number of attributes (if any) in the type

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
METHODS	NUMBER		Number of methods (if any) in the type
PREDEFINED	VARCHAR2(3)		Indicates whether the type is a predefined type (YES) or not (NO)
INCOMPLETE	VARCHAR2(3)		Indicates whether the type is an incomplete type (YES) or not (NO)
FINAL	VARCHAR2(3)		Indicates whether the type is a final type (YES) or not (NO)
INSTANTIABLE	VARCHAR2(3)		Indicates whether the type is an instantiable type (YES) or not (NO)
SUPERTYPE_OWNER	VARCHAR2(128)		Owner of the supertype (NULL if type is not a subtype)
SUPERTYPE_NAME	VARCHAR2(128)		Name of the supertype (NULL if type is not a subtype)
LOCAL_ATTRIBUTES	NUMBER		Number of local (not inherited) attributes (if any) in the subtype
LOCAL_METHODS	NUMBER		Number of local (not inherited) methods (if any) in the subtype

See Also

- "DBA_SQLJ_TYPES"
- "USER_SQLJ_TYPES"

3.80 ALL_SQLSET

ALL_SQLSET displays information about all SQL tuning sets accessible to the current user.

Related Views

- DBA_SQLSET displays information about all SQL tuning sets in the database.
- USER_SQLSET displays information about the SQL tuning sets owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
NAME	VARCHAR2(128)	NOT NULL	Name of the SQL tuning set
ID	NUMBER	NOT NULL	SQL tuning set identifier
CON_DBID	NUMBER	NOT NULL	The database ID of the PDB
OWNER	VARCHAR2(128)		Owner of the SQL tuning set
DESCRIPTION	VARCHAR2(256)		Description of the SQL tuning set
CREATED	DATE		Date the SQL tuning set was created
LAST_MODIFIED	DATE		Date the SQL tuning set was last modified
STATEMENT_COUNT	NUMBER		Number of statements in the SQL tuning set

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_SQLSET"
- "USER_SQLSET"

3.81 ALL_SQLSET_BINDS

ALL_SQLSET_BINDS displays the bind values associated with all SQL tuning sets accessible to the current user.

Related Views

- DBA_SQLSET_BINDS displays the bind values associated with all SQL tuning sets in the database.
- USER_SQLSET_BINDS displays the bind values associated with the SQL tuning sets owned by the current user. This view does not display the SQLSET_OWNER column.

Column	Datatype	NULL	Description
SQLSET_NAME	VARCHAR2(128)		Name of the SQL tuning set for the statement
SQLSET_OWNER	VARCHAR2(128)		User name of the SQL tuning set owner
SQLSET_ID	NUMBER		ID of the SQL tuning set for the statement
CON_DBID	NUMBER		The database ID of the PDB
SQL_ID	VARCHAR2(13)		SQL identifier of the parent cursor in the library cache
FORCE_MATCHING_SIGNATURE	NUMBER		The signature used when the CURSOR_SHARING parameter is set to FORCE
PLAN_HASH_VALUE	NUMBER		Numerical representation of the SQL plan for the cursor. Comparing one PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line-by-line).
POSITION	NUMBER		Bind position
VALUE	ANYDATA		Bind value. This column is NULL for PL/SQL bind types.
CAPTURED	CHAR(1)		Binds captured
SQL_SEQ	NUMBER		SQL sequence

See Also

- "DBA_SQLSET_BINDS"
- "USER_SQLSET_BINDS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.82 ALL_SQLSET_PLANS

ALL_SQLSET_PLANS describes captured plans for statements in the SQL tuning sets accessible to the current user.

Related Views

- DBA_SQLSET_PLANS describes captured plans in the SQL tuning sets in the database.
- USER_SQLSET_PLANS describes captured plans for statements in the SQL tuning sets owned by the current user. This view does not display the SQLSET_OWNER column.

Column	Datatype	NULL	Description
SQLSET_NAME	VARCHAR2(128)	NOT NULL	Name of SQL tuning set for the statement
SQLSET_OWNER	VARCHAR2(128)		User name of SQL tuning set owner
SQLSET_ID	NUMBER	NOT NULL	ID of SQL tuning set for the statement
CON_DBID	NUMBER	NOT NULL	The database ID of the PDB
SQL_ID	VARCHAR2(13)	NOT NULL	SQL identifier of the parent cursor in the library cache
FORCE_MATCHING_SIGNATURE	NUMBER	NOT NULL	The signature used when the CURSOR_SHARING parameter is set to FORCE
PLAN_HASH_VALUE	NUMBER	NOT NULL	Numerical representation of the SQL plan for the cursor. Comparing one PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line-by-line).
SQL_SEQ	NUMBER	NOT NULL	SQL sequence
STATEMENT_ID	VARCHAR2(128)		Statement ID
PLAN_ID	NUMBER		Plan ID
TIMESTAMP	DATE		Date and time timestamp
REMARKS	VARCHAR2(4000)		Remarks
OPERATION	VARCHAR2(128)		Name of the internal operation performed in this step (for example, TABLE ACCESS)
OPTIONS	VARCHAR2(255)		A variation on the operation described in the OPERATION column (for example, FULL)
OBJECT_NODE	VARCHAR2(128)		Name of the database link used to reference the object (a table name or view name). For local queries that use parallel execution, this column describes the order in which output from operations is consumed.
OBJECT_OWNER	VARCHAR2(128)		Name of the user who owns the schema containing the table or index
OBJECT_NAME	VARCHAR2(128)		Name of the table or index
OBJECT_ALIAS	VARCHAR2(261)		Alias for the object
OBJECT_INSTANCE	NUMBER(38)		Instance number for the object
OBJECT_TYPE	VARCHAR2(128)		Type of the object
OPTIMIZER	VARCHAR2(255)		Current mode of the optimizer for the first row in the plan (statement line), for example, CHOOSE. When the operation is a database access (for example, TABLE ACCESS), this column indicates whether or not the object is analyzed.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SEARCH_COLUMNS	NUMBER		Number of index columns with start and stop keys (that is, the number of columns with matching predicates)
ID	NUMBER(38)	NOT NULL	A number assigned to each step in the execution plan
PARENT_ID	NUMBER(38)		ID of the next execution step that operates on the output of the current step
DEPTH	NUMBER(38)		Depth (or level) of the operation in the tree. It is not necessary to issue a `CONNECT BY` statement to get the level information, which is generally used to indent the rows from the `PLAN_TABLE` table. The root operation (statement) is level 0.
POSITION	NUMBER(38)		Order of processing for all operations that have the same `PARENT_ID`.
COST	NUMBER(38)		Cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
CARDINALITY	NUMBER(38)		Estimate, made by the cost-based optimizer, of the number of rows produced by the operation
BYTES	NUMBER(38)		Estimate, made by the cost-based optimizer, of the number of bytes produced by the operation
OTHER_TAG	VARCHAR2(255)		Describes the contents of the `OTHER` column. For information about values, see *Oracle Database SQL Tuning Guide*.
PARTITION_START	VARCHAR2(255)		Start partition of a range of accessed partitions
PARTITION_STOP	VARCHAR2(255)		Stop partition of a range of accessed partitions
PARTITION_ID	NUMBER(38)		Step that computes the pair of values of the `PARTITION_START` and `PARTITION_STOP` columns
OTHER	LONG		Other information specific to the execution step that users may find useful. For information about values, see *Oracle Database SQL Tuning Guide*.
DISTRIBUTION	VARCHAR2(128)		Stores the method used to distribute rows from producer query servers to consumer query servers
CPU_COST	NUMBER(38)		CPU cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
IO_COST	NUMBER(38)		I/O cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
TEMP_SPACE	NUMBER(38)		Temporary space usage of the operation (sort or hash join) as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
ACCESS_PREDICATES	VARCHAR2(4000)		Predicates used to locate rows in an access structure. For example, start or stop predicates for an index range scan.
FILTER_PREDICATES	VARCHAR2(4000)		Predicates used to filter rows before producing them
PROJECTION	VARCHAR2(4000)		Expressions produced by the operation

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TIME	NUMBER(38)		Elapsed time (in seconds) of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
QBLOCK_NAME	VARCHAR2(128)		Name of the query block
OTHER_XML	CLOB		Provides extra information specific to an execution step of the execution plan. The content of this column is structured using XML since multiple pieces of information can be stored there. This includes: • Name of the schema against which the query was parsed • Release number of the Oracle Database that produced the explain plan • Hash value associated with the execution plan • Name (if any) of the outline or the SQL profile used to build the execution plan • Indication of whether or not dynamic statistics were used to produce the plan • The outline data, a set of optimizer hints that can be used to regenerate the same plan For further information about values, see *Oracle Database SQL Tuning Guide*.
EXECUTIONS	NUMBER		Number of times the plan has been executed
STARTS	NUMBER		Number of times this operation has been started, accumulated over the past executions
OUTPUT_ROWS	NUMBER		Number of rows produced by the row source, accumulated over the past executions
CR_BUFFER_GETS	NUMBER		Number of buffers received in consistent mode, accumulated over the past executions. Buffers are usually retrieved in consistent mode for queries.
CU_BUFFER_GETS	NUMBER		Number of buffers retrieved in current mode, accumulated over the past executions. Buffers are retrieved in current mode for statements such as INSERT, UPDATE, and DELETE.
DISK_READS	NUMBER		Number of physical disk reads performed by the operation, accumulated over the past executions
DISK_WRITES	NUMBER		Number of physical disk writes performed by the operation, accumulated over the past executions
ELAPSED_TIME	NUMBER		Elapsed time (in microseconds) corresponding to this operation, accumulated over the past executions
LAST_STARTS	NUMBER		Number of times this operation has been started, during the last execution
LAST_OUTPUT_ROWS	NUMBER		Number of rows produced by the row source, during the last execution
LAST_CR_BUFFER_GETS	NUMBER		Number of buffers retrieved in consistent mode, during the last execution. Buffers are usually retrieved in consistent mode for queries.
LAST_CU_BUFFER_GETS	NUMBER		Number of buffers retrieved in current mode, during the last execution. Buffers are retrieved in current mode for statements such as INSERT, UPDATE, and DELETE.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LAST_DISK_READS	NUMBER		Number of physical disk reads performed by the operation, during the last execution
LAST_DISK_WRITES	NUMBER		Number of physical disk writes performed by the operation, during the last execution
LAST_ELAPSED_TIME	NUMBER		Elapsed time (in microseconds) corresponding to this operation, during the last execution
POLICY	VARCHAR2(10)		Sizing policy for this work area: • MANUAL • AUTO
ESTIMATED_OPTIMAL_SIZE	NUMBER		Estimated size (in KB) required by this work area to execute the operation completely in memory (optimal execution). This is either derived from optimizer statistics or from previous executions.
ESTIMATED_ONEPASS_SIZE	NUMBER		Estimated size (in KB) required by this work area to execute the operation in a single pass. This is either derived from optimizer statistics or from previous executions.
LAST_MEMORY_USED	NUMBER		Memory size (in KB) used by this work area during the last execution of the cursor
LAST_EXECUTION	VARCHAR2(10)		Indicates whether this work area ran using OPTIMAL, ONE PASS, or under ONE PASS memory requirement (MULTI-PASS), during the last execution of the cursor
LAST_DEGREE	NUMBER		Degree of parallelism used, during the last execution of the cursor
TOTAL_EXECUTIONS	NUMBER		Number of times this work area was active
OPTIMAL_EXECUTIONS	NUMBER		Number of times this work area ran in optimal mode
ONEPASS_EXECUTIONS	NUMBER		Number of times this work area ran in one pass mode
MULTIPASSES_EXECUTIONS	NUMBER		Number of times this work area ran below the one pass memory requirement
ACTIVE_TIME	NUMBER		Average time this work area is active (in hundredths of a second)
MAX_TEMPSEG_SIZE	NUMBER		Maximum temporary segment size (in bytes) created by an instantiation of this work area. This column is null if this work area has never spilled to disk.
LAST_TEMPSEG_SIZE	NUMBER		Temporary segment size (in bytes) created in the last instantiation of this work area. This column is null if the last instantiation of this work area did not spill to disk.

See Also

- "DBA_SQLSET_PLANS"
- "USER_SQLSET_PLANS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.83 ALL_SQLSET_REFERENCES

ALL_SQLSET_REFERENCES describes whether or not the SQL tuning sets accessible to the current user are active.

Related Views

- DBA_SQLSET_REFERENCES describes whether or not all SQL tuning sets in the database are active. A SQL tuning set cannot be dropped if it is referenced.
- USER_SQLSET_REFERENCES describes whether or not the SQL tuning sets owned by the current user are active.

Column	Datatype	NULL	Description
SQLSET_NAME	VARCHAR2(128)	NOT NULL	Name of the SQL tuning set
SQLSET_OWNER	VARCHAR2(128)		User name of SQL tuning set owner
SQLSET_ID	NUMBER	NOT NULL	Identifier of the SQL tuning set
ID	NUMBER	NOT NULL	Reference identifier
OWNER	VARCHAR2(128)		User who registered to use the SQL tuning set
DESCRIPTION	VARCHAR2(256)		Description of the usage of the SQL tuning set
CREATED	DATE		Date the reference was created

See Also

- "DBA_SQLSET_REFERENCES"
- "USER_SQLSET_REFERENCES"

3.84 ALL_SQLSET_STATEMENTS

ALL_SQLSET_STATEMENTS displays information about the SQL statements, along with their statistics, that form all SQL tuning sets accessible to the current user.

Related Views

- DBA_SQLSET_STATEMENTS displays information about the SQL statements, along with their statistics, that form all SQL tuning sets in the database.
- USER_SQLSET_STATEMENTS displays information about the SQL statements, along with their statistics, that form the SQL tuning sets owned by the current user. This view does not display the SQLSET_OWNER column.

Column	Datatype	NULL	Description
SQLSET_NAME	VARCHAR2(128)	NOT NULL	Name of the SQL tuning set for the statement
SQLSET_OWNER	VARCHAR2(128)		User name of the SQL tuning set owner
SQLSET_ID	NUMBER	NOT NULL	ID of the SQL tuning set for the statement
CON_DBID	NUMBER	NOT NULL	The database ID of the PDB
SQL_ID	VARCHAR2(13)	NOT NULL	SQL identifier of the parent cursor in the library cache

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FORCE_MATCHING_SIGNATURE	NUMBER	NOT NULL	The signature used when the CURSOR_SHARING parameter is set to FORCE
SQL_TEXT	CLOB		Full text for the SQL statement exposed as a CLOB column.
PARSING_SCHEMA_NAME	VARCHAR2(128)		Name of the user in whose schema the statement was parsed
PLAN_HASH_VALUE	NUMBER	NOT NULL	Hash value for the plan corresponding to statistics in this record
BIND_DATA	RAW(2000)		Bind data
BINDS_CAPTURED	CHAR(1)		Binds captured
MODULE	VARCHAR2(64)		Contains the name of the module that was executing when the SQL statement was first parsed, which is set by calling DBMS_APPLICATION_INFO.SET_MODULE
ACTION	VARCHAR2(64)		Contains the name of the action that was executing when the SQL statement was first parsed, which is set by calling DBMS_APPLICATION_INFO.SET_ACTION
ELAPSED_TIME	NUMBER		Elapsed time (in microseconds) used by this cursor for parsing, executing, and fetching
CPU_TIME	NUMBER		CPU time (in microseconds) used by this cursor for parsing, executing, and fetching
BUFFER_GETS	NUMBER		Number of buffer gets for this child cursor
DISK_READS	NUMBER		Number of disk reads for this child cursor
DIRECT_WRITES	NUMBER		Number of direct writes for this child cursor
ROWS_PROCESSED	NUMBER		Total number of rows that the parsed SQL statement returns
FETCHES	NUMBER		Number of fetches associated with the SQL statement
EXECUTIONS	NUMBER		Number of executions that took place on this object since it was brought into the library cache
END_OF_FETCH_COUNT	NUMBER		Number of times this cursor was fully executed since the cursor was brought into the library cache. The value of this statistic in not incremented when the cursor is partially executed, either because it failed during the execution or because only the first few rows produced by this cursor are fetched before the cursor is closed or re-executed. By definition, the value of the END_OF_FETCH_COUNT column should be less than, or equal to, the value of the EXECUTIONS column.
OPTIMIZER_COST	NUMBER		Cost of this query, given by the optimizer
OPTIMIZER_ENV	RAW(2000)		Optimizer environment
PRIORITY	NUMBER		User-defined priority
COMMAND_TYPE	NUMBER		Oracle command type definition
FIRST_LOAD_TIME	VARCHAR2(19)		Timestamp of the parent creation time
STAT_PERIOD	NUMBER		Time (in seconds) during which the statistics of the SQL statement were collected
ACTIVE_STAT_PERIOD	NUMBER		Effective time (in seconds) during which the SQL statement was active
OTHER	CLOB		Client data, specified by the user, for this statement

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PLAN_TIMESTAMP	DATE		Timestamp for the plan corresponding to the statistics in this record
SQL_SEQ	NUMBER	NOT NULL	SQL sequence
LAST_EXEC_START_TIME	VARCHAR2(19)		For SQLs captured from the cursor cache, this is the time when the most recent execution of this SQL started

See Also

- "DBA_SQLSET_STATEMENTS"
- "USER_SQLSET_STATEMENTS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_MODULE procedure
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_ACTION procedure

3.85 ALL_STAT_EXTENSIONS

ALL_STAT_EXTENSIONS displays information about the optimizer statistics extensions accessible to the current user.

Related Views

- DBA_STAT_EXTENSIONS displays information about all optimizer statistics extensions in the database.
- USER_STAT_EXTENSIONS displays information about the optimizer statistics extensions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the extension
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table to which the extension belongs
EXTENSION_NAME	VARCHAR2(128)	NOT NULL	Name of the statistics extension
EXTENSION	CLOB		Extension (the expression or column group)
CREATOR	VARCHAR2(6)		Creator of the extension: • USER • SYSTEM
DROPPABLE	VARCHAR2(3)		Indicates whether the extension is droppable using DBMS_STATS.DROP_EXTENDED_STATS (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_STAT_EXTENSIONS"
- "USER_STAT_EXTENSIONS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_STATS.DROP_EXTENDED_STATS` procedure

3.86 ALL_STATEMENTS

`ALL_STATEMENTS` describes all SQL statements in stored PL/SQL objects accessible to the user.

Related Views

- `DBA_STATEMENTS` describes SQL statements in stored PL/SQL objects accessible to `SYS`.
- `USER_STATEMENTS` describes SQL statements in stored PL/SQL objects accessible to the user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`	`NOT NULL`	Owner of the statement
`SIGNATURE`	`VARCHAR2(32)`		Signature of the statement. Every statement type has a unique PL/Scope signature that identifies that instance of th statement.
`TYPE`	`VARCHAR2(17)`		Type of the statement. Statement types correspond to statements that can be used in PL/SQL to execute or otherwise interact with SQL: • `SELECT` • `UPDATE` • `INSERT` • `DELETE` • `MERGE` • `CLOSE` • `FETCH` • `OPEN` • `SAVEPOINT` • `COMMIT` • `SET_TRANSACTION` • `ROLLBACK` • `LOCK_TABLE` • `EXECUTE_IMMEDIATE`
`OBJECT_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the object where the statement usage occurred
`OBJECT_TYPE`	`VARCHAR2(12)`		Type of the object where the statement usage occurred
`USAGE_ID`	`NUMBER`		Unique key for a statement usage within the object
`LINE`	`NUMBER`		Line number of the statement usage
`COL`	`NUMBER`		Column number of the statement usage
`USAGE_CONTEXT_ID`	`NUMBER`		Context USAGE_ID of an statement usage

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SQL_ID	VARCHAR2(13)		SQL ID of the SQL statement. The value of this column is null for statements that do not have a SQL ID.
HAS_HINT	VARCHAR2(3)		YES if the SQL statement contains a hint, NO otherwise. If a hint appears inside of a subquery, then HAS_HINT will be YES for the containing statement or statements of the subquery as well as for the subquery itself.
HAS_INTO_BULK	VARCHAR2(3)		Indicates whether the statement contains a BULK_COLLECT clause (YES) or not (NO)
HAS_INTO_RETURNING	VARCHAR2(3)		Indicates whether the statement contains a RETURNING_INTO clause (YES) or not (NO)
HAS_INTO_RECORD	VARCHAR2(3)		Indicates whether the statement returns results into a PL/SQL record (YES) or not (NO)
HAS_CURRENT_OF	VARCHAR2(3)		Indicates whether the statement contains a HAS_CURRENT_OF clause (YES) or not (NO)
HAS_FOR_UPDATE	VARCHAR2(3)		Indicates whether the statement contains a HAS_FOR_UPDATE clause (YES) or not (NO)
HAS_IN_BINDS	VARCHAR2(3)		Indicates whether the statement contains an IN_BINDS clause (YES) or not (NO)
TEXT	VARCHAR2(4000)		The normalized form of the statement, when the statement has a normalized form. These are typically the same statements that have a SQL ID. The column value is null when the statement does not have a normalized form.
FULL_TEXT	CLOB		Clob text of the SQL statement
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)

See Also

- "DBA_STATEMENTS"
- "USER_STATEMENTS"

3.87 ALL_STORED_SETTINGS

ALL_STORED_SETTINGS provides information about the persistent parameter settings for stored PL/SQL units for which the current user has execute privileges.

Related Views

- DBA_STORED_SETTINGS lists information about the persistent parameter settings for stored PL/SQL units for which the current user has execute privileges. It also returns parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

information for all objects in the database and is accessible only to users with the SELECT_CATALOG_ROLE privilege.

- USER_STORED_SETTINGS lists information about the persistent parameter settings for stored PL/SQL units, but only shows information about PL/SQL units owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Name of the database user owning the stored PL/SQL unit
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the PL/SQL unit
OBJECT_ID	NUMBER	NOT NULL	Object number of the PL/SQL unit
OBJECT_TYPE	VARCHAR2(12)		The type of PL/SQL unit: PROCEDURE, FUNCTION, PACKAGE, PACKAGE BODY, TRIGGER, TYPE, or TYPE BODY
PARAM_NAME	VARCHAR2(128)	NOT NULL	The name of the parameter stored persistently with the PL/SQL unit
PARAM_VALUE	VARCHAR2(4000)		The TO_CHAR() representation of the value of the persistently stored parameter. The width of this column is operating system dependent; however, it is never less than 255.
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)

Note

This view is deprecated in favor of the ALL_PLSQL_OBJECT_SETTINGS view. Oracle recommends that you use ALL_PLSQL_OBJECT_SETTINGS instead. ALL_STORED_SETTINGS is retained for backward compatibility only.

See Also

"ALL_PLSQL_OBJECT_SETTINGS"

3.88 ALL_STREAMS_GLOBAL_RULES

ALL_STREAMS_GLOBAL_RULES displays information about rules.

ALL_STREAMS_GLOBAL_RULES displays information about the following types of rules:

- Global rules created for the capture processes that enqueue the captured changes into queues accessible to the current user
- Global rules created for the propagations that have a source queue accessible to the current user

Copyright © 2002, 2026, Oracle and/or its affiliates.

- Global rules created for the apply processes that dequeue events from queues accessible to the current user

This view does not contain information about rules created using the DBMS_RULE_ADM package.

Related View

DBA_STREAMS_GLOBAL_RULES displays information about the global rules created for all capture processes, propagations, and apply processes in the database.

Column	Datatype	NULL	Description
STREAMS_NAME	VARCHAR2(128)		Name of the Replication process or propagation
STREAMS_TYPE	VARCHAR2(11)		Type of the Replication process or propagation: • CAPTURE • PROPAGATION • APPLY
RULE_TYPE	VARCHAR2(9)		Type of the rule: • DML • DDL
INCLUDE_TAGGED_LCR	VARCHAR2(3)		Indicates whether a redo entry or logical change record (LCR) with a non-NULL tag is considered for capture, propagation, or apply (YES) or not (NO)
SOURCE_DATABASE	VARCHAR2(128)		Source database in the rule condition. The rule evaluates to true for a redo entry or logical change record (LCR) only if the redo entry or LCR contains this source database.
RULE_NAME	VARCHAR2(128)	NOT NULL	Name of the rule
RULE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule
RULE_CONDITION	VARCHAR2(4000)		First 4000 bytes of the system-generated rule condition evaluated by the rules engine

See Also

- "DBA_STREAMS_GLOBAL_RULES"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_RULE_ADM package

3.89 ALL_STREAMS_MESSAGE_CONSUMERS

ALL_STREAMS_MESSAGE_CONSUMERS displays information about the Replication messaging clients accessible to the current user.

Related View

DBA_STREAMS_MESSAGE_CONSUMERS displays information about all Replication messaging clients in the database.

Column	Datatype	NULL	Description
STREAMS_NAME	VARCHAR2(128)	NOT NULL	Name of the messaging client

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
QUEUE_NAME	VARCHAR2(128)	NOT NULL	Name of the queue
QUEUE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the queue
RULE_SET_NAME	VARCHAR2(128)		Name of the positive rule set
RULE_SET_OWNER	VARCHAR2(128)		Owner of the positive rule set
NEGATIVE_RULE_SET_NAME	VARCHAR2(128)		Name of the negative rule set
NEGATIVE_RULE_SET_OWNER	VARCHAR2(128)		Owner of the negative rule set
NOTIFICATION_TYPE	VARCHAR2(9)		Type of the notification action: • PROCEDURE • MAIL • HTTP
NOTIFICATION_ACTION	VARCHAR2(256)		Notification action
NOTIFICATION_CONTEXT	ANYDATA		Context for the notification action

See Also

"DBA_STREAMS_MESSAGE_CONSUMERS"

3.90 ALL_STREAMS_NEWLY_SUPPORTED

ALL_STREAMS_NEWLY_SUPPORTED displays information about the tables accessible to the current user that are newly supported by capture processes.

Related View

DBA_STREAMS_NEWLY_SUPPORTED displays information about all tables in the database that are newly supported by capture processes.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
REASON	VARCHAR2(39)		Reason why the table was not supported in a previous release: • IOT • column with user-defined type • unsupported column exists • object table • AQ queue table • temporary table • sub object • external table • materialized view • FILE column exists • materialized view log • materialized view container table • streams unsupported object • domain index
COMPATIBLE	CHAR(4)		Minimum database compatibility for capture processes to support the database object

See Also

"DBA_STREAMS_NEWLY_SUPPORTED"

3.91 ALL_STREAMS_SCHEMA_RULES

ALL_STREAMS_SCHEMA_RULES displays information about several types of schema rules.

ALL_STREAMS_SCHEMA_RULES displays information about these types of schema rules:

- Schema rules created for the capture processes that enqueue the captured changes into queues accessible to the current user
- Schema rules created for the propagations that have a source queue accessible to the current user
- Schema rules created for the apply processes that dequeue events from queues accessible to the current user

This view does not contain information about rules created using the DBMS_RULE_ADM package.

Related View

DBA_STREAMS_SCHEMA_RULES displays information about the schema rules created for all capture processes, propagations, and apply processes in the database.

Column	Datatype	NULL	Description
STREAMS_NAME	VARCHAR2(128)		Name of the Replication process or propagation

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STREAMS_TYPE	VARCHAR2(11)		Type of the Replication process or propagation: • CAPTURE • PROPAGATION • APPLY
SCHEMA_NAME	VARCHAR2(128)		Schema name in the rule condition. The rule evaluates to true for a redo entry or logical change record (LCR) only if the redo entry or LCR contains this schema name.
RULE_TYPE	VARCHAR2(7)		Type of the rule: • DML • DDL
INCLUDE_TAGGED_LCR	VARCHAR2(3)		Indicates whether a redo entry or logical change record (LCR) with a non-NULL tag is considered for capture, propagation, or apply (YES) or not (NO)
SOURCE_DATABASE	VARCHAR2(128)		Source database in the rule condition. The rule evaluates to true for a redo entry or logical change record (LCR) only if the redo entry or LCR contains this source database.
RULE_NAME	VARCHAR2(128)	NOT NULL	Name of the rule
RULE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule
RULE_CONDITION	VARCHAR2(4000)		First 4000 bytes of the system-generated rule condition evaluated by the rules engine

See Also

- "DBA_STREAMS_SCHEMA_RULES"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_RULE_ADM package

3.92 ALL_STREAMS_TABLE_RULES

ALL_STREAMS_TABLE_RULES displays information about several types of rules.

ALL_STREAMS_TABLE_RULES displays information about these types of table and subset rules:

- Table rules created for the capture processes that enqueue the captured changes into queues accessible to the current user
- Table rules created for the propagations that have a source queue accessible to the current user
- Table rules created for the apply processes that dequeue events from queues accessible to the current user
- Subset rules created for the apply processes that have a source queue accessible to the current user

This view does not contain information about rules created using the DBMS_RULE_ADM package.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Related View

DBA_STREAMS_TABLE_RULES displays information about the table rules created for all capture processes, propagations, and apply processes in the database.

Column	Datatype	NULL	Description
STREAMS_NAME	VARCHAR2(128)		Name of the Replication process or propagation
STREAMS_TYPE	VARCHAR2(12)		Type of the Replication process or propagation: • CAPTURE • PROPAGATION • APPLY • DEQUEUE
TABLE_OWNER	VARCHAR2(128)		Table owner in the rule condition. The rule evaluates to true for a redo entry or logical change record (LCR) only if the redo entry or LCR contains this table owner.
TABLE_NAME	VARCHAR2(128)		Table name in the rule condition. The rule evaluates to true for a redo entry or logical change record (LCR) only if the redo entry or LCR contains this table name.
RULE_TYPE	VARCHAR2(7)		Type of the rule: • DML • DDL
DML_CONDITION	VARCHAR2(4000)		Row subsetting condition, if the rule is a subset rule
SUBSETTING_OPERATION	VARCHAR2(6)		DML operation for row subsetting in the rule condition, if the rule is a subset rule: • INSERT • UPDATE • DELETE The rule evaluates to true for a logical change record (LCR) only if the LCR contains this command type after internal transformation.
INCLUDE_TAGGED_LCR	VARCHAR2(3)		Indicates whether a redo entry or logical change record (LCR) with a non-NULL tag is considered for capture, propagation, or apply (YES) or not (NO)
SOURCE_DATABASE	VARCHAR2(128)		Source database in the rule condition. The rule evaluates to true for a redo entry or logical change record (LCR) only if the redo entry or LCR contains this source database.
RULE_NAME	VARCHAR2(128)	NOT NULL	Name of the rule
RULE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule
RULE_CONDITION	VARCHAR2(4000)		First 4000 bytes of the system-generated rule condition evaluated by the rules engine

See Also

- "DBA_STREAMS_TABLE_RULES"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_RULE_ADM package

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.93 ALL_STREAMS_TRANSFORM_FUNCTION

ALL_STREAMS_TRANSFORM_FUNCTION displays information about the rule-based transformation functions accessible to the current user.

Related View

DBA_STREAMS_TRANSFORM_FUNCTION displays information about all rule-based transformation functions in the database.

Column	Datatype	NULL	Description
RULE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule associated with the transformation function
RULE_NAME	VARCHAR2(128)	NOT NULL	Name of the rule associated with the transformation function
VALUE_TYPE	VARCHAR2(4000)		Type of the transformation function name. This type must be VARCHAR2 for a rule-based transformation to work properly.
TRANSFORM_FUNCTION_NAME	VARCHAR2(4000)		Name of the transformation function (NULL if VALUE_TYPE is not VARCHAR2)
CUSTOM_TYPE	VARCHAR2(11)		Type of the transformation function: • ONE TO ONE - One-to-one transformations • ONE TO MANY - One-to-many transformations

See Also

"DBA_STREAMS_TRANSFORM_FUNCTION"

3.94 ALL_SUBPART_COL_STATISTICS

ALL_SUBPART_COL_STATISTICS describes column statistics and histogram information for subpartitions of partitioned objects accessible to the current user.

Related Views

- DBA_SUBPART_COL_STATISTICS provides this information for all subpartitions in the database.
- USER_SUBPART_COL_STATISTICS provides this information for subpartitions of all partitioned objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
SUBPARTITION_NAME	VARCHAR2(128)		Table subpartition name
COLUMN_NAME	VARCHAR2(4000)		Column name
NUM_DISTINCT	NUMBER		Number of distinct values in the column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LOW_VALUE	RAW(1000)		Low value in the column
HIGH_VALUE	RAW(1000)		High value in the column
DENSITY	NUMBER		If a histogram is available on COLUMN_NAME, then this column displays the selectivity of a value that spans fewer than 2 endpoints in the histogram. It does not represent the selectivity of values that span 2 or more endpoints. If a histogram is not available on COLUMN_NAME, then the value of this column is 1/NUM_DISTINCT.
NUM_NULLS	NUMBER		Number of NULLs in the column
NUM_BUCKETS	NUMBER		Number of buckets in histogram for the column
SAMPLE_SIZE	NUMBER		Sample size used in analyzing this column
LAST_ANALYZED	DATE		Date on which this column was most recently analyzed
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics have been gathered or NO if statistics have not been gathered
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
NOTES	VARCHAR2(41)		Describes some additional properties of the statistics. For example, if the value is INCREMENTAL, the global statistics are derived from synopses, that is, the global statistics are incrementally maintained.
AVG_COL_LEN	NUMBER		Average length of the column (in bytes)
HISTOGRAM	VARCHAR2(15)		Indicates existence/type of histogram: • NONE • FREQUENCY • HEIGHT BALANCED • HYBRID • TOP-FREQUENCY

See Also

- "DBA_SUBPART_COL_STATISTICS"
- "USER_SUBPART_COL_STATISTICS"

3.95 ALL_SUBPART_HISTOGRAMS

ALL_SUBPART_HISTOGRAMS displays the actual histogram data (end-points per histogram) for histograms on table subpartitions accessible to the current user.

Related Views

- DBA_SUBPART_HISTOGRAMS displays this information for all subpartitions in the database.
- USER_SUBPART_HISTOGRAMS displays this information for subpartitions of all partitioned objects owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> These views are populated only if you collect statistics on the index using the DBMS_STATS package.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table
SUBPARTITION_NAME	VARCHAR2(128)		Table subpartition name
COLUMN_NAME	VARCHAR2(4000)		Column name
BUCKET_NUMBER	NUMBER		Bucket number
ENDPOINT_VALUE	NUMBER		Normalized endpoint values for this bucket
ENDPOINT_ACTUAL_VALUE	VARCHAR2(4000)		Actual (not normalized) string value of the endpoint for this bucket
ENDPOINT_ACTUAL_VALUE_RAW	RAW(1000)		Endpoint actual value in raw format
ENDPOINT_REPEAT_COUNT	NUMBER		Frequency of the endpoint (applies only to hybrid histograms, and is set to 0 for other histogram types)

> **See Also**
>
> - "DBA_SUBPART_HISTOGRAMS"
> - "USER_SUBPART_HISTOGRAMS"
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

3.96 ALL_SUBPART_KEY_COLUMNS

ALL_SUBPART_KEY_COLUMNS displays subpartitioning key columns for composite-partitioned tables (and local indexes on composite-partitioned tables) accessible to the current user.

Related Views

- DBA_SUBPART_KEY_COLUMNS displays this information for all subpartitions in the database.
- USER_SUBPART_KEY_COLUMNS displays this information for subpartitions of all partitioned objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the partitioned table or index
NAME	VARCHAR2(128)		Name of the partitioned table or index
OBJECT_TYPE	CHAR(5)		Object type: • TABLE • INDEX
COLUMN_NAME	VARCHAR2(4000)		Column name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COLUMN_POSITION	NUMBER		Position of the column within the subpartitioning key
COLLATED_COLUMN_ID	NUMBER		Internal sequence number of the column for which this column provides linguistic ordering

See Also

- "DBA_SUBPART_KEY_COLUMNS"
- "USER_SUBPART_KEY_COLUMNS"

3.97 ALL_SUBPARTITION_TEMPLATES

ALL_SUBPARTITION_TEMPLATES describes the subpartition templates accessible to the current user.

Related Views

- DBA_SUBPARTITION_TEMPLATES describes all subpartition templates in the database.
- USER_SUBPARTITION_TEMPLATES describes the subpartition templates owned by the current user. This view does not display the USER_NAME column.

Column	Datatype	NULL	Description
USER_NAME	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
SUBPARTITION_NAME	VARCHAR2(132)	NOT NULL	Name of the subpartition
SUBPARTITION_POSITION	NUMBER		Position of the subpartition
TABLESPACE_NAME	VARCHAR2(30)		Tablespace name of the subpartition
HIGH_BOUND	LONG		Literal list values of the subpartition
COMPRESSION	VARCHAR2(4)		Compression values of COMPRESSION or NOCOMPRESSION can be specified in a subpartition template. The value in this column indicates whether the subpartition template specifies that for each new added composite partition, its subpartition data will be stored in compressed format (YES) or not (NO). If compression is not specified in the subpartition template, then the default is that data stored in newly-added subpartitions will not be stored in compressed format (NO).
INDEXING	VARCHAR2(4)		Indexing values of INDEXING ON or INDEXING OFF can be specified in a subpartition template. The value in this column indicates whether the subpartition template specifies that for each new added composite partition, its subpartition data will be considered for a partial index (ON) or not (OFF). If indexing is not specified in the subpartition template, then the default is that data stored in newly-added subpartitions will considered for a partial index (ON).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
READ_ONLY	VARCHAR2(4)		Values of READ ONLY or READ WRITE can be specified in a subpartition template. The value in this column indicates whether the subpartition template specifies that for each new added composite partition, its subpartition data will be read only (YES) or not (NO). If the read clause is not specified in the subpartition template, then the default is that data stored in newly-added subpartitions will be read/write (NO).

See Also

- "DBA_SUBPARTITION_TEMPLATES"
- "USER_SUBPARTITION_TEMPLATES"

3.98 ALL_SUMDELTA

ALL_SUMDELTA lists direct path load entries accessible to the current user.

Column	Datatype	NULL	Description
TABLEOBJ#	NUMBER	NOT NULL	Object number of the table
PARTITIONOBJ#	NUMBER	NOT NULL	Object number of table partitions (if the table is partitioned)
DMLOPERATION	VARCHAR2(1)		Type of DML operation applied to the table
SCN	NUMBER	NOT NULL	SCN when the bulk DML occurred
TIMESTAMP	DATE	NOT NULL	Timestamp of the log entry
LOWROWID	ROWID	NOT NULL	Start ROWID in the loaded rowid range
HIGHROWID	ROWID	NOT NULL	End ROWID in the loaded rowid range
SEQUENCE	NUMBER		Sequence number of the direct load
XID	NUMBER		Transaction ID

3.99 ALL_SYNC_CAPTURE

ALL_SYNC_CAPTURE displays information about the synchronous capture processes that store the captured changes in queues accessible to the current user.

Related View

DBA_SYNC_CAPTURE displays information about all synchronous capture processes in the database.

Column	Datatype	NULL	Description
CAPTURE_NAME	VARCHAR2(128)	NOT NULL	Name of the capture process
QUEUE_NAME	VARCHAR2(128)	NOT NULL	Name of the queue used for holding captured changes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
QUEUE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the queue used for holding captured changes
RULE_SET_NAME	VARCHAR2(128)		Rule set used by the capture process
RULE_SET_OWNER	VARCHAR2(128)		Owner of the rule set
CAPTURE_USER	VARCHAR2(128)		Current user who is enqueuing captured messages

See Also

"DBA_SYNC_CAPTURE"

3.100 ALL_SYNC_CAPTURE_PREPARED_TABS

ALL_SYNC_CAPTURE_PREPARED_TABS displays information about the tables accessible to the current user that are prepared for synchronous capture instantiation.

Related View

DBA_SYNC_CAPTURE_PREPARED_TABS displays information about all tables in the database that are prepared for synchronous capture instantiation.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the table prepared for synchronous capture instantiation
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table prepared for synchronous capture instantiation
SCN	NUMBER	NOT NULL	SCN from which changes can be captured
TIMESTAMP	DATE		Time at which the table was ready to be instantiated

See Also

"DBA_SYNC_CAPTURE_PREPARED_TABS"

3.101 ALL_SYNC_CAPTURE_TABLES

ALL_SYNC_CAPTURE_TABLES displays information about the tables accessible to the current user that are captured by synchronous captures.

Related View

DBA_SYNC_CAPTURE_TABLES displays information about all tables in the database that are captured by synchronous captures.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)		Owner of the synchronous capture table
TABLE_NAME	VARCHAR2(128)		Name of the synchronous capture table
ENABLED	VARCHAR2(3)		Indicates whether synchronous capture is enabled for the table (YES) or not (NO)

See Also

"DBA_SYNC_CAPTURE_TABLES"

3.102 ALL_SYNONYMS

ALL_SYNONYMS describes the synonyms accessible to the current user.

The following criteria determine the list of synonyms that ALL_SYNONYMS shows:

- All private synonyms owned by the logged-in user, even if the base object pointed to is not accessible.
- All public synonyms, even if the base object pointed to is not accessible.
- All private synonyms owned by a different user, where the ultimate base object pointed to by that synonym or by any chain of nested synonyms, is know to be accessible because of a grant to the logged-in user, or a grant to a role in effect for this session.
- If the current session has any of the following privileges, then all synonyms that point directly to local objects are shown because it is assumed that the session can access those objects:
  - LOCK ANY TABLE
  - READ ANY TABLE
  - SELECT ANY TABLE
  - INSERT ANY TABLE
  - UPDATE ANY TABLE
  - DELETE ANY TABLE

  Synonyms that point to remote objects are excluded because the system privileges just listed do not automatically convey access to those remote objects. Also, if the synonyms point to objects other than tables and views (such as sequences, PL/SQL procedures, and so on) then this rule may show synonyms that ultimately resolve to objects that this session cannot access.
- All private synonyms owned by a different user, where the synonym is via a database link, are excluded.

Related Views

- DBA_SYNONYMS describes all synonyms in the database.
- USER_SYNONYMS describes the synonyms owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the synonym
SYNONYM_NAME	VARCHAR2(128)		Name of the synonym
TABLE_OWNER	VARCHAR2(128)		Owner of the object referenced by the synonym, or creator of the referring synonym if the target is a public synonym (that is, the object referred to by TABLE_NAME). Although the column is called TABLE_OWNER, the object owned is not necessarily a table. It can be any general object such as a view, sequence, stored procedure, synonym, and so on.
TABLE_NAME	VARCHAR2(128)		Name of the object referenced by the synonym. Although the column is called TABLE_NAME, the object does not necessarily have to be a table. It can be any general object such as a view, sequence, stored procedure, synonym, and so on.
DB_LINK	VARCHAR2(128)		Name of the database link referenced, if any
ORIGIN_CON_ID	VARCHAR2(256)		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)

See Also

- "DBA_SYNONYMS"
- "USER_SYNONYMS"

3.103 ALL_TAB_COL_STATISTICS

ALL_TAB_COL_STATISTICS displays column statistics and histogram information extracted from ALL_TAB_COLUMNS.

Related Views

- DBA_TAB_COL_STATISTICS displays such information extracted from "DBA_TAB_COLUMNS".
- USER_TAB_COL_STATISTICS displays such information extracted from "USER_TAB_COLUMNS". This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table
COLUMN_NAME	VARCHAR2(128)		Column name
NUM_DISTINCT	NUMBER		Number of distinct values in the column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LOW_VALUE	RAW(1000)		Low value in the column
HIGH_VALUE	RAW(1000)		High value in the column
DENSITY	NUMBER		If a histogram is available on COLUMN_NAME, then this column displays the selectivity of a value that spans fewer than 2 endpoints in the histogram. It does not represent the selectivity of values that span 2 or more endpoints. If a histogram is not available on COLUMN_NAME, then the value of this column is 1/NUM_DISTINCT.
NUM_NULLS	NUMBER		Number of NULLs in the column
NUM_BUCKETS	NUMBER		Number of buckets in histogram for the column
LAST_ANALYZED	DATE		Date on which this column was most recently analyzed
SAMPLE_SIZE	NUMBER		Sample size used in analyzing this column
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics are gathered or incrementally maintained, otherwise it will be NO
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
NOTES	VARCHAR2(99)		Describes some additional properties of the statistics. For example: • A value of INCREMENTAL indicates that the global statistics are derived from synopses, that is, the global statistics are incrementally maintained. • A value of STATS_ON_CONVENTIONAL_LOAD indicates that the statistics are obtained by online statistics gathering for conventional DML.[1]
AVG_COL_LEN	NUMBER		Average length of the column (in bytes)
HISTOGRAM	VARCHAR2(15)		Indicates existence/type of histogram: • NONE • FREQUENCY • HEIGHT BALANCED • HYBRID • TOP-FREQUENCY
SCOPE	VARCHAR2(7)		The value is SHARED for statistics gathered on any table other than global temporary tables. For a global tempoary table, the possible values are: • SESSION - Indicates that the statistics are session-specific • SHARED - Indicates that the statistics are shared across all sessions See *Oracle Database PL/SQL Packages and Types Reference* for information about using the GLOBAL_TEMP_TABLE_STATS preference of the DBMS_STATS package to control whether to gather session or shared statistics for global temporary tables.

[1] The STATS_ON_CONVENTIONAL_LOAD value is available starting with Oracle Database 19c.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_TAB_COL_STATISTICS"
- "USER_TAB_COL_STATISTICS"
- "ALL_TAB_COLUMNS"

3.104 ALL_TAB_COLS

ALL_TAB_COLS describes the columns of the tables, views, and clusters accessible to the current user.

To gather statistics for this view, use the DBMS_STATS package.

This view differs from ALL_TAB_COLUMNS in that system-generated hidden columns are not filtered out.

Related Views

- DBA_TAB_COLS describes the columns of all tables, views, and clusters in the database.
- USER_TAB_COLS describes the columns of the tables, views, and clusters owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table, view, or cluster
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table, view, or cluster
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Column name
DATA_TYPE	VARCHAR2(128)		Data type of the column
DATA_TYPE_MOD	VARCHAR2(3)		Data type modifier of the column
DATA_TYPE_OWNER	VARCHAR2(128)		Owner of the data type of the column
DATA_LENGTH	NUMBER	NOT NULL	Length of the column (in bytes)
DATA_PRECISION	NUMBER		Decimal precision for NUMBER datatype; binary precision for FLOAT datatype; NULL for all other datatypes
DATA_SCALE	NUMBER		Digits to the right of the decimal point in a number
NULLABLE	VARCHAR2(1)		Indicates whether a column allows NULLs. The value is N if there is a NOT NULL constraint on the column or if the column is part of a PRIMARY KEY.
COLUMN_ID	NUMBER		Sequence number of the column as created
DEFAULT_LENGTH	NUMBER		Length of the default value for the column
DATA_DEFAULT	LONG		Default value for the column
NUM_DISTINCT	NUMBER		Number of distinct values in the column. This column remains for backward compatibility with Oracle7. This information is now in the {TAB|PART}_COL_STATISTICS views.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LOW_VALUE	RAW(1000)		Low value in the column. This column remains for backward compatibility with Oracle7. This information is now in the {TAB|PART}_COL_STATISTICS views.
HIGH_VALUE	RAW(1000)		High value in the column. This column remains for backward compatibility with Oracle7. This information is now in the {TAB|PART}_COL_STATISTICS views.
DENSITY	NUMBER		If a histogram is available on COLUMN_NAME, then this column displays the selectivity of a value that spans fewer than 2 endpoints in the histogram. It does not represent the selectivity of values that span 2 or more endpoints. If a histogram is not available on COLUMN_NAME, then the value of this column is 1/NUM_DISTINCT. This column remains for backward compatibility with Oracle7. This information is now in the {TAB|PART}_COL_STATISTICS views.
NUM_NULLS	NUMBER		Number of NULLs in the column
NUM_BUCKETS	NUMBER		Number of buckets in the histogram for the column **Note:** The number of buckets in a histogram is specified in the SIZE parameter of the ANALYZE SQL statement. However, Oracle Database does not create a histogram with more buckets than the number of rows in the sample. Also, if the sample contains any values that are very repetitious, Oracle Database creates the specified number of buckets, but the value indicated by this column may be smaller because of an internal compression algorithm.
LAST_ANALYZED	DATE		Date on which this column was most recently analyzed
SAMPLE_SIZE	NUMBER		Sample size used in analyzing this column
CHARACTER_SET_NAME	VARCHAR2(44)		Name of the character set: • CHAR_CS • NCHAR_CS
CHAR_COL_DECL_LENGTH	NUMBER		Declaration length of the character type column
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics are gathered or incrementally maintained, otherwise it will be NO
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
AVG_COL_LEN	NUMBER		Average length of the column (in bytes)
CHAR_LENGTH	NUMBER		Displays the length of the column in characters. This value only applies to the following datatypes: • CHAR • VARCHAR2 • NCHAR • NVARCHAR2

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CHAR_USED	VARCHAR2(1)		Indicates that the column uses BYTE length semantics (B) or CHAR length semantics (C), or whether the datatype is not any of the following (NULL): • CHAR • VARCHAR2 • NCHAR • NVARCHAR2
V80_FMT_IMAGE	VARCHAR2(3)		Indicates whether the column data is in release 8.0 image format (YES) or not (NO)
DATA_UPGRADED	VARCHAR2(3)		Indicates whether the column data has been upgraded to the latest type version format (YES) or not (NO)
HIDDEN_COLUMN	VARCHAR2(3)		Indicates whether the column is a hidden column (YES) or not (NO)
VIRTUAL_COLUMN	VARCHAR2(3)		Indicates whether the column is a virtual column (YES) or not (NO)
SEGMENT_COLUMN_ID	NUMBER		Sequence number of the column in the segment
INTERNAL_COLUMN_ID	NUMBER	NOT NULL	Internal sequence number of the column
HISTOGRAM	VARCHAR2(15)		Indicates existence/type of histogram: • NONE • FREQUENCY • TOP-FREQUENCY • HEIGHT BALANCED • HYBRID
QUALIFIED_COL_NAME	VARCHAR2(4000)		Qualified column name
USER_GENERATED	VARCHAR2(3)		Indicates whether the column is a user-generated column (YES) or a system-generated column (NO). Invisible columns are hidden columns that are also user- generated columns.
DEFAULT_ON_NULL	VARCHAR2(3)		Indicates whether the column has DEFAULT ON NULL semantics (YES) or not (NO)
IDENTITY_COLUMN	VARCHAR2(3)		Indicates whether this is an identity column (YES) or not (NO)
EVALUATION_EDITION	VARCHAR2(128)		Name of the edition in which editioned objects referenced in an expression column are resolved
UNUSABLE_BEFORE	VARCHAR2(128)		Name of the oldest edition in which the column is usable
UNUSABLE_BEGINNING	VARCHAR2(128)		Name of the oldest edition in which the column becomes perpetually unusable
COLLATION	VARCHAR2(100)		Collation for the column. Only applies to columns with character data types.
COLLATED_COLUMN_ID	NUMBER		Internal sequence number of a column, for which this virtual column generates a collation key.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_TAB_COLS"
- "USER_TAB_COLS"
- "ALL_TAB_COLUMNS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

3.105 ALL_TAB_COLUMNS

ALL_TAB_COLUMNS describes the columns of the tables, views, and clusters accessible to the current user.

To gather statistics for this view, use the DBMS_STATS package.

This view filters out system-generated hidden columns. The ALL_TAB_COLS view does not filter out system-generated hidden columns.

Related Views

- DBA_TAB_COLUMNS describes the columns of all tables, views, and clusters in the database.
- USER_TAB_COLUMNS describes the columns of the tables, views, and clusters owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table, view, or cluster
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table, view, or cluster
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Column name
DATA_TYPE	VARCHAR2(128)		Data type of the column
DATA_TYPE_MOD	VARCHAR2(3)		Data type modifier of the column
DATA_TYPE_OWNER	VARCHAR2(128)		Owner of the data type of the column
DATA_LENGTH	NUMBER	NOT NULL	Length of the column (in bytes)
DATA_PRECISION	NUMBER		Decimal precision for NUMBER data type; binary precision for FLOAT data type; NULL for all other data types
DATA_SCALE	NUMBER		Digits to the right of the decimal point in a number
NULLABLE	VARCHAR2(1)		Indicates whether a column allows NULLs. The value is N if there is a NOT NULL constraint on the column or if the column is part of a PRIMARY KEY. The constraint should be in an ENABLE VALIDATE state.
COLUMN_ID	NUMBER		Sequence number of the column as created
DEFAULT_LENGTH	NUMBER		Length of the default value for the column
DATA_DEFAULT	LONG		Default value for the column
NUM_DISTINCT	NUMBER		Number of distinct values in the column. This column remains for backward compatibility with Oracle7. This information is now in the {TAB|PART}_COL_STATISTICS views.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LOW_VALUE	RAW(1000)		Low value in the column. This column remains for backward compatibility with Oracle7. This information is now in the {TAB|PART}_COL_STATISTICS views.
HIGH_VALUE	RAW(1000)		High value in the column. This column remains for backward compatibility with Oracle7. This information is now in the {TAB|PART}_COL_STATISTICS views.
DENSITY	NUMBER		If a histogram is available on COLUMN_NAME, then this column displays the selectivity of a value that spans fewer than 2 endpoints in the histogram. It does not represent the selectivity of values that span 2 or more endpoints. If a histogram is not available on COLUMN_NAME, then the value of this column is 1/NUM_DISTINCT. This column remains for backward compatibility with Oracle7. This information is now in the {TAB|PART}_COL_STATISTICS views.
NUM_NULLS	NUMBER		Number of NULLs in the column
NUM_BUCKETS	NUMBER		Number of buckets in the histogram for the column **Note:** The number of buckets in a histogram is specified in the SIZE parameter of the ANALYZE SQL statement. However, Oracle Database does not create a histogram with more buckets than the number of rows in the sample. Also, if the sample contains any values that are very repetitious, Oracle Database creates the specified number of buckets, but the value indicated by this column may be smaller because of an internal compression algorithm.
LAST_ANALYZED	DATE		Date on which this column was most recently analyzed
SAMPLE_SIZE	NUMBER		Sample size used in analyzing this column
CHARACTER_SET_NAME	VARCHAR2(44)		Name of the character set: • CHAR_CS • NCHAR_CS
CHAR_COL_DECL_LENGTH	NUMBER		Declaration length of the character type column
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics are gathered or incrementally maintained, otherwise it will be NO
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
AVG_COL_LEN	NUMBER		Average length of the column (in bytes)
CHAR_LENGTH	NUMBER		Displays the length of the column in characters. This value only applies to the following datatypes: • CHAR • VARCHAR2 • NCHAR • NVARCHAR2

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CHAR_USED	VARCHAR2(1)		Indicates that the column uses BYTE length semantics (B) or CHAR length semantics (C), or whether the datatype is not any of the following (NULL): • CHAR • VARCHAR2 • NCHAR • NVARCHAR2
V80_FMT_IMAGE	VARCHAR2(3)		Indicates whether the column data is in release 8.0 image format (YES) or not (NO)
DATA_UPGRADED	VARCHAR2(3)		Indicates whether the column data has been upgraded to the latest type version format (YES) or not (NO)
HISTOGRAM	VARCHAR2(15)		Indicates existence/type of histogram: • NONE • FREQUENCY • TOP-FREQUENCY • HEIGHT BALANCED • HYBRID
DEFAULT_ON_NULL	VARCHAR2(3)		Indicates whether the column has DEFAULT ON NULL semantics (YES) or not (NO)
IDENTITY_COLUMN	VARCHAR2(3)		Indicates whether this is an identity column (YES) or not (NO)
EVALUATION_EDITION	VARCHAR2(128)		Name of the edition in which editioned objects referenced in an expression column are resolved
UNUSABLE_BEFORE	VARCHAR2(128)		Name of the oldest edition in which the column is usable
UNUSABLE_BEGINNING	VARCHAR2(128)		Name of the oldest edition in which the column becomes perpetually unusable
COLLATION	VARCHAR2(100)		Collation for the column. Only applies to columns with character data types.

See Also

- "DBA_TAB_COLUMNS"
- "USER_TAB_COLUMNS"
- "ALL_TAB_COLS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

3.106 ALL_TAB_COMMENTS

ALL_TAB_COMMENTS displays comments on the tables and views accessible to the current user.

Related Views

- DBA_TAB_COMMENTS displays comments on all tables and views in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_TAB_COMMENTS displays comments on the tables and views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the object
TABLE_TYPE	VARCHAR2(11)		Type of the object
COMMENTS	VARCHAR2(4000)		Comment on the object
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs • *n*: This value is used for rows containing data that originate in the container with the ID *n* (*n*=1 if the data originates in root)

See Also

- "DBA_TAB_COMMENTS"
- "USER_TAB_COMMENTS"

3.107 ALL_TAB_HISTGRM_PENDING_STATS

ALL_TAB_HISTGRM_PENDING_STATS describes pending statistics for tables, partitions, and subpartitions accessible to the current user.

Related Views

- DBA_TAB_HISTGRM_PENDING_STATS describes pending statistics for tables, partitions, and subpartitions in the database.
- USER_TAB_HISTGRM_PENDING_STATS describes pending statistics for tables, partitions, and subpartitions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table
PARTITION_NAME	VARCHAR2(128)		Name of the partition
SUBPARTITION_NAME	VARCHAR2(128)		Name of the subpartition
COLUMN_NAME	VARCHAR2(128)		Name of the column
ENDPOINT_NUMBER	NUMBER		Endpoint number
ENDPOINT_VALUE	NUMBER		Normalized endpoint value
ENDPOINT_ACTUAL_VALUE	VARCHAR2(4000)		Actual endpoint value
ENDPOINT_ACTUAL_VALUE_RAW	RAW(1000)		Endpoint actual value in raw format
ENDPOINT_REPEAT_COUNT	NUMBER		Frequency of the endpoint (applies only to hybrid histograms, and is set to 0 for other histogram types)

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_TAB_HISTGRM_PENDING_STATS"
- "USER_TAB_HISTGRM_PENDING_STATS"

3.108 ALL_TAB_HISTOGRAMS

ALL_TAB_HISTOGRAMS describes histograms on tables and views accessible to the current user.

The ALL_TAB_HISTOGRAMS view may contain a one-bucket histogram, which in fact signifies "No histogram" to the Oracle Database software. Therefore, it should not be queried to indicate the presence or absence of a histogram on a particular column. Instead, query the value of column HISTOGRAM in the ALL_TAB_COL_STATISTICS view.

Related Views

- DBA_TAB_HISTOGRAMS describes histograms on all tables and views in the database.
- USER_TAB_HISTOGRAMS describes histograms on all tables and views owned by the current user. This view does not display the OWNER column.

Note

These views are populated only if you collect statistics on the table using the DBMS_STATS package. For more information, see *Oracle Database PL/SQL Packages and Types Reference*.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table
COLUMN_NAME	VARCHAR2(4000)		Column name or attribute of the object type column
ENDPOINT_NUMBER	NUMBER		Histogram bucket number
ENDPOINT_VALUE	NUMBER		Normalized endpoint value for this bucket
ENDPOINT_ACTUAL_VALUE	VARCHAR2(4000)		Actual (not normalized) string value of the endpoint for this bucket
ENDPOINT_ACTUAL_VALUE_RAW	RAW(1000)		Endpoint actual value in raw format
ENDPOINT_REPEAT_COUNT	NUMBER		Frequency of the endpoint (applies only to hybrid histograms, and is set to 0 for other histogram types)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SCOPE	VARCHAR2(7)		The value is SHARED for statistics gathered on any table other than global temporary tables. For a global tempoary table, the possible values are: • SESSION - Indicates that the statistics are session-specific • SHARED - Indicates that the statistics are shared across all sessions See *Oracle Database PL/SQL Packages and Types Reference* for information about using the GLOBAL_TEMP_TABLE_STATS preference of the DBMS_STATS package to control whether to gather session or shared statistics for global temporary tables.

See Also

- "DBA_TAB_HISTOGRAMS"
- "USER_TAB_HISTOGRAMS"
- "ALL_TAB_COL_STATISTICS"

3.109 ALL_TAB_IDENTITY_COLS

ALL_TAB_IDENTITY_COLS describes all table identity columns.

Related Views

- DBA_TAB_IDENTITY_COLS describes all table identity columns.
- USER_TAB_IDENTITY_COLS describes all table identity columns. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the identity column
GENERATION_TYPE	VARCHAR2(10)		Generation type of the identity column. Possible values are ALWAYS or BY DEFAULT.
SEQUENCE_NAME	VARCHAR2(128)	NOT NULL	Name of the sequence associated with the identity column
IDENTITY_OPTIONS	VARCHAR2(298)		Options for the identity column sequence generator

See Also

- "DBA_TAB_IDENTITY_COLS"
- "USER_TAB_IDENTITY_COLS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- The ALTER TABLE statement in *Oracle Database SQL Language Reference* for more information about creating an identity column
- The CREATE TABLE statements in *Oracle Database SQL Language Reference* for more information about creating an identity column

3.110 ALL_TAB_MODIFICATIONS

`ALL_TAB_MODIFICATIONS` describes tables accessible to the current user that have been modified since the last time statistics were gathered on the tables.

Related Views

- `DBA_TAB_MODIFICATIONS` describes such information for all tables in the database.
- `USER_TAB_MODIFICATIONS` describes such information for tables owned by the current user. This view does not display the `TABLE_OWNER` column.

Column	Datatype	NULL	Description
`TABLE_OWNER`	`VARCHAR2(128)`		Owner of the modified table
`TABLE_NAME`	`VARCHAR2(128)`		Name of the modified table
`PARTITION_NAME`	`VARCHAR2(128)`		Name of the modified partition
`SUBPARTITION_NAME`	`VARCHAR2(128)`		Name of the modified subpartition
`INSERTS`	`NUMBER`		Approximate number of inserts since the last time statistics were gathered
`UPDATES`	`NUMBER`		Approximate number of updates since the last time statistics were gathered
`DELETES`	`NUMBER`		Approximate number of deletes since the last time statistics were gathered
`TIMESTAMP`	`DATE`		Indicates the last time the table was modified
`TRUNCATED`	`VARCHAR2(3)`		Indicates whether the table has been truncated since the last analyze (`YES`) or not (`NO`)
`DROP_SEGMENTS`	`NUMBER`		Number of partition and subpartition segments dropped since the last analyze

See Also

- "DBA_TAB_MODIFICATIONS"
- "USER_TAB_MODIFICATIONS"

3.111 ALL_TAB_PARTITIONS

`ALL_TAB_PARTITIONS` displays partition-level partitioning information, partition storage parameters, and partition statistics generated by the `DBMS_STATS` package for the partitions accessible to the current user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Related Views

- `DBA_TAB_PARTITIONS` displays such information for all partitions in the database.
- `USER_TAB_PARTITIONS` displays such information for the partitions of all partitioned objects owned by the current user. This view does not display the `TABLE_OWNER` column.

> **Note**
>
> Columns marked with an asterisk (*) are populated only if you collect statistics on the table with the `DBMS_STATS` package.

> **Note**
>
> The following is true for the columns below that include double asterisks (**) in the column description:
>
> The column can display information about segment-level attributes (for simple partitioned tables) or metadata (for composite partitioned tables). In a simple partitioned table, the partition physically contains the data (the segment) in the database. In a composite partitioned table, the partition is metadata and the data itself is stored in the subpartitions.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table
COMPOSITE	VARCHAR2(3)		Indicates whether the table is composite-partitioned (YES) or not (NO)
PARTITION_NAME	VARCHAR2(128)		Name of the partition
SUBPARTITION_COUNT	NUMBER		If this is a composite partitioned table, the number of subpartitions in the partition
HIGH_VALUE	LONG		Partition bound value expression
HIGH_VALUE_LENGTH	NUMBER		Length of the partition bound value expression
PARTITION_POSITION	NUMBER		Position of the partition within the table
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the partition**
PCT_FREE	NUMBER		Minimum percentage of free space in a block**
PCT_USED	NUMBER		Minimum percentage of used space in a block**
INI_TRANS	NUMBER		Initial number of transactions**
MAX_TRANS	NUMBER		Maximum number of transactions**
INITIAL_EXTENT	NUMBER		Size of the initial extent in bytes (for a range partition); size of the initial extent in blocks (for a composite partition)**
NEXT_EXTENT	NUMBER		Size of secondary extents in bytes (for a range partition); size of secondary extents in blocks (for a composite partition)**
MIN_EXTENT	NUMBER		Minimum number of extents allowed in the segment**

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MAX_EXTENT	NUMBER		Maximum number of extents allowed in the segment**
MAX_SIZE	NUMBER		Maximum number of blocks allowed in the segment**
PCT_INCREASE	NUMBER		Percentage increase in extent size**
FREELISTS	NUMBER		Number of process freelists allocated in this segment**
FREELIST_GROUPS	NUMBER		Number of freelist groups allocated in this segment**
LOGGING	VARCHAR2(7)		Indicates whether or not changes to the table are logged:** • NONE - Not specified **See Also:** the *_TAB_SUBPARTITIONS view • YES • NO
COMPRESSION	VARCHAR2(8)		Indicates the actual compression property for a partition of a simple partitioned table, or the default (if specified) for subpartitions for composite partitioned tables, otherwise NONE.** • NONE - The partition is composite, and a default setting is not specified for compression. **See Also:** the *_TAB_SUBPARTITIONS view • ENABLED - The setting for compression is enabled. • DISABLED - The setting for compression is disabled.
COMPRESS_FOR	VARCHAR2(30)		Default compression for what kind of operations:** • BASIC • ADVANCED • QUERY LOW • QUERY HIGH • ARCHIVE LOW • ARCHIVE HIGH • NULL The QUERY_LOW, QUERY_HIGH, ARCHIVE_LOW, and ARCHIVE_HIGH values are associated with Hybrid Columnar Compression, a feature of the Enterprise Edition of Oracle Database that is dependent on the underlying storage system. See *Oracle Database Concepts* for more information.
NUM_ROWS*	NUMBER		Number of rows in the partition
BLOCKS*	NUMBER		Number of used data blocks in the partition
EMPTY_BLOCKS	NUMBER		Number of empty (never used) data blocks in the partition. This column is populated only if you collect statistics on the table using the DBMS_STATS package.
AVG_SPACE*	NUMBER		Average amount of free space, in bytes, in a data block allocated to the partition
CHAIN_CNT*	NUMBER		Number of rows in the partition that are chained from one data block to another, or which have migrated to a new block, requiring a link to preserve the old ROWID
AVG_ROW_LEN*	NUMBER		Average length of a row in the partition (in bytes)
SAMPLE_SIZE	NUMBER		Sample size used in analyzing this partition
LAST_ANALYZED	DATE		Date on which this partition was most recently analyzed

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BUFFER_POOL	VARCHAR2(7)		Buffer pool to be used for the partition blocks:** • DEFAULT • KEEP • RECYCLE • NULL
FLASH_CACHE	VARCHAR2(7)		Database Smart Flash Cache hint to be used for partition blocks:** • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)		Cell flash cache hint to be used for partition blocks:** • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics have been gathered or NO if statistics have been aggregated from subpartitions or have not been gathered
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
IS_NESTED	VARCHAR2(3)		Indicates whether this is a nested table partition (YES) or not (NO) **See Also:** the *_NESTED_TABLES view for the parent table name/column
PARENT_TABLE_PARTITION	VARCHAR2(128)		Parent table's corresponding partition **See Also:** the *_NESTED_TABLES view for the parent table name/column
INTERVAL	VARCHAR2(3)		Indicates whether the partition is in the interval section of an interval partitioned table (YES) or whether the partition is in the range section (NO)
SEGMENT_CREATED	VARCHAR2(4)		Indicates the actual segment creation property for a partition of a simple partitioned table, or the default (if specified) for subpartitions for composite partitioned tables, otherwise NONE.** For a simple partitioned table, this column indicates whether a segment was created (YES) or not (NO). For composite partitioned tables, this column indicates whether or not a default segment creation property was explicitly specified. Possible values: • NONE - No default segment creation property was specified at the partition level. This value appears only for composite partitions, and is treated as an unspecified value. • YES - Immediate segment creation was explicitly specified at the partition level and will be used as the default for all of its subpartitions. • NO - Deferred segment creation was explicitly specified at the partition level and will be used as the default for all of its subpartitions.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INDEXING	VARCHAR2(4)		Indicates the actual indexing property for a partition of a simple partitioned table, or the default (if specified) for subpartitions for composite partitioned tables, otherwise NONE.** Possible values: • NONE - The partition is composite, and a default setting is not specified for indexing. This value appears only for composite partitions, and is treated as an unspecified value. When a user adds a subpartition to a table, since the defaults for the partition are unspecified, the ALL_PART_TABLES.DEF_INDEXING value is used for the newly created subpartition. • ON - INDEXING is on. • OFF - INDEXING is off.
READ_ONLY	VARCHAR2(4)		Indicates the default setting for the partition: • YES: The default setting for the partition is read-only. • NO: The default setting for the partition is read/write. • NONE: No default setting is specified for the partition.
INMEMORY	VARCHAR2(8)		Indicates whether the In-Memory Column Store (IM column store) is enabled (ENABLED) or disabled (DISABLED) for this partition
INMEMORY_PRIORITY	VARCHAR2(8)		Indicates the priority for In-Memory Column Store (IM column store) population. Possible values: • LOW • MEDIUM • HIGH • CRITICAL • NONE • NULL
INMEMORY_DISTRIBUTE	VARCHAR2(15)		Indicates how the IM column store is distributed in an Oracle Real Application Clusters (Oracle RAC) environment: • AUTO • BY ROWID RANGE • BY PARTITION • BY SUBPARTITION
INMEMORY_COMPRESSION	VARCHAR2(17)		Indicates the compression level for the IM column store: • NO MEMCOMPRESS • FOR DML • FOR QUERY [LOW | HIGH] • FOR CAPACITY [LOW | HIGH] • NULL This column has a value based on where the segments lie for a table. For example, if the table is partitioned and is enabled for the IM column store, the value is NULL for ALL_TABLES but non-NULL for ALL_TAB_PARTITIONS.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INMEMORY_DUPLICATE	VARCHAR2(13)		Indicates the duplicate setting for the IM column store in an Oracle RAC environment: • NO DUPLICATE • DUPLICATE • DUPLICATE ALL
CELLMEMORY	VARCHAR2(24)		The value for columnar compression in the storage cell flash cache. Possible values: • ENABLED: Oracle Exadata Storage will decide automatically whether to cache in columnar form • DISABLED: Oracle Exadata Storage is prevented from caching in columnar form • NO CACHECOMPRESS: Oracle Exadata Storage will cache in HCC format (no recompression) • FOR QUERY: Oracle Exadata Storage will recompress and cache in INMEMORY query high format • FOR CAPACITY: Oracle Exadata Storage will recompress and cache in INMEMORY capacity low format This column is intended for use with Oracle Exadata.
INMEMORY_SERVICE	VARCHAR2(12)		Indicates how the IM column store is populated on various instances. The possible values are: • DEFAULT: Data is populated on all instances specified with the PARALLEL_INSTANCE_GROUP initialization parameter. If that parameter is not set, then the data is populated on all instances. This is the default. • NONE: Data is not populated on any instance. • ALL: Data is populated on all instances, regardless of the value of the PARALLEL_INSTANCE_GROUP initialization parameter. • USER_DEFINED: Data is populated only on the instances on which the user-specified service is active. The service name corresponding to this is stored in the INMEMORY_SERVICE_NAME column.
INMEMORY_SERVICE_NAME	VARCHAR2(100)		Indicates the service name for the service on which the IM column store should be populated. This column has a value only when the corresponding INMEMORY_SERVICE is USER_DEFINED. In all other cases, this column is null.
MEMOPTIMIZE_READ	VARCHAR2(8)		Indicates whether the table is enabled for Fast Key Based Access (ENABLED) or not (DISABLED)
MEMOPTIMIZE_WRITE	VARCHAR2(8)		For internal use only

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_TAB_PARTITIONS"
- "USER_TAB_PARTITIONS"
- "PARALLEL_INSTANCE_GROUP"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

3.112 ALL_TAB_PENDING_STATS

ALL_TAB_PENDING_STATS describes pending statistics for tables, partitions, and subpartitions accessible to the current user.

Related Views

- DBA_TAB_PENDING_STATS describes pending statistics for tables, partitions, and subpartitions in the database.
- USER_TAB_PENDING_STATS describes pending statistics for tables, partitions, and subpartitions owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table
PARTITION_NAME	VARCHAR2(128)		Name of the partition
SUBPARTITION_NAME	VARCHAR2(128)		Name of the subpartition
NUM_ROWS	NUMBER		Number of rows
BLOCKS	NUMBER		Number of blocks
AVG_ROW_LEN	NUMBER		Average row length
IM_IMCU_COUNT	NUMBER		Number of In-Memory Compression Units (IMCUs) in the table.
IM_BLOCK_COUNT	NUMBER		Number of In-Memory blocks in the table.
SCAN_RATE	NUMBER		Scan rate for the table in megabytes per second. This statistic is only relevant or meaningful for external tables.
SAMPLE_SIZE	NUMBER		Sample size
LAST_ANALYZED	DATE		Time of last analyze operation

See Also

- "DBA_TAB_PENDING_STATS"
- "USER_TAB_PENDING_STATS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.113 ALL_TAB_PRIVS

ALL_TAB_PRIVS describes grants.

ALL_TAB_PRIVS describes the following types of grants:

- Object grants for which the current user is the object owner, grantor, or grantee
- Object grants for which an enabled role or PUBLIC is the grantee

Related Views

- DBA_TAB_PRIVS describes all object grants in the database.
- USER_TAB_PRIVS describes the object grants for which the current user is the object owner, grantor, or grantee.

Column	Datatype	NULL	Description
GRANTOR	VARCHAR2(128)		Name of the user who performed the grant
GRANTEE	VARCHAR2(128)		Name of the user or role to whom access was granted
TABLE_SCHEMA	VARCHAR2(128)		Schema of the object
TABLE_NAME	VARCHAR2(128)		Name of the object
PRIVILEGE	VARCHAR2(40)		Privilege on the object
GRANTABLE	VARCHAR2(3)		Indicates whether the privilege was granted with the GRANT OPTION (YES) or not (NO)
HIERARCHY	VARCHAR2(3)		Indicates whether the privilege was granted with the HIERARCHY OPTION (YES) or not (NO)
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the privilege was granted commonly (CONTAINER=ALL was used) • NO if the privilege was granted locally (CONTAINER=ALL was not used)
TYPE	VARCHAR2(24)		Type of the object
INHERITED	VARCHAR2(3)		Indicates whether the privilege grant was inherited from another container (YES) or not (NO)

See Also

- "DBA_TAB_PRIVS"
- "USER_TAB_PRIVS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.114 ALL_TAB_PRIVS_MADE

ALL_TAB_PRIVS_MADE describes the object grants for which the current user is the object owner or grantor.

Related View

USER_TAB_PRIVS_MADE describes the object grants for which the current user is the object owner. This view does not display the OWNER column.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)		Name of the user or role to whom access was granted
OWNER	VARCHAR2(128)		Owner of the object
TABLE_NAME	VARCHAR2(128)		Name of the object
GRANTOR	VARCHAR2(128)		Name of the user who performed the grant
PRIVILEGE	VARCHAR2(40)		Privilege on the object
GRANTABLE	VARCHAR2(3)		Indicates whether the privilege was granted with the GRANT OPTION (YES) or not (NO)
HIERARCHY	VARCHAR2(3)		Indicates whether the privilege was granted with the HIERARCHY OPTION (YES) or not (NO)
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the privilege was granted commonly (CONTAINER=ALL was used) • NO if the privilege was granted locally (CONTAINER=ALL was not used)
TYPE	VARCHAR2(24)		Type of the object
INHERITED	VARCHAR2(3)		Indicates whether the privilege grant was inherited from another container (YES) or not (NO)

See Also

"USER_TAB_PRIVS_MADE"

3.115 ALL_TAB_PRIVS_RECD

ALL_TAB_PRIVS_RECD describes object grants.

ALL_TAB_PRIVS_RECD describes the following types of grants:

- Object grants for which the current user is the grantee
- Object grants for which an enabled role or PUBLIC is the grantee

Related View

USER_TAB_PRIVS_RECD describes the object grants for which the current user is the grantee. This view does not display the GRANTEE column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)		Name of the user or role to whom access was granted
OWNER	VARCHAR2(128)		Owner of the object
TABLE_NAME	VARCHAR2(128)		Name of the object
GRANTOR	VARCHAR2(128)		Name of the user who performed the grant
PRIVILEGE	VARCHAR2(40)		Privilege on the object
GRANTABLE	VARCHAR2(3)		Indicates whether the privilege was granted with the GRANT OPTION (YES) or not (NO)
HIERARCHY	VARCHAR2(3)		Indicates whether the privilege was granted with the HIERARCHY OPTION (YES) or not (NO)
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the privilege was granted commonly (CONTAINER=ALL was used) • NO if the privilege was granted locally (CONTAINER=ALL was not used)
TYPE	VARCHAR2(24)		Type of the object
INHERITED	VARCHAR2(3)		Indicates whether the privilege grant was inherited from another container (YES) or not (NO)

See Also

"USER_TAB_PRIVS_RECD"

3.116 ALL_TAB_STAT_PREFS

ALL_TAB_STAT_PREFS displays information about statistics preferences for the tables accessible to the current user.

Related Views

- DBA_TAB_STAT_PREFS displays information about statistics preferences for all tables in the database.
- USER_TAB_STAT_PREFS displays information about statistics preferences for the tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
PREFERENCE_NAME	VARCHAR2(30)		Name of the preference
PREFERENCE_VALUE	VARCHAR2(4000)		Value of the preference

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_TAB_STAT_PREFS"
- "USER_TAB_STAT_PREFS"

3.117 ALL_TAB_STATISTICS

ALL_TAB_STATISTICS displays optimizer statistics for the tables accessible to the current user.

Related Views

- DBA_TAB_STATISTICS displays optimizer statistics for all tables in the database.
- USER_TAB_STATISTICS displays optimizer statistics for the tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the object
TABLE_NAME	VARCHAR2(128)		Name of the table
PARTITION_NAME	VARCHAR2(128)		Name of the partition
PARTITION_POSITION	NUMBER		Position of the partition within the table
SUBPARTITION_NAME	VARCHAR2(128)		Name of the subpartition
SUBPARTITION_POSITION	NUMBER		Position of the subpartition within the partition
OBJECT_TYPE	VARCHAR2(12)		Type of the object: • TABLE • PARTITION • SUBPARTITION
NUM_ROWS	NUMBER		Number of rows in the object
BLOCKS	NUMBER		Number of used blocks in the object
EMPTY_BLOCKS	NUMBER		Number of empty blocks in the object
AVG_SPACE	NUMBER		Average available free space in the object
CHAIN_CNT	NUMBER		Number of chained rows in the object
AVG_ROW_LEN	NUMBER		Average row length, including row overhead
AVG_SPACE_FREELIST_BLOCKS	NUMBER		Average freespace of all blocks on a freelist
NUM_FREELIST_BLOCKS	NUMBER		Number of blocks on the freelist
AVG_CACHED_BLOCKS	NUMBER		Average number of blocks in the buffer cache
AVG_CACHE_HIT_RATIO	NUMBER		Average cache hit ratio for the object
IM_IMCU_COUNT	NUMBER		Number of In-Memory Compression Units (IMCUs) in the table
IM_BLOCK_COUNT	NUMBER		Number of In-Memory blocks in the table
IM_STAT_UPDATE_TIME	TIMESTAMP(9)		The timestamp of the most recent update to the In-Memory statistics
SCAN_RATE	NUMBER		Scan rate for the object in megabytes per second. This statistic is only relevant or meaningful for external tables.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SAMPLE_SIZE	NUMBER		Sample size used in analyzing the table
LAST_ANALYZED	DATE		Date of the most recent time the table was analyzed
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics are gathered or incrementally maintained, otherwise it will be NO
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
STATTYPE_LOCKED	VARCHAR2(5)		Type of statistics lock: • DATA • CACHE • ALL
STALE_STATS	VARCHAR2(7)		Indicates whether statistics for the object are stale (YES) or not (NO)
NOTES[1]	VARCHAR2(25)		Describes some additional properties of the statistics. For example, a value of STATS_ON_CONVENTIONAL_LOAD indicates that the statistics are obtained by online statistics gathering for conventional DML.
SCOPE	VARCHAR2(7)		The value is SHARED for statistics gathered on any table other than global temporary tables. For a global temporary table, the possible values are: • SESSION - Indicates that the statistics are session-specific • SHARED - Indicates that the statistics are shared across all sessions See *Oracle Database PL/SQL Packages and Types Reference* for information about using the GLOBAL_TEMP_TABLE_STATS preference of the DBMS_STATS package to control whether to gather session or shared statistics for global temporary tables.

1 This column is available starting with Oracle Database 19c.

See Also

- "DBA_TAB_STATISTICS"
- "USER_TAB_STATISTICS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

3.118 ALL_TAB_STATS_HISTORY

ALL_TAB_STATS_HISTORY provides a history of table statistics modifications for all tables accessible to the current user.

Related Views

- DBA_TAB_STATS_HISTORY provides a history of table statistics modifications for all tables in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_TAB_STATS_HISTORY provides a history of table statistics modifications for all tables owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the object
TABLE_NAME	VARCHAR2(128)		Name of the table
PARTITION_NAME	VARCHAR2(128)		Name of the partition
SUBPARTITION_NAME	VARCHAR2(128)		Name of the subpartition
STATS_UPDATE_TIME	TIMESTAMP(6) WITH TIME ZONE		Time at which the statistics were updated

See Also

- "DBA_TAB_STATS_HISTORY"
- "USER_TAB_STATS_HISTORY"

3.119 ALL_TAB_SUBPARTITIONS

ALL_TAB_SUBPARTITIONS displays, for each table subpartition accessible to the current user, the subpartition name, name of the table and partition to which it belongs, its storage attributes, and statistics generated by the DBMS_STATS package.

Related Views

- DBA_TAB_SUBPARTITIONS displays such information for all subpartitions in the database.
- USER_TAB_SUBPARTITIONS displays such information for subpartitions of all partitioned objects owned by the current user. This view does not display the TABLE_OWNER column.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
PARTITION_NAME	VARCHAR2(128)		Name of the partition
SUBPARTITION_NAME	VARCHAR2(128)		Name of the subpartition
HIGH_VALUE	LONG		Subpartition bound value expression
HIGH_VALUE_LENGTH	NUMBER	NOT NULL	Length of the subpartition bound value expression
PARTITION_POSITION	NUMBER		Position of the partition within the table
SUBPARTITION_POSITION	NUMBER		Position of the subpartition within the partition
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of the tablespace containing the subpartition
PCT_FREE	NUMBER	NOT NULL	Minimum percentage of free space in a block
PCT_USED	NUMBER		Minimum percentage of used space in a block
INI_TRANS	NUMBER	NOT NULL	Initial number of transactions
MAX_TRANS	NUMBER	NOT NULL	Maximum number of transactions

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INITIAL_EXTENT	NUMBER		Size of the initial extent in bytes (for a range partition); size of the initial extent in blocks (for a composite partition)
NEXT_EXTENT	NUMBER		Size of secondary extents in bytes (for a range partition); size of secondary extents in blocks (for a composite partition)
MIN_EXTENT	NUMBER		Minimum number of extents allowed in the segment
MAX_EXTENT	NUMBER		Maximum number of extents allowed in the segment
MAX_SIZE	NUMBER		Maximum number of blocks allowed in the segment
PCT_INCREASE	NUMBER		Percentage increase in extent size
FREELISTS	NUMBER		Number of freelist groups allocated in this segment
FREELIST_GROUPS	NUMBER		Number of freelist groups allocated in this segment
LOGGING	VARCHAR2(3)		Indicates whether or not changes to the table are logged: • YES • NO
COMPRESSION	VARCHAR2(8)		Indicates whether this subpartition is compressed (ENABLED) or not (DISABLED)
COMPRESS_FOR	VARCHAR2(30)		Default compression for what kind of operations: • BASIC • ADVANCED • QUERY LOW • QUERY HIGH • ARCHIVE LOW • ARCHIVE HIGH • NULL The QUERY_LOW, QUERY_HIGH, ARCHIVE_LOW, and ARCHIVE_HIGH values are associated with Hybrid Columnar Compression, a feature of the Enterprise Edition of Oracle Database that is dependent on the underlying storage system. See *Oracle Database Concepts* for more information.
NUM_ROWS	NUMBER		Number of rows in the subpartition
BLOCKS	NUMBER		Number of blocks in the subpartition
EMPTY_BLOCKS	NUMBER		Number of empty blocks in the subpartition
AVG_SPACE	NUMBER		Average space in the subpartition
CHAIN_CNT	NUMBER		Chain count
AVG_ROW_LEN	NUMBER		Average row length
SAMPLE_SIZE	NUMBER		Sample size
LAST_ANALYZED	DATE		Date on which this table was most recently analyzed
BUFFER_POOL	VARCHAR2(7)		Buffer pool for this subpartition: • DEFAULT • KEEP • RECYCLE • NULL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FLASH_CACHE	VARCHAR2(7)		Database Smart Flash Cache hint to be used for subpartition blocks: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)		Cell flash cache hint to be used for subpartition blocks: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics have been gathered or NO if statistics have not been gathered
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
INTERVAL	VARCHAR2(3)		Indicates whether the partition is in the interval section of an interval partitioned table (YES) or whether the partition is in the range section (NO)
SEGMENT_CREATED	VARCHAR2(3)		Indicates whether the table subpartition segment has been created (YES) or not (NO); N/A indicates that this table is not subpartitioned
INDEXING	VARCHAR2(3)		Indicates the indexing property. Possible values: • ON - Indexing is on for this subpartition • OFF - Indexing is off for this subpartition
READ_ONLY	VARCHAR2(3)		Indicates whether a subpartition is read-only or read/write: • YES: The default setting for the subpartition is read-only. • NO: The default setting for the subpartition is read/write.
INMEMORY	VARCHAR2(8)		Indicates whether the In-Memory Column Store (IM column store) is enabled (ENABLED) or disabled (DISABLED) for this subpartition
INMEMORY_PRIORITY	VARCHAR2(8)		Indicates the priority for In-Memory Column Store (IM column store) population. Possible values: • LOW • MEDIUM • HIGH • CRITICAL • NONE • NULL
INMEMORY_DISTRIBUTE	VARCHAR2(15)		Indicates how the IM column store is distributed in an Oracle Real Application Clusters (Oracle RAC) environment: • AUTO • BY ROWID RANGE • DBY PARTITION • BY SUBPARTITION

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INMEMORY_COMPRESSION	VARCHAR2(17)		Indicates the compression level for the IM column store in an Oracle RAC environment: • NO MEMCOMPRESS • FOR DML • FOR QUERY [LOW | HIGH] • FOR CAPACITY [LOW | HIGH] • NULL This column has a value based on where the segments lie for a table. For example, if the table is partitioned and is enabled for IM column store, the value is NULL for ALL_TABLES but non-NULL for ALL_TAB_SUBPARTITIONS.
INMEMORY_DUPLICATE			Indicates the duplicate setting for the IM column store in an Oracle RAC environment: • DUPLICATE • NO DUPLICATE • DUPLICATE ALL
INMEMORY_SERVICE	VARCHAR2(12)		Indicates how the IM column store is populated on various instances. The possible values are: • DEFAULT: Data is populated on all instances specified with the PARALLEL_INSTANCE_GROUP initialization parameter. If that parameter is not set, then the data is populated on all instances. This is the default. • NONE: Data is not populated on any instance. • ALL: Data is populated on all instances, regardless of the value of the PARALLEL_INSTANCE_GROUP initialization parameter. • USER_DEFINED: Data is populated only on the instances on which the user-specified service is active. The service name corresponding to this is stored in the INMEMORY_SERVICE_NAME column.
INMEMORY_SERVICE_NAME	VARCHAR2(1000)		Indicates the service name for the service on which the IM column store should be populated. This column has a value only when the corresponding INMEMORY_SERVICE is USER_DEFINED. In all other cases, this column is null.
CELLMEMORY	VARCHAR2(24)		The value for columnar compression in the storage cell flash cache. Possible values: • ENABLED: Oracle Exadata Storage will decide automatically whether to cache in columnar form • DISABLED: Oracle Exadata Storage is prevented from caching in columnar form • NO CACHECOMPRESS: Oracle Exadata Storage will cache in HCC format (no recompression) • FOR QUERY: Oracle Exadata Storage will recompress and cache in INMEMORY query high format • FOR CAPACITY: Oracle Exadata Storage will recompress and cache in INMEMORY capacity low format This column is intended for use with Oracle Exadata.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MEMOPTIMIZE_READ	VARCHAR2(8)		Indicates whether the table is enabled for Fast Key Based Access (ENABLED) or not (DISABLED)
MEMOPTIMIZE_WRITE	VARCHAR2(8)		For internal use only

See Also

- "DBA_TAB_SUBPARTITIONS"
- "USER_TAB_SUBPARTITIONS"
- "PARALLEL_INSTANCE_GROUP"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

3.120 ALL_TABLES

ALL_TABLES describes the relational tables accessible to the current user. To gather statistics for this view, use the DBMS_STATS package.

Related Views

- DBA_TABLES describes all relational tables in the database.
- USER_TABLES describes the relational tables owned by the current user. This view does not display the OWNER column.

Note

Columns marked with an asterisk (*) are populated only if you collect statistics on the table with the DBMS_STATS package.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the table; NULL for partitioned, temporary, and index-organized tables
CLUSTER_NAME	VARCHAR2(128)		Name of the cluster, if any, to which the table belongs
IOT_NAME	VARCHAR2(128)		Name of the index-organized table, if any, to which the overflow or mapping table entry belongs. If the IOT_TYPE column is not NULL, then this column contains the base table name.
STATUS	VARCHAR2(8)		If a previous DROP TABLE operation failed, indicates whether the table is unusable (UNUSABLE) or valid (VALID)
PCT_FREE	NUMBER		Minimum percentage of free space in a block; NULL for partitioned tables

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PCT_USED	NUMBER		Minimum percentage of used space in a block; NULL for partitioned tables
INI_TRANS	NUMBER		Initial number of transactions; NULL for partitioned tables
MAX_TRANS	NUMBER		Maximum number of transactions; NULL for partitioned tables
INITIAL_EXTENT	NUMBER		Size of the initial extent (in bytes); NULL for partitioned tables
NEXT_EXTENT	NUMBER		Size of secondary extents (in bytes); NULL for partitioned tables
MIN_EXTENTS	NUMBER		Minimum number of extents allowed in the segment; NULL for partitioned tables
MAX_EXTENTS	NUMBER		Maximum number of extents allowed in the segment; NULL for partitioned tables
PCT_INCREASE	NUMBER		Percentage increase in extent size; NULL for partitioned tables
FREELISTS	NUMBER		Number of process freelists allocated to the segment; NULL for partitioned tables
FREELIST_GROUPS	NUMBER		Number of freelist groups allocated to the segment; NULL for partitioned tables
LOGGING	VARCHAR2(3)		Indicates whether or not changes to the table are logged; NULL for partitioned tables: • YES • NO
BACKED_UP	VARCHAR2(1)		Indicates whether the table has been backed up since the last modification (Y) or not (N)
NUM_ROWS*	NUMBER		Number of rows in the table
BLOCKS*	NUMBER		Number of used data blocks in the table
EMPTY_BLOCKS	NUMBER		Number of empty (never used) data blocks in the table. This column is populated only if you collect statistics on the table using the DBMS_STATS package.
AVG_SPACE*	NUMBER		Average amount of free space, in bytes, in a data block allocated to the table
CHAIN_CNT*	NUMBER		Number of rows in the table that are chained from one data block to another, or which have migrated to a new block, requiring a link to preserve the old ROWID
AVG_ROW_LEN*	NUMBER		Average length of a row in the table (in bytes)
AVG_SPACE_FREELIST_BLOCKS	NUMBER		Average freespace of all blocks on a freelist
NUM_FREELIST_BLOCKS	NUMBER		Number of blocks on the freelist
DEGREE	VARCHAR2(10)		Number of threads per instance for scanning the table, or DEFAULT
INSTANCES	VARCHAR2(10)		Number of instances across which the table is to be scanned, or DEFAULT
CACHE	VARCHAR2(5)		Indicates whether the table is to be cached in the buffer cache (Y) or not (N)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TABLE_LOCK	VARCHAR2(8)		Indicates whether table locking is enabled (ENABLED) or disabled (DISABLED)
SAMPLE_SIZE	NUMBER		Sample size used in analyzing the table
LAST_ANALYZED	DATE		Date on which the table was most recently analyzed
PARTITIONED	VARCHAR2(3)		Indicates whether the table is partitioned (YES) or not (NO)
IOT_TYPE	VARCHAR2(12)		If the table is an index-organized table, then IOT_TYPE is IOT, IOT_OVERFLOW, or IOT_MAPPING. If the table is not an index-organized table, then IOT_TYPE is NULL.
TEMPORARY	VARCHAR2(1)		Indicates whether the table is temporary (Y) or not (N)
SECONDARY	VARCHAR2(1)		Indicates whether the table is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge (Y) or not (N)
NESTED	VARCHAR2(3)		Indicates whether the table is a nested table (YES) or not (NO)
BUFFER_POOL	VARCHAR2(7)		Buffer pool for the table; NULL for partitioned tables: • DEFAULT • KEEP • RECYCLE • NULL
FLASH_CACHE	VARCHAR2(7)		Database Smart Flash Cache hint to be used for table blocks: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)		Cell flash cache hint to be used for table blocks: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
ROW_MOVEMENT	VARCHAR2(8)		Indicates whether partitioned row movement is enabled (ENABLED) or disabled (DISABLED)
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics are gathered or incrementally maintained, otherwise it will be NO
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
DURATION	VARCHAR2(15)		Indicates the duration of a temporary table: • SYS$SESSION - Rows are preserved for the duration of the session • SYS$TRANSACTION - Rows are deleted after COMMIT Null - Permanent table
SKIP_CORRUPT	VARCHAR2(8)		Indicates whether Oracle Database ignores blocks marked corrupt during table and index scans (ENABLED) or raises an error (DISABLED). To enable this feature, run the DBMS_REPAIR.SKIP_CORRUPT_BLOCKS procedure.
MONITORING	VARCHAR2(3)		This column is obsolete

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CLUSTER_OWNER	VARCHAR2(128)		Owner of the cluster, if any, to which the table belongs
DEPENDENCIES	VARCHAR2(8)		Indicates whether row-level dependency tracking is enabled (ENABLED) or disabled (DISABLED)
COMPRESSION	VARCHAR2(8)		Indicates whether table compression is enabled (ENABLED) or not (DISABLED); NULL for partitioned tables
COMPRESS_FOR	VARCHAR2(30)		Default compression for what kind of operations: • BASIC • ADVANCED • QUERY LOW • QUERY HIGH • ARCHIVE LOW • ARCHIVE HIGH • QUERY LOW ROW LEVEL LOCKING • QUERY HIGH ROW LEVEL LOCKING • ARCHIVE LOW ROW LEVEL LOCKING • ARCHIVE HIGH ROW LEVEL LOCKING • NO ROW LEVEL LOCKING • NULL The QUERY LOW, QUERY HIGH, ARCHIVE LOW, ARCHIVE HIGH, QUERY LOW ROW LEVEL LOCKING, QUERY HIGH ROW LEVEL LOCKING, ARCHIVE LOW ROW LEVEL LOCKING, ARCHIVE HIGH ROW LEVEL LOCKING, and NO ROW LEVEL LOCKING values are associated with Hybrid Columnar Compression, a feature of the Enterprise Edition of Oracle Database that is dependent on the underlying storage system. See *Oracle Database Concepts* for more information.
DROPPED	VARCHAR2(3)		Indicates whether the table has been dropped and is in the recycle bin (YES) or not (NO); NULL for partitioned tables This view does not return the names of tables that have been dropped.
READ_ONLY	VARCHAR2(3)		Indicates whether the table segment is READ-ONLY or not. Possible values: • YES - The table segment is READ-ONLY • NO - The table segment is not READ-ONLY • N/A - Not applicable. This value appears in a partitioned table, where there is no segment that relates to the logical table object.
SEGMENT_CREATED	VARCHAR2(3)		Indicates whether the table segment is created. Possible values: • YES - The table segment is created. • NO - The table segment is not created. • N/A - Not applicable. This value appears in a partitioned table, where there is no segment that relates to the logical table object.
RESULT_CACHE	VARCHAR2(7)		Result cache mode annotation for the table: • DEFAULT - Table has not been annotated • FORCE • MANUAL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CLUSTERING	VARCHAR2(3)		Indicates whether the table has the attribute clustering clause (YES) or not (NO)
ACTIVITY_TRACKING	VARCHAR2(23)		Indicates whether Heat Map tracking is enabled on the table
DML_TIMESTAMP	VARCHAR2(25)		Modification time, creation time, or both for Automatic Data Optimization
HAS_IDENTITY	VARCHAR2(3)		Indicates whether the table has an identity column (YES) or not (NO)
CONTAINER_DATA	VARCHAR2(3)		Indicates whether the table contains container-specific data. Possible values: • YES if the table was created with the CONTAINER_DATA clause • NO otherwise
INMEMORY	VARCHAR2(8)		Indicates whether the In-Memory Column Store (IM column store) is enabled (ENABLED) or disabled (DISABLED) for the table
INMEMORY_PRIORITY	VARCHAR2(8)		Indicates the priority for In-Memory Column Store (IM column store) population. Possible values: • LOW • MEDIUM • HIGH • CRITICAL • NONE • NULL
INMEMORY_DISTRIBUTE	VARCHAR2(15)		Indicates how the IM column store is distributed in an Oracle Real Application Clusters (Oracle RAC) environment: • AUTO • BY ROWID RANGE • DUPLICATE • NONE • NULL
INMEMORY_COMPRESSION	VARCHAR2(17)		Indicates the compression level for the IM column store: • BASIC • FOR CAPACITY [HIGH | LOW] • FOR QUERY [HIGH | LOW] • NULL This column has a value based on where the segments lie for a table. For example, if the table is partitioned and is enabled for the IM column store, the value is NULL for ALL_TABLES but non-NULL for ALL_TAB_PARTITIONS.
INMEMORY_DUPLICATE	VARCHAR2(13)		Indicates the duplicate setting for the IM column store in an Oracle RAC environment: • NO DUPLICATE • DUPLICATE • DUPLICATE ALL
DEFAULT_COLLATION	VARCHAR2(100)		Default collation for the table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DUPLICATED	VARCHAR2(1)		Indicates whether the table is duplicated on this shard (Y) or not (N)
SHARDED	VARCHAR2(1)		Indicates whether the table is sharded (Y) or not (N)
EXTERNAL	VARCHAR2(3)		Indicates whether the table is an external table (YES) or not (NO)
HYBRID[1]	VARCHAR2(3)		Indicates whether the table is a hybrid partitioned table (YES) or not (NO). A hybrid partitioned table can contain a mixture of partitions stored in segments and partitions stored externally.
CELLMEMORY	VARCHAR2(24)		The value for columnar compression in the storage cell flash cache. Possible values: • ENABLED: Oracle Exadata Storage will decide automatically whether to cache in columnar form • DISABLED: Oracle Exadata Storage is prevented from caching in columnar form • NO CACHECOMPRESS: Oracle Exadata Storage will cache in HCC format (no recompression) • FOR QUERY: Oracle Exadata Storage will recompress and cache in INMEMORY query high format • FOR CAPACITY: Oracle Exadata Storage will recompress and cache in INMEMORY capacity low format This column is intended for use with Oracle Exadata.
CONTAINERS_DEFAULT	VARCHAR2(3)		Indicates whether the table is enabled for CONTAINERS() by default (YES) or not (NO)
CONTAINER_MAP	VARCHAR2(3)		Indicates whether the table is enabled for use with the container_map database property (YES) or not (NO)
EXTENDED_DATA_LINK	VARCHAR2(3)		Indicates whether the table is enabled for fetching an extended data link from the root (YES) or not (NO)
EXTENDED_DATA_LINK_MAP	VARCHAR2(3)		For internal use only
INMEMORY_SERVICE	VARCHAR2(12)		Indicates how the IM column store is populated on various instances. The possible values are: • DEFAULT: Data is populated on all instances specified with the PARALLEL_INSTANCE_GROUP initialization parameter. If that parameter is not set, then the data is populated on all instances. This is the default. • NONE: Data is not populated on any instance. • ALL: Data is populated on all instances, regardless of the value of the PARALLEL_INSTANCE_GROUP initialization parameter. • USER_DEFINED: Data is populated only on the instances on which the user-specified service is active. The service name corresponding to this is stored in the INMEMORY_SERVICE_NAME column.
INMEMORY_SERVICE_NAME	VARCHAR2(1000)		Indicates the service name for the service on which the IM column store should be populated. This column has a value only when the corresponding INMEMORY_SERVICE is USER_DEFINED. In all other cases, this column is null.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CONTAINER_MAP_OBJECT	VARCHAR2(3)		Indicates whether the table is used as the value of the container_map database property (YES) or not (NO)
MEMOPTIMIZE_READ	VARCHAR2(8)		Indicates whether the table is enabled for Fast Key Based Access (ENABLED) or not (DISABLED)
MEMOPTIMIZE_WRITE	VARCHAR2(8)		For internal use only
HAS_SENSITIVE_COLUMN	VARCHAR2(3)		Indicates whether the table has one or more sensitive columns (YES) or not (NO)
ADMIT_NULL[1]	VARCHAR2(3)		Indicates whether the table admits null CON_ID data (YES) or not (NO)
DATA_LINK_DML_ENABLED[1]	VARCHAR2(3)		Indicates whether DML is permitted on the Data Link table (YES) or not (NO)
LOGICAL_REPLICATION[1]	VARCHAR2(8)		Indicates whether the table is enabled for logical replication (ENABLED) or not (DISABLED). This setting is ignored if database-wide column data supplemental logging is enabled.

1 This column is available starting with Oracle Database 19c.

Examples

This SQL query returns the names of the tables in the EXAMPLES tablespace:

```
SELECT table_name FROM all_tables
WHERE tablespace_name = 'EXAMPLE' ORDER BY table_name;
```

This SQL query returns the name of the tablespace that contains the HR schema:

```
SELECT DISTINCT tablespace_name FROM all_tables WHERE owner='HR';
```

See Also

- "DBA_TABLES"
- "USER_TABLES"
- "PARALLEL_INSTANCE_GROUP"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

3.121 ALL_TRANSFORMATIONS

ALL_TRANSFORMATIONS displays information about all transformations accessible to the current user.

These transformations can be specified with Advanced Queuing operations such as enqueue, dequeue, and subscribe to automatically integrate transformations in AQ messaging.

Related Views

- DBA_TRANSFORMATIONS displays information about all transformations in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_TRANSFORMATIONS displays information about transformations owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
TRANSFORMATION_ID	NUMBER	NOT NULL	Unique identifier for the transformation
OWNER	VARCHAR2(128)	NOT NULL	Owning user of the transformation
NAME	VARCHAR2(128)	NOT NULL	Transformation name
FROM_TYPE	VARCHAR2(128)		Source type name
TO_TYPE	VARCHAR2(256)		Target type name

See Also

- "DBA_TRANSFORMATIONS"
- "USER_TRANSFORMATIONS"

3.122 ALL_TRIGGER_COLS

ALL_TRIGGER_COLS describes the use of columns in the triggers accessible to the current user and in triggers on tables accessible to the current user.

If the user has the CREATE ANY TRIGGER privilege, then this view describes the use of columns in all triggers in the database.

Related Views

- DBA_TRIGGER_COLS describes the use of columns in all triggers in the database.
- USER_TRIGGER_COLS describes the use of columns in the triggers owned by the current user and in triggers on tables owned by the current user.

Column	Datatype	NULL	Description
TRIGGER_OWNER	VARCHAR2(128)		Owner of the trigger
TRIGGER_NAME	VARCHAR2(128)		Name of the trigger
TABLE_OWNER	VARCHAR2(128)		Owner of the table on which the trigger is defined
TABLE_NAME	VARCHAR2(128)		Table on which the trigger is defined
COLUMN_NAME	VARCHAR2(4000)		Name of the column used in the trigger
COLUMN_LIST	VARCHAR2(3)		Indicates whether the column is specified in the UPDATE clause (YES) or not (NO)
COLUMN_USAGE	VARCHAR2(17)		How the column is used in the trigger: • NEW IN • OLD IN • NEW IN OLD IN • NEW OUT • NEW IN OUT • NEW OUT OLD IN • NEW IN OUT OLD IN • PARENT IN

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_TRIGGER_COLS"
- "USER_TRIGGER_COLS"

3.123 ALL_TRIGGER_ORDERING

ALL_TRIGGER_ORDERING describes the triggers accessible to the current user that have FOLLOWS or PRECEDES ordering.

Related Views

- DBA_TRIGGER_ORDERING describes all triggers in the database that have FOLLOWS or PRECEDES ordering.
- USER_TRIGGER_ORDERING describes the triggers owned by the current user that have FOLLOWS or PRECEDES ordering. This view does not display the TRIGGER_OWNER column.

Column	Datatype	NULL	Description
TRIGGER_OWNER	VARCHAR2(128)	NOT NULL	Owner of the trigger
TRIGGER_NAME	VARCHAR2(128)	NOT NULL	Name of the trigger
REFERENCED_TRIGGER_OWNER	VARCHAR2(128)		Owner of the referenced trigger
REFERENCED_TRIGGER_NAME	VARCHAR2(128)		Name of the referenced trigger
ORDERING_TYPE	VARCHAR2(8)		Type of the ordering between the trigger and the referenced trigger: • FOLLOWS • PRECEDES

See Also

- "DBA_TRIGGER_ORDERING"
- "USER_TRIGGER_ORDERING"

3.124 ALL_TRIGGERS

ALL_TRIGGERS describes the triggers on tables accessible to the current user.

If the user has the CREATE ANY TRIGGER privilege, then this view describes all triggers in the database.

Related Views

- DBA_TRIGGERS describes all triggers in the database.
- USER_TRIGGERS describes the triggers owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the trigger
TRIGGER_NAME	VARCHAR2(128)		Name of the trigger
TRIGGER_TYPE	VARCHAR2(16)		When the trigger fires: • BEFORE STATEMENT • BEFORE EACH ROW • AFTER STATEMENT • AFTER EACH ROW • INSTEAD OF • COMPOUND
TRIGGERING_EVENT	VARCHAR2(246)		DML, DDL, or database event that fires the trigger **See Also:** *Oracle Database PL/SQL Language Reference* for additional information about triggers and triggering events.
TABLE_OWNER	VARCHAR2(128)		Owner of the table on which the trigger is defined
BASE_OBJECT_TYPE	VARCHAR2(18)		Base object on which the trigger is defined: • TABLE • VIEW • SCHEMA • DATABASE
TABLE_NAME	VARCHAR2(128)		If the base object type of the trigger is SCHEMA or DATABASE, then this column is NULL; if the base object type of the trigger is TABLE or VIEW, then this column indicates the table or view name on which the trigger is defined
COLUMN_NAME	VARCHAR2(4000)		Name of the nested table column (if a nested table trigger), else NULL
REFERENCING_NAMES	VARCHAR2(422)		Names used for referencing OLD and NEW column values from within the trigger
WHEN_CLAUSE	VARCHAR2(4000)		Must evaluate to TRUE for TRIGGER_BODY to execute
STATUS	VARCHAR2(8)		Indicates whether the trigger is enabled (ENABLED) or disabled (DISABLED); a disabled trigger will not fire
DESCRIPTION	VARCHAR2(4000)		Trigger description; useful for re-creating a trigger creation statement
ACTION_TYPE	VARCHAR2(11)		Action type of the trigger body: • CALL • PL/SQL
TRIGGER_BODY	LONG		Statements executed by the trigger when it fires
CROSSEDITION	VARCHAR2(7)		Type of crossedition trigger: • FORWARD • REVERSE • NO
BEFORE_STATEMENT	VARCHAR2(3)		Indicates whether the trigger has a BEFORE STATEMENT section (YES) or not (NO)
BEFORE_ROW	VARCHAR2(3)		Indicates whether the trigger has a BEFORE EACH ROW section (YES) or not (NO)
AFTER_ROW	VARCHAR2(3)		Indicates whether the trigger has an AFTER EACH ROW section (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
AFTER_STATEMENT	VARCHAR2(3)		Indicates whether the trigger has an AFTER STATEMENT section (YES) or not (NO)
INSTEAD_OF_ROW	VARCHAR2(3)		Indicates whether the trigger has an INSTEAD OF section (YES) or not (NO)
FIRE_ONCE	VARCHAR2(3)		Indicates whether the trigger will fire only for user processes making changes (YES) or whether the trigger will also fire for Replication Apply or SQL Apply processes (NO)
APPLY_SERVER_ONLY	VARCHAR2(3)		Indicates whether the trigger will only fire for a Replication Apply or SQL Apply process (YES) or not (NO). If set to YES, then the setting of FIRE_ONCE does not matter. **See Also:** the DBMS_DDL.SET_TRIGGER_FIRING_PROPERTY procedure

See Also

- "DBA_TRIGGERS"
- "USER_TRIGGERS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_DDL.SET_TRIGGER_FIRING_PROPERTY procedure

3.125 ALL_TRIGGERS_AE

ALL_TRIGGERS_AE describes the triggers on tables (across all editions) accessible to the current user.

If the user has the CREATE ANY TRIGGER privilege, then this view describes all triggers (across all editions) in the database.

Related Views

- DBA_TRIGGERS_AE describes all triggers (across all editions) in the database.
- USER_TRIGGERS_AE describes the triggers (across all editions) owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the trigger
TRIGGER_NAME	VARCHAR2(128)		Name of the trigger
TRIGGER_TYPE	VARCHAR2(16)		When the trigger fires: • BEFORE STATEMENT • BEFORE EACH ROW • AFTER STATEMENT • AFTER EACH ROW • INSTEAD OF • COMPOUND

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TRIGGERING_EVENT	VARCHAR2(246)		DML, DDL, or database event that fires the trigger **See Also:** *Oracle Database PL/SQL Language Reference* for additional information about triggers and triggering events.
TABLE_OWNER	VARCHAR2(128)		Owner of the table on which the trigger is defined
BASE_OBJECT_TYPE	VARCHAR2(18)		Base object on which the trigger is defined: • TABLE • VIEW • SCHEMA • DATABASE
TABLE_NAME	VARCHAR2(128)		If the base object type of the trigger is SCHEMA or DATABASE, then this column is NULL; if the base object type of the trigger is TABLE or VIEW, then this column indicates the table or view name on which the trigger is defined
COLUMN_NAME	VARCHAR2(4000)		Name of the nested table column (if a nested table trigger), else NULL
REFERENCING_NAMES	VARCHAR2(422)		Names used for referencing OLD and NEW column values from within the trigger
WHEN_CLAUSE	VARCHAR2(4000)		Must evaluate to TRUE for TRIGGER_BODY to execute
STATUS	VARCHAR2(8)		Indicates whether the trigger is enabled (ENABLED) or disabled (DISABLED); a disabled trigger will not fire
DESCRIPTION	VARCHAR2(4000)		Trigger description; useful for re-creating a trigger creation statement
ACTION_TYPE	VARCHAR2(11)		Action type of the trigger body: • CALL • PL/SQL
TRIGGER_BODY	LONG		Statements executed by the trigger when it fires
CROSSEDITION	VARCHAR2(7)		Type of crossedition trigger: • FORWARD • REVERSE • NO
BEFORE_STATEMENT	VARCHAR2(3)		Indicates whether the trigger has a BEFORE STATEMENT section (YES) or not (NO)
BEFORE_ROW	VARCHAR2(3)		Indicates whether the trigger has a BEFORE EACH ROW section (YES) or not (NO)
AFTER_ROW	VARCHAR2(3)		Indicates whether the trigger has an AFTER EACH ROW section (YES) or not (NO)
AFTER_STATEMENT	VARCHAR2(3)		Indicates whether the trigger has an AFTER STATEMENT section (YES) or not (NO)
INSTEAD_OF_ROW	VARCHAR2(3)		Indicates whether the trigger has an INSTEAD OF section (YES) or not (NO)
FIRE_ONCE	VARCHAR2(3)		Indicates whether the trigger will fire only for user processes making changes (YES) or whether the trigger will also fire for Replication Apply or SQL Apply processes (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
APPLY_SERVER_ONLY	VARCHAR2(3)		Indicates whether the trigger will only fire for a Replication Apply or SQL Apply process (YES) or not (NO). If set to YES, then the setting of FIRE_ONCE does not matter. **See Also:** the DBMS_DDL.SET_TRIGGER_FIRING_PROPERTY procedure
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the trigger is defined

Note

This view is available starting with Oracle Database 19c.

See Also

- "DBA_TRIGGERS_AE"
- "USER_TRIGGERS_AE"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_DDL.SET_TRIGGER_FIRING_PROPERTY procedure

3.126 ALL_TSTZ_TAB_COLS

ALL_TSTZ_TAB_COLS displays information about the columns of the tables accessible to the current user, which have columns defined on TIMESTAMP WITH TIME ZONE data types or object types containing attributes of TIMESTAMP WITH TIME ZONE data types.

Related Views

- DBA_TSTZ_TAB_COLS displays information about the columns of all tables in the database, which have columns defined on TIMESTAMP WITH TIME ZONE data types or object types containing attributes of TIMESTAMP WITH TIME ZONE data types. This view does not display the COLUMN_NAME, NESTED, and VIRTUAL_COLUMN columns.
- USER_TSTZ_TAB_COLS displays information about the columns of the tables owned by the current user, which have columns defined on TIMESTAMP WITH TIME ZONE data types or object types containing attributes of TIMESTAMP WITH TIME ZONE data types. This view does not display the OWNER, COLUMN_NAME, NESTED, and VIRTUAL_COLUMN columns.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Column name
QUALIFIED_COL_NAME	VARCHAR2(4000)		Qualified column name
NESTED	NUMBER		Indicates whether the column is a nested table (1) or not (0)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
VIRTUAL_COLUMN	NUMBER		Indicates whether the column is a virtual column (1) or not (0)
SCALAR_COLUMN	NUMBER		Indicates whether the column is a scalar column (1) or not (0)
UNUSED_COLUMN	NUMBER		Indicates whether the column is an unused column (1) or not (0)

See Also

- "DBA_TSTZ_TAB_COLS"
- "USER_TSTZ_TAB_COLS"

3.127 ALL_TSTZ_TABLES

ALL_TSTZ_TABLES displays information about the tables accessible to the current user, which have columns defined on TIMESTAMP WITH TIME ZONE data types or object types containing attributes of TIMESTAMP WITH TIME ZONE data types.

Related Views

- DBA_TSTZ_TABLES displays information about all tables in the database, which have columns defined on TIMESTAMP WITH TIME ZONE data types or object types containing attributes of TIMESTAMP WITH TIME ZONE data types.
- USER_TSTZ_TABLES displays information about the tables owned by the current user, which have columns defined on TIMESTAMP WITH TIME ZONE data types or object types containing attributes of TIMESTAMP WITH TIME ZONE data types. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
UPGRADE_IN_PROGRESS	VARCHAR2(3)		Indicates whether a table upgrade is in progress (YES) or not (NO)

See Also

- "DBA_TSTZ_TABLES"
- "USER_TSTZ_TABLES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.128 ALL_TYPE_ATTRS

ALL_TYPE_ATTRS describes the attributes of the object types accessible to the current user.

Related Views

- DBA_TYPE_ATTRS describes the attributes of all object types in the database. This view does not include the CHAR_USED column.
- USER_TYPE_ATTRS describes the attributes of the object types owned by the current user. This view does not display the OWNER or CHAR_USED column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the type
TYPE_NAME	VARCHAR2(128)		Name of the type
ATTR_NAME	VARCHAR2(128)		Name of the attribute
ATTR_TYPE_MOD	VARCHAR2(7)		Type modifier of the attribute: • REF • POINTER
ATTR_TYPE_OWNER	VARCHAR2(128)		Owner of the type of the attribute
ATTR_TYPE_NAME	VARCHAR2(128)		Name of the type of the attribute
LENGTH	NUMBER		Length of the CHAR attribute, or maximum length of the VARCHAR or VARCHAR2 attribute.
PRECISION	NUMBER		Decimal precision of the NUMBER or DECIMAL attribute, or binary precision of the FLOAT attribute.
SCALE	NUMBER		Scale of the NUMBER or DECIMAL attribute
CHARACTER_SET _NAME	VARCHAR2(44)		Character set name of the attribute (CHAR_CS or NCHAR_CS)
ATTR_NO	NUMBER		Syntactical order number or position of the attribute as specified in the type specification or CREATE TYPE statement (not to be used as an ID number)
INHERITED	VARCHAR2(3)		Indicates whether the attribute is inherited from a supertype (YES) or not (NO)
CHAR_USED	VARCHAR2(1)		Indicates whether the attribute uses BYTE length semantics (B) or CHAR length semantics (C). For NCHAR and NVARCHAR2 attribute types, this value is always C.

See Also

- "DBA_TYPE_ATTRS"
- "USER_TYPE_ATTRS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.129 ALL_TYPE_METHODS

ALL_TYPE_METHODS describes the methods of the object types accessible to the current user.

Related Views

- DBA_TYPE_METHODS describes the methods of all object types in the database.
- USER_TYPE_METHODS describes the methods of the object types owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the type
TYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the type
METHOD_NAME	VARCHAR2(128)	NOT NULL	Name of the method
METHOD_NO	NUMBER	NOT NULL	Method number for distinguishing overloaded methods (not to be used as ID number)
METHOD_TYPE	VARCHAR2(6)		Type of the method: • MAP • ORDER • PUBLIC
PARAMETERS	NUMBER	NOT NULL	Number of parameters to the method
RESULTS	NUMBER	NOT NULL	Number of results returned by the method
FINAL	VARCHAR2(3)		Indicates whether the method is final (YES) or not (NO)
INSTANTIABLE	VARCHAR2(3)		Indicates whether the method is instantiable (YES) or not (NO)
OVERRIDING	VARCHAR2(3)		Indicates whether the method is overriding a supertype method (YES) or not (NO)
INHERITED	VARCHAR2(3)		Indicates whether the method is inherited from a supertype (YES) or not (NO)

See Also

- "DBA_TYPE_METHODS"
- "USER_TYPE_METHODS"

3.130 ALL_TYPE_VERSIONS

ALL_TYPE_VERSIONS describes the versions of the object types accessible to the current user.

Related Views

- DBA_TYPE_VERSIONS describes the versions of all object types in the database.
- USER_TYPE_VERSIONS describes the versions of the object types owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the type
TYPE_NAME	VARCHAR2(128)	NOT NULL	Name of the type
VERSION#	NUMBER	NOT NULL	Internal version number of the type
TYPECODE	VARCHAR2(128)		Typecode of the type
STATUS	VARCHAR2(7)		Status of the type: • N/A • VALID • INVALID
LINE	NUMBER	NOT NULL	Line number of the type's spec
TEXT	VARCHAR2(4000)		Text of the type's spec
HASHCODE	RAW(17)		Hashcode of the type

See Also

- "DBA_TYPE_VERSIONS"
- "USER_TYPE_VERSIONS"

3.131 ALL_TYPES

ALL_TYPES describes the object types accessible to the current user.

Related Views

- DBA_TYPES describes all object types in the database.
- USER_TYPES describes the object types owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the type
TYPE_NAME	VARCHAR2(128)		Name of the type
TYPE_OID	RAW(16)		Object identifier (OID) of the type
TYPECODE	VARCHAR2(128)		Typecode of the type
ATTRIBUTES	NUMBER		Number of attributes (if any) in the type
METHODS	NUMBER		Number of methods (if any) in the type
PREDEFINED	VARCHAR2(3)		Indicates whether the type is a predefined type (YES) or not (NO)
INCOMPLETE	VARCHAR2(3)		Indicates whether the type is an incomplete type (YES) or not (NO)
FINAL	VARCHAR2(3)		Indicates whether the type is a final type (YES) or not (NO)
INSTANTIABLE	VARCHAR2(3)		Indicates whether the type is an instantiable type (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PERSISTABLE	VARCHAR2(3)		Indicates whether the type is a persistable type (YES) or not (NO)
SUPERTYPE_OWNER	VARCHAR2(128)		Owner of the supertype (NULL if type is not a subtype)
SUPERTYPE_NAME	VARCHAR2(128)		Name of the supertype (NULL if type is not a subtype)
LOCAL_ATTRIBUTES	NUMBER		Number of local (not inherited) attributes (if any) in the subtype
LOCAL_METHODS	NUMBER		Number of local (not inherited) methods (if any) in the subtype
TYPEID	RAW(16)		Type ID value of the type

See Also

- "DBA_TYPES"
- "USER_TYPES"

3.132 ALL_UNIFIED_AUDIT_ACTIONS

ALL_UNIFIED_AUDIT_ACTIONS describes unified audit trail actions.

The actions described in this view are valid for audit trail records from the UNIFIED_AUDIT_TRAIL view. Such records are generated only when unified auditing is enabled.

See Also

- *Oracle Database Security Guide* for more information about unified auditing.
- *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
TYPE	NUMBER	NOT NULL	Numeric component type for system wide actions
COMPONENT	VARCHAR2(64)	NOT NULL	Name of component for system wide actions
ACTION	NUMBER	NOT NULL	Numeric auditable action code for system wide actions **Note:** The action code values have changed from Oracle Database release 12.2 to the current release. If your applications have queries that include the ACTION column, and if these queries were written in release 12.2, then be aware that the output may be different if you are running these queries in the current release.
NAME	VARCHAR2(64)	NOT NULL	Name of auditable action

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "UNIFIED_AUDIT_TRAIL"

3.133 ALL_UNUSED_COL_TABS

ALL_UNUSED_COL_TABS describes the tables accessible to the current user that contain unused columns.

Related Views

- DBA_UNUSED_COL_TABS describes all tables in the database that contain unused columns.
- USER_UNUSED_COL_TABS describes the tables owned by the current user that contain unused columns. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
COUNT	NUMBER		Number of unused columns

> **See Also**
>
> - "DBA_UNUSED_COL_TABS"
> - "USER_UNUSED_COL_TABS"

3.134 ALL_UPDATABLE_COLUMNS

ALL_UPDATABLE_COLUMNS describes all columns in a join view that are updatable by the current user, subject to appropriate privileges.

Related Views

- DBA_UPDATABLE_COLUMNS describes all columns in a join view that are updatable by the database administrator, subject to appropriate privileges.
- USER_UPDATABLE_COLUMNS describes all columns owned by the current user that are in a join view and are updatable by the current user, subject to appropriate privileges.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Column name
UPDATABLE	VARCHAR2(3)		Indicates whether the column is updatable (YES) or not (NO)
INSERTABLE	VARCHAR2(3)		Indicates whether the column is insertable (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DELETABLE	VARCHAR2(3)		Indicates whether the column is deletable (YES) or not (NO)

> **Note**
>
> The values shown in the UPDATABLE, INSERTABLE, and DELETABLE columns are not instantly updated when a DDL operation that would affect these attributes occurs on one of the tables referenced in the FROM clause of the view definition. For example, these columns are not instantly updated when a primary key or unique constraint is added to or removed from the non-key-preserved table. To remedy this situation, recompile the view using the ALTER VIEW COMPILE statement to ensure that the latest information is displayed.

> **See Also**
>
> - "DBA_UPDATABLE_COLUMNS"
> - "USER_UPDATABLE_COLUMNS"
> - *Oracle Database Concepts* for information on updatable join views

3.135 ALL_USERS

ALL_USERS lists all users of the database visible to the current user.

This view does not describe the users (see the related views).

Related Views

- DBA_USERS describes all users of the database, and contains more columns than ALL_USERS.
- USER_USERS describes the current user, and contains more columns than ALL_USERS.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user
USER_ID	NUMBER	NOT NULL	ID number of the user
CREATED	DATE	NOT NULL	User creation date
COMMON	VARCHAR2(3)		Indicates whether a given user is common. Possible values: • YES if a user is common • NO if a user is local (not common)
ORACLE_MAINTAINED	VARCHAR2(1)		Denotes whether the user was created, and is maintained, by Oracle-supplied scripts (such as catalog.sql or catproc.sql). A user for which this column has the value Y must not be changed in any way except by running an Oracle-supplied script.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INHERITED	VARCHAR2(3)		Indicates whether the user definition was inherited from another container (YES) or not (NO)
DEFAULT_COLLATION	VARCHAR2(100)		Default collation for the user's schema
IMPLICIT	VARCHAR2(3)		Indicates whether this user is a common user created by an implicit application (YES) or not (NO)
ALL_SHARD	VARCHAR2(3)		In a sharded database, the value in this column indicates whether the user was created with shard DDL enabled. The possible values are: • YES: The user was created with shard DDL enabled. The user exists on all shards and the shard catalog. • NO: The user was created without shard DDL enabled. The user exists only in the database in which the user was created. In a non-sharded database, the value in this column is always NO.

See Also

- "DBA_USERS"
- "USER_USERS"
- *Using Oracle Sharding* for more information about sharded database management

3.136 ALL_USTATS

ALL_USTATS describes the user-defined statistics collected on the tables and indexes accessible to the current user.

Related Views

- DBA_USTATS describes the user-defined statistics collected on all tables and indexes in the database.
- USER_USTATS describes the user-defined statistics collected on the tables and indexes owned by the current user.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)		Owner of the table or index for which the statistics have been collected
OBJECT_NAME	VARCHAR2(128)		Name of the table or index for which the statistics have been collected
PARTITION_NAME	VARCHAR2(128)		Partition name of a table; NULL if the table is either nonpartitioned or the entry corresponds to the aggregate statistics for the table
OBJECT_TYPE	VARCHAR2(6)		Type of the object for which statistics have been collected: • INDEX • COLUMN

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ASSOCIATION	VARCHAR2(8)		Statistics type association: • DIRECT Direct association with the object for which the statistics have been collected • IMPLICIT - Association for which the statistics have been collected is with the column type or index type, and the object is an instance of that column type or index type
COLUMN_NAME	VARCHAR2(128)		Column name, if OBJECT_TYPE is COLUMN, for which statistics have been collected
STATSTYPE_SCHEMA	VARCHAR2(128)		Schema of the statistics type which was used to collect the statistics
STATSTYPE_NAME	VARCHAR2(128)		Name of the statistics type which was used to collect statistics
STATISTICS	RAW(2000)		User-collected statistics for the object

See Also

- "DBA_USTATS"
- "USER_USTATS"

3.137 ALL_VARRAYS

ALL_VARRAYS describes the varrays accessible to the current user.

Related Views

- DBA_VARRAYS describes all varrays in the database.
- USER_VARRAYS describes the varrays owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table containing the varray
PARENT_TABLE_NAME	VARCHAR2(128)		Name of the containing table
PARENT_TABLE_COLUMN	VARCHAR2(4000)		Name of the varray column or attribute
TYPE_OWNER	VARCHAR2(128)		Owner of the varray type
TYPE_NAME	VARCHAR2(128)		Name of the varray type
LOB_NAME	VARCHAR2(128)		Name of the LOB if the varray is stored in a LOB
STORAGE_SPEC	VARCHAR2(30)		Indicates whether the storage was defaulted (DEFAULT) or user-specified (USER_SPECIFIED)
RETURN_TYPE	VARCHAR2(20)		Return type of the column: • LOCATOR • VALUE
ELEMENT_SUBSTITUTABLE	VARCHAR2(25)		Indicates whether the varray element is substitutable (Y) or not (N)

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_VARRAYS"
- "USER_VARRAYS"

3.138 ALL_VIEWS

ALL_VIEWS describes the views accessible to the current user.

Related Views

- DBA_VIEWS describes all views in the database.
- USER_VIEWS describes the views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the view
VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the view
TEXT_LENGTH	NUMBER		Length of the view text
TEXT	LONG		View text. This column returns the correct value only when the row originates from the current container. The BEQUEATH clause will not appear as part of the TEXT column in this view.
TEXT_VC	VARCHAR2(4000)		View text. This column may truncate the view text. The BEQUEATH clause will not appear as part of the TEXT_VC column in this view.
TYPE_TEXT_LENGTH	NUMBER		Length of the type clause of the typed view
TYPE_TEXT	VARCHAR2(4000)		Type clause of the typed view
OID_TEXT_LENGTH	NUMBER		Length of the WITH OID clause of the typed view
OID_TEXT	VARCHAR2(4000)		WITH OID clause of the typed view
VIEW_TYPE_OWNER	VARCHAR2(128)		Owner of the type of the view if the view is a typed view
VIEW_TYPE	VARCHAR2(128)		Type of the view if the view is a typed view
SUPERVIEW_NAME	VARCHAR2(128)		Name of the superview
EDITIONING_VIEW	VARCHAR2(1)		Reserved for future use
READ_ONLY	VARCHAR2(1)		Indicates whether the view is read-only (Y) or not (N)
CONTAINER_DATA	VARCHAR2(1)		Indicates whether the view contains container-specific data. Possible values: • Y if the view was created with the CONTAINER_DATA clause • N otherwise

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BEQUEATH	VARCHAR2(12)		Possible values: • CURRENT_USER: When the view is a BEQUEATH CURRENT_USER view • DEFINER: When the view is a BEQUEATH DEFINER view For more information about the syntax and semantics of the BEQUEATH clause in the SQL CREATE VIEW statement, see *Oracle Database SQL Language Reference*.
ORIGIN_CON_ID	VARCHAR2(256)		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)
DEFAULT_COLLATION	VARCHAR2(100)		Default collation for the view
CONTAINERS_DEFAULT	VARCHAR2(3)		Indicates whether the view is enabled for CONTAINERS() by default (YES) or not (NO)
CONTAINER_MAP	VARCHAR2(3)		Indicates whether the view is enabled for use with the container_map database property (YES) or not (NO)
EXTENDED_DATA_LINK	VARCHAR2(3)		Indicates whether the view is enabled for fetching an extended data link from the root (YES) or not (NO)
EXTENDED_DATA_LINK_MAP	VARCHAR2(3)		For internal use only
HAS_SENSITIVE_COLUMN	VARCHAR2(3)		Indicates whether the view has one or more sensitive columns (YES) or not (NO)
ADMIT_NULL[1]	VARCHAR2(3)		Indicates whether the view admits null CON_ID data (YES) or not (NO)
PDB_LOCAL_ONLY[1]	VARCHAR2(3)		For internal use only

[1] This column is available starting with Oracle Database 19c.

See Also

- "DBA_VIEWS"
- "USER_VIEWS"

3.139 ALL_VIEWS_AE

ALL_VIEWS_AE describes the views (across all editions) accessible to the current user.

Related Views

- DBA_VIEWS_AE describes all views (across all editions) in the database.
- USER_VIEWS_AE describes the views (across all editions) owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the view
VIEW_NAME	VARCHAR2(128)	NOT NULL	Name of the view
TEXT_LENGTH	NUMBER		Length of the view text
TEXT	LONG		View text. The BEQUEATH clause will not appear as part of the TEXT column in this view.
TEXT_VC	VARCHAR2(4000)		View text. This column may truncate the view text. The BEQUEATH clause will not appear as part of the TEXT_VC column in this view.
TYPE_TEXT_LENGTH	NUMBER		Length of the type clause of the typed view
TYPE_TEXT	VARCHAR2(4000)		Type clause of the typed view
OID_TEXT_LENGTH	NUMBER		Length of the WITH OID clause of the typed view
OID_TEXT	VARCHAR2(4000)		WITH OID clause of the typed view
VIEW_TYPE_OWNER	VARCHAR2(128)		Owner of the type of the view if the view is an typed view
VIEW_TYPE	VARCHAR2(128)		Type of the view if the view is a typed view
SUPERVIEW_NAME	VARCHAR2(128)		Name of the superview, if the view is a subview
EDITIONING_VIEW	VARCHAR2(1)		Indicates whether the view is an editioning view (Y) or not (N)
READ_ONLY	VARCHAR2(1)		Indicates whether the view is read-only (Y) or not (N)
EDITION_NAME	VARCHAR2(128)		Name of the application edition where the object is defined
CONTAINER_DATA	VARCHAR2(1)		Indicates whether the view contains container-specific data. Possible values: • Y if the view was created with the CONTAINER_DATA clause • N otherwise
BEQUEATH	VARCHAR2(12)		Possible values: • CURRENT_USER: When the view is a BEQUEATH CURRENT_USER view • DEFINER: When the view is a BEQUEATH DEFINER view For more information about the syntax and semantics of the BEQUEATH clause in the SQL CREATE VIEW statement, see *Oracle Database SQL Language Reference*.
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)
DEFAULT_COLLATION	VARCHAR2(100)		Default collation for the view
CONTAINERS_DEFAULT	VARCHAR2(3)		Indicates whether the view is enabled for CONTAINERS() by default (YES) or not (NO)
CONTAINER_MAP	VARCHAR2(3)		Indicates whether the view is enabled for use with the container_map database property (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
EXTENDED_DATA_LINK	VARCHAR2(3)		Indicates whether the view is enabled for fetching an extended data link from the root (YES) or not (NO)
EXTENDED_DATA_LINK_MAP	VARCHAR2(3)		For internal use only
HAS_SENSITIVE_COLUMN	VARCHAR2(3)		Indicates whether the view has one or more sensitive columns (YES) or not (NO)
ADMIT_NULL[1]	VARCHAR2(3)		Indicates whether the view admits null CON_ID data (YES) or not (NO)
PDB_LOCAL_ONLY[1]	VARCHAR2(3)		For internal use only

1 This column is available starting with Oracle Database 19c.

See Also

- "DBA_VIEWS_AE"
- "USER_VIEWS_AE"

3.140 ALL_WARNING_SETTINGS

ALL_WARNING_SETTINGS displays information about the warning parameter settings for the objects accessible to the current user.

Related Views

- DBA_WARNING_SETTINGS displays information about the warning parameter settings for all objects in the database.
- USER_WARNING_SETTINGS displays information about the warning parameter settings for the objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object
OBJECT_ID	NUMBER	NOT NULL	Object number of the object
OBJECT_TYPE	VARCHAR2(12)		Type of the object: • PROCEDURE • FUNCTION • PACKAGE • PACKAGE BODY • TRIGGER • TYPE • TYPE BODY
WARNING	VARCHAR2(40)		Warning number or category: • INFORMATIONAL • PERFORMANCE • SEVERE • ALL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SETTING	VARCHAR2(7)		Value of the warning setting: • DISABLE • ENABLE • ERROR

See Also

- "DBA_WARNING_SETTINGS"
- "USER_WARNING_SETTINGS"

3.141 ALL_XML_INDEXES

ALL_XML_INDEXES describes the XML indexes accessible to the current user.

Related Views

- DBA_XML_INDEXES describes all XML indexes in the database.
- USER_XML_INDEXES describes the XML indexes owned by the current user. This view does not display the INDEX_OWNER column.

Column	Datatype	NULL	Description
INDEX_OWNER	VARCHAR2(128)	NOT NULL	Owner of the XML index
INDEX_NAME	VARCHAR2(128)	NOT NULL	Name of the XML index
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the indexed object
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the indexed object
TYPE	VARCHAR2(10)		Type of the indexed column: • REPOSITORY • BINARY • CLOB in OR • CLOB
INDEX_TYPE	VARCHAR2(27)		Type of the index: • STRUCTURED • STRUCTURED and UNSTRUCTURED • UNSTRUCTURED
PATH_TABLE_NAME	VARCHAR2(128)		Name of the path table
PARAMETERS	XMLTYPE		Indexed paths and Scheduler job information
ASYNC	VARCHAR2(9)		Asynchronous index type: • ON-COMMIT • MANUAL • EVERY • ALWAYS
STALE	VARCHAR2(5)		Indicates whether the index type is stale (TRUE) or not (FALSE)
PEND_TABLE_NAME	VARCHAR2(128)		Name of the pending table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
EX_OR_INCLUDE	VARCHAR2(8)		Path subsetting: • INCLUDE • EXCLUDE • FULLY IX

See Also

- "DBA_XML_INDEXES"
- "USER_XML_INDEXES"

3.142 ALL_XML_NESTED_TABLES

ALL_XML_NESTED_TABLES describes all the tables and their corresponding nested tables accessible to the current user.

Related Views

- DBA_XML_NESTED_TABLES describes all the tables and their corresponding nested tables in the database.
- USER_XML_NESTED_TABLES describes all the tables and their corresponding nested tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table
NESTED_TABLE_NAME	VARCHAR2(128)		Name of the nested table
PARENT_COLUMN_NAME	VARCHAR2(4000)		Name of the parent XML column

See Also

- "DBA_XML_NESTED_TABLES"
- "USER_XML_NESTED_TABLES"

3.143 ALL_XML_OUT_OF_LINE_TABLES

ALL_XML_OUT_OF_LINE_TABLES descibes all the out of line tables connected to a given root table for the same schema accessible to the current user.

Related Views

- DBA_XML_OUT_OF_LINE_TABLES describes all the out of line tables connected to a given root table for the same schema in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_XML_OUT_OF_LINE_TABLES describes all the out of line tables connected to a given root table for the same schema owned by the current user. This view does not display the TABLE_OWNER column.

Column	Datatype	NULL	Description
SCHEMA_URL	VARCHAR2(700)		The URL of the schema within which the out of line table is defined Refer to the **See Also** note below for links to more information about the schemaurl attribute for an XML schema.
SCHEMA_OWNER	VARCHAR2(128)		Owner of the schema
TABLE_NAME	VARCHAR2(128)		Name of the out of line table
TABLE_OWNER	VARCHAR2(128)		Owner of the out of line table

See Also

- "DBA_XML_OUT_OF_LINE_TABLES"
- "USER_XML_OUT_OF_LINE_TABLES"
- *Oracle XML DB Developer's Guide* for information about registering an XML schema with Oracle XML DB
- *Oracle XML DB Developer's Guide* for information about restrictions for an XML schema URL

3.144 ALL_XML_SCHEMA_ATTRIBUTES

ALL_XML_SCHEMA_ATTRIBUTES describes all the attributes and their properties accessible to the current user.

Related Views

- DBA_XML_SCHEMA_ATTRIBUTES describes all the attributes and their properties accessible to the current user in the database.
- USER_XML_SCHEMA_ATTRIBUTES describes all the attributes and their properties owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		The user who owns the attribute
SCHEMA_URL	VARCHAR2(700)		The URL of the schema within which the attribute is defined Refer to the **See Also** note below for links to more information about the schemaurl attribute for an XML schema.
TARGET_NAMESPACE	VARCHAR2(2000)		The namespace of the attribute
ATTRIBUTE_NAME	VARCHAR2(2000)		Name of the attribute
IS_REF	NUMBER		Indicates whether an attribute was defined using a reference in the XML schema (1) or not (0)
TYPE_NAME	VARCHAR2(2000)		Name of the type of the attribute

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
GLOBAL	RAW(1)		Indicates whether the attribute is global (1) or not (0)
ATTRIBUTE	XMLTYPE		Actual XMLType for the attribute
ELEMENT_ID	RAW(20)		Element ID of the element to which the attribute belongs
SQL_TYPE	VARCHAR2(128)		XDB annotation for sqlType
SQL_NAME	VARCHAR2(128)		XDB annotation value for sqlName

See Also

- "DBA_XML_SCHEMA_ATTRIBUTES"
- "USER_XML_SCHEMA_ATTRIBUTES"
- *Oracle XML DB Developer's Guide* for information about registering an XML schema with Oracle XML DB
- *Oracle XML DB Developer's Guide* for information about restrictions for an XML schema URL

3.145 ALL_XML_SCHEMA_COMPLEX_TYPES

ALL_XML_SCHEMA_COMPLEX_TYPES describes all complex types accessible to the current user.

Related Views

- DBA_XML_SCHEMA_COMPLEX_TYPES describes all complex types in the database.
- USER_XML_SCHEMA_COMPLEX_TYPES describes all complex types owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		The user who owns the type
SCHEMA_URL	VARCHAR2(700)		The URL of the schema within which the type is defined Refer to the **See Also** note below for links to more information about the schemaurl attribute for an XML schema.
TARGET_NAMESPACE	VARCHAR2(2000)		The namespace of the type
COMPLEX_TYPE_NAME	VARCHAR2(256)		Name of the complex type
COMPLEX_TYPE	XMLTYPE(XMLSchema "http://xmlns.oracle.com/xdb/XDBSchema.xsd" Element "complexType")		The actual XMLType of the type
BASE_NAME	VARCHAR2(256)		Name of the base type to which the complex type refers

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BASE_SCHEMA_URL	VARCHAR2(700)		The URL of the schema within which the complex type is defined
BASE_TARGET_NAMESPACE	VARCHAR2(2000)		The namespace of the type
MAINTAIN_DOM	RAW(1)		XDB annotation for maintainDOM
SQL_TYPE	VARCHAR2(128)		XDB annotation for sqlType
SQL_SCHEMA	VARCHAR2(128)		XDB annotation for sqlSchema

See Also

- "DBA_XML_SCHEMA_COMPLEX_TYPES"
- "USER_XML_SCHEMA_COMPLEX_TYPES"
- *Oracle XML DB Developer's Guide* for information about registering an XML schema with Oracle XML DB
- *Oracle XML DB Developer's Guide* for information about restrictions for an XML schema URL

3.146 ALL_XML_SCHEMA_ELEMENTS

ALL_XML_SCHEMA_ELEMENTS describes all the elements and their properties accessible to the current user.

Related Views

- DBA_XML_SCHEMA_ELEMENTS describes all the elements and their properties.
- USER_XML_SCHEMA_ELEMENTS describes all the elements and their properties owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		The user who owns the element
SCHEMA_URL	VARCHAR2(700)		The URL of the schema within which the element is defined Refer to the **See Also** note below for links to more information about the schemaurl attribute for an XML schema.
TARGET_NAMESPACE	VARCHAR2(2000)		The namespace of the element
ELEMENT_NAME	VARCHAR2(2000)		Name of the element
IS_REF	NUMBER		Indicates whether an attribute was defined using a reference in the XML schema (1) or not (0)
TYPE_NAME	VARCHAR2(2000)		Name of the type of the element
GLOBAL	RAW(1)		Indicates whether the attribute is global (1) or not (0)
ELEMENT	XMLTYPE		The actual XML fragment of the element
SQL_INLINE	RAW(1)		XDB annotation for sqlInline
SQL_TYPE	VARCHAR2(128)		XDB annotation value for sqlType

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SQL_SCHEMA	VARCHAR2(128)		XDB annotation value for sqlSchema
DEFAULT_TABLE	VARCHAR2(128)		XDB annotation value for default table
SQL_NAME	VARCHAR2(128)		XDB annotation value for sqlName
SQL_COL_TYPE	VARCHAR2(128)		XDB annotation value for sqlColType
MAINTAIN_DOM	RAW(1)		XDB annotation for maintainDOM
MAINTAIN_ORDER	RAW(1)		XDB annotation for maintainOrder
ELEMENT_ID	RAW(20)		Unique identifier for the element
PARENT_ELEMENT_ID	RAW(20)		Identies the parent of the element

See Also

- "DBA_XML_SCHEMA_ELEMENTS"
- "USER_XML_SCHEMA_ELEMENTS"
- *Oracle XML DB Developer's Guide* for information about registering an XML schema with Oracle XML DB
- *Oracle XML DB Developer's Guide* for information about restrictions for an XML schema URL

3.147 ALL_XML_SCHEMA_NAMESPACES

ALL_XML_SCHEMA_NAMESPACES describes all the available namespaces accessible to the current user.

Related Views

- DBA_XML_SCHEMA_NAMESPACES describes all the available namespaces.
- USER_XML_SCHEMA_NAMESPACES describes all the available namespaces owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		User who owns the namespace
TARGET_NAMESPACE	VARCHAR2(2000)		The target namespace
SCHEMA_URL	VARCHAR2(700)		The URL of the schema Refer to the **See Also** note below for links to more information about the schemaurl attribute for an XML schema.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_XML_SCHEMA_NAMESPACES"
- "USER_XML_SCHEMA_NAMESPACES"
- *Oracle XML DB Developer's Guide* for information about registering an XML schema with Oracle XML DB
- *Oracle XML DB Developer's Guide* for information about restrictions for an XML schema URL

3.148 ALL_XML_SCHEMA_SIMPLE_TYPES

`ALL_XML_SCHEMA_SIMPLE_TYPES` describes all simple types accessible to the current user.

Related Views

- `DBA_XML_SCHEMA_SIMPLE_TYPES` describes all simple types.
- `USER_XML_SCHEMA_SIMPLE_TYPES` describes all simple types owned by the current user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`		The user who owns the type
`SCHEMA_URL`	`VARCHAR2(700)`		The URL of the schema within which the type is defined Refer to the **See Also** note below for links to more information about the `schemaurl` attribute for an XML schema.
`TARGET_NAMESPACE`	`VARCHAR2(2000)`		The namespace of the type
`SIMPLE_TYPE_NAME`	`VARCHAR2(256)`		Name of the simple type
`SIMPLE_TYPE`	`XMLTYPE(XMLSchema "http://xmlns.oracle.com/xdb/XDBSchema.xsd" Element "simpleType")`		The actual XMLType of the type

See Also

- "DBA_XML_SCHEMA_SIMPLE_TYPES"
- "USER_XML_SCHEMA_SIMPLE_TYPES"
- *Oracle XML DB Developer's Guide* for information about registering an XML schema with Oracle XML DB
- *Oracle XML DB Developer's Guide* for information about restrictions for an XML schema URL

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.149 ALL_XML_SCHEMA_SUBSTGRP_HEAD

ALL_XML_SCHEMA_SUBSTGRP_HEAD describes the heads of substitution groups accessible to the current user.

Related Views

- DBA_XML_SCHEMA_SUBSTGRP_HEAD describes the heads of substitution groups.
- USER_XML_SCHEMA_SUBSTGRP_HEAD describes the heads of substitution groups owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		The user who owns the element
SCHEMA_URL	VARCHAR2(700)		The URL of the schema within which the element is defined Refer to the **See Also** note below for links to more information about the schemaurl attribute for an XML schema.
TARGET_NAMESPACE	VARCHAR2(2000)		The namespace of the element
ELEMENT_NAME	VARCHAR2(256)		Name of the element
ELEMENT	XMLTYPE(XMLSchema "http://xmlns.oracle.com/xdb/XDBSchema.xsd" Element "element")		The actual XML fragment of the element

See Also

- "DBA_XML_SCHEMA_SUBSTGRP_HEAD"
- "USER_XML_SCHEMA_SUBSTGRP_HEAD"
- *Oracle XML DB Developer's Guide* for information about registering an XML schema with Oracle XML DB
- *Oracle XML DB Developer's Guide* for information about restrictions for an XML schema URL

3.150 ALL_XML_SCHEMA_SUBSTGRP_MBRS

ALL_XML_SCHEMA_SUBSTGRP_MBRS describes all members of substitution groups accessible to the current user.

Related Views

- DBA_XML_SCHEMA_SUBSTGRP_MBRS describes all members of substitution groups.
- USER_XML_SCHEMA_SUBSTGRP_MBRS describes all members of substitution groups owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		The user who owns the element
SCHEMA_URL	VARCHAR2(700)		The URL of the schema within which the element is defined Refer to the **See Also** note below for links to more information about the schemaurl attribute for an XML schema.
TARGET_NAMESPACE	VARCHAR2(2000)		The namespace of the element
ELEMENT_NAME	VARCHAR2(256)		Name of the element
ELEMENT	XMLTYPE(XMLSchema "http://xmlns.oracle.com/xdb/XDBSchema.xsd" Element "element")		The actual XML fragment of the element
HEAD_OWNER	VARCHAR2(128)		The user who owns the head element for the current element
HEAD_SCHEMA_URL	VARCHAR2(700)		The URL of the schema within which the head element exists
HEAD_TARGET_NAMESPACE	VARCHAR2(2000)		The namespace of the head element
HEAD_ELEMENT_NAME	VARCHAR2(256)		Name of the head element
HEAD_ELEMENT	XMLTYPE(XMLSchema "http://xmlns.oracle.com/xdb/XDBSchema.xsd" Element "element")		The actual XMLType of the head element

See Also

- "DBA_XML_SCHEMA_SUBSTGRP_MBRS"
- "USER_XML_SCHEMA_SUBSTGRP_MBRS"
- *Oracle XML DB Developer's Guide* for information about registering an XML schema with Oracle XML DB
- *Oracle XML DB Developer's Guide* for information about restrictions for an XML schema URL

3.151 ALL_XML_SCHEMAS

ALL_XML_SCHEMAS describes the registered XML schemas accessible to the current user.

Related Views

- DBA_XML_SCHEMAS describes all registered XML schemas in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- USER_XML_SCHEMAS describes the registered XML schemas owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the XML schema
SCHEMA_URL	VARCHAR2(700)		Schema URL of the XML schema Refer to the **See Also** note below for links to more information about the schemaurl attribute for an XML schema.
LOCAL	VARCHAR2(3)		Indicates whether the XML schema is local (YES) or global (NO)
SCHEMA	XMLTYPE		XML schema document
INT_OBJNAME	VARCHAR2(4000)		Internal database object name for the schema
QUAL_SCHEMA_URL	VARCHAR2(865)		Fully qualified schema URL
HIER_TYPE	VARCHAR2(11)		Type of hierarchy for which the schema is enabled: • NONE • RESMETADATA • CONTENTS
BINARY	VARCHAR2(3)		Indicates whether the XML Schema is registered for binary encoding usage (YES) or not (NO)
SCHEMA_ID	RAW(16)		Opaque schema identifier (16 bytes)
HIDDEN	VARCHAR2(3)		Indicates whether the XML Schema has been deleted in hidden mode (YES) or not (NO)

See Also

- "DBA_XML_SCHEMAS"
- "USER_XML_SCHEMAS"
- *Oracle XML DB Developer's Guide* for information about registering an XML schema with Oracle XML DB
- *Oracle XML DB Developer's Guide* for information about restrictions for an XML schema URL

3.152 ALL_XML_TAB_COLS

ALL_XML_TAB_COLS describes the columns of the XML tables accessible to the current user.

Related Views

- DBA_XML_TAB_COLS describes the columns of all XML tables in the database.
- USER_XML_TAB_COLS describes the columns of the XML tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the XML table
TABLE_NAME	VARCHAR2(128)		Name of the XML table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COLUMN_NAME	VARCHAR2(4000)		Name of the XML table column
XMLSCHEMA	VARCHAR2(700)		Name of the XML Schema that is used for the table definition
SCHEMA_OWNER	VARCHAR2(128)		Owner of the XML Schema that is used for the table definition
ELEMENT_NAME	VARCHAR2(2000)		Name of the XML SChema element that is used for the table
STORAGE_TYPE	VARCHAR2(17)		Storage option for the XMLType data: • OBJECT-RELATIONAL • BINARY • CLOB **Note:** The CLOB storage option for XMLType data is deprecated in Oracle Database 12*c* Release 1 (12.1). Oracle recommends using the BINARY storage option, instead.
ANYSCHEMA	VARCHAR2(3)		If storage is BINARY, indicates whether the column allows ANYSCHEMA (YES) or not (NO), else NULL
NONSCHEMA	VARCHAR2(3)		If storage is BINARY, indicates whether the column allows NONSCHEMA (YES) or not (NO), else NULL
TOKENSETS	VARCHAR2(4000)		This column is for internal use only.

See Also

- "DBA_XML_TAB_COLS"
- "USER_XML_TAB_COLS"

3.153 ALL_XML_TABLES

ALL_XML_TABLES describes the XML tables accessible to the current user.

Related Views

- DBA_XML_TABLES describes all XML tables in the database.
- USER_XML_TABLES describes the XML tables owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the XML table
TABLE_NAME	VARCHAR2(128)		Name of the XML table
XMLSCHEMA	VARCHAR2(700)		Name of the XML Schema that is used for the table definition
SCHEMA_OWNER	VARCHAR2(128)		Owner of the XML Schema that is used for the table definition
ELEMENT_NAME	VARCHAR2(2000)		Name of the XML SChema element that is used for the table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STORAGE_TYPE	VARCHAR2(17)		Storage option for the XMLType data: • OBJECT-RELATIONAL • BINARY • CLOB **Note:** The CLOB storage option for XMLType data is deprecated in Oracle Database 12*c* Release 1 (12.1). Oracle recommends using the BINARY storage option, instead.
ANYSCHEMA	VARCHAR2(3)		If storage is BINARY, indicates whether the column allows ANYSCHEMA (YES) or not (NO), else NULL
NONSCHEMA	VARCHAR2(3)		If storage is BINARY, indicates whether the column allows NONSCHEMA (YES) or not (NO), else NULL
TOKENSETS	VARCHAR2(4000)		This column is for internal use only.

See Also

- "DBA_XML_TABLES"
- "USER_XML_TABLES"

3.154 ALL_XML_VIEW_COLS

ALL_XML_VIEW_COLS describes the columns of the XML views accessible to the current user.

Related Views

- DBA_XML_VIEW_COLS describes the columns of all XML views in the database.
- USER_XML_VIEW_COLS describes the columns of the XML views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the XML view
VIEW_NAME	VARCHAR2(128)		Name of the XML view
COLUMN_NAME	VARCHAR2(4000)		Name of the XML view column
XMLSCHEMA	VARCHAR2(700)		Name of the XML Schema that is used for the view definition
SCHEMA_OWNER	VARCHAR2(128)		Owner of the XML Schema that is used for the view definition
ELEMENT_NAME	VARCHAR2(2000)		Name of the XML SChema element that is used for the view

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_XML_VIEW_COLS"
- "USER_XML_VIEW_COLS"

3.155 ALL_XML_VIEWS

ALL_XML_VIEWS describes the XML views accessible to the current user.

Related Views

- DBA_XML_VIEWS describes all XML views the database.
- USER_XML_VIEWS describes the XML views owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the XML view
VIEW_NAME	VARCHAR2(128)		Name of the XML view
XMLSCHEMA	VARCHAR2(700)		Name of the XML Schema that is used for the view definition
SCHEMA_OWNER	VARCHAR2(128)		Owner of the XML Schema that is used for the view definition
ELEMENT_NAME	VARCHAR2(2000)		Name of the XML SChema element that is used for the view

See Also

- "DBA_XML_VIEWS"
- "USER_XML_VIEWS"

3.156 ALL_XSTREAM_ADMINISTRATOR

ALL_XSTREAM_ADMINISTRATOR displays information about the current users's granted privileges to be an XStream administrator by procedures in the DBMS_XSTREAM_AUTH package.

Related View

DBA_XSTREAM_ADMINISTRATOR displays information about the users who have been granted privileges to be XStream administrators by procedures in the DBMS_XSTREAM_AUTH package.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user who has been granted privileges to be an XStream administrator

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PRIVILEGE_TYPE	VARCHAR2(7)		Type of privilege granted: • APPLY • CAPTURE • * - Both APPLY and CAPTURE
GRANT_SELECT_PRIVILEGES	VARCHAR2(3)		Shows whether set of privileges granted to the user (grantee) includes the SELECT_CATALOG_ROLE role, which enables the user to manage other XStream servers that belong to other XStream users. Possible values: • YES: The administrator has the SELECT_CATALOG_ROLE role and other privileges, is considered a full privilege administrator, and can manage other users' XStream configuration • NO: The administrator is considered a minimum privilege administrator, and can only manage XStream configurations where the apply_user or capture_user (based on the PRIVILEGE_TYPE column) matches the username.
CREATE_TIME	TIMESTAMP(6)		Time at which the privilege was granted

See Also

- "DBA_XSTREAM_ADMINISTRATOR"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_XSTREAM_AUTH package

3.157 ALL_XSTREAM_INBOUND

ALL_XSTREAM_INBOUND displays information about the XStream inbound servers accessible to the current user.

Related View

DBA_XSTREAM_INBOUND displays information about all XStream inbound servers in the database.

Column	Datatype	NULL	Description
SERVER_NAME	VARCHAR2(128)	NOT NULL	Name of the inbound server
QUEUE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the queue associated with the inbound server
QUEUE_NAME	VARCHAR2(128)	NOT NULL	Name of the queue associated with the inbound server
APPLY_USER	VARCHAR2(128)		Name of the user who can connect to the inbound server and apply messages
USER_COMMENT	VARCHAR2(4000)		User comment
CREATE_DATE	TIMESTAMP(6)		Date when the inbound server was created

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(8)		Status of the inbound server: • DISABLED - The inbound server is not running. • DETACHED - The inbound server is running, but the XStream client application is not attached to it. • ATTACHED - The inbound server is running, and the XStream client application is attached to it. • ABORTED - The inbound server became disabled because it encountered an error.
COMMITTED_DATA_ONLY	VARCHAR2(3)		YES - means the inbound server can receive only LCRs in committed transactions from the XStream client application. A committed transaction is an assembled, noninterleaving transaction with no rollbacks.

See Also

"DBA_XSTREAM_INBOUND"

3.158 ALL_XSTREAM_INBOUND_PROGRESS

ALL_XSTREAM_INBOUND_PROGRESS displays information about the progress made by the XStream inbound servers accessible to the current user.

Related View

DBA_XSTREAM_INBOUND_PROGRESS displays information about the progress made by all XStream inbound servers in the database.

Column	Datatype	NULL	Description
SERVER_NAME	VARCHAR2(128)	NOT NULL	Name of the inbound server
PROCESSED_LOW_POSITION	RAW(64)		Position of the processed low transaction
APPLIED_LOW_POSITION	RAW(64)		All messages with a commit position less than this value have been applied
APPLIED_HIGH_POSITION	RAW(64)		Highest commit position of a transaction that has been applied
SPILL_POSITION	RAW(64)		Position of the spill low watermark of the transactions currently being applied
OLDEST_POSITION	RAW(64)		Earliest position of the transactions currently being applied
OLDEST_MESSAGE_NUMBER	NUMBER	NOT NULL	Earliest message number of the transactions currently being applied
APPLIED_MESSAGE_NUMBER	NUMBER	NOT NULL	Message number up to which all transactions have definitely been applied. This value is the low watermark for the inbound server. That is, messages with a commit message number less than or equal to this message number have definitely been applied, but some messages with a higher commit message number may also have been applied.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
APPLIED_TIME	DATE		Time at which the message with the message number displayed in the APPLIED_MESSAGE_NUMBER column was applied
APPLIED_MESSAGE_CREATE_TIME	DATE		Time at which the message with the message number displayed in the APPLIED_MESSAGE_NUMBER column was created at its source database
SPILL_MESSAGE_NUMBER	NUMBER		Spill low watermark. Any message with a lower SCN has either been applied or spilled to disk. The XStream client application does not need to send logical change records (LCRs) with a lower SCN than the spill low watermark. Spilled messages may not have been applied yet.
SOURCE_DATABASE	VARCHAR2(128)	NOT NULL	Database where the transaction originated
SOURCE_ROOT_NAME	VARCHAR2(128)		The global name of the source root database

See Also

"DBA_XSTREAM_INBOUND_PROGRESS"

3.159 ALL_XSTREAM_OUT_SUPPORT_MODE

ALL_XSTREAM_OUT_SUPPORT_MODE displays information about the level of XStream capture process support for the tables accessible to the current user in the database.

Related View

DBA_XSTREAM_OUT_SUPPORT_MODE displays information about the level of XStream capture process support for the tables in the database.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Table owner
OBJECT_NAME	VARCHAR2(128)		Table name
SUPPORT_MODE	VARCHAR2(6)		Capture process support level for the table: • FULL - A capture process can capture changes made to all of the columns in the table. • ID KEY - A capture process can capture changes made to the key columns and any other columns in the table that are supported by the capture process, except for LOB, LONG, LONG RAW, and XMLType columns. • NONE - A capture process cannot capture changes made to any columns in the table.

See Also

"DBA_XSTREAM_OUT_SUPPORT_MODE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.160 ALL_XSTREAM_OUTBOUND

ALL_XSTREAM_OUTBOUND displays information about the XStream outbound servers accessible to the current user.

Related View

DBA_XSTREAM_OUTBOUND displays information about all XStream outbound servers in the database.

Column	Datatype	NULL	Description
SERVER_NAME	VARCHAR2(128)	NOT NULL	Name of the outbound server
CONNECT_USER	VARCHAR2(128)		Name of the user who can connect to the outbound server and process the outbound LCRs
CAPTURE_NAME	VARCHAR2(128)		Name of the Replication capture process
SOURCE_DATABASE	VARCHAR2(128)		Database where the transaction originated
CAPTURE_USER	VARCHAR2(128)		Current user who is enqueuing captured messages
QUEUE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the queue associated with the outbound server
QUEUE_NAME	VARCHAR2(128)	NOT NULL	Name of the queue associated with the outbound server
USER_COMMENT	VARCHAR2(4000)		User comment
CREATE_DATE	TIMESTAMP(6)		Date when the outbound server was created
STATUS	VARCHAR2(8)		Status of the outbound server: • DISABLED - The outbound server is not running. • DETACHED - The outbound server is running, but the XStream client application is not attached to it. • ATTACHED - The outbound server is running, and the XStream client application is attached to it. • ABORTED - The outbound server became disabled because it encountered an error.
COMMITTED_DATA_ONLY	VARCHAR2(3)		YES if the outbound server can send only LCRs in committed transactions to the XStream client application. A committed transaction is an assembled, noninterleaving transaction with no rollbacks. NO if the outbound server can send LCRs in transactions that have not yet committed to the XStream client application. This mode is for internal Oracle use only.
START_SCN	NUMBER		The SCN from which the outbound server's capture process started capturing changes when it was last started
START_TIME	TIMESTAMP(6)		The time from which the outbound server's capture process started capturing changes when it was last started
SOURCE_ROOT_NAME	VARCHAR2(128)		The global name of the source root database
LCRID_VERSION	NUMBER		LCR ID format currently being used

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "DBA_XSTREAM_OUTBOUND"

3.161 ALL_XSTREAM_OUTBOUND_PROGRESS

ALL_XSTREAM_OUTBOUND_PROGRESS displays information about the progress made by the XStream outbound servers accessible to the current user.

Related View

DBA_XSTREAM_OUTBOUND_PROGRESS displays information about the progress made by all XStream outbound servers in the database.

Column	Datatype	NULL	Description
SERVER_NAME	VARCHAR2(128)	NOT NULL	Name of the outbound server
SOURCE_DATABASE	VARCHAR2(128)		Global name of the database where the transaction originated. For a PDB, this is the global name of the PDB.
PROCESSED_LOW_POSITION	RAW(64)		Position of the low-watermark transaction processed by the outbound server
PROCESSED_LOW_TIME	DATE		Time when the processed low position was last updated by the outbound server
OLDEST_POSITION	RAW(64)		The position of the earliest LCR that is required by the XStream client application
SOURCE_ROOT_NAME	VARCHAR2(128)		The global name of the source root database
PROCESSED_LOW_SCN	NUMBER	NOT NULL	SCN of the processed low transaction
OLDEST_SCN	NUMBER	NOT NULL	Oldest SCN of the transactions currently being captured

> **See Also**
>
> "DBA_XSTREAM_OUTBOUND_PROGRESS"

3.162 ALL_XSTREAM_RULES

ALL_XSTREAM_RULES displays information about the XStream rules accessible to the current user.

Related View

DBA_XSTREAM_RULES displays information about all XStream server rules in the database.

Column	Datatype	NULL	Description
STREAMS_NAME	VARCHAR2(128)		Name of the XStream process

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STREAMS_TYPE	VARCHAR2(12)		Type of the XStream process: • CAPTURE • APPLY
STREAMS_RULE_TYPE	VARCHAR2(9)		The XStream type of the rule: • TABLE • SCHEMA • GLOBAL
RULE_SET_OWNER	VARCHAR2(128)		Owner of the rule set
RULE_SET_NAME	VARCHAR2(128)		Name of the rule set
RULE_SET_TYPE	CHAR(8)		Type of the rule set: • POSITIVE • NEGATIVE
RULE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule
RULE_NAME	VARCHAR2(128)	NOT NULL	Name of the rule
RULE_TYPE	VARCHAR2(9)		The type of the rule: • DML • DDL
RULE_CONDITION	CLOB		Current rule condition
SCHEMA_NAME	VARCHAR2(128)		For table and schema rules, the schema name
OBJECT_NAME	VARCHAR2(128)		For table rules, the table name
INCLUDE_TAGGED_LCR	VARCHAR2(3)		Indicates whether to include tagged LCRs (YES) or not (NO)
SUBSETTING_OPERATION	VARCHAR2(6)		For subset rules, the type of operation: • INSERT • UPDATE • DELETE
DML_CONDITION	VARCHAR2(4000)		For subset rules, the row subsetting condition
SOURCE_DATABASE	VARCHAR2(128)		The global name of the database where the LCRs originated. In a PDB, this is the global name of the PDB.
ORIGINAL_RULE_CONDITION	VARCHAR2(4000)		For rules created by the XStream administrative APIs, the original rule condition when the rule was created
SAME_RULE_CONDITION	VARCHAR2(3)		For rules created by the XStream administrative APIs, indicates whether the current rule condition is the same as the original rule condition (YES) or not (NO)
SOURCE_ROOT_NAME	VARCHAR2(128)		The global name of the source root database
SOURCE_CONTAINER_NAME	VARCHAR2(128)		The container name of the database where the transactions originated

See Also

"DBA_XSTREAM_RULES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.163 ALL_XSTREAM_TRANSFORMATIONS

ALL_XSTREAM_TRANSFORMATIONS displays information about all XStream transformations accessible to the current user, in order of execution.

Related View

DBA_XSTREAM_TRANSFORMATIONS displays information about all XStream transformations available on a system, in order of execution.

Column	Datatype	NULL	Description
RULE_OWNER	VARCHAR2(128)		Owner of the rule which has an associated transformation
RULE_NAME	VARCHAR2(128)		Name of the rule which has an associated transformation
TRANSFORM_TYPE	VARCHAR2(26)		Type of the transformation: • DECLARATIVE TRANSFORMATION • SUBSET RULE • CUSTOM TRANSFORMATION
FROM_SCHEMA_NAME	VARCHAR2(128)		Schema to be renamed
TO_SCHEMA_NAME	VARCHAR2(128)		New schema name
FROM_TABLE_NAME	VARCHAR2(128)		Table to be renamed
TO_TABLE_NAME	VARCHAR2(128)		New table name
SCHEMA_NAME	VARCHAR2(128)		Schema of the column to be modified
TABLE_NAME	VARCHAR2(128)		Table of the column to be modified
FROM_COLUMN_NAME	VARCHAR2(4000)		Column to be renamed
TO_COLUMN_NAME	VARCHAR2(4000)		New column name
COLUMN_NAME	VARCHAR2(4000)		Column to add or delete
COLUMN_VALUE	ANYDATA		Value of the column to add
COLUMN_TYPE	VARCHAR2(4000)		Type of the new column
COLUMN_FUNCTION	VARCHAR2(128)		Name of the default function used to add a column
VALUE_TYPE	VARCHAR2(3)		Indicates whether to modify the old (OLD), new (NEW), or both (*) values of the LCR
USER_FUNCTION_NAME	VARCHAR2(4000)		Name of the user-defined transformation function to run
SUBSETTING_OPERATION	VARCHAR2(6)		DML operation for row subsetting: • INSERT • UPDATE • DELETE
DML_CONDITION	VARCHAR2(4000)		Row subsetting condition
DECLARATIVE_TYPE	VARCHAR2(13)		Type of declarative transform to run: • KEEP COLUMNS • DELETE COLUMN • RENAME COLUMN • ADD COLUMN • RENAME TABLE • RENAME SCHEMA

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PRECEDENCE	NUMBER		Execution order relative to other declarative transformations on the same STEP_NUMBER
STEP_NUMBER	NUMBER		Order in which this transformation should be executed

See Also

"DBA_XSTREAM_TRANSFORMATIONS"

3.164 ALL_XTERNAL_LOC_PARTITIONS

ALL_XTERNAL_LOC_PARTITIONS describes partition-level locations accessible to the current user.

If an external table is partitioned, then the existing ALL_EXTERNAL_LOCATIONS, DBA_EXTERNAL_LOCATIONS, and USER_EXTERNAL_LOCATIONS views will have no rows for that table. Instead, locations will be indicated in the ALL_XTERNAL_LOC_PARTITIONS, DBA_XTERNAL_LOC_PARTITIONS, USER_XTERNAL_LOC_PARTITIONS, ALL_XTERNAL_LOC_SUBPARTITIONS, DBA_XTERNAL_LOC_SUBPARTITIONS, and USER_XTERNAL_LOC_SUBPARTITIONS views.

Related Views

- DBA_XTERNAL_LOC_PARTITIONS describes partition-level locations in the database.
- USER_XTERNAL_LOC_PARTITIONS describes partition-level locations owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the partitioned external table
TABLE_NAME	VARCHAR2(128)		Name of the partitioned external table
PARTITION_NAME	VARCHAR2(128)		Name of the partition
LOCATION	VARCHAR2(4000)		External table location clause for the partition
DIRECTORY_OWNER	CHAR(3)		Owner of the directory containing the external table partition location
DIRECTORY_NAME	VARCHAR2(128)		Name of the directory containing the external table partition location

See Also

- "DBA_XTERNAL_LOC_PARTITIONS"
- "USER_XTERNAL_LOC_PARTITIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.165 ALL_XTERNAL_LOC_SUBPARTITIONS

ALL_XTERNAL_LOC_SUBPARTITIONS describes subpartition-level locations accessible to the current user.

If an external table is partitioned, then the existing ALL_EXTERNAL_LOCATIONS, DBA_EXTERNAL_LOCATIONS, and USER_EXTERNAL_LOCATIONS views will have no rows for that table. Instead, locations will be indicated in the ALL_XTERNAL_LOC_PARTITIONS, DBA_XTERNAL_LOC_PARTITIONS, USER_XTERNAL_LOC_PARTITIONS, ALL_XTERNAL_LOC_SUBPARTITIONS, DBA_XTERNAL_LOC_SUBPARTITIONS, and USER_XTERNAL_LOC_SUBPARTITIONS views.

Related Views

- DBA_XTERNAL_LOC_SUBPARTITIONS describes subpartition-level locations in the database.
- USER_XTERNAL_LOC_SUBPARTITIONS describes subpartition-level locations owned by the current user. This view does not display the TABLE_OWNER column.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the partitioned external table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the partitioned external table
PARTITION_NAME	VARCHAR2(128)		Name of the partition
SUBPARTITION_NAME	VARCHAR2(128)		Name of the subpartition
LOCATION	VARCHAR2(4000)		External table location clause for the subpartition
DIRECTORY_OWNER	CHAR(3)		Owner of the directory containing the external table subpartition location
DIRECTORY_NAME	VARCHAR2(128)		Name of the directory containing the external table subpartition location

See Also

- "DBA_XTERNAL_LOC_SUBPARTITIONS"
- "USER_XTERNAL_LOC_SUBPARTITIONS"

3.166 ALL_XTERNAL_PART_TABLES

ALL_XTERNAL_PART_TABLES describes object-level information for partitioned external tables accessible to the current user.

Related Views

- DBA_XTERNAL_PART_TABLES describes object-level information for partitioned external tables in the database
- USER_XTERNAL_PART_TABLES describes object-level information for partitioned external tables owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the partitioned external table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the partitioned external table
TYPE_OWNER	CHAR(3)		Owner of the implementation type for the external table access driver
TYPE_NAME	VARCHAR2(128)		Name of the implementation type for the external table access driver
DEFAULT_DIRECTORY_OWNER	CHAR(3)		Owner of the default directory for the external table
DEFAULT_DIRECTORY_NAME	VARCHAR2(128)		Name of the default directory for the external table
REJECT_LIMIT	VARCHAR2(40)		Reject limit for the external table, or UNLIMITED
ACCESS_TYPE	VARCHAR2(7)		Type of access parameters for the external table (BLOB, CLOB)
ACCESS_PARAMETERS	CLOB		Access parameters for the external table
PROPERTY	VARCHAR2(10)		Property of the external table (REFERENCED, ALL)

See Also

- "DBA_XTERNAL_PART_TABLES"
- "USER_XTERNAL_PART_TABLES"

3.167 ALL_XTERNAL_TAB_PARTITIONS

ALL_XTERNAL_TAB_PARTITIONS describes partition-level information for partitioned external tables accessible to the current user.

Related Views

- DBA_XTERNAL_TAB_PARTITIONS describes partition-level information for partitioned external tables in the databae
- USER_XTERNAL_TAB_PARTITIONS describes partition-level information for partitioned external tables owned by the current user. This view does not display the TABLE_OWNER column.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)		Owner of the partitioned external table
TABLE_NAME	VARCHAR2(128)		Name of the partitioned external table
PARTITION_NAME	VARCHAR2(128)		Name of the partition
DEFAULT_DIRECTORY_OWNER	CHAR(3)		Owner of the default directory for the external table partition
DEFAULT_DIRECTORY_NAME	VARCHAR2(128)		Name of the default directory for the external table partition
ACCESS_TYPE	VARCHAR2(7)		Type of access parameters for the partition (BLOB, CLOB)
ACCESS_PARAMETERS	CLOB		Access parameters for the external table partition

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_XTERNAL_TAB_PARTITIONS"
- "USER_XTERNAL_TAB_PARTITIONS"

3.168 ALL_XTERNAL_TAB_SUBPARTITIONS

ALL_XTERNAL_TAB_SUBPARTITIONS describes subpartition-level information for partitioned external tables accessible to the current user.

Related Views

- DBA_XTERNAL_TAB_SUBPARTITIONS describes subpartition-level information for partitioned external tables in the database.
- USER_XTERNAL_TAB_SUBPARTITIONS describes subpartition-level information for partitioned external tables owned by the current user. This view does not display the TABLE_OWNER column.

Column	Datatype	NULL	Description
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the partitioned external table
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the partitioned external table
PARTITION_NAME	VARCHAR2(128)		Name of the partition
SUBPARTITION_NAME	VARCHAR2(128)		Name of the subpartition
DEFAULT_DIRECTORY_OWNER	CHAR(3)		Owner of the default directory for the external table partition
DEFAULT_DIRECTORY_NAME	VARCHAR2(128)		Name of the default directory for the external table partition
ACCESS_TYPE	VARCHAR2(7)		Type of access parameters for the partition (BLOB, CLOB)
ACCESS_PARAMETERS	CLOB		Access parameters for the external table partition

See Also

- "DBA_XTERNAL_TAB_SUBPARTITIONS"
- "USER_XTERNAL_TAB_SUBPARTITIONS"

3.169 ALL_ZONEMAP_MEASURES

ALL_ZONEMAP_MEASURES describes the measures for all zone maps accessible to the user.

Related Views

- DBA_ZONEMAP_MEASURES describes the measures for all the zone maps in the database.
- USER_ZONEMAP_MEASURES describes the measures for all the zone maps owned by the user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the zone map
ZONEMAP_NAME	VARCHAR2(128)	NOT NULL	Name of the zone map
MEASURE	LONG		Column whose MIN/MAX value is computed
POSITION_IN_SELECT	NUMBER	NOT NULL	Original position of the mesaure aggregate on the SELECT list of zone map defining query
AGG_FUNCTION	VARCHAR2(13)		Name of aggregate in zone map table
AGG_COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column whose MIN/MAX per zone maintained

Note

This view is intended for use with Oracle Exadata release 12.1.2.1.1 or later.

See Also

- "DBA_ZONEMAP_MEASURES"
- "USER_ZONEMAP_MEASURES"
- *Oracle Database Data Warehousing Guide* for more information about zone maps

3.170 ALL_ZONEMAPS

ALL_ZONEMAPS describes all the zone maps accessible to the user.

Related Views

- DBA_ZONEMAPS describes all the zone maps in the database.
- USER_ZONEMAPS describes all the zone maps owned by the user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the zone map
ZONEMAP_NAME	VARCHAR2(128)		Name of the zone map
FACT_OWNER	VARCHAR2(128)		Owner of the fact table of the zone map
FACT_TABLE	VARCHAR2(128)		Name of the fact table on which the zone map is defined
SCALE	NUMBER		Scale factor of the zone map
HIERARCHICAL	VARCHAR2(12)		Indicates whether the zone map is hierarchical (YES) or not (NO)
WITH_CLUSTERING	VARCHAR2(15)		Indicates whether the zone map is created with the CLUSTERING clause (YES) or not (NO)
QUERY	LONG		Zone map defining query
QUERY_LEN	NUMBER(38)		Length of defining query in bytes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PRUNING	VARCHAR2(8)		Indicates whether the zone map is enabled for pruning (ENABLED) or not (DISABLED)
REFRESH_MODE	VARCHAR2(17)		Refresh mode for the zone map: • COMMIT • DEMAND • DATAMOVEMENT • LOAD • LOAD DATAMOVEMENT
REFRESH_METHOD	VARCHAR2(14)		Refresh method for the zone map • COMPLETE • FORCE • FAST
LAST_REFRESH_METHOD	VARCHAR2(19)		The last refresh method used for the zone map: • NA • COMPLETE • FAST • ERROR-UNKNOWN
LAST_REFRESH_TIME	TIMESTAMP(9)		Time of the last refresh
INVALID	VARCHAR2(7)		Indicates whether the zone map is invalid due to some DDL (YES) or not (NO)
STALE	VARCHAR2(7)		Indicates whether the zone map is stale because of DML operations and cannot be used for pruning (YES) or not (NO) or whether this cannot be determined (UNKNOWN)
UNUSABLE	VARCHAR2(8)		Indicates whether the zone map has been marked unusable by the owner (YES) or not (NO)
COMPILE_STATE	VARCHAR2(19)		Current compile state of the zone map: • VALID • AUTHORIZATION_ERROR • COMPILATION_ERROR • NEEDS_COMPILE • ERROR_UNKNOWN Similar to ALL_MVIEWS.COMPILE_STATE.

Note

This view is intended for use with Oracle Exadata release 12.1.2.1.1 or later.

See Also

- "DBA_ZONEMAPS"
- "USER_ZONEMAPS"
- *Oracle Database Data Warehousing Guide* for more information about zone maps

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.171 AUDIT_ACTIONS

AUDIT_ACTIONS describes audit trail action type codes. This table can be used to map action type numbers to action type names.

> **Note**
>
> The mapping explained in this view is valid for audit trail records from the following views only, and such audit records are generated only when unified auditing is not enabled:
>
> - DBA_AUDIT_TRAIL
> - DBA_COMMON_AUDIT_TRAIL
> - DBA_FGA_AUDIT_TRAIL
> - USER_AUDIT_TRAIL
> - V$XML_AUDIT_TRAIL

> **See Also**
>
> - *Oracle Database Security Guide* for more information about unified auditing.
> - *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
ACTION	NUMBER	NOT NULL	Numeric audit trail action type code.
NAME	VARCHAR2(28)	NOT NULL	Name of the type of audit trail action

3.172 AUDIT_UNIFIED_CONTEXTS

AUDIT_UNIFIED_CONTEXTS describes the application context's attributes, which are configured to be captured in the audit trail.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is enabled.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
NAMESPACE	VARCHAR2(128)		Application context namespace

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ATTRIBUTE	VARCHAR2(128)		Application context attribute
USER_NAME	VARCHAR2(128)		Username of database user for whom the application context's attribute is confiured to be captured in the audit trail

3.173 AUDIT_UNIFIED_ENABLED_POLICIES

AUDIT_UNIFIED_ENABLED_POLICIES describes all the audit policies that are enabled in the database.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is enabled.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
POLICY_NAME	VARCHAR2(128)		Name of the audit policy
ENABLED_OPTION	VARCHAR2(15)		Enabled option of the audit policy. Possible values: • BY USER: For policies that are enabled on users • EXCEPT USER: For policies that are enabled on users • BY GRANTED ROLE: For policies that are enabled on roles • INVALID: For policies that are not enabled on either users or roles
ENTITY_NAME	VARCHAR2(128)		Database entity (user name or role name) on which the audit policy is enabled. When an audit policy is enabled on all database users, ALL USERS is displayed in this column.
ENTITY_TYPE	VARCHAR2(7)		Database entity type. Possible values: • USER: Indicates that the policy is enabled on a user or users. • ROLE: Indicates that the policy is enabled on a role or roles.
SUCCESS	VARCHAR2(3)		Indicates whether the audit policy is enabled for auditing successful events (YES) or not (NO)
FAILURE	VARCHAR2(3)		Indicates whether the audit policy is enabled for auditing unsuccessful events (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.174 AUDIT_UNIFIED_POLICIES

AUDIT_UNIFIED_POLICIES describes all audit policies created in the database.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is enabled.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
POLICY_NAME	VARCHAR2(128)		Name of the audit policy
AUDIT_CONDITION	VARCHAR2(4000)		Condition associated with the audit policy
CONDITION_EVAL_OPT	VARCHAR2(9)		Evaluation option associated with the audit policy's condition. The possible values are STATEMENT, SESSION, INSTANCE, and NONE.
AUDIT_OPTION	VARCHAR2(128)		Auditing option defined in the audit policy
AUDIT_OPTION_TYPE	VARCHAR2(18)		Type of the auditing option. Possible values: • SYSTEM PRIVILEGE • STANDARD ACTION • SYSTEM ACTION • XS ACTION • OLS_ACTION • DATAPUMP ACTION • DIRECT LOAD ACTION • DV ACTION • INVALID • OBJECT ACTION • ROLE PRIVILEGE
OBJECT_SCHEMA	VARCHAR2(128)		Owner of the object, for an object-specific auditing option
OBJECT_NAME	VARCHAR2(128)		Name of the object, for an object-specific auditing option
OBJECT_TYPE	VARCHAR2(23)		Type of the object, for an object-specific auditing option

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COMMON	VARCHAR2(3)		Indicates whether the audit policy is a common audit policy or local audit policy. The value is NULL for a non-CDB. For local audit policies, the value of the COMMON column is always NO. For a CDB common policy: • If you query AUDIT_UNIFIED_POLICIES from the CDB root container, the value of the COMMON column will be YES and the value of the INHERITED column will be NO. • If you query AUDIT_UNIFIED_POLICIES from any other container besides the CDB root container, the value of the COMMON column and the INHERITED column will be YES. For an application container common policy: • If you query AUDIT_UNIFIED_POLICIES from the application root container, the value of the COMMON column will be YES and the value of the INHERITED column will be NO. • If you query AUDIT_UNIFIED_POLICIES from any other container besides the application root container, the value of the COMMON column and the INHERITED column will be YES.
INHERITED	VARCHAR2(3)		Indicates whether the audit policy was inherited from another container (YES) or not (NO). This value is NULL for non-CDBs.
AUDIT_ONLY_TOPLEVEL[1]	VARCHAR2(3)		Indicates whether the audit policy is defined to audit only top level SQL statements (YES) or both top level SQL statements and recursive SQL statements (NO)
ORACLE_SUPPLIED[2]	VARCHAR2(3)		Indicates whether the audit policy is an Oracle-supplied policy (YES) or not (NO) Oracle-supplied policies are also called predefined policies.

1 This column is available starting with Oracle Database 19c.

2 This column is available starting with Oracle Database 19c, Release Update 19.11.

3.175 AUDIT_UNIFIED_POLICY_COMMENTS

AUDIT_UNIFIED_POLICY_COMMENTS shows the description of each unified audit policy, if a description was entered for the unified audit policy using the COMMENT SQL statement.

Note

This view is populated only in an Oracle Database where unified auditing is enabled.

- See *Oracle Database Security Guide* for more information about unified auditing.
- See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
POLICY_NAME	VARCHAR2(128)	NOT NULL	Name of the unified audit policy
COMMENTS	VARCHAR2(4000)		Description of the unified audit policy, if one was entered using the COMMENT SQL statement

3.176 AUDITABLE_SYSTEM_ACTIONS

AUDITABLE_SYSTEM_ACTIONS maps the auditable system action numbers to the action names. These actions are configurable for audit when unified auditing is enabled.

The actions include:

- All standard RDBMS actions (from the V$SQLCOMMAND view) except the following, which are not configurable for auditing:
  - ALTER EDITION
  - ALTER REWRITE EQUIVALENCE
  - ALTER SUMMARY
  - ALTER TRACING
  - CREATE BITMAPFILE
  - CREATE CONTROL FILE
  - CREATE DATABASE
  - CREATE SUMMARY
  - DECLARE REWRITE EQUIVALENCE
  - DROP BITMAPFILE
  - DROP DATABASE
  - DROP REWRITE EQUIVALENCE
  - DROP SUMMARY
  - FLASHBACK DATABASE
  - MERGE
  - SAVEPOINT
  - SET CONSTRAINTS
  - UNDROP OBJECT
  - UPDATE INDEXES
  - UPDATE JOIN INDEX
  - VALIDATE INDEX
- Other actions:
  - ALL
  - LOGON
  - LOGOFF

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is enabled.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
TYPE	NUMBER		Numeric component type for system wide actions
COMPONENT	VARCHAR2(64)		Name of component for system wide actions
ACTION	NUMBER		Numeric auditable action code for system wide actions
NAME	VARCHAR2(64)		Name of auditable action

Some auditable actions in AUDITABLE_SYSTEM_ACTIONS have different names than their equivalent commands in V$SQLCOMMAND, as shown in the following table:

AUDITABLE_SYSTEM_ACTIONS Action Name	V$SQLCOMMAND Command Name
GRANT	GRANT OBJECT
REVOKE	REVOKE OBJECT
AUDIT	AUDIT OBJECT
NOAUDIT	NOAUDIT OBJECT
EXECUTE	PL/SQL EXECUTE
EXPLAIN PLAN	EXPLAIN
CALL	CALL METHOD
PURGE DBA_RECYCLEBIN	PURGE DBA RECYCLEBIN

> **See Also**
>
> "V$SQLCOMMAND"

3.177 CAT

CAT is a synonym for USER_CATALOG.

> **See Also**
>
> "USER_CATALOG"

3.178 CATALOG

CATALOG is included for compatibility. Oracle recommends that you not use this view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

3.179 CHAINED_ROWS

CHAINED_ROWS stores the output for the ANALYZE statement with the LIST CHAINED ROWS clause.

You must run the utlchain.sql or utlchn1.sql script to create this table.

Column	Description
OWNER_NAME	Table owner
TABLE_NAME	Table name
CLUSTER_NAME	Cluster the table is in, if any
PARTITION_NAME	The name of the partition
SUBPARTITION_NAME	The name of the subpartition
HEAD_ROWID	ROWID the chained row is accessed by
ANALYZE_TIMESTAMP	Date/time that the ANALYZE statement was issued

3.180 CLIENT_RESULT_CACHE_STATS$

CLIENT_RESULT_CACHE_STATS$ displays various Client Result Cache settings and usage statistics.

Statistics are stored as name, value pairs. For each client cache ID, there will be multiple rows for each statistic.

Column	Datatype	NULL	Description
CACHE_ID	NUMBER	NOT NULL	Unique ID per client cache
STAT_ID	NUMBER	NOT NULL	Statistic ID
NAME	VARCHAR2(128)		Name of the statistic (see Table 3-1)
VALUE	NUMBER		Value of the statistic

Table 3-1 CLIENT_RESULT_CACHE_STATS$ Statistics

Statistic Name	Description
Block Size	Size (in bytes) of each memory block in the result cache.
Block Count Max	Maximum number of blocks that can be allocated in the result cache based on the cache size configuration parameters on server and on client.
Block Count Current	Current number of blocks allocated by the client result cache.
Hash Bucket Count	Size of the hash table used for query matching.
Create Count Success	Number of cached result sets that did not get invalidated before caching all the rows of the result set.
Create Count Failure	Number of cached result sets that did not fetch all the rows in the result set.
Find Count	Number of cache hits.
Invalidation Count	Number of cached result sets that got invalidated due to database changes that could have affected the result set.
Delete Count Invalid	Number of cached result rests not invalidated whose memory was reclaimed by result cache.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 3-1 (Cont.) CLIENT_RESULT_CACHE_STATS$ Statistics

Statistic Name	Description
Delete Count Valid	Number of invalidated cached result rests whose memory was reclaimed by result cache.

3.181 CLU

`CLU` is a synonym for `USER_CLUSTERS`.

> **See Also**
>
> "USER_CLUSTERS"

3.182 COL

`COL` is included for compatibility. Oracle recommends that you not use this view.

3.183 COLS

`COLS` is a synonym for `USER_TAB_COLUMNS`.

> **See Also**
>
> "USER_TAB_COLUMNS"

3.184 DATABASE_EXPORT_OBJECTS

`DATABASE_EXPORT_OBJECTS` lists simple path names for some of the object types belonging to a full Data Pump export, which is invoked using the `FULL=Y` parameter on the `expdp` command.

Users of the Data Pump Export and Import utilities can query this view to determine valid values for the `EXCLUDE` and `INCLUDE` parameters.

Column	Datatype	NULL	Description
`OBJECT_PATH`	`VARCHAR2(200)`	`NOT NULL`	Simple path name for the object type
`COMMENTS`	`VARCHAR2(2000)`		Comment on the object type
`NAMED`	`CHAR(1)`		Do objects of this type have names? If yes (`Y`), then the name can be specified in the optional name_clause on the `EXCLUDE` and `INCLUDE` parameters.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "SCHEMA_EXPORT_OBJECTS"
- "TABLE_EXPORT_OBJECTS"
- *Oracle Database Utilities* for more information on performing a full Data Pump export using the `expdp` command

3.185 DATABASE_PROPERTIES

`DATABASE_PROPERTIES` lists permanent database properties.

Column	Datatype	NULL	Description
`PROPERTY_NAME`	`VARCHAR2(128)`	`NOT NULL`	Property name
`PROPERTY_VALUE`	`VARCHAR2(4000)`		Property value
`DESCRIPTION`	`VARCHAR2(4000)`		Property description

Note

The `CDB_PROPERTIES` view provides access to data visible to PDBs through the `DATABASE_PROPERTIES` view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

4

Static Data Dictionary Views: DBA_2PC_NEIGHBORS to DBA_HIST_JAVA_POOL_ADVICE

This chapter contains the static data dictionary views DBA_2PC_NEIGHBORS through DBA_HIST_JAVA_POOL_ADVICE.

4.1 DBA_2PC_NEIGHBORS

DBA_2PC_NEIGHBORS describes incoming and outgoing connections for pending transactions.

Column	Datatype	NULL	Description
LOCAL_TRAN_ID	VARCHAR2(22)		Local identifier of a transaction
IN_OUT	VARCHAR2(3)		IN for incoming connections, OUT for outgoing
DATABASE	VARCHAR2(128)		IN for client database name, OUT for outgoing database link
DBUSER_OWNER	VARCHAR2(128)		IN for name of local user, OUT for owner of database link
INTERFACE	VARCHAR2(1)		C for request commit, otherwise N for prepare or request read only commit
DBID	VARCHAR2(16)		Database ID at the other end of the connection
SESS#	NUMBER(38)		Session number of the connection at this database
BRANCH	VARCHAR2(128)		Transaction branch ID of the connection at this database

4.2 DBA_2PC_PENDING

DBA_2PC_PENDING describes distributed transactions awaiting recovery.

Column	Datatype	NULL	Description
LOCAL_TRAN_ID	VARCHAR2(22)	NOT NULL	String of form: n.n.n; n is a number
GLOBAL_TRAN_ID	VARCHAR2(169)		Globally unique transaction ID
STATE	VARCHAR2(16)	NOT NULL	Collecting, prepared, committed, forced commit, or forced rollback
MIXED	VARCHAR2(3)		YES indicates part of the transaction committed and part rolled back
ADVICE	VARCHAR2(1)		C for commit, R for rollback, else NULL
TRAN_COMMENT	VARCHAR2(255)		Text for commit work comment text
FAIL_TIME	DATE	NOT NULL	Value of SYSDATE when the row was inserted (transaction or system recovery)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FORCE_TIME	DATE		Time of manual force decision (null if not forced locally)
RETRY_TIME	DATE	NOT NULL	Time automatic recovery (RECO) last tried to recover the transaction
OS_USER	VARCHAR2(64)		Operating system-specific name for the end-user
OS_TERMINAL	VARCHAR2(255)		Operating system-specific name for the end-user terminal
HOST	VARCHAR2(128)		Name of the host system for the end-user
DB_USER	VARCHAR2(128)		Oracle user name of the end-user at the topmost database
COMMIT#	VARCHAR2(16)		Global commit number for committed transactions

4.3 DBA_ACCHK_EVENTS

DBA_ACCHK_EVENTS displays information about events that occurred during an Application Continuity Protection Check (ACCHK) workload run.

Each row in this view represents one trace record for an event. You can use this view in conjunction with the DBA_ACCHK_STATISTICS view. Join the SESSION_ID and SERIAL# columns in this view with the SESSION_ID and SERIAL# columns in DBA_ACCHK_STATISTICS to view Application Continuity protection statistics for a particular session.

The ACCHK_READ role allows users with no administrative privileges to query this view.

Column	Datatype	NULL	Description
INST_ID	NUMBER	NOT NULL	Identifier for the instance in which the trace record was generated
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
CON_UID[1]	NUMBER	NOT NULL	Unique ID of the container used by the session that generated the trace record
TIMESTAMP	TIMESTAMP(3) WITH TIME ZONE		Time at which the event occurred
SESSION_ID	NUMBER	NOT NULL	ID of the session that generated the trace record
SERIAL#	NUMBER	NOT NULL	Serial number of the session that generated the trace record
SERVICE_NAME	VARCHAR2(64)		Service name of the session that generated the trace record
PROGRAM	VARCHAR2(84)		Name of the operating system program that generated the trace record
MODULE	VARCHAR2(64)		Name of the module that generated the trace record

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ACTION	VARCHAR2(64)		Name of the action that generated the trace record
SQL_ID	VARCHAR2(13)		SQL identifier of the SQL statement that generated the trace record
CALL_NAME	VARCHAR2(20)		Name of the user call that generated the trace record
EVENT_TYPE	VARCHAR2(16)		Event type. Possible values: • DISABLED - The event caused capture or replay to be disabled. Refer to the ERROR_CODE column for more information. • NEVER ENABLED - Neither Application Continuity nor Transparent Application Continuity was enabled for the session when the event occurred. • NOT ENABLING - The event describes why the database could not reenable protection after protection was disabled. Refer to the ERROR_CODE column for more information. • REPLAY FAILED - The event describes why the session did not fail over. This event only occurs if Application Continuity could not fail over. Refer to the ERROR_CODE column for more information.
ERROR_CODE	NUMBER		If an error occurred, this column displays the error code: ORA-*number*. Otherwise, the value of this column is 0.
CON_ID_EXT[1]	NUMBER		Container ID used by the session for external recognition
INFO[1]	VARCHAR2(256)		Information or hint associated with the event

1 This column is available starting with Oracle Database 19c, Release Update 19.19.

Note

This view is available starting with Oracle Database 19c, Release Update 19.11.

For Oracle Database 19c, Release Update 19.11 through Release Update 19.18, this view is not created by default. To create this view, you must run the DBMS_APP_CONT_ADMIN.ACCHK_VIEWS procedure. See *Oracle Database PL/SQL Packages and Types Reference* for more information.

Starting with Oracle Database 19c, Release Update 19.19, this view is created by default.

See Also

"DBA_ACCHK_STATISTICS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.4 DBA_ACCHK_EVENTS_SUMMARY

DBA_ACCHK_EVENTS_SUMMARY displays summary information about events that occurred during an Application Continuity Protection Check (ACCHK) workload run.

This view describes the number of times a particular type of event occurred in a session. You can use this view in conjunction with the DBA_ACCHK_STATISTICS_SUMMARY view. Join the INST_ID and CON_ID columns in this view with the INST_ID and CON_ID columns in DBA_ACCHK_STATISTICS_SUMMARY to view Application Continuity protection statistics for a particular instance.

The ACCHK_READ role allows users with no administrative privileges to query this view.

Column	Datatype	NULL	Description
INST_ID	NUMBER	NOT NULL	Identifier for the instance
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
CON_UID[1]	NUMBER	NOT NULL	Unique ID of the container used by the session
CON_ID_EXT[1]	NUMBER		Container ID used by the session for external recognition
SERVICE_NAME	VARCHAR2(64)		Service name of the session
FAILOVER_TYPE	VARCHAR2(16)		Indicates the FAILOVER_TYPE service setting for the session. Possible values: • AUTO - Transparent Application Continuity (TAC) was enabled for the session • TRANSACTION - Application Continuity (AC) was enabled for the session • NONE - Failover was disabled for the session
FAILOVER_RESTORE	VARCHAR2(16)		Indicates the FAILOVER_RESTORE service setting for the session. Possible values: • AUTO - Transparent Application Continuity • LEVEL1 - Application Continuity • NONE - Session state was not restored at failover
RESET_STATE	VARCHAR2(16)		Indicates the RESET_STATE service setting for the session. Possible values: • LEVEL1 - The session state was cleaned at the end of a request • NONE - The session state was not cleaned at the end of a request
PROGRAM	VARCHAR2(84)		Name of the operating system program that generated the trace record
MODULE	VARCHAR2(64)		Name of the module that generated the trace record
ACTION	VARCHAR2(64)		Name of the action that generated the trace record

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SQL_ID	VARCHAR2(13)		SQL identifier of the SQL statement that generated the trace record
CALL_NAME	VARCHAR2(20)		Name of the user call that generated the trace record
EVENT_TYPE	VARCHAR2(16)		Event type. Possible values: • DISABLED - The event caused capture or replay to be disabled. Refer to the ERROR_CODE column for more information. • NEVER ENABLED - Neither Application Continuity nor Transparent Application Continuity was enabled for the session when the event occurred. • NOT ENABLING - The event describes why the database could not reenable protection after protection was disabled. Refer to the ERROR_CODE column for more information. • REPLAY FAILED - The event describes why the session did not fail over. This event only occurs if Application Continuity could not fail over. Refer to the ERROR_CODE column for more information.
ERROR_CODE	NUMBER		If an error occurred, this column displays the error code: ORA-*number*. Otherwise, the value of this column is 0.
FREQUENCY	NUMBER		Number of times the event occurred during the workload run
INFO[1]	VARCHAR2(256)		Information or hint associated with the event

1 This column is available starting with Oracle Database 19c, Release Update 19.19.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.11.
>
> For Oracle Database 19c, Release Update 19.11 through Release Update 19.18, this view is not created by default. To create this view, you must run the DBMS_APP_CONT_ADMIN.ACCHK_VIEWS procedure. See *Oracle Database PL/SQL Packages and Types Reference* for more information.
>
> Starting with Oracle Database 19c, Release Update 19.19, this view is created by default.

4.5 DBA_ACCHK_STATISTICS

DBA_ACCHK_STATISTICS displays Application Continuity protection statistics for each session that executed during an Application Continuity Protection Check (ACCHK) workload run.

ACCHK should be used on a running workload. Sessions that are terminated are not included in this view.

The ACCHK_READ role allows users with no administrative privileges to query this view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INST_ID	NUMBER	NOT NULL	Identifier for the instance for the session
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
CON_UID[1]	NUMBER	NOT NULL	Unique ID of the container used by the session
TIMESTAMP	TIMESTAMP(3) WITH TIME ZONE		Time at which the session was terminated and the statistics were reported
SESSION_ID	NUMBER	NOT NULL	ID of the session
SERIAL#	NUMBER	NOT NULL	Serial number of the session
STAT_TYPE	VARCHAR2(32)		Type of statistic The value of this column is always SESSION_STATISTICS.
SERVICE_NAME	VARCHAR2(64)		Service name of the session
FAILOVER_TYPE	VARCHAR2(16)		Indicates the FAILOVER_TYPE service setting for the session. Possible values: • AUTO - Transparent Application Continuity (TAC) was enabled for the session • TRANSACTION - Application Continuity (AC) was enabled for the session • NONE - Failover was disabled for the session
FAILOVER_RESTORE	VARCHAR2(16)		Indicates the FAILOVER_RESTORE service setting for the session. Possible values: • AUTO - Transparent Application Continuity • LEVEL1 - Application Continuity • NONE - Session state was not restored at failover
RESET_STATE	VARCHAR2(16)		Indicates the RESET_STATE service setting for the session. Possible values: • LEVEL1 - The session state was cleaned at the end of a request • NONE - The session state was not cleaned at the end of a request
PROGRAM	VARCHAR2(84)		Name of the operating system program
BEGIN_REQUESTS	NUMBER		Number of begin requests received for the session
END_REQUESTS	NUMBER		Number of end requests received for the session
USER_CALLS_IN_REQUESTS	NUMBER		Number of user calls received from the application within requests (between begin request and end request)
PROTECTED_CALLS_IN_REQUESTS	NUMBER		Number of user calls protected by Application Continuity within requests (between begin request and end request)
TIME_IN_REQUESTS	NUMBER		Time (in microseconds) spent on user calls within requests (between begin request and end request)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TIME_PROTECTED_IN_REQUESTS	NUMBER		Time (in microseconds) spent on user calls protected by Application Continuity within requests (between begin request and end request)
MAX_USER_CALLS_PER_REQUEST[1]	NUMBER		Maximum number of user calls received from the application per request (between begin request and end request)
MAX_PROT_CALLS_PER_REQUEST[1]	NUMBER		Maximum number of user calls protected by Application Continuity received from the application per request (between begin request and end request)
CON_ID_EXT[1]	NUMBER		Container ID used by the session for external recognition

1 This column is available starting with Oracle Database 19c, Release Update 19.19.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.11.
>
> For Oracle Database 19c, Release Update 19.11 through Release Update 19.18, this view is not created by default. To create this view, you must run the DBMS_APP_CONT_ADMIN.ACCHK_VIEWS procedure. See *Oracle Database PL/SQL Packages and Types Reference* for more information.
>
> Starting with Oracle Database 19c, Release Update 19.19, this view is created by default.

4.6 DBA_ACCHK_STATISTICS_SUMMARY

DBA_ACCHK_STATISTICS_SUMMARY displays a summary of Application Continuity protection statistics for each session that executed during an Application Continuity Protection Check (ACCHK) workload run.

ACCHK should be used on a running workload. Sessions that are terminated are not included in this view.

The ACCHK_READ role allows users with no administrative privileges to query this view.

Column	Datatype	NULL	Description
INST_ID	NUMBER	NOT NULL	Identifier for the instance for the session
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
CON_UID[1]	NUMBER	NOT NULL	Unique ID of the container used by the session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID_EXT[1]	NUMBER		Container ID used by the session for external recognition
SERVICE_NAME	VARCHAR2(64)		Service name of the session
FAILOVER_TYPE	VARCHAR2(16)		Indicates the FAILOVER_TYPE service setting for the session. Possible values: • AUTO - Transparent Application Continuity (TAC) was enabled for the session • TRANSACTION - Application Continuity (AC) was enabled for the session • NONE - Failover was disabled for the session
FAILOVER_RESTORE	VARCHAR2(16)		Indicates the FAILOVER_RESTORE service setting for the session. Possible values: • AUTO - Transparent Application Continuity • LEVEL1 - Application Continuity • NONE - Session state is not restored at failover
RESET_STATE	VARCHAR2(16)		Indicates the RESET_STATE service setting for the session. Possible values: • LEVEL1 - The session state was cleaned at the end of a request • NONE - The session state was not cleaned at the end of a request
TOTAL_REQUESTS	NUMBER		Number of requests received for this session
PROTECTED_CALLS_PERCENT	NUMBER		Percentage of user calls within requests protected by Application Continuity for failover
PROTECTED_TIME_PERCENT	NUMBER		Percentage of time spent within requests protected by Application Continuity for failover
AVG_USER_CALLS_IN_REQUESTS	NUMBER		Average number of user calls received from the application within requests (between begin request and end request)
AVG_PROT_CALLS_IN_REQUESTS	NUMBER		Average number of user calls protected by Application Continuity within requests (between begin request and end request)
AVG_TIME_IN_REQUESTS	NUMBER		Average time (in microseconds) spent on user calls within requests (between begin request and end request)
AVG_TIME_PROTECTED_IN_REQUESTS	NUMBER		Average time (in microseconds) spent on user calls protected by Application Continuity within requests (between begin request and end request)
MAX_USER_CALLS_PER_REQUEST[1]	NUMBER		Maximum number of user calls received from the application per request (between begin request and end request)
MAX_PROT_CALLS_PER_REQUEST[1]	NUMBER		Maximum number of user calls protected by Application Continuity received from the application per request (between begin request and end request)

[1] This column is available starting with Oracle Database 19c, Release Update 19.19.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.11.
>
> For Oracle Database 19c, Release Update 19.11 through Release Update 19.18, this view is not created by default. To create this view, you must run the `DBMS_APP_CONT_ADMIN.ACCHK_VIEWS` procedure. See *Oracle Database PL/SQL Packages and Types Reference* for more information.
>
> Starting with Oracle Database 19c, Release Update 19.19, this view is created by default.

4.7 DBA_ACL_NAME_MAP

`DBA_ACL_NAME_MAP` maps new names of the access control lists for PL/SQL network utility packages from old XDB names.

Column	Datatype	NULL	Description
`XDB_NAME`	`VARCHAR2(4000)`	`NOT NULL`	The old XDB name of the access control list
`ACL`	`VARCHAR2(128)`		The new name of the access control list
`ACL_OWNER`	`VARCHAR2(128)`		The owner of the access control list

4.8 DBA_ADDM_FDG_BREAKDOWN

`DBA_ADDM_FDG_BREAKDOWN` describes the contribution for each finding from the different instances.

There is one row for each finding and for each instance participating in the analysis. Rows are omitted if the impact from that instance is not sufficient to register a finding in a local ADDM analysis.

Related View

`USER_ADDM_FDG_BREAKDOWN` describes the contribution for each finding from the different instances owned by the current user.

Column	Datatype	NULL	Description
`TASK_ID`	`NUMBER`	`NOT NULL`	Identifies the task to which this finding belongs (see `DBA_ADVISOR_TASKS`)
`FINDING_ID`	`NUMBER`	`NOT NULL`	Identifies the finding (see `DBA_ADVISOR_FINDINGS`)
`INSTANCE_NUMBER`	`NUMBER`	`NOT NULL`	The number of the instance contributing to the finding
`DATABASE_TIME`	`NUMBER`		The database time, in microseconds, accumulated by this instance during the analysis period
`ACTIVE_SESSIONS`	`NUMBER`		The average number of active sessions of the finding in this instance
`PERC_ACTIVE_SESSIONS`	`NUMBER`		The percentage of contribution from this instance compared to the total impact of the finding

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"USER_ADDM_FDG_BREAKDOWN"

4.9 DBA_ADDM_FINDINGS

DBA_ADDM_FINDINGS displays the ADDM findings discovered by all advisors in the database.

Each row for ADDM tasks in the related DBA_ADVISOR_FINDINGS view has a corresponding row in this view.

Related View

USER_ADDM_FINDINGS displays the ADDM findings discovered by the advisors owned by the current user. Each row for ADDM tasks in the related USER_ADVISOR_FINDINGS view has a corresponding row in this view. The USER_ADDM_FINDINGS view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER	NOT NULL	Identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task
EXECUTION_NAME	VARCHAR2(128)		The name of the task execution with which this entry (row) is associated
FINDING_ID	NUMBER	NOT NULL	Identifier of the finding
FINDING_NAME	VARCHAR2(4000)		Name of the finding
TYPE	VARCHAR2(11)		Type of the finding: PROBLEM SYMPTOM ERROR INFORMATION
TYPE_ID	NUMBER	NOT NULL	Numeric ID for the value in column TYPE
PARENT	NUMBER	NOT NULL	Identifier of the parent finding
OBJECT_ID	NUMBER		Identifier of the associated object, if any
IMPACT_TYPE	VARCHAR2(4000)		Impact of the finding on the system
IMPACT	NUMBER		Impact value
MESSAGE	VARCHAR2(4000)		Message describing the finding
MORE_INFO	VARCHAR2(4000)		Additional info associated with the finding
FILTERED	VARCHAR2(1)		A value of Y means that the row in the view was filtered out by a directive (or a combination of directives). A value of N means that the row was not filtered.
FLAGS	NUMBER		For internal use only by advisor framework clients
DATABASE_TIME	NUMBER		The database time, in microseconds, accumulated by this finding
ACTIVE_SESSIONS	NUMBER		The average number of active sessions for the finding

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PERC_ACTIVE_SESS	NUMBER		The percentage of active sessions for this finding out of the total active sessions for the task
IS_AGGREGATE	CHAR(1)		A value of Y means that this finding is created for global/continental ADDM as an aggregate of local ADDM findings. Otherwise, the value is N.
METER_LEVEL	VARCHAR2(6)		Reserved for future use
QUERY_IS_APPROX	CHAR(1)		Indicates whether the ASH SQL associated with the finding is an approximate query (Y) or an exact query (N). To get the associated query, use the TASK_NAME and FINDING_ID columns from this view and call the PL/SQL function DBMS_ADDM.GET_ASH_QUERY(task_name, finding_id).

See Also

- "USER_ADDM_FINDINGS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_ADDM.GET_ASH_QUERY procedure

4.10 DBA_ADDM_INSTANCES

DBA_ADDM_INSTANCES displays instance-level information for ADDM tasks that finished executing.

For each instance that was supposed to be analyzed (whether it was or not) there is one row describing information about it.

Related View

USER_ADDM_INSTANCES displays instance-level information for ADDM tasks that finished executing in all instances owned by the current user.

Column	Datatype	NULL	Description
TASK_ID	NUMBER	NOT NULL	The ID of the main ADDM task
INSTANCE_NUMBER	NUMBER	NOT NULL	The number of the instance
INSTANCE_NAME	VARCHAR2(16)		The name of the instance
HOST_NAME	VARCHAR2(64)		The name of the system on which the instance was running

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(10)		How information from this instance was used by the ADDM task. A value of ANALYZED means that the instance participated fully in the analysis. For the following remaining values, the instance was not used during task execution, for the stated reason: BOUNCED - the instance was shut down or started during the analysis period NO_SNAPS - there were either begin or end snapshots missing for the instance NO_STATS - there were key statistics missing for the instance NOT_FOUND - no mention of this instance could be found in AWR during the analysis period
DATABASE_TIME	NUMBER		The database time, in microseconds, accumulated by this instance during the analysis period
ACTIVE_SESSIONS	NUMBER		The average number of active sessions for the instance during the analysis period
PERC_ACTIVE_SESS	NUMBER		The percentage of active sessions for this instance, out of the total active sessions for the task
METER_LEVEL	VARCHAR2(6)		Reserved for future use
LOCAL_TASK_ID	NUMBER		The ID of a local ADDM task that contained an analysis of the instance for the same analysis period as that of the main task. If the main task is a local ADDM, then this value is the same as the TASK_ID value.

See Also

"USER_ADDM_INSTANCES"

4.11 DBA_ADDM_SYSTEM_DIRECTIVES

DBA_ADDM_SYSTEM_DIRECTIVES displays information about global instances for ADDM system directives.

Column	Datatype	NULL	Description
INSTANCE_ID	NUMBER	NOT NULL	Unique ID for the directive instance. The directive management engine automatically generates ID numbers.
INSTANCE_NAME	VARCHAR2(128)	NOT NULL	User-assigned name for the directive instance.
DIRECTIVE_NAME	VARCHAR2(128)	NOT NULL	Any value that further classifies this directive within a domain. The domain and the name form a unique key for the directive.
DESCRIPTION	VARCHAR2(4000)		Description of the ADDM system directive, shown in the language used by the current session

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.12 DBA_ADDM_TASK_DIRECTIVES

DBA_ADDM_TASK_DIRECTIVES displays information about all ADDM task directives in the database.

Related View

USER_ADDM_TASK_DIRECTIVES displays information about ADDM task directives owned by the current user.

Column	Datatype	NULL	Description
TASK_ID	NUMBER		An ADDM advisor task identifier to which the directive instance is associated
TASK_NAME	VARCHAR2(128)		An ADDM advisor task to which the directive instance is associated
USERNAME	VARCHAR2(128)	NOT NULL	Database user who owns the ADDM task instance
SEQ_ID	NUMBER	NOT NULL	Unique ID for the directive instance. The directive management engine automatically generates ID numbers.
INSTANCE_NAME	VARCHAR2(128)	NOT NULL	A user-assigned name for the ADDM task directive instance
DIRECTIVE_NAME	VARCHAR2(128)	NOT NULL	Any value that further classifies this directive within a domain. The domain and the name form a unique key for the directive.
DESCRIPTION	VARCHAR2(4000)		Description of the ADDM task directive, shown in the language used by the current session

See Also

"USER_ADDM_TASK_DIRECTIVES"

4.13 DBA_ADDM_TASKS

DBA_ADDM_TASKS displays information about all ADDM tasks in the database.

The view contains one row for each row in the related DBA_ADVISOR_TASKS view that has ADVISOR_NAME=ADDM and STATUS=COMPLETED.

Related View

USER_ADDM_TASKS displays information about the ADDM tasks owned by the current user. The view contains one row for each row in the related USER_ADVISOR_TASKS view that has ADVISOR_NAME=ADDM and STATUS=COMPLETED. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER	NOT NULL	Unique identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DESCRIPTION	VARCHAR2(256)		User-supplied description of the task
ADVISOR_NAME	VARCHAR2(128)		Advisor associated with the task
CREATED	DATE	NOT NULL	Creation date of the task
LAST_MODIFIED	DATE	NOT NULL	Date on which the task was last modified
PARENT_TASK_ID	NUMBER		Identifier of the parent task (if the task was created because of the recommendation of another task)
PARENT_RXEC_ID	NUMBER		Identifier of the recommendation within the parent task that resulted in the creation of the task
LAST_EXECUTION	VARCHAR2(128)		Name of the current or last task execution
EXECUTION_TYPE	VARCHAR2(128)		Type of the last execution. This information is optional for single-execution tasks.
EXECUTION_TYPE#	NUMBER		Reserved for internal use
EXECUTION_DESCRIPTION	VARCHAR2(256)		Optional description of the last execution
EXECUTION_START	DATE		Execution start date and time of the task
EXECUTION_END	DATE		Execution end date and time of the task
STATUS	VARCHAR2(11)		Current operational status of the task: • INITIAL - Initial state of the task; no recommendations are present • EXECUTING - Task is currently running • INTERRUPTED - Task analysis was interrupted by the user. Recommendation data, if present, can be viewed and reported at this time. • COMPLETED - Task successfully completed the analysis operation. Recommendation data can be viewed and reported. • ERROR - An error occurred during the analysis operation. Recommendations, if present, can be viewed and reported at this time.
STATUS_MESSAGE	VARCHAR2(4000)		Informational message provided by the advisor, regarding the status
PCT_COMPLETION_TIME	NUMBER		Percent completion, in terms of time, of the task when it is executing
PROGRESS_METRIC	NUMBER		Metric that measures the progress of the task in terms of quality. Each advisor may have its own metric.
METRIC_UNITS	VARCHAR2(64)		Unit of the metric used to measure progress
ACTIVITY_COUNTER	NUMBER		Counter that is updated frequently by the advisor, denoting that useful work is being performed
RECOMMENDATION_COUNT	NUMBER		Number of recommendations produced
ERROR_MESSAGE	VARCHAR2(4000)		Informational message or an error message indicating the current operation or condition
SOURCE	VARCHAR2(128)		Optional name that identifies the creator of the task
HOW_CREATED	VARCHAR2(30)		Optional task or template on which the object was based
READ_ONLY	VARCHAR2(5)		Indicates whether the task is read-only (TRUE) or not (FALSE)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SYSTEM_TASK	VARCHAR2(5)		Indicates whether the task is a system task (TRUE) or not (FALSE). The automatic SQL tuning task, SYS_AUTO_SQL_TUNING_TASK, is one example of a system task.
ADVISOR_ID	NUMBER	NOT NULL	Unique identifier for the advisor
STATUS#	NUMBER		Reserved for internal use
DBID	NUMBER		The database ID that the task was using
DBNAME	VARCHAR2(9)		The name of the database that the task was analyzing
DBVERSION	VARCHAR2(17)		The version of the database that the task was analyzing
ANALYSIS_VERSION	VARCHAR2(17)		The version of the database that executed the task
BEGIN_SNAP_ID	NUMBER		The snapshot ID that starts the analysis period
BEGIN_TIME	TIMESTAMP(3)		The SYSDATE at the time the BEGIN_SNAP_ID was taken
END_SNAP_ID	NUMBER		The snapshot ID that ends the analysis period
END_TIME	TIMESTAMP(3)		The SYSDATE at the time the END_SNAP_ID was taken
REQUESTED_ANALYSIS	VARCHAR2(8)		The type of ADDM analysis that was requested before task execution, as follows: DATABASE - global ADDM INSTANCE - local ADDM PARTIAL - continental ADDM
ACTUAL_ANALYSIS	VARCHAR2(8)		The type of ADDM analysis that was actually performed when the task was executed (either DATABASE, INSTANCE, or PARTIAL)
DATABASE_TIME	NUMBER		The total database time accumulated in the analysis period (and analyzed instances) in microseconds
ACTIVE_SESSIONS	NUMBER		The average active sessions during the analysis period (and analyzed sessions)
METER_LEVEL	VARCHAR2(6)		Reserved for future use
FDG_COUNT	NUMBER		The number of findings for the ADDM task, which will appear in the list of findings in a default ADDM report
DB_TYPE_DETECTED[1]	VARCHAR2(10)		Indicates the open mode of the database that the task was analyzing, as detected by ADDM. Possible values: • READ-ONLY • READ-WRITE
DB_TYPE_ANALYZED[1]	VARCHAR2(10)		Indicates the assumed open mode of the database that the task was analyzing. If the value of this column is different from the value of the DB_TYPE_DETECTED column, then ADDM ignored the detected open mode of the database and analyzed the task as if it occurred on a database with the assumed open mode. Possible values: • READ-ONLY • READ-WRITE

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CDB_TYPE_DETECTED[1]	VARCHAR2(25)		Indicates the type of database that the task was analyzing, as detected by ADDM. Possible values: • NON-CDB - A non-CDB • CDB ROOT - The root container in a CDB • PDB - A pluggable database (PDB) in a CDB • AUTONOMOUS DATA WAREHOUSE - A PDB that hosts an Oracle Autonomous Data Warehouse Cloud service • AUTONOMOUS OLTP - A PDB that hosts an Oracle Autonomous Transaction Processing service
CDB_TYPE_ANALYZED[1]	VARCHAR2(25)		Indicates the assumed type of database that the task was analyzing. If the value of this column is different from the value of the CDB_TYPE_DETECTED column, then ADDM ignored the detected type of database and analyzed the task as if it occurred on the assumed type of database. Possible values: • NON-CDB - A non-CDB • CDB ROOT - The root container in a CDB • PDB - A pluggable database (PDB) in a CDB • AUTONOMOUS DATA WAREHOUSE - A PDB that hosts an Oracle Autonomous Data Warehouse Cloud service • AUTONOMOUS OLTP - A PDB that hosts an Oracle Autonomous Transaction Processing service

1 This column is available starting with Oracle Database 19c. If the task was analyzing a database at a release earlier than Oracle Database 19c, and that database was subsequently upgraded to Oracle Database 19c, then the value of this column is NULL.

See Also

"USER_ADDM_TASKS"

4.14 DBA_ADVISOR_ACTIONS

DBA_ADVISOR_ACTIONS displays information about the actions associated with all recommendations in the database.

Each action is specified by the COMMAND and ATTR1 through ATTR6 columns. Each command defines how the attribute columns will be used.

Related View

USER_ADVISOR_ACTIONS displays information about the actions associated with the recommendations owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER	NOT NULL	Identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
EXECUTION_NAME	VARCHAR2(128)		The name of the task execution with which this entry (row) is associated
REC_ID	NUMBER	NOT NULL	Recommendation associated with the action
ACTION_ID	NUMBER	NOT NULL	Unique identifier for the action
OBJECT_ID	NUMBER		Object associated with the action
COMMAND	VARCHAR2(64)		Command to be executed **See Also:** DBA_ADVISOR_COMMANDS for a list of commands
COMMAND_ID	NUMBER	NOT NULL	ID of the command to be executed **See Also:** DBA_ADVISOR_COMMANDS for a list of commands
FLAGS	NUMBER		Advisor-specific flags
ATTR1	VARCHAR2(4000)		Parameters defining the command
ATTR2	VARCHAR2(4000)		Parameters defining the command
ATTR3	VARCHAR2(4000)		Parameters defining the command
ATTR4	VARCHAR2(4000)		Parameters defining the command
ATTR5	CLOB		Parameters defining the command; to be used if the text is significantly large (for example, a SQL statement defining a materialized view)
ATTR6	CLOB		Parameters defining the command; to be used if the text is significantly large (for example, a SQL statement defining a materialized view)
NUM_ATTR1	NUMBER		General numeric attribute
NUM_ATTR2	NUMBER		General numeric attribute
NUM_ATTR3	NUMBER		General numeric attribute
NUM_ATTR4	NUMBER		General numeric attribute
NUM_ATTR5	NUMBER		General numeric attribute
MESSAGE	VARCHAR2(4000)		Message associated with the action
FILTERED	VARCHAR2(1)		A value of Y means that the row in the view was filtered out by a directive (or a combination of directives). A value of N means that the row was not filtered.
RESULT_STATUS[1]	VARCHAR2(25)		Result status. Possible values: • AUTOMATION PENDING • TIMED OUT • MISSING INFORMATION • COMPLETED WITH NO RESULTS • COMPLETED AND IMPLEMENTED • ENCOUNTERED ERROR • NOT AUTOMATED
RESULT_LAST_MODIFIED[1]	TIMESTAMP(3)		Time at which result status was last modified
RESULT_MESSAGE[1]	VARCHAR2(4000)		Result message text

[1] This column is available starting with Oracle Database 19c.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"USER_ADVISOR_ACTIONS"

4.15 DBA_ADVISOR_COMMANDS

DBA_ADVISOR_COMMANDS displays information about the commands used by all advisors in the database for specifying recommendation actions.

In addition to the set of commands in the COMMAND column of V$SESSION, the following additional commands are defined:

- RUN ADVISOR
- CHECK EXECUTION PLAN
- ALTER PARAMETER
- ENABLE TRACE

Column	Datatype	NULL	Description
COMMAND_ID	NUMBER		Identifier of the command
COMMAND_NAME	VARCHAR2(64)		Name of the command

4.16 DBA_ADVISOR_DEF_PARAMETERS

DBA_ADVISOR_DEF_PARAMETERS displays all default task parameters and their current values in the database.

When a task or object is created, the parameters and their values are copied into the private parameter table.

Column	Datatype	NULL	Description
ADVISOR_NAME	VARCHAR2(128)	NOT NULL	Name of the advisor that supports the parameter
PARAMETER_NAME	VARCHAR2(128)	NOT NULL	Name of the parameter
PARAMETER_VALUE	VARCHAR2(4000)	NOT NULL	Value of the parameter. Numeric parameter values are converted to a string equivalent. Possible keywords as values: • ALL • UNLIMITED • UNUSED

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PARAMETER_TYPE	VARCHAR2(10)		Datatype of the parameter: • NUMBER - Numeric value • STRING - String value. If the string contains special characters, then it will be enclosed in single quotes. • STRINGLIST - Comma-separated list of string elements. If a string element contains a comma or other special characters, then the element will be enclosed in single quotes. • TABLE - Single table reference. A reference will contain a schema name, followed by an optional table name. If the table name is omitted or is the character %, then the table name is interpreted as a wildcard. SQL quoted identifiers are supported. • TABLELIST - List of one or more comma-separated table references. A reference will contain schema name, followed by an optional table name. If the table name is omitted or is the character %, then the table name is interpreted as a wildcard. SQL quoted identifiers are supported.
IS_DEFAULT	VARCHAR2(1)		Indicates whether the parameter value is set to the advisor's default value (Y) or not (N)
IS_OUTPUT	VARCHAR2(1)		Indicates whether the task execution process sets the parameter value (Y) or not (N)
IS_MODIFIABLE_ANYTIME	VARCHAR2(1)		Indicates whether the parameter value can be modified when the task is not in its initial state (Y) or not (N)
IS_SYSTEM_TASK_ONLY	VARCHAR2(1)		Indicates whether the task is a system task (Y) or not (N)
DESCRIPTION	VARCHAR2(4000)		Optional description of the parameter
EXECUTION_TYPE	VARCHAR2(128)		Type of the last execution. This information is optional for single-execution tasks.

4.17 DBA_ADVISOR_DEFINITIONS

DBA_ADVISOR_DEFINITIONS displays the properties of all advisors in the database.

The view contains one row for each task, representing the current state of the task as well as execution-specific data such as progress monitoring and completion status.

Column	Datatype	NULL	Description
ADVISOR_ID	NUMBER	NOT NULL	Unique identifier for the advisor
ADVISOR_NAME	VARCHAR2(128)	NOT NULL	Name of the advisor

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PROPERTY	NUMBER	NOT NULL	Properties: • Bit 0: - Indicates whether the advisor runs in COMPREHENSIVE mode (1) or not (0) • Bit 1: - Indicates whether the advisor runs in LIMITED mode (1) or not (0) • Bit 2: - Indicates whether the advisor is resumable (1) or not (0) • Bit 3: - Indicates whether the advisor accepts user directives (1) or not (0)

4.18 DBA_ADVISOR_DIR_DEFINITIONS

DBA_ADVISOR_DIR_DEFINITIONS provides a definition of the base directive.

Column	Datatype	NULL	Description
ID	NUMBER	NOT NULL	Unique id for directive. The directive management engine automatically generates ID numbers. The identifier is unique among all directives regardless of the domain name and directive name.
ADVISOR_ID	NUMBER	NOT NULL	Identifier number of the owner advisor.
ADVISOR_NAME	VARCHAR2(128)	NOT NULL	The name of the advisor to which this directive belongs.
DIRECTIVE_NAME	VARCHAR2(128)	NOT NULL	Any value that further classifies this directive within a domain. The domain and the name form a unique key for the directive.
DOMAIN_NAME	VARCHAR2(128)	NOT NULL	Domain or namespace name.
DESCRIPTION	VARCHAR2(256)	NOT NULL	An optional description that documents the purpose of the directive.
TYPE	NUMBER	NOT NULL	Further describes the use of the directive. Possible values are: 1. **Filter** - An Xpath filter 2. **Single Value** - Evaluation returns a single string value 3. **Multiple Values** - Evaluation returns one to many string values 4. **Conditional** - Evaluation returns a single value based on an input key, similar to a CASE or SWITCH statement
TYPE_NAME	VARCHAR2(15)		A decoded version of the TYPE column.
TASK_STATUS	VARCHAR2(9)		The status of the directive instances when a task is not in its initial state. Possible values are: • IMMUTABLE • MUTABLE
INSTANCES	VARCHAR2(8)		Indicates whether a directive will permit multiple instances. Possible values are: • SINGLE • MULTIPLE

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
METADATA	CLOB	NOT NULL	A DTD that is used to process the directive.

4.19 DBA_ADVISOR_DIR_INSTANCES

DBA_ADVISOR_DIR_INSTANCES provides information about all global instances for a directive.

Column	Datatype	NULL	Description
DIRECTIVE_ID	NUMBER	NOT NULL	Unique id for directive. The directive management engine automatically generates ID numbers.
INSTANCE_ID	NUMBER	NOT NULL	Unique id for the directive instance. The directive management engine automatically generates ID numbers.
INSTANCE_NAME	VARCHAR2(128)	NOT NULL	A user-assigned name for the directive instance.
DATA	CLOB	NOT NULL	An XML document that gives meaningful default values for all parts of the directive.

4.20 DBA_ADVISOR_DIR_TASK_INST

DBA_ADVISOR_DIR_TASK_INST provides information about all task directive instances.

Related View

USER_ADVISOR_DIR_TASK_INST provides information about all task directive instances owned by the current user.

Column	Datatype	NULL	Description
DIRECTIVE_ID	NUMBER	NOT NULL	Unique id for directive. The directive management engine automatically generates ID numbers.
SEQ_ID	NUMBER	NOT NULL	Unique id for the directive instance. The directive management engine automatically generates ID numbers.
INSTANCE_NAME	VARCHAR2(128)	NOT NULL	A user-assigned name for the directive instance.
USERNAME	VARCHAR2(128)	NOT NULL	Database user who owns the task instance.
TASK_ID	NUMBER		An advisor task identifier to which the directive instance is associated
TASK_NAME	VARCHAR2(128)		An advisor task to which the directive instance is associated.
DATA	CLOB	NOT NULL	An XML document that gives meaningful default values for all parts of the directive.

See Also

"USER_ADVISOR_DIR_TASK_INST"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.21 DBA_ADVISOR_EXEC_PARAMETERS

DBA_ADVISOR_EXEC_PARAMETERS displays the parameter values used for past executions of tasks.

It is more useful for advisors supporting multiple executions, such as SQL Performance Analyzer, where a parameter can have different values for different executions.

Related View

USER_ADVISOR_EXEC_PARAMETERS displays the parameter values used for past executions of tasks owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER	NOT NULL	Unique identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task
EXECUTION_NAME	VARCHAR2(128)	NOT NULL	Name of the task execution with which this entry (row) is associated
EXECUTION_TYPE	VARCHAR2(128)		Type of the last execution. This information is optional for single-execution tasks.
PARAMETER_NAME	VARCHAR2(128)	NOT NULL	Name of the parameter
PARAMETER_VALUE	VARCHAR2(4000)		Value of the parameter. Numeric parameter values are converted to a string equivalent.
PARAMETER_TYPE	VARCHAR2(10)		Datatype of the parameter (see DBA_ADVISOR_PARAMETERS)
IS_DEFAULT	VARCHAR2(1)		Indicates whether the parameter value is set to the advisor's default value (Y) or not (N)
IS_OUTPUT	VARCHAR2(1)		Indicates whether the task execution process sets the parameter value (Y) or not (N)
IS_MODIFIABLE_ANYTIME	VARCHAR2(1)		Indicates whether the parameter value can be modified when the task is not in its initial state (Y) or not (N)
DESCRIPTION	VARCHAR2(4000)		Optional description of the parameter
PARAMETER_FLAGS	NUMBER	NOT NULL	Reserved for internal use
PARAMETER_TYPE#	NUMBER	NOT NULL	Reserved for internal use

See Also

"USER_ADVISOR_EXEC_PARAMETERS"

4.22 DBA_ADVISOR_EXECUTION_TYPES

DBA_ADVISOR_EXECUTION_TYPES displays possible execution action for a given advisor.

Only advisors that support multiple executions of their tasks have entries in this view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ADVISOR_NAME	VARCHAR2(128)	NOT NULL	Name of the advisor
EXECUTION_TYPE	VARCHAR2(128)	NOT NULL	Execution type supported by the advisor
EXECUTION_DESCRIPTION	VARCHAR2(4000)		Optional description of the execution type

4.23 DBA_ADVISOR_EXECUTIONS

DBA_ADVISOR_EXECUTIONS displays metadata information for task executions.

For example, the SQL Performance Analyzer creates a minimum of three executions to perform a change impact analysis on a SQL workload. The first one collects performance data for the version of the workload before the change, the second one collects data for the version of the workload after the change, and the third one performs impact analysis. All of these executions belong to the same task and are grouped into this view. Similarly, the automatic SQL tuning task, SYS_AUTO_SQL_TUNING_TASK, creates a new execution for each tuning run.

Related View

USER_ADVISOR_EXECUTIONS displays metadata information for task executions owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER	NOT NULL	Unique identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task
EXECUTION_NAME	VARCHAR2(128)	NOT NULL	Name of the task execution with which this entry (row) is associated
EXECUTION_ID	NUMBER	NOT NULL	Execution ID
DESCRIPTION	VARCHAR2(256)		User-supplied description of the task
EXECUTION_TYPE	VARCHAR2(128)		Type of the last execution (optional for single-execution tasks)
EXECUTION_TYPE#	NUMBER		Reserved for internal use
EXECUTION_START	DATE		Execution start date and time
EXECUTION_LAST_MODIFIED	DATE	NOT NULL	Last modified date and time for the execution
EXECUTION_END	DATE		Execution end date and time
ADVISOR_NAME	VARCHAR2(128)		Advisor associated with the task
REQUESTED_DOP	NUMBER		The degree of parallelism (DOP) value requested by the user (through the TEST_EXECUTE_DOP parameter). It can be any value greater or equal to zero.
ACTUAL_DOP	NUMBER		The actual degree of parallelism (DOP) with which the execution finished. If the requested DOP is greater than than what is available on the system, the ACTUAL_DOP value can be lower than the REQUESTED_DOP value.
CONCURRENT_EXECUTION	VARCHAR2(3)		Indicates whether concurrency was used for this execution (YES) or not (NO)
ADVISOR_ID	NUMBER	NOT NULL	Unique identifier for the advisor

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(11)		Current operational status of the task: • EXECUTING • COMPLETED • INTERRUPTED • CANCELLED • FATAL ERROR
STATUS#	NUMBER	NOT NULL	Reserved for internal use
STATUS_MESSAGE	VARCHAR2(4000)		Informational message provided by the advisor regarding the status
ERROR_MESSAGE	VARCHAR2(4000)		Informational message or an error message indicating the current operation or condition

See Also

"USER_ADVISOR_EXECUTIONS"

4.24 DBA_ADVISOR_FDG_BREAKDOWN

DBA_ADVISOR_FDG_BREAKDOWN describes the contribution from the different instances to the findings for each ADDM task.

This view is populated only with ADDM tasks that are analyzing multiple instances (that is, the ACTUAL_ANALYSIS column in the task's row in DBA_ADDM_TASKS is set to DATABASE or PARTIAL).

Related View

USER_ADVISOR_FDG_BREAKDOWN describes the contribution from the different instances to the findings for each ADDM task owned by the current user.

Column	Datatype	NULL	Description
TASK_ID	NUMBER	NOT NULL	Unique identifier of the task (see DBA_ADVISOR_TASKS and DBA_ADDM_TASKS)
FINDING_ID	NUMBER	NOT NULL	Identifier of the finding to which this breakdown applies (see DBA_ADVISOR_FINDINGS and DBA_ADDM_FINDINGS)
INSTANCE_NUMBER	NUMBER	NOT NULL	The number of the instance for the breakdown
IMPACT	NUMBER		The database time (in microseconds) of the finding in the instance
PERC_IMPACT	NUMBER		Percentage of the contribution of the instance to the overall finding's impact
EXECUTION_NAME	VARCHAR2(128)		The name of the task execution with which this entry (row) is associated

See Also

"USER_ADVISOR_FDG_BREAKDOWN"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.25 DBA_ADVISOR_FINDING_NAMES

DBA_ADVISOR_FINDING_NAMES provides a list of all finding names registered with the Advisor Framework.

Column	Datatype	NULL	Description
ID	NUMBER		ID of the finding name
ADVISOR_NAME	VARCHAR2(128)	NOT NULL	Advisor name
FINDING_NAME	VARCHAR2(4000)		Finding name

4.26 DBA_ADVISOR_FINDINGS

DBA_ADVISOR_FINDINGS displays the findings discovered by all advisors in the database.

Related View

USER_ADVISOR_FINDINGS displays the findings discovered by the advisors owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER	NOT NULL	Identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task
EXECUTION_NAME	VARCHAR2(128)		The name of the task execution with which this entry (row) is associated
FINDING_ID	NUMBER	NOT NULL	Identifier of the finding
FINDING_NAME	VARCHAR2(4000)		Name of the finding
TYPE	VARCHAR2(11)		Type of the finding: • PROBLEM • SYMPTOM • ERROR • INFORMATION
TYPE_ID	NUMBER	NOT NULL	Numeric ID for the value in column TYPE
PARENT	NUMBER	NOT NULL	Identifier of the parent finding
OBJECT_ID	NUMBER		Identifier of the associated object, if any
IMPACT_TYPE	VARCHAR2(4000)		Impact of the finding on the system
IMPACT	NUMBER		Impact value
MESSAGE	VARCHAR2(4000)		Message describing the finding
MORE_INFO	VARCHAR2(4000)		Additional info associated with the finding
FILTERED	VARCHAR2(1)		A value of Y means that the row in the view was filtered out by a directive (or a combination of directives). A value of N means that the row was not filtered.
FLAGS	NUMBER		For internal use only by advisor framework clients

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "USER_ADVISOR_FINDINGS"

4.27 DBA_ADVISOR_JOURNAL

DBA_ADVISOR_JOURNAL displays the journal entries for all tasks in the database.

Related View

USER_ADVISOR_JOURNAL displays the journal entries for the tasks owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER	NOT NULL	Identifier of the task or workload object
TASK_NAME	VARCHAR2(128)		Name of the task or workload object
EXECUTION_NAME	VARCHAR2(128)		The name of the task execution with which this entry (row) is associated
JOURNAL_ENTRY_SEQ	NUMBER	NOT NULL	Sequence number of the journal entry (unique for each task). This sequence number is used to order the data.
JOURNAL_ENTRY_TYPE	VARCHAR2(12)		Type of the task: • FATAL • ERROR • WARNING • INFORMATION • INFORMATION[2 | 3 | 4 | 5 | 6]
JOURNAL_ENTRY	VARCHAR2(4000)		Entry in the journal

> **See Also**
>
> "USER_ADVISOR_JOURNAL"

4.28 DBA_ADVISOR_LOG

DBA_ADVISOR_LOG displays information about the current state of all tasks in the database, as well as execution-specific data such as progress monitoring and completion status.

The view contains one row for each task.

Related View

USER_ADVISOR_LOG displays information about the current state of the tasks owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER	NOT NULL	Identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task
EXECUTION_START	DATE		Execution start date and time of the task
EXECUTION_END	DATE		Execution end date and time of the task
STATUS	VARCHAR2(11)		Current operational status of the task: • INITIAL - Initial state of the task; no recommendations are present • EXECUTING - Task is currently running • COMPLETED - Task successfully completed the analysis operation. Recommendation data can be viewed and reported. • INTERRUPTED - Task analysis was interrupted by the user. Recommendation data, if present, can be viewed and reported at this time. • CANCELLED • FATAL ERROR - An irrecoverable error occurred during the analysis operation. All recommendation data is unusable.
STATUS_MESSAGE	VARCHAR2(4000)		Informational message provided by the advisor regarding the status
PCT_COMPLETION_TIME	NUMBER		Percent completion, in terms of time, of the task when it is executing
PROGRESS_METRIC	NUMBER		Metric that measures the progress of the task in terms of quality. Each advisor could have its own metric.
METRIC_UNITS	VARCHAR2(64)		Unit of the metric used to measure progress
ACTIVITY_COUNTER	NUMBER		Counter that is updated frequently by the advisor, denoting that useful work is being performed
RECOMMENDATION_COUNT	NUMBER		Number of recommendations produced
ERROR_MESSAGE	VARCHAR2(4000)		Informational message or an error message indicating the current operation or condition

See Also

"USER_ADVISOR_LOG"

4.29 DBA_ADVISOR_OBJECT_TYPES

DBA_ADVISOR_OBJECT_TYPES displays information about the object types used by all advisors in the database.

In addition to the regular database object types (such as TABLE and INDEX), the following types are defined:

- SYSTEM
- I/O

Copyright © 2002, 2026, Oracle and/or its affiliates.

- SGA
- PGA
- SHARED POOL
- BUFFER CACHE
- LIBRARY CACHE
- PROCESS
- SESSION
- ENQUEUE
- LATCH
- ROLLBACK SEGMENT
- FILE
- PARAMETER
- CURSOR
- SQL
- SQL WORKLOAD

Column	Datatype	NULL	Description
OBJECT_TYPE_ID	NUMBER		Type identifier
OBJECT_TYPE	VARCHAR2(64)		Type name

4.30 DBA_ADVISOR_OBJECTS

DBA_ADVISOR_OBJECTS displays information about the objects currently referenced by all advisors in the database.

Each row in the view pertains to an object instantiation.

Related View

USER_ADVISOR_OBJECTS displays information about the objects currently referenced by the advisors owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the object
OBJECT_ID	NUMBER	NOT NULL	Identifier of the object
TYPE	VARCHAR2(64)		Name of the type
TYPE_ID	NUMBER	NOT NULL	Type identifier number
TASK_ID	NUMBER	NOT NULL	Task referencing the object
TASK_NAME	VARCHAR2(128)		Name of the task
EXECUTION_NAME	VARCHAR2(128)		The name of the task execution with which this entry (row) is associated
ATTR1	VARCHAR2(4000)		Attributes and identifier of the object
ATTR2	VARCHAR2(4000)		Attributes and identifier of the object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ATTR3	VARCHAR2(4000)		Attributes and identifier of the object
ATTR4	CLOB		Attributes and identifiers that cannot be expressed in the ATTR1, ATTR2, and ATTR3 columns
ATTR5	VARCHAR2(4000)		Attributes and identifier of the object
ATTR6	RAW(2000)		Attributes and identifier of the object
ATTR7	NUMBER		Attributes and identifier of the object
ATTR8	NUMBER		Attributes and identifier of the object
ATTR9	NUMBER		Attributes and identifier of the object
ATTR10	NUMBER		Attributes and identifier of the object
ATTR11	NUMBER		Attributes and identifier of the object
ATTR16	VARCHAR2(4000)		Attributes and identifier of the object
ATTR17	VARCHAR2(4000)		Attributes and identifier of the object
ATTR18	VARCHAR2(4000)		Attributes and identifier of the object
OTHER	CLOB		Other attributes and identifiers of the object
ADV_SQL_ID	VARCHAR2(13)		If the object type is SQL, then this column contains the SQL identifier of the SQL statement. Otherwise, the value of this column is null.

Note

The definition of the ATTRn columns depends on the advisors that are using the object. For example, the SQL object type defines the attribute columns as follows:

- ATTR1 contains the SQL ID
- ATTR2 contains the SQL address (in the cursor cache)
- ATTR4 contains the SQL text

See Also

"USER_ADVISOR_OBJECTS"

4.31 DBA_ADVISOR_PARAMETERS

DBA_ADVISOR_PARAMETERS displays all task parameters and their current values in the database.

This data is accessible by all tasks.

Related View

USER_ADVISOR_PARAMETERS displays the task parameters and their current values for the tasks owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task or workload object
TASK_ID	NUMBER	NOT NULL	Unique identifier number of the task or workload object
TASK_NAME	VARCHAR2(128)		Name of the task or workload object
PARAMETER_NAME	VARCHAR2(128)	NOT NULL	Name of the parameter
PARAMETER_VALUE	VARCHAR2(4000)	NOT NULL	Value of the parameter. Numeric parameter values are converted to a string equivalent. Possible keywords as values: • ALL • UNLIMITED • UNUSED
PARAMETER_TYPE	VARCHAR2(10)		Datatype of the parameter: • NUMBER - Numeric value • STRING - String value. If the string contains special characters, then it will be enclosed in single quotes. • STRINGLIST - Comma-separated list of string elements. If a string element contains a comma or other special characters, then the element will be enclosed in single quotes. • TABLE - Single table reference. A reference will contain a schema name, followed by an optional table name. If the table name is omitted or is the character %, then the table name is interpreted as a wildcard. SQL quoted identifiers are supported. • TABLELIST - List of one or more comma-separated table references. A reference will contain schema name, followed by an optional table name. If the table name is omitted or is the character %, then the table name is interpreted as a wildcard. SQL quoted identifiers are supported.
IS_DEFAULT	VARCHAR2(1)		Indicates whether the parameter value is set to the advisor's default value (Y) or not (N)
IS_OUTPUT	VARCHAR2(1)		Indicates whether the task execution process sets the parameter value (Y) or not (N)
IS_MODIFIABLE_ANYTIME	VARCHAR2(1)		Indicates whether the parameter value can be modified when the task is not in its initial state (Y) or not (N)
DESCRIPTION	VARCHAR2(4000)		Optional description of the parameter
EXECUTION_TYPE	VARCHAR2(128)		For advisors supporting multiple executions, the type of execution this parameter pertains to

See Also

"USER_ADVISOR_PARAMETERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.32 DBA_ADVISOR_RATIONALE

DBA_ADVISOR_RATIONALE displays information about the rationales for all recommendations in the database.

Related View

USER_ADVISOR_RATIONALE displays information about the rationales for the recommendations owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER	NOT NULL	Identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task
EXECUTION_NAME	VARCHAR2(128)		The name of the task execution with which this entry (row) is associated
REC_ID	NUMBER		Recommendation associated with the rationale
RATIONALE_ID	NUMBER	NOT NULL	Unique identifier for the rationale
IMPACT_TYPE	VARCHAR2(4000)		Impact on the system due to the problem described in the rationale. The impact can be described in terms of time, cost, or % degradation.
IMPACT	NUMBER		Calculated impact value
MESSAGE	VARCHAR2(4000)		Message containing an overview of the rationale
OBJECT_ID	NUMBER		Identifier of an object specified in the DBA_ADVISOR_OBJECTS view
TYPE	VARCHAR2(30)		Type of the rationale; defines what data exists in the attribute columns and how to interpret it: • TEXT - Text sentence for descriptive messages. The ATTR1 column contains the text. • CHART - Chart containing data to be displayed. The ATTR1 column contains the data.
ATTR1	VARCHAR2(4000)		Parameters defining the rationale
ATTR2	VARCHAR2(4000)		Parameters defining the rationale
ATTR3	VARCHAR2(4000)		Parameters defining the rationale
ATTR4	VARCHAR2(4000)		Parameters defining the rationale
ATTR5	CLOB		Parameters defining the rationale

See Also

"USER_ADVISOR_RATIONALE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.33 DBA_ADVISOR_RECOMMENDATIONS

DBA_ADVISOR_RECOMMENDATIONS displays the results of an analysis of all recommendations in the database.

A recommendation can have multiple actions associated with it. Actions are described in the DBA_ADVISOR_ACTIONS view. A recommendation also points to a set of rationales that present a justification/reasoning for that recommendation. These rationales are in the DBA_ADVISOR_RATIONALE view.

Related View

USER_ADVISOR_RECOMMENDATIONS displays the results of an analysis of the recommendations owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
REC_ID	NUMBER	NOT NULL	Unique identifier of the recommendation
TASK_ID	NUMBER	NOT NULL	Task that owns the recommendation
TASK_NAME	VARCHAR2(128)		Name of the task
EXECUTION_NAME	VARCHAR2(128)		The name of the task execution with which this entry (row) is associated
FINDING_ID	NUMBER		Unique identifier of the finding
TYPE	VARCHAR2(30)		Type of the recommendation
RANK	NUMBER		Ranking, in terms of importance, within the set of recommendations generated for the task
PARENT_REC_IDS	VARCHAR2(4000)		Comma-separated list of the recommendation IDs of the parent recommendations. If this column is nonzero, then the recommendation depends on the parents, and cannot be accepted if the parents are not accepted.
BENEFIT_TYPE	VARCHAR2(4000)		Describes the benefit obtained by carrying out the recommendation If there is a set of parents for the recommendation, then the benefit is the cumulative benefit (the improvement in system performance when this and all prior parent recommendations are accepted). If there are no parents, then this is the improvement when the recommendation is accepted, independent of other recommendations.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BENEFIT	NUMBER		Calculated benefit value This column displays an estimate of the savings in total activity (or load) on the system if all actions of the recommendation are implemented. Recommendations are not additive, that is, the benefit from one recommendation may overlap with the benefit from another. For example, Oracle might recommend increasing the shared pool size, or reducing hard parses (by not using literals), and both recommendations might reduce the same part of the workload related to parsing. The benefit is given in both pure active sessions and as a percent of the average active sessions of the analysis time period. Therefore, if the benefit is 20% of active sessions, Oracle estimates that if you apply the actions on the same workload, the average active sessions on the server will be reduced by 20%. The DBA_ADDM_TASKS view displays the average active sessions for an ADDM task.
ANNOTATION_STATUS	VARCHAR2(11)		When a task is complete, the recommendations are marked ACCEPT. The status can be changed later using the MARK_RECOMMENDATION procedure: • ACCEPT - Current recommendation is ready to implement. This recommendation can also be used as advice for future analysis operations. • REJECT - Current recommendation is not acceptable to the user, and therefore will be excluded from any implementation scripts. This recommendation can also be used as advice for future analysis operations. • IGNORE - Though not rejected, the current recommendation will be ignored when generating scripts and will never be used as advice to future analysis operations.
FLAGS	NUMBER		Advisor-specific flags
FILTERED	VARCHAR2(1)		A value of Y means that the row in the view was filtered out by a directive (or a combination of directives). A value of N means that the row was not filtered.
REC_TYPE_ID	NUMBER		Recommendation type ID

See Also

- "USER_ADVISOR_RECOMMENDATIONS"
- "DBA_ADDM_TASKS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.34 DBA_ADVISOR_SQLA_REC_SUM

DBA_ADVISOR_SQLA_REC_SUM displays recommendation rollup information for all workload objects in the database after an Access Advisor analysis operation.

Related View

USER_ADVISOR_SQLA_REC_SUM displays recommendation rollup information for the workload objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER		Unique identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task
REC_ID	NUMBER		Identifier of the recommendation
TOTAL_STMTS	NUMBER		Total number of statements processed during analysis
TOTAL_PRECOST	NUMBER		Total cost of executing the statements in which the recommended object will be used, before the recommendations
TOTAL_POSTCOST	NUMBER		Total cost of executing the statements in which the recommended object will be utilized, after the recommendations have been implemented

See Also

"USER_ADVISOR_SQLA_REC_SUM"

4.35 DBA_ADVISOR_SQLA_TABLES

DBA_ADVISOR_SQLA_TABLES displays cross references between the workload statements and the tables referenced in the statement.

Related View

USER_ADVISOR_SQLA_TABLES displays cross references between the workload statements and the tables referenced in the statement for the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the workload object
TASK_ID	NUMBER		Unique identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task
SQL_ID	VARCHAR2(13)		SQL identifier of the parent cursor in the library cache
STMT_ID	NUMBER		Statement ID
TABLE_OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Table name

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "USER_ADVISOR_SQLA_TABLES"

4.36 DBA_ADVISOR_SQLA_WK_MAP

DBA_ADVISOR_SQLA_WK_MAP displays the workload references for all tasks in the database.

Workload references are necessary to allow the SQL Access Advisor to find required workload data.

Related View

USER_ADVISOR_SQLA_WK_MAP displays the workload references for the tasks owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER		Unique identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task
WORKLOAD_ID	NUMBER		Unique identifier of the workload object
WORKLOAD_NAME	VARCHAR2(128)		Name of the workload
IS_STS	NUMBER		Type of workload source: • 0 - SQL workload object • 1 - SQL Tuning Set

> **See Also**
>
> "USER_ADVISOR_SQLA_WK_MAP"

4.37 DBA_ADVISOR_SQLA_WK_STMTS

DBA_ADVISOR_SQLA_WK_STMTS displays information about all workload objects in the database after an Access Advisor analysis operation.

Related View

USER_ADVISOR_SQLA_WK_STMTS displays information about the workload objects owned by the current user after an Access Advisor analysis operation. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_NAME	VARCHAR2(128)		Name of the task
TASK_ID	NUMBER	NOT NULL	Unique identifier of the task
SQLSET_ID	NUMBER		ID of the SQL tuning set for the statement

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SQLSET_NAME	VARCHAR2(128)		Name of the SQL tuning set for the statement
WORKLOAD_NAME	VARCHAR2(128)		Name of the workload
SQL_ID	VARCHAR(13)	NOT NULL	Generated identifier of the statement
SQL_SEQ	NUMBER	NOT NULL	SQL sequence
PLAN_HASH_VALUE	NUMBER	NOT NULL	Numerical representation of the SQL plan for the cursor. Comparing one PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line-by-line).
PARSING_SCHEMA_NAME	VARCHAR2(128)		Schema name that was used to originally build this child cursor
USERNAME	VARCHAR2(128)		Name of the user executing the statement
MODULE	VARCHAR2(64)		Name of the module issuing the statement
ACTION	VARCHAR2(64)		Module action for the statement
CPU_TIME	NUMBER		Total CPU count (in seconds) of the executing statement
BUFFER_GETS	NUMBER		Total number of buffer gets for the statement
DISK_READS	NUMBER		Total disk-read I/O count for the statement
ELAPSED_TIME	NUMBER		Total elapsed time (in seconds) of the executing statement
ROWS_PROCESSED	NUMBER		Total number of rows processed by the statement
EXECUTIONS	NUMBER		Total number of times the statement was executed
FIRST_LOAD_TIME	DATE		Load time of parent cursor
LAST_EXECUTION_DATE	DATE		Date on which the statement was last executed
PRIORITY	NUMBER		Business importance of the statement: • 1 - High • 2 - Medium • 3 - Low
COMMAND_TYPE	NUMBER		Type of the command
STAT_PERIOD	NUMBER		Unused
ACTIVE_STAT_PERIOD	NUMBER		Effective period of time (in seconds) during which the SQL statement was active
SQL_TEXT	CLOB		Text of the SQL statement
PRECOST	NUMBER		Cost of executing the statement in the workload prior to the recommendations
POSTCOST	NUMBER		Cost of executing the statement in the workload after the recommendations
IMPORTANCE	NUMBER		Advisor-calculated importance value
REC_ID	NUMBER		Associated recommendation identifier

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
VALIDATED	NUMBER		Indicates whether the statement is valid for analysis: • 0 - Statement will not be analyzed by the EXECUTE_TASK procedure. Typically, the statement references one or more tables that do not have valid statistics. To correct this problem, ensure that the tables have valid statistics and execute the RESET_SQLWKLD procedure on the current workload. • 1- Statement is eligible for analysis by the EXECUTE_TASK procedure

See Also

"USER_ADVISOR_SQLA_WK_STMTS"

4.38 DBA_ADVISOR_SQLPLANS

DBA_ADVISOR_SQLPLANS displays the different SQL execution plans generated as part of an advisor analysis.

Related View

USER_ADVISOR_SQLPLANS displays the different SQL execution plans owned by the current user generated as part of an advisor analysis.

Column	Datatype	NULL	Description
TASK_NAME	VARCHAR2(128)		Advisor task name in which the SQL plan was generated (see DBA_ADVISOR_TASKS)
TASK_ID	NUMBER(38)	NOT NULL	Advisor task ID in which the SQL plan was generated (see DBA_ADVISOR_TASKS)
EXECUTION_NAME	VARCHAR2(128)	NOT NULL	Advisor task execution in which the SQL plan was generated (see DBA_ADVISOR_EXECUTIONS)
SQL_ID	VARCHAR2(13)	NOT NULL	Identifier for the relevant SQL statement
OBJECT_ID	NUMBER(38)	NOT NULL	Advisor object ID identifying the relevant SQL statement (see DBA_ADVISOR_OBJECTS)
ATTRIBUTE	VARCHAR2(34)		Text string identifying the type of the execution plan. The following values are used by the SQL Tuning Advisor: • Original - Original plan of the query • Original with adjusted cost - Same as Original but with adjusted cost • Using SQL profile - Plan with SQL profile applied • Using new indices - Plan with indexes applied
STATEMENT_ID	VARCHAR2(30)		Optional statement identifier specified in the EXPLAIN PLAN statement
PLAN_HASH_VALUE	NUMBER	NOT NULL	Numerical representation of the execution plan

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PLAN_ID	NUMBER	NOT NULL	Plan identifier
TIMESTAMP	DATE		Date and time when the EXPLAIN PLAN statement was issued
REMARKS	VARCHAR2(4000)		Place for comments that can be added to the steps of the execution plan
OPERATION	VARCHAR2(30)		Name of the operation performed at this step
OPTIONS	VARCHAR2(255)		Options used for the operation performed at this step
OBJECT_NODE	VARCHAR2(128)		Name of the database link used to reference the object
OBJECT_OWNER	VARCHAR2(128)		Owner of the object
OBJECT_NAME	VARCHAR2(128)		Name of the object
OBJECT_ALIAS	VARCHAR2(261)		Object alias
OBJECT_INSTANCE	NUMBER(38)		Numbered position of the object name in the original SQL statement
OBJECT_TYPE	VARCHAR2(30)		Descriptive modifier that further describes the type of object
OPTIMIZER	VARCHAR2(255)		Current mode of the optimizer
SEARCH_COLUMNS	NUMBER		Number of index columns with start and stop keys (that is, the number of columns with matching predicates)
ID	NUMBER(38)	NOT NULL	Identification number for this step in the execution plan
PARENT_ID	NUMBER(38)		ID of the next step that operates on the results of this step
DEPTH	NUMBER(38)		Depth
POSITION	NUMBER(38)		Order of processing for steps with the same parent ID
COST	NUMBER(38)		Cost of the current operation estimated by the cost-based optimizer (CBO)
CARDINALITY	NUMBER(38)		Number of rows returned by the current operation (estimated by the CBO)
BYTES	NUMBER(38)		Number of bytes returned by the current operation

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OTHER_TAG	VARCHAR2(255)		Describes the function of the SQL text in the OTHER column. Values for OTHER_TAG are: • SERIAL - SQL is the text of a locally-executed, serial query plan. Currently, SQL is not loaded in OTHER for this case. • SERIAL_FROM_REMOTE - SQL text shown in the OTHER column will be executed at a remote site • PARALLEL_COMBINED_WITH_PARENT - Parent of this operation is a DFO that performs both operations in the parallel execution plan • PARALLEL_COMBINED_WITH_CHILD - Child of this operation is a DFO that performs both operations in the parallel execution plan. • PARALLEL_TO_SERIAL - SQL text shown in the OTHER column is the top-level of the parallel plan. • PARALLEL_TO_PARALLEL - SQL text shown in the OTHER column is executed and output in parallel • PARALLEL_FROM_SERIAL - Operation consumes data from a serial operation and outputs it in parallel
PARTITION_START	VARCHAR2(255)		Start partition of a range of accessed partitions
PARTITION_STOP	VARCHAR2(255)		Stop partition of a range of accessed partitions
PARTITION_ID	NUMBER(38)		Step that has computed the pair of values of the PARTITION_START and PARTITION_STOP columns
OTHER	LONG		Information about parallel execution servers and parallel queries
DISTRIBUTION	VARCHAR2(30)		Distribution method
CPU_COST	NUMBER(38)		User-defined CPU cost
IO_COST	NUMBER(38)		User-defined I/O cost
TEMP_SPACE	NUMBER(38)		Temporary space usage of the operation (sort or hash-join) as estimated by the CBO
ACCESS_PREDICATES	VARCHAR2(4000)		Predicates used to locate rows in an access structure. For example, start or stop predicates for an index range scan.
FILTER_PREDICATES	VARCHAR2(4000)		Predicates used to filter rows before producing them
PROJECTION	VARCHAR2(4000)		Expressions produced by the operation
TIME	NUMBER(38)		Elapsed time (in seconds) of the operation as estimated by the CBO
QBLOCK_NAME	VARCHAR2(128)		Name of the query block

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OTHER_XML	CLOB		Provides extra information specific to an execution step of the execution plan. The content of this column is structured using XML because it allows multiple pieces of information to be stored, including the following: • Name of the schema against which the query was parsed • Release number of the Oracle Database that produced the explain plan • Hash value associated with the execution plan • Name (if any) of the outline or the SQL profile used to build the execution plan • Indication of whether or not dynamic statistics were used to produce the plan • The outline data, a set of optimizer hints that can be used to regenerate the same plan

See Also

"USER_ADVISOR_SQLPLANS"

4.39 DBA_ADVISOR_SQLSTATS

DBA_ADVISOR_SQLSTATS displays execution statistics for the test-execution of different SQL plans during the advisor analysis.

Related View

USER_ADVISOR_SQLSTATS displays execution statistics owned by the current user for the test-execution of different SQL plans during the advisor analysis.

Column	Datatype	NULL	Description
TASK_NAME	VARCHAR2(128)		Advisor task name in which the SQL statement was executed (see DBA_ADVISOR_TASKS)
TASK_ID	NUMBER(38)	NOT NULL	Advisor task ID in which the SQL statement was executed (see DBA_ADVISOR_TASKS)
EXECUTION_NAME	VARCHAR2(128)	NOT NULL	Advisor task execution in which the SQL statement was executed (see DBA_ADVISOR_EXECUTIONS)
EXECUTION_TYPE	VARCHAR2(128)		Type of the advisor task execution in which the SQL statement was executed (see DBA_ADVISOR_EXECUTIONS)
OBJECT_ID	NUMBER(38)	NOT NULL	Advisor object ID identifying the relevant SQL statement (see DBA_ADVISOR_OBJECTS)
PLAN_ID	NUMBER	NOT NULL	Plan ID number generated to uniquely identify a plan for a particular SQL statement (foreign key to DBA_ADVISOR_SQLPLANS)
SQL_ID	VARCHAR2(13)	NOT NULL	Identifier for the SQL statement executed
PLAN_HASH_VALUE	NUMBER	NOT NULL	Hash value of the SQL execution plan

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ATTR1	NUMBER		For internal use only
CON_DBID	NUMBER		The database ID of the pluggable database (PDB)
PARSE_TIME	NUMBER		Parse time (in microseconds) measured for the SQL
ELAPSED_TIME	NUMBER		Elapsed time (in microseconds) to execute the SQL and fetch all of its rows, after parsing
CPU_TIME	NUMBER		CPU time (in microseconds) to execute the SQL and fetch all of its rows, after parsing
USER_IO_TIME	NUMBER		I/O time (in microseconds) to execute the SQL and fetch all of its rows, after parsing
BUFFER_GETS	NUMBER		Number of buffer gets measured for executing the SQL and fetching all of its rows
DISK_READS	NUMBER		Number of disk reads measured for executing the SQL and fetching all of its rows
DIRECT_WRITES	NUMBER		Number of direct writes measured for executing the SQL and fetching all of its rows
PHYSICAL_READ_REQUESTS	NUMBER		Number of physical read I/O requests issued by the monitored SQL
PHYSICAL_WRITE_REQUESTS	NUMBER		Number of physical write I/O requests issued by the monitored SQL
PHYSICAL_READ_BYTES	NUMBER		Number of bytes read from disks by the monitored SQL
PHYSICAL_WRITE_BYTES	NUMBER		Number of bytes written to disks by the monitored SQL
ROWS_PROCESSED	NUMBER		Number of rows returned by the SQL execution
FETCHES	NUMBER		Number of fetches for the SQL execution
EXECUTIONS	NUMBER		Execution count for the SQL. This column will always have a value of `1` or `0`.
END_OF_FETCH_COUNT	NUMBER		Indicates whether the SQL was executed to end-of-fetch (`1`) or not (`0`)
OPTIMIZER_COST	NUMBER		Optimizer cost for the execution plan
OTHER	CLOB		For internal use only
TESTEXEC_TOTAL_EXECS	NUMBER		Total number of executions during test execute
IO_INTERCONNECT_BYTES	NUMBER		Number of I/O bytes exchanged between Oracle Database and the storage system
TESTEXEC_FIRST_EXEC_IGNORED	VARCHAR2(1)		Indicates whether the first execution in test execute is ignored (`Y`) or not (`N`)
CON_DBID	NUMBER		The database ID of the PDB
ATTR2	NUMBER		For internal use only
ATTR3	NUMBER		For internal use only
CACHED_GETS	NUMBER		The total of the db block gets from cache statistic and the consistent gets from cache statistic
DIRECT_GETS	NUMBER		The total of the db block gets direct statistic and the consistent gets direct statistic

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "USER_ADVISOR_SQLSTATS"
- "Statistics Descriptions" for more information about statistics

4.40 DBA_ADVISOR_SQLW_JOURNAL

DBA_ADVISOR_SQLW_JOURNAL displays the journal entries for all workload objects in the database.

Related View

USER_ADVISOR_SQLW_JOURNAL displays the journal entries for the workload objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the workload
WORKLOAD_ID	NUMBER	NOT NULL	Identifier number of the workload object
WORKLOAD_NAME	VARCHAR2(128)		Name of the workload object
JOURNAL_ENTRY_SEQ	NUMBER	NOT NULL	Sequence number of the journal entry (unique for each workload). The sequence number is used to order the data.
JOURNAL_ENTRY_TYPE	VARCHAR2(12)		Type of the task: • FATAL • ERROR • WARNING • INFORMATION • INFORMATION[2 | 3 | 4 | 5 | 6]
JOURNAL_ENTRY	VARCHAR2(4000)		Entry in the journal

See Also

"USER_ADVISOR_SQLW_JOURNAL"

4.41 DBA_ADVISOR_SQLW_PARAMETERS

DBA_ADVISOR_SQLW_PARAMETERS displays all workload parameters and their current values in the database.

Related View

USER_ADVISOR_SQLW_PARAMETERS displays the workload parameters and their current values owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task or workload object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
WORKLOAD_ID	NUMBER	NOT NULL	Unique identifier number of the workload object
WORKLOAD_NAME	VARCHAR2(128)		Name of the workload object
PARAMETER_NAME	VARCHAR2(128)	NOT NULL	Name of the parameter
PARAMETER_VALUE	VARCHAR2(4000)	NOT NULL	Value of the parameter. Numeric parameter values are converted to a string equivalent. Possible keywords as values: • ALL • UNLIMITED • UNUSED
PARAMETER_TYPE	VARCHAR2(10)		Datatype of the parameter: • NUMBER - Numeric value • STRING - String value. If the string contains special characters, then it will be enclosed in single quotes. • STRINGLIST - Comma-separated list of string elements. If a string element contains a comma or other special characters, then the element will be enclosed in single quotes. • TABLE - Single table reference. A reference contains a schema name, followed by an optional table name. If the table name is omitted or is the character %, then the table name is interpreted as a wildcard. SQL quoted identifiers are supported. • TABLELIST - List of one or more comma-separated table references. A reference contains a schema name, followed by an optional table name. If the table name is omitted or is the character %, then the table name is interpreted as a wildcard. SQL quoted identifiers are supported.
DESCRIPTION	VARCHAR2(4000)		Parameter description

See Also

"USER_ADVISOR_SQLW_PARAMETERS"

4.42 DBA_ADVISOR_SQLW_STMTS

DBA_ADVISOR_SQLW_STMTS displays rows that correspond to all statements in the workload.

All columns are guaranteed to be non-null.

Related View

USER_ADVISOR_SQLW_STMTS displays rows that correspond to the statements in the workload owned by the current user. This view does not display the OWNER column.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the workload object
WORKLOAD_ID	NUMBER	NOT NULL	Unique identifier number of the workload object
WORKLOAD_NAME	VARCHAR2(128)		Name of the workload
SQL_ID	NUMBER	NOT NULL	Generated identifier of the statement
HASH_VALUE	NUMBER		Hash value for the parent statement in the cache
USERNAME	VARCHAR2(128)		Name of the user executing the statement
MODULE	VARCHAR2(64)		Name of the module issuing the statement
ACTION	VARCHAR2(64)		Module action for the statement
CPU_TIME	NUMBER		Total CPU count (in seconds) of the executing statement
BUFFER_GETS	NUMBER		Total number of buffer gets for the statement
DISK_READS	NUMBER		Total disk-read I/O count for the statement
ELAPSED_TIME	NUMBER		Total elapsed time (in seconds) of the executing statement
ROWS_PROCESSED	NUMBER		Total number of rows processed by the statement
EXECUTIONS	NUMBER		Total number of times the statement was executed
OPTIMIZER_COST	NUMBER		Cost of executing the statement in the workload prior to the recommendations
LAST_EXECUTION_DATE	DATE		Date on which the statement was last executed
PRIORITY	NUMBER		Priority of the statement: • 1 - High • 2 - Medium • 3 - Low
COMMAND_TYPE	NUMBER		Type of the command
STAT_PERIOD	NUMBER		Unused
SQL_TEXT	CLOB		Text of the SQL statement
VALID	NUMBER		Indicates whether the statement is valid for analysis: • 0 - Statement will not be analyzed by the EXECUTE_TASK procedure. Typically, the statement references one or more tables that do not have valid statistics. To correct this problem, ensure that the tables have valid statistics and execute the RESET_SQLWKLD procedure on the current workload. • 1 - Statement is eligible for analysis by the EXECUTE_TASK procedure.

See Also

"USER_ADVISOR_SQLW_STMTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.43 DBA_ADVISOR_SQLW_SUM

DBA_ADVISOR_SQLW_SUM displays an aggregated picture of all SQLWkld workload objects in the database.

Related View

USER_ADVISOR_SQLW_SUM displays an aggregated picture of the SQLWkld workload objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the workload object
WORKLOAD_ID	NUMBER	NOT NULL	Unique identifier number of the workload object
WORKLOAD_NAME	VARCHAR2(128)		Unique name of the workload
DESCRIPTION	VARCHAR2(256)		User-specified description of the workload
CREATE_DATE	DATE	NOT NULL	Date on which the workload object was created
MODIFY_DATE	DATE	NOT NULL	Date of last update to the current workload
NUM_SELECT_STMT	NUMBER		Number of SELECT statements in the workload
NUM_UPDATE_STMT	NUMBER		Number of UPDATE statements in the workload
NUM_DELETE_STMT	NUMBER		Number of DELETE statements in the workload
NUM_INSERT_STMT	NUMBER		Number of INSERT statements in the workload
NUM_MERGE_STMT	NUMBER		Number of MERGE statements in the workload
SOURCE	VARCHAR2(128)		Optional name that identifies the creator of the object
HOW_CREATED	VARCHAR2(30)		Optional object or template on which the object was based
DATA_SOURCE	VARCHAR2(2000)		Workload data source
READ_ONLY	VARCHAR2(5)		Indicates whether or not the workload can be modified or deleted (TRUE) or not (FALSE)

See Also

"USER_ADVISOR_SQLW_SUM"

4.44 DBA_ADVISOR_SQLW_TABLES

DBA_ADVISOR_SQLW_TABLES displays cross references between the workload statements and the tables referenced in the statement.

Related View

USER_ADVISOR_SQLW_TABLES displays cross references between the workload statements and the tables referenced in the statement. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the workload object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
WORKLOAD_ID	NUMBER		Unique identifier number of the workload object
WORKLOAD_NAME	VARCHAR2(128)		Name of the workload
SQL_ID	NUMBER		Identifier of the statement
TABLE_OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table

See Also

"USER_ADVISOR_SQLW_TABLES"

4.45 DBA_ADVISOR_SQLW_TEMPLATES

DBA_ADVISOR_SQLW_TEMPLATES displays an aggregated picture of all SQLWkld template objects in the database.

Related View

USER_ADVISOR_SQLW_TEMPLATES displays an aggregated picture of the SQLWkld template objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the workload object
WORKLOAD_ID	NUMBER	NOT NULL	Unique identifier number of the workload object
WORKLOAD_NAME	VARCHAR2(128)		Unique name of the workload
DESCRIPTION	VARCHAR2(256)		User-specified description of the workload
CREATE_DATE	DATE	NOT NULL	Date on which the workload object was created
MODIFY_DATE	DATE	NOT NULL	Date of last update to the current workload
SOURCE	VARCHAR2(128)		Optional object or template on which the object was based
READ_ONLY	VARCHAR2(5)		Indicates whether the workload template can be modified or deleted (TRUE) or not (FALSE)

See Also

"USER_ADVISOR_SQLW_TEMPLATES"

4.46 DBA_ADVISOR_TASKS

DBA_ADVISOR_TASKS displays information about all tasks in the database.

The view contains one row for each task. Each task has a name that is unique to the owner. Task names are just informational and no uniqueness is enforced within any other namespace.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Related View

USER_ADVISOR_TASKS displays information about the tasks owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER	NOT NULL	Unique identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task
DESCRIPTION	VARCHAR2(256)		User-supplied description of the task
ADVISOR_NAME	VARCHAR2(128)		Advisor associated with the task
CREATED	DATE	NOT NULL	Creation date of the task
LAST_MODIFIED	DATE	NOT NULL	Date on which the task was last modified
PARENT_TASK_ID	NUMBER		Identifier of the parent task (if the task was created because of the recommendation of another task)
PARENT_RXEC_ID	NUMBER		Identifier of the recommendation within the parent task that resulted in the creation of the task
LAST_EXECUTION	VARCHAR2(128)		Name of the current or last task execution
EXECUTION_TYPE	VARCHAR2(128)		Type of the last execution. This information is optional for single-execution tasks.
EXECUTION_TYPE#	NUMBER		Reserved for internal use
EXECUTION_DESCRIPTION	VARCHAR2(256)		Optional description of the last execution
EXECUTION_START	DATE		Execution start date and time of the task
EXECUTION_END	DATE		Execution end date and time of the task
STATUS	VARCHAR2(11)		Current operational status of the task: • INITIAL - Initial state of the task; no recommendations are present • EXECUTING - Task is currently running • INTERRUPTED - Task analysis was interrupted by the user. Recommendation data, if present, can be viewed and reported at this time. • COMPLETED - Task successfully completed the analysis operation. Recommendation data can be viewed and reported. • ERROR - An error occurred during the analysis operation. Recommendations, if present, can be viewed and reported at this time.
STATUS_MESSAGE	VARCHAR2(4000)		Informational message provided by the advisor regarding the status
PCT_COMPLETION_TIME	NUMBER		Percent completion, in terms of time, of the task when it is executing
PROGRESS_METRIC	NUMBER		Metric that measures the progress of the task in terms of quality. Each advisor may have its own metric.
METRIC_UNITS	VARCHAR2(64)		Unit of the metric used to measure progress
ACTIVITY_COUNTER	NUMBER		Counter that is updated frequently by the advisor, denoting that useful work is being performed
RECOMMENDATION_COUNT	NUMBER		Number of recommendations produced

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ERROR_MESSAGE	VARCHAR2(4000)		Informational message or an error message indicating the current operation or condition
SOURCE	VARCHAR2(128)		Optional name that identifies the creator of the task
HOW_CREATED	VARCHAR2(30)		Optional task or template on which the object was based
READ_ONLY	VARCHAR2(5)		Indicates whether the task is read-only (TRUE) or not (FALSE)
SYSTEM_TASK	VARCHAR2(5)		Indicates whether the task is a system task (TRUE) or not (FALSE). The automatic SQL tuning task, SYS_AUTO_SQL_TUNING_TASK, is one example of a system task.
ADVISOR_ID	NUMBER	NOT NULL	Unique identifier for the advisor
STATUS#	NUMBER		Reserved for internal use

See Also

"USER_ADVISOR_TASKS"

4.47 DBA_ADVISOR_TEMPLATES

DBA_ADVISOR_TEMPLATES displays information about all templates in the database.

Related View

USER_ADVISOR_TEMPLATES displays information about the templates owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_ID	NUMBER	NOT NULL	Unique identifier of the task
TASK_NAME	VARCHAR2(128)		Name of the task
DESCRIPTION	VARCHAR2(256)		User-supplied description of the task
ADVISOR_NAME	VARCHAR2(128)		Advisor associated with the task
CREATED	DATE	NOT NULL	Creation date of the task
LAST_MODIFIED	DATE	NOT NULL	Date on which the task was last modified
SOURCE	VARCHAR2(128)		Optional task or template on which the template was based
READ_ONLY	VARCHAR2(5)		Indicates whether the task can be modified or deleted (TRUE) or not (FALSE)

See Also

"USER_ADVISOR_TEMPLATES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.48 DBA_ADVISOR_USAGE

DBA_ADVISOR_USAGE displays the usage information for each type of advisor in the database.

Column	Datatype	NULL	Description
ADVISOR_ID	NUMBER	NOT NULL	Type of the advisor
ADVISOR_NAME	VARCHAR2(128)	NOT NULL	Name of the advisor
LAST_EXEC_TIME	DATE	NOT NULL	Date of the last execution
NUM_EXECS	NUMBER	NOT NULL	Cumulative number of executions
NUM_DB_REPORTS	NUMBER	NOT NULL	Cumulative number of reports
FIRST_REPORT_TIME	DATE		Time of the first report
LAST_REPORT_TIME	DATE		Time of the last report

4.49 DBA_ALERT_HISTORY

DBA_ALERT_HISTORY describes a time-limited history of alerts which are no longer outstanding.

Column	Datatype	NULL	Description
SEQUENCE_ID	NUMBER	NOT NULL	Alert sequence number
REASON_ID	NUMBER		ID of the alert reason
OWNER	VARCHAR2(128)		Owner of the object on which the alert was issued
OBJECT_NAME	VARCHAR2(513)		Name of the object
SUBOBJECT_NAME	VARCHAR2(128)		Name of the subobject (for example: partition)
OBJECT_TYPE	VARCHAR2(64)		Type of the object (for example: table, tablespace)
REASON	VARCHAR2(4000)		Reason for the alert
TIME_SUGGESTED	TIMESTAMP(6) WITH TIME ZONE		Time when the alert was last updated
CREATION_TIME	TIMESTAMP(6) WITH TIME ZONE		Time when the alert was first created
SUGGESTED_ACTION	VARCHAR2(4000)		Advice of the recommended action
ADVISOR_NAME	VARCHAR2(128)		Name of the advisor to be invoked for more information
METRIC_VALUE	NUMBER		Value of the related metrics
MESSAGE_TYPE	VARCHAR2(12)		Message type: • Notification • Warning
MESSAGE_GROUP	VARCHAR2(64)		Name of the message group to which the alert belongs
MESSAGE_LEVEL	NUMBER		Severity message level (1 to 32)
HOSTING_CLIENT_ID	VARCHAR2(64)		ID of the client or security group to which the alert relates
MODULE_ID	VARCHAR2(64)		ID of the module that originated the alert
PROCESS_ID	VARCHAR2(128)		Process ID
HOST_ID	VARCHAR2(256)		DNS host name of the originating host

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
HOST_NW_ADDR	VARCHAR2(256)		IP or other network address of the originating host
INSTANCE_NAME	VARCHAR2(16)		Originating instance name
INSTANCE_NUMBER	NUMBER		Originating instance number
USER_ID	VARCHAR2(128)		User ID
EXECUTION_CONTEXT_ID	VARCHAR2(128)		Execution Context ID
ERROR_INSTANCE_ID	VARCHAR2(142)		ID of an error instance plus a sequence number
RESOLUTION	VARCHAR2(7)		Resolution: • Cleared • N/A
PDB_NAME	VARCHAR2(128)		PDB name
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.50 DBA_ALERT_HISTORY_DETAIL

DBA_ALERT_HISTORY_DETAIL describes a time-limited history of cleared and outstanding alerts.

Column	Datatype	NULL	Description
SEQUENCE_ID	NUMBER	NOT NULL	Alert sequence number
REASON_ID	NUMBER		ID of the alert reason
OWNER	VARCHAR2(128)		Owner of the object on which alert is issued
OBJECT_NAME	VARCHAR2(513)		Name of the object
SUBOBJECT_NAME	VARCHAR2(128)		Name of the subobject (for example: partition)
OBJECT_TYPE	VARCHAR2(64)		Type of the object (for example: table, tablespace)
REASON	VARCHAR2(4000)		Reason for the alert
TIME_SUGGESTED	TIMESTAMP(6) WITH TIME ZONE		Time when the alert was last updated
CREATION_TIME	TIMESTAMP(6) WITH TIME ZONE		Time when the alert was first created
SUGGESTED_ACTION	VARCHAR2(4000)		Advice of the recommended action
ADVISOR_NAME	VARCHAR2(128)		Name of the advisor to be invoked for more information
METRIC_VALUE	NUMBER		Value of the related metrics
MESSAGE_TYPE	VARCHAR2(12)		Message type: • Notification • Warning
MESSAGE_GROUP	VARCHAR2(64)		Name of the message group to which the alert belongs

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MESSAGE_LEVEL	NUMBER		Severity level (1 to 32)
HOSTING_CLIENT_ID	VARCHAR2(64)		ID of the client or security group to which the alert relates
MODULE_ID	VARCHAR2(64)		ID of the module that originated the alert
PROCESS_ID	VARCHAR2(128)		Process id
HOST_ID	VARCHAR2(256)		DNS host name of the originating host
HOST_NW_ADDR	VARCHAR2(256)		IP or other network address of originating host
INSTANCE_NAME	VARCHAR2(16)		Originating instance name
INSTANCE_NUMBER	NUMBER		Originating instance number
USER_ID	VARCHAR2(128)		User ID
EXECUTION_CONTEXT_ID	VARCHAR2(128)		Execution Context ID
ERROR_INSTANCE_ID	VARCHAR2(142)		ID of an error instance plus a sequence number
RESOLUTION	VARCHAR2(11)		Resolution: • Cleared • Outstanding • N/A
STATE_TRANSITION_NUMBER	NUMBER	NOT NULL	Sequence number of the state transition for the alert
PDB_NAME	VARCHAR2(128)		PDB name
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.51 DBA_ALL_TABLES

DBA_ALL_TABLES describes all object tables and relational tables in the database. Its columns are the same as those in ALL_ALL_TABLES.

See Also

"ALL_ALL_TABLES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.52 DBA_ANALYTIC_VIEW_ATTR_CLASS

DBA_ANALYTIC_VIEW_ATTR_CLASS describes the attribute classifications of all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_ATTR_CLASS.

See Also

"ALL_ANALYTIC_VIEW_ATTR_CLASS"

4.53 DBA_ANALYTIC_VIEW_ATTR_CLS

DBA_ANALYTIC_VIEW_ATTR_CLS is identical to DBA_ANALYTIC_VIEW_ATTR_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"DBA_ANALYTIC_VIEW_ATTR_CLASS"

4.54 DBA_ANALYTIC_VIEW_ATTR_CLS_AE

DBA_ANALYTIC_VIEW_ATTR_CLS_AE describes the attribute classifications of all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_ATTR_CLS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_ATTR_CLS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.55 DBA_ANALYTIC_VIEW_BAS_MEAS

DBA_ANALYTIC_VIEW_BAS_MEAS is identical to DBA_ANALYTIC_VIEW_BASE_MEAS.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "DBA_ANALYTIC_VIEW_BASE_MEAS"

4.56 DBA_ANALYTIC_VIEW_BAS_MEAS_AE

DBA_ANALYTIC_VIEW_BAS_MEAS_AE describes the base measures in all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_BAS_MEAS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_BAS_MEAS_AE"

4.57 DBA_ANALYTIC_VIEW_BASE_MEAS

DBA_ANALYTIC_VIEW_BASE_MEAS describes the base measures in all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_BASE_MEAS.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_BASE_MEAS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.58 DBA_ANALYTIC_VIEW_CALC_MEAS

DBA_ANALYTIC_VIEW_CALC_MEAS describes the calculated measures in all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_CALC_MEAS.

See Also

"ALL_ANALYTIC_VIEW_CALC_MEAS"

4.59 DBA_ANALYTIC_VIEW_CLASS

DBA_ANALYTIC_VIEW_CLASS describes the classifications of all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_CLASS.

See Also

"ALL_ANALYTIC_VIEW_CLASS"

4.60 DBA_ANALYTIC_VIEW_CLASS_AE

DBA_ANALYTIC_VIEW_CLASS_AE describes the classifications of all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_CLASS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_CLASS_AE"

4.61 DBA_ANALYTIC_VIEW_CLC_MEAS

DBA_ANALYTIC_VIEW_CLC_MEAS is identical to DBA_ANALYTIC_VIEW_CALC_MEAS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"DBA_ANALYTIC_VIEW_CALC_MEAS"

4.62 DBA_ANALYTIC_VIEW_CLC_MEAS_AE

DBA_ANALYTIC_VIEW_CLC_MEAS_AE describes the calculated measures in all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_CLC_MEAS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_CLC_MEAS_AE"

4.63 DBA_ANALYTIC_VIEW_COLUMNS

DBA_ANALYTIC_VIEW_COLUMNS describes the columns of all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_COLUMNS.

See Also

"ALL_ANALYTIC_VIEW_COLUMNS"

4.64 DBA_ANALYTIC_VIEW_COLUMNS_AE

DBA_ANALYTIC_VIEW_COLUMNS_AE describes the columns of all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_COLUMNS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_COLUMNS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.65 DBA_ANALYTIC_VIEW_DIM_CLASS

DBA_ANALYTIC_VIEW_DIM_CLASS describes the classifications of the attribute dimensions in all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_DIM_CLASS.

See Also

"ALL_ANALYTIC_VIEW_DIM_CLASS"

4.66 DBA_ANALYTIC_VIEW_DIM_CLS

DBA_ANALYTIC_VIEW_DIM_CLS is identical to DBA_ANALYTIC_VIEW_DIM_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"DBA_ANALYTIC_VIEW_DIM_CLASS"

4.67 DBA_ANALYTIC_VIEW_DIM_CLS_AE

DBA_ANALYTIC_VIEW_DIM_CLS_AE describes the classifications of the attribute dimensions in all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_DIM_CLS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_DIM_CLS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.68 DBA_ANALYTIC_VIEW_DIMENSIONS

DBA_ANALYTIC_VIEW_DIMENSIONS describes the attribute dimensions in all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_DIMENSIONS.

See Also

"ALL_ANALYTIC_VIEW_DIMENSIONS"

4.69 DBA_ANALYTIC_VIEW_DIMS

DBA_ANALYTIC_VIEW_DIMS is identical to DBA_ANALYTIC_VIEW_DIMENSIONS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"DBA_ANALYTIC_VIEW_DIMENSIONS"

4.70 DBA_ANALYTIC_VIEW_DIMS_AE

DBA_ANALYTIC_VIEW_DIMS_AE describes the attribute dimensions in all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_DIMS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_DIMS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.71 DBA_ANALYTIC_VIEW_HIER_CLASS

DBA_ANALYTIC_VIEW_HIER_CLASS describes the classifications of the hierarchies in all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_HIER_CLASS.

See Also

"ALL_ANALYTIC_VIEW_HIER_CLASS"

4.72 DBA_ANALYTIC_VIEW_HIER_CLS

DBA_ANALYTIC_VIEW_HIER_CLS is identical to DBA_ANALYTIC_VIEW_HIER_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"DBA_ANALYTIC_VIEW_HIER_CLASS"

4.73 DBA_ANALYTIC_VIEW_HIER_CLS_AE

DBA_ANALYTIC_VIEW_HIER_CLS_AE describes the classifications of the hierarchies in all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_HIER_CLS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_HIER_CLS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.74 DBA_ANALYTIC_VIEW_HIERS

DBA_ANALYTIC_VIEW_HIERS describes the hierarchies in all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_HIERS.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_HIERS"

4.75 DBA_ANALYTIC_VIEW_HIERS_AE

DBA_ANALYTIC_VIEW_HIERS_AE describes the hierarchies in all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_HIERS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_HIERS_AE"

4.76 DBA_ANALYTIC_VIEW_KEYS

DBA_ANALYTIC_VIEW_KEYS describes the key columns of the attribute dimensions in all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_KEYS.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_KEYS"

4.77 DBA_ANALYTIC_VIEW_KEYS_AE

DBA_ANALYTIC_VIEW_KEYS_AE describes the key columns of the attribute dimensions in all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_KEYS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_KEYS_AE"

4.78 DBA_ANALYTIC_VIEW_LEVEL_CLASS

DBA_ANALYTIC_VIEW_LEVEL_CLASS describes the level classifications of all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_LEVEL_CLASS.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_LEVEL_CLASS"

4.79 DBA_ANALYTIC_VIEW_LEVELS

DBA_ANALYTIC_VIEW_LEVELS describes the levels in the hierarchies in all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_LEVELS.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_LEVELS"

4.80 DBA_ANALYTIC_VIEW_LEVELS_AE

DBA_ANALYTIC_VIEW_LEVELS_AE describes the levels in the hierarchies in all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_LEVELS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_LEVELS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.81 DBA_ANALYTIC_VIEW_LVL_CLS

DBA_ANALYTIC_VIEW_LVL_CLS is identical to DBA_ANALYTIC_VIEW_LEVEL_CLASS.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "DBA_ANALYTIC_VIEW_LEVEL_CLASS"

4.82 DBA_ANALYTIC_VIEW_LVL_CLS_AE

DBA_ANALYTIC_VIEW_LVL_CLS_AE describes the level classifications of all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_LVL_CLS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_LVL_CLS_AE"

4.83 DBA_ANALYTIC_VIEW_LVLGRPS

DBA_ANALYTIC_VIEW_LVLGRPS describes the analytic view measure and level groups of all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_LVLGRPS.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_LVLGRPS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.84 DBA_ANALYTIC_VIEW_LVLGRPS_AE

DBA_ANALYTIC_VIEW_LVLGRPS_AE describes the analytic view measure and level groups of all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_LVLGRPS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_LVLGRPS_AE"

4.85 DBA_ANALYTIC_VIEW_MEAS_CLASS

DBA_ANALYTIC_VIEW_MEAS_CLASS describes the classifications of the measures of all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_MEAS_CLASS.

See Also

"ALL_ANALYTIC_VIEW_MEAS_CLASS"

4.86 DBA_ANALYTIC_VIEW_MEAS_CLS

DBA_ANALYTIC_VIEW_MEAS_CLS is identical to DBA_ANALYTIC_VIEW_MEAS_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"DBA_ANALYTIC_VIEW_MEAS_CLASS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.87 DBA_ANALYTIC_VIEW_MEAS_CLS_AE

DBA_ANALYTIC_VIEW_MEAS_CLS_AE describes the classifications of the measures of all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEW_MEAS_CLS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_MEAS_CLS_AE"

4.88 DBA_ANALYTIC_VIEWS

DBA_ANALYTIC_VIEWS describes all analytic views in the database. Its columns are the same as those in ALL_ANALYTIC_VIEWS.

See Also

"ALL_ANALYTIC_VIEWS"

4.89 DBA_ANALYTIC_VIEWS_AE

DBA_ANALYTIC_VIEWS_AE describes all analytic views (across all editions) in the database. Its columns are the same as those in ALL_ANALYTIC_VIEWS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEWS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.90 DBA_ANNOTATION_VALUES

DBA_ANNOTATION_VALUES displays the values for all schema annotations in the database. Its columns are the same as those in ALL_ANNOTATION_VALUES.

Note

This view is available starting with Oracle Database 19c, Release Update 19.28.

See Also

"ALL_ANNOTATION_VALUES"

4.91 DBA_ANNOTATIONS

DBA_ANNOTATIONS displays all schema annotations in the database. Its columns are the same as those in ALL_ANNOTATIONS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.28.

See Also

"ALL_ANNOTATIONS"

4.92 DBA_ANNOTATIONS_USAGE

DBA_ANNOTATIONS_USAGE provides usage information about all schema annotations in the database. Its columns are the same as those in ALL_ANNOTATIONS_USAGE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.28.

See Also

"ALL_ANNOTATIONS_USAGE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.93 DBA_APP_ERRORS

DBA_APP_ERRORS displays errors raised when an application PDB synchronizes with an application in the application root.

This view displays errors raised during the last synchronization for each application.

Column	Datatype	NULL	Description
APP_NAME	VARCHAR2(128)		Name of the application whose statement was captured
APP_STATEMENT	CLOB		Application statement
ERRORNUM	NUMBER		Error number for the statement
ERRORMSG	VARCHAR2(4000)		Error message for the statement
SYNC_TIME	DATE	NOT NULL	Sync time for the statement
SYSTEM_IGNORABLE	VARCHAR2(1)		Indicates whether the error is a system-ignorable error (Y) or not (N)
USER_IGNORABLE	VARCHAR2(1)		Indicates whether the error is a user-ignorable error (Y) or not (N)

4.94 DBA_APP_PATCHES

DBA_APP_PATCHES describes all the application patches in the application container.

Column	Datatype	NULL	Description
APP_NAME	VARCHAR2(128)		Name of the application
PATCH_NUMBER	NUMBER		Patch number
PATCH_MIN_VERSION	VARCHAR2(30)		Minimum application version for the patch
PATCH_STATUS	VARCHAR2(10)		Status of the patch
PATCH_COMMENT	VARCHAR2(4000)		Comment associated with the patch
PATCH_CHECKSUM	NUMBER		Checksum for the patch

4.95 DBA_APP_PDB_STATUS

DBA_APP_PDB_STATUS provides information about applications in all the application PDBs in the current application container. It provides this information when queried in the application root.

The view should be queried in the application root.

This view can be used to show which version of an application has been synced to which application PDBs.

Column	Datatype	NULL	Description
CON_UID	NUMBER		Unique ID of the PDB
APP_NAME	VARCHAR2(128)		Name of the application
APP_ID	NUMBER		Id of the application

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
APP_VERSION	VARCHAR2(30)		Version of the application
APP_STATUS	VARCHAR2(12)		Status of the application

4.96 DBA_APP_STATEMENTS

DBA_APP_STATEMENTS describes all statements from all the applications in the application container.

Column	Datatype	NULL	Description
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n*=1 if the row originates in root)
STATEMENT_ID	NUMBER		Statement ID
CAPTURE_TIME	DATE	NOT NULL	Time of capture of the application statement
APP_STATEMENT	CLOB		Application statement
APP_NAME	VARCHAR2(128)		Name of the application whose statement was captured
APP_STATUS	VARCHAR2(12)		Status of the application when the statement was captured
PATCH_NUMBER	NUMBER		Patch number of patch installation when the statement was captured
VERSION_NUMBER	NUMBER		Version number when the statement was captured
SESSION_ID	NUMBER		Unique session ID when the statement was captured
OPCODE	NUMBER	NOT NULL	Operation code indicating the statement type

4.97 DBA_APP_VERSIONS

DBA_APP_VERSIONS displays information about all application versions installed in an application container.

Column	Datatype	NULL	Description
APP_NAME	VARCHAR2(128)		Name of the application
APP_VERSION	VARCHAR2(30)		Version of the application
APP_VERSION_COMMENT	VARCHAR2(4000)		Comment associated with the application version
APP_VERSION_CHECKSUM	NUMBER		Checksum for the application version
APP_ROOT_CLONE_NAME[1]	VARCHAR2(64)		Name of the application root clone that corresponds to the application
APP_VERSION_BEGIN_TIME[2]	DATE		Begin time of the most recent sync of the application
APP_VERSION_END_TIME[2]	DATE		End time of the most recent sync of the application

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
APP_VERSION_NUMBER[2]	NUMBER		Version number of the application version

1 This column is available starting with Oracle Database 19c, Release Update 19.7.

2 This column is available only on Oracle Autonomous Database Serverless.

> **See Also**
>
> *Oracle Multitenant Administrator's Guide* for more information about application containers

4.98 DBA_APPLICATION_ROLES

DBA_APPLICATION_ROLES describes all the roles that have authentication policy functions defined.

Column	Datatype	NULL	Description
ROLE	VARCHAR2(128)	NOT NULL	Name of the application role
SCHEMA	VARCHAR2(128)	NOT NULL	Schema of the authorized package
PACKAGE	VARCHAR2(128)	NOT NULL	Name of the authorized package

4.99 DBA_APPLICATIONS

DBA_APPLICATIONS provides information about the applications in the current application container.

Column	Datatype	NULL	Description
APP_NAME	VARCHAR2(128)		Name of the application
APP_ID	NUMBER		ID of the application
APP_VERSION	VARCHAR2(30)		Version of the application
APP_STATUS	VARCHAR2(12)		Status of the application
APP_IMPLICIT	VARCHAR2(1)		Indicates whether the application is implicit (Y) or not (N)
APP_CAPTURE_SERVICE	VARCHAR2(64)		Service name used for the capture
APP_CAPTURE_MODULE	VARCHAR2(64)		Module name used for the capture

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.100 DBA_APPLY

DBA_APPLY displays information about all apply processes in the database. Its columns are the same as those in ALL_APPLY.

See Also

"ALL_APPLY"

4.101 DBA_APPLY_CHANGE_HANDLERS

DBA_APPLY_CHANGE_HANDLERS displays information about the change handlers on all tables in the database. Its columns are the same as those in ALL_APPLY_CHANGE_HANDLERS.

See Also

"ALL_APPLY_CHANGE_HANDLERS"

4.102 DBA_APPLY_CONFLICT_COLUMNS

DBA_APPLY_CONFLICT_COLUMNS displays information about the conflict handlers on all tables in the database. Its columns are the same as those in ALL_APPLY_CONFLICT_COLUMNS.

See Also

"ALL_APPLY_CONFLICT_COLUMNS"

4.103 DBA_APPLY_DML_CONF_HANDLERS

DBA_APPLY_DML_CONF_HANDLERS provides details about DML conflict handlers. Its columns are the same as those in ALL_APPLY_DML_CONF_HANDLERS.

See Also

"ALL_APPLY_DML_CONF_HANDLERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.104 DBA_APPLY_DML_HANDLERS

DBA_APPLY_DML_HANDLERS displays information about the DML handlers on all tables in the database. Its columns are the same as those in ALL_APPLY_DML_HANDLERS.

See Also

"ALL_APPLY_DML_HANDLERS"

4.105 DBA_APPLY_ENQUEUE

DBA_APPLY_ENQUEUE displays information about the apply enqueue actions for all rules in the database. Its columns are the same as those in ALL_APPLY_ENQUEUE.

See Also

"ALL_APPLY_ENQUEUE"

4.106 DBA_APPLY_ERROR

DBA_APPLY_ERROR displays information about the error transactions generated by all apply processes in the database. Its columns are the same as those in ALL_APPLY_ERROR.

See Also

"ALL_APPLY_ERROR"

4.107 DBA_APPLY_ERROR_MESSAGES

DBA_APPLY_ERROR_MESSAGES displays information about the individual messages in all of the error transactions generated by all apply processes in the database. Its columns are the same as those in ALL_APPLY_ERROR_MESSAGES.

For XStream inbound servers, each message in an error transaction is an LCR.

Note

- Messages that were spilled from memory to hard disk do not appear in this view.
- This view does not contain information related to XStream outbound servers.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "ALL_APPLY_ERROR_MESSAGES"

4.108 DBA_APPLY_EXECUTE

DBA_APPLY_EXECUTE displays information about the apply execute actions for all rules in the database. Its columns are the same as those in ALL_APPLY_EXECUTE.

> **See Also**
>
> "ALL_APPLY_EXECUTE"

4.109 DBA_APPLY_HANDLE_COLLISIONS

DBA_APPLY_HANDLE_COLLISIONS provides details about apply handlers for collisions at the table level. Its columns are the same as those in ALL_APPLY_HANDLE_COLLISIONS.

> **See Also**
>
> "ALL_APPLY_HANDLE_COLLISIONS"

4.110 DBA_APPLY_INSTANTIATED_GLOBAL

DBA_APPLY_INSTANTIATED_GLOBAL displays information about databases for which an instantiation SCN has been set. Its columns are the same as those in ALL_APPLY_INSTANTIATED_GLOBAL.

> **See Also**
>
> "ALL_APPLY_INSTANTIATED_GLOBAL"

4.111 DBA_APPLY_INSTANTIATED_OBJECTS

DBA_APPLY_INSTANTIATED_OBJECTS displays information about objects for which an instantiation SCN has been set. Its columns are the same as those in ALL_APPLY_INSTANTIATED_OBJECTS.

> **See Also**
>
> "ALL_APPLY_INSTANTIATED_OBJECTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.112 DBA_APPLY_INSTANTIATED_SCHEMAS

DBA_APPLY_INSTANTIATED_SCHEMAS displays information about schemas for which an instantiation SCN has been set. Its columns are the same as those in ALL_APPLY_INSTANTIATED_SCHEMAS.

See Also

"ALL_APPLY_INSTANTIATED_SCHEMAS"

4.113 DBA_APPLY_KEY_COLUMNS

DBA_APPLY_KEY_COLUMNS displays information about the substitute key columns for all tables in the database. Its columns are the same as those in ALL_APPLY_KEY_COLUMNS.

See Also

"ALL_APPLY_KEY_COLUMNS"

4.114 DBA_APPLY_OBJECT_DEPENDENCIES

DBA_APPLY_OBJECT_DEPENDENCIES displays information about the object dependencies for all apply processes in the database.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object
PARENT_OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Parent of the object owner
PARENT_OBJECT_NAME	VARCHAR2(128)	NOT NULL	Parent of the named object

4.115 DBA_APPLY_PARAMETERS

DBA_APPLY_PARAMETERS displays information about the parameters for all apply processes in the database. Its columns are the same as those in ALL_APPLY_PARAMETERS.

See Also

"ALL_APPLY_PARAMETERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.116 DBA_APPLY_PROGRESS

DBA_APPLY_PROGRESS displays information about the progress made by all apply processes in the database. Its columns are the same as those in ALL_APPLY_PROGRESS.

> **See Also**
>
> "ALL_APPLY_PROGRESS"

4.117 DBA_APPLY_REPERROR_HANDLERS

DBA_APPLY_REPERROR_HANDLERS provides details about apply reperror handlers. Its columns are the same as those in ALL_APPLY_REPERROR_HANDLERS.

> **See Also**
>
> "ALL_APPLY_REPERROR_HANDLERS"

4.118 DBA_APPLY_SPILL_TXN

DBA_APPLY_SPILL_TXN displays information about the transactions spilled from memory to hard disk by all apply processes in the database.

Column	Datatype	NULL	Description
APPLY_NAME	VARCHAR2(128)	NOT NULL	Name of the apply process that spilled one or more transactions
XIDUSN	NUMBER	NOT NULL	Transaction ID undo segment number
XIDSLT	NUMBER	NOT NULL	Transaction ID slot number
XIDSQN	NUMBER	NOT NULL	Transaction ID sequence number
PDB_ID	NUMBER		PDB ID number
FIRST_SCN	NUMBER	NOT NULL	SCN of the first message in the transaction
MESSAGE_COUNT	NUMBER		Number of messages spilled for the transaction
FIRST_MESSAGE_CREATE_TIME	DATE		Source creation time of the first message in the transaction
SPILL_CREATION_TIME	DATE		Time the first message was spilled
FIRST_POSITION	RAW(64)		Position of the first message in this transaction. This column is populated only for an XStream inbound server.
TRANSACTION_ID	VARCHAR2(128)		Transaction ID of the spilled transaction

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.119 DBA_APPLY_TABLE_COLUMNS

DBA_APPLY_TABLE_COLUMNS displays, for all tables in the database, information about the nonkey table columns for which apply process conflict detection has been stopped for update and delete operations. Its columns are the same as those in ALL_APPLY_TABLE_COLUMNS.

See Also

"ALL_APPLY_TABLE_COLUMNS"

4.120 DBA_APPLY_VALUE_DEPENDENCIES

DBA_APPLY_VALUE_DEPENDENCIES displays information about the value dependencies for all apply processes in the database.

Column	Datatype	NULL	Description
DEPENDENCY_NAME	VARCHAR2(128)	NOT NULL	Name of the dependency
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column
COLUMN_POSITION	NUMBER		Position of the column

4.121 DBA_AQ_AGENT_PRIVS

DBA_AQ_AGENT_PRIVS displays information about the registered AQ agents that are mapped to all users in the database.

Related View

USER_AQ_AGENT_PRIVS displays information about the registered AQ agents that are mapped to the current user. This view does not display the DB_USERNAME column.

Column	Datatype	NULL	Description
AGENT_NAME	VARCHAR2(128)	NOT NULL	Name of the AQ agent
DB_USERNAME	VARCHAR2(128)		Name of the database user that the agent maps to
HTTP_ENABLED	VARCHAR2(4)		Indicates whether the agent is allowed to access AQ through HTTP (YES) or not (NO)
SMTP_ENABLED	VARCHAR2(4)		Indicates whether the agent is allowed to access AQ through SMTP (YES) or not (NO)

See Also

"USER_AQ_AGENT_PRIVS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.122 DBA_AQ_AGENTS

DBA_AQ_AGENTS displays information about all registered AQ agents in the database.

Column	Datatype	NULL	Description
AGENT_NAME	VARCHAR2(128)	NOT NULL	Name of the AQ agent
HTTP_ENABLED	VARCHAR2(4)		Indicates whether the agent is allowed to access AQ through HTTP (YES) or not (NO)
SMTP_ENABLED	VARCHAR2(4)		Indicates whether the agent is allowed to access AQ through SMTP (YES) or not (NO)

4.123 DBA_ARGUMENTS

DBA_ARGUMENTS lists the arguments of the functions and procedures that are available in the database. Its columns are the same as those in ALL_ARGUMENTS.

See Also

- "ALL_ARGUMENTS"
- "DBA_PROCEDURES" for information about the functions and procedures that are available in the database

4.124 DBA_ASSEMBLIES

DBA_ASSEMBLIES provides information about all assemblies in the database. Its columns are the same as those in ALL_ASSEMBLIES.

See Also

"ALL_ASSEMBLIES"

4.125 DBA_ASSOCIATIONS

DBA_ASSOCIATIONS describes all user-defined statistics in the database. Its columns are the same as those in ALL_ASSOCIATIONS.

See Also

"ALL_ASSOCIATIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.126 DBA_ATTRIBUTE_DIM_ATTR_CLASS

DBA_ATTRIBUTE_DIM_ATTR_CLASS describes the attribute classifications of all attribute dimensions in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_ATTR_CLASS.

See Also

"ALL_ATTRIBUTE_DIM_ATTR_CLASS"

4.127 DBA_ATTRIBUTE_DIM_ATTR_CLS

DBA_ATTRIBUTE_DIM_ATTR_CLS is identical to DBA_ATTRIBUTE_DIM_ATTR_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"DBA_ATTRIBUTE_DIM_ATTR_CLASS"

4.128 DBA_ATTRIBUTE_DIM_ATTR_CLS_AE

DBA_ATTRIBUTE_DIM_ATTR_CLS_AE describes the attribute classifications of all attribute dimensions (across all editions) in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_ATTR_CLS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_ATTR_CLS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.129 DBA_ATTRIBUTE_DIM_ATTRS

DBA_ATTRIBUTE_DIM_ATTRS describes the attributes of all attribute dimensions in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_ATTRS.

See Also

"ALL_ATTRIBUTE_DIM_ATTRS"

4.130 DBA_ATTRIBUTE_DIM_ATTRS_AE

DBA_ATTRIBUTE_DIM_ATTRS_AE describes the attributes of all attribute dimensions (across all editions) in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_ATTRS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_ATTRS_AE"

4.131 DBA_ATTRIBUTE_DIM_CLASS

DBA_ATTRIBUTE_DIM_CLASS describes the classifications of all attribute dimensions in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_CLASS.

See Also

"ALL_ATTRIBUTE_DIM_CLASS"

4.132 DBA_ATTRIBUTE_DIM_CLASS_AE

DBA_ATTRIBUTE_DIM_CLASS_AE describes the classifications of all attribute dimensions (across all editions) in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_CLASS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "ALL_ATTRIBUTE_DIM_CLASS_AE"

4.133 DBA_ATTRIBUTE_DIM_JN_PTHS

DBA_ATTRIBUTE_DIM_JN_PTHS is identical to DBA_ATTRIBUTE_DIM_JOIN_PATHS.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "DBA_ATTRIBUTE_DIM_JOIN_PATHS"

4.134 DBA_ATTRIBUTE_DIM_JN_PTHS_AE

DBA_ATTRIBUTE_DIM_JN_PTHS_AE describes the join paths for all attribute dimensions (across all editions) in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_JN_PTHS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ATTRIBUTE_DIM_JN_PTHS_AE"

4.135 DBA_ATTRIBUTE_DIM_JOIN_PATHS

DBA_ATTRIBUTE_DIM_JOIN_PATHS describes the join paths for all attribute dimensions in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_JOIN_PATHS.

> **See Also**
>
> "ALL_ATTRIBUTE_DIM_JOIN_PATHS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.136 DBA_ATTRIBUTE_DIM_KEYS

DBA_ATTRIBUTE_DIM_KEYS describes the keys of all attribute dimensions in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_KEYS.

See Also

"ALL_ATTRIBUTE_DIM_KEYS"

4.137 DBA_ATTRIBUTE_DIM_KEYS_AE

DBA_ATTRIBUTE_DIM_KEYS_AE describes the keys of all attribute dimensions (across all editions) in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_KEYS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_KEYS_AE"

4.138 DBA_ATTRIBUTE_DIM_LEVEL_ATTRS

DBA_ATTRIBUTE_DIM_LEVEL_ATTRS describes the attributes of the levels of all attribute dimensions in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_LEVEL_ATTRS.

See Also

"ALL_ATTRIBUTE_DIM_LEVEL_ATTRS"

4.139 DBA_ATTRIBUTE_DIM_LEVELS

DBA_ATTRIBUTE_DIM_LEVELS describes the levels of all attribute dimensions in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_LEVELS.

See Also

"ALL_ATTRIBUTE_DIM_LEVELS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.140 DBA_ATTRIBUTE_DIM_LEVELS_AE

DBA_ATTRIBUTE_DIM_LEVELS_AE describes the levels of all attribute dimensions (across all editions) in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_LEVELS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_LEVELS_AE"

4.141 DBA_ATTRIBUTE_DIM_LVL_ATRS

DBA_ATTRIBUTE_DIM_LVL_ATRS is identical to DBA_ATTRIBUTE_DIM_LEVEL_ATTRS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"DBA_ATTRIBUTE_DIM_LEVEL_ATTRS"

4.142 DBA_ATTRIBUTE_DIM_LVL_ATRS_AE

DBA_ATTRIBUTE_DIM_LVL_ATRS_AE describes the attributes of the levels of all attribute dimensions (across all editions) in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_LVL_ATRS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_LVL_ATRS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.143 DBA_ATTRIBUTE_DIM_LVL_CLASS

DBA_ATTRIBUTE_DIM_LVL_CLASS describes the level classifications of all attribute dimensions in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_LVL_CLASS.

See Also

"ALL_ATTRIBUTE_DIM_LVL_CLASS"

4.144 DBA_ATTRIBUTE_DIM_LVL_CLS

DBA_ATTRIBUTE_DIM_LVL_CLS is identical to DBA_ATTRIBUTE_DIM_LVL_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"DBA_ATTRIBUTE_DIM_LVL_CLASS"

4.145 DBA_ATTRIBUTE_DIM_LVL_CLS_AE

DBA_ATTRIBUTE_DIM_LVL_CLS_AE describes the level classifications of all attribute dimensions (across all editions) in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_LVL_CLS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_LVL_CLS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.146 DBA_ATTRIBUTE_DIM_ORD_ATRS

DBA_ATTRIBUTE_DIM_ORD_ATRS is identical to DBA_ATTRIBUTE_DIM_ORDER_ATTRS.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "DBA_ATTRIBUTE_DIM_ORDER_ATTRS"

4.147 DBA_ATTRIBUTE_DIM_ORD_ATRS_AE

DBA_ATTRIBUTE_DIM_ORD_ATRS_AE describes the order attributes of all attribute dimensions (across all editions) in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_ORD_ATRS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ATTRIBUTE_DIM_ORD_ATRS_AE"

4.148 DBA_ATTRIBUTE_DIM_ORDER_ATTRS

DBA_ATTRIBUTE_DIM_ORDER_ATTRS describes the order attributes of all attribute dimensions in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_ORDER_ATTRS.

> **See Also**
>
> "ALL_ATTRIBUTE_DIM_ORDER_ATTRS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.149 DBA_ATTRIBUTE_DIM_TABLES

DBA_ATTRIBUTE_DIM_TABLES describes the tables used by all attribute dimensions in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_TABLES.

See Also

"ALL_ATTRIBUTE_DIM_TABLES"

4.150 DBA_ATTRIBUTE_DIM_TABLES_AE

DBA_ATTRIBUTE_DIM_TABLES_AE describes the tables used by all attribute dimensions (across all editions) in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIM_TABLES_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_TABLES_AE"

4.151 DBA_ATTRIBUTE_DIMENSIONS

DBA_ATTRIBUTE_DIMENSIONS describes all attribute dimensions in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIMENSIONS.

See Also

"ALL_ATTRIBUTE_DIMENSIONS"

4.152 DBA_ATTRIBUTE_DIMENSIONS_AE

DBA_ATTRIBUTE_DIMENSIONS_AE describes all attribute dimensions (across all editions) in the database. Its columns are the same as those in ALL_ATTRIBUTE_DIMENSIONS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "ALL_ATTRIBUTE_DIMENSIONS_AE"

4.153 DBA_ATTRIBUTE_TRANSFORMATIONS

DBA_ATTRIBUTE_TRANSFORMATIONS describes the transformation functions for all transformations in the database. Its columns are the same as those in ALL_ATTRIBUTE_TRANSFORMATIONS.

> **See Also**
>
> "ALL_ATTRIBUTE_TRANSFORMATIONS"

4.154 DBA_AUDIT_EXISTS

DBA_AUDIT_EXISTS displays audit trail entries produced by AUDIT EXISTS and AUDIT NOT EXISTS.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from UNIFIED_AUDIT_TRAIL.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
OS_USERNAME	VARCHAR2(255)		Operating system login username of the user whose actions were audited
USERNAME	VARCHAR2(128)		Name (not ID number) of the user whose actions were audited
USERHOST	VARCHAR2(128)		Client host machine name
TERMINAL	VARCHAR2(255)		Identifier of the user's terminal
TIMESTAMP	DATE		Date and time of the creation of the audit trail entry (date and time of user login for entries created by AUDIT SESSION) in the local database session time zone
OWNER	VARCHAR2(128)		Intended creator of the non-existent object
OBJ_NAME	VARCHAR2(128)		Name of the object affected by the action
ACTION_NAME	VARCHAR2(28)		Name of the action type corresponding to the numeric code in the ACTION column in DBA_AUDIT_TRAIL
NEW_OWNER	VARCHAR2(128)		Owner of the object named in the NEW_NAME column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
NEW_NAME	VARCHAR2(128)		New name of an object after a RENAME or the name of the underlying object
OBJ_PRIVILEGE	VARCHAR2(32)		Object privileges granted or revoked by a GRANT or REVOKE statement. The value of this column is a 32-character string of Y and dash (-) characters. Each character corresponds to a numbered privilege in the following list. The left-most character corresponds to privilege 0, the next character corresponds to privilege 1, and so on. The right-most character corresponds to privilege 31. 0 - ALTER 1 - AUDIT 2 - COMMENT 3 - DELETE 4 - GRANT 5 - INDEX 6 - INSERT 7 - LOCK 8 - CREATE 9 - SELECT 10 - UPDATE 11 - REFERENCES 12 - EXECUTE 13 - VIEW 14 - DROP 15 - ANALYZE 16 - CREATE 17 - READ 18 - WRITE 19 - KEEP SEQUENCE 20 - ENQUEUE 21 - DEQUEUE 22 - UNDER 23 - ON COMMIT 24 - REWRITE 25 - UPSERT 26 - DEBUG 27 - FLASHBACK 28 - MERGE 29 - USE 30 - FLASHBACK ARCHIVE 31 - DIRECTORY EXECUTE A Y indicates that the privilege was granted or revoked by the statement. A dash indicates that the privilege was not affected by the statement. For example, the following value indicates that the MERGE privilege was granted or revoked by the statement: ----------------------------Y---
SYS_PRIVILEGE	VARCHAR2(40)		System privileges granted or revoked by a GRANT or REVOKE statement

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)		Name of the grantee specified in a GRANT or REVOKE statement
SESSIONID	NUMBER	NOT NULL	Numeric ID for each Oracle session
ENTRYID	NUMBER	NOT NULL	Numeric ID for each audit trail entry in the session
STATEMENTID	NUMBER	NOT NULL	Numeric ID for each statement run
RETURNCODE	NUMBER	NOT NULL	Oracle error code generated by the action. Some useful values: • 0 - Action succeeded • 2004 - Security violation
CLIENT_ID	VARCHAR2(128)		Client identifier in each Oracle session
ECONTEXT_ID	VARCHAR2(64)		Application execution context identifier
SESSION_CPU	NUMBER		Amount of CPU time used by each Oracle session
EXTENDED_TIMESTAMP	TIMESTAMP(6) WITH TIME ZONE		Timestamp of the creation of the audit trail entry (timestamp of user login for entries created by AUDIT SESSION) in UTC (Coordinated Universal Time) time zone
PROXY_SESSIONID	NUMBER		Proxy session serial number, if an enterprise user has logged in through the proxy mechanism
GLOBAL_UID	VARCHAR2(32)		Global user identifier for the user, if the user has logged in as an enterprise user
INSTANCE_NUMBER	NUMBER		Instance number as specified by the INSTANCE_NUMBER initialization parameter
OS_PROCESS	VARCHAR2(16)		Operating System process identifier of the Oracle process
TRANSACTIONID	RAW(8)		Transaction identifier of the transaction in which the object is accessed or modified
SCN	NUMBER		System change number (SCN) of the query
SQL_BIND	NVARCHAR2(2000)		Bind variable data of the query
SQL_TEXT	NVARCHAR2(2000)		SQL text of the query
OBJ_EDITION_NAME	VARCHAR2(128)		Name of the edition containing the audited object

Note

The SQL_BIND and SQL_TEXT columns are only populated if the AUDIT_TRAIL initialization parameter is set to db, extended.

See Also

"AUDIT_TRAIL"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.155 DBA_AUDIT_MGMT_CLEAN_EVENTS

DBA_AUDIT_MGMT_CLEAN_EVENTS displays information about the history of audit trail cleanup or purge events.

Periodically, you should delete the contents of this view so that it will not grow too large.

> **Note**
>
> This view is intended for use with traditional auditing (pre-Oracle Database 12c auditing) only, not for unified auditing.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
AUDIT_TRAIL	VARCHAR2(28)		Audit trail that was cleaned at the time of the event: • STANDARD AUDIT TRAIL • FGA AUDIT TRAIL • STANDARD AND FGA AUDIT TRAIL • OS AUDIT TRAIL • XML AUDIT TRAIL • OS AND XML AUDIT TRAIL • ALL AUDIT TRAILS
RAC_INSTANCE	NUMBER	NOT NULL	Instance number indicating the Oracle RAC instance that was cleaned up at the time of the event; 0 implies not applicable
CLEANUP_TIME	TIMESTAMP(6) WITH TIME ZONE		Timestamp when the cleanup event completed
DELETE_COUNT	NUMBER		Number of audit records or audit files that were deleted at the time of the event
WAS_FORCED	VARCHAR2(3)		Indicates whether a forced cleanup occurred (YES) or not (NO); forced cleanup bypasses the last archive timestamp set

> **Note**
>
> In a read-only database, including an Oracle Active Data Guard physical standby database, this view is not populated. Instead, a summary of cleanup events is written to the respective database instance's alert log file.

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.156 DBA_AUDIT_MGMT_CLEANUP_JOBS

DBA_AUDIT_MGMT_CLEANUP_JOBS displays information about the configured audit trail purge jobs.

> **Note**
>
> This view is populated in any Oracle Database where auditing is enabled, regardless of whether pre-Oracle Database 12*c* auditing or unified auditing is enabled for the database.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
JOB_NAME	VARCHAR2(100)	NOT NULL	Name of the audit trail purge job
JOB_STATUS	VARCHAR2(8)		Current status of the audit trail purge job (ENABLED) or (DISABLED)
AUDIT_TRAIL	VARCHAR2(28)		Audit trail for which the audit trail purge job is configured: • STANDARD AUDIT TRAIL • FGA AUDIT TRAIL • STANDARD AND FGA AUDIT TRAIL • OS AUDIT TRAIL • XML AUDIT TRAIL • OS AND XML AUDIT TRAIL • ALL AUDIT TRAILS • UNIFIED AUDIT TRAIL
JOB_FREQUENCY	VARCHAR2(100)		Frequency at which the audit trail purge job runs
USE_LAST_ARCHIVE_TIMESTAMP	VARCHAR2(3)		Indicates whether the audit trail purge job invocation uses the last archive timestamp. The last archive timestamp is shown in the LAST_ARCHIVE_TS column of the DBA_AUDIT_MGMT_LAST_ARCH_TS view. Possible values: • YES - Indicates that the audit trail purge job invocation uses the last archive timestamp • NO - Indicates that the audit trail purge job invocation does not use the last archive timestamp
JOB_CONTAINER	VARCHAR2(7)		In a CDB, indicates whether audit trail purge job will be performed only in the current container or in all the containers. Possible values: • CURRENT - Indicates that audit trail purge job will be performed only in the current container • ALL - Indicates that audit trail purge job will be performed in all the containers In a non-CDB, the value in this column is always CURRENT.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "DBA_AUDIT_MGMT_LAST_ARCH_TS"

4.157 DBA_AUDIT_MGMT_CONFIG_PARAMS

DBA_AUDIT_MGMT_CONFIG_PARAMS displays information about the currently configured audit trail properties that are used by the DBMS_AUDIT_MGMT PL/SQL package.

> **Note**
>
> This view is populated in any Oracle Database where auditing is enabled, regardless of whether pre-Oracle Database 12*c* auditing or unified auditing is enabled for the database.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
PARAMETER_NAME	VARCHAR2(1024)	NOT NULL	Name of the property
PARAMETER_VALUE	VARCHAR2(4000)		Value of the property
AUDIT_TRAIL	VARCHAR2(28)		Audit trails for which the property is configured: • STANDARD AUDIT TRAIL • FGA AUDIT TRAIL • STANDARD AND FGA AUDIT TRAIL • OS AUDIT TRAIL • XML AUDIT TRAIL • OS AND XML AUDIT TRAIL • ALL AUDIT TRAILS • UNIFIED AUDIT TRAIL

> **Note**
>
> In a read-only database, including an Oracle Active Data Guard physical standby database, this view is not populated when the DBMS_AUDIT_MGMT.SET_AUDIT_TRAIL PL/SQL procedure is invoked. If the procedure was used when the database was in read-only mode, use DBMS_AUDIT_MGMT.GET_AUDIT_TRAIL to check the value of the property.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database PL/SQL Packages and Types Reference* for more information about the parameters specified with the `DBMS_AUDIT_MGMT.SET_AUDIT_TRAIL_PROPERTY` procedure

4.158 DBA_AUDIT_MGMT_LAST_ARCH_TS

`DBA_AUDIT_MGMT_LAST_ARCH_TS` displays information about the last archive timestamps set for audit trail cleanup or purges.

> **Note**
>
> This view is populated in any Oracle Database where auditing is enabled, regardless of whether pre-Oracle Database 12*c* auditing or unified auditing is enabled for the database.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
`AUDIT_TRAIL`	`VARCHAR2(20)`		Audit trail for which the last archive timestamp applies: • `STANDARD AUDIT TRAIL` • `FGA AUDIT TRAIL` • `OS AUDIT TRAIL` • `XML AUDIT TRAIL` • `UNIFIED AUDIT TRAIL`
`RAC_INSTANCE`	`NUMBER`	`NOT NULL`	Oracle RAC instance number for which the last archive timestamp applies; `0` implies not applicable
`LAST_ARCHIVE_TS`	`TIMESTAMP(6) WITH TIME ZONE`		Timestamp of the last audit record or audit file that has been archived
`DATABASE_ID`	`NUMBER`	`NOT NULL`	Database ID of the audit records to clean up
`CONTAINER_GUID`	`VARCHAR2(33)`	`NOT NULL`	GUID of the container of the audit records to clean up

> **Note**
>
> In a read-only database, including an Oracle Active Data Guard physical standby database, this view is not populated when `DBMS_AUDIT_MGMT.SET_LAST_ARCHIVE_TIMESTAMP` is invoked. In such a case, use `DBMS_AUDIT_MGMT.GET_LAST_ARCHIVE_TIMESTAMP` to check for the timestamp, if it was configured for the database instance.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database PL/SQL Packages and Types Reference* for more information about `DBMS_AUDIT_MGMT` subprograms

4.159 DBA_AUDIT_OBJECT

`DBA_AUDIT_OBJECT` displays audit trail records for all objects in the database.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from `UNIFIED_AUDIT_TRAIL`.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Related View

`USER_AUDIT_OBJECT` displays audit trail records for the objects accessible to the current user.

Column	Datatype	NULL	Description
`OS_USERNAME`	`VARCHAR2(255)`		Operating system login username of the user whose actions were audited
`USERNAME`	`VARCHAR2(128)`		Name (not ID number) of the user whose actions were audited
`USERHOST`	`VARCHAR2(128)`		Client host machine name
`TERMINAL`	`VARCHAR2(255)`		Identifier of the user's terminal
`TIMESTAMP`	`DATE`		Date and time of the creation of the audit trail entry (date and time of user login for entries created by `AUDIT SESSION`) in the local database session time zone
`OWNER`	`VARCHAR2(128)`		Creator of the object affected by the action
`OBJ_NAME`	`VARCHAR2(128)`		Name of the object affected by the action
`ACTION_NAME`	`VARCHAR2(28)`		Name of the action type corresponding to the numeric code in the `ACTION` column in `DBA_AUDIT_TRAIL`
`NEW_OWNER`	`VARCHAR2(128)`		Owner of the object named in the `NEW_NAME` column
`NEW_NAME`	`VARCHAR2(128)`		New name of an object after a `RENAME` or the name of the underlying object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SES_ACTIONS	VARCHAR2(19)		Session summary (a string of 16 characters, one for each action type in the order ALTER, AUDIT, COMMENT, DELETE, GRANT, INDEX, INSERT, LOCK, RENAME, SELECT, UPDATE, REFERENCES, and EXECUTE). Positions 14, 15, and 16 are reserved for future use. The characters are: • - - None • S - Success • F - Failure • B - Both
COMMENT_TEXT	VARCHAR2(4000)		Text comment on the audit trail
SESSIONID	NUMBER	NOT NULL	Numeric ID for each Oracle session
ENTRYID	NUMBER	NOT NULL	Numeric ID for each audit trail entry in the session
STATEMENTID	NUMBER	NOT NULL	Numeric ID for each statement run
RETURNCODE	NUMBER	NOT NULL	Oracle error code generated by the action. Some useful values: • 0 - Action succeeded • 2004 - Security violation
PRIV_USED	VARCHAR2(40)		System privilege used to execute the action
CLIENT_ID	VARCHAR2(128)		Client identifier in each Oracle session
ECONTEXT_ID	VARCHAR2(64)		Application execution context identifier
SESSION_CPU	NUMBER		Amount of CPU time used by each Oracle session
EXTENDED_TIMESTAMP	TIMESTAMP(6) WITH TIME ZONE		Timestamp of the creation of the audit trail entry (timestamp of user login for entries created by AUDIT SESSION) in UTC (Coordinated Universal Time) time zone
PROXY_SESSIONID	NUMBER		Proxy session serial number, if an enterprise user has logged in through the proxy mechanism
GLOBAL_UID	VARCHAR2(32)		Global user identifier for the user, if the user has logged in as an enterprise user
INSTANCE_NUMBER	NUMBER		Instance number as specified by the INSTANCE_NUMBER initialization parameter
OS_PROCESS	VARCHAR2(16)		Operating System process identifier of the Oracle process
TRANSACTIONID	RAW(8)		Transaction identifier of the transaction in which the object is accessed or modified
SCN	NUMBER		System change number (SCN) of the query
SQL_BIND	NVARCHAR2(2000)		Bind variable data of the query
SQL_TEXT	NVARCHAR2(2000)		SQL text of the query
OBJ_EDITION_NAME	VARCHAR2(128)		Name of the edition containing the audited object

Note

The SQL_BIND and SQL_TEXT columns are only populated if the AUDIT_TRAIL initialization parameter is set to db, extended.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> - "AUDIT_TRAIL"
> - "USER_AUDIT_OBJECT"

4.160 DBA_AUDIT_POLICIES

DBA_AUDIT_POLICIES describes all fine-grained auditing policies in the database. Its columns are the same as those in ALL_AUDIT_POLICIES.

> **See Also**
>
> "ALL_AUDIT_POLICIES"

4.161 DBA_AUDIT_POLICY_COLUMNS

DBA_AUDIT_POLICY_COLUMNS describes all fine-grained auditing policy columns in the database. Its columns are the same as those in ALL_AUDIT_POLICY_COLUMNS.

> **See Also**
>
> "ALL_AUDIT_POLICY_COLUMNS"

4.162 DBA_AUDIT_SESSION

DBA_AUDIT_SESSION displays all audit trail records concerning CONNECT and DISCONNECT.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from UNIFIED_AUDIT_TRAIL.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Related View

USER_AUDIT_SESSION displays the audit trail records concerning connections and disconnections of the current user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OS_USERNAME	VARCHAR2(255)		Operating system login username of the user whose actions were audited
USERNAME	VARCHAR2(128)		Name (not ID number) of the user whose actions were audited
USERHOST	VARCHAR2(128)		Client host machine name
TERMINAL	VARCHAR2(255)		Identifier of the user's terminal
TIMESTAMP	DATE		Date and time of the creation of the audit trail entry (date and time of user login for entries created by AUDIT SESSION) in the local database session time zone
ACTION_NAME	VARCHAR2(28)		Name of the action type corresponding to the numeric code in the ACTION column in DBA_AUDIT_TRAIL
LOGOFF_TIME	DATE		Date and time of user log off
LOGOFF_LREAD	NUMBER		Logical reads for the session
LOGOFF_PREAD	NUMBER		Physical reads for the session
LOGOFF_LWRITE	NUMBER		Logical writes for the session
LOGOFF_DLOCK	VARCHAR2(40)		Deadlocks detected during the session
SESSIONID	NUMBER	NOT NULL	Numeric ID for each Oracle session
RETURNCODE	NUMBER	NOT NULL	Oracle error code generated by the action. Some useful values: • 0 - Action succeeded • 2004 - Security violation
CLIENT_ID	VARCHAR2(128)		Client identifier in each Oracle session
SESSION_CPU	NUMBER		Amount of CPU time used by each Oracle session (in centiseconds)
EXTENDED_TIMESTAMP	TIMESTAMP(6) WITH TIME ZONE		Timestamp of the creation of the audit trail entry (timestamp of user login for entries created by AUDIT SESSION) in UTC (Coordinated Universal Time) time zone
PROXY_SESSIONID	NUMBER		Proxy session serial number, if an enterprise user has logged in through the proxy mechanism
GLOBAL_UID	VARCHAR2(32)		Global user identifier for the user, if the user has logged in as an enterprise user
INSTANCE_NUMBER	NUMBER		Instance number as specified by the INSTANCE_NUMBER initialization parameter
OS_PROCESS	VARCHAR2(16)		Operating System process identifier of the Oracle process

See Also

"USER_AUDIT_SESSION"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.163 DBA_AUDIT_STATEMENT

DBA_AUDIT_STATEMENT displays audit trail records for all GRANT, REVOKE, AUDIT, NOAUDIT, and ALTER SYSTEM statements in the database.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from UNIFIED_AUDIT_TRAIL.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Related View

USER_AUDIT_STATEMENT displays audit trail records for the GRANT, REVOKE, AUDIT, NOAUDIT, and ALTER SYSTEM statements issued by the current user.

Column	Datatype	NULL	Description
OS_USERNAME	VARCHAR2(255)		Operating system login username of the user whose actions were audited
USERNAME	VARCHAR2(128)		Name (not ID number) of the user whose actions were audited
USERHOST	VARCHAR2(128)		Client host machine name
TERMINAL	VARCHAR2(255)		Identifier of the user's terminal
TIMESTAMP	DATE		Date and time of the creation of the audit trail entry (date and time of user login for entries created by AUDIT SESSION) in the local database session time zone
OWNER	VARCHAR2(128)		Creator of the object affected by the action
OBJ_NAME	VARCHAR2(128)		Name of the object affected by the action
ACTION_NAME	VARCHAR2(28)		Name of the action type corresponding to the numeric code in the ACTION column in DBA_AUDIT_TRAIL
NEW_NAME	VARCHAR2(128)		New name of an object after a RENAME or the name of the underlying object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OBJ_PRIVILEGE	VARCHAR2(32)		Object privileges granted or revoked by a GRANT or REVOKE statement. The value of this column is a 32-character string of Y and dash (-) characters. Each character corresponds to a numbered privilege in the following list. The left-most character corresponds to privilege 0, the next character corresponds to privilege 1, and so on. The right-most character corresponds to privilege 31. 0 - ALTER 1 - AUDIT 2 - COMMENT 3 - DELETE 4 - GRANT 5 - INDEX 6 - INSERT 7 - LOCK 8 - CREATE 9 - SELECT 10 - UPDATE 11 - REFERENCES 12 - EXECUTE 13 - VIEW 14 - DROP 15 - ANALYZE 16 - CREATE 17 - READ 18 - WRITE 19 - KEEP SEQUENCE 20 - ENQUEUE 21 - DEQUEUE 22 - UNDER 23 - ON COMMIT 24 - REWRITE 25 - UPSERT 26 - DEBUG 27 - FLASHBACK 28 - MERGE 29 - USE 30 - FLASHBACK ARCHIVE 31 - DIRECTORY EXECUTE A Y indicates that the privilege was granted or revoked by the statement. A dash indicates that the privilege was not affected by the statement. For example, the following value indicates that the MERGE privilege was granted or revoked by the statement: ----------------------------Y---
SYS_PRIVILEGE	VARCHAR2(40)		System privileges granted or revoked by a GRANT or REVOKE statement
ADMIN_OPTION	VARCHAR2(1)		Signifies the role or system privilege was granted with the ADMIN option

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)		Name of the grantee specified in a GRANT or REVOKE statement
AUDIT_OPTION	VARCHAR2(40)		Auditing option set with the AUDIT statement
SES_ACTIONS	VARCHAR2(19)		Session summary (a string of 16 characters, one for each action type in the order ALTER, AUDIT, COMMENT, DELETE, GRANT, INDEX, INSERT, LOCK, RENAME, SELECT, UPDATE, REFERENCES, and EXECUTE). Positions 14, 15, and 16 are reserved for future use. The characters are: • - - None • S - Success • F - Failure • B - Both
COMMENT_TEXT	VARCHAR2(4000)		Text comment on the audit trail, inserted by the application
SESSIONID	NUMBER	NOT NULL	Numeric ID for each Oracle session
ENTRYID	NUMBER	NOT NULL	Numeric ID for each audit trail entry in the session
STATEMENTID	NUMBER	NOT NULL	Numeric ID for each statement run
RETURNCODE	NUMBER	NOT NULL	Oracle error code generated by the action. Some useful values: • 0 - Action succeeded • 2004 - Security violation
PRIV_USED	VARCHAR2(40)		System privilege used to execute the action
CLIENT_ID	VARCHAR2(128)		Client identifier in each Oracle session
ECONTEXT_ID	VARCHAR2(64)		Application execution context identifier
SESSION_CPU	NUMBER		Amount of CPU time used by each Oracle session
EXTENDED_TIMESTAMP	TIMESTAMP(6) WITH TIME ZONE		Timestamp of the creation of the audit trail entry (timestamp of user login for entries created by AUDIT SESSION) in UTC (Coordinated Universal Time) time zone
PROXY_SESSIONID	NUMBER		Proxy session serial number, if an enterprise user has logged in through the proxy mechanism
GLOBAL_UID	VARCHAR2(32)		Global user identifier for the user, if the user has logged in as an enterprise user
INSTANCE_NUMBER	NUMBER		Instance number as specified by the INSTANCE_NUMBER initialization parameter
OS_PROCESS	VARCHAR2(16)		Operating System process identifier of the Oracle process
TRANSACTIONID	RAW(8)		Transaction identifier of the transaction in which the object is accessed or modified
SCN	NUMBER		System change number (SCN) of the query
SQL_BIND	NVARCHAR2(2000)		Bind variable data of the query
SQL_TEXT	NVARCHAR2(2000)		SQL text of the query
OBJ_EDITION_NAME	VARCHAR2(128)		Name of the edition containing the audited object

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The SQL_BIND and SQL_TEXT columns are only populated if the AUDIT_TRAIL initialization parameter is set to db, extended.

> **See Also**
>
> - "AUDIT_TRAIL"
> - "USER_AUDIT_STATEMENT"

4.164 DBA_AUDIT_TRAIL

DBA_AUDIT_TRAIL displays all standard audit trail entries.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from UNIFIED_AUDIT_TRAIL.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Related View

USER_AUDIT_TRAIL displays the standard audit trail entries related to the current user.

Column	Datatype	NULL	Description
OS_USERNAME	VARCHAR2(255)		Operating system login username of the user whose actions were audited
USERNAME	VARCHAR2(128)		Name (not ID number) of the user whose actions were audited
USERHOST	VARCHAR2(128)		Client host machine name
TERMINAL	VARCHAR2(255)		Identifier of the user's terminal
TIMESTAMP	DATE		Date and time of the creation of the audit trail entry (date and time of user login for entries created by AUDIT SESSION) in the local database session time zone
OWNER	VARCHAR2(128)		Creator of the object affected by the action
OBJ_NAME	VARCHAR2(128)		Name of the object affected by the action
ACTION	NUMBER	NOT NULL	Numeric action type code. The corresponding name of the action type is in the ACTION_NAME column.
ACTION_NAME	VARCHAR2(28)		Name of the action type corresponding to the numeric code in the ACTION column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
NEW_OWNER	VARCHAR2(128)		Owner of the object named in the NEW_NAME column
NEW_NAME	VARCHAR2(128)		New name of the object after a RENAME or the name of the underlying object
OBJ_PRIVILEGE	VARCHAR2(32)		Object privileges granted or revoked by a GRANT or REVOKE statement. The value of this column is a 32-character string of Y and dash (-) characters. Each character corresponds to a numbered privilege in the following list. The left-most character corresponds to privilege 0, the next character corresponds to privilege 1, and so on. The right-most character corresponds to privilege 31. 0 - ALTER 1 - AUDIT 2 - COMMENT 3 - DELETE 4 - GRANT 5 - INDEX 6 - INSERT 7 - LOCK 8 - CREATE 9 - SELECT 10 - UPDATE 11 - REFERENCES 12 - EXECUTE 13 - VIEW 14 - DROP 15 - ANALYZE 16 - CREATE 17 - READ 18 - WRITE 19 - KEEP SEQUENCE 20 - ENQUEUE 21 - DEQUEUE 22 - UNDER 23 - ON COMMIT 24 - REWRITE 25 - UPSERT 26 - DEBUG 27 - FLASHBACK 28 - MERGE 29 - USE 30 - FLASHBACK ARCHIVE 31 - DIRECTORY EXECUTE A Y indicates that the privilege was granted or revoked by the statement. A dash indicates that the privilege was not affected by the statement. For example, the following value indicates that the MERGE privilege was granted or revoked by the statement: ----------------------------Y---
SYS_PRIVILEGE	VARCHAR2(40)		System privileges granted or revoked by a GRANT or REVOKE statement

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ADMIN_OPTION	VARCHAR2(1)		Indicates whether the role or system privilege was granted with the ADMIN option
GRANTEE	VARCHAR2(128)		Name of the grantee specified in a GRANT or REVOKE statement
AUDIT_OPTION	VARCHAR2(40)		Auditing option set with the AUDIT statement
SES_ACTIONS	VARCHAR2(19)		Session summary (a string of 16 characters, one for each action type in the order ALTER, AUDIT, COMMENT, DELETE, GRANT, INDEX, INSERT, LOCK, RENAME, SELECT, UPDATE, REFERENCES, and EXECUTE). Positions 14, 15, and 16 are reserved for future use. The characters are: • - - None • S - Success • F - Failure • B - Both
LOGOFF_TIME	DATE		Date and time of user log off
LOGOFF_LREAD	NUMBER		Logical reads for the session
LOGOFF_PREAD	NUMBER		Physical reads for the session
LOGOFF_LWRITE	NUMBER		Logical writes for the session
LOGOFF_DLOCK	VARCHAR2(40)		Deadlocks detected during the session
COMMENT_TEXT	VARCHAR2(4000)		Text comment on the audit trail entry, providing more information about the statement audited Also indicates how the user or remote call was authenticated. The method can be one of the following: • DATABASE - Authentication was done by password • NETWORK - Authentication was done by Oracle Net Services or strong authentication • PROXY - Client was authenticated by another user; the name of the proxy user follows the method type When an object is accessed remotely over a database link, the COMMENT_TEXT column also captures the information about the database link. For example: `DBLINK_INFO: (SOURCE_GLOBAL_NAME=view02.regress.rdbms.dev.us.example.com, DBLINK_NAME=VIEW05_LINK.REGRESS.RDBMS.DEV.US.EXAMPLE.COM, SOURCE_AUDIT_SESSIONID=250805)`
SESSIONID	NUMBER	NOT NULL	Numeric ID for each Oracle session. Each user session gets a unique session ID.
ENTRYID	NUMBER	NOT NULL	Numeric ID for each audit trail entry in the session. The entry ID is an index of a session's audit entries that starts at 1 and increases to the number of entries that are written.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATEMENTID	NUMBER	NOT NULL	*n*th statement in the user session. The first SQL statement gets a value of 1 and the value is incremented for each subsequent SQL statement. Note that one SQL statement can create more than one audit trail entry (for example, when more than one object is audited from the same SQL statement), and in this case the statement ID remains the same for that statement and the entry ID increases for each audit trail entry created by the statement.
RETURNCODE	NUMBER	NOT NULL	Oracle error code generated by the action. Some useful values: • 0 - Action succeeded • 2004 - Security violation
PRIV_USED	VARCHAR2(40)		System privilege used to execute the action
CLIENT_ID	VARCHAR2(128)		Client identifier in each Oracle session
ECONTEXT_ID	VARCHAR2(64)		Application execution context identifier
SESSION_CPU	NUMBER		Amount of CPU time used by each Oracle session (in centiseconds)
EXTENDED_TIMESTAMP	TIMESTAMP(6) WITH TIME ZONE		Timestamp of the creation of the audit trail entry (timestamp of user login for entries created by AUDIT SESSION) in UTC (Coordinated Universal Time) time zone
PROXY_SESSIONID	NUMBER		Proxy session serial number, if an enterprise user has logged in through the proxy mechanism
GLOBAL_UID	VARCHAR2(32)		Global user identifier for the user, if the user has logged in as an enterprise user
INSTANCE_NUMBER	NUMBER		Instance number as specified by the INSTANCE_NUMBER initialization parameter
OS_PROCESS	VARCHAR2(16)		Operating System process identifier of the Oracle process
TRANSACTIONID	RAW(8)		Transaction identifier of the transaction in which the object is accessed or modified
SCN	NUMBER		System change number (SCN) of the creation of the audit trail entry
SQL_BIND	NVARCHAR2(2000)		Bind variable data of the query
SQL_TEXT	NVARCHAR2(2000)		SQL text of the query
OBJ_EDITION_NAME	VARCHAR2(128)		Name of the edition containing the audited object
DBID	NUMBER		Database identifier of the audited database
RLS_INFO	CLOB		Stores virtual private database (VPD) policy names and predicates separated by delimiter. To format the output into individual rows, use the DBMS_AUDIT_UTIL.DECODE_RLS_INFO_ATRAIL_STD function.
CURRENT_USER	VARCHAR2(128)		Effective user for the statement execution

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The SQL_BIND and SQL_TEXT columns are only populated if the AUDIT_TRAIL initialization parameter is set to db, extended.

> **See Also**
>
> - "AUDIT_TRAIL"
> - "USER_AUDIT_TRAIL"
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_AUDIT_UTIL.DECODE_RLS_INFO_ATRAIL_XML function.

4.165 DBA_AUTO_INDEX_CONFIG

DBA_AUTO_INDEX_CONFIG displays the current configuration parameter settings for automatic indexing.

You can set automatic indexing configuration parameters by using the DBMS_AUTO_INDEX.CONFIGURE procedure.

Column	Datatype	NULL	Description
PARAMETER_NAME	VARCHAR2(128)	NOT NULL	Name of the configuration parameter
PARAMETER_VALUE	VARCHAR2(4000)		Value of the configuration parameter
LAST_MODIFIED	TIMESTAMP(6)		Time at which the parameter value was last modified
MODIFIED_BY	VARCHAR2(128)		User who last modified the parameter value

> **Note**
>
> This view is available starting with Oracle Database 19c.

> **See Also**
>
> *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_AUTO_INDEX.CONFIGURE procedure

4.166 DBA_AUTO_SEGADV_CTL

DBA_AUTO_SEGADV_CTL exposes the control information used by the segment advisor.

This information gives the DBA an idea of what is happening in the auto advisor.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
AUTO_TASKID	NUMBER		Unique task ID generated by the auto advisor
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the segment
SEGMENT_OWNER	VARCHAR2(128)		Owner of the segment
SEGMENT_NAME	VARCHAR2(128)		Name of the segment
SEGMENT_TYPE	VARCHAR2(18)		The type of segment (TABLE, INDEX, and so on)
PARTITION_NAME	VARCHAR2(128)		Name of the subsegment (if partitioned)
STATUS	VARCHAR2(40)		Status of the analysis: • NEW - the segment/tablespace has not been analyzed • BEING-PROCESSED - the segment/tablespace is being processed • COMPLETE - the segment/tablespace has been analyzed • ERROR - problem with the segment/tablespace
REASON	VARCHAR2(40)		Reason why this segment was chosen
REASON_VALUE	NUMBER		A value associated with the reason
CREATION_TIME	TIMESTAMP(6)		Time when this entry was created
PROCESSED_TASKID	NUMBER		Auto advisor task that was used to process the segment/tablespace
END_TIME	TIMESTAMP(6)		Time at which the advisor task was completed

4.167 DBA_AUTO_SEGADV_SUMMARY

DBA_AUTO_SEGADV_SUMMARY provides a summary of the auto advisor task runs.

Column	Datatype	NULL	Description
AUTO_TASKID	NUMBER	NOT NULL	Unique auto task ID
SNAPID	NUMBER		Maximum AWR snapid used to process the segments
SEGMENTS_SELECTED	NUMBER		Number of segments chosen for analysis
SEGMENTS_PROCESSED	NUMBER		Number of segments actually processed
TABLESPACE_SELECTED	NUMBER		Number of tablespaces chosen for analysis
TABLESPACE_PROCESSED	NUMBER		Number of tablespaces actually processed
RECOMMENDATIONS_COUNT	NUMBER		Number of recommendations generated
START_TIME	TIMESTAMP(6)		Time at which the auto task was started
END_TIME	TIMESTAMP(6)		Time at which the auto task ended

4.168 DBA_AUTO_STAT_EXECUTIONS

DBA_AUTO_STAT_EXECUTIONS displays information about automatic optimizer statistics collection tasks, which are executed by the automated maintenance tasks infrastructure (known as AutoTask).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OPID	NUMBER		Unique identifier for the execution of the task
ORIGIN	VARCHAR2(19)		Origin of the execution of the task. Possible values: • AUTO_TASK: A standard automatic optimizer statistics collection task, which is executed automatically in an Oracle Scheduler window, known as maintenance window • HIGH_FREQ_AUTO_TASK: A high-frequency automatic optimizer statistics collection task, which is executed at frequent intervals and complements the standard automatic optimizer statistics collection tasks
STATUS	VARCHAR2(49)		Status of the execution of the task. Possible values: • IN PROGRESS: The operation is currently running • COMPLETED: The operation has completed successfully • FAILED: The operation has failed • TIMED OUT: The operation timed out because the time allocated for the maintenance window was not sufficient for the operation to complete
START_TIME	TIMESTAMP(6) WITH TIME ZONE		Start time for the execution of the task
END_TIME	TIMESTAMP(6) WITH TIME ZONE		End time for the execution of the task
COMPLETED	NUMBER		Number of objects for which statistics collection was completed during the execution of the task
FAILED	NUMBER		Number of objects for which statistics collection failed during the execution of the task
TIMED_OUT	NUMBER		Number of objects for which statistics collection timed out during the execution of the task
IN_PROGRESS	NUMBER		Number of objects for which statistics collection is in progress during the execution of the task

> **Note**
>
> This view is available starting with Oracle Database 19c.

4.169 DBA_AUTOTASK_CLIENT

DBA_AUTOTASK_CLIENT displays statistical data for each automated maintenance task over 7-day and 30-day periods.

Column	Datatype	NULL	Description
CLIENT_NAME	VARCHAR2(64)		Name of the client
STATUS	VARCHAR2(8)		Job status: • ENABLED • DISABLED

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CONSUMER_GROUP	VARCHAR2(128)		Consumer group used for normal priority jobs for this client
CLIENT_TAG	VARCHAR2(2)		Tag used to identify jobs for this client
PRIORITY_OVERRIDE	VARCHAR2(7)		User-specified priority at which the task executes: • URGENT • HIGH • MEDIUM • LOW
ATTRIBUTES	VARCHAR2(4000)		Attributes of the client
WINDOW_GROUP	VARCHAR2(64)		Window group used to schedule jobs
SERVICE_NAME	VARCHAR2(64)		Name of the service on which jobs will execute for the client
RESOURCE_PERCENTAGE	NUMBER		Percentage of maintenance resources for high priority maintenance tasks for this client
USE_RESOURCE_ESTIMATES	VARCHAR2(5)		Indicates whether resource estimates are used for this client (TRUE) or not (FALSE)
MEAN_JOB_DURATION	INTERVAL DAY(9) TO SECOND(9)		Average elapsed time for a job for this client (in seconds)
MEAN_JOB_CPU	INTERVAL DAY(9) TO SECOND(9)		Average CPU time for a job submitted by this client (in seconds)
MEAN_JOB_ATTEMPTS	NUMBER		Average number of attempts it takes to complete a task
MEAN_INCOMING_TASKS_7_DAYS	NUMBER		Average number of incoming tasks at the Maintenance Window Start over the last 7 days
MEAN_INCOMING_TASKS_30_DAYS	NUMBER		Average number of incoming tasks at the Maintenance Window Start over the last 30 days
TOTAL_CPU_LAST_7_DAYS	INTERVAL DAY(9) TO SECOND(9)		Cumulative CPU time used by the jobs for this client over the last 7 days (in seconds)
TOTAL_CPU_LAST_30_DAYS	INTERVAL DAY(9) TO SECOND(9)		Cumulative CPU time used by the jobs for this client over the last 30 days (in seconds)
MAX_DURATION_LAST_7_DAYS	INTERVAL DAY(3) TO SECOND(0)		Maximum elapsed time for a job over the last 7 days (in seconds)
MAX_DURATION_LAST_30_DAYS	INTERVAL DAY(3) TO SECOND(0)		Maximum elapsed time for a job over the last 30 days (in seconds)
WINDOW_DURATION_LAST_7_DAYS	INTERVAL DAY(9) TO SECOND(9)		Total time during which the client was active during the last 7 days
WINDOW_DURATION_LAST_30_DAYS	INTERVAL DAY(9) TO SECOND(9)		Total time during which the client was active during the last 30 days
LAST_CHANGE	TIMESTAMP(6) WITH TIME ZONE		Timestamp of last configuration change for the client

4.170 DBA_AUTOTASK_CLIENT_HISTORY

DBA_AUTOTASK_CLIENT_HISTORY displays per-window history of job execution counts for each automated maintenance task.

This information is viewable in the Job History page of Enterprise Manager.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CLIENT_NAME	VARCHAR2(64)		Name of the client
WINDOW_NAME	VARCHAR2(261)		Name of the maintenance window
WINDOW_START_TIME	TIMESTAMP(6) WITH TIME ZONE		Maintenance window start time
WINDOW_DURATION	INTERVAL DAY(9) TO SECOND(6)		Window duration (NULL for currently open window)
JOBS_CREATED	NUMBER		Number of jobs created on behalf of the client in this window
JOBS_STARTED	NUMBER		Number of jobs started on behalf of the client during the maintenance window
JOBS_COMPLETED	NUMBER		Number of jobs successfully completed on behalf of the client during the maintenance window
WINDOW_END_TIME	TIMESTAMP(6) WITH TIME ZONE		Window end time

4.171 DBA_AUTOTASK_CLIENT_JOB

DBA_AUTOTASK_CLIENT_JOB displays information about currently running Scheduler jobs created for automated maintenance tasks.

DBA_AUTOTASK_CLIENT_JOB provides information about some objects targeted by those jobs, as well as some additional statistics from previous instantiations of the same task. Some of this additional data is taken from generic Scheduler views.

Column	Datatype	NULL	Description
CLIENT_NAME	VARCHAR2(64)		Name of the client
JOB_NAME	VARCHAR2(65)		Name of the job
JOB_SCHEDULER_STATUS	VARCHAR2(15)		Job status: • DISABLED • RETRY SCHEDULED • SCHEDULED • RUNNING • COMPLETED • BROKEN • FAILED • REMOTE • SUCCEEDED • CHAIN_STALLED
TASK_NAME	VARCHAR2(64)		Name of the task being performed
TASK_TARGET_TYPE	VARCHAR2(64)		Type of the target being processed
TASK_TARGET_NAME	VARCHAR2(513)	NOT NULL	Name of the target
TASK_PRIORITY	VARCHAR2(7)		Priority of the task: • URGENT • HIGH • MEDIUM • LOW

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TASK_OPERATION	VARCHAR2(64)		Operation performed on the object

4.172 DBA_AUTOTASK_JOB_HISTORY

DBA_AUTOTASK_JOB_HISTORY displays the history of automated maintenance task job runs. Jobs are added to this view after they finish executing.

Column	Datatype	NULL	Description
CLIENT_NAME	VARCHAR2(64)		Name of the automated maintenance client
WINDOW_NAME	VARCHAR2(261)		Name of the maintenance window
WINDOW_START_TIME	TIMESTAMP(6) WITH TIME ZONE		Start time of the maintenance window
WINDOW_DURATION	INTERVAL DAY(9) TO SECOND(6)		Duration of the maintenance window
JOB_NAME	VARCHAR2(261)		Name of the maintenance job
JOB_STATUS	VARCHAR2(30)		Status of the maintenance job
JOB_START_TIME	TIMESTAMP(6) WITH TIME ZONE		Start time of the maintenance job
JOB_DURATION	INTERVAL DAY(3) TO SECOND(0)		Duration of the maintenance job
JOB_ERROR	NUMBER		Error code for the job (if any)
JOB_INFO	VARCHAR2(4000)		Additional information about the job

4.173 DBA_AUTOTASK_OPERATION

DBA_AUTOTASK_OPERATION displays all automated maintenance task operations for each client.

Column	Datatype	NULL	Description
CLIENT_NAME	VARCHAR2(64)		Name of the client
OPERATION_NAME	VARCHAR2(64)		Name of the operation
OPERATION_TAG	VARCHAR2(3)		Tag for the operation
PRIORITY_OVERRIDE	VARCHAR2(7)		User-specified priority at which the task executes: • URGENT • HIGH • MEDIUM • LOW
ATTRIBUTES	VARCHAR2(4000)		Attributes of the operation
USE_RESOURCE_ESTIMATES	VARCHAR2(5)		Indicates whether resource usage estimates are used for the operation (TRUE) or not (FALSE)
STATUS	VARCHAR2(8)		Job status: • ENABLED • DISABLED
LAST_CHANGE	TIMESTAMP(6) WITH TIME ZONE		Timestamp of the last change

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.174 DBA_AUTOTASK_SCHEDULE

DBA_AUTOTASK_SCHEDULE displays the schedule of maintenance windows for the next 32 days for each client.

Column	Datatype	NULL	Description
WINDOW_NAME	VARCHAR2(128)		Name of the maintenance window
START_TIME	TIMESTAMP(6) WITH TIME ZONE		Projected start time of the window
DURATION	INTERVAL DAY(3) TO SECOND(0)		Currently defined duration of the window (NULL for the currently open window)

4.175 DBA_AUTOTASK_STATUS

DBA_AUTOTASK_STATUS displays status information for automated maintenance.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(8)		Shows the status of automated maintenance. Possible values are: • ENABLED • DISABLED • ALLOWED • INVALID
LAST_CHANGE	TIMESTAMP(6) WITH TIME ZONE		Timestamp of last status change

4.176 DBA_AUTOTASK_TASK

DBA_AUTOTASK_TASK displays information about current and past automated maintenance tasks.

Column	Datatype	NULL	Description
CLIENT_NAME	VARCHAR2(64)		Name of the client
TASK_NAME	VARCHAR2(64)		Name of the task being performed
TASK_TARGET_TYPE	VARCHAR2(64)		Target type of the task
TASK_TARGET_NAME	VARCHAR2(513)	NOT NULL	Name of the target
OPERATION_NAME	VARCHAR2(64)		Operation performed on the object
ATTRIBUTES	VARCHAR2(4000)		Attributes of the task
TASK_PRIORITY	NUMBER		Task priority, relative to other tasks for this Client
PRIORITY_OVERRIDE	NUMBER		Task priority as overridden by the user
STATUS	VARCHAR2(8)		Status of the task: • DISABLED • DEFERRED • ENABLED
DEFERRED_WINDOW_NAME	VARCHAR2(65)		Appropriate window for this task

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CURRENT_JOB_NAME	VARCHAR2(65)		Name of the currently scheduled job, if any
JOB_SCHEDULER_STATUS	VARCHAR2(15)		Job status: • DISABLED • RETRY SCHEDULED • SCHEDULED • RUNNING • COMPLETED • BROKEN • FAILED • REMOTE • SUCCEEDED • CHAIN_STALLED
ESTIMATE_TYPE	VARCHAR2(7)		Type of resource estimates applied: • DERIVED • FORCED • LOCKED
ESTIMATED_WEIGHT	NUMBER		Task weight indicator
ESTIMATED_DURATION	NUMBER		Estimated elapsed time for the job (in seconds)
ESTIMATED_CPU_TIME	NUMBER		Estimated CPU time for the job (in seconds)
ESTIMATED_TEMP	NUMBER		Estimated temporary space usage for the job (in KB)
ESTIMATED_DOP	NUMBER		Estimated degree of parallelism for the job
ESTIMATED_IO_RATE	NUMBER		Estimated I/O utilization for the job (in KB per second)
ESTIMATED_UNDO_RATE	NUMBER		Estimated UNDO generation rate for the job (in KB per second)
RETRY_COUNT	NUMBER		Number of attempts to perform this task since the last successful attempt
LAST_GOOD_DATE	TIMESTAMP(6) WITH TIME ZONE		Timestamp of the last successful attempt to perform this task
LAST_GOOD_PRIORITY	NUMBER		Job priority of the last successful attempt to perform this task
LAST_GOOD_DURATION	NUMBER		Elapsed time (in seconds) of the last successful attempt to perform this task
LAST_GOOD_CPU_TIME	NUMBER		CPU time for the job (in seconds) of the last successful attempt to perform this task
LAST_GOOD_TEMP	NUMBER		Temporary space usage for the job (in KB) of the last successful attempt to perform this task
LAST_GOOD_DOP	NUMBER		Peak degree of parallelism for the job during the last successful attempt to perform this task
LAST_GOOD_IO_RATE	NUMBER		I/O utilization rate for the job (in KB per second) of the last successful attempt to perform this task
LAST_GOOD_UNDO_RATE	NUMBER		NDO generation rate (in KB per second) of the last successful attempt to perform this task
LAST_GOOD_CPU_WAIT	NUMBER		Resource Manager wait time (in seconds) of the last successful attempt to perform this task
LAST_GOOD_IO_WAIT	NUMBER		Resource Manager wait time (in seconds) of the last successful attempt to perform this task

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LAST_GOOD_UNDO_WAIT	NUMBER		Resource Manager wait time (in seconds) of the last successful attempt to perform this task
LAST_GOOD_TEMP_WAIT	NUMBER		Resource Manager wait time (in seconds) of the last successful attempt to perform this task
LAST_GOOD_CONCURRENCY	NUMBER		Concurrency wait time (in seconds) of the last successful attempt to perform this task
LAST_GOOD_CONTENTION	NUMBER		Contention wait time (in seconds) of the last successful attempt to perform this task
NEXT_TRY_DATE	TIMESTAMP(6) WITH TIME ZONE		Next projected start time for the deferred maintenance window
LAST_TRY_DATE	TIMESTAMP(6) WITH TIME ZONE		Time at which the task was last attempted
LAST_TRY_PRIORITY	NUMBER		Task priority at the time of the last attempt
LAST_TRY_RESULT	VARCHAR2(36)		Result code of the last execution of the task: • SUCCEEDED • FAILED • STOPPED BY USER ACTION • STOPPED AT END OF MAINTENANCE WINDOW • STOPPED AT INSTANCE SHUTDOWN • STOPPED
LAST_TRY_DURATION	NUMBER		Elapsed time of the last run (in seconds)
LAST_TRY_CPU_TIME	NUMBER		CPU time during the last run (in seconds)
LAST_TRY_TEMP	NUMBER		Temporary space usage for the job (in KB) for the last run
LAST_TRY_DOP	NUMBER		Peak degree of parallelism for the job during the last run
LAST_TRY_IO_RATE	NUMBER		I/O rate during the last run (in seconds)
LAST_TRY_UNDO_RATE	NUMBER		UNDO generation rate during the last run (in seconds)
LAST_TRY_CPU_WAIT	NUMBER		Time spent waiting for CPU during the last run (in seconds)
LAST_TRY_IO_WAIT	NUMBER		Time spent waiting for I/O during the last run (in seconds)
LAST_TRY_UNDO_WAIT	NUMBER		Time spent waiting for UNDO during the last run (in seconds)
LAST_TRY_TEMP_WAIT	NUMBER		Time spent waiting for temporary space during the last run (in seconds)
LAST_TRY_CONCURRENCY	NUMBER		Concurrency wait time during the last run (in seconds)
LAST_TRY_CONTENTION	NUMBER		Contention wait time during the last run (in seconds)
MEAN_GOOD_DURATION	NUMBER		Average elapsed time for successful executions of this task (in seconds)
MEAN_GOOD_CPU_TIME	NUMBER		Average CPU time for successful executions of this task (in seconds)
MEAN_GOOD_TEMP	NUMBER		Average temporary space usage for successful executions of this task (in KB)
MEAN_GOOD_DOP	NUMBER		Average peak degree of parallelism for successful executions of this task

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MEAN_GOOD_IO	NUMBER		Average I/O utilization for successful executions of this task (in KB per second)
MEAN_GOOD_UNDO	NUMBER		Average UNDO generation rate for this task (in KB per second)
MEAN_GOOD_CPU_WAIT	NUMBER		Average time waiting for CPU for successful executions of this task (in seconds)
MEAN_GOOD_IO_WAIT	NUMBER		Average time waiting for I/O for successful executions of this task (in seconds)
MEAN_GOOD_UNDO_WAIT	NUMBER		Average time waiting for UNDO for successful executions of this task (in seconds)
MEAN_GOOD_TEMP_WAIT	NUMBER		Average time waiting for temporary space for successful executions of this task (in seconds)
MEAN_GOOD_CONCURRENCY	NUMBER		Average concurrency wait time for successful executions of this task (in seconds)
MEAN_GOOD_CONTENTION	NUMBER		Average contention wait time for successful executions of this task (in seconds)
INFO_FIELD_1	VARCHAR2(4000)		Client-interpreted information
INFO_FIELD_2	CLOB		Client-interpreted information
INFO_FIELD_3	NUMBER		Client-interpreted information
INFO_FIELD_4	NUMBER		Client-interpreted information

4.177 DBA_AUTOTASK_WINDOW_CLIENTS

DBA_AUTOTASK_WINDOW_CLIENTS displays the windows that belong to MAINTENANCE_WINDOW_GROUP, along with the Enabled or Disabled status for the window for each maintenance task.

DBA_AUTOTASK_WINDOW_CLIENTS is primarily used by Enterprise Manager.

Column	Datatype	NULL	Description
WINDOW_NAME	VARCHAR2(128)	NOT NULL	Name of the maintenance window
WINDOW_NEXT_TIME	TIMESTAMP(6) WITH TIME ZONE		Next scheduled window open time unless the window is disabled
WINDOW_ACTIVE	VARCHAR2(5)		Indicates whether the window is currently active (open) (TRUE) or not (FALSE)
AUTOTASK_STATUS	VARCHAR2(8)		Status of the automated maintenance task subsystem: • ENABLED • DISABLED
OPTIMIZER_STATS	VARCHAR2(8)		Status of optimizer statistics gathering: • ENABLED • DISABLED
SEGMENT_ADVISOR	VARCHAR2(8)		Status of Segment Advisor: • ENABLED • DISABLED

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SQL_TUNE_ADVISOR	VARCHAR2(8)		Status of SQL Tuning Advisor: • ENABLED • DISABLED

4.178 DBA_AUTOTASK_WINDOW_HISTORY

DBA_AUTOTASK_WINDOW_HISTORY displays historical information for automated maintenance task windows.

Column	Datatype	NULL	Description
WINDOW_NAME	VARCHAR2(261)		Name of the maintenance window
WINDOW_START_TIME	TIMESTAMP(6) WITH TIME ZONE		Window start time
WINDOW_END_TIME	TIMESTAMP(6) WITH TIME ZONE		Window end time

4.179 DBA_AW_PS

DBA_AW_PS describes the page spaces in all analytic workspaces in the database. Its columns are the same as those in ALL_AW_PS.

See Also

"ALL_AW_PS"

4.180 DBA_AWS

DBA_AWS describes all analytic workspaces in the database. Its columns are the same as those in ALL_AWS.

See Also

"ALL_AWS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.181 DBA_BASE_TABLE_MVIEWS

DBA_BASE_TABLE_MVIEWS describes all materialized views using materialized view logs in the database. Its columns are the same as those in ALL_BASE_TABLE_MVIEWS.

See Also

"ALL_BASE_TABLE_MVIEWS"

4.182 DBA_BLOCKCHAIN_TABLES

DBA_BLOCKCHAIN_TABLES describes all blockchain tables in the database. Its columns are the same as those in ALL_BLOCKCHAIN_TABLES.

Note

This view is available starting with Oracle Database 19c, Release Update 19.10.

See Also

"ALL_BLOCKCHAIN_TABLES"

4.183 DBA_BLOCKERS

DBA_BLOCKERS displays a session if it is not waiting for a locked object but is holding a lock on an object for which another session is waiting.

In an Oracle RAC environment, this only applies if the blocker is on the same instance.

Column	Datatype	NULL	Description
HOLDING_SESSION	NUMBER		Session holding a lock
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.184 DBA_CAPTURE

DBA_CAPTURE displays information about all capture processes in the database. Its columns are the same as those in ALL_CAPTURE.

See Also

"ALL_CAPTURE"

4.185 DBA_CAPTURE_EXTRA_ATTRIBUTES

DBA_CAPTURE_EXTRA_ATTRIBUTES displays information about the extra attributes for all capture processes in the database. Its columns are the same as those in ALL_CAPTURE_EXTRA_ATTRIBUTES.

See Also

"ALL_CAPTURE_EXTRA_ATTRIBUTES"

4.186 DBA_CAPTURE_PARAMETERS

DBA_CAPTURE_PARAMETERS displays information about the parameters for all capture processes in the database. Its columns are the same as those in ALL_CAPTURE_PARAMETERS.

See Also

"ALL_CAPTURE_PARAMETERS"

4.187 DBA_CAPTURE_PREPARED_DATABASE

DBA_CAPTURE_PREPARED_DATABASE displays information about when the local database was prepared for instantiation. Its columns are the same as those in ALL_CAPTURE_PREPARED_DATABASE.

See Also

"ALL_CAPTURE_PREPARED_DATABASE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.188 DBA_CAPTURE_PREPARED_SCHEMAS

DBA_CAPTURE_PREPARED_SCHEMAS displays information about all schemas prepared for instantiation at the local database. Its columns are the same as those in ALL_CAPTURE_PREPARED_SCHEMAS.

See Also

"ALL_CAPTURE_PREPARED_SCHEMAS"

4.189 DBA_CAPTURE_PREPARED_TABLES

DBA_CAPTURE_PREPARED_TABLES displays information about all tables prepared for instantiation at the local database. Its columns are the same as those in ALL_CAPTURE_PREPARED_TABLES.

See Also

"ALL_CAPTURE_PREPARED_TABLES"

4.190 DBA_CATALOG

DBA_CATALOG lists all tables, views, clusters, synonyms, and sequences in the database. Its columns are the same as those in ALL_CATALOG.

See Also

"ALL_CATALOG"

4.191 DBA_CDB_RSRC_PLAN_DIRECTIVES

DBA_CDB_RSRC_PLAN_DIRECTIVES provides information about all the CDB resource plan directives.

Note

ORA$DEFAULT_PDB_DIRECTIVE is the default directive for PDBs. For more information about ORA$DEFAULT_PDB_DIRECTIVE, see *Oracle Multitenant Administrator's Guide*.

Column	Datatype	NULL	Description
PLAN	VARCHAR2(128)		Name of the CDB resource plan to which this directive belongs

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PLUGGABLE_DATABASE	VARCHAR2(128)		Name of the PDB referred to. NULL for profile directives
PROFILE	VARCHAR2(128)		For internal use only
DIRECTIVE_TYPE	VARCHAR2(30)		For internal use only
SHARES	NUMBER		Resource allocation, expressed in shares
UTILIZATION_LIMIT	NUMBER		Maximum resource utilization allowed, expressed in percentage
PARALLEL_SERVER_LIMIT	NUMBER		Maximum percentage of the parallel target used before queueing subsequent parallel queries
MEMORY_MIN	NUMBER		The percentage of Exadata Smart Flash Cache and Exadata PMEM cache that is guaranteed to the PDB This percentage is based on the total amount of space allocated to the CDB for Exadata Smart Flash Cache and Exadata PMEM cache. See *Oracle Exadata System Software User's Guide* for more information.
MEMORY_LIMIT	NUMBER		The maximum percentage of Exadata Smart Flash Cache and Exadata PMEM cache that the PDB can use This percentage is based on the total amount of space allocated to the CDB for Exadata Smart Flash Cache and Exadata PMEM cache. See *Oracle Exadata System Software User's Guide* for more information.
COMMENTS	VARCHAR2(2000)		Text comment on the resource plan directive
STATUS	VARCHAR2(128)		PENDING if it is part of the pending area, NULL otherwise
MANDATORY	VARCHAR2(3)		Whether the resource plan directive is mandatory. Mandatory plans cannot be deleted.

> **Note**
>
> Oracle recommends that you do not use the parallel_server_limit directive in a CDB resource plan.

4.192 DBA_CDB_RSRC_PLANS

DBA_CDB_RSRC_PLANS provides information about all the CDB resource plans.

Column	Datatype	NULL	Description
PLAN_ID	NUMBER	NOT NULL	CDB resource plan ID
PLAN	VARCHAR2(128)		CDB resource plan name
COMMENTS	VARCHAR2(2000)		Text comment on the CDB resource plan

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(128)		PENDING if it is part of the pending area, NULL otherwise
MANDATORY	VARCHAR2(3)		Whether the resource plan is mandatory. Mandatory plans cannot be deleted.

4.193 DBA_CERTIFICATES

DBA_CERTIFICATES displays all certificates in the database which are used for signature verification for blockchain tables. Its columns are the same as those in ALL_CERTIFICATES.

Note

This view is available starting with Oracle Database 19c, Release Update 19.10.

See Also

"ALL_CERTIFICATES"

4.194 DBA_CHANGE_NOTIFICATION_REGS

DBA_CHANGE_NOTIFICATION_REGS describes all change notification registrations in the database.

Related View

USER_CHANGE_NOTIFICATION_REGS describes the change notification registrations owned by the current user. This view does not display the USERNAME column.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(31)		For invoker's rights units, the user creating the registration For definer's rights units, the owner of the registration
REGID	NUMBER		Internal registration ID
REGFLAGS	NUMBER		Registration flags
CALLBACK	VARCHAR2(256)		Notification callback
OPERATIONS_FILTER	NUMBER		Operations filter (if specified)
CHANGELAG	NUMBER		Transaction lag between notifications (if specified)
TIMEOUT	NUMBER		Registration timeout (if specified)
TABLE_NAME	VARCHAR2(63)		Name of the registered table

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "USER_CHANGE_NOTIFICATION_REGS"

4.195 DBA_CHECKED_ROLES

DBA_CHECKED_ROLES lists the roles (without role grant paths) that are used for the role analysis policies reported by the DBMS__CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed role records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of a role analysis policy
SEQUENCE	NUMBER	NOT NULL	The sequence number of the role analysis run during which the role was reported
OS_USER	VARCHAR2(128)		Operating system login username
USERHOST	VARCHAR2(128)		Client host machine name
MODULE	VARCHAR2(64)		Module name
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user whose role was reported
CHECKED_ROLE	VARCHAR2(128)		Checked role
RUN_NAME	VARCHAR2(128)		The name of the run during which the role was reported

> **See Also**
>
> "DBA_CHECKED_ROLES_PATH"

4.196 DBA_CHECKED_ROLES_PATH

DBA_CHECKED_ROLES_PATH lists the roles that are used for the role analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed role records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of a role analysis policy
SEQUENCE	NUMBER	NOT NULL	The sequence number of the role analysis run during which the role was reported
OS_USER	VARCHAR2(128)		Operating system login username
USERHOST	VARCHAR2(128)		Client host machine name
MODULE	VARCHAR2(64)		Module name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user whose role was reported
CHECKED_ROLE	VARCHAR2(128)		Checked role
PATH	GRANT_PATH		Role grant paths
RUN_NAME	VARCHAR2(128)		The name of the run during which the role was reported

See Also

"DBA_CHECKED_ROLES"

4.197 DBA_CLU_COLUMNS

DBA_CLU_COLUMNS maps all table columns to related cluster columns.

Related View

USER_CLU_COLUMNS maps all table columns owned by the current user to related cluster columns. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the cluster
CLUSTER_NAME	VARCHAR2(128)	NOT NULL	Name of the cluster
CLU_COLUMN_NAME	VARCHAR2(128)	NOT NULL	Key column in the cluster
TABLE_NAME	VARCHAR2(128)	NOT NULL	Clustered table name
TAB_COLUMN_NAME	VARCHAR2(4000)		Key column or attribute of the object type column

See Also

"USER_CLU_COLUMNS"

4.198 DBA_CLUSTER_HASH_EXPRESSIONS

DBA_CLUSTER_HASH_EXPRESSIONS lists hash functions for all hash clusters in the database. Its columns are the same as those in ALL_CLUSTER_HASH_EXPRESSIONS.

See Also

"ALL_CLUSTER_HASH_EXPRESSIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.199 DBA_CLUSTERING_DIMENSIONS

`DBA_CLUSTERING_DIMENSIONS` describes dimension tables associated with all tables with an attribute clustering clause in the database. Its columns are the same as those in `ALL_CLUSTERING_DIMENSIONS`.

See Also

- "ALL_CLUSTERING_DIMENSIONS"
- *Oracle Database Data Warehousing Guide* for information about attribute clustering with zone maps

4.200 DBA_CLUSTERING_JOINS

`DBA_CLUSTERING_JOINS` describes joins to the dimension tables associated with all tables with an attribute clustering clause in the database. Its columns are the same as those in `ALL_CLUSTERING_JOINS`.

See Also

- "ALL_CLUSTERING_JOINS"
- *Oracle Database Data Warehousing Guide* for information about attribute clustering with zone maps

4.201 DBA_CLUSTERING_KEYS

`DBA_CLUSTERING_KEYS` describes clustering keys for all tables with an attribute clustering clause. Its columns are the same as those in `ALL_CLUSTERING_KEYS`.

See Also

- "ALL_CLUSTERING_KEYS"
- *Oracle Database Data Warehousing Guide* for information about attribute clustering with zone maps

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.202 DBA_CLUSTERING_TABLES

DBA_CLUSTERING_TABLES describes all the tables with an attribute clustering clause. Its columns are the same as those in ALL_CLUSTERING_TABLES.

See Also

- "ALL_CLUSTERING_TABLES"
- *Oracle Database Data Warehousing Guide* for information about attribute clustering with zone maps

4.203 DBA_CLUSTERS

DBA_CLUSTERS describes all clusters in the database. Its columns are the same as those in ALL_CLUSTERS.

Note

"ALL_CLUSTERS"

4.204 DBA_CODE_ROLE_PRIVS

DBA_CODE_ROLE_PRIVS describes all the roles that are associated with program units in the database. Its columns are the same as those in ALL_CODE_ROLE_PRIVS.

See Also

"ALL_CODE_ROLE_PRIVS"

4.205 DBA_COL_COMMENTS

DBA_COL_COMMENTS displays comments on the columns of all tables and views in the database. Its columns are the same as those in ALL_COL_COMMENTS.

See Also

"ALL_COL_COMMENTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.206 DBA_COL_PENDING_STATS

DBA_COL_PENDING_STATS describes the pending statistics of all columns in the database. Its columns are the same as those in ALL_COL_PENDING_STATS.

> **See Also**
>
> "ALL_COL_PENDING_STATS"

4.207 DBA_COL_PRIVS

DBA_COL_PRIVS describes all column object grants in the database.

Related View

USER_COL_PRIVS describes the column object grants for which the current user is the object owner, grantor, or grantee.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)		Name of the user or role to whom access was granted
OWNER	VARCHAR2(128)		Owner of the object
TABLE_NAME	VARCHAR2(128)		Name of the object
COLUMN_NAME	VARCHAR2(128)		Name of the column
GRANTOR	VARCHAR2(128)		Name of the user who performed the grant
PRIVILEGE	VARCHAR2(40)		Privilege on the column
GRANTABLE	VARCHAR2(3)		Indicates whether the privilege was granted with the GRANT OPTION (YES) or not (NO)
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the privilege was granted commonly (CONTAINER=ALL was used) • NO if the privilege was granted locally (CONTAINER=ALL was not used)
INHERITED	VARCHAR2(3)		Indicates whether the privilege grant was inherited from another container (YES) or not (NO)

> **See Also**
>
> "USER_COL_PRIVS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.208 DBA_COLL_TYPES

DBA_COLL_TYPES describes all named collection types (arrays, nested tables, object tables, and so on) in the database. Its columns are the same as those in ALL_COLL_TYPES.

> **Note**
>
> "ALL_COLL_TYPES"

4.209 DBA_COMMON_AUDIT_TRAIL

DBA_COMMON_AUDIT_TRAIL displays all standard and fine-grained audit trail entries, mandatory and SYS audit records written in XML format.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from UNIFIED_AUDIT_TRAIL.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
AUDIT_TYPE	VARCHAR2(22)		Audit trail type: • Standard Audit • Standard XML Audit • Fine Grained Audit • Fine Grained XML Audit • SYS XML Audit • Mandatory XML Audit
SESSION_ID	NUMBER		Numeric ID for the Oracle session
PROXY_SESSIONID	NUMBER		Proxy session serial number, if an enterprise user has logged in through the proxy mechanism
STATEMENTID	NUMBER		Numeric ID for the statement run; a statement may cause multiple audit records
ENTRYID	NUMBER		Numeric ID for the audit trail entry in the session
EXTENDED_TIMESTAMP	TIMESTAMP(6) WITH TIME ZONE		Timestamp of the audited operation (timestamp of user login for entries created by AUDIT SESSION) in the session's time zone
GLOBAL_UID	VARCHAR2(32)		Global user identifier for the user, if the user has logged in as an enterprise user
DB_USER	VARCHAR2(128)		Database user name of the user whose actions were audited

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CLIENT_ID	VARCHAR2(128)		Client identifier in the Oracle session
ECONTEXT_ID	VARCHAR2(64)		Application execution context identifier
EXT_NAME	VARCHAR2(4000)		User external name
OS_USER	VARCHAR2(255)		Operating system login user name of the user whose actions were audited
USERHOST	VARCHAR2(128)		Client host machine name
OS_PROCESS	VARCHAR2(16)		Operating system process identifier of the Oracle process
TERMINAL	VARCHAR2(255)		Identifier of the user's terminal
INSTANCE_NUMBER	NUMBER		Instance number as specified by the INSTANCE_NUMBER initialization parameter
OBJECT_SCHEMA	VARCHAR2(128)		Owner of the audited object
OBJECT_NAME	VARCHAR2(128)		Name of the object affected by the action
POLICY_NAME	VARCHAR2(128)		Name of the Fine-Grained Auditing Policy
NEW_OWNER	VARCHAR2(128)		Owner of the object named in the NEW_NAME column
NEW_NAME	VARCHAR2(128)		New name of the object after a RENAME or the name of the underlying object
ACTION	NUMBER		Numeric action type code. The corresponding name of the action type is in the STATEMENT_TYPE column.
STATEMENT_TYPE	VARCHAR2(28)		Name of the action type corresponding to the numeric code in the ACTION column
AUDIT_OPTION	VARCHAR2(40)		Auditing option set with the AUDIT statement
TRANSACTIONID	RAW(8)		Transaction identifier of the transaction in which the object was accessed or modified
RETURNCODE	NUMBER		Oracle error code generated by the action (0 if the action succeeded)
SCN	NUMBER		System change number (SCN) of the query
COMMENT_TEXT	VARCHAR2(4000)		Text comment on the audit trail entry, providing more information about the statement audited Also indicates how the user was authenticated: • DATABASE - Authentication was done by password • NETWORK - Authentication was done by Oracle Net Services or the Advanced Networking Option • PROXY - Client was authenticated by another user. The name of the proxy user follows the method type.
SQL_BIND	NVARCHAR2(2000)		Bind variable data of the query
SQL_TEXT	NVARCHAR2(2000)		SQL text of the query

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OBJ_PRIVILEGE	VARCHAR2(32)		Object privileges granted or revoked by a GRANT or REVOKE statement. The value of this column is a 32-character string of Y and dash (-) characters. Each character corresponds to a numbered privilege in the following list. The left-most character corresponds to privilege 0, the next character corresponds to privilege 1, and so on. The right-most character corresponds to privilege 31. 0 - ALTER 1 - AUDIT 2 - COMMENT 3 - DELETE 4 - GRANT 5 - INDEX 6 - INSERT 7 - LOCK 8 - CREATE 9 - SELECT 10 - UPDATE 11 - REFERENCES 12 - EXECUTE 13 - VIEW 14 - DROP 15 - ANALYZE 16 - CREATE 17 - READ 18 - WRITE 19 - KEEP SEQUENCE 20 - ENQUEUE 21 - DEQUEUE 22 - UNDER 23 - ON COMMIT 24 - REWRITE 25 - UPSERT 26 - DEBUG 27 - FLASHBACK 28 - MERGE 29 - USE 30 - FLASHBACK ARCHIVE 31 - DIRECTORY EXECUTE A Y indicates that the privilege was granted or revoked by the statement. A dash indicates that the privilege was not affected by the statement. For example, the following value indicates that the MERGE privilege was granted or revoked by the statement: ----------------------------Y---
SYS_PRIVILEGE	VARCHAR2(40)		System privileges granted or revoked by a GRANT or REVOKE statement
ADMIN_OPTION	VARCHAR2(1)		Indicates whether the role or system privilege was granted with the ADMIN option

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OS_PRIVILEGE	VARCHAR2(7)		Operating privilege (SYSDBA or SYSOPER), if any, used in the session. If no privilege is used, it will be NONE.
GRANTEE	VARCHAR2(128)		Name of the grantee specified in a GRANT or REVOKE statement
PRIV_USED	VARCHAR2(40)		System privilege used to execute the action
SES_ACTIONS	VARCHAR2(19)		Session summary (a string of 16 characters, one for each action type in the order ALTER, AUDIT, COMMENT, DELETE, GRANT, INDEX, INSERT, LOCK, RENAME, SELECT, UPDATE, REFERENCES, and EXECUTE). Positions 14, 15, and 16 are reserved for future use. The characters are: • - - None • S - Success • F - Failure • B - Both
LOGOFF_TIME	DATE		Timestamp of user log off
LOGOFF_LREAD	NUMBER		Number of logical reads in the session
LOGOFF_PREAD	NUMBER		Number of physical reads in the session
LOGOFF_LWRITE	NUMBER		Number of logical writes for the session
LOGOFF_DLOCK	VARCHAR2(40)		Number of deadlocks detected during the session
SESSION_CPU	NUMBER		Amount of CPU time used by the Oracle session
OBJ_EDITION_NAME	VARCHAR2(128)		Name of the edition containing the audited object
DBID	NUMBER		Database identifier of the audited database
RLS_INFO	CLOB		Stores virtual private database (VPD) policy names and predicates separated by delmiter
COMMON_USER	VARCHAR2(128)		Effective user for the statement execution

Note

The SQL_BIND and SQL_TEXT columns are only populated if the AUDIT_TRAIL initialization parameter is set to db, extended or xml, extended or if the AUDIT_SYS_OPERATIONS initialization parameter is set to TRUE.

See Also

- "AUDIT_SYS_OPERATIONS"
- "AUDIT_TRAIL"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.210 DBA_COMPARISON

DBA_COMPARISON displays information about all comparison objects in the database.

Related View

USER_COMPARISON displays information about the comparison objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the comparison
COMPARISON_NAME	VARCHAR2(128)	NOT NULL	Name of the comparison
COMPARISON_MODE	VARCHAR2(5)		Mode of the comparison: • TABLE
SCHEMA_NAME	VARCHAR2(128)		Schema name of the local object
OBJECT_NAME	VARCHAR2(128)		Name of the local object
OBJECT_TYPE	VARCHAR2(17)		Type of the local object: • TABLE • VIEW • SYNONYM • MATERIALIZED VIEW
REMOTE_SCHEMA_NAME	VARCHAR2(128)		Schema name of the remote object
REMOTE_OBJECT_NAME	VARCHAR2(128)		Name of the remote object
REMOTE_OBJECT_TYPE	VARCHAR2(17)		Type of the remote object: • TABLE • VIEW • SYNONYM • MATERIALIZED VIEW
DBLINK_NAME	VARCHAR2(128)		Database link name to the remote database
SCAN_MODE	VARCHAR2(9)		Scan mode of the comparison: FULL • FULL • RANDOM • CYCLIC • CUSTOM
SCAN_PERCENT	NUMBER		Scan percent of the comparison; applicable to random and cyclic modes
CYCLIC_INDEX_VALUE	VARCHAR2(4000)		Last index column value used in a cyclic scan
NULL_VALUE	VARCHAR2(4000)		Value to use for NULL columns
LOCAL_CONVERGE_TAG	RAW(2000)		Local Replication tag used while performing converge DMLs
REMOTE_CONVERGE_TAG	RAW(2000)		Remote Replication tag used while performing converge DMLs
MAX_NUM_BUCKETS	NUMBER		Suggested maximum number of buckets in a scan
MIN_ROWS_IN_BUCKET	NUMBER		Suggested minimum number of rows in a bucket
LAST_UPDATE_TIME	TIMESTAMP(6)		Time that this row was last updated

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "USER_COMPARISON"

4.211 DBA_COMPARISON_COLUMNS

DBA_COMPARISON_COLUMNS displays information about the columns for all comparison objects in the database.

Related View

USER_COMPARISON_COLUMNS displays information about the columns for the comparison objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the comparison
COMPARISON_NAME	VARCHAR2(128)	NOT NULL	Name of the comparison
COLUMN_POSITION	NUMBER	NOT NULL	Position of the column
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column
INDEX_COLUMN	VARCHAR2(1)		Indicates whether the column is an index column (Y) or not (N)

> **See Also**
>
> "USER_COMPARISON_COLUMNS"

4.212 DBA_COMPARISON_ROW_DIF

DBA_COMPARISON_ROW_DIF displays information about the differing rows in all comparison scans in the database.

Related View

USER_COMPARISON_ROW_DIF displays information about the differing rows in the comparison scans owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the comparison
COMPARISON_NAME	VARCHAR2(128)	NOT NULL	Name of the comparison
SCAN_ID	NUMBER	NOT NULL	Scan ID for the comparison scan
LOCAL_ROWID	ROWID		Local rowid of the differing row
REMOTE_ROWID	ROWID		Remote rowid of the differing row
INDEX_VALUE	VARCHAR2(4000)		Index column value of the differing row

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(3)		Status of the differing row: • SUC • DIF
LAST_UPDATE_TIME	TIMESTAMP(6)		Time that this row was last updated

See Also

"USER_COMPARISON_ROW_DIF"

4.213 DBA_COMPARISON_SCAN

DBA_COMPARISON_SCAN displays information about all comparison scans in the database.

Related View

USER_COMPARISON_SCAN displays information about the comparison scans owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the comparison scan
COMPARISON_NAME	VARCHAR2(128)	NOT NULL	Name of the comparison scan
SCAN_ID	NUMBER	NOT NULL	Scan ID
PARENT_SCAN_ID	NUMBER		Scan ID of the immediate parent scan
ROOT_SCAN_ID	NUMBER		Scan ID of the root (top-most) parent
STATUS	VARCHAR2(16)		Status of the scan: • SUC • BUCKET DIF • FINAL BUCKET DIF • ROW DIF
CURRENT_DIF_COUNT	NUMBER		Current cumulative (including children) diff count of the scan
INITIAL_DIF_COUNT	NUMBER		Initial cumulative (including children) diff count of the scan
COUNT_ROWS	NUMBER		Number of rows in the scan
SCAN_NULLS	VARCHAR2(1)		Indicates whether NULLs are part of this scan (Y) or not (N)
LAST_UPDATE_TIME	TIMESTAMP(6)		Time that this row was last updated

See Also

"USER_COMPARISON_SCAN"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.214 DBA_COMPARISON_SCAN_VALUES

DBA_COMPARISON_SCAN_VALUES displays information about the values for all comparison scans in the database.

Related View

USER_COMPARISON_SCAN_VALUES displays information about the values for the comparison scans owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the comparison scan
COMPARISON_SCAN	VARCHAR2(128)	NOT NULL	Name of the comparison scan
SCAN_ID	NUMBER	NOT NULL	Scan ID
COLUMN_POSITION	NUMBER	NOT NULL	Column position, as in DBA_COMPARISON_COLUMNS
MIN_VALUE	VARCHAR2(4000)		Minimum value of the scan
MAX_VALUE	VARCHAR2(4000)		Maximum value of the scan
LAST_UPDATE_TIME	TIMESTAMP(6)		Time that this row was last updated

See Also

"USER_COMPARISON_SCAN_VALUES"

4.215 DBA_CONNECT_ROLE_GRANTEES

DBA_CONNECT_ROLE_GRANTEES displays information about users who are granted the CONNECT privilege.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)		User or schema to which the CONNECT role is granted
PATH_OF_CONNECT_ROLE_GRANT	VARCHAR2(4000)		The path of role inheritance through which the grantee is granted the CONNECT role
ADMIN_OPT	VARCHAR2(3)		Whether or not the grantee was granted the ADMIN option for the CONNECT role

4.216 DBA_CONNECTION_TESTS

DBA_CONNECTION_TESTS provides information about connection tests in use for CDBs and PDBs.

This view shows SQL and non-SQL connection tests.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PREDEFINED	VARCHAR2(1)		Indicates whether the test is predefined or custom. Possible values: • Y: The test is predefined. • N: The test is added by the user. Predefined tests can be disabled, but not deleted.
CONNECTION_TEST_TYPE	VARCHAR2(15)		Indicates the test type. Possible values include: • ENDREQUEST_TEST: Used when request boundaries are received at the RDBMS. Oracle Connection Pools and all application servers using JDK9 send request boundaries to the RDBMS starting in Oracle Database 12*c* Release 2 (12.2.0.1). • PING_TEST: Used when you are using tests that use the ping function to test the connection, including the OCIPing, isValid, isUsable, connection.status, and PING_DATABASE connection tests. • SQL_TEST: Application servers and applications use SQL tests to check the validity of a connection. Use this value for SQL based connection tests, for example (SELECT 1 FROM DUAL;) • XA_START_TEST: Applies to services with applications that are using XA transactions. Allows XA tranasactions that are underway to complete and does not allow new XA transactions to start. The test type values are the CONNECTION_TEST_TYPE parameter values that can be specified for the ENABLE_CONNECTION_TEST and DISABLE_CONNECTION_TEST procedures for the DBMS_APP_CONT_ADMIN PL/SQL package.
SQL_CONNECTION_TEST	VARCHAR2(64)		SQL test. This column is null for non-SQL tests.
SERVICE_NAME	VARCHAR2(128)		Optional service name qualifier
ENABLED	VARCHAR2(1)		Indicates whether the SQL test is enabled. Possible values: • Y: The test is enabled. • N: The test is not enabled.

See Also

Oracle Database PL/SQL Packages and Types Reference for additional information about the ENABLE_CONNECTION_TEST procedure for the DBMS_APP_CONT_ADMIN PL/SQL package

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.217 DBA_CONS_COLUMNS

DBA_CONS_COLUMNS describes all columns in the database that are specified in constraints. Its columns are the same as those in ALL_CONS_COLUMNS.

> **See Also**
>
> "ALL_CONS_COLUMNS"

4.218 DBA_CONS_OBJ_COLUMNS

DBA_CONS_OBJ_COLUMNS displays information about the types that object columns (or attributes) or collection elements have been constrained to, in all tables in the database. Its columns are the same as those in ALL_CONS_OBJ_COLUMNS.

> **See Also**
>
> "ALL_CONS_OBJ_COLUMNS"

4.219 DBA_CONSTRAINTS

DBA_CONSTRAINTS describes all constraint definitions on all tables in the database. Its columns are the same as those in ALL_CONSTRAINTS.

> **See Also**
>
> "ALL_CONSTRAINTS"

4.220 DBA_CONTAINER_DATA

DBA_CONTAINER_DATA displays default (user-level) and object-specific CONTAINER_DATA attributes for container data objects.

Objects created with the CONTAINER_DATA clause include CONTAINER_DATA attributes.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(128)		Name of the user whose attribute is described by this row
DEFAULT_ATTR	CHAR2(1)		An indicator of whether the attribute is a default attribute
OWNER	VARCHAR2(128)		Name of the object owner if the attribute is object-specific
OBJECT_NAME	VARCHAR2(128)		Name of the object if the attribute is object-specific

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ALL_CONTAINERS	VARCHAR2(1)		An indicator of whether this attribute applies to all containers
CONTAINER_NAME	VARCHAR2(128)		Name of a container included in this attribute if it does not apply to all containers

See Also

For more information about container data objects:

- "CDB_* Views"
- "ALL_TABLES"
- "ALL_VIEWS"
- "ALL_VIEWS_AE"
- "V$ Views"
- "GV$ Views"
- *Oracle Multitenant Administrator's Guide*
- *Oracle Database Security Guide*

4.221 DBA_CONTEXT

DBA_CONTEXT provides all context namespace information in the database.

Related View

ALL_CONTEXT describes all context namespaces in the current session for which attributes and values have been specified using the DBMS_SESSION.SET_CONTEXT procedure. This view does not describe the TYPE and ORIGIN_CON_ID columns.

Column	Datatype	NULL	Description
NAMESPACE	VARCHAR2(128)	NOT NULL	Name of the context namespace
SCHEMA	VARCHAR2(128)	NOT NULL	Schema name of the designated package that can set attributes using this namespace
PACKAGE	VARCHAR2(128)	NOT NULL	Package name of the designated package that can set attributes using this namespace
TYPE	VARCHAR2(22)		Type of the context create
ORIGIN_CON_ID	NUMBER		The ID of the container where the data originates. Possible values include: • 0: This value is used for rows in non-CDBs. This value is not used for CDBs. • *n*: This value is used for rows containing data that originate in the container with container ID *n* (*n* = 1 if the row originates in root)

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"ALL_CONTEXT"

4.222 DBA_CPOOL_INFO

DBA_CPOOL_INFO displays configuration information about all Database Resident Connection Pools in the database.

Column	Datatype	NULL	Description
CONNECTION_POOL	VARCHAR2(128)		Name of the connection pool
STATUS	VARCHAR2(16)		Status of the pool: • ACTIVE • INACTIVE
MINSIZE	NUMBER		Minimum number of pooled servers that are always running in the pool
MAXSIZE	NUMBER		Maximum number of pooled servers in the pool
INCRSIZE	NUMBER		Number of pooled servers by which the pool is incremented if servers are unavailable when a client application request is received
SESSION_CACHED_CURSORS	NUMBER		Number of session cursors to cache in each pooled server
INACTIVITY_TIMEOUT	NUMBER		Maximum time (in seconds) that the pooled server can stay idle in the pool. After this time, the server is terminated.
MAX_THINK_TIME	NUMBER		Maximum time of inactivity (in seconds) for a client after obtaining a pooled server from the pool. After obtaining a pooled server from the pool, if the client application does not issue a database call for the time specified by this column, then the pooled server is freed and the client connection is terminated.
MAX_USE_SESSION	NUMBER		Maximum number of acquire-and-release cycles allowed for a pooled server before it is terminated and replaced The default value is 500000. A value of 0 indicates that the usage limit is disabled.
MAX_LIFETIME_SESSION	NUMBER		Maximum lifetime (in seconds) for a pooled server If the lifetime expires for a server that is being returned to the pool or is free in the pool, it is removed from the pool and cleaned up. The default value is 86400. A value of 0 indicates that the lifetime limit is disabled.
NUM_CBROK	NUMBER		Number of connection brokers spawned per instance
MAXCONN_CBROK	NUMBER		Maximum number of connections per connection broker

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MAX_TXN_THINK_TIME	NUMBER		Maximum time of inactivity (in seconds) for a client after it obtains a pooled server from the pool and starts a transaction. If the client application does not issue a database call for the time specified by MAX_TXN_THINK_TIME while in a transaction, the pooled server is freed, the transaction is rolled back, and the client connection is terminated. The default value is 0, which means MAX_THINK_TIME applies for all connections, irrespective of transactions being open or not in those connections. Care should be taken while setting the two parameters MAX_THINK_TIME and MAX_TXN_THINK_TIME to higher values, as it would mean the servers are not released to the pool for a longer time, even if clients are not responding for any reason, thereby making other users wait unnecessarily.
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- *Oracle Database Administrator's Guide* for more information about the configuration parameters for Database Resident Connection Pooling
- *Oracle Database PL/SQL Packages and Types Reference* for more information on the DBMS_CONNECTION_POOL package

4.223 DBA_CPU_USAGE_STATISTICS

DBA_CPU_USAGE_STATISTICS displays database CPU usage statistics.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
VERSION	VARCHAR2(17)	NOT NULL	Database version
TIMESTAMP	DATE	NOT NULL	Time at which the CPU usage changed
CPU_COUNT	NUMBER		CPU count of the database
CPU_CORE_COUNT	NUMBER		CPU core count of the database
CPU_SOCKET_COUNT	NUMBER		CPU socket count of the database

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.224 DBA_CQ_NOTIFICATION_QUERIES

DBA_CQ_NOTIFICATION_QUERIES describes the registered queries for all CQ notifications in the database.

Related View

USER_CQ_NOTIFICATION_QUERIES describes the registered queries for the CQ notifications owned by the current user. This view does not display the USERNAME column.

Column	Datatype	NULL	Description
QUERYID	NUMBER		ID of the query
QUERYTEXT	CLOB		Text of the query
REGID	NUMBER		Registration ID that the query is registered with
USERNAME	VARCHAR2(31)		Name of the user who registered the query

See Also

"USER_CQ_NOTIFICATION_QUERIES"

4.225 DBA_CREDENTIALS

DBA_CREDENTIALS lists all credentials in the database. Its columns are the same as those in ALL_CREDENTIALS.

See Also

"ALL_CREDENTIALS"

4.226 DBA_CUBE_ATTR_VISIBILITY

DBA_CUBE_ATTR_VISIBILITY describes all OLAP attributes visible for the dimensions, hierarchies, and levels in the database. Its columns are the same as those in ALL_CUBE_ATTR_VISIBILITY.

See Also

"ALL_CUBE_ATTR_VISIBILITY"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.227 DBA_CUBE_ATTRIBUTES

DBA_CUBE_ATTRIBUTES describes the attributes for all OLAP cube dimensions in the database. Its columns are the same as those in ALL_CUBE_ATTRIBUTES.

See Also

"ALL_CUBE_ATTRIBUTES"

4.228 DBA_CUBE_BUILD_PROCESSES

DBA_CUBE_BUILD_PROCESSES describes all OLAP build processes and maintenance scripts in the database. Its columns are the same as those in ALL_CUBE_BUILD_PROCESSES.

See Also

"ALL_CUBE_BUILD_PROCESSES"

4.229 DBA_CUBE_CALCULATED_MEMBERS

DBA_CUBE_CALCULATED_MEMBERS describes the calculated members for all OLAP cube dimensions in the database. Its columns are the same as those in ALL_CUBE_CALCULATED_MEMBERS.

See Also

"ALL_CUBE_CALCULATED_MEMBERS"

4.230 DBA_CUBE_DIM_LEVELS

DBA_CUBE_DIM_LEVELS describes all OLAP dimension levels in the database. Its columns are the same as those in ALL_CUBE_DIM_LEVELS.

See Also

"ALL_CUBE_DIM_LEVELS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.231 DBA_CUBE_DIM_MODELS

DBA_CUBE_DIM_MODELS describes the models for all OLAP dimensions in the database. Its columns are the same as those in ALL_CUBE_DIM_MODELS.

See Also

"ALL_CUBE_DIM_MODELS"

4.232 DBA_CUBE_DIM_VIEW_COLUMNS

DBA_CUBE_DIM_VIEW_COLUMNS describes the columns of the relational views of all OLAP cube dimensions in the database. Its columns are the same as those in ALL_CUBE_DIM_VIEW_COLUMNS.

See Also

"ALL_CUBE_DIM_VIEW_COLUMNS"

4.233 DBA_CUBE_DIM_VIEWS

DBA_CUBE_DIM_VIEWS describes the relational views of all OLAP dimensions in the database. Its columns are the same as those in ALL_CUBE_DIM_VIEWS.

See Also

"ALL_CUBE_DIM_VIEWS"

4.234 DBA_CUBE_DIMENSIONALITY

DBA_CUBE_DIMENSIONALITY describes the dimension order for all OLAP cubes in the database. Its columns are the same as those in ALL_CUBE_DIMENSIONALITY.

See Also

"ALL_CUBE_DIMENSIONALITY"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.235 DBA_CUBE_DIMENSIONS

DBA_CUBE_DIMENSIONS describes all OLAP cube dimensions in the database. Its columns are the same as those in ALL_CUBE_DIMENSIONS.

See Also

"ALL_CUBE_DIMENSIONS"

4.236 DBA_CUBE_HIER_LEVELS

DBA_CUBE_HIER_LEVELS describes the hierarchy levels for all OLAP cube dimensions in the database. Its columns are the same as those in ALL_CUBE_HIER_LEVELS.

See Also

"ALL_CUBE_HIER_LEVELS"

4.237 DBA_CUBE_HIER_VIEW_COLUMNS

DBA_CUBE_HIER_VIEW_COLUMNS describes the columns of the relational hierarchy views of all OLAP cube dimensions in the database. Its columns are the same as those in ALL_CUBE_HIER_VIEW_COLUMNS.

See Also

"ALL_CUBE_HIER_VIEW_COLUMNS"

4.238 DBA_CUBE_HIER_VIEWS

DBA_CUBE_HIER_VIEWS describes the hierarchies for all OLAP cube dimensions in the database. Its columns are the same as those in ALL_CUBE_HIER_VIEWS.

See Also

"ALL_CUBE_HIER_VIEWS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.239 DBA_CUBE_HIERARCHIES

DBA_CUBE_HIERARCHIES describes all OLAP dimension hierarchies in the database. Its columns are the same as those in ALL_CUBE_HIERARCHIES.

See Also

"ALL_CUBE_HIERARCHIES"

4.240 DBA_CUBE_MEASURES

DBA_CUBE_MEASURES describes the measures for all OLAP cubes in the database. Its columns are the same as those in ALL_CUBE_MEASURES.

See Also

"ALL_CUBE_MEASURES"

4.241 DBA_CUBE_NAMED_BUILD_SPECS

DBA_CUBE_NAMED_BUILD_SPECS describes the OLAP cube named build specifications in the database. Its columns are the same as those in ALL_CUBE_NAMED_BUILD_SPECS.

See Also

"ALL_CUBE_NAMED_BUILD_SPECS"

4.242 DBA_CUBE_SUB_PARTITION_LEVELS

DBA_CUBE_SUB_PARTITION_LEVELS describes the OLAP secondary partition levels in the database. Its columns are the same as those in ALL_CUBE_SUB_PARTITION_LEVELS.

See Also

"ALL_CUBE_SUB_PARTITION_LEVELS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.243 DBA_CUBE_VIEW_COLUMNS

DBA_CUBE_VIEW_COLUMNS describes the columns of relational views of all OLAP cubes in the database. Its columns are the same as those in ALL_CUBE_VIEW_COLUMNS.

> **See Also**
>
> "ALL_CUBE_VIEW_COLUMNS"

4.244 DBA_CUBE_VIEWS

DBA_CUBE_VIEWS describes the relational views of all OLAP cubes in the database. Its columns are the same as those in ALL_CUBE_VIEWS.

> **See Also**
>
> "ALL_CUBE_VIEWS"

4.245 DBA_CUBES

DBA_CUBES describes all OLAP cubes in the database. Its columns are the same as those in ALL_CUBES.

> **See Also**
>
> "ALL_CUBES"

4.246 DBA_DATA_FILES

DBA_DATA_FILES describes database files.

> **Note**
>
> When you query the DBA_DATA_FILES data dictionary view, Oracle must have access to all tablespaces and their data files if the requested information is not already available in the dictionary. If the tablespaces are encrypted, then you must open the Oracle wallet (keystore) before you can query DBA_DATA_FILES. You can use the ADMINISTER KEY MANAGEMENT SET KEYSTORE OPEN statement to open the keystore.

Column	Datatype	NULL	Description
FILE_NAME	VARCHAR2(513)		Name of the database file

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FILE_ID	NUMBER		Absolute file number of the database file
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace to which the file belongs
BYTES	NUMBER		Size of the file in bytes
BLOCKS	NUMBER		Size of the file in Oracle blocks
STATUS	VARCHAR2(9)		File status: AVAILABLE or INVALID (INVALID means that the file number is not in use, for example, a file in a tablespace that was dropped)
RELATIVE_FNO	NUMBER		Relative file number
AUTOEXTENSIBLE	VARCHAR2(3)		Autoextensible indicator
MAXBYTES	NUMBER		Maximum file size in bytes
MAXBLOCKS	NUMBER		Maximum file size in blocks
INCREMENT_BY	NUMBER		Number of Oracle blocks used as autoextension increment
USER_BYTES	NUMBER		The size of the file available for user data. The actual size of the file minus the USER_BYTES value is used to store file related metadata.
USER_BLOCKS	NUMBER		Number of blocks which can be used by the data
ONLINE_STATUS	VARCHAR2(7)		Online status of the file: • SYSOFF • SYSTEM • OFFLINE • ONLINE • RECOVER
LOST_WRITE_PROTECT	VARCHAR2(7)		Lost write protection status of the file. Possible values: • ENABLED: Indicates that lost write data is being collected • OFF: Indicates that lost write data is not being collected • SUSPEND: Indicates that lost write data is not currently being collected, but it can be enabled at a later date. The lost write data collected when the file was ENABLED remains in the lost write database, but it is not being checked or updated. If lost write protection is enabled for a single data file, it does not have to be enabled for another data file in the same tablespace. If lost write protection is enabled for a tablespace, it is enabled for all data files for that tablespace, including data files added later. You can check the lost write protection status for a tablespace by querying the LOST_WRITE_PROTECT column in the DBA_TABLESPACES view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database Advanced Security Guide* for information on opening a software keystore
- *Oracle Database Advanced Security Guide* for information on opening a hardware keystore
- "DBA_TABLESPACES"

4.247 DBA_DATAPUMP_JOBS

DBA_DATAPUMP_JOBS identifies all active Data Pump jobs in the database, regardless of their state, on an instance (or on all instances for Real Application Clusters). It also show all Data Pump control job tables not currently associated with an active job.

Related View

USER_DATAPUMP_JOBS displays the Data Pump jobs owned by the current user. This view does not display the OWNER_NAME column.

Column	Datatype	NULL	Description
OWNER_NAME	VARCHAR2(128)		User that initiated the job
JOB_NAME	VARCHAR2(128)		User-supplied name for the job (or the default name generated by the server)
OPERATION	VARCHAR2(128)		Type of job
JOB_MODE	VARCHAR2(128)		Mode of job
STATE	VARCHAR2(128)		Current job state
DEGREE	NUMBER		Number of worker processes performing the operation
ATTACHED_SESSIONS	NUMBER		Number of sessions attached to the job
DATAPUMP_SESSIONS	NUMBER		Number of Data Pump sessions participating in the job

See Also

"USER_DATAPUMP_JOBS"

4.248 DBA_DATAPUMP_SESSIONS

DBA_DATAPUMP_SESSIONS identifies the user sessions that are attached to a Data Pump job. The information in this view is useful for determining why a stopped Data Pump operation has not gone away.

Column	Datatype	NULL	Description
OWNER_NAME	VARCHAR2(128)		User that initiated the job
JOB_NAME	VARCHAR2(128)		User-supplied name for the job (or the default name generated by the server)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INSTANCE_ID	NUMBER		Instance ID
SADDR	RAW(4 | 8)		Address of the session attached to the job. Can be used with V$SESSION view.
SESSION_TYPE	VARCHAR2(14)		Data Pump session type: • DBMS_DATAPUMP - Data Pump interface process (one for each active instantiation of DBMS_DATAPUMP.OPEN and DBMS_DATAPUMP.ATTACH per job.) • MASTER - Data Pump control job process (one per job) • WORKER - worker process (1 to *n* per job, depending on degree of parallelism) • EXTERNAL TABLE - external table data access process (1 to *n*, depending on degree of parallelism, for jobs that use external tables as the data access method for some tables) • OTHER

See Also

Oracle Database PL/SQL Packages and Types Reference for more information about the DBMS_DATAPUMP package

4.249 DBA_DB_LINK_SOURCES

DBA_DB_LINK_SOURCES identifies all unique source databases that opened database links to the local database.

By default, only a DBA has access to this view. However, a DBA can grant access to this view to others.

This view is based on a persistent table that resides in the same system tablespace that is used by Database Auditing.

In a multitenant container database (CDB) environment, for every DBA_ view, there is a corresponding CDB_ view that contains data for all the pluggable databases (PDBs) in the CDB. A query on the CDB_DB_LINK_SOURCES view done in the CDB$ROOT container will show sources of all the database links recorded in all PDBs. A query on the corresponding DBA_DB_LINK_SOURCES view done in a PDB show information corresponding to that PDB only (that is, where that specific PDB was the target of an inbound database link).

Note that the CDB_ views would only show data from PDBs that are open at the time the query is issued. Therefore, when you are diagnosing sources of database links, Oracle recommends that you keep open any or all PDBs that might contain useful information for the diagnosis.

Column	Datatype	NULL	Description
SOURCE_ID	NUMBER	NOT NULL	Unique ID that identifies an incoming database link
DB_NAME	VARCHAR2(256)	NOT NULL	Global name of the source database

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database identifier of the source database. Maps to the DBID of the source database in V$DATABASE.
DB_UNIQUE_NAME	VARCHAR2(256)		Unique database name of the source database. Maps to the DB_UNIQUE_NAME of the source database in V$DATABASE. Null for source databases that do not provide this information.
HOST_NAME	VARCHAR2(256)		Resolved host name. Null if not available.
IP_ADDRESS	VARCHAR2(128)		IP address of source machine. Null if not available.
PROTOCOL	VARCHAR2(64)		One of supported protocols such as ipc, sdp, tcp, or tcps. Null if not available.
USERNAME	VARCHAR2(128)	NOT NULL	Oracle username of the user who logged into the local database. Maps to the USERNAME column in V$SESSION.
USER#	NUMBER	NOT NULL	Oracle user id of the user who logged into the local database. Maps to the USER# column in V$SESSION.
FIRST_LOGON_TIME	TIMESTAMP(6)	NOT NULL	The timestamp of the first connection on this database link in UTC
LAST_LOGON_TIME	TIMESTAMP(6)		The timestamp of the last connection on this database link in UTC
LOGON_COUNT	NUMBER		Number of times connection has been established through this database link

See Also

- "V$DATABASE"
- "V$SESSION"
- "DBA_EXTERNAL_SCN_ACTIVITY"
- "DBA_DB_LINKS"

4.250 DBA_DB_LINKS

DBA_DB_LINKS describes all database links in the database. Its columns are the same as those in ALL_DB_LINKS.

See Also

- "ALL_DB_LINKS"
- "DBA_DB_LINK_SOURCES"
- "DBA_EXTERNAL_SCN_ACTIVITY"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.251 DBA_DBFS_HS

DBA_DBFS_HS shows all Database File System (DBFS) hierarchical stores.

Related View

USER_DBFS_HS shows all Database File System hierarchical stores owned by the current user. This view does not display the STOREOWNER column.

Column	Datatype	NULL	Description
STORENAME	VARCHAR2(256)		Name of store
STOREOWNER	VARCHAR2(64)		Owner of store

> **See Also**
>
> "USER_DBFS_HS"

4.252 DBA_DBFS_HS_COMMANDS

DBA_DBFS_HS_COMMANDS shows all the registered store commands for all Database File System (DBFS) hierarchical stores.

Related View

USER_DBFS_HS_COMMANDS shows all the registered store commands for all DBFS hierarchical stores owned by current user. This view does not display the STOREOWNER column.

Column	Datatype	NULL	Description
STORENAME	VARCHAR2(256)		Name of store
STOREOWNER	VARCHAR2(64)		Owner of store
STORECOMMAND	VARCHAR2(512)		Store command
STOREFLAGS	NUMBER		Valid values are: • 1 - Indicates that the command is sent to the device before put • 2 - Indicates that the command is sent to the device before get

> **See Also**
>
> "USER_DBFS_HS_COMMANDS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.253 DBA_DBFS_HS_FIXED_PROPERTIES

DBA_DBFS_HS_FIXED_PROPERTIES shows non-modifiable properties of all Database File System (DBFS) hierarchical stores.

Related View

USER_DBFS_HS_FIXED_PROPERTIES shows non-modifiable properties of all DBFS hierarchical stores owned by current user. This view does not display the STORE_OWNER column.

Column	Datatype	NULL	Description
STORE_NAME	VARCHAR2(128)	NOT NULL	Name of store
STORE_OWNER	VARCHAR2(128)	NOT NULL	Owner of store
PROP_NAME	VARCHAR2(256)	NOT NULL	Property name
PROP_VALUE	VARCHAR2(256)	NOT NULL	Property value

See Also

"USER_DBFS_HS_FIXED_PROPERTIES"

4.254 DBA_DBFS_HS_PROPERTIES

DBA_DBFS_HS_PROPERTIES shows modifiable properties of all Database File System (DBFS) hierarchical stores.

Related View

USER_DBFS_HS_PROPERTIES shows modifiable properties of all DBFS hierarchical stores owned by current user. This view does not display the STOREOWNER column.

Column	Datatype	NULL	Description
STORENAME	VARCHAR2(256)		Name of store
STOREOWNER	VARCHAR2(64)		Owner of store
PROPERTYNAME	VARCHAR2(256)		Property name
PROPERTYVALUE	VARCHAR2(256)		Property value

See Also

"USER_DBFS_HS_PROPERTIES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.255 DBA_DDL_LOCKS

DBA_DDL_LOCKS lists all DDL locks held in the database and all outstanding requests for a DDL lock.

Column	Datatype	NULL	Description
SESSION_ID	NUMBER		Session identifier
OWNER	VARCHAR2(128)		Owner of the lock
NAME	VARCHAR2(1000)		Name of the lock
TYPE	VARCHAR2(40)		Lock type: • Cursor • Table/Procedure/Type • Body • Trigger • Index • Cluster • Java Source • Java Resource • Java Data
MODE_HELD	VARCHAR2(9)		Lock mode: • None • Null • Share • Exclusive
MODE_REQUESTED	VARCHAR2(9)		Lock request type: • None • Null • Share • Exclusive

See Also

Oracle Database Concepts for more information about DDL locks

4.256 DBA_DEPENDENCIES

DBA_DEPENDENCIES describes all dependencies in the database between procedures, packages, functions, package bodies, and triggers, including dependencies on views created without any database links. Its columns are the same as those in ALL_DEPENDENCIES.

See Also

"ALL_DEPENDENCIES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.257 DBA_DIGEST_VERIFIERS

DBA_DIGEST_VERIFIERS enables the database administrator to check which users have Digest verifiers stored on disk and the type of hashing algorithm used for the verifiers.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(128)		Name of the user
HAS_DIGEST_VERIFIERS	VARCHAR2(3)		YES if a Digest verifier exists, NO otherwise
DIGEST_TYPE	CHAR(3)		The type of hashing algorithm used for the Digest verifier. For instance, MD5 for users with MD5 Digest verifiers. If no Digest verifier exists, this column is NULL.

4.258 DBA_DIM_ATTRIBUTES

DBA_DIM_ATTRIBUTES describes the relationships between dimension levels and functionally dependent columns in the database. The level columns and the dependent column must be in the same table. This view's columns are the same as those in ALL_DIM_ATTRIBUTES.

See Also

"ALL_DIM_ATTRIBUTES"

4.259 DBA_DIM_CHILD_OF

DBA_DIM_CHILD_OF describes a hierarchical relationship of 1 to *n* between all the pairs of levels in the dimensions in the database. Its columns are the same as those in ALL_DIM_CHILD_OF.

See Also

"ALL_DIM_CHILD_OF"

4.260 DBA_DIM_HIERARCHIES

DBA_DIM_HIERARCHIES describes all the dimension hierarchies in the database. Its columns are the same as those in ALL_DIM_HIERARCHIES.

See Also

"ALL_DIM_HIERARCHIES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.261 DBA_DIM_JOIN_KEY

DBA_DIM_JOIN_KEY describes all joins in the database between two dimension tables. The join is always specified between a parent dimension level column and a child column. This view's columns are the same as those in ALL_DIM_JOIN_KEY.

See Also

"ALL_DIM_JOIN_KEY"

4.262 DBA_DIM_LEVEL_KEY

DBA_DIM_LEVEL_KEY describes the columns of all dimension levels in the database. This view's columns are the same as those in ALL_DIM_LEVEL_KEY.

See Also

"ALL_DIM_LEVEL_KEY"

4.263 DBA_DIM_LEVELS

DBA_DIM_LEVELS describes all dimension levels in the database. All columns of a dimension level must come from the same relation. This view's columns are the same as those in ALL_DIM_LEVELS.

See Also

"ALL_DIM_LEVELS"

4.264 DBA_DIMENSIONS

DBA_DIMENSIONS represents dimension objects. Its columns are the same as those in ALL_DIMENSIONS.

See Also

"ALL_DIMENSIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.265 DBA_DIRECTORIES

DBA_DIRECTORIES describes all directory objects in the database. Its columns are the same as those in ALL_DIRECTORIES.

> **See Also**
>
> "ALL_DIRECTORIES"

4.266 DBA_DISCOVERY_SOURCE

DBA_DISCOVERY_SOURCE describes sensitive data discovery import information.

Column	Datatype	NULL	Description
SOURCE_NAME	VARCHAR2(128)		The name of the discovery source. In the case of Application Data Model (ADM), this is the ADM instance name.
SOURCE_TYPE	VARCHAR2(6)		The type of the source: • ADM: import from ADM • CUSTOM: custom import • DB: discovered within the database
CTIME	TIMESTAMP(6)		The last time sensitive data was imported from this source

> **See Also**
>
> *Oracle Database Security Guide* for more information about transparent sensitive data protection

4.267 DBA_DML_LOCKS

DBA_DML_LOCKS lists all DML locks held in the database and all outstanding requests for a DML lock.

Column	Datatype	NULL	Description
SESSION_ID	NUMBER		Session holding or acquiring the lock
OWNER	VARCHAR2(128)	NOT NULL	Owner of the lock
NAME	VARCHAR2(128)	NOT NULL	Name of the lock
MODE_HELD	VARCHAR2(13)		The type of lock held. The values are: • ROWS_S (SS): row share lock • ROW-X (SX): row exclusive lock • SHARE (S): share lock • S/ROW-X (SSX): exclusive lock • NONE: lock requested but not yet obtained

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MODE_REQUESTED	VARCHAR2(13)		Lock request type. The values are: • ROWS_S (SS): row share lock • ROW-X (SX): row exclusive lock • SHARE (S): share lock • S/ROW-X (SSX): exclusive lock • NONE: Lock identifier obtained; lock not held or requested
LAST_CONVERT	NUMBER		Time since current mode was granted
BLOCKING_OTHERS	VARCHAR2(40)		Blocking others

See Also

Oracle Database Concepts for more information about lock modes for table locks

4.268 DBA_DMT_FREE_SPACE

DBA_DMT_FREE_SPACE describes the free extents in all dictionary managed tablespaces in the database.

Column	Datatype	NULL	Description
TABLESPACE_ID	NUMBER	NOT NULL	Identifier number of the tablespace containing the extent
FILE_ID	NUMBER	NOT NULL	File identifier number of the file containing the extent
BLOCK_ID	NUMBER	NOT NULL	Starting block number of the extent
BLOCKS	NUMBER	NOT NULL	Size of the extent (in Oracle blocks)

4.269 DBA_DMT_USED_EXTENTS

DBA_DMT_USED_EXTENTS describes the extents comprising the segments in all dictionary managed tablespaces in the database.

Column	Datatype	NULL	Description
SEGMENT_FILEID	NUMBER	NOT NULL	File number of the segment header of the extent
SEGMENT_BLOCK	NUMBER	NOT NULL	Block number of the segment header of the extent
TABLESPACE_ID	NUMBER	NOT NULL	Identifier number of the tablespace containing the extent
EXTENT_ID	NUMBER	NOT NULL	Extent number in the segment
FILEID	NUMBER	NOT NULL	File identifier number of the file containing the extent
BLOCK	NUMBER	NOT NULL	Starting block number of the extent
LENGTH	NUMBER	NOT NULL	Number of blocks in the extent

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.270 DBA_EDITION_COMMENTS

DBA_EDITION_COMMENTS describes the comments on all editions in the database. Its columns are the same as those in ALL_EDITION_COMMENTS.

See Also

"ALL_EDITION_COMMENTS"

4.271 DBA_EDITIONED_TYPES

DBA_EDITIONED_TYPES lists all types that are editioned by default for every user in the database.

Related View

USER_EDITIONED_TYPES lists the types that are editioned by default for the current user. This view does not display the SCHEMA column.

Column	Datatype	NULL	Description
SCHEMA	VARCHAR2(128)	NOT NULL	Schema in which the object types is editionable
OBJECT_TYPE	VARCHAR2(23)		Object type that is editionable

See Also

"USER_EDITIONED_TYPES"

4.272 DBA_EDITIONING_VIEW_COLS

DBA_EDITIONING_VIEW_COLS describes the relationship between the columns of all editioning views in the database and the table columns to which they map. Its columns are the same as those in ALL_EDITIONING_VIEW_COLS.

See Also

"ALL_EDITIONING_VIEW_COLS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.273 DBA_EDITIONING_VIEW_COLS_AE

DBA_EDITIONING_VIEW_COLS_AE describes the relationship between the columns of all editioning views (across all editions) in the database and the table columns to which they map. Its columns are the same as those in ALL_EDITIONING_VIEW_COLS_AE.

See Also

"ALL_EDITIONING_VIEW_COLS_AE"

4.274 DBA_EDITIONING_VIEWS

DBA_EDITIONING_VIEWS describes all editioning views in the database. Its columns are the same as those in ALL_EDITIONING_VIEWS.

See Also

"ALL_EDITIONING_VIEWS"

4.275 DBA_EDITIONING_VIEWS_AE

DBA_EDITIONING_VIEWS_AE describes all editioning views (across all editions) in the database. Its columns are the same as those in ALL_EDITIONING_VIEWS_AE.

See Also

"ALL_EDITIONING_VIEWS_AE"

4.276 DBA_EDITIONS

DBA_EDITIONS describes all editions in the database. Its columns are the same as those in ALL_EDITIONS.

See Also

- "ALL_EDITIONS"
- *Oracle Database Development Guide* for more information about editions

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.277 DBA_ENABLED_AGGREGATIONS

DBA_ENABLED_AGGREGATIONS displays information about enabled on-demand statistic aggregation.

Column	Datatype	NULL	Description
AGGREGATION_TYPE	VARCHAR2(21)		Type of the aggregation: • CLIENT_ID • SERVICE • SERVICE_MODULE • SERVICE_MODULE_ACTION
PRIMARY_ID	VARCHAR2(64)		Primary qualifier (specific client identifier or service name)
QUALIFIER_ID1	VARCHAR2(48)		Secondary qualifier (specific module name)
QUALIFIER_ID2	VARCHAR2(32)		Additional qualifier (specific action name)

4.278 DBA_ENABLED_TRACES

DBA_ENABLED_TRACES displays information about enabled SQL traces.

Column	Datatype	NULL	Description
TRACE_TYPE	VARCHAR2(21)		Type of the trace: • CLIENT_ID • SERVICE • SERVICE_MODULE • SERVICE_MODULE_ACTION • DATABASE
PRIMARY_ID	VARCHAR2(64)		Primary qualifier (specific client identifier or service name)
QUALIFIER_ID1	VARCHAR2(64)		Secondary qualifier (specific module name)
QUALIFIER_ID2	VARCHAR2(64)		Additional qualifier (specific action name)
WAITS	VARCHAR2(5)		Indicates whether waits are traced (TRUE) or not (FALSE)
BINDS	VARCHAR2(5)		Indicates whether binds are traced (TRUE) or not (FALSE)
PLAN_STATS	VARCHAR2(10)		Indicates whether cursor execution statistics are traced. Possible values include: • ALL_EXEC: Execution statistics are dumped at each cursor execution • NEVER: Execution statistics are never dumped • FIRST_EXEC: Execution statistics are dumped during the first execution of the cursor. This is the default behavior.
INSTANCE_NAME	VARCHAR2(16)		Instance name for tracing restricted to named instances

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_MONITOR` package

4.279 DBA_ENCRYPTED_COLUMNS

`DBA_ENCRYPTED_COLUMNS` maintains encryption algorithm information for all encrypted columns in the database. Its columns are the same as those in `ALL_ENCRYPTED_COLUMNS`.

> **See Also**
>
> "ALL_ENCRYPTED_COLUMNS"

4.280 DBA_EPG_DAD_AUTHORIZATION

`DBA_EPG_DAD_AUTHORIZATION` describes the DADs that are authorized to use different user's privileges.

Related View

`USER_EPG_DAD_AUTHORIZATION` describes the DADs that are authorized to use the user's privileges. This view does not display the `USERNAME` column.

Column	Datatype	NULL	Description
`DAD_NAME`	`VARCHAR2(64)`	NOT NULL	Name of DAD
`USERNAME`	`VARCHAR2(128)`	NOT NULL	Name of the user whose privileges the DAD is authorized to use

> **See Also**
>
> "USER_EPG_DAD_AUTHORIZATION"

4.281 DBA_ERROR_TRANSLATIONS

`DBA_ERROR_TRANSLATIONS` describes all error translations in the database. Its columns are the same as those in `ALL_ERROR_TRANSLATIONS`.

> **See Also**
>
> "ALL_ERROR_TRANSLATIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.282 DBA_ERRORS

DBA_ERRORS describes the current errors on all stored objects in the database. Its columns are the same as those in ALL_ERRORS.

See Also

"ALL_ERRORS"

4.283 DBA_ERRORS_AE

DBA_ERRORS_AE describes the current errors on all stored objects (across all editions) in the database. Its columns are the same as those in ALL_ERRORS_AE.

See Also

"ALL_ERRORS_AE"

4.284 DBA_EVALUATION_CONTEXT_TABLES

DBA_EVALUATION_CONTEXT_TABLES describes the tables in all rule evaluation contexts in the database. Its columns are the same as those in ALL_EVALUATION_CONTEXT_TABLES.

See Also

"ALL_EVALUATION_CONTEXT_TABLES"

4.285 DBA_EVALUATION_CONTEXT_VARS

DBA_EVALUATION_CONTEXT_VARS describes the variables in all rule evaluation contexts in the database. Its columns are the same as those in ALL_EVALUATION_CONTEXT_VARS.

See Also

"ALL_EVALUATION_CONTEXT_VARS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.286 DBA_EVALUATION_CONTEXTS

DBA_EVALUATION_CONTEXTS describes all rule evaluation contexts in the database. Its columns are the same as those in ALL_EVALUATION_CONTEXTS.

See Also

"ALL_EVALUATION_CONTEXTS"

4.287 DBA_EXP_FILES

DBA_EXP_FILES describes export files.

Column	Datatype	NULL	Description
EXP_VERSION	NUMBER(3)	NOT NULL	Version number of the export session
EXP_TYPE	VARCHAR2(11)		Type of export file: complete, cumulative, or incremental
FILE_NAME	VARCHAR2(100)	NOT NULL	Name of the export file
USER_NAME	VARCHAR2(128)	NOT NULL	Name of user who executed export
TIMESTAMP	DATE	NOT NULL	Timestamp of the export session

4.288 DBA_EXP_OBJECTS

DBA_EXP_OBJECTS describes objects that have been incrementally exported.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of exported object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of exported object
OBJECT_TYPE	VARCHAR2(13)		Type of exported object
CUMULATIVE	DATE		Timestamp of last cumulative export
INCREMENTAL	DATE	NOT NULL	Timestamp of last incremental export
EXPORT_VERSION	NUMBER(3)	NOT NULL	The ID of the export session

4.289 DBA_EXP_VERSION

DBA_EXP_VERSION displays the version number of the last export session.

Column	Datatype	NULL	Description
EXP_VERSION	NUMBER(3)	NOT NULL	Version number of the last export session

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.290 DBA_EXPRESSION_STATISTICS

DBA_EXPRESSION_STATISTICS provides expression usage tracking statistics for all the tables in the database. Its columns are the same as those in ALL_EXPRESSION_STATISTICS.

See Also

"ALL_EXPRESSION_STATISTICS"

4.291 DBA_EXTENTS

DBA_EXTENTS describes the extents comprising the segments in all tablespaces in the database.

Note that if a data file (or entire tablespace) is offline in a locally managed tablespace, you will not see any extent information. If an object has extents in an online file of the tablespace, you will see extent information about the offline data file. However, if the object is entirely in the offline file, a query of this view will not return any records.

Related View

USER_EXTENTS describes the extents comprising the segments owned by the current user's objects. This view does not display the OWNER, FILE_ID, BLOCK_ID, or RELATIVE_FNO columns.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the segment associated with the extent
SEGMENT_NAME	VARCHAR2(128)		Name of the segment associated with the extent
PARTITION_NAME	VARCHAR2(128)		Object Partition Name (Set to NULL for nonpartitioned objects)
SEGMENT_TYPE	VARCHAR2(18)		Type of the segment: INDEX PARTITION, TABLE PARTITION
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the extent
EXTENT_ID	NUMBER		Extent number in the segment
FILE_ID	NUMBER		Absolute file number of the data file containing the extent
BLOCK_ID	NUMBER		Starting block number of the extent
BYTES	NUMBER		Size of the extent in bytes
BLOCKS	NUMBER		Size of the extent in Oracle blocks
RELATIVE_FNO	NUMBER		Relative file number of the first extent block

See Also

"USER_EXTENTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.292 DBA_EXTERNAL_LOCATIONS

DBA_EXTERNAL_LOCATIONS describes the locations (data sources) of all external tables in the database. Its columns are the same as those in ALL_EXTERNAL_LOCATIONS.

See Also

"ALL_EXTERNAL_LOCATIONS"

4.293 DBA_EXTERNAL_SCN_ACTIVITY

DBA_EXTERNAL_SCN_ACTIVITY works in conjunction with the DBA_DB_LINK_SOURCES and DBA_DB_LINKS views to determine the source of high SCN activities.

If the SCN is increased by an inbound database link, then you can join the DBA_EXTERNAL_SCN_ACTIVITY view with the DBA_DB_LINK_SOURCES view on the INBOUND_DB_LINK_SOURCE_ID column to get details of the remote database where the SCN increase originated.

If the SCN is increased by an outbound database link, then the INBOUND_DB_LINK_SOURCE_ID column will be NULL, but the OUTBOUND_DB_LINK_NAME and OUTBOUND_DB_LINK_OWNER columns can be joined with the DB_LINK and OWNER columns respectively in the DBA_DB_LINKS view to determine the remote database that caused the SCN increase.

If neither of the above cases are true (the INBOUND_DB_LINK_SOURCE_ID, OUTBOUND_DB_LINK_NAME, and OUTBOUND_DB_LINK_OWNER are all NULL), then the SCN increase resulted from a client connection and not as a result of a database link to or from another database. You can join the SESSION_ID and SESSION_SERIAL# columns with the SID and SERIAL# columns in V$SESSION to get the client session details.

In a multitenant container database (CDB) environment, for every DBA_ view, there is a corresponding CDB_ view that contains data for all the pluggable databases (PDBs) in the CDB.

As the SCN is a property of the CDB (and not a PDB), a DBA interested in understanding large SCN jumps will likely find the CDB_EXTERNAL_SCN_ACTIVITY view more useful for diagnosing SCN jumps on a CDB. Querying the CDB_EXTERNAL_SCN_ACTIVITY view from CDB$ROOT ensures that external SCN jumps occurring on all PDBs are looked at and noticed. On the other hand, a query on the corresponding DBA_EXTERNAL_SCN_ACTIVITY view, or a query on the CDB_EXTERNAL_SCN_ACTIVITY view done from a PDB would only show data for that PDB (that is, details regarding any external activity that occurred on that specific PDB that resulted in large SCN jumps).

Note that the CDB_ views would only show data from PDBs that are open at the time the query is issued. Therefore, when you are diagnosing sources of external SCN activities, Oracle recommends that you keep open any or all PDBs that might contain useful information for the diagnosis.

Column	Datatype	NULL	Description
OPERATION_TIMESTAMP	TIMESTAMP(6)	NOT NULL	Timestamp when SCN was received in UTC

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SESSION_ID	NUMBER	NOT NULL	Session identifier of the local session that created this entry. Maps to V$SESSION.SID and to V$ACTIVE_SESSION_HISTORY.SESSION_ID.
SESSION_SERIAL#	NUMBER	NOT NULL	Session serial number of the local session that created this entry. Maps to V$SESSION.SERIAL# and to V$ACTIVE_SESSION_HISTORY.SESSION_SERIAL#.
AUDIT_SESSIONID	NUMBER		Session identifier that can be joined with DBA_AUDIT_TRAIL.SESSIONID or UNIFIED_AUDIT_TRAIL.SESSIONID (depending on which kind of auditing is enabled). Null if auditing is not enabled.
USERNAME	VARCHAR2(128)	NOT NULL	Oracle username of the user who logged into the local database. Maps to V$SESSION.USERNAME.
INBOUND_DB_LINK_SOURCE_ID	NUMBER		If the SCN was bumped by an inbound database link, then this is the inbound database link identified by the DBA_DB_LINK_SOURCES.SOURCE_ID database link. If the SCN was not increased by an inbound database link, then this value is null.
OUTBOUND_DB_LINK_NAME	VARCHAR2(128)		If the SCN was bumped by an outbound database link, then this is the outbound database link identified by the DBA_DB_LINKS.DB_LINK database link. Using this column and the OUTBOUND_DB_LINK_OWNER column, you can determine the source of the SCN increase for outbound links. If the SCN was not increased by an outbound database link, then this value is null.
OUTBOUND_DB_LINK_OWNER	VARCHAR2(128)		If the SCN was bumped by an outbound database link, then this is the owner of the outbound database link identified by DBA_DB_LINKS.OWNER. Using this column and the OUTBOUND_DB_LINK_NAME column, you can determine the source of the SCN increase for outbound links. If the SCN was not increased by an outbound database link, then this value is null.
RESULT	VARCHAR2(64)	NOT NULL	The following SCN activities are captured: • REJECTED_HIGH_SCN - SCN rejection due to unreasonable value • REJECTED_HIGH_DELTA - SCN rejection due to unreasonable rate of growth • ACCEPTED - SCN accepted with warning Regular SCN activities which do not result in errors or warnings are not captured . SCN errors and warnings also appear in alert.log.
EXTERNAL_SCN	NUMBER	NOT NULL	The external SCN received from an inbound database link, an outbound database link, or a client
SCN_ADJUSTMENT	NUMBER	NOT NULL	For ACCEPTED SCNs in the RESULT column, how much the local SCN was increased. For REJECTED SCNs in the RESULT column, the attempted SCN increase.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "V$SESSION"
- "V$ACTIVE_SESSION_HISTORY"
- "DBA_AUDIT_TRAIL"
- "UNIFIED_AUDIT_TRAIL"
- "DBA_DB_LINKS"
- "DBA_DB_LINK_SOURCES"

4.294 DBA_EXTERNAL_TABLES

DBA_EXTERNAL_TABLES describes all external tables in the database. Its columns are the same as those in ALL_EXTERNAL_TABLES.

See Also

"ALL_EXTERNAL_TABLES"

4.295 DBA_FEATURE_USAGE_STATISTICS

DBA_FEATURE_USAGE_STATISTICS displays information about database feature usage statistics.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database identifier of the database being tracked
NAME	VARCHAR2(64)	NOT NULL	Name of the feature
VERSION	VARCHAR2(17)	NOT NULL	Database version in which the feature was tracked
DETECTED_USAGES	NUMBER	NOT NULL	Number of times the system has detected usage for the feature
TOTAL_SAMPLES	NUMBER	NOT NULL	Number of times the system has woken up and checked for feature usage
CURRENTLY_USED	VARCHAR2(5)		Indicates whether usage was detected the last time the system checked (TRUE) or not (FALSE)
FIRST_USAGE_DATE	DATE		First sample time the system detected usage of the feature
LAST_USAGE_DATE	DATE		Last sample time the system detected usage of the feature
AUX_COUNT	NUMBER		This column stores feature-specific usage data in number format.
FEATURE_INFO	CLOB		This column stores feature-specific usage data in character format.
LAST_SAMPLE_DATE	DATE		The last time the system checked for usage
LAST_SAMPLE_PERIOD	NUMBER		Amount of time (in seconds) between the last two usage sample times

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SAMPLE_INTERVAL	NUMBER		Sample interval
DESCRIPTION	VARCHAR2(128)		Description of the feature and usage detection logic

Note

Use the following SQL query to list the database features and their descriptions in alphabetical order:

```
SELECT name, description FROM dba_feature_usage_statistics
ORDER BY name;
```

4.296 DBA_FGA_AUDIT_TRAIL

DBA_FGA_AUDIT_TRAIL displays all audit records for fine-grained auditing.

Note

This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from UNIFIED_AUDIT_TRAIL.

- See *Oracle Database Security Guide* for more information about unified auditing.
- See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
SESSION_ID	NUMBER	NOT NULL	Session id of the query
TIMESTAMP	DATE		Date and time of the query in the local database session time zone
DB_USER	VARCHAR2(128)		Database username who executed the query
OS_USER	VARCHAR2(255)		Operating system username who executed the query
USERHOST	VARCHAR2(128)		Client host machine name
CLIENT_ID	VARCHAR2(128)		Client identifier in each Oracle session
ECONTEXT_ID	VARCHAR2(64)		Application execution context identifier
EXT_NAME	VARCHAR2(4000)		External name
OBJECT_SCHEMA	VARCHAR2(128)		Owner of the table or view
OBJECT_NAME	VARCHAR2(128)		Name of the table or view
POLICY_NAME	VARCHAR2(128)		Name of the Fine-Grained Auditing Policy
SCN	NUMBER		System change number (SCN) of the query
SQL_TEXT	NVARCHAR2(2000)		SQL text of the query
SQL_BIND	NVARCHAR2(2000)		Bind variable data of the query
COMMENT$TEXT	VARCHAR2(4000)		Comments

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATEMENT_TYPE	VARCHAR2(7)		Statement type of the query: • SELECT • INSERT • UPDATE • DELETE
EXTENDED_TIMESTAMP	TIMESTAMP(6) WITH TIME ZONE		Timestamp of the query in UTC (Coordinated Universal Time) time zone
PROXY_SESSIONID	NUMBER		Proxy session serial number, if an enterprise user has logged in through the proxy mechanism
GLOBAL_UID	VARCHAR2(32)		Global user identifier for the user, if the user has logged in as an enterprise user
INSTANCE_NUMBER	NUMBER		Instance number as specified by the INSTANCE_NUMBER initialization parameter
OS_PROCESS	VARCHAR2(16)		Operating System process identifier of the Oracle process
TRANSACTIONID	RAW(8)		Transaction identifier of the transaction in which the object is accessed or modified
STATEMENTID	NUMBER		Numeric ID for each statement run (a statement may cause many actions)
ENTRYID	NUMBER		Numeric ID for each audit trail entry in the session
OBJ_EDITION_NAME	VARCHAR2(128)		Name of the edition containing the audited object
DBID	NUMBER		Database identifier of the audited database
RLS_INFO	CLOB		Stores virtual private database (VPD) policy names and predicates separated by delimiter. To format the output into individual rows, use the DBMS_AUDIT_UTIL.DECODE_RLS_INFO_ATRAIL_FGA function.
CURRENT_USER	VARCHAR2(128)		Effective user for the statement execution

Note

The SQL_BIND and SQL_TEXT columns are populated only if the policy has been created with the AUDIT_TRAIL parameter set to db, extended.

See Also

Oracle Database PL/SQL Packages and Types Reference for more information about the DBMS_AUDIT_UTIL.DECODE_RLS_INFO_ATRAIL_FGA function.

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.297 DBA_FILE_GROUP_EXPORT_INFO

DBA_FILE_GROUP_EXPORT_INFO shows export-related information for each version in the database that has a valid Data Pump dump file. Its columns are the same as those in ALL_FILE_GROUP_EXPORT_INFO.

See Also

"ALL_FILE_GROUP_EXPORT_INFO".

4.298 DBA_FILE_GROUP_FILES

DBA_FILE_GROUP_FILES shows the file set for each versioned file group in the database. Its columns are the same as those in ALL_FILE_GROUP_FILES.

See Also

"ALL_FILE_GROUP_FILES"

4.299 DBA_FILE_GROUP_TABLES

DBA_FILE_GROUP_TABLES shows information about all the tables in the database that can be imported using the file set. Its columns are the same as those in ALL_FILE_GROUP_TABLES.

See Also

"ALL_FILE_GROUP_TABLES"

4.300 DBA_FILE_GROUP_TABLESPACES

DBA_FILE_GROUP_TABLESPACES shows information about the transportable tablespaces present (partially or completely) in all file sets in the database (when the file set contains dump files). Its columns are the same as those in ALL_FILE_GROUP_TABLESPACES.

See Also

"ALL_FILE_GROUP_TABLESPACES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.301 DBA_FILE_GROUP_VERSIONS

DBA_FILE_GROUP_VERSIONS shows top-level version information for all file groups in the database. Its columns are the same as those in ALL_FILE_GROUP_VERSIONS.

See Also

"ALL_FILE_GROUP_VERSIONS"

4.302 DBA_FILE_GROUPS

DBA_FILE_GROUPS shows top-level metadata about all file groups in the database. Its columns are the same as those for ALL_FILE_GROUPS.

See Also

"ALL_FILE_GROUPS"

4.303 DBA_FLASHBACK_ARCHIVE

DBA_FLASHBACK_ARCHIVE describes all flashback archives available in the database.

Related View

USER_FLASHBACK_ARCHIVE describes the flashback archives available to the current user.

Column	Datatype	NULL	Description
OWNER_NAME	VARCHAR2(255)		Name of the creator of the flashback archive
FLASHBACK_ARCHIVE_NAME	VARCHAR2(255)	NOT NULL	Name of the flashback archive
FLASHBACK_ARCHIVE#	NUMBER	NOT NULL	Number of the flashback archive
RETENTION_IN_DAYS	NUMBER	NOT NULL	Maximum duration (in days) for which data is retained in the flashback archive
CREATE_TIME	TIMESTAMP(9)		Time at which the flashback archive was created
LAST_PURGE_TIME	TIMESTAMP(9)		Time at which the data in the flashback archive was last purged by the system
STATUS	VARCHAR2(7)		Indicates whether the flashback archive is a default flashback archive for the system (DEFAULT) or not (NULL)

See Also

"USER_FLASHBACK_ARCHIVE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.304 DBA_FLASHBACK_ARCHIVE_TABLES

DBA_FLASHBACK_ARCHIVE_TABLES displays information about all tables in the database that are enabled for Flashback Archive.

Related View

USER_FLASHBACK_ARCHIVE_TABLES displays information about the tables owned by the current user that are enabled for Flashback Archive.

Column	Datatype	NULL	Description
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table enabled for Flashback Archive
OWNER_NAME	VARCHAR2(128)	NOT NULL	Owner name of the table enabled for Flashback Archive
FLASHBACK_ARCHIVE_NAME	VARCHAR2(255)	NOT NULL	Name of the flashback archive
ARCHIVE_TABLE_NAME	VARCHAR2(53)		Name of the archive table containing the historical data for the user table
STATUS	VARCHAR2(13)		Status of whether flashback archive is enabled or being disabled on the table

See Also

"USER_FLASHBACK_ARCHIVE_TABLES"

4.305 DBA_FLASHBACK_ARCHIVE_TS

DBA_FLASHBACK_ARCHIVE_TS describes all tablespaces in the flashback archives available in the database.

Column	Datatype	NULL	Description
FLASHBACK_ARCHIVE_NAME	VARCHAR2(255)	NOT NULL	Name of the flashback archive
FLASHBACK_ARCHIVE#	NUMBER	NOT NULL	Number of the flashback archive
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of a tablespace in the flashback archive
QUOTA_IN_MB	VARCHAR2(40)		Maximum space (in MB) that can be used for Flashback Archive from the tablespace; NULL indicates no Quota restriction

4.306 DBA_FLASHBACK_TXN_REPORT

DBA_FLASHBACK_TXN_REPORT displays information about all compensating transactions that have been committed in the database.

Each row in this view is associated with one compensating transaction.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Related View

USER_FLASHBACK_TXN_REPORT displays information about the compensating transactions owned by the current user that have been committed in the database. This view does not display the USERNAME column.

Column	Datatype	NULL	Description
COMPENSATING_XID	RAW(8)	NOT NULL	Transaction responsible for backout
COMPENSATING_TXN_NAME	VARCHAR2(256)		Name of the compensating transaction
COMMIT_TIME	DATE		Timestamp when the compensating transaction committed
XID_REPORT	CLOB		An XML report describing the details of the transactions backed out by the compensating transaction
USERNAME	VARCHAR2(128)	NOT NULL	User who is executing the compensating transaction

See Also

"USER_FLASHBACK_TXN_REPORT"

4.307 DBA_FLASHBACK_TXN_STATE

DBA_FLASHBACK_TXN_STATE displays information about the compensating status of all transactions in the database.

For each compensating transaction, there could be multiple rows, where each row provides the dependency relation between the transactions that have been compensated by the compensating transaction.

Related View

USER_FLASHBACK_TXN_STATE displays information about the compensating status of the transactions owned by the current user. This view does not display the USERNAME column.

Column	Datatype	NULL	Description
COMPENSATING_XID	RAW(8)		Transaction ID of the compensating transaction
XID	RAW(8)		A transaction that has been compensated by the compensating transaction
DEPENDENT_XID	RAW(8)		A dependent transaction of XID **Note:** In the case of BACKOUT_MODE = CASCADE, there must be another row with XID = DEPENDENT_XID of this column.
BACKOUT_MODE	VARCHAR2(16)		Mode in which XID was backed out: • NOCASCADE • NOCASCADE_FORCE • NONCONFLICT_ONLY • CASCADE
USERNAME	VARCHAR2(128)	NOT NULL	User who is performing the compensating transaction

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "USER_FLASHBACK_TXN_STATE"

4.308 DBA_FREE_SPACE

DBA_FREE_SPACE describes the free extents in all tablespaces in the database.

If a data file (or entire tablespace) is offline in a locally managed tablespace, you will not see any extent information. If an object has extents in an online file of the tablespace, you will see extent information about the offline data file. However, if the object is entirely in the offline file, a query of this view will not return any records.

If a data file does not have any free space, you will not see a row for the data file in this view.

Related View

USER_FREE_SPACE describes the free extents in the tablespaces accessible to the current user.

Column	Datatype	NULL	Description
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the extent
FILE_ID	NUMBER		Absolute file number of the data file containing the extent
BLOCK_ID	NUMBER		Starting block number of the extent
BYTES	NUMBER		Size of the extent (in bytes)
BLOCKS	NUMBER		Size of the extent (in Oracle blocks)
RELATIVE_FNO	NUMBER		Relative file number of the file containing the extent

> **See Also**
>
> "USER_FREE_SPACE"

4.309 DBA_FREE_SPACE_COALESCED

DBA_FREE_SPACE_COALESCED describes statistics on coalesced space in all tablespaces in the database.

Column	Datatype	NULL	Description
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace
TOTAL_EXTENTS	NUMBER		Total number of free extents in the tablespace
EXTENTS_COALESCED	NUMBER		Total number of coalesced free extents in the tablespace
PERCENT_EXTENTS_COALESCED	NUMBER		Percentage of coalesced free extents in the tablespace
TOTAL_BYTES	NUMBER		Total number of free bytes in the tablespace

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BYTES_COALESCED	NUMBER		Total number of coalesced free bytes in the tablespace
TOTAL_BLOCKS	NUMBER		Total number of free Oracle blocks in the tablespace
BLOCKS_COALESCED	NUMBER		Total number of coalesced free Oracle blocks in the tablespace
PERCENT_BLOCKS_COALESCED	NUMBER		Percentage of coalesced free Oracle blocks in the tablespace

4.310 DBA_GG_AUTO_CDR_COLUMN_GROUPS

DBA_GG_AUTO_CDR_COLUMN_GROUPS provides details about all of the Oracle GoldenGate automatic conflict detection and resolution (CDR) column groups in the database.

Its columns are the same as those in ALL_GG_AUTO_CDR_COLUMN_GROUPS.

See Also

"ALL_GG_AUTO_CDR_COLUMN_GROUPS"

4.311 DBA_GG_AUTO_CDR_COLUMNS

DBA_GG_AUTO_CDR_COLUMNS provides details about all of the Oracle GoldenGate automatic conflict detection and resolution (CDR) columns in the database.

Its columns are the same as those in ALL_GG_AUTO_CDR_COLUMNS.

See Also

"ALL_GG_AUTO_CDR_COLUMNS"

4.312 DBA_GG_AUTO_CDR_TABLES

DBA_GG_AUTO_CDR_TABLES provides details about all the tables configured for Oracle GoldenGate automatic conflict detection and resolution (CDR).

Its columns are the same as those in ALL_GG_AUTO_CDR_TABLES.

See Also

"ALL_GG_AUTO_CDR_TABLES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.313 DBA_GG_INBOUND_PROGRESS

DBA_GG_INBOUND_PROGRESS displays information about the progress made by all GoldenGate inbound servers in the database. Its columns are the same as those in ALL_GG_INBOUND_PROGRESS.

See Also

"ALL_GG_INBOUND_PROGRESS"

4.314 DBA_GG_PROC_OBJECT_EXCLUSION

DBA_GG_PROC_OBJECT_EXCLUSION provides details about all tables that should be filtered when operating on given objects.

Column	Datatype	NULL	Description
PACKAGE_OWNER	VARCHAR2(384)		Procedure package owner
PACKAGE_NAME	VARCHAR2(384)		Procedure package name
OBJECT_OWNER	VARCHAR2(384)		Object owner to filter for the given procedure
OBJECT_NAME	VARCHAR2(384)		Object name to filter for the given procedure

4.315 DBA_GG_PROCEDURE_ANNOTATION

DBA_GG_PROCEDURE_ANNOTATION annotates the position of Owner and Object arguments in procedure calls.

Column	Datatype	NULL	Description
PACKAGE_OWNER	VARCHAR2(384)	NOT NULL	Procedure package owner
PACKAGE_NAME	VARCHAR2(384)	NOT NULL	Procedure package name
PROCEDURE_NAME	VARCHAR2(384)	NOT NULL	Procedure name
OBJECT_OWNER_ARGPOS	NUMBER	NOT NULL	Object owner name position in argument list, -1 if not present
OBJECT_ARGPOS	NUMBER	NOT NULL	Object name position in argument list, -1 if not present
MIN_DB_VERSION	VARCHAR2(100)		Minimum database version for the procedure
MAX_DB_VERSION	VARCHAR2(100)		Maximum database version for the procedure
MIN_REDO_COMPAT_LEVEL	VARCHAR2(100)		Minimum redo compatibility for the procedure
MAX_REDO_COMPAT_LEVEL	VARCHAR2(100)		Maximum redo compatibility for the procedure
FLAGS	NUMBER		Additional information about procedure arguments

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.316 DBA_GG_SUPPORTED_PACKAGES

DBA_GG_SUPPORTED_PACKAGES provides details about supported procedure packages for Oracle GoldenGate replication.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(384)	NOT NULL	Procedure package owner
NAME	VARCHAR2(384)	NOT NULL	Procedure package name
FEATURE	VARCHAR2(384)	NOT NULL	DBMS feature that the procedure package belongs to
MIN_DB_VERSION	VARCHAR2(100)		Minimum database version for the supported package
MAX_DB_VERSION	VARCHAR2(100)		Maximum database version for the supported package
MIN_REDO_COMPAT_LEVEL	VARCHAR2(100)		Minimum redo compatibility for the supported package
MAX_REDO_COMPAT_LEVEL	VARCHAR2(100)		Maximum redo compatibility for the supported package
SUPPORTED_LEVEL	VARCHAR2(100)		Supported level of the package

4.317 DBA_GG_SUPPORTED_PROCEDURES

DBA_GG_SUPPORTED_PROCEDURES provides details about all procedures that are supported for Oracle GoldenGate replication.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Procedure package owner
PACKAGE_NAME	VARCHAR2(128)		Procedure package name
PROCEDURE_NAME	VARCHAR2(128)		Procedure name
MIN_DB_VERSION	VARCHAR2(100)		Minimum database version for the procedure
MAX_DB_VERSION	VARCHAR2(100)		Maximum database version for the procedure
MIN_REDO_COMPAT_LEVEL	VARCHAR2(100)		Minimum redo compatibility for the procedure
MAX_REDO_COMPAT_LEVEL	VARCHAR2(100)		Maximum redo compatibility for the procedure
SUPPORTED_MODE	VARCHAR2(100)		Supported mode for the procedure: ALWAYS or DBMS_ROLLING
EXCLUSION_RULE_EXISTS	VARCHAR2(3)		Specifies whether an exclusion rule exists for the procedure (YES) or not (NO). **See Also:** "DBA_GG_PROC_OBJECT_EXCLUSION"

4.318 DBA_GLOBAL_CONTEXT

DBA_GLOBAL_CONTEXT displays the definition (name, schema, and package) of all global contexts created in the database.

This view is a subset of DBA_CONTEXT, which describes all contexts, including global contexts.

Column	Datatype	NULL	Description
NAMESPACE	VARCHAR2(128)	NOT NULL	Name of the context namespace

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SCHEMA	VARCHAR2(128)	NOT NULL	Schema of the package that administers the globally accessible context
PACKAGE	VARCHAR2(128)	NOT NULL	Package that administers the globally accessible context

See Also

- "DBA_CONTEXT"
- *Oracle Database Security Guide* for more information about using global application contexts
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SESSION.SET_CONTEXT procedure

4.319 DBA_GOLDENGATE_INBOUND

DBA_GOLDENGATE_INBOUND displays information about all GoldenGate inbound servers in the database. Its columns are the same as those in ALL_GOLDENGATE_INBOUND.

See Also

"ALL_GOLDENGATE_INBOUND"

4.320 DBA_GOLDENGATE_NOT_UNIQUE

DBA_GOLDENGATE_NOT_UNIQUE displays all tables that have no primary and no non-null unique indexes.

Most of the tables displayed by this view are supported because their columns contain enough information to be maintained by Oracle GoldenGate. Some tables, however, cannot be supported because their columns do not contain the necessary information. Unsupported tables usually contain a column defined using an unsupported data type.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Schema name of the non-unique table
TABLE_NAME	VARCHAR2(128)		Table name of the non-unique table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BAD_COLUMN	VARCHAR2(1)		Indicates that the table has a column not useful in the where clause. Possible values: • Y - Table column is defined using an unbounded data type, such as LONG or BLOB. If two rows in the table match except in their LOB columns, then the table cannot be maintained properly. Log apply services will attempt to maintain these tables, but you must ensure the application does not allow uniqueness only in the unbounded columns. • N - Enough column information is present to maintain the table in Oracle GoldenGate but the log transport services and log apply services would run more efficiently if you added a primary key. You should consider adding a disabled RELY constraint to these tables.

4.321 DBA_GOLDENGATE_PRIVILEGES

DBA_GOLDENGATE_PRIVILEGES displays details about Oracle GoldenGate privileges. Its columns are the same as those in ALL_GOLDENGATE_PRIVILEGES.

See Also

"ALL_GOLDENGATE_PRIVILEGES"

4.322 DBA_GOLDENGATE_RULES

DBA_GOLDENGATE_RULES displays information about all GoldenGate server rules in the database. Its columns are the same as those in ALL_GOLDENGATE_RULES.

See Also

"ALL_GOLDENGATE_RULES"

4.323 DBA_GOLDENGATE_SUPPORT_MODE

DBA_GOLDENGATE_SUPPORT_MODE displays information about the level of Oracle GoldenGate capture process support for the tables in the database.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Table owner
OBJECT_NAME	VARCHAR2(128)		Table name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SUPPORT_MODE	VARCHAR2(8)		Capture process support level for the table: • FULL - A capture process can capture changes made to all of the columns in the table • ID KEY - A capture process can capture changes made to the key columns and any other columns in the table supported by the capture process, except for LOB, LONG, LONG RAW, and XMLType columns. • INTERNAL - A capture process cannot capture changes made to any columns in the table because the table is secondary to a user-created table and is updated implicitly when changes are made to the user-created table. Such tables include mapping tables for index-organized tables, storage tables for nested tables, materialized view logs, secondary objects associated with domain indexes, and temporary tables. • NONE - A capture process cannot capture changes made to any columns in the table because the table is not supported for replication.

4.324 DBA_HANG_MANAGER_PARAMETERS

DBA_HANG_MANAGER_PARAMETERS shows the available user tunable Hang Manager parameters and their values.

Column	Datatype	NULL	Description
NAME	VARCHAR2(40)	NOT NULL	String representation of the parameter names
CURRENT_VALUE	VARCHAR2(20)		String representation of the parameter values
CURRENT_TIME	DATE		Time when the current value was set
PREVIOUS_VALUE	VARCHAR2(20)		String representation of the parameter value from the previous update
PREVIOUS_TIME	DATE		Time when the previous value was set

Note

- "V$HANG_INFO"
- "V$HANG_SESSION_INFO"
- "V$HANG_STATISTICS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.325 DBA_HEAT_MAP_SEG_HISTOGRAM

DBA_HEAT_MAP_SEG_HISTOGRAM displays segment access information for all segments. Its columns are the same as those in ALL_HEAT_MAP_SEG_HISTOGRAM.

> **See Also**
>
> "ALL_HEAT_MAP_SEG_HISTOGRAM"

4.326 DBA_HEAT_MAP_SEGMENT

DBA_HEAT_MAP_SEGMENT displays the latest segment access time for all segments. Its columns are the same as those in ALL_HEAT_MAP_SEGMENT.

> **See Also**
>
> "ALL_HEAT_MAP_SEGMENT"

4.327 DBA_HEATMAP_TOP_OBJECTS

DBA_HEATMAP_TOP_OBJECTS displays heat map information for the top 10000 objects by default.

If the database contains fewer than 10000 objects, then fewer than 10000 objects are returned by the view.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Object owner
OBJECT_NAME	VARCHAR2(128)		Object name
OBJECT_TYPE	VARCHAR2(18)		Object type
TABLESPACE_NAME	VARCHAR2(30)		Tablespace name
SEGMENT_COUNT	NUMBER		Segments in the tablespace
OBJECT_SIZE	NUMBER		Size of the object in MB
MIN_WRITETIME	DATE		Oldest modification time for a set of blocks
MAX_WRITETIME	DATE		Latest modification time for a set of blocks
AVG_WRITETIME	DATE		Average of the modification times for a set of blocks
MIN_READTIME	DATE		Oldest read time for a set of blocks
MAX_READTIME	DATE		Latest read time for a set of blocks
AVG_READTIME	DATE		Average of the read times for a set of blocks
MIN_FTSTIME	DATE		Minimum full table scan time of the object
MAX_FTSTIME	DATE		Maximum full table scan time of the object
AVG_FTSTIME	DATE		Average full table scan time of the object
MIN_LOOKUPTIME	DATE		Minimum lookup time of the object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MAX_LOOKUPTIME	DATE		Maximum lookup time of the object
AVG_LOOKUPTIME	DATE		Average lookup time of the object

4.328 DBA_HEATMAP_TOP_TABLESPACES

DBA_HEATMAP_TOP_TABLESPACES displays heat map information for the top 100 tablespaces.

Column	Datatype	NULL	Description
TABLESPACE_NAME	VARCHAR2(128)		Tablespace name
SEGMENT_COUNT	NUMBER		Segments in the tablespace
ALLOCATED_BYTES	NUMBER		Total bytes allocated to the objects in the tablespace
MIN_WRITETIME	DATE		Minimum write time of objects tracked
MAX_WRITETIME	DATE		Maximum write time of objects tracked
AVG_WRITETIME	DATE		Average write time of objects tracked
MIN_READTIME	DATE		Minimum read time of objects tracked
MAX_READTIME	DATE		Maximum read time of objects tracked
AVG_READTIME	DATE		Average read time of objects tracked
MIN_FTSTIME	DATE		Minimum full table scan time of objects tracked
MAX_FTSTIME	DATE		Maximum full table scan time of objects tracked
AVG_FTSTIME	DATE		Average full table scan time of objects tracked
MIN_LOOKUPTIME	DATE		Minimum lookup time of objects tracked
MAX_LOOKUPTIME	DATE		Maximum lookup time of objects tracked
AVG_LOOKUPTIME	DATE		Average lookup time of objects tracked

4.329 DBA_HIER_CLASS

DBA_HIER_CLASS describes the classifications of all hierarchies in the database. Its columns are the same as those in ALL_HIER_CLASS.

See Also

"ALL_HIER_CLASS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.330 DBA_HIER_CLASS_AE

DBA_HIER_CLASS_AE describes the classifications of all hierarchies (across all editions) in the database. Its columns are the same as those in ALL_HIER_CLASS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_HIER_CLASS_AE"

4.331 DBA_HIER_COLUMNS

DBA_HIER_COLUMNS describes the columns of all hierarchies in the database. Its columns are the same as those in ALL_HIER_COLUMNS.

See Also

"ALL_HIER_COLUMNS"

4.332 DBA_HIER_COLUMNS_AE

DBA_HIER_COLUMNS_AE describes the columns of all hierarchies (across all editions) in the database. Its columns are the same as those in ALL_HIER_COLUMNS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_HIER_COLUMNS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.333 DBA_HIER_HIER_ATTR_CLASS

DBA_HIER_HIER_ATTR_CLASS describes the classifications of the hierarchical attributes of all hierarchies in the database. Its columns are the same as those in ALL_HIER_HIER_ATTR_CLASS.

See Also

"ALL_HIER_HIER_ATTR_CLASS"

4.334 DBA_HIER_HIER_ATTR_CLASS_AE

DBA_HIER_HIER_ATTR_CLASS_AE describes the classifications of the hierarchical attributes of all hierarchies (across all editions) in the database. Its columns are the same as those in ALL_HIER_HIER_ATTR_CLASS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_HIER_HIER_ATTR_CLASS_AE"

4.335 DBA_HIER_HIER_ATTRIBUTES

DBA_HIER_HIER_ATTRIBUTES describes the hierarchical attributes of all hierarchies in the database. Its columns are the same as those in ALL_HIER_HIER_ATTRIBUTES.

See Also

"ALL_HIER_HIER_ATTRIBUTES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.336 DBA_HIER_HIER_ATTRIBUTES_AE

DBA_HIER_HIER_ATTRIBUTES_AE describes the hierarchical attributes of all hierarchies (across all editions) in the database. Its columns are the same as those in ALL_HIER_HIER_ATTRIBUTES_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_HIER_HIER_ATTRIBUTES_AE"

4.337 DBA_HIER_JOIN_PATHS

DBA_HIER_JOIN_PATHS describes the join paths for all hierarchies in the database. Its columns are the same as those in ALL_HIER_JOIN_PATHS.

> **See Also**
>
> "ALL_HIER_JOIN_PATHS"

4.338 DBA_HIER_JOIN_PATHS_AE

DBA_HIER_JOIN_PATHS_AE describes the join paths for all hierarchies (across all editions) in the database. Its columns are the same as those in ALL_HIER_JOIN_PATHS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_HIER_JOIN_PATHS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.339 DBA_HIER_LEVEL_ID_ATTRS

DBA_HIER_LEVEL_ID_ATTRS describes the attributes that uniquely identify members of the levels of all hierarchies in the database. Its columns are the same as those in ALL_HIER_LEVEL_ID_ATTRS.

> **See Also**
>
> "ALL_HIER_LEVEL_ID_ATTRS"

4.340 DBA_HIER_LEVEL_ID_ATTRS_AE

DBA_HIER_LEVEL_ID_ATTRS_AE describes the attributes that uniquely identify members of the levels of all hierarchies (across all editions) in the database. Its columns are the same as those in ALL_HIER_LEVEL_ID_ATTRS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_HIER_LEVEL_ID_ATTRS_AE"

4.341 DBA_HIER_LEVELS

DBA_HIER_LEVELS describes the levels of all hierarchies in the database. Its columns are the same as those in ALL_HIER_LEVELS.

> **See Also**
>
> "ALL_HIER_LEVELS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.342 DBA_HIER_LEVELS_AE

DBA_HIER_LEVELS_AE describes the levels of all hierarchies (across all editions) in the database. Its columns are the same as those in ALL_HIER_LEVELS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_HIER_LEVELS_AE"

4.343 DBA_HIERARCHIES

DBA_HIERARCHIES describes all hierarchies in the database. Its columns are the same as those in ALL_HIERARCHIES.

> **See Also**
>
> "ALL_HIERARCHIES"

4.344 DBA_HIERARCHIES_AE

DBA_HIERARCHIES_AE describes all hierarchies (across all editions) in the database. Its columns are the same as those in ALL_HIERARCHIES_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_HIERARCHIES_AE"

4.345 DBA_HIGH_WATER_MARK_STATISTICS

DBA_HIGH_WATER_MARK_STATISTICS displays information about database high-watermark statistics.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Identifier of the database for which the high-watermark statistics are tracked
NAME	VARCHAR2(64)	NOT NULL	Name of the high-watermark statistic (see Table 4-1)
VERSION	VARCHAR2(17)	NOT NULL	Database version in which the high watermarks were tracked
HIGHWATER	NUMBER		Highest value of the statistic seen at sampling time
LAST_VALUE	NUMBER		Value of the statistic at the last sample time
DESCRIPTION	VARCHAR2(128)		Description of the high-watermark statistics (see Table 4-1)

Table 4-1 DBA_HIGH_WATER_MARK_STATISTICS Statistics

Name	Description
ACTIVE_SESSIONS	Maximum Number of Active Sessions seen in the system
CPU_COUNT	Maximum Number of CPUs
DATAFILES	Maximum Number of Datafiles
DB_SIZE	Maximum Size of the Database (Bytes)
EXADATA_DISKS	Number of physical disks
INSTANCES	Oracle Database instances
PART_INDEXES	Maximum Number of Partitions belonging to an User Index
PART_TABLES	Maximum Number of Partitions belonging to an User Table
QUERY_LENGTH	Maximum Query Length
SEGMENT_SIZE	Size of Largest Segment (Bytes)
SESSIONS	Maximum Number of Concurrent Sessions seen in the database
SQL_NCHAR_COLUMNS	Maximum Number of SQL NCHAR Columns
TABLESPACES	Maximum Number of Tablespaces
USER_INDEXES	Number of User Indexes
USER_MV	Maximum Number of Materialized Views (User)
USER_TABLES	Number of User Tables

4.346 DBA_HIST_ACTIVE_SESS_HISTORY

DBA_HIST_ACTIVE_SESS_HISTORY displays the history of the contents of the in-memory active session history of recent system activity.

DBA_HIST_ACTIVE_SESS_HISTORY contains snapshots of V$ACTIVE_SESSION_HISTORY. See "V$ACTIVE_SESSION_HISTORY" for further interpretation details for many of these columns (except SNAP_ID, DBID, and INSTANCE_NUMBER).

> **Note**
>
> If you want to perform a join with the snapshots view, use the DBA_HIST_ASH_SNAPSHOT view instead of the DBA_HIST_SNAPSHOT view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
SAMPLE_ID	NUMBER	NOT NULL	ID of the sample
SAMPLE_TIME	TIMESTAMP(3)	NOT NULL	Time of the sample
SAMPLE_TIME_UTC	TIMESTAMP(3)		SAMPLE_TIME in UTC
USECS_PER_ROW	NUMBER		Time in microseconds since the last active session history sample
SESSION_ID	NUMBER	NOT NULL	Session identifier
SESSION_SERIAL#	NUMBER		Session serial number (used to uniquely identify a session's objects)
SESSION_TYPE	VARCHAR2(10)		Session type: • FOREGROUND • BACKGROUND
FLAGS	NUMBER		Reserved for future use
USER_ID	NUMBER		Oracle user identifier
SQL_ID	VARCHAR2(13)		SQL identifier of the SQL statement that is currently being executed
IS_SQLID_CURRENT	VARCHAR2(1)		Indicates whether the SQL identifier in the SQL_ID column is being executed (Y) or not (N)
SQL_CHILD_NUMBER	NUMBER		Child number of the SQL statement that is currently being executed
SQL_OPCODE	NUMBER		Indicates what phase of operation the SQL statement is in
SQL_OPNAME	VARCHAR2(64)		SQL command name
FORCE_MATCHING_SIGNATURE	NUMBER		Signature used when the CURSOR_SHARING parameter is set to FORCE
TOP_LEVEL_SQL_ID	VARCHAR2(13)		SQL identifier of the top level SQL statement
TOP_LEVEL_SQL_OPCODE	NUMBER		Indicates what phase of operation the top level SQL statement was in
SQL_PLAN_HASH_VALUE	NUMBER		Numerical representation of the SQL plan for the cursor
SQL_FULL_PLAN_HASH_VALUE	NUMBER		Numerical representation of the complete SQL plan for the cursor being executed by this session
SQL_ADAPTIVE_PLAN_RESOLVED	NUMBER		Indicates whether the SQL plan of the sampled database session is a resolved adaptive plan or not
SQL_PLAN_LINE_ID	NUMBER		SQL plan line ID
SQL_PLAN_OPERATION	VARCHAR2(64)		Plan operation name
SQL_PLAN_OPTIONS	VARCHAR2(64)		Plan operation options
SQL_EXEC_ID	NUMBER		SQL execution identifier
SQL_EXEC_START	DATE		Time when the execution of the SQL started
PLSQL_ENTRY_OBJECT_ID	NUMBER		Object ID of the top-most PL/SQL subprogram on the stack (or NULL if there is no PL/SQL subprogram on the stack)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PLSQL_ENTRY_SUBPROGRAM_ID	NUMBER		Subprogram ID of the top-most PL/SQL subprogram on the stack (or NULL if there is no PL/SQL subprogram on the stack)
PLSQL_OBJECT_ID	NUMBER		Object ID of the currently executing PL/SQL subprogram (or NULL if executing SQL)
PLSQL_SUBPROGRAM_ID	NUMBER		Subprogram ID of the currently executing PL/SQL object (or NULL if executing SQL)
QC_INSTANCE_ID	NUMBER		Query coordinator instance ID
QC_SESSION_ID	NUMBER		Query coordinator session ID
QC_SESSION_SERIAL#	NUMBER		Query coordinator session serial number
PX_FLAGS	NUMBER		Reserved for internal use
EVENT	VARCHAR2(64)		If SESSION_STATE = WAITING, then the event for which the session was waiting at the time of sampling. If SESSION_STATE = ON CPU, then this column will be NULL.
EVENT_ID	NUMBER		Identifier of the resource or event for which the session is waiting or for which the session last waited
SEQ#	NUMBER		Sequence number that uniquely identifies the wait (incremented for each wait)
P1TEXT	VARCHAR2(64)		Text of first additional parameter
P1	NUMBER		First additional parameter
P2TEXT	VARCHAR2(64)		Text of second additional parameter
P2	NUMBER		Second additional parameter
P3TEXT	VARCHAR2(64)		Text of third additional parameter
P3	NUMBER		Third additional parameter
WAIT_CLASS	VARCHAR2(64)		Wait class name of the event for which the session was waiting at the time of sampling. Interpretation is similar to that of the EVENT column. Maps to V$SESSION.WAIT_CLASS.
WAIT_CLASS_ID	NUMBER		Wait class identifier of the event for which the session was waiting at the time of sampling. Interpretation is similar to that of the EVENT column. Maps to V$SESSION.WAIT_CLASS_ID.
WAIT_TIME	NUMBER		Total wait time (in microseconds) for the event for which the session last waited (0 if currently waiting)
SESSION_STATE	VARCHAR2(7)		Session state: • WAITING • ON CPU
TIME_WAITED	NUMBER		Time that the current session actually spent waiting for the event (in microseconds). This column is set for waits that were in progress at the time the sample was taken.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BLOCKING_SESSION_STATUS	VARCHAR2(11)		Status of the blocking session: • VALID • NO HOLDER • GLOBAL • NOT IN WAIT • UNKNOWN
BLOCKING_SESSION	NUMBER		Session identifier of the blocking session. Populated only when the session was waiting for enqueues or a "buffer busy" wait. Maps to V$SESSION.BLOCKING_SESSION.
BLOCKING_SESSION_SERIAL#	NUMBER		Serial number of the blocking session
BLOCKING_INST_ID	NUMBER		Instance number of the blocker shown in BLOCKING_SESSION
BLOCKING_HANGCHAIN_INFO	VARCHAR2(1)		Indicates whether the information about BLOCKING_SESSION comes from the hang chain (Y) or not (N)
CURRENT_OBJ#	NUMBER		Object ID of the object that the session is currently referencing. This information is only available if the session was waiting for Application, Cluster, Concurrency, and User I/O wait events. Maps to V$SESSION.ROW_WAIT_OBJ#.
CURRENT_FILE#	NUMBER		File number of the file containing the block that the session is currently referencing. This information is only available if the session was waiting for Cluster, Concurrency, and User I/O wait events. Maps to V$SESSION.ROW_WAIT_FILE#.
CURRENT_BLOCK#	NUMBER		ID of the block that the session is currently referencing
CURRENT_ROW#	NUMBER		Row identifier that the session is referencing
TOP_LEVEL_CALL#	NUMBER		Oracle top level call number
TOP_LEVEL_CALL_NAME	VARCHAR2(64)		Oracle top level call name
CONSUMER_GROUP_ID	NUMBER		Consumer group ID
XID	RAW(8)		Transaction ID that the session was working on at the time of sampling. V$SESSION does not contain this information.
REMOTE_INSTANCE#	NUMBER		Remote instance identifier that will serve the block that this session is waiting for. This information is only available if the session was waiting for cluster events.
TIME_MODEL	NUMBER		Time model information
IN_CONNECTION_MGMT	VARCHAR2(1)		Indicates whether the session was doing connection management at the time of sampling (Y) or not (N)
IN_PARSE	VARCHAR2(1)		Indicates whether the session was parsing at the time of sampling (Y) or not (N)
IN_HARD_PARSE	VARCHAR2(1)		Indicates whether the session was hard parsing at the time of sampling (Y) or not (N)
IN_SQL_EXECUTION	VARCHAR2(1)		Indicates whether the session was executing SQL statements at the time of sampling (Y) or not (N)
IN_PLSQL_EXECUTION	VARCHAR2(1)		Indicates whether the session was executing PL/SQL at the time of sampling (Y) or not (N)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
IN_PLSQL_RPC	VARCHAR2(1)		Indicates whether the session was executing inbound PL/SQL RPC calls at the time of sampling (Y) or not (N)
IN_PLSQL_COMPILATION	VARCHAR2(1)		Indicates whether the session was compiling PL/SQL at the time of sampling (Y) or not (N)
IN_JAVA_EXECUTION	VARCHAR2(1)		Indicates whether the session was executing Java at the time of sampling (Y) or not (N)
IN_BIND	VARCHAR2(1)		Indicates whether the session was doing bind operations at the time of sampling (Y) or not (N)
IN_CURSOR_CLOSE	VARCHAR2(1)		Indicates whether the session was closing a cursor at the time of sampling (Y) or not (N)
IN_SEQUENCE_LOAD	VARCHAR2(1)		Indicates whether the session is loading in sequence (in sequence load code) (Y) or not (N)
IN_INMEMORY_QUERY	VARCHAR2(1)		Indicates whether the session was querying the In-Memory Column Store (IM column store) at the time of sampling (Y) or not (N)
IN_INMEMORY_POPULATE	VARCHAR2(1)		Indicates whether the session was populating the IM column store at the time of sampling (Y) or not (N)
IN_INMEMORY_PREPOPULATE	VARCHAR2(1)		Indicates whether the session was prepopulating the IM column store at the time of sampling (Y) or not (N)
IN_INMEMORY_REPOPULATE	VARCHAR2(1)		Indicates whether the session was repopulating the IM column store at the time of sampling (Y) or not (N)
IN_INMEMORY_TREPOPULATE	VARCHAR2(1)		Indicates whether the session was trickle repopulating the IM column store at the time of sampling (Y) or not (N)
IN_TABLESPACE_ENCRYPTION	VARCHAR2(1)		Indicates whether encryption or decryption of a tablespace occurred at the time of sampling (Y) or not (N)
CAPTURE_OVERHEAD	VARCHAR2(1)		Indicates whether the session is executing capture code (Y) or not (N)
REPLAY_OVERHEAD	VARCHAR2(1)		Indicates whether the session is executing replay code (Y) or not (N)
IS_CAPTURED	VARCHAR2(1)		Indicates whether the session is being captured (Y) or not (N)
IS_REPLAYED	VARCHAR2(1)		Indicates whether the session is being replayed (Y) or not (N)
IS_REPLAY_SYNC_TOKEN_HOLDER	VARCHAR2(1)		Indicates whether the session is holding a synchronization token (Y) or not (N) during workload replay
SERVICE_HASH	NUMBER		Hash that identifies the Service
PROGRAM	VARCHAR2(64)		Name of the operating system program
MODULE	VARCHAR2(64)		Name of the currently executing module as set by the DBMS_APPLICATION_INFO.SET_MODULE procedure
ACTION	VARCHAR2(64)		Name of the currently executing action as set by the DBMS_APPLICATION_INFO.SET_ACTION procedure
CLIENT_ID	VARCHAR2(64)		Client identifier of the session
MACHINE	VARCHAR2(64)		Client's operating system machine name
PORT	NUMBER		Client port number

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ECID	VARCHAR2(64)		Execution context identifier (sent by Application Server)
DBREPLAY_FILE_ID	NUMBER		If the session is being captured or replayed, then DBREPLAY_FILE_ID is the file ID for the workload capture or workload replay; otherwise it is NULL.
DBREPLAY_CALL_COUNTER	NUMBER		If the session is being captured or replayed, then DBREPLAY_CALL_COUNTER is the call counter of the user call that is being captured or replayed; otherwise it is NULL.
TM_DELTA_TIME	NUMBER		Time interval (in microseconds) over which TM_DELTA_CPU_TIME and TM_DELTA_DB_TIME are accumulated
TM_DELTA_CPU_TIME	NUMBER		Amount of time this session spent on CPU over the last TM_DELTA_TIME microseconds
TM_DELTA_DB_TIME	NUMBER		Amount of time spent by this session in database calls over the last TM_DELTA_TIME microseconds
DELTA_TIME	NUMBER		Time interval (in microseconds) since the last time this session was sampled or created, over which the next five statistics are accumulated
DELTA_READ_IO_REQUESTS	NUMBER		Number of read I/O requests made by this session over the last DELTA_TIME microseconds
DELTA_WRITE_IO_REQUESTS	NUMBER		Number of write I/O requests made by this session over the last DELTA_TIME microseconds
DELTA_READ_IO_BYTES	NUMBER		Number of I/O bytes read by this session over the last DELTA_TIME microseconds
DELTA_WRITE_IO_BYTES	NUMBER		Number of I/O bytes written by this session over the last DELTA_TIME microseconds
DELTA_INTERCONNECT_IO_BYTES	NUMBER		Number of I/O bytes sent over the I/O interconnect over the last DELTA_TIME microseconds
PGA_ALLOCATED	NUMBER		Amount of PGA memory (in bytes) consumed by this session at the time this sample was taken
TEMP_SPACE_ALLOCATED	NUMBER		Amount of TEMP memory (in bytes) consumed by this session at the time this sample was taken
DBOP_NAME	VARCHAR2(64)		Database operation name. If the type is SQL, the DBOP_NAME will be NULL.
DBOP_EXEC_ID	NUMBER		Database operation execution identifier for the current execution. If the type is SQL, the DBOP_EXEC_ID will be NULL.
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_HIST_ASH_SNAPSHOT"
- "DBA_HIST_SNAPSHOT"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_APPLICATION_INFO` package

4.347 DBA_HIST_APPLY_SUMMARY

`DBA_HIST_APPLY_SUMMARY` displays historical statistics information about each apply process for Oracle GoldenGate, and Oracle XStream. This view is intended for use with Automatic Workload Repository (AWR).

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
APPLY_NAME	VARCHAR2(128)	NOT NULL	Name of the apply process
STARTUP_TIME	DATE	NOT NULL	The time that the apply process was last started
READER_TOTAL_MESSAGES_DEQUEUED	NUMBER		Total number of messages dequeued since the apply process was last started
READER_LAG	NUMBER		For captured messages, the delay (in seconds) between the creation of the last message and it being received by the apply process. For user enqueued messages, the delay between the message being enqueued in the local database and being received by the apply process.
COORD_TOTAL_RECEIVED	NUMBER		Total number of transactions received by the coordinator process since the apply process was last started
COORD_TOTAL_APPLIED	NUMBER		Total number of transactions applied by the apply process since the apply process was last started
COORD_TOTAL_ROLLBACKS	NUMBER		Number of transactions which were rolled back due to unexpected contention
COORD_TOTAL_WAIT_DEPS	NUMBER		Number of times since the apply process was last started that an apply server waited to apply a logical change record (LCR) in a transaction until another apply server applied a transaction because of a dependency between the transactions
COORD_TOTAL_WAIT_CMTS	NUMBER		Number of times since the apply process was last started that an apply server waited to commit a transaction until another apply server committed a transaction to serialize commits
COORD_LWM_LAG	NUMBER		For captured messages, the delay (in seconds) between the creation of the message corresponding to the low watermark and it being applied by the apply process. For user enqueued messages, the delay between the message being enqueued in the local database and being applied by the apply process.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SERVER_TOTAL_MESSAGES_APPLIED	NUMBER		Total number of messages applied by all the apply servers since the apply process was last started
SERVER_ELAPSED_DEQUEUE_TIME	NUMBER		Time elapsed (in hundredths of a second) dequeuing messages by all the apply servers since the apply process was last started
SERVER_ELAPSED_APPLY_TIME	NUMBER		Time elapsed (in hundredths of a second) applying messages by all the apply servers since the apply process was last started
CON_DBID	NUMBER		The database ID of the PDB
REPLICAT_NAME	VARCHAR2(128)		The name of the replicat group created from GGSCI using GoldenGate
UNASSIGNED_COMPLETE_TXN	NUMBER		Total number of complete transactions that the coordinator has not assigned to any apply servers
TOTAL_LCRS_RETRIED	NUMBER		Total number of LCRs retried by this server
TOTAL_TRANSACTIONS_RETRIED	NUMBER		Total transactions retried by this server
TOTAL_ERRORS	NUMBER		Number of transactions applied by the apply process that resulted in an apply error since the apply process was last started
SESSION_MODULE	VARCHAR2(64)	NOT NULL	Session module. Valid values: • XStream • GoldenGate
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.348 DBA_HIST_ASH_SNAPSHOT

DBA_HIST_ASH_SNAPSHOT provides the list of snapshots that contains Active Session History (ASH) data.

This view differs from DBA_HIST_SNAPSHOT in that it provides snapshots which had errors flushing some Automatic Workload Repository (AWR) tables, but for which ASH data may be successfully flushed (DBA_HIST_SNAPSHOT filters out snapshots which had errors flushing AWR tables).

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
STARTUP_TIME	TIMESTAMP(3)	NOT NULL	Startup time of the instance
BEGIN_INTERVAL_TIME	TIMESTAMP(3)	NOT NULL	Time at the beginning of the snapshot interval

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
END_INTERVAL_TIME	TIMESTAMP(3)	NOT NULL	Time at the end of the snapshot interval; the actual time the snapshot was taken
FLUSH_ELAPSED	INTERVAL DAY(5) TO SECOND(1)		Amount of time to perform the snapshot
SNAP_LEVEL	NUMBER		Snapshot level
STATUS	NUMBER		Indicates if the snapshot was successfully flushed without any errors. Possible values: • 0 - No errors • 1 - Errors on some AWR tables
ERROR_COUNT	NUMBER		Number of errors occurring in the tables for the particular snapshot
BL_MOVED	NUMBER		Reserved for internal use
SNAP_FLAG	NUMBER		Condition under which the snapshot was inserted. Possible values are: • 1 - Manual snapshot created using a PL/SQL package • 2 - Imported snapshot • 4 - Snapshot taken while Diagnostic Pack or Tuning Pack was not enabled
SNAP_TIMEZONE	INTERVAL DAY(0) TO SECOND(0)		Snapshot time zone expressed as offset from UTC (Coordinated Universal Time) time zone
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"DBA_HIST_SNAPSHOT"

4.349 DBA_HIST_BASELINE

DBA_HIST_BASELINE displays information on baselines taken in the system.

For each baseline, this view displays the complete time range and whether the baseline is the default baseline.

Column	Datatype	NULL	Description
DBID	NUMBER		Database ID
BASELINE_ID	NUMBER		Internal ID for the baseline
BASELINE_NAME	VARCHAR2(64)		User-specified name for the baseline

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BASELINE_TYPE	VARCHAR2(13)		The baseline type, as follows: STATIC - baselines that are created manually by the user MOVING WINDOW - baselines that have dynamic start and end snapshot IDs GENERATED - baselines that are automatically generated by the system, using a template
START_SNAP_ID	NUMBER		Start snapshot ID for the baseline
START_SNAP_TIME	TIMESTAMP(3)		Time associated with the start snapshot ID
END_SNAP_ID	NUMBER		End snapshot ID for the baseline
END_SNAP_TIME	TIMESTAMP(3)		Time associated with the end snapshot ID
MOVING_WINDOW_SIZE	NUMBER		If BASELINE_TYPE is MOVING WINDOW, this field is the size of the moving window in number of days. If NULL, then the window size is the value of the AWR retention setting.
CREATION_TIME	DATE		Time the baseline was created
EXPIRATION	NUMBER		How long to keep the baseline, in number of days. A NULL value means that the baseline will be kept forever.
TEMPLATE_NAME	VARCHAR2(64)		Name of the template that created this baseline, if any.
LAST_TIME_COMPUTED	DATE		Last time that statistics were computed on the baseline.
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.350 DBA_HIST_BASELINE_DETAILS

DBA_HIST_BASELINE_DETAILS displays details about the baseline.

Column	Datatype	NULL	Description
DBID	NUMBER		Database ID
INSTANCE_NUMBER	NUMBER		Instance ID for the baseline data
BASELINE_ID	NUMBER		Internal ID for the baseline
BASELINE_NAME	VARCHAR2(64)		User-specified name for the baseline

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BASELINE_TYPE	VARCHAR2(13)		The baseline type, as follows: STATIC - baselines that are created manually by the user MOVING WINDOW - baselines that have dynamic start and end snapshot IDs GENERATED - baselines that are automatically generated by the system, using a template
START_SNAP_ID	NUMBER		Start snapshot ID for the baseline
START_SNAP_TIME	TIMESTAMP(3)		Start snapshot time for the baseline
END_SNAP_ID	NUMBER		End snapshot ID for the baseline
END_SNAP_TIME	TIMESTAMP(3)		End snapshot time for the baseline
SHUTDOWN	VARCHAR2(3)		Whether or not there is a database startup or shutdown in this interval (YES, NO, or NULL).
ERROR_COUNT	NUMBER		Number of errors in the snapshots in the baseline snapshot range
PCT_TOTAL_TIME	NUMBER		Amount of time captured in snapshots, divided by the total possible time for this baseline
LAST_TIME_COMPUTED	DATE		Last time that statistics were computed on the baseline
MOVING_WINDOW_SIZE	NUMBER		If BASELINE_TYPE is MOVING WINDOW, this field is the size of the moving window in number of days. If NULL, then the window size is the value of the AWR retention setting.
CREATION_TIME	DATE		Time the baseline was created
EXPIRATION	NUMBER		How long to keep the baseline, in number of days. A value of NULL indicates that the baseline will be kept forever.
TEMPLATE_NAME	VARCHAR2(64)		Name of the template that created this baseline, if any.
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.351 DBA_HIST_BASELINE_METADATA

DBA_HIST_BASELINE_METADATA displays metadata information for the baseline.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
BASELINE_ID	NUMBER	NOT NULL	Internal ID for the baseline
BASELINE_NAME	VARCHAR2(64)		User-specified name for the baseline

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BASELINE_TYPE	VARCHAR2(13)		The baseline type, as follows: STATIC - baselines that are created manually by the user MOVING WINDOW - baselines that have dynamic start and end snapshot IDs GENERATED - baselines that are automatically generated by the system, using a template
START_SNAP_ID	NUMBER		Start snapshot ID for the baseline
END_SNAP_ID	NUMBER		End snapshot ID for the baseline
MOVING_WINDOW_SIZE	NUMBER		If BASELINE_TYPE is MOVING WINDOW, this field is the size of the moving window in number of days. If NULL, then the window size is the value of the AWR retention setting.
CREATION_TIME	DATE		Time the baseline was created
EXPIRATION	NUMBER		How long to keep the baseline, in number of days. If the value is NULL, the baseline will be kept forever.
TEMPLATE_NAME	VARCHAR2(64)		Name of the template that created this baseline, if any
LAST_TIME_COMPUTED	DATE		Last time that statistics were computed on the baseline
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.352 DBA_HIST_BASELINE_TEMPLATE

DBA_HIST_BASELINE_TEMPLATE displays the templates used by the system for baseline generation.

The system uses this information to determine which baselines should be automatically created or removed.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
TEMPLATE_ID	NUMBER	NOT NULL	Internal ID for the template
TEMPLATE_NAME	VARCHAR2(128)	NOT NULL	Name of the template
TEMPLATE_TYPE	VARCHAR2(9)	NOT NULL	Type of the template, as follows: SINGLE - one time period REPEATING - maintain a time period

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BASELINE_NAME_PREFIX	VARCHAR2(128)	NOT NULL	Name to use for the baselines that are created: For a template type of SINGLE, the BASELINE_NAME_PREFIX is the name that will be used. For a template type of REPEATING, the BASELINE_NAME will be the prefix to the name.
START_TIME	DATE	NOT NULL	For a template type of SINGLE, this is the start time for future baselines For a template type of REPEATING, this is the effective start time that baselines should start being generated.
END_TIME	DATE	NOT NULL	For a template type of SINGLE, this is the end time for future baselines. For a template type of REPEATING, this is the effective end time that baselines should stop being generated.
DAY_OF_WEEK	VARCHAR2(9)		For a template type of REPEATING, this indicates the day of the week to create the baseline: SUNDAY, MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY, SATURDAY, ALL.
HOUR_IN_DAY	NUMBER		For a template type of REPEATING, a value from 0 - 23 to indicate the hour of the day to create the baseline for.
DURATION	NUMBER		For a template type of REPEATING, the length of time for the baseline to be created.
EXPIRATION	NUMBER		How long to keep the baseline, in number of days
REPEAT_INTERVAL	VARCHAR2(128)		String that represents the time repeating information in the format used by the DBMS_SCHEDULER package
LAST_GENERATED	DATE		Last time a baseline was generated for this template
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database PL/SQL Packages and Types Reference for more information about the DBMS_SCHEDULER package

4.353 DBA_HIST_BG_EVENT_SUMMARY

DBA_HIST_BG_EVENT_SUMMARY displays the historical summary background event activity.

This view contains snapshots from V$SESSION_EVENT.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
EVENT_ID	NUMBER	NOT NULL	Identifier of the wait event
EVENT_NAME	VARCHAR2(64)	NOT NULL	Name of the wait event
WAIT_CLASS_ID	NUMBER		Identifier of the class of the wait event
WAIT_CLASS	VARCHAR2(64)		Name of the class of the wait event
TOTAL_WAITS	NUMBER		Total number of waits for the event
TOTAL_TIMEOUTS	NUMBER		Total number of timeouts for the event
TIME_WAITED_MICRO	NUMBER		Total amount of time waited for the event (in microseconds)
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SESSION_EVENT"

4.354 DBA_HIST_BUFFER_POOL_STAT

DBA_HIST_BUFFER_POOL_STAT displays historical statistics about all buffer pools available for the instance.

This view contains snapshots of V$BUFFER_POOL_STATISTICS.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
ID	NUMBER	NOT NULL	Buffer pool identifier number
NAME	VARCHAR2(20)		Name of the buffer pool
BLOCK_SIZE	NUMBER		Block Size
SET_MSIZE	NUMBER		Buffer pool maximum set size
CNUM_REPL	NUMBER		Number of buffers on the replacement list
CNUM_WRITE	NUMBER		Number of buffers on the write list

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CNUM_SET	NUMBER		Number of buffers in the set
BUF_GOT	NUMBER		Number of buffers gotten by the set
SUM_WRITE	NUMBER		Number of buffers written by the set
SUM_SCAN	NUMBER		Number of buffers scanned in the set
FREE_BUFFER_WAIT	NUMBER		Free buffer wait statistic
WRITE_COMPLETE_WAIT	NUMBER		Write complete wait statistic
BUFFER_BUSY_WAIT	NUMBER		Buffer busy wait statistic
FREE_BUFFER_INSPECTED	NUMBER		Free buffer inspected statistic
DIRTY_BUFFERS_INSPECTED	NUMBER		Dirty buffers inspected statistic
DB_BLOCK_CHANGE	NUMBER		Database blocks changed statistic
DB_BLOCK_GETS	NUMBER		Database blocks gotten statistic
CONSISTENT_GETS	NUMBER		Consistent gets statistic
PHYSICAL_READS	NUMBER		Physical reads statistic
PHYSICAL_WRITES	NUMBER		Physical writes statistic
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$BUFFER_POOL_STATISTICS"

4.355 DBA_HIST_BUFFERED_QUEUES

DBA_HIST_BUFFERED_QUEUES displays historical information about all buffered queues available for the instance.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
QUEUE_SCHEMA	VARCHAR2(128)	NOT NULL	Owner of the queue
QUEUE_NAME	VARCHAR2(128)	NOT NULL	Name of the queue
STARTUP_TIME	DATE	NOT NULL	Startup time of the instance

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
QUEUE_ID	NUMBER	NOT NULL	ID of the queue
NUM_MSGS	NUMBER		Total number of outstanding messages currently enqueued in the buffered queue for the subscriber (includes the count of the messages overflowed to disk)
SPILL_MSGS	NUMBER		Current number of overflow messages spilled to disk from the buffered queue
CNUM_MSGS	NUMBER		Cumulative total number of messages enqueued into the buffered queue since the buffered queue was created.
CSPILL_MSGS	NUMBER		Cumulative total number of overflow messages spilled to disk from the buffered queue since the buffered queue was created
EXPIRED_MSGS	NUMBER		Number of expired messages
OLDEST_MSGID	RAW(16)		Message ID of the oldest message
OLDEST_MSG_ENQTM	TIMESTAMP(3)		Enqueue time of the oldest message
QUEUE_STATE	VARCHAR2(25)		Indicates whether the queue is in recovery mode (QUEUE IS IN RECOVERY MODE) or not (NORMAL)
ELAPSED_ENQUEUE_TIME	NUMBER		Total time spent in enqueue (in hundredths of a second)
ELAPSED_DEQUEUE_TIME	NUMBER		Total time spent in dequeue (in hundredths of a second)
ELAPSED_TRANSFORMATION_TIME	NUMBER		Total time for evaluating transformations (in hundredths of a second)
ELAPSED_RULE_EVALUATION_TIME	NUMBER		Total time for rule evaluations (in hundredths of a second)
ENQUEUE_CPU_TIME	NUMBER		Total CPU time for enqueue (in hundredths of a second)
DEQUEUE_CPU_TIME	NUMBER		Total CPU time for dequeue (in hundredths of a second)
LAST_ENQUEUE_TIME	TIMESTAMP(3)		Last message enqueue time
LAST_DEQUEUE_TIME	TIMESTAMP(3)		Last message dequeue time
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.356 DBA_HIST_BUFFERED_SUBSCRIBERS

DBA_HIST_BUFFERED_SUBSCRIBERS displays historical information about the subscribers for all buffered queues in the instance.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
QUEUE_SCHEMA	VARCHAR2(128)	NOT NULL	Owner of the queue
QUEUE_NAME	VARCHAR2(128)	NOT NULL	Name of the queue
SUBSCRIBER_ID	NUMBER	NOT NULL	Internal subscriber number (for identification)
SUBSCRIBER_NAME	VARCHAR2(128)		Name of the subscriber
SUBSCRIBER_ADDRESS	VARCHAR2(1024)		Address of the subscribing agent
SUBSCRIBER_TYPE	VARCHAR2(128)		Type of subscriber: • PROXY - Proxy subscriber • SUBSCRIBOR
STARTUP_TIME	DATE	NOT NULL	Startup time of the instance
LAST_BROWSED_SEQ	NUMBER		Sequence number of the most recently browsed message for the subscriber (comparable to the number of messages in the V$STREAMS_APPLY_READER view)
LAST_BROWSED_NUM	NUMBER		Internal Message number for the most recently browsed message for the subscriber
LAST_DEQUEUED_SEQ	NUMBER		Sequence number of the most recently dequeued message for the subscriber (comparable to the number of messages in the V$STREAMS_APPLY_COORDINATOR view)
LAST_DEQUEUED_NUM	NUMBER		Internal Message number for the most recently dequeued message for the subscriber
CURRENT_ENQ_SEQ	NUMBER		Current sequence number of the most recently enqueued message for the subscriber
NUM_MSGS	NUMBER		Total number of outstanding messages currently enqueued in the buffered queue for the subscriber (includes the count of the messages overflowed to disk)
CNUM_MSGS	NUMBER		Cumulative total number of messages enqueued for the subscriber since the creation of the buffered queue
TOTAL_DEQUEUED_MSG	NUMBER		Total number of messages dequeued by the subscriber
TOTAL_SPILLED_MSG	NUMBER		Total number of spilled messages for the subscriber
EXPIRED_MSGS	NUMBER		Number of expired messages
MESSAGE_LAG	NUMBER		Message lag of the subscriber
ELAPSED_DEQUEUE_TIME	NUMBER		Total time spent in dequeue (in hundredths of a second)
DEQUEUE_CPU_TIME	NUMBER		Total CPU time for dequeue (in hundredths of a second)
LAST_DEQUEUE_TIME	TIMESTAMP(3)		Last message dequeue time
OLDEST_MSGID	RAW(16)		Message ID of the oldest message
OLDEST_MSG_ENQTM	TIMESTAMP(3)		Enqueue time of the oldest message
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.357 DBA_HIST_CAPTURE

DBA_HIST_CAPTURE displays historical statistics information about each capture process for Oracle GoldenGate, and XStream capture operations.

This view is intended for use with Automatic Workload Repository (AWR).

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
CAPTURE_NAME	VARCHAR2(128)	NOT NULL	Name of the capture process
STARTUP_TIME	DATE	NOT NULL	Time that the capture process was last started
LAG	NUMBER		Delay (in seconds) between the creation and capture of the most recently captured message
TOTAL_MESSAGES_CAPTURED	NUMBER		Total changes captured since the capture process was last started
TOTAL_MESSAGES_ENQUEUED	NUMBER		Total number of messages enqueued since the capture process was last started
ELAPSED_RULE_TIME	NUMBER		Elapsed time (in hundredths of a second) evaluating rules since the capture process was last started
ELAPSED_ENQUEUE_TIME	NUMBER		Elapsed time (in hundredths of a second) enqueuing messages since the capture process was last started
ELAPSED_REDO_WAIT_TIME	NUMBER		Elapsed time (in hundredths of a second) spent by the capture process in the WAITING FOR REDO state since the capture process was last started.
ELAPSED_PAUSE_TIME	NUMBER		Elapsed pause time (in hundredths of a second) spent by the capture process since the capture process was last restarted
CON_DBID	NUMBER		The database ID of the PDB
EXTRACT_NAME	VARCHAR2(128)		Name of the extract process, if applicable
BYTES_REDO_MINED	NUMBER		The total amount of redo data mined (in bytes) since the capture process last started
BYTES_SENT	NUMBER		Total number of bytes sent by the capture process to the extract process since the last time the extract process attached to the capture process

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SESSION_MODULE	VARCHAR2(64)	NOT NULL	Session module. Valid values: • XStream • GoldenGate
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.358 DBA_HIST_CHANNEL_WAITS

DBA_HIST_CHANNEL_WAITS display the amount of messages broadcast on KSR and KSXR channels as well as the total time taken for the broadcast to complete.

KSR channels are local to an instance, that is, only processes within an instance subscribed to the channel can receive the message. KSXR channels allow messages to be broadcast across instances. The messages broadcast and the total time to broadcast are cumulative from the start of the instance. Channels with high overall average wait times could indicate potential problems with a subscriber on that channel which can lead to poor scaled performance.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER		Unique snapshot ID
DBID	NUMBER		Database ID for the snapshot
INSTANCE_NUMBER	NUMBER		Instance number for the snapshot
CHANNEL	VARCHAR2(64)		The name of the KSR or KSXR channel
MESSAGES_PUBLISHED	NUMBER		The cumulative count of messages published on the channel (from instance startup)
WAIT_COUNT	NUMBER		The total number of times a publisher has waited for a broadcast to complete. This metric is only pertinent for asynchronous broadcasts where the broadcast can be dispatched and publisher can wait for completion at a later point of time. A high wait count along with increased wait time can indicate a potential performance bottleneck.
WAIT_TIME_USEC	NUMBER		The cumulative amount of time in microseconds that publishers have waited for message broadcast to complete. Average time for broadcast on a channel can be computed by dividing WAIT_TIME_USEC by WAIT_COUNT. A high average time can indicate a potential performance bottleneck.
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.359 DBA_HIST_CLUSTER_INTERCON

DBA_HIST_CLUSTER_INTERCON displays information about the devices used by the instance to access the interconnect (that is, communicate with other instances).

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
NAME	VARCHAR2(256)	NOT NULL	Operating system name of the device
IP_ADDRESS	VARCHAR2(64)	NOT NULL	IP address of the device
IS_PUBLIC	VARCHAR2(3)		Indicates whether the device is a public interface (YES) or a private interface (NO) Public interfaces can be listened to by outside applications, which may be a security problem. Oracle recommends using private interfaces for interconnect.
SOURCE	VARCHAR2(31)		Describes the type of device
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.360 DBA_HIST_COLORED_SQL

DBA_HIST_COLORED_SQL displays the SQL IDs that have been marked for AWR SQL capture.

If a SQL statement is colored using the DBMS_WORKLOAD_REPOSITORY.ADD_COLORED_SQL procedure, then AWR will always capture the SQL statistics for the colored SQL ID. A SQL statement can be removed from coloring using the DBMS_WORKLOAD_REPOSITORY.REMOVE_COLORED_SQL procedure.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
SQL_ID	VARCHAR2(13)	NOT NULL	SQL ID of colored SQL statement
CREATE_TIME	DATE	NOT NULL	Time the SQL ID was colored
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database PL/SQL Packages and Types Reference for more information about the DBMS_WORKLOAD_REPOSITORY package.

4.361 DBA_HIST_COMP_IOSTAT

DBA_HIST_COMP_IOSTAT displays information about I/O statistics aggregated on the component level.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
COMPONENT	VARCHAR2(64)	NOT NULL	Component name
FILE_TYPE	VARCHAR2(64)	NOT NULL	File type
IO_TYPE	CHAR(5)	NOT NULL	The type of I/O performed
OPERATION	CHAR(5)	NOT NULL	Operation name
BYTES	NUMBER	NOT NULL	Number of bytes
IO_COUNT	NUMBER	NOT NULL	Number of I/Os that were performed
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.362 DBA_HIST_CON_SYS_TIME_MODEL

DBA_HIST_CON_SYS_TIME_MODEL displays historical system time model statistics, including OLAP timed stastistics. This view contains snapshots of V$SYS_TIME_MODEL.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER		Unique snapshot ID
DBID	NUMBER		Database ID for the snapshot
INSTANCE_NUMBER	NUMBER		Instance number for the snapshot
STAT_ID	NUMBER		Statistic ID
STAT_NAME	VARCHAR2(64)		Statistic name
VALUE	NUMBER		Statistic value
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • When queried from a non-CDB, the statistics for that instance are returned, and the CON_ID value is 0. • When queried from the root of a CDB, the statistics in every container are returned, and the CON_ID value indicates the container to which the statistics belong. • When queried from a PDB, statistics from that PDB are returned, and the CON_ID value is the container ID for that PDB.

See Also

- "V$CON_SYSMETRIC"
- "DBA_HIST_SYS_TIME_MODEL"
- "V$SYS_TIME_MODEL"

4.363 DBA_HIST_CON_SYSMETRIC_HIST

DBA_HIST_CON_SYSMETRIC_HIST externalizes all available history of the system metric values for the entire set of data kept in the database. This view contains snapshots of V$CON_SYSMETRIC_HISTORY.

Note

This view is not populated and is reserved for future use.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER		Unique snapshot ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DBID	NUMBER		Database ID for the snapshot
INSTANCE_NUMBER	NUMBER		Instance number for the snapshot
BEGIN_TIME	DATE		Begin time of the interval
END_TIME	DATE		End time of the interval
INTSIZE	NUMBER		Interval size (in hundredths of a second)
GROUP_ID	NUMBER		Group ID
METRIC_ID	NUMBER		Metric ID
METRIC_NAME	VARCHAR2(64)		Metric name
VALUE	NUMBER		Metric value
METRIC_UNIT	VARCHAR2(64)		Unit of measurement
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • When queried from a non-CDB, the statistics for that instance are returned, and the CON_ID value is 0. • When queried from the root of a CDB, the statistics in every container are returned, and the CON_ID value indicates the container to which the statistics belong. • When queried from a PDB, statistics from that PDB are returned, and the CON_ID value is the container ID for that PDB.

See Also

- "V$CON_SYSMETRIC_HISTORY"
- "DBA_HIST_SYSMETRIC_HISTORY"

4.364 DBA_HIST_CON_SYSMETRIC_SUMM

DBA_HIST_CON_SYSMETRIC_SUMM displays a history of statistical summary of all metric values in the system metrics long duration (60–second) group. This view contains snapshots of V$CON_SYSMETRIC_SUMMARY.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER		Unique snapshot ID
DBID	NUMBER		Database ID for the snapshot
INSTANCE_NUMBER	NUMBER		Instance number for the snapshot
BEGIN_TIME	DATE		Begin time of the interval
END_TIME	DATE		End time of the interval
INTSIZE	NUMBER		Interval size (in hundredths of a second)
GROUP_ID	NUMBER		Group ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
METRIC_ID	NUMBER		Metric ID
METRIC_NAME	VARCHAR2(64)		Metric name
METRIC_UNIT	VARCHAR2(64)		Unit of measurement
NUM_INTERVAL	NUMBER		Number of intervals observed
MINVAL	NUMBER		Minimum value observed
MAXVAL	NUMBER		Maximum value observed
AVERAGE	NUMBER		Average over the period
STANDARD_DEVIATION	NUMBER		One standard deviation
SUM_SQUARES	NUMBER		Sum of the squared deviations from the mean
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • When queried from a non-CDB, the statistics for that instance are returned, and the CON_ID value is 0. • When queried from the root of a CDB, the statistics in every container are returned, and the CON_ID value indicates the container to which the statistics belong. • When queried from a PDB, statistics from that PDB are returned, and the CON_ID value is the container ID for that PDB.

Note

- "V$CON_SYSMETRIC_SUMMARY"
- "DBA_HIST_SYSMETRIC_SUMMARY"

4.365 DBA_HIST_CON_SYSSTAT

DBA_HIST_CON_SYSSTAT displays historical system statistics information, including OLAP kernel statistics. This view contains snapshots of V$CON_SYSSTAT.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
STAT_ID	NUMBER	NOT NULL	Statistic identifier
STAT_NAME	VARCHAR2(64)	NOT NULL	Statistic name
VALUE	NUMBER		Statistic value
CON_DBID	NUMBER	NOT NULL	The database ID for the PDB of the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER	NOT NULL	The ID of the container that CON_DBID identifies. Possible values include: • When queried from a non-CDB, the statistics for that instance are returned, and the CON_ID value is 0. • When queried from the root of a CDB, the statistics in every container are returned, and the CON_ID value indicates the container to which the statistics belong. • When queried from a PDB, statistics from that PDB are returned, and the CON_ID value is the container ID for that PDB.

See Also

- "V$CON_SYSSTAT"
- "DBA_HIST_SYSSTAT"

4.366 DBA_HIST_CON_SYSTEM_EVENT

DBA_HIST_CON_SYSTEM_EVENT displays historical information on total waits for an event in a container. This view contains snapshots of V$CON_SYSTEM_EVENT.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
EVENT_ID	NUMBER	NOT NULL	Identifier of the wait event
EVENT_NAME	VARCHAR2(64)	NOT NULL	Name of the wait event
WAIT_CLASS_ID	NUMBER		Identifier of the Class of the Wait Event
WAIT_CLASS	VARCHAR2(64)		Name of the Class of the Wait Event
TOTAL_WAITS	NUMBER		Total number of waits for the event
TOTAL_TIMEOUTS	NUMBER		Total number of timeouts for the event
TIME_WAITED_MICRO	NUMBER		Total amount of time waited for the event (in microseconds)
TOTAL_WAITS_FG	NUMBER		Total number of waits for the event, from foreground sessions
TOTAL_TIMEOUTS_FG	NUMBER		Total number of timeouts for the event, from foreground sessions
TIME_WAITED_MICRO_FG	NUMBER		Amount of time waited for the event (in microseconds), from foreground sessions
CON_DBID	NUMBER	NOT NULL	The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$CON_SYSTEM_EVENT"

4.367 DBA_HIST_CR_BLOCK_SERVER

DBA_HIST_CR_BLOCK_SERVER displays historical statistics on the Global Cache Service processes (LMS) used in cache fusion.

This view contains snapshots of V$CR_BLOCK_SERVER.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
CR_REQUESTS	NUMBER		Number of CR blocks served due to remote CR block requests
CURRENT_REQUESTS	NUMBER		Number of current blocks served due to remote CR block requests CR_REQUESTS + CURRENT_REQUESTS = global cache CR blocks served (from V$SYSSTAT).
DATA_REQUESTS	NUMBER		Number of current or CR requests for data blocks
UNDO_REQUESTS	NUMBER		Number of CR requests for undo blocks
TX_REQUESTS	NUMBER		Number of CR requests for undo segment header blocks DATA_REQUESTS + UNDO_REQUESTS + TX_REQUESTS = total number of requests handled by the LMS processes
CURRENT_RESULTS	NUMBER		Number of requests for which no changes were rolled out of the block returned to the requesting instance
PRIVATE_RESULTS	NUMBER		Number of requests for which changes were rolled out of the block returned to the requesting instance, and only the requesting transaction can use the resulting CR block
ZERO_RESULTS	NUMBER		Number of requests for which changes were rolled out of the block returned to the requesting instance. Only zero-XID transactions can use the block.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DISK_READ_RESULTS	NUMBER		Number of requests for which the requesting instance had to read the requested block from disk
FAIL_RESULTS	NUMBER		Number of requests that failed; the requesting transaction must reissue the request
FAIRNESS_DOWN_CONVERTS	NUMBER		Number of times an instance receiving a request has down-converted an X lock on a block because it was not modifying the block
FLUSHES	NUMBER		Number of times the log has been flushed by an LMS process
FLUSHES	NUMBER		Number of times the log has been flushed by an LMS process
BUILDS	NUMBER		Number of requests for which the server had to fabricate a CR block
LIGHT_WORKS	NUMBER		Number of times the light-work rule was evoked. This rule prevents the LMS processes from going to disk while responding to CR requests for data, undo, or undo segment header blocks. This rule can prevent the LMS process from completing its response to the CR request.
ERRORS	NUMBER		Number of times an error was signalled by an LMS process
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$CR_BLOCK_SERVER"

4.368 DBA_HIST_CURRENT_BLOCK_SERVER

DBA_HIST_CURRENT_BLOCK_SERVER displays historical statistics on the Global Cache Service processes (LMS) used in cache fusion.

This view contains snapshots of V$CURRENT_BLOCK_SERVER.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PIN0	NUMBER		Pins taking less than 100 microseconds
PIN1	NUMBER		Pins taking 100 microseconds to 1 millisecond
PIN10	NUMBER		Pins taking 1 to 10 milliseconds
PIN100	NUMBER		Pins taking 10 to 100 milliseconds
PIN1000	NUMBER		Pins taking 100 to 1000 milliseconds
PIN10000	NUMBER		Pins taking 1000 to 10000 milliseconds
FLUSH0	NUMBER		Flushes taking less than 100 microseconds
FLUSH1	NUMBER		Flushes taking 100 microseconds to 1 millisecond
FLUSH10	NUMBER		Flushes taking 1 to 10 milliseconds
FLUSH100	NUMBER		Flushes taking 10 to 100 milliseconds
FLUSH1000	NUMBER		Flushes taking 100 to 1000 milliseconds
FLUSH10000	NUMBER		Flushes taking 1000 to 10000 milliseconds
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$CURRENT_BLOCK_SERVER"

4.369 DBA_HIST_DATABASE_INSTANCE

DBA_HIST_DATABASE_INSTANCE displays the databases and instances in the Workload Repository.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number
STARTUP_TIME	TIMESTAMP(3)	NOT NULL	Startup time of the instance
PARALLEL	VARCHAR2(3)	NOT NULL	Indicates whether the instance is running in an Oracle Real Application Clusters (Oracle RAC) environment (YES) or not (NO)
VERSION	VARCHAR2(17)	NOT NULL	Database version
DB_NAME	VARCHAR2(9)		Name of the database
INSTANCE_NAME	VARCHAR2(16)		Name of the instance

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
HOST_NAME	VARCHAR2(64)		Name of the host
LAST_ASH_SAMPLE_ID	NUMBER	NOT NULL	Last sample ID for the active session history
PLATFORM_NAME	VARCHAR2(101)		Platform on which the instance is running
CDB	VARCHAR2(3)		Possible values are: • YES if the database is a CDB • NO if the database is not a CDB
EDITION	VARCHAR2(7)		The edition of the database: • PO: Personal Edition • XE: Express Edition • SE2: Standard Edition or Standard Edition 2 • EE: Enterprise Edition • HP: Enterprise Edition - High Performance • XP: Enterprise Edition - Extreme Performance
DB_UNIQUE_NAME	VARCHAR2(30)		Unique database name
DATABASE_ROLE	VARCHAR2(16)		Current role of the database: • SNAPSHOT STANDBY • LOGICAL STANDBY • PHYSICAL STANDBY • PRIMARY • FAR SYNC
CDB_ROOT_DBID	NUMBER		The database ID of the CDB root for the sampled session
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
STARTUP_TIME_TZ	TIMESTAMP(3) WITH TIME ZONE		Startup time of the instance

See Also

"DB_UNIQUE_NAME"

4.370 DBA_HIST_DATAFILE

DBA_HIST_DATAFILE displays a history of the data file information from the control file.

This view contains snapshots of V$DATAFILE.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FILE#	NUMBER	NOT NULL	File identification number
CREATION_CHANGE#	NUMBER	NOT NULL	Change number at which the data file was created
FILENAME	VARCHAR2(513)	NOT NULL	Name of the data file
TS#	NUMBER	NOT NULL	Tablespace number
TSNAME	VARCHAR2(30)		Name of the tablespace
BLOCK_SIZE	NUMBER		Block size of the data file
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$DATAFILE"

4.371 DBA_HIST_DB_CACHE_ADVICE

DBA_HIST_DB_CACHE_ADVICE displays historical predictions of the number of physical reads for the cache size corresponding to each row.

This view contains snapshots of V$DB_CACHE_ADVICE.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
BPID	NUMBER	NOT NULL	Buffer Pool identifier (ranges from 1 to 8)
BUFFERS_FOR_ESTIMATE	NUMBER	NOT NULL	Cache size for prediction (in terms of buffers)
NAME	VARCHAR2(20)		Buffer pool name
BLOCK_SIZE	NUMBER		Block size in bytes for buffers in the pool (the standard block size, the power of 2 nonstandard block sizes, 2048, 4096, 8192, 16384, or 32768)
ADVICE_STATUS	VARCHAR2(3)		Status of the advisory: • ON - Currently running • OFF - Disabled (the estimates are historical and calculated when last enabled)
SIZE_FOR_ESTIMATE	NUMBER		Cache size for prediction (in megabytes)
SIZE_FACTOR	NUMBER		Size factor with respect to the current cache size

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PHYSICAL_READS	NUMBER		Physical reads for the cache size
BASE_PHYSICAL_READS	NUMBER		Base physical reads for the cache size
ACTUAL_PHYSICAL_READS	NUMBER		Actual physical reads for the cache size
ESTD_PHYSICAL_READ_TIME	NUMBER		Estimated physical read time for the cache size
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$DB_CACHE_ADVICE"

4.372 DBA_HIST_DISPATCHER

DBA_HIST_DISPATCHER displays historical information for each dispatcher process present at the time of the snapshot.

This view contains snapshots of information from V$DISPATCHER and V$QUEUE.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
NAME	VARCHAR2(4)	NOT NULL	Name of the dispatcher process
SERIAL#	NUMBER		Serial number of the dispatcher process
IDLE	NUMBER		Total idle time for the dispatcher (in hundredths of a second)
BUSY	NUMBER		Total busy time for the dispatcher (in hundredths of a second)
WAIT	NUMBER		Total time that all items in the dispatcher queue have waited (in hundredths of a second). Divide by TOTALQ for average wait per item.
TOTALQ	NUMBER		Total number of items that have ever been in the dispatcher queue

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SAMPLED_TOTAL_CONN	NUMBER		Cumulative sum of total number of connections to the dispatcher over all samples. To determine the average number of connections to the dispatcher between two snapshots, divide the difference in SAMPLED_TOTAL_CONN by the difference in NUM_SAMPLES (obtained from DBA_HIST_SHARED_SERVER_SUMMARY).
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$DISPATCHER"
- "V$QUEUE"

4.373 DBA_HIST_DLM_MISC

DBA_HIST_DLM_MISC displays miscellaneous Oracle Real Application Clusters (Oracle RAC) statistics.

This view contains snapshots of V$DLM_MISC.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER		Unique snapshot ID
DBID	NUMBER		Database ID for the snapshot
INSTANCE_NUMBER	NUMBER		Instance number for the snapshot
STATISTIC#	NUMBER		Statistic number
NAME	VARCHAR2(38)		Statistic name
VALUE	NUMBER		Statistic value
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.374 DBA_HIST_DYN_REMASTER_STATS

DBA_HIST_DYN_REMASTER_STATS displays historical statistical information about the dynamic remastering process.

All times are given in hundredths of a second, and total values reflect what has been collected since instance startup. This view contains snapshots of V$DYNAMIC_REMASTER_STATS.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
REMASTER_TYPE	VARCHAR2(11)	NOT NULL	Remaster process type. Possible values: • AFFINITY: This value is used for the row containing statistics that pertain to dynamic remastering activity on object affinity. • READ-MOSTLY: This value is used for the row containing statistics that pertain to dynamic remastering activity on read-mostly objects.
PERSISTENT_OBJECTS[1]	NUMBER		Current number of objects that are marked persistent read-mostly in the cluster
REMASTER_OPS	NUMBER		Total number of dynamic remastering operations
REMASTER_TIME	NUMBER		Total dynamic remastering time
REMASTERED_OBJECTS	NUMBER		Total number of objects dynamically remastered due to affinity
QUIESCE_TIME	NUMBER		Total quiesce step time
FREEZE_TIME	NUMBER		Total freeze step time
CLEANUP_TIME	NUMBER		Total cleanup step time
REPLAY_TIME	NUMBER		Total replay step time
FIXWRITE_TIME	NUMBER		Total fixwrite step time
SYNC_TIME	NUMBER		Total synchronization step time
RESOURCES_CLEANED	NUMBER		Total number of resources cleaned in the cleanup steps
REPLAYED_LOCKS_SENT	NUMBER		Total number of locks replayed to other instances in the replay steps
REPLAYED_LOCKS_RECEIVED	NUMBER		Total number of locks received from other instances in the replay steps
CURRENT_OBJECTS	NUMBER		Current number of objects remastered on this instance due to affinity or the current number of objects that are marked read-mostly in the cluster
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

[1] This column is available starting with Oracle Database 19c.

See Also

"V$DYNAMIC_REMASTER_STATS"

4.375 DBA_HIST_ENQUEUE_STAT

DBA_HIST_ENQUEUE_STAT displays historical statistics on the number of enqueue (lock) requests for each type of lock.

This view contains snapshots of V$ENQUEUE_STATISTICS.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
EQ_TYPE	VARCHAR2(2)	NOT NULL	Type of enqueue requested
REQ_REASON	VARCHAR2(64)	NOT NULL	Reason for the enqueue request
TOTAL_REQ#	NUMBER		Total number of enqueue requests or enqueue conversions for this type of enqueue
TOTAL_WAIT#	NUMBER		Total number of times an enqueue request or conversion resulted in a wait
SUCC_REQ#	NUMBER		Number of times an enqueue request or conversion was granted
FAILED_REQ#	NUMBER		Number of times an enqueue request or conversion failed
CUM_WAIT_TIME	NUMBER		Total amount of time (in milliseconds) spent waiting for the enqueue or enqueue conversion
EVENT#	NUMBER		Event number
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$ENQUEUE_STATISTICS"

4.376 DBA_HIST_EVENT_HISTOGRAM

DBA_HIST_EVENT_HISTOGRAM displays event histogram historical statistics information.

This view contains snapshots of V$EVENT_HISTOGRAM.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
EVENT_ID	NUMBER	NOT NULL	Identifier of the wait event
EVENT_NAME	VARCHAR2(64)	NOT NULL	Name of the wait event
WAIT_CLASS_ID	NUMBER		Identifier of the class of the wait event
WAIT_CLASS	VARCHAR2(64)		Name of the class of the wait event
WAIT_TIME_MILLI	NUMBER	NOT NULL	Wait time (in milliseconds)
WAIT_COUNT	NUMBER		Wait count
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$EVENT_HISTOGRAM"

Copyright © 2002, 2026, Oracle and/or its affiliates.

4.377 DBA_HIST_EVENT_NAME

DBA_HIST_EVENT_NAME displays information about wait events.

This view contains a snapshot of V$EVENT_NAME.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
EVENT_ID	NUMBER	NOT NULL	Identifier of the wait event
EVENT_NAME	VARCHAR2(64)	NOT NULL	Name of the wait event
PARAMETER1	VARCHAR2(64)		Description of the first parameter for the wait event
PARAMETER2	VARCHAR2(64)		Description of the second parameter for the wait event
PARAMETER3	VARCHAR2(64)		Description of the third parameter for the wait event
WAIT_CLASS_ID	NUMBER		Identifier of the class of the wait event
WAIT_CLASS	VARCHAR2(64)		Name of the class of the wait event
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$EVENT_NAME"

4.378 DBA_HIST_FILEMETRIC_HISTORY

DBA_HIST_FILEMETRIC_HISTORY displays the history of file metrics collected in the Workload Repository.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
FILEID	NUMBER	NOT NULL	File number
CREATIONTIME	NUMBER	NOT NULL	File creation time
BEGIN_TIME	DATE	NOT NULL	Begin time of the interval
END_TIME	DATE	NOT NULL	End time of the interval
INTSIZE	NUMBER	NOT NULL	Interval size (in hundredths of a second)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
GROUP_ID	NUMBER	NOT NULL	ID of the group to which the file belongs
AVGREADTIME	NUMBER	NOT NULL	Average file read time
AVGWRITETIME	NUMBER	NOT NULL	Average file write time
PHYSICALREAD	NUMBER	NOT NULL	Number of physical reads
PHYSICALWRITE	NUMBER	NOT NULL	Number of physical writes
PHYBLKREAD	NUMBER	NOT NULL	Number of physical block reads
PHYBLKWRITE	NUMBER	NOT NULL	Number of physical block writes
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.379 DBA_HIST_FILESTATXS

DBA_HIST_FILESTATXS displays information about file read/write statistics.

This view contains snapshots of V$FILESTAT.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
FILE#	NUMBER	NOT NULL	File identification number
CREATION_CHANGE#	NUMBER	NOT NULL	Change number at which the data file was created
FILENAME	VARCHAR2(513)	NOT NULL	Name of the data file
TS#	NUMBER	NOT NULL	Tablespace number
TSNAME	VARCHAR2(30)		Name of the tablespace
BLOCK_SIZE	NUMBER		Block size of the data file
PHYRDS	NUMBER		Number of physical reads done
PHYWRTS	NUMBER		Number of times DBWR is required to write
SINGLEBLKRDS	NUMBER		Number of single block reads
READTIM	NUMBER		Time (in hundredths of a second) spent doing reads if the TIMED_STATISTICS parameter is true; 0 if TIMED_STATISTICS is false
WRITETIM	NUMBER		Time (in hundredths of a second) spent doing writes if the TIMED_STATISTICS parameter is true; 0 if TIMED_STATISTICS is false
SINGLEBLKRDTIM	NUMBER		Cumulative single block read time (in hundredths of a second)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PHYBLKRD	NUMBER		Number of physical blocks read
PHYBLKWRT	NUMBER		Number of blocks written to disk, which may be the same as PHYWRTS if all writes are single blocks
WAIT_COUNT	NUMBER		Shows the number of waits at the file level for contended buffers. This value includes the individual wait events that are included in the buffer busy waits wait event. **See Also:** "buffer busy waits"
TIME	NUMBER		Time spent waiting for the wait events in the WAIT_COUNT column
OPTIMIZED_PHYBLKRD	NUMBER		Number of physical reads from Database Smart Flash Cache blocks
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$FILESTAT"

4.380 DBA_HIST_IC_CLIENT_STATS

DBA_HIST_IC_CLIENT_STATS displays information about the usage of an interconnect device by the instance.

The information is divided into several areas of the Oracle Database, each identified by the NAME value.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
NAME	VARCHAR2(9)	NOT NULL	Identifies the area of the Oracle Database: • ipq - Parallel query communications • dlm - Database lock management • cache - Global cache communications All other values are internal to Oracle and are not expected to have high usage.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BYTES_SENT	NUMBER		Number of bytes sent by the instance since instance startup for the software area identified by NAME. This information is aggregated across all devices used by the instance.
BYTES_RECEIVED	NUMBER		Number of bytes received by the instance since instance startup for the software area identified by NAME. This information is aggregated across all devices used by the instance.
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.381 DBA_HIST_IC_DEVICE_STATS

DBA_HIST_IC_DEVICE_STATS displays operating system information about the usage of interconnect devices by the machine.

This usage contains Oracle usage but is not limited to it. The quality of the information depends on the operating system.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
IF_NAME	VARCHAR2(256)	NOT NULL	Name of the device (same as NAME in DBA_HIST_CLUSTER_INTERCON)
IP_ADDR	VARCHAR2(64)	NOT NULL	IP address of the device (same as IP_ADDRESS in DBA_HIST_CLUSTER_INTERCON)
NET_MASK	VARCHAR2(16)		Network mask
FLAGS	VARCHAR2(32)		Flags
MTU	NUMBER		Maximum transmission unit
BYTES_RECEIVED	NUMBER		Number of bytes received since operating system start time
PACKETS_RECEIVED	NUMBER		Number of packets received since operating system start time
RECEIVE_ERRORS	NUMBER		Number of receive errors since operating system start time
RECEIVE_DROPPED	NUMBER		Number of receive messages that were dropped
RECEIVE_BUF_OR	NUMBER		Number of receive buffer overruns experienced
RECEIVE_FRAME_ERR	NUMBER		Number of receive errors due to frame error

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BYTES_SENT	NUMBER		Number of bytes sent since operating system start time
PACKETS_SENT	NUMBER		Number of packets sent since operating system start time
SEND_ERRORS	NUMBER		Number of send errors since operating system start time
SENDS_DROPPED	NUMBER		Number of send messages that were dropped
SEND_BUF_OR	NUMBER		Number of send buffer overruns experienced
SEND_CARRIER_LOST	NUMBER		Number of send errors due to carrier lost
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.382 DBA_HIST_IM_SEG_STAT

DBA_HIST_IM_SEG_STAT displays information about historical in-memory segment statistics.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
TS#	NUMBER	NOT NULL	Tablespace number
OBJ#	NUMBER	NOT NULL	Dictionary object number
DATAOBJ#	NUMBER	NOT NULL	Data object number
MEMBYTES	NUMBER		Size of in-memory version of the segment in bytes
SCANS	NUMBER		Count of segment statistics
SCANS_DELTA	NUMBER		Delta values for in-memory scans
DB_BLOCK_CHANGES	NUMBER		The total number of changes that were part of an update or delete operation that were made to segment blocks
DB_BLOCK_CHANGES_DELTA	NUMBER		Delta value for database block changes
POPULATE_CUS	NUMBER		Count of compression units (CUs) populated per segment
POPULATE_CUS_DELTA	NUMBER		Delta value for compression unit (CU) populate operations
REPOPULATE_CUS	NUMBER		Count of CUs repopulated per segment
REPOPULATE_CUS_DELTA	NUMBER		Delta value for compression unit (CU) repopulate operations

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.383 DBA_HIST_IM_SEG_STAT_OBJ

DBA_HIST_IM_SEG_STAT_OBJ displays information about object metadata for historical in-memory segments.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database id for the snapshot
TS#	NUMBER	NOT NULL	Tablespace number
OBJ#	NUMBER	NOT NULL	Dictionary object number
DATAOBJ#	NUMBER	NOT NULL	Data object number
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
OBJECT_NAME	VARCHAR2(128)		Name of the object
SUBOBJECT_NAME	VARCHAR2(128)		Name of the subobject
OBJECT_TYPE	VARCHAR2(128)		Type of the object
TABLESPACE_NAME	VARCHAR2(128)		Tablespace name for the object
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.384 DBA_HIST_INST_CACHE_TRANSFER

DBA_HIST_INST_CACHE_TRANSFER displays the historical statistics on the cache blocks transferred among instances.

This view contains snapshots of V$INSTANCE_CACHE_TRANSFER.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
INSTANCE	NUMBER	NOT NULL	Instance from which the blocks are transferred
CLASS	VARCHAR2(18)	NOT NULL	Class of the cache block
CR_BLOCK	NUMBER		CR block transfers not affected by remote processing delays
CR_BUSY	NUMBER		Current block transfers affected by remote contention
CR_CONGESTED	NUMBER		CR block transfers affected by remote system load
CURRENT_BLOCK	NUMBER		Current block transfers not affected by remote processing delays
CURRENT_BUSY	NUMBER		Current block transfers affected by remote contention
CURRENT_CONGESTED	NUMBER		Current block transfers affected by remote system load
LOST	NUMBER		The number of blocks that were sent by a particular instance but that never arrived in this instance
CR_2HOP	NUMBER		The count of CR blocks which were received by this instance from a particular instance after a 2-way round-trip
CR_3HOP	NUMBER		The count of CR blocks which were received by this instance from a particular instance after a 3-way round-trip
CURRENT_2HOP	NUMBER		The count of current blocks which were received by this instance from a particular instance after a 2-way round-trip
CURRENT_3HOP	NUMBER		The count of current blocks which were received by this instance from a particular instance after a 3-way round-trip
CR_BLOCK_TIME	NUMBER		Total time waited for CR blocks from a particular instance (includes the other times)
CR_BUSY_TIME	NUMBER		The time waited for CR blocks which were received by this instance from a particular instance and which were delayed by a log flushed on the sending instance
CR_CONGESTED_TIME	NUMBER		The time waited for CR blocks which were received by this instance from a particular instance and which were delayed because LMS was busy
CURRENT_BLOCK_TIME	NUMBER		Total time waited for CR blocks from a particular instance (includes the other times)
CURRENT_BUSY_TIME	NUMBER		The time waited for current blocks which were received by this instance from a particular instance and which were delayed by a log flushed on the sending instance
CURRENT_CONGESTED_TIME	NUMBER		The time waited for current blocks which were received by this instance from a particular instance and which were delayed because LMS was busy
LOST_TIME	NUMBER		The time waited for blocks that were sent by a particular instance but that never arrived in this instance
CR_2HOP_TIME	NUMBER		The time waited for CR blocks which were received by this instance from a particular instance after a 2-way round-trip

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CR_3HOP_TIME	NUMBER		The time waited for CR blocks which were received by this instance from a particular instance after a 3-way round-trip
CURRENT_2HOP_TIME	NUMBER		The time waited for current blocks which were received by this instance from a particular instance after a 2-way round-trip
CURRENT_3HOP_TIME	NUMBER		The time waited for current blocks which were received by this instance from a particular instance after a 3-way round-trip
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$INSTANCE_CACHE_TRANSFER"

4.385 DBA_HIST_INSTANCE_RECOVERY

DBA_HIST_INSTANCE_RECOVERY displays the historical monitoring of the mechanisms available to the user to limit recovery I/O.

This view contains snapshots of V$INSTANCE_RECOVERY.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
RECOVERY_ESTIMATED_IOS	NUMBER		Number of dirty buffers in the buffer cache.
ACTUAL_REDO_BLKS	NUMBER		Current actual number of redo blocks required for recovery
TARGET_REDO_BLKS	NUMBER		Current target number of redo blocks that must be processed for recovery. This value is the minimum value of the following 3 columns, and identifies which of the 3 user-defined limits determines checkpointing.
LOG_FILE_SIZE_REDO_BLKS	NUMBER		Maximum number of redo blocks required to guarantee that a log switch does not occur before the checkpoint completes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LOG_CHKPT_TIMEOUT_REDO_BLKS	NUMBER		Number of redo blocks that need to be processed during recovery to satisfy the LOG_CHECKPOINT_TIMEOUT parameter. The value displayed is not meaningful unless LOG_CHECKPOINT_TIMEOUT has been set.
LOG_CHKPT_INTERVAL_REDO_BLKS	NUMBER		Number of redo blocks that need to be processed during recovery to satisfy the LOG_CHECKPOINT_INTERVAL parameter. The value displayed is not meaningful unless LOG_CHECKPOINT_INTERVAL has been set.
FAST_START_IO_TARGET_REDO_BLKS	NUMBER		This column is obsolete and maintained for backward compatibility. The value of this column is always null.
TARGET_MTTR	NUMBER		Effective MTTR (mean time to recover) target value in seconds. The TARGET_MTTR value is calculated based on the value of the FAST_START_MTTR_TARGET parameter (the TARGET_MTTR value is used internally), and is usually an approximation of the parameter's value. However, if the FAST_START_MTTR_TARGET parameter value is very small (for example, one second), or very large (for example, 3600 seconds), then the calculation will produce a target value dictated by system limitations. In such cases, the TARGET_MTTR value will be the shortest calculated time, or the longest calculated time that recovery is expected to take. If FAST_START_MTTR_TARGET is not specified, then the value of this field is the current estimated MTTR.
ESTIMATED_MTTR	NUMBER		Current estimated mean time to recover (MTTR) based on the number of dirty buffers and log blocks (0 if FAST_START_MTTR_TARGET is not specified). This value tells you how long you can expect recovery to take based on the work the system is doing right now.
CKPT_BLOCK_WRITES	NUMBER		Number of blocks written by checkpoint writes
OPTIMAL_LOGFILE_SIZE	NUMBER		Redo log file size (in megabytes) that is considered optimal based on the current setting of FAST_START_MTTR_TARGET. It is recommended that all online redo logs be configured to be at least this value.
ESTD_CLUSTER_AVAILABLE_TIME	NUMBER		Estimated time (in seconds) that the cluster would become partially available should the instance fail. This column is only meaningful in an Oracle Real Application Clusters (Oracle RAC) environment. In a non-Oracle RAC environment, the value of this column is null.
WRITES_MTTR	NUMBER		Number of writes driven by the FAST_START_MTTR_TARGET parameter
WRITES_LOGFILE_SIZE	NUMBER		Number of writes driven by the smallest redo log file size
WRITES_LOG_CHECKPOINT_SETTINGS	NUMBER		Number of writes driven by the LOG_CHECKPOINT_INTERVAL parameter or the LOG_CHECKPOINT_TIMEOUT parameter
WRITES_OTHER_SETTINGS	NUMBER		Number of writes driven by other reasons (such as the deprecated FAST_START_IO_TARGET parameter)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
WRITES_AUTOTUNE	NUMBER		Number of writes due to auto-tune checkpointing
WRITES_FULL_THREAD_CKPT	NUMBER		Number of writes due to full thread checkpoints
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$INSTANCE_RECOVERY"

4.386 DBA_HIST_INTERCONNECT_PINGS

DBA_HIST_INTERCONNECT_PINGS displays information about measured latency of interconnect messages (round-trip) from instance to instance.

In Oracle Database 11*g* and later releases, the PING process assesses the latencies associated with communications for each pair of instances.

Every few seconds, the process in one instance (INSTANCE_NUMBER value) sends two messages to each instance (TARGET_INSTANCE value). One message has a size of 500 bytes and the other has a size of 8 KB. The message is received by the PING process on the target instance and is immediately acknowledged. The time for the round-trip is measured and collected.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
TARGET_INSTANCE	NUMBER	NOT NULL	Target instance number
CNT_500B	NUMBER		Number of pings of size 500 bytes from INSTANCE_NUMBER to TARGET_INSTANCE since the startup of the source instance (INSTANCE_NUMBER)
WAIT_500B	NUMBER		Sum of round-trip times for messages of size 500 bytes from INSTANCE_NUMBER to TARGET_INSTANCE since the startup of the source instance (INSTANCE_NUMBER). Dividing by CNT_500B gives the average latency.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
WAITSQ_500B	NUMBER		Sum of squares (divided by 1000) of round-trip times for messages of size 500 bytes from INSTANCE_NUMBER to TARGET_INSTANCE since the startup of the source instance (INSTANCE_NUMBER). When used with CNT_500B and WAIT_500B, the standard deviation of the latency can be calculated.
CNT_8K	NUMBER		Number of pings of size 8 KB from INSTANCE_NUMBER to TARGET_INSTANCE since the startup of the source instance (INSTANCE_NUMBER)
WAIT_8K	NUMBER		Sum of round-trip times for messages of size 8 KB from INSTANCE_NUMBER to TARGET_INSTANCE since the startup of the source instance (INSTANCE_NUMBER). Dividing by CNT_8K gives the average latency.
WAITSQ_8K	NUMBER		Sum of squares (divided by 1000) of round-trip times for messages of size 8 KB from INSTANCE_NUMBER to TARGET_INSTANCE since the startup of the source instance (INSTANCE_NUMBER). When used with CNT_8K and WAIT_8K, the standard deviation of the latency can be calculated.
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

4.387 DBA_HIST_IOSTAT_DETAIL

DBA_HIST_IOSTAT_DETAIL displays I/O statistics aggregated by combination of file type and function (component).

This view contains snapshots of V$IOSTAT_FILE and V$IOSTAT_FUNCTION.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
FUNCTION_ID	NUMBER	NOT NULL	Function ID
FUNCTION_NAME	VARCHAR2(30)	NOT NULL	Function name
FILETYPE_ID	NUMBER	NOT NULL	Type of file (for example, log file, data file, and so on)
FILETYPE_NAME	VARCHAR2(30)	NOT NULL	Name of the file, in the case of a data file or temp file. For all other files, a corresponding string to be displayed (for example, ARCHIVELOG).
SMALL_READ_MEGABYTES	NUMBER	NOT NULL	Number of single block megabytes read
SMALL_WRITE_MEGABYTES	NUMBER	NOT NULL	Number of single block megabytes written

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LARGE_READ_MEGABYTES	NUMBER	NOT NULL	Number of multiblock megabytes read
LARGE_WRITE_MEGABYTES	NUMBER	NOT NULL	Number of multiblock megabytes written
SMALL_READ_REQS	NUMBER	NOT NULL	Number of single block read requests
SMALL_WRITE_REQS	NUMBER	NOT NULL	Number of single block write requests
LARGE_READ_REQS	NUMBER	NOT NULL	Number of multiblock read requests
LARGE_WRITE_REQS	NUMBER	NOT NULL	Number of multiblock write requests
NUMBER_OF_WAITS	NUMBER	NOT NULL	Number of I/O waits by functionality
WAIT_TIME	NUMBER	NOT NULL	Total wait time (in milliseconds)
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$IOSTAT_FILE"
- "V$IOSTAT_FUNCTION"

4.388 DBA_HIST_IOSTAT_FILETYPE

DBA_HIST_IOSTAT_FILETYPE displays historical I/O statistics by file type.

This view contains snapshots of V$IOSTAT_FILE.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
FILETYPE_ID	NUMBER	NOT NULL	Type of file (for example, log file, data file, and so on)
FILETYPE_NAME	VARCHAR2(30)	NOT NULL	Name of the file, in the case of a data file or temp file. For all other files, a corresponding string to be displayed (for example, ARCHIVELOG).
SMALL_READ_MEGABYTES	NUMBER	NOT NULL	Number of single block megabytes read
SMALL_WRITE_MEGABYTES	NUMBER	NOT NULL	Number of single block megabytes written
LARGE_READ_MEGABYTES	NUMBER	NOT NULL	Number of multiblock megabytes read
LARGE_WRITE_MEGABYTES	NUMBER	NOT NULL	Number of multiblock megabytes written
SMALL_READ_REQS	NUMBER	NOT NULL	Number of single block read requests

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SMALL_WRITE_REQS	NUMBER	NOT NULL	Number of single block write requests
SMALL_SYNC_READ_REQS	NUMBER	NOT NULL	Number of synchronous single block read requests
LARGE_READ_REQS	NUMBER	NOT NULL	Number of multiblock read requests
LARGE_WRITE_REQS	NUMBER	NOT NULL	Number of multiblock write requests
SMALL_READ_SERVICETIME	NUMBER	NOT NULL	Total service time (in milliseconds) for single block read requests
SMALL_WRITE_SERVICETIME	NUMBER	NOT NULL	Total service time (in milliseconds) for single block write requests
SMALL_SYNC_READ_LATENCY	NUMBER	NOT NULL	Latency for single block synchronous reads (in milliseconds)
LARGE_READ_SERVICETIME	NUMBER	NOT NULL	Total service time (in milliseconds) for multiblock read requests
LARGE_WRITE_SERVICETIME	NUMBER	NOT NULL	Total service time (in milliseconds) for multiblock write requests
RETRIES_ON_ERROR	NUMBER	NOT NULL	Number of read retries on error
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$IOSTAT_FILE"

4.389 DBA_HIST_IOSTAT_FILETYPE_NAME

DBA_HIST_IOSTAT_FILETYPE_NAME displays historical I/O statistics for file type names.

This view contains snapshots of V$IOSTAT_FILE.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID for the snapshot
FILETYPE_ID	NUMBER	NOT NULL	Type of file (for example, log file, data file, and so on)
FILETYPE_NAME	VARCHAR2(30)	NOT NULL	Name of the file, in the case of a data file or temp file. For all other files, a corresponding string to be displayed (for example, ARCHIVELOG).
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$IOSTAT_FILE"

4.390 DBA_HIST_IOSTAT_FUNCTION

DBA_HIST_IOSTAT_FUNCTION displays historical I/O statistics by function.

This view contains snapshots of V$IOSTAT_FUNCTION.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
FUNCTION_ID	NUMBER	NOT NULL	Function ID
FUNCTION_NAME	VARCHAR2(128)	NOT NULL	Function name
SMALL_READ_MEGABYTES	NUMBER	NOT NULL	Number of single block megabytes read
SMALL_WRITE_MEGABYTES	NUMBER	NOT NULL	Number of single block megabytes written
LARGE_READ_MEGABYTES	NUMBER	NOT NULL	Number of multiblock megabytes read
LARGE_WRITE_MEGABYTES	NUMBER	NOT NULL	Number of multiblock megabytes written
SMALL_READ_REQS	NUMBER	NOT NULL	Number of single block read requests
SMALL_WRITE_REQS	NUMBER	NOT NULL	Number of single block write requests
LARGE_READ_REQS	NUMBER	NOT NULL	Number of multiblock read requests
LARGE_WRITE_REQS	NUMBER	NOT NULL	Number of multiblock write requests
NUMBER_OF_WAITS	NUMBER	NOT NULL	Number of I/O waits by functionality
WAIT_TIME	NUMBER	NOT NULL	Total wait time (in milliseconds)
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$IOSTAT_FUNCTION"

4.391 DBA_HIST_IOSTAT_FUNCTION_NAME

DBA_HIST_IOSTAT_FUNCTION_NAME displays historical I/O statistics by function names.

This view contains snapshots of V$IOSTAT_FUNCTION.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID for the snapshot
FUNCTION_ID	NUMBER	NOT NULL	Function ID
FUNCTION_NAME	VARCHAR2(128)	NOT NULL	Function name
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$IOSTAT_FUNCTION"

4.392 DBA_HIST_JAVA_POOL_ADVICE

DBA_HIST_JAVA_POOL_ADVICE displays historical information about estimated parse time in the Java pool for different pool sizes.

This view contains snapshots of V$JAVA_POOL_ADVICE.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
JAVA_POOL_SIZE_FOR_ESTIMATE	NUMBER	NOT NULL	Java pool size for the estimate (in megabytes)
JAVA_POOL_SIZE_FACTOR	NUMBER		Size factor with respect to the current Java pool size
ESTD_LC_SIZE	NUMBER		Estimated memory in use by the library cache (in megabytes)
ESTD_LC_MEMORY_OBJECTS	NUMBER		Estimated number of library cache memory objects in the Java pool of the specified size
ESTD_LC_TIME_SAVED	NUMBER		Estimated elapsed parse time saved (in seconds), owing to library cache memory objects being found in a Java pool of the specified size. This is the time that would have been spent in reloading the required objects in the Java pool had they been aged out due to insufficient amount of available free memory.
ESTD_LC_TIME_SAVED_FACTOR	NUMBER		Estimated parse time saved factor with respect to the current Java pool size
ESTD_LC_LOAD_TIME	NUMBER		Estimated elapsed time (in seconds) for parsing in a Java pool of the specified size.
ESTD_LC_LOAD_TIME_FACTOR	NUMBER		Estimated load time factor with respect to the current Java pool size
ESTD_LC_MEMORY_OBJECT_HITS	NUMBER		Estimated number of times a library cache memory object was found in a Java pool of the specified size
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$JAVA_POOL_ADVICE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5

Static Data Dictionary Views: DBA_HIST_LATCH to DBA_STORED_SETTINGS

This chapter contains the static data dictionary views DBA_HIST_LATCH through DBA_STORED_SETTINGS.

5.1 DBA_HIST_LATCH

DBA_HIST_LATCH displays historical aggregate latch statistics for both parent and child latches, grouped by latch name.

This view contains snapshots of V$LATCH.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
LATCH_HASH	NUMBER	NOT NULL	Latch hash
LATCH_NAME	VARCHAR2(64)	NOT NULL	Latch name
LEVEL#	NUMBER		Latch level
GETS	NUMBER		Number of times the latch was requested in willing-to-wait mode
MISSES	NUMBER		Number of times the latch was requested in willing-to-wait mode and the requester had to wait
SLEEPS	NUMBER		Number of times a willing-to-wait latch request resulted in a session sleeping while waiting for the latch
IMMEDIATE_GETS	NUMBER		Number of times a latch was requested in no-wait mode
IMMEDIATE_MISSES	NUMBER		Number of times a no-wait latch request did not succeed (that is, missed)
SPIN_GETS	NUMBER		Number of willing-to-wait latch requests which missed the first try but succeeded while spinning
SLEEP[1 | 2 | 3 | 4]	NUMBER		These columns have been deprecated and are present only for compatibility with previous releases of Oracle. No data is accumulated for these columns; they will always have a value of zero.
WAIT_TIME	NUMBER		Elapsed time spent waiting for the latch (in microseconds)
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$LATCH"

5.2 DBA_HIST_LATCH_CHILDREN

DBA_HIST_LATCH_CHILDREN displays historical statistics about child latches.

This view includes all columns of DBA_HIST_LATCH plus the CHILD# column. Note that child latches have the same parent if their LATCH# columns match each other. This view contains snapshots of V$LATCH_CHILDREN.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
LATCH_HASH	NUMBER	NOT NULL	Latch hash
LATCH_NAME	VARCHAR2(64)	NOT NULL	Latch name
CHILD#	NUMBER	NOT NULL	Child latch number (unique only to each parent latch)
GETS	NUMBER		Number of times the latch was requested in willing-to-wait mode
MISSES	NUMBER		Number of times the latch was requested in willing-to-wait mode and the requester had to wait
SLEEPS	NUMBER		Number of times a willing-to-wait latch request resulted in a session sleeping while waiting for the latch
IMMEDIATE_GETS	NUMBER		Number of times a latch was requested in no-wait mode
IMMEDIATE_MISSES	NUMBER		Number of times a no-wait latch request did not succeed (that is, missed)
SPIN_GETS	NUMBER		Number of willing-to-wait latch requests which missed the first try but succeeded while spinning

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SLEEP[1 | 2 | 3 | 4]	NUMBER		These columns have been deprecated and are present only for compatibility with previous releases of Oracle. No data is accumulated for these columns; they will always have a value of zero.
WAIT_TIME	NUMBER		Elapsed time spent waiting for the latch (in microseconds)
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "DBA_HIST_LATCH"
- "V$LATCH"

5.3 DBA_HIST_LATCH_MISSES_SUMMARY

DBA_HIST_LATCH_MISSES_SUMMARY displays historical summary statistics about missed attempts to acquire a latch.

This view contains snapshots of V$LATCH_MISSES.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
PARENT_NAME	VARCHAR2(50)	NOT NULL	Latch name of a parent latch
WHERE_IN_CODE	VARCHAR2(64)	NOT NULL	Location that attempted to acquire the latch
NWFAIL_COUNT	NUMBER		Number of times that no-wait acquisition of the latch failed
SLEEP_COUNT	NUMBER		Number of times that acquisition attempts caused sleeps
WTR_SLP_COUNT	NUMBER		Number of times a waiter slept
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$LATCH_MISSES"

5.4 DBA_HIST_LATCH_NAME

DBA_HIST_LATCH_NAME displays information about decoded latch names for the latches shown in DBA_HIST_LATCH.

This view contains a snapshot of V$LATCHNAME.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
LATCH_HASH	NUMBER	NOT NULL	Latch hash
LATCH_NAME	VARCHAR2(64)	NOT NULL	Latch name
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "DBA_HIST_LATCH"
- "V$LATCHNAME"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.5 DBA_HIST_LATCH_PARENT

DBA_HIST_LATCH_PARENT displays historical statistics about parent latches.

This view contains snapshots of V$LATCH_PARENT.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
LATCH_HASH	NUMBER	NOT NULL	Latch hash
LATCH_NAME	VARCHAR2(64)	NOT NULL	Latch name
LEVEL#	NUMBER	NOT NULL	Latch level
GETS	NUMBER		Number of times the latch was requested in willing-to-wait mode
MISSES	NUMBER		Number of times the latch was requested in willing-to-wait mode and the requester had to wait
SLEEPS	NUMBER		Number of times a willing-to-wait latch request resulted in a session sleeping while waiting for the latch
IMMEDIATE_GETS	NUMBER		Number of times a latch was requested in no-wait mode
IMMEDIATE_MISSES	NUMBER		Number of times a no-wait latch request did not succeed (that is, missed)
SPIN_GETS	NUMBER		Number of willing-to-wait latch requests which missed the first try but succeeded while spinning
SLEEP[1 | 2 | 3 | 4]	NUMBER		These columns have been deprecated and are present only for compatibility with previous releases of Oracle. No data is accumulated for these columns; they will always have a value of zero.
WAIT_TIME	NUMBER		Elapsed time spent waiting for the latch (in microseconds)
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$LATCH_PARENT"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.6 DBA_HIST_LIBRARYCACHE

DBA_HIST_LIBRARYCACHE displays historical statistics about library cache performance and activity.

This view contains snapshots of V$LIBRARYCACHE.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
NAMESPACE	VARCHAR2(15)	NOT NULL	Library cache namespace
GETS	NUMBER		Number of times a lock was requested for objects of the namespace
GETHITS	NUMBER		Number of times an object's handle was found in memory
PINS	NUMBER		Number of times a PIN was requested for objects of the namespace
PINHITS	NUMBER		Number of times all of the metadata pieces of the library object were found in memory
RELOADS	NUMBER		Any PIN of an object that is not the first PIN performed since the object handle was created, and which requires loading the object from disk
INVALIDATIONS	NUMBER		Total number of times objects in the namespace were marked invalid because a dependent object was modified
DLM_LOCK_REQUESTS	NUMBER		Number of GET requests lock instance locks
DLM_PIN_REQUESTS	NUMBER		Number of PIN requests lock instance locks
DLM_PIN_RELEASES	NUMBER		Number of release requests PIN instance locks
DLM_INVALIDATION_REQUESTS	NUMBER		Number of GET requests for invalidation instance locks
DLM_INVALIDATIONS	NUMBER		Number of invalidation pings received from other instances
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$LIBRARYCACHE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.7 DBA_HIST_LOG

DBA_HIST_LOG displays historical log file information from the control file. This view contains snapshots of V$LOG.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
GROUP#	NUMBER	NOT NULL	Log group number
THREAD#	NUMBER	NOT NULL	Log thread number
SEQUENCE#	NUMBER	NOT NULL	Log sequence number
BYTES	NUMBER		Size of the log (in bytes)
MEMBERS	NUMBER		Number of members in the log group
ARCHIVED	VARCHAR2(3)		Archive status (YES) or NO)
STATUS	VARCHAR2(16)		Log status: • UNUSED - Online redo log has never been written to. This is the state of a redo log that was just added, or just after a RESETLOGS, when it is not the current redo log. • CURRENT - Current redo log. This implies that the redo log is active. The redo log could be open or closed. • ACTIVE - Log is active but is not the current log. It is needed for crash recovery. It may be in use for block recovery. It may or may not be archived. • CLEARING - Log is being re-created as an empty log after an ALTER DATABASE CLEAR LOGFILE statement. After the log is cleared, the status changes to UNUSED. • CLEARING_CURRENT - Current log is being cleared of a closed thread. The log can stay in this status if there is some failure in the switch such as an I/O error writing the new log header. • INACTIVE - Log is no longer needed for instance recovery. It may be in use for media recovery. It may or may not be archived. • INVALIDATED - Archived the current redo log without a log switch.
FIRST_CHANGE#	NUMBER		Lowest system change number (SCN) in the log
FIRST_TIME	DATE		Time of the first SCN in the log
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$LOG"

5.8 DBA_HIST_MEM_DYNAMIC_COMP

DBA_HIST_MEM_DYNAMIC_COMP displays historical memory component sizes.

This view contains snapshots of V$MEMORY_DYNAMIC_COMPONENTS.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
COMPONENT	VARCHAR2(64)	NOT NULL	Component name
CURRENT_SIZE	NUMBER		Current size of the component
MIN_SIZE	NUMBER		Minimum size of the component since instance startup
MAX_SIZE	NUMBER		Maximum size of the component since instance startup
USER_SPECIFIED_SIZE	NUMBER		Value of the user parameter for the component
OPER_COUNT	NUMBER		Number of operations since instance startup
LAST_OPER_TYPE	VARCHAR2(13)		Last completed operation for the component: • STATIC • INITIALIZING • DISABLED • GROW • SHRINK • SHRINK_CANCEL
LAST_OPER_MODE	VARCHAR2(9)		Mode of the last completed operation: • MANUAL • DEFERRED • IMMEDIATE
LAST_OPER_TIME	DATE		Start time of the last completed operation
GRANULE_SIZE	NUMBER		Granularity of the GROW or SHRINK operation
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$MEMORY_DYNAMIC_COMPONENTS"

5.9 DBA_HIST_MEMORY_RESIZE_OPS

DBA_HIST_MEMORY_RESIZE_OPS displays memory resize operations history.

This view contains snapshots of V$MEMORY_RESIZE_OPS.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
COMPONENT	VARCHAR2(64)	NOT NULL	Component name
OPER_TYPE	VARCHAR2(13)	NOT NULL	Operation type: • STATIC • INITIALIZING • DISABLED • GROW • SHRINK • SHRINK_CANCEL
START_TIME	DATE	NOT NULL	Start time of the operation
END_TIME	DATE	NOT NULL	End time of the operation
TARGET_SIZE	NUMBER	NOT NULL	Requested value of the parameter after the resize
OPER_MODE	VARCHAR2(9)		Operation mode: • MANUAL • DEFERRED • IMMEDIATE
PARAMETER	VARCHAR2(80)		Name of the parameter for the resize operation
INITIAL_SIZE	NUMBER		Parameter value at the start of the operation
FINAL_SIZE	NUMBER		Real value of the parameter after the resize

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(9)		Completion status of the operation: • INACTIVE • PENDING • COMPLETE • CANCELLED • ERROR
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$MEMORY_RESIZE_OPS"

5.10 DBA_HIST_MEMORY_TARGET_ADVICE

DBA_HIST_MEMORY_TARGET_ADVICE displays memory target advice history.

This view contains snapshots of V$MEMORY_TARGET_ADVICE.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
MEMORY_SIZE	NUMBER	NOT NULL	If the MEMORY_SIZE_FACTOR column has a value of 1, then this column shows the current size of memory, as set by the MEMORY_TARGET initialization parameter. If the value of the MEMORY_SIZE_FACTOR column is less than or greater than 1, then this column shows a proposed memory size.
MEMORY_SIZE_FACTOR	NUMBER		A multiplier for the current memory size. Possible values are 0.25, 0.5, 0.75, 1, 1.5, 1.75 and 2. This multiplier times the current memory size equals the value of the MEMORY_SIZE column.
ESTD_DB_TIME	NUMBER		For current memory size (MEMORY_SIZE_FACTOR = 1), the amount of database time required to complete the current workload. For a proposed memory size, the estimated amount of database time that would be required if the MEMORY_TARGET parameter were changed to the proposed size.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ESTD_DB_TIME_FACTOR	NUMBER		For a proposed memory size, ratio of estimated database time to current database time
VERSION	NUMBER		Version number of this recommendation
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$MEMORY_TARGET_ADVICE"

5.11 DBA_HIST_METRIC_NAME

DBA_HIST_METRIC_NAME describes attributes of the set of RDBMS metrics.

This view contains a snapshot of V$METRICNAME.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
GROUP_ID	NUMBER	NOT NULL	Metric Group ID
GROUP_NAME	VARCHAR2(64)		Metric group name
METRIC_ID	NUMBER	NOT NULL	Metric ID
METRIC_NAME	VARCHAR2(64)	NOT NULL	Metric name
METRIC_UNIT	VARCHAR2(64)	NOT NULL	Unit of measurement
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"V$METRICNAME"

5.12 DBA_HIST_MTTR_TARGET_ADVICE

DBA_HIST_MTTR_TARGET_ADVICE displays historical predictions of the number of physical I/O requests for the MTTR corresponding to each row.

The data also includes a physical I/O factor, which is the ratio of the number of estimated I/O requests to the number of I/O requests actually performed by the current MTTR setting during the measurement interval. This view contains snapshots of V$MTTR_TARGET_ADVICE.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
MTTR_TARGET_FOR_ESTIMATE	NUMBER		MTTR setting being simulated (equal to the current MTTR setting if this is the first row of the view)
ADVICE_STATUS	VARCHAR2(5)		Current status of MTTR simulation: • ON SET • READY SET • OFF
DIRTY_LIMIT	NUMBER		Dirty buffer limit derived from the MTTR being simulated
ESTD_CACHE_WRITES	NUMBER		Estimated number of cache physical writes under the MTTR
ESTD_CACHE_WRITE_FACTOR	NUMBER		Estimated cache physical write ratio under the MTTR. It is the ratio of the estimated number of cache writes to the number of cache writes under the current MTTR setting.
ESTD_TOTAL_WRITES	NUMBER		Estimated total number of physical writes under the MTTR
ESTD_TOTAL_WRITE_FACTOR	NUMBER		Estimated total physical write ratio under the MTTR. It is the ratio of the estimated total number of physical writes to the total number of physical writes under the current MTTR setting.
ESTD_TOTAL_IOS	NUMBER		Estimated total number of I/O requests under the MTTR
ESTD_TOTAL_IO_FACTOR	NUMBER		Estimated total I/O ratio under the MTTR. It is the ratio of the estimated total number of I/O requests to the total number of I/O requests under the current MTTR setting.
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$MTTR_TARGET_ADVICE"

5.13 DBA_HIST_MUTEX_SLEEP

DBA_HIST_MUTEX_SLEEP displays mutex sleep summary historical statistics information.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database identifier for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
MUTEX_TYPE	VARCHAR2(32)	NOT NULL	Mutex type
LOCATION	VARCHAR2(40)	NOT NULL	The code location where the waiter slept for the mutex
SLEEPS	NUMBER		Number of sleeps for this MUTEX_TYPE and LOCATION
WAIT_TIME	NUMBER		Wait time in microseconds
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.14 DBA_HIST_OPTIMIZER_ENV

DBA_HIST_OPTIMIZER_ENV displays the optimizer environments that have been captured in the Workload Repository.

This view is used with the DBA_HIST_SQLSTAT view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
OPTIMIZER_ENV_HASH_VALUE	NUMBER	NOT NULL	Hash value for the optimizer environment
OPTIMIZER_ENV	RAW(2000)		Optimizer environment
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"DBA_HIST_SQLSTAT"

5.15 DBA_HIST_OSSTAT

DBA_HIST_OSSTAT displays historical operating system statistics.

This view contains snapshots of V$OSSTAT.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
STAT_ID	NUMBER	NOT NULL	Statistic ID
STAT_NAME	VARCHAR2(64)	NOT NULL	Statistic name
VALUE	NUMBER		Statistic value
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "V$OSSTAT"

5.16 DBA_HIST_OSSTAT_NAME

DBA_HIST_OSSTAT_NAME displays the names of the operating system statistics.

This view is used with DBA_HIST_OSSTAT.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
STAT_ID	NUMBER	NOT NULL	Statistic ID
STAT_NAME	VARCHAR2(64)	NOT NULL	Statistic name
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **See Also**
>
> "DBA_HIST_OSSTAT"

5.17 DBA_HIST_PARAMETER

DBA_HIST_PARAMETER displays historical information about the initialization parameters that were in effect for the instance.

This view contains snapshots of V$SYSTEM_PARAMETER.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
PARAMETER_HASH	NUMBER	NOT NULL	Parameter hash
PARAMETER_NAME	VARCHAR2(64)	NOT NULL	Name of the parameter
VALUE	VARCHAR2(512)		Parameter value for the session (if modified within the session); otherwise, the instance-wide parameter value

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ISDEFAULT	VARCHAR2(9)		Indicates whether the parameter is set to the default value (TRUE) or the parameter value was specified in the parameter file (FALSE)
ISMODIFIED	VARCHAR2(10)		Indicates whether the parameter has been modified after instance startup: • MODIFIED - Parameter has been modified with ALTER SESSION • SYSTEM_MOD - Parameter has been modified with ALTER SYSTEM (which causes all the currently logged in sessions' values to be modified) • FALSE - Parameter has not been modified after instance startup
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SYSTEM_PARAMETER"

5.18 DBA_HIST_PARAMETER_NAME

DBA_HIST_PARAMETER_NAME displays information about the parameter names captured in the workload repository.

This view is used with the DBA_HIST_PARAMETER view.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
PARAMETER_HASH	NUMBER	NOT NULL	Parameter hash
PARAMETER_NAME	VARCHAR2(64)	NOT NULL	Name of the parameter
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "DBA_HIST_PARAMETER"

5.19 DBA_HIST_PDB_IN_SNAP

DBA_HIST_PDB_IN_SNAP captures a list of open pluggable databases (PDBs) at the time of the Automatic Workload Repository (AWR) snapshot. This view can be used with other DBA_HIST_ views to construct the number of opened PDBs at the time of the snapshot.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	AWR snapshot ID
DBID	NUMBER	NOT NULL	Database ID of the database that took this snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number of the instance that took this snapshot
CON_DBID	NUMBER		DBID of an open PDB at the time of the snapshot
FLAG	NUMBER		Flag field in capture properties of the PDB. Not used at this time.
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
OPEN_TIME_TZ	TIMESTAMP(3) WITH TIME ZONE		Time the PDB was last opened

5.20 DBA_HIST_PDB_INSTANCE

DBA_HIST_PDB_INSTANCE displays the pluggable databases (PDBs) and instances in the Workload Repository.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number
STARTUP_TIME	TIMESTAMP(3)	NOT NULL	Startup time of the instance
CON_DBID	NUMBER	NOT NULL	The database ID of the PDB for the sampled session
OPEN_TIME	TIMESTAMP(3)	NOT NULL	Time the PDB was last opened
OPEN_MODE	VARCHAR2(16)		Open mode of the database
PDB_NAME	VARCHAR2(128)		PDB name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
SNAP_ID	NUMBER		The unique snapshot identifier of the snapshot that flushed the corresponding row
STARTUP_TIME_TZ	TIMESTAMP(3) WITH TIME ZONE		Startup time of the instance
OPEN_TIME_TZ	TIMESTAMP(3) WITH TIME ZONE		Time the PDB was last opened

5.21 DBA_HIST_PERSISTENT_QMN_CACHE

DBA_HIST_PERSISTENT_QMN_CACHE displays the historical summary background queue table activity.

This view contains snapshots from V$PERSISTENT_QMN_CACHE.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
QUEUE_TABLE_ID	NUMBER	NOT NULL	Queue table object ID
TYPE	VARCHAR2(32)		Type of the queue table's queue monitor cache
STATUS	NUMBER		Status of the queue table's queue monitor cache
NEXT_SERVICE_TIME	TIMESTAMP(3)		Time when the queue table should be serviced by QMON servers
WINDOW_END_TIME	TIMESTAMP(3)		Time manager activity period for non-owner queue table operations
TOTAL_RUNS	NUMBER		Total number of times this queue table is served
TOTAL_LATENCY	NUMBER		Cumulative latency in serving the queue table (in hundredths of a second)
TOTAL_ELAPSED_TIME	NUMBER		Total time spent in processing this queue table (in seconds)
TOTAL_CPU_TIME	NUMBER		Cumulative CPU time for serving the queue table (in hundredths of a second)
TMGR_ROWS_PROCESSED	NUMBER		Number of time manager entries processed
TMGR_ELAPSED_TIME	NUMBER		Cumulative time for time management activities (in hundredths of a second)
TMGR_CPU_TIME	NUMBER		Cumulative CPU time for time management activities (in hundredths of a second)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LAST_TMGR_PROCESSING_TIME	TIMESTAMP(3)		Last timer manager processing time
DEQLOG_ROWS_PROCESSED	NUMBER		Number of dequeue log entries processed
DEQLOG_PROCESSING_ELAPSED_TIME	NUMBER		Total time for processing dequeue log entries (in hundredths of a second)
DEQLOG_PROCESSING_CPU_TIME	NUMBER		Total CPU time for processing dequeue log entries (in hundredths of a second)
LAST_DEQLOG_PROCESSING_TIME	TIMESTAMP(3)		Last dequeue log processing time
DEQUEUE_INDEX_BLOCKS_FREED	NUMBER		Number of dequeue index blocks freed
HISTORY_INDEX_BLOCKS_FREED	NUMBER		Number of history index blocks freed
TIME_INDEX_BLOCKS_FREED	NUMBER		Number of time manager index blocks freed
INDEX_CLEANUP_COUNT	NUMBER		Number of times index block cleanup was attempted
INDEX_CLEANUP_ELAPSED_TIME	NUMBER		Total time for index block cleanup (in hundredths of a second)
INDEX_CLEANUP_CPU_TIME	NUMBER		Total CPU time for index block cleanup (in hundredths of a second)
LAST_INDEX_CLEANUP_TIME	TIMESTAMP(3)		Last index block cleanup time
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$PERSISTENT_QMN_CACHE"

5.22 DBA_HIST_PERSISTENT_QUEUES

DBA_HIST_PERSISTENT_QUEUES displays Oracle Database AQ persistent queues historical statistics information.

This view contains snapshots of V$PERSISTENT_QUEUES.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
QUEUE_SCHEMA	VARCHAR2(128)	NOT NULL	Owner of the queue
QUEUE_NAME	VARCHAR2(128)	NOT NULL	Name of the queue
QUEUE_ID	NUMBER	NOT NULL	Identifier for the queue
FIRST_ACTIVITY_TIME	TIMESTAMP(6)		First queue activity time since database startup
ENQUEUED_MSGS	NUMBER		Number of messages enqueued
DEQUEUED_MSGS	NUMBER		Number of messages dequeued **Note:** This column will not be incremented until all the subscribers of the message have dequeued the message and its retention time has elapsed.
BROWSED_MSGS	NUMBER		Number of messages that have been browsed
ELAPSED_ENQUEUE_TIME	NUMBER		Total time (in hundredths of a second) spent doing enqueue
ELAPSED_DEQUEUE_TIME	NUMBER		Total time (in hundredths of a second) spent doing dequeue
ENQUEUE_CPU_TIME	NUMBER		Total CPU time for enqueue (in hundredths of a second)
DEQUEUE_CPU_TIME	NUMBER		Total CPU time for dequeue (in hundredths of a second)
AVG_MSG_AGE	NUMBER		Average age of messages in the queue
DEQUEUED_MSG_LATENCY	NUMBER		Last dequeued message latency (in seconds)
ELAPSED_TRANSFORMATION_TIME	NUMBER		Total time (in hundredths of a second) spent doing transformation
ELAPSED_RULE_EVALUATION_TIME	NUMBER		Total time (in hundredths of a second) spent doing rule evaluation
ENQUEUED_EXPIRY_MSGS	NUMBER		Number of messages enqueued with expiry
ENQUEUED_DELAY_MSGS	NUMBER		Number of messages enqueued with delay
MSGS_MADE_EXPIRED	NUMBER		Number of messages expired by time manager
MSGS_MADE_READY	NUMBER		Number of messages made ready by time manager
LAST_ENQUEUE_TIME	TIMESTAMP(6)		Last message enqueue time
LAST_DEQUEUE_TIME	TIMESTAMP(6)		Last message dequeue time
LAST_TM_EXPIRY_TIME	TIMESTAMP(6)		Last time message was expired by time manager
LAST_TM_READY_TIME	TIMESTAMP(6)		Last time message was made ready by time manager
ENQUEUE_TRANSACTIONS	NUMBER		Number of enqueue transactions
DEQUEUE_TRANSACTIONS	NUMBER		Number of dequeue transactions
EXECUTION_COUNT	NUMBER		Number of executions of the dequeue cursor
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$PERSISTENT_QUEUES"

5.23 DBA_HIST_PERSISTENT_SUBS

DBA_HIST_PERSISTENT_SUBS displays Oracle Database AQ persistent queue subscribers historical statistics information.

This view contains snapshots of V$PERSISTENT_SUBSCRIBERS.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
QUEUE_SCHEMA	VARCHAR2(128)	NOT NULL	Owner of the queue
QUEUE_NAME	VARCHAR2(128)	NOT NULL	Name of the queue
SUBSCRIBER_ID	NUMBER	NOT NULL	Internal subscriber number
SUBSCRIBER_NAME	VARCHAR2(128)		Name of the subscriber
SUBSCRIBER_ADDRESS	VARCHAR2(1024)		Address of the subscribing agent
SUBSCRIBER_TYPE	VARCHAR2(128)		Type of the subscriber: • PROXY - Propagation subscriber • SUBSCRIBER - Normal subscriber • RECIPIENT - Recipient
FIRST_ACTIVITY_TIME	TIMESTAMP(6)		First subscriber activity time since database startup
ENQUEUED_MSGS	NUMBER		Number of messages enqueued since FIRST_ACTIVITY_TIME
DEQUEUED_MSGS	NUMBER		Number of messages dequeued since FIRST_ACTIVITY_TIME
AVG_MSG_AGE	NUMBER		Average age of messages in the queue
BROWSED_MSGS	NUMBER		Number of messages that have been browsed
EXPIRED_MSGS	NUMBER		Number of messages expired since FIRST_ACTIVITY_TIME
DEQUEUED_MSG_LATENCY	NUMBER		Last dequeued message latency (in seconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LAST_ENQUEUE_TIME	TIMESTAMP(6)		Timestamp of the last enqueued message
LAST_DEQUEUE_TIME	TIMESTAMP(6)		Timestamp of the last dequeued message
ELAPSED_DEQUEUE_TIME	NUMBER		Total time spent in dequeue (in hundredths of a second)
DEQUEUE_CPU_TIME	NUMBER		Total CPU time for dequeue (in hundredths of a second)
DEQUEUE_TRANSACTIONS	NUMBER		Number of dequeue transactions
EXECUTION_COUNT	NUMBER		Number of executions of the dequeue index cursor
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$PERSISTENT_SUBSCRIBERS"

5.24 DBA_HIST_PGA_TARGET_ADVICE

DBA_HIST_PGA_TARGET_ADVICE displays historical predictions of how the cache hit percentage and over allocation count statistics displayed by the V$PGASTAT performance view would be impacted if the value of the PGA_AGGREGATE_TARGET parameter is changed.

This view contains snapshots of V$PGA_TARGET_ADVICE.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
PGA_TARGET_FOR_ESTIMATE	NUMBER	NOT NULL	Value of PGA_AGGREGATE_TARGET for the prediction (in bytes)
PGA_TARGET_FACTOR	NUMBER		PGA_TARGET_FOR_ESTIMATE / the current value of the PGA_AGGREGATE_TARGET parameter
ADVICE_STATUS	VARCHAR2(3)		Indicates whether the advice is enabled (ON) or disabled (OFF) depending on the value of the STATISTICS_LEVEL parameter
BYTES_PROCESSED	NUMBER		Total bytes processed by all the work areas considered by this advice (in bytes)
ESTD_TIME	NUMBER		Time (in seconds) required to process the bytes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ESTD_EXTRA_BYTES_RW	NUMBER		Estimated number of extra bytes which would be read or written if PGA_AGGREGATE_TARGET was set to the value of the PGA_TARGET_FOR_ESTIMATE column. This number is derived from the estimated number and size of work areas which would run in one-pass (or multi-pass) for that value of PGA_AGGREGATE_TARGET.
ESTD_PGA_CACHE_HIT_PERCENTAGE	NUMBER		Estimated value of the cache hit percentage statistic when PGA_AGGREGATE_TARGET equals PGA_TARGET_FOR_ESTIMATE. This column is derived from the above two columns and is equal to BYTES_PROCESSED / (BYTES_PROCESSED + ESTD_EXTRA_BYTES_RW)
ESTD_OVERALLOC_COUNT	NUMBER		Estimated number of PGA memory over-allocations if the value of PGA_AGGREGATE_TARGET is set to PGA_TARGET_FOR_ESTIMATE. A nonzero value means that PGA_TARGET_FOR_ESTIMATE is not large enough to run the work area workload. Hence, PGA_AGGREGATE_TARGET should not be set to PGA_TARGET_FOR_ESTIMATE since Oracle will not be able to honor that target.
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$PGASTAT"
- "V$PGA_TARGET_ADVICE"
- "PGA_AGGREGATE_TARGET"

5.25 DBA_HIST_PGASTAT

DBA_HIST_PGASTAT displays historical PGA memory usage statistics as well as statistics about the automatic PGA memory manager when it is enabled.

This view contains snapshots of V$PGASTAT.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Database instance number

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
NAME	VARCHAR2(64)	NOT NULL	Name of the statistic: • aggregate PGA auto target • aggregate PGA target parameter • bytes processed • cache hit percentage • extra bytes read/written • global memory bound • max processes count • maximum PGA allocated • maximum PGA used for auto workareas • maximum PGA used for manual workareas • over allocation count • PGA memory freed back to OS • process count • recompute count (total) • total freeable PGA memory • total PGA allocated • total PGA inuse • total PGA used for auto workareas • total PGA used for manual workareas **See Also:** V$PGASTAT for descriptions of the statistics
VALUE	NUMBER		Statistic value
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$PGASTAT"

5.26 DBA_HIST_PLAN_OPERATION_NAME

DBA_HIST_PLAN_OPERATION_NAME displays historical information about SQL plan operation names.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database identifier
OPERATION_ID	NUMBER	NOT NULL	Plan operation identifier

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OPERATION_NAME	VARCHAR2(64)		Plan operation name. This value also appears in the SQL_PLAN_OPERATION column of the DBA_HIST_ACTIVE_SESS_HISTORY view.
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.27 DBA_HIST_PLAN_OPTION_NAME

DBA_HIST_PLAN_OPTION_NAME displays historical information about SQL plan option names.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database identifier
OPTION_ID	NUMBER	NOT NULL	Plan option identifier
OPTION_NAME	VARCHAR2(64)		Plan option name. This value also appears in the SQL_PLAN_OPTIONS column of the DBA_HIST_ACTIVE_SESS_HISTORY view.
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.28 DBA_HIST_PROCESS_MEM_SUMMARY

DBA_HIST_PROCESS_MEM_SUMMARY displays historical information about dynamic PGA memory usage by named component categories for each process.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CATEGORY	VARCHAR2(15)	NOT NULL	Category name. Categories include "SQL", "PL/SQL", "OLAP" and "JAVA". Special categories are "Freeable" and "Other". Freeable memory has been allocated to the process by the operating system, but has not been allocated to a category. "Other" memory has been allocated to a category, but not to one of the named categories
IS_INSTANCE_WIDE	NUMBER		This column shows whether the process memory detail is for only this container or for the whole instance. If the value is 1, the detail is for the whole instance. Any other value is the container ID for the container to which the detail pertains, as seen in the CON_ID column.
NUM_PROCESSES	NUMBER		Number of processes
NON_ZERO_ALLOCS	NUMBER		Number of processes with nonzero allocations
USED_TOTAL	NUMBER		Bytes of PGA memory used by the process for the category
ALLOCATED_TOTAL	NUMBER		Total number of bytes of PGA memory allocated by the process for the category.
ALLOCATED_AVG	NUMBER		Average number of bytes of PGA memory allocated by the process for the category
ALLOCATED_STDDEV	NUMBER		Standard deviation of the number of bytes of PGA memory allocated by the process for the category
ALLOCATED_MAX	NUMBER		Maximum bytes of PGA memory ever allocated by the process for the category
MAX_ALLOCATED_MAX	NUMBER		Maximum bytes of PGA memory that can be allocated by the process for the category
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.29 DBA_HIST_PROCESS_WAITTIME

DBA_HIST_PROCESS_WAITTIME displays CPU and wait time by process types.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
PROCESS_TYPE	VARCHAR2(5)	NOT NULL	Process type

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DESCRIPTION	VARCHAR2(64)	NOT NULL	Process description
WAIT_CLASS_TYPE	VARCHAR2(64)	NOT NULL	Type of wait class
VALUE	NUMBER		Wait time or CPU used time in milliseconds
CON_DBID	NUMBER		The database ID of the PDB for the process
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.30 DBA_HIST_RECOVERY_PROGRESS

DBA_HIST_RECOVERY_PROGRESS displays database recovery progress information for an instance.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
START_TIME	DATE	NOT NULL	Start time of the recovery operation
TYPE	VARCHAR2(64)	NOT NULL	Type of recovery operation being performed: • CRASH RECOVERY • INSTANCE RECOVERY • MEDIA RECOVERY
ITEM	VARCHAR2(32)	NOT NULL	Item being measured. When TYPE is CRASH RECOVERY or INSTANCE RECOVERY, the possible values are: • Log Files • Redo Blocks When TYPE is MEDIA RECOVERY, the possible values are: • Active Apply Rate • Average Apply Rate • Maximum Apply Rate • Redo Applied • Log Files • Last Applied Redo • Active Time • Elapsed Time • Apply Time per Log • Checkpoint Time per Log • Standby Apply Lag • Recovery ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
UNITS	VARCHAR2(32)		The units of measurement for each item
SOFAR	NUMBER		Amount of work done so far
TOTAL	NUMBER		Total amount of work expected
TIMESTAMP	DATE		Timestamp of the last redo record applied
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root. • *n*: Where *n* is the applicable container ID for the rows containing data

5.31 DBA_HIST_REPLICATION_TBL_STATS

DBA_HIST_REPLICATION_TBL_STATS displays replication table statistics for Oracle GoldenGate and XStream sessions. This view is intended for use with Automatic Workload Repository (AWR).

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
APPLY_NAME	VARCHAR2(128)	NOT NULL	Name of the apply process
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Owner of the table
SESSION_MODULE	VARCHAR2(64)	NOT NULL	Session module. Valid values: • XStream • GoldenGate
TOTAL_INSERTS	NUMBER		Number of insert operations on this table processed by this apply server
TOTAL_UPDATES	NUMBER		Number of update operations on this table processed by this apply server
TOTAL_DELETES	NUMBER		Number of delete operations on this table processed by this apply server
CDR_SUCCESSFUL	NUMBER		Number of successfully resolved conflicts
CDR_FAILED	NUMBER		Number of conflicts that could not be resolved due to an error during resolution
REPERR_CNT	NUMBER		The total number of errors for the replication operation
HANDLE_COLLISIONS	NUMBER		Number of collisions on this table handled by this apply server
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.32 DBA_HIST_REPLICATION_TXN_STATS

DBA_HIST_REPLICATION_TXN_STATS displays replication transaction statistics for Oracle GoldenGate and XStream sessions.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Object name
SESSION_TYPE	VARCHAR2(64)	NOT NULL	Type of session
SESSION_MODULE	VARCHAR2(64)	NOT NULL	Session module. Valid values: • XStream • GoldenGate
SOURCE_DATABASE	VARCHAR2(128)		Database where the transaction originated
SOURCE_TXN_ID	VARCHAR2(128)	NOT NULL	Original transaction ID at the source database
FIRST_LCR_TIME	DATE		Time of the first LCR (message in an error transaction)
TOTAL_LCRS_COUNT	NUMBER		Total number of LCRs for this replication transaction
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.33 DBA_HIST_REPORTS

DBA_HIST_REPORTS displays information about XML reports captured into Automatic Workload Repository (AWR).

The reports themselves belong to components such as SQL Monitor, DBOP, and Real-Time ADDM.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Each XML report contains details about some activity of a component. For example, a SQL Monitor report contains a detailed report about a particular execution of a SQL statement, or a Real-Time ADDM report contains system performance data analyzed by Real-Time ADDM.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER		ID of the first Automatic Workload Repository (AWR) snapshot that will be taken after this report is generated
DBID	NUMBER		Database ID of the current database for the report
INSTANCE_NUMBER	NUMBER		Instance number (for an Oracle RAC system)
REPORT_ID	NUMBER		ID of the captured report
COMPONENT_ID	NUMBER		ID of the component (for example, SQL Monitor) whose report is captured
SESSION_ID	NUMBER		ID of the session corresponding to the captured report (currently used only for SQL Monitor reports)
SESSION_SERIAL#	NUMBER		Session serial number corresponding to the captured report (currently used only for SQL Monitor reports)
PERIOD_START_TIME	DATE		Time when the activity period started
PERIOD_END_TIME	DATE		Time when the activity period ended
GENERATION_TIME	DATE		Time when this report was generated
COMPONENT_NAME	VARCHAR2(128)		Name of the component whose report this is
REPORT_NAME	VARCHAR2(128)		Name of this report
REPORT_PARAMETERS	VARCHAR2(1024)		Parameters associated with this report
KEY1	VARCHAR2(128)		Key1 associated with the captured report
KEY2	VARCHAR2(128)		Key2 associated with the captured report
KEY3	VARCHAR2(128)		Key3 associated with the captured report
KEY4	VARCHAR2(256)		Key4 associated with the captured report
GENERATION_COST_SECONDS	NUMBER		Time taken to generate this report (in seconds)
REPORT_SUMMARY	VARCHAR2(4000)		Summary of this report
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"DBA_HIST_REPORTS_DETAILS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.34 DBA_HIST_REPORTS_CONTROL

DBA_HIST_REPORTS_CONTROL contains control information about the report capture mechanism that automatically captures XML reports to Automatic Workload Repository (AWR).

Reports are captured automatically for components like SQL Monitor and Real-Time Automatic Database Diagnostic Monitor (Real-Time ADDM).

Column	Datatype	NULL	Description
DBID	NUMBER		Database ID of the current database for the report
EXECUTION_MODE	VARCHAR2(12)		Mode of execution of automatic report capture. Possible values: • REGULAR: Regular per-minute report capture subject to DBTIME budget • FULL_CAPTURE: Report capture will be run per minute without the DBTIME budget constraints and is provided to capture a more comprehensive set of reports **NOTE**: The FULL_CAPTURE mode can be started and ended respectively by executing the START_REPORT_CAPTURE and FINISH_REPORT_CAPTURE APIs in the DBMS_AUTO_REPORT package. At all other times, the execution mode should be REGULAR.

See Also

Oracle Database PL/SQL Packages and Types Reference for more information about the DBMS_AUTO_REPORT package

5.35 DBA_HIST_REPORTS_DETAILS

DBA_HIST_REPORTS_DETAILS displays details about each report captured in Automatic Workload Repository (AWR).

Metadata for each report appears in the DBA_HIST_REPORTS view while the actual report is available in the DBA_HIST_REPORTS_DETAILS view.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER		ID of the first AWR snapshot that will be taken after this report is generated
DBID	NUMBER		Database ID of the current database for the report
INSTANCE_NUMBER	NUMBER		Instance number (for an Oracle RAC system)
REPORT_ID	NUMBER		ID of the captured report
SESSION_ID	NUMBER		ID of the session corresponding to the captured report (currently used only for SQL Monitor reports)
SESSION_SERIAL#	NUMBER		Session serial number relevant to this report (currently used only for SQL Monitor reports)
GENERATION_TIME	DATE		Time when this report was generated

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
REPORT_COMPRESSED	BLOB		Actual XML report in compressed form
REPORT	CLOB		Full uncompressed report
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"DBA_HIST_REPORTS"

5.36 DBA_HIST_REPORTS_TIMEBANDS

DBA_HIST_REPORTS_TIMEBANDS contains bands of time with a new row created every day corresponding to a band of time.

Each band of time represents a period of time, and has a row for every report captured automatically into Automatic Workload Repository (AWR) during that time. If the activity period of a report spans across two bands of time (for example. the activity started before midnight and ended after midnight), then the view contains two rows for that report, with one row for each band of time. The view is partitioned to provide fast access to all reports captured in a given time frame.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER		The AWR snapshot id corresponding to the report
DBID	NUMBER		Database ID of the current database for the report
INSTANCE_NUMBER	NUMBER		Instance number (for an Oracle RAC system)
CON_DBID	NUMBER		CDB ID of the captured report
COMPONENT_ID	NUMBER		ID of the component (for example, SQL Monitor) whose report is captured
COMPONENT_NAME	VARCHAR2(128)		Name of the component whose report is captured
BAND_START_TIME	DATE		Starting time of the time band
BAND_LENGTH	NUMBER		Length of time band in days (currently unused)
REPORT_ID	NUMBER		ID of the captured report
REPORT_GENERATION_TIME	DATE		Time when the report was generated
PERIOD_START_TIME	DATE		Time when the activity period started
PERIOD_END_TIME	DATE		Time when the activity period ended
KEY1	VARCHAR2(128)		Key1 associated with the captured report

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
KEY2	VARCHAR2(128)		Key2 associated with the captured report
KEY3	VARCHAR2(128)		Key3 associated with the captured report
KEY4	VARCHAR2(256)		Key4 associated with the captured report
SESSION_ID	NUMBER		ID of the session corresponding to the captured report (currently used only for SQL Monitor reports)
SESSION_SERIAL#	NUMBER		Session serial number corresponding to the captured report (currently used only for SQL Monitor reports)
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.37 DBA_HIST_RESOURCE_LIMIT

DBA_HIST_RESOURCE_LIMIT displays historical information about global resource use for some of the system resource.

This view contains snapshots of V$RESOURCE_LIMIT.

If time is of interest, join this view with DBA_HIST_SNAPSHOT.END_INTERVAL_TIME.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
RESOURCE_NAME	VARCHAR2(30)	NOT NULL	Name of the resource
CURRENT_UTILIZATION	NUMBER		Number of (resources, locks, or processes) currently being used
MAX_UTILIZATION	NUMBER		Maximum consumption of the resource since the last instance start up
INITIAL_ALLOCATION	VARCHAR2(10)		Initial allocation. This will be equal to the value specified for the resource in the initialization parameter file (UNLIMITED for infinite allocation).
LIMIT_VALUE	VARCHAR2(10)		Unlimited for resources and locks. This can be greater than the initial allocation value (UNLIMITED for infinite limit).
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.38 DBA_HIST_ROWCACHE_SUMMARY

DBA_HIST_ROWCACHE_SUMMARY displays historical summary statistics for data dictionary activity.

This view contains snapshots of V$ROWCACHE.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
PARAMETER	VARCHAR2(32)	NOT NULL	Name of the initialization parameter that determines the number of entries in the data dictionary cache
TOTAL_USAGE	NUMBER		Sum of the total number of entries in the cache
USAGE	NUMBER		Number of cache entries that contain valid data
GETS	NUMBER		Total number of requests for information on the data object
GETMISSES	NUMBER		Number of data requests resulting in cache misses
SCANS	NUMBER		Number of scan requests
SCANMISSES	NUMBER		Number of times a scan failed to find the data in the cache
SCANCOMPLETES	NUMBER		For a list of subordinate entries, the number of times the list was scanned completely
MODIFICATIONS	NUMBER		Number of inserts, updates, and deletions
FLUSHES	NUMBER		Number of times flushed to disk
DLM_REQUESTS	NUMBER		Number of DLM requests
DLM_CONFLICTS	NUMBER		Number of DLM conflicts
DLM_RELEASES	NUMBER		Number of DLM releases
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"V$ROWCACHE"

5.39 DBA_HIST_RSRC_CONSUMER_GROUP

DBA_HIST_RSRC_CONSUMER_GROUP displays historical information about Resource Manager consumer groups.

This view contains snapshots of V$RSRC_CONS_GROUP_HISTORY.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
SEQUENCE#	NUMBER	NOT NULL	A sequential counter that uniquely describes the DBA_HIST_RSRC_PLAN entry to which these consumer group statistics apply. When the instance is restarted, this value is reset to zero.
CONSUMER_GROUP_ID	NUMBER	NOT NULL	Consumer group object ID (a unique number, consistent across database shutdowns and startups)
CONSUMER_GROUP_NAME	VARCHAR2(128)	NOT NULL	Name of the consumer group
REQUESTS	NUMBER	NOT NULL	Cumulative number of requests that were executed in the consumer group
CPU_WAIT_TIME	NUMBER	NOT NULL	Cumulative amount of time that sessions waited for CPU because of resource management. This does not include waits due to latch or enqueue contention, I/O waits, and so on.
CPU_WAITS	NUMBER	NOT NULL	Cumulative number of times all sessions in the consumer group had to wait for CPU because of resource management. This does not include waits due to latch or enqueue contention, I/O waits, and so on.
CONSUMED_CPU_TIME	NUMBER	NOT NULL	Cumulative amount of CPU time consumed by all sessions in the consumer group (in milliseconds)
YIELDS	NUMBER	NOT NULL	Cumulative number of times that sessions in the consumer group had to yield CPU to other sessions because of quantum expiration
ACTIVE_SESS_LIMIT_HIT	NUMBER	NOT NULL	Number of times that sessions in the consumer group were queued because the consumer group reached its active session limit
UNDO_LIMIT_HIT	NUMBER	NOT NULL	Number of times that queries in the consumer group were cancelled because the consumer group reached its UNDO_POOL limit
SWITCHES_IN_CPU_TIME	NUMBER	NOT NULL	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_TIME limit
SWITCHES_OUT_CPU_TIME	NUMBER	NOT NULL	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_TIME limit

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SWITCHES_IN_IO_MEGABYTES	NUMBER	NOT NULL	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_IO_MEGABYTES limit
SWITCHES_OUT_IO_MEGABYTES	NUMBER	NOT NULL	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_IO_MEGABYTES limit
SWITCHES_IN_IO_REQUESTS	NUMBER	NOT NULL	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_IO_REQS limit
SWITCHES_OUT_IO_REQUESTS	NUMBER	NOT NULL	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_IO_REQS limit
SWITCHES_IN_IO_LOGICAL	NUMBER		Number of switches into the consumer group because of the Resource Manager plan's SWITCH_IO_LOGICAL limit
SWITCHES_OUT_IO_LOGICAL	NUMBER		Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_IO_LOGICAL limit
SWITCHES_IN_ELAPSED_TIME	NUMBER		Number of switches into the consumer group because of the Resource Manager plan's SWITCH_ELAPSED_TIME limit
SWITCHES_OUT_ELAPSED_TIME	NUMBER		Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_ELAPSED_TIME limit
PGA_LIMIT_SESSIONS_KILLED	NUMBER		The number of sessions that were terminated because their PGA allocation exceeded the PGA limit specified in the Resource Plan’s SESSION_PGA_LIMIT directive
SQL_CANCELED	NUMBER	NOT NULL	Number of times that SQL queries running in the consumer group were terminated because they exceeded the Resource Manager plan's SWITCH_TIME limit and CANCEL_SQL was specified as the Resource Manager plan's SWITCH_GROUP
ACTIVE_SESS_KILLED	NUMBER	NOT NULL	Number of times that sessions running in the consumer group were terminated because they exceeded the Resource Manager plan's SWITCH_TIME limit and KILL_SESSION was specified as the Resource Manager plan's SWITCH_GROUP
IDLE_SESS_KILLED	NUMBER	NOT NULL	Number of times that sessions in the consumer group were terminated because they were idle for too long (reached MAX_IDLE_TIME)
IDLE_BLKR_SESS_KILLED	NUMBER	NOT NULL	Number of times that sessions in the consumer group were terminated because they were idle too long (reached MAX_IDLE_BLOCKER_TIME) and were blocking other sessions
QUEUED_TIME	NUMBER	NOT NULL	Total amount of time that sessions in the consumer group have spent in the QUEUED state because of the active session limit (in milliseconds)
QUEUE_TIME_OUTS	NUMBER	NOT NULL	Number of times that requests from sessions in the consumer group timed out because they were queued for too long (reached QUEUEING_P1)
IO_SERVICE_TIME	NUMBER	NOT NULL	Cumulative I/O wait time (in milliseconds)
IO_SERVICE_WAITS	NUMBER	NOT NULL	Total number of wait requests

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SMALL_READ_MEGABYTES	NUMBER	NOT NULL	Number of single block megabytes read
SMALL_WRITE_MEGABYTES	NUMBER	NOT NULL	Number of single block megabytes written
LARGE_READ_MEGABYTES	NUMBER	NOT NULL	Number of multiblock megabytes read
LARGE_WRITE_MEGABYTES	NUMBER	NOT NULL	Number of multiblock megabytes written
SMALL_READ_REQUESTS	NUMBER	NOT NULL	Number of single block read requests
SMALL_WRITE_REQUESTS	NUMBER	NOT NULL	Number of single block write requests
LARGE_READ_REQUESTS	NUMBER	NOT NULL	Number of multiblock read requests
LARGE_WRITE_REQUESTS	NUMBER	NOT NULL	Number of multiblock write requests
PQS_QUEUED	NUMBER		Number of times that sessions in the consumer group were queued when trying to run parallel statements
PQ_QUEUED_TIME	NUMBER		Total amount of time that sessions in the consumer group were queued when trying to run parallel statements (in milliseconds)
PQ_QUEUE_TIME_OUTS	NUMBER		Number of times that parallel statements from sessions in the consumer group timed out because their queue time exceeded the Resource Manager plan's PARALLEL_QUEUE_TIMEOUT limit
PQS_COMPLETED	NUMBER		Total number of completed parallel statements in the consumer group
PQ_SERVERS_USED	NUMBER		Total number of parallel servers used by completed parallel statements in the consumer group
PQ_ACTIVE_TIME	NUMBER		Cumulative sum of the parallel active times for all completed parallel statements in the consumer group (in milliseconds)
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$RSRC_CONS_GROUP_HISTORY"

5.40 DBA_HIST_RSRC_METRIC

DBA_HIST_RSRC_METRIC displays information about historical Resource Manager metrics for the past hour.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
BEGIN_TIME	TIMESTAMP(3)	NOT NULL	Begin time of the interval
END_TIME	TIMESTAMP(3)	NOT NULL	End time of the interval
INTSIZE_CSEC	NUMBER	NOT NULL	Interval size (in hundredths of a second)
SEQUENCE#	NUMBER	NOT NULL	A sequential counter that uniquely describes the V$RSRC_PLAN_HISTORY entry to which these consumer group statistics apply. When the instance is restarted, this value is reset to zero.
CONSUMER_GROUP_ID	NUMBER	NOT NULL	Consumer group object ID (a unique number, consistent across database shutdowns and startups)
CPU_CONSUMED_TIME	NUMBER	NOT NULL	Cumulative amount of CPU time consumed by all sessions in the consumer group, in milliseconds
CPU_WAIT_TIME	NUMBER	NOT NULL	Cumulative amount of time that sessions waited for CPU because of resource management, in milliseconds. This does not include waits due to latch or enqueue contention, I/O waits, and so on. When CPU resources are not being actively managed, this value is set to zero.
AVG_RUNNING_SESSIONS	NUMBER		Average number of sessions in the consumer group that are currently running
AVG_WAITING_SESSIONS	NUMBER		Average number of sessions in the consumer group that are waiting for CPU due to resource management. When CPU resources are not being actively managed, this value is set to zero.
AVG_CPU_UTILIZATION	NUMBER		Average percentage of CPU consumed by the consumer group, with respect to the total number of CPUs in the system
IO_REQUESTS	NUMBER	NOT NULL	I/O requests
IO_MEGABYTES	NUMBER	NOT NULL	I/O megabytes
IOPS	NUMBER		I/O operations per second during the previous minute for this PDB
IOMBPS	NUMBER		I/O megabytes per second during the previous minute for this PDB
AVG_ACTIVE_PARALLEL_STMTS	NUMBER		The average number of parallel statements that were running during the 1-minute metric window
AVG_QUEUED_PARALLEL_STMTS	NUMBER		The average number of parallel statements that were queued during the 1-minute metric window
AVG_ACTIVE_PARALLEL_SERVERS	NUMBER		The average number of parallel servers that were actively running as part of a parallel statement during the 1-minute metric window
AVG_QUEUED_PARALLEL_SERVERS	NUMBER		The average number of parallel servers that were requested by queued parallel statements during the 1-minute metric window
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.41 DBA_HIST_RSRC_PDB_METRIC

DBA_HIST_RSRC_PDB_METRIC displays information about historical Resource Manager metrics for the past hour by PDB.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
BEGIN_TIME	TIMESTAMP(3)	NOT NULL	Begin time of the interval
END_TIME	TIMESTAMP(3)	NOT NULL	End time of the interval
INTSIZE_CSEC	NUMBER	NOT NULL	Interval size (in hundredths of a second)
SEQUENCE#	NUMBER	NOT NULL	A sequential counter that uniquely describes the V$RSRC_PLAN_HISTORY entry to which these PDB statistics apply. When the instance is restarted, this value is reset to zero.
CPU_CONSUMED_TIME	NUMBER	NOT NULL	Cumulative amount of CPU time consumed by all sessions in the PDB, in milliseconds
CPU_WAIT_TIME	NUMBER	NOT NULL	Cumulative amount of time that sessions waited for CPU because of resource management, in milliseconds. This does not include waits due to latch or enqueue contention, I/O waits, and so on. When CPU resources are not being actively managed, this value is set to zero.
AVG_RUNNING_SESSIONS	NUMBER		Average number of sessions in the PDB that are currently running
AVG_WAITING_SESSIONS	NUMBER		Average number of sessions in the PDB that are waiting for CPU due to resource management. When CPU resources are not being actively managed, this value is set to zero.
AVG_CPU_UTILIZATION	NUMBER		Average percentage of CPU consumed by the PDB, with respect to the total number of CPUs in the system
IOPS	NUMBER		I/O operations per second during the previous minute for this PDB
IOMBPS	NUMBER		I/O megabytes per second during the previous minute for this PDB

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
IOPS_THROTTLE_EXEMPT	NUMBER		Indicates how much of the I/O per second in the current PDB was exempted from throttling. For example, if the value in the IOPS column is 20 I/Os and the value in the IOPS_THROTTLE_EXEMPT column is 5 I/Os, then 5 I/Os of the 20 I/Os in that second were exempted from throttling. I/O throttling is defined by the MAX_IOPS database initialization parameter.
IOMBPS_THROTTLE_EXEMPT	NUMBER		Indicates how many megabytes of I/O executed per second in the current PDB were exempted from throttling. For example, if the value in the IOMBPS column is 200 megabytes and the value in the IOMBPS_THROTTLE_EXEMPT column is 50 megabytes, then 50 megabytes of the 200 megabytes were exempt from throttling. I/O megabytes per second throttling is defined by the MAX_MBPS database initialization parameter.
AVG_IO_THROTTLE	NUMBER		Average throttle time per I/O operation in milliseconds during the previous minute for this PDB
AVG_ACTIVE_PARALLEL_STMTS	NUMBER		The average number of parallel statements that were running during the 1-minute metric window
AVG_QUEUED_PARALLEL_STMTS	NUMBER		The average number of parallel statements that were queued during the 1-minute metric window
AVG_ACTIVE_PARALLEL_SERVERS	NUMBER		The average number of parallel servers that were actively running as part of a parallel statement during the 1-minute metric window
AVG_QUEUED_PARALLEL_SERVERS	NUMBER		The average number of parallel servers that were requested by queued parallel statements during the 1-minute metric window
SGA_BYTES	NUMBER	NOT NULL	The current SGA usage for this PDB in bytes
BUFFER_CACHE_BYTES	NUMBER	NOT NULL	The current usage of the buffer cache by this PDB in bytes
SHARED_POOL_BYTES	NUMBER	NOT NULL	The current usage of the shared pool by this PDB in bytes
PGA_BYTES	NUMBER	NOT NULL	The current PGA usage for this PDB in bytes
PLAN_ID	NUMBER	NOT NULL	Resource Manager plan identifier
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
MAX_AVG_RUNNING_SESSIONS	NUMBER	NOT NULL	Maximum value of AVG_RUNNING_SESSIONS during the last hour

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> - "MAX_IOPS"
> - "MAX_MBPS"

5.42 DBA_HIST_RSRC_PLAN

DBA_HIST_RSRC_PLAN displays historical information about resource plans.

This view contains snapshots of V$RSRC_PLAN_HISTORY.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
SEQUENCE#	NUMBER	NOT NULL	A sequential counter that uniquely describes a row. When the instance is restarted, this value is reset to zero.
START_TIME	DATE	NOT NULL	Time that the resource plan was enabled
END_TIME	DATE		Time that the resource plan was disabled; NULL if the row contains the current resource plan information
PLAN_ID	NUMBER	NOT NULL	Resource plan ID; NULL if the Resource Manager was disabled
PLAN_NAME	VARCHAR2(128)	NOT NULL	Resource plan name; NULL if the Resource Manager was disabled
CPU_MANAGED	VARCHAR2(4)	NOT NULL	Indicates whether the resource plan has parameters that specify a policy for how the Resource Manager should schedule sessions to manage CPU usage (ON) or whether Resource Manager is not managing CPU usage (OFF)
PARALLEL_EXECUTION_MANAGED	VARCHAR2(4)		State of parallel statement queuing: • OFF - Parallel statement queuing is disabled • STARTUP - Parallel statement queuing is enabled. This is a temporary state that can occur when an Oracle RAC database is undergoing configuration changes • FIFO - Parallel statement queuing is enabled. All parallel statements are managed in a single Oracle RAC FIFO queue • FULL - Parallel statement queuing is enabled. All parallel statements are managed in per-consumer group queues according to the current resource plan. This state is used when a resource plan that contains resource allocation directives (MGMT_P*) is enabled.
INSTANCE_CAGING	VARCHAR2(4)		Indicates whether instance caging is enabled (ON) or disabled (OFF). Instance caging is enabled if the CPU_COUNT initialization parameter is explicitly modified to a value other than 0 and Resource Manager is enabled.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$RSRC_PLAN_HISTORY"

5.43 DBA_HIST_RULE_SET

DBA_HIST_RULE_SET displays historical information about rule set statistics.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
OWNER	VARCHAR2(128)	NOT NULL	Owner of the rule set
NAME	VARCHAR2(128)	NOT NULL	Name of the rule set
STARTUP_TIME	DATE	NOT NULL	Startup time of the instance
CPU_TIME	NUMBER		Total CPU time (in hundredths of a second) spent in evaluation of the rule set
ELAPSED_TIME	NUMBER		Total elapsed time (in hundredths of a second) spent in evaluation of the rule set
EVALUATIONS	NUMBER		Number of evaluations on the rule set
SQL_FREE_EVALUATIONS	NUMBER		Number of evaluations on the rule set which did not internally issue SQL to evaluate rules
SQL_EXECUTIONS	NUMBER		Total number of SQL statements executed during evaluation of the rule set
RELOADS	NUMBER		Number of times the rule set object was reloaded in shared memory
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.44 DBA_HIST_SEG_STAT

DBA_HIST_SEG_STAT displays historical information about segment-level statistics.

This view captures the top segments based on a set of criteria and captures information from V$SEGSTAT. The total value is the value of the statistics since instance startup. The delta value is the value of the statistics from the BEGIN_INTERVAL_TIME to the END_INTERVAL_TIME in the DBA_HIST_SNAPSHOT view.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
TS#	NUMBER	NOT NULL	Tablespace number
OBJ#	NUMBER	NOT NULL	Dictionary object number
DATAOBJ#	NUMBER	NOT NULL	Data object number
LOGICAL_READS_TOTAL	NUMBER		Cumulative value for logical reads
LOGICAL_READS_DELTA	NUMBER		Delta value for logical reads
BUFFER_BUSY_WAITS_TOTAL	NUMBER		Cumulative value for buffer busy waits
BUFFER_BUSY_WAITS_DELTA	NUMBER		Delta value for buffer busy waits
DB_BLOCK_CHANGES_TOTAL	NUMBER		Cumulative value for db block changes (in blocks)
DB_BLOCK_CHANGES_DELTA	NUMBER		Delta value for db block changes (in blocks)
PHYSICAL_READS_TOTAL	NUMBER		Cumulative value for physical reads (in blocks)
PHYSICAL_READS_DELTA	NUMBER		Delta value for physical reads (in blocks)
PHYSICAL_WRITES_TOTAL	NUMBER		Cumulative value for physical writes (in blocks)
PHYSICAL_WRITES_DELTA	NUMBER		Delta value for physical writes (in blocks)
PHYSICAL_READS_DIRECT_TOTAL	NUMBER		Cumulative value for physical reads direct (in blocks)
PHYSICAL_READS_DIRECT_DELTA	NUMBER		Delta value for physical reads direct (in blocks)
PHYSICAL_WRITES_DIRECT_TOTAL	NUMBER		Cumulative value for physical writes direct (in blocks)
PHYSICAL_WRITES_DIRECT_DELTA	NUMBER		Delta value for physical writes direct (in blocks)
ITL_WAITS_TOTAL	NUMBER		Cumulative value for ITL waits

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ITL_WAITS_DELTA	NUMBER		Delta value for ITL waits
ROW_LOCK_WAITS_TOTAL	NUMBER		Cumulative value for row lock waits
ROW_LOCK_WAITS_DELTA	NUMBER		Delta value for row lock waits
GC_CR_BLOCKS_SERVED_TOTAL	NUMBER		Cumulative value for global cache CR blocks served
GC_CR_BLOCKS_SERVED_DELTA	NUMBER		Delta value for global cache CR blocks served
GC_CU_BLOCKS_SERVED_TOTAL	NUMBER		Cumulative value for global cache current blocks served
GC_CU_BLOCKS_SERVED_DELTA	NUMBER		Delta value for global cache current blocks served
GC_BUFFER_BUSY_TOTAL	NUMBER		Cumulative value for global cache buffer busy
GC_BUFFER_BUSY_DELTA	NUMBER		Delta value for global cache buffer busy
GC_CR_BLOCKS_RECEIVED_TOTAL	NUMBER		Cumulative value for global cache CR blocks received
GC_CR_BLOCKS_RECEIVED_DELTA	NUMBER		Delta value for global cache CR blocks received
GC_CU_BLOCKS_RECEIVED_TOTAL	NUMBER		Cumulative value for global cache current blocks received
GC_CU_BLOCKS_RECEIVED_DELTA	NUMBER		Delta value for global cache current blocks received
SPACE_USED_TOTAL	NUMBER		Number of bytes used by user data
SPACE_USED_DELTA	NUMBER		Delta value for space used by user data (in bytes). A negative value indicates the number of bytes deleted in the segment.
SPACE_ALLOCATED_TOTAL	NUMBER		The number of bytes that are allocated
SPACE_ALLOCATED_DELTA	NUMBER		Delta value for the space allocated (in bytes). A negative value indicates the number of bytes deallocated to the tablespace.
TABLE_SCANS_TOTAL	NUMBER		Cumulative value for table scans
TABLE_SCANS_DELTA	NUMBER		Delta value for table scans
CHAIN_ROW_EXCESS_TOTAL	NUMBER		Cumulative value of number of chained row pieces that can be eliminated by table reorganization
CHAIN_ROW_EXCESS_DELTA	NUMBER		Delta value of number of chained row pieces that can be eliminated by table reorganization
PHYSICAL_READ_REQUESTS_TOTAL	NUMBER		Cumulative value of number of physical read I/O requests issued for the monitored segment
PHYSICAL_READ_REQUESTS_DELTA	NUMBER		Delta value of number of physical read I/O requests issued for the monitored segment
PHYSICAL_WRITE_REQUESTS_TOTAL	NUMBER		Cumulative value of number of physical write I/O requests issued for the monitored segment
PHYSICAL_WRITE_REQUESTS_DELTA	NUMBER		Delta value of number of physical write I/O requests issued for the monitored segment
OPTIMIZED_PHYSICAL_READS_TOTAL	NUMBER		Cumulative value of number of physical reads from Database Smart Flash Cache for the monitored segment

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OPTIMIZED_PHYSICAL_READS_DELTA	NUMBER		Delta value of number of physical reads from Database Smart Flash Cache for the monitored segment
IM_SCANS_TOTAL	NUMBER		Count of segment statistics
IM_SCANS_DELTA	NUMBER		Delta values for in-memory scans
POPULATE_CUS_TOTAL	NUMBER		Count of compression units (CUs) populated per segment
POPULATE_CUS_DELTA	NUMBER		Delta value for compression unit (CU) populate operations
REPOPULATE_CUS_TOTAL	NUMBER		Count of CUs repopulated per segment
REPOPULATE_CUS_DELTA	NUMBER		Delta value for compression unit (CU) repopulate operations
IM_DB_BLOCK_CHANGES_TOTAL	NUMBER		The total number of changes that were part of an update or delete operation that were made to segment blocks
IM_DB_BLOCK_CHANGES_DELTA	NUMBER		Delta value for database block changes
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SEGSTAT"
- "DBA_HIST_SNAPSHOT"

5.45 DBA_HIST_SEG_STAT_OBJ

DBA_HIST_SEG_STAT_OBJ displays all the names of the segments captured in the workload repository.

This view is used with the DBA_HIST_SEG_STAT view.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
TS#	NUMBER	NOT NULL	Tablespace number
OBJ#	NUMBER	NOT NULL	Dictionary object number
DATAOBJ#	NUMBER	NOT NULL	Data object number

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object
SUBOBJECT_NAME	VARCHAR2(128)		Name of the subobject (for example: partition)
OBJECT_TYPE	VARCHAR2(18)		Type of the object for example: table, tablespace)
TABLESPACE_NAME	VARCHAR2(30)		Tablespace Name for the object
PARTITION_TYPE	VARCHAR2(8)		Partition Type, if relevant
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"DBA_HIST_SEG_STAT"

5.46 DBA_HIST_SERVICE_NAME

DBA_HIST_SERVICE_NAME displays the names of the Services tracked by the Workload Repository.

This view contains information for V$SERVICES.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID for the snapshot
SERVICE_NAME_HASH	NUMBER	NOT NULL	Hash of the service name
SERVICE_NAME	VARCHAR2(64)	NOT NULL	Name of the service
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "V$SERVICES"

5.47 DBA_HIST_SERVICE_STAT

DBA_HIST_SERVICE_STAT displays the history of important service statistics tracked by the Workload Repository.

The call rate statistics in this view can be used for making run-time routing decisions, for tracking service levels, and for per-instance diagnostics per call rate.

The elapsed timing for each call provides a relative value across instances for how well a node is processing SQL calls issued under a service name. When aggregation is enabled for the service name, this view provides historical data on the timing and work done for calls issued for the whole service. This view contains information from V$SERVICE_STATS.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
SERVICE_NAME_HASH	NUMBER	NOT NULL	Hash of the service name
SERVICE_NAME	VARCHAR2(64)	NOT NULL	Name of the service
STAT_ID	NUMBER	NOT NULL	Statistic identifier
STAT_NAME	VARCHAR2(64)	NOT NULL	Statistic name
VALUE	NUMBER		Value of the statistic
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **See Also**
>
> "V$SERVICE_STATS"

5.48 DBA_HIST_SERVICE_WAIT_CLASS

DBA_HIST_SERVICE_WAIT_CLASS displays the history of wait class information for services as tracked by the Workload Repository.

This view contains information from V$SERVICE_WAIT_CLASS.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
SERVICE_NAME_HASH	NUMBER	NOT NULL	Hash of the service name
SERVICE_NAME	VARCHAR2(64)	NOT NULL	Name of the service
WAIT_CLASS_ID	NUMBER	NOT NULL	Identifier for the class of the wait event
WAIT_CLASS	VARCHAR2(64)		Name for the class of the wait event
TOTAL_WAITS	NUMBER		Total number of waits for this event
TIME_WAITED	NUMBER		Total amount of time waited for this event (in hundredths of a second)
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SERVICE_WAIT_CLASS"

5.49 DBA_HIST_SESS_SGA_STATS

DBA_HIST_SESS_SGA_STATS shows usage statistics for high utilization Oracle GoldenGate and XStream sessions.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Object name
SESSION_TYPE	VARCHAR2(64)	NOT NULL	Type of session
SESSION_MODULE	VARCHAR2(64)	NOT NULL	Session module. Valid values: • XStream • GoldenGate
SGA_USED	NUMBER		The total amount of shared memory (in bytes) currently used by the session out of the amount allocated (SGA_ALLOCATED)
SGA_ALLOCATED	NUMBER		The total amount of shared memory (in bytes) allocated from the pool for the session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.50 DBA_HIST_SESS_TIME_STATS

DBA_HIST_SESS_TIME_STATS displays information about CPU and I/O time for interesting Oracle GoldenGate and XStream sessions.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
SESSION_TYPE	VARCHAR2(64)	NOT NULL	Type of session
MIN_LOGON_TIME	DATE		Minimum logon time
SUM_CPU_TIME	NUMBER		Total CPU time
SUM_SYS_IO_WAIT	NUMBER		Total system I/O wait time
SUM_USER_IO_WAIT	NUMBER		Total user I/O wait time
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
SESSION_MODULE	VARCHAR2(64)	NOT NULL	Session module. Valid values: • XStream • GoldenGate
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.51 DBA_HIST_SESSMETRIC_HISTORY

DBA_HIST_SESSMETRIC_HISTORY displays the history of several important session metrics.

> **Note**
>
> This view is populated only if a session metric exceeds a server metric threshold that was configured using the DBMS_SERVER_ALERT package.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
BEGIN_TIME	DATE	NOT NULL	Begin time of the interval
END_TIME	DATE	NOT NULL	End time of the interval
SESSID	NUMBER	NOT NULL	Session ID
SERIAL#	NUMBER	NOT NULL	Session serial number
INTSIZE	NUMBER	NOT NULL	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	NOT NULL	Group ID
METRIC_ID	NUMBER	NOT NULL	Metric ID
METRIC_NAME	VARCHAR2(64)	NOT NULL	Metric name
VALUE	NUMBER	NOT NULL	Metric Value
METRIC_UNIT	VARCHAR2(64)	NOT NULL	Unit of measurement
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **See Also**
>
> The DBMS_SERVER_ALERT package in *Oracle Database PL/SQL Packages and Types Reference*

5.52 DBA_HIST_SGA

DBA_HIST_SGA displays historical summary information about the system global area (SGA).

This view contains snapshots of V$SGA.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
NAME	VARCHAR2(64)	NOT NULL	SGA component group
VALUE	NUMBER	NOT NULL	Memory size (in bytes)
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SGA"

5.53 DBA_HIST_SGA_TARGET_ADVICE

DBA_HIST_SGA_TARGET_ADVICE provides historical information about the SGA_TARGET initialization parameter.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
SGA_SIZE	NUMBER	NOT NULL	Size of the SGA
SGA_SIZE_FACTOR	NUMBER	NOT NULL	Ratio between the SGA_SIZE and the current size of the SGA
ESTD_DB_TIME	NUMBER	NOT NULL	Estimated DB_TIME for this SGA_SIZE
ESTD_PHYSICAL_READS	NUMBER		Estimated number of physical reads
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "SGA_TARGET"

5.54 DBA_HIST_SGASTAT

DBA_HIST_SGASTAT displays detailed historical information on the system global area (SGA).

This view contains snapshots of V$SGASTAT.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
NAME	VARCHAR2(64)		SGA component group
POOL	VARCHAR2(30)		Designates the pool in which the memory in NAME resides: • in-memory pool - Memory is allocated from the In-Memory pool • java pool - Memory is allocated from the Java pool • large pool - Memory is allocated from the large pool • numa pool - Memory is allocated from the NUMA pool • shared pool - Memory is allocated from the shared pool • streams pool - Memory is allocated from the Streams pool
BYTES	NUMBER		Memory size (in bytes)
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **See Also**
>
> "V$SGASTAT"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.55 DBA_HIST_SHARED_POOL_ADVICE

DBA_HIST_SHARED_POOL_ADVICE displays historical information about estimated parse time in the shared pool for different pool sizes.

This view contains snapshots of V$SHARED_POOL_ADVICE.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
SHARED_POOL_SIZE_FOR_ESTIMATE	NUMBER	NOT NULL	Shared pool size for the estimate (in megabytes)
SHARED_POOL_SIZE_FACTOR	NUMBER		Size factor with respect to the current shared pool size
ESTD_LC_SIZE	NUMBER		Estimated memory in use by the library cache (in megabytes)
ESTD_LC_MEMORY_OBJECTS	NUMBER		Estimated number of library cache memory objects in the shared pool of the specified size
ESTD_LC_TIME_SAVED	NUMBER		Estimated elapsed parse time saved (in seconds), owing to library cache memory objects being found in a shared pool of the specified size. This is the time that would have been spent in reloading the required objects in the shared pool had they been aged out due to insufficient amount of available free memory.
ESTD_LC_TIME_SAVED_FACTOR	NUMBER		Estimated parse time saved factor with respect to the current shared pool size
ESTD_LC_LOAD_TIME	NUMBER		Estimated elapsed time (in seconds) for parsing in a shared pool of the specified size.
ESTD_LC_LOAD_TIME_FACTOR	NUMBER		Estimated load time factor with respect to the current shared pool size
ESTD_LC_MEMORY_OBJECT_HITS	NUMBER		Estimated number of times a library cache memory object was found in a shared pool of the specified size
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SHARED_POOL_ADVICE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.56 DBA_HIST_SHARED_SERVER_SUMMARY

DBA_HIST_SHARED_SERVER_SUMMARY displays historical information for shared servers.

This includes information about shared server activity, the servers, common queues, and dispatcher queues. This view obtains information from V$SHARED_SERVER, V$DISPATCHER, V$CIRCUIT, and V$QUEUE, and is aggregated over all servers, dispatchers, queues, and circuits.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
NUM_SAMPLES	NUMBER		Total number of samples
SAMPLE_TIME	NUMBER		Last sample timestamp
SAMPLED_TOTAL_CONN	NUMBER		Cumulative sum of total number of connections over all samples. To determine the average number of connections between two snapshots, divide the difference in SAMPLED_TOTAL_CONN by the difference in NUM_SAMPLES.
SAMPLED_ACTIVE_CONN	NUMBER		Cumulative sum of active number of connections over all samples. To determine the average number of active connections between two snapshots, divide the difference in SAMPLED_ACTIVE_CONN by the difference in NUM_SAMPLES.
SAMPLED_TOTAL_SRV	NUMBER		Cumulative sum of total number of servers over all samples. To determine the average number of servers between two snapshots, divide the difference in SAMPLED_TOTAL_SRV by the difference in NUM_SAMPLES.
SAMPLED_ACTIVE_SRV	NUMBER		Cumulative sum of active number of servers over all samples. To determine the average number of active servers between two snapshots, divide the difference in SAMPLED_ACTIVE_SRV by the difference in NUM_SAMPLES.
SAMPLED_TOTAL_DISP	NUMBER		Cumulative sum of total number of dispatchers over all samples. To determine the average number of dispatchers between two snapshots, divide the difference in SAMPLED_TOTAL_DISP by the difference in NUM_SAMPLES.
SAMPLED_ACTIVE_DISP	NUMBER		Cumulative sum of active number of dispatchers over all samples. To determine the average number of active dispatchers between two snapshots, divide the difference in SAMPLED_ACTIVE_DISP by the difference in NUM_SAMPLES.
SRV_BUSY	NUMBER		Total shared server busy time (in hundredths of a second)
SRV_IDLE	NUMBER		Total shared server idle time (in hundredths of a second)
SRV_IN_NET	NUMBER		Total shared server incoming network wait time (in hundredths of a second). This includes waits for receives and resets. This time is also included in SRV_BUSY.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SRV_OUT_NET	NUMBER		Total shared server outgoing network wait time (in hundredths of a second). This includes waits for sends and outbound connection requests. This time is also included in SRV_BUSY.
SRV_MESSAGES	NUMBER		Number of messages processed
SRV_BYTES	NUMBER		Total number of bytes in all messages
CQ_WAIT	NUMBER		Total time that all items in the common queue have waited (in hundredths of a second)
CQ_TOTALQ	NUMBER		Total number of items that have ever been in the common queue
DQ_TOTALQ	NUMBER		Total number of items that have ever been in a dispatcher queue
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SHARED_SERVER"
- "V$DISPATCHER"
- "V$CIRCUIT"
- "V$QUEUE"

5.57 DBA_HIST_SNAP_ERROR

DBA_HIST_SNAP_ERROR displays information about the snapshot error information in the Workload Repository.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table in which the error occurred
ERROR_NUMBER	NUMBER	NOT NULL	Error number for the error encountered
STEP_ID	NUMBER		For internal use only

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.58 DBA_HIST_SNAPSHOT

DBA_HIST_SNAPSHOT displays information about the snapshots in the Workload Repository.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
STARTUP_TIME	TIMESTAMP(3)	NOT NULL	Startup time of the instance
BEGIN_INTERVAL_TIME	TIMESTAMP(3)	NOT NULL	Time at the beginning of the snapshot interval
END_INTERVAL_TIME	TIMESTAMP(3)	NOT NULL	Time at the end of the snapshot interval; the actual time the snapshot was taken
FLUSH_ELAPSED	INTERVAL DAY(5) TO SECOND(1)		Amount of time to perform the snapshot
SNAP_LEVEL	NUMBER		Snapshot level
ERROR_COUNT	NUMBER		Number of errors occurring in the tables for the particular snapshot
SNAP_FLAG	NUMBER		Condition under which the snapshot was inserted. Possible values are: 0 - Snapshot was taken automatically by the Manageability Monitor Process (MMON process) 1 - Manual snapshot created using a PL/SQL package 2 - Imported snapshot 4 - Snapshot taken while Diagnostic Pack or Tuning Pack was not enabled
SNAP_TIMEZONE	INTERVAL DAY(0) TO SECOND(0)		Snapshot time zone expressed as offset from UTC (Coordinated Universal Time) time zone
BEGIN_INTERVAL_TIME_TZ	TIMESTAMP(3) WITH TIME ZONE		Time at the beginning of the snapshot interval, with timezone
END_INTERVAL_TIME_TZ	TIMESTAMP(3) WITH TIME ZONE		Time at the end of the snapshot interval; the actual time the snapshot was taken, with timezone

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Table F-1 for more information about the MMON process

5.59 DBA_HIST_SNAPSHOT_DETAILS

DBA_HIST_SNAPSHOT_DETAILS displays table details for snapshots in the Workload Repository.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
TABLE_ID	NUMBER	NOT NULL	Dictionary object ID of the table
TABLE_NAME	VARCHAR2(64)		Name of the table
BEGIN_TIME	TIMESTAMP(3)		Time at the beginning of the snapshot interval for the table
END_TIME	TIMESTAMP(3)		Time at the end of the snapshot interval for the table; the actual time the snapshot was taken
ELAPSED	INTERVAL DAY(9) TO SECOND(3)		Amount of time to perform the snapshot for the table
STEP_ID	NUMBER	NOT NULL	For internal use only
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Note

This view is available starting with Oracle Database 19c, Release Update 19.28.

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.60 DBA_HIST_SQL_BIND_METADATA

DBA_HIST_SQL_BIND_METADATA displays historical information on metadata for bind variables used by SQL cursors.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID for the snapshot
SQL_ID	VARCHAR2(13)	NOT NULL	SQL identifier of the parent cursor in the library cache
NAME	VARCHAR2(128)		Name of the bind variable
POSITION	NUMBER	NOT NULL	Position of the bind variable in the SQL statement
DUP_POSITION	NUMBER		If the binding is performed by name and the bind variable is duplicated, then this column gives the position of the primary bind variable
DATATYPE	NUMBER		Internal identifier for the bind data type. Beginning in Oracle Database 12*c*, a number representing a PL/SQL data type can appear in this column.
DATATYPE_STRING	VARCHAR2(15)		Textual representation of the bind data type. Beginning in Oracle Database 12*c*, a text representation of a PL/SQL-only data type can appear in this column. If the actual data type is a PL/SQL sub type, the name of the data type, not the sub type will be displayed.
CHARACTER_SID	NUMBER		National character set identifier
PRECISION	NUMBER		Precision (for numeric binds)
SCALE	NUMBER		Scale (for numeric binds)
MAX_LENGTH	NUMBER		Maximum bind length
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.61 DBA_HIST_SQL_PLAN

DBA_HIST_SQL_PLAN displays the execution plan information for each child cursor in the workload repository.

This view captures information from V$SQL_PLAN and is used with the DBA_HIST_SQLSTAT view.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
SQL_ID	VARCHAR2(13)	NOT NULL	SQL identifier of the parent cursor in the library cache

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PLAN_HASH_VALUE	NUMBER	NOT NULL	Numerical representation of the SQL plan for the cursor. Comparing one PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line).
ID	NUMBER	NOT NULL	A number assigned to each step in the execution plan
OPERATION	VARCHAR2(30)		Name of the internal operation performed in this step (for example, TABLE ACCESS)
OPTIONS	VARCHAR2(30)		A variation on the operation described in the OPERATION column (for example, FULL)
OBJECT_NODE	VARCHAR2(128)		Name of the database link used to reference the object (a table name or view name). For local queries that use parallel execution, this column describes the order in which output from operations is consumed.
OBJECT#	NUMBER		Object number of the table or the index
OBJECT_OWNER	VARCHAR2(128)		Name of the user who owns the schema containing the table or index
OBJECT_NAME	VARCHAR2(128)		Name of the table or index
OBJECT_ALIAS	VARCHAR2(261)		Alias for the object
OBJECT_TYPE	VARCHAR2(20)		Type of the object
OPTIMIZER	VARCHAR2(20)		Current mode of the optimizer for the first row in the plan (statement line), for example, ALL_ROWS. When the operation is a database access (for example, TABLE ACCESS), this column indicates whether or not the object is analyzed.
PARENT_ID	NUMBER		ID of the next execution step that operates on the output of the current step
DEPTH	NUMBER		Depth (or level) of the operation in the tree. It is not necessary to issue a CONNECT BY statement to get the level information, which is generally used to indent the rows from the PLAN_TABLE table. The root operation (statement) is level 0.
POSITION	NUMBER		Order of processing for all operations that have the same PARENT_ID
SEARCH_COLUMNS	NUMBER		Number of index columns with start and stop keys (that is, the number of columns with matching predicates)
COST	NUMBER		Cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
CARDINALITY	NUMBER		Estimate, by the cost-based optimizer, of the number of rows produced by the operation
BYTES	NUMBER		Estimate, by the cost-based optimizer, of the number of bytes produced by the operation
OTHER_TAG	VARCHAR2(35)		Describes the contents of the OTHER column. See EXPLAIN PLAN for values.
PARTITION_START	VARCHAR2(64)		Start partition of a range of accessed partitions
PARTITION_STOP	VARCHAR2(64)		Stop partition of a range of accessed partitions
PARTITION_ID	NUMBER		Step that computes the pair of values of the PARTITION_START and PARTITION_STOP columns

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OTHER	VARCHAR2(4000)		Other information specific to the execution step that users may find useful. See EXPLAIN PLAN for values.
DISTRIBUTION	VARCHAR2(20)		Stores the method used to distribute rows from producer query servers to consumer query servers
CPU_COST	NUMBER		CPU cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
IO_COST	NUMBER		I/O cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
TEMP_SPACE	NUMBER		Temporary space usage of the operation (sort or hash-join) as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
ACCESS_PREDICATES	VARCHAR2(4000)		Predicates used to locate rows in an access structure. For example, start or stop predicates for an index range scan.
FILTER_PREDICATES	VARCHAR2(4000)		Predicates used to filter rows before producing them
PROJECTION	VARCHAR2(4000)		Expressions produced by the operation
TIME	NUMBER		Elapsed time (in seconds) of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
QBLOCK_NAME	VARCHAR2(128)		Name of the query block
REMARKS	VARCHAR2(4000)		Remarks
TIMESTAMP	DATE		Timestamp for when the plan was produced
OTHER_XML	CLOB		Provides extra information specific to an execution step of the execution plan. The content of this column is structured using XML because it allows multiple pieces of information to be stored, including the following: • Name of the schema against which the query was parsed • Release number of the Oracle Database that produced the explain plan • Hash value associated with the execution plan • Name (if any) of the outline or the SQL profile used to build the execution plan • Indication of whether or not dynamic statistics were used to produce the plan • The outline data, a set of optimizer hints that can be used to regenerate the same plan
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SQL_PLAN"
- "DBA_HIST_SQLSTAT"

5.62 DBA_HIST_SQL_SUMMARY

DBA_HIST_SQL_SUMMARY displays historical SQL summary information.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
TOTAL_SQL	NUMBER	NOT NULL	Total number of SQLs
TOTAL_SQL_MEM	NUMBER	NOT NULL	Total sharable memory in bytes for SQLs
SINGLE_USE_SQL	NUMBER	NOT NULL	Total number of single execution SQLs
SINGLE_USE_SQL_MEM	NUMBER	NOT NULL	Total sharable memory in bytes for single execution SQLs
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.63 DBA_HIST_SQL_WORKAREA_HSTGRM

DBA_HIST_SQL_WORKAREA_HSTGRM displays the historical cumulative work area execution statistics (cumulated since instance startup) for different work area groups.

This view contains snapshots of V$SQL_WORKAREA_HISTOGRAM.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
LOW_OPTIMAL_SIZE	NUMBER	NOT NULL	Lower bound for the optimal memory requirement of work areas included in the row (in bytes)
HIGH_OPTIMAL_SIZE	NUMBER	NOT NULL	Upper bound for the optimal memory requirement of work areas included in the row (in bytes)
OPTIMAL_EXECUTIONS	NUMBER		Number of work areas with an optimal memory requirement comprised between LOW_OPTIMAL_SIZE and HIGH_OPTIMAL_SIZE which have been executed in optimal mode since instance startup
ONEPASS_EXECUTIONS	NUMBER		Number of work areas with an optimal memory requirement comprised between LOW_OPTIMAL_SIZE and HIGH_OPTIMAL_SIZE which have been executed in one-pass mode since instance startup
MULTIPASSES_EXECUTIONS	NUMBER		Number of work areas with an optimal memory requirement comprised between LOW_OPTIMAL_SIZE and HIGH_OPTIMAL_SIZE which have been executed in multi-pass mode since instance startup
TOTAL_EXECUTIONS	NUMBER		Sum of OPTIMAL_EXECUTIONS, ONEPASS_EXECUTIONS, and MULTIPASSES_EXECUTIONS
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SQL_WORKAREA_HISTOGRAM"

5.64 DBA_HIST_SQLBIND

DBA_HIST_SQLBIND displays historical information on bind variables used by SQL cursors.

This view contains snapshots of V$SQL_BIND_CAPTURE.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SQL_ID	VARCHAR2(13)	NOT NULL	SQL identifier of the parent cursor in the library cache
NAME	VARCHAR2(128)		Name of the bind variable
POSITION	NUMBER	NOT NULL	Position of the bind variable in the SQL statement
DUP_POSITION	NUMBER		If the binding is performed by name and the bind variable is duplicated, then this column gives the position of the primary bind variable.
DATATYPE	NUMBER		Internal identifier for the bind data type. Beginning in Oracle Database 12*c*, a number representing a PL/SQL data type can appear in this column.
DATATYPE_STRING	VARCHAR2(15)		Textual representation of the bind data type. Beginning in Oracle Database 12*c*, a text representation of a PL/SQL-only data type can appear in this column. If the actual data type is a PL/SQL sub type, the name of the data type, not the sub type will be displayed.
CHARACTER_SID	NUMBER		National character set identifier
PRECISION	NUMBER		Precision (for numeric binds)
SCALE	NUMBER		Scale (for numeric binds)
MAX_LENGTH	NUMBER		Maximum bind length
WAS_CAPTURED	VARCHAR2(3)		Indicates whether the bind value was captured (YES) or not (NO)
LAST_CAPTURED	DATE		Date when the bind value was captured. Bind values are captured when SQL statements are executed. To limit the overhead, binds are captured at most every 15 minutes for a given cursor.
VALUE_STRING	VARCHAR2(4000)		Value of the bind represented as a string
VALUE_ANYDATA	ANYDATA		Value of the bind represented using the self-descriptive Sys.AnyData data type. This representation is useful to programmatically decode the value of the bind variable. This column is NULL if a PL/SQL-only data type appears in the DATATYPE column.
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SQL_BIND_CAPTURE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.65 DBA_HIST_SQLCOMMAND_NAME

DBA_HIST_SQLCOMMAND_NAME displays the mapping between SQL opcodes and names.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
COMMAND_TYPE	NUMBER	NOT NULL	SQL command number
COMMAND_NAME	VARCHAR2(64)		SQL command name
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.66 DBA_HIST_SQLSTAT

DBA_HIST_SQLSTAT displays historical information about SQL statistics.

This view captures the top SQL statements based on a set of criteria and captures the statistics information from V$SQL. The total value is the value of the statistics since instance startup. The delta value is the value of the statistics from the BEGIN_INTERVAL_TIME to the END_INTERVAL_TIME in the DBA_HIST_SNAPSHOT view.

This view is used with the DBA_HIST_OPTIMIZER_ENV, DBA_HIST_SQLTEXT, and DBA_HIST_SQL_PLAN views to provide a complete picture of historical SQL statistics.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
SQL_ID	VARCHAR2(13)	NOT NULL	SQL identifier of the parent cursor in the library cache
PLAN_HASH_VALUE	NUMBER	NOT NULL	Numerical representation of the SQL plan for the cursor. Comparing one PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line).
OPTIMIZER_COST	NUMBER		Cost of the query given by the optimizer
OPTIMIZER_MODE	VARCHAR2(10)		Mode under which the SQL statement is executed
OPTIMIZER_ENV_HASH_VALUE	NUMBER		Hash Value for the optimizer environment
SHARABLE_MEM	NUMBER		Amount of shared memory used by the child cursor (in bytes)
LOADED_VERSIONS	NUMBER		Indicates whether the context heap is loaded (1) or not (0)
VERSION_COUNT	NUMBER		Number of children associated with the cursor

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MODULE	VARCHAR2(64)		Contains the name of the module that was executing at the time that the SQL statement was first parsed, which is set by calling DBMS_APPLICATION_INFO.SET_MODULE
ACTION	VARCHAR2(64)		Contains the name of the action that was executing at the time that the SQL statement was first parsed, which is set by calling DBMS_APPLICATION_INFO.SET_ACTION
SQL_PROFILE	VARCHAR2(64)		Name of the applied SQL Profile
FORCE_MATCHING_SIGNATURE	NUMBER		The signature used when the CURSOR_SHARING parameter is set to FORCE
PARSING_SCHEMA_ID	NUMBER		Schema ID that was used to originally build the child cursor
PARSING_SCHEMA_NAME	VARCHAR2(128)		Schema name that was used to originally build the child cursor
PARSING_USER_ID	NUMBER		User ID that was used to originally build the child cursor
FETCHES_TOTAL	NUMBER		Cumulative number of fetches associated with the SQL statement
FETCHES_DELTA	NUMBER		Delta number of fetches associated with the SQL statement
END_OF_FETCH_COUNT_TOTAL	NUMBER		Cumulative number of times this cursor was fully executed since the cursor was brought into the library cache. The value of this statistic is not incremented when the cursor is partially executed, either because it failed during the execution or because only the first few rows produced by this cursor are fetched before the cursor is closed or re-executed. By definition, the value of the END_OF_FETCH_COUNT column should be less or equal to the value of the EXECUTIONS column.
END_OF_FETCH_COUNT_DELTA	NUMBER		Delta number of times this cursor was fully executed since the cursor was brought into the library cache. The value of this statistic is not incremented when the cursor is partially executed, either because it failed during the execution or because only the first few rows produced by this cursor are fetched before the cursor is closed or re-executed.
SORTS_TOTAL	NUMBER		Cumulative number of sorts that were done for this child cursor
SORTS_DELTA	NUMBER		Delta number of sorts that were done for this child cursor
EXECUTIONS_TOTAL	NUMBER		Cumulative number of executions that took place on this object since it was brought into the library cache
EXECUTIONS_DELTA	NUMBER		Delta number of executions that took place on this object since it was brought into the library cache
PX_SERVERS_EXECS_TOTAL	NUMBER		Cumulative number of PX server executions
PX_SERVERS_EXECS_DELTA	NUMBER		Delta number of PX server executions
LOADS_TOTAL	NUMBER		Cumulative number of times the object was either loaded or reloaded

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LOADS_DELTA	NUMBER		Delta number of times the object was either loaded or reloaded
INVALIDATIONS_TOTAL	NUMBER		Cumulative number of times this child cursor has been invalidated
INVALIDATIONS_DELTA	NUMBER		Delta number of times this child cursor has been invalidated
PARSE_CALLS_TOTAL	NUMBER		Cumulative number of parse calls for this child cursor
PARSE_CALLS_DELTA	NUMBER		Delta number of parse calls for this child cursor
DISK_READS_TOTAL	NUMBER		Cumulative number of disk reads for this child cursor
DISK_READS_DELTA	NUMBER		Delta number of disk reads for this child cursor
BUFFER_GETS_TOTAL	NUMBER		Cumulative number of buffer gets for this child cursor
BUFFER_GETS_DELTA	NUMBER		Delta number of buffer gets for this child cursor
ROWS_PROCESSED_TOTAL	NUMBER		Cumulative number of rows the parsed SQL statement returns
ROWS_PROCESSED_DELTA	NUMBER		Delta number of rows the parsed SQL statement returns
CPU_TIME_TOTAL	NUMBER		Cumulative value of CPU time (in microseconds) used by this cursor for parsing/executing/fetching
CPU_TIME_DELTA	NUMBER		Delta value of CPU time (in microseconds) used by this cursor for parsing/executing/fetching
ELAPSED_TIME_TOTAL	NUMBER		Cumulative value of elapsed time (in microseconds) used by this cursor for parsing/executing/fetching. If the cursor uses parallel execution, then ELAPSED_TIME_TOTAL is the cumulative time for the query coordinator, plus all parallel query worker processes.
ELAPSED_TIME_DELTA	NUMBER		Delta value of elapsed time (in microseconds) used by this cursor for parsing/executing/fetching
IOWAIT_TOTAL	NUMBER		Cumulative value of user I/O wait time (in microseconds)
IOWAIT_DELTA	NUMBER		Delta value of user I/O wait time (in microseconds)
CLWAIT_TOTAL	NUMBER		Cumulative value of cluster wait time (in microseconds)
CLWAIT_DELTA	NUMBER		Delta value of cluster wait time (in microseconds)
APWAIT_TOTAL	NUMBER		Cumulative value of application wait time (in microseconds)
APWAIT_DELTA	NUMBER		Delta value of application wait time (in microseconds)
CCWAIT_TOTAL	NUMBER		Cumulative value of concurrency wait time (in microseconds)
CCWAIT_DELTA	NUMBER		Delta value of concurrency wait time (in microseconds)
DIRECT_WRITES_TOTAL	NUMBER		Cumulative value of direct writes
DIRECT_WRITES_DELTA	NUMBER		Delta value of direct writes
PLSEXEC_TIME_TOTAL	NUMBER		Cumulative value of PL/SQL Execution Time (in microseconds)
PLSEXEC_TIME_DELTA	NUMBER		Delta value of PL/SQL Execution Time (in microseconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
JAVEXEC_TIME_TOTAL	NUMBER		Cumulative value of Java Execution Time (in microseconds)
JAVEXEC_TIME_DELTA	NUMBER		Delta value of Java Execution Time (in microseconds)
IO_OFFLOAD_ELIG_BYTES_TOTAL	NUMBER		Cumulative value of number of I/O bytes which can be filtered by the Exadata storage system **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_OFFLOAD_ELIG_BYTES_DELTA	NUMBER		Delta value of number of I/O bytes which can be filtered by the Exadata storage system **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_INTERCONNECT_BYTES_TOTAL	NUMBER		Cumulative value of number of I/O bytes exchanged between Oracle Database and the storage system
IO_INTERCONNECT_BYTES_DELTA	NUMBER		Delta value of number of I/O bytes exchanged between Oracle Database and the storage system
PHYSICAL_READ_REQUESTS_TOTAL	NUMBER		Cumulative value of number of physical read I/O requests issued by the monitored SQL
PHYSICAL_READ_REQUESTS_DELTA	NUMBER		Delta value of number of physical read I/O requests issued by the monitored SQL
PHYSICAL_READ_BYTES_TOTAL	NUMBER		Cumulative value of number of bytes read from disks by the monitored SQL
PHYSICAL_READ_BYTES_DELTA	NUMBER		Delta value of number of bytes read from disks by the monitored SQL
PHYSICAL_WRITE_REQUESTS_TOTAL	NUMBER		Cumulative value of number of physical write I/O requests issued by the monitored SQL
PHYSICAL_WRITE_REQUESTS_DELTA	NUMBER		Delta value of number of physical write I/O requests issued by the monitored SQL
PHYSICAL_WRITE_BYTES_TOTAL	NUMBER		Cumulative value of number of bytes written to disks by the monitored SQL
PHYSICAL_WRITE_BYTES_DELTA	NUMBER		Delta value of number of bytes written to disks by the monitored SQL
OPTIMIZED_PHYSICAL_READS_TOTAL	NUMBER		Cumulative value of number of physical reads from the Database Smart Flash Cache or the Exadata Smart Flash Cache by the monitored SQL
OPTIMIZED_PHYSICAL_READS_DELTA	NUMBER		Delta value of number of physical reads from the Database Smart Flash Cache or the Exadata Smart Flash Cache by the monitored SQL
CELL_UNCOMPRESSED_BYTES_TOTAL	NUMBER		Cumulative value of number of uncompressed bytes (that is, size after decompression) that are offloaded to the Exadata cells **See Also:** Oracle Exadata Storage Server Software documentation for more information
CELL_UNCOMPRESSED_BYTES_DELTA	NUMBER		Delta value of number of uncompressed bytes (that is, size after decompression) that are offloaded to the Exadata cells **See Also:** Oracle Exadata Storage Server Software documentation for more information

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
IO_OFFLOAD_RETURN_BYTES_TOTAL	NUMBER		Cumulative value of number of bytes that are returned by the Exadata cell for smart scan only (that is, not including bytes for other database I/O) **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_OFFLOAD_RETURN_BYTES_DELTA	NUMBER		Delta value of number of bytes that are returned by the Exadata cell for smart scan only (that is, not including bytes for other database I/O) **See Also:** Oracle Exadata Storage Server Software documentation for more information
BIND_DATA	RAW(2000)		Bind data
FLAG	NUMBER		Reserved for internal use
OBSOLETE_COUNT	NUMBER		Number of times that a parent cursor became obsolete
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "DBA_HIST_SNAPSHOT"
- "DBA_HIST_OPTIMIZER_ENV"
- "DBA_HIST_SQLTEXT"
- "DBA_HIST_SQL_PLAN"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO package

5.67 DBA_HIST_SQLTEXT

DBA_HIST_SQLTEXT displays the text of SQL statements belonging to shared SQL cursors captured in the Workload Repository.

This view captures information from V$SQL and is used with the DBA_HIST_SQLSTAT view.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
SQL_ID	VARCHAR2(13)	NOT NULL	SQL identifier of the parent cursor in the library cache
SQL_TEXT	CLOB		Full text for the SQL statement exposed as a CLOB column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COMMAND_TYPE	NUMBER		Oracle command type definition
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SQL"
- "DBA_HIST_SQLSTAT"

5.68 DBA_HIST_STAT_NAME

DBA_HIST_STAT_NAME displays decoded statistic names for the statistics captured in the Workload Repository.

This includes OLAP statisitics and OLAP timed events. This view captures information from V$STATNAME and is used with DBA_HIST_SYSSTAT and DBA_HIST_SYS_TIME_MODEL.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
STAT_ID	NUMBER	NOT NULL	Statistic identifier
STAT_NAME	VARCHAR2(64)	NOT NULL	Statistic name
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "V$STATNAME"
- "DBA_HIST_SYSSTAT"
- "DBA_HIST_SYS_TIME_MODEL"

5.69 DBA_HIST_STREAMS_APPLY_SUM

DBA_HIST_STREAMS_APPLY_SUM displays information about each apply process and its activities.

This view contains a snapshot of V$STREAMS_APPLY_COORDINATOR, V$STREAMS_APPLY_READER, and V$STREAMS_APPLY_SERVER. This view is intended for use with Automatic Workload Repository (AWR).

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
APPLY_NAME	VARCHAR2(128)	NOT NULL	Name of the apply process
STARTUP_TIME	DATE	NOT NULL	Time that the apply process was last started
READER_TOTAL_MESSAGES_DEQUEUED	NUMBER		Total number of messages dequeued since the apply process was last started
READER_LAG	NUMBER		For captured messages, the delay (in seconds) between the creation of the last message and it being received by the apply process. For user enqueued messages, the delay between the message being enqueued in the local database and being received by the apply process.
COORD_TOTAL_RECEIVED	NUMBER		Total number of transactions received by the coordinator process since the apply process was last started
COORD_TOTAL_APPLIED	NUMBER		Total number of transactions applied by the apply process since the apply process was last started
COORD_TOTAL_ROLLBACKS	NUMBER		Number of transactions which were rolled back due to unexpected contention
COORD_TOTAL_WAIT_DEPS	NUMBER		Number of times since the apply process was last started that an apply server waited to apply a logical change record (LCR) in a transaction until another apply server applied a transaction because of a dependency between the transactions
COORD_TOTAL_WAIT_CMTS	NUMBER		Number of times since the apply process was last started that an apply server waited to commit a transaction until another apply server committed a transaction to serialize commits
COORD_LWM_LAG	NUMBER		For captured messages, the delay (in seconds) between the creation of the message corresponding to the low watermark and it being applied by the apply process. For user enqueued messages, the delay between the message being enqueued in the local database and being applied by the apply process.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SERVER_TOTAL_MESSAGES_APPLIED	NUMBER		Total number of messages applied by all the apply servers since the apply process was last started
SERVER_ELAPSED_DEQUEUE_TIME	NUMBER		Time elapsed (in hundredths of a second) dequeuing messages by all the apply servers since the apply process was last started
SERVER_ELAPSED_APPLY_TIME	NUMBER		Time elapsed (in hundredths of a second) applying messages by all the apply servers since the apply process was last started
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$STREAMS_APPLY_COORDINATOR"
- "V$STREAMS_APPLY_READER"
- "V$STREAMS_APPLY_SERVER"

5.70 DBA_HIST_STREAMS_CAPTURE

DBA_HIST_STREAMS_CAPTURE displays information about each capture process.

This view is intended for use with Automatic Workload Repository (AWR).

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
CAPTURE_NAME	VARCHAR2(128)	NOT NULL	Name of the capture process
STARTUP_TIME	DATE	NOT NULL	Time that the capture process was last started
LAG	NUMBER		Delay (in seconds) between the creation and capture of the most recently captured message
TOTAL_MESSAGES_CAPTURED	NUMBER		Total changes captured since the capture process was last started
TOTAL_MESSAGES_ENQUEUED	NUMBER		Total number of messages enqueued since the capture process was last started
ELAPSED_RULE_TIME	NUMBER		Elapsed time (in hundredths of a second) evaluating rules since the capture process was last started

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ELAPSED_ENQUEUE_TIME	NUMBER		Elapsed time (in hundredths of a second) enqueuing messages since the capture process was last started
ELAPSED_REDO_WAIT_TIME	NUMBER		Elapsed time (in hundredths of a second) spent by the capture process in the WAITING FOR REDO state
ELAPSED_PAUSE_TIME	NUMBER		Elapsed pause time
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.71 DBA_HIST_STREAMS_POOL_ADVICE

DBA_HIST_STREAMS_POOL_ADVICE displays historical information about the estimated count of spilled or unspilled messages and the associated time spent in the spill or unspill activity for different Streams pool sizes.

This view is intended for use with Automatic Workload Repository (AWR).

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID of the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number of the snapshot
SIZE_FOR_ESTIMATE	NUMBER	NOT NULL	Pool size for the estimate (in megabytes)
SIZE_FACTOR	NUMBER		Size factor with respect to the current pool size
ESTD_SPILL_COUNT	NUMBER		Estimated count of messages spilled from the Streams pool
ESTD_SPILL_TIME	NUMBER		Estimated elapsed time (in seconds) to spill
ESTD_UNSPILL_COUNT	NUMBER		Estimated count of unspills (read back from disk)
ESTD_UNSPILL_TIME	NUMBER		Estimated elapsed time (in seconds) to unspill
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.72 DBA_HIST_SYS_TIME_MODEL

DBA_HIST_SYS_TIME_MODEL displays historical system time model statistics, including OLAP timed stastistics.

This view contains snapshots of V$SYS_TIME_MODEL.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
STAT_ID	NUMBER	NOT NULL	Statistic ID
STAT_NAME	VARCHAR2(64)	NOT NULL	Statistic name
VALUE	NUMBER		Statistic value
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SYS_TIME_MODEL"
- "DBA_HIST_CON_SYS_TIME_MODEL"
- "V$CON_SYSMETRIC"

5.73 DBA_HIST_SYSMETRIC_HISTORY

DBA_HIST_SYSMETRIC_HISTORY externalizes all available history of the system metric values for the entire set of data kept in the database.

This view contains snapshots of V$SYSMETRIC_HISTORY.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
BEGIN_TIME	DATE	NOT NULL	Begin time of the interval
END_TIME	DATE	NOT NULL	End time of the interval
INTSIZE	NUMBER	NOT NULL	Interval size (in hundredths of a second)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
GROUP_ID	NUMBER	NOT NULL	Group ID
METRIC_ID	NUMBER	NOT NULL	Metric ID
METRIC_NAME	VARCHAR2(64)	NOT NULL	Metric name
VALUE	NUMBER	NOT NULL	Metric Value
METRIC_UNIT	VARCHAR2(64)	NOT NULL	Unit of measurement
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SYSMETRIC_HISTORY"
- "DBA_HIST_CON_SYSMETRIC_HIST"
- "V$CON_SYSMETRIC_HISTORY"

5.74 DBA_HIST_SYSMETRIC_SUMMARY

DBA_HIST_SYSMETRIC_SUMMARY displays a history of statistical summary of all metric values in the System Metrics Long Duration group.

This view contains snapshots of V$SYSMETRIC_SUMMARY.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
BEGIN_TIME	DATE	NOT NULL	Begin time of the interval
END_TIME	DATE	NOT NULL	End time of the interval
INTSIZE	NUMBER	NOT NULL	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	NOT NULL	Group ID
METRIC_ID	NUMBER	NOT NULL	Metric ID
METRIC_NAME	VARCHAR2(64)	NOT NULL	Metric name
METRIC_UNIT	VARCHAR2(64)	NOT NULL	Unit of measurement
NUM_INTERVAL	NUMBER	NOT NULL	Number of intervals observed
MINVAL	NUMBER	NOT NULL	Minimum value observed

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MAXVAL	NUMBER	NOT NULL	Maximum value observed
AVERAGE	NUMBER	NOT NULL	Average over the period
STANDARD_DEVIATION	NUMBER	NOT NULL	One standard deviation
SUM_SQUARES	NUMBER		Sum of the squared deviations from the mean
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SYSMETRIC_SUMMARY"
- "DBA_HIST_CON_SYSMETRIC_SUMM"
- "V$CON_SYSMETRIC_SUMMARY"

5.75 DBA_HIST_SYSSTAT

DBA_HIST_SYSSTAT displays historical system statistics information, including OLAP kernel statistics.

This view contains snapshots of V$SYSSTAT.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
STAT_ID	NUMBER	NOT NULL	Statistic identifier
STAT_NAME	VARCHAR2(64)	NOT NULL	Statistic name
VALUE	NUMBER		Statistic value
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "V$SYSSTAT"

5.76 DBA_HIST_SYSTEM_EVENT

DBA_HIST_SYSTEM_EVENT displays historical information on total waits for an event.

This view contains snapshots of V$SYSTEM_EVENT.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
EVENT_ID	NUMBER	NOT NULL	Identifier of the wait event
EVENT_NAME	VARCHAR2(64)	NOT NULL	Name of the wait event
WAIT_CLASS_ID	NUMBER		Identifier of the Class of the Wait Event
WAIT_CLASS	VARCHAR2(64)		Name of the Class of the Wait Event
TOTAL_WAITS	NUMBER		Total number of waits for the event
TOTAL_TIMEOUTS	NUMBER		Total number of timeouts for the event
TIME_WAITED_MICRO	NUMBER		Total amount of time waited for the event (in microseconds)
TOTAL_WAITS_FG	NUMBER		Total number of waits for the event, from foreground sessions
TOTAL_TIMEOUTS_FG	NUMBER		Total number of timeouts for the event, from foreground sessions
TIME_WAITED_MICRO_FG	NUMBER		Amount of time waited for the event (in microseconds), from foreground sessions
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **See Also**
>
> "V$SYSTEM_EVENT"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.77 DBA_HIST_TABLESPACE

DBA_HIST_TABLESPACE displays tablespace information contained in the Workload Repository.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
TS#	NUMBER	NOT NULL	Tablespace number
TSNAME	VARCHAR2(30)	NOT NULL	Tablespace name
CONTENTS	VARCHAR2(30)		Tablespace contents: • UNDO • PERMANENT • TEMPORARY
SEGMENT_SPACE_MANAGEMENT	VARCHAR2(30)		Indicates whether the free and used space in the tablespace is managed using free lists (MANUAL) or bitmaps (AUTO)
EXTENT_MANAGEMENT	VARCHAR2(30)		Indicates whether the extents in the tablespace are dictionary managed (DICTIONARY) or locally managed (LOCAL)
BLOCK_SIZE	NUMBER		Block size of the tablespace
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.78 DBA_HIST_TABLESPACE_STAT

DBA_HIST_TABLESPACE_STAT displays tablespace information from the control file.

This view contains snapshots of V$TABLESPACE and DBA_TABLESPACES.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
TS#	NUMBER	NOT NULL	Tablespace number
TSNAME	VARCHAR2(30)		Tablespace name
CONTENTS	VARCHAR2(9)		Tablespace contents: • PERMANENT • TEMPORARY

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(9)		Tablespace status: • ONLINE • OFFLINE • READ ONLY
SEGMENT_SPACE_MANAGEMENT	VARCHAR2(6)		Indicates whether the free and used segment space in the tablespace is managed using free lists (MANUAL) or bitmaps (AUTO)
EXTENT_MANAGEMENT	VARCHAR2(10)		Indicates whether the extents in the tablespace are dictionary managed (DICTIONARY) or locally managed (LOCAL)
IS_BACKUP	VARCHAR2(5)		Indicates whether the tablespace is part of a backup
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$TABLESPACE"
- "DBA_TABLESPACES"

5.79 DBA_HIST_TBSPC_SPACE_USAGE

DBA_HIST_TBSPC_SPACE_USAGE displays historical tablespace usage statistics.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER		Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
TABLESPACE_ID	NUMBER		Tablespace ID
TABLESPACE_SIZE	NUMBER		Tablespace size (in database blocks)
TABLESPACE_MAXSIZE	NUMBER		Maximum size of the tablespace (in database blocks)
TABLESPACE_USEDSIZE	NUMBER		Used size of the tablespace (in database blocks)
RTIME	VARCHAR2(25)		Runtime
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.80 DBA_HIST_TEMPFILE

DBA_HIST_TEMPFILE displays a history of the temp file information from the control file. This view contains snapshots of V$TEMPFILE.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
FILE#	NUMBER	NOT NULL	File identification number
CREATION_CHANGE#	NUMBER	NOT NULL	Change number at which the temp file was created
FILENAME	VARCHAR2(513)	NOT NULL	Name of the temp file
TS#	NUMBER	NOT NULL	Tablespace number
TSNAME	VARCHAR2(30)		Name of the tablespace
BLOCK_SIZE	NUMBER		Block size of the temp file
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$TEMPFILE"

5.81 DBA_HIST_TEMPSTATXS

DBA_HIST_TEMPSTATXS displays information about temporary file read/write statistics.

This view contains snapshots of V$TEMPSTAT.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
FILE#	NUMBER	NOT NULL	File identification number
CREATION_CHANGE#	NUMBER	NOT NULL	Change number at which the temp file was created
FILENAME	VARCHAR2(513)	NOT NULL	Name of the temp file
TS#	NUMBER	NOT NULL	Tablespace number
TSNAME	VARCHAR2(30)		Name of the tablespace
BLOCK_SIZE	NUMBER		Block size of the temp file
PHYRDS	NUMBER		Number of physical reads done
PHYWRTS	NUMBER		Number of times DBWR is required to write
SINGLEBLKRDS	NUMBER		Number of single block reads
READTIM	NUMBER		Time (in hundredths of a second) spent doing reads if the TIMED_STATISTICS parameter is true; 0 if false
WRITETIM	NUMBER		Time (in hundredths of a second) spent doing writes if the TIMED_STATISTICS parameter is true; 0 if false
SINGLEBLKRDTIM	NUMBER		Cumulative single block read time (in hundredths of a second)
PHYBLKRD	NUMBER		Number of physical blocks read
PHYBLKWRT	NUMBER		Number of blocks written to disk, which may be the same as PHYWRTS if all writes are single blocks
WAIT_COUNT	NUMBER		Shows the number of waits at the file level for contended buffers. This value includes the individual wait events that are included in the buffer busy waits wait event. **See Also:** "buffer busy waits"
TIME	NUMBER		Time spent waiting for the wait events in the WAIT_COUNT column
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$TEMPSTAT"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.82 DBA_HIST_THREAD

DBA_HIST_THREAD displays historical thread information from the control file.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
THREAD#	NUMBER	NOT NULL	Thread number
THREAD_INSTANCE_NUMBER	NUMBER		Instance number of the thread
STATUS	VARCHAR2(6)		Thread status (OPEN) or (CLOSED)
OPEN_TIME	DATE		Last time the thread was opened
CURRENT_GROUP#	NUMBER		Current log group
SEQUENCE#	NUMBER		Sequence number of the current log
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.83 DBA_HIST_TOPLEVELCALL_NAME

DBA_HIST_TOPLEVELCALL_NAME displays the mapping between Oracle top level calls and names.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
TOP_LEVEL_CALL#	NUMBER	NOT NULL	Oracle top level call number
TOP_LEVEL_CALL_NAME	VARCHAR2(64)		Oracle top level call name
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.84 DBA_HIST_UNDOSTAT

DBA_HIST_UNDOSTAT displays the history of histograms of statistical data to show how well the system is working.

The available statistics include undo space consumption, transaction concurrency, and length of queries executed in the instance. This view contains snapshots of V$UNDOSTAT.

Column	Datatype	NULL	Description
BEGIN_TIME	DATE	NOT NULL	Identifies the beginning of the time interval
END_TIME	DATE	NOT NULL	Identifies the end of the time interval
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
UNDOTSN	NUMBER	NOT NULL	Represents the last active undo tablespace in the duration of time. The tablespace ID of the active undo tablespace is returned in this column. If more than one undo tablespace was active in that period, the active undo tablespace that was active at the end of the period is reported.
UNDOBLKS	NUMBER		Represents the total number of undo blocks consumed. You can use this column to obtain the consumption rate of undo blocks, and thereby estimate the size of the undo tablespace needed to handle the workload on your system.
TXNCOUNT	NUMBER		Identifies the total number of transactions executed within the period
MAXQUERYLEN	NUMBER		Identifies the length of the longest query (in number of seconds) executed in the instance during the period. You can use this statistic to estimate the proper setting of the UNDO_RETENTION initialization parameter. The length of a query is measured from the cursor open time to the last fetch/execute time of the cursor. Only the length of those cursors that have been fetched/ executed during the period are reflected in the view.
MAXQUERYSQLID	VARCHAR2(13)		SQL identifier of the longest running SQL statement in the period
MAXCONCURRENCY	NUMBER		Identifies the highest number of transactions executed concurrently within the period
UNXPSTEALCNT	NUMBER		Number of attempts to obtain undo space by stealing unexpired extents from other transactions
UNXPBLKRELCNT	NUMBER		Number of unexpired blocks removed from certain undo segments so they can be used by other transactions
UNXPBLKREUCNT	NUMBER		Number of unexpired undo blocks reused by transactions
EXPSTEALCNT	NUMBER		Number of attempts to steal expired undo blocks from other undo segments
EXPBLKRELCNT	NUMBER		Number of expired undo blocks stolen from other undo segments

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
EXPBLKREUCNT	NUMBER		Number of expired undo blocks reused within the same undo segments
SSOLDERRCNT	NUMBER		Identifies the number of times the error ORA-01555 occurred. You can use this statistic to decide whether the UNDO_RETENTION initialization parameter is set properly given the size of the undo tablespace. Increasing the value of UNDO_RETENTION can reduce the occurrence of this error.
NOSPACEERRCNT	NUMBER		Identifies the number of times space was requested in the undo tablespace and there was no free space available. That is, all of the space in the undo tablespace was in use by active transactions. The corrective action is to add more space to the undo tablespace.
ACTIVEBLKS	NUMBER		Total number of blocks in the active extents of the undo tablespace for the instance at the sampled time in the period
UNEXPIREDBLKS	NUMBER		Total number of blocks in the unexpired extents of the undo tablespace for the instance at the sampled time in the period
EXPIREDBLKS	NUMBER		Total number of blocks in the expired extents of the undo tablespace for the instance at the sampled time in the period
TUNED_UNDORETENTION	NUMBER		System tuned value indicating the period for which undo is being retained The value of this column is not meaningful on an Oracle Active Data Guard standby database instance, because the system does not tune this value on such instances.
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$UNDOSTAT"

5.85 DBA_HIST_WAITCLASSMET_HISTORY

DBA_HIST_WAITCLASSMET_HISTORY displays the history of the wait event class metric data kept by the Workload Repository.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID of the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number of the snapshot
WAIT_CLASS_ID	NUMBER	NOT NULL	Identifier of the class of the wait event
WAIT_CLASS	VARCHAR2(64)		Name of the class of the wait event
BEGIN_TIME	DATE	NOT NULL	Begin time of the interval
END_TIME	DATE	NOT NULL	End time of the interval
INTSIZE	NUMBER	NOT NULL	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	NOT NULL	Metric group ID
AVERAGE_WAITER_COUNT	NUMBER	NOT NULL	Average waiter count
DBTIME_IN_WAIT	NUMBER	NOT NULL	Percent of database time spent in the wait
TIME_WAITED	NUMBER	NOT NULL	Time waited during the interval (in microseconds)
WAIT_COUNT	NUMBER	NOT NULL	Number of times waited
TIME_WAITED_FG	NUMBER		Time waited (in hundredths of a second), from foreground sessions
WAIT_COUNT_FG	NUMBER		Number of times waited, from foreground sessions
CON_DBID	NUMBER		The database ID of the PDB for the sampled session
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.86 DBA_HIST_WAITSTAT

DBA_HIST_WAITSTAT displays historical block contention statistics. This view contains snapshots of V$WAITSTAT.

Column	Datatype	NULL	Description
SNAP_ID	NUMBER	NOT NULL	Unique snapshot ID
DBID	NUMBER	NOT NULL	Database ID for the snapshot
INSTANCE_NUMBER	NUMBER	NOT NULL	Instance number for the snapshot
CLASS	VARCHAR2(18)	NOT NULL	Class of the block
WAIT_COUNT	NUMBER		Number of waits by the OPERATION for this CLASS of block
TIME	NUMBER		Sum of all wait times for all the waits by the OPERATION for this CLASS of block
CON_DBID	NUMBER		The database ID of the PDB for the sampled session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$WAITSTAT"

5.87 DBA_HIST_WR_CONTROL

DBA_HIST_WR_CONTROL displays the control information for the Workload Repository.

Column	Datatype	NULL	Description
DBID	NUMBER	NOT NULL	Database ID
SNAP_INTERVAL	INTERVAL DAY(5) TO SECOND(1)	NOT NULL	Snapshot interval; how often to automatically take snapshots
RETENTION	INTERVAL DAY(5) TO SECOND(1)	NOT NULL	Retention setting for the snapshots; amount of time to keep the snapshots
TOPNSQL	VARCHAR2(10)		The number of Top SQL flushed for each SQL criteria (elapsed time, CPU time, parse calls, sharable memory, version count)
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
SRC_DBID	NUMBER		Database ID of the non-CDB, CDB, or PDB where the AWR snapshot data was collected
SRC_DBNAME	VARCHAR2(128)		Database name of the non-CDB, CDB, or PDB where the AWR snapshot data was collected

5.88 DBA_HIST_WR_SETTINGS

DBA_HIST_WR_SETTINGS displays the settings and metadata for the Workload Repository.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LOCAL_AWRDBID	NUMBER	NOT NULL	Database ID of the local database
VIEW_LOCATION	VARCHAR2(8)		Data source of the DBA_HIST dictionary views. Possible values include: • AWR_PDB: Views display AWR data stored in the PDB. • AWR_ROOT: Views display AWR data stored in the root.
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.89 DBA_HISTOGRAMS

DBA_HISTOGRAMS is a synonym for DBA_TAB_HISTOGRAMS.

See Also

"DBA_TAB_HISTOGRAMS"

5.90 DBA_HIVE_COLUMNS

DBA_HIVE_COLUMNS describes all Hive columns in a Hive metastore. Its columns are the same as those in ALL_HIVE_COLUMNS.

See Also

"ALL_HIVE_COLUMNS"

5.91 DBA_HIVE_DATABASES

DBA_HIVE_DATABASES describes all the Hive schemas in a Hadoop cluster. Its columns are the same as those in ALL_HIVE_DATABASES.

See Also

"ALL_HIVE_DATABASES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.92 DBA_HIVE_PART_KEY_COLUMNS

DBA_HIVE_PART_KEY_COLUMNS provides information about all Hive table partition columns in the database. Its columns are the same as those in ALL_HIVE_PART_KEY_COLUMNS.

See Also

"ALL_HIVE_PART_KEY_COLUMNS"

5.93 DBA_HIVE_TAB_PARTITIONS

DBA_HIVE_TAB_PARTITIONS provides information about all Hive table partitions in the database. Its columns are the same as those in ALL_HIVE_TAB_PARTITIONS.

See Also

"ALL_HIVE_TAB_PARTITIONS"

5.94 DBA_HIVE_TABLES

DBA_HIVE_TABLES provides information about all the Hive tables in the Hive metastore. Its columns are the same as those in ALL_HIVE_TABLES.

See Also

"ALL_HIVE_TABLES"

5.95 DBA_HOST_ACES

DBA_HOST_ACES describes access control entries defined in host access control lists.

Related View

USER_HOST_ACES describes the status of access control entries for the current user to access network hosts through PL/SQL network utility packages. This view does not display the ACE_ORDER, START_DATE, END_DATE, GRANT_TYPE, INVERTED_PRINCIPAL, PRINCIPAL, or PRINCIPAL_TYPE columns.

Column	Datatype	NULL	Description
HOST	VARCHAR2(1000)	NOT NULL	Network host
LOWER_PORT	NUMBER(5)		Lower bound of the port range
UPPER_PORT	NUMBER(5)		Upper bound of the port range
ACE_ORDER	NUMBER	NOT NULL	Order number of the access control entry

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
START_DATE	TIMESTAMP(6)		Start date of the access control entry
END_DATE	TIMESTAMP(6)		End date of the access control entry
GRANT_TYPE	VARCHAR2(5)		Indicates whether the access control entry grants or denies the privilege
INVERTED_PRINCIPAL	VARCHAR2(3)		Indicates whether the principal is inverted or not
PRINCIPAL	VARCHAR2(128)		Principal the privilege is applied to
PRINCIPAL_TYPE	VARCHAR2(16)		Type of the principal
PRIVILEGE	VARCHAR2(128)		Privilege

See Also

"USER_HOST_ACES"

5.96 DBA_HOST_ACLS

DBA_HOST_ACLS describes access control lists assigned to restrict access to network hosts through PL/SQL network utility packages.

Column	Datatype	NULL	Description
HOST	VARCHAR2(1000)	NOT NULL	Network host
LOWER_PORT	NUMBER(5)		Lower bound of the port range
UPPER_PORT	NUMBER(5)		Upper bound of the port range
ACL	VARCHAR2(4000)		The name of the access control list
ACLID	RAW(8)		The object ID of the access control list
ACL_OWNER	VARCHAR2(128)		The owner of the access control list

5.97 DBA_IDENTIFIERS

DBA_IDENTIFIERS displays information about the identifiers in all stored objects in the database. Its columns are the same as those in ALL_IDENTIFIERS.

See Also

"ALL_IDENTIFIERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.98 DBA_ILMDATAMOVEMENTPOLICIES

DBA_ILMDATAMOVEMENTPOLICIES contains information specific to data movement-related attributes of an Automatic Data Optimization policy in a database.

Related View

USER_ILMDATAMOVEMENTPOLICIES contains information specific to data movement-related attributes of an Automatic Data Optimization policy for a user.

> **Note**
>
> Automatic Data Optimization is supported in Oracle Database 12*c* Release 2 multitenant environments.

Column	Datatype	NULL	Description
POLICY_NAME	VARCHAR2(128)	NOT NULL	The Automatic Data Optimization policy name is autogenerated
ACTION_TYPE	VARCHAR2(11)		Type of data movement action performed by the Automatic Data Optimization policy: • ANNOTATE • COMPRESSION • EVICT • STORAGE
SCOPE	VARCHAR2(7)		Identifies the scope of the Automatic Data Optimization policy: • ROW • GROUP • SEGMENT
COMPRESSION_LEVEL	VARCHAR2(30)		Compression level to move selected rows or the entire segment to, for a compression Automatic Data Optimization policy
TIER_TABLESPACE	VARCHAR2(128)		Tablespace to move the object to, for a storage Automatic Data Optimization policy
TIER_STATUS	VARCHAR2(9)		This column is only valid for storage tiering policies, and indicates whether the storage tiering policy was specified with a READ ONLY clause. This column takes READ ONLY as a potential value. In all other cases, it is blank. In other words, this column indicates whether the tablespace the object is being moved to will be made READ ONLY after the movement.
CONDITION_TYPE	VARCHAR2(22)		Column on which the condition for the policy is based. Possible values: • LAST ACCESS TIME • LAST MODIFICATION TIME • CREATION TIME • USER DEFINED • LOW ACCESS
CONDITION_DAYS	NUMBER		Number of days in the condition for the policy

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CUSTOM_FUNCTION	VARCHAR2(128)		Optional function that evaluates the precondition on the policy
POLICY_SUBTYPE	VARCHAR2(10)		Storage tier on which the policy is specified
ACTION_CLAUSE	CLOB		Text of the action executed by the policy
TIER_TO	VARCHAR2(10)		Storage tier where the data is placed after the policy is executed

See Also

"USER_ILMDATAMOVEMENTPOLICIES"

5.99 DBA_ILMEVALUATIONDETAILS

DBA_ILMEVALUATIONDETAILS displays details on evaluation of Automatic Data Optimization policies considered for Automatic Data Optimization tasks.

It also shows the job name that executes the policy, in case the policy was selected for execution. If the policy was not selected for execution, this view provides a reason.

Related View

USER_ILMEVALUATIONDETAILS displays details on evaluation of Automatic Data Optimization policies considered for Automatic Data Optimization tasks for a user. It also shows the job name that executes the policy, in case the policy was selected for execution. If the policy was not selected for execution, this view provides a reason.

Note

Automatic Data Optimization is supported in Oracle Database 12*c* Release 2 multitenant environments.

Column	Datatype	NULL	Description
TASK_ID	NUMBER		Number that uniquely identifies a specific Automatic Data Optimization task
POLICY_NAME	VARCHAR2(128)	NOT NULL	Name of the Automatic Data Optimization policy
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the object associated with the Automatic Data Optimization policy
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object associated with the Automatic Data Optimization policy
SUBOBJECT_NAME	VARCHAR2(128)		Name of the subobject associated with the Automatic Data Optimization policy
OBJECT_TYPE	VARCHAR2(18)		Object type. Valid values include TABLE, TABLE PARTITION, and TABLE SUBPARTITION.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SELECTED_FOR_EXECUTION	VARCHAR2(42)		Indicates whether the policy has been selected for execution on the object. If not, the reason for not being selected is listed. Possible values: • POLICY DISABLED • SELECTED FOR EXECUTION • POLICY OVERRULED • INHERITED POLICY OVERRULED • PRECONDITION NOT SATISFIED • JOB ALREADY EXISTS • NO OPERATION SINCE LAST ILM ACTION • TABLE HAS MATERIALIZED VIEW • TARGET COMPRESSION NOT HIGHER THAN CURRENT • STATISTICS NOT AVAILABLE The value SELECTED FOR EXECUTION means a job was created for this policy on the object. The other values state the reason why the policy on the object was selected for execution.
JOB_NAME	VARCHAR2(128)		Name of the job in the case where the policy is selected for execution on this object
COMMENTS	VARCHAR2(4000)		Reserved for future use

See Also

"USER_ILMEVALUATIONDETAILS"

5.100 DBA_ILMOBJECTS

DBA_ILMOBJECTS displays all the Automatic Data Optimization policies and objects in the database.

Many objects inherit policies via their parent objects or because they were created in a particular tablespace. This view provides a mapping between the policies and objects and indicates whether a policy is inherited by an object or is directly specified on it.

Related View

USER_ILMOBJECTS displays all the Automatic Data Optimization policies and objects for a user.

Note

Automatic Data Optimization is supported in Oracle Database 12*c* Release 2 multitenant environments.

Column	Datatype	NULL	Description
POLICY_NAME	VARCHAR2(128)	NOT NULL	Policy name is auto-generated

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the object associated with the Automatic Data Optimization policy
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object associated with the Automatic Data Optimization policy
SUBOBJECT_NAME	VARCHAR2(128)		Name of the subobject associated with the Automatic Data Optimization policy
OBJECT_TYPE	VARCHAR2(18)		Object type. Valid values include: • INDEX • INDEX PARTITION • LOB • LOB PARTITION • TABLE • TABLE PARTITION • TABLE SUBPARTITION Direct policies on Index, Index Partition, LOB, and LOB Partition are not supported in Oracle Database 12*c*.
INHERITED_FROM	VARCHAR2(20)		Indicates if the policy is inherited by the object or subobject, or directly specified on the object or subobject. If the policy is inherited, the level from which the policy is inherited (TABLE, TABLE PARTITION, TABLESPACE) is identified.
TBS_INHERITED_FROM	VARCHAR2(30)		The tablespace name, if the policy is inherited from a tablespace
ENABLED	VARCHAR2(7)		Indicates if the Automatic Data Optimization policy is enabled for the object (YES or NO)
DELETED	VARCHAR2(7)		Possible values: • YES - Indicates that the policy is deleted for any objects that may be added in the future, but is active for those objects that are currently associated with that policy • NO - Indicates that the policy is active

See Also

"USER_ILMOBJECTS"

5.101 DBA_ILMPARAMETERS

DBA_ILMPARAMETERS can be queried to provide information on the Automatic Data Optimization parameters in the database and their values.

Note

Automatic Data Optimization is supported in Oracle Database 12*c* Release 2 multitenant environments.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
NAME	VARCHAR2(128)		Name of the Automatic Data Optimization environment parameter. The value is one of the constants defined in the DBMS_ILM_ADMIN package.
VALUE	NUMBER		Value of the parameter

See Also

- *Oracle Database PL/SQL Packages and Types Reference* for more information about the API interface for implementing Automatic Data Optimization strategies
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_ILM_ADMIN package

5.102 DBA_ILMPOLICIES

DBA_ILMPOLICIES displays details about Automatic Data Optimization policies in the database.

The view contains common details relevant to all types of Automatic Data Optimization policies, not just details relevant to the data movement-related Automatic Data Optimization policies.

Related View

USER_ILMPOLICIES displays details about Automatic Data Optimization policies owned by the user.

Note

Automatic Data Optimization is supported in Oracle Database 12*c* Release 2 multitenant environments.

Column	Datatype	NULL	Description
POLICY_NAME	VARCHAR2(128)		The name of the Automatic Data Optimization policy is auto-generated
POLICY_TYPE	VARCHAR2(13)		Type of the policy. Valid values include DATAMOVEMENT.
TABLESPACE	VARCHAR2(30)		Tablespace name, in the case of a tablespace-level policy
ENABLED	VARCHAR2(6)		Indicates whether the policy is enabled or not
DELETED	VARCHAR2(7)		Possible values: • YES - Indicates that the policy associated has been deleted (but shall remain active for this object) • NO - Indicates that the policy is active

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> Automatic Data Optimization policies cannot be set on tables with object types or materialized views.

> **See Also**
>
> "USER_ILMPOLICIES"

5.103 DBA_ILMRESULTS

DBA_ILMRESULTS displays information on data movement-related Automatic Data Optimization jobs in the database.

Related View

USER_ILMRESULTS displays information on data movement-related Automatic Data Optimization jobs for tasks created by the user.

> **Note**
>
> Automatic Data Optimization is supported in Oracle Database 12*c* Release 2 multitenant environments.

Column	Datatype	NULL	Description
TASK_ID	NUMBER		Number that uniquely identifies a specific Automatic Data Optimization task
JOB_NAME	VARCHAR2(128)		Name of the Automatic Data Optimization job
JOB_STATE	VARCHAR2(35)		State of the job. Possible values: • JOB CREATED • COMPLETED SUCCESSFULLY • FAILED • STOPPED • JOB CREATION FAILED • DEPENDENT OBJECTS BEING REBUILT • FAILED TO REBUILD DEPENDENT OBJECTS
START_TIME	TIMESTAMP(6)		Start time of the Automatic Data Optimization job
COMPLETION_TIME	TIMESTAMP(6)		Completion time of the Automatic Data Optimization job
COMMENTS	VARCHAR2(4000)		Additional information in cases where the Automatic Data Optimization job execution fails
STATISTICS	CLOB		Job specific statistics, such as space saved via compression. This column is in the form of comma separated name / value pairs, with each pair representing a particular statistic name and value.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "USER_ILMRESULTS"

5.104 DBA_ILMTASKS

DBA_ILMTASKS displays information on Automatic Data Optimization execution.

Related View

USER_ILMTASKS displays information on Automatic Data Optimization tasks created by a user.

> **Note**
>
> Automatic Data Optimization is supported in Oracle Database 12*c* Release 2 multitenant environments.

Column	Datatype	NULL	Description
TASK_ID	NUMBER		Number that uniquely identifies a specific Automatic Data Optimization task
TASK_OWNER	VARCHAR2(128)	NOT NULL	User who initiates the task
STATE	VARCHAR2(9)		Possible values: • INACTIVE: Indicates that the task was created for previewing • ACTIVE: Indicates that jobs have been created for the qualifying policies in the task • COMPLETE: Indicates that the task has completed
CREATION_TIME	TIMESTAMP(6)		The time that the task was created
START_TIME	TIMESTAMP(6)		Start time of a specific task
COMPLETION_TIME	TIMESTAMP(6)		Completion time of a specific task

> **See Also**
>
> "USER_ILMTASKS"

5.105 DBA_IM_EXPRESSIONS

DBA_IM_EXPRESSIONS provides information about the list of expressions (SYS_IME virtual columns) that are currently enabled for in-memory storage.

Typically, you can query this view after invoking the DBMS_INMEMORY_ADMIN.IME_CAPTURE_EXPRESSIONS PL/SQL procedure to see the list of hot expressions added to different tables across the database.

Based on this view, you can:

Copyright © 2002, 2026, Oracle and/or its affiliates.

- Populate expressions on a particular table immediately
- Drop certain expressions that are marked for in-memory but not desired by users

Related View

`USER_IM_EXPRESSIONS` provides information about the list of expressions (SYS_IME virtual columns) that are currently enabled for in-memory storage in schemas owned by the current user. This view does not display the `OWNER` column.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(129)`		Table owner
`TABLE_NAME`	`VARCHAR2(129)`		Table name
`OBJECT_NUMBER`	`NUMBER`		Object number of the table
`COLUMN_NAME`	`VARCHAR2(128)`	`NOT NULL`	Column name of the expression added to the table (with SYS_IME prefix)
`SQL_EXPRESSION`	`LONG`		SQL representation of the expression

See Also

"USER_IM_EXPRESSIONS"

5.106 DBA_IMMUTABLE_TABLES

`DBA_IMMUTABLE_TABLES` describes all immutable tables in the database. Its columns are the same as those in `ALL_IMMUTABLE_TABLES`.

Note

This view is available starting with Oracle Database 19c, Release Update 19.11.

See Also

"ALL_IMMUTABLE_TABLES"

5.107 DBA_IND_COLUMNS

`DBA_IND_COLUMNS` describes the columns of all the indexes on all tables and clusters in the database. Its columns are the same as those in `ALL_IND_COLUMNS`.

See Also

"ALL_IND_COLUMNS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.108 DBA_IND_EXPRESSIONS

DBA_IND_EXPRESSIONS lists expressions of function-based indexes on all tables and clusters in the database. Its columns are the same as those in ALL_IND_EXPRESSIONS.

See Also

"ALL_IND_EXPRESSIONS"

5.109 DBA_IND_PARTITIONS

DBA_IND_PARTITIONS displays, for each index partition in the database, the partition-level partitioning information, the storage parameters for the partition, and various partition statistics generated by the DBMS_STATS package. Its columns are the same as those in ALL_IND_PARTITIONS.

See Also

"ALL_IND_PARTITIONS"

5.110 DBA_IND_PENDING_STATS

DBA_IND_PENDING_STATS describes pending statistics for all tables, partitions, and subpartitions in the database collected using the DBMS_STATS package. Its columns are the same as those in ALL_IND_PENDING_STATS.

See Also

"ALL_IND_PENDING_STATS"

5.111 DBA_IND_STATISTICS

DBA_IND_STATISTICS displays optimizer statistics for all indexes in the database collected using the DBMS_STATS package. Its columns are the same as those in ALL_IND_STATISTICS.

See Also

"ALL_IND_STATISTICS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.112 DBA_IND_SUBPARTITIONS

DBA_IND_SUBPARTITIONS displays, for each index subpartition in the database, the subpartition-level partitioning information, the storage parameters for the subpartition, and various subpartition statistics generated by the DBMS_STATS package. Its columns are the same as those in ALL_IND_SUBPARTITIONS.

See Also

"ALL_IND_SUBPARTITIONS"

5.113 DBA_INDEX_USAGE

DBA_INDEX_USAGE displays cumulative statistics for each index.

Column	Datatype	NULL	Description
OBJECT_ID	NUMBER	NOT NULL	Object ID for the index
NAME	VARCHAR2(128)	NOT NULL	Index name
OWNER	VARCHAR2(128)	NOT NULL	Index owner
TOTAL_ACCESS_COUNT	NUMBER		Total number of times the index has been accessed
TOTAL_EXEC_COUNT	NUMBER		Total executions the index has participated in
TOTAL_ROWS_RETURNED	NUMBER		Total rows returned by the index.
			Index usage is categorized into buckets of different ranges. Each bucket has a range of values for access count and rows returned. An entry is placed into a bucket if the rows returned or access counts falls within the range of that bucket.
BUCKET_0_ACCESS_COUNT	NUMBER		The index has not been accessed
BUCKET_1_ACCESS_COUNT	NUMBER		The index has been accessed once
BUCKET_2_10_ACCESS_COUNT	NUMBER		The index has been accessed between 2 and 10 times
BUCKET_2_10_ROWS_RETURNED	NUMBER		The index has returned between 2 and 10 rows
BUCKET_11_100_ACCESS_COUNT	NUMBER		The index has been accessed between 11 and 100 times
BUCKET_11_100_ROWS_RETURNED	NUMBER		The index has returned between 11 and 100 rows
BUCKET_101_1000_ACCESS_COUNT	NUMBER		The index has been accessed between 101 and 1000 times
BUCKET_101_1000_ROWS_RETURNED	NUMBER		The index has returned between 101 and 1000 rows
BUCKET_1000_PLUS_ACCESS_COUNT	NUMBER		The index has been accessed more than 1000 times
BUCKET_1000_PLUS_ROWS_RETURNED	NUMBER		The index has returned more than 1000 rows
LAST_USED	DATE		Time that the index was last used

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "V$INDEX_USAGE_INFO"

5.114 DBA_INDEXES

DBA_INDEXES describes all indexes in the database. To gather statistics for this view, use the DBMS_STATS package. This view supports parallel partitioned index scans. Its columns (except for TABLE_TYPE) are the same as those in ALL_INDEXES.

> **See Also**
>
> "ALL_INDEXES"

5.115 DBA_INDEXTYPE_ARRAYTYPES

DBA_INDEXTYPE_ARRAYTYPES displays information about the array types specified by all indextypes in the database. Its columns are the same as those in ALL_INDEXTYPE_ARRAYTYPES.

> **See Also**
>
> "ALL_INDEXTYPE_ARRAYTYPES"

5.116 DBA_INDEXTYPE_COMMENTS

DBA_INDEXTYPE_COMMENTS displays comments for all user-defined indextypes in the database. Its columns are the same as those in ALL_INDEXTYPE_COMMENTS.

> **See Also**
>
> "ALL_INDEXTYPE_COMMENTS"

5.117 DBA_INDEXTYPE_OPERATORS

DBA_INDEXTYPE_OPERATORS lists all the operators supported by indextypes in the database. Its columns are the same as those in ALL_INDEXTYPE_OPERATORS.

> **See Also**
>
> "ALL_INDEXTYPE_OPERATORS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.118 DBA_INDEXTYPES

DBA_INDEXTYPES describes all indextypes in the database. Its columns are the same as those in ALL_INDEXTYPES.

> **See Also**
>
> "ALL_INDEXTYPES"

5.119 DBA_INMEMORY_AIMTASKDETAILS

DBA_INMEMORY_AIMTASKDETAILS displays details for an Automatic In-Memory management task.

Column	Datatype	NULL	Description
TASK_ID	NUMBER		Number that uniquely identifies a specific automatic IM column store management task
OBJECT_OWNER	VARCHAR2(128)		Owner of the object subject to automatic IM column store management task action
OBJECT_NAME	VARCHAR2(128)		Name of the object subject to automatic IM column store management task action
SUBOBJECT_NAME	VARCHAR2(128)		Name of the subobject subject to automatic IM column store management task action
ACTION	VARCHAR2(16)		Action taken on the object
STATE	VARCHAR2(10)		Status of the action on the object

> **See Also**
>
> *Oracle Database In-Memory Guide* for more information about configuring the Automatic In-Memory feature

5.120 DBA_INMEMORY_AIMTASKS

DBA_INMEMORY_AIMTASKS displays information about Automatic In-Memory management tasks.

Column	Datatype	NULL	Description
TASK_ID	NUMBER		Number that uniquely identifies a specific automatic IM column store management task
CREATION_TIME	TIMESTAMP(6)		Creation time of the task
STATE	VARCHAR2(7)		State of the task

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database In-Memory Guide* for more information about configuring the Automatic In-Memory feature

5.121 DBA_INTERNAL_TRIGGERS

`DBA_INTERNAL_TRIGGERS` describes internal triggers on all tables in the database. Its columns are the same as those in `ALL_INTERNAL_TRIGGERS`.

> **See Also**
>
> "ALL_INTERNAL_TRIGGERS"

5.122 DBA_INVALID_OBJECTS

`DBA_INVALID_OBJECTS` describes all invalid objects in the database. You can use this view to identify invalid objects before and after a database upgrade.

This view eliminates old versions of object types. It only includes the object type it if is the latest version.

Column	Datatype	NULL	Description
`OWNER`	`VARCHAR2(128)`		Owner of the object
`OBJECT_NAME`	`VARCHAR2(128)`		Name of the object
`SUBOBJECT_NAME`	`VARCHAR2(128)`		Name of the subobject (for example, partition)
`OBJECT_ID`	`NUMBER`		Dictionary object number of the object.
`DATA_OBJECT_ID`	`NUMBER`		Dictionary object number of the segment that contains the object. **Note:** `OBJECT_ID` and `DATA_OBJECT_ID` display data dictionary metadata. Do not confuse these numbers with the unique 16-byte object identifier (*object ID*) that Oracle Database assigns to row objects in object tables in the system.
`OBJECT_TYPE`	`VARCHAR2(23)`		Type of the object (such as `TABLE`, `INDEX`). The current version of the type is shown only if it is invalid. In other words, if prior versions of the type are invalid but the most recent version of the type is valid, it will not be in this list.
`CREATED`	`DATE`		Timestamp for the creation of the object
`LAST_DDL_TIME`	`DATE`		Timestamp for the last modification of the object and dependent objects resulting from a DDL statement (including grants and revokes)
`TIMESTAMP`	`VARCHAR2(19)`		Timestamp for the specification of the object (character data)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(7)		Status of the object: • VALID • INVALID • N/A
TEMPORARY	VARCHAR2(1)		Indicates whether the object is temporary (the current session can see only data that it placed in this object itself) (Y) or not (N)
GENERATED	VARCHAR2(1)		Indicates whether the name of this object was system-generated (Y) or not (N)
SECONDARY	VARCHAR2(1)		Indicates whether this is a secondary object created by the ODCIIndexCreate method of the Oracle Data Cartridge (Y) or not (N)
NAMESPACE	NUMBER		Namespace for the object
EDITION_NAME	VARCHAR2(128)		Name of the edition in which the object is actual
SHARING	VARCHAR2(18)		Values: • DATA LINK - If the object is data-linked or a data link to an object in the root • METADATA LINK - If the object is metadata-linked or a metadata link to an object in the root • EXTENDED DATA LINK - If the object is extended-data-linked or an extended data link to an object in the root • NONE - If none of the above applies
EDITIONABLE	VARCHAR2(1)		Values: • Y - For objects marked EDITIONABLE • N - For objects marked NONEDITIONABLE • NULL - For objects whose type is not editionable in the database
ORACLE_MAINTAINED	VARCHAR2(1)		Denotes whether the object was created, and is maintained, by Oracle-supplied scripts (such as catalog.sql or catproc.sql). An object for which this column has the value Y must not be changed in any way except by running an Oracle-supplied script.
APPLICATION	VARCHAR2(1)		Indicates whether the object is an Application common object (Y) or not (N)
DEFAULT_COLLATION	VARCHAR2(100)		Default collation for the object
DUPLICATED	VARCHAR2(1)		Indicates whether this object is duplicated on this shard (Y) or not (N)
SHARDED	VARCHAR2(1)		Indicates whether this object is sharded (Y) or not (N)
CREATED_APPID	NUMBER		ID of the Application that created the object
CREATED_VSNID	NUMBER		ID of the Application Version that created the object
MODIFIED_APPID	NUMBER		ID of the Application that last modified the object
MODIFIED_VSNID	NUMBER		ID of the Application Version that last modified the object

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.123 DBA_JAVA_ARGUMENTS

DBA_JAVA_ARGUMENTS displays argument information about all stored Java classes in the database. Its columns are the same as those in ALL_JAVA_ARGUMENTS.

See Also

"ALL_JAVA_ARGUMENTS"

5.124 DBA_JAVA_CLASSES

DBA_JAVA_CLASSES displays class level information about all stored Java classes in the database. Its columns are the same as those in ALL_JAVA_CLASSES.

See Also

"ALL_JAVA_CLASSES"

5.125 DBA_JAVA_COMPILER_OPTIONS

DBA_JAVA_COMPILER_OPTIONS displays information about all native compiler options in the database. Its columns are the same as those in ALL_JAVA_COMPILER_OPTIONS.

See Also

"ALL_JAVA_COMPILER_OPTIONS"

5.126 DBA_JAVA_DERIVATIONS

DBA_JAVA_DERIVATIONS displays mapping information about Java source objects and their derived Java class objects and Java resource objects for all Java classes in the database. Its columns are the same as those in ALL_JAVA_DERIVATIONS.

See Also

"ALL_JAVA_DERIVATIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.127 DBA_JAVA_FIELDS

DBA_JAVA_FIELDS displays field information about all stored Java classes in the database. Its columns are the same as those in ALL_JAVA_FIELDS.

See Also

"ALL_JAVA_FIELDS"

5.128 DBA_JAVA_IMPLEMENTS

DBA_JAVA_IMPLEMENTS describes interfaces implemented by all stored Java classes in the database. Its columns are the same as those in ALL_JAVA_IMPLEMENTS.

See Also

"ALL_JAVA_IMPLEMENTS"

5.129 DBA_JAVA_INNERS

DBA_JAVA_INNERS displays information about inner classes referred to by all stored Java classes in the database. Its columns are the same as those in ALL_JAVA_INNERS.

See Also

"ALL_JAVA_INNERS"

5.130 DBA_JAVA_LAYOUTS

DBA_JAVA_LAYOUTS displays class layout information about all stored Java classes in the database. Its columns are the same as those in ALL_JAVA_LAYOUTS.

See Also

"ALL_JAVA_LAYOUTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.131 DBA_JAVA_METHODS

DBA_JAVA_METHODS displays method information about all stored Java classes in the database. Its columns are the same as those in ALL_JAVA_METHODS.

> **See Also**
>
> "ALL_JAVA_METHODS"

5.132 DBA_JAVA_NCOMPS

DBA_JAVA_NCOMPS displays ncomp-related information about all Java classes in the database. Its columns are the same as those in ALL_JAVA_NCOMPS.

> **See Also**
>
> "ALL_JAVA_NCOMPS"

5.133 DBA_JAVA_POLICY

DBA_JAVA_POLICY describes Java security permissions for all users in the database.

Related View

USER_JAVA_POLICY describes Java security permissions for the current user.

Column	Datatype	NULL	Description
KIND	VARCHAR2(8)		Indicates whether the permission is a positive (GRANT) or a limitation (RESTRICT)
GRANTEE	VARCHAR2(128)	NOT NULL	Name of the user, schema, or role to which the permission object is assigned
TYPE_SCHEMA	VARCHAR2(128)	NOT NULL	Schema in which the permission object is loaded
TYPE_NAME	VARCHAR2(4000)		Permission class type, which is designated by a string containing the full class name, such as, java.io.FilePermission
NAME	VARCHAR2(4000)		Target attribute (name) of the permission object. This name is used when defining the permission.
ACTION	VARCHAR2(4000)		Action attribute for this permission. Many permissions expect a null value if no action is appropriate for the permission.
ENABLED	VARCHAR2(8)		Indicates whether the permission is enabled (ENABLED) or disabled (DISABLED)
SEQ	NUMBER		Sequence number used to identify this row. This number should be supplied when disabling, enabling, or deleting the permission.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "USER_JAVA_POLICY"

5.134 DBA_JAVA_RESOLVERS

DBA_JAVA_RESOLVERS displays information about resolvers of all Java classes in the database. Its columns are the same as those in ALL_JAVA_RESOLVERS.

> **See Also**
>
> "ALL_JAVA_RESOLVERS"

5.135 DBA_JAVA_THROWS

DBA_JAVA_THROWS displays information about exceptions thrown from methods of all Java classes in the database. Its columns are the same as those in ALL_JAVA_THROWSa.

> **See Also**
>
> "ALL_JAVA_THROWS"

5.136 DBA_JOBS

DBA_JOBS describes all jobs in the database.

Related View

USER_JOBS describes the jobs owned by the current user.

Column	Datatype	NULL	Description
JOB	NUMBER	NOT NULL	Identifier of job. Neither import/export nor repeated executions change this value.
LOG_USER	VARCHAR2(128)	NOT NULL	Login user when the job was submitted
PRIV_USER	VARCHAR2(128)	NOT NULL	User whose default privileges apply to this job
SCHEMA_USER	VARCHAR2(128)	NOT NULL	Default schema used to parse the job For example, if the SCHEMA_USER is SCOTT and you submit the procedure HIRE_EMP as a job, the Oracle Database looks for SCOTT.HIRE_EMP
LAST_DATE	DATE		Date on which this job last successfully executed
LAST_SEC	VARCHAR2(8)		Same as LAST_DATE. This is when the last successful execution started.
THIS_DATE	DATE		Date that this job started executing (usually null if not executing)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
THIS_SEC	VARCHAR2(8)		Same as THIS_DATE. This is when the last successful execution started.
NEXT_DATE	DATE	NOT NULL	Date that this job will next be executed
NEXT_SEC	VARCHAR2(8)		Same as NEXT_DATE. The job becomes due for execution at this time.
TOTAL_TIME	NUMBER		Total wall clock time spent by the system on this job (in seconds) when it last executed
BROKEN	VARCHAR2(1)		Y: no attempt is made to run this job N: an attempt is made to run this job
INTERVAL	VARCHAR2(200)	NOT NULL	A date function, evaluated at the start of execution, becomes next NEXT_DATE
FAILURES	NUMBER		Number of times the job has started and failed since its last success
WHAT	VARCHAR2(4000)		Body of the anonymous PL/SQL block that the job executes
NLS_ENV	VARCHAR2(4000)		Session parameters describing the NLS environment of the job
MISC_ENV	RAW(32)		Other session parameters of the session that created the job. The job is run using these parameters.
INSTANCE	NUMBER		ID of the instance that can execute or is executing the job. The default is 0.

See Also

"USER_JOBS"

5.137 DBA_JOBS_RUNNING

DBA_JOBS_RUNNING lists all jobs that are currently running in the instance.

Column	Datatype	NULL	Description
SID	NUMBER		Identifier of process that is executing the job. See "V$LOCK".
JOB	NUMBER		Identifier of job. This job is currently executing.
FAILURES	NUMBER		Number of times this job started and failed since its last success.
LAST_DATE	DATE		Date that this job last successfully executed.
LAST_SEC	VARCHAR2(8)		Same as LAST_DATE. This is when the last successful execution started.
THIS_DATE	DATE		Date that this job started executing.
THIS_SEC	VARCHAR2(8)		Same as THIS_DATE. This is when the last successful execution started.
INSTANCE	NUMBER		Indicates which instance can execute or is executing the job; the default is 0.

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.138 DBA_JOIN_IND_COLUMNS

DBA_JOIN_IND_COLUMNS describes all join conditions in the database. Its columns are the same as those in ALL_JOIN_IND_COLUMNS.

> **See Also**
>
> "ALL_JOIN_IND_COLUMNS"

5.139 DBA_JOINGROUPS

DBA_JOINGROUPS describes join groups in the database. A join group is a user-created object that consists of two or more columns that can be meaningfully joined. The maximum number of columns that can be included in a join group is 255.

In certain queries, join groups enable the database to eliminate the performance overhead of decompressing and hashing column values. Join groups require an In-Memory column store (IM column store).

Related View

USER_JOINGROUPS describes join groups belonging to the user. This view does not display the JOINGROUP_OWNER column.

Column	Datatype	NULL	Description
JOINGROUP_OWNER	VARCHAR2(128)	NOT NULL	Join group owner. This is the user that created the join group.
JOINGROUP_NAME	VARCHAR2(128)	NOT NULL	This is the user specified name of the join group. The join group name is specified when the join group is created as part of the CREATE INMEMORY JOIN GROUP statement.
TABLE_OWNER	VARCHAR2(128)	NOT NULL	Table owner
TABLE_NAME	VARCHAR2(128)	NOT NULL	Table name
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Column name
FLAGS	VARCHAR2(6)		Possible values: • MASTER: Indicates which column in the join group is mastering the global dictionary. A join group is a group of columns sharing a global dictionary; the global dictionary is associated with one column and the other columns share the same dictionary. The column with which the global dictionary is associated is called the mastering column. • NULL: Indicates that the column is not mastering the global dictionary.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
GD_ADDRESS	RAW(8)		The memory address of the global dictionary. Ideally, all the columns in one join group should have the same global dictionary address (that is, they share the same global structure). This might not always be the case (for example, a column might be added to a join group after it was populated into memory - in which case its GD_ADDRESS field will be NULL). In such cases, you should force re-populate the tables that are part of the join group and check the views after the repopulates complete.

See Also

- "USER_JOINGROUPS"
- *Oracle Database In-Memory Guide* for an introduction to join groups
- *Oracle Database SQL Language Reference* for information about creating a join group using the CREATE INMEMORY JOIN GROUP statement

5.140 DBA_JSON_COLUMNS

DBA_JSON_COLUMNS provides information on all JavaScript Object Notation (JSON) columns in the database. Its columns are the same as those in ALL_JSON_COLUMNS.

Each column that has an IS JSON check constraint in an AND condition appears in this view. This view enables a DBA to find all the JSON columns in the database

For example, if a check constraint combines the IS JSON condition with another condition using logical condition OR, then the column is not listed in this view. In this case, it is not certain that the data in the column is JSON data. For example, the following constraint does not ensure that the data in column jcol is JSON data:

```
jcol is json OR length(jcol) < 1000
```

See Also

- "ALL_JSON_COLUMNS"
- *Oracle Database JSON Developer's Guide* for more information about using JSON with Oracle Database

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.141 DBA_JSON_DATAGUIDE_FIELDS

DBA_JSON_DATAGUIDE_FIELDS extracts the path and type information from all the data guides in the database, which are the data guides returned by the DBA_JSON_DATAGUIDE view. Its columns are the same as those in ALL_JSON_DATAGUIDE_FIELDS.

See Also

"ALL_JSON_DATAGUIDE_FIELDS"

5.142 DBA_JSON_DATAGUIDES

DBA_JSON_DATAGUIDES provides information on all JavaScript Object Notation (JSON) columns in the database that have data guide enabled. Its columns are the same as those in ALL_JSON_DATAGUIDES.

See Also

"ALL_JSON_DATAGUIDES"

5.143 DBA_KGLLOCK

DBA_KGLLOCK lists all the locks and pins held on KGL objects (objects in the Kernel Generic Library cache).

Column	Datatype	NULL	Description
kgllkuse	RAW(4)		Address of the user session that holds the lock or pin
kgllkhdl	RAW(4)		Address of the handle for the KGL object
kgllkmod	NUMBER		Current mode of the lock or pin
kgllkreq	NUMBER		Mode in which the lock or pin was requested
kgllktype	VARCHAR2(4)		Whether this is a lock or a pin

5.144 DBA_LIBRARIES

DBA_LIBRARIES describes all libraries in the database. Its columns are the same as those in ALL_LIBRARIES.

See Also

"ALL_LIBRARIES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.145 DBA_LMT_FREE_SPACE

DBA_LMT_FREE_SPACE describes the free extents in all locally managed tablespaces in the database.

Column	Datatype	NULL	Description
TABLESPACE_ID	NUMBER		Identifier number of the tablespace containing the extent
FILE_ID	NUMBER		File identifier number of the file containing the extent
BLOCK_ID	NUMBER		Starting block number of the extent
BLOCKS	NUMBER		Size of the extent (in Oracle blocks)

5.146 DBA_LMT_USED_EXTENTS

DBA_LMT_USED_EXTENTS describes the extents comprising the segments in all locally managed tablespaces in the database.

Column	Datatype	NULL	Description
SEGMENT_FILEID	NUMBER		File number of the segment header of the extent
SEGMENT_BLOCK	NUMBER		Block number of the segment header of the extent
TABLESPACE_ID	NUMBER		Identifier number of the tablespace containing the extent
EXTENT_ID	NUMBER		Extent number in the segment
FILEID	NUMBER		File identifier number of the file containing the extent
BLOCK	NUMBER		Starting block number of the extent
LENGTH	NUMBER		Number of blocks in the extent

5.147 DBA_LOB_PARTITIONS

DBA_LOB_PARTITIONS displays all LOB partitions in the database. Its columns are the same as those in ALL_LOB_PARTITIONS.

See Also

"ALL_LOB_PARTITIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.148 DBA_LOB_SUBPARTITIONS

DBA_LOB_SUBPARTITIONS displays partition-level attributes of all LOB data subpartitions in the database. Its columns are the same as those in ALL_LOB_SUBPARTITIONS.

> **See Also**
>
> "ALL_LOB_SUBPARTITIONS"

5.149 DBA_LOB_TEMPLATES

DBA_LOB_TEMPLATES describes all LOB subpartition templates in the database. Its columns are the same as those in ALL_LOB_TEMPLATES.

> **See Also**
>
> "ALL_LOB_TEMPLATES"

5.150 DBA_LOBS

DBA_LOBS displays the BLOBs and CLOBs contained in all tables in the database. BFILEs are stored outside the database, so they are not described by this view. This view's columns are the same as those in ALL_LOBS.

> **See Also**
>
> "ALL_LOBS"

5.151 DBA_LOCK

DBA_LOCK lists all locks or latches held in the database, and all outstanding requests for a lock or latch.

Column	Datatype	NULL	Description
SESSION_ID	NUMBER		Session holding or acquiring the lock
LOCK_TYPE	VARCHAR2(26)		Lock type **See Also:** For a listing of lock types, see Oracle Enqueue Names
MODE HELD	VARCHAR2(40)		Lock mode
MODE REQUESTED	VARCHAR2(40)		Lock mode requested
LOCK_ID1	VARCHAR2(40)		Type-specific lock identifier, part 1
LOCK_ID2	VARCHAR2(40)		Type-specific lock identifier, part 2

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LAST_CONVERT	NUMBER		The last convert
BLOCKING_OTHERS	VARCHAR2(40)		Whether the lock is currently blocking others
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.152 DBA_LOCK_INTERNAL

DBA_LOCK_INTERNAL displays a row for each lock or latch that is being held, and one row for each outstanding request for a lock or latch.

Column	Datatype	NULL	Description
SESSION_ID	NUMBER		Session holding or acquiring the lock
LOCK_TYPE	VARCHAR2(56)		Lock type **See Also:** For a listing of lock types, see Oracle Enqueue Names
MODE HELD	VARCHAR2(40)		Lock mode
MODE REQUESTED	VARCHAR2(40)		Lock mode requested
LOCK_ID1	VARCHAR2(1130)		Type-specific lock identifier, part 1
LOCK_ID2	VARCHAR2(40)		Type-specific lock identifier, part 2
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.153 DBA_LOCKDOWN_PROFILES

DBA_LOCKDOWN_PROFILES provides information about lockdown profiles.

The PRIVATE_DBAAS, PUBLIC_DBAAS, and SAAS lockdown profiles are empty placeholder profiles for the lockdown profiles of their corresponding deployment type. You can modify and add restrictions to these profiles based on their deployment purpose. For example, if you have a Software as a Service (SAAS) application, you can modify the SAAS lockdown profile and use it. You can also delete and re-create these profiles.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PROFILE_NAME	VARCHAR2(128)	NOT NULL	Name of the lockdown profile
RULE_TYPE	VARCHAR2(128)		Rule type. A lockdown profile is used to restrict operations that can be performed by users connected to a given PDB. It provides the ability to add or remove different types of rules like STATEMENT, FEATURES or OPTIONS which will be restricted in the PDB.
RULE	VARCHAR2(128)		Rule to be enabled or disabled
CLAUSE	VARCHAR2(128)		Clause of the statement
CLAUSE_OPTION	VARCHAR2(128)		Option of the clause
OPTION_VALUE	VARCHAR2(4000)		Value of the option
MIN_VALUE	VARCHAR2(4000)		Minimum value allowed for the option
MAX_VALUE	VARCHAR2(4000)		Maximum value allowed for the option
LIST	VARCHAR2(4000)		List of allowed values for the option
STATUS	VARCHAR2(7)		Status of the lockdown profile: • ENABLE • DISABLE • EMPTY
USERS	VARCHAR2(6)		User type. Possible values: • COMMON • LOCAL • ALL
EXCEPT_USERS[1]	CLOB		For internal use only

[1] This column is available starting with Oracle Database 19c.

See Also

- "PDB_LOCKDOWN"
- *Oracle Multitenant Administrator's Guide* for an introduction to PDB lockdown profiles
- *Oracle Database SQL Language Reference* for more information about creating lockdown profiles
- *Oracle Database SQL Language Reference* for more information about dropping lockdown profiles
- *Oracle Database SQL Language Reference* for more information about altering lockdown profiles

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.154 DBA_LOCKS

DBA_LOCKS is a synonym for DBA_LOCK.

> **See Also**
>
> "DBA_LOCK"

5.155 DBA_LOG_GROUP_COLUMNS

DBA_LOG_GROUP_COLUMNS describes all columns in the database that are specified in log groups. Its columns are the same as those in ALL_LOG_GROUP_COLUMNS.

> **See Also**
>
> "ALL_LOG_GROUP_COLUMNS"

5.156 DBA_LOG_GROUPS

DBA_LOG_GROUPS describes log group definitions on all tables in the database. Its columns are the same as those in ALL_LOG_GROUPS.

> **See Also**
>
> "ALL_LOG_GROUPS"

5.157 DBA_LOGMNR_DICTIONARY_BUILDLOG

DBA_LOGMNR_DICTIONARY_BUILDLOG describes all successful LogMiner dictionary builds available for GoldenGate REGISTER EXTRACT.

When this view is queried from a PDB, it returns only LogMiner dictionary builds performed in that PDB.

Column	Datatype	NULL	Description
NAME	VARCHAR2(384)		User-supplied name of the LogMiner dictionary build
DATE_OF_BUILD	VARCHAR2(20)		Date and time at which the LogMiner dictionary build was performed
START_SCN	NUMBER		Begin SCN of the LogMiner dictionary build operation
END_SCN	NUMBER		End SCN of the LogMiner dictionary build operation
BUILD_TYPE	VARCHAR2(20)		Internal string that identifies how the LogMiner dictionary build was initiated

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BUILD_OP	NUMBER		Internal number that identifies how the LogMiner dictionary build was initiated
CONTAINER_ID	NUMBER		ID of the container in which the LogMiner dictionary build was performed (the CDB root or a PDB)
CONTAINER_UID	NUMBER		Unique number that identifies the container in which the LogMiner dictionary build was performed (the CDB root or a PDB)
CONTAINER_NAME	VARCHAR2(384)		Name of the container in which the LogMiner dictionary build was performed (the CDB root or a PDB)
RESETLOGS_CHANGE#	NUMBER		SCN that identifies the redo branch of the LogMiner dictionary build

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.10.

5.158 DBA_LOGMNR_LOG

DBA_LOGMNR_LOG displays all archived logs registered with active LogMiner persistent sessions in the database.

A persistent LogMiner session is created either by starting Data Guard SQL Apply on a logical standby database for the first time or by creating Replication capture.

Column	Datatype	NULL	Description
LOGMNR_SESSION_ID	NUMBER	NOT NULL	Unique identifier of the persistent session
NAME	VARCHAR2(513)		Name of the archived log
DBID	NUMBER	NOT NULL	Database identifier that produced the archived log
RESETLOGS_SCN	NUMBER	NOT NULL	SCN at which resetlogs operation was performed at the source database generating the archived log
RESETLOGS_TIME	NUMBER	NOT NULL	Timestamp at which resetlogs operation was performed at the source database generating the archived log
MODIFIED_TIME	DATE		Time at which the archived log was registered with LogMiner
THREAD#	NUMBER	NOT NULL	Redo thread at the source database that generated the archived log
SEQUENCE#	NUMBER	NOT NULL	Logfile sequence number
FIRST_SCN	NUMBER	NOT NULL	Lowest SCN of the redo record contained in the logfile
NEXT_SCN	NUMBER		Highest possible SCN of the redo record contained in the logfile
FIRST_TIME	DATE		Time of the first redo record contained in the logfile
NEXT_TIME	DATE		Time of the last redo record contained in the logfile

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DICTIONARY_BEGIN	VARCHAR2(3)		Indicates whether the archived log contains the beginning of a LogMiner dictionary (YES) or not (NO)
DICTIONARY_END	VARCHAR2(3)		Indicates whether the archived log contains the end of a LogMiner dictionary (YES) or not (NO)
KEEP	VARCHAR2(3)		Indicates whether the logfile is still required for this LogMiner session (YES) or not (NO)
SUSPECT	VARCHAR2(3)		Indicates whether the archived log content was deemed to be corrupt or the archived log is partially filled (YES) or not (NO)

5.159 DBA_LOGMNR_PURGED_LOG

DBA_LOGMNR_PURGED_LOG displays archived redo log files that have been applied to the logical standby database and can be deleted because they are no longer needed.

Files in this view are refreshed as a result of executing the DBMS_LOGSTDBY.PURGE_SESSION PL/SQL procedure for Oracle Data Guard SQL Apply:

Column	Datatype	NULL	Description
FILE_NAME	VARCHAR2(513)		Fully qualified names of the archived redo log files that are no longer needed by SQL Apply and can be deleted from the operating system

See Also

Oracle Database PL/SQL Packages and Types Reference for more information about the DBMS_LOGSTDBY.PURGE_SESSION procedure

5.160 DBA_LOGMNR_SESSION

DBA_LOGMNR_SESSION displays all active LogMiner persistent sessions in the database.

A persistent LogMiner session is created either by starting Data Guard SQL Apply on a logical standby database for the first time or by creating Replication capture.

Column	Datatype	NULL	Description
ID	NUMBER	NOT NULL	Unique session identifier
NAME	VARCHAR2(128)	NOT NULL	Unique session name
SOURCE_DATABASE	VARCHAR2(128)		Global name of the source database whose archived logs are to be mined in this persistent LogMiner session
SOURCE_DBID	NUMBER		Database ID of the source database
SOURCE_RESETLOGS_SCN	NUMBER		Resetlogs SCN associated with the incarnation of the source database whose archived logs are mined
SOURCE_RESETLOGS_TIME	NUMBER		Resetlogs time associated with the incarnation of the source database whose archived logs are mined

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FIRST_SCN	NUMBER		Only modifications that occurred on or after this SCN can be mined using this persistent session
END_SCN	NUMBER		No modifications that occurred on or after this SCN can be mined using this persistent session
BRANCH_SCN	NUMBER		SCN at which a branch will be taken in terms of the incarnation corresponding to the source database. This implies a point-in-time recovery was performed at the source database at this SCN.
WAIT_FOR_LOG	VARCHAR2(3)		Indicates whether the persistent session waits for RFS to register new archived logs or to fill gaps (YES) or not (NO)
HOT_MINE	VARCHAR2(3)		Indicates whether real-time mining is on (YES) or not (NO)
SAFE_PURGE_SCN	NUMBER		Persistent session can safely be purged up to this SCN
CHECKPOINT_SCN	NUMBER		SCN at which the latest checkpoint is taken by the persistent LogMiner session
PURGE_SCN	NUMBER		The session has been purged up to this SCN

5.161 DBA_LOGSTDBY_EVENTS

DBA_LOGSTDBY_EVENTS displays information about the activity of the logical standby database system.

It can be used to determine the cause of failures that occur when applying redo data to logical standby databases. This view is for logical standby databases only.

Column	Datatype	NULL	Description
EVENT_TIME	DATE		Time when the event was logged
EVENT_TIMESTAMP	TIMESTAMP(6)	NOT NULL	Timestamp when the event was logged
START_SCN	NUMBER		The SCN at which the associated transaction started at the primary database. This SCN refers to the system change number at the primary database.
CURRENT_SCN	NUMBER		SCN associated with the change at the primary database. If a failure occurred, then examine this column to determine which archived log file contains the source of the failure (for example, an unsupported record).
COMMIT_SCN	NUMBER		SCN value on which the change was committed at the primary database
XIDUSN	NUMBER		Transaction ID undo segment number at the primary database of the associated transaction
XIDSLT	NUMBER		Transaction ID slot number at the primary database of the associated transaction
XIDSQN	NUMBER		Transaction ID sequence number at the primary database of the associated transaction
EVENT	CLOB		Statement that was being processed when the failure occurred

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS_CODE	NUMBER		Status (or Oracle error code) belonging to the STATUS message
STATUS	VARCHAR2(2000)		Description of the current activity of the process or the reason why the apply operation stopped
SRC_CON_NAME	VARCHAR2(30)		Identifies the PDB name at the primary database where the transaction was executed
SRC_CON_ID	NUMBER		Contains the PDB ID (the PDB_ID column from the DBA_PDBS view) of the primary database where the associated change was generated.

> **Note**
>
> In a CDB, this view shows data only when queried in the root.

5.162 DBA_LOGSTDBY_HISTORY

DBA_LOGSTDBY_HISTORY displays the history of switchovers and failovers in a Data Guard configuration.

It does this by showing the complete sequence of redo log streams processed or created on the local system, across all role transitions. (After a role transition, a new log stream is started and the log stream sequence number is incremented by the new primary database.). This view is for logical standby databases only.

Column	Datatype	NULL	Description
STREAM_SEQUENCE#	NUMBER		Lists the sequence numbers for all log streams created or applied on the local system. **Note**: A value of 0 indicates an unknown sequence order; this is reserved for future log streams.
STATUS	VARCHAR2(16)		Description of the log stream processing: • Past - The log stream has already been processed • Immediate Past - This is the most recently processed log stream; its status is transitioning from Current to Past • Current - The log stream is currently being processed • Immediate Future - This is the next log stream to be processed; its status is transitioning from Future to Current • Future - The log stream will be processed

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SOURCE	VARCHAR2(5)		Describes how the log stream was started: • RFS - The RFS process created the log stream • User - A user registered the initial log file for the log stream • Synch - A user issued the ALTER DATABASE START LOGICAL STANDBY APPLY NEW PRIMARY DDL statement • Redo - The log stream information was recorded in the redo log
DBID	NUMBER		Database identifier of the primary database that created the log stream
FIRST_CHANGE#	NUMBER		Lowest system change number (SCN) in the current log file
LAST_CHANGE#	NUMBER		Highest system change number (SCN) in the current log file
FIRST_TIME	DATE		Time of the first SCN entry (FIRST_CHANGE#) in the current log file
LAST_TIME	DATE		Time of the last SCN entry (LAST_CHANGE#) in the current log file
DGNAME	VARCHAR2(255)		Unique database name (DB_UNIQUE_NAME) of the primary database that produced the log stream. See V$DATAGUARD_CONFIG to display all database DB_UNIQUE_NAME values defined in the Data Guard configuration.
MERGE_CHANGE#	NUMBER		SCN that should be used to flashback a failed primary (that created the log stream) or to flashback a bystander logical standby database following a failover, in the context of the associated redo log stream. It is the SCN up to which redo for the associated log stream can be merged safely in all databases using local copies of archived logs received from the primary database. In order to apply changes beyond this following a failover, you will need to fetch and mine the redo logs from the failover target.
PROCESSED_CHANGE#	NUMBER		Strict upper bound on the SCN up to which SQL Apply has applied redo records before it switched to a new log stream (either because it was activated and became the primary database, or in the case of a bystander logical standby database where it switched to a new log stream to accommodate a new primary database).

Note

In a CDB, this view shows data only when queried in the root.

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.163 DBA_LOGSTDBY_LOG

DBA_LOGSTDBY_LOG displays information about the logs registered for a logical standby database.

This view is for logical standby databases only.

Column	Datatype	NULL	Description
THREAD#	NUMBER	NOT NULL	Thread ID of the archive log. The THREAD number is 1 for a single instance. For Real Application Clusters, this column will contain different numbers.
RESETLOGS_CHANGE#	NUMBER	NOT NULL	Start SCN of the branch
RESETLOGS_ID	NUMBER	NOT NULL	Resetlogs identifier (a numeric form of the timestamp of the branch)
SEQUENCE#	NUMBER	NOT NULL	Sequence number of the archive log file
FIRST_CHANGE#	NUMBER	NOT NULL	System change number (SCN) of the current archive log
NEXT_CHANGE#	NUMBER		SCN of the next archive log
FIRST_TIME	DATE		Date of the current archive log
NEXT_TIME	DATE		Date of the next archive log
FILE_NAME	VARCHAR2(513)		Name of the archive log
TIMESTAMP	DATE		Time when the archive log was registered
DICT_BEGIN	VARCHAR2(3)		Indicates whether the beginning of the dictionary build is in this archive log (YES) or not (NO)
DICT_END	VARCHAR2(3)		Indicates whether the end of the dictionary build is in this archive log (YES) or not (NO)
APPLIED	VARCHAR2(8)		Indicates primarily whether a given foreign archived log has been applied fully by SQL Apply: • YES - SQL Apply has fully applied the foreign archived log and no longer needs it • CURRENT - SQL Apply is currently applying changes contained in the foreign archived log • NO - SQL Apply has not started applying any changes contained in the foreign archived log • FETCHING - SQL Apply encountered a corruption while reading redo records from this foreign archived log, and is currently using the automatic gap resolution to refetch a new copy of the log from the primary database • CORRUPT - SQL Apply encountered a corruption while reading redo records from this foreign archived log, and refetching a new copy of the archived log did not resolve the problem. SQL Apply will not refetch a new copy of this archived log automatically, and will require user intervention to manually register a new copy of the foreign archived log.
BLOCKS	NUMBER		Number of blocks in the log
BLOCK_SIZE	NUMBER		Size of each block in the log

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The SCN values in this view correlate to the SCN values shown in the DBA_LOGSTDBY_PROGRESS view.

> **Note**
>
> In a CDB, this view shows data only when queried in the root.

5.164 DBA_LOGSTDBY_NOT_UNIQUE

DBA_LOGSTDBY_NOT_UNIQUE displays all tables that have no primary and no non-null unique indexes.

Most of the tables displayed by this view are supported because their columns contain enough information to be maintained in a logical standby database. Some tables, however, cannot be supported because their columns do not contain the necessary information. Unsupported tables usually contain a column defined using an unsupported data type.

In a CDB, the data displayed pertains to the container in which the view is queried.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Schema name of the non-unique table
TABLE_NAME	VARCHAR2(128)		Table name of the non-unique table
BAD_COLUMN	VARCHAR2(1)		• Y - Table column is defined using an unbounded data type, such as LONG or BLOB. If two rows in the table match except in their LOB columns, then the table cannot be maintained properly. Log apply services will attempt to maintain these tables, but you must ensure the application does not allow uniqueness only in the unbounded columns. • N - Enough column information is present to maintain the table in the logical standby database but the log transport services and log apply services would run more efficiently if you added a primary key. You should consider adding a disabled RELY constraint to these tables.

5.165 DBA_LOGSTDBY_PARAMETERS

DBA_LOGSTDBY_PARAMETERS displays the list of parameters used by SQL apply for logical standby databases.

This view is for logical standby databases only.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
NAME	VARCHAR2(64)		Name of the parameter: • MAX_SGA - System global area (SGA) allocated for the log apply services cache (in megabytes) • MAX_SERVERS - Number of processes used by SQL Apply services • PREPARE_SERVERS - Controls the number of parallel execution servers used to prepare changes • APPLY_SERVERS - Controls the number of parallel execution servers used to apply changes • MAX_EVENTS_RECORDED - Number of events stored in the DBA_LOGSTDBY_EVENTS view • RECORD_SKIP_ERRORS - Indicates records that are skipped • RECORD_SKIP_DDL - Indicates skipped DDL statements • RECORD_APPLIED_DDL - Indicates applied DDL statements • RECORD_UNSUPPORTED_OPERATIONS - Shows whether SQL Apply will capture information about transactions that did unsupported operations at the primary database in the DBA_LOGSTDBY_EVENTS view • EVENT_LOG_DEST - Indicates where SQL Apply records the occurrence of an interesting event • LOG_AUTO_DELETE - Shows whether SQL Apply will automatically delete remote archived logs received from the primary database, once the contents of the logs are applied at the logical standby database. • LOG_AUTO_DEL_RETENTION_TARGET - How many minutes a remote archived log received from the primary database will be retained at the logical standby database, once the contents of the log are applied by SQL Apply. • PRESERVE_COMMIT_ORDER - Shows whether transactions are committed at the logical standby database in the same order that they were committed at the primary database
VALUE	VARCHAR2(2000)		Value of the parameter
UNIT	VARCHAR2(64)		Unit of the value, if applicable
SETTING	VARCHAR2(64)		Possible values are as follows: • SYSTEM - Parameter value was not explicitly set by the user. However, the user can change it with an appropriate call to the APPLY_SET procedure. • USER - Parameter value was explicitly set by the user
DYNAMIC	VARCHAR2(64)		YES if the parameter can be set dynamically (that is, without having to stop SQL Apply) NO if setting the parameter requires that SQL Apply be stopped

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> In a CDB, this view shows data when queried in the root.

5.166 DBA_LOGSTDBY_PLSQL_MAP

DBA_LOGSTDBY_PLSQL_MAP shows the mapping between a supported user invokable (/external) PL/SQL procedure to the corresponding replicated internal PL/SQL procedure.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner name of the procedure
PKG_NAME	VARCHAR2(128)		Package name of the user invokable procedure
PROC_NAME	VARCHAR2(128)		Procedure name of the user invokable procedure
INTERNAL_PKG_NAME	VARCHAR2(128)		Package name of the internal procedure
INTERNAL_PROC_NAME	VARCHAR2(128)		Procedure name of the internal procedure

> **Note**
>
> In a CDB, this view shows data when queried in the root or a PDB.

5.167 DBA_LOGSTDBY_PLSQL_SUPPORT

DBA_LOGSTDBY_PLSQL_SUPPORT shows the PL/SQL packages that are only supported during rolling operations.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner name of the package
PKG_NAME	VARCHAR2(128)		Package name of the user invokable procedure
SUPPORT_LEVEL	VARCHAR2(12)		Logical standby PL/SQL support level for the package: • ALWAYS - PL/SQL replication is always supported for this package, whether it is called inside or outside of DBMS_ROLLING • DBMS_ROLLING: PL/SQL replication is supported only when the procedure is called inside DBMS_ROLLING

> **Note**
>
> In a CDB, this view shows data when queried in the root or a PDB.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_ROLLING package

5.168 DBA_LOGSTDBY_PROGRESS

DBA_LOGSTDBY_PROGRESS is deprecated. The information that was provided in this view is now provided in the V$LOGSTDBY_PROGRESS view.

> **See Also**
>
> "V$LOGSTDBY_PROGRESS"

5.169 DBA_LOGSTDBY_SKIP

DBA_LOGSTDBY_SKIP displays the skip rules that are used by SQL Apply.

This view is for logical standby databases only.

Column	Datatype	NULL	Description
ERROR	VARCHAR2(1)		Indicates how the skip rule was created: • Y - For rules from DBMS_LOGSTDBY.SKIP_ERROR • N - For rules from DBMS_LOGSTDBY.SKIP
STATEMENT_OPT	VARCHAR2(128)		Specifies the type of statement that should be skipped
OWNER	VARCHAR2(128)		Name of the schema under which the skip option should be used
NAME	VARCHAR2(261)		Name of the object that is being skipped
USE_LIKE	VARCHAR2(1)		Indicates whether the statement should use a SQL wildcard search when matching names (Y) or not (N)
ESC	VARCHAR2(1)		Escape character used when performing wildcard matches
PROC	VARCHAR2(392)		Name of a stored procedure that will be executed when processing the skip option

> **See Also**
>
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_LOGSTDBY.SKIP_ERROR procedure
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_LOGSTDBY.SKIP procedure

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.170 DBA_LOGSTDBY_SKIP_TRANSACTION

DBA_LOGSTDBY_SKIP_TRANSACTION displays the skip settings chosen. This view is for logical standby databases only.

Column	Datatype	NULL	Description
XIDUSN	NUMBER		Transaction ID undo segment number
XIDSLT	NUMBER		Transaction ID slot number
XIDSQN	NUMBER		Transaction ID sequence number
CON_NAME	VARCHAR2(384)		Container name

5.171 DBA_LOGSTDBY_UNSUPPORTED

DBA_LOGSTDBY_UNSUPPORTED displays the schemas, tables, and columns in those tables that contain unsupported data types.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the unsupported table
TABLE_NAME	VARCHAR2(128)		Name of the unsupported table
COLUMN_NAME	VARCHAR2(128)		Name of the unsupported column
ATTRIBUTES	VARCHAR2(39)		When possible, displays the reason why the table is not supported by SQL Apply. The ATTRIBUTES column may be NULL if the table structure itself is not supported by SQL Apply (for example, the table is system-partitioned), or when the structure of the table is supported but certain columns in the table have an unsupported datatype.
DATA_TYPE	VARCHAR2(32)		Datatype of the unsupported column

Note

A rolling upgrade performed using the DBMS_ROLLING PL/SQL package supports more object types than a manual rolling upgrade performed using transient logical standby databases.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_ROLLING_UNSUPPORTED" for more information about determining unsupported data types for a rolling upgrade using the DBMS_ROLLING package
- *Oracle Data Guard Concepts and Administration* for more information about rolling operations
- *Oracle Data Guard Concepts and Administration* for more information about unsupported tables for rolling upgrade operations
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_ROLLING package

5.172 DBA_LOGSTDBY_UNSUPPORTED_TABLE

DBA_LOGSTDBY_UNSUPPORTED_TABLE displays the data tables that are not supported by Logical Standby.

The data displayed pertains to the container in which the view is queried. This view is for logical standby databases only.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the unsupported table
TABLE_NAME	VARCHAR2(128)		Name of the unsupported table

5.173 DBA_MEASURE_FOLDER_CONTENTS

DBA_MEASURE_FOLDER_CONTENTS describes the contents of all OLAP measure folders in the database. Its columns are the same as those in ALL_MEASURE_FOLDER_CONTENTS.

See Also

"ALL_MEASURE_FOLDER_CONTENTS"

5.174 DBA_MEASURE_FOLDER_SUBFOLDERS

DBA_MEASURE_FOLDER_SUBFOLDERS describes the OLAP measure folders contained within the database OLAP measure folders. Its columns are the same as those in ALL_MEASURE_FOLDER_SUBFOLDERS.

See Also

"ALL_MEASURE_FOLDER_SUBFOLDERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.175 DBA_MEASURE_FOLDERS

DBA_MEASURE_FOLDERS describes all OLAP measure folders in the database. Its columns are the same as those in ALL_MEASURE_FOLDERS.

See Also

"ALL_MEASURE_FOLDERS"

5.176 DBA_METADATA_PROPERTIES

DBA_METADATA_PROPERTIES describes OLAP metadata properties in the database. Its columns are the same as those in ALL_METADATA_PROPERTIES.

See Also

"ALL_METADATA_PROPERTIES"

5.177 DBA_METHOD_PARAMS

DBA_METHOD_PARAMS describes the method parameters of all object types in the database. Its columns are the same as those in ALL_METHOD_PARAMS.

See Also

"ALL_METHOD_PARAMS"

5.178 DBA_METHOD_RESULTS

DBA_METHOD_RESULTS describes the method results of all object types in the database. Its columns are the same as those in ALL_METHOD_RESULTS.

See Also

"ALL_METHOD_RESULTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.179 DBA_MINING_MODEL_ATTRIBUTES

DBA_MINING_MODEL_ATTRIBUTES describes all mining model attributes in the database. Its columns are the same as those in ALL_MINING_MODEL_ATTRIBUTES.

See Also

"ALL_MINING_MODEL_ATTRIBUTES"

5.180 DBA_MINING_MODEL_PARTITIONS

DBA_MINING_MODEL_PARTITIONS describes all the model partitions accessible to the system. Its columns are the same as those in ALL_MINING_MODEL_PARTITIONS.

See Also

"ALL_MINING_MODEL_PARTITIONS"

5.181 DBA_MINING_MODEL_SETTINGS

DBA_MINING_MODEL_SETTINGS describes all mining model settings in the database. Its columns are the same as those in ALL_MINING_MODEL_SETTINGS.

See Also

"ALL_MINING_MODEL_SETTINGS"

5.182 DBA_MINING_MODEL_TABLES

DBA_MINING_MODEL_TABLES describes the tables that contain metadata about the mining models in the database.

Mining models are schema objects created by Oracle Data Mining.

Model tables reside in the schema of the mining model owner. The metadata stored in the tables is controlled by Oracle Data Mining APIs. The tables are read-only. They should not be modified by users.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the mining model
MODEL_NAME	VARCHAR2(128)	NOT NULL	Name of the mining model
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
TABLE_TYPE	VARCHAR2(21)		The type of metadata stored in the table

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.183 DBA_MINING_MODEL_VIEWS

DBA_MINING_MODEL_VIEWS describes all model views in the database. Its columns are the same as those in ALL_MINING_MODEL_VIEWS.

See Also

"ALL_MINING_MODEL_VIEWS"

5.184 DBA_MINING_MODEL_XFORMS

DBA_MINING_MODEL_XFORMS describes the user-specified transformations embedded in all models accessible in the system. Its columns are the same as those in ALL_MINING_MODEL_XFORMS.

See Also

"ALL_MINING_MODEL_XFORMS"

5.185 DBA_MINING_MODELS

DBA_MINING_MODELS describes all mining models in the database. Its columns are the same as those in ALL_MINING_MODELS.

See Also

"ALL_MINING_MODELS"

5.186 DBA_MVIEW_AGGREGATES

DBA_MVIEW_AGGREGATES describes the grouping functions (aggregated measures) that appear in the SELECT list of all aggregated materialized view in the database. Its columns are the same as those in ALL_MVIEW_AGGREGATES.

See Also

"ALL_MVIEW_AGGREGATES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.187 DBA_MVIEW_ANALYSIS

DBA_MVIEW_ANALYSIS describes all materialized views in the database that potentially support query rewrite and that provide additional information for analysis by applications. Its columns are the same as those in ALL_MVIEW_ANALYSIS.

> **Note**
>
> This view excludes materialized views that reference remote tables or that include references to non-static values such as SYSDATE or USER. This view also excludes materialized views that were created as snapshots before Oracle8*i* and that were never altered to enable query rewrite.

> **See Also**
>
> "ALL_MVIEW_ANALYSIS"

5.188 DBA_MVIEW_COMMENTS

DBA_MVIEW_COMMENTS displays comments on all materialized views in the database. Its columns are the same as those in ALL_MVIEW_COMMENTS.

> **See Also**
>
> "ALL_MVIEW_COMMENTS"

5.189 DBA_MVIEW_DETAIL_PARTITION

DBA_MVIEW_DETAIL_PARTITION displays freshness information, with respect to partition change tracking (PCT) detail partitions, for all materialized views in the database. Its columns are the same as those in ALL_MVIEW_DETAIL_PARTITION.

> **See Also**
>
> "ALL_MVIEW_DETAIL_PARTITION".

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.190 DBA_MVIEW_DETAIL_RELATIONS

DBA_MVIEW_DETAIL_RELATIONS represents the named detail relations that are either in the FROM list of a materialized view, or that are indirectly referenced through views in the FROM list. Its columns are the same as those in ALL_MVIEW_DETAIL_RELATIONS.

See Also

"ALL_MVIEW_DETAIL_RELATIONS"

5.191 DBA_MVIEW_DETAIL_SUBPARTITION

DBA_MVIEW_DETAIL_SUBPARTITION displays freshness information, with respect to partition change tracking (PCT) detail subpartitions, for all materialized views in the database. Its columns are the same as those in ALL_MVIEW_DETAIL_SUBPARTITION.

See Also

"ALL_MVIEW_DETAIL_SUBPARTITION"

5.192 DBA_MVIEW_JOINS

DBA_MVIEW_JOINS describes a join between two columns in the WHERE clause of a subquery that defines a materialized view. Its columns are the same as those in ALL_MVIEW_JOINS.

See Also

"ALL_MVIEW_JOINS"

5.193 DBA_MVIEW_KEYS

DBA_MVIEW_KEYS describes the columns or expressions in the SELECT list upon which materialized views in the database are based. Its columns are the same as those in ALL_MVIEW_KEYS.

See Also

"ALL_MVIEW_KEYS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.194 DBA_MVIEW_LOG_FILTER_COLS

DBA_MVIEW_LOG_FILTER_COLS displays all columns (excluding primary key columns) being logged in the materialized view logs.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the master table being logged
NAME	VARCHAR2(128)		Name of the master table being logged
COLUMN_NAME	VARCHAR2(128)		Column being logged

5.195 DBA_MVIEW_LOGS

DBA_MVIEW_LOGS describes all materialized view logs in the database. Its columns are the same as those in ALL_MVIEW_LOGS.

See Also

"ALL_MVIEW_LOGS"

5.196 DBA_MVIEW_REFRESH_TIMES

DBA_MVIEW_REFRESH_TIMES describes refresh times of all materialized views in the database. Its columns are the same as those in ALL_MVIEW_REFRESH_TIMES.

See Also

"ALL_MVIEW_REFRESH_TIMES"

5.197 DBA_MVIEWS

DBA_MVIEWS describes all materialized views in the database. Its columns are the same as those in ALL_MVIEWS.

See Also

"ALL_MVIEWS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.198 DBA_MVREF_CHANGE_STATS

DBA_MVREF_CHANGE_STATS displays the change data load information on the base tables associated with a refresh run for all the materialized views for the database.

Related View

USER_MVREF_CHANGE_STATS displays the change data load information on the master tables associated with a refresh run for all the materialized views in the database that are accessible to the current user.

Column	Datatype	NULL	Description
TBL_OWNER	VARCHAR2(128)	NOT NULL	Owner of the master table for the materialized view
TBL_NAME	VARCHAR2(128)	NOT NULL	Name of the master table for the materialized view
MV_OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
MV_NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
REFRESH_ID	NUMBER	NOT NULL	The refresh ID of the refresh run
NUM_ROWS_INS	NUMBER		The number of inserts in the materialized view log of the table (applicable only if the table has a materialized view log)
NUM_ROWS_UPD	NUMBER		The number of updates in the materialized view log of the table (applicable only if the table has a materialized view log)
NUM_ROWS_DEL	NUMBER		The number of deletes in the materialized view log of the table (applicable only if the table has a materialized view log)
NUM_ROWS_DL_INS	NUMBER		The number of direct load inserts on the table
PMOPS_OCCURRED	CHAR(1)		Indicates whether a partition-maintenance operation (PMOP) occurred. Possible values: • Y • N • NULL: Indicates an unknown value
PMOP_DETAILS	VARCHAR2(4000)		Details of the PMOPs in the following format: • TRUNCATE (low_bound, high_bound) • EXECHANGE (low_bound, high_bound)
NUM_ROWS	NUMBER		The number of rows in the table at the start of the refresh operation

See Also

"USER_MVREF_CHANGE_STATS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.199 DBA_MVREF_RUN_STATS

DBA_MVREF_RUN_STATS has information about each refresh run for all the materialized views for the database, with each run being identified by the REFRESH_ID. The information includes timing statistics related to the run and the parameters specified in that run.

Related View

USER_MVREF_RUN_STATS has information about each refresh run for the materialized views accessible for the current database user, with each run being identified by the REFRESH_ID. The information includes timing statistics related to the run and the parameters specified in that run. This view does not display the RUN_OWNER column.

Column	Datatype	NULL	Description
RUN_OWNER	VARCHAR2(128)	NOT NULL	Owner of the refresh operation (the user who launched the operation)
REFRESH_ID	NUMBER	NOT NULL	The refresh ID of the refresh run
NUM_MVS	NUMBER	NOT NULL	The number of materialized views being refreshed in the run
MVIEWS	VARCHAR2(4000)		Shows the list of comma separated parameters specified in the API for the materialized view refresh operation
BASE_TABLES	VARCHAR2(4000)		For internal use only
METHOD	VARCHAR2(4000)		The METHOD parameter specified by the API
ROLLBACK_SEG	VARCHAR2(4000)		The ROLLBACK_SEG parameter specified by the API
PUSH_DEFERRED_RPC	CHAR(1)		The PUSH_DEFERRED_RPC parameter specified by the API
REFRESH_AFTER_ERRORS	CHAR(1)		The REFRESH_AFTER_ERRORS parameter specified by the API
PURGE_OPTION	NUMBER		The PURGE_OPTION parameter specified by the API
PARALLELISM	NUMBER		The PARALLELISM parameter specified by the API
HEAP_SIZE	NUMBER		The HEAP_SIZE parameter specified by the API
ATOMIC_REFRESH	CHAR(1)		The ATOMIC_REFRESH parameter specified by the API
NESTED	CHAR(1)		The NESTED parameter specified by the API
OUT_OF_PLACE	CHAR(1)		The OUT_OF_PLACE parameter specified by the API
NUMBER_OF_FAILURES	NUMBER		The number of failures that occurred in processing the API
START_TIME	TIMESTAMP(6)		Start time of the refresh run
END_TIME	TIMESTAMP(6)		End time of the refresh run
ELAPSED TIME	NUMBER		The length of time for the refresh run, in seconds
LOG_SETUP_TIME	NUMBER		Log setup time (in seconds) for the materialized view for a non-atomic refresh; NULL for an atomic refresh
LOG_PURGE_TIME	NUMBER		Log purge time (in seconds) for the materialized view in the case of atomic refresh; NULL in the case of non-atomic refresh

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COMPLETE_STATS_AVAILABLE	CHAR(1)		Indicates whether all the complete refresh statistics are available for this run: • Y: All the statistics are available • N: All the statistics are not available

See Also

"USER_MVREF_RUN_STATS"

5.200 DBA_MVREF_STATS

DBA_MVREF_STATS shows the REFRESH_ID associated with each refresh run of each materialized view for the database. It also provides some basic timing statistics related to that materialized view's refresh in that run.

Related View

USER_MVREF_STATS shows the REFRESH_ID associated with each refresh run of each materialized view for the database that is accessible to the current user. It also provides some basic timing statistics related to that materialized view's refresh in that run. This view does not display the MV_OWNER column.

Column	Datatype	NULL	Description
MV_OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
MV_NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
REFRESH_ID	NUMBER	NOT NULL	The refresh ID of the refresh run
REFRESH_METHOD	VARCHAR2(30)		The refresh method used to refresh the materialized view: • FAST • PCT • COMPLETE • OUT OF PLACE FAST • OUT OF PLACE PCT • OUT OF PLACE COMPLETE
REFRESH_OPTIMIZATIONS	VARCHAR2(4000)		The refresh optimization, for example, a null refresh, or a primary key/foreign key that is applied during refresh of the materialize view
ADDITIONAL_EXECUTIONS	VARCHAR2(4000)		The additional executions, for example, an index rebuild, or log operations involved during refresh of the materialized view
START_TIME	TIMESTAMP(6)		Start time of the refresh run
END_TIME	TIMESTAMP(6)		End time of the refresh run
ELAPSED_TIME	NUMBER		The length of time for the refresh run, in seconds
LOG_SETUP_TIME	NUMBER		Log setup time (in seconds) for the materialized view for a non-atomic refresh; NULL for an atomic refresh

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LOG_PURGE_TIME	NUMBER		Log purge time (in seconds) for the materialized view in the case of atomic refresh; NULL in the case of non-atomic refresh
INITIAL_NUM_ROWS	NUMBER		Initial number of rows in the materialized view (at the start of the refresh)
FINAL_NUM_ROWS	NUMBER		Final number of rows in the materialized view (at the end of the refresh)

See Also

"USER_MVREF_STATS"

5.201 DBA_MVREF_STATS_PARAMS

DBA_MVREF_STATS_PARAMS displays the refresh statistics properties associated with each materialized view. These properties can be modified with the DBMS_MVIEW_STATS.SET_MVREF_STATS_PARAMS procedure.

Related View

USER_MVREF_STATS_PARAMS displays the refresh statistics properties associated with each materialized view accessible to the current user. These properties can be modified with the DBMS_MVIEW_STATS.SET_MVREF_STATS_PARAMS procedure.

Column	Datatype	NULL	Description
MV_OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
MV_NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
COLLECTION_LEVEL	VARCHAR2(8)		The collection level for the materialized view
RETENTION_PERIOD	NUMBER		The retention period for the materialize view

See Also

"USER_MVREF_STATS_PARAMS"

5.202 DBA_MVREF_STATS_SYS_DEFAULTS

DBA_MVREF_STATS_SYS_DEFAULTS displays the system-wide defaults for the refresh history statistics properties for materialized views. These values can be altered with the SET_SYSTEM_DEFAULTS procedure by a database administrator.

This view contains exactly two rows corresponding to the collection-level and retention-period properties; their initial values are TYPICAL and 31 respectively.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Related View

USER_MVREF_STATS_SYS_DEFAULTS displays the system-wide defaults for the refresh history statistics properties for materialized views accessible to the current user. These values can be altered with the SET_SYSTEM_DEFAULTS procedure by a database administrator.

Column	Datatype	NULL	Description
PARAMETER_NAME	CHAR(16)		Value of the parameter_name parameter: • COLLECTION_LEVEL • RETENTION_PERIOD
VALUE	VARCHAR2(40)		The system-wide default value for the parameter

See Also

"USER_MVREF_STATS_SYS_DEFAULTS"

5.203 DBA_MVREF_STMT_STATS

DBA_MVREF_STMT_STATS shows information associated with each refresh statement of a materialized view in a refresh run.

Related View

USER_MVREF_STMT_STATS shows information associated with each refresh statement of a materialized view accessible to the current user in a refresh run.

Column	Datatype	NULL	Description
MV_OWNER	VARCHAR2(128)	NOT NULL	Owner of the materialized view
MV_NAME	VARCHAR2(128)	NOT NULL	Name of the materialized view
REFRESH_ID	NUMBER	NOT NULL	The refresh ID of the refresh run
STEP	NUMBER	NOT NULL	A number indicating the step in the refresh process in which the statement is executed for the materialized view. Steps are numbered consecutively starting at 1.
SQLID	VARCHAR2(14)	NOT NULL	The SQL ID of the statement
STMT	CLOB	NOT NULL	The text of the SQL statement
EXECUTION_TIME	NUMBER	NOT NULL	The time it took to execute the statement (in seconds)
EXECUTION_PLAN	XMLTYPE STORAGE BINARY		For internal use only

See Also

"USER_MVREF_STMT_STATS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.204 DBA_NESTED_TABLE_COLS

DBA_NESTED_TABLE_COLS describes the columns of all nested tables in the database. Its columns are the same as those in ALL_NESTED_TABLE_COLS.

To gather statistics for this view, use the DBMS_STATS package.

See Also

"ALL_NESTED_TABLE_COLS"

5.205 DBA_NESTED_TABLES

DBA_NESTED_TABLES describes all nested tables contained in all tables in the database. Its columns are the same as those in ALL_NESTED_TABLES.

See Also

"ALL_NESTED_TABLES"

5.206 DBA_NETWORK_ACL_PRIVILEGES

DBA_NETWORK_ACL_PRIVILEGES describes the network privileges defined in all access control lists that are currently assigned to network hosts.

Note

This DBA_NETWORK_ACL_PRIVILEGES view is deprecated in Oracle Database 12*c* Release 1 (12.1). Oracle recommends that you use the DBA_HOST_ACES view, instead.

Column	Datatype	NULL	Description
ACL	VARCHAR2(4000)		Path of the access control list
ACLID	RAW(8)	NOT NULL	Object ID of the access control list
PRINCIPAL	VARCHAR2(128)		Principal (database user or role) whom the privilege is granted to or denied from
PRIVILEGE	VARCHAR2(128)		Network privilege
IS_GRANT	VARCHAR2(5)		Indicates whether the privilege is granted (true) or denied (false)
INVERT	VARCHAR2(5)		Indicates whether the access control entry contains invert principal (true) or not (false)
START_DATE	TIMESTAMP(6)		Start date of the access control entry
END_DATE	TIMESTAMP(6)		End date of the access control entry

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ACL_OWNER	VARCHAR2(128)		Owner of the access control list

See Also

"DBA_HOST_ACES"

5.207 DBA_NETWORK_ACLS

DBA_NETWORK_ACLS describes the access control list assignments to network hosts.

Note

This DBA_NETWORK_ACLS view is deprecated in Oracle Database 12*c* Release 1 (12.1). Oracle recommends that you use the DBA_HOST_ACLS view, instead.

Column	Datatype	NULL	Description
HOST	VARCHAR2(1000)	NOT NULL	Network host
LOWER_PORT	NUMBER(5)		Lower bound of the port range
UPPER_PORT	NUMBER(5)		Upper bound of the port range
ACL	VARCHAR2(4000)		Path of the access control list
ACLID	RAW(8)		Object ID of the access control list
ACL_OWNER	VARCHAR2(128)		Owner of the access control list

See Also

"DBA_HOST_ACLS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.208 DBA_OBJ_AUDIT_OPTS

DBA_OBJ_AUDIT_OPTS describes auditing options on all objects.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from UNIFIED_AUDIT_TRAIL.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Related View

USER_OBJ_AUDIT_OPTS describes auditing options on all objects owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the object
OBJECT_NAME	VARCHAR2(128)		Name of the object
OBJECT_TYPE	VARCHAR2(23)		Type of the object
ALT	VARCHAR2(3)		Auditing ALTER WHENEVER SUCCESSFUL / UNSUCCESSFUL
AUD	VARCHAR2(3)		Auditing AUDIT WHENEVER SUCCESSFUL / UNSUCCESSFUL
COM	VARCHAR2(3)		Auditing COMMENT WHENEVER SUCCESSFUL / UNSUCCESSFUL
DEL	VARCHAR2(3)		Auditing DELETE WHENEVER SUCCESSFUL / UNSUCCESSFUL
GRA	VARCHAR2(3)		Auditing GRANT WHENEVER SUCCESSFUL / UNSUCCESSFUL
IND	VARCHAR2(3)		Auditing INDEX WHENEVER SUCCESSFUL / UNSUCCESSFUL
INS	VARCHAR2(3)		Auditing INSERT WHENEVER SUCCESSFUL / UNSUCCESSFUL
LOC	VARCHAR2(3)		Auditing LOCK WHENEVER SUCCESSFUL / UNSUCCESSFUL
REN	VARCHAR2(3)		Auditing RENAME WHENEVER SUCCESSFUL / UNSUCCESSFUL
SEL	VARCHAR2(3)		Auditing SELECT WHENEVER SUCCESSFUL / UNSUCCESSFUL
UPD	VARCHAR2(3)		Auditing UPDATE WHENEVER SUCCESSFUL / UNSUCCESSFUL
EXE	VARCHAR2(3)		Auditing EXECUTE WHENEVER SUCCESSFUL / UNSUCCESSFUL
CRE	VARCHAR2(3)		Auditing CREATE WHENEVER SUCCESSFUL / UNSUCCESSFUL
REA	VARCHAR2(3)		Auditing READ WHENEVER SUCCESSFUL / UNSUCCESSFUL
WRI	VARCHAR2(3)		Auditing WRITE WHENEVER SUCCESSFUL / UNSUCCESSFUL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FBK	VARCHAR2(3)		Auditing FLASHBACK WHENEVER SUCCESSFUL / UNSUCCESSFUL

See Also

- "USER_OBJ_AUDIT_OPTS"
- *Oracle Database SQL Language Reference* for more information about the SQL AUDIT statement for unified auditing
- *Oracle Database SQL Language Reference* for more information about the SQL AUDIT statement for traditional auditing
- *Oracle Database Security Guide* to learn how to find information about audited activities

5.209 DBA_OBJ_COLATTRS

DBA_OBJ_COLATTRS describes object columns and attributes contained in all tables in the database. Its columns are the same as those in ALL_OBJ_COLATTRS.

See Also

"ALL_OBJ_COLATTRS"

5.210 DBA_OBJECT_SIZE

DBA_OBJECT_SIZE lists the sizes, in bytes, of various PL/SQL objects.

Related View

- USER_OBJECT_SIZE lists the size of PL/SQL objects owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
NAME	VARCHAR2(128)	NOT NULL	Name of the object
TYPE	VARCHAR2(18)		Type of the object: TYPE, TYPE BODY, TABLE, VIEW, SYNONYM, SEQUENCE, PROCEDURE, FUNCTION, PACKAGE, PACKAGE BODY, JAVA SOURCE, JAVA CLASS or JAVA RESOURCE
SOURCE_SIZE	NUMBER		Size of the source in bytes. Must be in memory during compilation, or dynamic recompilation.
PARSED_SIZE	NUMBER		Size of the parsed form of the object, in bytes. Must be in memory when an object is being compiled that references this object.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CODE_SIZE	NUMBER		Code size, in bytes. Must be in memory when this object is executing.
ERROR_SIZE	NUMBER		Size of error messages, in bytes. In memory during the compilation of the object when there are compilation errors.

See Also

"USER_OBJECT_SIZE"

5.211 DBA_OBJECT_TABLES

DBA_OBJECT_TABLES describes all object tables in the database. Its columns are the same as those in ALL_OBJECT_TABLES.

See Also

"ALL_OBJECT_TABLES"

5.212 DBA_OBJECT_USAGE

DBA_OBJECT_USAGE displays statistics about index usage gathered from the database for all the indexes in the database.

You can use this view to monitor index usage. All indexes that have been used at least once can be monitored and displayed in this view.

Related View

- USER_OBJECT_USAGE displays statistics about index usage gathered from the database for the indexes owned by the current user.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Index owner
INDEX_NAME	VARCHAR2(128)	NOT NULL	Index name in sys.obj$.name
TABLE_NAME	VARCHAR2(128)	NOT NULL	Table name in sys.obj$.name
MONITORING	VARCHAR2(3)		Indicates whether the monitoring feature is turned on. Possible values: • YES • NO
USED	VARCHAR2(3)		Indicates whether the index has been accessed. Possible values: • YES • NO

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
START_MONITORING	VARCHAR2(19)		Start monitoring time in sys.object_stats.start_monitoring
END_MONITORING	VARCHAR2(19)		End monitoring time in sys.object_stats.end_monitoring

See Also

"USER_OBJECT_USAGE"

5.213 DBA_OBJECTS

DBA_OBJECTS describes all objects in the database. Its columns are the same as those in ALL_OBJECTS.

See Also

"ALL_OBJECTS"

5.214 DBA_OBJECTS_AE

DBA_OBJECTS_AE describes all objects (across all editions) in the database. Its columns are the same as those in ALL_OBJECTS_AE.

See Also

"ALL_OBJECTS_AE"

5.215 DBA_OPANCILLARY

DBA_OPANCILLARY provides ancillary information for all operator bindings in the database. Its columns are the same as those in ALL_OPANCILLARY.

See Also

"ALL_OPANCILLARY"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.216 DBA_OPARGUMENTS

DBA_OPARGUMENTS provides argument information for all operator bindings in the database. Its columns are the same as those in ALL_OPARGUMENTS.

See Also

"ALL_OPARGUMENTS"

5.217 DBA_OPBINDINGS

DBA_OPBINDINGS describes the binding functions and methods on all operators in the database. Its columns are the same as those in ALL_OPBINDINGS.

See Also

"ALL_OPBINDINGS"

5.218 DBA_OPERATOR_COMMENTS

DBA_OPERATOR_COMMENTS displays comments for all user-defined operators in the database. Its columns are the same as those in ALL_OPERATOR_COMMENTS.

See Also

"ALL_OPERATOR_COMMENTS"

5.219 DBA_OPERATORS

DBA_OPERATORS describes all operators in the database. Its columns are the same as those in ALL_OPERATORS.

See Also

"ALL_OPERATORS"

5.220 DBA_OPTSTAT_OPERATION_TASKS

DBA_OPTSTAT_OPERATION_TASKS displays the history of tasks that are performed as part of statistics operations (recorded in DBA_OPTSTAT_OPERATIONS). Each task represents a target object to be processed in the corresponding parent operation.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OPID	NUMBER		Internal identifier for the statistics operation that the task belongs to
TARGET	VARCHAR2(100)		Name of the object that this task operates on
TARGET_OBJN	NUMBER		Object number of the target object
TARGET_TYPE	VARCHAR2(40)		Type of the target object. Possible values are: • TABLE • TABLE (GLOBAL STATS ONLY): Task is created to gather only global statistics of a partitioned table • TABLE (COORDINATOR JOB): Coordinator task for a partitioned table when concurrency is on • TABLE PARTITION • TABLE SUBPARTITION • INDEX • INDEX PARTITION • INDEX SUBPARTITION
TARGET_SIZE	NUMBER		Target size (in number of blocks) when the task started
START_TIME	TIMESTAMP(6) WITH TIME ZONE		Task start time
END_TIME	TIMESTAMP(6) WITH TIME ZONE		Task end time
STATUS	VARCHAR2(49)		Current task status. Possible values are: • PENDING: Task is queued for processing • IN PROGRESS: Task is currently running • COMPLETED: Task has completed successfully • FAILED: Task has failed • SKIPPED: Task has been skipped, as it does not exist any more, or its stats are not stale (applies only to only automatic statistics gathering) • TIMED OUT: Maintenance window was not enough to complete this task (applies only to automatic statistics gathering)
JOB_NAME	VARCHAR2(50)		Name of the scheduler job that executes this task (for example, when concurrency is on)
ESTIMATED_COST	NUMBER		Estimated cost of the task (measured as elapsed time in seconds). This column is populated only when concurrency is on.
BATCHING_COEFF	NUMBER		For internal use only
ACTIONS	NUMBER		For internal use only
PRIORITY	NUMBER		Rank of the task among all target objects for the parent operation
FLAGS	NUMBER		For internal use only
NOTES	VARCHAR2(4000)		Notes about the underlying task, such as the failure message for tasks with status FAILED.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "DBA_OPTSTAT_OPERATIONS"

5.221 DBA_OPTSTAT_OPERATIONS

DBA_OPTSTAT_OPERATIONS contains a history of statistics operations performed at the schema and database level using the DBMS_STATS package.

Column	Datatype	NULL	Description
ID	NUMBER		Internal ID of the statistics operation
OPERATION	VARCHAR2(64)		Operation name
TARGET	VARCHAR2(64)		Target on which the operation was performed
START_TIME	TIMESTAMP(6) WITH TIME ZONE		Time at which the operation started
END_TIME	TIMESTAMP(6) WITH TIME ZONE		Time at which the operation ended
STATUS	VARCHAR2(49)		Current operation status. Possible values are: • IN PROGRESS: Operation is currently running • COMPLETED: Operation has completed successfully • FAILED: Operation has failed • TIMED OUT: Maintenance window was not enough to complete this operation (applies only to automatic statistics gathering)
JOB_NAME	VARCHAR2(32)		Name of the scheduler job that executes this operation (for example, a user scheduled statistics gathering job)
SESSION_ID	NUMBER		ID of the session in which this operation is invoked
NOTES	VARCHAR2(4000)		Notes about the operation, such as a failure message for operations with status FAILED

> **See Also**
>
> - "DBA_OPTSTAT_OPERATION_TASKS"
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_REPAIR.ADMIN_TABLES procedure
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_REPAIR.DUMP_ORPHAN_KEYS procedure

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.222 DBA_ORPHAN_KEY_TABLE

DBA_ORPHAN_KEY_TABLE reports key values from indexes where the underlying base table has block corruptions.

To create the view, run the DBMS_REPAIR.ADMIN_TABLES procedure. To populate the orphan key table for an index, run the DBMS_REPAIR.DUMP_ORPHAN_KEYS procedure on the index. For each key in the index that points to a corrupt data block, Oracle inserts a row into the orphan key table.

Column	Datatype	NULL	Description
SCHEMA_NAME	VARCHAR2(128)	NOT NULL	Schema name of the index
INDEX_NAME	VARCHAR2(128)	NOT NULL	Name of the index
IPART_NAME	VARCHAR2(128)	NULL	Name of the index partition or subpartition
INDEX_ID	NUMBER	NOT NULL	Dictionary object ID of the index
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the base table of the index
PART_NAME	VARCHAR2(128)	NULL	Name of the base table partition or subpartition
TABLE_ID	NUMBER	NOT NULL	Dictionary object ID of the base table
KEYROWID	ROWID	NOT NULL	Physical rowid of the corrupt data row
KEY	ROWID	NOT NULL	Key values for the index entry
DUMP_TIMESTAMP	DATE	NOT NULL	Timestamp when the entry was made into the orphan key table

5.223 DBA_OUTLINE_HINTS

DBA_OUTLINE_HINTS describes the set of hints stored in all outlines in the database.

Related View

USER_OUTLINE_HINTS describes the set of hints stored in the outlines owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
NAME	VARCHAR2(128)		Name of the outline
OWNER	VARCHAR2(128)		Name of the user who created the outline
NODE	NUMBER		ID of the query or subquery to which the hint applies. The top-level query is labeled 1. Subqueries are assigned sequentially numbered labels, starting with 2.
STAGE	NUMBER		Outline hints can be applied at three different stages during the compilation process. This column indicates the stage at which this hint was applied.
JOIN_POS	NUMBER		Position of the table in the join order. The value is 0 for all hints except access method hints, which identify a table to which the hint and the join position apply.
HINT	CLOB		Text of the hint

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "USER_OUTLINE_HINTS"

5.224 DBA_OUTLINES

DBA_OUTLINES describes all stored outlines in the database.

Related View

USER_OUTLINES describes the stored outlines owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
NAME	VARCHAR2(128)		User-specified or generated name of the stored outline. The name must be of a form that can be expressed in SQL.
OWNER	VARCHAR2(128)		Name of the user who created the outline
CATEGORY	VARCHAR2(128)		User-defined name of the category to which the outline belongs
USED	VARCHAR2(6)		Indicates whether the outline has ever been used (USED) or not (UNUSED)
TIMESTAMP	DATE		Timestamp of outline creation
VERSION	VARCHAR2(64)		Oracle version that created the outline
SQL_TEXT	LONG		SQL text of the query, including any hints that were a part of the original statement. If bind variables are included, the variable names are stored as SQL text, not the values that are assigned to the variables. **Note:** This field may contain sensitive information about your database or application. Therefore, use discretion when granting SELECT or VIEW object privileges on these views.
SIGNATURE	RAW(16)		Signature uniquely identifying the outline SQL text
COMPATIBLE	VARCHAR2(12)		Indicates whether the outline hints were compatible across a migration (COMPATIBLE) or not (INCOMPATIBLE)
ENABLED	VARCHAR2(8)		Indicates whether the outline is enabled (ENABLED) or disabled (DISABLED)
FORMAT	VARCHAR2(6)		Hint format: • NORMAL • LOCAL
MIGRATED	VARCHAR2(12)		Indicates whether the outline has been migrated to a SQL plan baseline (MIGRATED) or not (NOT-MIGRATED)

> **See Also**
>
> "USER_OUTLINES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.225 DBA_OUTSTANDING_ALERTS

DBA_OUTSTANDING_ALERTS describes alerts which the server considers to be outstanding.

Column	Datatype	NULL	Description
SEQUENCE_ID	NUMBER		Alert sequence number
REASON_ID	NUMBER	NOT NULL	ID of the alert reason
OWNER	VARCHAR2(128)		Owner of the object on which the alert was issued
OBJECT_NAME	VARCHAR2(513)		Name of the object
SUBOBJECT_NAME	VARCHAR2(128)		Name of the subobject (for example: partition)
OBJECT_TYPE	VARCHAR2(64)		Object type (for example: table, tablespace)
REASON	VARCHAR2(4000)		Reason for the alert
TIME_SUGGESTED	TIMESTAMP(6) WITH TIME ZONE		Time when the alert was last updated
CREATION_TIME	TIMESTAMP(6) WITH TIME ZONE		Time when the alert was first created
SUGGESTED_ACTION	VARCHAR2(4000)		Advice of the recommended action
ADVISOR_NAME	VARCHAR2(128)		Name of the advisor to be invoked for more information
METRIC_VALUE	NUMBER		Value of the related metrics
MESSAGE_TYPE	VARCHAR2(12)		Message type: • Notification • Warning
MESSAGE_GROUP	VARCHAR2(64)		Name of the message group to which the alert belongs
MESSAGE_LEVEL	NUMBER		Message severity level (1 to 32)
HOSTING_CLIENT_ID	VARCHAR2(64)		ID of the client or security group to which the alert relates
MODULE_ID	VARCHAR2(64)		ID of the module that originated the alert
PROCESS_ID	VARCHAR2(128)		Process ID
HOST_ID	VARCHAR2(256)		DNS host name of the originating host
HOST_NW_ADDR	VARCHAR2(256)		IP or other network address of the originating host
INSTANCE_NAME	VARCHAR2(16)		Originating instance name
INSTANCE_NUMBER	NUMBER		Originating instance number
USER_ID	VARCHAR2(128)		User ID
EXECUTION_CONTEXT_ID	VARCHAR2(128)		Execution Context ID
ERROR_INSTANCE_ID	VARCHAR2(142)		ID of an error instance plus a sequence number
STATE_TRANSITION_NUMBER	NUMBER		Sequence number of the state transition for the alert
PDB_NAME	VARCHAR2(128)		PDB name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER	NOT NULL	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

5.226 DBA_PARALLEL_EXECUTE_CHUNKS

DBA_PARALLEL_EXECUTE_CHUNKS displays the chunks for all tasks in the database.

Related View

USER_PARALLEL_EXECUTE_CHUNKS displays the chunks for tasks created by the current user. This view does not display the TASK_OWNER column.

Column	Datatype	NULL	Description
CHUNK_ID	NUMBER	NOT NULL	Unique ID for the chunk
TASK_OWNER	VARCHAR2(128)	NOT NULL	Owner of the task
TASK_NAME	VARCHAR2(128)	NOT NULL	Name of the task
STATUS	VARCHAR2(20)		Status of the chunk: • UNASSIGNED • ASSIGNED • PROCESSED • PROCESSED_WITH_ERROR
START_ROWID	ROWID		Rowid for the first row in the chunk
END_ROWID	ROWID		Rowid for the last row in the chunk
START_ID	NUMBER		Number column value of the first row in the chunk
END_ID	NUMBER		Number column value of the last row in the chunk
JOB_NAME	VARCHAR2(128)		Name of the job which processed this chunk
START_TS	TIMESTAMP(6)		Processing start time for the chunk
END_TS	TIMESTAMP(6)		Processing end time for the chunk
ERROR_CODE	NUMBER		Error code returned during the execution of the chunk if the STATUS column is PROCESSED_WITH_ERROR
ERROR_MESSAGE	VARCHAR2(4000)		Error message returned during the execution of the chunk if the STATUS column is PROCESSED_WITH_ERROR

See Also

"USER_PARALLEL_EXECUTE_CHUNKS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.227 DBA_PARALLEL_EXECUTE_TASKS

DBA_PARALLEL_EXECUTE_TASKS displays all tasks in the database.

Related View

USER_PARALLEL_EXECUTE_TASKS displays the tasks created by the current user. This view does not display the TASK_OWNER column.

Column	Datatype	NULL	Description
TASK_OWNER	VARCHAR2(128)	NOT NULL	Owner of the task
TASK_NAME	VARCHAR2(128)	NOT NULL	Name of the task
CHUNK_TYPE	VARCHAR2(12)		Type of parallel update: • UNDELARED • ROWID_RANGE • NUMBER_RANGE
STATUS	VARCHAR2(19)		Status of the task: • CREATED • CHUNKING • CHUNKING_FAILED • CHUNKED • PROCESSING • FINISHED • FINISHED_WITH_ERROR • CRASHED
TABLE_OWNER	VARCHAR2(128)		Owner of the table to be chunked
TABLE_NAME	VARCHAR2(128)		Name of the table to be chunked
NUMBER_COLUMN	VARCHAR2(128)		Name of the column holding IDs (only applicable to NUMBER_RANGE chunking type)
TASK_COMMENT	VARCHAR2(4000)		Comment field
JOB_PREFIX	VARCHAR2(128)		Prefix of the job name executing this task
SQL_STMT	CLOB		Argument used in the previous DBMS_PARALLEL_EXECUTE.RUN_TASK
LANGUAGE_FLAG	NUMBER		Argument used in the previous DBMS_PARALLEL_EXECUTE.RUN_TASK
EDITION	VARCHAR2(130)		Argument used in the previous DBMS_PARALLEL_EXECUTE.RUN_TASK
APPLY_CROSSEDITION_TRIGGER	VARCHAR2(130)		Argument used in the previous DBMS_PARALLEL_EXECUTE.RUN_TASK
FIRE_APPLY_TRIGGER	VARCHAR2(10)		Argument used in the previous DBMS_PARALLEL_EXECUTE.RUN_TASK
PARALLEL_LEVEL	NUMBER		Argument used in the previous DBMS_PARALLEL_EXECUTE.RUN_TASK
JOB_CLASS	VARCHAR2(128)		Argument used in the previous DBMS_PARALLEL_EXECUTE.RUN_TASK

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "USER_PARALLEL_EXECUTE_TASKS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_PARALLEL_EXECUTE.RUN_TASK` procedure

5.228 DBA_PART_COL_STATISTICS

`DBA_PART_COL_STATISTICS` displays column statistics and histogram information for all table partitions in the database. Its columns are the same as those in `ALL_PART_COL_STATISTICS`.

See Also

"ALL_PART_COL_STATISTICS"

5.229 DBA_PART_HISTOGRAMS

`DBA_PART_HISTOGRAMS` displays the histogram data (endpoints per histogram) for the histograms on all table partitions in the database. Its columns are the same as those in `ALL_PART_HISTOGRAMS`.

See Also

"ALL_PART_HISTOGRAMS"

5.230 DBA_PART_INDEXES

`DBA_PART_INDEXES` displays the object-level partitioning information for all partitioned indexes in the database. Its columns are the same as those in `ALL_PART_INDEXES`.

See Also

"ALL_PART_INDEXES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.231 DBA_PART_KEY_COLUMNS

DBA_PART_KEY_COLUMNS describes the partitioning key columns for all partitioned objects in the database. Its columns are the same as those in ALL_PART_KEY_COLUMNS.

See Also

"ALL_PART_KEY_COLUMNS"

5.232 DBA_PART_LOBS

DBA_PART_LOBS displays table-level information for all partitioned LOBs in the database, including default attributes for LOB data partitions. Its columns are the same as those in ALL_PART_LOBS.

See Also

"ALL_PART_LOBS"

5.233 DBA_PART_TABLES

DBA_PART_TABLES displays the object-level partitioning information for all partitioned tables in the database. Its columns are the same as those in ALL_PART_TABLES.

See Also

"ALL_PART_TABLES"

5.234 DBA_PARTIAL_DROP_TABS

DBA_PARTIAL_DROP_TABS describes all tables in the database that have partially completed DROP COLUMN operations. Its columns are the same as those in ALL_PARTIAL_DROP_TABS.

See Also

"ALL_PARTIAL_DROP_TABS"

5.235 DBA_PDB_HISTORY

DBA_PDB_HISTORY describes the lineage of the PDB to which it belongs.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PDB_NAME	VARCHAR2(128)	NOT NULL	Name of this PDB in one of its incarnations
PDB_ID	NUMBER	NOT NULL	Container ID of this PDB in one of its incarnations.
PDB_DBID	NUMBER	NOT NULL	Database ID of this PDB in one of its incarnations
PDB_GUID	RAW(16)	NOT NULL	Globally unique ID of this PDB in one of its incarnations
OP_SCNBAS	NUMBER	NOT NULL	SCN base when an operation was performed on one of the incarnations of this PDB
OP_SCNWRP	NUMBER	NOT NULL	SCN wrap when an operation was performed on one of incarnations of this PDB
OP_TIMESTAMP	DATE	NOT NULL	Timestamp of an operation performed on one of the incarnations of this PDB
OPERATION	VARCHAR2(16)	NOT NULL	Operation that was performed on one of the incarnations of this PDB
DB_VERSION	NUMBER	NOT NULL	Database version
CLONED_FROM_PDB_NAME	VARCHAR2(128)		Name of a PDB from which one of the incarnations of this PDB was cloned
CLONED_FROM_PDB_DBID	NUMBER		Database ID of a PDB from which one of the incarnations of this PDB was cloned
CLONED_FROM_PDB_GUID	RAW(16)		Globally unique ID of a PDB from which one of the incarnations of this PDB was cloned
DB_NAME	VARCHAR2(128)		Name of a CDB in which one of the incarnations of this PDB was created
DB_UNIQUE_NAME	VARCHAR2(128)		Unique name of a CDB in which one of the incarnations of this PDB was created
DB_DBID	NUMBER		Database ID of a CDB in which one of the incarnations of this PDB was created
CLONETAG	VARCHAR2(128)		Clone tag name for the PDB if the PDB was cloned using the snapshot copy mechanism
DB_VERSION_STRING	VARCHAR2(204)		Database version string

5.236 DBA_PDB_SAVED_STATES

DBA_PDB_SAVED_STATES shows information about the current saved PDB states in the CDB.

This view is a data link, so the data is also available within the PDB.

Column	Datatype	NULL	Description
CON_ID	NUMBER	NOT NULL	The ID of the PDB
CON_NAME	VARCHAR2(128)	NOT NULL	Name of the PDB
INSTANCE_NAME	VARCHAR2(128)	NOT NULL	Name of the instance for which the state is saved
CON_UID	NUMBER	NOT NULL	Unique ID assigned to the PDB at creation time
GUID	RAW(16)		Globally unique immutable ID assigned to the PDB at creation time
STATE	VARCHAR2(14)		Open state of the PDB
RESTRICTED	VARCHAR2(3)		Restricted mode of the PDB

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database SQL Language Reference* for more information about preserving a PDB's open mode across an instance restart

5.237 DBA_PDB_SNAPSHOTFILE

`DBA_PDB_SNAPSHOTFILE` displays the files associated with snapshots taken of pluggable databases (PDBs).

You can use this view in conjunction with the `DBA_PDB_SNAPSHOT` view. Join the `SNAPSHOT_SCN` column in this view with the `SNAPSHOT_SCN` column in `DBA_PDB_SNAPSHOT` to determine the files associated with a particular PDB snapshot. A PDB snapshot consists of an archive log file, one or more data files, and one or more XML files. A row is added to this view for each file associated with a PDB snapshot.

Column	Datatype	NULL	Description
`CON_ID`	`NUMBER`	`NOT NULL`	The ID of the PDB
`SNAPSHOT_SCN`	`NUMBER`	`NOT NULL`	SCN at which the snapshot was taken
`SNAPSHOT_FILENAME`	`VARCHAR2(513)`	`NOT NULL`	Snapshot file name
`SNAPSHOT_FILETYPE`	`VARCHAR2(8)`	`NOT NULL`	Snapshot file type. Possible values: • `ARCH`: Archive log file • `DATA`: Data file • `XML`: XML file

> **Note**
>
> This view is available starting with Oracle Database 19c.

> **See Also**
>
> "DBA_PDB_SNAPSHOTS"

5.238 DBA_PDB_SNAPSHOTS

`DBA_PDB_SNAPSHOTS` describes the snapshots taken of pluggable databases (PDBs).

Rows are added to this view when a snapshot of a PDB is taken by using the `ALTER PLUGGABLE DATABASE SNAPSHOT` SQL statement.

Column	Datatype	NULL	Description
`CON_ID`	`NUMBER`	`NOT NULL`	The ID of the PDB
`CON_UID`	`NUMBER`	`NOT NULL`	Unique ID assigned to the PDB at creation time
`CON_NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the PDB

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SNAPSHOT_NAME	VARCHAR2(128)	NOT NULL	Snapshot name of the PDB
SNAPSHOT_SCN	NUMBER	NOT NULL	SCN at which the snapshot was taken
PREVIOUS_SNAPSHOT_SCN	NUMBER	NOT NULL	SCN at which the previous snapshot for the PDB was taken
SNAPSHOT_TIME	NUMBER	NOT NULL	Timestamp at which the snapshot was taken
PREVIOUS_SNAPSHOT_TIME	NUMBER	NOT NULL	Timestamp of the previous snapshot for this PDB
FULL_SNAPSHOT_PATH	VARCHAR2(4000)	NOT NULL	Full path for the snapshot

Note

This view does not display snapshot copy PDBs, which are created by using the `CREATE PLUGGABLE DATABASE ... SNAPSHOT COPY` SQL statement.

See Also

"DBA_PDB_SNAPSHOTFILE" for information about the files associated with a particular PDB snapshot

5.239 DBA_PDBS

`DBA_PDBS` describes PDBs belonging to a given CDB.

When queried from a CDB root, this view describes all PDBs that belong to the CDB. When queried from an application root, it describes all PDBs that belong to the application container. When queried from a regular PDB or from an application PDB, it describes the regular PDB or the application PDB.

Column	Datatype	NULL	Description
PDB_ID	NUMBER	NOT NULL	Container ID of the PDB
PDB_NAME	VARCHAR2(128)	NOT NULL	Name of the PDB
DBID	NUMBER	NOT NULL	PDB identifier calculated when the PDB is created and stored in all file headers associated with the PDB
CON_UID	NUMBER	NOT NULL	Unique identifier associated with the container
GUID	RAW(16)		Globally unique immutable ID assigned to the PDB at creation time

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(10)		State of the PDB. Possible values: • NEW - The PDB has never been opened since it was created. It must be opened in READ WRITE mode for Oracle to perform processing needed to complete the integration of the PDB into the CDB and mark it NORMAL. An error will be thrown if an attempt is made to open the PDB read only. • NORMAL - The PDB is ready to be used. • UNPLUGGED - The PDB has been unplugged. The only operation that can be performed on it is DROP PLUGGABLE DATABASE. • RELOCATING: The PDB is in the process of being relocated to a different CDB. • RELOCATED: The PDB has been relocated to a different CDB. • REFRESHING: The PDB is a refresh PDB. • UNDEFINED: The PDB is in an undefined state. • UNUSABLE - The PDB is being created or an unrecoverable error was encountered during its creation. The PDB cannot be opened while its state is set to UNUSABLE. If the PDB remains in this state because of an error encountered during its creation, it can only be dropped. The alert log can be checked to determine if there was an error during PDB creation.
CREATION_SCN	NUMBER		Creation SCN
VSN	NUMBER		The version number of the PDB
LOGGING	VARCHAR2(9)		Shows the current logging mode for the PDB. Possible values: • LOGGING • NOLOGGING
FORCE_LOGGING	VARCHAR2(3)		Specifies whether force logging is turned on for the PDB. Possible values: • NO • YES
FORCE_NOLOGGING	VARCHAR2(3)		Specifies whether force nologging is turned on for the PDB. Possible values: • NO • YES
APPLICATION_ROOT	VARCHAR2(3)		Indicates whether the PDB is an application root.
APPLICATION_PDB	VARCHAR2(3)		Indicates whether a PDB is an application PDB
APPLICATION_SEED	VARCHAR2(3)		Indicates whether a PDB is an application seed (an application seed is also an application PDB)
APPLICATION_ROOT_CON_ID	NUMBER		If this PDB is an application PDB, the container ID of an application root to which this application PDB belongs. If this PDB is an application root clone, the container ID of an application root to which this application root clone belongs. Otherwise, NULL.
IS_PROXY_PDB	VARCHAR2(3)		Indicates whether this PDB is a proxy PDB

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER	NOT NULL	The ID of the container that CON_DBID identifies. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
UPGRADE_PRIORITY	NUMBER		The upgrade priority of the PDB.
APPLICATION_CLONE	VARCHAR2(3)		Indicates whether this PDB is an application root clone (YES) or not (NO)
FOREIGN_CDB_DBID	NUMBER		The foreign CDB's DBID
UNPLUG_SCN	NUMBER		SCN at which the PDB was unplugged
FOREIGN_PDB_ID	NUMBER		The foreign PDB ID
CREATION_TIME	DATE	NOT NULL	PDB creation timestamp
REFRESH_MODE	VARCHAR2(6)		PDB refresh mode. Possible values: • MANUAL • AUTO
REFRESH_INTERVAL	NUMBER		PDB refresh interval. This is applicable only when REFRESH_MODE is AUTO.
TEMPLATE	VARCHAR2(3)		For internal use only
LAST_REFRESH_SCN	NUMBER		System change number (SCN) of the last refresh operation
TENANT_ID	VARCHAR2(255)		Pluggable database tenant key
SNAPSHOT_MODE	VARCHAR2(6)		Pluggable database snapshot mode
SNAPSHOT_INTERVAL	NUMBER		Pluggable database snapshot interval, in minutes
CREDENTIAL_NAME[1]	VARCHAR2(262)		Credential object name associated with the PDB

1 This column is available starting with Oracle Database 19c.

5.240 DBA_PENDING_CONV_TABLES

DBA_PENDING_CONV_TABLES describes all pending conversion tables in the database. Its columns are the same as those in ALL_PENDING_CONV_TABLES.

See Also

"ALL_PENDING_CONV_TABLES"

5.241 DBA_PENDING_TRANSACTIONS

DBA_PENDING_TRANSACTIONS describes unresolved transactions (either due to failure or if the coordinator has not sent a commit/rollback).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FORMATID	NUMBER		The format identifier of the transaction identifier
GLOBALID	RAW(64)		The global part (gtrid) of the transaction identifier
BRANCHID	RAW(64)		The branch qualifier (bqual) of the transaction identifier

5.242 DBA_PLSQL_COLL_TYPES

DBA_PLSQL_COLL_TYPES describes all named PL/SQL collection types in the database. Its columns (except for CHAR_USED) are the same as those in ALL_PLSQL_COLL_TYPES.

See Also

"ALL_PLSQL_COLL_TYPES"

5.243 DBA_PLSQL_OBJECT_SETTINGS

DBA_PLSQL_OBJECT_SETTINGS displays information about the compiler settings for all stored objects in the database. Its columns are the same as those in ALL_PLSQL_OBJECT_SETTINGS.

See Also

"ALL_PLSQL_OBJECT_SETTINGS"

5.244 DBA_PLSQL_TYPE_ATTRS

DBA_PLSQL_TYPE_ATTRS describes the attributes of all PL/SQL types in the database. Its columns are the same as those in ALL_PLSQL_TYPE_ATTRS.

See Also

"ALL_PLSQL_TYPE_ATTRS"

5.245 DBA_PLSQL_TYPES

DBA_PLSQL_TYPES describes all PL/SQL types in the database. Its columns are the same as those in ALL_PLSQL_TYPES.

See Also

"ALL_PLSQL_TYPES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.246 DBA_POLICIES

DBA_POLICIES describes all Oracle Virtual Private Database (VPD) security policies in the database. Its columns are the same as those in ALL_POLICIES.

A security policy is a list of security requirements and rules that regulate row level access to database objects.

See Also

"ALL_POLICIES"

5.247 DBA_POLICY_ATTRIBUTES

DBA_POLICY_ATTRIBUTES lists the attribute associations {Namespaces, Attributes} of all context-sensitive and shared context-sensitive Oracle Virtual Private Database (VPD) policies in the database. Its columns are the same as those in ALL_POLICY_ATTRIBUTES.

See Also

"ALL_POLICY_ATTRIBUTES"

5.248 DBA_POLICY_CONTEXTS

DBA_POLICY_CONTEXTS describes all driving contexts in the database. Its columns are the same as those in ALL_POLICY_CONTEXTS.

See Also

"ALL_POLICY_CONTEXTS"

5.249 DBA_POLICY_GROUPS

DBA_POLICY_GROUPS describes all policy groups in the database. Its columns are the same as those in ALL_POLICY_GROUPS.

See Also

"ALL_POLICY_GROUPS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.250 DBA_PRIV_AUDIT_OPTS

DBA_PRIV_AUDIT_OPTS describes current system privileges being audited across the system and by user.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from UNIFIED_AUDIT_TRAIL.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
USER_NAME	VARCHAR2(128)		User name if by user auditing; ANY CLIENT if access by a proxy on behalf of a client is being audited; NULL for systemwide auditing
PROXY_NAME	VARCHAR2(128)		Name of the proxy user which is performing an operation for the client; NULL if the client is performing the operation directly
PRIVILEGE	VARCHAR2(40)	NOT NULL	Name of the system privilege being audited
SUCCESS	VARCHAR2(10)		Mode for WHENEVER SUCCESSFUL system auditing
FAILURE	VARCHAR2(10)		Mode for WHENEVER NOT SUCCESSFUL system auditing

5.251 DBA_PRIV_CAPTURES

DBA_PRIV_CAPTURES lists the privilege analysis policies in the database.

Column	Datatype	NULL	Description
NAME	VARCHAR2(128)	NOT NULL	Name of the privilege analysis policy
DESCRIPTION	VARCHAR2(1024)		Description of the privilege analysis
TYPE	VARCHAR2(16)		Type of the privilege analysis policy. Possible values: • G_DATABASE: Database wide privilege analysis • G_ROLE: Role privilege analysis • G_CONTEXT: Context privilege analysis • G_ROLE_AND_CONTEXT: Role and context privilege analysis
ENABLED	VARCHAR2(1)		Enabling status of the privilege analysis
ROLES	ROLE_ID_LIST		List of roles whose privileges to analyze if the privilege analysis type is G_ROLE or G_ROLE_AND_CONTEXT
CONTEXT	VARCHAR2(4000)		Context condition if the privilege analysis type is G_CONTEXT or G_ROLE_AND_CONTEXT

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
RUN_NAME	VARCHAR2(128)		Displays run name information for each run

See Also

Oracle Database Security Guide for more information about privilege analysis

5.252 DBA_PRIVATE_TEMP_TABLES

DBA_PRIVATE_TEMP_TABLES describes all of the private temporary tables in the database.

Related View

USER_PRIVATE_TEMP_TABLES describes the private temporary tables in the current session. This view does not display the INST_ID column.

Column	Datatype	NULL	Description
SID	NUMBER		Session ID of the session that created the private temporary table
SERIAL#	NUMBER		Session serial number of the session that created the private temporary table
INST_ID	NUMBER		Instance ID of the session that created the private temporary table
OWNER	VARCHAR2(128)		Owner name of the private temporary table
TABLE_NAME	VARCHAR2(128)		Private temporary table name
TABLESPACE_NAME	VARCHAR2(128)		Private temporary table's tablespace name
DURATION	VARCHAR2(128)		Private temporary table's duration (for example, SESSION or TRANSACTION)
NUM_ROWS	NUMBER		Number of rows in the private temporary table when analyzed
BLOCKS	NUMBER		Number of blocks used by private temporary table
AVG_ROW_LEN	NUMBER		Average row length
LAST_ANALYZED	DATE		Timestamp of last analyze
TXN_ID	RAW(8)		Transaction ID of the transaction duration private temporary table
SAVE_POINT_NUM	NUMBER		Save point number of the transaction duration private temporary table

See Also

- "USER_PRIVATE_TEMP_TABLES"
- "PRIVATE_TEMP_TABLE_PREFIX"
- *Oracle Database Administrator's Guide* for an introduction to private temporary tables

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.253 DBA_PROCEDURES

DBA_PROCEDURES lists all functions and procedures that are available in the database, along with their associated properties. Its columns are the same as those in ALL_PROCEDURES.

See Also

- "ALL_PROCEDURES"
- "DBA_ARGUMENTS" for information about the arguments of all of the functions and procedures that are available in the database

5.254 DBA_PROFILES

DBA_PROFILES displays all profiles and their limits.

Column	Datatype	NULL	Description
PROFILE	VARCHAR2(128)	NOT NULL	Profile name
RESOURCE_NAME	VARCHAR2(32)	NOT NULL	Resource name
RESOURCE_TYPE	VARCHAR2(8)		Indicates whether the resource profile is a KERNEL or a PASSWORD parameter
LIMIT	VARCHAR2(128)		Limit placed on this resource for this profile
COMMON	VARCHAR2(3)		Indicates whether a given profile is common. Possible values: • YES if a profile is common • NO if a profile is local (not common)
INHERITED	VARCHAR2(3)		Indicates whether the profile definition was inherited from another container (YES) or not (NO)
IMPLICIT	VARCHAR2(3)		Indicates whether this profile was created by an implicit application (YES) or not (NO)

5.255 DBA_PROPAGATION

DBA_PROPAGATION displays information about all propagations in the database. Its columns are the same as those in ALL_PROPAGATION.

See Also

"ALL_PROPAGATION"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.256 DBA_PROXIES

DBA_PROXIES displays information about all proxy connections in the database.

Related View

USER_PROXIES displays information about connections the current user is allowed to proxy. This view does not display the PROXY or PROXY_AUTHORITY columns.

Column	Datatype	NULL	Description
PROXY	VARCHAR2(128)		Name of the proxy user
CLIENT	VARCHAR2(128)	NOT NULL	Name of the client user who the proxy user can act on behalf of
AUTHENTICATION	VARCHAR2(3)		Indicates whether the proxy is required to supply the client's authentication credentials (YES) or not (NO)
AUTHORIZATION_CONSTRAINT	VARCHAR2(35)		Indicates the proxy's authority to exercise roles on the client's behalf: • NO CLIENT ROLES MAY BE ACTIVATED • PROXY MAY ACTIVATE ROLE • PROXY MAY ACTIVATE ALL CLIENT ROLES • PROXY MAY NOT ACTIVATE ROLE
ROLE	VARCHAR2(128)		Name of the role referenced in AUTHORIZATION_CONSTRAINT
PROXY_AUTHORITY	VARCHAR2(9)		Value is either: • DIRECTORY if an EUS proxy is enabled for the database user • DATABASE if this row describes a local proxy permission

See Also

- "USER_PROXIES"
- "PROXY_USERS" for a list of proxy users and the clients on whose behalf they can act

5.257 DBA_QUEUE_SCHEDULES

DBA_QUEUE_SCHEDULES describes all propagation schedules in the database. Its columns are the same as those in ALL_QUEUE_SCHEDULES.

See Also

"ALL_QUEUE_SCHEDULES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.258 DBA_QUEUE_SUBSCRIBERS

DBA_QUEUE_SUBSCRIBERS displays all subscribers on all queues in the database. Its columns are the same as those in ALL_QUEUE_SUBSCRIBERS.

See Also

"ALL_QUEUE_SUBSCRIBERS"

5.259 DBA_QUEUE_TABLES

DBA_QUEUE_TABLES contains information about the owner instance for a queue table.

A queue table can contain multiple queues. In this case, each queue in a queue table has the same owner instance as the queue table. Its columns are the same as those in ALL_QUEUE_TABLES.

See Also

"ALL_QUEUE_TABLES"

5.260 DBA_QUEUES

DBA_QUEUES describes the operational characteristics of every queue in a database. Its columns are the same as those in ALL_QUEUES.

See Also

- "ALL_QUEUES"
- *Oracle Database Advanced Queuing User's Guide* for more information about Advanced Queuing

5.261 DBA_RAT_CAPTURE_SCHEMA_INFO

DBA_RAT_CAPTURE_SCHEMA_INFO displays the login schema and current schema that were in effect when SQL statements were recorded in a workload capture.

This view is useful when you perform a workload replay in extended PL/SQL mode. This type of replay may include SQL statements that perform table operations such as SELECT, UPDATE, and DELETE. If the current schema was different from the login schema at the time of the workload capture, then those table operations may have been performed with the privileges of the current user, not the login user. During workload replay, all operations are performed with the privileges of the login user. Therefore, errors can occur during replay if the login user does not have the necessary privileges to perform the table operations.

Copyright © 2002, 2026, Oracle and/or its affiliates.

To resolve this issue, you can use this view in conjunction with the DBA_WORKLOAD_CAPTURE_SQLTEXT view. Join the CAPTURE_ID column in this view with the CAPTURE_ID column in DBA_WORKLOAD_CAPTURE_SQLTEXT to determine the login schema and current schema that were in effect when each SQL statement in DBA_WORKLOAD_CAPTURE_SQLTEXT was captured. Examine the SQL_TEXT column in DBA_WORKLOAD_CAPTURE_SQLTEXT to determine whether the SQL statement involved any table operations, and whether those table operations were performed with the privileges of the current user or the login user. You can then grant to the login user the necessary privileges for performing those table operations before performing a workload replay.

Column	Datatype	NULL	Description
CAPTURE_ID	NUMBER(38)		Internal key for the workload capture
SQL_ID	VARCHAR2(13)		SQL identifier for the parent cursor in the library cache
LOGIN_SCHEMA	VARCHAR2(128)		The schema of the user who logged on to the session in which the SQL statement was recorded in the workload capture. This value does not change during a session.
CURRENT_SCHEMA	VARCHAR2(128)		The currently active default schema for the session when the SQL statement was recorded in the workload capture. This value may change during a session through use of an ALTER SESSION SET CURRENT_SCHEMA statement. It may also change during a session to reflect the owner of any active definer's rights object.

Note

This view is available starting with Oracle Database 19c.

See Also

"DBA_WORKLOAD_CAPTURE_SQLTEXT"

5.262 DBA_RCHILD

DBA_RCHILD displays all the children in any refresh group.

Column	Datatype	NULL	Description
REFGROUP	NUMBER		Internal identifier of the refresh group
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object in the refresh group
NAME	VARCHAR2(128)	NOT NULL	Name of the object in the refresh group
TYPE#	VARCHAR2(128)		Type of the object in the refresh group

5.263 DBA_RECOVERABLE_SCRIPT

DBA_RECOVERABLE_SCRIPT provides details about recoverable operations.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SCRIPT_ID	RAW(16)		Unique ID of the operation
CREATION_TIME	DATE		Time the operation was invoked
INVOKING_PACKAGE_OWNER	VARCHAR2(128)		Invoking package owner of the operation
INVOKING_PACKAGE	VARCHAR2(128)		Invoking package of the operation
INVOKING_PROCEDURE	VARCHAR2(128)		Invoking procedure of the operation
INVOKING_USER	VARCHAR2(128)		Script owner
STATUS	VARCHAR2(12)		State of the recoverable script: GENERATING, NOT EXECUTED, EXECUTING, EXECUTED, or ERROR
TOTAL_BLOCKS	NUMBER		Total number of blocks for the recoverable script to be executed
DONE_BLOCK_NUM	NUMBER		Last block executed, thus far
SCRIPT_COMMENT	VARCHAR2(4000)		Comment for the recoverable script

5.264 DBA_RECOVERABLE_SCRIPT_BLOCKS

DBA_RECOVERABLE_SCRIPT_BLOCKS provides details about recoverable script blocks.

Column	Datatype	NULL	Description
SCRIPT_ID	RAW(16)		Global unique ID of the recoverable script to which this block belongs
BLOCK_NUM	NUMBER		The *n*th block in the recoverable script to be executed
FORWARD_BLOCK	CLOB		Forward block to be executed
FORWARD_BLOCK_DBLINK	VARCHAR2(128)		Database where the forward block is executed
UNDO_BLOCK	CLOB		Block to roll back the forward operation
UNDO_BLOCK_DBLINK	VARCHAR2(128)		Database where the undo block is executed
STATUS	VARCHAR2(12)		Status of the block execution: GENERATING, NOT EXECUTED, EXECUTING, EXECUTED, or ERROR
BLOCK_COMMENT	VARCHAR2(4000)		Comment for the block

5.265 DBA_RECOVERABLE_SCRIPT_ERRORS

DBA_RECOVERABLE_SCRIPT_ERRORS provides details about errors that occurred during script execution.

Column	Datatype	NULL	Description
SCRIPT_ID	RAW(16)		Global unique ID of the recoverable script
BLOCK_NUM	NUMBER		The *n*th block that failed
ERROR_NUMBER	NUMBER		Number of the error encountered while executing the block
ERROR_MESSAGE	VARCHAR2(4000)		Error message encountered while executing the block
ERROR_CREATION_TIME	DATE		Time that the error was created

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.266 DBA_RECOVERABLE_SCRIPT_HIST

DBA_RECOVERABLE_SCRIPT_HIST displays details about executed or purged recoverable operations.

Column	Datatype	NULL	Description
SCRIPT_ID	RAW(16)		Unique id of the operation
CREATION_TIME	DATE		Time the operation was invoked
INVOKING_PACKAGE_OWNER	VARCHAR2(128)		Invoking package owner of the operation
INVOKING_PACKAGE	VARCHAR2(128)		Invoking package of the operation
INVOKING_PROCEDURE	VARCHAR2(128)		Invoking procedure of the operation
INVOKING_USER	VARCHAR2(128)		Script owner
STATUS	VARCHAR2(8)		state of the recoverable script: EXECUTED, PURGED
TOTAL_BLOCKS	NUMBER		total number of blocks for the recoverable script to be executed
DONE_BLOCK_NUM	NUMBER		last block so far executed
SCRIPT_COMMENT	VARCHAR2(4000)		comment for the recoverable script

5.267 DBA_RECOVERABLE_SCRIPT_PARAMS

DBA_RECOVERABLE_SCRIPT_PARAMS provides details about recoverable operation parameters.

Column	Datatype	NULL	Description
SCRIPT_ID	RAW(16)		Unique ID of the operation
PARAMETER	VARCHAR2(128)		Name of the parameter
PARAM_INDEX	NUMBER		Index for multi-valued parameter
VALUE	VARCHAR2(4000)		Value of the parameter

5.268 DBA_RECYCLEBIN

DBA_RECYCLEBIN displays information about all recycle bins in the database.

Related View

USER_RECYCLEBIN displays information about the recycle bin owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Name of the original owner of the object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	New name of the object
ORIGINAL_NAME	VARCHAR2(128)		Original name of the object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OPERATION	VARCHAR2(9)		Operation carried out on the object: • DROP - Object was dropped • TRUNCATE - Object was truncated **Note:** The Oracle Database currently only supports recovering dropped objects from the recycle bin. The truncated objects cannot be recovered.
TYPE	VARCHAR2(25)		Type of the object: • TABLE • NORMAL INDEX • BITMAP INDEX • NESTED TABLE • LOB • LOB INDEX • DOMAIN INDEX • IOT TOP INDEX • IOT OVERFLOW SEGMENT • IOT MAPPING TABLE • TRIGGER • Table Partition • Table Composite Partition • Index Partition • Index Composite Partition • LOB Partition • LOB Composite Partition
TS_NAME	VARCHAR2(30)		Name of the tablespace to which the object belongs
CREATETIME	VARCHAR2(19)		Timestamp for the creation of the object
DROPTIME	VARCHAR2(19)		Timestamp for the dropping of the object
DROPSCN	NUMBER		System change number (SCN) of the transaction which moved the object to the recycle bin
PARTITION_NAME	VARCHAR2(128)		Name of the partition which was dropped
CAN_UNDROP	VARCHAR2(3)		Indicates whether the object can be undropped (YES) or not (NO)
CAN_PURGE	VARCHAR2(3)		Indicates whether the object can be purged (YES) or not (NO)
RELATED	NUMBER	NOT NULL	Object number of the parent object
BASE_OBJECT	NUMBER	NOT NULL	Object number of the base object
PURGE_OBJECT	NUMBER	NOT NULL	Object number for the object which gets purged
SPACE	NUMBER		Number of blocks used by the object

See Also

"USER_RECYCLEBIN"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.269 DBA_REDEFINITION_ERRORS

DBA_REDEFINITION_ERRORS is an online redefinition view. It displays the dependent objects for which errors were raised while attempting to create similar objects on the interim table of the redefinition.

Column	Datatype	NULL	Description
OBJECT_TYPE	VARCHAR2(12)		Type of the redefinition object: • TABLE • INDEX • CONSTRAINT • TRIGGER • NESTED TABLE • PARTITION • MV LOG
OBJECT_OWNER	VARCHAR2(4000)		Owner of the redefinition object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the redefinition object
BASE_TABLE_OWNER	VARCHAR2(128)		Owner of the base table of the redefinition object
BASE_TABLE_NAME	VARCHAR2(128)		Name of the base table of the redefinition object
DDL_TXT	CLOB		DDL used to create the corresponding interim redefinition object
EDITION_NAME	VARCHAR2(128)		Reserved for future use
ERR_NO	NUMBER(38)		Oracle error number corresponding to this error
ERR_TXT	VARCHAR2(1000)		Oracle error text corresponding to this error

See Also

Oracle Database Administrator's Guide for more information about online redefinition

5.270 DBA_REDEFINITION_OBJECTS

DBA_REDEFINITION_OBJECTS is an online redefinition view. It displays the objects involved in the current redefinitions.

Column	Datatype	NULL	Description
OBJECT_TYPE	VARCHAR2(12)		Type of the redefinition object: • TABLE • INDEX • CONSTRAINT • TRIGGER • NESTED TABLE • PARTITION • MV LOG
OBJECT_OWNER	VARCHAR2(4000)		Owner of the redefinition object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the redefinition object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BASE_TABLE_OWNER	VARCHAR2(128)		Owner of the base table of the redefinition object
BASE_TABLE_NAME	VARCHAR2(128)		Name of the base table of the redefinition object
INTERIM_OBJECT_OWNER	VARCHAR2(4000)		Owner of the corresponding interim redefinition object
INTERIM_OBJECT_NAME	VARCHAR2(128)		Name of the corresponding interim redefinition object
EDITION_NAME	VARCHAR2(128)		Reserved for future use

See Also

Oracle Database Administrator's Guide for more information about online redefinition

5.271 DBA_REDEFINITION_STATUS

DBA_REDEFINITION_STATUS is an online redefinition view. It provides information about the online redefinition status.

Column	Datatype	NULL	Description
REDEFINITION_ID	NUMBER(38)	NOT NULL	ID for the redefinition object
BASE_TABLE_OWNER	VARCHAR2(128)		Owner of the base table of the redefinition object
BASE_TABLE_NAME	VARCHAR2(128)		Name of the base table of the redefinition object
BASE_OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the base object of the redefinition object
BASE_OBJECT_TYPE	VARCHAR2(9)		Type of the base object of the redefinition object
INTERIM_OBJECT_OWNER	VARCHAR2(128)		Owner of the interim object of the redefinition object
INTERIM_OBJECT_NAME	VARCHAR2(128)		Name of the interim object of the redefinition object
OPERATION	VARCHAR2(128)	NOT NULL	The current redefinition operation: • START_REDEF_TABLE • SYNC_INTERIM_TABLE • COPY_TABLE_DEPENDENTS • FINISH_REDEF_TABLE
STATUS	VARCHAR2(128)	NOT NULL	Status of the previous redefinition operation: • FAILURE • SUCCESS
RESTARTABLE	VARCHAR2(1)	NOT NULL	Indicates whether the previous operation can be restarted
ERR_TXT	VARCHAR2(1000)		The error message raised from the previous operation
ACTION	VARCHAR2(400)		The suggested action
REFRESH_DEP_MVIEWS	VARCHAR2(1)		Indicates whether the online redefinition will also refresh dependent materialized views when syncing the interim table (Y) or not (N)

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Administrator's Guide* for more information about online redefinition

5.272 DBA_REFRESH

DBA_REFRESH describes all refresh groups in the database. Its columns are the same as those in ALL_REFRESH.

> **See Also**
>
> "ALL_REFRESH"

5.273 DBA_REFRESH_CHILDREN

DBA_REFRESH_CHILDREN lists all of the objects in all refresh groups in the database. Its columns are the same as those in ALL_REFRESH_CHILDREN.

> **See Also**
>
> "ALL_REFRESH_CHILDREN"

5.274 DBA_REFS

DBA_REFS describes the REF columns and REF attributes in object type columns of all the objects in the database. Its columns are the same as those in ALL_REFS.

> **See Also**
>
> "ALL_REFS"

5.275 DBA_REGISTERED_ARCHIVED_LOG

DBA_REGISTERED_ARCHIVED_LOG displays information about all registered archived logfiles in the database.

Column	Datatype	NULL	Description
CONSUMER_NAME	VARCHAR2(128)	NOT NULL	Consumer name of the archived logs
SOURCE_DATABASE	VARCHAR2(128)		Name of the database which generated the redo logs
THREAD#	NUMBER	NOT NULL	Thread number of the archived redo log. The thread number is 1 for a single instance. For Real Application Clusters, this column will contain different numbers.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SEQUENCE#	NUMBER	NOT NULL	Sequence number of the archived redo log file
FIRST_SCN	NUMBER	NOT NULL	System change number (SCN) of the current archived redo log
NEXT_SCN	NUMBER		System change number (SCN) of the next archived redo log
FIRST_TIME	DATE		Date and time of the current archived redo log
NEXT_TIME	DATE		Date and time of the next archived redo log
NAME	VARCHAR2(513)		Name of the archived redo log
MODIFIED_TIME	DATE		Time when the archived redo log was registered
DICTIONARY_BEGIN	VARCHAR2(3)		Indicates whether the beginning of the dictionary build is in the archived redo log (YES) or not (NO)
DICTIONARY_END	VARCHAR2(3)		Indicates whether the end of the dictionary build is in the archived redo log (YES) or not (NO)
PURGEABLE	VARCHAR2(3)		Indicates whether the redo log can be permanently removed (YES) or not (NO)
RESET_LOGS_CHANGE#	NUMBER	NOT NULL	Resetlogs change number of the database when the log was written
RESET_TIMESTAMP	NUMBER	NOT NULL	Resetlogs time of the database when the log was written

5.276 DBA_REGISTERED_MVIEWS

DBA_REGISTERED_MVIEWS describes all registered materialized views (registered at a master site or a master materialized view site) in the database. Its columns are the same as those in ALL_REGISTERED_MVIEWS.

See Also

"ALL_REGISTERED_MVIEWS"

5.277 DBA_REGISTRY

DBA_REGISTRY displays information about all components in the database that are loaded into the component registry.

The component registry tracks components that can be separately loaded into the Oracle Database. When a SQL script loads the PL/SQL packages and other database objects for a component into the database, the script records the component name, status, and version. If scripts are used to upgrade/downgrade the dictionary elements for the component, then those scripts also record status and version information.

Related View

USER_REGISTRY displays information about the components owned by the current user that are loaded into the component registry.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COMP_ID	VARCHAR2(30)	NOT NULL	Component identifier
COMP_NAME	VARCHAR2(255)		Component name
VERSION	VARCHAR2(30)		Component version loaded
VERSION_FULL	VARCHAR2(30)		Component full version
STATUS	VARCHAR2(11)		Component status: • INVALID • VALID • LOADING • LOADED • UPGRADING • UPGRADED • DOWNGRADING • DOWNGRADED • REMOVING • REMOVED
MODIFIED	VARCHAR2(20)		Time when the component was last modified
NAMESPACE	VARCHAR2(30)	NOT NULL	Component namespace
CONTROL	VARCHAR2(128)	NOT NULL	User that created the component entry
SCHEMA	VARCHAR2(128)	NOT NULL	User that contains the objects for the component
PROCEDURE	VARCHAR2(61)		Validation procedure
STARTUP	VARCHAR2(8)		Indicates whether the component requires a startup after the upgrade (REQUIRED) or not
PARENT_ID	VARCHAR2(30)		Parent component identifier
OTHER_SCHEMAS	VARCHAR2(4000)		A list of ancillary schema names associated with the component

See Also

"USER_REGISTRY"

5.278 DBA_REGISTRY_BACKPORTS

DBA_REGISTRY_BACKPORTS displays information about backported bug fixes recorded to support database upgrade and downgrade operations. It is not intended to provide a complete list of all bug fixes or all data dictionary changes in the database.

Column	Datatype	NULL	Description
BUGNO	NUMBER	NOT NULL	Bug number
VERSION_FULL	VARCHAR2(30)	NOT NULL	Component full version
COMP_ID	VARCHAR2(30)	NOT NULL	Component identifier
NAMESPACE	VARCHAR2(30)	NOT NULL	Component namespace

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BACKPORT_TYPE	VARCHAR2(30)	NOT NULL	Type of backported bug fix The value of this column is always DICTIONARY, which indicates that the bug fix changed the data dictionary of the database.
BACKPORT_TIME	TIMESTAMP(6)	NOT NULL	Date and time at which the backported bug fix was applied

> **Note**
>
> This view is available starting with Oracle Database 19c.

5.279 DBA_REGISTRY_HIERARCHY

DBA_REGISTRY_HIERARCHY displays information about the components loaded into the database, grouped by owner and organized in the component hierarchy.

Column	Datatype	NULL	Description
NAMESPACE	VARCHAR2(30)	NOT NULL	Component namespace
COMP_ID	VARCHAR2(4000)		Component identifier
VERSION	VARCHAR2(30)		Component version loaded
VERSION_FULL	VARCHAR2(30)		Component full version
STATUS	VARCHAR2(11)		Component status: • INVALID • VALID • LOADING • LOADED • UPGRADING • UPGRADED • DOWNGRADING • DOWNGRADED • REMOVING • REMOVED
MODIFIED	VARCHAR2(20)		Time when the component was last modified

5.280 DBA_REGISTRY_HISTORY

DBA_REGISTRY_HISTORY provides information about upgrades, downgrades, and critical patch updates that have been performed on the database.

Column	Datatype	NULL	Description
ACTION_TIME	TIMESTAMP(6)		The time the upgrade, downgrade, or patch action was completed

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ACTION	VARCHAR2(30)		The specific action (for example, UPGRADE or DOWNGRADE)
NAMESPACE	VARCHAR2(30)		The namespace of the components affected (for example, SERVER)
VERSION	VARCHAR2(30)		The version number of the server (for example, 10.2.0.1.0)
ID	NUMBER		Bundle ID
COMMENTS	VARCHAR2(255)		Additional comments about the action taken
BUNDLE_SERIES	VARCHAR2(30)		If a bundle patch, the series (for example, PSU or DBBP)

5.281 DBA_REGISTRY_LOG

DBA_REGISTRY_LOG displays operating information about components loaded into the database.

Column	Datatype	NULL	Description
OPTIME	VARCHAR2(20)		Operation time
NAMESPACE	VARCHAR2(30)		Component namespace
COMP_ID	VARCHAR2(30)		Component identifier
OPERATION	VARCHAR2(11)		Operation name
MESSAGE	VARCHAR2(1000)		Message

5.282 DBA_REGISTRY_SCHEMAS

DBA_REGISTRY_SCHEMAS lists the primary and ancillary schemas included in the component registry. The ancillary schemas that are listed in this view are the same schemas that would be included in the OTHER_SCHEMAS column of the DBA_REGISTRY view.

Column	Datatype	NULL	Description
NAMESPACE	VARCHAR2(30)		Component namespace
COMP_ID	VARCHAR2(30)		Component identifier
SCHEMA	VARCHAR2(128)		User that contains the objects for the component

See Also

- "DBA_REGISTRY"
- "USER_REGISTRY"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.283 DBA_REGISTRY_SQLPATCH

DBA_REGISTRY_SQLPATCH contains information about the SQL patches that have been installed in the database.

A SQL patch is a patch that contains SQL scripts which need to be run after OPatch completes. DBA_REGISTRY_SQLPATCH is updated by the datapatch utility. Each row contains information about an installation attempt (apply or roll back) for a given patch.

Column	Datatype	NULL	Description
INSTALL_ID	NUMBER	NOT NULL	Unique numeric identifier for this datapatch session. All patches installed in the same invocation of datapatch will have the same value for INSTALL_ID.
PATCH_ID	NUMBER	NOT NULL	ID associated with the patch
PATCH_UID	NUMBER	NOT NULL	UPI (Universal Patch ID) associated with the patch
PATCH_TYPE	VARCHAR2(10)	NOT NULL	Type of the patch. Possible values: • INTERIM: Interim patch • RU: Release Update • RUI: Release Update Increment • RUR: Release Update Revision • CU: Cumulative Update
ACTION	VARCHAR2(15)	NOT NULL	APPLY or ROLLBACK
STATUS	VARCHAR2(25)	NOT NULL	Possible values: • SUCCESS: Patch application has completed with no errors • WITH ERRORS: Patch application finished with errors
ACTION_TIME	TIMESTAMP(6)	NOT NULL	Timestamp when the install was performed
DESCRIPTION	VARCHAR2(100)		Description of this patch from OPatch metadata
LOGFILE	VARCHAR2(500)	NOT NULL	Location of the logfile for this apply or rollback attempt
RU_LOGFILE	VARCHAR2(500)		Logfile location for RU specific commands
FLAGS	VARCHAR2(10)		One or more of the following: • J: Patch is a JVM patch • M: Patch installation was merged with another patch • N: Patch requires normal mode • R: Patch installation has been retried • U: Patch requires upgrade mode
PATCH_DESCRIPTOR	XMLTYPE		Contents of the XML descriptor for the patch
PATCH_DIRECTORY	BLOB		Contents of the patch directory under ORACLE_HOME/sqlpatch
SOURCE_VERSION	VARCHAR2(15)		5 digit version (for example, 18.3.2.0.0) for the version on which the patch was applied
SOURCE_BUILD_DESCRIPTION	VARCHAR2(80)		Build description (for example, Release_Update or Release_Update_Revision) for the version on which the patch was applied
SOURCE_BUILD_TIMESTAMP	TIMESTAMP(6)		Build timestamp for the version on which the patch was applied

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TARGET_VERSION	VARCHAR2(15)		5 digit version (for example, 18.4.0.0.0) for the version to be installed
TARGET_BUILD_DESCRIPTION	VARCHAR2(80)		Build description (for example, Release_Update or Release_Update_Revision) for the version to be installed
TARGET_BUILD_TIMESTAMP	TIMESTAMP(6)		Build timestamp for the version to be installed

See Also

- *Oracle OPatch User's Guide for Windows and UNIX* for more information about OPatch and related patching utilities
- My Oracle Support note KB148594 "Datapatch: Database 12c Post Patch SQL Automation" at the following URL for more information about datapatch:

  https://support.oracle.com/support/?documentId=KB148594

5.284 DBA_REPAIR_TABLE

DBA_REPAIR_TABLE describes any corruptions found by the DBMS_REPAIR.CHECK_OBJECT procedure.

This information is used by the DBMS_REPAIR.FIX_CORRUPT_BLOCKS procedure on execution. To create this view, first run the DBMS_REPAIR.ADMIN_TABLES procedure. To populate the resulting repair table for an object, run the DBMS_REPAIR.CHECK_OBJECT procedure on the object.

Note

The table created by the DBMS_REPAIR.ADMIN_TABLES procedure is called REPAIR TABLE by default. If you specify a different name, this view will have the name you specify, preceded by "DBA_REPAIR_".

Column	Datatype	NULL	Description
OBJECT_ID	NUMBER	NOT NULL	Dictionary object number of the object with the corruption
TABLESPACE_ID	NUMBER	NOT NULL	Tablespace number of the corrupt object
RELATIVE_FILE_ID	NUMBER)	NOT NULL	Relative file number of the corrupt object
BLOCK_ID	NUMBER	NOT NULL	Block number of the corruption
CORRUPT_TYPE	NUMBER	NOT NULL	Type of corruption encountered
SCHEMA_NAME	VARCHAR2(128)	NOT NULL	Schema of the corrupt object
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the corrupt object
BASEOBJECT_NAME	VARCHAR2(128)	NULL	If the object is an index, the name of its base table
PARTITION_NAME	VARCHAR2(128)	NULL	Partition or subpartition name, if applicable
CORRUPT_DESCRIPTION	VARCHAR2(2000)	NULL	Description of corruption

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
REPAIR_DESCRIPTION	VARCHAR2(200)	NULL	Description of repair action
MARKED_CORRUPT	VARCHAR2(10)	NOT NULL	Whether the block is marked corrupt (TRUE | FALSE)
CHECK_TIMESTAMP	DATE	NOT NULL	Date and time when this row was inserted into the repair table
FIX_TIMESTAMP	DATE	NULL	Date and time when the block was modified by the FIX_CORRUPT_BLOCKS procedure, if applicable
REFORMAT_TIMESTAMP	DATE	NULL	Reserved for future use

See Also

Oracle Database PL/SQL Packages and Types Reference for more information about the DBMS_REPAIR package

5.285 DBA_REPL_DBNAME_MAPPING

DBA_REPL_DBNAME_MAPPING provides details about the database name mapping in replication. Its columns are the same as those in ALL_REPL_DBNAME_MAPPING.

See Also

"ALL_REPL_DBNAME_MAPPING"

5.286 DBA_REPLICATION_PROCESS_EVENTS

DBA_REPLICATION_PROCESS_EVENTS provides information about the replication processes events in the database. Its columns are the same as those in ALL_REPLICATION_PROCESS_EVENTS.

See Also

"ALL_REPLICATION_PROCESS_EVENTS"

5.287 DBA_RESOURCE_INCARNATIONS

DBA_RESOURCE_INCARNATIONS lists all resource incarnations that are running or eligible for HA status notification.

Column	Datatype	NULL	Description
RESOURCE_TYPE	VARCHAR2(30)	NOT NULL	Type of resource
RESOURCE_NAME	VARCHAR2(256)		Name of resource

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DB_UNIQUE_NAME	VARCHAR2(30)	NOT NULL	Database unique name
DB_DOMAIN	VARCHAR2(128)	NOT NULL	Database domain
INSTANCE_NAME	VARCHAR2(30)	NOT NULL	Name of instance at which resource is located
HOST_NAME	VARCHAR2(512)		Name of host at which resource is located
STARTUP_TIME	TIMESTAMP(9) WITH TIME ZONE		Resource startup date and time

5.288 DBA_RESUMABLE

DBA_RESUMABLE displays all resumable statements executed in the system.

Related View

USER_RESUMABLE displays the resumable statements executed by the current user. This view does not display the USER_ID column.

Column	Datatype	NULL	Description
USER_ID	NUMBER		User ID Number of the Resumable Statement Owner
SESSION_ID	NUMBER		Session Identifier of the Resumable Statement
INSTANCE_ID	NUMBER		Instance Number of the Resumable Statement
COORD_INSTANCE_ID	NUMBER		Instance Number on which the Parallel Coordinator is Running
COORD_SESSION_ID	NUMBER		Session Identifier of the Parallel Coordinator
STATUS	VARCHAR2(9)		Status of the resumable statement: • RUNNING • SUSPENDED • TIMEOUT • ERROR • ABORTED
TIMEOUT	NUMBER		Timeout of the resumable statement
START_TIME	VARCHAR2(20)		Start time of the resumable statement
SUSPEND_TIME	VARCHAR2(20)		Last time the resumable statement was suspended (initialized to NULL)
RESUME_TIME	VARCHAR2(20)		Last time the suspended resumable statement was resumed (initialized to NULL)
NAME	VARCHAR2(4000)		Name given in the resumable clause of the resumable statement
SQL_TEXT	VARCHAR2(1000)		Resumable statement, selected from the V$SQL view
ERROR_NUMBER	NUMBER		Error code of the last correctable error. When STATUS is set to RUNNING, its value will be 0.
ERROR_PARAMETER1	VARCHAR2(80)		First parameter for the error message (NULL if no error)
ERROR_PARAMETER2	VARCHAR2(80)		Second parameter for the error message (NULL if no error)
ERROR_PARAMETER3	VARCHAR2(80)		Third parameter for the error message (NULL if no error)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ERROR_PARAMETER4	VARCHAR2(80)		Forth parameter for the error message (NULL if no error)
ERROR_PARAMETER5	VARCHAR2(80)		Fifth parameter for the error message (NULL if no error)
ERROR_MSG	VARCHAR2(4000)		Error message corresponding to ERROR_NUMBER. It will be NULL when ERROR_NUMBER is 0.

See Also

"USER_RESUMABLE"

5.289 DBA_REWRITE_EQUIVALENCES

DBA_REWRITE_EQUIVALENCES describes all rewrite equivalences in the database. Its columns are the same as those in ALL_REWRITE_EQUIVALENCES.

See Also

"ALL_REWRITE_EQUIVALENCES"

5.290 DBA_RGROUP

DBA_RGROUP displays all refresh groups.

Column	Datatype	NULL	Description
REFGROUP	NUMBER		Internal identifier of the refresh group
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object in the refresh group
NAME	VARCHAR2(128)	NOT NULL	Name of the object in the refresh group
IMPLICIT_DESTROY	VARCHAR2(1)		Indicates whether the refresh group is destroyed when its last item is removed (Y) or not (N)
PUSH_DEFERRED_RPC	VARCHAR2(1)		Indicates whether changes are pushed from the snapshot to the master before refresh (Y) or not (N)
REFRESH_AFTER_ERRORS	VARCHAR2(1)		Indicates whether to proceed with refresh despite errors when pushing deferred RPCs (Y) or not (N)
ROLLBACK_SEG	VARCHAR2(128)		Name of the rollback segment to use while refreshing
JOB	NUMBER	NOT NULL	Identifier of the job used to refresh the group automatically
PURGE_OPTION	NUMBER(38)		Method for purging the transaction queue after each push. 1 indicates quick purge option; 2 indicates precise purge option
PARALLELISM	NUMBER(38)		Level of parallelism for transaction propagation
HEAP_SIZE	NUMBER(38)		Size of the heap

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
JOB_NAME	VARCHAR2(128)		The name of the job used to automatically refresh the group

5.291 DBA_ROLE_PRIVS

DBA_ROLE_PRIVS describes the roles granted to all users and roles in the database.

Related View

USER_ROLE_PRIVS describes the roles granted to the current user.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)		Name of the user or role receiving the grant
GRANTED_ROLE	VARCHAR2(128)		Granted role name
ADMIN_OPTION	VARCHAR2(3)		Indicates whether the grant was with the ADMIN OPTION (YES) or not (NO)
DELEGATE_OPTION	VARCHAR2(3)		Indicates whether the grant was with the DELEGATE OPTION (YES) or not (NO)
DEFAULT_ROLE	VARCHAR2(3)		Indicates whether the role is designated as a DEFAULT ROLE for the user (YES) or not (NO)
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the role was granted commonly (CONTAINER=ALL was used) • NO if the role was granted locally (CONTAINER=ALL was not used)
INHERITED	VARCHAR2(3)		Indicates whether the role grant was inherited from another container (YES) or not (NO)

See Also

"USER_ROLE_PRIVS"

5.292 DBA_ROLES

DBA_ROLES describes all roles in the database.

Column	Datatype	NULL	Description
ROLE	VARCHAR2(128)	NOT NULL	Name of the role
ROLE_ID	NUMBER	NOT NULL	ID number of the role
PASSWORD_REQUIRED	VARCHAR2(8)		This column is deprecated in favor of the AUTHENTICATION_TYPE column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
AUTHENTICATION_TYPE	VARCHAR2(11)		Indicates the authentication mechanism for the role: • NONE - CREATE ROLE *role1*; • EXTERNAL - CREATE ROLE *role2* IDENTIFIED EXTERNALLY; • GLOBAL - CREATE ROLE *role3* IDENTIFIED GLOBALLY; • APPLICATION - CREATE ROLE *role4* IDENTIFIED USING *schema.package*; • PASSWORD - CREATE ROLE *role5* IDENTIFIED BY *role5*;
COMMON	VARCHAR2(3)		Indicates whether a given role is common. Possible values: • YES if the role is common • NO if the role is local (not common)
ORACLE_MAINTAINED	VARCHAR2(1)		Denotes whether the role was created, and is maintained, by Oracle-supplied scripts (such as catalog.sql or catproc.sql). A role for which this column has the value Y must not be changed in any way except by running an Oracle-supplied script.
INHERITED	VARCHAR2(3)		Indicates whether the role was inherited from another container (YES) or not (NO)
IMPLICIT	VARCHAR2(3)		Indicates whether the role is a common role created by an implicit application (YES) or not (NO)
EXTERNAL_NAME	VARCHAR2(4000)		For a global role, the external name refers to the DN of a group from a directory service that is mapped to the global role. This is not applicable to a local role.

5.293 DBA_ROLLBACK_SEGS

DBA_ROLLBACK_SEGS describes rollback segments.

Column	Datatype	NULL	Description
SEGMENT_NAME	VARCHAR2(30)	NOT NULL	Name of the rollback segment
OWNER	VARCHAR2(6)		Owner of the rollback segment: • PUBLIC • SYS
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of the tablespace containing the rollback segment
SEGMENT_ID	NUMBER	NOT NULL	ID number of the rollback segment
FILE_ID	NUMBER	NOT NULL	Absolute file number of the data file containing the segment header
BLOCK_ID	NUMBER	NOT NULL	ID number of the block containing the segment header
INITIAL_EXTENT	NUMBER		Initial extent size in bytes
NEXT_EXTENT	NUMBER		Secondary extent size in bytes
MIN_EXTENTS	NUMBER	NOT NULL	Minimum number of extents
MAX_EXTENTS	NUMBER	NOT NULL	Maximum number of extent
PCT_INCREASE	NUMBER		Percent increase for extent size

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(16)		Rollback segment status: • OFFLINE • ONLINE • NEEDS RECOVERY • PARTLY AVAILABLE • UNDEFINED
INSTANCE_NUM	VARCHAR2(40)		Rollback segment owning Oracle Real Application Clusters instance number
RELATIVE_FNO	NUMBER	NOT NULL	Relative file number of the segment header

5.294 DBA_ROLLING_DATABASES

DBA_ROLLING_DATABASES lists all the databases eligible for configuration with rolling operations.

Column	Datatype	NULL	Description
RDBID	NUMBER		Rolling operation database identifier
DBID	NUMBER		Oracle database identifier
DBUN	VARCHAR2(128)		Database unique name
ROLE	VARCHAR2(8)		Database role
OPEN_MODE	VARCHAR2(15)		Open mode information
PARTICIPANT	VARCHAR2(3)		Indicates whether the database is participating in the rolling operation (YES) or not (NO)
VERSION	VARCHAR2(128)		RDBMS version number
ENGINE_STATUS	VARCHAR2(14)		Running status of the MRP-recovery or LSP-apply process
RAC	VARCHAR2(3)		Indicates whether the database is an Oracle Real Application Clusters (Oracle RAC) database
UPDATE_PROGRESS	VARCHAR2(11)		Upgrade status of the system catalog
PROD_RSCN	VARCHAR(40)		Resetlogs SCN at which redo is currently being produced
PROD_RID	VARCHAR(40)		Resetlogs ID at which redo is currently being produced
PROD_SCN	VARCHAR(40)		Last SCN at which redo was produced
REDO_SOURCE	VARCHAR2(128)		Database unique name of the producer of redo being consumed
CONS_RSCN	VARCHAR(40)		Resetlogs SCN at which redo is currently being consumed
CONS_RID	VARCHAR(40)		Resetlogs ID at which redo is currently being consumed
CONS_SCN	VARCHAR(40)		Last SCN at which redo was consumed
UPDATE_TIME	TIMESTAMP(6)		Time of the last record update

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Data Guard Concepts and Administration* for more information about rolling operations.

5.295 DBA_ROLLING_EVENTS

DBA_ROLLING_EVENTS lists all the events reported from the DBMS_ROLLING PL/SQL package.

Column	Datatype	NULL	Description
EVENTID	NUMBER		Event identifier which identifies event order
EVENT_TIME	TIMESTAMP(6)		Time associated with the event
TYPE	VARCHAR2(7)		Type of event: INFO, NOTICE, WARNING, or ERROR
MESSAGE	VARCHAR2(256)		Text describing the event details
STATUS	NUMBER		Status code associated with an event
INSTID	NUMBER		Instruction ID associated with an event
REVISION	NUMBER		Plan revision number associated with an event

> **See Also**
>
> - *Oracle Data Guard Concepts and Administration* for more information about rolling operations.
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_ROLLING package

5.296 DBA_ROLLING_PARAMETERS

DBA_ROLLING_PARAMETERS lists the available parameters of the DBMS_ROLLING PL/SQL package.

Column	Datatype	NULL	Description
SCOPE	VARCHAR2(128)		Database unique name associated with a parameter
TYPE	VARCHAR2(7)		Type of parameter
NAME	VARCHAR2(32)		Name of the parameter
DESCRIPTION	VARCHAR2(256)		Description of the parameter
CURVAL	VARCHAR2(256)		Current value of the parameter
LSTVAL	VARCHAR2(256)		Prior value of the parameter
DEFVAL	VARCHAR2(256)		Default value of the parameter
MINVAL	NUMBER		Minimum value of the parameter
MAXVAL	NUMBER		Maximum value of the parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Data Guard Concepts and Administration* for more information about rolling operations.
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_ROLLING package

5.297 DBA_ROLLING_PLAN

DBA_ROLLING_PLAN displays the instructions which constitute the active upgrade plan.

Each row in DBA_ROLLING_PLAN identifies a specific instruction scheduled to execute at a specific database. Instructions are created as a result of successful calls to the DBMS_ROLLING.BUILD_PLAN procedure.

During execution, groups of instructions are scheduled in batches to execute at remote databases. Groups of instructions are guaranteed to complete in BATCHID order.

Column	Datatype	NULL	Description
REVISION	NUMBER		Plan revision number associated with an instruction
BATCHID	NUMBER		Identifier for a batch of instructions which are requested together
INSTID	NUMBER		Identifier for a single instruction
SOURCE	VARCHAR2(128)		Database unique name where an instruction executes
TARGET	VARCHAR2(128)		The site where a given instruction will execute
PHASE	VARCHAR2(14)		rolling operation phase in which an instruction executes
STATUS	VARCHAR2(7)		Scheduling status of the instruction
PROGRESS	VARCHAR2(10)		Execution progress of the instruction
DESCRIPTION	VARCHAR2(256)		Description of the instruction
EXEC_STATUS	NUMBER		Status code returned from instruction execution
EXEC_INFO	VARCHAR2(256)		Supplemental information obtained during instruction execution
EXEC_TIME	TIMESTAMP(6)		Time of instruction execution
FINISH_TIME	TIMESTAMP(6)		Time of instruction completion

See Also

- *Oracle Data Guard Concepts and Administration* for more information about rolling operations.
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_ROLLING package

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.298 DBA_ROLLING_STATISTICS

DBA_ROLLING_STATISTICS provides a list of rolling operation statistics.

Column	Datatype	NULL	Description
NAME	VARCHAR2(256)		Name of the statistic. Possible values: • DBMS_ROLLING execution time • logical to primary switchover finish time • primary services offline for • primary to logical switchover start time • rolling upgrade finish time • rolling upgrade start time • time for former primary to fully recover upgrade redo • time for former primary to start upgrade redo recovery • total time former primary in physical role • transient logical creation finish time • transient logical creation start time • transient logical protection finish time • transient logical protection start time • user upgrade time
VALUE	VARCHAR2(256)		Value of the statistic
UPDATE_TIME	TIMESTAMP(6)		Time of last update

See Also

Oracle Data Guard Concepts and Administration for more information about rolling operations.

5.299 DBA_ROLLING_STATUS

DBA_ROLLING_STATUS displays the overall status of the rolling operation.

Column	Datatype	NULL	Description
REVISION	NUMBER		Revision number of the current upgrade plan
STATUS	VARCHAR2(12)		Readiness of the facility to begin or resume the rolling operation
PHASE	VARCHAR2(14)		Current phase of the plan
NEXT_INSTRUCTION	NUMBER		Instruction ID of the next pending instruction
REMAINING_INSTRUCTIONS	NUMBER		Number of remaining instructions to execute in the plan
COORDINATOR_INSTANCE	NUMBER		Instance number from which the rolling operation is being coordinated
COORDINATOR_PID	NUMBER		Process PID in which the rolling operation is being coordinated

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ORIGINAL_PRIMARY	VARCHAR2(128)		Database unique name of the original primary
FUTURE_PRIMARY	VARCHAR2(128)		Database unique name of the future primary
TOTAL_DATABASES	NUMBER		Number of total databases eligible to participate in the rolling operation
PARTICIPATING_DATABASES	NUMBER		Number of databases configured to participate in the rolling operation
INIT_TIME	TIMESTAMP(6)		Time of the last call to DBMS_ROLLING.INIT_PLAN
BUILD_TIME	TIMESTAMP(6)		Time of the last call to DBMS_ROLLING.BUILD
START_TIME	TIMESTAMP(6)		Time of the last call to DBMS_ROLLING.START_UPGRADE
SWITCH_TIME	TIMESTAMP(6)		Time of the last call to DBMS_ROLLING.SWITCHOVER
FINISH_TIME	TIMESTAMP(6)		Time of the last call to DBMS_ROLLING.FINISH

See Also

- *Oracle Data Guard Concepts and Administration* for more information about rolling operations.
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_ROLLING package

5.300 DBA_ROLLING_UNSUPPORTED

DBA_ROLLING_UNSUPPORTED displays the schemas, tables, and columns in those tables that contain unsupported data types for a rolling upgrade operation for a logical standby database using the DBMS_ROLLING PL/SQL package.

Use this view before you perform a rolling upgrade using DBMS_ROLLING to determine what is unsupported.

The data pertains to the container in which the view is queried.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Schema name of the unsupported column
TABLE_NAME	VARCHAR2(128)		Name of the table that the unsupported column belongs to
COLUMN_NAME	VARCHAR2(128)		Name of the unsupported column
ATTRIBUTES	VARCHAR2(39)		If not a data type issue, displays the reason why the table is unsupported
DATA_TYPE	VARCHAR2(106)		Data type of the unsupported column

Note

A rolling upgrade using DBMS_ROLLING supports more object types than a manual rolling upgrade using transient logical standby databases

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_LOGSTDBY_UNSUPPORTED" for more information about determining unsupported data types for a manual rolling upgrade operation using transient logical standby databases
- *Oracle Data Guard Concepts and Administration* for more information about rolling operations
- *Oracle Data Guard Concepts and Administration* for more information about unsupported tables for rolling upgrade operations
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_ROLLING package

5.301 DBA_RSRC_CATEGORIES

DBA_RSRC_CATEGORIES displays all resource consumer group categories.

Column	Datatype	NULL	Description
NAME	VARCHAR2(128)		Name of the consumer group category
COMMENTS	VARCHAR2(2000)		Text comment on the consumer group category
STATUS	VARCHAR2(128)		Indicates whether the consumer group category is part of the pending area (PENDING) or not (NULL)
MANDATORY	VARCHAR2(3)		Indicates whether the consumer group category is mandatory (YES) or not (NO)

5.302 DBA_RSRC_CONSUMER_GROUP_PRIVS

DBA_RSRC_CONSUMER_GROUP_PRIVS displays information about all resource consumer groups and the users and roles assigned to them.

The grant referred to in this view and the related view is the grant of the SWITCH_CONSUMER_GROUP object privilege, which is granted using the DBMS_RESOURCE_MANAGER_PRIVS package. This privilege is not granted through the GRANT SQL statement.

Related View

USER_RSRC_CONSUMER_GROUP_PRIVS displays information about the resource consumer groups to which the current user is assigned. This view does not display the GRANTEE column.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)	NOT NULL	User or role receiving the grant
GRANTED_GROUP	VARCHAR2(128)		Granted consumer group name
GRANT_OPTION	VARCHAR2(3)		Indicates whether the grant was with the GRANT option (YES) or not (NO)
INITIAL_GROUP	VARCHAR2(3)		Indicates whether the consumer group is designated as the default for this user or role (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "USER_RSRC_CONSUMER_GROUP_PRIVS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_RESOURCE_MANAGER_PRIVS package

5.303 DBA_RSRC_CONSUMER_GROUPS

DBA_RSRC_CONSUMER_GROUPS displays information about all resource consumer groups in the database.

Column	Datatype	NULL	Description
CONSUMER_GROUP_ID	NUMBER	NOT NULL	ID of the consumer group
CONSUMER_GROUP	VARCHAR2(128)		Name of the consumer group
CPU_METHOD	VARCHAR2(128)		CPU resource allocation method for the consumer group
MGMT_METHOD	VARCHAR2(128)		Resource allocation method for the consumer group
INTERNAL_USE	VARCHAR2(3)		Indicates whether the consumer group is for internal use only (YES) or not (NO)
COMMENTS	VARCHAR2(2000)		Text comment on the consumer group
CATEGORY	VARCHAR2(128)		Category of the consumer group
STATUS	VARCHAR2(128)		Indicates whether the consumer group is part of the pending area (PENDING) or not (NULL)
MANDATORY	VARCHAR2(3)		Indicates whether the consumer group is mandatory (YES) or not (NO)

5.304 DBA_RSRC_GROUP_MAPPINGS

DBA_RSRC_GROUP_MAPPINGS displays the mapping between session attributes and consumer groups in the database.

Column	Datatype	NULL	Description
ATTRIBUTE	VARCHAR2(128)		Session attribute to match
VALUE	VARCHAR2(128)		Attribute value
CONSUMER_GROUP	VARCHAR2(128)		Target consumer group name
STATUS	VARCHAR2(128)		Indicates whether the consumer group is part of the pending area (PENDING) or not (NULL)

5.305 DBA_RSRC_IO_CALIBRATE

DBA_RSRC_IO_CALIBRATE displays I/O calibration results for the latest calibration run.

Column	Datatype	NULL	Description
START_TIME	TIMESTAMP(6)		Start time of the most recent I/O calibration

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
END_TIME	TIMESTAMP(6)		End time of the most recent I/O calibration
MAX_IOPS	NUMBER		Maximum number of data block read requests that can be sustained per second
MAX_MBPS	NUMBER		Maximum megabytes per second of maximum-sized read requests that can be sustained
MAX_PMBPS	NUMBER		Maximum megabytes per second of large I/O requests that can be sustained by a single process
LATENCY	NUMBER		Latency for data block read requests
NUM_PHYSICAL_DISKS	NUMBER		Number of physical disks in the storage subsystem (as specified by the user)
ADDITIONAL_INFO	VARCHAR2(1024)		Additional information about the most recent calibration run

5.306 DBA_RSRC_MANAGER_SYSTEM_PRIVS

DBA_RSRC_MANAGER_SYSTEM_PRIVS displays information about all the users and roles that have been granted the ADMINISTER_RESOURCE_MANAGER system privilege, which is granted using the DBMS_RESOURCE_MANAGER_PRIVS package.

This privilege is not granted through the GRANT SQL statement.

Related View

USER_RSRC_MANAGER_SYSTEM_PRIVS displays information about the users who are granted system privileges for the DBMS_RESOURCE_MANAGER package. This view does not display the GRANTEE column.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)	NOT NULL	User or role receiving the grant
PRIVILEGE	VARCHAR2(40)	NOT NULL	Name of the system privilege
ADMIN_OPTION	VARCHAR2(3)		Indicates whether the grant was with the ADMIN option (YES) or not (NO)

See Also

- "USER_RSRC_MANAGER_SYSTEM_PRIVS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_RESOURCE_MANAGER package

5.307 DBA_RSRC_MAPPING_PRIORITY

DBA_RSRC_MAPPING_PRIORITY displays information about all consumer group mapping attribute priorities.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ATTRIBUTE	VARCHAR2(128)		Session attribute
PRIORITY	NUMBER		Priority (1 is the highest)
STATUS	VARCHAR2(128)		Indicates whether the consumer group is part of the pending area (PENDING) or not (NULL)

5.308 DBA_RSRC_PLAN_DIRECTIVES

DBA_RSRC_PLAN_DIRECTIVES displays information about all resource plan directives in the database.

Column	Datatype	NULL	Description
PLAN	VARCHAR2(128)		Name of the plan to which the directive belongs
GROUP_OR_SUBPLAN	VARCHAR2(128)		Name of the consumer group or subplan referred to
TYPE	VARCHAR2(14)		Indicates whether GROUP_OR_SUBPLAN refers to a consumer group (CONSUMER_GROUP) or a plan (PLAN)
CPU_P1	NUMBER		This column is deprecated. Use the MGMT_P1 column instead.
CPU_P2	NUMBER		This column is deprecated. Use the MGMT_P2 column instead.
CPU_P3	NUMBER		This column is deprecated. Use the MGMT_P3 column instead.
CPU_P4	NUMBER		This column is deprecated. Use the MGMT_P4 column instead.
CPU_P5	NUMBER		This column is deprecated. Use the MGMT_P5 column instead.
CPU_P6	NUMBER		This column is deprecated. Use the MGMT_P6 column instead.
CPU_P7	NUMBER		This column is deprecated. Use the MGMT_P7 column instead.
CPU_P8	NUMBER		This column is deprecated. Use the MGMT_P8 column instead.
MGMT_P1	NUMBER		Resource allocation at level 1. For share-based plans, indicates the number of shares.
MGMT_P2	NUMBER		Resource allocation at level 2.
MGMT_P3	NUMBER		Resource allocation at level 3.
MGMT_P4	NUMBER		Resource allocation at level 4.
MGMT_P5	NUMBER		Resource allocation at level 5.
MGMT_P6	NUMBER		Resource allocation at level 6.
MGMT_P7	NUMBER		Resource allocation at level 7.
MGMT_P8	NUMBER		Resource allocation at level 8.
ACTIVE_SESS_POOL_P1	NUMBER		Maximum number of calls this consumer group can run concurrently
QUEUEING_P1	NUMBER		Timeout in seconds for waits in the Active Session Limit queue

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PARALLEL_TARGET_PERCENTAGE	NUMBER		This column is deprecated. Use the PARALLEL_SERVER_LIMIT column instead.
PARALLEL_DEGREE_LIMIT_P1	NUMBER		Sessions in this consumer group are limited to this maximum degree of parallelism for all parallel operations
SWITCH_GROUP	VARCHAR2(128)		Group to switch to once the switch time is reached
SWITCH_FOR_CALL	VARCHAR2(5)		Indicates whether to switch back to the initial consumer group once the top call has completed (TRUE) or not (FALSE)
SWITCH_TIME	NUMBER		Amount of run time (in seconds) within a group before the session is automatically switched. As with other switch directives, if SWITCH_FOR_CALL is TRUE, the run time is accumulated from the start of a call. Otherwise, the run time is accumulated for the length of the session.
SWITCH_IO_MEGABYTES	NUMBER		The maximum megabytes of I/O within a group that will trigger the action specified by SWITCH_GROUP. As with other switch directives, if SWITCH_FOR_CALL is TRUE, the maximum megabytes of I/O is accumulated from the start of a call. Otherwise, the maximum megabytes of I/O is accumulated for the length of the session.
SWITCH_IO_REQS	NUMBER		The maximum I/O requests within a group that will trigger the action specified by SWITCH_GROUP. As with other switch directives, if SWITCH_FOR_CALL is TRUE, the maximum I/O requests is accumulated from the start of a call. Otherwise, the maximum I/O requests is accumulated for the length of the session.
SWITCH_ESTIMATE	VARCHAR2(5)		Indicates whether estimated execution time should be used for switch criteria (TRUE) or not (FALSE)
MAX_EST_EXEC_TIME	NUMBER		Maximum estimated execution time
UNDO_POOL	NUMBER		Undo pool size for the consumer group
MAX_IDLE_TIME	NUMBER		Maximum idle time for the session
MAX_IDLE_BLOCKER_TIME	NUMBER		Maximum idle time for the session when blocking other sessions
MAX_UTILIZATION_LIMIT	NUMBER		This column is deprecated. Use UTILIZATION_LIMIT instead.
PARALLEL_QUEUE_TIMEOUT	NUMBER		Time (in seconds) that a query can remain in the parallel statement queue for the consumer group before it is removed from the queue. The PQ_TIMEOUT_ACTION directive in a Resource Manager plan can be used in conjunction with the PARALLEL_QUEUE_TIMEOUT directive to either cancel or run the removed query. If the PQ_TIMEOUT_ACTION directive is not specified, the default behavior is to cancel the query with ORA-07454.
SWITCH_TIME_IN_CALL	NUMBER		This column is deprecated. Specify the time in the SWITCH_TIME directive and set SWITCH_FOR_CALL to TRUE.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SWITCH_IO_LOGICAL	NUMBER		The number of logical I/Os that will trigger the action specified by SWITCH_GROUP. As with other switch directives, if SWITCH_FOR_CALL is TRUE, the number of logical I/Os is accumulated from the start of a call. Otherwise, the number of logical IOs is accumulated for the length of the session.
SWITCH_ELAPSED_TIME	NUMBER		The elapsed time that will trigger the action specified by SWITCH_GROUP. As with other switch directives, if SWITCH_FOR_CALL is TRUE, the elapsed time is accumulated from the start of a call. Otherwise, the elapsed time is accumulated for the length of the session.
PARALLEL_SERVER_LIMIT	NUMBER		Maximum percentage of the parallel target used before queuing subsequent parallel queries
UTILIZATION_LIMIT	NUMBER		Maximum resource utilization allowed, expressed in percentage
PARALLEL_STMT_CRITICAL	VARCHAR2(12)		Indicates whether parallel statements from this consumer group are eligible for queuing in the parallel statement queue: • BYPASS QUEUE - Parallel statements in this consumer group are critical and should never be queued. • QUEUE - All parallel statements in this consumer group, irrespective of the PARALLEL_DEGREE_POLICY initialization parameter value, are eligible for being queued. • FALSE - Certain parallel statements are eligible for queuing, depending on the PARALLEL_DEGREE_POLICY initialization parameter value. This is the default.
SESSION_PGA_LIMIT	NUMBER		The maximum amount of untunable PGA in MB that sessions in this consumer group can allocate before being terminated
PQ_TIMEOUT_ACTION	VARCHAR2(6)		Indicates the action to be taken on a parallel query in the parallel queue when its queue time exceeds the limit set by the Resource Manager plan's PARALLEL_QUEUE_TIMEOUT directive: • CANCEL - The statement terminates with error ORA-07454 • RUN - The statement runs immediately, and may get downgraded if parallel servers are unavailable
COMMENTS	VARCHAR2(2000)		Text comment on the plan directive
STATUS	VARCHAR2(128)		Indicates whether the plan directive is part of the pending area (PENDING) or not (NULL). **Note:** PDB resource plans must be single-level, they cannot contain subplans, and they must have 8 or fewer consumer groups. If a resource plan is imported into a PDB and it violates any of these PDB requirements, then the import will automatically convert the resource plan to a compliant version. The original, unmodified resource plan will be stored with a STATUS of LEGACY.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MANDATORY	VARCHAR2(3)		Indicates whether the plan directive is mandatory (YES) or not (NO). Mandatory plans cannot be deleted.

See Also

- "PARALLEL_DEGREE_POLICY"
- *Oracle Database Administrator's Guide* for information on resource plans in general
- *Oracle Database PL/SQL Packages and Types Reference* for information about specifying Resource Manager directive values using the DBMS_RESOURCE_MANAGER.CREATE_PLAN_DIRECTIVE procedure

5.309 DBA_RSRC_PLANS

DBA_RSRC_PLANS displays information about all resource plans in the database.

For a list of currently active plans, see "V$RSRC_PLAN".

Column	Datatype	NULL	Description
PLAN_ID	NUMBER	NOT NULL	ID number of the resource plan
PLAN	VARCHAR2(128)		Name of the resource plan
NUM_PLAN_DIRECTIVES	NUMBER		Number of plan directives for the plan
CPU_METHOD	VARCHAR2(128)		CPU resource allocation method for the plan
MGMT_METHOD	VARCHAR2(128)		Resource allocation method for the plan
ACTIVE_SESS_POOL_MTH	VARCHAR2(128)		Active session pool resource allocation method for the plan
PARALLEL_DEGREE_LIMIT_MTH	VARCHAR2(128)		Parallel degree limit resource allocation method for the plan
QUEUING_MTH	VARCHAR2(128)		Queuing resource allocation method for the plan
SUB_PLAN	VARCHAR2(3)		Indicates whether the plan is a subplan (YES) or not (NO)
COMMENTS	VARCHAR2(2000)		Text comment on the plan
STATUS	VARCHAR2(128)		Indicates whether the plan is part of the pending area (PENDING) or not (NULL)
MANDATORY	VARCHAR2(3)		Indicates whether the plan is mandatory (YES) or not (NO). Mandatory plans cannot be deleted.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database Administrator's Guide* for information on resource plans in general
- *Oracle Database PL/SQL Packages and Types Reference* for more information on creating resource plans with the `DBMS_RESOURCE_MANAGER` package

5.310 DBA_RULE_SET_RULES

`DBA_RULE_SET_RULES` describes the rules in all rule sets in the database. Its columns are the same as those in `ALL_RULE_SET_RULES`.

See Also

"ALL_RULE_SET_RULES"

5.311 DBA_RULE_SETS

`DBA_RULE_SETS` describes all rule sets in the database. Its columns are the same as those in `ALL_RULE_SETS`.

See Also

"ALL_RULE_SETS"

5.312 DBA_RULES

`DBA_RULES` describes all rules in the database. Its columns are the same as those in `ALL_RULES`.

See Also

"ALL_RULES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.313 DBA_SCHEDULER_CHAIN_RULES

DBA_SCHEDULER_CHAIN_RULES displays information about the rules for all chains in the database. Its columns are the same as those in ALL_SCHEDULER_CHAIN_RULES.

See Also

"ALL_SCHEDULER_CHAIN_RULES"

5.314 DBA_SCHEDULER_CHAIN_STEPS

DBA_SCHEDULER_CHAIN_STEPS displays information about the defined steps of all chains in the database. Its columns are the same as those in ALL_SCHEDULER_CHAIN_STEPS.

See Also

"ALL_SCHEDULER_CHAIN_STEPS"

5.315 DBA_SCHEDULER_CHAINS

DBA_SCHEDULER_CHAINS displays information about all chains in the database. Its columns are the same as those in ALL_SCHEDULER_CHAINS.

See Also

"ALL_SCHEDULER_CHAINS"

5.316 DBA_SCHEDULER_CREDENTIALS

DBA_SCHEDULER_CREDENTIALS displays information about all credentials in the database. Its columns are the same as those in ALL_SCHEDULER_CREDENTIALS.

Note

This view is deprecated in favor of the DBA_CREDENTIALS view. Oracle recommends that you use DBA_CREDENTIALS instead. DBA_SCHEDULER_CREDENTIALS is retained for backward compatibility only.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_CREDENTIALS"
- "ALL_SCHEDULER_CREDENTIALS"

5.317 DBA_SCHEDULER_DB_DESTS

DBA_SCHEDULER_DB_DESTS displays information about all destination objects in the database pointing to remote databases. Its columns are the same as those in ALL_SCHEDULER_DB_DESTS.

See Also

"ALL_SCHEDULER_DB_DESTS"

5.318 DBA_SCHEDULER_DESTS

DBA_SCHEDULER_DESTS displays information about all destination objects for jobs in the database. Its columns are the same as those in ALL_SCHEDULER_DESTS.

See Also

"ALL_SCHEDULER_DESTS"

5.319 DBA_SCHEDULER_EXTERNAL_DESTS

DBA_SCHEDULER_EXTERNAL_DESTS displays information about all destination objects in the database pointing to remote agents. Its columns are the same as those in ALL_SCHEDULER_EXTERNAL_DESTS.

See Also

"ALL_SCHEDULER_EXTERNAL_DESTS"

5.320 DBA_SCHEDULER_FILE_WATCHERS

DBA_SCHEDULER_FILE_WATCHERS displays information about all scheduler file watch requests in the database. Its columns are the same as those in ALL_SCHEDULER_FILE_WATCHERS.

See Also

"ALL_SCHEDULER_FILE_WATCHERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.321 DBA_SCHEDULER_GLOBAL_ATTRIBUTE

DBA_SCHEDULER_GLOBAL_ATTRIBUTE displays the values of all scheduler attributes in the database (for example, DEFAULT_TIMEZONE and CURRENT_OPEN_WINDOW). Its columns are the same as those in ALL_SCHEDULER_GLOBAL_ATTRIBUTE.

See Also

"ALL_SCHEDULER_GLOBAL_ATTRIBUTE"

5.322 DBA_SCHEDULER_GROUP_MEMBERS

DBA_SCHEDULER_GROUP_MEMBERS displays information about the members of all Scheduler object groups in the database. Its columns are the same as those in ALL_SCHEDULER_GROUP_MEMBERS.

See Also

"ALL_SCHEDULER_GROUP_MEMBERS"

5.323 DBA_SCHEDULER_GROUPS

DBA_SCHEDULER_GROUPS displays information about all Scheduler object groups in the database. Its columns are the same as those in ALL_SCHEDULER_GROUPS.

See Also

"ALL_SCHEDULER_GROUPS"

5.324 DBA_SCHEDULER_INCOMPAT_MEMBER

DBA_SCHEDULER_INCOMPAT_MEMBER displays all Scheduler incompatibility resource objects members in the database. Its columns are the same as those in ALL_SCHEDULER_INCOMPAT_MEMBER.

See Also

"ALL_SCHEDULER_INCOMPAT_MEMBER"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.325 DBA_SCHEDULER_INCOMPATS

DBA_SCHEDULER_INCOMPATS displays all Scheduler incompatibility resource objects in the database. Its columns are the same as those in ALL_SCHEDULER_INCOMPATS.

See Also

"ALL_SCHEDULER_INCOMPATS"

5.326 DBA_SCHEDULER_JOB_ARGS

DBA_SCHEDULER_JOB_ARGS displays information about the arguments of all Scheduler jobs in the database. Its columns are the same as those in ALL_SCHEDULER_JOB_ARGS.

See Also

"ALL_SCHEDULER_JOB_ARGS"

5.327 DBA_SCHEDULER_JOB_CLASSES

DBA_SCHEDULER_JOB_CLASSES displays information about all Scheduler job classes in the database. Its columns are the same as those in ALL_SCHEDULER_JOB_CLASSES.

See Also

"ALL_SCHEDULER_JOB_CLASSES"

5.328 DBA_SCHEDULER_JOB_DESTS

DBA_SCHEDULER_JOB_DESTS displays information about the state of all jobs in the database at each of their destinations. Its columns are the same as those in ALL_SCHEDULER_JOB_DESTS.

See Also

"ALL_SCHEDULER_JOB_DESTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.329 DBA_SCHEDULER_JOB_LOG

DBA_SCHEDULER_JOB_LOG displays log information for all Scheduler jobs in the database. Its columns are the same as those in ALL_SCHEDULER_JOB_LOG.

> **See Also**
>
> "ALL_SCHEDULER_JOB_LOG"

5.330 DBA_SCHEDULER_JOB_ROLES

DBA_SCHEDULER_JOB_ROLES displays information about all Scheduler jobs in the database by database role.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the Scheduler job
JOB_NAME	VARCHAR2(128)	NOT NULL	Name of the Scheduler job
JOB_SUBNAME	VARCHAR2(128)		Subname of the Scheduler job (for a job running a chain step)
JOB_CREATOR	VARCHAR2(128)		Creator of the Scheduler job
DATABASE_ROLE	VARCHAR2(16)		Name of the database role
PROGRAM_OWNER	VARCHAR2(4000)		Owner of the program associated with the job
PROGRAM_NAME	VARCHAR2(4000)		Name of the program associated with the job
JOB_TYPE	VARCHAR2(16)		Inline job action type: • PLSQL_BLOCK • STORED_PROCEDURE • EXECUTABLE • CHAIN
JOB_ACTION	VARCHAR2(4000)		Inline job action
JOB_CLASS	VARCHAR2(128)		Name of the job class associated with the job
SCHEDULE_OWNER	VARCHAR2(4000)		Owner of the schedule that the job uses (can be a window or a window group)
SCHEDULE_NAME	VARCHAR2(4000)		Name of the schedule that the job uses (can be a window or a window group)
SCHEDULE_TYPE	VARCHAR2(12)		Type of the schedule that the job uses: • IMMEDIATE - Start date and repeat interval are NULL • ONCE - Repeat interval is NULL • PLSQL - PL/SQL expression used as schedule • CALENDAR - Oracle calendaring expression used as schedule • EVENT - Event schedule • NAMED - Named schedule • WINDOW - Window used as schedule • WINDOW_GROUP - Window group used as schedule

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
START_DATE	TIMESTAMP(6) WITH TIME ZONE		Original scheduled start date of the job (for an inline schedule)
REPEAT_INTERVAL	VARCHAR2(4000)		Inline schedule PL/SQL expression or calendar string
END_DATE	TIMESTAMP(6) WITH TIME ZONE		Date after which the job will no longer run (for an inline schedule)
LAST_START_DATE	TIMESTAMP(6) WITH TIME ZONE		Last date on which the job ran
ENABLED	VARCHAR2(5)		Indicates whether the job is enabled (TRUE) or disabled (FALSE)
STATE	VARCHAR2(15)		Current state of the job: • DISABLED • RETRY SCHEDULED • SCHEDULED • RUNNING • COMPLETED • BROKEN • FAILED • REMOTE • SUCCEEDED • CHAIN_STALLED
COMMENTS	VARCHAR2(4000)		Comments on the job

5.331 DBA_SCHEDULER_JOB_RUN_DETAILS

DBA_SCHEDULER_JOB_RUN_DETAILS displays log run details for all Scheduler jobs in the database. Its columns are the same as those in ALL_SCHEDULER_JOB_RUN_DETAILS.

See Also

"ALL_SCHEDULER_JOB_RUN_DETAILS"

5.332 DBA_SCHEDULER_JOBS

DBA_SCHEDULER_JOBS displays information about all Scheduler jobs in the database. Its columns are the same as those in ALL_SCHEDULER_JOBS.

See Also

"ALL_SCHEDULER_JOBS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.333 DBA_SCHEDULER_NOTIFICATIONS

DBA_SCHEDULER_NOTIFICATIONS displays information about the E-mail notifications for all jobs in the database. Its columns are the same as those in ALL_SCHEDULER_NOTIFICATIONS.

See Also

"ALL_SCHEDULER_NOTIFICATIONS"

5.334 DBA_SCHEDULER_PROGRAM_ARGS

DBA_SCHEDULER_PROGRAM_ARGS displays information about the arguments of all Scheduler programs in the database. Its columns are the same as those in ALL_SCHEDULER_PROGRAM_ARGS.

See Also

"ALL_SCHEDULER_PROGRAM_ARGS"

5.335 DBA_SCHEDULER_PROGRAMS

DBA_SCHEDULER_PROGRAMS displays information about all Scheduler programs in the database. Its columns are the same as those in ALL_SCHEDULER_PROGRAMS.

See Also

"ALL_SCHEDULER_PROGRAMS"

5.336 DBA_SCHEDULER_REMOTE_DATABASES

DBA_SCHEDULER_REMOTE_DATABASES displays information about all remote databases that have been registered as sources and destinations for remote database jobs. Its columns are the same as those in ALL_SCHEDULER_REMOTE_DATABASES.

See Also

"ALL_SCHEDULER_REMOTE_DATABASES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.337 DBA_SCHEDULER_REMOTE_JOBSTATE

DBA_SCHEDULER_REMOTE_JOBSTATE displays information about the state of all jobs at remote databases. Its columns are the same as those in ALL_SCHEDULER_REMOTE_JOBSTATE.

See Also

"ALL_SCHEDULER_REMOTE_JOBSTATE"

5.338 DBA_SCHEDULER_RESOURCES

DBA_SCHEDULER_RESOURCES displays all scheduler resource objects in the database. Its columns are the same as those in ALL_SCHEDULER_RESOURCES.

See Also

"ALL_SCHEDULER_RESOURCES"

5.339 DBA_SCHEDULER_RSC_CONSTRAINTS

DBA_SCHEDULER_RSC_CONSTRAINTS lists all Oracle Scheduler resource constraint members in the database. Its columns are the same as those in ALL_SCHEDULER_RSC_CONSTRAINTS.

See Also

"ALL_SCHEDULER_RSC_CONSTRAINTS"

5.340 DBA_SCHEDULER_RUNNING_CHAINS

DBA_SCHEDULER_RUNNING_CHAINS displays information about the chain steps of all running chains in the database. Its columns are the same as those in ALL_SCHEDULER_RUNNING_CHAINS.

See Also

"ALL_SCHEDULER_RUNNING_CHAINS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.341 DBA_SCHEDULER_RUNNING_JOBS

DBA_SCHEDULER_RUNNING_JOBS displays information about all running Scheduler jobs in the database. Its columns are the same as those in ALL_SCHEDULER_RUNNING_JOBS.

See Also

"ALL_SCHEDULER_RUNNING_JOBS"

5.342 DBA_SCHEDULER_SCHEDULES

DBA_SCHEDULER_SCHEDULES displays information about all Scheduler schedules in the database. Its columns are the same as those in ALL_SCHEDULER_SCHEDULES.

See Also

"ALL_SCHEDULER_SCHEDULES"

5.343 DBA_SCHEDULER_WINDOW_DETAILS

DBA_SCHEDULER_WINDOW_DETAILS displays log details for all Scheduler windows in the database. Its columns are the same as those in ALL_SCHEDULER_WINDOW_DETAILS.

See Also

"ALL_SCHEDULER_WINDOW_DETAILS"

5.344 DBA_SCHEDULER_WINDOW_GROUPS

DBA_SCHEDULER_WINDOW_GROUPS displays information about all Scheduler window groups in the database. Its columns are the same as those in ALL_SCHEDULER_WINDOW_GROUPS.

See Also

"ALL_SCHEDULER_WINDOW_GROUPS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.345 DBA_SCHEDULER_WINDOW_LOG

DBA_SCHEDULER_WINDOW_LOG displays log information for all Scheduler windows in the database. Its columns are the same as those in ALL_SCHEDULER_WINDOW_LOG.

See Also

"ALL_SCHEDULER_WINDOW_LOG"

5.346 DBA_SCHEDULER_WINDOWS

DBA_SCHEDULER_WINDOWS displays information about all Scheduler windows in the database. Its columns are the same as those in ALL_SCHEDULER_WINDOWS.

See Also

"ALL_SCHEDULER_WINDOWS"

5.347 DBA_SCHEDULER_WINGROUP_MEMBERS

DBA_SCHEDULER_WINGROUP_MEMBERS displays the members of all Scheduler window groups in the database. Its columns are the same as those in ALL_SCHEDULER_WINGROUP_MEMBERS.

See Also

"ALL_SCHEDULER_WINGROUP_MEMBERS"

5.348 DBA_SEC_RELEVANT_COLS

DBA_SEC_RELEVANT_COLS describes the security relevant columns of all security policies in the database. Its columns are the same as those in ALL_SEC_RELEVANT_COLS.

See Also

"ALL_SEC_RELEVANT_COLS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.349 DBA_SECONDARY_OBJECTS

DBA_SECONDARY_OBJECTS provides information about all secondary objects that are associated with domain indexes in the database.

This view is only relevant in the context of domain indexes. Its columns are the same as those in "ALL_SECONDARY_OBJECTS".

5.350 DBA_SEGMENTS

DBA_SEGMENTS describes the storage allocated for all segments in the database.

Related View

USER_SEGMENTS describes the storage allocated for the segments owned by the current user's objects. This view does not display the OWNER, HEADER_FILE, HEADER_BLOCK, or RELATIVE_FNO columns.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Username of the segment owner
SEGMENT_NAME	VARCHAR2(128)		Name, if any, of the segment
PARTITION_NAME	VARCHAR2(128)		Object Partition Name (Set to NULL for nonpartitioned objects)
SEGMENT_TYPE	VARCHAR2(18)		Type of segment: • NESTED TABLE • TABLE • TABLE PARTITION • CLUSTER • LOBINDEX • INDEX • INDEX PARTITION • LOBSEGMENT • TABLE SUBPARTITION • INDEX SUBPARTITION • LOB PARTITION • LOB SUBPARTITION • ROLLBACK • TYPE2 UNDO • DEFERRED ROLLBACK • TEMPORARY • CACHE • SPACE HEADER • UNDEFINED
SEGMENT_SUBTYPE	VARCHAR2(10)		Subtype of LOB segment: SECUREFILE, ASSM, MSSM, and NULL
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the segment
HEADER_FILE	NUMBER		Absolute file number of the data file containing the segment header
HEADER_BLOCK	NUMBER		ID of the block containing the segment header
BYTES	NUMBER		Size, in bytes, of the segment

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BLOCKS	NUMBER		Size, in Oracle blocks, of the segment
EXTENTS	NUMBER		Number of extents allocated to the segment
INITIAL_EXTENT	NUMBER		Size in bytes requested for the initial extent of the segment at create time. (Oracle rounds the extent size to multiples of 5 blocks if the requested size is greater than 5 blocks.)
NEXT_EXTENT	NUMBER		Size in bytes of the next extent to be allocated to the segment
MIN_EXTENTS	NUMBER		Minimum number of extents allowed in the segment
MAX_EXTENTS	NUMBER		Maximum number of extents allowed in the segment
MAX_SIZE	NUMBER		Maximum number of blocks allowed in the segment
RETENTION	VARCHAR2(7)		Retention option for SECUREFILE segment
MINRETENTION	NUMBER		Minimum retention duration for SECUREFILE segment
PCT_INCREASE	NUMBER		Percent by which to increase the size of the next extent to be allocated
FREELISTS	NUMBER		Number of process freelists allocated to this segment
FREELIST_GROUPS	NUMBER		Number of freelist groups allocated to this segment
RELATIVE_FNO	NUMBER		Relative file number of the segment header
BUFFER_POOL	VARCHAR2(7)		Buffer pool to be used for segment blocks: • DEFAULT • KEEP • RECYCLE
FLASH_CACHE	VARCHAR2(7)		Database Smart Flash Cache hint to be used for segment blocks: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)		Cell flash cache hint to be used for segment blocks: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
INMEMORY	VARCHAR2(8)		Indicates whether the In-Memory Column Store (IM column store) is enabled (ENABLED) or disabled (DISABLED) for this segment
INMEMORY_PRIORITY	VARCHAR2(8)		Indicates the priority for In-Memory Column Store (IM column store) population: • LOW • MEDIUM • HIGH • CRITICAL • NONE • NULL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INMEMORY_DISTRIBUTE	VARCHAR2(15)		Indicates how the IM column store is distributed in an Oracle Real Application Clusters (Oracle RAC) environment: • AUTO • BY ROWID RANGE • BY PARTITION • BY SUBPARTITION
INMEMORY_DUPLICATE	VARCHAR2(13)		Indicates the duplicate setting for the IM column store in an Oracle RAC environment: • NO DUPLICATE • DUPLICATE • DUPLICATE ALL
INMEMORY_COMPRESSION	VARCHAR2(17)		Indicates the compression level for the IM column store: • NO MEMCOMPRESS • FOR DML • FOR QUERY [LOW | HIGH] • FOR CAPACITY [LOW | HIGH] • NULL
CELLMEMORY[11]	VARCHAR2(24)		The value for columnar compression in the storage cell flash cache. Possible values: • ENABLED: Oracle Exadata Storage will decide automatically whether to cache in columnar form • DISABLED: Oracle Exadata Storage is prevented from caching in columnar form • NO CACHECOMPRESS: Oracle Exadata Storage will cache in HCC format (no recompression) • FOR QUERY: Oracle Exadata Storage will recompress and cache in INMEMORY query high format • FOR CAPACITY: Oracle Exadata Storage will recompress and cache in INMEMORY capacity low format

1 This column is intended for use with Oracle Exadata

See Also

"USER_SEGMENTS"

5.351 DBA_SEGMENTS_OLD

DBA_SEGMENTS_OLD lists information about storage allocated for all database segments.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Username of the segment owner
SEGMENT_NAME	VARCHAR2(128)		Name, if any, of the segment
PARTITION_NAME	VARCHAR2(128)		Name of the partition

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SEGMENT_TYPE	VARCHAR2(18)		Type of segment: INDEX PARTITION, TABLE PARTITION, TABLE, CLUSTER, INDEX, ROLLBACK, DEFERRED ROLLBACK, TEMPORARY, CACHE, LOBSEGMENT and LOBINDEX
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the segment
HEADER_FILE	NUMBER		ID of the file containing the segment header
HEADER_BLOCK	NUMBER		ID of the block containing the segment header
BYTES	NUMBER		Size, in bytes, of the segment
BLOCKS	NUMBER		Size, in Oracle blocks, of the segment
EXTENTS	NUMBER		Number of extents allocated to the segment
INITIAL_EXTENT	NUMBER		Size in bytes requested for the initial extent of the segment at create time. (Oracle rounds the extent size to multiples of 5 blocks if the requested size is greater than 5 blocks.)
NEXT_EXTENT	NUMBER		Size in bytes of the next extent to be allocated to the segment
MIN_EXTENTS	NUMBER		Minimum number of extents allowed in the segment
MAX_EXTENTS	NUMBER		Maximum number of extents allowed in the segment
PCT_INCREASE	NUMBER		Percent by which to increase the size of the next extent to be allocated
FREELISTS	NUMBER		Number of process freelists allocated to the segment
FREELIST_GROUPS	NUMBER		Number of freelist groups allocated to this segment
RELATIVE_FNO	NUMBER		Relative file number of the segment header
BUFFER_POOL	VARCHAR2(7)		Buffer pool for the object

5.352 DBA_SENSITIVE_COLUMN_TYPES

DBA_SENSITIVE_COLUMN_TYPES describes sensitive column types in the database.

Column	Datatype	NULL	Description
NAME	VARCHAR2(128)		The name of the sensitive column type
USER_COMMENT	VARCHAR2(4000)		User comment on the sensitive column type
SOURCE_NAME	VARCHAR2(128)		The name of the discovery source for the sensitive column type
SOURCE_TYPE	VARCHAR2(3)		The type of the discovery source: • ADM: import from ADM • DB: added within the database

See Also

Oracle Database Security Guide for more information about transparent sensitive data protection

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.353 DBA_SENSITIVE_DATA

DBA_SENSITIVE_DATA describes the sensitive columns in the database.

Column	Datatype	NULL	Description
SENSITIVE#	NUMBER	NOT NULL	Dictionary ID for the sensitive data
SCHEMA_NAME	VARCHAR2(128)	NOT NULL	The schema containing the sensitive data
TABLE_NAME	VARCHAR2(128)	NOT NULL	The table containing the sensitive data
COLUMN_NAME	VARCHAR2(128)	NOT NULL	The name of the column identified as sensitive
SENSITIVE_TYPE	VARCHAR2(128)		The sensitive column type of the data
SOURCE_NAME	VARCHAR2(128)		The name of the discovery source for the sensitive data
USER_COMMENT	VARCHAR2(4000)		User comment on the sensitive data
TS	TIMESTAMP(6)		The time when the data was identified as sensitive in the database

See Also

Oracle Database Security Guide for more information about transparent sensitive data protection

5.354 DBA_SEQUENCES

DBA_SEQUENCES describes all sequences in the database. Its columns are the same as those in ALL_SEQUENCES.

See Also

"ALL_SEQUENCES"

5.355 DBA_SERVER_REGISTRY

DBA_SERVER_REGISTRY displays information about all server components in the database that are loaded into the component registry.

Column	Datatype	NULL	Description
COMP_ID	VARCHAR2(30)	NOT NULL	Component identifier
COMP_NAME	VARCHAR2(255)		Component name
VERSION	VARCHAR2(30)		Component version loaded
VERSION_FULL	VARCHAR2(30)		Component full version

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(11)		Component status: • INVALID • VALID • LOADING • LOADED • UPGRADING • UPGRADED • DOWNGRADING • DOWNGRADED • REMOVING • REMOVED
MODIFIED	VARCHAR2(20)		Time when the component was last modified
CONTROL	VARCHAR2(128)	NOT NULL	User that created the component entry
SCHEMA	VARCHAR2(128)	NOT NULL	User that contains the objects for the component
PROCEDURE	VARCHAR2(61)		Validation procedure
STARTUP	VARCHAR2(8)		Indicates whether the component requires a startup after the upgrade (REQUIRED) or not
PARENT_ID	VARCHAR2(30)		Parent component identifier
OTHER_SCHEMAS	VARCHAR2(4000)		A list of ancillary schema names associated with the component

5.356 DBA_SERVICES

DBA_SERVICES displays all services in the database. The view excludes rows marked for deletion. Its columns are the same as those in ALL_SERVICES.

See Also

"ALL_SERVICES"

5.357 DBA_SODA_COLLECTIONS

DBA_SODA_COLLECTIONS describes all Simple Oracle Document Access (SODA) collections in the database.

Related View

USER_SODA_COLLECTIONS describes the SODA collections owned by the current user. This view does not display the OWNER column.

Column	Datatype	NULL	Description
URI_NAME	NVARCHAR2(255)	NOT NULL	Collection name
OWNER	VARCHAR2(128)	NOT NULL	Owner of the collection
OBJECT_TYPE	VARCHAR2(10)	NOT NULL	Indicates whether the collection is table-based (TABLE) or view-based (VIEW)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OBJECT_SCHEMA	VARCHAR2(128)	NOT NULL	Name of the schema that includes the table or view on which the collection is based
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the table or view on which the collection is based
CREATED_ON	TIMESTAMP(6)	NOT NULL	Collection creation time
CREATE_MODE	VARCHAR2(10)	NOT NULL	Creation mode. Possible values: • DDL: A new table was created at collection creation time • MAP: The collection was created by mapping a preexisting table or view **Note:** For view-based collections, the value of this column is always MAP. New views are not created for view-based collections. A view-based collection is always created by mapping a preexisting view.
JSON_DESCRIPTOR	VARCHAR2(4000)	NOT NULL	Collection metadata, expressed in JavaScript Object Notation (JSON)

Note

The DBA_SODA_COLLECTIONS view is available only in Oracle Autonomous Database. However, the USER_SODA_COLLECTIONS view is available in on-premises Oracle databases and Oracle Autonomous Database.

See Also

"USER_SODA_COLLECTIONS"

5.358 DBA_SOURCE

DBA_SOURCE describes the text source of all stored objects in the database. Its columns are the same as those in ALL_SOURCE.

See Also

"ALL_SOURCE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.359 DBA_SOURCE_AE

DBA_SOURCE_AE describes the text source of all stored objects (across all editions) in the database. Its columns are the same as those in ALL_SOURCE_AE.

See Also

"ALL_SOURCE_AE"

5.360 DBA_SQL_MANAGEMENT_CONFIG

DBA_SQL_MANAGEMENT_CONFIG displays the configuration parameters of the SQL management base.

You must have the DBA role in order to change the configuration parameter values.

Column	Datatype	NULL	Description
PARAMETER_NAME	VARCHAR2(128)	NOT NULL	Name of the configuration parameter: • AUTO_CAPTURE_ACTION • AUTO_CAPTURE_MODULE • AUTO_CAPTURE_PARSING_SCHEMA_NAME • AUTO_CAPTURE_SQL_TEXT • PLAN_RETENTION_WEEKS • SPACE_BUDGET_PERCENT
PARAMETER_VALUE	VARCHAR2(4000)		Value of the configuration parameter
LAST_MODIFIED	TIMESTAMP(6)		Time the parameter value was last updated
MODIFIED_BY	VARCHAR2(128)		User who last updated the parameter value

5.361 DBA_SQL_QUARANTINE

DBA_SQL_QUARANTINE displays information about SQL Quarantine configurations.

Each row in this view represents a quarantine configuration for a SQL plan.

Column	Datatype	NULL	Description
SIGNATURE	NUMBER	NOT NULL	Unique SQL identifier generated from normalized SQL text
NAME	VARCHAR2(128)	NOT NULL	Unique plan identifier in string form as a search key
SQL_TEXT	CLOB	NOT NULL	Un-normalized SQL text
PLAN_HASH_VALUE	NUMBER		Unique plan identifier in numeric form as a search key
CPU_TIME	VARCHAR2(4000)		CPU time threshold (in seconds)
IO_MEGABYTES	VARCHAR2(4000)		I/O threshold (in megabytes)
IO_REQUESTS	VARCHAR2(4000)		Physical I/O threshold (number of physical I/O requests)
ELAPSED_TIME	VARCHAR2(4000)		Elapsed time threshold (in seconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
IO_LOGICAL	VARCHAR2(4000)		Logical I/O threshold (number of logical I/O requests)
CREATOR	VARCHAR2(128)		User who created the quarantine configuration
ORIGIN	VARCHAR2(16)		Method by which the quarantine configuration was created. The only possible value is RESOURCE-MANAGER, which indicates that the quarantine configuration was created by the Resource Manager.
DESCRIPTION	VARCHAR2(500)		Text description
CREATED	TIMESTAMP(6)	NOT NULL	Time at which the quarantine configuration was created
LAST_EXECUTED	TIMESTAMP(6)		Time at which the quarantine configuration was last used
ENABLED	VARCHAR2(3)		Indicates whether the quarantine configuration is enabled (YES) or disabled (NO)
AUTOPURGE	VARCHAR2(3)		Indicates whether the quarantine configuration is auto-purged (YES) or not (NO)

Note

This view is available starting with Oracle Database 19c.

See Also

Oracle Database SQL Tuning Guide for more information about quarantined SQL plans

5.362 DBA_SQL_PATCHES

DBA_SQL_PATCHES displays the set of SQL patches.

Column	Datatype	NULL	Description
NAME	VARCHAR2(128)	NOT NULL	Name of the SQL patch
CATEGORY	VARCHAR2(128)	NOT NULL	Category of the SQL patch
SIGNATURE	NUMBER	NOT NULL	Unique identifier generated from normalized SQL text
SQL_TEXT	CLOB	NOT NULL	Un-normalized SQL text
CREATED	TIMESTAMP(6)	NOT NULL	Timestamp when the SQL patch was created
LAST_MODIFIED	TIMESTAMP(6)		Timestamp when the SQL patch was last modified
DESCRIPTION	VARCHAR2(500)		Text description provided for the SQL patch
STATUS	VARCHAR2(8)		Status of the SQL patch: • ENABLED • DISABLED
FORCE_MATCHING	VARCHAR2(3)		Indicates whether the signature is force matching (YES) or exact matching (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TASK_ID	NUMBER		Advisor task ID that generated the SQL patch
TASK_EXEC_NAME	VARCHAR2(128)		Advisor execution name for the SQL patch
TASK_OBJ_ID	NUMBER		Advisor object ID for the SQL patch
TASK_FND_ID	NUMBER		Advisor finding ID for the SQL patch
TASK_REC_ID	NUMBER		Advisor recommendation ID for the SQL patch

5.363 DBA_SQL_PLAN_BASELINES

DBA_SQL_PLAN_BASELINES displays information about the SQL plan baselines currently created for specific SQL statements.

Column	Datatype	NULL	Description
SIGNATURE	NUMBER	NOT NULL	Unique SQL identifier generated from normalized SQL text
SQL_HANDLE	VARCHAR2(30)	NOT NULL	Unique SQL identifier in string form as a search key
SQL_TEXT	CLOB	NOT NULL	Un-normalized SQL text
PLAN_NAME	VARCHAR2(128)	NOT NULL	Unique plan identifier in string form as a search key
CREATOR	VARCHAR2(128)		User who created the plan baseline
ORIGIN	VARCHAR2(29)		How the plan baseline was created: • ADDM-SQLTUNE • AUTO-CAPTURE • AUTO-SQLTUNE • EVOLVE-AUTO-INDEX-LOAD • EVOLVE-CREATE-FROM-ADAPTIVE • EVOLVE-LOAD-FROM-AWR • EVOLVE-LOAD-FROM-CURSOR-CACHE • EVOLVE-LOAD-FROM-STS • MANUAL-LOAD • MANUAL-LOAD-FROM-AWR • MANUAL-LOAD-FROM-CURSOR-CACHE • MANUAL-LOAD-FROM-STS • MANUAL-SQLTUNE • STORED-OUTLINE
PARSING_SCHEMA_NAME	VARCHAR2(128)		Name of the parsing schema
DESCRIPTION	VARCHAR2(500)		Text description provided for the plan baseline
VERSION	VARCHAR2(64)		Database version at the time of plan baseline creation
CREATED	TIMESTAMP(6)	NOT NULL	Timestamp when the plan baseline was created
LAST_MODIFIED	TIMESTAMP(6)		Timestamp when the plan baseline was last modified
LAST_EXECUTED	TIMESTAMP(6)		Timestamp when the plan baseline was last executed **Note:** For performance reasons, this column is not updated immediately after each execution of the plan baseline. Therefore, the plan baseline may have been executed more recently than the value of this column indicates.
LAST_VERIFIED	TIMESTAMP(6)		Timestamp when the plan baseline was last verified

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ENABLED	VARCHAR2(3)		Indicates whether the plan baseline is enabled (YES) or disabled (NO)
ACCEPTED	VARCHAR2(3)		Indicates whether the plan baseline is accepted (YES) or not (NO)
FIXED	VARCHAR2(3)		Indicates whether the plan baseline is fixed (YES) or not (NO)
REPRODUCED	VARCHAR2(3)		Indicates whether the optimizer was able to reproduce the plan (YES) or not (NO). The value of this column is set to YES when a plan is initially added to the plan baseline.
AUTOPURGE	VARCHAR2(3)		Indicates whether the plan baseline is auto-purged (YES) or not (NO)
ADAPTIVE	VARCHAR2(3)		Indicates whether a plan that is automatically captured by SQL plan management is marked adaptive or not. When a new adaptive plan is found for a SQL statement that has an existing SQL plan baseline, that new plan will be added to the SQL plan baseline as an unaccepted plan, and the ADAPTIVE column will be marked YES. When this new plan is verified (either manually or via the auto evolve task), the plan will be test executed and the final plan determined at execution will become an accepted plan if its performance is better than the existing plan baseline. At this point, the value of the ADAPTIVE column is set to NO since the plan is no longer adaptive, but resolved.
OPTIMIZER_COST	NUMBER		Optimizer cost at the time the plan baseline was created
MODULE	VARCHAR2(64)		Application module name
ACTION	VARCHAR2(64)		Application action
EXECUTIONS[1]	NUMBER		Number of executions at the time the plan baseline was created
ELAPSED_TIME[1]	NUMBER		Total elapsed time (in microseconds) at the time the plan baseline was created
CPU_TIME[1]	NUMBER		Total CPU time (in microseconds) at the time the plan baseline was created
BUFFER_GETS[1]	NUMBER		Total buffer gets at the time the plan baseline was created
DISK_READS[1]	NUMBER		Total disk reads at the time the plan baseline was created
DIRECT_WRITES[1]	NUMBER		Total direct writes at the time the plan baseline was created
ROWS_PROCESSED[1]	NUMBER		Total rows processed at the time the plan baseline was created
FETCHES[1]	NUMBER		Total number of fetches at the time the plan baseline was created
END_OF_FETCH_COUNT[1]	NUMBER		Total number of full fetches at the time the plan baseline was created

[1] If the value of the ORIGIN column is equal to AUTO-CAPTURE, then data for this column is not populated.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database SQL Tuning Guide* for more information about SQL plan baselines
- The `DBMS_SQLTUNE` package in *Oracle Database PL/SQL Packages and Types Reference*

5.364 DBA_SQL_PLAN_DIR_OBJECTS

`DBA_SQL_PLAN_DIR_OBJECTS` displays the objects created in the SQL plan directive.

Column	Datatype	NULL	Description
`DIRECTIVE_ID`	`NUMBER`		The identifier of the SQL plan directive
`OWNER`	`VARCHAR2(128)`		The username of the owner of the object in the SQL plan directive
`OBJECT_NAME`	`VARCHAR2(128)`		The name of the object in the SQL plan directive
`SUBOBJECT_NAME`	`VARCHAR2(128)`		The name of the subobject (for example, column) in the SQL plan directive
`OBJECT_TYPE`	`VARCHAR2(6)`		The type of the subobject in the SQL plan directive
`NUM_ROWS`	`NUMBER`		The number of rows in the object when the directive is created
`NOTES`	`XMLTYPE`		Other notes about the object

See Also

- "DBA_SQL_PLAN_DIRECTIVES"
- *Oracle Database SQL Tuning Guide* for more information about SQL plan directives

5.365 DBA_SQL_PLAN_DIRECTIVES

`DBA_SQL_PLAN_DIRECTIVES` displays information about the SQL plan directives in the system.

Column	Datatype	NULL	Description
`DIRECTIVE_ID`	`NUMBER`	`NOT NULL`	The identifier of the SQL plan directive
`TYPE`	`VARCHAR2(16)`		The type of the SQL plan directive: • `DYNAMIC SAMPLING`: SQL plan directive • `DYNAMIC SAMPLING RESULT`: Dynamic sampling query results. This value appears only in Oracle Database 12*c* Release 2 (12.2.0.1) and later releases. • `UNKNOWN`: Unknown

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ENABLED	VARCHAR2(3)		Indicates whether the SQL plan directive is enabled. Possible values: • YES: The SQL plan directive is enabled. • NO: The SQL plan directive is not enabled.
STATE	VARCHAR2(10)		The state of the SQL plan directive. Possible values include: • SUPERSEDED: This value indicates that the corresponding column or groups have an extension or histogram, or that another SQL plan directive exists that can be used for the directive. • USABLE: This value indicates that the SQL plan directive is usable for the optimizer.
AUTO_DROP	VARCHAR2(3)		If YES, the SQL plan directive gets dropped when unused beyond SPD_RETENTION_WEEKS
REASON	VARCHAR2(36)		The reason for creating the SQL plan directive
CREATED	TIMESTAMP(6)		The creation timestamp of the SQL plan directive
LAST_MODIFIED	TIMESTAMP(6)		The timestamp of most recent modification of the SQL plan directive
LAST_USED	TIMESTAMP(9)		The timestamp of most recent usage of the SQL plan directive
NOTES	XMLTYPE		Extra information about the SQL plan directive

See Also

- "DBA_SQL_PLAN_DIR_OBJECTS"
- *Oracle Database SQL Tuning Guide* for more information about SQL plan directives

5.366 DBA_SQL_PROFILES

DBA_SQL_PROFILES displays information about SQL profiles currently created for specific SQL statements.

Column	Datatype	NULL	Description
NAME	VARCHAR2(128)	NOT NULL	Name of the SQL profile
CATEGORY	VARCHAR2(128)	NOT NULL	Category of the SQL profile
SIGNATURE	NUMBER	NOT NULL	Unique identifier generated from normalized SQL text
SQL_TEXT	CLOB	NOT NULL	Un-normalized SQL text
CREATED	TIMESTAMP(6)	NOT NULL	Timestamp when the SQL profile was created
LAST_MODIFIED	TIMESTAMP(6)		Timestamp when the SQL profile was last modified
DESCRIPTION	VARCHAR2(500)		Text description provided for the SQL profile
TYPE	VARCHAR2(7)		Type of the SQL profile (how it was created): • MANUAL • AUTO

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATUS	VARCHAR2(8)		Status of the SQL profile: • ENABLED • DISABLED • VOID
FORCE_MATCHING	VARCHAR2(3)		If YES, this causes SQL Profiles to target all SQL statements which have the same text after normalizing all literal values to bind variables. If a combination of literal values and bind variables is used in the same SQL text, then no transformation occurs. This is analogous to the matching algorithm use by the FORCE option of the CURSOR_SHARING parameter. If NO, literals are not transformed. This is analogous to the matching algorithm used by the EXACT option of the CURSOR_SHARING parameter.
TASK_ID	NUMBER		Advisor task ID that generated the SQL profile
TASK_EXEC_NAME	VARCHAR2(128)		Advisor execution name for the SQL profile
TASK_OBJ_ID	NUMBER		Advisor object ID for the SQL profile
TASK_FND_ID	NUMBER		Advisor finding ID for the SQL profile
TASK_REC_ID	NUMBER		Advisor recommendation ID for the SQL profile
TASK_CON_DBID	NUMBER		Database ID for the PDB tuning task generating the SQL profile

See Also

The DBMS_SQLTUNE package in *Oracle Database PL/SQL Packages and Types Reference*

5.367 DBA_SQL_TRANSLATION_PROFILES

DBA_SQL_TRANSLATION_PROFILES describes all SQL translation profiles in the database. Its columns are the same as those in ALL_SQL_TRANSLATION_PROFILES.

See Also

"ALL_SQL_TRANSLATION_PROFILES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.368 DBA_SQL_TRANSLATIONS

DBA_SQL_TRANSLATIONS describes all SQL translations in the database. Its columns are the same as those in ALL_SQL_TRANSLATIONS.

See Also

"ALL_SQL_TRANSLATIONS"

5.369 DBA_SQLJ_TYPE_ATTRS

DBA_SQLJ_TYPE_ATTRS describes the attributes of all SQLJ object types in the database. Its columns are the same as those in ALL_SQLJ_TYPE_ATTRS.

See Also

"ALL_SQLJ_TYPE_ATTRS"

5.370 DBA_SQLJ_TYPE_METHODS

DBA_SQLJ_TYPE_METHODS describes the methods of all SQLJ object types in the database. Its columns are the same as those in ALL_SQLJ_TYPE_METHODS.

See Also

"ALL_SQLJ_TYPE_METHODS"

5.371 DBA_SQLJ_TYPES

DBA_SQLJ_TYPES describes all SQLJ object types in the database. Its columns are the same as those in ALL_SQLJ_TYPES.

See Also

"ALL_SQLJ_TYPES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.372 DBA_SQLSET

DBA_SQLSET displays information about all SQL tuning sets in the database. Its columns are the same as those in ALL_SQLSET.

See Also

"ALL_SQLSET"

5.373 DBA_SQLSET_BINDS

DBA_SQLSET_BINDS displays the bind values associated with all SQL tuning sets in the database. Its columns are the same as those in ALL_SQLSET_BINDS.

See Also

"ALL_SQLSET_BINDS"

5.374 DBA_SQLSET_PLANS

DBA_SQLSET_PLANS describes captured plans in the SQL tuning sets in the database. Its columns are the same as those in ALL_SQLSET_PLANS.

See Also

"ALL_SQLSET_PLANS"

5.375 DBA_SQLSET_REFERENCES

DBA_SQLSET_REFERENCES describes whether or not all SQL tuning sets in the database are active. A SQL tuning set cannot be dropped if it is referenced. Its columns are the same as those in ALL_SQLSET_REFERENCES.

See Also

"ALL_SQLSET_REFERENCES"

5.376 DBA_SQLSET_STATEMENTS

DBA_SQLSET_STATEMENTS displays information about the SQL statements, along with their statistics, that form all SQL tuning sets in the database. Its columns, except for PARSING_SCHEMA_ID, are the same as those in ALL_SQLSET_STATEMENTS.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SQLSET_NAME	VARCHAR2(128)	NOT NULL	Name of the SQL tuning set for the statement
SQLSET_OWNER	VARCHAR2(128)		User name of the SQL tuning set owner
SQLSET_ID	NUMBER	NOT NULL	ID of the SQL tuning set for the statement
CON_DBID	NUMBER	NOT NULL	The database ID of the PDB
SQL_ID	VARCHAR2(13)	NOT NULL	SQL identifier of the parent cursor in the library cache
FORCE_MATCHING_SIGNATURE	NUMBER	NOT NULL	The signature used when the CURSOR_SHARING parameter is set to FORCE
SQL_TEXT	CLOB		Full text for the SQL statement exposed as a CLOB column.
PARSING_SCHEMA_NAME	VARCHAR2(128)		Name of the user in whose schema the statement was parsed
PARSING_SCHEMA_ID	NUMBER		ID of the schema in which the statement was parsed
PLAN_HASH_VALUE	NUMBER	NOT NULL	Hash value for the plan corresponding to statistics in this record
BIND_DATA	RAW(2000)		Bind data
BINDS_CAPTURED	CHAR(1)		Binds captured
MODULE	VARCHAR2(64)		Contains the name of the module that was executing at the time that the SQL statement was first parsed, which is set by calling DBMS_APPLICATION_INFO.SET_MODULE
ACTION	VARCHAR2(64)		Contains the name of the action that was executing at the time that the SQL statement was first parsed, which is set by calling DBMS_APPLICATION_INFO.SET_ACTION
ELAPSED_TIME	NUMBER		Elapsed time (in microseconds) used by this cursor for parsing, executing, and fetching
CPU_TIME	NUMBER		CPU time (in microseconds) used by this cursor for parsing, executing, and fetching
BUFFER_GETS	NUMBER		Number of buffer gets for this child cursor
DISK_READS	NUMBER		Number of disk reads for this child cursor
DIRECT_WRITES	NUMBER		Number of direct writes for this child cursor
ROWS_PROCESSED	NUMBER		Total number of rows that the parsed SQL statement returns
FETCHES	NUMBER		Number of fetches associated with the SQL statement
EXECUTIONS	NUMBER		Number of executions that took place on this object since it was brought into the library cache
END_OF_FETCH_COUNT	NUMBER		Number of times this cursor was fully executed since the cursor was brought into the library cache. The value of this statistic in not incremented when the cursor is partially executed, either because it failed during the execution or because only the first few rows produced by this cursor are fetched before the cursor is closed or re-executed. By definition, the value of the END_OF_FETCH_COUNT column should be less than, or equal to, the value of the EXECUTIONS column.
OPTIMIZER_COST	NUMBER		Cost of this query, given by the optimizer
OPTIMIZER_ENV	RAW(2000)		Optimizer environment

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PRIORITY	NUMBER		User-defined priority
COMMAND_TYPE	NUMBER		Oracle command type definition
FIRST_LOAD_TIME	VARCHAR2(19)		Timestamp of the parent creation time
STAT_PERIOD	NUMBER		Period of time (in seconds) during which the statistics of the SQL statement were collected
ACTIVE_STAT_PERIOD	NUMBER		Effective period of time (in seconds) during which the SQL statement was active
OTHER	CLOB		Client data, specified by the user, for this statement
PLAN_TIMESTAMP	DATE		Timestamp for the plan corresponding to the statistics in this record
SQL_SEQ	NUMBER	NOT NULL	SQL sequence
LAST_EXEC_START_TIME	VARCHAR2(19)		For SQLs captured from the cursor cache, this is the time when the most recent execution of this SQL started

See Also

- "ALL_SQLSET_STATEMENTS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_MODULE procedure
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_ACTION procedure

5.377 DBA_SQLTUNE_BINDS

DBA_SQLTUNE_BINDS displays the bind values associated with all tuned SQL statements in the database.

Related View

USER_SQLTUNE_BINDS displays the bind values associated with the tuned SQL statements owned by the current user.

Column	Datatype	NULL	Description
TASK_ID	NUMBER(38)	NOT NULL	Tuning task identifier
OBJECT_ID	NUMBER(38)	NOT NULL	Advisor framework object identifier
POSITION	NUMBER(38)	NOT NULL	Bind position
VALUE	ANYDATA		Bind value. This column is NULL for PL/SQL bind types.

See Also

"USER_SQLTUNE_BINDS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.378 DBA_SQLTUNE_PLANS

DBA_SQLTUNE_PLANS displays information about the execution plans generated for all SQL statements in the database during a SQL tuning session.

Related View

USER_SQLTUNE_PLANS displays information about the execution plans generated for the SQL statements owned by the current user during a SQL tuning session.

Column	Datatype	NULL	Description
TASK_ID	NUMBER(38)	NOT NULL	Advisor task ID
EXECUTION_NAME	VARCHAR2(128)	NOT NULL	Advisor task execution
OBJECT_ID	NUMBER(38)	NOT NULL	Advisor object ID
ATTRIBUTE	VARCHAR2(34)		Text string identifying the type of the execution plan: • Original - Original plan of the query • Original with adjusted cost - Same as Original but with adjusted cost • Using SQL profile - Plan with SQL profile applied • Using new indices - Plan with indexes applied
STATEMENT_ID	VARCHAR2(30)		Optional statement identifier specified in the EXPLAIN PLAN statement
PLAN_HASH_VALUE	NUMBER	NOT NULL	Numerical representation of the execution plan
PLAN_ID	NUMBER		Plan identifier
TIMESTAMP	DATE		Date and time when the EXPLAIN PLAN statement was issued
REMARKS	VARCHAR2(4000)		Place for comments that can be added to the steps of the execution plan
OPERATION	VARCHAR2(30)		Name of the operation performed at this step
OPTIONS	VARCHAR2(255)		Options used for the operation performed at this step
OBJECT_NODE	VARCHAR2(128)		Name of the database link used to reference the object
OBJECT_OWNER	VARCHAR2(128)		Owner of the object
OBJECT_NAME	VARCHAR2(128)		Name of the object
OBJECT_ALIAS	VARCHAR2(261)		Object alias
OBJECT_INSTANCE	NUMBER(38)		Numbered position of the object name in the original SQL statement
OBJECT_TYPE	VARCHAR2(30)		Descriptive modifier that further describes the type of object
OPTIMIZER	VARCHAR2(255)		Current mode of the optimizer
SEARCH_COLUMNS	NUMBER		Number of index columns with start and stop keys (that is, the number of columns with matching predicates)
ID	NUMBER(38)	NOT NULL	Identification number for this step in the execution plan
PARENT_ID	NUMBER(38)		ID of the next step that operates on the results of this step

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEPTH	NUMBER(38)		Depth
POSITION	NUMBER(38)		Order of processing for steps with the same parent ID
COST	NUMBER(38)		Cost of the current operation estimated by the cost-based optimizer (CBO)
CARDINALITY	NUMBER(38)		Number of rows returned by the current operation (estimated by the CBO)
BYTES	NUMBER(38)		Number of bytes returned by the current operation
OTHER_TAG	VARCHAR2(255)		Describes the function of the SQL text in the OTHER column. Values for OTHER_TAG are: • SERIAL - SQL is the text of a locally-executed, serial query plan. Currently, SQL is not loaded in OTHER for this case. • SERIAL_FROM_REMOTE - SQL text shown in the OTHER column will be executed at a remote site • PARALLEL_COMBINED_WITH_PARENT - Parent of this operation is a DFO that performs both operations in the parallel execution plan • PARALLEL_COMBINED_WITH_CHILD - Child of this operation is a DFO that performs both operations in the parallel execution plan. • PARALLEL_TO_SERIAL - SQL text shown in the OTHER column is the top-level of the parallel plan. • PARALLEL_TO_PARALLEL - SQL text shown in the OTHER column is executed and output in parallel • PARALLEL_FROM_SERIAL - Operation consumes data from a serial operation and outputs it in parallel
PARTITION_START	VARCHAR2(255)		Start partition of a range of accessed partitions
PARTITION_STOP	VARCHAR2(255)		Stop partition of a range of accessed partitions
PARTITION_ID	NUMBER(38)		Step that has computed the pair of values of the PARTITION_START and PARTITION_STOP columns
OTHER	LONG		Information about parallel execution servers and parallel queries
DISTRIBUTION	VARCHAR2(30)		Distribution method
CPU_COST	NUMBER(38)		User-defined CPU cost
IO_COST	NUMBER(38)		User-defined I/O cost
TEMP_SPACE	NUMBER(38)		Temporary space usage of the operation (sort or hash-join) as estimated by the CBO
ACCESS_PREDICATES	VARCHAR2(4000)		Predicates used to locate rows in an access structure. For example, start or stop predicates for an index range scan.
FILTER_PREDICATES	VARCHAR2(4000)		Predicates used to filter rows before producing them
PROJECTION	VARCHAR2(4000)		Expressions produced by the operation
TIME	NUMBER(38)		Elapsed time (in seconds) of the operation as estimated by the CBO
QBLOCK_NAME	VARCHAR2(128)		Name of the query block

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OTHER_XML	CLOB		Provides extra information specific to an execution step of the execution plan. The content of this column is structured using XML because it allows multiple pieces of information to be stored, including the following: • Name of the schema against which the query was parsed • Release number of the Oracle Database that produced the explain plan • Hash value associated with the execution plan • Name (if any) of the outline or the SQL profile used to build the execution plan • Indication of whether or not dynamic statistics were used to produce the plan • The outline data, a set of optimizer hints that can be used to regenerate the same plan

See Also

"USER_SQLTUNE_PLANS"

5.379 DBA_SQLTUNE_RATIONALE_PLAN

DBA_SQLTUNE_RATIONALE_PLAN displays the association between rationales and operations in the execution plan of all SQL statements in the database.

Related View

USER_SQLTUNE_RATIONALE_PLAN displays the association between rationales and operations in the execution plan of the SQL statements owned by the current user.

Column	Datatype	NULL	Description
TASK_ID	NUMBER(38)	NOT NULL	Tuning task identifier
EXECUTION_NAME	VARCHAR2(128)	NOT NULL	The name of the task execution with which this entry (row) is associated
RATIONALE_ID	NUMBER(38)	NOT NULL	Rationale identifier
OBJECT_ID	NUMBER(38)	NOT NULL	Advisor framework object identifier
OPERATION_ID	NUMBER(38)	NOT NULL	Operation identifier
PLAN_ATTRIBUTE	VARCHAR2(34)		Type of the execution plan: • Original - Original plan of the query • Original with adjusted cost - Same as Original but with adjusted cost • Using SQL profile - Plan with SQL profile applied • Using new indices - Plan with indexes applied

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "USER_SQLTUNE_RATIONALE_PLAN"

5.380 DBA_SQLTUNE_STATISTICS

DBA_SQLTUNE_STATISTICS displays statistics associated with all SQL statements in the database.

Related View

USER_SQLTUNE_STATISTICS displays statistics associated with the SQL statements owned by the current user.

Column	Datatype	NULL	Description
TASK_ID	NUMBER(38)	NOT NULL	Tuning task identifier
OBJECT_ID	NUMBER(38)	NOT NULL	Advisor framework object identifier
PARSING_SCHEMA_ID	NUMBER		Schema under which the SQL is parsed
MODULE	VARCHAR2(64)		Last application module recorded for the SQL
ACTION	VARCHAR2(64)		Last application action recorded for the SQL
ELAPSED_TIME	NUMBER		Elapsed time for the SQL statement
CPU_TIME	NUMBER		CPU time for the SQL
BUFFER_GETS	NUMBER		Number of buffer gets
DISK_READS	NUMBER		Number of disk reads
DIRECT_WRITES	NUMBER		Number of disk writes
ROWS_PROCESSED	NUMBER		Number of rows processed by the SQL
FETCHES	NUMBER		Number of fetches
EXECUTIONS	NUMBER		Number of executions
END_OF_FETCH_COUNT	NUMBER		End of fetch count
OPTIMIZER_COST	NUMBER		Optimizer cost for the SQL
OPTIMIZER_ENV	RAW(2000)		Optimizer environment
COMMAND_TYPE	NUMBER		Command type

> **See Also**
>
> "USER_SQLTUNE_STATISTICS"

5.381 DBA_SR_GRP_STATUS

DBA_SR_GRP_STATUS provides information on the current refresh operations for the current synchronous refresh groups in the database. It has the same columns as the DBA_SR_GRP_STATUS_ALL view.

Refresh operations are controlled using the DBMS_SYNC_REFRESH package.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Related View

USER_SR_GRP_STATUS provides information on the current refresh operations for the current synchronous refresh groups in the database which are owned by the current user. Its columns are the same as those in DBA_SR_GRP_STATUS.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the refresh operation, which is the user who launched the operation
GROUP_ID	NUMBER	NOT NULL	Group ID of the synchronous refresh group
OPERATION	VARCHAR2(7)		The phase of the refresh operation performed: • PREPARE • EXECUTE
STATUS	VARCHAR2(10)		The status of the refresh operation: • RUNNING • NOT PROCESSED • COMPLETE • ERROR-SOFT • ERROR-HARD • ABORT • PARTIAL
NUM_TBLS	NUMBER	NOT NULL	The number of tables in the synchronous refresh group
NUM_MVS	NUMBER	NOT NULL	The number of materialized views in the synchronous refresh group
BASE_TBLS_REFR_STATUS	VARCHAR2(13)		Indicates the refresh status of base tables in the synchronous refresh group. The possible values are: • NOT PROCESSED • COMPLETE • ABORT
NUM_MVS_COMPLETED	NUMBER		The number of materialized views which have completed refresh in the synchronous refresh group
NUM_MVS_ABORTED	NUMBER		The number of materialized views which have terminated refresh in the synchronous refresh group
ERROR_NUMBER	NUMBER		Error number of the run (if any)
ERROR_MESSAGE	VARCHAR2(4000)		Error message of the run (if any)
PREPARE_START_TIME	DATE		Time that the PREPARE_REFRESH phase of the run started
PREPARE_END_TIME	DATE		Time that the PREPARE_REFRESH phase of the run ended
EXECUTE_START_TIME	DATE		Time that the EXECUTE_REFRESH phase of the run started
EXECUTE_END_TIME	DATE		Time that the EXECUTE_REFRESH phase of the run ended

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "USER_SR_GRP_STATUS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SYNC_REFRESH package

5.382 DBA_SR_GRP_STATUS_ALL

DBA_SR_GRP_STATUS_ALL provides information on the refresh operations on the synchronous refresh groups in the database.

A refresh operation is also called a run, and it has two phases: PREPARE_REFRESH and EXECUTE_REFRESH. These phases are controlled using the DBMS_SYNC_REFRESH package.

Each row in this view provides information on a run of a group, identified by its GROUP_ID. The view contains information on the status of the objects of both current and past runs of both current and defunct synchronous refresh groups. Therefore, this view can be used to examine the history of synchronous refresh operations.

The current run of a group is the most recent run of a group; a current group is a currently valid group, which is capable of being refreshed. A group becomes defunct when it is unregistered for any reason, either explicitly by the user or implicitly as a side-effect when the user registers materialized views related to the materialized views in the group.

To view the status of refresh operations for the most recent runs of only the current groups, use the DBA_SR_GRP_STATUS view.

Related View

USER_SR_GRP_STATUS_ALL provides information on the refresh operations on the synchronous refresh groups in the database which are owned by the current user. Its columns are the same as those in DBA_SR_GRP_STATUS_ALL.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the refresh operation, which is the user who launched the operation
GROUP_ID	NUMBER	NOT NULL	Group ID of the synchronous refresh group
OPERATION	VARCHAR2(7)		The phase of the refresh operation performed: • PREPARE • EXECUTE
STATUS	VARCHAR2(10)		The status of the refresh operation: • RUNNING • NOT PROCESSED • COMPLETE • ERROR-SOFT • ERROR-HARD • ABORT • PARTIAL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CURRENT_RUN	VARCHAR2(1)		Indicates whether the record is for the most recent refresh-operation on the group: • Y - Yes • N - No
CURRENT_GROUP	VARCHAR2(1)		Indicates whether the record is for a current group: • Y - Yes • N - No
NUM_TBLS	NUMBER	NOT NULL	The number of tables in the synchronous refresh group
NUM_MVS	NUMBER	NOT NULL	The number of materialized views in the synchronous refresh group
BASE_TBLS_REFR_STATUS	VARCHAR2(13)		Indicates the refresh status of base tables in the synchronous refresh group. The possible values are: • NOT PROCESSED • COMPLETE • ABORT
NUM_MVS_COMPLETED	NUMBER		The number of materialized views which have completed refresh in the synchronous refresh group
NUM_MVS_ABORTED	NUMBER		The number of materialized views which have terminated refresh in the synchronous refresh group
ERROR_NUMBER	NUMBER		Error number of the run (if any)
ERROR_MESSAGE	VARCHAR2(4000)		Error message of the run (if any)
PREPARE_START_TIME	DATE		Time that the PREPARE_REFRESH phase of the run started
PREPARE_END_TIME	DATE		Time that the PREPARE_REFRESH phase of the run ended
EXECUTE_START_TIME	DATE		Time that the EXECUTE_REFRESH phase of the run started
EXECUTE_END_TIME	DATE		Time that the EXECUTE_REFRESH phase of the run ended

See Also

- "USER_SR_GRP_STATUS_ALL"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SYNC_REFRESH package

5.383 DBA_SR_OBJ

DBA_SR_OBJ provides information on the objects registered for synchronous refresh for current groups.

Related View

USER_SR_OBJ provides information on the objects registered for synchronous refresh for current groups for the current user. Its columns are the same as those in DBA_SR_OBJ.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the synchronous refresh object
NAME	VARCHAR2(128)	NOT NULL	Name of the synchronous refresh object
TYPE	VARCHAR2(5)		Type of synchronous refresh object: • MVIEW • TABLE
GROUP_ID	NUMBER	NOT NULL	Group ID of the synchronous refresh group to which this object belongs
STAGING_LOG_NAME	VARCHAR2(128)		Name of the staging log for tables. This column has a value of NULL for materialized views.

See Also

"USER_SR_OBJ"

5.384 DBA_SR_OBJ_ALL

DBA_SR_OBJ_ALL provides information on the objects registered for synchronous refresh for current and defunct groups.

To see information on the objects registered for synchronous refresh for only the current groups, use the DBA_SR_OBJ view.

Related View

USER_SR_OBJ_ALL provides information on the objects registered for synchronous refresh for current and defunct groups for the current user. Its columns are the same as those in DBA_SR_OBJ_ALL.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the synchronous refresh object
NAME	VARCHAR2(128)	NOT NULL	Name of the synchronous refresh object
TYPE	VARCHAR2(5)		Type of synchronous refresh object: • MVIEW • TABLE
GROUP_ID	NUMBER	NOT NULL	Group ID of the synchronous refresh group to which this object belongs
CURRENT_GROUP	VARCHAR2(1)		Indicates whether the record is for a current group: • Y - Yes • N - No
STAGING_LOG_NAME	VARCHAR2(128)		Name of the staging log for tables. This column has a value of NULL for materialized views.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"USER_SR_OBJ_ALL"

5.385 DBA_SR_OBJ_STATUS

DBA_SR_OBJ_STATUS provides information on the status of objects registered for synchronous refresh for the current refresh operations for the current synchronous refresh groups in the database.

To see information on the status of objects registered for synchronous refresh, use the DBA_SR_OBJ_STATUS_ALL view.

Related View

USER_SR_OBJ_STATUS provides information on the status of objects registered for synchronous refresh for the current refresh operations for the current synchronous refresh groups in the database which are owned by the current user. Its columns are the same as those in DBA_SR_OBJ_STATUS.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the synchronous refresh object
NAME	VARCHAR2(128)	NOT NULL	Name of the synchronous refresh object
TYPE	VARCHAR2(5)		Type of synchronous refresh object: • MVIEW • TABLE
GROUP_ID	NUMBER	NOT NULL	Group ID of the synchronous refresh group to which this object belongs
STATUS	VARCHAR2(13)		Status of the synchronous refresh object: • NOT PROCESSED • COMPLETE • ABORT
ERROR_NUMBER	NUMBER		Error number of the run (if any)
ERROR_MESSAGE	VARCHAR2(4000)		Error message of the run (if any)
LAST_MODIFIED_TIME	DATE		Last modification time of the synchronous refresh object

See Also

"USER_SR_OBJ_STATUS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.386 DBA_SR_OBJ_STATUS_ALL

DBA_SR_OBJ_STATUS_ALL provides information on the status of objects registered for synchronous refresh.

The view contains information on the status of the objects of both the current and past runs of both current and defunct groups. Therefore, this view can be used to examine the history of synchronous refresh operations. The current run of a group is the most recent run of a group; a current group is a currently valid group, which is capable of being refreshed. A group becomes defunct when it is unregistered for any reason, either explicitly by the user or implicitly as a side-effect when the user registers materialized views related to the materialized views in the group.

To see information on the status of refresh operations for the most recent runs of only the current groups, use the DBA_SR_OBJ_STATUS view.

Related View

USER_SR_OBJ_STATUS_ALL provides information on the status of objects registered for synchronous refresh in the database which are owned by the current user. Its columns are the same as those in DBA_SR_OBJ_STATUS_ALL.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the synchronous refresh object
NAME	VARCHAR2(128)	NOT NULL	Name of the synchronous refresh object
TYPE	VARCHAR2(5)		Type of synchronous refresh object: • MVIEW • TABLE
GROUP_ID	NUMBER	NOT NULL	Group ID of the synchronous refresh group to which this object belongs
STATUS	VARCHAR2(13)		Status of the synchronous refresh object: • NOT PROCESSED • COMPLETE • ABORT
CURRENT_RUN	VARCHAR2(1)		Indicates whether the record is for the most recent refresh operation on the group: • Y - Yes • N - No
CURRENT_GROUP	VARCHAR2(1)		Indicates whether the record is for a current group: • Y - Yes • N - No
ERROR_NUMBER	NUMBER		Error number of the run (if any)
ERROR_MESSAGE	VARCHAR2(4000)		Error message of the run (if any)
LAST_MODIFIED_TIME	DATE		Last modification time of the synchronous refresh object

See Also

"USER_SR_OBJ_STATUS_ALL"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.387 DBA_SR_PARTN_OPS

DBA_SR_PARTN_OPS provides information on the partition operations registered on the base tables of the materialized views registered for synchronous refresh.

These rows last only as long as the registrations are active; that is, they disappear after EXECUTE_REFRESH or ABORT_REFRESH of the base table by the DBMS_SYNC_REFRESH package.

Related View

USER_SR_PARTN_OPS provides information on the partition operations registered on the base tables of the materialized views registered for synchronous refresh belonging to the current user. Its columns are the same as those in DBA_SR_PARTN_OPS.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the base table registered for synchronous refresh
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
PARTITION_OP	VARCHAR2(128)	NOT NULL	Type of partition operation: • DROP • EXCHANGE • TRUNCATE
PARTITION_NAME	VARCHAR2(128)	NOT NULL	Name of the partition to be changed
OUTSIDE_TABLE_SCHEMA	VARCHAR2(128)		Schema in which the outside table (for EXCHANGE PARTITION) was created
OUTSIDE_TABLE_NAME	VARCHAR2(128)		Name of the outside table (for EXCHANGE PARTITION)

See Also

- "USER_SR_PARTN_OPS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SYNC_REFRESH package

5.388 DBA_SR_STLOG_EXCEPTIONS

DBA_SR_STLOG_EXCEPTIONS provides information on the exceptions in the staging logs for the tables processed by DBMS_SYNC_REFRESH.PREPARE_STAGING_LOG.

Related View

USER_SR_STLOG_EXCEPTIONS provides information on the exceptions in the staging logs for the tables belonging to the current user processed by DBMS_SYNC_REFRESH.PREPARE_STAGING_LOG. Its columns are the same as those in DBA_SR_STLOG_EXCEPTIONS.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the base table registered for synchronous refresh

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the base table registered for synchronous refresh
STAGING_LOG_NAME	VARCHAR2(128)	NOT NULL	Name of the staging log for tables. This column has a value of NULL for materialized views.
BAD_ROWID	ROWID	NOT NULL	Row ID of the staging log row causing the exception for the synchronous refresh
ERROR_NUMBER	NUMBER		Error number of the exception for the synchronous refresh
ERROR_MESSAGE	VARCHAR2(4000)		Error message associated with the ERROR_NUMBER for the synchronous refresh

See Also

- "USER_SR_STLOG_EXCEPTIONS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SYNC_REFRESH package

5.389 DBA_SR_STLOG_STATS

DBA_SR_STLOG_STATS provides information on the statistics in the staging logs for the tables processed by DBMS_SYNC_REFRESH.PREPARE_STAGING_LOG.

These three statistics columns in the staging log are filled in PREPARE_STAGING_LOG:

- The number of inserts (NUM_INSERTS)
- The number of deletes (NUM_DELETES)
- The number of updates (NUM_UPDATES)

After the data in the staging logs of a synchronous refresh group have been processed by PREPARE_REFRESH and EXECUTE_REFRESH, the statistics columns for the tables in the group are cleared and appear as NULL.

Related View

USER_SR_STLOG_STATS provides information on the statistics in the staging logs for the tables belonging to the current user processed by DBMS_SYNC_REFRESH.PREPARE_STAGING_LOG.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the base table registered for synchronous refresh
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table
STAGING_LOG_NAME	VARCHAR2(128)	NOT NULL	Name of the staging log for tables. NULL for materialized views
NUM_INSERTS	NUMBER	NOT NULL	The number of inserts in the staging log
NUM_DELETES	NUMBER	NOT NULL	The number of deletes in the staging log
NUM_UPDATES	NUMBER	NOT NULL	The number of updates in the staging log

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PSL_MODE	VARCHAR2(33)		The mode specified by the user in running DBMS_SYNC_REFRESH.PREPARE_STAGING_LOG. Possible values: • DELETE_TRUSTED • DELETE_TRUSTED and UPDATE_TRUSTED • ENFORCED • INSERT_TRUSTED • INSERT_TRUSTED and DELETE_TRUSTED • TRUSTED • UPDATE_TRUSTED • UPDATE_TRUSTED and INSERT_TRUSTED

See Also

- "USER_SR_STLOG_STATS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SYNC_REFRESH package

5.390 DBA_SSCR_CAPTURE

DBA_SSCR_CAPTURE displays session state capture statistics.

Column	Datatype	NULL	Description
DB_NAME	VARCHAR2(4000)		Database name of captured session
INST_NAME	VARCHAR2(4000)		Instance name of captured session
INST_ID	NUMBER		Instance ID of captured session
SESSION_ID	NUMBER		Session ID of captured session
SESSION_SERIAL#	NUMBER		Session serial number of captured session
USER_NAME	VARCHAR2(128)	NOT NULL	User name of captured session
SCHEMA_NAME	VARCHAR2(128)	NOT NULL	Schema name of captured session
SEQUENCE#	NUMBER		Sequence number of captured session
CAPTURE_MODE	VARCHAR2(7)		Mode of capture operation
CAPTURE_SCOPE	VARCHAR2(7)		Scope of capture operation
CAPTURE_FORMAT	VARCHAR2(9)		Format of capture files
CAPTURE_DIR	VARCHAR2(128)		Directory object of capture files
CAPTURE_LOCATOR	RAW(64)		Locator of primary capture file
CAPTURE_TIME	TIMESTAMP(6)		Timestamp of capture operation

5.391 DBA_SSCR_RESTORE

DBA_SSCR_RESTORE displays session state restore statistics.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DB_NAME	VARCHAR2(4000)		Database name of restored session
INST_NAME	VARCHAR2(4000)		Instance name of restored session
INST_ID	NUMBER		Instance ID of restored session
SESSION_ID	NUMBER		Session ID of restored session
SESSION_SERIAL#	NUMBER		Session serial number of restored session
USER_NAME	VARCHAR2(128)	NOT NULL	User name of restored session
SCHEMA_NAME	VARCHAR2(128)	NOT NULL	Schema name of restored session
SEQUENCE#	NUMBER		Sequence number of restore operation
RESTORE_MODE	VARCHAR2(7)		Mode of restore operation
RESTORE_SCOPE	VARCHAR2(7)		Scope of restore operation
RESTORE_FORMAT	VARCHAR2(9)		Format of restore files
RESTORE_DIR	VARCHAR2(128)		Directory object of restore files
RESTORE_LOCATOR	RAW(64)		Locator of primary restore file
RESTORE_TIME	TIMESTAMP(6)		Timestamp of restore operation

5.392 DBA_STAT_EXTENSIONS

DBA_STAT_EXTENSIONS displays information about all optimizer statistics extensions in the database.

See Also

"ALL_STAT_EXTENSIONS"

5.393 DBA_STATEMENTS

DBA_STATEMENTS Statements in stored objects accessible to sys. Its columns are the same as those in ALL_STATEMENTS.

See Also

"ALL_STATEMENTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

5.394 DBA_STMT_AUDIT_OPTS

DBA_STMT_AUDIT_OPTS describes current system auditing options across the system and by user.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from UNIFIED_AUDIT_TRAIL.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	NULL	Description
USER_NAME	VARCHAR2(128)		User name if by user auditing; ANY CLIENT if access by a proxy on behalf of a client is being audited; NULL for system-wide auditing
PROXY_NAME	VARCHAR2(128)		Name of the proxy user which is performing an operation for the client; NULL if the client is performing the operation directly
AUDIT_OPTION	VARCHAR2(40)	NOT NULL	Name of the system auditing option
SUCCESS	VARCHAR2(10)		Mode for WHENEVER SUCCESSFUL system auditing
FAILURE	VARCHAR2(10)		Mode for WHENEVER NOT SUCCESSFUL system auditing

5.395 DBA_STORED_SETTINGS

DBA_STORED_SETTINGS lists information about the persistent parameter settings for stored PL/SQL units for which the current user has execute privileges.

It also returns parameter information for all objects in the database and is accessible only to users with the SELECT_CATALOG_ROLE privilege. Its columns are the same as those in ALL_STORED_SETTINGS.

> **See Also**
>
> "ALL_STORED_SETTINGS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6

Static Data Dictionary Views: DBA_STREAMS_ADD_COLUMN to USER_ZONEMAPS

This chapter contains the static data dictionary views DBA_STREAMS_ADD_COLUMN through USER_ZONEMAPS.

6.1 DBA_STREAMS_ADD_COLUMN

DBA_STREAMS_ADD_COLUMN displays information about declarative rule-based transformations that add a column to a row logical change record (LCR).

Column	Datatype	NULL	Description
RULE_OWNER	VARCHAR2(128)		Owner of the rule
RULE_NAME	VARCHAR2(128)		Name of the rule
SCHEMA_NAME	VARCHAR2(128)		Schema of the column to be added
TABLE_NAME	VARCHAR2(128)		Table of the column to be added
COLUMN_NAME	VARCHAR2(4000)		Name of the column to be added
COLUMN_VALUE	ANYDATA		Value of the column to be added
COLUMN_TYPE	VARCHAR2(4000)		Type of the column to be added
COLUMN_FUNCTION	VARCHAR2(128)		Name of the default function used to add a column
VALUE_TYPE	VARCHAR2(3)		Indicates whether to modify the old (OLD), new (NEW), or both (*) values of the LCR
PRECEDENCE	NUMBER		3 (the execution order relative to other transformations on the same STEP_NUMBER; the smaller number will be executed first)
STEP_NUMBER	NUMBER		Order in which this transformation should be executed

6.2 DBA_STREAMS_DELETE_COLUMN

DBA_STREAMS_DELETE_COLUMN displays information about declarative rule-based transformations that delete a column from a row logical change record (LCR).

Column	Datatype	NULL	Description
RULE_OWNER	VARCHAR2(128)		Owner of the rule
RULE_NAME	VARCHAR2(128)		Name of the rule
SCHEMA_NAME	VARCHAR2(128)		Schema of the column to be deleted
TABLE_NAME	VARCHAR2(128)		Table of the column to be deleted
COLUMN_NAME	VARCHAR2(4000)		Name of the column to delete

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
VALUE_TYPE	VARCHAR2(3)		Indicates whether to modify the old (OLD), new (NEW), or both (*) values of the LCR
PRECEDENCE	NUMBER		1 (the execution order relative to other transformations on the same STEP_NUMBER; the smaller number will be executed first)
STEP_NUMBER	NUMBER		Order in which this transformation should be executed

6.3 DBA_STREAMS_GLOBAL_RULES

DBA_STREAMS_GLOBAL_RULES displays information about the global rules created for all capture processes, propagations, and apply processes in the database. Its columns are the same as those in ALL_STREAMS_GLOBAL_RULES.

See Also

"ALL_STREAMS_GLOBAL_RULES"

6.4 DBA_STREAMS_KEEP_COLUMNS

DBA_STREAMS_KEEP_COLUMNS displays information about declarative rule-based transformations that keep a list of columns in a row logical change record (LCR).

Column	Datatype	NULL	Description
RULE_OWNER	VARCHAR2(128)		Owner of the rule which has an associated transformation
RULE_NAME	VARCHAR2(128)		Name of the rule which has an associated transformation
SCHEMA_NAME	VARCHAR2(128)		Schema of the column to be kept
TABLE_NAME	VARCHAR2(128)		Table of the column to be kept
COLUMN_NAME	VARCHAR2(4000)		Column to keep
VALUE_TYPE	VARCHAR2(3)		Indicates whether to keep the old (OLD), new (NEW), or both (*) value of the LCR
PRECEDENCE	NUMBER		0 (the execution order relative to other transformations on the same STEP_NUMBER; the smaller number will be executed first)
STEP_NUMBER	NUMBER		Order in which this transformation should be executed

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.5 DBA_STREAMS_MESSAGE_CONSUMERS

DBA_STREAMS_MESSAGE_CONSUMERS displays information about all Streams messaging clients in the database. Its columns are the same as those in ALL_STREAMS_MESSAGE_CONSUMERS.

See Also

"ALL_STREAMS_MESSAGE_CONSUMERS"

6.6 DBA_STREAMS_NEWLY_SUPPORTED

DBA_STREAMS_NEWLY_SUPPORTED displays information about all tables in the database that are newly supported by capture processes. Its columns are the same as those in ALL_STREAMS_NEWLY_SUPPORTED.

See Also

"ALL_STREAMS_NEWLY_SUPPORTED"

6.7 DBA_STREAMS_RENAME_COLUMN

DBA_STREAMS_RENAME_COLUMN displays information about declarative rule-based transformations that rename a column in a row logical change record (LCR).

Column	Datatype	NULL	Description
RULE_OWNER	VARCHAR2(128)		Owner of the rule
RULE_NAME	VARCHAR2(128)		Name of the rule
SCHEMA_NAME	VARCHAR2(128)		Schema of the column to be renamed
TABLE_NAME	VARCHAR2(128)		Table of the column to be renamed
FROM_COLUMN_NAME	VARCHAR2(4000)		Column to rename
TO_COLUMN_NAME	VARCHAR2(4000)		New column name
VALUE_TYPE	VARCHAR2(3)		Indicates whether to modify the old (OLD), new (NEW), or both (*) values of the LCR
PRECEDENCE	NUMBER		2 (the execution order relative to other transformations on the same STEP_NUMBER; the smaller number will be executed first)
STEP_NUMBER	NUMBER		Order in which this transformation should be executed

6.8 DBA_STREAMS_RENAME_SCHEMA

DBA_STREAMS_RENAME_SCHEMA displays information about declarative rule-based transformations that rename a schema in a row logical change record (LCR).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
RULE_OWNER	VARCHAR2(128)		Owner of the rule
RULE_NAME	VARCHAR2(128)		Name of the rule
FROM_SCHEMA_NAME	VARCHAR2(128)		Schema to be renamed
TO_SCHEMA_NAME	VARCHAR2(128)		New schema name
PRECEDENCE	NUMBER		5 (the execution order relative to other transformations on the same STEP_NUMBER; the smaller number will be executed first)
STEP_NUMBER	NUMBER		Order in which this transformation should be executed

6.9 DBA_STREAMS_RENAME_TABLE

DBA_STREAMS_RENAME_TABLE displays information about declarative rule-based transformations that rename a table in a row logical change record (LCR).

Column	Datatype	NULL	Description
RULE_OWNER	VARCHAR2(128)		Owner of the rule
RULE_NAME	VARCHAR2(128)		Name of the rule
FROM_SCHEMA_NAME	VARCHAR2(128)		Schema to be renamed
TO_SCHEMA_NAME	VARCHAR2(128)		New schema name
FROM_TABLE_NAME	VARCHAR2(128)		Table to be renamed
TO_TABLE_NAME	VARCHAR2(128)		New table name
PRECEDENCE	NUMBER		4 (the execution order relative to other transformations on the same STEP_NUMBER; the smaller number will be executed first)
STEP_NUMBER	NUMBER		Order in which this transformation should be executed

6.10 DBA_STREAMS_SCHEMA_RULES

DBA_STREAMS_SCHEMA_RULES displays information about the schema rules created for all capture processes, propagations, and apply processes in the database. Its columns are the same as those in ALL_STREAMS_SCHEMA_RULES.

See Also

"ALL_STREAMS_SCHEMA_RULES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.11 DBA_STREAMS_TABLE_RULES

DBA_STREAMS_TABLE_RULES displays information about the table rules created for all capture processes, propagations, and apply processes in the database. Its columns are the same as those in ALL_STREAMS_TABLE_RULES.

See Also

"ALL_STREAMS_TABLE_RULES"

6.12 DBA_STREAMS_TP_COMPONENT

DBA_STREAMS_TP_COMPONENT displays information about each Replication component at each database.

Column	Datatype	NULL	Description
COMPONENT_ID	NUMBER	NOT NULL	ID of the Replication component
COMPONENT_NAME	VARCHAR2(4000)		Name of the Replication component
COMPONENT_DB	VARCHAR2(128)		Database where the Replication component resides
COMPONENT_TYPE	VARCHAR2(20)		Type of the Replication component: • CAPTURE • PROPAGATION SENDER • PROPAGATION RECEIVER • APPLY • QUEUE
COMPONENT_CHANGED_TIME	DATE		Time that the Replication component was last changed

6.13 DBA_STREAMS_TP_COMPONENT_LINK

DBA_STREAMS_TP_COMPONENT_LINK displays information about how messages flow between Replication components.

Column	Datatype	NULL	Description
SOURCE_COMPONENT_ID	NUMBER	NOT NULL	ID of the source Replication component
SOURCE_COMPONENT_NAME	VARCHAR2(4000)		Name of the source Replication component
SOURCE_COMPONENT_DB	VARCHAR2(128)		Database where the source Replication component resides
SOURCE_COMPONENT_TYPE	VARCHAR2(20)		Type of the source Replication component: • CAPTURE • PROPAGATION SENDER • PROPAGATION RECEIVER • APPLY • QUEUE

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DESTINATION_COMPONENT_ID	NUMBER	NOT NULL	ID of the destination Replication component
DESTINATION_COMPONENT_NAME	VARCHAR2(4000)		Name of the destination Replication component
DESTINATION_COMPONENT_DB	VARCHAR2(128)		Database where the destination Replication component resides
DESTINATION_COMPONENT_TYPE	VARCHAR2(20)		Type of the destination Replication component: • CAPTURE • PROPAGATION SENDER • PROPAGATION RECEIVER • APPLY • QUEUE
PATH_ID	NUMBER	NOT NULL	ID of the stream path
POSITION	NUMBER		Position of the link within the stream path

6.14 DBA_STREAMS_TP_COMPONENT_STAT

DBA_STREAMS_TP_COMPONENT_STAT displays temporary performance statistics and session statistics about each Replication component.

Column	Datatype	NULL	Description
COMPONENT_ID	NUMBER	NOT NULL	ID of the Replication component
COMPONENT_NAME	VARCHAR2(4000)		Name of the Replication component
COMPONENT_DB	VARCHAR2(128)		Database where the Replication component resides
COMPONENT_TYPE	VARCHAR2(20)		Type of the Replication component: • CAPTURE • PROPAGATION SENDER • PROPAGATION RECEIVER • APPLY • QUEUE
SUB_COMPONENT_TYPE	VARCHAR2(27)		Type of the Replication subcomponent: • LOGMINER READER • LOGMINER PREPARER • LOGMINER BUILDER • CAPTURE SESSION • PROPAGATION SENDER+RECEIVER • APPLY READER • APPLY COORDINATOR • APPLY SERVER
SESSION_ID	NUMBER		ID of the Replication session for the Replication component
SESSION_SERIAL#	NUMBER		Serial number of the Replication session for the Replication component
STATISTIC_TIME	DATE		Time that the statistic was taken
STATISTIC_NAME	VARCHAR2(64)		Name of the statistic
STATISTIC_VALUE	VARCHAR2(4000)		Value of the statistic

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATISTIC_UNIT	VARCHAR2(64)		Unit of the statistic
ADVISOR_RUN_ID	NUMBER		Logical number (1-based) of the Advisor run
ADVISOR_RUN_TIME	DATE		Time that the Advisor was run

6.15 DBA_STREAMS_TP_DATABASE

DBA_STREAMS_TP_DATABASE displays information about each database that contains Replication components.

Column	Datatype	NULL	Description
GLOBAL_NAME	VARCHAR2(128)	NOT NULL	Global name of the database
LAST_QUERIED	DATE	NOT NULL	Time that the database was last queried
VERSION	VARCHAR2(128)		Database version of the database
COMPATIBILITY	VARCHAR2(128)		Compatible setting of the database
MANAGEMENT_PACK_ACCESS	VARCHAR2(128)		Management pack access of the database

6.16 DBA_STREAMS_TP_PATH_BOTTLENECK

DBA_STREAMS_TP_PATH_BOTTLENECK displays temporary information about Replication components that might be slowing down the flow of messages in a stream path.

Column	Datatype	NULL	Description
PATH_ID	NUMBER		ID of the stream path
COMPONENT_ID	NUMBER		ID of the bottleneck component
COMPONENT_NAME	VARCHAR2(4000)		Name of the bottleneck component
COMPONENT_DB	VARCHAR2(128)		Database where the bottleneck component resides
COMPONENT_TYPE	VARCHAR2(20)		Type of the bottleneck component: • CAPTURE • PROPAGATION SENDER • PROPAGATION RECEIVER • APPLY • QUEUE
TOP_SESSION_ID	NUMBER		ID of the top session for the bottleneck component
TOP_SESSION_SERIAL#	NUMBER		Serial number of the top session for the bottleneck component
ACTION_NAME	VARCHAR2(64)		Action name for the bottleneck process
BOTTLENECK_IDENTIFIED	VARCHAR2(30)		Indicates whether the bottleneck was identified (YES) or not (NO)
ADVISOR_RUN_ID	NUMBER		Logical number (1-based) of the Advisor run
ADVISOR_RUN_TIME	DATE		Time that the Advisor was run

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ADVISOR_RUN_REASON	VARCHAR2(4000)		Reason for the bottleneck analysis result.: • NULL - Bottleneck is identified • PRE-11.1 DATABASE EXISTS - A pre-release 11.1 database exists in the stream path • DIAGNOSTIC PACK REQUIRED - A database in the stream path does not have the diagnostic package installed • NO BOTTLENECK IDENTIFIED

6.17 DBA_STREAMS_TP_PATH_STAT

DBA_STREAMS_TP_PATH_STAT displays temporary performance statistics about each stream path that exists in the Replication topology.

Column	Datatype	NULL	Description
PATH_ID	NUMBER		ID of the stream path
STATISTIC_TIME	DATE		Time that the statistic was taken
STATISTIC_NAME	VARCHAR2(64)		Name of the statistic
STATISTIC_VALUE	NUMBER		Value of the statistic
STATISTIC_UNIT	VARCHAR2(64)		Unit of the statistic
ADVISOR_RUN_ID	NUMBER		Logical number (1-based) of the Advisor run
ADVISOR_RUN_TIME	DATE		Time that the Advisor was run

6.18 DBA_STREAMS_TRANSFORM_FUNCTION

DBA_STREAMS_TRANSFORM_FUNCTION displays information about all rule-based transformation functions in the database. Its columns are the same as those in ALL_STREAMS_TRANSFORM_FUNCTION.

See Also

"ALL_STREAMS_TRANSFORM_FUNCTION"

6.19 DBA_SUBPART_COL_STATISTICS

DBA_SUBPART_COL_STATISTICS provides column statistics and histogram information for all subpartitions in the database. Its columns are the same as those in ALL_SUBPART_COL_STATISTICS.

See Also

"ALL_SUBPART_COL_STATISTICS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.20 DBA_SUBPART_HISTOGRAMS

DBA_SUBPART_HISTOGRAMS lists actual histogram data (end-points per histogram) for histograms on all table subpartitions in the database. Its columns are the same as those in ALL_SUBPART_HISTOGRAMS.

> **See Also**
>
> "ALL_SUBPART_HISTOGRAMS"

6.21 DBA_SUBPART_KEY_COLUMNS

DBA_SUBPART_KEY_COLUMNS lists subpartitioning key columns for all composite-partitioned tables (and local indexes on composite-partitioned tables) in the database. Its columns are the same as those in ALL_SUBPART_KEY_COLUMNS.

> **See Also**
>
> "ALL_SUBPART_KEY_COLUMNS"

6.22 DBA_SUBPARTITION_TEMPLATES

DBA_SUBPARTITION_TEMPLATES describes all subpartition templates in the database. Its columns are the same as those in ALL_SUBPARTITION_TEMPLATES.

> **See Also**
>
> "ALL_SUBPARTITION_TEMPLATES"

6.23 DBA_SUBSCR_REGISTRATIONS

DBA_SUBSCR_REGISTRATIONS displays information about all subscription registrations in the database.

Related View

USER_SUBSCR_REGISTRATIONS displays information about the subscription registrations owned by the current user.

Column	Datatype	NULL	Description
REG_ID	NUMBER		Registration ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SUBSCRIPTION_NAME	VARCHAR2(128)	NOT NULL	Name of the subscription registration. The subscription name is of the form *schema*.*queue* if the registration is for a single consumer queue or *schema*.*queue*:*consumer_name* if the registration is for a multiconsumer queue.
LOCATION_NAME	VARCHAR2(256)	NOT NULL	Location endpoint of the registration
USER#	NUMBER	NOT NULL	Internally generated user ID
USER_CONTEXT	RAW(128)		Context the user provided during registration of PL/SQL registrations or an internally generated context for OCI registrations
CONTEXT_SIZE	NUMBER		Size of the context
NAMESPACE	VARCHAR2(9)		Namespace of the subscription registration: • ANONYMOUS • AQ • DBCHANGE
PRESENTATION	VARCHAR2(7)		Presentation format of notifications: • DEFAULT - Binary • XML
VERSION	VARCHAR2(8)		Database version: • 8.1.6 • 10.2 • 11.1
STATUS	VARCHAR2(8)		Status of the registration: • DB REG - Database registration • LDAP REG - LDAP registration
ANY_CONTEXT	ANYDATA		AnyData user context
CONTEXT_TYPE	NUMBER		Type of the user context
QOSFLAGS	VARCHAR2(64)		Quality of service of the registration: • RELIABLE - Reliable notifications persist across instance and database restarts • PAYLOAD - Payload delivery is required. It is only supported for client notification and only for RAW queues. • PURGE_ON_NTFN - Registration is to be purged automatically when the first notification is delivered to this registration location
PAYLOAD_CALLBACK	VARCHAR2(4000)		Any callback registered to serialize the notification payload
TIMEOUT	TIMESTAMP(6)		Registration timeout
REG_TIME	TIMESTAMP(6) WITH TIME ZONE		Time of the registration
NTFN_GROUPING_CLASS	VARCHAR2(4)		Notification grouping class
NTFN_GROUPING_VALUE	NUMBER		Notification grouping value
NTFN_GROUPING_TYPE	VARCHAR2(7)		Notification grouping type: • SUMMARY • LAST
NTFN_GROUPING_START_TIME	TIMESTAMP(6) WITH TIME ZONE		Notification grouping start time

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
NTFN_GROUPING_REPEAT_COUNT	VARCHAR2(40)		Notification grouping repeat count, or FOREVER

See Also

"USER_SUBSCR_REGISTRATIONS"

6.24 DBA_SUPPLEMENTAL_LOGGING

DBA_SUPPLEMENTAL_LOGGING provides information about supplemental logging for a pluggable database (PDB) in a multitenant container database (CDB).

Column	Datatype	NULL	Description
MINIMAL	VARCHAR2(3)		Identifies whether minimal supplemental logging is on (YES or NO)
PRIMARY_KEY	VARCHAR2(3)		Identifies whether primary key supplemental logging is on (YES or NO)
UNIQUE_INDEX	VARCHAR2(3)		Identifies whether unique column supplemental logging is on (YES or NO)
FOREIGN_KEY	VARCHAR2(3)		Identifies whether foreign key supplemental logging is on (YES or NO)
ALL_COLUMN	VARCHAR2(3)		Identifies whether all column supplemental logging is on (YES or NO)
PROCEDURAL	VARCHAR2(3)		Identifies whether supplemental logging for procedural replication is on (YES or NO)
SUBSET_REP[1]	VARCHAR2(3)		Indicates whether subset database replication is on (YES or NO)

1 This column is available starting with Oracle Database 19c.

See Also

- *Oracle Database Utilities* for more information about supplemental logging
- "V$DATABASE" for information about supplemental logging in a CDB

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.25 DBA_SYNC_CAPTURE

DBA_SYNC_CAPTURE displays information about all synchronous capture processes in the database. Its columns are the same as those in ALL_SYNC_CAPTURE.

See Also

"ALL_SYNC_CAPTURE"

6.26 DBA_SYNC_CAPTURE_PREPARED_TABS

DBA_SYNC_CAPTURE_PREPARED_TABS displays information about all tables in the database that are prepared for synchronous capture instantiation. Its columns are the same as those in ALL_SYNC_CAPTURE_PREPARED_TABS.

See Also

"ALL_SYNC_CAPTURE_PREPARED_TABS"

6.27 DBA_SYNC_CAPTURE_TABLES

DBA_SYNC_CAPTURE_TABLES displays information about all tables in the database that are captured by synchronous captures. Its columns are the same as those in ALL_SYNC_CAPTURE_TABLES.

See Also

"ALL_SYNC_CAPTURE_TABLES"

6.28 DBA_SYNONYMS

DBA_SYNONYMS describes all synonyms in the database. Its columns are the same as those in ALL_SYNONYMS.

See Also

"ALL_SYNONYMS"

6.29 DBA_SYS_PRIVS

DBA_SYS_PRIVS describes system privileges granted to users and roles.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)		Grantee name, user, or role receiving the grant
PRIVILEGE	VARCHAR2(40)		System privilege
ADMIN_OPTION	VARCHAR2(3)		Indicates whether the grant was with the ADMIN option (YES) or not (NO)
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the privilege was granted commonly (CONTAINER=ALL was used) • NO if the privilege was granted locally (CONTAINER=ALL was not used)
INHERITED	VARCHAR2(3)		Indicates whether the grant was inherited from another container (YES) or not (NO)

See Also

"USER_SYS_PRIVS"

6.30 DBA_TAB_COL_STATISTICS

DBA_TAB_COL_STATISTICS contains column statistics and histogram information extracted from DBA_TAB_COLUMNS. Its columns are the same as those in ALL_TAB_COL_STATISTICS.

See Also

- "DBA_TAB_COLUMNS"
- "ALL_TAB_COL_STATISTICS"

6.31 DBA_TAB_COLS

DBA_TAB_COLS describes the columns of all tables, views, and clusters in the database.

Its columns (except for SENSITIVE_COLUMN) are the same as those in ALL_TAB_COLS.

To gather statistics for this view, use the DBMS_STATS package.

This view differs from DBA_TAB_COLUMNS in that system-generated hidden columns are not filtered out.

Columns marked with an asterisk (*) in the table below remain for backward compatibility with Oracle7. This information is now in the [TAB|PART]_COL_STATISTICS views.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table, view, or cluster
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table, view, or cluster
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Column name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DATA_TYPE	VARCHAR2(128)		Data type of the column
DATA_TYPE_MOD	VARCHAR2(3)		Data type modifier of the column
DATA_TYPE_OWNER	VARCHAR2(128)		Owner of the data type of the column
DATA_LENGTH	NUMBER	NOT NULL	Length of the column (in bytes)
DATA_PRECISION	NUMBER		Decimal precision for NUMBER data type; binary precision for FLOAT data type; NULL for all other data types
DATA_SCALE	NUMBER		Digits to the right of the decimal point in a number
NULLABLE	VARCHAR2(1)		Indicates whether a column allows NULLs. The value is N if there is a NOT NULL constraint on the column or if the column is part of a PRIMARY KEY.
COLUMN_ID	NUMBER		Sequence number of the column as created
DEFAULT_LENGTH	NUMBER		Length of the default value for the column
DATA_DEFAULT	LONG		Default value for the column
NUM_DISTINCT*	NUMBER		Number of distinct values in the column
LOW_VALUE*	RAW(1000)		Low value in the column
HIGH_VALUE*	RAW(1000)		High value in the column
DENSITY*	NUMBER		If a histogram is available on COLUMN_NAME, then this column displays the selectivity of a value that spans fewer than 2 endpoints in the histogram. It does not represent the selectivity of values that span 2 or more endpoints. If a histogram is not available on COLUMN_NAME, then the value of this column is 1/NUM_DISTINCT.
NUM_NULLS	NUMBER		Number of NULLs in the column
NUM_BUCKETS	NUMBER		Number of buckets in the histogram for the column **Note:** The number of buckets in a histogram is specified in the SIZE parameter of the ANALYZE SQL statement. However, Oracle Database does not create a histogram with more buckets than the number of rows in the sample. Also, if the sample contains any values that are very repetitious, Oracle Database creates the specified number of buckets, but the value indicated by this column may be smaller because of an internal compression algorithm.
LAST_ANALYZED	DATE		Date on which this column was most recently analyzed
SAMPLE_SIZE	NUMBER		Sample size used in analyzing this column
CHARACTER_SET_NAME	VARCHAR2(44)		Name of the character set: • CHAR_CS • NCHAR_CS
CHAR_COL_DECL_LENGTH	NUMBER		Declaration length of the character type column
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics are gathered or incrementally maintained, otherwise it will be NO
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
AVG_COL_LEN	NUMBER		Average length of the column (in bytes)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CHAR_LENGTH	NUMBER		Displays the length of the column in characters. This value only applies to the following data types: • CHAR • VARCHAR2 • NCHAR • NVARCHAR2
CHAR_USED	VARCHAR2(1)		Indicates that the column uses BYTE length semantics (B) or CHAR length semantics (C), or whether the data type is not any of the following (NULL): • CHAR • VARCHAR2 • NCHAR • NVARCHAR2
V80_FMT_IMAGE	VARCHAR2(3)		Indicates whether the column data is in release 8.0 image format (YES) or not (NO)
DATA_UPGRADED	VARCHAR2(3)		Indicates whether the column data has been upgraded to the latest type version format (YES) or not (NO)
HIDDEN_COLUMN	VARCHAR2(3)		Indicates whether the column is a hidden column (YES) or not (NO)
VIRTUAL_COLUMN	VARCHAR2(3)		Indicates whether the column is a virtual column (YES) or not (NO)
SEGMENT_COLUMN_ID	NUMBER		Sequence number of the column in the segment
INTERNAL_COLUMN_ID	NUMBER	NOT NULL	Internal sequence number of the column
HISTOGRAM	VARCHAR2(15)		Indicates existence/type of histogram: • NONE • FREQUENCY • TOP-FREQUENCY • HEIGHT BALANCED • HYBRID
QUALIFIED_COL_NAME	VARCHAR2(4000)		Qualified column name
USER_GENERATED	VARCHAR2(3)		Indicates whether the column is a user-generated column (YES) or a system-generated column (NO)
DEFAULT_ON_NULL	VARCHAR2(3)		Indicates whether the column has DEFAULT ON NULL semantics (YES) or not (NO)
IDENTITY_COLUMN	VARCHAR2(3)		Indicates whether this is an identity column (YES) or not (NO)
SENSITIVE_COLUMN	VARCHAR2(3)		Indicates whether this is a sensitive column (YES) or not (NO)
EVALUATION_EDITION	VARCHAR2(128)		Name of the edition in which editioned objects referenced in an expression column are resolved
UNUSABLE_BEFORE	VARCHAR2(128)		Name of the oldest edition in which the index may be used as part of a query plan
UNUSABLE_BEGINNING	VARCHAR2(128)		Name of the edition for which the index may not be used as part of a query plan in this edition or any of its descendants
COLLATION	VARCHAR2(100)		Collation for the column. Only applies to columns with character data types.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COLLATED_COLUMN_ID	NUMBER		Internal sequence number of a column, for which this virtual column generates a collation key

See Also

- "ALL_TAB_COLS"
- "DBA_TAB_COLUMNS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

6.32 DBA_TAB_COLUMNS

DBA_TAB_COLUMNS describes the columns of all tables, views, and clusters in the database.

Its columns (except for SENSITIVE_COLUMN) are the same as those in ALL_TAB_COLUMNS.

To gather statistics for this view, use the DBMS_STATS package.

This view filters out system-generated hidden columns. The DBA_TAB_COLS view does not filter out system-generated hidden columns.

Columns marked with an asterisk (*) in the table below remain for backward compatibility with Oracle7. This information is now in the [TAB|PART]_COL_STATISTICS views.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	Owner of the table, view, or cluster
TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the table, view, or cluster
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Column name
DATA_TYPE	VARCHAR2(128)		Data type of the column
DATA_TYPE_MOD	VARCHAR2(3)		Data type modifier of the column
DATA_TYPE_OWNER	VARCHAR2(128)		Owner of the data type of the column
DATA_LENGTH	NUMBER	NOT NULL	Length of the column (in bytes)
DATA_PRECISION	NUMBER		Decimal precision for NUMBER data type; binary precision for FLOAT data type; NULL for all other data types
DATA_SCALE	NUMBER		Digits to the right of the decimal point in a number
NULLABLE	VARCHAR2(1)		Indicates whether a column allows NULLs. The value is N if there is a NOT NULL constraint on the column or if the column is part of a PRIMARY KEY. The constraint should be in an ENABLE VALIDATE state.
COLUMN_ID	NUMBER		Sequence number of the column as created
DEFAULT_LENGTH	NUMBER		Length of the default value for the column
DATA_DEFAULT	LONG		Default value for the column
NUM_DISTINCT*	NUMBER		Number of distinct values in the column
LOW_VALUE*	RAW(1000)		Low value in the column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
HIGH_VALUE*	RAW(1000)		High value in the column
DENSITY*	NUMBER		If a histogram is available on COLUMN_NAME, then this column displays the selectivity of a value that spans fewer than 2 endpoints in the histogram. It does not represent the selectivity of values that span 2 or more endpoints. If a histogram is not available on COLUMN_NAME, then the value of this column is 1/NUM_DISTINCT.
NUM_NULLS	NUMBER		Number of NULLs in the column
NUM_BUCKETS	NUMBER		Number of buckets in the histogram for the column **Note:** The number of buckets in a histogram is specified in the SIZE parameter of the ANALYZE SQL statement. However, Oracle Database does not create a histogram with more buckets than the number of rows in the sample. Also, if the sample contains any values that are very repetitious, Oracle Database creates the specified number of buckets, but the value indicated by this column may be smaller because of an internal compression algorithm.
LAST_ANALYZED	DATE		Date on which this column was most recently analyzed
SAMPLE_SIZE	NUMBER		Sample size used in analyzing this column
CHARACTER_SET_NAME	VARCHAR2(44)		Name of the character set: • CHAR_CS • NCHAR_CS
CHAR_COL_DECL_LENGTH	NUMBER		Declaration length of the character type column
GLOBAL_STATS	VARCHAR2(3)		GLOBAL_STATS will be YES if statistics are gathered or incrementally maintained, otherwise it will be NO
USER_STATS	VARCHAR2(3)		Indicates whether statistics were entered directly by the user (YES) or not (NO)
AVG_COL_LEN	NUMBER		Average length of the column (in bytes)
CHAR_LENGTH	NUMBER		Displays the length of the column in characters. This value only applies to the following data types: • CHAR • VARCHAR2 • NCHAR • NVARCHAR2
CHAR_USED	VARCHAR2(1)		Indicates that the column uses BYTE length semantics (B) or CHAR length semantics (C), or whether the data type is not any of the following (NULL): • CHAR • VARCHAR2 • NCHAR • NVARCHAR2
V80_FMT_IMAGE	VARCHAR2(3)		Indicates whether the column data is in release 8.0 image format (YES) or not (NO)
DATA_UPGRADED	VARCHAR2(3)		Indicates whether the column data has been upgraded to the latest type version format (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
HISTOGRAM	VARCHAR2(15)		Indicates existence/type of histogram: • NONE • FREQUENCY • TOP-FREQUENCY • HEIGHT BALANCED • HYBRID
DEFAULT_ON_NULL	VARCHAR2(3)		Indicates whether the column has DEFAULT ON NULL semantics (YES) or not (NO)
IDENTITY_COLUMN	VARCHAR2(3)		Indicates whether this is an identity column (YES) or not (NO)
SENSITIVE_COLUMN	VARCHAR2(3)		Indicates whether this is a sensitive column (YES) or not (NO)
EVALUATION_EDITION	VARCHAR2(128)		Name of the edition in which editioned objects referenced in an expression column are resolved
UNUSABLE_BEFORE	VARCHAR2(128)		Name of the oldest edition in which the index may be used as part of a query plan
UNUSABLE_BEGINNING	VARCHAR2(128)		Name of the edition for which the index may not be used as part of a query plan in this edition or any of its descendants
COLLATION	VARCHAR2(100)		Collation for the column. Only applies to columns with character data types.

See Also

- "ALL_TAB_COLUMNS"
- "DBA_TAB_COLS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

6.33 DBA_TAB_COMMENTS

DBA_TAB_COMMENTS displays comments on all tables and views in the database. Its columns are the same as those in ALL_TAB_COMMENTS.

See Also

"ALL_TAB_COMMENTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.34 DBA_TAB_HISTGRM_PENDING_STATS

DBA_TAB_HISTGRM_PENDING_STATS describes pending statistics for tables, partitions, and subpartitions in the database. Its columns are the same as those in ALL_TAB_HISTGRM_PENDING_STATS.

See Also

"ALL_TAB_HISTGRM_PENDING_STATS"

6.35 DBA_TAB_HISTOGRAMS

DBA_TAB_HISTOGRAMS describes histograms on columns of all tables in the database. Its columns are the same as those in ALL_TAB_HISTOGRAMS.

See Also

"ALL_TAB_HISTOGRAMS"

6.36 DBA_TAB_IDENTITY_COLS

DBA_TAB_IDENTITY_COLS describes all table identity columns. Its columns are the same as those in ALL_TAB_IDENTITY_COLS.

See Also

"ALL_TAB_IDENTITY_COLS"

6.37 DBA_TAB_MODIFICATIONS

DBA_TAB_MODIFICATIONS describes modifications to all tables in the database that have been modified since the last time statistics were gathered on the tables. Its columns are the same as those in ALL_TAB_MODIFICATIONS.

See Also

"ALL_TAB_MODIFICATIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.38 DBA_TAB_PARTITIONS

DBA_TAB_PARTITIONS displays partition-level partitioning information, partition storage parameters, and partition statistics generated by the DBMS_STATS package for all partitions in the database.

Its columns are the same as those in "ALL_TAB_PARTITIONS".

6.39 DBA_TAB_PENDING_STATS

DBA_TAB_PENDING_STATS describes pending statistics for tables, partitions, and subpartitions in the database. Its columns are the same as those in ALL_TAB_PENDING_STATS.

> **See Also**
>
> "ALL_TAB_PENDING_STATS"

6.40 DBA_TAB_PRIVS

DBA_TAB_PRIVS describes all object grants in the database.

Related View

USER_TAB_PRIVS describes the object grants for which the current user is the object owner, grantor, or grantee.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)		Name of the user or role to whom access was granted
OWNER	VARCHAR2(128)		Owner of the object
TABLE_NAME	VARCHAR2(128)		Name of the object. The object can be any object, including tables, packages, indexes, sequences, and so on.
GRANTOR	VARCHAR2(128)		Name of the user who performed the grant
PRIVILEGE	VARCHAR2(40)		Privilege on the object
GRANTABLE	VARCHAR2(3)		Indicates whether the privilege was granted with the GRANT OPTION (YES) or not (NO)
HIERARCHY	VARCHAR2(3)		Indicates whether the privilege was granted with the HIERARCHY OPTION (YES) or not (NO)
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the privilege was granted commonly (CONTAINER=ALL was used) • NO if the privilege was granted locally (CONTAINER=ALL was not used)
TYPE	VARCHAR2(24)		Type of the object
INHERITED	VARCHAR2(3)		Indicates whether the grant was inherited from another container (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "USER_TAB_PRIVS"

6.41 DBA_TAB_STATISTICS

DBA_TAB_STATISTICS displays optimizer statistics for all tables in the database. Its columns are the same as those in ALL_TAB_STATISTICS.

> **See Also**
>
> "ALL_TAB_STATISTICS"

6.42 DBA_TAB_STAT_PREFS

DBA_TAB_STAT_PREFS displays information about statistics preferences for all tables in the database. Its columns are the same as those in ALL_TAB_STAT_PREFS.

> **Note**
>
> In earlier releases, this view could be queried by any user with the PUBLIC role. Starting with Oracle Database 19c, Release Update 19.28, this view adheres to the convention for DBA_ views, that is, it can be queried only by users with the SYSDBA system privilege or SELECT ANY DICTIONARY privilege, or SELECT_CATALOG_ROLE role, or by users with direct privileges granted to them. This change improves access control for optimizer and statistics metadata and reduces information disclosure.

> **See Also**
>
> "ALL_TAB_STAT_PREFS".

6.43 DBA_TAB_STATS_HISTORY

DBA_TAB_STATS_HISTORY provides a history of table statistics modifications for all tables in the database. Its columns are the same as those in ALL_TAB_STATS_HISTORY.

> **See Also**
>
> "ALL_TAB_STATS_HISTORY"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.44 DBA_TAB_SUBPARTITIONS

DBA_TAB_SUBPARTITIONS displays, for each table subpartition, the subpartition name, name of the table and partition to which it belongs, its storage attributes, and statistics generated by the DBMS_STATS package.

Its columns are the same as those in "ALL_TAB_SUBPARTITIONS".

6.45 DBA_TABLES

DBA_TABLES describes all relational tables in the database. Its columns are the same as those in ALL_TABLES.

To gather statistics for this view, use the DBMS_STATS package.

> **See Also**
>
> "ALL_TABLES"

6.46 DBA_TABLESPACE_GROUPS

DBA_TABLESPACE_GROUPS describes all tablespace groups in the database.

Column	Datatype	NULL	Description
GROUP_NAME	VARCHAR2(30)	NOT NULL	Name of the tablespace group
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of the tablespace

6.47 DBA_TABLESPACE_THRESHOLDS

DBA_TABLESPACE_THRESHOLDS describes space utilization threshold settings for all tablespaces in the database.

Column	Datatype	NULL	Description
TABLESPACE_NAME	VARCHAR2(30)		Tablespace name
CONTENTS	VARCHAR2(9)		Tablespace contents: • UNDO • PERMANENT • TEMPORARY
EXTENT_MANAGEMENT	VARCHAR2(10)		Indicates whether the extents in the tablespace are dictionary managed (DICTIONARY) or locally managed (LOCAL)
THRESHOLD_TYPE	VARCHAR2(8)		Indicates whether the threshold value is derived from a DEFAULT threshold or an EXPLICIT threshold
METRICS_NAME	VARCHAR2(64)		Name of the metric for which the threshold is set

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
WARNING_OPERATOR	VARCHAR2(12)		Relational operator for warning thresholds: • GT • EQ • LT • LE • GE • CONTAINS • NE • DO_NOT_CHECK
WARNING_VALUE	VARCHAR2(256)		Warning threshold value
CRITICAL_OPERATOR	VARCHAR2(12)		Relational operator for critical thresholds: • GT • EQ • LT • LE • GE • CONTAINS • NE • DO_NOT_CHECK
CRITICAL_VALUE	VARCHAR2(256)		Critical threshold value

6.48 DBA_TABLESPACE_USAGE_METRICS

DBA_TABLESPACE_USAGE_METRICS describes tablespace usage metrics for all types of tablespaces, including permanent, temporary, and undo tablespaces.

Column	Datatype	NULL	Description
TABLESPACE_NAME	VARCHAR2(30)		Tablespace name
USED_SPACE	NUMBER		Total space consumed by all objects created in the tablespace, expressed as number of data blocks For undo tablespaces, the value of this column includes space consumed by both expired and unexpired undo segments.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TABLESPACE_SIZE	NUMBER		The maximum size of the tablespace, expressed as number of data blocks • If the tablespace contains any datafiles with autoextend enabled, then this column displays the maximum size to which the tablespace can grow. Underlying storage free space, such as Oracle ASM or file system storage, is taken into account when computing this value. In a CDB, if the value of the MAX_PDB_STORAGE property of the CDB_PROPERTIES view is non-zero, then that value is also taken into account. For example: – If a tablespace has a current size of 5 GB, the combined maximum size of its datafiles is 32 GB, and its underlying storage has 20 GB of free space, then this column will have a value of approximately 25 GB. – If a tablespace has a current size of 10 GB, the combined maximum size of its datafiles is 20 GB, and its underlying storage has 25 GB of free space, then this column will have a value of approximately 20 GB. – If a tablespace has a current size of 15 GB, the combined maximum size of its datafiles is 32 GB, its underlying storage has 90 GB of free space, the current PDB size is 40 GB, and MAX_PDB_STORAGE is 50 GB, then this column will have a value of approximately 25 GB. • If the tablespace contains only datafiles with autoextend disabled, then this column displays the combined size of all datafiles in the tablespace.
USED_PERCENT	NUMBER		Percentage of used space, as a function of the maximum possible tablespace size

6.49 DBA_TABLESPACES

DBA_TABLESPACES describes all tablespaces in the database.

Related View

USER_TABLESPACES describes the tablespaces accessible to the current user. This view does not display the PLUGGED_IN column.

Column	Datatype	NULL	Description
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of the tablespace
BLOCK_SIZE	NUMBER	NOT NULL	Tablespace block size (in bytes)
INITIAL_EXTENT	NUMBER		Default initial extent size (in bytes)
NEXT_EXTENT	NUMBER		Default incremental extent size (in bytes)
MIN_EXTENTS	NUMBER	NOT NULL	Default minimum number of extents
MAX_EXTENTS	NUMBER		Default maximum number of extents

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MAX_SIZE	NUMBER		Default maximum size of segments (in Oracle blocks)
PCT_INCREASE	NUMBER		Default percent increase for extent size
MIN_EXTLEN	NUMBER		Minimum extent size for this tablespace (in bytes)
STATUS	VARCHAR2(9)		Tablespace status: • ONLINE • OFFLINE • READ ONLY
CONTENTS	VARCHAR2(9)		Tablespace contents: • UNDO • LOST WRITE PROTECTION • PERMANENT • TEMPORARY
LOGGING	VARCHAR2(9)		Default logging attribute: • LOGGING • NOLOGGING
FORCE_LOGGING	VARCHAR2(3)		Indicates whether the tablespace is under force logging mode (YES) or not (NO)
EXTENT_MANAGEMENT	VARCHAR2(10)		Indicates whether the extents in the tablespace are dictionary managed (DICTIONARY) or locally managed (LOCAL)
ALLOCATION_TYPE	VARCHAR2(9)		Type of extent allocation in effect for the tablespace: • SYSTEM • UNIFORM • USER
PLUGGED_IN	VARCHAR2(3)		Indicates whether the tablespace is plugged in (YES) or not (NO)
SEGMENT_SPACE_MANAGEMENT	VARCHAR2(6)		Indicates whether the free and used segment space in the tablespace is managed using free lists (MANUAL) or bitmaps (AUTO)
DEF_TAB_COMPRESSION	VARCHAR2(8)		Indicates whether default table compression is enabled (ENABLED) or not (DISABLED) **Note:** Enabling default table compression indicates that all tables in the tablespace will be created with table compression enabled unless otherwise specified.
RETENTION	VARCHAR2(11)		Undo tablespace retention: • GUARANTEE - Tablespace is an undo tablespace with RETENTION specified as GUARANTEE A RETENTION value of GUARANTEE indicates that unexpired undo in all undo segments in the undo tablespace should be retained even if it means that forward going operations that need to generate undo in those segments fail. • NOGUARANTEE - Tablespace is an undo tablespace with RETENTION specified as NOGUARANTEE • NOT APPLY - Tablespace is not an undo tablespace
BIGFILE	VARCHAR2(3)		Indicates whether the tablespace is a bigfile tablespace (YES) or a smallfile tablespace (NO)
PREDICATE_EVALUATION	VARCHAR2(7)		Indicates whether predicates are evaluated by host (HOST) or by storage (STORAGE)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ENCRYPTED	VARCHAR2(3)		Indicates whether the tablespace is encrypted (YES) or not (NO)
COMPRESS_FOR	VARCHAR2(30)		Default compression for what kind of operations: • BASIC • ADVANCED • QUERY LOW • QUERY HIGH • ARCHIVE LOW • ARCHIVE HIGH • NULL The QUERY LOW, QUERY HIGH, ARCHIVE LOW, and ARCHIVE HIGH values are associated with Hybrid Columnar Compression, a feature of the Enterprise Edition of Oracle Database that is dependent on the underlying storage system. See *Oracle Database Concepts* for more information.
DEF_INMEMORY	VARCHAR2(8)		Indicates whether the In-Memory Column Store (IM column store) is by default enabled (ENABLED) or disabled (DISABLED) for tables in this tablespace
DEF_INMEMORY_PRIORITY	VARCHAR2(8)		Indicates the default priority for In-Memory Column Store (IM column store) population for this tablespace. Possible values: • LOW • MEDIUM • HIGH • CRITICAL • NONE • NULL
DEF_INMEMORY_DISTRIBUTE	VARCHAR2(15)		Indicates how the IM column store is distributed by default for this tablespace in an Oracle Real Application Clusters (Oracle RAC) environment: • AUTO • BY ROWID RANGE • BY PARTITION • BY SUBPARTITION
DEF_INMEMORY_COMPRESSION	VARCHAR2(17)		Indicates the default compression level for the IM column store for this tablespace: • NO MEMCOMPRESS • FOR DML • FOR QUERY [LOW | HIGH] • FOR CAPACITY [LOW | HIGH] • NULL
DEF_INMEMORY_DUPLICATE	VARCHAR2(13)		Indicates the duplicate setting for the IM column store in an Oracle RAC environment: • NO DUPLICATE • DUPLICATE • DUPLICATE ALL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
SHARED	VARCHAR2(12)		Tablespace type: • SHARED: For shared tablespace • LOCAL_ON_LEAF: For local temporary tablespace for leaf (read-only) instances • LOCAL_ON_ALL: For local temporary tablespace for all instance types
DEF_INDEX_COMPRESSION	VARCHAR2(8)		Indicates whether default index compression is enabled (ENABLED) or not (DISABLED) **Note:** Enabling default index compression indicates that all indexes in the tablespace will be created with index compression enabled unless otherwise specified.
INDEX_COMPRESS_FOR	VARCHAR2(13)		Valid values are: • ADVANCED LOW • ADVANCED HIGH • NULL No other values are allowed.
DEF_CELLMEMORY	VARCHAR2(14)		This specifies the default value for the CELLMEMORY attribute that tables created in the tablespace will inherit unless the behavior is overridden explicitly This column is intended for use with Oracle Exadata.
DEF_INMEMORY_SERVICE	VARCHAR2(12)		Indicates how the IM column store is populated on various instances by default for this tablespace. The possible values are: • DEFAULT: Data is populated on all instances specified with the PARALLEL_INSTANCE_GROUP initialization parameter. If that parameter is not set, then the data is populated on all instances. This is the default. • NONE: Data is not populated on any instance. • ALL: Data is populated on all instances, regardless of the value of the PARALLEL_INSTANCE_GROUP initialization parameter. • USER_DEFINED: Data is populated only on the instances on which the user-specified service is active. The service name corresponding to this is stored in the DEF_INMEMORY_SERVICE_NAME column.
DEF_INMEMORY_SERVICE_NAME	VARCHAR2(1000)		Indicates the service name for the service on which the IM column store should be populated by default for this tablespace. This column has a value only when the corresponding DEF_INMEMORY_SERVICE is USER_DEFINED. In all other cases, this column is null.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
LOST_WRITE_PROTECT	VARCHAR2(7)		The lost write protection setting for the tablespace. Possible values: • ENABLED: Indicates that lost write data is being collected • OFF: Indicates that lost write data is not being collected • SUSPEND: Indicates that lost write data is not currently being collected, but it can be enabled at a later date. The lost write data collected when the file was ENABLED remains in the lost write database, but it is not being checked or updated. If lost write protection is enabled for a tablespace, it is enabled for all data files for that tablespace, including data files added later. If lost write protection is enabled for a single data file, it does not have to be enabled for another data file in the same tablespace. You can check the lost write protection status for a data file by querying the LOST_WRITE_PROTECT column in the DBA_DATA_FILES view.
CHUNK_TABLESPACE	VARCHAR2(1)		Indicates whether this is a chunk tablespace (Y) or not (N)

See Also

- "USER_TABLESPACES"
- "PARALLEL_INSTANCE_GROUP"
- "DBA_DATA_FILES"

6.50 DBA_TEMP_FILES

DBA_TEMP_FILES describes all temporary files (tempfiles) in the database.

Column	Datatype	NULL	Description
FILE_NAME	VARCHAR2(513)		Name of the database temp file
FILE_ID	NUMBER		File identifier number of the database temp file
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of the tablespace to which the file belongs
BYTES	NUMBER		Size of the file (in bytes)
BLOCKS	NUMBER		Size of the file (in Oracle blocks)
STATUS	VARCHAR2(7)		File status: • OFFLINE • ONLINE • UNKNOWN
RELATIVE_FNO	NUMBER		Tablespace-relative file number
AUTOEXTENSIBLE	VARCHAR2(3)		Indicates whether the file is autoextensible (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MAXBYTES	NUMBER		maximum size of the file (in bytes)
MAXBLOCKS	NUMBER		Maximum size of the file (in Oracle blocks)
INCREMENT_BY	NUMBER		Default increment for autoextension (in Oracle blocks)
USER_BYTES	NUMBER		Size of the useful portion of the file (in bytes)
USER_BLOCKS	NUMBER		Size of the useful portion of the file (in Oracle blocks)
SHARED	VARCHAR2(12)		Type of tablespace this file belongs to: • SHARED: For shared tablespace • LOCAL_FOR_RIM: Local temporary tablespace for RIM (read-only) instances • LOCAL_FOR_ALL: Local temporary tablespace for all instance types
INST_ID	NUMBER		Instance ID of the instance to which the temp file belongs. This column has a NULL value for temp files that belong to shared tablespaces.

6.51 DBA_TEMP_FREE_SPACE

DBA_TEMP_FREE_SPACE displays temporary space usage information at tablespace level.

Column	Datatype	NULL	Description
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of the tablespace
TABLESPACE_SIZE	NUMBER		Total size of the tablespace, in bytes
ALLOCATED_SPACE	NUMBER		Total allocated space, in bytes, including space that is currently allocated and used and space that is currently allocated and available for reuse
FREE_SPACE	NUMBER		Total free space available, in bytes, including space that is currently allocated and available for reuse and space that is currently unallocated
SHARED	VARCHAR2(12)		Type of tablespace this file belongs to: • SHARED: For shared tablespace • LOCAL_FOR_RIM: Local temporary tablespace for RIM (read-only) instances • LOCAL_FOR_ALL: Local temporary tablespace for all instance types
INST_ID	NUMBER		Instance ID of the instance to which the tempfile belongs

6.52 DBA_THRESHOLDS

DBA_THRESHOLDS describes all thresholds.

Column	Datatype	NULL	Description
METRICS_NAME	VARCHAR2(64)		Metrics name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
WARNING_OPERATOR	VARCHAR2(12)		Relational operator for warning thresholds: • GT • EQ • LT • LE • GE • CONTAINS • NE • DO NOT CHECK • DO_NOT_CHECK
WARNING_VALUE	VARCHAR2(256)		Warning threshold value
CRITICAL_OPERATOR	VARCHAR2(12)		Relational operator for critical thresholds: • GT • EQ • LT • LE • GE • CONTAINS • NE • DO NOT CHECK • DO_NOT_CHECK
CRITICAL_VALUE	VARCHAR2(256)		Critical threshold value
OBSERVATION_PERIOD	NUMBER		Observation period length (in minutes)
CONSECUTIVE_OCCURRENCES	NUMBER		Number of occurrences before an alert is issued
INSTANCE_NAME	VARCHAR2(16)		Instance name; NULL for database-wide alerts
OBJECT_TYPE	VARCHAR2(64)		Object type: • SYSTEM • SERVICE • EVENT_CLASS • TABLESPACE • FILE
OBJECT_NAME	VARCHAR2(513)		Name of the object for which the threshold is set
STATUS	VARCHAR2(7)		Indicates whether the threshold is applicable on a valid object (VALID) or not (INVALID) Thresholds for non-tablespace metrics can only be set in ROOT and apply to a CDB as a whole. Any pre-existing non-tablespace thresholds that may exist in a PDB have a status of INVALID in the DBA_THRESHOLDS view. You can remove these threshold settings using the DBMS_SERVER_ALERT.SET_THRESHOLD API. See *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SERVER_ALERT.SET_THRESHOLD API.

6.53 DBA_TRANSFORMATIONS

DBA_TRANSFORMATIONS displays information about all transformations in the database. These transformations can be specified with Advanced Queuing operations such as enqueue,

Copyright © 2002, 2026, Oracle and/or its affiliates.

dequeue, and subscribe to automatically integrate transformations in AQ messaging. Its columns are the same as those in ALL_TRANSFORMATIONS.

See Also

"ALL_TRANSFORMATIONS"

6.54 DBA_TRIGGER_COLS

DBA_TRIGGER_COLS describes the use of columns in all triggers in the database. Its columns are the same as those in ALL_TRIGGER_COLS.

See Also

"ALL_TRIGGER_COLS"

6.55 DBA_TRIGGER_ORDERING

DBA_TRIGGER_ORDERING describes all triggers in the database that have FOLLOWS or PRECEDES ordering. Its columns are the same as those in ALL_TRIGGER_ORDERING.

See Also

"ALL_TRIGGER_ORDERING"

6.56 DBA_TRIGGERS

DBA_TRIGGERS describes all triggers in the database. Its columns are the same as those in ALL_TRIGGERS.

See Also

"ALL_TRIGGERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.57 DBA_TRIGGERS_AE

DBA_TRIGGERS_AE describes all triggers (across all editions) in the database. Its columns are the same as those in ALL_TRIGGERS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c.

> **See Also**
>
> "ALL_TRIGGERS_AE"

6.58 DBA_TS_QUOTAS

DBA_TS_QUOTAS describes tablespace quotas for all users.

Related View

USER_TS_QUOTAS describes tablespace quotas for the current user. This view does not display the USERNAME column.

Column	Datatype	NULL	Description
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Tablespace name
USERNAME	VARCHAR2(128)	NOT NULL	User with resource rights on the tablespace
BYTES	NUMBER		Number of bytes charged to the user
MAX_BYTES	NUMBER		User's quota in bytes, or -1 if no limit
BLOCKS	NUMBER		Number of Oracle blocks charged to the user
MAX_BLOCKS	NUMBER		User's quota in Oracle blocks, or -1 if no limit
DROPPED	VARCHAR2(3)		Whether the tablespace has been dropped

> **See Also**
>
> "USER_TS_QUOTAS"

6.59 DBA_TSDP_IMPORT_ERRORS

DBA_TSDP_IMPORT_ERRORS shows information about the errors encountered during import of the Transparent Sensitive Data Protection discovery result.

This error information corresponds to the last import of the discovery result done using the DBMS_TSDP_MANAGE.IMPORT_DISCOVERY_RESULT API.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ERROR_CODE	NUMBER	NOT NULL	The ORA error code of the error encountered
SCHEMA_NAME	VARCHAR2(128)		The schema corresponding to the error
TABLE_NAME	VARCHAR2(128)		The table corresponding to the error
COLUMN_NAME	VARCHAR2(128)		The column corresponding to the error
SENSITIVE_TYPE	VARCHAR2(128)		The sensitive type corresponding to the error

See Also

- *Oracle Database Security Guide* for more information about using Transparent Sensitive Data Protection
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_TSDP_MANAGE.IMPORT_DISCOVERY_RESULT procedure

6.60 DBA_TSDP_POLICY_CONDITION

DBA_TSDP_POLICY_CONDITION describes the Transparent Sensitive Data Protection policy and condition mapping. It also lists the property-value pairs for the condition.

Column	Datatype	NULL	Description
POLICY_NAME	VARCHAR2(128)		The name of the Transparent Sensitive Data Protection policy
SUB_POLICY	NUMBER	NOT NULL	The sub policy of the Transparent Sensitive Data Protection policy
PROPERTY	VARCHAR2(11)		The condition property. Possible values: • DATATYPE • LENGTH • SCHEMA_NAME • TABLE_NAME
VALUE	VARCHAR2(128)		The value of the condition property

See Also

Oracle Database Security Guide for more information about using Transparent Sensitive Data Protection

6.61 DBA_TSDP_POLICY_FEATURE

DBA_TSDP_POLICY_FEATURE shows the Transparent Sensitive Data Protection policy security feature mapping for all the TSDP policies in the database.

At this time, only Oracle Data Redaction is supported.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
POLICY_NAME	VARCHAR2(128)		The name of the Transparent Sensitive Data Protection policy
SECURITY_FEATURE	VARCHAR2(12)		The Oracle security feature with which the Transparent Sensitive Data Protection policy is associated

See Also

Oracle Database Security Guide for more information about using Transparent Sensitive Data Protection

6.62 DBA_TSDP_POLICY_PARAMETER

DBA_TSDP_POLICY_PARAMETER shows the parameter-value pairs for the condition of the Transparent Sensitive Data Protection policy.

Column	Datatype	NULL	Description
POLICY_NAME	VARCHAR2(128)		The name of the Transparent Sensitive Data Protection policy
SUB_POLICY	NUMBER	NOT NULL	The sub policy of the Transparent Sensitive Data Protection policy
PARAMETER	VARCHAR2(128)		The parameter for the Transparent Sensitive Data Protection sub policy
VALUE	VARCHAR2(4000)		The value of the parameter
DEFAULT_OPTION	VARCHAR2(5)		Indicates whether this is the default option for the policy: • TRUE: This is the default option for the policy • FALSE: This is not the default option for the policy

See Also

Oracle Database Security Guide for more information about using Transparent Sensitive Data Protection

6.63 DBA_TSDP_POLICY_PROTECTION

DBA_TSDP_POLICY_PROTECTION shows the list of columns that have been protected through Transparent Sensitive Data Protection.

Column	Datatype	NULL	Description
SCHEMA_NAME	VARCHAR2(128)	NOT NULL	The schema containing the sensitive data
TABLE_NAME	VARCHAR2(128)	NOT NULL	The table containing the sensitive column
COLUMN_NAME	VARCHAR2(128)	NOT NULL	The name of the sensitive column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TSDP_POLICY	VARCHAR2(128)		The TSDP policy name based on which the column protection was enabled
SECURITY_FEATURE	VARCHAR2(12)		The security feature enabled on the sensitive column
SECURITY_FEATURE_POLICY	VARCHAR2(128)	NOT NULL	Name of the underlying Oracle security feature policy
SUBPOLICY#	NUMBER	NOT NULL	The subpolicy of the Transparent Sensitive Data Protection policy based on which protection has been enabled

See Also

Oracle Database Security Guide for more information about using Transparent Sensitive Data Protection

6.64 DBA_TSDP_POLICY_TYPE

DBA_TSDP_POLICY_TYPE shows the Transparent Sensitive Data Protection policy to sensitive column type mapping.

Column	Datatype	NULL	Description
POLICY_NAME	VARCHAR2(128)		The Transparent Sensitive Data Protection policy name
SENSITIVE_TYPE	VARCHAR2(128)		The sensitive column type name

See Also

Oracle Database Security Guide for more information about using Transparent Sensitive Data Protection

6.65 DBA_TSM_DESTINATION

DBA_TSM_DESTINATION lists transparent session migration (TSM) destination session statistics.

Column	Datatype	NULL	Description
SOURCE_DATABASE_NAME	VARCHAR2(4000)		Database name of source session
DESTINATION_DATABASE_NAME	VARCHAR2(4000)		Database name of destination session
DESTINATION_INSTANCE_NAME	VARCHAR2(4000)		Instance name of destination session
DESTINATION_INSTANCE_ID	VARCHAR2(4000)		Instance ID of destination session
DESTINATION_INST_START_TIME	TIMESTAMP(6) WITH TIME ZONE		Instance start time of destination session
SEQUENCE#	NUMBER		Migration sequence number

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DESTINATION_SID	NUMBER		Session ID of destination session
DESTINATION_SERIAL#	NUMBER		Session serial number of destination session
DESTINATION_START_TIME	TIMESTAMP(6) WITH TIME ZONE		Start time for migration on destination session
DESTINATION_END_TIME	TIMESTAMP(6) WITH TIME ZONE		End time for migration on destination session
DESTINATION_USER_NAME	VARCHAR2(128)	NOT NULL	User associated with the destination session
DESTINATION_SCHEMA_NAME	VARCHAR2(128)	NOT NULL	Schema associated with the destination session
DESTINATION_STATE	VARCHAR2(24)		Migration state of destination session

6.66 DBA_TSM_SOURCE

DBA_TSM_SOURCE lists transparent session migration (TSM) source session statistics.

Column	Datatype	NULL	Description
SOURCE_DATABASE_NAME	VARCHAR2(4000)		Database name of source session
SOURCE_INSTANCE_NAME	VARCHAR2(4000)		Instance name of source session
SOURCE_INSTANCE_ID	VARCHAR2(4000)		Instance ID of source session
SOURCE_INSTANCE_START_TIME	TIMESTAMP(6) WITH TIME ZONE		Instance start time of source session
SEQUENCE#	NUMBER		Migration sequence number
SOURCE_SID	NUMBER		Session ID of source session
SOURCE_SERIAL#	NUMBER		Source serial number of source session
SOURCE_STATE	VARCHAR2(24)		Migration state of source session
CONNECT_STRING	VARCHAR2(4000)		Connect string specified for migration
SOURCE_START_TIME	TIMESTAMP(6) WITH TIME ZONE		Start time for migration on source session
COST	NUMBER		Estimate of migration cost
FAILURE_REASON	VARCHAR2(34)		Reason for migration failure, if any
SOURCE_END_TIME	TIMESTAMP(6) WITH TIME ZONE		End time for migration on source session
ROUNDTRIPS	NUMBER		Number of client/server round trips during migration
SOURCE_USER_NAME	VARCHAR2(128)	NOT NULL	User associated with the source session
SOURCE_SCHEMA_NAME	VARCHAR2(128)	NOT NULL	Schema associated with the source session
DESTINATION_DATABASE_NAME	VARCHAR2(4000)		Database name of the destination session

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.67 DBA_TSTZ_TAB_COLS

DBA_TSTZ_TAB_COLS displays information about the columns of all tables in the database, which have columns defined on TIMESTAMP WITH TIME ZONE data types or object types containing attributes of TIMESTAMP WITH TIME ZONE data types.

Its columns (except for COLUMN_NAME, NESTED, VIRTUAL_COLUMN, SCALAR_COLUMN, and UNUSED_COLUMN) are the same as those in ALL_TSTZ_TAB_COLS.

See Also

"ALL_TSTZ_TAB_COLS"

6.68 DBA_TSTZ_TABLES

DBA_TSTZ_TABLES displays information about all tables in the database, which have columns defined on TIMESTAMP WITH TIME ZONE data types or object types containing attributes of TIMESTAMP WITH TIME ZONE data types.

Its columns are the same as those in ALL_TSTZ_TABLES.

See Also

"ALL_TSTZ_TABLES"

6.69 DBA_TUNE_MVIEW

DBA_TUNE_MVIEW displays the result of executing the DBMS_ADVISOR.TUNE_MVIEW procedure.

Related View

USER_TUNE_MVIEW displays the result of executing the DBMS_ADVISOR.TUNE_MVIEW procedure. This view does not display the OWNER column.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)		Owner of the task
TASK_NAME	VARCHAR2(128)		Name of the task
ACTION_ID	NUMBER	NOT NULL	Identifier of the action
SCRIPT_TYPE	VARCHAR2(14)		Type of the script: • IMPLEMENTATION • UNDO
STATEMENT	CLOB		Action statement

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "USER_TUNE_MVIEW"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_ADVISOR.TUNE_MVIEW procedure

6.70 DBA_TYPE_ATTRS

DBA_TYPE_ATTRS describes the attributes of all object types in the database. Its columns (except for CHAR_USED) are the same as those in ALL_TYPE_ATTRS.

See Also

"ALL_TYPE_ATTRS"

6.71 DBA_TYPE_METHODS

DBA_TYPE_METHODS describes the methods of all object types in the database. Its columns are the same as those in ALL_TYPE_METHODS.

See Also

"ALL_TYPE_METHODS"

6.72 DBA_TYPE_VERSIONS

DBA_TYPE_VERSIONS describes the versions of all object types in the database. Its columns are the same as those in ALL_TYPE_VERSIONS.

See Also

"ALL_TYPE_VERSIONS"

6.73 DBA_TYPES

DBA_TYPES describes all object types in the database. Its columns are the same as those in ALL_TYPES.

See Also

"ALL_TYPES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.74 DBA_UMF_LINK

DBA_UMF_LINK displays information about the registered database links in the Remote Management Framework (RMF).

This view returns no rows if you are querying on an RMF source node. It returns all the registered database links in the topology if you are querying on a target node.

Column	Datatype	NULL	Description
TOPOLOGY_NAME	VARCHAR2(128)	NOT NULL	Topology name for the link
FROM_NODE_ID	NUMBER	NOT NULL	Node ID of the local node
TO_NODE_ID	NUMBER	NOT NULL	Node ID of the remote node
LINK_NAME	VARCHAR2(128)	NOT NULL	Fully qualified database link name

See Also

Oracle Database Performance Tuning Guide for information about configuring the Remote Management Framework (RMF) architecture

6.75 DBA_UMF_REGISTRATION

DBA_UMF_REGISTRATION displays information about the registered nodes in the Remote Management Framework (RMF).

This view returns no rows if you are querying on an RMF source node. It returns all the registered nodes in the topology if you are querying on a target node.

Column	Datatype	NULL	Description
TOPOLOGY_NAME	VARCHAR2(128)	NOT NULL	Topology name for the node
NODE_NAME	VARCHAR2(128)	NOT NULL	Unique node name in the topology
NODE_ID	NUMBER	NOT NULL	Unique node ID in the topology
NODE_TYPE	NUMBER	NOT NULL	Node type. Possible value: • 0: RDBMS node
AS_SOURCE	VARCHAR2(5)		Indicates whether the node is a source node. Possible values: • TRUE: The node is a source node, and it can provide remote services • FALSE: The node is not a source node, and it cannot provide remote services
AS_CANDIDATE_TARGET	VARCHAR2(5)		Node is a candidate target. Possible values: • TRUE: Node can be promoted to target role • FALSE: Node cannot be promoted to target role

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STATE	VARCHAR2(20)		Current state of the node. Possible values: • OK: Node is registered • REGISTRATION_PENDING: Node registration has started, but has not been completed • SYNC_FAILED: Unable to synchronize the topology with the node

See Also

Oracle Database Performance Tuning Guide for information about configuring the Remote Management Framework (RMF) architecture

6.76 DBA_UMF_SERVICE

DBA_UMF_SERVICE displays information about the registered services in the Remote Management Framework (RMF).

This view returns no rows if you are querying on an RMF source node. It returns all the registered services in the topology if you are querying on a target node

Column	Datatype	NULL	Description
TOPOLOGY_NAME	VARCHAR2(128)	NOT NULL	Topology name for the service
NODE_ID	NUMBER	NOT NULL	Node ID of the node providing the service
SERVICE_ID	VARCHAR2(7)		Service Identifier. Possible values: • 1: Automatic Workload Repository • 2: SQL Tuning

See Also

Oracle Database Performance Tuning Guide for information about configuring the Remote Management Framework (RMF) architecture

6.77 DBA_UMF_TOPOLOGY

DBA_UMF_TOPOLOGY displays information about the registered topologies in the Remote Management Framework (RMF).

This view returns no rows if you are querying on an RMF source node. It returns one row per registered topology if you are querying on a target node.

Column	Datatype	NULL	Description
TOPOLOGY_NAME	VARCHAR2(128)	NOT NULL	Unique topology name
TARGET_ID	NUMBER		Node ID of the target node

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
TOPOLOGY_VERSION	NUMBER	NOT NULL	Topology version number
TOPOLOGY_STATE	VARCHAR2(8)		Possible values: • ACTIVE: Topology can be used for RMF operations • INACTIVE: Topology cannot be used for RMF operations

See Also

Oracle Database Performance Tuning Guide for information about configuring the Remote Management Framework (RMF) architecture

6.78 DBA_UNDO_EXTENTS

DBA_UNDO_EXTENTS describes the extents comprising the segments in all undo tablespaces in the database.

Note

The status of the undo space distribution reported by DBA_UNDO_EXTENTS is correct for the undo tablespace that is active on the instance on which DBA_UNDO_EXTENTS is queried. However, due to the use of in-memory information that is different on each instance, there can be a discrepancy in the status of the undo space distribution of undo tablespaces active on other instances when queried from one instance. This does not affect undo functionality and is only a reporting discrepancy for other instances' undo tablespace space distribution status. As a best practice, query the space distribution for an undo tablespace from the instance on which it is active.

Column	Datatype	NULL	Description
OWNER	CHAR(3)		Owner of the undo tablespace
SEGMENT_NAME	VARCHAR2(128)	NOT NULL	Name of the undo segment
TABLESPACE_NAME	VARCHAR2(128)	NOT NULL	Name of the undo tablespace
EXTENT_ID	NUMBER		ID of the extent
FILE_ID	NUMBER	NOT NULL	Absolute file number of the data file containing the extent
BLOCK_ID	NUMBER		Start block number of the extent
BYTES	NUMBER		Size of the extent (in bytes)
BLOCKS	NUMBER		Size of the extent (in blocks)
RELATIVE_FNO	NUMBER		Relative number of the file containing the segment header
COMMIT_JTIME	NUMBER		Commit time of the undo in the extent expressed as Julian time. This column is deprecated, but retained for backward compatibility reasons.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
COMMIT_WTIME	VARCHAR2(20)		Commit time of the undo in the extent expressed as Wallclock time.This column is deprecated, but retained for backward compatibility reasons.
STATUS	VARCHAR2(9)		Transaction Status of the undo in the extent: • ACTIVE • EXPIRED • UNEXPIRED

6.79 DBA_UNUSED_COL_TABS

DBA_UNUSED_COL_TABS describes all tables in the database containing unused columns. Its columns are the same as those in ALL_UNUSED_COL_TABS.

See Also

"ALL_UNUSED_COL_TABS"

6.80 DBA_UNUSED_GRANTS

DBA_UNUSED_GRANTS shows all the grants that are not used during the privilege capture.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)		Name of the privilege analysis policy
RUN_NAME	VARCHAR2(128)		Name of the run of the privilege analysis policy
GRANTEE	VARCHAR2(128)		Name of the user who is granted with the privilege or role
ROLENAME	VARCHAR2(128)		Name of the role that is granted to the grantee
SYS_PRIV	VARCHAR2(40)		Name of the system privilege that is granted to the grantee
OBJ_PRIV	VARCHAR2(40)		Name of the object privilege that is granted to the grantee
USER_PRIV	VARCHAR2(25)		Name of the user privilege that is granted to the grantee
OBJECT_OWNER	VARCHAR2(128)		Name of the owner of the object for which the object privilege is granted
OBJECT_NAME	VARCHAR2(128)		Name of the object for which the object privilege is granted
OBJECT_TYPE	VARCHAR2(23)		Type of the object for which the object privilege is granted
COLUMN_NAME	VARCHAR2(128)		Name of the column in the table for which the object privilege is granted
OPTION$	NUMBER		Whether the grant option of the privilege is granted

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Security Guide* for more information about privilege analysis

6.81 DBA_UNUSED_OBJPRIVS

`DBA_UNUSED_OBJPRIVS` lists the object privileges (without privilege grant paths) that are not used for the privilege analysis policies reported by the `DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT` procedure.

This view provides access to analyzed privilege records in `SYS` tables.

You must have the `CAPTURE_ADMIN` role to access this view.

Column	Datatype	NULL	Description
`CAPTURE`	`VARCHAR2(128)`	`NOT NULL`	Name of the privilege analysis policy
`USERNAME`	`VARCHAR2(128)`		Name of the user whose privileges are reported
`ROLENAME`	`VARCHAR2(128)`		Name of the role whose unused privileges are reported (for `ROLE` type privilege analysis or `ROLE AND CONTEXT` privilege analysis)
`OBJ_PRIV`	`VARCHAR2(40)`		Unused object privilege
`OBJECT_OWNER`	`VARCHAR2(128)`		Object owner
`OBJECT_NAME`	`VARCHAR2(128)`		Name of the object that `USERNAME` has `OBJ_PRIV` on
`OBJECT_TYPE`	`VARCHAR2(23)`		Type of the object `USERNAME` has `OBJ_PRIV` on
`COLUMN_NAME`	`VARCHAR2(128)`		Name of the column that `USERNAME` has `OBJ_PRIV` on
`GRANT_OPTION`	`NUMBER`		Indicates whether the privilege is granted with the `GRANT` option: • `0` - Indicates that the privilege is granted without the `GRANT` option • `1` - Indicates that the privilege is granted with the `GRANT` option
`RUN_NAME`	`VARCHAR2(128)`		The name of the run during which the privilege was reported

> **See Also**
>
> - "DBA_UNUSED_OBJPRIVS_PATH" for privilege grant path information for unused object privileges
> - *Oracle Database Security Guide* for more information about privilege analysis
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT` procedure

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.82 DBA_UNUSED_OBJPRIVS_PATH

DBA_UNUSED_OBJPRIVS_PATH lists the object privileges that are not used for the privilege analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed privilege records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of the privilege analysis policy
USERNAME	VARCHAR2(128)		Name of the user whose privileges are reported
ROLENAME	VARCHAR2(128)		Name of the role whose unused privileges are reported (for ROLE type privilege analysis or ROLE AND CONTEXT privilege analysis)
OBJ_PRIV	VARCHAR2(40)		Unused object privilege
OBJECT_OWNER	VARCHAR2(128)		Object owner
OBJECT_NAME	VARCHAR2(128)		Name of the object that USERNAME has OBJ_PRIV on
OBJECT_TYPE	VARCHAR2(23)		Type of the object that USERNAME has OBJ_PRIV on
COLUMN_NAME	VARCHAR2(128)		Name of the column that USERNAME has OBJ_PRIV on
GRANT_OPTION	NUMBER		Indicates whether the privilege is granted with the GRANT option: • 0 - Indicates that the privilege is granted without the GRANT option • 1 - Indicates that the privilege is granted with the GRANT option
PATH	GRANT_PATH		Object privilege grant paths
RUN_NAME	VARCHAR2(128)		The name of the run during which the privilege was reported

See Also

- "DBA_USED_OBJPRIVS_PATH"
- "DBA_UNUSED_OBJPRIVS"
- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure

6.83 DBA_UNUSED_PRIVS

DBA_UNUSED_PRIVS lists the privileges that are not used for the privilege analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed privilege records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of the privilege analysis policy
USERNAME	VARCHAR2(128)		Name of the user whose unused privileges are reported
ROLENAME	VARCHAR2(128)		Name of the role whose unused privileges are reported (for ROLE type privilege analysis or ROLE AND CONTEXT privilege analysis)
SYS_PRIV	VARCHAR2(40)		Unused system privilege
OBJ_PRIV	VARCHAR2(40)		Unused object privilege
USER_PRIV	VARCHAR2(25)		Unused user privilege
OBJECT_OWNER	VARCHAR2(128)		Object owner
OBJECT_NAME	VARCHAR2(128)		Name of the object that USERNAME has OBJ_PRIV or USER_PRIV on
OBJECT_TYPE	VARCHAR2(23)		Type of the object that OBJ_PRIV has accessed or USER if USER_PRIV was used
COLUMN_NAME	VARCHAR2(128)		Name of the column that OBJ_PRIV has access on
OPTION$	NUMBER		Indicates whether the privilege is granted with the GRANT option or the ADMIN option: • 0 - Indicates that the privilege is granted without the GRANT option or ADMIN option • 1 - Indicates that the privilege is granted with the GRANT option or ADMIN option
PATH	GRANT_PATH		Privilege grant paths
RUN_NAME	VARCHAR2(128)		The name of the run during which the privilege was reported

See Also

- "DBA_USED_PRIVS"
- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure

6.84 DBA_UNUSED_SYSPRIVS

DBA_UNUSED_SYSPRIVS lists the system privileges (without privilege grant paths) that are not used for the privilege analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed privilege records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of a privilege analysis policy
USERNAME	VARCHAR2(128)		Name of the user whose privileges are reported

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ROLENAME	VARCHAR2(128)		Name of the role whose unused privileges are reported (for ROLE type privilege analysis or ROLE AND CONTEXT privilege analysis)
SYS_PRIV	VARCHAR2(40)		Unused system privilege
ADMIN_OPTION	NUMBER		Indicates whether the privilege is granted with the ADMIN option: • 0 - Indicates that the privilege is granted without the ADMIN option • 1 - Indicates that the privilege is granted with the ADMIN option
RUN_NAME	VARCHAR2(128)		The name of the run during which the privilege was reported

See Also

- "DBA_UNUSED_SYSPRIVS_PATH" for privilege grant path information for unused system privileges
- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure

6.85 DBA_UNUSED_SYSPRIVS_PATH

DBA_UNUSED_SYSPRIVS_PATH lists the system privileges that are not used for the privilege analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed privilege records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of a privilege analysis policy
USERNAME	VARCHAR2(128)		Name of the user whose privileges are reported
ROLENAME	VARCHAR2(128)		Name of the role whose unused privileges are reported (for ROLE type privilege analysis or ROLE AND CONTEXT privilege analysis)
SYS_PRIV	VARCHAR2(40)		Unused system privilege
ADMIN_OPTION	NUMBER		Indicates whether the privilege is granted with the ADMIN option: • 0 - Indicates that the privilege is granted without the ADMIN option • 1 - Indicates that the privilege is granted with the ADMIN option
PATH	GRANT_PATH		System privilege grant paths
RUN_NAME	VARCHAR2(128)		The name of the run during which the privilege was reported

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_USED_SYSPRIVS_PATH"
- "DBA_UNUSED_SYSPRIVS"
- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT` procedure

6.86 DBA_UNUSED_USERPRIVS

`DBA_UNUSED_USERPRIVS` lists the user privileges (without privilege grant paths) that are not used for the privilege analysis policies reported by the `DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT` procedure.

This view provides access to analyzed privilege records in `SYS` tables.

You must have the `CAPTURE_ADMIN` role to access this view.

Column	Datatype	NULL	Description
`CAPTURE`	`VARCHAR2(128)`	`NOT NULL`	Name of a privilege analysis policy
`USERNAME`	`VARCHAR2(128)`		Name of the user whose privileges are reported
`ROLENAME`	`VARCHAR2(128)`		Name of the role whose unused privileges are reported (for `ROLE` type privilege analysis or `ROLE AND CONTEXT` privilege analysis)
`USER_PRIV`	`VARCHAR2(25)`		Unused user privilege
`ONUSER`	`VARCHAR2(128)`		The user whose user privileges the grantee can exercise
`GRANT_OPTION`	`NUMBER`		Indicates whether the privilege is granted with the `GRANT` option: • `0` - Indicates that the privilege is granted without the `GRANT` option • `1` - Indicates that the privilege is granted with the `GRANT` option
`RUN_NAME`	`VARCHAR2(128)`		The name of the run during which the privilege was reported

See Also

- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT` procedure

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.87 DBA_UNUSED_USERPRIVS_PATH

DBA_UNUSED_USERPRIVS_PATH lists the user privileges that are not used for the privilege analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed privilege records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of a privilege analysis policy
USERNAME	VARCHAR2(128)		Name of the user whose privileges are reported
ROLENAME	VARCHAR2(128)		Name of the role whose unused privileges are reported (for ROLE type privilege analysis or ROLE AND CONTEXT privilege analysis)
USER_PRIV	VARCHAR2(25)		Unused user privilege
ONUSER	VARCHAR2(128)		The user whose user privileges the grantee can exercise
GRANT_OPTION	NUMBER		Indicates whether the privilege is granted with the GRANT option: • 0 - Indicates that the privilege is granted without the GRANT option • 1 - Indicates that the privilege is granted with the GRANT option
PATH	GRANT_PATH		User privilege grant paths
RUN_NAME	VARCHAR2(128)		The name of the run during which the privilege was reported

See Also

- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure

6.88 DBA_UPDATABLE_COLUMNS

DBA_UPDATABLE_COLUMNS describes all columns in a join view that can be updated by the database administrator, subject to appropriate privileges. Its columns are the same as those in ALL_UPDATABLE_COLUMNS.

See Also

- "ALL_UPDATABLE_COLUMNS"
- *Oracle Database Concepts* for information on updatable join views

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.89 DBA_USED_OBJPRIVS

DBA_USED_OBJPRIVS lists the object privileges (without privilege grant paths) that are used for the privilege analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed privilege records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of the privilege analysis policy
SEQUENCE	NUMBER	NOT NULL	The sequence number of the privilege analysis run during which the privilege was reported
OS_USER	VARCHAR2(128)		Operating system login username
USERHOST	VARCHAR2(128)		Client host machine name
MODULE	VARCHAR2(64)		Module name
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user whose privilege was reported
USED_ROLE	VARCHAR2(128)		Used role
OBJ_PRIV	VARCHAR2(40)		Used object privilege
OBJECT_OWNER	VARCHAR2(128)		Object owner
OBJECT_NAME	VARCHAR2(128)		Name of the object that OBJ_PRIV was used to access
OBJECT_TYPE	VARCHAR2(23)		Type of the object that OBJ_PRIV was used to access
COLUMN_NAME	VARCHAR2(128)		Name of the column that OBJ_PRIV was used to access
GRANT_OPTION	NUMBER		Indicates whether the GRANT option was used: • 0 - Indicates that the GRANT option was not used • 1 - Indicates that the GRANT option was used
RUN_NAME	VARCHAR2(128)		The name of the run during which the privilege was reported

See Also

- "DBA_USED_OBJPRIVS_PATH" for privilege grant path information for used object privileges
- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure

6.90 DBA_USED_OBJPRIVS_PATH

DBA_USED_OBJPRIVS_PATH lists the object privileges that are used for the privilege analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed privilege records in SYS tables.

Copyright © 2002, 2026, Oracle and/or its affiliates.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of the privilege analysis policy
SEQUENCE	NUMBER	NOT NULL	The sequence number of the privilege analysis run during which the privilege was reported
OS_USER	VARCHAR2(128)		Operating system login username
USERHOST	VARCHAR2(128)		Client host machine name
MODULE	VARCHAR2(64)		Module name
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user whose privilege was reported
USED_ROLE	VARCHAR2(128)		Used role
OBJ_PRIV	VARCHAR2(40)		Used object privilege
OBJECT_OWNER	VARCHAR2(128)		Object owner
OBJECT_NAME	VARCHAR2(128)		Name of the object that OBJ_PRIV is used to access
OBJECT_TYPE	VARCHAR2(23)		Type of the object that OBJ_PRIV is used to access
COLUMN_NAME	VARCHAR2(128)		Name of the column that OBJ_PRIV is used to access
GRANT_OPTION	NUMBER		Indicates whether the GRANT option was used: • 0 - Indicates that the GRANT option was not used • 1 - Indicates that the GRANT option was used
PATH	GRANT_PATH		Object privilege grant paths
RUN_NAME	VARCHAR2(128)		The name of the run during which the privilege was reported

See Also

- "DBA_UNUSED_OBJPRIVS_PATH"
- "DBA_USED_OBJPRIVS"
- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure

6.91 DBA_USED_PRIVS

DBA_USED_PRIVS lists the privileges that are used for the privilege analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed privilege records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of a privilege analysis policy
SEQUENCE	NUMBER	NOT NULL	The sequence number of the privilege analysis run during which the privilege was reported
OS_USER	VARCHAR2(128)		Operating system login username

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
USERHOST	VARCHAR2(128)		Client host machine name
MODULE	VARCHAR2(64)		Module name
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user whose privilege was reported
USED_ROLE	VARCHAR2(128)		Used role
SYS_PRIV	VARCHAR2(40)		Used system privilege
OBJ_PRIV	VARCHAR2(40)		Used object privilege
USER_PRIV	VARCHAR2(25)		Used user privilege
OBJECT_OWNER	VARCHAR2(128)		Object owner
OBJECT_NAME	VARCHAR2(128)		Name of the object or user that OBJ_PRIV or USER_PRIV is used to access
OBJECT_TYPE	VARCHAR2(23)		Type of the object or user that OBJ_PRIV or USER_PRIV is used to access
COLUMN_NAME	VARCHAR2(128)		Name of the column that OBJ_PRIV is used to access
OPTION$	NUMBER		Indicates whether the GRANT option or the ADMIN option was used: • 0 - Indicates that the GRANT option or ADMIN option was not used • 1 - Indicates that the GRANT option or ADMIN option was used
PATH	GRANT_PATH		Used privilege grant paths
RUN_NAME	VARCHAR2(128)		The name of the run during which the privilege was reported

See Also

- "DBA_UNUSED_PRIVS"
- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure

6.92 DBA_USED_PUBPRIVS

DBA_USED_PUBPRIVS lists the privileges that are used from the PUBLIC role for the privilege analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed privilege records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of a privilege analysis policy
SEQUENCE	NUMBER	NOT NULL	The sequence number of the privilege analysis run during which the privilege was reported
OS_USER	VARCHAR2(128)		Operating system login username

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
USERHOST	VARCHAR2(128)		Client host machine name
MODULE	VARCHAR2(64)		Module name
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user who used the privilege from the PUBLIC role
SYS_PRIV	VARCHAR2(40)		Used system privilege
OBJ_PRIV	VARCHAR2(40)		Used object privilege
OBJECT_OWNER	VARCHAR2(128)		Object owner
OBJECT_NAME	VARCHAR2(128)		Name of the object that OBJ_PRIV is used to access
OBJECT_TYPE	VARCHAR2(23)		Type of the object that OBJ_PRIV is used to access
OPTION$	NUMBER		Indicates whether the GRANT option or the ADMIN option was used: • 0 - Indicates that the GRANT option or ADMIN option was not used • 1 - Indicates that the GRANT option or ADMIN option was used
RUN_NAME	VARCHAR2(128)		The name of the run during which the privilege was reported

See Also

- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure

6.93 DBA_USED_SYSPRIVS

DBA_USED_SYSPRIVS lists the system privileges (without privilege grant paths) that are used for the privilege analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed privilege records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of a privilege analysis policy
SEQUENCE	NUMBER	NOT NULL	The sequence number of the privilege analysis run during which the privilege was used
OS_USER	VARCHAR2(128)		Operating system login username
USERHOST	VARCHAR2(128)		Client host machine name
MODULE	VARCHAR2(64)		Module name
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user whose privilege was reported
USED_ROLE	VARCHAR2(128)		Used role
SYS_PRIV	VARCHAR2(40)		Used system privilege

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ADMIN_OPTION	NUMBER		Indicates whether the ADMIN option was used: • 0 - Indicates that the ADMIN option was not used • 1 - Indicates that the ADMIN option was used
RUN_NAME	VARCHAR2(128)		The name of the run during which the privilege was reported

See Also

- "DBA_USED_SYSPRIVS_PATH" for privilege grant path information for used system privileges
- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure

6.94 DBA_USED_SYSPRIVS_PATH

DBA_USED_SYSPRIVS_PATH lists the system privileges that are used for the privilege analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed privilege records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of a privilege analysis policy
SEQUENCE	NUMBER	NOT NULL	The sequence number of the privilege analysis run during which the privilege was reported
OS_USER	VARCHAR2(128)		Operating system login username
USERHOST	VARCHAR2(128)		Client host machine name
MODULE	VARCHAR2(64)		Module name
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user whose privilege was reported
USED_ROLE	VARCHAR2(128)		Used role
SYS_PRIV	VARCHAR2(40)		Used system privilege
ADMIN_OPTION	NUMBER		Indicates whether the ADMIN option was used: • 0 - Indicates that the ADMIN option was not used • 1 - Indicates that the ADMIN option was used
PATH	GRANT_PATH		System privilege grant paths
RUN_NAME	VARCHAR2(128)		The name of the run during which the privilege was reported

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_UNUSED_SYSPRIVS_PATH"
- "DBA_USED_SYSPRIVS"
- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT` procedure

6.95 DBA_USED_USERPRIVS

`DBA_USED_USERPRIVS` lists the user privileges (without privilege grant paths) that are used for the privilege analysis policies reported by the `DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT` procedure.

This view provides access to analyzed privilege records in `SYS` tables.

You must have the `CAPTURE_ADMIN` role to access this view.

Column	Datatype	NULL	Description
`CAPTURE`	`VARCHAR2(128)`	`NOT NULL`	Name of a privilege analysis policy
`SEQUENCE`	`NUMBER`	`NOT NULL`	The sequence number of the privilege analysis run during which the privilege was reported
`OS_USER`	`VARCHAR2(128)`		Operating system login username
`USERHOST`	`VARCHAR2(128)`		Client host machine name
`MODULE`	`VARCHAR2(64)`		Module name
`USERNAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the user whose privilege was reported
`USED_ROLE`	`VARCHAR2(128)`		Used role
`USER_PRIV`	`VARCHAR2(25)`		Used user privilege
`ONUSER`	`VARCHAR2(128)`		The user whose user privileges the grantee can exercise
`GRANT_OPTION`	`NUMBER`		Indicates whether the privilege is granted with the `GRANT` option: • `0` - Indicates that the privilege is granted without the `GRANT` option • `1` - Indicates that the privilege is granted with the `GRANT` option
`RUN_NAME`	`VARCHAR2(128)`		The name of the run during which the privilege was reported

See Also

- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the `DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT` procedure

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.96 DBA_USED_USERPRIVS_PATH

DBA_USED_USERPRIVS_PATH lists the user privileges that are used for the privilege analysis policies reported by the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure.

This view provides access to analyzed privilege records in SYS tables.

You must have the CAPTURE_ADMIN role to access this view.

Column	Datatype	NULL	Description
CAPTURE	VARCHAR2(128)	NOT NULL	Name of a privilege analysis policy
SEQUENCE	NUMBER	NOT NULL	The sequence number of the privilege analysis run during which the privilege was reported
OS_USER	VARCHAR2(128)		Operating system login username
USERHOST	VARCHAR2(128)		Client host machine name
MODULE	VARCHAR2(64)		Module name
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user whose privilege was reported
USED_ROLE	VARCHAR2(128)		Used role
USER_PRIV	VARCHAR2(25)		Used user privilege
ONUSER	VARCHAR2(128)		The user whose user privileges the grantee can exercise
GRANT_OPTION	NUMBER		Indicates whether the privilege is granted with the GRANT option: • 0 - Indicates that the privilege is granted without the GRANT option • 1 - Indicates that the privilege is granted with the GRANT option
PATH	GRANT_PATH		User privilege grant paths
RUN_NAME	VARCHAR2(128)		The name of the run during which the privilege was reported

See Also

- *Oracle Database Security Guide* for more information about privilege analysis
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_PRIVILEGE_CAPTURE.GENERATE_RESULT procedure

6.97 DBA_USERS

DBA_USERS describes all users of the database.

Related View

USER_USERS describes the current user. This view does not display the PASSWORD, PROFILE, PASSWORD_VERSIONS, EDITIONS_ENABLED, AUTHENTICATION_TYPE, and LAST_LOGIN columns.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user
USER_ID	NUMBER	NOT NULL	ID number of the user
PASSWORD	VARCHAR2(4000)		This column is deprecated in favor of the AUTHENTICATION_TYPE column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ACCOUNT_STATUS	VARCHAR2(32)	NOT NULL	Account status: • OPEN The account is open. • EXPIRED The password for the account is expired, either because the PASSWORD_LIFE_TIME limit was reached or because the password was expired by the ALTER USER ... PASSWORD EXPIRE command. The user can log in with the expired password, then change the password. • EXPIRED(GRACE) The password for the account is expired because the PASSWORD_LIFE_TIME limit was reached, but the password change grace period (PASSWORD_GRACE_TIME) has not yet elapsed. The user can log in with the expired password, but will receive an ORA-28002 warning as a reminder that the password must soon be changed. If the PASSWORD_GRACE_TIME elapses, the user can log in with the expired password, then change the password. • LOCKED The account is locked, either by the ALTER USER ... ACCOUNT LOCK command, or because the number of consecutive failed login attempts exceeded the FAILED_LOGIN_ATTEMPTS limit and the value of PASSWORD_LOCK_TIME is UNLIMITED. The account can be unlocked by the ALTER USER ... ACCOUNT UNLOCK command. • LOCKED(TIMED) The account is locked because the number of consecutive failed login attempts exceeded the FAILED_LOGIN_ATTEMPTS limit and the PASSWORD_LOCK_TIME has not yet elapsed. The account can be unlocked either by the ALTER USER ... ACCOUNT UNLOCK command or by waiting until the PASSWORD_LOCK_TIME has elapsed. • EXPIRED & LOCKED The password for the account is expired, as described for the EXPIRED account status, and the account is locked as described for the LOCKED account status. The account can first be unlocked as described for the LOCKED account status, then the password can be changed as described for the EXPIRED account status. • EXPIRED(GRACE) & LOCKED The password for the account is expired, as described for the EXPIRED(GRACE) account status, and the account is locked as described for the LOCKED account status. The account can first be unlocked as described for the LOCKED account status, then the password can be changed as described for the EXPIRED(GRACE) account status. • EXPIRED & LOCKED(TIMED) The password for the account is expired, as described for the EXPIRED account status, and the

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
			account is locked as described for the `LOCKED(TIMED)` account status. The account can first be unlocked as described for the `LOCKED(TIMED)` account status, then the password can be changed as described for the `EXPIRED` account status. • `EXPIRED(GRACE) & LOCKED(TIMED)` The password for the account is expired, as described for the `EXPIRED(GRACE)` account status, and the account is locked as described for the `LOCKED(TIMED)` account status. The account can first be unlocked as described for the `LOCKED(TIMED)` account status, then the password can be changed as described for the `EXPIRED(GRACE)` account status. • `OPEN & IN ROLLOVER` The account is in the password rollover period. The user can log in with either the earlier password or the new password. However, at the time the user logs in, the server recalculates whether the account is still in its password rollover period. If the password rollover period has elapsed, then the login will succeed only if the new password was specified, and the account status will change to `OPEN`. • `EXPIRED & IN ROLLOVER` The account is in the password rollover period and the password is expired as described for the `EXPIRED` account status. The user can log in with either the earlier password or the new password. However, at the time the user logs in, the server recalculates whether the account is still in its password rollover period. If the password rollover period has elapsed, then the login will succeed only if the new password was specified, and the account status will change to `EXPIRED`. After logging in, the user will be prompted to change the password. • `LOCKED & IN ROLLOVER` The account is in the password rollover period and is also locked as described for the `LOCKED` account status. The account can be unlocked as described for the `LOCKED` account status, after which the user can log in as described for the `OPEN & IN ROLLOVER` account status. • `EXPIRED & LOCKED & IN ROLLOVER` The account is in the password rollover period, its password is expired as described for the `EXPIRED` account status, and the account is locked as described for the `LOCKED` account status. The account can be unlocked as described for the `LOCKED` account status, after which the user can log in as described for the `EXPIRED & IN ROLLOVER` account status. • `LOCKED(TIMED) & IN ROLLOVER` The account is in the password rollover period and is also locked as described for the

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
			`LOCKED(TIMED)` account status. The account can be unlocked as described for the `LOCKED(TIMED)` account status, after which the user can log in with either the earlier password or the new password. However, at the time the user logs in, the server recalculates whether the account is still in its password rollover period. If the password rollover period has elapsed, then the login will succeed only if the new password was specified. • `EXPIRED & LOCKED(TIMED) & IN ROL` The account is in the password rollover period, its password is expired as described for the `EXPIRED` account status, and the account is locked as described for the `LOCKED(TIMED)` account status. The account can be unlocked as described for the `LOCKED(TIMED)` account status, after which the user can log in as described for the `EXPIRED & IN ROLLOVER` account status.
`LOCK_DATE`	`DATE`		Date the account was locked if account status was `LOCKED`
`EXPIRY_DATE`	`DATE`		Date of expiration of the account
`DEFAULT_TABLESPACE`	`VARCHAR2(30)`	`NOT NULL`	Default tablespace for data
`TEMPORARY_TABLESPACE`	`VARCHAR2(30)`	`NOT NULL`	Name of the default tablespace for temporary tables or the name of a tablespace group
`LOCAL_TEMP_TABLESPACE`	`VARCHAR2(30)`		Default local temporary tablespace for the user
`CREATED`	`DATE`	`NOT NULL`	User creation date
`PROFILE`	`VARCHAR2(128)`	`NOT NULL`	User resource profile name
`INITIAL_RSRC_CONSUMER_GROUP`	`VARCHAR2(128)`		Initial resource consumer group for the user
`EXTERNAL_NAME`	`VARCHAR2(4000)`		User external name. For centrally managed users, if the database user mapping is an exclusive mapping, then this will be the directory service DN for the user. If this database user is a shared schema, it will be the DN of a group.
`PASSWORD_VERSIONS`	`VARCHAR2(12)`		Shows the list of versions of the password hashes (also known as "verifiers") existing for the account. The values for this column can include: • `10G`: If an old case-insensitive ORCL hash exists • `11G`: If a SHA-1 hash exists • `12C`: If a de-optimized PBKDF2-based hash exists • `HTTP`: If an MD5 hash (for HTTP Digest authentication) exists For more information about the 12C verifier, see *Oracle Database Concepts*. Note that any combination of these verifiers can exist for any given account.
`EDITIONS_ENABLED`	`VARCHAR2(1)`		Indicates whether editions have been enabled for the corresponding user (`Y`) or not (`N`)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
AUTHENTICATION_TYPE	VARCHAR2(8)		Indicates the authentication mechanism for the user: • NONE - The user has not been configured for an authentication method • EXTERNAL - CREATE USER *user1* IDENTIFIED EXTERNALLY; • GLOBAL - CREATE USER *user2* IDENTIFIED GLOBALLY; • PASSWORD - CREATE USER *user3* IDENTIFIED BY *password*;
PROXY_ONLY_CONNECT	VARCHAR2(1)		Indicates whether a user can connect directly (N) or whether the account can only be proxied (Y) by users who have proxy privileges for this account (that is, by users who have been granted the "connect through" privilege for this account). For more information about creating proxy user accounts and authorizing users to connect through them, see *Oracle Database Security Guide*.
COMMON	VARCHAR2(3)		Indicates whether a given user is common. Possible values • YES if a user is common • NO if a user is local (not common)
LAST_LOGIN	TIMESTAMP(9) WITH TIME ZONE		The time of the last user login This column is not populated when a user connects to the database with administrative privileges, that is, AS { SYSASM | SYSBACKUP | SYSDBA | SYSDG | SYSOPER | SYSRAC | SYSKM }.
ORACLE_MAINTAINED	VARCHAR2(1)		Denotes whether the user was created, and is maintained, by Oracle-supplied scripts (such as catalog.sql or catproc.sql). A user for which this column has the value Y must not be changed in any way except by running an Oracle-supplied script.
INHERITED	VARCHAR2(3)		Indicates whether the user definition was inherited from another container (YES) or not (NO)
DEFAULT_COLLATION	VARCHAR2(100)		Default collation for the user's schema
IMPLICIT	VARCHAR2(3)		Indicates whether this user is a common user created by an implicit application (YES) or not (NO)
ALL_SHARD	VARCHAR2(3)		In a sharded database, the value in this column indicates whether the user was created with shard DDL enabled. The possible values are: • YES: The user was created with shard DDL enabled. The user exists on all shards and the shard catalog. • NO: The user was created without shard DDL enabled. The user exists only in the database in which the user was created. In a non-sharded database, the value in this column is always NO.
PASSWORD_CHANGE_DATE[1]	DATE		Date on which the user's password was last set This column is populated only when the value of the AUTHENTICATION_TYPE column is PASSWORD. Otherwise, this column is null.

Copyright © 2002, 2026, Oracle and/or its affiliates.

[1] This column is available starting with Oracle Database 19c.

> **Note**
>
> In a CDB, common users should be connected to the CDB root when querying DBA_USERS or USER_USERS. This ensures that complete information is returned.

> **See Also**
>
> - "USER_USERS"
> - *Oracle Database Security Guide* for information about creating schema only accounts where the schema user has no password
> - *Using Oracle Sharding* for more information about sharded database management

6.98 DBA_USERS_WITH_DEFPWD

DBA_USERS_WITH_DEFPWD displays all users in the database that are still using their default passwords.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user
PRODUCT	VARCHAR2(4000)		Name of the product the user belongs to

> **Note**
>
> In a CDB, when DBA_USERS_WITH_DEFPWD is queried from a PDB, information about local users who are using their default passwords is displayed. To display information about common users, query DBA_USERS_WITH_DEFPWD from the CDB root.

> **Note**
>
> Starting with Oracle Database 19c, Release Update 19.32, this view displays additional default password account findings and associated product labels for SYS, SYSTEM, and ADMIN account combinations. This feature gives administrators a more complete view of default password risks and the related product context for remediation.

> **See Also**
>
> *Oracle Multitenant Administrator's Guide* for an introduction to local and common users in a CDB

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.99 DBA_USTATS

DBA_USTATS describes the user-defined statistics collected on all tables and indexes in the database. Its columns are the same as those in ALL_USTATS.

See Also

"ALL_USTATS"

6.100 DBA_VARRAYS

DBA_VARRAYS describes all varrays in the database. Its columns are the same as those in ALL_VARRAYS.

See Also

"ALL_VARRAYS"

6.101 DBA_VIEWS

DBA_VIEWS describes all views in the database. Its columns are the same as those in ALL_VIEWS.

See Also

"ALL_VIEWS"

6.102 DBA_VIEWS_AE

DBA_VIEWS_AE describes all views (across all editions) in the database. Its columns are the same as those in ALL_VIEWS_AE.

See Also

"ALL_VIEWS_AE"

6.103 DBA_WAITERS

DBA_WAITERS shows all the sessions that are waiting for a lock. In an Oracle RAC environment, this only applies if the waiter is on the same instance.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
WAITING_SESSION	NUMBER		The waiting session
WAITING_CON_ID	NUMBER		The container ID (CON_ID) of the waiting session
HOLDING_SESSION	NUMBER		The holding session
HOLDING_CON_ID	NUMBER		The container ID (CON_ID) of the holding session.
LOCK_TYPE	VARCHAR2(26)		The lock type
MODE_HELD	VARCHAR2(40)		The mode held
MODE_REQUESTED	VARCHAR2(40)		The mode requested
LOCK_ID1	NUMBER		Lock ID 1
LOCK_ID2	NUMBER		Lock ID 2

6.104 DBA_WALLET_ACES

DBA_WALLET_ACES describes access control entries defined in wallet access control lists.

Related View

USER_WALLET_ACES describes the status of access control entries for the current user to access wallets through PL/SQL network utility packages. This view does not display the ACE_ORDER, START_DATE, END_DATE, GRANT_TYPE, INVERTED_PRINCIPAL, PRINCIPAL, or PRINCIPAL_TYPE columns.

Column	Datatype	NULL	Description
WALLET_PATH	VARCHAR2(1000)	NOT NULL	Wallet path
ACE_ORDER	NUMBER	NOT NULL	Order number of the access control entry
START_DATE	TIMESTAMP(6)		Start date of the access control entry
END_DATE	TIMESTAMP(6)		End date of the access control entry
GRANT_TYPE	VARCHAR2(5)		Indicates whether the access control entry grants or denies the privilege
INVERTED_PRINCIPAL	VARCHAR2(3)		Indicates whether the principal is inverted or not
PRINCIPAL	VARCHAR2(128)		Principal the privilege is applied to
PRINCIPAL_TYPE	VARCHAR2(16)		Type of the principal
PRIVILEGE	VARCHAR2(128)		Privilege

See Also

"USER_WALLET_ACES"

6.105 DBA_WALLET_ACLS

DBA_WALLET_ACLS displays the access control lists assigned to restrict access to wallets through PL/SQL network utility packages.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
WALLET_PATH	VARCHAR2(1000)	NOT NULL	Wallet path
ACL	VARCHAR2(4000)		Path of the access control list
ACLID	RAW(8)		Object ID of the access control list
ACL_OWNER	VARCHAR2(128)		Owner of the access control list

6.106 DBA_WARNING_SETTINGS

DBA_WARNING_SETTINGS displays information about the warning parameter settings for all objects in the database. Its columns are the same as those in ALL_WARNING_SETTINGS.

See Also

"ALL_WARNING_SETTINGS"

6.107 DBA_WI_CAPTURE_FILES

Each row in DBA_WI_CAPTURE_FILES represents a capture file that belongs to the workload analyzed in the current Workload Intelligence job.

Column	Datatype	NULL	Description
JOB_ID	NUMBER	NOT NULL	The identifier of the job in the workload to which the given capture file belongs
FILE_ID	NUMBER	NOT NULL	The identifier of the current capture file
FILE_PATH	VARCHAR2(4000)	NOT NULL	The path of the current capture file

6.108 DBA_WI_JOBS

Each row in DBA_WI_JOBS describes a Workload Intelligence job, that is, a task that applies the algorithms of Workload Intelligence on a given capture directory.

Column	Datatype	NULL	Description
JOB_ID	NUMBER	NOT NULL	The job identifier
JOB_NAME	VARCHAR2(128)	NOT NULL	A name that uniquely identifies the given job
CAPTURE_DIRECTORY	VARCHAR2(4000)	NOT NULL	Path to the capture directory on which the given job has been applied
MODEL_ORDER	NUMBER		The order of the markov model that describes the workload associated with the current job. If NULL, the corresponding order has not been calculated yet.
THRESHOLD	NUMBER		A number in the range [0, 1] that represents the threshold that the user has given as an input parameter to the current job of Workload Intelligence for the identification of significant patterns. If NULL, the process of pattern identification has not been initiated yet.

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.109 DBA_WI_OBJECTS

Each row in DBA_WI_OBJECTS represents a database object (table) that is accessed by the given template in the given Workload Intelligence job.

Column	Datatype	NULL	Description
JOB_ID	NUMBER	NOT NULL	The identifier of the job in the workload of which the given object has been accessed
TEMPLATE_ID	NUMBER	NOT NULL	The identifier of the template in the given job by which the current object has been accessed
OBJECT_ID	NUMBER	NOT NULL	The identifier of the current object
ACCESS_TYPE	VARCHAR2(2)	NOT NULL	Possible values: • R - Indicates that the current object has been accessed for reading by the given template • W - Indicates that the current object has been accessed for writing by the given template • RW - Indicates that the current object has been accessed for both reading and writing by the given template

6.110 DBA_WI_PATTERN_ITEMS

Each row in DBA_WI_PATTERN_ITEMS represents a template that participates in a significant pattern that has been found by the given Workload Intelligence job.

Column	Datatype	NULL	Description
JOB_ID	NUMBER	NOT NULL	The identifier of the job in the workload of which the current pattern has been found
PATTERN_ID	NUMBER	NOT NULL	The identifier of the pattern to which the current item (template) belongs
SEQUENCE_NUMBER	NUMBER	NOT NULL	Number that indicates the position of the current item in the given pattern
TEMPLATE_ID	NUMBER	NOT NULL	The identifier of the template that participates in the given position of the current pattern
IS_FIRST_IN_LOOP	CHAR(1)	NOT NULL	A flag that indicates whether or not the current item marks the beginning of a loop in the given pattern. The possible values are Y and N.
IS_LAST_IN_LOOP	CHAR(1)	NOT NULL	A flag that indicates whether or not the current item marks the end of a loop in the given pattern. The possible values are Y and N.

6.111 DBA_WI_PATTERNS

Each row in DBA_WI_PATTERNS represents a pattern that has been identified by Workload Intelligence as significant in the workload associated with the given job. Such a pattern consists of one or more templates.

These templates that comprise the given pattern are described in the related view DBA_WI_PATTERN_ITEMS.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
JOB_ID	NUMBER	NOT NULL	The identifier of the job in the workload of which the current pattern has been found
PATTERN_ID	NUMBER	NOT NULL	The identifier of the current pattern
LENGTH	NUMBER	NOT NULL	The length of the pattern, that is, the number of items (templates) it consists of
NUMBER_OF_EXECUTIONS	NUMBER	NOT NULL	The number of times the current pattern has been executed in the given workload
DB_TIME	NUMBER	NOT NULL	The total time consumed in the database server by all the executions of the current pattern in the given workload

See Also

"DBA_WI_PATTERN_ITEMS"

6.112 DBA_WI_STATEMENTS

Each row in DBA_WI_STATEMENTS describes a statement (SQL or PL/SQL) that is part of the template with identifier TEMPLATE_ID, which has been found in the workload that is related to the Workload Intelligence job whose identifier is equal to JOB_ID.

A template may consist of multiple statements, for example, if it represents a transaction. In this case, there is one row in this view for every one of these statements. These statements are ordered, based on the order defined by the corresponding transaction. Column SEQUENCE_NUMBER is used to describe this order.

Column	Datatype	NULL	Description
JOB_ID	NUMBER	NOT NULL	The identifier of the job in the workload of which the given statement has been found
TEMPLATE_ID	NUMBER	NOT NULL	The identifier of the template in the given job to which the current statement belongs
SEQUENCE_NUMBER	NUMBER	NOT NULL	A number that indicates the order of the current statement in the given template
SQL_TEXT	CLOB	NOT NULL	The SQL text associated with the current statement. Note that although multiple SQL statements can be classified to the same template, only one row is stored that represents them all. This row corresponds to the first instance of the given template that is found during parsing of the workload.

6.113 DBA_WI_TEMPLATE_EXECUTIONS

Each row in DBA_WI_TEMPLATE_EXECUTIONS represents an execution of a template in a capture that belongs to the workload that is associated with the current Workload Intelligence job.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
JOB_ID	NUMBER	NOT NULL	The identifier of the job in the workload of which the current execution of the given template belongs
CAPTURE_FILE_ID	NUMBER	NOT NULL	The identifier of the capture file in which the current execution of the given template was found
SEQUENCE_NUMBER	NUMBER	NOT NULL	A number that indicates the order of the current execution in the given capture file
TEMPLATE_ID	NUMBER	NOT NULL	The identifier of the template that was executed in the execution represented by the current row
DB_TIME	NUMBER	NOT NULL	The time that the current execution consumed on the database server

6.114 DBA_WI_TEMPLATES

Each row in DBA_WI_TEMPLATES describes a template that has been found in the workload that is related to the Workload Intelligence job whose identifier is equal to JOB_ID.

A template can represent either a simple query, or an entire transaction. Two queries in the given workload belong to the same template, if they exhibit trivial differences, for example, if they contain different literal values, different bind variable names, different comments, or different white spaces.

Column	Datatype	NULL	Description
JOB_ID	NUMBER	NOT NULL	The identifier of the job in the workload of which the given template has been found
TEMPLATE_ID	NUMBER	NOT NULL	The identifier of a template in a given job
IS_TRANSACTION	CHAR(1)	NOT NULL	Flag that indicates whether the given template represents a transaction: • Y - indicates that the given template represents a transaction • N - indicates that the given template does not represent a transaction

6.115 DBA_WORKLOAD_ACTIVE_USER_MAP

DBA_WORKLOAD_ACTIVE_USER_MAP contains the mappings that are going to be valid for the next replay or are valid for the current replay.

Column	Datatype	NULL	Description
SCHEDULE_CAP_ID	NUMBER		The ID of a capture in the schedule
CAPTURE_USER	VARCHAR2(4000)	NOT NULL	The user name during the time of the workload capture
REPLAY_USER	VARCHAR2(4000)		The user name to which captured user should be remapped during replay

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.116 DBA_WORKLOAD_CAPTURE_SQLTEXT

DBA_WORKLOAD_CAPTURE_SQLTEXT displays all the SQL statements that have been recorded in a workload capture. For those SQL statements whose length exceeds 1000 characters, the full statements can be loaded to the DBA_WORKLOAD_LONG_SQLTEXT view using the DBMS_WORKLOAD_REPLAY.LOAD_LONG_SQLTEXT procedure.

Column	Datatype	NULL	Description
CAPTURE_ID	NUMBER(38)	NOT NULL	Internal key for the workload capture
SQL_ID	VARCHAR2(13)	NOT NULL	SQL identifier of the parent cursor in the library cache
SQL_TYPE	VARCHAR2(64)		Type of the SQL statement, which can include values such as INSERT, SELECT, and CREATE INDEX
SQL_TEXT	VARCHAR2(1000)		First thousand characters of the SQL text for the current cursor
SQL_LENGTH	NUMBER(38)		The length of the SQL statement
SQL_TEXT_COMPLETE	CHAR(1)		Indicates whether the SQL_TEXT column includes the full text of the SQL statement. Possible values: • Y: The column SQL_TEXT includes the full text of the SQL statement • N: The column SQL_TEXT contains only the first thousand characters of the SQL text

See Also

- "DBA_WORKLOAD_LONG_SQLTEXT"
- "DBA_RAT_CAPTURE_SCHEMA_INFO"
- *Oracle Database PL/SQL Packages and Types Reference* for information about the DBMS_WORKLOAD_REPLAY package

6.117 DBA_WORKLOAD_CAPTURES

DBA_WORKLOAD_CAPTURES displays all the workload captures that have been performed in the current database.

It also lists captures on which DBMS_WORKLOAD_CAPTURE.GET_CAPTURE_INFO() or DBMS_WORKLOAD_REPLAY.GET_REPLAY_INFO() have been called. Each row contains information about one workload capture.

Column	Datatype	NULL	Description
ID	NUMBER	NOT NULL	Internal key for the workload capture
NAME	VARCHAR2(128)	NOT NULL	Name for the workload capture
DBID	NUMBER	NOT NULL	ID of the database in which the workload was captured
DBNAME	VARCHAR2(10)	NOT NULL	Name of the database in which the workload was captured

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DBVERSION	VARCHAR2(17)	NOT NULL	Version of the database in which the workload was captured
PARALLEL	VARCHAR2(3)		Indicates whether the database in which the workload was captured is an Oracle RAC database (YES) or a single instance database (NO)
DIRECTORY	VARCHAR2(128)	NOT NULL	Name of the directory object for workload capture
STATUS	VARCHAR2(40)	NOT NULL	Current status of the workload capture: • IN PROGRESS - Workload capture is in progress • COMPLETED - Workload capture has completed successfully • FAILED - Workload capture was terminated due to errors encountered
START_TIME	DATE	NOT NULL	Datetime when the capture began
END_TIME	DATE		Datetime when the capture completed or failed; NULL if the capture is still in progress
DURATION_SECS	NUMBER		Duration of the workload capture (in seconds)
START_SCN	NUMBER	NOT NULL	Start SCN value for this capture
END_SCN	NUMBER		End SCN value for this capture; NULL if the capture is still in progress
DEFAULT_ACTION	VARCHAR2(30)	NOT NULL	Mode in which to apply workload capture filters: • INCLUDE - All the capture filters are treated as EXCLUSION filters, determining the workload that will not be captured. • EXCLUDE - All the capture filters are treated as INCLUSION FILTERS, determining the workload that will be captured.
FILTERS_USED	NUMBER		Number of filters that were used for this capture
CAPTURE_SIZE	NUMBER		Total size of workload capture
DBTIME	NUMBER		Total amount of database time (in microseconds) that has been recorded in this workload capture
DBTIME_TOTAL	NUMBER		Total amount of database time (in microseconds) across the entire database during the workload capture, including the part of the workload that was not captured.
USER_CALLS	NUMBER		Total number of user calls that have been recorded in this workload capture
USER_CALLS_TOTAL	NUMBER		Total number of user calls across the entire database during the workload capture, including the part of the workload that was not captured.
USER_CALLS_UNREPLAYABLE	NUMBER		Total number of user calls that will not be replayed in a subsequent replay of this workload capture
PLSQL_SUBCALL_SIZE	NUMBER		Total size of workload capture for SQL executed from PL/SQL
PLSQL_CALLS	NUMBER		Total number of PL/SQL calls recorded in the workload capture
PLSQL_SUBCALLS	NUMBER		Total number of calls recorded in the workload capture for SQL executed from PL/SQL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PLSQL_DBTIME	NUMBER		Total amount of database time (in microseconds) from PL/SQL calls that have been recorded in the workload capture
TRANSACTIONS	NUMBER		Total number of transactions that have been recorded in this workload capture
TRANSACTIONS_TOTAL	NUMBER		Total number of transactions across the entire database during the workload capture, including the part of the workload that was not captured.
CONNECTS	NUMBER		Total number of session connects that have been recorded in this workload capture
CONNECTS_TOTAL	NUMBER		Total number of session connects across the entire database during the workload capture, including the part of the workload that was not captured
ERRORS	NUMBER		Total number of errors that have been recorded in this workload capture
AWR_DBID	NUMBER		Database ID of the AWR snapshots that correspond to this workload capture. For captures that were performed in the current database, this value is equal to the current database's DBID. For captures that were performed in other databases, this value will either be NULL or will be populated by `DBMS_WORKLOAD_CAPTURE.IMPORT_AWR()`.
AWR_BEGIN_SNAP	NUMBER		Begin snapshot ID of the AWR snapshots that correspond to this workload capture
AWR_END_SNAP	NUMBER		End snapshot ID of the AWR snapshots that correspond to this workload capture
AWR_EXPORTED	VARCHAR2(12)		Indicates whether the AWR snapshots that correspond to this workload capture have been exported using `DBMS_WORKLOAD_CAPTURE.EXPORT_AWR()` (`YES`) or not (`NO`), or whether AWR snapshots cannot be exported because the capture is still in progress, has run to completion successfully, or was done in a different database from which it was not exported (`NOT POSSIBLE`)
ERROR_CODE	NUMBER		Error code for this workload capture
ERROR_MESSAGE	VARCHAR2(512)		Error message for this workload capture
DIR_PATH	VARCHAR2(4000)	NOT NULL	Full directory path for the workload capture directory object
DIR_PATH_SHARED	VARCHAR2(10)	NOT NULL	Indicates whether the workload capture directory is shared by all the instances of the recording database (applicable only for Oracle RAC databases)
LAST_PROCESSED_VERSION	VARCHAR2(128)		Database version in which this capture was preprocessed using `DBMS_WORKLOAD_REPLAY.PROCESS_CAPTURE()` last; NULL if the capture has never been preprocessed
SQLSET_OWNER	VARCHAR2(128)		User name of the SQL tuning set owner
SQLSET_NAME	VARCHAR2(128)		Name of the SQL tuning set for this workload capture

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PLSQL_MODE	VARCHAR2(12)		Capture options for PL/SQL calls. Possible values: • TOP_LEVEL: Top-level PL/SQL only • EXTENDED: Both top-level PL/SQL and SQL executed from PL/SQL
ENCRYPTION	VARCHAR2(128)		Indicates the encryption standard used for the given capture: • NULL - Capture files are not encrypted • AES128 – Capture files are encrypted using AES128 • AES192 – Capture files are encrypted using AES192 • AES256 – Capture files are encrypted using AES256
ENCRYPTION_VERIFIER[1]	RAW(512)		Data used internally for creating an encrypted capture

[1] This column is available starting with Oracle Database 19c, Release Update 19.7.

See Also

- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_WORKLOAD_CAPTURE package
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_WORKLOAD_REPLAY package

6.118 DBA_WORKLOAD_CONNECTION_MAP

DBA_WORKLOAD_CONNECTION_MAP displays the connection mapping information for workload replay. Each row defines one connection mapping for a particular workload replay.

Column	Datatype	NULL	Description
REPLAY_ID	NUMBER	NOT NULL	ID of the replay (corresponds to DBA_WORKLOAD_REPLAYS.ID)
CONN_ID	NUMBER	NOT NULL	Key (ID) of the connection mapping table
SCHEDULE_CAP_ID	NUMBER		Schedule capture ID (corresponds to DBA_WORKLOAD_SCHEDULE_CAPTURES.SCHEDULE_CAP_ID)
CAPTURE_CONN	VARCHAR2(4000)	NOT NULL	Connection string that was used during capture
REPLAY_CONN	VARCHAR2(4000)		Connection string that should be used during replay

6.119 DBA_WORKLOAD_DIV_SUMMARY

DBA_WORKLOAD_DIV_SUMMARY displays a summary of the replay divergence information in the DBA_WORKLOAD_REPLAY_DIVERGENCE view. DBA_WORKLOAD_REPLAY_DIVERGENCE may have

Copyright © 2002, 2026, Oracle and/or its affiliates.

duplicate entries, while DBA_WORKLOAD_DIV_SUMMARY keeps only one entry and tracks the number of occurrences of each duplicate entry.
Starting with Oracle Database 12.2.0.1, the replay report is generated from DBA_WORKLOAD_DIV_SUMMARY instead of from DBA_WORKLOAD_REPLAY_DIVERGENCE, which results in faster generation of the replay report.

Column	Datatype	NULL	Description
REPLAY_ID	NUMBER	NOT NULL	ID (key) for the workload replay
DIVERGENCE_TYPE	NUMBER	NOT NULL	Reserved for future use
IS_QUERY_DATA_DIVERGENCE	VARCHAR2(1)		Indicates whether the data divergence is from the number of rows fetched by SELECT queries (Y) or not (N)
IS_DML_DATA_DIVERGENCE	VARCHAR2(1)		Indicates whether the divergence is from the number of rows affected by INSERT, UPDATE, or DELETE SQL statements (Y) or not (N)
IS_ERROR_DIVERGENCE	VARCHAR2(1)		Indicates whether the divergence is from errors seen during capture or replay (Y) or not (N)
IS_THREAD_FAILURE	VARCHAR2(1)		Indicates whether the divergence is from sessions that failed during replay (Y) or not (N)
IS_DATA_MASKED	VARCHAR2(1)		Indicates whether the SQL call contains masked bind data (Y) or not (N). If data masking technology is used at the replay database, the workload capture files need to be masked. Otherwise, SQL statements generated from capture files that contain sensitive bind data will not match the database. When the replay client sends masked bind data to the server, it turns on the IS_DATA_MASKED flag for the current SQL call.
STREAM_ID	NUMBER	NOT NULL	Stream ID of the session that reported the divergence
SQL_ID	VARCHAR2(13)		SQL ID of the SQL that reported the divergence
EXPECTED_ERROR#	NUMBER		Error number that was seen during capture (0 if the capture ran successfully)
EXPECTED_ERROR_MESSAGE	VARCHAR2(4000)		Text of the error message whose number appears in the EXPECTED_ERROR# column
OBSERVED_ERROR#	NUMBER		Actual error number seen during replay (0 if the replay ran successfully, 15566 (corresponding to ORA-15566) if the captured call could not be replayed)
OBSERVED_ERROR_MESSAGE	VARCHAR2(4000)		Text of the error message whose number appears in the OBSERVED_ERROR# column
SERVICE	VARCHAR2(64)		Service name of the session that reported the divergence
MODULE	VARCHAR2(64)		Module name of the session that reported the divergence
OCCURRENCES	NUMBER		Number of times the divergence occurred during replay

Example

The following query prints the top 3 SQL statements that got error divergence during replay. This query shows the captured error number and the actual error number seen during replay.

```
SQL> SELECT * FROM
 (SELECT occurrences, sql_id, expected_error#, observed_error#
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

```
  FROM    dba_workload_div_summary
  WHERE   replay_id = 123
   AND    is_error_divergence = 'Y'
  ORDER BY occurrences DESC)
WHERE ROWNUM <= 3;

OCCURRENCES SQL_ID        EXPECTED_ERROR# OBSERVED_ERROR#
----------- ------------- --------------- ---------------
          8 0xrm2wjdqv17m               0               1
          4 8bzwdnnznspjd            1422               0
          3 6d8rwrac8dsk7               1            1400

SQL>
```

> **See Also**
>
> "DBA_WORKLOAD_REPLAY_DIVERGENCE"

6.120 DBA_WORKLOAD_FILTERS

DBA_WORKLOAD_FILTERS displays all the workload filters that have been defined in the current database.

In Oracle Database 11*g*, only workload filters of type CAPTURE are supported. Starting with Oracle Database 11*g*R2, filters of type REPLAY are supported.

Column	Datatype	NULL	Description
TYPE	VARCHAR2(30)		Type of the workload filter (CAPTURE or REPLAY)
ID	VARCHAR2(40)		Sequence number of the workload filter
STATUS	VARCHAR2(6)		Status of the workload filter: • NEW - This filter will be used by the next subsequent operation such as the next workload capture. • IN USE - This filter is currently being used by an operation that is in progress such as an active workload capture. • USED - This filter was used in the past by some operation such as a past workload capture.
SET_NAME	VARCHAR2(1000)		Name of the filter set to which the filter belongs
NAME	VARCHAR2(128)		Name of the workload filter
ATTRIBUTE	VARCHAR2(128)		Name of the attribute on which the filter is defined
VALUE	VARCHAR2(4000)		Value of the attribute on which the filter is defined. Wildcards such as % and _ are supported if the attribute is of string type.

6.121 DBA_WORKLOAD_GROUP_ASSIGNMENTS

DBA_WORKLOAD_GROUP_ASSIGNMENTS displays all the workload capture groups and their assigned instances. A workload capture group is a subset of the captured workload. Each group accesses its own set of recorded database objects.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
REPLAY_DIR_NUMBER	NUMBER(38)	NOT NULL	The value that is associated with the subdirectory under the replay directory. See REPLAY_DIR_NUMBER in DBA_WORKLOAD_REPLAYS.
GROUP_ID	NUMBER(38)	NOT NULL	The identifier of a workload capture group
INSTANCE_NUMBER	NUMBER(38)	NOT NULL	The instance a given group is assigned to

See Also

"DBA_WORKLOAD_REPLAYS"

6.122 DBA_WORKLOAD_LONG_SQLTEXT

DBA_WORKLOAD_LONG_SQLTEXT displays the captured SQL statements that are longer than 1000 characters. You can load SQL statements longer than 1000 characters to the DBA_WORKLOAD_LONG_SQLTEXT view using the DBMS_WORKLOAD_REPLAY.LOAD_LONG_SQLTEXT procedure.

Column	Datatype	NULL	Description
CAPTURE_ID	NUMBER(38)	NOT NULL	Internal key for the workload capture
SQL_ID	VARCHAR2(13)	NOT NULL	SQL identifier of the parent cursor in the library cache
SQL_FULLTEXT	CLOB		Full text for the SQL statement exposed as a CLOB column

See Also

- "DBA_WORKLOAD_CAPTURE_SQLTEXT"
- *Oracle Database PL/SQL Packages and Types Reference* for information about the DBMS_WORKLOAD_REPLAY package

6.123 DBA_WORKLOAD_REPLAY_CLIENTS

DBA_WORKLOAD_REPLAY_CLIENTS displays all workload replay clients and their assigned instances.

Column	Datatype	NULL	Description
WRC_ID	NUMBER(38)	NOT NULL	The identifier of a workload replay client
SCHEDULE_CAP_ID	NUMBER(38)	NOT NULL	A unique identifier for a workload capture added to a replay schedule. 0 for a non-consolidated replay
INSTANCE_NUMBER	NUMBER(38)	NOT NULL	The instance that the replay client connects to

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.124 DBA_WORKLOAD_REPLAY_DIVERGENCE

DBA_WORKLOAD_REPLAY_DIVERGENCE displays information about data/error divergence for a user call that has been replayed.

DBA_WORKLOAD_DIV_SUMMARY displays a summary of the replay divergence information in the DBA_WORKLOAD_REPLAY_DIVERGENCE view. DBA_WORKLOAD_REPLAY_DIVERGENCE may have duplicate entries, while DBA_WORKLOAD_DIV_SUMMARY keeps only one entry and tracks the number of occurrences of each duplicate entry.

Column	Datatype	NULL	Description
REPLAY_ID	NUMBER	NOT NULL	ID (key) for the workload replay
TIMESTAMP	TIMESTAMP(6) WITH TIME ZONE		Time that the divergence occurred
DIVERGENCE_TYPE	NUMBER	NOT NULL	Reserved for future use
IS_QUERY_DATA_DIVERGENCE	VARCHAR2(1)		Indicates whether the data divergence is from the number of rows fetched by SELECT queries (Y) or not (N)
IS_DML_DATA_DIVERGENCE	VARCHAR2(1)		Indicates whether the divergence is from the number of rows affected by INSERT, UPDATE, or DELETE SQL statements (Y) or not (N)
IS_ERROR_DIVERGENCE	VARCHAR2(1)		Indicates whether the divergence is from errors seen during capture or replay (Y) or not (N)
IS_THREAD_FAILURE	VARCHAR2(1)		Indicates whether the divergence is from sessions that failed during replay (Y) or not (N)
IS_DATA_MASKED	VARCHAR2(1)		Indicates whether the SQL call contains masked bind data (Y) or not (N). If data masking technology is used at the replay database, the workload capture files need to be masked. Otherwise, SQL statements generated from capture files that contain sensitive bind data will not match the database. When the replay client sends masked bind data to the server, it turns on the IS_DATA_MASKED flag for the current SQL call.
EXPECTED_ROW_COUNT	NUMBER		Number of rows fetched for SELECT queries or rows affected for INSERT, UPDATE, or DELETE SQL statements during capture
OBSERVED_ROW_COUNT	NUMBER		Actual number of rows fetched for SELECT queries or rows affected for INSERT, UPDATE, or DELETE SQL statements during replay
EXPECTED_ERROR#	NUMBER		Error number that was seen during capture (0 if the capture ran successfully)
EXPECTED_ERROR_MESSAGE	VARCHAR2(4000)		Text of the error message whose number appears in the EXPECTED_ERROR# column
OBSERVED_ERROR#	NUMBER		Actual error number seen during replay (0 if the replay ran successfully, 15566 (corresponding to ORA-15566) if the captured call could not be replayed)
OBSERVED_ERROR_MESSAGE	VARCHAR2(4000)		Text of the error message whose number appears in the OBSERVED_ERROR# column
STREAM_ID	NUMBER	NOT NULL	Stream ID of the session that reported the divergence

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CALL_COUNTER	NUMBER	NOT NULL	Call counter of the user call that reported the divergence
CAPTURE_STREAM_ID	NUMBER		Internal ID of the capture file whose replay produced the divergence
SQL_ID	VARCHAR2(13)		SQL ID of the SQL that reported the divergence
SESSION_ID	NUMBER	NOT NULL	Session ID of the session that reported the divergence
SESSION_SERIAL#	NUMBER	NOT NULL	Captured session serial number of the session that reported the divergence
SERVICE	VARCHAR2(64)		Service name of the session that reported the divergence
MODULE	VARCHAR2(64)		Module name of the session that reported the divergence
ACTION	VARCHAR2(64)		Action name of the session that reported the divergence

See Also

"DBA_WORKLOAD_DIV_SUMMARY"

6.125 DBA_WORKLOAD_REPLAY_SCHEDULES

DBA_WORKLOAD_REPLAY_SCHEDULES displays the names of replay schedules for the current replay directory.

A replay schedule defines one or multiple workload captures, and the order to start their replays. The current replay directory is set by DBMS_WORKLOAD_REPLAY.SET_REPLAY_DIRECTORY('replay_dir'). Each row in the view contains information about one replay schedule.

Column	Datatype	NULL	Description
SCHEDULE_NAME	VARCHAR2(128)	NOT NULL	The name of a schedule to be replayed. It defines one or multiple workload captures, and the order to start their replays.
DIRECTORY	VARCHAR2(128)	NOT NULL	Directory object name for the replay schedule name
STATUS	VARCHAR2(128)		NEW if the schedule is being created, CURRENT if the schedule is currently being used by a replay, otherwise NULL

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_WORKLOAD_SCHEDULE_CAPTURES" displays the workload captures in a replay schedule.
- "DBA_WORKLOAD_SCHEDULE_ORDERING" displays the order to start captures in a replay schedule.
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_WORKLOAD_REPLAY package

6.126 DBA_WORKLOAD_REPLAYS

DBA_WORKLOAD_REPLAYS displays all the workload replays that have been performed in the current database.

It also lists replays on which DBMS_WORKLOAD_REPLAY.GET_REPLAY_INFO() has been called. Each row contains information about one workload replay.

Column	Datatype	NULL	Description
ID	NUMBER	NOT NULL	Internal key for the workload replay
NAME	VARCHAR2(128)	NOT NULL	Name of the workload replay
DBID	NUMBER	NOT NULL	ID of the database in which the workload was replayed
DBNAME	VARCHAR2(10)	NOT NULL	Name of the database in which the workload was replayed
DBVERSION	VARCHAR2(17)	NOT NULL	Version of the database in which the workload was replayed
PARALLEL	VARCHAR2(3)		Indicates whether the database in which the workload was replayed was an Oracle RAC database (YES) or a single instance database (NO)
DIRECTORY	VARCHAR2(128)	NOT NULL	Name of the directory object for the workload replay
CAPTURE_ID	NUMBER		ID of the capture (DBA_WORKLOAD_CAPTURES.ID) that was replayed. If the replay involves a replay schedule, the CAPTURE_ID will be null.
STATUS	VARCHAR2(40)	NOT NULL	Current status of the workload replay: • PREPARE - Workload prepare has been started and is waiting for clients to join • IN PROGRESS - Workload replay is in progress • COMPLETED - Workload replay has successfully completed • CANCELLED - Workload replay or the workload prepare has been cancelled • FAILED - Workload replay was terminated due to errors encountered. See the COMMENTS column for further information.
PREPARE_TIME	DATE		Datetime at which the workload prepare started
START_TIME	DATE		Datetime when the replay began
END_TIME	DATE		Datetime when the replay completed or cancelled; NULL if the replay is still in progress
DURATION_SECS	NUMBER		Duration of the workload replay (in seconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
NUM_CLIENTS	NUMBER	NOT NULL	Number of workload replay client processes that were used in this workload replay
NUM_CLIENTS_DONE	NUMBER	NOT NULL	Number of workload replay client processes that have finished replay
FILTER_SET_NAME	VARCHAR2(128)		Name of the filter set used for the replay
DEFAULT_ACTION	VARCHAR2(30)	NOT NULL	Reserved for future use
SYNCHRONIZATION	VARCHAR2(9)		Indicates whether recorded transaction semantics should be maintained (TRUE) or not (FALSE) When synchronization is on, the commit order observed during the original workload capture will be preserved. Every action that is replayed will be executed only after all of its dependent commits have been executed. Dependent commits are commits that were issued before the given action in the original workload capture. **See Also:** DBMS_WORKLOAD_REPLAY.PREPARE_REPLAY() in *Oracle Database PL/SQL Packages and Types Reference* for a detailed explanation of this replay parameter
CONNECT_TIME_SCALE	NUMBER	NOT NULL	Connection time scaling factor for captured streams during replay. The value is interpreted as a percentage.The default value of 100 means 100 percent. **See Also:** DBMS_WORKLOAD_REPLAY.PREPARE_REPLAY() in *Oracle Database PL/SQL Packages and Types Reference* for a detailed explanation of this replay parameter
THINK_TIME_SCALE	NUMBER	NOT NULL	Think time scaling factor for captured streams during replay. It scales the thinking time elapsed between two successive user calls from the same captured stream. The input is interpreted as a percentage. The default value of 100 means 100 percent. **See Also:** DBMS_WORKLOAD_REPLAY.PREPARE_REPLAY() in *Oracle Database PL/SQL Packages and Types Reference* for a detailed explanation of this replay parameter
THINK_TIME_AUTO_CORRECT	VARCHAR2(5)		Indicates whether the think time should be automatically corrected between calls (TRUE) or not (FALSE) A value of TRUE reduces think time if replay goes slower than capture. A value of FALSE results in no action. **See Also:** DBMS_WORKLOAD_REPLAY.PREPARE_REPLAY() in *Oracle Database PL/SQL Packages and Types Reference* for a detailed explanation of this replay parameter
SCALE_UP_MULTIPLIER	NUMBER	NOT NULL	Before the multiple-capture replay, SCALE_UP_MULTIPLIER is used to scale up the query part of a workload capture. The queries from each captured session are replayed concurrently as many times as the value of SCALE_UP_MULTIPLIER.
USER_CALLS	NUMBER		Total number of user calls replayed

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DBTIME	NUMBER		Accumulated database time (in microseconds) for the replay
NETWORK_TIME	NUMBER		Accumulated network time for the replay (in microseconds)
THINK_TIME	NUMBER		Accumulated think time (in microseconds) for the replay
PAUSE_TIME	NUMBER		The total time (in seconds) that the replay has been paused (by calling the PAUSE_REPLAY procedure)
PLSQL_CALLS	NUMBER		Total number of replayed top-level PL/SQL calls
PLSQL_SUBCALLS	NUMBER		Total number of replayed calls for SQL executed from PL/SQL
PLSQL_DBTIME	NUMBER		Total amount of database time (in microseconds) from PL/SQL calls
ELAPSED_TIME_DIFF	NUMBER		Reserved for future use
REPLAY_DEADLOCKS	NUMBER		A workload replay uses either the timing information from the capture files or the commit-based synchronization. With commit-based synchronization, the capture-time commit order is preserved during replay, and sessions normally wait on the session that is to do the next commit; such waits are classified as “WCR: replay clock” waits. A replay deadlock occurs if the session that is to do the next commit is itself blocked by a session that is waiting on “WCR: replay clock.” Replay deadlocks are resolved by allowing the blocker to go ahead and commit out of order. Replay deadlocks are not detected as database deadlocks since “WCR: replay clock” is an idle wait, introduced specifically for DB Replay. **See Also**: "WCR: replay clock"
AWR_DBID	NUMBER		Database ID of the AWR snapshots that correspond to this workload replay. For replays that were performed in the current database, this value is equal to the current database's DBID. For replays that were performed in other databases, this value will either be NULL or will be populated by `DBMS_WORKLOAD_REPLAY.IMPORT_AWR()`. **See Also:** DBMS_WORKLOAD_REPLAY.IMPORT_AWR() in *Oracle Database PL/SQL Packages and Types Reference*
AWR_BEGIN_SNAP	NUMBER		Begin snapshot ID of the AWR snapshots that correspond to this workload replay
AWR_END_SNAP	NUMBER		End snapshot ID of the AWR snapshots that correspond to this workload replay

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
AWR_EXPORTED	VARCHAR2(12)		Indicates whether the AWR snapshots that correspond to this workload replay have been exported using DBMS_WORKLOAD_REPLAY.EXPORT_AWR( ) (YES) or not (NO), or whether AWR snapshots cannot be exported because the replay is still in progress, has run to completion successfully, or was done in a different database from which it was not exported (NOT POSSIBLE) **See Also:** DBMS_WORKLOAD_REPLAY.EXPORT_AWR() in *Oracle Database PL/SQL Packages and Types Reference*
ERROR_CODE	NUMBER		Error code for this workload replay
ERROR_MESSAGE	VARCHAR2(512)		Error message for this workload replay
DIR_PATH	VARCHAR2(4000)	NOT NULL	Full directory path for the replay directory object
REPLAY_DIR_NUMBER	NUMBER		A hash value computed based on the values of other columns in this view, such as the NAME, DBID, DBNAME, PREPARE_TIME, START_TIME, and END_TIME columns. It should be fairly unique for any replay. The value is used to create a subdirectory under the replay directory.
SQLSET_OWNER	VARCHAR2(128)		User name of the SQL tuning set owner
SQLSET_NAME	VARCHAR2(128)		Name of the SQL tuning set for this workload replay
SCHEDULE_NAME	VARCHAR2(128)		The name of a schedule to be replayed. It defines one or multiple workload captures, and the order to start their replays. If SCHEDULE_NAME is NULL, the replay is a regular replay introduced since Oracle Database 11*g*, done with existing APIs from DBMS_WORKLOAD_REPLAY: INITIALIZE_REPLAY, PREPARE_REPLAY, and START_REPLAY. If SCHEDULE_NAME is not NULL, the replay is a new consolidated replay introduced in Oracle Database 12*c*. That is, it is a replay of one or more workload captures done with new APIs at DBMS_WORKLOAD_REPLAY: INITIALIZE_CONSOLIDATED_REPLAY, PREPARE_CONSOLIDATED_REPLAY, and START_CONSOLIDATED_REPLAY. **See Also:** DBMS_WORKLOAD_REPLAY in *Oracle Database PL/SQL Packages and Types Reference*
DIVERGENCE_LOAD_STATUS	VARCHAR2(5)		Indicates whether replay divergence data have been loaded (TRUE) or not (FALSE)
PLSQL_MODE	VARCHAR2(12)		Replay options for PL/SQL calls. Possible values: • TOP_LEVEL: Top-level PL/SQL only • EXTENDED: SQL executed from PL/SQL or top-level PL/SQL if there is no SQL recorded inside the PL/SQL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CONNECT_TIME_AUTO_CORRECT	VARCHAR2(12)		Indicates whether the waiting time for a new session to be connected is automatically reduced when the replay proceeds faster than its capture. The reduced amount is determined by the elapsed-time difference between the replay and the capture of the slowest session. The default value is true. There is no impact when the replay proceeds slower than the capture.
RAC_MODE	VARCHAR2(19)		Replay options in an Oracle RAC environment: • GLOBAL_SYNC: Synchronization is across all instances. This is the default. Database connections from workload replay client (wrc) are done based on connection remapping. • PER_INSTANCE_CLIENT: Synchronization is across all instances. All database connections from one wrc are connected to only one instance. • PER_INSTANCE_SYNC: Synchronization is within one instance. All database connections from one wrc are connected to only one instance.
QUERY_ONLY	VARCHAR2(1)		Indicates whether only the query-only workload from the current workload capture will be replayed, skipping all the DML/DDL that might update the database (Y) or not (N)

See Also

Oracle Database PL/SQL Packages and Types Reference for more information about the DBMS_WORKLOAD_REPLAY package

6.127 DBA_WORKLOAD_SCHEDULE_CAPTURES

DBA_WORKLOAD_SCHEDULE_CAPTURES displays the workload captures used by replay schedules.

Each row in the view contains information about one workload capture.

Column	Datatype	NULL	Description
SCHEDULE_NAME	VARCHAR2(128)	NOT NULL	The name of a schedule to be replayed
SCHEDULE_CAP_ID	NUMBER	NOT NULL	Identifies a workload capture added to a replay schedule. It starts with 1. If the same capture is added multiple times to a schedule, there will be multiple rows with different SCHEDULE_CAP_ID columns and identical CAPTURE_ID columns.
CAPTURE_ID	NUMBER	NOT NULL	Points to the capture ID from DBA_WORKLOAD_CAPTURES.
CAPTURE_DIR	VARCHAR2(128)	NOT NULL	Name of the directory object for workload capture
OS_SUBDIR	VARCHAR2(4000)	NOT NULL	Name of the subdirectory under the replay directory for this workload capture

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
MAX_CONCURRENT_SESSIONS	NUMBER		The maximal number of concurrent sessions that was seen in this workload capture
NUM_CLIENTS_ASSIGNED	NUMBER		Number of clients assigned to this workload capture before replay starts
NUM_CLIENTS	NUMBER		Number of clients that are running for this workload capture during replay
NUM_CLIENTS_DONE	NUMBER		Number of clients that have finished the replay of this workload capture
STOP_REPLAY	VARCHAR2(1)	NOT NULL	Indicates whether the whole replay will stop once the replay of this workload capture is done (Y) or not (N)
TAKE_BEGIN_SNAPSHOT	VARCHAR2(1)	NOT NULL	Indicates whether an AWR snapshot will be taken when the replay of this capture starts (Y) or not (N)
TAKE_END_SNAPSHOT	VARCHAR2(1)	NOT NULL	Indicates whether an AWR snapshot will be taken when the replay of this capture finishes (Y) or not (N)
QUERY_ONLY	VARCHAR2(1)	NOT NULL	Indicates whether only the query-only workload from the current workload capture will be replayed, skipping all the DML/DDL that might update the database (Y) or not (N)
START_DELAY_SECS	NUMBER		Displays the wait time (in seconds) when the replay of a workload capture is ready to start. "Ready to start" means the capture does not wait for any other capture, or all the captures for which it should wait have already been replayed. The default value is 0.
START_TIME	DATE		Start time for the replay of this capture
END_TIME	DATE		Finish time for the replay of this capture
AWR_DBID	NUMBER		AWR database ID of the replay
AWR_BEGIN_SNAP	NUMBER		AWR snapshot ID when the replay starts
AWR_END_SNAP	NUMBER		AWR snapshot ID when the replay finishes

6.128 DBA_WORKLOAD_SCHEDULE_ORDERING

DBA_WORKLOAD_SCHEDULE_ORDERING displays the start ordering between workload captures in the replay schedule.

Each row in the view defines one start ordering between two workload captures in the same replay schedule.

Column	Datatype	NULL	Description
SCHEDULE_NAME	VARCHAR2(128)	NOT NULL	Name of a schedule to be replayed
SCHEDULE_CAP_ID	NUMBER	NOT NULL	Identifies the workload capture that will wait

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
WAITFOR_CAP_ID	NUMBER	NOT NULL	Identifies the workload capture for which the workload capture identified by SCHEDULE_CAP_ID needs to wait. The replay of capture SCHEDULE_CAP_ID will not start until capture WAITFOR_CAP_ID finishes its replay. If the view has multiple rows with the same SCHEDULE_CAP_ID but different WAITFOR_CAP_ID, it defines a schedule so that the replay of a capture specified by SCHEDULE_CAP_ID will not start unless all the replays of the waited captures run into completion. If the view has multiple rows with the same WAITFOR_CAP_ID but different SCHEDULE_CAP_ID, it defines a schedule so that the replay of multiple captures will not start unless the replay of the capture specified by WAITFOR_CAP_ID finishes.

6.129 DBA_WORKLOAD_SQL_MAP

DBA_WORKLOAD_SQL_MAP contains the mapping information for skipping or replacing a SQL statement based on its sql_id during workload replay.

Column	Datatype	NULL	Description
REPLAY_ID	NUMBER(38)	NOT NULL	A foreign key to the ID column in the DBA_WORKLOAD_REPLAYs view
SCHEDULE_CAP_ID	NUMBER(38)	NOT NULL	The ID of a capture in the schedule
SQL_ID	VARCHAR2(13)	NOT NULL	SQL identifier of the SQL statement at the time of capture
OPERATION	VARCHAR2(7)		SKIP or REPLACE
SQL_ID_NUMBER	NUMBER		Internal representation of SQL_ID
REPLACEMENT_SQL_TEXT	VARCHAR2(4000)		When the value in the OPERATION column is SKIP, this column is NULL. When the value in the OPERATION column is REPLACE, this column shows the SQL statement to be used.

6.130 DBA_WORKLOAD_TRACKED_COMMITS

DBA_WORKLOAD_TRACKED_COMMITS displays the commits tracked every second during a database replay.

Column	Datatype	NULL	Description
REPLAY_DIR_NUMBER	NUMBER	NOT NULL	The numerical value that is associated with the subdirectory under the replay directory. See REPLAY_DIR_NUMBER in DBA_WORKLOAD_REPLAYS
INSTANCE_NUMBER	NUMBER(38)		The instance where the commit is executed
FILE_ID	NUMBER(38)	NOT NULL	The file ID
CALL_CTR	NUMBER(38)	NOT NULL	The call counter of the commit
COMMIT_SCN	NUMBER(38)		The recorded commit SCN value

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PREV_GLOBAL_COMMIT_FILE_ID	NUMBER(38)		The file ID of the latest commit across all sessions
PREV_GLOBAL_COMMIT_SCN	NUMBER(38)		The recorded SCN of the latest commit across all sessions
PREV_LOCAL_COMMIT_CALL_CTR	NUMBER(38)		The call counter of the latest commit in the same session
CAPTURE_COMMIT_TIME	NUMBER(38)		The time in seconds since the capture started
CAPTURE_COMMIT_TIME_DELTA	NUMBER(38)		The elapsed time in seconds since the previous commit across all sessions during capture
REPLAY_COMMIT_TIME	NUMBER(38)		The time in seconds since the replay started
REPLAY_COMMIT_TIME_DELTA	NUMBER(38)		The elapsed time in seconds since the previous commit across all sessions during the replay

See Also

DBA_WORKLOAD_REPLAYS

6.131 DBA_WORKLOAD_USER_MAP

DBA_WORKLOAD_USER_MAP contains all the mappings ever done until they are removed at some point.

The mappings are stored in a table made public through this view.

To remove old mappings, execute this statement:

```
SQL> delete * from DBA_WORKLOAD_USER_MAP;
```

Column	Datatype	NULL	Description
REPLAY_ID	NUMBER		This is a foreign key to the ID column in the DBA_WORKLOAD_REPLAYS view
SCHEDULE_CAP_ID	NUMBER		The ID of a capture in the schedule
CAPTURE_USER	VARCHAR2(4000)	NOT NULL	The user name during the time of the workload capture
REPLAY_USER	VARCHAR2(4000)		The user name to which the captured user should be remapped during replay. If the REPLAY_USER is null, the CAPTURE_USER is used during replay. In other words, the original user is used.

See Also

"DBA_WORKLOAD_REPLAYS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.132 DBA_XML_INDEXES

DBA_XML_INDEXES describes all XML indexes in the database. Its columns are the same as those in ALL_XML_INDEXES.

See Also

"ALL_XML_INDEXES"

6.133 DBA_XML_NESTED_TABLES

DBA_XML_NESTED_TABLES describes all the tables and their corresponding nested tables. Its columns are the same as those in ALL_XML_NESTED_TABLES.

See Also

"ALL_XML_NESTED_TABLES"

6.134 DBA_XML_OUT_OF_LINE_TABLES

DBA_XML_OUT_OF_LINE_TABLES descibes all the out of line tables connected to a given root table for the same schema. Its columns are the same as those in ALL_XML_OUT_OF_LINE_TABLES.

See Also

"ALL_XML_OUT_OF_LINE_TABLES"

6.135 DBA_XML_SCHEMA_ATTRIBUTES

DBA_XML_SCHEMA_ATTRIBUTES describes all the attributes and their properties. Its columns are the same as those in ALL_XML_SCHEMA_ATTRIBUTES.

See Also

"ALL_XML_SCHEMA_ATTRIBUTES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.136 DBA_XML_SCHEMA_COMPLEX_TYPES

DBA_XML_SCHEMA_COMPLEX_TYPES describes all complex types in the database. Its columns are the same as those in ALL_XML_SCHEMA_COMPLEX_TYPES.

See Also

"ALL_XML_SCHEMA_COMPLEX_TYPES"

6.137 DBA_XML_SCHEMA_ELEMENTS

DBA_XML_SCHEMA_ELEMENTS describes all the elements and their properties. Its columns are the same as those in ALL_XML_SCHEMA_ELEMENTS.

See Also

"ALL_XML_SCHEMA_ELEMENTS"

6.138 DBA_XML_SCHEMA_NAMESPACES

DBA_XML_SCHEMA_NAMESPACES describes all the available namespaces. Its columns are the same as those in ALL_XML_SCHEMA_NAMESPACES.

See Also

"ALL_XML_SCHEMA_NAMESPACES"

6.139 DBA_XML_SCHEMA_SIMPLE_TYPES

DBA_XML_SCHEMA_SIMPLE_TYPES describes all simple types. Its columns are the same as those in ALL_XML_SCHEMA_SIMPLE_TYPES.

See Also

"ALL_XML_SCHEMA_SIMPLE_TYPES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.140 DBA_XML_SCHEMA_SUBSTGRP_HEAD

DBA_XML_SCHEMA_SUBSTGRP_HEAD describes the heads of substitution groups. Its columns are the same as those in ALL_XML_SCHEMA_SUBSTGRP_HEAD.

See Also

"ALL_XML_SCHEMA_SUBSTGRP_HEAD"

6.141 DBA_XML_SCHEMA_SUBSTGRP_MBRS

DBA_XML_SCHEMA_SUBSTGRP_MBRS describes all members of substitution groups. Its columns are the same as those in ALL_XML_SCHEMA_SUBSTGRP_MBRS.

See Also

"ALL_XML_SCHEMA_SUBSTGRP_MBRS"

6.142 DBA_XML_SCHEMAS

DBA_XML_SCHEMAS describes all registered XML schemas in the database. Its columns are the same as those in ALL_XML_SCHEMAS.

See Also

"ALL_XML_SCHEMAS"

6.143 DBA_XML_TAB_COLS

DBA_XML_TAB_COLS describes the columns of all XML tables in the database. Its columns are the same as those in ALL_XML_TAB_COLS.

See Also

"ALL_XML_TAB_COLS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.144 DBA_XML_TABLES

DBA_XML_TABLES describes all XML tables in the database. Its columns are the same as those in ALL_XML_TABLES.

See Also

"ALL_XML_TABLES"

6.145 DBA_XML_VIEW_COLS

DBA_XML_VIEW_COLS describes the columns of all XML views in the database. Its columns are the same as those in ALL_XML_VIEW_COLS.

See Also

"ALL_XML_VIEW_COLS"

6.146 DBA_XML_VIEWS

DBA_XML_VIEWS describes all XML views in the database. Its columns are the same as those in ALL_XML_VIEWS.

See Also

"ALL_XML_VIEWS"

6.147 DBA_XS_AUDIT_POLICY_OPTIONS

DBA_XS_AUDIT_POLICY_OPTIONS describes auditing options defined under all audit policies specific to Oracle Database Real Application Security.

Column	Datatype	NULL	Description
POLICY_NAME	VARCHAR2(128)		Name of the audit policy
AUDIT_CONDITION	VARCHAR2(4000)		Condition associated with the audit policy
AUDIT_OPTION	VARCHAR2(128)		Auditing option defined in the audit policy
CONDITION_EVAL_OPT	VARCHAR2(9)		Evaluation option associated with the audit policy's condition. The possible values are STATEMENT, SESSION, INSTANCE, NONE.
COMMON	VARCHAR2(3)		Indicates whether the audit policy is a common audit policy (YES) or local (NO). The value is NULL in non-CDBs.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

Oracle Database Security Guide for more information about auditing

6.148 DBA_XS_AUDIT_TRAIL

DBA_XS_AUDIT_TRAIL describes all audit records specific to Oracle Database Real Application Security.

Column	Datatype	NULL	Description
USERID	VARCHAR2(128)		Name of the database user whose actions were audited
ACTION	NUMBER		Numeric audit trail action type code. The corresponding name of the action type is in the ACTION_NAME column.
ACTION_NAME	VARCHAR2(64)		Name of the action type corresponding to the numeric code in the ACTION column
OBJ_OWNER	VARCHAR2(128)		Owner of the object affected by the action
OBJ_NAME	VARCHAR2(128)		Name of the object affected by the action
RETURN_CODE	NUMBER		Oracle error code generated by the action
XS_USER_NAME	VARCHAR2(128)		Name of the Real Application Security user
XS_SESSIONID	RAW(33)		Identifer of the Real Application Security session
XS_INACTIVITY_TIMEOUT	NUMBER		Inactivity timeout of the Real Application Security session
XS_ENTITY_TYPE	VARCHAR2(32)		Type of the Real Application Security entity. Possible values are USER, ROLE, ROLESET, SECURITYCLASS, ACL, DATASECURITY, and NSTEMPLATE.
XS_TARGET_PRINCIPAL_NAME	VARCHAR2(128)		Target principal name in Real Application Security operations. Possible operations are set verifier, set password, add proxy, remove proxy, switch user, assign user, create session, grant roles.
XS_PROXY_USER_NAME	VARCHAR2(128)		Name of the Real Application Security proxy user.
XS_DATASEC_POLICY_NAME	VARCHAR2(128)		Name of the Real Application Security data security policy enabled or disabled
XS_SCHEMA_NAME	VARCHAR2(128)		Name of the schema in enable, disable data security policy and global callback operation
XS_CALLBACK_EVENT_TYPE	VARCHAR2(32)		Real Application Security global callback event type
XS_PACKAGE_NAME	VARCHAR2(128)		Real Application Security callback package name for the global callback
XS_PROCEDURE_NAME	VARCHAR2(128)		Real Application Security callback procedure name for the global callback
XS_ENABLED_ROLE	VARCHAR2(128)		The role that is enabled
XS_COOKIE	VARCHAR2(1024)		Real Application Security session cookie
XS_NS_NAME	VARCHAR2(128)		Name of the Real Application Security session namespace
XS_NS_ATTRIBUTE	VARCHAR2(4000)		Name of the Real Application Security session namespace attribute

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
XS_NS_ATTRIBUTE_OLD_VAL	VARCHAR2(4000)		The old value of the Real Application Security session namespace attribute
XS_NS_ATTRIBUTE_NEW_VAL	VARCHAR2(4000)		The new value of the Real Application Security session namespace attribute
EVENT_TIMESTAMP	TIMESTAMP(6) WITH TIME ZONE		Timestamp of audit record

See Also

Oracle Database Security Guide for more information about auditing

6.149 DBA_XS_ENABLED_AUDIT_POLICIES

DBA_XS_ENABLED_AUDIT_POLICIES describes all the audit policies specific to Oracle Database Real Application Security that are enabled to users.

Note

This view was known as DBA_XS_ENB_AUDIT_POLICIES in Oracle Database 12*c* Release 1. It was renamed to DBA_XS_ENABLED_AUDIT_POLICIES in Oracle Database 12*c* Release 2 (12.2.0.1).

Column	Datatype	NULL	Description
POLICY_NAME	VARCHAR2(128)		Name of the audit policy
ENABLED_OPTION	VARCHAR2(15)		Enabled option of the audit policy. Possible values: • BY USER: For policies that are enabled on users • EXCEPT USERS: For policies that are enabled on users • BY GRANTED ROLE: For policies that are enabled on roles • INVALID: For policies that are enabled on roles
ENTITY_NAME	VARCHAR2(128)		Database entity (user name or role name) on which the audit policy is enabled
ENTITY_TYPE	VARCHAR2(7)		Database entity type. Possible values: • USER: Indicates that the policy is enabled on a user or users. • ROLE: Indicates that the policy is enabled on a role or roles.
SUCCESS	VARCHAR2(3)		Indicates whether the audit policy is enabled for auditing successful events (YES) or not (NO)
FAILURE	VARCHAR2(3)		Indicates whether the audit policy is enabled for auditing unsuccessful events (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Security Guide* for more information about auditing

6.150 DBA_XS_ENB_AUDIT_POLICIES

`DBA_XS_ENB_AUDIT_POLICIES` is a synonym for the `DBA_XS_ENABLED_AUDIT_POLICIES` view.

> **See Also**
>
> - "DBA_XS_ENABLED_AUDIT_POLICIES"
> - *Oracle Database Security Guide* for more information about auditing
> - *Oracle Database Security Guide* for more information about auditing

6.151 DBA_XSTREAM_ADMINISTRATOR

`DBA_XSTREAM_ADMINISTRATOR` displays information about the users who have been granted privileges to be XStream administrators by procedures in the `DBMS_XSTREAM_AUTH` package. Its columns are the same as those in `ALL_XSTREAM_ADMINISTRATOR`.

> **See Also**
>
> "ALL_XSTREAM_ADMINISTRATOR"

6.152 DBA_XSTREAM_INBOUND

`DBA_XSTREAM_INBOUND` displays information about all XStream inbound servers in the database. Its columns are the same as those in `ALL_XSTREAM_INBOUND`.

> **See Also**
>
> "ALL_XSTREAM_INBOUND"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.153 DBA_XSTREAM_INBOUND_PROGRESS

DBA_XSTREAM_INBOUND_PROGRESS displays information about the progress made by all XStream inbound servers in the database. Its columns are the same as those in ALL_XSTREAM_INBOUND_PROGRESS.

See Also

"ALL_XSTREAM_INBOUND_PROGRESS"

6.154 DBA_XSTREAM_OUT_SUPPORT_MODE

DBA_XSTREAM_OUT_SUPPORT_MODE displays information about the level of XStream capture process support for the tables in the database. Its columns are the same as those in ALL_XSTREAM_OUT_SUPPORT_MODE.

See Also

"ALL_XSTREAM_OUT_SUPPORT_MODE"

6.155 DBA_XSTREAM_OUTBOUND

DBA_XSTREAM_OUTBOUND displays information about all XStream outbound servers in the database. Its columns are the same as those in ALL_XSTREAM_OUTBOUND.

See Also

"ALL_XSTREAM_OUTBOUND"

6.156 DBA_XSTREAM_OUTBOUND_PROGRESS

DBA_XSTREAM_OUTBOUND_PROGRESS displays information about the progress made by all XStream outbound servers in the database. Its columns are the same as those in ALL_XSTREAM_OUTBOUND_PROGRESS.

See Also

"ALL_XSTREAM_OUTBOUND_PROGRESS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.157 DBA_XSTREAM_RULES

DBA_XSTREAM_RULES displays information about all XStream rules in the database. Its columns are the same as those in ALL_XSTREAM_RULES.

> **See Also**
>
> "ALL_XSTREAM_RULES"

6.158 DBA_XSTREAM_SPLIT_MERGE

DBA_XSTREAM_SPLIT_MERGE displays information about XStream current automatic split and merge operations.

Column	Datatype	NULL	Description
ORIGINAL_CAPTURE_NAME	VARCHAR2(128)	NOT NULL	Name of the original capture process
CLONED_CAPTURE_NAME	VARCHAR2(128)		Name of the cloned capture process
ORIGINAL_CAPTURE_STATUS	VARCHAR2(8)		Status of the original capture process: • DISABLED • ENABLED • ABORTED
CLONED_CAPTURE_STATUS	VARCHAR2(8)		Status of the cloned capture process: • DISABLED • ENABLED • ABORTED
ORIGINAL_XSTREAM_NAME	VARCHAR2(128)		Name of the original XStream component that receives database changes directly from the original capture process. The component is either a progagation or a local apply process.
CLONED_XSTREAM_NAME	VARCHAR2(128)		Name of the cloned XStream component that receives database changes directly from the cloned capture process. The component is either a progagation or a local apply process.
XSTREAM_TYPE	VARCHAR2(11)		Type of the component in ORIGINAL_XSTREAM_NAME and CLONED_XSTREAM_NAME: • PROPAGATION • APPLY
RECOVERABLE_SCRIPT_ID	RAW(16)		Unique ID of the script to split or merge operation
SCRIPT_STATUS	VARCHAR2(12)		Status of the recoverable script: • GENERATING • NOT EXECUTING • EXECUTING • EXECUTED • ERROR

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ACTION_TYPE	VARCHAR2(7)		type of action performed by the script: • SPLIT • MERGE • MONITOR
ACTION_THRESHOLD	VARCHAR2(40)		For SPLIT actions, the threshold set by the split_threshold capture process parameter. For MERGE actions, the threshold set by the merge_threshold capture process parameter.
STATUS	VARCHAR2(16)		Status of the action: • NOTHING TO SPLIT - Not ready to split or does not need to split • ABOUT TO SPLIT • SPLITTING - A split is in progress • SPLIT DONE - A split is done • NOTHING TO MERGE - Not ready to merge • ABOUT TO MERGE • MERGING - A merge is in progress • MERGE DONE - A merge is done • ERROR - An error was returned during a split or merge • NONSPLITTABLE - The original capture is not splittable either because it is disabled, it has more than one publisher to its queue, or it has only one destination for captured messages
STATUS_UPDATE_TIME	TIMESTAMP(6)		Time when status was last updated
CREATION_TIME	TIMESTAMP(6)		Time when the action started
LAG	NUMBER		Time (in seconds) that the cloned capture process lags behind the original capture process
JOB_OWNER	VARCHAR2(128)		Owner of the job that performs the split or merge operation
JOB_NAME	VARCHAR2(128)		Name of the job that performs the split or merge operation
JOB_STATE	VARCHAR2(15)		Current state of the job: • DISABLED • RETRY SCHEDULED • SCHEDULED • RUNNING • COMPLETED • BROKEN • FAILED • REMOTE • SUCCEEDED • CHAIN_STALLED
JOB_NEXT_RUN_DATE	VARCHAR2(64)		Next time the job will run
ERROR_NUMBER	NUMBER		Error number if the capture process was terminated
ERROR_MESSAGE	VARCHAR2(4000)		Error message if the capture process was terminated

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.159 DBA_XSTREAM_SPLIT_MERGE_HIST

DBA_XSTREAM_SPLIT_MERGE_HIST displays information about past XStream automatic split and merge operations.

Column	Datatype	NULL	Description
ORIGINAL_CAPTURE_NAME	VARCHAR2(128)	NOT NULL	Name of the original capture process
CLONED_CAPTURE_NAME	VARCHAR2(128)		Name of the cloned capture process
ORIGINAL_QUEUE_OWNER	VARCHAR2(128)		Owner of the queue used by the original capture process
ORIGINAL_QUEUE_NAME	VARCHAR2(128)		Name of the queue used by the original capture process
CLONED_QUEUE_OWNER	VARCHAR2(128)		Owner of the queue used by the cloned capture process
CLONED_QUEUE_NAME	VARCHAR2(128)		Name of the queue used by the cloned capture process
ORIGINAL_CAPTURE_STATUS	VARCHAR2(8)		Status of the original capture process: • DISABLED • ENABLED • ABORTED
CLONED_CAPTURE_STATUS	VARCHAR2(8)		Status of the cloned capture process: • DISABLED • ENABLED • ABORTED
ORIGINAL_XSTREAM_NAME	VARCHAR2(128)		Name of the original XStream component that receives database changes directly from the original capture process. The component is either a progagation or a local apply process.
CLONED_XSTREAM_NAME	VARCHAR2(128)		Name of the cloned XStream component that receives database changes directly from the cloned capture process. The component is either a progagation or a local apply process.
XSTREAM_TYPE	VARCHAR2(11)		Type of the component in ORIGINAL_XSTREAM_NAME and CLONED_XSTREAM_NAME: • PROPAGATION • APPLY
RECOVERABLE_SCRIPT_ID	RAW(16)		Unique ID of the script to split or merge operation
SCRIPT_STATUS	VARCHAR2(12)		Status of the recoverable script: • GENERATING • NOT EXECUTING • EXECUTING • EXECUTED • ERROR
ACTION_TYPE	VARCHAR2(7)		type of action performed by the script: • SPLIT • MERGE • MONITOR

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ACTION_THRESHOLD	VARCHAR2(40)		For SPLIT actions, the threshold set by the split_threshold capture process parameter. For MERGE actions, the threshold set by the merge_threshold capture process parameter.
STATUS	VARCHAR2(16)		Status of the action: • NOTHING TO SPLIT - Not ready to split or does not need to split • ABOUT TO SPLIT • SPLITTING - A split is in progress • SPLIT DONE - A split is done • NOTHING TO MERGE - Not ready to merge • ABOUT TO MERGE • MERGING - A merge is in progress • MERGE DONE - A merge is done • ERROR - An error was returned during a split or merge • NONSPLITTABLE - The original capture is not splittable either because it is disabled, it has more than one publisher to its queue, or it has only one destination for captured messages
STATUS_UPDATE_TIME	TIMESTAMP(6)		Time when status was last updated
CREATION_TIME	TIMESTAMP(6)		Time when the action started
LAG	NUMBER		Time (in seconds) that the cloned capture process lags behind the original capture process
JOB_OWNER	VARCHAR2(128)		Owner of the job that performs the split or merge operation
JOB_NAME	VARCHAR2(128)		Name of the job that performs the split or merge operation
ERROR_NUMBER	NUMBER		Error number if the capture process was terminated
ERROR_MESSAGE	VARCHAR2(4000)		Error message if the capture process was terminated

6.160 DBA_XSTREAM_STMT_HANDLERS

DBA_XSTREAM_STMT_HANDLERS displays information about all XStream statement DML handlers in the database.

Column	Datatype	NULL	Description
HANDLER_NAME	VARCHAR2(128)	NOT NULL	Name of the statement handler
HANDLER_COMMENT	VARCHAR2(4000)		Comment of the statement handler
CREATION_TIME	TIMESTAMP(6)		Timestamp for script creation
MODIFICATION_TIME	TIMESTAMP(6)		Timestamp for script modification

6.161 DBA_XSTREAM_STMTS

DBA_XSTREAM_STMTS displays information about the statements in all XStream statement DML handlers in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
HANDLER_NAME	VARCHAR2(128)	NOT NULL	Name of the statement handler
EXECUTION_SEQUENCE	NUMBER	NOT NULL	Execution sequence of the statement
STATEMENT	CLOB		Text of the SQL statement
CREATION_TIME	TIMESTAMP(6)		Timestamp for statement creation
MODIFICATION_TIME	TIMESTAMP(6)		Timestamp for statement modification

6.162 DBA_XSTREAM_TRANSFORMATIONS

DBA_XSTREAM_TRANSFORMATIONS displays information about all XStream transformations available on a system, in order of execution. Its columns are the same as those in ALL_XSTREAM_TRANSFORMATIONS.

See Also

"ALL_XSTREAM_TRANSFORMATIONS"

6.163 DBA_XTERNAL_LOC_PARTITIONS

DBA_XTERNAL_LOC_PARTITIONS describes partition-level locations in the database. Its columns are the same as those in ALL_XTERNAL_LOC_PARTITIONS.

See Also

"ALL_XTERNAL_LOC_PARTITIONS"

6.164 DBA_XTERNAL_LOC_SUBPARTITIONS

DBA_XTERNAL_LOC_SUBPARTITIONS describes subpartition-level locations in the database. Its columns are the same as those in ALL_XTERNAL_LOC_SUBPARTITIONS.

See Also

"ALL_XTERNAL_LOC_SUBPARTITIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.165 DBA_XTERNAL_PART_TABLES

DBA_XTERNAL_PART_TABLES describes object-level information for partitioned external tables in the database. Its columns are the same as those in ALL_XTERNAL_PART_TABLES.

See Also

"ALL_XTERNAL_PART_TABLES"

6.166 DBA_XTERNAL_TAB_PARTITIONS

DBA_XTERNAL_TAB_PARTITIONS describes partition-level information for partitioned external tables in the database. Its columns are the same as those in ALL_XTERNAL_TAB_PARTITIONS.

See Also

"ALL_XTERNAL_TAB_PARTITIONS"

6.167 DBA_XTERNAL_TAB_SUBPARTITIONS

DBA_XTERNAL_TAB_SUBPARTITIONS describes subpartition-level information for partitioned external tables in the database. Its columns are the same as those in ALL_XTERNAL_TAB_SUBPARTITIONS.

See Also

"ALL_XTERNAL_TAB_SUBPARTITIONS"

6.168 DBA_ZONEMAP_MEASURES

DBA_ZONEMAP_MEASURES describes the measures for all the zone maps in the database. Its columns are the same as those in ALL_ZONEMAP_MEASURES.

Note

This view is intended for use with Oracle Exadata release 12.1.2.1.1 or later.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> - "ALL_ZONEMAP_MEASURES"
> - *Oracle Database Data Warehousing Guide* for more information about zone maps

6.169 DBA_ZONEMAPS

DBA_ZONEMAPS describes all the zone maps in the database. Its columns are the same as those in ALL_ZONEMAPS.

> **Note**
>
> This view is intended for use with Oracle Exadata release 12.1.2.1.1 or later.

> **See Also**
>
> - "ALL_ZONEMAPS"
> - *Oracle Database Data Warehousing Guide* for more information about zone maps

6.170 DBFS_CONTENT

DBFS_CONTENT displays all the path items from all available content stores in the system.

Column	Datatype	NULL	Description
STORE	VARCHAR2(256)		Name of store
MOUNT	VARCHAR2(256)		Location at which instance of store is mounted
PATHNAME	VARCHAR2(1024)		Name of path to item
PATHTYPE	VARCHAR2(32)		Type of path item (see DBMS_DBFS_CONTENT Constants - Path Name Types)
FILEDATA	BLOB		BLOB locator that can be used to access data in the path item
STD_ACCESS_TIME	TIMESTAMP(6)		Time of last access of a path name's contents
STD_ACL	VARCHAR2(1024)		Access Control List (in standard ACL syntax)
STD_CHANGE_TIME	TIMESTAMP(6)		Time of last change to the path name
STD_CHILDREN	NUMBER		Number of child directories/folders a directory/folder path has (this property should be available in providers that support the FEATURE_FOLDERS feature)
STD_CONTENT_TYPE	VARCHAR2(1024)		One or more client-supplied mime-types (in standard RFC syntax) describing the path name which is typically of TYPE_FILE. The content type is not necessarily interpreted by the store.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
STD_CREATION_TIME	TIMESTAMP(6)		Time at which the item was created. Once set, this value remains the same for the lifetime of the path name.
STD_DELETED	NUMBER		Set to a nonzero number if the path name has been soft-deleted but not yet purged (see DBMS_DBFS_CONTENT Constants - Store Features)
STD_GUID	NUMBER		Store-specific unique identifier for a path name. Clients must not depend on the GUID being unique across different stores, but a given (store-name, store-specific-path name) has a stable and unique GUID for its lifetime.
STD_MODIFICATION_TIME	TIMESTAMP(6)		Time of last change to the data associated with a path name. Changes to the content of a TYPE_FILE or TYPE_REFERENCE path, the referent of the TYPE_LINK path, and addition or deletion of immediate children in a TYPE_DIRECTORY path, all constitute data changes.
STD_OWNER	VARCHAR2(32)		Client-supplied (or implicit) owner name for the path name
STD_PARENT_GUID	NUMBER		Store-specific unique identifier for the parent of a path name. Clients must not depend on the GUID being unique across different stores, but a given (store-name, store-specific-path name) has a stable and unique GUID for its lifetime. The GUID of the parent of this path name (that is that std_parent_guid(pathname) == std_guid(parent(pathname))).
STD_REFERENT	VARCHAR2(1024)		Content of the symbolic link of a TYPE_LINK path, otherwise NULL. The STD_REFERENT can be an arbitrary string and must not necessarily be interpreted as path name by clients (or such interpretation should be done with great care).
OPT_HASH_TYPE	VARCHAR2(32)		Type of hash provided in the OPT_HASH_VALUE property (see DBMS_CRYPTO for possible options)
OPT_HASH_VALUE	VARCHAR2(128)		Hash value of type OPT_HASH_TYPE describing the content of the path name
OPT_LOCK_COUNT	NUMBER		Number of compatible locks placed on a path name. If different principals are allowed to place compatible (read) locks on a path, the OPT_LOCKER must specify all lockers with repeats so that lock counts can be correctly maintained.
OPT_LOCK_DATA	VARCHAR2(128)		Client-supplied user-data associated with a user-lock, uninterpreted by the store
OPT_LOCKER	VARCHAR2(128)		One or more implicit or client-specified principals that applied a user-lock on a path name
OPT_LOCK_STATUS	NUMBER		One of the (LOCK_READ_ONLY, LOCK_WRITE_ONLY, LOCK_READ_WRITE) values describing the type of lock currently applied on a path name
OPT_VERSION	NUMBER		Sequence number for linear versioning of a path name
OPT_VERSION_PATH	VARCHAR2(1024)		Version path name for hierarchical versioning of a path name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OPT_CONTENT_ID	RAW(128)		A provider-generated store-specific unique contentID in the form of a string for a file element (that may optionally not be associated with a path (see FEATURE_CONTENT_ID and FEATURE_LAZY_PATH in DBMS_DBFS_CONTENT Constants - Store Features)

See Also

- *Oracle Database PL/SQL Packages and Types Reference* for more information about DBMS_DBFS_CONTENT Constants - Path Name Types and DBMS_DBFS_CONTENT Constants - Store Features
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_CRYPTO package

6.171 DBFS_CONTENT_PROPERTIES

DBFS_CONTENT_PROPERTIES displays all the property/value pairs for all path items in all content stores in the system.

Column	Datatype	NULL	Description
STORE	VARCHAR2(256)		Name of store
MOUNT	VARCHAR2(256)		Location at which instance of store is mounted
PATHNAME	VARCHAR2(1024)		Name of path to item
PROPERTY_NAME	VARCHAR2(32)		Name of the property
PROPERTY_VALUE	VARCHAR2(1024)		Value of the property
PROPERTY_TYPE	NUMBER		PL/SQL typecode for the property value

6.172 DBMS_ALERT_INFO

DBMS_ALERT_INFO describes registered alerts.

Column	Datatype	NULL	Description
NAME	VARCHAR2(30)	NOT NULL	Name of the alert
SID	VARCHAR2(30)	NOT NULL	Session ID of a session waiting for this alert
CHANGED	VARCHAR2(1)		Boolean flag to indicate that an alert has been signaled. Y: alert signaled, N: no alert.
MESSAGE	VARCHAR2(1800)		Optional message passed by signaler

6.173 DBMS_LOCK_ALLOCATED

DBMS_LOCK_ALLOCATED describes user-allocated locks.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
NAME	VARCHAR2(128)	NOT NULL	Name of the lock
LOCKID	NUMBER(38)		Lock identifier number
EXPIRATION	DATE		Planned lock expiration date (updates whenever the allocation procedure is run)

6.174 DBMS_METADATA_PARSE_ITEMS

DBMS_METADATA_PARSE_ITEMS documents the valid parse items.

Column	Datatype	NULL	Description
OBJECT_TYPE	VARCHAR2(128)	NOT NULL	Object type name
PARSE_ITEM	VARCHAR2(124)		Parse item name
ALTER_XML	VARCHAR2(1)		Y: Can be used when generating ALTER_XML document
FETCH_XML_CLOB	VARCHAR2(1)		Y: Can be returned by FETCH_XML_CLOB
CONVERT	VARCHAR2(1)		Y: Can be returned by CONVERT or FETCH_DDL
DESCRIPTION	VARCHAR2(4000)		Description of the parse item

See Also

- "DBMS_METADATA_TRANSFORM_PARAMS"
- "DBMS_METADATA_TRANSFORMS"

6.175 DBMS_METADATA_TRANSFORM_PARAMS

DBMS_METADATA_TRANSFORM_PARAMS documents the valid transform parameters for each transform.

Column	Datatype	NULL	Description
OBJECT_TYPE	VARCHAR2(128)	NOT NULL	Object type name
TRANSFORM	VARCHAR2(128)	NOT NULL	Transform name
PARAM	VARCHAR2(128)	NOT NULL	Parameter name
DATATYPE	VARCHAR2(11)		BOOLEAN, TEXT, NUMBER or UNSPECIFIED
DEFAULT_VAL	VARCHAR2(2000)		Default value of transform parameter
DESCRIPTION	VARCHAR2(4000)		Description of the transform parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBMS_METADATA_PARSE_ITEMS"
- "DBMS_METADATA_TRANSFORMS"

6.176 DBMS_METADATA_TRANSFORMS

DBMS_METADATA_TRANSFORMS documents the valid Oracle-supplied transforms.

Column	Datatype	NULL	Description
OBJECT_TYPE	VARCHAR2(128)	NOT NULL	Object type name
TRANSFORM	VARCHAR2(128)	NOT NULL	Transform name
INPUT_TYPE	VARCHAR2(24)		Type of input document
OUTPUT_TYPE	VARCHAR2(24)		Type of output document
DESCRIPTION	VARCHAR2(71)		Description of the transform

See Also

- "DBMS_METADATA_PARSE_ITEMS"
- "DBMS_METADATA_TRANSFORM_PARAMS"

6.177 DEPTREE

This view, created by utldtree.sql, contains information on the object dependency tree.

For user SYS, this view displays shared cursors (and only shared cursors) that depend on the object. For all other users, it displays objects other than shared cursors. Other users can access SYS.DEPTREE for information on shared cursors.

Column	Datatype	NULL	Description
NESTED_LEVEL	NUMBER		Nesting level in the dependency tree
TYPE	VARCHAR2(23)		Object type
SCHEMA	VARCHAR2(128)		Object schema
NAME	VARCHAR2(1002)		Object name
SEQ#	NUMBER		Sequence number in the dependency tree. Used for ordering queries. **See Also:** "IDEPTREE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.178 DICT

`DICT` is a synonym for `DICTIONARY`.

> **See Also**
>
> "DICTIONARY"

6.179 DICT_COLUMNS

`DICT_COLUMNS` contains descriptions of columns in data dictionary tables and views.

Column	Datatype	NULL	Description
`TABLE_NAME`	`VARCHAR2(128)`		Name of the object that contains the column
`COLUMN_NAME`	`VARCHAR2(128)`		Name of the column
`COMMENTS`	`VARCHAR2(4000)`		Text comment on the column

6.180 DICTIONARY

`DICTIONARY` contains descriptions of data dictionary tables and views.

Column	Datatype	NULL	Description
`TABLE_NAME`	`VARCHAR2(128)`		Name of the object
`COMMENTS`	`VARCHAR2(4000)`		Text comment on the object

6.181 DICTIONARY_CREDENTIALS_ENCRYPT

`DICTIONARY_CREDENTIALS_ENCRYPT` indicates whether encryption of dictionary credentials is enforced or not. You can encrypt sensitive credential information, such as passwords that are stored in the data dictionary.

Column	Datatype	NULL	Description
`ENFORCEMENT`	`VARCHAR2(8)`		Enforcement status for encryption of dictionary credentials. Possible values: • `ENABLED`: Encryption of dictionary credentials is enforced • `DISABLED`: Encryption of dictionary credentials is not enforced

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Security Guide* for information about encrypting sensitive credential information in the data dictionary

6.182 DM_USER_MODELS

`DM_USER_MODELS` displays information about the models in the user's schema.

Column	Datatype	NULL	Description
`NAME`	`VARCHAR2(128)`	`NOT NULL`	Name of the model
`FUNCTION_NAME`	`VARCHAR2(30)`		Model function: • `association` - Association is a descriptive mining function. An association model identifies relationships and the probability of their occurrence within a data set. • `attribute_importance` - Attribute Importance is a predictive mining function. An attribute importance model identifies the relative importance of an attribute in predicting a given outcome. • `classification` - Classification is a predictive mining function. A classification model uses historical data to predict new discrete or categorical data. The `classification` function can also be used for anomaly detection. In this case, the SVM algorithm with a null target is used (One-Class SVM). • `clustering` - Clustering is a descriptive mining function. A clustering model identifies natural groupings within a data set. • `feature_extraction` - Feature Extraction is a descriptive mining function. A feature extraction model creates an optimized data set on which to base a model. • `regression` - Regression is a predictive mining function. A regression model uses historical data to predict new continuous, numeric data.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ALGORITHM_NAME	VARCHAR2(30)		Algorithm used by the model: • algo_name - Setting that specifies the algorithm used by the model. • asso_max_rule_length - Setting that specifies the maximum length of a rule used by an association model. • asso_min_confidence - Setting that specifies the minimum confidence for an association model. • asso_min_support - Setting that specifies the minimum support for an association model. • clas_cost_table_name - Setting that specifies the name of the cost matrix table for a classification model. • clas_priors_table_name - Setting that specifies the name of the prior probability table for NB and ABN models. Decision Tree is the only classification algorithm that does not use priors. For SVM classification models, this setting specifies the name of a table of weights. • clus_num_clusters - Setting that specifies the number of clusters for a clustering model. • feat_num_features - Setting that specifies the number of features for a feature selection model.
CREATION_DATE	DATE	NOT NULL	Date on which the model was created
BUILD_DURATION	NUMBER		Duration of the model build process
TARGET_ATTRIBUTE	VARCHAR2(128)		Attribute designated as the target of a classification model
MODEL_SIZE	NUMBER		Size of the model (in megabytes)

6.183 DOCUMENT_LINKS

DOCUMENT_LINKS provides system information about Oracle XML DB document links in Oracle XML DB Repository documents.

When an XML document that includes XLink or XInclude links is added to the repository, these links can be mapped to document links, which are tracked using view DOCUMENT_LINKS.

Column	Datatype	NULL	Description
SOURCE_ID	RAW(16)		The source resource OID
TARGET_ID	RAW(16)		The target resource OID
TARGET_PATH	VARCHAR2(4000)		This column is always NULL. It is reserved for future use.
LINK_TYPE	VARCHAR2(8)		The document link type: Hard or Weak
LINK_FORM	VARCHAR2(8)		Whether the original link was of form XLink or XInclude
SOURCE_TYPE	VARCHAR2(17)		Whether the link is contained in Resource Content or Resource Metadata

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

Oracle XML DB Developer's Guide for information about using this view

6.184 ERROR_SIZE

ERROR_SIZE is accessed to create DBA_OBJECT_SIZE and USER_OBJECT_SIZE.

See Also

- "DBA_OBJECT_SIZE"
- "USER_OBJECT_SIZE"

6.185 EXCEPTIONS

EXCEPTIONS contains information on violations of integrity constraints. This table is created by the utlexcpt.sql script.

Column	Datatype	NULL	Description
ROW_ID	ROWID		Row that caused the violation
OWNER	VARCHAR2(128)		Owner of the table
TABLE_NAME	VARCHAR2(128)		Name of the table
CONSTRAINT	VARCHAR2(128)		Integrity constraint that was violated

6.186 FLASHBACK_TRANSACTION_QUERY

FLASHBACK_TRANSACTION_QUERY displays information about all flashback transaction queries in the database.

The database must have at least minimal supplemental logging enabled to avoid unpredictable behavior.

Column	Datatype	NULL	Description
XID	RAW(8)		Transaction identifier
START_SCN	NUMBER		Transaction start system change number (SCN)
START_TIMESTAMP	DATE		Transaction start timestamp
COMMIT_SCN	NUMBER		Transaction commit system change number; NULL for active transactions
COMMIT_TIMESTAMP	DATE		Transaction commit timestamp; NULL for active transactions
LOGON_USER	VARCHAR2(128)		Logon user for the transaction
UNDO_CHANGE#	NUMBER		Undo system change number (1 or higher)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OPERATION	VARCHAR2(32)		Forward-going DML operation performed by the transaction: • D - Delete • I - Insert • U - Update • B • UNKNOWN
TABLE_NAME	VARCHAR2(256)		Name of the table to which the DML applies
TABLE_OWNER	VARCHAR2(386)		Owner of the table to which the DML applies
ROW_ID	VARCHAR2(19)		Rowid of the row that was modified by the DML
UNDO_SQL	VARCHAR2(4000)		SQL to undo the DML indicated by OPERATION

Note

Starting with Oracle Database 19c, Release Update 19.32, unified auditing records the use of the SELECT ANY TRANSACTION privilege when accessing FLASHBACK_TRANSACTION_QUERY. This feature provides administrators with more complete audit visibility for privileged access to historical transaction data.

See Also

Oracle Database Utilities for information on how to enable minimal supplemental logging

6.187 GLOBAL_CONTEXT

GLOBAL_CONTEXT displays the values of global context attributes, which are accessible for the current session, based on the CLIENT_IDENTIFIER value.

GLOBAL_CONTEXT is similar to SESSION_CONTEXT, which lists the values of session (or local) context attributes set under the current session.

Column	Datatype	NULL	Description
NAMESPACE	VARCHAR2(31)		Namespace of the globally accessible context
ATTRIBUTE	VARCHAR2(31)		Attribute of the globally accessible context
VALUE	VARCHAR2(4000)		Value of the attribute of the globally accessible context
USERNAME	VARCHAR2(31)		Username for which globally accessible context value is applicable
CLIENT_IDENTIFIER	VARCHAR2(65)		Client identifier of the globally accessible context

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "SESSION_CONTEXT"
- *Oracle Database Security Guide* for more information about using global application contexts

6.188 GLOBAL_NAME

GLOBAL_NAME contains one row that displays the global name of the current database.

Column	Datatype	NULL	Description
GLOBAL_NAME	VARCHAR2(4000)		Global name of the database

6.189 HS_ALL_CAPS

HS_ALL_CAPS contains information about all of the capabilities (that is, features) associated with non-Oracle (FDS) data stores.

Column	Datatype	NULL	Description
CAP_NUMBER	NUMBER		Capability number
CONTEXT	NUMBER		Context in which this capability is applicable
TRANSLATION	VARCHAR2(255)		Valid for functions; contains translation to FDS dialect
ADDITIONAL_INFO	NUMBER		Flag for internal use
FDS_CLASS_NAME	VARCHAR2(30)		Name of the FDS Class
FDS_INST_NAME	VARCHAR2(30)		Name of the FDS instance

6.190 HS_ALL_DD

HS_ALL_DD contains data dictionary information about non-Oracle (FDS) data stores.

Column	Datatype	NULL	Description
DD_TABLE_NAME	VARCHAR2(128)		Data dictionary table name
TRANSLATION_TYPE	CHAR(1)		T = Translation, M = Mimic
TRANSLATION_TEXT	VARCHAR2(4000)		SQL statement containing the mapping
FDS_CLASS_NAME	VARCHAR2(30)		Name of the FDS Class
FDS_INST_NAME	VARCHAR2(30)		Name of the FDS instance
DD_TABLE_DESC	VARCHAR2(255)		Description of the Oracle data dictionary table

6.191 HS_ALL_INITS

HS_ALL_INITS contains initialization parameter information about non-Oracle (FDS) data stores.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INIT_VALUE_NAME	VARCHAR2(64)		Name of the initialization parameter
INIT_VALUE	VARCHAR2(255)		Value of the initialization parameter
INIT_VALUE_TYPE	VARCHAR2(1)		Environment variable (T or F). T means this is an environment variable; F means do not set as an environment variable
FDS_CLASS_NAME	VARCHAR2(30)		Name of the FDS Class
FDS_INST_NAME	VARCHAR2(30)		Name of the FDS instance

6.192 HS_BASE_CAPS

HS_BASE_CAPS contains information about base capability (that is, base features) of the non-Oracle (FDS) data store.

Column	Datatype	NULL	Description
CAP_NUMBER	NUMBER	NOT NULL	Capability number
CAP_DESCRIPTION	VARCHAR2(255)		Description of the capability

6.193 HS_BASE_DD

HS_BASE_DD displays information from the base data dictionary translation table.

Column	Datatype	NULL	Description
DD_TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the Oracle data dictionary table
DD_TABLE_DESC	VARCHAR2(255)		Description of the Oracle data dictionary table
DD_TABLE_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)

6.194 HS_CLASS_CAPS

HS_CLASS_CAPS contains information about the class-specific (driver) capabilities belonging to the non-Oracle (FDS) data store.

Column	Datatype	NULL	Description
CAP_NUMBER	NUMBER	NOT NULL	Capability number
CAP_DESCRIPTION	VARCHAR2(255)		Capability description
CONTEXT	NUMBER		Flag indicating the context in which the capability is enabled
TRANSLATION	VARCHAR2(255)		Valid for functions; contains translation to FDS dialect
ADDITIONAL_INFO	NUMBER		Additional flags for internal use
FDS_CLASS_NAME	VARCHAR2(30)	NOT NULL	Name of the FDS Class
FDS_CLASS_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.195 HS_CLASS_DD

HS_CLASS_DD displays information from the non-Oracle data store (FDS) class-specific data dictionary translations.

Column	Datatype	NULL	Description
DD_TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the Oracle data dictionary table
DD_TABLE_DESC	VARCHAR2(255)		Description of the Oracle data dictionary table
TRANSLATION_TYPE	CHAR(1)	NOT NULL	T = Translation, M = Mimic
TRANSLATION_TEXT	VARCHAR2(4000)		SQL statement containing the mapping
FDS_CLASS_NAME	VARCHAR2(30)	NOT NULL	Name of the FDS Class
DD_TABLE_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)
FDS_CLASS_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)

6.196 HS_CLASS_INIT

HS_CLASS_INIT displays information about the non-Oracle (FDS) class-specific initialization parameters.

Column	Datatype	NULL	Description
INIT_VALUE_NAME	VARCHAR2(64)	NOT NULL	Name of the initialization parameter
INIT_VALUE	VARCHAR2(255)	NOT NULL	Value of the initialization parameter
INIT_VALUE_TYPE	VARCHAR2(1)	NOT NULL	Environment variable (T or F). T means this is an environment variable; F means do not set as an environment variable
FDS_CLASS_NAME	VARCHAR2(30)	NOT NULL	Name of the FDS Class
FDS_CLASS_INIT_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)
FDS_CLASS_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)

6.197 HS_FDS_CLASS

HS_FDS_CLASS contains information about legal non-Oracle (FDS) classes.

Column	Datatype	NULL	Description
FDS_CLASS_NAME	VARCHAR2(30)	NOT NULL	Name of the FDS class (for example, ODBC, DB2)
FDS_CLASS _COMMENTS	VARCHAR2(255)		Text description of the non-Oracle class
FDS_CLASS_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.198 HS_FDS_INST

HS_FDS_INST contains information about non-Oracle (FDS) instances.

Column	Datatype	NULL	Description
FDS_INST_NAME	VARCHAR2(30)	NOT NULL	Name of the FDS instance
FDS_INST_COMMENTS	VARCHAR2(255)		Text description of the non-Oracle instance
FDS_CLASS_NAME	VARCHAR2(30)	NOT NULL	Name of the FDS class
FDS_INST_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)
FDS_CLASS_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)

6.199 HS_INST_CAPS

HS_INST_CAPS contains information about instance-specific capabilities (that is, features).

Column	Datatype	NULL	Description
CAP_NUMBER	NUMBER	NOT NULL	Capability number
CAP_DESCRIPTION	VARCHAR2(255)		Capability description
CONTEXT	NUMBER		Context in which this capability is applicable
TRANSLATION	VARCHAR2(255)		Valid for functions; contains translation to FDS dialect
ADDITIONAL_INFO	NUMBER		Additional flags for internal use
FDS_CLASS_NAME	VARCHAR2(30)	NOT NULL	Name of the FDS class (for example, ODBC, DB2)
FDS_INST_NAME	VARCHAR2(30)	NOT NULL	Name of the FDS instance
FDS_CLASS_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)
FDS_INST_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)

6.200 HS_INST_DD

HS_INST_DD displays information from the non-Oracle (FDS) instance-specific data dictionary translations.

Column	Datatype	NULL	Description
DD_TABLE_NAME	VARCHAR2(128)	NOT NULL	Name of the Oracle data dictionary table
DD_TABLE_DESC	VARCHAR2(255)		Description of the Oracle data dictionary table
TRANSLATION_TYPE	CHAR(1)	NOT NULL	T = Translation, M = Mimic
TRANSLATION_TEXT	VARCHAR2(4000)		SQL statement containing the mapping
FDS_CLASS_NAME	VARCHAR2(30)	NOT NULL	Name of the FDS class (for example, ODBC, DB2)
FDS_INST_NAME	VARCHAR2(30)	NOT NULL	Name of the FDS instance
DD_TABLE_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
FDS_CLASS_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)
FDS_INST_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)

6.201 HS_INST_INIT

HS_INST_INIT contains information about the non-Oracle (FDS) instance-specific initialization parameters.

Column	Datatype	NULL	Description
INIT_VALUE_NAME	VARCHAR2(64)	NOT NULL	Name of the initialization parameter
INIT_VALUE	VARCHAR2(255)	NOT NULL	Value of the initialization parameter
INIT_VALUE_TYPE	VARCHAR2(1)	NOT NULL	Environment variable (T or F). T means this is an environment variable; F means do not set as an environment variable
FDS_CLASS_NAME	VARCHAR2(30)	NOT NULL	Name of the FDS class (for example: ODBC, DB2)
FDS_INST_NAME	VARCHAR2(30)	NOT NULL	Name of the FDS instance
FDS_INST_INIT_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)
FDS_CLASS_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)
FDS_INST_ID	NUMBER	NOT NULL	Sequence: a counter that is incremented for every row inserted (used internally)

6.202 IDEPTREE

This view, created by utldtree.sql, lists the indented dependency tree. It is a pre-sorted, pretty-print version of DEPTREE.

Column	Datatype	NULL	Description
NESTED_LEVEL	NUMBER		Nesting level in the dependency tree
TYPE	VARCHAR2(23)		Object type
OWNER	VARCHAR2(28)		Object schema
NAME	VARCHAR2(1002)		Object name

6.203 IND

IND is a synonym for USER_INDEXES.

See Also

"USER_INDEXES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.204 INDEX_HISTOGRAM

INDEX_HISTOGRAM contains information from the ANALYZE INDEX ... VALIDATE STRUCTURE statement.

> **Note**
>
> The ANALYZE INDEX ... VALIDATE STRUCTURE OFFLINE statement must be used to collect statistics.

Column	Datatype	NULL	Description
REPEAT_COUNT	NUMBER		Number of times that one or more index keys is repeated in the table
KEYS_WITH_REPEAT_COUNT	NUMBER		Number of index keys that are repeated that many times

6.205 INDEX_STATS

INDEX_STATS stores information from the last ANALYZE INDEX ... VALIDATE STRUCTURE statement.

> **Note**
>
> The ANALYZE INDEX ... VALIDATE STRUCTURE OFFLINE statement must be used in order to collect statistics

Column	Datatype	NULL	Description
HEIGHT	NUMBER		Height of the B-Tree
BLOCKS	NUMBER	NOT NULL	Blocks allocated to the segment
NAME	VARCHAR2(128)	NOT NULL	Name of the index
PARTITION_NAME	VARCHAR2(128)		Name of the partition of the index which was analyzed. If the index is not partitioned, NULL is returned.
LF_ROWS	NUMBER		Number of leaf rows (values in the index)
LF_BLKS	NUMBER		Number of leaf blocks in the B-Tree
LF_ROWS_LEN	NUMBER		Sum of the lengths of all the leaf rows
LF_BLK_LEN	NUMBER		Usable space in a leaf block
BR_ROWS	NUMBER		Number of branch rows in the B-Tree
BR_BLKS	NUMBER		Number of branch blocks in the B-Tree
BR_ROWS_LEN	NUMBER		Sum of the lengths of all the branch blocks in the B-Tree
BR_BLK_LEN	NUMBER		Usable space in a branch block
DEL_LF_ROWS	NUMBER		Number of deleted leaf rows in the index

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEL_LF_ROWS_LEN	NUMBER		Total length of all deleted rows in the index
DISTINCT_KEYS	NUMBER		Number of distinct keys in the index (may include rows that have been deleted)
MOST_REPEATED_KEY	NUMBER		How many times the most repeated key is repeated (may include rows that have been deleted)
BTREE_SPACE	NUMBER		Total space currently allocated in the B-Tree
USED_SPACE	NUMBER		Total space that is currently being used in the B-Tree
PCT_USED	NUMBER		Percent of space allocated in the B-Tree that is being used
ROWS_PER_KEY	NUMBER		Average number of rows per distinct key (this figure is calculated without consideration of deleted rows)
BLKS_GETS_PER_ACCESS	NUMBER		Expected number of consistent mode block reads per row, assuming that a randomly chosen row is accessed using the index. Used to calculate the number of consistent reads that will occur during an index scan.
PRE_ROWS	NUMBER		Number of prefix rows (values in the index)
PRE_ROWS_LEN	NUMBER		Sum of lengths of all prefix rows
OPT_CMPR_COUNT	NUMBER		Optimal index compression length
OPT_CMPR_PCTSAVE	NUMBER		Corresponding space savings after an ANALYZE
DEL_LF_CMP_ROWS	NUMBER		Number of deleted rows that are within a compression unit (CU)
PRG_LF_CMP_ROWS	NUMBER		Number of purged rows that are within a CU
LF_CMP_ROWS	NUMBER		Number of rows that are in a CU or prefix compressed
LF_CMP_ROWS_LEN	NUMBER		Sum of lengths of all prefix rows and CUs
LF_UNCMP_ROWS	NUMBER		Number of rows that are neither in a CU nor prefix compressed
LF_UNCMP_ROWS_LEN	NUMBER		Sum of lengths of rows that are neither in a CU nor prefix compressed
LF_SUF_ROWS_LEN	NUMBER		Sum of lengths of suffix rows
LF_CMP_ROWS_UNCMP_LEN	NUMBER		Sum of the uncompressed lengths of rows that are in a CU or prefix compressed
LF_CMP_RECMP_COUNT	NUMBER		Sum of CU recompression counts
LF_CMP_LOCK_VEC_LEN	NUMBER		Sum of CU lock vector lengths
LF_CMP_BLKS	NUMBER		Number of blocks that have a CU or nonzero prefix column count
LF_UNCMP_BLKS	NUMBER		Number of blocks that do not have a CU and have a zero prefix column count

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.206 LOGSTDBY_UNSUPPORTED_TABLES

LOGSTDBY_UNSUPPORTED_TABLES is a synonym for DBA_LOGSTDBY_UNSUPPORTED_TABLE.

See Also

"DBA_LOGSTDBY_UNSUPPORTED_TABLE"

6.207 MAP_OBJECT

MAP_OBJECT is a global temporary table that displays the hierarchical arrangement of storage containers for objects. Each row in the table represents a level in the hierarchy.

Column	Datatype	NULL	Description
OBJECT_NAME	VARCHAR2(2000)		Name of the object
OBJECT_OWNER	VARCHAR2(2000)		Owner of the object
OBJECT_TYPE	VARCHAR2(2000)		Object type
FILE_MAP_IDX	NUMBER		File index (corresponds to FILE_MAP_IDX in V$MAP_FILE)
DEPTH	NUMBER		Element depth within the I/O stack
ELEM_IDX	NUMBER		Index corresponding to the element
CU_SIZE	NUMBER		Contiguous set of logical blocks of the file (in HKB units) that is resident contiguously on the element
STRIDE	NUMBER		Number of HKB between contiguous units (CU) in the file that are contiguous on this element. Used in RAID5 and striped files.
NUM_CU	NUMBER		Number of contiguous units that are adjacent to each other on this element that are separated by STRIDE HKB in the file. In RAID5, the number of contiguous units also include the parity stripes.
ELEM_OFFSET	NUMBER		Element offset (in HKB units)
FILE_OFFSET	NUMBER		Offset (in HKB units) from the start of the file to the first byte of the contiguous units
DATA_TYPE	VARCHAR2(2000)		Data type (DATA, PARITY, or DATA AND PARITY)
PARITY_POS	NUMBER		Position of the parity. Only for RAID5. This field is needed to distinguish the parity from the data part.
PARITY_PERIOD	NUMBER		Parity period. Only for RAID5.

6.208 NLS_DATABASE_PARAMETERS

NLS_DATABASE_PARAMETERS lists permanent NLS parameters of the database.

Column	Datatype	NULL	Description
PARAMETER	VARCHAR2(128)		Parameter name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
VALUE	VARCHAR2(64)		Parameter value

6.209 NLS_INSTANCE_PARAMETERS

NLS_INSTANCE_PARAMETERS lists NLS parameters of the instance.

Column	Datatype	NULL	Description
PARAMETER	VARCHAR2(30)		Parameter name
VALUE	VARCHAR2(64)		Parameter value

6.210 NLS_SESSION_PARAMETERS

NLS_SESSION_PARAMETERS lists NLS parameters of the user session.

Column	Datatype	NULL	Description
PARAMETER	VARCHAR2(30)		Parameter name
VALUE	VARCHAR2(64)		Parameter value

6.211 OBJ

OBJ is a synonym for USER_OBJECTS.

See Also

"USER_OBJECTS"

6.212 PATH_VIEW

PATH_VIEW contains one row for each unique path to access a resource in the Oracle XML DB repository.

Column	Datatype	NULL	Description
PATH	VARCHAR2(1024)		An (absolute) path to repository resource RES
RES	XMLTYPE(XMLSchema "http://xmlns.oracle.com/xdb/XDBResource.xsd" Element "Resource")		The resource referred to by the PATH column
LINK	XMLTYPE		Link property
RESID	RAW(16)		Resource OID

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

Oracle XML DB Developer's Guide for information about using this view

6.213 PDB_ALERTS

PDB_ALERTS contains descriptions of reasons for PDB alerts.

Column	Datatype	NULL	Description
TIME	TIMESTAMP(6)	NOT NULL	Time when the violation happened
NAME	VARCHAR2(128)	NOT NULL	A name of a PDB or non-CDB to which this record applies
CAUSE_NO	NUMBER	NOT NULL	Number identifying a specific reason for a PDB alert
TYPE_NO	NUMBER	NOT NULL	Type of the violation
ERROR	NUMBER		Oracle error, if any, for this violation
LINE	NUMBER	NOT NULL	Line number for the violation message
MESSAGE	VARCHAR2(4000)	NOT NULL	Description of the violation
STATUS	NUMBER		Status of the violation
ACTION	VARCHAR2(4000)		Actions to take to resolve the violations

6.214 PDB_PLUG_IN_VIOLATIONS

PDB_PLUG_IN_VIOLATIONS displays information about incompatibilities between a PDB and the CDB to which it belongs.

This view is also used to display information generated by executing DBMS_PDB.CHECK_PLUG_COMPATIBILITY.

Column	Datatype	NULL	Description
TIME	TIMESTAMP(6)	NOT NULL	Time when a violation described by this row was discovered
NAME	VARCHAR2(128)	NOT NULL	The name of an existing PDB or a PDB intended to be created (if a row was entered as a result of running DBMS_PDB.CHECK_PLUG_COMPATIBILITY)
CAUSE	VARCHAR2(64)	NOT NULL	Attribute which was being checked
TYPE	VARCHAR2(9)	NOT NULL	ERROR or WARNING
ERROR_NUMBER	NUMBER		Oracle error number, if any, encountered during a check
LINE	NUMBER	NOT NULL	Used to differentiate between violations which share a cause
MESSAGE	VARCHAR2(4000)	NOT NULL	Description of a violation
STATUS	VARCHAR2(9)		PENDING, RESOLVED, or IGNORE
ACTION	VARCHAR2(4000)		Description of an action to take to correct the violation

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database PL/SQL Packages and Types Reference for more information about the DBMS_PDB package, which provides an interface for examining and manipulating data about PDBs

6.215 PLAN_TABLE

PLAN_TABLE is automatically created as a global temporary table to hold the output of an EXPLAIN PLAN statement for all users.

PLAN_TABLE is the default sample output table into which the EXPLAIN PLAN statement inserts rows describing execution plans.

While a PLAN_TABLE table is automatically set up for each user, you can use the SQL script utlxplan.sql to manually create a local PLAN_TABLE in your schema.

Column	Datatype	NULL	Description
STATEMENT_ID	VARCHAR2(30)		Value of the optional STATEMENT_ID parameter specified in the EXPLAIN PLAN statement
PLAN_ID	NUMBER		Unique identifier of a plan in the database
TIMESTAMP	DATE		Date and time when the EXPLAIN PLAN statement was generated
REMARKS	VARCHAR2(4000)		Any comment (of up to 4000 bytes) you want to associate with each step of the explained plan. This column is used to indicate whether an outline or SQL Profile was used for the query. If you need to add or change a remark on any row of the PLAN_TABLE, then use the UPDATE statement to modify the rows of the PLAN_TABLE.
OPERATION	VARCHAR2(30)		Name of the internal operation performed in this step. In the first row generated for a statement, the column contains one of the following values: • DELETE STATEMENT • INSERT STATEMENT • SELECT STATEMENT • UPDATE STATEMENT

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OPTIONS	VARCHAR2(255)		A variation on the operation described in the OPERATION column
OBJECT_NODE	VARCHAR2(128)		Name of the database link used to reference the object (a table name or view name). For local queries using parallel execution, this column describes the order in which output from operations is consumed.
OBJECT_OWNER	VARCHAR2(128)		Owner of the table or index
OBJECT_NAME	VARCHAR2(128)		Name of the table or index
OBJECT_ALIAS	VARCHAR2(261)		Unique alias of a table or view in a SQL statement. For indexes, it is the object alias of the underlying table.
OBJECT_INSTANCE	NUMBER(38)		Number corresponding to the ordinal position of the object as it appears in the original statement. The numbering proceeds from left to right, outer to inner with respect to the original statement text. View expansion results in unpredictable numbers.
OBJECT_TYPE	VARCHAR2(30)		Modifier that provides descriptive information about the object; for example, NON-UNIQUE for indexes
OPTIMIZER	VARCHAR2(255)		Current mode of the optimizer
SEARCH_COLUMNS	NUMBER		Not currently used
ID	NUMBER(38)		A number assigned to each step in the execution plan
PARENT_ID	NUMBER(38)		ID of the next execution step that operates on the output of the ID step
DEPTH	NUMBER(38)		Depth of the operation in the row source tree that the plan represents. The value can be used for indenting the rows in a plan table report.
POSITION	NUMBER(38)		For the first row of output, this indicates the optimizer's estimated cost of executing the statement. For the other rows, it indicates the position relative to the other children of the same parent.
COST	NUMBER(38)		Cost of the operation as estimated by the optimizer's query approach. Cost is not determined for table access operations. The value of this column does not have any particular unit of measurement; it is merely a weighted value used to compare costs of execution plans. The value of this column is a function of the CPU_COST and IO_COST columns.
CARDINALITY	NUMBER(38)		Estimate by the query optimization approach of the number of rows accessed by the operation
BYTES	NUMBER(38)		Estimate by the query optimization approach of the number of bytes accessed by the operation

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OTHER_TAG	VARCHAR2(255)		Describes the contents of the OTHER column: • SERIAL - Serial execution. Currently, SQL is not loaded in the OTHER column for this case. • SERIAL_FROM_REMOTE - Serial execution at a remote site. • PARALLEL_FROM_SERIAL - Serial execution. Output of step is partitioned or broadcast to parallel execution servers. • PARALLEL_TO_SERIAL - Parallel execution. Output of step is returned to serial query coordinator (QC) process. • PARALLEL_TO_PARALLEL - Parallel execution. Output of step is repartitioned to second set of parallel execution servers. • PARALLEL_COMBINED_WITH_PARENT - Parallel execution; Output of step goes to next step in same parallel process. No interprocess communication to parent. • PARALLEL_COMBINED_WITH_CHILD - Parallel execution. Input of step comes from prior step in same parallel process. No interprocess communication from child.
PARTITION_START	VARCHAR2(255)		Start partition of a range of accessed partitions: • *number* - Start partition has been identified by the SQL compiler, and its partition number is given by *number* • KEY - Start partition will be identified at run time from partitioning key values • ROW LOCATION - Start partition (same as the stop partition) will be computed at run time from the location of each record being retrieved. The record location is obtained from a user-specified ROWID or from a global index. • INVALID - Range of accessed partitions is empty
PARTITION_STOP	VARCHAR2(255)		Stop partition of a range of accessed partitions: • *number* - Stop partition has been identified by the SQL compiler, and its partition number is given by *number* • KEY - Stop partition will be identified at run time from partitioning key values • ROW LOCATION - Stop partition (same as the start partition) will be computed at run time from the location of each record being retrieved. The record location is obtained from a user-specified ROWID or from a global index. • INVALID - Range of accessed partitions is empty
PARTITION_ID	NUMBER(38)		Step that has computed the pair of values of the PARTITION_START and PARTITION_STOP columns
OTHER	LONG		Other information that is specific to the execution step that a user might find useful (see the OTHER_TAG column)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OTHER_XML	CLOB		Provides extra information specific to an execution step of the execution plan. The content of this column is structured using XML since multiple pieces of information can be stored there. This includes: • Name of the schema against which the query was parsed • Release number of the Oracle Database that produced the explain plan • Hash value associated with the execution plan • Name (if any) of the outline or the SQL profile used to build the execution plan • Indication of whether or not dynamic statistics were used to produce the plan • The outline data, a set of optimizer hints that can be used to regenerate the same plan
DISTRIBUTION	VARCHAR2(30)		Method used to distribute rows from producer query servers to consumer query servers
CPU_COST	NUMBER(38)		CPU cost of the operation as estimated by the query optimizer's approach. The value of this column is proportional to the number of machine cycles required for the operation. For statements that use the rule-based approach, this column is NULL.
IO_COST	NUMBER(38)		I/O cost of the operation as estimated by the query optimizer's approach. The value of this column is proportional to the number of data blocks read by the operation. For statements that use the rule-based approach, this column is NULL.
TEMP_SPACE	NUMBER(38)		Temporary space (in bytes) used by the operation as estimated by the query optimizer's approach. For statements that use the rule-based approach, or for operations that do not use any temporary space, this column is NULL.
ACCESS_PREDICATES	VARCHAR2(4000)		Predicates used to locate rows in an access structure. For example, start or stop predicates for an index range scan.
FILTER_PREDICATES	VARCHAR2(4000)		Predicates used to filter rows before producing them
PROJECTION	VARCHAR2(4000)		Expressions produced by the operation
TIME	NUMBER(38)		Elapsed time (in seconds) of the operation as estimated by query optimization. For statements that use the rule-based approach, this column is NULL.
QBLOCK_NAME	VARCHAR2(128)		Name of the query block (either system-generated or defined by the user with the QB_NAME hint)

6.216 PLUGGABLE_SET_CHECK

PLUGGABLE_SET_CHECK contains pluggable set checks.

Column	Datatype	NULL	Description
OBJ1_OWNER	VARCHAR2(128)		Owner of the object
OBJ1_NAME	VARCHAR2(128)		Object 1

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OBJ1_SUBNAME	VARCHAR2(128)		SubObject1Name
OBJ1_TYPE	VARCHAR2(18)		Object Type
TS1_NAME	VARCHAR2(30)		Tablespace containing Object 1
OBJ2_NAME	VARCHAR2(128)		Object Name
OBJ2_SUBNAME	VARCHAR2(128)		SubObject2Name
OBJ2_TYPE	VARCHAR2(18)		Object Type
OBJ2_OWNER	VARCHAR2(128)		Object owner of second object
TS2_NAME	VARCHAR2(30)		Tablespace containing Object 1
CONSTRAINT_NAME	VARCHAR2(128)		Name of dependent constraint
REASON	VARCHAR2(86)		Reason for Pluggable check violation
MESG_ID	NUMBER		The message ID

6.217 PRODUCT_COMPONENT_VERSION

PRODUCT_COMPONENT_VERSION contains version and status information for component products.

Column	Datatype	NULL	Description
PRODUCT	VARCHAR2(80)		Product name
VERSION	VARCHAR2(80)		Version number
VERSION_FULL	VARCHAR2(160)		The version number with the new Oracle Database version scheme introduced in Oracle Database 18c. The version number is displayed only for the database component. All other components return a null value for this column.
STATUS	VARCHAR2(80)		Status of release

6.218 PROXY_USERS

PROXY_USERS describes the list of proxy users and the clients on whose behalf they can act.

Column	Datatype	NULL	Description
PROXY	VARCHAR2(128)	NOT NULL	Name of a proxy user
CLIENT	VARCHAR2(128)	NOT NULL	Name of the client user who the proxy user can act as
AUTHENTICATION	VARCHAR2(3)		Indicates whether the proxy is required to supply the client's authentication credentials (YES) or not (NO)
FLAGS	VARCHAR2(35)		Flags associated with the proxy/client pair: • PROXY MAY ACTIVATE ALL CLIENT ROLES • NO CLIENT ROLES MAY BE ACTIVATED • PROXY MAY ACTIVATE ROLE • PROXY MAY ACTIVATE ALL CLIENT ROLES • PROXY MAY NOT ACTIVATE ROLE

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "DBA_PROXIES" for information about all proxy connections in the database

6.219 PSTUBTBL

This table contains information on stubs generated by the PSTUB utility so that an Oracle Forms 3.0 client can call stored procedures in Oracle Database.

> **Note**
>
> The contents of this table are intended only for use by the PSTUB utility.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(128)		Schema part of the identifier of a stored procedure
DBNAME	VARCHAR2(128)		Database link part of the identifier of a stored procedure
LUN	VARCHAR2(128)		Library unit name part of the identifier of a stored procedure
LUTYPE	VARCHAR2(3)		Type of the stored procedure
LINENO	NUMBER		Line number of the stub
LINE	VARCHAR2(1800)		Text of the stub

6.220 PUBLIC_DEPENDENCY

PUBLIC_DEPENDENCY lists dependencies to and from objects, by object number.

Column	Datatype	NULL	Description
OBJECT_ID	NUMBER	NOT NULL	Object number
REFERENCED_OBJECT_ID	NUMBER	NOT NULL	Referenced object (the parent object)

6.221 PUBLICSYN

PUBLICSYN contains information on public synonyms.

Column	Datatype	NULL	Description
SNAME	VARCHAR2(128)		Name of the synonym
CREATOR	VARCHAR2(128)		Owner of the synonym
TNAME	VARCHAR2(128)		Table of which this is a synonym
DATABASE	VARCHAR2(128)		Database in which the table resides
TABTYPE	VARCHAR2(10)		Type of table

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.222 QUEUE_PRIVILEGES

QUEUE_PRIVILEGES shows all Advanced Queuing object privileges granted to the session.

Column	Datatype	NULL	Description
GRANTEE	VARCHAR2(128)	NOT NULL	Name of the user or role to whom access was granted
OWNER	VARCHAR2(128)	NOT NULL	Owner of the object
NAME	VARCHAR2(128)	NOT NULL	Name of the object
GRANTOR	VARCHAR2(128)	NOT NULL	Name of the user who performed the grant
ENQUEUE_PRIVILEGE	NUMBER		Permission to ENQUEUE to the queue
DEQUEUE_PRIVILEGE	NUMBER		Permission to DEQUEUE from the queue

6.223 RECYCLEBIN

RECYCLEBIN is a synonym for USER_RECYCLEBIN.

See Also

"USER_RECYCLEBIN"

6.224 REDACTION_COLUMNS

REDACTION_COLUMNS describes all redacted columns in the database, giving the owner of the table or view within which the column resides, the object name, the column name, the type of redaction function, the parameters to the redaction function (if any), and an optional user-provided description of the redaction function that is performed on the column.

Column	Datatype	NULL	ullDescription
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the object that is redacted
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object that is redacted
COLUMN_NAME	VARCHAR2(128)	NOT NULL	Name of the column that is redacted
FUNCTION_TYPE	VARCHAR2(27)		Redaction function for this column
FUNCTION_PARAMETERS	VARCHAR2(1000)		Redaction function_parameters for this column
REGEXP_PATTERN	VARCHAR2(512)		Regular expression pattern to search for
REGEXP_REPLACE_STRING	VARCHAR2(4000)		Replacement string (up to 4000 characters in length) with up to 500 back-references to subexpressions in the form \n, (where n is a number from 1 to 9)
REGEXP_POSITION	NUMBER		Integer counting from 1, giving the position where the search should begin
REGEXP_OCCURRENCE	NUMBER		Either 0 (to replace all occurrences of the match), or a positive integer n (to replace the nth occurrence of the match)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	ullDescription
REGEXP_MATCH_PARAMETER	VARCHAR2(10)		To change the default matching behavior, possible values are a combination of i, c, n, m, and x. See the documentation of the match_parameter in the REGEXP_REPLACE section of the *Oracle Database SQL Language Reference*.
COLUMN_DESCRIPTION	VARCHAR2(4000)		User-provided description of the redaction function that is performed on the column. For example, for a Social Security Number column, the description might be: "redact SSN to XXX-XX-(last 4 digits)".

See Also

Oracle Database Advanced Security Guide for more information about Oracle Data Redaction

6.225 REDACTION_EXPRESSIONS

REDACTION_EXPRESSIONS shows all the Data Redaction named Policy Expressions in the database.

Column	Datatype	NULL	Description
POLICY_EXPRESSION_NAME	VARCHAR2(256)		Customer-specified name of the named Policy Expression
EXPRESSION	VARCHAR2(4000)		The SQL expression defined for this Data Redaction named Policy Expression
OBJECT_OWNER	VARCHAR2(128)		Owner of the table or view which this named Policy Expression is associated with
OBJECT_NAME	VARCHAR2(128)		Name of the table or view which this named Policy Expression is associated with
COLUMN_NAME	VARCHAR2(128)		Name of the column which this named Policy Expression is associated with
POLICY_EXPRESSION_DESCRIPTION	VARCHAR2(4000)		Description of this named Policy Expression

See Also

Oracle Database Advanced Security Guide for more information about Oracle Data Redaction

6.226 REDACTION_POLICIES

REDACTION_POLICIES displays all redaction policies in the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OBJECT_OWNER	VARCHAR2(128)	NOT NULL	Owner of the object with the policy
OBJECT_NAME	VARCHAR2(128)	NOT NULL	Name of the object with the policy
POLICY_NAME	VARCHAR2(128)	NOT NULL	Name of the policy
EXPRESSION	VARCHAR2(4000)	NOT NULL	Expression for this policy
ENABLE	VARCHAR2(7)		Indicates whether the policy is enabled (YES) or not (NO)
POLICY_DESCRIPTION	VARCHAR2(4000)		Description of the policy

See Also

Oracle Database Advanced Security Guide for more information about Oracle Data Redaction

6.227 REDACTION_VALUES_FOR_TYPE_FULL

REDACTION_VALUES_FOR_TYPE_FULL shows all of the current values for full redaction.

For example, if a redaction policy is applied to a column of type BINARY_DOUBLE and the redaction type is full redaction, that column will be redacted with the value shown in the BINARY_DOUBLE_VALUE column of this view.

Column	Datatype	NULL	Description
NUMBER_VALUE	NUMBER	NOT NULL	Redaction result for full redaction on NUMBER columns
BINARY_FLOAT_VALUE	BINARY_FLOAT	NOT NULL	Redaction result for full redaction on BINARY_FLOAT columns
BINARY_DOUBLE_VALUE	BINARY_DOUBLE	NOT NULL	Redaction result for full redaction on BINARY_DOUBLE columns
CHAR_VALUE	VARCHAR2(1)		Redaction result for full redaction on CHAR columns
VARCHAR_VALUE	VARCHAR2(1)		Redaction result for full redaction on VARCHAR2 columns
NCHAR_VALUE	NCHAR(1)		Redaction result for full redaction on NCHAR columns
NVARCHAR_VALUE	NVARCHAR2(1)		Redaction result for full redaction on NVARCHAR2 columns
DATE_VALUE	DATE	NOT NULL	Redaction result for full redaction on DATE columns
TIMESTAMP_VALUE	TIMESTAMP(6)	NOT NULL	Redaction result for full redaction on TIMESTAMP columns
TIMESTAMP_WITH_TIME_ZONE_VALUE	TIMESTAMP(6) WITH TIME ZONE	NOT NULL	Redaction result for full redaction on TIMESTAMP WITH TIME ZONE columns
BLOB_VALUE	BLOB		Redaction result for full redaction on BLOB columns
CLOB_VALUE	CLOB		Redaction result for full redaction on CLOB columns
NCLOB_VALUE	NCLOB		Redaction result for full redaction on NCLOB columns

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Advanced Security Guide* for more information about Oracle Data Redaction

6.228 REPORT_COMPONENTS

REPORT_COMPONENTS displays metadata about different database components offering reports in XML, HTML, or Text formats.

Reports are first generated in XML and can then be translated into HTML or Text formats, for supported report types. Each component generates one or more reports containing different types of content. You can request reports using the component's own PL/SQL interface (for example, DBMS_SQLTUNE for SQL Tuning Advisor).

Column	Datatype	NULL	Description
COMPONENT_ID	NUMBER	NOT NULL	ID number of the database component building the report
COMPONENT_NAME	VARCHAR2(128)	NOT NULL	Name of the database component building the report (for example, sqltune for SQL Tuning Advisor)
COMPONENT_DESCRIPTION	VARCHAR2(256)		Component description
REPORT_ID	NUMBER	NOT NULL	ID number of the report type
REPORT_NAME	VARCHAR2(128)	NOT NULL	Name of the report type
REPORT_DESCRIPTION	VARCHAR2(256)		Description of the report type
SCHEMA_FILENAME	VARCHAR2(500)		Filename of the XML schema for the report (optional)
SCHEMA_DATA	XMLTYPE		XML schema for this report (optional)

> **See Also**
>
> *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SQLTUNE package

6.229 REPORT_FILES

REPORT_FILES displays data for all of the XML schema and XSLT files associated with reports for a given component.

Column	Datatype	NULL	Description
FILENAME	VARCHAR2(500)	NOT NULL	Name of the XSLT/XML schema file
DATA	XMLTYPE		XSLT/XML schema data

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.230 REPORT_FORMATS

REPORT_FORMATS displays metadata about the different output formats supported for reports.

Some reports are generated in XML only, in which case no data will appear in this view. When reports support HTML or Text formats (for example, the SQL Performance Analyzer reports), metadata about the format will appear in this view.

Column	Datatype	NULL	Description
COMPONENT_ID	NUMBER	NOT NULL	ID number of the database component building the report
COMPONENT_NAME	VARCHAR2(128)	NOT NULL	Name of the database component building the report (for example, sqltune for SQL Tuning Advisor)
REPORT_ID	NUMBER	NOT NULL	ID number of the report type
REPORT_NAME	VARCHAR2(128)	NOT NULL	Name of the report type
FORMAT_NAME	VARCHAR2(128)	NOT NULL	Name of the report format
DESCRIPTION	VARCHAR2(256)		Description of the report format
TYPE	VARCHAR2(6)		Format type: • XSLT - Reports generated by applying an XSLT style sheet to XML data (for example, HTML reports) • Text - Reports generated by first applying an XSLT style sheet to convert XML data to HTML, and then converting the HTML to formatted Text using the internal report HTML-to-text translation engine. • Custom - Custom formats implemented natively by report clients
XSLT_FILENAME	VARCHAR2(500)		Name of the XSLT used for this format (XSLT and Text format types only)
XSLT_DATA	XMLTYPE		XSLT data (XSLT and Text format types only)
TEXT_LINESIZE	NUMBER		Maximum line size of the formatted text report (Text format types only)

6.231 RESOURCE_COST

RESOURCE_COST lists the cost for each resource.

Column	Datatype	NULL	Description
RESOURCE_NAME	VARCHAR2(32)	NOT NULL	Name of the resource
UNIT_COST	NUMBER	NOT NULL	Cost of the resource

6.232 RESOURCE_MAP

RESOURCE_MAP describes resources. This table can be used to map resource names to resource numbers.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
RESOURCE#	NUMBER	NOT NULL	Numeric resource code
TYPE#	NUMBER	NOT NULL	Numeric type code
NAME	VARCHAR2(32)	NOT NULL	Name of the resource

6.233 RESOURCE_VIEW

RESOURCE_VIEW contains one row for each resource in the Oracle XML DB repository.

Column	Datatype	NULL	Description
RES	XMLTYPE(XMLSchema "http://xmlns.oracle.com/xdb/XDBResource.xsd" Element "Resource")		A resource in the repository
ANY_PATH	VARCHAR2(4000)		An (absolute) path to the resource
RESID	RAW(16)		Resource OID, which is a unique handle to the resource

See Also

Oracle XML DB Developer's Guide for information about using this view

6.234 ROLE_ROLE_PRIVS

ROLE_ROLE_PRIVS describes the roles granted to other roles.

Information is provided only about roles to which the user has access.

Column	Datatype	NULL	Description
ROLE	VARCHAR2(128)		Name of the role
GRANTED_ROLE	VARCHAR2(128)		Role that was granted
ADMIN_OPTION	VARCHAR2(3)		Signifies that the role was granted with ADMIN option
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the role was granted commonly (CONTAINER=ALL was used) • NO if the role was granted locally (CONTAINER=ALL was not used)
INHERITED	VARCHAR2(3)		Indicates whether the role grant was inherited from another container (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.235 ROLE_SYS_PRIVS

ROLE_SYS_PRIVS describes system privileges granted to roles.

Information is provided only about roles to which the user has access.

Column	Datatype	NULL	Description
ROLE	VARCHAR2(128)		Name of the role
PRIVILEGE	VARCHAR2(40)		System privilege granted to the role
ADMIN_OPTION	VARCHAR2(3)		Indicates whether the grant was with the ADMIN option (YES) or not (NO)
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the privilege was granted commonly (CONTAINER=ALL was used) • NO if the privilege was granted locally (CONTAINER=ALL was not used)
INHERITED	VARCHAR2(3)		Indicates whether the role grant was inherited from another container (YES) or not (NO)

6.236 ROLE_TAB_PRIVS

ROLE_TAB_PRIVS describes table privileges granted to roles. Information is provided only about roles to which the user has access.

> **Note**
>
> When a user with the SYSDBA privilege queries this view, information is provided only about roles to which the user has been explicitly granted access. To view information about all roles, a user with the SYSDBA privilege can query the DBA_TAB_PRIVS view.

Column	Datatype	NULL	Description
ROLE	VARCHAR2(128)		Name of the role
OWNER	VARCHAR2(128)		Owner of the object
TABLE_NAME	VARCHAR2(128)		Name of the object
COLUMN_NAME	VARCHAR2(128)		Name of the column, if applicable
PRIVILEGE	VARCHAR2(40)		Object privilege granted to the role
GRANTABLE	VARCHAR2(3)		YES if the role was granted with ADMIN OPTION; otherwise NO
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the privilege was granted commonly (CONTAINER=ALL was used) • NO if the privilege was granted locally (CONTAINER=ALL was not used)
INHERITED	VARCHAR2(3)		Indicates whether the role grant was inherited from another container (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.237 SCHEDULER_BATCH_ERRORS

SCHEDULER_BATCH_ERRORS displays the errors caused by each call in the batch after a Scheduler batch call (when the COMMIT_SEMANTICS argument has been set to ABSORB_ERRORS).

Column	Datatype	NULL	Description
ARRAY_INDEX	NUMBER		Index of the job in the batch
OBJECT_TYPE	VARCHAR2(30)		Object type: • JOB • LIGHTWEIGHT JOB • UNKNOWN
OBJECT_NAME	VARCHAR2(100)		Full name of the object (including schema)
ATTR_NAME	VARCHAR2(30)		Name of the attribute being set (if this is a batch set attribute call); NULL otherwise
ERROR_CODE	NUMBER		Top level error code
ERROR_MESSAGE	VARCHAR2(4000)		Complete error stack
ADDITIONAL_INFO	VARCHAR2(4000)		Additional information (currently unused)

6.238 SCHEMA_EXPORT_OBJECTS

SCHEMA_EXPORT_OBJECTS lists simple path names for some of the object types belonging to a Data Pump schema export, which is invoked using the SCHEMAS parameter on the expdp command.

Users of the Data Pump Export and Import utilities can query this view to determine valid values for the EXCLUDE and INCLUDE parameters.

Column	Datatype	NULL	Description
OBJECT_PATH	VARCHAR2(200)	NOT NULL	Simple path name for the object type
COMMENTS	VARCHAR2(2000)		Comment on the object type
NAMED	CHAR(1)		Do objects of this type have names? If yes (Y), then the name can be specified in the optional name_clause on the EXCLUDE and INCLUDE parameters.

See Also

- "DATABASE_EXPORT_OBJECTS"
- "TABLE_EXPORT_OBJECTS"
- *Oracle Database Utilities* for more information on performing a full Data Pump export using the expdp command

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.239 SEQ

SEQ is a synonym for USER_SEQUENCES.

> **See Also**
>
> "USER_SEQUENCES"

6.240 SESSION_CONTEXT

SESSION_CONTEXT describes the context attributes and their values set for the current session.

Column	Datatype	NULL	Description
NAMESPACE	VARCHAR2(128)		Namespace that the active attribute is in
ATTRIBUTE	VARCHAR2(128)		Name of the active attribute
VALUE	VARCHAR2(4000)		Value of the active attribute

6.241 SESSION_PRIVS

SESSION_PRIVS describes the system privileges that are currently available to the user.

Column	Datatype	NULL	Description
PRIVILEGE	VARCHAR2(40)	NOT NULL	Name of the privilege

6.242 SESSION_ROLES

SESSION_ROLES describes the roles that are currently enabled to the user.

Column	Datatype	NULL	Description
ROLE	VARCHAR2(128)	NOT NULL	Name of the role

6.243 SOURCE_SIZE

Oracle accesses this view to create views about object size.

> **See Also**
>
> "DBA_OBJECT_SIZE" and "USER_OBJECT_SIZE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.244 STMT_AUDIT_OPTION_MAP

STMT_AUDIT_OPTION_MAP describes auditing option type codes. This table can be used to map auditing option type numbers to type names.

> **Note**
>
> This view is populated in any Oracle Database where auditing is enabled, regardless of whether pre-Oracle Database 12*c* auditing or unified auditing is enabled for the database.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

> **Note**
>
> The mapping explained in this view is valid for audit configuration from the DBA_STMT_AUDIT_OPTS view only, and such audit configurations can be made when unified auditing is not enabled.

Column	Datatype	NULL	Description
OPTION#	NUMBER	NOT NULL	Numeric auditing option type code
NAME	VARCHAR2(40)	NOT NULL	Name of the type of auditing option
PROPERTY	NUMBER	NOT NULL	Property flag of the auditing option

> **See Also**
>
> "DBA_STMT_AUDIT_OPTS"

6.245 SYN

SYN is a synonym for USER_SYNONYMS.

> **See Also**
>
> "ALL_SYNONYMS"

6.246 SYNONYMS

SYNONYMS is included for compatibility. Oracle recommends that you do not use this view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.247 SYS_OBJECTS

SYS_OBJECTS maps object IDs to object types and segment data block addresses.

Column	Datatype	NULL	Description
OBJECT_TYPE	VARCHAR2(18)		Type of the object
OBJECT_TYPE_ID	NUMBER		Type ID of the object
SEGMENT_TYPE_ID	NUMBER		Type of segment: TABLE, CLUSTER, INDEX, ROLLBACK, DEFERRED ROLLBACK, TEMPORARY, CACHE
OBJECT_ID	NUMBER		Object identifier
HEADER_FILE	NUMBER		ID of the file containing the segment header
HEADER_BLOCK	NUMBER		ID of the block containing the segment header
TS_NUMBER	NUMBER		The tablespace number

6.248 SYSCATALOG

SYSCATALOG is included for compatibility.

Oracle recommends that you do not use this view.

6.249 SYSFILES

SYSFILES is included for compatibility.

Oracle recommends that you do not use this view.

6.250 SYSSEGOBJ

SYSSEGOBJ is included for compatibility.

Oracle recommends that you do not use this view.

6.251 SYSTEM_PRIVILEGE_MAP

SYSTEM_PRIVILEGE_MAP describes privilege (auditing option) type codes.

This table can be used to map privilege (auditing option) type numbers to type names.

Column	Datatype	NULL	Description
PRIVILEGE	NUMBER	NOT NULL	Numeric privilege (auditing option) type code
NAME	VARCHAR2(40)	NOT NULL	Name of the type of privilege (auditing option). See *Oracle Database SQL Language Reference* for a list of valid system privileges.
PROPERTY	NUMBER	NOT NULL	Property flag of the privilege (auditing option): • 0 - Indicates a privilege that can be granted with a SQL GRANT statement • 1 - Indicates a privilege that can only be granted using a PL/SQL package

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.252 TAB

TAB is included for compatibility.

Oracle recommends that you do not use this view.

6.253 TABLE_EXPORT_OBJECTS

TABLE_EXPORT_OBJECTS lists simple path names for some of the object types belonging to a Data Pump schema export, which is invoked using the TABLES parameter on the expdp command.

Users of the Data Pump Export and Import utilities can query this view to determine valid values for the EXCLUDE and INCLUDE parameters.

Column	Datatype	NULL	Description
OBJECT_PATH	VARCHAR2(200)	NOT NULL	Simple path name for the object type
COMMENTS	VARCHAR2(2000)		Comment on the object type
NAMED	CHAR(1)		Do objects of this type have names? If yes (Y), then the name can be specified in the optional name_clause on the EXCLUDE and INCLUDE parameters.

See Also

- "DATABASE_EXPORT_OBJECTS"
- "SCHEMA_EXPORT_OBJECTS"
- *Oracle Database Utilities* for more information on performing a full Data Pump export using the expdp command

6.254 TABLE_PRIVILEGE_MAP

TABLE_PRIVILEGE_MAP describes privilege (auditing option) type codes.

This table can be used to map privilege (auditing option) type numbers to type names.

Column	Datatype	NULL	Description
PRIVILEGE	NUMBER	NOT NULL	Numeric privilege (auditing option) type code
NAME	VARCHAR2(40)	NOT NULL	Name of the type of privilege (auditing option)

6.255 TABQUOTAS

TABQUOTAS is included for compatibility.

Oracle recommends that you do not use this view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.256 TABS

TABS is a synonym for USER_TABLES.

> **See Also**
>
> "USER_TABLES"

6.257 TRUSTED_SERVERS

TRUSTED_SERVERS displays whether a server is trusted or untrusted.

Column	Datatype	NULL	Description
TRUST	VARCHAR2(9)		Trustedness of the server listed. Values can be TRUSTED or UNTRUSTED servers which are not listed in the NAME column have opposite trustedness.
NAME	VARCHAR2(128)		Server name. Can be a specific server name or ALL for all servers.

Table 6-1 shows examples of the values returned depending on the status of the servers.

Table 6-1 TRUSTED_SERVERS Values

Condition (If . . .)	TRUSTED column	NAME column
... all servers are trusted	Trusted	ALL
... no servers are trusted	Untrusted	ALL
... all servers except DB1 are trusted	Untrusted	DB1
... all servers except DB1 are untrusted	Trusted	DB1

> **See Also**
>
> - *Oracle Database PL/SQL Packages and Types Reference*
> - *Oracle Database Enterprise User Security Administrator's Guide*

6.258 TS_PITR_CHECK

This view, created by catpitr.sql, provides information on any dependencies or restrictions that might prevent tablespace point-in-time recovery from proceeding.

This view applies only to the tablespace point-in-time recovery feature.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OBJ1_OWNER	VARCHAR2(128)		The owner of the object preventing tablespace point-in-time recovery. See the REASON column for details.
OBJ1_NAME	VARCHAR2(128)		The name of the object preventing tablespace point-in-time recovery
OBJ1_SUBNAME	VARCHAR2(128)		Subordinate to OBJ1_NAME
OBJ1_TYPE	VARCHAR2(16)		The object type for the object preventing tablespace point-in-time recovery
TS1_NAME	VARCHAR2(30)		Name of the tablespace containing the object preventing tablespace point-in-time recovery
OBJ2_NAME	VARCHAR2(128)		The name of a second object which may be preventing tablespace point-in-time recovery. If NULL, object 1 is the only object preventing recovery.
OBJ2_SUBNAME	VARCHAR2(128)		Subordinate to OBJ2_NAME
OBJ2_TYPE	VARCHAR2(15)		The object type for the second object (will be NULL if OBJ2_NAME is NULL)
OBJ2_OWNER	VARCHAR2(128)		The owner of the second object (will be NULL if OBJ2_NAME is NULL)
TS2_NAME	VARCHAR2(30)		Name of the tablespace containing second object which may be preventing tablespace point-in-time recovery (-1 indicates not applicable)
CONSTRAINT_NAME	VARCHAR2(128)		Name of the constraint
REASON	VARCHAR2(81)		Reason why tablespace point-in-time recovery cannot proceed

See Also

Oracle Database Backup and Recovery User's Guide for more information about tablespace point-in-time recovery

6.259 TS_PITR_OBJECTS_TO_BE_DROPPED

TS_PITR_OBJECTS_TO_BE_DROPPED lists all objects lost as a result of performing tablespace point-in-time recovery.

This view applies only to the tablespace point-in-time recovery feature.

Column	Datatype	NULL	Description
OWNER	VARCHAR2(128)	NOT NULL	The owner of the object
NAME	VARCHAR2(128)	NOT NULL	The name of the object that will be lost as a result of undergoing tablespace point-in-time recovery
CREATION_TIME	DATE	NOT NULL	Creation timestamp of the object
TABLESPACE_NAME	VARCHAR2(30)		Name of the tablespace containing the object

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Backup and Recovery User's Guide* for more information about tablespace point-in-time recovery

6.260 UNI_PLUGGABLE_SET_CHECK

`UNI_PLUGGABLE_SET_CHECK` contains pluggable check information.

Column	Datatype	NULL	Description
`OBJ1_OWNER`	`VARCHAR2(128)`		Owner of object
`OBJ1_NAME`	`VARCHAR2(128)`		Object 1
`OBJ1_SUBNAME`	`VARCHAR2(128)`		SubObject1Name
`OBJ1_TYPE`	`VARCHAR2(18)`		Object Type
`TS1_NAME`	`VARCHAR2(30)`		Tablespace containing Object 1
`OBJ2_NAME`	`VARCHAR2(128)`		Object Name
`OBJ2_SUBNAME`	`VARCHAR2(128)`		SubObject2Name
`OBJ2_TYPE`	`VARCHAR2(18)`		Object Type
`OBJ2_OWNER`	`VARCHAR2(128)`		Object owner of second object
`TS2_NAME`	`VARCHAR2(30)`		Tablespace containing Object 1
`CONSTRAINT_NAME`	`VARCHAR2(128)`		Name of dependent constraint
`REASON`	`VARCHAR2(86)`		Reason for Pluggable check violation
`MESG_ID`	`NUMBER`		The message ID

6.261 UNIFIED_AUDIT_TRAIL

When unified auditing is enabled in Oracle Database, the audit records are populated in this new audit trail.

This view displays audit records in tabular form by retrieving the audit records from the audit trail.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is enabled.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
AUDIT_TYPE	VARCHAR2(64)		Type of auditing: • Database Vault • Datapump • Direct path API • FineGrainedAudit • KACL_AUDIT • Label Security • Protocol • RMAN_AUDIT • Standard • XS
SESSIONID	NUMBER		Audit session identifier
PROXY_SESSIONID	NUMBER		Audit session identifier of proxying session
OS_USERNAME	VARCHAR2(128)		Name of the operating system user for the database session
USERHOST	VARCHAR2(128)		Name of the host machine from which the session was spawned
TERMINAL	VARCHAR2(30)		The operating system terminal of the user session
INSTANCE_ID	NUMBER		Instance number as specified in the initialization parameter file, init.ora
DBID	NUMBER		Database identifier of the audited database
AUTHENTICATION_TYPE	VARCHAR2(1024)		Authentication information for the session user **See:** "AUTHENTICATION_TYPE Description"
DBUSERNAME	VARCHAR2(128)		Database user name of the user whose actions were audited
DBPROXY_USERNAME	VARCHAR2(128)		Proxying user name, in the case of proxy authentication
EXTERNAL_USERID	VARCHAR2(1024)		External user name, in the case of network or external authentication For OCI IAM users, the user OCID (Oracle Cloud Identifier)
GLOBAL_USERID	VARCHAR2(32)		For a user logged in as an enterprise user, the global user identifier for the user For OCI IAM users, the value of this column is null
CLIENT_PROGRAM_NAME	VARCHAR2(48)		Name of the program used for the database session
DBLINK_INFO	VARCHAR2(4000)		Value of SYS_CONTEXT('USERENV','DBLINK_INFO'). Valid if the connection was via a database link.
XS_USER_NAME	VARCHAR2(128)		Name of the Real Application Security user
XS_SESSIONID	RAW(33)		Identifer of the Real Application Security session
ENTRY_ID	NUMBER		Numeric ID for each audit trail entry in the session
STATEMENT_ID	NUMBER		Numeric ID for each statement run (a statement may cause many actions)
EVENT_TIMESTAMP	TIMESTAMP(6)		Timestamp of the creation of the audit trail entry in the local time zone
EVENT_TIMESTAMP_UTC	TIMESTAMP(6)		Timestamp of the creation of the audit trail entry in UTC (Coordinated Universal Time)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ACTION_NAME	VARCHAR2(64)		Name of the action executed by the user. The name should be read in conjunction with the AUDIT_TYPE to understand the real action.
RETURN_CODE	NUMBER		Oracle error code generated by the action. Zero if the action succeeded
OS_PROCESS	VARCHAR2(16)		Operating system process identifier of the Oracle database process
TRANSACTION_ID	RAW(8)		Transaction identifier of the transaction in which the object is modified
SCN	NUMBER		System change number (SCN) of the creation of the audit trail entry
EXECUTION_ID	VARCHAR2(64)		Execution context identifier for each action
OBJECT_SCHEMA	VARCHAR2(128)		Schema name of object affected by the action
OBJECT_NAME	VARCHAR2(128)		Name of the object affected by the action
SQL_TEXT	CLOB		SQL associated with the event
SQL_BINDS	CLOB		List of bind variables, if any, associated with SQL_TEXT
APPLICATION_CONTEXTS	VARCHAR2(4000)		Semicolon-seperated list of Application Context Namespace, Attribute, Value information in (APPCTX_NSPACE,APPCTX_ATTRIBUTE=<value>) format
CLIENT_IDENTIFIER	VARCHAR2(64)		Client identifier in each Oracle session
NEW_SCHEMA	VARCHAR2(128)		The schema of the object named in the NEW_NAME column This column is populated only when AUDIT_TYPE = Standard.
NEW_NAME	VARCHAR2(128)		New name of object after RENAME, or name of underlying object (for example, CREATE INDEX owner.obj_name ON new_owner.new_name) This column is populated only when AUDIT_TYPE = Standard.
OBJECT_EDITION	VARCHAR2(128)		Name of the edition containing the audited object
SYSTEM_PRIVILEGE_USED	VARCHAR2(1024)		Comma-separated list of system privileges used to execute the action This column is populated only when AUDIT_TYPE = Standard.
SYSTEM_PRIVILEGE	VARCHAR2(40)		System privilege granted/revoked by a GRANT/ REVOKE statement This column is populated only when AUDIT_TYPE = Standard.
AUDIT_OPTION	VARCHAR2(40)		AUDIT/NOAUDIT SQL command This column is populated only when AUDIT_TYPE = Standard.
OBJECT_PRIVILEGES	VARCHAR2(35)		Object privileges granted/revoked by a GRANT/ REVOKE statement This column is populated only when AUDIT_TYPE = Standard.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ROLE	VARCHAR2(128)		Roles granted or revoked or set by GRANT/ REVOKE/SET ROLE command This column is populated only when AUDIT_TYPE = Standard.
TARGET_USER	VARCHAR2(128)		User on whom the GRANT/REVOKE/AUDIT/NOAUDIT statement was executed This column is populated only when AUDIT_TYPE = Standard.
EXCLUDED_USER	VARCHAR2(128)		User who was excluded when the AUDIT/NOAUDIT statement was executed This column is populated only when AUDIT_TYPE = Standard.
EXCLUDED_SCHEMA	VARCHAR2(128)		Displays the schema of the excluded objects This column is populated only when AUDIT_TYPE = Standard.
EXCLUDED_OBJECT	VARCHAR2(128)		Displays object excluded from the action This column is populated only when AUDIT_TYPE = Standard.
CURRENT_USER	VARCHAR2(128)		Effective user for the statement execution
ADDITIONAL_INFO	VARCHAR2(4000)		Text comment on the audit trail entry, if any This column is populated only when AUDIT_TYPE = Standard.
UNIFIED_AUDIT_POLICIES	VARCHAR2(4000)		Lists the audit policies that caused the current audit record. For example, if SELECT ON SCOTT.EMP was configured from policy SCOTT_EMP_POL, for the SELECT event this column will display SCOTT_EMP_POL. If more than one policy was configured, the list of policies that caused the event to be recorded in the audit trail are displayed in a comma-separated list. This column has a NULL value for mandatory audit records. See *Oracle Database Security Guide* for information on activities that are mandatorily audited.
FGA_POLICY_NAME	VARCHAR2(128)		Fine-grained auditing (FGA) policy name that generated this FGA audit record This column is populated only when AUDIT_TYPE = FineGrainedAudit.
XS_INACTIVITY_TIMEOUT	NUMBER		Inactivity timeout of the Real Application Security session This column is populated only when AUDIT_TYPE = XS.
XS_ENTITY_TYPE	VARCHAR2(32)		Type of the Real Application Security entity. Possible values are USER, ROLE, ROLESET, SECURITYCLASS, ACL, DATASECURITY, and NSTEMPLATE. This column is populated only when AUDIT_TYPE = XS.
XS_TARGET_PRINCIPAL_NAME	VARCHAR2(128)		Target principal name in Real Application Security operations. Possible operations are set verifier, set password, add proxy, remove proxy, switch user, assign user, create session,and grant roles. This column is populated only when AUDIT_TYPE = XS.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
XS_PROXY_USER_NAME	VARCHAR2(128)		Name of the Real Application Security proxy user This column is populated only when AUDIT_TYPE = XS.
XS_DATASEC_POLICY_NAME	VARCHAR2(128)		Name of the Real Application Security data security policy enabled or disabled This column is populated only when AUDIT_TYPE = XS.
XS_SCHEMA_NAME	VARCHAR2(128)		Name of the schema in enable, disable data security policy and global callback operation This column is populated only when AUDIT_TYPE = XS.
XS_CALLBACK_EVENT_TYPE	VARCHAR2(32)		Real Application Security global callback event type This column is populated only when AUDIT_TYPE = XS.
XS_PACKAGE_NAME	VARCHAR2(128)		Real Application Security callback package name for the global callback This column is populated only when AUDIT_TYPE = XS.
XS_PROCEDURE_NAME	VARCHAR2(128)		Real Application Security callback procedure name for the global callback This column is populated only when AUDIT_TYPE = XS.
XS_ENABLED_ROLE	VARCHAR2(128)		The role that is enabled This column is populated only when AUDIT_TYPE = XS.
XS_COOKIE	VARCHAR2(1024)		Real Application Security session cookie This column is populated only when AUDIT_TYPE = XS.
XS_NS_NAME	VARCHAR2(128)		Name of the Real Application Security session namespace This column is populated only when AUDIT_TYPE = XS.
XS_NS_ATTRIBUTE	VARCHAR2(4000)		Name of the Real Application Security session namespace attribute This column is populated only when AUDIT_TYPE = XS.
XS_NS_ATTRIBUTE_OLD_VAL	VARCHAR2(4000)		The old value of the Real Application Security session namespace attribute This column is populated only when AUDIT_TYPE = XS.
XS_NS_ATTRIBUTE_NEW_VAL	VARCHAR2(4000)		The new value of the Real Application Security session namespace This column is populated only when AUDIT_TYPE = XS.
DV_ACTION_CODE	NUMBER		Numeric action type code for Database Vault This column is populated only when AUDIT_TYPE = Database Vault.
DV_ACTION_NAME	VARCHAR2(30)		Name of the action whose numeric code appears in the DV_ACTION_CODE column This column is populated only when AUDIT_TYPE = Database Vault.
DV_EXTENDED_ACTION_CODE	NUMBER		Numeric action type code for Database Vault administration This column is populated only when AUDIT_TYPE = Database Vault.
DV_GRANTEE	VARCHAR2(128)		Name of the user whose Database Vault authorization was modified This column is populated only when AUDIT_TYPE = Database Vault.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DV_RETURN_CODE	NUMBER		Database Vault specific error code This column is populated only when AUDIT_TYPE = Database Vault.
DV_ACTION_OBJECT_NAME	VARCHAR2(128)		The unique name of the Database Vault object that was modified This column is populated only when AUDIT_TYPE = Database Vault.
DV_RULE_SET_NAME	VARCHAR2(128)		The unique name of the rule set that was executing and caused the audit event to trigger This column is populated only when AUDIT_TYPE = Database Vault.
DV_COMMENT	VARCHAR2(4000)		Text comment on the audit trail entry, providing more information about the statement audited This column is populated only when AUDIT_TYPE = Database Vault.
DV_FACTOR_CONTEXT	VARCHAR2(4000)		An XML document that contains all of the factor identifiers for the current session at the point when the audit event was triggered This column is populated only when AUDIT_TYPE = Database Vault.
DV_OBJECT_STATUS	VARCHAR2(1)		Indicates whether a particular Database Vault object is enabled or disabled. For example, if a Database Vault administrator enables or disables a realm, then this event will be audited and the DV_OBJECT_STATUS value will show the status of the realm after the event occurred. Possible values for this column are: • Y - The object is enabled • N - The object is disabled This column is populated only when AUDIT_TYPE = Database Vault.
OLS_POLICY_NAME	VARCHAR2(128)		Name of the Oracle Label Security (OLS) policy for which this audit record is generated This column is populated only when AUDIT_TYPE = Label Security.
OLS_GRANTEE	VARCHAR2(1024)		Name of the user whose OLS authorization was modified This column is populated only when AUDIT_TYPE = Label Security.
OLS_MAX_READ_LABEL	VARCHAR2(4000)		Maximum read label assigned to a user This column is populated only when AUDIT_TYPE = Label Security.
OLS_MAX_WRITE_LABEL	VARCHAR2(4000)		Maximum write label assigned to a user This column is populated only when AUDIT_TYPE = Label Security.
OLS_MIN_WRITE_LABEL	VARCHAR2(4000)		Minimum write label assigned to a user This column is populated only when AUDIT_TYPE = Label Security.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
OLS_PRIVILEGES_GRANTED	VARCHAR2(128)		OLS privileges assigned to a user or a trusted stored procedure This column is populated only when AUDIT_TYPE = Label Security.
OLS_PROGRAM_UNIT_NAME	VARCHAR2(128)		Name of the trusted stored procedure whose authorization was modified or was executed This column is populated only when AUDIT_TYPE = Label Security.
OLS_PRIVILEGES_USED	VARCHAR2(128)		OLS privileges used for an event This column is populated only when AUDIT_TYPE = Label Security.
OLS_STRING_LABEL	VARCHAR2(4000)		String representation of the OLS label This column is populated only when AUDIT_TYPE = Label Security.
OLS_LABEL_COMPONENT_TYPE	VARCHAR2(12)		Type of the OLS label component This column is populated only when AUDIT_TYPE = Label Security.
OLS_LABEL_COMPONENT_NAME	VARCHAR2(30)		Name of the OLS label component This column is populated only when AUDIT_TYPE = Label Security.
OLS_PARENT_GROUP_NAME	VARCHAR2(30)		Name of the parent of the OLS group This column is populated only when AUDIT_TYPE = Label Security.
OLS_OLD_VALUE	VARCHAR2(4000)		Old value for OLS ALTER events This column is populated only when AUDIT_TYPE = Label Security.
OLS_NEW_VALUE	VARCHAR2(4000)		New value for OLS ALTER events This column is populated only when AUDIT_TYPE = Label Security.
RMAN_SESSION_RECID	NUMBER		RMAN session identifier. Together with RMAN_SESSION_STAMP uniquely identifies an RMAN job (note that this is not same as user session ID; the value is a recid in controlfile that identifies RMAN job) This column is populated only when AUDIT_TYPE = RMAN_AUDIT.
RMAN_SESSION_STAMP	NUMBER		Timestamp for the session This column is populated only when AUDIT_TYPE = RMAN_AUDIT.
RMAN_OPERATION	VARCHAR2(20)		The RMAN operation executed by the job. One row will be added for each distinct operation within an RMAN session. For example, a backup job would contain BACKUP in the RMAN_OPERATION column. This column is populated only when AUDIT_TYPE = RMAN_AUDIT.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
RMAN_OBJECT_TYPE	VARCHAR2(20)		Type of objects involved for backup or restore/recover or change/delete/crosscheck commands. It contains one of the following values. If RMAN command does not satisfy one of them, then preference is given in order, from top to bottom of the list: • DB FULL • RECVR AREA • DB INCR • DATAFILE FULL • DATAFILE INCR • ARCHIVELOG • CONTROLFILE • SPFILE This column is populated only when AUDIT_TYPE = RMAN_AUDIT.
RMAN_DEVICE_TYPE	VARCHAR2(5)		Device involved in the RMAN job. It may be DISK or SBT_TAPE or * (An * indicates that more than one location is involved). For a backup job, it will be the output device type. For other commands (such as restore or crosscheck), it will be the input device type. This column is populated only when AUDIT_TYPE = RMAN_AUDIT.
DP_TEXT_PARAMETERS1	VARCHAR2(512)		Parameters during a Data Pump operation that have a text/string value. This may contain the values for: • ACCESS METHODS • DATA OPTIONS • DUMPER DIRECTORY • JOB_TYPE • JOB VERSION • MASTER TABLE • METADATA_JOB_MODE • PARTITION OPTIONS • REMOTE LINK • TABLE EXISTS For descriptions and more information about the settings that can appear for these Data Pump text parameters, see Table 6-2. This column is populated only when AUDIT_TYPE = Datapump.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DP_BOOLEAN_PARAMETERS1	VARCHAR2(512)		Parameters during a Data Pump operation that have a boolean value. This may contain the values for: • DATA_ONLY - Boolean value for whether or not the operation processed data only (as opposed to metadata only, or metadata and data combined) • DUMPFILE_PRESENT - Denotes whether a dump file exists. Typically, it indicates whether a network export in which no dumpfile is required. • JOB_RESTARTED - Boolean that indicates if the export or import job had to be restarted • MASTER_ONLY - Indicates whether the import job imported just the control job table and then stopped the job so that the contents of the control job table can be examined • METADATA_ONLY - Boolean value for whether or not the operation processed metadata only (as opposed to data only, or metadata and data combined) This column is populated only when AUDIT_TYPE = Datapump.
DIRECT_PATH_NUM_COLUMNS_LOADED	NUMBER		Shows the number of columns that were loaded using the SQL*Loader direct path load method This column is populated only when AUDIT_TYPE = Direct path API.
RLS_INFO	CLOB		Stores virtual private database (VPD) policy names and predicates separated by delimiter. To format the output into individual rows, use the DBMS_AUDIT_UTIL.DECODE_RLS_INFO_ATRAIL_UNI function.
KSACL_USER_NAME	VARCHAR2(128)		The connecting user name The value in this column is meaningful only when the UNIFIED_AUDIT_TRAIL.RETURN_CODE is 46981, which is the denial-of-service (DoS) error code. This column is populated only when AUDIT_TYPE = KACL_AUDIT.
KSACL_SERVICE_NAME	VARCHAR2(512)		The target database service name The value in this column is meaningful only when the UNIFIED_AUDIT_TRAIL.RETURN_CODE is 46981, which is the denial-of-service (DoS) error code. This column is populated only when AUDIT_TYPE = KACL_AUDIT.
KSACL_SOURCE_LOCATION	VARCHAR2(48)		The source location of the initiating connection The value in this column is meaningful only when the UNIFIED_AUDIT_TRAIL.RETURN_CODE is 46981, which is the denial-of-service (DoS) error code. This column is populated only when AUDIT_TYPE = KACL_AUDIT.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
PROTOCOL_SESSION_ID	NUMBER		Oracle XML DB session ID The PROTOCOL_* columns in this view are meaningful when auditing Oracle XML DB protocol (HTTP or FTP) messages. You can use this column to identify audit records from the same Oracle XML DB session. This column is populated only when AUDIT_TYPE = Protocol.
PROTOCOL_RETURN_CODE	NUMBER		Return code for the Oracle XML DB protocol request This value is logged in the audit records for both the reply message containing the return code and its corresponding request message. For HTTP requests, a return code of 200 (OKAY) or 304 (NOT-MODIFIED) is considered successful, and a return code of 207 means the reply may have multiple components with separate return codes. All other HTTP return codes are considered unsuccessful. This column is populated only when AUDIT_TYPE = Protocol.
PROTOCOL_ACTION_NAME	VARCHAR2(32)		Indicates the protocol and method for the Oracle XML DB protocol message This value is of the form *protocol-method*. Possible values for *protocol* are HTTP or FTP. Examples: HTTP-GET and FTP-RETR. Note that the HTTP GET and HEAD methods are both logged as HTTP-GET. Unrecognized methods are logged as HTTP-UNKNOWN or FTP-UNKNOWN. This column is populated only when AUDIT_TYPE = Protocol.
PROTOCOL_USERHOST	VARCHAR2(128)		IP address of the client This column is populated only when AUDIT_TYPE = Protocol.
PROTOCOL_MESSAGE	VARCHAR2(4000)		Text of the Oracle XML DB protocol message This column is populated only when AUDIT_TYPE = Protocol.
SOURCE[1]	VARCHAR2(8)		Source of the audit record: • FILE - Refers to records from the operating system spillover files in each database instance • DATABASE - Refers to records from the AUDSYS.AUD$UNIFIED table

[1] This column is available starting with Oracle Database 19c, Release Update 19.21.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 6-2 Data Pump Text Parameter Descriptions

Parameter	Description
ACCESS METHOD	The method used to load the data. Settings can be: • AUTOMATIC: Enables Oracle Data Pump to determine the optimal load method • DIRECT_PATH : Uses the direct path API to pass the data to be loaded. • EXTERNAL_TABLE: Loads data using the external tables option. • CONVENTIONAL: Loads the data using SQL INSERT statements
DATA OPTIONS	Indicates how certain types of data were handled during import operations. Settings are in bit-mask format, which are as follows: • 1 (SKIP_CONSTRAINT_ERRORS): Specifies that the import operation proceeded even if non-deferred constraint violations were encountered. • 8 (DISABLE_APPEND_HINT): The import operation did not use the APPEND hint while loading a data object. • 16 (REJECT_ROWS_WITH_REPL_CHAR): Warnings are issued when the replacement character may be used and an option was added to reject data rows where the replacement character was used during a Data Pump import. This situation can occur if different character sets are used for the export/import process
DUMPER DIRECTORY	Not in use
JOB_TYPE	Is either EXPORT or IMPORT
JOB VERSION	Specifies the version of database objects that were imported
MASTER TABLE	Indicates the name of the control job table. By default, it appears as follows for export operations: *schema_name*.SYS_EXPORT_TABLE_*n* For import operations, it appears as follows: *schema_name*.SYS_IMPORT_TABLE_*n* The *n* represents a numeric value of 01. If 01 is in use, the number is incremented with 02, 03, and so on.
METADATA_JOB_MODE	Type of export or import operation. For example a table export would be TABLE_EXPORT
PARTITION OPTIONS	Indicates how table partitions were created during an import operation. Settings can be: • NONE: The tables were created as they existed on the system from which the export operation was performed. • DEPARTITION: Each partition or subpartition was promoted to a new individual table. • MERGE: All partitions and subpartitions were merged into one table
REMOTE LINK	Indicates that the export was performed from a (source) database identified by a valid database link. The data from the source database instance was written to a dump file set on the connected database instance.
TABLE EXISTS	Indicates the action that was taken on an import operation when the target table already existed. The values are as follows: • REPLACE • TRUNCATE • SKIP • APPEND

Copyright © 2002, 2026, Oracle and/or its affiliates.

AUTHENTICATION_TYPE Description

The AUTHENTICATION_TYPE column of UNIFIED_AUDIT_TRAIL displays authentication information for the session user.

The value of this column is a string with the following syntax:

```
(TYPE=(auth_string));(CLIENT ADDRESS=((PROTOCOL=protocol)
(HOST=client_ip_address)(PORT=client_port_number)));
[(LOGON_INFO=((VERIFIER=%s-%s)(CLIENT_CAPABILITIES=%s));]
```

- `(TYPE=(auth_string));`

  Indicates the type of authentication for the session user.

  Possible values for *auth_string*:

  - DATABASE - Username/password authentication
  - DIRECTORY PASSWORD - Directory-based user authentication
  - NETWORK SERVICE - Authentication was performed by Oracle Net Services or strong authentication
  - OS - Operating system external user authentication
  - PROXY - OCI proxy connection authentication

- `(CLIENT ADDRESS=((PROTOCOL=`*protocol*`)(HOST=`*client_ip_address*`)(PORT=`*client_port_number*`)));`

  Displays the protocol used by the client, such as ipc, sdp, tcp, or tcps, the client IP address, and the client port number.

- `(LOGON_INFO=((VERIFIER=`*version-rollover_state*`)(CLIENT_CAPABILITIES=`*capability_list*`));`

  This syntax is displayed only if authentication was completed during gradual database password rollover.

  The value of VERIFIER comprises the following two values, separated by a hyphen:

  - *version* - Indicates the password version (11G or 12C)
  - *rollover_state* - Indicates whether the user was authenticated with the OLD password or the NEW password

  For CLIENT_CAPABILITIES, the value of *capabilitiy_list* is a comma-separated list of one or more of the following client capabilities: O5L_NP, O7L_MR, or O8L_LI. See *Oracle Database Net Services Reference* for more information about client capabilities.

See Also

- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_AUDIT_MGMT package
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_AUDIT_UTIL.DECODE_RLS_INFO_ATRAIL_UNI function

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.262 USABLE_EDITIONS

USABLE_EDITIONS describes the usable editions of the current user.

Column	Datatype	NULL	Description
EDITION_NAME	VARCHAR2(128)	NOT NULL	Name of the edition
PARENT_EDITION_NAME	VARCHAR2(128)		Name of the parent edition for this edition

> **See Also**
>
> *Oracle Database Development Guide* for more information about editions

6.263 USER_ADDM_FDG_BREAKDOWN

USER_ADDM_FDG_BREAKDOWN describes the contribution for each finding from the different instances owned by the current user. Its columns are the same as those in DBA_ADDM_FDG_BREAKDOWN.

> **See Also**
>
> "DBA_ADDM_FDG_BREAKDOWN"

6.264 USER_ADDM_FINDINGS

USER_ADDM_FINDINGS displays the ADDM findings discovered by the advisors owned by the current user.

Each row for ADDM tasks in the related USER_ADVISOR_FINDINGS view has a corresponding row in this view. Its columns (except for OWNER) are the same as those in DBA_ADDM_FINDINGS.

> **See Also**
>
> "DBA_ADDM_FINDINGS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.265 USER_ADDM_INSTANCES

USER_ADDM_INSTANCES provides instance-level information for ADDM tasks that finished executing in all instances owned by the current user. Its columns are the same as those in DBA_ADDM_INSTANCES.

See Also

"DBA_ADDM_INSTANCES"

6.266 USER_ADDM_TASK_DIRECTIVES

USER_ADDM_TASK_DIRECTIVES displays information about ADDM task directives owned by the current user. Its columns (except for USERNAME and SEQ_ID) are the same as those in DBA_ADDM_TASK_DIRECTIVES.

See Also

"DBA_ADDM_TASK_DIRECTIVES"

6.267 USER_ADDM_TASKS

USER_ADDM_TASKS displays information about the ADDM tasks owned by the current user.

The view contains one row for each row in the related USER_ADVISOR_TASKS view that has ADVISOR_NAME=ADDM and STATUS=COMPLETED. Its columns (except for OWNER) are the same as those in DBA_ADDM_TASKS.

See Also

"DBA_ADDM_TASKS"

6.268 USER_ADVISOR_ACTIONS

USER_ADVISOR_ACTIONS displays information about the actions associated with the recommendations owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_ACTIONS.

See Also

"DBA_ADVISOR_ACTIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.269 USER_ADVISOR_DIR_TASK_INST

USER_ADVISOR_DIR_TASK_INST displays information about all task directive instances owned by the current user. Its columns (except for SEQ_ID and USERNAME) are the same as those in DBA_ADVISOR_DIR_TASK_INST.

See Also

"DBA_ADVISOR_DIR_TASK_INST"

6.270 USER_ADVISOR_EXEC_PARAMETERS

USER_ADVISOR_EXEC_PARAMETERS displays the parameter values used for past executions of tasks owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_EXEC_PARAMETERS.

See Also

"DBA_ADVISOR_EXEC_PARAMETERS"

6.271 USER_ADVISOR_EXECUTIONS

USER_ADVISOR_EXECUTIONS displays metadata information for tasks owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_EXECUTIONS.

See Also

"DBA_ADVISOR_EXECUTIONS"

6.272 USER_ADVISOR_FDG_BREAKDOWN

USER_ADVISOR_FDG_BREAKDOWN describes the contribution from the different instances to the findings for each ADDM task owned by the current user. Its columns are the same as those in DBA_ADVISOR_FDG_BREAKDOWN.

See Also

"DBA_ADVISOR_FDG_BREAKDOWN"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.273 USER_ADVISOR_FINDINGS

USER_ADVISOR_FINDINGS displays the findings discovered by the advisors owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_FINDINGS.

See Also

"DBA_ADVISOR_FINDINGS"

6.274 USER_ADVISOR_JOURNAL

USER_ADVISOR_JOURNAL displays the journal entries for the tasks owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_JOURNAL.

See Also

"DBA_ADVISOR_JOURNAL"

6.275 USER_ADVISOR_LOG

USER_ADVISOR_LOG displays information about the current state of the tasks owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_LOG.

See Also

"DBA_ADVISOR_LOG"

6.276 USER_ADVISOR_OBJECTS

USER_ADVISOR_OBJECTS displays information about the objects currently referenced by the advisors owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_OBJECTS.

See Also

"DBA_ADVISOR_OBJECTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.277 USER_ADVISOR_PARAMETERS

USER_ADVISOR_PARAMETERS displays the task parameters and their current values for the tasks owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_PARAMETERS.

See Also

"DBA_ADVISOR_PARAMETERS"

6.278 USER_ADVISOR_RATIONALE

USER_ADVISOR_RATIONALE displays information about the rationales for the recommendations owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_RATIONALE.

See Also

"DBA_ADVISOR_RATIONALE"

6.279 USER_ADVISOR_RECOMMENDATIONS

USER_ADVISOR_RECOMMENDATIONS displays the results of an analysis of the recommendations owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_RECOMMENDATIONS.

See Also

"DBA_ADVISOR_RECOMMENDATIONS"

6.280 USER_ADVISOR_SQLA_REC_SUM

USER_ADVISOR_SQLA_REC_SUM displays recommendation rollup information for the workload objects owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_SQLA_REC_SUM.

See Also

"DBA_ADVISOR_SQLA_REC_SUM"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.281 USER_ADVISOR_SQLA_TABLES

USER_ADVISOR_SQLA_TABLES displays cross references between the workload statements and the tables referenced in the statement for the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_SQLA_TABLES.

See Also

"DBA_ADVISOR_SQLA_TABLES"

6.282 USER_ADVISOR_SQLA_WK_MAP

USER_ADVISOR_SQLA_WK_MAP displays the workload references for the tasks owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_SQLA_WK_MAP.

See Also

"DBA_ADVISOR_SQLA_WK_MAP"

6.283 USER_ADVISOR_SQLA_WK_STMTS

USER_ADVISOR_SQLA_WK_STMTS displays information about the workload objects owned by the current user after an Access Advisor analysis operation. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_SQLA_WK_STMTS.

See Also

"DBA_ADVISOR_SQLA_WK_STMTS"

6.284 USER_ADVISOR_SQLPLANS

USER_ADVISOR_SQLPLANS displays the different SQL execution plans owned by the current user generated as part of an advisor analysis. Its columns are the same as those in DBA_ADVISOR_SQLPLANS.

See Also

"DBA_ADVISOR_SQLPLANS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.285 USER_ADVISOR_SQLSTATS

USER_ADVISOR_SQLSTATS displays execution statistics owned by the current user for the test-execution of different SQL plans during the advisor analysis. Its columns are the same as those in DBA_ADVISOR_SQLSTATS.

See Also

"DBA_ADVISOR_SQLSTATS"

6.286 USER_ADVISOR_SQLW_JOURNAL

USER_ADVISOR_SQLW_JOURNAL displays the journal entries for the workload objects owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_SQLW_JOURNAL.

See Also

"DBA_ADVISOR_SQLW_JOURNAL"

6.287 USER_ADVISOR_SQLW_PARAMETERS

USER_ADVISOR_SQLW_PARAMETERS displays the workload parameters and their current values owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_SQLW_PARAMETERS.

See Also

"DBA_ADVISOR_SQLW_PARAMETERS"

6.288 USER_ADVISOR_SQLW_STMTS

USER_ADVISOR_SQLW_STMTS displays rows that correspond to the statements in the workload owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_SQLW_STMTS.

See Also

"DBA_ADVISOR_SQLW_STMTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.289 USER_ADVISOR_SQLW_SUM

USER_ADVISOR_SQLW_SUM displays an aggregated picture of the SQLWkld workload objects owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_SQLW_SUM.

See Also

"DBA_ADVISOR_SQLW_SUM"

6.290 USER_ADVISOR_SQLW_TABLES

USER_ADVISOR_SQLW_TABLES displays cross references between the workload statements and the tables referenced in the statement. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_SQLW_TABLES.

See Also

"DBA_ADVISOR_SQLW_TABLES"

6.291 USER_ADVISOR_SQLW_TEMPLATES

USER_ADVISOR_SQLW_TEMPLATES displays an aggregated picture of the SQLWkld template objects owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_SQLW_TEMPLATES.

See Also

"DBA_ADVISOR_SQLW_TEMPLATES"

6.292 USER_ADVISOR_TASKS

USER_ADVISOR_TASKS displays information about the tasks owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_TASKS.

See Also

"DBA_ADVISOR_TASKS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.293 USER_ADVISOR_TEMPLATES

USER_ADVISOR_TEMPLATES displays information about the templates owned by the current user. Its columns (except for OWNER) are the same as those in DBA_ADVISOR_TEMPLATES.

See Also

"DBA_ADVISOR_TEMPLATES"

6.294 USER_ALL_TABLES

USER_ALL_TABLES describes the object tables and relational tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ALL_TABLES.

See Also

"ALL_ALL_TABLES"

6.295 USER_ANALYTIC_VIEW_ATTR_CLASS

USER_ANALYTIC_VIEW_ATTR_CLASS describes the attribute classifications of the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_ATTR_CLASS.

See Also

"ALL_ANALYTIC_VIEW_ATTR_CLASS"

6.296 USER_ANALYTIC_VIEW_ATTR_CLS

USER_ANALYTIC_VIEW_ATTR_CLS is identical to USER_ANALYTIC_VIEW_ATTR_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"USER_ANALYTIC_VIEW_ATTR_CLASS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.297 USER_ANALYTIC_VIEW_ATTR_CLS_AE

USER_ANALYTIC_VIEW_ATTR_CLS_AE describes the attribute classifications of the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_ATTR_CLS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_ATTR_CLS_AE"

6.298 USER_ANALYTIC_VIEW_BAS_MEAS

USER_ANALYTIC_VIEW_BAS_MEAS is identical to USER_ANALYTIC_VIEW_BASE_MEAS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"USER_ANALYTIC_VIEW_BASE_MEAS"

6.299 USER_ANALYTIC_VIEW_BAS_MEAS_AE

USER_ANALYTIC_VIEW_BAS_MEAS_AE describes the base measures in the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_BAS_MEAS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_BAS_MEAS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.300 USER_ANALYTIC_VIEW_BASE_MEAS

USER_ANALYTIC_VIEW_BASE_MEAS describes the base measures in the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_BASE_MEAS.

See Also

"ALL_ANALYTIC_VIEW_BASE_MEAS"

6.301 USER_ANALYTIC_VIEW_CALC_MEAS

USER_ANALYTIC_VIEW_CALC_MEAS describes the calculated measures in the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_CALC_MEAS.

See Also

"ALL_ANALYTIC_VIEW_CALC_MEAS"

6.302 USER_ANALYTIC_VIEW_CLASS

USER_ANALYTIC_VIEW_CLASS describes the classifications of the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in USER_ANALYTIC_VIEW_CLASS.

See Also

"ALL_ANALYTIC_VIEW_CLASS"

6.303 USER_ANALYTIC_VIEW_CLASS_AE

USER_ANALYTIC_VIEW_CLASS_AE describes the classifications of the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_CLASS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"ALL_ANALYTIC_VIEW_CLASS_AE"

6.304 USER_ANALYTIC_VIEW_CLC_MEAS

USER_ANALYTIC_VIEW_CLC_MEAS is identical to USER_ANALYTIC_VIEW_CALC_MEAS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"USER_ANALYTIC_VIEW_CALC_MEAS"

6.305 USER_ANALYTIC_VIEW_CLC_MEAS_AE

USER_ANALYTIC_VIEW_CLC_MEAS_AE describes the calculated measures in the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_CLC_MEAS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_CLC_MEAS_AE"

6.306 USER_ANALYTIC_VIEW_COLUMNS

USER_ANALYTIC_VIEW_COLUMNS describes the columns of the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_COLUMNS.

See Also

"ALL_ANALYTIC_VIEW_COLUMNS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.307 USER_ANALYTIC_VIEW_COLUMNS_AE

USER_ANALYTIC_VIEW_COLUMNS_AE describes the columns of the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_COLUMNS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_COLUMNS_AE"

6.308 USER_ANALYTIC_VIEW_DIM_CLASS

USER_ANALYTIC_VIEW_DIM_CLASS describes the classifications of the attribute dimensions in the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_DIM_CLASS.

See Also

"ALL_ANALYTIC_VIEW_DIM_CLASS"

6.309 USER_ANALYTIC_VIEW_DIM_CLS

USER_ANALYTIC_VIEW_DIM_CLS is identical to USER_ANALYTIC_VIEW_DIM_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"USER_ANALYTIC_VIEW_DIM_CLASS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.310 USER_ANALYTIC_VIEW_DIM_CLS_AE

USER_ANALYTIC_VIEW_DIM_CLS_AE describes the classifications of the attribute dimensions in the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_DIM_CLS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_DIM_CLS_AE"

6.311 USER_ANALYTIC_VIEW_DIMENSIONS

USER_ANALYTIC_VIEW_DIMENSIONS describes the attribute dimensions in the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_DIMENSIONS.

> **See Also**
>
> "ALL_ANALYTIC_VIEW_DIMENSIONS"

6.312 USER_ANALYTIC_VIEW_DIMS

USER_ANALYTIC_VIEW_DIMS is identical to USER_ANALYTIC_VIEW_DIMENSIONS.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "USER_ANALYTIC_VIEW_DIMENSIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.313 USER_ANALYTIC_VIEW_DIMS_AE

USER_ANALYTIC_VIEW_DIMS_AE describes the attribute dimensions in the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_DIMS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_DIMS_AE"

6.314 USER_ANALYTIC_VIEW_HIER_CLASS

USER_ANALYTIC_VIEW_HIER_CLASS describes the classifications of the hierarchies in the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_HIER_CLASS.

See Also

"ALL_ANALYTIC_VIEW_HIER_CLASS"

6.315 USER_ANALYTIC_VIEW_HIER_CLS

USER_ANALYTIC_VIEW_HIER_CLS is identical to USER_ANALYTIC_VIEW_HIER_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"USER_ANALYTIC_VIEW_HIER_CLASS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.316 USER_ANALYTIC_VIEW_HIER_CLS_AE

USER_ANALYTIC_VIEW_HIER_CLS_AE describes the classifications of the hierarchies in the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_HIER_CLS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_HIER_CLS_AE"

6.317 USER_ANALYTIC_VIEW_HIERS

USER_ANALYTIC_VIEW_HIERS describes the hierarchies in the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_HIERS.

See Also

"ALL_ANALYTIC_VIEW_HIERS"

6.318 USER_ANALYTIC_VIEW_HIERS_AE

USER_ANALYTIC_VIEW_HIERS_AE describes the hierarchies in the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_HIERS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_HIERS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.319 USER_ANALYTIC_VIEW_KEYS

USER_ANALYTIC_VIEW_KEYS describes the key columns of the attribute dimensions in the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_KEYS.

See Also

"ALL_ANALYTIC_VIEW_KEYS"

6.320 USER_ANALYTIC_VIEW_KEYS_AE

USER_ANALYTIC_VIEW_KEYS_AE describes the key columns of the attribute dimensions in the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_KEYS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_KEYS_AE"

6.321 USER_ANALYTIC_VIEW_LEVEL_CLASS

USER_ANALYTIC_VIEW_LEVEL_CLASS describes the level classifications of the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_LEVEL_CLASS.

See Also

"ALL_ANALYTIC_VIEW_LEVEL_CLASS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.322 USER_ANALYTIC_VIEW_LEVELS

USER_ANALYTIC_VIEW_LEVELS describes the levels in the hierarchies in the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_LEVELS.

See Also

"ALL_ANALYTIC_VIEW_LEVELS"

6.323 USER_ANALYTIC_VIEW_LEVELS_AE

USER_ANALYTIC_VIEW_LEVELS_AE describes the levels in the hierarchies in the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_LEVELS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_LEVELS_AE"

6.324 USER_ANALYTIC_VIEW_LVL_CLS

USER_ANALYTIC_VIEW_LVL_CLS is identical to USER_ANALYTIC_VIEW_LEVEL_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"USER_ANALYTIC_VIEW_LEVEL_CLASS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.325 USER_ANALYTIC_VIEW_LVL_CLS_AE

USER_ANALYTIC_VIEW_LVL_CLS_AE describes the level classifications of the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_LVL_CLS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_LVL_CLS_AE"

6.326 USER_ANALYTIC_VIEW_LVLGRPS

USER_ANALYTIC_VIEW_LVLGRPS describes the analytic view measure and level groups of the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_LVLGRPS.

See Also

"ALL_ANALYTIC_VIEW_LVLGRPS"

6.327 USER_ANALYTIC_VIEW_LVLGRPS_AE

USER_ANALYTIC_VIEW_LVLGRPS_AE describes the analytic view measure and level groups of the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_LVLGRPS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_LVLGRPS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.328 USER_ANALYTIC_VIEW_MEAS_CLASS

USER_ANALYTIC_VIEW_MEAS_CLASS describes the classifications of the measures of the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_MEAS_CLASS.

See Also

"ALL_ANALYTIC_VIEW_MEAS_CLASS"

6.329 USER_ANALYTIC_VIEW_MEAS_CLS

USER_ANALYTIC_VIEW_MEAS_CLS is identical to USER_ANALYTIC_VIEW_MEAS_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"USER_ANALYTIC_VIEW_MEAS_CLASS"

6.330 USER_ANALYTIC_VIEW_MEAS_CLS_AE

USER_ANALYTIC_VIEW_MEAS_CLS_AE describes the classifications of the measures of the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEW_MEAS_CLS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEW_MEAS_CLS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.331 USER_ANALYTIC_VIEWS

USER_ANALYTIC_VIEWS describes the analytic views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEWS.

See Also

"ALL_ANALYTIC_VIEWS"

6.332 USER_ANALYTIC_VIEWS_AE

USER_ANALYTIC_VIEWS_AE describes the analytic views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ANALYTIC_VIEWS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ANALYTIC_VIEWS_AE"

6.333 USER_ANNOTATION_VALUES

USER_ANNOTATION_VALUES displays the values for schema annotations owned by the current user. Its columns (except for ANNOTATION_OWNER) are the same as those in ALL_ANNOTATION_VALUES.

The annotation owner is the same as the owner of the annotated object.

Note

This view is available starting with Oracle Database 19c, Release Update 19.28.

See Also

"ALL_ANNOTATION_VALUES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.334 USER_ANNOTATIONS

USER_ANNOTATIONS displays schema annotations owned by the current user. Its columns (except for ANNOTATION_OWNER) are the same as those in ALL_ANNOTATIONS.

The annotation owner is the same as the owner of the annotated object.

Note

This view is available starting with Oracle Database 19c, Release Update 19.28.

See Also

"ALL_ANNOTATIONS"

6.335 USER_ANNOTATIONS_USAGE

USER_ANNOTATIONS_USAGE provides usage information about schema annotations owned by the current user. Its columns (except for ANNOTATION_OWNER) are the same as those in ALL_ANNOTATIONS_USAGE.

The annotation owner is the same as the owner of the annotated object.

Note

This view is available starting with Oracle Database 19c, Release Update 19.28.

See Also

"ALL_ANNOTATIONS_USAGE"

6.336 USER_APPLY_ERROR

USER_APPLY_ERROR displays information about the error transactions generated by apply processes visible to the current user. Its columns (except for SOURCE_ROOT_NAME) are the same as those in ALL_APPLY_ERROR.

See Also

"ALL_APPLY_ERROR"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.337 USER_AQ_AGENT_PRIVS

USER_AQ_AGENT_PRIVS displays information about the registered AQ agents that are mapped to the current user. Its columns (except for DB_USERNAME) are the same as those in DBA_AQ_AGENT_PRIVS.

See Also

"DBA_AQ_AGENT_PRIVS"

6.338 USER_ARGUMENTS

USER_ARGUMENTS lists the arguments of the functions and procedures that are owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ARGUMENTS.

See Also

- "ALL_ARGUMENTS"
- "USER_PROCEDURES" for information about the functions and procedures that are owned by the current user

6.339 USER_ASSEMBLIES

USER_ASSEMBLIES provides information about all assemblies owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ASSEMBLIES.

See Also

"ALL_ASSEMBLIES"

6.340 USER_ASSOCIATIONS

USER_ASSOCIATIONS describes user-defined statistics associated with objects owned by the current user. Its columns are the same as those in ALL_ASSOCIATIONS.

See Also

"ALL_ASSOCIATIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.341 USER_ATTRIBUTE_DIM_ATTR_CLASS

USER_ATTRIBUTE_DIM_ATTR_CLASS describes the attribute classifications of the attribute dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_ATTR_CLASS.

See Also

"ALL_ATTRIBUTE_DIM_ATTR_CLASS"

6.342 USER_ATTRIBUTE_DIM_ATTR_CLS

USER_ATTRIBUTE_DIM_ATTR_CLS is identical to USER_ATTRIBUTE_DIM_ATTR_CLASS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"USER_ATTRIBUTE_DIM_ATTR_CLASS"

6.343 USER_ATTRIBUTE_DIM_ATTR_CLS_AE

USER_ATTRIBUTE_DIM_ATTR_CLS_AE describes the attribute classifications of the attribute dimensions (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_ATTR_CLS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_ATTR_CLS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.344 USER_ATTRIBUTE_DIM_ATTRS

USER_ATTRIBUTE_DIM_ATTRS describes the attributes of the attribute dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_ATTRS.

See Also

"ALL_ATTRIBUTE_DIM_ATTRS"

6.345 USER_ATTRIBUTE_DIM_ATTRS_AE

USER_ATTRIBUTE_DIM_ATTRS_AE describes the attributes of the attribute dimensions (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_ATTRS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_ATTRS_AE"

6.346 USER_ATTRIBUTE_DIM_CLASS

USER_ATTRIBUTE_DIM_CLASS describes the classifications of the attribute dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_CLASS.

See Also

"ALL_ATTRIBUTE_DIM_CLASS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.347 USER_ATTRIBUTE_DIM_CLASS_AE

USER_ATTRIBUTE_DIM_CLASS_AE describes the classifications of the attribute dimensions (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_CLASS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_CLASS_AE"

6.348 USER_ATTRIBUTE_DIM_JN_PTHS

USER_ATTRIBUTE_DIM_JN_PTHS is identical to USER_ATTRIBUTE_DIM_JOIN_PATHS.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"USER_ATTRIBUTE_DIM_JOIN_PATHS"

6.349 USER_ATTRIBUTE_DIM_JN_PTHS_AE

USER_ATTRIBUTE_DIM_JN_PTHS_AE describes the join paths for the attribute dimensions (across all editions) owned by the current user. Its columns are the same as those in ALL_ATTRIBUTE_DIM_JN_PTHS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_JN_PTHS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.350 USER_ATTRIBUTE_DIM_JOIN_PATHS

USER_ATTRIBUTE_DIM_JOIN_PATHS describes the join paths for the attribute dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_JOIN_PATHS.

See Also

"ALL_ATTRIBUTE_DIM_JOIN_PATHS"

6.351 USER_ATTRIBUTE_DIM_KEYS

USER_ATTRIBUTE_DIM_KEYS describes the keys of the attribute dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_KEYS.

See Also

"ALL_ATTRIBUTE_DIM_KEYS"

6.352 USER_ATTRIBUTE_DIM_KEYS_AE

USER_ATTRIBUTE_DIM_KEYS_AE describes the keys of the attribute dimensions (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_KEYS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_KEYS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.353 USER_ATTRIBUTE_DIM_LEVEL_ATTRS

USER_ATTRIBUTE_DIM_LEVEL_ATTRS describes the attributes of the levels of the attribute dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_LEVEL_ATTRS.

See Also

"ALL_ATTRIBUTE_DIM_LEVEL_ATTRS"

6.354 USER_ATTRIBUTE_DIM_LEVELS

USER_ATTRIBUTE_DIM_LEVELS describes the levels of the attribute dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_LEVELS.

See Also

"ALL_ATTRIBUTE_DIM_LEVELS"

6.355 USER_ATTRIBUTE_DIM_LEVELS_AE

USER_ATTRIBUTE_DIM_LEVELS_AE describes the levels of the attribute dimensions (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_LEVELS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_LEVELS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.356 USER_ATTRIBUTE_DIM_LVL_ATRS

USER_ATTRIBUTE_DIM_LVL_ATRS is identical to USER_ATTRIBUTE_DIM_LEVEL_ATTRS.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "USER_ATTRIBUTE_DIM_LEVEL_ATTRS"

6.357 USER_ATTRIBUTE_DIM_LVL_ATRS_AE

USER_ATTRIBUTE_DIM_LVL_ATRS_AE describes the attributes of the levels of the attribute dimensions (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_LVL_ATRS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ATTRIBUTE_DIM_LVL_ATRS_AE"

6.358 USER_ATTRIBUTE_DIM_LVL_CLASS

USER_ATTRIBUTE_DIM_LVL_CLASS describes the level classifications of the attribute dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_LVL_CLASS.

> **See Also**
>
> "ALL_ATTRIBUTE_DIM_LVL_CLASS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.359 USER_ATTRIBUTE_DIM_LVL_CLS

USER_ATTRIBUTE_DIM_LVL_CLS is identical to USER_ATTRIBUTE_DIM_LVL_CLASS.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "USER_ATTRIBUTE_DIM_LVL_CLASS"

6.360 USER_ATTRIBUTE_DIM_LVL_CLS_AE

USER_ATTRIBUTE_DIM_LVL_CLS_AE describes the level classifications of the attribute dimensions (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_LVL_CLS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_ATTRIBUTE_DIM_LVL_CLS_AE"

6.361 USER_ATTRIBUTE_DIM_ORD_ATRS

USER_ATTRIBUTE_DIM_ORD_ATRS is identical to USER_ATTRIBUTE_DIM_ORDER_ATTRS.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "USER_ATTRIBUTE_DIM_ORDER_ATTRS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.362 USER_ATTRIBUTE_DIM_ORD_ATRS_AE

USER_ATTRIBUTE_DIM_ORD_ATRS_AE describes the order attributes of the attribute dimensions (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_ORD_ATRS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_ORD_ATRS_AE"

6.363 USER_ATTRIBUTE_DIM_ORDER_ATTRS

USER_ATTRIBUTE_DIM_ORDER_ATTRS describes the order attributes of the attribute dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_ORDER_ATTRS.

See Also

"ALL_ATTRIBUTE_DIM_ORDER_ATTRS"

6.364 USER_ATTRIBUTE_DIM_TABLES

USER_ATTRIBUTE_DIM_TABLES describes the tables used by the attribute dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_TABLES.

See Also

"ALL_ATTRIBUTE_DIM_TABLES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.365 USER_ATTRIBUTE_DIM_TABLES_AE

USER_ATTRIBUTE_DIM_TABLES_AE describes the tables used by the attribute dimensions (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIM_TABLES_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIM_TABLES_AE"

6.366 USER_ATTRIBUTE_DIMENSIONS

USER_ATTRIBUTE_DIMENSIONS describes the attribute dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIMENSIONS.

See Also

"ALL_ATTRIBUTE_DIMENSIONS"

6.367 USER_ATTRIBUTE_DIMENSIONS_AE

USER_ATTRIBUTE_DIMENSIONS_AE describes the attribute dimensions (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_DIMENSIONS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_ATTRIBUTE_DIMENSIONS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.368 USER_ATTRIBUTE_TRANSFORMATIONS

USER_ATTRIBUTE_TRANSFORMATIONS describes the transformation functions for the transformations owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ATTRIBUTE_TRANSFORMATIONS.

See Also

"ALL_ATTRIBUTE_TRANSFORMATIONS"

6.369 USER_AUDIT_OBJECT

USER_AUDIT_OBJECT displays audit trail records for the objects accessible to the current user. Its columns are the same as those in DBA_AUDIT_OBJECT.

Note

This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from UNIFIED_AUDIT_TRAIL.

- "DBA_AUDIT_OBJECT"
- See *Oracle Database Security Guide* for more information about unified auditing.
- See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

6.370 USER_AUDIT_POLICIES

USER_AUDIT_POLICIES describes the fine-grained auditing policies on the tables and views owned by the current user. Its columns (except for OBJECT_SCHEMA) are the same as those in ALL_AUDIT_POLICIES.

See Also

"ALL_AUDIT_POLICIES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.371 USER_AUDIT_POLICY_COLUMNS

`USER_AUDIT_POLICY_COLUMNS` describes the fine-grained auditing policy columns on the tables and views owned by the current user. Its columns are the same as those in `ALL_AUDIT_POLICY_COLUMNS`.

See Also

"ALL_AUDIT_POLICY_COLUMNS"

6.372 USER_AUDIT_SESSION

`USER_AUDIT_SESSION` displays the audit trail records concerning connections and disconnections of the current user. Its columns are the same as those in `DBA_AUDIT_SESSION`.

Note

This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from `UNIFIED_AUDIT_TRAIL`.

- "DBA_AUDIT_SESSION"
- See *Oracle Database Security Guide* for more information about unified auditing.
- See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

6.373 USER_AUDIT_STATEMENT

`USER_AUDIT_STATEMENT` displays audit trail entries for the `GRANT`, `REVOKE`, `AUDIT`, `NOAUDIT`, and `ALTER SYSTEM` statements issued by the current user.

Its columns are the same as those in "DBA_AUDIT_STATEMENT".

Note

This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from `UNIFIED_AUDIT_TRAIL`.

- See *Oracle Database Security Guide* for more information about unified auditing.
- See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.374 USER_AUDIT_TRAIL

`USER_AUDIT_TRAIL` displays the standard audit trail entries related to the current user.

Its columns are the same as those in "DBA_AUDIT_TRAIL".

The view displays audit records generated by actions performed by the user and audit records generated by actions performed on the user's schema objects.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from `UNIFIED_AUDIT_TRAIL`.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

6.375 USER_AW_PS

`USER_AW_PS` describes the page spaces in the analytic workspaces owned by the current user. Its columns (except for `OWNER`) are the same as those in `ALL_AW_PS`.

> **See Also**
>
> "ALL_AW_PS"

6.376 USER_AWS

`USER_AWS` describes the analytic workspaces owned by the current user. Its columns (except for `OWNER`) are the same as those in `ALL_AWS`.

> **See Also**
>
> "ALL_AWS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.377 USER_BASE_TABLE_MVIEWS

USER_BASE_TABLE_MVIEWS describes the materialized views using materialized view logs owned by the current user. Its columns are the same as those in ALL_BASE_TABLE_MVIEWS.

See Also

"ALL_BASE_TABLE_MVIEWS"

6.378 USER_BLOCKCHAIN_TABLES

USER_BLOCKCHAIN_TABLES describes the blockchain tables owned by the current user. Its columns (except for SCHEMA_NAME) are the same as those in ALL_BLOCKCHAIN_TABLES.

Note

This view is available starting with Oracle Database 19c, Release Update 19.10.

See Also

"ALL_BLOCKCHAIN_TABLES"

6.379 USER_CATALOG

USER_CATALOG lists tables, views, clusters, synonyms, and sequences owned by the current user. Its columns are the same as those in ALL_CATALOG.

See Also

"ALL_CATALOG"

6.380 USER_CERTIFICATES

USER_CERTIFICATES displays the certificates added by the current user which are used for signature verification for blockchain tables. Its columns (except USER_NAME) are the same as those in ALL_CERTIFICATES.

Note

This view is available starting with Oracle Database 19c, Release Update 19.10.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "ALL_CERTIFICATES"

6.381 USER_CHANGE_NOTIFICATION_REGS

USER_CHANGE_NOTIFICATION_REGS describes the change notification registrations owned by the current user. Its columns (except for USERNAME) are the same as those in DBA_CHANGE_NOTIFICATION_REGS.

> **See Also**
>
> "DBA_CHANGE_NOTIFICATION_REGS"

6.382 USER_CLU_COLUMNS

USER_CLU_COLUMNS maps columns in the current user's tables to cluster columns. Its columns are the same as those in DBA_CLU_COLUMNS.

> **See Also**
>
> "DBA_CLU_COLUMNS"

6.383 USER_CLUSTER_HASH_EXPRESSIONS

USER_CLUSTER_HASH_EXPRESSIONS lists hash functions for the hash clusters owned by the current user. Its columns are the same as those in ALL_CLUSTER_HASH_EXPRESSIONS.

> **See Also**
>
> "ALL_CLUSTER_HASH_EXPRESSIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.384 USER_CLUSTERING_DIMENSIONS

`USER_CLUSTERING_DIMENSIONS` describes dimension tables associated with tables with an attribute clustering clause owned by the user. Its columns (except for `OWNER`) are the same as those in `ALL_CLUSTERING_DIMENSIONS`.

See Also

- "ALL_CLUSTERING_DIMENSIONS"
- *Oracle Database Data Warehousing Guide* for information about attribute clustering with zone maps

6.385 USER_CLUSTERING_JOINS

`USER_CLUSTERING_JOINS` describes joins to the dimension tables associated with tables with an attribute clustering clause owned by the user. Its columns (except for `OWNER`) are the same as those in `ALL_CLUSTERING_JOINS`.

See Also

- "ALL_CLUSTERING_JOINS"
- *Oracle Database Data Warehousing Guide* for information about attribute clustering with zone maps

6.386 USER_CLUSTERING_KEYS

`USER_CLUSTERING_KEYS` describes clustering keys for tables with an attribute clustering clause owned by the user. Its columns are the same as those in `ALL_CLUSTERING_KEYS`.

See Also

- "ALL_CLUSTERING_KEYS"
- *Oracle Database Data Warehousing Guide* for information about attribute clustering with zone maps

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.387 USER_CLUSTERING_TABLES

USER_CLUSTERING_TABLES describes the tables with an attribute clustering clause owned by the user. Its columns are the same as those in ALL_CLUSTERING_TABLES.

See Also

- "ALL_CLUSTERING_TABLES"
- *Oracle Database Data Warehousing Guide* for information about attribute clustering with zone maps

6.388 USER_CLUSTERS

USER_CLUSTERS describes all the clusters owned by the current user. Its columns are the same as those in ALL_CLUSTERS.

See Also

"ALL_CLUSTERS"

6.389 USER_CODE_ROLE_PRIVS

USER_CODE_ROLE_PRIVS describes all the roles that are associated with program units owned by current user. Its columns (except for OWNER) are the same as those in ALL_CODE_ROLE_PRIVS.

See Also

"ALL_CODE_ROLE_PRIVS"

6.390 USER_COL_COMMENTS

USER_COL_COMMENTS displays comments on the columns of the tables and views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_COL_COMMENTS.

See Also

"ALL_COL_COMMENTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.391 USER_COL_PENDING_STATS

USER_COL_PENDING_STATS describes the pending statistics of the columns owned by the current user. Its columns (except for OWNER) are the same as those in ALL_COL_PENDING_STATS.

See Also

"ALL_COL_PENDING_STATS"

6.392 USER_COL_PRIVS

USER_COL_PRIVS describes the column object grants for which the current user is the object owner, grantor, or grantee. Its columns are the same as those in DBA_COL_PRIVS.

See Also

"DBA_COL_PRIVS"

6.393 USER_COL_PRIVS_MADE

USER_COL_PRIVS_MADE describes the column object grants for which the current user is the object owner. Its columns (except for OWNER) are the same as those in ALL_COL_PRIVS_MADE.

See Also

"ALL_COL_PRIVS_MADE"

6.394 USER_COL_PRIVS_RECD

USER_COL_PRIVS_RECD describes the column object grants for which the current user is the grantee. Its columns (except for GRANTEE) are the same as those in ALL_COL_PRIVS_RECD.

See Also

"ALL_COL_PRIVS_RECD"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.395 USER_COLL_TYPES

USER_COLL_TYPES describes named collection types (VARRAYs, nested tables, object tables, and so on) in the current user's schema. Its columns are the same as those in ALL_COLL_TYPES.

See Also

"ALL_COLL_TYPES"

6.396 USER_COMPARISON

USER_COMPARISON displays information about the comparison objects owned by the current user. Its columns (except for OWNER) are the same as those in DBA_COMPARISON.

See Also

"DBA_COMPARISON"

6.397 USER_COMPARISON_COLUMNS

USER_COMPARISON_COLUMNS displays information about the columns for the comparison objects owned by the current user. Its columns (except for OWNER) are the same as those in DBA_COMPARISON_COLUMNS.

See Also

"DBA_COMPARISON_COLUMNS"

6.398 USER_COMPARISON_ROW_DIF

USER_COMPARISON_ROW_DIF displays information about the differing rows in the comparison scans owned by the current user. Its columns are the same as those in DBA_COMPARISON_ROW_DIF.

See Also

"DBA_COMPARISON_ROW_DIF"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.399 USER_COMPARISON_SCAN

USER_COMPARISON_SCAN displays information about the comparison scans owned by the current user. Its columns (except for OWNER) are the same as those in DBA_COMPARISON_SCAN.

See Also

"DBA_COMPARISON_SCAN"

6.400 USER_COMPARISON_SCAN_VALUES

USER_COMPARISON_SCAN_VALUES displays information about the values for the comparison scans owned by the current user. Its columns (except for OWNER) are the same as those in DBA_COMPARISON_SCAN_VALUES.

See Also

"DBA_COMPARISON_SCAN_VALUES"

6.401 USER_CONS_COLUMNS

USER_CONS_COLUMNS describes columns that are owned by the current user and that are specified in constraint definitions. Its columns are the same as those in ALL_CONS_COLUMNS.

See Also

"ALL_CONS_COLUMNS"

6.402 USER_CONS_OBJ_COLUMNS

USER_CONS_OBJ_COLUMNS displays information about the types that object columns (or attributes) or collection elements have been constrained to, in the tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CONS_OBJ_COLUMNS.

See Also

"ALL_CONS_OBJ_COLUMNS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.403 USER_CONSTRAINTS

USER_CONSTRAINTS describes all constraint definitions on tables owned by the current user. Its columns are the same as those in ALL_CONSTRAINTS.

See Also

"ALL_CONSTRAINTS"

6.404 USER_CQ_NOTIFICATION_QUERIES

USER_CQ_NOTIFICATION_QUERIES describes the registered queries for the CQ notifications owned by the current user. Its columns (except for USERNAME) are the same as those in DBA_CQ_NOTIFICATION_QUERIES.

See Also

"DBA_CQ_NOTIFICATION_QUERIES"

6.405 USER_CREDENTIALS

USER_CREDENTIALS displays credentials owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CREDENTIALS.

See Also

"ALL_CREDENTIALS"

6.406 USER_CUBE_ATTR_VISIBILITY

USER_CUBE_ATTR_VISIBILITY describes the OLAP attributes visible for the dimensions, hierarchies, and levels owned by the current user Its columns (except for OWNER) are the same as those in ALL_CUBE_ATTR_VISIBILITY.

See Also

"ALL_CUBE_ATTR_VISIBILITY"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.407 USER_CUBE_ATTRIBUTES

USER_CUBE_ATTRIBUTES describes the attributes for the OLAP cube dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_ATTRIBUTES.

See Also

"ALL_CUBE_ATTRIBUTES"

6.408 USER_CUBE_BUILD_PROCESSES

USER_CUBE_BUILD_PROCESSES describes the OLAP build processes and maintenance scripts owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_BUILD_PROCESSES.

See Also

"ALL_CUBE_BUILD_PROCESSES"

6.409 USER_CUBE_CALCULATED_MEMBERS

USER_CUBE_CALCULATED_MEMBERS describes the calculated members for the OLAP cube dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_CALCULATED_MEMBERS.

See Also

"ALL_CUBE_CALCULATED_MEMBERS"

6.410 USER_CUBE_DIM_LEVELS

USER_CUBE_DIM_LEVELS describes the OLAP dimension levels owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_DIM_LEVELS.

See Also

"ALL_CUBE_DIM_LEVELS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.411 USER_CUBE_DIM_MODELS

USER_CUBE_DIM_MODELS describes the models for the OLAP dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_DIM_MODELS.

See Also

"ALL_CUBE_DIM_MODELS"

6.412 USER_CUBE_DIM_VIEW_COLUMNS

USER_CUBE_DIM_VIEW_COLUMNS describes the columns of the relational views of the OLAP cube dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_DIM_VIEW_COLUMNS.

See Also

"ALL_CUBE_DIM_VIEW_COLUMNS"

6.413 USER_CUBE_DIM_VIEWS

USER_CUBE_DIM_VIEWS describes the relational views of the OLAP dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_DIM_VIEWS.

See Also

"ALL_CUBE_DIM_VIEWS"

6.414 USER_CUBE_DIMENSIONALITY

USER_CUBE_DIMENSIONALITY describes the dimension order for the OLAP cubes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_DIMENSIONALITY.

See Also

"ALL_CUBE_DIMENSIONALITY"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.415 USER_CUBE_DIMENSIONS

USER_CUBE_DIMENSIONS describes the OLAP cube dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_DIMENSIONS.

See Also

"ALL_CUBE_DIMENSIONS"

6.416 USER_CUBE_HIER_LEVELS

USER_CUBE_HIER_LEVELS describes the hierarchy levels for the OLAP cube dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_HIER_LEVELS.

See Also

"ALL_CUBE_HIER_LEVELS"

6.417 USER_CUBE_HIER_VIEW_COLUMNS

USER_CUBE_HIER_VIEW_COLUMNS describes the columns of the relational hierarchy views of the OLAP cube dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_HIER_VIEW_COLUMNS.

See Also

"ALL_CUBE_HIER_VIEW_COLUMNS"

6.418 USER_CUBE_HIER_VIEWS

USER_CUBE_HIER_VIEWS describes the hierarchies for the OLAP cube dimensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_HIER_VIEWS.

See Also

"ALL_CUBE_HIER_VIEWS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.419 USER_CUBE_HIERARCHIES

USER_CUBE_HIERARCHIES describes the OLAP dimension hierarchies owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_HIERARCHIES.

See Also

"ALL_CUBE_HIERARCHIES"

6.420 USER_CUBE_MEASURES

USER_CUBE_MEASURES describes the measures for the OLAP cubes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_MEASURES.

See Also

"ALL_CUBE_MEASURES"

6.421 USER_CUBE_NAMED_BUILD_SPECS

USER_CUBE_NAMED_BUILD_SPECS describes the OLAP cube named build specifications in the database that are owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_NAMED_BUILD_SPECS.

See Also

"ALL_CUBE_NAMED_BUILD_SPECS"

6.422 USER_CUBE_SUB_PARTITION_LEVELS

USER_CUBE_SUB_PARTITION_LEVELS describes the OLAP secondary partition levels in the database that are owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_SUB_PARTITION_LEVELS.

See Also

"ALL_CUBE_SUB_PARTITION_LEVELS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.423 USER_CUBE_VIEW_COLUMNS

USER_CUBE_VIEW_COLUMNS describes the columns of relational views of OLAP cubes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_VIEW_COLUMNS.

See Also

"ALL_CUBE_VIEW_COLUMNS"

6.424 USER_CUBE_VIEWS

USER_CUBE_VIEWS describes the relational views of the OLAP cubes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBE_VIEWS.

See Also

"ALL_CUBE_VIEWS"

6.425 USER_CUBES

USER_CUBES describes the OLAP cubes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_CUBES.

See Also

"ALL_CUBES"

6.426 USER_DATAPUMP_JOBS

USER_DATAPUMP_JOBS displays the Data Pump jobs owned by the current user. Its columns (except for OWNER_NAME) are the same as those in DBA_DATAPUMP_JOBS.

See Also

"DBA_DATAPUMP_JOBS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.427 USER_DB_LINKS

USER_DB_LINKS describes the database links owned by the current user. Its columns are the same as those in ALL_DB_LINKS except that it does not have the OWNER column.

It also displays an additional column, PASSWORD, which is no longer used and for which nothing is returned. The PASSWORD column is maintained for backward compatibility only.

See Also

"ALL_DB_LINKS"

6.428 USER_DBFS_HS

USER_DBFS_HS shows all Database File System (DBFS) hierarchical stores owned by the current user. Its columns (except for STOREOWNER) are the same as those in DBA_DBFS_HS.

See Also

"DBA_DBFS_HS"

6.429 USER_DBFS_HS_COMMANDS

USER_DBFS_HS_COMMANDS shows all the registered store commands for all Database File system (DBFS) hierarchical stores owned by current user. Its columns (except for STOREOWNER) are the same as those in DBA_DBFS_HS_COMMANDS.

See Also

"DBA_DBFS_HS_COMMANDS"

6.430 USER_DBFS_HS_FILES

USER_DBFS_HS_FILES displays files in the Database File System (DBFS) hierarchical store owned by the current user and their location on the back-end device.

Column	Datatype	NULL	Description
PATH	VARCHAR2(1024)		Path name of the file
SEQUENCENUMBER	NUMBER		Sequence number of this piece of the file
STARTOFFSET	NUMBER		Begin offset of this piece in the tarball
ENDOFFSET	NUMBER		End offset of this piece in the tarball
TARBALLID	NUMBER		Tarball ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BACKUPFILENAME	VARCHAR2(256)		File on back end in which this tarball is located
TARSTARTOFFSET	NUMBER		Begin offset of this tarball in the backup file
TARENDOFFSET	NUMBER		End offset of this tarball in the backup file

6.431 USER_DBFS_HS_FIXED_PROPERTIES

USER_DBFS_HS_FIXED_PROPERTIES shows non-modifiable properties of all Databaase File System (DBFS) hierarchical stores owned by current user. Its columns (except for STORE_OWNER) are the same as those in DBA_DBFS_HS_FIXED_PROPERTIES.

See Also

"DBA_DBFS_HS_FIXED_PROPERTIES"

6.432 USER_DBFS_HS_PROPERTIES

USER_DBFS_HS_PROPERTIES shows modifiable properties of all Database File System (DBFS) hierarchical stores owned by current user. Its columns (except for STOREOWNER) are the same as those in DBA_DBFS_HS_PROPERTIES.

See Also

"DBA_DBFS_HS_PROPERTIES"

6.433 USER_DEPENDENCIES

USER_DEPENDENCIES describes dependencies between procedures, packages, functions, package bodies, and triggers owned by the current user, including dependencies on views created without any database links. Its columns are the same as those in ALL_DEPENDENCIES.

See Also

"ALL_DEPENDENCIES"

6.434 USER_DIM_ATTRIBUTES

USER_DIM_ATTRIBUTES describes the relationship between dimension levels and functionally dependent columns in the current user's schema.

The level columns and the dependent column must be in the same table. This view's columns are the same as those in ALL_DIM_ATTRIBUTES.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "ALL_DIM_ATTRIBUTES"

6.435 USER_DIM_CHILD_OF

USER_DIM_CHILD_OF describes a hierarchical relationship of 1 to *n* between pairs of levels in dimensions owned by the current user. Its columns are the same as those in ALL_DIM_CHILD_OF.

> **See Also**
>
> "ALL_DIM_CHILD_OF"

6.436 USER_DIM_HIERARCHIES

USER_DIM_HIERARCHIES describes the dimension hierarchies owned by the current user. Its columns are the same as those in ALL_DIM_HIERARCHIES.

> **See Also**
>
> "ALL_DIM_HIERARCHIES"

6.437 USER_DIM_JOIN_KEY

USER_DIM_JOIN_KEY describes the join between two dimension tables owned by the current user. The join is always specified between a parent dimension level column and a child column. This view's columns are the same as those in ALL_DIM_JOIN_KEY.

> **See Also**
>
> "ALL_DIM_JOIN_KEY"

6.438 USER_DIM_LEVEL_KEY

USER_DIM_LEVEL_KEY describes columns of dimension levels owned by the current user. This view's columns are the same as those in ALL_DIM_LEVEL_KEY.

> **See Also**
>
> "ALL_DIM_LEVEL_KEY"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.439 USER_DIM_LEVELS

USER_DIM_LEVELS describes the levels of dimensions owned by the current user. All columns of a dimension level must come from the same relation. This view's columns are the same as those in ALL_DIM_LEVELS.

See Also

"ALL_DIM_LEVELS"

6.440 USER_DIMENSIONS

USER_DIMENSIONS describes dimension objects in the user's schema. Its columns are the same as those in ALL_DIMENSIONS.

See Also

"ALL_DIMENSIONS"

6.441 USER_EDITIONED_TYPES

USER_EDITIONED_TYPES lists the types that are editioned by default for the current user. Its columns (except for SCHEMA) are the same as those in DBA_EDITIONED_TYPES.

See Also

"DBA_EDITIONED_TYPES"

6.442 USER_EDITIONING_VIEW_COLS

USER_EDITIONING_VIEW_COLS describes the relationship between the columns of the editioning views owned by the current user and the table columns to which they map. Its columns (except for OWNER) are the same as those in ALL_EDITIONING_VIEW_COLS.

See Also

"ALL_EDITIONING_VIEW_COLS"

6.443 USER_EDITIONING_VIEW_COLS_AE

USER_EDITIONING_VIEW_COLS_AE describes the relationship between the columns of the editioning views (across all editions) owned by the current user and the table columns to which

Copyright © 2002, 2026, Oracle and/or its affiliates.

they map. Its columns (except for OWNER) are the same as those in ALL_EDITIONING_VIEW_COLS_AE.

See Also

"ALL_EDITIONING_VIEW_COLS_AE"

6.444 USER_EDITIONING_VIEWS

USER_EDITIONING_VIEWS describes the editioning views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_EDITIONING_VIEWS.

See Also

"ALL_EDITIONING_VIEWS"

6.445 USER_EDITIONING_VIEWS_AE

USER_EDITIONING_VIEWS_AE describes the editioning views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_EDITIONING_VIEWS_AE.

See Also

"ALL_EDITIONING_VIEWS_AE"

6.446 USER_ENCRYPTED_COLUMNS

USER_ENCRYPTED_COLUMNS maintains encryption algorithm information for all encrypted columns in all tables in the user's schema. Its columns (except for OWNER) are the same as those in ALL_ENCRYPTED_COLUMNS.

See Also

"ALL_ENCRYPTED_COLUMNS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.447 USER_EPG_DAD_AUTHORIZATION

USER_EPG_DAD_AUTHORIZATION describes the DADs that are authorized to use the user's privileges. Its columns (except for USERNAME) are the same as those in DBA_EPG_DAD_AUTHORIZATION.

See Also

"DBA_EPG_DAD_AUTHORIZATION"

6.448 USER_ERROR_TRANSLATIONS

USER_ERROR_TRANSLATIONS describes all error translations owned by the user. Its columns (except for OWNER) are the same as those in ALL_ERROR_TRANSLATIONS.

See Also

"ALL_ERROR_TRANSLATIONS"

6.449 USER_ERRORS

USER_ERRORS describes the current errors on the stored objects owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ERRORS.

See Also

"ALL_ERRORS"

6.450 USER_ERRORS_AE

USER_ERRORS_AE describes the current errors on the stored objects (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_ERRORS_AE.

See Also

"ALL_ERRORS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.451 USER_EVALUATION_CONTEXT_TABLES

USER_EVALUATION_CONTEXT_TABLES describes the tables in the rule evaluation contexts owned by the current user. Its columns (except for EVALUATION_CONTEXT_OWNER) are the same as those in ALL_EVALUATION_CONTEXT_TABLES.

See Also

"ALL_EVALUATION_CONTEXT_TABLES"

6.452 USER_EVALUATION_CONTEXT_VARS

USER_EVALUATION_CONTEXT_VARS describes the variables in the rule evaluation contexts owned by the current user. Its columns (except for EVALUATION_CONTEXT_OWNER) are the same as those in ALL_EVALUATION_CONTEXT_VARS.

See Also

"ALL_EVALUATION_CONTEXT_VARS"

6.453 USER_EVALUATION_CONTEXTS

USER_EVALUATION_CONTEXTS describes the rule evaluation contexts owned by the current user. Its columns (except for EVALUATION_CONTEXT_OWNER) are the same as those in ALL_EVALUATION_CONTEXTS.

See Also

"ALL_EVALUATION_CONTEXTS"

6.454 USER_EXPRESSION_STATISTICS

USER_EXPRESSION_STATISTICS provides expression usage tracking statistics for tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_EXPRESSION_STATISTICS.

See Also

"ALL_EXPRESSION_STATISTICS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.455 USER_EXTENTS

USER_EXTENTS describes the extents comprising the segments owned by the current user's objects. Its columns (except for OWNER, FILE_ID, BLOCK_ID, and RELATIVE_FNO) are the same as those in DBA_EXTENTS.

See Also

"DBA_EXTENTS"

6.456 USER_EXTERNAL_LOCATIONS

USER_EXTERNAL_LOCATIONS describes the locations (data sources) of the external tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_EXTERNAL_LOCATIONS.

See Also

"ALL_EXTERNAL_LOCATIONS"

6.457 USER_EXTERNAL_TABLES

USER_EXTERNAL_TABLES describes the external tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_EXTERNAL_TABLES.

See Also

"ALL_EXTERNAL_TABLES"

6.458 USER_FILE_GROUP_EXPORT_INFO

USER_FILE_GROUP_EXPORT_INFO shows export-related information for all file groups that the current user can manage. Its columns (except for FILE_GROUP_OWNER) are the same as those in ALL_FILE_GROUP_EXPORT_INFO.

See Also

"ALL_FILE_GROUP_EXPORT_INFO"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.459 USER_FILE_GROUP_FILES

USER_FILE_GROUP_FILES shows the file set for each versioned group owned by the current user. Its columns (except for FILE_GROUP_OWNER) are the same as those in ALL_FILE_GROUP_FILES.

See Also

"ALL_FILE_GROUP_FILES"

6.460 USER_FILE_GROUP_TABLES

USER_FILE_GROUP_TABLES shows information about tables owned by the current user that can be imported using the file set. Its columns (except for FILE_GROUP_OWNER) are the same as those in ALL_FILE_GROUP_TABLES.

See Also

"ALL_FILE_GROUP_TABLES"

6.461 USER_FILE_GROUP_TABLESPACES

USER_FILE_GROUP_TABLESPACES shows information about the transportable tablespaces present (partially or completely) in the file set owned by the current user (when the file set contains dump files). Its columns (except for FILE_GROUP_OWNER) are the same as those in ALL_FILE_GROUP_TABLESPACES.

See Also

"ALL_FILE_GROUP_TABLESPACES"

6.462 USER_FILE_GROUP_VERSIONS

USER_FILE_GROUP_VERSIONS shows top-level version information for all file groups owned by the current user. Its columns (except for FILE_GROUP_OWNER) are the same as those in ALL_FILE_GROUP_VERSIONS.

See Also

"ALL_FILE_GROUP_VERSIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.463 USER_FILE_GROUPS

USER_FILE_GROUPS shows top-level metadata about file groups owned by the current user. Its columns (except for FILE_GROUP_OWNER) are the same as those in ALL_FILE_GROUPS.

See Also

"ALL_FILE_GROUPS".

6.464 USER_FLASHBACK_ARCHIVE

USER_FLASHBACK_ARCHIVE describes flashback data archives, which consist of multiple tablespaces and historic data from all transactions against tracked tables.

The content of this view depends on the privileges of the user who queries it, as follows:

- If the user has the FLASHBACK ARCHIVE ADMINISTER system privilege, then USER_FLASHBACK_ARCHIVE describes the flashback archives for all users who have been granted the FLASHBACK ARCHIVE object privilege.
- If the user does not have the FLASHBACK ARCHIVE ADMINISTER system privilege, then USER_FLASHBACK_ARCHIVE describes flashback archives for which the current user has been granted the FLASHBACK ARCHIVE object privilege.

The columns of the USER_FLASHBACK_ARCHIVE view are the same as those in DBA_FLASHBACK_ARCHIVE.

See Also

"DBA_FLASHBACK_ARCHIVE"

6.465 USER_FLASHBACK_ARCHIVE_TABLES

USER_FLASHBACK_ARCHIVE_TABLES displays information about the tables owned by the current user that are enabled for Flashback Archive. Its columns are the same as those in DBA_FLASHBACK_ARCHIVE_TABLES.

See Also

"DBA_FLASHBACK_ARCHIVE_TABLES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.466 USER_FLASHBACK_TXN_REPORT

USER_FLASHBACK_TXN_REPORT displays information about the compensating transactions owned by the current user that have been committed in the database. Its columns (except for USERNAME) are the same as those in DBA_FLASHBACK_TXN_REPORT.

See Also

"DBA_FLASHBACK_TXN_REPORT"

6.467 USER_FLASHBACK_TXN_STATE

USER_FLASHBACK_TXN_STATE displays information about the compensating status of the transactions owned by the current user. Its columns (except for USERNAME) are the same as those in DBA_FLASHBACK_TXN_STATE.

See Also

"DBA_FLASHBACK_TXN_STATE"

6.468 USER_FREE_SPACE

USER_FREE_SPACE describes the free extents in the tablespaces accessible to the current user. Its columns are the same as those in DBA_FREE_SPACE.

See Also

"DBA_FREE_SPACE"

6.469 USER_GOLDENGATE_PRIVILEGES

USER_GOLDENGATE_PRIVILEGES displays details about Oracle GoldenGate privileges. Its columns (except for USERNAME) are the same as those in ALL_GOLDENGATE_PRIVILEGES.

See Also

"ALL_GOLDENGATE_PRIVILEGES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.470 USER_HEAT_MAP_SEG_HISTOGRAM

USER_HEAT_MAP_SEG_HISTOGRAM displays segment access information for segments owned by the user. Its columns (except for OWNER) are the same as those in ALL_HEAT_MAP_SEG_HISTOGRAM.

See Also

"ALL_HEAT_MAP_SEG_HISTOGRAM"

6.471 USER_HEAT_MAP_SEGMENT

USER_HEAT_MAP_SEGMENT displays the latest segment access time for all segments owned by the user. Its columns (except for OWNER) are the same as those in ALL_HEAT_MAP_SEGMENT.

See Also

"ALL_HEAT_MAP_SEGMENT"

6.472 USER_HIER_CLASS

USER_HIER_CLASS describes the classifications of the hierarchies owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_CLASS.

See Also

"ALL_HIER_CLASS"

6.473 USER_HIER_CLASS_AE

USER_HIER_CLASS_AE describes the classifications of the hierarchies (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_CLASS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "ALL_HIER_CLASS_AE"

6.474 USER_HIER_COLUMNS

USER_HIER_COLUMNS describes the columns of the hierarchies owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_COLUMNS.

> **See Also**
>
> "ALL_HIER_COLUMNS"

6.475 USER_HIER_COLUMNS_AE

USER_HIER_COLUMNS_AE describes the columns of the hierarchies (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_COLUMNS_AE.

> **Note**
>
> This view is available starting with Oracle Database 19c, Release Update 19.13.

> **See Also**
>
> "ALL_HIER_COLUMNS_AE"

6.476 USER_HIER_HIER_ATTR_CLASS

USER_HIER_HIER_ATTR_CLASS describes the classifications of the hierarchical attributes of the hierarchies owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_HIER_ATTR_CLASS.

> **See Also**
>
> "ALL_HIER_HIER_ATTR_CLASS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.477 USER_HIER_HIER_ATTR_CLASS_AE

USER_HIER_HIER_ATTR_CLASS_AE describes the classifications of the hierarchical attributes of the hierarchies (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_HIER_ATTR_CLASS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_HIER_HIER_ATTR_CLASS_AE"

6.478 USER_HIER_HIER_ATTRIBUTES

USER_HIER_HIER_ATTRIBUTES describes the hierarchical attributes of the hierarchies owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_HIER_ATTRIBUTES.

See Also

"ALL_HIER_HIER_ATTRIBUTES"

6.479 USER_HIER_HIER_ATTRIBUTES_AE

USER_HIER_HIER_ATTRIBUTES_AE describes the hierarchical attributes of the hierarchies (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_HIER_ATTRIBUTES_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_HIER_HIER_ATTRIBUTES_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.480 USER_HIER_JOIN_PATHS

USER_HIER_JOIN_PATHS describes the join paths for the hierarchies owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_JOIN_PATHS.

See Also

"ALL_HIER_JOIN_PATHS"

6.481 USER_HIER_JOIN_PATHS_AE

USER_HIER_JOIN_PATHS_AE describes the join paths for the hierarchies (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_JOIN_PATHS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_HIER_JOIN_PATHS_AE"

6.482 USER_HIER_LEVEL_ID_ATTRS

USER_HIER_LEVEL_ID_ATTRS describes the attributes that uniquely identify members of the levels of the hierarchies owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_LEVEL_ID_ATTRS.

See Also

"ALL_HIER_LEVEL_ID_ATTRS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.483 USER_HIER_LEVEL_ID_ATTRS_AE

USER_HIER_LEVEL_ID_ATTRS_AE describes the attributes that uniquely identify members of the levels of the hierarchies (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_LEVEL_ID_ATTRS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_HIER_LEVEL_ID_ATTRS_AE"

6.484 USER_HIER_LEVELS

USER_HIER_LEVELS describes the levels of the hierarchies owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_LEVELS.

See Also

"ALL_HIER_LEVELS"

6.485 USER_HIER_LEVELS_AE

USER_HIER_LEVELS_AE describes the levels of the hierarchies (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIER_LEVELS_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_HIER_LEVELS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.486 USER_HIERARCHIES

USER_HIERARCHIES describes the hierarchies owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIERARCHIES.

See Also

"ALL_HIERARCHIES"

6.487 USER_HIERARCHIES_AE

USER_HIERARCHIES_AE describes the hierarchies (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_HIERARCHIES_AE.

Note

This view is available starting with Oracle Database 19c, Release Update 19.13.

See Also

"ALL_HIERARCHIES_AE"

6.488 USER_HISTOGRAMS

USER_HISTOGRAMS is a synonym for USER_TAB_HISTOGRAMS.

See Also

"USER_TAB_HISTOGRAMS"

6.489 USER_HIVE_COLUMNS

USER_HIVE_COLUMNS describes all Hive columns owned by the current user in a Hive metastore. Its columns are the same as those in ALL_HIVE_COLUMNS.

See Also

"ALL_HIVE_COLUMNS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.490 USER_HIVE_DATABASES

USER_HIVE_DATABASES describes all the Hive schemas owned by the current user in a Hadoop cluster. Its columns are the same as those in ALL_HIVE_DATABASES.

See Also

"ALL_HIVE_DATABASES"

6.491 USER_HIVE_PART_KEY_COLUMNS

USER_HIVE_PART_KEY_COLUMNS provides information about all Hive table partition columns owned by the current user in the database. Its columns are the same as those in ALL_HIVE_PART_KEY_COLUMNS.

See Also

"ALL_HIVE_PART_KEY_COLUMNS"

6.492 USER_HIVE_TAB_PARTITIONS

USER_HIVE_TAB_PARTITIONS provides information about all Hive table partitions owned by the current user in the database. Its columns are the same as those in ALL_HIVE_TAB_PARTITIONS.

See Also

"ALL_HIVE_TAB_PARTITIONS"

6.493 USER_HIVE_TABLES

USER_HIVE_TABLES provides information about all the Hive tables owned by the current user in the Hive metastore. Its columns are the same as those in ALL_HIVE_TABLES.

See Also

"ALL_HIVE_TABLES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.494 USER_HOST_ACES

USER_HOST_ACES describes the status of access control entries for the current user to access network hosts through PL/SQL network utility packages.

Its columns (except for ACE_ORDER, START_DATE, END_DATE, GRANT_TYPE, INVERTED_PRINCIPAL, PRINCIPAL, PRINCIPAL_TYPE, and STATUS) are the same as those in DBA_HOST_ACES.

See Also

"DBA_HOST_ACES"

6.495 USER_IDENTIFIERS

USER_IDENTIFIERS displays information about the identifiers in the stored objects owned by the current user. Its columns (except for OWNER) are the same as those in ALL_IDENTIFIERS.

See Also

"ALL_IDENTIFIERS"

6.496 USER_ILMDATAMOVEMENTPOLICIES

USER_ILMDATAMOVEMENTPOLICIES contains information specific to data movement-related attributes of an Automatic Data Optimization policy for a user. Its columns are the same as those in DBA_ILMDATAMOVEMENTPOLICIES.

Note

Automatic Data Optimization is supported in Oracle Database 12*c* Release 2 multitenant environments.

See Also

"DBA_ILMDATAMOVEMENTPOLICIES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.497 USER_ILMEVALUATIONDETAILS

`USER_ILMEVALUATIONDETAILS` displays details on evaluation of Automatic Data Optimization policies considered for Automatic Data Optimization tasks for a user.

It also shows the job name that executes the policy, in case the policy was selected for execution. If the policy was not selected for execution, this view provides a reason. Its columns are the same as those in `DBA_ILMEVALUATIONDETAILS`.

> **Note**
>
> Automatic Data Optimization is supported in Oracle Database 12*c* Release 2 multitenant environments.

> **See Also**
>
> "DBA_ILMEVALUATIONDETAILS"

6.498 USER_ILMOBJECTS

`USER_ILMOBJECTS` displays all the Automatic Data Optimization policies and objects for a user.

Many objects inherit policies via their parent objects or because they were created in a particular tablespace. This view provides a mapping between the policies and objects and indicates whether a policy is inherited by an object or is directly specified on it. Its columns are the same as those in `DBA_ILMOBJECTS`.

> **Note**
>
> Automatic Data Optimization is supported in Oracle Database 12*c* Release 2 multitenant environments.

> **See Also**
>
> "DBA_ILMOBJECTS"

6.499 USER_ILMPOLICIES

`USER_ILMPOLICIES` displays details about Automatic Data Optimization policies owned by the user.

The view contains common details relevant to all types of Automatic Data Optimization policies, not just details relevant to the data movement-related Automatic Data Optimization policies. Its columns are the same as those in `DBA_ILMPOLICIES`.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> Automatic Data Optimization is supported in Oracle Database 12c Release 2 multitenant environments.

> **See Also**
>
> "DBA_ILMPOLICIES"

6.500 USER_ILMRESULTS

`USER_ILMRESULTS` displays information on data movement-related Automatic Data Optimization jobs for tasks created by the user.

Its columns are the same as those in `DBA_ILMRESULTS`.

> **Note**
>
> Automatic Data Optimization is supported in Oracle Database 12*c* Release 2 multitenant environments.

> **See Also**
>
> "DBA_ILMRESULTS"

6.501 USER_ILMTASKS

`USER_ILMTASKS` displays information on Automatic Data Optimization tasks created by a user. Its columns are the same as those in `DBA_ILMTASKS`.

> **Note**
>
> Automatic Data Optimization is supported in Oracle Database 12*c* Release 2 multitenant environments.

> **See Also**
>
> "DBA_ILMTASKS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.502 USER_IM_EXPRESSIONS

`USER_IM_EXPRESSIONS` provides information about the list of expressions (SYS_IME virtual columns) that are currently enabled for in-memory storage in schemas owned by the current user. Its columns (except for `OWNER`) are the same as those in `DBA_IM_EXPRESSIONS`.

Typically, you can query this view after invoking the `DBMS_INMEMORY_ADMIN.IME_CAPTURE_EXPRESSIONS` PL/SQL procedure to see the list of hot expressions added to tables owned by you across the database.

Based on this view, you can:

- Populate expressions on a particular table immediately
- Drop certain expressions that are marked for in-memory but not desired by you

See Also

"DBA_IM_EXPRESSIONS"

6.503 USER_IMMUTABLE_TABLES

`USER_IMMUTABLE_TABLES` describes the immutable tables owned by the current user. Its columns (except for `SCHEMA_NAME`) are the same as those in `ALL_IMMUTABLE_TABLES`.

Note

This view is available starting with Oracle Database 19c, Release Update 19.11.

See Also

"ALL_IMMUTABLE_TABLES"

6.504 USER_IND_COLUMNS

`USER_IND_COLUMNS` describes the columns of the indexes owned by the current user and columns of indexes on tables owned by the current user.

Its columns (except for `INDEX_OWNER` and `TABLE_OWNER`) are the same as those in "ALL_IND_COLUMNS".

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.505 USER_IND_EXPRESSIONS

USER_IND_EXPRESSIONS describes expressions of function-based indexes on tables owned by the current user. Its columns (except for INDEX_OWNER and TABLE_OWNER) are the same as those in ALL_IND_EXPRESSIONS.

See Also

"ALL_IND_EXPRESSIONS"

6.506 USER_IND_PARTITIONS

USER_IND_PARTITIONS displays, for each index partition owned by the current user, the partition-level partitioning information, the storage parameters for the partition, and various partition statistics generated by the DBMS_STATS package. Its columns are the same as those in ALL_IND_PARTITIONS.

See Also

"ALL_IND_PARTITIONS"

6.507 USER_IND_PENDING_STATS

USER_IND_PENDING_STATS describes pending statistics for all tables, partitions, and subpartitions owned by the current user and collected using the DBMS_STATS package. Its columns (except for OWNER) are the same as those in ALL_IND_PENDING_STATS.

See Also

"ALL_IND_PENDING_STATS"

6.508 USER_IND_STATISTICS

USER_IND_STATISTICS displays optimizer statistics for the indexes on the tables owned by the current user and collected using the DBMS_STATS package. Its columns (except for OWNER) are the same as those in ALL_IND_STATISTICS.

See Also

"ALL_IND_STATISTICS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.509 USER_IND_SUBPARTITIONS

USER_IND_SUBPARTITIONS displays, for each index subpartition owned by the current user, the subpartition-level partitioning information, the storage parameters for the subpartition, and various subpartition statistics generated by the DBMS_STATS package.

Its columns are the same as those in "ALL_IND_SUBPARTITIONS".

6.510 USER_INDEXES

USER_INDEXES describes indexes owned by the current user. To gather statistics for this view, use the DBMS_STATS package. This view supports parallel partitioned index scans. Its columns (except for OWNER and TABLE_TYPE) are the same as those in ALL_INDEXES.

See Also

"ALL_INDEXES"

6.511 USER_INDEXTYPE_ARRAYTYPES

USER_INDEXTYPE_ARRAYTYPES displays information about the array types specified by the indextypes owned by the current user. Its columns are the same as those in ALL_INDEXTYPE_ARRAYTYPES.

See Also

"ALL_INDEXTYPE_ARRAYTYPES"

6.512 USER_INDEXTYPE_COMMENTS

USER_INDEXTYPE_COMMENTS displays comments for the user-defined indextypes owned by the current user. Its columns are the same as those in ALL_INDEXTYPE_COMMENTS.

See Also

"ALL_INDEXTYPE_COMMENTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.513 USER_INDEXTYPE_OPERATORS

USER_INDEXTYPE_OPERATORS lists all the operators supported by indextypes owned by the current user. Its columns are the same as those in ALL_INDEXTYPE_OPERATORS.

See Also

"ALL_INDEXTYPE_OPERATORS"

6.514 USER_INDEXTYPES

USER_INDEXTYPES describes the indextypes owned by the current user. Its columns are the same as those in ALL_INDEXTYPES.

See Also

"ALL_INDEXTYPES"

6.515 USER_INTERNAL_TRIGGERS

USER_INTERNAL_TRIGGERS describes the internal triggers on all tables owned by the current user. Its columns are the same as those in ALL_INTERNAL_TRIGGERS.

See Also

"ALL_INTERNAL_TRIGGERS".

6.516 USER_JAVA_ARGUMENTS

USER_JAVA_ARGUMENTS displays argument information about the stored Java classes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_JAVA_ARGUMENTS.

See Also

"ALL_JAVA_ARGUMENTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.517 USER_JAVA_CLASSES

USER_JAVA_CLASSES displays class level information about the stored Java classes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_JAVA_CLASSES.

See Also

"ALL_JAVA_CLASSES"

6.518 USER_JAVA_COMPILER_OPTIONS

USER_JAVA_COMPILER_OPTIONS displays information about the native compiler options owned by the current user. Its columns (except for OWNER) are the same as those in ALL_JAVA_COMPILER_OPTIONS.

See Also

"ALL_JAVA_COMPILER_OPTIONS"

6.519 USER_JAVA_DERIVATIONS

USER_JAVA_DERIVATIONS displays mapping information about Java source objects and their derived Java class objects and Java resource objects for the Java classes owned by the current user.

Its columns (except for OWNER) are the same as those in ALL_JAVA_DERIVATIONS.

See Also

"ALL_JAVA_DERIVATIONS"

6.520 USER_JAVA_FIELDS

USER_JAVA_FIELDS displays field information about the stored Java classes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_JAVA_FIELDS.

See Also

"ALL_JAVA_FIELDS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.521 USER_JAVA_IMPLEMENTS

USER_JAVA_IMPLEMENTS describes interfaces implemented by the stored Java classes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_JAVA_IMPLEMENTS.

See Also

"ALL_JAVA_IMPLEMENTS"

6.522 USER_JAVA_INNERS

USER_JAVA_INNERS displays information about inner classes referred to by the stored Java classes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_JAVA_INNERS.

See Also

"ALL_JAVA_INNERS"

6.523 USER_JAVA_LAYOUTS

USER_JAVA_LAYOUTS displays class layout information about the stored Java classes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_JAVA_LAYOUTS.

See Also

"ALL_JAVA_LAYOUTS"

6.524 USER_JAVA_METHODS

USER_JAVA_METHODS displays method information about the stored Java classes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_JAVA_METHODS.

See Also

"ALL_JAVA_METHODS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.525 USER_JAVA_NCOMPS

USER_JAVA_NCOMPS displays ncomp-related information about the Java classes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_JAVA_NCOMPS.

See Also

"ALL_JAVA_NCOMPS"

6.526 USER_JAVA_POLICY

USER_JAVA_POLICY describes Java security permissions for the current user. Its columns are the same as those in DBA_JAVA_POLICY.

See Also

"DBA_JAVA_POLICY"

6.527 USER_JAVA_RESOLVERS

USER_JAVA_RESOLVERS displays information about resolvers of the Java classes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_JAVA_RESOLVERS.

See Also

"ALL_JAVA_RESOLVERS"

6.528 USER_JAVA_THROWS

USER_JAVA_THROWS displays information about exceptions thrown from methods of the Java classes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_JAVA_THROWS.

See Also

"ALL_JAVA_THROWS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.529 USER_JOBS

USER_JOBS describes the jobs owned by the current user. Its columns are the same as those in DBA_JOBS.

See Also

"DBA_JOBS"

6.530 USER_JOIN_IND_COLUMNS

USER_JOIN_IND_COLUMNS describes all join conditions owned by the current user. Its columns are the same as those in ALL_JOIN_IND_COLUMNS.

See Also

"ALL_JOIN_IND_COLUMNS"

6.531 USER_JOINGROUPS

USER_JOINGROUPS describes join groups belonging to the user. A join group is a user-created object that consists of two or more columns that can be meaningfully joined. The maximum number of columns that can be included in a join group is 255. The USER_JOINGROUPS columns (except for JOINGROUP_OWNER) are the same as those in DBA_JOINGROUPS.

In certain queries, join groups enable the database to eliminate the performance overhead of decompressing and hashing column values. Join groups require an In-Memory column store (IM column store).

See Also

- "DBA_JOINGROUPS"
- *Oracle Database In-Memory Guide* for an introduction to join groups
- *Oracle Database SQL Language Reference* for information about creating a join group using the CREATE INMEMORY JOIN GROUP statement

6.532 USER_JSON_COLUMNS

USER_JSON_COLUMNS provides information on the JavaScript Object Notation (JSON) columns for which the user is the owner. Its columns (except for OWNER) are the same as those in ALL_JSON_COLUMNS.

Each column owned by the user that has an IS JSON check constraint in an AND condition appears in this view. This view enables a user to find all the JSON columns that he or she owns.

Copyright © 2002, 2026, Oracle and/or its affiliates.

For example, if a check constraint combines the IS JSON condition with another condition using logical condition OR, then the column is not listed in this view. In this case, it is not certain that the data in the column is JSON data. For example, the following constraint does not ensure that the data in column jcol is JSON data:

```
jcol is json OR length(jcol) < 1000
```

See Also

- "ALL_JSON_COLUMNS"
- *Oracle Database JSON Developer's Guide* for more information about using JSON with Oracle Database

6.533 USER_JSON_DATAGUIDE_FIELDS

USER_JSON_DATAGUIDE_FIELDS extracts the path and type information from all the data guides in the current user's schema, which are the data guides returned to the user by the USER_JSON_DATAGUIDE view. Its columns (except for OWNER) are the same as those in ALL_JSON_DATAGUIDE_FIELDS.

See Also

"ALL_JSON_DATAGUIDE_FIELDS"

6.534 USER_JSON_DATAGUIDES

USER_JSON_DATAGUIDES provides information on the JavaScript Object Notation (JSON) columns owned by the current user that have data guide enabled. Its columns (except for OWNER) are the same as those in ALL_JSON_DATAGUIDES.

See Also

"ALL_JSON_DATAGUIDES"

6.535 USER_LIBRARIES

USER_LIBRARIES describes the libraries owned by the current user. Its columns (except for OWNER) are the same as those in ALL_LIBRARIES.

See Also

"ALL_LIBRARIES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.536 USER_LOB_PARTITIONS

USER_LOB_PARTITIONS displays the LOB partitions contained in the tables owned by the current user. Its columns are the same as those in ALL_LOB_PARTITIONS.

See Also

"ALL_LOB_PARTITIONS"

6.537 USER_LOB_SUBPARTITIONS

USER_LOB_SUBPARTITIONS displays partition-level attributes of the LOB data subpartitions owned by the current user. Its columns are the same as those in ALL_LOB_SUBPARTITIONS.

DBA_LOB_TEMPLATES describes all LOB subpartition templates in the database.

See Also

"ALL_LOB_SUBPARTITIONS"

6.538 USER_LOB_TEMPLATES

USER_LOB_TEMPLATES describes the LOB subpartition templates owned by the current user. Its columns (except for USER_NAME) are the same as those in ALL_LOB_TEMPLATES.

See Also

"ALL_LOB_TEMPLATES"

6.539 USER_LOBS

USER_LOBS displays the user's CLOBs and BLOBs contained in the user's tables. BFILEs are stored outside the database, so they are not described by this view. This view's columns are the same as those in ALL_LOBS.

See Also

"ALL_LOBS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.540 USER_LOG_GROUP_COLUMNS

USER_LOG_GROUP_COLUMNS describes columns that are owned by the current user and that are specified in log groups. Its columns are the same as those in ALL_LOG_GROUP_COLUMNS.

See Also

"ALL_LOG_GROUP_COLUMNS"

6.541 USER_LOG_GROUPS

USER_LOG_GROUPS describes log group definitions on tables owned by the current user. Its columns are the same as those in ALL_LOG_GROUPS.

See Also

"ALL_LOG_GROUPS"

6.542 USER_MEASURE_FOLDER_CONTENTS

USER_MEASURE_FOLDER_CONTENTS describes the contents of the OLAP measure folders owned by the current user. Its columns (except for OWNER) are the same as those in ALL_MEASURE_FOLDER_CONTENTS.

See Also

"ALL_MEASURE_FOLDER_CONTENTS"

6.543 USER_MEASURE_FOLDER_SUBFOLDERS

USER_MEASURE_FOLDER_SUBFOLDERS describes the OLAP measure folders contained within the OLAP measure folders owned by the current user. Its columns (except for OWNER) are the same as those in ALL_MEASURE_FOLDER_SUBFOLDERS.

See Also

"ALL_MEASURE_FOLDER_SUBFOLDERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.544 USER_MEASURE_FOLDERS

USER_MEASURE_FOLDERS describes the OLAP measure folders owned by the current user. Its columns (except for OWNER) are the same as those in ALL_MEASURE_FOLDERS.

See Also

"ALL_MEASURE_FOLDERS"

6.545 USER_METADATA_PROPERTIES

USER_METADATA_PROPERTIES describes OLAP metadata properties in the current user's schema. Its columns (except for OWNER) are the same as those in ALL_METADATA_PROPERTIES.

See Also

"ALL_METADATA_PROPERTIES"

6.546 USER_METHOD_PARAMS

USER_METHOD_PARAMS describes the method parameters of the object types owned by the current user. Its columns (except for OWNER) are the same as those in ALL_METHOD_PARAMS.

See Also

"ALL_METHOD_PARAMS"

6.547 USER_METHOD_RESULTS

USER_METHOD_RESULTS describes the method results of the object types owned by the current user. Its columns (except for OWNER) are the same as those in ALL_METHOD_RESULTS.

See Also

"ALL_METHOD_RESULTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.548 USER_MINING_MODEL_ATTRIBUTES

USER_MINING_MODEL_ATTRIBUTES describes the mining model attributes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_MINING_MODEL_ATTRIBUTES.

See Also

"ALL_MINING_MODEL_ATTRIBUTES"

6.549 USER_MINING_MODEL_PARTITIONS

USER_MINING_MODEL_PARTITIONS describes the user's own model partitions. Its columns (except OWNER) are the same as those in ALL_MINING_MODEL_PARTITIONS.

Note

The USER_MINING_MODEL_PARTITIONS view is available in Oracle Database 12*c* Release 2 and later.

See Also

"ALL_MINING_MODEL_PARTITIONS"

6.550 USER_MINING_MODEL_SETTINGS

USER_MINING_MODEL_SETTINGS describes the mining model settings owned by the current user. Its columns (except for OWNER) are the same as those in ALL_MINING_MODEL_SETTINGS.

See Also

"ALL_MINING_MODEL_SETTINGS"

6.551 USER_MINING_MODEL_VIEWS

USER_MINING_MODEL_VIEWS describes the user's own model views. Its columns (except OWNER) are the same as those in ALL_MINING_MODEL_VIEWS.

See Also

"ALL_MINING_MODEL_VIEWS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.552 USER_MINING_MODEL_XFORMS

USER_MINING_MODEL_XFORMS describes the user-specified transformations embedded with the user's own models. Its columns (except OWNER) are the same as those in ALL_MINING_MODEL_XFORMS.

> **See Also**
>
> "ALL_MINING_MODEL_XFORMS"

6.553 USER_MINING_MODELS

USER_MINING_MODELS describes the mining models owned by the current user. Its columns (except for OWNER) are the same as those in ALL_MINING_MODELS.

> **See Also**
>
> "ALL_MINING_MODELS"

6.554 USER_MVIEW_AGGREGATES

USER_MVIEW_AGGREGATES describes the grouping functions (aggregated measures) that appear in the SELECT list of aggregated materialized views owned by the current user. Its columns are the same as those in ALL_MVIEW_AGGREGATES.

> **See Also**
>
> "ALL_MVIEW_AGGREGATES"

6.555 USER_MVIEW_ANALYSIS

USER_MVIEW_ANALYSIS describes all materialized views owned by the current user that potentially support query rewrite and that provide additional information for analysis by applications. Its columns are the same as those in ALL_MVIEW_ANALYSIS.

> **Note**
>
> This view excludes materialized views that reference remote tables or that include references to non-static values such as SYSDATE or USER. This view also excludes materialized views that were created as snapshots before Oracle8*i* and that were never altered to enable query rewrite.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "ALL_MVIEW_ANALYSIS"

6.556 USER_MVIEW_COMMENTS

USER_MVIEW_COMMENTS displays comments on the materialized views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_MVIEW_COMMENTS.

> **See Also**
>
> "ALL_MVIEW_COMMENTS"

6.557 USER_MVIEW_DETAIL_PARTITION

USER_MVIEW_DETAIL_PARTITION displays freshness information, with respect to partition change tracking (PCT) detail partitions, for the materialized views owned by the current user. Its columns are the same as those in ALL_MVIEW_DETAIL_PARTITION.

> **See Also**
>
> "ALL_MVIEW_DETAIL_PARTITION"

6.558 USER_MVIEW_DETAIL_RELATIONS

USER_MVIEW_DETAIL_RELATIONS represents the named detail relations that are either in the FROM list of a materialized view, or that are indirectly referenced through views in the FROM list. Its columns are the same as those in ALL_MVIEW_DETAIL_RELATIONS.

> **See Also**
>
> "ALL_MVIEW_DETAIL_RELATIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.559 USER_MVIEW_DETAIL_SUBPARTITION

USER_MVIEW_DETAIL_SUBPARTITION displays freshness information, with respect to partition change tracking (PCT) detail subpartitions, for the materialized views owned by the current user. Its columns are the same as those in ALL_MVIEW_DETAIL_SUBPARTITION.

See Also

"ALL_MVIEW_DETAIL_SUBPARTITION"

6.560 USER_MVIEW_JOINS

USER_MVIEW_JOINS describes a join between two columns in the WHERE clause of a subquery that defines a materialized view. Its columns are the same as those in ALL_MVIEW_JOINS.

See Also

"ALL_MVIEW_JOINS"

6.561 USER_MVIEW_KEYS

USER_MVIEW_KEYS describes the columns or expressions in the SELECT list upon which materialized views in the current user's schema are based. Its columns are the same as those in ALL_MVIEW_KEYS.

See Also

"ALL_MVIEW_KEYS"

6.562 USER_MVIEW_LOGS

USER_MVIEW_LOGS describes all materialized view logs owned by the current user. Its columns are the same as those in ALL_MVIEW_LOGS.

See Also

"ALL_MVIEW_LOGS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.563 USER_MVIEW_REFRESH_TIMES

USER_MVIEW_REFRESH_TIMES describes refresh times of the materialized views owned by the current user. Its columns are the same as those in ALL_MVIEW_REFRESH_TIMES.

See Also

"ALL_MVIEW_REFRESH_TIMES"

6.564 USER_MVIEWS

USER_MVIEWS describes all materialized views owned by the current user. Its columns are the same as those in ALL_MVIEWS.

See Also

"ALL_MVIEWS"

6.565 USER_MVREF_CHANGE_STATS

USER_MVREF_CHANGE_STATS displays the change data load information on the master tables associated with a refresh run for all the materialized views in the database that are accessible to the current user. Its columns are the same as those in DBA_MVREF_CHANGE_STATS.

See Also

"DBA_MVREF_CHANGE_STATS"

6.566 USER_MVREF_RUN_STATS

USER_MVREF_RUN_STATS has information about each refresh run for the materialized views accessible for the current database user, with each run being identified by the REFRESH_ID. The information includes timing statistics related to the run and the parameters specified in that run.

Its columns (except for RUN_OWNER) are the same as those in DBA_MVREF_RUN_STATS.

See Also

"DBA_MVREF_RUN_STATS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.567 USER_MVREF_STATS

USER_MVREF_STATS shows the REFRESH_ID associated with each refresh run of each materialized view for the database that is accessible to the current user. It also provides some basic timing statistics related to that materialized view's refresh in that run.

Its columns (except for MV_OWNER) are the same as those in DBA_MVREF_STATS.

See Also

"DBA_MVREF_STATS"

6.568 USER_MVREF_STATS_PARAMS

USER_MVREF_STATS_PARAMS displays the refresh statistics properties associated with each materialized view accessible to the current user. These properties can be modified with the DBMS_MVIEW_STATS.SET_MVREF_STATS_PARAMS procedure.

Its columns are the same as those in DBA_MVREF_STATS_PARAMS.

See Also

"DBA_MVREF_STATS_PARAMS"

6.569 USER_MVREF_STATS_SYS_DEFAULTS

USER_MVREF_STATS_SYS_DEFAULTS displays the system-wide defaults for the refresh history statistics properties for materialized views accessible to the current user. These values can be altered with the SET_SYSTEM_DEFAULTS procedure by a database administrator.

Its columns are the same as those in DBA_MVREF_STATS_SYS_DEFAULTS.

This view contains exactly two rows corresponding to the collection-level and retention-period properties; their initial values are TYPICAL and 31 respectively.

See Also

"DBA_MVREF_STATS_SYS_DEFAULTS"

6.570 USER_MVREF_STMT_STATS

USER_MVREF_STMT_STATS shows information associated with each refresh statement of a materialized view accessible to the current user in a refresh run.

Its columns are the same as those in DBA_MVREF_STMT_STATS.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "DBA_MVREF_STMT_STATS"

6.571 USER_NESTED_TABLE_COLS

USER_NESTED_TABLE_COLS describes the columns of the nested tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_NESTED_TABLE_COLS.

To gather statistics for this view, use the DBMS_STATS package.

> **See Also**
>
> "ALL_NESTED_TABLE_COLS"

6.572 USER_NESTED_TABLES

USER_NESTED_TABLES describes the nested tables in tables owned by the current user. Its columns are the same as those in ALL_NESTED_TABLES.

> **See Also**
>
> "ALL_NESTED_TABLES"

6.573 USER_NETWORK_ACL_PRIVILEGES

USER_NETWORK_ACL_PRIVILEGES describes the status of the network privileges for the current user to access network hosts.

> **Note**
>
> This USER_NETWORK_ACL_PRIVILEGES view is deprecated in Oracle Database 12*c* Release 1 (12.1). Oracle recommends that you use the USER_HOST_ACES view, instead.

Column	Datatype	NULL	Description
HOST	VARCHAR2(1000)	NOT NULL	Network host
LOWER_PORT	NUMBER(5)		Lower bound of the port range
UPPER_PORT	NUMBER(5)		Upper bound of the port range
PRIVILEGE	CHAR(128)		Network privilege
STATUS	VARCHAR2(7)		Privilege status: • DENIED • GRANTED

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"USER_HOST_ACES"

6.574 USER_OBJ_AUDIT_OPTS

USER_OBJ_AUDIT_OPTS describes auditing options on all objects owned by the current user. Its columns (except for OWNER) are the same as those in DBA_OBJ_AUDIT_OPTS.

Note

This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from UNIFIED_AUDIT_TRAIL.

- See *Oracle Database Security Guide* for more information about unified auditing.
- See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

See Also

"DBA_OBJ_AUDIT_OPTS"

6.575 USER_OBJ_COLATTRS

USER_OBJ_COLATTRS describes object columns and attributes contained in the tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_OBJ_COLATTRS.

See Also

"ALL_OBJ_COLATTRS"

6.576 USER_OBJECT_SIZE

USER_OBJECT_SIZE lists the sizes, in bytes, of various PL/SQL objects. Its columns are the same as those in DBA_OBJECT_SIZE.

See Also

"DBA_OBJECT_SIZE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.577 USER_OBJECT_TABLES

USER_OBJECT_TABLES describes the object tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_OBJECT_TABLES.

See Also

"ALL_OBJECT_TABLES"

6.578 USER_OBJECT_USAGE

USER_OBJECT_USAGE displays statistics about index usage gathered from the database for the indexes owned by the current user.

You can use this view to monitor index usage. All indexes owned by the current user that have been used at least once can be monitored and displayed in this view. Its columns (except for OWNER) are the same as those in DBA_OBJECT_USAGE.

See Also

"DBA_OBJECT_USAGE"

6.579 USER_OBJECTS

USER_OBJECTS describes all objects owned by the current user. Its columns (except for OWNER), are the same as those in ALL_OBJECTS.

See Also

"ALL_OBJECTS"

6.580 USER_OBJECTS_AE

USER_OBJECTS_AE describes the objects (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_OBJECTS_AE.

See Also

"ALL_OBJECTS_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.581 USER_OPANCILLARY

USER_OPANCILLARY provides ancillary information for operators owned by the current user. Its columns are the same as those in ALL_OPANCILLARY.

See Also

"ALL_OPANCILLARY"

6.582 USER_OPARGUMENTS

USER_OPARGUMENTS provides argument information for operator bindings owned by the current user. Its columns are the same as those in ALL_OPARGUMENTS.

See Also

"ALL_OPARGUMENTS"

6.583 USER_OPBINDINGS

USER_OPBINDINGS describes the binding functions and methods on the operators owned by the current user. Its columns are the same as those in ALL_OPBINDINGS.

See Also

"ALL_OPBINDINGS"

6.584 USER_OPERATOR_COMMENTS

USER_OPERATOR_COMMENTS displays comments for the user-defined operators owned by the current user. Its columns are the same as those in ALL_OPERATOR_COMMENTS.

See Also

"ALL_OPERATOR_COMMENTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.585 USER_OPERATORS

USER_OPERATORS describes all operators owned by the current user. Its columns are the same as those in ALL_OPERATORS.

See Also

"ALL_OPERATORS"

6.586 USER_OUTLINE_HINTS

USER_OUTLINE_HINTS describes the set of hints stored in the outlines owned by the current user. Its columns (except for OWNER) are the same as those in DBA_OUTLINE_HINTS.

See Also

"DBA_OUTLINE_HINTS"

6.587 USER_OUTLINES

USER_OUTLINES describes the stored outlines owned by the current user. Its columns (except for OWNER) are the same as those in DBA_OUTLINES.

See Also

"DBA_OUTLINES"

6.588 USER_PARALLEL_EXECUTE_CHUNKS

USER_PARALLEL_EXECUTE_CHUNKS displays the chunks for tasks created by the current user. Its columns (except for TASK_OWNER) are the same as those in DBA_PARALLEL_EXECUTE_CHUNKS.

See Also

"DBA_PARALLEL_EXECUTE_CHUNKS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.589 USER_PARALLEL_EXECUTE_TASKS

USER_PARALLEL_EXECUTE_TASKS displays the tasks created by the current user. Its columns (except for TASK_OWNER) are the same as those in DBA_PARALLEL_EXECUTE_TASKS.

See Also

"DBA_PARALLEL_EXECUTE_TASKS"

6.590 USER_PART_COL_STATISTICS

USER_PART_COL_STATISTICS displays column statistics and histogram information for the table partitions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_PART_COL_STATISTICS.

See Also

"ALL_PART_COL_STATISTICS"

6.591 USER_PART_HISTOGRAMS

USER_PART_HISTOGRAMS displays the histogram data (endpoints per histogram) for the histograms on the table partitions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_PART_HISTOGRAMS.

See Also

"ALL_PART_HISTOGRAMS"

6.592 USER_PART_INDEXES

USER_PART_INDEXES displays the object-level partitioning information for the partitioned indexes owned by the current user. Its columns (except for OWNER) are the same as those in ALL_PART_INDEXES.

See Also

"ALL_PART_INDEXES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.593 USER_PART_KEY_COLUMNS

USER_PART_KEY_COLUMNS describes the partitioning key columns for the partitioned objects owned by the current user. Its columns (except for OWNER) are the same as those in ALL_PART_KEY_COLUMNS.

See Also

"ALL_PART_KEY_COLUMNS"

6.594 USER_PART_LOBS

USER_PART_LOBS displays table-level information about the partitioned LOBs owned by the current user, including default attributes for LOB data partitions. Its columns (except for TABLE_OWNER) are the same as those in ALL_PART_LOBS.

See Also

"ALL_PART_LOBS"

6.595 USER_PART_TABLES

USER_PART_TABLES displays the object-level partitioning information for the partitioned tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_PART_TABLES.

See Also

"ALL_PART_TABLES"

6.596 USER_PARTIAL_DROP_TABS

USER_PARTIAL_DROP_TABS describes all tables in the schema of the current user that have partially completed DROP COLUMN operations. Its columns are the same as those in ALL_PARTIAL_DROP_TABS.

See Also

"ALL_PARTIAL_DROP_TABS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.597 USER_PASSWORD_LIMITS

USER_PASSWORD_LIMITS describes the password profile parameters that are assigned to the user.

Column	Datatype	NULL	Description
RESOURCE_NAME	VARCHAR2(32)	NOT NULL	Name of the password resource
LIMIT	VARCHAR2(40)		Value of the resource limit

6.598 USER_PENDING_CONV_TABLES

USER_PENDING_CONV_TABLES describes the pending conversion tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_PENDING_CONV_TABLES.

See Also

"ALL_PENDING_CONV_TABLES"

6.599 USER_PLSQL_COLL_TYPES

USER_PLSQL_COLL_TYPES describes the user's own named PL/SQL collection types. Its columns (except for OWNER and CHAR_USED) are the same as those in ALL_PLSQL_COLL_TYPES.

See Also

"ALL_PLSQL_COLL_TYPES"

6.600 USER_PLSQL_OBJECT_SETTINGS

USER_PLSQL_OBJECT_SETTINGS displays information about the compiler settings for the stored objects owned by the current user. Its columns (except for OWNER) are the same as those in ALL_PLSQL_OBJECT_SETTINGS.

See Also

"ALL_PLSQL_OBJECT_SETTINGS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.601 USER_PLSQL_TYPE_ATTRS

USER_PLSQL_TYPE_ATTRS describes the attributes of the user's own PL/SQL types. Its columns (except for OWNER and CHAR_USED) are the same as those in ALL_PLSQL_TYPE_ATTRS.

See Also

"ALL_PLSQL_TYPE_ATTRS"

6.602 USER_PLSQL_TYPES

USER_PLSQL_TYPES describes the user's own PL/SQL types. Its columns (except for OWNER) are the same as those in ALL_PLSQL_TYPES.

See Also

"ALL_PLSQL_TYPES"

6.603 USER_POLICIES

USER_POLICIES describes all Oracle Virtual Private Database (VPD) security policies associated with objects owned by the current user. Its columns (except for OBJECT_OWNER) are the same as those in ALL_POLICIES.

See Also

"ALL_POLICIES"

6.604 USER_POLICY_ATTRIBUTES

USER_POLICY_ATTRIBUTES lists the attribute associations {Namespaces, Attributes} of all context-sensitive and shared-context sensitive Oracle Virtual Private Database (VPD) policies for synonyms, tables, or views owned by the user.

Its columns are the same as those in ALL_POLICY_ATTRIBUTES.

See Also

"ALL_POLICY_ATTRIBUTES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.605 USER_POLICY_CONTEXTS

USER_POLICY_CONTEXTS describes the driving contexts defined for the synonyms, tables, and views owned by the current user. Its columns (except for OBJECT_OWNER) are the same as those in ALL_POLICY_CONTEXTS.

See Also

"ALL_POLICY_CONTEXTS"

6.606 USER_POLICY_GROUPS

USER_POLICY_GROUPS describes the policy groups defined for the synonyms, tables, and views owned by the current user. Its columns (except for OBJECT_OWNER) are the same as those in ALL_POLICY_GROUPS.

See Also

"ALL_POLICY_GROUPS"

6.607 USER_PRIVATE_TEMP_TABLES

USER_PRIVATE_TEMP_TABLES describes the private temporary tables in the current session. Its columns (except for INST_ID) are the same as those in DBA_PRIVATE_TEMP_TABLES.

See Also

"DBA_PRIVATE_TEMP_TABLES"

6.608 USER_PRIVILEGE_MAP

USER_PRIVILEGE_MAP shows privilege (auditing option) type codes for object privileges that can be granted on a user.

This table can be used to map privilege type numbers to type names.

Column	Datatype	NULL	Description
PRIVILEGE	NUMBER	NOT NULL	A numeric privilege (auditing option) type code
NAME	VARCHAR2(40)	NOT NULL	Name of the type of privilege (auditing option)

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.609 USER_PROCEDURES

USER_PROCEDURES lists all functions and procedures that are owned by the current user, along with their associated properties. Its columns (except OWNER) are the same as those in ALL_PROCEDURES.

See Also

- "ALL_PROCEDURES"
- "USER_ARGUMENTS" for information about the arguments of the functions and procedures owned by the current user

6.610 USER_PROXIES

USER_PROXIES displays information about connections the current user is allowed to proxy. Its columns (except for PROXY) are the same as those in DBA_PROXIES.

See Also

"DBA_PROXIES"

6.611 USER_QUEUE_SCHEDULES

USER_QUEUE_SCHEDULES describes the propagation schedules whose source queues are owned by the current user. Its columns (except for SCHEMA) are the same as those in ALL_QUEUE_SCHEDULES.

See Also

"ALL_QUEUE_SCHEDULES"

6.612 USER_QUEUE_SUBSCRIBERS

USER_QUEUE_SUBSCRIBERS displays the list of subscribers on queues that are under the current user's schema. Its columns (except for OWNER) are the same as those in ALL_QUEUE_SUBSCRIBERS.

See Also

"ALL_QUEUE_SUBSCRIBERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.613 USER_QUEUE_TABLES

USER_QUEUE_TABLES describes the queues in the queue tables created in the current user's schema. Its columns (except for OWNER) are the same as those in ALL_QUEUE_TABLES.

See Also

"ALL_QUEUE_TABLES"

6.614 USER_QUEUES

USER_QUEUES describes the operational characteristics of every queue in the user's schema. Its columns (except for OWNER) are the same as those in ALL_QUEUES.

See Also

- "ALL_QUEUES"
- *Oracle Database Advanced Queuing User's Guide* for more information about these views and Advanced Queuing

6.615 USER_RECYCLEBIN

USER_RECYCLEBIN displays information about the recycle bin owned by the current user. Its columns (except for OWNER) are the same as those in DBA_RECYCLEBIN.

See Also

"DBA_RECYCLEBIN"

6.616 USER_REFRESH

USER_REFRESH describes all refresh groups owned by the current user. Its columns are the same as those in ALL_REFRESH.

See Also

"ALL_REFRESH"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.617 USER_REFRESH_CHILDREN

USER_REFRESH_CHILDREN lists all the objects in refresh groups owned by the current user. Its columns are the same as those in ALL_REFRESH_CHILDREN.

See Also

"ALL_REFRESH_CHILDREN"

6.618 USER_REFS

USER_REFS describes the REF columns and REF attributes in the object type columns of tables owned by the current user. Its columns are the same as those in ALL_REFS.

See Also

"ALL_REFS"

6.619 USER_REGISTERED_MVIEWS

USER_REGISTERED_MVIEWS describes all registered materialized views (registered at a master site or a master materialized view site) owned by the current user. Its columns are the same as those in ALL_REGISTERED_MVIEWS.

See Also

"ALL_REGISTERED_MVIEWS"

6.620 USER_REGISTRY

USER_REGISTRY displays information about the components owned by the current user that are loaded into the component registry. Its columns are the same as those in DBA_REGISTRY.

See Also

"DBA_REGISTRY"

6.621 USER_RESOURCE_LIMITS

USER_RESOURCE_LIMITS displays the resource limits for the current user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
RESOURCE_NAME	VARCHAR2(32)	NOT NULL	Name of the resource
LIMIT	VARCHAR2(40)		Limit placed on this resource

6.622 USER_RESUMABLE

USER_RESUMABLE displays the resumable statements executed by the current user. Its columns (except for USER_ID) are the same as those in DBA_RESUMABLE.

See Also

"DBA_RESUMABLE"

6.623 USER_REWRITE_EQUIVALENCES

USER_REWRITE_EQUIVALENCES describes the rewrite equivalences owned by the current user. Its columns are the same as those in ALL_REWRITE_EQUIVALENCES.

See Also

"ALL_REWRITE_EQUIVALENCES"

6.624 USER_ROLE_PRIVS

USER_ROLE_PRIVS describes the roles granted to the current user.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(128)		Name of the user, or PUBLIC
GRANTED_ROLE	VARCHAR2(128)		Name of the role granted to the user
ADMIN_OPTION	VARCHAR2(3)		Indicates whether the grant was with the ADMIN OPTION (YES) or not (NO)
DELEGATE_OPTION	VARCHAR2(3)		Indicates whether the grant was with the DELEGATE OPTION (YES) or not (NO)
DEFAULT_ROLE	VARCHAR2(3)		Indicates whether the role is designated as a DEFAULT ROLE for the user (YES) or not (NO)
OS_GRANTED	VARCHAR2(3)		Indicates whether the role was granted by the operating system (YES) or not (NO); occurs if the OS_ROLES initialization parameter is true
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the role was granted commonly (CONTAINER=ALL was used) • NO if the role was granted locally (CONTAINER=ALL was not used)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
INHERITED	VARCHAR2(3)		Indicates whether the grant was inherited from another container (YES) or not (NO)

See Also

"DBA_ROLE_PRIVS"

6.625 USER_RSRC_CONSUMER_GROUP_PRIVS

USER_RSRC_CONSUMER_GROUP_PRIVS displays information about the resource consumer groups to which the current user is assigned. Its columns (except for GRANTEE) are the same as those in DBA_RSRC_CONSUMER_GROUP_PRIVS.

See Also

"DBA_RSRC_CONSUMER_GROUP_PRIVS"

6.626 USER_RSRC_MANAGER_SYSTEM_PRIVS

USER_RSRC_MANAGER_SYSTEM_PRIVS displays information about the users who are granted system privileges for the DBMS_RESOURCE_MANAGER package. Its columns (except for GRANTEE) are the same as those in DBA_RSRC_MANAGER_SYSTEM_PRIVS.

See Also

"DBA_RSRC_MANAGER_SYSTEM_PRIVS"

6.627 USER_RULE_SET_RULES

USER_RULE_SET_RULES describes the rules in the rule sets owned by the current user. Its columns (except for RULE_SET_OWNER) are the same as those in ALL_RULE_SET_RULES.

See Also

"ALL_RULE_SET_RULES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.628 USER_RULE_SETS

USER_RULE_SETS describes the rule sets owned by the current user. Its columns (except for RULE_SET_OWNER) are the same as those in ALL_RULE_SETS.

See Also

"ALL_RULE_SETS"

6.629 USER_RULES

USER_RULES describes the rules owned by the current user. Its columns (except for RULE_OWNER) are the same as those in ALL_RULES.

See Also

"ALL_RULES"

6.630 USER_SCHEDULER_CHAIN_RULES

USER_SCHEDULER_CHAIN_RULES displays information about the rules for the chains owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_CHAIN_RULES.

See Also

"ALL_SCHEDULER_CHAIN_RULES"

6.631 USER_SCHEDULER_CHAIN_STEPS

USER_SCHEDULER_CHAIN_STEPS displays information about the defined steps of the chains owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_CHAIN_STEPS.

See Also

"ALL_SCHEDULER_CHAIN_STEPS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.632 USER_SCHEDULER_CHAINS

USER_SCHEDULER_CHAINS displays information about the chains owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_CHAINS.

See Also

"ALL_SCHEDULER_CHAINS"

6.633 USER_SCHEDULER_CREDENTIALS

USER_SCHEDULER_CREDENTIALS displays information about the credentials owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_CREDENTIALS.

Note

This view is deprecated in favor of the USER_CREDENTIALS view. Oracle recommends that you use USER_CREDENTIALS instead. USER_SCHEDULER_CREDENTIALS is retained for backward compatibility only.

See Also

- "USER_CREDENTIALS"
- "ALL_SCHEDULER_CREDENTIALS"

6.634 USER_SCHEDULER_DB_DESTS

USER_SCHEDULER_DB_DESTS displays information about the destination objects owned by the current user pointing to remote databases. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_DB_DESTS.

See Also

"ALL_SCHEDULER_DB_DESTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.635 USER_SCHEDULER_DESTS

USER_SCHEDULER_DESTS displays information about the destination objects for jobs owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_DESTS.

See Also

"ALL_SCHEDULER_DESTS"

6.636 USER_SCHEDULER_FILE_WATCHERS

USER_SCHEDULER_FILE_WATCHERS displays information about the Scheduler file watch requests owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_FILE_WATCHERS.

See Also

"ALL_SCHEDULER_FILE_WATCHERS"

6.637 USER_SCHEDULER_GROUP_MEMBERS

USER_SCHEDULER_GROUP_MEMBERS displays information about the members of the Scheduler object groups owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_GROUP_MEMBERS.

See Also

"ALL_SCHEDULER_GROUP_MEMBERS"

6.638 USER_SCHEDULER_GROUPS

USER_SCHEDULER_GROUPS displays information about the Scheduler object groups owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_GROUPS.

See Also

"ALL_SCHEDULER_GROUPS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.639 USER_SCHEDULER_INCOMPAT_MEMBER

USER_SCHEDULER_INCOMPAT_MEMBER displays all Scheduler incompatibility resource objects members owned by the current user. Its columns are the same as those in ALL_SCHEDULER_INCOMPAT_MEMBER.

See Also

"ALL_SCHEDULER_INCOMPAT_MEMBER"

6.640 USER_SCHEDULER_INCOMPATS

USER_SCHEDULER_INCOMPATS displays all Scheduler incompatibility resource objects owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_INCOMPATS.

See Also

"ALL_SCHEDULER_INCOMPATS"

6.641 USER_SCHEDULER_JOB_ARGS

USER_SCHEDULER_JOB_ARGS displays information about the arguments of the Scheduler jobs owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_JOB_ARGS.

See Also

"ALL_SCHEDULER_JOB_ARGS"

6.642 USER_SCHEDULER_JOB_DESTS

USER_SCHEDULER_JOB_DESTS displays information about the state of the jobs owned by the current user at each of their destinations. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_JOB_DESTS.

See Also

"ALL_SCHEDULER_JOB_DESTS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.643 USER_SCHEDULER_JOB_LOG

USER_SCHEDULER_JOB_LOG displays log information for the Scheduler jobs owned by the current user. Its columns are the same as those in ALL_SCHEDULER_JOB_LOG.

See Also

"ALL_SCHEDULER_JOB_LOG"

6.644 USER_SCHEDULER_JOB_RUN_DETAILS

USER_SCHEDULER_JOB_RUN_DETAILS displays log run details for the Scheduler jobs owned by the current user. Its columns are the same as those in ALL_SCHEDULER_JOB_RUN_DETAILS.

See Also

"ALL_SCHEDULER_JOB_RUN_DETAILS"

6.645 USER_SCHEDULER_JOBS

USER_SCHEDULER_JOBS displays information about the Scheduler jobs owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_JOBS.

See Also

"ALL_SCHEDULER_JOBS"

6.646 USER_SCHEDULER_NOTIFICATIONS

USER_SCHEDULER_NOTIFICATIONS displays information about the E-mail notifications for the jobs owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_NOTIFICATIONS.

See Also

"ALL_SCHEDULER_NOTIFICATIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.647 USER_SCHEDULER_PROGRAM_ARGS

USER_SCHEDULER_PROGRAM_ARGS displays information about the arguments of the Scheduler programs owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_PROGRAM_ARGS.

See Also

"ALL_SCHEDULER_PROGRAM_ARGS"

6.648 USER_SCHEDULER_PROGRAMS

USER_SCHEDULER_PROGRAMS displays information about the Scheduler programs owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_PROGRAMS.

See Also

"ALL_SCHEDULER_PROGRAMS"

6.649 USER_SCHEDULER_REMOTE_JOBSTATE

USER_SCHEDULER_REMOTE_JOBSTATE displays information about the state of the jobs owned by the current user at remote databases. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_REMOTE_JOBSTATE.

See Also

"ALL_SCHEDULER_REMOTE_JOBSTATE"

6.650 USER_SCHEDULER_RESOURCES

USER_SCHEDULER_RESOURCES displays all scheduler resource objects in the database from the schema of the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_RESOURCES.

See Also

"ALL_SCHEDULER_RESOURCES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.651 USER_SCHEDULER_RSC_CONSTRAINTS

USER_SCHEDULER_RSC_CONSTRAINTS lists all Oracle Scheduler resource constraint members owned by the current user. Its columns are the same as those in ALL_SCHEDULER_RSC_CONSTRAINTS.

See Also

"ALL_SCHEDULER_RSC_CONSTRAINTS"

6.652 USER_SCHEDULER_RUNNING_CHAINS

USER_SCHEDULER_RUNNING_CHAINS displays information about the chain steps of the running chains owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_RUNNING_CHAINS.

See Also

"ALL_SCHEDULER_RUNNING_CHAINS"

6.653 USER_SCHEDULER_RUNNING_JOBS

USER_SCHEDULER_RUNNING_JOBS displays information about the running Scheduler jobs owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_RUNNING_JOBS.

See Also

"ALL_SCHEDULER_RUNNING_JOBS"

6.654 USER_SCHEDULER_SCHEDULES

USER_SCHEDULER_SCHEDULES displays information about the Scheduler schedules owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SCHEDULER_SCHEDULES.

See Also

"ALL_SCHEDULER_SCHEDULES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.655 USER_SEC_RELEVANT_COLS

USER_SEC_RELEVANT_COLS describes the security relevant columns of the security policies for the tables and views owned by the current user. Its columns (except for OBJECT_OWNER) are the same as those in ALL_SEC_RELEVANT_COLS.

See Also

"ALL_SEC_RELEVANT_COLS"

6.656 USER_SECONDARY_OBJECTS

USER_SECONDARY_OBJECTS provides information about secondary objects associated with domain indexes owned by the current user. Its columns are the same as those in ALL_SECONDARY_OBJECTS.

This view is only relevant in the context of domain indexes.

See Also

"ALL_SECONDARY_OBJECTS"

6.657 USER_SEGMENTS

USER_SEGMENTS describes the storage allocated for the segments owned by the current user's objects. Its columns (except for OWNER, HEADER_FILE, HEADER_BLOCK, and RELATIVE_FNO) are the same as those in DBA_SEGMENTS.

See Also

"DBA_SEGMENTS"

6.658 USER_SEQUENCES

USER_SEQUENCES describes all sequences owned by the current user. Its columns are the same as those in ALL_SEQUENCES.

See Also

"ALL_SEQUENCES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.659 USER_SODA_COLLECTIONS

USER_SODA_COLLECTIONS describes the Simple Oracle Document Access (SODA) collections owned by the current user. Its columns (except for OWNER) are the same as those in DBA_SODA_COLLECTIONS.

> **Note**
>
> The USER_SODA_COLLECTIONS view is available in on-premises Oracle databases and Oracle Autonomous Database. However, the DBA_SODA_COLLECTIONS view is available only in Oracle Autonomous Database.

> **See Also**
>
> "DBA_SODA_COLLECTIONS"

6.660 USER_SOURCE

USER_SOURCE describes the text source of the stored objects owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SOURCE.

> **See Also**
>
> "ALL_SOURCE"

6.661 USER_SOURCE_AE

USER_SOURCE_AE describes the text source of the stored objects (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SOURCE_AE.

> **See Also**
>
> "ALL_SOURCE_AE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.662 USER_SQL_TRANSLATION_PROFILES

USER_SQL_TRANSLATION_PROFILES describes all SQL translation profiles owned by the user. Its columns (except for OWNER) are the same as those in ALL_SQL_TRANSLATION_PROFILES.

See Also

"ALL_SQL_TRANSLATION_PROFILES"

6.663 USER_SQL_TRANSLATIONS

USER_SQL_TRANSLATIONS describes all SQL translations owned by the user. Its columns (except for OWNER) are the same as those in ALL_SQL_TRANSLATIONS.

See Also

"ALL_SQL_TRANSLATIONS"

6.664 USER_SQLJ_TYPE_ATTRS

USER_SQLJ_TYPE_ATTRS describes the attributes of the SQLJ object types owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SQLJ_TYPE_ATTRS.

See Also

"ALL_SQLJ_TYPE_ATTRS"

6.665 USER_SQLJ_TYPE_METHODS

USER_SQLJ_TYPE_METHODS describes the methods of the SQLJ object types owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SQLJ_TYPE_METHODS.

See Also

"ALL_SQLJ_TYPE_METHODS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.666 USER_SQLJ_TYPES

USER_SQLJ_TYPES describes the SQLJ object types owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SQLJ_TYPES.

See Also

"ALL_SQLJ_TYPES"

6.667 USER_SQLSET

USER_SQLSET displays information about the SQL tuning sets owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SQLSET.

See Also

"ALL_SQLSET"

6.668 USER_SQLSET_BINDS

USER_SQLSET_BINDS displays the bind values associated with the SQL tuning sets owned by the current user. Its columns (except for SQLSET_OWNER) are the same as those in ALL_SQLSET_BINDS.

See Also

"ALL_SQLSET_BINDS"

6.669 USER_SQLSET_PLANS

USER_SQLSET_PLANS describes captured plans for statements in the SQL tuning sets owned by the current user. Its columns (except for SQLSET_OWNER) are the same as those in ALL_SQLSET_PLANS.

See Also

"ALL_SQLSET_PLANS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.670 USER_SQLSET_REFERENCES

USER_SQLSET_REFERENCES describes whether or not the SQL tuning sets owned by the current user are active. Its columns (except for SQLSET_OWNER) are the same as those in ALL_SQLSET_REFERENCES.

See Also

"ALL_SQLSET_REFERENCES"

6.671 USER_SQLSET_STATEMENTS

USER_SQLSET_STATEMENTS displays information about the SQL statements, along with their statistics, that form the SQL tuning sets owned by the current user. Its columns (except for SQLSET_OWNER) are the same as those in ALL_SQLSET_STATEMENTS.

See Also

"ALL_SQLSET_STATEMENTS"

6.672 USER_SQLTUNE_BINDS

USER_SQLTUNE_BINDS displays the bind values associated with the tuned SQL statements owned by the current user. Its columns are the same as those in DBA_SQLTUNE_BINDS.

See Also

"DBA_SQLTUNE_BINDS"

6.673 USER_SQLTUNE_PLANS

USER_SQLTUNE_PLANS displays information about the execution plans generated for the SQL statements owned by the current user during a SQL tuning session. Its columns are the same as those in DBA_SQLTUNE_PLANS.

See Also

"DBA_SQLTUNE_PLANS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.674 USER_SQLTUNE_RATIONALE_PLAN

USER_SQLTUNE_RATIONALE_PLAN displays the association between rationales and operations in the execution plan of the SQL statements owned by the current user. Its columns are the same as those in DBA_SQLTUNE_RATIONALE_PLAN.

See Also

"DBA_SQLTUNE_RATIONALE_PLAN"

6.675 USER_SQLTUNE_STATISTICS

USER_SQLTUNE_STATISTICS displays statistics associated with the SQL statements owned by the current user. Its columns are the same as those in DBA_SQLTUNE_STATISTICS.

See Also

"DBA_SQLTUNE_STATISTICS"

6.676 USER_SR_GRP_STATUS

USER_SR_GRP_STATUS provides information on the current refresh operations for the current synchronous refresh groups in the database which are owned by the current user. Its columns are the same as those in DBA_SR_GRP_STATUS.

See Also

"DBA_SR_GRP_STATUS"

6.677 USER_SR_GRP_STATUS_ALL

USER_SR_GRP_STATUS_ALL provides information on the refresh operations on the synchronous refresh groups in the database which are owned by the current user. Its columns are the same as those in DBA_SR_GRP_STATUS_ALL.

See Also

"DBA_SR_GRP_STATUS_ALL"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.678 USER_SR_OBJ

USER_SR_OBJ provides information on the objects currently registered for synchronous refresh for current groups for the current user. Its columns are the same as those in DBA_SR_OBJ.

See Also

"DBA_SR_OBJ"

6.679 USER_SR_OBJ_ALL

USER_SR_OBJ_ALL provides information on the objects registered for synchronous refresh for current and defunct groups for the current user. Its columns are the same as those in DBA_SR_OBJ_ALL.

See Also

"DBA_SR_OBJ_ALL"

6.680 USER_SR_OBJ_STATUS

USER_SR_OBJ_STATUS provides information on the status of objects registered for synchronous refresh for the current refresh operations for the current synchronous refresh groups in the database which are owned by the current user. Its columns are the same as those in DBA_SR_OBJ_STATUS.

See Also

"DBA_SR_OBJ_STATUS"

6.681 USER_SR_OBJ_STATUS_ALL

USER_SR_OBJ_STATUS_ALL provides information on the status of objects registered for synchronous refresh in the database which are owned by the current user. Its columns are the same as those in DBA_SR_OBJ_STATUS_ALL.

See Also

"DBA_SR_OBJ_STATUS_ALL"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.682 USER_SR_PARTN_OPS

USER_SR_PARTN_OPS provides information on the partition operations registered on the base tables of the materialized views registered for synchronous refresh belonging to the current user. Its columns are the same as those in DBA_SR_PARTN_OPS.

See Also

"DBA_SR_PARTN_OPS"

6.683 USER_SR_STLOG_EXCEPTIONS

USER_SR_STLOG_EXCEPTIONS provides information on the exceptions in the staging logs for the tables belonging to the current user processed by DBMS_SYNC_REFRESH.PREPARE_STAGING_LOG. Its columns are the same as those in DBA_SR_STLOG_EXCEPTIONS.

See Also

"DBA_SR_STLOG_EXCEPTIONS"

6.684 USER_SR_STLOG_STATS

USER_SR_STLOG_STATS provides information on the statistics in the staging logs for the tables belonging to the current user processed by DBMS_SYNC_REFRESH.PREPARE_STAGING_LOG. Its columns are the same as those in DBA_SR_STLOG_STATS.

See Also

"DBA_SR_STLOG_STATS"

6.685 USER_STAT_EXTENSIONS

USER_STAT_EXTENSIONS displays information about the optimizer statistics extensions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_STAT_EXTENSIONS.

See Also

"ALL_STAT_EXTENSIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.686 USER_STATEMENTS

USER_STATEMENTS SQL statements in stored PL/SQL objects accessible to the user. Its columns (except for OWNER) are the same as those in ALL_STATEMENTS.

See Also

"ALL_STATEMENTS"

6.687 USER_STORED_SETTINGS

USER_STORED_SETTINGS lists information about the persistent parameter settings for stored PL/SQL units, but only shows information about PL/SQL units owned by the current user.

USER_STORED_SETTINGS does not display the OWNER column. The rest of its columns are the same as those in ALL_STORED_SETTINGS.

See Also

"ALL_STORED_SETTINGS"

6.688 USER_SUBPART_COL_STATISTICS

USER_SUBPART_COL_STATISTICS provides column statistics and histogram information for subpartitions of subpartitioned objects owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SUBPART_COL_STATISTICS.

See Also

"ALL_SUBPART_COL_STATISTICS"

6.689 USER_SUBPART_HISTOGRAMS

USER_SUBPART_HISTOGRAMS lists actual histogram data (end-points per histogram) for histograms on table subpartitions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_SUBPART_HISTOGRAMS.

See Also

"ALL_SUBPART_HISTOGRAMS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.690 USER_SUBPART_KEY_COLUMNS

USER_SUBPART_KEY_COLUMNS lists subpartitioning key columns for composite-partitioned tables (and local indexes on composite-partitioned tables) owned by the current user. Its columns are the same as those in ALL_SUBPART_KEY_COLUMNS.

See Also

"ALL_SUBPART_KEY_COLUMNS"

6.691 USER_SUBPARTITION_TEMPLATES

USER_SUBPARTITION_TEMPLATES describes the subpartition templates owned by the current user. Its columns (except for USER_NAME) are the same as those in ALL_SUBPARTITION_TEMPLATES.

See Also

"ALL_SUBPARTITION_TEMPLATES"

6.692 USER_SUBSCR_REGISTRATIONS

USER_SUBSCR_REGISTRATIONS displays information about the subscription registrations owned by the current user. Its columns are the same as those in DBA_SUBSCR_REGISTRATIONS.

See Also

"DBA_SUBSCR_REGISTRATIONS"

6.693 USER_SYNONYMS

USER_SYNONYMS describes the private synonyms (synonyms owned by the current user). Its columns (except for OWNER) are the same as those in ALL_SYNONYMS.

See Also

"ALL_SYNONYMS"

6.694 USER_SYS_PRIVS

USER_SYS_PRIVS describes system privileges granted to the current user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(128)		Name of the user, or PUBLIC
PRIVILEGE	VARCHAR2(40)		System privilege
ADMIN_OPTION	VARCHAR2(3)		Indicates whether the grant was with the ADMIN option (YES) or not (NO)
COMMON	VARCHAR2(3)		Indicates how the grant was made. Possible values: • YES if the privilege was granted commonly (CONTAINER=ALL was used) • NO if the privilege was granted locally (CONTAINER=ALL was not used)
INHERITED	VARCHAR2(3)		Indicates whether the grant was inherited from another container (YES) or not (NO)

See Also

"DBA_SYS_PRIVS"

6.695 USER_TAB_COL_STATISTICS

USER_TAB_COL_STATISTICS contains column statistics and histogram information extracted from USER_TAB_COLUMNS.

Its columns (except for OWNER) are the same as those in ALL_TAB_COL_STATISTICS.

See Also

- "USER_TAB_COLUMNS"
- "ALL_TAB_COL_STATISTICS"

6.696 USER_TAB_COLS

USER_TAB_COLS describes the columns of the tables, views, and clusters owned by the current user.

Its columns (except for OWNER) are the same as those in ALL_TAB_COLS.

To gather statistics for this view, use the DBMS_STATS package.

This view differs from USER_TAB_COLUMNS in that system-generated hidden columns are not filtered out.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "ALL_TAB_COLS"
- "USER_TAB_COLUMNS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

6.697 USER_TAB_COLUMNS

USER_TAB_COLUMNS describes the columns of the tables, views, and clusters owned by the current user.

Its columns (except for OWNER) are the same as those in ALL_TAB_COLUMNS.

To gather statistics for this view, use the DBMS_STATS package.

This view filters out system-generated hidden columns. The USER_TAB_COLS view does not filter out system-generated hidden columns.

See Also

- "ALL_TAB_COLUMNS"
- "USER_TAB_COLS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

6.698 USER_TAB_COMMENTS

USER_TAB_COMMENTS displays comments on the tables and views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TAB_COMMENTS.

See Also

"ALL_TAB_COMMENTS"

6.699 USER_TAB_HISTGRM_PENDING_STATS

USER_TAB_HISTGRM_PENDING_STATS describes pending statistics for tables, partitions, and subpartitions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TAB_HISTGRM_PENDING_STATS.

See Also

"ALL_TAB_HISTGRM_PENDING_STATS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.700 USER_TAB_HISTOGRAMS

USER_TAB_HISTOGRAMS describes histograms on columns of tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TAB_HISTOGRAMS.

See Also

"ALL_TAB_HISTOGRAMS"

6.701 USER_TAB_IDENTITY_COLS

USER_TAB_IDENTITY_COLS describes all table identity columns. Its columns (except for OWNER) are the same as those in ALL_TAB_IDENTITY_COLS.

See Also

"ALL_TAB_IDENTITY_COLS"

6.702 USER_TAB_MODIFICATIONS

USER_TAB_MODIFICATIONS describes modifications to all tables owned by the current user that have been modified since the last time statistics were gathered on the tables. Its columns are the same as those in ALL_TAB_MODIFICATIONS.

Note

For performance reasons, Oracle Database does not populate this view immediately when the actual modifications occur.

See Also

"ALL_TAB_MODIFICATIONS"

6.703 USER_TAB_PARTITIONS

USER_TAB_PARTITIONS describes partition-level partitioning information, partition storage parameters, and partition statistics generated by the DBMS_STATS package for all partitions owned by the current user.

Its columns are the same as those in ALL_TAB_PARTITIONS.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "ALL_TAB_PARTITIONS"

6.704 USER_TAB_PENDING_STATS

USER_TAB_PENDING_STATS describes pending statistics for tables, partitions, and subpartitions owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TAB_PENDING_STATS.

> **See Also**
>
> "ALL_TAB_PENDING_STATS"

6.705 USER_TAB_PRIVS

USER_TAB_PRIVS describes the object grants for which the current user is the object owner, grantor, or grantee. Its columns are the same as those in DBA_TAB_PRIVS.

> **See Also**
>
> "DBA_TAB_PRIVS"

6.706 USER_TAB_PRIVS_MADE

USER_TAB_PRIVS_MADE describes the object grants for which the current user is the object owner. Its columns (except for OWNER) are the same as those in ALL_TAB_PRIVS_MADE.

> **See Also**
>
> "ALL_TAB_PRIVS_MADE"

6.707 USER_TAB_PRIVS_RECD

USER_TAB_PRIVS_RECD describes the object grants for which the current user is the grantee. Its columns (except for GRANTEE) are the same as those in ALL_TAB_PRIVS_RECD.

> **See Also**
>
> "ALL_TAB_PRIVS_RECD"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.708 USER_TAB_STAT_PREFS

USER_TAB_STAT_PREFS displays information about statistics preferences for the tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TAB_STAT_PREFS.

See Also

"ALL_TAB_STAT_PREFS"

6.709 USER_TAB_STATISTICS

USER_TAB_STATISTICS displays optimizer statistics for the tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TAB_STATISTICS.

See Also

"ALL_TAB_STATISTICS"

6.710 USER_TAB_STATS_HISTORY

USER_TAB_STATS_HISTORY provides a history of table statistics modifications for all tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TAB_STATS_HISTORY.

See Also

"ALL_TAB_STATS_HISTORY"

6.711 USER_TAB_SUBPARTITIONS

USER_TAB_SUBPARTITIONS describes, for each table subpartition owned by the current user, the subpartition name, name of the table and partition to which it belongs, and its storage attributes. Its columns are the same as those in ALL_TAB_SUBPARTITIONS.

See Also

"ALL_TAB_SUBPARTITIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.712 USER_TABLES

USER_TABLES describes the relational tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TABLES.

To gather statistics for this view, use the DBMS_STATS package.

See Also

"ALL_TABLES"

6.713 USER_TABLESPACES

USER_TABLESPACES describes the tablespaces accessible to the current user. Its columns (except for PLUGGED_IN) are the same as those in DBA_TABLESPACES.

See Also

"DBA_TABLESPACES"

6.714 USER_TRANSFORMATIONS

USER_TRANSFORMATIONS displays information about the transformations owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TRANSFORMATIONS.

See Also

"ALL_TRANSFORMATIONS"

6.715 USER_TRIGGER_COLS

USER_TRIGGER_COLS describes the use of columns in the triggers owned by the current user and in triggers on tables owned by the current user. Its columns are the same as those in ALL_TRIGGER_COLS.

See Also

"ALL_TRIGGER_COLS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.716 USER_TRIGGER_ORDERING

USER_TRIGGER_ORDERING describes the triggers owned by the current user that have FOLLOWS or PRECEDES ordering. Its columns (except for TRIGGER_OWNER) are the same as those in ALL_TRIGGER_ORDERING.

See Also

"ALL_TRIGGER_ORDERING"

6.717 USER_TRIGGERS

USER_TRIGGERS describes the triggers owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TRIGGERS.

See Also

"ALL_TRIGGERS"

6.718 USER_TRIGGERS_AE

USER_TRIGGERS_AE describes the triggers (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TRIGGERS_AE.

Note

This view is available starting with Oracle Database 19c.

See Also

"ALL_TRIGGERS_AE"

6.719 USER_TS_QUOTAS

USER_TS_QUOTAS contains information about tablespace quotas for the current user. Its columns (except for USERNAME) the same as those in DBA_TS_QUOTAS.

See Also

"DBA_TS_QUOTAS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.720 USER_TSTZ_TAB_COLS

USER_TSTZ_TAB_COLS displays information about the columns of the tables owned by the current user, which have columns defined on TIMESTAMP WITH TIME ZONE data types or object types containing attributes of TIMESTAMP WITH TIME ZONE data types.

Its columns (except for OWNER, COLUMN_NAME, NESTED, VIRTUAL_COLUMN, SCALAR_COLUMN, and UNUSED_COLUMN) are the same as those in ALL_TSTZ_TAB_COLS.

See Also

"ALL_TSTZ_TAB_COLS"

6.721 USER_TSTZ_TABLES

USER_TSTZ_TABLES displays information about the tables owned by the current user, which have columns defined on TIMESTAMP WITH TIME ZONE data types or object types containing attributes of TIMESTAMP WITH TIME ZONE data types.

Its columns (except for OWNER) are the same as those in ALL_TSTZ_TABLES.

See Also

"ALL_TSTZ_TABLES"

6.722 USER_TUNE_MVIEW

USER_TUNE_MVIEW displays the result of executing the DBMS_ADVISOR.TUNE_MVIEW procedure. Its columns (except for OWNER) are the same as those in DBA_TUNE_MVIEW.

See Also

"DBA_TUNE_MVIEW"

6.723 USER_TYPE_ATTRS

USER_TYPE_ATTRS describes the attributes of the object types owned by the current user. Its columns (except for OWNER and CHAR_USED) are the same as those in ALL_TYPE_ATTRS.

See Also

"ALL_TYPE_ATTRS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.724 USER_TYPE_METHODS

USER_TYPE_METHODS describes the methods of the object types owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TYPE_METHODS.

See Also

"ALL_TYPE_METHODS"

6.725 USER_TYPE_VERSIONS

USER_TYPE_VERSIONS describes the versions of the object types owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TYPE_VERSIONS.

See Also

"ALL_TYPE_VERSIONS"

6.726 USER_TYPES

USER_TYPES describes the object types owned by the current user. Its columns (except for OWNER) are the same as those in ALL_TYPES.

See Also

"ALL_TYPES"

6.727 USER_UNUSED_COL_TABS

USER_UNUSED_COL_TABS describes the tables owned by the current user that contain unused columns. Its columns (except for OWNER) are the same as those in ALL_UNUSED_COL_TABS.

See Also

"ALL_UNUSED_COL_TABS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.728 USER_UPDATABLE_COLUMNS

USER_UPDATABLE_COLUMNS describes columns in a join view that can be updated by the current user, subject to appropriate privileges. Its columns are the same as those in ALL_UPDATABLE_COLUMNS.

See Also

- "ALL_UPDATABLE_COLUMNS"
- *Oracle Database Concepts* for information on updatable join views

6.729 USER_USERS

USER_USERS describes the current user.

Column	Datatype	NULL	Description
USERNAME	VARCHAR2(128)	NOT NULL	Name of the user
USER_ID	NUMBER	NOT NULL	ID number of the user

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
ACCOUNT_STATUS	VARCHAR2(32)	NOT NULL	Account status: • OPEN The account is open. • EXPIRED The password for the account is expired, either because the PASSWORD_LIFE_TIME limit was reached or because the password was expired by the ALTER USER ... PASSWORD EXPIRE command. The user can log in with the expired password, then change the password. • EXPIRED(GRACE) The password for the account is expired because the PASSWORD_LIFE_TIME limit was reached, but the password change grace period (PASSWORD_GRACE_TIME) has not yet elapsed. The user can log in with the expired password, but will receive an ORA-28002 warning as a reminder that the password must soon be changed. If the PASSWORD_GRACE_TIME elapses, the user can log in with the expired password, then change the password. • LOCKED The account is locked, either by the ALTER USER ... ACCOUNT LOCK command, or because the number of consecutive failed login attempts exceeded the FAILED_LOGIN_ATTEMPTS limit and the value of PASSWORD_LOCK_TIME is UNLIMITED. The account can be unlocked by the ALTER USER ... ACCOUNT UNLOCK command. • LOCKED(TIMED) The account is locked because the number of consecutive failed login attempts exceeded the FAILED_LOGIN_ATTEMPTS limit and the PASSWORD_LOCK_TIME has not yet elapsed. The account can be unlocked either by the ALTER USER ... ACCOUNT UNLOCK command or by waiting until the PASSWORD_LOCK_TIME has elapsed. • EXPIRED & LOCKED The password for the account is expired, as described for the EXPIRED account status, and the account is locked as described for the LOCKED account status. The account can first be unlocked as described for the LOCKED account status, then the password can be changed as described for the EXPIRED account status. • EXPIRED(GRACE) & LOCKED The password for the account is expired, as described for the EXPIRED(GRACE) account status, and the account is locked as described for the LOCKED account status. The account can first be unlocked as described for the LOCKED account status, then the password can be changed as described for the EXPIRED(GRACE) account status. • EXPIRED & LOCKED(TIMED) The password for the account is expired, as described for the EXPIRED account status, and the

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
			account is locked as described for the `LOCKED(TIMED)` account status. The account can first be unlocked as described for the `LOCKED(TIMED)` account status, then the password can be changed as described for the `EXPIRED` account status. • `EXPIRED(GRACE) & LOCKED(TIMED)` The password for the account is expired, as described for the `EXPIRED(GRACE)` account status, and the account is locked as described for the `LOCKED(TIMED)` account status. The account can first be unlocked as described for the `LOCKED(TIMED)` account status, then the password can be changed as described for the `EXPIRED(GRACE)` account status. • `OPEN & IN ROLLOVER` The account is in the password rollover period. The user can log in with either the earlier password or the new password. However, at the time the user logs in, the server recalculates whether the account is still in its password rollover period. If the password rollover period has elapsed, then the login will succeed only if the new password was specified, and the account status will change to `OPEN`. • `EXPIRED & IN ROLLOVER` The account is in the password rollover period and the password is expired as described for the `EXPIRED` account status. The user can log in with either the earlier password or the new password. However, at the time the user logs in, the server recalculates whether the account is still in its password rollover period. If the password rollover period has elapsed, then the login will succeed only if the new password was specified, and the account status will change to `EXPIRED`. After logging in, the user will be prompted to change the password. • `LOCKED & IN ROLLOVER` The account is in the password rollover period and is also locked as described for the `LOCKED` account status. The account can be unlocked as described for the `LOCKED` account status, after which the user can log in as described for the `OPEN & IN ROLLOVER` account status. • `EXPIRED & LOCKED & IN ROLLOVER` The account is in the password rollover period, its password is expired as described for the `EXPIRED` account status, and the account is locked as described for the `LOCKED` account status. The account can be unlocked as described for the `LOCKED` account status, after which the user can log in as described for the `EXPIRED & IN ROLLOVER` account status. • `LOCKED(TIMED) & IN ROLLOVER` The account is in the password rollover period and is also locked as described for the

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
			LOCKED(TIMED) account status. The account can be unlocked as described for the LOCKED(TIMED) account status, after which the user can log in with either the earlier password or the new password. However, at the time the user logs in, the server recalculates whether the account is still in its password rollover period. If the password rollover period has elapsed, then the login will succeed only if the new password was specified. • EXPIRED & LOCKED(TIMED) & IN ROL The account is in the password rollover period, its password is expired as described for the EXPIRED account status, and the account is locked as described for the LOCKED(TIMED) account status. The account can be unlocked as described for the LOCKED(TIMED) account status, after which the user can log in as described for the EXPIRED & IN ROLLOVER account status.
LOCK_DATE	DATE		Date the account was locked if account status was LOCKED
EXPIRY_DATE	DATE		Date of expiration of the account
DEFAULT_TABLESPACE	VARCHAR2(30)	NOT NULL	Default tablespace for data
TEMPORARY_TABLESPACE	VARCHAR2(30)	NOT NULL	Name of the default tablespace for temporary tables or the name of a tablespace group
LOCAL_TEMP_TABLESPACE	VARCHAR2(30)		Default local temporary tablespace for the user
CREATED	DATE	NOT NULL	User creation date
INITIAL_RSRC_CONSUMER_GROUP	VARCHAR2(128)		Initial resource consumer group for the user
EXTERNAL_NAME	VARCHAR2(4000)		User external name. For centrally managed users, if the database user mapping is an exclusive mapping, then this will be the directory service DN for the user. If this database user is a shared schema, it will be the DN of a group.
PROXY_ONLY_CONNECT	VARCHAR2(1)		Indicates whether a user can connect directly (N) or whether the account can only be proxied (Y) by users who have proxy privileges for this account (that is, by users who have been granted the "connect through" privilege for this account). For more information about creating proxy user accounts and authorizing users to connect through them, see *Oracle Database Security Guide*.
COMMON	VARCHAR2(3)		Indicates whether a given user is common. Possible values • YES if a user is common • NO if a user is local (not common)
ORACLE_MAINTAINED	VARCHAR2(1)		Denotes whether the user was created, and is maintained, by Oracle-supplied scripts (such as catalog.sql or catproc.sql). A user for which this column has the value Y must not be changed in any way except by running an Oracle-supplied script.
INHERITED	VARCHAR2(3)		Indicates whether the user definition was inherited from another container (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
DEFAULT_COLLATION	VARCHAR2(100)		Default collation for the user's schema
IMPLICIT	VARCHAR2(3)		Indicates whether this user is a common user created by an implicit application (YES) or not (NO)
ALL_SHARD	VARCHAR2(3)		In a sharded database, the value in this column indicates whether the user was created with shard DDL enabled. The possible values are: • YES: The user was created with shard DDL enabled. The user exists on all shards and the shard catalog. • NO: The user was created without shard DDL enabled. The user exists only in the database in which the user was created. In a non-sharded database, the value in this column is always NO.
PASSWORD_CHANGE_DATE[1]	DATE		Date on which the user's password was last set This column is populated only when the value of the AUTHENTICATION_TYPE column is PASSWORD. Otherwise, this column is null.

1 This column is available starting with Oracle Database 19c.

Note

In a CDB, common users should be connected to the CDB root when querying DBA_USERS or USER_USERS. This ensures that complete information is returned.

See Also

- "DBA_USERS"
- *Using Oracle Sharding* for more information about sharded database management

6.730 USER_USTATS

USER_USTATS describes the user-defined statistics collected on the tables and indexes owned by the current user. Its columns are the same as those in ALL_USTATS.

See Also

"ALL_USTATS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.731 USER_VARRAYS

USER_VARRAYS describes the varrays owned by the current user. Its columns (except for OWNER) are the same as those in ALL_VARRAYS.

See Also

"ALL_VARRAYS"

6.732 USER_VIEWS

USER_VIEWS describes the views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_VIEWS.

See Also

"ALL_VIEWS"

6.733 USER_VIEWS_AE

USER_VIEWS_AE describes the views (across all editions) owned by the current user. Its columns (except for OWNER) are the same as those in ALL_VIEWS_AE.

See Also

"ALL_VIEWS_AE"

6.734 USER_WALLET_ACES

USER_WALLET_ACES describes the status of access control entries for the current user to access wallets through PL/SQL network utility packages.

Its columns (except for ACE_ORDER, START_DATE, END_DATE, GRANT_TYPE, INVERTED_PRINCIPAL, PRINCIPAL, PRINCIPAL_TYPE, and STATUS) are the same as those in DBA_WALLET_ACES.

See Also

"DBA_WALLET_ACES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.735 USER_WARNING_SETTINGS

USER_WARNING_SETTINGS displays information about the warning parameter settings for the objects owned by the current user. Its columns (except for OWNER) are the same as those in ALL_WARNING_SETTINGS.

See Also

"ALL_WARNING_SETTINGS"

6.736 USER_XML_INDEXES

USER_XML_INDEXES describes the XML indexes owned by the current user. Its columns (except for INDEX_OWNER) are the same as those in ALL_XML_INDEXES.

See Also

"ALL_XML_INDEXES"

6.737 USER_XML_NESTED_TABLES

USER_XML_NESTED_TABLES describes all the tables and their corresponding nested tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_NESTED_TABLES.

See Also

"ALL_XML_NESTED_TABLES"

6.738 USER_XML_OUT_OF_LINE_TABLES

USER_XML_OUT_OF_LINE_TABLES describes all the out of line tables connected to a given root table for the same schema owned by the current user. Its columns (except for TABLE_OWNER) are the same as those in ALL_XML_OUT_OF_LINE_TABLES.

See Also

"ALL_XML_OUT_OF_LINE_TABLES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.739 USER_XML_SCHEMA_ATTRIBUTES

USER_XML_SCHEMA_ATTRIBUTES describes all the attributes and their properties owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_SCHEMA_ATTRIBUTES.

See Also

"ALL_XML_SCHEMA_ATTRIBUTES"

6.740 USER_XML_SCHEMA_COMPLEX_TYPES

USER_XML_SCHEMA_COMPLEX_TYPES describes all complex types owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_SCHEMA_COMPLEX_TYPES.

See Also

"ALL_XML_SCHEMA_COMPLEX_TYPES"

6.741 USER_XML_SCHEMA_ELEMENTS

USER_XML_SCHEMA_ELEMENTS describes all the elements and their properties owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_SCHEMA_ELEMENTS.

See Also

"ALL_XML_SCHEMA_ELEMENTS"

6.742 USER_XML_SCHEMA_NAMESPACES

USER_XML_SCHEMA_NAMESPACES describes all the available namespaces owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_SCHEMA_NAMESPACES.

See Also

"ALL_XML_SCHEMA_NAMESPACES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.743 USER_XML_SCHEMA_SIMPLE_TYPES

USER_XML_SCHEMA_SIMPLE_TYPES describes all simple types owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_SCHEMA_SIMPLE_TYPES.

See Also

"ALL_XML_SCHEMA_SIMPLE_TYPES"

6.744 USER_XML_SCHEMA_SUBSTGRP_HEAD

USER_XML_SCHEMA_SUBSTGRP_HEAD describes the heads of substitution groups owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_SCHEMA_SUBSTGRP_HEAD.

See Also

"ALL_XML_SCHEMA_SUBSTGRP_HEAD"

6.745 USER_XML_SCHEMA_SUBSTGRP_MBRS

USER_XML_SCHEMA_SUBSTGRP_MBRS describes all members of substitution groups owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_SCHEMA_SUBSTGRP_MBRS.

See Also

"ALL_XML_SCHEMA_SUBSTGRP_MBRS"

6.746 USER_XML_SCHEMAS

USER_XML_SCHEMAS describes the registered XML schemas owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_SCHEMAS.

See Also

"ALL_XML_SCHEMAS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.747 USER_XML_TAB_COLS

USER_XML_TAB_COLS describes the columns of the XML tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_TAB_COLS.

See Also

"ALL_XML_TAB_COLS"

6.748 USER_XML_TABLES

USER_XML_TABLES describes the XML tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_TABLES.

See Also

"ALL_XML_TABLES"

6.749 USER_XML_VIEW_COLS

USER_XML_VIEW_COLS describes the columns of the XML views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_VIEW_COLS.

See Also

"ALL_XML_VIEW_COLS"

6.750 USER_XML_VIEWS

USER_XML_VIEWS describes the XML views owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XML_VIEWS.

See Also

"ALL_XML_VIEWS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.751 USER_XTERNAL_LOC_PARTITIONS

USER_XTERNAL_LOC_PARTITIONS describes partition-level locations owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XTERNAL_LOC_PARTITIONS.

See Also

"ALL_XTERNAL_LOC_PARTITIONS"

6.752 USER_XTERNAL_LOC_SUBPARTITIONS

USER_XTERNAL_LOC_SUBPARTITIONS describes subpartition-level locations owned by the current user. Its columns (except for TABLE_OWNER) are the same as those in ALL_XTERNAL_LOC_SUBPARTITIONS.

See Also

"ALL_XTERNAL_LOC_SUBPARTITIONS"

6.753 USER_XTERNAL_PART_TABLES

USER_XTERNAL_PART_TABLES describes object-level information for partitioned external tables owned by the current user. Its columns (except for OWNER) are the same as those in ALL_XTERNAL_PART_TABLES.

See Also

"ALL_XTERNAL_PART_TABLES"

6.754 USER_XTERNAL_TAB_PARTITIONS

USER_XTERNAL_TAB_PARTITIONS describes partition-level information for partitioned external tables owned by the current user. Its columns (except for TABLE_OWNER) are the same as those in ALL_XTERNAL_TAB_PARTITIONS.

See Also

"ALL_XTERNAL_TAB_PARTITIONS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

6.755 USER_XTERNAL_TAB_SUBPARTITIONS

USER_XTERNAL_TAB_SUBPARTITIONS describes subpartition-level information for partitioned external tables owned by the current user. Its columns (except for TABLE_OWNER) are the same as those in ALL_XTERNAL_TAB_SUBPARTITIONS.

See Also

"ALL_XTERNAL_TAB_SUBPARTITIONS"

6.756 USER_ZONEMAP_MEASURES

USER_ZONEMAP_MEASURES describes the measures for all the zone maps owned by the user. Its columns are the same as those in ALL_ZONEMAP_MEASURES.

Note

This view is intended for use with Oracle Exadata release 12.1.2.1.1 or later.

See Also

- "ALL_ZONEMAP_MEASURES"
- *Oracle Database Data Warehousing Guide* for more information about zone maps

6.757 USER_ZONEMAPS

USER_ZONEMAPS describes the zone maps owned by the user. Its columns are the same as those in ALL_ZONEMAPS.

Note

This view is intended for use with Oracle Exadata release 12.1.2.1.1 or later.

See Also

- "ALL_ZONEMAPS"
- *Oracle Database Data Warehousing Guide* for more information about zone maps

Copyright © 2002, 2026, Oracle and/or its affiliates.

Part III
Dynamic Performance Views

This part describes the **dynamic performance views**, which are often referred to as V$ views.

This part contains the following chapters:

- Dynamic Performance (V$) Views: V$ACCESS to V$HVMASTER_INFO
- Dynamic Performance (V$) Views: V$IM_COLUMN_LEVEL to V$RULE_SET_AGGREGATE_STATS
- Dynamic Performance (V$) Views: V$SCHEDULER_RUNNING_JOBS to V$ZONEMAP_USAGE_STATS

> **Note**
>
> Oracle also maintains tables and views that change only when a change is made to the data dictionary. These **static** tables and views are described in Static Data Dictionary Views .

Copyright © 2002, 2026, Oracle and/or its affiliates.

7

Dynamic Performance (V$) Views: V$ACCESS to V$HVMASTER_INFO

This chapter describes the first set (in alphabetical order) of dynamic performance views.

The remaining dynamic performance views appear in alphabetical order in Dynamic Performance (V$) Views: V$IM_COLUMN_LEVEL to V$RULE_SET_AGGREGATE_STATS through Dynamic Performance (V$) Views: V$SCHEDULER_RUNNING_JOBS to V$ZONEMAP_USAGE_STATS.

This chapter contains the following topics:

- About Dynamic Performance Views
- Oracle Exadata Database Machine Views
- Dynamic Performance View Descriptions

7.1 About Dynamic Performance Views

Oracle contains a set of underlying views that are maintained by the database server and accessible to the database administrator user SYS. These views are called **dynamic performance views** because they are continuously updated while a database is open and in use, and their contents relate primarily to performance.
Although these views appear to be regular database tables, they are not. These views provide data on internal disk structures and memory structures. You can select from these views, but you can never update or alter them.

> **Note**
>
> - You can query the dynamic performance views to extract information from them. However, only simple queries are supported. If sorts, joins, `GROUP BY` clauses and the like are needed, then you should copy the information from each V$ view into a table (for example, using a `CREATE TABLE ... AS SELECT` statement), and then query from those tables.
> - Because the information in the V$ views is dynamic, read consistency is not guaranteed for `SELECT` operations on these views.

The `catalog.sql` script contains definitions of the views and public synonyms for the dynamic performance views. You must run `catalog.sql` to create these views and synonyms. After installation, only user SYS or anyone with SYSDBA privilege has access to the dynamic performance tables. See *Oracle Database Administrator's Guide* for more information about running `catalog.sql`.

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.1.1 V$ Views

The actual dynamic performance views are identified by the prefix V_$. Public synonyms for these views have the prefix V$. Database administrators and other users should access only the V$ objects, not the V_$ objects.

The dynamic performance views are used by Oracle Enterprise Manager, which is the primary interface for accessing information about system performance. After an instance is started, the V$ views that read from memory are accessible. Views that read data from disk require that the database be mounted, and some require that the database be open.

V$ views are CONTAINER_DATA objects. When a user connected to the root queries a V$ view, the query results will depend on the CONTAINER_DATA attribute for users for the view. The CONTAINER_DATA clause of the SQL ALTER USER statement is used to set and modify users' CONTAINER_DATA attribute.

V$ views can return data from different containers in a CDB when queried from the root container. These objects will implicitly convert data to the character set of the root container (AL32UTF8) and then return the result to the user. Some character sets may have character expansion (more bytes needed to represent a character) when converted to AL32UTF8, so there may be data truncation if the view column width is not able to accommodate data from a given PDB.

See Also

- *Oracle Database Security Guide* for more information about container data objects
- *Oracle Database SQL Language Reference* for more information about the CONTAINER_DATA clause for the SQL ALTER USER statement

7.1.2 GV$ Views

For almost every V$ view described in this chapter, Oracle has a corresponding GV$ (global V$) view. In Oracle Real Application Clusters, querying a GV$ view retrieves the V$ view information from all qualified instances. In addition to the V$ information, each GV$ view contains an extra column named INST_ID of data type NUMBER. The INST_ID column displays the instance number from which the associated V$ view information was obtained. The INST_ID column can be used as a filter to retrieve V$ information from a subset of available instances. For example, the following query retrieves the information from the V$LOCK view on instances 2 and 5:

```
SQL> SELECT * FROM GV$LOCK WHERE INST_ID = 2 OR INST_ID = 5;
```

See Also

Oracle Real Application Clusters Installation and Configuration Guide for your operating system

GV$ views are CONTAINER_DATA objects. When a user connected to the root queries a GV$ view, the query results will depend on the CONTAINER_DATA attribute for users for the view. The CONTAINER_DATA clause of the SQL ALTER USER statement is used to set and modify users' CONTAINER_DATA attribute.

Copyright © 2002, 2026, Oracle and/or its affiliates.

GV$ views can return data from different containers in a CDB when queried from the root container. These objects will implicitly convert data to the character set of the root container (AL32UTF8) and then return the result to the user. Some character sets may have character expansion (more bytes needed to represent a character) when converted to AL32UTF8, so there may be data truncation if the view column width is not able to accommodate data from a given PDB.

See Also

- *Oracle Database Security Guide* for more information about container data objects
- *Oracle Database SQL Language Reference* for more information about the `CONTAINER_DATA` clause for the SQL `ALTER USER` statement

7.2 Oracle Exadata Database Machine Views

Descriptions of Oracle Exadata Database Machine views are not provided in this manual.

See *Oracle Exadata System Software User's Guide* for descriptions of Oracle Exadata Database Machine views.

Note

Oracle Exadata Database Machine view names begin with `V$CELL`.

7.3 Dynamic Performance View Descriptions

The remainder of this chapter describes the dynamic performance views in alphabetical order.

7.4 V$ACCESS

`V$ACCESS` displays information about locks that are currently imposed on library cache objects.

The locks are imposed to ensure that they are not aged out of the library cache while they are required for SQL execution.

Column	Datatype	Description
`SID`	`NUMBER`	Session number that is accessing an object
`OWNER`	`VARCHAR2(64)`	Owner of the object
`OBJECT`	`VARCHAR2(1000)`	Name of the object
`TYPE`	`VARCHAR2(64)`	Type identifier for the object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.5 V$ACTIVE_INSTANCES

V$ACTIVE_INSTANCES displays the mapping between instance names and instance numbers for all instances that have the database currently mounted.

Column	Datatype	Description
INST_NUMBER	NUMBER	Instance number
INST_NAME	VARCHAR2(256)	Instance name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.6 V$ACTIVE_SERVICES

V$ACTIVE_SERVICES displays information about the active services in the database.

Column	Datatype	Description
SERVICE_ID	NUMBER	Service ID
NAME	VARCHAR2(64)	Name of the service
NAME_HASH	NUMBER	Service name hash
NETWORK_NAME	VARCHAR2(512)	Network name
CREATION_DATE	DATE	Creation date
CREATION_DATE_HASH	NUMBER	Creation date hash
GOAL	VARCHAR2(12)	Runtime Load Balancing Goal being used to create run-time load balancing and connection load balancing advice: • NONE • SERVICE_TIME - Connections are balanced by response time • THROUGHPUT - Connections are balanced by throughput

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DTP	VARCHAR2(1)	Indicates whether or not Distributed Transaction Processing is enabled for this service. When Distributed Transaction Processing is set to Y (YES), it means that the service is offered at exactly one instance at a time for XA affinity. Possible values: • Y - YES • N - NO
BLOCKED	CHAR(2)	Indicates whether a service on the specified instance is blocked from accepting new connections altogether (YES) or not (NO). If a service is blocked, then all connections will be directed to other instances (if any) that are hosting the desired service.
AQ_HA_NOTIFICATION	VARCHAR2(3)	Indicates whether FAN - Fast Application Notification for OCI connections is set (YES) or not (NO)
CLB_GOAL	VARCHAR2(5)	Connection load balancing goal used with statistics that are sent to the listeners to determine how new connections are distributed: • LONG • SHORT
COMMIT_OUTCOME	VARCHAR2(3)	For Transaction Guard, indicates whether the database service associated with the user session has the COMMIT_OUTCOME service attribute enabled (YES) or not (NO). This attribute applies on a per session basis and is set at connect time. When COMMIT_OUTCOME = YES: • Transaction Guard manages the commit status for all supported transaction types. The outcome of a COMMIT transaction is known. If there is an outage, the application can use DBMS_APP_CONT.GET_LTXID_OUTCOME to return a reliable status for the last in-flight work. • A logical transaction ID (LTXID) is set for each user session at login and at each successful commit. **See Also:** For information about preserving the commit outcome, see *Oracle Database Development Guide*. For information about logical transaction IDs, see *Oracle Database Development Guide*
RETENTION_TIME	NUMBER	For Transaction Guard, when COMMIT_OUTCOME = YES, this value indicates the amount of time (in seconds) that the commit outcome is retained in the database.
REPLAY_INITIATION_TIMEOUT	NUMBER	For Application Continuity, this option specifies the difference between the time (in seconds) of original execution of the first operation of a request and the time that the replay is ready to start after a successful reconnect. Application Continuity will not replay after the specified amount of time has passed. This option is intended to avoid the unintentional execution of a transaction when a system is recovered after a long period of time. The default is 5 minutes (300 seconds).
SESSION_STATE_CONSISTENCY	VARCHAR2(128)	Describes how non-transactional is changed during a request. This parameter is considered only if failover_type is set to TRANSACTION for Application Continuity. Examples of session state are NLS settings, optimizer preferences, event settings, PL/SQL global variables, temporary tables, advanced queues, LOBs, and result cache. If non-transactional values change after the request starts, the default value of DYNAMIC should be set. Almost all applications should use DYNAMIC mode. If you are unsure, use DYNAMIC mode.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
GLOBAL	VARCHAR2(3)	Indicates whether the service is global. A global service is managed by Global Service Manager (GSM) and can be provided by multiple databases that contain replicated data. Possible values: • YES: Indicates the service is global • NO: Indicates the service is not global
CON_NAME	VARCHAR2(128)	Container name of the object. The value of this column is NULL in non-CDBs.
SQL_TRANSLATION_PROFILE	VARCHAR2(261)	A non-NULL value specifies the initial SQL translation profile for subsequent database connections that use the service and do not specify a SQL translation profile. A NULL value has no effect.
MAX_LAG_TIME	VARCHAR2(128)	The maximum replication lag (in seconds) that is acceptable for a data replica to be used for providing the database service. Can only be specified for global services.
STOP_OPTION	VARCHAR2(128)	Stop option for sessions of this service for planned maintenance
FAILOVER_RESTORE	VARCHAR2(128)	Indicates whether sessions restore their commonly used session state (such as NLS and schema settings) when they are failed over. Possible values: • AUTO - Sessions restore their commonly used session state when they are failed over with Transparent Application Continuity • LEVEL1 - Sessions restore their commonly used session state when they are failed over with Application Continuity • NONE - Sessions do not restore their commonly used session state when they are failed over
DRAIN_TIMEOUT	NUMBER	Number of seconds to wait for sessions to be drained
TABLE_FAMILY_ID[1]	NUMBER	Sharded table family ID associated with the service
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

1 This column is available starting with Oracle Database 19c.

7.7 V$ACTIVE_SESS_POOL_MTH

V$ACTIVE_SESS_POOL_MTH displays available active session pool resource allocation methods.

Column	Datatype	Description
NAME	VARCHAR2(40)	Name of the active session pool resource allocation method
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.8 V$ACTIVE_SESSION_HISTORY

V$ACTIVE_SESSION_HISTORY displays sampled session activity in the database.

It contains snapshots of active database sessions taken once a second. A database session is considered active if it was on the CPU or was waiting for an event that didn't belong to the Idle wait class. Refer to the V$EVENT_NAME view for more information on wait classes.

This view contains one row for each active session per sample and returns the latest session sample rows first. A majority of the columns describing the session in the active session history are present in the V$SESSION view.

Column	Datatype	Description
SAMPLE_ID	NUMBER	ID of the sample
SAMPLE_TIME	TIMESTAMP(3)	Time at which the sample was taken
SAMPLE_TIME_UTC	TIMESTAMP(3)	Time at which the sample was taken in UTC (Coordinated Universal Time) time zone
USECS_PER_ROW	NUMBER	Time in microseconds since the last sample was taken
IS_AWR_SAMPLE	VARCHAR2(1)	Indicates whether this sample has been flushed or will be flushed to the Automatic Workload Repository (DBA_HIST_ACTIVE_SESS_HISTORY) (Y) or not (N)
SESSION_ID	NUMBER	Session identifier; maps to V$SESSION.SID
SESSION_SERIAL#	NUMBER	Session serial number (used to uniquely identify a session's objects); maps to V$SESSION.SERIAL#
SESSION_TYPE	VARCHAR2(10)	Session type: • FOREGROUND • BACKGROUND
FLAGS	NUMBER	Reserved for future use
USER_ID	NUMBER	Oracle user identifier; maps to V$SESSION.USER#
SQL_ID	VARCHAR2(13)	SQL identifier of the SQL statement that the session was executing at the time of sampling
IS_SQLID_CURRENT	VARCHAR2(1)	Indicates whether the SQL identifier in the SQL_ID column is being executed (Y) or not (N)
SQL_CHILD_NUMBER	NUMBER	Child number of the SQL statement that the session was executing at the time of sampling
SQL_OPCODE	NUMBER	Indicates what phase of operation the SQL statement was in; maps to V$SESSION.COMMAND **See Also:** "V$SESSION" for information on interpreting this column
SQL_OPNAME	VARCHAR2(64)	SQL command name
FORCE_MATCHING_SIGNATURE	NUMBER	Signature used when the CURSOR_SHARING parameter is set to FORCE
TOP_LEVEL_SQL_ID	VARCHAR2(13)	SQL identifier of the top level SQL statement
TOP_LEVEL_SQL_OPCODE	NUMBER	Indicates what phase of operation the top level SQL statement was in
SQL_ADAPTIVE_PLAN_RESOLVED	NUMBER	Indicates whether the SQL plan of the sampled database session is a resolved adaptive plan or not
SQL_FULL_PLAN_HASH_VALUE	NUMBER	Numerical representation of the complete SQL plan for the cursor being executed by this session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SQL_PLAN_HASH_VALUE	NUMBER	Numeric representation of the SQL plan for the cursor. This information might not be available for all session samples. V$SESSION does not contain this information.
SQL_PLAN_LINE_ID	NUMBER	SQL plan line ID
SQL_PLAN_OPERATION	VARCHAR2(30)	Plan operation name
SQL_PLAN_OPTIONS	VARCHAR2(30)	Plan operation options
SQL_EXEC_ID	NUMBER	SQL execution identifier
SQL_EXEC_START	DATE	Time when the execution of the SQL started
PLSQL_ENTRY_OBJECT_ID	NUMBER	Object ID of the top-most PL/SQL subprogram on the stack; NULL if there is no PL/SQL subprogram on the stack. Maps to DBA_OBJECTS.OBJECT_ID.
PLSQL_ENTRY_SUBPROGRAM_ID	NUMBER	Subprogram ID of the top-most PL/SQL subprogram on the stack. Maps to DBA_OBJECTS.DATA_OBJECT_ID.
PLSQL_OBJECT_ID	NUMBER	Object ID of the currently executing PL/SQL subprogram. Maps to DBA_OBJECTS.OBJECT_ID.
PLSQL_SUBPROGRAM_ID	NUMBER	Subprogram ID of the currently executing PL/SQL object; NULL if executing SQL. Maps to DBA_OBJECTS.DATA_OBJECT_ID.
QC_INSTANCE_ID	NUMBER	Query coordinator instance ID. This information is only available if the sampled session is a parallel query worker. For all other sessions, the value is 0.
QC_SESSION_ID	NUMBER	Query coordinator session ID. This information is only available if the sampled session is a parallel query worker. For all other sessions, the value is 0.
QC_SESSION_SERIAL#	NUMBER	Query coordinator session serial number. This information is only available if the sampled session is a parallel query worker. For all other sessions, the value is 0.
PX_FLAGS	NUMBER	Reserved for internal use
EVENT	VARCHAR2(64)	If SESSION_STATE = WAITING, then the event for which the session was waiting for at the time of sampling. If SESSION_STATE = ON CPU, then this column is NULL. **See Also:** Oracle Wait Events
EVENT_ID	NUMBER	Identifier of the resource or event for which the session is waiting or for which the session last waited. Interpretation is similar to that of the EVENT column.
EVENT#	NUMBER	Number of the resource or event for which the session is waiting or for which the session last waited. Interpretation is similar to that of the EVENT column.
SEQ#	NUMBER	Sequence number that uniquely identifies the wait (incremented for each wait)
P1TEXT	VARCHAR2(64)	Text of the first additional parameter
P1	NUMBER	First additional parameter
P2TEXT	VARCHAR2(64)	Text of the second additional parameter
P2	NUMBER	Second additional parameter
P3TEXT	VARCHAR2(64)	Text of the third additional parameter
P3	NUMBER	Third additional parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
WAIT_CLASS	VARCHAR2(64)	Wait class name of the event for which the session was waiting at the time of sampling. Interpretation is similar to that of the EVENT column. Maps to V$SESSION.WAIT_CLASS.
WAIT_CLASS_ID	NUMBER	Wait class identifier of the event for which the session was waiting at the time of sampling. Interpretation is similar to that of the EVENT column. Maps to V$SESSION.WAIT_CLASS_ID.
WAIT_TIME	NUMBER	Total wait time for the event for which the session last waited if the session was on the CPU when sampled; 0 if the session was waiting at the time of sampling **Note:** Whether or not WAIT_TIME = 0 is what is useful to find the SESSION_STATE at the time of sampling, rather than the actual value of WAIT_TIME itself. Maps to V$SESSION.WAIT_TIME.
SESSION_STATE	VARCHAR2(7)	Session state: • WAITING • ON CPU
TIME_WAITED	NUMBER	If SESSION_STATE = WAITING, then the time that the session actually spent waiting for that event (in microseconds). This column is set for waits that were in progress at the time the sample was taken. If a wait event lasted for more than a second and was caught waiting in more than one session sample row, then the actual time spent waiting for that wait event will be populated in the last of those session sample rows. At any given time, this information will not be available for the latest session sample.
BLOCKING_SESSION_STATUS	VARCHAR2(11)	Status of the blocking session: • VALID • NO HOLDER • GLOBAL • NOT IN WAIT • UNKNOWN
BLOCKING_SESSION	NUMBER	Session identifier of the blocking session. Populated only if the blocker is on the same instance and the session was waiting for enqueues or a "buffer busy" wait. Maps to V$SESSION.BLOCKING_SESSION.
BLOCKING_SESSION_SERIAL#	NUMBER	Serial number of the blocking session
BLOCKING_INST_ID	NUMBER	Instance number of the blocker shown in BLOCKING_SESSION
BLOCKING_HANGCHAIN_INFO	VARCHAR2(1)	Indicates whether the information about BLOCKING_SESSION comes from the hang chain (Y) or not (N)
CURRENT_OBJ#	NUMBER	Object ID of the object that the session is referencing. This information is only available if the session was waiting for application, cluster, concurrency, and user I/O wait events. Maps to V$SESSION.ROW_WAIT_OBJ#.
CURRENT_FILE#	NUMBER	File number of the file containing the block that the session is referencing. This information is only available if the session was waiting for cluster, concurrency, and user I/O wait events. Maps to V$SESSION.ROW_WAIT_FILE#.
CURRENT_BLOCK#	NUMBER	ID of the block that the session is referencing. This information is only available if the session was waiting for cluster, concurrency, and user I/O wait events. Maps to V$SESSION.ROW_WAIT_BLOCK#.
CURRENT_ROW#	NUMBER	Row identifier that the session is referencing. This information is only available if the session was waiting for cluster, concurrency, and user I/O wait events. Maps to V$SESSION.ROW_WAIT_ROW#.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOP_LEVEL_CALL#	NUMBER	Oracle top level call number
TOP_LEVEL_CALL_NAME	VARCHAR2(64)	Oracle top level call name
CONSUMER_GROUP_ID	NUMBER	Consumer group ID
XID	RAW(8)	Transaction ID that the session was working on at the time of sampling. V$SESSION does not contain this information.
REMOTE_INSTANCE#	NUMBER	Remote instance identifier that will serve the block that this session is waiting for. This information is only available if the session was waiting for cluster events.
TIME_MODEL	NUMBER	Time model information
IN_CONNECTION_MGMT	VARCHAR2(1)	Indicates whether the session was doing connection management at the time of sampling (Y) or not (N)
IN_PARSE	VARCHAR2(1)	Indicates whether the session was parsing at the time of sampling (Y) or not (N)
IN_HARD_PARSE	VARCHAR2(1)	Indicates whether the session was hard parsing at the time of sampling (Y) or not (N)
IN_SQL_EXECUTION	VARCHAR2(1)	Indicates whether the session was executing SQL statements at the time of sampling (Y) or not (N)
IN_PLSQL_EXECUTION	VARCHAR2(1)	Indicates whether the session was executing PL/SQL at the time of sampling (Y) or not (N)
IN_PLSQL_RPC	VARCHAR2(1)	Indicates whether the session was executing inbound PL/SQL RPC calls at the time of sampling (Y) or not (N)
IN_PLSQL_COMPILATION	VARCHAR2(1)	Indicates whether the session was compiling PL/SQL at the time of sampling (Y) or not (N)
IN_JAVA_EXECUTION	VARCHAR2(1)	Indicates whether the session was executing Java at the time of sampling (Y) or not (N)
IN_BIND	VARCHAR2(1)	Indicates whether the session was doing bind operations at the time of sampling (Y) or not (N)
IN_CURSOR_CLOSE	VARCHAR2(1)	Indicates whether the session was closing a cursor at the time of sampling (Y) or not (N)
IN_SEQUENCE_LOAD	VARCHAR2(1)	Indicates whether the session is loading in sequence (in sequence load code) (Y) or not (N)
IN_INMEMORY_QUERY	VARCHAR2(1)	Indicates whether the session was querying the In-Memory Column Store (IM column store) at the time of sampling (Y) or not (N)
IN_INMEMORY_POPULATE	VARCHAR2(1)	Indicates whether the session was populating the IM column store at the time of sampling (Y) or not (N)
IN_INMEMORY_PREPOPULATE	VARCHAR2(1)	Indicates whether the session was prepopulating the IM column store at the time of sampling (Y) or not (N)
IN_INMEMORY_REPOPULATE	VARCHAR2(1)	Indicates whether the session was repopulating the IM column store at the time of sampling (Y) or not (N)
IN_INMEMORY_TREPOPULATE	VARCHAR2(1)	Indicates whether the session was trickle repopulating the IM column store at the time of sampling (Y) or not (N)
IN_TABLESPACE_ENCRYPTION	VARCHAR2(1)	Indicates whether encryption or decryption of a tablespace occurred at the time of sampling (Y) or not (N)
CAPTURE_OVERHEAD	VARCHAR2(1)	Indicates whether the session is executing capture code (Y) or not (N)
REPLAY_OVERHEAD	VARCHAR2(1)	Indicates whether the session is executing replay code (Y) or not (N)
IS_CAPTURED	VARCHAR2(1)	Indicates whether the session is being captured (Y) or not (N)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
IS_REPLAYED	VARCHAR2(1)	Indicates whether the session is being replayed (Y) or not (N)
IS_REPLAY_SYNC_TOKEN_HOLDER	VARCHAR2(1)	Indicates whether the session is holding a synchronization token (Y) or not (N) during workload replay
SERVICE_HASH	NUMBER	Hash that identifies the Service; maps to V$ACTIVE_SERVICES.NAME_HASH
PROGRAM	VARCHAR2(48)	Name of the operating system program
MODULE	VARCHAR2(64)	Name of the executing module when sampled, as set by the DBMS_APPLICATION_INFO.SET_MODULE procedure
ACTION	VARCHAR2(64)	Name of the executing module when sampled, as set by the DBMS_APPLICATION_INFO.SET_ACTION procedure
CLIENT_ID	VARCHAR2(64)	Client identifier of the session; maps to V$SESSION.CLIENT_IDENTIFIER
MACHINE	VARCHAR2(64)	Client's operating system machine name
PORT	NUMBER	Client port number
ECID	VARCHAR2(64)	Execution context identifier (sent by Application Server)
DBREPLAY_FILE_ID	NUMBER	If the session is being captured or replayed, then DBREPLAY_FILE_ID is the file ID for the workload capture or workload replay; otherwise it is NULL.
DBREPLAY_CALL_COUNTER	NUMBER	If the session is being captured or replayed, then DBREPLAY_CALL_COUNTER is the call counter of the user call that is being captured or replayed; otherwise it is NULL.
TM_DELTA_TIME	NUMBER	Time interval (in microseconds) over which TM_DELTA_CPU_TIME and TM_DELTA_DB_TIME are accumulated
TM_DELTA_CPU_TIME	NUMBER	Amount of time this session spent on CPU over the last TM_DELTA_TIME microseconds
TM_DELTA_DB_TIME	NUMBER	Amount of time spent by this session in database calls over the last TM_DELTA_TIME microseconds
DELTA_TIME	NUMBER	Time interval (in microseconds) since the last time this session was sampled or created, over which the next five statistics are accumulated
DELTA_READ_IO_REQUESTS	NUMBER	Number of read I/O requests made by this session over the last DELTA_TIME microseconds
DELTA_WRITE_IO_REQUESTS	NUMBER	Number of write I/O requests made by this session over the last DELTA_TIME microseconds
DELTA_READ_IO_BYTES	NUMBER	Number of I/O bytes read by this session over the last DELTA_TIME microseconds
DELTA_WRITE_IO_BYTES	NUMBER	Number of I/O bytes written by this session over the last DELTA_TIME microseconds
DELTA_INTERCONNECT_IO_BYTES	NUMBER	Number of I/O bytes sent over the I/O interconnect over the last DELTA_TIME microseconds
DELTA_READ_MEM_BYTES	NUMBER	Number of read bytes through the buffer cache
PGA_ALLOCATED	NUMBER	Amount of PGA memory (in bytes) consumed by this session at the time this sample was taken
TEMP_SPACE_ALLOCATED	NUMBER	Amount of TEMP memory (in bytes) consumed by this session at the time this sample was taken
CON_DBID	NUMBER	The database ID of the pluggable database (PDB)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
DBOP_NAME	VARCHAR2(30)	Database operation name. If the type is SQL, the DBOP_NAME will be NULL.
DBOP_EXEC_ID	NUMBER	Database operation execution identifier for the current execution. If the type is SQL, the DBOP_EXEC_ID will be NULL.

See Also

- "V$SESSION"
- "V$ALL_ACTIVE_SESSION_HISTORY"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_MODULE procedure
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_ACTION procedure

7.9 V$ADVISOR_PROGRESS

V$ADVISOR_PROGRESS displays information about the progress of advisor execution.

Column	Datatype	Description
SID	NUMBER	Session ID
SERIAL#	NUMBER	Session serial number
USERNAME	VARCHAR2(128)	Oracle user name
OPNAME	VARCHAR2(64)	Operation name
ADVISOR_NAME	VARCHAR2(64)	Advisor name
TASK_ID	NUMBER	Task ID
TARGET_DESC	VARCHAR2(32)	Description of the target of the advisor
SOFAR	NUMBER	Amount of work done so far
TOTALWORK	NUMBER	Total work to be done
UNITS	VARCHAR2(32)	Units that the work is measured in
BENEFIT_SOFAR	NUMBER	Benefit obtained so far
BENEFIT_MAX	NUMBER	Estimate of maximum benefit that could be obtained
FINDINGS	NUMBER	Number of findings so far
RECOMMENDATIONS	NUMBER	Number of recommendations so far
TIME_REMAINING	NUMBER	Estimate of time remaining for the completion of the task (in seconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
START_TIME	DATE	Start time of the task
LAST_UPDATE_TIME	DATE	Last time progress was posted
ELAPSED_SECONDS	NUMBER	Elapsed time so far
ADVISOR_METRIC1	NUMBER	Value of the advisor-specific metric
METRIC1_DESC	VARCHAR2(64)	Description of the advisor-specific metric
EXECUTION_TYPE	VARCHAR2(64)	Type of the last execution. This information is optional for single-execution tasks.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.10 V$ALERT_TYPES

V$ALERT_TYPES displays information about server alert types.

Column	Datatype	Description
REASON_ID	NUMBER	ID of the alert reason
OBJECT_TYPE	VARCHAR2(64)	Object type
TYPE	VARCHAR2(9)	Alert type: • Stateful • Stateless
GROUP_NAME	VARCHAR2(64)	Group name
SCOPE	VARCHAR2(8)	Scope: • Database • Instance
INTERNAL_METRIC_CATEGORY	VARCHAR2(64)	Internal metric category
INTERNAL_METRIC_NAME	VARCHAR2(64)	Internal metric name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.11 V$ALL_ACTIVE_SESSION_HISTORY

V$ALL_ACTIVE_SESSION_HISTORY displays sampled session activity in the database.

> **Note**
>
> V$ALL_ACTIVE_SESSION_HISTORY allows application developers to view their own session activity. It is similar to the V$ACTIVE_SESSION_HISTORY view, with the following exception:
>
> - When queried by a user with the SYSDBA privilege, the V$ALL_ACTIVE_SESSION_HISTORY view displays the same data as the V$ACTIVE_SESSION_HISTORY view.
> - When queried by any other user, the V$ALL_ACTIVE_SESSION_HISTORY view displays only data about sessions owned by that user.
>
> See also "V$ACTIVE_SESSION_HISTORY"

Column	Datatype	Description
SAMPLE_ID	NUMBER	ID of the sample
SAMPLE_TIME	TIMESTAMP(3)	Time at which the sample was taken
SAMPLE_TIME_UTC	TIMESTAMP(3)	Time at which the sample was taken in UTC (Coordinated Universal Time) time zone
USECS_PER_ROW	NUMBER	Time in microseconds since the last sample was taken
IS_AWR_SAMPLE	VARCHAR2(1)	Indicates whether this sample has been flushed or will be flushed to the Automatic Workload Repository (DBA_HIST_ACTIVE_SESS_HISTORY) (Y) or not (N)
SESSION_ID	NUMBER	Session identifier; maps to V$SESSION.SID
SESSION_SERIAL#	NUMBER	Session serial number (used to uniquely identify a session's objects); maps to V$SESSION.SERIAL#
SESSION_TYPE	VARCHAR2(10)	Session type: • FOREGROUND • BACKGROUND
FLAGS	NUMBER	Reserved for future use
USER_ID	NUMBER	Oracle user identifier; maps to V$SESSION.USER#
SQL_ID	VARCHAR2(13)	SQL identifier of the SQL statement that the session was executing at the time of sampling
IS_SQLID_CURRENT	VARCHAR2(1)	Indicates whether the SQL identifier in the SQL_ID column is being executed (Y) or not (N)
SQL_CHILD_NUMBER	NUMBER	Child number of the SQL statement that the session was executing at the time of sampling
SQL_OPCODE	NUMBER	Indicates what phase of operation the SQL statement was in; maps to V$SESSION.COMMAND **See Also:** "V$SESSION" for information on interpreting this column
SQL_OPNAME	VARCHAR2(64)	SQL command name
FORCE_MATCHING_SIGNATURE	NUMBER	Signature used when the CURSOR_SHARING parameter is set to FORCE

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOP_LEVEL_SQL_ID	VARCHAR2(13)	SQL identifier of the top level SQL statement
TOP_LEVEL_SQL_OPCODE	NUMBER	Indicates what phase of operation the top level SQL statement was in
SQL_ADAPTIVE_PLAN_RESOLVED	NUMBER	Indicates whether the SQL plan of the sampled database session is a resolved adaptive plan or not
SQL_FULL_PLAN_HASH_VALUE	NUMBER	Numerical representation of the complete SQL plan for the cursor being executed by this session
SQL_PLAN_HASH_VALUE	NUMBER	Numeric representation of the SQL plan for the cursor. This information might not be available for all session samples. V$SESSION does not contain this information.
SQL_PLAN_LINE_ID	NUMBER	SQL plan line ID
SQL_PLAN_OPERATION	VARCHAR2(30)	Plan operation name
SQL_PLAN_OPTIONS	VARCHAR2(30)	Plan operation options
SQL_EXEC_ID	NUMBER	SQL execution identifier
SQL_EXEC_START	DATE	Time when the execution of the SQL started
PLSQL_ENTRY_OBJECT_ID	NUMBER	Object ID of the top-most PL/SQL subprogram on the stack; NULL if there is no PL/SQL subprogram on the stack. Maps to DBA_OBJECTS.OBJECT_ID.
PLSQL_ENTRY_SUBPROGRAM_ID	NUMBER	Subprogram ID of the top-most PL/SQL subprogram on the stack. Maps to DBA_OBJECTS.DATA_OBJECT_ID.
PLSQL_OBJECT_ID	NUMBER	Object ID of the currently executing PL/SQL subprogram. Maps to DBA_OBJECTS.OBJECT_ID.
PLSQL_SUBPROGRAM_ID	NUMBER	Subprogram ID of the currently executing PL/SQL object; NULL if executing SQL. Maps to DBA_OBJECTS.DATA_OBJECT_ID.
PLSQL_CURRENT_ENTRY_OBJECT_ID	NUMBER	Object ID of the currently executing PL/SQL subprogram; NULL if executing SQL.
QC_INSTANCE_ID	NUMBER	Query coordinator instance ID. This information is only available if the sampled session is a parallel query worker. For all other sessions, the value is 0.
QC_SESSION_ID	NUMBER	Query coordinator session ID. This information is only available if the sampled session is a parallel query worker. For all other sessions, the value is 0.
QC_SESSION_SERIAL#	NUMBER	Query coordinator session serial number. This information is only available if the sampled session is a parallel query worker. For all other sessions, the value is 0.
PX_FLAGS	NUMBER	Reserved for internal use
EVENT	VARCHAR2(64)	If SESSION_STATE = WAITING, then the event for which the session was waiting for at the time of sampling. If SESSION_STATE = ON CPU, then this column is NULL. **See Also:** Oracle Wait Events
EVENT_ID	NUMBER	Identifier of the resource or event for which the session is waiting or for which the session last waited. Interpretation is similar to that of the EVENT column.
EVENT#	NUMBER	Number of the resource or event for which the session is waiting or for which the session last waited. Interpretation is similar to that of the EVENT column.
SEQ#	NUMBER	Sequence number that uniquely identifies the wait (incremented for each wait)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
P1TEXT	VARCHAR2(64)	Text of the first additional parameter
P1	NUMBER	First additional parameter
P2TEXT	VARCHAR2(64)	Text of the second additional parameter
P2	NUMBER	Second additional parameter
P3TEXT	VARCHAR2(64)	Text of the third additional parameter
P3	NUMBER	Third additional parameter
WAIT_CLASS	VARCHAR2(64)	Wait class name of the event for which the session was waiting at the time of sampling. Interpretation is similar to that of the EVENT column. Maps to V$SESSION.WAIT_CLASS.
WAIT_CLASS_ID	NUMBER	Wait class identifier of the event for which the session was waiting at the time of sampling. Interpretation is similar to that of the EVENT column. Maps to V$SESSION.WAIT_CLASS_ID.
WAIT_TIME	NUMBER	Total wait time for the event for which the session last waited if the session was on the CPU when sampled; 0 if the session was waiting at the time of sampling **Note:** Whether or not WAIT_TIME = 0 is what is useful to find the SESSION_STATE at the time of sampling, rather than the actual value of WAIT_TIME itself. Maps to V$SESSION.WAIT_TIME.
SESSION_STATE	VARCHAR2(7)	Session state: • WAITING • ON CPU
TIME_WAITED	NUMBER	If SESSION_STATE = WAITING, then the time that the session actually spent waiting for that event (in microseconds). This column is set for waits that were in progress at the time the sample was taken. If a wait event lasted for more than a second and was caught waiting in more than one session sample row, then the actual time spent waiting for that wait event will be populated in the last of those session sample rows. At any given time, this information will not be available for the latest session sample.
BLOCKING_SESSION_STATUS	VARCHAR2(11)	Status of the blocking session: • VALID • NO HOLDER • GLOBAL • NOT IN WAIT • UNKNOWN
BLOCKING_SESSION	NUMBER	Session identifier of the blocking session. Populated only if the blocker is on the same instance and the session was waiting for enqueues or a "buffer busy" wait. Maps to V$SESSION.BLOCKING_SESSION.
BLOCKING_SESSION_SERIAL#	NUMBER	Serial number of the blocking session
BLOCKING_INST_ID	NUMBER	Instance number of the blocker shown in BLOCKING_SESSION
BLOCKING_HANGCHAIN_INFO	VARCHAR2(1)	Indicates whether the information about BLOCKING_SESSION comes from the hang chain (Y) or not (N)
CURRENT_OBJ#	NUMBER	Object ID of the object that the session is referencing. This information is only available if the session was waiting for application, cluster, concurrency, and user I/O wait events. Maps to V$SESSION.ROW_WAIT_OBJ#.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CURRENT_FILE#	NUMBER	File number of the file containing the block that the session is referencing. This information is only available if the session was waiting for cluster, concurrency, and user I/O wait events. Maps to V$SESSION.ROW_WAIT_FILE#.
CURRENT_BLOCK#	NUMBER	ID of the block that the session is referencing. This information is only available if the session was waiting for cluster, concurrency, and user I/O wait events. Maps to V$SESSION.ROW_WAIT_BLOCK#.
CURRENT_ROW#	NUMBER	Row identifier that the session is referencing. This information is only available if the session was waiting for cluster, concurrency, and user I/O wait events. Maps to V$SESSION.ROW_WAIT_ROW#.
TOP_LEVEL_CALL#	NUMBER	Oracle top level call number
TOP_LEVEL_CALL_NAME	VARCHAR2(64)	Oracle top level call name
CONSUMER_GROUP_ID	NUMBER	Consumer group ID
XID	RAW(8)	Transaction ID that the session was working on at the time of sampling. V$SESSION does not contain this information.
REMOTE_INSTANCE#	NUMBER	Remote instance identifier that will serve the block that this session is waiting for. This information is only available if the session was waiting for cluster events.
TIME_MODEL	NUMBER	Time model information
IN_CONNECTION_MGMT	VARCHAR2(1)	Indicates whether the session was doing connection management at the time of sampling (Y) or not (N)
IN_PARSE	VARCHAR2(1)	Indicates whether the session was parsing at the time of sampling (Y) or not (N)
IN_HARD_PARSE	VARCHAR2(1)	Indicates whether the session was hard parsing at the time of sampling (Y) or not (N)
IN_SQL_EXECUTION	VARCHAR2(1)	Indicates whether the session was executing SQL statements at the time of sampling (Y) or not (N)
IN_PLSQL_EXECUTION	VARCHAR2(1)	Indicates whether the session was executing PL/SQL at the time of sampling (Y) or not (N)
IN_PLSQL_RPC	VARCHAR2(1)	Indicates whether the session was executing inbound PL/SQL RPC calls at the time of sampling (Y) or not (N)
IN_PLSQL_COMPILATION	VARCHAR2(1)	Indicates whether the session was compiling PL/SQL at the time of sampling (Y) or not (N)
IN_JAVA_EXECUTION	VARCHAR2(1)	Indicates whether the session was executing Java at the time of sampling (Y) or not (N)
IN_BIND	VARCHAR2(1)	Indicates whether the session was doing bind operations at the time of sampling (Y) or not (N)
IN_CURSOR_CLOSE	VARCHAR2(1)	Indicates whether the session was closing a cursor at the time of sampling (Y) or not (N)
IN_SEQUENCE_LOAD	VARCHAR2(1)	Indicates whether the session is loading in sequence (in sequence load code) (Y) or not (N)
IN_INMEMORY_QUERY	VARCHAR2(1)	Indicates whether the session was querying the In-Memory Column Store (IM column store) at the time of sampling (Y) or not (N)
IN_INMEMORY_POPULATE	VARCHAR2(1)	Indicates whether the session was populating the IM column store at the time of sampling (Y) or not (N)
IN_INMEMORY_PREPOPULATE	VARCHAR2(1)	Indicates whether the session was prepopulating the IM column store at the time of sampling (Y) or not (N)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
IN_INMEMORY_REPOPULATE	VARCHAR2(1)	Indicates whether the session was repopulating the IM column store at the time of sampling (Y) or not (N)
IN_INMEMORY_TREPOPULATE	VARCHAR2(1)	Indicates whether the session was trickle repopulating the IM column store at the time of sampling (Y) or not (N)
IN_TABLESPACE_ENCRYPTION	VARCHAR2(1)	Indicates whether encryption or decryption of a tablespace occurred at the time of sampling (Y) or not (N)
CAPTURE_OVERHEAD	VARCHAR2(1)	Indicates whether the session is executing capture code (Y) or not (N)
REPLAY_OVERHEAD	VARCHAR2(1)	Indicates whether the session is executing replay code (Y) or not (N)
IS_CAPTURED	VARCHAR2(1)	Indicates whether the session is being captured (Y) or not (N)
IS_REPLAYED	VARCHAR2(1)	Indicates whether the session is being replayed (Y) or not (N)
IS_REPLAY_SYNC_TOKEN_HOLDER	VARCHAR2(1)	Indicates whether the session is holding a synchronization token (Y) or not (N) during workload replay
SERVICE_HASH	NUMBER	Hash that identifies the Service; maps to V$ACTIVE_SERVICES.NAME_HASH
PROGRAM	VARCHAR2(48)	Name of the operating system program
MODULE	VARCHAR2(64)	Name of the executing module when sampled, as set by the DBMS_APPLICATION_INFO.SET_MODULE procedure
ACTION	VARCHAR2(64)	Name of the executing module when sampled, as set by the DBMS_APPLICATION_INFO.SET_ACTION procedure
CLIENT_ID	VARCHAR2(64)	Client identifier of the session; maps to V$SESSION.CLIENT_IDENTIFIER
MACHINE	VARCHAR2(64)	Client's operating system machine name
PORT	NUMBER	Client port number
ECID	VARCHAR2(64)	Execution context identifier (sent by Application Server)
DBREPLAY_FILE_ID	NUMBER	If the session is being captured or replayed, then DBREPLAY_FILE_ID is the file ID for the workload capture or workload replay; otherwise it is NULL.
DBREPLAY_CALL_COUNTER	NUMBER	If the session is being captured or replayed, then DBREPLAY_CALL_COUNTER is the call counter of the user call that is being captured or replayed; otherwise it is NULL.
TM_DELTA_TIME	NUMBER	Time interval (in microseconds) over which TM_DELTA_CPU_TIME and TM_DELTA_DB_TIME are accumulated
TM_DELTA_CPU_TIME	NUMBER	Amount of time this session spent on CPU over the last TM_DELTA_TIME microseconds
TM_DELTA_DB_TIME	NUMBER	Amount of time spent by this session in database calls over the last TM_DELTA_TIME microseconds
DELTA_TIME	NUMBER	Time interval (in microseconds) since the last time this session was sampled or created, over which the next five statistics are accumulated
DELTA_READ_IO_REQUESTS	NUMBER	Number of read I/O requests made by this session over the last DELTA_TIME microseconds
DELTA_WRITE_IO_REQUESTS	NUMBER	Number of write I/O requests made by this session over the last DELTA_TIME microseconds
DELTA_READ_IO_BYTES	NUMBER	Number of I/O bytes read by this session over the last DELTA_TIME microseconds
DELTA_WRITE_IO_BYTES	NUMBER	Number of I/O bytes written by this session over the last DELTA_TIME microseconds

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DELTA_INTERCONNECT_IO_BYTES	NUMBER	Number of I/O bytes sent over the I/O interconnect over the last DELTA_TIME microseconds
DELTA_READ_MEM_BYTES	NUMBER	Number of read bytes through the buffer cache
PGA_ALLOCATED	NUMBER	Amount of PGA memory (in bytes) consumed by this session at the time this sample was taken
TEMP_SPACE_ALLOCATED	NUMBER	Amount of TEMP memory (in bytes) consumed by this session at the time this sample was taken
CON_DBID	NUMBER	The database ID of the pluggable database (PDB)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
DBOP_NAME	VARCHAR2(30)	Database operation name. If the type is SQL, the DBOP_NAME will be NULL.
DBOP_EXEC_ID	NUMBER	Database operation execution identifier for the current execution. If the type is SQL, the DBOP_EXEC_ID will be NULL.

7.12 V$ALL_SQL_MONITOR

V$ALL_SQL_MONITOR displays SQL statements whose execution have been (or are being) monitored by Oracle.

> **Note**
>
> V$ALL_SQL_MONITOR allows application developers to view information about their own SQL statements. It is similar to the V$SQL_MONITOR view, with the following exception:
>
> - When queried by a user with the SYSDBA privilege, the V$ALL_SQL_MONITOR view displays the same data as the V$SQL_MONITOR view.
> - When queried by any other user, the V$ALL_SQL_MONITOR view displays data only about SQL statements that were executed in sessions owned by that user.
>
> See also "V$SQL_MONITOR"

Column	Datatype	Description
KEY	NUMBER	Artificial join key to efficiently join V$ALL_SQL_MONITOR with its corresponding plan level monitoring statistics stored in V$ALL_SQL_PLAN_MONITOR
REPORT_ID	NUMBER	Unique ID of the XML report stored in Automatic Workload Repository (AWR) for this monitored entity

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(19)	SQL execution status: • QUEUED - SQL statement is queued • EXECUTING - SQL statement is still executing • DONE (ERROR) - Execution terminated with an error • DONE (FIRST N ROWS) - Execution terminated by the application before all rows were fetched • DONE (ALL ROWS) - Execution terminated and all rows were fetched • DONE - Execution terminated (parallel execution)
USER#	NUMBER	User ID of the database user who issued the SQL being monitored
USERNAME	VARCHAR2(128)	User name of the database user who issued the SQL being monitored
MODULE	VARCHAR2(64)	Name of the executing module when sampled, as set by the DBMS_APPLICATION_INFO.SET_MODULE procedure
ACTION	VARCHAR2(64)	Name of the executing action when sampled, as set by the DBMS_APPLICATION_INFO.SET_ACTION procedure
SERVICE_NAME	VARCHAR2(64)	Service name of the user session
CLIENT_IDENTIFIER	VARCHAR2(64)	Client identifier from the user session
CLIENT_INFO	VARCHAR2(64)	Client information for the user session
PROGRAM	VARCHAR2(48)	Name of the operating system program that issued the monitored SQL
PLSQL_ENTRY_OBJECT_ID	NUMBER	Object ID of the top-most PL/SQL subprogram on the stack; NULL if there is no PL/SQL subprogram on the stack
PLSQL_ENTRY_SUBPROGRAM_ID	NUMBER	Subprogram ID of the top-most PL/SQL subprogram on the stack; NULL if there is no PL/SQL subprogram on the stack
PLSQL_OBJECT_ID	NUMBER	Object ID of the currently executing PL/SQL subprogram; NULL if executing SQL
PLSQL_OWNER_ID	NUMBER	ID of the owner of the currently executing PL/SQL subprogram; NULL if executing SQL
PLSQL_SUBPROGRAM_ID	NUMBER	Subprogram ID of the currently executing PL/SQL object; NULL if executing SQL
FIRST_REFRESH_TIME	DATE	Time when monitoring of the SQL statement started, generally a few seconds after execution start time
LAST_REFRESH_TIME	DATE	Time when statistics in V$ALL_SQL_MONITOR were last updated for the SQL statement. Statistics are generally refreshed every second when the statement executes.
REFRESH_COUNT	NUMBER	Number of times V$ALL_SQL_MONITOR statistics have been refreshed (generally once every second when the SQL statement executes)
DBOP_EXEC_ID	NUMBER	Database operation execution identifier for the current execution. If the type is SQL, the DBOP_EXEC_ID will be NULL.
DBOP_NAME	VARCHAR2(30)	Database operation name. If the type is SQL, the DBOP_NAME will be NULL.
SID	NUMBER	Session identifier executing (or having executed) the SQL statement being monitored
PROCESS_NAME	VARCHAR2(5)	Process name identifier executing (or having executed)the statement; ora if the process is foreground, else the background process name (for example, p001 for PX server p001)
SQL_ID	VARCHAR2(13)	SQL identifier of the statement being monitored
SQL_TEXT	VARCHAR2(2000)	Up to the first 2000 characters of the text of the SQL being monitored

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
IS_FULL_SQLTEXT	VARCHAR2(1)	Indicates whether the SQL_TEXT column has the entire SQL text (Y) or not (N)
SQL_EXEC_START	DATE	Time when the execution started
SQL_EXEC_ID	NUMBER	Execution identifier. Together, the three columns SQL_ID, SQL_EXEC_START, and SQL_EXEC_ID represent the execution key. The execution key is used to uniquely identify one execution of the SQL statement.
SQL_PLAN_HASH_VALUE	NUMBER	SQL plan hash value
SQL_FULL_PLAN_HASH_VALUE	NUMBER	Numeric representation of the complete SQL plan for this cursor. Comparing one SQL_FULL_PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line). Note that the SQL_FULL_PLAN_HASH_VALUE cannot be compared across databases releases. It is not backward compatible.
EXACT_MATCHING_SIGNATURE	NUMBER	Signature calculated on the normalized SQL text. The normalization includes the removal of white space and the uppercasing of all non-literal strings.
FORCE_MATCHING_SIGNATURE	NUMBER	Same as EXACT_MATCHING_SIGNATURE but literals in the SQL text are replaced by binds
SQL_CHILD_ADDRESS	RAW(8)	Address of the child cursor (can be used with SQL_ID to join with V$SQL)
SESSION_SERIAL#	NUMBER	Session serial number executing the statement being monitored
PX_IS_CROSS_INSTANCE	VARCHAR2(1)	Indicates whether the SQL statement ran parallel across multiple instances (Y) or not (N)
PX_MAXDOP	NUMBER	Maximum degree of parallelism for any plan operation executed on behalf of the monitored SQL
PX_MAXDOP_INSTANCES	NUMBER	Number of database instances touched at the maximum degree of parallelism
PX_SERVERS_REQUESTED	NUMBER	Total number of parallel execution servers requested to execute the monitored SQL
PX_SERVERS_ALLOCATED	NUMBER	Actual number of parallel execution servers allocated to execute the query
PX_SERVER#	NUMBER	Logical parallel execution server process number executing (or having executed) the statement being monitored; NULL if this monitoring entry is not associated with an execution server. This is a logical number within the parallel server set (see SERVER# in V$PX_SESSION).
PX_SERVER_GROUP	NUMBER	Logical parallel execution server group number to which PX_SERVER# belongs (see SERVER_GROUP in V$PX_SESSION); NULL if this monitoring entry is not associated with a parallel execution server. This value is generally 1 unless the SQL statement has one or more parallel sub-queries.
PX_SERVER_SET	NUMBER	Number (1 or 2) of the logical set of parallel execution servers to which PX_SERVER# belongs (see SERVER_SET in V$PX_SESSION); NULL if this monitoring entry is not associated with a parallel execution server
PX_QCINST_ID	NUMBER	Instance identifier where the parallel execution coordinator runs; NULL if PX_SERVER# is NULL
PX_QCSID	NUMBER	Session identifier for the parallel execution coordinator; NULL if PX_SERVER# is NULL
ERROR_NUMBER	VARCHAR2(40)	Error number encountered in case a SQL fails to execute successfully (for example, 932 in case of ORA-00932)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ERROR_FACILITY	VARCHAR2(4)	Error facility in case a SQL fails to execute successfully (for example, ORA in case of ORA-00932)
ERROR_MESSAGE	VARCHAR2(256)	Detailed error message displayed corresponding to the error number and error facility when a SQL fails to execute successfully
BINDS_XML	CLOB	Information about bind variables used with the SQL, such as name, position, value, data type, and so on (stored in XML format)
OTHER_XML	CLOB	Additional information about SQL execution stored in XML format
ELAPSED_TIME	NUMBER	Elapsed time (in microseconds); updated as the statement executes
QUEUING_TIME	NUMBER	Duration of time (in microseconds) spent by SQL in the statement queue
CPU_TIME	NUMBER	CPU time (in microseconds); updated as the statement executes
FETCHES	NUMBER	Number of fetches associated with the SQL statement; updated as the statement executes
BUFFER_GETS	NUMBER	Number of buffer get operations; updated as the statement executes
DISK_READS	NUMBER	Number of disk reads; updated as the statement executes
DIRECT_WRITES	NUMBER	Number of direct writes; updated as the statement executes
IO_INTERCONNECT_BYTES	NUMBER	Number of I/O bytes exchanged between Oracle Database and the storage system
PHYSICAL_READ_REQUESTS	NUMBER	Number of physical read I/O requests issued by the monitored SQL
PHYSICAL_READ_BYTES	NUMBER	Number of bytes read from disks by the monitored SQL
PHYSICAL_WRITE_REQUESTS	NUMBER	Number of physical write I/O requests issued by the monitored SQL
PHYSICAL_WRITE_BYTES	NUMBER	Number of bytes written to disks by the monitored SQL
APPLICATION_WAIT_TIME	NUMBER	Application wait time (in microseconds); updated as the statement executes
CONCURRENCY_WAIT_TIME	NUMBER	Concurrency wait time (in microseconds); updated as the statement executes
CLUSTER_WAIT_TIME	NUMBER	Cluster wait time (in microseconds); updated as the statement executes
USER_IO_WAIT_TIME	NUMBER	User I/O Wait Time (in microseconds); updated as the statement executes
PLSQL_EXEC_TIME	NUMBER	PL/SQL execution time (in microseconds); updated as the statement executes
JAVA_EXEC_TIME	NUMBER	Java execution time (in microseconds); updated as the statement executes
RM_LAST_ACTION	VARCHAR2(48)	The most recent action that was taken on this SQL operation by Resource Manager. Its value is one of the following: • CANCEL_SQL • KILL_SESSION • LOG_ONLY • SWITCH TO <CG NAME> For the last value, <CG NAME> is the name of the consumer group that the SQL operation was switched to. If the Resource Plan has since been changed then <CG NAME> is the ID of the consumer group.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RM_LAST_ACTION_REASON	VARCHAR2(128)	The reason for the most recent action that was taken on this SQL operation by Resource Manager. Its value is one of the following: • SWITCH_CPU_TIME • SWITCH_IO_REQS • SWITCH_IO_MBS • SWITCH_ELAPSED_TIME • SWITCH_IO_LOGICAL
RM_LAST_ACTION_TIME	DATE	The time of the most recent action that was taken on this SQL operation by Resource Manager
RM_CONSUMER_GROUP	VARCHAR2(128)	The current consumer group for this SQL operation
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
CON_NAME	VARCHAR2(128)	Container name of the object. The value of this column is NULL in non-CDBs.
ECID	VARCHAR2(64)	Execution context identifier (sent by Application Server)
IS_ADAPTIVE_PLAN	VARCHAR2(1)	Indicates whether the statistics are from an adaptive plan (Y) or not (N).
IS_FINAL_PLAN	VARCHAR2(1)	Indicates whether the statistics are from the final plan (Y) or not (N).
IN_DBOP_NAME	VARCHAR2(30)	If the SQL that is monitored was executed by a session that was also monitored by a database operation (DBOP), then this column specifies the name of that DBOP
IN_DBOP_EXEC_ID	NUMBER	If the SQL that is monitored was executed by a session that was also monitored by a database operation (DBOP), then this column specifies the execution ID of that DBO
IO_CELL_UNCOMPRESSED_BYTES	NUMBER	Number of uncompressed bytes (that is, size after decompression) that are offloaded to the Exadata cells **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_CELL_OFFLOAD_ELIGIBLE_BYTES	NUMBER	Number of I/O bytes which can be filtered by the Exadata storage system **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_CELL_OFFLOAD_RETURNED_BYTES	NUMBER	Number of filtered bytes returned by Exadata cells (that is, the number of bytes returned after processing has been offloaded on the Exadata cells) **See Also:** Oracle Exadata Storage Server Software documentation for more information
CURRENT_USER#	NUMBER	Unique number identifying the current user
CURRENT_USERNAME	VARCHAR2(128)	Username for the current user

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.13 V$ALL_SQL_PLAN_MONITOR

V$ALL_SQL_PLAN_MONITOR displays plan level monitoring statistics for each SQL statement found in V$ALL_SQL_MONITOR.

> **Note**
>
> V$ALL_SQL_PLAN_MONITOR allows application developers to view statistics about their own SQL statements. It is similar to the V$SQL_PLAN_MONITOR view, with the following exception:
>
> - When queried by a user with the SYSDBA privilege, the V$ALL_SQL_PLAN_MONITOR view displays the same data as the V$SQL_PLAN_MONITOR view.
> - When queried by any other user, the V$ALL_SQL_PLAN_MONITOR view displays data only about SQL statements that were executed in sessions owned by that user.
>
> See also "V$SQL_PLAN_MONITOR"

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
KEY	NUMBER	Foreign key to efficiently join V$ALL_SQL_PLAN_MONITOR with V$ALL_SQL_MONITOR (see V$ALL_SQL_MONITOR)
STATUS	VARCHAR2(19)	SQL execution status: • EXECUTING - SQL statement is still executing • DONE (ERROR) - Execution terminated with an error • DONE (FIRST N ROWS) - Execution terminated by the application before all rows were fetched • DONE (ALL ROWS) - Execution terminated and all rows were fetched • DONE - Execution terminated (parallel execution)
FIRST_REFRESH_TIME	DATE	Time when monitoring of the SQL statement started
LAST_REFRESH_TIME	DATE	Time when statistics were last updated for the SQL statement
FIRST_CHANGE_TIME	DATE	First time a row was produced by this operation
LAST_CHANGE_TIME	DATE	Last time a row was produced by this operation
REFRESH_COUNT	NUMBER	Number of times statistics have been refreshed
SID	NUMBER	Session identifier executing (or having executed) the SQL statement being monitored
PROCESS_NAME	VARCHAR2(5)	Process name identifier
SQL_ID	VARCHAR2(13)	SQL identifier
SQL_EXEC_START	DATE	Time when the execution started
SQL_EXEC_ID	NUMBER	Execution identifier

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SQL_PLAN_HASH_VALUE	NUMBER	SQL plan hash value
SQL_CHILD_ADDRESS	RAW(8)	Address of the child cursor
PLAN_PARENT_ID	NUMBER	ID of the next execution step that operates on the output of the current step
PLAN_LINE_ID	NUMBER	Plan line number for the entry
PLAN_OPERATION	VARCHAR2(30)	Plan operation name (from V$SQL_PLAN)
PLAN_OPTIONS	VARCHAR2(30)	Plan option name (from V$SQL_PLAN)
PLAN_OBJECT_OWNER	VARCHAR2(128)	Name of the user who owns the schema containing the table or index
PLAN_OBJECT_NAME	VARCHAR2(128)	Name of the table or index
PLAN_OBJECT_TYPE	VARCHAR2(80)	Type of the object
PLAN_DEPTH	NUMBER	Depth (or level) of the operation in the tree. It is not necessary to issue a CONNECT BY statement to get the level information, which is generally used to indent the rows from the PLAN_TABLE table. The root operation (statement) is level 0.
PLAN_POSITION	NUMBER	Order of processing for all operations that have the same PARENT_ID
PLAN_COST	NUMBER	Cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
PLAN_CARDINALITY	NUMBER	Estimate, by the cost-based optimizer, of the number of rows produced by the operation
PLAN_BYTES	NUMBER	Estimate, by the cost-based optimizer, of the number of bytes produced by the operation
PLAN_TIME	NUMBER	Elapsed time (in seconds) of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
PLAN_PARTITION_START	VARCHAR2(256)	Start partition of a range of accessed partitions
PLAN_PARTITION_STOP	VARCHAR2(256)	Stop partition of a range of accessed partitions
PLAN_CPU_COST	NUMBER	CPU cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
PLAN_IO_COST	NUMBER	I/O cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
PLAN_TEMP_SPACE	NUMBER	Temporary space usage of the operation (sort or hash-join) as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
STARTS	NUMBER	Number of times this operation was executed. For example, an operation is executed multiple times when it is on the right side of a nested-loop join (once for each row of the left input of that nested-loop join).
OUTPUT_ROWS	NUMBER	Number of rows produced by this operation since the execution started. This number is cumulated for all executions of this operation. Divide by the value of the STARTS column to compute the average number of rows per execution of the operation. Note that the value in the STARTS column is equal to or higher than the value in the OUTPUT_ROWS column. The value will usually be equal, but depending on internal optimizations a higher value may be seen.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
IO_INTERCONNECT_BYTES	NUMBER	Number of I/O bytes exchanged between Oracle Database and the storage system. Maintained only after Oracle starts to monitor the execution.
PHYSICAL_READ_REQUESTS	NUMBER	Number of physical read I/O requests issued by the monitored SQL. Maintained only after Oracle starts to monitor the execution.
PHYSICAL_READ_BYTES	NUMBER	Number of bytes read from disks by the monitored SQL. Maintained only after Oracle starts to monitor the execution.
PHYSICAL_WRITE_REQUESTS	NUMBER	Number of physical write I/O requests issued by the monitored SQL. Maintained only after Oracle starts to monitor the execution.
PHYSICAL_WRITE_BYTES	NUMBER	Number of bytes written to disks by the monitored SQL. Maintained only after Oracle starts to monitor the execution.
WORKAREA_MEM	NUMBER	Amount of memory (in bytes) used by the operation when the query is executing; NULL if the execution is done. This applies only to operations using a work area, such as sort, hash-join, group-by, and so on.
WORKAREA_MAX_MEM	NUMBER	Maximum value (in bytes) for WORKAREA_MEM; NULL if the operation is not using a work area. When the execution is finished, this value will hold the maximum amount of memory consumed by this operation during the execution of the statement.
WORKAREA_TEMPSEG	NUMBER	Amount of temporary space (in bytes) used by the operation when the query is executing; NULL if the operation has not spilled to disk or if the execution is finished
WORKAREA_MAX_TEMPSEG	NUMBER	Maximum value (in bytes) for WORKAREA_TEMPSEG; NULL if this operation never spilled to disk. When the execution is done, this value will hold the maximum amount of temporary space consumed by this operation during the entire execution.
OTHERSTAT_GROUP_ID	NUMBER	Plan line statistic group identifier (see GROUP_ID column in V$SQL_MONITOR_STATNAME)
OTHERSTAT_1_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 1 of that plan line
OTHERSTAT_1_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_1_VALUE	NUMBER	Value of statistic number 1 of that plan line
OTHERSTAT_2_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 2 of that plan line
OTHERSTAT_2_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_2_VALUE	NUMBER	Value of statistic number 2 of that plan line
OTHERSTAT_3_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 3 of that plan line
OTHERSTAT_3_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_3_VALUE	NUMBER	Value of statistic number 3 of that plan line
OTHERSTAT_4_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 4 of that plan line
OTHERSTAT_4_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_4_VALUE	NUMBER	Value of statistic number 4 of that plan line
OTHERSTAT_5_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 5 of that plan line
OTHERSTAT_5_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_5_VALUE	NUMBER	Value of statistic number 5 of that plan line

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OTHERSTAT_6_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 6 of that plan line
OTHERSTAT_6_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_6_VALUE	NUMBER	Value of statistic number 6 of that plan line
OTHERSTAT_7_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 7 of that plan line
OTHERSTAT_7_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_7_VALUE	NUMBER	Value of statistic number 7 of that plan line
OTHERSTAT_8_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 8 of that plan line
OTHERSTAT_8_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_8_VALUE	NUMBER	Value of statistic number 8 of that plan line
OTHERSTAT_9_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 9 of that plan line
OTHERSTAT_9_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_9_VALUE	NUMBER	Value of statistic number 9 of that plan line
OTHERSTAT_10_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 10 of that plan line
OTHERSTAT_10_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_10_VALUE	NUMBER	Value of statistic number 10 of that plan line
OTHER_XML	CLOB	Provides extra information specific to an execution step of the execution plan. The content of this column is structured using XML because it allows multiple pieces of information to be stored, including the following: • Name of the schema against which the query was parsed • Release number of the Oracle Database that produced the explain plan • Hash value associated with the execution plan • Name (if any) of the outline or the SQL profile used to build the execution plan • Indication of whether or not dynamic statistics were used to produce the plan • The outline data, a set of optimizer hints that can be used to regenerate the same plan
PLAN_OPERATION_INACTIVE	NUMBER	Indicates whether this plan operation was part of the final resolved plan

7.14 V$AQ

V$AQ displays statistics for the queues in the database.

Column	Datatype	Description
QID	NUMBER	Unique queue identifier
WAITING	NUMBER	Number of messages in the queue in the state 'WAITING'
READY	NUMBER	Number of messages in the queue in the state 'READY'
EXPIRED	NUMBER	Number of messages in the queue the state 'EXPIRED'

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
AVERAGE_MSG_AGE	NUMBER	Average age of the messages in the queue'
TOTAL_WAIT	NUMBER	Total wait time of all 'READY' messages in the queue
AVERAGE_WAIT	NUMBER	Average wait time of 'READY' messages in the queue
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Advanced Queuing User's Guide for more information about Oracle Database Advanced Queueing

7.15 V$AQ_BACKGROUND_COORDINATOR

V$AQ_BACKGROUND_COORDINATOR lists performance statistics for the Oracle Database Advanced Queueing primary background coordinator process (AQPC).

Column	Datatype	Description
PROCESS_ID	VARCHAR2(24)	Operating system process ID of the primary
PROCESS_NAME	VARCHAR2(48)	Operating system name of the primary
NUM_JOBS	NUMBER	Number of jobs started
JOB_LATENCY	NUMBER	Job start latency
NUM_COORDINATORS	NUMBER	Number of primaries started
CON_ID	NUMBER	The ID of the container to which the data pertains. The CON_ID value in this view is always 0. The rows pertain to the entire CDB or to the non-CDB.

See Also

Oracle Database Advanced Queuing User's Guide for more information about Oracle Database Advanced Queueing

7.16 V$AQ_BMAP_NONDUR_SUBSCRIBERS

V$AQ_BMAP_NONDUR_SUBSCRIBERS can be used to get the available bit positions. The view is queried to get the free bit position during creation of a non-durable subscriber.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
QUEUE_ID	NUMBER	Queue ID
USED_POS	RAW(128)	Stream of bits to identify used and available bit positions
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Advanced Queuing User's Guide for more information about Oracle Database Advanced Queueing

7.17 V$AQ_CROSS_INSTANCE_JOBS

V$AQ_CROSS_INSTANCE_JOBS describes each of the cross process jobs.

Each job serves to forward messages for a shard from a source instance to a destination instance for a set of subscribers of a sharded queue.

Column	Datatype	Description
JOB_ID	NUMBER	Job ID within this coordinator
SCHEMA_NAME	VARCHAR2(128)	Source schema of the job
QUEUE_NAME	VARCHAR2(128)	Source queue name of the cross job
SHARD_ID	NUMBER	Source shard ID
START_SUBSHARD_ID	NUMBER	Start subshard ID of the job
DESTINATION_INSTANCE_ID	NUMBER	Destination instance of the cross job
COORDINATOR_ID	NUMBER	Index of the coordinator serving the job
DEST_SERVER_PROCESS_ID	NUMBER	Process ID of the destination server
JOB_STATE	VARCHAR2(28)	State of the job: • REQUESTED • RUNNING • STOPPED • PAUSED • CRASHED • INACTIVE
FLOW_CONTROL	NUMBER	Indicates whether the job is flow controlled: • 0 - The job is not flow controlled • 1 - The job is flow controlled
MSGS_SENT	NUMBER	Messages sent during the job
BYTES_SENT	NUMBER	Bytes sent during the job

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ACK_LATENCY	NUMBER	Latency for receiving ACK for the job
JOB_TYPE	VARCHAR2(26)	Cross instance job type. Values: • CROSS_STREAM : This job type is responsible for forwarding a shard from its owner instance to a destination dequeue instance for all subscribers performing dequeue from that shard at the destination dequeue instance. • DEQUEUE_AFFINITY_TO_REMOTE: This job type is responsible for switching a subscriber's dequeue affinity from a shard's owner instance to a remote dequeue instance. • DEQUEUE_AFFINITY_TO_LOCAL: This job type is responsible for switching back a subscriber's dequeue affinity from a remote dequeue instance to a shard's owner instance.
PRIORITY0_CROSS_LWM	NUMBER	Last priority 0 subshard received at DESTINATION_INSTANCE_ID
PRIORITY1_CROSS_LWM	NUMBER	Last priority 1 subshard received at DESTINATION_INSTANCE_ID
PRIORITY2_CROSS_LWM	NUMBER	Last priority 2 subshard received at DESTINATION_INSTANCE_ID
PRIORITY3_CROSS_LWM	NUMBER	Last priority 3 subshard received at DESTINATION_INSTANCE_ID
PRIORITY4_CROSS_LWM	NUMBER	Last priority 4 subshard received at DESTINATION_INSTANCE_ID
PRIORITY5_CROSS_LWM	NUMBER	Last priority 5 subshard received at DESTINATION_INSTANCE_ID
PRIORITY6_CROSS_LWM	NUMBER	Last priority 6 subshard received at DESTINATION_INSTANCE_ID
PRIORITY7_CROSS_LWM	NUMBER	Last priority 7 subshard received at DESTINATION_INSTANCE_ID
PRIORITY8_CROSS_LWM	NUMBER	Last priority 8 subshard received at DESTINATION_INSTANCE_ID
PRIORITY9_CROSS_LWM	NUMBER	Last priority 9 subshard received at DESTINATION_INSTANCE_ID
JOB_START_TIME	TIMESTAMP(3) WITH TIME ZONE	Start time of this job
SUBSCRIBER_ID	NUMBER	Subscriber ID whose affinity is being switched for the DEQUEUE_AFFINITY_TO_REMOTE and DEQUEUE_AFFINITY_TO_LOCAL job types
SUBSCRIBER_NAME	VARCHAR2(512)	Subscriber name whose affinity is being switched for the DEQUEUE_AFFINITY_TO_REMOTE and DEQUEUE_AFFINITY_TO_LOCAL job types. This column is NULL for the CROSS_STREAM job type.
OWNER_INSTANCE_ID	NUMBER	Owner instance of the shard
QUEUE_ID	NUMBER	Queue ID
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Advanced Queuing User's Guide for more information about Oracle Database Advanced Queueing

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.18 V$AQ_IPC_ACTIVE_MSGS

V$AQ_IPC_ACTIVE_MSGS displays the information related to active IPC messages being processed by AQ background processes.

Column	Datatype	Description
PROCESS_NAME	VARCHAR2(48)	Worker process name
PROCESS_ID	NUMBER	Oracle process ID for this worker
SLAVE_STATE	NUMBER	Worker status: • 3: Picked task • 4: Running • 7: Idle
SLAVEOBJ_STATE	NUMBER	Worker object status: • 0: Invalid • 1: Valid
SEQUENCE_NUMBER	NUMBER	Message sequence number
MSG_CLASS_NAME	VARCHAR2(30)	Message class name
MSG_FLAGS	NUMBER	Message flags: • 1: This message needs ack • 2: Short • 4: Long • 8: Priority • 10: Special ack
MSG_SUBMT_TIME	NUMBER	Time this message was added into the IPC primary's list (in seconds)
MSG_PICKD_TIME	NUMBER	Time this message was picked by worker or primary for processing (in seconds)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.19 V$AQ_IPC_MSG_STATS

V$AQ_IPC_MSG_STATS displays the statistics of each IPC message class, such as the total number of invocations of a message class, total pending message/processed message count, and last failure related data. Information like total processed message count, average pending time/average processing time gives a real-time outline of AQ IPC background state.

Column	Datatype	Description
MSG_CLASS_NAME	VARCHAR2(30)	Message class name
TOTAL_MSG_CALLS	NUMBER	Total number of calls for this message class

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOTAL_ACTIVE_MSGS	NUMBER	Total number of active messages presently processed by workers and primary
TOTAL_PENDING_MSGS	NUMBER	Total number of pending messages in the primary's local context
TOTAL_PROCESSED_MSGS	NUMBER	Total number of processed messages
LAST_RECEIVED_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when the last message of this type was received in the primary's context list
LAST_PROCESS_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when the last message was picked for processing
LAST_DONE_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when the last message was done processing
AVERAGE_PENDING_TIME	NUMBER	Average pending time for this message class (in seconds)
AVERAGE_PROCESSING_TIME	NUMBER	Average processing time for this message class (in seconds)
LAST_FAILURE_TIME	TIMESTAMP(3) WITH TIME ZONE	Time of the last failure for this message class
LAST_ERROR_MSG	VARCHAR2(512)	Last error message for this message class
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.20 V$AQ_IPC_PENDING_MSGS

V$AQ_IPC_PENDING_MSGS displays information about pending messages, present in the local primary context.

Column	Datatype	Description
SEQUENCE_NUMBER	NUMBER	Message sequence number
MSG_CLASS_NAME	VARCHAR2(30)	Message class name
MSG_FLAGS	NUMBER	Message flags: • 1: This message needs ack • 2: Short • 4: Long • 8: Priority • 10: Special ack
MSG_SUBMT_TIME	NUMBER	Time this message was added into the IPC primary's list (in seconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.21 V$AQ_JOB_COORDINATOR

V$AQ_JOB_COORDINATOR lists performance statistics per coordinator, for every AQ coordinator controlled by the Oracle Database Advanced Queueing primary coordinator.

Column	Datatype	Description
COORDINATOR_ID	NUMBER	ID of the coordinator
PROCESS_ID	VARCHAR2(24)	Operating system process ID of the coordinator
PROCESS_NAME	VARCHAR2(48)	Operating system process name of the coordinator
JOB_NAME	VARCHAR2(32)	Name of the job handled
JOB_TYPE	NUMBER	Type of job handled
SERVER_COUNT	NUMBER	Number of servers active
MAX_SERVER_COUNT	NUMBER	Maximum server fanout achieved
CON_ID	NUMBER	The ID of the container to which the data pertains. The CON_ID value in this view is always 0. The rows pertain to the entire CDB or to the non-CDB.

See Also

Oracle Database Advanced Queuing User's Guide for more information about Oracle Database Advanced Queueing

7.22 V$AQ_MESSAGE_CACHE

V$AQ_MESSAGE_CACHE provides performance statistics of the message cache for sharded queues at the subshard level in the instance.

Column	Datatype	Description
QUEUE_ID	NUMBER	Queue ID
SHARD_ID	NUMBER	Shard ID
PRIORITY	NUMBER	Priority of the subshard for dequeue, range 0-9
SUBSHARD_ID	NUMBER	Subshard ID in the shard
PARTITION_ID	NUMBER	Partition id for the particular subshard

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MAX_MSGS	NUMBER	Maximum number of messages of subshard
ENQUEUED_MSGS	NUMBER	Number of messages enqueued for the subshard
MSGS_MADE_EXPIRED	NUMBER	Number of messages made expired
CHUNK_SIZE	NUMBER	The size of the memory chunk for storing messages
NUM_CHUNKS	NUMBER	Number of chunks for the subshard
NUM_FREE_CHUNKS	NUMBER	Number of free chunks for the subshard
USED_MEMORY_SIZE	NUMBER	Total estimated size of memory in use (in bytes) for the subshard
STATE	VARCHAR2(13)	Subshard state. Possible values: • CACHED (in memory) • UNCACHED (on disk) • UNCACHED_FREE • CACHED_FREE • UNKNOWN
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Advanced Queuing User's Guide for more information about Oracle Database Advanced Queueing

7.23 V$AQ_MESSAGE_CACHE_ADVICE

V$AQ_MESSAGE_CACHE_ADVICE shows simulated metrics for a range of potential message cache sizes. This view assists in cache sizing by providing information in the form of metrics as described below.

Column	Datatype	Description
SIZE_FOR_ESTIMATE	NUMBER	Cache size for simulation (in megabytes)
SIZE_FACTOR	NUMBER	Size factor with respect to the current cache size
ESTD_SIZE_TYPE	VARCHAR2(9)	Possible values: • MINIMUM: This cache size is required to have all dequeues in-memory (no uncached). • PREFERRED: This is the smallest message cache size required to eliminate 80% of unevictions that occur with a message cache size of MINIMUM for subscribers who keep up, that is, subscribers whose dequeue rate matches the enqueue rate. • MAXIMUM: This cache size is required to have zero evictions. • NULL: This is the value in all other cases.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ESTD_CACHED_SUBSHARDS	NUMBER	Estimated number of cached subshards for this size
ESTD_UNCACHED_SUBSHARDS	NUMBER	Estimated number of uncached subshards for this size
ESTD_EVICTIONS	NUMBER	Estimated number of subshards evicted for this size
ESTD_EVICTION_RATE	NUMBER	Estimated number of subshards getting evicted per minute
ESTD_FG_UNEVICTIONS	NUMBER	Estimated number of subshards unevicted by foreground processes
ESTD_FG_UNEVICTION_RATE	NUMBER	Estimated number of subshards getting unevicted by foreground processes
ESTD_BG_UNEVICTIONS	NUMBER	Estimated number of subshards unevicted by background processes
ESTD_BG_UNEVICTION_RATE	NUMBER	Estimated number of subshards getting unevicted by background processes
ESTD_BG_PROCESSES	NUMBER	Estimated number of background processes required for this size
TOTAL_ENQUEUE_RATE	NUMBER	Simulated number of messages being enqueued per second
TOTAL_DEQUEUE_RATE	NUMBER	Simulated number of messages being dequeued per second
AVG_SUBSHARD_SIZE	NUMBER	Simulated average number of messages per cached subshard
AVG_SUBSHARD_MEMORY	NUMBER	Simulated average memory per cached subshard (in megabytes)
AVG_EVICTION_TIME	NUMBER	Simulated average time to evict a cached subshard (in milliseconds)
AVG_UNEVICTION_TIME	NUMBER	Simulated average time to unevict a cached subshard (in milliseconds)
FLAGS	NUMBER	Reserved for internal use
SIMULATION_TIME	NUMBER	Amount of time that was simulated for (in minutes)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.24 V$AQ_MESSAGE_CACHE_STAT

V$AQ_MESSAGE_CACHE_STAT displays statistics about memory management for sharded queues in the Streams pool within the System Global Area (SGA). Sharded queues use the Streams pool in units of subshards. Thus, columns of this view shows statistics at subshard level. This view shows statistics across all sharded queues.

Column	Datatype	Description
NUM_EVICTED	NUMBER	Number of evicted subshards across all sharded queues
NUM_PREFETCHED	NUMBER	Number of subshards pre-fetched by AQ background process
NUM_UNEVICTION	NUMBER	Number of subshards un-evicted by foreground process (like dequeue process)
NUM_UNCACHED	NUMBER	Number of subshards stored as uncached
NUM_TRACKED	NUMBER	Number of subshards which are actively tracking dequeue rates
NUM_CACHED	NUMBER	Number of subshards stored as cached

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MAX_SUBSH_SIZE	NUMBER	Maximum subshard size seen till now, in terms of number of messages per subshard
MIN_SUBSH_SIZE	NUMBER	Minimum subshard size seen till now, in terms of number of messages per subshard
MEAN_SUBSH_SIZE	NUMBER	Mean subshard size seen till now, in terms of number of messages per subshard
AVG_EVICTION_RATE	NUMBER	Average number subshards evicted per second
AVG_LOAD_RATE	NUMBER	Average number of subshards pre-fetched or un-evicted per second
AVG_EVICTION_TIME	NUMBER	Average time taken to evict one subshard (in milliseconds)
AVG_LOAD_TIME	NUMBER	Average time taken to un-evict one subshard (in milliseconds)
AVG_MISS_RATIO	NUMBER	Average ratio of number of foreground un-evictions versus background pre-fetch
AVG_THRASH_RATIO	NUMBER	Average ratio of number of subshard pre-fetched by background without dequeue attempt versus total number of subshards prefetched
MANDATORY_AFF_SWITCH_ATTEMPTS	NUMBER	An affinity switch is a change in dequeue instance for a shard-subscriber pair. A mandatory affinity switch is when there are local enqueues in the queue at the instance but no local dequeues present, so the dequeue affinity is switched to another instance for that shard-subscriber pair. This column shows the number of times mandatory affinity switches were attempted across all instances. Populated at the smallest instance id only.
OPTIONAL_AFF_SWITCH_ATTEMPTS	NUMBER	Optional affinity switches are affinity switches that are not mandatory. Optional affinity switches are done for global load balancing across the Oracle Real Application Clusters (Oracle RAC) database. This column shows the number of times optional affinity switches were attempted across all instances. Populated at the smallest instance id only.
MIN_EVICT_PERCENT	NUMBER	Percentage of `streams_pool` memory beyond which sharded queue subshard eviction is triggered
LAST_AVG_CACHED_HORIZON	NUMBER	Last average number of cached subshards seen in sharded queue memory manager horizon
LAST_AVG_MEMORY_HORIZON	NUMBER	Last average memory of cached subshards seen in sharded queue memory manager horizon
LAST_AVG_SUBSHARD_HORIZON	NUMBER	Last average number of subshards seen in sharded queue memory manager horizon
LAST_LEEWAY_SHIFT	NUMBER	Internal leeway for memory threshold
AVG_OPTTIME_DRIFT	NUMBER	Average drift of opt_time for all subshards. A drift is defined as a difference between opt_time as set on a subshard and the actual time at which subshard is first dequeueed after prefetch/uneviction.
NUM_THRESHOLD_DRIFT	NUMBER	Number of times drift value has gone over threshold time. A typical threshold time can be horizon_time/2.
MAX_OPT_TIME_DRIFT	NUMBER	Maximum drift till now
MIN_OPT_TIME_DRIFT	NUMBER	Minimum drift till now
AVG_OPT_TIME_ERROR	NUMBER	An opt_time error occurs when a cached subshard is unevicted by a foreground process (instead of being prefetched by the aq background). Thus, opt_time error is the difference between the actual opt_time set for a sharded queue subshard and the absolute time at which foreground unevicts the same subshard. This column represents average time of this error.
MAX_OPT_TIME_ERROR	NUMBER	Maximum opt_time error

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MIN_OPT_TIME_ERROR	NUMBER	Minimum opt_time error
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.25 V$AQ_NONDUR_REGISTRATIONS

V$AQ_NONDUR_REGISTRATIONS provides information about non-durable subscriptions.

Column	Datatype	Description
REG_ID	NUMBER	Registration ID
SUBSCRIPTION	VARCHAR2(128)	Subscription name
LOCATION	VARCHAR2(256)	Location name
USER#	NUMBER	User ID
USER_CONTEXT	RAW(32)	Context the user provided
CONTEXT_SIZE	NUMBER	Size of the context
NAMESPACE	NUMBER	Subscription namespace
VERSION	NUMBER	Database version number
STATE	NUMBER	State of the registration: • ENABLED: The registration is enabled for notification • STOPPING: The registration is in a transient state before it becomes DISABLED • DISABLED: The registration is disabled for notification • DEAD: The registration does not exist any more, and it is marked for deferred cleanup.
QOS	NUMBER	Quality of service
REG_TIME	TIMESTAMP(3) WITH TIME ZONE	Time of registration
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

Oracle Database Advanced Queuing User's Guide for more information about Oracle Database Advanced Queueing

7.26 V$AQ_NONDUR_SUBSCRIBER

V$AQ_NONDUR_SUBSCRIBER provides information about the non-durable subscriptions on sharded queues.

Column	Datatype	Description
QUEUE_ID	NUMBER	Queue ID
SUBSCRIBER_ID	NUMBER	ID of the non-durable subscriber
SUBSCRIBER_NAME	VARCHAR2(128)	Subscriber name
RULE_CONDITION	VARCHAR2(4000)	Rule condition of the subscriber
TRANSFORMATION_OWNER	VARCHAR2(128)	Owner of the transformation (for JMS queues)
TRANSFORMATION_NAME	VARCHAR2(128)	Name of the transformation (for JMS queues)
CREATION_TIME	TIMESTAMP(1) WITH TIME ZONE	Non-durable subscriber creation time
FLAGS	NUMBER	Property of the subscriber: • 0x0001 – Persistent subscriber • 0x0002 – Buffered subscriber • 0x0004 – JMS subscriber • 0x0008 – Rule-based subscriber • 0x0010 – Subscriber has transformation • 0x0020 – Notification-only non-durable subscriber subscriber
SUBSCRIBER_TYPE	NUMBER	Type of subscriber: • 1 – JMS non-durable subscriber • 2 – Service layer dummy non-durable subscriber • 3 – Notification-only non-durable subscriber
BITPOS	NUMBER	Position of subscriber in subscriber bitmap
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Advanced Queuing User's Guide for more information about Oracle Database Advanced Queueing

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.27 V$AQ_NONDUR_SUBSCRIBER_LWM

V$AQ_NONDUR_SUBSCRIBER_LWM projects the low watermarks (LWMs) of non-durable subscribers in a sharded queue. The LWM of a non-durable subscriber is a combination of shard, priority and LWM (subshard).

Column	Datatype	Description
QUEUE_ID	NUMBER	Queue ID
SUBSCRIBER_ID	NUMBER	ID of the non-durable subscriber
SHARD_ID	NUMBER	Shard ID
PRIORITY	NUMBER	Priority of the shard
LWM	NUMBER	Lower watermark (in a subshard) of the non-durable subscriber
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Advanced Queuing User's Guide for more information about Oracle Database Advanced Queueing

7.28 V$AQ_NOTIFICATION_CLIENTS

V$AQ_NOTIFICATION_CLIENTS displays performance statistics for secure OCI client connections.

Column	Datatype	Description
CLIENT_ID	VARCHAR2(29)	Internally generated client ID for secure notification clients
EMON_ID	NUMBER	Emon ID serving the client
NOTIFICATION_STATE	NUMBER	Notification state: • ACTIVE – Sending notification • WAIT_FOR_ACK – Waiting for client acknowledgment • INACTIVE - Idle connection
NUM_MESSAGE_SENT	NUMBER	Number of messages sent on the connection
NUM_BYTES_SENT	NUMBER	Number of bytes sent on the connection
NUM_MESSAGE_RECEIVED	NUMBER	Number of messages successfully received by the client
LAST_SEND_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when the last message was sent on the connection
LAST_RECEIVE_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when the last message was received over the connection

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CONNECT_TIME	TIMESTAMP(3) WITH TIME ZONE	Time at which the client connected
DISCONNECT_TIME	TIMESTAMP(3) WITH TIME ZONE	Time at which the client disconnected
LAST_ERROR	NUMBER	The last error that occurred on the client connection
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Advanced Queuing User's Guide for more information about Oracle Database Advanced Queueing

7.29 V$AQ_PARTITION_STATS

V$AQ_PARTITION_STATS displays usage statistics for the queue partition cache and the dequeue log partition cache.

Column	Datatype	Description
INST_ID	NUMBER	Current instance ID
QUEUE_ID	NUMBER	Queue ID
QUEUE_TABLE_ID	NUMBER	Queue table object ID
QUEUE_SCHEMA	VARCHAR2(128)	Queue schema name
QUEUE_NAME	VARCHAR2(128)	Queue name
PT_TUNED_SIZ_QT	NUMBER	Current tuned size of the queue partition cache, expressed in number of partition cache elements
PT_CACHED_PTNS_QT	NUMBER	Current number of cached partitions in the queue partition cache
PT_OVER_CACHED_PTNS_QT	NUMBER	Current number of over-cached partitions in the queue partition cache, that is, the number of cached partitions whose partition cache elements exceed the current tuned size of the queue partition cache
PT_TOTAL_UPTUNE_QT	NUMBER	Total amount of space added to the queue partition cache due to tune-ups, since the cache was initialized, expressed in number of partition cache elements
PT_NO_OF_UPTUNES_QT	NUMBER	Total number of times a tune-up was triggered on the queue partition cache, since the cache was initialized
PT_TOTAL_DOWNTUNE_QT	NUMBER	Total amount of space removed from the queue partition cache due to tune-downs, since the cache was initialized, expressed in number of partition cache elements

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PT_NO_OF_DOWNTUNES_QT	NUMBER	Total number of times a tune-down was triggered on the queue partition cache, since the cache was initialized
PT_CACHE_MISS_QT	NUMBER	Total number of cache misses during partition lookups on the queue partition cache
PT_CACHE_HIT_QT	NUMBER	Total number of cache hits during partition lookups on the queue partition cache
PT_TOTAL_CACH_GET_QT	NUMBER	Total number of successful fetches from the queue partition cache
PT_TOTAL_CACH_PUT_QT	NUMBER	Total number of caching operations that occurred on the queue partition cache
PT_UNBOUNDINGS_QT	NUMBER	Total number of times queue partitions were unbounded
PT_TUNED_SIZ_DQ	NUMBER	Current tuned size of the dequeue log partition cache, expressed in number of partition cache elements
PT_CACHED_PTNS_DQ	NUMBER	Current number of cached partitions in the dequeue log partition cache
PT_OVER_CACHED_PTNS_DQ	NUMBER	Current number of over-cached partitions in the dequeue log partition cache, that is, the number of cached dequeue log partitions whose partition cache elements exceed the current tuned size of the dequeue log partition cache
PT_TOTAL_UPTUNE_DQ	NUMBER	Total amount of space added to the dequeue log partition cache due to tune-ups, since the cache was initialized, expressed in number of partition cache elements
PT_NO_OF_UPTUNES_DQ	NUMBER	Total number of times a tune-up was triggered on the dequeue log partition cache, since the cache was initialized
PT_TOTAL_DOWNTUNE_DQ	NUMBER	Total amount of space removed from the dequeue log partition cache due to tune-downs, since the cache was initialized, expressed in number of partition cache elements
PT_NO_OF_DOWNTUNES_DQ	NUMBER	Total number of times a tune-down was triggered on the dequeue log partition cache, since the cache was initialized
PT_CACHE_MISS_DQ	NUMBER	Total number of cache misses during partition lookups on the dequeue log partition cache
PT_CACHE_HIT_DQ	NUMBER	Total number of cache hits during partition lookups on the dequeue log partition cache
PT_TOTAL_CACH_GET_DQ	NUMBER	Total number of successful fetches from the dequeue log partition cache
PT_TOTAL_CACH_PUT_DQ	NUMBER	Total number of caching operations that occurred on the dequeue log partition cache
PT_UNBOUNDINGS_DQ	NUMBER	Total number of times dequeue log partitions were unbounded
ADD_PARTITION_FG_QT	NUMBER	Total number of queue partitions created inline during foreground AQ enqueue operations
ADD_PARTITION_BG_QT	NUMBER	Total number of queue partitions created asynchronously by the AQ partitioning background process
ADD_PARTITION_FG_DQLOG	NUMBER	Total number of dequeue log partitions created inline during foreground AQ dequeue operations
ADD_PARTITION_BG_DQLOG	NUMBER	Total number of dequeue log partitions created asynchronously by the AQ partitioning background process
TRUNC_PARTITION_QT	NUMBER	Total number of truncated and recycled queue partitions
TRUNC_PARTITION_DQLOG	NUMBER	Total number of truncated and recycled dequeue log partitions

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **Note**
>
> This view is available starting with Oracle Database 19c.

7.30 V$AQ_REMOTE_DEQUEUE_AFFINITY

V$AQ_REMOTE_DEQUEUE_AFFINITY lists the dequeue affinity instance of the subscribers not dequeuing locally from the shard's owner instance. Cross instance message forwarding is used for these subscribers.

Column	Datatype	Description
QUEUE_ID	NUMBER	Queue ID
SHARD_ID	NUMBER	Shard ID which is being forwarded from SOURCE_INSTANCE to INST_ID for the subscriber
SOURCE_INSTANCE_ID	NUMBER	Owner instance ID from where the shard is being forwarded
SUBSCRIBER_ID	NUMBER	Subscriber ID
QUEUE_SCHEMA	VARCHAR2(128)	Queue schema name
QUEUE_NAME	VARCHAR2(128)	Queue name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.31 V$AQ_SERVER_POOL

V$AQ_SERVER_POOL lists performance statistics for all the servers in the pool.

Column	Datatype	Description
COORDINATOR_ID	NUMBER	ID of the coordinator
COORDINATOR_INSTANCE_ID	NUMBER	Instance of the coordinator
PROCESS_ID	VARCHAR2(24)	Operating system process ID of the server

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PROCESS_NAME	VARCHAR2(48)	Operating system process name of the server
JOB_NAME	VARCHAR2(32)	Name of the job handled
POOL_STATE	VARCHAR2(20)	State of the pool: • REMOTE • IDLE • ACTIVE
CON_ID	NUMBER	The ID of the container to which the data pertains. The CON_ID value in this view is always 0. The rows pertain to the entire CDB or to the non-CDB.

See Also

Oracle Database Advanced Queuing User's Guide for more information about Oracle Database Advanced Queueing

7.32 V$AQ_SHARDED_SUBSCRIBER_STAT

V$AQ_SHARDED_SUBSCRIBER_STAT displays basic statistical information about the subscribers of sharded queues. There is one row per queue per shard per subscriber.

Column	Datatype	Description
QUEUE_ID	NUMBER	Queue identifier
SUBSCRIBER_ID	NUMBER	Subscriber identifier
SHARD_ID	NUMBER	Shard identifier
PRIORITY	NUMBER	Priority value of the shard
DEQUEUE_SUBSHARD	NUMBER	Last known dequeue position in this shard
ENQUEUED_MSGS	NUMBER	Number of enqueued messages
DEQUEUED_MSGS	NUMBER	Number of dequeued messages
ELAPSED_DEQUEUE_TIME	NUMBER	Amount of time spent performing dequeues (in seconds)
CPU_DEQUEUE_TIME	NUMBER	Actual amount of CPU time spent performing dequeues (in seconds)
DEQUEUE_RATE	NUMBER	Number of messages dequeued per second
TIME_SINCE_LAST_DEQUEUE	NUMBER	Time since last dequeue activity (in seconds)
ESTD_TIME_TO_DRAIN	NUMBER	Estimated amount of time to drain the shard (in seconds) with current enqueue and dequeue rates. Null, if the enqueue rate is greater than the dequeue rate.
ESTD_TIME_TO_DRAIN_NO_ENQ	NUMBER	Estimated amount of time to drain the shard (in seconds) with no new enqueues

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.33 V$AQ_SUBSCRIBER_LOAD

V$AQ_SUBSCRIBER_LOAD describes the load of all subscribers of sharded queues in terms of latency at every instance in an Oracle RAC environment.

Latency denotes the predicted amount of time (in seconds) required from the current time to drain all the messages for that subscriber at each respective instance. The latency calculation considers past enqueue/dequeue rates and future enqueue/dequeue rates based on history.

Column	Datatype	Description
QUEUE_ID	NUMBER	Queue ID
QUEUE_SCHEMA	VARCHAR2(128)	Queue schema
QUEUE_NAME	VARCHAR2(128)	Queue name
SUBSCRIBER_ID	NUMBER	Subscriber ID
SUBSCRIBER_NAME	VARCHAR2(128)	Subscriber name
LATENCY_STATE	VARCHAR2(8)	Possible values: • FINITE - The subscriber will be able to dequeue all the messages in a finite amount of time • INFINITE - The subscriber's dequeue rate may not catch up to the enqueue rate • UNKNOWN - Latency is not yet known
LATENCY	NUMBER	Latency (in seconds). Valid only when LATENCY_STATE is FINITE.
DEQUEUE_REQUESTS	NUMBER	Approximate number of dequeue requests noted recently
ACTIVE_SHARDS	NUMBER	Number of queue shards that have messages for this subscriber
ACTIVE_LISTENER	VARCHAR2(5)	Indicates whether the subscriber is actively listening at this instance for messages (TRUE) or not (FALSE)
DEQUEUE_SESSIONS[1]	NUMBER	Displays the number of active dequeue sessions for this subscriber
FLAGS	NUMBER	For internal use only
MANDATORY_AFF_SWITCHES	NUMBER	An affinity switch is a change in the dequeue instance for a shard-subscriber pair. A mandatory affinity switch is when there are local enqueues in the queue at the instance but no local dequeues present, so the dequeue affinity is switched to another instance for that shard-subscriber pair. This column shows the number of times mandatory affinity switches were needed from this instance to another for this subscriber.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OPTIONAL_AFF_SWITCHES	NUMBER	Optional affinity switches are affinity switches that are not mandatory. Optional affinity switches are done for global load balancing across the Oracle Real Application Clusters (Oracle RAC) database. This column shows the number of times optional affinity switches were needed from this instance to another for this subscriber.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

[1] This column is available starting with Oracle Database 19c.

> **See Also**
>
> *Oracle Database Advanced Queuing User's Guide* for more information about Oracle Database Advanced Queueing

7.34 V$ARCHIVE

V$ARCHIVE displays information about redo log files in need of archiving.

Each row displays information for one thread. This information is also available in V$LOG. Oracle recommends that you use V$LOG.

> **See Also**
>
> "V$LOG"

Column	Datatype	Description
GROUP#	NUMBER	Log file group number
THREAD#	NUMBER	Log file thread number
SEQUENCE#	NUMBER	Log file sequence number
ISCURRENT	VARCHAR2(3)	This is the current online redo log
CURRENT	VARCHAR2(3)	This column is obsolete and maintained for backward compatibility. The value of this column is always equal to the value in ISCURRENT.
FIRST_CHANGE#	NUMBER	First SCN stored in the current log

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.35 V$ARCHIVE_DEST

V$ARCHIVE_DEST displays, for the current instance, all of the destinations in the Data Guard configuration, including each destination's current value, mode, and status.

Column	Datatype	Description
DEST_ID	NUMBER	Log archive destination parameter identifier (1 to 31)
DEST_NAME	VARCHAR2(256)	Log archive destination parameter name
STATUS	VARCHAR2(9)	Identifies the current status of the destination: • VALID - Initialized and available • INACTIVE - No destination information • DEFERRED - Manually disabled by the user • ERROR - Error during open or copy • DISABLED - Disabled after error • BAD PARAM - Parameter has errors • ALTERNATE - Destination is available as an alternate • FULL - Exceeded quota size for the destination
BINDING	VARCHAR2(9)	Specifies how failure will affect the archival operation: • MANDATORY - Successful archival is required • OPTIONAL - Successful archival is not required (depends on LOG_ARCHIVE_MIN_SUCCEED_DEST)
NAME_SPACE	VARCHAR2(7)	Identifies the scope of parameter setting: • SYSTEM - System definition • SESSION - Session definition
TARGET	VARCHAR2(7)	For a primary database, this column specifies whether the archive destination is local or remote to the primary database: • PRIMARY - local • STANDBY - remote For a standby database, this column specifies whether the archive destination is local or remote to the standby database: • LOCAL - local • REMOTE - remote
ARCHIVER	VARCHAR2(10)	Identifies the archiver process relative to the database where the query is issued: • ARC*n* • FOREGROUND • LGWR • RFS

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SCHEDULE	VARCHAR2(8)	Indicates whether the archival of this destination is INACTIVE, PENDING, ACTIVE, or LATENT
DESTINATION	VARCHAR2(256)	Specifies the location where the archived redo logs are to be archived
LOG_SEQUENCE	NUMBER	Identifies the sequence number of the last archived redo log to be archived
REOPEN_SECS	NUMBER	Identifies the retry time (in seconds) after error
DELAY_MINS	NUMBER	Identifies the delay interval (in minutes) before the archived redo log is automatically applied to a standby database
MAX_CONNECTIONS[1]	NUMBER	Maximum number of connections
NET_TIMEOUT	NUMBER	Number of seconds the log writer process will wait for status from the network server of a network operation issued by the log writer process
PROCESS	VARCHAR2(10)	Identifies the archiver process relative to the primary database, even if the query is issued on the standby database: • ARC*n* • FOREGROUND • LGWR
REGISTER	VARCHAR2(3)	Indicates whether the archived redo log is registered in the remote destination control file (YES) or not (NO). If the archived redo log is registered, it is available to log apply services.
FAIL_DATE	DATE	Date and time of last error
FAIL_SEQUENCE	NUMBER	Sequence number of the archived redo log being archived when the last error occurred
FAIL_BLOCK	NUMBER	Block number of the archived redo log being archived when the last error occurred
FAILURE_COUNT	NUMBER	Current number of contiguous archival operation failures that have occurred for the destination
MAX_FAILURE	NUMBER	Allows you to control the number of times log transport services will attempt to reestablish communication and resume archival operations with a failed destination
ERROR	VARCHAR2(256)	Displays the error text
ALTERNATE	VARCHAR2(256)	Alternate destination, if any
DEPENDENCY	VARCHAR2(256)	Reserved for future use
REMOTE_TEMPLATE	VARCHAR2(256)	Specifies the template to be used to derive the location to be recorded
QUOTA_SIZE	NUMBER	Destination quotas, expressed in bytes
QUOTA_USED	NUMBER	Size of all the archived redo logs currently residing on the specified destination
MOUNTID	NUMBER	Instance mount identifier
TRANSMIT_MODE	VARCHAR2(12)	Specifies network transmission mode: • SYNCHRONOUS • PARALLELSYNC • ASYNCHRONOUS
ASYNC_BLOCKS	NUMBER	Number of blocks specified for the ASYNC attribute
AFFIRM	VARCHAR2(3)	Specifies disk I/O mode

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TYPE	VARCHAR2(7)	Indicates whether the archived log destination definition is PUBLIC or PRIVATE. Only PUBLIC destinations can be modified at run time using the ALTER SYSTEM SET or ALTER SESSION SET statements. By default, all archived log destinations are PUBLIC.
VALID_NOW	VARCHAR2(16)	Indicates whether the destination is valid right now for archival operations: • YES - Redo log type and database role for this destination are valid for the current database • WRONG VALID_TYPE - Redo log type specified for this destination is not valid for the current database role. For example, WRONG VALID_TYPE would be returned if a destination specified with the VALID_FOR=(STANDBY_LOGFILE,STANDBY_ROLE) attribute is running in the standby database role but does not have standby redo logs implemented. • WRONG VALID_ROLE - Database role specified for this destination is not the role in which the database is currently running. For example, the WRONG VALID_ROLE would be returned when a destination defined with the VALID_FOR=(ONLINE_LOGFILE,STANDBY_ROLE) attribute is running in the primary database role. • INACTIVE - Destination is inactive, probably due to an error
VALID_TYPE	VARCHAR2(15)	Redo log type or types that are valid for the destination: • ONLINE_LOGFILE • STANDBY_LOGFILE • ALL_LOGFILES
VALID_ROLE	VARCHAR2(12)	Database role or roles that are valid for the destination: • PRIMARY_ROLE • STANDBY_ROLE • ALL_ROLES
DB_UNIQUE_NAME	VARCHAR2(30)	Unique database name
VERIFY	VARCHAR2(3)	Indicates whether the value of the VERIFY attribute on the LOG_ARCHIVE_DEST_*n* parameter is verified (YES) or not verified (NO)
COMPRESSION	VARCHAR2(7)	Indicates whether network compression is ENABLED or DISABLED, or whether the ZLIB or LZO algorithm is in use. These values correspond to the values accepted by the COMPRESSION attribute for the database initialization parameter LOG_ARCHIVE_DEST_n. The possible values include: • DISABLE: Compression is disabled. • ENABLE: Compression is enabled. The ZLIB compression algorithm is used. • ZLIB: ZLIB compression is used. • LZO: LZO compression is used.
APPLIED_SCN	NUMBER	For a destination that corresponds to a physical or logical standby database, the SCN of the last applied redo. For a destination that corresponds to a snapshot standby database, the SCN of the last redo applied before conversion to a snapshot standby database. This column is only valid for enabled and active standby database destinations on a primary or cascading standby database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
ENCRYPTION	VARCHAR2(7)	Indicates whether encryption of the redo stream sent to Zero Data Loss Recovery Appliance (Recovery Appliance) is enabled (ENABLED) or not (DISABLED)

[1] Starting with Oracle Database 19c, the data in this column is not meaningful. This column may be removed in a future release.

See Also

- *Zero Data Loss Recovery Appliance Administrator's Guide* for introductory information about Recovery Appliance
- "LOG_ARCHIVE_DEST" and "LOG_ARCHIVE_DEST_*n*"
- "LOG_ARCHIVE_DUPLEX_DEST" and "LOG_ARCHIVE_DEST_STATE_*n*"
- "LOG_ARCHIVE_MIN_SUCCEED_DEST"

7.36 V$ARCHIVE_DEST_STATUS

V$ARCHIVE_DEST_STATUS displays run time and configuration information for the archived redo log destinations.

The information in this view does not persist across an instance shutdown.

Column	Datatype	Description
DEST_ID	NUMBER	Identifies the log archive destination parameter (1 to 31)
DEST_NAME	VARCHAR2(256)	Log archive destination parameter name
STATUS	VARCHAR2(9)	Current status of the destination: • ALTERNATE - Destination is in an alternate state • BAD PARAM - Parameter has errors • DEFERRED - Manually disabled by the user • DISABLED - Disabled after error • ERROR - Error during open or copy • FULL - Exceeded quota size for the destination • IDLE - Destination is idle • INACTIVE - No destination information • VALID - Initialized and available

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TYPE	VARCHAR2(16)	Type of archival destination database: • BACKUP APPLIANCE - Backup appliance • CROSS-INSTANCE - An instance of the primary • DOWNSTREAM - Streams downstream capture database • FAR SYNC - Far Sync Instance • LOCAL - Local to primary database • LOGICAL - Logical standby • PHYSICAL - Physical standby • SNAPSHOT - Snapshot standby database
DATABASE_MODE	VARCHAR2(15)	Current mode of the archival destination database: • STARTED - Instance started, not mounted • MOUNTED - Mounted • MOUNTED-STANDBY - Mounted standby • OPEN - Open read/write • OPEN_READ-ONLY - Open read-only
RECOVERY_MODE	VARCHAR2(34)	Current apply mode at the archival destination: • IDLE - Managed recovery is not active • LOGICAL APPLY - SQL Apply • LOGICAL REAL TIME APPLY - Real time SQL Apply • MANAGED - Managed recovery is active. This value is used when the standby database is mounted. In this recovery mode, users cannot query data on the standby. • MANAGED REAL TIME APPLY - In this recovery mode, log apply services recover redo data from standby redo logs at the same time the logs are being written to, as opposed to recovering redo from archived redo logs when a log switch occurs. In this recovery mode, users cannot query data on the standby. • MANAGED REAL TIME APPLY WITH QUERY - In this recovery mode, log apply services recover redo data from standby redo logs at the same time the logs are being written to, as opposed to recovering redo from archived redo logs when a log switch occurs. In a non-CDB, this mode is used when the database is open, and users can query data on the standby. In a CDB, this value is used when one or more PDBs in the CDB are open, and users can query data on the standby. • MANAGED WITH QUERY - Managed recovery is active in this recovery mode. In a non-CDB, this value is used when the database is open, and users can query data on the standby. In a CDB, this value is used when one or more PDBs in the CDB are open, and users can query data on the standby.
PROTECTION_MODE	VARCHAR2(20)	Indicates how the database is protected: • MAXIMUM AVAILABILITY • MAXIMUM PERFORMANCE • MAXIMUM PROTECTION • RESYNCHRONIZATION
DESTINATION	VARCHAR2(256)	Specifies the location where the redo data is to be archived
STANDBY_LOGFILE_COUNT	NUMBER	Indicates the total number of standby redo logs created on the standby database
STANDBY_LOGFILE_ACTIVE	NUMBER	Indicates the total number of standby redo logs on the standby database that are active and contain primary database online redo log information

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ARCHIVED_THREAD#	NUMBER	Identifies the thread number of the most recent archived redo log received at the destination
ARCHIVED_SEQ#	NUMBER	Identifies the log sequence number of the most recent archived redo log received at the destination
APPLIED_THREAD#	NUMBER	Identifies the thread number of the most recent applied redo log received at the destination
APPLIED_SEQ#	NUMBER	Identifies the log sequence number of the most recent applied redo log received at the destination
ERROR	VARCHAR2(256)	Displays the error text
SRL	VARCHAR2(3)	Indicates whether standby redo logfiles are used on the standby database (YES) or not (NO)
DB_UNIQUE_NAME	VARCHAR2(30)	Specifies the unique database name of the current instance that was defined with the DB_UNIQUE_NAME attribute on the LOG_ARCHIVE_DEST_*n* parameter
SYNCHRONIZATION_STATUS	VARCHAR2(22)	Possible values for this column are as follows: • CHECK CONFIGURATION - Synchronization with this destination is not possible because this database is either not in MAXIMUM PROTECTION or MAXIMUM AVAILABILITY data protection mode, or the LOG_ARCHIVE_DEST_*n* parameter associated with this destination has not been configured with the SYNC and AFFIRM attributes. • CHECK CONNECTIVITY - One or more instances of this database cannot send redo data to this destination. • CHECK STANDBY REDO LOG - The standby redo log at this destination is configured improperly. • DESTINATION HAS A GAP - This destination is missing redo data needed for synchronization with this database. • OK - This destination is synchronized with this database. • STATUS NOT AVAILABLE - Synchronization status is not available. **See Also:** *Oracle Data Guard Concepts and Administration* for more information about redo transport configuration
SYNCHRONIZED	VARCHAR2(3)	Possible values are: • YES - This destination is synchronized with the primary database. • NO - The destination is not synchronized with the primary database. • UNKNOWN - The synchronization status of this destination cannot be determined.
GAP_STATUS	VARCHAR2(24)	Redo gap status: • LOCALLY UNRESOLVABLE GAP - Destination has a redo gap that cannot be automatically resolved by fetching the missing redo from this database. It may be possible to resolve the gap by fetching the missing redo from another destination. • LOG SWITCH GAP - Destination has not yet received all of the redo from the previous log file. • NO GAP - Destination does not have a redo gap. • RESOLVABLE GAP - Destination has a redo gap that can be automatically resolved by fetching the missing redo from this database. • UNRESOLVABLE GAP - Destination has a redo gap that cannot be automatically resolved by fetching the missing redo from this database and there are no other destinations from which redo can be fetched.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.37 V$ARCHIVE_GAP

V$ARCHIVE_GAP displays information about archive gaps on a standby database. This view can be used to find out the current archive gap that is blocking recovery for the current recovery incarnation.

Column	Datatype	Description
THREAD#	NUMBER	Thread number of the missing archived redo log files. The number is 1 for a single instance. For Oracle Real Application Clusters, this column will contain different numbers.
LOW_SEQUENCE#	NUMBER	Lowest sequence number of the log files received on the standby system
HIGH_SEQUENCE#	NUMBER	Highest sequence number of the log files received on the standby system
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.38 V$ARCHIVE_PROCESSES

V$ARCHIVE_PROCESSES displays the state of the various ARCH processes for the instance.

Column	Datatype	Description
PROCESS	NUMBER	Identifier for the ARCH process for the instance, numbered from 0-9
STATUS	VARCHAR2(10)	Status of the ARCH process, displayed as a keyword. Possible values are: STOPPED, SCHEDULED, STARTING, ACTIVE, STOPPING, and TERMINATED.
LOG_SEQUENCE	NUMBER	This is the online redo log sequence number currently being archived, if STATE="BUSY"
STATE	VARCHAR2(4)	This is the current state of the ARCH process, displayed as a keyword. Possible keywords are IDLE or BUSY.
ROLES	VARCHAR2(36)	The list of roles assigned to the archive process. The roles include: HEART_BEAT, NO_FAL, NO_SRL, CLEAR_LOGS

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.39 V$ARCHIVED_LOG

V$ARCHIVED_LOG displays archived log information from the control file, including archive log names.

An archive log record is inserted after the online redo log is successfully archived or cleared (name column is NULL if the log was cleared). If the log is archived twice, there will be two archived log records with the same THREAD#, SEQUENCE#, and FIRST_CHANGE#, but with a different name. An archive log record is also inserted when an archive log is restored from a backup set or a copy and whenever a copy of a log is made with the RMAN COPY command.

Column	Datatype	Description
RECID	NUMBER	Archived log record ID
STAMP	NUMBER	Archived log record stamp
NAME	VARCHAR2(513)	Archived log file name. If set to NULL, either the log file was cleared before it was archived or an RMAN backup command with the "delete input" option was executed to back up archivelog all (RMAN> backup archivelog all delete input;).
DEST_ID	NUMBER	Original destination from which the archive log was generated. The value is 0 if the destination identifier is not available.
THREAD#	NUMBER	Redo thread number
SEQUENCE#	NUMBER	Redo log sequence number
RESETLOGS_CHANGE#	NUMBER	Resetlogs change number of the database when the log was written
RESETLOGS_TIME	DATE	Resetlogs time of the database when the log was written
RESETLOGS_ID	NUMBER	Resetlogs identifier associated with the archived redo log
FIRST_CHANGE#	NUMBER	First change number in the archived log
FIRST_TIME	DATE	Timestamp of the first change
NEXT_CHANGE#	NUMBER	First change in the next log
NEXT_TIME	DATE	Timestamp of the next change
BLOCKS	NUMBER	Size of the archived log (in blocks)
BLOCK_SIZE	NUMBER	Redo log block size. This is the logical block size of the archived log, which is the same as the logical block size of the online log from which the archived log was copied. The online log logical block size is a platform-specific value that is not adjustable by the user.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CREATOR	VARCHAR2(7)	Creator of the archivelog: • ARCH - Archiver process • FGRD - Foreground process • RMAN - Recovery Manager • SRMN - RMAN at standby • LGWR - Logwriter process
REGISTRAR	VARCHAR2(7)	Registrar of the entry: • RFS - Remote File Server process • ARCH - Archiver process • FGRD - Foreground process • RMAN - Recovery manager • SRMN - RMAN at standby • LGWR - Logwriter process
STANDBY_DEST	VARCHAR2(3)	Indicates whether the entry is an archivelog destination (YES) or not (NO)
ARCHIVED	VARCHAR2(3)	Indicates whether the online redo log was archived (YES) or whether RMAN only inspected the log and created a record for future application of redo logs during recovery (NO).
APPLIED	VARCHAR2(9)	Indicates whether an archived redo log file has been applied to the corresponding physical standby database. The value is always NO for local destinations. This column is meaningful on a physical standby database for rows where REGISTRAR = RFS: • If REGISTRAR = RFS and APPLIED = NO, then the log file has been received but has not yet been applied. • If REGISTRAR = RFS and APPLIED = IN-MEMORY, then the log file has been applied in memory, but the data files have not yet been updated. • If REGISTRAR = RFS and APPLIED = YES, then the log file has been applied and the data files have been updated. This column can be used to identify log files that can be backed up and deleted. When used for this purpose, the value IN-MEMORY should be treated as if it were NO.
DELETED	VARCHAR2(3)	Indicates whether an RMAN DELETE command has physically deleted the archived log file from disk, as well as logically removing it from the control file of the target database and from the recovery catalog (YES) or not (NO)
STATUS	VARCHAR2(1)	Status of the archived log: A - Available D - Deleted U - Unavailable X - Expired
COMPLETION_TIME	DATE	Time when the archiving completed
DICTIONARY_BEGIN	VARCHAR2(3)	Indicates whether the log contains the start of a LogMiner dictionary (YES) or not (NO)
DICTIONARY_END	VARCHAR2(3)	Indicates whether the log contains the end of a LogMiner dictionary (YES) or not (NO)
END_OF_REDO	VARCHAR2(3)	Indicates whether the archived redo log contains the end of all redo information from the primary database (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BACKUP_COUNT	NUMBER	Indicates the number of times this file has been backed up. Values range from 0-15. If the file has been backed up more than 15 times, the value remains 15.
ARCHIVAL_THREAD#	NUMBER	Redo thread number of the instance that performed the archival operation. This column differs from the THREAD# column only when a closed thread is archived by another instance.
ACTIVATION#	NUMBER	Number assigned to the database instantiation
IS_RECOVERY_DEST_FILE	VARCHAR2(3)	Indicates whether the file was created in the fast recovery area (YES) or not (NO)
COMPRESSED	VARCHAR2(3)	Reserved for internal use
FAL	VARCHAR2(3)	Indicates whether the archive log was generated as the result of a FAL request (YES) or not (NO)
END_OF_REDO_TYPE	VARCHAR2(10)	Possible values are as follows: • SWITCHOVER - Shows archived redo log files that are produced at the end of a switchover • TERMINAL - Shows archived redo log files produced after a failover • RESETLOGS - Shows online redo log files archived on the primary database after an ALTER DATABASE OPEN RESETLOGS statement is issued • ACTIVATION - Shows any log files archived on a physical standby database after an ALTER DATABASE ACTIVATE STANDBY DATABASE statement is issued • "empty string" - Any empty string implies that the log is just a normal archival and was not archived due to any of the other events
BACKED_BY_VSS	VARCHAR2(3)	Whether or not the file has been backed up by Volume Shadow Copy Service (VSS). This column is reserved for internal use.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.40 V$ASM_ACFS_ENCRYPTION_INFO

V$ASM_ACFS_ENCRYPTION_INFO displays encryption information for every mounted Oracle ACFS.

Column	Datatype	Description
FS_NAME	VARCHAR2(1024)	File system mount point (foreign key to V$ASM_FILESYSTEM)
VOL_DEVICE	VARCHAR2(256)	Name of the Oracle ADVM device
SET_STATUS	VARCHAR2(7)	Indicates whether encryption parameters have been set on the file system (YES) or not (NO); otherwise UNKNOWN.
ENABLED_STATUS	VARCHAR2(8)	Indicates whether file system level encryption is enabled (ENABLED) or not (DISABLED); otherwise UNKNOWN.
ALGORITHM	VARCHAR2(7)	Encryption algorithm used. AES is the only supported algorithm.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
KEY_LENGTH	VARCHAR2(7)	Key length used for the encryption key
LAST_REKEY_TIME	DATE	Time that the volume was last rekeyed
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Note

To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.41 V$ASM_ACFS_SEC_ADMIN

V$ASM_ACFS_SEC_ADMIN contains Oracle ACFS security administrators in the cluster. This view also supports GV$ global views.

Column	Datatype	Description
ADMIN_NAME	VARCHAR2(32)	User name of the security administrator
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Note

To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.42 V$ASM_ACFS_SEC_CMDRULE

`V$ASM_ACFS_SEC_CMDRULE` contains one row for every Oracle ACFS security command rule for each Oracle ACFS file system. This view also supports GV$ global views.

Column	Datatype	Description
`CMD_RULE_NAME`	`VARCHAR2(32)`	Name of the command rule
`FS_NAME`	`VARCHAR2(1024)`	Mount path name to identify the file system
`VOL_DEVICE`	`VARCHAR2(256)`	ADVM volume name
`CON_ID`	`NUMBER`	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always `0`.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.43 V$ASM_ACFS_SEC_REALM

`V$ASM_ACFS_SEC_REALM` contains an entry for every realm in the Oracle ACFS security file system in an Oracle Automatic Storage Management (Oracle ASM) instance. This view also supports GV$ global views.

Column	Datatype	Description
`REALM_NAME`	`VARCHAR2(255)`	Realm name
`REALM_STATUS`	`NUMBER`	Realm status: • `0`: disable • `1`: enable
`REALM_ENCR_STATUS`	`NUMBER`	Realm encryption status: • `0`: disable • `1`: enable Will be provided only when realm encryption status is `1`. The values will be among 128, 192, 256.
`REALM_ENCR_KEYLEN`	`NUMBER`	Encryption key length. Will be provided only when realm encryption status is `1`. The values will be among 128, 192, 256.
`REALM_ENCR_ALGO`	`VARCHAR2(7)`	Encryption algorithm. Will be provided only when realm encryption status is `1`.
`REALM_DESC`	`VARCHAR2(1024)`	Realm description
`FS_NAME`	`VARCHAR2(1024)`	Mount path name to identify the file system
`VOL_DEVICE`	`VARCHAR2(256)`	ADVM volume name
`CON_ID`	`NUMBER`	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always `0`.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Note

To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.44 V$ASM_ACFS_SEC_REALM_FILTER

`V$ASM_ACFS_SEC_REALM_FILTER` contains an entry for every filter in the Oracle ACFS security realm for each Oracle ACFS file system. A filter is a defined as a command rule/rule set pair in a realm. This view also supports GV$ global views.

Column	Datatype	Description
`REALM_NAME`	`VARCHAR2(255)`	Realm name
`CMD_RULE_NAME`	`VARCHAR2(32)`	Command rule name
`RULESET_NAME`	`VARCHAR2(256)`	Ruleset name
`FS_NAME`	`VARCHAR2(1024)`	Mount path name to identify the file system
`VOL_DEVICE`	`VARCHAR2(256)`	ADVM volume name
`CON_ID`	`NUMBER`	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always `0`.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Note

To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.45 V$ASM_ACFS_SEC_REALM_GROUP

`V$ASM_ACFS_SEC_REALM_GROUP` contains an entry for every group in the Oracle ACFS security realm for each Oracle ACFS file system. This view also supports GV$ global views.

Column	Datatype	Description
`REALM_NAME`	`VARCHAR2(255)`	Realm name
`GROUP_NAME`	`VARCHAR2(256)`	Name of the group
`FS_NAME`	`VARCHAR2(1024)`	Mount path name to identify the file system
`VOL_DEVICE`	`VARCHAR2(256)`	ADVM volume name
`CON_ID`	`NUMBER`	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always `0`.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.46 V$ASM_ACFS_SEC_REALM_USER

`V$ASM_ACFS_SEC_REALM_USER` contains an entry for every user in the Oracle ACFS security realm for each Oracle ACFS file system. This view also supports GV$ global views.

Column	Datatype	Description
`REALM_NAME`	`VARCHAR2(255)`	Realm name
`USER_NAME`	`VARCHAR2(32)`	Name of the user
`FS_NAME`	`VARCHAR2(1024)`	Mount path name to identify the file system
`VOL_DEVICE`	`VARCHAR2(256)`	ADVM volume name
`CON_ID`	`NUMBER`	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always `0`.

> **See Also**
>
> - *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
> - *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

> **Note**
>
> To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.47 V$ASM_ACFS_SEC_RULE

V$ASM_ACFS_SEC_RULE contains information about all Oracle ACFS security rules for each Oracle ACFS file system. This view also supports GV$ global views.

Column	Datatype	Description
RULE_NAME	VARCHAR2(256)	Name of the rule
RULE_TYPE	VARCHAR2(32)	Type of the rule: • time • username • groupname • hostname • application
RULE_VALUE	VARCHAR2(512)	Value of the rule, which could include values such as 23:10:00 for time, Bob for username, dba for groupname, host147 for hostname, /sbin/cat for application.
RULE_VALUE2	VARCHAR2(512)	The value will be populated only if the rule type is time. When the rule type is time, the start time is entered in RULE_VALUE and the end time is entered in RULE_VALUE2.
RULE_OPTION	VARCHAR2(7)	Possible values are ALLOW or DENY
FS_NAME	VARCHAR2(1024)	Mount path name to identify the file system
VOL_DEVICE	VARCHAR2(256)	ADVM volume name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.48 V$ASM_ACFS_SEC_RULESET

`V$ASM_ACFS_SEC_RULESET` contains information about all Oracle ACFS security rule sets for each Oracle ACFS file system. This view also supports GV$ global views.

Column	Datatype	Description
`RULESET_NAME`	`VARCHAR2(256)`	Name of the ruleset
`RULESET_OPTION`	`VARCHAR2(8)`	The possible values are `ANY_TRUE` or `ALL_TRUE`
`FS_NAME`	`VARCHAR2(1024)`	Mount path name to identify the file system
`VOL_DEVICE`	`VARCHAR2(256)`	ADVM volume name
`CON_ID`	`NUMBER`	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always `0`.

> **See Also**
>
> - *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
> - *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

> **Note**
>
> To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.49 V$ASM_ACFS_SEC_RULESET_RULE

V$ASM_ACFS_SEC_RULESET_RULE contains an entry for every rule in the Oracle ACFS security rule set for each Oracle ACFS file system. This view also supports GV$ global views.

Column	Datatype	Description
RULESET_NAME	VARCHAR2(256)	Rule set name
RULE_NAME	VARCHAR2(256)	Rule name
FS_NAME	VARCHAR2(1024)	Mount path name to identify the file system
VOL_DEVICE	VARCHAR2(256)	ADVM volume name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Note

To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.50 V$ASM_ACFS_SECURITY_INFO

V$ASM_ACFS_SECURITY_INFO displays security information for every mounted Oracle ACFS.

Column	Datatype	Description
FS_NAME	VARCHAR2(1024)	File system mount point (foreign key to V$ASM_FILESYSTEM)
VOL_DEVICE	VARCHAR2(256)	Name of the Oracle ADVM device

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PREPARED_STATUS	VARCHAR2(7)	Indicates whether the file system is prepared for security (YES) or not (NO); otherwise UNKNOWN.
ENABLED_STATUS	VARCHAR2(8)	Indicates whether security is enabled for the file system (ENABLED) or not (DISABLED); otherwise UNKNOWN.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Note

To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.51 V$ASM_ACFSAUTORESIZE

V$ASM_ACFSAUTORESIZE displays the auto-resize settings for each mounted Oracle ACFS file system.

Column	Datatype	Description
FS_NAME	VARCHAR2(1024)	Full path name of the Oracle ACFS file system mount point
RESIZE_INCREMENT	NUMBER	Auto-resize increment (in megabytes)
RESIZE_MAXIMUM	NUMBER	Auto-resize maximum (in megabytes)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Note

This view is available starting with Oracle Database 19c.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Note

To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.52 V$ASM_ACFSREPL

V$ASM_ACFSREPL displays information for Oracle ACFS file systems that are initialized for replication.

Column	Datatype	Description
FSNAME	VARCHAR2(1024)	File system mount point (foreign key to V$ASM_FILESYSTEM)
VOLDEV	VARCHAR2(256)	Name of the Oracle ADVM device
SITE	VARCHAR2(7)	Replication site role: • PRIMARY - File system is initialized as a primary • STANDBY - File system is initialized as a standby
LAG	VARCHAR2(128)	Amount of time standby is behind primary in *hours*:*minutes*:*seconds* (primary only)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(12)	Replication primary or standby status: • ONLINE - Replication is initialized and processing real-time changes • PAUSED - The primary is not sending logs to the standby or the standby is not applying logged changes
INITSTATUS	VARCHAR2(11)	This column is not maintained.
DIRSSCAN	NUMBER	This column is not maintained.
PERCENT	NUMBER	This column is not maintained.
LASTSYNC	DATE	Time of last apply on the standby site (standby only)
CRSRUN	NUMBER	Number of Cluster Ready Services replication daemons currently running
CRSTOTAL	NUMBER	Number of Cluster Ready Services replication daemons expected to be running
PMNTPT	VARCHAR2(1024)	Replication primary site mount point
SMNTPT	VARCHAR2(1024)	Replication standby site mount point
PSVCNAME	VARCHAR2(1024)	This column is not maintained.
SSVCNAME	VARCHAR2(1024)	This column is not maintained.
PHOST	VARCHAR2(1024)	A replication primary cluster host name
SHOST	VARCHAR2(1024)	A replication standby cluster host name
REMALIAS	VARCHAR2(4000)	This column is not maintained.
TAGS	NUMBER	TRUE (1) if replicating tagged files
COMPRESSION	VARCHAR2(3)	This column is not maintained.
DBGLVL	NUMBER	Replication trace log debug level
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

Note

For Oracle ASM releases 12.2 or later, this view contains records only for storage locations that are file systems (and not for storage locations that are snapshots). The following columns in this view are no longer maintained: INITSTATUS, DIRSCAN, PERCENT, PSVCNAME, SSVCNAME, REMALIAS, and COMPRESSION.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Note

To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.53 V$ASM_ACFSREPLTAG

`V$ASM_ACFSREPLTAG` displays replicated tag information for Oracle ACFS file systems that are initialized for replication.

Column	Datatype	Description
`FSNAME`	`VARCHAR2(1024)`	File system mount point (foreign key to `V$ASM_FILESYSTEM`)
`VOLDEV`	`VARCHAR2(256)`	Name of the Oracle ADVM device
`TAG`	`VARCHAR2(32)`	Tag name that is being replicated
`CON_ID`	`NUMBER`	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always `0`.

Note

This view only contains records for Oracle ASM releases prior to Oracle Database 12*c* Release 2 (12.2.0.1). To display Oracle ACFS replication information for Oracle Database 12*c* Release 2 (12.2.0.1) or higher, use the `acfsutil repl info` command.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Note

To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.54 V$ASM_ACFSSNAPSHOTS

`V$ASM_ACFSSNAPSHOTS` displays snapshot information for every mounted Oracle ACFS.

Column	Datatype	Description
`FS_NAME`	`VARCHAR2(1024)`	File system mount point (foreign key to `V$ASM_FILESYSTEM`)
`VOL_DEVICE`	`VARCHAR2(256)`	Name of the Oracle ADVM device
`SNAP_NAME`	`VARCHAR2(1024)`	Name of the snapshot
`CREATE_TIME`	`DATE`	Time when the snapshot was created
`TYPE`	`VARCHAR2(2)`	Snapshot type. Possible types are read-only (RO) or read/write (RW).
`PARENT`	`VARCHAR2(1024)`	Parent name used to create the snapshot. If the snapshot was created using the mount point as a base, parent name will be NULL. If the snapshot was created using an existing snapshot as the base, the parent name will be the name of the snapshot passed to the `acfsutil snap create -p` operation.
`LINK`[1]	`VARCHAR2(1024)`	If link has been set, the path of the link. If unset, NULL.
`ADDITIONAL_STORAGE`[1]	`NUMBER`	Equivalent to "storage added to snapshot" output
`QUOTA`[1]	`NUMBER`	Value of quota. If unset, NULL.
`REPL`[1]	`VARCHAR2(5)`	Indicates whether the snapshot is being used for replication (`TRUE`) or not (`FALSE`)
`STATE`[1]	`VARCHAR2(14)`	The state of the snapshot: • `AVAILABLE` - Default • `DELETE_WAITING` - Maps to "delete waiting for last close" • `DELETE_IN_PROGRESS` - Maps to "delete in progress"

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

1 This column is available starting with Oracle Database 19c.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Note

To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.55 V$ASM_ACFSTAG

V$ASM_ACFSTAG displays every file or directory that has a tag and its tag name for every mounted Oracle ACFS.

Column	Datatype	Description
TAG_NAME	VARCHAR2(32)	Tag name on file or directory
FS_NAME	VARCHAR2(1024)	File system mount point
PATH_NAME	VARCHAR2(4096)	Complete file path name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Note

To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.56 V$ASM_ACFSVOLUMES

V$ASM_ACFSVOLUMES displays information about mounted Oracle ACFS volumes, correlated with V$ASM_FILESYSTEM.

Column	Datatype	Description
FS_NAME	VARCHAR2(1024)	File system mount point (foreign key to V$ASM_FILESYSTEM)
VOL_DEVICE	VARCHAR2(256)	Name of the Oracle ADVM device
VOL_LABEL	VARCHAR2(64)	Volume label (optional name) assigned through mkfs/acfsformat; NULL if no name exists
PRIMARY_VOL	VARCHAR2(5)	Indicates whether the volume is the primary volume for the file system (TRUE) or not (FALSE)
TOTAL_MB	NUMBER	Size of the volume device (in megabytes)
FREE_MB	NUMBER	Available space on the volume device (in megabytes)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

- "V$ASM_FILESYSTEM"
- *Oracle Automatic Storage Management Administrator's Guide* for more information about Oracle Automatic Storage Management Cluster File System (Oracle ACFS)
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle Automatic Storage Management Cluster File System (Oracle ACFS) and Oracle ASM Dynamic Volume Manager (Oracle ADVM) information

Note

To display information about Oracle ACFS file systems or volumes that are located on nodes in an Oracle Flex ASM configuration, you must connect to the Oracle ASM proxy instance instead of the local Oracle ASM instance. For information about Oracle Flex ASM, refer to *Oracle Automatic Storage Management Administrator's Guide*.

7.57 V$ASM_ALIAS

In an Oracle Automatic Storage Management (Oracle ASM) instance, V$ASM_ALIAS displays one row for every alias present in every disk group mounted by the Oracle ASM instance.

Column	Datatype	Description
NAME	VARCHAR2(70)	Oracle ASM alias or alias directory name
GROUP_NUMBER	NUMBER	Owning disk group number of the alias (foreign key to the V$ASM_DISKGROUP view)
FILE_NUMBER	NUMBER	Oracle ASM file number of the alias (foreign key to the V$ASM_FILE view)
FILE_INCARNATION	NUMBER	Oracle ASM file incarnation number for the alias
ALIAS_INDEX	NUMBER	Alias entry number for the alias
ALIAS_INCARNATION	NUMBER	Incarnation number for the parent of the alias

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PARENT_INDEX	NUMBER	A 32-bit number consisting of a disk group number in the high-order 8 bits and an alias entry number in the low-order 24 bits (number of the directory containing the alias)
REFERENCE_INDEX	NUMBER	A 32-bit number consisting of a disk group number in the high-order 8 bits and an alias entry number in the low-order 24 bits (number of the directory describing the current entry)
ALIAS_DIRECTORY	VARCHAR2(1)	Indicates whether the alias is to a directory (Y) or to an Oracle ASM file (N)
SYSTEM_CREATED	VARCHAR2(1)	Indicates whether the alias is system created (Y) or user created (N)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

Oracle Automatic Storage Management Administrator's Guide for additional information about using views to display Oracle ASM information

7.58 V$ASM_ATTRIBUTE

V$ASM_ATTRIBUTE displays one row for each attribute defined. In addition to attributes specified by CREATE DISKGROUP and ALTER DISKGROUP statements, the view may show other attributes that are created automatically.

Note that attributes are only displayed for disk groups where COMPATIBLE.ASM is set to 11.1 or higher.

Column	Datatype	Description
NAME	VARCHAR2(256)	Full name of the attribute
VALUE	VARCHAR2(256)	Value of the attribute
GROUP_NUMBER	NUMBER	Number of the disk group in which this attribute exists (composite primary key)
ATTRIBUTE_INDEX	NUMBER	Number of this attribute in the disk group (composite primary key)
ATTRIBUTE_INCARNATION	NUMBER	Incarnation number for this attribute (composite primary key)
READ_ONLY	VARCHAR2(7)	Indicates whether the attribute is read-only (Y) or not (N)
SYSTEM_CREATED	VARCHAR2(7)	Indicates whether this is a system-created attribute (Y) or not (N)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

- *Oracle Automatic Storage Management Administrator's Guide* for more information about viewing Oracle ASM disk group attributes using this view
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle ASM information

7.59 V$ASM_AUDIT_CLEANUP_JOBS

In an Oracle Automatic Storage Management (Oracle ASM) instance, V$ASM_AUDIT_CLEANUP_JOBS displays information about the configured audit trail purge jobs.

In a database instance, V$ASM_AUDIT_CLEANUP_JOBS displays no rows.

Column	Datatype	Description
JOB_NAME	VARCHAR2(64)	Name of the audit trail purge job
JOB_STATUS	VARCHAR2(64)	Current status of the audit trail purge job (ENABLED) or (DISABLED)
AUDIT_TRAIL	VARCHAR2(64)	Audit trail for which the audit trail purge job is configured: OS AUDIT TRAIL and UNIFIED AUDIT TRAIL
JOB_FREQUENCY	NUMBER	Frequency at which the audit trail purge job runs
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

Oracle Automatic Storage Management Administrator's Guide for additional information about using views to display Oracle ASM information

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.60 V$ASM_AUDIT_CONFIG_PARAMS

In an Oracle Automatic Storage Management (Oracle ASM) instance, V$ASM_AUDIT_CONFIG_PARAMS displays information about the currently configured audit trail properties that are used by the DBMS_AUDIT_MGMT package.

In a database instance, V$ASM_AUDIT_CONFIG_PARAMS displays no rows.

Column	Datatype	Description
PARAMETER_NAME	VARCHAR2(64)	Name of the property
PARAMETER_VALUE	VARCHAR2(64)	Value of the property
AUDIT_TRAIL	VARCHAR2(64)	Audit trails for which the property is configured: OS AUDIT TRAIL and UNIFIED AUDIT TRAIL
PARAMETER_UNIT[1]	VARCHAR2(64)	Provides description of units of audit properties (file size in KB and file age in day(s))
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

[1] This column is available starting with Oracle Database 19c.

See Also

- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_AUDIT_MGMT package
- *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle ASM information

7.61 V$ASM_AUDIT_LAST_ARCH_TS

In an Oracle Automatic Storage Management (Oracle ASM) instance, V$ASM_AUDIT_LAST_ARCH_TS displays information about the last archive timestamps set for audit trail cleanup or purges.

In a database instance, V$ASM_AUDIT_LAST_ARCH_TS displays no rows.

Column	Datatype	Description
AUDIT_TRAIL	VARCHAR2(64)	Audit trail for which the last archive timestamp applies: OS AUDIT TRAIL and UNIFIED AUDIT TRAIL
LAST_ARCHIVE_TS	TIMESTAMP(6) WITH TIME ZONE	Timestamp of the last audit file that has been archived

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

> **See Also**
>
> *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle ASM information

7.62 V$ASM_CLIENT

In an Oracle Automatic Storage Management (Oracle ASM) instance, V$ASM_CLIENT identifies databases using disk groups managed by the Oracle ASM instance.

In a database instance, V$ASM_CLIENT displays information about the Oracle ASM instance if the database has any open Oracle ASM files.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Number of the disk group being used by the database instance (foreign key to the V$ASM_DISKGROUP view). When a client is connected to an Oracle ASM instance, but is not currently using any disk groups, V$ASM_CLIENT.GROUP_NUMBER contains 0. When the Cluster Ready Services Daemon (CRSD) or Cluster Synchronization Services (OCSSD) connects to the Oracle ASM instance, this will represent the number of the disk group used for their OCR and voting files respectively.
INSTANCE_NAME	VARCHAR2(64)	Identifier for the database instance client When CRSD or OCSSD has opened its files in the disk group mounted by the Oracle ASM instance, this will contain the node name where CRSD or OCSSD is running.
DB_NAME	VARCHAR2(8)	Unique database name of the database client instance When CRSD has opened the Oracle Cluster Registry (OCR) file in the disk group mounted by the Oracle ASM instance, DB_NAME will contain _OCR. When OCSSD has opened the voting file in the disk group mounted by the Oracle ASM instance, DB_NAME will contain _CSS.
CLUSTER_NAME	VARCHAR2(31)	Name of the cluster

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(12)	Status of the client connection: • CONNECTED - Database instance client has an active connection to the Oracle ASM instance • DISCONNECTED - Database instance client normally ended its connection to the Oracle ASM instance • BROKEN - Connection with the database instance client terminated in a nonstandard way When CRSD or OCSSD has opened its files in the disk group mounted by the Oracle ASM instance, it will report Connected as status.
SOFTWARE_VERSION	VARCHAR2(60)	Software version number of the database or Oracle ASM instance for the selected disk group connection
COMPATIBLE_VERSION	VARCHAR2(60)	Compatibility setting of the database or Oracle ASM instance for the selected disk group connection The value will be - when CRSD or OCSSD has opened its files in the disk group mounted by the Oracle ASM instance.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

Oracle Automatic Storage Management Administrator's Guide for additional information about using views to display Oracle ASM information

7.63 V$ASM_DBCLONE_INFO

V$ASM_DBCLONE_INFO shows the relationship between the parent database and point-in-time database clones.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Disk group number
DBCLONE_NAME	VARCHAR2(128)	Name of the point-in-time database clone
MIRRORCOPY_NAME	VARCHAR2(128)	Name of the mirror copy associated with the database clone

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DBCLONE_STATUS	VARCHAR2(128)	Status of the database clone. Possible values: • PREPARING — The process to prepare the database for splitting has started, but the database is not yet ready to be split. • PREPARED — The preparation process is complete and the database is ready to be split. • SPLITTING — The process to split the database files has started. • SPLIT COMPLETED — The database has been split. • DROPPING — The process to drop the prepared database clone has started. • FAILED — The clone operation has failed. The incomplete clone will be dropped.
PARENT_DBNAME	VARCHAR2(128)	Name of the parent database
PARENT_FILEGROUP_NAME	VARCHAR2(128)	Name of the file group associated with the parent database
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Automatic Storage Management Administrator's Guide for information about point-in-time database clones

7.64 V$ASM_DISK

V$ASM_DISK displays one row for every disk discovered by the Oracle Automatic Storage Management (Oracle ASM) instance, including disks that are not part of any disk group.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Number of the disk group containing the disk (foreign key to the V$ASM_DISKGROUP view)
DISK_NUMBER	NUMBER	Number assigned to the disk within its disk group
COMPOUND_INDEX	NUMBER	A 32-bit number consisting of a disk group number in the high-order 8 bits and a disk number in the low-order 24 bits (for efficient access to the view)
INCARNATION	NUMBER	Incarnation number for the disk

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MOUNT_STATUS	VARCHAR2(7)	Per-instance status of the disk relative to group mounts: • MISSING - Oracle ASM metadata indicates that the disk is known to be part of the Oracle ASM disk group but no disk in the storage system was found with the indicated name • CLOSED - Disk is present in the storage system but is not being accessed by Oracle ASM • OPENED - Disk is present in the storage system and is being accessed by Oracle ASM. This is the normal state for disks in a database instance which are part of a disk group being actively used by the instance. • CACHED - Disk is present in the storage system and is part of a disk group being accessed by the Oracle ASM instance. This is the normal state for disks in an Oracle ASM instance which are part of a mounted disk group. • IGNORED - Disk is present in the system but is ignored by Oracle ASM because of one of the following: – The disk is detected by the system library but is ignored because an Oracle ASM library discovered the same disk – Oracle ASM has determined that the membership claimed by the disk header is no longer valid • CLOSING - Oracle ASM is in the process of closing this disk
HEADER_STATUS	VARCHAR2(12)	Per-instance status of the disk as seen by discovery: • UNKNOWN - Oracle ASM disk header has not been read • CANDIDATE - Disk is not part of a disk group and may be added to a disk group with the ALTER DISKGROUP statement • INCOMPATIBLE - Version number in the disk header is not compatible with the Oracle ASM software version • PROVISIONED - Disk is not part of a disk group and may be added to a disk group with the ALTER DISKGROUP statement. The PROVISIONED header status is different from the CANDIDATE header status in that PROVISIONED implies that an additional platform-specific action has been taken by an administrator to make the disk available for Oracle ASM. • MEMBER - Disk is a member of an existing disk group. No attempt should be made to add the disk to a different disk group. The ALTER DISKGROUP statement will reject such an addition unless overridden with the FORCE option. • FORMER - Disk was once part of a disk group but has been dropped cleanly from the group. It may be added to a new disk group with the ALTER DISKGROUP statement. • CONFLICT - Oracle ASM disk was not mounted due to a conflict • FOREIGN - Disk contains data created by an Oracle product other than Oracle ASM. This includes data files, logfiles, and OCR disks.
MODE_STATUS	VARCHAR2(7)	Global status about which kinds of I/O requests are allowed to the disk: • ONLINE - Disk is online and operating normally. Reads and writes are attempted when the disk is mounted. Reads are attempted as part of disk discovery. • OFFLINE - Disk is offline and access to data is not permitted. Reads and writes are not attempted. An offline disk remains logically part of its disk group. • SYNCING - Disk is offline for reads, but online for writes. Oracle ASM is resyncing stale data on the disk. See V$ASM_OPERATION to view resync progress.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATE	VARCHAR2(8)	Global state of the disk with respect to the disk group: • UNKNOWN - Oracle ASM disk state is not known (typically the disk is not mounted) • NORMAL - Disk is online and operating normally • ADDING - Disk is being added to a disk group, and is pending validation by all instances that have the disk group mounted • DROPPING - Disk has been manually taken offline and space allocation or data access for the disk halts. Rebalancing will commence to relocate data off the disks to other disks in the disk group. Upon completion of the rebalance, the disk is expelled from the group. • HUNG - Disk drop operation cannot continue because there is insufficient space to relocate the data from the disk being dropped • FORCING - Disk is being removed from the disk group without attempting to offload its data. The data will be recovered from redundant copies, where possible.
REDUNDANCY	VARCHAR2(7)	Hardware redundancy of the disk: • UNKNOWN • UNPROT • MIRROR • PARITY **Note:** This column is valid only if an ASMLIB is present that supports returning hardware redundancy information. This column is not related to the redundancy of the disk group of which the disk is a member.
LIBRARY	VARCHAR2(64)	Name of the library that discovered the disk
OS_MB	NUMBER	Size of the disk (in megabytes) as reported by the host operating system
TOTAL_MB	NUMBER	Total capacity of the disk (in megabytes)
FREE_MB	NUMBER	Unused capacity of the disk (in megabytes)
HOT_USED_MB	NUMBER	Number of used megabytes in the hot region
COLD_USED_MB	NUMBER	Number of used megabytes in the cold region
NAME	VARCHAR2(30)	Name of the disk
FAILGROUP	VARCHAR2(30)	Name of the failure group containing the disk
LABEL	VARCHAR2(31)	Disk label portion of the name returned by discovery
PATH	VARCHAR2(256)	Operating system path name portion of the name returned by discovery
UDID	VARCHAR2(64)	Unique Device ID portion of the name returned by discovery
PRODUCT	VARCHAR2(32)	Name of the manufacturer and the name of the product. All disks with the same product id will have the same performance and reliability characteristics.
CREATE_DATE	DATE	Date and time when the disk was added to the disk group
MOUNT_DATE	DATE	Date and time when the disk was mounted by the first instance
REPAIR_TIMER	NUMBER	Seconds remaining until the disk is automatically dropped (0 if not failed)
READS	NUMBER	Total number of I/O read requests for the disk
WRITES	NUMBER	Total number of I/O write requests for the disk
READ_ERRS	NUMBER	Total number of failed I/O read requests for the disk
WRITE_ERRS	NUMBER	Total number of failed I/O write requests for the disk

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
READ_TIMEOUT	NUMBER	Number of read I/Os that are timed out
WRITE_TIMEOUT	NUMBER	Number of write I/Os that are timed out
READ_TIME	NUMBER	Total I/O time (in seconds) for read requests for the disk if the TIMED_STATISTICS initialization parameter is set to true (0 if set to false)
WRITE_TIME	NUMBER	Total I/O time (in seconds) for write requests for the disk if the TIMED_STATISTICS initialization parameter is set to true (0 if set to false)
BYTES_READ	NUMBER	Total number of bytes read from the disk
BYTES_WRITTEN	NUMBER	Total number of bytes written to the disk
PREFERRED_READ	VARCHAR2(1)	Status of the preferred read failure group: • U - Disk group has no preferred read failure group • Y - Disk is a preferred read disk • N - Disk is not a preferred read disk For a disk group with one or more preferred read failure groups, if the disk is in one of the preferred read failure groups, the value of this column is Y; otherwise it is N.
HASH_VALUE	NUMBER	A unique hash value for an Oracle ASM disk, computed using the Oracle ASM disk name (as specified by NAME column in v$asm_disk) and the Oracle ASM disk group name (as specified by NAME column in v$asm_diskgroup).
HOT_READS	NUMBER	Number of reads from the hot region on disk
HOT_WRITES	NUMBER	Number of writes to the hot region on disk
HOT_BYTES_READ	NUMBER	Number of bytes read from the hot region on disk
HOT_BYTES_WRITTEN	NUMBER	Number of bytes written to the hot region on disk
COLD_READS	NUMBER	Number of reads from the cold region on disk
COLD_WRITES	NUMBER	Number of writes to the cold region on disk
COLD_BYTES_READ	NUMBER	Number of bytes read from the cold region on disk
COLD_BYTES_WRITTEN	NUMBER	Number of bytes written to the cold region on disk
VOTING_FILE	VARCHAR2(1)	Indicates whether the disk contains a voting file (Y) or not (N)
SECTOR_SIZE	NUMBER	Physical block size (in bytes)
LOGICAL_SECTOR_SIZE	NUMBER	Shows the logical sector size value of the disk in bytes. This is the smallest possible I/O that can be done by the disk. If the value of the LOGICAL_SECTOR_SIZE column is 512, while the value in the SECTOR_SIZE column is 4096, then the disk supports 512 sector size emulation. This means the disk can be used for both 512 native and 4096 native disk groups.
FAILGROUP_TYPE	VARCHAR2(7)	Type of the failure group: • REGULAR • QUORUM

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.
THIN_PROVISION_CAPABLE	VARCHAR2(1)	Indicates whether the disk supports Storage Thin Provisioning (Y) or not (N)
DATA_INTEGRITY_CAPABLE	VARCHAR2(1)	Indicates whether the disk supports the Data Integrity feature (Y) or not (N)
SITE_NAME	VARCHAR2(30)	The name of the site to which the specific disk belongs
SITE_GUID	VARCHAR2(33)	The GUID for the site to which the specific disk belongs
FAILGROUP_LABEL	VARCHAR2(30)	This is the suggested name for the failure group to which the disk belongs, as returned by discovery
SITE_LABEL	VARCHAR2(30)	This is the suggested name for the site to which the disk belongs, as returned by discovery
SITE_STATUS	VARCHAR2(11)	This column is populated only for disks in Oracle ASM extended disk groups. The site status is computed after every Partnership and Status Table (PST) refresh or when the PST is read from disks into memory. The possible states include: • UNKNOWN: This is the state before any checks have been performed. • UNAVAILABLE: This is the state when all the disks in the site are offline. • COMPROMISED: The state when enough disks or PST copies are offline, such that the site can no longer provide data availability. • AVAILABLE: The state when there is enough redundancy within the site to provide data availability in case another site becomes compromised or unavailable.

Note

The GROUP_NUMBER and DISK_NUMBER columns will only be valid if the disk is part of a disk group which is currently mounted by the instance. Otherwise, GROUP_NUMBER will be 0, and DISK_NUMBER will be a unique value with respect to the other disks that also have a group number of 0.

Note

This view performs disk discovery every time it is queried. Because performing disk discovery is very resource intensive, Oracle recommends against using this view for monitoring scripts. Instead, use the less expensive view V$ASM_DISK_STAT.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle ASM information

7.65 V$ASM_DISK_IOSTAT

V$ASM_DISK_IOSTAT displays information about disk I/O statistics for each Oracle Automatic Storage Management (Oracle ASM) client.

If this view is queried from the database instance, only the rows for that instance are shown.

Column	Datatype	Description
INSTNAME	VARCHAR2(64)	Identifier for the DB/ADVM instance client
DBNAME	VARCHAR2(8)	Unique database name (DB_UNIQUE_NAME)
CLUSTERNAME	VARCHAR2(31)	Name of the cluster
GROUP_NUMBER	NUMBER	Number of the disk group containing the disk
DISK_NUMBER	NUMBER	Number assigned to the disk within its disk group
FAILGROUP	VARCHAR2(30)	Name of the failure group to which the disk belongs
SITE_NAME	VARCHAR2(30)	This is the name of the site to which the disk belongs
READS	NUMBER	Total number of I/O read requests for the disk
WRITES	NUMBER	Total number of I/O write requests for the disk
READ_ERRS	NUMBER	Total number of failed I/O read requests for the disk
WRITE_ERRS	NUMBER	Total number of failed I/O write requests for the disk
READ_TIMEOUT	NUMBER	Number of read I/Os that are timed out
WRITE_TIMEOUT	NUMBER	Number of write I/Os that are timed out
READ_TIME	NUMBER	Total I/O time (in seconds) for read requests for the disk if the TIMED_STATISTICS initialization parameter is set to TRUE (0 if set to FALSE).
WRITE_TIME	NUMBER	Total I/O time (in seconds) for write requests for the disk if the TIMED_STATISTICS initialization parameter is set to TRUE (0 if set to FALSE)
BYTES_READ	NUMBER	Total number of bytes read from the disk
BYTES_WRITTEN	NUMBER	Total number of bytes written from the disk
HOT_READS	NUMBER	Number of reads from the hot region on disk
HOT_WRITES	NUMBER	Number of writes to the hot region on disk
HOT_BYTES_READ	NUMBER	Number of bytes read from the hot region on disk
HOT_BYTES_WRITTEN	NUMBER	Number of bytes written to the hot region on disk
COLD_READS	NUMBER	Number of reads from the cold region on disk
COLD_WRITES	NUMBER	Number of writes to the cold region on disk
COLD_BYTES_READ	NUMBER	Number of bytes read from the cold region on disk
COLD_BYTES_WRITTEN	NUMBER	Number of bytes written to the cold region on disk

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

> **See Also**
>
> *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle ASM information

7.66 V$ASM_DISK_STAT

V$ASM_DISK_STAT displays performance statistics in the same way that V$ASM_DISK does, but without performing discovery of new disks

This results in a less expensive operation. However, since discovery is not performed, the output of this view does not include any data about disks that are new to the system.

The columns for V$ASM_DISK_STAT are the same as those for V$ASM_DISK.

> **See Also**
>
> "V$ASM_DISK"

7.67 V$ASM_DISKGROUP

V$ASM_DISKGROUP displays one row for every Oracle Automatic Storage Management (Oracle ASM) disk group discovered by the Oracle ASM instance on the node.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Cluster-wide number assigned to the disk group (primary key)
NAME	VARCHAR2(30)	Name of the disk group
SECTOR_SIZE	NUMBER	Physical block size (in bytes)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LOGICAL_SECTOR_SIZE	NUMBER	This column represents the logical sector size value of the disk group in bytes. This is the smallest possible I/O that can be done in this disk group. Any I/O smaller than the logical sector size will cause an assert in the code. If the value of the LOGICAL_SECTOR_SIZE column is 512, while the value in the SECTOR_SIZE column is 4096, then the disk group supports 512 sector size emulation. This means that I/O operations can be both 512 or 4096 aligned. For example, this means that you can have redo log files with a file block size of 512 bytes in this disk group that can be both read and written to, as well as create new redo log files with a file block size of 4096 bytes.
BLOCK_SIZE	NUMBER	Oracle ASM metadata block size (in bytes)
ALLOCATION_UNIT_SIZE	NUMBER	Size of the allocation unit (in bytes)
STATE	VARCHAR2(11)	State of the disk group relative to the instance: • BROKEN - Database instance lost connectivity to the Oracle ASM instance that mounted the disk group • CONNECTED - Disk group is in use by the database instance • DISMOUNTED - Disk group was cleanly dismounted by the Oracle ASM instance following a successful mount • MOUNTED - Instance is successfully serving the disk group to its database clients • QUIESCING - CRSCTL utility attempted to dismount a disk group that contains the Oracle Cluster Registry (OCR). The disk group cannot be dismounted until Cluster Ready Services (CRS) exits, because the disk group contains the OCR. • RESTRICTED - A disk group mounted in this mode can only be seen by this Oracle ASM instance for maintenance purposes • UNKNOWN - Oracle ASM instance has never attempted to mount the disk group
TYPE	VARCHAR2(6)	Redundancy type for the disk group: • EXTEND • EXTERN • FLEX • HIGH • NORMAL
TOTAL_MB	NUMBER	Total capacity of the disk group, excluding that of quorum disks (in megabytes)
FREE_MB	NUMBER	Unused capacity of the disk group (in megabytes)
HOT_USED_MB	NUMBER	Number of used megabytes in the hot region
COLD_USED_MB	NUMBER	Number of used megabytes in the cold region
REQUIRED_MIRROR_FREE_MB	NUMBER	Amount of space that is required to be available in a given disk group in order to restore redundancy after one or more disk failures. The amount of space displayed in this column takes mirroring effects into account.
USABLE_FILE_MB	NUMBER	Amount of free space that can be safely utilized taking mirroring into account and yet be able to restore redundancy after a disk failure
OFFLINE_DISKS	NUMBER	Number of disks in the disk group that are currently offline
COMPATIBILITY	VARCHAR2(60)	Minimum software version required for an Oracle ASM instance to mount this disk group
DATABASE_COMPATIBILITY	VARCHAR2(60)	Minimum software version required for a database instance to use files in this disk group

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
VOTING_FILES	VARCHAR2(1)	Indicates whether the disk contains voting files (Y) or not (N)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

Note

The GROUP_NUMBER, TOTAL_MB, and FREE_MB columns are only meaningful if the disk group is mounted by the instance. Otherwise, their values will be 0.

Note

This view performs disk discovery every time it is queried. Because performing disk discovery is very resource intensive, Oracle recommends against using this view for monitoring scripts. Instead, use the less expensive view V$ASM_DISKGROUP_STAT.

See Also

Oracle Automatic Storage Management Administrator's Guide for additional information about using views to display Oracle ASM information

7.68 V$ASM_DISKGROUP_STAT

V$ASM_DISKGROUP_STAT displays performance statistics in the same way that V$ASM_DISKGROUP does, but without performing discovery of new disk groups.

This results in a less expensive operation. However, since discovery is not performed, the output of this view does not include any data about disk groups that are new to the system.

The columns for V$ASM_DISKGROUP_STAT are the same as those for V$ASM_DISKGROUP.

See Also

"V$ASM_DISKGROUP"

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.69 V$ASM_ESTIMATE

V$ASM_ESTIMATE displays an estimate of the work involved in execution plans for Oracle Automatic Storage Management (Oracle ASM) disk group rebalance and resync operations.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Oracle ASM disk group number
STATEMENT_ID	VARCHAR2(30)	Value of the optional STATEMENT_ID parameter specified in the EXPLAIN WORK statement
TIMESTAMP	DATE	Date and time when the EXPLAIN WORK statement was issued
EST_WORK	NUMBER	Estimated number of Allocation Units that have to be moved by the operation
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

Oracle Automatic Storage Management Administrator's Guide for additional information about using views to display Oracle ASM information

7.70 V$ASM_FILE

In an Oracle Automatic Storage Management (Oracle ASM) instance, V$ASM_FILE displays one row for each file in each disk group mounted by the Oracle ASM instance.

For example, if there are three disk groups and five files in each, fifteen rows are displayed (unless the query is qualified with a WHERE clause).

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Number of the disk group containing the file (composite primary key)
FILE_NUMBER	NUMBER	Number of the file within the disk group (composite primary key)
COMPOUND_INDEX	NUMBER	A 32-bit number consisting of a disk group number in the high-order 8 bits and a file number in the low-order 24 bits (for efficient access to the view)
INCARNATION	NUMBER	Incarnation number for the file (composite primary key)
BLOCK_SIZE	NUMBER	Block size of the file (in bytes)
BLOCKS	NUMBER	Number of blocks in the file
BYTES	NUMBER	Number of bytes in the file
SPACE	NUMBER	Number of bytes allocated to the file

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TYPE	VARCHAR2(64)	Type of the file. Possible values are as follows: • ARCHIVELOG • AUTOBACKUP • BACKUPSET • CHANGETRACKING • CONTROLFILE • DATAFILE • DATAGUARDCONFIG • DUMPSET • FLASHBACK • ONLINELOG • PARAMETERFILE • TEMPFILE • XTRANSPORT
REDUNDANCY	VARCHAR2(6)	Redundancy of the file: • HIGH • MIRROR • PARITY • UNPROT
STRIPED	VARCHAR2(6)	Indicates how the file is striped: • FINE • COARSE
CREATION_DATE	DATE	Date on which the file was created
MODIFICATION_DATE	DATE	Date of the last open/close for writing, rounded back to the nearest hour
REDUNDANCY_LOWERED	VARCHAR2(1)	Indicates whether a file has lower redundancy than what was expected (Y) or not (N). Redundancy is said to have been lowered for a file when one or more data extents in that file are not mirrored at the level specified by the administrator. In case of unprotected files, data extents could be missing altogether. Another possible value for this column is (U), which means that it is unknown. This column is deprecated, and it always displays a value of U.
PERMISSIONS	VARCHAR2(16)	Access permissions of the file, in order of user, group, and other
USER_NUMBER	NUMBER	User number
USER_INCARNATION	NUMBER	Incarnation number of the user
USERGROUP_NUMBER	NUMBER	User group number
USERGROUP_INCARNATION	NUMBER	Incarnation number of the user group
PRIMARY_REGION	VARCHAR2(4)	Region used for allocating primary extents: • HOT • COLD
MIRROR_REGION	VARCHAR2(4)	Region for allocating mirrored extents: • HOT • COLD
HOT_READS	NUMBER	Number of reads from the hot region for the file
HOT_WRITES	NUMBER	Number of writes to the hot region for the file
HOT_BYTES_READ	NUMBER	Number of bytes read from the hot region for the file
HOT_BYTES_WRITTEN	NUMBER	Number of bytes written to the hot region for the file
COLD_READS	NUMBER	Number of reads from the cold region for the file

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
COLD_WRITES	NUMBER	Number of writes to the cold region for the file
COLD_BYTES_READ	NUMBER	Number of bytes read from the cold region for the file
COLD_BYTES_WRITTEN	NUMBER	Number of bytes written to the cold region for the file
FILEGROUP_NUMBER	NUMBER	Shows the number of the associated file group
FILEGROUP_INCARNATION	NUMBER	The incarnation number for the file group the file belongs to in a flex redundancy disk group
REMIRROR	VARCHAR2(1)	This column has the value Y when rebalance is needed for the file after a redundancy change in a flex redundancy disk group, N otherwise. A redundancy change can occur when the file is moved to a file group with a different redundancy, or when the redundancy property of the file group is changed. After rebalance is run for the file, the value changes to N.
PARENT_FILNUM	NUMBER	For internal use only.
PARENT_FILNUMINC	NUMBER	For internal use only.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

Oracle Automatic Storage Management Administrator's Guide for additional information about using views to display Oracle ASM information

7.71 V$ASM_FILEGROUP

V$ASM_FILEGROUP describes the properties of the Oracle Automatic Storage Management (Oracle ASM) File Groups.

In both Oracle ASM and Oracle Database instances, V$ASM_FILEGROUP displays one row for every File Group present in every Disk Group mounted by the Oracle ASM instance. File Groups are only displayed for Disk Groups where COMPATIBLE.ASM is set to 12.2 or higher.

Note

There will not be an entry for the default File Group.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Number of the Disk Group in which this File Group exists (composite primary key)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FILEGROUP_NUMBER	NUMBER	Number associated to the File Group within its Disk Group (composite primary key)
INCARNATION	NUMBER	Incarnation number for the File Group (composite primary key)
COMPOUND_INDEX	NUMBER	A 32-bit number consisting of a Disk Group number in the high-order 8 bits and a File Group number in the low-order 24 bits (for efficient access to the view)
NAME	VARCHAR2(128)	Name of the File Group
CLIENT_TYPE	NUMBER	Type of client that the File Group is associated to: • DATABASE • CLUSTER • VOLUME
CLIENT_NAME	VARCHAR2(128)	Name of the client (database, PDB, CDB, cluster, or volume) that the File Group is associated to
GUID	VARCHAR2(32)	If the CLIENT_TYPE is DATABASE: • In a CDB environment, it is the GUID of the PDB or CDB associated with the file group, the same value as the GUID in V$CONTAINERS. • In a non-CDB environment it is the database identifier (DBID) of the database associated with that file group. If the CLIENT_TYPE is VOLUME or CLUSTER, the GUID column is empty.
QUOTAGROUP_NUMBER	NUMBER	Number of the quota group associated with this File Group (foreign key to the V$ASM_QUOTAGROUP view)
QUOTAGROUP_INCARNATION	NUMBER	Incarnation number for the Quota Group
USED_QUOTA_MB	NUMBER	Used quota of the quota group in GB
USER_NUMBER	NUMBER	User number
USER_INCARNATION	NUMBER	Incarnation number of the user
USERGROUP_NUMBER	NUMBER	User group number
USERGROUP_INCARNATION	NUMBER	Incarnation number of the user group
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$ASM_FILEGROUP_PROPERTY"

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.72 V$ASM_FILEGROUP_FILE

V$ASM_FILEGROUP_FILE lists all the Oracle Automatic Storage Management (Oracle ASM) files associated with each File Group.

In both Oracle ASM and Oracle Database instances, V$ASM_FILEGROUP_FILE will display one row for every file associated with a File Group contained in every Disk Group mounted by the instance.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Number of the Disk Group in which this File Group exists
FILEGROUP_NUMBER	NUMBER	Number associated to the File Group within its Disk Group
FILEGROUP_INCARN	NUMBER	Incarnation number for the File Group
FILE_NUMBER	NUMBER	Number associated to the ASM File (same file number as in V$ASM_FILE)
INCARNATION	NUMBER	Incarnation number for the ASM File
COMPOUND_INDEX	NUMBER	A 32-bit number consisting of a Disk Group number in the high-order 8 bits and a File number in the low-order 24 bits (for efficient access to the view)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$ASM_FILE"
- "V$ASM_FILEGROUP"

7.73 V$ASM_FILEGROUP_PROPERTY

V$ASM_FILEGROUP_PROPERTY describes all the properties of every Oracle Automatic Storage Management (Oracle ASM) File Group.

In both Oracle ASM and Oracle Database instances, V$ASM_FILEGROUP_PROPERTY will display one row for every property of every file type of every File Group contained in every Disk Group mounted by the instance.

File Group properties are only displayed for File Groups on Disk Groups where COMPATIBLE.ASM is set to 12.2 or higher.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> There will not be an entry for the default File Group.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Number of the Disk Group in which this File Group exists (composite primary key)
FILEGROUP_NUMBER	NUMBER	Number associated to the File Group within its Disk Group (composite primary key)
COMPOUND_INDEX	NUMBER	A 32-bit number consisting of a Disk Group number in the high-order 8 bits and a Property Number in the low-order 24 bits (for efficient access to the view)
PROPERTY_INDEX	NUMBER	Number of this property in the Disk Group for the File Group (composite primary key)
INCARNATION	NUMBER	Incarnation number for this property (composite primary key)
FILE_TYPE	VARCHAR2(30)	Type of file the property will be applied to: • ARCHIVELOG • ASMPARAMETERFILE • ASMVDRL • ASMVOL • AUDIT_SPILLFILES • AUTOBACKUP • BACKUPSET • CHANGETRACKING • CONTAINER • CONTROLFILE • DATAFILE • DATAGUARDCONFIG • DUMPSET • FLASHBACK • FLASHFILE • KEY_STORE • OCRBACKUP • OCRFILE • ONLINELOG • PARAMETERFILE • TEMPFILE • VOTINGFILE • XTRANSPORT The value can be NULL for properties with a File Group granularity.
NAME	VARCHAR2(64)	Full name of the property. Possible values: • COMPATIBLE.CLIENT • DBCLONE_STATUS • OWNER • PARENT_FILEGROUP_NUMBER • POWER_LIMIT • PRIORITY • REDUNDANCY • STRIPING • USER_GROUP

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
VALUE	VARCHAR2(256)	Value of the property See *Oracle Automatic Storage Management Administrator's Guide* for information about file group property values.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$ASM_FILEGROUP"

7.74 V$ASM_FILESYSTEM

V$ASM_FILESYSTEM displays information for every mounted Oracle ACFS.

Column	Datatype	Description
FS_NAME	VARCHAR2(1024)	Mount point (primary key)
AVAILABLE_TIME	DATE	Mount time or the time that the file system became available again; NULL if the file system is not available
BLOCK_SIZE	NUMBER	File system block size (in kilobytes)
STATE	VARCHAR2(13)	File system status:: • NOT AVAILABLE • AVAILABLE • OFFLINE - Either the Oracle ASM instance is down, the disk group has been forced dismounted, or less commonly, a metadata I/O failure occurred or serious metadata corruption was detected. In the case of a metadata I/O failure, the file system is also marked as corrupt. An offline file system can only be unmounted; other attempts at access result in errors.
CORRUPT	VARCHAR2(5)	Indicates whether the file system needs the fixer (fsck, acfschkdsk) to be run (TRUE) or not (FALSE)
NUM_VOL	NUMBER	Number of volumes in the file system
TOTAL_SIZE	NUMBER	Total capacity of the file system (in megabytes)
TOTAL_FREE	NUMBER	Total free space in the file system (in megabytes)
TOTAL_SNAP_SPACE_USAGE	NUMBER	Total space used by snapshots (in megabytes)
REPLSTATE	VARCHAR2(7)	Replication status: • NO REPL - Replication is not initialized • PRIMARY - File system is initialized for replication as a primary • STANDBY - File system is initialized for replication as a standby

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RESIZE_STATE[1]	VARCHAR2(5)	Possible resize states: • NONE - AutoResize not configured • AUTO - Equivalent to AutoResizeEnabled flag • ERROR - Equivalent to AutoResizeError flag
COMPRESS_STATE[1]	VARCHAR2(7)	Compression status of the file system. Possible values: • DISABLED - Compression is not active • ENABLED - Compression is active • PARTIAL - Compression is disabled, but the file system may contain compressed files
FROZEN_STATE[1]	VARCHAR2(5)	Indicates whether the filesystem is frozen (TRUE) or not (FALSE)
ACFS_COMPATIBILITY[1]	VARCHAR2(60)	Oracle Release streams where Oracle ACFS has shipped
METADATA_BLOCK_SIZE[1]	NUMBER	Metadata block size of the file system. Possible values: • 512 • 4096
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

1 This column is available starting with Oracle Database 19c.

> **See Also**
>
> *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle ASM information

7.75 V$ASM_OPERATION

In an Oracle Automatic Storage Management (Oracle ASM) instance, V$ASM_OPERATION displays one row for every active Oracle ASM long running operation executing in the Oracle ASM instance.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Disk group number (primary key). This is the foreign key to the V$ASM_DISKGROUP view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OPERATION	CHAR(5)	Type of the operation: • REBAL - Rebalance pending for this group. The disk group is rebalancing. • REMIRROR - Remirror is pending for this group. • SCRUB - Scrubbing is pending for this group. Starting with Oracle Database 12*c*, new queries should use the PASS column instead of this column.
PASS	VARCHAR2(9)	Type of the operation: • COMPACT - Oracle ASM is moving user data closer together, which improves performance by reducing seek distance • PREPARE - Completing work corresponding to the prepare SQL operation. This phase is enabled only for FLEX or EXTENDED redundancy disk groups and COMPATIBLE.ASM must be set to 12.2 or higher. • REBALANCE - Rebalance pending for this group. The disk group is rebalancing. • REBUILD - Restoring the redundancy of forcing disks. Forcing disks are those disks that have been dropped with the FORCE option. • RESILVER - This value appears in Oracle Exadata environments when WriteBack FlashCache is enabled • RESYNC - Resync operation in progress to bring one or more Oracle ASM disks online • SCRUBBING - The disk group is scrubbing. Starting with Oracle Database 12*c*, new queries should use this column instead of the OPERATION column.
STATE	VARCHAR2(4)	State of the operation: • WAIT - No operations running for the group • EST - An estimate is computed on the amount of work to be done for the rebalance • RUN - Operation running for the group • REAP - Operation is being run down • DONE - Displayed for a pass that is complete • ERRS - Operation halted by errors Estimates are computed in parallel in the background while the work is being executed. Thus, the transition from the EST to the RUN state may be extremely fast.
POWER	NUMBER	Power requested for the operation as specified by the ASM_POWER_LIMIT initialization parameter or command syntax. Or, power requested for the operation as specified by the power option of the scrubbing SQL syntax.
ACTUAL	NUMBER	Power allocated to the operation
SOFAR	NUMBER	Number of Allocation Units that are being moved per minute by the operation. Or, the number of Allocation Units that have been scrubbed so far.
EST_WORK	NUMBER	Estimated number of Allocation Units that have to be moved by the operation. Or, the estimated number of Allocation Units that have to be scrubbed by the scrubbing operation.
EST_RATE	NUMBER	Estimated number of Allocation Units that are being moved per minute by the operation
EST_MINUTES	NUMBER	Estimated amount of time (in minutes) that the remainder of the operation is expected to take

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ERROR_CODE	VARCHAR2(44)	Oracle external error code; NULL if no error
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

Oracle Automatic Storage Management Administrator's Guide for additional information about using views to display Oracle ASM information

7.76 V$ASM_QUOTAGROUP

V$ASM_QUOTAGROUP displays one row for every Oracle Automatic Storage Management (Oracle ASM) quota group discovered by the Oracle ASM instance on the node.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Number of the disk group containing the quota group (composite primary key)
QUOTAGROUP_NUMBER	NUMBER	Number of the quota group within the disk group (composite primary key)
INCARNATION	NUMBER	Incarnation of the quota group (composite primary key)
NAME	VARCHAR2(30)	Name of the quota group
USED_QUOTA_MB	NUMBER	Used quota of the quota group in GB
QUOTA_LIMIT_MB	NUMBER	Quota limit (capacity) of the quota group in GB
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.77 V$ASM_TEMPLATE

In an Oracle Automatic Storage Management (Oracle ASM) instance, V$ASM_TEMPLATE displays one row for every template present in every disk group mounted by the Oracle ASM instance.

In a database instance, V$ASM_TEMPLATE displays one row for every template present in every disk group mounted by the Oracle ASM instance with which the database instance communicates.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Owning disk group number (foreign key to the V$ASM_DISKGROUP view)
ENTRY_NUMBER	NUMBER	Template number (primary key)
REDUNDANCY	VARCHAR2(6)	Redundancy of the template: • HIGH • MIRROR • PARITY • UNPROT
STRIPE	VARCHAR2(6)	Indicates how the template is striped: • FINE • COARSE
SYSTEM	VARCHAR2(1)	Indicates whether the template is a system template (Y) or not (N)
NAME	VARCHAR2(30)	Name of the template
PRIMARY_REGION	VARCHAR2(4)	Region used for allocating primary extents: • HOT • COLD
MIRROR_REGION	VARCHAR2(4)	Region for allocating mirrored extents: • HOT • COLD
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

Oracle Automatic Storage Management Administrator's Guide for additional information about using views to display Oracle ASM information

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.78 V$ASM_USER

V$ASM_USER displays the effective operating system user names of connected database instances and of file owners.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Oracle ASM disk group number
USER_NUMBER	NUMBER	Oracle ASM internal unique user number
COMPOUND_INDEX	NUMBER	A 32-bit number consisting of a disk group number in the high-order 8 bits and a user number in the low-order 24 bits (for efficient access to the view)
INCARNATION	NUMBER	Incarnation number of the user
CLUSTER_ID	VARCHAR2(128)	Oracle ASM cluster ID. The CLUSTER_ID and OS_NAME pair can be used to uniquely identify a user.
OS_ID	VARCHAR2(128)	Operating system user ID
OS_NAME	VARCHAR2(33)	Operating system user name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

Oracle Automatic Storage Management Administrator's Guide for additional information about using views to display Oracle ASM information

7.79 V$ASM_USERGROUP

V$ASM_USERGROUP displays the creator for each Oracle Automatic Storage Management (Oracle ASM) File Access Control group.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Oracle ASM disk group number
USERGROUP_NUMBER	NUMBER	Number of the user group
COMPOUND_INDEX	NUMBER	A 32-bit number consisting of a disk group number in the high-order 8 bits and a user group number in the low-order 24 bits (for efficient access to the view)
INCARNATION	NUMBER	Incarnation number of the user group
OWNER_NUMBER	NUMBER	User group owner identified by a unique number
OWNER_INCARNATION	NUMBER	Incarnation number of the user group owner

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NAME	VARCHAR2(64)	User group name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

Oracle Automatic Storage Management Administrator's Guide for additional information about using views to display Oracle ASM information

7.80 V$ASM_USERGROUP_MEMBER

V$ASM_USERGROUP_MEMBER displays the members for each Oracle Automatic Storage Management (Oracle ASM) File Access Control group.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Oracle ASM disk group number
MEMBER_NUMBER	NUMBER	Number of the user group member
MEMBER_INCARNATION	NUMBER	Incarnation number of the user group member
USERGROUP_NUMBER	NUMBER	User group number
USERGROUP_INCARNATION	NUMBER	Incarnation number of the user group
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

See Also

Oracle Automatic Storage Management Administrator's Guide for additional information about using views to display Oracle ASM information

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.81 V$ASM_VOLUME

In an Oracle Automatic Storage Management (Oracle ASM) instance, V$ASM_VOLUME displays information about each Oracle ADVM volume.

Column	Datatype	Description
GROUP_NUMBER	NUMBER	Cluster-wide number assigned to the disk group (composite primary key)
VOLUME_NAME	VARCHAR2(30)	Name of the volume
COMPOUND_INDEX	NUMBER	A 32-bit number consisting of a disk group number in the high-order 8 bits and a volume number in the low-order 24 bits (for efficient access to the view)
SIZE_MB	NUMBER	Size of the volume (in megabytes)
VOLUME_NUMBER	NUMBER	Number of the Volume within the disk group (composite primary key)
REDUNDANCY	VARCHAR2(6)	Redundancy type for the volume: • UNPROT • HIGH • MIRROR
STRIPE_COLUMNS	NUMBER	Number of columns in a stripe set
STRIPE_WIDTH_K	NUMBER	Stripe width of the volume (in kilobytes)
STATE	VARCHAR2(8)	Indicates whether the volume is enabled (ENABLED), disabled (DISABLED), or remote (REMOTE)
FILE_NUMBER	NUMBER	Volume file number
INCARNATION	NUMBER	Volume file incarnation number
DRL_FILE_NUMBER	NUMBER	Volume Dirty Region Logging (DRL) file used for mirrored volumes
RESIZE_UNIT_MB	NUMBER	Volume allocation unit (in megabytes) that a volume can be created
USAGE	VARCHAR2(30)	Optional usage string for the volume
VOLUME_DEVICE	VARCHAR2(256)	OSD path for the volume device
MOUNTPATH	VARCHAR2(1024)	Optional mount path string for the volume
PRIMARY_REGION	VARCHAR2(4)	Region used for allocating primary extents: • HOT • COLD
MIRROR_REGION	VARCHAR2(4)	Region used for allocating mirrored extents: • HOT • COLD
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always 0.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle ASM information

7.82 V$ASM_VOLUME_STAT

In an Oracle Automatic Storage Management (Oracle ASM) instance, `V$ASM_VOLUME_STAT` displays information about statistics for each Oracle ADVM volume.

Column	Datatype	Description
`GROUP_NUMBER`	`NUMBER`	Cluster-wide number assigned to the disk group (composite primary key)
`VOLUME_NAME`	`VARCHAR2(30)`	Name of the volume
`COMPOUND_INDEX`	`NUMBER`	A 32-bit number consisting of a disk group number in the high-order 8 bits and a volume number in the low-order 24 bits (for efficient access to the view)
`VOLUME_NUMBER`	`NUMBER`	Number of the Volume within the disk group (composite primary key)
`READS`	`NUMBER`	Total number of read requests for this volume
`WRITES`	`NUMBER`	Total number of write requests for this volume
`READ_ERRS`	`NUMBER`	Total number of failed read I/O operations for this volume
`WRITE_ERRS`	`NUMBER`	Total number of failed write I/O operations for this volume
`READ_TIME`	`NUMBER`	Total I/O time (in seconds) for read requests for this volume
`WRITE_TIME`	`NUMBER`	Total I/O time (in seconds) for write requests for this volume
`BYTES_READ`	`NUMBER`	Total number of bytes read for this volume
`BYTES_WRITTEN`	`NUMBER`	Total number of bytes written for this volume
`CON_ID`	`NUMBER`	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data For this view, the value is always `0`.

> **See Also**
>
> *Oracle Automatic Storage Management Administrator's Guide* for additional information about using views to display Oracle ASM information

7.83 V$AW_AGGREGATE_OP

`V$AW_AGGREGATE_OP` displays the aggregation operators available in analytic workspaces. You can use this view in an application to provide a list of choices.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NAME	VARCHAR2(14)	Keyword for the aggregation operator
LONGNAME	VARCHAR2(30)	Descriptive name for the operator
DEFAULT_WEIGHT	NUMBER	Default weight factor for weighted operators
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.84 V$AW_ALLOCATE_OP

V$AW_ALLOCATE_OP displays the allocation operators available in analytic workspaces. You can use this view in an application to provide a list of choices.

Column	Datatype	Description
NAME	VARCHAR2(14)	Keyword for the allocation operator
LONGNAME	VARCHAR2(30)	Descriptive name for the operator
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.85 V$AW_CALC

V$AW_CALC reports on the effectiveness of various caches used by Oracle OLAP during dynamic aggregation.

Column	Datatype	Description
SESSION_ID	NUMBER	A unique numeric identifier for the session.
AGGREGATE_CACHE_HITS	NUMBER	The number of times a dimension member is found in the aggregate cache (a hit). The number of hits for run-time aggregation can be increased by fetching data across the dense dimension
AGGREGATE_CACHE_MISSES	NUMBER	The number of times a dimension member is not found in the aggregate cache and must be read from disk (a miss).
SESSION_CACHE_HITS	NUMBER	The number of times the data is found in the session cache (a hit).
SESSION_CACHE_MISSES	NUMBER	The number of times the data is not found in the session cache (a miss).
POOL_HITS	NUMBER	The number of times the data is found in a page in the OLAP page pool (a hit).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
POOL_MISSES	NUMBER	The number of times the data is not found in the OLAP page pool (a miss).
POOL_NEW_PAGES	NUMBER	The number of newly created pages in the OLAP page pool that have not yet been written to the workspace LOB.
POOL_RECLAIMED_PAGES	NUMBER	The number of previously unused pages that have been recycled with new data.
CACHE_WRITES	NUMBER	The number of times the data from the OLAP page pool has been written to the database cache.
POOL_SIZE	NUMBER	The number of kilobytes in the OLAP page pool.
CURR_DML_COMMAND	VARCHAR2(64)	The command currently being executed.
PREV_DML_COMMAND	VARCHAR2(64)	The command most recently completed.
AGGR_FUNC_LOGICAL_NA	NUMBER	The number of times the aggregation engine returns a logical NA because the AGGINDEX option is on and the composite tuple does not exist.
AGGR_FUNC_PRECOMPUTE	NUMBER	The number of times the aggregation engine finds a value in a position that it was called to calculate.
AGGR_FUNC_CALCS	NUMBER	The number of times the aggregation engine calculates a parent value based on the values of its children.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.86 V$AW_LONGOPS

V$AW_LONGOPS displays status information about active SQL cursors initiated in an analytic workspace.

Column	Datatype	Description
SESSION_ID	NUMBER	Identifier for the session in which the fetch is executing. This table can be joined with V$SESSION to obtain the user name.
CURSOR_NAME	VARCHAR2(64)	Name assigned to the cursor
COMMAND	VARCHAR2(17)	Command that is actively fetching data from relational tables: • QUERY • FETCH • IMPORT • EXECUTE • UPDATE • SOLVE • CLEAR • LOAD CUBE • LOAD DIMENSION • DIMENSION COMPILE

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(9)	Status of the current operation: • EXECUTING - Command has begun executing • FETCHING - Data is being fetched into the analytic workspace • FINISHED - Command has finished executing. This status appears very briefly before the record disappears from the table.
ROWS_PROCESSED	NUMBER	Number of rows already inserted, updated, or deleted
SEQ_NUMBER	NUMBER	Sequence number in the Cube Build log
SQL_ID	VARCHAR2(13)	SQL ID of the statement
TARGET	VARCHAR2(64)	Operated object name
TARGET_DESC	VARCHAR2(64)	A brief description of the operated object
START_TIME	DATE	Time the command started executing
LAST_UPDATE_TIME	DATE	Last updated time
ELAPSED_SECONDS	NUMBER	Number of seconds between START_TIME and LAST_UPDATE_TIME
SOFAR	NUMBER	Number of units so far
TOTALWORK	NUMBER	Total number of units
UNITS	VARCHAR2(6)	Units description: • ROWS • NODES • VALUES
MESSAGE	VARCHAR2(512)	Message for the user
USERNAME	VARCHAR2(32)	User name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.87 V$AW_OLAP

V$AW_OLAP provides a record of active sessions and their use with analytic workspaces.

A row is generated whenever an analytic workspace is created or attached. The first row for a session is created when the first command is issued. It identifies the SYS.EXPRESS workspace, which is attached automatically to each session. Rows related to a particular analytic workspace are deleted when the workspace is detached from the session or the session ends.

Column	Datatype	Description
SESSION_ID	NUMBER	A unique numeric identifier for a session
AW_NUMBER	NUMBER	A unique numeric identifier for an analytic workspace. To get the name of the analytic workspace, join this column to the AW_NUMBER column of the USER_AWS view or to the AWSEQ# column of the AW$ table.
ATTACH_MODE	VARCHAR2(10)	READ ONLY or READ WRITE

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
GENERATION	NUMBER	The generation of an analytic workspace. Each UPDATE creates a new generation. Sessions attaching the same workspace between UPDATE commands share the same generation.
TEMP_SPACE_PAGES	NUMBER	The number of pages stored in temporary segments for the analytic workspace.
TEMP_SPACE_READS	NUMBER	The number of times data has been read from a temporary segment and not from the page pool.
LOB_READS	NUMBER	The number of times data has been read from the table where the analytic workspace is stored (the permanent LOB).
POOL_CHANGED_PAGES	NUMBER	The number of pages in the page pool that have been modified in this analytic workspace.
POOL_UNCHANGED_PAGES	NUMBER	The number of pages in the page pool that have not been modified in this analytic workspace.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.88 V$AW_SESSION_INFO

V$AW_SESSION_INFO provides information about each active session.

A transaction is a single exchange between a client session and Oracle OLAP. Multiple commands can execute within a single transaction.

Column	Datatype	Description
SESSION_ID	NUMBER	A unique numeric identifier for a session
CLIENT_TYPE	VARCHAR2(64)	OLAP
SESSION_STATE	VARCHAR2(64)	TRANSACTING, NOT_TRANSACTING, EXCEPTION_HANDLING, CONSTRUCTING, CONSTRUCTED, DECONSTRUCTING, or DECONSTRUCTED
SESSION_HANDLE	NUMBER	The session identifier
USERID	VARCHAR2(64)	The database user name under which the session opened
TOTAL_TRANSACTION	NUMBER	The total number of transactions executed within the session; this number provides a general indication of the level of activity in the session
TRANSACTION_TIME	NUMBER	The elapsed time in milliseconds of the mostly recently completed transaction
TOTAL_TRANSACTION_TIME	NUMBER	The total elapsed time in milliseconds in which transactions were being executed
AVERAGE_TRANSACTION_TIME	NUMBER	The average elapsed time in milliseconds to complete a transaction
TRANSACTION_CPU_TIME	NUMBER	The total CPU time in milliseconds used to complete the most recent transaction

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOTAL_TRANSACTION_CPU_TIME	NUMBER	The total CPU time used to execute all transactions in this session; this total does not include transactions that are currently in progress
AVERAGE_TRANSACTION_CPU_TIME	NUMBER	The average CPU time to complete a transaction; this average does not include transactions that are currently in progress
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.89 V$BACKUP

V$BACKUP displays the backup status of all online data files.

Column	Datatype	Description
FILE#	NUMBER	File identifier
STATUS	VARCHAR2(18)	File status: NOT ACTIVE, ACTIVE (backup in progress), OFFLINE NORMAL, or description of an error. NOT ACTIVE indicates that the file is not currently in backup mode (that is, an ALTER TABLESPACE ... BEGIN BACKUP or ALTER DATABASE BEGIN BACKUP statement has not been issued), whereas ACTIVE indicates that the file is currently in backup mode.
CHANGE#	NUMBER	System change number when backup started
TIME	DATE	Time the backup started
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.90 V$BACKUP_ARCHIVELOG_DETAILS

V$BACKUP_ARCHIVELOG_DETAILS contains information about all restorable archive logs.

It will include all archived logs backed up in a backup set or proxy copies.

Column	Datatype	Description
BTYPE	CHAR(9)	Backup type container, either BACKUPSET or PROXYCOPY
BTYPE_KEY	NUMBER	Unique identifier for the backup type. For BACKUPSET, it is BS_KEY.
SESSION_KEY	NUMBER	Session identifier
SESSION_RECID	NUMBER	Session recid

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SESSION_STAMP	NUMBER	Session stamp
ID1	NUMBER	If BACKUPSET, it contains SET_STAMP If PROXYCOPY, it is RECID from the control file
ID2	NUMBER	If BACKUPSET, it contains SET_COUNT If PROXYCOPY, it is STAMP
THREAD#	NUMBER	Thread number
SEQUENCE#	NUMBER	Sequence number
RESETLOGS_CHANGE#	NUMBER	Resetlogs change SCN
RESETLOGS_TIME	DATE	Resetlogs change time
FIRST_CHANGE#	NUMBER	First change SCN
FIRST_TIME	DATE	First change time
NEXT_CHANGE#	NUMBER	Next change SCN
NEXT_TIME	DATE	Next change time
FILESIZE	NUMBER	File size
COMPRESSION_RATIO	NUMBER	The ratio between the total blocks in the archive log and the blocks that RMAN backed up. This is *not* the ratio from the AS COMPRESSED BACKUPSET clause of the BACKUP command.
FILESIZE_DISPLAY	VARCHAR2(4000)	Same value as the FILESIZE column, but converted to a user-displayable format, for example *n*M, *n*G, *n*T, *n*P, and so on
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.91 V$BACKUP_ARCHIVELOG_SUMMARY

V$BACKUP_ARCHIVELOG_SUMMARY provides archive log summary information based on archive logs in the backup set or on proxy copies.

Column	Datatype	Description
NUM_FILES_BACKED	NUMBER	Number of files backed up
NUM_DISTINCT_FILES_BACKED	NUMBER	Number of distinct archive log files backed up
MIN_FIRST_CHANGE#	NUMBER	Lowest SCN range value
MAX_NEXT_CHANGE#	NUMBER	Highest SCN range value
MIN_FIRST_TIME	DATE	Lowest SCN range time
MAX_NEXT_TIME	DATE	Highest SCN range time
INPUT_BYTES	NUMBER	Total input bytes read
OUTPUT_BYTES	NUMBER	Output size of backups

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
COMPRESSION_RATIO	NUMBER	The ratio between the total blocks in the archive log and the blocks that RMAN backed up. This is *not* the ratio from the AS COMPRESSED BACKUPSET clause of the BACKUP command.
INPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for input bytes
OUTPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for output bytes
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.92 V$BACKUP_ASYNC_IO

V$BACKUP_ASYNC_IO displays performance information about ongoing and recently completed RMAN backups and restores.

For each backup, it contains one row for each input data file, one row for the aggregate total performance of all data files, and one row for the output backup piece. This data is not stored persistently, and is not preserved when the instance is re-started.

Column	Datatype	Description
SID	NUMBER	Oracle SID of the session doing the backup or restore
SERIAL	NUMBER	Use count for the SID doing the backup or restore
USE_COUNT	NUMBER	A counter that can be used to identify rows from different backup sets
RMAN_STATUS_RECID	NUMBER	Owning V$RMAN_STATUS record ID
RMAN_STATUS_STAMP	NUMBER	Owning V$RMAN_STATUS record stamp
DEVICE_TYPE	VARCHAR2(17)	Device type where the file is located
TYPE	VARCHAR2(9)	INPUT, OUTPUT, or AGGREGATE
STATUS	VARCHAR2(11)	NOT STARTED, IN PROGRESS, or FINISHED
FILENAME	VARCHAR2(513)	Name of the backup file being read or written
SET_COUNT	NUMBER	Set count of the backup set being read or written
SET_STAMP	NUMBER	Set stamp of the backup set being read or written
BUFFER_SIZE	NUMBER	Size of the buffers being used to read/write the file, in bytes
BUFFER_COUNT	NUMBER	Number of buffers being used to read/write the file
TOTAL_BYTES	NUMBER	Total number of bytes that will be read or written for the file, if known. If not known, this column will be null
OPEN_TIME	DATE	Time the file was opened. If TYPE='AGGREGATE', then this is the time that the first file in the aggregate was opened
CLOSE_TIME	DATE	Time the file was closed. If TYPE='AGGREGATE', then this is the time that the last file in the aggregate was closed
ELAPSED_TIME	NUMBER	Time, in hundredths of a second, that the file was open

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MAXOPENFILES	NUMBER	Number of concurrently open DISK files. This value is only present in rows where TYPE='AGGREGATE'.
BYTES	NUMBER	Number of bytes read or written so far
EFFECTIVE_BYTES_PER_SECOND	NUMBER	I/O rate that was achieved with this device during this backup
IO_COUNT	NUMBER	Number of I/Os that were performed to this file
READY	NUMBER	Number of asynchronous requests for which a buffer was immediately ready for use
SHORT_WAITS	NUMBER	Number of times that a buffer was not immediately available, but a buffer became available after doing a nonblocking poll for I/O completion
SHORT_WAIT_TIME_TOTAL	NUMBER	Total time, in hundredths of a second, taken by nonblocking polls for I/O completion
SHORT_WAIT_TIME_MAX	NUMBER	Maximum time taken for a nonblocking poll for I/O completion, in hundredths of a second
LONG_WAITS	NUMBER	The number of times that a buffer was not immediately available, and only became available after a blocking wait was issued
LONG_WAIT_TIME_TOTAL	NUMBER	The total time, in hundredths of a second, taken by blocking waits for I/O completion
LONG_WAIT_TIME_MAX	NUMBER	The maximum time taken for a blocking wait for I/O completion, in hundredths of a second
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.93 V$BACKUP_CONTROLFILE_DETAILS

V$BACKUP_CONTROLFILE_DETAILS contains information about restorable control files.

It will include all the control files backed up in the backup set, image copies, and proxy copies.

Column	Datatype	Description
BTYPE	CHAR(9)	Backup type container. Possible values are BACKUPSET, IMAGECOPY, PROXYCOPY.
BTYPE_KEY	NUMBER	Unique identifier for the backup type, either BS_KEY or COPY_KEY.
SESSION_KEY	NUMBER	Session identifier
SESSION_RECID	NUMBER	Session recid
SESSION_STAMP	NUMBER	Session stamp
ID1	NUMBER	If BACKUPSET, it contains SET_STAMP If IMAGECOPY or PROXYCOPY, it is RECID from the control file
ID2	NUMBER	If BACKUPSET, it contains SET_COUNT If IMAGECOPY or PROXYCOPY, it is STAMP

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CREATION_TIME	DATE	File creation time
RESETLOGS_CHANGE#	NUMBER	Resetlogs change SCN
RESETLOGS_TIME	DATE	Resetlogs change time
CHECKPOINT_CHANGE#	NUMBER	Checkpoint change SCN
CHECKPOINT_TIME	DATE	Checkpoint change time
FILESIZE	NUMBER	File size, in bytes, for the output of backing up this control file
COMPRESSION_RATIO	NUMBER	The ratio between the total blocks in the datafile and the blocks that RMAN backed up. This is *not* the the ratio from the AS COMPRESSED BACKUPSET clause of the BACKUP command.
FILESIZE_DISPLAY	VARCHAR2(4000)	Same value as the FILESIZE column, but converted to a user-displayable format, for example *n*M, *n*G, *n*T, *n*P, and so on
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.94 V$BACKUP_CONTROLFILE_SUMMARY

V$BACKUP_CONTROLFILE_SUMMARY provides control file summary information, based on either a backup set of files, image copies, or proxy copies.

Column	Datatype	Description
NUM_FILES_BACKED	NUMBER	Number of files backed up for specific criteria
NUM_DISTINCT_FILES_BACKED	NUMBER	Number of distinct files backed up
MIN_CHECKPOINT_CHANGE#	NUMBER	Minimum checkpoint change number of the data file for specified criteria
MAX_CHECKPOINT_CHANGE#	NUMBER	Maximum checkpoint change number of the data file for specified criteria
MIN_CHECKPOINT_TIME	DATE	Minimum checkpoint time of the data file for specified criteria
MAX_CHECKPOINT_TIME	DATE	Maximum checkpoint time of the data file for specified criteria
INPUT_BYTES	NUMBER	Total input bytes of files read
OUTPUT_BYTES	NUMBER	Total output bytes written
COMPRESSION_RATIO	NUMBER	The ratio between the total blocks in the datafile and the blocks that RMAN backed up. This is *not* the the ratio from the AS COMPRESSED BACKUPSET clause of the BACKUP command.
INPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for input bytes
OUTPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for output bytes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.95 V$BACKUP_COPY_DETAILS

V$BACKUP_COPY_DETAILS contains information about all available control file and data file copies.

Column	Datatype	Description
SESSION_KEY	NUMBER	Session identifier
SESSION_RECID	NUMBER	Session recid
SESSION_STAMP	NUMBER	Session stamp
COPY_KEY	NUMBER	Unique identifier for this data file or control file copy
FILE#	NUMBER	Absolute data file number
NAME	VARCHAR2(513)	File name of the data file copy. The maximum length of the name is dependent on your operating system.
TAG	VARCHAR2(32)	Data file copy tag
CREATION_CHANGE#	NUMBER	Data file creation change number
CREATION_TIME	DATE	Data file creation timestamp
CHECKPOINT_CHANGE#	NUMBER	Checkpoint change number of the data file when the copy was made
CHECKPOINT_TIME	DATE	Checkpoint timestamp of the data file when the copy was made
MARKED_CORRUPT	NUMBER	Number of blocks marked corrupt by this copy operation. That is, blocks that were not marked corrupted in the source data file, but were detected and marked as corrupted during the copy operation.
OUTPUT_BYTES	NUMBER	Total output bytes written
COMPLETION_TIME	DATE	Time when the copy was completed
CONTROLFILE_TYPE	VARCHAR2(1)	Type of control file. B indicates normal copies. S indicates standby copies.
KEEP	VARCHAR2(3)	(YES|NO) Indicates whether or not this backup set has a retention policy that is different than the value for the configure retention policy
KEEP_UNTIL	DATE	If specified, this is the date after which the backup becomes obsolete. If this column is null, then the backup never expires.
KEEP_OPTIONS	VARCHAR2(11)	Lists additional retention options for this backup set. Possible values are: • LOGS - The logs needed to recover this backup set are kept • NOLOGS - The logs needed to recover this backup set are not kept
IS_RECOVERY_DEST_FILE	VARCHAR2(3)	Indicates whether the file was created in the fast recovery area (YES) or not (NO)
SPARSE_BACKUP	VARCHAR2(3)	Indicates whether the file is sparse (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OUTPUT_BYTES_DISPLAY	VARCHAR2(4000)	Size of backup set to display
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.96 V$BACKUP_COPY_SUMMARY

V$BACKUP_COPY_SUMMARY provides summary information for the output data file and control file copy.

Column	Datatype	Description
NUM_COPIES	NUMBER	Number of copies created
NUM_DISTINCT_COPIES	NUMBER	Number of distinct copies (that contain data files with different checkpoints)
MIN_CHECKPOINT_CHANGE#	NUMBER	Minimum checkpoint change SCN
MAX_CHECKPOINT_CHANGE#	NUMBER	Maximum checkpoint change SCN
MIN_CHECKPOINT_TIME	DATE	Minimum checkpoint change time
MAX_CHECKPOINT_TIME	DATE	Maximum checkpoint change time
OUTPUT_BYTES	NUMBER	Total number of output bytes
OUTPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for output bytes
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.97 V$BACKUP_CORRUPTION

V$BACKUP_CORRUPTION displays information about corrupt block ranges in data file backups from the control file.

Note that corruptions are not tolerated in the control file and archived redo log backups.

Column	Datatype	Description
RECID	NUMBER	Backup corruption record ID
STAMP	NUMBER	Backup corruption record stamp
SET_STAMP	NUMBER	Backup set stamp
SET_COUNT	NUMBER	Backup set count

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PIECE#	NUMBER	backup piece that contains this corrupt block
FILE#	NUMBER	Absolute file number of the data file that contains the corrupt blocks
BLOCK#	NUMBER	Block number of the first corrupt block in the range of corrupted blocks
BLOCKS	NUMBER	Number of corrupted blocks found starting with BLOCK#
CORRUPTION_CHANGE#	NUMBER	Change number at which the logical corruption was detected. Set to 0 to indicate media corruption.
MARKED_CORRUPT	VARCHAR2(3)	Indicates whether this corruption was not previously detected by the Oracle Database (YES) or the Oracle Database had already discovered this corrupt block and marked it as corrupt (NO). Note that when a corrupt block is encountered in a backup, and was not already marked corrupt by the Oracle Database, then the backup process does not mark the block as corrupt in the production data file. Thus, this field may be YES for the same block in more than one backup set.
CORRUPTION_TYPE	VARCHAR2(9)	Type of block corruption in the data file: • ALL ZERO - Block header on disk contained only zeros. The block may be valid if it was never filled and if it is in an Oracle7 file. The buffer will be reformatted to the Oracle8 standard for an empty block. • FRACTURED - Block header looks reasonable, but the front and back of the block are different versions. • CHECKSUM - optional check value shows that the block is not self-consistent. It is impossible to determine exactly why the check value fails, but it probably fails because sectors in the middle of the block are from different versions. • CORRUPT - Block is wrongly identified or is not a data block (for example, the data block address is missing) • LOGICAL - Block is logically corrupt • NOLOGGING - Block does not have redo log entries (for example, NOLOGGING operations on primary database can introduce this type of corruption on a physical standby)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.98 V$BACKUP_DATAFILE

V$BACKUP_DATAFILE displays information about control files and data files in backup sets from the control file.

Column	Datatype	Description
RECID	NUMBER	Backup data file record ID
STAMP	NUMBER	Backup data file record stamp
SET_STAMP	NUMBER	Backup set stamp
SET_COUNT	NUMBER	Backup set count

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FILE#	NUMBER	Data file number; set to 0 for control file
CREATION_CHANGE#	NUMBER	Creation system change number (SCN) of the data file
CREATION_TIME	DATE	Creation timestamp of the data file
RESETLOGS_CHANGE#	NUMBER	Resetlogs system change number (SCN) of the data file when it was backed up
RESETLOGS_TIME	DATE	Resetlogs timestamp of the data file when it was backed up
INCREMENTAL_LEVEL	NUMBER	Normal full backups have a NULL value, level 0 incremental backups have a value of 0, and level 1 incremental backups have a value of 1
INCREMENTAL_CHANGE#	NUMBER	All blocks changed after the incremental change number is included in this backup; set to 0 for a full backup
CHECKPOINT_CHANGE#	NUMBER	All changes up to the checkpoint change number are included in this backup
CHECKPOINT_TIME	DATE	Timestamp of the checkpoint
ABSOLUTE_FUZZY_CHANGE#	NUMBER	Highest change number in this backup
MARKED_CORRUPT	NUMBER	Number of blocks marked corrupt
MEDIA_CORRUPT	NUMBER	Number of blocks media corrupt
LOGICALLY_CORRUPT	NUMBER	Number of blocks logically corrupt
DATAFILE_BLOCKS	NUMBER	Size of the data file in blocks at backup time. This value is also the number of blocks taken by the data file restarted from this backup.
BLOCKS	NUMBER	Size of the backup data file (in blocks). Unused blocks are not copied to the backup.
BLOCK_SIZE	NUMBER	Block size
OLDEST_OFFLINE_RANGE	NUMBER	RECID of the oldest offline range record in this backup control file. 0 for data file backups.
COMPLETION_TIME	DATE	Time completed
CONTROLFILE_TYPE	VARCHAR2(1)	B - Normal copies S - Standby copies
USED_CHANGE_TRACKING	VARCHAR2(3)	Indicates whether change tracking data was used to accelerate this incremental backup (YES) or whether change tracking data was not used (NO)
BLOCKS_READ	NUMBER	Number of blocks that were scanned while taking this backup. If this was an incremental backup, and change tracking was used to optimize the backup, then the value of this column will be smaller than DATAFILE_BLOCKS. Otherwise, the value of this column will be the same as DATAFILE_BLOCKS. Even when change tracking data is used, the value of this column may be larger than BLOCKS, because the data read by change tracking is further refined during the process of creating an incremental backup.
USED_OPTIMIZATION	VARCHAR2(3)	Indicates whether backup optimization was applied (YES) or not (NO)
FOREIGN_DBID	NUMBER	Foreign DBID of the database from which this data file was transported. The value is 0 if the file backed up is not a foreign database file.
PLUGGED_READONLY	VARCHAR2(3)	YES if this is a backup of a transported read-only foreign file; otherwise NO.
PLUGIN_CHANGE#	NUMBER	SCN at which the foreign data file was transported into the database. The value is 0 if this file is not a foreign database file.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PLUGIN_RESETLOGS_CHANGE#	NUMBER	The SCN of the RESETLOGS operation for the incarnation into which this foreign file was transported. The value is 0 if this file is not a foreign database file.
PLUGIN_RESETLOGS_TIME	DATE	The time of the RESETLOGS operation for the incarnation into which this foreign file was transported. The value is 0 if this file is not a foreign database file.
SECTION_SIZE	NUMBER	Specifies the number of blocks in each section of a multisection backup. Value is 0 for whole file backups.
UNDO_OPTIMIZED	VARCHAR2(3)	Indicates whether undo blocks were ignored when creating the backup data file (YES) or not (NO)
BLOCKS_SKIPPED_IN_CELL	NUMBER	Number of blocks that were not backed up because they were skipped by the Exadata cell **See Also:** Oracle Exadata Storage Server Software documentation for more information
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
BACKED_BY_PDB	VARCHAR2(3)	Recovery Manager (RMAN) allows a PDB to be backed up in two ways. The value in this column indicates how the PDB backup was taken: • YES: The backup was taken when connected to the PDB • NO: The backup was taken when connected to the root container
SPARSE_BACKUP	VARCHAR2(3)	Indicates whether the file is sparse (YES) or not (NO)
GUID	RAW(16)	The GUID of the PDB to which the backup belongs. This is useful after the PDB is dropped to identify which PDB the backup belongs to.

7.99 V$BACKUP_DATAFILE_DETAILS

V$BACKUP_DATAFILE_DETAILS contains information about restorable data files.

It will include all data files backed in the backup set, image copies, and proxy copies.

Column	Datatype	Description
BTYPE	CHAR(9)	Backup type container. Possible values are: BACKUPSET, IMAGECOPY, PROXYCOPY.
BTYPE_KEY	NUMBER	Unique identifier for the backup type. For BACKUPSET, it is BS_KEY.
SESSION_KEY	NUMBER	Session identifier
SESSION_RECID	NUMBER	Session record ID
SESSION_STAMP	NUMBER	Session stamp
ID1	NUMBER	If BACKUPSET, it contains SET_STAMP. If IMAGECOPY or PROXYCOPY, it is RECID from the control file.
ID2	NUMBER	If BACKUPSET, it contains SET_COUNT. If IMAGECOPY or PROXYCOPY, it is STAMP.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FILE#	NUMBER	File number
CREATION_CHANGE#	NUMBER	File creation change SCN
CREATION_TIME	DATE	File creation time
RESETLOGS_CHANGE#	NUMBER	Resetlogs change SCN
RESETLOGS_TIME	DATE	Resetlogs change time
INCREMENTAL_LEVEL	NUMBER	Normal full backups have a NULL value, level 0 incremental backups have a value of 0, and level 1 incremental backups have a value of 1
INCREMENTAL_CHANGE#	NUMBER	Incremental change SCN
CHECKPOINT_CHANGE#	NUMBER	Checkpoint change SCN
CHECKPOINT_TIME	DATE	Checkpoint change time
MARKED_CORRUPT	NUMBER	Number of blocks marked corrupt
FILESIZE	NUMBER	File size, in bytes
COMPRESSION_RATIO	NUMBER	The ratio between the total blocks in the datafile and the blocks that RMAN backed up. This is *not* the the ratio from the AS COMPRESSED BACKUPSET clause of the BACKUP command.
SPARSE_BACKUP	VARCHAR2(3)	Indicates whether the file is sparse (YES) or not (NO)
TS#	NUMBER	Tablespace number
TSNAME	VARCHAR2(30)	Tablespace name
FILESIZE_DISPLAY	VARCHAR2(4000)	Same value as the FILESIZE column, but converted to a user-displayable format, for example *n*M, *n*G, *n*T, *n*P, and so on
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.100 V$BACKUP_DATAFILE_SUMMARY

V$BACKUP_DATAFILE_SUMMARY provides summary information for a specific criteria set, based on a backup job, a time range applicable to jobs, or a specific data file).

Column	Datatype	Description
NUM_FILES_BACKED	NUMBER	Number of files backed up for specified criteria
NUM_DISTINCT_FILES_BACKED	NUMBER	Number of distinct files backed up
NUM_DISTINCT_TS_BACKED	NUMBER	Number of distinct tablespaces backed up
MIN_CHECKPOINT_CHANGE#	NUMBER	Minimum checkpoint change number of the data file for specified criteria
MAX_CHECKPOINT_CHANGE#	NUMBER	Maximum checkpoint change number of the data file for specified criteria
MIN_CHECKPOINT_TIME	DATE	Minimum checkpoint time of the data file for specified criteria
MAX_CHECKPOINT_TIME	DATE	Maximum checkpoint time of the data file for specified criteria

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
INPUT_BYTES	NUMBER	Total input bytes of files read
OUTPUT_BYTES	NUMBER	Total output bytes written
COMPRESSION_RATIO	NUMBER	The ratio between the total blocks in the datafile and the blocks that RMAN backed up. This is *not* the the ratio from the AS COMPRESSED BACKUPSET clause of the BACKUP command.
INPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for input bytes
OUTPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for output bytes
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.101 V$BACKUP_DEVICE

V$BACKUP_DEVICE displays information about supported backup devices.

If a device type does not support named devices, then one row with the device type and a null device name is returned for that device type. If a device type supports named devices then one row is returned for each available device of that type. The special device type DISK is not returned by this view because it is always available.

Column	Datatype	Description
DEVICE_TYPE	VARCHAR2(17)	Type of the backup device
DEVICE_NAME	VARCHAR2(513)	Name of the backup device
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.102 V$BACKUP_FILES

V$BACKUP_FILES displays information about all RMAN backups (both image copies and backup sets) and archived logs.

This view simulates the LIST BACKUP and LIST COPY RMAN commands. This view requires that the database be set using the DBMS_RCVMAN.SETDATABASE procedure.

Column	Datatype	Description
PKEY	NUMBER	Primary key for the backup

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BACKUP_TYPE	VARCHAR2(32)	Type of the backup: • BACKUP SET • COPY • PROXY COPY
FILE_TYPE	VARCHAR2(32)	Type of the file: • DATAFILE • CONTROLFILE • SPFILE • REDO LOG • ARCHIVED LOG • COPY (for an image copy backup) • PIECE (for a backup piece)
KEEP	VARCHAR2(3)	Indicates whether the backup has a retention policy different from the value for CONFIGURE RETENTION POLICY (YES) or not (NO)
KEEP_UNTIL	DATE	If the KEEP UNTIL TIME clause of the BACKUP command was specified, then this column shows the date after which the backup becomes obsolete. If the column is null and KEEP_OPTIONS is not null, the backup never becomes obsolete.
KEEP_OPTIONS	VARCHAR2(13)	KEEP options for the backup: • LOGS - RMAN keeps the logs needed to recover the backup • NOLOGS - RMAN does not keep the logs needed to recover the backup If this column is null, then the backup has no KEEP options and will be made obsolete based on the retention policy.
STATUS	VARCHAR2(16)	Status of the backup: • AVAILABLE • UNAVAILABLE • EXPIRED • OTHER
FNAME	VARCHAR2(1024)	Name of the file
TAG	VARCHAR2(32)	Tag of the piece, copy, or proxy copy
MEDIA	VARCHAR2(80)	Media ID of the piece or proxy copy
RECID	NUMBER	Recid of the record in the control file
STAMP	NUMBER	Stamp of the record in the control file
DEVICE_TYPE	VARCHAR2(255)	Type of media device that stores the backup
BLOCK_SIZE	NUMBER	Block size for the backup (in bytes)
COMPLETION_TIME	DATE	Time when the backup completed
COMPRESSED	VARCHAR2(3)	Indicates whether the backup piece is compressed (YES) or not (NO); valid only if FILE_TYPE is PIECE. Image copies cannot be compressed.
OBSOLETE	VARCHAR2(3)	Indicates whether the backup piece or copy is obsolete (YES) or not (NO); valid only if FILE_TYPE is PIECE or COPY
BYTES	NUMBER	Size of the file (in bytes)
BS_KEY	NUMBER	Primary key of the backup set (valid only if BACKUP_TYPE is BACKUP SET)
BS_COUNT	NUMBER	Count of the backup set from the control file record (valid only if BACKUP_TYPE is BACKUP SET)
BS_STAMP	NUMBER	Stamp of the backup set from the control file record (valid only if BACKUP_TYPE is BACKUP SET)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BS_TYPE	VARCHAR2(32)	Type of the backup set (valid only if BACKUP_TYPE is BACKUP SET): • DATAFILE • ARCHIVED LOG
BS_INCR_TYPE	VARCHAR2(32)	Incremental type of the backup set (valid only if BACKUP_TYPE is BACKUP SET)
BS_PIECES	NUMBER	Number of backup pieces in the backup set (valid only if BACKUP_TYPE is BACKUP SET)
BS_COPIES	NUMBER	Number of copies of the backup set (valid only if FILE_TYPE is PIECE and BACKUP_TYPE is BACKUP SET)
BS_COMPLETION_TIME	DATE	Completion time of the backup set (valid only if BACKUP_TYPE is BACKUP SET)
BS_STATUS	VARCHAR2(16)	Status of the backup set (valid only if BACKUP_TYPE is BACKUP SET): • AVAILABLE • UNAVAILABLE • EXPIRED • OTHER - Pieces which are part of the backup set do not have uniform status (that is, some of them are available, some not)
BS_BYTES	NUMBER	Sum of all backup piece sizes in the backup set (valid only if BACKUP_TYPE is BACKUP SET)
BS_COMPRESSED	VARCHAR2(3)	Indicates whether the backup pieces of the backup set are compressed (YES) or not (NO); valid only if BACKUP_TYPE is BACKUP SET
BS_TAG	VARCHAR2(1024)	Tags of the backup set. If pieces have different tags, then all piece tags are concatenated and separated by commas. Valid only if BACKUP_TYPE is BACKUP SET)
BS_DEVICE_TYPE	VARCHAR2(255)	Device type of the backup set. If there is more than one device type, then they are separated by commas. Valid only if BACKUP_TYPE is BACKUP SET
BP_PIECE#	NUMBER	Number of pieces inside the backup set (valid only if FILE_TYPE is PIECE and BACKUP_TYPE is BACKUP SET)
BP_COPY#	NUMBER	Number of copies of the backup set (valid only if FILE_TYPE is PIECE and BACKUP_TYPE is BACKUP SET)
DF_FILE#	NUMBER	Absolute file number of the data file (valid only if FILE_TYPE is DATAFILE)
DF_TABLESPACE	VARCHAR2(30)	Tablespace name of the data file (valid only if FILE_TYPE is DATAFILE)
DF_RESETLOGS_CHANGE#	NUMBER	System change number (SCN) of the most recent RESETLOGS when the control file or data file was created (valid only if FILE_TYPE is DATAFILE)
DF_CREATION_CHANGE#	NUMBER	Creation SCN of the control file or data file (valid only if FILE_TYPE is CONTROLFILE or DATAFILE)
DF_CHECKPOINT_CHANGE#	NUMBER	System change number (SCN) of the most recent control file or data file checkpoint (valid only if FILE_TYPE is CONTROLFILE or DATAFILE)
DF_CKP_MOD_TIME	DATE	Modification time in case of SPFILE, otherwise time when the control file or data file was checkpointed (valid only if FILE_TYPE is SPFILE, CONTROLFILE, or DATAFILE)
RL_THREAD#	NUMBER	Redo log thread number of the archived log (valid only if FILE_TYPE is REDO LOG)
RL_SEQUENCE#	NUMBER	Redo log sequence number of the archived log (valid only if FILE_TYPE is REDO LOG)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RL_RESETLOGS_CHANGE#	NUMBER	System change number (SCN) of the most recent RESETLOGS when the record was created (valid only if FILE_TYPE is REDO LOG)
RL_FIRST_CHANGE#	NUMBER	First SCN of the redo log (valid only if FILE_TYPE is REDO LOG)
RL_FIRST_TIME	DATE	Time when the Oracle Database switched into the redo log (valid only if FILE_TYPE is REDO LOG)
RL_NEXT_CHANGE#	NUMBER	First SCN of the next redo log in the thread (valid only if FILE_TYPE is REDO LOG)
RL_NEXT_TIME	DATE	First timestamp of the next redo log in the thread (valid only if FILE_TYPE is REDO LOG)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Backup and Recovery User's Guide for more information about the DBMS_RCVMAN.SETDATABASE procedure

7.103 V$BACKUP_NONLOGGED

V$BACKUP_NONLOGGED displays information about nonlogged block ranges in data file backups, recorded in the control file.

Column	Datatype	Description
RECID	NUMBER	Nologged backup record ID
STAMP	NUMBER	Nonlogged backup record stamp
SET_STAMP	NUMBER	Backup set stamp
SET_COUNT	NUMBER	Backup set count
PIECE#	NUMBER	Backup piece that contains the range of nonlogged blocks
FILE#	NUMBER	Absolute file number of the data file that contains this range of nonlogged blocks
BLOCK#	NUMBER	Block number of the first nonlogged block in the range of nologged blocks
BLOCKS	NUMBER	Number of nonlogged blocks found starting with BLOCK#
NONLOGGED_CHANGE#	NUMBER	The smallest SCN on which any block in this block range became nonlogged. NULL if unknown.
NONLOGGED_TIME	VARCHAR2	The time that corresponds to NONLOGGED_CHANGE#. NULL if unknown.
RESETLOGS_CHANGE#	VARCHAR2	The resetlogs SCN of the incarnation on which this block range was first marked as nonlogged. NULL if unknown.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RESETLOGS_TIME	VARCHAR2	The resetlogs time of the incarnation on which this block range was first marked as nologged. NULL if unknown.
OBJECT#	VARCHAR2	The object ID this range belongs to. If this field is NULL, the object number is unknown.
REASON	CHAR(7)	The reason why this block range appears in this list, for example, primary file offline, could not talk to primary, non-standby recovery, and so on. For Oracle Database 12*c* and later releases, it is always UNKNOWN.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.104 V$BACKUP_PIECE

V$BACKUP_PIECE displays information about backup pieces from the control file. Each backup set consists of one or more backup pieces.

Column	Datatype	Description
RECID	NUMBER	Backup piece record ID
STAMP	NUMBER	Backup piece record stamp
SET_STAMP	NUMBER	Backup set stamp
SET_COUNT	NUMBER	Backup set count
PIECE#	NUMBER	Backup piece number (1-N)
COPY#	NUMBER	Indicates the copy number for backup pieces created with duplex enabled. 1 if the backup piece is not duplexed.
DEVICE_TYPE	VARCHAR2(17)	Type of the device on which the backup piece resides. Set to DISK for backup sets on disk. **See Also:** V$BACKUP_DEVICE
HANDLE	VARCHAR2(513)	Backup piece handle identifies the backup piece on restore
COMMENTS	VARCHAR2(64)	Comment returned by the operating system or storage subsystem. Set to NULL for backup pieces on disk. This value is informational only; not needed for restore.
MEDIA	VARCHAR2(65)	Name of the media on which the backup piece resides. This value is informational only; not needed for restore.
MEDIA_POOL	NUMBER	The media pool in which the copy resides. This is the same value that was entered in the POOL operand of the Recovery Manager BACKUP command.
CONCUR	VARCHAR2(3)	(YES | NO) Indicates whether the piece on a media that can be accessed concurrently
TAG	VARCHAR2(32)	Backup piece tag. The tag is specified at backup set level, but stored at piece level.
STATUS	VARCHAR2(1)	Indicates the status of the piece: A (available), D (deleted), or X (expired)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
START_TIME	DATE	Starting time
COMPLETION_TIME	DATE	Completion time
ELAPSED_SECONDS	NUMBER	Number of elapsed seconds
DELETED	VARCHAR2(3)	(YES/NO) NO indicates that the file still exists. YES indicates the file no longer exists because it has been deleted.
BYTES	NUMBER	Size of the backup piece (in bytes)
IS_RECOVERY_DEST_FILE	VARCHAR2(3)	Indicates whether the file was created in the fast recovery area (YES) or not (NO)
RMAN_STATUS_RECID	NUMBER	Owning V$RMAN_STATUS record ID
RMAN_STATUS_STAMP	NUMBER	Owning V$RMAN_STATUS record stamp
COMPRESSED	VARCHAR2(3)	Indicates whether the backup piece is compressed (YES) or not (NO)
BACKED_BY_VSS	VARCHAR2(3)	Whether or not the file has been backed up by Volume Shadow Copy Service (VSS). This column is reserved for internal use.
ENCRYPTED	VARCHAR2(3)	A value of YES means an encrypted backup, otherwise not an encrypted backup.
BACKED_BY_OSB	VARCHAR2(3)	A value of YES means the backup was done to Oracle Secure Backup. Otherwise, backed up by other third party tape library.
FOR_XTTS	VARCHAR2(3)	Indicates whether this is a cross platform backup piece: • YES: This is a cross platform backup piece. • NO: This is not a cross platform backup piece.
SAME_ENDIAN	VARCHAR2(3)	If the value of FOR_XTTS is YES, then this column indicates whether the backup piece has the same endianess as the current database (YES) or not (NO); otherwise NULL.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
GUID	RAW(16)	The GUID of the PDB to which the backup belongs. This is useful after the PDB is dropped to identify which PDB the backup belongs to.

7.105 V$BACKUP_PIECE_DETAILS

V$BACKUP_PIECE_DETAILS displays information about all available backup pieces.

Column	Datatype	Description
SESSION_KEY	NUMBER	Session identifier
SESSION_RECID	NUMBER	Session recid
SESSION_STAMP	NUMBER	Session stamp
BS_KEY	NUMBER	Backup set identifier
BP_KEY	NUMBER	Backup piece key
RECID	NUMBER	Backup piece record ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STAMP	NUMBER	Backup piece record stamp
SET_STAMP	NUMBER	Backup set stamp
SET_COUNT	NUMBER	Backup set count
PIECE#	NUMBER	Backup piece number (1-N)
COPY#	NUMBER	Indicates the copy number for backup pieces created with duplex enabled. The value is 1 if the backup piece is not duplexed.
DEVICE_TYPE	VARCHAR2(17)	Type of device on which the backup piece resides. Set to DISK for backup sets on disk.
HANDLE	VARCHAR2(513)	Backup piece handle identifies the backup piece on restore
COMMENTS	VARCHAR2(64)	Comments returned by the operating system or storage subsystem. Set to NULL for backup pieces on disk. This value is informational only. It is not needed for restore.
MEDIA	VARCHAR2(65)	Name of the media on which the backup piece resides. This value is informational only. It is not needed for restore.
MEDIA_POOL	NUMBER	The media pool in which the copy resides. This is the same value that was entered in the POOL operand of the Recovery Manager BACKUP command.
CONCUR	VARCHAR2(3)	(YES|NO) indicates whether or not the piece is on a media that can be accessed concurrently
TAG	VARCHAR2(32)	Backup piece tag. The tag is specified at backup set level, but stored at piece level.
STATUS	VARCHAR2(1)	Indicates the status of the piece: A (available), D (deleted), or X (expired)
START_TIME	DATE	Starting time
COMPLETION_TIME	DATE	Completion time
ELAPSED_SECONDS	NUMBER	Number of elapsed seconds
DELETED	VARCHAR2(3)	NO indicates that the file still exists. YES indicates that the file no longer exists because it has been deleted.
BYTES	NUMBER	Size of the backup piece, in bytes
IS_RECOVERY_DEST_FILE	VARCHAR2(3)	Indicates whether or not the file was created in the fast recovery area (YES) or not (NO)
RMAN_STATUS_RECID	NUMBER	Owning V$RMAN_STATUS record ID
RMAN_STATUS_STAMP	NUMBER	Owning V$RMAN_STATUS record stamp
COMPRESSED	VARCHAR2(3)	Indicates whether the backup piece is compressed (YES) or not (NO)
BACKED_BY_VSS	VARCHAR2(3)	Whether or not the file has been backed up by Volume Shadow Copy Service (VSS). This column is reserved for internal use.
ENCRYPTED	VARCHAR2(3)	A value of YES means an encrypted backup, otherwise not an encrypted backup.
BACKED_BY_OSB	VARCHAR2(3)	A value of YES means the backup was done to Oracle Secure Backup. Otherwise, backed up by other third party tape library.
FOR_XTTS	VARCHAR2(3)	Indicates whether this is a cross platform backup piece: • YES: This is a cross platform backup piece. • NO: This is not a cross platform backup piece.
SAME_ENDIAN	VARCHAR2(3)	If the value of FOR_XTTS is YES, then this column indicates whether the backup piece has the same endianess as the current database (YES) or not (NO); otherwise NULL.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
PIECES_PER_SET	NUMBER	Number of backup pieces per set
SIZE_BYTES_DISPLAY	VARCHAR2(4000)	Size (in bytes) of the backup piece to display

7.106 V$BACKUP_REDOLOG

V$BACKUP_REDOLOG displays information about archived logs in backup sets from the control file.

Note that online redo logs cannot be backed up directly; they must be archived first to disk and then backed up. An archive log backup set can contain one or more archived logs.

Column	Datatype	Description
RECID	NUMBER	Record ID for this row; it is an integer that identifies this row
STAMP	NUMBER	Timestamp used with RECID to uniquely identify this row
SET_STAMP	NUMBER	One of the foreign keys for the row of the V$BACKUP_SET table that identifies this backup set
SET_COUNT	NUMBER	One of the foreign keys for the row of the V$BACKUP_SET table that identifies this backup set
THREAD#	NUMBER	Thread number for the log
SEQUENCE#	NUMBER	Log sequence number
RESETLOGS_CHANGE#	NUMBER	Change number of the last resetlogs before the log was written
RESETLOGS_TIME	DATE	Change time of the last resetlogs before the log was written. These will be the same for all logs in a backup set.
FIRST_CHANGE#	NUMBER	SCN when the log was switched into. The redo in the log is at this SCN and greater.
FIRST_TIME	DATE	Time allocated when the log was switched into
NEXT_CHANGE#	NUMBER	SCN when the next log in this thread was switched into. The redo in the log is below this SCN.
NEXT_TIME	DATE	Time when the next log in this thread was switched into
BLOCKS	NUMBER	Size of the log in logical blocks including the header block
BLOCK_SIZE	NUMBER	Size of the log blocks in bytes
TERMINAL	VARCHAR2(3)	Indicates whether this record corresponds to a terminal archived redo log, as defined in V$ARCHIVED_LOG (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.107 V$BACKUP_SET

V$BACKUP_SET displays information about backup sets from the control file.

A backup set record is inserted after the backup set is successfully completed.

Column	Datatype	Description
RECID	NUMBER	Backup set record ID
STAMP	NUMBER	Backup set record stamp
SET_STAMP	NUMBER	Backup set stamp. The backup set stamp and count uniquely identify the backup set. Primary key for the V$BACKUP_SET table, and the foreign key for the following tables: V$BACKUP_PIECE, V$BACKUP_DATAFILE, V$BACKUP_REDOLOG, V$BACKUP_CORRUPTION.
SET_COUNT	NUMBER	Backup set count. The backup set count is incremented by one every time a new backup set is started (if the backup set is never completed the number is "lost"). If the control file is re-created then the count is reset to 1. Therefore the count must be used with the stamp to uniquely identify a backup set. Primary key for the V$BACKUP_SET table, and the foreign key for the following tables: V$BACKUP_PIECE, V$BACKUP_DATAFILE, V$BACKUP_REDOLOG, V$BACKUP_CORRUPTION
BACKUP_TYPE	VARCHAR2(1)	Type of files that are in this backup. If the backup contains archived redo logs, the value is L. If this is a data file full backup, the value is D. If this is an incremental backup, the value is I.
CONTROLFILE_INCLUDED	VARCHAR2(3)	Provides information about the control file in the backup set: • YES: The control file included is a primary control file. • SBY: The control file included is a standby control file. • NO: A control file is not included.
INCREMENTAL_LEVEL	NUMBER	Location where this backup set fits into the database's backup strategy. Set to NULL for full datafile, archivelog, controlfile, and spfile backups, set to 0 for incremental level 0 datafile backups, and set to 1 for incremental level 1 datafile backups.
PIECES	NUMBER	Number of distinct backup pieces in the backup set
START_TIME	DATE	Starting time
COMPLETION_TIME	DATE	Time that this backup set completed
ELAPSED_SECONDS	NUMBER	The number of elapsed seconds
BLOCK_SIZE	NUMBER	Block size of the backup set

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
INPUT_FILE_SCAN_ONLY	VARCHAR2(3)	YES indicates no actual backup is performed, but the data files are read. NO indicates a normal backup is performed.
KEEP	VARCHAR2(3)	(YES/NO) Indicates whether or not this backup set has a retention policy that is different than the value for the configure retention policy
KEEP_UNTIL	DATE	If KEEP_UNTIL_TIME is specified, this is the date after which the backup becomes obsolete. If this column is null, then the backup never expires.
KEEP_OPTIONS	VARCHAR2(11)	Lists additional retention options for this backup set. Possible values are: LOGS - The logs need to recover this backup are kept NOLOGS - The logs needed to recover this backup will not be kept BACKUP_LOGS - An archive log backup exists to support this backup set
MULTI_SECTION	VARCHAR2(3)	Indicates whether or not this backup set is a multi-section backup. Valid values are YES and NO. A multi-section backup is a backup in which multiple backup pieces are produced independently in parallel by multiple channels.
FOR_XTTS	VARCHAR2(3)	Indicates whether this is a cross platform backup set: • YES: This is a cross platform backup set. • NO: This is not a cross platform backup set.
SAME_ENDIAN	VARCHAR2(3)	If the value of FOR_XTTS is YES, then this column indicates whether the backup set has the same endianess as the current database (YES) or not (NO); otherwise NULL.
INC_DMPFILE	VARCHAR2(3)	If the value of FOR_XTTS is YES, then this column indicates whether the backup set includes a Data Pump export file of the backed up data files (YES) or not (NO), otherwise NULL.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
GUID	RAW(16)	The GUID of the PDB to which the backup belongs. This is useful after the PDB is dropped to identify which PDB the backup belongs to.

7.108 V$BACKUP_SET_DETAILS

V$BACKUP_SET_DETAILS provides detailed information about the backup set.

This view will contain an extra row for each backup session that invokes BACKUP BACKUPSET (that is, creates new copies for the same backup set or copies backup set information from disk to tape). However, the remaining values of other columns belong to the complete backup set.

Column	Datatype	Description
SESSION_KEY	NUMBER	Session identifier
SESSION_RECID	NUMBER	Identifies the job, together with SESSION_STAMP.
SESSION_STAMP	NUMBER	Identifies the job, together with SESSION_RECID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BS_KEY	NUMBER	Backup set identifier
RECID	NUMBER	RECID from V$BACKUP_SET record
STAMP	NUMBER	Stamp from V$BACKUP_SET record
SET_STAMP	NUMBER	Backup set stamp
SET_COUNT	NUMBER	Backup count number
BACKUP_TYPE	VARCHAR2(1)	Type of backup (same as in V$BACKUP_SET)
CONTROLFILE_INCLUDED	VARCHAR2(3)	Control file included in backup set (same as in V$BACKUP_SET)
INCREMENTAL_LEVEL	NUMBER	Incremental level (same as in V$BACKUP_SET)
PIECES	NUMBER	Number of pieces (same as in V$BACKUP_SET)
START_TIME	DATE	Start time of the backup set (same as in V$BACKUP_SET)
COMPLETION_TIME	DATE	Completion time of the backup set (same as in V$BACKUP_SET)
ELAPSED_SECONDS	NUMBER	Time taken for backup set creation (same as in V$BACKUP_SET)
BLOCK_SIZE	NUMBER	Block size (same as in V$BACKUP_SET)
KEEP	VARCHAR2(3)	Keep value (same as in V$BACKUP_SET)
KEEP_UNTIL	DATE	Keep Until time (same as in V$BACKUP_SET)
KEEP_OPTIONS	VARCHAR2(11)	Keep options (same as in V$BACKUP_SET)
DEVICE_TYPE	VARCHAR2(17)	Type of device. If the backup set exists on more than one device type, an * is indicated here.
COMPRESSED	VARCHAR2(3)	YES, if backup is compressed
NUM_COPIES	NUMBER	Number of identical copies
OUTPUT_BYTES	NUMBER	Size of the backup set
ORIGINAL_INPUT_BYTES	NUMBER	Amount of data backed up when the backup set was created
COMPRESSION_RATIO	NUMBER	The ratio between the total blocks in the datafile and the blocks that RMAN backed up. This is *not* the the ratio from the AS COMPRESSED BACKUPSET clause of the BACKUP command.
STATUS	CHAR(1)	The status of the backup set. It is always A (all backup pieces available), because this view only reflects available backup sets.
ORIGINAL_INPRATE_BYTES	NUMBER	Number of bytes read per second when backup set was initially created
OUTPUT_RATE_BYTES	NUMBER	Number of bytes written per second when the backup set was initially created
ORIGINAL_INPUT_BYTES_DISPLAY	VARCHAR2(4000)	Input rate to display
OUTPUT_BYTES_DISPLAY	VARCHAR2(4000)	Size of backup set to display
ORIGINAL_INPRATE_BYTES_DISPLAY	VARCHAR2(4000)	Input rate per second for display
OUTPUT_RATE_BYTES_DISPLAY	VARCHAR2(4000)	Output rate per second for display
TIME_TAKEN_DISPLAY	VARCHAR2(4000)	Elapsed time in hh:mm:ss format
ENCRYPTED	VARCHAR2(3)	A value of YES means an encrypted backup, otherwise not an encrypted backup.
BACKED_BY_OSB	VARCHAR2(3)	A value of YES means the backup was done to Oracle Secure Backup. Otherwise, backed up by other third party tape library.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$BACKUP_SET"

7.109 V$BACKUP_SET_SUMMARY

V$BACKUP_SET_SUMMARY provides summary information for a backup set.

Column	Datatype	Description
NUM_BACKUPSETS	NUMBER	Total number of backup sets created
OLDEST_BACKUP_TIME	DATE	Oldest backup start time
NEWEST_BACKUP_TIME	DATE	Newest backup start time
OUTPUT_BYTES	NUMBER	Number of output bytes (not including multiple copies)
ORIGINAL_INPUT_BYTES	NUMBER	Number of input bytes when backup sets were created
ORIGINAL_INPRATE_BYTES	NUMBER	Average input rate
OUTPUT_RATE_BYTES	NUMBER	Average output rate
COMPRESSION_RATIO	NUMBER	The ratio between the total blocks in the datafile and the blocks that RMAN backed up. This is *not* the the ratio from the AS COMPRESSED BACKUPSET clause of the BACKUP command.
ORIGINAL_INPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for input bytes
OUTPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for output bytes
ORIGINAL_INPRATE_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for input rate
OUTPUT_RATE_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for output rate
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.110 V$BACKUP_SPFILE

V$BACKUP_SPFILE displays information about server parameter files in backup sets from the control file.

Column	Datatype	Description
RECID	NUMBER	Backup SPFILE record ID
STAMP	NUMBER	Backup SPFILE record stamp
SET_STAMP	NUMBER	Backup set stamp (of the set which contains this SPFILE backup)
SET_COUNT	NUMBER	Backup set count (of the set which contains this SPFILE backup)
MODIFICATION_TIME	DATE	Time when the SPFILE was last modified (this also includes creation time)
BYTES	NUMBER	Size of the SPFILE (in bytes)
COMPLETION_TIME	DATE	Time when the backup of the SPFILE completed
DB_UNIQUE_NAME	VARCHAR2(30)	Unique database name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
GUID	RAW(16)	The GUID of the PDB to which the backup belongs. This is useful after the PDB is dropped to identify which PDB the backup belongs to.

7.111 V$BACKUP_SPFILE_DETAILS

V$BACKUP_SPFILE_DETAILS displays information about all restorable SP files backed up in the backup set.

Column	Datatype	Description
SESSION_KEY	NUMBER	Session identifier
SESSION_RECID	NUMBER	Session recid
SESSION_STAMP	NUMBER	Session stamp
BS_KEY	NUMBER	Unique backup set identifier
SET_STAMP	NUMBER	Backup set stamp
SET_COUNT	NUMBER	Backup set count
MODIFICATION_TIME	DATE	Time the backup set was modified
FILESIZE	NUMBER	Size, in bytes, of the SPFILE that was backed up
FILESIZE_DISPLAY	VARCHAR2(4000)	Same value as the FILESIZE column, but converted to a user-displayable format, for example *n*M, *n*G, *n*T, *n*P, and so on

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.112 V$BACKUP_SPFILE_SUMMARY

V$BACKUP_SPFILE_SUMMARY provides summary information for input SP file, based on either a backup job or time range applicable to jobs.

Column	Datatype	Description
NUM_FILES_BACKED	NUMBER	Number of files backed up
NUM_DISTINCT_FILES_BACKED	NUMBER	Number of distinct SP files backed up (with modification timestamp)
MIN_MODIFICATION_TIME	DATE	Minimum modification time
MAX_MODIFICATION_TIME	DATE	Maximum modification time
INPUT_BYTES	NUMBER	Total input bytes for all SP files backed up
INPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for all input bytes
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.113 V$BACKUP_SYNC_IO

V$BACKUP_SYNC_IO displays performance information about ongoing and recently completed RMAN backups and restores.

For each backup, it contains one row for each input data file, one row for the aggregate total performance of all data files, and one row for the output backup piece. This data is not stored persistently, and is not preserved when the instance is re-started.

Column	Datatype	Description
SID	NUMBER	The Oracle SID of the session doing the backup or restore
SERIAL	NUMBER	The use count for the SID doing the backup or restore
USE_COUNT	NUMBER	A counter that can be used to identify rows from different backup sets
RMAN_STATUS_RECID	NUMBER	Owning V$RMAN_STATUS record ID
RMAN_STATUS_STAMP	NUMBER	Owning V$RMAN_STATUS record stamp

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DEVICE_TYPE	VARCHAR2(17)	The device type where the file is located
TYPE	VARCHAR2(9)	INPUT, OUTPUT, or AGGREGATE
STATUS	VARCHAR2(11)	NOT STARTED, IN PROGRESS, or FINISHED
FILENAME	VARCHAR2(513)	The name of the backup file being read or written
SET_COUNT	NUMBER	The set count of the backup set being read or written
SET_STAMP	NUMBER	The set stamp of the backup set being read or written
BUFFER_SIZE	NUMBER	The size of the buffers being used to read/write this file, in bytes
BUFFER_COUNT	NUMBER	The number of buffers being used to read/write this file
TOTAL_BYTES	NUMBER	The total number of bytes that will be read or written for this file, if known. If not known, this column will be null.
OPEN_TIME	DATE	The time this file was opened. If TYPE='AGGREGATE', then this is the time that the first file in the aggregate was opened.
CLOSE_TIME	DATE	The time this file was closed. If TYPE='AGGREGATE', then this is the time that the last file in the aggregate was closed.
ELAPSED_TIME	NUMBER	The time, in hundredths of a second, that the file was open
MAXOPENFILES	NUMBER	The number of concurrently open DISK files. This value is only present in rows where TYPE='AGGREGATE'.
BYTES	NUMBER	The number of bytes read or written so far
EFFECTIVE_BYTES_PER_SECOND	NUMBER	The I/O rate that was achieved with this device during this backup
IO_COUNT	NUMBER	The number of I/Os that were performed to this file
IO_TIME_TOTAL	NUMBER	The total time, in hundredths of a second, taken to do I/O for this file
IO_TIME_MAX	NUMBER	The maximum time taken for a single I/O request
DISCRETE_BYTES_PER_SECOND	NUMBER	The average transfer rate for this file
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.114 V$BGPROCESS

V$BGPROCESS displays information about the background processes.

Column	Datatype	Description
PADDR	RAW(4 | 8)	Address of the process state object
PSERIAL#	NUMBER	Process state object serial number
NAME	VARCHAR2(5)	Name of this background process
DESCRIPTION	VARCHAR2(64)	Description of the background process

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ERROR	NUMBER	Error encountered
TYPE	VARCHAR2(5)	This column has a value of SLAVE for all background worker processes, otherwise it is null.
PRIORITY	VARCHAR2(8)	Lists the current priority with which the process is running. This column has string values based on the operating system. On Linux, the values that can appear in this column are: • TS: Time sharing • RT: Real time priority
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.115 V$BH

V$BH displays the status and number of pings for every buffer in the SGA. This is an Oracle Real Application Clusters view.

Column	Datatype	Description
FILE#	NUMBER	Data file identifier number (to find the file name, query DBA_DATA_FILES or V$DBFILE)
BLOCK#	NUMBER	Block number
CLASS#	NUMBER	Class number
STATUS	VARCHAR2(10)	Status of the buffer: • free - Not currently in use • xcur - Exclusive • scur - Shared current • cr - Consistent read • read - Being read from disk • mrec - In media recovery mode • irec - In instance recovery mode • pi - A past image in RAC mode • securefile - A secured file buffer • flashfree - A free flash cache buffer • flashcur - A current flash cache buffer
XNC	NUMBER	This column is obsolete. Its value is hard-coded to 0.
FORCED_READS	NUMBER	This column is obsolete. Its value is hard-coded to 0.
FORCED_WRITES	NUMBER	This column is obsolete. Its value is hard-coded to 0.
LOCK_ELEMENT_ADDR	RAW(4 | 8)	Address of the lock element that contains the PCM lock that is covering the buffer. If more than one buffer has the same address, then these buffers are covered by the same PCM lock.
LOCK_ELEMENT_NAME	NUMBER	The address of the lock element that contains the PCM lock that is covering the buffer. If more than one buffer has the same address, then these buffers are covered by the same PCM lock.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LOCK_ELEMENT_CLASS	NUMBER	The address of the lock element that contains the PCM lock that is covering the buffer. If more than one buffer has the same address, then these buffers are covered by the same PCM lock.
DIRTY	VARCHAR2(1)	Y - block modified
TEMP	VARCHAR2(1)	Y - temporary block
PING	VARCHAR2(1)	Y - block pinged
STALE	VARCHAR2(1)	Y - block is stale
DIRECT	VARCHAR2(1)	Y - direct block
NEW	CHAR(1)	Always set to N. This column is obsolete and maintained for backward compatibility.
OBJD	NUMBER	Database object number of the block that the buffer represents
TS#	NUMBER	Tablespace number of block
LOBID	NUMBER	If the buffer belongs to a SecureFiles object, the value in this column is the unique identifier for the SecureFiles object. For other buffer types, the value in this column is meaningless.
CACHEHINT	NUMBER	Numeric representation of the values in the FLASH_CACHE and CELL_FLASH_CACHE columns. See the descriptions and possible values for the FLASH_CACHE and CELL_FLASH_CACHE columns in this view.
FLASH_CACHE	VARCHAR2(7)	Database Smart Flash Cache hint to be used for segment blocks: • DEFAULT • KEEP • NONE Solaris and Oracle Linux functionality only.
CELL_FLASH_CACHE	VARCHAR2(7)	Cell flash cache hint to be used for segment blocks: • DEFAULT • KEEP • NONE **See Also:** Oracle Exadata Storage Server Software documentation for more information
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.116 V$BLOCK_CHANGE_TRACKING

V$BLOCK_CHANGE_TRACKING displays the status of block change tracking for the database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(10)	Status of block change tracking in the database: • DISABLED - Block change tracking is disabled • TRANSITION - Block change tracking is in the process of transitioning between the enabled and disabled states. The TRANSITION state should usually never be observed, because it only exists while enabling or disabling block change tracking. This state might be observed if the instance crashed while enabling or disabling block change tracking, in which case it will be cleaned up automatically the next time that the database is opened. • ENABLED - Block change tracking is enabled
FILENAME	VARCHAR2(513)	Name of the block change tracking file for the database
BYTES	NUMBER	Size of the block change tracking file (in bytes)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Backup and Recovery User's Guide for information on setting up block change tracking

7.117 V$BLOCKING_QUIESCE

V$BLOCKING_QUIESCE indicates if a session is blocking, or would block, a quiesce operation.

Column	Datatype	Description
SID	NUMBER	Session identifier
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.118 V$BT_SCAN_CACHE

V$BT_SCAN_CACHE shows the parameters and status of the big table cache section.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BT_CACHE_ALLOC	NUMBER	Current ratio of the big table cache section to the buffer cache
BT_CACHE_TARGET	NUMBER	Target ratio of the big table cache section to the buffer cache
OBJECT_COUNT	NUMBER	Number of objects tracked by the big table cache section
MEMORY_BUF_ALLOC	NUMBER	Number of memory buffers allocated by the big table cache section to objects
MIN_CACHED_TEMP	NUMBER	Minimum temperature of any object that is allowed to be cached by the big table cache section
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"DB_BIG_TABLE_CACHE_PERCENT_TARGET" for more information about enabling the big table cache

7.119 V$BT_SCAN_OBJ_TEMPS

V$BT_SCAN_OBJ_TEMPS shows the active objects currently tracked by the big table cache.

Column	Datatype	Description
TS#	NUMBER	Tablespace number
DATAOBJ#	NUMBER	Data object number (objd)
SIZE_IN_BLKS	NUMBER	Size of the object being scanned on this instance, in blocks
TEMPERATURE	NUMBER	Temperature of this object
POLICY	VARCHAR2(10)	Caching policy of this object. Possible values: • MEM_ONLY: This object will be fully cached in memory. • MEM_PART: This object will be partially cached in memory and some portion will remain on disk and will not be cached. • DISK: this object will not be cached in memory or flash for the scan at all. • INVALID: The caching policy is not valid.
CACHED_IN_MEM	NUMBER	The number of blocks that are cached/allocated in memory for this object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"DB_BIG_TABLE_CACHE_PERCENT_TARGET" for more information about enabling the big table cache

7.120 V$BUFFER_POOL

V$BUFFER_POOL displays information about all buffer pools available for the instance.

Column	Datatype	Description
ID	NUMBER	Buffer pool identifier number
NAME	VARCHAR2(20)	Name of the buffer pool: • DEFAULT • KEEP • RECYCLE **Note:** Currently, KEEP and RECYCLE pools only exist for the standard block size. All nonstandard block size pools are DEFAULT.
BLOCK_SIZE	NUMBER	Block size (in bytes) for buffers in this pool. Possible values: the standard block size, the power of 2 nonstandard block sizes, 2048, 4096, 8192, 16384, 32768.
RESIZE_STATE	VARCHAR2(10)	Current state of the resize operation: STATIC - Not being resized ALLOCATING - Memory is being allocated (can be canceled by the user) ACTIVATING - New buffers are being created (user cannot cancel) SHRINKING - Buffers are being deleted (can be canceled by the user)
CURRENT_SIZE	NUMBER	Present size of the sub-cache (in megabytes)
BUFFERS	NUMBER	Current instantaneous number of buffers
TARGET_SIZE	NUMBER	If a resize is in progress (state is not STATIC), records new target size (in megabytes). If the pool is STATIC, the value in this column is the same as the current size of the pool.
TARGET_BUFFERS	NUMBER	If a resize is in progress, records new target size in terms of buffers. Otherwise, the value in this column is the same as the current number of buffers.
PREV_SIZE	NUMBER	Previous buffer pool size. If the buffer pool has never been resized, the previous size is zero.
PREV_BUFFERS	NUMBER	Previous number of buffers in the buffer pool. Value is zero if the buffer pool has never been resized.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LO_BNUM	NUMBER	Obsolete column
HI_BNUM	NUMBER	Obsolete column
LO_SETID	NUMBER	Obsolete column
HI_SETID	NUMBER	Obsolete column
SET_COUNT	NUMBER	Obsolete column
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"DB_BLOCK_SIZE"

7.121 V$BUFFER_POOL_STATISTICS

V$BUFFER_POOL_STATISTICS displays statistics about all buffer pools available for the instance.

Column	Datatype	Description
ID	NUMBER	Buffer pool identifier number
NAME	VARCHAR2(20)	Name of the buffer pool
BLOCK_SIZE	NUMBER	Block size (in bytes) for buffers in this pool. Possible values: the standard block size, the power of 2 nonstandard block sizes, 2048, 4096, 8192, 16384, 32768.
SET_MSIZE	NUMBER	Buffer pool maximum set size
CNUM_REPL	NUMBER	Number of buffers on replacement list
CNUM_WRITE	NUMBER	Number of buffers on write list
CNUM_SET	NUMBER	Number of buffers in set
BUF_GOT	NUMBER	Number of buffers gotten by the set
SUM_WRITE	NUMBER	Number of buffers written by the set
SUM_SCAN	NUMBER	Number of buffers scanned in the set
FREE_BUFFER_WAIT	NUMBER	Free buffer wait statistic
WRITE_COMPLETE_WAIT	NUMBER	Write complete wait statistic
BUFFER_BUSY_WAIT	NUMBER	Buffer busy wait statistic
FREE_BUFFER_INSPECTED	NUMBER	Free buffer inspected statistic
DIRTY_BUFFERS_INSPECTED	NUMBER	Dirty buffers inspected statistic
DB_BLOCK_CHANGE	NUMBER	Database blocks changed statistic
DB_BLOCK_GETS	NUMBER	Database blocks gotten statistic

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CONSISTENT_GETS	NUMBER	Consistent gets statistic
PHYSICAL_READS	NUMBER	Physical reads statistic
PHYSICAL_WRITES	NUMBER	Physical writes statistic
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"DB_CACHE_SIZE"

7.122 V$BUFFERED_PUBLISHERS

V$BUFFERED_PUBLISHERS displays information about all buffered publishers in the instance.

There is one row per queue per sender. The values are reset to zero when the database (or instance in an Oracle RAC environment) restarts.

Column	Datatype	Description
QUEUE_ID	NUMBER	Identifier for the queue
QUEUE_SCHEMA	VARCHAR2(128)	Owner of the queue
QUEUE_NAME	VARCHAR2(128)	Name of the queue
SENDER_NAME	VARCHAR2(128)	Name of the agent enqueuing the message (the Streams name assigned for a capture process)
SENDER_ADDRESS	VARCHAR2(1024)	Queue name and database name of the source (last propagating) queue; database name is not specified if the source queue is in the local database
SENDER_PROTOCOL	NUMBER	Protocol used by the sender's address
NUM_MSGS	NUMBER	Current number of messages that have yet to be dequeued from the buffer queue
CNUM_MSGS	NUMBER	Cumulative total number of messages enqueued into the buffered queue since the database last started
LAST_ENQUEUED_MSG	NUMBER	Most recently enqueued message identifier
UNBROWSED_MSGS	NUMBER	Number of messages that have been enqueued but not browsed
OVERSPILLED_MSGS	NUMBER	Number of messages that have been spilled but not browsed
MEMORY_USAGE	NUMBER	Percentage of the Streams pool that is being used (or 0 if there is no Streams pool)
ELAPSED_ENQUEUE_TIME	NUMBER	Total time spent in enqueue (in hundredths of a second)
ENQUEUE_CPU_TIME	NUMBER	Total CPU time for enqueue (in hundredths of a second)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LAST_ENQUEUE_TIME	TIMESTAMP(3) WITH TIME ZONE	Last message enqueue time
PUBLISHER_STATE	VARCHAR2(59)	State of the publisher: • IN FLOW CONTROL: TOO MANY UNBROWSED MESSAGES • IN FLOW CONTROL: OVERSPILLED MESSAGES • IN FLOW CONTROL: INSUFFICIENT MEMORY AND UNBROWSED MESSAGES • PUBLISHING MESSAGES - Normal
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.123 V$BUFFERED_QUEUES

V$BUFFERED_QUEUES displays information about all buffered queues in the instance. There is one row per queue.

Column	Datatype	Description
QUEUE_ID	NUMBER	Identifier for the queue
QUEUE_SCHEMA	VARCHAR2(128)	Owner of the queue
QUEUE_NAME	VARCHAR2(128)	Name of the queue
STARTUP_TIME	DATE	Startup time
NUM_MSGS	NUMBER	Total number of messages currently in the buffered queue
SPILL_MSGS	NUMBER	Current number of overflow messages spilled to disk from the buffered queue
CNUM_MSGS	NUMBER	Cumulative total number of messages enqueued into the buffered queue since the database last started
CSPILL_MSGS	NUMBER	Cumulative total number of overflow messages spilled to disk from the buffered queue since the database last started
EXPIRED_MSGS	NUMBER	Number of expired messages
OLDEST_MSGID	RAW(16)	Message ID of the oldest message
OLDEST_MSG_ENQTM	TIMESTAMP(3)	Enqueue time of the oldest message
QUEUE_STATE	VARCHAR2(25)	Indicates whether the queue is in recovery mode (QUEUE IS IN RECOVERY MODE) or not (NORMAL)
ELAPSED_ENQUEUE_TIME	NUMBER	Total time spent in enqueue (in hundredths of a second)
ELAPSED_DEQUEUE_TIME	NUMBER	Total time spent in dequeue (in hundredths of a second)
ELAPSED_TRANSFORMATION_TIME	NUMBER	Total time for evaluating transformations (in hundredths of a second)
ELAPSED_RULE_EVALUATION_TIME	NUMBER	Total time for rule evaluations (in hundredths of a second)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ENQUEUE_CPU_TIME	NUMBER	Total CPU time for enqueue (in hundredths of a second)
DEQUEUE_CPU_TIME	NUMBER	Total CPU time for dequeue (in hundredths of a second)
AVG_MSG_AGE	NUMBER	Average age of messages in the queue
LAST_ENQUEUE_TIME	TIMESTAMP(3) WITH TIME ZONE	Last message enqueue time
LAST_DEQUEUE_TIME	TIMESTAMP(3) WITH TIME ZONE	Last message dequeue time
QUEUE_SIZE	NUMBER	Size of queue, which is the total number of bytes allocated for all messages and metadata
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.124 V$BUFFERED_SUBSCRIBERS

V$BUFFERED_SUBSCRIBERS displays information about the subscribers for all buffered queues in the instance. There is one row per subscriber per queue.

Column	Datatype	Description
QUEUE_ID	NUMBER	Identifier for the queue
QUEUE_SCHEMA	VARCHAR2(128)	Owner of the queue
QUEUE_NAME	VARCHAR2(128)	Name of the queue
SUBSCRIBER_ID	NUMBER	Internal subscriber number (for identification)
SUBSCRIBER_NAME	VARCHAR2(512)	Name of the subscriber
SUBSCRIBER_ADDRESS	VARCHAR2(1024)	Address of the subscribing agent
PROTOCOL	NUMBER	Protocol of the subscribing agent
SUBSCRIBER_TYPE	VARCHAR2(128)	Type of the subscriber: • PROXY - Proxy subscriber • SUBSCRIBER
STARTUP_TIME	DATE	Startup time
LAST_BROWSED_SEQ	NUMBER	Sequence number of the most recently browsed message for the subscriber (comparable to the number of messages in the V$STREAMS_APPLY_READER view)
LAST_BROWSED_NUM	NUMBER	Internal Message number for the most recently browsed message for the subscriber
LAST_DEQUEUED_SEQ	NUMBER	Sequence number of the most recently dequeued message for the subscriber (comparable to the number of messages in the V$STREAMS_APPLY_COORDINATOR view)
LAST_DEQUEUED_NUM	NUMBER	Internal Message number for the most recently dequeued message for the subscriber

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CURRENT_ENQ_SEQ	NUMBER	Current sequence number of the most recently enqueued message for the subscriber
NUM_MSGS	NUMBER	Total number of outstanding messages currently enqueued in the buffered queue for the subscriber (includes the count of the messages overflowed to disk)
CNUM_MSGS	NUMBER	Cumulative total number of messages enqueued for the subscriber since the database last started
TOTAL_DEQUEUED_MSG	NUMBER	Total number of messages dequeued by the subscriber
TOTAL_SPILLED_MSG	NUMBER	Total number of spilled messages for the subscriber
EXPIRED_MSGS	NUMBER	Number of expired messages
MESSAGE_LAG	NUMBER	Message lag of the subscriber
ELAPSED_DEQUEUE_TIME	NUMBER	Total time spent in dequeue (in hundredths of a second)
DEQUEUE_CPU_TIME	NUMBER	Total CPU time for dequeue (in hundredths of a second)
AVG_MSG_AGE	NUMBER	Average age of messages currently enqueued in the buffered queue for the subscriber
LAST_DEQUEUE_TIME	TIMESTAMP(3) WITH TIME ZONE	Last message dequeue time
OLDEST_MSGID	RAW(16)	Message ID of the oldest message
OLDEST_MSG_ENQTM	TIMESTAMP(3)	Enqueue time of the oldest message
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.125 V$CACHE

V$CACHE displays information from the block header of each block in the SGA of the current instance as related to particular database objects. This is an Oracle Real Application Clusters view.

Column	Datatype	Description
FILE#	NUMBER	Data file identifier number (to find the file name, query DBA_DATA_FILES or V$DBFILE)
BLOCK#	NUMBER	Block number
CLASS#	NUMBER	Class number

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(10)	Status of the block: • free - Not currently in use • xcur - Exclusive • scur - Shared current • cr - Consistent read • read - Being read from disk • mrec - In media recovery mode • irec - In instance recovery mode
XNC	NUMBER	Number of PCM x to null lock conversions due to contention with another instance. This column is obsolete and maintained for backward compatibility.
FORCED_READS	NUMBER	Number of times the block had to be reread from the cache because another instance has forced it out of this instance's cache by requesting the lock on the block in exclusive mode
FORCED_WRITES	NUMBER	Number of times GCS had to write this block to cache because this instance had used the block and another instance had requested the lock on the block in a conflicting mode
NAME	VARCHAR2(128)	Name of the database object containing the block
PARTITION_NAME	VARCHAR2(128)	Name of the partition (null for nonpartitioned objects)
KIND	VARCHAR2(15)	Type of the database object: • INDEX • TABLE • CLUSTER • VIEW • SYNONYM • SEQUENCE • PROCEDURE • FUNCTION • PACKAGE • NON-EXISTENT • PACKAGE BODY • TRIGGER • TYPE • TYPE BODY • TABLE PARTITION • INDEX PARTITION • LOB • LIBRARY • JAVA SOURCE • JAVA CLASS • JAVA RESOURCE • JAVA DATA • UNDO
OWNER#	NUMBER	Owner number
LOCK_ELEMENT_ADDR	RAW(4 | 8)	Address of the lock element that contains the PCM lock that is covering the buffer. If more than one buffer has the same address, then these buffers are covered by the same PCM lock.
LOCK_ELEMENT_NAME	NUMBER	Name of the lock element that contains the PCM lock that is covering the buffer. If more than one buffer has the same address, then these buffers are covered by the same PCM lock.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.126 V$CACHE_LOCK

V$CACHE_LOCK is deprecated. The information that was provided in this view is now provided in the V$INSTANCE_CACHE_TRANSFER and V$SEGMENT_STATISTICS views.

Column	Datatype	Description
FILE#	NUMBER	Data file identifier number (to find file name, query DBA_DATA_FILES or V$DBFILE)
BLOCK#	NUMBER	Block number
STATUS	VARCHAR2(10)	Status of the block: • free - Not currently in use • xcur - Exclusive • scur - Shared current • cr - Consistent read • read - Being read from disk • mrec - In media recovery mode • irec - In instance recovery mode
XNC	NUMBER	Number of parallel cache management (PCM) lock conversions due to contention with another instance
FORCED_READS	NUMBER	Number of times the block had to be reread from the cache because another instance has forced it out of this instance's cache by requesting the lock on the block in exclusive mode
FORCED_WRITES	NUMBER	Number of times GCS had to write this block to cache because this instance had used the block and another instance had requested the lock on the block in a conflicting mode
NAME	VARCHAR2(128)	Name of the database object containing the block

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
KIND	VARCHAR2(15)	Type of database object: • 1 - Index • 2 - Table • 3 - Cluster • 4 - View • 5 - Synonym • 6 - Sequence • 7 - Procedure • 8 - Function • 9 - Package • 10 - Nonexistent • 11 - Package body • 12 - Trigger • 13 - Type • 14 - Type body • 19 - Table partition • 20 - Index partition • 21 - LOB • 22 - Library • Null - Unknown
OWNER#	NUMBER	Owner number
LOCK_ELEMENT_ADDR	RAW(4 | 8)	Address of the lock element that contains the PCM lock that is covering the buffer. If more than one buffer has the same address, then these buffers are covered by the same PCM lock.
LOCK_ELEMENT_NAME	NUMBER	Address of the lock element that contains the PCM lock that is covering the buffer. If more than one buffer has the same address, then these buffers are covered by the same PCM lock.
INDX	NUMBER	Platform-specific lock manager identifier
CLASS	NUMBER	Platform-specific lock manager identifier
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.127 V$CACHE_TRANSFER

V$CACHE_TRANSFER is identical to the V$CACHE view but only displays blocks that have been pinged at least once.

This view contains information from the block header of each block in the SGA of the current instance as related to particular database objects. This is an Oracle Real Application Clusters view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FILE#	NUMBER	Data file identifier number (to find the file name, query DBA_DATA_FILES or V$DBFILE)
BLOCK#	NUMBER	Block number
CLASS#	NUMBER	Class number
STATUS	VARCHAR2(10)	Status of the block: • free - Not currently in use • xcur - Exclusive • scur - Shared current • cr - Consistent read • read - Being read from disk • mrec - In media recovery mode • irec - In instance recovery mode
XNC	NUMBER	Number of PCM lock conversions due to contention with another instance. This column is obsolete and maintained for backward compatibility.
FORCED_READS	NUMBER	Number of times the block had to be reread from the cache because another instance has forced it out of this instance's cache by requesting the lock on the block in exclusive mode
FORCED_WRITES	NUMBER	Number of times GCS had to write this block to cache because this instance had used the block and another instance had requested the lock on the block in a conflicting mode
NAME	VARCHAR2(128)	Name of the database object containing the block
PARTITION_NAME	VARCHAR2(128)	NULL for nonpartitioned objects
KIND	VARCHAR2(15)	Type of database object **See Also:** Table 8-1
OWNER#	NUMBER	Owner number
LOCK_ELEMENT_ADDR	RAW(4 | 8)	Address of the lock element that contains the PCM lock that is covering the buffer. If more than one buffer has the same address, then these buffers are covered by the same PCM lock.
LOCK_ELEMENT_NAME	NUMBER	The name of the lock that contains the PCM lock that is covering the buffer
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$CACHE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.128 V$CHUNK_METRIC

V$CHUNK_METRIC displays the metric values captured for the most recent 30-second intervals for the workload against each chunk available on the database.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
CHUNK_ID	NUMBER	Chunk number (internal)
CALLSPERSEC	NUMBER	Number of user calls per second to the chunks
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SERVICE_REGION_METRIC"

7.129 V$CIRCUIT

V$CIRCUIT contains information about virtual circuits, which are user connections to the database through dispatchers and servers.

Column	Datatype	Description
CIRCUIT	RAW(4 | 8)	Circuit address
DISPATCHER	RAW(4 | 8)	Current dispatcher process address
SERVER	RAW(4 | 8)	Current server process address
WAITER	RAW(4 | 8)	Address of the server process that is waiting for the (currently busy) circuit to become available
SADDR	RAW(4 | 8)	Address of the session bound to the circuit
STATUS	VARCHAR2(16)	Status of the circuit: • BREAK - currently interrupted • EOF - about to be removed • OUTBOUND - an outward link to a remote database • NORMAL - normal circuit into the local database

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
QUEUE	VARCHAR2(16)	Queue the circuit is currently on: • COMMON - on the common queue, waiting to be picked up by a server process • DISPATCHER - waiting for the dispatcher • SERVER - currently being serviced • NONE - idle circuit
MESSAGE0	NUMBER	Size in bytes of the messages in the first message buffer
MESSAGE1	NUMBER	Size in bytes of the messages in the second message buffer
MESSAGE2	NUMBER	Size in bytes of the messages in the third message buffer
MESSAGE3	NUMBER	Size in bytes of the messages in the fourth message buffer
MESSAGES	NUMBER	Total number of messages that have gone through this circuit
BYTES	NUMBER	Total number of bytes that have gone through this circuit
BREAKS	NUMBER	Total number of breaks (interruptions) for this circuit
PRESENTATION	VARCHAR2(257)	Presentation protocol used by the client and server
PCIRCUIT	RAW(4 | 8)	Address of the parent circuit
BOUND_TIME[1]	NUMBER	Time that a circuit and shared server have been bound (in centiseconds)
BOUND_REASON[1]	VARCHAR2(32)	Provides a reason (a short explanation) for why a shared server and circuit could not be unbound. This column is empty when a circuit is not bound to a server. When the server starts serving a circuit, BOUND_REASON is empty and remains empty unless the server tries unsuccessfully to unbind the circuit (after it finishes serving the current request). When this column is not empty, it will be cleared once the server and circuit are unbound (that is, once the resources preventing the session migration to another shared server are released).
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

[1] This column is available starting with Oracle Database 19c.

7.130 V$CLASS_CACHE_TRANSFER

V$CLASS_CACHE_TRANSFER is deprecated. The information that was provided in this view is now provided in the V$INSTANCE_CACHE_TRANSFER and V$SEGMENT_STATISTICS views.

Column	Datatype	Description
CLASS	CHAR(10)	Block class; always data block
X_2_NULL	NUMBER	Number of blocks with Exclusive-to-NULL conversions; always 0
X_2_NULL_FORCED_WRITE	NUMBER	Number of Exclusive-to-NULL forced writes; always 0

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
X_2_NULL_FORCED_STALE	NUMBER	Number of Exclusive-to-NULL blocks converted to CR; always 0
X_2_S	NUMBER	Number of blocks with Exclusive-to-Shared conversions; always 0
X_2_S_FORCED_WRITE	NUMBER	Number of Exclusive-to-Shared forced writes; always 0
S_2_NULL	NUMBER	Number of blocks with Shared-to-NULL conversions; always 0
S_2_NULL_FORCED_STALE	NUMBER	Number of Shared-to-NULL blocks converted to CR; always 0
NULL_2_X	NUMBER	Number of blocks with NULL-to-Exclusive conversions; always 0
S_2_X	NUMBER	Number of blocks with Shared-to-Exclusive conversions; always 0
NULL_2_S	NUMBER	Number of blocks with NULL-to-Shared conversions; always 0
CR_TRANSFER	NUMBER	Number of CR blocks transferred; always 0
CURRENT_TRANSFER	NUMBER	Number of current blocks transferred; always 0
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$INSTANCE_CACHE_TRANSFER"
- "V$SEGMENT_STATISTICS"

7.131 V$CLEANUP_PROCESS

V$CLEANUP_PROCESS provides information on the PMON processes.

Column	Datatype	Description
NAME	VARCHAR2(5)	Name of the cleanup process (PMON, CLMN, CL**)
PADDR	RAW(8)	Process pointer for the cleanup process (can join with V$PROCESS)
SADDR	RAW(8)	Session pointer for the cleanup process (can join with V$SESSION)
STATE	VARCHAR2(4)	Cleanup process state: • IDLE: Not currently performing cleanup • BUSY: Currently performing cleanup
DEAD_IN_CLEANUP	RAW(8)	Pointer to the root of the tree in cleanup (can join with ROOT_ADDR in V$DEAD_CLEANUP)
CLEANUP_TIME	NUMBER	If STATE = BUSY, the time spent in the current cleanup attempt (in seconds). Otherwise, 0.
TIME_SINCE_LAST_CLEANUP	NUMBER	If STATE = IDLE, time since last needed to perform cleanup (in seconds). Otherwise, 0.
NUM_CLEANED	NUMBER	Number of trees cleaned up by the cleanup process. Increased by one every time a root tree is attempted.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$DEAD_CLEANUP"

7.132 V$CLIENT_SECRETS

V$CLIENT_SECRETS lists the secrets that are present in the keystore.

Only SYS, SYSKM, and users with the ADMINISTER KEY MANAGEMENT privilege can access this view.

Column	Datatype	Description
CLIENT	VARCHAR2(2000)	Name provided by the client
SECRET_TAG	VARCHAR2(4000)	Associated information with the client
CREATION_TIME	TIMESTAMP(6) WITH TIME ZONE	Time when the secret was created
ACTIVATION_TIME	TIMESTAMP(6) WITH TIME ZONE	Time when the secret was actually put to use
OWNER	VARCHAR2(128)	User who created the secret
OWNER_ID	NUMBER	User ID of the user who created the secret
KEYSTORE_TYPE	VARCHAR2(17)	Type of keystore in which the secret is stored: • OKV - Oracle Key Vault • SOFTWARE KEYSTORE • UNDEFINED - This value is shown if the database has no information about the type of keystore where the master key resides
BACKED_UP	VARCHAR2(9)	Indicates whether the secret has been backed up or not
OWNER_DBNAME	VARCHAR2(128)	Database that created the secret
OWNER_DBID	NUMBER	Database ID where the secret was created
OWNER_INSTANCE_NAME	VARCHAR2(30)	Instance name of the instance where the secret was created
OWNER_INSTANCE_NUMBER	NUMBER	Instance number of the instance where the secret was created
OWNER_INSTANCE_SERIAL	NUMBER	Serial number of the instance where the secret was created
OWNER_PDBNAME	VARCHAR2(128)	Pluggable database (PDB) where the secret was created
OWNER_PDBID	NUMBER	PDB ID where the secret was created

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OWNER_PDBUID	NUMBER	PDB UID where the secret was created
OWNER_PDBGUID	RAW(16)	PDB GUID where the secret was created
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Advanced Security Guide for information about keystore management

7.133 V$CLIENT_STATS

V$CLIENT_STATS displays measures for all sessions that are active for the client identifier per instance.

The statistics available in this view are a subset of those available in V$SESSTAT and V$SESS_TIME_MODEL.

Column	Datatype	Description
CLIENT_IDENTIFIER	VARCHAR2(64)	Client identifier
STAT_ID	NUMBER	Statistic identifier
STAT_NAME	VARCHAR2(64)	Derived statistic name from V$STATNAME and V$SESS_TIME_MODEL
VALUE	NUMBER	Cumulative value (in microseconds)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SESSTAT"
- "V$SESS_TIME_MODEL"

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.134 V$CLONEDFILE

V$CLONEDFILE provides CloneDB file information.

Column	Datatype	Description
SNAPSHOTFILENAME	VARCHAR2(513)	Snapshot file name. This file is the snapshot file from the snapshot database. The snapshot file is also sometimes referred to as the srcfile.
CLONEFILENAME	VARCHAR2(513)	CloneDB file name. This file is the cloned file (of the snapshot file) which resides in the current (cloned database). The cloned file is sometimes referred to as the destfile.
SNAPSHOTBLKREAD	NUMBER	Number of blocks reads to the snapshot file
SNAPSHOTREQUEST	NUMBER	Number of read requests to the snapshot file
FILENUMBER	NUMBER	File number of the cloned file
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
BLOCKS_ALLOCATED	NUMBER	Amount of space allocated in blocks in the file system. This is less than or equal to the actual file size.

Note

When this view is queried in an Oracle Database environment, rows are returned for every opened file, even those without a parent file backing them (in those cases the column is empty/NULL).

In an Oracle ASM environement, rows are returned for files that an Oracle ASM instance has mounted in disk groups only if those files are children (a clonefile) of a parent snapshot file.

7.135 V$CLUSTER_INTERCONNECTS

V$CLUSTER_INTERCONNECTS displays one or more interconnects that are being used for cluster communication.

Column	Datatype	Description
NAME	VARCHAR2(15)	Name of the interconnect (such as eth0)
IP_ADDRESS	VARCHAR2(64)	IP address of the interconnect

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
IS_PUBLIC	VARCHAR2(3)	If the value is YES, the interface is known to the public. If the value is NO, the interface is known to be private. Note that if the CLUSTER_INTERCONNETS initialization parameter is also specified, then it is expected that the interconnect is private. Oracle expects cluster traffic to be run on private interconnects only. If the value is empty, it is unknown whether the interface is public or private. Oracle recommends that you set the interface for Oracle Real Application Clusters (Oracle RAC) communication in the Oracle Cluster Registry (OCR).
SOURCE	VARCHAR2(31)	Indicates where this interface was picked up from: • Oracle Cluster Registry - Interface was configured in the OCR and Oracle Database found the interface in the OCR • Operating-system dependent software - Oracle Database automatically detects this • CLUSTER_INTERCONNECTS parameter - This initialization parameter was set
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.136 V$CODE_CLAUSE

V$CODE_CLAUSE contains the supported clause names and parameter names for the Oracle Data Vault ALTER SYSTEM and ALTER SESSION commands.

Column	Datatype	Description
CODE_ID#	NUMBER	The OCT code ID: • 42: For ALTER SESSION • 49: For ALTER SYSTEM
CLAUSE_ID#	NUMBER	Clause ID
CLAUSE_NAME	VARCHAR2(100)	The clause name that is supported by the Oracle Data Vault command rule (for example, SET or ADVICE)
PARAMETER_NAME	VARCHAR2(100)	The parameter name that is supported by the Oracle Data Vault command rule. For example, for ALTER SYSTEM SET EVENTS, EVENTS is the parameter.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.137 V$CON_EVENT_HISTOGRAM_MICRO

V$CON_EVENT_HISTOGRAM_MICRO displays a histogram of the number of waits, the maximum wait, and total wait time on an event basis for a container, in microseconds. The histogram has buckets of time intervals from < 1 us, < 2 us, < 4 us, < 8 us, ... < 2^{31} us, < 2^{32} us, and >= 2^{32} us.

The histogram will not be filled unless the TIMED_STATISTICS initialization parameter is set to true.

Column	Datatype	Description
EVENT#	NUMBER	Event number
EVENT	VARCHAR2(64)	Name of the event
WAIT_TIME_FORMAT	VARCHAR2(30)	A human readable time string which is converted from WAIT_TIME_MICRO. When WAIT_TIME_MICRO < 1 millisecond, WAIT_TIME_FORMAT is shown in microseconds. When WAIT_TIME_MICRO < 1 second, WAIT_TIME_FORMAT is shown in milliseconds. When WAIT_TIME_MICRO < 1 minute, WAIT_TIME_FORMAT is shown in seconds. When WAIT_TIME_MICRO > 1 minute, WAIT_TIME_FORMAT is shown in minutes and seconds.
WAIT_TIME_MICRO	NUMBER	Amount of time the bucket represents (in microseconds). If the duration = num, then this column represents waits of duration < num that are not included in any smaller bucket.
WAIT_COUNT	NUMBER	Number of waits of the duration belonging to the bucket of the histogram
LAST_UPDATE_TIME	VARCHAR2(73)	Indicates the last time the bucket was updated (the ending timestamp of the last wait falling into the bucket's duration)
CON_ID	NUMBER	When queried from a non-CDB, the wait event data in that instance are returned, and the CON_ID value is 0. When queried from the root of a CDB, the wait event data in every container is returned, and the CON_ID value indicates the container to which the wait event data belong. When queried from a PDB, wait event data in that PDB are returned, and the CON_ID value is the container ID for that PDB.

7.138 V$CON_SYS_TIME_MODEL

V$CON_SYS_TIME_MODEL displays the systemwide accumulated times for various operations for the container from which it is queried.

The time reported is the total elapsed or CPU time (in microseconds). Any timed operation will buffer at most 5 seconds of time data. Specifically, this means that if a timed operation (such as SQL execution) takes a long period of time to perform, the data published to this view is at most missing 5 seconds of the time accumulated for the operation.

The time values are 8-byte integers and can therefore hold approximately 580,000 years worth of time before wrapping. Background process time is not included in a statistic value unless the statistic is specifically for background processes.

Column	Datatype	Description
STAT_ID	NUMBER	Statistic identifier for the time statistic

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STAT_NAME	VARCHAR2(64)	Name of the statistic (see Table 9-1)
VALUE	NUMBER	Amount of time (in microseconds) that the system has spent in this operation
CON_ID	NUMBER	When queried from a non-CDB, the accumulated times for operations in that instance are returned, and the CON_ID value is 0. When queried from the root of a CDB, accumulated times for operations in every container are returned, and the CON_ID value indicates the container to which the times belong. When queried from a PDB, accumulated times for operations in that PDB are returned, and the CON_ID value is the container ID for that PDB.

7.139 V$CON_SYSMETRIC

V$CON_SYSMETRIC displays the system metric values captured for the most current time interval for the PDB long duration (60-second) system metrics.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	Metric group ID
METRIC_ID	NUMBER	Metric ID
METRIC_NAME	VARCHAR2(64)	Metric name
VALUE	NUMBER	Metric value
METRIC_UNIT	VARCHAR2(64)	Metric unit description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • When queried from a non-CDB, the statistics for that instance are returned, and the CON_ID value is 0. • When queried from the root of a CDB, the statistics for every container are returned, and the CON_ID value indicates the container to which the statistics belong. • When queried from a PDB, statistics from that PDB are returned, and the CON_ID value is the container ID for that PDB.

See Also

- "DBA_HIST_CON_SYS_TIME_MODEL"
- "DBA_HIST_SYS_TIME_MODEL"
- "V$SYSMETRIC"

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.140 V$CON_SYSMETRIC_HISTORY

V$CON_SYSMETRIC_HISTORY displays all PDB long duration (60-second with 1 hour history) system metric values available in the database.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	Metric group ID
METRIC_ID	NUMBER	Metric ID
METRIC_NAME	VARCHAR2(64)	Metric name
VALUE	NUMBER	Metric value
METRIC_UNIT	VARCHAR2(64)	Metric unit description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • When queried from a non-CDB, the statistics for that instance are returned, and the CON_ID value is 0. • When queried from the root of a CDB, the statistics for every container are returned, and the CON_ID value indicates the container to which the statistics belong. • When queried from a PDB, statistics from that PDB are returned, and the CON_ID value is the container ID for that PDB.

See Also

- "DBA_HIST_CON_SYSMETRIC_HIST"
- "DBA_HIST_SYSMETRIC_HISTORY"
- "V$SYSMETRIC_HISTORY"

7.141 V$CON_SYSMETRIC_SUMMARY

V$CON_SYSMETRIC_SUMMARY displays a summary of all system metric values for the PDB long-duration system metrics. The average, maximum value, minimum value, and the value of one standard deviation for the last hour are displayed for each metric item.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	Metric group ID
METRIC_ID	NUMBER	Metric ID
METRIC_NAME	VARCHAR2(64)	Metric name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NUM_INTERVAL	NUMBER	Number of intervals observed
MAXVAL	NUMBER	Maximum value observed
MINVAL	NUMBER	Minimum value observed
AVERAGE	NUMBER	Average value over the period
STANDARD_DEVIATION	NUMBER	One standard deviation
METRIC_UNIT	VARCHAR2(64)	Metric unit description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • When queried from a non-CDB, the statistics for that instance are returned, and the CON_ID value is 0. • When queried from the root of a CDB, the statistics for every container are returned, and the CON_ID value indicates the container to which the statistics belong. • When queried from a PDB, statistics from that PDB are returned, and the CON_ID value is the container ID for that PDB.

See Also

- "V$SYSMETRIC_HISTORY"
- "DBA_HIST_SYSMETRIC_SUMMARY"
- "V$SYSMETRIC_SUMMARY"

7.142 V$CON_SYSSTAT

V$CON_SYSSTAT displays system statistics, including OLAP kernel statistics for the container from which it is queried. To find the name of the statistic associated with each statistic number (STATISTIC#), query the V$STATNAME view.

Column	Datatype	Description
STATISTIC#	NUMBER	Statistic number **Note:** Statistics numbers are not guaranteed to remain constant from one release to another. Therefore, you should rely on the statistics name rather than its number in your applications.
NAME	VARCHAR2(64)	Statistic name. You can get a complete listing of statistic names by querying the V$STATNAME view.
CLASS	NUMBER	A number representing one or more statistics class. The following class numbers are additive: • 1 - User • 2 - Redo • 4 - Enqueue • 8 - Cache • 16 - OS • 32 - Real Application Clusters • 64 - SQL • 128 - Debug

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
VALUE	NUMBER	Statistic value
STAT_ID	NUMBER	Identifier of the statistic
CON_ID	NUMBER	When queried from a non-CDB, the statistics in that instance are returned, and the CON_ID value is 0. When queried from the root of a CDB, the statistics in every container are returned, and the CON_ID value indicates the container to which the statistics belong. When queried from a PDB, statistics in that PDB are returned, and the CON_ID value is the container ID for that PDB.

See Also

"V$STATNAME" and " Statistics Descriptions"

7.143 V$CON_SYSTEM_EVENT

V$CON_SYSTEM_EVENT displays information on total waits for an event in a container.

Note that the TIME_WAITED and AVERAGE_WAIT columns will contain a value of zero on those platforms that do not support a fast timing mechanism. If you are running on one of these platforms and you want this column to reflect true wait times, then you must set TIMED_STATISTICS to TRUE in the parameter file; doing this will have a small negative effect on system performance.

See Also

"TIMED_STATISTICS"

Column	Datatype	Description
EVENT	VARCHAR2(64)	Name of the wait event; derived statistic name from V$EVENT_NAME **See Also:** " Oracle Wait Events"
TOTAL_WAITS	NUMBER	Total number of waits for the event
TOTAL_TIMEOUTS	NUMBER	Total number of timeouts for the event
TIME_WAITED	NUMBER	Total amount of time waited for the event (in hundredths of a second)
AVERAGE_WAIT	NUMBER	Average amount of time waited for the event (in hundredths of a second)
TIME_WAITED_MICRO	NUMBER	Total amount of time waited for the event (in microseconds)
TOTAL_WAITS_FG	NUMBER	Total number of waits for the event, from foreground sessions
TOTAL_TIMEOUTS_FG	NUMBER	Total number of timeouts for the event, from foreground sessions
TIME_WAITED_FG	NUMBER	Total amount of time waited for the event, from foreground sessions (in hundredths of a second)
AVERAGE_WAIT_FG	NUMBER	Average amount of time waited for the event, from foreground sessions (in hundredths of a second)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TIME_WAITED_MICRO_FG	NUMBER	Total amount of time waited for the event, from foreground sessions (in microseconds)
EVENT_ID	NUMBER	Identifier of the wait event
WAIT_CLASS_ID	NUMBER	Identifier of the wait class of the wait event
WAIT_CLASS#	NUMBER	Number of the wait class of the wait event
WAIT_CLASS	VARCHAR2(64)	Name of the wait class of the wait event
CON_ID	NUMBER	When queried from a non-CDB, the wait event data in that instance are returned, and the CON_ID value is 0. When queried from the root of a CDB, the wait event data in every container is returned, and the CON_ID value indicates the container to which the wait event data belong. When queried from a PDB, wait event data in that PDB are returned, and the CON_ID value is the container ID for that PDB.

7.144 V$CON_SYSTEM_WAIT_CLASS

V$CON_SYSTEM_WAIT_CLASS displays the time totals for each registered wait class in a container.

Column	Datatype	Description
WAIT_CLASS_ID	NUMBER	Identifier of the wait class
WAIT_CLASS#	NUMBER	Number of the wait class
WAIT_CLASS	VARCHAR2(64)	Name of the wait class
TOTAL_WAITS	NUMBER	Total number of waits from this wait class
TIME_WAITED	NUMBER	Total amount of time spent in waits from this wait class (in hundredths of a second)
TOTAL_WAITS_FG	NUMBER	Total number of waits from this wait class, from foreground sessions
TIME_WAITED_FG	NUMBER	Total amount of time spent in waits from this wait class, from foreground sessions (in hundredths of a second)
CON_ID	NUMBER	When queried from a non-CDB, time totals for each registered wait class in that instance are returned, and the CON_ID value is 0. When queried from the root of a CDB, time totals for each registered wait class in every container is returned, and the CON_ID value indicates the container to which the time total belong. When queried from a PDB, time totals for each registered wait class in that PDB are returned, and the CON_ID value is the container ID for that PDB.

7.145 V$CONFIGURED_INTERCONNECTS

V$CONFIGURED_INTERCONNECTS displays all the interconnects that Oracle is aware of. This view attempts to answer the question of where Oracle found the information about a specific interconnect.

Column	Datatype	Description
NAME	VARCHAR2(15)	Name of the interconnect (such as eth0)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
IP_ADDRESS	VARCHAR2(64)	IP address of the interconnect
IS_PUBLIC	VARCHAR2(3)	If the value is YES, the interface is known to the public. If the value is NO, the interface is known to be private. Note that if the CLUSTER_INTERCONNECTS initialization parameter is also specified, then it is expected that the interconnect is private. Oracle expects cluster traffic to be run on private interconnects only. If the value is empty, it is unknown whether the interface is public or private. Oracle recommends that you set the interface for Oracle Real Application Clusters (Oracle RAC) communication in the Oracle Cluster Registry (OCR).
SOURCE	VARCHAR2(31)	Indicates where this interface was picked up from: • Oracle Cluster Registry - Interface was configured in the OCR and Oracle Database found the interface in the OCR • Operating-system dependent software - Oracle Database automatically detects this • CLUSTER_INTERCONNECTS parameter - This initialization parameter was set
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.146 V$CONTAINERS

V$CONTAINERS displays information about PDBs and the root associated with the current instance.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
DBID	NUMBER	PDB identifier calculated when the PDB is created and stored in all file headers associated with the PDB
CON_UID	NUMBER	Unique identifier associated with the PDB
GUID	RAW(16)	Globally unique identifier (GUID) of this PDB
NAME	VARCHAR2(128)	Name of the PDB

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OPEN_MODE	VARCHAR2(10)	Open mode information. Possible values: • MOUNTED • READ WRITE • READ ONLY • MIGRATE
RESTRICTED	VARCHAR2(3)	Indicates whether only users possessing RESTRICTED SESSION privilege can connect to the PDB
OPEN_TIME	TIMESTAMP(3)	Date and time when the database was last opened
CREATE_SCN	NUMBER	System change number (SCN) for the creation of this PDB
TOTAL_SIZE	NUMBER	If a PDB is opened, disk space (in bytes) used by the container, including both data and temp files. If a PDB is closed, will be set to 0.
BLOCK_SIZE	NUMBER	The current block size for the PDB
RECOVERY_STATUS	VARCHAR2(8)	Shows whether recovery is enabled or disabled for the PDB. Possible values: • ENABLED • DISABLED
SNAPSHOT_PARENT_CON_ID	NUMBER	This column shows the container ID of the PDB that this PDB is a snapshot clone of. This column shows a nonzero value only if the PDB is a snapshot clone. For all other cases, it shows a value of 0.
APPLICATION_ROOT	VARCHAR2(3)	Indicates whether the PDB is an application root
APPLICATION_PDB	VARCHAR2(3)	Indicates whether the PDB is an application PDB
APPLICATION_SEED	VARCHAR2(3)	Indicates whether the PDB is an application seed (an application seed is also an application PDB)
APPLICATION_ROOT_CON_ID	NUMBER	If this PDB is an application PDB, the container ID of an application root to which this application PDB belongs. If this PDB is an application root clone, the container ID of an application root to which this application root clone belongs. Otherwise, NULL.
APPLICATION_ROOT_CLONE	VARCHAR2(3)	Indicates whether this PDB is an application root clone (YES) or not (NO)
PROXY_PDB	VARCHAR2(3)	Indicates whether this PDB is a proxy PDB (YES) or not (NO)
LOCAL_UNDO	NUMBER	Shows whether the PDB is in local undo. Possible values: • 1 – PDB is in local undo mode • 0 – PDB is in shared undo mode This column is not relevant for CDB$ROOT.
UNDO_SCN	NUMBER	System change number (SCN) at which the PDB was last converted from shared to local undo, or from local to shared undo. This column is not relevant for CDB$ROOT.
UNDO_TIMESTAMP	DATE	Date and time at which the PDB was last converted from shared to local undo, or from local to shared undo. This column is not relevant for CDB$ROOT.
CREATION_TIME	DATE	Date and time at which the PDB was created.
PDB_COUNT	NUMBER	The number of user-created PDBs belonging to a given application root or CDB$ROOT. For all other containers, its value is 0.
AUDIT_FILES_SIZE	NUMBER	Shows the current disk space usage (in bytes) by Unified Audit files (.bin format) in the container
MAX_SIZE	NUMBER	Shows the maximum amount of disk space (in bytes) that can be used by data and temp files in the container

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MAX_DIAGNOSTICS_SIZE	NUMBER	Shows the maximum amount of disk space (in bytes) that can be used by diagnostic traces generated in the container
MAX_AUDIT_SIZE	NUMBER	Shows the maximum amount of disk space (in bytes) that can be used by Unified Audit files (.bin format) in the container
LAST_CHANGED_BY	VARCHAR2(11)	Indicates what type of user last changed the PDB. Possible values: • COMMON USER • LOCAL USER
MEMBER_CDB	VARCHAR2(3)	Indicates whether the row corresponds to a Member CDB part of the CDB Fleet (YES) or not (NO). This column is only meaningful in the Lead CDB of a CDB Fleet, which keeps track of other CDBs as containers in the CDB Fleet.
TENANT_ID	VARCHAR2(256)	Pluggable database tenant key
UPGRADE_LEVEL	NUMBER	For internal use only
GUID_BASE64	VARCHAR2(30)	The GUID of the PDB, encoded in base64

7.147 V$CONTEXT

V$CONTEXT displays set attributes in the current session.

Column	Datatype	Description
NAMESPACE	VARCHAR2(31)	Namespace that the attribute is in
ATTRIBUTE	VARCHAR2(31)	Name of the attribute
VALUE	VARCHAR2(4000)	Value of the attribute
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.148 V$CONTROLFILE

V$CONTROLFILE displays the names of the control files.

Column	Datatype	Description
STATUS	VARCHAR2(7)	INVALID if the name cannot be determined (which should not occur); NULL if the name can be determined
NAME	VARCHAR2(513)	Name of the control file
IS_RECOVERY_DEST_FILE	VARCHAR2(3)	Indicates whether the file was created in the fast recovery area (YES) or not (NO)
BLOCK_SIZE	NUMBER	Control file block size
FILE_SIZE_BLKS	NUMBER	Control file size (in blocks)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.149 V$CONTROLFILE_RECORD_SECTION

V$CONTROLFILE_RECORD_SECTION displays information about the control file record sections.

Column	Datatype	Description
TYPE	VARCHAR2(28)	Identifies the type of record section: • DATABASE • CKPT PROGRESS • REDO THREAD • REDO LOG • DATAFILE • FILENAME • TABLESPACE • TEMPORARY FILENAME • RMAN CONFIGURATION • LOG HISTORY • OFFLINE RANGE • ARCHIVED LOG • BACKUP SET • BACKUP PIECE • BACKUP DATAFILE • BACKUP REDOLOG • DATAFILE COPY • BACKUP CORRUPTION • COPY CORRUPTION • DELETED OBJECT • PROXY COPY • BACKUP SPFILE • DATABASE INCARNATION • FLASHBACK LOG • RECOVERY DESTINATION • INSTANCE SPACE RESERVATION • REMOVABLE RECOVERY FILES • RMAN STATUS • THREAD INSTANCE NAME MAPPING • MTTR • DATAFILE HISTORY • PLUGGED IN DATAFILE
RECORD_SIZE	NUMBER	Record size in bytes
RECORDS_TOTAL	NUMBER	Number of records allocated for the section

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RECORDS_USED	NUMBER	Number of records used in the section
FIRST_INDEX	NUMBER	Index (position) of the first record
LAST_INDEX	NUMBER	Index of the last record
LAST_RECID	NUMBER	Record ID of the last record
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.150 V$COPY_CORRUPTION

V$COPY_CORRUPTION displays information about data file copy corruptions from the control file.

Column	Datatype	Description
RECID	NUMBER	Copy corruption record ID
STAMP	NUMBER	Copy corruption record stamp
COPY_RECID	NUMBER	Data file copy record ID
COPY_STAMP	NUMBER	Data file copy record stamp
FILE#	NUMBER	Data file number
BLOCK#	NUMBER	First block of the corrupted range
BLOCKS	NUMBER	Number of contiguous blocks in the corrupted range
CORRUPTION_CHANGE#	NUMBER	Change number at which the logical corruption was detected. Set to 0 to indicate media corruption.
MARKED_CORRUPT	VARCHAR2(3)	(YES | NO) If set to YES the blocks were not marked corrupted in the data file, but were detected and marked as corrupted while making the data file copy
CORRUPTION_TYPE	VARCHAR2(9)	Type of block corruption in the data file: • ALL ZERO - Block header on disk contained only zeros. The block may be valid if it was never filled and if it is in an Oracle7 file. The buffer will be reformatted to the Oracle8 standard for an empty block. • FRACTURED - Block header looks reasonable, but the front and back of the block are different versions. • CHECKSUM - optional check value shows that the block is not self-consistent. It is impossible to determine exactly why the check value fails, but it probably fails because sectors in the middle of the block are from different versions. • CORRUPT - Block is wrongly identified or is not a data block (for example, the data block address is missing) • LOGICAL - Block is logically corrupt • NOLOGGING - Block does not have redo log entries (for example, NOLOGGING operations on primary database can introduce this type of corruption on a physical standby)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.151 V$COPY_NONLOGGED

V$COPY_NONLOGGED displays information about nonlogged block ranges in data file copy blocks, recorded in the control file.

Column	Datatype	Description
INST_ID	NUMBER	Instance ID
RECID	NUMBER	Nonlogged copy record ID
STAMP	NUMBER	Nonlogged copy record stamp
COPY_RECID	NUMBER	Data file copy record ID
COPY_STAMP	NUMBER	Data file copy record stamp
FILE#	NUMBER	Absolute file number of the data file that contains this range of nonlogged blocks
BLOCK#	NUMBER	Block number of the first nonlogged block in the range of nologged blocks
BLOCKS	NUMBER	Number of nonlogged blocks found starting with BLOCK#
NONLOGGED_CHANGE#	NUMBER	The smallest SCN on which any block in this block range became nonlogged. NULL if unknown.
NONLOGGED_TIME	VARCHAR2	The time that corresponds to NONLOGGED_CHANGE#. NULL if unknown.
RESETLOGS_CHANGE#	VARCHAR2	The resetlogs SCN of the incarnation on which this block range was first marked as nonlogged. NULL if unknown.
RESETLOGS_TIME	VARCHAR2	The resetlogs time of the incarnation on which this block range was first marked as nologged. NULL if unknown.
OBJECT#	VARCHAR2	The object ID this range belongs to. If this field is NULL, the object number is unknown.
REASON	CHAR(7)	The reason why this block range appears in this list, for example, primary file offline, could not talk to primary, non-standby recovery, and so on. For Oracle Database 12*c* and later releases, it is always UNKNOWN.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.152 V$CORRUPT_XID_LIST

V$CORRUPT_XID_LIST displays all corrupted XIDs.

Column	Datatype	Description
CORRUPT_XID	VARCHAR2(256)	Name of corrupt XID
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.153 V$CPOOL_CC_INFO

V$CPOOL_CC_INFO displays information about the pool-to-connection class mapping for the Database Resident Connection Pool per instance.

Column	Datatype	Description
POOL_NAME	VARCHAR2(1024)	Name of the Database Resident Connection Pool
CCLASS_NAME	VARCHAR2(1024)	Name of the connection class
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.154 V$CPOOL_CC_STATS

V$CPOOL_CC_STATS displays information about the connection class level statistics for the Database Resident Connection Pool per instance.

Column	Datatype	Description
CCLASS_NAME	VARCHAR2(1024)	Name of the connection class
NUM_REQUESTS	NUMBER	Number of session requests
NUM_HITS	NUMBER	Total number of times a session that matches with the request was found in the pool
NUM_MISSES	NUMBER	Total number of times an exact match to the request was not found in the pool and a new session had to be created
NUM_WAITS	NUMBER	Total number of times session requests had to wait before getting served
WAIT_TIME	NUMBER	Reserved for future use

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CLIENT_REQ_TIMEOUTS	NUMBER	Reserved for future use
NUM_AUTHENTICATIONS	NUMBER	Total number of authentications of clients done by the pool
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **Note**
>
> In a multitenant container database (CDB), this view returns data only when queried from a CDB root. When queried from a PDB, this view returns 0 rows.

7.155 V$CPOOL_CONN_INFO

V$CPOOL_CONN_INFO displays connection information about each connection to the connection broker.

Column	Datatype	Description
CMON_ADDR	RAW(4 | 8)	Address of the connection broker
SESSION_ADDR	RAW(4 | 8)	Address of the session associated with the connection; NULL if the connection does not have an associated session
CONNECTION_ADDR	RAW(4 | 8)	Address of the connection
USERNAME	VARCHAR2(1024)	Name of the user associated with the connection
PROXY_USER	VARCHAR2(1024)	Name of the proxy user
CCLASS_NAME	VARCHAR2(1024)	Connection class associated with the connection
PURITY	VARCHAR2(1024)	Purity used to create the connection (can be SELF or NEW)
TAG	VARCHAR2(1024)	Tag, if specified, at connection creation time
SERVICE	VARCHAR2(64)	TNS service name for the connection
PROCESS_ID	VARCHAR2(24)	Client process ID of the process which created the connection
PROGRAM	VARCHAR2(48)	Program name of the client process which created the connection
MACHINE	VARCHAR2(64)	Machine name of the client process which created the connection
TERMINAL	VARCHAR2(30)	Terminal identifier of the client process which created the connection
CONNECTION_MODE	VARCHAR2(1024)	Reserved for internal use

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CONNECTION_STATUS	VARCHAR2(10)	Status of the connection: • NONE • CONNECTING • ACTIVE: A pooled server has been mapped to this connection. • WAITING: The connection is waiting for a pooled server based on the client request. • IDLE: No pooled server has been mapped to this connection and there is no request to map one. • CLOSING
CLIENT_REGID	NUMBER	Query cache registration ID sent by the client
CURSTATUS_TIME	NUMBER	Time in microseconds spent in the current state. See CONNECTION_STATUS above.
IDLE_TIME	NUMBER	Total time in IDLE state for the connection (in microseconds)
ACTIVE_TIME	NUMBER	Total time in ACTIVE state for the connection (in microseconds)
WAIT_TIME	NUMBER	Total time in WAITING state for the connection (in microseconds)
THINK_TIME	NUMBER	Total think time for the connection assigned to the pooled server, but not doing any database activity (in microseconds)
LAST_IDLE_TIME	NUMBER	Time in last IDLE state for this connection (in microseconds)
LAST_ACTIVE_TIME	NUMBER	Time in last ACTIVE state for this connection (in microseconds)
LAST_WAIT_TIME	NUMBER	Time in last WAITING state for this connection (in microseconds)
LAST_THINK_TIME	NUMBER	Time for the connection assigned to the pooled server, but not doing any database activity during the last ACTIVE state (in microseconds)
NUMGETS	NUMBER	Total number of requests at the connection level
NUMHITS	NUMBER	Total number of hits at the connection level
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.156 V$CPOOL_STATS

V$CPOOL_STATS displays information about the Database Resident Connection Pool statistics for an instance.

Column	Datatype	Description
POOL_NAME	VARCHAR2(1024)	Name of the Database Resident Connection Pool
NUM_OPEN_SERVERS	NUMBER	Total number of busy and free servers in the pool (including the authentication servers)
NUM_BUSY_SERVERS	NUMBER	Total number of busy servers in the pool (not including the authentication servers)
NUM_AUTH_SERVERS	NUMBER	Number of authentication servers in the pool
NUM_REQUESTS	NUMBER	Number of client requests

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NUM_HITS	NUMBER	Total number of times client requests found matching pooled servers in the pool
NUM_MISSES	NUMBER	Total number of times client requests could not find a matching pooled server in the pool
NUM_WAITS	NUMBER	Total number of client requests that had to wait due to non-availability of free pooled servers
WAIT_TIME	NUMBER	Reserved for future use
CLIENT_REQ_TIMEOUTS	NUMBER	Reserved for future use
NUM_AUTHENTICATIONS	NUMBER	Total number of authentications of clients done by the pool
NUM_PURGED	NUMBER	Total number of sessions purged by the pool
HISTORIC_MAX	NUMBER	Maximum size that the pool has ever reached
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **Note**
>
> In a multitenant container database (CDB), this view returns data only when queried from a CDB root. When queried from a PDB, this view returns 0 rows.

7.157 V$CR_BLOCK_SERVER

V$CR_BLOCK_SERVER displays statistics on the Global Cache Service processes (LMS) used in cache fusion.

Column	Datatype	Description
CR_REQUESTS	NUMBER	Number of CR blocks served due to remote CR block requests
CURRENT_REQUESTS	NUMBER	Number of current blocks served due to remote CR block requests CR_REQUESTS + CURRENT_REQUESTS = global cache CR clocks served (from V$SYSSTAT).
DATA_REQUESTS	NUMBER	Number of current or CR requests for data blocks
UNDO_REQUESTS	NUMBER	Number of CR requests for undo blocks
TX_REQUESTS	NUMBER	Number of CR requests for undo segment header blocks
OTHER_REQUESTS	NUMBER	Number of CR requests for other types of blocks DATA_REQUESTS + UNDO_REQUESTS + TX_REQUESTS + OTHER_REQUESTS = total number of requests handled by the LMS processes
CURRENT_RESULTS	NUMBER	Number of requests for which no changes were rolled out of the block returned to the requesting instance

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PRIVATE_RESULTS	NUMBER	Number of requests for which changes were rolled out of the block returned to the requesting instance, and only the requesting transaction can use the resulting CR block
ZERO_RESULTS	NUMBER	Number of requests for which changes were rolled out of the block returned to the requesting instance. Only zero-XID transactions can use the block.
DISK_READ_RESULTS	NUMBER	Number of requests for which the requesting instance had to read the requested block from disk
FAIL_RESULTS	NUMBER	Number of requests that failed; the requesting transaction must reissue the request
STALE	NUMBER	Number of requests for which the disk read of the requested block was stale
FAIRNESS_DOWN_CONVERTS	NUMBER	Number of times an instance receiving a request has down-converted an X lock on a block because it was not modifying the block
FAIRNESS_CLEARS	NUMBER	Number of times the "fairness counter" was cleared. This counter tracks the number of times a block was modified after it was served.
FREE_GC_ELEMENTS	NUMBER	Number of times a request was received from another instance and the X lock had no buffers
FLUSHES	NUMBER	Number of times the log has been flushed by an LMS process
FLUSHES_QUEUED	NUMBER	Number of flushes queued by an LMS process
FLUSH_QUEUE_FULL	NUMBER	Number of times the flush queue was full
FLUSH_MAX_TIME	NUMBER	Maximum time for flush
LIGHT_WORKS	NUMBER	Number of times the light-work rule was evoked. This rule prevents the LMS processes from going to disk while responding to CR requests for data, undo, or undo segment header blocks. This rule can prevent the LMS process from completing its response to the CR request.
ERRORS	NUMBER	Number of times an error was signalled by an LMS process
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **Note**
>
> This view contains internal diagnostic information for use by Oracle Support Services. It is subject to change without notice.

7.158 V$CURRENT_BLOCK_SERVER

V$CURRENT_BLOCK_SERVER displays statistics on the Global Cache Service processes (LMS) used in cache fusion.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PIN0	NUMBER	Pins taking less than 100 microseconds
PIN1	NUMBER	Pins taking 100 microseconds to 1 millisecond
PIN10	NUMBER	Pins taking 1 to 10 milliseconds
PIN100	NUMBER	Pins taking 10 to 100 milliseconds
PIN1000	NUMBER	Pins taking 100 to 1000 milliseconds
PIN10000	NUMBER	Pins taking 1000 to 10000 milliseconds
FLUSH0	NUMBER	Flushes taking less than 100 microseconds
FLUSH1	NUMBER	Flushes taking 100 microseconds to 1 millisecond
FLUSH10	NUMBER	Flushes taking 1 to 10 milliseconds
FLUSH100	NUMBER	Flushes taking 10 to 100 milliseconds
FLUSH1000	NUMBER	Flushes taking 100 to 1000 milliseconds
FLUSH10000	NUMBER	Flushes taking 1000 to 10000 milliseconds
WRITE1[1]	NUMBER	Writes taking less than 1 millisecond
WRITE10[1]	NUMBER	Writes taking 1 to 10 milliseconds
WRITE100[1]	NUMBER	Writes taking 10 to 100 milliseconds
WRITE1000[1]	NUMBER	Writes taking 100 to 1000 milliseconds
WRITE10000[1]	NUMBER	Writes taking 1000 to 10000 milliseconds
CLEANDC	NUMBER	Reserved for internal use
RCVDC	NUMBER	Number of lock down-converts to S (shared) caused by instance recovery
QUEUEDC	NUMBER	Number of queued lock down-converts to NULL
EVICTDC	NUMBER	Number of lock down-converts to NULL caused by an SGA shrink
WRITEDC	NUMBER	Number of dirty blocks in read-mostly objects which were written and the X (exclusive) lock down-converted to S (shared) locks
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

1 This column is deprecated in Oracle Database 12*c* Release 2 (12.2.0.1) and may be removed in a future release.

7.159 V$DATABASE

V$DATABASE displays information about the database from the control file.

Column	Datatype	Description
DBID	NUMBER	Database identifier calculated when the database is created and stored in all file headers
NAME	VARCHAR2(9)	Name of the database

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CREATED	DATE	Creation date of the database. If the control file was re-created using the CREATE CONTROLFILE statement, then this column displays the date that the control file was re-created.
RESETLOGS_CHANGE#	NUMBER	System change number (SCN) at open resetlogs
RESETLOGS_TIME	DATE	Timestamp of open resetlogs
PRIOR_RESETLOGS_CHANGE#	NUMBER	SCN at prior resetlogs
PRIOR_RESETLOGS_TIME	DATE	Timestamp of prior resetlogs
LOG_MODE	VARCHAR2(12)	Archive log mode: • NOARCHIVELOG • ARCHIVELOG • MANUAL
CHECKPOINT_CHANGE#	NUMBER	Last SCN checkpointed
ARCHIVE_CHANGE#	NUMBER	Database force archiving SCN. Any redo log with a start SCN below this will be forced to archive out.
CONTROLFILE_TYPE	VARCHAR2(7)	Type of control file: • BACKUP | CREATED - Indicates the database is being recovered using a backup or created control file • CLONE - Indicates a clone database • CURRENT - Indicates that the database is available for general use • FARSYNC - Indicates that the instance is a Far Sync instance • STANDBY - Indicates that the database is in standby mode
CONTROLFILE_CREATED	DATE	Creation date of the control file
CONTROLFILE_SEQUENCE#	NUMBER	Control file sequence number incremented by control file transactions
CONTROLFILE_CHANGE#	NUMBER	Last SCN in backup control file; null if the control file is not a backup
CONTROLFILE_TIME	DATE	Last timestamp in backup control file; null if the control file is not a backup
OPEN_RESETLOGS	VARCHAR2(11)	(NOT ALLOWED | ALLOWED | REQUIRED) Indicates whether the next database open allows or requires the resetlogs option
VERSION_TIME	DATE	Version time
OPEN_MODE	VARCHAR2(20)	Open mode information: • MOUNTED • READ WRITE • READ ONLY • READ ONLY WITH APPLY - A physical standby database is open in real-time query mode
PROTECTION_MODE	VARCHAR2(20)	Protection mode currently in effect for the database: • MAXIMUM PROTECTION - Database is running in maximized protection mode • MAXIMUM AVAILABILITY - Database is running in maximized availability mode • RESYNCHRONIZATION - Database is running in resynchronization mode • MAXIMUM PERFORMANCE - Database is running in maximized performance mode • UNPROTECTED - Database is unprotected (this normally occurs when the primary database is mounted and not open)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PROTECTION_LEVEL	VARCHAR2(20)	Aggregated protection mode currently in effect for the database: • MAXIMUM PROTECTION - Database is running in maximized protection mode • MAXIMUM AVAILABILITY - Database is running in maximized availability mode • RESYNCHRONIZATION - Database is running in resynchronization mode • MAXIMUM PERFORMANCE - Database is running in maximized performance mode • UNPROTECTED - Database is unprotected (this normally occurs when the primary database is mounted and not open) **Note:** This column is an aggregation of the PROTECTION_MODE of all standby archive log destinations.
REMOTE_ARCHIVE	VARCHAR2(8)	Value of the REMOTE_ARCHIVE_ENABLE initialization parameter
ACTIVATION#	NUMBER	Number assigned to the database instantiation
SWITCHOVER#	NUMBER	Number assigned to the database switchover
DATABASE_ROLE	VARCHAR2(16)	Current role of the database: • SNAPSHOT STANDBY • LOGICAL STANDBY • PHYSICAL STANDBY • PRIMARY • FAR SYNC
ARCHIVELOG_CHANGE#	NUMBER	Highest NEXT_CHANGE# (from the V$ARCHIVED_LOG view) for an archive log
ARCHIVELOG_COMPRESSION	VARCHAR2(8)	Status of the archive log compression (ENABLED) or (DISABLED)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SWITCHOVER_STATUS	VARCHAR2(20)	Indicates whether switchover is allowed: • NOT ALLOWED - On a primary database, this status indicates that there are no valid and enabled standby databases. On a standby database, this status indicates that a switchover request has not been received from the primary database. • SESSIONS ACTIVE - The database has active sessions. On a physical standby database, the WITH SESSION SHUTDOWN SQL clause must be specified to perform a role transition while in this state. On a logical standby database, a role transition can be performed while in this state, but the role transition will not complete until all current transactions have committed. • SWITCHOVER PENDING - On a physical standby database, this status indicates that a switchover request has been received from the primary database and is being processed. A physical standby database cannot switch to the primary role while in this transient state. • SWITCHOVER LATENT - On a physical standby database, this status indicates that a switchover request was pending, but the original primary database has been switched back to the primary role. • TO PRIMARY - The database is ready to switch to the primary role. • TO STANDBY - The database is ready to switch to either the physical or logical standby role. • TO LOGICAL STANDBY - The database has received a data dictionary from a logical standby database and is ready to switch to the logical standby role. • RECOVERY NEEDED - On a physical standby database, this status indicates that additional redo must be applied before the database can switch to the primary role. • PREPARING SWITCHOVER - On a primary database, this status indicates that a data dictionary is being received from a logical standby database in preparation for switching to the logical standby role. On a logical standby database, this status indicates that the data dictionary has been sent to the primary database and other standby databases. • PREPARING DICTIONARY - On a logical standby database, this status indicates that the data dictionary is being sent to the primary database and other standby databases in preparation for switching to the primary role. • FAILED DESTINATION - On a primary database, this status indicates that one or more standby destinations are in an error state. • RESOLVABLE GAP - On a primary database, this status indicates that one or more standby databases have a redo gap that can be automatically resolved by fetching the missing redo from the primary database or from another standby database. • UNRESOLVABLE GAP - On a primary database, this status indicates that one or more standby databases have a redo gap that cannot be automatically resolved by fetching the missing redo from the primary database or from another standby database. • LOG SWITCH GAP - On a primary database, this status indicates that one or more standby databases are missing redo due to a recent log switch.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DATAGUARD_BROKER	VARCHAR2(8)	Data Guard broker information: • ENABLED - Database is part of a broker configuration and broker management of the database is enabled • DISABLED - Database is part of a broker configuration and broker management of the database is disabled. This value is displayed if the user disabled broker management of the database or configuration, or if broker management was disabled due to a role change (for example, the old primary was disabled after a failover operation).
GUARD_STATUS	VARCHAR2(7)	Protects data from being changed: • ALL - Indicates all users other than SYS are prevented from making changes to any data in the database. • STANDBY - Indicates all users other than SYS are prevented from making changes to any database object being maintained by logical standby. • NONE - Indicates normal security for all data in the database.
SUPPLEMENTAL_LOG_DATA_MIN	VARCHAR2(8)	Ensures that LogMiner (and any products building on LogMiner technology) will have sufficient information to support chained rows and various storage arrangements such as cluster tables: • NO - None of the database-wide supplemental logging directives are enabled. In a CDB, a value of NO means that minimal supplemental logging is not enabled in all of the PDBs in the CDB. • IMPLICIT - Minimal supplemental logging is enabled because all or a combination of primary key, unique key, and foreign key supplemental logging is enabled • YES - Minimal supplemental logging is enabled through an ALTER DATABASE ADD SUPPLEMENTAL LOG DATA statement. In a CDB, a value of YES means that minimal supplemental logging is enabled in all of the PDBs in the CDB. **See Also:** *Oracle Database SQL Language Reference* for additional information about the ALTER DATABASE ADD SUPPLEMENTAL LOG DATA statement
SUPPLEMENTAL_LOG_DATA_PK	VARCHAR2(3)	For all tables with a primary key, indicates whether all columns of the primary key are placed into the redo log whenever an update is performed (YES) or not (NO). When a value of YES appears in a CDB, it means that primary key supplemental logging is enabled in all of the PDBs in the CDB. When a value of NO appears in a CDB, query the PRIMARY_KEY column in the DBA_SUPPLEMENTAL_LOGGING view for each PDB in the CDB to see whether primary key supplemental logging is enabled in the PDB. **See Also:** *Oracle Database SQL Language Reference* for more information about the ALTER DATABASE ADD SUPPLEMENTAL LOG *supplemental_id_key_clause* statement

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SUPPLEMENTAL_LOG_DATA_UI	VARCHAR2(3)	For all tables with a unique key, indicates whether all other columns belonging to the unique key are placed into the redo log if any of the unique key columns are modified (YES) or not (NO). When a value of YES appears in a CDB, it means that this value is enabled in all of the PDBs in the CDB. When a value of NO appears in a CDB, query the UNIQUE_INDEX column in the DBA_SUPPLEMENTAL_LOGGING view for each PDB in the CDB to see whether unique column supplemental logging is enabled in the PDB. **See Also:** *Oracle Database SQL Language Reference* for more information about the ALTER DATABASE ADD SUPPLEMENTAL LOG *supplemental_id_key_clause* statement
FORCE_LOGGING	VARCHAR2(39)	Indicates the type of logging mode that is currently in force. The valid values and their meanings are: • NO - This value means that no logging mode has been enabled for the database • YES - This value means that FORCE LOGGING mode has been enabled for the database • STANDBY NOLOGGING FOR LOAD PERFORMANCE - This value is used when this is the current mode for the database • STANDBY NOLOGGING FOR DATA AVAILABILITY - This value is used when this is the current mode for the database
PLATFORM_ID	NUMBER	Platform identification number of the database
PLATFORM_NAME	VARCHAR2(101)	Platform name of the database
RECOVERY_TARGET_INCARNATION#	NUMBER	Incarnation number where all data files are recovered by the RECOVER DATABASE command
LAST_OPEN_INCARNATION#	NUMBER	Record number of the incarnation in V$DATABASE_INCARNATION that was last opened successfully
CURRENT_SCN	NUMBER	Current SCN; null if the database is not currently open. For a standby database, it is the checkpoint SCN of the mounted physical standby database during media recovery and is always less than the last applied SCN tracked in V$RECOVERY_PROGRESS.
FLASHBACK_ON	VARCHAR2(18)	Possible values are as follows: • YES - Flashback is on • NO - Flashback is off • RESTORE POINT ONLY - Flashback is on but one can only flashback to guaranteed restore points
SUPPLEMENTAL_LOG_DATA_FK	VARCHAR2(3)	For all tables with a foreign key, indicates whether all other columns belonging to the foreign key are placed into the redo log if any foreign key columns are modified (YES) or not (NO). When a value of YES appears in a CDB, it means that foreign key supplemental logging is enabled in all of the PDBs in the CDB. When a value of NO appears in a CDB, query the FOREIGN_KEY column in the DBA_SUPPLEMENTAL_LOGGING view for each PDB in the CDB to see whether foreign key supplemental logging is enabled in the PDB. **See Also:** *Oracle Database SQL Language Reference* for more information about the ALTER DATABASE ADD SUPPLEMENTAL LOG *supplemental_id_key_clause* statement

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SUPPLEMENTAL_LOG_DATA_ALL	VARCHAR2(3)	For all columns, indicates whether all the fixed-length maximum size columns of that row are placed into the redo log (YES) or not (NO). When a value of YES appears in a CDB, it means that all column supplemental logging is enabled in all of the PDBs in the CDB. When a value of NO appears in a CDB, query the ALL_COLUMN column in the DBA_SUPPLEMENTAL_LOGGING view for each PDB in the CDB to see whether all column supplemental logging is enabled in the PDB. **See Also:** *Oracle Database SQL Language Reference* for more information about the ALTER DATABASE ADD SUPPLEMENTAL LOG *supplemental_id_key_clause* statement
DB_UNIQUE_NAME	VARCHAR2(30)	Unique database name
STANDBY_BECAME_PRIMARY_SCN	NUMBER	SCN at which a physical standby database became a primary database. This SCN is useful for converting a failed primary database into a physical standby database after a forced failover. **See Also:** *Oracle Data Guard Concepts and Administration* for more information about Oracle Data Guard.
FS_FAILOVER_MODE[1]	VARCHAR2(19)	Displays the current fast-start failover mode. Possible values are: • DISABLED - Fast-start failover is disabled. • OBSERVE-ONLY - Fast-start failover is enabled in test drive mode. • ZERO DATA LOSS - Fast-start failover is enabled and a fast-start failover cannot incur any data loss. • POTENTIAL DATA LOSS - Fast-start failover is enabled and a fast-start failover can incur data loss within FastStartFailoverLagLimit seconds. **See Also:** *Oracle Data Guard Broker* for more information about the FastStartFailoverLagLimit configuration property
FS_FAILOVER_STATUS	VARCHAR2(22)	Fast-start failover status: • DISABLED • BYSTANDER • SYNCHRONIZED • UNSYNCHRONIZED • SUSPENDED • STALLED • LOADING DICTIONARY • PRIMARY UNOBSERVED • REINSTATE REQUIRED • REINSTATE FAILED • TARGET OVER LAG LIMIT • TARGET UNDER LAG LIMIT **See Also:** *Oracle Data Guard Broker* for detailed descriptions of these values **Note:** If the value of this column is DISABLED, then the values for the FS_FAILOVER_CURRENT_TARGET, FS_FAILOVER_THRESHOLD, FS_FAILOVER_OBSERVER_PRESENT, and FS_FAILOVER_OBSERVER_HOST columns in this table are not meaningful.
FS_FAILOVER_CURRENT_TARGET	VARCHAR2(30)	DB_UNIQUE_NAME of the standby that is the current fail-safe failover observer target standby for the Data Guard configuration
FS_FAILOVER_THRESHOLD	NUMBER	Time (in seconds) that the observer will attempt to reconnect with a disconnected primary before attempting fail-safe failover observer with the target standby

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FS_FAILOVER_OBSERVER_PRESENT	VARCHAR2(7)	Indicates whether the master observer is currently connected to the local database (YES) or not (NO) **Note:** This column is consistent throughout an Oracle RAC environment; that is, if the observer is connected to any instance, then all instances will show a value of YES.
FS_FAILOVER_OBSERVER_HOST	VARCHAR2(512)	Machine name that is currently hosting the master observer process, if fast-start failover is enabled. If fast-start failover is not enabled, this column returns a NULL string.
CONTROLFILE_CONVERTED	VARCHAR2(3)	Indicates whether the control file was implicitly converted from its original type during restore (YES) or not (NO) This column will be set to YES when RMAN restores a standby control file from a backup of the control file taken at the primary database or restores a backup control file from a backup taken at the physical standby database. This column will change to NO when the file names are fixed using information in the recovery catalog schema.
PRIMARY_DB_UNIQUE_NAME	VARCHAR2(30)	For any Standby database (Physical, Logical, or Snapshot), this column will contain the DB_UNIQUE_NAME of the Primary database that this Standby last received current redo from. If this standby has not received any current redo since last being started, then this column will be null. For a Primary database that had previously been a Standby, this column will contain the DB_UNIQUE_NAME of the last Primary that this database received current redo from while acting as a Standby. For a Primary database that has never been a Standby, this column will be null.
SUPPLEMENTAL_LOG_DATA_PL	VARCHAR2(3)	Indicates whether additional information is logged in the redo log (YES) or not (NO) during invocation of procedures in Oracle-supplied packages for which procedural replication is supported. When a value of YES appears in a CDB, it means that supplemental logging for procedural replication is enabled in all of the PDBs in the CDB. When a value of NO appears in a CDB, query the PROCEDURAL column in the DBA_SUPPLEMENTAL_LOGGING view for each PDB in the CDB to see whether supplemental logging for procedural replication is supported in the PDB. **See Also:** *Oracle Data Guard Concepts and Administration* for a list of Oracle-supplied packages that are procedurally replicated to a logical standby database
MIN_REQUIRED_CAPTURE_CHANGE#	NUMBER	Minimum REQUIRED_CHECKPOINT_SCN for all local capture processes on the database
CDB	VARCHAR2(3)	Possible values are: • YES if the database is a CDB • NO if the database is not a CDB
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PENDING_ROLE_CHANGE_TASKS	VARCHAR2(512)	Tasks remaining after an Oracle Data Guard role change. Possible values: • NONE: No pending tasks remain • NOT APPLICABLE: The database is either standby or is not a DG_CONFIG member • BUILD_PENDING: The database was formerly a logical standby and has not yet taken a snapshot of its data dictionary into the redo stream • SRL_ARCHIVE_PENDING: The database was formerly a logical standby and the standby redo logs associated with the earlier failover operation have not yet been archived • ERROR: The database was formerly a logical standby and the snapshot of the dictionary failed • UNKNOWN: The database is not open or the query failed
CON_DBID	NUMBER	The database ID of the PDB
FORCE_FULL_DB_CACHING	VARCHAR2(3)	Indicates the status of the force full database caching feature in the database. Possible values: • YES - The database is in force full database caching mode. • NO - The database is not in force full database caching mode. **See Also**: *Oracle Database SQL Language Reference* for information about the FORCE FULL DATABASE CACHING clause for the ALTER DATABASE statement
SUPPLEMENTAL_LOG_DATA_SR[1]	VARCHAR2(3)	Indicates whether the database is enabled for subset database replication (YES) or not (NO). If the database is enabled for subset database replication, then redo overhead and feature restriction for tables without column data supplemental logging will be reduced.

[1] This column is available starting with Oracle Database 19c.

> **See Also**
>
> "DBA_SUPPLEMENTAL_LOGGING" for more information about supplemental logging in a PDB

7.160 V$DATABASE_BLOCK_CORRUPTION

V$DATABASE_BLOCK_CORRUPTION displays information about database blocks that were corrupted after the last backup.

Column	Datatype	Description
FILE#	NUMBER	Absolute file number of the data file that contains the corrupt blocks
BLOCK#	NUMBER	Block number of the first corrupt block in the range of corrupted blocks
BLOCKS	NUMBER	Number of corrupted blocks found starting with BLOCK#
CORRUPTION_CHANGE#	NUMBER	Change number at which the logical corruption was detected. Set to 0 to indicate media corruption.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CORRUPTION_TYPE	VARCHAR2(9)	Type of block corruption in the data file: • ALL ZERO - Block header on disk contained only zeros. The block may be valid if it was never filled and if it is in an Oracle7 file. The buffer will be reformatted to the Oracle8 standard for an empty block. • FRACTURED - Block header looks reasonable, but the front and back of the block are different versions. • CHECKSUM - optional check value shows that the block is not self-consistent. It is impossible to determine exactly why the check value fails, but it probably fails because sectors in the middle of the block are from different versions. • CORRUPT - Block is wrongly identified or is not a data block (for example, the data block address is missing) • LOGICAL - Block is logically corrupt
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$NONLOGGED_BLOCK" for information about nonlogged blocks

7.161 V$DATABASE_INCARNATION

V$DATABASE_INCARNATION displays information about all database incarnations.

Oracle creates a new incarnation whenever a database is opened with the RESETLOGS option. Records about the current and immediately previous incarnation are also contained in the V$DATABASE view.

Column	Datatype	Description
INCARNATION#	NUMBER	Record ID for the branch record in the control file
RESETLOGS_CHANGE#	NUMBER	Resetlogs system change number (SCN) for the incarnation of the current row
RESETLOGS_TIME	DATE	Resetlogs timestamp for the incarnation of the current row
PRIOR_RESETLOGS_CHANGE#	NUMBER	Resetlogs SCN for the previous incarnation
PRIOR_RESETLOGS_TIME	DATE	Resetlogs timestamp for the previous incarnation
STATUS	VARCHAR2(7)	Incarnation status: • ORPHAN - Orphan incarnation • CURRENT - Current incarnation of the database • PARENT - Parent of the current incarnation

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RESETLOGS_ID	NUMBER	Branch ID for the incarnation of the current row (used by user-managed recovery/RMAN restore to get unique names for archived logs across incarnations)
PRIOR_INCARNATION#	NUMBER	Parent incarnation record ID if nonzero
FLASHBACK_DATABASE_ALLOWED	VARCHAR2(26)	Indicate whether or not Flashback Database can be performed into SCNs or timestamps in the incarnation. A value of YES means that you can flashback to some point in that incarnation. A value of NO indicates that you cannot flashback into the incarnation.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$DATABASE"

7.162 V$DATABASE_KEY_INFO

V$DATABASE_KEY_INFO provides the information of the default database key used to encrypt data blocks. Oracle uses the database key to encrypt sensitive information in SYSTEM, UNDO, and TEMP tablespaces when such data has dependency from encrypted tablespaces or encrypted columns

V$DATABASE_KEY_INFO reflects the database key information stored in the control file. Generally it is the same as the system tablespace key recorded in the system tablespace. If the system tablespace is encrypted, it will also appear in V$ENCRYPTED_TABLESPACES. If the system tablespace is not encrypted, this is the only view providing this information.

Column	Datatype	Description
ENCRYPTIONALG	VARCHAR2(7)	Encryption algorithm: • NONE • 3DES168 • AES128 • AES192 • AES256 AES128 is the default encryption algorithm.
ENCRYPTEDKEY	RAW(48)	Encrypted version of the database key
MASTERKEYID	RAW(16)	ID of the master key that was used to encrypt the database key
MASTERKEY_ACTIVATED	VARCHAR2(3)	Indicates whether the master key has been set (YES) or not (NO) for this database or container in the Oracle Key Store. In other words, indicates whether Transparent Data Encryption (TDE) has been activated for this database or container

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$ENCRYPTED_TABLESPACES"

7.163 V$DATAFILE

V$DATAFILE displays datafile information from the control file.

See Also

"V$DATAFILE_HEADER", which displays information from data file headers

Column	Datatype	Description
FILE#	NUMBER	Absolute file number
CREATION_CHANGE#	NUMBER	Change number at which the data file was created
CREATION_TIME	DATE	Timestamp of the data file creation
TS#	NUMBER	Tablespace number
RFILE#	NUMBER	Tablespace relative data file number
STATUS	VARCHAR2(7)	Type of file (system or user) and its status. Values: OFFLINE, ONLINE, SYSTEM, RECOVER, SYSOFF (an offline file from the SYSTEM tablespace)
ENABLED	VARCHAR2(10)	Describes how accessible the file is from SQL: • DISABLED - No SQL access allowed. The tablespace might be offline. • READ ONLY - No SQL updates allowed • READ WRITE - Full access allowed • UNKNOWN - Unknown whether SQL updates would be allowed or not
CHECKPOINT_CHANGE#	NUMBER	SCN at last checkpoint
CHECKPOINT_TIME	DATE	Timestamp of the checkpoint#
UNRECOVERABLE_CHANGE#	NUMBER	Last unrecoverable change number made to this data file. If the database is in ARCHIVELOG mode, then this column is updated when an unrecoverable operation completes. If the database is not in ARCHIVELOG mode, this column does not get updated.
UNRECOVERABLE_TIME	DATE	Timestamp of the last unrecoverable change. This column is updated only if the database is in ARCHIVELOG mode.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LAST_CHANGE#	NUMBER	Last change number made to this data file (null if the data file is being changed)
LAST_TIME	DATE	Timestamp of the last change
OFFLINE_CHANGE#	NUMBER	Offline change number of the last offline range. This column is updated only when the data file is brought online.
ONLINE_CHANGE#	NUMBER	Online change number of the last offline range
ONLINE_TIME	DATE	Online timestamp of the last offline range
BYTES	NUMBER	Current data file size (in bytes); 0 if inaccessible
BLOCKS	NUMBER	Current data file size (in blocks); 0 if inaccessible
CREATE_BYTES	NUMBER	Size when created (in bytes)
BLOCK_SIZE	NUMBER	Block size of the data file
NAME	VARCHAR2(513)	Name of the data file
PLUGGED_IN	NUMBER	Describes whether the tablespace is plugged in. The value is 1 if the tablespace is plugged in and has not been made read/write, 0 if not.
BLOCK1_OFFSET	NUMBER	Offset from the beginning of the file to where the Oracle generic information begins. The exact length of the file can be computed as follows: BYTES + BLOCK1_OFFSET.
AUX_NAME	VARCHAR2(513)	Auxiliary name that has been set for this file via CONFIGURE AUXNAME
FIRST_NONLOGGED_SCN	NUMBER	First nonlogged SCN (check in standby database)
FIRST_NONLOGGED_TIME	DATE	First nonlogged time (check in standby database)
FOREIGN_DBID	NUMBER	Foreign DBID from which this data file came from. The value is 0 if this file is not a foreign database file.
FOREIGN_CREATION_CHANGE#	NUMBER	Creation SCN of a foreign data file. The value is 0 if this file is not a foreign database file.
FOREIGN_CREATION_TIME	DATE	Creation time of a foreign data file. The value is 0 if this file is not a foreign database file.
PLUGGED_READONLY	VARCHAR2(3)	YES if this is a transported read-only foreign file; otherwise NO.
PLUGIN_CHANGE#	NUMBER	SCN at which the foreign data file was transported into the database. The value is 0 if this file is not a foreign database file.
PLUGIN_RESETLOGS_CHANGE#	NUMBER	The SCN of the RESETLOGS operation for the incarnation into which this foreign file was transported. The value is 0 if this file is not a foreign database file.
PLUGIN_RESETLOGS_TIME	DATE	The time of the RESETLOGS operation for the incarnation into which this foreign file was transported. The value is 0 if this file is not a foreign database file.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.164 V$DATAFILE_COPY

V$DATAFILE_COPY displays data file copy information from the control file.

Column	Datatype	Description
RECID	NUMBER	Data file copy record ID
STAMP	NUMBER	Data file copy record stamp
NAME	VARCHAR2(513)	File name of the data file copy. The maximum length of the name is operating system dependent.
TAG	VARCHAR2(32)	Data file copy tag
FILE#	NUMBER	Absolute data file number
RFILE#	NUMBER	Tablespace relative data file number
CREATION_CHANGE#	NUMBER	Data file creation change#
CREATION_TIME	DATE	Data file creation timestamp
RESETLOGS_CHANGE#	NUMBER	Resetlogs change number of the data file when the copy was made
RESETLOGS_TIME	DATE	Resetlogs timestamp of the data file when the copy was made
INCREMENTAL_LEVEL	NUMBER	Normal full backups have a NULL value, level 0 incremental backups have a value of 0, and level 1 incremental backups have a value of 1
CHECKPOINT_CHANGE#	NUMBER	Checkpoint change number of the data file when the copy was made
CHECKPOINT_TIME	DATE	Checkpoint timestamp of the data file when the copy was made
ABSOLUTE_FUZZY_CHANGE#	NUMBER	Highest change seen when the data file was copied
RECOVERY_FUZZY_CHANGE#	NUMBER	Highest change written to the file by media recovery
RECOVERY_FUZZY_TIME	DATE	Timestamp of the highest change written to the file by media recovery
ONLINE_FUZZY	VARCHAR2(3)	(YES|NO) If set to YES, this is a copy taken using an operating system utility after a crash or offline immediate (or an invalid copy taken while data file was online and the database open). Recovery will need to apply all redo up to the next crash recovery marker to make the file consistent.
BACKUP_FUZZY	VARCHAR2(3)	(YES|NO) If set to YES, this is a copy taken using the BEGIN BACKUP/END BACKUP technique. Recovery will need to apply all redo up to the end backup marker to make this copy consistent.
MARKED_CORRUPT	NUMBER	Number of blocks marked corrupt by this copy operation. That is, blocks that were not marked corrupted in the source data file, but were detected and marked as corrupted during the copy operation.
MEDIA_CORRUPT	NUMBER	Total number of media corrupt blocks. For example, blocks with checksum errors are marked media corrupt.
LOGICALLY_CORRUPT	NUMBER	Total number of logically corrupt blocks. For example, applying redo for unrecoverable operations will mark affected blocks logically corrupt.
BLOCKS	NUMBER	Size of the data file copy in blocks (also the size of the data file when the copy was made)
BLOCK_SIZE	NUMBER	Block size of the data file
OLDEST_OFFLINE_RANGE	NUMBER	RECID of the oldest offline range record in this control file copy; 0 for data file copies
DELETED	VARCHAR2(3)	(YES|NO) If set to YES the data file copy has been deleted or overwritten

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(1)	Identifies the status of this data file copy. Possible values are: A - Available D - Deleted U - Unavailable X - Expired
COMPLETION_TIME	DATE	Time when the copy was completed
CONTROLFILE_TYPE	VARCHAR2(1)	B indicates normal copies S indicates standby copies
KEEP	VARCHAR2(3)	(YES/NO) Indicates whether or not this backup set has a retention policy that is different than the value for the configure retention policy
KEEP_UNTIL	DATE	If KEEP_UNTIL is specified, this is the date after which the backup becomes obsolete. If this column is null, then the backup never expires.
KEEP_OPTIONS	VARCHAR2(11)	Lists additional retention options for this backup set. Possible values are: LOGS - The logs need to recover this backup are kept NOLOGS - The logs needed to recover this backup will not be kept
SCANNED	VARCHAR2(3)	Indicates whether RMAN scanned the file (YES) or not (NO)
IS_RECOVERY_DEST_FILE	VARCHAR2(3)	Indicates whether the file was created in the fast recovery area (YES) or not (NO)
RMAN_STATUS_RECID	NUMBER	Owning V$RMAN_STATUS record ID
RMAN_STATUS_STAMP	NUMBER	Owning V$RMAN_STATUS record stamp
CONVERTED_FILE	VARCHAR2(3)	(YES/NO) Indicates whether or not the data file copy was created using the RMAN CONVERT command
SAME_ENDIAN	VARCHAR2(3)	If the value of the CONVERTED_FILE column is YES, then this column indicates whether the data file copy has the same endianness as the source data file (YES) or not (NO); otherwise NULL
FOREIGN_DBID	NUMBER	Foreign DBID from which this data file came from. The value is 0 if this file is not a foreign database file.
PLUGGED_READONLY	VARCHAR2(3)	YES if this is a transported read-only foreign file; otherwise NO.
PLUGIN_CHANGE#	NUMBER	SCN at which the foreign data file was transported into the database. The value is 0 if this file is not a foreign database file.
PLUGIN_RESETLOGS_CHANGE#	NUMBER	The SCN of the RESETLOGS operation for the incarnation into which this foreign file was transported. The value is 0 if this file is not a foreign database file.
PLUGIN_RESETLOGS_TIME	DATE	The time of the RESETLOGS operation for the incarnation into which this foreign file was transported. The value is 0 if this file is not a foreign database file.
BACKED_BY_VSS	VARCHAR2(3)	Whether or not the file has been backed up by Volume Shadow Copy Service (VSS). This column is reserved for internal use.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BACKED_BY_PDB	VARCHAR2(3)	Recovery Manager (RMAN) allows a PDB to be backed up in two ways. The value in this column indicates how the PDB backup was taken: • YES: The backup was taken when connected to the PDB • NO: The backup was taken when connected to the root container
SPARSE_BACKUP	VARCHAR2(3)	Indicates whether the file is sparse (YES) or not (NO)
GUID	RAW(16)	The GUID of the PDB to which the backup belongs. This is useful after the PDB is dropped to identify which PDB the backup belongs to.

7.165 V$DATAFILE_HEADER

V$DATAFILE_HEADER displays data file information from the data file headers.

Column	Datatype	Description
FILE#	NUMBER	Data file number (from control file)
STATUS	VARCHAR2(7)	ONLINE | OFFLINE (from control file)
ERROR	VARCHAR2(18)	NULL if the data file header read and validation were successful. If the read failed then the rest of the columns are NULL. If the validation failed then the rest of columns may display invalid data. If there is an error then usually the data file must be restored from a backup before it can be recovered or used.
FORMAT	NUMBER	Indicates the format for the header block. The possible values are 6, 7, 8, 10 or 0. 6 - indicates Oracle Version 6 7 - indicates Oracle Version 7 8 - indicates Oracle Version 8 10 - indicates Oracle Version 10 0 - indicates the format could not be determined (for example, the header could not be read)
RECOVER	VARCHAR2(3)	File needs media recovery (YES | NO)
FUZZY	VARCHAR2(3)	File is fuzzy (YES | NO)
CREATION_CHANGE#	NUMBER	Data file creation change#
CREATION_TIME	DATE	Data file creation timestamp
TABLESPACE_NAME	VARCHAR2(30)	Tablespace name
TS#	NUMBER	Tablespace number
RFILE#	NUMBER	Tablespace relative data file number
RESETLOGS_CHANGE#	NUMBER	Resetlogs change#
RESETLOGS_TIME	DATE	Resetlogs timestamp
CHECKPOINT_CHANGE#	NUMBER	Data file checkpoint change#
CHECKPOINT_TIME	DATE	Data file checkpoint timestamp
CHECKPOINT_COUNT	NUMBER	Data file checkpoint count
BYTES	NUMBER	Current data file size in bytes
BLOCKS	NUMBER	Current data file size in blocks
NAME	VARCHAR2(513)	Data file name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SPACE_HEADER	VARCHAR2(40)	Represents the block address of a space file header block of a locally managed data file
LAST_DEALLOC_CHANGE#	VARCHAR2(16)	Last deallocated SCN
UNDO_OPT_CURRENT_CHANGE#	VARCHAR2(40)	For internal use only
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
IS_SPARSE	VARCHAR2(3)	Indicates whether the file is sparse (YES) or not (NO)
ENCRYPTED	VARCHAR2(3)	After file encryption is completed, this column indicates whether the file is encrypted (YES) or not (NO). When file encryption is still in progress, this column has a value of NO.

7.166 V$DATAGUARD_CONFIG

V$DATAGUARD_CONFIG displays the unique database names defined with the DB_UNIQUE_NAME and LOG_ARCHIVE_CONFIG initialization parameters, providing a view of the Oracle Data Guard environment from any database in the configuration.

The first row of the view lists the unique database name of the current database that was specified with the DB_UNIQUE_NAME initialization parameter. Additional rows reflect the unique database names of the other databases in the configuration that were specified with the DG_CONFIG keyword of the LOG_ARCHIVE_CONFIG initialization parameter.

Column	Datatype	Description
DB_UNIQUE_NAME	VARCHAR2(30)	Unique database name
PARENT_DBUN	VARCHAR2(30)	DB_UNIQUE_NAME of the parent database, also known as the database that supplies live redo to the destination. For example, suppose Boston is the DB_UNIQUE_NAME of the primary database, Chicago is the DB_UNIQUE_NAME of the Far Sync Instance, and Seattle is the DB_UNIQUE_NAME of the terminal standby database. Since Boston is the primary database, it has no parent database so the PARENT_DBUN for Boston will be NULL. Since Boston services Chicago, the PARENT_DBUN of Chicago will be Boston. Since Chicago services Seattle, Chicago will be the PARENT_DBUN for Seattle.
DEST_ROLE	VARCHAR2(17)	Type of archival destination database: • BACKUP APPLIANCE • FAR SYNC STANDBY • LOGICAL STANDBY • PHYSICAL STANDBY • PRIMARY DATABASE • SNAPSHOT STANDBY
CURRENT_SCN	NUMBER	The SCN up to which the specified database has applied redo

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "DB_UNIQUE_NAME"
- "LOG_ARCHIVE_CONFIG"

7.167 V$DATAGUARD_PROCESS

V$DATAGUARD_PROCESS displays one row for each Oracle Data Guard process that is currently running.

Column	Datatype	Description
NAME	VARCHAR2(5)	Name of the process whose information is being reported. Some of the possible values include: • ARC*n* - Archiver process • DTS - Data transport process • FAL - File/announce process • LGWR - Log Writer Process • MRP0 - Detached recovery server process • NSS*n* - SYNC Redo Transport process • ORA - Foreground process • RFS - Remote file server • RMI - Remote message process • TMON - Redo Transport Process monitor • TT*nn* - Redo Transport Worker Process
PID	VARCHAR2(24)	Operating system process identifier of the process
TYPE	VARCHAR2(3)	Indicates which Oracle subsystem created the process. Possible values: • KSB • KSV • NET • UNK

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ROLE	VARCHAR2(24)	Role of the process. Possible values: • test • async ORL multi • async ORL single • async SRL multi • async SRL single • log writer • sync • archive redo • archive local • archive gap • RFS async • RFS sync • RFS archive • RFS gap • RFS SMON • data transport • data receive • redo transport monitor • heartbeat redo informer • process kill • post role transition • gap manager • update TMI • RFS ping • FAL gap • FAL announce • failover • switchover • remote failover • remote switchover • redo transport timer • announce request • managed recovery • recovery • controlfile update • broker monitor • broker worker • broker net slave • broker instance slave • fast-start failover ping • recovery coordinator • recovery logmerger • recovery apply slave • recovery sending slave • recovery receiving slave • redo transport test • temporary • datafile pre-create • redo log management • UNKNOWN

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PROC_TIME	TIMESTAMP(0)	Timestamp of when the process started or registered for inclusion in this fixed view
TASK_TIME	TIMESTAMP(0)	Timestamp of when the first task of the process was requested
TASK_DONE	VARCHAR2(1)	Indicates whether the task performed by the process is done (Y) or not (N)
ACTION	VARCHAR2(12)	Current action of the process. Possible values can include: • ANNOUNCING • APPLYING_LOG • ATTACHED • CLOSING • CONNECTED • ERROR • IDLE • OPENING • PROCESSING • RECEIVING • REGISTERING • STARTING • TERMINATING • UNKNOWN • UNUSED • WAIT_FOR_GAP • WAIT_FOR_LOG • WRITING
CLIENT_PID	NUMBER	For RFS and DTS processes, the PID of the process communicating with this process
CLIENT_ROLE	VARCHAR2(16)	For RFS and DTS processes, the role of the process communicating with this process. Possible values can include: • none • async ORL multi • async ORL single • async SRL multi • async SRL single • log writer • sync • archive redo • archive gap • data transport • gap manager • failover • switchover • announce request • managed recovery • recovery • UNKNOWN
GROUP#	NUMBER	Group number of the log that the process is operating upon
RESETLOG_ID	NUMBER	Resetlog ID (branch) of the log that the process is operating upon
THREAD#	NUMBER	Thread number that the process is operating upon

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SEQUENCE#	NUMBER	Sequence number that the process is operating upon
BLOCK#	NUMBER	Starting block number that the process is operating upon
BLOCK_COUNT	NUMBER	Number of blocks that the process is operating upon
DELAY_MINS	NUMBER	Archived redo log delay interval in minutes
DEST_ID	NUMBER	Destination ID that the process is currently operating upon
DEST_MASK	NUMBER	Mask of all destination IDs that the process will operate upon, where: Bit 0 is LOG_ARCHIVE_DEST1 Bit 1 is LOG_ARCHIVE_DEST2 Bit 2 is LOG_ARCHIVE_DEST3 and so on.
DBID	NUMBER	Database ID of the redo that the process is operating upon
DGID	NUMBER	Data Guard ID that the process must communicate with
INSTANCE	NUMBER	Instance number that the process must communicate with at the specified DGID
STOP_STATE	VARCHAR2(7)	Indicates the method by which the process has been requested to stop: • NOW - Requested to stop immediately and exit • SOON - Requested to stop on next log and exit • COMMIT - Requested to write remaining redo and exit • N/A - No request to stop
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **Note**
>
> Oracle recommends that you use this view instead of V$MANAGED_STANDBY.

7.168 V$DATAGUARD_STATS

V$DATAGUARD_STATS displays information about Oracle Data Guard metrics when queried on a target database. No rows are returned when queried on a primary database. In the descriptions in this section, the originating database that is generating the redo is either a primary database or a logical standby database.

Column	Datatype	Description
SOURCE_DBID	NUMBER	Database ID of the originating database
SOURCE_DB_UNIQUE_NAME	VARCHAR2(32)	DB unique name of the originating database

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NAME	VARCHAR2(32)	Name of the metric: • apply finish time - An estimate of the time needed to apply all received, but unapplied redo from the originating database. If there are one or more redo gaps on the target database, an estimate of the time needed to apply all received, but unapplied redo up to the end of the last archived redo log before the beginning of the earliest redo gap. • apply lag - Apply lag is a measure of the degree to which the data in a target database lags behind the data in the originating database, due to delays in propagating and applying redo to the target database. This value is relevant only to the applying instance. • estimated startup time - An estimate of the time needed to start and open the database. • transport lag - Transport lag is a measure of the degree to which the transport of redo to the target database lags behind the generation of redo on the originating database. If there are one or more redo gaps on the target database, the transport lag is calculated as if no redo has been received after the beginning of the earliest redo gap.
VALUE	VARCHAR2(64)	Value of the metric
UNIT	VARCHAR2(30)	Unit of measurement
TIME_COMPUTED	VARCHAR2(30)	Local time at the target database when the metric was computed
DATUM_TIME	VARCHAR2(30)	Local time at the target database when the data used to compute the metric was received The apply lag and transport lag metrics are computed based on data that is periodically received from the source database. An unchanging value in this column across multiple queries indicates that the target database is not receiving data from the source database.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.169 V$DATAGUARD_STATUS

V$DATAGUARD_STATUS displays messages recently written to the alert log or server process trace files that concern physical standby databases or redo transport services for all standby database types.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FACILITY	VARCHAR2(24)	Facility that encountered the event: • Crash Recovery • Log Transport Services • Log Apply Services • Role Management Services • Remote File Server • Fetch Archive Log • Data Guard • Network Services
SEVERITY	VARCHAR2(13)	Severity of the event: • Informational - Informational message • Warning - Warning message • Error - Indicates the process has failed • Fatal • Control - An expected change in state such as the start or completion of an archival, log recovery, or switchover operation
DEST_ID	NUMBER	Destination ID number to which the event pertains. If the event does not pertain to a particular destination, then the value is 0.
MESSAGE_NUM	NUMBER	A chronologically increasing number giving each event a unique number
ERROR_CODE	NUMBER	Error ID pertaining to the event
CALLOUT	VARCHAR2(3)	Reserved for future use
TIMESTAMP	DATE	Message date
MESSAGE	VARCHAR2(256)	A text message describing the event
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.170 V$DB_CACHE_ADVICE

V$DB_CACHE_ADVICE contains rows that predict the number of physical reads for the cache size corresponding to each row.

The rows also compute a "physical read factor," which is the ratio of the number of estimated reads to the number of reads actually performed by the real buffer cache during the measurement interval.

See Also

"DB_CACHE_ADVICE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ID	NUMBER	Buffer pool identifier (ranges from 1 to 8)
NAME	VARCHAR2(20)	Buffer pool name
BLOCK_SIZE	NUMBER	Block size in bytes for buffers in this pool. Possible values: the standard block size, the power of 2 nonstandard block sizes, 2048, 4096, 8192, 16384, 32768.
ADVICE_STATUS	VARCHAR2(3)	Status of the advisory. ON indicates it is currently running; OFF indicates it is disabled (in this case the estimates are historical and calculated when last enabled).
SIZE_FOR_ESTIMATE	NUMBER	Cache size for prediction (in megabytes)
SIZE_FACTOR	NUMBER	Size factor with respect to the current cache size
BUFFERS_FOR_ESTIMATE	NUMBER	Cache size for prediction (in terms of buffers)
ESTD_PHYSICAL_READ_FACTOR	NUMBER	Physical read factor for this cache size, which is the ratio of the number of estimated physical reads to the number of reads in the real cache. If there are no physical reads in the real cache, the value of this column is null.
ESTD_PHYSICAL_READS	NUMBER	Estimated number of physical reads for this cache size
ESTD_PHYSICAL_READ_TIME	NUMBER	Estimated disk read time (in seconds)
ESTD_PCT_OF_DB_TIME_FOR_READS	NUMBER	Estimated disk time as a percentage of the total time
ESTD_CLUSTER_READS	NUMBER	Estimated total number of blocks foreground processes read from the global cache (Oracle Real Application Clusters only)
ESTD_CLUSTER_READ_TIME	NUMBER	Estimated total amount of time, in seconds, foreground processes read from global cache (Oracle Real Application Clusters only)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.171 V$DB_OBJECT_CACHE

V$DB_OBJECT_CACHE displays database objects that are cached in the library cache. Objects include tables, indexes, clusters, synonym definitions, PL/SQL procedures and packages, and triggers.

Column	Datatype	Description
OWNER	VARCHAR2(64)	Owner of the object
NAME	VARCHAR2(1000)	Name of the object
DB_LINK	VARCHAR2(64)	Database link name, if any
NAMESPACE	VARCHAR2(64)	Library cache namespace of the object: TABLE/PROCEDURE, BODY, TRIGGER, INDEX, CLUSTER, OBJECT
TYPE	VARCHAR2(64)	Type of the object: INDEX, TABLE, CLUSTER, VIEW, SET, SYNONYM, SEQUENCE, PROCEDURE, FUNCTION, PACKAGE, PACKAGE BODY, TRIGGER, CLASS, OBJECT, USER, DBLINK

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SHARABLE_MEM	NUMBER	Amount of sharable memory in the shared pool consumed by the object
LOADS	NUMBER	Number of times the object has been loaded. This count also increases when an object has been invalidated.
EXECUTIONS	NUMBER	Not used **See Also:** "V$SQLAREA" to see actual execution counts
LOCKS	NUMBER	Number of users currently locking this object
PINS	NUMBER	Number of users currently pinning this object
KEPT	VARCHAR2(3)	(YES | NO) Depends on whether this object has been "kept" (permanently pinned in memory) with the PL/SQL procedure DBMS_SHARED_POOL.KEEP
CHILD_LATCH	NUMBER	Child latch number that is protecting the object. This column is obsolete and maintained for backward compatibility.
INVALIDATIONS	NUMBER	Total number of times objects in the namespace were marked invalid because a dependent object was modified
HASH_VALUE	NUMBER	Hash value of the object
LOCK_MODE	VARCHAR2(9)	Current lock mode of the object
PIN_MODE	VARCHAR2(9)	Current pin mode of the object
STATUS	VARCHAR2(19)	Status of the object
TIMESTAMP	VARCHAR2(19)	Timestamp for the specification of the object
PREVIOUS_TIMESTAMP	VARCHAR2(19)	Previous timestamp for the specification of the object
LOCKED_TOTAL	NUMBER	Total number of times the object has been locked
PINNED_TOTAL	NUMBER	Total number of times the object has been pinned
PROPERTY	VARCHAR2(80)	Property of the object. Possible values include HOT or HOTCOPY when the library cache hot copy feature is used using DBMS_SHARED_POOL.MARKHOT.
FULL_HASH_VALUE	VARCHAR2(32)	Full hash value of the object
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
CON_NAME	VARCHAR2(64)	Container name of the object. The value of this column is NULL in non-CDBs.
ADDR	RAW(8)	Address of the handle for this object
EDITION	VARCHAR2(138)	Edition name

See Also

- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SHARED_POOL.KEEP procedure
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SHARED_POOL.MARKHOT procedure

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.172 V$DB_PIPES

V$DB_PIPES displays the pipes that are currently represented in the shared pool for this instance.

Column	Datatype	Description
OWNERID	NUMBER	Owner ID of the owner (if this is a private pipe), else NULL
NAME	VARCHAR2(1000)	Name of the pipe (for example, SCOTT.PIPE)
TYPE	VARCHAR2(7)	Type of the pipe: • PUBLIC • PRIVATE
PIPE_SIZE	NUMBER	Amount of memory the pipe uses **Note:** The value of this column may be larger than maxpipesize because of an internal algorithm.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
CON_NAME	VARCHAR2(64)	Container name of the object. The value of this column is NULL in non-CDBs.

7.173 V$DB_TRANSPORTABLE_PLATFORM

V$DB_TRANSPORTABLE_PLATFORM displays all platforms to which the database can be transported using the RMAN CONVERT DATABASE command.

The transportable database feature only supports transports of the same endian platform. Therefore, V$DB_TRANSPORTABLE_PLATFORM displays fewer rows than V$TRANSPORTABLE_PLATFORM.

Column	Datatype	Description
PLATFORM_ID	NUMBER	Platform identification number
PLATFORM_NAME	VARCHAR2(101)	Platform name
ENDIAN_FORMAT	VARCHAR2(14)	Platform endian format: • Big • Little • UNKNOWN FORMAT
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "V$TRANSPORTABLE_PLATFORM"

7.174 V$DBFILE

V$DBFILE displays all data files making up the database. This view is retained for historical compatibility. Use of V$DATAFILE is recommended instead.

Column	Datatype	Description
FILE#	NUMBER	File identifier
NAME	VARCHAR2(513)	Name of the file
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **See Also**
>
> "V$DATAFILE"

7.175 V$DBLINK

V$DBLINK describes all database links (links with IN_TRANSACTION = YES) opened by the session issuing the query on V$DBLINK. These database links must be committed or rolled back before being closed.

Column	Datatype	Description
DB_LINK	VARCHAR2(128)	Name of the database link
OWNER_ID	NUMBER	Owner of the database link UID
LOGGED_ON	VARCHAR2(3)	Whether the database link is currently logged on
HETEROGENEOUS	VARCHAR2(3)	Whether the database link is heterogeneous
PROTOCOL	VARCHAR2(6)	Communication protocol for the database link
OPEN_CURSORS	NUMBER	Whether there are open cursors for the database link
IN_TRANSACTION	VARCHAR2(3)	Whether the database link is currently in a transaction
UPDATE_SENT	VARCHAR2(3)	Whether there has been an update on the database link
COMMIT_POINT_STRENGTH	NUMBER	Commit point strength of the transactions on the database link

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.176 V$DEAD_CLEANUP

V$DEAD_CLEANUP shows the terminated processes and sessions present in the instance and their cleanup status.

Column	Datatype	Description
TYPE	VARCHAR2(64)	Indicates whether a row contains a DEAD PROCESS or KILLED SESSION
PADDR	RAW(8)	Process pointer. Can be joined with V$PROCESS. For a KILLED SESSION, this is the owner of the session, which can be NULL.
SADDR	RAW(8)	Session pointer. Can be joined with V$SESSION. For a DEAD PROCESS, this is the user session.
ROOT_ADDR	RAW(8)	Pointer to the root of the tree
CLEANUP_OWNER	VARCHAR2(64)	Indicates which process is responsible for cleanup of this tree: • CLEANUP PROCESS - a cleanup process is responsible • OWNER PROCESS - the root owner is responsible
STATE	VARCHAR2(64)	Cleanup state: • CLEANUP PENDING - Occurs for a DEAD PROCESS or KILLED SESSION that can be cleaned up, but PMON has not yet made an attempt • IN PROGRESS - A cleanup attempt is currently in progress • RESOURCES FREED - Occurs for a DEAD PROCESS or KILLED SESSION where all children have been freed, but the process or session itself is not yet freed • PARTIAL CLEANUP - Occurs if some of the children have been cleaned up
DEAD_TIME	NUMBER	Time since the DEAD PROCESS or KILLED SESSION was marked as such (in seconds)
CLEANUP_ATTEMPTS	NUMBER	Number of times PMON has attempted cleanup
LAST_ATTEMPT	NUMBER	How long ago the last cleanup attempt occurred (in seconds)
CLEANUP_PROCESS	RAW(8)	Process pointer for the cleanup process currently cleaning up this tree (can join with V$CLEANUP_PROCESS). It will be NULL if cleanup is not currently in progress or if the owner is responsible for cleanup.
CLEANUP_TIME	NUMBER	Total amount of time PMON has spent on cleanup of the process/ session (in seconds)
NUM_BLOCKED	NUMBER	Number of sessions blocked on cleanup of this session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$CLEANUP_PROCESS"

7.177 V$DELETED_OBJECT

V$DELETED_OBJECT displays information about deleted archived logs, data file copies and backup pieces from the control file.

The only purpose of this view is to optimize the recovery catalog resync operation. When an archived log, data file copy, or backup piece is deleted, the corresponding record is marked deleted.

Column	Datatype	Description
RECID	NUMBER	Deleted object record ID
STAMP	NUMBER	Deleted object record stamp

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TYPE	VARCHAR2(26)	Identifies the type of deleted object: • ARCHIVED LOG • BACKUP PIECE • DATAFILE COPY • PROXY COPY • BACKUP PIECE AVAILABLE • BACKUP PIECE EXPIRED • PROXY COPY AVAILABLE • PROXY COPY EXPIRED • BACKUP PIECE UNAVAILABLE • PROXY COPY UNAVAILABLE • DATAFILE COPY AVAILABLE • DATAFILE COPY EXPIRED • DATAFILE COPY UNAVAILABLE • ARCHIVED LOG AVAILABLE • ARCHIVED LOG EXPIRED • ARCHIVED LOG UNAVAILABLE • BACKUP SET KEEP OPTIONS • BACKUP SET KEEP UNTIL • PROXY COPY KEEP OPTIONS • PROXY COPY KEEP UNTIL • DATAFILE COPY KEEP OPTIONS • DATAFILE COPY KEEP UNTIL • DATAFILE RENAME ON RESTORE
OBJECT_RECID	NUMBER	Record ID of the deleted object
OBJECT_STAMP	NUMBER	Record timestamp of the deleted object
OBJECT_DATA	NUMBER	Displays additional internal information related to this deleted object. For internal Oracle use only.
SET_STAMP	NUMBER	Set stamp of the deleted object
SET_COUNT	NUMBER	Set count of the deleted object
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.178 V$DG_BROKER_CONFIG

V$DG_BROKER_CONFIG provides a summary of an Oracle Data Guard broker configuration.

This is similar to the DGMGRL CLI's SHOW CONFIGURATION command. It provides a view of the entire Oracle Data Guard broker configuration from any database in the configuration.

Column	Datatype	Description
DATABASE	VARCHAR2(512)	Database unique name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CONNECT_IDENTIFIER	VARCHAR2(512)	Net connect identifier used to reach the database
DATAGUARD_ROLE	VARCHAR2(27)	Oracle Data Guard role of the database: • PRIMARY • PHYSICAL STANDBY • LOGICAL STANDBY • SNAPSHOT STANDBY • FAR SYNC INSTANCE • RECOVERY APPLIANCE
REDO_SOURCE	VARCHAR(30)	The database unique name of the redo source
ENABLED	VARCHAR2(5)	TRUE or FALSE to denote whether or not the database is managed by Oracle Data Guard broker
STATUS	NUMBER	An Oracle error number denoting the database's current status
VERSION	VARCHAR2(30)	Version of the broker configuration
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.179 V$DIAG_ALERT_EXT

V$DIAG_ALERT_EXT shows the contents of the XML-based alert log in the Automatic Diagnostic Repository (ADR) for the current container (PDB).

Column	Datatype	Description
ORIGINATING_TIMESTAMP	TIMESTAMP(9) WITH TIME ZONE	Date and time when the message was generated
NORMALIZED_TIMESTAMP	TIMESTAMP(9) WITH TIME ZONE	Date and time when the message originated, normalized for clock drift to the Oracle Enterprise Manager repository time
ORGANIZATION_ID	VARCHAR2(67)	ID of the organization that wrote the originating component, usually the domain of the organization
COMPONENT_ID	VARCHAR2(67)	ID of the product or component that originated the message
HOST_ID	VARCHAR2(67)	DNS hostname of originating host
HOST_ADDRESS	VARCHAR2(49)	IP of other network address of the originating host for the mesaage

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MESSAGE_TYPE	NUMBER	Type of the message, indicating that a different type of response is required. Possible values include: • 1: UNKNOWN - Essentially the NULL type • 2: INCIDENT_ERROR - The program has encountered an error for some internal or unexpected reason, and it must be reported to Oracle Support • 3: ERROR - An error of some kind has occurred • 4: WARNING: An action occurred or a condition was discovered that should be reviewed and may require action • 5: NOTIFICATION: reports a normal action or event. This could be a user action such as "logon completed" • 6: TRACE: Output of a diagnostic trace
MESSAGE_LEVEL	NUMBER	Level the message belongs to. Lower level values imply higher severity for errors. Possible values include: • 1: CRITICAL: critical errors • 2: SEVERE: severe errors • 8: IMPORTANT: important message • 16: NORMAL: normal message
MESSAGE_ID	VARCHAR2(67)	ID of the message
MESSAGE_GROUP	VARCHAR2(67)	Name of the group to which the message belongs
CLIENT_ID	VARCHAR2(67)	ID of the client or security group that the message relates to
MODULE_ID	VARCHAR2(67)	ID of the module that originated the message. This value is unique within a component.
PROCESS_ID	VARCHAR2(35)	ID of the process that originated the message
THREAD_ID	VARCHAR2(67)	ID of the thread of the process that originated the message
USER_ID	VARCHAR2(131)	ID of the user that originated the message
INSTANCE_ID	VARCHAR2(67)	For internal use only
DETAILED_LOCATION	VARCHAR2(163)	Absolute pathname of supplemental detail file on the originating host
UPSTREAM_COMP_ID	VARCHAR2(103)	ID of a component that the originating component is working with on the upstream (client) side
DOWNSTREAM_COMP_ID	VARCHAR2(103)	ID of a component that the originating component is working with on the downstream (server) side
EXECUTION_CONTEXT_ID	VARCHAR2(103)	Identifies the thread of execution that the originating component participates in
EXECUTION_CONTEXT_SEQUENCE	NUMBER	Execution sequence of the thread that the originating component participates in
ERROR_INSTANCE_ID	NUMBER	ID of the instance where error occurred
ERROR_INSTANCE_SEQUENCE	NUMBER	Instance sequence where error occurred
MESSAGE_TEXT	VARCHAR2(2051)	Fully formed and localized text of the message
MESSAGE_ARGUMENTS	VARCHAR2(515)	Arguments to be bound with the generic text of the message
SUPPLEMENTAL_ATTRIBUTES	VARCHAR2(515)	Supplemental attributes that are specific to a message. This field contains the impacts for an incident type error message.
SUPPLEMENTAL_DETAILS	VARCHAR2(515)	Supplemental data that is specific to a particular program and error message required to complete the diagnosis. Similar to the extra detail referred to in DETAILED_LOCATION but short enough to simply provide in the message itself
PARTITION	NUMBER	Segment number of physical file

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RECORD_ID	NUMBER	Record number for the message (this value is same as the row number)
FILENAME	VARCHAR2(515)	Physical file on disk
LOG_NAME	VARCHAR2(67)	For internal use only
PROBLEM_KEY	VARCHAR2(553)	Describes the key for the current problem that the message is associated with
VERSION	NUMBER	ARB version number for the message
CON_UID	NUMBER	The unique ID of the container to which the data pertains
CONTAINER_ID	NUMBER	The ID of the container to which the data pertains
CONTAINER_NAME	VARCHAR2(33)	The name of the container to which the data pertains
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.180 V$DIAG_APP_TRACE_FILE

V$DIAG_APP_TRACE_FILE contains information about all trace files present in the Automatic Diagnostic Repository (ADR) for the current container (PDB) which contain application trace data (SQL_TRACE or OPTIMIZER_TRACE event data). This view also supports GV$ global views.

Column	Datatype	Description
ADR_HOME	VARCHAR2(444)	Path to the current ADR home
TRACE_FILENAME	VARCHAR2(68)	Displays the name of the process trace file
CHANGE_TIME	TIMESTAMP(3) WITH TIME ZONE	Displays the change time timestamp of the process trace fiile
MODIFY_TIME	TIMESTAMP(3) WITH TIME ZONE	Displays the last modification timestamp of the process trace file
SQL_TRACE	VARCHAR2(1)	Identifies if the process trace file contains SQL_TRACE data. Possible values include: • Y: This value is used when the process trace file contains SQL_TRACE event data • N: This value indicates that the process trace does not contain SQL_TRACE data
OPTIMIZER_TRACE	VARCHAR2(1)	Identifies if the process trace file contains OPTIMIZER_TRACE data. Possible values include: • Y: This value is used when the process trace file contains OPTIMIZER_TRACE event data • N: This value indicates that the process trace does not contain OPTIMIZER_TRACE data

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.181 V$DIAG_INCIDENT

V$DIAG_INCIDENT contains information about all incident metadata records present in the Automatic Diagnostic Repository (ADR) for the current container (PDB).

Column	Datatype	Description
INCIDENT_ID	NUMBER	ID for the current incident
PROBLEM_ID	NUMBER	ID for the problem that the incident is associated with
CREATE_TIME	TIMESTAMP(9) WITH TIME ZONE	Displays the date and time when the incident was created
CLOSE_TIME	TIMESTAMP(9) WITH TIME ZONE	Displays the date and time when the incident was closed
STATUS	NUMBER	Describes the current status for the incident. Possible values include: • 1: Incident is inflight • 2: Incident is ready • 3: Incident is tracked • 4: Incident is closed • 5: Incident data is removed • 6: Incident has been purged • 7: Incident is incomplete
FLAGS	NUMBER	For internal use only
FLOOD_CONTROLLED	NUMBER	Describes the flood control status for the current incident. Possible values include: • 0: Incident is not flood-controlled • 1: Incident is fully flood controlled (no dumps)
ERROR_FACILITY	VARCHAR2(12)	Displays the error facility for the current incident
ERROR_NUMBER	NUMBER	Displays the error number for the current incident
ERROR_ARG1	VARCHAR2(66)	Displays error-arguments associated with the given incident
ERROR_ARG2	VARCHAR2(66)	Displays error-arguments associated with the given incident
ERROR_ARG3	VARCHAR2(66)	Displays error-arguments associated with the given incident
ERROR_ARG4	VARCHAR2(66)	Displays error-arguments associated with the given incident
ERROR_ARG5	VARCHAR2(66)	Displays error-arguments associated with the given incident
ERROR_ARG6	VARCHAR2(66)	Displays error-arguments associated with the given incident
ERROR_ARG7	VARCHAR2(66)	Displays error-arguments associated with the given incident
ERROR_ARG8	VARCHAR2(66)	Displays error-arguments associated with the given incident
SIGNALLING_COMPONENT	VARCHAR2(66)	Signalling component for the given incident

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SIGNALLING_SUBCOMPONENT	VARCHAR2(66)	Signalling sub-component for the given incident
SUSPECT_COMPONENT	VARCHAR2(66)	Displays the suspect component for the given incident
SUSPECT_SUBCOMPONENT	VARCHAR2(66)	Displays the suspect sub-component for the given incident
ECID	VARCHAR2(66)	Execution context ID for the current incident
IMPACT	NUMBER	Describes the internal representation of the incident impact for the given incident
ERROR_ARG9	VARCHAR2(66)	Displays error-arguments associated with the given incident
ERROR_ARG10	VARCHAR2(66)	Displays error-arguments associated with the given incident
ERROR_ARG11	VARCHAR2(66)	Displays error-arguments associated with the given incident
ERROR_ARG12	VARCHAR2(66)	Displays error-arguments associated with the given incident
CON_UID	NUMBER	Describes the container unique ID to which the data pertains
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.182 V$DIAG_INFO

V$DIAG_INFO describes the state of Automatic Diagnostic Repository (ADR) functionality using NAME=VALUE pairs.

Column	Datatype	Description
INST_ID	NUMBER	Instance ID
NAME	VARCHAR2(64)	Identifies a piece of data that reflects the state of ADR, such as whether it is enabled, where the directories and files are located, and how many ongoing issues (incidents and problems) there are. Possible values include: • Diag Enabled: Indicates whether ADR is enabled or not • ADR Base through Health Monitor: Display different directories (ADR base, ADR home, and then subdirectories within the ADR home) • Default Trace File: Specifies the current default trace file for the current process • Active Problem Count and Active Incident Count: Specify how many problems/incidents there are in this ADR that *either* happened in the last 24 hours *or* have a piece of metadata set indicating that it is a persistent error (like a disk corruption)
VALUE	VARCHAR2(512)	Describes the current state of the piece of data identified in the NAME column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.183 V$DIAG_OPT_TRACE_RECORDS

V$DIAG_OPT_TRACE_RECORDS contains all optimizer trace event data that is present in the trace files that are part of the current Automatic Diagnostic Repository (ADR). This view also supports GV$ global views.

Column	Datatype	Description
ADR_HOME	VARCHAR2(444)	Path to current ADR home
TRACE_FILENAME	VARCHAR2(68)	Displays the name of the process trace file
RECORD_LEVEL	NUMBER	Displays the level of the trace record
PARENT_LEVEL	NUMBER	Displays the top parent level of the trace record
RECORD_TYPE	NUMBER	Displays the type of the trace record. Possible values include: • 1: Regular trace record • 2: Freeform trace record • 3: Begin Section trace record • 4: Begin Dump trace record • 5: Bucket Dump Begin trace record • 6: Section End trace record • 7: Dump End trace record • 8: Bucket Dump End trace record
TIMESTAMP	TIMESTAMP(3) WITH TIME ZONE	Displays the timestamp when the trace record was produced
PAYLOAD	VARCHAR2(4000)	Describes the trace record payload (contents)
SECTION_ID	NUMBER	Displays the section ID / dump ID of the trace record
SECTION_NAME	VARCHAR2(64)	Displays the section name / dump name of the trace record
COMPONENT_NAME	VARCHAR2(64)	Displays the component name which produced the trace record
OPERATION_NAME	VARCHAR2(64)	Displays the operation name which produced the trace record
FILE_NAME	VARCHAR2(64)	Displays the name of the code file where this trace record is produced
FUNCTION_NAME	VARCHAR2(64)	Displays the function which produced this trace record
LINE_NUMBER	NUMBER	Displays the line number in the code file which produced this trace record
THREAD_ID	VARCHAR2(64)	Displays the operating system thread ID of the process which produced the trace record
SESSION_ID	NUMBER	Displays the user session ID which generated the trace record
SERIAL#	NUMBER	Displays the user session serial number which produced the trace record

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_UID	NUMBER	Displays the container unique ID where the trace record was produced
CONTAINER_NAME	VARCHAR2(64)	Displays the container name where the trace record was produced
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.184 V$DIAG_PROBLEM

V$DIAG_PROBLEM contains information about all problem metadata records present in the Automatic Diagnostic Repository (ADR) for the current container (PDB).

Column	Datatype	Description
PROBLEM_ID	NUMBER	Displays the ID for the current problem
PROBLEM_KEY	VARCHAR2(552)	Displays the problem key for the current problem
FIRST_INCIDENT	NUMBER	Displays the first incident ID for the current problem
FIRSTINC_TIME	TIMESTAMP(9) WITH TIME ZONE	Displays the timestamp when the first incident occurred for the current problem
LAST_INCIDENT	NUMBER	Displays the last incident ID for the current problem
LASTINC_TIME	TIMESTAMP(9) WITH TIME ZONE	Displays the timestamp when the last incident occurred for the current problem
IMPACT1	NUMBER	Displays the first impact for the current problem
IMPACT2	NUMBER	Displays the second impact for the current problem
IMPACT3	NUMBER	Displays the third impact for the current problem
IMPACT4	NUMBER	Displays the fourth impact for the current problem
SERVICE_REQUEST	VARCHAR2(66)	Displays the service request number for the current problem, if a number has been entered in Support Workbench
BUG_NUMBER	VARCHAR2(66)	Displays the bug number for the current problem, if a number has been entered in Support Workbench
CON_UID	NUMBER	Displays the container unique ID to which the data pertains
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.185 V$DIAG_SESS_OPT_TRACE_RECORDS

V$DIAG_SESS_OPT_TRACE_RECORDS contains all optimizer trace event data that is present in the trace files for the current user session that is part of the current Automatic Diagnostic Repository (ADR).

Column	Datatype	Description
ADR_HOME	VARCHAR2(444)	Path to current ADR home
TRACE_FILENAME	VARCHAR2(68)	Displays the name of the process trace file
RECORD_LEVEL	NUMBER	Displays the level of the trace record
PARENT_LEVEL	NUMBER	Displays the top parent level of the trace record
RECORD_TYPE	NUMBER	Displays the type of the trace record. Possible values include: • 1: Regular trace record • 2: Freeform trace record • 3: Begin Section trace record • 4: Begin Dump trace record • 5: Bucket Dump Begin trace record • 6: Section End trace record • 7: Dump End trace record • 8: Bucket Dump End trace record
TIMESTAMP	TIMESTAMP(3) WITH TIME ZONE	Displays the timestamp when the trace record was produced
PAYLOAD	VARCHAR2(4000)	Displays the trace record payload (contents)
SECTION_ID	NUMBER	Displays the section ID / dump ID of the trace record
SECTION_NAME	VARCHAR2(64)	Displays the section name / dump name of the trace record
COMPONENT_NAME	VARCHAR2(64)	Displays the component name which produced the trace record
OPERATION_NAME	VARCHAR2(64)	Displays the operation name which produced the trace record
FILE_NAME	VARCHAR2(64)	Displays the name of the code file where this trace record is produced
FUNCTION_NAME	VARCHAR2(64)	Displays the function which produced this trace record
LINE_NUMBER	NUMBER	Displays the line number in the code file which produced this trace record
THREAD_ID	VARCHAR2(64)	Displays the operating system thread ID of the process which produced the trace record
SESSION_ID	NUMBER	Displays the user session ID which generated the trace record
SERIAL#	NUMBER	Displays the user session serial number which produced the trace record
CON_UID	NUMBER	Displays the container unique ID where the trace record was produced
CONTAINER_NAME	VARCHAR2(64)	Displays the container name where the trace record was produced
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.186 V$DIAG_SESS_SQL_TRACE_RECORDS

V$DIAG_SESS_SQL_TRACE_RECORDS contains all SQL_TRACE data that is present in the trace files for the current user session that is part of the current Automatic Diagnostic Repository (ADR).

Column	Datatype	Description
ADR_HOME	VARCHAR2(444)	Path to current ADR home
TRACE_FILENAME	VARCHAR2(68)	Displays the name of the process trace file
RECORD_LEVEL	NUMBER	Displays the level of the trace record
PARENT_LEVEL	NUMBER	Displays the top parent level of the trace record
RECORD_TYPE	NUMBER	Displays the type of the trace record. Possible values include: • 1: Regular trace record • 2: Freeform trace record • 3: Begin Section trace record • 4: Begin Dump trace record • 5: Bucket Dump Begin trace record • 6: Section End trace record • 7: Dump End trace record • 8: Bucket Dump End trace record
TIMESTAMP	TIMESTAMP(3) WITH TIME ZONE	Displays the timestamp when the trace record was produced
PAYLOAD	VARCHAR2(4000)	Displays the trace record payload (contents)
SECTION_ID	NUMBER	Displays the section ID / dump ID of the trace record
SECTION_NAME	VARCHAR2(64)	Displays the section name / dump name of the trace record
COMPONENT_NAME	VARCHAR2(64)	Displays the component name which produced the trace record
OPERATION_NAME	VARCHAR2(64)	Displays the operation name which produced the trace record
FILE_NAME	VARCHAR2(64)	Displays the name of the code file where this trace record is produced
FUNCTION_NAME	VARCHAR2(64)	Displays the function which produced this trace record
LINE_NUMBER	NUMBER	Displays the line number in the code file which produced this trace record
THREAD_ID	VARCHAR2(64)	Displays the operating system thread ID of the process which produced the trace record
SESSION_ID	NUMBER	Displays the user session ID which generated the trace record
SERIAL#	NUMBER	Displays the user session serial number which produced the trace record
CON_UID	NUMBER	Displays the container unique ID where the trace record was produced
CONTAINER_NAME	VARCHAR2(64)	Displays the container name where the trace record was produced
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.187 V$DIAG_SQL_TRACE_RECORDS

V$DIAG_SQL_TRACE_RECORDS contains all SQL_TRACE data that is present in the trace files that are part of the current Automatic Diagnostic Repository (ADR). This view also supports GV$ global views.

Column	Datatype	Description
ADR_HOME	VARCHAR2(444)	Path to current ADR home
TRACE_FILENAME	VARCHAR2(68)	Displays the name of the process trace file
RECORD_LEVEL	NUMBER	Displays the level of the trace record
PARENT_LEVEL	NUMBER	Displays the top parent level of the trace record
RECORD_TYPE	NUMBER	Displays the type of the trace record. Possible values include: • 1: Regular trace record • 2: Freeform trace record • 3: Begin Section trace record • 4: Begin Dump trace record • 5: Bucket Dump Begin trace record • 6: Section End trace record • 7: Dump End trace record • 8: Bucket Dump End trace record
TIMESTAMP	TIMESTAMP(3) WITH TIME ZONE	Displays the timestamp when the trace record was produced
PAYLOAD	VARCHAR2(4000)	Displays the trace record payload (contents)
SECTION_ID	NUMBER	Displays the section ID / dump ID of the trace record
SECTION_NAME	VARCHAR2(64)	Displays the section name / dump name of the trace record
COMPONENT_NAME	VARCHAR2(64)	Displays the component name which produced the trace record
OPERATION_NAME	VARCHAR2(64)	Displays the operation name which produced the trace record
FILE_NAME	VARCHAR2(64)	Displays the name of the code file where this trace record is produced
FUNCTION_NAME	VARCHAR2(64)	Displays the function which produced this trace record
LINE_NUMBER	NUMBER	Displays the line number in the code file which produced this trace record
THREAD_ID	VARCHAR2(64)	Displays the operating system thread ID of the process which produced the trace record
SESSION_ID	NUMBER	Displays the user session ID which generated the trace record
SERIAL#	NUMBER	Displays the user session serial number which produced the trace record
CON_UID	NUMBER	Displays the container unique ID where the trace record was produced
CONTAINER_NAME	VARCHAR2(64)	Displays the container name where the trace record was produced
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.188 V$DIAG_TRACE_FILE

V$DIAG_TRACE_FILE contains information about all trace files present in the Automatic Diagnostic Repository (ADR) for the current container (PDB). This view also supports GV$ global views.

Column	Datatype	Description
ADR_HOME	VARCHAR2(444)	Path to the current ADR home
TRACE_FILENAME	VARCHAR2(68)	Displays the name of the process trace file
CHANGE_TIME	TIMESTAMP(3) WITH TIME ZONE	Displays the change time timestamp of the process trace file
MODIFY_TIME	TIMESTAMP(3) WITH TIME ZONE	Displays the last modification timestamp of the process trace file
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.189 V$DIAG_TRACE_FILE_CONTENTS

V$DIAG_TRACE_FILE_CONTENTS contains trace data that is present in the trace files that are part of the current Automatic Diagnostic Repository (ADR). This view also supports GV$ global views.

Column	Datatype	Description
ADR_HOME	VARCHAR2(444)	Path to the current ADR home
TRACE_FILENAME	VARCHAR2(68)	Displays the name of the process trace file
RECORD_LEVEL	NUMBER	Displays the level of the trace record
PARENT_LEVEL	NUMBER	Displays the top parent level of the trace record
RECORD_TYPE	NUMBER	Displays the type of the trace record. Possible values include: • 1: Regular trace record • 2: Freeform trace record • 3: Begin Section trace record • 4: Begin Dump trace record • 5: Bucket Dump Begin trace record • 6: Section End trace record • 7: Dump End trace record • 8: Bucket Dump End trace record
TIMESTAMP	TIMESTAMP(3) WITH TIME ZONE	Displays the timestamp when the trace record was produced
PAYLOAD	VARCHAR2(4000)	Displays the trace record payload (contents)
SECTION_ID	NUMBER	Displays the section ID / dump ID of the trace record

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SECTION_NAME	VARCHAR2(64)	Displays the section name / dump name of the trace record
COMPONENT_NAME	VARCHAR2(64)	Displays the component name which produced the trace record
OPERATION_NAME	VARCHAR2(64)	Displays the operation name which produced the trace record
FILE_NAME	VARCHAR2(64)	Displays the name of the code file where this trace record is produced
FUNCTION_NAME	VARCHAR2(64)	Displays the function which produced this trace record
LINE_NUMBER	NUMBER	Displays the line number in the code file which produced this trace record
THREAD_ID	VARCHAR2(64)	Displays the operating system thread ID of the process which produced the trace record
SESSION_ID	NUMBER	Displays the user session ID which generated the trace record
SERIAL#	NUMBER	Displays the user session serial number which produced the trace record
CON_UID	NUMBER	Displays the container unique ID where the trace record was produced
CONTAINER_NAME	VARCHAR2(64)	Displays the container name where the trace record was produced
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.190 V$DISPATCHER

V$DISPATCHER displays information about the dispatcher processes.

Column	Datatype	Description
NAME	VARCHAR2(4)	Name of the dispatcher process
NETWORK	VARCHAR2(1024)	Network address of the dispatcher
PADDR	RAW(4 | 8)	Process address
STATUS	VARCHAR2(16)	Status of the dispatcher: • WAIT - Idle • SEND - Sending a message • RECEIVE - Receiving a message • CONNECT - Establishing a connection • DISCONNECT - Handling a disconnect request • BREAK - Handling a break • TERMINATE - In the process of terminating • ACCEPT - Accepting connections (no further information available) • REFUSE - Rejecting connections (no further information available)
ACCEPT	VARCHAR2(3)	Indicates whether the dispatcher is accepting new connections (YES) or not (NO)
MESSAGES	NUMBER	Number of messages processed by the dispatcher
BYTES	NUMBER	Size (in bytes) of messages processed by the dispatcher

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BREAKS	NUMBER	Number of breaks occurring in the connection
OWNED	NUMBER	Number of circuits owned by the dispatcher
CREATED	NUMBER	Number of circuits created by the dispatcher
IDLE	NUMBER	Total idle time for the dispatcher (in hundredths of a second)
BUSY	NUMBER	Total busy time for the dispatcher (in hundredths of a second)
CPU	NUMBER	Total CPU time for the dispatcher (in millionths of a second)
LISTENER	NUMBER	Most recent Oracle error number the dispatcher received from the listener
CONF_INDX	NUMBER	Zero-based index of the DISPATCHERS configuration used by the dispatcher
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.191 V$DISPATCHER_CONFIG

V$DISPATCHER_CONFIG displays information about the dispatcher configurations and their attributes.

Column	Datatype	Description
CONF_INDX	NUMBER	Zero-based index of the DISPATCHERS configuration
NETWORK	VARCHAR2(1024)	Network protocol or listening address of dispatchers (may be truncated)
DISPATCHERS	NUMBER	Number of dispatchers to maintain for the configuration
CONNECTIONS	NUMBER	Maximum number of concurrent connections per dispatcher
SESSIONS	NUMBER	Maximum number of concurrent sessions per dispatcher
MULTIPLEX	VARCHAR2(4)	Indicates whether Session Multiplexing is on: • IN • OUT • BOTH • OFF
LISTENER	VARCHAR2(1200)	Listeners to register dispatchers with (may be truncated)
SERVICE	VARCHAR2(512)	Service names supported (may be truncated)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.192 V$DISPATCHER_RATE

V$DISPATCHER_RATE displays rate statistics for a number of activities performed by the dispatcher processes.

Collected samples have an activity-specific "time-to-live" (TTL_* columns). Statistics are reported over the following two types of time intervals:

- Current statistics (CUR_ columns)

  Current statistics use samples collected over the most recent time-to-live interval.

- Historical statistics (AVG_ and most of the MAX_ columns)

  Historical statistics make use of all samples that are no longer current.

At the time of collection, a sample is current. After the time-to-live has elapsed, the sample becomes historical. Each type of activity has a specific scale (represented by the SCALE_* columns) at which the statistics are reported.

Column	Datatype	Description
NAME	VARCHAR2(4)	Name of the dispatcher process
PADDR	RAW(4 | 8)	Address of the dispatcher process
CUR_LOOP_RATE	NUMBER	Rate at which the dispatcher has been iterating through its dispatching loop, reported over the past TTL_LOOPS, in iterations/SCALE_LOOPS
CUR_EVENT_RATE	NUMBER	Rate at which the dispatcher has been processing dispatcher events, reported over the past TTL_LOOPS, in events/SCALE_LOOPS. Such dispatcher events include network events and shared server requests.
CUR_EVENTS_PER_LOOP	NUMBER	Average number of events the dispatcher has been processing in each iteration through its dispatching loop, reported over the past TTL_LOOPS, in events/iteration
CUR_MSG_RATE	NUMBER	Rate at which the dispatcher has been relaying messages between clients and shared servers, reported over the past TTL_MSG, in messages/SCALE_MSG
CUR_SVR_BUF_RATE	NUMBER	Rate at which the dispatcher has been relaying buffers to shared servers, reported over the past TTL_SVR_BUF, in messages/ SCALE_SVR_BUF
CUR_SVR_BYTE_RATE	NUMBER	Rate at which the dispatcher has been relaying data to shared servers, reported over the past TTL_SVR_BUF, in bytes/SCALE_SVR_BUF
CUR_SVR_BYTE_PER_BUF	NUMBER	Average number of data types in each buffer relayed to shared servers, reported over the past TTL_SVR_BUF, in bytes/buffer
CUR_CLT_BUF_RATE	NUMBER	Rate at which the dispatcher has been relaying buffers to clients, reported over the past TTL_CLT_BUF, in buffers/SCALE_CLT_BUF
CUR_CLT_BYTE_RATE	NUMBER	Rate at which the dispatcher has been relaying data to clients, reported over the past TTL_CLT_BUF, in bytes/SCALE_CLT_BUF
CUR_CLT_BYTE_PER_BUF	NUMBER	Average number of data bytes in each buffer relayed to clients, reported over the past TTL_CLT_BUF, in bytes/buffer
CUR_BUF_RATE	NUMBER	Rate at which the dispatcher has been relaying buffers to either clients or shared servers, reported over the past TTL_BUF, in bytes/SCALE_BUF
CUR_BYTE_RATE	NUMBER	Rate at which the dispatcher has been relaying data to either clients or shared servers, reported over the past TTL_BUF, in bytes/SCALE_BUF
CUR_BYTE_PER_BUF	NUMBER	Average number of data bytes in each buffer relayed to either clients or shared servers, reported over the past TTL_BUF, in bytes/buffer

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CUR_IN_CONNECT_RATE	NUMBER	Rate at which the dispatcher has been accepting incoming client connections, reported over the past TTL_IN_CONNECT, in connections/SCALE_IN_CONNECT
CUR_OUT_CONNECT_RATE	NUMBER	Rate at which the dispatcher has been establishing outbound connections, reported over the past TTL_OUT_CONNECT, in connections/SCALE_OUT_CONNECT
CUR_RECONNECT_RATE	NUMBER	In a connection pooling setup, the rate at which clients have been reconnecting to the dispatcher, reported over the past TTL_RECONNECT, in reconnections/SCALE_RECONNECT
MAX_LOOP_RATE	NUMBER	Maximum rate at which the dispatcher has ever iterated through its dispatching loop, reported in iterations/SCALE_LOOPS, over the dispatcher's lifetime excluding the past TTL_LOOPS
MAX_EVENT_RATE	NUMBER	Maximum rate at which the dispatcher has ever processed dispatcher events, reported in events/SCALE_LOOPS, over the dispatcher's lifetime excluding the past TTL_LOOPS
MAX_EVENTS_PER_LOOP	NUMBER	Maximum number of events the dispatcher has ever processed in one iteration through its dispatching loop, reported in events/iteration, over the dispatcher's lifetime
MAX_MSG_RATE	NUMBER	Maximum rate at which the dispatcher has ever relayed messages between clients and shared servers, reported in messages/SCALE_MSG, over the dispatcher's lifetime excluding the past TTL_MSG
MAX_SVR_BUF_RATE	NUMBER	Maximum rate at which the dispatcher has ever relayed buffers to shared servers, reported in buffers/SCALE_SVR_BUF, over the dispatcher's lifetime excluding the past TTL_SVR_BUF
MAX_SVR_BYTE_RATE	NUMBER	Maximum rate at which the dispatcher has ever relayed data to shared servers, reported in bytes/SCALE_SVR_BUF, over the dispatcher's lifetime excluding the past TTL_SVR_BUF
MAX_SVR_BYTE_PER_BUF	NUMBER	Maximum number of data bytes the dispatcher has ever relayed in one buffer to a client, reported in bytes/buffer, over the dispatcher's lifetime
MAX_CLT_BUF_RATE	NUMBER	Maximum rate at which the dispatcher has ever relayed buffers to either clients or shared servers, reported in buffers/SCALE_CLT_BUF, over the dispatcher's life time excluding the past TTL_CLT_BUF
MAX_CLT_BYTE_RATE	NUMBER	Maximum rate at which the dispatcher has ever relayed buffers to clients, reported in bytes/SCALE_CLT_BUF, over the dispatcher's lifetime excluding the last TTL_CLT_BUF
MAX_CLT_BYTE_PER_BUF	NUMBER	Maximum number of data bytes the dispatcher has ever relayed in one buffer to a client, reported in bytes/buffer, over the dispatcher's lifetime
MAX_BUF_RATE	NUMBER	Maximum rate at which the dispatcher has ever relayed buffers to either clients or shared servers, reported in buffers/SCALE_BUF, over the dispatcher's lifetime, excluding the past TTL_BUF
MAX_BYTE_RATE	NUMBER	Maximum rate at which the dispatcher has ever relayed data to either clients or shared servers, reported in bytes/SCALE_BUF, over the dispatcher's lifetime excluding the past TTL_BUF
MAX_BYTE_PER_BUF	NUMBER	Maximum number of data bytes the dispatcher has ever relayed in one buffer to either a client or a shared server, reported in bytes/buffer, over the dispatcher's lifetime
MAX_IN_CONNECT_RATE	NUMBER	Maximum rate at which the dispatcher has ever accepted incoming client connections, reported in connections/SCALE_IN_CONNECT, over the dispatcher's lifetime excluding the past TTL_IN_CONNECT

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MAX_OUT_CONNECT_RATE	NUMBER	Maximum rate at which the dispatcher has ever established outbound connections, reported in connections/SCALE_OUT_CONNECT, over the dispatcher's lifetime excluding the past TTL_OUT_CONNECT
MAX_RECONNECT_RATE	NUMBER	In a connection pooling setup, the maximum rate at which clients have ever reconnected to this dispatcher, reported in reconnections/SCALE_RECONNECT, over the dispatcher's lifetime excluding the past TTL_RECONNECT
AVG_LOOP_RATE	NUMBER	Historical average rate at which the dispatcher has iterated through its dispatching loop, reported in iterations/SCALE_LOOPS, over the dispatcher's lifetime excluding the past TTL_LOOPS
AVG_EVENT_RATE	NUMBER	Historical average rate at which the dispatcher has processed dispatcher events, reported in events/SCALE_LOOPS, over the dispatcher's lifetime excluding the past TTL_LOOPS
AVG_EVENTS_PER_LOOP	NUMBER	Historical average number of events the dispatcher has processed in one iteration through its dispatching loop, reported in events/iteration, over the dispatcher's lifetime excluding the past TTL_LOOPS
AVG_MSG_RATE	NUMBER	Historical average rate at which the dispatcher has relayed messages between clients and shared servers, reported in messages/SCALE_MSG, over the dispatcher's lifetime excluding the past TTL_MSG
AVG_SVR_BUF_RATE	NUMBER	Historical average rate at which the dispatcher has relayed buffers to shared servers, reported in buffers/SCALE_SVR_BUF, over the dispatcher's lifetime excluding the past TTL_SVR_BUF
AVG_SVR_BYTE_RATE	NUMBER	Historical average rate at which the dispatcher has relayed data to shared servers, reported in bytes/SCALE_SVR_BUF, over the dispatcher's lifetime excluding the past TTL_SVR_BUF
AVG_SVR_BYTE_PER_BUF	NUMBER	Historical average number of data bytes per buffer the dispatcher has relayed to shared servers, reported in bytes/buffer, over the dispatcher's lifetime excluding the past TTL_SVR_BUF
AVG_CLT_BUF_RATE	NUMBER	Historical average rate at which the dispatcher has relayed buffers to clients, reported in buffers/SCALE_CLT_BUF, over the dispatcher's lifetime excluding the past TTL_CLT_BUF
AVG_CLT_BYTE_RATE	NUMBER	Historical average rate at which the dispatcher has relayed data to clients, reported in bytes/SCALE_CLT_BUF, over the dispatcher's lifetime excluding the past TTL_CLT_BUF
AVG_CLT_BYTE_PER_BUF	NUMBER	Historical average number of data bytes per buffer the dispatcher has relayed to clients, reported in bytes/buffer, over the dispatcher's lifetime excluding the past TTL_CLT_BUF
AVG_BUF_RATE	NUMBER	Historical average rate at which the dispatcher has relayed buffers to either clients or shared servers, reported in buffers/SCALE_BUF, over the dispatcher's lifetime excluding the past TTL_BUF
AVG_BYTE_RATE	NUMBER	Historical average rate at which the dispatcher has relayed data to either clients or shared servers, reported in bytes/SCALE_BUF, over the dispatcher's lifetime excluding the past TTL_BUF
AVG_BYTE_PER_BUF	NUMBER	Historical average number of data bytes per buffer the dispatcher has relayed to either clients or shared servers, reported in bytes/buffer, over the dispatcher's lifetime excluding the past TTL_BUF
AVG_IN_CONNECT_RATE	NUMBER	Historical average rate at which the dispatcher has accepted incoming client connections, reported in connections/SCALE_IN_CONNECT, over the dispatcher's lifetime excluding the past TTL_IN_CONNECT

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
AVG_OUT_CONNECT_RATE	NUMBER	Historical average rate at which the dispatcher has established outbound connections, reported in connections/SCALE_OUT_CONNECT, over the dispatcher's lifetime excluding the past TTL_OUT_CONNECT
AVG_RECONNECT_RATE	NUMBER	In a connection pooling setup, the historical average rate at which clients have reconnected to this dispatcher, reported in reconnections/ SCALE_RECONNECT, over the dispatcher's lifetime excluding the past TTL_RECONNECT
TTL_LOOPS	NUMBER	Time-to-live for "loops" samples, reported in hundredths of a second. Default is 10 minutes.
TTL_MSG	NUMBER	Time-to-live for "messages" samples, reported in hundredths of a second. Default is 10 seconds.
TTL_SVR_BUF	NUMBER	Time-to-live for "buffers to servers" samples, reported in hundredths of a second. Default is 1 second.
TTL_CLT_BUF	NUMBER	Time-to-live for "buffers to clients" samples, reported in hundredths of a second. Default is 1 second.
TTL_BUF	NUMBER	Time-to-live for "buffers to clients/servers" samples, reported in hundredths of a second. Default is 1 second.
TTL_IN_CONNECT	NUMBER	Time-to-live for "inbound connections" samples, reported in hundredths of a second. Default is 10 minutes.
TTL_OUT_CONNECT	NUMBER	Time-to-live for "outbound connections" samples, reported in hundredths of a second. Default is 10 minutes.
TTL_RECONNECT	NUMBER	Time-to-live for "reconnections" samples, reported in hundredths of a second. Default is 10 minutes.
SCALE_LOOPS	NUMBER	Scale for "loops" statistics, reported in hundredths of a second. Default is 1 minute.
SCALE_MSG	NUMBER	Scale for "messages" statistics, reported in hundredths of a second. Default is 1 second.
SCALE_SVR_BUF	NUMBER	Scale for "buffers to servers" statistics, reported in hundredths of a second. Default is 1/10 second.
SCALE_CLT_BUF	NUMBER	Scale for "buffers to clients" statistics, reported in hundredths of a second. Default is 1/10 second.
SCALE_BUF	NUMBER	Scale for "buffers to clients/servers" statistics, reported in hundredths of a second. Default is 1/10 second.
SCALE_IN_CONNECT	NUMBER	Scale for "inbound connections" statistics, reported in hundredths of a second. Default is 1 minute.
SCALE_OUT_CONNECT	NUMBER	Scale for "outbound connections" statistics, reported in hundredths of a second. Default is 1 minute.
SCALE_RECONNECT	NUMBER	Scale for "reconnections" statistics, reported in hundredths of a second. Default is 1 minute.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.193 V$DNFS_CHANNELS

V$DNFS_CHANNELS displays information about the Oracle process connections (channels) open to NFS servers.

Column	Datatype	Description
PNUM	NUMBER	Oracle process number
SVRNAME	VARCHAR2(255)	NFS server name
PATH	VARCHAR2(255)	Network path to the NFS server specified by IP address or by name
LOCAL	VARCHAR2(255)	Local path on the database host specified by IP address or by name
CH_ID	NUMBER	Direct NFS channel identifier
SVR_ID	NUMBER	Direct NFS server identifier
SENDS	NUMBER	Send operations over the channel since the last select
RECVS	NUMBER	Receive operations over the channel since the last select
PINGS	NUMBER	Ping operations over the channel since the last select
SPRECO	NUMBER	Reconnects for the channel on the same port since the last select, given that the reconnect occurred during the first series of reconnect attempts. If the first series of reconnect attempts fails and the reconnect process on that channel times out for 5 minutes, the next reconnect on that channel will not increment either the SPRECO or DPRECO columns, regardless of the port connected to.
DPRECO	NUMBER	Reconnects for the channel on a different port since the last select, given that the reconnect occurred during the first series of reconnect attempts. If the first series of reconnect attempts fails and the reconnect process on that channel times out for 5 minutes, the next reconnect on that channel will not increment either the SPRECO or DPRECO columns, regardless of the port connected to.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
RDMA	NUMBER	Indicates whether RDMA is enabled for the channel or not. Possible values: • 0: RDMA is not enabled for the channel • 1: RDMA is enabled for the channel
RDMA_CREDITS	NUMBER	The number of RDMA credits supported by the server
CLIENTPORT	NUMBER	The client port to which the channel is bounded
ACTIVE_SPEED	NUMBER	The active speed of the HCA card present (in Gb/second)
PEAK_FMR	NUMBER	The size of fast memory registration (FMR) memory registered by the channel (in bytes)
CURRENT_FMR	NUMBER	The current FMR memory registered (in bytes)
FMRREG_COUNT	NUMBER	The number of FMR memory registration calls for the channel

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> RDMA functionality is enabled only for the Exadata environment

7.194 V$DNFS_FILES

V$DNFS_FILES displays information about the Oracle process files open through Direct NFS.

Column	Datatype	Description
FILENAME	VARCHAR2(513)	File name
FILESIZE	NUMBER	File Size
PNUM	NUMBER	Oracle process number which opened the file
SVR_ID	NUMBER	Direct NFS server identifier which identifies the server the file is open on
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.195 V$DNFS_SERVERS

V$DNFS_SERVERS displays information about the Direct NFS servers accessed by Direct NFS.

Column	Datatype	Description
ID	NUMBER	Direct NFS server identifier
SVRNAME	VARCHAR2(255)	NFS server name
DIRNAME	VARCHAR2(1024)	Mounted directory
MNTPORT	NUMBER	NFS mount port
NFSPORT	NUMBER	NFS port
NFSVERSION	VARCHAR2(8)	NFS version
WTMAX	NUMBER	WTMAX exported by the NFS server
RTMAX	NUMBER	RTMAX exported by the NFS server
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RDMAENABLE	VARCHAR2(16)	Indicates whether the server is RDMA enabled. Possible values: • YES: The server is RDMA enabled. • NO: The server is not RDMA enabled.
RDMAPORT	NUMBER	The port number for RDMA communications on the server. The value is always 20049.
SECURITY	VARCHAR2(32)	NFS RPC authentication method used for the given NFS export

> **Note**
>
> RDMA functionality is enabled only for the Exadata environment

7.196 V$DNFS_STATS

V$DNFS_STATS displays information about the Oracle process NFS operation statistics issued by Direct NFS.

Column	Datatype	Description
PNUM	NUMBER	Oracle process number that statistics are relevant to
NFS_NULL	NUMBER	Null
NFS_GETATTR	NUMBER	Get attributes
NFS_SETATTR	NUMBER	Set attributes
NFS_LOOKUP	NUMBER	Lookup object
NFS_ACCESS	NUMBER	Access object
NFS_READLINK	NUMBER	Read link
NFS_READ	NUMBER	Read file
NFS_WRITE	NUMBER	Write file
NFS_CREATE	NUMBER	Create file
NFS_MKDIR	NUMBER	Make directory
NFS_SYMLINK	NUMBER	Symbolic link
NFS_MKNOD	NUMBER	Make node
NFS_REMOVE	NUMBER	Remove file
NFS_RMDIR	NUMBER	Remove directory
NFS_RENAME	NUMBER	Rename
NFS_LINK	NUMBER	Link
NFS_READDIR	NUMBER	Read directory
NFS_READDIRPLUS	NUMBER	Read directory plus
NFS_FSSTAT	NUMBER	File system status
NFS_FSINFO	NUMBER	File system information
NFS_PATHCONF	NUMBER	Path configuration
NFS_COMMIT	NUMBER	Commit

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NFS_MOUNT	NUMBER	Mount
NFS_READBYTES	NUMBER	Number of bytes read from NFS server
NFS_WRITEBYTES	NUMBER	Number of bytes written to NFS server
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.197 V$DYNAMIC_REMASTER_STATS

V$DYNAMIC_REMASTER_STATS displays statistical information about the dynamic remastering process of object affinity and read-mostly. All times are given in hundredths of a second, and total values reflect what has been collected since instance startup.

Column	Datatype	Description
REMASTER_TYPE	VARCHAR2(11)	Remaster process type. Possible values: • AFFINITY: This value is used for the row containing statistics that pertain to dynamic remastering activity on object affinity. • READ-MOSTLY: This value is used for the row containing statistics that pertain to dynamic remastering activity on read-mostly objects.
REMASTER_OPS	NUMBER	Total number of dynamic remastering operations
REMASTER_TIME	NUMBER	Total dynamic remastering time
REMASTERED_OBJECTS	NUMBER	Total number of objects dynamically remastered due to affinity
PERSISTENT_OBJECTS[1]	NUMBER	Current number of objects that are marked persistent read-mostly in the cluster
QUIESCE_TIME	NUMBER	Total quiesce step time
FREEZE_TIME	NUMBER	Total freeze step time
CLEANUP_TIME	NUMBER	Total cleanup step time
REPLAY_TIME	NUMBER	Total replay step time
FIXWRITE_TIME	NUMBER	Total fixwrite step time
SYNC_TIME	NUMBER	Total synchronization step time
RESOURCES_CLEANED	NUMBER	Total number of resources cleaned in the cleanup steps
REPLAYED_LOCKS_SENT	NUMBER	Total number of locks replayed to other instances in the replay steps
REPLAYED_LOCKS_RECEIVED	NUMBER	Total number of locks received from other instances in the replay steps
CURRENT_OBJECTS	NUMBER	Current number of objects remastered on this instance due to affinity or the current number of objects that are marked read-mostly in the cluster

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

[1] This column is available starting with Oracle Database 19c.

7.198 V$EDITIONABLE_TYPES

V$EDITIONABLE_TYPES lists all the editionable types based on the current compatibility setting. The SELECT privilege on V$EDITIONABLE_TYPES will be granted to PUBLIC.

The database compatibility setting will determine the set of editionable types.

With compatibility set to 11.2 or 12, this set includes FUNCTION, LIBRARY, PACKAGE, PACKAGE BODY, PROCEDURE, SYNONYM, TRIGGER, TYPE, TYPE BODY, and VIEW. With compatibility set to 12, the set will include these types as well as the SQL TRANSLATION PROFILE.

Column	Datatype	Description
EDITIONABLE_TYPE	VARCHAR2(64)	The name of the type that is editionable
TYPE#	NUMBER	The number of the type that is editionable
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

For more information about edition-based redefinitions, see *Oracle Database Development Guide*.

7.199 V$EMON

V$EMON displays performance statistics per event monitor (EMON) worker for diagnosability of notifications. All processing time and latency is in seconds.

Column	Datatype	Description
EMON#	NUMBER	EMON identifier (0 - 9)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SID	NUMBER	Session identifier
STARTUP_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when this EMON worker was started
SERVER_TYPE	VARCHAR2(8)	Notification quality of the service provided by EMON: • REGULAR • RELIABLE
STATUS	VARCHAR2(6)	EMON status: • IDLE • ACTIVE
STATUS_CHANGE_TIME	TIMESTAMP(3) WITH TIME ZONE	Time at which EMON switched to the current STATUS
NUM_NTFNS	NUMBER	Total number of notifications (including grouping notifications)
NUM_GROUPING_NTFNS	NUMBER	Number of grouping notifications
NUM_NTFNS_ALL_GROUPS	NUMBER	Total number of events in all notification groups
NUM_OCI_NTFNS	NUMBER	Number of OCI notifications
NUM_PLSQL_NTFNS	NUMBER	Number of PL/SQL notifications
NUM_EMAIL_NTFNS	NUMBER	Number of E-mail notifications
NUM_HTTP_NTFNS	NUMBER	Number of HTTP notifications
NUM_EVENTS_PROCESSED	NUMBER	Number of events posted by a publisher for which notifications have been delivered
NUM_EVENTS_PENDING	NUMBER	Number of events posted by a publisher for which notifications are not yet delivered
NUM_ANONYMOUS_NTFNS	NUMBER	Number of anonymous notifications
NUM_AQ_NTFNS	NUMBER	Number of AQ notifications
NUM_DBCHANGE_NTFNS	NUMBER	Number of DBChange notifications
TOTAL_ANONYMOUS_NTFN_TIME	NUMBER	Total time to process Anonymous notifications
TOTAL_AQ_NTFN_TIME	NUMBER	Total time to process AQ notifications
TOTAL_DBCHANGE_NTFN_TIME	NUMBER	Total time to process dbchange notifications
TOTAL_PLSQL_NTFN_TIME	NUMBER	Total time to process PL/SQL notifications
TOTAL_OCI_NTFN_TIME	NUMBER	Total time to process OCI notifications
TOTAL_EMAIL_NTFN_TIME	NUMBER	Total time to process E-mail notifications
TOTAL_HTTP_NTFN_TIME	NUMBER	Total time to process HTTP notifications
TOTAL_EMON_LATENCY	NUMBER	Total latency in processing events
REGISTRATIONS_EXPIRED	NUMBER	Number of expired registrations
REGISTRATIONS_PURGED	NUMBER	Number of purged registrations
REGISTRATIONS_INVALID	NUMBER	Number of registrations invalidated due to notification delivery failure
LAST_UPDATE_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when statistics were last updated
CON_ID	NUMBER	The ID of the container to which the data pertains. The CON_ID value in this view is always 0. The rows pertain to the entire CDB or to the non-CDB.

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.200 V$EMX_USAGE_STATS

V$EMX_USAGE_STATS is used to track how often each report in Oracle Enterprise Manager Database Express (EM Express) is used and how long the EM Express servlet takes to serve these reports to the client.

This view contains statistics such as the number of requests captured in the EM Express servlet for each report, total elapsed time for the EM Express servlet to render each report, as well as a detailed time breakdown including database login time, request initialization time, time to run the SQL query, and time to stream and send the query result back to the client.

All statistics are accumulated over all requests for each EM Express report since the last time the database instance was restarted.

It also includes the timestamp of the last request for each report.

Column	Datatype	Description
REPORT	VARCHAR2(400)	Name of the EM Express report. All EM Express reports are XML reports with this format: %/orarep/*component*/*report*
COUNT	NUMBER	Number of requests captured by the EM Express servlet for this report since the last time the database instance was restarted
LOGIN_ELAPSED_TIME	NUMBER	Accumulated elapsed time for logging into the database across all requests for this report since the last time the database instance was restarted
INITREQ_ELAPSED_TIME	NUMBER	Accumulated elapsed time for initializing and preparing requests across all requests for this report since the last time the database instance was restarted
SQL_ELAPSED_TIME	NUMBER	Accumulated elapsed time for all requests that ran SQL queries for this report since the last time the database instance was restarted
SEND_ELAPSED_TIME	NUMBER	Accumulated elapsed time for streaming and sending the query result back to the client browser across all requests for this report since the last time the database instance was restarted
TOTAL_ELAPSED_TIME	NUMBER	Accumulated total elapsed time across all requests for this report since the last time the database instance was restarted
LAST_REQ_TIME	DATE	Timestamp of the last request for this report since the last time the database instance was restarted
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.201 V$ENABLEDPRIVS

V$ENABLEDPRIVS displays the system privileges that have been granted to the current user directly or through the currently enabled roles.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PRIV_NUMBER	NUMBER	Numeric identifier of the system privilege The matching privilege name for each PRIV_NUMBER can be found in the table SYSTEM_PRIVILEGE_MAP.
SCOPE	VARCHAR2(11)	Indicates the scope with which a privilege was granted. Possible values: • COMMON • APPLICATION • LOCAL • NULL (if connected to a non-CDB)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"SYSTEM_PRIVILEGE_MAP"

7.202 V$ENCRYPTED_TABLESPACES

V$ENCRYPTED_TABLESPACES displays information about tablespaces that are encrypted.

In a non-CDB, the information displayed by this view is meaningful only when the database is open and for data files that are online. In a CDB, the information displayed by this view is meaningful only for tablespaces in open containers. This is because the information is derived after the file headers making up a tablespace have been examined during the open operation.

During an encryption or decryption operation, the encryption or decryption progress can be obtained by comparing the values of the BLOCKS_ENCRYPTED and BLOCKS_DECRYPTED columns in this view with the value of the BLOCKS column in the DBA_DATA_FILES view.

Column	Datatype	Description
TS#	NUMBER	Tablespace number
ENCRYPTIONALG	VARCHAR2(7)	Encryption algorithm: • NONE • 3DES168 • AES128 • AES192 • AES256
ENCRYPTEDTS	VARCHAR2(3)	Indicates whether the tablespace is encrypted (YES) or not (NO)
ENCRYPTEDKEY	RAW(32)	Encrypted version of the tablespace key for the encrypted tablespace
MASTERKEYID	RAW(16)	ID of the master key that was used to encrypt the tablespace key
BLOCKS_ENCRYPTED	NUMBER	Number of tablespace blocks that have been encrypted during the lifetime of this instance

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BLOCKS_DECRYPTED	NUMBER	Number of tablespace blocks that have been decrypted during the lifetime of this instance
KEY_VERSION	NUMBER	Every encrypt/decrypt/rekey of a tablespace adds a carnation/version of the tablespace key, and the key version is incremented. A decrypted tablespace could still have a none-zero key version. In certain scenarios, however, the tablespace key version might reset to 0; for example, when a tablespace or a pluggable database (PDB) is plugged into a foreign database, or if the control file is recreated.
STATUS	VARCHAR2(10)	Shows the status of a tablespace. Possible values: • NORMAL: Used when the tablespace is not in one of the other statuses. • REKEYING: Used when a rekey operation is taking place • ENCRYPTING: Used when an encrypt operation is taking place • DECRYPTING: Used when a decrypt operation is taking place • UNKNOWN: Used when the database is mounted but not open (before the datafile is online and the key is known to the database) **Note:** The REKEYING, ENCRYPTING, and DECRYPTING values can also be reported after a datafile, whose encryption state is different from the tablespace definition, is restored. In such cases, the value represents the operation that is required to make all datafiles in the tablespace consistent by using the FINISH clause of the ALTER TABLESPACE ENCRYPTION statement. See *Oracle Database Advanced Security Guide* for information about changing the encryption state of a tablespace.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$DATABASE_KEY_INFO"

7.203 V$ENCRYPTION_KEYS

V$ENCRYPTION_KEYS displays master key description attributes.

Note that, even if Transparent Data Encryption (TDE) is not configured in the database, a query of V$ENCRYPTION_KEYS checks for TDE configuration validity and may print warnings in alert log.

Column	Datatype	Description
KEY_ID	VARCHAR2(78)	Master key identifier
TAG	VARCHAR2(4000)	Associated user-defined Information with the master key
CREATION_TIME	TIMESTAMP(6) WITH TIME ZONE	Time that the master key was created

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ACTIVATION_TIME	TIMESTAMP(6) WITH TIME ZONE	Time that the master key was put into use
CREATOR	VARCHAR2(128)	User that created the master key
CREATOR_ID	NUMBER	User ID that created the master key
USER	VARCHAR2(128)	User that activated the master key
USER_ID	NUMBER	User ID that activated the master key
KEY_USE	VARCHAR2(10)	Indicates whether the master key is used for TDE operations in a PDB or not
KEYSTORE_TYPE	VARCHAR2(17)	Master key is in: • OKV - Oracle Key Vault • SOFTWARE KEYSTORE • UNDEFINED - This value is shown if the database has no information about the type of keystore where the master key resides
ORIGIN	VARCHAR2(41)	Provides information about the origin of the master key: • LOCAL - The master key was created locally in this database • IMPORTED - The master key was imported from another database • IMPORTED BUT KEY METADATA CREATED LOCALLY - The master key was imported from another database, but the key metadata was created locally due to activation • UNKNOWN BUT KEY METADATA CREATED LOCALLY - It is unknown whether the master key was imported from another database or created locally, but the key metadata was created locally due to activation • UNDEFINED - The status of the master key is unknown
BACKED_UP	VARCHAR2(9)	Indicates whether the key has been backed up or not
CREATOR_DBNAME	VARCHAR2(128)	Database that created the key
CREATOR_DBID	NUMBER	Database ID where the key was created
CREATOR_INSTANCE_NAME	VARCHAR2(30)	Instance name of the instance where the key was created
CREATOR_INSTANCE_NUMBER	NUMBER	Instance number of the instance where the key was created
CREATOR_INSTANCE_SERIAL	NUMBER	Serial number of the instance where the key was created
CREATOR_PDBNAME	VARCHAR2(128)	PDB where the key was created
CREATOR_PDBID	NUMBER	PDB ID where the key was created
CREATOR_PDBUID	NUMBER	PDB UID where the key was created
CREATOR_PDBGUID	RAW(16)	PDB GUID where the key was created
ACTIVATING_DBNAME	VARCHAR2(128)	Database that activated the key
ACTIVATING_DBID	NUMBER	Database ID where the key was activated
ACTIVATING_INSTANCE_NAME	VARCHAR2(30)	Instance name of the instance where the key was activated
ACTIVATING_INSTANCE_NUMBER	NUMBER	Instance number of the instance where the key was activated
ACTIVATING_INSTANCE_SERIAL	NUMBER	Serial number of the instance where the key was activated
ACTIVATING_PDBNAME	VARCHAR2(128)	PDB where the key was activated
ACTIVATING_PDBID	NUMBER	PDB ID where the key was activated
ACTIVATING_PDBUID	NUMBER	PDB UID where the key was activated
ACTIVATING_PDBGUID	RAW(16)	PDB GUID where the key was activated

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Advanced Security Guide for information about keystore management

7.204 V$ENCRYPTION_WALLET

V$ENCRYPTION_WALLET displays information on the status of the wallet and the wallet location for Transparent Data Encryption (TDE).

In a multitenant container database (CDB), this view displays information on the wallets for all pluggable database (PDBs) when queried from CDB$ROOT. When queried from a PDB, this view only displays wallet details of that PDB.

Note that, even if TDE is not configured in the database, a query of V$ENCRYPTION_WALLET checks for TDE configuration validity and may print warnings in alert log.

Column	Datatype	Description
WRL_TYPE	VARCHAR2(20)	Type of the wallet resource locator (for example, FILE)
WRL_PARAMETER	VARCHAR2(4000)	Parameter of the wallet resource locator (for example, absolute directory location of the wallet or keystore, if WRL_TYPE = FILE)
STATUS	VARCHAR2(30)	Status of the wallet. Possible values: • CLOSED: The wallet is closed • NOT_AVAILABLE: The wallet is not available in the location specified by the WALLET_ROOT initialization parameter • OPEN: The wallet is open • OPEN_NO_MASTER_KEY: The wallet is open, but no master key is set • OPEN_UNKNOWN_MASTER_KEY_STATUS: The wallet is open, but the database could not determine whether the master key is set. This situation can occur when the database is in the mounted state and cannot check if the master key for a hardware keystore is set because the data dictionary is not available. • UNDEFINED: The database could not determine the status of the wallet
WALLET_TYPE	VARCHAR2(20)	Displays the type of keystore being used, FILE or OKV If the keystore was created with the mkstore utility, then the WALLET_TYPE is UNKNOWN. Oracle recommends that you create keystores with the ADMINISTER KEY MANAGEMENT statement.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
WALLET_ORDER	VARCHAR2(9)	Possible values: • SINGLE - This value is displayed: – When only a single wallet is configured – In an auto-login OKV configuration. In this case, TDE_CONFIGURATION = OKV|FILE, and the SINGLE value is displayed for both the OKV and FILE keystores. • PRIMARY - When more than one wallet is configured, this value indicates that the wallet is primary (holds the current master key). • SECONDARY - When more than one wallet is configured, this value indicates that the wallet is secondary (holds old keys). The lookup of master keys happens in the primary keystore first, and then in the secondary keystore, if required. If there is only one type of keystore (Hardware Security Module or Software Keystore) being used, then SINGLE will appear. If both types are used, then the value in this column shows the order in which each keystore will be looked up.
KEYSTORE_MODE	VARCHAR2(8)	Displays the keystore mode: • NONE: This value is seen when this column is queried from the CDB$ROOT, or when the database is a non-CDB. The keystore mode does not apply in these cases. • UNITED: The PDB is configured to use the wallet of the CDB$ROOT. To open the wallet in this configuration, the password of the wallet of the CDB$ROOT must be used. • ISOLATED: The PDB is configured to use its own wallet. To open the wallet in this configuration, the password of the isolated wallet must be used.
FULLY_BACKED_UP	VARCHAR2(9)	Indicates whether all the keys in the keystore have been backed up
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "TDE_CONFIGURATION"
- "WALLET_ROOT"
- *Oracle Database Advanced Security Guide* for information about creating user-defined master encryption keys
- *Oracle Database Advanced Security Guide* for information about opening hardware keystores

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.205 V$ENQUEUE_LOCK

V$ENQUEUE_LOCK displays all locks owned by enqueue state objects. The columns in this view are identical to the columns in V$LOCK.

See Also

"V$LOCK"

Column	Datatype	Description
ADDR	RAW(4 | 8)	Address of lock state object
KADDR	RAW(4 | 8)	Address of lock
SID	NUMBER	Identifier for session holding or acquiring the lock
TYPE	VARCHAR2(2)	Type of lock. Lists user and system types that can have locks.
ID1	NUMBER	Lock identifier #1 (depends on type)
ID2	NUMBER	Lock identifier #2 (depends on type)
LMODE	NUMBER	Lock mode in which the session holds the lock: • 0 - none • 1 - null (NULL) • 2 - row-S (SS) • 3 - row-X (SX) • 4 - share (S) • 5 - S/Row-X (SSX) • 6 - exclusive (X)
REQUEST	NUMBER	Lock mode in which the process requests the lock: • 0 - none • 1 - null (NULL) • 2 - row-S (SS) • 3 - row-X (SX) • 4 - share (S) • 5 - S/Row-X (SSX) • 6 - exclusive (X)
CTIME	NUMBER	Time since current mode was granted
BLOCK	NUMBER	The lock is blocking another lock
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.206 V$ENQUEUE_STAT

V$ENQUEUE_STAT displays statistics on the number of enqueue (lock) requests for each type of lock.

Column	Datatype	Description
INST_ID	NUMBER	ID of the instance
EQ_TYPE	VARCHAR2(2)	Type of enqueue requested
TOTAL_REQ#	NUMBER	Total number of enqueue requests or enqueue conversions for this type of enqueue
TOTAL_WAIT#	NUMBER	Total number of times an enqueue request or conversion resulted in a wait
SUCC_REQ#	NUMBER	Number of times an enqueue request or conversion was granted
FAILED_REQ#	NUMBER	Number of times an enqueue request or conversion failed
CUM_WAIT_TIME	NUMBER	Total amount of time (in milliseconds) spent waiting for the enqueue or enqueue conversion
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.207 V$ENQUEUE_STATISTICS

V$ENQUEUE_STATISTICS displays statistics on the number of enqueue (lock) requests for each type of lock.

V$ENQUEUE_STATISTICS encompasses V$ENQUEUE_STAT and gives more detailed information (several rows for same enqueues with different reasons).

Column	Datatype	Description
EQ_NAME	VARCHAR2(64)	Name of the enqueue request
EQ_TYPE	VARCHAR2(2)	Type of enqueue requested
REQ_REASON	VARCHAR2(64)	Reason for the enqueue request
TOTAL_REQ#	NUMBER	Total number of enqueue requests or enqueue conversions for this type of enqueue
TOTAL_WAIT#	NUMBER	Total number of times an enqueue request or conversion resulted in a wait
SUCC_REQ#	NUMBER	Number of times an enqueue request or conversion was granted
FAILED_REQ#	NUMBER	Number of times an enqueue request or conversion failed
CUM_WAIT_TIME	NUMBER	Total amount of time (in milliseconds) spent waiting for the enqueue or enqueue conversion
REQ_DESCRIPTION	VARCHAR2(4000)	Description of the enqueue request
EVENT#	NUMBER	Event number

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$ENQUEUE_STAT"

7.208 V$EVENT_HISTOGRAM

V$EVENT_HISTOGRAM displays a histogram of the number of waits, the maximum wait, and total wait time on an event basis, in milliseconds. The histogram has buckets of time intervals from < 1 ms, < 2 ms, < 4 ms, < 8 ms, ... < 2^{21} ms, < 2^{22} ms, and >= 2^{22} ms.

The histogram will not be filled unless the TIMED_STATISTICS initialization parameter is set to true.

Column	Datatype	Description
EVENT#	NUMBER	Event number
EVENT	VARCHAR2(64)	Name of the event
WAIT_TIME_MILLI	NUMBER	Amount of time the bucket represents (in milliseconds). If the duration = *num*, then this column represents waits of duration < *num* that are not included in any smaller bucket.
WAIT_COUNT	NUMBER	Number of waits of the duration belonging to the bucket of the histogram
LAST_UPDATE_TIME	VARCHAR2(64)	Indicates the last time the bucket was updated (the ending timestamp of the last wait falling into the bucket's duration)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"TIMED_STATISTICS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.209 V$EVENT_HISTOGRAM_MICRO

V$EVENT_HISTOGRAM_MICRO displays a histogram of the number of waits, the maximum wait, and total wait time on an event basis, in microseconds. The histogram has buckets of time intervals from < 1 us, < 2 us, < 4 us, < 8 us, ... < 2^{31} us, < 2^{32} us, and >= 2^{32} us.

The histogram will not be filled unless the TIMED_STATISTICS initialization parameter is set to true.

Column	Datatype	Description
EVENT#	NUMBER	Event number
EVENT	VARCHAR2(64)	Name of the event
WAIT_TIME_FORMAT	VARCHAR2(30)	A human readable time string which is converted from WAIT_TIME_MICRO. When WAIT_TIME_MICRO < 1 millisecond, WAIT_TIME_FORMAT is shown in microseconds. When WAIT_TIME_MICRO < 1 second, WAIT_TIME_FORMAT is shown in milliseconds. When WAIT_TIME_MICRO < 1 minute, WAIT_TIME_FORMAT is shown in seconds. When WAIT_TIME_MICRO > 1 minute, WAIT_TIME_FORMAT is shown in minutes and seconds.
WAIT_TIME_MICRO	NUMBER	Amount of time the bucket represents (in microseconds). If the duration = num, then this column represents waits of duration < num that are not included in any smaller bucket.
WAIT_COUNT	NUMBER	Number of waits of the duration belonging to the bucket of the histogram
LAST_UPDATE_TIME	VARCHAR2(64)	Indicates the last time the bucket was updated (the ending timestamp of the last wait falling into the bucket's duration)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.210 V$EVENT_NAME

V$EVENT_NAME displays information about wait events.

Column	Datatype	Description
EVENT#	NUMBER	Number of the wait event
EVENT_ID	NUMBER	Identifier of the wait event
NAME	VARCHAR2(64)	Name of the wait event. Names that appear in this column remain stable across Oracle Database releases, and they can be relied on by customer scripts.
PARAMETER1	VARCHAR2(64)	Description of the first parameter for the wait event
PARAMETER2	VARCHAR2(64)	Description of the second parameter for the wait event
PARAMETER3	VARCHAR2(64)	Description of the third parameter for the wait event
WAIT_CLASS_ID	NUMBER	Identifier of the class of the wait event

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
WAIT_CLASS#	NUMBER	Number of the class of the wait event
WAIT_CLASS	VARCHAR2(64)	Name of the class of the wait event **See Also:** "Classes of Wait Events" for a description of the different wait event classes
DISPLAY_NAME	VARCHAR2(64)	A clearer and more descriptive name for the wait event that appears in the NAME column. Names that appear in the DISPLAY_NAME column can change across Oracle Database releases, therefore customer scripts should not rely on names that appear in the DISPLAY_NAME column across releases.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.211 V$EVENTMETRIC

V$EVENTMETRIC displays values of wait event metrics for the most recent 60-second interval.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
EVENT#	NUMBER	Number of the event
EVENT_ID	NUMBER	Identifier of the event
NUM_SESS_WAITING	NUMBER	Number of sessions waiting at the end of the interval
TIME_WAITED	NUMBER	Time waited (in hundredths of a second)
WAIT_COUNT	NUMBER	Number of times waited
TIME_WAITED_FG	NUMBER	Time waited (in hundredths of a second), from foreground sessions
WAIT_COUNT_FG	NUMBER	Number of times waited, from foreground sessions
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.212 V$EXADIRECT_ACL

V$EXADIRECT_ACL monitors current ACLs propagated to the database instance.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SERVICE_NAME	VARCHAR2(512)	Database service name
SGID	VARCHAR2(39)	Identifier of the VM allowed access to the specified service name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.213 V$EXECUTION

V$EXECUTION displays information on parallel execution.

Column	Datatype	Description
PID	NUMBER	Session ID
DEPTH	NUMBER	The depth
FUNCTION	VARCHAR2(10)	Session serial number
TYPE	VARCHAR2(7)	Name of the OBJECT_NODE in plan table
NVALS	NUMBER	Elapsed time for OBJECT_NODE
VAL1	NUMBER	The value for number 1
VAL2	NUMBER	The value for number 2
SEQH	NUMBER	A sequence
SEQL	NUMBER	A sequence
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.214 V$EXP_STATS

V$EXP_STATS stores the expression tracking statistics of recently executed queries.

Column	Datatype	Description
EXPID	NUMBER	Expression ID of the current expression
OBJNUM	NUMBER	The object number contained in the expression
DYNCOST	NUMBER	Optimizer dynamic cost of evaluating the expression
EVALCNT	NUMBER	Number of times the expression has been evaluated

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "ALL_EXPRESSION_STATISTICS"
- "DBA_EXPRESSION_STATISTICS"
- "ALL_EXPRESSION_STATISTICS"

7.215 V$FALSE_PING

V$FALSE_PING is deprecated. The information that was provided in this view is now provided in the V$INSTANCE_CACHE_TRANSFER and V$SEGMENT_STATISTICS views.

Column	Datatype	Description
FILE#	NUMBER	Data file identifier number (to find the file name, query DBA_DATA_FILES or V$DBFILE)
BLOCK#	NUMBER	Block number
STATUS	VARCHAR2(10)	Status of the block: • free - Not currently in use • xcur - Exclusive • scur - Shared current • cr - Consistent read • read - Being read from disk • mrec - In media recovery mode • irec - In instance recovery mode
XNC	NUMBER	Number of PCM lock conversions from Exclusive mode due to contention with another instance. This column is obsolete and maintained for backward compatibility.
FORCED_READS	NUMBER	Number of times the block had to be reread from the cache because another instance has forced it out of this instance's cache by requesting the lock on the block in exclusive mode
FORCED_WRITES	NUMBER	Number of times GCS had to write this block to cache because this instance had used the block and another instance had requested the lock on the block in a conflicting mode
NAME	VARCHAR2(128)	Name of the database object containing the block
PARTITION_NAME	VARCHAR2(128)	NULL for nonpartitioned objects
KIND	VARCHAR2(15)	Type of database object
OWNER#	NUMBER	Owner number

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LOCK_ELEMENT_ADDR	RAW(4 | 8)	Address of the lock element that contains the PCM lock that is covering the buffer. If more than one buffer has the same address, then these buffers are covered by the same PCM lock.
LOCK_ELEMENT_NAME	NUMBER	Name of the lock that contains the PCM lock that is covering the buffer
CLASS#	NUMBER	Lock element class
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$INSTANCE_CACHE_TRANSFER"
- "V$SEGMENT_STATISTICS"

7.216 V$FAST_START_SERVERS

V$FAST_START_SERVERS provides information about all the recovery workers performing parallel transaction recovery.

Column	Datatype	Description
STATE	VARCHAR2(11)	State of the server (IDLE or RECOVERING)
UNDOBLOCKSDONE	NUMBER	Number of undo blocks done so far
PID	NUMBER	Process ID
XID	RAW(8)	Transaction ID
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.217 V$FAST_START_TRANSACTIONS

V$FAST_START_TRANSACTIONS displays information about the progress of the transactions that Oracle is recovering.

This view displays only transactions recovered in parallel. Recovery progress for serial transactions is not displayed.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
USN	NUMBER	Undo segment number of the transaction
SLT	NUMBER	Slot within the rollback segment
SEQ	NUMBER	Incarnation number of the slot
STATE	VARCHAR2(16)	State of the transaction (may be TO BE RECOVERED, RECOVERED, or RECOVERING)
UNDOBLOCKSDONE	NUMBER	Number of undo blocks completed on the transaction
UNDOBLOCKSTOTAL	NUMBER	Total number of undo blocks that need recovery
PID	NUMBER	ID of the current server it has been assigned to
CPUTIME	NUMBER	Time for which recovery has progressed (in seconds)
PARENTUSN	NUMBER	Undo segment number of the parent transaction in PDML
PARENTSLT	NUMBER	Slot of the parent transaction in PDML
PARENTSEQ	NUMBER	Sequence number of the parent transaction in PDML
XID	RAW(8)	Transaction ID
PXID	RAW(8)	Parent transaction ID
RCVSERVERS	NUMBER	Number of servers used in the last recovery
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.218 V$FILE_CACHE_TRANSFER

V$FILE_CACHE_TRANSFER is deprecated. The information that was provided in this view is now provided in the V$INSTANCE_CACHE_TRANSFER and V$SEGMENT_STATISTICS views.

Column	Datatype	Description
FILE_NUMBER	NUMBER	Number of the data file
X_2_NULL	NUMBER	Number of blocks with Exclusive-to-NULL conversions; always 0
X_2_NULL_FORCED_WRITE	NUMBER	Number of Exclusive-to-NULL forced writes; always 0
X_2_NULL_FORCED_STALE	NUMBER	Number of Exclusive-to-NULL blocks converted to CR; always 0
X_2_S	NUMBER	Number of blocks with Exclusive-to-Shared conversions; always 0
X_2_S_FORCED_WRITE	NUMBER	Number of Exclusive-to-Shared forced writes; always 0
S_2_NULL	NUMBER	Number of blocks with Shared-to-NULL conversions; always 0
S_2_NULL_FORCED_STALE	NUMBER	Number of Shared-to-NULL blocks converted to CR; always 0
RBR	NUMBER	Number of reuse blocks cross-instance calls; always 0
RBR_FORCED_WRITE	NUMBER	Number of blocks written due to reuse blocks cross-instance calls; always 0
RBR_FORCED_STALE	NUMBER	Number of blocks marked as flushed due to reuse blocks cross-instance calls; always 0

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NULL_2_X	NUMBER	Number of blocks with NULL-to-Exclusive conversions; always 0
S_2_X	NUMBER	Number of blocks with Shared-to-Exclusive conversions; always 0
NULL_2_S	NUMBER	Number of blocks with NULL-to-Shared conversions; always 0
CR_TRANSFERS	NUMBER	Number of CR blocks received; always 0
CUR_TRANSFERS	NUMBER	Number of current blocks received; always 0
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$INSTANCE_CACHE_TRANSFER"
- "V$SEGMENT_STATISTICS"

7.219 V$FILE_HISTOGRAM

V$FILE_HISTOGRAM displays a histogram of all synchronous single block reads on a per-file basis (for data files). The histogram has buckets of time intervals from < 1 ms, < 2 ms, < 4 ms, < 8 ms, ... < 2^{21} ms, < 2^{22} ms, and >= 2^{22} ms.

The histogram will not be filled unless the STATISTICS_LEVEL initialization parameter is set to ALL.

Column	Datatype	Description
FILE#	NUMBER	File number
SINGLEBLKRDTIM_MILLI	NUMBER	Amount of time the bucket represents (in milliseconds). If the duration = *num*, then this column represents waits of duration < *num* that are not included in any smaller bucket.
SINGLEBLKRDS	NUMBER	Number of waits of the duration belonging to the bucket of the histogram
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.220 V$FILEMETRIC

V$FILEMETRIC displays values of file metrics for the most recent 10-minute interval. A history of the last one hour will be kept in the system.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
FILE_ID	NUMBER	File number
CREATION_TIME	NUMBER	Timestamp of the file creation
AVERAGE_READ_TIME	NUMBER	Average file read time (in hundredths of a second)
AVERAGE_WRITE_TIME	NUMBER	Average file write time (in hundredths of a second)
PHYSICAL_READS	NUMBER	Number of physical reads
PHYSICAL_WRITES	NUMBER	Number of physical writes
PHYSICAL_BLOCK_READS	NUMBER	Number of physical block reads
PHYSICAL_BLOCK_WRITES	NUMBER	Number of physical block writes
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.221 V$FILEMETRIC_HISTORY

V$FILEMETRIC_HISTORY displays values of file metrics for all intervals in the last one hour.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
FILE_ID	NUMBER	File number
CREATION_TIME	NUMBER	Timestamp of the file creation
AVERAGE_READ_TIME	NUMBER	Average file read time (in hundredths of a second)
AVERAGE_WRITE_TIME	NUMBER	Average file write time (in hundredths of a second)
PHYSICAL_READS	NUMBER	Number of physical reads
PHYSICAL_WRITES	NUMBER	Number of physical writes
PHYSICAL_BLOCK_READS	NUMBER	Number of physical block reads
PHYSICAL_BLOCK_WRITES	NUMBER	Number of physical block writes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.222 V$FILESPACE_USAGE

V$FILESPACE_USAGE summarizes space allocation information of each data file and temp file.

Column	Datatype	Description
TABLESPACE_ID	NUMBER	ID of the tablespace to which the file belongs
RFNO	NUMBER	Relative file number of the file
ALLOCATED_SPACE	NUMBER	Total allocated space in the file
FILE_SIZE	NUMBER	Current file size
FILE_MAXSIZE	NUMBER	Maximum file size
CHANGESCN_BASE	NUMBER	SCN base of the last change to the file
CHANGESCN_WRAP	NUMBER	SCN wrap of the last change to the file
CHANGESCN8	NUMBER	The 8–byte representation of the SCN at which the last change happened to the file
FLAG	NUMBER	Flags for file attributes
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.223 V$FILESTAT

V$FILESTAT displays the number of physical reads and writes done and the total number of single-block and multiblock I/Os done at file level.

As of Oracle Database 10g Release 2 (10.2), this view also includes reads done by RMAN processes for backup operations.

Column	Datatype	Description
FILE#	NUMBER	Number of the file
PHYRDS	NUMBER	Number of physical reads done
PHYWRTS	NUMBER	Number of times DBWR is required to write
PHYBLKRD	NUMBER	Number of physical blocks read

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OPTIMIZED_PHYBLKRD	NUMBER	Number of physical reads from Database Smart Flash Cache blocks
PHYBLKWRT	NUMBER	Number of blocks written to disk, which may be the same as PHYWRTS if all writes are single blocks
SINGLEBLKRDS	NUMBER	Number of single block reads
READTIM	NUMBER	Time (in hundredths of a second) spent doing reads if the TIMED_STATISTICS parameter is true; 0 if false
WRITETIM	NUMBER	Time (in hundredths of a second) spent doing writes if the TIMED_STATISTICS parameter is true; 0 if false
SINGLEBLKRDTIM	NUMBER	Cumulative single block read time (in hundredths of a second)
AVGIOTIM	NUMBER	Average time (in hundredths of a second) spent on I/O, if the TIMED_STATISTICS parameter is true; 0 if false
LSTIOTIM	NUMBER	Time (in hundredths of a second) spent doing the last I/O, if the TIMED_STATISTICS parameter is true; 0 if false
MINIOTIM	NUMBER	Minimum time (in hundredths of a second) spent on a single I/O, if the TIMED_STATISTICS parameter is true; 0 if false
MAXIORTM	NUMBER	Maximum time (in hundredths of a second) spent doing a single read, if the TIMED_STATISTICS parameter is true; 0 if false
MAXIOWTM	NUMBER	Maximum time (in hundredths of a second) spent doing a single write, if the TIMED_STATISTICS parameter is true; 0 if false
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.224 V$FIXED_TABLE

V$FIXED_TABLE displays all dynamic performance tables, views, and derived tables in the database.

Some V$ tables (for example, V$ROLLNAME) refer to real tables and are therefore not listed.

Column	Datatype	Description
NAME	VARCHAR2(128)	Name of the object
OBJECT_ID	NUMBER	Identifier of the fixed object
TYPE	VARCHAR2(5)	Object type (TABLE | VIEW)
TABLE_NUM	NUMBER	Number that identifies the dynamic performance table if it is of type TABLE

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.225 V$FIXED_VIEW_DEFINITION

V$FIXED_VIEW_DEFINITION contains the definitions of all the fixed views (views beginning with V$).

Use this table with caution. Oracle tries to keep the behavior of fixed views the same from release to release, but the definitions of the fixed views can change without notice. Use these definitions to optimize your queries by using indexed columns of the dynamic performance tables.

Column	Datatype	Description
VIEW_NAME	VARCHAR2(128)	Name of the fixed view
VIEW_DEFINITION	VARCHAR2(4000)	Definition of the fixed view
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.226 V$FLASHBACK_DATABASE_LOG

V$FLASHBACK_DATABASE_LOG displays information about the flashback data. Use this view to help estimate the amount of flashback space required for the current workload.

Column	Datatype	Description
OLDEST_FLASHBACK_SCN	NUMBER	Lowest system change number (SCN) in the flashback data, for any incarnation
OLDEST_FLASHBACK_TIME	DATE	Time of the lowest SCN in the flashback data, for any incarnation
RETENTION_TARGET	NUMBER	Target retention time (in minutes)
FLASHBACK_SIZE	NUMBER	Current size (in bytes) of the flashback data
ESTIMATED_FLASHBACK_SIZE	NUMBER	Estimated size of flashback data needed (in bytes) for the current target retention

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.227 V$FLASHBACK_DATABASE_LOGFILE

V$FLASHBACK_DATABASE_LOGFILE displays information about the flashback log files.

Column	Datatype	Description
NAME	VARCHAR2(513)	Name of the log file
LOG#	NUMBER	Log file number
THREAD#	NUMBER	Log file thread number
SEQUENCE#	NUMBER	Log file sequence number
BYTES	NUMBER	Log file size (in bytes)
FIRST_CHANGE#	NUMBER	Lowest system change number (SCN) in the log file
FIRST_TIME	DATE	Time of the first SCN in the log file
TYPE	VARCHAR2(9)	Log type: • NORMAL • RESERVED • FREE • TO DELETE
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.228 V$FLASHBACK_DATABASE_STAT

V$FLASHBACK_DATABASE_STAT displays statistics for monitoring the I/O overhead of logging flashback data. This view also displays the estimated flashback space needed based on previous workloads.

Column	Datatype	Description
BEGIN_TIME	DATE	Beginning of the time interval
END_TIME	DATE	End of the time interval
FLASHBACK_DATA	NUMBER	Number of bytes of flashback data written during the interval
DB_DATA	NUMBER	Number of bytes of database data read and written during the interval

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
REDO_DATA	NUMBER	Number of bytes of redo data written during the interval
ESTIMATED_FLASHBACK_SIZE	NUMBER	Value of ESTIMATED_FLASHBACK_SIZE in V$FLASHBACK_DATABASE_LOG at the end of the time interval
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.229 V$FLASHBACK_TXN_GRAPH

V$FLASHBACK_TXN_GRAPH displays a tabular representation of the transaction dependency graph. For each dependency edge, there could be multiple rows, one for each conflicting operation.

This view is relevant AFTER a compensating transaction has been started through the DBMS_FLASHBACK.TRANSACTION_BACKOUT() set of functions, and is no longer relevant once the compensating transaction is either committed or rolled back. It also provides a tabular representation of the undo SQL that is not available through the CLOB XML construct in the DBA_FLASHBACK_TXN_REPORT view.

Column	Datatype	Description
COMPENSATING_XID	RAW(8)	Transaction ID of the compensating transaction
COMPENSATING_TXN_NAME	VARCHAR2(255)	Name of the compensating transaction
XID	RAW(8)	Transaction ID of a relevant transaction found in memory
TXN_NAME	VARCHAR2(255)	Name of the transaction with XID as the transaction ID; NULL if none
PARENT_XID	RAW(8)	Parent transaction ID (for a PDML transaction)
INTERESTING	NUMBER	If the transaction is in the transaction dependency graph
ORIGINAL	NUMBER	If the transaction is part of the input set provided
BACKOUT_SEQ	NUMBER	Order in which the transaction has been backed out
NUM_PREDS	NUMBER	Number of predecessors of the transaction specified by XID in the transaction graph
NUM_SUCCS	NUMBER	Number of successors of the transaction specified by XID in the transaction graph
DEP_XID	RAW(8)	One dependent transaction ID of the transaction specified by XID. This is a particular child of XID.
DEP_TXN_NAME	VARCHAR2(255)	Transaction name, if any, for the transaction specified by DEP_XID
TXN_CONF_SQL_ID	NUMBER	SQL ID of undo SQL executed in the context of XID which conflicts with the dependent transaction
DEP_TXN_CONF_SQL_ID	NUMBER	SQL ID of undo SQL executed in the context of DEP_XID which conflicts with XID
CONFLICT_TYPE	VARCHAR2(32)	The type of conflict that the conflict resolution method is used to resolve: delete, uniqueness, or update

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "DBA_FLASHBACK_TXN_REPORT"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_FLASHBACK.TRANSACTION_BACKOUT procedures

7.230 V$FLASHBACK_TXN_MODS

V$FLASHBACK_TXN_MODS displays the individual modifications of all the transactions in memory.

This view is relevant AFTER a compensating transaction has been started through the DBMS_FLASHBACK.TRANSACTION_BACKOUT( ) set of functions, and is no longer relevant once the compensating transaction is either committed or rolled back. It also provides a tabular representation of the undo SQL that is not available through the CLOB XML construct in the DBA_FLASHBACK_TXN_REPORT view.

Column	Datatype	Description
COMPENSATING_XID	RAW(8)	Transaction ID of the compensating transaction
COMPENSATING_TXN_NAME	VARCHAR2(255)	Name of the compensating transaction
XID	RAW(8)	Transaction ID of a relevant transaction found in memory
TXN_NAME	VARCHAR2(255)	Name of the transaction with XID as the transaction ID; NULL if none
PARENT_XID	RAW(8)	Parent transaction ID (for a PDML transaction)
INTERESTING	NUMBER	If the transaction is in the transaction dependency graph
ORIGINAL	NUMBER	If the transaction is part of the input set provided
BACKOUT_SEQ	NUMBER	Order in which the transaction has been backed out
UNDO_SQL	VARCHAR2(4000)	Undo SQL for the modification
UNDO_SQL_SQN	NUMBER	Order in which the given SQL has been executed to back out this transaction
UNDO_SQL_SUB_SQN	NUMBER	If the undo SQL is greater than 4000 bytes, then a sequence number, starting from 1, of a 4000-byte division of the undo SQL
BACKOUT_SQL_ID	NUMBER	SQL ID of the undo SQL (used only for this compensating transaction)
OPERATION	VARCHAR2(128)	Operation (such as insert/update/delete) performed by the forward-going operation
BACKEDOUT	NUMBER	Indicates whether the transaction has been backed out as of now
CONFLICT_MOD	NUMBER	If the concerned modification is causing a conflict

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MODS_PER_LCR	NUMBER	Sometimes an LCR could cause multiple modifications (for example, an update of an IOT could actually be a delete followed by an insert)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "DBA_FLASHBACK_TXN_REPORT"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_FLASHBACK.TRANSACTION_BACKOUT procedures

7.231 V$FLASHFILESTAT

V$FLASHFILESTAT displays statistics about Database Smart Flash Cache.

By taking snapshots of SINGLEBLKRDS and SINGLEBLKRDTIM_MICRO, you can easily calculate the average latency of all the flash files in a given time period

Column	Datatype	Description
FLASHFILE#	NUMBER	The file number of the flash file
NAME	VARCHAR2(513)	Name and path of the flash file
BYTES	NUMBER	Size of the flash file, in bytes
ENABLED	NUMBER	Indicates whether this flash file is enabled or not
SINGLEBLKRDS	NUMBER	Number of reads to the flash file
SINGLEBLKRDTIM_MICRO	NUMBER	Cumulative latency of reading blocks from this particular flash file/ device, in microseconds
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Administrator's Guide for more information about configuring Database Smart Flash Cache

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.232 V$FOREIGN_ARCHIVED_LOG

V$FOREIGN_ARCHIVED_LOG can be queried on a logical standby database to find out the list of foreign archived logs received by a database.

No rows are returned for this view on a physical standby database.

Column	Datatype	Description
RECID	NUMBER	Archived log record ID
STAMP	NUMBER	Archived log record stamp
NAME	VARCHAR2(513)	Archived log file name. If set to NULL, either the log file was cleared before it was archived or an RMAN BACKUP command with the delete input option was executed to back up archivelog all (RMAN> backup archivelog all delete input;).
DEST_ID	NUMBER	Original destination from which the archive log was generated. The value is 0 if the destination identifier is not available
THREAD#	NUMBER	Redo thread number
SEQUENCE#	NUMBER	Redo log sequence number
RESETLOGS_CHANGE#	NUMBER	Resetlogs change number of the database when the log was written
RESETLOGS_TIME	DATE	Resetlogs time of the database when the log was written
RESETLOGS_ID	NUMBER	Resetlogs identifier associated with the archived redo log
FIRST_CHANGE#	NUMBER	First change number in the archived log
FIRST_TIME	DATE	Timestamp of the first change
NEXT_CHANGE#	NUMBER	First change in the next log
NEXT_TIME	DATE	Timestamp of the next change
BLOCKS	NUMBER	Size of the archived log (in blocks)
BLOCK_SIZE	NUMBER	Redo log block size. This is the logical block size of the archived log, which is the same as the logical block size of the online log from which the archived log was copied. The online log logical block size is a platform-specific value that is not adjustable by the user.
CREATOR	VARCHAR2(7)	Creator of the archive log: ARCH - Archiver process FGRD - Foreground process RMAN - Recovery Manager SRMN - RMAN at standby LGWR - Logwriter process
REGISTRAR	VARCHAR2(7)	Registrar of the entry: RFS - Remote File Server process ARCH - Archiver process FGRD - Foreground process RMAN - Recovery Manager SRMN - RMAN at standby LGWR - Logwriter process
ARCHIVED	VARCHAR2(3)	Indicates whether the online redo log was archived (YES) or whether RMAN only inspected the log and created a record for future application of redo logs during recovery (NO).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
APPLIED	VARCHAR2(3)	Indicates whether the archivelog has been applied to its corresponding standby database (YES) or not (NO). The value is always NO for local destinations. This column is meaningful at the standby site for the ARCHIVED_LOG entries with REGISTRAR='RFS' (which means this log is shipped from the primary to the standby database). If REGISTRAR='RFS' and APPLIED is NO, then the log has arrived at the standby but has not yet been applied. If REGISTRAR='RFS' and APPLIED is YES, the log has arrived and been applied at the standby database. You can use this field to identify archivelogs that can be backed up and removed from disk.
DELETED	VARCHAR2(3)	Indicates whether an RMAN DELETE command has physically deleted the archived log file from disk, as well as logically removing it from the control file of the target database and from the recovery catalog (YES) or not (NO)
STATUS	VARCHAR2(1)	Status of the archived log: A - Available D - Deleted U - Unavailable X - Expired
COMPLETION_TIME	DATE	Time when the archiving completed
DICTIONARY_BEGIN	VARCHAR2(3)	Indicates whether the log contains the start of a LogMiner dictionary (YES) or not (NO)
DICTIONARY_END	VARCHAR2(3)	Indicates whether the log contains the end of a LogMiner dictionary (YES) or not (NO)
END_OF_REDO	VARCHAR2(3)	Indicates whether the archived redo log contains the end of all redo information from the primary database () or not ()
ARCHIVAL_THREAD#	NUMBER	Redo thread number of the instance that performed the archival operation. This column differs from the THREAD# column only when a closed thread is archived by another instance.
IS_RECOVERY_DEST_FILE	VARCHAR2(3)	Indicates whether the file was created in the fast recovery area (YES) or not (NO)
COMPRESSED	VARCHAR2(3)	Reserved for internal use
FAL	VARCHAR2(3)	Indicates whether the archive log was generated as the result of a FAL request (YES) or not (NO)
END_OF_REDO_TYPE	VARCHAR2(10)	Possible values are as follows: SWITCHOVER - Shows archived redo log files that are produced at the end of a switchover TERMINAL - Shows archived redo log files produced after a failover RESETLOGS - Shows online redo log files archived on the primary database after an ALTER DATABASE OPEN RESETLOGS statement is issued ACTIVATION - Shows any log files archived on a physical standby database after an ALTER DATABASE ACTIVATE STANDBY DATABASE statement is issued "empty string" - Any empty string implies that the log is just a normal archival and was not archived due to any of the other events
SOURCE_DBID	NUMBER	Database ID of the source database that generated this archived log

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.233 V$FS_FAILOVER_OBSERVERS

V$FS_FAILOVER_OBSERVERS provides information about fast-start failover observers.

If you are querying on the primary database, this view returns three rows, each describing one observer. However, only a row having an non-empty value in column NAME corresponds to a started observer. If you are querying on a non-primary database, the behavior of this view is not defined.

Column	Datatype	Description
NAME	VARCHAR2(513)	The fast-start failover observer name
REGISTERED	VARCHAR2(4)	Indicates if this observer is registered (YES) or not (NO). Note that the observer is registered *only* if HOST is not NULL.
HOST	VARCHAR2(513)	The name of the host where this observer is running
ISMASTER	VARCHAR2(4)	Indicates if this observer is the master observer (YES) or not (NO)
TIME_SELECTED	TIMESTAMP(9)	Shows when this observer became master observer, if ISMASTER is YES; otherwise the following constant appears: 1990-01-01 00:00:00.00
PINGING_PRIMARY	VARCHAR2(4)	Possible values: • YES: Observer is currently connected to the primary database • NO: Observer is not connected to the primary database **Note**: This field is consistent throughout an Oracle Real Application Clusters (Oracle RAC) environment; that is, if the observer is connected to any instance of the primary database in the Oracle RAC environment, all instances will show a value of YES.
PINGING_TARGET	VARCHAR2(4)	Possible values: • YES: Observer is currently connected to the target standby database • NO: Observer is not connected to the target standby database **Note**: This field is consistent throughout an Oracle Real Application Clusters (Oracle RAC) environment; that is, if the observer is connected to any instance of the target standby database in the Oracle RAC environment, all instances will show a value of YES.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.234 V$FS_FAILOVER_STATS

V$FS_FAILOVER_STATS displays statistics about fast-start failovers occurring on the system.

Column	Datatype	Description
LAST_FAILOVER_TIME	VARCHAR2(20)	Timestamp of the last fast-start failover
LAST_FAILOVER_REASON	VARCHAR2(255)	Reason for the last fast-start failover
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.235 V$FS_OBSERVER_HISTOGRAM

V$FS_OBSERVER_HISTOGRAM displays statistics that are based on the frequency of successful pings between the observer and primary database for different time intervals. The wait event in this histogram is the observer's wait until pings to the primary succeed.

The histogram displays only when there were ping failures between the observer and the primary database.

No rows are shown in this view for unregistered observers.

These statistics can be used to select an appropriate value for the FastStartFailoverThreshold configuration property for your environment.

Column	Datatype	Description
OBSERVER_NAME	VARCHAR2(513)	The Fast-Start Failover observer name
OBSERVER_HOST	VARCHAR2(513)	The name of the host where this observer is running
WAIT_TIME	NUMBER	The time interval between a pair of successful observer pings (ping-pairs) to this instance. Note that the values in this column are the upper bound of the inter-ping interval samples represented by a given histogram bucket. If WAIT_TIME = *number*, then this column represents inter-ping intervals <= *number* that are not included in any smaller bucket. The value in this column is the time (in seconds) that a ping failure lasted.
WAIT_COUNT	NUMBER	The number of ping-pairs with an inter-ping interval that corresponds with this histogram bucket. If all pings have been successful so far, this column has a value of 0.
LAST_UPDATE_TIME	VARCHAR2(20)	The last time this row was updated

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **See Also**
>
> *Oracle Data Guard Broker* for more information about the `FastStartFailoverThreshold` configuration property

Example

Assume that the following shows the status of observer's pings to the primary:

```
Ping time   Ping result
---------  -----------
1:00:00    SUCCESS
1:00:03    FAIL
1:00:06    FAIL
1:00:09    SUCCESS   => Wait time of 6 seconds
1:00:12    SUCCESS
1:00:15    FAIL
1:00:18    FAIL
1:00:21    SUCCESS   => Wait time of 6 seconds
1:00:24    SUCCESS
1:00:27    FAIL
1:00:30    SUCCESS   => Wait time of 3 seconds
```

These ping results will result in the histogram view below:

```
WAIT_TIME   WAIT_COUNT   LAST_UPDATE_TIME
  3         1            1:00:30

  6         2            1:00:21

  9         0

 12         0

...
```

In this case, the `FastStartFailoverThreshold` value should be set to larger than 6 because communication between the observer and the primary sometimes fails for 6 seconds.

7.236 V$GC_ELEMENT

`V$GC_ELEMENT` displays one entry for each global cache resource that is used by the buffer cache. The name of the global cache resource that corresponds to a lock element is {'BL', indx, class}. This is an Oracle Real Application Clusters view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
GC_ELEMENT_ADDR	RAW(4 | 8)	Address of the lock element that contains the PCM lock that is covering the buffer. If more than one buffer has the same address, then these buffers are covered by the same PCM lock.
INDX	NUMBER	Platform specific lock manager identifier
CLASS	NUMBER	Platform specific lock manager identifier
GC_ELEMENT_NAME	NUMBER	Name of the lock that contains the PCM lock that is covering the buffer
MODE_HELD	NUMBER	Numeric global-cache lock mode for the element: • 0 - Null (N): no shared/exclusive grant held • 1 - Shared (S): shared/read-compatible grant • 2 - Exclusive (X): exclusive grant
BLOCK_COUNT	NUMBER	Number of blocks covered by PCM lock
RELEASING	NUMBER	Nonzero if PCM lock is being downgraded
ACQUIRING	NUMBER	Nonzero if PCM lock is being upgraded
WRITING	NUMBER	If the GC_ELEMENT is being written, the write status
RECOVERING	NUMBER	If the GC_ELEMENT is being recovered, the recovery status
LOCAL	NUMBER	Zero if the GC_ELEMENT is local, one if it is global
FLAGS	NUMBER	Process level flags for the lock element
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.237 V$GC_ELEMENTS_WITH_COLLISIONS

V$GC_ELEMENTS_WITH_COLLISIONS is deprecated. The information that was provided in this view is now provided in the V$INSTANCE_CACHE_TRANSFER and V$SEGMENT_STATISTICS views.

Column	Datatype	Description
GC_ELEMENT_ADDR	RAW(4 | 8)	Address of the lock element that contains the PCM lock covering the buffer. If more than one buffer has the same address, then these buffers are covered by the same PCM lock.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "V$INSTANCE_CACHE_TRANSFER"
- "V$SEGMENT_STATISTICS"

7.238 V$GCR_ACTIONS

V$GCR_ACTIONS displays information about the current status of the actions defined to the GCR component that runs under the LMHB background process to detect and mitigate potential issues in the cluster instances.

Column	Datatype	Description
ACTION_ID	NUMBER	Action identification number as used internally to refer to the metric
ACTION_NAME	VARCHAR2(40)	Name of the action
ENVIRONMENT	NUMBER	Environment that the action should run in: • 0: Nowhere (do not run) • 1: Oracle ASM instance only • 2: Oracle Database instance only • 3: Oracle ASM and Oracle Database instance • 4: Standby instance • 5: Oracle ASM and Standby instance • 6: Oracle Database and Standby instance • 7: Oracle Database, Oracle ASM, and Standby instance • 15: All instance types
FLAGS	NUMBER	Various flags describing the action. Used only by Oracle for diagnosis.
ACTIVE	VARCHAR2(9)	Whether the action is active or not: • ACTIVE: Action is active. • INACTIVE: Action is disabled.
STATUS_CHANGE_TIME	TIMESTAMP(6)	Time of last action change of status
LAST_RAN_ITERATION	NUMBER	Iteration of the last time the action ran
LAST_RAN_TIME	TIMESTAMP(6)	Time of last run
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.239 V$GCR_LOG

V$GCR_LOG provides information on the last 30 significant events that have occurred in GCR in the recent past.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
INST_ID	NUMBER	Instance identier
ITERATION	NUMBER	GCR internal iteration count
TIME	TIMESTAMP(6)	Timestamp of record
TYPE	VARCHAR2(14)	Type of record: • METRIC • METRIC ENABLE • METRIC DISABLE • GLOBAL METRIC • ACTION RAN • RESET RAN
DESCRIPTION	VARCHAR2(40)	Description of function
RESULT	VARCHAR2(7)	Result: • OK • FAIL
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data.

7.240 V$GCR_METRICS

V$GCR_METRICS displays information about the current status of the metrics defined to the GCR component that runs under the LMHB background process to detect and mitigate potential issues in the cluster instances.

Column	Datatype	Description
METRIC_ID	NUMBER	Metric identification number as used internally to refer to the metric
METRIC_NAME	VARCHAR2(40)	Name of the metric
ENVIRONMENT	NUMBER	Environment that the metric should run in: • 0: Nowhere (do not run) • 1: Oracle ASM instance only • 2: Oracle Database instance only • 3: Oracle ASM and Oracle Database instance • 4: Standby instance • 5: Oracle ASM and Standby instance • 6: Oracle Database and Standby instance • 7: Oracle Database, Oracle ASM, and Standby instance • 15: All instance types
FREQUENCY	NUMBER	How often the metric runs, in seconds

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SCOPE	NUMBER	The scope of sharing of the metric: • 0: The metric does not have associated data • 1: Global - shared with all other instances in the cluster • 2: Local - used only locally (not shared) • 4: Node Global - shared with other instances in the cluster on the same node only • 8: DB Only - Shared with other instances of the same database only
DATA_TYPE	VARCHAR2(7)	The type of the data associated with this metric: • 0: None • 1: Boolean • 2: Numeric • 3: Process
STATUS	VARCHAR2(9)	Indicates the status of the metric: • ACTIVE: Indicates that the metric is active. • INACTIVE: Indicates that the metric is turned off.
STATUS_CHANGE_TIME	TIMESTAMP(6)	Time of last metric change of status
LAST_RAN_ITERATION	NUMBER	Iteration of the last time the metric ran
LAST_RAN_TIME	TIMESTAMP(6)	Time of last run
LAST_PASS_ITERATION	NUMBER	Iteration of last time the metric passed
LAST_PASS_TIME	TIMESTAMP(6)	Time of last metric pass
TOTAL_PASSES	NUMBER	Total number of times the metric passed since start of LMHB
LAST_FAIL_ITERATION	NUMBER	Iteration of last time the metric failed
LAST_FAIL_TIME	TIMESTAMP(6)	Time of last metric failure
TOTAL_FAILS	NUMBER	Total number of times the metric has failed since start of LMHB
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.241 V$GCR_STATUS

V$GCR_STATUS provides information on the current GCR status, what metrics ran and their result. It records the last 100 events.

Column	Datatype	Description
INST_ID	NUMBER	Instance identifier
ITERATION	NUMBER	GCR internal iteration count
TIME	TIMESTAMP(6)	Timestamp of record

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TYPE	VARCHAR2(14)	Type of record: • METRIC • GLOBAL METRIC • ACTION RAN • RESET RAN
DESCRIPTION	VARCHAR2(40)	Description of function
RESULT	VARCHAR2(7)	Result: • OK • FAIL
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data.

7.242 V$GCSHVMASTER_INFO

V$GCSHVMASTER_INFO describes the current and previous master instances and the number of re-masterings of Global Cache Service resources except those belonging to files mapped to a particular master.

Column	Datatype	Description
HV_ID	NUMBER	PCM hash value ID
CURRENT_MASTER	NUMBER	Master instance of this PCM hash value ID
PREVIOUS_MASTER	NUMBER	Previous master instance of this PCM hash value ID
REMASTER_CNT	NUMBER	Number of times this has been remastered
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.243 V$GCSPFMASTER_INFO

V$GCSPFMASTER_INFO describes the current and previous master instances and the number of re-masterings of Global Cache Service resources belonging to files mapped to instances.

Column	Datatype	Description
FILE_ID	NUMBER	File number
DATA_OBJECT_ID	NUMBER	Data object ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
GC_MASTERING_POLICY	VARCHAR2(11)	Data object type. The possible values are Affinity or Read mostly.
CURRENT_MASTER	NUMBER	Master instance of this file
PREVIOUS_MASTER	NUMBER	Previous master instance of this file
REMASTER_CNT	NUMBER	Number of times this has been remastered
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.244 V$GES_BLOCKING_ENQUEUE

V$GES_BLOCKING_ENQUEUE describes all locks currently known to lock manager that are being blocked or blocking others.

The output of this view is a subset of the output from V$GES_ENQUEUE. This is an Oracle Real Application Clusters view.

See Also

"V$GES_ENQUEUE" for a description of all locks known to the lock manager

Column	Datatype	Description
HANDLE	RAW(4 | 8)	Lock pointer
GRANT_LEVEL	VARCHAR2(9)	Granted level of the lock
REQUEST_LEVEL	VARCHAR2(9)	Requested level of the lock
RESOURCE_NAME1	VARCHAR2(30)	Resource name for the lock
RESOURCE_NAME2	VARCHAR2(30)	Resource name for the lock
PID	NUMBER	Process identifier which holds the lock
TRANSACTION_ID0	NUMBER	Lower 4 bytes of the transaction identifier where the lock belongs to
TRANSACTION_ID1	NUMBER	Upper 4 bytes of the transaction identifier where the lock belongs to
GROUP_ID	NUMBER	Group identifier for the lock
OPEN_OPT_DEADLOCK	NUMBER	1 if DEADLOCK open option is set, otherwise 0
OPEN_OPT_PERSISTENT	NUMBER	1 if PERSISTENT open option is set, otherwise 0
OPEN_OPT_PROCESS_OWNED	NUMBER	1 if PROCESS_OWNED open option is set, otherwise 0
OPEN_OPT_NO_XID	NUMBER	1 if NO_XID open option is set, otherwise 0
CONVERT_OPT_GETVALUE	NUMBER	1 if GETVALUE convert option is set, otherwise 0
CONVERT_OPT_PUTVALUE	NUMBER	1 if PUTVALUE convert option is set, otherwise 0
CONVERT_OPT_NOVALUE	NUMBER	1 if NOVALUE convert option is set, otherwise 0

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CONVERT_OPT_DUBVALUE	NUMBER	1 if DUBVALUE convert option is set, otherwise 0
CONVERT_OPT_NOQUEUE	NUMBER	1 if NOQUEUE convert option is set, otherwise 0
CONVERT_OPT_EXPRESS	NUMBER	1 if EXPRESS convert option is set, otherwise 0
CONVERT_OPT_NODEADLOCKWAIT	NUMBER	1 if NODEADLOCKWAIT convert option is set, otherwise 0
CONVERT_OPT_NODEADLOCKBLOCK	NUMBER	1 if NODEADLOCKBLOCK convert option is set, otherwise 0
WHICH_QUEUE	NUMBER	Which queue the lock is currently located. 0 for NULL queue; 1 for GRANTED queue; 2 for CONVERT queue.
STATE	VARCHAR2(64)	State of lock as owner sees it
AST_EVENT0	NUMBER	Last AST event
OWNER_NODE	NUMBER	Node identifier
BLOCKED	NUMBER	1 if this lock request is blocked by others, otherwise 0
BLOCKER	NUMBER	1 if this lock is blocking others, otherwise 0
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.245 V$GES_CONVERT_LOCAL

V$GES_CONVERT_LOCAL displays statistics for local GES enqueue operations. This view records average convert times, count information, and timed statistics for global enqueue requests.

Column	Datatype	Description
INST_ID	NUMBER	ID of the instance
CONVERT_TYPE	VARCHAR2(16)	Conversion type: • NULL -> SS - NULL mode to subshared mode • NULL -> SX - NULL mode to shared exclusive mode • NULL -> S - NULL mode to shared mode • NULL -> SSX - NULL mode to subshared exclusive mode • NULL -> X - NULL mode to exclusive mode • SS -> SX - Subshared mode to shared exclusive mode • SS -> S - Subshared mode to shared mode • SS -> SSX - Subshared mode to subshared exclusive mode • SS -> X - Subshared mode to exclusive mode • SX -> S - Shared exclusive mode to shared mode • SX -> SSX - Shared exclusive mode to subshared exclusive mode • SX -> X - Shared exclusive mode to exclusive mode • S -> SX - Shared mode to shared exclusive mode • S -> SSX - Shared mode to subshared exclusive mode • S -> X - Shared mode to exclusive mode • SSX -> X - Sub-shared exclusive mode to exclusive mode

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
AVERAGE_CONVERT_TIME	NUMBER	Average conversion time for each type of lock operation (in hundredths of a second)
CONVERT_COUNT	NUMBER	Number of operations
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.246 V$GES_CONVERT_REMOTE

V$GES_CONVERT_REMOTE displays values for remote GES enqueue conversions. This view records average convert times, count information, and timed statistics for global enqueue requests.

Column	Datatype	Description
INST_ID	NUMBER	ID of the instance
CONVERT_TYPE	VARCHAR2(16)	Conversion type: • NULL -> SS - NULL mode to subshared mode • NULL -> SX - NULL mode to shared exclusive mode • NULL -> S - NULL mode to shared mode • NULL -> SSX - NULL mode to subshared exclusive mode • NULL -> X - NULL mode to exclusive mode • SS -> SX - Subshared mode to shared exclusive mode • SS -> S - Subshared mode to shared mode • SS -> SSX - Subshared mode to subshared exclusive mode • SS -> X - Subshared mode to exclusive mode • SX -> S - Shared exclusive mode to shared mode • SX -> SSX - Shared exclusive mode to subshared exclusive mode • SX -> X - Shared exclusive mode to exclusive mode • S -> SX - Shared mode to shared exclusive mode • S -> SSX - Shared mode to subshared exclusive mode • S -> X - Shared mode to exclusive mode • SSX -> X - Sub-shared exclusive mode to exclusive mode
AVERAGE_CONVERT_TIME	NUMBER	Average conversion time for each type of lock operation (in hundredths of a second)
CONVERT_COUNT	NUMBER	Number of operations
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.247 V$GES_DEADLOCK_SESSIONS

V$GES_DEADLOCK_SESSIONS displays information about sessions waiting for global enqueue service deadlocks detected by the distributed lock manager. This is an Oracle Real Application Clusters view.

This view displays a maximum of 20 sessions per deadlock.

Column	Datatype	Description
DID	VARCHAR2(30)	Deadlock ID
SID	NUMBER	Waiting session ID
SERIAL#	NUMBER	Waiting session serial number
INSTANCE	NUMBER	Waiting session instance
BLOCKER_SID	NUMBER	Blocker session ID
BLOCKER_SERIAL#	NUMBER	Blocker session serial number
BLOCKER_INSTANCE	NUMBER	Blocker session instance
OSUSER	VARCHAR2(128)	Waiting OS user
APP_NAME	VARCHAR2(64)	Waiting application name
BLOCKED_RES_NAME	VARCHAR2(64)	Resource on which the waiting session is blocked
REQUEST_LEVEL	VARCHAR2(16)	Waiting session requested level
BLOCKER_RES_NAME	VARCHAR2(64)	Resource on which the blocker is blocked
GRANT_LEVEL	VARCHAR2(16)	Granted level of the blocker
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.248 V$GES_DEADLOCKS

V$GES_DEADLOCKS displays information about global enqueue service deadlocks detected by the distributed lock manager. This is an Oracle Real Application Clusters view.

This view displays the 20 most recent deadlocks.

Column	Datatype	Description
DID	VARCHAR2(30)	Deadlock ID
DD_TIME	DATE	Deadlock timestamp
NUMBER_OF_SESSIONS	NUMBER	Number of sessions involved in the deadlock
MASTER_INSTANCE	NUMBER	Master instance of the resource
VICTIM_INSTANCE	NUMBER	Victim instance
VICTIM_PID	VARCHAR2(24)	OS process ID of the victim
VICTIM_RESNM	VARCHAR2(64)	Resource involved in the deadlock

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
VICTIM_SID	NUMBER	Session ID of the victim
VICTIM_SERIAL#	NUMBER	Session serial number of the victim
VICTIM_REQ_LEVEL	VARCHAR2(16)	Level of the lock being requested by the victim
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.249 V$GES_ENQUEUE

V$GES_ENQUEUE describes all locks currently known to lock manager. This is an Oracle Real Application Clusters view.

See Also

"V$GES_BLOCKING_ENQUEUE" for a description of all such locks that are currently blocking or being blocked

Column	Datatype	Description
HANDLE	RAW(4 | 8)	Lock pointer
GRANT_LEVEL	VARCHAR2(9)	Granted level of the lock
REQUEST_LEVEL	VARCHAR2(9)	Requested level of the lock
RESOURCE_NAME1	VARCHAR2(30)	Resource name for the lock
RESOURCE_NAME2	VARCHAR2(30)	Resource name for the lock
PID	NUMBER	Process identifier which holds the lock
TRANSACTION_ID0	NUMBER	Lower 4 bytes of the transaction identifier to which the lock belongs
TRANSACTION_ID1	NUMBER	Upper 4 bytes of the transaction identifier to which the lock belongs
GROUP_ID	NUMBER	Group identifier for the lock
OPEN_OPT_DEADLOCK	NUMBER	1 if DEADLOCK open option is set, otherwise 0
OPEN_OPT_PERSISTENT	NUMBER	1 if PERSISTENT open option is set, otherwise 0
OPEN_OPT_PROCESS_OWNED	NUMBER	1 if PROCESS_OWNED open option is set, otherwise 0
OPEN_OPT_NO_XID	NUMBER	1 if NO_XID open option is set, otherwise 0
CONVERT_OPT_GETVALUE	NUMBER	1 if GETVALUE convert option is set, otherwise 0
CONVERT_OPT_PUTVALUE	NUMBER	1 if PUTVALUE convert option is set, otherwise 0
CONVERT_OPT_NOVALUE	NUMBER	1 if NOVALUE convert option is set, otherwise 0
CONVERT_OPT_DUBVALUE	NUMBER	1 if DUBVALUE convert option is set, otherwise 0
CONVERT_OPT_NOQUEUE	NUMBER	1 if NOQUEUE convert option is set, otherwise 0
CONVERT_OPT_EXPRESS	NUMBER	1 if EXPRESS convert option is set, otherwise 0

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CONVERT_OPT_NODEADLOCKWAIT	NUMBER	1 if NODEADLOCKWAIT convert option is set, otherwise 0
CONVERT_OPT_NODEADLOCKBLOCK	NUMBER	1 if NODEADLOCKBLOCK convert option is set, otherwise 0
WHICH_QUEUE	NUMBER	Which queue the lock is currently located. 0 for NULL queue; 1 for GRANTED queue; 2 for CONVERT queue.
STATE	VARCHAR2(64)	State of the lock as the owner sees it
AST_EVENT0	NUMBER	Last AST event
OWNER_NODE	NUMBER	Node identifier
BLOCKED	NUMBER	1 if this lock request is blocked by others, otherwise 0
BLOCKER	NUMBER	1 if this lock is blocking others, otherwise 0
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.250 V$GES_LATCH

V$GES_LATCH is deprecated.

See Also

"V$LATCH" for statistics about GES latch performance

7.251 V$GES_RESOURCE

V$GES_RESOURCE is an Oracle Real Application Clusters view. It displays information of all resources currently known to the lock manager.

Column	Datatype	Description
RESP	RAW(4 | 8)	Resource pointer
RESOURCE_NAME	VARCHAR2(30)	Resource name in hexadecimal for the lock
ON_CONVERT_Q	NUMBER	1 if on convert queue, 0 otherwise
ON_GRANT_Q	NUMBER	1 if on granted queue, 0 otherwise
PERSISTENT_RES	NUMBER	1 if it is a persistent resource, 0 otherwise
MASTER_NODE	NUMBER	Primary node ID
NEXT_CVT_LEVEL	VARCHAR2(9)	Next lock level to convert on global convert queue
VALUE_BLK_STATE	VARCHAR2(32)	State of the value block
VALUE_BLK	VARCHAR2(64)	First 64 bytes of the value block

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.252 V$GES_STATISTICS

V$GES_STATISTICS displays miscellaneous GES statistics.

Column	Datatype	Description
STATISTIC#	NUMBER	Statistic number
NAME	VARCHAR2(38)	Name of the statistic
VALUE	NUMBER	Value associated with the statistic
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.253 V$GG_APPLY_COORDINATOR

V$GG_APPLY_COORDINATOR displays information about each GoldenGate apply process coordinator.

The coordinator for an apply process gets transactions from the apply process reader and passes them to apply servers. An apply process coordinator is a subcomponent of an apply process used by Oracle GoldenGate Integrated Replicat.

Column	Datatype	Description
SID	NUMBER	Session ID of the coordinator's session
SERIAL#	NUMBER	Serial number of the coordinator's session
STATE	VARCHAR2(21)	State of the coordinator: • INITIALIZING - Starting up • IDLE - Performing no work • APPLYING - Passing transactions to apply servers • SHUTTING DOWN CLEANLY - Stopping without an error • ABORTING - Stopping because of an apply error
APPLY#	NUMBER	Apply process number. An apply process coordinator is an Oracle background process, prefixed by ap.
APPLY_NAME	VARCHAR2(128)	Name of the apply process

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOTAL_APPLIED	NUMBER	Total number of transactions applied by the apply process since the apply process was last started
TOTAL_WAIT_DEPS	NUMBER	Number of times since the apply process was last started that an apply server waited to apply a logical change record (LCR) in a transaction until another apply server applied a transaction because of a dependency between the transactions
TOTAL_WAIT_COMMITS	NUMBER	Number of times since the apply process was last started that an apply server waited to commit a transaction until another apply server committed a transaction to serialize commits
TOTAL_ADMIN	NUMBER	Number of administrative jobs issued since the apply process was last started
TOTAL_ASSIGNED	NUMBER	Number of transactions assigned to apply servers since the apply process was last started
TOTAL_RECEIVED	NUMBER	Total number of transactions received by the coordinator process since the apply process was last started
TOTAL_IGNORED	NUMBER	Number of transactions which were received by the coordinator but were ignored because they had been previously applied
TOTAL_ROLLBACKS	NUMBER	Number of transactions which were rolled back due to unexpected contention
TOTAL_ERRORS	NUMBER	Number of transactions applied by the apply process that resulted in an apply error since the apply process was last started
UNASSIGNED_COMPLETE_TXNS	NUMBER	Total number of complete transactions that the coordinator has not assigned to any apply servers
LWM_TIME	DATE	Time when the message with the lowest message number was recorded. The creation time of the message with the lowest message number was also recorded at this time.
LWM_MESSAGE_NUMBER	NUMBER	Number of the message corresponding to the low watermark. That is, messages with a commit message number less than or equal to this message number have definitely been applied, but some messages with a higher commit message number also may have been applied.
LWM_MESSAGE_CREATE_TIME	DATE	For captured messages, creation time at the source database of the message corresponding to the low watermark. For user-enqueued messages, time when the message corresponding to the low watermark was enqueued into the queue at the local database.
HWM_TIME	DATE	Time when the message with the highest message number was recorded. The creation time of the message with the highest message number was also recorded at this time.
HWM_MESSAGE_NUMBER	NUMBER	Number of the message corresponding to the high watermark. That is, no messages with a commit message number greater than this message number have been applied.
HWM_MESSAGE_CREATE_TIME	DATE	For captured messages, creation time at the source database of the message corresponding to the high watermark. For user-enqueued messages, time when the message corresponding to the high watermark was enqueued into the queue at the local database.
STARTUP_TIME	DATE	Time when the apply process was last started
ELAPSED_SCHEDULE_TIME	NUMBER	Time elapsed (in hundredths of a second) scheduling messages since the apply process was last started
ELAPSED_IDLE_TIME	NUMBER	Elapsed idle time
LWM_POSITION	RAW(64)	Position of the low-watermark LCR

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
HWM_POSITION	RAW(64)	Position of the high-watermark LCR
PROCESSED_MESSAGE_NUMBER	NUMBER	Message number currently processed by the apply coordinator
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
ACTIVE_SERVER_COUNT	NUMBER	Active server count

Note

The ELAPSED_SCHEDULE_TIME column is only populated if the TIMED_STATISTICS initialization parameter is set to true, or if the STATISTICS_LEVEL initialization parameter is set to TYPICAL or ALL.

See Also

- "TIMED_STATISTICS"
- "STATISTICS_LEVEL"

7.254 V$GG_APPLY_READER

V$GG_APPLY_READER displays information about each GoldenGate apply reader.

The apply reader is a process which reads (dequeues) messages from the queue, computes message dependencies, and builds transactions. It passes the transactions on to the coordinator in commit order for assignment to the apply servers. An apply reader is a subcomponent of an apply process used by Oracle GoldenGate Integrated Replicat.

Column	Datatype	Description
SID	NUMBER	Session ID of the reader's session
SERIAL#	NUMBER	Serial number of the reader's session
APPLY#	NUMBER	Apply process number. An apply process is an Oracle background process, prefixed by ap.
APPLY_NAME	VARCHAR2(128)	Name of the apply process

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATE	VARCHAR2(36)	Shows the state of the apply reader and the hash server. The possible values include: • INITIALIZING - Starting up • IDLE - Performing no work • DEQUEUE MESSAGES - Dequeuing messages from the queue • SCHEDULE MESSAGES - Computing dependencies between messages and assembling messages into transactions • SPILLING - Spilling unapplied messages from memory to hard disk • PAUSED - WAITING FOR DDL TO COMPLETE - Waiting for a data definition language (DDL) logical change record (LCR) to be applied The state of the apply reader is displayed first, followed by the state of the hash server. A semicolon separates the apply reader state from the hash server state.
TOTAL_MESSAGES_DEQUEUED	NUMBER	Total number of messages dequeued since the apply process was last started
TOTAL_MESSAGES_SPILLED	NUMBER	Number of messages spilled by the reader since the apply process was last started
DEQUEUE_TIME	DATE	Time when the last message was received
DEQUEUED_MESSAGE_CREATE_TIME	DATE	For captured messages, creation time at the source database of the last message received. For user-enqueued messages, time when the message was enqueued into the queue at the local database.
SGA_USED	NUMBER	Amount (in bytes) of SGA memory used by the apply process since it was last started
ELAPSED_DEQUEUE_TIME	NUMBER	Time elapsed (in hundredths of a second) dequeuing messages since the apply process was last started
ELAPSED_SCHEDULE_TIME	NUMBER	Time elapsed (in hundredths of a second) scheduling messages since the apply process was last started. Scheduling includes computing dependencies between messages and assembling messages into transactions.
ELAPSED_SPILL_TIME	NUMBER	Elapsed time (in hundredths of a second) spent spilling messages since the apply process was last started
SPILL_LWM_SCN	NUMBER	Spill low-watermark SCN
PROXY_SID	NUMBER	When the apply process uses combined capture and apply, the session ID of the propagation receiver that is responsible for direct communication between capture and apply. If the apply process does not use combined capture and apply, then this column is 0.
PROXY_SERIAL	NUMBER	When the apply process uses combined capture and apply, the serial number of the propagation receiver that is responsible for direct communication between capture and apply. If the apply process does not use combined capture and apply, then this column is 0.
PROXY_SPID	VARCHAR2(12)	When the apply process uses combined capture and apply, the process identification number of the propagation receiver that is responsible for direct communication between capture and apply. If the apply process does not use combined capture and apply, then this column is 0.
BYTES_RECEIVED	NUMBER	When the apply process uses combined capture and apply, the number of bytes received by the apply process from the capture process since the apply process last started. If the apply process does not use combined capture and apply, then this column is not populated.
DEQUEUED_POSITION	RAW(64)	Dequeued position. This column is populated only for an apply process that is functioning as a GoldenGate inbound server.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SPILL_LWM_POSITION	RAW(64)	Spill low-watermark position. This column is populated only for an apply process that is functioning as a GoldenGate inbound server.
OLDEST_TRANSACTION_ID	VARCHAR2(128)	Oldest transaction ID
TOTAL_LCRS_WITH_DEP	NUMBER	Total number of LCRs with row-level dependencies since the apply process last started
TOTAL_LCRS_WITH_WMDEP	NUMBER	Total number of LCRs with watermark dependencies since the apply process last started. A watermark dependency occurs when an apply process must wait until the apply process's low watermark reaches a particular threshold.
TOTAL_IN_MEMORY_LCRS	NUMBER	Total number of LCRs currently in memory
SGA_ALLOCATED	NUMBER	The total amount of shared memory (in bytes) allocated from the GoldenGate pool for the apply process since the apply process last started
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Note

The ELAPSED_SCHEDULE_TIME column is only populated if the TIMED_STATISTICS initialization parameter is set to true, or if the STATISTICS_LEVEL initialization parameter is set to TYPICAL or ALL.

See Also

- "TIMED_STATISTICS"
- "STATISTICS_LEVEL"

7.255 V$GG_APPLY_RECEIVER

V$GG_APPLY_RECEIVER displays information about the message receiver of the Replicat process.

The values are reset to zero when the database (or instance in an Oracle Real Application Clusters (Oracle RAC) environment) restarts and when the Replicat process is stopped.

Column	Datatype	Description
SID	NUMBER	Session ID of the apply receiver
SERIAL#	NUMBER	Serial number of the apply receiver
APPLY_NAME	VARCHAR2(128)	Name of the apply process

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STARTUP_TIME	DATE	Startup time of the apply process
SOURCE_DATABASE_NAME	VARCHAR2(128)	Name of the source database
ACKNOWLEDGEMENT	NUMBER	acknowledgment of the messages received by the receiver
LAST_RECEIVED_MSG	NUMBER	Last received message
TOTAL_MESSAGES_RECEIVED	NUMBER	Total number of messages received
TOTAL_AVAILABLE_MESSAGES	NUMBER	Number of available messages
STATE	VARCHAR2(46)	State of the apply receiver: • Initializing • Sending unapplied txns • Waiting for message from client • Receiving LCRs • Evaluating rules • Enqueueing LCRS • Waiting for memory • Waiting for apply to read • Waiting for message from Replicat • Waiting for Replicat flush request to complete • Waiting for Replicat commit to complete
LAST_RECEIVED_MSG_POSITION	VARCHAR2(64)	Last received message position
ACKNOWLEDGEMENT_POSITION	VARCHAR2(64)	acknowledgment position of the messages received by the receiver. Corresponds to ACKNOWLEDGEMENT, except the value is in position rather than SCN.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
OS_PROCESS_ID[1]	VARCHAR2(12)	The OS process ID of the apply receiver process
CURRENT_POSITION[1]	VARCHAR2(81)	The trail position of the current record processed by the apply receiver
TOTAL_TRANSACTIONS[1]	NUMBER	The total number of transactions processed by the apply receiver
TOTAL_COMMITS[1]	NUMBER	The total number of commits executed by the apply receiver
TOTAL_ERRORS[1]	NUMBER	The total number of errors encountered by the apply receiver

1 This column is available starting with Oracle Database 19c.

7.256 V$GG_APPLY_SERVER

V$GG_APPLY_SERVER displays information about each GoldenGate apply server and its activities.

An apply server receives messages from the apply coordinator for an apply process. For each message received, an apply server either applies the message or sends the message to the appropriate apply handler. An apply server is a subcomponent of an apply process used by Oracle GoldenGate Integrated Replicat.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SID	NUMBER	Session ID of the apply server's session
SERIAL#	NUMBER	Serial number of the apply server's session
APPLY#	NUMBER	Apply process number. An apply process is an Oracle background process, prefixed by ap.
APPLY_NAME	VARCHAR2(128)	Name of the apply process
SERVER_ID	NUMBER	Parallel execution server number of the apply server
STATE	VARCHAR2(20)	State of the apply server: • INITIALIZING - Starting up • IDLE - Performing no work • RECORD LOW-WATERMARK - Performing an administrative job that maintains information about the apply progress, which is used in the ALL_APPLY_PROGRESS and DBA_APPLY_PROGRESS data dictionary views • ADD PARTITION - Performing an administrative job that adds a partition that is used for recording information about in-progress transactions • DROP PARTITION - Performing an administrative job that purges rows that were used to record information about in-progress transactions • EXECUTE TRANSACTION - Applying a transaction • WAIT COMMIT - Waiting to commit a transaction until all other transactions with a lower commit SCN are applied. This state is possible only if the COMMIT_SERIALIZATION apply process parameter is set to a value other than DEPENDENT_TRANSACTIONS and the PARALLELISM apply process parameter is set to a value greater than 1. • WAIT DEPENDENCY - Waiting to apply a logical change record (LCR) in a transaction until another transaction, on which it has a dependency, is applied. This state is possible only if the PARALLELISM apply process parameter is set to a value greater than 1. • ROLLBACK TRANSACTION - Rolling back a transaction • TRANSACTION CLEANUP - Cleaning up an applied transaction, which includes removing LCRs from the apply process's queue • WAIT FOR CLIENT - Waiting for an XStream client application to request more LCRs • WAIT FOR NEXT CHUNK - Waiting for the next set of LCRs for a large transaction
XIDUSN	NUMBER	Transaction ID undo segment number of the transaction currently being applied
XIDSLT	NUMBER	Transaction ID slot number of the transaction currently being applied
XIDSQN	NUMBER	Transaction ID sequence number of the transaction currently being applied
COMMITSCN	NUMBER	Commit system change number (SCN) of the transaction currently being applied
DEP_XIDUSN	NUMBER	Transaction ID undo segment number of a transaction on which the transaction being applied by this apply server depends
DEP_XIDSLT	NUMBER	Transaction ID slot number of a transaction on which the transaction being applied by this apply server depends
DEP_XIDSQN	NUMBER	Transaction ID sequence number of a transaction on which the transaction being applied by this apply server depends

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DEP_COMMITSCN	NUMBER	Commit system change number (SCN) of the transaction on which this apply server depends
MESSAGE_SEQUENCE	NUMBER	Number of the current message being applied by the apply server. This value is reset to 1 at the beginning of each transaction.
TOTAL_ASSIGNED	NUMBER	Total number of transactions assigned to the apply server since the apply process was last started
TOTAL_ADMIN	NUMBER	Total number of administrative jobs done by the apply server since the apply process was last started. See the STATE information in this view for the types of administrative jobs.
TOTAL_ROLLBACKS	NUMBER	Number of transactions assigned to this server which were rolled back
TOTAL_MESSAGES_APPLIED	NUMBER	Total number of messages applied by this apply server since the apply process was last started
APPLY_TIME	DATE	Time the last message was applied
ELAPSED_APPLY_TIME	NUMBER	Time elapsed (in hundredths of a second) applying messages since the apply process was last started
COMMIT_POSITION	RAW(64)	Commit position of the transaction. This column is populated only for an apply process that is functioning as a GoldenGate Integrated Replicat.
DEP_COMMIT_POSITION	RAW(64)	Commit position of the transaction the worker depends on. This column is populated only for an apply process that is functioning as a GoldenGate inbound server.
LAST_APPLY_POSITION	RAW(64)	For inbound servers, the position of the last message applied; for outbound servers, the position of the last message sent to the XStream client application. This column is populated only for an apply process that is functioning as a GoldenGate outbound server or inbound server.
TRANSACTION_ID	VARCHAR2(128)	Transaction ID that the worker is applying. This column is populated only for an apply process that is functioning as a GoldenGate inbound server.
DEP_TRANSACTION_ID	VARCHAR2(128)	Transaction ID of the transaction the worker depends on. This column is populated only for an apply process that is functioning as a GoldenGate inbound server.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
TOTAL_LCRS_RETRIED	NUMBER	Total number of LCRs retried by this server
LCR_RETRY_ITERATION	NUMBER	Retry iteration for this transaction by this server
TOTAL_TXNS_RETRIED	NUMBER	Total transactions retried by this server
TXN_RETRY_ITERATION	NUMBER	Retry iteration for this transaction by this server
TOTAL_TXNS_RECORDED	NUMBER	Total transactions recorded in error queue by this server

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The ELAPSED_SCHEDULE_TIME column is only populated if the TIMED_STATISTICS initialization parameter is set to true, or if the STATISTICS_LEVEL initialization parameter is set to TYPICAL or ALL.

> **See Also**
>
> - "TIMED_STATISTICS"
> - "STATISTICS_LEVEL"

7.257 V$GLOBAL_BLOCKED_LOCKS

V$GLOBAL_BLOCKED_LOCKS displays global blocked locks.

Column	Datatype	Description
ADDR	RAW(4 | 8)	Address of lock state object
KADDR	RAW(4 | 8)	Address of lock
SID	NUMBER	Identifier of session holding the lock (number)
TYPE	VARCHAR2(2)	Resource type (char)
ID1	NUMBER	Resource identifier #1 (number)
ID2	NUMBER	Resource identifier #2 (number)
LMODE	NUMBER	Lock mode held (number)
REQUEST	NUMBER	Lock mode requested (number)
CTIME	NUMBER	Time since current mode was granted
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.258 V$GLOBAL_TRANSACTION

V$GLOBAL_TRANSACTION displays information on the currently active global transactions.

Column	Datatype	Description
FORMATID	NUMBER	Format identifier of the global transaction
GLOBALID	RAW(64)	Global transaction identifier of the global transaction
BRANCHID	RAW(64)	Branch qualifier of the global transaction
BRANCHES	NUMBER	Total number of branches in the global transaction

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
REFCOUNT	NUMBER	Number of siblings for the global transaction (must be the same as branches)
PREPARECOUNT	NUMBER	Number of branches of the global transaction that have prepared
STATE	VARCHAR2(38)	State of the branch of the global transaction: • ACTIVE • COLLECTING • FINALIZED • FAILED • RECOVERING • UNASSOCIATED • FORGOTTEN • READY FOR RECOVERY • NO-READONLY FAILED • SIBLING INFO WRITTEN • [ORACLE COORDINATED]ACTIVE • [ORACLE COORDINATED]COLLECTING • [ORACLE COORDINATED]FINALIZED • [ORACLE COORDINATED]FAILED • [ORACLE COORDINATED]RECOVERING • [ORACLE COORDINATED]UNASSOCIATED • [ORACLE COORDINATED]FORGOTTEN • [ORACLE COORDINATED]READY FOR RECOVERY • [ORACLE COORDINATED]NO-READONLY FAILED • [MULTINODE]ACTIVE • [MULTINODE]COLLECTING • [MULTINODE]FINALIZED • [MULTINODE]FAILED • [MULTINODE]RECOVERING • [MULTINODE]UNASSOCIATED • [MULTINODE]FORGOTTEN • [MULTINODE]READY FOR RECOVERY • [MULTINODE]NO-READONLY FAILED • [MULTINODE]SIBLING INFO WRITTEN • COMBINATION
FLAGS	NUMBER	The numeric representation of the state
COUPLING	VARCHAR2(15)	Indicates whether the branches are free (FREE), loosely coupled (LOOSELY COUPLED), or tightly coupled (TIGHTLY COUPLED)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.259 V$GOLDENGATE_CAPABILITIES

V$GOLDENGATE_CAPABILITIES displays information about the usage of various Oracle GoldenGate features, as well as database features used by Oracle GoldenGate.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NAME	VARCHAR2(22)	Feature name. Possible values: • AUTOCDR • DBENCRYPTION • DBLOGREADER • DDLTRIGGEROPTIMIZATION • DELETECASCADEHINT • GGSESSION • PROCREPLICATION • SUPPLEMENTALLOG • TRANSIENTDUPLICATE • TRIGGERSUPPRESSION
COUNT	NUMBER	Number of times the feature has been used since instance startup
LAST_USED	DATE	Date on which the feature was last used
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.260 V$GOLDENGATE_CAPTURE

V$GOLDENGATE_CAPTURE displays information about each capture process that sends LCRs to an Oracle GoldenGate outbound server.

Column	Datatype	Description
SID	NUMBER	Session identifier of the capture process
SERIAL#	NUMBER	Session serial number of the capture process session
CAPTURE#	NUMBER	Capture process number. A capture process is an Oracle background process prefixed by cp
CAPTURE_NAME	VARCHAR2(128)	Name of the capture process
LOGMINER_ID	NUMBER	Session ID of the Oracle LogMiner session associated with the capture process
STARTUP_TIME	DATE	Time when the capture process was last started

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATE	VARCHAR2(589)	State of the capture process and state of the network. The two states are separated by a semicolon. The possible values are: • ABORTING - Terminating. • CAPTURING CHANGES - Scanning the redo log for changes that satisfy the capture process rule sets. • CREATING LCR - Converting a change into an LCR. • DICTIONARY INITIALIZATION - Processing a dictionary build. • ENQUEUING MESSAGE - Enqueuing an LCR that satisfies the capture process rule sets into the capture process queue. • EVALUATING RULE - Evaluating a change against a capture process rule set. • INITIALIZING - Starting up. • PAUSED FOR FLOW CONTROL - Unable to enqueue LCRs either because of low memory or because propagations and outbound servers are consuming messages slower than the capture process is creating them. This state indicates flow control that is used to reduce spilling of captured LCRs when propagation or apply has fallen behind or is unavailable. • SHUTTING DOWN - Stopping. • SUSPENDED FOR AUTO SPLIT/MERGE - Waiting for a merge operation to complete. • WAITING FOR BUFFERED QUEUE TO SHRINK - Waiting for the buffered queue to change to a smaller size. The buffered queue shrinks when there is a memory limitation or when an administrator reduces its size. • WAITING FOR CLIENT REQUESTS - Waiting for a response from the extract process. • WAITING FOR DICTIONARY REDO - Waiting for redo log files containing the dictionary build related to the first SCN to be added to the capture process session. A capture process cannot begin to scan the redo log files until all of the log files containing the dictionary build have been added. • WAITING FOR INACTIVE DEQUEUERS - Waiting for the capture process's queue subscribers to start. The capture process stops enqueuing LCRs if there are no active subscribers to the queue. • WAITING FOR *n* SUBSCRIBER(S) INITIALIZING - Waiting for outbound servers that receive LCRs from the capture process to start, where *n* is the number of apply processes. • WAITING FOR REDO - Waiting for new redo log files to be added to the capture process session. The capture process has finished processing all of the redo log files added to its session. This state is possible if there is no activity at a source database. For a downstream capture process, this state is possible if the capture process is waiting for new log files to be added to its session. • WAITING FOR TRANSACTION - Waiting for LogMiner to provide more transactions.
TOTAL_PREFILTER_DISCARDED	NUMBER	Total number of prefiltered messages discarded
TOTAL_PREFILTER_KEPT	NUMBER	Total number of prefiltered messages kept
TOTAL_PREFILTER_EVALUATIONS	NUMBER	Total number of prefilter evaluations

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOTAL_MESSAGES_CAPTURED	NUMBER	Total number of redo entries passed by LogMiner to the capture process for detailed rule evaluation since the capture process last started. A capture process converts a redo entry into a message and performs detailed rule evaluation on the message when capture process prefiltering cannot discard the change.
CAPTURE_TIME	DATE	Time when the most recent message was captured
CAPTURE_MESSAGE_NUMBER	NUMBER	Number of the most recently captured message
CAPTURE_MESSAGE_CREATE_TIME	DATE	Creation time of the most recently captured message
TOTAL_MESSAGES_CREATED	NUMBER	Count associated with ELAPSED_LCR_TIME to calculate rate
TOTAL_FULL_EVALUATIONS	NUMBER	Count associated with ELAPSED_RULE_TIME to calculate rate
TOTAL_MESSAGES_ENQUEUED	NUMBER	Total number of messages enqueued since the capture process was last started
ENQUEUE_TIME	DATE	Time when the last message was enqueued
ENQUEUE_MESSAGE_NUMBER	NUMBER	Number of the last enqueued message
ENQUEUE_MESSAGE_CREATE_TIME	DATE	Creation time of the last enqueued message
AVAILABLE_MESSAGE_NUMBER	NUMBER	For local capture, the last redo SCN flushed to the log files. For downstream capture, the last SCN added to LogMiner through the archived redo log files.
AVAILABLE_MESSAGE_CREATE_TIME	DATE	For local capture, the time the SCN was written to the log file. For downstream capture, the time the most recent archived redo log file (containing the most recent SCN) was added to LogMiner.
ELAPSED_CAPTURE_TIME	NUMBER	Elapsed time (in hundredths of a second) scanning for changes in the redo log since the capture process was last started
ELAPSED_RULE_TIME	NUMBER	Elapsed time (in hundredths of a second) evaluating rules since the capture process was last started
ELAPSED_ENQUEUE_TIME	NUMBER	Elapsed time (in hundredths of a second) enqueuing messages since the capture process was last started
ELAPSED_LCR_TIME	NUMBER	Elapsed time (in hundredths of a second) creating LCRs since the capture process was last started
ELAPSED_REDO_WAIT_TIME	NUMBER	Elapsed time (in hundredths of a second) spent by the capture process in the WAITING FOR REDO state
ELAPSED_PAUSE_TIME	NUMBER	Elapsed flow control pause time (in hundredths of a second)
STATE_CHANGED_TIME	DATE	Time at which the state of the capture process changed
SGA_USED	NUMBER	The total amount of shared memory used (in bytes) by the capture process
SGA_ALLOCATED	NUMBER	The total amount of shared memory (in bytes) allocated from the Streams pool for the capture process
BYTES_OF_REDO_MINED	VARCHAR2(64)	The total amount of redo data mined (in bytes) since the capture process last started
SESSION_RESTART_SCN	VARCHAR2(64)	The SCN from which the capture process started mining redo data when it was last started

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
SPID	VARCHAR2(12)	Operating system process identifier of the capture process
EXTRACT_NAME	VARCHAR2(128)	Name of the extract process
SERVER_SID	NUMBER	Session ID of the capture server
SERVER_SERIAL#	NUMBER	Session serial number of the capture server
SERVER_SPID	VARCHAR2(12)	Operating system process identifier of the capture server
TOTAL_MESSAGES_SENT	NUMBER	Total number of LCRs sent by the capture process to the GoldenGate extract process since the last time the extract attached to the capture
SEND_TIME	DATE	Time the last LCR was sent by the capture process to the extract process
LAST_SENT_MESSAGE_NUMBER	NUMBER	Message number of the last LCR sent by the capture process to the extract process
LAST_SENT_MESSAGE_CREATE_TIME	DATE	Creation time at the source database of the last LCR sent by the capture process to the extract process
ELAPSED_SEND_TIME	NUMBER	Time elapsed (in hundredths of a second) sending LCRs to the extract process since the last time the extract process attached to the capture process
BYTES_SENT	NUMBER	Total number of bytes sent by the capture process to the extract process since the last time the extract process attached to the capture process

7.261 V$GOLDENGATE_MESSAGE_TRACKING

V$GOLDENGATE_MESSAGE_TRACKING displays information about LCRs tracked through the stream that are processed by Oracle GoldenGate components.

You can track an LCR through a stream using one of the following methods:

- Set the message_tracking_frequency apply process parameter to 1 or another relatively low value via the Oracle GoldenGate DBOPTIONS INTEGRATEDPARAMS parameter.
- Use the DBMS_XSTREAM_ADM.SET_MESSAGE_TRACKING procedure to specify a tracking label that becomes part of each LCR generated by the current session.

  When the actions parameter in the DBMS_XSTREAM_ADM.SET_MESSAGE_TRACKING procedure is set to DBMS_XSTREAM_ADM.ACTION_MEMORY, information about the LCRs is tracked in memory, and this view is populated with information about the LCRs. Currently, DBMS_XSTREAM_ADM.ACTION_MEMORY is the only valid setting for the actions parameter in the procedure.

Column	Datatype	Description
TRACKING_LABEL	VARCHAR2(128)	User-specified tracking label

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TAG	RAW(128)	First 128 bytes of the tag of the LCR
COMPONENT_NAME	VARCHAR2(128)	Name of the component that processed the LCR
COMPONENT_TYPE	VARCHAR2(128)	Type of the component that processed the LCR
ACTION	VARCHAR2(50)	Action performed on the LCR
ACTION_DETAILS	VARCHAR2(312)	Details of the action
TIMESTAMP	TIMESTAMP(9) WITH TIME ZONE	Time when the action was performed
MESSAGE_CREATION_TIME	DATE	Time when the message was created
MESSAGE_NUMBER	NUMBER	SCN of the message
TRACKING_ID	RAW(16)	Globally unique OID of the LCR
SOURCE_DATABASE_NAME	VARCHAR2(128)	Name of the source database
OBJECT_OWNER	VARCHAR2(128)	Owner of the object
OBJECT_NAME	VARCHAR2(128)	Name of the object
XID	VARCHAR2(128)	Transaction ID
COMMAND_TYPE	VARCHAR2(128)	Command type of the LCR
MESSAGE_POSITION	RAW(64)	Position of the message
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database PL/SQL Packages and Types Reference for more information about the DBMS_XSTREAM_ADM.SET_MESSAGE_TRACKING procedure

7.262 V$GOLDENGATE_PROCEDURE_STATS

V$GOLDENGATE_PROCEDURE_STATS displays procedural replication statistics processed by each Oracle GoldenGate apply server.

Column	Datatype	Description
APPLY_NAME	VARCHAR2(128)	Name of the apply process
SERVER_ID	NUMBER	Parallel apply server worker ID
PROCEDURE_OWNER	VARCHAR2(128)	Owner of the replicated procedure
PACKAGE_NAME	VARCHAR2(128)	Replicated procedure package
PROCEDURE_NAME	VARCHAR2(128)	Replicated procedure name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LAST_UPDATE	DATE	Time of the last update of the statistics
TOTAL_EXECUTIONS	NUMBER	Number of executions of the procedure by this apply server
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.263 V$GOLDENGATE_TABLE_STATS

V$GOLDENGATE_TABLE_STATS displays table statistics for all the tables used by each Oracle GoldenGate apply server.

Column	Datatype	Description
APPLY_NAME	VARCHAR2(128)	Name of the apply process
SERVER_ID	NUMBER	Parallel apply server worker ID
SOURCE_TABLE_OWNER	VARCHAR2(128)	Source owner of the captured or replicated table
SOURCE_TABLE_NAME	VARCHAR2(128)	Source name of the captured or replicated table
DESTINATION_TABLE_OWNER	VARCHAR2(128)	Target owner of the captured or replicated table
DESTINATION_TABLE_NAME	VARCHAR2(30)	Target name of the captured or replicated table
LAST_UPDATE	DATE	Time of the last update of the statistics
TOTAL_INSERTS	NUMBER	Number of insert operations on this table processed by this apply server
TOTAL_UPDATES	NUMBER	Number of update operations on this table processed by this apply server
TOTAL_DELETES	NUMBER	Number of delete operations on this table processed by this apply server
INSERT_COLLISIONS	NUMBER	Number of insert collisions on this table encountered by this apply server
UPDATE_COLLISIONS	NUMBER	Number of update collisions on this table encountered by this apply server
DELETE_COLLISIONS	NUMBER	Number of delete collisions on this table encountered by this apply server
REPERROR_RECORDS	NUMBER	Number of change records that resulted in an error that were recorded on this table by this apply server
REPERROR_IGNORES	NUMBER	Number of change records that resulted in an error that were ignored on this table by this apply server
WAIT_DEPENDENCIES	NUMBER	Number of waits for this table due to dependency

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
CDR_INSERT_ROW_EXISTS	NUMBER	Number of conflicts where an insert gets an ORA-00001 error
CDR_UPDATE_ROW_EXISTS	NUMBER	Number of conflicts where an update gets an ORA-26786 error
CDR_UPDATE_ROW_MISSING	NUMBER	Number of conflicts where an update gets an ORA-26787 error
CDR_DELETE_ROW_EXISTS	NUMBER	Number of conflicts where a delete gets an ORA-26786 error
CDR_DELETE_ROW_MISSING	NUMBER	Number of conflicts where a delete gets an ORA-26787 error
CDR_SUCCESSFUL_RESOLUTIONS	NUMBER	Number of successfully resolved conflicts
CDR_FAILED_RESOLUTIONS	NUMBER	Number of conflicts that could not be resolved due to an error during resolution
LOB_OPERATIONS	NUMBER	The number of LOB updates (LOB writes, LOB trims, and LOB erases) applied by the inbound server.

7.264 V$GOLDENGATE_TRANSACTION

V$GOLDENGATE_TRANSACTION displays information about transactions that are being processed by Oracle GoldenGate capture processes, outbound servers, and inbound servers.

This view can identify long running transactions and display how many LCRs are being processed in each transaction. This view only contains information about captured LCRs. It does not contain information about user-enqueued LCRs or user messages.

This view only shows information about LCRs that are being processed because they satisfied the rule sets for the component at the time of the query. For capture processes, this view only shows information about changes in transactions that the capture process has converted into LCRs. It does not show information about all the active transactions present in the redo log.

For outbound servers, this view only shows information about LCRs that the outbound server has dequeued. It does not show information about LCRs in the outbound server's queue. For outbound servers, information about a transaction remains in the view until the transaction is sent to the Oracle GoldenGate client application.

For inbound servers, information about a transaction remains in the view until the transaction commits or until the entire transaction is rolled back.

Column	Datatype	Description
COMPONENT_NAME	VARCHAR2(138)	Name of the component
COMPONENT_TYPE	VARCHAR2(20)	Type of component • CAPTURE - Capture process • APPLY - Apply reader subcomponent in an outbound server or inbound server • PROPAGATION_SENDER - Propagation sender that sends LCRs from a capture process to an outbound server

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
XIDUSN	NUMBER	Transaction ID undo segment number of the transaction
XIDSLT	NUMBER	Transaction ID slot number of the transaction
XIDSQN	NUMBER	Transaction ID sequence number of the transaction
BATCH_XIDUSN	NUMBER	Transaction ID undo segment number of the batch parent transaction. When using BATCHSQL mode in Oracle GoldenGate Integrated Replicat the BATCH_XID columns identify the parent transaction of the batch the current transaction belongs to. The parent transaction is the first transaction of each batch. This column is populated only if COMPONENT_TYPE is APPLY and BATCHSQL mode is enabled.
BATCH_XIDSLT	NUMBER	Transaction ID slot number of the batch parent transaction. When using BATCHSQL mode in Oracle GoldenGate Integrated Replicat the BATCH_XID columns identify the parent transaction of the batch the current transaction belongs to. The parent transaction is the first transaction of each batch. This column is populated only if COMPONENT_TYPE is APPLY and BATCHSQL mode is enabled.
BATCH_XIDSQN	NUMBER	Transaction ID sequence number of the batch parent transaction. When using BATCHSQL mode in Oracle GoldenGate Integrated Replicat the BATCH_XID columns identify the parent transaction of the batch the current transaction belongs to. The parent transaction is the first transaction of each batch. This column is populated only if COMPONENT_TYPE is APPLY and BATCHSQL mode is enabled.
CUMULATIVE_MESSAGE_COUNT	NUMBER	Number of LCRs processed in the transaction. If a component is restarted while the transaction is being processed, then this column shows the number of LCRs processed in the transaction since the component was started.
TOTAL_MESSAGE_COUNT	NUMBER	Total number of LCRs processed in the transaction by an outbound server or inbound server. This column does not pertain to capture processes.
FIRST_MESSAGE_TIME	DATE	Time stamp of the first LCR processed in the transaction. If a capture process is restarted while the transaction is being processed, then this column shows the time stamp of the first LCR processed after the capture process was started.
FIRST_MESSAGE_NUMBER	NUMBER	SCN of the first message in the transaction. If a capture process is restarted while the transaction is being processed, then this column shows the SCN of the first message processed after the capture process was started.
LAST_MESSAGE_TIME	DATE	Time stamp of the last LCR processed in the transaction
LAST_MESSAGE_NUMBER	NUMBER	SCN of the most recent message encountered for the transaction
FIRST_MESSAGE_POSITION	RAW(64)	Position of the first message seen by an XStream inbound server This column is populated only for an apply process that is functioning as an Oracle GoldenGate inbound server.
LAST_MESSAGE_POSITION	RAW(64)	Position of the last message seen by an Oracle GoldenGate inbound server This column is populated only for an apply process that is functioning as an Oracle GoldenGate inbound server.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TRANSACTION_ID	VARCHAR2(128)	Transaction ID for an Oracle GoldenGate inbound server This column is populated only for an apply process that is functioning as an Oracle GoldenGate inbound server.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.265 V$HANG_INFO

V$HANG_INFO displays information about hangs found on the cluster.

A hang can be an open wait chain or closed wait chain (cycle or deadlock). A wait chain is a series of sessions that are blocking one another. Each row represents a hang and describes how severe the hang is. This view also includes the victim or final blocker of the hang.

Column	Datatype	Description
HANG_ID	NUMBER	A number identifying the hang
HANG_CHAIN_SESSIONS	NUMBER	Indicates how many sessions are in the main wait chain of the hang
TOTAL_HUNG_SESSIONS	NUMBER	Indicates how many total sessions are affected by the hang, including the main wait chain and any wait chains branching off of the main wait chain
HANG_TYPE	VARCHAR2(18)	Is set to Hang if this is an open wait chain or Deadlock if this is a cycle or closed wait chain
HANG_CREATE_TIME	VARCHAR2(20)	Date and time that the hang was detected
HANG_RESOLVE_TIME	VARCHAR2(20)	Time that the hang may be automatically resolved
IGNORED_HANG	VARCHAR2(1)	Y - Hang was ignored and will not be automatically resolved at this time N - The hang has not yet been verified
RESOLUTION_ATTEMPTED	VARCHAR2(1)	Y - An attempt was made to resolve the hang N - No attempt was made to resolve the hang
GLOBAL_HANG	VARCHAR2(1)	Y - The hang spans multiple nodes in the cluster N - All of the sessions in the hang reside on a single instance
ESCALATED_HANG	VARCHAR2(1)	Y - An attempt was made to resolve the hang but was unsuccessful, and the hang is becoming more severe. Another resolution attempt will be made. N - The first resolution attempt was successful and it was not necessary to escalate the hang to a higher resolution level.
RESOLUTION_STATUS	VARCHAR2(45)	Contains the reason why the hang was resolved or ignored
VICTIM_INSTANCE	NUMBER	Instance number of the instance on which the victim or final blocker session resides
VICTIM_SESSION_ID	NUMBER	Victim or final blocker's Oracle session ID
VICTIM_SERIAL#	NUMBER	Victim or final blocker's Oracle session serial number

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
VICTIM_OSPID	VARCHAR2(24)	Victim or final blocker's operating system process ID
FATAL_BACKGROUND	VARCHAR2(1)	Y - Victim or final blocker is a critical background process N - Victim or final blocker is not a critical background process
PNAME	VARCHAR2(5)	Name of the victim or final blocker process
WAIT_EVENT_TEXT	VARCHAR2(64)	Resource or event for which the victim or final blocker is waiting; set to not in a wait if the session is not waiting on a resource
VICTIM_QOS_PC_RANK	VARCHAR2(12)	Oracle Database Quality of Service (QoS) Management Performance Class rank of the victim or final blocker of the hang.
VICTIM_QOS_PC_ITT	NUMBER	Oracle Database QoS Management Performance Class In-Trouble Threshold of the victim or final blocker of the hang
VICTIM_QOS_PC_RTT	NUMBER	Oracle Database QoS Management Performance Class Root Termination allowed Threshold of the victim or final blocker of the hang
VICTIM_QOS_PC_KEY	NUMBER	Oracle Database QoS Management Performance Class Key of the victim or final blocker of the hang
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$HANG_SESSION_INFO"
- "V$HANG_STATISTICS"
- "DBA_HANG_MANAGER_PARAMETERS"

7.266 V$HANG_SESSION_INFO

V$HANG_SESSION_INFO displays information about sessions involved in hangs described by V$HANG_INFO.

Each row with the same HANG_ID describes a session that is in the hang wait chain described by the row with the same HANG_ID in V$HANG_INFO. This session is blocked by the victim or final blocker of that hang.

Column	Datatype	Description
HANG_ID	NUMBER	A number identifying the hang including this session. This column can be used to join V$HANG_INFO with V$HANG_SESSION_INFO.
INSTANCE	NUMBER	Instance number of the instance on which this session resides
SID	NUMBER	Oracle session ID of this session
SERIAL#	NUMBER	Oracle session serial number of this session
OSPID	VARCHAR2(24)	Operating system process ID of this session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FATAL_BACKGROUND	VARCHAR2(1)	Y - Session is a critical background process N - Session is not a critical background process
ROOT	VARCHAR2(1)	Y - This session is the victim or final blocker of the hang N - This session is not the victim or final blocker of the hang
PNAME	VARCHAR2(5)	Name of the victim or final blocker process
WAIT_EVENT_TEXT	VARCHAR2(64)	Resource or event for which this session is waiting; set to `not in a wait` if the session is not waiting on a resource
QOS_PC_RANK	VARCHAR2(12)	Session's Oracle Database Quality of Service (QoS) Management Performance Class rank
QOS_PC_ITT	NUMBER	Session's Oracle Database Quality of Service (QoS) Management Performance Class In-Trouble Threshold
QOS_PC_RTT	NUMBER	Session's Oracle Database Quality of Service (QoS) Management Performance Class Root Termination allowed Threshold
QOS_PC_KEY	NUMBER	Session's Oracle Database Quality of Service (QoS) Management Performance Class Key
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$HANG_INFO"
- "V$HANG_STATISTICS"
- "DBA_HANG_MANAGER_PARAMETERS"

7.267 V$HANG_STATISTICS

V$HANG_STATISTICS displays statistics about hangs found on the cluster.

Column	Datatype	Description
STATISTIC#	NUMBER	Statistic number
NAME	VARCHAR2(45)	Name of the statistic
VALUE	NUMBER	Value associated with the statistic

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$HANG_INFO"
- "V$HANG_SESSION_INFO"
- "DBA_HANG_MANAGER_PARAMETERS"

7.268 V$HEAT_MAP_SEGMENT

V$HEAT_MAP_SEGMENT displays real-time segment access information.

Column	Datatype	Description
OBJECT_NAME	VARCHAR2(128)	Name of the object
SUBOBJECT_NAME	VARCHAR2(128)	Name of the subobject
OBJ#	NUMBER	Object number
DATAOBJ#	NUMBER	Data object number
TS#	NUMBER	ID of the tablespace containing the segment whose heat map is tracked
TRACK_TIME	DATE	Timestamp of current activity tracking
SEGMENT_WRITE	VARCHAR2(3)	Indicates whether the segment has write access: (YES or NO)
SEGMENT_READ	VARCHAR2(3)	Indicates whether the segment has read access: (YES or NO)
FULL_SCAN	VARCHAR2(3)	Indicates whether the segment has full table scan: (YES or NO)
LOOKUP_SCAN	VARCHAR2(3)	Indicates whether the segment has lookup scan: (YES or NO)
N_SEGMENT_WRITE	NUMBER	Number of segment writes
N_FULL_SCAN	NUMBER	Number of table scans
N_LOOKUP_SCAN	NUMBER	Number of lookup scans
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"ALL_HEAT_MAP_SEGMENT"

7.269 V$HM_CHECK

V$HM_CHECK displays information about all the checks registered with Health Monitor. Each check is uniquely identified by a name or an ID.

Column	Datatype	Description
ID	NUMBER	Health check ID
NAME	VARCHAR2(64)	Health check name
NAME_NLS	VARCHAR2(1024)	Internationalized names of the health checks
CLSID	NUMBER	Class ID to which this check belongs
CLS_NAME	VARCHAR2(15)	Class name of the check: • GENERIC • PERSISTENT_DATA
FLAGS	NUMBER	Reserved for internal use
INTERNAL_CHECK	VARCHAR2(1)	Internal check
OFFLINE_CAPABLE	VARCHAR2(1)	Ability to run when database is not open yet (Y or N)
DESCRIPTION	VARCHAR2(1024)	Description of what the check does
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.270 V$HM_CHECK_PARAM

V$HM_CHECK_PARAM displays information about the input parameters of all Health Monitor checks.

Column	Datatype	Description
ID	NUMBER	Parameter ID
NAME	VARCHAR2(64)	Parameter name
CHECK_ID	NUMBER	ID of the check to which this parameter belongs

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TYPE	VARCHAR2(20)	Data type of the input parameter: • DBKH_PARAM_UB4 • DBKH_PARAM_UB8 • DBKH_PARAM_TEXT • DBKH_PARAM_DATE • DBKH_PARAM_UB4_LIST • DBKH_PARAM_UB8_LIST • DBKH_PARAM_TEXT_LIST • DBKH_PARAM_DATE_LIST
DEFAULT_VALUE	VARCHAR2(64)	Default value for this parameter
FLAGS	NUMBER	Reserved for internal use
DESCRIPTION	VARCHAR2(1024)	Description of the parameter
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.271 V$HM_FINDING

V$HM_FINDING displays information about all the findings of various Health Monitor runs.

Column	Datatype	Description
FINDING_ID	NUMBER	Unique ID to represent the finding
RUN_ID	NUMBER	ID of the run that created this finding
NAME	VARCHAR2(32)	Name of the finding
PARENT_ID	NUMBER	Parent finding ID for this finding
CHILD_COUNT	NUMBER	Number of active (open) child findings, if this finding is a parent finding type
CLASS_NAME	VARCHAR2(32)	Name of the class to which this finding belongs
TIME_DETECTED	TIMESTAMP(6)	Time this finding was detected
MODIFIED	TIMESTAMP(6)	Time that this finding was last modified
PRIORITY	VARCHAR2(8)	Priority of the finding: • LOW, • HIGH • CRITICAL
STATUS	VARCHAR2(12)	Status of the finding: • OPEN • CLOSED
TYPE	VARCHAR2(13)	Type of the finding: • INFORMATIONAL • FAILURE
DESCRIPTION	VARCHAR2(1024)	Description of the finding

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DAMAGE_DESCRIPTION	VARCHAR2(1024)	Possible damage description of the finding
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.272 V$HM_INFO

V$HM_INFO displays information about Health Monitor runs, findings, and recommendations.

The information for a run/finding/recommendation is organized as a name, value pair.

If the type of information is RUN, then the data represents the input parameters for that run. If the type of information is FINDING or RECOMMENDATION, then the data represents the information about that particular finding/recommendation.

Column	Datatype	Description
ID	NUMBER	Unique identifier of the information
TYPE	VARCHAR2(14)	Type of the information: • RUN • RUN-RESUME • FINDING • RECOMMENDATION
NAME	VARCHAR2(32)	Information parameter name
VALUE	VARCHAR2(513)	Information parameter value
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.273 V$HM_RECOMMENDATION

V$HM_RECOMMENDATION displays information about all the recommendations made to various Health Monitor findings.

Column	Datatype	Description
RECOMMENDATION_ID	NUMBER	Unique ID to represent the recommendation
FDG_ID	NUMBER	Unique ID to represent either the finding ID or the finding set ID for which the recommendation was made

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RUN_ID	NUMBER	ID of the run that may have generated the recommendation If the ID is 0, then the recommendations were generated by RMAN.
NAME	VARCHAR2(32)	Name of the recommendation
TYPE	VARCHAR2(7)	Type of the recommendation: • MANUAL • REPAIR
RANK	NUMBER	Rank of the recommendation
TIME_DETECTED	TIMESTAMP(6)	Time that the recommendation was made
EXECUTED	TIMESTAMP(6)	Time that the recommendation (repair) was implemented
STATUS	VARCHAR2(7)	Status of the recommendation (repair) execution: • NOT RUN • RUNNING • SUCCESS • FAILED
DESCRIPTION	VARCHAR2(1024)	Description of the recommendation
REPAIR_SCRIPT	VARCHAR2(512)	Location of the repair script file
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.274 V$HM_RUN

V$HM_RUN displays information about all Health Monitor checks and their status.

Column	Datatype	Description
RUN_ID	NUMBER	Unique ID to represent the run
NAME	VARCHAR2(32)	Unique name used to identify the run
CHECK_NAME	VARCHAR2(32)	Name of the check
RUN_MODE	VARCHAR2(8)	Mode of the run: • MANUAL • REACTIVE • AUTO
TIMEOUT	NUMBER	Number of seconds allowed for this run to complete before it is terminated
START_TIME	TIMESTAMP(6)	Start time of the run
LAST_RESUME_TIME	TIMESTAMP(6)	Last resumed time for the run
END_TIME	TIMESTAMP(6)	End time of the run
MODIFIED_TIME	TIMESTAMP(6)	Last modified time of the run record

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(11)	Status of the run: • INITIAL • EXECUTING • INTERRUPTED • TIMEDOUT • CANCELLED • COMPLETED • ERROR
SRC_INCIDENT	NUMBER	Source incident ID that activated this run
NUM_INCIDENT	NUMBER	Number of incidents created by this run
ERROR_NUMBER	NUMBER	Error number if the run failed to complete because of an error
PROBLEM_ID	NUMBER	Problem ID of the source incident ID that may have activated this Run
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.275 V$HS_AGENT

V$HS_AGENT displays the set of HS agents currently running on a given host. There is one row per agent process.

Column	Datatype	Description
AGENT_ID	NUMBER	Oracle Net session identifier used for connections to the agent (the identifier used in the LISTENER.ORA file)
MACHINE	VARCHAR2(64)	Operating system machine name
PROCESS	VARCHAR2(9)	Operating system process identifier of the agent
PROGRAM	VARCHAR2(48)	Program name of the agent
OSUSER	VARCHAR2(128)	Operating system user
STARTTIME	DATE	Starting time
AGENT_TYPE	NUMBER	Type of the agent
FDS_CLASS_ID	NUMBER	ID of the Foreign Data Store class
FDS_INST_ID	NUMBER	Instance name of the Foreign Data Store
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.276 V$HS_PARAMETER

V$HS_PARAMETER describes the initialization parameters in use by the server and agent.

Column	Datatype	Description
HS_SESSION_ID	NUMBER	Unique HS session identifier (maps to the HS_SESSION_ID column of V$HS_SESSION)
PARAMETER	VARCHAR2(128)	Name of the parameter
VALUE	VARCHAR2(64)	Value of the parameter
SOURCE	VARCHAR2(1)	Indicates whether the parameter was defined in the agent (A) or server (S)
ENV	VARCHAR2(1)	Indicates whether the parameter was set as an environment variable in the agent (T) or elsewhere (F)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

7.277 V$HS_SESSION

V$HS_SESSION describes the current HS session.

Column	Datatype	Description
HS_SESSION_ID	NUMBER	Unique HS session identifier
AGENT_ID	NUMBER	Oracle Net session identifier used for connections to the agent (maps to the AGENT_ID column of V$HS_AGENT)
SID	NUMBER	User session identifier (maps to the SID column of V$SESSION)
DB_LINK	VARCHAR2(128)	Server database link name used to access the agent; blank if no database link is used (for example, when using external procedures)
DB_LINK_OWNER	NUMBER	Owner of the database link in DB_LINK
STARTTIME	DATE	Time the connection was initiated
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

7.278 V$HVMASTER_INFO

V$HVMASTER_INFO describes the current and previous master instances and the number of re-masterings of Global Enqueue Service resources.

Column	Datatype	Description
HV_ID	NUMBER	Hash value ID
CURRENT_MASTER	NUMBER	Master instance of this hash value ID
PREVIOUS_MASTER	NUMBER	Previous master instance of this hash value ID
REMASTER_CNT	NUMBER	Number of times this has been remastered
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8

Dynamic Performance (V$) Views: V$IM_COLUMN_LEVEL to V$RULE_SET_AGGREGATE_STATS

This chapter contains the dynamic performance views V$IM_COLUMN_LEVEL to V$RULE_SET_AGGREGATE_STATS.

8.1 V$IM_COLUMN_LEVEL

V$IM_COLUMN_LEVEL presents the selective column compression levels that are defined using the inmemory_memcompress clause of the inmemory_column_clause of the CREATE TABLE statement.

This view returns no rows for a table which has no associated selective column compression levels.

Column	Datatype	Description
INST_ID	NUMBER	Instance ID
OWNER	VARCHAR2(31)	Username of the table owner
OBJ_NUM	NUMBER	Table object number
TABLE_NAME	VARCHAR2(31)	Table name
SEGMENT_COLUMN_ID	NUMBER	Segment column number
COLUMN_NAME	VARCHAR2(31)	Column name
INMEMORY_COMPRESSION	VARCHAR2(26)	Column compression level. Possible values: • DEFAULT: This value appears for physical columns of a table that do not have a selective column clause. It also appears when a virtual column has been explicitly enabled for in-memory storage using ALTER TABLE *table-name* INMEMORY(VC). • UNSPECIFIED: This value appears for virtual columns of a table that do not have a selective column clause. If the INMEMORY_VIRTUAL_COLUMNS initialization parameter is set to MANUAL, a virtual column with a DEFAULT INMEMORY_COMPRESSION clause will be materialized while a virtual column with an UNSPECIFIED value will not be.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The SYS_IME hidden virtual columns automatically added by the In-Memory Expression (IME) infrastructure will not be shown in this view.

> **See Also**
>
> - "INMEMORY_CLAUSE_DEFAULT"
> - "V$IM_SEGMENTS"
> - "V$IM_USER_SEGMENTS"
> - *Oracle Database In-Memory Guide* for an introduction to the IM column store
> - *Oracle Database SQL Language Reference* for more information about the `inmemory_column_clause` of the SQL `CREATE TABLE` statement

8.2 V$IM_SEGMENTS

`V$IM_SEGMENTS` presents information about all the in-memory segments in the database.

Only segments that have an in-memory representation are displayed. If a segment is marked for the In-Memory Column Store (IM column store) but is not populated, no corresponding row for that segment is displayed in this view.

Column	Datatype	Description
`OWNER`	`VARCHAR2(128)`	User name of the segment owner
`SEGMENT_NAME`	`VARCHAR2(128)`	Name of the segment, if any
`PARTITION_NAME`	`VARCHAR2(128)`	Object partition name (set to NULL for non-partitioned objects)
`SEGMENT_TYPE`	`VARCHAR2(18)`	Type of segment: • `TABLE` • `TABLE PARTITION` • `TABLE SUBPARTION`
`TABLESPACE_NAME`	`VARCHAR2(30)`	Name of the tablespace containing the segment
`INMEMORY_SIZE`	`NUMBER`	Size of the in-memory version of the segment, in bytes
`BYTES`	`NUMBER`	Number of on-disk data bytes for the segment that could be represented in memory (no space metadata blocks)
`BYTES_NOT_POPULATED`	`NUMBER`	Size of the portion of the on-disk segment that is not populated in memory, in bytes.
`POPULATE_STATUS`	`VARCHAR2(13)`	Status of segment population: • `STARTED`: A populate task for the segment has started • `COMPLETED`: There are no populate tasks pending for the segment • `OUT OF MEMORY`: A populate task for the segment failed due to lack of space in the IM column store

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
INMEMORY_PRIORITY	VARCHAR2(8)	Indicates the priority for IM column store population: • LOW • MEDIUM • HIGH • CRITICAL • NONE
INMEMORY_DISTRIBUTE	VARCHAR2(15)	Indicates how the IM column store is distributed in an Oracle Real Application Clusters (Oracle RAC) environment: • AUTO • BY ROWID RANGE • BY PARTITION • BY SUBPARTITION
INMEMORY_DUPLICATE	VARCHAR2(13)	Indicates the duplicate setting for the IM column store in an Oracle RAC environment: • NO DUPLICATE • DUPLICATE • DUPLICATE ALL
INMEMORY_COMPRESSION	VARCHAR2(17)	Compression level for the IM column store: • NO MEMCOMPRESS • FOR DML • FOR QUERY [LOW | HIGH] • FOR CAPACITY [LOW | HIGH]
INMEMORY_SERVICE	VARCHAR2(12)	Specifies how the IM-enabled table is populated on various instances: • DEFAULT: Pre-Oracle Database 12c Release 2 (12.2.0.1) behavior • NONE: Do not populate on any instance • ALL: Populate on all instances • USER_DEFINED: Populate only on the instances on which the user-specified service is active. The service name corresponding to this is stored in the INMEMORY_SERVICE_NAME column.
INMEMORY_SERVICE_NAME	VARCHAR2(129)	Specifies the service name on which the IM-enabled table should be populated. This column has a value only when the corresponding INMEMORY_SERVICE column has a value of USER_DEFINED.
IS_EXTERNAL	VARCHAR2(5)	Indicates whether the IM segment is for an external table. Possible values: • TRUE: The IM segment is for an external table. • FALSE: The IM segment is not for an external table.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "INMEMORY_CLAUSE_DEFAULT"
- "V$IM_COLUMN_LEVEL"
- "V$IM_USER_SEGMENTS"
- *Oracle Database In-Memory Guide* for an introduction to the IM column store

8.3 V$IM_USER_SEGMENTS

V$IM_USER_SEGMENTS presents information about the in-memory segments for the current user in the database.

Only segments that have an in-memory representation are displayed. If a segment is marked for the In-Memory Column Store (IM column store) but is not populated, no corresponding row for that segment is displayed in this view.

Column	Datatype	Description
SEGMENT_NAME	VARCHAR2(128)	Name of the segment, if any
PARTITION_NAME	VARCHAR2(128)	Object partition name (set to NULL for non-partitioned objects)
SEGMENT_TYPE	VARCHAR2(18)	Type of segment: • TABLE • TABLE PARTITION • TABLE SUBPARTION
TABLESPACE_NAME	VARCHAR2(30)	Name of the tablespace containing the segment
INMEMORY_SIZE	NUMBER	Size of the in-memory version of the segment, in bytes
BYTES	NUMBER	Total size of the on-disk segment, in bytes
BYTES_NOT_POPULATED	NUMBER	Size of the portion of the on-disk segment that is not populated in memory, in bytes
POPULATE_STATUS	VARCHAR2(13)	Status of segment population: • STARTED: A populate task for the segment has started • COMPLETED: There are no populate tasks pending for the segment • OUT OF MEMORY: A populate task for the segment failed due to lack of space in the IM column store
INMEMORY_PRIORITY	VARCHAR2(8)	Indicates the priority for IM column store population: • LOW • MEDIUM • HIGH • CRITICAL • NONE
INMEMORY_DISTRIBUTE	VARCHAR2(15)	Indicates how the IM column store is distributed in an Oracle Real Application Clusters (Oracle RAC) environment: • AUTO • BY ROWID RANGE • BY PARTITION • BY SUBPARTITION

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
INMEMORY_DUPLICATE	VARCHAR2(13)	Indicates the duplicate setting for the IM column store in an Oracle RAC environment: • NO DUPLICATE • DUPLICATE • DUPLICATE ALL
INMEMORY_COMPRESSION	VARCHAR2(17)	Compression level for the IM column store: • NO MEMCOMPRESS • FOR DML • FOR QUERY [LOW | HIGH] • FOR CAPACITY [LOW | HIGH]
INMEMORY_SERVICE	VARCHAR2(12)	Specifies how the IM-enabled table is populated on various instances: • DEFAULT: Pre-Oracle Database 12c Release 2 (12.2.0.1) behavior • NONE: Do not populate on any instance • ALL: Populate on all instances • USER_DEFINED: Populate only on the instances on which the user-specified service is active. The service name corresponding to this is stored in the INMEMORY_SERVICE_NAME column.
INMEMORY_SERVICE_NAME	VARCHAR2(129)	Specifies the service name on which the IM-enabled table should be populated. This column has a value only when the corresponding INMEMORY_SERVICE column has a value of USER_DEFINED.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "INMEMORY_CLAUSE_DEFAULT"
- "V$IM_COLUMN_LEVEL"
- "V$IM_SEGMENTS"
- *Oracle Database In-Memory Guide* for an introduction to the IM column store

8.4 V$INDEX_USAGE_INFO

V$INDEX_USAGE_INFO keeps track of index usage since the last flush. A flush occurs every 15 minutes. After each flush, ACTIVE_ELEM_COUNT is reset to 0 and LAST_FLUSH_TIME is updated to the current time.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
INDEX_STATS_ENABLED	NUMBER	Indicates whether the index usage statistics are enabled. Possible values: • 0: Index statistics are disabled • 1: Index statistics are enabled
INDEX_STATS_COLLECTION_TYPE	NUMBER	Indicates the type of collection used for the index usage statistics. Possible values: • 0: Indicates the ALL collection type. With this type of collection, the statistics are collected for each execution that has index access. Selecting this statistics collection type may have some impact on performance. • 1: Indicates the SAMPLED collection type. With this type of collection, the statistics are collected based on sampling (only a few of the executions are considered when collecting the statistics). This is the default statistics collection type. Index statistics collected with the SAMPLED collection type are less accurate than index statistics collected with the ALL collection type.
ACTIVE_ELEM_COUNT	NUMBER	The number of active indexes since the last flush
ALLOC_ELEM_COUNT	NUMBER	The number of index entries allocated
MAX_ELEM_COUNT	NUMBER	The maximum number of active indexes that can be tracked
FLUSH_COUNT	NUMBER	Number of successful flushes since the database started
TOTAL_FLUSH_DURATION	NUMBER	Cumulative elapsed time taken to complete the index usage statistics flush since the database start
LAST_FLUSH_TIME	TIMESTAMP(3)	The time of the last flush
STATUS_MSG	VARCHAR2(256)	Status messages, if any. Flush errors are reported here.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This row is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"DBA_INDEX_USAGE"

8.5 V$INDEXED_FIXED_COLUMN

V$INDEXED_FIXED_COLUMN displays the columns in dynamic performance tables that are indexed (X$ tables).

The X$ tables can change without notice. Use this view only to write queries against fixed views (V$ views) more efficiently.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TABLE_NAME	VARCHAR2(128)	Name of the dynamic performance table that is indexed
INDEX_NUMBER	NUMBER	Number that distinguishes to which index a column belongs
COLUMN_NAME	VARCHAR2(128)	Name of the column that is being indexed
COLUMN_POSITION	NUMBER	Position of the column in the index key (this is mostly relevant for multicolumn indexes)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.6 V$INMEMORY_AREA

V$INMEMORY_AREA contains information on the space allocation inside the In-Memory area.

The In-Memory area is sub-divided into two pools: a 1MB pool used to store the actual column formatted data populated into memory, and a 64K pool used to store metadata about the objects that are populated into the In-Memory Column Store (IM column store). The amount of available memory in each pool is visible in the V$INMEMORY_AREA view. The relative size of the two pools is determined by internal heuristics. The majority of the In-Memory area memory is allocated to the 1MB pool.

Column	Datatype	Description
POOL	VARCHAR2(26)	Name of the pools in the In-Memory area
ALLOC_BYTES	NUMBER	Total amount of memory allocated to this pool
USED_BYTES	NUMBER	Amount of memory currently used in this pool
POPULATE_STATUS	VARCHAR2(26)	Shows the status of the IM column store, for example, whether it is currently being populated or if it is done
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "INMEMORY_CLAUSE_DEFAULT"
- "V$IM_COLUMN_LEVEL"
- "V$IM_SEGMENTS"
- "V$IM_USER_SEGMENTS"
- *Oracle Database In-Memory Guide* for an introduction to the IM column store

8.7 V$INMEMORY_FASTSTART_AREA

V$INMEMORY_FASTSTART_AREA provides information about the In-Memory FastStart (IM FastStart) area.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
TABLESPACE_NAME	VARCHAR2(128)	IM FastStart tablespace name. When IM FastStart is not enabled, the value of TABLESPACE_NAME is INVALID_TABLESPACE and the value of STATUS is DISABLE.
STATUS	VARCHAR2(10)	IM FastStart status. Possible values include: • ENABLE: An IM FastStart tablespace has been specified and the content of the IM column is being periodically checkpointed to disk. • ENABLING: An IM FastStart tablespace has been specified and the database is creating the IM FastStart area. • DISABLE: An IM FastStart tablespace has not been specified. This is the default. • MIGRATING: A user has requested the IM FastStart area be migrated from one tablespace to another. • DISABLING: A user has requested IM FastStart to be disabled.
ALLOCATED_SIZE	NUMBER	The allocated size of an IM FastStart tablespace in bytes
USED_SIZE	NUMBER	The currently used size of an IM FastStart area (in bytes) within the tablespace.
LAST_CHECKPOINT_TIME	TIMESTAMP(6)	The time when the last IMCU was checkpointed to the IM FastStart area
LAST_POPULATE_TIME	TIMESTAMP(6)	The time of the last population from the IM FastStart area
NUM_DEFERRED_WRITES	NUMBER	The number of pending deferred writes to the IM FastStart area

8.8 V$INSTANCE

V$INSTANCE displays the state of the current instance.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
INSTANCE_NUMBER	NUMBER	Instance number used for instance registration (corresponds to the INSTANCE_NUMBER initialization parameter) **See Also:** "INSTANCE_NUMBER"
INSTANCE_NAME	VARCHAR2(16)	Name of the instance
HOST_NAME	VARCHAR2(64)	Name of the host machine
VERSION	VARCHAR2(17)	Database version
VERSION_LEGACY	VARCHAR2(17)	The legacy database version used before Oracle Database 18c. This column displays the same value as the VERSION column
VERSION_FULL	VARCHAR2(17)	The version string with the new Oracle Database version scheme introduced in Oracle Database 18c.
STARTUP_TIME	DATE	Time when the instance was started
STATUS	VARCHAR2(12)	Status of the instance: • STARTED - After STARTUP NOMOUNT • MOUNTED - After STARTUP MOUNT or ALTER DATABASE CLOSE • OPEN - After STARTUP or ALTER DATABASE OPEN • OPEN MIGRATE - After ALTER DATABASE OPEN { UPGRADE | DOWNGRADE }
PARALLEL	VARCHAR2(3)	Indicates whether the instance is mounted in cluster database mode (YES) or not (NO)
THREAD#	NUMBER	Redo thread opened by the instance
ARCHIVER	VARCHAR2(7)	Automatic archiving status: • STOPPED • STARTED • FAILED - Archiver failed to archive a log last time but will try again within 5 minutes
LOG_SWITCH_WAIT	VARCHAR2(15)	Event that log switching is waiting for: • ARCHIVE LOG • CLEAR LOG • CHECKPOINT • NULL - ALTER SYSTEM SWITCH LOGFILE is hung but there is room in the current online redo log
LOGINS	VARCHAR2(10)	Indicates whether the instance is in unrestricted mode, allowing logins by all users (ALLOWED, or in restricted mode, allowing logins by database administrators only (RESTRICTED)
SHUTDOWN_PENDING	VARCHAR2(3)	Indicates whether a shutdown is pending (YES) or not (NO)
DATABASE_STATUS	VARCHAR2(17)	Status of the database: • ACTIVE • SUSPENDED • INSTANCE RECOVERY
INSTANCE_ROLE	VARCHAR2(18)	Indicates whether the instance is an active instance (PRIMARY_INSTANCE) or an inactive secondary instance (SECONDARY_INSTANCE), or UNKNOWN if the instance has been started but not mounted

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ACTIVE_STATE	VARCHAR2(9)	Quiesce state of the instance: • NORMAL - Database is in a normal state. • QUIESCING - ALTER SYSTEM QUIESCE RESTRICTED has been issued: no new user transactions, queries, or PL/SQL statements are processed in this instance. User transactions, queries, or PL/SQL statements issued before the ALTER SYSTEM QUIESCE RESTRICTED statement are unaffected. DBA transactions, queries, or PL/SQL statements are also unaffected. • QUIESCED - ALTER SYSTEM QUIESCE RESTRICTED has been issued: no user transactions, queries, or PL/SQL statements are processed. DBA transactions, queries, or PL/SQL statements are unaffected. User transactions, queries, or PL/ SQL statements issued after the ALTER SYSTEM QUIESCE RESTRICTED statement are not processed. A single ALTER SYSTEM QUIESCE RESTRICTED statement quiesces all instances in an Oracle RAC environment. After this statement has been issued, some instances may enter into a quiesced state before other instances; the system is quiesced when all instances enter the quiesced state.
BLOCKED	VARCHAR2(3)	Indicates whether all services are blocked (YES) or not (NO)
CON_ID	NUMBER	The ID of the container to which the data pertains: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
INSTANCE_MODE	VARCHAR2(11)	The instance mode of the current instance: • REGULAR: A regular Oracle RAC instance. This value is also always used for any non-Oracle RAC instance. • READ MOSTLY: An Oracle RAC instance that performs very few database writes • READ ONLY: A read-only Oracle RAC instance
EDITION	VARCHAR2(7)	The edition of the database: • PO: Personal Edition • XE: Express Edition • SE2: Standard Edition or Standard Edition 2 • EE: Enterprise Edition • HP: Enterprise Edition - High Performance • XP: Enterprise Edition - Extreme Performance
FAMILY	VARCHAR2(80)	For internal use only.
DATABASE_TYPE	VARCHAR2(15)	Database type: • RAC: If the database is a regular Oracle RAC database which may have multiple instances. • RACONENODE: If the database is Oracle RAC, but allows only one instance to run at any time - the RAC One Node mode. • SINGLE: If the database is running as a single instance. • UNKNOWN: If the database's type can't be determined. This might happen when the database is registered as a DB resource with CRS but the CRS service has failed to return valid database type information. Typically, this indicates that either the CRS service is down or it is in a faulty state.

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.9 V$INSTANCE_CACHE_TRANSFER

V$INSTANCE_CACHE_TRANSFER displays statistics for the cache blocks transferred among instances.

Oracle keeps multiple versions of data buffered in the buffer cache. The current buffer (or block), CURRENT_BLOCK, is the most up-to-date copy, containing all recent modifications. A consistent read buffer (or block), CR_BLOCK, contains the version of the data at a particular time prior to the current buffer. It is read-consistent (that is, all the data shown in that buffer are consistent for the start time of a query).

Therefore, for the same data block there can be multiple copies in the buffer cache: one current copy, and one or more consistent read copies with data consistent as of different snapshot times.

Column	Datatype	Description
INSTANCE	NUMBER	Instance from which the blocks are transferred
CLASS	VARCHAR2(18)	Class of the cache block
LOST	NUMBER	The number of blocks that were sent by a particular instance but that never arrived in this instance
LOST_TIME	NUMBER	The time waited for blocks that were sent by a particular instance but that never arrived in this instance
CR_BLOCK	NUMBER	CR Block transfers not affected by remote processing delays
CR_BLOCK_TIME	NUMBER	Total time waited for CR blocks from a particular instance (includes the other times)
CR_2HOP	NUMBER	The count of CR blocks which were received by this instance from a particular instance after a 2-way round-trip
CR_2HOP_TIME	NUMBER	The time waited for CR blocks which were received by this instance from a particular instance after a 2-way round-trip
CR_3HOP	NUMBER	The count of CR blocks which were received by this instance from a particular instance after a 3-way round-trip
CR_3HOP_TIME	NUMBER	The time waited for CR blocks which were received by this instance from a particular instance after a 3-way round-trip
CR_BUSY	NUMBER	CR Block transfers affected by remote contention
CR_BUSY_TIME	NUMBER	The time waited for CR blocks which were received by this instance from a particular instance and which were delayed by a log flushed on the sending instance
CR_CONGESTED	NUMBER	CR Block transfers affected by remote system load
CR_CONGESTED_TIME	NUMBER	The time waited for CR blocks which were received by this instance from a particular instance and which were delayed because LMS was busy
CURRENT_BLOCK	NUMBER	Current block transfers not affected by remote processing delays
CURRENT_BLOCK_TIME	NUMBER	Total time waited for current blocks from a particular instance (includes the other times
CURRENT_2HOP	NUMBER	The count of current blocks which were received by this instance from a particular instance after a 2-way round-trip
CURRENT_2HOP_TIME	NUMBER	The time waited for current blocks which were received by this instance from a particular instance after a 2-way round-trip

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CURRENT_3HOP	NUMBER	The count of current blocks which were received by this instance from a particular instance after a 3-way round-trip
CURRENT_3HOP_TIME	NUMBER	The time waited for current blocks which were received by this instance from a particular instance after a 3-way round-trip
CURRENT_BUSY	NUMBER	Current block transfers affected by remote contention
CURRENT_BUSY_TIME	NUMBER	The time waited for current blocks which were received by this instance from a particular instance and which were delayed by a log flushed on the sending instance
CURRENT_CONGESTED	NUMBER	Current block transfers affected by remote system load
CURRENT_CONGESTED_TIME	NUMBER	The time waited for current blocks which were received by this instance from a particular instance and which were delayed because LMS was busy
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.10 V$INSTANCE_PING

V$INSTANCE_PING provides information about measured latency of the interconnect for all instances in an Oracle Real Application Clusters (Oracle RAC) environment.

In an Oracle RAC environment, every few seconds the PING process of each instance checks the response of the interconnect to all instances of the same database.

It sends two messages. One message is 500 bytes in size (referred to in the column descriptions as 500B), and the other is 8 kilobytes in size (referred to in column descriptions as 8K).

For each message sent to each instance, the amount of time it took to get a response back is measured (in microseconds). The view records the latest measurements as well as cumulative data since instance startup.

Column	Datatype	Description
INSTANCE	NUMBER	The instance communicated with. In the V$ view for each instance, there is one row for that instance, and in the GV$ view for each instance, there is also one row for that instance, so there are n^2 rows for *n* instances.
CURRENT_500B	NUMBER	The latest response time for the 500B message (in microseconds)
AVERAGE_500B	NUMBER	The average response time for the 500B messages since instance startup (in microseconds). Note that AVERAGE_500B should be the same as WAIT_TIME_500B/COUNT_500B.
MAX_500B	NUMBER	The maximal response time for the 500B messages since instance startup (in microseconds)
COUNT_500B	NUMBER	The number of measurements for the 500B messages since instance startup

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
WAIT_TIME_500B	NUMBER	The sum of all response times for the 500B messages since instance startup (in microseconds)
WAIT_TIME_SQUARED_500B	NUMBER	The sum of the response time squared for 500B messages since instance startup. The unit is in microseconds squared and divided by 1000.
CURRENT_8K	NUMBER	The latest response time for the 8K message (in microseconds)
AVERAGE_8K	NUMBER	The average response time for the 8K messages since instance startup (in microseconds). Note that AVERAGE_8K should be the same as WAIT_TIME_8K/COUNT_8K.
MAX_8K	NUMBER	The maximal response time for the 8K messages since instance startup (in microseconds)
COUNT_8K	NUMBER	The number of measurements for the 8K messages since instance startup
WAIT_TIME_8K	NUMBER	The sum of all response times for the 8K messages since instance startup (in microseconds)
WAIT_TIME_SQUARED_8K	NUMBER	The sum of the response time squared for 8K messages since instance startup. The unit is in microseconds squared and divided by 16.
CON_ID	NUMBER	The ID of the container to which the data pertains. For this view, the possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs.

8.11 V$INSTANCE_RECOVERY

V$INSTANCE_RECOVERY monitors the mechanisms available to users to limit recovery I/O. Those mechanisms are:

- Set the LOG_CHECKPOINT_TIMEOUT initialization parameter
- Set the LOG_CHECKPOINT_INTERVAL initialization parameter
- Set the FAST_START_MTTR_TARGET initialization parameter
- Set the size of the smallest redo log

Column	Datatype	Description
RECOVERY_ESTIMATED_IOS	NUMBER	Number of dirty buffers in the buffer cache.
ACTUAL_REDO_BLKS	NUMBER	Current actual number of redo blocks required for recovery
TARGET_REDO_BLKS	NUMBER	Current target number of redo blocks that must be processed for recovery. This value is the minimum value of the following 3 columns, and identifies which of the 3 user-defined limits determines checkpointing.
LOG_FILE_SIZE_REDO_BLKS	NUMBER	Maximum number of redo blocks required to guarantee that a log switch does not occur before the checkpoint completes.
LOG_CHKPT_TIMEOUT_REDO_BLKS	NUMBER	Number of redo blocks that need to be processed during recovery to satisfy the LOG_CHECKPOINT_TIMEOUT parameter. The value displayed is not meaningful unless that parameter has been set.
LOG_CHKPT_INTERVAL_REDO_BLKS	NUMBER	Number of redo blocks that need to be processed during recovery to satisfy the LOG_CHECKPOINT_INTERVAL parameter. The value displayed is not meaningful unless that parameter has been set.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FAST_START_IO_TARGET_REDO_BLKS	NUMBER	This column is obsolete and maintained for backward compatibility. The value of this column is always null.
TARGET_MTTR	NUMBER	Effective MTTR (mean time to recover) target value in seconds. The TARGET_MTTR value is calculated based on the value of the FAST_START_MTTR_TARGET parameter (the TARGET_MTTR value is used internally), and is usually an approximation of the parameter's value. However, if the FAST_START_MTTR_TARGET parameter value is very small (for example, one second), or very large (for example, 3600 seconds), the calculation will produce a target value dictated by system limitations. In such cases, the TARGET_MTTR value will be the shortest calculated time, or the longest calculated time that recovery is expected to take.
ESTIMATED_MTTR	NUMBER	Current estimated mean time to recover (MTTR) based on the number of dirty buffers and log blocks. Basically, this value tells you how long you could expect recovery to take based on the work your system is doing right now.
CKPT_BLOCK_WRITES	NUMBER	Number of blocks written by checkpoint writes
OPTIMAL_LOGFILE_SIZE	NUMBER	Redo log file size (in megabytes) that is considered optimal based on the current setting of FAST_START_MTTR_TARGET. It is recommended that the user configure all online redo logs to be at least this value. Note that redo log files must be at least 4 megabytes in size; otherwise an error is generated.
ESTD_CLUSTER_AVAILABLE_TIME	NUMBER	Estimated time (in seconds) that the cluster would become partially available should this instance fail. This column is only meaningful in an Oracle Real Application Clusters (Oracle RAC) environment. In a non-Oracle RAC environment, the value of this column is null.
WRITES_MTTR	NUMBER	Number of writes driven by the FAST_START_MTTR_TARGET initialization parameter
WRITES_LOGFILE_SIZE	NUMBER	Number of writes driven by the smallest redo log file size
WRITES_LOG_CHECKPOINT_SETTINGS	NUMBER	Number of writes driven by the LOG_CHECKPOINT_INTERVAL or LOG_CHECKPOINT_TIMEOUT initialization parameter
WRITES_OTHER_SETTINGS	NUMBER	Number of writes driven by other reasons (such as the deprecated FAST_START_IO_TARGET initialization parameter)
WRITES_AUTOTUNE	NUMBER	Number of writes due to auto-tune checkpointing
WRITES_FULL_THREAD_CKPT	NUMBER	Number of writes due to full thread checkpoints
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.12 V$IO_CALIBRATION_STATUS

V$IO_CALIBRATION_STATUS displays the status of I/O calibration in the instance.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(13)	Calibration status: • IN PROGRESS - Calibration in Progress (Results from a previous calibration run are displayed, if available) • READY Results are ready and available from an earlier run • NOT AVAILABLE Calibration results are not available
CALIBRATION_TIME	TIMESTAMP(3)	End time of the last calibration run
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.13 V$IO_OUTLIER

V$IO_OUTLIER contains entries corresponding to I/Os that have taken a long time (more than 500 ms) to complete. Use this view to see if there any occasional delays in serving disk I/O requests by the storage subsystem.

Column	Datatype	Description
FUNCTION_NAME	VARCHAR2(18)	I/O function name of the delayed I/O
IO_SIZE	NUMBER	Size of the I/O in bytes
WAIT_EVENT	VARCHAR2(64)	Wait event name that was used to track the I/O
FILE_NAME	VARCHAR2(513)	Name of the file to which the I/O was targeted
IO_LATENCY	NUMBER	Time taken to complete the I/O (in milliseconds)
DISK1_NAME	VARCHAR2(255)	For Oracle ASM, the name of the first disk to which the I/O was issued
DISK1_LATENCY	NUMBER	Latency seen on the first disk (in milliseconds)
DISK2_NAME	VARCHAR2(255)	For Oracle ASM, the name of the second disk to which the I/O was issued
DISK2_LATENCY	NUMBER	Latency seen on the second disk (in milliseconds)
DISK3_NAME	VARCHAR2(255)	For Oracle ASM, the name of the third disk to which the I/O was issued
DISK3_LATENCY	NUMBER	Latency seen on the third disk (in milliseconds)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> - "V$KERNEL_IO_OUTLIER"
> - "V$LGWRIO_OUTLIER"

8.14 V$IOFUNCMETRIC

`V$IOFUNCMETRIC` displays I/O statistics information by database function for the most recent time interval period.

Column	Datatype	Description
`BEGIN_TIME`	`DATE`	Begin time of the interval
`END_TIME`	`DATE`	End time of the interval
`INTSIZE_CSEC`	`NUMBER`	Interval size (in hundredths of a second)
`FUNCTION_ID`	`NUMBER`	Function ID
`FUNCTION_NAME`	`VARCHAR2(18)`	Function name
`SMALL_READ_MBPS`	`NUMBER`	Single block megabytes read per second
`SMALL_WRITE_MBPS`	`NUMBER`	Single block megabytes written per second
`LARGE_READ_MBPS`	`NUMBER`	Multiblock megabytes read per second
`LARGE_WRITE_MBPS`	`NUMBER`	Multiblock megabytes written per second
`SMALL_READ_IOPS`	`NUMBER`	Single block read requests per second
`SMALL_WRITE_IOPS`	`NUMBER`	Single block write requests per second
`LARGE_READ_IOPS`	`NUMBER`	Multiblock read requests per second
`LARGE_WRITE_IOPS`	`NUMBER`	Multiblock write requests per second
`AVG_WAIT_TIME`	`NUMBER`	Average wait time (in milliseconds)
`CON_ID`	`NUMBER`	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.15 V$IOFUNCMETRIC_HISTORY

`V$IOFUNCMETRIC_HISTORY` displays a recent history of the I/O statistics information by database function for the most recent time interval period.

The columns for `V$IOFUNCMETRIC_HISTORY` are the same as those for `V$IOFUNCMETRIC`.

> **See Also**
>
> "V$IOFUNCMETRIC"

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.16 V$IOS_CLIENT

V$IOS_CLIENT provides more information about IO Server clients.

Column	Datatype	Description
CLIENT_ID	NUMBER	The unique ID for this client instance
CLUSTER_ID	VARCHAR2(33)	The GUID of the cluster where the client instance is running
CLUSTER_NAME	VARCHAR2(16)	When the query is executed on an Oracle IOServer (IOS) instance, this column shows the name of the cluster where the client (database) instance is running. When the query is executed on a database instance, this column shows the name of the cluster where the IOS instance that the database is connected to is running.
NODE	NUMBER	Number of the node within the cluster where the client instance is running
INSTANCE_NAME	VARCHAR2(64)	Instance name of the database client instance
DB_NAME	VARCHAR2(64)	Database name of the database client instance
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.17 V$IOSTAT_CONSUMER_GROUP

V$IOSTAT_CONSUMER_GROUP displays disk I/O statistics for consumer groups.

If the resource manager is enabled, then I/O statistics for all consumer groups that are part of the currently enabled resource plan are captured.

Column	Datatype	Description
CONSUMER_GROUP_ID	NUMBER	Consumer group ID
SMALL_READ_MEGABYTES	NUMBER	Number of single block megabytes read
SMALL_WRITE_MEGABYTES	NUMBER	Number of single block megabytes written
LARGE_READ_MEGABYTES	NUMBER	Number of multiblock megabytes read
LARGE_WRITE_MEGABYTES	NUMBER	Number of multiblock megabytes written
SMALL_READ_REQS	NUMBER	Number of single block read requests
SMALL_WRITE_REQS	NUMBER	Number of single block write requests
LARGE_READ_REQS	NUMBER	Number of multiblock read requests
LARGE_WRITE_REQS	NUMBER	Number of multiblock write requests
NUMBER_OF_WAITS	NUMBER	Number of I/O waits by consumer group
WAIT_TIME	NUMBER	Total wait time (in milliseconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.18 V$IOSTAT_FILE

V$IOSTAT_FILE displays information about disk I/O statistics of database files (including data files, temp files, and other types of database files).

I/O statistics for Data files and Temp files are provided for each file. All other file types (for example, control files, log files, archive logs, and so on) have their statistics consolidated into one entry in the view.

Column	Datatype	Description
FILE_NO	NUMBER	File identification number
FILETYPE_ID	NUMBER	Type of file (for example, log file, data file, and so on)
FILETYPE_NAME	VARCHAR2(28)	Name of the file, in the case of a data file or temp file. For all other files, a corresponding string to be displayed (for example, ARCHIVELOG).
SMALL_READ_MEGABYTES	NUMBER	Number of single block megabytes read
SMALL_WRITE_MEGABYTES	NUMBER	Number of single block megabytes written
LARGE_READ_MEGABYTES	NUMBER	Number of multiblock megabytes read
LARGE_WRITE_MEGABYTES	NUMBER	Number of multiblock megabytes written
SMALL_READ_REQS	NUMBER	Number of single block read requests
SMALL_WRITE_REQS	NUMBER	Number of single block write requests
SMALL_SYNC_READ_REQS	NUMBER	Number of synchronous single block read requests
LARGE_READ_REQS	NUMBER	Number of multiblock read requests
LARGE_WRITE_REQS	NUMBER	Number of multiblock write requests
SMALL_READ_SERVICETIME	NUMBER	Total service time (in milliseconds) for single block read requests
SMALL_WRITE_SERVICETIME	NUMBER	Total service time (in milliseconds) for single block write requests
SMALL_SYNC_READ_LATENCY	NUMBER	Latency for single block synchronous reads (in milliseconds)
LARGE_READ_SERVICETIME	NUMBER	Total service time (in milliseconds) for multiblock read requests
LARGE_WRITE_SERVICETIME	NUMBER	Total service time (in milliseconds) for multiblock write requests
ASYNCH_IO	VARCHAR2(9)	Indicates whether asynchronous I/O is available for the file (ASYNC_ON) or not (ASYNC_OFF)
ACCESS_METHOD	VARCHAR2(11)	I/O library used to access the file. Possible values include: • OS_LIB - Operating system calls are used to access the file • ODM_LIB - Oracle Disk Manager library is used to access the file • ASM_MANAGED - The file is managed and accessed though ASM • DNFS_LIB - The file is accessed through Direct NFS library
RETRIES_ON_ERROR	NUMBER	Number of read retries on error

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.19 V$IOSTAT_FUNCTION

V$IOSTAT_FUNCTION displays disk I/O statistics for database functions (such as the LGWR and DBWR).

Column	Datatype	Description
FUNCTION_ID	NUMBER	Function ID
FUNCTION_NAME	VARCHAR2(18)	Function name: • RMAN • DBWR • LGWR • ARCH • XDB • Streams AQ • Data Pump • Recovery • Buffer Cache Reads • Direct Reads • Direct Writes • Smart Scan • Archive Manager • Others
SMALL_READ_MEGABYTES	NUMBER	Number of megabytes read via single block read requests
SMALL_WRITE_MEGABYTES	NUMBER	Number of megabytes written via single block write requests
LARGE_READ_MEGABYTES	NUMBER	Number of megabytes read via multiblock read requests
LARGE_WRITE_MEGABYTES	NUMBER	Number of megabytes written via multiblock write requests
SMALL_READ_REQS	NUMBER	Number of single block read requests
SMALL_WRITE_REQS	NUMBER	Number of single block write requests
LARGE_READ_REQS	NUMBER	Number of multiblock read requests
LARGE_WRITE_REQS	NUMBER	Number of multiblock write requests
NUMBER_OF_WAITS	NUMBER	Number of synchronous I/O waits by functionality
WAIT_TIME	NUMBER	Total synchronous I/O wait time (in milliseconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.20 V$IOSTAT_FUNCTION_DETAIL

V$IOSTAT_FUNCTION_DETAIL displays disk I/O statistics for database functions (such as the LGWR and DBWR), broken down by file type.

Column	Datatype	Description
FUNCTION_ID	NUMBER	Function ID
FUNCTION_NAME	VARCHAR2(18)	Function name: • RMAN • DBWR • LGWR • ARCH • XDB • Streams AQ • Data Pump • Recovery • Buffer Cache Reads • Direct Reads • Direct Writes • Smart Scan • Archive Manager • Others
FILETYPE_ID	NUMBER	File type ID
FILETYPE_NAME	VARCHAR2(28)	File type name: • Control File • Data File • Log File • Archive Log • Temp File • Data File Backup • Data File Incremental Backup • Archive Log Backup • Data File Copy • Flashback Log • Data Pump Dump File • Other
SMALL_READ_MEGABYTES	NUMBER	Number of megabytes read via single block read requests
SMALL_WRITE_MEGABYTES	NUMBER	Number of megabytes written via single block write requests
LARGE_READ_MEGABYTES	NUMBER	Number of megabytes read via multiblock read requests

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LARGE_WRITE_MEGABYTES	NUMBER	Number of megabytes written via multiblock write requests
SMALL_READ_REQS	NUMBER	Number of single block read requests
SMALL_WRITE_REQS	NUMBER	Number of single block write requests
LARGE_READ_REQS	NUMBER	Number of multiblock read requests
LARGE_WRITE_REQS	NUMBER	Number of multiblock write requests
NUMBER_OF_WAITS	NUMBER	Number of synchronous I/O waits by functionality
WAIT_TIME	NUMBER	Total synchronous I/O wait time (in milliseconds)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.21 V$IOSTAT_NETWORK

V$IOSTAT_NETWORK displays information about network I/O statistics that were caused by accessing files on a remote database instance.

Column	Datatype	Description
CLIENT	VARCHAR2(32)	Database client name initiating the network I/O (for example, RMAN or PL/SQL)
READS#	NUMBER	Number of read operations issued
WRITES#	NUMBER	Number of write operations issued
KBYTES_READ	NUMBER	Total number of kilobytes read
KBYTES_WRITTEN	NUMBER	Total number of kilobytes written
READ_LATENCY	NUMBER	Total read wait time (in milliseconds)
WRITE_LATENCY	NUMBER	Total write wait time (in milliseconds)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.22 V$IP_ACL

V$IP_ACL provides information about access control to database services from network hosts.

Column	Datatype	Description
SERVICE_NAME	VARCHAR2(512)	The database service name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
HOST	VARCHAR2(64)	The host being granted access to the SERVICE_NAME. The host can be a hostname, dotted-decimal IPv4 or heximal IPv6 address. Wildcard "*" for IPv4 and Classless Inter-Domain Routing (CIDR) format is allowed.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs • 1: This value is used for rows containing data that pertain only to the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.23 V$JAVA_LIBRARY_CACHE_MEMORY

V$JAVA_LIBRARY_CACHE_MEMORY displays information about memory allocated to library cache memory objects in different namespaces for Java objects.

A memory object is an internal grouping of memory for efficient management. A library cache object may consist of one or more memory objects.

Column	Datatype	Description
LC_NAMESPACE	VARCHAR2(15)	Library cache namespace
LC_INUSE_MEMORY_OBJECTS	NUMBER	Number of library cache memory objects currently in use in the Java pool
LC_INUSE_MEMORY_SIZE	NUMBER	Total size of library cache in-use memory objects (in megabytes)
LC_FREEABLE_MEMORY_OBJECTS	NUMBER	Number of freeable library cache memory objects in the Java pool
LC_FREEABLE_MEMORY_SIZE	NUMBER	Size of library cache freeable memory objects (in megabytes)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.24 V$JAVA_POOL_ADVICE

V$JAVA_POOL_ADVICE displays information about estimated parse time in the Java pool for different pool sizes.

The sizes range from 10% of the current Java pool size or the amount of pinned Java library cache memory (whichever is higher) to 200% of the current Java pool size, in equal intervals. The value of the interval depends on the current size of the Java pool.

Parse time saved refers to the amount of time saved by keeping library cache memory objects in the Java pool, as opposed to having to reload these objects.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
JAVA_POOL_SIZE_FOR_ESTIMATE	NUMBER	Java pool size for the estimate (in megabytes)
JAVA_POOL_SIZE_FACTOR	NUMBER	Size factor with respect to the current Java pool size
ESTD_LC_SIZE	NUMBER	Estimated memory in use by the library cache (in megabytes)
ESTD_LC_MEMORY_OBJECTS	NUMBER	Estimated number of library cache memory objects in the Java pool of the specified size
ESTD_LC_TIME_SAVED	NUMBER	Estimated elapsed parse time saved (in seconds), owing to library cache memory objects being found in a Java pool of the specified size. This is the time that would have been spent in reloading the required objects in the Java pool had they been aged out due to insufficient amount of available free memory.
ESTD_LC_TIME_SAVED_FACTOR	NUMBER	Estimated parse time saved factor with respect to the current Java pool size
ESTD_LC_LOAD_TIME	NUMBER	Estimated elapsed time (in seconds) for parsing in a Java pool of the specified size
ESTD_LC_LOAD_TIME_FACTOR	NUMBER	Estimated load time factor with respect to the current Java pool size
ESTD_LC_MEMORY_OBJECT_HITS	NUMBER	Estimated number of times a library cache memory object was found in a Java pool of the specified size
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.25 V$KERNEL_IO_OUTLIER

V$KERNEL_IO_OUTLIER contains entries corresponding to I/Os that have taken a long time (more than 500 ms) to complete.

Use this view to see the individual kernel components of I/Os for which there are any occasional delays in serving disk I/O requests by the storage subsystem.

> **Note**
>
> Although this view exists on all platforms in Oracle Database 12*c*, it is only populated on the Solaris platform.

Column	Datatype	Description
TIMESTAMP	NUMBER	Number of seconds elapsed since 00:00 UTC, January 1, 1970
IO_SIZE	NUMBER	Size of the I/O, in KB.
IO_OFFSET	NUMBER	Offset into the device of the I/O
DEVICE_NAME	VARCHAR2(513)	Name of the device to which the I/O was targeted
PROCESS_NAME	VARCHAR2(64)	Name of the process that issued the I/O

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOTAL_LATENCY	NUMBER	Total time the I/O spent in the kernel (in milliseconds)
SETUP_LATENCY	NUMBER	Time spent during initial I/O setup before sending to SCSI target device driver (in milliseconds)
QUEUE_TO_HBA_LATENCY	NUMBER	Time spent in the SCSI target device driver before being sent to the Host Bus Adaptor (in milliseconds)
TRANSFER_LATENCY	NUMBER	Time spent in the Host Bus Adaptor and physically transferring the I/O to the storage device (in milliseconds)
CLEANUP_LATENCY	NUMBER	Time spent freeing resources used by the completed I/O (in milliseconds)
PID	NUMBER	Process ID that issued the I/O
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$IO_OUTLIER"
- "V$LGWRIO_OUTLIER"

8.26 V$KEY_VECTOR

V$KEY_VECTOR provides debugging information related to the data structures used by in-memory aggregation for current and recent queries using key vectors.

Column	Datatype	Description
SESSION_ID	NUMBER	Session ID
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
TRANSLATE_ID	NUMBER	Translation vector ID
SQL_ID	VARCHAR2(13)	SQL ID that uses the translation vector
SQL_EXEC_START	DATE	Time when the execution of the SQL started
SQL_EXEC_ID	NUMBER	SQL execution identifier
PROCESS	VARCHAR2(64)	Operating system client process ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATE	VARCHAR2(9)	State of the in-memory aggregation operation: • BYPASS Refers to key vectors that are currently in use by queries that have undergone the vector transform. • BYPASSED This is a historical entry for key vectors whose queries have already finished. • FINISHED This is a historical entry for key vectors whose queries have already finished. • RUNNING Refers to key vectors that are currently in use by queries that have undergone the vector transform. One query will probably have multiple key vectors if it has gone through the vector transform. Also, if the query is operating in parallel, there may be multiple entries per PQ worker. The PROCESS column can be joined to V$SESSION to differentiate these cases.
TYPE	VARCHAR2(9)	Type of translation vector created: • DOUBLEIND • PAGED • SIMPLE • INDIRECT • HASH • OFFSET
WIDTH	NUMBER	Width (number of bits) of DOUBLEIND, SIMPLE, INDIRECT, or OFFSETvector translation array: • 1 • 4 • 8 • 16 • 32
KEY_DATA_TYPE	VARCHAR2(13)	The internal join column key data type of either DOUBLEIND, SIMPLE, INDIRECT, or OFFSET translation vectors: • BINARY • BINARY_FLOAT • NUMBER • PACKED_BINARY • PACKED_DATE • PACKED_NUMBER Also, a value of NULL is displayed for HASH translation vectors with non-numeric data types and for HASH translation vectors with more than one join key column.
JOIN_COLUMN_COUNT	NUMBER	Number of join columns. PAGED and HASHED are the only types which can have greater than one join column.
JOIN_KEY_COUNT	NUMBER	Number of join key values used to build translation vector
DUP_JOIN_KEY_COUNT	NUMBER	Indicates the number of key values in the key vector that have more than one parent value
MIN_JOIN_KEY	NUMBER	Min join key value in translation vector

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MAX_JOIN_KEY	NUMBER	Max join key value in translation vector
GROUP_KEY_COUNT	NUMBER	Max dense grouping key value. This value is computed while the data that creates the translation vector is processed.
FILTERED	NUMBER	Number of rows filtered by translation vector
PROBED	NUMBER	Number of rows that probed the translation vector in key vector use row source
ACTIVE	NUMBER	Number of active translation vectors used as filters across all workers
DISABLED	NUMBER	Number of translation vector filters across all workers that were disabled
MEMORY_ALLOCATED	NUMBER	Amount of memory allocated for the key vector
JOIN_STRUCTURE_SIZE	NUMBER	Amount of memory used out of the allocated space
FACT_OWNER	VARCHAR2(128)	The owner of the fact table. Null when the fact table is null.
FACT_NAME	VARCHAR2(128)	The table that contains measure data. May be null if more than one fact table is used in the query.
DIMENSION_OWNER	VARCHAR2(128)	When a single table is used to construct the key vector, the table owner will appear in this column. If multiple tables are joined to serve as the dimension (more of a snowflake than a star schema shape, for one example), then this column will be null.
DIMENSION_NAME	VARCHAR2(128)	The table that contains attribute data and is joined to the fact table. May be null if two or more tables are used (for example, snowflake style dimension tables).
CREATION_DURATION	NUMBER	Records the total time in seconds that it took to create the key vector
PAYLOAD_COLUMN_COUNT	NUMBER	The number of columns being carried from the dimension table to the fact table scan for processing without joinback

See Also

Oracle Database SQL Tuning Guide for more information about in-memory aggregation

8.27 V$LATCH

V$LATCH displays aggregate latch statistics for both parent and child latches, grouped by latch name.

Individual parent and child latch statistics are broken down in the views V$LATCH_PARENT and V$LATCH_CHILDREN.

Column	Datatype	Description
ADDR	RAW(4 | 8)	Address of the latch object
LATCH#	NUMBER	Latch number
LEVEL#	NUMBER	Latch level
NAME	VARCHAR2(64)	Latch name
HASH	NUMBER	Latch hash
GETS	NUMBER	Number of times the latch was requested in willing-to-wait mode

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MISSES	NUMBER	Number of times the latch was requested in willing-to-wait mode and the requester had to wait
SLEEPS	NUMBER	Number of times a willing-to-wait latch request resulted in a session sleeping while waiting for the latch
IMMEDIATE_GETS	NUMBER	Number of times a latch was requested in no-wait mode
IMMEDIATE_MISSES	NUMBER	Number of times a no-wait latch request did not succeed (that is, missed)
WAITERS_WOKEN	NUMBER	This column has been deprecated and is present only for compatibility with previous releases of Oracle. No data is accumulated for this column; it will always have a value of zero.
WAITS_HOLDING_LATCH	NUMBER	This column has been deprecated and is present only for compatibility with previous releases of Oracle. No data is accumulated for this column; it will always have a value of zero.
SPIN_GETS	NUMBER	Willing-to-wait latch requests which missed the first try but succeeded while spinning
SLEEP[1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11]	NUMBER	These columns have been deprecated and are present only for compatibility with previous releases of Oracle. No data is accumulated for these columns; they will always have a value of zero. As a substitute for these columns you can query the appropriate rows of the V$EVENT_HISTOGRAM view where the EVENT column has a value of latch free or latch:%.
WAIT_TIME	NUMBER	Elapsed time spent waiting for the latch (in microseconds)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$LATCH_CHILDREN"
- "V$LATCH_PARENT"

8.28 V$LATCH_CHILDREN

V$LATCH_CHILDREN displays statistics about child latches.

This view includes all columns of V$LATCH plus the CHILD# column. Note that child latches have the same parent if their LATCH# columns match each other.

Column	Datatype	Description
ADDR	RAW(4 | 8)	Address of the latch object
LATCH#	NUMBER	Latch number of the parent latch

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CHILD#	NUMBER	Child latch number (unique only to each parent latch)
LEVEL#	NUMBER	Latch level
NAME	VARCHAR2(64)	Latch name
HASH	NUMBER	Latch hash
GETS	NUMBER	Number of times the latch was requested in willing-to-wait mode
MISSES	NUMBER	Number of times the latch was requested in willing-to-wait mode and the requester had to wait
SLEEPS	NUMBER	Number of times a willing-to-wait latch request resulted in a session sleeping while waiting for the latch
IMMEDIATE_GETS	NUMBER	Number of times a latch was requested in no-wait mode
IMMEDIATE_MISSES	NUMBER	Number of times a no-wait latch request did not succeed (that is, missed)
WAITERS_WOKEN	NUMBER	This column has been deprecated and is present only for compatibility with previous releases of Oracle. No data is accumulated for this column; it will always have a value of zero.
WAITS_HOLDING_LATCH	NUMBER	This column has been deprecated and is present only for compatibility with previous releases of Oracle. No data is accumulated for this column; it will always have a value of zero.
SPIN_GETS	NUMBER	Willing-to-wait latch requests which missed the first try but succeeded while spinning
SLEEP[1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11]	NUMBER	These columns have been deprecated and are present only for compatibility with previous releases of Oracle. No data is accumulated for these columns; they will always have a value of zero. As a substitute for these columns you can query the appropriate rows of the V$EVENT_HISTOGRAM view where the EVENT column has a value of latch free or latch:%.
WAIT_TIME	NUMBER	Elapsed time spent waiting for the latch (in microseconds)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$LATCH"

8.29 V$LATCH_MISSES

V$LATCH_MISSES displays statistics about missed attempts to acquire a latch.

Column	Datatype	Description
PARENT_NAME	VARCHAR2(64)	Latch name of a parent latch

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
WHERE	VARCHAR2(80)	This column is obsolete and maintained for backward compatibility. The value of this column is always equal to the value in LOCATION.
NWFAIL_COUNT	NUMBER	Number of times that no-wait acquisition of the latch failed
SLEEP_COUNT	NUMBER	Number of times that acquisition attempts caused sleeps
WTR_SLP_COUNT	NUMBER	Number of times a waiter slept
LONGHOLD_COUNT	NUMBER	Number of times someone held a latch for the entire duration of someone else's sleep
LOCATION	VARCHAR2(80)	Location that attempted to acquire the latch
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.30 V$LATCH_PARENT

V$LATCH_PARENT displays statistics about parent latches.

\ The columns for V$LATCH_PARENT are the same as those for V$LATCH.

See Also

"V$LATCH"

8.31 V$LATCHHOLDER

V$LATCHHOLDER displays information about the current latch holders.

Column	Datatype	Description
PID	NUMBER	Identifier of the process holding the latch
SID	NUMBER	Identifier of the session that owns the latch
LADDR	RAW(4 | 8)	Latch address
NAME	VARCHAR2(64)	Name of the latch being held
GETS	NUMBER	Number of times that the latch was obtained in either wait mode or no-wait mode

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.32 V$LATCHNAME

V$LATCHNAME displays information about decoded latch names for the latches shown in V$LATCH.

The rows of V$LATCHNAME have a one-to-one correspondence to the rows of V$LATCH.

Column	Datatype	Description
LATCH#	NUMBER	Latch number
NAME	VARCHAR2(64)	Latch name
DISPLAY_NAME	VARCHAR2(64)	A clearer and more descriptive name for the latch that appears in the NAME column. Names that appear in the DISPLAY_NAME column can change across Oracle Database releases, therefore customer scripts should not rely on names that appear in the DISPLAY_NAME column across releases.
HASH	NUMBER	Latch hash
TYPE	VARCHAR2(4)	Type of the latch (SGA, PDB, or OSP):
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$LATCH"

8.33 V$LGWRIO_OUTLIER

V$LGWRIO_OUTLIER contains entries corresponding to Log Writer (LGWR) process I/Os that have taken a long time (more than 500 ms) to complete. Use this view to see if there any occasional delays in serving disk I/O requests by the storage subsystem.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FUNCTION_NAME	VARCHAR2(18)	I/O function name of the delayed I/O
IO_SIZE	NUMBER	Size of the I/O in bytes
WAIT_EVENT	VARCHAR2(64)	Wait event name that was used to track the I/O
FILE_NAME	VARCHAR2(513)	Name of the file to which the I/O was targeted
IO_LATENCY	NUMBER	Time taken to complete the I/O (in milliseconds)
DISK1_NAME	VARCHAR2(255)	For Oracle ASM, the name of the first disk to which the I/O was issued
DISK1_LATENCY	NUMBER	Latency seen on the first disk (in milliseconds)
DISK2_NAME	VARCHAR2(255)	For Oracle ASM, the name of the second disk to which the I/O was issued
DISK2_LATENCY	NUMBER	Latency seen on the second disk (in milliseconds)
DISK3_NAME	VARCHAR2(255)	For Oracle ASM, the name of the third disk to which the I/O was issued
DISK3_LATENCY	NUMBER	Latency seen on the third disk (in milliseconds)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$IO_OUTLIER"
- "V$KERNEL_IO_OUTLIER"

8.34 V$LIBCACHE_LOCKS

V$LIBCACHE_LOCKS displays information about the library cache locks and pins. Locks and pins are distinguished based on the value of the TYPE column.

Column	Datatype	Description
TYPE	VARCHAR2(4)	LOCK or PIN
ADDR	RAW(4 | 8)	Address of the lock/pin
HOLDING_USER_SESSION	RAW(4 | 8)	User session holding this lock/pin
HOLDING_SESSION	RAW(4 | 8)	Session holding this lock/pin
OBJECT_HANDLE	RAW(4 | 8)	Handle address for which the lock/pin is acquired
LOCK_HELD	RAW(4 | 8)	If the type is LOCK, then LOCK_HELD represents the pin that is pinning the object. If the type is PIN, then LOCK_HELD represents the lock that is locking the object.
REFCOUNT	NUMBER	Reference count for this lock/pin

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MODE_HELD	NUMBER	Lock/pin mode held: • 0 - No lock/pin held • 1 - Null mode • 2 - Share mode • 3 - Exclusive mode
MODE_REQUESTED	NUMBER	Lock/pin mode requested: • 0 - No lock/pin requested • 1 - Null mode • 2 - Share mode • 3 - Exclusive mode
SAVEPOINT_NUMBER	NUMBER	Kernel transaction savepoint number at the time the lock/pin was acquired
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.35 V$LIBRARY_CACHE_MEMORY

V$LIBRARY_CACHE_MEMORY displays information about memory allocated to library cache memory objects in different namespaces.

A memory object is an internal grouping of memory for efficient management. A library cache object may consist of one or more memory objects.

Column	Datatype	Description
LC_NAMESPACE	VARCHAR2(15)	Library cache namespace
LC_INUSE_MEMORY_OBJECTS	NUMBER	Number of library cache memory objects currently in use in the shared pool
LC_INUSE_MEMORY_SIZE	NUMBER	Total size of library cache in-use memory objects (in megabytes)
LC_FREEABLE_MEMORY_OBJECTS	NUMBER	Number of freeable library cache memory objects in the shared pool
LC_FREEABLE_MEMORY_SIZE	NUMBER	Size of library cache freeable memory objects (in megabytes)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.36 V$LIBRARYCACHE

V$LIBRARYCACHE contains statistics about library cache performance and activity.

Column	Datatype	Description
NAMESPACE	VARCHAR2(64)	Library cache namespace
GETS	NUMBER	Number of times a lock was requested for objects of this namespace
GETHITS	NUMBER	Number of times an object's handle was found in memory
GETHITRATIO	NUMBER	Ratio of GETHITS to GETS
PINS	NUMBER	Number of times a PIN was requested for objects of this namespace
PINHITS	NUMBER	Number of times all of the metadata pieces of the library object were found in memory
PINHITRATIO	NUMBER	Ratio of PINHITS to PINS
RELOADS	NUMBER	Any PIN of an object that is not the first PIN performed since the object handle was created, and which requires loading the object from disk
INVALIDATIONS	NUMBER	Total number of times objects in this namespace were marked invalid because a dependent object was modified
DLM_LOCK_REQUESTS	NUMBER	Number of GET requests lock instance locks
DLM_PIN_REQUESTS	NUMBER	Number of PIN requests lock instance locks
DLM_PIN_RELEASES	NUMBER	Number of release requests PIN instance locks
DLM_INVALIDATION_REQUESTS	NUMBER	Number of GET requests for invalidation instance locks
DLM_INVALIDATIONS	NUMBER	Number of invalidation pings received from other instances
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.37 V$LICENSE

V$LICENSE displays information about license limits.

Column	Datatype	Description
SESSIONS_MAX	NUMBER	Maximum number of concurrent user sessions allowed for the instance
SESSIONS_WARNING	NUMBER	Warning limit for concurrent user sessions for the instance
SESSIONS_CURRENT	NUMBER	Current number of concurrent user sessions
SESSIONS_HIGHWATER	NUMBER	Highest number of concurrent user sessions since the instance started
USERS_MAX	NUMBER	Maximum number of named users allowed for the database
CPU_COUNT_CURRENT	NUMBER	Current number of logical CPUs or processors on the system
CPU_CORE_COUNT_CURRENT	NUMBER	Current number of CPU cores on the system (includes subcores of multicore CPUs, as well as single-core CPUs)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CPU_SOCKET_COUNT_CURRENT	NUMBER	Current number of CPU sockets on the system (represents an absolute count of CPU chips on the system, regardless of multithreading or multicore architectures)
CPU_COUNT_HIGHWATER	NUMBER	Highest number of logical CPUs or processors on the system since the instance started
CPU_CORE_COUNT_HIGHWATER	NUMBER	Highest number of CPU cores on the system since the instance started (includes subcores of multicore CPUs, as well as single-core CPUs)
CPU_SOCKET_COUNT_HIGHWATER	NUMBER	Highest number of CPU sockets on the system since the instance started (represents an absolute count of CPU chips on the system, regardless of multithreading or multicore architectures)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **Note**
>
> The availability of the CPU core count and CPU socket count statistics is subject to the operating system platform on which the Oracle Database is running. If a statistic is unavailable, the view will return NULL for the statistic value.

8.38 V$LOADISTAT

V$LOADISTAT contains errors that occurred when updating indexes on a table during a load using the Direct Path API.

Column	Datatype	Description
OWNER	VARCHAR2(31)	Schema name
TABNAME	VARCHAR2(31)	Table name
INDEXNAME	VARCHAR2(31)	Index name
SUBNAME	VARCHAR2(31)	Index sub name
MESSAGE_NUM	NUMBER	Error message number
MESSAGE	VARCHAR2(4000)	Error message
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.39 V$LOADPSTAT

V$LOADPSTAT contains statistics about the number of rows loaded into a partition, or subpartition, during a load using the Direct Path API.

Column	Datatype	Description
OWNER	VARCHAR2(31)	Schema name
TABNAME	VARCHAR2(31)	Table name
PARTNAME	VARCHAR2(31)	Partition name
LOADED	NUMBER	Number of rows loaded
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.40 V$LOCK

V$LOCK lists the locks currently held by the Oracle Database and outstanding requests for a lock or latch.

Column	Datatype	Description
ADDR	RAW(4 | 8)	Address of lock state object
KADDR	RAW(4 | 8)	Address of lock
SID	NUMBER	Identifier for session holding or acquiring the lock
TYPE	VARCHAR2(2)	Type of user or system lock The locks on the user types are obtained by user applications. Any process that is blocking others is likely to be holding one of these locks. The user type locks are: TM - DML enqueue TX - Transaction enqueue UL - User supplied The system type locks are listed in Table 8-1. Be aware that not all types of locks are documented. To find a complete list of locks for the current release, query the V$LOCK_TYPE data dictionary view, described on "V$LOCK_TYPE".
ID1	NUMBER	Lock identifier #1 (depends on type)
ID2	NUMBER	Lock identifier #2 (depends on type)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LMODE	NUMBER	Lock mode in which the session holds the lock: • 0 - none • 1 - null (NULL) • 2 - row-S (SS) • 3 - row-X (SX) • 4 - share (S) • 5 - S/Row-X (SSX) • 6 - exclusive (X)
REQUEST	NUMBER	Lock mode in which the process requests the lock: • 0 - none • 1 - null (NULL) • 2 - row-S (SS) • 3 - row-X (SX) • 4 - share (S) • 5 - S/Row-X (SSX) • 6 - exclusive (X)
CTIME	NUMBER	Time since current mode was granted
BLOCK	NUMBER	Indicates whether the lock in question is blocking other processes. The possible values are: • 0 - The lock is not blocking any other processes • 1 - The lock is blocking other processes • 2 - The lock is not blocking any blocked processes on the local node, but it may or may not be blocking processes on remote nodes. This value is used only in Oracle Real Application Clusters (Oracle RAC) configurations (not in single instance configurations).
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Table 8-1 Values for the TYPE Column: System Types

System Type	Description	System Type	Description
AE	Edition enqueue	MR	Media recovery
AT	Lock held for the ALTER TABLE statement	NA..NZ	Library cache pin instance (A..Z = namespace)
BL	Buffer hash table instance	PF	Password File
CF	Control file schema global enqueue	PI, PS	Parallel operation
CI	Cross-instance function invocation instance	PR	Process startup
CU	Cursor bind	QA..QZ	Row cache instance (A..Z = cache)
DF	datafile instance	RT	Redo thread global enqueue
DL	Direct loader parallel index create	SC	System change number instance

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 8-1 (Cont.) Values for the TYPE Column: System Types

System Type	Description	System Type	Description
DM	Mount/startup db primary/secondary instance	SM	SMON
DR	Distributed recovery process	SN	Sequence number instance
DX	Distributed transaction entry	SQ	Sequence number enqueue
FS	File set	SS	Sort segment
HW	Space management operations on a specific segment	ST	Space transaction enqueue
IN	Instance number	SV	Sequence number value
IR	Instance recovery serialization global enqueue	TA	Generic enqueue
IS	Instance state	TS	Temporary segment enqueue (ID2=0)
IV	Library cache invalidation instance	TS	New block allocation enqueue (ID2=1)
JQ	Job queue	TT	Temporary table enqueue
KK	Thread kick	UN	User name
LA .. LP	Library cache lock instance lock (A..P = namespace)	US	Undo segment DDL
MM	Mount definition global enqueue	WL	Being-written redo log instance

8.41 V$LOCK_ACTIVITY

V$LOCK_ACTIVITY is deprecated. The information that was provided in this view is now provided in the V$INSTANCE_CACHE_TRANSFER and V$SEGMENT_STATISTICS views.

Column	Datatype	Description
FROM_VAL	CHAR(4)	Global Cache Resource initial state; always NULL
TO_VAL	CHAR(1)	Global Cache Resource initial state; always S
ACTION_VAL	CHAR(21)	Description of the conversion; always Lock buffers for read
COUNTER	NUMBER	Number of times the lock operation executed
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$INSTANCE_CACHE_TRANSFER"
- "V$SEGMENT_STATISTICS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.42 V$LOCK_TYPE

V$LOCK_TYPE describes the type of locks available.

Column	Datatype	Description
TYPE	VARCHAR2(64)	A two-letter internal resource identifier
NAME	VARCHAR2(64)	Resource type name. This is a short (less than 32 characters) enqueue type name.
ID1_TAG	VARCHAR2(64)	Description of the enqueue type.
ID2_TAG	VARCHAR2(64)	Further description of the enqueue type.
IS_USER	VARCHAR2(3)	User enqueue. These are enqueues that are acquired as a direct result of a SQL statement. Applications may get deadlocks on these enqueues. Such deadlocks are considered application errors.
IS_RECYCLE	VARCHAR2(3)	Indicates whether the enqueue requires caching of the associated resource in the DLM's resource cache. Possible values: • NO: Means the enqueue requires caching of the associated resource in the DLM's resource cache • YES: Means the enqueue does not require caching of the associated resource in the DLM's resource cache
DESCRIPTION	VARCHAR2(4000)	Explanation of how or for what purpose the enqueue is used.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.43 V$LOCKDOWN_RULES

V$LOCKDOWN_RULES displays information about lockdown profile rules that are applicable in the pluggable database (PDB) where this view is queried.

Column	Datatype	Description
RULE_TYPE	VARCHAR2(128)	Type of the rule: • STATEMENT • FEATURE • OPTION
RULE	VARCHAR2(128)	Rule to be enabled or disabled
CLAUSE	VARCHAR2(128)	Clause of the statement
CLAUSE_OPTION	VARCHAR2(128)	Option of the clause
STATUS	VARCHAR2(7)	Status of the lockdown profile: • ENABLE • DISABLE • EMPTY

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
USERS	VARCHAR2(6)	Type of users affected by the rule: • ALL • LOCAL • COMMON
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs • 1: This value is used for rows containing data that pertain to only the root. • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Security Guide for more information about PDB lockdown profiles

8.44 V$LOCKED_OBJECT

V$LOCKED_OBJECT lists all locks acquired by every transaction on the system. It shows which sessions are holding DML locks (that is, TM-type enqueues) on what objects and in what mode.

Column	Datatype	Description
XIDUSN	NUMBER	Undo segment number
XIDSLOT	NUMBER	Slot number
XIDSQN	NUMBER	Sequence number
OBJECT_ID	NUMBER	Object ID being locked
SESSION_ID	NUMBER	Session ID
ORACLE_USERNAME	VARCHAR2(128)	Oracle user name
OS_USER_NAME	VARCHAR2(128)	Operating system user name
PROCESS	VARCHAR2(24)	Operating system process ID
LOCKED_MODE	NUMBER	Lock mode. The numeric values for this column map to these text values for the lock modes for table locks: • 0 - NONE: lock requested but not yet obtained • 1 - NULL • 2 - ROWS_S (SS): Row Share Lock • 3 - ROW_X (SX): Row Exclusive Table Lock • 4 - SHARE (S): Share Table Lock • 5 - S/ROW-X (SSX): Share Row Exclusive Table Lock • 6 - Exclusive (X): Exclusive Table Lock **See Also:** *Oracle Database Concepts* for more information about lock modes for table locks

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.45 V$LOG

V$LOG displays log file information from the control file.

Column	Datatype	Description
GROUP#	NUMBER	Log group number
THREAD#	NUMBER	Log thread number
SEQUENCE#	NUMBER	Log sequence number
BYTES	NUMBER	Size of the log (in bytes)
BLOCKSIZE	NUMBER	Block size of the logfile (512 or 4096)
MEMBERS	NUMBER	Number of members in the log group
ARCHIVED	VARCHAR2(3)	Archive status (YES) or (NO)
STATUS	VARCHAR2(16)	Log status: • UNUSED - Online redo log has never been written to. This is the state of a redo log that was just added, or just after a RESETLOGS, when it is not the current redo log. • CURRENT - Current redo log. This implies that the redo log is active. The redo log could be open or closed. • ACTIVE - Log is active but is not the current log. It is needed for crash recovery. It may be in use for block recovery. It may or may not be archived. • CLEARING - Log is being re-created as an empty log after an ALTER DATABASE CLEAR LOGFILE statement. After the log is cleared, the status changes to UNUSED. • CLEARING_CURRENT - Current log is being cleared of a closed thread. The log can stay in this status if there is some failure in the switch such as an I/O error writing the new log header. • INACTIVE - Log is no longer needed for instance recovery. It may be in use for media recovery. It may or may not be archived.
FIRST_CHANGE#	NUMBER	Lowest system change number (SCN) in the log
FIRST_TIME	DATE	Time of the first SCN in the log
NEXT_CHANGE#	NUMBER	Highest change number (SCN) in the log When STATUS=CURRENT, NEXT_CHANGE# is set to the highest possible SCN. Starting with Oracle Database 19c, Release Update 19.5, the highest possible SCN is 9295429630892703743. In prior versions, the highest possible SCN is 18446744073709551615.
NEXT_TIME	DATE	Time of the highest SCN in the log. When STATUS=CURRENT, NEXT_TIME is set to NULL.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.46 V$LOG_HISTORY

V$LOG_HISTORY displays log history information from the control file.

Column	Datatype	Description
RECID	NUMBER	Control file record ID
STAMP	NUMBER	Control file record stamp
THREAD#	NUMBER	Thread number of the archived log
SEQUENCE#	NUMBER	Sequence number of the archived log
FIRST_CHANGE#	NUMBER	Lowest system change number (SCN) in the log
FIRST_TIME	DATE	Time of the first entry (lowest SCN) in the log
NEXT_CHANGE#	NUMBER	Highest SCN in the log
RESETLOGS_CHANGE#	NUMBER	Resetlogs change number of the database when the log was written
RESETLOGS_TIME	DATE	Resetlogs time of the database when the log was written
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.47 V$LOGFILE

V$LOGFILE contains information about redo log files.

Column	Datatype	Description
GROUP#	NUMBER	Redo log group identifier number
STATUS	VARCHAR2(7)	Status of the log member: • INVALID - File is inaccessible • STALE - File's contents are incomplete • DELETED - File is no longer used • null - File is in use
TYPE	VARCHAR2(7)	Type of the logfile: • ONLINE • STANDBY

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MEMBER	VARCHAR2(513)	Redo log member name
IS_RECOVERY_DEST_FILE	VARCHAR2(3)	Indicates whether the file was created in the fast recovery area (YES) or not (NO)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.48 V$LOGHIST

V$LOGHIST contains log history information from the control file. This view is retained for historical compatibility. Oracle recommends that you use V$LOG_HISTORY instead.

See Also

"V$LOG_HISTORY"

Column	Datatype	Description
THREAD#	NUMBER	Log thread number
SEQUENCE#	NUMBER	Log sequence number
FIRST_CHANGE#	NUMBER	Lowest SCN in the log
FIRST_TIME	DATE	Time of first SCN in the log
SWITCH_CHANGE#	NUMBER	SCN at which the log switch occurred; one more than highest SCN in the log
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.49 V$LOGMNR_CONTENTS

V$LOGMNR_CONTENTS contains log history information. To query this view, you must have the LOGMINING privilege.

When a SELECT statement is executed against the V$LOGMNR_CONTENTS view, the archive redo log files are read sequentially. Translated records from the redo log files are returned as rows in the V$LOGMNR_CONTENTS view. This continues until either the filter criteria specified at startup (EndTime or endScn) are met or the end of the archive log file is reached.

Copyright © 2002, 2026, Oracle and/or its affiliates.

When this view is queried from a PDB, it returns only redo generated by that PDB.

Column	Datatype	Description
SCN	NUMBER	System change number (SCN) when the database change was made
START_SCN	NUMBER	System change number (SCN) when the transaction that contains this change started; only meaningful if the COMMITTED_DATA_ONLY option was chosen in a DBMS_LOGMNR.START_LOGMNR() invocation, NULL otherwise. This column may also be NULL if the query is run over a time/SCN range that does not contain the start of the transaction.
COMMIT_SCN	NUMBER	System change number (SCN) when the transaction committed; only meaningful if the COMMITTED_DATA_ONLY option was chosen in a DBMS_LOGMNR.START_LOGMNR() invocation
TIMESTAMP	DATE	Timestamp when the database change was made
START_TIMESTAMP	DATE	Timestamp when the transaction that contains this change started; only meaningful if the COMMITTED_DATA_ONLY option was chosen in a DBMS_LOGMNR.START_LOGMNR() invocation, NULL otherwise. This column may also be NULL if the query is run over a time/SCN range that does not contain the start of the transaction.
COMMIT_TIMESTAMP	DATE	Timestamp when the transaction committed; only meaningful if the COMMITTED_DATA_ONLY option was chosen in a DBMS_LOGMNR.START_LOGMNR() invocation
XIDUSN	NUMBER	Transaction ID undo segment number of the transaction that generated the change
XIDSLT	NUMBER	Transaction ID slot number of the transaction that generated the change
XIDSQN	NUMBER	Transaction ID sequence number of the transaction that generated the change
XID	RAW(8)	Raw representation of the transaction identifier
PXIDUSN	NUMBER	Parent transaction ID undo segment number of a parallel transaction
PXIDSLT	NUMBER	Parent transaction ID slot number of a parallel transaction
PXIDSQN	NUMBER	Parent transaction ID sequence number of a parallel transaction
PXID	RAW(8)	Raw representation of the parent transaction identifier
TX_NAME	VARCHAR2(256)	Name of the transaction that made the change; only meaningful if the transaction is a named transaction

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OPERATION	VARCHAR2(32)	User level SQL operation that made the change: • INTERNAL - Change was caused by internal operations initiated by the database • INSERT - Change was caused by an insert statement • DELETE - Change was caused by a delete statement • UPDATE - Change was caused by an update statement • DDL - Change was caused by a DDL statement • START - Change was caused by the start of a transaction • COMMIT - Change was caused by the commit of a transaction • SEL_LOB_LOCATOR - Operation was a SELECT statement that returned a LOB locator • LOB_WRITE - Change was caused by an invocation of DBMS_LOB.WRITE • LOB_TRIM - Change was caused by an invocation of DBMS_LOB.TRIM • LOB_ERASE - Change was caused by an invocation of DBMS_LOB.ERASE • MISSING_SCN - LogMiner encountered a gap in the redo records. This is most likely because not all redo logs were registered with LogMiner. • ROLLBACK - Change was caused by a full rollback of a transaction • XML DOC BEGIN - Beginning of a change to an XMLType column or table • XML DOC WRITE - Data for an XML document • XML DOC END - End of the Data for an XML document • UNSUPPORTED - Change was caused by operations not currently supported by LogMiner (for example, changes made to nested tables) The OPERATION and OPERATION_CODE columns in this view are available for top-level user operations, for example, DML and DDL. Values that are not documented for these columns are internal to LogMiner or the RDBMS and do not reflect user operations.
OPERATION_CODE	NUMBER	Number of the operation code: • 0 - INTERNAL • 1 - INSERT • 2 - DELETE • 3 - UPDATE • 5 - DDL • 6 - START • 7 - COMMIT • 9 - SELECT_LOB_LOCATOR • 10 - LOB_WRITE • 11 - LOB_TRIM • 29 - LOB_ERASE • 34 - MISSING_SCN • 36 - ROLLBACK • 68 - XML DOC BEGIN • 70 = XML DOC WRITE • 71 = XML DOC END • 255 - UNSUPPORTED

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ROLLBACK	NUMBER	1 = if the redo record was generated because of a partial or a full rollback of the associated transaction 0 = otherwise
SEG_OWNER	VARCHAR2(386)	Owner of the modified data segment
SEG_NAME	VARCHAR2(256)	Name of the modified data segment
TABLE_NAME	VARCHAR2(386)	Name of the modified table (in case the redo pertains to a table modification)
SEG_TYPE	NUMBER	Type of the modified data segment: • 0 - UNKNOWN • 1 - INDEX • 2 - TABLE • 19 - TABLE PARTITION • 20 - INDEX PARTITION • 34 - TABLE SUBPARTITION • All other values - UNSUPPORTED
SEG_TYPE_NAME	VARCHAR2(32)	Segment type name: • INDEX • INDEX PARTITION • TABLE • TABLE PARTITION • TABLE SUBPARTITION • UNKNOWN • UNSUPPORTED
TABLE_SPACE	VARCHAR2(92)	Name of the tablespace containing the modified data segment. This column is not populated for rows where the value of the OPERATION column is DDL. This is because DDL may operate on more than one tablespace.
ROW_ID	VARCHAR2(18)	Row ID of the row modified by the change (only meaningful if the change pertains to a DML). This will be NULL if the redo record is not associated with a DML.
USERNAME	VARCHAR2(384)	Name of the user who executed the transaction
OS_USERNAME	VARCHAR2(4000)	Name of the operating system user
MACHINE_NAME	VARCHAR2(4000)	Machine from which the user connected to the database
AUDIT_SESSIONID	NUMBER	Audit session ID associated with the user session making the change
SESSION#	NUMBER	Session number of the session that made the change
SERIAL#	NUMBER	Serial number of the session that made the change
SESSION_INFO	VARCHAR2(4000)	Information about the database session that executed the transaction. Contains process information, machine name from which the user logged in, and so on. A possible SESSION_INFO column may contain the following: • login_username = HR • client_info = • OS_username = jkundu • Machine_name = nirvan • OS_terminal = pts/31 • OS_program_name = sqlplus@nirvan (TNS V1-V3)
THREAD#	NUMBER	Number of the thread that made the change to the database

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SEQUENCE#	NUMBER	Sequence number of the SQL statement within the transaction. If you are mining without the COMMITED_DATA_ONLY option set, then this value is 1.
RBASQN	NUMBER	Sequence# associated with the Redo Block Address (RBA) of the redo record associated with the change
RBABLK	NUMBER	RBA block number within the log file
RBABYTE	NUMBER	RBA byte offset within the block
UBAFIL	NUMBER	Undo Block Address (UBA) file number identifying the file containing the undo block
UBABLK	NUMBER	UBA block number for the undo block
UBAREC	NUMBER	UBA record index within the undo block
UBASQN	NUMBER	UBA undo block sequence number
ABS_FILE#	NUMBER	Data block absolute file number of the block changed by the transaction
REL_FILE#	NUMBER	Data block relative file number. The file number is relative to the tablespace of the object.
DATA_BLK#	NUMBER	Data block number within the file
DATA_OBJ#	NUMBER	Data block object number identifying the object
DATA_OBJV#	NUMBER	Version number of the table being modified
DATA_OBJD#	NUMBER	Data block data object number identifying the object within the tablespace
SQL_REDO	VARCHAR2(4000)	Reconstructed SQL statement that is equivalent to the original SQL statement that made the change. Refer to *Oracle Database Utilities* before executing SQL_REDO to your database. LogMiner does not generate SQL redo for temporary tables. In such a case, this column will contain the string "/* No SQL_REDO for temporary tables */".
SQL_UNDO	VARCHAR2(4000)	Reconstructed SQL statement that can be used to undo the effect of the original statement that made the change. DDL statements have no corresponding SQL_UNDO. Refer to *Oracle Database Utilities* before executing SQL_UNDO to your database. LogMiner does not generate SQL undo for temporary tables. In such a case, this column will contain the string "/* No SQL_UNDO for temporary tables */".
RS_ID	VARCHAR2(32)	Record set ID. The tuple (RS_ID, SSN) together uniquely identifies a logical row change. This will usually mean one row from V$LOGMNR_CONTENTS, but could be more than one row if a single SQL statement for either the Redo or Undo would be too large to fit within the respective columns SQL_UNDO or SQL_REDO. RS_ID uniquely identifies the redo record that generated the row.
SSN	NUMBER	SQL sequence number. Used in conjunction with RS_ID, this uniquely identifies a logical row change, shown as one or more rows from the V$LOGMNR_CONTENTS view.
CSF	NUMBER	Continuation SQL flag. Possible values are: • 0 - Indicates SQL_REDO and SQL_UNDO is contained within the same row • 1 - Indicates that either SQL_REDO or SQL_UNDO is greater than 4000 bytes in size and is continued in the next row returned by the view

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
INFO	VARCHAR2(64)	Informational message about the row. For instance, the string "USER DDL" indicates that the DDL statement returned in the SQL_REDO column was the top-level DDL executed by the user and the string "INTERNAL DDL" indicates that the DDL statement returned in the SQL_REDO column was executed internally by the RDBMS.
STATUS	NUMBER	A value of 0 indicates that the reconstructed SQL statements as shown in the SQL_REDO and SQL_UNDO columns are valid executable SQL statements. Otherwise, the reconstructed SQL statements are not executable. This may be because no data dictionary was provided to LogMiner for the analysis, or that the data dictionary provided did not have the definition of the object being mined. A value of 5 indicates that this row is part of a change to an XMLType column or table and the XML document must be assembled before being applied.
REDO_VALUE	NUMBER	Used as input to the DBMS_LOGMNR.MINE_VALUE() and DBMS_LOGMNR.COLUMN_PRESENT() functions
UNDO_VALUE	NUMBER	Used as input to the DBMS_LOGMNR.MINE_VALUE() and DBMS_LOGMNR.COLUMN_PRESENT() functions
SAFE_RESUME_SCN	NUMBER	Reserved for future use
CSCN	NUMBER	This column is deprecated in favor of the COMMIT_SCN column
OBJECT_ID	RAW(16)	Object identifier for DMLs to XMLType tables. For changes to non-typed tables, this column is NULL.
EDITION_NAME	VARCHAR2(384)	Identifies the edition in which a DDL statement was executed
CLIENT_ID	VARCHAR2(64)	Client identifier in the session that performed the operation, if available.
SRC_CON_NAME	VARCHAR2(384)	Contains the pluggable database (PDB) name. This information will only be available when mining with a current LogMiner dictionary.
SRC_CON_ID	NUMBER	Contains the PDB ID (the PDB_ID column from the DBA_PDBS view). This information will be available only with a current LogMiner dictionary.
SRC_CON_UID	NUMBER	Contains the PDB UID (the CON_UID column from the DBA_PDBS view). This information will be available with or without a current LogMiner dictionary.
SRC_CON_DBID	NUMBER	Contains the PDB identifier (the DBID column from the DBA_PDBS view). This information will only be available when mining with a current LogMiner dictionary.
SRC_CON_GUID	RAW(16)	Contains the GUID associated with the PDB (the GUID column from the DBA_PDBS view). This information will only be available when mining with a current LogMiner dictionary.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_LOGMNR package
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_LOB package

8.50 V$LOGMNR_DICTIONARY

V$LOGMNR_DICTIONARY contains log history information.

Column	Datatype	Description
DB_NAME	VARCHAR2(9)	Name of the database
DB_ID	NUMBER	Database ID
DB_CREATED	DATE	Creation date of the source database (corresponds to the CREATED column in the V$DATABASE view)
TIMESTAMP	DATE	Date when the dictionary was created
RESET_SCN	NUMBER	Reset log SCN when the dictionary was created
RESET_SCN_TIME	DATE	Timestamp of the reset log SCN when the dictionary was created
DB_VERSION_TIME	DATE	Version time for the source database (corresponds to the VERSION_TIME column in the V$DATABASE view)
DB_CHARACTER_SET	VARCHAR2(30)	Character set of the source database
DB_VERSION	VARCHAR2(64)	This column is deprecated.
DB_STATUS	VARCHAR2(64)	This column is deprecated.
DICTIONARY_SCN	NUMBER	Database checkpoint SCN at which the dictionary was created
ENABLED_THREAD_MAP	RAW(16)	This column is deprecated.
DB_TXN_SCN	NUMBER	SCN at which the dictionary was created
FILENAME	VARCHAR2(512)	Dictionary file name
INFO	VARCHAR2(32)	Informational/Status message BAD_DATE indicates that the SCN of the dictionary file does not match the SCN range of the log files
STATUS	NUMBER	A NULL indicates a valid dictionary file for the list of log files. A non-NULL value indicates further information is contained in the INFO column as a text string.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.51 V$LOGMNR_DICTIONARY_LOAD

V$LOGMNR_DICTIONARY_LOAD displays information about LogMiner dictionaries for all active LogMiner sessions on the system.

Each query of this view will return one row for each attached LogMiner session. This view will not show valid information for LogMiner adhoc query clients.

Column	Datatype	Description
SESSION_ID	NUMBER	LogMiner session ID
LOGMNR_UID	NUMBER	LogMiner dictionary UID
ACTION#	NUMBER	Reserved for internal use
OPCODE	NUMBER	Reserved for internal use
COMMAND	VARCHAR2(161)	Current command being executed
CURRENT_STATE	VARCHAR2(32)	Summary information if LOADED=ACTIVE
COMPLETED_ACTIONS	NUMBER	The steps completed so far
TOTAL_ACTIONS	NUMBER	Total steps to complete
LOADED	VARCHAR2(7)	Status of the dictionary load: • YES - dictionary has been loaded • NO - dictionary has not been loaded • ACTIVE - dictionary is in the process of being loaded
PERCENT_DONE	NUMBER	Percentage of the dictionary that has been loaded
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.52 V$LOGMNR_LATCH

V$LOGMNR_LATCH can be joined with the V$LATCH and the V$LATCH_CHILDREN views to obtain statistics about different latches used by active LogMiner persistent sessions.

A persistent LogMiner session is created either by starting Data Guard SQL Apply on a logical standby database for the first time or by creating a Replication capture.

Column	Datatype	Description
SESSION_ID	NUMBER	Unique identifier of the LogMiner persistent session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NAME	VARCHAR2(32)	Name of the Latch: • LogMiner internal state - Identifies the latch that protects communications between the LogMiner READER, PREPARER, and BUILDER processes • LogMiner memory allocation - Identifies the latch that protects all memory allocation and deallocation inside LogMiner • LogMiner transaction list - Identifies the latch that protects interaction between the LogMiner layer and its clients during Data Guard SQL Apply on a logical standby database
CHILD_ADDR	RAW(4 | 8)	Address of the child latch object. This column matches the corresponding ADDR column in the V$LATCH_CHLDREN view
STATE	VARCHAR2(6)	State of the Latch: • UNINIT - The latch structure is uninitialized; that is, it is not currently assigned to any LogMiner persistent session • READY - The latch structure is being used by a persistent LogMiner session
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$LATCH"
- "V$LATCH_CHILDREN"

8.53 V$LOGMNR_LOGS

V$LOGMNR_LOGS contains log information.

Column	Datatype	Description
LOG_ID	NUMBER	This column is deprecated.
FILENAME	VARCHAR2(512)	Name of the log file
LOW_TIME	DATE	Oldest date of any records in the file
HIGH_TIME	DATE	Most recent date of any records in the file
DB_ID	NUMBER	Database ID
DB_NAME	VARCHAR2(8)	Name of the database
RESET_SCN	NUMBER	Resetlogs SCN of the database incarnation that generated the log file
RESET_SCN_TIME	DATE	Resetlogs timestamp of the database incarnation that generated the log file
COMPATIBLE	VARCHAR2(17)	The setting of the database COMPATIBLE initialization parameter at the time the log file was generated

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
THREAD_ID	NUMBER	Thread number
THREAD_SQN	NUMBER	Thread sequence number
LOW_SCN	NUMBER	SCN allocated when log switched into
NEXT_SCN	NUMBER	SCN after this log. Low SCN of the next log.
DICTIONARY_BEGIN	VARCHAR2(3)	Indicates whether dictionary dumped to redo logs starts in this redo log (YES) or not (NO)
DICTIONARY_END	VARCHAR2(3)	Indicates whether dictionary dumped to redo logs ends in this redo log (YES) or not (NO)
TYPE	VARCHAR2(7)	Redo log file type: • ARCHIVED • ONLINE
BLOCKSIZE	NUMBER	Database block size
FILESIZE	NUMBER	Size of the redo file (in bytes)
INFO	VARCHAR2(32)	Informational message. A value of MISSING_LOGFILE will be assigned to a row entry where a needed log file is missing from the list of log files.
STATUS	NUMBER	Status of the redo log file: • 0 - Will be read • 1 - First to be read • 2 - Not needed • 4 - Missing log file
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.54 V$LOGMNR_PARAMETERS

V$LOGMNR_PARAMETERS contains log information.

Column	Datatype	Description
START_DATE	DATE	Date to start the search
REQUIRED_START_DATE	DATE	Required date to start the search if DDL tracking is enabled
END_DATE	DATE	Date to end the search
START_SCN	NUMBER	System change number to start the search
REQUIRED_START_SCN	NUMBER	Required system change number to start the search if DDL tracking is enabled
END_SCN	NUMBER	System change number to end the search
OPTIONS	NUMBER	Options specified for the current LogMiner session
INFO	VARCHAR2(32)	This column is always null.
STATUS	NUMBER	This column is always 0.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.55 V$LOGMNR_PROCESS

V$LOGMNR_PROCESS identifies all processes attached to an active LogMiner persistent session.

(A persistent LogMiner session is created either by starting Data Guard SQL Apply on a logical standby database for the first time or by creating Replication capture.) This view can be joined with either the V$SESSION view or the V$PROCESS view to gather process-specific information.

Column	Datatype	Description
SESSION_ID	NUMBER	Unique identifier for the LogMiner persistent session
PID	NUMBER	Oracle process identifier for the SQL Apply or Replication capture process (same as the V$PROCESS.PID)
SPID	VARCHAR2(24)	Operating system process identifier (same as the V$PROCESS.SPID)
ROLE	VARCHAR2(32)	Identifies the role of the active LogMiner process: READER, PREPARER, BUILDER, COORDINATOR, or APPLY SERVER
USERNAME	VARCHAR2(15)	Operating system process user name that is connected to the database
SID	NUMBER	Session identifier for the V$SESSION.SID process
SERIAL#	NUMBER	Session serial number associated with the V$SESSION.SERIAL process
LATCHWAIT	VARCHAR2(16)	Address of the latch the process is waiting for; NULL if none
LATCHSPIN	VARCHAR2(16)	This column is obsolete
WORK_MICROSEC	VARCHAR2(21)	Microseconds spent by the process doing useful work
OVERHEAD_MICROSEC	VARCHAR2(21)	Microseconds spent by the process doing overhead tasks or simply waiting/idling
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SESSION"
- "V$PROCESS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.56 V$LOGMNR_SESSION

V$LOGMNR_SESSION displays information about active LogMiner persistent sessions.

(A LogMiner persistent session is created either by starting Data Guard SQL Apply on a logical standby database for the first time or by creating Replication capture.) Transient LogMiner sessions (those created as a result of querying the V$LOGMNR_CONTENTS view) do not show up in the V$LOGMNR_SESSION view. The statistics shown in this view correspond to snapshots of the system and are not cumulative in nature.

Column	Datatype	Description
SESSION_ID	NUMBER	Unique identifier for the LogMiner persistent session
SESSION_NAME	VARCHAR2(32)	Unique session name
SESSION_STATE	VARCHAR2(9)	Current state of the LogMiner persistent session: • READY - Client has created the LogMiner persistent session and added the relevant archived redo log files, but has not loaded the initial LogMiner dictionary • STARTED - The LogMiner dictionary has been loaded • ACTIVE - The LogMiner persistent session is mining the redo stream • DETACHED - The LogMiner persistent session is not currently active, and it is in the process of becoming inactive • DISCARDED - Client is about to destroy the LogMiner persistent session
DB_NAME	VARCHAR2(128)	Global database name for the source database
DB_ID	NUMBER	Database identifier of the source database
RESET_SCN	NUMBER	System change number (SCN) when the session started
RESET_TIMESTAMP	NUMBER	Time of the RESETLOGs when the LogMiner persistent session started
RESET_TIME	DATE	The time when the session started
NUM_PROCESS	NUMBER	Number of processes allocated to this session
CHUNK_SIZE	NUMBER	Amount of memory allocated for this chunk
START_SCN	NUMBER	System change number (SCN) at start of the session
END_SCN	NUMBER	System change number (SCN) at end of the session
SPILL_SCN	NUMBER	In the event of a restart, redo records with an SCN lower than this will not be read from the archived redo log files
PROCESSED_SCN	NUMBER	The BUILDER process has successfully mined redo records up to this SCN
PROCESSED_TIME	DATE	The BUILDER process has successfully mined redo records up to this time
PREPARED_SCN	NUMBER	The PREPARER processes have successfully transformed all redo records below this SCN into logical change records (LCRs). However the LCRs may not have been grouped into transactions or merged in case they pertain to chained rows or LOB updates.
READ_SCN	NUMBER	The READER process has read all redo records below this SCN
LOW_MARK_SCN	NUMBER	LogMiner has delivered all transactions that committed below this SCN to the client
CONSUMED_SCN	NUMBER	Client has consumed and released all transactions that committed below this SCN.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MAX_MEMORY_SIZE	NUMBER	Maximum amount of shared memory (in bytes) that LogMiner is allowed to consume
USED_MEMORY_SIZE	NUMBER	Amount of shared memory (in bytes) actually consumed by LogMiner
BUILDER_WORK_SIZE	NUMBER	Amount of redo (in bytes) in the current work unit being processed by the BUILDER process.
PREPARED_WORK_SIZE	NUMBER	Amount of redo (in bytes) that has been prepared by LogMiner
AVAILABLE_WORK_SIZE	NUMBER	Amount of redo records (in bytes) that are ready, but are yet to be consumed by the client
AVAILABLE_TXN	NUMBER	Number of transaction chunks ready for consumption
AVAILABLE_COMMITTED_TXN	NUMBER	Number of committed transactions ready for consumption. This is less than, or equal to, AVAILABLE_TXN.
DELIVERED_TXN	NUMBER	Number of transaction chunks that the client currently has in its possession
DELIVERED_COMMITTED_TXN	NUMBER	Number of committed transaction chunks that the client is currently working on. This is less than, or equal to, the value of the DELIVERED_TXN column.
PINNED_TXN	NUMBER	Number of transactions pinned (the client is actively working on)
PINNED_COMMITTED_TXN	NUMBER	Number of committed transactions pinned (the client is actively working on)
CHECKPOINT_INTERVAL	NUMBER	Checkpoint interval
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$LOGMNR_CONTENTS"

8.57 V$LOGMNR_STATS

V$LOGMNR_STATS displays the activity currently being performed by the active LogMiner persistent sessions.

Column	Datatype	Description
SESSION_ID	NUMBER	Unique identifier for the LogMiner persistent sessions

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NAME	VARCHAR2(64)	Name of the LogMiner statistic, state, or status, including: • Bytes of Redo Processed - Cumulative count of bytes processed by SQL Apply • Redo Records Processed - Count of redo records processed by SQL Apply • Txns Delivered to Client - Count of SQL transactions processed by SQL Apply • DML txns delivered - Count of DML transactions processed by SQL Apply • DDL txns delivered - Count of DDL transactions processed by SQL Apply • CTAS txns delivered - Count of `CREATE TABLE AS SELECT` (CTAS) transactions processed by SQL Apply • Recursive txns delivered - Count of recursive transactions processed by SQL Apply • Rolled back txns seen • LCRs delivered to client - Number of logical change records (LCRs) processed by SQL Apply • Bytes paged out - Cumulative count of bytes that have been paged out. LogMiner pages out memory from the LCR cache to accommodate certain ill-behaved workloads or under-configured systems. The ratio of bytes paged out to bytes of redo processed should be low. If this ratio is high (10% or higher), try increasing the `MAX_SGA` allocated to SQL Apply. • Microsecs spent in pageout - Time spent by LogMiner paging out memory from the LCR cache • Bytes checkpointed - Keeps track of the amount of bytes checkpointed. The mining engine takes periodic checkpoints, whereby it writes out logical change records (LCRs) pertaining to long-running transactions. The ratio of `Bytes Checkpointed` to `Bytes of Redo Processed` should be low. A high ratio (10% or higher) indicates an ill-behaved workload. • Microsecs spent in checkpoint - Time spent by the mining engine taking checkpoints, whereby it writes out logical change records (LCRs) pertaining to long-running transactions. • Bytes rolled back - Cumulative value of the number of bytes rolled back by LogMiner. There are times that LogMiner needs to backtrack and reprocess a section of the redo stream. In this case, it will roll back work it has already done. The ratio of `Bytes Rolled Back` to `Bytes of Redo Processed` should be low. If this ratio is high (10% or higher), reduce the number of `PREPARER` processes allocated to SQL Apply. • Microsecs spent in rollback - Time spent rolling back transactions already applied to the logical standby database
VALUE	VARCHAR2(64)	The corresponding metric value
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.58 V$LOGSTDBY

V$LOGSTDBY is deprecated.

Column	Datatype	Description
SERIAL#	NUMBER	SQL Session serial number. This data is used when joining this view with the V$SESSION and V$PX_SESSION views.
LOGSTDBY_ID	NUMBER	Parallel query worker ID
PID	VARCHAR2(24)	Process ID of the SQL apply process
TYPE	VARCHAR2(128)	Indicates the task being performed by the process (COORDINATOR, APPLIER, ANALYZER, READER, PREPARER, or BUILDER)
STATUS_CODE	NUMBER	Status number (or Oracle error code) belonging to the STATUS message
STATUS	VARCHAR2(256)	Description of the current activity of the process
HIGH_SCN	NUMBER	Highest system change number (SCN) seen by the process. This column is used to confirm the progress of the individual process.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.59 V$LOGSTDBY_PROCESS

V$LOGSTDBY_PROCESS displays dynamic information about what is happening to the Data Guard log apply services.

This view is helpful when diagnosing performance problems during the logical application of archived redo logs to the standby database, and it can be helpful for other problems. This view is for logical standby databases only.

Column	Datatype	Description
SID	NUMBER	Session id of the associated session. This matches the SID column of the corresponding row in the V$SESSION view.
SERIAL#	NUMBER	Serial number of the associated session. Together, (SID,SERIAL#) uniquely identify the session in the current database instance.
LOGSTDBY_ID	NUMBER	Parallel query worker ID
SPID	VARCHAR2(24)	This corresponds to the SPID value of the row corresponding to this process in the V$PROCESS view

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TYPE	VARCHAR2(128)	Role that the process plays in the context of SQL Apply: • COORDINATOR • APPLIER • ANALYZER • READER • PREPARER • BUILDER
STATUS_CODE	NUMBER	Operation code identifying the current action of the process: • 16111 - SQL Apply process is initializing • 16112 - SQL Apply process is cleaning up as apply and mining processes are stopping based on a user command • 16116 - SQL Apply process is idle • 16117 - SQL Apply process is busy and is not waiting on any interesting event • 16110 - APPLIER process has invoked a user-provided stored procedure in order to inspect a DDL statement prior to it being processed • 16113 - APPLIER process is applying DML changes to some user object or to a sequence • 16114 - APPLIER process is applying a DDL change • 16115 - COORDINATOR process is loading the LogMiner dictionary from the redo stream • 16243 - BUILDER process is paging out memory to free up space in lcr cache • 16240 - READER process idle waiting for additional logfile to be available • 16241 - READER process is idle waiting for the logfile to fill the log sequence gap • 16242 - READER process is processing a logfile
STATUS	VARCHAR2(256)	Description of the current action of the process
HIGH_SCN	NUMBER	Identifies the highest redo record/LCR processed by this process
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.60 V$LOGSTDBY_PROGRESS

V$LOGSTDBY_PROGRESS displays the progress of log apply services on the logical standby database. This view is for logical standby databases only.

Column	Datatype	Description
APPLIED_SCN	NUMBER	All the transactions with COMMIT SCN lower than or equal to this SCN have been applied
APPLIED_TIME	DATE	The time and date of APPLIED_SCN

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RESTART_SCN	NUMBER	During an apply restart, LogMiner does not read any log file with a NEXT_CHANGE# lower than this SCN.
RESTART_TIME	DATE	The time and date of RESTART_SCN
LATEST_SCN	NUMBER	The highest SCN of all redo records that Logical Standby has encountered
LATEST_TIME	DATE	The time and date of LATEST_SCN
MINING_SCN	NUMBER	The SCN of the latest redo record processed by the builder process
MINING_TIME	DATE	The time and date of MINING_SCN
RESETLOGS_ID	NUMBER	A redo branch is identified by resetlogs SCN and resetlogs timestamp. The RESETLOGS_ID column contents are the same as resetlogs timestamp converted to a number.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.61 V$LOGSTDBY_STATE

V$LOGSTDBY_STATE provides consolidated information from V$LOGSTDBY and V$LOGSTDBY_STATS about the running state of Logical Standby.

Column	Datatype	Description
PRIMARY_DBID	NUMBER	Database ID (DBID) of the primary database
PRIMARY_CON_DBID	NUMBER	This column indicates the DBID of the source database or source PDB corresponding to the database (or CDB) from which this column is queried.For a non-CDB and the root of a CDB, this column matches the PRIMARY_DBID column.For a maintained PDB, this column indicates the DBID for the corresponding PDB at the source.For a skipped PDB or a local PDB, this column would be NULL.
SESSION_ID	NUMBER	LogMiner session ID allocated to SQL Apply.
REALTIME_APPLY	VARCHAR2(64)	Y indicates that SQL Apply is running in real-time apply mode. If a standby redo log is configured, SQL Apply applies changes as they are written to the standby redo log files. N indicates that SQL Apply applies changes as each archived redo log file is received.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATE	VARCHAR2(64)	• INITIALIZING: LogMiner session has been created and coordinator has attached to it • LOADING DICTIONARY: SQL Apply is loading the LogMiner dictionary • WAITING ON GAP: SQL Apply is waiting for a log file to be sent from the primary database • APPLYING: SQL Apply is actively mining or applying transactions • WAITING FOR DICTIONARY LOGS: SQL Apply is waiting for the archived logs containing the LogMiner dictionary to be shipped from the primary database • IDLE: SQL Apply has applied all changes available at the logical standby, and is caught up with the primary database
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$LOGSTDBY"
- "V$LOGSTDBY_STATS"

8.62 V$LOGSTDBY_STATS

V$LOGSTDBY_STATS displays statistics, current state, and status information related to SQL Apply.

No rows are returned from this view when SQL Apply is not running. This view is only meaningful in the context of a logical standby database.

All statistics shown in this view are reinitialized at each SQL Apply start.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NAME	VARCHAR2(64)	Name of the statistic, state, or status: **Note:** Many of the following statistics are subject to change or deletion; programmers should write application code to tolerate missing or extra statistics. • id of the logminer session used by SQL Apply to mine the redo logs. • number of preparers • number of appliers • server processes in use for SQL Apply • maximum SGA (in MBytes) for LCR cache • whether SQL Apply is preserving the commit order seen at the primary database while applying changes • maximum events recorded in the DBA_LOGSTDBY_EVENTS table • whether SQL Apply is logging errors that are skipped in the DBA_LOGSTDBY_EVENTS table • whether SQL Apply is logging DDLs that are skipped in the DBA_LOGSTDBY_EVENTS table • whether SQL Apply is logging DDLs that are applied in the DBA_LOGSTDBY_EVENTS table • whether SQL Apply is logging unsupported operations that are encountered in the DBA_LOGSTDBY_EVENTS table • whether or not real time apply is on • value of apply delay (in minutes) • coordinator state • coordinator uptime in seconds • time of the most recent start of SQL Apply • number of transactions mined and made available for apply • number of transactions applied • number of rolled back transactions mined • number of DDL txns mined • number of CTAS (Create Table as Select) txns mined • number of thread enable events encountered in the redo stream • number of thread disable events encountered in the redo stream • bytes of redo records mined • bytes paged out • seconds spent in pageout activity • bytes checkpointed • seconds spent in checkpointing activity • seconds SQL Apply is idle • number of times a complete standby redo logs are mined without having to mine the corresponding archived log • number of times SQL Apply had to switch from a standby redo log to the corresponding archived log • number of times SQL Apply mined redo from the archived logs • number of archived logs that arrived at the standby via gap fetch mechanism (gap fetched logs mined) • number of failed attempts to open a logfile • amount of time spent in waiting for the current gap to resolve if SQL Apply is running in real time mode (current logfile wait)[1] • time spent in waiting for gap to resolve if SQL Apply is running in real time mode (total logfile wait)[2]
VALUE	VARCHAR2(64)	Value of the statistic or state information

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

1 In case SQL Apply is not running in real time mode, this may not reflect time spent in gap resolution, but simply the time spent waiting for the most recent archived log to show up at the logical standby.
2 In case SQL Apply is not running in real time mode, this will include time that SQL Apply spent every time it finished processing all archived logs registered with it, and waited for the next log to be archived.

8.63 V$LOGSTDBY_TRANSACTION

V$LOGSTDBY_TRANSACTION displays all transactions that are actively being processed by SQL Apply.

The transaction identifiers shown in this view are those mined from the redo stream and correspond to transaction identifiers assigned at the primary database, and do not correspond to the transactions that are active at the logical standby database. For information regarding transactions active in the logical standby database, including those created as part of SQL Apply, query the V$TRANSACTION view at the logical standby database.

Column	Datatype	Description
PRIMARY_XIDUSN	NUMBER	Undo segment number of the transaction
PRIMARY_XIDSLT	NUMBER	Slot number of the transaction
PRIMARY_XIDSQN	NUMBER	Sequence number of the transaction
PRIMARY_XID	RAW(8)	Transaction ID
PRIMARY_START_SCN	NUMBER	Start system change number (SCN) base
PRIMARY_START_TIME	DATE	Start time
PRIMARY_PARENT_XIDUSN	NUMBER	Undo segment number of the parent transaction
PRIMARY_PARENT_XIDSLT	NUMBER	Slot number of the parent transaction
PRIMARY_PARENT_XIDSQN	NUMBER	Sequence number of the parent transaction
PRIMARY_PARENT_XID	RAW(8)	Transaction ID of the parent transaction (PDML)
TYPE	VARCHAR2(32)	Type: • PL/SQL - Transaction was done as part of a supported PL/SQL procedure • Direct Path Load - Transaction is a direct path load • CTAS - Transaction contains at least one CREATE TABLE ... AS SELECT operation • DDL - Transaction contains one or more DDL operations • PDML Child - Transaction is a child transaction • DML - Transaction contains only DML operations

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MINING_STATUS	VARCHAR2(32)	Mining status: • ACTIVE - Transaction is still being mined by LogMiner. At least part of this transaction is ready to be applied or has already been applied. • COMPLETE - Transaction is complete and ready to be applied. LogMiner has finished mining.
SRC_CON_ID	NUMBER	Contains the PDB ID (the PDB_ID column from the DBA_PDBS view) of the source database that generated the change for this transaction.
APPLY_STATUS	VARCHAR2(6)	• ACTIVE - Transaction has been assigned to an apply server. It is in one of the following states: - The transaction is being actively applied - The transaction is being held by an apply server waiting for certain events to occur - The transaction is being held by an apply server waiting for subsequent parts of this transaction • NONE - Transaction has not yet been assigned to an apply server
SID	NUMBER	Session ID of the apply server's session; Null if APPLY_STATUS is NONE
SERIAL#	NUMBER	Serial number of the apply server's session; Null if APPLY_STATUS is NONE
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$TRANSACTION"

8.64 V$MANAGED_STANDBY

V$MANAGED_STANDBY displays current status information for some Oracle Database processes related to physical standby databases in the Data Guard environment. This view does not persist after an instance shutdown.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PROCESS	VARCHAR2(9)	Type of the process whose information is being reported: • RFS - Remote file server • MRP0 - Detached recovery server process • MR(fg) - Foreground recovery session • ARCH - Archiver process • DGRD - Generic Oracle Data Guard process • FGRD • LGWR • RFS(FAL) • RFS(NEXP) • LNS - ASYNC Redo Transport process
PID	VARCHAR2(24)	Operating system process identifier of the process
STATUS	VARCHAR2(12)	Current process status: • ALLOCATED - Process is active but not currently connected to a primary database • ANNOUNCING - Process is announcing the existence of a potential dependent archived redo log • APPLYING_LOG - Process is actively applying the archived redo log to the standby database • ATTACHED - Process is actively attached and communicating to a primary database • CLOSING - Process has completed archival and is closing the archived redo log • CONNECTED - Network connection established to a primary database • ERROR - Process has failed • IDLE - Process is not performing any activities • OPENING - Process is opening the archived redo log • RECEIVING - Process is receiving network communication • REGISTERING - Process is registering the existence of a completed dependent archived redo log • UNUSED - No active process • WAIT_FOR_GAP - Process is waiting for the archive gap to be resolved • WAIT_FOR_LOG - Process is waiting for the archived redo log to be completed • WRITING - Process is actively writing redo data to the archived redo log
CLIENT_PROCESS	VARCHAR2(8)	Identifies the corresponding primary database process: • Archival - Foreground (manual) archival process (SQL) • ARCH - Background ARC*n* process • LGWR - Background LGWR process
CLIENT_PID	VARCHAR2(40)	Operating system process identifier of the client process
CLIENT_DBID	VARCHAR2(40)	Database identifier of the primary database
GROUP#	VARCHAR2(40)	Standby redo log group
RESETLOG_ID	NUMBER	Resetlogs identifier of the archived redo log
THREAD#	NUMBER	Archived redo log thread number
SEQUENCE#	NUMBER	Archived redo log sequence number
BLOCK#	NUMBER	Last processed archived redo log block number

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BLOCKS	NUMBER	Count (in 512-byte blocks) of the last write to a redo log, or for a recovery process, the expected final read count
DELAY_MINS	NUMBER	Archived redo log delay interval in minutes
KNOWN_AGENTS	NUMBER	Total number of standby database agents processing an archived redo log
ACTIVE_AGENTS	NUMBER	Number of standby database agents actively processing an archived redo log
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Note

This view is deprecated in Oracle Database 12*c* Release 2 (12.2.0.1) and may be desupported in a future release. The V$DATAGUARD_PROCESS view should be used, instead.

See Also

"V$DATAGUARD_PROCESS"

8.65 V$MAP_COMP_LIST

V$MAP_COMP_LIST displays supplementary information for all element mapping structures.

Column	Datatype	Description
ELEM_IDX	NUMBER	Index corresponding to the element
NUM_COMP	NUMBER	Number of components (maximum is 5)
COMP1_NAME	VARCHAR2(256)	Name of the first component
COMP1_VAL	VARCHAR2(256)	Value of the first component
COMP2_NAME	VARCHAR2(256)	Name of the second component
COMP2_VAL	VARCHAR2(256)	Value of the second component
COMP3_NAME	VARCHAR2(256)	Name of the third component
COMP3_VAL	VARCHAR2(256)	Value of the third component
COMP4_NAME	VARCHAR2(256)	Name of the fourth component
COMP4_VAL	VARCHAR2(256)	Value of the fourth component
COMP5_NAME	VARCHAR2(256)	Name of the fifth component

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
COMP5_VAL	VARCHAR2(256)	Value of the fifth component
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.66 V$MAP_ELEMENT

V$MAP_ELEMENT displays a list of all element mapping structures in the SGA of the instance.

Column	Datatype	Description
ELEM_NAME	VARCHAR2(256)	Element name
ELEM_IDX	NUMBER	Index corresponding to the element
ELEM_CFGID	VARCHAR2(256)	Configuration ID (N/A if configuration ID is not supported)
ELEM_TYPE	VARCHAR2(12)	Element type: • MIRROR • STRIPE • RAID5 • CONCATENATED • PARTITION • DISK • NONE
ELEM_SIZE	NUMBER	Element Size in HKB
ELEM_NSUBELEM	NUMBER	Number of Subelements
ELEM_DESCR	VARCHAR2(256)	Element Description
STRIPE_SIZE	NUMBER	Stripe Size in HKB for RAID-5 and STRIPE elements, 0 for the remaining types
LIB_IDX	NUMBER	Index of the library which claims ownership of the element
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.67 V$MAP_EXT_ELEMENT

V$MAP_EXT_ELEMENT displays supplementary information for all element mapping structures.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ELEM_IDX	NUMBER	Index corresponding to the element
NUM_ATTRB	NUMBER	Number of Attributes (maximum is 5)
ATTRB1_NAME	VARCHAR2(256)	Name of the first Attribute
ATTRB1_VAL	VARCHAR2(256)	Value of the first attribute
ATTRB2_NAME	VARCHAR2(256)	Name of the second attribute
ATTRB2_VAL	VARCHAR2(256)	Value of the second attribute
ATTRB3_NAME	VARCHAR2(256)	Name of the third attribute
ATTRB3_VAL	VARCHAR2(256)	Value of the third attribute
ATTRB4_NAME	VARCHAR2(256)	Name of the fourth attribute
ATTRB4_VAL	VARCHAR2(256)	Value of the fourth attribute
ATTRB5_NAME	VARCHAR2(256)	Name of the fifth attribute
ATTRB5_VAL	VARCHAR2(256)	Value of the fifth attribute
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.68 V$MAP_FILE

V$MAP_FILE displays a list of all file mapping structures in the shared memory of the instance.

Column	Datatype	Description
FILE_MAP_IDX	NUMBER	Index corresponding to the file
FILE_CFGID	VARCHAR2(256)	Configuration ID (N/A if configuration ID is not supported)
FILE_STATUS	VARCHAR2(7)	Status of the mapping information: • VALID - File mapping information is latest • INVALID - Mapping must be refreshed
FILE_NAME	VARCHAR2(256)	Absolute file name
FILE_TYPE	VARCHAR2(11)	File type: • DATAFILE • SPFILE • TEMPFILE • CONTROLFILE • LOGFILE • ARCHIVEFILE
FILE_STRUCTURE	VARCHAR2(9)	File structure: • FILE • RAWVOLUME • RAWDEVICE • NONE
FILE_SIZE	NUMBER	File size in HKB (half KB)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FILE_NEXTS	NUMBER	Number of file extents in the file (not necessarily the same as the number of file extents mapped)
LIB_IDX	NUMBER	Index of mapping library claiming ownership of the file
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.69 V$MAP_FILE_EXTENT

V$MAP_FILE_EXTENT displays a list of all file extent mapping structures in the shared memory of the instance.

Column	Datatype	Description
FILE_MAP_IDX	NUMBER	File index (corresponds to FILE_MAP_IDX in V$MAP_FILE)
EXT_NUM	NUMBER	File extent number
EXT_ELEM_OFF	NUMBER	Element offset in HKB
EXT_SIZE	NUMBER	File extent size in HKB
EXT_FILE_OFF	NUMBER	File Offset in HKB
EXT_TYPE	VARCHAR2(6)	File Extent Type: • DATA • PARITY • NONE
ELEM_IDX	NUMBER	Index in V$MAP_ELEMENT corresponding to the element where the file extent resides
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.70 V$MAP_FILE_IO_STACK

V$MAP_FILE_IO_STACK displays the hierarchical arrangement of storage containers for files. Each row in the view represents a level in the hierarchy.

Column	Datatype	Description
FILE_MAP_IDX	NUMBER	File index (corresponds to FILE_MAP_IDX in V$MAP_FILE)
DEPTH	NUMBER	Element depth within the I/O stack

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ELEM_IDX	NUMBER	Index corresponding to the element
CU_SIZE	NUMBER	Contiguous set of logical blocks of the file (in HKB units) that is resident contiguously on the element
STRIDE	NUMBER	Number of HKB between contiguous units (CU) in the file that are contiguous on this element. Used in RAID5 and striped files.
NUM_CU	NUMBER	Number of contiguous units that are adjacent to each other on this element that are separated by STRIDE HKB in the file. In RAID5, the number of contiguous units also include the parity stripes.
ELEM_OFFSET	NUMBER	Element offset in HKB units
FILE_OFFSET	NUMBER	Offset in HKB units from the start of the file to the first byte of the contiguous units
DATA_TYPE	VARCHAR2(15)	Datatype: • DATA • PARITY • DATA AND PARITY
PARITY_POS	NUMBER	Position of the parity. Only for RAID5. This column is needed to distinguish the parity from the data part.
PARITY_PERIOD	NUMBER	Parity period. Only for RAID5.
ID	NUMBER	Unique identifier
PARENT_ID	NUMBER	Parent identifier
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.71 V$MAP_LIBRARY

V$MAP_LIBRARY displays a list of all mapping libraries dynamically loaded by the external process.

Column	Datatype	Description
LIB_IDX	NUMBER	Index corresponding to the library
LIB_NAME	VARCHAR2(256)	Absolute library name
VENDOR_NAME	VARCHAR2(64)	Name of the vendor implementing the library
PROTOCOL_NUM	NUMBER	Mapping protocol that the library supports
VERSION_NUM	VARCHAR2(32)	Version number
PATH_NAME	VARCHAR2(4000)	Path name
MAP_FILE	VARCHAR2(1)	Indicates whether the library supports mapping files (Y) or not (N)
FILE_CFGID	VARCHAR2(13)	Type of configuration ID supported for files: • NONE - Not supported • PERSISTENT • NONPERSISTENT

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MAP_ELEM	VARCHAR2(1)	Indicates whether the library supports mapping elements (Y) or not (N)
ELEM_CFGID	VARCHAR2(13)	Type of configuration id supported for elements: • NONE - Not supported • PERSISTENT • NONPERSISTENT
MAP_SYNC	VARCHAR2(1)	Indicates whether the library needs to be explicitly synchronized so that future mappings reflect the most recent changes (Y) or not (N). Note that configuration IDs cannot be supported if the library needs to be explicitly synced.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.72 V$MAP_SUBELEMENT

V$MAP_SUBELEMENT displays a list of all subelement mapping structures in the shared memory of the instance.

Column	Datatype	Description
CHILD_IDX	NUMBER	Index in V$MAP_ELEMENT corresponding to the child element
PARENT_IDX	NUMBER	Index in V$MAP_ELEMENT corresponding to the parent element
SUB_NUM	NUMBER	Subelement number
SUB_SIZE	NUMBER	Subelement size in HKB
ELEM_OFFSET	NUMBER	Offset in HKB on the child element
SUB_FLAGS	NUMBER	Subelement flags (currently unused)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.73 V$MAPPED_SQL

V$MAPPED_SQL lists the SQL statements that are translated and mapped in memory to a different SQL statement for execution.

Column	Datatype	Description
SQL_TEXT	VARCHAR2(1000)	First 1000 characters of the original SQL text

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SQL_FULLTEXT	CLOB	Full text for the original SQL statement
SQL_ID	VARCHAR2(13)	SQL identifier of the original SQL statement
HASH_VALUE	NUMBER	Hash value of the original SQL statement
MAPPED_SQL_TEXT	VARCHAR2(1000)	First 1000 characters of the mapped SQL text
MAPPED_SQL_FULLTEXT	CLOB	Full text for the mapped SQL statement
MAPPED_SQL_ID	VARCHAR2(13)	SQL identifier of the mapped SQL statement
MAPPED_HASH_VALUE	NUMBER	Hash value of the mapped SQL statement
SQL_TRANSLATION_PROFILE_ID	NUMBER	A non-NULL value specifies the object ID of the SQL translation profile used to translate the SQL statement into the mapped SQL statement
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
TRANSLATION_TIMESTAMP	DATE	Time this SQL statement was translated
TRANSLATION_CPU_TIME	NUMBER	CPU time used to translate this SQL statement
TRANSLATION_ELAPSED_TIME	NUMBER	Elapsed time used to translate this SQL statement
TRANSLATION_METHOD	VARCHAR2(10)	Method used to translate this SQL statement
DICTIONARY_SQL_ID	VARCHAR2(13)	SQL ID of SQL text in custom translation dictionary used to translate this SQL statement
USE_COUNT	NUMBER	Number of times this translation has been used

8.74 V$MEMOPTIMIZE_WRITE_AREA

V$MEMOPTIMIZE_WRITE_AREA displays information about fast ingest data in the large pool.

Column	Datatype	Description
TOTAL_SIZE	NUMBER	Total amount of memory allocated for fast ingest data in the large pool (in bytes)
USED_SPACE	NUMBER	Total amount of memory currently used by fast ingest data in the large pool (in bytes)
FREE_SPACE	NUMBER	Total amount of memory currently free for storing fast ingest data in the large pool (in bytes)
NUM_WRITES	NUMBER	Number of fast ingest insert operations for which data is still in the large pool and is yet to be written to disk
NUM_WRITERS	NUMBER	Number of clients currently using fast ingest for inserting data into the database

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **Note**
>
> This view is available starting with Oracle Database 19c.

8.75 V$MEMORY_CURRENT_RESIZE_OPS

V$MEMORY_CURRENT_RESIZE_OPS displays information about memory resize operations (both automatic and manual) which are currently in progress.

An operation can be a grow or a shrink of a dynamic memory component. All sizes are expressed in bytes.

Column	Datatype	Description
COMPONENT	VARCHAR2(64)	Component name
OPER_TYPE	VARCHAR2(13)	Operation type: • STATIC • INITIALIZING • DISABLED • GROW • SHRINK • SHRINK_CANCEL
OPER_MODE	VARCHAR2(9)	Operation mode: • MANUAL • DEFERRED • IMMEDIATE
PARAMETER	VARCHAR2(80)	Name of the parameter for the resize operation
INITIAL_SIZE	NUMBER	Parameter value at the start of the operation
TARGET_SIZE	NUMBER	Desired value of the parameter after the resize
CURRENT_SIZE	NUMBER	Current value of the parameter
START_TIME	DATE	Start time of the operation
LAST_UPDATE_TIME	DATE	Last time progress was made for the operation

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.76 V$MEMORY_DYNAMIC_COMPONENTS

V$MEMORY_DYNAMIC_COMPONENTS displays information about the dynamic SGA components.

This view summarizes information based on all completed SGA resize operations since instance startup. All sizes are expressed in bytes.

Column	Datatype	Description
COMPONENT	VARCHAR2(64)	Component name
CURRENT_SIZE	NUMBER	Current size of the component
MIN_SIZE	NUMBER	Minimum size of the component since instance startup
MAX_SIZE	NUMBER	Maximum size of the component since instance startup
USER_SPECIFIED_SIZE	NUMBER	Value of the user parameter for the component
OPER_COUNT	NUMBER	Number of operations since instance startup
LAST_OPER_TYPE	VARCHAR2(13)	Last completed operation for the component: • STATIC • INITIALIZING • DISABLED • GROW • SHRINK • SHRINK_CANCEL
LAST_OPER_MODE	VARCHAR2(9)	Mode of the last completed operation: • MANUAL • DEFERRED • IMMEDIATE
LAST_OPER_TIME	DATE	Start time of the last completed operation
GRANULE_SIZE	NUMBER	Granularity of the GROW or SHRINK operation
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.77 V$MEMORY_RESIZE_OPS

V$MEMORY_RESIZE_OPS displays information about the last 800 completed memory resize operations (both automatic and manual). This does not include in-progress operations. All sizes are expressed in bytes.

Column	Datatype	Description
COMPONENT	VARCHAR2(64)	Component name
OPER_TYPE	VARCHAR2(13)	Operation type: • STATIC • INITIALIZING • DISABLED • GROW • SHRINK • SHRINK_CANCEL
OPER_MODE	VARCHAR2(9)	Operation mode: • MANUAL • DEFERRED • IMMEDIATE
PARAMETER	VARCHAR2(80)	Name of the parameter for the resize operation
INITIAL_SIZE	NUMBER	Parameter value at the start of the operation
TARGET_SIZE	NUMBER	Requested value of the parameter after the resize
FINAL_SIZE	NUMBER	Real value of the parameter after the resize
STATUS	VARCHAR2(9)	Completion status of the operation: • INACTIVE • PENDING • COMPLETE • CANCELLED • ERROR
START_TIME	DATE	Start time of the operation
END_TIME	DATE	End time of the operation
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.78 V$MEMORY_TARGET_ADVICE

V$MEMORY_TARGET_ADVICE provides information about how the MEMORY_TARGET parameter should be sized based on current sizing and satisfaction metrics.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MEMORY_SIZE	NUMBER	If the MEMORY_SIZE_FACTOR column has a value of 1, then this column shows the current size of memory, as set by the MEMORY_TARGET initialization parameter. If the value of the MEMORY_SIZE_FACTOR column is less than or greater than 1, then this column shows a proposed memory size.
MEMORY_SIZE_FACTOR	NUMBER	A multiplier for the current memory size. Possible values are 0.25, 0.5, 0.75, 1, 1.5, 1.75, and 2. This multiplier times the current memory size equals the value of the MEMORY_SIZE column.
ESTD_DB_TIME	NUMBER	For current memory size (MEMORY_SIZE_FACTOR = 1), the amount of database time required to complete the current workload. For a proposed memory size, the estimated amount of database time that would be required if the MEMORY_TARGET parameter were changed to the proposed size.
ESTD_DB_TIME_FACTOR	NUMBER	For a proposed memory size, ratio of estimated database time to current database time
VERSION	NUMBER	Version number of this recommendation (this snapshot of the V$MEMORY_TARGET_ADVICE view)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Table 8-2 shows how the information provided in V$MEMORY_TARGET_ADVICE could be used to improve performance. The data indicates that if current memory size is 380M, and you were to increase it to 760M (2x), the current workload would take 80525 units of DBtime as opposed to 115475 units of DBtime, which is a significant improvement in performance.

Table 8-2 Example of Using V$MEMORY_TARGET_ADVICE

MEMORY_SIZE	MEMORY_SIZE_FACTOR	ESTD_DB_TIME	ESTD_DB_TIME_FACTOR	VERSION
380	1	115475	1	3
95	.25	200500	1.7	3
190	.5	125600	1.1	3
760	2	80525	0.7	3

See Also

"MEMORY_TARGET"

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.79 V$METRIC

V$METRIC displays the most recent statistic values for the complete set of metrics captured by the Automatic Workload Repository (AWR) infrastructure.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	Metric Group ID. Refer to the V$METRICNAME view for the name of the group.
ENTITY_ID	NUMBER	Entity ID for the metric in question. The value of the Entity ID depends upon the metric group. (See the following table for an explanation of possible values.)
ENTITY_SEQUENCE	NUMBER	Entity Sequence number for the metric in question. The value of the Entity Sequence depends upon the metric group. (See the following table for an explanation of possible values.)
METRIC_ID	NUMBER	Metric ID
METRIC_NAME	VARCHAR2(64)	Metric Name. This is the statistic that is captured for the entity.
VALUE	NUMBER	Value of the statistic between BEGIN_TIME and END_TIME
METRIC_UNIT	VARCHAR2(64)	Unit for the VALUE
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

The following table describes what the ENTITY_ID and ENTITY_SEQUENCE are for each metric group:

GID	Group NAME	Entity ID	Entity Sequence
0	Event Metrics	Event#	N/A
1	Event Class Metrics	Wait Class ID	N/A
2	System Metrics Long Duration	N/A	N/A
3	System Metrics Short Duration	N/A	N/A
4	Session Metrics Long Duration	Session ID	Serial#
5	Session Metrics Short Duration	Session ID	Serial#
6	Service Metrics	N/A	Service Hash
7	File Metrics Long Duration	File#	Creation Change#
9	Tablespace Metrics Long Duration	Tablespace#	N/A
10	Service Metrics (Short)	N/A	Service Hash

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.80 V$METRICGROUP

V$METRICGROUP displays information about the metric group for each of the four major Replication components: capture, propagation, apply, and queue.

Column	Datatype	Description
GROUP_ID	NUMBER	Internal ID associated with each group
NAME	VARCHAR2(64)	External name of the group
INTERVAL_SIZE	NUMBER	How often to collect statistics
MAX_INTERVAL	NUMBER	Total number of intervals over which statistics should be collected
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.81 V$METRIC_HISTORY

V$METRIC_HISTORY displays all the available statistic values for the complete set of metrics captured by the Automatic Workload Repository (AWR) infrastructure.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	Metric Group ID. Refer to the V$METRICNAME view for the name of the group.
ENTITY_ID	NUMBER	Entity ID for the metric in question. The value of the Entity ID depends upon the metric group. (See the table in the description of V$METRIC for an explanation of possible values.)
ENTITY_SEQUENCE	NUMBER	Entity Sequence number for the metric in question. The value of the Entity Sequence depends upon the metric group. (See the table in the description of V$METRIC for an explanation of possible values.)
METRIC_ID	NUMBER	Metric ID
METRIC_NAME	VARCHAR2(64)	Metric Name. This is the statistic that is captured for the entity.
VALUE	NUMBER	Value of the statistic between BEGIN_TIME and END_TIME
METRIC_UNIT	VARCHAR2(64)	Unit for the VALUE

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.82 V$METRICNAME

V$METRICNAME displays the mapping of the name of metrics to their metric ID.

Column	Datatype	Description
GROUP_ID	NUMBER	Metric group ID
GROUP_NAME	VARCHAR2(64)	Metric group name
METRIC_ID	NUMBER	Metric ID
METRIC_NAME	VARCHAR2(64)	Metric name
METRIC_UNIT	VARCHAR2(64)	Unit of measurement
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.83 V$MTTR_TARGET_ADVICE

V$MTTR_TARGET_ADVICE displays rows that predict the number of physical I/Os for the MTTR corresponding to each row.

The rows also compute a physical I/O factor, which is the ratio of the number of estimated I/Os to the number of I/Os actually performed by the current MTTR setting during the measurement interval.

The content of the view is empty if MTTR advisory has not been turned on since database startup. Otherwise, it returns the advisory information collected. If advisory is currently off, then this information comes from the last time MTTR advisory was on. FAST_START_MTTR_TARGET must be set to a nonzero value if the STATISTICS_LEVEL parameter is dynamically modified to turn MTTR advisory on.

If the FAST_START_MTTR_TARGET parameter is changed while MTTR advisory is on, then MTTR advisory is temporarily turned off until the new FAST_START_MTTR_TARGET setting takes effect. During this transition period, the contents of V$MTTR_TARGET_ADVICE reflect the simulation result for the old MTTR setting.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MTTR_TARGET_FOR_ESTIMATE	NUMBER	MTTR setting being simulated. Equal to the current MTTR setting if this is the first row of the view.
ADVICE_STATUS	VARCHAR2(5)	Current status of MTTR simulation: • ON • READY • OFF
DIRTY_LIMIT	NUMBER	Dirty buffer limit derived from the MTTR being simulated
ESTD_CACHE_WRITES	NUMBER	Estimated number of cache physical writes under this MTTR
ESTD_CACHE_WRITE_FACTOR	NUMBER	Estimated cache physical write ratio under this MTTR. It is the ratio of the estimated number of cache writes to the number of cache writes under the current MTTR setting.
ESTD_TOTAL_WRITES	NUMBER	Estimated total number of physical writes under this MTTR
ESTD_TOTAL_WRITE_FACTOR	NUMBER	Estimated total physical write ratio under this MTTR. It is the ratio of the estimated total number of physical writes to the total number of physical writes under the current MTTR setting.
ESTD_TOTAL_IOS	NUMBER	Estimated total number of I/Os under this MTTR
ESTD_TOTAL_IO_FACTOR	NUMBER	Estimated total I/O ratio under this MTTR. It is the ratio of the estimated total number of I/Os to the total number of I/Os under the current MTTR setting.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "FAST_START_MTTR_TARGET"
- "STATISTICS_LEVEL"

8.84 V$MUTEX_SLEEP

V$MUTEX_SLEEP shows the wait time, and the number of sleeps for each combination of mutex type and location.

Column	Datatype	Description
MUTEX_TYPE	VARCHAR2(32)	Type of action/object the mutex protects
LOCATION	VARCHAR2(40)	The code location where the waiter slept for the mutex
SLEEPS	NUMBER	Number of sleeps for this MUTEX_TYPE and LOCATION
WAIT_TIME	NUMBER	Wait time in microseconds

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.85 V$MUTEX_SLEEP_HISTORY

V$MUTEX_SLEEP_HISTORY displays time-series data.

Each row in this view is for a specific time, mutex type, location, requesting session and blocking session combination. That is, it shows data related to a specific session (requesting session) that slept while requesting a specific mutex type and location, because it was being held by a specific blocking session. The data in this view is contained within a circular buffer, with the most recent sleeps shown.

Column	Datatype	Description
MUTEX_IDENTIFIER	NUMBER	Mutex ID
SLEEP_TIMESTAMP	TIMESTAMP(6)	The last date/time this MUTEX_TYPE and LOCATION was slept for by the REQUESTING_SESSION, while being held by the BLOCKING_SESSION.
MUTEX_TYPE	VARCHAR2(32)	Type of action/object the mutex protects
GETS	NUMBER	The total number of gets since the mutex was created and up until the time of the wait (and from all sessions past and present)
SLEEPS	NUMBER	The number of times the requester had to sleep before obtaining the mutex
REQUESTING_SESSION	NUMBER	The SID of a session requesting the mutex
BLOCKING_SESSION	NUMBER	The SID of a session holding the mutex
LOCATION	VARCHAR2(40)	The code location where the waiter slept for the mutex
MUTEX_VALUE	RAW(4 | 8)	If the mutex is held in exclusive (X) mode, this column shows the SID of the blocking session, else it shows the number of sessions referencing the mutex in S mode.
P1	NUMBER	Reserved for internal use
P1RAW	RAW(4 | 8)	Reserved for internal use
P2	NUMBER	Reserved for internal use
P3	NUMBER	Reserved for internal use
P4	NUMBER	Reserved for internal use
P5	VARCHAR2(64)	Reserved for internal use

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.86 V$MVREFRESH

V$MVREFRESH displays information about the materialized views currently being refreshed.

Column	Datatype	Description
SID	NUMBER	Session identifier
SERIAL#	NUMBER	Session serial number, which is used to uniquely identify a session's objects. Guarantees that session-level commands are applied to the correct session objects if the session ends with, and another session begins with, the same session ID.
CURRMVOWNER	VARCHAR2(31)	Owner of the materialized view currently being refreshed. The materialized view resides in this user's schema.
CURRMVNAME	VARCHAR2(31)	Name of the materialized view currently being refreshed
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.87 V$MYSTAT

V$MYSTAT contains statistics on the current session.

Column	Datatype	Description
SID	NUMBER	ID of the current session
STATISTIC#	NUMBER	Number of the statistic
VALUE	NUMBER	Value of the statistic
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.88 V$NFS_CLIENTS

V$NFS_CLIENTS displays information about NFS clients currently connected to the XML DB NFS Server.

Column	Datatype	Description
CLIENTID	NUMBER	A number identifying the client
PRINCIPAL	VARCHAR2(2000)	User string denoting the principal that set the client ID (SetClientId)
CLIENTOPAQUEIDENTIFIER	VARCHAR2(1000)	Opaque string presented as identification by the client to the NFS server
VERIFIER	RAW(8)	Verifier presented by the client
LEASEEXPIRY	NUMBER	Number of seconds in which the lease expires for the client
CLIENTADDR	VARCHAR2(2000)	Address of the client
CONFIRMED	VARCHAR2(5)	TRUE if the client is confirmed; otherwise FALSE
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.89 V$NFS_LOCKS

V$NFS_LOCKS displays information about byte range locks held on different files by NFS clients.

Column	Datatype	Description
OPENSTATEID	RAW(16)	Open state ID of the open owner
OPENSEQUENCEID	NUMBER	Open Sequence ID of the open owner
LOCKSTATEID	RAW(16)	Lock state ID of the lock owner
LOCKSEQUENCEID	NUMBER	Lock sequence ID of the lock owner
LOCKOWNER	VARCHAR2(2000)	Opaque string presented as identification by the lock owner to the NFS server
OFFSET	NUMBER	Byte Offset from which lock starts
LENGTH	NUMBER	Length of the lock
LOCKTYPE	VARCHAR2(20)	Type of the lock
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.90 V$NFS_OPEN_FILES

V$NFS_OPEN_FILES displays information about all the files currently opened by clients at the NFS server.

Column	Datatype	Description
CLIENTID	NUMBER	Number identifying the client
OPENOWNEROPAQUE	VARCHAR2(2000)	All the files currently opened by clients at the NFS server
OPENSTATEID	RAW(16)	Open state ID of the open owner
FILEHANDLE	RAW(32)	FileHandle of the file that has been opened
OPENSEQUENCEID	NUMBER	Open sequence ID of open owner
OPENREAD	VARCHAR2(5)	TRUE if the file is open for READ operations; otherwise FALSE
OPENWRITE	VARCHAR2(5)	TRUE if the file is open for WRITE operations; otherwise FALSE
SHAREACCESS	VARCHAR2(15)	Sharing mode of the file (SharedReadWrite, SharedRead, SharedWrite)
SHAREDENY	VARCHAR2(13)	Deny mode of the file (DenyReadWrite, DenyRead, DenyWrite)
CONFIRMED	VARCHAR2(5)	TRUE if open is confirmed; otherwise FALSE
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.91 V$NLS_PARAMETERS

V$NLS_PARAMETERS contains current values of NLS parameters.

Column	Datatype	Description
PARAMETER	VARCHAR2(64)	Parameter names are as follows: NLS_CALENDAR, NLS_CHARACTERSET, NLS_COMP, NLS_CURRENCY, NLS_DATE_FORMAT, NLS_DATE_LANGUAGE, NLS_DUAL_CURRENCY, NLS_ISO_CURRENCY, NLS_LANGUAGE, NLS_LENGTH_SEMANTICS, NLS_NCHAR_CHARACTERSET, NLS_NCHAR_CONV_EXCP, NLS_NUMERIC_CHARACTERS, NLS_SORT, NLS_TERRITORY, NLS_TIMESTAMP_FORMAT, and NLS_TIMESTAMP_TZ_FORMAT. Two additional parameters, NLS_TIME_FORMAT and NLS_TIME_TZ_FORMAT, are currently used for internal purposes only.
VALUE	VARCHAR2(64)	NLS parameter value
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.92 V$NLS_VALID_VALUES

V$NLS_VALID_VALUES lists all valid values for NLS parameters.

Column	Datatype	Description
PARAMETER	VARCHAR2(64)	Parameter name (LANGUAGE | SORT | TERRITORY | CHARACTERSET)
VALUE	VARCHAR2(64)	NLS parameter value
ISDEPRECATED	VARCHAR2(5)	Indicates whether the parameter has been deprecated (TRUE) or not (FALSE)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.93 V$NONLOGGED_BLOCK

V$NONLOGGED_BLOCK displays ranges of nonlogged datafile blocks recorded in the control file.

Prior to Oracle Database 12*c*, the presence of any nonlogged blocks in a data file was recorded in the file header via the FIRST_NONLOGGED_SCN column of the V$DATAFILE view. Now with 12*c*, in addition to the file header data, the ranges themselves are recorded in the control file. A control file range is a superset of the actual nonlogged blocks, meaning that small ranges can be merged to form larger ranges, even when there are some valid blocks between the smaller ranges.

The information in the view is maintained by RMAN VALIDATE, RMAN RESTORE, RMAN RECOVER, and Flashback Database and Media Recovery. A non RMAN-based restore will cause the data to become invalid, and it will be purged the next time any of those tasks are invoked and involve the file. As a result of space reuse, it is possible for ranges to no longer contain any nonlogged blocks. An RMAN VALIDATE command can be used to synchronize the ranges with the actual nonlogged blocks found from a scan of the data file.

Column	Datatype	Description
FILE#	NUMBER	Absolute file number of the data file that contains the nonlogged blocks
BLOCK#	NUMBER	Block number of the first nonlogged block in the range of nologged blocks
BLOCKS	NUMBER	Number of nonlogged blocks found starting with BLOCK#
NONLOGGED_START_CHANGE#	NUMBER	The smallest SCN on which any block in this block range became nonlogged. NULL if unknown.
NONLOGGED_START_TIME	DATE	The time that corresponds to NONLOGGED_START_CHANGE#. NULL if unknown.
NONLOGGED_END_CHANGE#	NUMBER	The largest SCN on which any block in this block range became nonlogged. NULL if unknown.
NONLOGGED_END_TIME	DATE	The time that corresponds to NONLOGGED_END_CHANGE#. NULL if unknown.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RESETLOGS_CHANGE#	NUMBER	The resetlogs SCN of the incarnation on which this block range was first marked as nonlogged. NULL if unknown.
RESETLOGS_TIME	DATE	The resetlogs time of the incarnation on which this block range was first marked as nologged. NULL if unknown.
OBJECT#	VARCHAR2(40)	The object ID this range belongs to. If this field is NULL, the object number is unknown.
REASON	VARCHAR2(9)	The reason why this block range appears in this list, for example, primary file offline, could not talk to primary, non-standby recovery, and so on. For Oracle Database 12*c* and later releases, it is always UNKNOWN.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.94 V$OBJECT_DEPENDENCY

V$OBJECT_DEPENDENCY displays the objects depended on by a package, procedure, or cursor that is currently loaded in the shared pool. For example, together with V$SESSION and V$SQL, this view can be used to determine which tables are used in the SQL statement that a user is currently executing.

See Also

"V$SESSION" and "V$SQL"

Column	Datatype	Description
FROM_ADDRESS	RAW(4 | 8)	Address of a procedure, package, or cursor that is currently loaded in the shared pool
FROM_HASH	NUMBER	Hash value of a procedure, package, or cursor that is currently loaded in the shared pool
TO_OWNER	VARCHAR2(64)	Owner of the object that is depended on
TO_NAME	VARCHAR2(1000)	Name of the object that is depended on
TO_ADDRESS	RAW(4 | 8)	Address of the object that is depended on. These can be used to look up more information on the object in V$DB_OBJECT_CACHE.
TO_HASH	NUMBER	Hash value of the object that is depended on. These can be used to look up more information on the object in V$DB_OBJECT_CACHE.
TO_TYPE	NUMBER	Type of the object that is depended on

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.95 V$OBJECT_PRIVILEGE

V$OBJECT_PRIVILEGE displays information about privileges associated with an object.

Column	Datatype	Description
OBJECT_TYPE_NAME	VARCHAR2(64)	Name of the object type
OBJECT_TYPE_ID	NUMBER	ID of the object type
PRIVILEGE_ID	NUMBER	ID of the privilege
PRIVILEGE_NAME	VARCHAR2(64)	Name of the privilege
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.96 V$OBJECT_USAGE

V$OBJECT_USAGE displays statistics about index usage gathered from the database for the indexes owned by the current user. You can use this view to monitor index usage. All indexes that have been used at least once can be monitored and displayed in this view.

> **Note**
>
> The V$OBJECT_USAGE view is deprecated in Oracle Database 12*c* Release 1 (12.1) and maintained for backward compatibility. Support for this view may be removed in a future release. Oracle recommends that you use the USER_OBJECT_USAGE view instead of the V$OBJECT_USAGE view.

Column	Datatype	NULL	Description
INDEX_NAME	VARCHAR2	NOT NULL	Index name in sys.obj$.name
TABLE_NAME	VARCHAR2	NOT NULL	Table name in sys.obj$.name
MONITORING	VARCHAR2		YES| NO
USED	VARCHAR2		YES| NO

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
START_MONITORING	VARCHAR2		Start monitoring time in sys.object_stats.start_monitoring
END_MONITORING	VARCHAR2		End monitoring time in sys.object_stats.end_monitoring

See Also

- "USER_OBJECT_USAGE"
- "DBA_OBJECT_USAGE"

8.97 V$OBSOLETE_BACKUP_FILES

V$OBSOLETE_BACKUP_FILES displays all obsolete backups, copies, and archived logs according to the current retention policy.

This view requires that the database is set using the DBMS_RCVMAN.SETDATABASE procedure.

Column	Datatype	Description
PKEY	NUMBER	Primary key for the backup
BACKUP_TYPE	VARCHAR2(32)	Type of the backup: • BACKUP SET • COPY • PROXY COPY
FILE_TYPE	VARCHAR2(32)	Type of the file: • DATAFILE • CONTROLFILE • SPFILE • REDO LOG • PIECE
KEEP	VARCHAR2(3)	Indicates whether the backup has a retention policy different from the value for CONFIGURE RETENTION POLICY (YES) or not (NO)
KEEP_UNTIL	DATE	If the KEEP UNTIL TIME clause of the BACKUP command was specified, then this column shows the date after which the backup becomes obsolete. If the column is null and KEEP_OPTIONS is not null, the backup never becomes obsolete.
KEEP_OPTIONS	VARCHAR2(13)	KEEP options for the backup: • LOGS - RMAN keeps the logs needed to recover the backup • NOLOGS - RMAN does not keep the logs needed to recover the backup If this column is null, then the backup has no KEEP options and will be made obsolete based on the retention policy.
STATUS	VARCHAR2(16)	Status of the backup: • AVAILABLE • UNAVAILABLE • EXPIRED • OTHER

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FNAME	VARCHAR2(1024)	Name of the file
TAG	VARCHAR2(32)	Tag of the piece, copy, or proxy copy
MEDIA	VARCHAR2(80)	Media ID of the piece or proxy copy
RECID	NUMBER	Recid of the record in the controlfile
STAMP	NUMBER	Stamp of the record in the controlfile
DEVICE_TYPE	VARCHAR2(255)	Type of media device that stores the backup
BLOCK_SIZE	NUMBER	Block size for the backup (in bytes)
COMPLETION_TIME	DATE	Time when the backup completed
BS_KEY	NUMBER	Primary key of the backup set (valid only when BACKUP_TYPE is BACKUP SET)
BS_COUNT	NUMBER	Count of the backup set from the controlfile record (valid only when BACKUP_TYPE is BACKUP SET)
BS_STAMP	NUMBER	Stamp of the backup set from the controlfile record (valid only when BACKUP_TYPE is BACKUP SET)
BS_TYPE	VARCHAR2(32)	Type of the backup set (valid only when BACKUP_TYPE is BACKUP SET): • DATAFILE • ARCHIVED LOG
BS_INCR_TYPE	VARCHAR2(32)	Incremental level of the backup set (valid only when BACKUP_TYPE is BACKUP SET)
BS_PIECES	NUMBER	Number of backup pieces in the backup set (valid only when BACKUP_TYPE is BACKUP SET)
BS_COMPLETION_TIME	DATE	Completion time of the backup set (valid only when BACKUP_TYPE is BACKUP SET)
BP_PIECE#	NUMBER	Number of the backup piece (valid only when FILE_TYPE is PIECE and BACKUP_TYPE is BACKUP SET)
BP_COPY#	NUMBER	Copy number of the backup piece (valid only when FILE_TYPE is PIECE and BACKUP_TYPE is BACKUP SET)
DF_FILE#	NUMBER	Absolute file number of the datafile (valid only when FILE_TYPE is DATAFILE)
DF_RESETLOGS_CHANGE#	NUMBER	System change number (SCN) of the most recent RESETLOGS when the control file or datafile was created (valid only when FILE_TYPE is DATAFILE)
DF_CREATION_CHANGE#	NUMBER	Creation SCN of the control file or datafile (valid only when FILE_TYPE is DATAFILE)
DF_CHECKPOINT_CHANGE#	NUMBER	System change number (SCN) of the most recent control file or datafile checkpoint (valid only when FILE_TYPE is DATAFILE)
DF_CKP_MOD_TIME	DATE	Modification time in case of SPFILE, otherwise time when the control file or datafile was checkpointed (valid only when FILE_TYPE is DATAFILE)
RL_THREAD#	NUMBER	Number of the redo thread (valid only when FILE_TYPE is REDO LOG)
RL_SEQUENCE#	NUMBER	Log sequence number (valid only when FILE_TYPE is REDO LOG)
RL_RESETLOGS_CHANGE#	NUMBER	System change number (SCN) of the most recent RESETLOGS when the record was created (valid only when FILE_TYPE is REDO LOG)
RL_FIRST_CHANGE#	NUMBER	First SCN of the redo log (valid only when FILE_TYPE is REDO LOG)
RL_FIRST_TIME	DATE	Time when Oracle switched into the redo log (valid only when FILE_TYPE is REDO LOG)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RL_NEXT_CHANGE#	NUMBER	First SCN of the next redo log in the thread (valid only when FILE_TYPE is REDO LOG)
RL_NEXT_TIME	DATE	First timestamp of the next redo log in the thread (valid only when FILE_TYPE is REDO LOG)

See Also

Oracle Database Backup and Recovery User's Guide for more information about the DBMS_RCVMAN.SETDATABASE procedure

8.98 V$OBSOLETE_PARAMETER

V$OBSOLETE_PARAMETER displays information about obsolete initialization parameters. If any row of the view contains TRUE in the ISSPECIFIED column, then you should examine why.

Column	Datatype	Description
NAME	VARCHAR2(64)	Name of the parameter
ISSPECIFIED	VARCHAR2(5)	Indicates whether the parameter was specified in the parameter file (TRUE) or not (FALSE)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.99 V$OFFLINE_RANGE

V$OFFLINE_RANGE displays datafile offline information from the control file.

Note that the last offline range of each datafile is kept in the DATAFILE record.

An offline range is created for a datafile when its tablespace is first altered to be OFFLINE NORMAL or READ ONLY, and then subsequently altered to be ONLINE or read/write. Note that no offline range is created if the datafile itself is altered to be OFFLINE or if the tablespace is altered to be OFFLINE IMMEDIATE.

See Also

"V$DATAFILE"

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RECID	NUMBER	Record ID
STAMP	NUMBER	Record stamp
FILE#	NUMBER	Datafile number
OFFLINE_CHANGE#	NUMBER	SCN at which offlined
ONLINE_CHANGE#	NUMBER	SCN at which onlined
ONLINE_TIME	DATE	Time of offline SCN
RESETLOGS_CHANGE#	NUMBER	Resetlogs change number of the record
RESETLOGS_TIME	DATE	Resetlogs timestamp of the record
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.100 V$OFS_STATS

V$OFS_STATS displays performance statistics for various Oracle File System operations. These statistics are maintained at per mount level.

Column	Datatype	Description
OFS_MNTPNT	VARCHAR2(4096)	Mount point
OFS_LOOKUP	NUMBER	Number of lookup operations performed
OFS_FORGET	NUMBER	Number of forget operations performed
OFS_GETATTR	NUMBER	Number of getattr operations
OFS_SETATTR	NUMBER	Number of setattr operations
OFS_READLINK	NUMBER	Number of readlink operations
OFS_SYMLINK	NUMBER	Number of symlink operations
OFS_MKNOD	NUMBER	Number of mknod operations
OFS_MKDIR	NUMBER	Number of mkdir operations
OFS_UNLINK	NUMBER	Number of remove file operations
OFS_RMDIR	NUMBER	Number of remove directory operations
OFS_RENAME	NUMBER	Number of file rename operations
OFS_LINK	NUMBER	Number of hard link operations
OFS_OPEN	NUMBER	Number of file open operations
OFS_READ	NUMBER	Number of file read operations
OFS_WRITE	NUMBER	Number of file write operations
OFS_STATFS	NUMBER	Number of statfs operations performed
OFS_RELEASE	NUMBER	Number of release operations
OFS_FSYNC	NUMBER	Number of file sync operations

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OFS_SETXATTR	NUMBER	Number of set extended attributes operations
OFS_GETXATTR	NUMBER	Number of get extended attributes
OFS_LISTXATTR	NUMBER	Number of list extended attributes
OFS_REMOVEXATTR	NUMBER	Number of remove extended attributes
OFS_FLUSH	NUMBER	Number of flush operations
OFS_INIT	NUMBER	Number of init operations
OFS_OPENDIR	NUMBER	Number of opendir operations
OFS_READDIR	NUMBER	Number of readdir operations
OFS_RELEASEDIR	NUMBER	Number of releasedir operations
OFS_FSYNCDIR	NUMBER	Number of directory sync operations
OFS_GETLK	NUMBER	Number of file get lock operations
OFS_SETLK	NUMBER	Number of file lock operations
OFS_SETLKW	NUMBER	Number of file lock operation with wait option
OFS_ACCESS	NUMBER	Number of access operations
OFS_CREATE	NUMBER	Number of create operations
OFS_INTERRUPT	NUMBER	Number of interrupt operations received on the mount
OFS_BMAP	NUMBER	Number of block map operations received
OFS_DESTROY	NUMBER	Number of destroy operations
OFS_BYTES_READ	NUMBER	Total number of bytes read on the mount point
OFS_BYTES_WRITTEN	NUMBER	Total number of bytes written to the mount point
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Note

This database view is supported only on the Linux operating system.

See Also

"V$OFSMOUNT"

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.101 V$OFSMOUNT

V$OFSMOUNT provides information about the file systems that are mounted by Oracle File System.

Column	Datatype	Description
OFS_MNTPATH	VARCHAR2(1024)	Mount path where the file system is mounted
OFS_FSPATH	VARCHAR2(1024)	File system path
OFS_MNTOPTS	VARCHAR2(1024)	Mount options used to mount the file system
OFS_MNTFLAGS	VARCHAR2(7)	Flags to specify if the file system is mounted. A value of 1 indicates that the file system is mounted.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
OFS_NODENM	VARCHAR2(255)	Node name
OFS_FSID	NUMBER	Unique ID that identifies the mounted file system
OFS_FSTYPE	VARCHAR2(255)	Oracle file system type. This is the value that is passed to the dbms_fs.mount_oracle_fs() procedure. Some of the expected values are dbfs and ofs.

Note

This database view is supported only on the Linux operating system.

See Also

- "V$OFS_STATS"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_FS.MOUNT_ORACLE_FS procedure

8.102 V$ONLINE_REDEF

V$ONLINE_REDEF provides information about the status of currently running online redefinitions.

Column	Datatype	Description
SID	NUMBER	Session identifier
SERIAL#	NUMBER	Session serial number, which is used to uniquely identify a session's objects

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
REDEFINITION_ID	NUMBER	Redefinition identifier
TABLE_OWNER	VARCHAR2(129)	Owner of the table currently being redefined. The table resides in this user's schema.
ORIGINAL_TABLE_NAME	VARCHAR2(129)	Name of the original table
INTERIM_TABLE_NAME	VARCHAR2(1024)	Interim table currently being redefined
PARTITION_NAME	VARCHAR2(1024)	Partition of the table currently being redefined. The table resides in this user's schema.
OPERATION	VARCHAR2(128)	Operations during the redefining process: • start_redef_table • sync_interim_table • copy_table_dependents • finish_redef_table • abort_redef_table
SUBOPERATION	VARCHAR2(128)	Sub operation during the redefining process
DETAILED_MESSAGE	VARCHAR2(1024)	Details of operations during redefining process. This can include details such as the number of DML being processed, execution start_time, and partition_name
PROGRESS	VARCHAR2(128)	Percentage of completion of each operation
REFRESH_STATEMENT_SQL_ID	VARCHAR2(128)	The SQL ID for the statement in the REFRESH_STATEMENT column
REFRESH_STATEMENT	VARCHAR2(4000)	A refresh statement executed during a refresh operation in some online redefinition procedures. When refresh is a sub operation in the SUBOPERATION column, there will be a refresh statement in this REFRESH_STATEMENT column.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.103 V$OPEN_CURSOR

V$OPEN_CURSOR lists cursors that each user session currently has opened and parsed, or cached.

Column	Datatype	Description
SADDR	RAW(4 | 8)	Session address
SID	NUMBER	Session identifier
USER_NAME	VARCHAR2(128)	User that is logged in to the session
ADDRESS	RAW(4 | 8)	Used with HASH_VALUE to uniquely identify the SQL statement being executed in the session
HASH_VALUE	NUMBER	Used with ADDRESS to uniquely identify the SQL statement being executed in the session
SQL_ID	VARCHAR2(13)	SQL identifier of the SQL statement being executed in the session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SQL_TEXT	VARCHAR2(60)	First 60 characters of the SQL statement that is parsed into the open cursor
LAST_SQL_ACTIVE_TIME	DATE	Time when this cursor was last executed
SQL_EXEC_ID	NUMBER	If the open cursor is executing, then the SQL execution identifier for that execution (see V$SQL_MONITOR)
CURSOR_TYPE	VARCHAR2(64)	Type of cursor: • OPEN PL/SQL - Open PL/SQL cursors • OPEN - Other open cursors • SESSION CURSOR CACHED - Cursors cached in the generic session cursor cache • OPEN-RECURSIVE - Open recursive cursors • DICTIONARY LOOKUP CURSOR CACHED - Cursors cached in the dictionary lookup cursor cache • BUNDLE DICTIONARY LOOKUP CACHED - Cursors cached in the bundled dictionary lookup cursor cache • JAVA NAME TRANSLATION CURSOR CACHED - Cursors cached in the Java name translation cursor cache • REPLICATION TRIGGER CURSOR CACHED - Cursors cached in the replication trigger cursor cache • CONSTRAINTS CURSOR CACHED - Cursors cached in the constraints cursor cache • PL/SQL CURSOR CACHED - Cursors cached in the PL/SQL cursor cache
CHILD_ADDRESS	RAW(4 | 8)	Address of the child cursor
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.104 V$OPTIMIZER_PROCESSING_RATE

V$OPTIMIZER_PROCESSING_RATE displays the processing rates used by the optimizer to compute degree of parallelism.

> **Note**
>
> You can manipulate these rates using these procedures for the DBMS_STATS package:
>
> - SET_PROCESSING_RATE
> - DELETE_PROCESSING_RATE
> - GATHER_PROCESSING_RATE
>
> See *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ADDRESS	RAW(4| 8)	Address of the handle to the parent for this cursor
OPERATION_NAME	VARCHAR2(64)	Name of the operation. The possible values are AGGR, ALL, CPU, CPU_ACCESS, CPU_AGGR, CPU_BYTES_PER_SEC, CPU_FILTER, CPU_GBY, CPU_HASH_JOIN, CPU_JOIN, CPU_NL_JOIN, CPU_RANDOM_ACCESS, CPU_ROWS_PER_SEC, CPU_SEQUENTIAL_ACCESS, CPU_SM_JOIN, CPU_SORT, HASH, IO, IO_ACCESS, IO_BYTES_PER_SEC, IO_RANDOM_ACCESS, IO_ROWS_PER_SEC, IO_SEQUENTIAL_ACCESS, MEMCMP, MEMCPY
MANUAL_VALUE	VARCHAR2(10)	Value of the operation set manually by the user
CALIBRATION_VALUE	VARCHAR2(10)	Value of the operation obtained from calibration (by running the GATHER_PROCESSING_RATE procedure)
DEFAULT_VALUE	VARCHAR2(10)	Default value of the operation
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.105 V$OPTION

V$OPTION displays Oracle Database options and features.

Typically, although not always, options must be separately licensed, whereas features come with the product and are enabled based on the product that is running (for example, Enterprise Edition).

Column	Datatype	Description
PARAMETER	VARCHAR2(64)	Name of the option (or feature)
VALUE	VARCHAR2(64)	Indicates whether the option (or feature) is installed (TRUE) or not (FALSE)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Licensing Information User Manual for more information about Oracle Database options and features

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.106 V$OSSTAT

V$OSSTAT displays system utilization statistics from the operating system. One row is returned for each system statistic.

Column	Datatype	Description
STAT_NAME	VARCHAR2(64)	Name of the statistic (see Table 8-3)
VALUE	NUMBER	Instantaneous statistic value
OSSTAT_ID	NUMBER	Statistic ID
COMMENTS	VARCHAR2(64)	Any additional operating system-specific clarifications for the statistic
CUMULATIVE	VARCHAR2(3)	Indicates whether the statistic is cumulative (that is, accumulates over time) (YES) or not (NO)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Table 8-3 V$OSSTAT Statistics

Statistic Name	Description
NUM_CPUS	Number of CPUs or processors available
IDLE_TIME	Number of hundredths of a second that a processor has been idle, totalled over all processors
BUSY_TIME	Number of hundredths of a second that a processor has been busy executing user or kernel code, totalled over all processors
USER_TIME	Number of hundredths of a second that a processor has been busy executing user code, totalled over all processors
SYS_TIME	Number of hundredths of a second that a processor has been busy executing kernel code, totalled over all processors
IOWAIT_TIME	Number of hundredths of a second that a processor has been waiting for I/O to complete, totalled over all processors
NICE_TIME	Number of hundredths of a second that a processor has been busy executing low-priority user code, totalled over all processors
AVG_IDLE_TIME	Number of hundredths of a second that a processor has been idle, averaged over all processors
AVG_BUSY_TIME	Number of hundredths of a second that a processor has been busy executing user or kernel code, averaged over all processors
AVG_USER_TIME	Number of hundredths of a second that a processor has been busy executing user code, averaged over all processors
AVG_SYS_TIME	Number of hundredths of a second that a processor has been busy executing kernel code, averaged over all processors
AVG_IOWAIT_TIME	Number of hundredths of a second that a processor has been waiting for I/O to complete, averaged over all processors

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 8-3 (Cont.) V$OSSTAT Statistics

Statistic Name	Description
AVG_NICE_TIME	Number of hundredths of a second that a processor has been busy executing low-priority user code, averaged over all processors
OS_CPU_WAIT_TIME	Total number of hundredths of a second that processes have been in a ready state, waiting to be selected by the operating system scheduler to run
RSRC_MGR_CPU_WAIT_TIME	Total number of hundredths of a second that Oracle processes have been in a ready state, waiting for CPU to be available for their consumer group in the currently active resource plan
VM_IN_BYTES	Total number of bytes of data that have been paged in due to virtual memory paging
VM_OUT_BYTES	Total number of bytes of data that have been paged out due to virtual memory paging
PHYSICAL_MEMORY_BYTES	Total number of bytes of physical memory
LOAD	Current number of processes that are either running or in the ready state, waiting to be selected by the operating-system scheduler to run. On many platforms, this statistic reflects the average load over the past minute.
NUM_CPU_CORES	Number of CPU cores available (includes subcores of multicore CPUs as well as single-core CPUs)
NUM_CPU_SOCKETS	Number of CPU sockets available (represents an absolute count of CPU chips on the system, regardless of multithreading or multi-core architectures)
NUM_VCPUS	Number of virtual CPUs available
NUM_LCPUS	Number of logical CPUs available (includes hardware threads if hardware threading is turned on)
TCP_SEND_SIZE_MIN	Minimum size of the TCP send buffer
TCP_SEND_SIZE_DEFAULT	Default size of the TCP send buffer
TCP_SEND_SIZE_MAX	Maximum size of the TCP send buffer
TCP_RECEIVE_SIZE_MIN	Minimum size of the TCP receive buffer
TCP_RECEIVE_SIZE_DEFAULT	Default size of the TCP receive buffer
TCP_RECEIVE_SIZE_MAX	Maximum size of the TCP receive buffer
GLOBAL_SEND_SIZE_MAX	Maximum size of the global send buffer
GLOBAL_RECEIVE_SIZE_MAX	Maximum size of the global receive buffer

> **Note**
>
> The availability of all statistics except for `NUM_CPUS` and `RSRC_MGR_CPU_WAIT_TIME` is subject to the operating system platform on which the Oracle Database is running.

8.107 V$PARALLEL_DEGREE_LIMIT_MTH

`V$PARALLEL_DEGREE_LIMIT_MTH` displays all available parallel degree limit resource allocation methods.

Column	Datatype	Description
`NAME`	`VARCHAR2(40)`	Name of the parallel degree limit resource allocation method

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.108 V$PARAMETER

V$PARAMETER displays information about the initialization parameters that are currently in effect for the session. A new session inherits parameter values from the instance-wide values displayed by the V$SYSTEM_PARAMETER view.

Column	Datatype	Description
NUM	NUMBER	Parameter number
NAME	VARCHAR2(80)	Name of the parameter
TYPE	NUMBER	Parameter type: • 1 - Boolean • 2 - String • 3 - Integer • 4 - Parameter file • 5 - Reserved • 6 - Big integer
VALUE	VARCHAR2(4000)	Parameter value for the session (if modified within the session); otherwise, the instance-wide parameter value
DISPLAY_VALUE	VARCHAR2(4000)	Parameter value in a user-friendly format. For example, if the VALUE column shows the value 262144 for a big integer parameter, then the DISPLAY_VALUE column will show the value 256K.
DEFAULT_VALUE	VARCHAR2(255)	The default value for this parameter. This is the value of the parameter if a value is not explicitly specified for the parameter.
ISDEFAULT	VARCHAR2(9)	Indicates whether the parameter is set to the default value (TRUE) or the parameter value was specified in the parameter file (FALSE). The database sets the value of the ISDEFAULT column to TRUE for parameters that are not specified in the init.ora or server parameter file (SPFILE).
ISSES_MODIFIABLE	VARCHAR2(5)	Indicates whether the parameter can be changed with ALTER SESSION (TRUE) or not (FALSE)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ISSYS_MODIFIABLE	VARCHAR2(9)	Indicates whether the parameter can be changed with ALTER SYSTEM and when the change takes effect: • IMMEDIATE - Parameter can be changed with ALTER SYSTEM regardless of the type of parameter file used to start the instance. The change takes effect immediately. • DEFERRED - Parameter can be changed with ALTER SYSTEM regardless of the type of parameter file used to start the instance. The change takes effect in subsequent sessions. • FALSE - Parameter cannot be changed with ALTER SYSTEM unless a server parameter file was used to start the instance. The change takes effect in subsequent instances.
ISPDB_MODIFIABLE	VARCHAR2(5)	Indicates whether the parameter can be modified inside a PDB (TRUE) or not (FALSE)
ISINSTANCE_MODIFIABLE	VARCHAR2(5)	For parameters that can be changed with ALTER SYSTEM, indicates whether the value of the parameter can be different for every instance (TRUE) or whether the parameter must have the same value for all Real Application Clusters instances (FALSE). If the ISSYS_MODIFIABLE column is FALSE, then this column is always FALSE.
ISMODIFIED	VARCHAR2(10)	Indicates whether the parameter has been modified after instance startup: • MODIFIED - Parameter has been modified with ALTER SESSION • SYSTEM_MOD - Parameter has been modified with ALTER SYSTEM (which causes all the currently logged in sessions' values to be modified) • FALSE - Parameter has not been modified after instance startup
ISADJUSTED	VARCHAR2(5)	Indicates whether Oracle adjusted the input value to a more suitable value (for example, the parameter value should be prime, but the user input a non-prime number, so Oracle adjusted the value to the next prime number)
ISDEPRECATED	VARCHAR2(5)	Indicates whether the parameter has been deprecated (TRUE) or not (FALSE)
ISBASIC	VARCHAR2(5)	Indicates whether the parameter is a basic parameter (TRUE) or not (FALSE)
DESCRIPTION	VARCHAR2(255)	Description of the parameter
UPDATE_COMMENT	VARCHAR2(255)	Comments associated with the most recent update
HASH	NUMBER	Hash value for the parameter name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Examples

The following query returns the default value for the ALLOW_GLOBAL_DBLINKS initialization parameter:

```
SQL> SELECT name, default_value FROM v$parameter
  2  WHERE name = 'allow_global_dblinks';
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

```
NAME
----------------------------------------------------------
DEFAULT_VALUE
----------------------------------------------------------
allow_global_dblinks
FALSE


SQL>
```

The following query shows that the `ALLOW_GLOBAL_DBLINKS` initialization parameter is not modifiable in a PDB:

```
SQL> SELECT name, ispdb_modifiable FROM v$parameter
  2  WHERE name = 'allow_global_dblinks';

NAME
----------------------------------------------------------
ISPDB
-----
allow_global_dblinks
FALSE


SQL>
```

> **See Also**
>
> "V$SYSTEM_PARAMETER"

8.109 V$PARAMETER_VALID_VALUES

`V$PARAMETER_VALID_VALUES` displays a list of valid values for list parameters.

Column	Datatype	Description
`NUM`	`NUMBER`	Parameter number
`NAME`	`VARCHAR2(64)`	Parameter name
`ORDINAL`	`NUMBER`	Ordinal number in the list (1-based)
`VALUE`	`VARCHAR2(255)`	Parameter value at ordinal
`ISDEFAULT`	`VARCHAR2(64)`	Indicates whether the given ordinal value is the default value for the parameter

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.110 V$PARAMETER2

V$PARAMETER2 displays information about the initialization parameters that are currently in effect for the session, with each list parameter value appearing as a row in the view. A new session inherits parameter values from the instance-wide values displayed in the V$SYSTEM_PARAMETER2 view.

Presenting the list parameter values in this format enables you to quickly determine the values for a list parameter. For example, if a parameter value is a, b, then the V$PARAMETER view does not tell you if the parameter has two values (both a and b) or one value (a, b). V$PARAMETER2 makes the distinction between the list parameter values clear.

Column	Datatype	Description
NUM	NUMBER	Parameter number
NAME	VARCHAR2(80)	Name of the parameter
TYPE	NUMBER	Parameter type: • 1 - Boolean • 2 - String • 3 - Integer • 4 - Parameter file • 5 - Reserved • 6 - Big integer
VALUE	VARCHAR2(4000)	Parameter value for the session (if modified within the session); otherwise, the instance-wide parameter value
DISPLAY_VALUE	VARCHAR2(4000)	Parameter value in a user-friendly format. For example, if the VALUE column shows the value 262144 for a big integer parameter, then the DISPLAY_VALUE column will show the value 256K.
ISDEFAULT	VARCHAR2(6)	Indicates whether the parameter is set to the default value (TRUE) or the parameter value was specified in the parameter file (FALSE)
ISSES_MODIFIABLE	VARCHAR2(5)	Indicates whether the parameter can be changed with ALTER SESSION (TRUE) or not (FALSE)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ISSYS_MODIFIABLE	VARCHAR2(9)	Indicates whether the parameter can be changed with ALTER SYSTEM and when the change takes effect: • IMMEDIATE - Parameter can be changed with ALTER SYSTEM regardless of the type of parameter file used to start the instance. The change takes effect immediately. • DEFERRED - Parameter can be changed with ALTER SYSTEM regardless of the type of parameter file used to start the instance. The change takes effect in subsequent sessions. • FALSE - Parameter cannot be changed with ALTER SYSTEM unless a server parameter file was used to start the instance. The change takes effect in subsequent instances.
ISPDB_MODIFIABLE	VARCHAR2(5)	Indicates whether the parameter can be modified inside a PDB (TRUE) or not (FALSE). In a non-CDB, the value of this column is NULL.
ISINSTANCE_MODIFIABLE	VARCHAR2(5)	For parameters that can be changed with ALTER SYSTEM, indicates whether the value of the parameter can be different for every instance (TRUE) or whether the parameter must have the same value for all Real Application Clusters instances (FALSE). If the ISSYS_MODIFIABLE column is FALSE, then this column is always FALSE.
ISMODIFIED	VARCHAR2(10)	Indicates whether the parameter has been modified after instance startup: • MODIFIED - Parameter has been modified with ALTER SESSION • SYSTEM_MOD - Parameter has been modified with ALTER SYSTEM (which causes all the currently logged in sessions' values to be modified) • FALSE - Parameter has not been modified after instance startup
ISADJUSTED	VARCHAR2(5)	Indicates whether Oracle adjusted the input value to a more suitable value (for example, the parameter value should be prime, but the user input a non-prime number, so Oracle adjusted the value to the next prime number)
ISDEPRECATED	VARCHAR2(5)	Indicates whether the parameter has been deprecated (TRUE) or not (FALSE)
ISBASIC	VARCHAR2(5)	Indicates whether the parameter is a basic parameter (TRUE) or not (FALSE)
DESCRIPTION	VARCHAR2(255)	Description of the parameter
ORDINAL	NUMBER	Position (ordinal number) of the parameter value. Useful only for parameters whose values are lists of strings.
UPDATE_COMMENT	VARCHAR2(255)	Comments associated with the most recent update
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "V$SYSTEM_PARAMETER"
- "V$PARAMETER"

8.111 V$PASSWORDFILE_INFO

V$PASSWORDFILE_INFO provides information about the database password file.

This view can be queried from the root or from a pluggable database (PDB) in a multitenant container database (CDB). When queried, this view always returns one row.

Column	Datatype	Description
FILE_NAME	VARCHAR2(513)	Fully qualified password file name/location.
FORMAT	VARCHAR2(6)	Shows the format of the password file: • LEGACY • 12 • 12.2
IS_ASM	VARCHAR2(5)	Indicates where the password file is stored. Possible values: • TRUE: The password file is stored in Oracle ASM. • FALSE: The password file is stored in the operating system file system.
CON_ID	NUMBER	The ID of the container to which the data pertains. Because there is only one password file for a CDB that is common for the entire CDB, the only value possible for CON_ID is 0 for this view.

Note

If the database password file name or location was recently changed, and you do not see the change reflected in this view, you can run the following SQL statement:

```
SQL> ALTER SYSTEM FLUSH PASSWORDFILE_METADATA_CACHE;
```

This statement flushes the metadata cache and updates the database to use the new password file. It also updates this view with the current password file information.

8.112 V$PATCHES

V$PATCHES shows the patches applied on an Oracle ASM instance and the list of patches applied to an Oracle Grid infrastructure home directory.

Note

If this view is queried in an RDBMS instance, no patch information is returned.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PATCH_ID	NUMBER	Patch identifier
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.113 V$PDB_INCARNATION

V$PDB_INCARNATION displays information about all PDB incarnations. Oracle creates a new PDB incarnation whenever a PDB is opened with the RESETLOGS option.

Column	Datatype	Description
DB_INCARNATION#	NUMBER	Database incarnation number
PDB_INCARNATION#	NUMBER	PDB incarnation number
STATUS	VARCHAR2(7)	Incarnation status: • ORPHAN: Orphan incarnation • CURRENT: Current incarnation of the PDB • PARENT: Parent of the current incarnation
INCARNATION_SCN	NUMBER	The SCN to flashback or recover to for this PDB incarnation
INCARNATION_TIME	DATE	The point in time recovered to for this PDB incarnation
BEGIN_RESETLOGS_SCN	NUMBER	The SCN at the beginning of PDB resetlogs
BEGIN_RESETLOGS_TIME	DATE	The time at the beginning of PDB resetlogs
END_RESETLOGS_SCN	NUMBER	The SCN at the end of PDB resetlogs
END_RESETLOGS_TIME	DATE	The time at the end of PDB resetlogs
PRIOR_DB_INCARNATION#	NUMBER	Parent database incarnation number
PRIOR_PDB_INCARNATION#	VARCHAR2(40)	Parent PDB incarnation number
FLASHBACK_DATABASE_ALLOWED	VARCHAR2(3)	Indicates whether the PDB can be flashbacked to this incarnation as part of a flashback database operation for the CDB
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.114 V$PDBS

V$PDBS displays information about PDBs associated with the current instance.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
DBID	NUMBER	PDB identifier calculated when the PDB is created and stored in all file headers associated with the PDB
CON_UID	NUMBER	Unique identifier associated with the PDB
GUID	RAW(16)	Globally unique identifier (GUID) of this PDB
NAME	VARCHAR2(128)	Name of the PDB
OPEN_MODE	VARCHAR2(10)	Open mode information. Possible values: • MOUNTED • READ WRITE • READ ONLY • MIGRATE
RESTRICTED	VARCHAR2(3)	Indicates whether only users possessing RESTRICTED SESSION privilege can connect to the PDB. Possible values: • YES • NO • NULL
OPEN_TIME	TIMESTAMP(3) WITH TIME ZONE	Date and time when the database was last opened
CREATE_SCN	NUMBER	System change number (SCN) for the creation of this PDB
TOTAL_SIZE	NUMBER	Shows the disk space (in bytes) used by the PDB, including both data and temp files.
BLOCK_SIZE	NUMBER	The current block size for the PDB
RECOVERY_STATUS	VARCHAR2(8)	Shows whether recovery is enabled or disabled for the PDB. Possible values: • ENABLED • DISABLED
SNAPSHOT_PARENT_CON_ID	NUMBER	This column shows the container ID of the PDB that this PDB is a snapshot clone of. This column shows a nonzero value only if the PDB is a snapshot clone. For all other cases, it shows a value of 0.
APPLICATION_ROOT	VARCHAR2(3)	Indicates whether the PDB is an application root
APPLICATION_PDB	VARCHAR2(3)	Indicates whether the PDB is an application PDB
APPLICATION_SEED	VARCHAR2(3)	Indicates whether the PDB is an application seed (an application seed is also an application PDB)
APPLICATION_ROOT_CON_ID	NUMBER	If this PDB is an application PDB, the container ID of an application root to which this application PDB belongs. If this PDB is an application root clone, the container ID of an application root to which this application root clone belongs. Otherwise, NULL.
APPLICATION_ROOT_CLONE	VARCHAR2(3)	Indicates whether this PDB is an application root clone (YES) or not (NO)
PROXY_PDB	VARCHAR2(3)	Indicates whether this PDB is a proxy PDB (YES) or not (NO)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LOCAL_UNDO	NUMBER	Shows whether the PDB is in local undo. Possible values: • 1 – PDB is in local undo mode • 0 – PDB is in shared undo mode This column is not relevant for CDB$ROOT.
UNDO_SCN	NUMBER	System change number (SCN) at which the PDB was last converted from shared to local undo, or from local to shared undo. This column is not relevant for CDB$ROOT.
UNDO_TIMESTAMP	DATE	Date and time at which the PDB was last converted from shared to local undo, or from local to shared undo. This column is not relevant for CDB$ROOT.
CREATION_TIME	DATE	Date and time at which the PDB was created.
DIAGNOSTICS_SIZE	NUMBER	Shows the current disk space usage (in bytes) of the diagnostic traces generated in the PDB, which is represented by the CON_ID column of the row
PDB_COUNT	NUMBER	The number of user-created PDBs belonging to a given application root or CDB$ROOT. For all other containers, its value is 0.
AUDIT_FILES_SIZE	NUMBER	Shows the current disk space usage (in bytes) by Unified Audit files (.bin format) in the current PDB.
MAX_SIZE	NUMBER	Shows the maximum amount of disk space (in bytes) that can be used by data and temp files in the PDB
MAX_DIAGNOSTICS_SIZE	NUMBER	Shows the maximum amount of disk space (in bytes) that can be used by diagnostic traces generated in the PDB
MAX_AUDIT_SIZE	NUMBER	Shows the maximum amount of disk space (in bytes) that can be used by Unified Audit files (.bin format) in the PDB
LAST_CHANGED_BY	VARCHAR2(11)	Indicates what type of user last changed the PDB. Possible values: • COMMON USER • LOCAL USER
TEMPLATE	VARCHAR2(3)	For internal use only
TENANT_ID	VARCHAR2(256)	Pluggable database tenant key
UPGRADE_LEVEL	NUMBER	For internal use only
GUID_BASE64	VARCHAR2(30)	The GUID of the PDB, encoded in base64

8.115 V$PERSISTENT_PUBLISHERS

V$PERSISTENT_PUBLISHERS displays information about all active publishers of the persistent queues in the database. There is one row per instance per queue per publisher. The rows are deleted when the database (or instance in an Oracle RAC environment) restarts.

Column	Datatype	Description
QUEUE_ID	NUMBER	Identifier for the queue
QUEUE_SCHEMA	VARCHAR2(128)	Owner of the queue
QUEUE_NAME	VARCHAR2(128)	Name of the queue
PUBLISHER_NAME	VARCHAR2(128)	Name of the agent enqueuing the message
PUBLISHER_ADDRESS	VARCHAR2(1024)	Address of the publisher agent
PROTOCOL	NUMBER	Protocol used by the publisher's address

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ENQUEUED_MSGS	NUMBER	Number of messages that have been enqueued
ELAPSED_ENQUEUE_TIME	NUMBER	Total time spent doing enqueue (in hundredths of a second)
ENQUEUE_CPU_TIME	NUMBER	Total CPU time for enqueue (in hundredths of a second)
LAST_ENQUEUE_TIME	TIMESTAMP(6)	Last enqueue message timestamp
ENQUEUE_TRANSACTIONS	NUMBER	Number of enqueue transactions
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.116 V$PERSISTENT_QMN_CACHE

V$PERSISTENT_QMN_CACHE displays detailed information and statistics about the background activities for all queue tables in the system. There is one row per queue table. The rows are deleted when the database (or instance in an Oracle RAC environment) restarts.

Column	Datatype	Description
QUEUE_TABLE_ID	NUMBER	Queue table object ID
TYPE	VARCHAR2(32)	Type of the queue table's queue monitor cache
STATUS	NUMBER	Status of the queue table's queue monitor cache
NEXT_SERVICE_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when the queue table should be serviced by QMON servers
WINDOW_END_TIME	TIMESTAMP(3) WITH TIME ZONE	Time manager activity period for non-owner queue table operations
TOTAL_RUNS	NUMBER	Total number of times this queue table is served
TOTAL_LATENCY	NUMBER	Cumulative latency in serving the queue table (in hundredths of a second)
TOTAL_ELAPSED_TIME	NUMBER	Total time spent in processing this queue table (in seconds)
TOTAL_CPU_TIME	NUMBER	Cumulative CPU time for serving the queue table (in hundredths of a second)
TMGR_ROWS_PROCESSED	NUMBER	Number of time manager entries processed
TMGR_ELAPSED_TIME	NUMBER	Cumulative time for time management activities (in hundredths of a second)
TMGR_CPU_TIME	NUMBER	Cumulative CPU time for time management activities (in hundredths of a second)
LAST_TMGR_PROCESSING_TIME	TIMESTAMP(3) WITH TIME ZONE	Last timer manager processing time
DEQLOG_ROWS_PROCESSED	NUMBER	Number of dequeue log entries processed
DEQLOG_PROCESSING_ELAPSED_TIME	NUMBER	Total time for processing dequeue log entries (in hundredths of a second)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DEQLOG_PROCESSING_CPU_TIME	NUMBER	Total CPU time for processing dequeue log entries (in hundredths of a second)
LAST_DEQLOG_PROCESSING_TIME	TIMESTAMP(3) WITH TIME ZONE	Last dequeue log processing time
DEQUEUE_INDEX_BLOCKS_FREED	NUMBER	Number of dequeue index blocks freed
HISTORY_INDEX_BLOCKS_FREED	NUMBER	Number of history index blocks freed
TIME_INDEX_BLOCKS_FREED	NUMBER	Number of time manager index blocks freed
INDEX_CLEANUP_COUNT	NUMBER	Number of times index block cleanup was attempted
INDEX_CLEANUP_ELAPSED_TIME	NUMBER	Total time for index block cleanup (in hundredths of a second)
INDEX_CLEANUP_CPU_TIME	NUMBER	Total CPU time for index block cleanup (in hundredths of a second)
LAST_INDEX_CLEANUP_TIME	TIMESTAMP(3) WITH TIME ZONE	Last index block cleanup time
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.117 V$PERSISTENT_QUEUES

V$PERSISTENT_QUEUES displays information about all active persistent queues in the database since the queues' first activity time. There is one row per queue. The rows are deleted when the database (or instance in an Oracle RAC environment) restarts.

Column	Datatype	Description
QUEUE_ID	NUMBER	Identifier for the queue
QUEUE_TABLE_ID	NUMBER	Queue table identifier
QUEUE_SCHEMA	VARCHAR2(128)	Owner of the queue
QUEUE_NAME	VARCHAR2(128)	Name of the queue
FIRST_ACTIVITY_TIME	TIMESTAMP(6)	First queue activity time since database startup
ENQUEUED_MSGS	NUMBER	Number of messages enqueued
DEQUEUED_MSGS	NUMBER	Number of messages dequeued **Note:** This column will not be incremented until all the subscribers of the message have dequeued the message and its retention time has elapsed.
BROWSED_MSGS	NUMBER	Number of messages that have been browsed
ELAPSED_ENQUEUE_TIME	NUMBER	Total time (in hundredths of a second) spent doing enqueue
ELAPSED_DEQUEUE_TIME	NUMBER	Total time (in hundredths of a second) spent doing dequeue
ENQUEUE_CPU_TIME	NUMBER	Total CPU time for enqueue (in hundredths of a second)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DEQUEUE_CPU_TIME	NUMBER	Total CPU time for dequeue (in hundredths of a second)
AVG_MSG_AGE	NUMBER	Average age of messages in the queue
DEQUEUED_MSG_LATENCY	NUMBER	Last dequeued message latency (in seconds)
ELAPSED_TRANSFORMATION_TIME	NUMBER	Total time (in hundredths of a second) spent doing transformation
ELAPSED_RULE_EVALUATION_TIME	NUMBER	Total time (in hundredths of a second) spent doing rule evaluation
ENQUEUED_EXPIRY_MSGS	NUMBER	Number of messages enqueued with expiry
ENQUEUED_DELAY_MSGS	NUMBER	Number of messages enqueued with delay
MSGS_MADE_EXPIRED	NUMBER	Number of messages expired by time manager
MSGS_MADE_READY	NUMBER	Number of messages made ready by time manager
LAST_ENQUEUE_TIME	TIMESTAMP(6)	Last message enqueue time
LAST_DEQUEUE_TIME	TIMESTAMP(6)	Last message dequeue time
LAST_TM_EXPIRY_TIME	TIMESTAMP(6)	Last time message was expired by time manager
LAST_TM_READY_TIME	TIMESTAMP(6)	Last time message was made ready by time manager
ENQUEUE_TRANSACTIONS	NUMBER	Number of enqueue transactions
DEQUEUE_TRANSACTIONS	NUMBER	Number of dequeue transactions
EXECUTION_COUNT	NUMBER	Number of executions of the dequeue cursor
OLDEST_MSGID	RAW(16)	Message ID of the oldest message in the queue
OLDEST_MSG_ENQTM	TIMESTAMP(6)	Enqueue time of the oldest message in the queue
MANDATORY_AFF_SWITCHES_OUT	NUMBER	An affinity switch is a change in dequeue instance for a shard-subscriber pair. A mandatory affinity switch is when there are local enqueues in the queue at the instance but no local dequeues present, so the dequeue affinity is switched to another instance for that shard-subscriber pair. This column shows the number of times mandatory affinity switches were needed from this instance to another for this queue.
OPTIONAL_AFF_SWITCHES_OUT	NUMBER	Optional affinity switches are affinity switches that are not mandatory. Optional affinity switches are done for global load balancing across the Oracle Real Application Clusters (Oracle RAC) database. This column shows the number of times optional affinity switches were needed from this instance to another for this queue.
AFF_SWITCHES_BACK_IN	NUMBER	The number of times dequeue affinities have come back from other instances to this instance. (MANDATORY_AFF_SWITCHES_OUT + OPTIONAL_AFF_SWITCHES_OUT - AFF_SWITCHES_BACK_IN) is the number of cross instance affinities present across all shard-subscriber pair for shards owned by this instance for this queue.
CROSS_STREAM_JOBS	NUMBER	The number of times a shard is being forwarded to another instance due to cross instance dequeues for this queue
RESTORE_BITMAP_JOBS	NUMBER	The number of times subscribers used existing shard forwarding to have cross instance dequeues for this queue
SHADOW_AFF_SWITCHES_IN	NUMBER	The number of affinity switches for this queue where this instance is the dequeue instance for a shard-subscriber pair where the shard is being enqueued at another instance

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SHADOW_AFF_SWITCHES_OUT	NUMBER	The number of affinity switches for this queue where shadow affinity is switched back to source instance of the shard. (SHADOW_AFF_SWITCHES_IN - SHADOW_AFF_SWITCHES_OUT) is the number of dequeue affinities which are performing cross instance dequeues from non-local shards.
SHADOW_SHARDS_RECEIVED	NUMBER	The number of times a shard is being forwarded from another instance to this instance due to cross instance dequeues for this queue
SHADOW_SHARDS_FREED	NUMBER	The number of times a forwarded shard to this instance was stopped due to removal of cross instance dequeues for this queue
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **Note**
>
> For sharded queues, only the following columns in this view contain accurate information: FIRST_ACTIVITY_TIME, BROWSED_MSGS, LAST_ENQUEUE_TIME, LAST_DEQUEUE_TIME, ENQUEUE_TRANSACTIONS, and DEQUEUE_TRANSACTIONS. The rest of the columns in this view should be ignored when querying information about sharded queues.

8.118 V$PERSISTENT_SUBSCRIBERS

V$PERSISTENT_SUBSCRIBERS displays information about all active subscribers of the persistent queues in the database.

There is one row per instance per queue per subscriber. The rows are deleted when the database (or instance in an Oracle RAC environment) restarts.

> **Note**
>
> This view does not display information about sharded queues. For information about sharded queues, refer to the "V$AQ_SHARDED_SUBSCRIBER_STAT" view.

Column	Datatype	Description
QUEUE_ID	NUMBER	Identifier for the queue
QUEUE_SCHEMA	VARCHAR2(128)	Owner of the queue
QUEUE_NAME	VARCHAR2(128)	Name of the queue
SUBSCRIBER_ID	NUMBER	Internal subscriber number
SUBSCRIBER_NAME	VARCHAR2(512)	Name of the subscriber
SUBSCRIBER_ADDRESS	VARCHAR2(1024)	Address of the subscribing agent

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PROTOCOL	NUMBER	Protocol of the subscribing agent
SUBSCRIBER_TYPE	VARCHAR2(128)	Type of the subscriber: • PROXY - Propagation subscriber • SUBSCRIBER - Normal subscriber • RECIPIENT - Recipient
FIRST_ACTIVITY_TIME	TIMESTAMP(6)	First subscriber activity time since database startup
ENQUEUED_MSGS	NUMBER	Number of messages enqueued since FIRST_ACTIVITY_TIME
DEQUEUED_MSGS	NUMBER	Number of messages dequeued since FIRST_ACTIVITY_TIME
AVG_MSG_AGE	NUMBER	Average age of messages in the queue
BROWSED_MSGS	NUMBER	Number of messages that have been browsed
EXPIRED_MSGS	NUMBER	Number of messages expired since FIRST_ACTIVITY_TIME
DEQUEUED_MSG_LATENCY	NUMBER	Last dequeued message latency (in seconds)
LAST_ENQUEUE_TIME	TIMESTAMP(6)	Timestamp of the last enqueued message
LAST_DEQUEUE_TIME	TIMESTAMP(6)	Timestamp of the last dequeued message
ELAPSED_DEQUEUE_TIME	NUMBER	Total time spent in dequeue (in hundredths of a second)
DEQUEUE_CPU_TIME	NUMBER	Total CPU time for dequeue (in hundredths of a second)
DEQUEUE_TRANSACTIONS	NUMBER	Number of dequeue transactions
EXECUTION_COUNT	NUMBER	Number of executions of the dequeue index cursor
DEQUEUE_MEMORY_LOCKS	NUMBER	Number of dequeue transactions that obtained memory locks
DEQUEUE_DISK_LOCKS	NUMBER	Number of dequeue transactions that obtained disk locks
DEQUEUE_DISK_DELETES	NUMBER	Number of dequeue transactions that deleted index-organized table entries
OLDEST_MSGID	RAW(16)	Message ID of the oldest message
OLDEST_MSG_ENQTM	TIMESTAMP(6)	Enqueue time of the oldest message
PARENT_SUBSCRIBER_ID	NUMBER	Subscriber ID of the parent durable subscriber
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.119 V$PGA_TARGET_ADVICE

V$PGA_TARGET_ADVICE predicts how the cache hit percentage and over allocation count statistics displayed by the V$PGASTAT performance view would be impacted if the value of the PGA_AGGREGATE_TARGET parameter is changed.

The prediction is performed for various values of the PGA_AGGREGATE_TARGET parameter, selected around its current value. The advice statistic is generated by simulating the past workload run by the instance.

The content of the view is empty if PGA_AGGREGATE_TARGET is not set. In addition, the content of this view is not updated if the STATISTICS_LEVEL parameter is set to BASIC. Base statistics for

Copyright © 2002, 2026, Oracle and/or its affiliates.

this view are reset at instance startup and when the value of the PGA_AGGREGATE_TARGET initialization parameter is dynamically modified.

Column	Datatype	Description
PGA_TARGET_FOR_ESTIMATE	NUMBER	Value of PGA_AGGREGATE_TARGET for this prediction (in bytes)
PGA_TARGET_FACTOR	NUMBER	PGA_TARGET_FOR_ESTIMATE / the current value of the PGA_AGGREGATE_TARGET parameter
ADVICE_STATUS	VARCHAR2(3)	Indicates whether the advice is enabled (ON) or disabled (OFF) depending on the value of the STATISTICS_LEVEL parameter
BYTES_PROCESSED	NUMBER	Total bytes processed by all the work areas considered by this advice (in bytes)
ESTD_TIME	NUMBER	Time (in seconds) required to process the bytes
ESTD_EXTRA_BYTES_RW	NUMBER	Estimated number of extra bytes which would be read or written if PGA_AGGREGATE_TARGET was set to the value of the PGA_TARGET_FOR_ESTIMATE column. This number is derived from the estimated number and size of work areas which would run in one-pass (or multi-pass) for that value of PGA_AGGREGATE_TARGET.
ESTD_PGA_CACHE_HIT_PERCENTAGE	NUMBER	Estimated value of the cache hit percentage statistic when PGA_AGGREGATE_TARGET equals PGA_TARGET_FOR_ESTIMATE. This column is derived from the above two columns and is equal to BYTES_PROCESSED / (BYTES_PROCESSED + ESTD_EXTRA_BYTES_RW)
ESTD_OVERALLOC_COUNT	NUMBER	Estimated number of PGA memory over-allocations if the value of PGA_AGGREGATE_TARGET is set to PGA_TARGET_FOR_ESTIMATE. A nonzero value means that PGA_TARGET_FOR_ESTIMATE is not large enough to run the work area workload. Hence, the DBA should not set PGA_AGGREGATE_TARGET to PGA_TARGET_FOR_ESTIMATE since Oracle will not be able to honor that target.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$PGASTAT"
- "PGA_AGGREGATE_TARGET"
- "STATISTICS_LEVEL"
- *Oracle Database Performance Tuning Guide* for information on tuning the PGA_AGGREGATE_TARGET initialization parameter using the PGA advice views

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.120 V$PGA_TARGET_ADVICE_HISTOGRAM

V$PGA_TARGET_ADVICE_HISTOGRAM predicts how statistics displayed by the V$SQL_WORKAREA_HISTOGRAM dynamic view would be impacted if the value of the PGA_AGGREGATE_TARGET parameter is changed.

This prediction is performed for various values of the PGA_AGGREGATE_TARGET parameter, selected around its current value. The advice statistic is generated by simulating the past workload run by the instance.

The content of the view is empty if PGA_AGGREGATE_TARGET is not set. In addition, the content of this view is not updated when the STATISTICS_LEVEL initialization parameter is set to BASIC. Base statistics for this view are reset at instance startup or when the value of the PGA_AGGREGATE_TARGET initialization parameter is dynamically modified.

Column	Datatype	Description
PGA_TARGET_FOR_ESTIMATE	NUMBER	Value of PGA_AGGREGATE_TARGET for this prediction (in bytes)
PGA_TARGET_FACTOR	NUMBER	PGA_TARGET_FOR_ESTIMATE / the current value of the PGA_AGGREGATE_TARGET parameter
ADVICE_STATUS	VARCHAR2(3)	Indicates whether the advice is enabled (ON) or disabled (OFF) depending on the value of the STATISTICS_LEVEL parameter
LOW_OPTIMAL_SIZE	NUMBER	Lower bound for the optimal memory requirement of work areas included in this row (in bytes)
HIGH_OPTIMAL_SIZE	NUMBER	Upper bound for the optimal memory requirement of work areas included in this row (in bytes)
ESTD_OPTIMAL_EXECUTIONS	NUMBER	Number of work areas with an optimal memory requirement comprised between LOW_OPTIMAL_SIZE and HIGH_OPTIMAL_SIZE which are predicted to run optimal given a value of PGA_AGGREGATE_TARGET equal to PGA_TARGET_FOR_ESTIMATE
ESTD_ONEPASS_EXECUTIONS	NUMBER	Number of work areas with an optimal memory requirement comprised between LOW_OPTIMAL_SIZE and HIGH_OPTIMAL_SIZE which are predicted to run one-pass given a value of PGA_AGGREGATE_TARGET equal to PGA_TARGET_FOR_ESTIMATE
ESTD_MULTIPASSES_EXECUTIONS	NUMBER	Number of work areas with an optimal memory requirement comprised between LOW_OPTIMAL_SIZE and HIGH_OPTIMAL_SIZE which are predicted to run multi-pass given a value of PGA_AGGREGATE_TARGET equal to PGA_TARGET_FOR_ESTIMATE
ESTD_TOTAL_EXECUTIONS	NUMBER	Sum of ESTD_OPTIMAL_EXECUTIONS, ESTD_ONEPASS_EXECUTIONS, and ESTD_MULTIPASSES_EXECUTIONS
IGNORED_WORKAREAS_COUNT	NUMBER	Number of work areas with optimal memory requirement between LOW_OPTIMAL_SIZE and HIGH_OPTIMAL_SIZE ignored in the advice generation due to memory and CPU constraints
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "V$SQL_WORKAREA_HISTOGRAM"
- "PGA_AGGREGATE_TARGET"
- "STATISTICS_LEVEL"
- *Oracle Database Performance Tuning Guide* for information on tuning the PGA_AGGREGATE_TARGET initialization parameter using the PGA advice views

8.121 V$PGASTAT

V$PGASTAT displays PGA memory usage statistics as well as statistics about the automatic PGA memory manager when it is enabled (that is, when PGA_AGGREGATE_TARGET is set). Cumulative values in V$PGASTAT are accumulated since instance startup.

Column	Datatype	Description
NAME	VARCHAR2(64)	Name of the statistic (see Table 8-4)
VALUE	NUMBER	Statistic value
UNIT	VARCHAR2(12)	Unit for the value: • bytes • microseconds • percent
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Table 8-4 V$PGASTAT Statistics

Statistic Name	Description
aggregate PGA auto target	Amount of PGA memory the Oracle Database can use for work areas running in automatic mode. This amount is dynamically derived from the value of the PGA_AGGREGATE_TARGET initialization parameter and the current work area workload, and continuously adjusted by the Oracle Database. If this value is small compared to the value of PGA_AGGREGATE_TARGET, then a large amount of PGA memory is used by other components of the system (for example, PL/SQL or Java memory) and little is left for work areas. The DBA must ensure that enough PGA memory is left for work areas running in automatic mode.
aggregate PGA target parameter	Current value of the PGA_AGGREGATE_TARGET initialization parameter. If this parameter is not set, then its value is 0 and automatic management of PGA memory is disabled.
bytes processed	Number of bytes processed by memory intensive SQL operators, cumulated since instance startup.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 8-4 (Cont.) V$PGASTAT Statistics

Statistic Name	Description
cache hit percentage	A metric computed by the Oracle Database to reflect the performance of the PGA memory component, cumulative since instance startup. A value of 100% means that all work areas executed by the system since instance startup have used an optimal amount of PGA memory. When a work area cannot run optimal, one or more extra passes is performed over the input data. This will reduce the cache hit percentage in proportion to the size of the input data and the number of extra passes performed.
extra bytes read/written	Number of bytes processed during extra passes of the input data, cumulated since instance startup. When a work area cannot run optimal, one or more of these extra passes is performed.
global memory bound	Maximum size of a work area executed in automatic mode. This value is continuously adjusted by the Oracle Database to reflect the current state of the work area workload. The global memory bound generally decreases when the number of active work areas is increasing in the system. If the value of the global bound decreases below 1 MB, then the value of PGA_AGGREGATE_TARGET should be increased.
max processes count	Maximum number of processes active at any one time since instance startup.
maximum PGA allocated	Maximum number of bytes of PGA memory allocated at one time since instance startup.
maximum PGA used for auto workareas	Maximum amount of PGA memory consumed at one time by work areas running under the automatic memory management mode since instance startup.
maximum PGA used for manual workareas	Maximum amount of PGA memory consumed at one time by work areas running under the manual memory management mode since instance startup.
over allocation count	This statistic is cumulative since instance startup. Over allocating PGA memory can happen if the value of PGA_AGGREGATE_TARGET is too small. When this happens, the Oracle Database cannot honor the value of PGA_AGGREGATE_TARGET and extra PGA memory needs to be allocated. If over allocation occurs, then increase the value of PGA_AGGREGATE_TARGET using the information provided by the V$PGA_TARGET_ADVICE view.
PGA memory freed back to OS	Number of bytes of PGA memory freed back to the operating system, cumulated since instance startup.
process count	Number of processes active within up to the last 3 seconds.
recompute count (total)	Number of times the instance bound, which is a cap on the maximum size of each active work area, has been recomputed since instance startup. Generally, the instance bound is recomputed in the background every 3 seconds, but it could be recomputed by a foreground process when the number of work areas changes rapidly in a short period of time.
total freeable PGA memory	Number of bytes of PGA memory in all processes that could be freed back to the operating system.
total PGA allocated	Current amount of PGA memory allocated by the instance. The Oracle Database attempts to keep this number below the value of the PGA_AGGREGATE_TARGET initialization parameter. However, it is possible for the PGA allocated to exceed that value by a small percentage and for a short period of time when the work area workload is increasing very rapidly or when PGA_AGGREGATE_TARGET is set to a small value.
total PGA inuse	Indicates how much PGA memory is currently consumed by work areas. This number can be used to determine how much memory is consumed by other consumers of the PGA memory (for example, PL/SQL or Java).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 8-4 (Cont.) V$PGASTAT Statistics

Statistic Name	Description
total PGA used for auto workareas	Indicates how much PGA memory is currently consumed by work areas running under the automatic memory management mode. This number can be used to determine how much memory is consumed by other consumers of the PGA memory (for example, PL/SQL or Java).
total PGA used for manual workareas	Indicates how much PGA memory is currently consumed by work areas running under the manual memory management mode. This number can be used to determine how much memory is consumed by other consumers of the PGA memory (for example, PL/SQL or Java).

See Also

"PGA_AGGREGATE_TARGET"

8.122 V$PLSQL_DEBUGGABLE_SESSIONS

`V$PLSQL_DEBUGGABLE_SESSIONS` shows the current sessions of all users that the current user has privileges to debug with a PL/SQL debugger.

Column	Datatype	Description
`SID`	`NUMBER`	Session identifier
`SERIAL#`	`NUMBER`	Session serial number
`LOGON_TIME`	`DATE`	Time of logon
`USER#`	`NUMBER`	Oracle user identifier
`USERNAME`	`VARCHAR2(128)`	Oracle user name
`OSUSER`	`VARCHAR2(128)`	Operating system client user name
`PROCESS`	`VARCHAR2(24)`	Operating system client process ID
`MACHINE`	`VARCHAR2(64)`	Operating system client machine name
`PORT`	`NUMBER`	Operating system client port number
`TERMINAL`	`VARCHAR2(30)`	Operating system client terminal name
`PROGRAM`	`VARCHAR2(48)`	Operating system client program name
`TYPE`	`VARCHAR2(10)`	Session type
`SERVICE_NAME`	`VARCHAR2(64)`	Service name of the session
`PLSQL_DEBUGGER_CONNECTED`	`VARCHAR2(5)`	Indicates whether the session is connected to a PL/SQL debugger. Possible values: • `TRUE` • `FALSE`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.123 V$PQ_SESSTAT

V$PQ_SESSTAT lists session statistics for parallel queries. After you have run a query or DML operation, you can use the information derived from V$PQ_SESSTAT to view the number of worker processes used, and other information for the session and system.

Column	Datatype	Description
STATISTIC	VARCHAR2(30)	Name of the statistic: • Allocation Height - Requested number of servers per instance • Allocation Width - Requested number of instances • DDL Parallelized - Number of DDL operations run in parallel • DFO Trees - Number of executed DFO trees • DML Parallelized - Number of DML operations run in parallel • DOP - Degree of parallelism used for the last statement • Distr Msgs Recv'd - Number of remote (inter-instance) messages received • Distr Msgs Sent - Number of remote (inter-instance) messages sent • Local Msgs Recv'd - Number of local (intra-instance) messages received • Local Msgs Sent - Number of local (intra-instance) messages sent • Queries Parallelized - Number of queries run in parallel • Server Threads - Number of PX servers used • Slave Sets - Total number of worker sets used for the last statement. For multiple parallelizers in the parallel plan, it is the cumulative count.
LAST_QUERY	NUMBER	Value of the statistic for the last operation
SESSION_TOTAL	NUMBER	Value of the statistic for the entire session to this point in time. The value of the statistic for the entire session to this point in time is not maintained for the DOP, Server Threads, Allocation Height, Allocation Width, and Slave Set statistics.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.124 V$PQ_SLAVE

V$PQ_SLAVE lists statistics for each of the active parallel execution servers on an instance.

Column	Datatype	Description
SLAVE_NAME	VARCHAR2(4)	Name of the parallel execution server
STATUS	VARCHAR2(4)	Current status of the parallel execution server: • BUSY • IDLE
SESSIONS	NUMBER	Number of sessions that have used this parallel execution server
IDLE_TIME_CUR	NUMBER	Amount of time spent idle while processing statements in the current session
BUSY_TIME_CUR	NUMBER	Amount of time spent busy while processing statements in the current session
CPU_SECS_CUR	NUMBER	Amount of CPU time spent on the current session
MSGS_SENT_CUR	NUMBER	Number of messages sent while processing statements for the current session
MSGS_RCVD_CUR	NUMBER	Number of messages received while processing statements for the current session
IDLE_TIME_TOTAL	NUMBER	Total amount of time this query server has been idle
BUSY_TIME_TOTAL	NUMBER	Total amount of time this query server has been active
CPU_SECS_TOTAL	NUMBER	Total amount of CPU time this query server has used to process statements
MSGS_SENT_TOTAL	NUMBER	Total number of messages this query server has sent
MSGS_RCVD_TOTAL	NUMBER	Total number of messages this query server has received
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.125 V$PQ_SYSSTAT

V$PQ_SYSSTAT lists system statistics for parallel queries.

After you have run a query or DML operation, you can use the information derived from V$PQ_SYSSTAT to view the number of PX (Parallel Execution) servers used, and other information for the system.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATISTIC	VARCHAR2(30)	Name of the statistic: • DDL Initiated - Total number of parallel DDL operations that were initiated • DFO Trees - Total number of DFO trees executed on this instance • DML Initiated - Total number of parallel DML operations that were initiated • Distr Msgs Recv'd - Total number of remote (inter-instance) messages received on this instance • Distr Msgs Sent - Total number of remote (inter-instance) messages sent on this instance • Local Msgs Recv'd - Total number of remote (inter-instance) messages received on this instance • Local Msgs Sent - Total number of local (intra-instance) messages sent on this instance • Queries Initiated - Total number of parallel queries initiated on this instance • Queries Queued - Total number of queued queries on this instance • Server Sessions - Total number of operations executed in all PX servers on this instance • Servers Busy - Number of currently busy PX servers on this instance • Servers Cleaned Up - Total number of PX servers on this instance cleaned up due to process death • Servers Highwater - Number of active PX servers on this instance that have partaken in >= 1 operation so far • Servers Idle - Number of currently idle PX servers on this instance • Servers Shutdown - Total number of PX servers shutdown on this instance • Servers Started - Total number of PX servers started on this instance • Sessions Active - Total number of active sessions on this instance
VALUE	NUMBER	Value of the statistic
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.126 V$PQ_TQSTAT

V$PQ_TQSTAT contains statistics on parallel execution operations. The statistics are compiled after the query completes and only remain for the duration of the session. It displays the number of rows processed through each parallel execution server at each stage of the execution tree. This view can help determine skew problems in a query's execution. (Note that for PDML, information from V$PQ_TQSTAT is available only after a commit or rollback operation.)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DFO_NUMBER	NUMBER	Data flow operator (DFO) tree number to differentiate queries
TQ_ID	NUMBER	Table queue ID within the query, which represents the connection between two DFO nodes in the query execution tree
SERVER_TYPE	VARCHAR2(10)	The role in table queue - producer/consumer/ranger
NUM_ROWS	NUMBER	The number of rows produced/consumed
BYTES	NUMBER	The number of bytes produced/consumed
OPEN_TIME	NUMBER	Time (seconds) the table queue remained open
AVG_LATENCY	NUMBER	Time (minutes) for a message to be dequeued after it enters the queue
WAITS	NUMBER	The number of waits encountered during dequeue
TIMEOUTS	NUMBER	The number of timeouts when waiting for a message
PROCESS	VARCHAR2(6)	Process ID
INSTANCE	NUMBER	Instance ID
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.127 V$PROCESS

V$PROCESS displays information about the currently active processes.

Column	Datatype	Description
ADDR	RAW(4 | 8)	Address of the process state object
PID	NUMBER	Oracle process identifier
SOSID	VARCHAR2(24)	Operating system (process, thread) identifier. This identifier is unique whether the Oracle multiprocess/multithread feature is enabled or not.
SPID	VARCHAR2(24)	Operating system process identifier. The Oracle multiprocess/multithread feature is available for UNIX systems. When the Oracle multiprocess/multithread feature is enabled, RDBMS processes are mapped to threads running in operating system processes, and the SPID identifier is not unique for RDBMS processes. When the Oracle multiprocess/multithread feature is not enabled on UNIX systems, the SPID identifier is unique for RDBMS processes.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STID	VARCHAR2(24)	Operating system thread identifier. The Oracle multiprocess/multithread feature is available for UNIX systems. When the Oracle multiprocess/multithread feature is enabled, RDBMS processes are mapped to threads running in operating system processes, and the SPID and STID together uniquely identify an RDBMS process. The STID is not unique on Solaris. The STID is unique on Linux and AIX.
EXECUTION_TYPE	VARCHAR2(10)	Operating system execution type
PNAME	VARCHAR2(5)	Name of this process
USERNAME	VARCHAR2(15)	Operating system process username
SERIAL#	NUMBER	Process serial number
TERMINAL	VARCHAR2(30)	Operating system terminal identifier
PROGRAM	VARCHAR2(48)	Program in progress
TRACEID	VARCHAR2(255)	Trace file identifier
TRACEFILE	VARCHAR2(513)	Trace file name of the process
BACKGROUND	VARCHAR2(1)	1 for a SYSTEM background process; NULL for foreground processes or non-SYSTEM background processes
LATCHWAIT	VARCHAR2(16)	Address of the latch the process is waiting for; NULL if none
LATCHSPIN	VARCHAR2(16)	This column is obsolete
PGA_USED_MEM	NUMBER	PGA memory currently used by the process (in bytes)
PGA_ALLOC_MEM	NUMBER	PGA memory currently allocated by the process (including free PGA memory not yet released to the operating system by the server process), in bytes
PGA_FREEABLE_MEM	NUMBER	Allocated PGA memory which can be freed (in bytes)
PGA_MAX_MEM	NUMBER	Maximum PGA memory ever allocated by the process (in bytes)
NUMA_DEFAULT	NUMBER	The NUMA processor group of this process at initialization time
NUMA_CURR	NUMBER	The NUMA processor group of this process currently
CPU_USED	NUMBER	Tracks the CPU used by this process (in microseconds) from the time it was spawned
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.128 V$PROCESS_MEMORY

V$PROCESS_MEMORY displays dynamic PGA memory usage by named component categories for each process.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PID	NUMBER	Oracle process identifier
SERIAL#	NUMBER	Oracle process serial number
CATEGORY	VARCHAR2(15)	Category name. Categories include "SQL", "PL/SQL", "OLAP" and "JAVA". Special categories are "Freeable" and "Other". Freeable memory has been allocated to the process by the operating system, but has not been allocated to a category. "Other" memory has been allocated to a category, but not to one of the named categories.
ALLOCATED	NUMBER	Bytes of PGA memory allocated by the process for the category. For the "Freeable" category, it is the amount of free PGA memory eligible to be released to the operating system.
USED	NUMBER	Bytes of PGA memory used by the process for the category. For "Freeable", the value is zero. For "Other", the value is NULL for performance reasons.
MAX_ALLOCATED	NUMBER	Maximum bytes of PGA memory ever allocated by the process for the category.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.129 V$PROCESS_MEMORY_DETAIL

V$PROCESS_MEMORY_DETAIL provides detailed information on dynamic PGA memory usage for each automatically captured snapshot.

Column	Datatype	Description
PID	NUMBER	Oracle process identifier
SERIAL#	NUMBER	Oracle process serial number
CATEGORY	VARCHAR2(15)	Category name. Categories include: • SQL • PL/SQL • OLAP • JAVA • Freeable • Other
NAME	VARCHAR2(26)	PGA memory allocation comment. Small allocations may be grouped together with NAME set to Miscellaneous for performance reasons.
HEAP_NAME	VARCHAR2(15)	Name of heap or heaps (if same name) containing the allocations
BYTES	NUMBER	Bytes of PGA memory allocated in the process from heaps with the given heap name and with the given allocation comment
ALLOCATION_COUNT	NUMBER	Number of allocations with the comment found in the process inside heaps with the given heap name
HEAP_DESCRIPTOR	RAW(4 | 8)	If all the allocations are from one heap, then this is the address of the heap descriptor for that heap. Otherwise, this column is NULL.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PARENT_HEAP_DESCRIPTOR	RAW(4 | 8)	If all the allocations are from one heap, then this is the address of the parent heap descriptor for that heap. Otherwise, this column is NULL. If the heap has no parent, the value is zero.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.130 V$PROCESS_POOL

V$PROCESS_POOL provides information about process pools.

Column	Datatype	Description
POOL_NAME	VARCHAR2(512)	Pool name
ENABLED	VARCHAR2(5)	Indicates whether the pool is active (TRUE) or not (FALSE)
MIN_COUNT	NUMBER	The default or configured minimum value
BATCH_COUNT	NUMBER	The default or configured batch count
INIT_COUNT	NUMBER	The default or configured initial count
CUR_COUNT	NUMBER	The number of spawned processes available in this pool
MAX_COUNT	NUMBER	When INIT_COUNT is set, MAX_COUNT shows the number of processes yet to be spawned. Note that the current process count (CUR_COUNT) cannot reach INIT_COUNT in cases where processes are consumed during the spawn.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This row is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.131 V$PROPAGATION_RECEIVER

V$PROPAGATION_RECEIVER displays information about buffer queue propagation schedules on the receiving (destination) side. The values are reset to zero when the database (or instance in an Oracle Real Application Clusters (Oracle RAC) environment) restarts, when propagation migrates to another instance, or when an unscheduled propagation is attempted.

Column	Datatype	Description
SRC_QUEUE_SCHEMA	VARCHAR2(128)	Name of the source schema
SRC_QUEUE_NAME	VARCHAR2(128)	Name of the source queue

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SRC_DBNAME	VARCHAR2(395)	Name of the source database
DST_QUEUE_SCHEMA	VARCHAR2(128)	Name of the destination schema
DST_QUEUE_NAME	VARCHAR2(128)	Name of the destination queue
STARTUP_TIME	DATE	Startup time of this schedule. This time changes when the source/ destination database gets restarted.
HIGH_WATER_MARK	NUMBER	High watermark of the messages received
ACKNOWLEDGEMENT	NUMBER	Acknowledgement of the messages received by the receiver
LAST_RECEIVED_MSG	NUMBER	Last received message
TOTAL_MSGS	NUMBER	Total number of messages
ELAPSED_UNPICKLE_TIME	NUMBER	Elapsed unpickle time
ELAPSED_RULE_TIME	NUMBER	Elapsed rule time
ELAPSED_ENQUEUE_TIME	NUMBER	Elapsed enqueue time
SESSION_ID	NUMBER	Session ID of the propagation receiver
SERIAL#	NUMBER	Serial number of the propagation receiver
SPID	VARCHAR2(24)	Process identification number of the propagation receiver
PROPAGATION_NAME	VARCHAR2(128)	Name of the propagation on the source database
STATE	VARCHAR2(43)	State of the propagation receiver: • Initializing • Sending unapplied txns • Waiting for message from client • Receiving LCRs • Evaluating rules • Enqueueing LCRS • Waiting for memory • Waiting for apply to read • Waiting for message from propagation sender When the propagation schedule is not optimized, the state is Normal.
LAST_RECEIVED_MSG_POSITION	RAW(64)	Last received message position. Corresponds to LAST_RECEIVED_MSG, except the value is in position rather than SCN. Position is used by XStream to determine ordering.
ACKNOWLEDGEMENT_POSITION	RAW(64)	Acknowledgement position of the messages received by the receiver. Corresponds to ACKNOWLEDGEMENT, except the value is in position rather than SCN. Position is used by XStream to determine ordering.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.132 V$PROPAGATION_SENDER

V$PROPAGATION_SENDER displays information about buffer queue propagation schedules on the sending (source) side. The values are reset to zero when the database (or instance in an

Copyright © 2002, 2026, Oracle and/or its affiliates.

Oracle Real Application Clusters (Oracle RAC) environment) restarts, when propagation migrates to another instance, or when an unscheduled propagation is attempted.

Column	Datatype	Description
QUEUE_ID	NUMBER	Queue identifier of the queue
QUEUE_SCHEMA	VARCHAR2(128)	Schema of the queue
QUEUE_NAME	VARCHAR2(128)	Name of the queue
DST_QUEUE_SCHEMA	VARCHAR2(128)	Destination schema of the queue
DST_QUEUE_NAME	VARCHAR2(128)	Name of the destination queue
STARTUP_TIME	DATE	Time at which the propagation started
DBLINK	VARCHAR2(395)	Name of the destination database link
HIGH_WATER_MARK	NUMBER	High watermark of the messages sent
ACKNOWLEDGEMENT	NUMBER	Acknowledgement of the messages received by the receiver
SCHEDULE_STATUS	VARCHAR2(128)	Status of the propagation schedule
TOTAL_MSGS	NUMBER	Total messages propagated
TOTAL_BYTES	NUMBER	Total bytes propagated
ELAPSED_DEQUEUE_TIME	NUMBER	Elapsed dequeue time (in hundredths of a second)
ELAPSED_PICKLE_TIME	NUMBER	Elapsed pickle time (time taken to linearize a logical change record (LCR) into a stream of bytes that can be sent over the network) (in hundredths of a second)
ELAPSED_PROPAGATION_TIME	NUMBER	Elapsed propagation time (in hundredths of a second)
ELAPSED_RULE_TIME	NUMBER	Elapsed rule time (in hundredths of a second)
MAX_NUM_PER_WIN	NUMBER	Maximum bytes per window
MAX_SIZE	NUMBER	Maximum bytes sent per window
LAST_MSG_LATENCY	NUMBER	Last propagated message latency
LAST_MSG_ENQUEUE_TIME	TIMESTAMP(6)	Last propagated message enqueue time
LAST_MSG_PROPAGATION_TIME	TIMESTAMP(6)	Last time when the message was propagated
LAST_LCR_LATENCY	NUMBER	Last propagated LCR latency
LAST_LCR_CREATION_TIME	DATE	Last propagated LCR timestamp
LAST_LCR_PROPAGATION_TIME	DATE	Last time when the LCR was propagated
DST_DATABASE_NAME	VARCHAR2(395)	Global name of the destination database
SESSION_ID	NUMBER	Session ID of the propagation sender process
SERIAL#	NUMBER	Serial number of the propagation sender process
SPID	VARCHAR2(24)	Process identification number of the propagation sender process
PROPAGATION_NAME	VARCHAR2(128)	Name of the propagation

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATE	VARCHAR2(128)	State of the propagation sender process: • Initializing • Initializing propagation receiver • Browsing LCRs • Evaluating rules • Dequeueing LCRs • Sending LCRs • Waiting for apply to be enabled • Waiting for apply database to start • Waiting for propagation to be enabled • Waiting for capture to terminate • Waiting for a subscriber to be added • Suspended due to a dropped subscriber • Suspended for auto split/merge • Waiting on empty queue When the SCHEDULE_STATUS column is not SCHEDULE OPTIMIZED, the state is the value of the SCHEDULE_STATUS column.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.133 V$PROXY_ARCHIVEDLOG

V$PROXY_ARCHIVEDLOG contains descriptions of archived log backups that were taken using the proxy copy functionality.

In a proxy copy, the media manager takes over the operations of backing up and restoring data. Each row represents a backup of one control file.

Column	Datatype	Description
RECID	NUMBER	Proxy copy record identifier
STAMP	NUMBER	Proxy copy stamp
DEVICE_TYPE	VARCHAR2(17)	Type of media device that stores the proxy copy
HANDLE	VARCHAR2(513)	Name or "handle" for the proxy copy
COMMENTS	VARCHAR2(81)	Comments about the proxy copy
MEDIA	VARCHAR2(65)	A comment that contains further information about the media manager that created this backup
MEDIA_POOL	NUMBER	Number of the media pool in which the proxy copy is stored
TAG	VARCHAR2(32)	Tag for the proxy copy

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(1)	Status of the backup set: • A - Available • U - Unavailable • X - Expired • D - Deleted
DELETED	VARCHAR2(3)	Indicates whether this record has been deleted (YES) or not (NO)
THREAD#	NUMBER	Number of the redo thread
SEQUENCE#	NUMBER	Log sequence number
RESETLOGS_CHANGE#	NUMBER	RESETLOGS SCN of the database incarnation to which this archived log belongs
RESETLOGS_TIME	DATE	RESETLOGS time stamp of the database incarnation to which this archived log belongs
FIRST_CHANGE#	NUMBER	First SCN of this redo log
FIRST_TIME	DATE	Time when Oracle switched into the redo log
NEXT_CHANGE#	NUMBER	First SCN of the next redo log in the thread
NEXT_TIME	DATE	First time stamp of the next redo log in the thread
BLOCKS	NUMBER	Size of this archived redo log (in operating system blocks)
BLOCK_SIZE	NUMBER	Block size for the copy (in bytes)
START_TIME	DATE	Time when the proxy copy was initiated
COMPLETION_TIME	DATE	Time when the proxy copy was completed
ELAPSED_SECONDS	NUMBER	Duration of the proxy copy
RMAN_STATUS_RECID	NUMBER	Owning V$RMAN_STATUS record ID
RMAN_STATUS_STAMP	NUMBER	Owning V$RMAN_STATUS stamp
TERMINAL	VARCHAR2(3)	Indicates whether this record corresponds to a terminal archived redo log, as defined in V$ARCHIVED_LOG (YES) or not (NO)
KEEP	VARCHAR2(3)	Indicates whether this backup set has a retention policy that is different than the value for the configure retention policy (YES) or not (NO)
KEEP_UNTIL	DATE	If specified, then this is the date after which the backup becomes obsolete. If this column is NULL, then the backup never expires.
KEEP_OPTIONS	VARCHAR2(11)	Additional retention options for this backup set: • LOGS - Indicates a long-term backup made with the LOGS keyword, which is now deprecated • BACKUP_LOGS - Indicates that the backup was made in open mode, so archived log backups must be applied to make it consistent • NOLOGS - Indicates a consistent backup made when the database was mounted • NULL - Indicates that this backup has no KEEP options and becomes obsolete based on the retention policy
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.134 V$PROXY_ARCHIVELOG_DETAILS

V$PROXY_ARCHIVELOG_DETAILS contains information about all available archive log proxy copies.

Column	Datatype	Description
SESSION_KEY	NUMBER	Session identifier
SESSION_RECID	NUMBER	Session recid
SESSION_STAMP	NUMBER	Session stamp
COPY_KEY	NUMBER	Copy identifier
THREAD#	NUMBER	Redo thread number
SEQUENCE#	NUMBER	Redo log sequence number
RESETLOGS_CHANGE#	NUMBER	Resetlogs change number of the database when this log was written
RESETLOGS_TIME	DATE	Resetlogs time of the database when this log was written
HANDLE	VARCHAR2(513)	Proxy copy handle identifies the copy for restore
MEDIA	VARCHAR2(65)	Name of the media on which the copy resides. This value is informational only. It is not needed for restore.
MEDIA_POOL	NUMBER	Media pool in which the copy resides. This is the same value that was entered in the POOL operand of the Recovery Manager BACKUP command.
TAG	VARCHAR2(32)	Proxy copy tag
FIRST_CHANGE#	NUMBER	First change number in the archived log
NEXT_CHANGE#	NUMBER	First change number in the next log
FIRST_TIME	DATE	Timestamp of the first change
NEXT_TIME	DATE	Timestamp of the next change
OUTPUT_BYTES	NUMBER	Total output bytes written
COMPLETION_TIME	DATE	Completion time
OUTPUT_BYTES_DISPLAY	VARCHAR2(3)	Displayable format for output bytes
KEEP	DATE	Indicates whether this backup set has a retention policy that is different than the value for the configure retention policy (YES) or not (NO)
KEEP_UNTIL	VARCHAR2(11)	If specified, then this is the date after which the backup becomes obsolete. If this column is NULL, then the backup never expires.
KEEP_OPTIONS	VARCHAR2(4000)	Additional retention options for this backup set: • LOGS - Indicates a long-term backup made with the LOGS keyword, which is now deprecated • BACKUP_LOGS - Indicates that the backup was made in open mode, so archived log backups must be applied to make it consistent • NOLOGS - Indicates a consistent backup made when the database was mounted • NULL - Indicates that this backup has no KEEP options and becomes obsolete based on the retention policy

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.135 V$PROXY_ARCHIVELOG_SUMMARY

V$PROXY_ARCHIVELOG_SUMMARY provides summary information about the output proxy archive log file.

Column	Datatype	Description
NUM_FILES_BACKED	NUMBER	Number of archived log files backed up
NUM_DISTINCT_FILES_BACKED	NUMBER	Number of distinct archived log files backed up
MIN_FIRST_CHANGE#	NUMBER	Minimum first change SCN
MAX_NEXT_CHANGE#	NUMBER	Maximum first change SCN
MIN_FIRST_TIME	DATE	Minimum first change time. Forms the redo range, along with MAX_NEXT_TIME.
MAX_NEXT_TIME	DATE	Maximum next change time
OUTPUT_BYTES	NUMBER	Total output size, in bytes
OUTPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for output bytes
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.136 V$PROXY_COPY_DETAILS

V$PROXY_COPY_DETAILS contains information about all available control file and datafile proxy copies.

Column	Datatype	Description
SESSION_KEY	NUMBER	Session identifier
SESSION_RECID	NUMBER	Session recid
SESSION_STAMP	NUMBER	Session stamp
COPY_KEY	NUMBER	Copy identifier
FILE#	NUMBER	Absolute datafile number, or 0 if this is a control file backup

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
HANDLE	VARCHAR2(513)	Proxy copy handle identifies the copy for restore
MEDIA	VARCHAR2(65)	Name of the media on which the copy resides. This value is informational only. It is not needed for restore.
MEDIA_POOL	NUMBER	Media pool in which the copy resides. This is the same value that was entered in the POOL operand of the Recovery Manager BACKUP command.
TAG	VARCHAR2(32)	Proxy copy tag
CREATION_CHANGE#	NUMBER	Datafile creation change number
CREATION_TIME	DATE	Datafile creation timestamp
CHECKPOINT_CHANGE#	NUMBER	Checkpoint change number of the datafile when the copy was made
CHECKPOINT_TIME	DATE	Checkpoint timestamp of the datafile when the copy was made
OUTPUT_BYTES	NUMBER	Total output bytes written
COMPLETION_TIME	DATE	Completion time
CONTROLFILE_TYPE	VARCHAR2(1)	Type of control file: • B - Normal control file • S - Standby control file
KEEP	VARCHAR2(3)	Indicates whether this backup set has a retention policy that is different than the value for the configure retention policy (YES) or not (NO)
KEEP_UNTIL	DATE	If specified, then this is the date after which the backup becomes obsolete. If this column is NULL, then the backup never expires.
KEEP_OPTIONS	VARCHAR2(11)	Additional retention options for this backup set: • LOGS - Indicates a long-term backup made with the LOGS keyword, which is now deprecated • BACKUP_LOGS - Indicates that the backup was made in open mode, so archived log backups must be applied to make it consistent • NOLOGS - Indicates a consistent backup made when the database was mounted. • NULL - Indicates that this backup has no KEEP options and becomes obsolete based on the retention policy
OUTPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for output bytes
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.137 V$PROXY_COPY_SUMMARY

V$PROXY_COPY_SUMMARY provides summary information about the output proxy datafile and control file.

Column	Datatype	Description
NUM_COPIES	NUMBER	Number of copies created

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NUM_DISTINCT_COPIES	NUMBER	Number of distinct copies (that contain datafiles with different checkpoints)
MIN_CHECKPOINT_CHANGE#	NUMBER	Minimum checkpoint change SCN
MAX_CHECKPOINT_CHANGE#	NUMBER	Maximum checkpoint change SCN
MIN_CHECKPOINT_TIME	DATE	Minimum checkpoint change time
MAX_CHECKPOINT_TIME	DATE	Maximum checkpoint change time
OUTPUT_BYTES	NUMBER	Total output bytes
OUTPUT_BYTES_DISPLAY	VARCHAR2(4000)	Displayable format for output bytes
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.138 V$PROXY_DATAFILE

V$PROXY_DATAFILE contains descriptions of datafile and control file backups that are taken with Proxy Copy. Each row represents a backup of one database file.

Column	Datatype	Description
RECID	NUMBER	Proxy copy record ID
STAMP	NUMBER	Proxy copy record stamp
DEVICE_TYPE	VARCHAR2(17)	Type of the device on which the copy resides
HANDLE	VARCHAR2(513)	Proxy copy handle identifies the copy for restore
COMMENTS	VARCHAR2(81)	Comment returned by the operating system or storage subsystem. This value is informational only. It is not needed for restore.
MEDIA	VARCHAR2(65)	Name of the media on which the copy resides. This value is informational only. It is not needed for restore.
MEDIA_POOL	NUMBER	Media pool in which the copy resides. This is the same value that was entered in the POOL operand of the Recovery Manager BACKUP command
TAG	VARCHAR2(32)	Proxy copy tag
STATUS	VARCHAR2(1)	Status of the backup set: • A - Available • U - Unavailable • X - Expired • D - Deleted
DELETED	VARCHAR2(3)	Indicates whether this record has been deleted (YES) or not (NO)
FILE#	NUMBER	Absolute datafile number, or 0 if this is a control file backup
CREATION_CHANGE#	NUMBER	Datafile creation change number
CREATION_TIME	DATE	Datafile creation Timestamp
RESETLOGS_CHANGE#	NUMBER	Resetlogs change number of the datafile when the copy was made

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RESETLOGS_TIME	DATE	Resetlogs timestamp of the datafile when the copy was made
CHECKPOINT_CHANGE#	NUMBER	Checkpoint change number of the datafile when the copy was made
CHECKPOINT_TIME	DATE	Checkpoint timestamp of the datafile when the copy was made
ABSOLUTE_FUZZY_CHANGE#	NUMBER	Highest change in any block of the file, if known
RECOVERY_FUZZY_CHANGE#	NUMBER	Highest change written to the file by media recovery
RECOVERY_FUZZY_TIME	DATE	Timestamp of the highest change written to the file by media recovery
INCREMENTAL_LEVEL	NUMBER	If this backup is part of an incremental backup strategy, then 0. Otherwise null.
ONLINE_FUZZY	VARCHAR2(3)	Indicates whether this copy was made after a crash or offline immediate (or is a copy of a copy which was taken improperly while the database was open) (YES) or not (NO). Recovery will need to apply all redo up to the next crash recovery marker to make the file consistent.
BACKUP_FUZZY	VARCHAR2(3)	Indicates whether this is a copy taken using the BEGIN BACKUP | END BACKUP technique (YES) or not (NO). The BEGIN BACKUP | END BACKUP technique is used internally when proxy copies of open files are created. Recovery will need to apply all redo up to the end backup marker to make this copy consistent.
BLOCKS	NUMBER	Size of the copy (in blocks). Also the size of the datafile when the copy was made.
BLOCK_SIZE	NUMBER	Block size of the datafile
OLDEST_OFFLINE_RANGE	NUMBER	If the file number is 0 (that is, this is a control file backup), the RECID of the oldest offline range record in this control file copy. 0 for datafile copies.
START_TIME	DATE	Starting time
COMPLETION_TIME	DATE	Completion time
ELAPSED_SECONDS	NUMBER	Number of elapsed seconds
CONTROLFILE_TYPE	VARCHAR2(1)	Type of control file: • B - Normal control file • S - Standby control file
KEEP	VARCHAR2(3)	Indicates whether this backup set has a retention policy that is different than the value for the configure retention policy (YES) or not (NO)
KEEP_UNTIL	DATE	If specified, then this is the date after which the backup becomes obsolete. If this column is NULL, then the backup never expires.
KEEP_OPTIONS	VARCHAR2(11)	Additional retention options for this backup set: • LOGS - Indicates a long-term backup made with the LOGS keyword, which is now deprecated • BACKUP_LOGS - Indicates that the backup was made in open mode, so archived log backups must be applied to make it consistent • NOLOGS - Indicates a consistent backup made when the database was mounted. • NULL - Indicates that this backup has no KEEP options and becomes obsolete based on the retention policy
RMAN_STATUS_RECID	NUMBER	Owning V$RMAN_STATUS record ID
RMAN_STATUS_STAMP	NUMBER	Owning V$RMAN_STATUS stamp
FOREIGN_DBID	NUMBER	Foreign DBID of the database from which this datafile was transported. The value is 0 if the file backed up is not a foreign database file.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PLUGGED_READONLY	VARCHAR2(3)	Indicates whether this is a proxy copy of a transported read-only foreign file (YES) or not (NO)
PLUGIN_CHANGE#	NUMBER	SCN at which the foreign datafile was transported into the database. The value is 0 if this file is not a foreign database file.
PLUGIN_RESETLOGS_CHANGE#	NUMBER	SCN of the RESETLOGS operation for the incarnation into which this foreign file was transported. The value is 0 if this file is not a foreign database file.
PLUGIN_RESETLOGS_TIME	DATE	Time of the RESETLOGS operation for the incarnation into which this foreign file was transported. The value is 0 if this file is not a foreign database file.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
BACKED_BY_PDB	VARCHAR2(3)	Recovery Manager (RMAN) allows a PDB to be backed up in two ways. The value in this column indicates how the PDB backup was taken: • YES: The backup was taken when connected to the PDB • NO: The backup was taken when connected to the root container
GUID	RAW(16)	The GUID of the PDB to which the backup belongs. This is useful after the PDB is dropped to identify which PDB the backup belongs to.

8.139 V$PROXY_PDB_TARGETS

V$PROXY_PDB_TARGETS provides information about the target of a proxy PDB.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: *n*: Where *n* is the container ID of a proxy PDB
TARGET_PORT	NUMBER	Port number that the target of the proxy PDB has registered with the listener, and which will be used by the proxy PDB to connect to the target
TARGET_HOST	VARCHAR2(128)	Host name where the target of the proxy PDB is running, and which will be used by the proxy PDB to connect to the target
TARGET_SERVICE	VARCHAR2(128)	Service name that the target of the proxy PDB has registered with the listener, and which will be used by the proxy PDB to connect to the target
TARGET_USER	VARCHAR2(128)	User name used by the proxy PDB to connect to the target of the proxy PDB. If null, the name of the connected user will be used.

See Also

Oracle Multitenant Administrator's Guide for information about creating proxy PDBs

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.140 V$PWFILE_USERS

V$PWFILE_USERS lists all users in the password file, and indicates whether the user has been granted the SYSDBA, SYSOPER, SYSASM, SYSBACKUP, SYSDG, and SYSKM privileges.

Column	Datatype	Description
USERNAME	VARCHAR2(128)	Name of the user that is contained in the password file
SYSDBA	VARCHAR2(5)	Indicates whether the user can connect with SYSDBA privileges (TRUE) or not (FALSE)
SYSOPER	VARCHAR2(5)	Indicates whether the user can connect with SYSOPER privileges (TRUE) or not (FALSE)
SYSASM	VARCHAR2(5)	Indicates whether the user can connect with SYSASM privileges (TRUE) or not (FALSE)
SYSBACKUP	VARCHAR2(5)	Indicates whether the user can connect with SYSBACKUP privileges (TRUE) or not (FALSE)
SYSDG	VARCHAR2(5)	Indicates whether the user can connect with SYSDG privileges (TRUE) or not (FALSE)
SYSKM	VARCHAR2(5)	Indicates whether the user can connect with SYSKM privileges (TRUE) or not (FALSE)
ACCOUNT_STATUS	VARCHAR2(30)	Account status: • OPEN • EXPIRED • EXPIRED (GRACE) • LOCKED (TIMED) • LOCKED • EXPIRED & LOCKED (TIMED) • EXPIRED & LOCKED • EXPIRED (GRACE) & LOCKED
PASSWORD_PROFILE	VARCHAR2(128)	Password profile name
LAST_LOGIN	TIMESTAMP(9) WITH TIME ZONE	The time of the last user login This column is not populated when a user connects to the database with administrative privileges, that is, AS { SYSASM | SYSBACKUP | SYSDBA | SYSDG | SYSOPER | SYSRAC | SYSKM }. The exception is as follows: If a user connects to the database with administrative privileges from a *remote* session, and the database password file format is 12.2 (V$PASSWORDFILE_INFO.FORMAT = 12.2), then this column is populated.
LOCK_DATE	DATE	Date the account was locked if account status was LOCKED
EXPIRY_DATE	DATE	Date of expiration of the account
EXTERNAL_NAME	VARCHAR2(1024)	Shows Certificate DN or Principal Name of externally authenticated users
AUTHENTICATION_TYPE	VARCHAR2(8)	Indicates the authentication mechanism for the user: • EXTERNAL - CREATE USER *user1* IDENTIFIED EXTERNALLY; • GLOBAL - CREATE USER *user2* IDENTIFIED GLOBALLY; • PASSWORD - CREATE USER *user3* IDENTIFIED BY *password*;
COMMON	VARCHAR2(3)	This column has a value of YES if an administrative privilege (for example, SYSDBA) was granted with CONTAINER=ALL. Otherwise, the column has a value of NO.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.141 V$PX_INSTANCE_GROUP

V$PX_INSTANCE_GROUP provides information about the instance groups being used for parallel operations by the current session.

Column	Datatype	Description
QC_INSTANCE_GROUP	VARCHAR2(64)	The instance group being used by this session for parallel operations. This value repeats for every row returned.
WHY	VARCHAR2(23)	Where the current instance group name comes from, as follows: SERVICE - the instance group being used is from the session's service name PARALLEL_INSTANCE_GROUP - the instance group being used is being used because the PARALLEL_INSTANCE_GROUP initialization parameter has been set. The value repeats for every row returned.
INSTANCE_NUMBER	NUMBER	Instance number of the instance providing this instance group. There will be one row for each instance in the instance group that the Query Coordinator is using.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"PARALLEL_INSTANCE_GROUP"

8.142 V$PX_PROCESS

V$PX_PROCESS contains information about the sessions running parallel execution.

Column	Datatype	Description
SERVER_NAME	VARCHAR2(4)	The name of the PX server (P000, P001, and so on)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(9)	The state of the PX server (IN USE or AVAILABLE)
PID	NUMBER	The process identifier
SPID	VARCHAR2(24)	Operating system process ID
SID	NUMBER	The session ID of the PX server, if in use
SERIAL#	NUMBER	The session serial number of the PX server, if in use
IS_GV	VARCHAR2(5)	Indicates whether a PX server process in parallel is a normal process or a GV$ process. Possible values: • FALSE: The PX server process in parallel is a normal process • TRUE: The PX server process in parallel is a GV$ process
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.143 V$PX_PROCESS_SYSSTAT

V$PX_PROCESS_SYSSTAT contains information about the sessions running parallel execution.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATISTIC	VARCHAR2(30)	Name of the statistic: • Servers In Use - Number of PX servers currently performing parallel operations • Servers Available - Number of PX servers available to perform parallel operations • Servers Started - Number of times the system has had to create a PX server process • Servers Shutdown - Number of times a PX server process has been shut down. A PX server process will be shut down if it has not been used recently. If this value is large, then consider increasing the parameter. This will improve performance by avoiding the latency of PX server process creation. • Servers HWM - Maximum number of concurrent PX server processes If this number is equal to the PARALLEL_MAX_SERVERS initialization parameter, then consider increasing the parameter. This could allow you to increase your throughput, especially if your system is under-utilized and the V$SYSSTAT statistic "Parallel operations downgraded to serial" is large. • Servers Cleaned Up - Number of times PMON had to clean up a PX server. This should only happen during nonstandard termination of a parallel operation. If this number is large, then you should determine the cause. • Server Sessions - Total number of sessions created by all PX servers • Memory Chunks Allocated - Number of large memory chunks allocated by PX servers • Memory Chunks Freed - Number of large memory chunks freed • Memory Chunks Current - Number of large memory chunks currently being used • Memory Chunks HWM - Maximum number of concurrently allocated chunks • Buffers allocated - Number of times a message buffer has been allocated • Buffers freed - Number of times a message buffer has been freed • Buffers Current - Number of message buffers currently being used • Buffers HWM - Maximum number of concurrently allocated message buffers
VALUE	NUMBER	Value of the statistic
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.144 V$PX_SESSION

V$PX_SESSION contains information about the sessions running parallel execution.

Column	Datatype	Description
SADDR	RAW(4 | 8)	Session address
SID	NUMBER	Session identifier
SERIAL#	NUMBER	Session serial number
QCSID	NUMBER	Session identifier of the parallel coordinator
QCSERIAL#	NUMBER	Session serial number of the parallel coordinator
QCINST_ID	NUMBER	Instance number on which the parallel coordinator is running
SERVER_GROUP	NUMBER	The logical group of servers to which this cluster database process belongs
SERVER_SET	NUMBER	The logical set of servers to which this cluster database process belongs. A single server group will have at most two server sets.
SERVER#	NUMBER	The logical number of the cluster database process within a server set
DEGREE	NUMBER	The degree of parallelism being used by the server set
REQ_DEGREE	NUMBER	The degree of parallelism that was requested by the user when the statement was issued and before any resource, multiuser, or load balancing reductions
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.145 V$PX_SESSTAT

V$PX_SESSTAT contains information about the sessions running parallel execution.

Column	Datatype	Description
SADDR	RAW(4 | 8)	Session address
SID	NUMBER	Session identifier
SERIAL#	NUMBER	Session serial number
QCSID	NUMBER	Session identifier of the parallel coordinator
QCSERIAL#	NUMBER	Session serial number of the parallel coordinator
QCINST_ID	NUMBER	Instance number on which the parallel coordinator is running
SERVER_GROUP	NUMBER	The logical group of servers to which this cluster database process belongs
SERVER_SET	NUMBER	The logical set of servers that this cluster database process belongs to. A single server group will have at most two server sets.
SERVER#	NUMBER	The logical number of the cluster database process within a server set

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DEGREE	NUMBER	The degree of parallelism being used by the server set
REQ_DEGREE	NUMBER	The degree of parallelism that was requested by the user when the statement was issued and before any resource, multiuser, or load balancing reductions
STATISTIC#	NUMBER	Statistic number
VALUE	NUMBER	Statistic value
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.146 V$QMON_COORDINATOR_STATS

V$QMON_COORDINATOR_STATS displays statistics of the non-sharded queue primary process. There is one row per instance. The rows are deleted when the database (or instance in an Oracle RAC environment) restarts.

Column	Datatype	Description
QMNC_PID	VARCHAR2(24)	Non-sharded queue primary process ID
STATUS	VARCHAR2(24)	Current status of the coordinator: • DEAD • INITIALIZING • RUNNING TASK COORDINATOR • WAITING • ADDING SERVER
NUM_SERVERS	NUMBER	Number of QMON servers currently running
LAST_SERVER_START_TIME	TIMESTAMP(3) WITH TIME ZONE	Last server startup time
LAST_SERVER_PID	VARCHAR2(24)	Process ID of the last server process (Qnnn) created
NEXT_WAKEUP_TIME	TIMESTAMP(3) WITH TIME ZONE	Next wakeup time of the coordinator
NEXT_READY_TIME	TIMESTAMP(3) WITH TIME ZONE	Ready time of the first delayed task
NEXT_EXPIRY_TIME	TIMESTAMP(3) WITH TIME ZONE	Expiry time of the next ready task
LAST_WAIT_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when the coordinator went to sleep
LAST_FAILURE	VARCHAR2(32)	Last failure encountered
LAST_FAILURE_TIME	TIMESTAMP(3) WITH TIME ZONE	Last failure time
MAX_TASK_LATENCY	VARCHAR2(40)	Maximum task latency across all the servers (in seconds)
MIN_TASK_LATENCY	VARCHAR2(40)	Minimum task latency across all the servers (in seconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOTAL_TASK_LATENCY	NUMBER	Cumulative latency across all the tasks (in seconds)
TOTAL_TASKS_EXECUTED	NUMBER	Cumulative number of tasks serviced by all the servers
MAX_SERVERS	NUMBER	Maximum number of servers present at any point of time
CON_ID	NUMBER	The ID of the container to which the data pertains. The CON_ID value in this view is always 0. The rows pertain to the entire CDB or to the non-CDB.

8.147 V$QMON_SERVER_STATS

V$QMON_SERVER_STATS displays information and statistics about the active queue monitor server processes. There is one row per live queue monitor server process. The rows are deleted when the database (or instance in an Oracle RAC environment) restarts.

Column	Datatype	Description
QMNC_PID	VARCHAR2(24)	Non-sharded queue primary process ID
SERVER_PID	VARCHAR2(24)	Process ID of the server
SERVER_NAME	VARCHAR2(48)	Name of the server
STATUS	VARCHAR2(40)	Current state of the server: • UNUSED • RUNNING • IDLE WAIT • EXITING • NOT ACTIVE
SERVER_START_TIME	TIMESTAMP(3) WITH TIME ZONE	Start time of the server
TASK_NAME	VARCHAR2(32)	Current executing task
TASK_NUMBER	NUMBER	Unique task number of the running task
TASK_START_TIME	TIMESTAMP(3) WITH TIME ZONE	Start time of the running task
LAST_WAIT_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when the server last waited
MAX_LATENCY	NUMBER	Maximum task latency for this server (in seconds)
MIN_LATENCY	NUMBER	Minimum task latency for this server (in seconds)
TOTAL_LATENCY	NUMBER	Cumulative task latency for this server (in seconds)
NUM_TASKS	NUMBER	Number of tasks processed by the server
LAST_FAILURE	VARCHAR2(128)	Last failure encountered by the server
LAST_FAILURE_TIME	TIMESTAMP(3) WITH TIME ZONE	Last failure time
LAST_FAILURE_TASK	VARCHAR2(32)	Task being run at the time of the last failure
LAST_FAILURE_TASKNUM	NUMBER	Unique task number of the failed task
CON_ID	NUMBER	The ID of the container to which the data pertains. The CON_ID value in this view is always 0. The rows pertain to the entire CDB or to the non-CDB.

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.148 V$QMON_TASK_STATS

V$QMON_TASK_STATS displays information and statistics based on different queue monitor tasks in the system (spilling, time manager activity, and so on). There is one row per kind of task. The rows are deleted when the database (or instance in an Oracle RAC environment) restarts.

Column	Datatype	Description
TASK_NAME	VARCHAR2(32)	Name of the task
TASK_TYPE	VARCHAR2(40)	Type of the task
LAST_CREATED_TASKNUM	NUMBER	Unique task number last created for this task
NUM_TASKS	NUMBER	Number of tasks currently present
TOTAL_TASK_RUN_TIME	NUMBER	Cumulative task run time
TOTAL_TASK_RUNS	NUMBER	Cumulative task runs
TOTAL_TASK_FAILURES	NUMBER	Cumulative failures
METRIC_TYPE	VARCHAR2(50)	Type of metric gathered for this task type
METRIC_VALUE	NUMBER	Value of this metric
LAST_FAILURE	VARCHAR2(32)	Last failure encountered while executing this type of task
LAST_FAILURE_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when the last failure occurred
LAST_FAILURE_TASKNUM	NUMBER	Task number of the last failed task for this task
REMARK	VARCHAR2(64)	Remarks about the task
CON_ID	NUMBER	The ID of the container to which the data pertains. The CON_ID value in this view is always 0. The rows pertain to the entire CDB or to the non-CDB.

8.149 V$QMON_TASKS

V$QMON_TASKS displays information and statistics about all queue background tasks in the system, which would be served by queue monitor servers. There is one row per task. The rows are deleted when the database (or instance in an Oracle RAC environment) restarts.

Column	Datatype	Description
TASK_NAME	VARCHAR2(32)	Task name
TASK_NUMBER	NUMBER	Unique task number
TASK_TYPE	VARCHAR2(40)	Task type
TASK_SUBMIT_TIME	TIMESTAMP(3) WITH TIME ZONE	Task submit time
TASK_READY_TIME	TIMESTAMP(3) WITH TIME ZONE	Task ready time
TASK_EXPIRY_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when this task expires
TASK_START_TIME	TIMESTAMP(3) WITH TIME ZONE	Last actual start time for the task
TASK_STATUS	VARCHAR2(32)	Status of the task
SERVER_NAME	VARCHAR2(48)	Name of the QMON server running this task

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MAX_RETRIES	NUMBER	Maximum retry count for the task
NUM_RUNS	NUMBER	Number of runs of the task if repeatable
NUM_FAILURES	NUMBER	Number of failures encountered while running the task
CON_ID	NUMBER	The ID of the container to which the data pertains. The CON_ID value in this view is always 0. The rows pertain to the entire CDB or to the non-CDB.

8.150 V$QUARANTINE

V$QUARANTINE provides information about quarantined objects.

Column	Datatype	Description
OBJECT	VARCHAR2(64)	Type of quarantined object (description of the memory)
ADDRESS	RAW(8)	Address of the object
BYTES	NUMBER	Amount of memory used by the object (this is the minimum amount; the actual amount may be unknown)
ERROR	VARCHAR2(128)	Oracle error causing quarantine
TIMESTAMP	TIMESTAMP(6) WITH TIME ZONE	Time the object was quarantined
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.151 V$QUARANTINE_SUMMARY

V$QUARANTINE_SUMMARY provides a summary of quarantine for each pluggable database (PDB).

Column	Datatype	Description
OBJECT_COUNT	NUMBER	Number of objects in quarantine
OBJECT_LIMIT	NUMBER	Limit on the number of objects that can be in quarantine before a termination will occur
MEMORY_TOTAL	NUMBER	Number of bytes in quarantine
MEMORY_LIMIT	NUMBER	Limit on the amount of memory that can be in quarantine before a termination will occur
RECENT_COUNT	NUMBER	Number of objects recently placed in quarantine (within the last hour)
RECENT_LIMIT	NUMBER	Limit on the number of recently quarantined objects before a termination will occur

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$QUARANTINE" for additional details on objects recently placed into quarantine

8.152 V$QUEUE

V$QUEUE contains information on the shared server message queues.

Column	Datatype	Description
PADDR	RAW(4 | 8)	Address of the process that owns the queue
TYPE	VARCHAR2(10)	Type of queue: • COMMON - Processed by servers • DISPATCHER
QUEUED	NUMBER	Number of items in the queue
WAIT	NUMBER	Total time that all items in this queue have waited (in hundredths of a second). Divide by TOTALQ for average wait per item.
TOTALQ	NUMBER	Total number of items that have ever been in the queue
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.153 V$QUEUEING_MTH

V$QUEUEING_MTH displays all available queuing resource allocation methods.

Column	Datatype	Description
NAME	VARCHAR2(40)	Name of the queuing resource allocation method

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.154 V$RECOVER_FILE

V$RECOVER_FILE displays the status of files needing media recovery.

Column	Datatype	Description
FILE#	NUMBER	File identifier number
ONLINE	VARCHAR2(7)	This column is obsolete and maintained for backward compatibility. The value of this column is always equal to the value in ONLINE_STATUS.
ONLINE_STATUS	VARCHAR2(7)	Online status (ONLINE, OFFLINE)
ERROR	VARCHAR2(18)	Why the file must be recovered: NULL if reason unknown, or OFFLINE NORMAL if recovery not needed
CHANGE#	NUMBER	SCN where recovery must start
TIME	DATE	Time of SCN when recovery must start
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.155 V$RECOVERY_AREA_USAGE

V$RECOVERY_AREA_USAGE displays usage information about recovery areas.

Column	Datatype	Description
FILE_TYPE	VARCHAR2(23)	File type: • CONTROL FILE • REDO LOG • ARCHIVED LOG • BACKUP PIECE • IMAGE COPY • FLASHBACK LOG • REMOTE ARCHIVED LOG
PERCENT_SPACE_USED	NUMBER	Percent of the recovery area that is in use
PERCENT_SPACE_RECLAIMABLE	NUMBER	Percent of the recovery area that is reclaimable

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NUMBER_OF_FILES	NUMBER	Number of files in the recovery area
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.156 V$RECOVERY_FILE_DEST

V$RECOVERY_FILE_DEST displays information about the disk quota and current disk usage in the fast recovery area.

Column	Datatype	Description
NAME	VARCHAR2(513)	Location name. This is the value specified in the DB_RECOVERY_FILE_DEST initialization parameter.
SPACE_LIMIT	NUMBER	Maximum amount of disk space (in bytes) that the database can use for the fast recovery area. This is the value specified in the DB_RECOVERY_FILE_DEST_SIZE initialization parameter.
SPACE_USED	NUMBER	Amount of disk space (in bytes) used by fast recovery area files created in current and all previous fast recovery areas. Changing fast recovery areas does not reset SPACE_USED to 0.
SPACE_RECLAIMABLE	NUMBER	Total amount of disk space (in bytes) that can be created by deleting obsolete, redundant, and other low priority files from the fast recovery area
NUMBER_OF_FILES	NUMBER	Number of files in the fast recovery area
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "DB_RECOVERY_FILE_DEST"
- "DB_RECOVERY_FILE_DEST_SIZE"

8.157 V$RECOVERY_FILE_STATUS

V$RECOVERY_FILE_STATUS contains one row for each datafile for each RECOVER statement. This view contains useful information only for the Oracle process doing the recovery. When Recovery Manager directs a server process to perform recovery, only Recovery Manager can

Copyright © 2002, 2026, Oracle and/or its affiliates.

view the relevant information in this view. V$RECOVERY_FILE_STATUS will be empty to all other Oracle users.

Column	Datatype	Description
FILENUM	NUMBER	Number of the file being recovered
FILENAME	VARCHAR2(513)	Filename of the datafile being recovered
STATUS	VARCHAR2(13)	Status of the recovery: • IN RECOVERY • CURRENT • NOT RECOVERED
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Backup and Recovery User's Guide

8.158 V$RECOVERY_LOG

V$RECOVERY_LOG lists information about archived logs that are needed to complete media recovery. This information is derived from the log history view, V$LOG_HISTORY.

V$RECOVERY_LOG contains useful information only for the Oracle process doing the recovery. When Recovery Manager directs a server process to perform recovery, only Recovery Manager can view the relevant information in this view. V$RECOVERY_LOG will be empty to all other Oracle users.

Column	Datatype	Description
THREAD#	NUMBER	Thread number of the archived log
SEQUENCE#	NUMBER	Sequence number of the archived log
TIME	DATE	Time of the first entry (lowest SCN) in the log
ARCHIVE_NAME	VARCHAR2(513)	Name of the file when archived, using the naming convention specified by the LOG_ARCHIVE_FORMAT initialization parameter **See Also:** "LOG_ARCHIVE_FORMAT"
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "V$LOG_HISTORY" and *Oracle Database Backup and Recovery User's Guide*

8.159 V$RECOVERY_PROGRESS

`V$RECOVERY_PROGRESS` can be used to track database recovery operations to ensure that they are not stalled, and also to estimate the time required to complete the operation in progress.

On non-coordinator instances, `V$RECOVERY_PROGRESS` is not populated.

On the coordinator instance (the instance where MRP0 was started to start recovery), `V$RECOVERY_PROGRESS` has the same set of rows as before, except the following rows in the `ITEM` column are always 0 (not used) with Multi-Instance Redo Apply:

- Active Apply
- Maximum Apply Rate
- Apply Time per Log
- Checkpoint Time per Log
- Recovery ID

`V$RECOVERY_PROGRESS` is a subview of `V$SESSION_LONGOPS`.

> **Note**
>
> This view is populated on the instance where the MRP0 process is started if recovery is running in Multi-Instance Redo Apply mode. Not all the columns will be populated.

Column	Datatype	Description
`START_TIME`	`DATE`	Start time of the recovery operation
`TYPE`	`VARCHAR2(64)`	Type of recovery operation being performed: • `CRASH RECOVERY` • `INSTANCE RECOVERY` • `MEDIA RECOVERY`

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ITEM	VARCHAR2(32)	Item being measured. When TYPE is CRASH RECOVERY or INSTANCE RECOVERY, the possible values are: • Log Files • Redo Blocks When TYPE is MEDIA RECOVERY, the possible values are: • Active Apply Rate • Average Apply Rate • Maximum Apply Rate • Redo Applied • Log Files • Last Applied Redo • Active Time • Elapsed Time • Apply Time per Log • Checkpoint Time per Log • Standby Apply Lag • Recovery ID
UNITS	VARCHAR2(32)	The units of measurement for each item
SOFAR	NUMBER	Amount of work done so far
TOTAL	NUMBER	Total amount of work expected
TIMESTAMP	DATE	Timestamp of the last redo record applied
COMMENTS	VARCHAR2(248)	Miscellaneous notes; currently displays the SCN for the last applied redo
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SESSION_LONGOPS"
- "Background Processes" for more information about the MRP0 process
- *Oracle Database Backup and Recovery User's Guide* for more information about performing database recovery

8.160 V$RECOVERY_SLAVE

V$RECOVERY_SLAVE is used to track database media recovery processes to monitor their performance statistics and analyze a media recovery session.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
START_TIME	DATE	Start time of the recovery process
TYPE	VARCHAR2(64)	Type of recovery process being performed: • MEDIA RECOVERY Apply SLAVE • MEDIA RECOVERY Merge Slave • SERIAL MEDIA RECOVERY
ITEM	VARCHAR2(32)	Item being measured. When TYPE is MEDIA RECOVERY Apply SLAVE, the possible values are: • Recovery ID • Process ID • Number of Redo Cache Full • Number of Redo Cache Copy • Number of CV Cached • CV Applied OK • CV Applied Stuck • CV Applied Repair • CV Applied Corrupt • CV Applied Ckpt • CV Applied Reapplied • Total CV Processed Size • Total CV Applied • Number of Buffer Cache Full • Number of Buffer Retries • Number of Max Reads Issued • Number of Unrcv Condition • Number of Influx Buffer Flushed • Number of Reap Request • Number of Reap Wait IO • Number of Reap No Buffer • Number of Wait All Read • Number of Buffer Pinged • Buffer Ping Time • Read Issue Time • Number of Read Request Issued When TYPE is MEDIA RECOVERY Merge Slave, the possible values are: • Recovery ID • Process ID • Number of Redo Cache Full • Number of Redo Cache Copy • Number of CV Cached • Total Redo Read Bytes • Total CV Parsed When TYPE is SERIAL MEDIA RECOVERY, the possible values include values from the other two TYPEs.
UNITS	VARCHAR2(32)	The units of measurement for each item
SOFAR	NUMBER	Amount of work done so far
TOTAL	NUMBER	Total amount of work expected
COMMENTS	VARCHAR2(248)	Miscellaneous notes, which may display the recovery ID and process ID for current recovery session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **See Also**
>
> *Oracle Database Backup and Recovery User's Guide*

8.161 V$RECOVERY_STATUS

V$RECOVERY_STATUS contains statistics of the current recovery process. This view contains useful information only for the Oracle process doing the recovery. When Recovery Manager directs a server process to perform recovery, only Recovery Manager can view the relevant information in this view. V$RECOVERY_STATUS will be empty to all other Oracle users.

Column	Datatype	Description
RECOVERY_CHECKPOINT	DATE	Point in time to which the recovery has occurred. If no logs have been applied, this is the point in time the recovery starts.
THREAD	NUMBER	Number of the redo thread currently being processed
SEQUENCE_NEEDED	NUMBER	Log sequence number of the log needed by the recovery process. The value is 0 if no log is needed.
SCN_NEEDED	VARCHAR2(16)	Low SCN of the log needed by recovery. The value is 0 if unknown or no log is needed.
TIME_NEEDED	DATE	Time when the log was created. The value is midnight on 1/1/88 if the time is unknown or if no log is needed.
PREVIOUS_LOG_NAME	VARCHAR2(513)	Filename of the log
PREVIOUS_LOG_STATUS	VARCHAR2(13)	Status of the previous log. Contains one of the following values: RELEASE; WRONG NAME; MISSING NAME; UNNEEDED NAME; NONE
REASON	VARCHAR2(13)	Reason recovery is returning control to the user (NEED LOG|LOG REUSED|THREAD DISABLED)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Backup and Recovery User's Guide*

8.162 V$REDO_DEST_RESP_HISTOGRAM

V$REDO_DEST_RESP_HISTOGRAM provides statistical information for each redo transport destination.

Column	Datatype	Description
DEST_ID	NUMBER	A nonnegative integer value representing a SYNC standby destination
TIME	VARCHAR2(20)	A text string that shows the last wall-clock time that a bucket was hit
DURATION	NUMBER	A positive integer value that represents a bucket of seconds, 1, 2, 3, up to 300 seconds, followed by 5 additional buckets that represent 600, 1200, 2400, 4800, and 9600 (>= 4801) seconds
FREQUENCY	NUMBER	A nonnegative integer that shows the number of times a particular bucket was hit by the destination specified by DEST_ID
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.163 V$REQDIST

V$REQDIST lists statistics for the histogram of shared server dispatcher request times, divided into 12 buckets, or ranges of time. The time ranges grow exponentially as a function of the bucket number.

Column	Datatype	Description
BUCKET	NUMBER	Bucket number: 0 - 11; the maximum time for each bucket is (4 * 2^N)/100 seconds
COUNT	NUMBER	Count of requests whose total time to complete (excluding wait time) falls in this range
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.164 V$RESERVED_WORDS

V$RESERVED_WORDS displays a list of all SQL keywords. To determine whether a particular keyword is reserved in any way, check the RESERVED, RES_TYPE, RES_ATTR, and RES_SEMI columns.

Column	Datatype	Description
KEYWORD	VARCHAR2(128)	Name of the keyword
LENGTH	NUMBER	Length of the keyword
RESERVED	VARCHAR2(1)	Indicates whether the keyword cannot be used as an identifier (Y) or whether the keyword is not reserved (N)
RES_TYPE	VARCHAR2(1)	Indicates whether the keyword cannot be used as a type name (Y) or whether the keyword is not reserved (N)
RES_ATTR	VARCHAR2(1)	Indicates whether the keyword cannot be used as an attribute name (Y) or whether the keyword is not reserved (N)
RES_SEMI	VARCHAR2(1)	Indicates whether the keyword is not allowed as an identifier in certain situations, such as in DML (Y) or whether the keyword is not reserved (N)
DUPLICATE	VARCHAR2(1)	Indicates whether the keyword is a duplicate of another keyword (Y) or whether the keyword is not a duplicate (N)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.165 V$RESOURCE

V$RESOURCE contains resource name and address information.

Column	Datatype	Description
ADDR	RAW(4 | 8)	Address of the resource object
TYPE	VARCHAR2(2)	Resource type; the resource types are listed in Table 8-1
ID1	NUMBER	Resource identifier #1
ID2	NUMBER	Resource identifier #2
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.166 V$RESOURCE_LIMIT

V$RESOURCE_LIMIT displays information about global resource use for some of the system resources. Use this view to monitor the consumption of resources so that you can take corrective action, if necessary. Many of the resources correspond to initialization parameters listed in Table 8-5.
Some resources, those used by DLM for example, have an initial allocation (soft limit), and the hard limit, which is theoretically infinite (although in practice it is limited by SGA size). During SGA reservation/initialization, a place is reserved in SGA for the INITIAL_ALLOCATION of resources, but if this allocation is exceeded, additional resources are allocated up to the value indicated by LIMIT_VALUE. The CURRENT_UTILIZATION column indicates whether the initial allocation has been exceeded. When the initial allocation value is exceeded, the additional required resources are allocated from the shared pool, where they must compete for space with other resources.

A good choice for the value of INITIAL_ALLOCATION will avoid the contention for space. For most resources, the value for INITIAL_ALLOCATION is the same as the LIMIT_VALUE. Exceeding LIMIT_VALUE results in an error.

Column	Datatype	Description
RESOURCE_NAME	VARCHAR2(128)	Name of the resource (see Table 8-5)
CURRENT_UTILIZATION	NUMBER	Number of (resources, locks, or processes) currently being used
MAX_UTILIZATION	NUMBER	Maximum consumption of this resource since the last instance start-up
INITIAL_ALLOCATION	VARCHAR2(40)	Initial allocation. This will be equal to the value specified for the resource in the initialization parameter file (UNLIMITED for infinite allocation).
LIMIT_VALUE	VARCHAR2(40)	Unlimited for resources and locks. This can be greater than the initial allocation value (UNLIMITED for infinite limit).
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Table 8-5 Values for the RESOURCE_NAME Column

Resource Name	Corresponds to
DML_LOCKS	See "DML_LOCKS"
ENQUEUE_LOCKS	This value is computed by the Oracle Database. See V$ENQUEUE_LOCK to obtain more information about the enqueue locks.
GES_LOCKS	Global Enqueue Service locks
GES_PROCS	Global Enqueue Service processes
GES_RESS	Global Enqueue Service resources
MAX_SHARED_SERVERS	See "MAX_SHARED_SERVERS"
PARALLEL_MAX_SERVERS	See "PARALLEL_MAX_SERVERS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 8-5 (Cont.) Values for the RESOURCE_NAME Column

Resource Name	Corresponds to
PROCESSES	See "PROCESSES"
SESSIONS	See "SESSIONS"
SORT_SEGMENT_LOCKS	This value is computed by the Oracle Database
TEMPORARY_LOCKS	This value is computed by the Oracle Database
TRANSACTIONS	See "TRANSACTIONS"

8.167 V$RESTORE_POINT

V$RESTORE_POINT displays information about restore points.

Column	Datatype	Description
SCN	NUMBER	Database SCN when the restore point was created
DATABASE_INCARNATION#	NUMBER	Database incarnation number when the restore point was created
GUARANTEE_FLASHBACK_DATABASE	VARCHAR2(3)	Indicates whether flashback log files will be kept to ensure a flashback to this point (YES) or not (NO)
STORAGE_SIZE	NUMBER	Approximate number of bytes of disk space currently tied up supporting this restore point, and which would no longer be tied up if this restore point is the oldest restore point and it is dropped. This will always be zero for non-guaranteed restore points.
TIME	TIMESTAMP(9)	Wall-clock time when the restore point was created
RESTORE_POINT_TIME	TIMESTAMP(9)	Time that was specified when the restore point was created. If a time was not specified, this value is NULL.
PRESERVED	VARCHAR2(3)	Indicates whether the restore point must be explicitly deleted (YES) or not (NO)
NAME	VARCHAR2(128)	Name of the restore point
PDB_RESTORE_POINT	VARCHAR2(3)	Indicates whether there is a PDB restore point for this PDB (YES) or not (NO)
CLEAN_PDB_RESTORE_POINT	VARCHAR2(3)	Indicates whether there is a clean PDB restore point for this PDB (YES) or not (NO)
PDB_INCARNATION#	NUMBER	Pluggable database (PDB) incarnation number. This value is meaningful only for PDB restore points.
REPLICATED[1]	VARCHAR2(3)	This column is useful in Oracle Data Guard environments. It indicates the method by which a restore point was created. Possible values: • YES - The restore point was automatically replicated from the primary database to this database when this database was a standby database. The string _PRIMARY is appended to the name of such a restore point. • NO - The restore point was created by a user and was not replicated from the primary database.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

[1] This column is available starting with Oracle Database 19c.

8.168 V$RESULT_CACHE_DEPENDENCY

V$RESULT_CACHE_DEPENDENCY displays the depends-on relationship between cached results and dependencies.

Column	Datatype	Description
RESULT_ID	NUMBER	Cached result
DEPEND_ID	NUMBER	Dependency object
OBJECT_NO	NUMBER	Dictionary object number of the dependency object
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.169 V$RESULT_CACHE_MEMORY

V$RESULT_CACHE_MEMORY displays all the memory blocks and their status.

Column	Datatype	Description
ID	NUMBER	Unique block identifier (that is, the block number)
CHUNK	NUMBER	Chunk to which the block belongs (the upper 27 bits of the ID)
OFFSET	NUMBER	Offset of the block within its chunk (the lower 5 bits of the ID)
FREE	VARCHAR2(3)	Indicates whether the block is free (YES) or not (NO)
OBJECT_ID	NUMBER	Cache object to which the memory block belongs; NULL if the memory block is not allocated to a cache object (FREE = YES)
POSITION	NUMBER	Position of the block in the cached object; NULL if the memory block is not allocated to a cache object (FREE = YES)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.170 V$RESULT_CACHE_OBJECTS

V$RESULT_CACHE_OBJECTS displays all the objects (both cached results and dependencies) and their attributes.

Column	Datatype	Description
ID	NUMBER	Identifier for the cache object (also the ID of the first block)
TYPE	VARCHAR2(10)	Type of the cache object: • Result • Dependency
STATUS	VARCHAR2(9)	Status of the object: • New - Result is still under construction • Published - Result is available for use • Bypass - Result will be bypassed from use • Expired - Result has exceeded expiration time • Invalid - Result is no longer available for use
BUCKET_NO	NUMBER	Internal hash bucket for the object
HASH	NUMBER	Hash value for the object
NAME	VARCHAR2(387)	Name (for example, SQL prefix or PL/SQL function name)
NAMESPACE	VARCHAR2(10)	Namespace: • SQL • PLSQL • KEY VECTOR
CREATION_TIMESTAMP	DATE	Time when the object was created
CREATOR_UID	NUMBER	UID that created the object
DEPEND_COUNT	NUMBER	Number of dependencies (TYPE = Result) or dependents (TYPE = Dependency)
BLOCK_COUNT	NUMBER	Total number of blocks in the cached object
SCN	NUMBER	Build SCN (TYPE = Result) or invalidation SCN (TYPE = Dependency)
COLUMN_COUNT	NUMBER	Number of columns in the cached result[1]
PIN_COUNT	NUMBER	Number of active scans on this result[1]
SCAN_COUNT	NUMBER	Total number of scans initiated on the cached result[1]
ROW_COUNT	NUMBER	Total number of rows in the cached result[1]
ROW_SIZE_MAX	NUMBER	Size of the largest row (in bytes)[1]
ROW_SIZE_MIN	NUMBER	Size of the smallest row (in bytes)[1]

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ROW_SIZE_AVG	NUMBER	Average size of a row (in bytes)[1]
BUILD_TIME	NUMBER	Amount of time (in hundredths of a second) it took to build the cached result[1]
LRU_NUMBER	NUMBER	LRU list position (the larger the value, the more recent the usage)[1]
OBJECT_NO	NUMBER	Dictionary object number of the dependency object[2]
INVALIDATIONS	NUMBER	Number of times the object has invalidated its dependents[2]
SPACE_OVERHEAD	NUMBER	Overhead (in bytes) for the result[1]
SPACE_UNUSED	NUMBER	Unused space (in bytes) for the result[1]
CACHE_ID	VARCHAR2(387)	CacheId for the result (object name if it's a dependency)
CACHE_KEY	VARCHAR2(387)	CacheKey for the result (object name if it's a dependency)
CHECKSUM	NUMBER	Checksum for the result object. The checksum is computed over all the blocks in the result cache object minus the object header.
EDITION_ID	NUMBER	Shows the edition's object ID that was in use when the result was calculated
DB_LINK	VARCHAR2(3)	Possible values: • YES: If the result cache object references a remote database object • NO: If the result cache object does not reference a remote database object
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

1 This column is valid only for TYPE = Result; otherwise, its value is 0.
2 This column is valid only for TYPE = Dependency; otherwise, its value is 0.

8.171 V$RESULT_CACHE_STATISTICS

V$RESULT_CACHE_STATISTICS displays various Result Cache settings and usage statistics.

Column	Datatype	Description
ID	NUMBER	Statistic number
NAME	VARCHAR2(128)	Name of the statistic (see Table 8-6)
VALUE	VARCHAR2(81)	Value of the statistic
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 8-6 V$RESULT_CACHE_STATISTICS Statistics

Statistic Name	Description
Block Size (Bytes)	Size of each memory block
Block Count Maximum	Maximum number of memory blocks allowed
Block Count Current	Number of memory blocks currently allocated
Result Size Maximum (Blocks)	Maximum number of blocks allowed for a single result
Create Count Success	Number of cache results successfully created
Create Count Failure	Number of cache results that failed to create
Find Count	Number of cached results that were successfully found
Invalidation Count	Total number of invalidations
Delete Count Invalid	Number of invalid cached results deleted
Delete Count Valid	Number of valid cached results deleted
Hash Chain Length	Average length of items in the hash chain
Find Copy Count	Number of results copied directly out of the cache

8.172 V$RMAN_BACKUP_JOB_DETAILS

V$RMAN_BACKUP_JOB_DETAILS displays details about backup jobs.

Column	Datatype	Description
SESSION_KEY	NUMBER	Session identifier
SESSION_RECID	NUMBER	Together, with SESSION_KEY and SESSION_STAMP, used to uniquely identify job output from V$RMAN_OUTPUT
SESSION_STAMP	NUMBER	Together, with SESSION_KEY and SESSION_RECID, used to uniquely identify job output from V$RMAN_OUTPUT
COMMAND_ID	VARCHAR2(33)	Either a user-specified SET COMMAND ID or a unique command ID generated by RMAN
START_TIME	DATE	Start time of the first BACKUP command in the job
END_TIME	DATE	End time of the last BACKUP command in the job
INPUT_BYTES	NUMBER	Sum of all input file sizes backed up by this job
OUTPUT_BYTES	NUMBER	Output size of all pieces generated by this job
STATUS_WEIGHT	NUMBER	Used internally by Enterprise Manager
OPTIMIZED_WEIGHT	NUMBER	Used internally by Enterprise Manager
OBJECT_TYPE_WEIGHT	NUMBER	Used internally by Enterprise Manager
OUTPUT_DEVICE_TYPE	VARCHAR2(17)	Can be DISK, SBT, or *. An * indicates more than one device (in most cases, it will be DISK or SBT).
AUTOBACKUP_COUNT	NUMBER	Number of autobackups performed by this job
BACKED_BY_OSB	VARCHAR2(3)	A value of YES means the backup was done to Oracle Secure Backup. Otherwise, backed up by other third party tape library.
AUTOBACKUP_DONE	VARCHAR2(3)	YES or NO, depending upon whether or not a control file autobackup was done as part of this backup job

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(23)	One of the following values: • RUNNING WITH WARNINGS • RUNNING WITH ERRORS • COMPLETED • COMPLETED WITH WARNINGS • COMPLETED WITH ERRORS • FAILED
INPUT_TYPE	VARCHAR2(13)	Contains one of the following values. If the user command does not satisfy one of them, then preference is given in order, from top to bottom of the list. • DB FULL • RECVR AREA • DB INCR • DATAFILE FULL • DATAFILE INCR • ARCHIVELOG • CONTROLFILE • SPFILE
OPTIMIZED	VARCHAR2(3)	YES or NO, depending on whether optimization was applied. Applicable to backup jobs only.
ELAPSED_SECONDS	NUMBER	Number of elapsed seconds
COMPRESSION_RATIO	NUMBER	Compression ratio
INPUT_BYTES_PER_SEC	NUMBER	Input read-rate-per-second
OUTPUT_BYTES_PER_SEC	NUMBER	Output write-rate-per-second
INPUT_BYTES_DISPLAY	VARCHAR2(4000)	Values in user-displayable form. They will be converted to a format of nM, nG, nT, nP, and so on.
OUTPUT_BYTES_DISPLAY	VARCHAR2(4000)	Values in user-displayable form. They will be converted to a format of nM, nG, nT, nP, and so on
INPUT_BYTES_PER_SEC_DISPLAY	VARCHAR2(4000)	Input read-rate-per-second. These values are in user-displayable form. They will be converted to a format of nM, nG, nT, nP, and so on.
OUTPUT_BYTES_PER_SEC_DISPLAY	VARCHAR2(4000)	Output write-rate-per-second. These values are in user-displayable form. They will be converted to a format of nM, nG, nT, nP, and so on.
TIME_TAKEN_DISPLAY	VARCHAR2(4000)	Time taken, shown in user-displayable format <nn>h:<nn>m:<nn>s
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.173 V$RMAN_BACKUP_SUBJOB_DETAILS

V$RMAN_BACKUP_SUBJOB_DETAILS merges similar operations within an RMAN session into a single row. For example, if there are four BACKUP DATAFILE <n> commands, three RECOVERY COPY OF DATAFILE commands, and one BACKUP RECOVERY AREA command, this view will contain three rows - one each for BACKUP, ROLLFORWARD, and COPY_DISK_TO_TAPE operation.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SESSION_KEY	NUMBER	Session identifier
SESSION_RECID	NUMBER	Together with SESSION_KEY and SESSION_STAMP, used to uniquely identify job output from V$RMAN_OUTPUT
SESSION_STAMP	NUMBER	Together with SESSION_KEY and SESSION_RECID, used to uniquely identify job output from V$RMAN_OUTPUT
OPERATION	VARCHAR2(33)	Can be BACKUP, ROLLFORWARD, VALIDATE, or COPY_DISK_TO_TAPE. A row for each suboperation type for the session will be in the output view.
COMMAND_ID	VARCHAR2(33)	Either a user-specified SET COMMAND ID or a unique command ID generated by RMAN
START_TIME	DATE	Start time of the first BACKUP command in the job
END_TIME	DATE	End time of the last BACKUP command in the job
INPUT_BYTES	NUMBER	Sum of all input file sizes backed up by this job
OUTPUT_BYTES	NUMBER	Output size of all pieces generated by this job
STATUS_WEIGHT	NUMBER	Used internally by Enterprise Manager
OBJECT_TYPE_WEIGHT	NUMBER	Used internally by Enterprise Manager
OPTIMIZED_WEIGHT	NUMBER	Used internally by Enterprise Manager
OUTPUT_DEVICE_TYPE	VARCHAR2(17)	Can be DISK, SBT, or *. An * indicates more than one device (in most cases, it will be DISK or SBT).
BACKED_BY_OSB	VARCHAR2(3)	A value of YES means the backup was done to Oracle Secure Backup. Otherwise, backed up by other third party tape library.
AUTOBACKUP_DONE	VARCHAR2(3)	YES or NO, depending upon whether or not a control file autobackup was done as part of this job
STATUS	VARCHAR2(23)	One of the following values: • RUNNING WITH WARNINGS • RUNNING WITH ERRORS • COMPLETED • COMPLETED WITH WARNINGS • COMPLETED WITH ERRORS • FAILED
INPUT_TYPE	VARCHAR2(13)	Contains one of the following values. If the user command does not satisfy one of them, then preference is given in order, from top to bottom of the list. • DB FULL • RECVR AREA • DB INCR • DATAFILE FULL • DATAFILE INCR • ARCHIVELOG • CONTROLFILE • SPFILE
OPTIMIZED	VARCHAR2(3)	YES or NO, depending on whether optimization was applied. Applicable to backup jobs only.
AUTOBACKUP_COUNT	NUMBER	Number of autobackups performed by this job
COMPRESSION_RATIO	NUMBER	Compression ratio
INPUT_BYTES_DISPLAY	VARCHAR2(4000)	Values in user-displayable form. They will be converted to a format of nM, nG, nT, nP, and so on.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OUTPUT_BYTES_DISPLAY	VARCHAR2(4000)	Values in user-displayable form. They will be converted to a format of nM, nG, nT, nP, and so on.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.174 V$RMAN_BACKUP_TYPE

V$RMAN_BACKUP_TYPE displays information about RMAN backup types.

Column	Datatype	Description
WEIGHT	NUMBER	Used to set precedence order of different backup types in reports.
INPUT_TYPE	VARCHAR2(13)	Used to represent possible filters used in creating various reporting screens.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.175 V$RMAN_COMPRESSION_ALGORITHM

V$RMAN_COMPRESSION_ALGORITHM provides descriptions of supported compression algorithms. It is used by the RMAN client.

Column	Datatype	Description
ALGORITHM_ID	NUMBER	Algorithm ID
ALGORITHM_NAME	VARCHAR2(64)	Name of the algorithm (for example, LOW, MEDIUM, DEFAULT, or HIGH)
INITIAL_RELEASE	VARCHAR2(18)	First Oracle Database release when this compression algorithm was available
TERMINAL_RELEASE	VARCHAR2(18)	Last Oracle Database release that supported using this compression algorithm to create new backups. Existing backups can always be restored, even if they use a deprecated compression algorithm.
ALGORITHM_DESCRIPTION	VARCHAR2(64)	Description of the algorithm
ALGORITHM_COMPATIBILITY	VARCHAR2(18)	Required database compatibility level for the algorithm (for example, 11.2.0 for DEFAULT)
IS_VALID	VARCHAR2(3)	Indicates whether the algorithm is valid with regard to the compatibility setting (YES) or not (NO). The value is YES if ALGORITHM_COMPATIBILITY <= DATABASE_COMPATIBILITY.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
REQUIRES_ACO	VARCHAR2(3)	Indicates whether the algorithm requires the Advanced Compression Option (YES) or not (NO)
IS_DEFAULT	VARCHAR2(3)	Indicates whether the algorithm is the default compression algorithm that RMAN uses to create compressed backup sets (YES) or not (NO)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.176 V$RMAN_CONFIGURATION

V$RMAN_CONFIGURATION lists information about RMAN persistent configuration settings.

Column	Datatype	Description
CONF#	NUMBER	A unique key identifying this configuration record within the target database that owns it.
NAME	VARCHAR2(65)	The type of configuration. All options of the CONFIGURE command are valid types except: • CONFIGURE EXCLUDE (described in RC_TABLESPACE) • CONFIGURE AUXNAME (described in RC_DATAFILE) • CONFIGURE SNAPSHOT CONTROLFILE (stored only in control file)
VALUE	VARCHAR2(1025)	The CONFIGURE command setting. Example: RETENTION POLICY TO RECOVERY WINDOW OF 10 DAYS
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.177 V$RMAN_ENCRYPTION_ALGORITHMS

V$RMAN_ENCRYPTION_ALGORITHMS displays supported encryption algorithms. It is used by the RMAN client to validate user-requested algorithms. This view will list AES128, AES192, and AES256 encryption algorithms for the current release. The default algorithm is AES128.

Column	Datatype	Description
ALGORITHM_ID	NUMBER	Number to identify the algorithm
ALGORITHM_NAME	VARCHAR2(64)	Name of the algorithm (for example, AES128, AES192, or AES256)
ALGORITHM_DESCRIPTION	VARCHAR2(64)	Description of the algorithm

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
IS_DEFAULT	VARCHAR2(3)	Indicates whether this is the default encryption algorithm (YES) or not (NO). This value is set by Oracle Database and may vary for each Release (that is, it is not dependent on user-specified RMAN configuration).
RESTORE_ONLY	VARCHAR2(3)	Indicates whether this algorithm can be used for restore only (YES) or not (NO). If the value is NO, then the algorithm is also allowed for backup. This column is useful in determining whether an encryption algorithm is deprecated for backup purpose.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.178 V$RMAN_OUTPUT

V$RMAN_OUTPUT displays messages reported by RMAN.

This is an in-memory view and is not recorded in the controlfile. The view can hold 32768 rows.

Column	Datatype	Description
SID	NUMBER	Session ID of the session which is running this RMAN operation
RECID	NUMBER	Record ID of the corresponding V$RMAN_STATUS row
STAMP	NUMBER	Timestamp of the corresponding V$RMAN_STATUS row
SESSION_RECID	NUMBER	Record ID of the session (corresponding V$RMAN_STATUS row with ROW_LEVEL = 0)
SESSION_STAMP	NUMBER	Timestamp of the session (corresponding V$RMAN_STATUS row with ROW_LEVEL = 0)
OUTPUT	VARCHAR2(130)	Output text reported by RMAN
RMAN_STATUS_RECID	NUMBER	Owning V$RMAN_STATUS record ID
RMAN_STATUS_STAMP	NUMBER	Owning V$RMAN_STATUS record stamp
SESSION_KEY	NUMBER	Session identifier
GUID[1]	NUMBER	The guid of the pluggable database (PDB) that did the RMAN operation. V$RMAN_OUTPUT captures the rman output. If a PDB sysdba did the backup, the guid of the PDB is displayed in V$RMAN_OUTPUT. If root did the backup, then guid of the root is displayed in V$RMAN_OUTPUT. Root A root user can see all of V$RMAN_OUTPUT rows (that is, the rows owned by all PDBs), but a PDB user can only see that PDB's V$RMAN_OUTPUT rows (that is, no rows from root or other PDBs).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

[1] This column is available starting with Oracle Database 19c.

8.179 V$RMAN_STATUS

V$RMAN_STATUS displays the finished and on-going RMAN jobs. For on-going jobs, this view displays progress and status. The jobs which are in progress are stored only in memory while the finished jobs are stored in the controlfile.

Column	Datatype	Description
SID	NUMBER	Session ID of the session which is running this RMAN operation
RECID	NUMBER	Record ID of the row in the controlfile
STAMP	NUMBER	Timestamp of the row (RECID + STAMP is unique)
PARENT_RECID	NUMBER	Record ID of the parent row of this row (corresponding V$RMAN_STATUS row with ROW_LEVEL = ROW_LEVEL - 1)
PARENT_STAMP	NUMBER	Timestamp of the parent row of this row (corresponding V$RMAN_STATUS row with ROW_LEVEL = ROW_LEVEL - 1)
SESSION_RECID	NUMBER	Record ID of the session (corresponding V$RMAN_STATUS row with ROW_LEVEL = 0)
SESSION_STAMP	NUMBER	Timestamp of the session (corresponding V$RMAN_STATUS row with ROW_LEVEL = 0)
ROW_LEVEL	NUMBER	Level of the row. The session has level 0.
ROW_TYPE	VARCHAR2(19)	Type of the row: • SESSION • COMMAND • RECURSIVE OPERATION
COMMAND_ID	VARCHAR2(33)	Command ID set by the RMAN SET COMMAND ID command. If not set, then RMAN will create a unique number.
OPERATION	VARCHAR2(33)	Name of the command in the execution explained by this row
STATUS	VARCHAR2(23)	Status of the operation: • RUNNING • RUNNING WITH WARNINGS • RUNNING WITH ERRORS • COMPLETED • COMPLETED WITH WARNINGS • COMPLETED WITH ERRORS • FAILED
MBYTES_PROCESSED	NUMBER	Percentage of the job completed; null if not applicable for the operation
START_TIME	DATE	Start time of the job

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
END_TIME	DATE	End time of the job
INPUT_BYTES	NUMBER	Number of input bytes read
OUTPUT_BYTES	NUMBER	Number of output bytes written
OPTIMIZED	VARCHAR2(3)	YES, if backup optimization was applied during the backup job. Otherwise, NO.
OBJECT_TYPE	VARCHAR2(13)	Identifies types of objects backed up
OUTPUT_DEVICE_TYPE	VARCHAR2(17)	DISK, SBT_TAPE, or *. An * indicates that output was written to more than one device type.
OSB_ALLOCATED	VARCHAR2(3)	A value of YES means an Oracle Secure Backup channel was allocated during the specified operation identified by the V$RMAN_STATUS view.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.180 V$RO_USER_ACCOUNT

V$RO_USER_ACCOUNT is populated only on Oracle databases that are open in read-only mode. When a database is read-only, security data cannot be stored in normal catalogue tables. Instead, the security data is stored in an in-memory table that is queried through this view. In an Oracle Data Guard environment, some of the security information for user accounts on the standby is inherited from the primary server. For example, if the account is locked out unlimited on the primary, then it will be locked on the standby database(s). The information stored on the standby is volatile information that user actions on the standby database(s) can affect, such as the number of failed logins, and the time the account was locked on the standby due to failed access attempts. Note that failed login attempts on standbys do not affect the account status on primaries.

If this view is queried from the root in a multitenent container database (CDB), then only common users and the SYS user are returned.

If this view is queried from a pluggable database (PDB), only rows that pertain to the current PDB are returned.

Column	Datatype	Description
USERID	NUMBER	User ID number
PASSW_EXPIRED	NUMBER	Indicates whether the password has expired (1) or not (0)
PASSW_IN_GRACE	NUMBER	Indicates whether the account is in grace (1) or not (0)
PASSW_LOCKED	NUMBER	Indicates whether the account is locked (1) or not (0)
PASSW_LOCK_UNLIM	NUMBER	Indicates whether the account is locked for an unlimited time (1) or not (0)
FAILED_LOGINS	NUMBER	The number of failed login attempts. The count is not cumulative; it is reset upon successful logon to the account
EXPIRATION_AFTER_GRACE	TIMESTAMP(3)	The expiration time after grace

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PASSW_LOCK_TIME	TIMESTAMP(3)	The time the account was locked out
CON_ID	NUMBER	The ID of the container where the failed login occurred. For users that are not common users, the CON_IDis the PDB ID where the failed login attempt occurred. For common users, the CON_ID is 0. The login attempts that occurred on a PDB are not displayed when you query V$RO_USER_ACCOUNT from another PDB. You only see the failed login attempts of any users (that are not common users) if those failed login attempts occurred on the same PDB from which you are querying V$RO_USER_ACCOUNT. The failed login attempts of common users (and of the SYS user) are only displayed when V$RO_USER_ACCOUNT is queried from the root of a CDB, not when it is queried from a PDB. In a non-CDB, the value is always 0.
USERNAME	VARCHAR2(128)	User name

8.181 V$ROLLNAME

V$ROLLNAME lists the names of all online rollback segments. It can only be accessed when the database is open.

Column	Datatype	NULL	Description
USN	NUMBER		Rollback (undo) segment number
NAME	VARCHAR2(30)	NOT NULL	Rollback segment name
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.182 V$ROLLSTAT

V$ROLLSTAT contains rollback segment statistics.

Column	Datatype	Description
USN	NUMBER	Rollback segment number
LATCH	NUMBER	Latch for the rollback segment
EXTENTS	NUMBER	Number of extents in the rollback segment
RSSIZE	NUMBER	Size (in bytes) of the rollback segment. This value differs by the number of bytes in one database block from the value of the BYTES column of the *_SEGMENTS view. **See Also:** *Oracle Database Administrator's Guide* for more information about space management for rollback segments

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
WRITES	NUMBER	Number of bytes written to the rollback segment
XACTS	NUMBER	Number of active transactions
GETS	NUMBER	Number of header gets
WAITS	NUMBER	Number of header waits
OPTSIZE	NUMBER	Optimal size of the rollback segment
HWMSIZE	NUMBER	High watermark of the rollback segment size
SHRINKS	NUMBER	Number of times the size of a rollback segment decreases
WRAPS	NUMBER	Number of times rollback segment is wrapped
EXTENDS	NUMBER	Number of times rollback segment size is extended
AVESHRINK	NUMBER	Average shrink size
AVEACTIVE	NUMBER	Current size of active extents, averaged over time.
STATUS	VARCHAR2(15)	Rollback segment status: • ONLINE • PENDING OFFLINE • OFFLINE • FULL
CUREXT	NUMBER	Current extent
CURBLK	NUMBER	Current block
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.183 V$ROWCACHE

V$ROWCACHE displays statistics for data dictionary activity. Each row contains statistics for one data dictionary cache.

Column	Datatype	Description
CACHE#	NUMBER	Row cache ID number
TYPE	VARCHAR2(11)	Parent or subordinate row cache type
SUBORDINATE#	NUMBER	Subordinate set number
PARAMETER	VARCHAR2(32)	Name of the initialization parameter that determines the number of entries in the data dictionary cache
COUNT	NUMBER	Total number of entries in the cache
USAGE	NUMBER	Number of cache entries that contain valid data
FIXED	NUMBER	Number of fixed entries in the cache
GETS	NUMBER	Total number of requests for information on the data object
FASTGETS	NUMBER	Reserved for internal use

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
GETMISSES	NUMBER	Number of data requests resulting in cache misses
SCANS	NUMBER	Number of scan requests
SCANMISSES	NUMBER	Number of times a scan failed to find the data in the cache
SCANCOMPLETES	NUMBER	For a list of subordinate entries, the number of times the list was scanned completely
MODIFICATIONS	NUMBER	Number of inserts, updates, and deletions
FLUSHES	NUMBER	Number of times flushed to disk
DLM_REQUESTS	NUMBER	Number of DLM requests
DLM_CONFLICTS	NUMBER	Number of DLM conflicts This column is obsolete and is maintained for backward compatibility.
DLM_RELEASES	NUMBER	Number of DLM releases This column is obsolete and is maintained for backward compatibility.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.184 V$ROWCACHE_PARENT

V$ROWCACHE_PARENT displays information for parent objects in the data dictionary. There is one row per lock owner, and one waiter for each object. This row shows the mode held or requested. For objects with no owners or waiters, a single row is displayed.

Column	Datatype	Description
INDX	NUMBER	Index of the row
HASH	NUMBER	Hash value
ADDRESS	RAW(4 | 8)	Address of the parent object
CACHE#	NUMBER	Parent cache ID
CACHE_NAME	VARCHAR2(64)	Parent cache name
EXISTENT	VARCHAR2(1)	Indicates whether the object is an existing object
LOCK_MODE	NUMBER	Mode the lock is held in
LOCK_REQUEST	NUMBER	Mode the lock is requested in
TXN	RAW(4 | 8)	Transaction currently locking the object
SADDR	RAW(4 | 8)	Address of the session
INST_LOCK_REQUEST	NUMBER	Mode in which instance lock is being requested. This column is only relevant for Real Application Clusters.
INST_LOCK_RELEASE	NUMBER	Whether the instance lock needs to be released. This column is only relevant for Real Application Clusters.
INST_LOCK_TYPE	VARCHAR2(2)	Type of instance lock. This column is only relevant for Real Application Clusters.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
INST_LOCK_ID1	RAW(4)	ID associated with the instance lock. This column is only relevant for Real Application Clusters.
INST_LOCK_ID2	RAW(4)	ID associated with the instance lock. This column is only relevant for Real Application Clusters.
KEY	RAW(100)	Contents of the key. This column is only relevant for Real Application Clusters.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.185 V$ROWCACHE_SUBORDINATE

V$ROWCACHE_SUBORDINATE displays information for subordinate objects in the data dictionary.

Column	Datatype	Description
INDX	NUMBER	The index
HASH	NUMBER	The hash value
ADDRESS	RAW(4 | 8)	Address of the subordinate object
CACHE#	NUMBER	The parent cache ID
SUBCACHE#	NUMBER	The subcache ID
SUBCACHE_NAME	VARCHAR2(64)	The subcache name
EXISTENT	VARCHAR2(1)	Whether the object is an existing object
PARENT	RAW(4 | 8)	Address of the parent object
KEY	RAW(100)	The contents of the key
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.186 V$RSRC_CONS_GROUP_HISTORY

V$RSRC_CONS_GROUP_HISTORY displays a history of consumer group statistics for each entry in V$RSRC_PLAN_HISTORY that has a non-NULL plan.

When the STATISTICS_LEVEL is set to TYPICAL or ALL, this view contains information about CPU utilization and wait times even when no Resource Manager plan is set or when the Resource Manager plan does not monitor CPU or session resources.

Copyright © 2002, 2026, Oracle and/or its affiliates.

A new window is created in V$RSRC_CON_GROUP_HISTORY when a pluggable database (PDB) changes its resource plan. The plan windows inside a PDB are not impacted by a multitenant container database (CDB) resource plan change.

Since PDB plans can be set independently across different PDBs, V$RSRC_CON_GROUP_HISTORY will not cover the same time period across different PDBs. Therefore, this view is not useful for comparing statistics across different PDBs.

Column	Datatype	Description
SEQUENCE#	NUMBER	A sequential counter that uniquely describes the V$RSRC_PLAN_HISTORY entry to which these consumer group statistics apply. When the instance is restarted, this value is reset to zero.
ID	NUMBER	Consumer group object ID (a unique number, consistent across database shutdowns and startups)
NAME	VARCHAR2(30)	Name of the consumer group
REQUESTS	NUMBER	Cumulative number of requests that were executed in the consumer group
CPU_WAIT_TIME	NUMBER	Cumulative amount of time that sessions waited for CPU on the resmgr: cpu quantum wait event because of resource management (in milliseconds). This does not include waits due to latch or enqueue contention, I/O waits, and so on. When CPU resources are not being actively managed, this value is set to zero.
CPU_WAITS	NUMBER	Cumulative number of times all sessions in the consumer group had to wait for CPU on the resmgr: cpu quantum wait event because of resource management. This does not include waits due to latch or enqueue contention, I/O waits, and so on. When CPU resources are not being actively managed, this value is set to zero.
CONSUMED_CPU_TIME	NUMBER	Cumulative amount of CPU time consumed by all sessions in the consumer group (in milliseconds)
YIELDS	NUMBER	Cumulative number of times that sessions in the consumer group had to yield CPU to other sessions because of quantum expiration. When CPU resources are not being actively managed, this value is set to zero.
CPU_DECISIONS	NUMBER	Percentage of CPU decisions for which the consumer group was present. When CPU resources are not being actively managed, this value is set to zero.
CPU_DECISIONS_EXCLUSIVE	NUMBER	Percentage of the CPU decisions for which the consumer group was present and was the only consumer group present. When CPU resources are not being actively managed, this value is set to zero.
CPU_DECISIONS_WON	NUMBER	Percentage of the CPU decisions that the consumer group won. When CPU resources are not being actively managed, this value is set to zero.
ACTIVE_SESS_LIMIT_HIT	NUMBER	Number of times that sessions in the consumer group were queued because the consumer group reached its active session limit
UNDO_LIMIT_HIT	NUMBER	Number of times that queries in the consumer group were canceled because the consumer group reached its UNDO_POOL limit
SWITCHES_IN_CPU_TIME	NUMBER	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_TIME limit
SWITCHES_OUT_CPU_TIME	NUMBER	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_TIME limit
SWITCHES_IN_IO_MEGABYTES	NUMBER	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_IO_MEGABYTES limit
SWITCHES_OUT_IO_MEGABYTES	NUMBER	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_IO_MEGABYTES limit

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SWITCHES_IN_IO_REQUESTS	NUMBER	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_IO_REQS limit
SWITCHES_OUT_IO_REQUESTS	NUMBER	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_IO_REQS limit
SWITCHES_IN_IO_LOGICAL	NUMBER	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_IO_LOGICAL limit
SWITCHES_OUT_IO_LOGICAL	NUMBER	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_IO_LOGICAL limit
SWITCHES_IN_ELAPSED_TIME	NUMBER	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_ELAPSED_TIME limit
SWITCHES_OUT_ELAPSED_TIME	NUMBER	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_ELAPSED_TIME limit
SQL_CANCELED	NUMBER	Number of times that SQL queries running in the consumer group were terminated because they exceeded one of the Resource Manager plan's SWITCH limits and CANCEL_SQL was specified as the Resource Manager plan's SWITCH_GROUP
ACTIVE_SESS_KILLED	NUMBER	Number of times that sessions running in the consumer group were terminated because they exceeded one of the Resource Manager plan's SWITCH limits and KILL_SESSION was specified as the Resource Manager plan's SWITCH_GROUP
IDLE_SESS_KILLED	NUMBER	Number of times that sessions in the consumer group were terminated because they were idle for too long (reached MAX_IDLE_TIME)
IDLE_BLKR_SESS_KILLED	NUMBER	Number of times that sessions in the consumer group were terminated because they were idle too long (reached MAX_IDLE_BLOCKER_TIME) and were blocking other sessions
QUEUED_TIME	NUMBER	Total amount of time that sessions in the consumer group have spent in the QUEUED state because of the active session limit (in milliseconds)
QUEUE_TIME_OUTS	NUMBER	Number of times that requests from sessions in the consumer group timed out because they were queued for too long (reached QUEUEING_P1)
IO_SERVICE_TIME	NUMBER	Cumulative I/O wait time (in milliseconds)
IO_SERVICE_WAITS	NUMBER	Total number of wait requests
SMALL_READ_MEGABYTES	NUMBER	Number of single block megabytes read
SMALL_WRITE_MEGABYTES	NUMBER	Number of single block megabytes written
LARGE_READ_MEGABYTES	NUMBER	Number of multiblock megabytes read
LARGE_WRITE_MEGABYTES	NUMBER	Number of multiblock megabytes written
SMALL_READ_REQUESTS	NUMBER	Number of single block read requests
SMALL_WRITE_REQUESTS	NUMBER	Number of single block write requests
LARGE_READ_REQUESTS	NUMBER	Number of multiblock read requests
LARGE_WRITE_REQUESTS	NUMBER	Number of multiblock write requests
PQS_COMPLETED	NUMBER	Total number of completed parallel statements in the consumer group
PQ_SERVERS_USED	NUMBER	Total number of parallel servers used by completed parallel statements in the consumer group
PQS_QUEUED	NUMBER	Number of times that sessions in the consumer group were queued when trying to run parallel statements
PQ_ACTIVE_TIME	NUMBER	Cumulative sum of the parallel active times for all completed parallel statements in the consumer group (in milliseconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PQ_QUEUED_TIME	NUMBER	Total amount of time that sessions in the consumer group were queued when trying to run parallel statements (in milliseconds)
PQ_QUEUE_TIME_OUTS	NUMBER	Number of times that parallel statements from sessions in the consumer group timed out because their queue time exceeded the Resource Manager plan's PARALLEL_QUEUE_TIMEOUT limit
PGA_LIMIT_SESSIONS_KILLED	NUMBER	Number of times that sessions in the consumer group were terminated because their untunable PGA usage exceeded the SESSION_PGA_LIMIT limit
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$RSRC_PDB_HISTORY"
- "V$RSRC_PLAN_HISTORY"
- "STATISTICS_LEVEL"

8.187 V$RSRC_CONSUMER_GROUP

V$RSRC_CONSUMER_GROUP displays data related to currently active resource consumer groups.

When the STATISTICS_LEVEL is set to TYPICAL or ALL, this view contains information about CPU utilization and wait times even when no Resource Manager plan is set or when the Resource Manager plan does not monitor CPU or session resources.

Statistics in V$RSRC_CONSUMER_GROUP are reset when a pluggable database (PDB) changes its resource plan. They are not impacted by multitenant container database (CDB) resource plan changes.

Since PDB plans can be set independently across different PDBs, V$RSRC_CONSUMER_GROUP will not cover the same time period across different PDBs. Therefore, this view is not useful for comparing statistics across different PDBs.

Column	Datatype	Description
ID	NUMBER	Consumer group object ID (a unique number, consistent across database shutdowns and startups)
NAME	VARCHAR2(32)	Name of the consumer group
ACTIVE_SESSIONS	NUMBER	Number of currently active sessions in the consumer group
EXECUTION_WAITERS	NUMBER	Number of currently active sessions waiting for an execution time slice in which they will be able to use CPU

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
REQUESTS	NUMBER	Cumulative number of requests that were executed in the consumer group
CPU_WAIT_TIME	NUMBER	Cumulative amount of time that sessions waited for CPU on the resmgr: cpu quantum wait event because of resource management (in milliseconds). This does not include waits due to latch or enqueue contention, I/O waits, and so on. When CPU resources are not being actively managed, this value is set to zero.
CPU_WAITS	NUMBER	Cumulative number of times all sessions in the consumer group had to wait for CPU on the resmgr: cpu quantum wait event because of resource management. This does not include waits due to latch or enqueue contention, I/O waits, and so on. When CPU resources are not being actively managed, this value is set to zero.
CONSUMED_CPU_TIME	NUMBER	Cumulative amount of CPU time consumed by all sessions in the consumer group (in milliseconds)
YIELDS	NUMBER	Cumulative number of times that sessions in the consumer group had to yield CPU to other sessions because of quantum expiration. When CPU resources are not being actively managed, this value is set to zero.
CPU_DECISIONS	NUMBER	Percentage of CPU decisions for which the consumer group was present. When CPU resources are not being actively managed, this value is set to zero.
CPU_DECISIONS_EXCLUSIVE	NUMBER	Percentage of the CPU decisions for which the consumer group was present and was the only consumer group present. When CPU resources are not being actively managed, this value is set to zero.
CPU_DECISIONS_WON	NUMBER	Percentage of the CPU decisions that the consumer group won. When CPU resources are not being actively managed, this value is set to zero.
QUEUE_LENGTH	NUMBER	Number of sessions waiting in the queue
CURRENT_UNDO_CONSUMPTION	NUMBER	Current amount (in KB) of undo consumed by the consumer group
ACTIVE_SESSION_LIMIT_HIT	NUMBER	Number of times that sessions in the consumer group were queued because the consumer group reached its active session limit
UNDO_LIMIT_HIT	NUMBER	Number of times that queries in the consumer group were canceled because the consumer group reached its UNDO_POOL limit
SWITCHES_IN_CPU_TIME	NUMBER	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_TIME limit
SWITCHES_OUT_CPU_TIME	NUMBER	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_TIME limit
SWITCHES_IN_IO_MEGABYTES	NUMBER	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_IO_MEGABYTES limit
SWITCHES_OUT_IO_MEGABYTES	NUMBER	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_IO_MEGABYTES limit
SWITCHES_IN_IO_REQUESTS	NUMBER	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_IO_REQS limit
SWITCHES_OUT_IO_REQUESTS	NUMBER	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_IO_REQS limit
SWITCHES_IN_IO_LOGICAL	NUMBER	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_IO_LOGICAL limit
SWITCHES_OUT_IO_LOGICAL	NUMBER	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_IO_LOGICAL limit

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SWITCHES_IN_ELAPSED_TIME	NUMBER	Number of switches into the consumer group because of the Resource Manager plan's SWITCH_ELAPSED_TIME limit
SWITCHES_OUT_ELAPSED_TIME	NUMBER	Number of switches out of the consumer group because of the Resource Manager plan's SWITCH_ELAPSED_TIME limit
SQL_CANCELED	NUMBER	Number of times that SQL queries running in the consumer group were terminated because they exceeded one of the Resource Manager plan's SWITCH limits and CANCEL_SQL was specified as the Resource Manager plan's SWITCH_GROUP
ACTIVE_SESSIONS_KILLED	NUMBER	Number of times that sessions running in the consumer group were terminated because they exceeded one of the Resource Manager plan's SWITCH limits and KILL_SESSION was specified as the Resource Manager plan's SWITCH_GROUP
IDLE_SESSIONS_KILLED	NUMBER	Number of times that sessions in the consumer group were terminated because they were idle for too long (reached MAX_IDLE_TIME)
IDLE_BLKR_SESSIONS_KILLED	NUMBER	Number of times that sessions in the consumer group were terminated because they were idle too long (reached MAX_IDLE_BLOCKER_TIME) and were blocking other sessions
QUEUED_TIME	NUMBER	Total amount of time that sessions in the consumer group have spent in the QUEUED state because of the active session limit (in milliseconds)
QUEUE_TIME_OUTS	NUMBER	Number of times that requests from sessions in the consumer group timed out because they were queued for too long (reached QUEUEING_P1)
IO_SERVICE_TIME	NUMBER	Cumulative I/O wait time (in milliseconds)
IO_SERVICE_WAITS	NUMBER	Total number of wait requests
SMALL_READ_MEGABYTES	NUMBER	Number of single block megabytes read
SMALL_WRITE_MEGABYTES	NUMBER	Number of single block megabytes written
LARGE_READ_MEGABYTES	NUMBER	Number of multiblock megabytes read
LARGE_WRITE_MEGABYTES	NUMBER	Number of multiblock megabytes written
SMALL_READ_REQUESTS	NUMBER	Number of single block read requests
SMALL_WRITE_REQUESTS	NUMBER	Number of single block write requests
LARGE_READ_REQUESTS	NUMBER	Number of multiblock read requests
LARGE_WRITE_REQUESTS	NUMBER	Number of multiblock write requests
CURRENT_PQS_ACTIVE	NUMBER	Number of active parallel statements in the consumer group. This value does not include parallel statements that are never queued, such as GV$ queries.
CURRENT_PQ_SERVERS_ACTIVE	NUMBER	Number of active parallel servers in the consumer group. This value does not include servers running parallel statements that are never queued, such as GV$ queries.
PQS_QUEUED	NUMBER	Number of times that sessions in the consumer group were queued when trying to run parallel statements
PQS_COMPLETED	NUMBER	Total number of completed parallel statements in the consumer group
PQ_SERVERS_USED	NUMBER	Total number of parallel servers used by completed parallel statements in the consumer group
PQ_ACTIVE_TIME	NUMBER	Cumulative sum of the parallel active times for all completed parallel statements in the consumer group (in milliseconds)
CURRENT_PQS_QUEUED	NUMBER	Number of sessions in the consumer group that are waiting in the parallel statement queue trying to run parallel statements

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PQ_QUEUED_TIME	NUMBER	Total amount of time that sessions in the consumer group were queued when trying to run parallel statements (in milliseconds)
PQ_QUEUE_TIME_OUTS	NUMBER	Number of times that parallel statements from sessions in the consumer group timed out because their queue time exceeded the Resource Manager plan's PARALLEL_QUEUE_TIMEOUT limit
PGA_LIMIT_SESSIONS_KILLED	NUMBER	Number of times that sessions in the consumer group were terminated because their untunable PGA usage exceeded the SESSION_PGA_LIMIT limit
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "STATISTICS_LEVEL"
- "V$RSRC_PDB"
- *Oracle Database Administrator's Guide* for information on resource groups
- *Oracle Database PL/SQL Packages and Types Reference* for information on creating resource groups with the DBMS_RESOURCE_MANAGER package

8.188 V$RSRC_CONSUMER_GROUP_CPU_MTH

V$RSRC_CONSUMER_GROUP_CPU_MTH displays all resource allocation methods defined for resource consumer groups.

Column	Datatype	Description
NAME	VARCHAR2(40)	Name of the CPU resource allocation method
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- *Oracle Database Administrator's Guide* for information on resource allocation methods
- *Oracle Database PL/SQL Packages and Types Reference* on defining resource allocation methods for consumer groups with the `DBMS_RESOURCE_MANAGER` package
- "V$RSRC_PLAN_CPU_MTH" for a listing of all resource allocation methods defined for resource plans

8.189 V$RSRC_PDB

`V$RSRC_PDB` displays data related to currently active resource consumer groups by pluggable database (PDB).

When the `STATISTICS_LEVEL` is set to `TYPICAL` or `ALL`, this view contains information about CPU utilization and wait times even when no Resource Manager plan is set or when the Resource Manager plan does not monitor CPU or session resources.

Statistics in `V$RSRC_PDB` are reset when a multitenant container database (CDB) changes its resource plan. They are not impacted by PDB resource plan changes.

`V$RSRC_PDB` covers the same time period for all PDBs. This view is specifically designed for comparing statistics across different PDBs.

Since `V$RSRC_PDB` does not contain information at the consumer group level, it is not useful for comparing consumer groups inside a PDB.

Column	Datatype	Description
`PDB_NAME`	`VARCHAR2(32)`	PDB name
`CPU_WAIT_TIME`	`NUMBER`	Cumulative amount of time that sessions waited for CPU because of resource management. This does not include waits due to latch or enqueue contention, I/O waits, and so on. When CPU resources are not being actively managed, this value is set to zero.
`CPU_WAITS`	`NUMBER`	Cumulative number of times all sessions in the consumer group had to wait for CPU because of resource management. This does not include waits due to latch or enqueue contention, I/O waits, and so on. When CPU resources are not being actively managed, this value is set to zero.
`CONSUMED_CPU_TIME`	`NUMBER`	Cumulative amount of CPU time consumed by all sessions in the consumer group (in milliseconds)
`YIELDS`	`NUMBER`	Cumulative number of times that sessions in the consumer group had to yield CPU to other sessions because of quantum expiration. When CPU resources are not being actively managed, this value is set to zero.
`IO_SERVICE_TIME`	`NUMBER`	Cumulative I/O wait time (in milliseconds)
`IO_SERVICE_WAITS`	`NUMBER`	Total number of wait requests
`SMALL_READ_MEGABYTES`	`NUMBER`	Number of single block megabytes read
`SMALL_WRITE_MEGABYTES`	`NUMBER`	Number of single block megabytes written
`LARGE_READ_MEGABYTES`	`NUMBER`	Number of multiblock megabytes read
`LARGE_WRITE_MEGABYTES`	`NUMBER`	Number of multiblock megabytes written

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SMALL_READ_REQUESTS	NUMBER	Number of single block read requests
SMALL_WRITE_REQUESTS	NUMBER	Number of single block write requests
LARGE_READ_REQUESTS	NUMBER	Number of multiblock read requests
LARGE_WRITE_REQUESTS	NUMBER	Number of multiblock write requests
PQS_COMPLETED	NUMBER	Total number of completed parallel statements in the consumer group
PQ_SERVERS_USED	NUMBER	Total number of parallel servers used by completed parallel statements in the consumer group
PQS_QUEUED	NUMBER	Number of times that sessions in the consumer group were queued when trying to run parallel statements
PQ_ACTIVE_TIME	NUMBER	Cumulative sum of the parallel active times for all completed parallel statements in the consumer group (in milliseconds)
PQ_QUEUED_TIME	NUMBER	Total amount of time that sessions in the consumer group were queued when trying to run parallel statements (in milliseconds)
PQ_QUEUE_TIME_OUTS	NUMBER	Number of times that parallel statements from sessions in the consumer group timed out because their queue time exceeded the Resource Manager plan's PARALLEL_QUEUE_TIMEOUT limit
CURRENT_PQS_ACTIVE	NUMBER	Number of active parallel statements in the consumer group. This value does not include parallel statements that are never queued, such as GV$ queries.
CURRENT_PQ_SERVERS_ACTIVE	NUMBER	Number of active parallel servers in the consumer group. This value does not include servers running parallel statements that are never queued, such as GV$ queries.
CURRENT_PQS_QUEUED	NUMBER	Number of sessions in the consumer group that are waiting in the parallel statement queue trying to run parallel statements
SGA_BYTES	NUMBER	The current SGA usage by this PDB in bytes
BUFFER_CACHE_BYTES	NUMBER	The current usage of buffer cache by this PDB in bytes
SHARED_POOL_BYTES	NUMBER	The current usage of shared pool by this PDB in bytes
PGA_BYTES	NUMBER	The current usage of PGA by this PDB in bytes
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$RSRC_CONSUMER_GROUP"

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.190 V$RSRC_PDB_HISTORY

V$RSRC_PDB_HISTORY displays a history of consumer group statistics for each entry in V$RSRC_PDB that has a non-NULL plan by pluggable database (PDB).

When the STATISTICS_LEVEL is set to TYPICAL or ALL, this view contains information about CPU utilization and wait times even when no Resource Manager plan is set or when the Resource Manager plan does not monitor CPU or session resources.

A new window is created in V$RSRC_PDB_HISTORY when a multitenant container database (CDB) changes its resource plan. The plan windows for the CDB are not impacted by a PDB resource plan change.

V$RSRC_PDB_HISTORY covers the same time period for all PDBs. This view is specifically designed for comparing statistics across different PDBs.

Since V$RSRC_PDB_HISTORY does not contain information at the consumer group level, it is not useful for comparing consumer groups inside a PDB.

Column	Datatype	Description
SEQUENCE#	NUMBER	A sequential counter that uniquely describes the V$RSRC_PDB entry to which these consumer group statistics apply. When the instance is restarted, this value is reset to zero.
PDB_NAME	VARCHAR2(32)	PBD name
CPU_WAIT_TIME	NUMBER	Cumulative amount of time that sessions waited for CPU because of resource management. This does not include waits due to latch or enqueue contention, I/O waits, and so on. When CPU resources are not being actively managed, this value is set to zero.
CPU_WAITS	NUMBER	Cumulative number of times all sessions in the consumer group had to wait for CPU because of resource management. This does not include waits due to latch or enqueue contention, I/O waits, and so on. When CPU resources are not being actively managed, this value is set to zero.
CONSUMED_CPU_TIME	NUMBER	Cumulative amount of CPU time consumed by all sessions in the consumer group (in milliseconds)
YIELDS	NUMBER	Cumulative number of times that sessions in the consumer group had to yield CPU to other sessions because of quantum expiration. When CPU resources are not being actively managed, this value is set to zero.
IO_SERVICE_TIME	NUMBER	Cumulative I/O wait time (in milliseconds)
IO_SERVICE_WAITS	NUMBER	Total number of wait requests
SMALL_READ_MEGABYTES	NUMBER	Number of single block megabytes read
SMALL_WRITE_MEGABYTES	NUMBER	Number of single block megabytes written
LARGE_READ_MEGABYTES	NUMBER	Number of multiblock megabytes read
LARGE_WRITE_MEGABYTES	NUMBER	Number of multiblock megabytes written
SMALL_READ_REQUESTS	NUMBER	Number of single block read requests
SMALL_WRITE_REQUESTS	NUMBER	Number of single block write requests
LARGE_READ_REQUESTS	NUMBER	Number of multiblock read requests
LARGE_WRITE_REQUESTS	NUMBER	Number of multiblock write requests
PQS_COMPLETED	NUMBER	Total number of completed parallel statements in the consumer group

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PQ_SERVERS_USED	NUMBER	Total number of parallel servers used by completed parallel statements in the consumer group
PQS_QUEUED	NUMBER	Number of times that sessions in the consumer group were queued when trying to run parallel statements
PQ_ACTIVE_TIME	NUMBER	Cumulative sum of the parallel active times for all completed parallel statements in the consumer group (in milliseconds)
PQ_QUEUED_TIME	NUMBER	Total amount of time that sessions in the consumer group were queued when trying to run parallel statements (in milliseconds)
PQ_QUEUE_TIME_OUTS	NUMBER	Number of times that parallel statements from sessions in the consumer group timed out because their queue time exceeded the Resource Manager plan's PARALLEL_QUEUE_TIMEOUT limit
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$RSRC_CONS_GROUP_HISTORY"

8.191 V$RSRC_PLAN

V$RSRC_PLAN displays the names of all currently active resource plans.

Column	Datatype	Description
ID	NUMBER	Resource plan ID (a unique number, consistent across database shutdowns and startups). This is also the data dictionary object ID.
NAME	VARCHAR2(32)	Name of the resource plan
IS_TOP_PLAN	VARCHAR2(5)	Indicates whether the resource plan is the current top plan (TRUE) or whether the resource plan is a subplan of the current top plan (FALSE)
CPU_MANAGED	VARCHAR2(3)	Indicates whether the resource plan has parameters that specify a policy for how the Resource Manager should schedule sessions to manage CPU usage (ON) or whether Resource Manager is not managing CPU usage (OFF)
INSTANCE_CAGING	VARCHAR2(3)	Indicates whether instance caging is enabled (ON) or disabled (OFF). Instance caging is enabled if the CPU_COUNT initialization parameter is explicitly modified to a value other than 0 and Resource Manager is enabled.
PARALLEL_SERVERS_ACTIVE	NUMBER	Total number of active parallel servers on the instance
PARALLEL_SERVERS_TOTAL	NUMBER	The value of PARALLEL_SERVERS_TARGET for the instance. Parallel statements are queued if the total number of active parallel servers exceeds this value.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PARALLEL_EXECUTION_MANAGED	VARCHAR2(32)	State of parallel statement queuing: • OFF - Parallel statement queuing is disabled • STARTUP - Parallel statement queuing is enabled. This is a temporary state that can occur when an Oracle RAC database is undergoing configuration changes • FIFO - Parallel statement queuing is enabled. All parallel statements are managed in a single Oracle RAC FIFO queue • FULL - Parallel statement queuing is enabled. All parallel statements are managed in per-consumer group queues according to the current resource plan. This state is used when a resource plan that contains resource allocation directives (MGMT_P*) is enabled. • DISABLED - Parallel statement queuing is disabled. This state can occur when memory is unavailable for use by parallel statement queuing in the System Global Area (SGA). Restart the Oracle instance to re-enable parallel statement queuing. For an Oracle RAC database, only the instance running as primary Database Resource Manager (DBRM) shows the correct state of parallel statement queuing. All other instances default to the value FULL. A single instance database always shows the correct value for this field.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
DIRECTIVE_TYPE	VARCHAR2(32)	The type of directive used by this PDB: • DEFAULT_DIRECTIVE: The default plan directive • PDB: A PDB directive • PROFILE: A profile directive
SHARES	NUMBER	Resource allocation for this PDB, expressed in shares
UTILIZATION_LIMIT	NUMBER	Maximum percentage of the container's resources allowed for this PDB
PARALLEL_SERVER_LIMIT	NUMBER	Maximum percentage of the parallel target the PDB can use before queuing subsequent parallel queries
MEMORY_MIN	NUMBER	The percentage of Exadata Smart Flash Cache and Exadata PMEM cache that is guaranteed to the PDB This percentage is based on the total amount of space allocated to the CDB for Exadata Smart Flash Cache and Exadata PMEM cache. See *Oracle Exadata System Software User's Guide* for more information.
MEMORY_LIMIT	NUMBER	The maximum percentage of Exadata Smart Flash Cache and Exadata PMEM cache that the PDB can use This percentage is based on the total amount of space allocated to the CDB for Exadata Smart Flash Cache and Exadata PMEM cache. See *Oracle Exadata System Software User's Guide* for more information.
PROFILE	VARCHAR2(32)	The value of the DB_PERFORMANCE_PROFILE initialization parameter for this PDB

The resource plan with CON_ID=ROOT is the CDB resource plan.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "DBA_RSRC_PLANS" for a listing of all plans in the database
- "DBA_CDB_RSRC_PLANS" for information about CDB resource plans
- "DBA_CDB_RSRC_PLAN_DIRECTIVES" for information about CDB resource plan directives
- "DB_PERFORMANCE_PROFILE"
- *Oracle Database Administrator's Guide* for information on resource plans
- *Oracle Database PL/SQL Packages and Types Reference* for information on defining resource allocation methods for consumer groups with the `DBMS_RESOURCE_MANAGER` package

8.192 V$RSRC_PLAN_CPU_MTH

`V$RSRC_PLAN_CPU_MTH` displays all available CPU resource allocation methods defined for resource plans.

Column	Datatype	Description
`NAME`	`VARCHAR2(40)`	Name of the resource allocation method
`CON_ID`	`NUMBER`	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$RSRC_CONSUMER_GROUP_CPU_MTH" for a listing of resource allocation methods defined for consumer groups
- *Oracle Database Administrator's Guide* for information on resource plans
- *Oracle Database PL/SQL Packages and Types Reference* for information on defining resource allocation methods for consumer plans with the `DBMS_RESOURCE_MANAGER` package

8.193 V$RSRC_PLAN_HISTORY

`V$RSRC_PLAN_HISTORY` displays a history of when a resource plan was enabled, disabled, or modified on the instance. Up to 15 of the most recent entries are shown. Once the database is opened, this view shows at least one row. The row with the most recent `START_TIME` and with `END_TIME` equal to NULL gives information about the current resource plan.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SEQUENCE#	NUMBER	A sequential counter that uniquely describes a row. When the instance is restarted, this value is reset to zero.
ID	NUMBER	Resource plan ID; NULL if the Resource Manager was disabled
NAME	VARCHAR2(30)	Resource plan name; NULL if the Resource Manager was disabled
START_TIME	DATE	Time that the resource plan was enabled
END_TIME	DATE	Time that the resource plan was disabled; NULL if the row contains the current resource plan information
ENABLED_BY_SCHEDULER	VARCHAR2(5)	Indicates whether the plan was enabled by a Job Scheduler window (TRUE) or not (FALSE)
WINDOW_NAME	VARCHAR2(128)	Job Scheduler window that triggered the resource plan event; NULL if a user triggered the resource plan event
ALLOWED_AUTOMATED_SWITCHES	VARCHAR2(5)	Indicates whether automated plan switches were allowed after this resource plan event (TRUE) or whether automated plan switches were disabled after this resource plan event (FALSE) **See Also:** the SWITCH_PLAN procedure
CPU_MANAGED	VARCHAR2(3)	Indicates whether the resource plan has parameters that specify a policy for how the Resource Manager should schedule sessions to manage CPU usage (ON) or whether Resource Manager is not managing CPU usage (OFF)
INSTANCE_CAGING	VARCHAR2(3)	Indicates whether instance caging is enabled (ON) or disabled (OFF). Instance caging is enabled if the CPU_COUNT initialization parameter is explicitly modified to a value other than 0 and Resource Manager is enabled.
PARALLEL_EXECUTION_MANAGED	VARCHAR2(32)	State of parallel statement queuing: • OFF - Parallel statement queuing is disabled • STARTUP - Parallel statement queuing is enabled. This is a temporary state that can occur when an Oracle RAC database is undergoing configuration changes • FIFO - Parallel statement queuing is enabled. All parallel statements are managed in a single Oracle RAC FIFO queue • FULL - Parallel statement queuing is enabled. All parallel statements are managed in per-consumer group queues according to the current resource plan. This state is used when a resource plan that contains resource allocation directives (MGMT_P*) is enabled. • DISABLED - Parallel statement queuing is disabled. This state can occur when memory is unavailable for use by parallel statement queuing in the System Global Area (SGA). Restart the Oracle instance to re-enable parallel statement queuing. For an Oracle RAC database, only the instance running as primary Database Resource Manager (DBRM) shows the correct state of parallel statement queuing. All other instances default to the value FULL. A single instance database always shows the correct value for this field.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

8.194 V$RSRC_SESSION_INFO

V$RSRC_SESSION_INFO displays Resource Manager statistics per session.

Column	Datatype	Description
SID	NUMBER	Session identifier
CURRENT_CONSUMER_GROUP_ID	NUMBER	Object ID of the consumer group in which the session currently belongs; NULL if the session has not yet logged in
CURRENT_CONSUMER_GROUP	VARCHAR2(32)	The name of the consumer group in which the session currently belongs
ORIG_CONSUMER_GROUP_ID	NUMBER	Object ID of the consumer group in which the session was placed by the consumer group mappings; NULL if the session has not yet logged in This group may not be the current group because the SWITCH_GROUP directive in the current plan may have changed the session's current group. This group may not equal the MAPPED_CONSUMER_GROUP because the MAPPED_CONSUMER_GROUP may not be part of the current plan.
MAPPING_ATTRIBUTE	VARCHAR2(32)	Session attribute that was used to map the session into its original consumer group; NULL if no mapping was used **See Also:** "DBA_RSRC_GROUP_MAPPINGS" for more details
MAPPED_CONSUMER_GROUP	VARCHAR2(32)	Consumer group to which the session was originally mapped; NULL if no mapping was used This may not correspond to the original consumer group, because the mapped group may not be part of the current plan. **See Also:** "DBA_RSRC_GROUP_MAPPINGS" for more details
STATE	VARCHAR2(32)	Current state of the session: • NOT MANAGED - Session has not logged in or the current Resource Manager plan does not require the session to be managed at this point in time • RUNNING - Session is currently running on the CPU • WAITING FOR CPU - Session is ready to run. It is waiting for a CPU quantum to run. • QUEUED - Session is queued because the active session limit was reached • IDLE - Session is idle • IDLE BLKR - Session is idle and blocking another session • WAITING - Session is currently in a wait **See Also:** "V$SESSION_WAIT" for the wait type • WAITING_FOR_IO - Session is waiting to submit an I/O request • UNBOUND - Session is not bound to any process • EXITING - Session is about to terminate
ACTIVE	VARCHAR2(5)	Indicates whether the session is currently active (TRUE) or not (FALSE). This includes when one of the following conditions is true: • Session is in the top call • Session has a transaction in progress • Session is using temporary space objects • Session holds user enqueues
CURRENT_IDLE_TIME	NUMBER	Number of milliseconds the session has been idle (in states IDLE or IDLE BLKR) while in this consumer group; NULL if the current Resource Manager plan does not require updating this statistic. This value is reset to zero when the session becomes active.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CURRENT_CPU_WAIT_TIME	NUMBER	Amount of time (in milliseconds) the session has waited for CPU because of resource management (in state WAIT FOR CPU) while in the current consumer group; NULL if the current Resource Manager plan does not require updating this statistic. This does not include waits due to latch or enqueue contention, I/O waits, and so on. If SWITCH_TIME_IN_CALL is used, then this will be reset at the end of every call.
CPU_WAIT_TIME	NUMBER	Cumulative amount of time (in milliseconds) the session has waited for CPU (in its lifetime) because of resource management. This does not include waits due to latch or enqueue contention, I/O waits, and so on.
CURRENT_CPU_WAITS	NUMBER	Number of times the session had to wait for CPU because of resource management while in this consumer group; NULL if the current Resource Manager plan does not require updating this statistic. This does not include waits due to latch or enqueue contention, I/O waits, and so on. If SWITCH_TIME_IN_CALL is used, then this will be reset at the end of every call.
CPU_WAITS	NUMBER	Cumulative number of times the session had to wait for CPU (in its lifetime) because of resource management. This does not include waits due to latch or enqueue contention, I/O waits, and so on.
CURRENT_CONSUMED_CPU_TIME	NUMBER	Amount of CPU time (in milliseconds) consumed by the session while in the current consumer group; NULL if the current Resource Manager plan does not require updating this statistic. If SWITCH_TIME_IN_CALL is used, then this will be reset at the end of every call.
CONSUMED_CPU_TIME	NUMBER	Cumulative amount of CPU time consumed by the session (in its lifetime) (in milliseconds)
CURRENT_ACTIVE_TIME	NUMBER	Amount of time (in milliseconds) the session has been active while in the current consumer group; NULL if the current Resource Manager plan does not require updating this statistic. Active time is time spent running and waiting while executing a call. It does not include the time a session waited for CPU resources. If SWITCH_TIME_IN_CALL is used, then this is reset at the end of every call.
ACTIVE_TIME	NUMBER	Cumulative amount of active time (in milliseconds) consumed by the session (in its lifetime)
CURRENT_QUEUED_TIME	NUMBER	Amount of time (in milliseconds) the current request from the session has been queued (in state QUEUED). If the session does not have a request currently queued up, then this number will be zero.
QUEUED_TIME	NUMBER	Total amount of time (in milliseconds) the session has spent in the QUEUED state (in its lifetime)
CURRENT_YIELDS	NUMBER	Number of times the session had to yield the CPU to other sessions (due to quantum expiration) while in the current consumer group; NULL if the current Resource Manager plan does not require updating this statistic. If SWITCH_TIME_IN_CALL is used, then this is reset at the end of every call.
YIELDS	NUMBER	Cumulative number of times the session had to yield CPU to other sessions due to quantum expiration (in its lifetime)
CURRENT_UNDO_CONSUMPTION	NUMBER	Current amount (in KB) of undo consumed by the session; NULL if the current Resource Manager plan does not have an UNDO_POOL directive
MAX_UNDO_CONSUMPTION	NUMBER	Maximum amount of undo consumed (in KB) during the session's lifetime. This value may not be updated because the current Resource Manager plan may not have an UNDO_POOL directive.
SQL_CANCELED	NUMBER	Number of times SQL queries running in the session were canceled due to exceeding the Resource Manager plan's SWITCH_TIME limit.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
QUEUE_TIME_OUTS	NUMBER	Number of times requests from the session timed out because they queued longer than the Resource Manager plan's limit
ESTIMATED_EXECUTION_LIMIT_HIT	NUMBER	Number of times requests from the session were not run because the optimizer's estimated time to execute the query exceeded the MAX_EST_EXEC_TIME limit
CURRENT_IO_SERVICE_TIME	NUMBER	Current I/O wait time of the session (in milliseconds) for the current SQL operation
IO_SERVICE_TIME	NUMBER	Cumulative amount of I/O wait time by the session (in its lifetime) (in milliseconds)
CURRENT_IO_SERVICE_WAITS	NUMBER	Current I/O waits by session for the current SQL operation
IO_SERVICE_WAITS	NUMBER	Cumulative I/O waits by session (in its lifetime)
CURRENT_SMALL_READ_MEGABYTES	NUMBER	Number of single block megabytes read by the session for the current SQL operation
SMALL_READ_MEGABYTES	NUMBER	Total number of single block megabytes read by the session (in its lifetime)
CURRENT_LARGE_READ_MEGABYTES	NUMBER	Number of multiblock megabytes read by the session for the current SQL operation
LARGE_READ_MEGABYTES	NUMBER	Total number of multiblock megabytes read by the session (in its lifetime)
CURRENT_SMALL_WRITE_MEGABYTES	NUMBER	Number of single block megabytes written by the session for the current SQL operation
SMALL_WRITE_MEGABYTES	NUMBER	Total number of single block megabytes written by the session (in its lifetime)
CURRENT_LARGE_WRITE_MEGABYTES	NUMBER	Number of multiblock megabytes written by the session for the current SQL operation
LARGE_WRITE_MEGABYTES	NUMBER	Total number of multiblock megabytes written by the session (in its lifetime)
CURRENT_SMALL_READ_REQUESTS	NUMBER	Number of single block read requests by the session for the current SQL operation
SMALL_READ_REQUESTS	NUMBER	Total number of single block read requests by the session (in its lifetime)
CURRENT_SMALL_WRITE_REQUESTS	NUMBER	Number of single block write requests by the session for the current SQL operation
SMALL_WRITE_REQUESTS	NUMBER	Total number of single block write requests by the session (in its lifetime)
CURRENT_LARGE_READ_REQUESTS	NUMBER	Number of multiblock read requests by the session for the current SQL operation
LARGE_READ_REQUESTS	NUMBER	Total number of multiblock read requests by the session (in its lifetime)
CURRENT_LARGE_WRITE_REQUESTS	NUMBER	Number of multiblock write requests by the session for the current SQL operation
LARGE_WRITE_REQUESTS	NUMBER	Total number of multiblock write requests by the session (in its lifetime)
CURRENT_PQ_ACTIVE_TIME	NUMBER	Amount of time that the current active parallel statement has been executing for the current SQL operation, not including the amount of time that the statement has been queued (in milliseconds). If the parallel statement is queued, then the value is 0.
PQ_ACTIVE_TIME	NUMBER	Cumulative amount of time that parallel statements have been executed over the lifetime of the session (in milliseconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DOP	NUMBER	Degree of parallelism for the active or queued parallel statement, if there are any in the session
PQ_SERVERS	NUMBER	The number of active parallel servers if the session is active and running the parallel query. If the query is queued, the number of parallel servers that this query is trying to run with is shown.
ESTIMATED_EXECUTION_TIME	NUMBER	Estimated execution time for the parallel statement, as estimated by the optimizer (in milliseconds). You can compare this value to CURRENT_PQ_ACTIVE_TIME to estimate how much longer the parallel statement will run.
CURRENT_PQ_QUEUED_TIME	NUMBER	Amount of time that the current parallel statement in the session has been queued (in milliseconds) for the current SQL operation. If the session does not have a queued parallel statement, then the value is 0.
PQ_QUEUED_TIME	NUMBER	Total amount of time that the session has spent in the PQ_QUEUED state in its lifetime (in milliseconds)
PQ_QUEUED	NUMBER	Number of times that parallel statements in the session have been queued
PQ_QUEUE_TIME_OUTS	NUMBER	Number of times that parallel statements in the session timed out because their queue time exceeded the Resource Manager plan's PARALLEL_QUEUE_TIMEOUT limit
PQ_ACTIVE	VARCHAR2(5)	Indicates whether the session is actively running a parallel statement (TRUE) or not (FALSE).
PQ_STATUS	VARCHAR2(44)	The status of the parallel statement running in this session: • Active: The parallel statement is currently active and running • Queued: The parallel statement is queued and is not a possible candidate to be run next • Queue head: The parallel statement is queued due to the global systemwide limit and is the next parallel statement to be dequeued and run • Queue head - waiting on CG limit: The parallel statement is queued due to the consumer group limit and is the one to be run next from the session's consumer group • Queue head - waiting on service availability: The parallel statement is queued since there are not enough parallel servers available for the service this session is running on. This session is the one to be run next from the session's service • Queue head - waiting on PDB limit: The parallel statement is queued due to the PDB limit and is the one to be run next from the session's PDB • NULL: The session currently does not involve a parallel execution
CURRENT_LOGICAL_IOS	NUMBER	Number of logical I/O requests by the session for the current SQL operation
LOGICAL_IOS	NUMBER	Total number of logical I/O requests in this session's lifetime
CURRENT_ELAPSED_TIME	NUMBER	Elapsed time of the session's current SQL operation
ELAPSED_TIME	NUMBER	Total elapsed time for all of this session's SQL operations in its lifetime

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LAST_ACTION	VARCHAR2(48)	The most recent action that was taken on this SQL operation by Resource Manager. Its value is one of the following: • CANCEL_SQL • KILL_SESSION • LOG_ONLY • SWITCH TO <CG NAME> For the last value, <CG NAME> is the name of the consumer group that the SQL operation was switched to. If the Resource Plan has since been changed then <CG NAME> is the ID of the consumer group.
LAST_ACTION_REASON	VARCHAR2(30)	The reason for the most recent action that was taken on this SQL operation by Resource Manager. Its value is one of the following: • SWITCH_CPU_TIME • SWITCH_IO_REQS • SWITCH_IO_MBS • SWITCH_ELAPSED_TIME • SWITCH_IO_LOGICAL
LAST_ACTION_TIME	DATE	The time of the most recent action that was taken on this SQL operation by Resource Manager
CURRENT_PGA	NUMBER	Amount of PGA memory (in bytes) used by the session for the current SQL operation
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.195 V$RSRCMGRMETRIC

V$RSRCMGRMETRIC displays information about resources consumed and wait times per consumer group.

When the STATISTICS_LEVEL is set to TYPICAL or ALL, this view contains information about CPU utilization and wait times even when no Resource Manager plan is set or when the Resource Manager plan does not monitor CPU or session resources. Metrics are collected and stored every minute when CPU utilization is not being monitored.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
SEQUENCE#	NUMBER	A sequential counter that uniquely describes the V$RSRC_PLAN_HISTORY entry to which these consumer group statistics apply. When the instance is restarted, this value is reset to zero.
CONSUMER_GROUP_ID	NUMBER	Consumer group object ID (a unique number, consistent across database shutdowns and startups)
CONSUMER_GROUP_NAME	VARCHAR2(30)	Name of the consumer group

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CPU_CONSUMED_TIME	NUMBER	Cumulative amount of CPU time consumed by all sessions in the consumer group, in milliseconds
CPU_WAIT_TIME	NUMBER	Cumulative amount of time that sessions waited for CPU because of resource management, in milliseconds. This does not include waits due to latch or enqueue contention, I/O waits, and so on. When CPU resources are not being actively managed, this value is set to zero.
NUM_CPUS	NUMBER	Number of CPUs that the Resource Manager is utilizing. If instance caging is enabled, then this column is equal to the value of the CPU_COUNT initialization parameter. If instance caging is not enabled, then this column is equal to the total number of CPUs in the system.
RUNNING_SESSIONS_LIMIT	NUMBER	Maximum number of sessions in the consumer group that can run simultaneously. The value of this column is NUM_CPUS multiplied by the consumer group's MAX_UTILIZATION_LIMIT directive in the current Resource Manager plan.
AVG_RUNNING_SESSIONS	NUMBER	Average number of sessions in the consumer group that are currently running
AVG_WAITING_SESSIONS	NUMBER	Average number of sessions in the consumer group that are waiting for CPU due to resource management. When CPU resources are not being actively managed, this value is set to zero.
CPU_UTILIZATION_LIMIT	NUMBER	Maximum percentage of CPU that the consumer group can use at any time, with respect to the total number of CPUs in the system. The value of this column is RUNNING_SESSIONS_LIMIT divided by the number of CPUs in the system. If instance caging is enabled, then this value is derived using the CPU_COUNT initialization parameter.
AVG_CPU_UTILIZATION	NUMBER	Average percentage of CPU consumed by the consumer group, with respect to the total number of CPUs in the system
CPU_DECISIONS	NUMBER	Percentage of CPU decisions for which the consumer group was present. When CPU resources are not being actively managed, this value is set to zero. This column is deprecated.
CPU_DECISIONS_EXCLUSIVE	NUMBER	Percentage of the CPU decisions for which the consumer group was present and was the only consumer group present. When CPU resources are not being actively managed, this value is set to zero. This column is deprecated.
CPU_DECISIONS_WON	NUMBER	Percentage of the CPU decisions that the consumer group won. When CPU resources are not being actively managed, this value is set to zero. This column is deprecated.
IO_REQUESTS	NUMBER	I/O requests
IO_MEGABYTES	NUMBER	I/O megabytes
AVG_ACTIVE_PARALLEL_STMTS	NUMBER	The average number of parallel statements that were running during the 1-minute metric window
AVG_QUEUED_PARALLEL_STMTS	NUMBER	The average number of parallel statements that were queued during the 1-minute metric window
AVG_ACTIVE_PARALLEL_SERVERS	NUMBER	The average number of parallel servers that were actively running as part of a parallel statement during the 1-minute metric window
AVG_QUEUED_PARALLEL_SERVERS	NUMBER	The average number of parallel servers that were requested by queued parallel statements during the 1-minute metric window
PARALLEL_SERVERS_LIMIT	NUMBER	The number of parallel servers allowed to be used by this consumer group during the 1-minute metric window
PLAN_NAME	VARCHAR2(30)	Resource Manager plan name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"STATISTICS_LEVEL"

8.196 V$RSRCMGRMETRIC_HISTORY

V$RSRCMGRMETRIC_HISTORY displays a history (the last one hour) of resource manager metrics, taken from V$RSRCMGRMETRIC. When a resource plan is set, this history is cleared and restarted. This view provides information about resources consumed and wait times per consumer group. The columns for V$RSRCMGRMETRIC_HISTORY are the same as those for V$RSRCMGRMETRIC.

See Also

"V$RSRCMGRMETRIC"

8.197 V$RSRCPDBMETRIC

V$RSRCPDBMETRIC displays information about resources consumed and wait times per PDB.

When the STATISTICS_LEVEL is set to TYPICAL or ALL, this view contains information about CPU utilization and wait times even when no Resource Manager plan is set or when the Resource Manager plan does not monitor CPU or session resources. Metrics are collected and stored every minute when CPU utilization is not being monitored.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
SEQUENCE#	NUMBER	A sequential counter that uniquely describes the V$RSRC_PLAN_HISTORY entry to which these PDB statistics apply. When the instance is restarted, this value is reset to zero.
CPU_CONSUMED_TIME	NUMBER	Cumulative amount of CPU time consumed by all sessions in the PDB, in milliseconds

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CPU_WAIT_TIME	NUMBER	Cumulative amount of time that sessions waited for CPU because of resource management, in milliseconds. This does not include waits due to latch or enqueue contention, I/O waits, and so on. When CPU resources are not being actively managed, this value is set to zero.
NUM_CPUS	NUMBER	Number of CPUs that the Resource Manager is utilizing. If instance caging is enabled, then this column is equal to the value of the CPU_COUNT initialization parameter. If instance caging is not enabled, then this column is equal to the total number of CPUs in the system.
RUNNING_SESSIONS_LIMIT	NUMBER	Maximum number of sessions in the PDB that can run simultaneously. The value of this column is NUM_CPUS multiplied by the PDB's MAX_UTILIZATION_LIMIT directive in the current Resource Manager plan.
AVG_RUNNING_SESSIONS	NUMBER	Average number of sessions in the PDB that are currently running
AVG_WAITING_SESSIONS	NUMBER	Average number of sessions in the PDB that are waiting for CPU due to resource management. When CPU resources are not being actively managed, this value is set to zero.
CPU_UTILIZATION_LIMIT	NUMBER	Maximum percentage of CPU that the PDB can use at any time, with respect to the total number of CPUs in the system. The value of this column is RUNNING_SESSIONS_LIMIT divided by the number of CPUs in the system. If instance caging is enabled, then this value is derived using the CPU_COUNT initialization parameter.
AVG_CPU_UTILIZATION	NUMBER	Average percentage of CPU consumed by the PDB, with respect to the total number of CPUs in the system
IOPS	NUMBER	I/O operations per second during the previous minute for this PDB
IOMBPS	NUMBER	I/O megabytes per second during the previous minute for this PDB
IOPS_THROTTLE_EXEMPT	NUMBER	Indicates how much of the I/O per second in the current PDB was exempted from throttling. For example, if the value in the IOPS column is 20 I/Os and the value in the IOPS_THROTTLE_EXEMPT column is 5 I/Os, then 5 I/Os of the 20 I/Os in that second were exempted from throttling. I/O throttling is defined by the MAX_IOPS database initialization parameter.
IOMBPS_THROTTLE_EXEMPT	NUMBER	Indicates how many megabytes of I/O executed per second in the current PDB were exempted from throttling. For example, if the value in the IOMBPS column is 200 megabytes and the value in the IOMBPS_THROTTLE_EXEMPT column is 50 megabytes, then 50 megabytes of the 200 megabytes were exempt from throttling. I/O megabytes per second throttling is defined by the MAX_MBPS database initialization parameter.
AVG_IO_THROTTLE	NUMBER	Average throttle time per I/O operation in milliseconds during the previous minute for this PDB
AVG_ACTIVE_PARALLEL_STMTS	NUMBER	The average number of parallel statements that were running during the 1-minute metric window
AVG_QUEUED_PARALLEL_STMTS	NUMBER	The average number of parallel statements that were queued during the 1-minute metric window
AVG_ACTIVE_PARALLEL_SERVERS	NUMBER	The average number of parallel servers that were actively running as part of a parallel statement during the 1-minute metric window
AVG_QUEUED_PARALLEL_SERVERS	NUMBER	The average number of parallel servers that were requested by queued parallel statements during the 1-minute metric window

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PARALLEL_SERVERS_LIMIT	NUMBER	The number of parallel servers allowed to be used by this PDB during the 1-minute metric window
SGA_BYTES	NUMBER	The current SGA usage by this PDB in bytes
BUFFER_CACHE_BYTES	NUMBER	The current usage of the buffer cache by this PDB in bytes
SHARED_POOL_BYTES	NUMBER	The current usage of the shared pool by this PDB in bytes
PGA_BYTES	NUMBER	The current PGA usage by this PDB in bytes
PLAN_NAME	VARCHAR2(30)	Resource Manager plan name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"STATISTICS_LEVEL"

8.198 V$RSRCPDBMETRIC_HISTORY

V$RSRCPDBMETRIC_HISTORY displays a history (the last one hour) of resource manager metrics for a PDB, taken from V$RSRCPDBMETRIC. When a resource plan is set, this history is cleared and restarted. This view provides information about resources consumed and wait times per consumer group.
The columns for V$RSRCPDBMETRIC_HISTORY are the same as those for V$RSRCPDBMETRIC.

See Also

"V$RSRCPDBMETRIC"

8.199 V$RULE

V$RULE displays rule statistics. This view has a row for every rule loaded into shared memory.

Column	Datatype	Description
RULE_SET_OBJECT_ID	NUMBER	Rule set object ID
EVALUATION_CONTEXT_OBJECT_ID	NUMBER	Evaluation context object ID
RULE_OWNER	VARCHAR2(128)	Owner of the rule
RULE_NAME	VARCHAR2(128)	Name of the rule

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
RULE_CONDITION	VARCHAR2(200)	Rule condition
TRUE_HITS	NUMBER	Number of times the rule evaluated to TRUE
MAYBE_HITS	NUMBER	Number of times the rule evaluated to MAYBE
SQL_EVALUATIONS	NUMBER	Number of evaluations of the rule that were performed by issuing SQL
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

8.200 V$RULE_SET

V$RULE_SET displays rule set statistics. This view has a row for every rule set loaded into shared memory.

> **Note**
>
> Querying the V$RULE_SET view may have a negative impact on performance if a database has a large library cache.

Column	Datatype	Description
OWNER	VARCHAR2(128)	Owner of the rule set
NAME	VARCHAR2(128)	Name of the rule set
CPU_TIME	NUMBER	Total CPU time (in hundredths of a second) spent in evaluation of the rule set
ELAPSED_TIME	NUMBER	Total elapsed time (in hundredths of a second) spent in evaluation of the rule set
FIRST_LOAD_TIME	DATE	First time the current cached copy of the rule set was loaded
LAST_LOAD_TIME	DATE	Last time the current cached copy of the rule set was loaded
LAST_LOADING_TIME	NUMBER	Total elapsed time (in hundredths of a second) spent to load the rule set the last time it was loaded
SHARABLE_MEM	NUMBER	Shared memory (in bytes) used by the rule set
RELOADS	NUMBER	Number of times the rule set object was reloaded in shared memory
INVALIDATIONS	NUMBER	Number of times the rule set object was invalidated
EVALUATIONS	NUMBER	Number of evaluations on the rule set
FIRST_HIT_EVALUATIONS	NUMBER	Number of evaluations on the rule set, with stop_on_first_hit set to TRUE
SIMPLE_RULES_ONLY_EVALUATIONS	NUMBER	Number of evaluations on the rule set, with simple_rules_only set to TRUE
SQL_FREE_EVALUATIONS	NUMBER	Number of evaluations on the rule set which did not internally issue SQL to evaluate rules

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SQL_EXECUTIONS	NUMBER	Total number of SQL statements executed during evaluation of the rule set
CONDITIONS_PROCESSED	NUMBER	Total number of fast (indexed) conditions processed during evaluation of the rule set
TRUE_RULES	NUMBER	Total number of TRUE rules returned during evaluation of the rule set
MAYBE_RULES	NUMBER	Total number of MAYBE rules returned during evaluation of the rule set
VARIABLE_VALUE_FUNCTION_CALLS	NUMBER	Total number of calls made to user-defined functions to retrieve variable values (specified by the variable_value_function field in RE$VARIABLE_TYPE) made during evaluation of the rule set
VARIABLE_METHOD_FUNCTION_CALLS	NUMBER	Total number of calls made to user-defined functions to retrieve variable method values (specified by the variable_method_function field in RE$VARIABLE_TYPE) made during evaluation of the rule set
EVALUATION_FUNCTION_CALLS	NUMBER	Total number of calls made to user-defined evaluation functions (specified as the evaluation_function argument to the DBMS_RULE_ADM.CREATE_EVALUATION_CONTEXT procedure) made during evaluation of the rule set
RESULT_CACHE_HITS	NUMBER	Number of result cache hits across all the sessions evaluating this rule set
IS_RESULT_CACHE	VARCHAR2(3)	Indicates whether this rule set result can be cached (YES) or not (NO)
RESULT_CACHE_ELEMENTS	NUMBER	Number of elements cached within the result cache
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database PL/SQL Packages and Types Reference for more information about the DBMS_RULE_ADM.CREATE_EVALUATION_CONTEXT procedure

8.201 V$RULE_SET_AGGREGATE_STATS

V$RULE_SET_AGGREGATE_STATS displays statistics aggregated over all evaluations on all rule sets. This view has a row for each type of statistic.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NAME	VARCHAR2(80)	Name of the statistic: • rule set evaluations (all) - Total number of evaluations on all rule sets • rule set evaluations (first_hit) - Total number of evaluations on rule sets with stop_on_first_hit set to TRUE • rule set evaluations (simple_rules_only) - Total number of evaluations on rule sets with simple_rules_only set to TRUE • rule set evaluations (SQL free) - Total number of evaluations on rule sets which did not internally issue SQL to evaluate rules • rule set evaluation time (CPU) - Total CPU time (in hundredths of a second) spent in evaluations on rule sets • rule set evaluation time (elapsed) - Total elapsed time (in hundredths of a second) spent in evaluations on rule sets • rule set SQL executions - Total number of SQL statements executed during evaluations on rule sets • rule set conditions processed - Total number of fast (indexed) conditions processed during evaluations on rule sets • rule set true rules - Total number of TRUE rules returned during evaluations on rule sets • rule set maybe rules - Total number of MAYBE rules returned during evaluations on rule sets • rule set user function calls (variable value function) - Total number of calls made to user-defined functions to retrieve variable values (specified by the variable_value_function field in RE$VARIABLE_TYPE) made during evaluations on rule sets • rule set user function calls (variable method function) - Total number of calls made to user-defined functions to retrieve variable method values (specified by the variable_method_function field in RE$VARIABLE_TYPE) made during evaluations on rule sets • rule set user function calls (evaluation function) - Total number of calls made to user-defined evaluation functions (specified as the evaluation_function argument to the DBMS_RULE_ADM.CREATE_EVALUATION_CONTEXT procedure) made during evaluations on rule sets
VALUE	NUMBER	Statistic value
CON_ID	NUMBER	The ID of the container to which the data pertains. The CON_ID value in this view is always 0. The rows pertain to the entire CDB or to the non-CDB.

See Also

Oracle Database PL/SQL Packages and Types Reference for more information about the DBMS_RULE_ADM.CREATE_EVALUATION_CONTEXT procedure

Copyright © 2002, 2026, Oracle and/or its affiliates.

9

Dynamic Performance (V$) Views: V$SCHEDULER_RUNNING_JOBS to V$ZONEMAP_USAGE_STATS

This chapter contains the dynamic performance views V$SCHEDULER_RUNNING_JOBS to V$ZONEMAP_USAGE_STATS.

9.1 V$SCHEDULER_RUNNING_JOBS

V$SCHEDULER_RUNNING_JOBS displays information about running Scheduler jobs.

Column	Datatype	Description
SESSION_ID	NUMBER	Identifier of the session running the Scheduler job
SESSION_SERIAL_NUM	NUMBER	Session serial number
JOB_ID	NUMBER	ID of the running Scheduler job
PADDR	RAW(4 | 8)	Process address
OS_PROCESS_ID	VARCHAR2(12)	Operating system process ID
SESSION_STAT_CPU	INTERVAL DAY(2) TO SECOND(3)	CPU statistics for the session
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire multitenant container database (CDB). This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.2 V$SECUREFILE_TIMER

V$SECUREFILE_TIMER displays information about time taken (in microseconds) by functions of SecureFiles. These timer values are collected per session.

Column	Datatype	Description
NAME	VARCHAR2(50)	Name of the function

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LAYER_ID	NUMBER	ID of the layer that the function belongs to: • 0 - Entire Subtree • 1 - Delta Updates • 2 - Write gather cache • 3 - Deduplication • 4 - Compression & Encryption • 5 - Inode • 6 - Space • 7 - Utilities • 8 - Row-Column Intersection
OWNTIME	NUMBER	Total time taken by the function
MAXTIME	NUMBER	Maximum time taken by a single call
MINTIME	NUMBER	Minimum time taken by a single call
INVOCATIONS	NUMBER	Number of times the function was invoked
LAYER_NAME	VARCHAR2(50)	Name of the layer to which the function belongs
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.3 V$SEGMENT_STATISTICS

V$SEGMENT_STATISTICS displays information about segment-level statistics.

Column	Datatype	Description
OWNER	VARCHAR2(128)	Owner of the object
OBJECT_NAME	VARCHAR2(128)	Name of the object
SUBOBJECT_NAME	VARCHAR2(128)	Name of the subobject
TABLESPACE_NAME	VARCHAR2(30)	Name of the table space to which the object belongs
TS#	NUMBER	Tablespace number
OBJ#	NUMBER	Dictionary object number of the object
DATAOBJ#	NUMBER	Data object number of the object
OBJECT_TYPE	VARCHAR2(18)	Type of the object
STATISTIC_NAME	VARCHAR2(64)	Name of the statistic
STATISTIC#	NUMBER	Statistic number
VALUE	NUMBER	Statistic value

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.4 V$SEGSTAT

V$SEGSTAT displays information about segment-level statistics.

Column	Datatype	Description
TS#	NUMBER	Tablespace number
OBJ#	NUMBER	Dictionary object number
DATAOBJ#	NUMBER	Data object number
STATISTIC_NAME	VARCHAR2(64)	Name of the statistic
STATISTIC#	NUMBER	Statistic number
VALUE	NUMBER	Statistic value
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.5 V$SEGSTAT_NAME

V$SEGSTAT_NAME displays information about segment-level statistics properties.

Column	Datatype	Description
STATISTIC#	NUMBER	Statistic number
NAME	VARCHAR2(64)	Name of the statistic
SAMPLED	VARCHAR2(3)	Indicates whether the statistic was collected by sampling (YES) or not (NO)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.6 V$SERV_MOD_ACT_STATS

V$SERV_MOD_ACT_STATS displays the same set of performance statistics as V$SERVICE_STATS except for a specific combination of service/module/action names.

When aggregation is enabled for the service name, module, and action name, then this view provides the timing and work done for calls issued for the business transaction.

Column	Datatype	Description
AGGREGATION_TYPE	VARCHAR2(21)	Aggregation statistic type: • SERVICE_MODULE - Action value is NULL and the entry is an aggregate for all actions within a given module • SERVICE_MODULE_ACTION - Action value is NULL only for an empty action, and the aggregation is on the level of service/module/action
SERVICE_NAME	VARCHAR2(64)	Service name from V$SERVICES
MODULE	VARCHAR2(65)	Module name from DBA_ENABLED_AGGREGATIONS
ACTION	VARCHAR2(65)	Action name from DBA_ENABLED_AGGREGATIONS
STAT_ID	NUMBER	Statistic identifier
STAT_NAME	VARCHAR2(64)	Derived statistic name from V$STATNAME and V$SESS_TIME_MODEL
VALUE	NUMBER	Cumulative value (in microseconds)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SERVICE_STATS"
- "V$STATNAME"
- "V$SESS_TIME_MODEL"

9.7 V$SERVICE_EVENT

V$SERVICE_EVENT displays aggregated wait counts and wait times for each wait statistic.

Column	Datatype	Description
SERVICE_NAME	VARCHAR2(64)	Service name from V$SERVICES
SERVICE_NAME_HASH	NUMBER	Service name hash from V$SERVICES
EVENT	VARCHAR2(64)	Name of the wait event; derived statistic name from V$EVENT_NAME **See Also:** " Oracle Wait Events"
EVENT_ID	NUMBER	Identifier of the wait event

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOTAL_WAITS	NUMBER	Total number of waits for the event by the service
TOTAL_TIMEOUTS	NUMBER	Total number of timeouts for the event by the service
TIME_WAITED	NUMBER	Total amount of time waited for the event by the service (in hundredths of a second)
AVERAGE_WAIT	NUMBER	Average amount of time waited for the event by the service (in hundredths of a second)
MAX_WAIT	NUMBER	Maximum amount of time waited for the event by the service (in hundredths of a second)
TIME_WAITED_MICRO	NUMBER	Total amount of time waited for the event by the service (in microseconds)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.8 V$SERVICE_REGION_METRIC

V$SERVICE_REGION_METRIC displays the metric values captured for the most recent 30-second intervals for the workload against each service region available on the database.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
SERVICE_ID	NUMBER	Service number (internal)
SERVICE_NETWORK_NAME	VARCHAR2(512)	Network name for the service
REGION_NAME	VARCHAR2(30)	Region name
CALLSPERSEC	NUMBER	Number of user calls per second to the services
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$CHUNK_METRIC"

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.9 V$SERVICE_STATS

V$SERVICE_STATS displays a minimal set of performance statistics. These call rate statistics are used for making run-time routing decisions, for tracking service levels, and for per-instance diagnostics per call rate. The elapsed timing for each call provides a relative value across instances for how well a node is processing SQL calls issued under a service name. When aggregation is enabled for the Service Name, then this view provides the timing and work done for calls issued for the whole service.

Column	Datatype	Description
SERVICE_NAME_HASH	NUMBER	Service name hash from V$SERVICES
SERVICE_NAME	VARCHAR2(64)	Service name from V$SERVICES
STAT_ID	NUMBER	Statistic identifier
STAT_NAME	VARCHAR2(64)	Derived statistic name from V$STATNAME and V$SESS_TIME_MODEL
VALUE	NUMBER	For statistics that measure time (such as the DB CPU, background elapsed time, or parse time elapsed statistics), this column displays a cumulative value in microseconds. For other statistics that do not measure time (such as the db block changes, execute count, or logons cumulative statistics), this column displays the appropriate numeric value for the statistic.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SERVICES"
- "V$STATNAME"
- "V$SESS_TIME_MODEL"
- *Oracle Database Performance Tuning Guide* for more information about using database statistics to manage the performance of Oracle Database

9.10 V$SERVICE_WAIT_CLASS

V$SERVICE_WAIT_CLASS displays aggregated wait counts and wait times for each wait statistic. An aggregation of these wait classes is used when thresholds are imported.

Column	Datatype	Description
SERVICE_NAME	VARCHAR2(64)	Service name from V$SERVICES
SERVICE_NAME_HASH	NUMBER	Service name hash from V$SERVICES
WAIT_CLASS_ID	NUMBER	Identifier of the wait class

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
WAIT_CLASS#	NUMBER	Number of the wait class
WAIT_CLASS	VARCHAR2(64)	Name of the wait class
TOTAL_WAITS	NUMBER	Total number waits from this wait class by the service
TIME_WAITED	NUMBER	Total amount of time spent in waits from this wait class by the service (in hundredths of a second)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.11 V$SERVICEMETRIC

V$SERVICEMETRIC displays metric values measured on the most recent time interval period for services executing inside the database. Service metrics are measured in 5-second and 1-minute intervals.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin timestamp for the interval period
END_TIME	DATE	End timestamp for the interval period
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	Group ID for the service metric group
SERVICE_NAME_HASH	NUMBER	Service name hash
SERVICE_NAME	VARCHAR2(64)	Service name
CTMHASH	NUMBER	Service create timestamp hash value
ELAPSEDPERCALL	NUMBER	Elapsed time per call (in microseconds). This column is deprecated in favor of the DBTIMEPERCALL column.
CPUPERCALL	NUMBER	CPU time per call (in microseconds)
DBTIMEPERCALL	NUMBER	Elapsed time per call (in microseconds)
CALLSPERSEC	NUMBER	Number of user calls per second
DBTIMEPERSEC	NUMBER	Database time per second
GOODNESS	NUMBER	Indicates how attractive a given instance is with respect to processing the workload that is presented to the service. A lower number is better. This number is internally computed based on the GOAL (LONG or SHORT) that is specified for the particular service.
DELTA	NUMBER	Indicates the predicted increase in the goodness for every additional session that is routed to this instance
FLAGS	NUMBER	Flags that can be any of the following values: • 0x01 - Service is BLOCKED from accepting new connections • 0x02 - Service is VIOLATING the set threshold on some metric • 0x04 - Goodness is UNKNOWN

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.12 V$SERVICEMETRIC_HISTORY

V$SERVICEMETRIC_HISTORY displays a recent history of the metric values measured in predefined time interval periods for services executing inside the database. Service metrics are measured in 5-second and 1-minute intervals.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin timestamp for the interval period
END_TIME	DATE	End timestamp for the interval period
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	Group ID for the service metric group
SERVICE_NAME_HASH	NUMBER	Service name hash
SERVICE_NAME	VARCHAR2(64)	Service name
CTMHASH	NUMBER	Service create timestamp hash value
ELAPSEDPERCALL	NUMBER	Elapsed time per call (in microseconds). This column is deprecated in favor of the DBTIMEPERCALL column.
CPUPERCALL	NUMBER	CPU time per call (in microseconds)
DBTIMEPERCALL	NUMBER	Elapsed time per call (in microseconds)
CALLSPERSEC	NUMBER	Number of user calls per second
DBTIMEPERSEC	NUMBER	Database time per second
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.13 V$SERVICES

V$SERVICES displays information about the services in the database.

Column	Datatype	Description
SERVICE_ID	NUMBER	Service ID
NAME	VARCHAR2(64)	Name of the service

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NAME_HASH	NUMBER	Service name hash
NETWORK_NAME	VARCHAR2(512)	Network name
CREATION_DATE	DATE	Creation date
CREATION_DATE_HASH	NUMBER	Creation date hash
GOAL	VARCHAR2(12)	Runtime Load Balancing Goal being used to create run-time load balancing and connection load balancing advice: • NONE • SERVICE_TIME - Connections are balanced by response time • THROUGHPUT - Connections are balanced by throughput
DTP	VARCHAR2(1)	Indicates whether or not Distributed Transaction Processing is enabled for this service. When Distributed Transaction Processing is set to Y (YES), it means that the service is offered at exactly one instance at a time for XA affinity. Possible values: • Y - YES • N - NO
AQ_HA_NOTIFICATION	VARCHAR2(3)	Indicates whether FAN - Fast Application Notification for OCI connections is set (YES) or not (NO)
CLB_GOAL	VARCHAR2(5)	Connection load balancing goal used with statistics that are sent to the listeners to determine how new connections are distributed: • LONG • SHORT
COMMIT_OUTCOME	VARCHAR2(3)	For Transaction Guard, indicates whether the database service associated with the user session has the COMMIT_OUTCOME service attribute enabled (YES) or not (NO). This attribute applies on a per session basis and is set at connect time. When COMMIT_OUTCOME = YES: • Transaction Guard manages the commit status for all supported transaction types. The outcome of a COMMIT transaction is known. If there is an outage, the application can use DBMS_APP_CONT.GET_LTXID_OUTCOME to return a reliable status for the last in-flight work. • A logical transaction ID (LTXID) is set for each user session at login and at each successful commit. **See Also:** For information about preserving the commit outcome, see *Oracle Database Development Guide*. For information about logical transaction IDs, see *Oracle Database Development Guide*
RETENTION_TIME	NUMBER	For Transaction Guard, when COMMIT_OUTCOME = YES, this value indicates the amount of time (in seconds) that the commit outcome is retained in the database.
REPLAY_INITIATION_TIMEOUT	NUMBER	For Application Continuity, this option specifies the difference between the time (in seconds) of original execution of the first operation of a request and the time that the replay is ready to start after a successful reconnect. Application Continuity will not replay after the specified amount of time has passed. This option is intended to avoid the unintentional execution of a request when a system is recovered after a long period of time. The default is 5 minutes (300 seconds).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SESSION_STATE_CONSISTENCY	VARCHAR2(128)	Describes how non-transactional is changed during a request. This parameter is considered only if failover_type is set to TRANSACTION for Application Continuity. Examples of session state are NLS settings, optimizer preferences, event settings, PL/SQL global variables, temporary tables, advanced queues, LOBs, and result cache. If non-transactional values change after the request starts, the default value of DYNAMIC should be set. Almost all applications should use DYNAMIC mode. If you are unsure, use DYNAMIC mode.
GLOBAL	VARCHAR2(3)	Indicates whether the service is global. A global service is managed by Global Service Manager (GSM) and can be provided by multiple databases that contain replicated data. Possible values: • YES: Indicates the service is global • NO: Indicates the service is not global
PDB	VARCHAR2(128)	Name of a pluggable database (PDB) associated with a given service. Will contain NULL if a database is a non-CDB or if the service is not associated with a PDB (that is, connecting to a CDB using this service will cause a user to connect to the root.)
SQL_TRANSLATION_PROFILE	VARCHAR2(261)	A non-NULL value specifies the initial SQL translation profile for subsequent database connections that use the service and do not specify a SQL translation profile. A NULL value has no effect.
MAX_LAG_TIME	VARCHAR2(128)	The maximum replication lag (in seconds) that is acceptable for a data replica to be used for providing the database service. Can only be specified for global services.
STOP_OPTION	VARCHAR2(128)	Stop option for sessions of this service for planned maintenance
FAILOVER_RESTORE	VARCHAR2(128)	Indicates whether sessions restore their commonly used session state (such as NLS and schema settings) when they are failed over. Possible values: • AUTO - Sessions restore their commonly used session state when they are failed over with Transparent Application Continuity • LEVEL1 - Sessions restore their commonly used session state when they are failed over with Application Continuity • NONE - Sessions do not restore their commonly used session state when they are failed over
DRAIN_TIMEOUT	NUMBER	Number of seconds to wait for sessions to be drained
TABLE_FAMILY_ID[1]	NUMBER	Sharded table family ID associated with the service
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

[1] This column is available starting with Oracle Database 19c.

9.14 V$SES_OPTIMIZER_ENV

V$SES_OPTIMIZER_ENV displays the contents of the optimizer environment used by each session. When a new session is first created, it automatically inherits its optimizer environment from the optimizer environment defined at the instance level by V$SYS_OPTIMIZER_ENV. The

Copyright © 2002, 2026, Oracle and/or its affiliates.

value of certain parameters can be dynamically modified by issuing an ALTER SESSION statement.

Column	Datatype	Description
SID	NUMBER	Session identifier. This column can be used to join with V$SESSION on the SID column.
ID	NUMBER	Unique identifier of the parameter in the optimizer environment
NAME	VARCHAR2(40)	Name of the parameter
SQL_FEATURE	VARCHAR2(64)	Associated feature control ID
ISDEFAULT	VARCHAR2(3)	Indicates whether the parameter is set to the default value (YES) or not (NO)
VALUE	VARCHAR2(25)	Value of the parameter for the session
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SYS_OPTIMIZER_ENV"

9.15 V$SESS_IO

V$SESS_IO displays I/O statistics for each user session.

Column	Datatype	Description
SID	NUMBER	Session identifier
BLOCK_GETS	NUMBER	Block gets for this session
CONSISTENT_GETS	NUMBER	Consistent gets for this session
PHYSICAL_READS	NUMBER	Physical reads for this session
BLOCK_CHANGES	NUMBER	Block changes for this session
CONSISTENT_CHANGES	NUMBER	Consistent changes for this session
OPTIMIZED_PHYSICAL_READS	NUMBER	Number of physical reads from Database Smart Flash Cache for this session
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.16 V$SESS_TIME_MODEL

V$SESS_TIME_MODEL displays the session-accumulated time for various operations. The time reported is the total elapsed or CPU time (in microseconds). Any timed operation will buffer at most 5 seconds of time data. Specifically, this means that if a timed operation (such as SQL execution) takes a long period of time to perform, the data published to this view is at most missing 5 seconds of the time accumulated for the operation.
The time values are 8-byte integers and can therefore hold approximately 580,000 years of time before wrapping. Background process time is not included in a statistic value unless the statistic is specifically for background processes.

Column	Datatype	Description
SID	NUMBER	Session ID (same as in V$SESSION)
STAT_ID	NUMBER	Statistic identifier for the time statistic
STAT_NAME	VARCHAR2(64)	Name of the statistic (see Table 9-1)
VALUE	NUMBER	Amount of time (in microseconds) that the session has spent in this operation
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Table 9-1 V$SESS_TIME_MODEL and V$SYS_TIME_MODEL Statistics

Statistic Name	Description
DB time	Amount of elapsed time (in microseconds) spent performing Database user-level calls. This does not include the elapsed time spent on instance background processes such as PMON.
DB CPU	Amount of CPU time (in microseconds) spent on database user-level calls. This does not include the CPU time spent on instance background processes such as PMON.
background elapsed time	Amount of elapsed time (in microseconds) consumed by database background processes.
background CPU time	Amount of CPU time (in microseconds) consumed by database background processes.
sequence load elapsed time	Amount of elapsed time spent getting the next sequence number from the data dictionary. If a sequence is cached, then this is the amount of time spent replenishing the cache when it runs out. No time is charged when a sequence number is found in the cache. For non-cached sequences, some time will be charged for every nextval call.
parse time elapsed	Amount of elapsed time spent parsing SQL statements. It includes both soft and hard parse time.
hard parse elapsed time	Amount of elapsed time spent hard parsing SQL statements.
SQL execute elapsed time	Amount of elapsed time SQL statements are executing. Note that for select statements this also includes the amount of time spent performing fetches of query results.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table 9-1 (Cont.) V$SESS_TIME_MODEL and V$SYS_TIME_MODEL Statistics

Statistic Name	Description
connection management call elapsed time	Amount of elapsed time spent performing session connect and disconnect calls.
failed parse elapsed time	Amount of time spent performing SQL parses which ultimately fail with some parse error.
failed parse (out of shared memory) elapsed time	Amount of time spent performing SQL parses which ultimately fail with error `ORA-04031`.
hard parse (sharing criteria) elapsed time	Amount of elapsed time spent performing SQL hard parses when the hard parse resulted from not being able to share an existing cursor in the SQL cache.
hard parse (bind mismatch) elapsed time	Amount of elapsed time spent performing SQL hard parses when the hard parse resulted from bind type or bind size mismatch with an existing cursor in the SQL cache.
PL/SQL execution elapsed time	Amount of elapsed time spent running the PL/SQL interpreter. This does not include time spent recursively executing/parsing SQL statements or time spent recursively executing the Java VM.
PL/SQL compilation elapsed time	Amount of elapsed time spent running the PL/SQL compiler.
inbound PL/SQL rpc elapsed time	Time inbound PL/SQL remote procedure calls have spent executing. It includes all time spent recursively executing SQL and JAVA, and therefore is not easily related to "PL/SQL execution elapsed time".
Java execution elapsed time	Amount of elapsed time spent running the Java VM. This does not include time spent recursively executing/parsing SQL statements or time spent recursively executing PL/SQL.
RMAN cpu time (backup/restore)	Amount of CPU time (in microseconds) spent in RMAN backup and restore operations.
repeated bind elapsed time	Amount of elapsed time spent giving new values to bind variables (rebinding).
OLAP engine elapsed time	Amount of time spent performing OLAP session transactions. This includes time spent on database user-level calls, SQL statement execution, and PL/SQL execution within the OLAP transaction.
OLAP engine CPU time	Amount of CPU time spent on OLAP session transactions. This includes time spent on database user-level calls, SQL statement execution, and PL/SQL execution within the OLAP transaction.

The relationships between the statistics listed in Table 9-1 form two trees in which all the time reported by a child in the tree is contained within the parent in the tree. The following are the relationship trees; the number is the level in the given tree.

```
1) background elapsed time
    2) background cpu time
          3) RMAN cpu time (backup/restore)
1) DB time
    2) DB CPU
    2) connection management call elapsed time
    2) sequence load elapsed time
    2) sql execute elapsed time
    2) parse time elapsed
          3) hard parse elapsed time
                4) hard parse (sharing criteria) elapsed time
                    5) hard parse (bind mismatch) elapsed time
          3) failed parse elapsed time
                4) failed parse (out of shared memory) elapsed time
    2) PL/SQL execution elapsed time
    2) inbound PL/SQL rpc elapsed time
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

```
2) PL/SQL compilation elapsed time
2) Java execution elapsed time
2) repeated bind elapsed time
```

The relationship between a parent and a child in the tree indicates containment only. Keep the following in mind regarding the tree:

- Children do not necessarily add up to the parent.
- Children are not necessarily exclusive (that is, they may overlap).
- The union of children does not necessarily cover the whole of the parent.

> **See Also**
>
> "V$SYS_TIME_MODEL"

9.17 V$SESSION

V$SESSION displays session information for each current session.

Column	Datatype	Description
SADDR	RAW(4 | 8)	Session address
SID	NUMBER	Session identifier
SERIAL#	NUMBER	Session serial number. Used to uniquely identify a session's objects. Guarantees that session-level commands are applied to the correct session objects if the session ends and another session begins with the same session ID.
AUDSID	NUMBER	Auditing session ID • If the database is configured for unified auditing, then this column displays the unified auditing session ID • If the database is configured for mixed mode auditing, then this column displays the traditional auditing session ID
PADDR	RAW(4 | 8)	Address of the process that owns the session
USER#	NUMBER	Oracle user identifier
USERNAME	VARCHAR2(128)	Oracle username
COMMAND	NUMBER	Command in progress (last statement parsed). You can find the command name for any value *n* returned in this COMMAND column by running this SQL query: `SELECT command_name FROM v$sqlcommand WHERE command_type = n;` A value of 0 in this COMMAND column means the command is not recorded in V$SESSION.
OWNERID	NUMBER	Identifier of the user who owns the migratable session; the column contents are invalid if the value is 2147483644 For operations using parallel worker processes, interpret this value as a 4-byte value. The low-order 2 bytes represent the session number and the high-order bytes represent the instance ID of the query coordinator.
TADDR	VARCHAR2(16)	Address of the transaction state object

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LOCKWAIT	VARCHAR2(16)	Address of the lock the session is waiting for; NULL if none
STATUS	VARCHAR2(8)	Status of the session: • ACTIVE - Session currently executing SQL • INACTIVE - Session which is inactive and either has no configured limits or has not yet exceeded the configured limits • KILLED - Session marked to be terminated • CACHED - Session temporarily cached for use by Oracle*XA • SNIPED - An inactive session that has exceeded some configured limits (for example, resource limits specified for the resource manager consumer group or idle_time specified in the user's profile). Such sessions will not be allowed to become active again.
SERVER	VARCHAR2(9)	Server type: • DEDICATED • SHARED • PSEUDO • POOLED • NONE
SCHEMA#	NUMBER	Schema user identifier
SCHEMANAME	VARCHAR2(128)	Schema user name
OSUSER	VARCHAR2(128)	Operating system client user name
PROCESS	VARCHAR2(24)	Operating system client process ID
MACHINE	VARCHAR2(64)	Operating system machine name
PORT	NUMBER	Client port number
TERMINAL	VARCHAR2(30)	Operating system terminal name
PROGRAM	VARCHAR2(48)	Operating system program name
TYPE	VARCHAR2(10)	Session type
SQL_ADDRESS	RAW(4 | 8)	Used with SQL_HASH_VALUE to identify the SQL statement that is currently being executed
SQL_HASH_VALUE	NUMBER	Used with SQL_ADDRESS to identify the SQL statement that is currently being executed
SQL_ID	VARCHAR2(13)	SQL identifier of the SQL statement that is currently being executed
SQL_CHILD_NUMBER	NUMBER	Child number of the SQL statement that is currently being executed
SQL_EXEC_START	DATE	Time when the execution of the SQL currently executed by this session started; NULL if SQL_ID is NULL
SQL_EXEC_ID	NUMBER	SQL execution identifier; NULL if SQL_ID is NULL or if the execution of that SQL has not yet started (see V$SQL_MONITOR)
PREV_SQL_ADDR	RAW(4 | 8)	Used with PREV_HASH_VALUE to identify the last SQL statement executed
PREV_HASH_VALUE	NUMBER	Used with SQL_HASH_VALUE to identify the last SQL statement executed
PREV_SQL_ID	VARCHAR2(13)	SQL identifier of the last SQL statement executed
PREV_CHILD_NUMBER	NUMBER	Child number of the last SQL statement executed
PREV_EXEC_START	DATE	SQL execution start of the last executed SQL statement
PREV_EXEC_ID	NUMBER	SQL execution identifier of the last executed SQL statement
PLSQL_ENTRY_OBJECT_ID	NUMBER	Object ID of the top-most PL/SQL subprogram on the stack; NULL if there is no PL/SQL subprogram on the stack

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PLSQL_ENTRY_SUBPROGRAM_ID	NUMBER	Subprogram ID of the top-most PL/SQL subprogram on the stack; NULL if there is no PL/SQL subprogram on the stack
PLSQL_OBJECT_ID	NUMBER	Object ID of the currently executing PL/SQL subprogram; NULL if executing SQL
PLSQL_SUBPROGRAM_ID	NUMBER	Subprogram ID of the currently executing PL/SQL object; NULL if executing SQL
MODULE	VARCHAR2(64)	Name of the currently executing module as set by calling the DBMS_APPLICATION_INFO.SET_MODULE procedure
MODULE_HASH	NUMBER	Hash value of the MODULE column
ACTION	VARCHAR2(64)	Name of the currently executing action as set by calling the DBMS_APPLICATION_INFO.SET_ACTION procedure
ACTION_HASH	NUMBER	Hash value of the ACTION column
CLIENT_INFO	VARCHAR2(64)	Information set by the DBMS_APPLICATION_INFO.SET_CLIENT_INFO procedure
FIXED_TABLE_SEQUENCE	NUMBER	This contains a number that increases every time the session completes a call to the database and there has been an intervening select from a dynamic performance table. This column can be used by performance monitors to monitor statistics in the database. Each time the performance monitor looks at the database, it only needs to look at sessions that are currently active or have a higher value in this column than the highest value that the performance monitor saw the last time. All the other sessions have been idle since the last time the performance monitor looked at the database.
ROW_WAIT_OBJ#	NUMBER	Object ID for the table containing the row specified in ROW_WAIT_ROW#
ROW_WAIT_FILE#	NUMBER	Identifier for the data file containing the row specified in ROW_WAIT_ROW#. This column is valid only if the session is currently waiting for another transaction to commit and the value of ROW_WAIT_OBJ# is not -1.
ROW_WAIT_BLOCK#	NUMBER	Identifier for the block containing the row specified in ROW_WAIT_ROW#. This column is valid only if the session is currently waiting for another transaction to commit and the value of ROW_WAIT_OBJ# is not -1.
ROW_WAIT_ROW#	NUMBER	Current row being locked. This column is valid only if the session is currently waiting for another transaction to commit and the value of ROW_WAIT_OBJ# is not -1.
TOP_LEVEL_CALL#	NUMBER	Oracle top level call number
LOGON_TIME	DATE	Time of logon
LAST_CALL_ET	NUMBER	If the session STATUS is currently ACTIVE, then the value represents the elapsed time (in seconds) since the session has become active. If the session STATUS is currently INACTIVE, then the value represents the elapsed time (in seconds) since the session has become inactive.
PDML_ENABLED	VARCHAR2(3)	This column has been replaced by the PDML_STATUS column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FAILOVER_TYPE	VARCHAR2(13)	Indicates whether failover is enabled for this session: • NONE - Failover is disabled for this session. • SESSION - Transparent application failover (TAF) is enabled for this session and the client can fail over its session following a disconnect. • SELECT - TAF is enabled for this session and the client can fail over queries. The application must not set session state after the initial setup. (Transactions are not included at failover.) • TRANSACTION - Application Continuity (AC) is enabled for this session. This feature is available with the Oracle Real Application Clusters (RAC), Oracle RAC One Node, and Oracle Active Data Guard options. AC hides outages from end users and applications by recovering the database sessions following recoverable outages and planned maintenance. • AUTO - Transparent Application Continuity (TAC) is enabled for this session. This feature is the automated version of Application Continuity (AC) that makes decisions on your behalf. TAC is on by default for Oracle Autonomous Database Cloud and can also be on by default for your applications. **See Also:** • *Oracle Database Net Services Administrator's Guide* for more information on TAF • *Oracle Real Application Clusters Administration and Deployment Guide* for more information on AC • *Oracle Real Application Clusters Administration and Deployment Guide* for more information on TAC
FAILOVER_METHOD	VARCHAR2(10)	Indicates the transparent application failover method for the session: • NONE - Failover is disabled for this session • BASIC - Client itself reconnects following a disconnect • PRECONNECT - Backup instance can support all connections from every instance for which it is backed up
FAILED_OVER	VARCHAR2(3)	Indicates whether the session is running in failover mode and failover has occurred (YES) or not (NO)
RESOURCE_CONSUMER_GROUP	VARCHAR2(32)	Name of the session's current resource consumer group
PDML_STATUS	VARCHAR2(8)	If ENABLED, the session is in a PARALLEL DML enabled mode. If DISABLED, PARALLEL DML enabled mode is not supported for the session. If FORCED, the session has been altered to force PARALLEL DML.
PDDL_STATUS	VARCHAR2(8)	If ENABLED, the session is in a PARALLEL DDL enabled mode. If DISABLED, PARALLEL DDL enabled mode is not supported for the session. If FORCED, the session has been altered to force PARALLEL DDL.
PQ_STATUS	VARCHAR2(8)	If ENABLED, the session is in a PARALLEL QUERY enabled mode. If DISABLED, PARALLEL QUERY enabled mode is not supported for the session. If FORCED, the session has been altered to force PARALLEL QUERY.
CURRENT_QUEUE_DURATION	NUMBER	If queued (1), the current amount of time the session has been queued. If not currently queued, the value is 0.
CLIENT_IDENTIFIER	VARCHAR2(64)	Client identifier of the session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BLOCKING_SESSION_STATUS	VARCHAR2(11)	This column provides details on whether there is a blocking session: • VALID - there is a blocking session, and it is identified in the BLOCKING_INSTANCE and BLOCKING_SESSION columns • NO HOLDER - there is no session blocking this session • NOT IN WAIT - this session is not in a wait • UNKNOWN - the blocking session is unknown
BLOCKING_INSTANCE	NUMBER	Instance identifier of the blocking session. This column is valid only if BLOCKING_SESSION_STATUS has the value VALID.
BLOCKING_SESSION	NUMBER	Session identifier of the blocking session. This column is valid only if BLOCKING_SESSION_STATUS has the value VALID.
FINAL_BLOCKING_SESSION_STATUS	VARCHAR2(11)	The final blocking session is the final element in the wait chain constructed by following the sessions that are blocked by one another starting with this session. In the case of a cyclical wait chain, one of the sessions in the wait chain will be chosen as the final blocker. This column provides details on whether there is a final blocking session: • VALID - there is a final blocking session and it is identified in the FINAL_BLOCKING_INSTANCE and FINAL_BLOCKING_SESSION columns • NO HOLDER - there is no session blocking this session • NOT IN WAIT - this session is not in a wait • UNKNOWN - the final blocking session is unknown
FINAL_BLOCKING_INSTANCE	NUMBER	Instance identifier of the final blocking session. This column is valid only if FINAL_BLOCKING_SESSION_STATUS has the value VALID.
FINAL_BLOCKING_SESSION	NUMBER	Session identifier of the blocking session. This column is valid only if FINAL_BLOCKING_SESSION_STATUS has the value VALID.
SEQ#	NUMBER	A number that uniquely identifies the current or last wait (incremented for each wait)
EVENT#	NUMBER	If the session is currently waiting, then the number of the resource or event for which the session is waiting. If the session is not in a wait, then the number of the resource or event for which the session most recently waited.
EVENT	VARCHAR2(64)	If the session is currently waiting, then the resource or event for which the session is waiting. If the session is not in a wait, then the resource or event for which the session most recently waited. **See Also:** " Oracle Wait Events"
P1TEXT	VARCHAR2(64)	Description of the first wait event parameter
P1	NUMBER	First wait event parameter (in decimal)
P1RAW	RAW(8)	First wait event parameter (in hexadecimal)[1]
P2TEXT	VARCHAR2(64)	Description of the second wait event parameter
P2	NUMBER	Second wait event parameter (in decimal)
P2RAW	RAW(8)	Second wait event parameter (in hexadecimal)[1]
P3TEXT	VARCHAR2(64)	Description of the third wait event parameter
P3	NUMBER	Third wait event parameter (in decimal)
P3RAW	RAW(8)	Third wait event parameter (in hexadecimal)[1]
WAIT_CLASS_ID	NUMBER	Identifier of the class of the wait event
WAIT_CLASS#	NUMBER	Number of the class of the wait event
WAIT_CLASS	VARCHAR2(64)	Name of the class of the wait event

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
WAIT_TIME	NUMBER	If the session is currently waiting, then the value is 0. If the session is not in a wait, then the value is as follows: • > 0 - Value is the duration of the last wait in hundredths of a second • -1 - Duration of the last wait was less than a hundredth of a second • -2 - Parameter TIMED_STATISTICS was set to false This column has been deprecated in favor of the columns WAIT_TIME_MICRO and STATE.
SECONDS_IN_WAIT	NUMBER	If the session is currently waiting, then the value is the amount of time waited for the current wait. If the session is not in a wait, then the value is the amount of time since the start of the last wait. This column has been deprecated in favor of the columns WAIT_TIME_MICRO and TIME_SINCE_LAST_WAIT_MICRO.
STATE	VARCHAR2(19)	Wait state: • WAITING - Session is currently waiting • WAITED UNKNOWN TIME - Duration of the last wait is unknown; this is the value when the parameter TIMED_STATISTICS is set to false • WAITED SHORT TIME - Last wait was less than a hundredth of a second • WAITED KNOWN TIME - Duration of the last wait is specified in the WAIT_TIME column
WAIT_TIME_MICRO	NUMBER	Amount of time waited (in microseconds). If the session is currently waiting, then the value is the time spent in the current wait. If the session is currently not in a wait, then the value is the amount of time waited in the last wait. A wait includes code within its beginning and end that can encounter other waits as part of its implementation. For example, a wait might need to acquire a latch to check state to determine when the wait is satisfied, and the latch itself might enter a wait. In this case, when the latch wait ends and returns to the outer wait, the value of WAIT_TIME_MICRO is reset to start after the latch wait.
TIME_REMAINING_MICRO	NUMBER	Value is interpreted as follows: • > 0 - Amount of time remaining for the current wait (in microseconds) • 0 - Current wait has timed out • -1 - Session can indefinitely wait in the current wait • NULL - Session is not currently waiting
TIME_SINCE_LAST_WAIT_MICRO	NUMBER	Time elapsed since the end of the last wait (in microseconds). If the session is currently in a wait, then the value is 0.
SERVICE_NAME	VARCHAR2(64)	Service name of the session
SQL_TRACE	VARCHAR2(8)	Indicates whether SQL tracing is enabled (ENABLED) or disabled (DISABLED)
SQL_TRACE_WAITS	VARCHAR2(5)	Indicates whether wait tracing is enabled (TRUE) or not (FALSE)
SQL_TRACE_BINDS	VARCHAR2(5)	Indicates whether bind tracing is enabled (TRUE) or not (FALSE)
SQL_TRACE_PLAN_STATS	VARCHAR2(10)	Frequency at which row source statistics are dumped in the trace files for each cursor: • never • first_execution • all_executions
SESSION_EDITION_ID	NUMBER	Shows the value that, in the session, would be reported by sys_context('USERENV', 'SESSION_EDITION_ID')

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CREATOR_ADDR	RAW(4 | 8)	Address of the creating process or circuit
CREATOR_SERIAL#	NUMBER	Serial number of the creating process or circuit
ECID	VARCHAR2(64)	Execution context identifier (sent by Application Server)
SQL_TRANSLATION_PROFILE_ID	NUMBER	Object number of the SQL translation profile
PGA_TUNABLE_MEM	NUMBER	The amount of tunable PGA memory (in bytes). Untunable memory is PGA_ALLOC_MEM from V$PROCESS minus PGA_TUNABLE_MEM from V$SESSION.
SHARD_DDL_STATUS	VARCHAR2(8)	Indicates whether shard DDL is enabled in the current session (ENABLED) or not (DISABLED). This value is only relevant for the shard catalog database.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
EXTERNAL_NAME	VARCHAR2(1024)	External name of the database user. For enterprise users, returns the Oracle Internet Directory DN.
PLSQL_DEBUGGER_CONNECTED	VARCHAR2(5)	Indicates whether the session is connected to a PL/SQL debugger. Possible values: • TRUE • FALSE

[1] The P1RAW, P2RAW, and P3RAW columns display the same values as the P1, P2, and P3 columns, except that the numbers are displayed in hexadecimal.

See Also

- *Oracle Database Performance Tuning Guide* for an example of using V$SESSION to help identify an object that is waiting for buffer busy waits
- *Oracle Database Performance Tuning Guide* for an example of using V$SESSION to determine which sessions are waiting for I/O
- *Oracle Database PL/SQL Packages and Types Reference* for more information on the DBMS_APPLICATION_INFO package

9.18 V$SESSION_BLOCKERS

V$SESSION_BLOCKERS displays the blocker sessions for each blocked session. Each row represents a blocked and blocker session pair. If a session is blocked by multiple sessions there will be multiple rows for that blocked session. The maximum number of blocker sessions displayed for a single blocked session is 30. If a session is not blocked by other sessions, then there will be no row in this view for that session.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SID	NUMBER	Blocked session's Oracle session identifier
SESS_SERIAL#	NUMBER	Blocked session's Oracle session serial number
WAIT_ID	NUMBER	A number identifying the wait for the blocked session
WAIT_EVENT	NUMBER	Resource or event number for which the blocked session is waiting
WAIT_EVENT_TEXT	VARCHAR2(64)	Resource or event for which the blocked session is waiting
BLOCKER_INSTANCE_ID	NUMBER	Blocker session's instance identifier
BLOCKER_SID	NUMBER	Blocker session's Oracle session identifier
BLOCKER_SESS_SERIAL#	NUMBER	Blocker session's Oracle session serial number
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.19 V$SESSION_CONNECT_INFO

V$SESSION_CONNECT_INFO displays information about network connections for all currently logged in sessions.

Column	Datatype	Description
SID	NUMBER	Session identifier (can be used to join this view with V$SESSION)
SERIAL#	NUMBER	Session serial number. Used to uniquely identify a session's objects. Guarantees that session-level commands are applied to the correct session objects if the session ends and another session begins with the same session ID. (Can be used to join this view with V$SESSION.)
AUTHENTICATION_TYPE	VARCHAR2(26)	How the user was authenticated: • DATABASE - Username/password authentication • OS - Operating system external user authentication • NETWORK - Network protocol or ANO authentication • PROXY - OCI proxy connection authentication • SERVER • PASSWORD • EXTERNAL ADAPTERS • INTERNAL • GLOBAL • EXTERNAL • PASSWORD BASED GLOBAL USER
OSUSER	VARCHAR2(128)	External username for this database user
NETWORK_SERVICE_BANNER	VARCHAR2(4000)	Product banners for each Oracle Net service used for this connection (one row per banner)
CLIENT_CHARSET	VARCHAR2(40)	Client character set as specified in the NLS_LANG environment variable or in the OCIEnvNlsCreate() call; Unknown if the Oracle client is older than release 11.1 or the connection is through the JDBC thin driver

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CLIENT_CONNECTION	VARCHAR2(13)	Client server connection flags: • Heterogeneous • Homogeneous
CLIENT_OCI_LIBRARY	VARCHAR2(27)	OCI client library: • Home-based • Full Instant Client • Light Weight Instant Client • OCI • Unknown
CLIENT_VERSION	VARCHAR2(40)	Client library version number
CLIENT_DRIVER	VARCHAR2(30)	Client driver name For pooled connections, the value of this column is null.
CLIENT_LOBATTR	VARCHAR2(23)	Client LOB flags: • Client Temp Lob Rfc On • Client Temp Lob Rfc Off
CLIENT_REGID	NUMBER	Query cache registration ID sent by the client
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SESSION"

9.20 V$SESSION_CURSOR_CACHE

V$SESSION_CURSOR_CACHE displays information on cursor usage for the current session.

Note

The V$SESSION_CURSOR_CACHE view is not a measure of the effectiveness of the SESSION_CACHED_CURSORS initialization parameter.

Column	Datatype	Description
MAXIMUM	NUMBER	Maximum number of cursors to cache. Once you hit this number, some cursors will need to be closed in order to open more. The value in this column is derived from the initialization parameter SESSION_CACHED_CURSORS.
COUNT	NUMBER	Current number of cursors (whether they are in use or not)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OPENS	NUMBER	Cumulative total of cursor opens minus one. This is because the cursor that is currently open and being used for this query is not counted in the OPENS statistic.
HITS	NUMBER	Cumulative total of cursor open hits
HIT_RATIO	NUMBER	Ratio of the number of times an open cursor was found divided by the number of times a cursor was sought
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"SESSION_CACHED_CURSORS"

9.21 V$SESSION_EVENT

V$SESSION_EVENT displays information on waits for an event by a session. Note that the TIME_WAITED and AVERAGE_WAIT columns will contain a value of zero on those platforms that do not support a fast timing mechanism. If you are running on one of these platforms and you want this column to reflect true wait times, then you must set TIMED_STATISTICS to true in the parameter file; doing this will have a small negative effect on system performance.

See Also

"TIMED_STATISTICS"

Column	Datatype	Description
SID	NUMBER	Session identifier
EVENT	VARCHAR2(64)	Name of the wait event; derived statistic name from V$EVENT_NAME **See Also:** " Oracle Wait Events"
TOTAL_WAITS	NUMBER	Total number of waits for the event by the session
TOTAL_TIMEOUTS	NUMBER	Total number of timeouts for the event by the session
TIME_WAITED	NUMBER	Total amount of time waited for the event by the session (in hundredths of a second)
AVERAGE_WAIT	NUMBER	Average amount of time waited for the event by the session (in hundredths of a second)
MAX_WAIT	NUMBER	Maximum amount of time waited for the event by the session (in hundredths of a second)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TIME_WAITED_MICRO	NUMBER	Total amount of time waited for the event by the session (in microseconds)
EVENT_ID	NUMBER	Identifier of the wait event
WAIT_CLASS_ID	NUMBER	Identifier of the wait class of the wait event
WAIT_CLASS#	NUMBER	Number of the wait class of the wait event
WAIT_CLASS	VARCHAR2(64)	Name of the wait class of the wait event
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.22 V$SESSION_FIX_CONTROL

V$SESSION_FIX_CONTROL displays information about Fix Control (enabled/disabled) for the current session.

Column	Datatype	Description
SESSION_ID	NUMBER	Session identifier (can be used to join this view with V$SESSION)
BUGNO	NUMBER	Bug number (as fix control identifier)
VALUE	NUMBER	Current value set for the fix control
SQL_FEATURE	VARCHAR2(64)	Feature control ID
DESCRIPTION	VARCHAR2(64)	Description of the fix control
OPTIMIZER_FEATURE_ENABLE	VARCHAR2(25)	Version on (and after) which the fix is enabled by default
EVENT	NUMBER	Event formerly used to control the fix
IS_DEFAULT	NUMBER	Indicates whether the current value is the same as the default (1) or not (0)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.23 V$SESSION_LONGOPS

V$SESSION_LONGOPS displays the status of various operations that run for longer than 6 seconds (in absolute time). These operations currently include many backup and recovery functions, statistics gathering, and query execution, and more operations are added for every Oracle release.
To monitor query execution progress, you must be using the cost-based optimizer and you must:

Copyright © 2002, 2026, Oracle and/or its affiliates.

- Set the TIMED_STATISTICS or SQL_TRACE parameters to true
- Gather statistics for your objects with the DBMS_STATS package

You can add information to this view about application-specific long-running operations by using the DBMS_APPLICATION_INFO.SET_SESSION_LONGOPS procedure.

Column	Datatype	Description
SID	NUMBER	Identifier of the session processing the long-running operation. If multiple sessions are cooperating in the long-running operation, then SID corresponds to the main session.
SERIAL#	NUMBER	Serial number of the session processing the long-running operation. If multiple sessions are cooperating in the long-running operation, then SERIAL# corresponds to the main session. SERIAL# is used to uniquely identify a session's objects. Guarantees that session-level commands are applied to the correct session objects if the session ends and another session begins with the same session ID.
OPNAME	VARCHAR2(64)	Brief description of the operation
TARGET	VARCHAR2(64)	Object on which the operation is carried out
TARGET_DESC	VARCHAR2(32)	Description of the target
SOFAR	NUMBER	Units of work done so far for the operation specified in the OPNAME column
TOTALWORK	NUMBER	Total units of work for the operation specified in the OPNAME column
UNITS	VARCHAR2(32)	Units of measurement
START_TIME	DATE	Starting time of the operation
LAST_UPDATE_TIME	DATE	Time when statistics were last updated for the operation
TIMESTAMP	DATE	Timestamp specific to the operation
TIME_REMAINING	NUMBER	Estimate (in seconds) of time remaining for the operation to complete
ELAPSED_SECONDS	NUMBER	Number of elapsed seconds from the start of the operations
CONTEXT	NUMBER	Context
MESSAGE	VARCHAR2(512)	Statistics summary message
USERNAME	VARCHAR2(30)	User ID of the user performing the operation
SQL_ADDRESS	RAW(4 | 8)	Used with the value of the SQL_HASH_VALUE column to identify the SQL statement associated with the operation
SQL_HASH_VALUE	NUMBER	Used with the value of the SQL_ADDRESS column to identify the SQL statement associated with the operation
SQL_ID	VARCHAR2(13)	SQL identifier of the SQL statement associated with the long operation, if any
SQL_PLAN_HASH_VALUE	NUMBER	SQL plan hash value; NULL if SQL_ID is NULL
SQL_EXEC_START	DATE	Time when the execution of the SQL started; NULL if SQL_ID is NULL
SQL_EXEC_ID	NUMBER	SQL execution identifier (see V$SQL_MONITOR)
SQL_PLAN_LINE_ID	NUMBER	SQL plan line ID corresponding to the long operation; NULL if the long operation is not associated with a line of the execution plan
SQL_PLAN_OPERATION	VARCHAR2(30)	Plan operation name; NULL if SQL_PLAN_LINE_ID is NULL
SQL_PLAN_OPTIONS	VARCHAR2(30)	Plan operation options; NULL if SQL_PLAN_LINE_ID is NULL
QCSID	NUMBER	Session identifier of the parallel coordinator

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "TIMED_STATISTICS"
- "SQL_TRACE"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_SESSION_LONGOPS procedure
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_STATS package

9.24 V$SESSION_OBJECT_CACHE

V$SESSION_OBJECT_CACHE displays object cache statistics for the current user session on the local server (instance).

Column	Datatype	Description
PINS	NUMBER	Number of object pins or look-ups in the cache
HITS	NUMBER	Number of object pins that found the object already in the cache
TRUE_HITS	NUMBER	Number of object pins that found the object already in the cache and in the desired state (thus, not requiring refresh from the database)
HIT_RATIO	NUMBER	Ratio of HITS / PINS
TRUE_HIT_RATIO	NUMBER	Ratio of TRUE_HITS/PINS
OBJECT_REFRESHES	NUMBER	Number of objects in the cache that were refreshed with a new value from the database
CACHE_REFRESHES	NUMBER	Number of times the whole cache (all objects) were refreshed
OBJECT_FLUSHES	NUMBER	Number of objects in the cache that were flushed to the database
CACHE_FLUSHES	NUMBER	Number of times the whole cache (all objects) were flushed to the database
CACHE_SHRINKS	NUMBER	Number of times the cache was shrunk to the optimal size
CACHED_OBJECTS	NUMBER	Number of objects currently cached
PINNED_OBJECTS	NUMBER	Number of objects currently pinned
CACHE_SIZE	NUMBER	Current size of the cache (in bytes)
OPTIMAL_SIZE	NUMBER	Optimal size of the cache (in bytes)
MAXIMUM_SIZE	NUMBER	Maximum size of the cache (in bytes)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.25 V$SESSION_WAIT

V$SESSION_WAIT displays the current or last wait for each session.

Column	Datatype	Description
SID	NUMBER	Session identifier; maps to V$SESSION.SID
SEQ#	NUMBER	A number that uniquely identifies the current or last wait (incremented for each wait)
EVENT	VARCHAR2(64)	If the session is currently waiting, then the resource or event for which the session is waiting. If the session is not in a wait, then the resource or event for which the session most recently waited. **See Also:** " Oracle Wait Events"
P1TEXT	VARCHAR2(64)	Description of the first wait event parameter
P1	NUMBER	First wait event parameter (in decimal)
P1RAW	RAW(8)	First wait event parameter (in hexadecimal)[1]
P2TEXT	VARCHAR2(64)	Description of the second wait event parameter
P2	NUMBER	Second wait event parameter (in decimal)
P2RAW	RAW(8)	Second wait event parameter (in hexadecimal)[1]
P3TEXT	VARCHAR2(64)	Description of the third wait event parameter
P3	NUMBER	Third wait event parameter (in decimal)
P3RAW	RAW(8)	Third wait event parameter (in hexadecimal)[1]
WAIT_CLASS_ID	NUMBER	Identifier of the class of the wait event
WAIT_CLASS#	NUMBER	Number of the class of the wait event
WAIT_CLASS	VARCHAR2(64)	Name of the class of the wait event
WAIT_TIME	NUMBER	If the session is currently waiting, then the value is 0. If the session is not in a wait, then the value is as follows: • > 0 - Value is the duration of the last wait in hundredths of a second • -1 - Duration of the last wait was less than a hundredth of a second • -2 - Parameter TIMED_STATISTICS was set to false This column has been deprecated in favor of the columns WAIT_TIME_MICRO and STATE.
SECONDS_IN_WAIT	NUMBER	If the session is currently waiting, then the value is the amount of time waited for the current wait. If the session is not in a wait, then the value is the amount of time since the start of the last wait. This column has been deprecated in favor of the columns WAIT_TIME_MICRO and TIME_SINCE_LAST_WAIT_MICRO.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATE	VARCHAR2(19)	Wait state: • WAITING - Session is currently waiting • WAITED UNKNOWN TIME - Duration of the last wait is unknown; this is the value when the parameter TIMED_STATISTICS is set to false • WAITED SHORT TIME - Last wait was less than a hundredth of a second • WAITED KNOWN TIME - Duration of the last wait is specified in the WAIT_TIME column
WAIT_TIME_MICRO	NUMBER	Amount of time waited (in microseconds). If the session is currently waiting, then the value is the time spent in the current wait. If the session is currently not in a wait, then the value is the amount of time waited in the last wait.
TIME_REMAINING_MICRO	NUMBER	Value is interpreted as follows: • > 0 - Amount of time remaining for the current wait (in microseconds) • 0 - Current wait has timed out • -1 - Session can indefinitely wait in the current wait • NULL - Session is not currently waiting
TIME_SINCE_LAST_WAIT_MICRO	NUMBER	Time elapsed since the end of the last wait (in microseconds). If the session is currently in a wait, then the value is 0.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

[1] The P1RAW, P2RAW, and P3RAW columns display the same values as the P1, P2, and P3 columns, except that the numbers are displayed in hexadecimal.

> **See Also**
>
> "TIMED_STATISTICS" and " Oracle Wait Events"

9.26 V$SESSION_WAIT_CLASS

V$SESSION_WAIT_CLASS displays the time spent in various wait event operations on a per-session basis.

Column	Datatype	Description
SID	NUMBER	Session identifier
SERIAL#	NUMBER	Serial number
WAIT_CLASS_ID	NUMBER	Identifier of the wait class
WAIT_CLASS#	NUMBER	Number of the wait class
WAIT_CLASS	VARCHAR2(64)	Name of the wait class

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOTAL_WAITS	NUMBER	Total number waits from this wait class by the session
TIME_WAITED	NUMBER	Total amount of time spent in waits from this wait class by the session (in hundredths of a second)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.27 V$SESSION_WAIT_HISTORY

V$SESSION_WAIT_HISTORY displays the last 10 wait events for each active session.

Column	Datatype	Description
SID	NUMBER	Session identifier
SEQ#	NUMBER	Sequence of wait events; 1 is the most recent
EVENT#	NUMBER	Event number
EVENT	VARCHAR2(64)	Resource or event for which the session is waiting
P1TEXT	VARCHAR2(64)	Description of the first wait event parameter
P1	NUMBER	First wait event parameter (in decimal)
P2TEXT	VARCHAR2(64)	Description of the second wait event parameter
P2	NUMBER	Second wait event parameter (in decimal)
P3TEXT	VARCHAR2(64)	Description of the third wait event parameter
P3	NUMBER	Third wait event parameter (in decimal)
WAIT_TIME	NUMBER	Amount of time waited (in hundredths of a second)
WAIT_TIME_MICRO	NUMBER	Amount of time waited (in microseconds)
TIME_SINCE_LAST_WAIT_MICRO	NUMBER	Time elapsed (in microseconds) since the end of the previous wait in the wait history
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.28 V$SESSIONS_COUNT

V$SESSIONS_COUNT displays the current number of sessions for each PDB.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
USER_SESSION_COUNT	NUMBER	Displays the current number of user sessions for the PDB
RECURSIVE_SESSION_COUNT	NUMBER	Displays the current number of recursive sessions for the PDB

9.29 V$SESSMETRIC

V$SESSMETRIC displays the metric values for all sessions.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
SESSION_ID	NUMBER	Session ID
SESSION_SERIAL_NUM	NUMBER	Session serial number
CPU	NUMBER	CPU usage
PHYSICAL_READS	NUMBER	Number of physical reads
LOGICAL_READS	NUMBER	Number of logical reads
PGA_MEMORY	NUMBER	PGA size at the end of the interval
HARD_PARSES	NUMBER	Number of hard parses
SOFT_PARSES	NUMBER	Number of soft parses
PHYSICAL_READ_PCT	NUMBER	Physical read ratio
LOGICAL_READ_PCT	NUMBER	Logical read ratio
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.30 V$SESSTAT

V$SESSTAT displays user session statistics. To find the name of the statistic associated with each statistic number (STATISTIC#), query the V$STATNAME view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SID	NUMBER	Session identifier
STATISTIC#	NUMBER	Statistic number **Note:** Statistics numbers are not guaranteed to remain constant from one release to another. Therefore, you should rely on the statistics name rather than its number in your applications.
VALUE	NUMBER	Statistic value
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$STATNAME" and " Statistics Descriptions"

9.31 V$SGA

V$SGA displays summary information about the system global area (SGA).

Column	Datatype	Description
NAME	VARCHAR2(20)	SGA component group
VALUE	NUMBER	Memory size (in bytes)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.32 V$SGA_CURRENT_RESIZE_OPS

V$SGA_CURRENT_RESIZE_OPS displays information about SGA resize operations which are currently in progress. An operation can be a grow or a shrink of a dynamic SGA component. All sizes are expressed in bytes.

Column	Datatype	Description
COMPONENT	VARCHAR2(64)	Component name

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OPER_TYPE	VARCHAR2(13)	Operation type: • STATIC • INITIALIZING • DISABLED • GROW • SHRINK • SHRINK_CANCEL
OPER_MODE	VARCHAR2(9)	Operation mode: • MANUAL • DEFERRED • IMMEDIATE
PARAMETER	VARCHAR2(80)	Name of the parameter for the resize operation
INITIAL_SIZE	NUMBER	Parameter value at the start of the operation
TARGET_SIZE	NUMBER	Desired value of the parameter after the resize
CURRENT_SIZE	NUMBER	Current value of the parameter
START_TIME	DATE	Start time of the operation
LAST_UPDATE_TIME	DATE	Last time progress was made for the operation
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.33 V$SGA_DYNAMIC_COMPONENTS

V$SGA_DYNAMIC_COMPONENTS displays information about the dynamic SGA components. This view summarizes information based on all completed SGA resize operations since instance startup. All sizes are expressed in bytes.

Column	Datatype	Description
COMPONENT	VARCHAR2(64)	Component name
CURRENT_SIZE	NUMBER	Current size of the component
MIN_SIZE	NUMBER	Minimum size of the component since instance startup
MAX_SIZE	NUMBER	Maximum size of the component since instance startup
USER_SPECIFIED_SIZE	NUMBER	Value of the user parameter for the component
OPER_COUNT	NUMBER	Number of operations since instance startup
LAST_OPER_TYPE	VARCHAR2(13)	Last completed operation for the component: • STATIC • INITIALIZING • DISABLED • GROW • SHRINK • SHRINK_CANCEL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LAST_OPER_MODE	VARCHAR2(9)	Mode of the last completed operation: • MANUAL • DEFERRED • IMMEDIATE
LAST_OPER_TIME	DATE	Start time of the last completed operation
GRANULE_SIZE	NUMBER	Granularity of the grow or the shrink operation
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.34 V$SGA_DYNAMIC_FREE_MEMORY

V$SGA_DYNAMIC_FREE_MEMORY displays information about the amount of SGA memory available for future dynamic SGA resize operations.

Column	Datatype	Description
CURRENT_SIZE	NUMBER	Amount of available memory (in bytes)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.35 V$SGA_RESIZE_OPS

V$SGA_RESIZE_OPS displays information about the last 800 completed SGA resize operations. This does not include in-progress operations. All sizes are expressed in bytes.

Column	Datatype	Description
COMPONENT	VARCHAR2(64)	Component name
OPER_TYPE	VARCHAR2(13)	Operation type: • STATIC • INITIALIZING • DISABLED • GROW • SHRINK • SHRINK_CANCEL

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OPER_MODE	VARCHAR2(9)	Operation mode: • MANUAL • DEFERRED • IMMEDIATE
PARAMETER	VARCHAR2(80)	Name of the parameter for the resize operation
INITIAL_SIZE	NUMBER	Parameter value at the start of the operation
TARGET_SIZE	NUMBER	Requested value of the parameter after the resize
FINAL_SIZE	NUMBER	Real value of the parameter after the resize
STATUS	VARCHAR2(9)	Completion status of the operation: • INACTIVE • PENDING • COMPLETE • CANCELLED • ERROR
START_TIME	DATE	Start time of the operation
END_TIME	DATE	End time of the operation
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.36 V$SGA_TARGET_ADVICE

V$SGA_TARGET_ADVICE displays information about the SGA_TARGET initialization parameter.

Column	Datatype	Description
SGA_SIZE	NUMBER	Size of the SGA
SGA_SIZE_FACTOR	NUMBER	Ratio between the SGA_SIZE and the current size of the SGA
ESTD_DB_TIME	NUMBER	Estimated DB_TIME for this SGA_SIZE
ESTD_DB_TIME_FACTOR	NUMBER	Ratio between ESTD_DB_TIME and DB_TIME for the current size of the SGA
ESTD_PHYSICAL_READS	NUMBER	Estimated number of physical reads
ESTD_BUFFER_CACHE_SIZE	NUMBER	Estimated size of the buffer cache
ESTD_SHARED_POOL_SIZE	NUMBER	Estimated size of the shared pool
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"SGA_TARGET"

9.37 V$SGAINFO

V$SGAINFO displays size information about the SGA, including the sizes of different SGA items, the granule size, and free memory.

Column	Datatype	Description
NAME	VARCHAR2(32)	Name of the SGA size item
BYTES	NUMBER	Size of the item (in bytes)
RESIZEABLE	VARCHAR2(3)	Indicates whether the item is resizeable (Yes) or not (No)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.38 V$SGASTAT

V$SGASTAT displays detailed information on the system global area (SGA).

Column	Datatype	Description
POOL	VARCHAR2(14)	Designates the pool in which the memory in NAME resides: • in-memory pool - Memory is allocated from the In-Memory pool • java pool - Memory is allocated from the Java pool • large pool - Memory is allocated from the large pool • numa pool - Memory is allocated from the NUMA pool • shared pool - Memory is allocated from the shared pool • streams pool - Memory is allocated from the Streams pool
NAME	VARCHAR2(26)	SGA component name
BYTES	NUMBER	Memory size (in bytes)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.39 V$SHARED_POOL_ADVICE

V$SHARED_POOL_ADVICE displays information about estimated parse time in the shared pool for different pool sizes. The sizes range from 10% of the current shared pool size or the amount of pinned library cache memory (whichever is higher) to 200% of the current shared pool size, in equal intervals. The value of the interval depends on the current size of the shared pool.

Column	Datatype	Description
SHARED_POOL_SIZE_FOR_ESTIMATE	NUMBER	Shared pool size for the estimate (in megabytes)
SHARED_POOL_SIZE_FACTOR	NUMBER	Size factor with respect to the current shared pool size
ESTD_LC_SIZE	NUMBER	Estimated memory in use by the library cache (in megabytes)
ESTD_LC_MEMORY_OBJECTS	NUMBER	Estimated number of library cache memory objects in the shared pool of the specified size
ESTD_LC_TIME_SAVED	NUMBER	Estimated elapsed parse time saved (in seconds), owing to library cache memory objects being found in a shared pool of the specified size. This is the time that would have been spent in reloading the required objects in the shared pool had they been aged out due to insufficient amount of available free memory.
ESTD_LC_TIME_SAVED_FACTOR	NUMBER	Estimated parse time saved factor with respect to the current shared pool size
ESTD_LC_LOAD_TIME	NUMBER	Estimated elapsed time (in seconds) for parsing in a shared pool of the specified size
ESTD_LC_LOAD_TIME_FACTOR	NUMBER	Estimated load time factor with respect to the current shared pool size
ESTD_LC_MEMORY_OBJECT_HITS	NUMBER	Estimated number of times a library cache memory object was found in a shared pool of the specified size
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.40 V$SHARED_POOL_RESERVED

V$SHARED_POOL_RESERVED displays statistics that help you tune the reserved pool and space within the shared pool.

Column	Datatype	Description
FREE_SPACE	NUMBER	Total amount of free space on the reserved list in bytes[1]
AVG_FREE_SIZE	NUMBER	Average size of the free memory on the reserved list in bytes[1]
FREE_COUNT	NUMBER	Number of free pieces of memory on the reserved list[1]
MAX_FREE_SIZE	NUMBER	Size of the largest free piece of memory on the reserved list in bytes[1]
USED_SPACE	NUMBER	Total amount of used memory on the reserved list in bytes[1]
AVG_USED_SIZE	NUMBER	Average size of the used memory on the reserved list in bytes[1]

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
USED_COUNT	NUMBER	Number of used pieces of memory on the reserved list[1]
MAX_USED_SIZE	NUMBER	Size of the largest used piece of memory on the reserved list in bytes[1]
REQUESTS	NUMBER	Number of times that the reserved list was searched for a free piece of memory[1]
REQUEST_MISSES	NUMBER	Number of times the reserved list did not have a free piece of memory to satisfy the request, and started flushing objects from the LRU list[1]
LAST_MISS_SIZE	NUMBER	Request size of the last request miss in bytes, when the reserved list did not have a free piece of memory to satisfy the request and started flushing objects from the LRU list[1]
MAX_MISS_SIZE	NUMBER	Request size of the largest request miss in bytes, when the reserved list did not have a free piece of memory to satisfy the request and started flushing objects from the LRU list[1]
REQUEST_FAILURES	NUMBER	Number of times that no memory was found to satisfy a request. If an internal flush (used to free up memory) does not meet a memory need, then the error ORA-04031 occurs.[2]
LAST_FAILURE_SIZE	NUMBER	Request size of the last failed request in bytes. If an internal flush (used to free up memory) does not meet a memory need, then the error ORA-04031 occurs.[2]
ABORTED_REQUEST_THRESHOLD	NUMBER	Minimum size of a request in bytes which signals an ORA-04031 error without flushing objects[2]
ABORTED_REQUESTS	NUMBER	Number of requests that signalled an ORA-04031 error without flushing objects[2]
LAST_ABORTED_SIZE	NUMBER	Last size of the request in bytes that returned an ORA-04031 error without flushing objects from the LRU list[2]
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

[1] These columns are valid only if the initialization parameter SHARED_POOL_RESERVED_SIZE is set to a valid value.

[2] These columns contain values which are valid even if SHARED_POOL_RESERVED_SIZE is not set.

9.41 V$SHARED_SERVER

V$SHARED_SERVER displays information on the shared server processes.

Column	Datatype	Description
NAME	VARCHAR2(4)	Name of the server
PADDR	RAW(4 | 8)	Server's process address

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(16)	Server status: • EXEC - Executing SQL • WAIT (ENQ) - Waiting for a lock • WAIT (SEND) - Waiting to send data to user • WAIT (COMMON) - Idle; waiting for a user request • WAIT (RECEIVE) - Waiting for records to be shown in the client application • WAIT (RESET) - Waiting for a circuit to reset after a break • QUIT - Terminating
MESSAGES	NUMBER	Number of messages processed
BYTES	NUMBER	Total number of bytes in all messages
BREAKS	NUMBER	Number of breaks
CIRCUIT	RAW(4 | 8)	Address of the circuit currently being serviced
IDLE	NUMBER	Total idle time (in hundredths of a second)
BUSY	NUMBER	Total busy time (in hundredths of a second)
IN_NET	NUMBER	Total incoming network wait time (in hundredths of a second)
OUT_NET	NUMBER	Total outgoing network wait time (in hundredths of a second)
REQUESTS	NUMBER	Total number of requests taken from the common queue in this server's lifetime
BOUND_TIME[1]	NUMBER	Time that a circuit and shared server have been bound (in centiseconds)
BOUND_REASON[1]	VARCHAR2(32)	Provides a reason (a short explanation) for why a shared server and circuit could not be unbound. This column is empty when a circuit is not bound to a server. When the server starts serving a circuit, BOUND_REASON is empty and remains empty unless the server tries unsuccessfully to unbind the circuit (after it finishes serving the current request). When this column is not empty, it will be cleared once the server and circuit are unbound (that is, once the resources preventing the session migration to another shared server are released).
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

[1] This column is available starting with Oracle Database 19c.

9.42 V$SHARED_SERVER_MONITOR

V$SHARED_SERVER_MONITOR displays information for tuning the shared server.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MAXIMUM_CONNECTIONS	NUMBER	Highest number of virtual circuits in use at one time since the instance started. If this value reaches the value set for the CIRCUITS initialization parameter, then consider raising the value of CIRCUITS. **See Also:** "CIRCUITS"
MAXIMUM_SESSIONS	NUMBER	Highest number of shared server sessions in use at one time since the instance started. If this reaches the value set for the SHARED_SERVER_SESSIONS initialization parameter, then consider raising the value of SHARED_SERVER_SESSIONS. **See Also:** "SHARED_SERVER_SESSIONS"
SERVERS_STARTED	NUMBER	Total number of shared servers started since the instance started (but not including those started during startup)
SERVERS_TERMINATED	NUMBER	Total number of shared servers stopped by Oracle since the instance started
SERVERS_HIGHWATER	NUMBER	Highest number of servers running at one time since the instance started. If this value reaches the value set for the MAX_SHARED_SERVERS initialization parameter, then consider raising the value of SHARED_SERVERS. **See Also:** "SHARED_SERVERS"
DISPATCHERS_CPU	NUMBER	Total CPU time of all dispatchers since the instance started (in millionths of a second)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.43 V$SHARED_SERVER_STAT

V$SHARED_SERVER_STAT displays statistics for all shared server processes. The statistics are cumulative from the start of the instance.

Column	Datatype	Description
MESSAGES	NUMBER	Number of messages processed
BYTES	NUMBER	Total number of bytes in all messages
IDLE	NUMBER	Total idle time (in hundredths of a second)
BUSY	NUMBER	Total busy time (in hundredths of a second)
IN_NET	NUMBER	Total incoming network wait time (in hundredths of a second)
OUT_NET	NUMBER	Total outgoing network wait time (in hundredths of a second)
REQUESTS	NUMBER	Total number of requests taken from the common queue

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **Note**
>
> This view is available starting with Oracle Database 19c.

9.44 V$SORT_SEGMENT

V$SORT_SEGMENT displays information about every sort segment in a given instance. The view is only updated when the tablespace is of the TEMPORARY type.

Column	Datatype	Description
TABLESPACE_NAME	VARCHAR2(30)	Name of the tablespace
SEGMENT_FILE	NUMBER	File number of the first extent
SEGMENT_BLOCK	NUMBER	Block number of the first extent
EXTENT_SIZE	NUMBER	Extent size
CURRENT_USERS	NUMBER	Number of active users of the segment
TOTAL_EXTENTS	NUMBER	Total number of extents in the segment
TOTAL_BLOCKS	NUMBER	Total number of blocks in the segment
USED_EXTENTS	NUMBER	Extents allocated to active sorts
USED_BLOCKS	NUMBER	Blocks allocated to active sorts
FREE_EXTENTS	NUMBER	Extents not allocated to any sort
FREE_BLOCKS	NUMBER	Blocks not allocated to any sort
ADDED_EXTENTS	NUMBER	Number of extent allocations
EXTENT_HITS	NUMBER	Number of times an unused extent was found in the pool
FREED_EXTENTS	NUMBER	Number of deallocated extents
FREE_REQUESTS	NUMBER	Number of requests to deallocate
MAX_SIZE	NUMBER	Maximum number of extents ever used
MAX_BLOCKS	NUMBER	Maximum number of blocks ever used
MAX_USED_SIZE	NUMBER	Maximum number of extents used by all sorts
MAX_USED_BLOCKS	NUMBER	Maximum number of blocks used by all sorts
MAX_SORT_SIZE	NUMBER	Maximum number of extents used by an individual sort
MAX_SORT_BLOCKS	NUMBER	Maximum number of blocks used by an individual sort
RELATIVE_FNO	NUMBER	Relative file number of the sort segment header

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TS#	NUMBER	Tablespace number
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
IS_LOCAL_TEMP	NUMBER	Indicates whether the sort segment is allocated from a local temporary tablespace (1) or not (0).

9.45 V$SPPARAMETER

V$SPPARAMETER displays information about the contents of the server parameter file. If a server parameter file was not used to start the instance, then each row of the view will contain FALSE in the ISSPECIFIED column.

Column	Datatype	Description
FAMILY	VARCHAR2(80)	For internal use only
SID	VARCHAR2(80)	SID for which the parameter is defined.
NAME	VARCHAR2(80)	Name of the parameter
TYPE	VARCHAR2(11)	Parameter type
VALUE	VARCHAR2(255)	Parameter value (null if a server parameter file was not used to start the instance)
DISPLAY_VALUE	VARCHAR2(255)	Parameter value in a user-friendly format. For example, if the VALUE column shows the value 262144 for a big integer parameter, then the DISPLAY_VALUE column will show the value 256K.
ISSPECIFIED	VARCHAR2(6)	Indicates whether the parameter was specified in the server parameter file (TRUE) or not (FALSE)
ORDINAL	NUMBER	Position (ordinal number) of the parameter value (0 if a server parameter file was not used to start the instance). Useful only for parameters whose values are lists of strings.
UPDATE_COMMENT	VARCHAR2(255)	Comments associated with the most recent update (null if a server parameter file was not used to start the instance)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.46 V$SQL

V$SQL lists statistics on shared SQL areas without the GROUP BY clause and contains one row for each child of the original SQL text entered. Statistics displayed in V$SQL are normally

Copyright © 2002, 2026, Oracle and/or its affiliates.

updated at the end of query execution. However, for long running queries, they are updated every 5 seconds. This makes it easy to see the impact of long running SQL statements while they are still in progress.

Column	Datatype	Description
SQL_TEXT	VARCHAR2(1000)	First thousand characters of the SQL text for the current cursor
SQL_FULLTEXT	CLOB	Full text for the SQL statement exposed as a CLOB column. The full text of a SQL statement can be retrieved using this column instead of joining with the V$SQLTEXT dynamic performance view.
SQL_ID	VARCHAR2(13)	SQL identifier of the parent cursor in the library cache
SHARABLE_MEM	NUMBER	Amount of shared memory used by the child cursor (in bytes)
PERSISTENT_MEM	NUMBER	Fixed amount of memory used for the lifetime of the child cursor (in bytes)
RUNTIME_MEM	NUMBER	Fixed amount of memory required during the execution of the child cursor
SORTS	NUMBER	Number of sorts that were done for the child cursor
LOADED_VERSIONS	NUMBER	Indicates whether the context heap is loaded (1) or not (0)
OPEN_VERSIONS	NUMBER	Indicates whether the child cursor is locked (1) or not (0)
USERS_OPENING	NUMBER	Number of users that have any of the child cursors open
FETCHES	NUMBER	Number of fetches associated with the SQL statement
EXECUTIONS	NUMBER	Number of executions that took place on this object since it was brought into the library cache
PX_SERVERS_EXECUTIONS	NUMBER	Total number of executions performed by parallel execution servers (0 when the statement has never been executed in parallel)
END_OF_FETCH_COUNT	NUMBER	Number of times this cursor was fully executed since the cursor was brought into the library cache. The value of this statistic is not incremented when the cursor is partially executed, either because it failed during the execution or because only the first few rows produced by this cursor are fetched before the cursor is closed or re-executed. By definition, the value of the END_OF_FETCH_COUNT column should be less or equal to the value of the EXECUTIONS column.
USERS_EXECUTING	NUMBER	Number of users executing the statement
LOADS	NUMBER	Number of times the object was either loaded or reloaded
FIRST_LOAD_TIME	VARCHAR2(76)	Timestamp of the parent creation time
INVALIDATIONS	NUMBER	Number of times this child cursor has been invalidated
PARSE_CALLS	NUMBER	Number of parse calls for this child cursor
DISK_READS	NUMBER	Number of disk reads for this child cursor
DIRECT_WRITES	NUMBER	Number of direct writes for this child cursor
DIRECT_READS	NUMBER	Number of direct reads for this child cursor
BUFFER_GETS	NUMBER	Number of buffer gets for this child cursor
APPLICATION_WAIT_TIME	NUMBER	Application wait time (in microseconds)
CONCURRENCY_WAIT_TIME	NUMBER	Concurrency wait time (in microseconds)
CLUSTER_WAIT_TIME	NUMBER	Cluster wait time (in microseconds)
USER_IO_WAIT_TIME	NUMBER	User I/O Wait Time (in microseconds)
PLSQL_EXEC_TIME	NUMBER	PL/SQL execution time (in microseconds)
JAVA_EXEC_TIME	NUMBER	Java execution time (in microseconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ROWS_PROCESSED	NUMBER	Total number of rows the parsed SQL statement returns
COMMAND_TYPE	NUMBER	Oracle command type definition
OPTIMIZER_MODE	VARCHAR2(10)	Mode under which the SQL statement was executed
OPTIMIZER_COST	NUMBER	Cost of this query given by the optimizer
OPTIMIZER_ENV	RAW(2000)	Optimizer environment
OPTIMIZER_ENV_HASH_VALUE	NUMBER	Hash value for the optimizer environment
PARSING_USER_ID	NUMBER	User ID of the user who originally built this child cursor
PARSING_SCHEMA_ID	NUMBER	Schema ID that was used to originally build this child cursor
PARSING_SCHEMA_NAME	VARCHAR2(128)	Schema name that was used to originally build this child cursor
KEPT_VERSIONS	NUMBER	Indicates whether this child cursor has been marked to be kept pinned in the cache using the DBMS_SHARED_POOL package
ADDRESS	RAW(4 | 8)	Address of the handle to the parent for this cursor
TYPE_CHK_HEAP	RAW(4 | 8)	Descriptor of the type check heap for this child cursor
HASH_VALUE	NUMBER	Hash value of the parent statement in the library cache
OLD_HASH_VALUE	NUMBER	Old SQL hash value
PLAN_HASH_VALUE	NUMBER	Numeric representation of the current SQL plan for this cursor. Comparing one PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line).
FULL_PLAN_HASH_VALUE	NUMBER	Numeric representation of the complete SQL plan for this cursor. Comparing one FULL_PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line). Note that the FULL_PLAN_HASH_VALUE cannot be compared across databases releases. It is not backward compatible.
CHILD_NUMBER	NUMBER	Number of this child cursor
SERVICE	VARCHAR2(64)	Service name
SERVICE_HASH	NUMBER	Hash value for the name listed in the SERVICE column
MODULE	VARCHAR2(64)	Contains the name of the module that was executing when the SQL statement was first parsed, which is set by calling DBMS_APPLICATION_INFO.SET_MODULE
MODULE_HASH	NUMBER	Hash value of the module listed in the MODULE column
ACTION	VARCHAR2(64)	Contains the name of the action that was executing when the SQL statement was first parsed, which is set by calling DBMS_APPLICATION_INFO.SET_ACTION
ACTION_HASH	NUMBER	Hash value of the action listed in the ACTION column
SERIALIZABLE_ABORTS	NUMBER	Number of times the transaction failed to serialize, producing ORA-08177 errors, per cursor
OUTLINE_CATEGORY	VARCHAR2(64)	If an outline was applied during construction of the cursor, then this column displays the category of that outline. Otherwise the column is left blank.
CPU_TIME	NUMBER	CPU time (in microseconds) used by this cursor for parsing, executing, and fetching

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ELAPSED_TIME	NUMBER	Elapsed database time (in microseconds) used by this cursor for parsing, executing, and fetching. If the cursor uses parallel execution, then ELAPSED_TIME is the cumulative time for the query coordinator, plus all parallel query worker processes. Note that this column displays database time, not wall clock time. In some cases, ELAPSED_TIME can exceed the duration of the query. For example, this can occur for parallel queries.
OUTLINE_SID	NUMBER	Outline session identifier
CHILD_ADDRESS	RAW(4 | 8)	Address of the child cursor
SQLTYPE	NUMBER	Denotes the version of the SQL language used for this statement
REMOTE	VARCHAR2(1)	Indicates whether the cursor is remote mapped (Y) or not (N)
OBJECT_STATUS	VARCHAR2(19)	Status of the cursor: • VALID - Valid, authorized without errors • VALID_AUTH_ERROR - Valid, authorized with authorization errors • VALID_COMPILE_ERROR - Valid, authorized with compilation errors • VALID_UNAUTH - Valid, unauthorized • INVALID_UNAUTH - Invalid, unauthorized • INVALID - Invalid, unauthorized but keep the timestamp
LITERAL_HASH_VALUE	NUMBER	Hash value of the literals which are replaced with system-generated bind variables and are to be matched, when CURSOR_SHARING is used. This is not the hash value for the SQL statement. If CURSOR_SHARING is not used, then the value is 0.
LAST_LOAD_TIME	VARCHAR2(76)	Time at which the query plan was loaded into the library cache
IS_OBSOLETE	VARCHAR2(1)	Indicates whether the cursor has become obsolete (Y) or not (N). This can happen if the number of child cursors is too large.
IS_BIND_SENSITIVE	VARCHAR2(1)	Indicates whether the cursor is bind sensitive (Y) or not (N). A query is considered bind-sensitive if the optimizer peeked at one of its bind variable values when computing predicate selectivities and where a change in a bind variable value may cause the optimizer to generate a different plan.
IS_BIND_AWARE	VARCHAR2(1)	Indicates whether the cursor is bind aware (Y) or not (N). A query is considered bind-aware if it has been marked to use extended cursor sharing. The query would already have been marked as bind-sensitive.
IS_SHAREABLE	VARCHAR2(1)	Indicates whether the cursor can be shared (Y) or not (N)
CHILD_LATCH	NUMBER	Child latch number that is protecting the cursor. This column is obsolete and maintained for backward compatibility.
SQL_PROFILE	VARCHAR2(64)	SQL profile used for this statement, if any
SQL_PATCH	VARCHAR2(128)	SQL patch used for this statement, if any
SQL_PLAN_BASELINE	VARCHAR2(128)	SQL plan baseline used for this statement, if any
PROGRAM_ID	NUMBER	Program identifier
PROGRAM_LINE#	NUMBER	Program line number
EXACT_MATCHING_SIGNATURE	NUMBER	Signature calculated on the normalized SQL text. The normalization includes the removal of white space and the uppercasing of all non-literal strings.
FORCE_MATCHING_SIGNATURE	NUMBER	Signature used when the CURSOR_SHARING parameter is set to FORCE
LAST_ACTIVE_TIME	DATE	Time at which the query plan was last active
BIND_DATA	RAW(2000)	Bind data

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TYPECHECK_MEM	NUMBER	Typecheck memory
IO_CELL_OFFLOAD_ELIGIBLE_BYTES	NUMBER	Number of I/O bytes which can be filtered by the Exadata storage system **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_INTERCONNECT_BYTES	NUMBER	Number of I/O bytes exchanged between Oracle Database and the storage system
PHYSICAL_READ_REQUESTS	NUMBER	Number of physical read I/O requests issued by the monitored SQL
PHYSICAL_READ_BYTES	NUMBER	Number of bytes read from disks by the monitored SQL
PHYSICAL_WRITE_REQUESTS	NUMBER	Number of physical write I/O requests issued by the monitored SQL
PHYSICAL_WRITE_BYTES	NUMBER	Number of bytes written to disks by the monitored SQL
OPTIMIZED_PHY_READ_REQUESTS	NUMBER	Number of physical read I/O requests from Database Smart Flash Cache issued by the monitored SQL
LOCKED_TOTAL	NUMBER	Total number of times the child cursor has been locked
PINNED_TOTAL	NUMBER	Total number of times the child cursor has been pinned
IO_CELL_UNCOMPRESSED_BYTES	NUMBER	Number of uncompressed bytes (that is, size after decompression) that are offloaded to the Exadata cells **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_CELL_OFFLOAD_RETURNED_BYTES	NUMBER	Number of filtered bytes returned by Exadata cells (that is, the number of bytes returned after processing has been offloaded on the Exadata cells) **See Also:** Oracle Exadata Storage Server Software documentation for more information
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
IS_REOPTIMIZABLE	VARCHAR2(1)	This column shows whether the next execution matching this child cursor will trigger a reoptimization. The values are: • Y: If the next execution will trigger a reoptimization • R: If the child cursor contains reoptimization information, but will not trigger reoptimization because the cursor was compiled in reporting mode • N: If the child cursor has no reoptimization information
IS_RESOLVED_ADAPTIVE_PLAN	VARCHAR2(1)	This column shows whether all of the adaptive parts of a plan have been resolved to the final plan. Once the plan is resolved, the plan hash value and the plan displayed by DBMS_XPLAN will not change through the end of execution. The values for this column are: • NULL: If the plan is not adaptive • Y: If the plan is fully resolved • N: If the plan is not yet fully resolved **See Also:** *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_XPLAN package
IM_SCANS	NUMBER	Number of In-Memory Column Store (IM column store) segment scans

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
IM_SCAN_BYTES_UNCOMPRESSED	NUMBER	Uncompressed size of data scanned from the IM column store
IM_SCAN_BYTES_INMEMORY	NUMBER	In-memory size of data scanned from the IM column store
DDL_NO_INVALIDATE	VARCHAR2(1)	Indicates if a DDL statement updated a dependent object and did not invalidate this cursor. The values are: • N: There has not been a DDL statement that updated a dependent object without invalidating this cursor. • Y: A DDL statement updated a dependent object and did not invalidate this cursor, but the cursor has not executed since this happened. • X: A DDL statement updated a dependent object and did not invalidate this cursor, and the cursor has executed since this happened.
IS_ROLLING_INVALID	VARCHAR2(1)	Indicates if this cursor is rolling invalidated. The values are: • N: This cursor is not rolling invalidated. • Y: This cursor is rolling invalidated, but the cursor has not executed in this state. • X: This cursor is rolling invalidated, and the cursor has executed in this state.
IS_ROLLING_REFRESH_INVALID	VARCHAR2(1)	Indicates if this cursor is rolling invalidated and requires execution time refresh. The values are: • N: This cursor is not a cursor that is rolling invalidated and requires execution time refresh. • Y: This cursor is rolling invalidated and requires execution time refresh, but the cursor has not executed in this state. • X: This cursor is rolling invalidated and requires execution time refresh, and the cursor has executed in this state.
RESULT_CACHE	VARCHAR2(1)	Indicates whether the SQL statement used the result cache (Y) or not (N)
SQL_QUARANTINE[1]	VARCHAR2(128)	If the execution plan for this cursor is quarantined, then this column contains the name of the SQL quarantine configuration (corresponds to the NAME column in the DBA_SQL_QUARANTINE view). Otherwise, this column is null.
AVOIDED_EXECUTIONS[1]	NUMBER	Number of times this cursor was prevented from being used due to the plan being quarantined

[1] This column is available starting with Oracle Database 19c.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "V$SQLTEXT"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SHARED_POOL package
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_MODULE procedure
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_ACTION procedure
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_XPLAN package

9.47 V$SQL_BIND_CAPTURE

V$SQL_BIND_CAPTURE displays information on bind variables used by SQL cursors. Each row in the view contains information for one bind variable defined in a cursor. This includes:

- Reference to the cursor defining the bind variable

  (hash_value, address) for the parent cursor and (hash_value, child_address) for the child cursor.

- Bind metadata

  Name, position, data type, character set ID, precision, scale, and maximum length of the bind variable.

- Bind data

  One of the bind values used for the bind variable during a past execution of its associated SQL statement. Bind values are not always captured for this view. Bind values are displayed by this view only when the type of the bind variable is simple (this excludes LONG, LOB, and ADT data types) and when the bind variable is used in the WHERE or HAVING clauses of the SQL statement.

Bind capture is disabled when the STATISTICS_LEVEL initialization parameter is set to BASIC. This view can be joined with V$SQLAREA on (HASH_VALUE, ADDRESS) and with V$SQL on (HASH_VALUE, CHILD_ADDRESS).

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Address of the parent cursor
HASH_VALUE	NUMBER	Hash value of the parent cursor in the library cache. The hash value is a fixed index for the view and should always be used to speed up access to the view.
SQL_ID	VARCHAR2(13)	SQL identifier of the parent cursor in the library cache
CHILD_ADDRESS	RAW(4 | 8)	Address of the child cursor
CHILD_NUMBER	NUMBER	Child cursor number
NAME	VARCHAR2(128)	Name of the bind variable
POSITION	NUMBER	Position of the bind variable in the SQL statement
DUP_POSITION	NUMBER	If the binding is performed by name and the bind variable is duplicated, then this column gives the position of the primary bind variable.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DATATYPE	NUMBER	Internal identifier for the bind data type. Beginning in Oracle Database 12*c*, a number representing a PL/SQL data type can appear in this column.
DATATYPE_STRING	VARCHAR2(15)	Textual representation of the bind data type. Beginning in Oracle Database 12*c*, a text representation of a PL/SQL-only data type can appear in this column. If the actual data type is a PL/SQL sub type, the name of the data type, not the sub type will be displayed.
CHARACTER_SID	NUMBER	National character set identifier
PRECISION	NUMBER	Precision (for numeric binds)
SCALE	NUMBER	Scale (for numeric binds)
MAX_LENGTH	NUMBER	Maximum bind length
WAS_CAPTURED	VARCHAR2(3)	Indicates whether the bind value was captured (YES) or not (NO)
LAST_CAPTURED	DATE	Date when the bind value was captured. Bind values are captured when SQL statements are executed. To limit the overhead, binds are captured at most every 15 minutes for a given cursor.
VALUE_STRING	VARCHAR2(4000)	Value of the bind represented as a string
VALUE_ANYDATA	ANYDATA	Value of the bind represented using the ANYDATA data type. This representation is useful to programmatically decode the value of the bind variable. This column is NULL if a PL/SQL-only data type appears in the DATATYPE column.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "STATISTICS_LEVEL"
- "V$SQLAREA"
- "V$SQL"

9.48 V$SQL_BIND_DATA

V$SQL_BIND_DATA describes information related to bind variables.

V$SQL_BIND_DATA describes, for each distinct bind variable in each cursor owned by the session querying this view:

- Actual bind data, if the bind variable is user defined
- The underlying literal, if the CURSOR_SHARING parameter is set to FORCE and the bind variable is system generated. (System-generated binds have a value of 256 in the SHARED_FLAG2 column.)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CURSOR_NUM	NUMBER	Cursor number for this bind
POSITION	NUMBER	Bind position
DATATYPE	NUMBER	Internal identifier for the bind data type. Beginning in Oracle Database 12*c*, a number representing a PL/SQL data type can appear in this column.
SHARED_MAX_LEN	NUMBER	Shared maximum length for this bind from the shared cursor object associated with this bind
PRIVATE_MAX_LEN	NUMBER	Private maximum length for this bind sent from the client
ARRAY_SIZE	NUMBER	Maximum number of array elements (for array binds only)
PRECISION	NUMBER	Precision (for numeric binds)
SCALE	NUMBER	Scale (for numeric binds)
SHARED_FLAG	NUMBER	Shared bind data flags
SHARED_FLAG2	NUMBER	Shared bind data flags (continued)
BUF_ADDRESS	RAW(4 | 8)	Bind buffer memory address
BUF_LENGTH	NUMBER	Bind buffer length
VAL_LENGTH	NUMBER	Actual bind value length
BUF_FLAG	NUMBER	Bind buffer flags
INDICATOR	NUMBER	Bind indicator
VALUE	VARCHAR2(4000)	Contents of the bind buffer
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"CURSOR_SHARING"

9.49 V$SQL_BIND_METADATA

V$SQL_BIND_METADATA describes metadata related to bind variables.

V$SQL_BIND_METADATA describes, for each distinct bind variable in each cursor owned by the session querying this view:

- Bind metadata provided by the client, if the bind variable is user defined
- Metadata based on the underlying literal, if the CURSOR_SHARING parameter is set to FORCE and the bind variable is system-generated.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Memory address of the child cursor that owns this bind variable
POSITION	NUMBER	Bind position
DATATYPE	NUMBER	Internal identifier for the bind data type. Beginning in Oracle Database 12*c*, a number representing a PL/SQL data type can appear in this column.
MAX_LENGTH	NUMBER	Maximum length of the bind value
ARRAY_LEN	NUMBER	Maximum number of array elements (for array binds only)
BIND_NAME	VARCHAR2(128)	User-defined or system-generated bind variable name (if used)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"CURSOR_SHARING"

9.50 V$SQL_CS_HISTOGRAM

V$SQL_CS_HISTOGRAM summarizes the monitoring information stored by adaptive cursor sharing. This information is used to decide whether to enable extended cursor sharing for a query. It is stored in a histogram, whose bucket's contents are exposed by this view.

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Address of the handle to the parent for this cursor
HASH_VALUE	NUMBER	Hash value of the parent statement in the library cache
SQL_ID	VARCHAR2(13)	SQL identifier of the parent cursor in the library cache
CHILD_NUMBER	NUMBER	Number of the child cursor being monitored
BUCKET_ID	NUMBER	Bucket number of the monitoring histogram
COUNT	NUMBER	Value in this bucket of the histogram
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.51 V$SQL_CS_SELECTIVITY

V$SQL_CS_SELECTIVITY exposes the valid selectivity ranges for a child cursor in extended cursor sharing mode. A valid range consists of a low and high value for each predicate containing binds. Each predicate's selectivity (with the current bind value) must fall between the corresponding low and high values in order for the child cursor to be shared.

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Address of the handle to the parent for this cursor
HASH_VALUE	NUMBER	Hash value of the parent statement in the library cache
SQL_ID	VARCHAR2(13)	SQL identifier of the parent cursor in the library cache
CHILD_NUMBER	NUMBER	Number of the child cursor
PREDICATE	VARCHAR2(40)	Predicate whose selectivity must fall between low and high values
RANGE_ID	NUMBER	Identifier for the range used to match up the low and high values for multiple predicates
LOW	VARCHAR2(10)	Lower bound for allowable selectivity
HIGH	VARCHAR2(10)	Upper bound for allowable selectivity
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.52 V$SQL_CS_STATISTICS

V$SQL_CS_STATISTICS contains the raw execution statistics used by the monitoring component of adaptive cursor sharing. A sample of the executions is monitored. This view exposes which executions were sampled, and what the statistics were for those executions. The statistics are cumulative for each distinct set of bind values.

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Address of the handle to the parent for this cursor
HASH_VALUE	NUMBER	Hash value of the parent statement in the library cache
SQL_ID	VARCHAR2(13)	SQL identifier of the parent cursor in the library cache
CHILD_NUMBER	NUMBER	Number of the child cursor being monitored
BIND_SET_HASH_VALUE	NUMBER	Hash of the values of the binds
PEEKED	VARCHAR2(1)	Indicates if this is the bind set used to build the cursor (Y) or not (N)
EXECUTIONS	NUMBER	Number of times this bind set has been executed and monitored
ROWS_PROCESSED	NUMBER	Cumulative number of rows processed by all row sources in the plan over all monitored executions with this bind set
BUFFER_GETS	NUMBER	Cumulative number of buffer gets over all monitored executions with this bind set

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CPU_TIME	NUMBER	Cumulative CPU time (in microseconds) used by this cursor for monitored executions with this bind set
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.53 V$SQL_CURSOR

V$SQL_CURSOR displays debugging information for each cursor associated with the session querying this view.

Column	Datatype	Description
CURNO	NUMBER	Cursor number
FLAG	NUMBER	Flags set in the cursor
STATUS	VARCHAR2(9)	Status of the cursor; that is, what state the cursor is in
PARENT_HANDLE	RAW(4 | 8)	Pointer to the parent cursor handle
PARENT_LOCK	RAW(4 | 8)	Pointer to the parent cursor lock
CHILD_LOCK	RAW(4 | 8)	Pointer to the child cursor lock
CHILD_PIN	RAW(4 | 8)	Pointer to the child cursor pin
PERS_HEAP_MEM	NUMBER	Total amount of memory allocated from persistent heap for this cursor
WORK_HEAP_MEM	NUMBER	Total amount of memory allocated from the work heap for this cursor
BIND_VARS	NUMBER	Total number of bind positions in the query currently parsed into this cursor
DEFINE_VARS	NUMBER	Total number of define variables in the query currently parsed into this cursor
BIND_MEM_LOC	VARCHAR2(64)	Which memory heap the bind variables are stored in: either the UGA or the CGA
INST_FLAG	VARCHAR2(64)	Instantiation object flags
INST_FLAG2	VARCHAR2(64)	Instantiation object flags (continued)
CHILD_HANDLE	RAW(4 | 8)	Pointer to the child cursor handle
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.54 V$SQL_JOIN_FILTER

V$SQL_JOIN_FILTER displays performance information about the characteristics of join filters when they are used for a parallel cursor. (A join filter is a bitmap filter applied to table rows before a join operation in order to avoid parallel communication.)

Column	Datatype	Description
QC_SESSION_ID	NUMBER	QC (Query Coordinator) session ID of the given cursor for the given parallel query
QC_INSTANCE_ID	NUMBER	QC (Query Coordinator) instance ID of the given cursor for the given parallel query
SQL_PLAN_HASH_VALUE	NUMBER	SQL plan hash value of the given cursor for the given parallel query
FILTER_ID	NUMBER	An ID that identifies the join filter in the given cursor and corresponds to the filter ID in the execution plan
LENGTH	NUMBER	Total size of the join filter field
BITS_SET	NUMBER	Number of bits set in this filter
FILTERED	NUMBER	Number of rows seen by the join filter
PROBED	NUMBER	Number of rows of the right table that have been tested against the bitmap filter. This is the sum of the filtered rows plus the non-filtered rows.
ACTIVE	NUMBER	Whether the filter is active (Yes) or not (No)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.55 V$SQL_MONITOR

V$SQL_MONITOR displays SQL statements whose execution have been (or are being) monitored by Oracle.

This view contains global, high-level information about simple and composite database operations.

Oracle Database monitors simple database operations, which are top SQL statements and PL/SQL subprograms, when any of the following conditions is true:

- The operations run in parallel.
- The operations have consumed at least 5 seconds of CPU or I/O time in a single execution.
- Tracking for the operations is forced by the /*+ MONITOR */ hint.

For simple database operations, monitoring statistics are not cumulative over several executions. In this case, one entry in V$SQL_MONITOR is dedicated to a single execution of a SQL statement. If the database monitors two executions of the same SQL statement, then each execution has a separate entry in V$SQL_MONITOR.

Copyright © 2002, 2026, Oracle and/or its affiliates.

For simple database operations, V$SQL_MONITOR has one entry for the parallel execution coordinator process and one entry for each parallel execution server process. Each entry has corresponding entries in V$SQL_PLAN_MONITOR. Because the processes allocated for the parallel execution of a SQL statement are cooperating for the same execution, these entries share the same execution key (the combination of SQL_ID, SQL_EXEC_START, and SQL_EXEC_ID).

Oracle Database monitors composite database operations when either of the following conditions is true:

- A database operation was started with DBMS_SQL_MONITOR.BEGIN_OPERATION and the operation has consumed at least 5 seconds of CPU or I/O time.
- Tracking for the operation is forced by setting FORCE_TRACKING to Y in DBMS_SQL_MONITOR.BEGIN_OPERATION.

For composite database operations, each row contains an operation whose statistics are accumulated over the SQL statements and PL/SQL subprograms that run in the same session as part of the operation.

The V$SQL_MONITOR view contains a subset of the statistics available in V$SQL. However, unlike V$SQL, monitoring statistics are not cumulative over several executions. Instead, one entry in V$SQL_MONITOR is dedicated to a single execution of a SQL statement. If the database monitors two executions of the same SQL statement, then each execution has a separate entry in V$SQL_MONITOR.

The primary key is the combination of the columns SQL_ID, SQL_EXEC_START, and SQL_EXEC_ID.

V$SQL_MONITOR has one entry for the parallel execution coordinator process, and one entry for each parallel execution server process. Each entry has corresponding entries in V$SQL_PLAN_MONITOR. Because the processes allocated for the parallel execution of a SQL statement are cooperating for the same execution, these entries share the same execution key (the composite SQL_ID, SQL_EXEC_START, and SQL_EXEC_ID). You can aggregate the execution key to determine the overall statistics for a parallel execution.

When the SQL statement being monitored is executing, statistics in V$SQL_MONITOR are generally refreshed in near real time, once every second. Once the execution ends, monitoring information is not deleted immediately. Instead, it is kept in V$SQL_MONITOR for at least one minute. The entry will eventually be deleted to reclaim its space as new statements are monitored.

Column	Datatype	Description
KEY	NUMBER	Artificial join key to efficiently join V$SQL_MONITOR with its corresponding plan level monitoring statistics stored in V$SQL_PLAN_MONITOR
REPORT_ID	NUMBER	Unique ID of the XML report stored in Automatic Workload Repository (AWR) for this monitored entity
STATUS	VARCHAR2(19)	SQL execution status: • QUEUED - SQL statement is queued • EXECUTING - SQL statement is still executing • DONE (ERROR) - Execution terminated with an error • DONE (FIRST N ROWS) - Execution terminated by the application before all rows were fetched • DONE (ALL ROWS) - Execution terminated and all rows were fetched • DONE - Execution terminated (parallel execution)
USER#	NUMBER	User ID of the database user who issued the SQL being monitored
USERNAME	VARCHAR2(128)	User name of the database user who issued the SQL being monitored

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
MODULE	VARCHAR2(64)	Name of the executing module when sampled, as set by the DBMS_APPLICATION_INFO.SET_MODULE procedure
ACTION	VARCHAR2(64)	Name of the executing action when sampled, as set by the DBMS_APPLICATION_INFO.SET_ACTION procedure
SERVICE_NAME	VARCHAR2(64)	Service name of the user session
CLIENT_IDENTIFIER	VARCHAR2(64)	Client identifier from the user session
CLIENT_INFO	VARCHAR2(64)	Client information for the user session
PROGRAM	VARCHAR2(48)	Name of the operating system program that issued the monitored SQL
PLSQL_ENTRY_OBJECT_ID	NUMBER	Object ID of the top-most PL/SQL subprogram on the stack; NULL if there is no PL/SQL subprogram on the stack
PLSQL_ENTRY_SUBPROGRAM_ID	NUMBER	Subprogram ID of the top-most PL/SQL subprogram on the stack; NULL if there is no PL/SQL subprogram on the stack
PLSQL_OBJECT_ID	NUMBER	Object ID of the currently executing PL/SQL subprogram; NULL if executing SQL
PLSQL_SUBPROGRAM_ID	NUMBER	Subprogram ID of the currently executing PL/SQL object; NULL if executing SQL
FIRST_REFRESH_TIME	DATE	Time when monitoring of the SQL statement started, generally a few seconds after execution start time
LAST_REFRESH_TIME	DATE	Time when statistics in V$SQL_MONITOR were last updated for the SQL statement. Statistics are generally refreshed every second when the statement executes.
REFRESH_COUNT	NUMBER	Number of times V$SQL_MONITOR statistics have been refreshed (generally once every second when the SQL statement executes)
DBOP_EXEC_ID	NUMBER	Database operation execution identifier for the current execution. If the type is SQL, the DBOP_EXEC_ID will be NULL.
DBOP_NAME	VARCHAR2(30)	Database operation name. If the type is SQL, the DBOP_NAME will be NULL.
SID	NUMBER	Session identifier executing (or having executed) the SQL statement being monitored
PROCESS_NAME	VARCHAR2(5)	Process name identifier executing (or having executed)the statement; ora if the process is foreground, else the background process name (for example, p001 for PX server p001)
SQL_ID	VARCHAR2(13)	SQL identifier of the statement being monitored
SQL_TEXT	VARCHAR2(2000)	Up to the first 2000 characters of the text of the SQL being monitored
IS_FULL_SQLTEXT	VARCHAR2(1)	Indicates whether the SQL_TEXT column has the entire SQL text (Y) or not (N)
SQL_EXEC_START	DATE	Time when the execution started
SQL_EXEC_ID	NUMBER	Execution identifier. Together, the three columns SQL_ID, SQL_EXEC_START, and SQL_EXEC_ID represent the execution key. The execution key is used to uniquely identify one execution of the SQL statement.
SQL_PLAN_HASH_VALUE	NUMBER	SQL plan hash value
SQL_FULL_PLAN_HASH_VALUE	NUMBER	Numeric representation of the complete SQL plan for this cursor. Comparing one SQL_FULL_PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line). Note that the SQL_FULL_PLAN_HASH_VALUE cannot be compared across databases releases. It is not backward compatible.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
EXACT_MATCHING_SIGNATURE	NUMBER	Signature calculated on the normalized SQL text. The normalization includes the removal of white space and the uppercasing of all non-literal strings.
FORCE_MATCHING_SIGNATURE	NUMBER	Same as EXACT_MATCHING_SIGNATURE but literals in the SQL text are replaced by binds
SQL_CHILD_ADDRESS	RAW(4 | 8)	Address of the child cursor (can be used with SQL_ID to join with V$SQL)
SESSION_SERIAL#	NUMBER	Session serial number executing the statement being monitored
PX_IS_CROSS_INSTANCE	VARCHAR2(1)	Indicates whether the SQL statement ran parallel across multiple instances (Y) or not (N)
PX_MAXDOP	NUMBER	Maximum degree of parallelism for any plan operation executed on behalf of the monitored SQL
PX_MAXDOP_INSTANCES	NUMBER	Number of database instances touched at the maximum degree of parallelism
PX_SERVERS_REQUESTED	NUMBER	Total number of parallel execution servers requested to execute the monitored SQL
PX_SERVERS_ALLOCATED	NUMBER	Actual number of parallel execution servers allocated to execute the query
PX_SERVER#	NUMBER	Logical parallel execution server process number executing (or having executed) the statement being monitored; NULL if this monitoring entry is not associated with an execution server. This is a logical number within the parallel server set (see SERVER# in V$PX_SESSION).
PX_SERVER_GROUP	NUMBER	Logical parallel execution server group number to which PX_SERVER# belongs (see SERVER_GROUP in V$PX_SESSION); NULL if this monitoring entry is not associated with a parallel execution server. This value is generally 1 unless the SQL statement has one or more parallel sub-queries.
PX_SERVER_SET	NUMBER	Number (1 or 2) of the logical set of parallel execution servers to which PX_SERVER# belongs (see SERVER_SET in V$PX_SESSION); NULL if this monitoring entry is not associated with a parallel execution server
PX_QCINST_ID	NUMBER	Instance identifier where the parallel execution coordinator runs; NULL if PX_SERVER# is NULL
PX_QCSID	NUMBER	Session identifier for the parallel execution coordinator; NULL if PX_SERVER# is NULL
ERROR_NUMBER	VARCHAR2(40)	Error number encountered in case a SQL fails to execute successfully (for example, 932 in case of ORA-00932)
ERROR_FACILITY	VARCHAR2(4)	Error facility in case a SQL fails to execute successfully (for example, ORA in case of ORA-00932)
ERROR_MESSAGE	VARCHAR2(256)	Detailed error message displayed corresponding to the error number and error facility when a SQL fails to execute successfully
BINDS_XML	CLOB	Information about bind variables used with the SQL, such as name, position, value, data type, and so on (stored in XML format)
OTHER_XML	CLOB	Additional information about SQL execution stored in XML format
ELAPSED_TIME	NUMBER	Elapsed time (in microseconds); updated as the statement executes
QUEUING_TIME	NUMBER	Duration of time (in microseconds) spent by SQL in the statement queue
CPU_TIME	NUMBER	CPU time (in microseconds); updated as the statement executes
FETCHES	NUMBER	Number of fetches associated with the SQL statement; updated as the statement executes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BUFFER_GETS	NUMBER	Number of buffer get operations; updated as the statement executes
DISK_READS	NUMBER	Number of disk reads; updated as the statement executes
DIRECT_WRITES	NUMBER	Number of direct writes; updated as the statement executes
IO_INTERCONNECT_BYTES	NUMBER	Number of I/O bytes exchanged between Oracle Database and the storage system
PHYSICAL_READ_REQUESTS	NUMBER	Number of physical read I/O requests issued by the monitored SQL
PHYSICAL_READ_BYTES	NUMBER	Number of bytes read from disks by the monitored SQL
PHYSICAL_WRITE_REQUESTS	NUMBER	Number of physical write I/O requests issued by the monitored SQL
PHYSICAL_WRITE_BYTES	NUMBER	Number of bytes written to disks by the monitored SQL
APPLICATION_WAIT_TIME	NUMBER	Application wait time (in microseconds); updated as the statement executes
CONCURRENCY_WAIT_TIME	NUMBER	Concurrency wait time (in microseconds); updated as the statement executes
CLUSTER_WAIT_TIME	NUMBER	Cluster wait time (in microseconds); updated as the statement executes
USER_IO_WAIT_TIME	NUMBER	User I/O Wait Time (in microseconds); updated as the statement executes
PLSQL_EXEC_TIME	NUMBER	PL/SQL execution time (in microseconds); updated as the statement executes
JAVA_EXEC_TIME	NUMBER	Java execution time (in microseconds); updated as the statement executes
RM_LAST_ACTION	VARCHAR2(48)	The most recent action that was taken on this SQL operation by Resource Manager. Its value is one of the following: • CANCEL_SQL • KILL_SESSION • LOG_ONLY • SWITCH TO <CG NAME> For the last value, <CG NAME> is the name of the consumer group that the SQL operation was switched to. If the Resource Plan has since been changed then <CG NAME> is the ID of the consumer group.
RM_LAST_ACTION_REASON	VARCHAR2(128)	The reason for the most recent action that was taken on this SQL operation by Resource Manager. Its value is one of the following: • SWITCH_CPU_TIME • SWITCH_IO_REQS • SWITCH_IO_MBS • SWITCH_ELAPSED_TIME • SWITCH_IO_LOGICAL
RM_LAST_ACTION_TIME	DATE	The time of the most recent action that was taken on this SQL operation by Resource Manager
RM_CONSUMER_GROUP	VARCHAR2(128)	The current consumer group for this SQL operation
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_NAME	VARCHAR2(128)	Container name of the object. The value of this column is NULL in non-CDBs.
ECID	VARCHAR2(64)	Execution context identifier (sent by Application Server)
IS_ADAPTIVE_PLAN	VARCHAR2(1)	Indicates whether the statistics are from an adaptive plan (Y) or not (N).
IS_FINAL_PLAN	VARCHAR2(1)	Indicates whether the statistics are from the final plan (Y) or not (N).
IN_DBOP_NAME	VARCHAR2(30)	If the SQL that is monitored was executed by a session that was also monitored by a database operation (DBOP), then this column specifies the name of that DBOP
IN_DBOP_EXEC_ID	NUMBER	If the SQL that is monitored was executed by a session that was also monitored by a database operation (DBOP), then this column specifies the execution ID of that DBO
IO_CELL_UNCOMPRESSED_BYTES	NUMBER	Number of uncompressed bytes (that is, size after decompression) that are offloaded to the Exadata cells **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_CELL_OFFLOAD_ELIGIBLE_BYTES	NUMBER	Number of I/O bytes which can be filtered by the Exadata storage system **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_CELL_OFFLOAD_RETURNED_BYTES	NUMBER	Number of filtered bytes returned by Exadata cells (that is, the number of bytes returned after processing has been offloaded on the Exadata cells) **See Also:** Oracle Exadata Storage Server Software documentation for more information
CURRENT_USER#[1]	NUMBER	Unique number identifying the current user
CURRENT_USERNAME[1]	VARCHAR2(128)	Username for the current user

[1] This column is available starting with Oracle Database 19c.

See Also

- "V$SQL_PLAN_MONITOR"
- "V$ALL_SQL_MONITOR"
- *Oracle Database SQL Tuning Guide* for more information about monitoring database operations
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_MODULE procedure
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_ACTION procedure

9.56 V$SQL_MONITOR_SESSTAT

V$SQL_MONITOR_SESSTAT displays information about statistics that are exposed in V$SESSTAT.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
KEY	NUMBER	Artificial join key to efficiently join with other database operation views
STATISTIC#	NUMBER	Statistic number
VALUE	NUMBER	Value associated with the statistic

See Also

"V$SESSTAT"

9.57 V$SQL_MONITOR_STATNAME

V$SQL_MONITOR_STATNAME provides information about the plan line statistics exposed in V$SQL_PLAN_MONITOR. A plan line statistic is identified by its group ID (column GROUP_ID) and its ID (column ID).

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
ID	NUMBER	Plan line statistic ID
GROUP_ID	NUMBER	Plan line statistic group ID
NAME	VARCHAR2(40)	Short name for the statistic
DESCRIPTION	VARCHAR2(200)	Short description for the statistic
TYPE	NUMBER	Reserved for internal use
FLAGS	NUMBER	Reserved for internal use

See Also

"V$SQL_PLAN_MONITOR"

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.58 V$SQL_OPTIMIZER_ENV

V$SQL_OPTIMIZER_ENV displays the contents of the optimizer environment used to build the execution plan of a SQL cursor. This view can be joined with V$SQLAREA on (HASH_VALUE, ADDRESS) and with V$SQL on (HASH_VALUE, CHILD_ADDRESS).

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Address of the parent cursor
HASH_VALUE	NUMBER	Hash value of the parent cursor in the library cache. The hash value is the fixed index for the view and should be used when querying V$SQL_OPTIMIZER_ENV to avoid scanning the entire library cache.
SQL_ID	VARCHAR2(13)	SQL identifier
CHILD_ADDRESS	RAW(4 | 8)	Address of the child cursor
CHILD_NUMBER	NUMBER	Child cursor number
ID	NUMBER	Unique identifier of the parameter in the optimizer environment
NAME	VARCHAR2(40)	Name of the parameter
ISDEFAULT	VARCHAR2(3)	Indicates whether the parameter is set to the default value (YES) or not (NO)
VALUE	VARCHAR2(25)	Value of the parameter
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.59 V$SQL_PLAN

V$SQL_PLAN contains the execution plan information for each child cursor loaded in the library cache.

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Address of the handle to the parent for this cursor
HASH_VALUE	NUMBER	Hash value of the parent statement in the library cache. The two columns ADDRESS and HASH_VALUE can be used to join with V$SQLAREA to add the cursor-specific information.
SQL_ID	VARCHAR2(13)	SQL identifier of the parent cursor in the library cache
PLAN_HASH_VALUE	NUMBER	Numerical representation of the current SQL plan for this cursor. Comparing one PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line).
FULL_PLAN_HASH_VALUE	NUMBER	Numeric representation of the complete SQL plan for this cursor. Comparing one FULL_PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line). Note that the FULL_PLAN_HASH_VALUE cannot be compared across databases releases. It is not backward compatible.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CHILD_ADDRESS	RAW(4 | 8)	Address of the child cursor
CHILD_NUMBER	NUMBER	Number of the child cursor that uses this execution plan. The columns ADDRESS, HASH_VALUE, and CHILD_NUMBER can be used to join with V$SQL to add the child cursor-specific information.
TIMESTAMP	DATE	Date and time when the execution plan was generated
OPERATION	VARCHAR2(120)	Name of the internal operation performed in this step (for example, TABLE ACCESS)
OPTIONS	VARCHAR2(120)	A variation on the operation described in the OPERATION column (for example, FULL)
OBJECT_NODE	VARCHAR2(160)	Name of the database link used to reference the object (a table name or view name). For local queries that use parallel execution, this column describes the order in which output from operations is consumed.
OBJECT#	NUMBER	Object number of the table or the index
OBJECT_OWNER	VARCHAR2(128)	Name of the user who owns the schema containing the table or index
OBJECT_NAME	VARCHAR2(128)	Name of the table or index
OBJECT_ALIAS	VARCHAR2(261)	Alias for the object
OBJECT_TYPE	VARCHAR2(80)	Type of the object
OPTIMIZER	VARCHAR2(80)	Current mode of the optimizer for the first row in the plan (statement line), for example, ALL_ROWS. When the operation is a database access (for example, TABLE ACCESS), this column indicates whether or not the object is analyzed.
ID	NUMBER	A number assigned to each step in the execution plan
PARENT_ID	NUMBER	ID of the next execution step that operates on the output of the current step
DEPTH	NUMBER	Depth (or level) of the operation in the tree. It is not necessary to issue a CONNECT BY statement to get the level information, which is generally used to indent the rows from the PLAN_TABLE table. The root operation (statement) is level 0.
POSITION	NUMBER	Order of processing for all operations that have the same PARENT_ID
SEARCH_COLUMNS	NUMBER	Number of index columns with start and stop keys (that is, the number of columns with matching predicates)
COST	NUMBER	Cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
CARDINALITY	NUMBER	Estimate, by the cost-based optimizer, of the number of rows produced by the operation
BYTES	NUMBER	Estimate, by the cost-based optimizer, of the number of bytes produced by the operation
OTHER_TAG	VARCHAR2(140)	Describes the contents of the OTHER column. See EXPLAIN PLAN for values.
PARTITION_START	VARCHAR2(256)	Start partition of a range of accessed partitions
PARTITION_STOP	VARCHAR2(256)	Stop partition of a range of accessed partitions
PARTITION_ID	NUMBER	Step that computes the pair of values of the PARTITION_START and PARTITION_STOP columns
OTHER	VARCHAR2(4000)	Other information specific to the execution step that users may find useful. See EXPLAIN PLAN for values.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DISTRIBUTION	VARCHAR2(80)	Stores the method used to distribute rows from producer query servers to consumer query servers
CPU_COST	NUMBER	CPU cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
IO_COST	NUMBER	I/O cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
TEMP_SPACE	NUMBER	Temporary space usage of the operation (sort or hash-join) as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
ACCESS_PREDICATES	VARCHAR2(4000)	Predicates used to locate rows in an access structure. For example, start or stop predicates for an index range scan.
FILTER_PREDICATES	VARCHAR2(4000)	Predicates used to filter rows before producing them
PROJECTION	VARCHAR2(4000)	Expressions produced by the operation
TIME	NUMBER	Elapsed time (in seconds) of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
QBLOCK_NAME	VARCHAR2(128)	Name of the query block
REMARKS	VARCHAR2(4000)	Remarks
OTHER_XML	CLOB	Provides extra information specific to an execution step of the execution plan. The content of this column is structured using XML since multiple pieces of information can be stored there. This includes: • Name of the schema against which the query was parsed • Release number of the Oracle Database that produced the explain plan • Hash value associated with the execution plan • Name (if any) of the outline or the SQL profile used to build the execution plan • Indication of whether or not dynamic statistics were used to produce the plan • The outline data, a set of optimizer hints that can be used to regenerate the same plan • Additional data that describes the relationship between rows in the plan table and subplans of adaptive plans. Note that in Oracle Database 12*c*, there will be extra rows in the plan table and V$SQL_PLAN.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.60 V$SQL_PLAN_MONITOR

V$SQL_PLAN_MONITOR displays plan level monitoring statistics for each SQL statement found in V$SQL_MONITOR. Each row in V$SQL_PLAN_MONITOR corresponds to an operation of the

Copyright © 2002, 2026, Oracle and/or its affiliates.

execution plan being monitored. As with V$SQL_MONITOR, statistics exposed in V$SQL_PLAN_MONITOR are generally updated every second when the statement executes. These statistics are recycled on the same basis as V$SQL_MONITOR.
To eliminate the overhead of SQL plan monitoring, statistics collected for each operation of the plan do not record timing information such as elapsed time, CPU time, or I/O time. Instead, this timing information can be estimated quite accurately by joining V$SQL_PLAN_MONITOR with V$ACTIVE_SESSION_HISTORY on SQL_ID, SQL_EXEC_START, SQL_EXEC_ID, and SQL_PLAN_LINE_ID (simply named PLAN_LINE_ID in V$SQL_PLAN_MONITOR). The result of that join is a sample of the activity performed by each operation in the plan, from which an estimate of CPU time and wait time can be derived. This can be achieved by breaking statement level monitoring time statistics found in V$SQL_MONITOR in proportion to the number of samples found in V$ACTIVE_SESSION_HISTORY for the corresponding activity type.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
KEY	NUMBER	Foreign key to efficiently join V$SQL_PLAN_MONITOR with V$SQL_MONITOR (see V$SQL_MONITOR)
STATUS	VARCHAR2(19)	SQL execution status: • EXECUTING - SQL statement is still executing • DONE (ERROR) - Execution terminated with an error • DONE (FIRST N ROWS) - Execution terminated by the application before all rows were fetched • DONE (ALL ROWS) - Execution terminated and all rows were fetched • DONE - Execution terminated (parallel execution)
FIRST_REFRESH_TIME	DATE	Time when monitoring of the SQL statement started
LAST_REFRESH_TIME	DATE	Time when statistics were last updated for the SQL statement
FIRST_CHANGE_TIME	DATE	First time a row was produced by this operation
LAST_CHANGE_TIME	DATE	Last time a row was produced by this operation
REFRESH_COUNT	NUMBER	Number of times statistics have been refreshed
SID	NUMBER	Session identifier executing (or having executed) the SQL statement being monitored
PROCESS_NAME	VARCHAR2(5)	Process name identifier
SQL_ID	VARCHAR2(13)	SQL identifier
SQL_EXEC_START	DATE	Time when the execution started
SQL_EXEC_ID	NUMBER	Execution identifier
SQL_PLAN_HASH_VALUE	NUMBER	SQL plan hash value
SQL_CHILD_ADDRESS	RAW(4 | 8)	Address of the child cursor
PLAN_PARENT_ID	NUMBER	ID of the next execution step that operates on the output of the current step
PLAN_LINE_ID	NUMBER	Plan line number for the entry
PLAN_OPERATION	VARCHAR2(30)	Plan operation name (from V$SQL_PLAN)
PLAN_OPTIONS	VARCHAR2(30)	Plan option name (from V$SQL_PLAN)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PLAN_OBJECT_OWNER	VARCHAR2(128)	Name of the user who owns the schema containing the table or index
PLAN_OBJECT_NAME	VARCHAR2(128)	Name of the table or index
PLAN_OBJECT_TYPE	VARCHAR2(80)	Type of the object
PLAN_DEPTH	NUMBER	Depth (or level) of the operation in the tree. It is not necessary to issue a CONNECT BY statement to get the level information, which is generally used to indent the rows from the PLAN_TABLE table. The root operation (statement) is level 0.
PLAN_POSITION	NUMBER	Order of processing for all operations that have the same PARENT_ID
PLAN_COST	NUMBER	Cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
PLAN_CARDINALITY	NUMBER	Estimate, by the cost-based optimizer, of the number of rows produced by the operation
PLAN_BYTES	NUMBER	Estimate, by the cost-based optimizer, of the number of bytes produced by the operation
PLAN_TIME	NUMBER	Elapsed time (in seconds) of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
PLAN_PARTITION_START	VARCHAR2(256)	Start partition of a range of accessed partitions
PLAN_PARTITION_STOP	VARCHAR2(256)	Stop partition of a range of accessed partitions
PLAN_CPU_COST	NUMBER	CPU cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
PLAN_IO_COST	NUMBER	I/O cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
PLAN_TEMP_SPACE	NUMBER	Temporary space usage of the operation (sort or hash-join) as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is NULL.
STARTS	NUMBER	Number of times this operation was executed. For example, an operation is executed multiple times when it is on the right side of a nested-loop join (once for each row of the left input of that nested-loop join).
OUTPUT_ROWS	NUMBER	Number of rows produced by this operation since the execution started. This number is cumulated for all executions of this operation. Divide by the value of the STARTS column to compute the average number of rows per execution of the operation. Note that the value in the STARTS column is equal to or higher than the value in the OUTPUT_ROWS column. The value will usually be equal, but depending on internal optimizations a higher value may be seen.
IO_INTERCONNECT_BYTES	NUMBER	Number of I/O bytes exchanged between Oracle Database and the storage system. Maintained only after Oracle starts to monitor the execution.
PHYSICAL_READ_REQUESTS	NUMBER	Number of physical read I/O requests issued by the monitored SQL. Maintained only after Oracle starts to monitor the execution.
PHYSICAL_READ_BYTES	NUMBER	Number of bytes read from disks by the monitored SQL. Maintained only after Oracle starts to monitor the execution.
PHYSICAL_WRITE_REQUESTS	NUMBER	Number of physical write I/O requests issued by the monitored SQL. Maintained only after Oracle starts to monitor the execution.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PHYSICAL_WRITE_BYTES	NUMBER	Number of bytes written to disks by the monitored SQL. Maintained only after Oracle starts to monitor the execution.
WORKAREA_MEM	NUMBER	Amount of memory (in bytes) used by the operation when the query is executing; NULL if the execution is done. This applies only to operations using a work area, such as sort, hash-join, group-by, and so on.
WORKAREA_MAX_MEM	NUMBER	Maximum value (in bytes) for WORKAREA_MEM; NULL if the operation is not using a work area. When the execution is finished, this value will hold the maximum amount of memory consumed by this operation during the execution of the statement.
WORKAREA_TEMPSEG	NUMBER	Amount of temporary space (in bytes) used by the operation when the query is executing; NULL if the operation has not spilled to disk or if the execution is finished
WORKAREA_MAX_TEMPSEG	NUMBER	Maximum value (in bytes) for WORKAREA_TEMPSEG; NULL if this operation never spilled to disk. When the execution is done, this value will hold the maximum amount of temporary space consumed by this operation during the entire execution.
OTHERSTAT_GROUP_ID	NUMBER	Plan line statistic group identifier (see GROUP_ID column in V$SQL_MONITOR_STATNAME)
OTHERSTAT_1_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 1 of that plan line
OTHERSTAT_1_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_1_VALUE	NUMBER	Value of statistic number 1 of that plan line
OTHERSTAT_2_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 2 of that plan line
OTHERSTAT_2_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_2_VALUE	NUMBER	Value of statistic number 2 of that plan line
OTHERSTAT_3_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 3 of that plan line
OTHERSTAT_3_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_3_VALUE	NUMBER	Value of statistic number 3 of that plan line
OTHERSTAT_4_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 4 of that plan line
OTHERSTAT_4_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_4_VALUE	NUMBER	Value of statistic number 4 of that plan line
OTHERSTAT_5_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 5 of that plan line
OTHERSTAT_5_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_5_VALUE	NUMBER	Value of statistic number 5 of that plan line
OTHERSTAT_6_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 6 of that plan line
OTHERSTAT_6_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_6_VALUE	NUMBER	Value of statistic number 6 of that plan line
OTHERSTAT_7_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 7 of that plan line
OTHERSTAT_7_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_7_VALUE	NUMBER	Value of statistic number 7 of that plan line

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OTHERSTAT_8_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 8 of that plan line
OTHERSTAT_8_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_8_VALUE	NUMBER	Value of statistic number 8 of that plan line
OTHERSTAT_9_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 9 of that plan line
OTHERSTAT_9_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_9_VALUE	NUMBER	Value of statistic number 9 of that plan line
OTHERSTAT_10_ID	NUMBER	Statistic identifier (see ID column in V$SQL_MONITOR_STATNAME) for statistic number 10 of that plan line
OTHERSTAT_10_TYPE	NUMBER	Reserved for internal use
OTHERSTAT_10_VALUE	NUMBER	Value of statistic number 10 of that plan line
OTHER_XML	CLOB	Provides extra information specific to an execution step of the execution plan. The content of this column is structured using XML because it allows multiple pieces of information to be stored, including the following: • Name of the schema against which the query was parsed • Release number of the Oracle Database that produced the explain plan • Hash value associated with the execution plan • Name (if any) of the outline or the SQL profile used to build the execution plan • Indication of whether or not dynamic statistics were used to produce the plan • The outline data, a set of optimizer hints that can be used to regenerate the same plan
PLAN_OPERATION_INACTIVE	NUMBER	Indicates whether this plan operation was part of the final resolved plan

See Also

- "V$SQL_MONITOR_STATNAME"
- "V$ALL_SQL_PLAN_MONITOR"

9.61 V$SQL_PLAN_STATISTICS

V$SQL_PLAN_STATISTICS provides execution statistics at the row source level for each child cursor.

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Address of the handle to the parent for this cursor
HASH_VALUE	NUMBER	Hash value of the parent statement in the library cache. The two columns ADDRESS and HASH_VALUE can be used to join with V$SQLAREA to locate the parent cursor.
SQL_ID	VARCHAR2(13)	SQL identifier of the parent statement in the library cache

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PLAN_HASH_VALUE	NUMBER	Numerical representation of the current SQL plan for this cursor. Comparing one PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line)
FULL_PLAN_HASH_VALUE	NUMBER	Numeric representation of the complete SQL plan for this cursor. Comparing one FULL_PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line). Note that the FULL_PLAN_HASH_VALUE cannot be compared across databases releases. It is not backward compatible.
CHILD_ADDRESS	RAW(4 | 8)	Address of the child cursor
CHILD_NUMBER	NUMBER	Number of the child cursor that uses this work area. The columns ADDRESS, HASH_VALUE, and CHILD_NUMBER can be used to join with V$SQL to locate the child cursor using this area.
OPERATION_ID	NUMBER	A number assigned to each step in the execution plan
EXECUTIONS	NUMBER	Number of times this cursor has been executed
LAST_STARTS	NUMBER	Number of times this operation has been started, during the last execution
STARTS	NUMBER	Number of times this operation has been started, accumulated over the past executions
LAST_OUTPUT_ROWS	NUMBER	Number of rows produced by the row source, during the last execution
OUTPUT_ROWS	NUMBER	Number of rows produced by the row source, accumulated over the past executions
LAST_CR_BUFFER_GETS	NUMBER	Number of buffers retrieved in consistent mode, during the last execution. Buffers are usually retrieved in consistent mode for queries.
CR_BUFFER_GETS	NUMBER	Number of buffers retrieved in consistent mode, accumulated over the past executions. Buffers are usually retrieved in consistent mode for queries.
LAST_CU_BUFFER_GETS	NUMBER	Number of buffers retrieved in current mode, during the last execution. Buffers are retrieved in current mode for statements such as INSERT, UPDATE, and DELETE.
CU_BUFFER_GETS	NUMBER	Number of buffers retrieved in current mode, accumulated over the past executions. Buffers are retrieved in current mode for statements such as INSERT, UPDATE, and DELETE.
LAST_DISK_READS	NUMBER	Number of physical disk reads performed by the operation, during the last execution
DISK_READS	NUMBER	Number of physical disk reads performed by the operation, accumulated over the past executions
LAST_DISK_WRITES	NUMBER	Number of physical disk writes performed by the operation, during the last execution
DISK_WRITES	NUMBER	Number of physical disk writes performed by the operation, accumulated over the past executions
LAST_ELAPSED_TIME	NUMBER	Elapsed time (in microseconds) corresponding to this operation, during the last execution
ELAPSED_TIME	NUMBER	Elapsed time (in microseconds) corresponding to this operation, accumulated over the past executions

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SQLAREA"

9.62 V$SQL_PLAN_STATISTICS_ALL

V$SQL_PLAN_STATISTICS_ALL contains memory usage statistics for row sources that use SQL memory (sort or hash-join). This view concatenates information in V$SQL_PLAN with execution statistics from V$SQL_PLAN_STATISTICS and V$SQL_WORKAREA.

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Address of the handle to the parent for this cursor
HASH_VALUE	NUMBER	Hash value of the parent statement in the library cache. The two columns ADDRESS and HASH_VALUE can be used to join with V$SQLAREA to add the cursor-specific information.
SQL_ID	VARCHAR2(13)	SQL identifier of the parent statement in the library cache
PLAN_HASH_VALUE	NUMBER	Numerical representation of the current SQL plan for this cursor. Comparing one PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line)
FULL_PLAN_HASH_VALUE	NUMBER	Numeric representation of the complete SQL plan for this cursor. Comparing one FULL_PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line). Note that the FULL_PLAN_HASH_VALUE cannot be compared across databases releases. It is not backward compatible.
CHILD_ADDRESS	RAW(4 | 8)	Address of the child cursor
CHILD_NUMBER	NUMBER	Number of the child cursor that uses this execution plan. The columns ADDRESS, HASH_VALUE, and CHILD_NUMBER can be used to join with V$SQL to add the child cursor-specific information.
TIMESTAMP	DATE	Date and time when the execution plan was generated
OPERATION	VARCHAR2(120)	Name of the internal operation performed in this step (for example, TABLE ACCESS)
OPTIONS	VARCHAR2(120)	A variation on the operation described in the OPERATION column (for example, FULL)
OBJECT_NODE	VARCHAR2(160)	Name of the database link used to reference the object (a table name or view name). For local queries that use parallel execution, this column describes the order in which output from operations is consumed.
OBJECT#	NUMBER	Object number of the table or the index

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OBJECT_OWNER	VARCHAR2(128)	Name of the user who owns the schema containing the table or index
OBJECT_NAME	VARCHAR2(128)	Name of the table or index
OBJECT_ALIAS	VARCHAR2(261)	Alias for the object
OBJECT_TYPE	VARCHAR2(80)	Type of the object
OPTIMIZER	VARCHAR2(80)	Current mode of the optimizer for the first row in the plan (statement line), for example, CHOOSE. When the operation is a database access (for example, TABLE ACCESS), this column indicates whether or not the object is analyzed.
ID	NUMBER	A number assigned to each step in the execution plan
PARENT_ID	NUMBER	ID of the next execution step that operates on the output of the current step
DEPTH	NUMBER	Depth (or level) of the operation in the tree. It is not necessary to issue a CONNECT BY statement to get the level information, which is generally used to indent the rows from the PLAN_TABLE table. The root operation (statement) is level 0.
POSITION	NUMBER	Order of processing for all operations that have the same PARENT_ID
SEARCH_COLUMNS	NUMBER	Number of index columns with start and stop keys (that is, the number of columns with matching predicates)
COST	NUMBER	Cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
CARDINALITY	NUMBER	Estimate, by the cost-based optimizer, of the number of rows produced by the operation
BYTES	NUMBER	Estimate, by the cost-based optimizer, of the number of bytes produced by the operation
OTHER_TAG	VARCHAR2(140)	Describes the contents of the OTHER column. See EXPLAIN PLAN for values.
PARTITION_START	VARCHAR2(256)	Start partition of a range of accessed partitions
PARTITION_STOP	VARCHAR2(256)	Stop partition of a range of accessed partitions
PARTITION_ID	NUMBER	Step that computes the pair of values of the PARTITION_START and PARTITION_STOP columns
OTHER	VARCHAR2(4000)	Other information specific to the execution step that users may find useful. See EXPLAIN PLAN for values.
DISTRIBUTION	VARCHAR2(80)	Stores the method used to distribute rows from producer query servers to consumer query servers
CPU_COST	NUMBER	CPU cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
IO_COST	NUMBER	I/O cost of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
TEMP_SPACE	NUMBER	Temporary space usage of the operation (sort or hash-join) as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
ACCESS_PREDICATES	VARCHAR2(4000)	Predicates used to locate rows in an access structure. For example, start or stop predicates for an index range scan.
FILTER_PREDICATES	VARCHAR2(4000)	Predicates used to filter rows before producing them

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PROJECTION	VARCHAR2(4000)	Expressions produced by the operation
TIME	NUMBER	Elapsed time (in seconds) of the operation as estimated by the optimizer's cost-based approach. For statements that use the rule-based approach, this column is null.
QBLOCK_NAME	VARCHAR2(128)	Name of the query block
REMARKS	VARCHAR2(4000)	Remarks
OTHER_XML	CLOB	Provides extra information specific to an execution step of the execution plan. The content of this column is structured using XML since multiple pieces of information can be stored there. This includes: • Name of the schema against which the query was parsed • Release number of the Oracle Database that produced the explain plan • Hash value associated with the execution plan • Name (if any) of the outline or the SQL profile used to build the execution plan • Indication of whether or not dynamic statistics were used to produce the plan • The outline data, a set of optimizer hints that can be used to regenerate the same plan • Additional data that describes the relationship between rows in the plan table and subplans of adaptive plans. Note that in Oracle Database 12*c*, there will be extra rows in the plan table and V$SQL_PLAN.
EXECUTIONS	NUMBER	Number of times this cursor has been executed
LAST_STARTS	NUMBER	Number of times this operation has been started, during the last execution
STARTS	NUMBER	Number of times this operation has been started, accumulated over the past executions
LAST_OUTPUT_ROWS	NUMBER	Number of rows produced by the row source, during the last execution
OUTPUT_ROWS	NUMBER	Number of rows produced by the row source, accumulated over the past executions
LAST_CR_BUFFER_GETS	NUMBER	Number of buffers retrieved in consistent mode, during the last execution. Buffers are usually retrieved in consistent mode for queries.
CR_BUFFER_GETS	NUMBER	Number of buffers retrieved in consistent mode, accumulated over the past executions. Buffers are usually retrieved in consistent mode for queries.
LAST_CU_BUFFER_GETS	NUMBER	Number of buffers retrieved in current mode, during the last execution. Buffers are retrieved in current mode for statements such as INSERT, UPDATE, and DELETE.
CU_BUFFER_GETS	NUMBER	Number of buffers retrieved in current mode, accumulated over the past executions. Buffers are retrieved in current mode for statements such as INSERT, UPDATE, and DELETE.
LAST_DISK_READS	NUMBER	Number of physical disk reads performed by the operation, during the last execution
DISK_READS	NUMBER	Number of physical disk reads performed by the operation, accumulated over the past executions
LAST_DISK_WRITES	NUMBER	Number of physical disk writes performed by the operation, during the last execution
DISK_WRITES	NUMBER	Number of physical disk writes performed by the operation, accumulated over the past executions

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LAST_ELAPSED_TIME	NUMBER	Elapsed time (in microseconds) corresponding to this operation, during the last execution
ELAPSED_TIME	NUMBER	Elapsed time (in microseconds) corresponding to this operation, accumulated over the past executions
POLICY	VARCHAR2(40)	Sizing policy for this work area: • MANUAL • AUTO
ESTIMATED_OPTIMAL_SIZE	NUMBER	Estimated size (in KB) required by this work area to execute the operation completely in memory (optimal execution). This is either derived from optimizer statistics or from previous executions.
ESTIMATED_ONEPASS_SIZE	NUMBER	Estimated size (in KB) required by this work area to execute the operation in a single pass. This is either derived from optimizer statistics or from previous executions.
LAST_MEMORY_USED	NUMBER	Memory size (in KB) used by this work area during the last execution of the cursor
LAST_EXECUTION	VARCHAR2(40)	Indicates whether this work area ran using OPTIMAL, ONE PASS, or under ONE PASS memory requirement (MULTI-PASS), during the last execution of the cursor
LAST_DEGREE	NUMBER	Degree of parallelism used, during the last execution of the cursor
TOTAL_EXECUTIONS	NUMBER	Number of times this work area was active
OPTIMAL_EXECUTIONS	NUMBER	Number of times this work area ran in optimal mode
ONEPASS_EXECUTIONS	NUMBER	Number of times this work area ran in one pass mode
MULTIPASSES_EXECUTIONS	NUMBER	Number of times this work area ran below the one pass memory requirement
ACTIVE_TIME	NUMBER	Average time this work area is active (in hundredths of a second)
MAX_TEMPSEG_SIZE	NUMBER	Maximum temporary segment size (in bytes) created by an instantiation of this work area. This column is null if this work area has never spilled to disk.
LAST_TEMPSEG_SIZE	NUMBER	Temporary segment size (in bytes) created in the last instantiation of this work area. This column is null if the last instantiation of this work area did not spill to disk.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
CON_DBID	NUMBER	The database ID of the PDB

See Also

- "V$SQL_PLAN_STATISTICS"
- "V$SQL_WORKAREA"

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.63 V$SQL_REDIRECTION

V$SQL_REDIRECTION displays SQL statements that are redirected.

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Address of the cursor handle
PARENT_HANDLE	RAW(4 | 8)	Address of the parent cursor handle
HASH_VALUE	NUMBER	Hash value of the SQL statement
SQL_ID	VARCHAR2(13)	SQL identifier of the SQL statement
CHILD_NUMBER	NUMBER	Number of the child (instance) for the hash
PARSING_USER_ID	NUMBER	Parsing user ID
PARSING_SCHEMA_ID	NUMBER	Parsing schema ID
COMMAND_TYPE	NUMBER	SELECT, UPDATE, INSERT, MERGE
REASON	VARCHAR2(14)	Reason for redirection ('INVALID OBJECT', 'ROWID', 'QUERY REWRITE', 'READ ONLY')
ERROR_CODE	NUMBER	Error code for local parse
POSITION	NUMBER	Error position, 0 if unknown
SQL_TEXT_PIECE	VARCHAR2(1000)	SQL Text containing position, usually a (qualified) identifier
ERROR_MESSAGE	VARCHAR2(1000)	Error code's corresponding error message resolved in the database language, no arguments resolved
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.64 V$SQL_SHARD

V$SQL_SHARD displays the shard information for a shard query's previous execution. This view uniquely maps a shard SQL fragment of a cross shard query to the target shard database.

Column	Datatype	Description
SQL_ID	VARCHAR2(13)	SQL identifier of a cross shard query on the coordinator
CHILD_NUMBER	NUMBER	Cursor child number of a cross shard query on the coordinator
OPERATION_ID	NUMBER	Operation ID of a remote node for a shard SQL fragment of a cross shard query
SHARD_SQL_ID	VARCHAR2(13)	SQL ID of the SQL segment associated with a remote operation identified by the operation ID in the OPERATION_ID column
SHARD_ID	NUMBER	IDs of shards where the shard SQL fragment was executed
SHARD_CHILD_NUMBER	NUMBER	Cursor child number of a shard SQL fragment on a shard. The default value is 0.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Using Oracle Sharding for more information about the V$SQL_SHARD view

9.65 V$SQL_SHARED_CURSOR

V$SQL_SHARED_CURSOR explains why a particular child cursor is not shared with existing child cursors. Each column identifies a specific reason why the cursor cannot be shared.

Column	Datatype	Description
SQL_ID	VARCHAR2(13)	SQL identifier
ADDRESS	RAW(4 | 8)	Address of the parent cursor
CHILD_ADDRESS	RAW(4 | 8)	Address of the child cursor
CHILD_NUMBER	NUMBER	Child number
UNBOUND_CURSOR	VARCHAR2(1)	(Y|N) The existing child cursor was not fully built (in other words, it was not optimized)
SQL_TYPE_MISMATCH	VARCHAR2(1)	(Y|N) The SQL type does not match the existing child cursor
OPTIMIZER_MISMATCH	VARCHAR2(1)	(Y|N) The optimizer environment does not match the existing child cursor
OUTLINE_MISMATCH	VARCHAR2(1)	(Y|N) The outlines do not match the existing child cursor
STATS_ROW_MISMATCH	VARCHAR2(1)	(Y|N) The existing statistics do not match the existing child cursor
LITERAL_MISMATCH	VARCHAR2(1)	(Y|N) Non-data literal values do not match the existing child cursor
FORCE_HARD_PARSE	VARCHAR2(1)	(Y|N) For internal use
EXPLAIN_PLAN_CURSOR	VARCHAR2(1)	(Y|N) The child cursor is an explain plan cursor and should not be shared
BUFFERED_DML_MISMATCH	VARCHAR2(1)	(Y|N) Buffered DML does not match the existing child cursor
PDML_ENV_MISMATCH	VARCHAR2(1)	(Y|N) PDML environment does not match the existing child cursor
INST_DRTLD_MISMATCH	VARCHAR2(1)	(Y|N) Insert direct load does not match the existing child cursor
SLAVE_QC_MISMATCH	VARCHAR2(1)	(Y|N) The existing child cursor is a worker cursor and the new one was issued by the coordinator (or, the existing child cursor was issued by the coordinator and the new one is a worker cursor)
TYPECHECK_MISMATCH	VARCHAR2(1)	(Y|N) The existing child cursor is not fully optimized
AUTH_CHECK_MISMATCH	VARCHAR2(1)	(Y|N) Authorization/translation check failed for the existing child cursor
BIND_MISMATCH	VARCHAR2(1)	(Y|N) The bind metadata does not match the existing child cursor

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DESCRIBE_MISMATCH	VARCHAR2(1)	(Y|N) The typecheck heap is not present during the describe for the child cursor
LANGUAGE_MISMATCH	VARCHAR2(1)	(Y|N) The language handle does not match the existing child cursor
TRANSLATION_MISMATCH	VARCHAR2(1)	(Y|N) The base objects of the existing child cursor do not match
BIND_EQUIV_FAILURE	VARCHAR2(1)	(Y|N) The bind value's selectivity does not match that used to optimize the existing child cursor
INSUFF_PRIVS	VARCHAR2(1)	(Y|N) Insufficient privileges on objects referenced by the existing child cursor
INSUFF_PRIVS_REM	VARCHAR2(1)	(Y|N) Insufficient privileges on remote objects referenced by the existing child cursor
REMOTE_TRANS_MISMATCH	VARCHAR2(1)	(Y|N) The remote base objects of the existing child cursor do not match
LOGMINER_SESSION_MISMATCH	VARCHAR2(1)	(Y|N) LogMiner Session parameters mismatch
INCOMP_LTRL_MISMATCH	VARCHAR2(1)	(Y|N) Cursor might have some binds (literals) which may be unsafe/non-data. Value mismatch.
OVERLAP_TIME_MISMATCH	VARCHAR2(1)	(Y|N) Mismatch caused by setting session parameter ERROR_ON_OVERLAP_TIME
EDITION_MISMATCH	VARCHAR2(1)	(Y|N) Cursor edition mismatch
MV_QUERY_GEN_MISMATCH	VARCHAR2(1)	(Y|N) Internal, used to force a hard-parse when analyzing materialized view queries
USER_BIND_PEEK_MISMATCH	VARCHAR2(1)	(Y|N) Cursor is not shared because value of one or more user binds is different and this has a potential to change the execution plan
TYPCHK_DEP_MISMATCH	VARCHAR2(1)	(Y|N) Cursor has typecheck dependencies
NO_TRIGGER_MISMATCH	VARCHAR2(1)	(Y|N) Cursor and child have no trigger mismatch
FLASHBACK_CURSOR	VARCHAR2(1)	(Y|N) Cursor non-shareability due to flashback
ANYDATA_TRANSFORMATION	VARCHAR2(1)	(Y|N) Is criteria for opaque type transformation and does not match
PDDL_ENV_MISMATCH	VARCHAR2(1)	(Y|N) Environment setting mismatch for parallel DDL cursor (that is, one or more of the following parameter values have changed: PARALLEL_EXECUTION_ENABLED, PARALLEL_DDL_MODE, PARALLEL_DDL_FORCED_DEGREE, or PARALLEL_DDL_FORCED_INSTANCES)
TOP_LEVEL_RPI_CURSOR	VARCHAR2(1)	(Y|N) Is top level RPI cursor
DIFFERENT_LONG_LENGTH	VARCHAR2(1)	(Y|N) Value of LONG does not match
LOGICAL_STANDBY_APPLY	VARCHAR2(1)	(Y|N) Logical standby apply context does not match
DIFF_CALL_DURN	VARCHAR2(1)	(Y|N) If worker SQL cursor/single call
BIND_UACS_DIFF	VARCHAR2(1)	(Y|N) One cursor has bind UACs and one does not
PLSQL_CMP_SWITCHS_DIFF	VARCHAR2(1)	(Y|N) PL/SQL anonymous block compiled with different PL/SQL compiler switches
CURSOR_PARTS_MISMATCH	VARCHAR2(1)	(Y|N) Cursor was compiled with subexecution (cursor parts were executed)
STB_OBJECT_MISMATCH	VARCHAR2(1)	(Y|N) STB is an internal name for a SQL Management Object Mismatch. A SQL Management Object Mismatch means that either a SQL plan baseline, or a SQL profile, or a SQL patch has been created for your SQL statement between the executions. Because a cursor is a read-only entity, a hard parse is forced to be able to create a new cursor that contains information about the new SQL management object related to this SQL statement.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CROSSEDITION_TRIGGER_MISMATCH	VARCHAR2(1)	(Y|N) The set of crossedition triggers to execute might differ
PQ_SLAVE_MISMATCH	VARCHAR2(1)	(Y|N) Top-level worker decides not to share cursor
TOP_LEVEL_DDL_MISMATCH	VARCHAR2(1)	(Y|N) Is top-level DDL cursor
MULTI_PX_MISMATCH	VARCHAR2(1)	(Y|N) Cursor has multiple parallelizers and is worker-compiled
BIND_PEEKED_PQ_MISMATCH	VARCHAR2(1)	(Y|N) Cursor based around bind peeked values
MV_REWRITE_MISMATCH	VARCHAR2(1)	(Y|N) Cursor needs recompilation because an SCN was used during compile time due to being rewritten by materialized view
ROLL_INVALID_MISMATCH	VARCHAR2(1)	(Y|N) Marked for rolling invalidation and invalidation window exceeded
OPTIMIZER_MODE_MISMATCH	VARCHAR2(1)	(Y|N) Parameter OPTIMIZER_MODE mismatch (for example, all_rows versus first_rows_1)
PX_MISMATCH	VARCHAR2(1)	(Y|N) Mismatch in one parameter affecting the parallelization of a SQL statement. For example, one cursor was compiled with parallel DML enabled while the other was not.
MV_STALEOBJ_MISMATCH	VARCHAR2(1)	(Y|N) Cursor cannot be shared because there is a mismatch in the list of materialized views which were stale when the cursor was built
FLASHBACK_TABLE_MISMATCH	VARCHAR2(1)	(Y|N) Cursor cannot be shared because there is a mismatch with triggers being enabled and/or referential integrity constraints being deferred
LITREP_COMP_MISMATCH	VARCHAR2(1)	(Y|N) Mismatch in use of literal replacement
PLSQL_DEBUG	VARCHAR2(1)	(Y|N) Value of the PLSQL_DEBUG parameter for the current session does not match the value used to build the cursor
LOAD_OPTIMIZER_STATS	VARCHAR2(1)	(Y|N) A hard parse is forced to initialize extended cursor sharing
ACL_MISMATCH	VARCHAR2(1)	(Y|N) Cached ACL evaluation result stored in the child cursor is not valid for the current session or user
FLASHBACK_ARCHIVE_MISMATCH	VARCHAR2(1)	(Y|N) Value of the FLASHBACK_DATA_ARCHIVE_INTERNAL_CURSOR parameter for the current session does not match the value used to build the cursor
LOCK_USER_SCHEMA_FAILED	VARCHAR2(1)	(Y|N) User or schema used to build the cursor no longer exists **Note:** This sharing criterion is deprecated
REMOTE_MAPPING_MISMATCH	VARCHAR2(1)	(Y|N) Reloaded cursor was previously remote-mapped and is currently not remote-mapped. Therefore, the cursor must be reparsed.
LOAD_RUNTIME_HEAP_FAILED	VARCHAR2(1)	(Y|N) Loading of run-time heap for the new cursor (or reload of aged out cursor) failed
HASH_MATCH_FAILED	VARCHAR2(1)	(Y|N) No existing child cursors have the unsafe literal bind hash values required by the current cursor
PURGED_CURSOR	VARCHAR2(1)	(Y|N) Child cursor is marked for purging
BIND_LENGTH_UPGRADEABLE	VARCHAR2(1)	(Y|N) Bind length(s) required for the current cursor are longer than the bind length(s) used to build the child cursor
USE_FEEDBACK_STATS	VARCHAR2(1)	(Y|N) A hard parse is forced so that the optimizer can reoptimize the query with improved optimizer inputs (for example, cardinality estimates)
REASON	CLOB	Child number, id, and reason the cursor is not shared. The content of this column is structured using XML. If this column is null, the FORCE_HARD_PARSE, LOAD_RUNTIME_HEAP_FAILED, and HASH_MATCH_FAILED columns may enable you to determine why the child cursor is not shared.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.66 V$SQL_SHARED_MEMORY

V$SQL_SHARED_MEMORY displays information about the cursor shared memory snapshot. Each SQL statement stored in the shared pool has one or more child objects associated with it. Each child object has several parts, one of which is the context heap, which holds, among other things, the query plan.

Column	Datatype	Description
SQL_TEXT	VARCHAR2(1000)	SQL text of the shared cursor child object for which this row is displaying information
SQL_FULLTEXT	CLOB	Full text for the SQL statement exposed as a CLOB column. The full text of a SQL statement can be retrieved using this column instead of joining with the V$SQLTEXT dynamic performance view.
HASH_VALUE	NUMBER	Hash value of the above SQL text in the shared pool
SQL_ID	VARCHAR2(13)	SQL identifier of the above SQL text in the shared pool
HEAP_DESC	RAW(4 | 8)	Address of the descriptor for the context heap of the child cursor described in this row
STRUCTURE	VARCHAR2(64)	If the memory chunk described in this row was allocated using a comment of the form "X : Y", then this is the "X" part of the comment
FUNCTION	VARCHAR2(64)	Similar to the STRUCTURE column, this is the "Y" field of the comment
CHUNK_COM	VARCHAR2(16)	Whole comment field that was supplied when this memory chunk was allocated
CHUNK_PTR	RAW(4 | 8)	Starting address of the allocated memory chunk
CHUNK_SIZE	NUMBER	Amount of memory allocated for this chunk
ALLOC_CLASS	VARCHAR2(8)	Class of memory that this chunk of memory belongs to. It will usually be either FREEABLE or PERMANENT.
CHUNK_TYPE	NUMBER	An index into a table of callback functions that tell the server how to re-create this chunk of memory should it need to be removed from the shared pool based on an LRU algorithm
SUBHEAP_DESC	RAW(4 | 8)	If the parent heap of this context heap is itself a subheap, then this is the address of the descriptor of the parent heap
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.67 V$SQL_TESTCASES

V$SQL_TESTCASES displays information about test cases exported by SQL Test Case Builder.

You can use this view in conjunction with the V$DIAG_INCIDENT view. Join the INCIDENT_ID column in this view with the INCIDENT_ID column in V$DIAG_INCIDENT to view information about the test case associated with a particular incident.

The V$SQL_TESTCASES view requires the existence of a TCB root directory named SQL_TCB_DIR. This view will not contain any rows if a TCB root directory does not exist, or if the TCB root directory exists with a name other than SQL_TCB_DIR. The operating system directory to which the TCB root directory refers must be writable by the owner of the Oracle Database binaries.

- In Oracle Autonomous Database environments, the TCB root directory is created automatically on each POD during provisioning.
- For on-premises databases, a user who has been granted the DBA role must explicitly create the TCB root directory. See *Oracle Database Administrator's Guide* for more information.

Column	Datatype	Description
TESTCASE_NAME	VARCHAR2(512)	Test case name
SQL_ID	VARCHAR2(13)	SQL identifier of the SQL statement in the test case
SQL_TEXT	VARCHAR2(1000)	First 1000 characters of text for the SQL statement in the test case
SQL_TEXT_FULL	CLOB	Full text for the SQL statement in the test case, exposed as a CLOB column
INCIDENT_ID	NUMBER	ID for the incident associated with the test case
PROBLEM_TYPE	NUMBER	Type of problem for the incident associated with the test case. Possible values: • 1: Performance problem (PROBLEM_TYPE_PERFORMANCE) • 2: Inconsistent results (PROBLEM_TYPE_WRONG_RESULTS) • 3: Crash in compilation (PROBLEM_TYPE_COMPILATION_ERROR) • 4: Crash in execution (PROBLEM_TYPE_EXECUTION_ERROR) These numeric values, and their associated constant values shown in parentheses, correspond to the numeric and constant values for problem type constants in the DBMS_SQLDIAG package. See *Oracle Database PL/SQL Packages and Types Reference* for more information.
CREATION_DATE	TIMESTAMP(6)	Creation time for the incident associated with the test case
STATUS	VARCHAR2(10)	Current status for the incident associated with the test case. Possible values: • COMPLETE: The test case export completed successfully • INCOMPLETE: The test case export failed due to an error
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This view is available starting with Oracle Database 19c.

> **See Also**
>
> "V$DIAG_INCIDENT"

9.68 V$SQL_WORKAREA

V$SQL_WORKAREA displays information about work areas used by SQL cursors. Each SQL statement stored in the shared pool has one or more child cursors that are listed in the V$SQL view. V$SQL_WORKAREA lists all work areas needed by these child cursors; V$SQL_WORKAREA can be joined with V$SQLAREA on (ADDRESS, HASH_VALUE) and with V$SQL on (ADDRESS, HASH_VALUE, CHILD_NUMBER).
You can use this view to find out answers to the following questions:

- What are the top 10 work areas that require the most cache area?
- For work areas allocated in AUTO mode, what percentage of work areas are running using maximum memory?

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Address of the parent cursor handle
HASH_VALUE	NUMBER	Hash value of the parent statement in the library cache. Two columns PARENT_HANDLE and HASH_VALUE can be used to join with V$SQLAREA to locate the parent cursor.
SQL_ID	VARCHAR2(13)	SQL identifier of the parent statement in the library cache
CHILD_NUMBER	NUMBER	Number of the child cursor that uses this work area. The columns PARENT_HANDLE, HASH_VALUE, and CHILD_NUMBER can be used to join with V$SQL to locate the child cursor using this area.
WORKAREA_ADDRESS	RAW(4 | 8)	Address of the work area handle. This is the primary key for the view.
OPERATION_TYPE	VARCHAR2(160)	Type of operation using the work area. Can include values such as SORT, HASH JOIN, GROUP BY, BUFFER, BITMAP MERGE, and BITMAP CONSTRUCTION.
OPERATION_ID	NUMBER	A unique number used to identify the operation in the execution plan. This identifier can be joined to V$SQL_PLAN to locate the operation that uses this work area.
POLICY	VARCHAR2(40)	Sizing policy for this work area (MANUAL or AUTO)
ESTIMATED_OPTIMAL_SIZE	NUMBER	Estimated size (in bytes) required by this work area to execute the operation completely in memory (optimal execution). Derived from either optimizer statistics or previous executions.
ESTIMATED_ONEPASS_SIZE	NUMBER	Estimated size (in bytes) required by this work area to execute the operation in a single pass. Derived from either optimizer statistics or previous executions.
LAST_MEMORY_USED	NUMBER	Memory (in bytes) used by this work area during the last execution of the cursor

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LAST_EXECUTION	VARCHAR2(40)	Indicates whether this work area runs using OPTIMAL, ONE PASS, or ONE PASS memory requirement (or MULTI-PASS), during the last execution of the cursor
LAST_DEGREE	NUMBER	Degree of parallelism used during the last execution of this operation
TOTAL_EXECUTIONS	NUMBER	Number of times this work area was active
OPTIMAL_EXECUTIONS	NUMBER	Number of times this work area ran in optimal mode
ONEPASS_EXECUTIONS	NUMBER	Number of times this work area ran in one-pass mode
MULTIPASSES_EXECUTIONS	NUMBER	Number of times this work area ran below the one-pass memory requirement
ACTIVE_TIME	NUMBER	Average time this work area is active (in hundredths of a second)
MAX_TEMPSEG_SIZE	NUMBER	Maximum temporary segment size (in bytes) created by an instantiation of this work area. This column is NULL if this work area has never spilled to disk.
LAST_TEMPSEG_SIZE	NUMBER	Temporary segment size (in bytes) created in the last instantiation of this work area. This column is NULL if the last instantiation of this work area did not spill to disk.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SQLAREA"
- "V$SQL"

9.69 V$SQL_WORKAREA_ACTIVE

V$SQL_WORKAREA_ACTIVE contains an instantaneous view of the work areas currently allocated by the system. You can join this view against V$SQL_WORKAREA on WORKAREA_ADDRESS to access the definition of that work area. If a work area spills to disk, then this view contains information for the temporary segment created on behalf of this work area.
The last three columns are included to enable joining V$SQL_WORKAREA_ACTIVE with V$TEMPSEG_USAGE to retrieve more information on this temporary segment.

You can use this view to answer the following:

- What are the top 10 largest work areas currently allocated in the system?
- What percentage of memory is over-allocated (EXPECTED_SIZE < ACTUAL_MEM_USED) and under-allocated (EXPECTED_SIZE > ACTUAL_MEM_USED)?
- What are the active work areas using more memory than what is expected by the memory manager?

Copyright © 2002, 2026, Oracle and/or its affiliates.

- What are the active work areas that have spilled to disk?

Column	Datatype	Description
SQL_HASH_VALUE	NUMBER	Hash value of the SQL statement that is currently being executed
SQL_ID	VARCHAR2(13)	SQL identifier of the SQL statement that is currently being executed
SQL_EXEC_START	DATE	Time when the execution of the SQL currently executed by this session started
SQL_EXEC_ID	NUMBER	SQL execution identifier (see V$SQL_MONITOR)
WORKAREA_ADDRESS	RAW(4 | 8)	Address of the work area handle. This is the primary key for the view.
OPERATION_TYPE	VARCHAR2(160)	Type of operation using the work area. Can include values such as SORT, HASH JOIN, GROUP BY, BUFFER, BITMAP MERGE, and BITMAP CONSTRUCTION.
OPERATION_ID	NUMBER	A unique number used to identify the operation in the execution plan. This identifier can be joined to V$SQL_PLAN to locate the operation that uses this work area.
POLICY	VARCHAR2(24)	Sizing policy for this work area (MANUAL or AUTO)
SID	NUMBER	Session identifier
QCINST_ID	NUMBER	Query coordinator instance identifier. Along with QCSID, enables you to uniquely identify the query coordinator.
QCSID	NUMBER	Query coordinator session identifier. This is the same as the SID if the work area is allocated by a serial cursor.
ACTIVE_TIME	NUMBER	Average time this work area is active (in microseconds)
WORK_AREA_SIZE	NUMBER	Maximum size (in bytes) of the work area as it is currently used by the operation
EXPECTED_SIZE	NUMBER	Expected size (in bytes) for this work area. EXPECTED_SIZE is set on behalf of the operation by the memory manager. Memory can be over-allocated when WORK_AREA_SIZE has a higher value than EXPECTED_SIZE. This can occur when the operation using this work area takes a long time to resize it.
ACTUAL_MEM_USED	NUMBER	Amount of PGA memory (in bytes) currently allocated on behalf of this work area. This value should range between 0 and WORK_AREA_SIZE.
MAX_MEM_USED	NUMBER	Maximum memory amount (in bytes) used by this work area
NUMBER_PASSES	NUMBER	Number of passes corresponding to this work area (0 if running in OPTIMAL mode)
TEMPSEG_SIZE	NUMBER	Size (in bytes) of the temporary segment used on behalf of this work area. This column is NULL if this work area has not (yet) spilled to disk.
TABLESPACE	VARCHAR2(128)	Tablespace name for the temporary segment created on behalf of this work area. This column is NULL if this work area has not (yet) spilled to disk.
SEGRFNO#	NUMBER	Relative file number within the tablespace for the temporary segment created on behalf of this work area. This column is NULL if this work area has not (yet) spilled to disk.
SEGBLK#	NUMBER	Block number for the temporary segment created on behalf of this work area. This column is NULL if this work area has not (yet) spilled to disk.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$SQL_WORKAREA"
- *Oracle Database Concepts* for more information about SQL work areas

9.70 V$SQL_WORKAREA_HISTOGRAM

V$SQL_WORKAREA_HISTOGRAM displays the cumulative work area execution statistics (cumulated since instance startup) for different work area groups. The work areas are split into 33 groups based on their optimal memory requirements with the requirements increasing in powers of two. That is, work areas whose optimal requirement varies from 0 KB to 1 KB, 1 KB to 2 KB, 2 KB to 4 KB, ... and 2 TB to 4 TB.
For each work area group, the V$SQL_WORKAREA_HISTOGRAM view shows how many work areas in that group were able to run in optimal mode, how many were able to run in one-pass mode, and finally how many ran in multi-pass mode. The DBA can take a snapshot at the beginning and the end of a desired time interval to derive the same statistics for that interval.

Column	Datatype	Description
LOW_OPTIMAL_SIZE	NUMBER	Lower bound for the optimal memory requirement of work areas included in this row (bytes)
HIGH_OPTIMAL_SIZE	NUMBER	Upper bound for the optimal memory requirement of work areas included in this row (bytes)
OPTIMAL_EXECUTIONS	NUMBER	Number of work areas with an optimal memory requirement comprised between LOW_OPTIMAL_SIZE and HIGH_OPTIMAL_SIZE which have been executed in optimal mode since instance startup
ONEPASS_EXECUTIONS	NUMBER	Number of work areas with an optimal memory requirement comprised between LOW_OPTIMAL_SIZE and HIGH_OPTIMAL_SIZE which have been executed in one-pass mode since instance startup
MULTIPASSES_EXECUTIONS	NUMBER	Number of work areas with an optimal memory requirement comprised between LOW_OPTIMAL_SIZE and HIGH_OPTIMAL_SIZE which have been executed in multi-pass mode since instance startup
TOTAL_EXECUTIONS	NUMBER	Sum of OPTIMAL_EXECUTIONS, ONEPASS_EXECUTIONS, and MULTIPASSES_EXECUTIONS

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Performance Tuning Guide for detailed information on how to monitor automatic PGA memory performance using this view

9.71 V$SQLAREA

V$SQLAREA displays statistics on shared SQL areas and contains one row per SQL string. It provides statistics on SQL statements that are in memory, parsed, and ready for execution.

Column	Datatype	Description
SQL_TEXT	VARCHAR2(1000)	First thousand characters of the SQL text for the current cursor
SQL_FULLTEXT	CLOB	All characters of the SQL text for the current cursor
SQL_ID	VARCHAR2(13)	SQL identifier of the parent cursor in the library cache
SHARABLE_MEM	NUMBER	Amount of shared memory used by a cursor. If multiple child cursors exist, then the sum of all shared memory used by all child cursors.
PERSISTENT_MEM	NUMBER	Fixed amount of memory used for the lifetime of an open cursor. If multiple child cursors exist, then the fixed sum of memory used for the lifetime of all the child cursors.
RUNTIME_MEM	NUMBER	Fixed amount of memory required during execution of a cursor. If multiple child cursors exist, then the fixed sum of all memory required during execution of all the child cursors.
SORTS	NUMBER	Sum of the number of sorts that were done for all the child cursors
VERSION_COUNT	NUMBER	Number of child cursors that are present in the cache under this parent
LOADED_VERSIONS	NUMBER	Number of child cursors that are present in the cache and have their context heap loaded
OPEN_VERSIONS	NUMBER	Number of child cursors that are currently open under this current parent
USERS_OPENING	NUMBER	Number of users that have any of the child cursors open
FETCHES	NUMBER	Number of fetches associated with the SQL statement
EXECUTIONS	NUMBER	Total number of executions, totalled over all the child cursors
PX_SERVERS_EXECUTIONS	NUMBER	Total number of executions performed by parallel execution servers (0 when the statement has never been executed in parallel)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
END_OF_FETCH_COUNT	NUMBER	Number of times this cursor was fully executed since the cursor was brought into the library cache. The value of this statistic is not incremented when the cursor is partially executed, either because it failed during the execution or because only the first few rows produced by this cursor are fetched before the cursor is closed or re-executed. By definition, the value of the END_OF_FETCH_COUNT column should be less or equal to the value of the EXECUTIONS column.
USERS_EXECUTING	NUMBER	Total number of users executing the statement over all child cursors
LOADS	NUMBER	Number of times the object was loaded or reloaded
FIRST_LOAD_TIME	VARCHAR2(76)	Timestamp of the parent creation time
INVALIDATIONS	NUMBER	Total number of invalidations over all the child cursors
PARSE_CALLS	NUMBER	Sum of all parse calls to all the child cursors under this parent
DISK_READS	NUMBER	Sum of the number of disk reads over all child cursors
DIRECT_WRITES	NUMBER	Sum of the number of direct writes over all child cursors
DIRECT_READS	NUMBER	Sum of the number of direct reads over all child cursors
BUFFER_GETS	NUMBER	Sum of buffer gets over all child cursors
APPLICATION_WAIT_TIME	NUMBER	Application wait time (in microseconds)
CONCURRENCY_WAIT_TIME	NUMBER	Concurrency wait time (in microseconds)
CLUSTER_WAIT_TIME	NUMBER	Cluster wait time (in microseconds)
USER_IO_WAIT_TIME	NUMBER	User I/O Wait Time (in microseconds)
PLSQL_EXEC_TIME	NUMBER	PL/SQL execution time (in microseconds)
JAVA_EXEC_TIME	NUMBER	Java execution time (in microseconds)
ROWS_PROCESSED	NUMBER	Total number of rows processed on behalf of this SQL statement
COMMAND_TYPE	NUMBER	Oracle command type definition
OPTIMIZER_MODE	VARCHAR2(10)	Mode under which the SQL statement was executed
OPTIMIZER_COST	NUMBER	Cost of this query given by the optimizer
OPTIMIZER_ENV	RAW(2000)	Optimizer environment
OPTIMIZER_ENV_HASH_VALUE	NUMBER	Hash value for the optimizer environment
PARSING_USER_ID	NUMBER	User ID of the user that has parsed the very first cursor under this parent
PARSING_SCHEMA_ID	NUMBER	Schema ID that was used to parse this child cursor
PARSING_SCHEMA_NAME	VARCHAR2(128)	Schema name that was used to parse this child cursor
KEPT_VERSIONS	NUMBER	Number of child cursors that have been marked to be kept using the DBMS_SHARED_POOL package
ADDRESS	RAW(4 | 8)	Address of the handle to the parent for this cursor
HASH_VALUE	NUMBER	Hash value of the parent statement in the library cache
OLD_HASH_VALUE	NUMBER	Old SQL hash value
PLAN_HASH_VALUE	NUMBER	Numeric representation of the current SQL plan for this cursor. Comparing one PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line).

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FULL_PLAN_HASH_VALUE	NUMBER	Numeric representation of the complete SQL plan for this cursor. Comparing one FULL_PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line). Note that the FULL_PLAN_HASH_VALUE cannot be compared across databases releases. It is not backward compatible.
MODULE	VARCHAR2(64)	Contains the name of the module that was executing when the SQL statement was first parsed as set by calling DBMS_APPLICATION_INFO.SET_MODULE
MODULE_HASH	NUMBER	Hash value of the module that is named in the MODULE column
ACTION	VARCHAR2(64)	Contains the name of the action that was executing when the SQL statement was first parsed as set by calling DBMS_APPLICATION_INFO.SET_ACTION
ACTION_HASH	NUMBER	Hash value of the action that is named in the ACTION column
SERIALIZABLE_ABORTS	NUMBER	Number of times the transaction failed to serialize, producing ORA-08177 errors, totalled over all the child cursors
OUTLINE_CATEGORY	VARCHAR2(64)	If an outline was applied during construction of the cursor, then this column displays the category of that outline. Otherwise the column is left blank.
CPU_TIME	NUMBER	CPU time (in microseconds) used by this cursor for parsing, executing, and fetching
ELAPSED_TIME	NUMBER	Elapsed time (in microseconds) used by this cursor for parsing, executing, and fetching. If the cursor uses parallel execution, then ELAPSED_TIME is the cumulative time for the query coordinator, plus all parallel query worker processes.
OUTLINE_SID	VARCHAR2(40)	Outline session identifier
LAST_ACTIVE_CHILD_ADDRESS	RAW(4 | 8)	Address (identifier) of the child cursor that was the last to be active in the group (that is, the child cursor on behalf of which statistics in V$SQL were updated)
REMOTE	VARCHAR2(1)	Indicates whether the cursor is remote mapped (Y) or not (N)
OBJECT_STATUS	VARCHAR2(19)	Status of the cursor: • VALID - Valid, authorized without errors • VALID_AUTH_ERROR - Valid, authorized with authorization errors • VALID_COMPILE_ERROR - Valid, authorized with compilation errors • VALID_UNAUTH - Valid, unauthorized • INVALID_UNAUTH - Invalid, unauthorized • INVALID - Invalid, unauthorized but keep the timestamp
LITERAL_HASH_VALUE	NUMBER	Hash value of the literals which are replaced with system-generated bind variables and are to be matched, when CURSOR_SHARING is used. This is not the hash value for the SQL statement. If CURSOR_SHARING is not used, then the value is 0.
LAST_LOAD_TIME	DATE	Time at which the query plan was loaded into the library cache
IS_OBSOLETE	VARCHAR2(1)	Indicates whether the cursor has become obsolete (Y) or not (N). This can happen if the number of child cursors is too large.
IS_BIND_SENSITIVE	VARCHAR2(1)	Indicates whether the cursor is bind sensitive (Y) or not (N). A query is considered bind-sensitive if the optimizer peeked at one of its bind variable values when computing predicate selectivities and where a change in a bind variable value may cause the optimizer to generate a different plan.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
IS_BIND_AWARE	VARCHAR2(1)	Indicates whether the cursor is bind aware (Y) or not (N). A query is considered bind-aware if it has been marked to use extended cursor sharing. The query would already have been marked as bind-sensitive.
CHILD_LATCH	NUMBER	Child latch number that is protecting the cursor. This column is obsolete and maintained for backward compatibility.
SQL_PROFILE	VARCHAR2(64)	SQL profile used for this statement, if any
SQL_PATCH	VARCHAR2(128)	SQL patch used for this statement, if any
SQL_PLAN_BASELINE	VARCHAR2(128)	SQL plan baseline used for this statement, if any
PROGRAM_ID	NUMBER	Program identifier
PROGRAM_LINE#	NUMBER	Program line number
EXACT_MATCHING_SIGNATURE	NUMBER	Signature used when the CURSOR_SHARING parameter is set to EXACT
FORCE_MATCHING_SIGNATURE	NUMBER	Signature used when the CURSOR_SHARING parameter is set to FORCE
LAST_ACTIVE_TIME	DATE	Time at which the query plan was last active
BIND_DATA	RAW(2000)	Bind data
TYPECHECK_MEM	NUMBER	Typecheck memory
IO_CELL_OFFLOAD_ELIGIBLE_BYTES	NUMBER	Number of I/O bytes which can be filtered by the Exadata storage system **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_INTERCONNECT_BYTES	NUMBER	Number of I/O bytes exchanged between Oracle Database and the storage system
PHYSICAL_READ_REQUESTS	NUMBER	Number of physical read I/O requests issued by the monitored SQL
PHYSICAL_READ_BYTES	NUMBER	Number of bytes read from disks by the monitored SQL
PHYSICAL_WRITE_REQUESTS	NUMBER	Number of physical write I/O requests issued by the monitored SQL
PHYSICAL_WRITE_BYTES	NUMBER	Number of bytes written to disks by the monitored SQL
OPTIMIZED_PHY_READ_REQUESTS	NUMBER	Number of physical read I/O requests from Database Smart Flash Cache issued by the monitored SQL
LOCKED_TOTAL	NUMBER	Total number of times the child cursor has been locked
PINNED_TOTAL	NUMBER	Total number of times the child cursor has been pinned
IO_CELL_UNCOMPRESSED_BYTES	NUMBER	Number of uncompressed bytes (that is, size after decompression) that are offloaded to the Exadata cells **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_CELL_OFFLOAD_RETURNED_BYTES	NUMBER	Number of bytes that are returned by Exadata cell through the regular I/O path **See Also:** Oracle Exadata Storage Server Software documentation for more information
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
IS_REOPTIMIZABLE	VARCHAR2(1)	This column shows whether the next execution matching this child cursor will trigger a reoptimization. The values are: • Y: If the next execution will trigger a reoptimization • R: If the child cursor contains reoptimization information, but will not trigger reoptimization because the cursor was compiled in reporting mode • N: If the child cursor has no reoptimization information
IS_RESOLVED_ADAPTIVE_PLAN	VARCHAR2(1)	This column shows whether all of the adaptive parts of a plan have been resolved to the final plan. Once the plan is resolved, the plan hash value and the plan displayed by DBMS_XPLAN will not change through the end of execution. The values for this column are: • NULL: If the plan is not adaptive • Y: If the plan is fully resolved • N: If the plan is not yet fully resolved **See Also:** *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_XPLAN package

See Also

- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SHARED_POOL package
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_MODULE procedure
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_ACTION procedure
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_XPLAN package

9.72 V$SQLAREA_PLAN_HASH

V$SQLAREA_PLAN_HASH displays statistics on shared SQL areas (V$SQL) by grouping on the SQL_ID and PLAN_HASH_VALUE columns. It can potentially create several rows for one parent cursor, one for each distinct value of the column PLAN_HASH_VALUE.

Column	Datatype	Description
SQL_TEXT	VARCHAR2(1000)	First thousand characters of the SQL text for the current cursor
SQL_FULLTEXT	CLOB	All characters of the SQL text for the current cursor
ADDRESS	RAW(4 | 8)	Address of the handle to the parent for this cursor
HASH_VALUE	NUMBER	Hash value of the parent statement in the library cache
SQL_ID	VARCHAR2(13)	SQL identifier of the parent cursor in the library cache
PLAN_HASH_VALUE	NUMBER	Numeric representation of the current SQL plan for this cursor. Comparing one PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line).
VERSION_COUNT	NUMBER	Number of child cursors that are present in the cache under this parent

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
LAST_ACTIVE_CHILD_ADDRESS	RAW(4 | 8)	Address (identifier) of the child cursor that was the last to be active in the group (that is, the child cursor on behalf of which statistics in V$SQL were updated)
SHARABLE_MEM	NUMBER	Amount of shared memory used by a cursor. If multiple child cursors exist, then it is the sum of all shared memory used by all child cursors.
PERSISTENT_MEM	NUMBER	Fixed amount of memory used for the lifetime of an open cursor. If multiple child cursors exist, then it is the fixed sum of memory used for the lifetime of all the child cursors.
RUNTIME_MEM	NUMBER	Fixed amount of memory required during execution of a cursor. If multiple child cursors exist, then the fixed sum of all memory required during execution of all the child cursors.
SORTS	NUMBER	Sum of the number of sorts that were done for all the child cursors
LOADED_VERSIONS	NUMBER	Number of child cursors that are present in the cache and that have their context heap loaded
OPEN_VERSIONS	NUMBER	Number of child cursors that are currently open under this parent
USERS_OPENING	NUMBER	Number of users that have any of the child cursors open
USERS_EXECUTING	NUMBER	Total number of users executing the statement over all child cursors
FETCHES	NUMBER	Number of fetches associated with the SQL statement
EXECUTIONS	NUMBER	Total number of executions, totalled over all the child cursors
PX_SERVERS_EXECUTIONS	NUMBER	Total number of executions performed by parallel execution servers (0 when the statement has never been executed in parallel)
END_OF_FETCH_COUNT	NUMBER	Number of times this cursor was fully executed since the cursor was brought into the library cache. The value of this statistic is not incremented when the cursor is partially executed, either because it failed during the execution or because only the first few rows produced by this cursor are fetched before the cursor is closed or re-executed. By definition, the value of the END_OF_FETCH_COUNT column should be less or equal to the value of the EXECUTIONS column.
LOADS	NUMBER	Number of times the object was loaded or reloaded
FIRST_LOAD_TIME	DATE	Timestamp of the parent creation time
LAST_LOAD_TIME	DATE	Time at which the query plan was loaded into the library cache
LAST_ACTIVE_TIME	DATE	Time at which the query plan was last active
LAST_EXEC_START_TIME	DATE	Time at which the most recent execution of this SQL started
INVALIDATIONS	NUMBER	Total number of invalidations over all the child cursors
PARSE_CALLS	NUMBER	Sum of all parse calls to all the child cursors under this parent
DISK_READS	NUMBER	Sum of the number of disk reads over all child cursors
DIRECT_WRITES	NUMBER	Sum of the number of direct writes over all child cursors
BUFFER_GETS	NUMBER	Sum of buffer gets over all child cursors
CPU_TIME	NUMBER	CPU time (in microseconds) used by this cursor for parsing, executing, and fetching
ELAPSED_TIME	NUMBER	Elapsed time (in microseconds) used by this cursor for parsing, executing, and fetching
APPLICATION_WAIT_TIME	NUMBER	Application wait time (in microseconds)
CONCURRENCY_WAIT_TIME	NUMBER	Concurrency wait time (in microseconds)
CLUSTER_WAIT_TIME	NUMBER	Cluster wait time (in microseconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
USER_IO_WAIT_TIME	NUMBER	User I/O wait time (in microseconds)
PLSQL_EXEC_TIME	NUMBER	PL/SQL execution time (in microseconds)
JAVA_EXEC_TIME	NUMBER	Java execution time (in microseconds)
ROWS_PROCESSED	NUMBER	Total number of rows processed on behalf of this SQL statement
COMMAND_TYPE	NUMBER	Oracle command type definition
OPTIMIZER_MODE	VARCHAR2(10)	Mode under which the SQL statement was executed
OPTIMIZER_COST	NUMBER	Cost of this query given by the optimizer
OPTIMIZER_ENV	RAW(2000)	Optimizer environment
OPTIMIZER_ENV_HASH_VALUE	NUMBER	Hash value for the optimizer environment
PARSING_USER_ID	NUMBER	User ID of the user that has parsed the very first cursor under this parent
PARSING_SCHEMA_ID	NUMBER	Schema ID that was used to parse this child cursor
PARSING_SCHEMA_NAME	VARCHAR2(128)	Schema name that was used to parse this child cursor
KEPT_VERSIONS	NUMBER	Number of child cursors that have been marked to be kept using the DBMS_SHARED_POOL package
MODULE	VARCHAR2(64)	Contains the name of the module that was executing when the SQL statement was first parsed as set by calling DBMS_APPLICATION_INFO.SET_MODULE
MODULE_HASH	NUMBER	Hash value of the module that is named in the MODULE column
ACTION	VARCHAR2(64)	Contains the name of the action that was executing when the SQL statement was first parsed as set by calling DBMS_APPLICATION_INFO.SET_ACTION
ACTION_HASH	NUMBER	Hash value of the action that is named in the ACTION column
SERIALIZABLE_ABORTS	NUMBER	Number of times the transaction failed to serialize, producing ORA-08177 errors, totalled over all the child cursors
OUTLINE_CATEGORY	VARCHAR2(64)	If an outline was applied during construction of the cursor, then this column displays the category of that outline. Otherwise, the column is left blank.
OUTLINE_SID	VARCHAR2(40)	Outline session identifier
REMOTE	VARCHAR2(1)	Indicates whether the cursor is remote mapped (Y) or not (N)
OBJECT_STATUS	VARCHAR2(19)	Status of the cursor: • VALID - Valid, authorized without errors • VALID_AUTH_ERROR - Valid, authorized with authorization errors • VALID_COMPILE_ERROR - Valid, authorized with compilation errors • VALID_UNAUTH - Valid, unauthorized • INVALID_UNAUTH - Invalid, unauthorized • INVALID - Invalid, unauthorized but keep the timestamp
LITERAL_HASH_VALUE	NUMBER	Hash value of the literals which are replaced with system-generated bind variables and are to be matched, when CURSOR_SHARING is used. This is not the hash value for the SQL statement. If CURSOR_SHARING is not used, then the value is 0.
SQL_PROFILE	VARCHAR2(64)	SQL profile used for this statement, if any
PROGRAM_ID	NUMBER	Program identifier
PROGRAM_LINE#	NUMBER	Program line number
EXACT_MATCHING_SIGNATURE	NUMBER	Signature used when the CURSOR_SHARING parameter is set to EXACT

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FORCE_MATCHING_SIGNATURE	NUMBER	Signature used when the CURSOR_SHARING parameter is set to FORCE
BIND_DATA	RAW(2000)	Bind data
TYPECHECK_MEM	NUMBER	Typecheck memory
IO_CELL_OFFLOAD_ELIGIBLE_BYTES	NUMBER	Number of I/O bytes which can be filtered by the Exadata storage system **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_INTERCONNECT_BYTES	NUMBER	Number of I/O bytes exchanged between Oracle Database and the storage system
PHYSICAL_READ_REQUESTS	NUMBER	Number of physical read I/O requests issued by the monitored SQL
PHYSICAL_READ_BYTES	NUMBER	Number of bytes read from disks by the monitored SQL
PHYSICAL_WRITE_REQUESTS	NUMBER	Number of physical write I/O requests issued by the monitored SQL
PHYSICAL_WRITE_BYTES	NUMBER	Number of bytes written to disks by the monitored SQL
OPTIMIZED_PHY_READ_REQUESTS	NUMBER	Number of physical read I/O requests from Database Smart Flash Cache issued by the monitored SQL
IO_CELL_UNCOMPRESSED_BYTES	NUMBER	Number of uncompressed bytes (that is, size after decompression) that are offloaded to the Exadata cells **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_CELL_OFFLOAD_RETURNED_BYTES	NUMBER	Number of bytes that are returned by Exadata cell through the regular I/O path **See Also:** Oracle Exadata Storage Server Software documentation for more information
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
CON_DBID	NUMBER	The database ID of the PDB

See Also

- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_SHARED_POOL package
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_MODULE procedure
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_APPLICATION_INFO.SET_ACTION procedure

9.73 V$SQLCOMMAND

V$SQLCOMMAND displays the mapping between SQL opcodes and names.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
COMMAND_TYPE	NUMBER	SQL command number
COMMAND_NAME	VARCHAR2(64)	SQL command name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.74 V$SQLFN_ARG_METADATA

V$SQLFN_ARG_METADATA contains metadata about function arguments. There is one row for each argument of every function found in V$SQLFN_METADATA. There are no rows for functions that do not have any arguments.

Column	Datatype	Description
FUNC_ID	NUMBER	Internal function identification number. This column can be used to join with the V$SQLFN_METADATA view.
ARGNUM	NUMBER	Argument number
DATATYPE	VARCHAR2(8)	Data type of the argument. The value is NULL if this argument is not used. Otherwise, it can take values of any Oracle data type, family data type, or EXPR data type.
DESCR	VARCHAR2(128)	This column is reserved for future use.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SQLFN_METADATA"

9.75 V$SQLFN_METADATA

V$SQLFN_METADATA contains metadata about operators and built-in functions. Note that this view does not contain information about arguments because the number of arguments will be different for various functions. Information about arguments is contained in V$SQLFN_ARG_METADATA, which can be joined with V$SQLFN_METADATA to get information about any function and its arguments.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FUNC_ID	NUMBER	Internal function identification number
NAME	VARCHAR2(128)	Name of the built-in function
MINARGS	NUMBER	Minimum number of arguments for the function
MAXARGS	NUMBER	Maximum number of arguments for the function
DATATYPE	VARCHAR2(8)	Return data type, which can take any Oracle data type values, data type family values, or ARG[n] data types
VERSION	VARCHAR2(12)	Minimum version of the database that has this function
ANALYTIC	VARCHAR2(3)	Indicates whether the function is an analytic function (YES) or not (NO)
AGGREGATE	VARCHAR2(3)	Indicates whether the function is an aggregate function (YES) or not (NO)
OFFLOADABLE	VARCHAR2(3)	Indicates whether execution of the function can be offloaded to the Oracle Exadata Storage Server (YES) or not (NO) **See Also:** Oracle Exadata Storage Server Software documentation for more information
DISP_TYPE	VARCHAR2(13)	Function display type: • NORMAL • ARITHMETIC • PARENTHESIS • REL-OP • CASELIKE • NOPARENTHESIS
USAGE	VARCHAR2(128)	A text explanation of how to use this function. The text is based on the syntax diagram for the function in the *Oracle Database SQL Language Reference*.
DESCR	VARCHAR2(4000)	Description of the function
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SQLFN_ARG_METADATA"

9.76 V$SQLSTATS

V$SQLSTATS displays basic performance statistics for SQL cursors and contains one row per SQL statement (that is, one row per unique value of SQL_ID). The column definitions for columns in V$SQLSTATS are identical to those in the V$SQL and V$SQLAREA views. However, the V$SQLSTATS view differs from V$SQL and V$SQLAREA in that it is faster, more scalable, and has a greater data retention (the statistics may still appear in this view, even after the cursor has been aged out of the shared pool). Note that V$SQLSTATS contains a subset of columns that appear in V$SQL and V$SQLAREA.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SQL_TEXT	VARCHAR2(1000)	First thousand characters of the SQL text for the current cursor
SQL_FULLTEXT	CLOB	Full text for the SQL statement exposed as a CLOB column. THe full text of a SQL statement can be retrieved using this column instead of joining with the V$SQLTEXT view.
SQL_ID	VARCHAR2(13)	SQL identifier of the parent cursor in the library cache
LAST_ACTIVE_TIME	DATE	Last time the statistics of a contributing cursor were updated
LAST_ACTIVE_CHILD_ADDRESS	RAW(4 | 8)	Address of the contributing cursor that last updated these statistics
PLAN_HASH_VALUE	NUMBER	Numeric representation of the current SQL plan for this cursor. Comparing one PLAN_HASH_VALUE to another easily identifies whether or not two plans are the same (rather than comparing the two plans line by line).
PARSE_CALLS	NUMBER	Number of parse calls for all cursors with this SQL text and plan
DISK_READS	NUMBER	Number of disk reads for all cursors with this SQL text and plan
DIRECT_WRITES	NUMBER	Number of direct writes for all cursors with this SQL text and plan
DIRECT_READS	NUMBER	Number of direct reads for all cursors with this SQL text and plan
BUFFER_GETS	NUMBER	Number of buffer gets for all cursors with this SQL text and plan
ROWS_PROCESSED	NUMBER	Total number of rows the parsed SQL statement returns
SERIALIZABLE_ABORTS	NUMBER	Number of times the transaction failed to serialize, producing ORA-08177 errors, per cursor
FETCHES	NUMBER	Number of fetches associated with the SQL statement
EXECUTIONS	NUMBER	Number of executions that took place on this object since it was brought into the library cache
END_OF_FETCH_COUNT	NUMBER	Number of times this cursor was fully executed since the cursor was brought into the library cache. The value of this statistic is not incremented when the cursor is partially executed, either because it failed during the execution or because only the first few rows produced by this cursor are fetched before the cursor is closed or re-executed. By definition, the value of the END_OF_FETCH_COUNT column should be less or equal to the value of the EXECUTIONS column.
LOADS	NUMBER	Number of times the object was either loaded or reloaded
VERSION_COUNT	NUMBER	number of cursors present in the cache with this SQL text and plan
INVALIDATIONS	NUMBER	Number of times this child cursor has been invalidated
PX_SERVERS_EXECUTIONS	NUMBER	Total number of executions performed by parallel execution servers (0 when the statement has never been executed in parallel)
CPU_TIME	NUMBER	CPU time (in microseconds) used by this cursor for parsing, executing, and fetching
ELAPSED_TIME	NUMBER	Elapsed time (in microseconds) used by this cursor for parsing, executing, and fetching. If the cursor uses parallel execution, then ELAPSED_TIME is the cumulative time for the query coordinator, plus all parallel query worker processes.
AVG_HARD_PARSE_TIME	NUMBER	Average hard parse time (in microseconds) used by this cursor
APPLICATION_WAIT_TIME	NUMBER	Application wait time (in microseconds)
CONCURRENCY_WAIT_TIME	NUMBER	Concurrency wait time (in microseconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CLUSTER_WAIT_TIME	NUMBER	Cluster wait time (in microseconds). This value is specific to Oracle RAC. It shows the total time spent waiting for all waits that are categorized under the cluster class of wait events. The value is this column is an accumulated wait time spent waiting for Oracle RAC cluster resources.
USER_IO_WAIT_TIME	NUMBER	User I/O wait time (in microseconds)
PLSQL_EXEC_TIME	NUMBER	PL/SQL execution time (in microseconds)
JAVA_EXEC_TIME	NUMBER	Java execution time (in microseconds)
SORTS	NUMBER	Number of sorts that were done for the child cursor
SHARABLE_MEM	NUMBER	Total shared memory (in bytes) currently occupied by all cursors with this SQL text and plan
TOTAL_SHARABLE_MEM	NUMBER	Total shared memory (in bytes) occupied by all cursors with this SQL text and plan if they were to be fully loaded in the shared pool (that is, cursor size)
TYPECHECK_MEM	NUMBER	Typecheck memory
IO_CELL_OFFLOAD_ELIGIBLE_BYTES	NUMBER	Number of I/O bytes which can be filtered by the Exadata storage system **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_INTERCONNECT_BYTES	NUMBER	Number of I/O bytes exchanged between Oracle Database and the storage system. Typically used for Cache Fusion or parallel queries.
PHYSICAL_READ_REQUESTS	NUMBER	Number of physical read I/O requests issued by the monitored SQL. The requests may not be disk reads.
PHYSICAL_READ_BYTES	NUMBER	Number of bytes read from disks by the monitored SQL
PHYSICAL_WRITE_REQUESTS	NUMBER	Number of physical write I/O requests issued by the monitored SQL
PHYSICAL_WRITE_BYTES	NUMBER	Number of bytes written to disks by the monitored SQL
EXACT_MATCHING_SIGNATURE	NUMBER	Signature used when the CURSOR_SHARING parameter is set to EXACT
FORCE_MATCHING_SIGNATURE	NUMBER	Signature used when the CURSOR_SHARING parameter is set to FORCE
IO_CELL_UNCOMPRESSED_BYTES	NUMBER	Number of uncompressed bytes (that is, size after decompression) that are offloaded to the Exadata cells **See Also:** Oracle Exadata Storage Server Software documentation for more information
IO_CELL_OFFLOAD_RETURNED_BYTES	NUMBER	Number of bytes that are returned by Exadata cell through the regular I/O path **See Also:** Oracle Exadata Storage Server Software documentation for more information
DELTA_PARSE_CALLS	NUMBER	Number of parse calls for the cursor since the last Automatic Workload Repository (AWR) snapshot **See Also:** *Oracle Database Concepts* for an introduction to AWR
DELTA_DISK_READS	NUMBER	Number of disk reads for the cursor since the last AWR snapshot
DELTA_DIRECT_WRITES	NUMBER	Number of direct writes for the cursor since the last AWR snapshot
DELTA_DIRECT_READS	NUMBER	Number of direct reads for the cursor since the last AWR snapshot
DELTA_BUFFER_GETS	NUMBER	Number of buffer gets for the cursor since the last AWR snapshot
DELTA_ROWS_PROCESSED	NUMBER	Number of rows returned by the cursor since the last AWR snapshot
DELTA_FETCH_COUNT	NUMBER	Number of fetches for the cursor since the last AWR snapshot
DELTA_EXECUTION_COUNT	NUMBER	Number of executions for the cursor since the last AWR snapshot

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DELTA_PX_SERVERS_EXECUTIONS	NUMBER	Number of executions performed by parallel execution servers since the last AWR snapshot
DELTA_END_OF_FETCH_COUNT	NUMBER	Number of times the cursor was fully executed since the last AWR snapshot
DELTA_CPU_TIME	NUMBER	CPU time (in microseconds) for the cursor since the last AWR snapshot
DELTA_ELAPSED_TIME	NUMBER	Database time (in microseconds) for the cursor since the last AWR snapshot
DELTA_APPLICATION_WAIT_TIME	NUMBER	Time spent by the cursor (in microseconds) in the Application wait class since the last AWR snapshot
DELTA_CONCURRENCY_TIME	NUMBER	Time spent by the cursor (in microseconds) in the Concurrency wait class since the last AWR snapshot
DELTA_CLUSTER_WAIT_TIME	NUMBER	Time spent by the cursor (in microseconds) in the Cluster wait class since the last AWR snapshot
DELTA_USER_IO_WAIT_TIME	NUMBER	Time spent by the cursor (in microseconds) in the User I/O wait class since the last AWR snapshot
DELTA_PLSQL_EXEC_TIME	NUMBER	Time spent by the cursor (in microseconds) executing PL/SQL since the last AWR snapshot
DELTA_JAVA_EXEC_TIME	NUMBER	Time spent by the cursor (in microseconds) executing Java since the last AWR snapshot
DELTA_SORTS	NUMBER	Number of sorts for the cursor since the last AWR snapshot
DELTA_LOADS	NUMBER	Number of times the cursor was loaded since the last AWR snapshot
DELTA_INVALIDATIONS	NUMBER	Number of times the cursor was invalidated since the last AWR snapshot
DELTA_PHYSICAL_READ_REQUESTS	NUMBER	Number of physical read I/O requests for the cursor since the last AWR snapshot
DELTA_PHYSICAL_READ_BYTES	NUMBER	Number of bytes read from disk for the cursor since the last AWR snapshot
DELTA_PHYSICAL_WRITE_REQUESTS	NUMBER	Number of physical write I/O requests for the cursor since the last AWR snapshot
DELTA_PHYSICAL_WRITE_BYTES	NUMBER	Number of bytes written to disk for the cursor since the last AWR snapshot
DELTA_IO_INTERCONNECT_BYTES	NUMBER	Number of I/O bytes exchanged between the Oracle database and the storage system for the cursor since the last AWR snapshot
DELTA_CELL_OFFLOAD_ELIG_BYTES	NUMBER	Number of I/O bytes which can be filtered by the Exadata storage system for the cursor since the last AWR snapshot **See Also:** Oracle Exadata Storage Server Software documentation for more information
DELTA_CELL_UNCOMPRESSED_BYTES	NUMBER	Number of uncompressed bytes that are offloaded to the Exadata cell for the cursor since the last AWR snapshot **See Also:** Oracle Exadata Storage Server Software documentation for more information

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
CON_DBID	NUMBER	The database ID of the PDB
OBSOLETE_COUNT	NUMBER	Number of times that a parent cursor became obsolete
AVOIDED_EXECUTIONS[1]	NUMBER	Number of executions attempted on this object, but prevented due to the SQL statement being in quarantine
DELTA_AVOIDED_EXECUTIONS[1]	NUMBER	Number of executions attempted on this object, but prevented due to the SQL statement being in quarantine, since the last AWR snapshot

[1] This column is available starting with Oracle Database 19c.

See Also

- "V$SQL"
- "V$SQLAREA"

9.77 V$SQLSTATS_PLAN_HASH

V$SQLSTATS_PLAN_HASH displays basic performance statistics for SQL cursors and contains one row per execution plan of a SQL statement (that is, one row per unique combination of SQL_ID and PLAN_HASH_VALUE).
The columns for V$SQLSTATS_PLAN_HASH are the same as those for V$SQLSTATS.

See Also

"V$SQLSTATS"

9.78 V$SQLTEXT

V$SQLTEXT displays the text of SQL statements belonging to shared SQL cursors in the SGA.

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Used with HASH_VALUE to uniquely identify a cached cursor
HASH_VALUE	NUMBER	Used with ADDRESS to uniquely identify a cached cursor
SQL_ID	VARCHAR2(13)	SQL identifier of a cached cursor
COMMAND_TYPE	NUMBER	Code for the type of SQL statement (SELECT, INSERT, and so on)
PIECE	NUMBER	Number used to order the pieces of SQL text

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SQL_TEXT	VARCHAR2(64)	A column containing one piece of the SQL text
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.79 V$SQLTEXT_WITH_NEWLINES

V$SQLTEXT_WITH_NEWLINES is identical to the V$SQLTEXT view except that, to improve legibility, V$SQLTEXT_WITH_NEWLINES does not replace newlines and tabs in the SQL statement with spaces.

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Used with HASH_VALUE to uniquely identify a cached cursor
HASH_VALUE	NUMBER	Used with ADDRESS to uniquely identify a cached cursor
SQL_ID	VARCHAR2(13)	SQL identifier of a cached cursor
COMMAND_TYPE	NUMBER	Code for the type of SQL statement (SELECT, INSERT, and so on)
PIECE	NUMBER	Number used to order the pieces of SQL text
SQL_TEXT	VARCHAR2(64)	A column containing one piece of the SQL text
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SQLTEXT"

9.80 V$STANDBY_EVENT_HISTOGRAM

V$STANDBY_EVENT_HISTOGRAM displays the histogram of apply lag on the physical standby. Each distinct value of apply lag has its own bucket and the count in the corresponding bucket represents the number of occurrences so far. The physical standby samples the apply lag every second and increments the corresponding bucket in the histogram.

Column	Datatype	Description
NAME	VARCHAR2(64)	Name of the event (currently APPLY LAG is the only valid value)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TIME	NUMBER	Time duration that the bucket represents
UNIT	VARCHAR2(16)	Time unit (seconds, minutes, hours, or days)
COUNT	NUMBER	Each row is a histogram bucket for apply lag. COUNT is the number of occurrences the apply lag falls into the histogram bucket.
LAST_TIME_UPDATED	VARCHAR2(20)	Last time the bucket was updated by an event falling into that time duration
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.81 V$STANDBY_LOG

V$STANDBY_LOG displays information about standby redo logs. Standby redo logs are similar to online redo logs, but standby redo logs are only used on a standby database that is receiving redo data from the primary database.

Column	Datatype	Description
GROUP#	NUMBER	Log group number
DBID	VARCHAR2(40)	Database ID of the primary database to which the standby redo logfile is assigned. If the standby redo logfile is unassigned, the value UNASSIGNED will be displayed.
THREAD#	NUMBER	Log thread number
SEQUENCE#	NUMBER	Log sequence number
BYTES	NUMBER	Size of the log (in bytes)
BLOCKSIZE	NUMBER	Block size of the logfile (512 or 4096)
USED	NUMBER	Number of bytes used in the log
ARCHIVED	VARCHAR2(3)	Archive status (YES) or (NO). See the STATUS column for further details.
STATUS	VARCHAR2(10)	Log status: • UNASSIGNED - If ARCHIVED is NO, then the standby redo log has been archived and is again available. If ARCHIVED is YES, then the standby redo log has never been used and is available. • ACTIVE - If ARCHIVED is NO, then the standby redo log is complete and waiting to be archived. If ARCHIVED is YES, then the standby redo log is currently being written to and is therefore not ready to be archived. For a given thread, there should be only one such log.
FIRST_CHANGE#	NUMBER	Lowest SCN in the log
FIRST_TIME	DATE	Time of the first SCN in the log
NEXT_CHANGE#	NUMBER	All redo records contained within this log will have an SCN lower than NEXT_CHANGE#. Only filled in once the log is complete. Also the lowest SCN of any redo record in the next log.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NEXT_TIME	DATE	All redo records contained within this log will have a timestamp lower than NEXT_TIME. Only filled in once the log is complete. Also the lowest timestamp of any redo record in the next log.
LAST_CHANGE#	NUMBER	Redo in the standby redo log that recovery can possibly apply from the log has an SCN lower than LAST_CHANGE#
LAST_TIME	DATE	Timestamp of LAST_CHANGE#
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.82 V$STATISTICS_LEVEL

V$STATISTICS_LEVEL displays the status of the statistics/advisories controlled by STATISTICS_LEVEL.

Column	Datatype	Description
STATISTICS_NAME	VARCHAR2(64)	Name of the statistic/advisory
DESCRIPTION	VARCHAR2(4000)	Description of the statistic/advisory
SESSION_STATUS	VARCHAR2(8)	Status of the statistic/advisory for the session: • ENABLED • DISABLED
SYSTEM_STATUS	VARCHAR2(8)	System-wide status of the statistic/advisory: • ENABLED • DISABLED
ACTIVATION_LEVEL	VARCHAR2(7)	Indicates the level of STATISTICS_LEVEL that enables the statistic/advisory: • BASIC • TYPICAL • ALL
STATISTICS_VIEW_NAME	VARCHAR2(64)	If there is a single view externalizing the statistic/advisory, then this column contains the name of that view. If there is no such view, then this column is null. If there are multiple views involved, then the DESCRIPTION column mentions the view names.
SESSION_SETTABLE	VARCHAR2(3)	Indicates whether the statistic/advisory can be set at the session level (YES) or not (NO)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

"STATISTICS_LEVEL"

9.83 V$STATNAME

V$STATNAME displays decoded statistic names for the statistics shown in the V$SESSTAT and V$SYSSTAT tables.

On some platforms, the NAME and CLASS columns contain additional operating system-specific statistics.

Column	Datatype	Description
STATISTIC#	NUMBER	Statistic number **Note:** Statistics numbers are not guaranteed to remain constant from one release to another. Therefore, you should rely on the statistics name rather than its number in your applications.
NAME	VARCHAR2(64)	Statistic name. Names that appear in this column remain stable across Oracle Database releases, and they can be relied on by customer scripts.
CLASS	NUMBER	A number representing one or more statistics classes. The following class numbers are additive: • 1 - User • 2 - Redo • 4 - Enqueue • 8 - Cache • 16 - OS • 32 - Real Application Clusters • 64 - SQL • 128 - Debug
STAT_ID	NUMBER	Identifier of the statistic
DISPLAY_NAME	VARCHAR2(64)	A clearer and more descriptive name for the statistic that appears in the NAME column. Names that appear in the DISPLAY_NAME column can change across Oracle Database releases, therefore customer scripts should not rely on names that appear in the DISPLAY_NAME column across releases.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "V$SESSTAT" and "V$SYSSTAT"
- " Statistics Descriptions" for statistic descriptions
- Your operating system-specific Oracle documentation

9.84 V$STATS_ADVISOR_RULES

V$STATS_ADVISOR_RULES displays the rule definition information for each Optimizer Statistics Advisor rule.

Column	Datatype	Description
RULE_ID	NUMBER	ID of the rule
NAME	VARCHAR2(64)	Name of the rule
RULE_TYPE	VARCHAR2(9)	Type of the rule: • OBJECT • OPERATION • SYSTEM
DESCRIPTION	VARCHAR2(64)	Description of the rule
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database SQL Tuning Guide for more information about Optimizer Statistics Advisor.

9.85 V$STREAMS_APPLY_COORDINATOR

V$STREAMS_APPLY_COORDINATOR displays information about each apply process coordinator. The coordinator for an apply process gets transactions from the apply process reader and passes them to apply servers. An apply process coordinator is a subcomponent of an apply process, outbound server, or inbound server.

Column	Datatype	Description
SID	NUMBER	Session ID of the coordinator's session
SERIAL#	NUMBER	Serial number of the coordinator's session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATE	VARCHAR2(21)	State of the coordinator: • INITIALIZING - Starting up • IDLE - Performing no work • APPLYING - Passing transactions to apply servers • SHUTTING DOWN CLEANLY - Stopping without an error • ABORTING - Stopping because of an apply error
APPLY#	NUMBER	Apply process number. An apply process coordinator is an Oracle background process, prefixed by ap.
APPLY_NAME	VARCHAR2(128)	Name of the apply process
TOTAL_APPLIED	NUMBER	Total number of transactions applied by the apply process since the apply process was last started
TOTAL_WAIT_DEPS	NUMBER	Number of times since the apply process was last started that an apply server waited to apply a logical change record (LCR) in a transaction until another apply server applied a transaction because of a dependency between the transactions
TOTAL_WAIT_COMMITS	NUMBER	Number of times since the apply process was last started that an apply server waited to commit a transaction until another apply server committed a transaction to serialize commits
TOTAL_ADMIN	NUMBER	Number of administrative jobs issued since the apply process was last started
TOTAL_ASSIGNED	NUMBER	Number of transactions assigned to apply servers since the apply process was last started
TOTAL_RECEIVED	NUMBER	Total number of transactions received by the coordinator process since the apply process was last started
TOTAL_IGNORED	NUMBER	Number of transactions which were received by the coordinator but were ignored because they had been previously applied
TOTAL_ROLLBACKS	NUMBER	Number of transactions which were rolled back due to unexpected contention
TOTAL_ERRORS	NUMBER	Number of transactions applied by the apply process that resulted in an apply error since the apply process was last started
UNASSIGNED_COMPLETE_TXNS	NUMBER	Total number of complete transactions that the coordinator has not assigned to any apply servers
AUTO_TXN_BUFFER_SIZE	NUMBER	Current value of transaction buffer size. Transaction buffer size refers to the number of transactions that the apply reader can assemble ahead of apply servers. The apply process periodically adjusts the transaction buffer size.
LWM_TIME	DATE	Time when the message with the lowest message number was recorded. The creation time of the message with the lowest message number was also recorded at this time.
LWM_MESSAGE_NUMBER	NUMBER	Number of the message corresponding to the low watermark. That is, messages with a commit message number less than or equal to this message number have definitely been applied, but some messages with a higher commit message number also may have been applied.
LWM_MESSAGE_CREATE_TIME	DATE	For captured messages, creation time at the source database of the message corresponding to the low watermark. For user-enqueued messages, time when the message corresponding to the low watermark was enqueued into the queue at the local database.
HWM_TIME	DATE	Time when the message with the highest message number was recorded. The creation time of the message with the highest message number was also recorded at this time.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
HWM_MESSAGE_NUMBER	NUMBER	Number of the message corresponding to the high watermark. That is, no messages with a commit message number greater than this message number have been applied.
HWM_MESSAGE_CREATE_TIME	DATE	For captured messages, creation time at the source database of the message corresponding to the high watermark. For user-enqueued messages, time when the message corresponding to the high watermark was enqueued into the queue at the local database.
STARTUP_TIME	DATE	Time when the apply process was last started
ELAPSED_SCHEDULE_TIME	NUMBER	Time elapsed (in hundredths of a second) scheduling messages since the apply process was last started
ELAPSED_IDLE_TIME	NUMBER	Elapsed idle time
LWM_POSITION	RAW(64)	Position of the low-watermark LCR
HWM_POSITION	RAW(64)	Position of the high-watermark LCR
PROCESSED_MESSAGE_NUMBER	NUMBER	Message number currently processed by the apply coordinator
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
ACTIVE_SERVER_COUNT	NUMBER	Active server count

Note

The ELAPSED_SCHEDULE_TIME column is only populated if the TIMED_STATISTICS initialization parameter is set to true, or if the STATISTICS_LEVEL initialization parameter is set to TYPICAL or ALL.

See Also

- "TIMED_STATISTICS"
- "STATISTICS_LEVEL"

9.86 V$STREAMS_APPLY_READER

V$STREAMS_APPLY_READER displays information about each apply reader. The apply reader is a process which reads (dequeues) messages from the queue, computes message dependencies, and builds transactions. It passes the transactions on to the coordinator in commit order for assignment to the apply servers. An apply reader is a subcomponent of an apply process, outbound server, or inbound server.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SID	NUMBER	Session ID of the reader's session
SERIAL#	NUMBER	Serial number of the reader's session
APPLY#	NUMBER	Apply process number. An apply process is an Oracle background process, prefixed by ap.
APPLY_NAME	VARCHAR2(128)	Name of the apply process
STATE	VARCHAR2(36)	State of the reader: • INITIALIZING - Starting up • IDLE - Performing no work • DEQUEUE MESSAGES - Dequeuing messages from the queue • SCHEDULE MESSAGES - Computing dependencies between messages and assembling messages into transactions • SPILLING - Spilling unapplied messages from memory to hard disk • PAUSED - WAITING FOR DDL TO COMPLETE - Waiting for a data definition language (DDL) logical change record (LCR) to be applied
TOTAL_MESSAGES_DEQUEUED	NUMBER	Total number of messages dequeued since the apply process was last started
TOTAL_MESSAGES_SPILLED	NUMBER	Number of messages spilled by the reader since the apply process was last started
DEQUEUE_TIME	DATE	Time when the last message was received
DEQUEUED_MESSAGE_NUMBER	NUMBER	Number of the last message received
DEQUEUED_MESSAGE_CREATE_TIME	DATE	For captured messages, creation time at the source database of the last message received. For user-enqueued messages, time when the message was enqueued into the queue at the local database.
SGA_USED	NUMBER	Amount (in bytes) of SGA memory used by the apply process since it was last started
ELAPSED_DEQUEUE_TIME	NUMBER	Time elapsed (in hundredths of a second) dequeuing messages since the apply process was last started
ELAPSED_SCHEDULE_TIME	NUMBER	Time elapsed (in hundredths of a second) scheduling messages since the apply process was last started. Scheduling includes computing dependencies between messages and assembling messages into transactions.
ELAPSED_SPILL_TIME	NUMBER	Elapsed time (in hundredths of a second) spent spilling messages since the apply process was last started
LAST_BROWSE_NUM	NUMBER	Reserved for internal use
OLDEST_SCN_NUM	NUMBER	Oldest SCN
LAST_BROWSE_SEQ	NUMBER	Reserved for internal use
LAST_DEQ_SEQ	NUMBER	Last dequeue sequence number
OLDEST_XIDUSN	NUMBER	Transaction ID undo segment number of the oldest transaction that either has been applied or is being applied
OLDEST_XIDSLT	NUMBER	Transaction ID slot number of the oldest transaction that either has been applied or is being applied
OLDEST_XIDSQN	NUMBER	Transaction ID sequence number of the oldest transaction that either has been applied or is being applied
SPILL_LWM_SCN	NUMBER	Spill low-watermark SCN

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
PROXY_SID	NUMBER	When the apply process uses combined capture and apply, the session ID of the propagation receiver that is responsible for direct communication between capture and apply. If the apply process does not use combined capture and apply, then this column is 0.
PROXY_SERIAL	NUMBER	When the apply process uses combined capture and apply, the serial number of the propagation receiver that is responsible for direct communication between capture and apply. If the apply process does not use combined capture and apply, then this column is 0.
PROXY_SPID	VARCHAR2(12)	When the apply process uses combined capture and apply, the process identification number of the propagation receiver that is responsible for direct communication between capture and apply. If the apply process does not use combined capture and apply, then this column is 0.
CAPTURE_BYTES_RECEIVED	NUMBER	When the apply process uses combined capture and apply, the number of bytes received by the apply process from the capture process since the apply process last started. If the apply process does not use combined capture and apply, then this column is not populated.
DEQUEUED_POSITION	RAW(64)	Dequeued position. This column is populated only for an apply process that is functioning as an XStream inbound server.
LAST_BROWSE_POSITION	RAW(64)	Reserved for internal use
OLDEST_POSITION	RAW(64)	The earliest position of the transactions currently being dequeued and applied. This column is populated only for an apply process that is functioning as an XStream inbound server.
SPILL_LWM_POSITION	RAW(64)	Spill low-watermark position. This column is populated only for an apply process that is functioning as an XStream inbound server.
OLDEST_TRANSACTION_ID	VARCHAR2(128)	Oldest transaction ID
TOTAL_LCRS_WITH_DEP	NUMBER	Total number of LCRs with row-level dependencies since the apply process last started
TOTAL_LCRS_WITH_WMDEP	NUMBER	Total number of LCRs with watermark dependencies since the apply process last started. A watermark dependency occurs when an apply process must wait until the apply process's low watermark reaches a particular threshold.
TOTAL_IN_MEMORY_LCRS	NUMBER	Total number of LCRs currently in memory
SGA_ALLOCATED	NUMBER	The total amount of shared memory (in bytes) allocated from the Streams pool for the apply process since the apply process last started
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Note

The ELAPSED_DEQUEUE_TIME and ELAPSED_SCHEDULE_TIME columns are only populated if the TIMED_STATISTICS initialization parameter is set to true, or if the STATISTICS_LEVEL initialization parameter is set to TYPICAL or ALL.

Copyright © 2002, 2026, Oracle and/or its affiliates.

See Also

- "TIMED_STATISTICS"
- "STATISTICS_LEVEL"

9.87 V$STREAMS_APPLY_SERVER

V$STREAMS_APPLY_SERVER displays information about each apply server and its activities. An apply server receives messages from the apply coordinator for an apply process. For each message received, an apply server either applies the message or sends the message to the appropriate apply handler. An apply server is a subcomponent of an apply process, outbound server, or inbound server.

Column	Datatype	Description
SID	NUMBER	Session ID of the apply server's session
SERIAL#	NUMBER	Serial number of the apply server's session
APPLY#	NUMBER	Apply process number. An apply process is an Oracle background process, prefixed by ap.
APPLY_NAME	VARCHAR2(128)	Name of the apply process
SERVER_ID	NUMBER	Parallel execution server number of the apply server

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATE	VARCHAR2(20)	State of the apply server: • INITIALIZING - Starting up • IDLE - Performing no work • RECORD LOW-WATERMARK - Performing an administrative job that maintains information about the apply progress, which is used in the ALL_APPLY_PROGRESS and DBA_APPLY_PROGRESS data dictionary views • ADD PARTITION - Performing an administrative job that adds a partition that is used for recording information about in-progress transactions • DROP PARTITION - Performing an administrative job that purges rows that were used to record information about in-progress transactions • EXECUTE TRANSACTION - Applying a transaction • WAIT COMMIT - Waiting to commit a transaction until all other transactions with a lower commit SCN are applied. This state is possible only if the COMMIT_SERIALIZATION apply process parameter is set to a value other than DEPENDENT_TRANSACTIONS and the PARALLELISM apply process parameter is set to a value greater than 1. • WAIT DEPENDENCY - Waiting to apply a logical change record (LCR) in a transaction until another transaction, on which it has a dependency, is applied. This state is possible only if the PARALLELISM apply process parameter is set to a value greater than 1. • ROLLBACK TRANSACTION - Rolling back a transaction • TRANSACTION CLEANUP - Cleaning up an applied transaction, which includes removing LCRs from the apply process's queue • WAIT FOR CLIENT - Waiting for an XStream client application to request more LCRs • WAIT FOR NEXT CHUNK - Waiting for the next set of LCRs for a large transaction
XIDUSN	NUMBER	Transaction ID undo segment number of the transaction currently being applied
XIDSLT	NUMBER	Transaction ID slot number of the transaction currently being applied
XIDSQN	NUMBER	Transaction ID sequence number of the transaction currently being applied
COMMITSCN	NUMBER	Commit system change number (SCN) of the transaction currently being applied
DEP_XIDUSN	NUMBER	Transaction ID undo segment number of a transaction on which the transaction being applied by this apply server depends
DEP_XIDSLT	NUMBER	Transaction ID slot number of a transaction on which the transaction being applied by this apply server depends
DEP_XIDSQN	NUMBER	Transaction ID sequence number of a transaction on which the transaction being applied by this apply server depends
DEP_COMMITSCN	NUMBER	Commit system change number (SCN) of the transaction on which this apply server depends
MESSAGE_SEQUENCE	NUMBER	Number of the current message being applied by the apply server. This value is reset to 1 at the beginning of each transaction.
TOTAL_ASSIGNED	NUMBER	Total number of transactions assigned to the apply server since the apply process was last started

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOTAL_ADMIN	NUMBER	Total number of administrative jobs done by the apply server since the apply process was last started. See the STATE information in this view for the types of administrative jobs.
TOTAL_ROLLBACKS	NUMBER	Number of transactions assigned to this server which were rolled back
TOTAL_MESSAGES_APPLIED	NUMBER	Total number of messages applied by this apply server since the apply process was last started
APPLY_TIME	DATE	Time the last message was applied
APPLIED_MESSAGE_NUMBER	NUMBER	Number of the last message applied
APPLIED_MESSAGE_CREATE_TIME	DATE	Creation time at the source database of the last captured message applied. No information about user-enqueued messages is recorded in this column.
ELAPSED_DEQUEUE_TIME	NUMBER	Time elapsed (in hundredths of a second) dequeuing messages since the apply process was last started
ELAPSED_APPLY_TIME	NUMBER	Time elapsed (in hundredths of a second) applying messages since the apply process was last started
COMMIT_POSITION	RAW(64)	Commit position of the transaction. This column is populated only for an apply process that is functioning as an XStream outbound server or inbound server.
DEP_COMMIT_POSITION	RAW(64)	Commit position of the transaction the worker depends on. This column is populated only for an apply process that is functioning as an XStream inbound server.
LAST_APPLY_POSITION	RAW(64)	For inbound servers, the position of the last message applied; for outbound servers, the position of the last message sent to the XStream client application. This column is populated only for an apply process that is functioning as an XStream outbound server or inbound server.
TRANSACTION_ID	VARCHAR2(128)	Transaction ID that the worker is applying. This column is populated only for an apply process that is functioning as an XStream inbound server.
DEP_TRANSACTION_ID	VARCHAR2(128)	Transaction ID of the transaction the worker depends on. This column is populated only for an apply process that is functioning as an XStream inbound server.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
TOTAL_LCRS_RETRIED	NUMBER	Total number of LCRs retried by this server
LCR_RETRY_ITERATION	NUMBER	Retry iteration for this transaction by this server
TOTAL_TXNS_RETRIED	NUMBER	Total transactions retried by this server
TXN_RETRY_ITERATION	NUMBER	Total retry iterations by this server
TOTAL_TXNS_RECORDED	NUMBER	Total transactions recorded in error queue by this server

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The ELAPSED_DEQUEUE_TIME and ELAPSED_APPLY_TIME columns are only populated if the TIMED_STATISTICS initialization parameter is set to true, or if the STATISTICS_LEVEL initialization parameter is set to TYPICAL or ALL.

> **See Also**
>
> - "TIMED_STATISTICS"
> - "STATISTICS_LEVEL"

9.88 V$STREAMS_POOL_ADVICE

V$STREAMS_POOL_ADVICE displays information about the estimated count of spilled or unspilled messages and the associated time spent in the spill or unspill activity for different Streams pool sizes. The sizes range from 10% to 200% of the current Streams pool size, in equal intervals. The value of the interval depends on the current size of the Streams pool.

Column	Datatype	Description
STREAMS_POOL_SIZE_FOR_ESTIMATE	NUMBER	Streams pool size (in megabytes) for the estimate. The size ranges from values smaller than the current Streams pool size to values larger than the current Streams pool size, and there is a separate row for each increment. There is always an entry that shows the current Streams pool size, and there are always 20 increments. The range and the size of the increments depend on the current size of the Streams pool.
STREAMS_POOL_SIZE_FACTOR	NUMBER	Size factor with respect to the current Streams pool size
ESTD_SPILL_COUNT	NUMBER	Estimated count of messages spilled from the Streams pool
ESTD_SPILL_TIME	NUMBER	Estimated elapsed time (in seconds) to spill
ESTD_UNSPILL_COUNT	NUMBER	Estimated count of unspills (read back from disk)
ESTD_UNSPILL_TIME	NUMBER	Estimated elapsed time (in seconds) to unspill
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.89 V$STREAMS_POOL_STATISTICS

V$STREAMS_POOL_STATISTICS displays information about the current Streams pool usage percentage.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOTAL_MEMORY_ALLOCATED	NUMBER	Total memory allocated to the Streams pool (in bytes). It should always be less than the current size of the Streams pool. You can get the percentage of the Streams pool used by dividing TOTAL_MEMORY_ALLOCATED by CURRENT_SIZE.
CURRENT_SIZE	NUMBER	Current size of the Streams pool (in bytes)
SGA_TARGET_VALUE	NUMBER	Value of SGA_TARGET. Used to determine whether or not streams pool automatic tuning is enabled. This should be set even if MEMORY_TARGET is set and SGA_TARGET is not set.
SHRINK_PHASE	NUMBER	This only pertains to the Streams pool in an automatic tuning environment (SGA_TARGET and MEMORY_TARGET set). In this case, this shows whether or not the Streams pool is being asked to shrink. During the shrink phase, enqueues are blocked, flow control is enabled for all components, and cached memory is returned to the SGA.
ADVICE_DISABLED	NUMBER	This determines whether or not Streams pool advice in V$STREAMS_POOL_ADVICE as well as all statistics gathering related to auto-tuning the Streams pool have been disabled.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.90 V$SUBCACHE

V$SUBCACHE displays information about the subordinate caches currently loaded into library cache memory. The view walks through the library cache, printing out a row for each loaded subordinate cache per library cache object.

Column	Datatype	Description
OWNER_NAME	VARCHAR2(64)	Owner of the object containing these cache entries
NAME	VARCHAR2(1000)	Object Name
TYPE	NUMBER	Object Type
HEAP_NUM	NUMBER	Heap number containing this subordinate cache
CACHE_ID	NUMBER	Subordinate cache ID
CACHE_CNT	NUMBER	Number of entries for this cache in this object
HEAP_SZ	NUMBER	Amount of extent space allocated to this heap
HEAP_ALOC	NUMBER	Amount of extent space allocated from this heap
HEAP_USED	NUMBER	Amount of space utilized in this heap

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.91 V$SUBSCR_REGISTRATION_STATS

V$SUBSCR_REGISTRATION_STATS displays information for diagnosability of notifications.

Column	Datatype	Description
REG_ID	NUMBER	Registration identifier
NUM_NTFNS	NUMBER	Number of notifications
NUM_GROUPING_NTFNS	NUMBER	Number of grouping notifications
NUM_NTFNS_CURRENT_GROUP	NUMBER	Number of events received in the current group
LAST_NTFN_START_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when the last notification was started
LAST_NTFN_SENT_TIME	TIMESTAMP(3) WITH TIME ZONE	Time when the last notification was sent
TOTAL_EMON_LATENCY	NUMBER	Total EMON latency (time taken by the EMON worker to process notifications)
EMON#	NUMBER	Active EMON worker serving the registration
ALL_EMON_SERVERS	RAW(2000)	EMON workers that served the registration
TOTAL_PAYLOAD_BYTES_SENT	NUMBER	Total payload bytes sent
SHARD_ID[1]	NUMBER	Shard number for current registration used for Key Based Messaging
NUM_RETRIES	NUMBER	Number of retries in sending notifications
TOTAL_PLSQL_EXEC_TIME	NUMBER	Total PL/SQL callback execution time (relevant only for PL/SQL notifications)
LAST_ERR	VARCHAR2(90)	Last error message
LAST_ERR_TIME	TIMESTAMP(3) WITH TIME ZONE	Time of the last error
LAST_UPDATE_TIME	TIMESTAMP(3) WITH TIME ZONE	Time of the last update
NUM_PENDING_NTFNS	NUMBER	Number of notifications pending to be sent
TOTAL_PENDING_NTFN_BYTES	NUMBER	Total number of bytes for notifications pending to be sent

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

[1] This column is available starting with Oracle Database 19c.

9.92 V$SYS_OPTIMIZER_ENV

V$SYS_OPTIMIZER_ENV displays the contents of the optimizer environment for the instance. The optimizer environment stores the value of the main parameters used by the Oracle optimizer when building the execution plan of a SQL statement. Hence, modifying the value of one or more of these parameters (for example, by issuing an ALTER SYSTEM statement) could lead to plan changes.
The parameters displayed by this view are either regular initialization parameters (such as OPTIMIZER_FEATURES_ENABLE) or pseudo parameters (such as ACTIVE_INSTANCE_COUNT).

Column	Datatype	Description
ID	NUMBER	Unique identifier of the parameter in the optimizer environment
NAME	VARCHAR2(40)	Name of the parameter
SQL_FEATURE	VARCHAR2(64)	Associated feature control ID
ISDEFAULT	VARCHAR2(3)	Indicates whether the parameter is set to the default value (YES) or not (NO)
VALUE	VARCHAR2(25)	Value of the parameter
DEFAULT_VALUE	VARCHAR2(25)	Default value of the parameter
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "OPTIMIZER_FEATURES_ENABLE"
- "ACTIVE_INSTANCE_COUNT"

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.93 V$SYS_TIME_MODEL

V$SYS_TIME_MODEL displays the system-wide accumulated times for various operations. The time reported is the total elapsed or CPU time (in microseconds). Any timed operation will buffer at most 5 seconds of time data. Specifically, this means that if a timed operation (such as SQL execution) takes a long period of time to perform, the data published to this view is at most missing 5 seconds of the time accumulated for the operation.
The time values are 8-byte integers and can therefore hold approximately 580,000 years worth of time before wrapping. Background process time is not included in a statistic value unless the statistic is specifically for background processes.

Column	Datatype	Description
STAT_ID	NUMBER	Statistic identifier for the time statistic
STAT_NAME	VARCHAR2(64)	Name of the statistic (see Table 9-1)
VALUE	NUMBER	Amount of time (in microseconds) that the system has spent in this operation
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **Note**
>
> This view returns instance-wide data and a value of 0 in the CON_ID column when queried from the root of a CDB.

9.94 V$SYSAUX_OCCUPANTS

V$SYSAUX_OCCUPANTS displays SYSAUX tablespace occupant information.

Column	Datatype	Description
OCCUPANT_NAME	VARCHAR2(64)	Occupant name
OCCUPANT_DESC	VARCHAR2(64)	Occupant description
SCHEMA_NAME	VARCHAR2(64)	Schema name for the occupant
MOVE_PROCEDURE	VARCHAR2(64)	Name of the move procedure; null if not applicable
MOVE_PROCEDURE_DESC	VARCHAR2(64)	Description of the move procedure
SPACE_USAGE_KBYTES	NUMBER	Current space usage of the occupant (in KB)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.95 V$SYSMETRIC

V$SYSMETRIC displays the system metric values captured for the most current time interval for both the long duration (60-second) and short duration (15-second) system metrics.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	Metric group ID
METRIC_ID	NUMBER	Metric ID
METRIC_NAME	VARCHAR2(64)	Metric name
VALUE	NUMBER	Metric value
METRIC_UNIT	VARCHAR2(64)	Metric unit description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "DBA_HIST_SYS_TIME_MODEL"
- "V$CON_SYSMETRIC"
- "DBA_HIST_CON_SYS_TIME_MODEL"

9.96 V$SYSMETRIC_HISTORY

V$SYSMETRIC_HISTORY displays all system metric values available in the database. Both long duration (60-second with 1 hour history) and short duration (15-second with one-interval only) metrics are displayed by this view.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	Metric group ID
METRIC_ID	NUMBER	Metric ID
METRIC_NAME	VARCHAR2(64)	Metric name
VALUE	NUMBER	Metric value
METRIC_UNIT	VARCHAR2(64)	Metric unit description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "DBA_HIST_SYSMETRIC_HISTORY"
- "V$CON_SYSMETRIC_HISTORY"
- "DBA_HIST_CON_SYSMETRIC_HIST"

9.97 V$SYSMETRIC_SUMMARY

V$SYSMETRIC_SUMMARY displays a summary of all system metric values for the long-duration system metrics. The average, maximum value, minimum value, and the value of one standard deviation for the last hour are displayed for each metric item.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
GROUP_ID	NUMBER	Metric group ID
METRIC_ID	NUMBER	Metric ID
METRIC_NAME	VARCHAR2(64)	Metric name
NUM_INTERVAL	NUMBER	Number of intervals observed
MAXVAL	NUMBER	Maximum value observed
MINVAL	NUMBER	Minimum value observed
AVERAGE	NUMBER	Average value over the period
STANDARD_DEVIATION	NUMBER	One standard deviation
METRIC_UNIT	VARCHAR2(64)	Metric unit description

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "DBA_HIST_SYSMETRIC_SUMMARY"
- "V$CON_SYSMETRIC_SUMMARY"
- "DBA_HIST_CON_SYSMETRIC_SUMM"

9.98 V$SYSSTAT

V$SYSSTAT displays system statistics. To find the name of the statistic associated with each statistic number (STATISTIC#), query the V$STATNAME view.

Column	Datatype	Description
STATISTIC#	NUMBER	Statistic number **Note:** Statistics numbers are not guaranteed to remain constant from one release to another. Therefore, you should rely on the statistics name rather than its number in your applications.
NAME	VARCHAR2(64)	Statistic name. You can get a complete listing of statistic names by querying the V$STATNAME view.
CLASS	NUMBER	A number representing one or more statistics class. The following class numbers are additive: • 1 - User • 2 - Redo • 4 - Enqueue • 8 - Cache • 16 - OS • 32 - Real Application Clusters • 64 - SQL • 128 - Debug
VALUE	NUMBER	Statistic value
STAT_ID	NUMBER	Identifier of the statistic
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> This view returns instance-wide data and a value of `0` in the `CON_ID` column when queried from the root of a CDB.

> **See Also**
>
> "V$STATNAME" and " Statistics Descriptions"

9.99 V$SYSTEM_CURSOR_CACHE

`V$SYSTEM_CURSOR_CACHE` displays system wide information on cursor usage.

> **See Also**
>
> "V$SESSION_CURSOR_CACHE"

Column	Datatype	Description
`OPENS`	`NUMBER`	Cumulative total of cursor opens
`HITS`	`NUMBER`	Cumulative total of cursor open hits
`HIT_RATIO`	`NUMBER`	Ratio of the number of times an open cursor was found divided by the number of times a cursor was sought
`CON_ID`	`NUMBER`	The ID of the container to which the data pertains. Possible values include: • `0`: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • `1`: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.100 V$SYSTEM_EVENT

`V$SYSTEM_EVENT` displays information on total waits for an event. Note that the `TIME_WAITED` and `AVERAGE_WAIT` columns will contain a value of zero on those platforms that do not support a fast timing mechanism. If you are running on one of these platforms and you want this column to reflect true wait times, then you must set `TIMED_STATISTICS` to `TRUE` in the parameter file; doing this will have a small negative effect on system performance.

> **See Also**
>
> "TIMED_STATISTICS"

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
EVENT	VARCHAR2(64)	Name of the wait event; derived statistic name from V$EVENT_NAME **See Also:** " Oracle Wait Events"
TOTAL_WAITS	NUMBER	Total number of waits for the event
TOTAL_TIMEOUTS	NUMBER	Total number of timeouts for the event
TIME_WAITED	NUMBER	Total amount of time waited for the event (in hundredths of a second)
AVERAGE_WAIT	NUMBER	Average amount of time waited for the event (in hundredths of a second)
TIME_WAITED_MICRO	NUMBER	Total amount of time waited for the event (in microseconds)
TOTAL_WAITS_FG	NUMBER	Total number of waits for the event, from foreground sessions
TOTAL_TIMEOUTS_FG	NUMBER	Total number of timeouts for the event, from foreground sessions
TIME_WAITED_FG	NUMBER	Total amount of time waited for the event, from foreground sessions (in hundredths of a second)
AVERAGE_WAIT_FG	NUMBER	Average amount of time waited for the event, from foreground sessions (in hundredths of a second)
TIME_WAITED_MICRO_FG	NUMBER	Total amount of time waited for the event, from foreground sessions (in microseconds)
EVENT_ID	NUMBER	Identifier of the wait event
WAIT_CLASS_ID	NUMBER	Identifier of the wait class of the wait event
WAIT_CLASS#	NUMBER	Number of the wait class of the wait event
WAIT_CLASS	VARCHAR2(64)	Name of the wait class of the wait event
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **Note**
>
> This view returns instance-wide data and a value of 0 in the CON_ID column when queried from the root of a CDB.

9.101 V$SYSTEM_FIX_CONTROL

V$SYSTEM_FIX_CONTROL displays information about Fix Control (enabled/disabled) at the system level.

Column	Datatype	Description
BUGNO	NUMBER	Bug number (as fix control identifier)
VALUE	NUMBER	Current value set for the fix control
SQL_FEATURE	VARCHAR2(64)	Feature control ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DESCRIPTION	VARCHAR2(64)	Description of the fix control
OPTIMIZER_FEATURE_ENABLE	VARCHAR2(25)	Version on (and after) which the fix is enabled by default
EVENT	NUMBER	Event formerly used to control this fix
IS_DEFAULT	NUMBER	Indicates whether the current value is the same as the default (1) or not (0)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.102 V$SYSTEM_PARAMETER

V$SYSTEM_PARAMETER displays information about the initialization parameters that are currently in effect for the instance. A new session inherits parameter values from the instance-wide values.

Column	Datatype	Description
NUM	NUMBER	Parameter number
NAME	VARCHAR2(80)	Name of the parameter
TYPE	NUMBER	Parameter type: • 1 - Boolean • 2 - String • 3 - Integer • 4 - Parameter file • 5 - Reserved • 6 - Big integer
VALUE	VARCHAR2(4000)	Instance-wide parameter value
DISPLAY_VALUE	VARCHAR2(4000)	Parameter value in a user-friendly format. For example, if the VALUE column shows the value 262144 for a big integer parameter, then the DISPLAY_VALUE column will show the value 256K.
DEFAULT_VALUE	VARCHAR2(255)	The default value for this parameter. This is the value of the parameter if a value is not explicitly specified for the parameter.
ISDEFAULT	VARCHAR2(9)	Indicates whether the parameter is set to the default value (TRUE) or the parameter value was specified in the parameter file (FALSE) The database sets the value of the ISDEFAULT column to TRUE for parameters that are not specified in the init.ora or server parameter file (SPFILE).
ISSES_MODIFIABLE	VARCHAR2(5)	Indicates whether the parameter can be changed with ALTER SESSION (TRUE) or not (FALSE)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ISSYS_MODIFIABLE	VARCHAR2(9)	Indicates whether the parameter can be changed with ALTER SYSTEM and when the change takes effect: • IMMEDIATE - Parameter can be changed with ALTER SYSTEM regardless of the type of parameter file used to start the instance. The change takes effect immediately. • DEFERRED - Parameter can be changed with ALTER SYSTEM regardless of the type of parameter file used to start the instance. The change takes effect in subsequent sessions. • FALSE - Parameter cannot be changed with ALTER SYSTEM unless a server parameter file was used to start the instance. The change takes effect in subsequent instances.
ISPDB_MODIFIABLE	VARCHAR2(5)	Indicates whether the parameter can be modified inside a PDB (TRUE) or not (FALSE). In a non-CDB, the value of this column is NULL.
ISINSTANCE_MODIFIABLE	VARCHAR2(5)	For parameters that can be changed with ALTER SYSTEM, indicates whether the value of the parameter can be different for every instance (TRUE) or whether the parameter must have the same value for all Real Application Clusters instances (FALSE). If the ISSYS_MODIFIABLE column is FALSE, then this column is always FALSE.
ISMODIFIED	VARCHAR2(8)	Indicates how the parameter was modified. If an ALTER SYSTEM was performed, the value will be MODIFIED.
ISADJUSTED	VARCHAR2(5)	Indicates whether Oracle adjusted the input value to a more suitable value (for example, the parameter value should be prime, but the user input a non-prime number, so Oracle adjusted the value to the next prime number)
ISDEPRECATED	VARCHAR2(5)	Indicates whether the parameter has been deprecated (TRUE) or not (FALSE)
ISBASIC	VARCHAR2(5)	Indicates whether the parameter is a basic parameter (TRUE) or not (FALSE)
DESCRIPTION	VARCHAR2(255)	Description of the parameter
UPDATE_COMMENT	VARCHAR2(255)	Comments associated with the most recent update
HASH	NUMBER	Hash value for the parameter name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$PARAMETER" for information about initialization parameters that are currently in effect for a session

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.103 V$SYSTEM_PARAMETER2

V$SYSTEM_PARAMETER2 displays information about the initialization parameters that are currently in effect for the instance, with each list parameter value appearing as a row in the view. A new session inherits parameter values from the instance-wide values.

Presenting the list parameter values in this format enables you to quickly determine the values for a list parameter. For example, if a parameter value is a, b, then the V$SYSTEM_PARAMETER view does not tell you if the parameter has two values (both a and b) or one value (a, b). V$SYSTEM_PARAMETER2 makes the distinction between the list parameter values clear.

Column	Datatype	Description
NUM	NUMBER	Parameter number
NAME	VARCHAR2(80)	Name of the parameter
TYPE	NUMBER	Parameter type: • 1 - Boolean • 2 - String • 3 - Integer • 4 - Parameter file • 5 - Reserved • 6 - Big integer
VALUE	VARCHAR2(4000)	Parameter value
DISPLAY_VALUE	VARCHAR2(4000)	Parameter value in a user-friendly format. For example, if the VALUE column shows the value 262144 for a big integer parameter, then the DISPLAY_VALUE column will show the value 256K.
ISDEFAULT	VARCHAR2(6)	Indicates whether the parameter is set to the default value (TRUE) or the parameter value was specified in the parameter file (FALSE)
ISSES_MODIFIABLE	VARCHAR2(5)	Indicates whether the parameter can be changed with ALTER SESSION (TRUE) or not (FALSE)
ISSYS_MODIFIABLE	VARCHAR2(9)	Indicates whether the parameter can be changed with ALTER SYSTEM and when the change takes effect: • IMMEDIATE - Parameter can be changed with ALTER SYSTEM regardless of the type of parameter file used to start the instance. The change takes effect immediately. • DEFERRED - Parameter can be changed with ALTER SYSTEM regardless of the type of parameter file used to start the instance. The change takes effect in subsequent sessions. • FALSE - Parameter cannot be changed with ALTER SYSTEM unless a server parameter file was used to start the instance. The change takes effect in subsequent instances.
ISPDB_MODIFIABLE	VARCHAR2(5)	Indicates whether the parameter can be modified on a per-PDB basis (TRUE) or not (FALSE). In a non-CDB, the value of this column is NULL.
ISINSTANCE_MODIFIABLE	VARCHAR2(5)	For parameters that can be changed with ALTER SYSTEM, indicates whether the value of the parameter can be different for every instance (TRUE) or whether the parameter must have the same value for all Real Application Clusters instances (FALSE). If the ISSYS_MODIFIABLE column is FALSE, then this column is always FALSE.
ISMODIFIED	VARCHAR2(8)	Indicates how the parameter was modified. If an ALTER SYSTEM was performed, the value will be MODIFIED.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ISADJUSTED	VARCHAR2(5)	Indicates whether Oracle adjusted the input value to a more suitable value (for example, the parameter value should be prime, but the user input a non-prime number, so Oracle adjusted the value to the next prime number)
ISDEPRECATED	VARCHAR2(5)	Indicates whether the parameter has been deprecated (TRUE) or not (FALSE)
ISBASIC	VARCHAR2(5)	Indicates whether the parameter is a basic parameter (TRUE) or not (FALSE)
DESCRIPTION	VARCHAR2(255)	Description of the parameter
ORDINAL	NUMBER	Position (ordinal number) of the parameter value. Useful only for parameters whose values are lists of strings.
UPDATE_COMMENT	VARCHAR2(255)	Comments associated with the most recent update
HASH	NUMBER	Hash value for the parameter name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$SYSTEM_PARAMETER"

9.104 V$SYSTEM_WAIT_CLASS

V$SYSTEM_WAIT_CLASS displays the instance-wide time totals for each registered wait class.

Column	Datatype	Description
WAIT_CLASS_ID	NUMBER	Identifier of the wait class
WAIT_CLASS#	NUMBER	Number of the wait class
WAIT_CLASS	VARCHAR2(64)	Name of the wait class
TOTAL_WAITS	NUMBER	Total number of waits from this wait class
TIME_WAITED	NUMBER	Total amount of time spent in waits from this wait class (in hundredths of a second)
TOTAL_WAITS_FG	NUMBER	Total number of waits for this wait class, from foreground sessions
TIME_WAITED_FG	NUMBER	Total amount of time spent in waits from this wait class, from foreground sessions (in hundredths of a second)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **Note**
>
> This view returns instance-wide data and a value of 0 in the CON_ID column when queried from the root of a CDB.

9.105 V$TABLESPACE

V$TABLESPACE displays tablespace information from the control file.

Column	Datatype	Description
TS#	NUMBER	Tablespace number
NAME	VARCHAR2(30)	Tablespace name
INCLUDED_IN_DATABASE_BACKUP	VARCHAR2(3)	Indicates whether the tablespace is included in full database backups using the BACKUP DATABASE RMAN command (YES) or not (NO); NO only if the CONFIGURE EXCLUDE RMAN command was used for this tablespace
BIGFILE	VARCHAR2(3)	Indicates whether the tablespace is a bigfile tablespace (YES) or a smallfile tablespace (NO)
FLASHBACK_ON	VARCHAR2(3)	Indicates whether the tablespace participates in FLASHBACK DATABASE operations (YES) or not (NO)
ENCRYPT_IN_BACKUP	VARCHAR2(3)	Indicates whether encryption is turned ON or off at the tablespace level: • ON - Encryption is turned ON at the tablespace level • OFF - Encryption is turned OFF at the tablespace level • NULL - Encryption is neither explicitly turned on nor off at the tablespace level (default or when cleared)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.106 V$TEMP_CACHE_TRANSFER

V$TEMP_CACHE_TRANSFER is deprecated. The information that was provided in this view is now provided in the V$INSTANCE_CACHE_TRANSFER and V$SEGMENT_STATISTICS views.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
FILE_NUMBER	NUMBER	Number of the temp file
X_2_NULL	NUMBER	Number of blocks with Exclusive-to-NULL conversions; always 0
X_2_NULL_FORCED_WRITE	NUMBER	Number of Exclusive-to-NULL forced writes; always 0
X_2_NULL_FORCED_STALE	NUMBER	Number of Exclusive-to-NULL blocks converted to CR; always 0
X_2_S	NUMBER	Number of blocks with Exclusive-to-Shared conversions; always 0
X_2_S_FORCED_WRITE	NUMBER	Number of Exclusive-to-Shared forced writes; always 0
S_2_NULL	NUMBER	Number of blocks with Shared-to-NULL conversions; always 0
S_2_NULL_FORCED_STALE	NUMBER	Number of Shared-to-NULL blocks converted to CR; always 0
RBR	NUMBER	Number of reuse blocks cross-instance calls; always 0
RBR_FORCED_WRITE	NUMBER	Number of blocks written due to reuse blocks cross-instance calls; always 0
NULL_2_X	NUMBER	Number of blocks with NULL-to-Exclusive conversions; always 0
S_2_X	NUMBER	Number of blocks with Shared-to-Exclusive conversions; always 0
NULL_2_S	NUMBER	Number of blocks with NULL-to-Shared conversions; always 0
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$INSTANCE_CACHE_TRANSFER"
- "V$SEGMENT_STATISTICS"

9.107 V$TEMP_EXTENT_MAP

V$TEMP_EXTENT_MAP displays the status of each unit for all LOCALLY MANAGED temporary tablespaces.

Column	Datatype	NULL	Description
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of the tablespace this unit belongs to
FILE_ID	NUMBER		Absolute file number
BLOCK_ID	NUMBER		Begin block number for this unit
BYTES	NUMBER		Bytes in the extent
BLOCKS	NUMBER		Blocks in the extent
OWNER	NUMBER		Instance which owns this unit
RELATIVE_FNO	NUMBER		Relative file number

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.108 V$TEMP_EXTENT_POOL

V$TEMP_EXTENT_POOL displays the state of temporary space cached and used for the instance. Note that loading of the temporary space cache is lazy and that instances can be dormant.

Column	Datatype	NULL	Description
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of the tablespace
FILE_ID	NUMBER		Absolute file number
EXTENTS_CACHED	NUMBER		Number of extents that have been cached
EXTENTS_USED	NUMBER		Number of extents that are actually being used
BLOCKS_CACHED	NUMBER		Number of blocks that are cached
BLOCKS_USED	NUMBER		Number of blocks that are used
BYTES_CACHED	NUMBER		Number of bytes that are cached
BYTES_USED	NUMBER		Number of bytes that are used
RELATIVE_FNO	NUMBER		Relative file number
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.109 V$TEMP_SPACE_HEADER

V$TEMP_SPACE_HEADER displays aggregate information per file per LOCALLY MANAGED temporary tablespace regarding how much space is currently being used and how much is free as identified in the space header.

Column	Datatype	NULL	Description
TABLESPACE_NAME	VARCHAR2(30)	NOT NULL	Name of the temporary tablespace
FILE_ID	NUMBER		Absolute file number
BYTES_USED	NUMBER		How many bytes are in use
BLOCKS_USED	NUMBER		How many blocks are in use

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	NULL	Description
BYTES_FREE	NUMBER		How many bytes are free
BLOCKS_FREE	NUMBER		How many blocks are free
RELATIVE_FNO	NUMBER		The relative file number for the file
CON_ID	NUMBER		The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.110 V$TEMPFILE

V$TEMPFILE displays temp file information.

Column	Datatype	Description
FILE#	NUMBER	Absolute file number
CREATION_CHANGE#	NUMBER	Creation System Change Number (SCN)
CREATION_TIME	DATE	Creation time
TS#	NUMBER	Tablespace number
RFILE#	NUMBER	Relative file number in the tablespace
STATUS	VARCHAR2(7)	Status of the file (OFFLINE|ONLINE)
ENABLED	VARCHAR2(10)	Enabled for read and/or write
BYTES	NUMBER	Size of the file in bytes (from the file header)
BLOCKS	NUMBER	Size of the file in blocks (from the file header)
CREATE_BYTES	NUMBER	Creation size of the file (in bytes)
BLOCK_SIZE	NUMBER	Block size for the file
NAME	VARCHAR2(513)	Name of the file
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.111 V$TEMPORARY_LOBS

V$TEMPORARY_LOBS displays information about temporary and abstract LOBs.

Column	Datatype	Description
SID	NUMBER	Session ID

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CACHE_LOBS	NUMBER	Number of cache temp LOBs
NOCACHE_LOBS	NUMBER	Number of nocache temp LOBs
ABSTRACT_LOBS	NUMBER	Number of abstract LOBs
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.112 V$TEMPSEG_USAGE

V$TEMPSEG_USAGE describes temporary segment usage.

Column	Datatype	Description
USERNAME	VARCHAR2(128)	User who requested temporary space
USER	VARCHAR2(128)	This column is obsolete and maintained for backward compatibility. The value of this column is always equal to the value in USERNAME.
SESSION_ADDR	RAW(4 | 8)	Session address
SESSION_NUM	NUMBER	Serial number of session
SQLADDR	RAW(4 | 8)	Address of SQL statement
SQLHASH	NUMBER	Hash value of SQL statement
SQL_ID	VARCHAR2(13)	SQL identifier of SQL statement
TABLESPACE	VARCHAR2(30)	Tablespace in which space is allocated
CONTENTS	VARCHAR2(9)	Indicates whether tablespace is TEMPORARY or PERMANENT
SEGTYPE	VARCHAR2(9)	Type of sort segment: • SORT • HASH • DATA • INDEX • LOB_DATA • LOB_INDEX
SEGFILE#	NUMBER	File number of initial extent
SEGBLK#	NUMBER	Block number of the initial extent
EXTENTS	NUMBER	Extents allocated to the sort
BLOCKS	NUMBER	Extents in blocks allocated to the sort
SEGRFNO#	NUMBER	Relative file number of initial extent
TS#	NUMBER	Tablespace number

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
SQL_ID_TEMPSEG	VARCHAR2(13)	SQL identifier of the SQL statement that created the temporary segment

9.113 V$TEMPSTAT

V$TEMPSTAT displays information about file read/write statistics.

Column	Datatype	Description
FILE#	NUMBER	Number of the file
PHYRDS	NUMBER	Number of physical reads done
PHYWRTS	NUMBER	Number of times DBWR is required to write
PHYBLKRD	NUMBER	Number of physical blocks read
PHYBLKWRT	NUMBER	Number of blocks written to disk, which may be the same as PHYWRTS if all writes are single blocks
SINGLEBLKRDS	NUMBER	Number of single block reads
READTIM	NUMBER	Time (in hundredths of a second) spent doing reads if the TIMED_STATISTICS parameter is true; 0 if false
WRITETIM	NUMBER	Time (in hundredths of a second) spent doing writes if the TIMED_STATISTICS parameter is true; 0 if false
SINGLEBLKRDTIM	NUMBER	Cumulative single block read time (in hundredths of a second)
AVGIOTIM	NUMBER	Average time (in hundredths of a second) spent on I/O, if the TIMED_STATISTICS parameter is true; 0 if false
LSTIOTIM	NUMBER	Time (in hundredths of a second) spent doing the last I/O, if the TIMED_STATISTICS parameter is true; 0 if false
MINIOTIM	NUMBER	Minimum time (in hundredths of a second) spent on a single I/O, if the TIMED_STATISTICS parameter is true; 0 if false
MAXIORTM	NUMBER	Maximum time (in hundredths of a second) spent doing a single read, if the TIMED_STATISTICS parameter is true; 0 if false
MAXIOWTM	NUMBER	Maximum time (in hundredths of a second) spent doing a single write, if the TIMED_STATISTICS parameter is true; 0 if false
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "TIMED_STATISTICS"

9.114 V$TEMPUNDOSTAT

V$TEMPUNDOSTAT shows various statistics related to the temporary undo log for this database instance. It displays a histogram of statistical data to show how the system is working. Each row in the view keeps statistics collected in the instance for a 10-minute interval. The rows are in the descending order of the BEGIN_TIME column value. This view contains a total of 576 rows, spanning a 4-day cycle. This view is similar to the V$UNDOSTAT view.

Column	Datatype	Description
BEGIN_TIME	DATE	Identifies the beginning of the time interval
END_TIME	DATE	Identifies the end of the time interval
UNDOTSN	NUMBER	Represents the last active undo tablespace in the duration of time. The tablespace ID of the active undo tablespace is returned in this column. If more than one undo tablespace was active in that period, the active undo tablespace that was active at the end of the period is reported.
TXNCOUNT	NUMBER	Total number of transaction that have bound to the temp undo segment contained in above tablespace within the interval period
MAXCONCURRENCY	NUMBER	Highest number of transactions executed concurrently which modified temporary objects within the interval period
MAXQUERYLEN	NUMBER	Reserved for future use
MAXQUERYID	VARCHAR2(13)	Reserved for future use
UNDOBLKCNT	NUMBER	Total number of temporary undo blocks consumed
EXTCNT	NUMBER	Total number of extents consumed
USCOUNT	NUMBER	Temp undo segments created in this period
SSOLDERRCNT	NUMBER	Identifies the number of times the error ORA-01555 occurred. You can use this statistic to decide whether or not the UNDO_RETENTION initialization parameter is set properly given the size of the undo tablespace. Increasing the value of UNDO_RETENTION can reduce the occurrence of this error.
NOSPACEERRCNT	NUMBER	Total number of times the error 'no space left for temporary undo' was raised
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

> **See Also**
>
> "V$UNDOSTAT"

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.115 V$THREAD

V$THREAD displays thread information from the control file. This view does not return meaningful results on a physical standby database.

Column	Datatype	Description
THREAD#	NUMBER	Thread number
STATUS	VARCHAR2(6)	Thread status (OPEN| CLOSED)
ENABLED	VARCHAR2(8)	Enabled status: DISABLED, (enabled) PRIVATE, or (enabled) PUBLIC
GROUPS	NUMBER	Number of log groups assigned to this thread
INSTANCE	VARCHAR2(80)	Instance name, if available
OPEN_TIME	DATE	Last time the thread was opened
CURRENT_GROUP#	NUMBER	Current log group
SEQUENCE#	NUMBER	Sequence number of current log
CHECKPOINT_CHANGE#	NUMBER	SCN at last checkpoint
CHECKPOINT_TIME	DATE	Time of last checkpoint
ENABLE_CHANGE#	NUMBER	SCN at which thread was enabled
ENABLE_TIME	DATE	Time of enable SCN
DISABLE_CHANGE#	NUMBER	SCN at which thread was disabled
DISABLE_TIME	DATE	Time of disable SCN
LAST_REDO_SEQUENCE#	NUMBER	Last redo sequence number written by LGWR
LAST_REDO_BLOCK	NUMBER	Last redo block written by LGWR
LAST_REDO_CHANGE#	NUMBER	SCN of last redo for the thread
LAST_REDO_TIME	DATE	Time of last redo for the thread
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.116 V$THRESHOLD_TYPES

V$THRESHOLD_TYPES displays information about threshold types.

Column	Datatype	Description
METRICS_ID	NUMBER	Metrics ID
METRICS_GROUP_ID	NUMBER	Metrics group ID
OPERATOR_MASK	NUMBER	Operator mask

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
OBJECT_TYPE	VARCHAR2(64)	Object type: • SYSTEM • FILE • SERVICE • EVENT_CLASS • TABLESPACE • SESSION
ALERT_REASON_ID	NUMBER	ID of the alert reason
METRIC_VALUE_TYPE	NUMBER	Metric value type
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.117 V$TIMER

V$TIMER displays the elapsed time in hundredths of a second. Time is measured since the beginning of the epoch, which is operating system specific, and wraps around to 0 again whenever the value overflows four bytes (roughly 497 days).

Column	Datatype	Description
HSECS	NUMBER	Elapsed time (in hundredths of a second)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.118 V$TIMEZONE_FILE

V$TIMEZONE_FILE describes the time zone file that is currently in use by the database.

Column	Datatype	Description
FILENAME	VARCHAR2(20)	Time zone file name
VERSION	NUMBER	Time zone file version

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.119 V$TIMEZONE_NAMES

V$TIMEZONE_NAMES displays valid time zone names.

Column	Datatype	Description
TZNAME	VARCHAR2(64)	Time zone region (for example, US/Pacific)
TZABBREV	VARCHAR2(64)	Corresponding daylight abbreviation (for example, PDT)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.120 V$TOPLEVELCALL

V$TOPLEVELCALL displays the mapping between Oracle top level calls and names.

Column	Datatype	Description
TOP_LEVEL_CALL#	NUMBER	Oracle top level call number
TOP_LEVEL_CALL_NAME	VARCHAR2(64)	Oracle top level call name
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.121 V$TRANSACTION

V$TRANSACTION lists the active transactions in the system.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
ADDR	RAW(4 | 8)	Address of the transaction state object
XIDUSN	NUMBER	Undo segment number
XIDSLOT	NUMBER	Slot number
XIDSQN	NUMBER	Sequence number
UBAFIL	NUMBER	Undo block address (UBA) filenum
UBABLK	NUMBER	UBA block number
UBASQN	NUMBER	UBA sequence number
UBAREC	NUMBER	UBA record number
STATUS	VARCHAR2(16)	Status
START_TIME	VARCHAR2(20)	Start time (wall clock)
START_SCNB	NUMBER	Start system change number (SCN) base
START_SCNW	NUMBER	Start SCN wrap
START_UEXT	NUMBER	Start extent number
START_UBAFIL	NUMBER	Start UBA file number
START_UBABLK	NUMBER	Start UBA block number
START_UBASQN	NUMBER	Start UBA sequence number
START_UBAREC	NUMBER	Start UBA record number
SES_ADDR	RAW(4 | 8)	User session object address
FLAG	NUMBER	Flag
SPACE	VARCHAR2(3)	YES if a space transaction
RECURSIVE	VARCHAR2(3)	YES if a recursive transaction
NOUNDO	VARCHAR2(3)	YES if a no undo transaction
PTX	VARCHAR2(3)	YES if parallel transaction
NAME	VARCHAR2(256)	Name of a named transaction
PRV_XIDUSN	NUMBER	Previous transaction undo segment number
PRV_XIDSLT	NUMBER	Previous transaction slot number
PRV_XIDSQN	NUMBER	Previous transaction sequence number
PTX_XIDUSN	NUMBER	Rollback segment number of the parent XID
PTX_XIDSLT	NUMBER	Slot number of the parent XID
PTX_XIDSQN	NUMBER	Sequence number of the parent XID
DSCN-B	NUMBER	This column is obsolete and maintained for backward compatibility. The value of this column is always equal to the value in DSCN_BASE.
DSCN-W	NUMBER	This column is obsolete and maintained for backward compatibility. The value of this column is always equal to the value in DSCN_WRAP.
USED_UBLK	NUMBER	Number of undo blocks used
USED_UREC	NUMBER	Number of undo records used
LOG_IO	NUMBER	Logical I/O
PHY_IO	NUMBER	Physical I/O
CR_GET	NUMBER	Consistent gets
CR_CHANGE	NUMBER	Consistent changes

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
START_DATE	DATE	Start time (wall clock)
DSCN_BASE	NUMBER	Dependent SCN base
DSCN_WRAP	NUMBER	Dependent SCN wrap
START_SCN	NUMBER	Start SCN
DEPENDENT_SCN	NUMBER	Dependent SCN
XID	RAW(8)	Transaction XID
PRV_XID	RAW(8)	Previous transaction XID
PTX_XID	RAW(8)	Parent transaction XID
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.122 V$TRANSACTION_ENQUEUE

V$TRANSACTION_ENQUEUE displays locks owned by transaction state objects.

Column	Datatype	Description
ADDR	RAW(4 | 8)	Address of lock state object
KADDR	RAW(4 | 8)	Address of lock
SID	NUMBER	Identifier for session holding or acquiring the lock
TYPE	VARCHAR2(2)	Type of lock. TX indicates transaction enqueue.
ID1	NUMBER	Lock identifier #1 (depends on type)
ID2	NUMBER	Lock identifier #2 (depends on type)
LMODE	NUMBER	Lock mode in which the session holds the lock: • 0 - none • 1 - null (NULL) • 2 - row-S (SS) • 3 - row-X (SX) • 4 - share (S) • 5 - S/Row-X (SSX) • 6 - exclusive (X)
REQUEST	NUMBER	Lock mode in which the process requests the lock: • 0 - none • 1 - null (NULL) • 2 - row-S (SS) • 3 - row-X (SX) • 4 - share (S) • 5 - S/Row-X (SSX) • 6 - exclusive (X)
CTIME	NUMBER	Time since current mode was granted

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BLOCK	NUMBER	The lock is blocking another lock
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.123 V$TRANSPORTABLE_PLATFORM

V$TRANSPORTABLE_PLATFORM displays all platforms that support cross-platform tablespace transport. Specifically, it lists all platforms supported by the RMAN CONVERT TABLESPACE command, along with the endianness of each platform.

Column	Datatype	Description
PLATFORM_ID	NUMBER	Platform identification number
PLATFORM_NAME	VARCHAR2(101)	Platform name
ENDIAN_FORMAT	VARCHAR2(14)	Platform endian format: • Big • Little • UNKNOWN FORMAT
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.124 V$TSDP_SUPPORTED_FEATURE

V$TSDP_SUPPORTED_FEATURE displays information about the features supported by Transparent Sensitive Data Protection (TSDP).

Column	Datatype	Description
FEATURE_NAME	VARCHAR2(200)	The name of the supported feature
FUNCTIONALITY	VARCHAR2(200)	The functionality that is supported within the feature. If all of the functionality of the feature is supported, the value is ALL, otherwise the value will show the specific functionality that is supported.
COMMENTS$	VARCHAR2(4000)	More information regarding the support for the feature and the specific functionality

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

Oracle Database Security Guide for more information about using Transparent Sensitive Data Protection.

9.125 V$TYPE_SIZE

V$TYPE_SIZE displays the sizes of various database components for use in estimating data block capacity.

Column	Datatype	Description
COMPONENT	VARCHAR2(8)	Component name, such as segment or buffer header
TYPE	VARCHAR2(8)	Component type
DESCRIPTION	VARCHAR2(32)	Description of the component
TYPE_SIZE	NUMBER	Size of the component
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.126 V$UNDOSTAT

V$UNDOSTAT displays a histogram of statistical data to show how well the system is working. The available statistics include undo space consumption, transaction concurrency, and length of queries executed in the instance. You can use this view to estimate the amount of undo space required for the current workload. Oracle uses this view to tune undo usage in the system. The view returns NULL values if the system is in manual undo management mode. Each row in the view keeps statistics collected in the instance for a 10-minute interval. The rows are in descending order by the BEGIN_TIME column value. Each row belongs to the time interval marked by (BEGIN_TIME, END_TIME). Each column represents the data collected for the particular statistic in that time interval. The first row of the view contains statistics for the (partial) current time period. The view contains a total of 576 rows, spanning a 4 day cycle.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BEGIN_TIME	DATE	Identifies the beginning of the time interval
END_TIME	DATE	Identifies the end of the time interval
UNDOTSN	NUMBER	Represents the last active undo tablespace in the duration of time. The tablespace ID of the active undo tablespace is returned in this column. If more than one undo tablespace was active in that period, the active undo tablespace that was active at the end of the period is reported.
UNDOBLKS	NUMBER	Represents the total number of undo blocks consumed. You can use this column to obtain the consumption rate of undo blocks, and thereby estimate the size of the undo tablespace needed to handle the workload on your system.
TXNCOUNT	NUMBER	Identifies the total number of transactions executed within the period
MAXQUERYLEN	NUMBER	Identifies the length of the longest query (in seconds) executed in the instance during the period. You can use this statistic to estimate the proper setting of the UNDO_RETENTION initialization parameter. The length of a query is measured from the cursor open time to the last fetch/execute time of the cursor. Only the length of those cursors that have been fetched/executed during the period are reflected in the view.
MAXQUERYID	VARCHAR2(13)	SQL identifier of the longest running SQL statement in the period
MAXCONCURRENCY	NUMBER	Identifies the highest number of transactions executed concurrently within the period
UNXPSTEALCNT	NUMBER	Number of attempts to obtain undo space by stealing unexpired extents from other transactions
UNXPBLKRELCNT	NUMBER	Number of unexpired blocks removed from certain undo segments so they can be used by other transactions
UNXPBLKREUCNT	NUMBER	Number of unexpired undo blocks reused by transactions
EXPSTEALCNT	NUMBER	Number of attempts to steal expired undo blocks from other undo segments
EXPBLKRELCNT	NUMBER	Number of expired undo blocks stolen from other undo segments
EXPBLKREUCNT	NUMBER	Number of expired undo blocks reused within the same undo segments
SSOLDERRCNT	NUMBER	Identifies the number of times the error ORA-01555 occurred. You can use this statistic to decide whether or not the UNDO_RETENTION initialization parameter is set properly given the size of the undo tablespace. Increasing the value of UNDO_RETENTION can reduce the occurrence of this error.
NOSPACEERRCNT	NUMBER	Identifies the number of times space was requested in the undo tablespace and there was no free space available. That is, all of the space in the undo tablespace was in use by active transactions. The corrective action is to add more space to the undo tablespace.
ACTIVEBLKS	NUMBER	Total number of blocks in the active extents of the undo tablespace for the instance at the sampled time in the period
UNEXPIREDBLKS	NUMBER	Total number of blocks in the unexpired extents of the undo tablespace for the instance at the sampled time in the period
EXPIREDBLKS	NUMBER	Total number of blocks in the expired extents of the undo tablespace for the instance at the sampled time in the period

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TUNED_UNDORETENTION	NUMBER	Amount of time (in seconds) for which undo will not be recycled from the time it was committed. At any point in time, the latest value of TUNED_UNDORETENTION is used to determine whether data committed at a particular time in the past can be recycled. The value of this column is not meaningful on an Oracle Active Data Guard standby database instance, because the system does not tune this value on such instances.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"UNDO_RETENTION"

9.127 V$UNUSABLE_BACKUPFILE_DETAILS

V$UNUSABLE_BACKUPFILE_DETAILS displays information about all backup files (backup pieces, proxy copies, or copies) that are marked unavailable and expired. You can select one of the rows and use BTYPE_KEY or FILETYPE_KEY to change the status of a backup file set or a specific file to available.

Column	Datatype	Description
SESSION_KEY	NUMBER	Session identifier
SESSION_RECID	NUMBER	Session record ID
SESSION_STAMP	NUMBER	Session stamp
RMAN_STATUS_RECID	NUMBER	Record ID of the corresponding row in the control file
RMAN_STATUS_STAMP	NUMBER	Timestamp of the row in the controlfile
BTYPE	CHAR(9)	Backup type container. Possible values are: BACKUPSET, IMAGECOPY, PROXYCOPY.
BTYPE_KEY	NUMBER	Unique identifier for the backup type, either BS_KEY or COPY_KEY.
ID1	NUMBER	If BACKUPSET, it contains SET_STAMP. If IMAGECOPY or PROXYCOPY, it is RECID from the control file.
ID2	NUMBER	If BACKUPSET, it contains SET_COUNT. If IMAGECOPY or PROXYCOPY, it is STAMP.
FILETYPE	VARCHAR2(15)	Type of file. Possible values are: BACKUPPIECE, COPY, PROXYCOPY.
FILETYPE_KEY	NUMBER	Backup piece key if the file is a backup piece; otherwise COPY_KEY.
STATUS	VARCHAR2(1)	Status of the backup file, either U (unavailable) or X (expired)
FILESIZE	NUMBER	Size of the file

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DEVICE_TYPE	VARCHAR2(17)	Type of device on which the file resides
FILENAME	VARCHAR2(513)	Name of the file
MEDIA	VARCHAR2(65)	Name of the media on which the copy resides. This value is informational only. It is not needed for restore.
MEDIA_POOL	NUMBER	Media pool in which the copy resides. This is the same value that was entered in the POOL operand of the Recovery Manager BACKUP command.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.128 V$VERSION

V$VERSION displays the version number of Oracle Database. The database components have the same version number as the database, so the version number is returned only once.

Column	Datatype	Description
BANNER	VARCHAR2(80)	Component name and version number
BANNER_FULL	VARCHAR2(160)	The new 2 line banner format introduced in Oracle Database 18c. The banner displays the database release and version number.
BANNER_LEGACY	VARCHAR2(80)	The legacy 1 line banner used before Oracle Database 18c. This column displays the same value as the BANNER column.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.129 V$VPD_POLICY

V$VPD_POLICY displays all the fine-grained security policies and predicates associated with the cursors currently in the library cache.

Column	Datatype	Description
ADDRESS	RAW(4 | 8)	Cursor address
PARADDR	RAW(4 | 8)	Parent cursor address
SQL_HASH	NUMBER	SQL hash number

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SQL_ID	VARCHAR2(13)	SQL identifier
CHILD_NUMBER	NUMBER	Cursor's child number under the parent
OBJECT_OWNER	VARCHAR2(128)	Owner of the object with the policy
OBJECT_NAME	VARCHAR2(128)	Name of the object with the policy
POLICY_GROUP	VARCHAR2(128)	Name of the policy group
POLICY	VARCHAR2(128)	Name of the policy
POLICY_FUNCTION_OWNER	VARCHAR2(128)	Owner of the policy function
PREDICATE	VARCHAR2(4000)	Predicate for the policy (truncated to 4000 bytes in length)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.130 V$WAIT_CHAINS

V$WAIT_CHAINS displays information about blocked sessions. A wait chain is composed of sessions that are blocked by one another. Each row represents a blocked and blocker session pair. If a wait chain is not a cyclical wait chain, then the last row for the chain does not have a blocker.

Column	Datatype	Description
CHAIN_ID	NUMBER	A number identifying the wait chain
CHAIN_IS_CYCLE	VARCHAR2(5)	Indicates whether the final blocked session in the wait chain is blocked by the initial blocked session (TRUE) or not (FALSE)
CHAIN_SIGNATURE	VARCHAR2(801)	An Oracle-specific text signature of the wait chain. This signature can be used to identify similar wait chains.
CHAIN_SIGNATURE_HASH	NUMBER	A numeric representation of CHAIN_SIGNATURE
INSTANCE	NUMBER	Blocked session's instance identifier
OSID	VARCHAR2(25)	Blocked session's operating system process identifier
PID	NUMBER	Blocked session's Oracle process identifier
SID	NUMBER	Blocked session's Oracle session identifier
SESS_SERIAL#	NUMBER	Blocked session's Oracle session serial number
PDB_ID	NUMBER	Blocked session's PDB identifier
PDB_NAME	VARCHAR2(31)	Blocked session's PDB name
BLOCKER_IS_VALID	VARCHAR2(5)	Indicates whether the blocked session has a blocker (TRUE) or not (FALSE)
BLOCKER_INSTANCE	NUMBER	Blocker session's instance identifier; NULL if BLOCKER_IS_VALID is FALSE
BLOCKER_OSID	VARCHAR2(25)	Blocker session's operating system process identifier; NULL if BLOCKER_IS_VALID is FALSE

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
BLOCKER_PID	NUMBER	Blocker session's Oracle process identifier; NULL if BLOCKER_IS_VALID is FALSE
BLOCKER_SID	NUMBER	Blocker session's Oracle session identifier; NULL if BLOCKER_IS_VALID is FALSE
BLOCKER_SESS_SERIAL#	NUMBER	Blocker session's Oracle session serial number; NULL if BLOCKER_IS_VALID is FALSE
BLOCKER_PDB_ID	NUMBER	Blocker session's PDB identifier
BLOCKER_PDB_NAME	VARCHAR2(31)	Blocker session's PDB name
BLOCKER_CHAIN_ID	NUMBER	If not NULL, then the blocker session is a member of another chain specified by this chain identifier. For the remaining wait chain information, see the wait chain with the specified CHAIN_ID.
IN_WAIT	VARCHAR2(5)	Indicates whether the blocked session is in a wait (TRUE) or not (FALSE)
TIME_SINCE_LAST_WAIT_SECS	NUMBER	Number of seconds since the last time the blocked session waited; NULL if IN_WAIT is TRUE
WAIT_ID	NUMBER	A number identifying the wait; NULL if IN_WAIT is FALSE
WAIT_EVENT	NUMBER	Resource or event number for which the blocked session is waiting; NULL if IN_WAIT is FALSE
WAIT_EVENT_TEXT	VARCHAR2(64)	Resource or event for which the blocked session is waiting; NULL if IN_WAIT is FALSE
P1	NUMBER	First additional wait parameter; NULL if IN_WAIT is FALSE
P1_TEXT	VARCHAR2(64)	Description of the first additional wait parameter; NULL if IN_WAIT is FALSE
P2	NUMBER	Second additional wait parameter; NULL if IN_WAIT is FALSE
P2_TEXT	VARCHAR2(64)	Description of the second additional wait parameter; NULL if IN_WAIT is FALSE
P3	NUMBER	Third additional wait parameter; NULL if IN_WAIT is FALSE
P3_TEXT	VARCHAR2(64)	Description of the third additional wait parameter; NULL if IN_WAIT is FALSE
IN_WAIT_SECS	NUMBER	Seconds the blocked session has been in the current wait; NULL if IN_WAIT is FALSE
TIME_REMAINING_SECS	NUMBER	Seconds remaining until the blocked session ends its wait (-1 if the blocked session can indefinitely wait); NULL if IN_WAIT is FALSE
NUM_WAITERS	NUMBER	Number of sessions waiting for the blocked session
ROW_WAIT_OBJ#	NUMBER	Object ID for the table containing the row specified in ROW_WAIT_ROW#; NULL if IN_WAIT is FALSE
ROW_WAIT_FILE#	NUMBER	Identifier for the data file containing the row specified in ROW_WAIT_ROW#; NULL if IN_WAIT is FALSE. This column is valid only if the blocked session is currently waiting for another transaction to commit and the value of ROW_WAIT_OBJ# is not -1.
ROW_WAIT_BLOCK#	NUMBER	Identifier for the block containing the row specified in ROW_WAIT_ROW#; NULL if IN_WAIT is FALSE. This column is valid only if the blocked session is currently waiting for another transaction to commit and the value of ROW_WAIT_OBJ# is not -1.
ROW_WAIT_ROW#	NUMBER	Current row being locked; NULL if IN_WAIT is FALSE. This column is valid only if the blocked session is currently waiting for another transaction to commit and the value of ROW_WAIT_OBJ# is not -1.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.131 V$WAITCLASSMETRIC

V$WAITCLASSMETRIC displays metric values of wait classes for the most recent 60-second interval. A history of the last one hour will be kept in the system.

Column	Datatype	Description
BEGIN_TIME	DATE	Begin time of the interval
END_TIME	DATE	End time of the interval
INTSIZE_CSEC	NUMBER	Interval size (in hundredths of a second)
WAIT_CLASS#	NUMBER	Number of the class of the wait event
WAIT_CLASS_ID	NUMBER	Identifier of the class of the wait event
AVERAGE_WAITER_COUNT	NUMBER	Average waiter count
DBTIME_IN_WAIT	NUMBER	Percent of database time spent in the wait
TIME_WAITED	NUMBER	Time waited during the interval (in hundredths of a second)
WAIT_COUNT	NUMBER	Number of times waited
TIME_WAITED_FG	NUMBER	Amount of time (in hundredths of a second) spent in waits from this wait class in foreground sessions
WAIT_COUNT_FG	NUMBER	Number of times foreground processes waited
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.132 V$WAITCLASSMETRIC_HISTORY

V$WAITCLASSMETRIC_HISTORY displays metric values of wait classes for all intervals in the last one hour.

The columns for V$WAITCLASSMETRIC_HISTORY are the same as those for V$WAITCLASSMETRIC.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "V$WAITCLASSMETRIC"

9.133 V$WAITSTAT

V$WAITSTAT displays block contention statistics. This table is only updated when timed statistics are enabled.

Column	Datatype	Description
CLASS	VARCHAR2(18)	Class of the block
COUNT	NUMBER	Number of waits by this OPERATION for this CLASS of block
TIME	NUMBER	Sum of all wait times for all the waits by this OPERATION for this CLASS of block (in centiseconds)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.134 V$WALLET

V$WALLET displays metadata of certificates that may be used as a master key for Transparent Data Encryption.

> **Note**
>
> The use of PKI encryption with Transparent Data Encryption is deprecated. To configure Transparent Data Encryption, use the ADMINISTER KEY MANAGEMENT SQL statement. See *Oracle Database Advanced Security Guide* for more information.

Column	Datatype	Description
CERT_ID	VARCHAR2(52)	A unique certificate identifier value used to specify a particular PKI certificate for use as the master key
DN	VARCHAR2(255)	Distinguished name of a particular PKI certificate
SERIAL_NUM	VARCHAR2(40)	Unique serial number assigned to a certificate by the issuer or signer
ISSUER	VARCHAR2(255)	Distinguished name of the Certificate Authority or issuer that issued and signed the certificate
KEYSIZE	NUMBER	Size of the PKI key associated with the certificate

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATUS	VARCHAR2(16)	Current status of the certificate: • UNUSED • IN USE • USED This column allows the user to identify whether a certificate is currently in use or has already been used for Transparent Data Encryption.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.135 V$WORKLOAD_REPLAY_THREAD

V$WORKLOAD_REPLAY_THREAD displays information for all the different types of replay sessions.

Column	Datatype	Description
CLOCK	NUMBER	Current simulated SCN based on the SCN recorded at capture time
NEXT_TICKER	NUMBER	Next stream ID that will increment the CLOCK after a commit action
SID	NUMBER	Session ID of the replay session
SERIAL#	NUMBER	Session serial number of the replay session
SPID	VARCHAR2(24)	The server process ID of the replay session
LOGON_USER	VARCHAR2(128)	Logon username of the replay session
LOGON_TIME	DATE	Logon time of the replay session
EVENT	VARCHAR2(64)	Event name
EVENT_ID	NUMBER	Event identifier
EVENT#	NUMBER	Event operation code
P1TEXT	VARCHAR2(64)	Text for event parameter 1
P1	NUMBER	Value of event parameter 1
P2TEXT	VARCHAR2(64)	Text for event parameter 2
P2	NUMBER	Value of event parameter 2
P3TEXT	VARCHAR2(64)	Text for event parameter 3
P3	NUMBER	Value of event parameter 3
WAIT_FOR_SCN	NUMBER	The captured SCN for which the current user call should wait
FILE_ID	NUMBER	The stream ID that is being replayed
CALL_COUNTER	NUMBER	The call counter of the user call that is being replayed
DEPENDENT_SCN	NUMBER	The dependent SCN, captured to order the commit actions using block-level dependencies
STATEMENT_SCN	NUMBER	Statement SCN
COMMIT_WAIT_SCN	NUMBER	The (maximum) SCN that the current commit should wait for

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
POST_COMMIT_SCN	NUMBER	The next SCN after the current commit
ACTION_TYPE	NUMBER	The type of commits. The possible values are: COMMIT, ROLLBACK, FAKED_COMMIT, and NULL. A value of NULL means it is not a commit operation.
SESSION_TYPE	VARCHAR2(13)	The type of replay session: LOGON ADMIN DISPATCHER REPLAY
WRC_ID	NUMBER	Unique replay client ID assigned by the server to all participating replay clients when replay starts
SCHEDULE_CAP_ID	NUMBER	An unique identifier for a workload capture added to a replay schedule. A value of 0 is used for a non-consolidated replay.
FILE_NAME	VARCHAR2(51)	File name of the captured stream
SKIP_IT	VARCHAR2(1)	Whether or not the current replayed user call is skipped or not
DIRTY_BUFFERS	VARCHAR2(1)	Reserved for internal use
DBTIME	NUMBER	Accumulated database time for the replay session
NETWORK_TIME	NUMBER	Accumulated network time for the replay session
THINK_TIME	NUMBER	Accumulated think time for the replay session
TIME_GAIN	NUMBER	If nonzero, the accumulated time in macro seconds indicating how fast the replay is
TIME_LOSS	NUMBER	If nonzero, the accumulated time in macro seconds indicating how slow the replay is
USER_CALLS	NUMBER	Total number of user calls
PLSQL_CALLS	NUMBER	Total number of PL/SQL calls recorded in the workload capture
PLSQL_SUBCALLS	NUMBER	Total number of calls recorded in the workload capture for SQL executed from PL/SQL
PLSQL_DBTIME	NUMBER	Total amount of database time in microseconds from PL/SQL calls that have been recorded in the workload capture
CLIENT_OS_USER	VARCHAR2(15)	Operating system username of the replay client
CLIENT_HOST	VARCHAR2(64)	Host name of the replay client
CLIENT_PID	VARCHAR2(24)	Process ID of the replay client
PROGRAM	VARCHAR2(48)	Program name of the replay client
CAPTURE_ELAPSED_TIME	NUMBER	Total amount of elapsed time of a session since capture starts (in seconds)
REPLAY_ELAPSED_TIME	NUMBER	Total amount of elapsed time of a session since replay starts (in seconds)
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.136 V$XML_AUDIT_TRAIL

V$XML_AUDIT_TRAIL shows standard, fine-grained, SYS, and mandatory audit records written in XML format files.

> **Note**
>
> This view is populated only in an Oracle Database where unified auditing is not enabled. When unified auditing is enabled in Oracle Database, the audit records are populated in the new audit trail and can be viewed from UNIFIED_AUDIT_TRAIL.
>
> - See *Oracle Database Security Guide* for more information about unified auditing.
> - See *Oracle Database Upgrade Guide* for more information about migrating to unified auditing.

Column	Datatype	Description
AUDIT_TYPE	NUMBER	Type of audit row: • 1 = Standard XML Audit • 2 = Fine Grained XML Audit • 4 = SYS XML Audit • 8 = Mandatory XML Audit
SESSION_ID	NUMBER	Numeric ID for the Oracle session
PROXY_SESSIONID	NUMBER	Proxy session serial number, if an enterprise user has logged in through a proxy mechanism
STATEMENTID	NUMBER	Numeric ID for the statement run (a statement may cause multiple audit records)
ENTRYID	NUMBER	Numeric ID for the audit trail entry in the session
EXTENDED_TIMESTAMP	TIMESTAMP(6) WITH TIME ZONE	Timestamp of the audited operation (the timestamp of the user's logon for entries is created by AUDIT SESSION)
GLOBAL_UID	VARCHAR2(32)	Global user identifier for the user, if the user has logged in as an enterprise user
DB_USER	VARCHAR2(128)	Database username of the user whose actions were audited
CLIENTIDENTIFIER	VARCHAR2(64)	Client identifier in the Oracle session
EXT_NAME	VARCHAR2(1024)	User's external name
OS_USER	VARCHAR2(128)	Operating system logon user name of the user whose actions were audited
OS_HOST	VARCHAR2(128)	Client host system name
OS_PROCESS	VARCHAR2(16)	Operating system process identifier of the Oracle server process
TERMINAL	VARCHAR2(30)	Identifier for the user's terminal
INSTANCE_NUMBER	NUMBER	Instance number as specified by the INSTANCE_NUMBER initialization parameter
OBJECT_SCHEMA	VARCHAR2(128)	Owner of the audited object
OBJECT_NAME	VARCHAR2(128)	Name of the object affected by the action
POLICY_NAME	VARCHAR2(128)	Name of the fine-grained auditing policy
NEW_OWNER	VARCHAR2(128)	Owner of the object named in the NEW_NAME column

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
NEW_NAME	VARCHAR2(128)	New name of object after renaming, or the name of an underlying object (for example, CREATE INDEX owner.obj_name ON new_owner.new_name)
ACTION	NUMBER	Numeric code for the action type
STATEMENT_TYPE	NUMBER	Description of the action
TRANSACTIONID	RAW(8)	Identifier of the transaction in which the object is accessed or modified
RETURNCODE	NUMBER	Oracle error code generated by the action. Zero if the action succeeded.
SCN	NUMBER	System change number (SCN) of the query
COMMENT_TEXT	VARCHAR2(4000)	Text comments on standard audit entries. Also indicates how the user was authenticated - the method can be one of the following: • DATABASE - authentication was done by password • NETWORK - authentication was done by Oracle Net Services or the Advanced Networking Option • PROXY - the client was authenticated by another user. The name of the proxy user follows the method type.
AUTH_PRIVILEGES	VARCHAR2(32)	Privileges granted and revoked in GRANT and REVOKE statements recorded for standard audit trail entry
GRANTEE	VARCHAR2(128)	User who granted or revoked the privilege
PRIV_USED	NUMBER	Numerical code of privileges, if any, used in the action
SES_ACTIONS	VARCHAR2(16)	Session summary for standard audit records. A string of 12 characters, one for each action type, in the following order: Alter, Audit, Comment, Delete, Grant, Index, Insert, Lock, Rename, Select, Update, Flashback. Values: - = None, S=Success, F=Failure, B=Both
OS_PRIVILEGE	VARCHAR2(7)	This column is populated only for administrative authentication audit records. It contains the administrative privilege used (for example, SYSDBA, SYSOPER, NONE). For all other types of audit records, the value of this column is null.
ECONTEXT_ID	VARCHAR2(64)	Application execution context identifier
SQL_BIND	VARCHAR2(4000)	List of bind variables used in the statement
SQL_TEXT	VARCHAR2(4000)	The statement or command that triggered the audit event
OBJ_EDITION_NAME	VARCHAR2(128)	Name of the edition containing the audited object
DBID	NUMBER	Database identifier of the audited database
RLS_INFO	VARCHAR2(4000)	Stores virtual private database (VPD) policy names and predicates separated by delimiter. To format the output into individual rows, use the DBMS_AUDIT_UTIL.DECODE_RLS_INFO_ATRAIL_XML function.
CURRENT_USER	VARCHAR2(128)	Effective user for the statement execution
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The SQL_BIND and SQL_TEXT columns are only populated if the AUDIT_TRAIL initialization parameter is set to xml, extended or if the AUDIT_SYS_OPERATIONS initialization parameter is set to TRUE.

> **See Also**
>
> - "UNIFIED_AUDIT_TRAIL"
> - "AUDIT_SYS_OPERATIONS"
> - "AUDIT_TRAIL"
> - *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_AUDIT_UTIL.DECODE_RLS_INFO_ATRAIL_XML function.

9.137 V$XSTREAM_APPLY_COORDINATOR

V$XSTREAM_APPLY_COORDINATOR displays information about each XStream apply process coordinator. The coordinator for an apply process gets transactions from the apply process reader and passes them to apply servers. An apply process coordinator is a subcomponent of an apply process, outbound server, or inbound server.

Column	Datatype	Description
SID	NUMBER	Session ID of the coordinator's session
SERIAL#	NUMBER	Serial number of the coordinator's session
STATE	VARCHAR2(21)	State of the coordinator: • INITIALIZING - Starting up • IDLE - Performing no work • APPLYING - Passing transactions to apply servers • SHUTTING DOWN CLEANLY - Stopping without an error • ABORTING - Stopping because of an apply error
APPLY#	NUMBER	Apply process number. An apply process coordinator is an Oracle background process, prefixed by ap.
APPLY_NAME	VARCHAR2(128)	Name of the apply process
TOTAL_APPLIED	NUMBER	Total number of transactions applied by the apply process since the apply process was last started
TOTAL_WAIT_DEPS	NUMBER	Number of times since the apply process was last started that an apply server waited to apply a logical change record (LCR) in a transaction until another apply server applied a transaction because of a dependency between the transactions
TOTAL_WAIT_COMMITS	NUMBER	Number of times since the apply process was last started that an apply server waited to commit a transaction until another apply server committed a transaction to serialize commits
TOTAL_ADMIN	NUMBER	Number of administrative jobs issued since the apply process was last started
TOTAL_ASSIGNED	NUMBER	Number of transactions assigned to apply servers since the apply process was last started

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOTAL_RECEIVED	NUMBER	Total number of transactions received by the coordinator process since the apply process was last started
TOTAL_IGNORED	NUMBER	Number of transactions which were received by the coordinator but were ignored because they had been previously applied
TOTAL_ROLLBACKS	NUMBER	Number of transactions which were rolled back due to unexpected contention
TOTAL_ERRORS	NUMBER	Number of transactions applied by the apply process that resulted in an apply error since the apply process was last started
UNASSIGNED_COMPLETE_TXNS	NUMBER	Total number of complete transactions that the coordinator has not assigned to any apply servers
LWM_TIME	DATE	Time when the message with the lowest message number was recorded. The creation time of the message with the lowest message number was also recorded at this time.
LWM_MESSAGE_NUMBER	NUMBER	Number of the message corresponding to the low watermark. That is, messages with a commit message number less than or equal to this message number have definitely been applied, but some messages with a higher commit message number also may have been applied.
LWM_MESSAGE_CREATE_TIME	DATE	For captured messages, creation time at the source database of the message corresponding to the low watermark. For user-enqueued messages, time when the message corresponding to the low watermark was enqueued into the queue at the local database.
HWM_TIME	DATE	Time when the message with the highest message number was recorded. The creation time of the message with the highest message number was also recorded at this time.
HWM_MESSAGE_NUMBER	NUMBER	Number of the message corresponding to the high watermark. That is, no messages with a commit message number greater than this message number have been applied.
HWM_MESSAGE_CREATE_TIME	DATE	For captured messages, creation time at the source database of the message corresponding to the high watermark. For user-enqueued messages, time when the message corresponding to the high watermark was enqueued into the queue at the local database.
STARTUP_TIME	DATE	Time when the apply process was last started
ELAPSED_SCHEDULE_TIME	NUMBER	Time elapsed (in hundredths of a second) scheduling messages since the apply process was last started
ELAPSED_IDLE_TIME	NUMBER	Elapsed idle time
LWM_POSITION	RAW(64)	Position of the low-watermark LCR
HWM_POSITION	RAW(64)	Position of the high-watermark LCR
PROCESSED_MESSAGE_NUMBER	NUMBER	Message number currently processed by the apply coordinator
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
ACTIVE_SERVER_COUNT	NUMBER	Active server count

Copyright © 2002, 2026, Oracle and/or its affiliates.

Note

The ELAPSED_SCHEDULE_TIME column is only populated if the TIMED_STATISTICS initialization parameter is set to true, or if the STATISTICS_LEVEL initialization parameter is set to TYPICAL or ALL.

See Also

- "TIMED_STATISTICS"
- "STATISTICS_LEVEL"

9.138 V$XSTREAM_APPLY_READER

V$XSTREAM_APPLY_READER displays information about each XStream apply reader. The apply reader is a process which reads (dequeues) messages from the queue, computes message dependencies, and builds transactions. It passes the transactions on to the coordinator in commit order for assignment to the apply servers. An apply reader is a subcomponent of an apply process, outbound server, or inbound server.

Column	Datatype	Description
SID	NUMBER	Session ID of the reader's session
SERIAL#	NUMBER	Serial number of the reader's session
APPLY#	NUMBER	Apply process number. An apply process is an Oracle background process, prefixed by ap.
APPLY_NAME	VARCHAR2(128)	Name of the apply process
STATE	VARCHAR2(36)	Shows the state of the apply reader and the hash server. The possible values include: • INITIALIZING - Starting up • IDLE - Performing no work • DEQUEUE MESSAGES - Dequeuing messages from the queue • SCHEDULE MESSAGES - Computing dependencies between messages and assembling messages into transactions • SPILLING - Spilling unapplied messages from memory to hard disk • PAUSED - WAITING FOR DDL TO COMPLETE - Waiting for a data definition language (DDL) logical change record (LCR) to be applied The state of the apply reader is displayed first, followed by the state of the hash server. A semicolon separates the apply reader state from the hash server state.
TOTAL_MESSAGES_DEQUEUED	NUMBER	Total number of messages dequeued since the apply process was last started
TOTAL_MESSAGES_SPILLED	NUMBER	Number of messages spilled by the reader since the apply process was last started
DEQUEUE_TIME	DATE	Time when the last message was received
DEQUEUED_MESSAGE_NUMBER	NUMBER	Number of the last message received

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
DEQUEUED_MESSAGE_CREATE_TIME	DATE	For captured messages, creation time at the source database of the last message received. For user-enqueued messages, time when the message was enqueued into the queue at the local database.
SGA_USED	NUMBER	Amount (in bytes) of SGA memory used by the apply process since it was last started
ELAPSED_DEQUEUE_TIME	NUMBER	Time elapsed (in hundredths of a second) dequeuing messages since the apply process was last started
ELAPSED_SCHEDULE_TIME	NUMBER	Time elapsed (in hundredths of a second) scheduling messages since the apply process was last started. Scheduling includes computing dependencies between messages and assembling messages into transactions.
ELAPSED_SPILL_TIME	NUMBER	Elapsed time (in hundredths of a second) spent spilling messages since the apply process was last started
OLDEST_SCN_NUM	NUMBER	Oldest SCN
OLDEST_XIDUSN	NUMBER	Transaction ID undo segment number of the oldest transaction that either has been applied or is being applied
OLDEST_XIDSLT	NUMBER	Transaction ID slot number of the oldest transaction that either has been applied or is being applied
OLDEST_XIDSQN	NUMBER	Transaction ID sequence number of the oldest transaction that either has been applied or is being applied
SPILL_LWM_SCN	NUMBER	Spill low-watermark SCN
PROXY_SID	NUMBER	When the apply process uses combined capture and apply, the session ID of the propagation receiver that is responsible for direct communication between capture and apply. If the apply process does not use combined capture and apply, then this column is `0`.
PROXY_SERIAL	NUMBER	When the apply process uses combined capture and apply, the serial number of the propagation receiver that is responsible for direct communication between capture and apply. If the apply process does not use combined capture and apply, then this column is `0`.
PROXY_SPID	VARCHAR2(12)	When the apply process uses combined capture and apply, the process identification number of the propagation receiver that is responsible for direct communication between capture and apply. If the apply process does not use combined capture and apply, then this column is `0`.
BYTES_RECEIVED	NUMBER	When the apply process uses combined capture and apply, the number of bytes received by the apply process from the capture process since the apply process last started. If the apply process does not use combined capture and apply, then this column is not populated.
DEQUEUED_POSITION	RAW(64)	Dequeued position. This column is populated only for an apply process that is functioning as an XStream inbound server.
SPILL_LWM_POSITION	RAW(64)	Spill low-watermark position. This column is populated only for an apply process that is functioning as an XStream inbound server.
OLDEST_TRANSACTION_ID	VARCHAR2(128)	Oldest transaction ID
TOTAL_LCRS_WITH_DEP	NUMBER	Total number of LCRs with row-level dependencies since the apply process last started
TOTAL_LCRS_WITH_WMDEP	NUMBER	Total number of LCRs with watermark dependencies since the apply process last started. A watermark dependency occurs when an apply process must wait until the apply process's low watermark reaches a particular threshold.
TOTAL_IN_MEMORY_LCRS	NUMBER	Total number of LCRs currently in memory

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SGA_ALLOCATED	NUMBER	The total amount of shared memory (in bytes) allocated from the XStreams pool for the apply process since the apply process last started
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Note

The ELAPSED_SCHEDULE_TIME column is only populated if the TIMED_STATISTICS initialization parameter is set to true, or if the STATISTICS_LEVEL initialization parameter is set to TYPICAL or ALL.

See Also

- "TIMED_STATISTICS"
- "STATISTICS_LEVEL"

9.139 V$XSTREAM_APPLY_RECEIVER

V$XSTREAM_APPLY_RECEIVER displays information about the message receiver of the apply process. The values are reset to zero when the database (or instance in an Oracle Real Application Clusters (Oracle RAC) environment) restarts, when apply migrates to another instance, or when the XStream process is stopped.

Column	Datatype	Description
SID	NUMBER	Session ID of the apply receiver
SERIAL#	NUMBER	Serial number of the apply receiver
APPLY_NAME	VARCHAR2(128)	Name of the apply process
STARTUP_TIME	DATE	Startup time of the apply process
SOURCE_DATABASE_NAME	VARCHAR2(128)	Name of the source database
ACKNOWLEDGEMENT	NUMBER	Acknowledgement SCN of the messages received by the receiver
LAST_RECEIVED_MSG	NUMBER	Last received message
TOTAL_MESSAGES_RECEIVED	NUMBER	Total number of messages received
TOTAL_AVAILABLE_MESSAGES	NUMBER	Number of available messages

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATE	VARCHAR2(44)	State of the apply receiver: • Initializing • Sending unapplied txns • Waiting for message from client • Waiting for LCR from client • Receiving LCRs • Evaluating rules • Enqueueing LCRS • Waiting for memory • Waiting for apply to read • Waiting for client flush request to complete • Waiting for client commit to complete
LAST_RECEIVED_MSG_POSITION	RAW(64)	Last received message position
ACKNOWLEDGEMENT_POSITION	RAW(64)	Acknowledgement position of the messages received by the receiver. Corresponds to ACKNOWLEDGEMENT, except the value is in position rather than SCN.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

9.140 V$XSTREAM_APPLY_SERVER

V$XSTREAM_APPLY_SERVER displays information about each XStream apply server and its activities. An apply server receives messages from the apply coordinator for an apply process. For each message received, an apply server either applies the message or sends the message to the appropriate apply handler. An apply server is a subcomponent of an apply process, outbound server, or inbound server.

Column	Datatype	Description
SID	NUMBER	Session ID of the apply server's session
SERIAL#	NUMBER	Serial number of the apply server's session
APPLY#	NUMBER	Apply process number. An apply process is an Oracle background process, prefixed by ap.
APPLY_NAME	VARCHAR2(128)	Name of the apply process
SERVER_ID	NUMBER	Parallel execution server number of the apply server

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATE	VARCHAR2(20)	State of the apply server: • INITIALIZING - Starting up • IDLE - Performing no work • RECORD LOW-WATERMARK - Performing an administrative job that maintains information about the apply progress, which is used in the ALL_APPLY_PROGRESS and DBA_APPLY_PROGRESS data dictionary views • ADD PARTITION - Performing an administrative job that adds a partition that is used for recording information about in-progress transactions • DROP PARTITION - Performing an administrative job that purges rows that were used to record information about in-progress transactions • EXECUTE TRANSACTION - Applying a transaction • WAIT COMMIT - Waiting to commit a transaction until all other transactions with a lower commit SCN are applied. This state is possible only if the COMMIT_SERIALIZATION apply process parameter is set to a value other than DEPENDENT_TRANSACTIONS and the PARALLELISM apply process parameter is set to a value greater than 1. • WAIT DEPENDENCY - Waiting to apply a logical change record (LCR) in a transaction until another transaction, on which it has a dependency, is applied. This state is possible only if the PARALLELISM apply process parameter is set to a value greater than 1. • ROLLBACK TRANSACTION - Rolling back a transaction • TRANSACTION CLEANUP - Cleaning up an applied transaction, which includes removing LCRs from the apply process's queue • WAIT FOR CLIENT - Waiting for an XStream client application to request more LCRs • WAIT FOR NEXT CHUNK - Waiting for the next set of LCRs for a large transaction
XIDUSN	NUMBER	Transaction ID undo segment number of the transaction currently being applied
XIDSLT	NUMBER	Transaction ID slot number of the transaction currently being applied
XIDSQN	NUMBER	Transaction ID sequence number of the transaction currently being applied
COMMITSCN	NUMBER	Commit system change number (SCN) of the transaction currently being applied
DEP_XIDUSN	NUMBER	Transaction ID undo segment number of a transaction on which the transaction being applied by this apply server depends
DEP_XIDSLT	NUMBER	Transaction ID slot number of a transaction on which the transaction being applied by this apply server depends
DEP_XIDSQN	NUMBER	Transaction ID sequence number of a transaction on which the transaction being applied by this apply server depends
DEP_COMMITSCN	NUMBER	Commit system change number (SCN) of the transaction on which this apply server depends
MESSAGE_SEQUENCE	NUMBER	Number of the current message being applied by the apply server. This value is reset to 1 at the beginning of each transaction.
TOTAL_ASSIGNED	NUMBER	Total number of transactions assigned to the apply server since the apply process was last started

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TOTAL_ADMIN	NUMBER	Total number of administrative jobs done by the apply server since the apply process was last started. See the STATE information in this view for the types of administrative jobs.
TOTAL_ROLLBACKS	NUMBER	Number of transactions assigned to this server which were rolled back
TOTAL_MESSAGES_APPLIED	NUMBER	Total number of messages applied by this apply server since the apply process was last started
APPLY_TIME	DATE	Time the last message was applied
APPLIED_MESSAGE_NUMBER	NUMBER	Number of the last message applied
APPLIED_MESSAGE_CREATE_TIME	DATE	Creation time at the source database of the last captured message applied. No information about user-enqueued messages is recorded in this column.
ELAPSED_DEQUEUE_TIME	NUMBER	Time elapsed (in hundredths of a second) dequeuing messages since the apply process was last started
ELAPSED_APPLY_TIME	NUMBER	Time elapsed (in hundredths of a second) applying messages since the apply process was last started
COMMIT_POSITION	RAW(64)	Commit position of the transaction. This column is populated only for an apply process that is functioning as an XStream inbound server.
DEP_COMMIT_POSITION	RAW(64)	Commit position of the transaction the worker depends on. This column is populated only for an apply process that is functioning as an XStream inbound server.
LAST_APPLY_POSITION	RAW(64)	For inbound servers, the position of the last message applied; for outbound servers, the position of the last message sent to the XStream client application. This column is populated only for an apply process that is functioning as an XStream outbound server or inbound server.
TRANSACTION_ID	VARCHAR2(128)	Transaction ID that the worker is applying. This column is populated only for an apply process that is functioning as an XStream inbound server.
DEP_TRANSACTION_ID	VARCHAR2(128)	Transaction ID of the transaction the worker depends on. This column is populated only for an apply process that is functioning as an XStream inbound server.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
TOTAL_LCRS_RETRIED	NUMBER	Total number of LCRs retried by this server
LCR_RETRY_ITERATION	NUMBER	Retry iteration for this transaction by this server
TOTAL_TXNS_RETRIED	NUMBER	Total transactions retried by this server
TXN_RETRY_ITERATION	NUMBER	Retry iteration for this transaction by this server
TOTAL_TXNS_RECORDED	NUMBER	Total transactions recorded in error queue by this server

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **Note**
>
> The ELAPSED_SCHEDULE_TIME column is only populated if the TIMED_STATISTICS initialization parameter is set to true, or if the STATISTICS_LEVEL initialization parameter is set to TYPICAL or ALL.

> **See Also**
>
> - "TIMED_STATISTICS"
> - "STATISTICS_LEVEL"

9.141 V$XSTREAM_CAPTURE

V$XSTREAM_CAPTURE displays information about each capture process that sends LCRs to an XStream outbound server.

Column	Datatype	Description
SID	NUMBER	Session identifier of the capture process
SERIAL#	NUMBER	Session serial number of the capture process session
CAPTURE#	NUMBER	Capture process number. A capture process is an Oracle background process prefixed by cp
CAPTURE_NAME	VARCHAR2(128)	Name of the capture process
LOGMINER_ID	NUMBER	Session ID of the Oracle LogMiner session associated with the capture process
STARTUP_TIME	DATE	Time when the capture process was last started

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
STATE	VARCHAR2(551)	State of the capture process: • ABORTING - Terminating. • CAPTURING CHANGES - Scanning the redo log for changes that satisfy the capture process rule sets. • CREATING LCR - Converting a change into an LCR. • DICTIONARY INITIALIZATION - Processing a dictionary build. • ENQUEUING MESSAGE - Enqueuing an LCR that satisfies the capture process rule sets into the capture process queue. • EVALUATING RULE - Evaluating a change against a capture process rule set. • INITIALIZING - Starting up. • PAUSED FOR FLOW CONTROL - Unable to enqueue LCRs either because of low memory or because propagations and outbound servers are consuming messages slower than the capture process is creating them. This state indicates flow control that is used to reduce spilling of captured LCRs when propagation or apply has fallen behind or is unavailable. • SHUTTING DOWN - Stopping. • SUSPENDED FOR AUTO SPLIT/MERGE - Waiting for a merge operation to complete. • WAITING FOR BUFFERED QUEUE TO SHRINK - Waiting for the buffered queue to change to a smaller size. The buffered queue shrinks when there is a memory limitation or when an administrator reduces its size. • WAITING FOR DICTIONARY REDO - Waiting for redo log files containing the dictionary build related to the first SCN to be added to the capture process session. A capture process cannot begin to scan the redo log files until all of the log files containing the dictionary build have been added. • WAITING FOR INACTIVE DEQUEUERS - Waiting for the capture process's queue subscribers to start. The capture process stops enqueuing LCRs if there are no active subscribers to the queue. • WAITING FOR *n* SUBSCRIBER(S) INITIALIZING - Waiting for outbound servers that receive LCRs from the capture process to start, where *n* is the number of apply processes. • WAITING FOR REDO - Waiting for new redo log files to be added to the capture process session. The capture process has finished processing all of the redo log files added to its session. This state is possible if there is no activity at a source database. For a downstream capture process, this state is possible if the capture process is waiting for new log files to be added to its session. • WAITING FOR TRANSACTION - Waiting for LogMiner to provide more transactions.
TOTAL_PREFILTER_DISCARDED	NUMBER	Total number of prefiltered messages discarded
TOTAL_PREFILTER_KEPT	NUMBER	Total number of prefiltered messages kept
TOTAL_PREFILTER_EVALUATIONS	NUMBER	Total number of prefilter evaluations
TOTAL_MESSAGES_CAPTURED	NUMBER	Total number of redo entries passed by LogMiner to the capture process for detailed rule evaluation since the capture process last started. A capture process converts a redo entry into a message and performs detailed rule evaluation on the message when capture process prefiltering cannot discard the change.
CAPTURE_TIME	DATE	Time when the most recent message was captured

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
CAPTURE_MESSAGE_NUMBER	NUMBER	Number of the most recently captured message
CAPTURE_MESSAGE_CREATE_TIME	DATE	Creation time of the most recently captured message
TOTAL_MESSAGES_CREATED	NUMBER	Count associated with ELAPSED_LCR_TIME to calculate rate
TOTAL_FULL_EVALUATIONS	NUMBER	Count associated with ELAPSED_RULE_TIME to calculate rate
TOTAL_MESSAGES_ENQUEUED	NUMBER	Total number of messages enqueued since the capture process was last started
ENQUEUE_TIME	DATE	Time when the last message was enqueued
ENQUEUE_MESSAGE_NUMBER	NUMBER	Number of the last enqueued message
ENQUEUE_MESSAGE_CREATE_TIME	DATE	Creation time of the last enqueued message
AVAILABLE_MESSAGE_NUMBER	NUMBER	For local capture, the last redo SCN flushed to the log files. For downstream capture, the last SCN added to LogMiner through the archived redo log files.
AVAILABLE_MESSAGE_CREATE_TIME	DATE	For local capture, the time the SCN was written to the log file. For downstream capture, the time the most recent archived redo log file (containing the most recent SCN) was added to LogMiner.
ELAPSED_CAPTURE_TIME	NUMBER	Elapsed time (in hundredths of a second) scanning for changes in the redo log since the capture process was last started
ELAPSED_RULE_TIME	NUMBER	Elapsed time (in hundredths of a second) evaluating rules since the capture process was last started
ELAPSED_ENQUEUE_TIME	NUMBER	Elapsed time (in hundredths of a second) enqueuing messages since the capture process was last started
ELAPSED_LCR_TIME	NUMBER	Elapsed time (in hundredths of a second) creating LCRs since the capture process was last started
ELAPSED_REDO_WAIT_TIME	NUMBER	Elapsed time (in hundredths of a second) spent by the capture process in the WAITING FOR REDO state
ELAPSED_PAUSE_TIME	NUMBER	Elapsed flow control pause time (in hundredths of a second)
STATE_CHANGED_TIME	DATE	Time at which the state of the capture process changed
SGA_USED	NUMBER	The total amount of shared memory (in bytes) currently used by the capture process out of the amount allocated (SGA_ALLOCATED)
SGA_ALLOCATED	NUMBER	The total amount of shared memory (in bytes) allocated from the Streams pool for the capture process
BYTES_OF_REDO_MINED	VARCHAR2(64)	The total amount of redo data mined (in bytes) since the capture process last started
SESSION_RESTART_SCN	VARCHAR2(64)	The SCN from which the capture process started mining redo data when it was last started
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

Note

The ELAPSED_CAPTURE_TIME, ELAPSED_RULE_TIME, ELAPSED_ENQUEUE_TIME, ELAPSED_LCR_TIME, and ELAPSED_REDO_WAIT_TIME columns are only populated if the TIMED_STATISTICS initialization parameter is set to true, or if the STATISTICS_LEVEL initialization parameter is set to TYPICAL or ALL.

See Also

- "TIMED_STATISTICS"
- "STATISTICS_LEVEL"

9.142 V$XSTREAM_MESSAGE_TRACKING

V$XSTREAM_MESSAGE_TRACKING displays information about LCRs tracked through the stream that are processed by XStream components.

You can track an LCR through a stream using one of the following methods:

- Set the message_tracking_frequency capture process parameter to 1 or another relatively low value for XStream Outbound processes.
- Set the message_tracking_frequency apply process parameter to 1 or another relatively low value for XStream Inbound processes.
- Use the DBMS_XSTREAM_ADM.SET_MESSAGE_TRACKING procedure to specify a tracking label that becomes part of each LCR generated by the current session.

  When the actions parameter in the DBMS_XSTREAM_ADM.SET_MESSAGE_TRACKING procedure is set to DBMS_XSTREAM_ADM.ACTION_MEMORY, information about the LCRs is tracked in memory, and this view is populated with information about the LCRs. Currently, DBMS_XSTREAM_ADM.ACTION_MEMORY is the only valid setting for the actions parameter in the procedure.

Note

This view does not display information about messages flowing in an Oracle GoldenGate configuration. To view information about such message streams, query the V$GOLDENGATE_MESSAGE_TRACKING view.

Column	Datatype	Description
TRACKING_LABEL	VARCHAR2(128)	User-specified tracking label
TAG	RAW(30)	First 30 bytes of the tag of the LCR
COMPONENT_NAME	VARCHAR2(128)	Name of the component that processed the LCR
COMPONENT_TYPE	VARCHAR2(128)	Type of the component that processed the LCR
ACTION	VARCHAR2(50)	Action performed on the LCR
ACTION_DETAILS	VARCHAR2(100)	Details of the action

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
TIMESTAMP	TIMESTAMP(9) WITH TIME ZONE	Time when the action was performed
MESSAGE_CREATION_TIME	DATE	Time when the message was created
MESSAGE_NUMBER	NUMBER	SCN of the message
TRACKING_ID	RAW(16)	Globally unique OID of the LCR
SOURCE_DATABASE_NAME	VARCHAR2(128)	Name of the source database
OBJECT_OWNER	VARCHAR2(128)	Owner of the object
OBJECT_NAME	VARCHAR2(128)	Name of the object
XID	VARCHAR2(128)	Transaction ID
COMMAND_TYPE	VARCHAR2(128)	Command type of the LCR
MESSAGE_POSITION	RAW(64)	Position of the message
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

- "V$GOLDENGATE_MESSAGE_TRACKING"
- *Oracle Database PL/SQL Packages and Types Reference* for more information about the DBMS_XSTREAM_ADM package
- *Oracle Database XStream Guide* for more information about XStream concepts

9.143 V$XSTREAM_OUTBOUND_SERVER

V$XSTREAM_OUTBOUND_SERVER displays statistics about an outbound server. An outbound server sends LCRs to the XStream client application.

Note

When the COMMITTED_DATA_ONLY column is YES in the V$XSTREAM_OUTBOUND_SERVER view, the V$STREAMS_APPLY_SERVER view provides additional information about the outbound server process, and information about the apply server background processes used by the outbound server.

Column	Datatype	Description
SID	NUMBER	Session ID of the outbound server's session
SERIAL#	NUMBER	Serial number of the outbound server's session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
SPID	VARCHAR2(12)	Process identification number of the operating-system process that sends LCRs to the client application
SERVER_NAME	VARCHAR2(128)	Name of the outbound server
STARTUP_TIME	DATE	Time when the client application attached to the outbound server
STATE	VARCHAR2(19)	State of the outbound server When the COMMITTED_DATA_ONLY column shows YES, the following states are possible: • INITIALIZING - Starting up the outbound server. • IDLE - Performing no work because there are no LCRs to send to the XStream client application. • GET TRANSACTIONS - Receiving transactions from the outbound server's apply coordinator. • SEND TRANSACTION - Sending a transaction to an XStream client application. • WAIT FOR NEXT CHUNK - Waiting for the next set of LCRs for a large transaction. • TRANSACTION CLEANUP - Cleaning up an applied transaction, which includes removing LCRs from the outbound server's queue. • WAIT FOR CLIENT - Waiting for an XStream client application to request more LCRs. When the COMMITTED_DATA_ONLY column shows NO, the following states are possible: • INITIALIZING - Starting up the outbound server. • INITIALIZING RULE EVALUATION CONTEXT - Initializing the context to evaluate the outbound server's rules. • IDLE - Performing no work because there is no LCR to send to the XStream client application. • BROWSING LCR - Browsing the outbound server's queue for next LCR. • EVALUATING RULES - Evaluating an LCR against a rule set. • DEQUEUING LCR - Dequeuing an LCR from the outbound server's queue. • SENDING LCR - Sending an LCR to an XStream client application. • WAITING FOR CAPTURE TO TERMINATE - Waiting for the capture process to become disabled. • SUSPENDED DUE TO A DROPPED SUBSCRIBER - Suspended because a connected subscriber was dropped. For example, a subscriber can be dropped during a split or merge operation. • SUSPENDED FOR AUTO SPLIT/MERGE - Suspended because an automatic split or merge operation is being performed. • WAITING ON EMPTY QUEUE - Waiting for more LCRs from the capture process. • WAITING FOR CLIENT - Waiting for the XStream client application to request more LCRs. • WAITING FOR CAPTURE TO INITIALIZE - Waiting for the capture process to finish the data dictionary build. • WAITING TO ATTACH TO CAPTURE - Waiting for the outbound server to attach to the capture process. When a state refers to a capture process, it is the capture process that captures changes for the outbound server. When a state refers to a propagation, it is the outbound server that sends LCRs to the XStream client application.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
XIDUSN	NUMBER	Transaction ID undo segment number of the transaction currently being processed. This column is populated only if the COMMITTED_DATA_ONLY column shows YES. When the COMMITTED_DATA_ONLY column shows NO, this column is NULL.
XIDSLT	NUMBER	Transaction ID slot number of the transaction currently being processed. This column is populated only if the COMMITTED_DATA_ONLY column shows YES. When the COMMITTED_DATA_ONLY column shows NO, this column is NULL.
XIDSQN	NUMBER	Transaction ID sequence number of the transaction currently being processed. This column is populated only if the COMMITTED_DATA_ONLY column shows YES. When the COMMITTED_DATA_ONLY column shows NO, this column is NULL.
COMMITSCN	NUMBER	Commit SCN of the transaction currently being processed. This column is populated only if the COMMITTED_DATA_ONLY column shows YES. When the COMMITTED_DATA_ONLY column shows NO, this column is NULL.
TOTAL_TRANSACTIONS_SENT	NUMBER	Total number of transactions sent by the outbound server to the XStream client application since the last time the client application attached to the outbound server. This column is populated only if the COMMITTED_DATA_ONLY column shows YES. When the COMMITTED_DATA_ONLY column shows NO, this column is NULL.
MESSAGE_SEQUENCE	NUMBER	Number of the current LCR being processed by the outbound server. This value is reset to 1 at the beginning of each transaction. This column is populated only if the COMMITTED_DATA_ONLY column shows YES. When the COMMITTED_DATA_ONLY column shows NO, this column is NULL.
TOTAL_MESSAGES_SENT	NUMBER	Total number of LCRs sent by the outbound server to the XStream client application since the last time the client application attached to the outbound server
SEND_TIME	DATE	Time the last LCR was sent by the outbound server to the XStream client application
LAST_SENT_MESSAGE_NUMBER	NUMBER	Message number of the last LCR sent by the outbound server to the XStream client application
LAST_SENT_MESSAGE_CREATE_TIME	DATE	Creation time at the source database of the last LCR sent by the outbound server to the client application
ELAPSED_SEND_TIME	NUMBER	Time elapsed (in hundredths of a second) sending LCRs to the XStream client application since the last time the client application attached to the outbound server
COMMIT_POSITION	RAW(64)	Commit position of the transaction currently being processed. This column is populated only if the COMMITTED_DATA_ONLY column shows YES. When the COMMITTED_DATA_ONLY column shows NO, this column is NULL.
LAST_SENT_POSITION	RAW(64)	Position of the last LCR sent to the XStream client application. This column is populated only if the COMMITTED_DATA_ONLY column shows YES. When the COMMITTED_DATA_ONLY column shows NO, this column is NULL.
BYTES_SENT	NUMBER	Total number of bytes sent by the outbound server to the XStream client application since the last time the client application attached to the outbound server

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
COMMITTED_DATA_ONLY	CHAR(3)	YES if the outbound server can send only LCRs in committed transactions to the XStream client application. A committed transaction is an assembled, noninterleaving transaction with no rollbacks. NO if the outbound server can send LCRs in transactions that have not yet committed to the XStream client application. This mode is for internal Oracle use only.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

See Also

"V$STREAMS_APPLY_SERVER"

9.144 V$XSTREAM_TABLE_STATS

V$XSTREAM_TABLE_STATS shows the statistics for all the tables processed by each apply server for the XStream session.

Column	Datatype	Description
APPLY_NAME	VARCHAR2(128)	Name of XStream Out or XStream In process
SERVER_ID	NUMBER	Parallel apply server worker ID. If the server ID is 0, then this is an aggregate of statistics for servers that may have been automatically shutdown due to session inactivity.
SOURCE_TABLE_OWNER	VARCHAR2(128)	Source owner of the captured or replicated table
SOURCE_TABLE_NAME	VARCHAR2(128)	Source name of the captured or replicated table
DESTINATION_TABLE_OWNER	VARCHAR2(128)	Target owner of the captured or replicated table
DESTINATION_TABLE_NAME	VARCHAR2(128)	Target name of the captured or replicated table
LAST_UPDATE	DATE	Time of last update
TOTAL_INSERTS	NUMBER	Number of insert operations on this table processed by this apply server for the current session
TOTAL_UPDATES	NUMBER	Number of update operations on this table processed by this apply server for the current session
TOTAL_DELETES	NUMBER	Number of delete operations on this table processed by this apply server for the current session
INSERT_COLLISIONS	NUMBER	Number of insert collisions on this table encountered by this apply server for the current session
UPDATE_COLLISIONS	NUMBER	Number of update collisions on this table encountered by this apply server for the current session
DELETE_COLLISIONS	NUMBER	Number of delete collisions on this table encountered by this apply server for the current session

Copyright © 2002, 2026, Oracle and/or its affiliates.

Column	Datatype	Description
REPERROR_RECORDS	NUMBER	Number of change records that were recorded on this table by this apply server for the current session
REPERROR_IGNORES	NUMBER	Number of ignored change records on this table by this apply server for the current session
WAIT_DEPENDENCIES	NUMBER	Number of waits for this table due to dependency for the current session
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data
CDR_INSERT_ROW_EXISTS	NUMBER	Number of conflicts where an insert gets an ORA-00001 error
CDR_UPDATE_ROW_EXISTS	NUMBER	Number of conflicts where an update gets an ORA-26786 error
CDR_UPDATE_ROW_MISSING	NUMBER	Number of conflicts where an update gets an ORA-26787 error
CDR_DELETE_ROW_EXISTS	NUMBER	Number of conflicts where a delete gets an ORA-26786 error
CDR_DELETE_ROW_MISSING	NUMBER	Number of conflicts where a delete gets an ORA-26787 error
CDR_SUCCESSFUL_RESOLUTIONS	NUMBER	Number of successfully resolved conflicts
CDR_FAILED_RESOLUTIONS	NUMBER	Number of conflicts that could not be resolved due to an error during resolution
LOB_OPERATIONS	NUMBER	The number of LOB updates (LOB writes, LOB trims, and LOB erases) applied by the inbound server.

See Also

Oracle Database XStream Guide for more information about XStream conflict detection and resolution

9.145 V$XSTREAM_TRANSACTION

V$XSTREAM_TRANSACTION displays information about transactions that are being processed by capture processes, outbound servers, and inbound servers. This view can identify long running transactions and to display how many LCRs are being processed in each transaction. This view only contains information about captured LCRs. It does not contain information about user-enqueued LCRs or user messages.
This view only shows information about LCRs that are being processed because they satisfied the rule sets for the component at the time of the query. For capture processes, this view only shows information about changes in transactions that the capture process has converted into LCRs. It does not show information about all the active transactions present in the redo log.

For outbound servers, this view only shows information about LCRs that the outbound server has dequeued. It does not show information about LCRs in the outbound server's queue. For outbound servers, information about a transaction remains in the view until the transaction is sent to the XStream client application.

Copyright © 2002, 2026, Oracle and/or its affiliates.

For inbound servers, information about a transaction remains in the view until the transaction commits or until the entire transaction is rolled back.

Column	Datatype	Description
COMPONENT_NAME	VARCHAR2(128)	Name of the component
COMPONENT_TYPE	VARCHAR2(20)	Type of component • CAPTURE - Capture process • APPLY - Apply reader subcomponent in an outbound server or inbound server • PROPAGATION_SENDER - Propagation sender that sends LCRs from a capture process to an outbound server
XIDUSN	NUMBER	Transaction ID undo segment number of the transaction
XIDSLT	NUMBER	Transaction ID slot number of the transaction
XIDSQN	NUMBER	Transaction ID sequence number of the transaction
CUMULATIVE_MESSAGE_COUNT	NUMBER	Number of LCRs processed in the transaction. If a component is restarted while the transaction is being processed, then this column shows the number of LCRs processed in the transaction since the component was started.
TOTAL_MESSAGE_COUNT	NUMBER	Total number of LCRs processed in the transaction by an outbound server or inbound server. This column does not pertain to capture processes.
FIRST_MESSAGE_TIME	DATE	Time stamp of the first LCR processed in the transaction. If a capture process is restarted while the transaction is being processed, then this column shows the time stamp of the first LCR processed after the capture process was started.
FIRST_MESSAGE_NUMBER	NUMBER	SCN of the first message in the transaction. If a capture process is restarted while the transaction is being processed, then this column shows the SCN of the first message processed after the capture process was started.
LAST_MESSAGE_TIME	DATE	Time stamp of the last LCR processed in the transaction
LAST_MESSAGE_NUMBER	NUMBER	SCN of the most recent message encountered for the transaction
FIRST_MESSAGE_POSITION	RAW(64)	Position of the first message seen by an XStream inbound server This column is populated only for an apply process that is functioning as an XStream inbound server.
LAST_MESSAGE_POSITION	RAW(64)	Position of the last message seen by an XStream inbound server. This column is populated only for an apply process that is functioning as an XStream inbound server.
TRANSACTION_ID	VARCHAR2(128)	Transaction ID for an XStream inbound server. This column is populated only for an apply process that is functioning as an XStream inbound server.
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

Copyright © 2002, 2026, Oracle and/or its affiliates.

9.146 V$ZONEMAP_USAGE_STATS

V$ZONEMAP_USAGE_STATS displays zone map usage statistics.

Column	Datatype	Description
ZONEMAP	VARCHAR2(257)	Name of the zone map in the form of *owner*.*name*
PRUNING_TYPE	VARCHAR2(11)	Type of data pruning performed using the zone map
EXECUTIONS	NUMBER	Number of executions, including parallel worker executions
BASE_COUNT	NUMBER	Base count accumulated over number of executions
PRUNED_COUNT	NUMBER	Pruned count accumulated over number of executions
CON_ID	NUMBER	The ID of the container to which the data pertains. Possible values include: • 0: This value is used for rows containing data that pertain to the entire CDB. This value is also used for rows in non-CDBs. • 1: This value is used for rows containing data that pertain to only the root • *n*: Where *n* is the applicable container ID for the rows containing data

The ratio (PRUNED_COUNT / BASE_COUNT) shows the fraction of data pruned by the zone map.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Part IV
Appendixes

This part includes the following appendixes:

- Database Limits
- SQL Scripts
- Oracle Wait Events
- Oracle Enqueue Names
- Statistics Descriptions
- Background Processes

Copyright © 2002, 2026, Oracle and/or its affiliates.

A
Database Limits

This appendix lists the limits of values associated with database functions and objects. Limits exist on several levels in the database. There is usually a hard-coded limit in the database that cannot be exceeded. This value may be further restricted for any given operating system. Database limits are divided into four categories:

- Datatype Limits
- Physical Database Limits
- Logical Database Limits
- Process and Runtime Limits

> **See Also**
>
> Your operating system-specific Oracle documentation

A.1 Datatype Limits

This table documents the limits for datatypes, and includes comments about the datatypes.

Datatypes	Limit	Comments
BFILE	Maximum size: 2^{64} bytes (18.44 exabytes) or OS file size limit, whichever is the lower value Maximum size of a file name: 255 characters Maximum size of a directory name: 128 bytes Maximum number of open BFILEs: see Comments	The maximum number of BFILEs is limited by the value of the SESSION_MAX_OPEN_FILES initialization parameter, which is itself limited by the maximum number of open files the operating system will allow.
BLOB	Maximum size: (4 GB - 1) * DB_BLOCK_SIZE initialization parameter (8 TB to 128 TB)	The number of LOB columns per table is limited only by the maximum number of columns per table (that is, 1000[1]).
CHAR	Maximum size: 2000 bytes	None
CLOB	Maximum size: (4 GB - 1) * DB_BLOCK_SIZE initialization parameter (8 TB to 128 TB)	The number of LOB columns per table is limited only by the maximum number of columns per table (that is, 1000[1]).
Literals (characters or numbers in SQL or PL/SQL)	Maximum size: 4000 characters	None
LONG[2]	Maximum size: 2 GB - 1	Only one LONG column is allowed per table.
NCHAR	Maximum size: 2000 bytes	None

Copyright © 2002, 2026, Oracle and/or its affiliates.

Datatypes	Limit	Comments
NCLOB	Maximum size: (4 GB - 1) * DB_BLOCK_SIZE initialization parameter (8 TB to 128 TB)	The number of LOB columns per table is limited only by the maximum number of columns per table (that is, 1000[1]).
NUMBER	999...(38 9's) x10^{125} maximum value -999...(38 9's) x10^{125} minimum value	Can be represented to full 38-digit precision (the mantissa) Can be represented to full 38-digit precision (the mantissa)
NVARCHAR2	Maximum size: 4000 bytes, or 32767 bytes if the MAX_STRING_SIZE initialization parameter is set to EXTENDED **See Also:** "MAX_STRING_SIZE" initialization parameter for additional details	None
Precision	38 significant digits	None
RAW	Maximum size: 2000 bytes, or 32767 bytes if the MAX_STRING_SIZE initialization parameter is set to EXTENDED **See Also:** "MAX_STRING_SIZE" initialization parameter for additional details	None
VARCHAR2	Maximum size: 4000 bytes, or 32767 bytes if the MAX_STRING_SIZE initialization parameter is set to EXTENDED **See Also:** "MAX_STRING_SIZE" initialization parameter for additional details	None

1 The absolute maximum number of columns in a table is 1000. However, when you create an object table (or a relational table with columns of object, nested table, varray, or REF type), Oracle maps the columns of the user-defined types to relational columns, creating in effect hidden columns that count toward the 1000-column limit. For details on how Oracle calculates the total number of columns in such a table, refer to *Oracle Database Administrator's Guide*.

2 All forms of LONG data types (LONG, LONG RAW, LONG VARCHAR, LONG VARRAW) were deprecated in Oracle8*i* Release 8.1.6. For succeeding releases, the LONG data type was provided for backward compatibility with existing applications. In new applications developed with later releases, Oracle strongly recommends that you use the CLOB or NCLOB data type for storing large amounts of character data.

See Also

- *Oracle Database SQL Language Reference* for more information about datatypes
- *Oracle Database SQL Language Reference* for more information on naming database objects

A.2 Physical Database Limits

This table describes limit types and limit values for physical database items.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Item	Type of Limit	Limit Value
Controlfiles	Number of control files	1 minimum; 2 or more (on separate devices) strongly recommended
Controlfiles	Size of a control file	Maximum of 201031680 logical blocks
Database Block Size	Minimum	2048 bytes; must be a multiple of operating system physical block size
Database Block Size	Maximum	Operating system dependent; never more than 32 KB
Database Blocks	Minimum in initial extent of a segment	2 blocks
Database Blocks	Maximum per datafile	Platform dependent; typically 2^{22} - 1 blocks
Database extents	Maximum per dictionary managed tablespace	4 GB * physical block size (with K/M modifier); 4 GB (without K/M modifier)
Database extents	Maximum per locally managed (uniform) tablespace	256 KB * physical block size
Database file size	Maximum	Operating system dependent. Limited by maximum operating system file size. See the **Bigfile Tablespaces** and **Smallfile (traditional) Tablespaces** rows for more information about the maximum database file size in these types of tablespaces.
Database files	Maximum per tablespace	Operating system dependent; usually 1023
Database files	Maximum per database	65533 May be less on some operating systems Limited also by size of database blocks and by the DB_FILES initialization parameter for a particular instance
External Tables file	Maximum size	Dependent on the operating system An external table can be composed of multiple files.
MAXEXTENTS	Default value	Derived from tablespace default storage or DB_BLOCK_SIZE initialization parameter
MAXEXTENTS	Maximum	Not defined
Redo Log File Size	Minimum size	4 MB
Redo Log File Size	Maximum Size	Operating system limit; typically 2 TB
Redo Log Files	Maximum number of logfiles	Limited by value of MAXLOGFILES parameter in the CREATE DATABASE statement Control file can be resized to allow more entries; ultimately an operating system limit
Redo Log Files	Maximum number of logfiles per group	Not defined
Tablespaces	Maximum number per database	64 K Number of tablespaces cannot exceed the number of database files because each tablespace must include at least one file
Bigfile Tablespaces	Number of blocks	A bigfile tablespace contains only one datafile or tempfile, which can contain up to approximately 4 billion (2^{32}) blocks. The maximum size of the single datafile or tempfile is 128 terabytes (TB) for a tablespace with 32 K blocks and 32 TB for a tablespace with 8 K blocks.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Item	Type of Limit	Limit Value
Smallfile (traditional) Tablespaces	Number of blocks	A smallfile tablespace is a traditional Oracle tablespace, which can contain 1023 datafiles or tempfiles, each of which can contain up to approximately 4 million (2^{22}) blocks.

A.3 Logical Database Limits

This table describes limit types and limit values for logical database items.

Item	Type of Limit	Limit Value
Columns	Maximum per table	1000
Columns	Maximum per index (or clustered index)	32
Columns	Maximum per bitmapped index	30
Constraints	Maximum per column	Not defined Oracle does not define a limit on the number of constraints per column. However, constraints are subject to the limit on the maximum number of constraints in the database. See the entry for "Constraints - Maximum per database" in this table.
Constraints	Maximum per database	4,294,967,293 Constraints are numbered internally from 1 to 4,294,967,294. The internal number is incremented each time a constraint is created, and numbers are never recycled. One number is always consumed by a special internal user called `_NEXT_CONSTRAINT`, which represents the next available number for a new constraint. Therefore, the maximum number of constraints that can be created during the lifetime of a database is 4,294,967,293. You can view constraint numbers by querying the `CON#` column of the `SYS.CON$` table. If your database has reached the constraint limit and you would like to create more constraints, contact Oracle Support.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Item	Type of Limit	Limit Value
Dictionary-managed database objects	Maximum per database	4,254,950,911 - overhead • The value of 4,254,950,911 is derived as follows: The data type used for object identifiers can store a maximum value of 4,294,967,294. However, approximately 40 million of these values are reserved at the high end for other purposes, which results in a maximum object identifier value of 4,254,950,911. • Overhead includes the following: – Multiple object identifiers for one database object Database objects consume one object identifier at the time of creation. This identifier is displayed in both the `OBJECT_ID` and `DATA_OBJECT_ID` columns of the ALL_OBJECTS view. Over the lifetime of an object, the value of `OBJECT_ID` remains unchanged, but the value of `DATA_OBJECT_ID` changes when the object is altered, for example, with a DDL statement or a table truncation. Therefore, one object can consume more than one object identifier over its lifetime. – Object identifiers for lightweight jobs Lightweight jobs consume object identifiers, even though they are not permanent objects. Therefore, if many lightweight jobs are running in the database, the number of objects that can be created in the database is dependent upon the remaining pool of identifiers.
Indexes	Maximum per table	Not defined Oracle does not define a limit on the number of indexes per table. However, indexes are subject to the limit on the maximum number of dictionary-managed database objects allowed per database. See the entry for "Dictionary-managed database objects" in this table.
Indexes	Total size of indexed column(s)	Approximately 75% of the database block size minus some overhead
Partitions	Maximum length of linear partitioning key	4 KB - overhead
Partitions	Maximum number of columns in partition key	16 columns
Partitions	Maximum number of partitions allowed per table or index	1024K - 1
Rows	Maximum number per table	Not defined
Stored Packages	Maximum size	Approximately `6,000,000` lines of code. **See Also:** *Oracle Database PL/SQL Language Reference* for details
Subpartitions	Maximum number of subpartitions in a composite partitioned table	1024K - 1

Copyright © 2002, 2026, Oracle and/or its affiliates.

Item	Type of Limit	Limit Value
Subqueries	Maximum levels of subqueries in a SQL statement	255 subqueries in the WHERE clause A limit is not defined for subqueries in the FROM clause of the top-level query.
System Change Numbers (SCNs)	Maximum	2^{63} - 2^{48} = 9,223,090,561,878,065,152 SCNs
Tables	Maximum per clustered table	32 tables
Tables	Maximum per database	Not defined Oracle does not define a limit on the number of tables per database. However, tables are subject to the limit on the maximum number of dictionary-managed database objects allowed per database. See the entry for "Dictionary-managed database objects" in this table.
Temporary Tablespaces	Maximum used per session	16
Users and Roles	Maximum	2,147,483,638 Users and roles are numbered internally from 0 to 2,147,483,638. The internal number is incremented each time a user or role is created, and numbers are never recycled. One number is always consumed by a special internal user called _NEXT_USER, which represents the next available number for a new user or role. Therefore, the maximum number of users and roles that can be created during the lifetime of a database is 2,147,483,638. You can view user and role numbers by querying the USER# column of the SYS.USER$ table.

Note

The limit on how long a SQL statement can be depends on many factors, including database configuration, disk space, and memory

Note

When an object instance exists in memory, there is no fixed limit on the number of attributes in the object. But the maximum total amount of memory consumed by an object instance is 4 GB. When an object instance is inserted into a table, the attributes are exploded into separate columns in the table, and the Oracle 1000-column limit applies.

A.4 Process and Runtime Limits

This table describes limit types and limit values for process and runtime items.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Item	Type of Limit	Limit Value
Advanced Queuing Processes	Maximum per instance	10
Backup Sessions	Maximum per instance	Limited by the PROCESSES and SESSIONS initialization parameters, for instance
Dispatchers	Maximum per instance	Limited by the PROCESSES and SESSIONS initialization parameters, for instance
Global Cache Service Processes	Maximum per instance	10
Instances per database	Maximum number of cluster database instances per database	Operating system-dependent
I/O Worker Processes	Maximum per background process (DBWR, LGWR, and so on)	15
I/O Worker Processes	Maximum per Backup session	15
Job Queue Processes	Maximum per instance	1000
Locks	Distributed Lock Manager	Operating system dependent
Locks	Row-level	Not defined
Parallel Execution Workers	Maximum per instance	Limited by the PROCESSES and SESSIONS initialization parameters, for instance
Services	Maximum per instance	8200
Sessions	Maximum per instance	2^{18} - 1, which is 262143; limited by the PROCESSES and SESSIONS initialization parameters.
SGA size	Maximum value	Operating system-dependent; typically 2 to 4 GB for 32-bit operating systems, and > 4 GB for 64-bit operating systems
Shared Servers	Maximum per instance	Limited by the PROCESSES and SESSIONS initialization parameters, for instance

Copyright © 2002, 2026, Oracle and/or its affiliates.

B
SQL Scripts

This appendix describes SQL scripts that are required for optimal operation of the Oracle Database.

The SQL scripts are described in the following sections:

- Creating the Data Dictionary
- Creating Additional Data Dictionary Structures
- The "NO" Scripts
- Upgrade Scripts
- Java Scripts

> **Note**
>
> Comments within the SQL scripts themselves contain more detailed information and examples.

> **Note**
>
> To run scripts in a pluggable database (PDB), connect to the PDB before running the script. For more information about running scripts in a multitenant container database (CDB), see *Oracle Multitenant Administrator's Guide*.

B.1 Creating the Data Dictionary

When you use the Database Configuration Assistant to create a database, Oracle automatically creates the data dictionary. Thereafter, whenever the database is in operation, Oracle updates the data dictionary in response to every DDL statement.
The data dictionary base tables are the first objects created in any Oracle database. They are created in the `SYSTEM` tablespace and must remain there. The data dictionary base tables store information about all user-defined objects in the database.

Table B-1 lists required scripts, which are run automatically when you create a database using the Database Configuration Assistant. They are described here because you might need to run them if you create a database manually. To run these scripts, you must be connected to Oracle as a user with `SYSDBA` privileges.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table B-1 Creating the Data Dictionary Scripts

Script Name	Needed For	Description
`catalog.sql`	All databases	Creates the data dictionary and public synonyms for many of its views Grants `PUBLIC` access to the synonyms
`catproc.sql`	All databases	Runs all scripts required for, or used with, PL/SQL
`catpcat.sql`[1]	All databases	Runs the `catalog.sql` and `catproc.sql` scripts as parallel processes, which can speed up database creation **Note:** You must run `catpcat.sql` by using the `catctl.pl` program. See *Oracle Database Administrator's Guide* for more information.
`catclust.sql`	Oracle Real Application Clusters	Creates Oracle Real Application Clusters data dictionary views

1 This script is available starting with Oracle Database 19c.

See Also

- Your operating system-specific Oracle documentation for the exact names and locations of these scripts on your operating system
- *Oracle Database Administrator's Guide* for more information about creating a database
- *Oracle Real Application Clusters Administration and Deployment Guide* for more information about creating Oracle Real Application Cluster views

B.2 Creating Additional Data Dictionary Structures

Oracle Database supplies other scripts that create additional structures you can use in managing your database and creating database applications. These scripts are listed in Table B-2.

See Also

Your operating system-specific Oracle documentation for the exact names and locations of these scripts on your operating system

Table B-2 Creating Additional Data Dictionary Structures

Script Name	Needed For	Run By	Description
`catblock.sql`	Performance management	SYS	Creates views that can dynamically display lock dependency graphs **See Also:** *Oracle Database Administrator's Guide*

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table B-2 (Cont.) Creating Additional Data Dictionary Structures

Script Name	Needed For	Run By	Description
`caths.sql`	Heterogeneous Services	SYS	Installs packages for administering heterogeneous services **See Also:** *Oracle Database Gateway for APPC Installation and Configuration Guide for Microsoft Windows*
`catio.sql`	Performance management	SYS	Allows I/O to be traced on a table-by-table basis
`catqueue.sql`	Advanced Queuing		Creates the dictionary objects required for Advanced Queuing
`catrep.sql`	Oracle Replication	SYS	Runs all SQL scripts for enabling database replication
`catwrr.sql`	Database Replay	SYS	Main script that creates the entire schema related to Database Replay - calls the create schema scripts for Workload Capture, Database Replay, Workload Replay, and Workload Intelligence
`catwrrwitb.sql`	Database Replay	SYS	Creates the schema for Workload Intelligence
`dbmsiotc.sql`	Storage management	Any user	Analyzes chained rows in index-organized tables
`dbmspool.sql`	Performance management	SYS or SYSDBA	Enables DBA to lock PL/SQL packages, SQL statements, and triggers into the shared pool **See Also:** *Oracle Database PL/SQL Packages and Types Reference*
`userlock.sql`	Concurrency control	SYS or SYSDBA	Provides a facility for user-named locks that can be used in a local or clustered environment to aid in sequencing application actions
`utlbstat.sql` and `utlestat.sql`	Performance monitoring	SYS	Respectively start and stop collecting performance tuning statistics
`utlchn1.sql`	Storage management	Any user	For use with the Oracle Database. Creates tables for storing the output of the `ANALYZE` command with the `CHAINED ROWS` option. Can handle both physical and logical rowids. **See Also:** *Oracle Database Administrator's Guide*
`utlconst.sql`	Year 2000 compliance	Any user	Provides functions to validate that `CHECK` constraints on date columns are year 2000 compliant
`utldtree.sql`	Metadata management	Any user	Creates tables and views that show dependencies between objects **See Also:** *Oracle Database SecureFiles and Large Objects Developer's Guide*
`utlexpt1.sql`	Constraints	Any user	For use with the Oracle Database. Creates the default table (`EXCEPTIONS`) for storing exceptions from enabling constraints. Can handle both physical and logical rowids. **See Also:** *Oracle Database Administrator's Guide*

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table B-2 (Cont.) Creating Additional Data Dictionary Structures

Script Name	Needed For	Run By	Description
utlfixdirs.sql[1]	Moving a database to a new Oracle home	SYS	Used after moving a database to a new Oracle home. Updates directory objects to use the new path names for the Oracle home and Oracle base directories, as defined by the new values for the ORACLE_HOME and ORACLE_BASE environment variables. When you run this script in the root container of a CDB, it updates directory objects in the root, as well as any Oracle-maintained directory objects in the PDBs; you must manually update any PDB directory objects that are not Oracle-maintained.
utlip.sql	PL/SQL	SYS	Used primarily for upgrade and downgrade operations. It invalidates all existing PL/SQL modules by altering certain dictionary tables so that subsequent recompilations will occur in the format required by the database. It also reloads the packages STANDARD and DBMS_STANDARD, which are necessary for any PL/SQL compilations.
utlirp.sql	PL/SQL	SYS	Used to change from 32-bit to 64-bit word size or vice versa. This script recompiles existing PL/SQL modules in the format required by the new database. It first alters some data dictionary tables. Then it reloads the packages STANDARD and DBMS_STANDARD, which are necessary for using PL/SQL. Finally, it triggers a recompilation of all PL/SQL modules, such as packages, procedures, and types.
utllockt.sql	Performance monitoring	SYS or SYSDBA	Displays a lock wait-for graph, in tree structure format **See Also:** *Oracle Database Administrator's Guide*
utlpwdmg.sql	Security	SYS or SYSDBA	Creates PL/SQL functions for default password complexity verification. Sets the default password profile parameters and enables password management features. **See Also:** *Oracle Database Security Guide*
utlrp.sql	PL/SQL	SYS	Recompiles all existing PL/SQL modules that were previously in an INVALID state, such as packages, procedures, and types.
utlsampl.sql	Examples	SYS or any user with DBA role	Creates sample tables, such as emp and dept, and users, such as scott
utlscln.sql	Oracle Replication	Any user	Copies a snapshot schema from another snapshot site
utltkprf.sql	Performance management	SYS	Creates the TKPROFER role to allow the TKPROF profiling utility to be run by non-DBA users
utlvalid.sql	Partitioned tables	Any user	Creates tables required for storing output of ANALYZE TABLE ... VALIDATE STRUCTURE of a partitioned table

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table B-2 (Cont.) Creating Additional Data Dictionary Structures

Script Name	Needed For	Run By	Description
`utlxplan.sql`	Performance management	Any user	Creates the table `PLAN_TABLE`, which holds output from the `EXPLAIN PLAN` statement See Also: *Oracle Database Data Warehousing Guide*

[1] This script is available starting with Oracle Database 19c.

B.3 The "NO" Scripts

These scripts are used to remove dictionary information for various optional services or components.

Table B-3 The NO Scripts

Script Name	Needed For	Run By	Description
`catnoadt.sql`	Objects	SYS	Drops views and synonyms on dictionary metadata that relate to object types
`catnoaud.sql`	Security	SYS	Drops views and synonyms on auditing metadata
`catnoclust.sql`	Real Application Clusters	SYS	Drops views that are specific to Oracle Real Application Clusters (Oracle RAC)
`catnodrdaas.sql`	DRDA Application Server	SYS	Drops all DRDA Application Server objects from the database
`catnogwm.sql`	Global Data Services	SYS	Drops users, roles, and other objects created for Global Data Services.
`catnohs.sql`	Heterogeneous Services	SYS	Removes Heterogeneous Services dictionary metadata
`catnojav.sql`	Java	SYS	Drops the RDBMS Java classes and system triggers created by the `catjava.sql` script.
`catnoprt.sql`	Partitioning	SYS	Drops views and synonyms on dictionary metadata that relate to partitioned tables and indexes
`catnosvm.sql`	Server Manager	SYS	Removes Oracle7 Server Manager views and synonyms
`catnowrr.sql`	Database Replay	SYS	Main script that drops the entire schema related to Database Replay - calls the drop schema scripts for Workload Capture, Database Replay, Workload Replay, and Workload Intelligence
`catnowrrc.sql`	Database Replay	SYS	Drops the schema related to Workload Capture
`catnowrrp.sql`	Database Replay	SYS	Drops the schema related to Workload Replay
`catnowrrwitb.sql`	Database Replay	SYS	Drops the schema related to Workload Intelligence
`catnsnmp.sql`	Distributed management	SYS	Drops the DBSNMP user and SNMPAGENT role

Copyright © 2002, 2026, Oracle and/or its affiliates.

B.4 Upgrade Scripts

Upgrade scripts are used when upgrading to another release of Oracle.

> **See Also**
>
> *Oracle Database Upgrade Guide* for information about upgrade scripts

B.5 Java Scripts

The Java scripts are useful only if the JServer option is installed.

Table B-4 Java Scripts

Script Name	Description
`initjvm.sql`	Initializes JServer by installing core Java class libraries and Oracle-specific Java classes
`rmjvm.sql`	Removes all elements of the JServer **See Also:** *Oracle Database Java Developer's Guide*
`catjava.sql`	Installs Java-related packages and classes

Copyright © 2002, 2026, Oracle and/or its affiliates.

C

Oracle Wait Events

This appendix describes Oracle wait events.

It includes these topics:

- Classes of Wait Events
- Descriptions of Common Wait Event Parameters
- Descriptions of Wait Events

Information about wait events is displayed in three dynamic performance views:

- `V$SESSION_WAIT` displays the events for which sessions have just completed waiting or are currently waiting.
- `V$SYSTEM_EVENT` displays the total number of times all the sessions have waited for the events in that view.
- `V$SESSION_EVENT` is similar to `V$SYSTEM_EVENT`, but displays all waits for each session.

> **See Also**
>
> "V$SESSION_WAIT", "V$SYSTEM_EVENT", and "V$SESSION_EVENT"

Many of these wait events are tied to the internal implementation of Oracle and therefore are subject to change or deletion without notice. Application developers should be aware of this and write their code to tolerate missing or extra wait events.

The following SQL statement displays an alphabetical list of all Oracle wait events and the wait class to which they belong:

```
SQL> SELECT name, wait_class FROM V$EVENT_NAME ORDER BY name;
```

C.1 Classes of Wait Events

Every wait event belongs to a class of wait event.

The following list describes each of the wait classes.

Administrative

Waits resulting from DBA commands that cause users to wait (for example, an index rebuild)

Application

Waits resulting from user application code (for example, lock waits caused by row level locking or explicit lock commands)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Cluster

Waits related to Oracle Real Application Clusters resources (for example, global cache resources such as 'gc cr block busy')

Commit

This wait class only comprises one wait event - wait for redo log write confirmation after a commit (that is, 'log file sync')

Concurrency

Waits for internal database resources (for example, latches)

Configuration

Waits caused by inadequate configuration of database or instance resources (for example, undersized log file sizes, shared pool size)

Idle

Waits that signify the session is inactive, waiting for work (for example, 'SQL*Net message from client')

Network

Waits related to network messaging (for example, 'SQL*Net more data to dblink')

Other

Waits which should not typically occur on a system (for example, 'wait for EMON to spawn')

Queueing

Contains events that signify delays in obtaining additional data in a pipelined environment. The time spent on these wait events indicates inefficiency or other problems in the pipeline. It affects features such as parallel queries or `DBMS_PIPE` PL/SQL packages.

Scheduler

Resource Manager related waits (for example, 'resmgr: become active')

System I/O

Waits for background process I/O (for example, DBWR wait for 'db file parallel write')

User I/O

Waits for user I/O (for example 'db file sequential read')

C.2 Descriptions of Common Wait Event Parameters

This section describes some of the common wait event parameters.

block#

This is the block number of the block for which Oracle needs to wait. The block number is relative to the start of the file. To find the object to which this block belongs, issue the following SQL statement:

Copyright © 2002, 2026, Oracle and/or its affiliates.

```
select segment_name, segment_type, owner, tablespace_name
from dba_extents
where file_id = file#
     and block#
     between block_id and block_id + blocks - 1;
```

blocks

The number of blocks that is being either read from or written to the file. The block size depends on the file type:

- Database files have a block size of `DB_BLOCK_SIZE`
- Logfiles and control files have a block size that is equivalent to the physical block size of the platform

break?

If the value for this parameter equals 0, a reset was sent to the client. A nonzero value indicates that a break was sent to the client.

class

The class of the block describes how the contents of the block are used. For example, class 1 represents data block, and class 2 represents sort block. Use this SQL query to see the block classes:

```
select view_definition
from v$fixed_view_definition
where view_name='GV$WAITSTAT';
```

dba

The initials "dba" represents the data block address, which consists of a file number and a block number.

driver id

The address of the disconnect function of the driver that is currently being used.

file#

The following query returns the name of the database file:

```
select *
from v$datafile
where file# = file#;
```

id1

The first identifier (*id1*) of the enqueue or global lock takes its value from P2 or P2RAW. The meaning of the identifier depends on the name (P1).

id2

The second identifier (*id2*) of the enqueue or global lock takes its value from P3 or P3RAW. The meaning of the identifier depends on the name (P1).

Copyright © 2002, 2026, Oracle and/or its affiliates.

le

The relative index number into V$GC_ELEMENT.

mode

The *mode* is usually stored in the low order bytes of P1 or P1RAW and indicates the mode of the enqueue or global lock request. This parameter has one of the following values:

Table C-1 Lock Mode Values

Mode Value	Description
1	Null mode
2	Sub-Share
3	Sub-Exclusive
4	Share
5	Share/Sub-Exclusive
6	Exclusive

Use the following SQL statement to retrieve the name of the lock and the mode of the lock request:

```
select chr(bitand(p1,-16777216)/16777215)||
       chr(bitand(p1, 16711680)/65535) "Lock",
       bitand(p1, 65535) "Mode"
from v$session_wait
where event = 'DFS enqueue lock acquisition';
```

name and type

The name or "type" of the enqueue or global lock can be determined by looking at the two high order bytes of P1 or P1RAW. The name is always two characters. Use the following SQL statement to retrieve the lock name.

```
select chr(bitand(p1,-16777216)/16777215)||
      chr(bitand(p1,16711680)/65535) "Lock"
from v$session_wait
where event = 'DFS enqueue lock acquisition';
```

namespace

The name of the object namespace as it is displayed in V$DB_OBJECT_CACHE view.

requests

The number of I/Os that are "requested." This differs from the number of blocks in that one request could potentially contain multiple blocks.

session#

The number of the inactive session. Use the following SQL statement to find more information about the session:

```
select *
from v$session
where sid = session#;
```


Copyright © 2002, 2026, Oracle and/or its affiliates.

waited

This is the total amount of time the session has waited for this session to terminate.

C.3 Descriptions of Wait Events

This section provides detailed descriptions for those wait events of greatest interest.

Where appropriate, pointers are provided to further information elsewhere in Oracle Database documentation. For a complete listing of wait events, in alphabetical order, you can issue the following SQL statement:

```
SQL> SELECT name FROM V$EVENT_NAME ORDER BY name;
```

C.3.1 alter system set dispatcher

A session has issued a statement `ALTER SYSTEM SET DISPATCHER = string` and is waiting for the dispatchers to get started.

Wait Time: The session will wait 1 / 100 of a second and verify if the new dispatchers have started, else the session will wait again

Parameter	Description
waited	Number of times that the session has waited 1 / 100 of a second

C.3.2 ARCH Remote Write

Used to track the time (in centiseconds) that ARC*n* background processes spend blocked waiting for network write operations to complete.

C.3.3 ASYNC Remote Write

Used to track the time (in centiseconds) for asynchronous streaming RFSWRITE operations. This includes stall reaps and streaming network submission time. This time is accumulated by TT*nn* (Redo Transport Worker) background processes.

> **See Also**
>
> *Oracle Data Guard Concepts and Administration* for more information about using RFS to manage standby redo logs.

C.3.4 BFILE check if exists

The session waits to check if an external large object (LOB) exists.

Wait Time: The total elapsed time for the **exists** call

Parameter	Description
session#	See "session#"

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
waited	See "waited"

C.3.5 BFILE check if open

The session waits for an external large object (LOB) to open.

Wait Time: The total elapsed time for the **isopen** call

Parameter	Description
session#	See "session#"
waited	See "waited"

C.3.6 BFILE closure

The session waits for an external large object (LOB) to close.

Wait Time: The total elapsed time for the **close** call

Parameter	Description
session#	See "session#"
waited	See "waited"

C.3.7 BFILE get length

The session waits on a call to check the size of an external large object (LOB).

Wait Time: The total elapsed time for the call to check the LOB size

Parameter	Description
session#	See "session#"
waited	See "waited"

C.3.8 BFILE get name object

The session waits on a call to find or generate the external name of a external large object.

Wait Time: The total elapsed time for **make external file name** to complete

Parameter	Description
session#	See "session#"
waited	See "waited"

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.9 BFILE get path object

The session is waiting on a call to find or generate the external path name of an external large object (LOB).

Wait Time: The total elapsed time for **make external path** to complete

Parameter	Description
session#	See "session#"
waited	See "waited"

C.3.10 BFILE internal seek

The session waits for a positioning call within the external large object (LOB) to complete.

Wait Time: The total elapsed time for the **seek** to complete

Parameter	Description
session#	See "session#"
waited	See "waited"

C.3.11 BFILE open

The session waits for an external large object (LOB) to open.

Wait Time: The total elapsed time for the **isopen** call

Parameter	Description
session#	See "session#"
waited	See "waited"

C.3.12 BFILE read

The session waits for a read from a external large object (LOB) to complete.

Wait Time: The total elapsed time for the **read** to complete

Parameter	Description
session#	See "session#"
waited	See "waited"

C.3.13 broadcast mesg queue transition

Processes enter "wait for broadcast mesg queue transition" when cleaning up a publisher channel handle to a RELIABLE broadcast channel. The publisher is responsible for moving the message to the free queue, but it cannot do so until the message is in the done queue. If the message is still not in the done queue, process enters this wait. This wait event will most likely

Copyright © 2002, 2026, Oracle and/or its affiliates.

appear when an Oracle process is about to exit normally, or when PMON cleans up a terminated process.

Wait Time: Varies

Parameter	Description
channel handle	publisher channel handle pointer
message	broadcast message pointer
location	A number indicating the function in KSR where the process is waiting

C.3.14 broadcast mesg recovery queue transition

Processes enter "wait for broadcast mesg recovery queue transition" when cleaning up a publisher channel handle to a RELIABLE broadcast channel. The broadcasted message is in the recovery queue of another channel handle (for example, ch2). Process enters this wait, if the message is yet to be removed from the recovery queue of the ch2 channel handle. This wait event will most likely appear when an Oracle process is about to exit normally, or when PMON cleans up a terminated process.

Wait Time: Varies

Parameter	Description
channel handle	Publisher channel handle pointer
message	Broadcast message pointer
location	A number indicating the function in KSR where the process is waiting

C.3.15 buffer busy waits

Wait until a buffer becomes available.

There are four reasons that a session cannot pin a buffer in the buffer cache, and a separate wait event exists for each reason:

1. "buffer busy waits": A session cannot pin the buffer in the buffer cache because another session has the buffer pinned.
2. "read by other session": A session cannot pin the buffer in the buffer cache because another session is reading the buffer from disk.
3. "gc buffer busy acquire": A session cannot pin the buffer in the buffer cache because another session is reading the buffer from the cache of another instance.
4. "gc buffer busy release": A session cannot pin the buffer in the buffer cache because another session on another instance is taking the buffer from this cache into its own cache so it can pin it.

Prior to release 10.1, all four reasons were covered by "buffer busy waits." In release 10.1, the "gc buffer busy" wait event covered both the "gc buffer busy acquire" and "gc buffer busy release" wait events.

Wait Time: Normal wait time is 1 second. If the session was waiting for a buffer during the last wait, then the next wait will be 3 seconds.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about the potential causes of the buffer busy waits wait event

C.3.16 buffer deadlock

Oracle does not really wait on this event; the foreground only yields the CPU. Thus, the chances of catching this event are very low. This is not an application induced deadlock, but an assumed deadlock by the cache layer. The cache layer cannot get a buffer in a certain mode within a certain amount of time.

Wait Time: 0 seconds. The foreground process only yields the CPU and will usually be placed at the end of the CPU run queue.

Parameter	Description
class	See "class"
mode	See "mode"
flag	The flag points to the internal flags used by the session to get this block
dba	See "dba"

C.3.17 buffer latch

The session waits on the buffer hash chain latch. Primarily used in the dump routines.

Wait Time: 1 second

Parameter	Description

latch addr	The virtual address in the SGA where this latch is located. Use the following statement to find the name of this latch:
	`select *`
	`from v$latch a, v$latchname b`
	`where addr = latch addr`
	`and a.latch# = b.latch#;`

chain#	The index into array of buffer hash chains. When the chain is 0xffffffff, the foreground waits on the LRU latch.

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.18 buffer read retry

This event occurs only if the instance is mounted in shared mode (Oracle Real Application Cluster).
During the read of the buffer, the contents changed. This means that either:

- The version number, dba, or the incarnation and sequence number stored in the block no longer match
- The checksum on the block does not match the checksum in the block

The block will be reread (this may fail up to 3 times), then corruption is assumed and the corrupt block is dumped in the trace file.

Wait Time: The wait time is the elapsed time of the read

Parameter	Description
file#	See "file#"
block#	See "block#"

C.3.19 checkpoint completed

A session waits for a checkpoint to complete. This could happen, for example, during a close database or a local checkpoint.

Wait Time: 5 seconds

Parameters: None

C.3.20 cleanup of aborted process

When a process spawn is terminated while the process spawning background is in the middle of spawning, the current session must wait until the pid of the new process is filled in. Once the pid is filled in, then the process spawn can be actually terminated.

Wait Time: Usually 3 seconds

Parameter	Description
location	Location of the wait

C.3.21 control file parallel write

This event occurs while the session is writing physical blocks to all control files. This happens when:

- The session starts a control file transaction (to ensure the control files are up to date in case the session crashes before committing the control file transaction)
- The session commits a transaction to a control file
- Changing a generic entry in the control file, the new value is being written to all control files

Wait Time: The wait time is the time it takes to finish all writes to all control files

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
files	The number of control files to which the session is writing
blocks	The number of blocks that the session is writing to the control file
requests	The number of I/O requests which the session wants to write

C.3.22 control file sequential read

Reading from the control file. This happens often. For example, while:

- Making a backup of the control files
- Sharing information (between instances) from the control file
- Reading other blocks from the control files
- Reading the header block

Wait Time: The wait time is the elapsed time of the read

Parameter	Description
file#	The control file from which the session is reading
block#	Block number in the control file from where the session starts to read. The block size is the physical block size of the port (usually 512 bytes, some UNIX ports have 1 or 2 Kilobytes).
blocks	The number of blocks that the session is trying to read

C.3.23 control file single write

This wait is signaled while the control file's shared information is written to disk. This is an atomic operation protected by an enqueue (CF), so that only one session at a time can write to the entire database.

Wait Time: The wait time is the elapsed time of the write

Parameter	Description
file#	This identifies the control file to which the session is currently writing
block#	Block number in the control file where the write begins. The block size is the as the physical block size of the port (usually 512 bytes, some UNIX ports have 1 or 2 Kilobytes).
blocks	The number of blocks that the session is trying to write

C.3.24 cursor: mutex S

A session waits on this event when it is requesting a mutex in shared mode, when another session is currently holding a this mutex in exclusive mode on the same cursor object.

Parameter	Description
P1	Hash value of cursor
P2	Mutex value (top 2 bytes contain SID holding mutex in exclusive mode, and bottom two bytes usually hold the value 0)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
P3	Mutex where (an internal code locator) OR'd with Mutex Sleeps

C.3.25 cursor: mutex X

The session requests the mutex for a cursor object in exclusive mode, and it must wait because the resource is busy. The mutex is busy because either the mutex is being held in exclusive mode by another session or the mutex is being held shared by one or more sessions. The existing mutex holder(s) must release the mutex before the mutex can be granted exclusively.

Parameter	Description
P1	Hash value of cursor
P2	Mutex value (top 2 bytes contain SID holding mutex in exclusive mode, and bottom two bytes usually hold the value 0)
P3	Mutex where (an internal code locator) OR'd with Mutex Sleeps

C.3.26 cursor: pin S

A session waits on this event when it wants to update a shared mutex pin and another session is currently in the process of updating a shared mutex pin for the same cursor object. This wait event should rarely be seen because a shared mutex pin update is very fast.

Wait Time: Microseconds

Parameter	Description
P1	Hash value of cursor
P2	Mutex value (top 2 bytes contains SID holding mutex in exclusive mode, and bottom two bytes usually hold the value 0)
P3	Mutex where (an internal code locator) OR'd with Mutex Sleeps

C.3.27 cursor: pin S wait on X

A session waits for this event when it is requesting a shared mutex pin and another session is holding an exclusive mutex pin on the same cursor object.

Wait Time: Microseconds

Parameter	Description
P1	Hash value of cursor
P2	Mutex value (top 2 bytes contains SID holding mutex in exclusive mode, and bottom two bytes usually hold the value 0)
P3	Mutex where (an internal code locator) OR'd with Mutex Sleeps

C.3.28 cursor: pin X

A session waits on this event when it is requesting an exclusive mutex pin for a cursor object and it must wait because the resource is busy. The mutex pin for a cursor object can be busy

Copyright © 2002, 2026, Oracle and/or its affiliates.

either because a session is already holding it exclusive, or there are one or more sessions which are holding shared mutex pin(s). The exclusive waiter must wait until all holders of the pin for that cursor object have released it, before it can be granted.

Wait Time: Microseconds

Parameter	Description
P1	Hash value of cursor
P2	Mutex value (top 2 bytes contains SID holding mutex in exclusive mode, and bottom two bytes usually hold the value 0)
P3	Mutex where (an internal code locator) OR'd with Mutex Sleeps

C.3.29 db file async I/O submit

When asynchronous I/O is available, this wait event captures any time spent in submitting I/Os to the underlying storage.

> **See Also**
>
> "db file parallel write"

C.3.30 db file parallel read

This happens during recovery. It can also happen during buffer prefetching, as an optimization (rather than performing multiple single-block reads). Database blocks that must be changed as part of recovery are read in parallel from the database.

Wait Time: Wait until all of the I/Os are completed

Parameter	Description
files	This indicates the number of files to which the session is reading
blocks	This indicates the total number of blocks to be read
requests	This indicates the total number of I/O requests, which will be the same as blocks

C.3.31 db file parallel write

This event occurs in the DBWR. It indicates the time that DBWR spends waiting for I/O completion.

If asynchronous I/O is available, then the db file async I/O submit wait event captures any time spent in submitting I/Os to the underlying storage.

When asynchronous I/O is not available, db file parallel write captures the time spent during submit and reap.

Wait Time: While there are outstanding I/Os, DBWR waits for some of the writes to complete. DBWR does not wait for all of the outstanding I/Os to complete.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
requests	This indicates the total number of I/O requests, which will be the same as blocks
interrupt	
timeout	This indicates the timeout value in hundredths of a second to wait for the I/O completion.

See Also

- *Oracle Database Performance Tuning Guide* about how this wait event can help identify I/O problems
- "db file async I/O submit"

C.3.32 db file scattered read

Similar to the “db file sequential read” wait event, except that the session is reading multiple data blocks.

Wait Time: The wait time is the actual time it takes to do all of the I/Os

Parameter	Description
file#	See "file#"
block#	See "block#"
blocks	The number of blocks that the session is trying to read from the *file#* starting at *block#*

See Also

- *Oracle Database Performance Tuning Guide* for more information about this wait event
- *Oracle Database Performance Tuning Guide* about how this wait event can help identify I/O problems
- *Oracle Database Performance Tuning Guide* about potential causes for this wait event

C.3.33 db file sequential read

The session waits while a sequential read from the database is performed. This event is also used for rebuilding the control file, dumping data file headers, and getting the database file headers.

Wait Time: The wait time is the actual time it takes to do the I/O

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
file#	See "file#"
block#	See "block#"
blocks	This is the number of blocks that the session is trying to read (should be 1)

See Also

- *Oracle Database Performance Tuning Guide* for more information about this wait event
- *Oracle Database Performance Tuning Guide* for more information about how this wait event can help identify I/O problems
- *Oracle Database Performance Tuning Guide* about potential causes for this wait event

C.3.34 db file single write

This event is used to wait for the writing of the file headers.

Wait Time: The wait time is the actual time it takes to do the I/O

Parameter	Description
file#	See "file#"
block#	See "block#"
blocks	This is the number of blocks that the session is trying to write in *file# starting at block#*

See Also

- *Oracle Database Performance Tuning Guide* about how this wait event can help identify I/O problems

C.3.35 DFS db file lock

This event occurs only for the DBWR in Real Application Clusters. Each DBWR of every instance holds a global lock on each file in shared mode. The instance that is trying to offline the file will escalate the global lock from shared to exclusive. This signals the other instances to synchronize their SGAs with the control file before the file can be taken offline.
The name of this lock is **DF** (see Oracle Enqueue Names for more information).

Wait Time: 1 second in loop. The DBWR is waiting in a loop (sleep, check) for the other instances to downgrade to NULL mode. During this time, the DBWR cannot perform other tasks such as writing buffers.

Parameter	Description
file	See "file#"

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.36 DFS lock handle

The session waits for the lock handle of a global lock request. The lock handle identifies a global lock. With this lock handle, other operations can be performed on this global lock (to identify the global lock in future operations such as conversions or release). The global lock is maintained by the DLM.

Wait Time: The session waits in a loop until it has obtained the lock handle from the DLM. Inside the loop there is a wait of 0.5 seconds.

Parameter	Description
name	See "name and type"
mode	See "mode"
id1	See "id1"
id2	See "id2"

C.3.37 direct path read

During Direct Path operations the data is asynchronously read from the database files. At some stage the session needs to make sure that all outstanding asynchronous I/O have been completed to disk. This can also happen if during a direct read no more slots are available to store outstanding load requests (a load request could consist of multiple I/Os).

Wait Time: 10 seconds. The session will be posted by the completing asynchronous I/O. It will never wait the entire 10 seconds. The session waits in a tight loop until all outstanding I/Os have completed.

Parameter	Description
descriptor address	This is a pointer to the I/O context of outstanding direct I/Os on which the session is currently waiting
first dba	The dba of the oldest I/O in the context referenced by the descriptor address
block cnt	Number of valid buffers in the context referenced by the descriptor address

See Also

Oracle Database Performance Tuning Guide for more information about this wait event

C.3.38 direct path sync

During Direct Path write operations the data is asynchronously written to the database files. At some point the session needs to ensure that all outstanding asynchronous I/O have been completed to disk. On UNIX the `fsync` command, which synchronizes data to disk, is issued to confirm that all the writes have completed and the data is all on disk.

Wait Time: The time taken for the fsync operation to complete, which normally is the time taken to complete the outstanding I/Os.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameters	Description
file#	See "file#"
flags	Flags used for debugging purposes

C.3.39 direct path write

During Direct Path operations, the data is asynchronously written to the database files. At some stage the session needs to make sure that all outstanding asynchronous I/O have been completed to disk. This can also happen if, during a direct write, no more slots are available to store outstanding load requests (a load request could consist of multiple I/Os).

Wait Time: 10 seconds. The session will be posted by the completing asynchronous I/O. It will never wait the entire 10 seconds. The session waits in a tight loop until all outstanding I/Os have completed.

Parameter	Description
descriptor address	This is a pointer to the I/O context of outstanding direct I/Os on which the session is currently waiting
first dba	The dba of the oldest I/O in the context referenced by the descriptor address
block cnt	Number of valid buffers in the context referenced by the descriptor address

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this parameter

C.3.40 Disk file operations I/O

This event is used to wait for disk file operations (for example, open, close, seek, and resize). It is also used for miscellaneous I/O operations such as block dumps and password file accesses.

Wait Time: The wait time is the actual time it takes to do the I/O

Parameter	Description
FileOperation	Type of file operation
fileno	File identification number
filetype	Type of file (for example, log file, data file, and so on)

C.3.41 dispatcher shutdown

During shutdown immediate or normal, the shutdown process must wait for all the dispatchers to shutdown. As each dispatcher is signaled, the session that causes the shutdown waits on this event until the requested dispatcher is no longer running.

Wait Time: 1 second

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
waited	Indicates the cumulative wait time. After 5 minutes, the session writes to the alert and trace files to indicate that there might be a problem.

> **See Also**
>
> *Oracle Database SQL Language Reference* for information about shutting down a dispatcher using `SHUTDOWN` clause of the SQL `ALTER SYSTEM` statement

C.3.42 dispatcher timer

This basically means that the dispatcher is idle and waiting for some work to arrive.

Wait Time: 60 seconds

Parameter	Description
sleep time	The intended sleep time. The dispatcher will return to work sooner if it is posted by either data arriving on the network or by a post from a shared server process to send data back to the client.

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this wait event

C.3.43 dupl. cluster key

It is possible for a race condition to occur when creating a new cluster key. If it is found that another process has put the cluster key into the data/index block, then the session waits and retries. The retry should then find a valid cluster key.

Wait Time: 0.01 seconds

Parameter	Description
dba	The dba of the block into which the session is trying to insert a cluster key

C.3.44 enq: OW - initialization

A session will wait on this event if it is trying to initialize the database wallet, and another session has already begun an initialization.

Wait Time: Total time necessary to initialize the wallet context

Parameters: None

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about the potential causes of an enqueue wait event

C.3.45 enq: OW - termination

A session will wait on this event if it is trying to terminate the database wallet, and another session has already begun a termination.

Wait Time: Total time necessary to deallocate memory used by the wallet context and terminate the context.

Parameters: None

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about the potential causes of an enqueue wait event

C.3.46 enq: SV - contention

A session is accessing a sequence number from an ordered sequence object.

Wait Time: The wait time is the actual time waited until the requested sequence number is acquired.

Parameter	Description
id1	Sequence object#

> **See Also**
>
> *Oracle Database SQL Language Reference* for more information about sequences

C.3.47 enq: TX - index contention

Occurs when a transaction inserting a row in an index has to wait for the end of an index block split being done by another transaction.

C.3.48 enq: TX - row lock contention

This wait event can occur for several reasons.

- If one user is wanting to update or delete a row or rows that another session is modifying. The session holding the lock will release it when it performs a COMMIT or ROLLBACK.

Copyright © 2002, 2026, Oracle and/or its affiliates.

- If a session is waiting due to potential duplicates in a UNIQUE index. If two sessions try to insert the same key value, the second session has to wait to see if an ORA-0001 should be raised or not. The session holding the lock will release it when it performs a COMMIT or ROLLBACK.
- If a session is waiting due to a shared bitmap index fragment. Bitmap indexes index key values and a range of rowids. Each entry in a bitmap index can cover many rows in the actual table. If two sessions want to update rows covered by the same bitmap index fragment, then the second session waits for the first transaction to either COMMIT or ROLLBACK by waiting for the TX lock.

C.3.49 enqueue

The session is waiting for a local enqueue.

The wait depends on the name of the enqueue (see Oracle Enqueue Names).

Wait Time: Depends on the enqueue name

Parameter	Description
name	See "name and type"
mode	See "mode"

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about the potential causes of an enqueue wait event

C.3.50 flashback buf free by RVWR

This wait event only occurs when Flashback Database is turned on. A session waits for recovery writer (RVWR) to write flashback data to the flashback logs on disk because the buffers are full. Until RVWR can free up the buffers, the session may need to wait.

If this event becomes a top wait event for the database, it is typically because the file system or storage system for the Fast Recovery Area does not support enough bandwidth for Oracle to write the flashback database logs. Refer to the Flashback Database section in *Oracle Database Backup and Recovery User's Guide* for tuning considerations.

Wait Time: 1 second

Parameters: None

C.3.51 flashback log file sync

Waits for flashback database data to be written to disk.

Wait Time: Includes RVWR writing the flashback database data and posting this process

Parameters: None

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.52 free buffer waits

This wait event can occur for several reasons.

- All buffer gets have been suspended. This could happen when a file was read-only and is now read/write. All the existing buffers must be invalidated since they are not linked to lock elements (needed when mounted parallel (shared)). So cache buffers are not assigned to data block addresses until the invalidation is finished.
- The session moved some dirty buffers to the dirty queue and now this dirty queue is full. The dirty queue must written first. The session will wait on this event and try again to find a free buffer
- This also happens after inspecting **free buffer inspected** buffers. If no free buffer is found, Oracle waits for one second, and then tries to get the buffer again (depends on the context). For more information, see free buffer inspected.

Wait Time: 1 second

Parameter	Description
file#	See "file#"
block#	See "block#"

See Also

- *Oracle Database Performance Tuning Guide* for more information about this wait event
- *Oracle Database Performance Tuning Guide* about potential causes of this wait event

C.3.53 free global transaction table entry

The session is waiting for a free slot in the global transaction table (used by the Distributed Database option). It will wait for 1 second and try again.

Wait Time: 1 second

Parameter	Description
tries	The number of times the session tried to find a free slot in the global transaction table

C.3.54 free process state object

Used during the creation of a process. The session will scan the process table and look for a free process slot. If none can be found, PMON is posted to check if all the processes currently in the process table are still running. If there are terminated processes, then PMON will clean them and make the process slot available to new processes. The waiting process will then rescan the process table to find the new slot.

Wait Time: 1 second

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameters: None

C.3.55 gc buffer busy acquire

This event indicates that the requested buffer was globally busy in the cluster, because it had already been requested from a remote instance by another local client.

Wait Time: The wait time is the actual time it took to resolve the buffer contention.

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

C.3.56 gc buffer busy release

This event indicates that the requested buffer was globally busy in the cluster, because the requested buffer had been transferred away by a remote instance and needed to be requested again.

Wait Time: The wait time is the actual time it took to resolve the buffer contention.

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

C.3.57 gc cr block 2-way

This event indicates that the requested consistent read (CR) buffer was transferred from another instance in the cluster, and 2 network hops were involved in the request. The Cache Fusion protocol guarantees that requests are fulfilled in 3 hops or less, regardless of the number of instances in the cluster.

Wait Time: The wait time is the actual time it took to receive the requested buffer.

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

C.3.58 gc cr block 3-way

This event indicates that the requested consistent read (CR) buffer was transferred from another instance in the cluster, and 3 network hops were involved in the request. The Cache Fusion protocol guarantees that requests are fulfilled in 3 hops or less, regardless of the number of instances in the cluster.

Wait Time: The wait time is the actual time it took to receive the requested buffer.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

C.3.59 gc cr block busy

This event indicates that the requested consistent read (CR) buffer was transferred from another instance in the cluster, but the request had to be put on hold at the serving instance.

This could happen when a dirty buffer had to be transferred, in which case REDO needs to be flushed before the Cache Fusion transfer could be processed. This could also happen if the requested buffer was being held by another process on the remote instance. The serving instance would transfer the requested buffer once the REDO flush is complete, or once the buffer is released by the holding process.

Wait Time: The wait time is the actual time it took to receive the requested buffer.

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

C.3.60 gc cr block congested

This event indicates that the requested consistent read (CR) buffer was transferred from another instance in the cluster, but the request was queued at the serving instance for too long.

This could happen when the GCS server processes (LMS*n* and LM*nn*) are very busy due to having a high volume of incoming Cache Fusion requests. Increasing the GCS server processes (`GCS_SERVER_PROCESSES`) is usually the remedy to alleviate congestion for the GCS server processes.

Wait Time: The wait time is the actual time it took to receive the requested buffer.

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

See Also

"GCS_SERVER_PROCESSES"

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.61 gc cr block lost

This event indicates that a Cache Fusion consistent read (CR) buffer request resulted in a potentially lost block. The request will be re-tried automatically.

This is typically caused by unreliable interconnect network protocols, usually when the network is congested and packets are dropped. It can also be the result of interconnect hardware errors or simply the result of an overloaded system, or a sign that a public network path is used for Cache Fusion instead of a private network.

Wait Time: The wait time is the actual time it took to detect the potentially lost block.

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

C.3.62 gc cr grant read-mostly invalidation

The client is performing a DML on a read-mostly object, which requires a read lock invalidation on all other instances.

Wait Time: The wait time is the actual time it takes to complete the invalidation on all other instances.

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

C.3.63 gc cr multi block grant

This event indicates that a client requested multiple consistent read (CR) buffers from another instance in the cluster, and the request resulted in multiple lock grants from the remote instance.

Wait Time: The wait time is the actual time waited for the multi-block request to complete.

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

C.3.64 gc cr multi block mixed

This event indicates that a client requested multiple consistent read (CR) buffers from another instance in the cluster, and the remote instance returned a mixture of blocks and grants.

Wait Time: The wait time is the actual time waited for the multi-block request to complete.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

C.3.65 gc cr request

This event is used for ongoing Cache Fusion consistent read (CR) buffer requests. The event is renamed to reflect the actual outcome of the request (example: "gc cr block 2-way") upon completion of the wait. Therefore, these events will not be reported in AWR reports, but may appear in ASH data.

Wait Time: The wait time is the actual time waited for the request.

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

C.3.66 gc current block 2-way

This event indicates that the requested current buffer was transferred from another instance in the cluster, and 2 network hops were involved in the request. The Cache Fusion protocol guarantees that requests are fulfilled in 3 hops or less, regardless of the number of instances in the cluster.

Wait Time: The wait time is the actual time it took to receive the requested buffer.

Parameter	Description
file#	See "file#"
block#	See "block#"
id#	Block class and global access mode held and requested

C.3.67 gc current block 3-way

This event indicates that the requested current buffer was transferred from another instance in the cluster, and 3 network hops were involved in the request. The Cache Fusion protocol guarantees that requests are fulfilled in 3 hops or less, regardless of the number of instances in the cluster.

Wait Time: The wait time is the actual time it took to receive the requested buffer.

Parameter	Description
file#	See "file#"
block#	See "block#"
id#	Block class and global access mode held and requested

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.68 gc current block busy

This event indicates that the requested current buffer was transferred from another instance in the cluster, but the request had to be put on hold at the serving instance.

This could happen when a dirty buffer had to be transferred, in which case REDO needs to be flushed before the Cache Fusion transfer could be processed. This could also happen if the requested buffer was being held by another process on the remote instance. The serving instance would transfer the requested buffer once the REDO flush is complete, or once the buffer is released by the holding process.

Wait Time: The wait time is the actual time it took to receive the requested buffer.

Parameter	Description
file#	See "file#"
block#	See "block#"
id#	Block class and global access mode held and requested

C.3.69 gc current block congested

This event indicates that the requested current buffer was transferred from another instance in the cluster, but the request was queued at the serving instance for too long.

This could happen when the GCS server processes (LMS) are very busy due to having a high volume of incoming Cache Fusion requests. Increasing the GCS server processes (`GCS_SERVER_PROCESSES`) is usually the remedy to alleviate congestion for the GCS server processes.

Wait Time: The wait time is the actual time it took to receive the requested buffer.

Parameter	Description
file#	See "file#"
block#	See "block#"
id#	Block class and global access mode held and requested

> **See Also**
>
> "GCS_SERVER_PROCESSES"

C.3.70 gc current block lost

This event indicates that a Cache Fusion current buffer request resulted in a potentially lost block. The request will be re-tried automatically.

This is typically caused by unreliable interconnect network protocols, usually when the network is congested and packets are dropped. It can also be the result of interconnect hardware errors or simply the result of an overloaded system, or a sign that a public network path is used for Cache Fusion instead of a private network.

Wait Time: The wait time is the actual time it took to detect the potentially lost block.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
file#	See "file#"
block#	See "block#"
id#	Block class and global access mode held and requested

C.3.71 gc current request

This event is used for ongoing Cache Fusion current buffer requests. The event is renamed to reflect the actual outcome of the request (example: "gc current block 2-way") upon completion of the wait. Therefore, these events will not be reported in AWR reports, but may appear in ASH data.

Wait Time: The wait time is the actual time waited for the request.

Parameter	Description
file#	See "file#"
block#	See "block#"
id#	Block class and global access mode held and requested

C.3.72 gc index operation

This event occurs when a transaction inserting a row into an index must wait for the end of an index block split being performed by another transaction. In an Oracle RAC database, some index split waits will use this wait event in lieu of the "enq: TX - index contention" wait event.

Wait Time: The wait time is the actual time waited for the split to complete.

Parameter	Description
undo seg#	The rollback segment ID
slot#	The slot ID inside the rollback segment
wrap#	The sequence number that is incremented for each transaction
count	The number of times that the session has waited on this transaction

C.3.73 gc recovery quiesce

Instance recovery is waiting for a global cache operation to complete.

Wait Time: The total elapsed time for the global cache operation to complete

Parameter	Description
file#	See "file#"
block#	See "block#"
class	See "class"

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.74 GCS lock open S

The session waits for a resource get in `SHARED` mode on the block identified by file# and block#.

Wait Time: 1 second

Parameter	Description
file#	See "file#"
block#	See "block#"
class	See "class"

C.3.75 GCS lock open X

The session waits for a resource get in `EXCLUSIVE` mode on the block identified by file# and block#.

Wait Time: 1 second

Parameter	Description
file#	See "file#"
block#	See "block#"
lenum	See "le"

C.3.76 global cache busy

The session waits to convert a buffer from Shared Current to Exclusive Current status.

Wait Time: 1 second

Parameter	Description
file#	See "file#"
block#	See "block#"
le	See "le"

C.3.77 inactive session

This event is used for switching and terminating sessions.

- Switching sessions

  If a timeout period has been specified, then wait that amount of time for the session to be detached.

- Terminating sessions

  From either `KILL SESSION` or internal request. Having posted a session that it should terminate itself, wait for up to 1 minute for the session to terminate.

Wait Time: 1 second

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
session#	See "session#"
waited	See "waited"

C.3.78 inactive transaction branch

The session waits for a transaction branch that is currently used by another session.

Wait Time: 1 second

Parameter	Description
branch#	The serial number of the transaction for which the session is waiting
waited	See "waited"

C.3.79 index block split

While trying to find an index key in an index block, Oracle noticed that the index block was being split. Oracle will wait for the split to finish and try to find the key again.

Wait Time: The session will yield the CPU, so there is no actual waiting time

Parameter	Description
rootdba	The root of the index
level	This is the level of the block that the session is trying to split in the index. The leaf blocks are level 0. If the level is > 0, it is a branch block. (The root block can be considered a special branch block).
childdba	The block that the session is trying to split

C.3.80 instance state change

The session waits for SMON to enable or disable cache or transaction recovery. This usually happens during `ALTER DATABASE OPEN` or `CLOSE`.

Wait Time: Wait time depends on the amount of time the action takes (that is, the amount of recovery needed)

Parameter	Description
layer	This value can be 1 or 2. If 1, it means that the transaction layer wants transaction recovery to be performed. If 2, it means that cache recovery will be performed.
value	This value can be 0 (disable) or 1 (enable)
waited	The number of seconds waited so far

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.81 io done

The session waits for an I/O to complete or it waits for a worker process to become available to submit the I/O request. This event occurs on platforms that do not support asynchronous I/O.

Wait Time: 50 milliseconds

Parameter	Description
msg ptr	A pointer to the I/O request

C.3.82 ksxr wait for mount shared

The cross instance broadcast facility of this Oracle instance is waiting for the database mount in shared mode to complete.

Wait Time: The time taken for the instance to mount. An indefinite wait on this event implies that the instance startup is hanging.

C.3.83 ktm: instance recovery

The session waits for SMON to finish the instance, transaction recovery, or sort segment cleanup.

Wait Time: The wait time can vary and depends on the amount of recovery needed

Parameter	Description

undo segment#	If the value is 0, SMON is probably performing instance recovery. If P1 > 0, use this query to find the undo segment:
	`select *`
	`from v$rollstat`
	`where usn = undo segment#;`

C.3.84 latch activity

This event is used as part of the process of determining whether a latch must be cleaned.

Wait Time: 0.05 to 0.1 seconds

Parameter	Description
address	The address of the latch that is being checked

number	The latch number of the latch that has activity. To find more information on the latch, use the following SQL statement:
	`select *`
	`from v$latchname`
	`where latch# = number;`

process#	If this is 0, it is the first phase of the in-flux tests

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about latch wait events

C.3.85 latch free

The process waits for a latch that is currently busy (held by another process).

Wait Time: The wait time increases exponentially and does not include spinning on the latch (active waiting). The maximum wait time also depends on the number of latches that the process is holding. There is an incremental wait of up to 2 seconds.

Parameter	Description
address	The address of the latch for which the process is waiting

number	The latch number that indexes in the V$LATCHNAME view.To find more information on the latch, use the following SQL statement:
	`select *`
	`from v$latchname`
	`where latch# = number;`

tries	A count of the number of times the process tried to get the latch (slow with spinning) and the process has to sleep

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about latch wait events

C.3.86 latch: redo copy

Redo copy latches are acquired by sessions in order to write changes into the redo log buffer and by the LGWR process to ensure that there are no current writers before it copies data from the buffer to the redo log.

C.3.87 latch: row cache objects

This event occurs when another session is modifying the contents of part of the dictionary cache (or row cache). This event is often associated with high parse activity but may occur for other reasons, such as manipulation of rollback segments. V$ROWCACHE may indicate which part of the row cache is being contended.

C.3.88 library cache load lock

The session tries to find the load lock for the database object so that it can load the object. The load lock is always obtained in Exclusive mode, so that no other process can load the same

Copyright © 2002, 2026, Oracle and/or its affiliates.

object. If the load lock is busy the session will wait on this event until the lock becomes available.

Wait Time: 3 seconds (1 second for PMON)

Parameter	Description
object address	Address of the object being loaded
lock address	Address of load lock being used
mask	Indicates which data pieces of the object that must loaded

C.3.89 library cache lock

This event controls the concurrency between clients of the library cache. It acquires a lock on the object handle so that either:

- One client can prevent other clients from accessing the same object
- The client can maintain a dependency for a long time (for example, no other client can change the object)

This lock is also obtained to locate an object in the library cache.

Wait Time: 3 seconds (1 second for PMON)

Parameter	Description
handle address	Address of the object being loaded
lock address	Address of the load lock being used. This is not the same thing as a latch or an enqueue, it is a State Object.
mode	Indicates the data pieces of the object which must be loaded
namespace	See "namespace"

See Also

- *Oracle Database Performance Tuning Guide* for more information about this wait event
- *Oracle Database Performance Tuning Guide* for more information about potential causes of this wait event

C.3.90 library cache pin

This event manages library cache concurrency. Pinning an object causes the heaps to be loaded into memory. If a client wants to modify or examine the object, the client must acquire a pin after the lock.

Wait Time: 3 seconds (1 second for PMON)

Parameter	Description
handle address	Address of the object being loaded
pin address	Address of the load lock being used. This is not the same thing as a latch or an enqueue, it is basically a State Object.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
mode	Indicates which data pieces of the object that must be loaded
namespace	See "namespace"

See Also

- *Oracle Database Performance Tuning Guide* for more information about this wait event
- *Oracle Database Performance Tuning Guide* for more information about potential causes of this wait event

C.3.91 library cache shutdown

The process shutting down the instance waits for sessions to complete before proceeding with library cache shutdown.

C.3.92 library cache: mutex X

Library cache mutexes protect hash buckets within the shared pool. Each hash bucket contains a number of cursors. The mutex must be held in exclusive mode before any of the structures that it protects can be changed. This wait event is often associated with high parse activity.

C.3.93 LMON global data update

The rolling migration operation is waiting for a response from LMON to acknowledge the global data was updated.

Wait Time: The time it takes for LMON to publish/retrieve the global data

Parameters: None

C.3.94 Log archive I/O

Used local archiving of online redo logs (for a production database) or standby redo logs (for a standby database). When the archiving process exhausts its I/O buffers because all of them are being used for on-going I/O's, the wait for an available I/O buffer is captured in this system wait event.

Wait Time: Depends on the speed of the disks

Parameters: None

C.3.95 log buffer space

Waiting for space in the log buffer because the session is writing data into the log buffer faster than LGWR can write it out. Consider making the log buffer bigger if it is small, or moving the log files to faster disks such as striped disks.

Wait Time: Usually 1 second, but 5 seconds if it is waiting for a Switch Logfile to complete

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameters: None

> **See Also**
>
> - *Oracle Database Performance Tuning Guide* for more information about this wait event
> - *Oracle Database Performance Tuning Guide* for more information about potential causes of this wait event

C.3.96 log file parallel write

Writing redo records to the redo log files from the log buffer.

Wait Time: Time it takes for the I/Os to complete. Even though redo records are written in parallel, the parallel write is not complete until the last I/O is on disk.

Parameter	Description
files	Number of files to be written
blocks	Number of blocks to be written
requests	Number of I/O requests

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for information about how this wait event can help identify I/O problems

C.3.97 log file sequential read

Waiting for the read from this logfile to return. This is used to read redo records from the log file.

Wait Time: Time it takes to complete the physical I/O (read)

Parameter	Description
log#	The relative sequence number of the logfiles within a log group (used only when dumping the logfiles)
block#	See "block#"
blocks	The number of blocks to read

C.3.98 log file single write

Waiting for the write to this logfile to complete. This event is used while updating the header of the logfile. It is signaled when adding a log file member and when incrementing sequence numbers.

Wait Time: Time it takes for the physical I/O (write) to complete

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
log#	This is the number of the group/log to which the session is currently writing
block#	See "block#"
blocks	The number of blocks to write

C.3.99 log file switch (archiving needed)

Waiting for a log switch because the log that the LGWR will be switching into has not been archived yet. Check the alert log to ensure that archiving has not stopped due to a failed archive write. To speed archiving, consider adding more archive processes or putting the archive files on striped disks.

Wait Time: 1 second

Parameters: None

C.3.100 log file switch (checkpoint incomplete)

Waiting for a log switch because the session cannot wrap into the next log. Wrapping cannot be performed because the checkpoint for that log has not completed.

Wait Time: 1 second

Parameters: None

C.3.101 log file switch (clearing log file)

Waiting for a log switch because the log is being cleared due to a `CLEAR LOGFILE` command or implicit clear logfile executed by recovery.

Wait Time: 1 second

Parameters: None

C.3.102 log file switch (private strand flush incomplete)

User sessions trying to generate redo, wait on this event when LGWR waits for DBWR to complete flushing redo from IMU buffers into the log buffer; when DBWR is complete LGWR can then finish writing the current log, and then switch log files.

Wait Time: 1 second

Parameters: None

C.3.103 log file switch completion

Waiting for a log switch to complete.

Wait Time: 1 second

Parameters: None

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.104 log file sync

When a user session commits, the session's redo information must be flushed to the redo logfile. The user session will post the LGWR to write the log buffer to the redo log file. When the LGWR has finished writing, it will post the user session.

Wait Time: The wait time includes the writing of the log buffer and the post.

Parameter	Description
buffer#	The number of the physical buffer in the redo log buffer that must be synchronized

See Also

- *Oracle Database Performance Tuning Guide* for more information about this wait event
- *Oracle Database Performance Tuning Guide* for information about potential causes of this wait event

C.3.105 log switch/archive

Used as part of the `ALTER SYSTEM ARCHIVE LOG CHANGE` *scn* statement. The session waits for the current log from all open threads to be archived.

Wait Time: Wait for up to 10 seconds

Parameter	Description
thread#	The thread number of the thread that is currently archiving its current log

C.3.106 optimizer stats update retry

When concurrent sessions try to update optimizer statistics for the same object, all of them except the one that successfully acquired all necessary locks/pins on the library/row cache entries, wait on this wait event and then retry locking after a short period of time. In addition to explicit statistics gathering and maintenance operations using the `DBMS_STATS` package, Oracle database itself may try to update statistics for some objects, either on behalf of the user or for its own maintenance purposes.

Wait Time: 10 ms

Parameters: None

C.3.107 parallel recovery change buffer free

The parallel recovery coordinator is waiting for a change mapping buffer to be released by one of the recovery workers.

Wait Time: 100ms

Parameters: None

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.108 parallel recovery control message reply

The parallel recovery coordinator is waiting for all recovery workers to respond to a synchronous control message.

Wait Time: 100ms

Parameters: None

C.3.109 parallel recovery coord send blocked

The parallel recovery coordinator cannot send a redo change message because the recovery worker is still actively applying redo that it has already received and has not yet released the channel.

Wait Time: 100ms

Parameters: None

C.3.110 parallel recovery coord wait for reply

The parallel recovery coordinator is waiting for all recovery workers to exit.

Wait Time: 100ms

Parameters: None

C.3.111 parallel recovery coordinator waits for slave cleanup

The parallel recovery coordinator is waiting for all recovery workers to exit gracefully.

Wait Time: 10ms

Parameters: None

C.3.112 parallel recovery read buffer free

The parallel recovery coordinator is waiting for a log read buffer to be released by all recovery workers. Only after every recovery worker finishes applying redo from a log read buffer, can the buffer be used by the coordinator to issue the next log read.

Wait Time: 100ms

Parameters: None

C.3.113 parallel recovery slave next change

A parallel recovery worker is idle and waiting for the next change message from the coordinator.

Wait Time: 100ms

Parameters: None

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.114 pending global transaction(s)

This event should happen only during testing. The session waits for pending transactions to clear.

Wait Time: 30 seconds

Parameter	Description
scans	Number of times the session has scanned the `PENDING_TRANS$` table

C.3.115 pipe get

The session waits for a message to be received on the pipe or for the pipe timer to expire.

Wait Time: There is a 5 second wake up (check) and the pipe timer set by the user

Parameter	Description
handle address	The library cache object handle for this pipe
buffer length	The length of the buffer
timeout	The pipe timer set by the user

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this wait event

C.3.116 pipe put

The session waits for the pipe send timer to expire or for space to be made available in the pipe.

Wait Time: There is the 5 second wakeup (check) and the user-supplied timeout value

Parameter	Description
handle address	The library cache object handle for this pipe
record length	The length of the record or buffer that has been put into the pipe
timeout	The pipe timer set by the user

C.3.117 PL/SQL lock timer

This event is called through the `DBMS_LOCK.SLEEP` procedure. This event will most likely originate from procedures written by a user.

Wait Time: The wait time is in hundredths of seconds and depends on the user context

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
duration	The duration that the user specified in the DBMS_LOCK.SLEEP procedure

C.3.118 pmon timer

This is the main wait event for PMON. When PMON is idle, it is waiting on this event.

Wait Time: Up to 3 seconds, if not posted before

Parameter	Description
duration	The actual amount of time that the PMON is trying to sleep

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this wait event

C.3.119 prior spawner clean up

When a prior process has terminated while spawning a background, the current process which is trying to spawn new a background must wait until the prior process state is cleaned up.

Wait Time: Usually 3 - 10 seconds

Parameter	Description
process_pid	process identifier (see V$PROCESS.PID) of the process whose state must be cleaned up.
process_sno	process serial number (see V$PROCESS.SERIAL#) of the process whose state must be cleaned up.

C.3.120 process startup

Wait for a shared server, Dispatcher, or other background process to start.

Wait Time: Wait up to 1 second for a background process to start. If timed out, then re-wait until 5 minutes have passed and signal an error. If the process has started, the event will acknowledge this.

Parameter	Description
type	The process type that was started
process#	The process number of the process being started
waited	Cumulative time waited for the process to start

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.121 PX Deque wait

The process is waiting for a message during a parallel execute.

Wait Time: The wait time depends on how quickly the message arrives. Wait times can vary, but it will normally be a short period of time.

Parameter	Description
reason	The reason for dequeuing
sleeptime	The amount of time that the session slept
loop	The total number of times that the session has slept

C.3.122 PX qref latch

Each parallel execution process has a parallel execution qref latch, which must be acquired before the queue buffers can be manipulated.

Wait Time: Wait up to 1 second

Parameter	Description
function	Indicates the type of wait that the session is doing
sleeptime	The amount of time that the session waits (in hundredths of a second)
qref	The address of the process queue for which the session is waiting

C.3.123 PX server shutdown

During normal or immediate shutdown the parallel execution servers are posted to shutdown cleanly. If any parallel execution servers are still running after 10 seconds, they are terminated.

Wait Time: Wait up to 0.5 seconds

Parameter	Description
nalive	The number of parallel execution servers that are still running
sleeptime	The total sleeptime since the session started to wait on this event
loop	The number of times the session waited for this event

C.3.124 PX signal server

This event occurs only in Exclusive mode. The query coordinator is signaling the Query Workers that an error has occurred.

Wait Time: 0.5 seconds

Parameter	Description
serial	The serial number of the worker process queue
error	The error that has occurred
nbusy	The number of worker processes that are still busy

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.125 rdbms ipc message

The background processes (LGWR, DBWR, LMS0) use this event to indicate that they are idle and are waiting for the foreground processes to send them an IPC message to do some work.

Wait Time: Up to 3 seconds. The parameter *timeout* shows the true sleep time.

Parameter	Description
timeout	The amount of time that the session waits for an IPC message

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this wait event

C.3.126 rdbms ipc message block

This event indicates that all message blocks are in use and that the session had to wait for a message block to become available.

Wait Time: Wait up to 60 seconds

Parameters: None

C.3.127 rdbms ipc reply

This event is used to wait for a reply from one of the background processes.

Wait Time: The wait time is specified by the user and is indicated by the parameter *timeout*.

Parameter	Description
from_process	The background process for which the session is waiting. The wait is for a reply to an IPC message sent by the session.
timeout	The amount of time in seconds that this process will wait for a reply

C.3.128 read by other session

This event occurs when a session requests a buffer that is currently being read into the buffer cache by another session. Prior to release 10.1, waits for this event were grouped with the other reasons for waiting for buffers under the 'buffer busy waits' event.

Wait Time: Time waited for the buffer to be read by the other session (in microseconds)

Parameter	Description
file#	See "file#"
block#	See "block#"
class#	See "class"

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.129 recovery active instance mapping setup

This event is used to track the time taken (in centiseconds) to set up recovery context on remote instances.

C.3.130 recovery apply pending

This event tracks the time the logmerger process waited (in centiseconds) for apply workers to apply all pending changes up to a certain SCN.

C.3.131 recovery cancel

This wait event is used to track the time taken (in centiseconds) to cancel multi instance media recovery.

C.3.132 recovery checkpoint

This wait event tracks the time (in centiseconds) spent waiting for DBWR processes on all instances to write dirty buffers to disk up to a certain SCN.

C.3.133 recovery file header update for checkpoint

This wait event tracks the time (in centiseconds) spent by MRP0 process waiting on logmerger to update file headers to record the latest checkpoint.

C.3.134 recovery file header update for fuzziness

This wait event tracks the time taken (in centiseconds) to update file headers to record media recovery fuzziness.

C.3.135 recovery marker apply

This wait event is used to track the time (in centiseconds) taken to process a recovery marker.

C.3.136 recovery merge pending

This wait event is used to track the time spent (in centiseconds) by the MRP0 process waiting for remote logmergers to merge changes up to a certain SCN.

C.3.137 recovery metadata latch

This wait event tracks the time taken (in centiseconds) to get the active DataGuard metadata update latch to advance a query SCN.

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.138 recovery move influx buffers

This wait event is used to track the time spent (in centiseconds) waiting for all apply workers to move their influx buffers to the dirty queue.

C.3.139 recovery read

A parallel recovery worker (or serial recovery process) is waiting for a batch of data block reads to complete.

Wait Time: Time it takes to complete the physical I/O (read)

Parameters: None

C.3.140 recovery receive buffer free

This wait event is used to track the time (in centiseconds) spent by the receiver process on instance waiting for apply workers to apply changes from received buffers so that they can be freed for the next change.

C.3.141 recovery remote file verification

This wait event tracks the time spent (in centiseconds) by the MRP0 process waiting for remote logmergers to finish file verification on remote instances.

C.3.142 recovery send buffer free

This wait event is used to track the time spent (in centiseconds) by the sender process waiting for apply workers to apply changes from the local instance so that the buffer can be reused for new changes.

C.3.143 recovery shutdown

This wait event is used to track the time (in centiseconds) taken for remote logmergers to shut down.

C.3.144 Redo Transport Attach

Used to track the time spent (in centiseconds) doing Connect, Logon, and RFSATTACH for *any* network process.

> **See Also**
>
> *Oracle Data Guard Concepts and Administration* for more information about using RFS to manage standby redo logs.

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.145 Redo Transport Close

Used to track the time spent (in centiseconds) by ARC*n*, NSS*n*, and TT*nn* processes doing RFSCLOSE and RFSRGSTR operations.

> **See Also**
>
> *Oracle Data Guard Concepts and Administration* for more information about using RFS to manage standby redo logs.

C.3.146 Redo Transport Detach

Used to track the time spent (in centiseconds) doing RFSDETACH and Disconnect for *any* network process.

> **See Also**
>
> *Oracle Data Guard Concepts and Administration* for more information about using RFS to manage standby redo logs.

C.3.147 Redo Transport Open

Used to track the time spent (in centiseconds) by ARC*n*, NSS*n*, and TT*nn* background processes doing RFSCREAT and RFSANNCE operations.

> **See Also**
>
> *Oracle Data Guard Concepts and Administration* for more information about using RFS to manage standby redo logs.

C.3.148 Redo Transport Ping

Used to track the time spent (in centiseconds) by ARC*n* background processes doing RFSPING operations.

> **See Also**
>
> *Oracle Data Guard Concepts and Administration* for more information about using RFS to manage standby redo logs.

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.149 Redo Transport Slave Shutdown

Used to track the time spent (in centiseconds) by LGWR doing NSS*n* and TT*nn* process shutdown and termination.

C.3.150 Redo Transport Slave Startup

Used to track the time spent (in centiseconds) by LGWR doing NSS*n* and TT*nn* process startup and initialization.

C.3.151 Redo Writer Remote Sync Complete

Used to track the time spent (in centiseconds) by LGWR reaping completed network writes to remote destinations.

C.3.152 Redo Writer Remote Sync Notify

Used to track the time spent (in centiseconds) by LGWR issuing network writes to remote destinations.

C.3.153 Remote SYNC Ping

Used to track the time spent (in centiseconds) by the LGWR and NSS*n* background processes doing synchronous PING operations.

C.3.154 resmgr:become active

The session is waiting for a resource manager active session slot. This event occurs when the resource manager is enabled and the number of active sessions in the session's current consumer group exceeds the current resource plan's active session limit for the consumer group. To reduce the occurrence of this wait event, increase the active session limit for the session's current consumer group.

Wait Time: The time the session waited to be allocated an active session slot

Parameter	Description
location	location of the wait

C.3.155 resmgr:cpu quantum

The session is waiting to be allocated a quantum of CPU. This event occurs when the resource manager is enabled and is throttling CPU consumption. To reduce the occurrence of this wait event, increase the CPU allocation for the sessions's current consumer group.

Wait Time: The time the session waited to acquire a CPU quantum

Parameter	Description
location	Location of the wait
Consumer Group ID	Consumer group ID of the session waiting. This value is from the `CONSUMER_GROUP_ID` column in the `DBA_RSRC_CONSUMER_GROUPS` view.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> "DBA_RSRC_CONSUMER_GROUPS"

C.3.156 resmgr: I/O rate limit

The session is waiting for I/O bandwidth to be available to issue I/Os. This event occurs when PDB I/O rate limits are enabled by setting the `MAX_IOPS` and/or `MAX_MBPS` parameters.

To reduce the occurrence of this wait event, the PDB I/O rate limit values need to be increased. This can be achieved by increasing the values of `MAX_IOPS` and `MAX_MBPS` or by removing the limits altogether by setting the parameter value to 0.

Wait Time: The time the session waited for the I/O bandwidth to become available.

Parameter	Description
P1: PDB_ID	This the PDB ID of the PDB that has its I/O throttled because the `MAX_IOPS` or `MAX_MBPS` parameter was set for it.

> **See Also**
>
> - "MAX_IOPS"
> - "MAX_MBPS"

C.3.157 resmgr:pq queued

The session is waiting in the parallel statement queue.

Wait Time: The time the session waited for sufficient parallel query processes to become available to run this session with the requested degree of parallelism

Parameter	Description
location	Location of the wait

> **See Also**
>
> *Oracle Database VLDB and Partitioning Guide* for more information about this wait event

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.158 rolling migration: cluster quiesce

This is the wait event that instances wait on when cluster is about to start a rolling migration. The instances are waiting for any privileged operations that blocks rolling migration to complete before allowing rolling migration.

Wait Time: 1 second

Parameter	Description
location	Its value will be 1 if the wait is for completion of the privileged operations so that a rolling upgrade/downgrade can start. Its value will be 2 if the wait is for completion of the rolling upgrade/downgrade on all the nodes in the cluster.
waits	The number of seconds spent waiting at the current location.

C.3.159 row cache lock

The session is trying to get a data dictionary lock.

Wait Time: Wait up to 60 seconds.

Parameter	Description
cache id	The CACHE# column value in the V$ROWCACHE view
mode	See "mode"
request	The pipe timer set by the user

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this wait event

C.3.160 RVWR wait for flashback copy

Waits for a process to copy flashback database data into the flashback buffer, in order to write out the requested flashback data.

Wait Time: 10 milliseconds

Parameter	Description
copy latch #	The number of flashback copy latch RVWR is waiting on

C.3.161 sbtbufinfo

This function is called when Oracle needs to discover the size, and number, of I/O buffers that have been allocated by the SBT layer. It should be very fast and never block.

Wait Time: Less than one millisecond

Parameters: None

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.162 sbtgetbuf

This function obtains one I/O buffer that Oracle will use for I/O during a backup job.

Wait Time: Less than one millisecond

Parameters: None

C.3.163 sbtmapbuf

This is an internal function used to facilitate multi-process buffer management. It should be very fast and never block.

Wait Time: Less than one millisecond

Parameters: None

C.3.164 sbtrelbuf

This function releases an I/O buffer that has been already processed during a restore job, so that the SBT layer can fill it with more data. It should be very fast and never block.

Wait Time: Less than one millisecond

Parameters: None

C.3.165 scginq AST call

Called by the session to find the highest lock mode that is held on a resource.

Wait Time: Wait up to 0.2 seconds, but the wait will continue until the NULL mode Acquisition AST has fired.

Parameters: None

C.3.166 SGA: allocation forcing component growth

Process waiting on an immediate mode memory transfer with auto-tune SGA after a 4031 for MMAN to get the memory and post it.

Wait Time: 10 msec

Parameters: None

C.3.167 SGA: MMAN sleep for component shrink

MMAN to wait and post itself for satisfying an auto-tuned memory request while trying to fully free a component's quiesced granules. In Release 10.1, the name of this event was 'wait for SGA component shrink'.

Wait Time: 10 msec

Parameter	Description
P1	component_id (corresponding to the memory pool)
P2	Current size in granules

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
P3	Target size in granules

C.3.168 SGA: sga_target resize

Memory resize requests wait while sga target is being resized. In Release 10.1, the name of this event was 'wait for sga_target resize'.

Wait Time: 10 msec

Parameters: None

C.3.169 Shared IO Pool Memory

Wait until a shared I/O pool buffer becomes available. This happens when processes are using these buffers for I/O and the current process needs to wait for the release of any one of the buffers to the shared I/O pool.

Wait Time: 10msec

Parameters: None

C.3.170 shared server idle wait

Idle wait event for a shared server.

The server waits on the common queue for a virtual circuit. (See also "virtual circuit wait".)

Wait Time: 30 seconds

Parameters: None

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this wait event

C.3.171 single-task message

When running single task, this event indicates that the session waits for the client side of the executable.

Wait Time: Total elapsed time that this session spent in the user application

Parameters: None

C.3.172 smon timer

This is the main idle event for SMON. SMON will be waiting on this event most of the time until it times out or is posted by another process.

Wait Time: 5 minutes (300 seconds)

Copyright © 2002, 2026, Oracle and/or its affiliates.

Parameter	Description
sleeptime	The amount of time that SMON tries to wait on this event in seconds
failed	The number of times SMON was posted when there some kind of error

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this wait event

C.3.173 SQL*Net break/reset to client

The server sends a break or reset message to the client. The session running on the server waits for a reply from the client.

Wait Time: The actual time it takes for the break or reset message to return from the client

Parameter	Description
driver id	See "driver id"
break?	See "break?"

C.3.174 SQL*Net break/reset to dblink

Same as **SQL*Net break/reset to client**, but in this case, the break/reset message is sent to another server process over a database link.

Wait Time: The actual time it takes for the break or reset message to return from the other server process

Parameter	Description
driver id	See "driver id"
break?	See "break?"

C.3.175 SQL*Net message from client

The server process (foreground process) waits for a message from the client process to arrive.

Wait Time: The time it took for a message to arrive from the client since the last message was sent to the client

Parameter	Description
driver id	See "driver id"
#bytes	The number of bytes received by the server (foreground process) from the client.

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this wait event

C.3.176 SQL*Net message from dblink

The session waits while the server process (foreground process) receives messages over a database link from another server process.

Wait Time: The time it took for a message to arrive from another server (foreground process) since a message was sent to the other foreground process.

Parameter	Description
driver id	See "driver id"
#bytes	The number of bytes received by the server (foreground process) from another foreground process over a database link.

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this wait event

C.3.177 SQL*Net message to client

The server (foreground process) is sending a message to the client.

Wait Time: The actual time the **send** takes

Parameter	Description
driver id	See "driver id"
#bytes	The number of bytes sent by the server process to the client

C.3.178 SQL*Net message to dblink

The server process (foreground process) is sending a message over a database link to another server process.

Wait Time: The actual time the **send** takes

Parameter	Description
driver id	See "driver id"
#bytes	The number of bytes sent by the server process to another server process over a database link

Copyright © 2002, 2026, Oracle and/or its affiliates.

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this wait event

C.3.179 SQL*Net more data from client

The server is waiting on the client to send more data to its client shadow process, in an already initiated operation.

Wait Time: The time waited depends on the time it took to receive the data (including the waiting time)

Parameter	Description
driver id	See "driver id"
#bytes	The number of bytes received from the client

C.3.180 SQL*Net more data from dblink

The foreground process is expecting more data from a data base link.

Wait Time: The total time it takes to read the data from the database link (including the waiting time for the data to arrive)

Parameter	Description
driver id	See "driver id"
#bytes	The number of bytes received

C.3.181 SQL*Net more data to client

The server process is sending more data/messages to the client. The previous operation to the client was also a **send**.

Wait Time: The actual time it took for the **send** to complete

Parameter	Description
driver id	See "driver id"
#bytes	The number of bytes that are being sent to the client

> **See Also**
>
> *Oracle Database Performance Tuning Guide* for more information about this wait event

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.182 SQL*Net more data to dblink

The event indicates that the server is sending data over a database link again. The previous operation over this database link was also a **send**.

Wait Time: The actual time it takes to send the data to the other server

Parameter	Description
driver id	See "driver id"
#bytes	The number of bytes that are sent over the database link to the other server process

C.3.183 Streams AQ: waiting for messages in the queue

The session is waiting on an empty OLTP queue (Advanced Queuing) for a message to arrive so that the session can dequeue that message.

Wait Time: The amount of time that the session wants to wait is determined by the parameter *wait time*

Parameter	Description
queue id	The ID of the OLTP queue for which this session is waiting
process#	The process number of the process in which this session runs
wait time	The intended wait time for this session

C.3.184 switch logfile command

The session waits on the user command `SWITCH LOGFILE` to complete.

Wait Time: 5 seconds

Parameters: None

C.3.185 SYNC Remote Write

Used to track the time spent by LGWR doing SYNC RFSWRITE operations.

> **See Also**
>
> *Oracle Data Guard Concepts and Administration* for more information about using RFS to manage standby redo logs.

C.3.186 TCP Socket (KGAS)

A session is waiting for an external host to provide requested data over a network socket. The time that this wait event tracks does not indicate a problem, and even a long wait time is not a

Copyright © 2002, 2026, Oracle and/or its affiliates.

reason to contact Oracle Support. It naturally takes time for data to flow between hosts over a network, and for the remote aspect of an application to process any request made to it.

An application that communicates with a remote host must wait until the data it will read has arrived. In addition, on Microsoft Windows, a separate thread monitors the arrival of traffic. This thread spends most of its life in waits tracked by the TCP Socket (KGAS) wait event.

Wait Time: The total elapsed time for the network connection to be established or for data to arrive from over the network

Parameter	Description
P0	For Oracle internal use only. Values 8, 9, and 10 occur within the special thread present on Microsoft Windows; other P0 values occur in normal user sessions.
P1	For Oracle internal use only

C.3.187 timer in sksawat

The session waits for the Archiver (ARC*n*) asynchronous I/O to complete.

Wait Time: 0.01 seconds

Parameters: None

C.3.188 transaction

Wait for a blocking transaction to be rolled back. Continue waiting until the transaction has been rolled back.

Wait Time: 1 second

Parameter	Description
undo seg#	The rollback segment ID
slot#	The slot ID inside the rollback segment
wrap#	The sequence number that is incremented for each transaction
count	The number of times that the session has waited on this transaction

C.3.189 unbound tx

The session waits to see if there are any transactions that have been started but do not have a Rollback Segment associated with them.

Wait Time: 1 second

Parameters: None

C.3.190 undo_retention publish retry

This wait can occur for two reasons. A session issuing an `ALTER SYSTEM SET UNDO_RETENTION` may wait on this event wait while a cluster reconfiguration takes place. Or the background

Copyright © 2002, 2026, Oracle and/or its affiliates.

process MMNL may wait for cluster reconfiguration while attempting to determine the max `UNDO_RETENTION`.

Wait time: 1 second

Parameter	Description
P1	Identifies where the retry is happening. Id = 1 retry while publishing into the max undo_retention namespace. Id = 2 retry while iterator accessing the max undo_retention namespace.
P2	Retry count (maximum number of retries is 5)

C.3.191 undo segment extension

The undo segment is being extended or shrunk. The session must wait until the operation on the undo segment has finished.

Wait Time: 0.01 seconds

Parameter	Description
segment#	The ID of the rollback segment that is being extended or shrunk

C.3.192 undo segment recovery

PMON is rolling back a terminated transaction. The wait continues until rollback finishes.

Wait Time: 3 seconds

Parameter	Description
segment#	The ID of the rollback segment that contains the transaction that is being rolled back
tx flags	The transaction flags (options) set for the transaction that is being rolled back

C.3.193 undo segment tx slot

Wait for a transaction slot to become available within the selected rollback segment. Continue waiting until the slot is available.

Wait Time: 1 second

Parameter	Description
segment#	The ID of the rollback segment that contains the transaction that is being rolled back

C.3.194 utl_file I/O

Waits associated with operations performed using the `UTL_FILE` package.

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.195 virtual circuit wait

The session waits for a virtual circuit operation to complete.

Wait Time: 30 seconds

Parameter	Description
circuit#	Indicates the virtual circuit# being waited on
type	Indicates the type of operation the session is waiting for

C.3.196 WCR: replay client notify

During replay, the Workload Replay Client always keeps an open connection to the database to detect some special errors. This session will normally be in the wait state until some specific replay-related exception occurs.

Wait Time: Waits until notified. Times out every 30 seconds.

Parameter	Description
who am I	Identifies the reason for wait for the admin thread. 1 - waiting for REPLAY.START() to be issued. 2 - waiting for the replay to run to completion or REPLAY.CANCEL() to be issued.

C.3.197 WCR: replay clock

A session will wait on this event during replay if it has some logical dependencies on another session that has not yet committed its work.

Wait Time: Depends upon the amount of row lock contention in the original capture

Parameter	Description
wait for scn's hi 4 bytes	High 4 bytes of the 8-byte replay SCN that the session is waiting on
wait for scn's lo 4 bytes	Low 4 bytes of the 8-byte replay SCN that the session is waiting on

C.3.198 WCR: replay lock order

A session will wait on this event during replay if it saw some lock contention during capture.

Wait Time: Depends upon the amount of row lock contention in the original capture

Parameter	Description
wait for scn's hi 4 bytes	High 4 bytes of the 8-byte replay SCN that the session is waiting on
wait for scn's lo 4 bytes	Low 4 bytes of the 8-byte replay SCN that the session is waiting on

Copyright © 2002, 2026, Oracle and/or its affiliates.

C.3.199 WCR: replay paused

When the user issues a DBMS_WORKLOAD_REPLAY.PAUSE_REPLAY command, all the replayed sessions are waiting on this wait event until replay is resumed.

Parameters: None

C.3.200 WCR: Sync context busy

In the default replay mode, commits during replay are synchronized to follow the same order as capture. All commits are serialized using the Sync context. A replayed session will wait on this event if it is trying to commit at the same time as another replayed session.

Parameters: None

C.3.201 write complete waits

The session waits for a buffer to be written. The write is caused by normal aging or by a cross-instance call.

Wait Time: 1 second

Parameter	Description
file#	The rollback segment id that contains the transaction that is being rolled back
block#	The transaction flags (options) set for the transaction that is being rolled back
id	Identifies the reason for waiting

C.3.202 writes stopped by instance recovery or database suspension

The session is blocked until the instance that started Instance Recovery is finished.

Wait Time: 5 seconds

Parameter	Description
bythread#	The rollback segment id that contains the transaction that is being rolled back
ourthread#	The current instance thread number

Copyright © 2002, 2026, Oracle and/or its affiliates.

D

Oracle Enqueue Names

This appendix lists Oracle enqueues. **Enqueues** are shared memory structures (locks) that serialize access to database resources. They can be associated with a session or transaction. Enqueue names are displayed in the `LOCK_TYPE` column of the `DBA_LOCK` and `DBA_LOCK_INTERNAL` data dictionary views.
A **resource** uniquely identifies an object that can be locked by different sessions within an instance (local resource) or between instances (global resource). Each session that tries to lock the resource will have an **enqueue** on the resource.

> **Note**
>
> The names of enqueues and their definitions may change from release to release.

> **See Also**
>
> "DBA_LOCK_INTERNAL" and "DBA_LOCK"

The Oracle enqueues are:

- BL, Buffer Cache Management
- BR, Backup/Restore
- CF, Controlfile Transaction
- CI, Cross-instance Call Invocation
- CU, Bind Enqueue
- DF, Datafile
- DL, Direct Loader Index Creation
- DM, Database Mount
- DR, Distributed Recovery Process
- DW, SecureFiles
- DX, Distributed Transaction
- FP, File Object
- FS, File Set
- HW, High-Water Lock
- IN, Instance Number
- IR, Instance Recovery
- IS, Instance State
- IV, Library Cache Invalidation

Copyright © 2002, 2026, Oracle and/or its affiliates.

- JI, Enqueue used during AJV snapshot refresh
- JQ, Job Queue
- KK, Redo Log "Kick"
- KP, contention in Oracle Data Pump startup and shutdown processes
- KO, Multiple Object Checkpoint
- L[A-P], Library Cache Lock
- LS, Log Start or Switch
- MM, Mount Definition
- MR, Media Recovery
- N[A-Z], Library Cache Pin
- PE, ALTER SYSTEM SET PARAMETER = VALUE
- PF, Password File
- PI, Parallel Slaves
- PR, Process Startup
- PS, Parallel Slave Synchronization
- Q[A-Z], Row Cache
- RO, Object Reuse
- RT, Redo Thread
- RW, Row Wait
- SC, System Change Number
- SM, SMON
- SN, Sequence Number
- SQ, Sequence Number Enqueue
- SR, Synchronized Replication
- SS, Sort Segment
- ST, Space Management Transaction
- SV, Sequence Number Value
- TA, Transaction Recovery
- TC, Thread Checkpoint
- TE, Extend Table
- TM, DML Enqueue
- TO, Temporary Table Object Enqueue
- TS, Temporary Segment (also TableSpace)
- TT, Tablespace
- TX, Transaction
- UL, User-defined Locks
- UN, User Name
- US, Undo Segment, Serialization

Copyright © 2002, 2026, Oracle and/or its affiliates.

- WL, Being Written Redo Log
- XA, Instance Attribute Lock
- XI, Instance Registration Lock
- ZA, Exclusive Lock When Moving Audit Table

Copyright © 2002, 2026, Oracle and/or its affiliates.

E

Statistics Descriptions

This appendix describes the statistics stored in the V$SESSTAT and V$SYSSTAT dynamic performance tables. These statistics are useful in identifying and correcting performance problems.

This appendix contains the following topics:

- Displaying Statistics
- Statistics Descriptions

E.1 Displaying Statistics

The V$SESSTAT view displays statistics on a per-session basis and is valid only for the session currently connected. When a session disconnects, all statistics for the session are updated in V$SYSSTAT. The values for the statistics are cleared until the next session uses them.
The V$STATNAME view contains all of the statistics for an Oracle release.

Many of these statistics are tied to the internal implementation of Oracle and therefore are subject to change or deletion without notice, even between patch releases. Application developers should be aware of this and write their code to tolerate missing or extra statistics.

> **See Also**
>
> "V$SESSTAT", "V$STATNAME", and "V$SYSSTAT" for more information on these views

E.2 Statistics Descriptions

This section describes some of the statistics stored in the V$SESSTAT and V$SYSSTAT views. The statistics are listed in alphabetical order.
The CLASS column contains a number representing one or more statistics classes. The following class numbers are additive:

- 1, User
- 2, Redo
- 4, Enqueue
- 8, Cache
- 16, OS
- 32, Real Application Clusters
- 64, SQL
- 128, Debug

Copyright © 2002, 2026, Oracle and/or its affiliates.

For example, a class value of 72 represents a statistic that relates to SQL statements and caching.

Some statistics are populated only if the `TIMED_STATISTICS` initialization parameter is set to `true`. Those statistics are flagged with a `Y` in the right-hand column.

Table E-1 Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
application wait time	1	Total wait time (in centiseconds) for waits that belong to the Application wait class	
background checkpoints completed	8	Number of checkpoints completed by the background process. This statistic is incremented when the background process successfully advances the thread checkpoint.	
background checkpoints started	8	Number of checkpoints started by the background process. This statistic can be larger than "background checkpoints completed" if a new checkpoint overrides an incomplete checkpoint or if a checkpoint is currently under way. This statistic includes only checkpoints of the redo thread. It does not include: • Individual file checkpoints for operations such as offline or begin backup • Foreground (user-requested) checkpoints (for example, performed by `ALTER SYSTEM CHECKPOINT LOCAL` statements)	
background timeouts	128	Number of times a background process has set an alarm for itself and the alarm has timed out rather than the background process being posted by another process to do some work	
branch node splits	128	Number of times an index branch block was split because of the insertion of an additional value	
buffer is not pinned count	72	Number of times a buffer was free when visited. Useful only for internal debugging purposes.	
buffer is pinned count	72	Number of times a buffer was pinned when visited. Useful only for internal debugging purposes.	
bytes received via SQL*Net from client	1	Total number of bytes received from the client over Oracle Net Services	
bytes received via SQL*Net from dblink	1	Total number of bytes received from a database link over Oracle Net Services	
bytes sent via SQL*Net to client	1	Total number of bytes sent to the client from the foreground processes	
bytes sent via SQL*Net to dblink	1	Total number of bytes sent over a database link	
Cached Commit SCN referenced	128	Useful only for internal debugging purposes	
calls to get snapshot scn: kcmgss	32	Number of times a snapshot system change number (SCN) was allocated. The SCN is allocated at the start of a transaction.	
calls to kcmgas	128	Number of calls to routine kcmgas to get a new SCN	
calls to kcmgcs	128	Number of calls to routine kcmgcs to get a current SCN	
calls to kcmgrs	128	Number of calls to routine kcsgrs to get a recent SCN	
cell flash cache read hits	8	Number of read requests that were satisfied by the cache	
cell flash cache read hits for smart IO	8	Number of read requests for smart IO that were satisfied by the cache	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
cell flash cache read hits for temp IO	8	Number of read requests for temp IO that were satisfied by the cache	
cell IO uncompressed bytes	64	Total size of uncompressed data that is processed on the cell For operations on segments compressed using Exadata Hybrid Columnar Compression, this statistic is the size of the data after decompression.	
cell num bytes in passthru due to quarantine	8	Number of bytes that were not offloaded and sent back to the database for processing due to a quarantine on the cell	
cell num bytes in passthru during predicate offload	64	Number of bytes that were not offloaded and sent back to the database for processing	
cell overwrites in flash cache	8	Total number of write requests that overwrote an existing cacheline in Exadata Smart Flash Cache that had not been written to disk In effect, this is the amount of disk I/O saved by using Write-Back mode. This statistic is incremented once per mirror write.	
cell partial writes in flash cache	8	Total number of write requests written to both Exadata Smart Flash Cache and disk Part of the data was written to flash, and the rest was written to disk. This statistic is incremented once per mirror write.	
cell physical IO bytes eligible for predicate offload	64	Number of bytes on-disk eligible for predicate offload	
cell physical IO bytes eligible for smart IOs	8	Number of actual bytes eligible for predicate offload For example, when using columnar cache, this is the size of columnar cache instead of the on-disk size.	
cell physical IO bytes processed for IM capacity	64	Number of bytes read from the columnar cache in `memcompress for capacity` format	
cell physical IO bytes processed for IM query	64	Number of bytes read from the columnar cache in `memcompress for query` format	
cell physical IO bytes processed for no memcompress	64	Number of bytes read from the columnar cache in `no memcompress` format	
cell physical IO bytes saved by columnar cache	8	Number of bytes saved by columnar cache, that is, the number of bytes of reading that were avoided	
cell physical IO bytes saved by storage index	8	Number of bytes saved by the storage index	
cell physical IO bytes saved during optimized file creation	64	Number of I/O bytes saved by the database host by offloading the file creation operation to the cells This statistic shows the benefit of optimized file creation operations.	
cell physical IO bytes saved during optimized RMAN file restore	64	Number of I/O bytes saved by the database host by offloading the RMAN file restore operation to the cells This statistic shows the benefit of optimized RMAN file restore operations.	
cell physical IO bytes sent directly to DB node to balance CPU	64	Number of I/O bytes sent back to the database server for processing due to high storage server CPU usage	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
cell physical IO interconnect bytes	64	Number of I/O bytes exchanged over the interconnect between the database host and the cells	
cell physical IO interconnect bytes returned by smart scan	64	Number of I/O bytes that are returned by the cell for Smart Scan operations This value does not include bytes for other database I/O.	
cell pmem cache read hits	8	Number of non-RDMA read requests processed by `cellsrv` resulting in a PMEM cache hit	
cell pmem cache writes	264	Number of non-RDMA write requests processed by `cellsrv` resulting in a PMEM cache write	
cell ram cache read hits	8	Number of read requests that hit the RAM cache on the cell	
cell RDMA reads	8	Number of PMEM cache read requests using RDMA	
cell RDMA writes	264	Number of PMEM cache write requests using RDMA	
cell writes to flash cache	8	Total number of write requests written entirely to Exadata Smart Flash Cache This statistic is incremented once per mirror write.	
cell writes to flash cache for temp IO	8	Number of write requests for temporary segments that were absorbed by Exadata Smart Flash Cache	
change write time	8	Elapsed redo write time for changes made to `CURRENT` blocks in 10s of milliseconds	Y
cleanouts and rollbacks - consistent read gets	128	Number of "consistent gets" that require both block rollbacks and block cleanouts	
cleanouts only - consistent read gets	128	Number of "consistent gets" that require only block cleanouts, no rollbacks	
cluster key scan block gets	64	Number of blocks obtained in a cluster scan	
cluster key scans	64	Number of cluster scans that were started	
cluster wait time	1	Total wait time (in centiseconds) for waits that belong to the Cluster wait class	
cold recycle reads	8	Number of buffers that were read through the least recently used end of the recycle cache with fast aging strategy	
commit cleanout failures: block lost	8	Number of times Oracle attempted a cleanout at commit but could not find the correct block due to forced write, replacement, or switch `CURRENT`	
commit cleanout failures: buffer being written	8	Number of times Oracle attempted a cleanout at commit, but the buffer was currently being written	
commit cleanout failures: callback failure	8	Number of times the cleanout callback function returns `FALSE`	
commit cleanout failures: cannot pin	8	Total number of times a commit cleanout was performed but failed because the block could not be pinned	
commit cleanout failures: hot backup in progress	8	Number of times Oracle attempted block cleanout at commit during hot backup. The image of the block must be logged before the buffer can be made dirty.	
commit cleanout failures: write disabled	8	Number of times a cleanout block at commit was performed but the writes to the database had been temporarily disabled	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
commit cleanouts	8	Total number of times the cleanout block at commit function was performed	
commit cleanouts successfully completed	8	Number of times the cleanout block at commit function completed successfully	
commit nowait performed	1	Number of asynchronous commits that were actually performed. These commits did not wait for the commit redo to be flushed and be present on disk before returning.	
commit nowait requested	1	Number of no-wait commit or asynchronous commit requests that were made either using SQL or the OCI transaction control API	
Commit SCN cached	128	Number of times the system change number of a commit operation was cached	
commit wait/nowait performed	1	Number of asynchronous/synchronous commits that were actually performed	
commit wait/nowait requested	1	Number of no-wait or wait commits that were made either using SQL or the OCI transaction control API	
commit wait performed	1	Number of synchronous commits that were actually performed. These commits waited for the commit redo to be flushed and be present on disk before returning.	
commit wait requested	1	Number of waiting or synchronous commit requests that were made either using SQL or the OCI transaction control API	
concurrency wait time	1	Total wait time (in centiseconds) for waits that belong to the Concurrency wait class	
consistent changes	8	Number of times a user process has applied rollback entries to perform a consistent read on the block Work loads that produce a great deal of consistent changes can consume a great deal of resources. The value of this statistic should be small in relation to "consistent gets".	
consistent gets	8	Number of times a consistent read was requested for a block **See Also:** "consistent changes" and "session logical reads"	
consistent gets direct	8	Number of times a consistent read was requested for a block bypassing the buffer cache (for example, direct load operation). This is a subset of "consistent gets".	
consistent gets from cache	8	Number of times a consistent read was requested for a block from buffer cache. This is a subset of "consistent gets".	
CPU used by this session	1	Amount of CPU time (in 10s of milliseconds) used by a session from the time a user call starts until it ends. If a user call completes within 10 milliseconds, the start and end user-call time are the same for purposes of this statistic, and 0 milliseconds are added. A similar problem can exist in the reporting by the operating system, especially on systems that suffer from many context switches.	Y
CPU used when call started	128	Amount of CPU time (in 10s of milliseconds) used when the call is started **See Also:** "CPU used by this session"	Y

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
CR blocks created	8	Number of CURRENT blocks cloned to create CR (consistent read) blocks. The most common reason for cloning is that the buffer is held in a incompatible mode.	
cumulative begin requests	32	Total number of begin requests received by the Oracle database allowing the database to detect and process requests	
cumulative DB time in requests	128	Total DB Time (in centiseconds) within requests to the Oracle database, measuring DB time used from the beginning to the end of each request. **See Also:** "DB time"	
cumulative DB time protected in requests	128	Total DB Time (in centiseconds) within requests to the Oracle database protected by Application Continuity or Transparent Application Continuity, measuring DB time used from the beginning to the end of each request. **See Also:** "DB time" This is a subset of "cumulative DB time in requests", which enables you to calculate the ratio of total time in requests to protected time in requests. See also Application Continuity Protection Check (ACCHK).	
cumulative end requests	32	Total number of end requests received by the Oracle database allowing the database to detect and process requests	
cumulative time in requests	128	Total wall clock time (in microseconds) within requests to the Oracle database, measuring time spent from the beginning to the end of each request. This statistic is used by draining and planned failover.	
cumulative user calls in requests	32	Total number of user calls received by the Oracle database within requests to the Oracle database	
cumulative user calls protected by Application Continuity	32	Total number of user calls received by the Oracle database within requests to the Oracle database that were protected by Application Continuity or Transparent Application Continuity	
current blocks converted for CR	8	Number CURRENT blocks converted to CR state	
cursor authentications	128	Number of privilege checks conducted during execution of an operation	
data blocks consistent reads - undo records applied	128	Number of undo records applied to data blocks that have been rolled back for consistent read purposes	
data warehousing cooling action	8	Number of times that cooling occurred on this instance	
data warehousing evicted objects	8	Number of times that objects got evicted by automatic big table caching on this instance	
data warehousing evicted objects - cooling	8	Number of times that objects got evicted on this instance due to a cooling action	
data warehousing evicted objects - replace	8	Number of times that objects got evicted due to caching replacement, that is, when an object is evicted because a hotter object forces it to be evicted from the cache	
data warehousing scanned blocks	8	Number of blocks scanned by automatic big table caching on this instance using parallel query	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
data warehousing scanned blocks - disk	8	Number of blocks scanned by automatic big table caching on this instance by direct read from disk	
data warehousing scanned blocks - memory	8	Number of blocks scanned by automatic big table caching on this instance by cache read from memory	
data warehousing scanned blocks - offload	8	Number of blocks scanned by automatic big table caching on this instance by Exadata offloading	
data warehousing scanned objects	8	Number of times the objects in automatic big table caching are scanned using parallel query	
db block changes	8	Closely related to "consistent changes", this statistic counts the total number of changes that were part of an update or delete operation that were made to all blocks in the SGA. Such changes generate redo log entries and hence become permanent changes to the database if the transaction is committed. This approximates total database work. This statistic indicates the rate at which buffers are being dirtied (on a per-transaction or per-second basis, for example).	
db block gets	8	Number of times a CURRENT block was requested **See Also:** "consistent gets"	
db block gets direct	8	Number of times a CURRENT block was requested bypassing the buffer cache (for example, a direct load operation). This is a subset of "db block gets".	
db block gets from cache	8	Number of times a CURRENT block was requested from the buffer cache. This is a subset of "db block gets".	
DB time	1	Total time (in centiseconds) spent in database calls for foreground sessions. This statistic is an indicator of the total instance workload. It is measured cumulatively from the time of instance startup and is calculated by aggregating the CPU and wait times of all foreground sessions not waiting on idle wait events (non-idle user sessions).	
DB time of LWTs for this session	1	Total DB time (in centiseconds) used by lightweight threads (LWTs) in a session **See Also:** "DB time"	
DBWR checkpoint buffers written	8	Number of buffers that were written for checkpoints	
DBWR checkpoints	8	Number of times the DBWR was asked to scan the cache and write all blocks marked for a checkpoint or the end of recovery. This statistic is always larger than "background checkpoints completed".	
DBWR lru scans	8	Number of times that DBWR scans the LRU queue looking for buffers to write. This count includes scans to fill a batch being written for another purpose (such as a checkpoint).	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
DBWR revisited being-written buffer	8	Number of times that DBWR tried to save a buffer for writing and found that it was already in the write batch. This statistic measures the amount of "useless" work that DBWR had to do in trying to fill the batch. Many sources contribute to a write batch. If the same buffer from different sources is considered for adding to the write batch, then all but the first attempt will be "useless" because the buffer is already marked as being written.	
DBWR transaction table writes	8	Number of rollback segment headers written by DBWR. This statistic indicates how many "hot" buffers were written, causing a user process to wait while the write completed.	
DBWR undo block writes	8	Number of rollback segment blocks written by DBWR	
DDL statements parallelized	32	Number of DDL statements that were executed in parallel	
deferred (CURRENT) block cleanout applications	128	Number of times cleanout records are deferred, piggyback with changes, always current get	
DFO trees parallelized	32	Number of times a serial execution plan was converted to a parallel plan	
dirty buffers inspected	8	Number of dirty buffers found by the user process while it is looking for a buffer to reuse	
DML statements parallelized	32	Number of DML statements that were executed in parallel	
DML statements retried	64	When a long-running DML is executing, the cursor may get invalidated due to some concurrent DDL on one of the cursor's dependencies. In this case, an internal ORA-14403 error is thrown and is caught and cleared in one of the calling functions. The current work is rolled back and the DML is restarted without the user being notified of this. The statistic counts the number of times that the thrown, caught, and cleared (ORA-14403) sequence occurred for DML statements. Should a DML vary widely in execution time, check this statistic to see if it increments during the DML execution. If so, then concurrent DDL may be the cause of the extra elapsed time.	
enqueue conversions	4	Total number of conversions of the state of table or row lock	
enqueue deadlocks	4	Total number of deadlocks between table or row locks in different sessions	
enqueue releases	4	Total number of table or row locks released	
enqueue requests	4	Total number of table or row locks acquired	
enqueue timeouts	4	Total number of table and row locks (acquired and converted) that timed out before they could complete	
enqueue waits	4	Total number of waits that occurred during an enqueue convert or get because the enqueue get was deferred	
exchange deadlocks	8	Number of times that a process detected a potential deadlock when exchanging two buffers and raised an internal, restartable error. Index scans are the only operations that perform exchanges.	
execute count	64	Total number of calls (user and recursive) that executed SQL statements	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
fbda woken up	128	Number of times the flashback data archive background process was woken up to do archiving	
file io wait time	1	Total time spent in wait (in microseconds) for I/O to datafiles, excluding the service time for such I/O. This is cumulative for all I/Os for all datafiles. The service time for one I/O operation is estimated as the minimum time spent in the I/O call seen so far. This service time is subtracted from the time spent in each I/O call to get the wait time for that I/O.	
flash cache eviction: aged out	8	Flash cache buffer is aged out of the Database Smart Flash Cache	
flash cache eviction: buffer pinned	8	Database Smart Flash Cache buffer is invalidated due to object or range reuse, and so on. The Database Flash Cache Buffer was in use at the time of eviction.	
flash cache eviction: invalidated	8	Database Smart Flash Cache buffer is invalidated due to object or range reuse, and so on. The Database Smart Flash Cache buffer was not in use at the time of eviction.	
flash cache insert skip: corrupt	8	In-memory buffer was skipped for insertion into the Database Smart Flash Cache because the buffer was corrupted	
flash cache insert skip: DBWR overloaded	8	In-memory buffer was skipped for insertion into the Database Smart Flash Cache because DBWR was busy writing other buffers	
flash cache insert skip: exists	8	In-memory buffer was skipped for insertion into the Database Smart Flash Cache because it was already in the flash cache	
flash cache insert skip: modification	8	In-memory buffer was skipped for insertion into the Database Smart Flash Cache because it was being modified	
flash cache insert skip: not current	8	In-memory buffer was skipped for insertion into the Database Smart Flash Cache because it was not current	
flash cache insert skip: not useful	8	In-memory buffer was skipped for insertion into the Database Smart Flash Cache because the type of buffer was not useful to keep	
flash cache inserts	8	Total number of in-memory buffers inserted into the Database Smart Flash Cache	
flashback log write bytes	2	Total size in bytes of flashback database data written by RVWR to flashback database logs	
flashback log writes	2	Total number of writes by RVWR to flashback database logs	
foreground propagated tracked transactions	128	Number of transactions modifying tables enabled for flashback data archive which were archived by a foreground process	
free buffer inspected	8	Number of buffers skipped over from the end of an LRU queue to find a reusable buffer. The difference between this statistic and "dirty buffers inspected" is the number of buffers that could not be used because they had a user, a waiter, or were being read or written, or because they were busy or needed to be written after rapid aging out.	
free buffer requested	8	Number of times a reusable buffer or a free buffer was requested to create or load a block	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
gc read wait failures	40	A read wait is when a CR server waits for a disk read to complete before serving a block to another instance. This statistic displays the number of times a read wait ended in failure, that is, after waiting it was unable to serve a block.	
gc read wait timeouts	40	A read wait is when a CR server waits for a disk read to complete before serving a block to another instance. This statistic displays the number of times a read wait timed out, that is, the disk read did not complete in time, so the wait was terminated.	
gc read waits	40	Number of times a CR server waited for a disk read, and then successfully served a block	
global enqueue CPU used by this session	32	Amount of CPU time (in 10s of milliseconds) used by synchronous and asynchronous global enqueue activity in a session from the time a user call starts until it ends. If a user call completes within 10 milliseconds, the start and end user-call time are the same for purposes of this statistic, and 0 milliseconds are added.	
global enqueue get time	32	Total elapsed time in 10s of milliseconds of all synchronous and asynchronous global enqueue gets and converts	
global enqueue gets async	32	Total number of asynchronous global enqueue gets and converts	
global enqueue gets sync	32	Total number of synchronous global enqueue gets and converts	
global enqueue releases	32	Total number of synchronous global enqueue releases	
hot buffers moved to head of LRU	8	When a hot buffer reaches the tail of its replacement list, Oracle moves it back to the head of the list to keep it from being reused. This statistic counts such moves.	
HCC analyze table CUs	128	Number of Compression Units that were decompressed for `ANALYZE TABLE`	
HCC analyzer calls	128	Number of times `ANALYZE TABLE` has been run on Hybrid Columnar Compression (HCC) compressed tables	
HCC block compressions attempted	128	Number of times the system failed to recompress an OLTP compressed block in HCC format	
HCC block compressions completed	128	Number of times an OLTP compressed block was successfully recompressed in HCC format	
HCC DML conventional	128	Number of searched DML statements (`UPDATE` and `DELETE`) issued against HCC compressed tables	
HCC DML CUs	128	Number of Compression Units that were decompressed for searched DMLs (`UPDATE` and `DELETE`)	
HCC fetch by rowid CUs	128	Number of Compression Units that were decompressed by single row fetch	
HCC load conventional bytes compressed	128	Length of the data after compression stored in Compression Units by conventional loads. The amount of space used on disk is the sum of the data stored plus the metadata needed to locate the stored data.	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
HCC load conventional bytes uncompressed	128	Length of the data before compression stored in Compression Units by conventional loads. The amount of space used on disk is the sum of the data stored plus the metadata needed to locate the stored data.	
HCC load conventional CUs	128	Number of Compression Units that were successfully compressed in any format from conventional loads	
HCC load conventional CUs archive high	128	Number of Compression Units that were successfully compressed in HCC `ARCHIVE HIGH` format from conventional loads	
HCC load conventional CUs archive low	128	Number of Compression Units that were successfully compressed in HCC `ARCHIVE LOW` format from conventional loads	
HCC load conventional CUs query high	128	Number of Compression Units that were successfully compressed in HCC `QUERY HIGH` format from conventional loads	
HCC load conventional CUs query low	128	Number of Compression Units that were successfully compressed in HCC `QUERY LOW` format from conventional loads	
HCC load conventional rows	128	Number of rows that were successfully compressed in HCC format from conventional loads	
HCC load conventional rows not compressed	128	Number of rows that were unable to be compressed in HCC format from conventional loads because the compression resulted in taking more space, not less	
HCC load direct bytes compressed	128	Length of the data after compression stored in Compression Units by Direct Path Loads. The amount of space used on disk is the sum of the data stored plus the metadata needed to locate the stored data.	
HCC load direct bytes uncompressed	128	Length of the data before compression stored in Compression Units by Direct Path Loads. The amount of space used on disk is the sum of the data stored plus the metadata needed to locate the stored data.	
HCC load direct CUs	128	Number of Compression Units that were successfully HCC compressed by Direct Path Loads	
HCC load direct CUs archive high	128	Number of rows from Direct Path Loads that were HCC compressed with `ARCHIVE HIGH` successfully	
HCC load direct CUs archive low	128	Number of rows from Direct Path Loads that were HCC compressed with `ARCHIVE LOW` successfully	
HCC load direct CUs query high	128	Number of rows from Direct Path Loads that were HCC compressed with `QUERY HIGH` successfully	
HCC load direct CUs query low	128	Number of rows from Direct Path Loads that were HCC compressed with `QUERY LOW` successfully	
HCC load direct rows	128	Number of rows from Direct Path Loads that were HCC compressed successfully	
HCC load direct rows not compressed	128	Number of rows from Direct Path Loads that were OLTP compressed instead of HCC compressed due to negative compression	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
HCC scan cell bytes compressed	128	Length of the compressed data stored in Compression Units prior to decompression. This is the length of all columns, not just the columns accessed.	
HCC scan cell bytes decompressed	128	Length of the data after decompression stored in Compression Units. This is the length of all columns, not just the columns accessed.	
HCC scan cell CUs archive high	128	Number of Compression Units scanned that had been compressed with `ARCHIVE HIGH`	
HCC scan cell CUs archive low	128	Number of Compression Units scanned that had been compressed with `ARCHIVE LOW`	
HCC scan cell CUs columns accessed	128	Number of columns that needed to be decompressed on the storage server to process the Compression Unit	
HCC scan cell CUs decompressed	128	Number of Compression Units that needed to be decompressed on the storage server	
HCC scan cell CUs decompression time	128	Total time spent decompressing the columns that were needed on the storage server	
HCC scan cell CUs optimized read	128	Number of Compression Units where all the rows were within the stored Min/Max values. Evaluation of those predicates was optimized out.	
HCC scan cell CUs pruned	128	Number of Compression Units that were pruned based on the Min/Max value checks	
HCC scan cell CUs query high	128	Number of Compression Units scanned that had been compressed with `QUERY HIGH`	
HCC scan cell CUs query low	128	Number of Compression Units scanned that had been compressed with `QUERY LOW`	
HCC scan cell CUs sent compressed	64	Number of Compression Units that the storage server returned to the RDBMS still compressed after predicate evaluation was done	
HCC scan cell CUs sent head piece	64	Number of Compression Units that the storage server was unable to process and the block header was returned to the RDBMS to be processed	
HCC scan cell CUs sent uncompressed	64	Number of Compression Units that the storage server returned to the RDBMS in uncompressed columnar format	
HCC scan cell rows	128	Number of rows from columnar formatted data that were returned by the storage server to the RDBMS	
HCC scan CUs pcode aggregation pushdown	128	Number of Compression Units that were aggregated on the cell using P(ortable)-byte code	
HCC scan CUs pcode pred evaled	128	Number of predicates that were evaluated using P(ortable)-byte code	
HCC scan CUs pcode pred evaled using rowsets	128	Number of predicates that were evaluated using P(ortable)-byte code using vectorized data	
HCC scan CUs predicates applied	128	Number of predicates where at least some value passed the Min/Max check	
HCC scan CUs predicates optimized	128	Number of predicates that could be evaluated directly against columnar data	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
HCC scan CUs predicates received	128	Number of predicates sent to the storage server that could be evaluated directly against HCC format data after decompression	
HCC scan rdbms bytes compressed	128	Length of the compressed data stored in Compression Units prior to decompression. This is the length of all columns, not just the columns accessed.	
HCC scan rdbms bytes decompressed	128	Length of the data after decompression stored in Compression Units. This is the length of all columns, not just the columns accessed.	
HCC scan rdbms CUs archive high	128	Number of HCC Compression Units decompressed by the RDBMS table scan that had been compressed with `ARCHIVE HIGH`	
HCC scan rdbms CUs archive low	128	Number of HCC Compression Units decompressed by the RDBMS table scan that had been compressed with `ARCHIVE LOW`	
HCC scan rdbms CUs columns accessed	128	Number of columns that were decompressed	
HCC scan rdbms CUs decompressed	128	Number of Compression Units that were decompressed on the RDBMS server	
HCC scan rdbms CUs decompression time	128	Total time spent decompressing the columns that were needed on the RDBMS server	
HCC scan rdbms CUs normal	128	Number of HCC Compression Units decompressed by the RDBMS regular table scan, i.e., 'Table Access Full' in a query plan. This is typically used in specialized scans that, for example, use row versions, row SCN, or use the `LONG RAW` datatype.	
HCC scan rdbms CUs pruned	128	Number of Compression Units that were eliminated by comparing the predicates to the stored Min and Max values for that Comprssion Unit	
HCC scan rdbms CUs query high	128	Number of HCC Compression Units decompressed by the RDBMS table scan that had been compressed with `QUERY HIGH` (which is the default)	
HCC scan rdbms CUs query low	128	Number of HCC Compression Units decompressed by the RDBMS table scan that had been compressed with `QUERY LOW`	
HCC scan rdbms CUs turbo	128	Number of HCC Compression Units decompressed by the RDBMS fast table scan (TurboScan). This is separate to decompression that happens on an Exadata cell.	
HCC scan rdbms rows	128	Number of rows returned from HCC blocks or that were returned from an Exadata cell in columnar format that passed the predicates on that table	
HCC scan rows pcode aggregated	128	Number of rows from HCC blocks or from columnar format blocks returned by an Exadata cell that could be aggregated using the P(ortable)-code style of query compilation	
HCC usage cloud	128	Internal count of how often TurboScan is invoked for HCC data stored in an Oracle Cloud environment. It does not include Oracle on other vendor's clouds. This may or may not correspond with the number of queries or granules scanned.	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
HCC usage pillar	128	Internal count of how often TurboScan is invoked for HCC data stored on Pillar storage. This may or may not correspond with the number of queries or granules scanned.	
HCC usage ZFS	128	Internal count of how often TurboScan is invoked for HCC data stored on ZFS storage. This may or may not correspond with the number of queries or granules scanned.	
immediate (CR) block cleanout applications	128	Number of times cleanout records are applied immediately during consistent-read requests	
immediate (CURRENT) block cleanout applications	128	Number of times cleanout records are applied immediately during current gets. Compare this statistic with "deferred (CURRENT) block cleanout applications".	
IM default area resized	128	Amount of memory by which the column store got resized	
IM populate accumulated time (ms)	128	Total amount of DB time (in milliseconds) spent populating CUs into the IM column store due to segment scans	
IM populate bytes in-memory EU data	128	Size in bytes of in-memory EU data populated due to segment scans	
IM populate bytes uncompressed EU data	128	Uncompressed size in bytes of in-memory EU data populated due to segment scans	
IM populate CUs	128	Number of CUs populated in the IM column store due to segment scans	
IM populate CUs memcompress for capacity high	128	Number of CUs populated in the IM column store due to segment scans using memcompress for capacity high	
IM populate CUs memcompress for capacity low	128	Number of CUs populated in the IM column store due to segment scans using memcompress for capacity low	
IM populate CUs memcompress for dml	128	Number of CUs populated in the IM column store due to segment scans using memcompress for DML	
IM populate CUs memcompress for query high	128	Number of CUs populated in the IM column store due to segment scans using memcompress for query high	
IM populate CUs memcompress for query low	128	Number of CUs populated in the IM column store due to segment scans using memcompress for query low	
IM populate CUs no memcompress	128	Number of CUs populated in the IM column store due to segment scans without compression	
IM populate CUs requested	128	Number of CUs requested to be populated due to segment scans	
IM populate EUs	128	Number of EUs populated in the IM column store due to segment scans	
IM populate EUs accumulated time (ms)	128	Total amount of DB time (in milliseconds) spent populating EUs into the IM column store due to segment scans	
IM populate EUs columns	128	Number of columns populated in EUs due to segment scans	
IM populate EUs memcompress for capacity high	128	Number of EUs populated in the IM column store due to segment scans at memcompress for capacity high	
IM populate EUs memcompress for capacity low	128	Number of EUs populated in the IM column store due to segment scans at memcompress for capacity low	
IM populate EUs memcompress for dml	128	Number of EUs populated in the IM column store due to segment scans at memcompress for DML	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
IM populate EUs memcompress for query high	128	Number of EUs populated in the IM column store due to segment scans at memcompress for query high	
IM populate EUs memcompress for query low	128	Number of EUs populated in the IM column store due to segment scans at memcompress for query low	
IM populate EUs no memcompress	128	Number of EUs populated in the IM column store without compression due to segment scans	
IM populate EUs requested	128	Number of EUs requested to be populated in the IM column store due to segment scans	
IM populate no contiguous inmemory space	128	Number of CUs that fail to populate due to lack of contiguous space in In-Memory area	
IM populate segments	128	Number of segments populated due to segment scan	
IM populate segments requested	128	Number of segments requested to be populated due to segment scan	
IM populate segments wall clock time (ms)	128	Total amount of wall clock time (in milliseconds) spent populating CUs into the IM column store due to segment scans	
IM prepopulate accumulated time (ms)	128	Total amount of DB time (in milliseconds) spent prepopulating CUs into the IM column store priority	
IM prepopulate bytes in-memory EU data	128	Size in bytes of in-memory EU data populated due to priority	
IM prepopulate bytes uncompressed EU data	128	Uncompressed size in bytes of in-memory EU data populated due to priority	
IM prepopulate CUs	128	Number of CUs prepopulated in the IM column store due to priority	
IM prepopulate CUs memcompress for capacity high	128	Number of CUs prepopulated in the IM column store due to priority using memcompress for capacity high	
IM prepopulate CUs memcompress for capacity low	128	Number of CUs prepopulated in the IM column store due to priority using memcompress for capacity low	
IM prepopulate CUs memcompress for dml	128	Number of CUs prepopulated in the IM column store due to priority using memcompress for DML	
IM prepopulate CUs memcompress for query high	128	Number of CUs prepopulated in the IM column store due to priority using memcompress for query high	
IM prepopulate CUs memcompress for query low	128	Number of CUs prepopulated in the IM column store due to priority using memcompress for query low	
IM prepopulate CUs no memcompress	128	Number of CUs prepopulated in the IM column store due to priority without compression	
IM prepopulate CUs requested	128	Number of CUs requested to be prepopulated due to priority	
IM prepopulate EUs	128	Number of EUs populated in the IM column store due to priority	
IM prepopulate EUs accumulated time (ms)	128	Total amount of DB time (in milliseconds) spent populating EUs into the IM column store due to priority	
IM prepopulate EUs columns	128	Number of columns populated in EUs due to priority	
IM prepopulate EUs memcompress for capacity high	128	Number of EUs populated in the IM column store due to priority at memcompress for capacity high	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
IM prepopulate EUs memcompress for capacity low	128	Number of EUs populated in the IM column store due to priority at memcompress for capacity low	
IM prepopulate EUs memcompress for dml	128	Number of EUs populated in the IM column store due to priority at memcompress for DML	
IM prepopulate EUs memcompress for query high	128	Number of EUs populated in the IM column store due to priority at memcompress for query high	
IM prepopulate EUs memcompress for query low	128	Number of EUs populated in the IM column store due to priority at memcompress for query low	
IM prepopulate EUs no memcompress	128	Number of EUs populated in the IM column store without compression due to priority	
IM prepopulate EUs requested	128	Number of EUs requested to be populated in the IM column store due to priority	
IM prepopulate segments	128	Number of segments prepopulated due to priority	
IM prepopulate segments requested	128	Number of segments requested to be prepopulated due to priority	
IM repopulate accumulated time (ms)	128	Total amount of DB time (in milliseconds) spent repopulating CUs into the IM column store due to DML changes	
IM repopulate bytes in-memory EU data	128	Size in bytes of in-memory EU data repopulated due to EU reaching staleness threshold	
IM repopulate CUs	128	Total number of CUs requested to be repopulated due to CU reaching staleness threshold	
IM repopulate CUs memcompress for capacity high	128	Number of CUs repopulated in the IM column store using memcompress for capacity high due to CU reaching staleness threshold	
IM repopulate CUs memcompress for capacity low	128	Number of CUs repopulated in the IM column store using memcompress for capacity low due to CU reaching staleness threshold	
IM repopulate CUs memcompress for dml	128	Number of CUs repopulated in the IM column store using memcompress for DML due to CU reaching staleness threshold	
IM repopulate CUs memcompress for query high	128	Number of CUs repopulated in the IM column store using memcompress for query high due to CU reaching staleness threshold	
IM repopulate CUs memcompress for query low	128	Number of CUs repopulated in the IM column store using memcompress for query low due to CU reaching staleness threshold	
IM repopulate CUs no memcompress	128	Number of CUs repopulated in the IM column store without compression due to CU reaching staleness threshold	
IM repopulate CUs requested	128	Total number of CUs requested to be repopulated due to CU reaching staleness threshold	
IM repopulate (doublebuffering) CUs	128	Number of CUs repopulated with double-buffering enabled on the earlier version of the CUs	
IM repopulate (doublebuffering) CUs requested	128	Number of CUs requested to be repopulated with double-buffering enabled on the earlier version of the CUs	
IM repopulate EUs	128	Number of EUs requested to be repopulated due to EU reaching staleness threshold	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
IM repopulate EUs accumulated time (ms)	128	Total amount of DB time (in milliseconds) spent repopulating EUs into the IM column store due to DML changes	
IM repopulate EUs columns	128	Number of columns repopulated in EUs due to EU reaching staleness threshold	
IM repopulate EUs memcompress for capacity high	128	Number of EUs repopulated in the IM column store at memcompress for capacity high due to EU reaching staleness threshold	
IM repopulate EUs memcompress for capacity low	128	Number of EUs repopulated in the IM column store at memcompress for capacity low due to EU reaching staleness threshold	
IM repopulate EUs memcompress for dml	128	Number of EUs repopulated in the IM column store at memcompress for DML due to EU reaching staleness threshold	
IM repopulate EUs memcompress for query high	128	Number of EUs repopulated in the IM column store at memcompress for query high due to EU reaching staleness threshold	
IM repopulate EUs memcompress for query low	128	Number of EUs repopulated in the IM column store at memcompress for query low due to EU reaching staleness threshold	
IM repopulate EUs no memcompress	128	Number of EUs repopulated in the IM column store without compression due to EU reaching staleness threshold	
IM repopulate EUs requested	128	Total number of EUs requested to be repopulated due to EU reaching staleness threshold	
IM repopulate (incremental) CUs	128	Number of CUs repopulated incrementally from earlier versions of the CUs	
IM repopulate (incremental) CUs requested	128	Number of CUs requested to be repopulated incrementally from earlier versions of the CUs	
IM repopulate (incremental) EUs	128	Number of EUs repopulated using unchanged data from the current EU due to EU reaching staleness threshold	
IM repopulate (incremental) EUs requested	128	Number of EUs requested to be repopulated using unchanged data from the current EU due to EU reaching staleness threshold	
IM repopulate no contiguous inmemory space	128	Number of CUs that failed to repopulate due to lack of contigunous space in In-Memory area	
IM repopulate (scan) CUs	128	Number of CUs repopulated in the IM column store due to scans	
IM repopulate (scan) CUs requested	128	Number of CUs requested to be repopulated in the IM column store due to scans	
IM repopulate (scan) EUs	128	Number of EUs repopulated in the IM column store that were triggered by scans on the EU	
IM repopulate (scan) EUs requested	128	Number of EUs requested for repopulation in the IM column store that were triggered by scans on the EU	
IM repopulate segments	128	Number of segments repopulated	
IM repopulate segments requested	128	Indicates the number of segments requested to be repopulated	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
IM repopulate (trickle) accumulated time (ms)	128	Total amount of DB time (in milliseconds) spent trickle repopulating CUs into the IM column store due to DML changes	
IM repopulate (trickle) bytes in-memory EU data	128	Size in bytes of in-memory EU data repopulated due to DML changes	
IM repopulate (trickle) bytes uncompressed EU data	128	Uncompressed size in bytes of in-memory EU data repopulated due to EU reaching staleness threshold	
IM repopulate (trickle) CUs	128	Number of CUs trickle repopulated in the IM column store due to DML changes	
IM repopulate (trickle) CUs memcompress for capacity high	128	Number of CUs trickle repopulated in the IM column store using memcompress for capacity high due to DML changes	
IM repopulate (trickle) CUs memcompress for capacity low	128	Number of CUs trickle repopulated in the IM column store using memcompress for capacity low due to DML changes	
IM repopulate (trickle) CUs memcompress for dml	128	Number of CUs trickle repopulated in the IM column store using memcompress for DML due to DML changes	
IM repopulate (trickle) CUs memcompress for query high	128	Number of CUs trickle repopulated in the IM column store using memcompress for query high due to DML changes	
IM repopulate (trickle) CUs memcompress for query low	128	Number of CUs trickle repopulated in the IM column store using memcompress for query low due to DML changes	
IM repopulate (trickle) CUs no memcompress	128	Number of CUs trickle repopulated in the IM column store without compression due to DML changes	
IM repopulate (trickle) CUs requested	128	Total number of CUs requested to be trickle repopulated due to DML changes	
IM repopulate (trickle) CUs resubmitted	128	Number of CUs trickle repopulate tasks submitted	
IM repopulate (trickle) EUs	128	Number of EUs trickle repopulated in the IM column store due to DML changes	
IM repopulate (trickle) EUs accumulated time (ms)	128	Total amount of DB time (in milliseconds) spent trickle repopulating EUs into the IM column store due to DML changes	
IM repopulate (trickle) EUs columns	128	Number of columns repopulated in EUs due to DML changes	
IM repopulate (trickle) EUs memcompress for capacity high	128	Number of EUs trickle repopulated in the IM column store due to DML changes at memcompress for capacity high	
IM repopulate (trickle) EUs memcompress for capacity low	128	Number of EUs trickle repopulated in the IM column store due to DML changes at memcompress for capacity low	
IM repopulate (trickle) EUs memcompress for dml	128	Number of EUs trickle repopulated in the IM column store due to DML changes at memcompress for DML	
IM repopulate (trickle) EUs memcompress for query high	128	Number of EUs trickle repopulated in the IM column store due to DML changes at memcompress for query high	
IM repopulate (trickle) EUs memcompress for query low	128	Number of EUs trickle repopulated in the IM column store due to DML changes at memcompress for query low	
IM repopulate (trickle) EUs no memcompress	128	Number of EUs trickle repopulated in the IM column store without compression due to DML changes	
IM repopulate (trickle) EUs requested	128	Number of EUs requested to be trickle repopulated in the IM column store due to DML changes	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
IM scan CUs column not in memory	128	Number of extents that could not be read from the IM column store because one of the columns required was not in memory	
IM scan CUs invalid or missing revert to on disk extent	128	Number of extents where no IMCU exists	
IM scan CUs memcompress for query low	128	Number of memcompress for query high CUs scanned in the IM column store	
IM scan CUs memcompress for query high	128	Number of memcompress for query high CUs scanned in the IM column store	
IM scan CUs memcompress for capacity low	128	Number of memcompress for capacity low CUs scanned in the IM column store	
IM scan CUs memcompress for capacity high	128	Number of memcompress for capacity high CUs scanned in the IM column store	
IM scan CUs memcompress for dml	128	Number of memcompress for DML CUs scanned in the IM column store	
IM scan CUs predicates applied	128	Number of where clause predicates applied to the In-Memory storage index	
IM scan CUs predicates optimized	128	Number of where clause predicates applied to the IM column store for which either all rows pass min/max pruning via an In-Memory storage index or no rows pass min/max pruning	
IM scan CUs pruned	128	Number of CUs pruned by the storage index	
IM scan (dynamic) multi-threaded scans	128	Number of In-Memory table scans which benefited from In-Memory dynamic scans	
IM scan (dynamic) tasks processed by parent	128	Number of IMCUs processed normally because of Resource Manager limit	
IM scan (dynamic) tasks processed by thread	128	Number of IMCUs processed in parallel by a worker thread	
IM scan (dynamic) rows	128	Number of rows processed by In-Memory dynamic scans	
IM scan EU bytes in-memory	128	Size in bytes of in-memory EU data accessed by scans	
IM scan EU bytes uncompressed	128	Uncompressed size in bytes of in-memory EU data accessed by scans	
IM scan EU rows	128	Number of rows scanned from EUs in the IM column store before where clause predicate applied	
IM scan EUs columns accessed	128	Number of columns in the EUs accessed by scans	
IM scan EUs columns decompressed	128	Number of columns in the EUs decompressed by scans	
IM scan EUs columns theoretical max	128	Number of columns that would have been accessed from the EU if the scans looked at all columns	
IM scan EUs memcompress for capacity high	128	Number of memcompress for capacity high EUs scanned in the IM column store	
IM scan EUs memcompress for capacity low	128	Number of memcompress for capacity low EUs scanned in the IM column store	
IM scan EUs memcompress for dml	128	Number of memcompress for DML EUs scanned in the IM column store	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
IM scan EUs memcompress for query high	128	Number of memcompress for query high EUs scanned in the IM column store	
IM scan EUs memcompress for query low	128	Number of memcompress for query low EUs scanned in the IM column store	
IM scan EUs no memcompress	128	Number of uncompressed EUs scanned in the IM column store	
IM scan EUs split pieces	128	Number of split EU pieces among all IM EUs	
IM scan rows	128	Number of rows in scanned In-Memory Compression Units (IMCUs)	
IM scan rows optimized	128	Number of rows that were not scanned in the IM column store as they were pruned via a number of optimizations such as min/max pruning via In-Memory storage indexes	
IM scan rows projected	128	Number of rows returned from the IM column store	
IM scan rows valid	128	Number of rows scanned from the IM column store after applying valid vector	
IM scan segments minmax eligible	128	Number of CUs that are eligible for min/max pruning via storage index	
IM space CU bytes allocated	128	Number of In-Memory bytes allocated	
IM space CU creations initiated	128	Number of space requests for CUs	
IM space CU extents allocated	128	Number of In-Memory extents allocated	
IM space segments allocated	128	Number of snapshot segments created	
IM space segments freed	128	Number of snapshot segments deleted	
IM transactions	128	Number of transactions that triggered data to be journaled in the IM column store	
IM transactions CUs invalid	128	Number of CUs in the IM column store invalidated by transactions	
IM transactions rows invalidated	128	Number of rows in the IM column store invalidated by transactions	
IM transactions rows journaled	128	Number of rows logged in the transaction journal	
in call idle wait time	1	Total wait time (in centiseconds) for waits that belong to the Idle wait class. **See Also:** "non-idle wait count" and "non-idle wait time"	
index cmph cu, uncomp sentinels	128	Number of CUs created with uncompressed sentinels	
index cmph dm, cu lock expand	128	Number of times CU lock structure expanded	
index cmph dm, cu migrate row	128	Number of times a row migrated from a CU	
index cmph dm, insert unpurge CU row	128	Number of times a CU row was unpurged during insert	
index cmph dm, purge dummy CU	128	Number of times dummy CU purged from leaf block	
index cmph dm, split for cu lock expand	128	Number of times leaf block split for CU lock expansion	
index cmph dm, split for cu migrate row	128	Number of leaf block splits due to CU row migration	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
index cmph ld, CU fit	128	Number of times load created a well sized CU, no space for uncompressed rows	
index cmph ld, CU fit, add rows	128	Number of times load created a well sized CU, with space for uncompressed rows	
index cmph ld, CU negative comp	128	Number of times load CU gave negative compression	
index cmph ld, CU over-est	128	Number of times load created an oversized CU	
index cmph ld, CU under-est	128	Number of times load created a small CU	
index cmph ld, infinite loop	128	Number of times shrink CU attempts resulted in uncompressed rows	
index cmph ld, lf blks flushed	128	Number of leaf blocks flushed by load	
index cmph ld, lf blks w/ und CU	128	Number of leaf blocks flushed with small CU	
index cmph ld, lf blks w/o CU	128	Number of leaf blocks flushed without a CU	
index cmph ld, lf blks w/o unc r	128	Number of leaf blocks flushed without uncompressed rows	
index cmph ld, retry in over-est	128	Number of times CU was resized after creating an oversized CU	
index cmph ld, rows compressed	128	Number of rows compressed by load	
index cmph ld, rows uncompressed	128	Number of rows left uncompressed by load	
index cmph sc, ffs decomp buffers	128	Number of blocks decompressed for fast scan	
index cmph sc, ffs decomp buffers released and found valid	128	Number of times decompressed CU buffer was reused by fast scan	
index cmph sc, ffs decomp buffers rows avail	128	Number of rows in decompressed buffer for fast scan	
index cmph sc, ffs decomp buffers rows used	128	Number of rows used from decompressed buffer for fast scan	
index cmph sc, ffs decomp failures	128	Number of time decompress CU was not possible for fast scan	
	128	Number of times 90-10 leaf block CU splits were made 50-50	
index cmph sp, leaf norecomp limit	128	Number of times leaf block recompression reached the recompression limit	
index cmph sp, leaf norecomp negcomp	128	Number of times leaf block recompression returned negative compression	
index cmph sp, leaf norecomp nospace	128	Number of times leaf block recompression returned not enough space	
index cmph sp, leaf norecomp notry	128	Number of times leaf block recompression not attempted	
index cmph sp, leaf norecomp oversize	128	Number of times leaf block recompression returned an oversized CU	
index cmph sp, leaf norecomp zerocur	128	Number of times leaf block recompression returned a CU with 0 rows	
index cmph sp, leaf recomp fewer ucs	128	Number of CUs created with reduced number of sentinels	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
index cmph sp, leaf recomp zero ucs	128	Number of CUs created with zero sentinels	
index cmph sp, leaf recompress	128	Number of times a leaf block CU was recompressed	
index cmpl co, prefix mismatch	128	Number of times reorg found a neighboring block prefix count mismatch	
index cmpl ro, blocks not compressed	128	Number of times prefix compression was not applied to avoid negative compression	
index cmpl ro, prefix change at block	128	Number of times prefix count was changed to an optimal value	
index cmpl ro, prefix no change at block	128	Number of times prefix count was already the optimal value	
index cmpl ro, reorg avoid load new block	128	Number of times a block reorg avoided a new block being created during load	
index cmpl ro, reorg avoid split	128	Number of times a block reorg avoided a block split during DML changes	
index fast full scans (direct read)	64	Number of fast full scans initiated using direct read	
index fast full scans (full)	64	Number of fast full scans initiated for full segments	
index fast full scans (rowid ranges)	64	Number of fast full scans initiated with rowid endpoints specified	
large tracked transactions	128	For tables tracked by flashback data archive, the number of transactions modifying rows in those tables which are large in terms of size or number of changes	
leaf node splits	128	Number of times an index leaf node was split because of the insertion of an additional value	
lob reads	8	Number of LOB API read operations performed in the session/system. A single LOB API read may correspond to multiple physical/logical disk block reads.	
lob writes	8	Number of LOB API write operations performed in the session/system. A single LOB API write may correspond to multiple physical/logical disk block writes.	
lob writes unaligned	8	Number of LOB API write operations whose start offset or buffer size is not aligned to the internal chunk size of the LOB. Writes aligned to chunk boundaries are the most efficient write operations. The internal chunk size of a LOB is available through the LOB API (for example, `DBMS_LOB.GETCHUNKSIZE()`).	
logons cumulative	1	Total number of logons since the instance started. Useful only in V$SYSSTAT. It gives an instance overview of all processes that logged on.	
logons current	1	Total number of current logons. Useful only in V$SYSSTAT.	
memopt r failed puts	128	Total failed puts on hash index	
memopt r failed reads on blocks	128	Total lookup failures due to read failure on blocks because of concurrent changes	
memopt r failed reads on buckets	128	Total lookup failures due to concurrent hash bucket changes	
memopt r hits	128	Total hits on hash index – primary key found	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
memopt r lookup detected CR buffer	128	Total lookup failures due to block pointed to by hash index being no longer the current version	
memopt r lookups	128	Total number of lookups on hash index	
memopt r misses	128	Total misses on hash index due to primary key not found	
memopt r puts	128	Total puts on hash index	
memopt r successful puts	128	Total successful puts on hash index	
messages received	128	Number of messages sent and received between background processes	
messages sent	128	Number of messages sent and received between background processes	
no buffer to keep pinned count	72	Number of times a visit to a buffer attempted, but the buffer was not found where expected. Like "buffer is not pinned count" and "buffer is pinned count", this statistic is useful only for internal debugging purposes.	
no work - consistent read gets	128	Number of "consistent gets" that require neither block cleanouts nor rollbacks	
non-idle wait count	1	Total number of waits performed with wait events that were not part of the Idle wait class. **See Also:** "in call idle wait time" and "non-idle wait time"	
non-idle wait time	1	Total wait time (in microseconds) for waits that do not belong to the Idle wait class. **See Also:** "in call idle wait time" and "non-idle wait count"	
OLAP Aggregate Function Calc	64	Number of times the `AGGREGATE` function computes a parent value based on the values of its children	
OLAP Aggregate Function Logical NA	64	Number of times an `AGGREGATE` function evaluates to a logical NA value. This could be because the `AGGINDEX` is on and the composite tuple does not exist.	
OLAP Aggregate Function Precompute	64	Number of times the `AGGREGATE` function is to compute a value and finds it precomputed in the cube	
OLAP Custom Member Limit	64	Number of times an OLAP table function issues a custom member limit	
OLAP Engine Calls	64	Total number of OLAP transactions executed within the session. This value provides a general indication of the level of OLAP activity in the session.	
OLAP Fast Limit	64	Number of times an OLAP table function issues a fast limit	
OLAP Full Limit	64	Number of times an OLAP table function issues a full limit	
OLAP GID Limit	64	Number of times an OLAP table function issues a Cube Grouping ID (CGID) limit. Typically, this type of limit occurs for query rewrite transformations that resolve to a cube organized materialized view.	
OLAP Import Rows Loaded	64	Number of OLAP import rows loaded. This statistic provides the number of rows of the source cursor that are actually loaded into an Analytic Workspace (AW). The difference between the OLAP Import Rows Pushed and OLAP Import Rows Loaded provides the number of rejected rows.	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
OLAP Import Rows Pushed	64	Number of OLAP import rows pushed. This statistic refers to the number of rows encountered from a source cursor and is useful during cube build operations.	
OLAP INHIER Limit	64	Number of times an OLAP table function issues an in-hierarchy limit. This type of limit can occur when you use cube dimension hierarchy views.	
OLAP Limit Time	64	Total time taken by all the OLAP Limit operations that were performed during the last call to the OLAP table function	
OLAP Paging Manager Cache Changed Page	64	Number of times the OLAP page pool is changed for any attached AW	
OLAP Paging Manager Cache Hit	64	Number of times a requested page is found in the OLAP page pool. Use this statistic in conjunction with "OLAP Paging Manager Cache Miss" to determine the OLAP page pool efficiency ratio.	
OLAP Paging Manager Cache Miss	64	Number of times a requested page is not found in the OLAP page pool. Use this statistic in conjunction with "OLAP Paging Manager Cache Hit" to determine the OLAP page pool efficiency ratio.	
OLAP Paging Manager Cache Write	64	Number of times the OLAP paging manager writes to a page in the OLAP page pool	
OLAP Paging Manager New Page	64	Number of newly-created pages in the OLAP page pool that have not yet been written to the workspace LOB	
OLAP Paging Manager Pool Size	64	Size (in bytes) of the OLAP page pool allocated to a session and the sum of all OLAP page pools in the system	
OLAP Perm LOB Read	64	Number of times data was read from the table where the AW is stored. These are permanent LOB reads.	
OLAP Row Id Limit	64	Number of times an OLAP table function issues a row ID limit	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
OLAP Row Load Time	64	Total time spent loading rows into an AW during cube build and OLAP SQL import operations Use this statistic along with the "OLAP engine elapsed time" to measure time spent running OLAP engine routines that involve loading data into AWs from a SQL source. This statistic has the following levels of precision: • Low precision timer This captures the elapsed time of the entire fetch phase of the SQL cursor that is being loaded into AWs. It includes the SQL execution time that occurs during a fetch operation from a source cursor and time taken by the OLAP engine to populate AWs. • High precision timer This captures the elapsed time, excluding the SQL processing of the cursor being loaded. It records the time spent in the OLAP engine only. • Default timer precision: This is based on the STATISTIC_LEVEL parameter. If the low precision is used, then STATISTICS_LEVEL is TYPICAL. The high precision timer is used when STATISTIC_LEVEL is set to ALL. No timing is captured when STATISTICS_LEVEL is BASIC.	
OLAP Row Source Rows Processed	64	Number of rows processed by the OLAP row source	
OLAP Session Cache Hit	64	Number of times the requested, dynamically-aggregated value of an AW object, was found in the OLAP session cache	
OLAP Session Cache Miss	64	Number of times the requested, dynamically-aggregated value of an AW object, was not found in the OLAP session cache	
OLAP Temp Segment Read	64	Number of times data was read from a temporary segment and not from the OLAP page pool	
OLAP Temp Segments	64	Number of OLAP pages stored in temporary segments for analytic workspaces	
OLAP Unique Key Attribute Limit	64	Number of times an OLAP table function issues a unique key attribute limit	
opened cursors cumulative	1	In V$SYSSTAT: Total number of cursors opened since the instance started In V$SESSTAT: Total number of cursors opened since the start of the session	
opened cursors current	1	Total number of current open cursors	
OS CPU Qt wait time	1	Amount of time a session spends on the CPU run queue (in microseconds), waiting to get the CPU to run	
OS Involuntary context switches	16	Number of context switches that were enforced by the operating system	
OS Signals received	16	Number of signals received	
OS Swaps	16	Number of swap pages	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
OS Voluntary context switches	16	Number of voluntary context switches (for example, when a process gives up the CPU by a SLEEP() system call)	
Parallel operations downgraded 1 to 25 pct	32	Number of times parallel execution was requested and the degree of parallelism was reduced because of insufficient parallel execution servers	
Parallel operations downgraded 25 to 50 pct	32	Number of times parallel execution was requested and the degree of parallelism was reduced because of insufficient parallel execution servers	
Parallel operations downgraded 50 to 75 pct	32	Number of times parallel execution was requested and the degree of parallelism was reduced because of insufficient parallel execution servers	
Parallel operations downgraded 75 to 99 pct	32	Number of times parallel execution was requested and the degree of parallelism was reduced because of insufficient parallel execution servers	
Parallel operations downgraded to serial	32	Number of times parallel execution was requested but execution was serial because of insufficient parallel execution servers	
Parallel operations not downgraded	32	Number of times parallel execution was executed at the requested degree of parallelism	
parse count (describe)	64	Total number of parse calls on a describe cursor. This operation is a less expensive than a hard parse and more expensive than a soft parse.	
parse count (hard)	64	Total number of parse calls (real parses). A hard parse is a very expensive operation in terms of memory use, because it requires Oracle to allocate a workheap and other memory structures and then build a parse tree.	
parse count (total)	64	Total number of parse calls (hard, soft, and describe). A soft parse is a check on an object already in the shared pool, to verify that the permissions on the underlying object have not changed.	
parse time cpu	64	Total CPU time used for parsing (hard and soft) in 10s of milliseconds	Y
parse time elapsed	64	Total elapsed time for parsing, in 10s of milliseconds. Subtract "parse time cpu" from this statistic to determine the total waiting time for parse resources.	Y
physical read bytes	8	Total size in bytes of all disk reads by application activity (and not other instance activity) only.	
physical read flash cache hits	8	Total number of reads from flash cache instead of disk	
physical read IO requests	8	Number of read requests for application activity (mainly buffer cache and direct load operation) which read one or more database blocks per request. This is a subset of "physical read total IO requests".	
physical read requests optimized	8	Number of read requests that read one or more database blocks from the Database Smart Flash Cache or the Exadata Smart Flash Cache.	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
physical read total bytes	8	Total size in bytes of disk reads by all database instance activity including application reads, backup and recovery, and other utilities. The difference between this value and "physical read bytes" gives the total read size in bytes by non-application workload.	
physical read total IO requests	8	Number of read requests which read one or more database blocks for all instance activity including application, backup and recovery, and other utilities. The difference between this value and "physical read total multi block requests" gives the total number of small I/O requests which are less than 128 kilobytes down to single block read requests.	
physical read total multi block requests	8	Total number of Oracle instance read requests which read 128 kilobytes or more in two or more database blocks per request for all instance activity including application, backup and recovery, and other utilities	
physical reads	8	Total number of data blocks read from disk. This value can be greater than the value of "physical reads direct" plus "physical reads cache" as reads into process private buffers also included in this statistic.	
physical reads cache	8	Total number of data blocks read from disk into the buffer cache. This is a subset of "physical reads".	
physical reads cache prefetch	8	Number of contiguous and noncontiguous blocks that were prefetched	
physical reads direct	8	Number of reads directly from disk, bypassing the buffer cache. For example, in high bandwidth, data-intensive operations such as parallel query, reads of disk blocks bypass the buffer cache to maximize transfer rates and to prevent the premature aging of shared data blocks resident in the buffer cache.	
physical reads direct (lob)	8	Number of buffers that were read directly for LOBs	
physical reads direct temporary tablespace	8	Number of buffers that were read directly from temporary tablespaces	
physical reads for flashback new	8	Number of blocks read for newing (that is, preparing a data block for a completely new change) blocks while flashback database is enabled	
physical reads prefetch warmup	8	Number of data blocks that were read from the disk during the automatic prewarming of the buffer cache	
physical write bytes	8	Total size in bytes of all disk writes from the database application activity (and not other kinds of instance activity)	
physical write IO requests	8	Number of write requests for application activity (mainly buffer cache and direct load operation) which wrote one or more database blocks per request	
physical write total bytes	8	Total size in bytes of all disk writes for the database instance including application activity, backup and recovery, and other utilities. The difference between this value and "physical write bytes" gives the total write size in bytes by non-application workload.	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
physical write total IO requests	8	Number of write requests which wrote one or more database blocks from all instance activity including application activity, backup and recovery, and other utilities. The difference between this stat and "physical write total multi block requests" gives the number of single block write requests.	
physical write total multi block requests	8	Total number of Oracle instance write requests which wrote two or more blocks per request to the disk for all instance activity including application activity, recovery and backup, and other utilities	
physical writes	8	Total number of data blocks written to disk. This value equals the sum of "physical writes direct" and "physical writes from cache".	
physical writes direct	8	Number of writes directly to disk, bypassing the buffer cache (as in a direct load operation)	
physical writes direct (lob)	8	Number of buffers that were directly written for LOBs	
physical writes direct temporary tablespace	8	Number of buffers that were directly written for temporary tablespaces	
physical writes from cache	8	Total number of data blocks written to disk from the buffer cache. This is a subset of "physical writes".	
physical writes non checkpoint	8	Number of times a buffer is written for reasons other than advancement of the checkpoint. Used as a metric for determining the I/O overhead imposed by setting the FAST_START_IO_TARGET parameter to limit recovery I/Os. (Note that FAST_START_IO_TARGET is a deprecated parameter.) Essentially this statistic measures the number of writes that would have occurred had there been no checkpointing. Subtracting this value from "physical writes" gives the extra I/O for checkpointing.	
pinned buffers inspected	8	Number of times a user process, when scanning the tail of the replacement list looking for a buffer to reuse, encountered a cold buffer that was pinned or had a waiter that was about to pin it. This occurrence is uncommon, because a cold buffer should not be pinned very often.	
prefetched blocks aged out before use	8	Number of contiguous and noncontiguous blocks that were prefetched but aged out before use	
process last non-idle time	128	The last time this process executed	Y
PX local messages recv'd	32	Number of local messages received for parallel execution within the instance local to the current session	
PX local messages sent	32	Number of local messages sent for parallel execution within the instance local to the current session	
PX remote messages recv'd	32	Number of remote messages received for parallel execution within the instance local to the current session	
PX remote messages sent	32	Number of remote messages sent for parallel execution within the instance local to the current session	
queries parallelized	32	Number of SELECT statements executed in parallel	
recovery array read time	8	Elapsed time of I/O during recovery	
recovery array reads	8	Number of reads performed during recovery	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
recovery blocks read	8	Number of blocks read during recovery	
recovery blocks read for lost write detection	8	Number of blocks read for lost write checks during recovery	
recovery blocks skipped lost write checks	8	Number of Block Read Records that skipped the lost write check during recovery	
recursive calls	1	Number of recursive calls generated at both the user and system level. Oracle maintains tables used for internal processing. When Oracle needs to make a change to these tables, it internally generates an internal SQL statement, which in turn generates a recursive call.	
recursive cpu usage	1	Total CPU time used by non-user calls (recursive calls). Subtract this value from "CPU used by this session" to determine how much CPU time was used by the user calls.	
redo blocks checksummed by FG (exclusive)	2	Number of exclusive redo blocks that were checksummed by the generating foreground processes. An exclusive redo block is the one whose entire redo content belongs to a single redo entry.	
redo blocks checksummed by LGWR	2	Number of redo blocks that were checksummed by the LGWR	
redo blocks written	2	Total number of redo blocks written. This statistic divided by "redo writes" equals number of blocks per write.	
redo buffer allocation retries	2	Total number of retries necessary to allocate space in the redo buffer. Retries are needed either because the redo writer has fallen behind or because an event such as a log switch is occurring.	
redo entries	2	Number of times a redo entry is copied into the redo log buffer	
redo entries for lost write detection	2	Number of times a Block Read Record is copied into the log buffer	
redo log space requests	2	Number of times the active log file is full and Oracle must wait for disk space to be allocated for the redo log entries. Such space is created by performing a log switch. Log files that are small in relation to the size of the SGA or the commit rate of the work load can cause problems. When the log switch occurs, Oracle must ensure that all committed dirty buffers are written to disk before switching to a new log file. If you have a large SGA full of dirty buffers and small redo log files, a log switch must wait for DBWR to write dirty buffers to disk before continuing. Also examine the **log file space** and **log file space switch** wait events in `V$SESSION_WAIT`.	
redo log space wait time	2	Total time waited in centiseconds for available space in the redo log buffer **See Also:** "redo log space requests"	Y
redo ordering marks	2	Number of times that a system change number was allocated to force a redo record to have a higher SCN than a record generated in another thread using the same block	
redo size	2	Total amount of redo generated in bytes	
redo size for lost write detection	2	Total amount of Block Read Records generated in bytes	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
redo synch time	8	Elapsed time of all "redo synch writes" calls in 10s of milliseconds	Y
redo synch writes	8	Number of times the redo is forced to disk, usually for a transaction commit. The log buffer is a circular buffer that LGWR periodically flushes. Usually, redo that is generated and copied into the log buffer need not be flushed out to disk immediately.	
redo wastage	2	Number of bytes wasted because redo blocks needed to be written before they are completely full. Early writing may be needed to commit transactions, to be able to write a database buffer, or to switch logs.	
redo write broadcast ack count	2	Number of times a commit broadcast acknowledgment has not been received by the time when the corresponding log write is completed. This is only for Oracle RAC.	
redo write broadcast ack time	2	Total amount of the latency associated with broadcast on commit beyond the latency of the log write (in microseconds). This is only for Oracle RAC.	Y
redo write time	2	Total elapsed time of the write from the redo log buffer to the current redo log file in 10s of milliseconds	Y
redo writes	2	Total number of writes by LGWR to the redo log files. "redo blocks written" divided by this statistic equals the number of blocks per write	
rollback changes - undo records applied	128	Number of undo records applied to user-requested rollback changes (not consistent-read rollbacks)	
rollbacks only - consistent read gets	128	Number of "consistent gets" that require only block rollbacks, no block cleanouts	
rows fetched via callback	64	Rows fetched via callback. Useful primarily for internal debugging purposes.	
scheduler wait time	1	Total wait time (in microseconds) for waits that belong to the Scheduler wait class	
SCN increments due to another database	128	SCN increments due to communication with another database	
serializable aborts	1	Number of times a SQL statement in a serializable isolation level had to terminate	
session connect time	1	The connect time for the session in 10s of milliseconds. This value is useful only in V$SESSTAT. It is the wall clock time since the logon to this session occurred.	Y
session cursor cache count	64	Total number of cursors cached. This statistic is incremented only if SESSION_CACHED_CURSORS > 0. This statistic is the most useful in V$SESSTAT. If the value for this statistic in V$SESSTAT is close to the setting of the SESSION_CACHED_CURSORS parameter, the value of the parameter should be increased.	
session cursor cache hits	64	Number of hits in the session cursor cache. A hit means that the SQL (including recursive SQL) or PL/SQL statement did not have to be reparsed. Subtract this statistic from "parse count (total)" to determine the real number of parses that occurred.	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
session logical reads	1	The sum of "db block gets" plus "consistent gets". This includes logical reads of database blocks from either the buffer cache or process private memory.	
session logical reads - IM	128	Number of database blocks read from the IM column store (number of blocks in IMCU - number of blocks with invalid rows)	
session pga memory	1	Current PGA size for the session (in bytes). Useful only in V$SESSTAT; it has no meaning in V$SYSSTAT.	
session pga memory max	1	Peak PGA size for the session (in bytes). Useful only in V$SESSTAT; it has no meaning in V$SYSSTAT.	
session stored procedure space	1	Amount of memory this session is using for stored procedures	
session uga memory	1	Current UGA size for the session (in bytes). Useful only in V$SESSTAT; it has no meaning in V$SYSSTAT.	
session uga memory max	1	Peak UGA size for a session (in bytes). Useful only in V$SESSTAT; it has no meaning in V$SYSSTAT.	
shared hash latch upgrades - no wait	8	A shared hash latch upgrade is when a hash latch is upgraded from shared mode to exclusive mode. This statistic displays the number of times the upgrade completed immediately.	
shared hash latch upgrades - wait	8	A shared hash latch upgrade is when a hash latch is upgraded from shared mode to exclusive mode. This statistic displays the number of times the upgrade did not complete immediately.	
shared io pool buffer get failure	128	Number of unsuccessful buffer gets from the shared I/O pool from instance startup time	
shared io pool buffer get success	128	Number of successful buffer gets from the shared I/O pool from instance startup time	
slave propagated tracked transactions	128	Number of transactions modifying tables enabled for flashback data archive which were archived by a worker process	
sorts (disk)	64	Number of sort operations that required at least one disk write Sorts that require I/O to disk are quite resource intensive. Try increasing the size of the initialization parameter SORT_AREA_SIZE.	
sorts (memory)	64	Number of sort operations that were performed completely in memory and did not require any disk writes You cannot do much better than memory sorts, except maybe no sorts at all. Sorting is usually caused by selection criteria specifications within table join SQL operations.	
sorts (rows)	64	Total number of rows sorted	
SQL*Net roundtrips to/from client	1	Total number of Oracle Net Services messages sent to and received from the client	
SQL*Net roundtrips to/from dblink	1	Total number of Oracle Net Services messages sent over and received from a database link	
summed dirty queue length	8	The sum of the dirty LRU queue length after every write request. Divide by write requests to get the average queue length after write completion.	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
switch current to new buffer	8	Number of times the CURRENT block moved to a different buffer, leaving a CR block in the original buffer	
table fetch by rowid	64	Number of rows that are fetched using a ROWID (usually recovered from an index) This occurrence of table scans usually indicates either non-optimal queries or tables without indexes. Therefore, this statistic should increase as you optimize queries and provide indexes in the application.	
table fetch continued row	64	Number of times a chained or migrated row is encountered during a fetch Retrieving rows that span more than one block increases the logical I/O by a factor that corresponds to the number of blocks than need to be accessed. Exporting and re-importing may eliminate this problem. Evaluate the settings for the storage parameters PCTFREE and PCTUSED. This problem cannot be fixed if rows are larger than database blocks (for example, if the LONG data type is used and the rows are extremely large).	
table scan blocks gotten	64	During scanning operations, each row is retrieved sequentially by Oracle. This statistic counts the number of blocks encountered during the scan. This statistic tells you the number of database blocks that you had to get from the buffer cache for the purpose of scanning. Compare this value with the value of "consistent gets" to determine how much of the consistent read activity can be attributed to scanning.	
table scan disk IMC fallback	128	Number of rows fetched from the buffer cache because they were not present in the IM column store (in a scan that was otherwise performed in memory)	
table scan disk non-IMC rows gotten	128	Number of rows fetched during non-In-Memory scan	
table scan rows gotten	64	Number of rows that are processed during scanning operations	
table scans (cache partitions)	64	Number of range scans performed on tables that have the CACHE option enabled	
table scans (direct read)	64	Number of table scans performed with direct read (bypassing the buffer cache)	
table scans (IM)	128	Number of segments / granules scanned using In-Memory	
table scans (long tables)	64	Long (or conversely short) tables can be defined as tables that do not meet the short table criteria as described in "table scans (short tables)"	
table scans (rowid ranges)	64	During parallel query, the number of table scans conducted with specified ROWID ranges	
table scans (short tables)	64	Long (or conversely short) tables can be defined by optimizer hints coming down into the row source access layer of Oracle. The table must have the CACHE option set.	
tracked rows	128	Number of rows modified in tables enabled for flashback data archive	

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table E-1 (Cont.) Database Statistics Descriptions

Name	Class	Description	TIMED_STATISTICS
tracked transactions	128	Number of transactions which modified a table enabled for flashback data archive	
transaction lock background get time	128	Useful only for internal debugging purposes	
transaction lock background gets	128	Useful only for internal debugging purposes	
transaction lock foreground requests	128	Useful only for internal debugging purposes	
transaction lock foreground wait time	128	Useful only for internal debugging purposes	
transaction rollbacks	128	Number of transactions being successfully rolled back	
transaction tables consistent read rollbacks	128	Number of times rollback segment headers are rolled back to create consistent read blocks	
transaction tables consistent reads - undo records applied	128	Number of undo records applied to transaction tables that have been rolled back for consistent read purposes	
user calls	1	Number of user calls such as login, parse, fetch, or execute When determining activity, the ratio of user calls to RPI calls, give you an indication of how much internal work gets generated because of the type of requests the user is sending to Oracle.	
user commits	1	Number of user commits. When a user commits a transaction, the redo generated that reflects the changes made to database blocks must be written to disk. Commits often represent the closest thing to a user transaction rate.	
user I/O wait time	1	Total wait time (in centiseconds) for waits that belong to the User I/O wait class	
user rollbacks	1	Number of times users manually issue the `ROLLBACK` statement or an error occurs during a user's transactions	
very large tracked transactions	128	For tables tracked by flashback data archive, number of transactions modifying those tables which are very large in terms of size or number of changes	
write clones created in background	8	Number of times a background or foreground process clones a `CURRENT` buffer that is being written. The clone becomes the new, accessible `CURRENT` buffer, leaving the original buffer (now the clone) to complete writing.	
write clones created in foreground	8	Number of times a background or foreground process clones a `CURRENT` buffer that is being written. The clone becomes the new, accessible `CURRENT` buffer, leaving the original buffer (now the clone) to complete writing.	

Copyright © 2002, 2026, Oracle and/or its affiliates.

F

Background Processes

An Oracle Database **background process** is defined as any process that is listed in `V$PROCESS` and has a non-null value in the `PNAME` column.

Table F-1 describes Oracle Database background processes.

The External Properties column lists the type of instance in which the process runs. If the process is specific to a particular feature, then the column names the feature.

> **Note**
>
> When the `THREADED_EXECUTION` initialization parameter is set to `TRUE` on Linux and UNIX, the `DBW`, `PMON`, `PSP`, and `VKTM` background processes run as operating system processes, and the other background processes run as operating system threads.
>
> See "THREADED_EXECUTION" for more information about the `THREADED_EXECUTION` initialization parameter.

Table F-1 Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
ABMR	Auto BMR Background Process	Coordinates execution of tasks such as filtering duplicate block media recovery requests and performing flood control	When a process submits a block media recovery request to ABMR, it dynamically spawns worker processes (BMR*n*) to perform the recovery. ABMR and BMR*n* terminate after being idle for a long time. **See Also:** *Oracle Database Backup and Recovery User's Guide*	Database instances
ACFS	Oracle Automatic Storage Management Cluster File System (Oracle ACFS) CSS Process	Tracks the cluster membership in CSS and informs the file system driver of membership changes	The ACFS process delivers CSS membership changes to the cluster file system. These membership changes are required for the file system to maintain file system consistency within the cluster.	Oracle ASM instances, Oracle RAC
ACMS	Atomic Control File to Memory Service Process	Coordinates consistent updates to a control file resource with its SGA counterpart on all instances in an Oracle RAC environment	The ACMS process works with a coordinating caller to ensure that an operation is executed on every instance in Oracle RAC despite failures. ACMS is the process in which a distributed operation is called. As a result, this process can exhibit a variety of behaviors. In general, ACMS is limited to small, nonblocking state changes for a limited set of cross-instance operations.	Database instances, Oracle RAC
AMB*n*	**See** ASMB, AMB*n*			

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
AP*nn*	Database Apply Process Coordinator Process	Obtains transactions from the reader server and passes them to apply servers	The coordinator process name is AP*nn*, where *nn* can include letters and numbers. For more information about the coordinator process, see `V$XSTREAM_APPLY_COORDINATOR` for XStream and `V$GG_APPLY_COORDINATOR` for Oracle GoldenGate. **See Also:** *Oracle Database XStream Guide*	Database instances, Logical Standby, XStream Inbound servers, XStream Outbound servers, GoldenGate Integrated Replicat
AQPC	AQ Process Coordinator	Per instance AQ global coordinator	AQPC is responsible for performing administrative tasks for AQ Primary Class Processes including commands like starting, stopping, and other administrative tasks. This process is automatically started on instance startup.	Database instances Advanced Queueing
ARB0	ASM Rebalance Process	Rebalances data extents within an Oracle ASM disk group	ARB0 uses the value of the `ASM_POWER_LIMIT` initialization parameter for the Oracle ASM instance as the maximum power for disk rebalancing.	Oracle ASM instances
ARBA	ASM Rebalance Estimate Process	Computes estimated ASM Rebalance work within an Oracle ASM disk group	The ARBA process displays the estimate rebalance work results in the `V$ASM_OPERATION` view.	Oracle ASM instances
ARC*n*	Archiver Process	Copies the redo log files to archival storage when they are full or an online redo log switch occurs	ARC*n* processes exist only when the database is in `ARCHIVELOG` mode and automatic archiving is enabled, in which case ARC*n* automatically archives online redo log files. LGWR cannot reuse and overwrite an online redo log group until it has been archived. The database starts multiple archiver processes as needed to ensure that the archiving of filled online redo logs does not fall behind. Possible processes are ARC0-ARC9 and ARCa-ARCt. The `LOG_ARCHIVE_MAX_PROCESSES` initialization parameter specifies the number of ARC*n* processes that the database initially invokes. **See Also:** *Oracle Database Concepts* and *Oracle Database Administrator's Guide*	Database instances
ARS*n*	ASM Recovery Worker Process	Recovers ASM transactional operations	The ASM RBAL background process coordinates and spawns one or more of these worker processes to recover terminated ASM transactional operations. These processes run only in the Oracle ASM instance. Possible processes are ARS0-ARS9.	Oracle ASM instances

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
ASMB, AMB*n*	ASM Background Process	Communicates with an Oracle ASM instance, managing storage and providing statistics	In a database instance, the ASMB and AMB*n* processes enable the database instance to connect to an Oracle ASM instance in order to access Oracle ASM disk groups. Possible processes are ASMB and AMB1-AMB3. In an Oracle ASM instance, the ASMB process runs when the ASMCMD `cp` command runs, or when a database instance first starts if the server parameter file is stored in Oracle ASM. ASMB also runs with Oracle Cluster Registry on Oracle ASM. The only possible process is ASMB; AMB*n* processes do not run in Oracle ASM instances. In an Oracle IOServer (IOS) instance, the ASMB process enables the IOS instance to connect to an Oracle ASM instance in order to access Oracle ASM disk groups. The only possible process is ASMB; AMB*n* processes do not run in IOS instances.	Database instances, Oracle ASM instances, Oracle IOServer (IOS) instances
AS*nn*	Database Apply Reader or Apply Server	Computes dependencies between logical change records (LCRs) and assembles messages into transactions (Reader Server) Applies LCRs to database objects or passes LCRs and user messages to their appropriate apply handlers (Apply Server)	When the reader server finishes computing dependencies between LCRs and assembling transactions, it returns the assembled transactions to the coordinator process. Query `V$STREAMS_APPLY_READER`, `V$XSTREAM_APPLY_READER`, and `V$GG_APPLY_READER` for information about the reader server background process. An apply server receives the transactions from the coordinator background process, and either applies database changes in LCRs or sends LCRs or messages to apply handlers. Apply servers can also enqueue a queue. If an apply server encounters an error, then it then tries to resolve the error with a user-specified conflict handler or error handler. If an apply server cannot resolve an error, then it rolls back the transaction and places the entire transaction, including all of its messages, in the error queue. When an apply server commits a completed transaction, this transaction has been applied. When an apply server places a transaction in the error queue and commits, this transaction also has been applied. Query `V$STREAMS_APPLY_SERVER` for information about the apply server background process. For XStream Inbound servers, query `V$XSTREAM_APPLY_SERVER`. For GoldenGate Integrated Replicat, query `V$GG_APPLY_SERVER`. The coordinator process name is AS*nn*, where *nn* can include letters and numbers.	Database instances, XStream Outbound servers, XStream Inbound servers, GoldenGate Integrated Replicat

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
BMR*n*	Automatic Block Media Recovery Worker Pool Process	Fetches blocks from a real-time readable standby database	When a process submits a block media recovery request to ABMR, it dynamically spawns worker processes (BMR*n*) to perform the recovery. BMR*n* processes fetch blocks from a real-time readable standby database. ABMR and BMR*n* terminate after being idle for a long time. **See Also:** *Oracle Database Backup and Recovery User's Guide*	Database instances
B*nnn*	ASM Blocking Worker Process for GMON	Performs maintenance actions on Oracle ASM disk groups	B*nnn* performs actions that require waiting for resources on behalf of GMON. GMON must be highly available and cannot wait. A B*nnn* worker is spawned when a disk is taken offline in an Oracle ASM disk group. Offline timer processing and drop of the disk are performed in this worker. Up to five process (B000 to B004) can exist depending on the load.	Oracle ASM instances
BW*nn*	Database Writer Process	Writes modified blocks from the database buffer cache to the data files	See the **Long Description** for the DBW*n* process in this table for more information about the BW*nn* process.	Database instances
CJQ0	Job Queue Coordinator Process	Selects jobs that need to be run from the data dictionary and spawns job queue worker processes (J*nnn*) to run the jobs	CJQ0 is automatically started and stopped as needed by Oracle Scheduler. The `JOB_QUEUE_PROCESSES` initialization parameter specifies the maximum number of processes that can be created for the execution of jobs. CJQ0 starts only as many job queue processes as required by the number of jobs to run and available resources. **See Also:** *Oracle Database Concepts* and *Oracle Database Administrator's Guide*	Database instances
CKPT	Checkpoint Process	Signals DBW*n* at checkpoints and updates all the data files and control files of the database to indicate the most recent checkpoint	At specific times CKPT starts a checkpoint request by messaging DBW*n* to begin writing dirty buffers. On completion of individual checkpoint requests, CKPT updates data file headers and control files to record most recent checkpoint. CKPT checks every three seconds to see whether the amount of memory exceeds the value of the `PGA_AGGREGATE_LIMIT` initialization parameter, and if so, takes the action described in "PGA_AGGREGATE_LIMIT". **See Also:** *Oracle Database Concepts*	Database instances, Oracle ASM instances
CL*nn*	Cleanup Worker Process	Performs cleanup of terminated processes	Cleanup workers assist in the cleanup of terminated processes and terminated sessions. The number of workers will be proportional to the amount of cleanup work to be done and the current efficiency of cleanup.	Database instances, Oracle ASM instances
CLG	Persistent Cluster Flash Cache Background Process	For Oracle Data Appliance only, this process performs actions related to recovery of a terminated instance's database flash cache	For Oracle Data Appliance only, in the event of an instance crash, the running instance will recover the crashed instance's database flash cache. The CLG process will perform actions related to scanning the crashed instance's database flash cache and claim flash blocks mastered by the crashed instance.	Database instances, Oracle RAC

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
CLMN	Cleanup Main Process	Performs cleanup of terminated processes, terminated sessions, terminated transactions, and terminated network connections	CLMN periodically performs cleanup of all the following: terminated processes, terminated sessions, transactions, network connections, idle sessions, detached transactions, and detached network connections that have exceeded their idle timeout.	Database instances, Oracle ASM instances
CP*nn*	Database Capture Process	Captures database changes from the redo log by using the infrastructure of LogMiner	The capture process name is CP*nn*, where *nn* can include letters and numbers. The underlying LogMiner process name is MS*nn*, where *nn* can include letters and numbers. The capture process includes one reader server that reads the redo log and divides it into regions, one or more preparer servers that scan the redo log, and one builder server that merges redo records from the preparer servers. Each reader server, preparer server, and builder server is a process. Query the `V$XSTREAM_CAPTURE` and `V$GOLDENGATE_CAPTURE` views for information about this background process. **See Also:***Oracle Database XStream Guide*	Database instances, XStream Outbound Servers
CR*nn*	LMS CR Worker Process	Offloads the work from LMS so that blocks that require lots of UNDO to be applied do not block the LMS. Such requests are passed on to the worker so that the LMS is not stalled	There can be a maximum of eight CR processes per LMS process, with names from CR00 to CR07. Each LMS has its own set with similar name. The CR*nn* processes are threads and the process ID part will be the same as the owning LMS's process ID. The names for CRnn processes will have the format CR0*n*_<spawning process id>_<thread id>.	Oracle RAC
CS*nn*	I/O Calibration Process	Issues I/Os to storage as part of storage calibration.	CS*nn* worker processes are started on execution of the `DBMS_RESOURCE_MANAGER.CALIBRATE_IO()` procedure. There is one worker process per CPU on each node of the database.	Database instances, Oracle RAC
CTWR	Change Tracking Writer Process	Tracks changed data blocks as part of the Recovery Manager block change tracking feature	CTWR tracks changed blocks as redo is generated at a primary database and as redo is applied at a standby database. The process is slightly different depending on the type of database. **See Also:** *Oracle Database Backup and Recovery User's Guide*	Database instances
CX*nn*	Propagation Sender Process	Sends LCRs to a propagation receiver	The propagation sender process name is CX*nn*, where *nn* can include letters and numbers. Query `V$PROPAGATION_SENDER` for information about a propagation sender.	Database instances, XStream Outbound Server
DBRM	Database Resource Manager Process	Sets resource plans and performs other tasks related to the Database Resource Manager	If a resource plan is not enabled, then this process is idle. **See Also:** *Oracle Database Administrator's Guide*	Database instances

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
DBW*n*	Database Writer Process	Writes modified blocks from the database buffer cache to the data files	The primary responsibility of the Database Writer Process is to write data blocks to disk. It also handles checkpoints, file open synchronization, and logging of Block Written records. In many cases the blocks that the Database Writer Process writes are scattered throughout the disk. Thus, the writes tend to be slower than the sequential writes performed by LGWR. The Database Writer Process performs multiblock writes when possible to improve efficiency. The number of blocks written in a multiblock write varies by operating system. The `DB_WRITER_PROCESSES` initialization parameter specifies the number of Database Writer Processes. There can be 1 to 100 Database Writer Processes. The names of the first 36 Database Writer Processes are DBW0-DBW9 and DBWa-DBWz. The names of the 37th through 100th Database Writer Processes are BW36-BW99. The database selects an appropriate default setting for the `DB_WRITER_PROCESSES` parameter or adjusts a user-specified setting based on the number of CPUs and processor groups. **See Also**: "DB_WRITER_PROCESSES"	Database instances, Oracle ASM instances
DIA0	Diagnostic Process	Detects and resolves hangs and deadlocks		Database instances, Oracle ASM instances
DIAG	Diagnostic Capture Process	Performs diagnostic dumps	DIAG performs diagnostic dumps requested by other processes and dumps triggered by process or instance termination. In Oracle RAC, DIAG performs global diagnostic dumps requested by remote instances.	Database instances, Oracle ASM instances
DM*nn*	Data Pump Control Job Process	Coordinates the Data Pump job tasks performed by Data Pump worker processes and handles client interactions	The Data Pump control job process is started during job creation and coordinates all tasks performed by the Data Pump job. It handles all client interactions and communication, establishes all job contexts, and coordinates all worker process activities on behalf of the job.	Database instances, Data Pump

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
DMON	Data Guard Broker Monitor Process	Manages and monitors a database that is part of a Data Guard broker configuration	When you start the Data Guard broker, a DMON process is created. DMON runs for every database instance that is managed by the broker. DMON interacts with the local database and the DMON processes of the other databases to perform the requested function. DMON also monitors the health of the broker configuration and ensures that every database has a consistent description of the configuration. DMON maintains profiles about all database objects in the broker configuration in a binary configuration file. A copy of this file is maintained by the DMON process for each of the databases that belong to the broker configuration. The process is created when the `DG_BROKER_START` initialization parameter is set to `true`. **See Also:** *Oracle Data Guard Broker*	Database instances, Data Guard
D*nnn*	Dispatcher Process	Performs network communication in the shared server architecture	In the shared server architecture, clients connect to a dispatcher process, which creates a virtual circuit for each connection. When the client sends data to the server, the dispatcher receives the data into the virtual circuit and places the active circuit on the common queue to be picked up by an idle shared server. The shared server then reads the data from the virtual circuit and performs the database work necessary to complete the request. When the shared server must send data to the client, the server writes the data back into the virtual circuit and the dispatcher sends the data to the client. After the shared server completes the client request, the server releases the virtual circuit back to the dispatcher and is free to handle other clients. Several initialization parameters relate to shared servers. The principal parameters are: `DISPATCHERS`, `SHARED_SERVERS`, `MAX_SHARED_SERVERS`, `LOCAL_LISTENER`, `REMOTE_LISTENER`. **See Also:** *Oracle Database Concepts*	Database instances, shared servers
DSKM	Worker Diskmon Process	Acts as the conduit between the database, Oracle ASM instances, and the Primary Diskmon daemon to communicate information to Exadata storage	This process is active only if Exadata Storage is used. DSKM performs operations related to Exadata I/O fencing and Exadata cell failure handling.	Oracle ASM instances, Exadata
DW*nn*	Data Pump Worker Process	Performs Data Pump tasks as assigned by the Data Pump control job process	The Data Pump worker process is responsible for performing tasks that are assigned by the Data Pump control job process, such as the loading and unloading of metadata and data.	Database instances

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
EMNC	EMON Coordinator Process	Coordinates database event management and notifications	EMNC is a primary background process that coordinates event management and notification activity in the database, including Streams Event Notifications, Continuous Query Notifications, and Fast Application Notifications.	Database instances
E*nnn*	EMON Worker Process	Performs database event management and notifications	The database event management and notification load is distributed among the EMON worker processes. These processes work on the system notifications in parallel, offering a capability to process a larger volume of notifications, a faster response time, and a lower shared memory use for staging notifications.	Database instances
FBDA	Flashback Data Archiver Process	Archives historical rows for tracked tables into flashback data archives and manages archive space, organization, and retention	When a transaction that modifies a tracked table commits, FBDA stores the pre-image of the rows in the archive. FBDA maintains metadata on the current rows and tracks how much data has been archived. FBDA is also responsible for automatically managing the flashback data archive for space, organization (partitioning tablespaces), and retention. FBDA also keeps track of how far the archiving of tracked transactions has progressed. **See Also:** *Oracle Database Development Guide*	Database instances
FD*nn*	Oracle ASM Stale FD Cleanup Worker Process	Cleans up Oracle ASM stale file descriptors on foreground processes	This process cleans up Oracle ASM stale file descriptors on foreground processes if an Oracle ASM disk is globally closed.	Database and Oracle ASM instances
FENC	Fence Monitor Process	Processes fence requests for RDBMS instances which are using Oracle ASM instances	CSS monitors RDBMS instances which are connected to the Oracle ASM instance and constantly doing I/Os. When the RDBMS instance terminates due to a failure, all the outstanding I/O's from the RDBMS instance should be drained and any new I/O's rejected. FENC receives and processes the fence request from CSSD.	Oracle ASM instances
FMON	File Mapping Monitor Process	Manages mapping information for the Oracle Database file mapping interface	The `DBMS_STORAGE_MAP` package enables you to control the mapping operations. When instructed by the user, FMON builds mapping information and stores it in the SGA, refreshes the information when a change occurs, saves the information to the data dictionary, and restores it to the SGA at instance startup. FMON is started by the database whenever the `FILE_MAPPING` initialization parameter is set to `true`.	Database instances, Oracle ASM instances
FSFP	Data Guard Broker Fast Start Failover Pinger Process	Maintains fast-start failover state between the primary and target standby databases	FSFP is created when fast-start failover is enabled.	Database instances, Data Guard
GCR*n*	Global Conflict Resolution Worker Process	Performs synchronous tasks on behalf of LMHB	GCR*n* processes are transient worker processes that are started and stopped as required by LMHB to perform synchronous or resource intensive tasks.	Database instances, Oracle ASM instances, Oracle RAC

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
GEN0	General Task Execution Process	Performs required tasks including SQL and DML		Database instances, Oracle ASM instances, Oracle ASM Proxy instances
GMON	ASM Disk Group Monitor Process	Monitors all mounted Oracle ASM disk groups	GMON monitors all the disk groups mounted in an Oracle ASM instance and is responsible for maintaining consistent disk membership and status information. Membership changes result from adding and dropping disks, whereas disk status changes result from taking disks offline or bringing them online.	Oracle ASM instances
GTX*n*	Global Transaction Process	Provides transparent support for XA global transactions in an Oracle RAC environment	These processes help maintain the global information about XA global transactions throughout the cluster. Also, the processes help perform two-phase commit for global transactions anywhere in the cluster so that an Oracle RAC database behaves as a single system to the externally coordinated distributed transactions. The `GLOBAL_TXN_PROCESSES` initialization parameter specifies the number of GTX*n* processes, where *n* is 0-9 or a-j. The database automatically tunes the number of these processes based on the workload of XA global transactions. You can disable these processes by setting the parameter to 0. If you try to run XA global transactions with these processes disabled, an error is returned. **See Also:** *Oracle Real Application Clusters Administration and Deployment Guide*	Database instances, Oracle RAC
I*nnn*	Disk and Tape I/O Worker Process	Serves as an I/O worker process spawned on behalf of DBWR, LGWR, or an RMAN backup session	I/O worker process can be configured on platforms where asynchronous I/O support is not available. These workers are started by setting the corresponding worker enable parameter in the server parameter file. The I/O workers simulate the asynchronous I/O behavior when the underlying platform does not have native support for asynchronous I/O.	Database instances
IMCO	In-Memory Coordinator	Initiates background population and repopulation of in-memory enabled objects	The IMCO background process initiates population (prepopulation) of in-memory enabled objects with priority LOW/MEDIUM/HIGH/CRITICAL. In-memory enabled objects with priority NONE will not be prepopulated but will be populated on demand via W*nnn* processes when queried. The IMCO background process can also initiate repopulation of in-memory objects. Starting with Oracle Database 19c, IMXT (In-Memory External Table) segments are dropped by the IMCO background process. In previous releases, IMXT segments were dropped by foreground processes.	Database instances

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
IMR0	Instance Membership Recovery Worker Process	Performs synchronous tasks on behalf of LMON	The IMR0 background process performs the Instance Member Recovery synchronous operations on behalf of LMON	Oracle RAC, Database instances, Oracle ASM instances
INSV	Data Guard Broker Instance Worker Process	Performs Data Guard broker communication among instances in an Oracle RAC environment	INSV is created when the `DG_BROKER_START` initialization parameter is set to `true`.	Database instances, Data Guard
IPC0	IPC Service Background Process	Common background server for basic messaging and RDMA primitives based on IPC (Inter-process communication) methods.	IPC0 handles very high rates of incoming connect requests, as well as, completing reconfigurations to support basic messaging and RDMA primitives over several transports such as UDP, RDS, InfiniBand and RC.	Oracle RAC
J*nnn*	Job Queue Worker Process	Executes jobs assigned by the job coordinator	Job worker processes are created or awakened by the job coordinator when it is time for a job to be executed. Job workers gather all the metadata required to run the job from the data dictionary. The worker processes start a database session as the owner of the job, execute triggers, and then execute the job. After the job is complete, the worker processes commit and then execute appropriate triggers and close the session. The worker can repeat this operation in case additional jobs need to be run.	Database instances
JP*n*	Java Patching Worker Process	Patches and updates the Java in the database classes	JP*n* patches and updates the Java in the database classes. It is only started for Oracle Real Application Clusters (Oracle RAC) databases, and one of the database instances is responsible for patching the Java in the database objects. For mulitenant container databases (CDBs), the process updates each pluggable database (PDB) individually. JP*n* is started automatically and does not require user intervention.	Oracle RAC
LCK*n*	Lock Process	Manages global enqueue requests and cross-instance broadcasts	The process handles all requests for resources other than data blocks. For examples, LCK*n* manages library and row cache requests. Possible processes are LCK0 and LCK1.	Database instances, Oracle ASM instances, Oracle RAC
LDD*n*	Global Enqueue Service Daemon Helper Worker	Helps the LMD*n* processes with various tasks	LDD*n* processes are worker processes spawned on demand by LMD*n* processes. They are spawned to help the dedicated LMD*n* processes with various tasks when certain workloads start creating performance bottlenecks. These worker processes are transient as they are started on demand and they can be shutdown when no longer needed. There can be up to 36 of these worker processes (LDD0-LDDz).	Database instances, Oracle ASM instances, Oracle RAC

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
LG*nn*	Log Writer Worker	Writes redo log	On multiprocessor systems, LGWR creates worker processes to improve the performance of writing to the redo log. LGWR workers are not used when there is a SYNC standby destination. Possible processes are LG00-LG99.	Database instances
LGWR	Log Writer Process	Writes redo entries to the online redo log	Redo log entries are generated in the redo log buffer of the system global area (SGA). LGWR writes the redo log entries sequentially into a redo log file. If the database has a multiplexed redo log, then LGWR writes the redo log entries to a group of redo log files. **See Also:** *Oracle Database Concepts* and *Oracle Database Administrator's Guide*	Database instances, Oracle ASM instances
LM*nn*	**See** LMSn, LMnn			
LMD*n*	Global Enqueue Service Daemon Process	Manages incoming remote resource requests from other instances	LMD*n* processes enqueue resources managed under Global Enqueue Service. In particular, they process incoming enqueue request messages and control access to global enqueues. They also perform distributed deadlock detections. There can be up to 36 of these processes (LMD0-LMDz).	Database instances, Oracle ASM instances, Oracle RAC
LMFC	Lock Manager Flash Cache Process	For Oracle Database Appliance only, performs actions related to recovery of a terminated instance's database flash cache.	For Oracle Database Appliance only, in the event of a instance crash, the running instance will recover the terminated instance's database flash cache. The LMFC process will perform actions related to scanning the terminated instance's database flash cache and claim flash blocks mastered by the terminated instance.	Database instances, Oracle RAC
LMHB	Global Cache/ Enqueue Service Heartbeat Monitor	Monitor the heartbeat of several processes	LMHB monitors the CKPT, DIA*n*, LCK*n*, LG*nn*, LGWR, LMD*n*, LMON, LMS*n* , and RMS*n* processes to ensure they are running normally without blocking or spinning.	Database instances, Oracle ASM instances, Oracle RAC
LMON	Global Enqueue Service Monitor Process	Monitors an Oracle RAC cluster to manage global resources	LMON maintains instance membership within Oracle RAC. The process detects instance transitions and performs reconfiguration of GES and GCS resources. **See Also:** *Oracle Real Application Clusters Administration and Deployment Guide*	Database instances, Oracle ASM instances, Oracle RAC
LMS*n*, LM*nn*	Global Cache Service Process	Manages resources and provides resource control among Oracle RAC instances	LMS*n* and LM*nn* processes maintain a lock database for Global Cache Service (GCS) and buffer cache resources. These processes receive, process, and send GCS requests, block transfers, and other GCS-related messages. There can be up to 100 of these processes, named as follows: LMS0-LMS9 LMSA-LMSZ LM10-LM19 LM1A-LM1Z LM20-LM29 LM2A-LM2R **See Also:** *Oracle Real Application Clusters Administration and Deployment Guide*	Database instances, Oracle ASM instances, Oracle RAC

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
LREG	Listener Registration Process	Registers the instance with the listeners	LREG notifies the listeners about instances, services, handlers, and endpoint.	Database instances, Oracle ASM instances, Oracle RAC
LSP0	Logical Standby Coordinator Process	Schedules transactions for Data Guard SQL Apply	LSP0 is the initial process created upon startup of Data Guard SQL Apply. In addition to managing LogMiner and Apply processes, LSP0 is responsible for maintaining inter-transaction dependencies and appropriately scheduling transactions with applier processes. LSP0 is also responsible for detecting and enabling run-time parameter changes for the SQL Apply product as a whole.	Database instances, Data Guard
LSP1	Logical Standby Dictionary Build Process	Performs a logical standby dictionary build on a primary database	The LSP1 process is spawned on a logical standby database that is intended to become the new primary database. A logical standby database becomes a primary database because of switchover or failover. The dictionary is necessary for logical standby databases to interpret the redo of the new primary database.	Database instances, Data Guard
LSP2	Logical Standby Set Guard Process	Determines which database objects will be protected by the database guard	The LSP2 process is created as needed during startup of SQL Apply to update the list of objects that are protected by the database guard.	Database instances, Data Guard
L*nnn*	Pooled Server Process	Handles client requests in Database Resident Connection Pooling	In Database Resident Connection Pooling, clients connect to a connection broker process. When a connection becomes active, the connection broker hands off the connection to a compatible pooled server process. The pooled server process performs network communication directly on the client connection and processes requests until the client releases the server. After being released, the connection is returned to the broker for monitoring, leaving the server free to handle other clients. **See Also:** *Oracle Database Concepts*	Database instances, Database Resident Connection Pooling
MARK	Mark AU for Resynchronization Coordinator Process	Marks ASM allocation units as stale following a missed write to an offline disk	MARK essentially tracks which extents require resynchronization for offline disks. This process runs in the database instance and is started when the database instance first begins using the Oracle ASM instance. If required, MARK can also be started on demand when disks go offline in the Oracle ASM redundancy disk group.	Database instances, Oracle ASM instances
MMAN	Memory Manager Process	Serves as the instance memory manager	This process performs the resizing of memory components on the instance.	Database instances, Oracle ASM instances
MMNL	Manageability Monitor Lite Process	Performs tasks relating to manageability, including active session history sampling and metrics computation	MMNL performs many tasks relating to manageability, including session history capture and metrics computation.	Database instances, Oracle ASM instances

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
MMON	Manageability Monitor Process	Performs or schedules many manageability tasks	MMON performs many tasks related to manageability, including taking Automatic Workload Repository snapshots and performing Automatic Database Diagnostic Monitor analysis.	Database instances, Oracle ASM instances
M*nnn*	Shared MMON Worker Process	Performs manageability tasks on behalf of MMON	The M*nnn* processes are a pool of worker processes that can be shared by multiple MZ*nn* processes. See the **Long Description** for MZ*nn* in this table for more information about the MZ*nn* processes.	Database instances, Oracle ASM instances
MRP0	Managed Standby Recovery Process	Coordinates the application of redo on a physical standby database	MRP0 is spawned at the start of redo apply on a physical standby database. This process handles the extraction of redo and coordinates the application of that redo on a physical standby database. **See Also:** *Oracle Data Guard Concepts and Administration*	Database instances, Data Guard
MS*nn*	LogMiner Worker Process	Reads redo log files and translates and assembles into transactions	Multiple MS*nn* processes can exists, where *n* is 0-9 or a-Z. A minimum of three MS*nn* processes work as a group to provide transactions to a LogMiner client, for example, a logical standby database or a database capture. There may be more than one such group, for example, multiple capture processes configured for either local or downstream capture in a database.	Database instances, Logical Standby, XStream Outbound servers, Oracle GoldenGate
MZ*nn*	Dedicated MMON Worker Process	Performs manageability tasks on behalf of MMON	MZ*nn* is a dedicated process for a single MMON worker action. It performs manageability tasks dispatched by MMON, which include taking Automatic Workload Repository snapshots and performing Automatic Database Diagnostic Monitor analysis.	Database instances, Oracle ASM instances
N*nnn*	Connection Broker Process	Monitors idle connections and hands off active connections in Database Resident Connection Pooling	In Database Resident Connection Pooling, clients connect to a connection broker process. When a connection becomes active, the connection broker hands off the connection to a compatible pooled server process. The pooled server process performs network communication directly on the client connection and processes requests until the client releases the server. After being released, the connection is returned to the broker for monitoring, leaving the server free to handle other clients. **See Also:** *Oracle Database Concepts*	Database instances, Database Resident Connection Pooling
NFS*n*	Direct NFS Dispatcher IO Worker Process	Performs direct NFS I/O for database processes	The dispatcher worker processes enable scaling of Direct NFS connections to a clustered NAS storage. These dedicated set of workers will be used to perform Direct NFS I/Os on behalf of database processes. The dispatcher processes are enabled by the `ENABLE_DNFS_DISPATCHER` initialization parameter. NFSn is spawned only if Direct NFS library is enabled for I/O to NFS servers. The number of worker processes spawned is based on the `CPU_COUNT` value. **See Also:** "ENABLE_DNFS_DISPATCHER"	Database instances

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
NSS*n*	Network Server SYNC Process	Transfers redo from current online redo logs to remote standby destinations configured for SYNC transport	NSS*n* can run as multiple processes, where *n* is 1-9 or A. **See Also:** *Oracle Data Guard Concepts and Administration*	Database instances, Data Guard
NSV*n*	Data Guard Broker NetSlave Process	Performs broker network communications between databases in a Data Guard environment	NSV*n* is created when a Data Guard broker configuration is enabled. There can be as many NSV*n* processes (where *n* is 0- 9 and A-U) created as there are databases in the Data Guard broker configuration.	Database instances, Data Guard
OCF*n*	ASM CF Connection Pool Process	Maintains a connection to the Oracle ASM instance for metadata operations		Database instances, Oracle ASM instances
OFSD	Oracle File Server Background Process	Serves file system requests submitted to an Oracle instance	This background process listens for new file system requests, both management (like mount, unmount, and export) and I/O requests, and executes them using Oracle threads.	Database instances, Oracle RAC
OF*nn*	Oracle File Server Background Process Thread	Serves file system requests submitted to an Oracle instance	This is a thread for the OFSD background process This background process thread is available only on Linux systems.	Database instances, Oracle RAC
O*nnn*	ASM Connection Pool Process	Maintains a connection to the Oracle ASM instance for metadata operations	O*nnn* worker processes are spawned on demand. These processes communicate with the Oracle ASM instance.	Database instances, Oracle ASM instances
PING	Interconnect Latency Measurement Process	Assesses latencies associated with communications for each pair of cluster instances	Every few seconds, the process in one instance sends messages to each instance. The message is received by PING on the target instance. The time for the round trip is measured and collected.	Database instances, Oracle ASM instances, Oracle RAC
PMAN	Process Manager	Manages several background processes including shared servers, pooled servers, and job queue processes	PMAN monitors, spawns, and stops the following as needed: • dispatcher and shared server processes • connection broker and pooled server processes for database resident connection pools • job queue processes • restartable background processes	Database instances, Oracle ASM instances, Oracle ASM Proxy instances
PMON	Process Monitor	Scans for terminated processes and coordinates cleanup	PMON periodically scans all processes to find any that have terminated in a nonstandard way. PMON is then responsible for coordinating cleanup performed by the CLMN process and the CL*nn* workers. **See Also:** *Oracle Database Concepts* and *Oracle Database Net Services Administrator's Guide*	Database instances, Oracle ASM instances, Oracle ASM Proxy instances

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
P*nnn*	Parallel Query Worker Process	Perform parallel execution of a SQL statement (query, DML, or DDL)	Parallel Query has two components: a foreground process that acts as query coordinator and a set of parallel workers (P*nnn*) that are background processes. These background processes are spawned or reused during the start of a parallel statement. They receive and perform units of work sent from the query coordinator. The maximum number of P*nnn* processes is controlled by the initialization parameter `PARALLEL_MAX_SERVERS`. Worker processes are numbered from 0 to the `PARALLEL_MAX_SERVERS` setting. If the query is a `GV$` query, then these background processes are numbered backward, starting from PPA7.	Database instances, Oracle ASM instances
PR*nn*	Parallel Recovery Process	Performs tasks assigned by the coordinator process performing parallel recovery	PR*nn* serves as a worker process for the coordinator process performing parallel media recovery and carries out tasks assigned by the coordinator. The default number of these processes is based on number of CPUs.	Database instances
PSP0	Process Spawner Process	Spawns Oracle background processes after initial instance startup		Database instances, Oracle ASM instances
PXMN	Parallel Execution Monitor	Spawns parallel server processes on local instances in an Oracle RAC environment for Query Coordinator in remote instances.		Database instances
QMNC	Non-sharded queue primary process	Monitors AQ	QMNC is the non-sharded queue primary process responsible for facilitating various background activities required by AQ: time management of messages, management of nonpersistent queues, cleanup of resources, and so on. QMNC dynamically spawns Q*nnn* processes as needed for performing these tasks. Note that if the `AQ_TM_PROCESSES` initialization parameter is set to 0, this process will not start. The database writes the following message to the alert log: `WARNING: AQ_TM_PROCESSES is set to 0. System might be adversely affected`.	Database instances Advanced Queueing
QM*nn*	AQ Primary Class Process	Per instance per AQ Primary Class Process	Each of this type of process represents a single class of work item such as AQ notification, queue monitors, and cross process.	Database instances Advanced Queueing
Q*nnn*	AQ Server Class Process	Per AQ Primary Class server process	Each server class process acts on behalf of an AQ primary class process. This relationship is maintained until the primary requires services of a particular service process. Once released, the server class processes are moved to a free server pool.	Database instances Advanced Queueing

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
R*nnn*	ASM Block Remap Worker Process	Remaps a block with a read error	A database instance reading from an Oracle ASM disk group can encounter an error during a read. If possible, Oracle ASM asynchronously schedules a R*nnn* worker process to remap this bad block from a mirror copy.	Oracle ASM instances
RBAL	ASM Rebalance Primary Process	Coordinates rebalance activity	In an Oracle ASM instance, it coordinates rebalance activity for disk groups. In a database instance, it manages Oracle ASM disk groups.	Database instances, Oracle ASM instances
RCBG	Result Cache Background Process	Handles result cache messages	This process is used for handling invalidation and other messages generated by server processes attached to other instances in Oracle RAC.	Database instances, Oracle RAC
RECO	Recoverer Process	Resolves distributed transactions that are pending because of a network or system failure in a distributed database	RECO uses the information in the pending transaction table to finalize the status of in-doubt transactions. At timed intervals, the local RECO attempts to connect to remote databases and automatically complete the commit or rollback of the local portion of any pending distributed transactions. All transactions automatically resolved by RECO are removed from the pending transaction table. **See Also:** *Oracle Database Concepts* and *Oracle Database Net Services Administrator's Guide*	Database instances
RL*nn*	ResetLogs Process	Clear online redo logs when performing open resetlogs and converting to physical standby	RL*nn* processes are spawned to clear online redo logs. These workers are terminated after the online redo logs are cleared, and the session does not persist. Possible processes are RL00-RL31.	Database instances
RM	RAT Masking Worker Process	Extracts and masks bind values from workloads like SQL tuning sets and DB Replay capture files	This background process is used with Data Masking and Real Application Testing.	Database instances
RMON	Rolling Migration Monitor Process	Manages the rolling migration procedure for an Oracle ASM cluster	The RMON process is spawned on demand to run the protocol for transitioning an ASM cluster in and out of rolling migration mode.	Oracle ASM instance, Oracle RAC
RMS*n*	Oracle RAC Management Process	Performs manageability tasks for Oracle RAC	RMS*n* performs a variety of tasks, including creating resources related to Oracle RAC when new instances are added to a cluster. **See Also:** *Oracle Real Application Clusters Administration and Deployment Guide*	Database instances, Oracle RAC

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
RP*nn*	Capture Processing Worker Process	Processes a set of workload capture files	RP*nn* are worker processes spawned by calling `DBMS_WORKLOAD_REPLAY.PROCESS_CAPTURE(capture_dir,parallel_level)`. Each worker process is assigned a set of workload capture files to process. Worker processes execute in parallel without needing to communicate with each other. After each process is finished processing its assigned files, it exits and informs its parent process. The number of worker processes is controlled by the `parallel_level` parameter of `DBMS_WORKLOAD_REPLAY.PROCESS_CAPTURE`. By default, `parallel_level` is null. Then, the number of worker processes is computed as follows: `SELECT VALUE FROM V$PARAMETER WHERE NAME='cpu_count';` When `parallel_level` is `1`, no worker processes are spawned.	Database instances
RPOP	Instant Recovery Repopulation Daemon	Responsible for re-creating and/or repopulating data files from snapshot files and backup files	The RPOP process is responsible for re-creating and repopulating data files from snapshots files. It works with the instant recovery feature to ensure immediate data file access. The local instance has immediate access to the remote snapshot file's data, while repopulation of the recovered primary data files happens concurrently. Any changes in the data are managed between the instance's DBW processes and RPOP to ensure the latest copy of the data is returned to the user.	Database instances
RS*nn*	Global Cache Service Remaster Process	Performs remastering for cluster reconfiguration and dynamic remastering	Each RS*nn* process is a worker process for LMS*n* to handle remastering work. They are also helper processes for LMS to handle non-critical work from global cache service. The RS*nn* processes were named RMV*n* in Oracle Database 12c and earlier releases.	Database instances, Oracle RAC
RSM0	Data Guard Broker Worker Process	Performs monitoring management tasks related to Data Guard on behalf of DMON	The process is created when a Data Guard broker configuration is enabled.	Database instances, Data Guard
RSMN	Remote Worker Monitor Process	Manages background worker process creation and communication on remote instances in Oracle RAC	This background process manages the creation of worker processes and the communication with their coordinators and peers. These background worker processes perform tasks on behalf of a coordinating process running in another cluster instance.	Database instances, Oracle RAC
RVWR	Recovery Writer Process	Writes flashback data to the flashback logs in the fast recovery area	RVWR writes flashback data from the flashback buffer in the SGA to the flashback logs. RVWR also creates flashback logs and performs some tasks for flashback log automatic management.	Database instances, Flashback Database

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
S*nnn*	Shared Server Process	Handles client requests in the shared server architecture	In the shared server architecture, clients connect to a dispatcher process, which creates a virtual circuit for each connection. When the client sends data to the server, the dispatcher receives the data into the virtual circuit and places the active circuit on the common queue to be picked up by an idle shared server. The shared server then reads the data from the virtual circuit and performs the database work necessary to complete the request. When the shared server must send data to the client, the server writes the data back into the virtual circuit and the dispatcher sends the data to the client. After the shared server completes the client request, the server releases the virtual circuit back to the dispatcher and is free to handle other clients. Several initialization parameters relate to shared servers. The principal parameters are: `DISPATCHERS`, `SHARED_SERVERS`, `MAX_SHARED_SERVERS`, `LOCAL_LISTENER`, `REMOTE_LISTENER`. **See Also:** *Oracle Database Concepts*	Database instances, shared servers
SA*nn*	SGA Allocator	Allocates SGA	A small fraction of SGA is allocated during instance startup. The SA*nn* process allocates the rest of SGA in small chunks. The process exits upon completion of SGA allocation. Possible processes are SA00-SAzz.	Database instances
SCC*n*	ASM Disk Scrubbing Worker Check Process	Performs Oracle ASM disk scrubbing check operation	SCC*n* acts as a worker process for SCRB and performs the checking operations. Possible processes are SCC0-SCC9.	Oracle ASM instances
SCM0	DLM Statistics Collection and Management Worker	Collects and manages statistics related to global enqueue service (GES) and global cache service (GCS)	The DLM Statistics Collection and Management worker (SCM0) is responsible for collecting and managing the statistics related to global enqueue service (GES) and global cache service (GCS). This worker exists only if DLM statistics collection is enabled.	Database instances
SCRB	ASM Disk Scrubbing Primary Process	Coordinates Oracle ASM disk scrubbing operations	SCRB runs in an Oracle ASM instance and coordinates Oracle ASM disk scrubbing operations.	Oracle ASM instances
SCR*n*	ASM Disk Scrubbing Worker Repair Process	Performs Oracle ASM disk scrubbing repair operation	SCR*n* acts as a worker process for SCRB and performs the repairing operations. Possible processes are SCR0-SCR9.	Oracle ASM instances
SCV*n*	ASM Disk Scrubbing Worker Verify Process	Performs Oracle ASM disk scrubbing verify operation	SCV*n* acts as a worker process for SCRB and performs the verifying operations. Possible processes are SCV0-SCV9.	Oracle ASM instances
SMCO	Space Management Coordinator Process	Coordinates the execution of various space management tasks	This background process coordinates the execution of various space management tasks, including proactive space allocation and space reclamation. SMCO dynamically spawns worker processes (W*nnn*) to implement these tasks.	Database instances

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
SMON	System Monitor Process	Performs critical tasks such as instance recovery and terminated transaction recovery, and maintenance tasks such as temporary space reclamation, data dictionary cleanup, and undo tablespace management	SMON performs many database maintenance tasks, including the following: • Creates and manages the temporary tablespace metadata • Reclaims space used by orphaned temporary segments • Maintains the undo tablespace by onlining, offlining, and shrinking the undo segments based on undo space usage statistics • Cleans up the data dictionary when it is in a transient and inconsistent state • Maintains the SCN to time mapping table used to support Oracle Flashback features In an Oracle RAC database, the SMON process of one instance can perform instance recovery for other instances that have failed. SMON is resilient to internal and external errors raised during background activities. **See Also:** *Oracle Database Concepts*	Database instances
SP	SPA Exec Worker	Analyzes single SQL statements sent from SQL Performance Analyzer (SPA)	Executions of SPA tasks created from a SQL tuning set use this worker to analyze the SQL statements of the SQL tuning set concurrently.	Database instances
SVCB	Service Background Process	Provides database service run-time load balancing and topology information to clients.	Every 30 seconds the process processes and publishes run-time load-balancing information and keeps the topology information current. This process is started only if Oracle Real Application Clusters (Oracle RAC) is enabled.	Oracle RAC
TEM*n*	ASM disk Test Error Emulation Process	Emulates I/O errors on Oracle ASM disks through named events	I/O errors can be emulated on Oracle ASM disk I/O through named events. The scope can be the process, instance, or even cluster. Optionally, a set of AUs can be chosen for error emulation.	Oracle ASM instances
TT*nn*	Redo Transport Worker Process	Ships redo from current online and standby redo logs to remote standby destinations configured for ASYNC transport	TT*nn* can run as multiple processes, where *nn* is 00 to ZZ. **See Also:** *Oracle Data Guard Concepts and Administration*	Database instances, Data Guard

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
U*nnn*	Container process for threads	Host processes where database processes execute as threads.	U*nnn* processes are database container operating system processes where database backgrounds processes like SMON, CJQ0, and database foreground processes run. The V$PROCESS view lists database processes running in these container processes. These container processes are created only when the THREADED_EXECUTION initialization parameter is set to TRUE. The number of these processes vary depending on the active database processes. On a host with multiple NUMA nodes, there will be at least one U*nnn* process per NUMA node. These processes are irrecoverable processes, if any of them is terminated, it will result in instance termination. These processes exit when the instance is shut down or terminated.	Database instances
VBG*n*	Volume Background Process	Communicates between the Oracle ASM instance and the operating system volume driver	VBG*n* handles messages originating from the volume driver in the operating system and sends them to the Oracle ASM instance. VBG*n* can run as multiple processes, where *n* is 0-9.	Oracle ASM instances, Oracle ASM Proxy instances
VDBG	Volume Driver Process	Forwards Oracle ASM requests to perform various volume-related tasks	VDBG handles requests to lock or unlock an extent for rebalancing, volume resize, disk offline, add or drop a disk, force and dismount disk group to the Dynamic Volume Manager driver.	Oracle ASM instances, Oracle ASM Proxy instances
VI*nn*	Volume I/O	Route ADVM volume I/O for ASM instances on compute nodes within an Exadata	These processes handle requests for I/Os targeted at storage not locally accessible. They are used for Exadata targeted storage as well. These background processes only start when an ASM Volume is created and set up to be used. One process will start for each NUMA node on target machines. Under normal operation on non-Exadata hardware and on Exadata hardware that is not utilizing ASM volumes, these processes will not be started. There can be up to 32 VI processes, and they are named sequentially from VI00 to VI31.	Oracle ASM Proxy instances
VKRM	Virtual Scheduler for Resource Manager Process	Serves as centralized scheduler for Resource Manager activity	VKRM manages the CPU scheduling for all managed Oracle processes. The process schedules managed processes in accordance with an active resource plan.	Database instances
VKTM	Virtual Keeper of Time Process	Provides a wall clock time and reference time for time interval measurements	VKTM acts as a time publisher for an Oracle instance. VKTM publishes two sets of time: a wall clock time using a seconds interval and a higher resolution time (which is not wall clock time) for interval measurements. The VKTM timer service centralizes time tracking and offloads multiple timer calls from other clients.	Database instances, Oracle ASM instances
VMB0	Volume Membership Process	Maintains cluster membership on behalf of the Oracle ASM volume driver	This process membership in the cluster as an I/O-capable client on behalf of the Oracle ASM volume driver.	Oracle ASM instances, Oracle ASM Proxy instances

Copyright © 2002, 2026, Oracle and/or its affiliates.

Table F-1 (Cont.) Background Processes

Name	Expanded Name	Short Description	Long Description	External Properties
VUBG	Volume drive Umbilicus Background	Relays messages between Oracle ASM instance and Oracle ASM Proxy instance that is used by ADVM (for ACFS)		Oracle ASM instances, Oracle ASM Proxy instances
W*nnn*	Space Management Worker Process	Performs various background space management tasks, including proactive space allocation and space reclamation	W*nnn* worker processes perform work on behalf of Space Management and on behalf of the Oracle Database In-Memory option. • When performing work on behalf of Space Management, W*nnn* processes are worker processes dynamically spawned by SMCO to perform space management tasks in the background. These tasks include preallocating space into locally managed tablespace and SecureFiles segments based on space usage growth analysis, and reclaiming space from dropped segments. After being started, the worker acts as an autonomous agent. After it finishes task execution, it automatically picks up another task from the queue. The process terminates itself after being idle for a long time. • When performing work on behalf of the Oracle Database In-Memory option, W*nnn* processes execute tasks for population or repopulation of objects that are enabled for the In-Memory column store (IM columns store), and tasks that drop in-memory segments when an object is disabled for the IM columns store. For in-memory population and repopulation, both the IMCO background process and foreground processes will utilize W*nnn* workers. W*nnn* processes are utilized by the IMCO background process for prepopulation of in-memory enabled objects with priority LOW/MEDIUM/HIGH/CRITICAL, and for repopulation of in-memory objects. In-memory populate and repopulate tasks running on W*nnn* workers are also initiated from foreground processes in response to queries and DMLs that reference in-memory enabled objects.	Database instances
XDMG	Exadata Automation Manager	Initiates automation tasks involved in managing Exadata storage	XDMG monitors all configured Exadata cells for state changes, such as a bad disk getting replaced, and performs the required tasks for such events. Its primary tasks are to watch for when inaccessible disks and cells become accessible again, and to initiate the ASM ONLINE operation. The ONLINE operation is handled by XDWK.	Oracle ASM instances, Exadata
XDWK	Exadata Automation Manager	Performs automation tasks requested by XDMG	XDWK gets started when asynchronous actions such as ONLINE, DROP, and ADD an Oracle ASM disk are requested by XDMG. After a 5 minute period of inactivity, this process will shut itself down.	Oracle ASM instances, Exadata
X*nnn*	ASM Disk Expel Worker Process	Performs Oracle ASM post-rebalance activities	This process expels dropped disks after an Oracle ASM rebalance.	Oracle ASM instances

Copyright © 2002, 2026, Oracle and/or its affiliates.

Copyright © 2002, 2026, Oracle and/or its affiliates.

Index

A

ACTIVE_INSTANCE_COUNT initialization parameter, *26*
ADG_ACCOUNT_INFO_TRACKING initialization parameter, *27*
ADG_REDIRECT_DML initialization parameter, *27*
ALL_ALL_TABLES view, *7*
ALL_ANALYTIC_VIEW_ATTR_CLASS view, *13*
ALL_ANALYTIC_VIEW_ATTR_CLS view, *14*
ALL_ANALYTIC_VIEW_ATTR_CLS_AE view, *14*
ALL_ANALYTIC_VIEW_BAS_MEAS view, *15*
ALL_ANALYTIC_VIEW_BAS_MEAS_AE view, *15*
ALL_ANALYTIC_VIEW_BASE_MEAS view, *16*
ALL_ANALYTIC_VIEW_CALC_MEAS view, *17*
ALL_ANALYTIC_VIEW_CLASS view, *18*
ALL_ANALYTIC_VIEW_CLASS_AE view, *19*
ALL_ANALYTIC_VIEW_CLC_MEAS view, *20*
ALL_ANALYTIC_VIEW_CLC_MEAS_AE view, *20*
ALL_ANALYTIC_VIEW_COLUMNS view, *21*
ALL_ANALYTIC_VIEW_COLUMNS_AE view, *22*
ALL_ANALYTIC_VIEW_DIM_CLASS view, *24*
ALL_ANALYTIC_VIEW_DIM_CLS view, *25*
ALL_ANALYTIC_VIEW_DIM_CLS_AE view, *25*
ALL_ANALYTIC_VIEW_DIMENSIONS view, *26*
ALL_ANALYTIC_VIEW_DIMS view, *27*
ALL_ANALYTIC_VIEW_DIMS_AE view, *27*
ALL_ANALYTIC_VIEW_HIER_CLASS view, *29*
ALL_ANALYTIC_VIEW_HIER_CLS view, *29*
ALL_ANALYTIC_VIEW_HIER_CLS_AE view, *30*
ALL_ANALYTIC_VIEW_HIERS view, *31*
ALL_ANALYTIC_VIEW_HIERS_AE view, *32*
ALL_ANALYTIC_VIEW_KEYS view, *33*
ALL_ANALYTIC_VIEW_KEYS_AE view, *33*
ALL_ANALYTIC_VIEW_LEVEL_CLASS view, *34*
ALL_ANALYTIC_VIEW_LEVELS view, *35*
ALL_ANALYTIC_VIEW_LEVELS_AE view, *36*
ALL_ANALYTIC_VIEW_LVL_CLS view, *37*
ALL_ANALYTIC_VIEW_LVL_CLS_AE view, *37*
ALL_ANALYTIC_VIEW_LVLGRPS view, *38*
ALL_ANALYTIC_VIEW_LVLGRPS_AE view, *39*
ALL_ANALYTIC_VIEW_MEAS_CLASS view, *40*
ALL_ANALYTIC_VIEW_MEAS_CLS view, *41*
ALL_ANALYTIC_VIEW_MEAS_CLS_AE view, *41*
ALL_ANALYTIC_VIEWS view, *42*
ALL_ANALYTIC_VIEWS_AE view, *43*
ALL_ANNOTATION_VALUES view, *44*
ALL_ANNOTATIONS view, *45*
ALL_ANNOTATIONS_USAGE view, *45*
ALL_APPLY view, *46*
ALL_APPLY_CHANGE_HANDLERS view, *48*
ALL_APPLY_CONFLICT_COLUMNS view, *49*
ALL_APPLY_DML_CONF_HANDLERS view, *50*
ALL_APPLY_DML_HANDLERS view, *50*
ALL_APPLY_ENQUEUE view, *51*
ALL_APPLY_ERROR view, *52*
ALL_APPLY_ERROR_MESSAGES view, *53*
ALL_APPLY_EXECUTE view, *55*
ALL_APPLY_HANDLE_COLLISIONS view, *55*
ALL_APPLY_INSTANTIATED_GLOBAL view, *56*
ALL_APPLY_INSTANTIATED_OBJECTS view, *57*
ALL_APPLY_INSTANTIATED_SCHEMAS view, *57*
ALL_APPLY_KEY_COLUMNS view, *58*
ALL_APPLY_PARAMETERS view, *58*
ALL_APPLY_PROGRESS view, *59*
ALL_APPLY_REPERROR_HANDLERS view, *60*
ALL_APPLY_TABLE_COLUMNS view, *61*
ALL_ARGUMENTS view, *62*
ALL_ASSEMBLIES view, *64*
ALL_ASSOCIATIONS view, *65*
ALL_ATTRIBUTE_DIM_ATTR_CLASS view, *66*
ALL_ATTRIBUTE_DIM_ATTR_CLS view, *66*
ALL_ATTRIBUTE_DIM_ATTR_CLS_AE view, *67*
ALL_ATTRIBUTE_DIM_ATTRS view, *68*
ALL_ATTRIBUTE_DIM_ATTRS_AE view, *68*
ALL_ATTRIBUTE_DIM_CLASS view, *69*
ALL_ATTRIBUTE_DIM_CLASS_AE view, *70*
ALL_ATTRIBUTE_DIM_JN_PTHS view, *71*
ALL_ATTRIBUTE_DIM_JN_PTHS_AE view, *71*
ALL_ATTRIBUTE_DIM_JOIN_PATHS view, *72*
ALL_ATTRIBUTE_DIM_KEYS view, *73*
ALL_ATTRIBUTE_DIM_KEYS_AE view, *74*
ALL_ATTRIBUTE_DIM_LEVEL_ATTRS view, *75*
ALL_ATTRIBUTE_DIM_LEVELS view, *76*
ALL_ATTRIBUTE_DIM_LEVELS_AE view, *77*
ALL_ATTRIBUTE_DIM_LVL_ATRS view, *78*
ALL_ATTRIBUTE_DIM_LVL_ATRS_AE view, *78*
ALL_ATTRIBUTE_DIM_LVL_CLASS view, *79*
ALL_ATTRIBUTE_DIM_LVL_CLS view, *80*
ALL_ATTRIBUTE_DIM_LVL_CLS_AE view, *80*

Copyright © 2002, 2026, Oracle and/or its affiliates.

ALL_ATTRIBUTE_DIM_ORD_ATRS view, *81*
ALL_ATTRIBUTE_DIM_ORD_ATRS_AE view, *81*
ALL_ATTRIBUTE_DIM_ORDER_ATTRS view, *82*
ALL_ATTRIBUTE_DIM_TABLES view, *83*
ALL_ATTRIBUTE_DIM_TABLES_AE view, *84*
ALL_ATTRIBUTE_DIMENSIONS view, *85*
ALL_ATTRIBUTE_DIMENSIONS_AE view, *86*
ALL_ATTRIBUTE_TRANSFORMATIONS view, *87*
ALL_AUDIT_POLICIES view, *88*
ALL_AUDIT_POLICY_COLUMNS view, *89*
ALL_AW_PS view, *90*
ALL_AWS view, *90*
ALL_BASE_TABLE_MVIEWS view, *91*
ALL_BLOCKCHAIN_TABLES view, *92*
ALL_CAPTURE view, *93*
ALL_CAPTURE_EXTRA_ATTRIBUTES view, *95*
ALL_CAPTURE_PARAMETERS view, *96*
ALL_CAPTURE_PREPARED_DATABASE view, *96*
ALL_CAPTURE_PREPARED_SCHEMAS view, *97*
ALL_CAPTURE_PREPARED_TABLES view, *98*
ALL_CATALOG view, *99*
ALL_CERTIFICATES view, *100*
ALL_CLUSTER_HASH_EXPRESSIONS view, *100*
ALL_CLUSTERING_DIMENSIONS view, *101*
ALL_CLUSTERING_JOINS view, *102*
ALL_CLUSTERING_KEYS view, *103*
ALL_CLUSTERING_TABLES view, *104*
ALL_CLUSTERS view, *105*
ALL_CODE_ROLE_PRIVS view, *107*
ALL_COL_COMMENTS view, *107*
ALL_COL_PENDING_STATS view, *108*
ALL_COL_PRIVS view, *109*
ALL_COL_PRIVS_MADE view, *110*
ALL_COL_PRIVS_RECD view, *110*
ALL_COLL_TYPES view, *111*
ALL_CONS_COLUMNS view, *112*
ALL_CONS_OBJ_COLUMNS view, *113*
ALL_CONSTRAINTS view, *114*
ALL_CONTEXT view, *116*
ALL_CREDENTIALS view, *116*
ALL_CUBE_ATTR_VISIBILITY view, *117*
ALL_CUBE_ATTRIBUTES view, *118*
ALL_CUBE_BUILD_PROCESSES view, *120*
ALL_CUBE_CALCULATED_MEMBERS view, *120*
ALL_CUBE_DIM_LEVELS view, *121*
ALL_CUBE_DIM_MODELS view, *122*
ALL_CUBE_DIM_VIEW_COLUMNS view, *122*
ALL_CUBE_DIM_VIEWS view, *123*
ALL_CUBE_DIMENSIONALITY view, *124*
ALL_CUBE_DIMENSIONS view, *124*
ALL_CUBE_HIER_LEVELS view, *125*
ALL_CUBE_HIER_VIEW_COLUMNS view, *126*
ALL_CUBE_HIER_VIEWS view, *127*
ALL_CUBE_HIERARCHIES view, *128*
ALL_CUBE_MEASURES view, *129*
ALL_CUBE_NAMED_BUILD_SPECS view, *130*
ALL_CUBE_SUB_PARTITION_LEVELS view, *130*
ALL_CUBE_VIEW_COLUMNS view, *131*
ALL_CUBE_VIEWS view, *132*
ALL_CUBES view, *132*
ALL_DB_LINKS view, *133*
ALL_DEF_AUDIT_OPTS view, *134*
ALL_DEPENDENCIES view, *135*
ALL_DEQUEUE_QUEUES view, *136*
ALL_DIM_ATTRIBUTES view, *137*
ALL_DIM_CHILD_OF view, *137*
ALL_DIM_HIERARCHIES view, *138*
ALL_DIM_JOIN_KEY view, *138*
ALL_DIM_LEVEL_KEY view, *139*
ALL_DIM_LEVELS view, *139*
ALL_DIMENSIONS view, *140*
ALL_DIRECTORIES view, *141*
ALL_EDITION_COMMENTS view, *141*
ALL_EDITIONING_VIEW_COLS view, *142*
ALL_EDITIONING_VIEW_COLS_AE view, *142*
ALL_EDITIONING_VIEWS view, *143*
ALL_EDITIONING_VIEWS_AE view, *143*
ALL_EDITIONS view, *144*
ALL_ENCRYPTED_COLUMNS view, *144*
ALL_ERROR_TRANSLATIONS view, *145*
ALL_ERRORS view, *146*
ALL_ERRORS_AE view, *147*
ALL_EVALUATION_CONTEXT_TABLES view, *148*
ALL_EVALUATION_CONTEXT_VARS view, *148*
ALL_EVALUATION_CONTEXTS view, *149*
ALL_EXPRESSION_STATISTICS view, *150*
ALL_EXTERNAL_LOCATIONS view, *150*
ALL_EXTERNAL_TABLES view, *151*
ALL_FILE_GROUP_EXPORT_INFO view, *152*
ALL_FILE_GROUP_FILES view, *153*
ALL_FILE_GROUP_TABLES view, *153*
ALL_FILE_GROUP_TABLESPACES view, *154*
ALL_FILE_GROUP_VERSIONS view, *155*
ALL_FILE_GROUPS view, *155*
ALL_GG_AUTO_CDR_COLUMN_GROUPS view, *156*
ALL_GG_AUTO_CDR_COLUMNS view, *157*
ALL_GG_AUTO_CDR_TABLES view, *157*
ALL_GG_INBOUND_PROGRESS view, *158*
ALL_GOLDENGATE_INBOUND view, *159*
ALL_GOLDENGATE_PRIVILEGES view, *160*
ALL_GOLDENGATE_RULES view, *160*
ALL_HEAT_MAP_SEG_HISTOGRAM view, *162*
ALL_HEAT_MAP_SEGMENT view, *162*
ALL_HIER_CLASS view, *163*

Copyright © 2002, 2026, Oracle and/or its affiliates.

ALL_HIER_CLASS_AE view, *164*
ALL_HIER_COLUMNS view, *165*
ALL_HIER_COLUMNS_AE view, *166*
ALL_HIER_HIER_ATTR_CLASS view, *167*
ALL_HIER_HIER_ATTR_CLASS_AE view, *168*
ALL_HIER_HIER_ATTRIBUTES view, *169*
ALL_HIER_HIER_ATTRIBUTES_AE view, *170*
ALL_HIER_JOIN_PATHS view, *171*
ALL_HIER_JOIN_PATHS_AE view, *172*
ALL_HIER_LEVEL_ID_ATTRS view, *172*
ALL_HIER_LEVEL_ID_ATTRS_AE view, *173*
ALL_HIER_LEVELS view, *174*
ALL_HIER_LEVELS_AE view, *175*
ALL_HIERARCHIES view, *176*
ALL_HIERARCHIES_AE view, *176*
ALL_HISTOGRAMS synonym for ALL_TAB_HISTOGRAMS, *177*
ALL_HIVE_COLUMNS view, *177*
ALL_HIVE_DATABASES view, *178*
ALL_HIVE_PART_KEY_COLUMNS view, *179*
ALL_HIVE_TAB_PARTITIONS view, *179*
ALL_HIVE_TABLES view, *180*
ALL_IDENTIFIERS view, *181*
ALL_IMMUTABLE_TABLES view, *183*
ALL_IND_COLUMNS view, *184*
ALL_IND_EXPRESSIONS view, *185*
ALL_IND_PARTITIONS view, *185*
ALL_IND_PENDING_STATS view, *188*
ALL_IND_STATISTICS view, *189*
ALL_IND_SUBPARTITIONS view, *191*
ALL_INDEXES view, *193*
ALL_INDEXTYPE_ARRAYTYPES view, *198*
ALL_INDEXTYPE_COMMENTS view, *198*
ALL_INDEXTYPE_OPERATORS view, *199*
ALL_INDEXTYPES view, *199*
ALL_INTERNAL_TRIGGERS view, *200*
ALL_JAVA_ARGUMENTS view, *201*
ALL_JAVA_CLASSES view, *202*
ALL_JAVA_COMPILER_OPTIONS view, *203*
ALL_JAVA_DERIVATIONS view, *203*
ALL_JAVA_FIELDS view, *204*
ALL_JAVA_IMPLEMENTS view, *205*
ALL_JAVA_INNERS view, *205*
ALL_JAVA_LAYOUTS view, *206*
ALL_JAVA_METHODS view, *207*
ALL_JAVA_NCOMPS view, *209*
ALL_JAVA_RESOLVERS view, *209*
ALL_JAVA_THROWS view, *210*
ALL_JOBS synonym for USER_JOBS view, *210*
ALL_JOIN_IND_COLUMNS view, *210*
ALL_JSON_COLUMNS view, *211*
ALL_JSON_DATAGUIDE_FIELDS view, *212*
ALL_JSON_DATAGUIDES view, *213*
ALL_LIBRARIES view, *213*
ALL_LOB_PARTITIONS view, *214*
ALL_LOB_SUBPARTITIONS view, *217*
ALL_LOB_TEMPLATES view, *219*
ALL_LOBS view, *220*
ALL_LOG_GROUP_COLUMNS view, *222*
ALL_LOG_GROUPS view, *223*
ALL_MEASURE_FOLDER_CONTENTS view, *223*
ALL_MEASURE_FOLDER_SUBFOLDERS view, *224*
ALL_MEASURE_FOLDERS view, *225*
ALL_METADATA_PROPERTIES view, *225*
ALL_METHOD_PARAMS view, *226*
ALL_METHOD_RESULTS view, *227*
ALL_MINING_ALGORITHMS view, *227*
ALL_MINING_MODEL_ATTRIBUTES view, *228*
ALL_MINING_MODEL_PARTITIONS view, *230*
ALL_MINING_MODEL_SETTINGS view, *230*
ALL_MINING_MODEL_VIEWS view, *231*
ALL_MINING_MODEL_XFORMS view, *231*
ALL_MINING_MODELS view, *232*
ALL_MVIEW_AGGREGATES view, *233*
ALL_MVIEW_ANALYSIS view, *234*
ALL_MVIEW_COMMENTS view, *235*
ALL_MVIEW_DETAIL_PARTITION view, *236*
ALL_MVIEW_DETAIL_RELATIONS view, *237*
ALL_MVIEW_DETAIL_SUBPARTITION view, *238*
ALL_MVIEW_JOINS view, *238*
ALL_MVIEW_KEYS view, *239*
ALL_MVIEW_LOGS view, *240*
ALL_MVIEW_REFRESH_TIMES view, *241*
ALL_MVIEWS view, *242*
ALL_NESTED_TABLE_COLS view, *246*
ALL_NESTED_TABLES view, *248*
ALL_OBJ_COLATTRS view, *249*
ALL_OBJECT_TABLES view, *250*
ALL_OBJECTS view, *255*
ALL_OBJECTS_AE view, *257*
ALL_OPANCILLARY view, *259*
ALL_OPARGUMENTS view, *259*
ALL_OPBINDINGS view, *260*
ALL_OPERATOR_COMMENTS view, *261*
ALL_OPERATORS view, *261*
ALL_OUTLINE_HINTS synonym for USER_OUTLINE_HINTS view, *262*
ALL_OUTLINES synonym for USER_OUTLINES view, *262*
ALL_PART_COL_STATISTICS view, *1*
ALL_PART_HISTOGRAMS view, *2*
ALL_PART_INDEXES view, *3*
ALL_PART_KEY_COLUMNS view, *6*
ALL_PART_LOBS view, *6*
ALL_PART_TABLES view, *9*
ALL_PARTIAL_DROP_TABS view, *14*
ALL_PENDING_CONV_TABLES view, *14*
ALL_PLSQL_COLL_TYPES view, *15*
ALL_PLSQL_OBJECT_SETTINGS view, *16*
ALL_PLSQL_TYPE_ATTRS view, *17*

Copyright © 2002, 2026, Oracle and/or its affiliates.

ALL_PLSQL_TYPES view, *17*
ALL_POLICIES view, *18*
ALL_POLICY_ATTRIBUTES view, *19*
ALL_POLICY_CONTEXTS view, *20*
ALL_POLICY_GROUPS view, *21*
ALL_PROCEDURES view, *21*
ALL_PROPAGATION view, *23*
ALL_QUEUE_SCHEDULES view, *24*
ALL_QUEUE_SUBSCRIBERS view, *26*
ALL_QUEUE_TABLES view, *27*
ALL_QUEUES view, *28*
ALL_REFRESH view, *29*
ALL_REFRESH_CHILDREN view, *30*
ALL_REFRESH_DEPENDENCIES view, *31*
ALL_REFS view, *31*
ALL_REGISTERED_MVIEWS view, *32*
ALL_REGISTRY_BANNERS view, *33*
ALL_REPL_DBNAME_MAPPING view, *33*
ALL_REPLICATION_PROCESS_EVENTS view, *33*
ALL_REWRITE_EQUIVALENCES view, *34*
ALL_RULE_SET_RULES view, *35*
ALL_RULE_SETS view, *36*
ALL_RULES view, *36*
ALL_SCHEDULER_CHAIN_RULES view, *37*
ALL_SCHEDULER_CHAIN_STEPS view, *37*
ALL_SCHEDULER_CHAINS view, *39*
ALL_SCHEDULER_CREDENTIALS view, *39*
ALL_SCHEDULER_DB_DESTS view, *40*
ALL_SCHEDULER_DESTS view, *41*
ALL_SCHEDULER_EXTERNAL_DESTS view, *42*
ALL_SCHEDULER_FILE_WATCHERS view, *42*
ALL_SCHEDULER_GLOBAL_ATTRIBUTE view, *43*
ALL_SCHEDULER_GROUP_MEMBERS view, *44*
ALL_SCHEDULER_GROUPS view, *44*
ALL_SCHEDULER_INCOMPAT_MEMBER view, *45*
ALL_SCHEDULER_INCOMPATS view, *45*
ALL_SCHEDULER_JOB_ARGS view, *46*
ALL_SCHEDULER_JOB_CLASSES view, *47*
ALL_SCHEDULER_JOB_DESTS view, *47*
ALL_SCHEDULER_JOB_LOG view, *49*
ALL_SCHEDULER_JOB_RUN_DETAILS view, *50*
ALL_SCHEDULER_JOBS view, *51*
ALL_SCHEDULER_NOTIFICATIONS view, *55*
ALL_SCHEDULER_PROGRAM_ARGS view, *56*
ALL_SCHEDULER_PROGRAMS view, *57*
ALL_SCHEDULER_REMOTE_DATABASES view, *58*
ALL_SCHEDULER_REMOTE_JOBSTATE view, *59*
ALL_SCHEDULER_RESOURCES view, *60*
ALL_SCHEDULER_RSC_CONSTRAINTS view, *60*
ALL_SCHEDULER_RUNNING_CHAINS view, *61*
ALL_SCHEDULER_RUNNING_JOBS view, *62*
ALL_SCHEDULER_SCHEDULES view, *63*
ALL_SCHEDULER_WINDOW_DETAILS view, *64*
ALL_SCHEDULER_WINDOW_GROUPS view, *65*
ALL_SCHEDULER_WINDOW_LOG view, *66*
ALL_SCHEDULER_WINDOWS view, *66*
ALL_SCHEDULER_WINGROUP_MEMBERS view, *68*
ALL_SEC_RELEVANT_COLS view, *68*
ALL_SECONDARY_OBJECTS view, *69*
ALL_SEQUENCES view, *69*
ALL_SERVICES view, *70*
ALL_SOURCE view, *74*
ALL_SOURCE_AE view, *75*
ALL_SQL_TRANSLATION_PROFILES view, *75*
ALL_SQL_TRANSLATIONS view, *76*
ALL_SQLJ_TYPE_ATTRS view, *77*
ALL_SQLJ_TYPE_METHODS view, *78*
ALL_SQLJ_TYPES view, *79*
ALL_SQLSET view, *80*
ALL_SQLSET_BINDS view, *81*
ALL_SQLSET_PLANS view, *82*
ALL_SQLSET_REFERENCES view, *86*
ALL_SQLSET_STATEMENTS view, *86*
ALL_STAT_EXTENSIONS view, *88*
ALL_STATEMENTS view, *89*
ALL_STORED_SETTINGS view, *90*
ALL_STREAMS_GLOBAL_RULES view, *91*
ALL_STREAMS_MESSAGE_CONSUMERS view, *92*
ALL_STREAMS_NEWLY_SUPPORTED view, *93*
ALL_STREAMS_SCHEMA_RULES view, *94*
ALL_STREAMS_TABLE_RULES view, *95*
ALL_STREAMS_TRANSFORM_FUNCTION view, *97*
ALL_SUBPART_COL_STATISTICS view, *97*
ALL_SUBPART_HISTOGRAMS view, *98*
ALL_SUBPART_KEY_COLUMNS view, *99*
ALL_SUBPARTITION_TEMPLATES view, *100*
ALL_SUMDELTA view, *101*
ALL_SYNC_CAPTURE view, *101*
ALL_SYNC_CAPTURE_PREPARED_TABS view, *102*
ALL_SYNC_CAPTURE_TABLES view, *102*
ALL_SYNONYMS view, *103*
ALL_TAB_COL_STATISTICS view, *104*
ALL_TAB_COLS view, *106*
ALL_TAB_COLUMNS view, *109*
ALL_TAB_COMMENTS view, *111*
ALL_TAB_HISTGRM_PENDING_STATS view, *112*

Copyright © 2002, 2026, Oracle and/or its affiliates.

ALL_TAB_HISTOGRAMS view, *113*
ALL_HISTOGRAMS synonym, *177*
ALL_TAB_IDENTITY_COLS view, *114*
ALL_TAB_MODIFICATIONS view, *115*
ALL_TAB_PARTITIONS view, *115*
ALL_TAB_PENDING_STATS view, *121*
ALL_TAB_PRIVS view, *122*
ALL_TAB_PRIVS_MADE view, *123*
ALL_TAB_PRIVS_RECD view, *123*
ALL_TAB_STAT_PREFS view, *124*
ALL_TAB_STATISTICS view, *125*
ALL_TAB_STATS_HISTORY view, *126*
ALL_TAB_SUBPARTITIONS view, *127*
ALL_TABLES view, *131*
ALL_TRANSFORMATIONS view, *137*
ALL_TRIGGER_COLS view, *138*
ALL_TRIGGER_ORDERING view, *139*
ALL_TRIGGERS view, *139*
ALL_TRIGGERS_AE view, *141*
ALL_TSTZ_TAB_COLS view, *143*
ALL_TSTZ_TABLES view, *144*
ALL_TYPE_ATTRS view, *145*
ALL_TYPE_METHODS view, *146*
ALL_TYPE_VERSIONS view, *146*
ALL_TYPES view, *147*
ALL_UNIFIED_AUDIT_ACTIONS table, *148*
ALL_UNUSED_COL_TABS view, *149*
ALL_UPDATABLE_COLUMNS view, *149*
ALL_USERS view, *150*
ALL_USTATS view, *151*
ALL_VARRAYS view, *152*
ALL_VIEWS view, *153*
ALL_VIEWS_AE view, *154*
ALL_WARNING_SETTINGS view, *156*
ALL_XML_INDEXES view, *157*
ALL_XML_NESTED_TABLES view, *158*
ALL_XML_OUT_OF_LINE_TABLES view, *158*
ALL_XML_SCHEMA_ATTRIBUTES view, *159*
ALL_XML_SCHEMA_COMPLEX_TYPES view, *160*
ALL_XML_SCHEMA_ELEMENTS view, *161*
ALL_XML_SCHEMA_NAMESPACES view, *162*
ALL_XML_SCHEMA_SIMPLE_TYPES view, *163*
ALL_XML_SCHEMA_SUBSTGRP_HEAD view, *164*
ALL_XML_SCHEMA_SUBSTGRP_MBRS view, *164*
ALL_XML_SCHEMAS view, *165*
ALL_XML_TAB_COLS view, *166*
ALL_XML_TABLES view, *167*
ALL_XML_VIEW_COLS view, *168*
ALL_XML_VIEWS view, *169*
ALL_XSTREAM_ADMINISTRATOR view, *169*
ALL_XSTREAM_INBOUND view, *170*
ALL_XSTREAM_INBOUND_PROGRESS view, *171*
ALL_XSTREAM_OUT_SUPPORT_MODE view, *172*
ALL_XSTREAM_OUTBOUND view, *173*
ALL_XSTREAM_OUTBOUND_PROGRESS view, *174*
ALL_XSTREAM_RULES view, *174*
ALL_XSTREAM_TRANSFORMATIONS view, *176*
ALL_XTERNAL_LOC_PARTITIONS view, *177*
ALL_XTERNAL_LOC_SUBPARTITIONS view, *178*
ALL_XTERNAL_PART_TABLES view, *178*
ALL_XTERNAL_TAB_PARTITIONS view, *179*
ALL_XTERNAL_TAB_SUBPARTITIONS view, *180*
ALL_ZONEMAP_MEASURES view, *180*
ALL_ZONEMAPS view, *181*
ALLOCATE EXTENT clause
of ALTER TABLE
instance number, *155*
ALLOW_GLOBAL_DBLINKS initialization parameter, *28*
ALLOW_GROUP_ACCESS_TO_SGA initialization parameter, *29*
ALLOW_ROWID_COLUMN_TYPE initialization parameter, *29*
ALTER DATABASE statement
ADD LOGFILE, *330*
ENABLE THREAD, *330*
THREAD, *330*
ALTER TABLE statement
allocating extents, *155*
analytic views
columns, *21*, *22*
hierarchical attributes, *169*, *170*
hierarchies, *31*, *32*
hierarchy classifications, *29*, *30*
hierarchy join paths, *171*, *172*
hierarchy level ID attributes, *172*, *173*
hierarchy levels, *174*, *175*
level classifications, *34*, *37*
levels, *35*, *36*
measure and level groups, *38*, *39*
measure classifications, *40*, *41*
APPROX_FOR_AGGREGATION initialization parameter, *30*
APPROX_FOR_COUNT_DISTINCT initialization parameter, *31*
APPROX_FOR_PERCENTILE initialization parameter, *31*
AQ_TM_PROCESSES initialization parameter, *32*
ARCHIVE LOG START statement
automatic archiving, *171*
ARCHIVE_LAG_TARGET initialization parameter, *33*
archived redo logs
destination file, *170*

Copyright © 2002, 2026, Oracle and/or its affiliates.

archived redo logs *(continued)*
storage device, *170*, *178*
ARCHIVELOG mode, *170*
archiving destination, *170*, *178*
ASM_DISKGROUPS initialization parameter, *34*
ASM_DISKSTRING initialization parameter, *35*
ASM_IO_PROCESSES initialization parameter, *36*
ASM_POWER_LIMIT initialization parameter, *36*
ASM_PREFERRED_READ_FAILURE_GROUPS initialization parameter, *37*
attribute dimension
join paths, *71*, *72*
attribute dimensions, *85*, *86*
attribute classifications, *66*, *67*
attributes, *68*
keys, *33*, *73*, *74*
level attributes, *75*, *78*
level classifications, *79*, *80*
levels, *76*, *77*
order attributes, *81*, *82*
tables, *83*, *84*
AUDIT_ACTIONS table, *183*
AUDIT_FILE_DEST initialization parameter, *38*
AUDIT_SYS_OPERATIONS initialization parameter, *39*
AUDIT_SYSLOG_LEVEL initialization parameter, *40*
AUDIT_TRAIL initialization parameter, *41*
AUDIT_UNIFIED_CONTEXTS view, *183*
AUDIT_UNIFIED_ENABLED_POLICIES view, *184*
AUDIT_UNIFIED_POLICIES view, *185*
AUDIT_UNIFIED_POLICY_COMMENTS view, *186*
AUDITABLE_SYSTEM_ACTIONS view, *187*
authenticating remote clients, *284*
authenticating users, *244*
AUTOTASK_MAX_ACTIVE_PDBS initialization parameter, *43*
AWR_PDB_* Views, *2*
AWR_PDB_AUTOFLUSH_ENABLED initialization parameter, *43*
AWR_PDB_MAX_PARALLEL_SLAVES initialization parameter, *45*
AWR_ROOT_* Views, *2*
AWR_SNAPSHOT_TIME_OFFSET initialization parameter, *45*

B

BACKGROUND_CORE_DUMP initialization parameter, *46*
BACKGROUND_DUMP_DEST initialization parameter, *47*
BACKUP_TAPE_IO_SLAVES initialization parameter, *47*
BITMAP_MERGE_AREA_SIZE initialization parameter, *48*
BLANK_TRIMMING initialization parameter, *49*
BLOCKCHAIN_TABLE_MAX_NO_DROP initialization parameter, *50*
blocks
redo log, *183*
size, *78*, *83*, *94*
buffer cache management, *D-1*

C

cache
dictionary, *166–168*
statistics, *183*
CAT synonym for USER_CATALOG view, *188*
CATALOG view, *188*
CATALOG.SQL script, *B-2*
creating V$ views, *1*
CATBLOCK.SQL script, *B-2*
CATCLUST.SQL script, *B-2*
CATHS.SQL script, *B-3*
CATIO.SQL script, *B-3*
CATJAVA.SQL script, *B-6*
CATNOCLUST.SQL script, *B-5*
CATNOJAV.SQL script, *B-5*
CATNOPRT.SQL script, *B-5*
CATNOSVM.SQL script, *B-5*
CATNSNMP.SQL script, *B-5*
CATPCAT.SQL script, *B-2*
CATPROC.SQL script, *B-2*
CATQUEUE.SQL script, *B-3*
CATREP.SQL script, *B-3*
CATWRR.SQL script, *B-3*
CATWRRWITB.SQL script, *B-3*
CDB_* views, *3*
CDBs, *38*
CHAINED_ROWS table, *189*
characters
numeric group separators, *219*
checkpoints
checkpoint interval, *183*
statistics, *183*
CIRCUITS initialization parameter, *51*
classifications
analytic view, *18*, *19*
analytic view attribute, *13*, *14*
analytic view attribute dimension, *24*, *25*
analytic view hierarchy, *29*, *30*
analytic view level, *34*, *37*
analytic view measure, *40*, *41*
attribute dimension, *69*, *70*
attribute dimension attribute, *66*, *67*
attribute dimension level, *79*, *80*

Copyright © 2002, 2026, Oracle and/or its affiliates.

classifications *(continued)*
hierarchy, *163*, *164*
hierarchy hierarchical attribute, *167*, *168*
CLIENT_RESULT_CACHE_LAG initialization parameter, *51*
CLIENT_RESULT_CACHE_SIZE initialization parameter, *52*
CLIENT_RESULT_CACHE_STATS$ view, *189*
CLIENT_STATISTICS_LEVEL initialization parameter, *52*
CLONEDB initialization parameter, *54*
CLONEDB_DIR initialization parameter, *54*
CLU synonym for USER_CLUSTERS view, *190*
CLUSTER_DATABASE initialization parameter, *55*
CLUSTER_DATABASE_INSTANCES initialization parameter, *55*
CLUSTER_INTERCONNECTS initialization parameter, *56*
COL view, *190*
COLS synonym for USER_TAB_COLUMNS view, *190*
commit wait/nowait performed, *E-5*
COMMIT_LOGGING initialization parameter, *57*
COMMIT_POINT_STRENGTH initialization parameter, *57*
COMMIT_WAIT initialization parameter, *58*
COMMIT_WRITE initialization parameter, *59*
COMMON_USER_PREFIX initialization parameter, *59*
COMPATIBLE initialization parameter, *61*
CONNECTION_BROKERS initialization parameter, *62*
CONTAINER_DATA initialization parameter, *63*
CONTAINERS_PARALLEL_DEGREE initialization parameter, *64*
contention
block-level, *142*
control files
names, *66*
specifying, *66*
CONTROL_FILE_RECORD_KEEP_TIME initialization parameter, *65*
CONTROL_FILES initialization parameter, *66*
CONTROL_MANAGEMENT_PACK_ACCESS initialization parameter, *66*
controlfile transaction, *D-1*
CORE_DUMP_DEST initialization parameter, *67*
cost-based optimization, *238*
favoring IN-list iterators, *237*
CPU_COUNT initialization parameter, *68*
CPU_MIN_COUNT initialization parameter, *69*
CREATE TABLE statement
FREELIST GROUPS clause, *155*
CREATE_BITMAP_AREA_SIZE initialization parameter, *70*
CREATE_STORED_OUTLINES initialization parameter, *71*
creating
a database
setting block size, *78*, *83*, *94*
currency, *211*, *212*
international currency symbol, *215*
CURSOR_BIND_CAPTURE_DESTINATION initialization parameter, *71*
CURSOR_INVALIDATION initialization parameter, *72*
CURSOR_SHARING initialization parameter, *73*
CURSOR_SPACE_FOR_TIME initialization parameter, *73*
cursors
OPEN_CURSORS initialization parameter, *228*
shared pool, *308*

D

data blocks
reading multiple, *88*
size of, *78*, *83*, *94*
data dictionary, *1*
cache, *166–168*
scripts, *B-2*
SQL scripts, *B-1*
tables, *1*
views
overview, *1*
user views, *1*
DATA_GUARD_MAX_IO_TIME initialization parameter, *75*
DATA_GUARD_MAX_LONGIO_TIME initialization parameter, *75*
DATA_GUARD_SYNC_LATENCY initialization parameter, *76*
DATA_TRANSFER_CACHE_SIZE initialization parameter, *77*
database writer process (DBWR)
checkpoint, *183*
DATABASE_EXPORT_OBJECTS view, *190*
DATABASE_PROPERTIES view, *191*
databases
database limits, *A-1*
default language, *216*
resource limits, *286*, *289*
datafiles
maximum number, *90*
parameter, *90*
dates
language used for, *214*, *221*, *222*
setting SYSDATE, *128*
TO_CHAR function, *213*
TO_DATE function, *213*

Copyright © 2002, 2026, Oracle and/or its affiliates.

DB_BIG_TABLE_CACHE_PERCENT_TARGET initialization parameter, 78
DB_BLOCK_BUFFERS initialization parameter, 79
DB_BLOCK_CHECKING initialization parameter, 80
DB_BLOCK_CHECKSUM initialization parameter, 82
DB_BLOCK_SIZE initialization parameter, 83
DB_CACHE_ADVICE initialization parameter, 83
DB_CACHE_SIZE initialization parameter, 84
DB_CREATE_FILE_DEST initialization parameter, 85
DB_CREATE_ONLINE_LOG_DEST_n initialization parameter, 86
DB_DOMAIN initialization parameter, 86
DB_FILE_MULTIBLOCK_READ_COUNT initialization parameter, 87
DB_FILE_NAME_CONVERT initialization parameter, 88
DB_FILES initialization parameter, 90
DB_FLASH_CACHE_FILE initialization parameter, 90
DB_FLASH_CACHE_SIZE initialization parameter, 91
DB_FLASHBACK_RETENTION_TARGET initialization parameter, 92
DB_INDEX_COMPRESSION_INHERITANCE initialization parameter, 92
DB_KEEP_CACHE_SIZE initialization parameter, 94
DB_LOST_WRITE_PROTECT initialization parameter, 94
DB_NAME initialization parameter, 95
DB_nK_CACHE_SIZE initialization parameter, 77
DB_PERFORMANCE_PROFILE initialization parameter, 96
DB_RECOVERY_FILE_DEST initialization parameter, 97
DB_RECOVERY_FILE_DEST_SIZE initialization parameter, 98
DB_RECYCLE_CACHE_SIZE initialization parameter, 98
DB_SECUREFILE initialization parameter, 99
DB_ULTRA_SAFE initialization parameter, 100
DB_UNIQUE_NAME initialization parameter, 101
DB_UNRECOVERABLE_SCN_TRACKING initialization parameter, 102
DB_WRITER_PROCESSES initialization parameter, 102
DBA_2PC_NEIGHBORS view, 1
DBA_2PC_PENDING view, 1
DBA_ACCHK_EVENTS view, 2
DBA_ACCHK_EVENTS_SUMMARY view, 4
DBA_ACCHK_STATISTICS view, 5
DBA_ACCHK_STATISTICS_SUMMARY view, 7
DBA_ACL_NAME_MAP view, 9
DBA_ADDM_FDG_BREAKDOWN view, 9
DBA_ADDM_FINDINGS view, 10
DBA_ADDM_INSTANCES view, 11
DBA_ADDM_SYSTEM_DIRECTIVES view, 12
DBA_ADDM_TASK_DIRECTIVES view, 13
DBA_ADDM_TASKS view, 13
DBA_ADVISOR_ACTIONS view, 16
DBA_ADVISOR_COMMANDS view, 18
DBA_ADVISOR_DEF_PARAMETERS view, 18
DBA_ADVISOR_DEFINITIONS view, 19
DBA_ADVISOR_DIR_DEFINITIONS view, 20
DBA_ADVISOR_DIR_INSTANCES view, 21
DBA_ADVISOR_DIR_TASK_INST view, 21
DBA_ADVISOR_EXEC_PARAMETERS view, 22
DBA_ADVISOR_EXECUTION_TYPES view, 22
DBA_ADVISOR_EXECUTIONS view, 23
DBA_ADVISOR_FDG_BREAKDOWN view, 24
DBA_ADVISOR_FINDING_NAMES view, 25
DBA_ADVISOR_FINDINGS view, 25
DBA_ADVISOR_JOURNAL view, 26
DBA_ADVISOR_LOG view, 26
DBA_ADVISOR_OBJECT_TYPES view, 27
DBA_ADVISOR_OBJECTS view, 28
DBA_ADVISOR_PARAMETERS view, 29
DBA_ADVISOR_RATIONALE view, 31
DBA_ADVISOR_RECOMMENDATIONS view, 32
DBA_ADVISOR_SQLA_REC_SUM view, 34
DBA_ADVISOR_SQLA_TABLES view, 34
DBA_ADVISOR_SQLA_WK_MAP view, 35
DBA_ADVISOR_SQLA_WK_STMTS view, 35
DBA_ADVISOR_SQLPLANS view, 37
DBA_ADVISOR_SQLSTATS view, 40
DBA_ADVISOR_SQLW_JOURNAL view, 42
DBA_ADVISOR_SQLW_PARAMETERS view, 42
DBA_ADVISOR_SQLW_STMTS view, 43
DBA_ADVISOR_SQLW_SUM view, 45
DBA_ADVISOR_SQLW_TABLES view, 45
DBA_ADVISOR_SQLW_TEMPLATES view, 46
DBA_ADVISOR_TASKS view, 46
DBA_ADVISOR_TEMPLATES view, 48
DBA_ADVISOR_USAGE view, 49
DBA_ALERT_HISTORY view, 49
DBA_ALERT_HISTORY_DETAIL view, 50
DBA_ALL_TABLES view, 51
DBA_ANALYTIC_VIEW_ATTR_CLASS view, 52
DBA_ANALYTIC_VIEW_ATTR_CLS view, 52
DBA_ANALYTIC_VIEW_ATTR_CLS_AE view, 52
DBA_ANALYTIC_VIEW_BAS_MEAS view, 53
DBA_ANALYTIC_VIEW_BAS_MEAS_AE view, 53
DBA_ANALYTIC_VIEW_BASE_MEAS view, 53
DBA_ANALYTIC_VIEW_CALC_MEAS view, 54
DBA_ANALYTIC_VIEW_CLASS view, 54
DBA_ANALYTIC_VIEW_CLASS_AE view, 54
DBA_ANALYTIC_VIEW_CLC_MEAS view, 54

Copyright © 2002, 2026, Oracle and/or its affiliates.

DBA_ANALYTIC_VIEW_CLC_MEAS_AE view, 55
DBA_ANALYTIC_VIEW_COLUMNS view, 55
DBA_ANALYTIC_VIEW_COLUMNS_AE view, 55
DBA_ANALYTIC_VIEW_DIM_CLASS view, 56
DBA_ANALYTIC_VIEW_DIM_CLS view, 56
DBA_ANALYTIC_VIEW_DIM_CLS_AE view, 56
DBA_ANALYTIC_VIEW_DIMENSIONS view, 57
DBA_ANALYTIC_VIEW_DIMS view, 57
DBA_ANALYTIC_VIEW_DIMS_AE view, 57
DBA_ANALYTIC_VIEW_HIER_CLASS view, 58
DBA_ANALYTIC_VIEW_HIER_CLS view, 58
DBA_ANALYTIC_VIEW_HIER_CLS_AE view, 58
DBA_ANALYTIC_VIEW_HIERS view, 59
DBA_ANALYTIC_VIEW_HIERS_AE view, 59
DBA_ANALYTIC_VIEW_KEYS view, 59
DBA_ANALYTIC_VIEW_KEYS_AE view, 59
DBA_ANALYTIC_VIEW_LEVEL_CLASS view, 60
DBA_ANALYTIC_VIEW_LEVELS view, 60
DBA_ANALYTIC_VIEW_LEVELS_AE view, 60
DBA_ANALYTIC_VIEW_LVL_CLS view, 61
DBA_ANALYTIC_VIEW_LVL_CLS_AE view, 61
DBA_ANALYTIC_VIEW_LVLGRPS view, 61
DBA_ANALYTIC_VIEW_LVLGRPS_AE view, 62
DBA_ANALYTIC_VIEW_MEAS_CLASS view, 62
DBA_ANALYTIC_VIEW_MEAS_CLS view, 62
DBA_ANALYTIC_VIEW_MEAS_CLS_AE view, 63
DBA_ANALYTIC_VIEWS view, 63
DBA_ANALYTIC_VIEWS_AE view, 63
DBA_ANNOTATION_VALUES view, 64
DBA_ANNOTATIONS view, 64
DBA_ANNOTATIONS_USAGE view, 64
DBA_APP_ERRORS view, 65
DBA_APP_PATCHES view, 65
DBA_APP_PDB_STATUS view, 65
DBA_APP_STATEMENTS view, 66
DBA_APP_VERSIONS view, 66
DBA_APPLICATION_ROLES view, 67
DBA_APPLICATIONS view, 67
DBA_APPLY view, 68
DBA_APPLY_CHANGE_HANDLERS view, 68
DBA_APPLY_CONFLICT_COLUMNS view, 68
DBA_APPLY_DML_CONF_HANDLERS view, 68
DBA_APPLY_DML_HANDLERS view, 69
DBA_APPLY_ENQUEUE view, 69
DBA_APPLY_ERROR view, 69
DBA_APPLY_ERROR_MESSAGES view, 69
DBA_APPLY_EXECUTE view, 70
DBA_APPLY_HANDLE_COLLISIONS view, 70
DBA_APPLY_INSTANTIATED_GLOBAL view, 70
DBA_APPLY_INSTANTIATED_OBJECTS view, 70
DBA_APPLY_INSTANTIATED_SCHEMAS view, 71
DBA_APPLY_KEY_COLUMNS view, 71
DBA_APPLY_OBJECT_DEPENDENCIES view, 71
DBA_APPLY_PARAMETERS view, 71
DBA_APPLY_PROGRESS view, 72
DBA_APPLY_REPERROR_HANDLERS view, 72
DBA_APPLY_SPILL_TXN view, 72
DBA_APPLY_TABLE_COLUMNS view, 73
DBA_APPLY_VALUE_DEPENDENCIES view, 73
DBA_AQ_AGENT_PRIVS view, 73
DBA_AQ_AGENTS view, 74
DBA_ARGUMENTS view, 74
DBA_ASSEMBLIES view, 74
DBA_ASSOCIATIONS view, 74
DBA_ATTRIBUTE_DIM_ATTR_CLASS view, 75
DBA_ATTRIBUTE_DIM_ATTR_CLS view, 75
DBA_ATTRIBUTE_DIM_ATTR_CLS_AE view, 75
DBA_ATTRIBUTE_DIM_ATTRS view, 76
DBA_ATTRIBUTE_DIM_ATTRS_AE view, 76
DBA_ATTRIBUTE_DIM_CLASS view, 76
DBA_ATTRIBUTE_DIM_CLASS_AE view, 76
DBA_ATTRIBUTE_DIM_JN_PTHS view, 77
DBA_ATTRIBUTE_DIM_JN_PTHS_AE view, 77
DBA_ATTRIBUTE_DIM_JOIN_PATHS view, 77
DBA_ATTRIBUTE_DIM_KEYS view, 78
DBA_ATTRIBUTE_DIM_KEYS_AE view, 78
DBA_ATTRIBUTE_DIM_LEVEL_ATTRS view, 78
DBA_ATTRIBUTE_DIM_LEVELS view, 78
DBA_ATTRIBUTE_DIM_LEVELS_AE view, 79
DBA_ATTRIBUTE_DIM_LVL_ATRS view, 79
DBA_ATTRIBUTE_DIM_LVL_ATRS_AE view, 79
DBA_ATTRIBUTE_DIM_LVL_CLASS view, 80
DBA_ATTRIBUTE_DIM_LVL_CLS view, 80
DBA_ATTRIBUTE_DIM_LVL_CLS_AE view, 80
DBA_ATTRIBUTE_DIM_ORD_ATRS view, 81
DBA_ATTRIBUTE_DIM_ORD_ATRS_AE view, 81
DBA_ATTRIBUTE_DIM_ORDER_ATTRS view, 81
DBA_ATTRIBUTE_DIM_TABLES view, 82
DBA_ATTRIBUTE_DIM_TABLES_AE view, 82
DBA_ATTRIBUTE_DIMENSIONS view, 82
DBA_ATTRIBUTE_DIMENSIONS_AE view, 82
DBA_ATTRIBUTE_TRANSFORMATIONS view, 83
DBA_AUDIT_EXISTS view, 83
DBA_AUDIT_MGMT_CLEAN_EVENTS view, 86
DBA_AUDIT_MGMT_CLEANUP_JOBS view, 87
DBA_AUDIT_MGMT_CONFIG_PARAMS view, 88
DBA_AUDIT_MGMT_LAST_ARCH_TS view, 89
DBA_AUDIT_OBJECT view, 90
DBA_AUDIT_POLICIES view, 92
DBA_AUDIT_POLICY_COLUMNS view, 92
DBA_AUDIT_SESSION view, 92
DBA_AUDIT_STATEMENT view, 94
DBA_AUDIT_TRAIL view, 97
DBA_AUTO_INDEX_CONFIG view, 101

Copyright © 2002, 2026, Oracle and/or its affiliates.

DBA_AUTO_SEGADV_CTL view, *101*
DBA_AUTO_SEGADV_SUMMARY view, *102*
DBA_AUTO_STAT_EXECUTIONS view, *102*
DBA_AUTOTASK_CLIENT view, *103*
DBA_AUTOTASK_CLIENT_HISTORY view, *104*
DBA_AUTOTASK_CLIENT_JOB view, *105*
DBA_AUTOTASK_JOB_HISTORY view, *106*
DBA_AUTOTASK_OPERATION view, *106*
DBA_AUTOTASK_SCHEDULE view, *107*
DBA_AUTOTASK_STATUS view, *107*
DBA_AUTOTASK_TASK view, *107*
DBA_AUTOTASK_WINDOW_CLIENTS view, *110*
DBA_AUTOTASK_WINDOW_HISTORY view, *111*
DBA_AW_PS view, *111*
DBA_AWS view, *111*
DBA_BASE_TABLE_MVIEWS view, *112*
DBA_BLOCKCHAIN_TABLES view, *112*
DBA_BLOCKERS view, *112*
DBA_CAPTURE view, *113*
DBA_CAPTURE_EXTRA_ATTRIBUTES view, *113*
DBA_CAPTURE_PARAMETERS view, *113*
DBA_CAPTURE_PREPARED_DATABASE view, *113*
DBA_CAPTURE_PREPARED_SCHEMAS view, *114*
DBA_CAPTURE_PREPARED_TABLES view, *114*
DBA_CATALOG view, *114*
DBA_CDB_RSRC_PLAN_DIRECTIVES view, *114*
DBA_CDB_RSRC_PLANS view, *115*
DBA_CERTIFICATES view, *116*
DBA_CHANGE_NOTIFICATION_REGS view, *116*
DBA_CHECKED_ROLES view, *117*
DBA_CHECKED_ROLES_PATH view, *117*
DBA_CLU_COLUMNS view, *118*
DBA_CLUSTER_HASH_EXPRESSIONS view, *118*
DBA_CLUSTERING_DIMENSIONS view, *119*
DBA_CLUSTERING_JOINS view, *119*
DBA_CLUSTERING_KEYS view, *119*
DBA_CLUSTERING_TABLES view, *120*
DBA_CLUSTERS view, *120*
DBA_CODE_ROLE_PRIVS view, *120*
DBA_COL_COMMENTS view, *120*
DBA_COL_PENDING_STATS view, *121*
DBA_COL_PRIVS view, *121*
DBA_COLL_TYPES view, *122*
DBA_COMMON_AUDIT_TRAIL view, *122*
DBA_COMPARISON view, *126*
DBA_COMPARISON_COLUMNS view, *127*
DBA_COMPARISON_ROW_DIF view, *127*
DBA_COMPARISON_SCAN view, *128*
DBA_COMPARISON_SCAN_VALUES view, *129*
DBA_CONNECT_ROLE_GRANTEES view, *129*
DBA_CONNECTION_TESTS view, *129*
DBA_CONS_COLUMNS view, *131*
DBA_CONS_OBJ_COLUMNS view, *131*
DBA_CONSTRAINTS view, *131*
DBA_CONTAINER_DATA view, *131*
DBA_CONTEXT view, *132*
DBA_CPOOL_INFO view, *133*
DBA_CPU_USAGE_STATISTICS view, *134*
DBA_CQ_NOTIFICATION_QUERIES view, *135*
DBA_CREDENTIALS view, *135*
DBA_CUBE_ATTR_VISIBILITY view, *135*
DBA_CUBE_ATTRIBUTES view, *136*
DBA_CUBE_BUILD_PROCESSES view, *136*
DBA_CUBE_CALCULATED_MEMBERS view, *136*
DBA_CUBE_DIM_LEVELS view, *136*
DBA_CUBE_DIM_MODELS view, *137*
DBA_CUBE_DIM_VIEW_COLUMNS view, *137*
DBA_CUBE_DIM_VIEWS view, *137*
DBA_CUBE_DIMENSIONALITY view, *137*
DBA_CUBE_DIMENSIONS view, *138*
DBA_CUBE_HIER_LEVELS view, *138*
DBA_CUBE_HIER_VIEW_COLUMNS view, *138*
DBA_CUBE_HIER_VIEWS view, *138*
DBA_CUBE_HIERARCHIES view, *139*
DBA_CUBE_MEASURES view, *139*
DBA_CUBE_NAMED_BUILD_SPECS view, *139*
DBA_CUBE_SUB_PARTITION_LEVELS view, *139*
DBA_CUBE_VIEW_COLUMNS view, *140*
DBA_CUBE_VIEWS view, *140*
DBA_CUBES view, *140*
DBA_DATA_FILES view, *140*
DBA_DATAPUMP_JOBS view, *142*
DBA_DATAPUMP_SESSIONS view, *142*
DBA_DB_LINK_SOURCES view, *143*
DBA_DB_LINKS view, *144*
DBA_DBFS_HS view, *145*
DBA_DBFS_HS_COMMANDS view, *145*
DBA_DBFS_HS_FIXED_PROPERTIES view, *146*
DBA_DBFS_HS_PROPERTIES view, *146*
DBA_DDL_LOCKS view, *147*
DBA_DEPENDENCIES view, *147*
DBA_DIGEST_VERIFIERS view, *148*
DBA_DIM_ATTRIBUTES view, *148*
DBA_DIM_CHILD_OF view, *148*
DBA_DIM_HIERARCHIES view, *148*
DBA_DIM_JOIN_KEY view, *149*
DBA_DIM_LEVEL_KEY view, *149*
DBA_DIM_LEVELS view, *149*
DBA_DIMENSIONS view, *149*
DBA_DIRECTORIES view, *150*
DBA_DISCOVERY_SOURCE view, *150*
DBA_DML_LOCKS view, *150*
DBA_DMT_FREE_SPACE view, *151*
DBA_DMT_USED_EXTENTS view, *151*
DBA_EDITION_COMMENTS view, *152*
DBA_EDITIONED_TYPES view, *152*

Copyright © 2002, 2026, Oracle and/or its affiliates.

DBA_EDITIONING_VIEW_COLS view, *152*
DBA_EDITIONING_VIEW_COLS_AE view, *153*
DBA_EDITIONING_VIEWS view, *153*
DBA_EDITIONING_VIEWS_AE view, *153*
DBA_EDITIONS view, *153*
DBA_ENABLED_AGGREGATIONS view, *154*
DBA_ENABLED_TRACES view, *154*
DBA_ENCRYPTED_COLUMNS view, *155*
DBA_EPG_DAD_AUTHORIZATION view, *155*
DBA_ERROR_TRANSLATIONS view, *155*
DBA_ERRORS view, *156*
DBA_ERRORS_AE view, *156*
DBA_EVALUATION_CONTEXT_TABLES view, *156*
DBA_EVALUATION_CONTEXT_VARS view, *156*
DBA_EVALUATION_CONTEXTS view, *157*
DBA_EXP_FILES view, *157*
DBA_EXP_OBJECTS view, *157*
DBA_EXP_VERSION view, *157*
DBA_EXPRESSION_STATISTICS view, *158*
DBA_EXTENTS view, *158*
DBA_EXTERNAL_LOCATIONS view, *159*
DBA_EXTERNAL_SCN_ACTIVITY view, *159*
DBA_EXTERNAL_TABLES view, *161*
DBA_FEATURE_USAGE_STATISTICS view, *161*
DBA_FGA_AUDIT_TRAIL view, *162*
DBA_FILE_GROUP_EXPORT_INFO view, *164*
DBA_FILE_GROUP_FILES view, *164*
DBA_FILE_GROUP_TABLES view, *164*
DBA_FILE_GROUP_TABLESPACES view, *164*
DBA_FILE_GROUP_VERSIONS view, *165*
DBA_FILE_GROUPS view, *165*
DBA_FLASHBACK_ARCHIVE view, *165*
DBA_FLASHBACK_ARCHIVE_TABLES view, *166*
DBA_FLASHBACK_ARCHIVE_TS view, *166*
DBA_FLASHBACK_TXN_REPORT view, *166*
DBA_FLASHBACK_TXN_STATE view, *167*
DBA_FREE_SPACE view, *168*
DBA_FREE_SPACE_COALESCED view, *168*
DBA_GG_AUTO_CDR_COLUMN_GROUPS view, *169*
DBA_GG_AUTO_CDR_COLUMNS view, *169*
DBA_GG_AUTO_CDR_TABLES view, *169*
DBA_GG_INBOUND_PROGRESS view, *170*
DBA_GG_PROC_OBJECT_EXCLUSION view, *170*
DBA_GG_PROCEDURE_ANNOTATION view, *170*
DBA_GG_SUPPORTED_PACKAGES view, *171*
DBA_GG_SUPPORTED_PROCEDURES view, *171*
DBA_GLOBAL_CONTEXT view, *171*
DBA_GOLDENGATE_INBOUND view, *172*
DBA_GOLDENGATE_NOT_UNIQUE view, *172*
DBA_GOLDENGATE_PRIVILEGES view, *173*
DBA_GOLDENGATE_RULES view, *173*
DBA_GOLDENGATE_SUPPORT_MODE view, *173*
DBA_HANG_MANAGER_PARAMETERS view, *174*
DBA_HEAT_MAP_SEG_HISTOGRAM view, *175*
DBA_HEAT_MAP_SEGMENT view, *175*
DBA_HEATMAP_TOP_OBJECTS view, *175*
DBA_HEATMAP_TOP_TABLESPACES view, *176*
DBA_HIER_CLASS view, *176*
DBA_HIER_CLASS_AE view, *177*
DBA_HIER_COLUMNS view, *177*
DBA_HIER_COLUMNS_AE view, *177*
DBA_HIER_HIER_ATTR_CLASS view, *178*
DBA_HIER_HIER_ATTR_CLASS_AE view, *178*
DBA_HIER_HIER_ATTRIBUTES view, *178*
DBA_HIER_HIER_ATTRIBUTES_AE view, *179*
DBA_HIER_JOIN_PATHS view, *179*
DBA_HIER_JOIN_PATHS_AE view, *179*
DBA_HIER_LEVEL_ID_ATTRS view, *180*
DBA_HIER_LEVEL_ID_ATTRS_AE view, *180*
DBA_HIER_LEVELS view, *180*
DBA_HIER_LEVELS_AE view, *181*
DBA_HIERARCHIES view, *181*
DBA_HIERARCHIES_AE view, *181*
DBA_HIGH_WATER_MARK_STATISTICS view, *181*
DBA_HIST_ACTIVE_SESS_HISTORY view, *182*
DBA_HIST_APPLY_SUMMARY view, *188*
DBA_HIST_ASH_SNAPSHOT view, *189*
DBA_HIST_BASELINE view, *190*
DBA_HIST_BASELINE_DETAILS view, *191*
DBA_HIST_BASELINE_METADATA view, *192*
DBA_HIST_BASELINE_TEMPLATE view, *193*
DBA_HIST_BG_EVENT_SUMMARY view, *194*
DBA_HIST_BUFFER_POOL_STAT view, *195*
DBA_HIST_BUFFERED_QUEUES view, *196*
DBA_HIST_BUFFERED_SUBSCRIBERS view, *197*
DBA_HIST_CAPTURE view, *199*
DBA_HIST_CHANNEL_WAITS view, *200*
DBA_HIST_CLUSTER_INTERCON view, *201*
DBA_HIST_COLORED_SQL view, *201*
DBA_HIST_COMP_IOSTAT view, *202*
DBA_HIST_CON_SYS_TIME_MODEL view, *203*
DBA_HIST_CON_SYSMETRIC_HIST view, *203*
DBA_HIST_CON_SYSMETRIC_SUMM view, *204*
DBA_HIST_CON_SYSSTAT view, *205*
DBA_HIST_CON_SYSTEM_EVENT view, *206*
DBA_HIST_CR_BLOCK_SERVER view, *207*
DBA_HIST_CURRENT_BLOCK_SERVER view, *208*
DBA_HIST_DATABASE_INSTANCE view, *209*
DBA_HIST_DATAFILE view, *210*
DBA_HIST_DB_CACHE_ADVICE view, *211*
DBA_HIST_DISPATCHER view, *212*

Copyright © 2002, 2026, Oracle and/or its affiliates.

DBA_HIST_DLM_MISC view, *213*
DBA_HIST_DYN_REMASTER_STATS view, *214*
DBA_HIST_ENQUEUE_STAT view, *215*
DBA_HIST_EVENT_HISTOGRAM view, *216*
DBA_HIST_EVENT_NAME view, *217*
DBA_HIST_FILEMETRIC_HISTORY view, *217*
DBA_HIST_FILESTATXS view, *218*
DBA_HIST_IC_CLIENT_STATS view, *219*
DBA_HIST_IC_DEVICE_STATS view, *220*
DBA_HIST_IM_SEG_STAT view, *221*
DBA_HIST_IM_SEG_STAT_OBJ view, *222*
DBA_HIST_INST_CACHE_TRANSFER view, *222*
DBA_HIST_INSTANCE_RECOVERY view, *224*
DBA_HIST_INTERCONNECT_PINGS view, *226*
DBA_HIST_IOSTAT_DETAIL view, *227*
DBA_HIST_IOSTAT_FILETYPE view, *228*
DBA_HIST_IOSTAT_FILETYPE_NAME view, *229*
DBA_HIST_IOSTAT_FUNCTION view, *230*
DBA_HIST_IOSTAT_FUNCTION_NAME view, *231*
DBA_HIST_JAVA_POOL_ADVICE view, *231*
DBA_HIST_LATCH view, *1*
DBA_HIST_LATCH_CHILDREN view, *2*
DBA_HIST_LATCH_MISSES_SUMMARY view, *3*
DBA_HIST_LATCH_NAME view, *4*
DBA_HIST_LATCH_PARENT view, *5*
DBA_HIST_LIBRARYCACHE view, *6*
DBA_HIST_LOG view, *7*
DBA_HIST_MEM_DYNAMIC_COMP view, *8*
DBA_HIST_MEMORY_RESIZE_OPS view, *9*
DBA_HIST_MEMORY_TARGET_ADVICE view, *10*
DBA_HIST_METRIC_NAME view, *11*
DBA_HIST_MTTR_TARGET_ADVICE view, *12*
DBA_HIST_MUTEX_SLEEP view, *13*
DBA_HIST_OPTIMIZER_ENV view, *13*
DBA_HIST_OSSTAT view, *14*
DBA_HIST_OSSTAT_NAME view, *15*
DBA_HIST_PARAMETER view, *15*
DBA_HIST_PARAMETER_NAME view, *16*
DBA_HIST_PDB_IN_SNAP view, *17*
DBA_HIST_PDB_INSTANCE view, *17*
DBA_HIST_PERSISTENT_QMN_CACHE view, *18*
DBA_HIST_PERSISTENT_QUEUES view, *19*
DBA_HIST_PERSISTENT_SUBS view, *21*
DBA_HIST_PGA_TARGET_ADVICE view, *22*
DBA_HIST_PGASTAT view, *23*
DBA_HIST_PLAN_OPERATION_NAME view, *24*
DBA_HIST_PLAN_OPTION_NAME view, *25*
DBA_HIST_PROCESS_MEM_SUMMARY view, *25*
DBA_HIST_PROCESS_WAITTIME view, *26*
DBA_HIST_RECOVERY_PROGRESS view, *27*
DBA_HIST_REPLICATION_TBL_STATS view, *28*
DBA_HIST_REPLICATION_TXN_STATS view, *29*
DBA_HIST_REPORTS view, *29*
DBA_HIST_REPORTS_CONTROL view, *31*
DBA_HIST_REPORTS_DETAILS view, *31*
DBA_HIST_REPORTS_TIMEBANDS view, *32*
DBA_HIST_RESOURCE_LIMIT view, *33*
DBA_HIST_ROWCACHE_SUMMARY view, *34*
DBA_HIST_RSRC_CONSUMER_GROUP view, *35*
DBA_HIST_RSRC_METRIC view, *37*
DBA_HIST_RSRC_PDB_METRIC view, *39*
DBA_HIST_RSRC_PLAN view, *41*
DBA_HIST_RULE_SET view, *42*
DBA_HIST_SEG_STAT view, *43*
DBA_HIST_SEG_STAT_OBJ view, *45*
DBA_HIST_SERVICE_NAME view, *46*
DBA_HIST_SERVICE_STAT view, *47*
DBA_HIST_SERVICE_WAIT_CLASS view, *47*
DBA_HIST_SESS_SGA_STATS view, *48*
DBA_HIST_SESS_TIME_STATS view, *49*
DBA_HIST_SESSMETRIC_HISTORY view, *50*
DBA_HIST_SGA view, *50*
DBA_HIST_SGA_TARGET_ADVICE view, *51*
DBA_HIST_SGASTAT view, *52*
DBA_HIST_SHARED_POOL_ADVICE view, *53*
DBA_HIST_SHARED_SERVER_SUMMARY view, *54*
DBA_HIST_SNAP_ERROR view, *55*
DBA_HIST_SNAPSHOT view, *56*
DBA_HIST_SNAPSHOT_DETAILS view, *57*
DBA_HIST_SQL_BIND_METADATA view, *58*
DBA_HIST_SQL_PLAN view, *58*
DBA_HIST_SQL_SUMMARY view, *61*
DBA_HIST_SQL_WORKAREA_HSTGRM view, *61*
DBA_HIST_SQLBIND view, *62*
DBA_HIST_SQLCOMMAND_NAME view, *64*
DBA_HIST_SQLSTAT view, *64*
DBA_HIST_SQLTEXT view, *68*
DBA_HIST_STAT_NAME view, *69*
DBA_HIST_STREAMS_APPLY_SUM view, *70*
DBA_HIST_STREAMS_CAPTURE view, *71*
DBA_HIST_STREAMS_POOL_ADVICE view, *72*
DBA_HIST_SYS_TIME_MODEL view, *73*
DBA_HIST_SYSMETRIC_HISTORY view, *73*
DBA_HIST_SYSMETRIC_SUMMARY view, *74*
DBA_HIST_SYSSTAT view, *75*
DBA_HIST_SYSTEM_EVENT view, *76*
DBA_HIST_TABLESPACE view, *77*
DBA_HIST_TABLESPACE_STAT view, *77*
DBA_HIST_TBSPC_SPACE_USAGE view, *78*
DBA_HIST_TEMPFILE view, *79*
DBA_HIST_TEMPSTATXS view, *79*
DBA_HIST_THREAD view, *81*
DBA_HIST_TOPLEVELCALL_NAME view, *81*
DBA_HIST_UNDOSTAT view, *82*
DBA_HIST_WAITCLASSMET_HISTORY view, *83*

Copyright © 2002, 2026, Oracle and/or its affiliates.

DBA_HIST_WAITSTAT view, *84*
DBA_HIST_WR_CONTROL view, *85*
DBA_HIST_WR_SETTINGS view, *85*
DBA_HISTOGRAMS synonym for DBA_TAB_HISTOGRAMS, *86*
DBA_HIVE_COLUMNS view, *86*
DBA_HIVE_DATABASES view, *86*
DBA_HIVE_PART_KEY_COLUMNS view, *87*
DBA_HIVE_TAB_PARTITIONS view, *87*
DBA_HIVE_TABLES view, *87*
DBA_HOST_ACES view, *87*
DBA_HOST_ACLS view, *88*
DBA_IDENTIFIERS view, *88*
DBA_ILMDATAMOVEMENTPOLICIES view, *89*
DBA_ILMEVALUATIONDETAILS view, *90*
DBA_ILMOBJECTS view, *91*
DBA_ILMPARAMETERS view, *92*
DBA_ILMPOLICIES view, *93*
DBA_ILMRESULTS view, *94*
DBA_ILMTASKS view, *95*
DBA_IM_EXPRESSIONS view, *95*
DBA_IMMUTABLE_TABLES view, *96*
DBA_IND_COLUMNS view, *96*
DBA_IND_EXPRESSIONS view, *97*
DBA_IND_PARTITIONS view, *97*
DBA_IND_PENDING_STATS view, *97*
DBA_IND_STATISTICS view, *97*
DBA_IND_SUBPARTITIONS view, *98*
DBA_INDEX_USAGE view, *98*
DBA_INDEXES view, *99*
  collecting statistics for, *98*
DBA_INDEXTYPE_ARRAYTYPES view, *99*
DBA_INDEXTYPE_COMMENTS view, *99*
DBA_INDEXTYPE_OPERATORS view, *99*
DBA_INDEXTYPES view, *100*
DBA_INMEMORY_AIMTASKDETAILS view, *100*
DBA_INMEMORY_AIMTASKS view, *100*
DBA_INTERNAL_TRIGGERS view, *101*
DBA_INVALID_OBJECTS view, *101*
DBA_JAVA_ARGUMENTS view, *103*
DBA_JAVA_CLASSES view, *103*
DBA_JAVA_COMPILER_OPTIONS view, *103*
DBA_JAVA_DERIVATIONS view, *103*
DBA_JAVA_FIELDS view, *104*
DBA_JAVA_IMPLEMENTS view, *104*
DBA_JAVA_INNERS view, *104*
DBA_JAVA_LAYOUTS view, *104*
DBA_JAVA_METHODS view, *105*
DBA_JAVA_NCOMPS view, *105*
DBA_JAVA_POLICY view, *105*
DBA_JAVA_RESOLVERS view, *106*
DBA_JAVA_THROWS view, *106*
DBA_JOBS view, *106*
DBA_JOBS_RUNNING view, *107*
DBA_JOIN_IND_COLUMNS view, *108*
DBA_JOINGROUPS view, *108*
DBA_JSON_COLUMNS view, *109*
DBA_JSON_DATAGUIDE_FIELDS view, *110*
DBA_JSON_DATAGUIDES view, *110*
DBA_KGLLOCK view, *110*
DBA_LIBRARIES view, *110*
DBA_LMT_FREE_SPACE view, *111*
DBA_LMT_USED_EXTENTS view, *111*
DBA_LOB_PARTITIONS view, *111*
DBA_LOB_SUBPARTITIONS view, *112*
DBA_LOB_TEMPLATES view, *112*
DBA_LOBS view, *112*
DBA_LOCK view, *112*
  DBA_LOCKS synonym, *115*
DBA_LOCK_INTERNAL view, *113*
DBA_LOCKDOWN_PROFILES view, *113*
DBA_LOCKS synonym for DBA_LOCK view, *115*
DBA_LOG_GROUP_COLUMNS view, *115*
DBA_LOG_GROUPS view, *115*
DBA_LOGMNR_DICTIONARY_BUILDLOG view, *115*
DBA_LOGMNR_LOG view, *116*
DBA_LOGMNR_PURGED_LOG view, *117*
DBA_LOGMNR_SESSION view, *117*
DBA_LOGSTDBY_EVENTS view, *118*
DBA_LOGSTDBY_HISTORY view, *119*
DBA_LOGSTDBY_LOG view, *121*
DBA_LOGSTDBY_NOT_UNIQUE view, *122*
DBA_LOGSTDBY_PARAMETERS view, *122*
DBA_LOGSTDBY_PLSQL_MAP view, *124*
DBA_LOGSTDBY_PLSQL_SUPPORT view, *124*
DBA_LOGSTDBY_PROGRESS view, *125*
DBA_LOGSTDBY_SKIP view, *125*
DBA_LOGSTDBY_SKIP_TRANSACTION view, *126*
DBA_LOGSTDBY_UNSUPPORTED view, *126*
DBA_LOGSTDBY_UNSUPPORTED_TABLE view, *127*
  LOGSTDBY_UNSUPPORTED_TABLES synonym, *116*
DBA_MEASURE_FOLDER_CONTENTS view, *127*
DBA_MEASURE_FOLDER_SUBFOLDERS view, *127*
DBA_MEASURE_FOLDERS view, *128*
DBA_METADATA_PROPERTIES view, *128*
DBA_METHOD_PARAMS view, *128*
DBA_METHOD_RESULTS view, *128*
DBA_MINING_MODEL_ATTRIBUTES view, *129*
DBA_MINING_MODEL_PARTITIONS view, *129*
DBA_MINING_MODEL_SETTINGS view, *129*
DBA_MINING_MODEL_TABLES view, *129*
DBA_MINING_MODEL_VIEWS view, *130*
DBA_MINING_MODEL_XFORMS view, *130*
DBA_MINING_MODELS view, *130*
DBA_MVIEW_AGGREGATES view, *130*
DBA_MVIEW_ANALYSIS view, *131*

Copyright © 2002, 2026, Oracle and/or its affiliates.

DBA_MVIEW_COMMENTS view, *131*
DBA_MVIEW_DETAIL_PARTITION view, *131*
DBA_MVIEW_DETAIL_RELATIONS view, *132*
DBA_MVIEW_DETAIL_SUBPARTITION view, *132*
DBA_MVIEW_JOINS view, *132*
DBA_MVIEW_KEYS view, *132*
DBA_MVIEW_LOG_FILTER_COLS view, *133*
DBA_MVIEW_LOGS view, *133*
DBA_MVIEW_REFRESH_TIMES view, *133*
DBA_MVIEWS view, *133*
DBA_MVREF_CHANGE_STATS view, *134*
DBA_MVREF_RUN_STATS view, *135*
DBA_MVREF_STATS view, *136*
DBA_MVREF_STATS_PARAMS view, *137*
DBA_MVREF_STATS_SYS_DEFAULTS view, *137*
DBA_MVREF_STMT_STATS view, *138*
DBA_NESTED_TABLE_COLS view, *139*
DBA_NESTED_TABLES view, *139*
DBA_NETWORK_ACL_PRIVILEGES view, *139*
DBA_NETWORK_ACLS view, *140*
DBA_OBJ_AUDIT_OPTS view, *141*
DBA_OBJ_COLATTRS view, *142*
DBA_OBJECT_SIZE view, *142*
DBA_OBJECT_TABLES view, *143*
DBA_OBJECT_USAGE view, *143*
DBA_OBJECTS view, *144*
DBA_OBJECTS_AE view, *144*
DBA_OPANCILLARY view, *144*
DBA_OPARGUMENTS view, *145*
DBA_OPBINDINGS view, *145*
DBA_OPERATOR_COMMENTS view, *145*
DBA_OPERATORS view, *145*
DBA_OPTSTAT_OPERATION_TASKS view, *145*
DBA_OPTSTAT_OPERATIONS view, *147*
DBA_ORPHAN_KEY_TABLE view, *148*
DBA_OUTLINE_HINTS view, *148*
DBA_OUTLINES view, *149*
DBA_OUTSTANDING_ALERTS view, *150*
DBA_PARALLEL_EXECUTE_CHUNKS view, *151*
DBA_PARALLEL_EXECUTE_TASKS view, *152*
DBA_PART_COL_STATISTICS view, *153*
DBA_PART_HISTOGRAMS view, *153*
DBA_PART_INDEXES view, *153*
DBA_PART_KEY_COLUMNS view, *154*
DBA_PART_LOBS view, *154*
DBA_PART_TABLES view, *154*
DBA_PARTIAL_DROP_TABS view, *154*
DBA_PDB_HISTORY view, *154*
DBA_PDB_SAVED_STATES view, *155*
DBA_PDB_SNAPSHOTFILE view, *156*
DBA_PDB_SNAPSHOTS view, *156*
DBA_PDBS view, *157*
DBA_PENDING_CONV_TABLES view, *159*
DBA_PENDING_TRANSACTIONS view, *159*
DBA_PLSQL_COLL_TYPES view, *160*
DBA_PLSQL_OBJECT_SETTINGS view, *160*
DBA_PLSQL_TYPE_ATTRS view, *160*
DBA_PLSQL_TYPES view, *160*
DBA_POLICIES view, *161*
DBA_POLICY_ATTRIBUTES view, *161*
DBA_POLICY_CONTEXTS view, *161*
DBA_POLICY_GROUPS view, *161*
DBA_PRIV_AUDIT_OPTS view, *162*
DBA_PRIV_CAPTURES view, *162*
DBA_PRIVATE_TEMP_TABLES view, *163*
DBA_PROCEDURES view, *164*
DBA_PROFILES view, *164*
DBA_PROPAGATION view, *164*
DBA_PROXIES view, *165*
DBA_QUEUE_SCHEDULES view, *165*
DBA_QUEUE_SUBSCRIBERS view, *166*
DBA_QUEUE_TABLES view, *166*
DBA_QUEUES view, *166*
DBA_RAT_CAPTURE_SCHEMA_INFO view, *166*
DBA_RCHILD view, *167*
DBA_RECOVERABLE_SCRIPT view, *167*
DBA_RECOVERABLE_SCRIPT_BLOCKS view, *168*
DBA_RECOVERABLE_SCRIPT_ERRORS view, *168*
DBA_RECOVERABLE_SCRIPT_HIST view, *169*
DBA_RECOVERABLE_SCRIPT_PARAMS view, *169*
DBA_RECYCLEBIN view, *169*
DBA_REDEFINITION_ERRORS view, *171*
DBA_REDEFINITION_OBJECTS view, *171*
DBA_REDEFINITION_STATUS view, *172*
DBA_REFRESH view, *173*
DBA_REFRESH_CHILDREN view, *173*
DBA_REFS view, *173*
DBA_REGISTERED_ARCHIVED_LOG view, *173*
DBA_REGISTERED_MVIEWS view, *174*
DBA_REGISTRY view, *174*
DBA_REGISTRY_BACKPORTS view, *175*
DBA_REGISTRY_HIERARCHY view, *176*
DBA_REGISTRY_HISTORY view, *176*
DBA_REGISTRY_LOG view, *177*
DBA_REGISTRY_SCHEMAS view, *177*
DBA_REGISTRY_SQLPATCH view, *178*
DBA_REPAIR_TABLE view, *179*
DBA_REPL_DBNAME_MAPPING view, *180*
DBA_REPLICATION_PROCESS_EVENTS view, *180*
DBA_RESOURCE_INCARNATIONS view, *180*
DBA_RESUMABLE view, *181*
DBA_REWRITE_EQUIVALENCES view, *182*
DBA_RGROUP view, *182*
DBA_ROLE_PRIVS view, *183*
DBA_ROLES view, *183*
DBA_ROLLBACK_SEGS view, *184*

Copyright © 2002, 2026, Oracle and/or its affiliates.

DBA_ROLLING_DATABASES view, *185*
DBA_ROLLING_EVENTS view, *186*
DBA_ROLLING_PARAMETERS view, *186*
DBA_ROLLING_PLAN view, *187*
DBA_ROLLING_STATISTICS view, *188*
DBA_ROLLING_STATUS view, *188*
DBA_ROLLING_UNSUPPORTED view, *189*
DBA_RSRC_CATEGORIES view, *190*
DBA_RSRC_CONSUMER_GROUP_PRIVS view, *190*
DBA_RSRC_CONSUMER_GROUPS view, *191*
DBA_RSRC_GROUP_MAPPINGS view, *191*
DBA_RSRC_IO_CALIBRATE view, *191*
DBA_RSRC_MANAGER_SYSTEM_PRIVS view, *192*
DBA_RSRC_MAPPING_PRIORITY view, *192*
DBA_RSRC_PLAN_DIRECTIVES view, *193*
DBA_RSRC_PLANS view, *196*
DBA_RULE_SET_RULES view, *197*
DBA_RULE_SETS view, *197*
DBA_RULES view, *197*
DBA_SCHEDULER_CHAIN_RULES view, *198*
DBA_SCHEDULER_CHAIN_STEPS view, *198*
DBA_SCHEDULER_CHAINS view, *198*
DBA_SCHEDULER_CREDENTIALS view, *198*
DBA_SCHEDULER_DB_DESTS view, *199*
DBA_SCHEDULER_DESTS view, *199*
DBA_SCHEDULER_EXTERNAL_DESTS view, *199*
DBA_SCHEDULER_FILE_WATCHERS view, *199*
DBA_SCHEDULER_GLOBAL_ATTRIBUTE view, *200*
DBA_SCHEDULER_GROUP_MEMBERS view, *200*
DBA_SCHEDULER_GROUPS view, *200*
DBA_SCHEDULER_INCOMPAT_MEMBER view, *200*
DBA_SCHEDULER_INCOMPATS view, *201*
DBA_SCHEDULER_JOB_ARGS view, *201*
DBA_SCHEDULER_JOB_CLASSES view, *201*
DBA_SCHEDULER_JOB_DESTS view, *201*
DBA_SCHEDULER_JOB_LOG view, *202*
DBA_SCHEDULER_JOB_ROLES view, *202*
DBA_SCHEDULER_JOB_RUN_DETAILS view, *203*
DBA_SCHEDULER_JOBS view, *203*
DBA_SCHEDULER_NOTIFICATIONS view, *204*
DBA_SCHEDULER_PROGRAM_ARGS view, *204*
DBA_SCHEDULER_PROGRAMS view, *204*
DBA_SCHEDULER_REMOTE_DATABASES view, *204*
DBA_SCHEDULER_REMOTE_JOBSTATE view, *205*
DBA_SCHEDULER_RESOURCES view, *205*
DBA_SCHEDULER_RSC_CONSTRAINTS view, *205*
DBA_SCHEDULER_RUNNING_CHAINS view, *205*
DBA_SCHEDULER_RUNNING_JOBS view, *206*
DBA_SCHEDULER_SCHEDULES view, *206*
DBA_SCHEDULER_WINDOW_DETAILS view, *206*
DBA_SCHEDULER_WINDOW_GROUPS view, *206*
DBA_SCHEDULER_WINDOW_LOG view, *207*
DBA_SCHEDULER_WINDOWS view, *207*
DBA_SCHEDULER_WINGROUP_MEMBERS view, *207*
DBA_SEC_RELEVANT_COLS view, *207*
DBA_SECONDARY_OBJECTS view, *208*
DBA_SEGMENTS view, *208*
DBA_SEGMENTS_OLD view, *210*
DBA_SENSITIVE_COLUMN_TYPES view, *211*
DBA_SENSITIVE_DATA view, *212*
DBA_SEQUENCES view, *212*
DBA_SERVER_REGISTRY view, *212*
DBA_SERVICES view, *213*
DBA_SODA_COLLECTIONS view, *213*
DBA_SOURCE view, *214*
DBA_SOURCE_AE view, *215*
DBA_SQL_MANAGEMENT_CONFIG view, *215*
DBA_SQL_PATCHES view, *216*
DBA_SQL_PLAN_BASELINES view, *217*
DBA_SQL_PLAN_DIR_OBJECTS view, *219*
DBA_SQL_PLAN_DIRECTIVES view, *219*
DBA_SQL_PROFILES view, *220*
DBA_SQL_QUARANTINE view, *215*
DBA_SQL_TRANSLATION_PROFILES view, *221*
DBA_SQL_TRANSLATIONS view, *222*
DBA_SQLJ_TYPE_ATTRS view, *222*
DBA_SQLJ_TYPE_METHODS view, *222*
DBA_SQLJ_TYPES view, *222*
DBA_SQLSET view, *223*
DBA_SQLSET_BINDS view, *223*
DBA_SQLSET_PLANS view, *223*
DBA_SQLSET_REFERENCES view, *223*
DBA_SQLSET_STATEMENTS view, *223*
DBA_SQLTUNE_BINDS view, *225*
DBA_SQLTUNE_PLANS view, *226*
DBA_SQLTUNE_RATIONALE_PLAN view, *228*
DBA_SQLTUNE_STATISTICS view, *229*
DBA_SR_GRP_STATUS view, *229*
DBA_SR_GRP_STATUS_ALL view, *231*
DBA_SR_OBJ view, *232*
DBA_SR_OBJ_ALL view, *233*
DBA_SR_OBJ_STATUS view, *234*
DBA_SR_OBJ_STATUS_ALL view, *235*
DBA_SR_PARTN_OPS view, *236*
DBA_SR_STLOG_EXCEPTIONS view, *236*
DBA_SR_STLOG_STATS view, *237*

Copyright © 2002, 2026, Oracle and/or its affiliates.

DBA_SSCR_CAPTURE view, *238*
DBA_SSCR_RESTORE view, *238*
DBA_STAT_EXTENSIONS view, *239*
DBA_STATEMENTS view, *239*
DBA_STMT_AUDIT_OPTS view, *240*
DBA_STORED_SETTINGS view, *240*
DBA_STREAMS_ADD_COLUMN view, *1*
DBA_STREAMS_DELETE_COLUMN view, *1*
DBA_STREAMS_GLOBAL_RULES view, *2*
DBA_STREAMS_KEEP_COLUMNS view, *2*
DBA_STREAMS_MESSAGE_CONSUMERS view, *3*
DBA_STREAMS_NEWLY_SUPPORTED view, *3*
DBA_STREAMS_RENAME_COLUMN view, *3*
DBA_STREAMS_RENAME_SCHEMA view, *3*
DBA_STREAMS_RENAME_TABLE view, *4*
DBA_STREAMS_SCHEMA_RULES view, *4*
DBA_STREAMS_TABLE_RULES view, *5*
DBA_STREAMS_TP_COMPONENT view, *5*
DBA_STREAMS_TP_COMPONENT_LINK view, *5*
DBA_STREAMS_TP_COMPONENT_STAT view, *6*
DBA_STREAMS_TP_DATABASE view, *7*
DBA_STREAMS_TP_PATH_BOTTLENECK view, *7*
DBA_STREAMS_TP_PATH_STAT view, *8*
DBA_STREAMS_TRANSFORM_FUNCTION view, *8*
DBA_SUBPART_COL_STATISTICS view, *8*
DBA_SUBPART_HISTOGRAMS view, *9*
DBA_SUBPART_KEY_COLUMNS view, *9*
DBA_SUBPARTITION_TEMPLATES view, *9*
DBA_SUBSCR_REGISTRATIONS view, *9*
DBA_SUPPLEMENTAL_LOGGING view, *11*
DBA_SYNC_CAPTURE view, *12*
DBA_SYNC_CAPTURE_PREPARED_TABS view, *12*
DBA_SYNC_CAPTURE_TABLES view, *12*
DBA_SYNONYMS view, *12*
DBA_SYS_PRIVS view, *12*
DBA_TAB_COL_STATISTICS view, *13*
DBA_TAB_COLS view, *13*
DBA_TAB_COLUMNS view, *16*
DBA_TAB_COMMENTS view, *18*
DBA_TAB_HISTGRM_PENDING_STATS view, *19*
DBA_TAB_HISTOGRAMS view, *19*
  DBA_HISTOGRAMS synonym, *86*
DBA_TAB_IDENTITY_COLS view, *19*
DBA_TAB_MODIFICATIONS view, *19*
DBA_TAB_PARTITIONS view, *20*
DBA_TAB_PENDING_STATS view, *20*
DBA_TAB_PRIVS view, *20*
DBA_TAB_STAT_PREFS view, *21*
DBA_TAB_STATISTICS view, *21*
DBA_TAB_STATS_HISTORY view, *21*
DBA_TAB_SUBPARTITIONS view, *22*
DBA_TABLES view, *22*
DBA_TABLESPACE_GROUPS view, *22*
DBA_TABLESPACE_THRESHOLDS view, *22*
DBA_TABLESPACE_USAGE_METRICS view, *23*
DBA_TABLESPACES view, *24*
DBA_TEMP_FILES view, *28*
DBA_TEMP_FREE_SPACE view, *29*
DBA_THRESHOLDS view, *29*
DBA_TRANSFORMATIONS view, *30*
DBA_TRIGGER_COLS view, *31*
DBA_TRIGGER_ORDERING view, *31*
DBA_TRIGGERS view, *31*
DBA_TRIGGERS_AE view, *32*
DBA_TS_QUOTAS view, *32*
DBA_TSDP_IMPORT_ERRORS view, *32*
DBA_TSDP_POLICY_CONDITION view, *33*
DBA_TSDP_POLICY_FEATURE view, *33*
DBA_TSDP_POLICY_PARAMETER view, *34*
DBA_TSDP_POLICY_PROTECTION view, *34*
DBA_TSDP_POLICY_TYPE view, *35*
DBA_TSM_DESTINATION view, *35*
DBA_TSM_SOURCE view, *36*
DBA_TSTZ_TAB_COLS view, *37*
DBA_TSTZ_TABLES view, *37*
DBA_TUNE_MVIEW view, *37*
DBA_TYPE_ATTRS view, *38*
DBA_TYPE_METHODS view, *38*
DBA_TYPE_VERSIONS view, *38*
DBA_TYPES view, *38*
DBA_UMF_LINK view, *39*
DBA_UMF_REGISTRATION view, *39*
DBA_UMF_SERVICE view, *40*
DBA_UMF_TOPOLOGY view, *40*
DBA_UNDO_EXTENTS view, *41*
DBA_UNUSED_COL_TABS view, *42*
DBA_UNUSED_GRANTS view, *42*
DBA_UNUSED_OBJPRIVS view, *43*
DBA_UNUSED_OBJPRIVS_PATH view, *44*
DBA_UNUSED_PRIVS view, *44*
DBA_UNUSED_SYSPRIVS view, *45*
DBA_UNUSED_SYSPRIVS_PATH view, *46*
DBA_UNUSED_USERPRIVS view, *47*
DBA_UNUSED_USERPRIVS_PATH view, *48*
DBA_UPDATABLE_COLUMNS view, *48*
DBA_USED_OBJPRIVS view, *49*
DBA_USED_OBJPRIVS_PATH view, *49*
DBA_USED_PRIVS view, *50*
DBA_USED_PUBPRIVS view, *51*
DBA_USED_SYSPRIVS view, *52*
DBA_USED_SYSPRIVS_PATH view, *53*
DBA_USED_USERPRIVS view, *54*
DBA_USED_USERPRIVS_PATH view, *55*
DBA_USERS view, *55*
DBA_USERS_WITH_DEFPWD view, *61*

Copyright © 2002, 2026, Oracle and/or its affiliates.

DBA_USTATS view, *62*
DBA_VARRAYS view, *62*
DBA_VIEWS view, *62*
DBA_VIEWS_AE view, *62*
DBA_WAITERS view, *62*
DBA_WALLET_ACES view, *63*
DBA_WALLET_ACLS view, *63*
DBA_WARNING_SETTINGS view, *64*
DBA_WI_CAPTURE_FILES view, *64*
DBA_WI_JOBS view, *64*
DBA_WI_OBJECTS view, *65*
DBA_WI_PATTERN_ITEMS view, *65*
DBA_WI_PATTERNS view, *65*
DBA_WI_STATEMENTS view, *66*
DBA_WI_TEMPLATE_EXECUTIONS view, *66*
DBA_WI_TEMPLATES view, *67*
DBA_WORKLOAD_ACTIVE_USER_MAP view, *67*
DBA_WORKLOAD_CAPTURE_SQLTEXT view, *68*
DBA_WORKLOAD_CAPTURES view, *68*
DBA_WORKLOAD_CONNECTION_MAP view, *71*
DBA_WORKLOAD_DIV_SUMMARY view, *71*
DBA_WORKLOAD_FILTERS view, *73*
DBA_WORKLOAD_GROUP_ASSIGNMENTS view, *73*
DBA_WORKLOAD_LONG_SQLTEXT view, *74*
DBA_WORKLOAD_REPLAY_CLIENTS view, *74*
DBA_WORKLOAD_REPLAY_DIVERGENCE view, *75*
DBA_WORKLOAD_REPLAY_SCHEDULES view, *76*
DBA_WORKLOAD_REPLAYS view, *77*
DBA_WORKLOAD_SCHEDULE_CAPTURES view, *81*
DBA_WORKLOAD_SCHEDULE_ORDERING view, *82*
DBA_WORKLOAD_SQL_MAP view, *83*
DBA_WORKLOAD_TRACKED_COMMITS view, *83*
DBA_WORKLOAD_USER_MAP view, *84*
DBA_XML_INDEXES view, *85*
DBA_XML_NESTED_TABLES view, *85*
DBA_XML_OUT_OF_LINE_TABLES view, *85*
DBA_XML_SCHEMA_ATTRIBUTES view, *85*
DBA_XML_SCHEMA_COMPLEX_TYPES view, *86*
DBA_XML_SCHEMA_ELEMENTS view, *86*
DBA_XML_SCHEMA_NAMESPACES view, *86*
DBA_XML_SCHEMA_SIMPLE_TYPES view, *86*
DBA_XML_SCHEMA_SUBSTGRP_HEAD view, *87*
DBA_XML_SCHEMA_SUBSTGRP_MBRS view, *87*
DBA_XML_SCHEMAS view, *87*
DBA_XML_TAB_COLS view, *87*
DBA_XML_TABLES view, *88*
DBA_XML_VIEW_COLS view, *88*
DBA_XML_VIEWS view, *88*
DBA_XS_AUDIT_POLICY_OPTIONS view, *88*
DBA_XS_AUDIT_TRAIL view, *89*
DBA_XS_ENABLED_AUDIT_POLICIES view, *90*
DBA_XS_ENB_AUDIT_POLICIES view, *91*
DBA_XSTREAM_ADMINISTRATOR view, *91*
DBA_XSTREAM_INBOUND view, *91*
DBA_XSTREAM_INBOUND_PROGRESS view, *92*
DBA_XSTREAM_OUT_SUPPORT_MODE view, *92*
DBA_XSTREAM_OUTBOUND view, *92*
DBA_XSTREAM_OUTBOUND_PROGRESS view, *92*
DBA_XSTREAM_RULES view, *93*
DBA_XSTREAM_SPLIT_MERGE view, *93*
DBA_XSTREAM_SPLIT_MERGE_HIST view, *95*
DBA_XSTREAM_STMT_HANDLERS view, *96*
DBA_XSTREAM_STMTS view, *96*
DBA_XSTREAM_TRANSFORMATIONS view, *97*
DBA_XTERNAL_LOC_PARTITIONS view, *97*
DBA_XTERNAL_LOC_SUBPARTITIONS view, *97*
DBA_XTERNAL_PART_TABLES view, *98*
DBA_XTERNAL_TAB_PARTITIONS view, *98*
DBA_XTERNAL_TAB_SUBPARTITIONS view, *98*
DBA_ZONEMAP_MEASURES view, *98*
DBA_ZONEMAPS view, *99*
DBFIPS_140 initialization parameter, *103*
DBFS_CONTENT view, *99*
DBFS_CONTENT_PROPERTIES view, *101*
DBMS_ALERT_INFO view, *101*
DBMS_LOCK_ALLOCATED view, *101*
DBMS_METADATA_PARSE_ITEMS view, *102*
DBMS_METADATA_TRANSFORM_PARAMS view, *102*
DBMS_METADATA_TRANSFORMS view, *103*
DBMS_STATS package, *193*, *246*, *106*, *109*, *131*, *98*, *139*, *13*, *16*, *22*, *238*, *270*, *271*, *275*
DBMSIOTC.SQL script, *B-3*
DBMSPOOL.SQL script, *B-3*
DBWR_IO_SLAVES initialization parameter, *104*
DDL_LOCK_TIMEOUT initialization parameter, *105*
DEFAULT_SHARING initialization parameter, *105*
DEFERRED_SEGMENT_CREATION initialization parameter, *106*
DEPTREE view, *103*
destination
  archiving redo log files, *170*
  overriding default, *171*
  USER_DUMP_DEST, *345*

Copyright © 2002, 2026, Oracle and/or its affiliates.

DG_BROKER_CONFIG_FILEn initialization parameter, *106*
DG_BROKER_START initialization parameter, *107*
DIAGNOSTIC_DEST initialization parameter, *107*
DICT synonym for DICTIONARY, *104*
DICT_COLUMNS view, *104*
DICTIONARY view, *104*
  DICT synonym, *104*
DICTIONARY_CREDENTIALS_ENCRYPT view, *104*
disk drives
  archiving destination, *170*
DISK_ASYNCH_IO initialization parameter, *108*
dispatcher processes
  maximum number, *190*
DISPATCHERS initialization parameter, *109*
distinguished name, *278*
DISTRIBUTED_LOCK_TIMEOUT initialization parameter, *111*
DM_USER_MODELS view, *105*
DML_LOCKS initialization parameter, *112*
DNFS_BATCH_SIZE initialization parameter, *113*
DOCUMENT_LINKS view, *106*
DRCP_DEDICATED_OPT initialization parameter, *113*
DST_UPGRADE_INSERT_CONV initialization parameter, *114*
dump files, *190*
dynamic performance tables
  CATALOG.SQL script, *1*
  public synonyms (V$), *2*
  views (V_$), *2*

E

embedded initialization parameter files, *136*
ENABLE_AUTOMATIC_MAINTENANCE_PDB initialization parameter, *115*
ENABLE_DDL_LOGGING initialization parameter, *116*
ENABLE_DNFS_DISPATCHER initialization parameter, *117*
ENABLE_GOLDENGATE_REPLICATION initialization parameter, *118*
ENABLE_IMC_WITH_MIRA initialization parameter, *119*
ENABLE_PLUGGABLE_DATABASE initialization parameter, *119*
ENABLED_PDBS_ON_STANDBY initialization parameter, *120*
ENCRYPT_NEW_TABLESPACES initialization parameter, *121*
enqueues, *D-1*
  AJV snapshot refresh, *D-2*
enqueues *(continued)*
  ALTER SYSTEM SET PARAMETER = VALUE, *D-2*
  backup/restore, *D-1*
  being written redo log, *D-3*
  bind, *D-1*
  cross-instance call invocation, *D-1*
  database mount, *D-1*
  datafile, *D-1*
  direct loader index creation, *D-1*
  disabling, *112*
  distributed recovery process, *D-1*
  distributed transaction, *D-1*
  DML, *D-2*
  exclusive lock when moving audit table, *D-3*
  extend table, *D-2*
  file set, *D-1*
  high-water lock, *D-1*
  instance attribute lock, *D-3*
  instance number, *D-1*
  instance recovery, *D-1*
  instance registration lock, *D-3*
  instance state, *D-1*
  job queue, *D-2*
  library cache invalidation, *D-1*
  library cache lock, *D-2*
  library cache pin, *D-2*
  log start or switch, *D-2*
  media recovery, *D-2*
  mount definition, *D-2*
  object reuse, *D-2*
  parallel slave synchronization, *D-2*
  parallel slaves, *D-2*
  password file, *D-2*
  process startup, *D-2*
  redo log "kick", *D-2*
  redo thread, *D-2*
  row cache, *D-2*
  row wait, *D-2*
  sequence number, *D-2*
  sequence number value, *D-2*
  SMON, *D-2*
  sort segment, *D-2*
  space management transaction, *D-2*
  synchronized replication, *D-2*
  system commit number, *D-2*
  temporary segment, *D-2*
  temporary table, *D-2*
  temporary table object, *D-2*
  thread checkpoint, *D-2*
  transaction, *D-2*
  transaction recovery, *D-2*
  undo segment, serialization, *D-2*
  user name, *D-2*
  user-defined locks, *D-2*
enterprise roles, *278*

Copyright © 2002, 2026, Oracle and/or its affiliates.

ERROR_SIZE view, *107*
EVENT initialization parameter, *122*
EXCEPTIONS table, *107*
extent
  allocating to instance, *155*
EXTERNAL_KEYSTORE_CREDENTIAL_LOCATION initialization parameter, *122*

F

FAL_CLIENT initialization parameter, *123*
FAL_SERVER initialization parameter, *124*
FAST_START_MTTR_TARGET initialization parameter, *124*
FAST_START_PARALLEL_ROLLBACK initialization parameter, *125*
FILE_MAPPING initialization parameter, *126*
FILEIO_NETWORK_ADAPTERS initialization parameter, *126*
filenames
  case significance, *5*
FILESYSTEMIO_OPTIONS initialization parameter, *127*
FIXED_DATE initialization parameter, *127*
FLASHBACK_TRANSACTION_QUERY view, *107*
FORWARD_LISTENER initialization parameter, *128*
free space list
  locating space, *155*
FREELIST GROUPS clause, *155*

G

GCS_SERVER_PROCESSES initialization parameter, *129*
GLOBAL_CONTEXT view, *108*
GLOBAL_NAME view, *109*
GLOBAL_NAMES initialization parameter, *130*
GLOBAL_TXN_PROCESSES initialization parameter, *130*
GV$ views
  for Real Application Clusters, *2*

H

HASH_AREA_SIZE initialization parameter, *131*
HEARTBEAT_BATCH_SIZE initialization parameter, *131*
HEAT_MAP initialization parameter, *132*
Heterogeneous Services
  agents, *133*
HI_SHARED_MEMORY_ADDRESS initialization parameter, *132*
hierarchies
  analytic view, *31*, *32*, *176*
  classifications, *163*, *164*
hierarchies *(continued)*
  columns, *165*, *166*
  hierarchical attribute classifications, *167*, *168*
  hierarchical attributes, *169*, *170*
  join paths, *171*, *172*
  level ID attributes, *172*, *173*
  levels, *174*, *175*
hints
  precedence over initialization parameter settings, *311*
HS_ALL_CAPS view, *109*
HS_ALL_DD view, *109*
HS_ALL_INITS view, *109*
HS_AUTOREGISTER initialization parameter, *133*
HS_BASE_CAPS view, *110*
HS_BASE_DD view, *110*
HS_CLASS_CAPS view, *110*
HS_CLASS_DD view, *111*
HS_CLASS_INIT view, *111*
HS_FDS_CLASS view, *111*
HS_FDS_INST view, *112*
HS_INST_CAPS view, *112*
HS_INST_DD view, *112*
HS_INST_INIT view, *113*

I

I/O
  reading multiple blocks, *88*
  redo log files, *183*
IDENTITY_PROVIDER_CONFIG initialization parameter, *134*
IDENTITY_PROVIDER_TYPE initialization parameter, *135*
IDEPTREE view, *113*
IFILE initialization parameter, *6*, *136*
IGNORE_SESSION_SET_PARAM_ERRORS initialization parameter, *137*
IM column store. See In-Memory Column Store, *140*
IM scan EUs memcompress for capacity high, *E-19*
IN-list iterators
  optimizer use of, *237*
In-Memory Column Store, *140*, *145*, *148*, *149*
IND synonym for USER_INDEXES view, *113*
INDEX_HISTOGRAM view, *114*
INDEX_STATS view, *114*
INIT.ORA file
  See initialization parameter file
initialization parameter file,
  character set to use in, *4*
  common file, *136*
  embedded, *136*
  INITDW.ORA, *4*

Copyright © 2002, 2026, Oracle and/or its affiliates.

initialization parameter file *(continued)*
line continuation character, *5*
overview, *3*
initialization parameters
ACTIVE_INSTANCE_COUNT, *26*
ADG_ACCOUNT_INFO_TRACKING, *27*
ADG_REDIRECT_DML, *27*
ALLOW_GLOBAL_DBLINKS, *28*
ALLOW_GROUP_ACCESS_TO_SGA, *29*
ALLOW_ROWID_COLUMN_TYPE, *29*
altering, *7*
APPROX_FOR_AGGREGATION, *30*
APPROX_FOR_COUNT_DISTINCT, *31*
APPROX_FOR_PERCENTILE, *31*
AQ_TM_PROCESSES, *32*
ARCHIVE_LAG_TARGET, *33*
ASM_DISKGROUPS, *34*
ASM_DISKSTRING, *35*
ASM_IO_PROCESSES, *36*
ASM_POWER_LIMIT, *36*
ASM_PREFERRED_READ_FAILURE_GROUPS, *37*
AUDIT_FILE_DEST, *38*
AUDIT_SYS_OPERATIONS, *39*
AUDIT_SYSLOG_LEVEL, *40*
AUDIT_TRAIL, *41*
AUTOTASK_MAX_ACTIVE_PDBS, *43*
AWR_PDB_AUTOFLUSH_ENABLED, *43*
AWR_PDB_MAX_PARALLEL_SLAVES, *45*
AWR_SNAPSHOT_TIME_OFFSET, *45*
BACKGROUND_CORE_DUMP, *46*
BACKGROUND_DUMP_DEST, *47*
BACKUP_TAPE_IO_SLAVES, *47*
basic, *3*
BITMAP_MERGE_AREA_SIZE, *48*
BLANK_TRIMMING, *49*
BLOCKCHAIN_TABLE_MAX_NO_DROP, *50*
case significance in filenames, *5*
CDB_COMPATIBLE, *223*
CIRCUITS, *51*
CLIENT_RESULT_CACHE_LAG, *51*
CLIENT_RESULT_CACHE_SIZE, *52*
CLIENT_STATISTICS_LEVEL, *52*
CLONEDB, *54*
CLONEDB_DIR, *54*
CLUSTER_DATABASE, *55*
CLUSTER_DATABASE_INSTANCES, *55*
CLUSTER_INTERCONNECTS, *56*
COMMIT_LOGGING, *57*
COMMIT_POINT_STRENGTH, *57*
COMMIT_WAIT, *58*
COMMIT_WRITE, *59*
COMMON_USER_PREFIX, *59*
COMPATIBLE, *61*
CONNECTION_BROKERS, *62*
CONTAINER_DATA, *63*
initialization parameters *(continued)*
CONTAINERS_PARALLEL_DEGREE, *64*
CONTROL_FILE_RECORD_KEEP_TIME, *65*
CONTROL_FILES, *66*
CONTROL_MANAGEMENT_PACK_ACCESS, *66*
CORE_DUMP_DEST, *67*
CPU_COUNT, *68*
CPU_MIN_COUNT, *69*
CREATE_BITMAP_AREA_SIZE, *70*
CREATE_STORED_OUTLINES, *71*
CURSOR_BIND_CAPTURE_DESTINATION, *71*
CURSOR_INVALIDATION, *72*
CURSOR_SHARING, *73*
CURSOR_SPACE_FOR_TIME, *73*
DATA_GUARD_MAX_IO_TIME, *75*
DATA_GUARD_MAX_LONGIO_TIME, *75*
DATA_GUARD_SYNC_LATENCY, *76*
DATA_TRANSFER_CACHE_SIZE, *77*
DB_BIG_TABLE_CACHE_PERCENT_TARGET, *78*
DB_BLOCK_BUFFERS, *79*
DB_BLOCK_CHECKING, *80*
DB_BLOCK_CHECKSUM, *82*
DB_BLOCK_SIZE, *83*
DB_CACHE_ADVICE, *83*
DB_CACHE_SIZE, *84*
DB_CREATE_FILE_DEST, *85*
DB_CREATE_ONLINE_LOG_DEST_n, *86*
DB_DOMAIN, *86*
DB_FILE_MULTIBLOCK_READ_COUNT, *87*
DB_FILE_NAME_CONVERT, *88*
DB_FILES, *90*
DB_FLASH_CACHE_FILE, *90*
DB_FLASH_CACHE_SIZE, *91*
DB_FLASHBACK_RETENTION_TARGET, *92*
DB_INDEX_COMPRESSION_INHERITANCE, *92*
DB_KEEP_CACHE_SIZE, *94*
DB_LOST_WRITE_PROTECT, *94*
DB_NAME, *95*
DB_nK_CACHE_SIZE, *77*
DB_PERFORMANCE_PROFILE, *96*
DB_RECOVERY_FILE_DEST, *97*
DB_RECOVERY_FILE_DEST_SIZE, *98*
DB_RECYCLE_CACHE_SIZE, *98*
DB_SECUREFILE, *99*
DB_ULTRA_SAFE, *100*
DB_UNIQUE_NAME, *101*
DB_UNRECOVERABLE_SCN_TRACKING, *102*
DB_WRITER_PROCESSES, *102*
DBFIPS_140, *103*
DBWR_IO_SLAVES, *104*
DDL_LOCK_TIMEOUT, *105*
DEFAULT_SHARING, *105*
DEFERRED_SEGMENT_CREATION, *106*
derived, *2*

Copyright © 2002, 2026, Oracle and/or its affiliates.

initialization parameters *(continued)*
DG_BROKER_CONFIG_FILEn, *106*
DG_BROKER_START, *107*
DIAGNOSTIC_DEST, *107*
DISK_ASYNCH_IO, *108*
DISPATCHERS, *109*
display current settings, *23*
DISTRIBUTED_LOCK_TIMEOUT, *111*
DML_LOCKS, *112*
DNFS_BATCH_SIZE, *113*
DRCP_DEDICATED_OPT, *113*
DST_UPGRADE_INSERT_CONV, *114*
embedded initialization parameter files, *6*
ENABLE_AUTOMATIC_MAINTENANCE_PDB, *115*
ENABLE_DDL_LOGGING, *116*
ENABLE_DNFS_DISPATCHER, *117*
ENABLE_GOLDENGATE_REPLICATION, *118*
ENABLE_IMC_WITH_MIRA, *119*
ENABLE_PLUGGABLE_DATABASE, *119*
ENABLED_PDBS_ON_STANDBY, *120*
ENCRYPT_NEW_TABLESPACES, *121*
EVENT, *122*
EXTERNAL_KEYSTORE_CREDENTIAL_LOCATION, *122*
FAL_CLIENT, *123*
FAL_SERVER, *124*
FAST_START_MTTR_TARGET, *124*
FAST_START_PARALLEL_ROLLBACK, *125*
FILE_MAPPING, *126*
FILEIO_NETWORK_ADAPTERS, *126*
FILESYSTEMIO_OPTIONS, *127*
FIXED_DATE, *127*
FORWARD_LISTENER, *128*
functional category, *7*
GCS_SERVER_PROCESSES, *129*
GLOBAL_NAMES, *130*
GLOBAL_TXN_PROCESSES, *130*
HASH_AREA_SIZE, *131*
HEARTBEAT_BATCH_SIZE, *131*
HEAT_MAP, *132*
HI_SHARED_MEMORY_ADDRESS, *132*
HS_AUTOREGISTER, *133*
IDENTITY_PROVIDER_CONFIG, *134*
IDENTITY_PROVIDER_TYPE, *135*
IFILE, *6*, *136*
IGNORE_SESSION_SET_PARAM_ERRORS, *137*
INMEMORY_ADG_ENABLED, *138*
INMEMORY_AUTOMATIC_LEVEL, *139*
INMEMORY_CLAUSE_DEFAULT, *140*
INMEMORY_EXPRESSIONS_USAGE, *144*
INMEMORY_FORCE, *145*
INMEMORY_MAX_POPULATE_SERVERS, *146*
INMEMORY_OPTIMIZED_ARITHMETIC, *148*
INMEMORY_QUERY, *148*

initialization parameters *(continued)*
INMEMORY_SIZE, *149*
INMEMORY_TRICKLE_REPOPULATE_SERVERS_PERCENT, *150*
INMEMORY_VIRTUAL_COLUMNS, *151*
INSTANCE_ABORT_DELAY_TIME, *152*
INSTANCE_GROUPS, *153*
INSTANCE_MODE, *153*
INSTANCE_NAME, *154*
INSTANCE_NUMBER, *155*
INSTANCE_TYPE, *155*
JAVA_JIT_ENABLED, *156*
JAVA_MAX_SESSIONSPACE_SIZE, *157*
JAVA_POOL_SIZE, *157*
JAVA_SOFT_SESSIONSPACE_LIMIT, *158*
JOB_QUEUE_PROCESSES, *158*
LARGE_POOL_SIZE, *160*
LDAP_DIRECTORY_ACCESS, *161*
LDAP_DIRECTORY_SYSAUTH, *163*
LICENSE_MAX_SESSIONS, *163*
LICENSE_MAX_USERS, *164*
LICENSE_SESSIONS_WARNING, *165*
LISTENER_NETWORKS, *166*
LOB_SIGNATURE_ENABLE, *167*
LOCAL_LISTENER, *167*
LOCK_NAME_SPACE, *168*
LOCK_SGA, *168*
LOG_ARCHIVE_CONFIG, *169*
LOG_ARCHIVE_DEST, *170*
LOG_ARCHIVE_DEST_n, *171*
LOG_ARCHIVE_DEST_STATE_n, *175*
LOG_ARCHIVE_DUPLEX_DEST, *176*
LOG_ARCHIVE_FORMAT, *177*
LOG_ARCHIVE_MAX_PROCESSES, *179*
LOG_ARCHIVE_MIN_SUCCEED_DEST, *179*
LOG_ARCHIVE_TRACE, *180*
LOG_BUFFER, *183*
LOG_CHECKPOINT_INTERVAL, *183*
LOG_CHECKPOINT_TIMEOUT, *184*
LOG_CHECKPOINTS_TO_ALERT, *185*
LOG_FILE_NAME_CONVERT, *185*
LONG_MODULE_ACTION, *187*
MAIN_WORKLOAD_TYPE, *187*
MAX_AUTH_SERVERS, *188*
MAX_DATAPUMP_JOBS_PER_PDB, *188*
MAX_DATAPUMP_PARALLEL_PER_JOB, *189*
MAX_DISPATCHERS, *190*
MAX_DUMP_FILE_SIZE, *190*
MAX_IDLE_BLOCKER_TIME, *191*
MAX_IDLE_TIME, *192*
MAX_IOPS, *193*
MAX_MBPS, *194*
MAX_PDBS, *195*
MAX_SHARED_SERVERS, *195*
MAX_STRING_SIZE, *196*
MEMOPTIMIZE_POOL_SIZE, *201*

Copyright © 2002, 2026, Oracle and/or its affiliates.

initialization parameters *(continued)*
MEMORY_MAX_TARGET, *202*
MEMORY_TARGET, *203*
MFA_DUO_API_HOST, *204*
MFA_OMA_IAM_DOMAIN_URL, *205*
MFA_SENDER_EMAIL_DISPLAYNAME, *206*
MFA_SENDER_EMAIL_ID, *206*
MFA_SMTP_HOST, *207*
MFA_SMTP_PORT, *208*
MIN_AUTH_SERVERS, *209*
modifiable, *16*
MULTISHARD_QUERY_DATA_CONSISTENCY, *209*
NLS_CALENDAR, *210*
NLS_COMP, *211*
NLS_CURRENCY, *212*
NLS_DATE_FORMAT, *213*
NLS_DATE_LANGUAGE, *214*
NLS_DUAL_CURRENCY, *215*
NLS_ISO_CURRENCY, *215*
NLS_LANGUAGE, *216*
NLS_LENGTH_SEMANTICS, *217*
NLS_NCHAR_CONV_EXCP, *218*
NLS_NUMERIC_CHARACTERS, *219*
NLS_SORT, *219*
NLS_TERRITORY, *221*
NLS_TIMESTAMP_FORMAT, *221*
NLS_TIMESTAMP_TZ_FORMAT, *222*
OBJECT_CACHE_MAX_SIZE_PERCENT, *224*
OBJECT_CACHE_OPTIMAL_SIZE, *225*
OFS_THREADS, *225*
OGG_ONLINE_DDL, *226*
OLAP_PAGE_POOL_SIZE, *226*
ONE_STEP_PLUGIN_FOR_PDB_WITH_TDE, *227*
OPEN_CURSORS, *228*
OPEN_LINKS, *228*
OPEN_LINKS_PER_INSTANCE, *229*
operating system-dependent values, *2*
OPTIMIZER_ADAPTIVE_PLANS, *230*
OPTIMIZER_ADAPTIVE_REPORTING_ONLY, *230*
OPTIMIZER_ADAPTIVE_STATISTICS, *231*
OPTIMIZER_CAPTURE_SQL_PLAN_BASELINES, *232*
OPTIMIZER_CAPTURE_SQL_QUARANTINE, *232*
OPTIMIZER_DYNAMIC_SAMPLING, *233*
OPTIMIZER_FEATURES_ENABLE, *234*
OPTIMIZER_IGNORE_HINTS, *236*
OPTIMIZER_IGNORE_PARALLEL_HINTS, *236*
OPTIMIZER_INDEX_CACHING, *237*
OPTIMIZER_INDEX_COST_ADJ, *237*
OPTIMIZER_INMEMORY_AWARE, *238*
OPTIMIZER_MODE, *238*
OPTIMIZER_REAL_TIME_STATISTICS, *239*
initialization parameters *(continued)*
OPTIMIZER_SECURE_VIEW_MERGING, *240*
OPTIMIZER_SESSION_TYPE, *241*
OPTIMIZER_USE_INVISIBLE_INDEXES, *241*
OPTIMIZER_USE_PENDING_STATISTICS, *242*
OPTIMIZER_USE_SQL_PLAN_BASELINES, *242*
OPTIMIZER_USE_SQL_QUARANTINE, *243*
OS_AUTHENT_PREFIX, *244*
OS_ROLES, *245*
OUTBOUND_DBLINK_PROTOCOLS, *245*
PARALLEL_ADAPTIVE_MULTI_USER, *246*
PARALLEL_DEGREE_LIMIT, *247*
PARALLEL_DEGREE_POLICY, *248*
PARALLEL_EXECUTION_MESSAGE_SIZE, *249*
PARALLEL_FORCE_LOCAL, *250*
PARALLEL_INSTANCE_GROUP, *250*
PARALLEL_MAX_SERVERS, *251*
PARALLEL_MIN_DEGREE, *252*
PARALLEL_MIN_PERCENT, *253*
PARALLEL_MIN_SERVERS, *254*
PARALLEL_MIN_TIME_THRESHOLD, *255*
PARALLEL_SERVERS_TARGET, *255*
PARALLEL_THREADS_PER_CPU, *257*
parameter files, *3*
PDB_FILE_NAME_CONVERT, *258*
PDB_LOCKDOWN, *259*
PDB_OS_CREDENTIAL, *261*
performance tuning, *2*
PERMIT_92_WRAP_FORMAT, *262*
PGA_AGGREGATE_LIMIT, *262*
PGA_AGGREGATE_TARGET, *264*
PLSCOPE_SETTINGS, *266*
PLSQL_CCFLAGS, *267*
PLSQL_CODE_TYPE, *268*
PLSQL_DEBUG, *269*
PLSQL_OPTIMIZE_LEVEL, *270*
PLSQL_V2_COMPATIBILITY, *271*
PLSQL_WARNINGS, *271*
PRE_PAGE_SGA, *273*
PRIVATE_TEMP_TABLE_PREFIX, *274*
PROCESSES, *275*
PROCESSOR_GROUP_NAME, *275*
QUERY_REWRITE_ENABLED, *276*
QUERY_REWRITE_INTEGRITY, *277*
RDBMS_SERVER_DN, *278*
READ_ONLY_OPEN_DELAYED, *279*
RECOVERY_PARALLELISM, *280*
RECYCLEBIN, *280*
REDO_TRANSPORT_USER, *281*
REMOTE_DEPENDENCIES_MODE, *281*
REMOTE_LISTENER, *282*
REMOTE_LOGIN_PASSWORDFILE, *282*
REMOTE_OS_AUTHENT, *284*
REMOTE_OS_ROLES, *284*
REMOTE_RECOVERY_FILE_DEST, *285*

Copyright © 2002, 2026, Oracle and/or its affiliates.

initialization parameters *(continued)*
REPLICATION_DEPENDENCY_TRACKING, *285*
RESOURCE_LIMIT, *286*
RESOURCE_MANAGE_GOLDENGATE, *286*
RESOURCE_MANAGER_CPU_ALLOCATION, *287*
RESOURCE_MANAGER_PLAN, *288*
RESULT_CACHE_MAX_RESULT, *289*
RESULT_CACHE_MAX_SIZE, *290*
RESULT_CACHE_MODE, *290*
RESULT_CACHE_REMOTE_EXPIRATION, *291*
RESUMABLE_TIMEOUT, *292*
ROLLBACK_SEGMENTS, *292*
SEC_CASE_SENSITIVE_LOGON, *293*
SEC_MAX_FAILED_LOGIN_ATTEMPTS, *294*
SEC_PROTOCOL_ERROR_FURTHER_ACTION, *295*
SEC_PROTOCOL_ERROR_TRACE_ACTION, *295*
SEC_RETURN_SERVER_RELEASE_BANNER, *296*
SERIAL_REUSE, *297*
SERVICE_NAMES, *298*
SESSION_CACHED_CURSORS, *299*
SESSION_EXIT_ON_PACKAGE_STATE_ERROR, *300*
SESSION_MAX_OPEN_FILES, *300*
SESSIONS, *301*
SGA_MAX_SIZE, *302*
SGA_MIN_SIZE, *303*
SGA_TARGET, *304*
SHADOW_CORE_DUMP, *306*
SHARED_MEMORY_ADDRESS, *307*
SHARED_POOL_RESERVED_SIZE, *307*
SHARED_POOL_SIZE, *308*
SHARED_SERVER_SESSIONS, *309*
SHARED_SERVERS, *310*
SHRD_DUPL_TABLE_REFRESH_RATE, *311*
SKIP_UNUSABLE_INDEXES, *311*
SMTP_OUT_SERVER, *312*
SORT_AREA_RETAINED_SIZE, *313*
SORT_AREA_SIZE, *314*
SPATIAL_VECTOR_ACCELERATION, *315*
specifying in initialization parameter files, *5*
SPFILE, *315*
SQL_TRACE, *316*
SQL92_SECURITY, *317*
SQLTUNE_CATEGORY, *318*
STANDBY_DB_PRESERVE_STATES, *318*
STANDBY_FILE_MANAGEMENT, *319*
STANDBY_PDB_SOURCE_FILE_DBLINK, *320*
STANDBY_PDB_SOURCE_FILE_DIRECTORY, *321*
STAR_TRANSFORMATION_ENABLED, *321*
STATISTICS_LEVEL, *322*
STREAMS_POOL_SIZE, *323*
initialization parameters *(continued)*
TABLESPACE_ENCRYPTION, *324*
TAPE_ASYNCH_IO, *326*
TDE_CONFIGURATION, *327*
TEMP_UNDO_ENABLED, *329*
THREAD, *330*
THREADED_EXECUTION, *331*
TIMED_OS_STATISTICS, *332*
TIMED_STATISTICS, *333*
TRACE_ENABLED, *333*
TRACEFILE_IDENTIFIER, *334*
TRANSACTIONS, *335*
TRANSACTIONS_PER_ROLLBACK_SEGMENT, *335*
UNDO_MANAGEMENT, *336*
UNDO_RETENTION, *336*
UNDO_TABLESPACE, *338*
UNIFIED_AUDIT_COMMON_SYSTEMLOG, *339*
UNIFIED_AUDIT_SGA_QUEUE_SIZE, *340*
UNIFIED_AUDIT_SYSTEMLOG, *341*
UNIFORM_LOG_TIMESTAMP_FORMAT, *342*
USE_DEDICATED_BROKER, *342*
USE_LARGE_PAGES, *343*
USER_DUMP_DEST, *345*
variable, *2*
WALLET_ROOT, *346*
WORKAREA_SIZE_POLICY, *349*
XML_DB_EVENTS, *350*
INITJVM.SQL script, *B-6*
INMEMORY_ADG_ENABLED initialization parameter, *138*
INMEMORY_AUTOMATIC_LEVEL initialization parameter, *139*
INMEMORY_CLAUSE_DEFAULT initialization parameter, *140*
INMEMORY_EXPRESSIONS_USAGE initialization parameter, *144*
INMEMORY_FORCE initialization parameter, *145*
INMEMORY_MAX_POPULATE_SERVERS initialization parameter, *146*
INMEMORY_OPTIMIZED_ARITHMETIC initialization parameter, *148*
INMEMORY_QUERY initialization parameter, *148*
INMEMORY_SIZE initialization parameter, *149*
INMEMORY_TRICKLE_REPOPULATE_SERVERS_PERCENT initialization parameter, *150*
INMEMORY_VIRTUAL_COLUMNS initialization parameter, *151*
inserts
locating free space, *155*
INST_ID column, *2*
INSTANCE clause
of ALTER TABLE
allocating extents, *155*
INSTANCE_ABORT_DELAY_TIME initialization parameter, *152*

Copyright © 2002, 2026, Oracle and/or its affiliates.

INSTANCE_GROUPS initialization parameter, *153*
INSTANCE_MODE initialization parameter, *153*
INSTANCE_NAME initialization parameter, *154*
INSTANCE_NUMBER initialization parameter, *155*
INSTANCE_TYPE initialization parameter, *155*
instances
  checkpoint, *183*
  startup order, *155*

J

Java initialization parameters
  JAVA_MAX_SESSIONSPACE_SIZE, *157*
  JAVA_POOL_SIZE, *157*
  JAVA_SOFT_SESSIONSPACE_LIMIT, *158*
Java scripts, *B-6*
JAVA_JIT_ENABLED initialization parameter, *156*
JAVA_MAX_SESSIONSPACE_SIZE initialization parameter, *157*
JAVA_POOL_SIZE initialization parameter, *157*
JAVA_SOFT_SESSIONSPACE_LIMIT initialization parameter, *158*
JOB_QUEUE_PROCESSES initialization parameter, *158*

L

language
  database default language, *216*
LARGE_POOL_SIZE initialization parameter, *160*
LDAP_DIRECTORY_ACCESS initialization parameter, *161*
LDAP_DIRECTORY_SYSAUTH initialization parameter, *163*
LICENSE_MAX_SESSIONS initialization parameter, *163*
LICENSE_MAX_USERS initialization parameter, *164*
LICENSE_SESSIONS_WARNING initialization parameter, *165*
licenses
  maximum sessions, *164*
  maximum users, *165*
  warning for concurrent user sessions, *165*
limits on the database, *A-1*
linguistic sorts, *219*
LISTENER_NETWORKS initialization parameter, *166*
LOB_SIGNATURE_ENABLE initialization parameter, *167*
local currency, *211*, *212*
LOCAL_LISTENER initialization parameter, *167*
LOCK_NAME_SPACE initialization parameter, *168*
LOCK_SGA initialization parameter, *168*
locks
  names, *C-4*
LOG_ARCHIVE_CONFIG initialization parameter, *169*
LOG_ARCHIVE_DEST initialization parameter, *170*
LOG_ARCHIVE_DEST_n initialization parameter, *171*
LOG_ARCHIVE_DEST_STATE_n initialization parameter, *175*
LOG_ARCHIVE_DUPLEX_DEST initialization parameter, *176*
LOG_ARCHIVE_FORMAT initialization parameter, *177*
LOG_ARCHIVE_MAX_PROCESSES initialization parameter, *179*
LOG_ARCHIVE_MIN_SUCCEED_DEST initialization parameter, *179*
LOG_ARCHIVE_TRACE initialization parameter, *180*
LOG_BUFFER initialization parameter, *183*
LOG_CHECKPOINT_INTERVAL initialization parameter, *183*
LOG_CHECKPOINT_TIMEOUT initialization parameter, *184*
LOG_CHECKPOINTS_TO_ALERT initialization parameter, *185*
LOG_FILE_NAME_CONVERT initialization parameter, *185*
LOGSTDBY_UNSUPPORTED_TABLES synonym for DBA_LOGSTDBY_UNSUPPORTED_TABLE view, *116*
LONG_MODULE_ACTION initialization parameter, *187*
lowercase significance, *5*

M

MAIN_WORKLOAD_TYPE initialization parameter, *187*
MAP_OBJECT table, *116*
MAX_AUTH_SERVERS initialization parameter, *188*
MAX_DATAPUMP_JOBS_PER_PDB initialization parameter, *188*
MAX_DATAPUMP_PARALLEL_PER_JOB initialization parameter, *189*
MAX_DISPATCHERS initialization parameter, *190*
MAX_DUMP_FILE_SIZE initialization parameter, *190*
MAX_IDLE_BLOCKER_TIME initialization parameter, *191*
MAX_IDLE_TIME initialization parameter, *192*
MAX_IOPS initialization parameter, *193*

Copyright © 2002, 2026, Oracle and/or its affiliates.

MAX_MBPS initialization parameter, *194*
MAX_PDBS initialization parameter, *195*
MAX_SHARED_SERVERS initialization parameter, *195*
MAX_STRING_SIZE initialization parameter, *196*
MEMOPTIMIZE_POOL_SIZE initialization parameter, *201*
memory
  amount used for sorting, *313*
  virtual, *2*
MEMORY_MAX_TARGET initialization parameter, *202*
MEMORY_TARGET initialization parameter, *203*
MFA_DUO_API_HOST initialization parameter, *204*
MFA_OMA_IAM_DOMAIN_URL initialization parameter, *205*
MFA_SENDER_EMAIL_DISPLAYNAME initialization parameter, *206*
MFA_SENDER_EMAIL_ID initialization parameter, *206*
MFA_SMTP_HOST initialization parameter, *207*
MFA_SMTP_PORT initialization parameter, *208*
MIN_AUTH_SERVERS initialization parameter, *209*
monitor
  performance, *183*
multiple-process systems
  number of processes, *275*
MULTISHARD_QUERY_DATA_CONSISTENCY initialization parameter, *209*
multitenant container databases
  See CDBs.

N

NLS_CALENDAR initialization parameter, *210*
NLS_COMP initialization parameter, *211*
NLS_CURRENCY initialization parameter, *212*
  defined by NLS_TERRITORY, *221*
NLS_DATABASE_PARAMETERS view, *116*
NLS_DATE_FORMAT initialization parameter, *213*
  defined by NLS_NUMERIC_CHARACTERS, *219*
  defined by NLS_TERRITORY, *221*
NLS_DATE_LANGUAGE initialization parameter, *214*
NLS_DUAL_CURRENCY initialization parameter, *215*
NLS_INSTANCE_PARAMETERS view, *117*
NLS_ISO_CURRENCY initialization parameter, *215*
  defined by NLS_TERRITORY, *221*
NLS_LANGUAGE initialization parameter, *216*
NLS_LENGTH_SEMANTICS initialization parameter, *217*
NLS_NCHAR_CONV_EXCP initialization parameter, *218*
NLS_NUMERIC_CHARACTERS initialization parameter, *219*
NLS_SESSION_PARAMETERS view, *117*
NLS_SORT initialization parameter, *219*
NLS_TERRITORY initialization parameter, *221*
NLS_TIME_FORMAT, *82*
NLS_TIME_TZ_FORMAT, *82*
NLS_TIMESTAMP_FORMAT initialization parameter, *221*
NLS_TIMESTAMP_TZ_FORMAT initialization parameter, *222*
NONCDB_COMPATIBLE initialization parameter, *223*
numeric group separators, *219*

O

OBJ synonym for USER_OBJECTS view, *117*
OBJECT_CACHE_MAX_SIZE_PERCENT initialization parameter, *224*
OBJECT_CACHE_OPTIMAL_SIZE initialization parameter, *225*
objects
  data dictionary, *1*
OFS_THREADS initialization parameter, *225*
OGG_ONLINE_DDL initialization parameter, *226*
OLAP_PAGE_POOL_SIZE initialization parameter, *226*
ONE_STEP_PLUGIN_FOR_PDB_WITH_TDE initialization parameter, *227*
online redo log
  archiving mode, *170*
  block, *183*
  file size, *183*
  setting checkpoint interval, *183*
OPEN_CURSORS initialization parameter, *228*
OPEN_LINKS initialization parameter, *228*
OPEN_LINKS_PER_INSTANCE initialization parameter, *229*
operating system
  authenticating remote clients, *284*
  authenticating users and passwords, *244*
  dependent parameters, *2*
  roles for remote clients, *284*
optimization
  trading cursor space for time, *74*
OPTIMIZER_ADAPTIVE_PLANS initialization parameter, *230*
OPTIMIZER_ADAPTIVE_REPORTING_ONLY initialization parameter, *230*
OPTIMIZER_ADAPTIVE_STATISTICS initialization parameter, *231*

Copyright © 2002, 2026, Oracle and/or its affiliates.

OPTIMIZER_CAPTURE_SQL_PLAN_BASELINES initialization parameter, *232*
OPTIMIZER_CAPTURE_SQL_QUARANTINE initialization parameter, *232*
OPTIMIZER_DYNAMIC_SAMPLING initialization parameter, *233*
OPTIMIZER_FEATURES_ENABLE initialization parameter, *234*
OPTIMIZER_IGNORE_HINTS initialization parameter, *236*
OPTIMIZER_IGNORE_PARALLEL_HINTS initialization parameter, *236*
OPTIMIZER_INDEX_CACHING initialization parameter, *237*
OPTIMIZER_INDEX_COST_ADJ initialization parameter, *237*
OPTIMIZER_INMEMORY_AWARE initialization parameter, *238*
OPTIMIZER_MODE initialization parameter, *238*
OPTIMIZER_REAL_TIME_STATISTICS initialization parameter, *239*
OPTIMIZER_SECURE_VIEW_MERGING initialization parameter, *240*
OPTIMIZER_SESSION_TYPE initialization parameter, *241*
OPTIMIZER_USE_INVISIBLE_INDEXES initialization parameter, *241*
OPTIMIZER_USE_PENDING_STATISTICS initialization parameter, *242*
OPTIMIZER_USE_SQL_PLAN_BASELINES initialization parameter, *242*
OPTIMIZER_USE_SQL_QUARANTINE initialization parameter, *243*
Oracle Database Real Application Security views, *4*
Oracle Database Vault views, *4*
Oracle Exadata Database Machine views, *3*
Oracle Label Security views, *4*
Oracle Workspace Manager views, *5*
ORPHAN_KEY_TABLE table
  See DBA_ORPHAN_KEY_TABLE view
OS_AUTHENT_PREFIX initialization parameter, *244*
OS_ROLES initialization parameter, *245*
OUTBOUND_DBLINK_PROTOCOLS initialization parameter, *245*

P

parallel query
  maximum number of instances, *249*
  maximum number of servers, *251*
  minimum number of queries, *254*
PARALLEL_ADAPTIVE_MULTI_USER initialization parameter, *246*
PARALLEL_DEGREE_LIMIT initialization parameter, *247*
PARALLEL_DEGREE_POLICY initialization parameter, *248*
PARALLEL_EXECUTION_MESSAGE_SIZE initialization parameter, *249*
PARALLEL_FORCE_LOCAL initialization parameter, *250*
PARALLEL_INSTANCE_GROUP initialization parameter, *250*
PARALLEL_MAX_SERVERS initialization parameter, *251*
PARALLEL_MIN_DEGREE initialization parameter, *252*
PARALLEL_MIN_PERCENT initialization parameter, *253*
PARALLEL_MIN_SERVERS initialization parameter, *254*
PARALLEL_MIN_TIME_THRESHOLD initialization parameter, *255*
PARALLEL_SERVERS_TARGET initialization parameter, *255*
PARALLEL_THREADS_PER_CPU initialization parameter, *257*
passwords
  authenticating, *244*, *284*
  REMOTE_LOGIN_PASSWORDFILE initialization parameter, *282*
PATH_VIEW view, *117*
PDB_ALERTS view, *118*
PDB_FILE_NAME_CONVERT initialization parameter, *258*
PDB_LOCKDOWN initialization parameter, *259*
PDB_OS_CREDENTIAL initialization parameter, *261*
PDB_PLUG_IN_VIOLATIONS view, *118*
PDBs, *38*
performance, *2*
  shared pool, *308*
PERMIT_92_WRAP_FORMAT initialization parameter, *262*
PGA_AGGREGATE_LIMIT initialization parameter, *262*
PGA_AGGREGATE_TARGET initialization parameter, *264*
PLAN_TABLE table, *119*
PLSCOPE_SETTINGS initialization parameter, *266*
PLSQL_CCFLAGS initialization parameter, *267*
PLSQL_CODE_TYPE initialization parameter, *268*
PLSQL_DEBUG initialization parameter, *269*
PLSQL_OPTIMIZE_LEVEL initialization parameter, *270*
PLSQL_V2_COMPATIBILITY initialization parameter, *271*

Copyright © 2002, 2026, Oracle and/or its affiliates.

PLSQL_WARNINGS initialization parameter, *271*
pluggable databases
See PDBs.
PLUGGABLE_SET_CHECK view, *122*
PRE_PAGE_SGA initialization parameter, *273*
PRIVATE_TEMP_TABLE_PREFIX initialization parameter, *274*
privileges
remote login, *282*
RESTRICTED_SESSION privilege, *164*, *165*
table-level select privileges, *317*
procedures
shared pool, *308*
processes
dispatcher process maximum number, *190*
maximum shared server processes, *195*
number of server processes, *157*, *158*, *309*, *310*
recovery, *280*
trace files, *345*
user processes, *275*
PROCESSES initialization parameter, *275*
PROCESSOR_GROUP_NAME initialization parameter, *275*
PRODUCT_COMPONENT_VERSION view, *123*
profiles
resource limits, *286*, *289*
PROXY_USERS view, *123*
PSTUBTBL table, *124*
PUBLIC clause
of ALTER DATABASE
thread of redo, *330*
PUBLIC_DEPENDENCY view, *124*
PUBLICSYN view, *124*

Q

queries
hints, *238*
QUERY_REWRITE_ENABLED initialization parameter, *276*
QUERY_REWRITE_INTEGRITY initialization parameter, *277*
QUEUE_PRIVILEGES view, *125*

R

RC_ARCHIVED_LOG view, *6*
RC_BACKUP_ARCHIVELOG_DETAILS view, *6*
RC_BACKUP_ARCHIVELOG_SUMMARY view, *6*
RC_BACKUP_CONTROLFILE view, *6*
RC_BACKUP_CONTROLFILE_DETAILS view, *6*
RC_BACKUP_CONTROLFILE_SUMMARY view, *6*
RC_BACKUP_COPY_DETAILS view, *6*
RC_BACKUP_COPY_SUMMARY view, *6*
RC_BACKUP_CORRUPTION view, *6*
RC_BACKUP_DATAFILE view, *6*
RC_BACKUP_DATAFILE_DETAILS view, *6*
RC_BACKUP_DATAFILE_SUMMARY view, *6*
RC_BACKUP_FILES view, *6*
RC_BACKUP_PIECE view, *6*
RC_BACKUP_PIECE_DETAILS view, *6*
RC_BACKUP_REDOLOG view, *6*
RC_BACKUP_SET view, *6*
RC_BACKUP_SET_DETAILS view, *6*
RC_BACKUP_SET_SUMMARY view, *6*
RC_BACKUP_SPFILE view, *6*
RC_BACKUP_SPFILE_DETAILS view, *6*
RC_BACKUP_SPFILE_SUMMARY view, *6*
RC_CHECKPOINT view, *6*
RC_CONTROLFILE_COPY view, *6*
RC_COPY_CORRUPTION view, *6*
RC_DATABASE view, *6*
RC_DATABASE_BLOCK_CORRUPTION view, *6*
RC_DATABASE_INCARNATION view, *6*
RC_DATAFILE view, *6*
RC_DATAFILE_COPY view, *6*
RC_LOG_HISTORY view, *6*
RC_OFFLINE_RANGE view, *6*
RC_PROXY_ARCHIVEDLOG view, *6*
RC_PROXY_ARCHIVELOG_DETAILS view, *6*
RC_PROXY_ARCHIVELOG_SUMMARY view, *6*
RC_PROXY_CONTROLFILE view, *6*
RC_PROXY_COPY_DETAILS view, *6*
RC_PROXY_COPY_SUMMARY view, *7*
RC_PROXY_DATAFILE view, *7*
RC_REDO_LOG view, *7*
RC_REDO_THREAD view, *7*
RC_RESTORE_POINT view, *7*
RC_RESYNC view, *7*
RC_RMAN_BACKUP_JOB_DETAILS view, *7*
RC_RMAN_BACKUP_SUBJOB_DETAILS view, *7*
RC_RMAN_BACKUP_TYPE view, *7*
RC_RMAN_CONFIGURATION view, *7*
RC_RMAN_OUTPUT view, *7*
RC_RMAN_STATUS view, *7*
RC_SITE view, *7*
RC_STORED_SCRIPT view, *7*
RC_STORED_SCRIPT_LINE view, *7*
RC_TABLESPACE view, *7*
RC_TEMPFILE view, *7*
RC_UNUSABLE_BACKUPFILE_DETAILS view, *7*
RDBMS_SERVER_DN initialization parameter, *278*
READ_ONLY_OPEN_DELAYED initialization parameter, *279*
Real Application Clusters
dynamic performance views, *2*
initialization parameters
CIRCUITS, *51*
CLUSTER_DATABASE, *55*

Copyright © 2002, 2026, Oracle and/or its affiliates.

Real Application Clusters *(continued)*
initialization parameters *(continued)*
CLUSTER_DATABASE_INSTANCES, *55*
CLUSTER_INTERCONNECTS, *56*
DISPATCHERS, *109*
FAST_START_PARALLEL_ROLLBACK, *125*
PARALLEL_ADAPTIVE_MULTI_USER, *246*
PARALLEL_EXECUTION_MESSAGE_SIZE, *249*
PARALLEL_INSTANCE_GROUP, *250*
PARALLEL_MAX_SERVERS, *251*
PARALLEL_MIN_PERCENT, *253*
PARALLEL_MIN_SERVERS, *254*
PARALLEL_THREADS_PER_CPU, *257*
recovery
number of required processes, *280*
Recovery Catalog views, *6*
RECOVERY_PARALLELISM initialization parameter, *280*
RECYCLEBIN initialization parameter, *280*
RECYCLEBIN synonym for USER_RECYCLEBIN view, *125*
REDACTION_COLUMNS view, *125*
REDACTION_EXPRESSIONS view, *126*
REDACTION_POLICIES view, *126*
REDACTION_VALUES_FOR_TYPE_FULL view, *127*
redo logs
buffer size, *183*
redo thread, *330*
REDO_TRANSPORT_USER initialization parameter, *281*
remote clients
authenticating, *284*
remote logins, *282*
REMOTE_DEPENDENCIES_MODE initialization parameter, *281*
REMOTE_LISTENER initialization parameter, *282*
REMOTE_LOGIN_PASSWORDFILE initialization parameter, *282*
REMOTE_OS_AUTHENT initialization parameter, *284*
REMOTE_OS_ROLES initialization parameter, *284*
REMOTE_RECOVERY_FILE_DEST initialization parameter, *285*
REPAIR_TABLE table
See DBA_REPAIR_TABLE view
REPLICATION_DEPENDENCY_TRACKING initialization parameter, *285*
REPORT_COMPONENTS view, *128*
REPORT_FILES view, *128*
REPORT_FORMATS view, *129*
resource limits for profiles, *286*, *289*
RESOURCE_COST view, *129*
RESOURCE_LIMIT initialization parameter, *286*
RESOURCE_MANAGE_GOLDENGATE initialization parameter, *286*
RESOURCE_MANAGER_CPU_ALLOCATION initialization parameter, *287*
RESOURCE_MANAGER_PLAN initialization parameter, *288*
RESOURCE_MAP table, *129*
RESOURCE_VIEW view, *130*
RESULT_CACHE_MAX_RESULT initialization parameter, *289*
RESULT_CACHE_MAX_SIZE initialization parameter, *290*
RESULT_CACHE_MODE initialization parameter, *290*
RESULT_CACHE_REMOTE_EXPIRATION initialization parameter, *291*
RESUMABLE_TIMEOUT initialization parameter, *292*
RMJVM.SQL script, *B-6*
ROLE_ROLE_PRIVS view, *130*
ROLE_SYS_PRIVS view, *131*
ROLE_TAB_PRIVS view, *131*
roles, *245*
remote clients, *284*
ROLLBACK_SEGMENTS initialization parameter, *292*

S

SCHEDULER_BATCH_ERRORS view, *132*
SCHEMA_EXPORT_OBJECTS view, *132*
scripts
CATALOG.SQL, *B-2*
CATBLOCK.SQL, *B-2*
CATCLUST.SQL, *B-2*
CATHS.SQL, *B-3*
CATIO.SQL, *B-3*
CATJAVA.SQL, *B-6*
CATNOCLUST.SQL, *B-5*
CATNOJAV.SQL, *B-5*
CATNOPRT.SQL, *B-5*
CATNOSVM.SQL, *B-5*
CATNSNMP.SQL, *B-5*
CATPCAT.SQL, *B-2*
CATPROC.SQL, *B-2*
CATQUEUE.SQL, *B-3*
CATREP.SQL, *B-3*
CATWRR.SQL, *B-3*
CATWRRWITB.SQL, *B-3*
data dictionary, *B-2*
DBMSIOTC.SQL, *B-3*
DBMSPOOL.SQL, *B-3*
INITJVM.SQL, *B-6*
Java, *B-6*
RMJVM.SQL, *B-6*
SQL scripts, *B-1*

Copyright © 2002, 2026, Oracle and/or its affiliates.

scripts *(continued)*
upgrading, *B-6*
USERLOCK.SQL, *B-3*
UTLBSTAT.SQL, *B-3*
UTLCHN1.SQL, *B-3*
UTLCONST.SQL, *B-3*
UTLDTREE.SQL, *B-3*
UTLESTAT.SQL, *B-3*
UTLEXPT1.SQL, *B-3*
UTLFIXDIRS.SQL, *B-4*
UTLIP.SQL, *B-4*
UTLIRP.SQL, *B-4*
UTLLOCKT.SQL, *B-4*
UTLPWDMG.SQL, *B-4*
UTLRP.SQL, *B-4*
UTLSAMPL.SQL, *B-4*
UTLSCLN.SQL, *B-4*
UTLTKPRF.SQL, *B-4*
UTLVALID.SQL, *B-4*
UTLXPLAN.SQL, *B-5*
SEC_CASE_SENSITIVE_LOGON initialization parameter, *293*
SEC_MAX_FAILED_LOGIN_ATTEMPTS initialization parameter, *294*
SEC_PROTOCOL_ERROR_FURTHER_ACTION initialization parameter, *295*
SEC_PROTOCOL_ERROR_TRACE_ACTION initialization parameter, *295*
SEC_RETURN_SERVER_RELEASE_BANNER initialization parameter, *296*
SEQ synonym for USER_SEQUENCES view, *133*
SERIAL_REUSE initialization parameter, *297*
server parameter file
overview, *3*
server processes
number, *157*, *158*, *309*, *310*
SERVICE_NAMES initialization parameter, *298*
SESSION_CACHED_CURSORS initialization parameter, *299*
SESSION_CONTEXT view, *133*
SESSION_EXIT_ON_PACKAGE_STATE_ERRO R initialization parameter, *300*
SESSION_MAX_OPEN_FILES initialization parameter, *300*
SESSION_PRIVS view, *133*
SESSION_ROLES view, *133*
SESSIONS initialization parameter, *301*
SGA_MAX_SIZE initialization parameter, *302*
SGA_MIN_SIZE initialization parameter, *303*
SGA_TARGET initialization parameter, *304*
SHADOW_CORE_DUMP initialization parameter, *306*
shared server
SESSIONS initialization parameter, *301*
shared server processes
maximum number, *195*
SHARED_MEMORY_ADDRESS initialization parameter, *307*
SHARED_POOL_RESERVED_SIZE initialization parameter, *307*
SHARED_POOL_SIZE initialization parameter, *308*
SHARED_SERVER_SESSIONS initialization parameter, *309*
SHARED_SERVERS initialization parameter, *310*
SHOW PARAMETERS statement, *23*
SHRD_DUPL_TABLE_REFRESH_RATE initialization parameter, *311*
SKIP_UNUSABLE_INDEXES initialization parameter, *311*
SMTP_OUT_SERVER initialization parameter, *312*
SORT_AREA_RETAINED_SIZE initialization parameter, *313*
SORT_AREA_SIZE initialization parameter, *314*
sorting
maximum memory, *313*
ORDER BY queries, *219*
SOURCE_SIZE view, *133*
SPATIAL_VECTOR_ACCELERATION initialization parameter, *315*
SPFILE initialization parameter, *315*
SQL scripts
creating the data dictionary, *B-1*
optional, *B-2*
required, *B-1*
SQL_TRACE initialization parameter, *316*
SQL92_SECURITY initialization parameter, *317*
SQLTUNE_CATEGORY initialization parameter, *318*
STANDBY_DB_PRESERVE_STATES initialization parameter, *318*
STANDBY_FILE_MANAGEMENT initialization parameter, *319*
STANDBY_PDB_SOURCE_FILE_DBLINK initialization parameter, *320*
STANDBY_PDB_SOURCE_FILE_DIRECTORY initialization parameter, *321*
STAR_TRANSFORMATION_ENABLED initialization parameter, *321*
starting up
instance number, *155*
startup order, *155*
statistics
application wait time, *E-2*
background checkpoints completed, *E-2*
background checkpoints started, *E-2*
background timeouts, *E-2*
branch node splits, *E-2*
buffer is not pinned count, *E-2*
buffer is pinned count, *E-2*
bytes received via SQL*Net from client, *E-2*

Copyright © 2002, 2026, Oracle and/or its affiliates.

statistics *(continued)*
bytes received via SQL*Net from dblink, *E-2*
bytes sent via SQL*Net to client, *E-2*
bytes sent via SQL*Net to dblink, *E-2*
Cached Commit SCN referenced, *E-2*
calls to get snapshot scn
kcmgss, *E-2*
calls to kcmgas, *E-2*
calls to kcmgcs, *E-2*
calls to kcmgrs, *E-2*
cell flash cache read hits, *E-2*
cell flash cache read hits for smart IO, *E-2*
cell flash cache read hits for temp IO, *E-3*
cell IO uncompressed bytes, *E-3*
cell num bytes in passthru due to quarantine, *E-3*
cell num bytes in passthru during predicate offload, *E-3*
cell overwrites in flash cache, *E-3*
cell partial writes in flash cache, *E-3*
cell physical IO bytes eligible for predicate offload, *E-3*
cell physical IO bytes eligible for smart IOs, *E-3*
cell physical IO bytes processed for IM capacity, *E-3*
cell physical IO bytes processed for IM query, *E-3*
cell physical IO bytes processed for no memcompress, *E-3*
cell physical IO bytes saved by columnar cache, *E-3*
cell physical IO bytes saved by storage index, *E-3*
cell physical IO bytes saved during optimized file creation, *E-3*
cell physical IO bytes saved during optimized RMAN file restore, *E-3*
cell physical IO bytes sent directly to DB node to balance CPU, *E-3*
cell physical IO interconnect bytes, *E-4*
cell physical IO interconnect bytes returned by smart scan, *E-4*
cell pmem cache read hits, *E-4*
cell pmem cache writes, *E-4*
cell ram cache read hits, *E-4*
cell RDMA reads, *E-4*
cell RDMA writes, *E-4*
cell writes to flash cache, *E-4*
cell writes to flash cache for temp IO, *E-4*
change write time, *E-4*
cleanouts and rollbacks - consistent read gets, *E-4*
cleanouts only - consistent read gets, *E-4*
cluster key scan block gets, *E-4*
cluster key scans, *E-4*

statistics *(continued)*
cluster wait time, *E-4*
cold recycle reads, *E-4*
commit cleanout failures
block lost, *E-4*
buffer being written, *E-4*
callback failure, *E-4*
cannot pin, *E-4*
hot backup in progress, *E-4*
write disabled, *E-4*
commit cleanouts, *E-5*
commit cleanouts successfully completed, *E-5*
commit nowait performed, *E-5*
commit nowait requested, *E-5*
Commit SCN cached, *E-5*
commit wait performed, *E-5*
commit wait requested, *E-5*
commit wait/nowait requested, *E-5*
concurrency wait time, *E-5*
consistent changes, *E-5*
consistent gets, *E-5*
consistent gets direct, *E-5*
consistent gets from cache, *E-5*
CPU used by this session, *E-5*
CPU used when call started, *E-5*
CR blocks created, *E-6*
cumulative begin requests, *E-6*
cumulative DB time in requests, *E-6*
cumulative DB time protected in requests, *E-6*
cumulative end requests, *E-6*
cumulative time in requests, *E-6*
cumulative user calls in requests, *E-6*
cumulative user calls protected by Application Continuity, *E-6*
current blocks converted for CR, *E-6*
cursor authentications, *E-6*
data blocks consistent reads - undo records applied, *E-6*
data warehousing cooling action, *E-6*
data warehousing evicted objects, *E-6*
data warehousing evicted objects - cooling, *E-6*
data warehousing evicted objects - replace, *E-6*
data warehousing scanned blocks, *E-6*
data warehousing scanned blocks - disk, *E-7*
data warehousing scanned blocks - memory, *E-7*
data warehousing scanned blocks - offload, *E-7*
data warehousing scanned objects, *E-7*
db block changes, *E-7*
db block gets, *E-7*
db block gets direct, *E-7*
db block gets from cache, *E-7*
DB time, *E-7*

Copyright © 2002, 2026, Oracle and/or its affiliates.

statistics *(continued)*
DB time of LWTs for this session, *E-7*
DBWR checkpoint buffers written, *E-7*
DBWR checkpoints, *183*, *E-7*
DBWR lru scans, *E-7*
DBWR revisited being-written buffer, *E-8*
DBWR transaction table writes, *E-8*
DBWR undo block writes, *E-8*
DDL statements parallelized, *E-8*
deferred (CURRENT) block cleanout applications, *E-8*
DFO trees parallelized, *E-8*
dirty buffers inspected, *E-8*
DML statements parallelized, *E-8*
DML statements retried, *E-8*
enqueue conversions, *E-8*
enqueue deadlocks, *E-8*
enqueue releases, *E-8*
enqueue requests, *E-8*
enqueue timeouts, *E-8*
enqueue waits, *E-8*
exchange deadlocks, *E-8*
execute count, *E-8*
fdba woken up, *E-9*
file io wait time, *E-9*
flash cache eviction: aged out, *E-9*
flash cache eviction: buffer pinned, *E-9*
flash cache eviction: nvalidated, *E-9*
flash cache insert skip: corrupt, *E-9*
flash cache insert skip: DBWR overloaded, *E-9*
flash cache insert skip: exists, *E-9*
flash cache insert skip: modification, *E-9*
flash cache insert skip: not current, *E-9*
flash cache insert skip: not useful, *E-9*
flash cache inserts, *E-9*
flashback log write bytes, *E-9*
flashback log writes, *E-9*
foreground propagated tracked transactions, *E-9*
free buffer inspected, *E-9*
free buffer requested, *E-9*
gc read wait failure, *E-10*
gc read wait timeouts, *E-10*
gc read waits, *E-10*
global enqueue CPU used by this session, *E-10*
global enqueue get time, *E-10*
global enqueue gets async, *E-10*
global enqueue gets sync, *E-10*
global enqueue releases, *E-10*
HCC analyze table CUs, *E-10*
HCC analyzer calls, *E-10*
HCC block compressions attempted, *E-10*
HCC block compressions completed, *E-10*
HCC DML conventional, *E-10*
statistics *(continued)*
HCC DML CUs, *E-10*
HCC fetch by rowid CUs, *E-10*
HCC load conventional bytes compressed, *E-10*
HCC load conventional bytes uncompressed, *E-11*
HCC load conventional CUs, *E-11*
HCC load conventional CUs archive high, *E-11*
HCC load conventional CUs archive low, *E-11*
HCC load conventional CUs query high, *E-11*
HCC load conventional CUs query low, *E-11*
HCC load conventional rows, *E-11*
HCC load conventional rows not compressed, *E-11*
HCC load direct bytes compressed, *E-11*
HCC load direct bytes uncompressed, *E-11*
HCC load direct CUs, *E-11*
HCC load direct CUs archive high, *E-11*
HCC load direct CUs archive low, *E-11*
HCC load direct CUs query high, *E-11*
HCC load direct CUs query low, *E-11*
HCC load direct rows, *E-11*
HCC load direct rows not compressed, *E-11*
HCC scan cell bytes compressed, *E-12*
HCC scan cell bytes decompressed, *E-12*
HCC scan cell CUs archive high, *E-12*
HCC scan cell CUs archive low, *E-12*
HCC scan cell CUs columns accessed, *E-12*
HCC scan cell CUs decompressed, *E-12*
HCC scan cell CUs decompression time, *E-12*
HCC scan cell CUs optimized read, *E-12*
HCC scan cell CUs pruned, *E-12*
HCC scan cell CUs query high, *E-12*
HCC scan cell CUs query low, *E-12*
HCC scan cell CUs sent compressed, *E-12*
HCC scan cell CUs sent head piece, *E-12*
HCC scan cell CUs sent uncompressed, *E-12*
HCC scan cell rows, *E-12*
HCC scan CUs pcode aggregation pushdown, *E-12*
HCC scan CUs pcode pred evaled, *E-12*
HCC scan CUs pcode pred evaled using rowsets, *E-12*
HCC scan CUs predicates applied, *E-12*
HCC scan CUs predicates optimized, *E-12*
HCC scan CUs predicates received, *E-13*
HCC scan rdbms bytes compressed, *E-13*
HCC scan rdbms bytes decompressed, *E-13*
HCC scan rdbms CUs archive high, *E-13*
HCC scan rdbms CUs archive low, *E-13*
HCC scan rdbms CUs columns accessed, *E-13*
HCC scan rdbms CUs decompressed, *E-13*

Copyright © 2002, 2026, Oracle and/or its affiliates.

statistics *(continued)*
HCC scan rdbms CUs decompression time, *E-13*
HCC scan rdbms CUs normal, *E-13*
HCC scan rdbms CUs pruned, *E-13*
HCC scan rdbms CUs query high, *E-13*
HCC scan rdbms CUs query low, *E-13*
HCC scan rdbms CUs turbo, *E-13*
HCC scan rdbms rows, *E-13*
HCC scan rows pcode aggregated, *E-13*
HCC usage cloud, *E-13*
HCC usage pillar, *E-14*
HCC usage ZFS, *E-14*
hot buffers moved to head of LRU, *E-10*
IM default area resized, *E-14*
IM populate accumulated time (ms), *E-14*
IM populate bytes in-memory EU data, *E-14*
IM populate bytes uncompressed EU data, *E-14*
IM populate CUs, *E-14*
IM populate CUs memcompress for capacity high, *E-14*
IM populate CUs memcompress for capacity low, *E-14*
IM populate CUs memcompress for dml, *E-14*
IM populate CUs memcompress for query high, *E-14*
IM populate CUs memcompress for query low, *E-14*
IM populate CUs no memcompress, *E-14*
IM populate CUs requested, *E-14*
IM populate EUs, *E-14*
IM populate EUs accumulated time (ms), *E-14*
IM populate EUs columns, *E-14*
IM populate EUs memcompress for capacity high, *E-14*
IM populate EUs memcompress for capacity low, *E-14*
IM populate EUs memcompress for dml, *E-14*
IM populate EUs memcompress for query high, *E-15*
IM populate EUs memcompress for query low, *E-15*
IM populate EUs no memcompress, *E-15*
IM populate EUs requested, *E-15*
IM populate no contiguous inmemory space, *E-15*
IM populate segments, *E-15*
IM populate segments requested, *E-15*
IM populate segments wall clock time (ms), *E-15*
IM prepopulate accumulated time (ms), *E-15*
IM prepopulate bytes in-memory EU data, *E-15*
IM prepopulate bytes uncompressed EU data, *E-15*

statistics *(continued)*
IM prepopulate CUs, *E-15*
IM prepopulate CUs memcompress for capacity high, *E-15*
IM prepopulate CUs memcompress for capacity low, *E-15*
IM prepopulate CUs memcompress for dml, *E-15*
IM prepopulate CUs memcompress for query high, *E-15*
IM prepopulate CUs memcompress for query low, *E-15*
IM prepopulate CUs no memcompress, *E-15*
IM prepopulate CUs requested, *E-15*
IM prepopulate EUs, *E-15*
IM prepopulate EUs accumulated time (ms), *E-15*
IM prepopulate EUs columns, *E-15*
IM prepopulate EUs memcompress for capacity high, *E-15*
IM prepopulate EUs memcompress for capacity low, *E-16*
IM prepopulate EUs memcompress for dml, *E-16*
IM prepopulate EUs memcompress for query high, *E-16*
IM prepopulate EUs memcompress for query low, *E-16*
IM prepopulate EUs no memcompress, *E-16*
IM prepopulate EUs requested, *E-16*
IM prepopulate segments, *E-16*
IM prepopulate segments requested, *E-16*
IM repopulate (doublebuffering) CUs, *E-16*
IM repopulate (doublebuffering) CUs requested, *E-16*
IM repopulate (incremental) CUs, *E-17*
IM repopulate (incremental) CUs requested, *E-17*
IM repopulate (incremental) EUs, *E-17*
IM repopulate (incremental) EUs requested, *E-17*
IM repopulate (scan) CUs, *E-17*
IM repopulate (scan) CUs requested, *E-17*
IM repopulate (scan) EUs, *E-17*
IM repopulate (scan) EUs requested, *E-17*
IM repopulate (trickle) accumulated time (ms), *E-18*
IM repopulate (trickle) bytes in-memory EU data, *E-18*
IM repopulate (trickle) bytes uncompressed EU data, *E-18*
IM repopulate (trickle) CUs, *E-18*
IM repopulate (trickle) CUs memcompress for capacity high, *E-18*
IM repopulate (trickle) CUs memcompress for capacity low, *E-18*

Copyright © 2002, 2026, Oracle and/or its affiliates.

statistics *(continued)*
IM repopulate (trickle) CUs memcompress for dml, *E-18*
IM repopulate (trickle) CUs memcompress for query high, *E-18*
IM repopulate (trickle) CUs memcompress for query low, *E-18*
IM repopulate (trickle) CUs no memcompress, *E-18*
IM repopulate (trickle) CUs requested, *E-18*
IM repopulate (trickle) CUs resubmitted, *E-18*
IM repopulate (trickle) EUs, *E-18*
IM repopulate (trickle) EUs accumulated time (ms), *E-18*
IM repopulate (trickle) EUs columns, *E-18*
IM repopulate (trickle) EUs memcompress for capacity high, *E-18*
IM repopulate (trickle) EUs memcompress for capacity low, *E-18*
IM repopulate (trickle) EUs memcompress for dml, *E-18*
IM repopulate (trickle) EUs memcompress for query high, *E-18*
IM repopulate (trickle) EUs memcompress for query low, *E-18*
IM repopulate (trickle) EUs no memcompress, *E-18*
IM repopulate (trickle) EUs requested, *E-18*
IM repopulate accumulated time (ms), *E-16*
IM repopulate bytes in-memory EU data, *E-16*
IM repopulate CUs, *E-16*
IM repopulate CUs memcompress for capacity high, *E-16*
IM repopulate CUs memcompress for capacity low, *E-16*
IM repopulate CUs memcompress for dml, *E-16*
IM repopulate CUs memcompress for query high, *E-16*
IM repopulate CUs memcompress for query low, *E-16*
IM repopulate CUs no memcompress, *E-16*
IM repopulate CUs requested, *E-16*
IM repopulate EUs, *E-16*
IM repopulate EUs accumulated time (ms), *E-17*
IM repopulate EUs columns, *E-17*
IM repopulate EUs memcompress for capacity high, *E-17*
IM repopulate EUs memcompress for capacity low, *E-17*
IM repopulate EUs memcompress for dml, *E-17*
IM repopulate EUs memcompress for query high, *E-17*
statistics *(continued)*
IM repopulate EUs memcompress for query low, *E-17*
IM repopulate EUs no memcompress, *E-17*
IM repopulate EUs requested, *E-17*
IM repopulate no contiguous inmemory space, *E-17*
IM repopulate segments, *E-17*
IM repopulate segments requested, *E-17*
IM scan (dynamic) multi-threaded scans, *E-19*
IM scan (dynamic) rows, *E-19*
IM scan (dynamic) tasks processed by parent, *E-19*
IM scan (dynamic) tasks processed by thread, *E-19*
IM scan CUs column not in memory, *E-19*
IM scan CUs invalid or missing revert to on disk extent, *E-19*
IM scan CUs memcompress for capacity high, *E-19*
IM scan CUs memcompress for capacity low, *E-19*
IM scan CUs memcompress for dml, *E-19*
IM scan CUs memcompress for query high, *E-19*
IM scan CUs memcompress for query low, *E-19*
IM scan CUs predicates applied, *E-19*
IM scan CUs predicates optimized, *E-19*
IM scan CUs pruned, *E-19*
IM scan EU bytes in-memory, *E-19*
IM scan EU bytes uncompressed, *E-19*
IM scan EU rows, *E-19*
IM scan EUs columns accessed, *E-19*
IM scan EUs columns decompressed, *E-19*
IM scan EUs columns theoretical max, *E-19*
IM scan EUs memcompress for capacity low, *E-19*
IM scan EUs memcompress for dml, *E-19*
IM scan EUs memcompress for query high, *E-20*
IM scan EUs memcompress for query low, *E-20*
IM scan EUs no memcompress, *E-20*
IM scan EUs split pieces, *E-20*
IM scan rows, *E-20*
IM scan rows optimized, *E-20*
IM scan rows projected, *E-20*
IM scan rows valid, *E-20*
IM scan segments minmax eligible, *E-20*
IM space CU bytes allocated, *E-20*
IM space CU creations initiated, *E-20*
IM space CU extents allocated, *E-20*
IM space segments allocated, *E-20*
IM space segments freed, *E-20*
IM transactions, *E-20*

Copyright © 2002, 2026, Oracle and/or its affiliates.

statistics *(continued)*
IM transactions CUs invalid, *E-20*
IM transactions rows invalidated, *E-20*
IM transactions rows journaled, *E-20*
immediate (CR) block cleanout applications, *E-14*
immediate (CURRENT) block cleanout applications, *E-14*
in call idle wait time, *E-20*
index cmph cu, uncomp sentinels, *E-20*
index cmph dm, cu lock expand, *E-20*
index cmph dm, cu migrate row, *E-20*
index cmph dm, insert unpurge CU row, *E-20*
index cmph dm, purge dummy CU, *E-20*
index cmph dm, split for cu lock expand, *E-20*
index cmph dm, split for cu migrate row, *E-20*
index cmph ld, CU fit, *E-21*
index cmph ld, CU fit, add rows, *E-21*
index cmph ld, CU negative comp, *E-21*
index cmph ld, CU over-est, *E-21*
index cmph ld, CU under-est, *E-21*
index cmph ld, infinite loop, *E-21*
index cmph ld, lf blks flushed, *E-21*
index cmph ld, lf blks w/ und CU, *E-21*
index cmph ld, lf blks w/o CU, *E-21*
index cmph ld, lf blks w/o unc r, *E-21*
index cmph ld, retry in over-est, *E-21*
index cmph ld, rows compressed, *E-21*
index cmph ld, rows uncompressed, *E-21*
index cmph sc, ffs decomp buffers, *E-21*
index cmph sc, ffs decomp buffers released and found valid, *E-21*
index cmph sc, ffs decomp buffers rows avail, *E-21*
index cmph sc, ffs decomp buffers rows used, *E-21*
index cmph sc, ffs decomp failures, *E-21*
index cmph sp, leaf 90_10 failed, *E-21*
index cmph sp, leaf norecomp limit, *E-21*
index cmph sp, leaf norecomp negcomp, *E-21*
index cmph sp, leaf norecomp nospace, *E-21*
index cmph sp, leaf norecomp notry, *E-21*
index cmph sp, leaf norecomp oversize, *E-21*
index cmph sp, leaf norecomp zerocur, *E-21*
index cmph sp, leaf recomp fewer ucs, *E-21*
index cmph sp, leaf recomp zero ucs, *E-22*
index cmph sp, leaf recompress, *E-22*
index cmpl co, prefix mismatch, *E-22*
index cmpl ro, blocks not compressed, *E-22*
index cmpl ro, prefix change at block, *E-22*
index cmpl ro, prefix no change at block, *E-22*
index cmpl ro, reorg avoid load new block, *E-22*
index cmpl ro, reorg avoid split, *E-22*
index fast full scans (direct read), *E-22*
index fast full scans (full), *E-22*
statistics *(continued)*
index fast full scans (rowid ranges), *E-22*
large tracked transactions, *E-22*
leaf node splits, *E-22*
lob reads, *E-22*
lob writes, *E-22*
lob writes unaligned, *E-22*
logons cumulative, *E-22*
logons current, *E-22*
memopt r failed puts, *E-22*
memopt r failed reads on blocks, *E-22*
memopt r failed reads on buckets, *E-22*
memopt r hits, *E-22*
memopt r lookup detected CR buffer, *E-23*
memopt r lookups, *E-23*
memopt r misses, *E-23*
memopt r puts, *E-23*
memopt r successful puts, *E-23*
messages received, *E-23*
messages sent, *E-23*
no buffer to keep pinned count, *E-23*
no work - consistent read gets, *E-23*
non-idle wait count, *E-23*
non-idle wait time, *E-23*
OLAP Aggregate Function Calc, *E-23*
OLAP Aggregate Function Logical NA, *E-23*
OLAP Aggregate Function Precompute, *E-23*
OLAP Custom Member Limit, *E-23*
OLAP Engine Calls, *E-23*
OLAP Fast Limit, *E-23*
OLAP Full Limit, *E-23*
OLAP GID Limit, *E-23*
OLAP Import Rows Loaded, *E-23*
OLAP Import Rows Pushed, *E-24*
OLAP INHIER Limit, *E-24*
OLAP Limit Time, *E-24*
OLAP Paging Manager Cache Changed Page, *E-24*
OLAP Paging Manager Cache Hit, *E-24*
OLAP Paging Manager Cache Miss, *E-24*
OLAP Paging Manager Cache Write, *E-24*
OLAP Paging Manager New Page, *E-24*
OLAP Paging Manager Pool Size, *E-24*
OLAP Perm LOB Read, *E-24*
OLAP Row Id Limit, *E-24*
OLAP Row Load Time, *E-25*
OLAP Row Source Rows Processed, *E-25*
OLAP Session Cache Hit, *E-25*
OLAP Session Cache Miss, *E-25*
OLAP Temp Segment Read, *E-25*
OLAP Temp Segments, *E-25*
OLAP Unique Key Attribute Limit, *E-25*
opened cursors cumulative, *E-25*
opened cursors current, *E-25*
operating system
OS CPU Qt wait time, *E-25*

Copyright © 2002, 2026, Oracle and/or its affiliates.

statistics *(continued)*
operating system *(continued)*
OS Involuntary context switches, *E-25*
OS Signals received, *E-25*
OS Swaps, *E-25*
OS Voluntary context switches, *E-26*
Parallel operations downgraded 1 to 25 pct, *E-26*
Parallel operations downgraded 25 to 50 pct, *E-26*
Parallel operations downgraded 50 to 75 pct, *E-26*
Parallel operations downgraded 75 to 99 pct, *E-26*
Parallel operations downgraded to serial, *E-26*
Parallel operations not downgraded, *E-26*
parse count (describe), *E-26*
parse count (hard), *E-26*
parse count (total), *E-26*
parse time cpu, *E-26*
parse time elapsed, *E-26*
physical read bytes, *E-26*
physical read flash cache hits, *E-26*
physical read IO requests, *E-26*
physical read requests optimized, *E-26*
physical read total bytes, *E-27*
physical read total IO requests, *E-27*
physical read total multi block requests, *E-27*
physical reads, *E-27*
physical reads cache, *E-27*
physical reads cache prefetch, *E-27*
physical reads direct, *E-27*
physical reads direct (lob), *E-27*
physical reads direct temporary tablespace, *E-27*
physical reads for flashback new, *E-27*
physical reads prefetch warmup, *E-27*
physical write bytes, *E-27*
physical write IO requests, *E-27*
physical write total bytes, *E-27*
physical write total IO requests, *E-28*
physical write total multi block requests, *E-28*
physical writes, *E-28*
physical writes direct, *E-28*
physical writes direct (lob), *E-28*
physical writes direct temporary tablespace, *E-28*
physical writes from cache, *E-28*
physical writes non checkpoint, *E-28*
pinned buffers inspected, *E-28*
prefetched blocks aged out before use, *E-28*
process last non-idle time, *E-28*
PX local messages recv'd, *E-28*
PX local messages sent, *E-28*
PX remote messages recv'd, *E-28*
statistics *(continued)*
PX remote messages sent, *E-28*
queries parallelized, *E-28*
recovery array read time, *E-28*
recovery array reads, *E-28*
recovery blocks read, *E-29*
recovery blocks read for lost write detection, *E-29*
recovery blocks skipped lost write checks, *E-29*
recursive calls, *E-29*
recursive cpu usage, *E-29*
redo blocks checksummed by FG (exclusive), *E-29*
redo blocks checksummed by LGWR, *E-29*
redo blocks written, *E-29*
redo buffer allocation retries, *E-29*
redo entries, *E-29*
redo entries for lost write detection, *E-29*
redo log space requests, *E-29*
redo log space wait time, *E-29*
redo ordering marks, *E-29*
redo size, *E-29*
redo size for lost write detection, *E-29*
redo synch time, *E-30*
redo synch writes, *E-30*
redo wastage, *E-30*
redo write time, *E-30*
redo writer latching time, *E-30*
redo writes, *E-30*
rollback changes - undo records applied, *E-30*
rollbacks only - consistent read gets, *E-30*
rows fetched via callback, *E-30*
scheduler wait time, *E-30*
SCN increments due to another database, *E-30*
serializable aborts, *E-30*
session connect time, *E-30*
session cursor cache count, *E-30*
session cursor cache hits, *E-30*
session logical reads, *E-31*
session logical reads - IM, *E-31*
session pga memory, *E-31*
session pga memory max, *E-31*
session stored procedure space, *E-31*
session uga memory, *E-31*
session uga memory max, *E-31*
shared hash latch upgrades - no wait, *E-31*
shared hash latch upgrades - wait, *E-31*
shared io pool buffer get failure, *E-31*
shared io pool buffer get success, *E-31*
slave propagated tracked transactions, *E-31*
sorts (disk), *E-31*
sorts (memory), *E-31*
sorts (rows), *E-31*
SQL*Net roundtrips to/from client, *E-31*

Copyright © 2002, 2026, Oracle and/or its affiliates.

statistics *(continued)*
SQL*Net roundtrips to/from dblink, *E-31*
summed dirty queue length, *E-31*
switch current to new buffer, *E-32*
table fetch by rowid, *E-32*
table fetch continued row, *E-32*
table scan blocks gotten, *E-32*
table scan disk IMC fallback, *E-32*
table scan disk non-IMC rows gotten, *E-32*
table scan rows gotten, *E-32*
table scans (cache partitions), *E-32*
table scans (direct read), *E-32*
table scans (IM), *E-32*
table scans (long tables), *E-32*
table scans (rowid ranges), *E-32*
table scans (short tables), *E-32*
timed, *333*
tracked rows, *E-32*
tracked transactions, *E-33*
transaction lock background get time, *E-33*
transaction lock background gets, *E-33*
transaction lock foreground requests, *E-33*
transaction lock foreground wait time, *E-33*
transaction rollbacks, *E-33*
transaction tables consistent read rollbacks, *E-33*
transaction tables consistent reads - undo records applied, *E-33*
user calls, *E-33*
user commits, *E-33*
user I/O wait time, *E-33*
user rollbacks, *E-33*
very large tracked transactions, *E-33*
write clones created in background, *E-33*
write clones created in foreground, *E-33*
STATISTICS_LEVEL initialization parameter, *322*
STMT_AUDIT_OPTION_MAP table, *134*
STREAMS_POOL_SIZE initialization parameter, *323*
switch redo log file, *183*
SYN synonym for USER_SYNONYMS view, *134*
SYNONYMS view, *134*
SYS_OBJECTS view, *135*
SYSCATALOG view, *135*
SYSFILES view, *135*
SYSSEGOBJ view, *135*
system global area, *2*
buffer areas, *183*
system performance, *2*
SYSTEM_PRIVILEGE_MAP table, *135*

T

TAB view, *136*
TABLE_EXPORT_OBJECTS view, *136*
TABLE_PRIVILEGE_MAP table, *136*
tables
data dictionary, *1*
DML locks, *112*
locating free space, *155*
TABLESPACE_ENCRYPTION initialization parameter, *324*
TABQUOTAS view, *136*
TABS synonym for USER_TABLES view, *137*
tape archiving destination, *170*
TAPE_ASYNCH_IO initialization parameter, *326*
TDE_CONFIGURATION initialization parameter, *327*
TEMP_UNDO_ENABLED initialization parameter, *329*
territory, *215*, *221*
THREAD initialization parameter, *330*
THREADED_EXECUTION initialization parameter, *331*
TIMED_OS_STATISTICS initialization parameter, *332*
TIMED_STATISTICS initialization parameter, *333*
file read/write statistics, *240*, *127*
TO_CHAR function, *213*
TO_DATE function, *213*
trace files
destination, *345*
maximum size, *190*
TRACE_ENABLED initialization parameter, *333*
TRACEFILE_IDENTIFIER initialization parameter, *334*
transactions
data locking, *112*
TRANSACTIONS initialization parameter, *335*
DML locks, *112*
TRANSACTIONS_PER_ROLLBACK_SEGMENT initialization parameter, *335*
TRUSTED_SERVERS view, *137*
TS_PITR_CHECK view, *137*
TS_PITR_OBJECTS TO_BE_DROPPED view, *138*

U

UNDO_MANAGEMENT initialization parameter, *336*
UNDO_RETENTION initialization parameter, *336*
UNDO_TABLESPACE initialization parameter, *338*
UNI_PLUGGABLE_SET_CHECK view, *139*
UNIFIED_AUDIT_COMMON_SYSTEMLOG initialization parameter, *339*
UNIFIED_AUDIT_SGA_QUEUE_SIZE initialization parameter, *340*
UNIFIED_AUDIT_SYSTEMLOG initialization parameter, *341*
UNIFIED_AUDIT_TRAIL view, *139*

Copyright © 2002, 2026, Oracle and/or its affiliates.

UNIFORM_LOG_TIMESTAMP_FORMAT initialization parameter, *342*
updates
  locating free space, *155*
upgrading
  scripts, *B-6*
uppercase characters, *5*
USABLE_EDITIONS view, *151*
USE_DEDICATED_BROKER initialization parameter, *342*
USE_LARGE_PAGES initialization parameter, *343*
user processes
  trace files, *345*
USER_ADDM_FDG_BREAKDOWN view, *151*
USER_ADDM_FINDINGS view, *151*
USER_ADDM_INSTANCES view, *152*
USER_ADDM_TASK_DIRECTIVES view, *152*
USER_ADDM_TASKS view, *152*
USER_ADVISOR_ACTIONS view, *152*
USER_ADVISOR_DIR_TASK_INST view, *153*
USER_ADVISOR_EXEC_PARAMETERS view, *153*
USER_ADVISOR_EXECUTIONS view, *153*
USER_ADVISOR_FDG_BREAKDOWN view, *153*
USER_ADVISOR_FINDINGS view, *154*
USER_ADVISOR_JOURNAL view, *154*
USER_ADVISOR_LOG view, *154*
USER_ADVISOR_OBJECTS view, *154*
USER_ADVISOR_PARAMETERS view, *155*
USER_ADVISOR_RATIONALE view, *155*
USER_ADVISOR_RECOMMENDATIONS view, *155*
USER_ADVISOR_SQLA_REC_SUM view, *155*
USER_ADVISOR_SQLA_TABLES view, *156*
USER_ADVISOR_SQLA_WK_MAP view, *156*
USER_ADVISOR_SQLA_WK_STMTS view, *156*
USER_ADVISOR_SQLPLANS view, *156*
USER_ADVISOR_SQLSTATS view, *157*
USER_ADVISOR_SQLW_JOURNAL view, *157*
USER_ADVISOR_SQLW_PARAMETERS view, *157*
USER_ADVISOR_SQLW_STMTS view, *157*
USER_ADVISOR_SQLW_SUM view, *158*
USER_ADVISOR_SQLW_TABLES view, *158*
USER_ADVISOR_SQLW_TEMPLATES view, *158*
USER_ADVISOR_TASKS view, *158*
USER_ADVISOR_TEMPLATES view, *159*
USER_ALL_TABLES view, *159*
USER_ANALYTIC_VIEW_ATTR_CLASS view, *159*
USER_ANALYTIC_VIEW_ATTR_CLS view, *159*
USER_ANALYTIC_VIEW_ATTR_CLS_AE view, *160*
USER_ANALYTIC_VIEW_BAS_MEAS view, *160*
USER_ANALYTIC_VIEW_BAS_MEAS_AE view, *160*
USER_ANALYTIC_VIEW_BASE_MEAS view, *161*
USER_ANALYTIC_VIEW_CALC_MEAS view, *161*
USER_ANALYTIC_VIEW_CLASS view, *161*
USER_ANALYTIC_VIEW_CLASS_AE view, *161*
USER_ANALYTIC_VIEW_CLC_MEAS view, *162*
USER_ANALYTIC_VIEW_CLC_MEAS_AE view, *162*
USER_ANALYTIC_VIEW_COLUMNS view, *162*
USER_ANALYTIC_VIEW_COLUMNS_AE view, *163*
USER_ANALYTIC_VIEW_DIM_CLASS view, *163*
USER_ANALYTIC_VIEW_DIM_CLS view, *163*
USER_ANALYTIC_VIEW_DIM_CLS_AE view, *164*
USER_ANALYTIC_VIEW_DIMENSIONS view, *164*
USER_ANALYTIC_VIEW_DIMS view, *164*
USER_ANALYTIC_VIEW_DIMS_AE view, *165*
USER_ANALYTIC_VIEW_HIER_CLASS view, *165*
USER_ANALYTIC_VIEW_HIER_CLS view, *165*
USER_ANALYTIC_VIEW_HIER_CLS_AE view, *166*
USER_ANALYTIC_VIEW_HIERS view, *166*
USER_ANALYTIC_VIEW_HIERS_AE view, *166*
USER_ANALYTIC_VIEW_KEYS view, *167*
USER_ANALYTIC_VIEW_KEYS_AE view, *167*
USER_ANALYTIC_VIEW_LEVEL_CLASS view, *167*
USER_ANALYTIC_VIEW_LEVELS view, *168*
USER_ANALYTIC_VIEW_LEVELS_AE view, *168*
USER_ANALYTIC_VIEW_LVL_CLS view, *168*
USER_ANALYTIC_VIEW_LVL_CLS_AE view, *169*
USER_ANALYTIC_VIEW_LVLGRPS view, *169*
USER_ANALYTIC_VIEW_LVLGRPS_AE view, *169*
USER_ANALYTIC_VIEW_MEAS_CLASS view, *170*
USER_ANALYTIC_VIEW_MEAS_CLS view, *170*
USER_ANALYTIC_VIEW_MEAS_CLS_AE view, *170*
USER_ANALYTIC_VIEWS view, *171*
USER_ANALYTIC_VIEWS_AE view, *171*
USER_ANNOTATION_VALUES view, *171*
USER_ANNOTATIONS view, *172*
USER_ANNOTATIONS_USAGE view, *172*
USER_APPLY_ERROR view, *172*
USER_AQ_AGENT_PRIVS view, *173*
USER_ARGUMENTS view, *173*
USER_ASSEMBLIES view, *173*
USER_ASSOCIATIONS view, *173*

Copyright © 2002, 2026, Oracle and/or its affiliates.

USER_ATTRIBUTE_DIM_ATTR_CLASS view, *174*
USER_ATTRIBUTE_DIM_ATTR_CLS view, *174*
USER_ATTRIBUTE_DIM_ATTR_CLS_AE view, *174*
USER_ATTRIBUTE_DIM_ATTRS view, *175*
USER_ATTRIBUTE_DIM_ATTRS_AE view, *175*
USER_ATTRIBUTE_DIM_CLASS view, *175*
USER_ATTRIBUTE_DIM_CLASS_AE view, *176*
USER_ATTRIBUTE_DIM_JN_PTHS view, *176*
USER_ATTRIBUTE_DIM_JN_PTHS_AE view, *176*
USER_ATTRIBUTE_DIM_JOIN_PATHS view, *177*
USER_ATTRIBUTE_DIM_KEYS view, *177*
USER_ATTRIBUTE_DIM_KEYS_AE view, *177*
USER_ATTRIBUTE_DIM_LEVEL_ATTRS view, *178*
USER_ATTRIBUTE_DIM_LEVELS view, *178*
USER_ATTRIBUTE_DIM_LEVELS_AE view, *178*
USER_ATTRIBUTE_DIM_LVL_ATRS view, *179*
USER_ATTRIBUTE_DIM_LVL_ATRS_AE view, *179*
USER_ATTRIBUTE_DIM_LVL_CLASS view, *179*
USER_ATTRIBUTE_DIM_LVL_CLS view, *180*
USER_ATTRIBUTE_DIM_LVL_CLS_AE view, *180*
USER_ATTRIBUTE_DIM_ORD_ATRS view, *180*
USER_ATTRIBUTE_DIM_ORD_ATRS_AE view, *181*
USER_ATTRIBUTE_DIM_ORDER_ATTRS view, *181*
USER_ATTRIBUTE_DIM_TABLES view, *181*
USER_ATTRIBUTE_DIM_TABLES_AE view, *182*
USER_ATTRIBUTE_DIMENSIONS view, *182*
USER_ATTRIBUTE_DIMENSIONS_AE view, *182*
USER_ATTRIBUTE_TRANSFORMATIONS view, *183*
USER_AUDIT_OBJECT view, *183*
USER_AUDIT_POLICIES view, *183*
USER_AUDIT_POLICY_COLUMNS view, *184*
USER_AUDIT_SESSION view, *184*
USER_AUDIT_STATEMENT view, *184*
USER_AUDIT_TRAIL view, *185*
USER_AW_PS view, *185*
USER_AWS view, *185*
USER_BASE_TABLE_MVIEWS view, *186*
USER_BLOCKCHAIN_TABLES view, *186*
USER_CATALOG view, *186*
  CAT synonym, *188*
USER_CERTIFICATES view, *186*
USER_CHANGE_NOTIFICATION_REGS view, *187*
USER_CLU_COLUMNS view, *187*
USER_CLUSTER_HASH_EXPRESSIONS view, *187*
USER_CLUSTERING_DIMENSIONS view, *188*
USER_CLUSTERING_JOINS view, *188*
USER_CLUSTERING_KEYS view, *188*
USER_CLUSTERING_TABLES view, *189*
USER_CLUSTERS view, *189*
  CLU synonym, *190*
USER_CODE_ROLE_PRIVS view, *189*
USER_COL_COMMENTS view, *189*
USER_COL_PENDING_STATS view, *190*
USER_COL_PRIVS view, *190*
USER_COL_PRIVS_MADE view, *190*
USER_COL_PRIVS_RECD view, *190*
USER_COLL_TYPES view, *191*
USER_COMPARISON view, *191*
USER_COMPARISON_COLUMNS view, *191*
USER_COMPARISON_ROW_DIF view, *191*
USER_COMPARISON_SCAN view, *192*
USER_COMPARISON_SCAN_VALUES view, *192*
USER_CONS_COLUMNS view, *192*
USER_CONS_OBJ_COLUMNS view, *192*
USER_CONSTRAINTS view, *193*
USER_CQ_NOTIFICATION_QUERIES view, *193*
USER_CREDENTIALS view, *193*
USER_CUBE_ATTR_VISIBILITY view, *193*
USER_CUBE_ATTRIBUTES view, *194*
USER_CUBE_BUILD_PROCESSES view, *194*
USER_CUBE_CALCULATED_MEMBERS view, *194*
USER_CUBE_DIM_LEVELS view, *194*
USER_CUBE_DIM_MODELS view, *195*
USER_CUBE_DIM_VIEW_COLUMNS view, *195*
USER_CUBE_DIM_VIEWS view, *195*
USER_CUBE_DIMENSIONALITY view, *195*
USER_CUBE_DIMENSIONS view, *196*
USER_CUBE_HIER_LEVELS view, *196*
USER_CUBE_HIER_VIEW_COLUMNS view, *196*
USER_CUBE_HIER_VIEWS view, *196*
USER_CUBE_HIERARCHIES view, *197*
USER_CUBE_MEASURES view, *197*
USER_CUBE_NAMED_BUILD_SPECS view, *197*
USER_CUBE_SUB_PARTITION_LEVELS view, *197*
USER_CUBE_VIEW_COLUMNS view, *198*
USER_CUBE_VIEWS view, *198*
USER_CUBES view, *198*
USER_DATAPUMP_JOBS view, *198*
USER_DB_LINKS view, *199*
USER_DBFS_HS view, *199*
USER_DBFS_HS_COMMANDS view, *199*
USER_DBFS_HS_FILES view, *199*
USER_DBFS_HS_FIXED_PROPERTIES view, *200*
USER_DBFS_HS_PROPERTIES view, *200*
USER_DEPENDENCIES view, *200*
USER_DIM_ATTRIBUTES view, *200*

Copyright © 2002, 2026, Oracle and/or its affiliates.

USER_DIM_CHILD_OF view, *201*
USER_DIM_HIERARCHIES view, *201*
USER_DIM_JOIN_KEY view, *201*
USER_DIM_LEVEL_KEY view, *201*
USER_DIM_LEVELS view, *202*
USER_DIMENSIONS view, *202*
USER_DUMP_DEST initialization parameter, *345*
USER_EDITIONED_TYPES view, *202*
USER_EDITIONING_VIEW_COLS view, *202*
USER_EDITIONING_VIEW_COLS_AE view, *202*
USER_EDITIONING_VIEWS view, *203*
USER_EDITIONING_VIEWS_AE view, *203*
USER_ENCRYPTED_COLUMNS view, *203*
USER_EPG_DAD_AUTHORIZATION view, *204*
USER_ERROR_TRANSLATIONS view, *204*
USER_ERRORS view, *204*
USER_ERRORS_AE view, *204*
USER_EVALUATION_CONTEXT_TABLES view, *205*
USER_EVALUATION_CONTEXT_VARS view, *205*
USER_EVALUATION_CONTEXTS view, *205*
USER_EXPRESSION_STATISTICS view, *205*
USER_EXTENTS view, *206*
USER_EXTERNAL_LOCATIONS view, *206*
USER_EXTERNAL_TABLES view, *206*
USER_FILE_GROUP_EXPORT_INFO view, *206*
USER_FILE_GROUP_FILES view, *207*
USER_FILE_GROUP_TABLES view, *207*
USER_FILE_GROUP_TABLESPACES view, *207*
USER_FILE_GROUP_VERSIONS view, *207*
USER_FILE_GROUPS view, *208*
USER_FLASHBACK_ARCHIVE view, *208*
USER_FLASHBACK_ARCHIVE_TABLES view, *208*
USER_FLASHBACK_TXN_REPORT view, *209*
USER_FLASHBACK_TXN_STATE view, *209*
USER_FREE_SPACE view, *209*
USER_GOLDENGATE_PRIVILEGES view, *209*
USER_HEAT_MAP_SEG_HISTOGRAM view, *210*
USER_HEAT_MAP_SEGMENT view, *210*
USER_HIER_CLASS view, *210*
USER_HIER_CLASS_AE view, *210*
USER_HIER_COLUMNS view, *211*
USER_HIER_COLUMNS_AE view, *211*
USER_HIER_HIER_ATTR_CLASS view, *211*
USER_HIER_HIER_ATTR_CLASS_AE view, *212*
USER_HIER_HIER_ATTRIBUTES view, *212*
USER_HIER_HIER_ATTRIBUTES_AE view, *212*
USER_HIER_JOIN_PATHS view, *213*
USER_HIER_JOIN_PATHS_AE view, *213*
USER_HIER_LEVEL_ID_ATTRS view, *213*
USER_HIER_LEVEL_ID_ATTRS_AE view, *214*
USER_HIER_LEVELS view, *214*
USER_HIER_LEVELS_AE view, *214*
USER_HIERARCHIES view, *215*
USER_HIERARCHIES_AE view, *215*
USER_HISTOGRAMS synonym for USER_TAB_HISTOGRAMS, *215*
USER_HIVE_COLUMNS view, *215*
USER_HIVE_DATABASES view, *216*
USER_HIVE_PART_KEY_COLUMNS view, *216*
USER_HIVE_TAB_PARTITIONS view, *216*
USER_HIVE_TABLES view, *216*
USER_HOST_ACES view, *217*
USER_IDENTIFIERS view, *217*
USER_ILMDATAMOVEMENTPOLICIES view, *217*
USER_ILMEVALUATIONDETAILS view, *218*
USER_ILMOBJECTS view, *218*
USER_ILMPOLICIES view, *218*
USER_ILMRESULTS view, *219*
USER_ILMTASKS view, *219*
USER_IM_EXPRESSIONS view, *220*
USER_IMMUTABLE_TABLES view, *220*
USER_IND_COLUMNS view, *220*
USER_IND_EXPRESSIONS view, *221*
USER_IND_PARTITIONS view, *221*
USER_IND_PENDING_STATS view, *221*
USER_IND_STATISTICS view, *221*
USER_IND_SUBPARTITIONS view, *222*
USER_INDEXES view, *222*
  IND synonym, *113*
USER_INDEXTYPE_ARRAYTYPES view, *222*
USER_INDEXTYPE_COMMENTS view, *222*
USER_INDEXTYPE_OPERATORS view, *223*
USER_INDEXTYPES view, *223*
USER_INTERNAL_TRIGGERS view, *223*
USER_JAVA_ARGUMENTS view, *223*
USER_JAVA_CLASSES view, *224*
USER_JAVA_COMPILER_OPTIONS view, *224*
USER_JAVA_DERIVATIONS view, *224*
USER_JAVA_FIELDS view, *224*
USER_JAVA_IMPLEMENTS view, *225*
USER_JAVA_INNERS view, *225*
USER_JAVA_LAYOUTS view, *225*
USER_JAVA_METHODS view, *225*
USER_JAVA_NCOMPS view, *226*
USER_JAVA_POLICY view, *226*
USER_JAVA_RESOLVERS view, *226*
USER_JAVA_THROWS view, *226*
USER_JOBS view, *227*
  ALL_JOBS synonym, *210*
USER_JOIN_IND_COLUMNS view, *227*
USER_JOINGROUPS view, *227*
USER_JSON_COLUMNS view, *227*
USER_JSON_DATAGUIDE_FIELDS view, *228*
USER_JSON_DATAGUIDES view, *228*
USER_LIBRARIES view, *228*
USER_LOB_PARTITIONS view, *229*
USER_LOB_SUBPARTITIONS view, *229*

Copyright © 2002, 2026, Oracle and/or its affiliates.

USER_LOB_TEMPLATES view, *229*
USER_LOBS view, *229*
USER_LOG_GROUP_COLUMNS view, *230*
USER_LOG_GROUPS view, *230*
USER_MEASURE_FOLDER_CONTENTS view, *230*
USER_MEASURE_FOLDER_SUBFOLDERS view, *230*
USER_MEASURE_FOLDERS view, *231*
USER_METADATA_PROPERTIES view, *231*
USER_METHOD_PARAMS view, *231*
USER_METHOD_RESULTS view, *231*
USER_MINING_MODEL_ATTRIBUTES view, *232*
USER_MINING_MODEL_PARTITIONS view, *232*
USER_MINING_MODEL_SETTINGS view, *232*
USER_MINING_MODEL_VIEWS view, *232*
USER_MINING_MODEL_XFORMS view, *233*
USER_MINING_MODELS view, *233*
USER_MVIEW_AGGREGATES view, *233*
USER_MVIEW_ANALYSIS view, *233*
USER_MVIEW_COMMENTS view, *234*
USER_MVIEW_DETAIL_PARTITION view, *234*
USER_MVIEW_DETAIL_RELATIONS view, *234*
USER_MVIEW_DETAIL_SUBPARTITION view, *235*
USER_MVIEW_JOINS view, *235*
USER_MVIEW_KEYS view, *235*
USER_MVIEW_LOGS view, *235*
USER_MVIEW_REFRESH_TIMES view, *236*
USER_MVIEWS view, *236*
USER_MVREF_CHANGE_STATS view, *236*
USER_MVREF_RUN_STATS view, *236*
USER_MVREF_STATS view, *237*
USER_MVREF_STATS_PARAMS view, *237*
USER_MVREF_STATS_SYS_DEFAULTS view, *237*
USER_MVREF_STMT_STATS view, *237*
USER_NESTED_TABLE_COLS view, *238*
USER_NESTED_TABLES view, *238*
USER_NETWORK_ACL_PRIVILEGES view, *238*
USER_OBJ_AUDIT_OPTS view, *239*
USER_OBJ_COLATTRS view, *239*
USER_OBJECT_SIZE view, *239*
USER_OBJECT_TABLES view, *240*
USER_OBJECT_USAGE view, *240*
USER_OBJECTS view, *240*
  OBJ synonym, *117*
USER_OBJECTS_AE view, *240*
USER_OPANCILLARY view, *241*
USER_OPARGUMENTS view, *241*
USER_OPBINDINGS view, *241*
USER_OPERATOR_COMMENTS view, *241*
USER_OPERATORS view, *242*
USER_OUTLINE_HINTS view, *242*
  ALL_OUTLINE_HINTS synonym, *262*
USER_OUTLINES view, *242*
  ALL_OUTLINES synonym, *262*
USER_PARALLEL_EXECUTE_CHUNKS view, *242*
USER_PARALLEL_EXECUTE_TASKS view, *243*
USER_PART_COL_STATISTICS view, *243*
USER_PART_HISTOGRAMS view, *243*
USER_PART_INDEXES view, *243*
USER_PART_KEY_COLUMNS view, *244*
USER_PART_LOBS view, *244*
USER_PART_TABLES view, *244*
USER_PARTIAL_DROP_TABS view, *244*
USER_PASSWORD_LIMITS view, *245*
USER_PENDING_CONV_TABLES view, *245*
USER_PLSQL_COLL_TYPES view, *245*
USER_PLSQL_OBJECT_SETTINGS view, *245*
USER_PLSQL_TYPE_ATTRS view, *246*
USER_PLSQL_TYPES view, *246*
USER_POLICIES view, *246*
USER_POLICY_ATTRIBUTES view, *246*
USER_POLICY_CONTEXTS view, *247*
USER_POLICY_GROUPS view, *247*
USER_PRIVATE_TEMP_TABLES view, *247*
USER_PRIVILEGE_MAP view, *247*
USER_PROCEDURES view, *248*
USER_PROXIES view, *248*
USER_QUEUE_SCHEDULES view, *248*
USER_QUEUE_SUBSCRIBERS view, *248*
USER_QUEUE_TABLES view, *249*
USER_QUEUES view, *249*
USER_RECYCLEBIN view, *249*
  RECYCLEBIN synonym, *125*
USER_REFRESH view, *249*
USER_REFRESH_CHILDREN view, *250*
USER_REFS view, *250*
USER_REGISTERED_MVIEWS view, *250*
USER_REGISTRY view, *250*
USER_RESOURCE_LIMITS view, *250*
USER_RESUMABLE view, *251*
USER_REWRITE_EQUIVALENCES view, *251*
USER_ROLE_PRIVS view, *251*
USER_RSRC_CONSUMER_GROUP_PRIVS view, *252*
USER_RSRC_MANAGER_SYSTEM_PRIVS view, *252*
USER_RULE_SET_RULES view, *252*
USER_RULE_SETS view, *253*
USER_RULES view, *253*
USER_SCHEDULER_CHAIN_RULES view, *253*
USER_SCHEDULER_CHAIN_STEPS view, *253*
USER_SCHEDULER_CHAINS view, *254*
USER_SCHEDULER_CREDENTIALS view, *254*
USER_SCHEDULER_DB_DESTS view, *254*
USER_SCHEDULER_DESTS view, *255*
USER_SCHEDULER_FILE_WATCHERS view, *255*

Copyright © 2002, 2026, Oracle and/or its affiliates.

USER_SCHEDULER_GROUP_MEMBERS view, *255*
USER_SCHEDULER_GROUPS view, *255*
USER_SCHEDULER_INCOMPAT_MEMBER view, *256*
USER_SCHEDULER_INCOMPATS view, *256*
USER_SCHEDULER_JOB_ARGS view, *256*
USER_SCHEDULER_JOB_DESTS view, *256*
USER_SCHEDULER_JOB_LOG view, *257*
USER_SCHEDULER_JOB_RUN_DETAILS view, *257*
USER_SCHEDULER_JOBS view, *257*
USER_SCHEDULER_NOTIFICATIONS view, *257*
USER_SCHEDULER_PROGRAM_ARGS view, *258*
USER_SCHEDULER_PROGRAMS view, *258*
USER_SCHEDULER_REMOTE_JOBSTATE view, *258*
USER_SCHEDULER_RESOURCES view, *258*
USER_SCHEDULER_RSC_CONSTRAINTS view, *259*
USER_SCHEDULER_RUNNING_CHAINS view, *259*
USER_SCHEDULER_RUNNING_JOBS view, *259*
USER_SCHEDULER_SCHEDULES view, *259*
USER_SEC_RELEVANT_COLS view, *260*
USER_SECONDARY_OBJECTS view, *260*
USER_SEGMENTS view, *260*
USER_SEQUENCES view, *260*
SEQ synonym, *133*
USER_SODA_COLLECTIONS view, *261*
USER_SOURCE view, *261*
USER_SOURCE_AE view, *261*
USER_SQL_TRANSLATION_PROFILES view, *262*
USER_SQL_TRANSLATIONS view, *262*
USER_SQLJ_TYPE_ATTRS view, *262*
USER_SQLJ_TYPE_METHODS view, *262*
USER_SQLJ_TYPES view, *263*
USER_SQLSET view, *263*
USER_SQLSET_BINDS view, *263*
USER_SQLSET_PLANS view, *263*
USER_SQLSET_REFERENCES view, *264*
USER_SQLSET_STATEMENTS view, *264*
USER_SQLTUNE_BINDS view, *264*
USER_SQLTUNE_PLANS view, *264*
USER_SQLTUNE_RATIONALE_PLAN view, *265*
USER_SQLTUNE_STATISTICS view, *265*
USER_SR_GRP_STATUS view, *265*
USER_SR_GRP_STATUS_ALL view, *265*
USER_SR_OBJ view, *266*
USER_SR_OBJ_ALL view, *266*
USER_SR_OBJ_STATUS view, *266*
USER_SR_OBJ_STATUS_ALL view, *266*
USER_SR_PARTN_OPS view, *267*
USER_SR_STLOG_EXCEPTIONS view, *267*
USER_SR_STLOG_STATS view, *267*
USER_STAT_EXTENSIONS view, *267*
USER_STATEMENTS view, *268*
USER_STORED_SETTINGS view, *268*
USER_SUBPART_COL_STATISTICS view, *268*
USER_SUBPART_HISTOGRAMS view, *268*
USER_SUBPART_KEY_COLUMNS view, *269*
USER_SUBPARTITION_TEMPLATES view, *269*
USER_SUBSCR_REGISTRATIONS view, *269*
USER_SYNONYMS view, *269*
SYN synonym, *134*
USER_SYS_PRIVS view, *269*
USER_TAB_COL_STATISTICS view, *270*
USER_TAB_COLS view, *270*
USER_TAB_COLUMNS view, *271*
COLS synonym, *190*
USER_TAB_COMMENTS view, *271*
USER_TAB_HISTGRM_PENDING_STATS view, *271*
USER_TAB_HISTOGRAMS view, *272*
USER_HISTOGRAMS synonym, *215*
USER_TAB_IDENTITY_COLS view, *272*
USER_TAB_MODIFICATIONS view, *272*
USER_TAB_PARTITIONS view, *272*
USER_TAB_PENDING_STATS view, *273*
USER_TAB_PRIVS view, *273*
USER_TAB_PRIVS_MADE view, *273*
USER_TAB_PRIVS_RECD view, *273*
USER_TAB_STAT_PREFS view, *274*
USER_TAB_STATISTICS view, *274*
USER_TAB_STATS_HISTORY view, *274*
USER_TAB_SUBPARTITIONS view, *274*
USER_TABLES view, *275*
TABS synonym, *137*
USER_TABLESPACES view, *275*
USER_TRANSFORMATIONS view, *275*
USER_TRIGGER_COLS view, *275*
USER_TRIGGER_ORDERING view, *276*
USER_TRIGGERS view, *276*
USER_TRIGGERS_AE view, *276*
USER_TS_QUOTAS view, *276*
USER_TSTZ_TAB_COLS view, *277*
USER_TSTZ_TABLES view, *277*
USER_TUNE_MVIEW view, *277*
USER_TYPE_ATTRS view, *277*
USER_TYPE_METHODS view, *278*
USER_TYPE_VERSIONS view, *278*
USER_TYPES view, *278*
USER_UNUSED_COL_TABS view, *278*
USER_UPDATABLE_COLUMNS view, *279*
USER_USERS view, *279*
USER_USTATS view, *283*
USER_VARRAYS view, *284*
USER_VIEWS view, *284*
USER_VIEWS_AE view, *284*

Copyright © 2002, 2026, Oracle and/or its affiliates.

USER_WALLET_ACES view, *284*
USER_WARNING_SETTINGS view, *285*
USER_XML_INDEXES view, *285*
USER_XML_NESTED_TABLES view, *285*
USER_XML_OUT_OF_LINE_TABLES view, *285*
USER_XML_SCHEMA_ATTRIBUTES view, *286*
USER_XML_SCHEMA_COMPLEX_TYPES view, *286*
USER_XML_SCHEMA_ELEMENTS view, *286*
USER_XML_SCHEMA_NAMESPACES view, *286*
USER_XML_SCHEMA_SIMPLE_TYPES view, *287*
USER_XML_SCHEMA_SUBSTGRP_HEAD view, *287*
USER_XML_SCHEMA_SUBSTGRP_MBRS view, *287*
USER_XML_SCHEMAS view, *287*
USER_XML_TAB_COLS view, *288*
USER_XML_TABLES view, *288*
USER_XML_VIEW_COLS view, *288*
USER_XML_VIEWS view, *288*
USER_XTERNAL_LOC_PARTITIONS view, *289*
USER_XTERNAL_LOC_SUBPARTITIONS view, *289*
USER_XTERNAL_PART_TABLES view, *289*
USER_XTERNAL_TAB_PARTITIONS view, *289*
USER_XTERNAL_TAB_SUBPARTITIONS view, *290*
USER_ZONEMAP_MEASURES view, *290*
USER_ZONEMAPS view, *290*
USERLOCK.SQL script, *B-3*
users
  authenticating, *244*
  multiple, *112*
UTLBSTAT.SQL script, *B-3*
UTLCHN1.SQL script, *B-3*
UTLCONST.SQL script, *B-3*
UTLDTREE.SQL script, *103*, *113*, *B-3*
UTLESTAT.SQL script, *B-3*
UTLEXPT1.SQL script, *B-3*
UTLFIXDIRS.SQL script, *B-4*
UTLIP.SQL script, *B-4*
UTLIRP.SQL script, *B-4*
UTLLOCKT.SQL script, *B-4*
UTLPWDMG.SQL script, *B-4*
UTLRP.SQL script, *B-4*
UTLSAMPL.SQL script, *B-4*
UTLSCLN.SQL script, *B-4*
UTLTKPRF.SQL script, *B-4*
UTLVALID.SQL script, *B-4*
UTLXPLAN.SQL script, *B-5*

V

V$ACCESS view, *3*
V$ACTIVE_INSTANCES view, *4*
V$ACTIVE_SERVICES view, *4*
V$ACTIVE_SESS_POOL_MTH view, *6*
V$ACTIVE_SESSION_HISTORY view, *7*
V$ADVISOR_PROGRESS view, *12*
V$ALERT_TYPES view, *13*
V$ALL_ACTIVE_SESSION_HISTORY view, *14*
V$ALL_SQL_MONITOR view, *19*
V$ALL_SQL_PLAN_MONITOR view, *24*
V$AQ view, *27*
V$AQ_BACKGROUND_COORDINATOR view, *28*
V$AQ_BMAP_NONDUR_SUBSCRIBERS view, *28*
V$AQ_CROSS_INSTANCE_JOBS view, *29*
V$AQ_IPC_ACTIVE_MSGS view, *31*
V$AQ_IPC_MSG_STATS view, *31*
V$AQ_IPC_PENDING_MSGS view, *32*
V$AQ_JOB_COORDINATOR view, *33*
V$AQ_MESSAGE_CACHE view, *33*
V$AQ_MESSAGE_CACHE_ADVICE view, *34*
V$AQ_MESSAGE_CACHE_STAT view, *35*
V$AQ_NONDUR_REGISTRATIONS view, *37*
V$AQ_NONDUR_SUBSCRIBER view, *38*
V$AQ_NONDUR_SUBSCRIBER_LWM view, *39*
V$AQ_NOTIFICATION_CLIENTS view, *39*
V$AQ_PARTITION_STATS view, *40*
V$AQ_REMOTE_DEQUEUE_AFFINITY view, *42*
V$AQ_SERVER_POOL view, *42*
V$AQ_SHARDED_SUBSCRIBER_STAT view, *43*
V$AQ_SUBSCRIBER_LOAD view, *44*
V$ARCHIVE view, *45*
V$ARCHIVE_DEST view, *46*
V$ARCHIVE_DEST_STATUS view, *49*
V$ARCHIVE_GAP view, *52*
V$ARCHIVE_PROCESSES view, *52*
V$ARCHIVED_LOG view, *53*
V$ASM_ACFS_ENCRYPTION_INFO view, *55*
V$ASM_ACFS_SEC_ADMIN view, *56*
V$ASM_ACFS_SEC_CMDRULE view, *57*
V$ASM_ACFS_SEC_REALM view, *58*
V$ASM_ACFS_SEC_REALM_FILTER view, *59*
V$ASM_ACFS_SEC_REALM_GROUP view, *60*
V$ASM_ACFS_SEC_REALM_USER view, *61*
V$ASM_ACFS_SEC_RULE view, *62*
V$ASM_ACFS_SEC_RULESET view, *63*
V$ASM_ACFS_SEC_RULESET_RULE view, *64*
V$ASM_ACFS_SECURITY_INFO view, *64*
V$ASM_ACFSAUTORESIZE view, *65*
V$ASM_ACFSREPL view, *66*
V$ASM_ACFSREPLTAG view, *68*
V$ASM_ACFSSNAPSHOTS view, *69*
V$ASM_ACFSTAG view, *70*
V$ASM_ACFSVOLUMES view, *71*
V$ASM_ALIAS view, *72*
V$ASM_ATTRIBUTE view, *73*
V$ASM_AUDIT_CLEANUP_JOBS view, *74*

Copyright © 2002, 2026, Oracle and/or its affiliates.

V$ASM_AUDIT_CONFIG_PARAMS view, *75*
V$ASM_AUDIT_LAST_ARCH_TS view, *75*
V$ASM_CLIENT view, *76*
V$ASM_DBCLONE_INFO view, *77*
V$ASM_DISK view, *78*
V$ASM_DISK_IOSTAT view, *83*
V$ASM_DISK_STAT view, *84*
V$ASM_DISKGROUP view, *84*
V$ASM_DISKGROUP_STAT view, *86*
V$ASM_ESTIMATE view, *87*
V$ASM_FILE view, *87*
V$ASM_FILEGROUP view, *89*
V$ASM_FILEGROUP_FILE view, *91*
V$ASM_FILEGROUP_PROPERTY view, *91*
V$ASM_FILESYSTEM view, *93*
V$ASM_OPERATION view, *94*
V$ASM_QUOTAGROUP view, *96*
V$ASM_TEMPLATE view, *97*
V$ASM_USER view, *98*
V$ASM_USERGROUP view, *98*
V$ASM_USERGROUP_MEMBER view, *99*
V$ASM_VOLUME view, *100*
V$ASM_VOLUME_STAT view, *101*
V$AW_AGGREGATE_OP view, *101*
V$AW_ALLOCATE_OP view, *102*
V$AW_CALC view, *102*
V$AW_LONGOPS view, *103*
V$AW_OLAP view, *104*
V$AW_SESSION_INFO view, *105*
V$BACKUP view, *106*
V$BACKUP_ARCHIVELOG_DETAILS view, *106*
V$BACKUP_ARCHIVELOG_SUMMARY view, *107*
V$BACKUP_ASYNC_IO view, *108*
V$BACKUP_CONTROLFILE_DETAILS view, *109*
V$BACKUP_CONTROLFILE_SUMMARY view, *110*
V$BACKUP_COPY_DETAILS view, *111*
V$BACKUP_COPY_SUMMARY view, *112*
V$BACKUP_CORRUPTION view, *112*
V$BACKUP_DATAFILE view, *113*
V$BACKUP_DATAFILE_DETAILS view, *115*
V$BACKUP_DATAFILE_SUMMARY view, *116*
V$BACKUP_DEVICE view, *117*
V$BACKUP_FILES view, *117*
V$BACKUP_NONLOGGED view, *120*
V$BACKUP_PIECE view, *121*
V$BACKUP_PIECE_DETAILS view, *122*
V$BACKUP_REDOLOG view, *124*
V$BACKUP_SET view, *125*
V$BACKUP_SET_DETAILS view, *126*
V$BACKUP_SET_SUMMARY view, *128*
V$BACKUP_SPFILE view, *129*
V$BACKUP_SPFILE_DETAILS view, *129*
V$BACKUP_SPFILE_SUMMARY view, *130*
V$BACKUP_SYNC_IO view, *130*
V$BGPROCESS view, *131*
V$BH view, *132*
V$BLOCK_CHANGE_TRACKING view, *133*
V$BLOCKING_QUIESCE view, *134*
V$BT_SCAN_CACHE view, *134*
V$BT_SCAN_OBJ_TEMPS view, *135*
V$BUFFER_POOL view, *136*
V$BUFFER_POOL_STATISTICS view, *137*
V$BUFFERED_PUBLISHERS view, *138*
V$BUFFERED_QUEUES view, *139*
V$BUFFERED_SUBSCRIBERS view, *140*
V$CACHE view, *141*
V$CACHE_LOCK view, *143*
V$CACHE_TRANSFER view, *144*
V$CELL_* views, *3*
V$CHUNK_METRIC view, *146*
V$CIRCUIT view, *146*
V$CLASS_CACHE_TRANSFER view, *147*
V$CLEANUP_PROCESS view, *148*
V$CLIENT_SECRETS view, *149*
V$CLIENT_STATS view, *150*
V$CLONEDFILE view, *151*
V$CLUSTER_INTERCONNECTS view, *151*
V$CODE_CLAUSE view, *152*
V$CON_EVENT_HISTOGRAM_MICRO view, *153*
V$CON_SYS_TIME_MODEL view, *153*
V$CON_SYSMETRIC view, *154*
V$CON_SYSMETRIC_HISTORY view, *155*
V$CON_SYSMETRIC_SUMMARY view, *155*
V$CON_SYSSTAT view, *156*
V$CON_SYSTEM_EVENT view, *157*
V$CON_SYSTEM_WAIT_CLASS view, *158*
V$CONFIGURED_INTERCONNECTS view, *158*
V$CONTAINERS view, *159*
V$CONTEXT view, *161*
V$CONTROLFILE view, *161*
V$CONTROLFILE_RECORD_SECTION view, *162*
V$COPY_CORRUPTION view, *163*
V$COPY_NONLOGGED view, *164*
V$CORRUPT_XID_LIST view, *165*
V$CPOOL_CC_INFO view, *165*
V$CPOOL_CC_STATS view, *165*
V$CPOOL_CONN_INFO view, *166*
V$CPOOL_STATS view, *167*
V$CR_BLOCK_SERVER view, *168*
V$CURRENT_BLOCK_SERVER view, *169*
V$DATABASE view, *170*
V$DATABASE_BLOCK_CORRUPTION view, *178*
V$DATABASE_INCARNATION view, *179*
V$DATABASE_KEY_INFO view, *180*
V$DATAFILE view, *181*
V$DATAFILE_COPY view, *183*
V$DATAFILE_HEADER view, *185*
V$DATAGUARD_CONFIG view, *186*

Copyright © 2002, 2026, Oracle and/or its affiliates.

V$DATAGUARD_PROCESS view, *187*
V$DATAGUARD_STATS view, *190*
V$DATAGUARD_STATUS view, *191*
V$DB_CACHE_ADVICE view, *192*
V$DB_OBJECT_CACHE view, *193*
V$DB_PIPES view, *195*
V$DB_TRANSPORTABLE_PLATFORM view, *195*
V$DBFILE view, *196*
V$DBLINK view, *196*
V$DEAD_CLEANUP view, *197*
V$DELETED_OBJECT view, *198*
V$DG_BROKER_CONFIG view, *199*
V$DIAG_ALERT_EXT view, *200*
V$DIAG_APP_TRACE_FILE view, *202*
V$DIAG_INCIDENT view, *203*
V$DIAG_INFO view, *204*
V$DIAG_OPT_TRACE_RECORDS view, *205*
V$DIAG_PROBLEM view, *206*
V$DIAG_SESS_OPT_TRACE_RECORDS view, *207*
V$DIAG_SESS_SQL_TRACE_RECORDS view, *208*
V$DIAG_SQL_TRACE_RECORDS view, *209*
V$DIAG_TRACE_FILE view, *210*
V$DIAG_TRACE_FILE_CONTENTS view, *210*
V$DISPATCHER view, *211*
V$DISPATCHER_CONFIG view, *212*
V$DISPATCHER_RATE view, *213*
V$DNFS_CHANNELS view, *217*
V$DNFS_FILES view, *218*
V$DNFS_SERVERS view, *218*
V$DNFS_STATS view, *219*
V$DYNAMIC_REMASTER_STATS view, *220*
V$EDITIONABLE_TYPES view, *221*
V$EMON view, *221*
V$EMX_USAGE_STATS view, *223*
V$ENABLEDPRIVS view, *223*
V$ENCRYPTED_TABLESPACES view, *224*
V$ENCRYPTION_KEYS view, *225*
V$ENCRYPTION_WALLET view, *227*
V$ENQUEUE_LOCK view, *229*
V$ENQUEUE_STAT view, *230*
V$ENQUEUE_STATISTICS view, *230*
V$EVENT_HISTOGRAM view, *231*
V$EVENT_HISTOGRAM_MICRO view, *232*
V$EVENT_NAME view, *232*
V$EVENTMETRIC view, *233*
V$EXADIRECT_ACL view, *233*
V$EXECUTION view, *234*
V$EXP_STATS view, *234*
V$FALSE_PING view, *235*
V$FAST_START_SERVERS view, *236*
V$FAST_START_TRANSACTIONS view, *236*
V$FILE_CACHE_TRANSFER view, *237*
V$FILE_HISTOGRAM view, *238*
V$FILEMETRIC view, *239*
V$FILEMETRIC_HISTORY view, *239*
V$FILESPACE_USAGE view, *240*
V$FILESTAT view, *240*
V$FIXED_TABLE view, *241*
V$FIXED_VIEW_DEFINITION view, *242*
V$FLASHBACK_DATABASE_LOG view, *242*
V$FLASHBACK_DATABASE_LOGFILE view, *243*
V$FLASHBACK_DATABASE_STAT view, *243*
V$FLASHBACK_TXN_GRAPH view, *244*
V$FLASHBACK_TXN_MODS view, *245*
V$FLASHFILESTAT view, *246*
V$FOREIGN_ARCHIVED_LOG view, *247*
V$FS_FAILOVER_OBSERVERS view, *249*
V$FS_FAILOVER_STATS view, *250*
V$FS_OBSERVER_HISTOGRAM view, *250*
V$GC_ELEMENT view, *251*
V$GC_ELEMENTS_WITH_COLLISIONS view, *252*
V$GCR_ACTIONS view, *253*
V$GCR_LOG view, *253*
V$GCR_METRICS view, *254*
V$GCR_STATUS view, *255*
V$GCSHVMASTER_INFO view, *256*
V$GCSPFMASTER_INFO view, *256*
V$GES_BLOCKING_ENQUEUE view, *257*
V$GES_CONVERT_LOCAL view, *258*
V$GES_CONVERT_REMOTE view, *259*
V$GES_DEADLOCK_SESSIONS view, *260*
V$GES_DEADLOCKS view, *260*
V$GES_ENQUEUE view, *261*
V$GES_LATCH view, *262*
V$GES_RESOURCE view, *262*
V$GES_STATISTICS view, *263*
V$GG_APPLY_COORDINATOR view, *263*
V$GG_APPLY_READER view, *265*
V$GG_APPLY_RECEIVER view, *267*
V$GG_APPLY_SERVER view, *268*
V$GLOBAL_BLOCKED_LOCKS view, *271*
V$GLOBAL_TRANSACTION view, *271*
V$GOLDENGATE_CAPABILITIES view, *272*
V$GOLDENGATE_CAPTURE view, *273*
V$GOLDENGATE_MESSAGE_TRACKING view, *276*
V$GOLDENGATE_PROCEDURE_STATS view, *277*
V$GOLDENGATE_TABLE_STATS view, *278*
V$GOLDENGATE_TRANSACTION view, *279*
V$HANG_INFO view, *281*
V$HANG_SESSION_INFO view, *282*
V$HANG_STATISTICS view, *283*
V$HEAT_MAP_SEGMENT view, *284*
V$HM_CHECK view, *285*
V$HM_CHECK_PARAM view, *285*
V$HM_FINDING view, *286*
V$HM_INFO view, *287*
V$HM_RECOMMENDATION view, *287*

Copyright © 2002, 2026, Oracle and/or its affiliates.

V$HM_RUN view, *288*
V$HS_AGENT view, *289*
V$HS_PARAMETER view, *290*
V$HS_SESSION view, *290*
V$HVMASTER_INFO view, *291*
V$IM_COLUMN_LEVEL view, *1*
V$IM_SEGMENTS view, *2*
V$IM_USER_SEGMENTS view, *4*
V$INDEX_USAGE_INFO view, *5*
V$INDEXED_FIXED_COLUMN view, *6*
V$INMEMORY_AREA view, *7*
V$INMEMORY_FASTSTART_AREA view, *8*
V$INSTANCE view, *8*
V$INSTANCE_CACHE_TRANSFER view, *11*
V$INSTANCE_PING view, *12*
V$INSTANCE_RECOVERY view, *13*
V$IO_CALIBRATION_STATUS view, *14*
V$IO_OUTLIER view, *15*
V$IOFUNCMETRIC view, *16*
V$IOFUNCMETRIC_HISTORY view, *16*
V$IOS_CLIENT view, *17*
V$IOSTAT_CONSUMER_GROUP view, *17*
V$IOSTAT_FILE view, *18*
V$IOSTAT_FUNCTION view, *19*
V$IOSTAT_FUNCTION_DETAIL view, *20*
V$IOSTAT_NETWORK view, *21*
V$IP_ACL view, *21*
V$JAVA_LIBRARY_CACHE_MEMORY view, *22*
V$JAVA_POOL_ADVICE view, *22*
V$KERNEL_IO_OUTLIER view, *23*
V$KEY_VECTOR view, *24*
V$LATCH view, *26*
V$LATCH_CHILDREN view, *27*
V$LATCH_MISSES view, *28*
V$LATCH_PARENT view, *29*
V$LATCHHOLDER view, *29*
V$LATCHNAME view, *30*
V$LGWRIO_OUTLIER view, *30*
V$LIBCACHE_LOCKS view, *31*
V$LIBRARY_CACHE_MEMORY view, *32*
V$LIBRARYCACHE view, *33*
V$LICENSE view, *33*
V$LOADISTAT view, *34*
V$LOADPSTAT view, *35*
V$LOCK view, *35*
V$LOCK_ACTIVITY view, *37*
V$LOCK_TYPE view, *38*
V$LOCKDOWN_RULES view, *38*
V$LOCKED_OBJECT view, *39*
V$LOG view, *40*
V$LOG_HISTORY view, *41*
V$LOGFILE view, *41*
V$LOGHIST view, *42*
V$LOGMNR_CONTENTS view, *42*
V$LOGMNR_DICTIONARY view, *48*
V$LOGMNR_DICTIONARY_LOAD view, *49*
V$LOGMNR_LATCH view, *49*
V$LOGMNR_LOGS view, *50*
V$LOGMNR_PARAMETERS view, *51*
V$LOGMNR_PROCESS view, *52*
V$LOGMNR_SESSION view, *53*
V$LOGMNR_STATS view, *54*
V$LOGSTDBY view, *56*
V$LOGSTDBY_PROCESS view, *56*
V$LOGSTDBY_PROGRESS view, *57*
V$LOGSTDBY_STATE view, *58*
V$LOGSTDBY_STATS view, *59*
V$LOGSTDBY_TRANSACTION view, *61*
V$MANAGED_STANDBY view, *62*
V$MAP_COMP_LIST view, *64*
V$MAP_ELEMENT view, *65*
V$MAP_EXT_ELEMENT view, *65*
V$MAP_FILE view, *66*
V$MAP_FILE_EXTENT view, *67*
V$MAP_FILE_IO_STACK view, *67*
V$MAP_LIBRARY view, *68*
V$MAP_SUBELEMENT view, *69*
V$MAPPED_SQL view, *69*
V$MEMOPTIMIZE_WRITE_AREA view, *70*
V$MEMORY_CURRENT_RESIZE_OPS view, *71*
V$MEMORY_DYNAMIC_COMPONENTS view, *72*
V$MEMORY_RESIZE_OPS view, *73*
V$MEMORY_TARGET_ADVICE view, *73*
V$METRIC view, *75*
V$METRIC_HISTORY view, *76*
V$METRICGROUP view, *76*
V$METRICNAME view, *77*
V$MTTR_TARGET_ADVICE view, *77*
V$MUTEX_SLEEP view, *78*
V$MUTEX_SLEEP_HISTORY view, *79*
V$MVREFRESH view, *80*
V$MYSTAT view, *80*
V$NFS_CLIENTS view, *81*
V$NFS_LOCKS view, *81*
V$NFS_OPEN_FILES view, *82*
V$NLS_PARAMETERS view, *82*
V$NLS_VALID_VALUES view, *83*
V$NONLOGGED_BLOCK view, *83*
V$OBJECT_DEPENDENCY view, *84*
V$OBJECT_PRIVILEGE view, *85*
V$OBJECT_USAGE view, *85*
V$OBSOLETE_BACKUP_FILES view, *86*
V$OBSOLETE_PARAMETER view, *88*
V$OFFLINE_RANGE view, *88*
V$OFS_STATS view, *89*
V$OFSMOUNT view, *91*
V$ONLINE_REDEF view, *91*
V$OPEN_CURSOR view, *92*
V$OPTIMIZER_PROCESSING_RATE view, *93*
V$OPTION view, *94*
V$OSSTAT view, *95*

Copyright © 2002, 2026, Oracle and/or its affiliates.

V$PARALLEL_DEGREE_LIMIT_MTH view, *96*
V$PARAMETER view, *97*
V$PARAMETER_VALID_VALUES view, *99*
V$PARAMETER2 view, *100*
V$PASSWORDFILE_INFO view, *102*
V$PATCHES view, *102*
V$PDB_INCARNATION view, *103*
V$PDBS view, *103*
V$PERSISTENT_PUBLISHERS view, *105*
V$PERSISTENT_QMN_CACHE view, *106*
V$PERSISTENT_QUEUES view, *107*
V$PERSISTENT_SUBSCRIBERS view, *109*
V$PGA_TARGET_ADVICE view, *110*
V$PGA_TARGET_ADVICE_HISTOGRAM view, *112*
V$PGASTAT view, *113*
V$PLSQL_DEBUGGABLE_SESSIONS view, *115*
V$PQ_SESSTAT view, *116*
V$PQ_SLAVE view, *117*
V$PQ_SYSSTAT view, *117*
V$PQ_TQSTAT view, *118*
V$PROCESS view, *119*
V$PROCESS_MEMORY view, *120*
V$PROCESS_MEMORY_DETAIL view, *121*
V$PROCESS_POOL view, *122*
V$PROPAGATION_RECEIVER view, *122*
V$PROPAGATION_SENDER view, *123*
V$PROXY_ARCHIVEDLOG view, *125*
V$PROXY_ARCHIVELOG_DETAILS view, *127*
V$PROXY_ARCHIVELOG_SUMMARY view, *128*
V$PROXY_COPY_DETAILS view, *128*
V$PROXY_COPY_SUMMARY view, *129*
V$PROXY_DATAFILE view, *130*
V$PROXY_PDB_TARGETS view, *132*
V$PWFILE_USERS view, *133*
V$PX_INSTANCE_GROUP view, *134*
V$PX_PROCESS view, *134*
V$PX_PROCESS_SYSSTAT view, *135*
V$PX_SESSION view, *137*
V$PX_SESSTAT view, *137*
V$QMON_COORDINATOR_STATS view, *138*
V$QMON_SERVER_STATS view, *139*
V$QMON_TASK_STATS view, *140*
V$QMON_TASKS view, *140*
V$QUARANTINE view, *141*
V$QUARANTINE_SUMMARY view, *141*
V$QUEUE view, *142*
V$QUEUEING_MTH view, *142*
V$RECOVER_FILE view, *143*
V$RECOVERY_AREA_USAGE view, *143*
V$RECOVERY_FILE_DEST view, *144*
V$RECOVERY_FILE_STATUS view, *144*
V$RECOVERY_LOG view, *145*
V$RECOVERY_PROGRESS view, *146*
V$RECOVERY_SLAVE view, *147*
V$RECOVERY_STATUS view, *149*
V$REDO_DEST_RESP_HISTOGRAM view, *150*
V$REQDIST view, *150*
V$RESERVED_WORDS view, *151*
V$RESOURCE view, *151*
V$RESOURCE_LIMIT view, *152*
V$RESTORE_POINT view, *153*
V$RESULT_CACHE_DEPENDENCY view, *154*
V$RESULT_CACHE_MEMORY view, *154*
V$RESULT_CACHE_OBJECTS view, *155*
V$RESULT_CACHE_STATISTICS view, *156*
V$RMAN_BACKUP_JOB_DETAILS view, *157*
V$RMAN_BACKUP_SUBJOB_DETAILS view, *158*
V$RMAN_BACKUP_TYPE view, *160*
V$RMAN_COMPRESSION_ALGORITHM view, *160*
V$RMAN_CONFIGURATION view, *161*
V$RMAN_ENCRYPTION_ALGORITHMS view, *161*
V$RMAN_OUTPUT view, *162*
V$RMAN_STATUS view, *163*
V$RO_USER_ACCOUNT view, *164*
V$ROLLNAME view, *165*
V$ROLLSTAT view, *165*
V$ROWCACHE view, *166*
V$ROWCACHE_PARENT view, *167*
V$ROWCACHE_SUBORDINATE view, *168*
V$RSRC_CONS_GROUP_HISTORY view, *168*
V$RSRC_CONSUMER_GROUP view, *171*
V$RSRC_CONSUMER_GROUP_CPU_MTH view, *174*
V$RSRC_PDB view, *175*
V$RSRC_PDB_HISTORY view, *177*
V$RSRC_PLAN view, *178*
V$RSRC_PLAN_CPU_MTH view, *180*
V$RSRC_PLAN_HISTORY view, *180*
V$RSRC_SESSION_INFO view, *182*
V$RSRCMGRMETRIC view, *186*
V$RSRCMGRMETRIC_HISTORY view, *188*
V$RSRCPDBMETRIC view, *188*
V$RSRCPDBMETRIC_HISTORY view, *190*
V$RULE view, *190*
V$RULE_SET view, *191*
V$RULE_SET_AGGREGATE_STATS view, *192*
V$SCHEDULER_RUNNING_JOBS view, *1*
V$SECUREFILE_TIMER view, *1*
V$SEGMENT_STATISTICS view, *2*
V$SEGSTAT view, *3*
V$SEGSTAT_NAME view, *3*
V$SERV_MOD_ACT_STATS view, *4*
V$SERVICE_EVENT view, *4*
V$SERVICE_REGION_METRIC view, *5*
V$SERVICE_STATS view, *6*
V$SERVICE_WAIT_CLASS view, *6*
V$SERVICEMETRIC view, *7*
V$SERVICEMETRIC_HISTORY view, *8*

Copyright © 2002, 2026, Oracle and/or its affiliates.

V$SERVICES view, *8*
V$SES_OPTIMIZER_ENV view, *10*
V$SESS_IO view, *11*
V$SESS_TIME_MODEL view, *12*
V$SESSION view, *14*
V$SESSION_BLOCKERS view, *20*
V$SESSION_CONNECT_INFO view, *21*
V$SESSION_CURSOR_CACHE view, *22*
V$SESSION_EVENT view, *23*, *C-1*
V$SESSION_FIX_CONTROL view, *24*
V$SESSION_LONGOPS view, *24*
V$SESSION_OBJECT_CACHE view, *26*
V$SESSION_WAIT view, *27*, *C-1*
V$SESSION_WAIT_CLASS view, *28*
V$SESSION_WAIT_HISTORY view, *29*
V$SESSIONS_COUNT view, *29*
V$SESSMETRIC view, *30*
V$SESSTAT view, *30*, *E-1*
V$SGA view, *31*
V$SGA_CURRENT_RESIZE_OPS view, *31*
V$SGA_DYNAMIC_COMPONENTS view, *32*
V$SGA_DYNAMIC_FREE_MEMORY view, *33*
V$SGA_RESIZE_OPS view, *33*
V$SGA_TARGET_ADVICE view, *34*
V$SGAINFO view, *35*
V$SGASTAT view, *35*
V$SHARED_POOL_ADVICE view, *36*
V$SHARED_POOL_RESERVED view, *36*
V$SHARED_SERVER view, *37*
V$SHARED_SERVER_MONITOR view, *38*
V$SHARED_SERVER_STAT view, *39*
V$SORT_SEGMENT view, *40*
V$SPPARAMETER view, *41*
V$SQL view, *41*
V$SQL_BIND_CAPTURE view, *47*
V$SQL_BIND_DATA view, *48*
V$SQL_BIND_METADATA view, *49*
V$SQL_CS_HISTOGRAM view, *50*
V$SQL_CS_SELECTIVITY view, *51*
V$SQL_CS_STATISTICS view, *51*
V$SQL_CURSOR view, *52*
V$SQL_JOIN_FILTER view, *53*
V$SQL_MONITOR view, *53*
V$SQL_MONITOR_SESSTAT view, *58*
V$SQL_MONITOR_STATNAME view, *59*
V$SQL_OPTIMIZER_ENV view, *60*
V$SQL_PLAN view, *60*
V$SQL_PLAN_MONITOR view, *62*
V$SQL_PLAN_STATISTICS view, *66*
V$SQL_PLAN_STATISTICS_ALL view, *68*
V$SQL_REDIRECTION view, *72*
V$SQL_SHARD view, *72*
V$SQL_SHARED_CURSOR view, *73*
V$SQL_SHARED_MEMORY view, *76*
V$SQL_TESTCASES view, *77*
V$SQL_WORKAREA view, *78*
V$SQL_WORKAREA_ACTIVE view, *79*
V$SQL_WORKAREA_HISTOGRAM view, *81*
V$SQLAREA view, *82*
V$SQLAREA_PLAN_HASH view, *86*
V$SQLCOMMAND view, *89*
V$SQLFN_ARG_METADATA view, *90*
V$SQLFN_METADATA view, *90*
V$SQLSTATS view, *91*
V$SQLSTATS_PLAN_HASH view, *95*
V$SQLTEXT view, *95*
V$SQLTEXT_WITH_NEWLINES view, *96*
V$STANDBY_EVENT_HISTOGRAM view, *96*
V$STANDBY_LOG view, *97*
V$STATISTICS_LEVEL view, *98*
V$STATNAME view, *99*, *E-1*
V$STATS_ADVISOR_RULES view, *100*
V$STREAMS_APPLY_COORDINATOR view, *100*
V$STREAMS_APPLY_READER view, *102*
V$STREAMS_APPLY_SERVER view, *105*
V$STREAMS_POOL_ADVICE view, *108*
V$STREAMS_POOL_STATISTICS view, *108*
V$SUBCACHE view, *109*
V$SUBSCR_REGISTRATION_STATS view, *110*
V$SYS_OPTIMIZER_ENV view, *111*
V$SYS_TIME_MODEL view, *112*
V$SYSAUX_OCCUPANTS view, *112*
V$SYSMETRIC view, *113*
V$SYSMETRIC_HISTORY view, *113*
V$SYSMETRIC_SUMMARY view, *114*
V$SYSSTAT view, *115*, *E-1*
V$SYSTEM_CURSOR_CACHE view, *116*
V$SYSTEM_EVENT view, *116*, *C-1*
V$SYSTEM_FIX_CONTROL view, *117*
V$SYSTEM_PARAMETER view, *118*
V$SYSTEM_PARAMETER2 view, *120*
V$SYSTEM_WAIT_CLASS view, *121*
V$TABLESPACE view, *122*
V$TEMP_CACHE_TRANSFER view, *122*
V$TEMP_EXTENT_MAP view, *123*
V$TEMP_EXTENT_POOL view, *124*
V$TEMP_SPACE_HEADER view, *124*
V$TEMPFILE view, *125*
V$TEMPORARY_LOBS view, *125*
V$TEMPSEG_USAGE view, *126*
V$TEMPSTAT view, *127*
V$TEMPUNDOSTAT view, *128*
V$THREAD view, *129*
V$THRESHOLD_TYPES view, *129*
V$TIMER view, *130*
V$TIMEZONE_FILE view, *130*
V$TIMEZONE_NAMES view, *131*
V$TOPLEVELCALL view, *131*
V$TRANSACTION view, *131*
V$TRANSACTION_ENQUEUE view, *133*
V$TRANSPORTABLE_PLATFORM view, *134*
V$TSDP_SUPPORTED_FEATURE view, *134*

Copyright © 2002, 2026, Oracle and/or its affiliates.

V$TYPE_SIZE view, *135*
V$UNDOSTAT view, *135*
V$UNUSABLE_BACKUPFILE_DETAILS view, *137*
V$VERSION view, *138*
V$VPD_POLICY view, *138*
V$WAIT_CHAINS view, *139*
V$WAITCLASSMETRIC view, *141*
V$WAITCLASSMETRIC_HISTORY view, *141*
V$WAITSTAT view, *142*
V$WALLET view, *142*
V$WORKLOAD_REPLAY_THREAD view, *143*
V$XML_AUDIT_TRAIL view, *145*
V$XSTREAM_APPLY_COORDINATOR view, *147*
V$XSTREAM_APPLY_READER view, *149*
V$XSTREAM_APPLY_RECEIVER view, *151*
V$XSTREAM_APPLY_SERVER view, *152*
V$XSTREAM_CAPTURE view, *155*
V$XSTREAM_MESSAGE_TRACKING view, *158*
V$XSTREAM_OUTBOUND_SERVER view, *159*
V$XSTREAM_TABLE_STATS view, *162*
V$XSTREAM_TRANSACTION view, *163*
V$ZONEMAP_USAGE_STATS view, *165*
views
  data dictionary views, *1*

W

wait events
  alter system set dispatcher, *C-5*
  ARCH Remote Write, *C-5*
  ASYNC Remote Write, *C-5*
  BFILE check if exists, *C-5*
  BFILE check if open, *C-6*
  BFILE closure, *C-6*
  BFILE get length, *C-6*
  BFILE get name object, *C-6*
  BFILE get path object, *C-7*
  BFILE internal seek, *C-7*
  BFILE open, *C-7*
  BFILE read, *C-7*
  broadcast mesg queue transition, *C-7*
  broadcast mesg recovery queue transition, *C-8*
  buffer busy waits, *C-8*
  buffer deadlock, *C-9*
  buffer latch, *C-9*
  buffer read retry, *C-10*
  checkpoint completed, *C-10*
  cleanup of aborted process, *C-10*
  controlfile parallel write, *C-10*
  controlfile sequential read, *C-11*
  controlfile single write, *C-11*
  cursor: mutex S, *C-11*
  cursor: mutex X, *C-12*
  cursor: pin S, *C-12*
wait events *(continued)*
  cursor: pin S wait on X, *C-12*
  cursor: pin X, *C-12*
  db file async I/O submit, *C-13*
  db file parallel read, *C-13*
  db file parallel write, *C-13*
  db file scattered read, *C-14*
  db file sequential read, *C-14*
  db file single write, *C-15*
  DFS db file lock, *C-15*
  DFS lock handle, *C-16*
  direct path read, *C-16*
  direct path sync, *C-16*
  direct path write, *C-17*
  Disk file operations I/O, *C-17*
  dispatcher shutdown, *C-17*
  dispatcher timer, *C-18*
  dupl. cluster key, *C-18*
  enq: OW - initialization, *C-18*
  enq: OW - termination, *C-19*
  enq: SV - contention, *C-19*
  enq: TX - index contention, *C-19*
  enq: TX - row lock contention, *C-19*
  enqueue, *C-20*
  flashback buf free by RVWR, *C-20*
  flashback log file sync, *C-20*
  free buffer waits, *C-21*
  free global transaction table entry, *C-21*
  free process state object, *C-21*
  gc buffer busy acquire, *C-22*
  gc buffer busy release, *C-22*
  gc cr block 2-way, *C-22*
  gc cr block 3-way, *C-22*
  gc cr block busy, *C-23*
  gc cr block congested, *C-23*
  gc cr block lost, *C-24*
  gc cr grant read-mostly invalidation, *C-24*
  gc cr multi block grant, *C-24*
  gc cr multi block mixed, *C-24*
  gc cr request, *C-25*
  gc current block 2-way, *C-25*
  gc current block 3-way, *C-25*
  gc current block busy, *C-26*
  gc current block congested, *C-26*
  gc current block lost, *C-26*
  gc current request, *C-27*
  gc index operation, *C-27*
  gc recovery quiesce, *C-27*
  GCS lock open S, *C-28*
  GCS lock open X, *C-28*
  global cache busy, *C-28*
  inactive session, *C-28*
  inactive transaction branch, *C-29*
  index block split, *C-29*
  instance state change, *C-29*
  io done, *C-30*

Copyright © 2002, 2026, Oracle and/or its affiliates.

wait events *(continued)*
ksxr wait for mount shared, *C-30*
ktm: instance recovery, *C-30*
latch activity, *C-30*
latch free, *C-31*
latch redo copy, *C-31*
latch: row cache objects, *C-31*
library cache load lock, *C-31*
library cache lock, *C-32*
library cache pin, *C-32*
library cache: mutex X, *C-33*
LMON global data update, *C-33*
Log archive I/O, *C-33*
log buffer space, *C-33*
log file parallel write, *C-34*
log file sequential read, *C-34*
log file single write, *C-34*
log file switch (archiving needed), *C-35*
log file switch (checkpoint incomplete), *C-35*
log file switch (clearing log file), *C-35*
log file switch (private strand flush incomplete), *C-35*
log file switch completion, *C-35*
log file sync, *C-36*
log switch/archive, *C-36*
optimizer stats update retry, *C-36*
parallel recovery change buffer free, *C-36*
parallel recovery control message reply, *C-37*
parallel recovery coord send blocked, *C-37*
parallel recovery coord wait for reply, *C-37*
parallel recovery coordinator waits for slave cleanup, *C-37*
parallel recovery read buffer free, *C-37*
parallel recovery slave next change, *C-37*
pending global transaction(s), *C-38*
pipe get, *C-38*
pipe put, *C-38*
PL/SQL lock timer, *C-38*
pmon timer, *C-39*
prior spawner clean up, *C-39*
process startup, *C-39*
PX dequeue wait, *C-40*
PX qref latch, *C-40*
PX server shutdown, *C-40*
PX signal server, *C-40*
rdbms ipc message, *C-41*
rdbms ipc message block, *C-41*
rdbms ipc reply, *C-41*
read by other session, *C-41*
recovery active instance mapping setup, *C-42*
recovery apply pending, *C-42*
recovery cancel, *C-42*
recovery checkpoint, *C-42*
recovery file header update for checkpoint, *C-42*
wait events *(continued)*
recovery file header update for fuzziness, *C-42*
recovery image pending, *C-42*
recovery marker apply, *C-42*
recovery metadata latch, *C-42*
recovery move influx buffers, *C-43*
recovery read, *C-43*
recovery receive buffer free, *C-43*
recovery remote file verification, *C-43*
recovery send buffer free, *C-43*
recovery shutdown, *C-43*
Redo Transport Attach, *C-43*
Redo Transport Close, *C-44*
Redo Transport Detach, *C-44*
Redo Transport Open, *C-44*
Redo Transport Ping, *C-44*
Redo Transport Slave Shutdown, *C-45*
Redo Transport Slave Startup, *C-45*
Redo Writer Remote Sync Complete, *C-45*
Redo Writer Remote Sync Notify, *C-45*
Remote SYNC Ping, *C-45*
resmgr: I/O rate limit, *C-46*
resmgr:become active, *C-45*
resmgr:cpu quantum, *C-45*
resmgr:pq queued, *C-46*
rolling migration: cluster quiesce, *C-47*
row cache lock, *C-47*
RVWR wait for flashback copy, *C-47*
sbtbufinfo, *C-47*
sbtgetbuf, *C-48*
sbtmapbuf, *C-48*
sbtrelbuf, *C-48*
scginq AST call, *C-48*
SGA: allocation forcing component growth, *C-48*
SGA: MMAN sleep for component shrink, *C-48*
SGA: sga_target resize, *C-49*
Shared IO Pool Memory, *C-49*
shared server idle event, *C-49*
single-task message, *C-49*
smon timer, *C-49*
SQL*Net break/reset to client, *C-50*
SQL*Net break/reset to dblink, *C-50*
SQL*Net message from client, *C-50*
SQL*Net message from dblink, *C-51*
SQL*Net message to client, *C-51*
SQL*Net message to dblink, *C-51*
SQL*Net more data from client, *C-52*
SQL*Net more data from dblink, *C-52*
SQL*Net more data to client, *C-52*
SQL*Net more data to dblink, *C-53*
Streams AQ: waiting for messages in the queue, *C-53*
switch logfile command, *C-53*

Copyright © 2002, 2026, Oracle and/or its affiliates.

wait events *(continued)*
SYNC Remote Write, *C-53*
timer in sksawat, *C-54*
transaction, *C-54*
unbound tx, *C-54*
undo segment extension, *C-55*
undo segment recovery, *C-55*
undo segment tx slot, *C-55*
undo_retention publish retry, *C-54*
utl_file I/O, *C-55*
virtual circuit wait, *C-56*
WCR: replay client notify, *C-56*
WCR: replay clock, *C-56*
WCR: replay lock order, *C-56*
wait events *(continued)*
WCR: replay paused, *C-57*
WCR: Sync context busy, *C-57*
write complete waits, *C-57*
writes stopped by instance recovery or database suspension, *C-57*
WALLET_ROOT initialization parameter, *346*
WORKAREA_SIZE_POLICY initialization parameter, *349*

X

XML_DB_EVENTS initialization parameter, *350*

Copyright © 2002, 2026, Oracle and/or its affiliates.